

PART TWO -CARDINAL MANIFESTATIONS AND PRESENTATION OF DISEASE

SECTION 1 -PAIN

12. PAIN: PATHOPHYSIOLOGY AND MANAGEMENT - Howard L. Fields, Joseph B. Martin

The task of medicine is to preserve and restore health and to relieve suffering. Understanding pain is essential to both these goals. Because pain is universally understood as a signal of disease, it is the most common symptom that brings a patient to a physician's attention. The function of the pain sensory system is to detect, localize, and identify tissue-damaging processes. Since different diseases produce characteristic patterns of tissue damage, the quality, time course, and location of a patient's pain complaint and the location of tenderness provide important diagnostic clues and are used to evaluate the response to treatment.

THE PAIN SENSORY SYSTEM

Pain is an unpleasant sensation localized to a part of the body. It is often described in terms of a penetrating or tissue-destructive process (e.g., stabbing, burning, twisting, tearing, squeezing) and/or of a bodily or emotional reaction (e.g., terrifying, nauseating, sickening). Furthermore, any pain of moderate or higher intensity is accompanied by anxiety and the urge to escape or terminate the feeling. These properties illustrate the duality of pain: it is both sensation and emotion. When acute, pain is characteristically associated with behavioral arousal and a stress response consisting of increased blood pressure, heart rate, pupil diameter, and plasma cortisol levels. In addition, local muscle contraction (e.g., limb flexion, abdominal wall rigidity) is often present.

THE PRIMARY AFFERENT NOCICEPTOR

A peripheral nerve consists of the axons of three different types of neurons: primary sensory afferents, motor neurons, and sympathetic postganglionic neurons ([Fig. 12-1](#)). The cell bodies of primary afferents are located in the dorsal root ganglia in the vertebral foramina. The primary afferent axon bifurcates to send one process into the spinal cord and the other to innervate tissues. Primary afferents are classified by their diameter, degree of myelination, and conduction velocity. The largest-diameter fibers, A-beta (Ab), respond maximally to light touch and/or moving stimuli; they are present primarily in nerves that innervate the skin. In normal individuals, the activity of these fibers does not produce pain. There are two other classes of primary afferents: the small-diameter myelinated A-delta (Ad) and the unmyelinated (C fiber) axons ([Fig. 12-1](#)). These fibers are present in nerves to the skin and to deep somatic and visceral structures. Some tissues, such as the cornea, are innervated only by Ad and C afferents. Most Ad and C afferents respond maximally only to intense (painful) stimuli and produce the subjective experience of pain when they are electrically stimulated; this defines them as *primary afferent nociceptors (pain receptors)*. The ability to detect painful stimuli is completely abolished when Ad and C axons are blocked.

Individual primary afferent nociceptors can respond to several different types of noxious stimuli. For example, most nociceptors respond to heating, intense mechanical stimuli

such as a pinch, and application of irritating chemicals.

Sensitization When intense, repeated, or prolonged stimuli are applied in the presence of damaged tissue or inflammation, the threshold for activating primary afferent nociceptors is lowered and the frequency of firing is higher for all stimulus intensities. Inflammatory mediators such as bradykinin, some prostaglandins, and leukotrienes contribute to this process, which is called *sensitization*. In sensitized tissues normally innocuous stimuli can produce pain. Sensitization is a clinically important process that contributes to tenderness, soreness, and hyperalgesia. A striking example of sensitization is sunburned skin, in which severe pain can be produced by a gentle slap on the back or a warm shower.

Under normal conditions, viscera are relatively insensitive to noxious mechanical and thermal stimuli. Hollow viscera do generate significant discomfort when distended. Furthermore, when affected by a disease process with an inflammatory component, deep structures such as joints or hollow viscera characteristically become exquisitely sensitive to mechanical stimulation.

A large proportion of Ad and C afferents innervating viscera are completely insensitive in normal noninjured, noninflamed tissue. That is, they cannot be activated by known mechanical or thermal stimuli and are not spontaneously active. However, in the presence of inflammatory mediators, these afferents become sensitive to mechanical stimuli. Such afferents have been termed *silent nociceptors*, and their characteristic properties may explain how under pathologic conditions the relatively insensitive deep structures can become the source of severe and debilitating pain and tenderness.

Nociceptor-Induced Inflammation One important concept to emerge in recent years is that afferent nociceptors also have a neuroeffector function. Most nociceptors contain polypeptide mediators that are released from their peripheral terminals when they are activated ([Fig. 12-2](#)). An example is substance P, an 11-amino-acid peptide. Substance P is released from primary afferent nociceptors and has multiple biologic activities. It is a potent vasodilator, degranulates mast cells, is a chemoattractant for leukocytes, and increases the production and release of inflammatory mediators. Interestingly, depletion of substance P from joints reduces the severity of experimental arthritis. Primary afferent nociceptors are not simply passive messengers of threats to tissue injury but also play an active role in tissue protection through these neuroeffector functions.

CENTRAL PATHWAYS FOR PAIN

The Spinal Cord and Referred Pain The axons of primary afferent nociceptors enter the spinal cord via the dorsal root. They terminate in the dorsal horn of the spinal gray matter ([Fig. 12-3](#)). The terminals of primary afferent axons contact spinal neurons that transmit the pain signal to brain sites involved in pain perception. The axon of each primary afferent contacts many spinal neurons, and each spinal neuron receives convergent inputs from many primary afferents.

From a clinical standpoint, the convergence of many sensory inputs to a single spinal pain-transmission neuron is of great importance because it underlies the phenomenon of referred pain. All spinal neurons that receive input from the viscera and deep

musculoskeletal structures also receive input from the skin. The convergence patterns are determined by the spinal segment of the dorsal root ganglion that supplies the afferent innervation of a structure. For example, the afferents that supply the central diaphragm are derived from the third and fourth cervical dorsal root ganglia. Primary afferents with cell bodies in these same ganglia supply the skin of the shoulder and lower neck. Thus sensory inputs from both the shoulder skin and the central diaphragm converge on pain-transmission neurons in the third and fourth cervical spinal segments. *Because of this convergence and the fact that the spinal neurons are most often activated by inputs from the skin, activity evoked in spinal neurons by input from deep structures is mislocalized by the patient to a place that is roughly coextensive with the region of skin innervated by the same spinal segment.* Thus inflammation near the central diaphragm is usually reported as discomfort near the shoulder. This spatial displacement of pain sensation from the site of the injury that produces it is known as *referred pain*.

Ascending Pathways for Pain A majority of spinal neurons contacted by primary afferent nociceptors send their axons to the contralateral thalamus. These axons form the contralateral spinothalamic tract which lies in the anterolateral white matter of the spinal cord, the lateral edge of the medulla, and the lateral pons and midbrain. The spinothalamic pathway is crucial for pain sensation in humans. Interruption of this pathway produces permanent deficits in pain and temperature discrimination.

Spinothalamic tract axons connect to thalamic neurons that project to somatosensory cortex ([Fig. 12-4](#)). This pathway from spinal cord to thalamus to somatosensory cortex appears to be particularly important for the sensory aspects of pain, i.e., its location, intensity, and quality. Spinothalamic tract axons also connect to thalamic and cortical regions linked to emotional responses, such as the cingulate gyrus and frontal lobe. This pathway is thought to subserve the affective or unpleasant emotional dimension of pain.

PAIN MODULATION

The pain produced by similar injuries is remarkably variable in different situations and in different people. For example, athletes have been known to sustain serious fractures with only minor pain, and Beecher's classic World War II survey revealed that many men were unbothered by battle injuries that would have produced agonizing pain in civilian patients. Furthermore, even the suggestion of relief can have a significant analgesic effect (placebo). On the other hand, many patients find even minor injuries (such as venipuncture) unbearable, and the expectation of pain has been demonstrated to induce pain *without a noxious stimulus*.

The powerful effect of expectation and other psychological variables on the perceived intensity of pain implies the existence of brain circuits that can modulate the activity of the pain-transmission pathways. Although there are probably several circuits that can modulate pain, only one has been studied extensively. This circuit has links in the hypothalamus, midbrain, and medulla, and it selectively controls spinal pain-transmission neurons through a descending pathway ([Fig. 12-4](#)).

There is good evidence that this pain-modulating circuit contributes to the pain-relieving

effect of opioid analgesic medications. Each of the component structures of the pathway contains opioid receptors and is sensitive to the direct application of opioid drugs. Furthermore, lesions of the system reduce the analgesic effect of systemically administered opioids such as morphine. Along with the opioid receptor, the component nuclei of this pain-modulating circuit contain endogenous opioid peptides such as the enkephalins and δ -endorphin.

The most reliable way to activate this endogenous opioid-mediated modulating system is by prolonged pain and/or fear. There is evidence that pain-relieving endogenous opioids are released following operative procedures and in patients given a placebo for pain relief.

Pain modulation is bidirectional. Pain-modulating circuits not only produce analgesia but are also capable of increasing pain. Both pain-inhibiting and pain-facilitating neurons in the medulla project to and control spinal pain-transmission neurons. Since pain-transmission neurons can be activated by modulatory neurons, it is theoretically possible to generate a pain signal with no peripheral noxious stimulus. Some such mechanism could account for the finding that pain can be induced by suggestion alone and may provide a framework for understanding how psychological factors can contribute to chronic pain.

NEUROPATHIC PAIN

The normal nervous system transmits coded signals that result in pain. Thus lesions of the peripheral or central nervous system may result in a loss or impairment of pain sensation. Paradoxically, damage or dysfunction of the nervous system can produce pain. For example, damage to peripheral nerves, as occurs in diabetic neuropathy, or to primary afferents, as in herpes zoster, can result in pain that is referred to the body region innervated by the damaged nerves. Though rare, pain may also be produced by damage to the central nervous system, particularly the spinothalamic pathway or thalamus. Such neuropathic pains are often severe and are notoriously intractable to standard treatments for pain.

Neuropathic pains typically have an unusual burning, tingling, or electric shock-like quality and may be triggered by very light touch. These features are rare in other types of pain. On examination, a sensory deficit is characteristically present in the area of the patient's pain.

A variety of mechanisms contribute to neuropathic pain. As with sensitized primary afferent nociceptors, damaged primary afferents, including nociceptors, become highly sensitive to mechanical stimulation and begin to generate impulses in the absence of stimulation. There is evidence that this increased sensitivity and spontaneous activity is due to an increased concentration of sodium channels. Damaged primary afferents may also develop sensitivity to norepinephrine. Interestingly, spinal pain-transmission neurons cut off from their normal input may also become spontaneously active. Thus both central and peripheral nervous system changes may contribute to neuropathic pain.

Sympathetically Maintained Pain A certain percentage of patients with peripheral

nerve injury develop a severe burning pain (causalgia) in the region innervated by the nerve. The pain typically begins after a delay of hours to days or even weeks. The pain is accompanied by swelling of the extremity, periarticular osteoporosis, and arthritic changes in the distal joints. A similar syndrome called *reflex sympathetic dystrophy* can be produced without obvious nerve damage by a variety of injuries, including fractures of bone, soft tissue trauma, myocardial infarction, and stroke ([Chap. 366](#)). Although the pathophysiology of this condition is poorly understood, the pain can be relieved within minutes by blocking the sympathetic nervous system. This implies that sympathetic activity activates nociceptors even if they are not obviously damaged. These results also suggest that the sympathetic nervous system can, under some circumstances, play an active role in inflammation.

TREATMENT

The ideal treatment for any pain is to remove the cause. Sometimes this is possible, but more often after diagnosis and initiation of appropriate treatments for the cause, there is a lag period before the pain subsides. Furthermore, some conditions are so painful that rapid and effective analgesia is essential (e.g., the postoperative state, burns, trauma, cancer, sickle cell crisis). Analgesic medications are a first line of treatment in these cases, and their use should be familiar to all practitioners.

Aspirin, Acetaminophen, and Nonsteroidal Anti-Inflammatory Agents (NSAIDs)

These drugs are considered together because they are used for similar problems and may have a similar mechanism of action ([Table 12-1](#)). All these compounds inhibit cyclooxygenase (COX), and, except for acetaminophen, all have anti-inflammatory actions, especially at higher dosages. They are particularly effective for mild to moderate headache and for pain of musculoskeletal origin.

Since they are effective for these common types of pains and are available without prescription, [COX](#) inhibitors are by far the most commonly used analgesics. They are absorbed well from the gastrointestinal tract and, with occasional use, side effects are minimal. With chronic use, gastric irritation is a common side effect of aspirin and [NSAIDs](#) and is the problem that most frequently limits the dose that can be given. Gastric irritation is most severe with aspirin, which may cause erosion of the gastric mucosa, and because aspirin irreversibly acetylates platelets and interferes with coagulation of the blood, gastrointestinal bleeding is a risk. The NSAIDs are less problematic in this regard. Although toxic to the liver when taken in a high dose, acetaminophen rarely produces gastric irritation and does not interfere with platelet function. [Table 12-1](#) lists the dosages and durations of action of the commonly used drugs of this class.

The introduction of a parenteral form of [NSAID](#), ketorolac, extends the usefulness of this class of compounds in the management of acute severe pain. Ketorolac is sufficiently potent and rapid in onset to supplant opioids for many patients with acute severe headache and musculoskeletal pain.

There are two major classes of [COX](#): COX 1 is constitutively expressed, and COX 2 is induced in the inflammatory state. COX 2-selective drugs have recently been introduced for the treatment of arthritis and are associated with a significant reduction of gastric

irritation. Whether COX 2-selective drugs have analgesic actions equivalent to other [NSAIDs](#) remains to be demonstrated.

Opioid Analgesics Opioids are the most potent pain-relieving drugs currently available. Furthermore, of all analgesics, they have the broadest range of efficacy, providing the most reliable method for rapidly relieving pain. Although side effects are common, they are usually not serious except for respiratory depression and can be reversed rapidly with the narcotic antagonist naloxone. The physician should not hesitate to use opioid analgesics in patients with acute severe pain. [Table 12-1](#) lists the most commonly used opioid analgesics.

Opioids produce analgesia by actions in the central nervous system. They activate pain-inhibitory neurons and directly inhibit pain-transmission neurons. Most of the commercially available opioid analgesics act at the same opioid receptor (mu receptor), differing mainly in potency, speed of onset, duration of action, and optimal route of administration. Although the dose-related side effects (sedation, respiratory depression, pruritus, constipation) are similar among the different opioids, some side effects are due to accumulation of nonopioid metabolites that are unique to individual drugs. One striking example of this is normeperidine, a metabolite of meperidine. Normeperidine produces hyperexcitability and seizures that are not reversible with naloxone. Normeperidine accumulation is much greater in patients with renal failure.

The most rapid relief with opioids is obtained by intravenous administration; relief with oral administration is significantly slower. Common acute side effects include nausea, vomiting, and sedation. The most serious side effect is respiratory depression. Patients with any form of respiratory compromise must be kept under close observation following opioid administration; an oxygen saturation monitor may be useful. The opioid antagonist, naloxone, should be readily available. These effects are dose-related, and there is great variability among patients in the doses that relieve pain and produce side effects. Because of this, initiation of therapy requires titration to optimal dose and interval. The most important principle is to provide adequate pain relief. This requires asking the patient whether the drug has relieved the pain and, if so, when the relief wears off. *The most common error made by physicians in managing severe pain with opioids is to prescribe an inadequate dose. Since many patients are reluctant to complain, this practice leads to needless suffering.* In the absence of sedation at the expected time of peak effect, a physician should not hesitate to repeat the initial dose to achieve satisfactory pain relief.

An innovative approach to the problem of achieving adequate pain relief is the use of patient-controlled analgesia (PCA). PCA requires a device that delivers a baseline continuous dose of an opioid drug, and preprogrammed additional doses whenever the patient pushes a button. The device can be programmed to limit the total hourly dose so that overdosing is impossible. The patient can then titrate the dose to the optimal level. This approach is used most extensively for the management of postoperative pain, but there is no reason why it should not be used for any hospitalized patient with persistent severe pain. PCA is also used for short-term home care of patients with intractable pain, such as is caused by metastatic cancer.

Many physicians, nurses, and patients have a certain trepidation about using opioids

that is based on an exaggerated fear of patients becoming addicted. In fact, there is a vanishingly small chance of patients becoming addicted to narcotics as a result of their appropriate medical use.

The availability of new routes of administration has extended the usefulness of opioid analgesics. Most important is the availability of spinal administration. Opioids can be infused through a spinal catheter placed either intrathecally or epidurally. By applying opioids directly to the spinal cord, regional analgesia can be obtained using a relatively low total dose. In this way, such side effects as sedation, nausea, and respiratory depression can be minimized. This approach has been used extensively in obstetric procedures and for lower-body postoperative pain. Opioids can also be given intranasally (butorphanol), rectally, and transdermally (fentanyl), thus avoiding the discomfort of frequent injections in patients who cannot be given oral medication.

Opioid and Cyclooxygenase Inhibitor Combinations When used in combination, opioids and COX inhibitors have additive effects. Because a lower dose of each can be used to achieve the same degree of pain relief and their side effects are nonadditive, such combinations can be used to lower the severity of dose-related side effects. Fixed-ratio combinations of an opioid with acetaminophen carry a special risk. Dose escalation as a result of increased severity of pain or decreased opioid effect as a result of tolerance may lead to levels of acetaminophen that are toxic to the liver.

CHRONIC PAIN

PATIENT EVALUATION

Managing patients with chronic pain is intellectually and emotionally challenging. The patient's problem is often difficult to diagnose: such patients are demanding of the physician's time and often appear emotionally distraught. The traditional medical approach of seeking an obscure organic pathology is usually unhelpful. On the other hand, psychological evaluation and behaviorally based treatment paradigms are frequently helpful, particularly in the setting of a multidisciplinary pain-management center.

There are several factors that can cause, perpetuate, or exacerbate chronic pain. First, of course, the patient may simply have a disease that is characteristically painful for which there is presently no cure. Arthritis, cancer, migraine headaches, fibromyalgia, and diabetic neuropathy are examples of this. Second, there may be secondary perpetuating factors that are initiated by disease and persist after that disease has resolved. Examples include damaged sensory nerves, sympathetic efferent activity, and painful reflex muscle contraction. Finally, a variety of psychological conditions can exacerbate or even cause pain.

There are certain areas to which special attention should be paid in the medical history. Because depression is the most common emotional disturbance in patients with chronic pain, patients should be questioned about their mood, appetite, sleep patterns, and daily activity. A simple standardized questionnaire, such as the Beck Depression Inventory, can be a useful screening device. It is important to remember that major depression is a common, treatable, and potentially fatal illness.

Other clues that a significant emotional disturbance is contributing to a patient's chronic pain complaint include: pain that occurs in multiple unrelated sites; a pattern of recurrent, but separate, pain problems beginning in childhood or adolescence; pain beginning at a time of emotional trauma, such as the loss of a parent or spouse; a history of physical or sexual abuse; and past or present substance abuse.

On examination, special attention should be paid to whether the patient guards the painful area and whether certain movements or postures are avoided because of pain. Discovering a mechanical component to the pain can be useful both diagnostically and therapeutically. Painful areas should be examined for deep tenderness, noting whether this is localized to muscle, ligamentous structures, or joints. Chronic myofascial pain is very common, and in these patients deep palpation may reveal highly localized trigger points that are firm bands or knots in muscle. If injection of local anesthetic into these trigger points relieves the pain, it supports the diagnosis. A neuropathic component to the pain is indicated by evidence of nerve damage, such as sensory impairment, exquisitely sensitive skin, weakness and muscle atrophy, or loss of deep tendon reflexes. Evidence suggesting sympathetic nervous system involvement includes the presence of diffuse swelling, changes in skin color and temperature, and hypersensitive skin and joint tenderness compared with the normal side. Relief of the pain with a sympathetic block is diagnostic.

A guiding principle in evaluating patients with chronic pain is to assess both emotional and organic factors before initiating therapy. Addressing these issues together, rather than waiting to "rule out" organic causes of the pain, improves compliance in part because it assures patients that a psychological evaluation does not mean that the physician is questioning the validity of their complaint. Even when an organic cause for a patient's pain can be found, it is still wise to look for other factors. For example, cancer patients with painful bony metastases may also have pain due to nerve damage and significant depression. Optimal therapy requires that each of these factors be looked for and treated.

TREATMENT

Once the evaluation process has been completed and the likely causative and exacerbating factors identified, an explicit treatment plan should be developed. An important part of this process is to identify specific and realistic functional goals for therapy, such as getting a good night's sleep, being able to go shopping, or returning to work. A multidisciplinary approach that utilizes medications, counseling, physical therapy, nerve blocks, and even surgery may be required to improve the patient's quality of life. This may require referral to a pain clinic; however, this is not necessary for all chronic pain patients. For some, pharmacologic management alone can provide significant help.

Antidepressant Medications The tricyclic antidepressants ([Table 12-1](#)) are extremely useful for the management of patients with chronic pain. Although developed for the treatment of depression, the tricyclics have a spectrum of dose-related biologic activities that include the production of analgesia in a variety of clinical conditions. Although the mechanism is unknown, the analgesic effect of tricyclics has a more rapid onset and

occurs at a lower dose than is typically required for the treatment of depression. Furthermore, patients with chronic pain who are not depressed obtain pain relief with antidepressants. There is evidence that tricyclic drugs potentiate opioid analgesia, so they are useful adjuncts for the treatment of severe persistent pain such as occurs with malignant tumors. [Table 12-2](#) lists some of the painful conditions that respond to tricyclics. Tricyclics are of particular value in the management of neuropathic pain such as occurs in diabetic neuropathy and postherpetic neuralgia, for which there are few other therapeutic options.

The tricyclics that have been shown to relieve pain have significant side effects ([Table 12-1](#); [Chap 385](#)). Unfortunately, some of the serotonin-selective reuptake inhibitors such as fluoxetine (Prozac) that have fewer and less serious side effects have not been shown to provide pain relief. On the other hand, venlafaxine (Effexor), a nontricyclic antidepressant that blocks both serotonin and norepinephrine reuptake, appears to be useful in patients who cannot tolerate tricyclics.

Anticonvulsants and Antiarrhythmics ([Table 12-1](#)) These drugs are useful primarily for patients with neuropathic pain. Phenytoin (Dilantin) and carbamazepine (Tegretol) were first shown to relieve the pain of trigeminal neuralgia. This pain has a characteristic brief, shooting, electric shock-like quality. In fact, anticonvulsants seem to be helpful largely for pains that have such a lancinating quality. A new-generation anticonvulsant, gabapentin (Neurontin), which increases brain γ -aminobutyric acid levels, is effective for a broad range of neuropathic pains.

Antiarrhythmic drugs such as low-dose lidocaine and mexiletine (Mexilitil) are also effective for neuropathic pains. These drugs block the spontaneous activity of primary afferent nociceptors that appears when they are damaged.

Chronic Opioid Medication The long-term use of opioids is accepted for patients with pain due to malignant disease. Although its use for chronic pain of nonmalignant origin is controversial, it is clear that for many such patients opioid analgesics are the only option available for obtaining effective relief. This is understandable since opioids are the most potent and have the broadest range of efficacy of any analgesic medications. Although addiction is rare in patients who first use opioids for pain relief, some degree of tolerance and physical dependence are likely to occur with long-term use. Therefore, before embarking on opioid therapy, other options should be explored, and the limitations and risks of opioids should be explained to the patient. It is also important to point out that some opioid analgesic medications have mixed agonist-antagonist properties (e.g., pentazocine and butorphanol). From a practical standpoint, this means that they may worsen pain by inducing an abstinence syndrome in patients who are physically dependent on other opioid analgesics.

With long-term outpatient use of orally administered opioids it is desirable to use long-acting compounds such as levorphanol, methadone, or sustained-release morphine ([Table 12-1](#)). The pharmacokinetic profile of these drugs enables prolonged pain relief, minimizes side effects such as sedation that are associated with high peak plasma levels, and, perhaps, reduces the likelihood of rebound pain associated with a rapid fall in plasma opioid concentration. Constipation is a virtually universal side effect of opioid use and should be treated expectantly.

It is worth emphasizing, in conclusion, that many patients, especially those with chronic pain, seek medical attention primarily because they are suffering and because only physicians can provide the medications required for their relief. A primary responsibility of all physicians is to minimize the physical and emotional discomfort of their patients. Familiarity with pain mechanisms and analgesic medications is an important step toward accomplishing this aim.

(Bibliography omitted in Palm version)

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13. CHEST DISCOMFORT AND PALPITATIONS - Thomas H. Lee

CHEST DISCOMFORT

Chest discomfort is one of the most common challenges for clinicians in the office or emergency department. The differential diagnosis includes conditions affecting organs throughout the thorax and abdomen, with prognostic implications that vary from benign to life-threatening ([Table 13-1](#)). Failure to recognize potentially serious conditions such as acute ischemic heart disease, aortic dissection, or pulmonary embolism can lead to serious complications, including death. Conversely, overly conservative management of low-risk patients leads to unnecessary hospital admissions, tests, and procedures.

CAUSES OF CHEST DISCOMFORT

Myocardial Ischemia and Injury (See also [Chap. 244](#)) Myocardial ischemia occurs when the oxygen supply to the heart is not sufficient to meet metabolic needs. This mismatch can result from a decrease in oxygen supply, a rise in demand, or both. The most common underlying cause of myocardial ischemia is obstruction of coronary arteries by atherosclerosis; in the presence of such obstruction, transient ischemic episodes are usually precipitated by an increase in oxygen demand as a result of physical exertion. However, ischemia can also result from psychological stress, fever, or large meals or from compromised oxygen delivery due to anemia, hypoxia, or hypotension. Ventricular hypertrophy due to valvular heart disease, hypertrophic cardiomyopathy, or hypertension can predispose the myocardium to ischemia because of impaired penetration of blood flow from epicardial coronary arteries to the endocardium.

Angina Pectoris The chest discomfort of myocardial ischemia is a visceral discomfort that is usually described as a heaviness, pressure, or squeezing ([Table 13-2](#)). Other common adjectives for anginal pain are burning and aching. Some patients deny any "pain" but may admit to dyspnea or a vague sense of anxiety. The word "sharp" is sometimes used by patients to describe intensity rather than quality.

The location of angina pectoris is usually retrosternal; most patients do not localize the pain to any small area. The discomfort may radiate to the neck, jaw, teeth, arms, or shoulders, reflecting the common origin in the posterior horn of the spinal cord of sensory neurons supplying the heart and these areas. Some patients present with aching in sites of radiated pain as their only symptoms of ischemia. Occasional patients report epigastric distress with ischemic episodes. Less common is radiation to below the umbilicus or to the back.

Stable angina pectoris usually develops gradually with exertion, emotional excitement, or after heavy meals. Rest or treatment with sublingual nitroglycerin typically leads to relief within several minutes. In contrast, pain that is fleeting (lasting only a few seconds) is rarely ischemic in origin. Similarly, pain that lasts for several hours is unlikely to represent angina, particularly if the patient's electrocardiogram does not show evidence of ischemia.

Anginal episodes can be precipitated by any physiologic or psychological stress that

induces tachycardia. Most myocardial perfusion occurs during diastole, when there is minimal pressure opposing coronary artery flow from within the left ventricle. Since tachycardia decreases the percentage of the time in which the heart is in diastole, it decreases myocardial perfusion.

Unstable angina and myocardial infarction (See also [Chaps. 243 and 244](#)) Patients with these acute ischemic syndromes usually complain of symptoms similar in quality to angina pectoris, but more prolonged and severe. The onset of these syndromes may occur with the patient at rest, and sublingual nitroglycerin may lead to transient or no relief. Accompanying symptoms may include diaphoresis, dyspnea, nausea, and light-headedness.

The physical examination may be completely normal in patients with chest discomfort due to ischemic heart disease. Careful auscultation during ischemic episodes may reveal a third or fourth heart sound, reflecting myocardial systolic or diastolic dysfunction. A transient murmur of mitral regurgitation suggests ischemic papillary muscle dysfunction. Severe episodes of ischemia can lead to pulmonary congestion and even pulmonary edema.

Other Cardiac Causes Myocardial ischemia caused by hypertrophic cardiomyopathy, aortic stenosis, or other conditions leads to angina pectoris similar to that caused by coronary atherosclerosis. In such cases, a systolic murmur or other findings usually suggest abnormalities other than coronary atherosclerosis that may be contributing to the patient's symptoms.

Pericarditis (See also [Chap. 239](#)) The pain in pericarditis is believed to be due to inflammation of the adjacent parietal pleura, since most of the pericardium is believed to be insensitive to pain. Thus, infectious pericarditis, which usually involves adjoining pleura surfaces, tends to be associated with pain, while conditions that cause only local inflammation (e.g., myocardial infarction or uremia) and cardiac tamponade tend to result in mild or no chest pain.

The adjacent parietal pleura receives its sensory supply from several sources, so the pain of pericarditis can be experienced in areas ranging from the shoulder and neck to the abdomen and back. Most typically, the pain is retrosternal and is aggravated by coughing, deep breaths, or changes in position -- all of which lead to movements of pleural surfaces. The pain is often worse in the supine position and relieved by sitting upright and leaning forward. Less common is a steady aching discomfort that mimics acute myocardial infarction.

Diseases of the Aorta (See also [Chap. 247](#)) *Aortic dissection* is a potentially catastrophic condition that is due to spread within the wall of the aorta of a subintimal hematoma. The hematoma may begin with a tear in the intima of the aorta or with rupture of the vasa vasorum within the aortic media. This syndrome can occur with trauma to the aorta, including motor vehicle accidents or medical procedures in which catheters or intraaortic balloon pumps damage the intima of the aorta. Nontraumatic aortic dissections are rare in the absence of hypertension and/or conditions associated with deterioration of the elastic or muscular components of the media within the aorta's wall. Cystic medial degeneration is a feature of several inherited connective tissue

diseases, including Marfan and Ehlers-Danlos syndromes. About half of all aortic dissections in women under 40 years of age occur during pregnancy.

Almost all patients with acute dissections present with severe chest pain, although some patients with chronic dissections are identified without associated symptoms. Unlike the pain of ischemic heart disease, symptoms of aortic dissection tend to reach peak severity immediately, often causing the patient to collapse from its intensity. The adjectives used to describe the pain reflect the process occurring within the wall of the aorta -- "ripping" and "tearing" -- and the location usually correlates with the site and extent of the dissection. Thus, dissections that begin in the ascending aorta and extend to the descending aorta tend to cause pain in the front of the chest that extends into the back, between the shoulder blades.

Physical findings may also reflect extension of the aortic dissection that compromises flow into arteries branching off the aorta. Thus, loss of a pulse in one or both arms, cerebrovascular accident, or paraplegia can all be catastrophic consequences of aortic dissection. Hematomas that extend proximally and undermine the coronary arteries or aortic valve apparatus may lead to acute myocardial infarction or acute aortic insufficiency. Rupture of the hematoma into the pericardial space leads to pericardial tamponade.

Another abnormality of the aorta that can cause chest pain is a *thoracic aortic aneurysm*. Aortic aneurysms are frequently asymptomatic but can cause chest pain and other symptoms by compressing adjacent structures. This pain tends to be steady, deep, and sometimes severe.

Pulmonary Embolism (See also [Chap. 261](#)) Chest pain due to pulmonary embolism is believed to be due to distention of the pulmonary artery or infarction of a segment of the lung adjacent to the pleura. Massive pulmonary emboli may lead to substernal pain that is suggestive of acute myocardial infarction. More commonly, smaller emboli lead to focal pulmonary infarctions that cause pain that is lateral and pleuritic. Associated symptoms include dyspnea and, occasionally, hemoptysis. Tachycardia is usually present.

Pneumonia or Pleuritis Lung diseases that damage and cause inflammation of the pleura of the lung usually cause a sharp, knifelike pain that is aggravated by inspiration or coughing.

Gastrointestinal Conditions Esophageal pain from acid reflux from the stomach, spasm, obstruction, or injury can be difficult to discern from myocardial syndromes. Acid reflux typically causes a deep burning discomfort that may be exacerbated by alcohol, aspirin, or some foods; this discomfort is often relieved by antacid or other acid-reducing therapies. Acid reflux tends to be exacerbated by lying down and may be worse in early morning when the stomach is empty of food that might otherwise absorb gastric acid.

Esophageal spasm may occur in the presence or absence of acid reflux, and leads to a squeezing pain indistinguishable from angina. Prompt relief of esophageal spasm is often provided by antianginal therapies such as sublingual nifedipine, further promoting confusion between these syndromes. Chest pain can also result from injury to the

esophagus, such as a Mallory-Weiss tear caused by severe vomiting.

Chest pain can result from diseases of the gastrointestinal tract below the diaphragm, including *peptic ulcer disease*, *biliary disease*, and *pancreatitis*. These conditions usually cause abdominal pain as well as chest discomfort; symptoms are not likely to be associated with exertion. The pain of ulcer disease typically occurs 60 to 90 min after meals, when postprandial acid production is no longer neutralized by food in the stomach. Cholecystitis usually causes a pain that is described as aching, occurring an hour or more after meals.

Neuromusculoskeletal Conditions *Cervical disk disease* can cause chest pain by compression of nerve roots. Pain in a dermatomal distribution can also be caused by *intercostal muscle cramps* or by *herpes zoster*. Chest pain symptoms due to herpes zoster may occur before skin lesions are apparent.

Costochondral and chondrosternal syndromes are the most common causes of anterior chest musculoskeletal pain. Only occasionally are physical signs of costochondritis such as swelling, redness, and warmth (Tietze's syndrome) present. The pain of such syndromes is usually fleeting and sharp, but some patients experience a dull ache that lasts for hours. Direct pressure on the chondrosternal and costochondral junctions may reproduce the pain from these and other musculoskeletal syndromes. Arthritis of the shoulder and spine and bursitis may also cause chest pain. Some patients who have these conditions and myocardial ischemia blur and confuse symptoms of these syndromes.

Emotional and Psychiatric Conditions As many as 10% of patients who present to emergency departments with acute chest pain have panic disorder or other emotional conditions. The symptoms in these populations are highly variable, but frequently the discomfort is described as visceral tightness or aching that lasts more than 30 min. Some patients offer other atypical descriptions, such as pain that is fleeting, sharp, and/or localized to a small region. The electrocardiogram in patients with emotional conditions may be difficult to interpret if hyperventilation causes ST-T-wave abnormalities. A careful history may elicit clues of depression, prior panic attacks, somatization, agoraphobia, or other phobias.

Approach to the Patient

The evaluation of the patient with chest discomfort must accommodate two goals -- determining the diagnosis and assessing the safety of the immediate management plan. The latter issue is often dominant when the patient has acute chest discomfort, such as patients seen in the emergency department. In such settings, the clinician must focus on questions such as the safety of discharge to home, admission to a non-coronary care unit facility, or immediate exercise testing. [Table 13-3](#) displays a sequence of questions that can be used in the evaluation of the patient with chest discomfort, with the diagnostic entities that are most important for consideration at each stage of the evaluation.

Acute Chest Discomfort In patients with acute chest discomfort, the clinician must first assess the patient's respiratory and hemodynamic status. If either is compromised,

initial management should focus on stabilizing the patient before the diagnostic evaluation is pursued. If, however, the patient does not require emergent interventions, then a focused history, physical examination, and laboratory evaluation should be performed to assess the patient's risk of life-threatening conditions, including acute ischemic heart disease, aortic dissection, and pulmonary embolism.

The *history* should include questions about the quality and location of the chest discomfort ([Table 13-2](#)). The patient should also be asked about the nature of onset of the pain and its duration. Myocardial ischemia is usually associated with a gradual intensification of symptoms over a period of minutes. Pain that is fleeting or that lasts hours without being associated with electrocardiographic changes is not likely to be ischemic in origin.

The *physical examination* should include evaluation of blood pressure in both arms and of pulses in both legs. Poor perfusion of a limb may be due to an aortic dissection that has compromised flow to an artery branching from the aorta. Chest auscultation may reveal diminished breath sounds; a pleural rub; or evidence of pneumothorax, pulmonary embolism, pneumonia, or pleurisy. The cardiac examination should seek pericardial rubs, systolic and diastolic murmurs, and third or fourth heart sounds.

An *electrocardiogram* is an essential test for adults with chest discomfort that is not due to an obvious traumatic cause. The presence of electrocardiographic changes consistent with ischemia or infarction ([Chap. 226](#)) is associated with high risks of acute myocardial infarction or unstable angina ([Table 13-4](#)); such patients should be admitted to a unit with electrocardiographic monitoring and the capacity to respond to a cardiac arrest. The absence of such changes does not exclude acute ischemic heart disease, but the risk of life-threatening complications is low for patients with normal electrocardiograms or only nonspecific ST-T-wave changes. If these patients are not considered appropriate for immediate discharge, they are often candidates for early or immediate exercise testing.

Markers of myocardial injury are often obtained in the emergency department evaluation of acute chest discomfort. The most commonly used markers are creatine kinase (CK), CK-MB, and the cardiac troponins (I and T). Single values of these markers do not have high sensitivity for acute myocardial infarction or for prediction of complications. Hence, decisions to discharge patients home should not be made on the basis of single negative values of these tests.

Provocative tests for coronary artery disease are not appropriate for patients with ongoing chest pain. In such patients, rest myocardial perfusion scans can be considered; a normal scan reduces the likelihood of coronary artery disease. Clinicians frequently employ therapeutic trials with sublingual nitroglycerin or antacids, and a common error is to assume that a response to either of these interventions clarifies the diagnosis. While such information is often helpful, the patient's response may be due to the placebo effect. Hence, myocardial ischemia should never be considered excluded solely because of a response to antacid therapy. Similarly, failure of nitroglycerin to relieve pain does not exclude the diagnosis of coronary disease.

If the patient's history or examination is consistent with aortic dissection, imaging studies

to evaluate the aorta must be pursued promptly because of the high risk of catastrophic complications with this condition. A chest x-ray is not sufficient to exclude this diagnosis. Appropriate tests include a chest computed tomography scan with contrast or a magnetic resonance imaging scan in patients who are hemodynamically stable, or a transesophageal echocardiogram in patients who are less stable. Aortic angiography is no longer a first test at most institutions.

Acute pulmonary embolism should be considered in patients with respiratory symptoms, pleuritic chest pain, hemoptysis, or a history of venous thromboembolism or coagulation abnormalities. Initial tests usually include a lung scan and/or pulmonary arteriography.

If patients with acute chest discomfort show no evidence of life-threatening conditions, the clinician should then focus on serious chronic conditions with the potential to cause major complications, the most common of which is stable angina. Early use of treadmill exercise testing for such patients, whether in the office or the emergency department, is now an accepted management strategy for low-risk patients. Exercise testing is not appropriate, however, for patients who (1) report pain that is believed to be ischemic occurring at rest or (2) have electrocardiographic changes consistent with ischemia not known to be old.

Patients with sustained chest discomfort who do not have evidence for life-threatening conditions should be evaluated for evidence of conditions likely to benefit from acute treatment ([Table 13-3](#)). Pericarditis may be suggested by the history, physical examination, and electrocardiogram ([Table 13-2](#)). Clinicians should carefully assess blood pressure patterns and consider echocardiography in such patients to detect evidence of impending pericardial tamponade. Chest x-rays can be used to evaluate the possibility of pulmonary disease.

GUIDELINES AND CRITICAL PATHWAYS FOR ACUTE CHEST PAIN

Guidelines for the initial evaluation for patients with acute chest pain have been developed by the American College of Emergency Physicians (ACEP) and other organizations. The ACEP statement describes *rules* and *guidelines* about the data that should be recorded as part of the evaluation, and the actions that should follow from certain findings ([Table 13-5](#)). In the ACEP framework, *rules* are actions that are general principles of good practice, while *guidelines* are actions that should be considered but are not always followed. Hence, failure to follow a guideline is not necessarily improper care.

Other organizations, including the Agency for Health Care Policy and Research (AHCPR) and the National Heart Attack Alert Program, have also issued guidelines for management of patients with a high probability of acute ischemic heart disease. In these and other guidelines, patients with possible or probable acute myocardial infarction as suggested by the description of their pain or electrocardiographic findings are expected to be admitted to the hospital. The AHCPR guidelines for unstable angina note that not all patients with that syndrome require admission but recommend that patients with unstable angina be monitored electrocardiographically during their evaluation; that those with ongoing rest pain should be placed at bed rest during the initial phase of stabilization. The [ACEP](#) policy statement indicates that patients who are discharged

should be given a referral for follow-up care and instructions regarding treatment and circumstances that require a return to the emergency department.

Many medical centers have adopted critical pathways and other forms of guidelines to increase efficiency. These guidelines emphasize two strategies:

- Triage to non-coronary care unit monitored facilities such as intermediate care units or chest pain units of patients with a low risk for complications, such as patients without new ischemic changes on their electrocardiograms and without ongoing chest pain. Such patients can usually be safely observed in non-coronary care unit settings, undergo early exercise testing, or be discharged home. Risk stratification can be assisted through use of prospectively validated multivariate algorithms that have been published for acute ischemic heart disease and its complications.
- Shortening lengths of stay in the coronary care unit and hospital. Recommendations regarding the minimum length of stay in a monitored bed for a patient who has no further symptoms have decreased in recent years to 12 h or less if exercise testing or other risk stratification technologies are available.

NONACUTE CHEST DISCOMFORT

The management of patients who do not require admission to the hospital or who no longer require inpatient observation should seek to identify the cause of the symptoms and the likelihood of major complications. Cost-effectiveness analyses support use of noninvasive testing for coronary disease, such as exercise electrocardiography and stress echocardiography. These tests serve both to diagnose coronary disease and to identify patients with high-risk forms of coronary disease who may benefit from revascularization. Gastrointestinal causes of chest pain can be evaluated via endoscopy or radiology studies. Emotional and psychiatric conditions warrant appropriate evaluation and treatment; randomized trial data indicate that cognitive therapy and group interventions lead to decreases in symptoms for such patients.

PALPITATIONS

Palpitations are characterized by an awareness of the beating of the heart. Patients commonly describe "pounding" or "fluttering" heart beats or report a sensation that the heart is stopping or skipping beats. These symptoms may be caused by a change in the heart's rhythm or rate or by an increase in the force of its contractions. In many cases, this awareness reflects lack of competing sensory stimuli, such as when a person is lying in bed, unable to sleep.

Palpitations are often manifestations of psychiatric conditions, the most common of which are depression and panic disorder. For example, in one study of outpatients referred for ambulatory electrocardiographic monitoring to evaluate palpitations, 19% were found to have a psychiatric disorder. Patients with psychiatric disorders were more likely than other patients to report that their palpitations lasted longer than 15 min or were accompanied by ancillary symptoms. In this study, physicians usually recognized the emotional basis of the patients' symptoms but frequently did not refer the patient for specific therapy.

Palpitations can also be caused by virtually any cardiac arrhythmia as well as by other cardiac and noncardiac conditions. A markedly enlarged left ventricle can cause awareness of the heart beat by contact with the chest wall. Any condition associated with increased catecholamine levels can lead to palpitations both by increasing the forcefulness of cardiac contractions and by increasing the rate of premature beats.

Palpitations can be intermittent or sustained and regular or irregular. Patients with this complaint should be asked to describe their palpitations' onset, duration, associated symptoms and the circumstances in which they occur. Abrupt onset and termination after several minutes may reflect a sustained ventricular or supraventricular tachyarrhythmia. Gradual onset and termination of a pounding heart beat is more consistent with sinus tachycardia. Patients should try to replicate the rhythm of their palpitations by tapping on a table. This maneuver can help the physician determine the nature of any cardiac arrhythmia. Patients should also be taught to take their pulse so that they can more accurately report their approximate heart rate and whether the rhythm was regular.

DIFFERENTIAL DIAGNOSIS

Patients who report "skipped" beats or a "flopping" sensation often have atrial or ventricular extrasystoles ([Chap. 230](#)). These premature beats are followed by a compensatory pause, and the first heart beat after the pause may be unusually strong due to increased left ventricular volume and enhanced contractility (a phenomenon called *postextrasystolic potentiation*). Sustained bursts of rapid heart beats may be due to ventricular or supraventricular tachyarrhythmias. A sustained irregular rhythm suggests atrial fibrillation.

Conditions that cause marked left ventricular enlargement such as aortic regurgitation can cause an awareness of the heart beat that is sometimes positional. Presumably because of associated arrhythmias, hypertrophic cardiomyopathy, mitral valve prolapse, and other cardiac structural abnormalities are also associated with palpitations.

Palpitations can also be a prominent symptom in noncardiac conditions, including thyrotoxicosis, hypoglycemia, pheochromocytoma, and fever. The physiologic basis of palpitations with these conditions is either arrhythmia or increased catecholamine levels leading to greater myocardial contractility. Drugs that can precipitate arrhythmias and palpitations include tobacco, coffee, tea, alcohol, epinephrine, ephedrine, aminophylline, and atropine.

Approach to the Patient

The first goal in the evaluation of patients with palpitations is to exclude the possibility of life-threatening arrhythmias. The risk for such arrhythmias is highest in patients with coronary artery disease, congestive heart failure, or other structural cardiac abnormalities. The history, physical examination, and electrocardiogram should therefore be focused on stratifying patients according to the risk of such conditions. Palpitations are also more likely to reflect serious arrhythmias if they are associated with symptoms that suggest hemodynamic compromise, such as syncope, light-headedness,

dizziness, or shortness of breath.

The most common first test after the initial evaluation of palpitations is continuous electrocardiographic (Holter) monitoring. This test is especially useful if patients have palpitations on a daily basis. For patients with more sporadic palpitations, a variety of new technologies have become available to allow capture of electrocardiographic tracings at the time of their symptoms. These technologies include loop recorders, that can freeze the last several minutes of data when the patient presses a button, and telephonic monitors, which can be used to "call in" tracings when symptoms occur. If episodes are associated with physical stress, exercise electrocardiography can be used in an attempt to elicit an arrhythmia.

Most patients with palpitations do not have evidence of major arrhythmias or abnormal physiologic conditions associated with increased catecholamine levels. Patients with emotional or psychological causes of palpitations should be evaluated for possible cognitive and pharmaceutical therapy. Drugs and medications that may precipitate palpitations should be eliminated or reduced. A trial of beta blockers is often successful in reducing premature beats and symptoms. Regardless of the cause and treatment, the clinician should remain aware that palpitations are extremely bothersome symptoms for patients. Reassurance that a comprehensive evaluation has been performed and that the palpitations do not adversely affect the patient's prognosis is a critical part of the patient's care.

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14. ABDOMINAL PAIN - William Silen

The correct interpretation of acute abdominal pain is challenging. Since proper therapy may require urgent action, the unhurried approach suitable for the study of other conditions is sometimes denied. Few other clinical situations demand greater judgment, because the most catastrophic of events may be forecast by the subtlest of symptoms and signs. A meticulously executed, detailed history and physical examination is of great importance. The etiologic classification in [Table 14-1](#), although not complete, forms a useful basis for the evaluation of patients with abdominal pain.

The diagnosis of "acute or surgical abdomen" is not an acceptable one because of its often misleading and erroneous connotation. The most obvious of "acute abdomens" may not require operative intervention, and the mildest of abdominal pains may herald an urgently correctable lesion. Any patient with abdominal pain of recent onset requires early and thorough evaluation and accurate diagnosis.

SOME MECHANISMS OF PAIN ORIGINATING IN THE ABDOMEN

Inflammation of the Parietal Peritoneum The pain of parietal peritoneal inflammation is steady and aching in character and is located directly over the inflamed area, its exact reference being possible because it is transmitted by somatic nerves supplying the parietal peritoneum. The intensity of the pain is dependent on the type and amount of material to which the peritoneal surfaces are exposed in a given time period. For example, the sudden release into the peritoneal cavity of a small quantity of *sterile* acid gastric juice causes much more pain than the same amount of grossly contaminated neutral feces. Enzymatically active pancreatic juice incites more pain and inflammation than does the same amount of sterile bile containing no potent enzymes. Blood and urine are often so bland as to go undetected if exposure of the peritoneum has not been sudden and massive. In the case of bacterial contamination, such as in pelvic inflammatory disease, the pain is frequently of low intensity early in the illness until bacterial multiplication has caused the elaboration of irritating substances.

The rate at which the irritating material is applied to the peritoneum is important. Perforated peptic ulcer may be associated with entirely different clinical pictures dependent only on the rapidity with which the gastric juice enters the peritoneal cavity.

The pain of peritoneal inflammation is invariably accentuated by pressure or changes in tension of the peritoneum, whether produced by palpation or by movement, as in coughing or sneezing. The patient with peritonitis lies quietly in bed, preferring to avoid motion, in contrast to the patient with colic, who may writhe incessantly.

Another characteristic feature of peritoneal irritation is tonic reflex spasm of the abdominal musculature, localized to the involved body segment. The intensity of the tonic muscle spasm accompanying peritoneal inflammation is dependent on the location of the inflammatory process, the rate at which it develops, and the integrity of the nervous system. Spasm over a perforated retrocecal appendix or perforated ulcer into the lesser peritoneal sac may be minimal or absent because of the protective effect of overlying viscera. A slowly developing process often greatly attenuates the degree of muscle spasm. Catastrophic abdominal emergencies such as a perforated ulcer may be

associated with minimal or no detectable pain or muscle spasm in obtunded, seriously ill, debilitated elderly patients or in psychotic patients.

Obstruction of Hollow Viscera The pain of obstruction of hollow abdominal viscera is classically described as intermittent, or colicky. Yet the lack of a truly cramping character should not be misleading, because distention of a hollow viscus may produce steady pain with only very occasional exacerbations. It is not nearly as well localized as the pain of parietal peritoneal inflammation.

The colicky pain of obstruction of the small intestine is usually periumbilical or supraumbilical and is poorly localized. As the intestine becomes progressively dilated with loss of muscular tone, the colicky nature of the pain may diminish. With superimposed strangulating obstruction, pain may spread to the lower lumbar region if there is traction on the root of the mesentery. The colicky pain of colonic obstruction is of lesser intensity than that of the small intestine and is often located in the infraumbilical area. Lumbar radiation of pain is common in colonic obstruction.

Sudden distention of the biliary tree produces a steady rather than colicky type of pain; hence the term *biliary colic* is misleading. Acute distention of the gallbladder usually causes pain in the right upper quadrant with radiation to the right posterior region of the thorax or to the tip of the right scapula, and distention of the common bile duct is often associated with pain in the epigastrium radiating to the upper part of the lumbar region. Considerable variation is common, however, so that differentiation between these may be impossible. The typical subscapular pain or lumbar radiation is frequently absent. Gradual dilatation of the biliary tree, as in carcinoma of the head of the pancreas, may cause no pain or only a mild aching sensation in the epigastrium or right upper quadrant. The pain of distention of the pancreatic ducts is similar to that described for distention of the common bile duct but, in addition, is very frequently accentuated by recumbency and relieved by the upright position.

Obstruction of the urinary bladder results in dull suprapubic pain, usually low in intensity. Restlessness without specific complaint of pain may be the only sign of a distended bladder in an obtunded patient. In contrast, acute obstruction of the intravesicular portion of the ureter is characterized by severe suprapubic and flank pain that radiates to the penis, scrotum, or inner aspect of the upper thigh. Obstruction of the ureteropelvic junction is felt as pain in the costovertebral angle, whereas obstruction of the remainder of the ureter is associated with flank pain that often extends into the same side of the abdomen.

Vascular Disturbances A frequent misconception, despite abundant experience to the contrary, is that pain associated with intraabdominal vascular disturbances is sudden and catastrophic in nature. The pain of embolism or thrombosis of the superior mesenteric artery or that of impending rupture of an abdominal aortic aneurysm certainly may be severe and diffuse. Yet, just as frequently, the patient with occlusion of the superior mesenteric artery has only mild continuous diffuse pain for 2 or 3 days before vascular collapse or findings of peritoneal inflammation appear. The early, seemingly insignificant discomfort is caused by hyperperistalsis rather than peritoneal inflammation. Indeed, absence of tenderness and rigidity in the presence of continuous, diffuse pain in a patient likely to have vascular disease is quite characteristic of

occlusion of the superior mesenteric artery. Abdominal pain with radiation to the sacral region, flank, or genitalia should always signal the possible presence of a rupturing abdominal aortic aneurysm. This pain may persist over a period of several days before rupture and collapse occur.

Abdominal Wall Pain arising from the abdominal wall is usually constant and aching. Movement, prolonged standing, and pressure accentuate the discomfort and muscle spasm. In the case of hematoma of the rectus sheath, now most frequently encountered in association with anticoagulant therapy, a mass may be present in the lower quadrants of the abdomen. Simultaneous involvement of muscles in other parts of the body usually serves to differentiate myositis of the abdominal wall from an intraabdominal process that might cause pain in the same region.

REFERRED PAIN IN ABDOMINAL DISEASES

Pain referred to the abdomen from the thorax, spine, or genitalia may prove a vexing diagnostic problem, because diseases of the upper part of the abdominal cavity such as acute cholecystitis or perforated ulcer are frequently associated with intrathoracic complications. A most important, yet often forgotten, dictum is that the possibility of intrathoracic disease must be considered in every patient with abdominal pain, especially if the pain is in the upper part of the abdomen. Systematic questioning and examination directed toward detecting myocardial or pulmonary infarction, pneumonia, pericarditis, or esophageal disease (the intrathoracic diseases that most often masquerade as abdominal emergencies) will often provide sufficient clues to establish the proper diagnosis. Diaphragmatic pleuritis resulting from pneumonia or pulmonary infarction may cause pain in the right upper quadrant and pain in the supraclavicular area, the latter radiation to be distinguished from the referred subscapular pain caused by acute distention of the extrahepatic biliary tree. The ultimate decision as to the origin of abdominal pain may require deliberate and planned observation over a period of several hours, during which repeated questioning and examination will provide the diagnosis.

Referred pain of thoracic origin is often accompanied by splinting of the involved hemithorax with respiratory lag and decrease in excursion more marked than that seen in the presence of intraabdominal disease. In addition, apparent abdominal muscle spasm caused by referred pain will diminish during the inspiratory phase of respiration, whereas it is persistent throughout both respiratory phases if it is of abdominal origin. Palpation over the area of referred pain in the abdomen also does not usually accentuate the pain and in many instances actually seems to relieve it. Thoracic and abdominal disease frequently coexist and may be difficult or impossible to differentiate. For example, the patient with known biliary tract disease often has epigastric pain during myocardial infarction, or biliary colic may be referred to the precordium or left shoulder in a patient who has suffered previously from angina pectoris. **For an explanation of the radiation of pain to a previously diseased area, see Chap. 12.*

Referred pain from the spine, which usually involves compression or irritation of nerve roots, is characteristically intensified by certain motions such as cough, sneeze, or strain and is associated with hyperesthesia over the involved dermatomes. Pain referred to the abdomen from the testicles or seminal vesicles is generally accentuated by the

slightest pressure on either of these organs. The abdominal discomfort is of dull aching character and is poorly localized.

METABOLIC ABDOMINAL CRISES

Pain of metabolic origin may simulate almost any other type of intraabdominal disease. Several mechanisms may be at work. In certain instances, such as hyperlipidemia, the metabolic disease itself may be accompanied by an intraabdominal process such as pancreatitis, which can lead to unnecessary laparotomy unless recognized. C¶I esterase deficiency associated with angioneurotic edema is often associated with episodes of severe abdominal pain. Whenever the cause of abdominal pain is obscure, a metabolic origin always must be considered. Abdominal pain is also the hallmark of familial Mediterranean fever ([Chap. 289](#)).

The problem of differential diagnosis is often not readily resolved. The pain of porphyria and of lead colic is usually difficult to distinguish from that of intestinal obstruction, because severe hyperperistalsis is a prominent feature of both. The pain of uremia or diabetes is nonspecific, and the pain and tenderness frequently shift in location and intensity. Diabetic acidosis may be precipitated by acute appendicitis or intestinal obstruction, so if prompt resolution of the abdominal pain does not result from correction of the metabolic abnormalities, an underlying organic problem should be suspected. Black widow spider bites produce intense pain and rigidity of the abdominal muscles and back, an area infrequently involved in intraabdominal disease.

NEUROGENIC CAUSES

Causalgic pain may occur in diseases that injure sensory nerves. It has a burning character and is usually limited to the distribution of a given peripheral nerve. Normal stimuli such as touch or change in temperature may be transformed into this type of pain, which is frequently present in a patient at rest. The demonstration of irregularly spaced cutaneous pain spots may be the only indication of an old nerve lesion underlying causalgic pain. Even though the pain may be precipitated by gentle palpation, rigidity of the abdominal muscles is absent, and the respirations are not disturbed. Distention of the abdomen is uncommon, and the pain has no relationship to the intake of food.

Pain arising from spinal nerves or roots comes and goes suddenly and is of a lancinating type ([Chap. 16](#)). It may be caused by herpes zoster, impingement by arthritis, tumors, herniated nucleus pulposus, diabetes, or syphilis. It is not associated with food intake, abdominal distention, or changes in respiration. Severe muscle spasm, as in the gastric crises of tabes dorsalis, is common but is either relieved or is not accentuated by abdominal palpation. The pain is made worse by movement of the spine and is usually confined to a few dermatomes. Hyperesthesia is very common.

Psychogenic pain conforms to none of the aforementioned patterns. Mechanism is hard to define. The most common problem is the hysterical adolescent or young person who develops abdominal pain and who frequently loses an appendix or other organs because of it. Ovulation or some other natural event that causes brief mild abdominal discomfort may be experienced as an abdominal catastrophe.

Psychogenic pain varies enormously in type and location but usually has no relation to meals. It is often markedly accentuated during the night. Nausea and vomiting are rarely observed. Spasm is seldom induced in the abdominal musculature and, if present, does not persist, especially if the attention of the patient can be distracted. Persistent localized tenderness is rare, and if found, the muscle spasm in the area is inconsistent or absent. Shallow respiration is the most common breathing abnormality; anxiety may produce a smothering or choking sensation. It occurs in the absence of thoracic splinting or change in the respiratory rate.

Approach to the Patient

Few abdominal conditions require such urgent operative intervention that an orderly approach need be abandoned, no matter how ill the patient. Only those patients with exsanguinating hemorrhage must be rushed to the operating room immediately, but in such instances, only a few minutes are required to assess the critical nature of the problem. Under these circumstances, all obstacles must be swept aside, adequate venous access for fluid replacement obtained, and the operation begun. Many patients of this type have died in the radiology department or the emergency room while awaiting such unnecessary examinations as electrocardiograms or abdominal films. *There are no contraindications to operation when massive hemorrhage is present.* This situation fortunately is relatively rare.

Nothing will supplant an orderly, painstakingly *detailed history*, which is far more valuable than any laboratory or radiographic examination. This kind of history is laborious and time-consuming, making it not especially popular, even though a reasonably accurate diagnosis can be made on the basis of the history alone in the majority of cases. Computer-aided diagnosis of abdominal pain provides no advantage over clinical assessment alone. In cases of *acute* abdominal pain, a diagnosis is readily established in most instances, whereas success is not so frequent in patients with *chronic* pain. Irritable bowel syndrome is one of the most common causes of abdominal pain and must always be kept in mind ([Chap. 288](#)). The *chronological sequence of events* in the patient's history is often more important than emphasis on the location of pain. If the examiner is sufficiently open-minded and unhurried, asks the proper questions, and listens, the patient will usually provide the diagnosis. Careful attention should be paid to the extraabdominal regions that may be responsible for abdominal pain. An accurate menstrual history in a female patient is essential. Narcotics or analgesics should *not* be withheld until a definitive diagnosis or a definitive plan has been formulated; obfuscation of the diagnosis by adequate analgesia is unlikely.

In the examination, simple critical inspection of the patient, e.g., of facies, position in bed, and respiratory activity, may provide valuable clues. The amount of information to be gleaned is directly proportional to the *gentleness* and thoroughness of the examiner. Once a patient with peritoneal inflammation has been examined brusquely, accurate assessment by the next examiner becomes almost impossible. Eliciting rebound tenderness by sudden release of a deeply palpating hand in a patient with suspected peritonitis is cruel and unnecessary. The same information can be obtained by gentle percussion of the abdomen (rebound tenderness on a miniature scale), a maneuver that can be far more precise and localizing. Asking the patient to cough will elicit true

rebound tenderness without the need for placing a hand on the abdomen. Furthermore, the forceful demonstration of rebound tenderness will startle and induce protective spasm in a nervous or worried patient in whom true rebound tenderness is not present. A palpable gallbladder will be missed if palpation is so brusque that voluntary muscle spasm becomes superimposed on involuntary muscular rigidity.

As in history taking, there is no substitute for sufficient time spent in the examination. Abdominal signs may be minimal but nevertheless, if accompanied by consistent symptoms, may be exceptionally meaningful. Abdominal signs may be virtually or totally absent in cases of pelvic peritonitis, so careful *pelvic and rectal examinations are mandatory in every patient with abdominal pain*. Tenderness on pelvic or rectal examination in the absence of other abdominal signs can be caused by operative indications such as perforated appendicitis, diverticulitis, twisted ovarian cyst, and many others.

Much attention has been paid to the presence or absence of peristaltic sounds, their quality, and their frequency. Auscultation of the abdomen is one of the least revealing aspects of the physical examination of a patient with abdominal pain. Catastrophes such as strangulating small intestinal obstruction or perforated appendicitis may occur in the presence of normal peristalsis. Conversely, when the proximal part of the intestine above an obstruction becomes markedly distended and edematous, peristaltic sounds may lose the characteristics of borborygmi and become weak or absent, even when peritonitis is not present. It is usually the severe chemical peritonitis of sudden onset that is associated with the truly silent abdomen. Assessment of the patient's state of hydration is important.

Laboratory examinations may be of great value in assessment of the patient with abdominal pain, yet with few exceptions they rarely establish a diagnosis. Leukocytosis should never be the single deciding factor as to whether or not operation is indicated. A white blood cell count greater than 20,000/uL may be observed with perforation of a viscus, but pancreatitis, acute cholecystitis, pelvic inflammatory disease, and intestinal infarction may be associated with marked leukocytosis. A normal white blood cell count is not rare in cases of perforation of abdominal viscera. The diagnosis of anemia may be more helpful than the white blood cell count, especially when combined with the history.

The urinalysis may reveal the state of hydration or rule out severe renal disease, diabetes, or urinary infection. Blood urea nitrogen, glucose, and serum bilirubin levels may be helpful. Serum amylase levels may be increased by many diseases other than pancreatitis, e.g., perforated ulcer, strangulating intestinal obstruction, and acute cholecystitis; thus, elevations of serum amylase do not rule out the need for an operation. The determination of the serum lipase may have greater accuracy than that of the serum amylase.

Plain and upright or lateral decubitus radiographs of the abdomen may be of value in cases of intestinal obstruction, perforated ulcer, and a variety of other conditions. They are usually unnecessary in patients with acute appendicitis or strangulated external hernias. In rare instances, barium or water-soluble contrast study of the upper part of the gastrointestinal tract may demonstrate partial intestinal obstruction that may elude diagnosis by other means. If there is any question of obstruction of the colon, oral

administration of barium sulfate should be avoided. On the other hand, in cases of suspected colonic obstruction (with perforation), contrast enema may be diagnostic.

In the absence of trauma, peritoneal lavage has been replaced as a diagnostic tool by ultrasound, computed tomography (CT), and laparoscopy. Ultrasonography has proved to be useful in detecting an enlarged gallbladder or pancreas, the presence of gallstones, an enlarged ovary, or a tubal pregnancy. Laparoscopy is especially helpful in diagnosing pelvic conditions, such as ovarian cysts, tubal pregnancies, salpingitis, and acute appendicitis. Radioisotopic scans (HIDA) may help differentiate acute cholecystitis from acute pancreatitis. A CT scan may demonstrate an enlarged pancreas, ruptured spleen, or thickened colonic or appendiceal wall and streaking of the mesocolon or mesoappendix characteristic of diverticulitis or appendicitis.

Sometimes, even under the best circumstances with all available aids and with the greatest of clinical skill, a definitive diagnosis cannot be established at the time of the initial examination. Nevertheless, despite lack of a clear anatomic diagnosis, it may be abundantly clear to an experienced and thoughtful physician and surgeon that on clinical grounds alone operation is indicated. Should that decision be questionable, watchful waiting with repeated questioning and examination will often elucidate the true nature of the illness and indicate the proper course of action.

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15. HEADACHE, INCLUDING MIGRAINE AND CLUSTER HEADACHE - Neil H. Raskin, Stephen J. Peroutka

Few of us are spared the experience of head pain. As many as 90% of individuals have at least one headache per year. Severe, disabling headache is reported to occur at least annually by 40% of individuals worldwide. A useful classification of the many causes of headache is shown in [Table 15-1](#). Headache is usually a benign symptom, but occasionally it is the manifestation of a serious illness such as brain tumor, subarachnoid hemorrhage, meningitis, or giant cell arteritis. In emergency settings, approximately 5% of patients with headache are found to have a serious underlying neurologic disorder. Therefore, it is imperative that the serious causes of headache be diagnosed rapidly and accurately.

PAIN-SENSITIVE STRUCTURES OF THE HEAD

Pain is most commonly due to tissue injury resulting in stimulation of peripheral nociceptors in an intact nervous system. Pain can also result from damage to or anomalous activation of pain-sensitive pathways of the peripheral or central nervous system. Headache may originate from either or both mechanisms. Relatively few cranial structures are pain-sensitive: the scalp, middle meningeal artery, dural sinuses, falx cerebri, and the proximal segments of the large pial arteries. The ventricular ependyma, choroid plexus, pial veins, and much of the brain parenchyma are pain-insensitive. Electrical stimulation of the midbrain in the region of the dorsal raphe has resulted in migraine-like headaches. Thus, whereas most of the brain is insensitive to electrode probing, a site in the midbrain represents a possible source of headache generation. Sensory stimuli from the head are conveyed to the central nervous system via the trigeminal nerves for structures above the tentorium in the anterior and middle fossae of the skull and via the first three cervical nerves for those in the posterior fossa and the inferior surface of the tentorium.

Headache can occur as the result of (1) distention, traction, or dilation of intracranial or extracranial arteries; (2) traction or displacement of large intracranial veins or their dural envelope; (3) compression, traction, or inflammation of cranial and spinal nerves; (4) spasm, inflammation, or trauma to cranial and cervical muscles; (5) meningeal irritation and raised intracranial pressure; or (6) other possible mechanisms such as activation of brainstem structures.

GENERAL CLINICAL CONSIDERATIONS

The quality, location, duration, and time course of the headache and the conditions that produce, exacerbate, or relieve it should be carefully reviewed. Ascertaining the *quality* of cephalic pain is occasionally helpful for diagnosis. Most tension-type headaches are described as tight "bandlike" pain or as dull, deeply located, and aching pain. Jabbing, brief, sharp cephalic pain, often occurring multifocally (ice pick-like pain), is the signature of a benign, nondescript disorder. A throbbing quality and tight muscles about the head, neck, and shoulder girdle are common nonspecific accompaniments of vascular headaches.

Pain *intensity* rarely has diagnostic value, although from the patient's perspective, it is

the single aspect of pain that is most important. Although meningitis, subarachnoid hemorrhage, and cluster headache produce intense cranial pain, most patients entering emergency departments with the most severe headache of their lives usually have migraine. Contrary to common belief, the headache produced by a brain tumor is not usually distinctive or severe.

Data regarding *location* of headache may be informative. If the source is an extracranial structure, as in giant cell arteritis, the correspondence with the site of pain is fairly precise. Inflammation of an extracranial artery causes pain and exquisite tenderness localized to the site of the vessel. Lesions of paranasal sinuses, teeth, eyes, and upper cervical vertebrae induce less sharply localized pain, but pain that is still referred in a regional distribution. Intracranial lesions in the posterior fossa cause pain that is usually occipitotemporal, and supratentorial lesions most often induce frontotemporal pain.

Duration and *time-intensity* curves of headaches are diagnostically useful. A ruptured aneurysm results in head pain that peaks in an instant, thunderclap-like; much less often, unruptured aneurysms may signal their presence in the same way. Cluster headache attacks reach their peak over 3 to 5 min, remain at maximal levels for about 45 min, and then taper off. Migraine attacks build up over hours, are maintained for several hours to days, and are characteristically relieved by sleep. Sleep disruption and early morning headaches that improve during the day are characteristics of headaches produced by brain tumors.

The analysis of facial pain requires a different approach. Trigeminal and, less commonly, glossopharyngeal neuralgia are frequent causes of facial pain ([Chap. 367](#)). "Neuralgias" are painful disorders characterized by paroxysmal, fleeting, often electric shock-like episodes that are frequently caused by demyelinating lesions of nerves (the trigeminal or glossopharyngeal nerves in cranial neuralgias). Certain maneuvers characteristically trigger paroxysms of pain. However, the most common cause of facial pain by far is dental; provocation by hot, cold, or sweet foods is typical. The application of a cold stimulus will repeatedly induce dental pain, whereas in neuralgic disorders, a refractory period usually occurs after the initial response so that pain cannot be repeatedly induced.

The effect of eating on facial pain may provide insight into its cause. Is it the chewing, swallowing, or taste of the food that elicits pain? Chewing points toward trigeminal neuralgia, temporomandibular joint dysfunction, or giant cell arteritis ("jaw claudication"), whereas swallowing *and* taste provocation point toward glossopharyngeal neuralgia. Pain upon swallowing is common among patients with carotidynia (see below) because the inflamed, tender carotid artery abuts the esophagus during deglutition.

Many patients with facial pain do not experience stereotypic neuralgias; the term *atypical facial pain* has been used in this setting. Vague, poorly localized, continuous facial pain is characteristic of nasopharyngeal carcinoma; a burning pain often develops as deafferentation occurs and evidence of cranial neuropathy appears. Burning facial pain may also occur with tumors of the fifth cranial nerve (meningioma or schwannoma) or with lesions of the pons that interrupt the dorsal root entry zone of the nerve (multiple sclerosis). In patients with facial pain, the finding of objective sensory loss is an important clue to a serious underlying disorder. Occasionally, the cause of a pain

problem cannot be resolved promptly, necessitating periodic follow- up until further signs appear.

CLINICAL EVALUATION OF ACUTE, NEW-ONSET HEADACHE

Patients who present with their first severe headache raise entirely different diagnostic possibilities than those with recurrent headaches over many years. In new-onset and severe headaches, the probability of finding a potentially serious cause is considerably greater than in recurrent headache. When a patient complains of an acute, new-onset headache, a number of causes should be considered including meningitis, subarachnoid hemorrhage, epidural or subdural hematoma, glaucoma, and purulent sinusitis. Clinical features of acute, new-onset headache caused by serious underlying conditions are summarized in [Table 15-2](#).

A complete neurologic examination is an essential first step in the evaluation. In most cases, an abnormal examination should be followed by a computed tomography (CT) or a magnetic resonance imaging (MRI) study. As a screening procedure for intracranial pathology in this setting, CT and MRI methods appear to be equally sensitive. A general evaluation of acute headache might include the investigation of cardiovascular and renal status by blood pressure monitoring and urine examination; eyes by fundoscopy, intraocular pressure measurement, and refraction; cranial arteries by palpation; and cervical spine by the effect of passive movement of the head and imaging.

The psychological state of the patient should also be evaluated since a relationship exists between head pain and depression. Many patients in chronic daily pain cycles become depressed; moreover, there is a greater-than-chance coincidence of migraine with both bipolar (manic depressive) and unipolar major depressive disorders. Drugs with antidepressant actions are also effective in the prophylactic treatment of both tension-type headache and migraine.

Underlying recurrent headache disorders may be activated by pain that follows otologic or endodontic surgical procedures. Treatment of the headache problem is largely ineffective until the cause of the primary problem is addressed. Thus, pain about the head as the result of diseased tissue or trauma may reawaken an otherwise quiescent migrainous syndrome.

Serious underlying conditions that are associated with headache are described below and in [Table 15-3](#).

MENINGITIS

In general, acute, severe headache with stiff neck and fever suggests meningitis. Lumbar puncture is mandatory. Often there is striking accentuation of pain with eye movement. Meningitis is particularly easy to mistake for migraine in that the cardinal symptoms of pounding headache, photophobia, nausea, and vomiting are present. **A detailed discussion of meningitis can be found in [Chaps. 372 to 374](#).*

INTRACRANIAL HEMORRHAGE

In general, acute, severe headache with stiff neck but without fever suggests subarachnoid hemorrhage. A ruptured aneurysm, arteriovenous malformation, or intraparenchymal hemorrhage may also present with only headache. Rarely, if the hemorrhage is small or below the foramen magnum, the head [CT](#) scan can be normal. Therefore, a lumbar puncture may be required to make the definitive diagnosis of a subarachnoid hemorrhage. **A detailed discussion of intracranial hemorrhage can be found in [Chap. 361](#).*

BRAIN TUMOR

Approximately 30% of patients with brain tumors consider headache to be their chief complaint. The head pain is usually nondescript -- an intermittent deep, dull aching of moderate intensity, which may worsen with exertion or change in position and may be associated with nausea and vomiting. This pattern of symptoms results from migraine far more often than from brain tumor. Headache of brain tumor disturbs sleep in about 10% of patients. Vomiting that precedes the appearance of headache by weeks is highly characteristic of posterior fossa brain tumors. A history of amenorrhea or galactorrhea should lead one to question whether a prolactin-secreting pituitary adenoma (or the polycystic ovary syndrome) is the source of headache. Headache arising de novo in a patient with known malignancy suggests either cerebral metastases and/or carcinomatous meningitis. Head pain appearing abruptly after bending, lifting, or coughing can be the clue to a posterior fossa mass (or a Chiari malformation). **A detailed discussion of brain tumors can be found in [Chap. 370](#).*

TEMPORAL ARTERITIS (See also [Chaps. 28](#) and [317](#))

Temporal (giant cell) arteritis is an inflammatory disorder of arteries that frequently involves the extracranial carotid circulation. This is a common disorder of the elderly; its annual incidence is 77:100,000 in individuals aged 50 and older. The average age of onset is 70 years, and women account for 65% of cases. About half of patients with untreated temporal arteritis develop blindness due to involvement of the ophthalmic artery and its branches; indeed, the ischemic optic neuropathy induced by giant cell arteritis is the major cause of rapidly developing bilateral blindness in patients over 60 years of age. Because treatment with glucocorticoids is effective in preventing this complication, prompt recognition of this disorder is important.

Typical presenting symptoms include headache, polymyalgia rheumatica ([Chap. 317](#)), jaw claudication, fever, and weight loss. Headache is the dominant symptom and often appears in association with malaise and muscle aches. Head pain may be unilateral or bilateral and is located temporally in 50% of patients but may involve any and all aspects of the cranium. Pain usually appears gradually over a few hours before peak intensity is reached; occasionally, it is explosive in onset. The quality of pain is only seldom throbbing; it is almost invariably described as dull and boring with superimposed episodic ice pick-like lancinating pains similar to the sharp pains that appear in migraine. Most patients can recognize that the origin of their head pain is superficial, external to the skull, rather than originating deep within the cranium (the pain site for migraineurs). Scalp tenderness is present, often to a marked degree; brushing the hair or resting the head on a pillow may be impossible because of pain. Headache is usually worse at night and is often aggravated by exposure to cold. Reddened, tender nodules or red

streaking of the skin overlying the temporal arteries may be found in patients with headache, as is tenderness of the temporal or, less commonly, the occipital arteries.

The erythrocyte sedimentation rate (ESR) is often, though not always, elevated; a normal ESR does not exclude giant cell arteritis. A temporal artery biopsy and the initiation of prednisone at 80 mg daily for the first 4 to 6 weeks should be instituted when clinical suspicion is high. The prevalence of migraine among the elderly is substantial, considerably higher than that of giant cell arteritis. Migraineurs often report amelioration of their headaches with prednisone, so that one must be cautious about interpreting the therapeutic response.

GLAUCOMA

Glaucoma may present with a prostrating headache associated with nausea and vomiting. The history will usually reveal that the headache started with severe eye pain. On physical examination, the eye is often red with a fixed, moderately dilated pupil. **A detailed discussion of glaucoma can be found in [Chap. 28](#).*

OTHER CAUSES OF HEADACHE

Systemic Illness There is hardly any illness that is never manifested by headache; however, some illnesses are frequently associated with headache. These include infectious mononucleosis, systemic lupus erythematosus, chronic pulmonary failure with hypercapnia (early morning headaches), Hashimoto's thyroiditis, inflammatory bowel disease, many of the illnesses associated with HIV, and the acute blood pressure elevations that occur in pheochromocytoma and in malignant hypertension. The last two examples are the exceptions to the generalization that hypertension per se is a very uncommon cause of headache; diastolic pressures of at least 120 mmHg are requisite for hypertension to cause headache. Persistent headache and fever are often the manifestations of an acute systemic viral infection; if the neck is supple in such a patient, lumbar puncture may be deferred. Some drugs and drug-withdrawal states, e.g., oral contraceptives, ovulation-promoting medications, and glucocorticoid withdrawal, are also associated with headache in some individuals.

Idiopathic Intracranial Hypertension (Pseudotumor Cerebri) Headache, clinically resembling that of brain tumor, is a common presenting symptom of pseudotumor cerebri, a disorder of raised intracranial pressure probably resulting from impaired cerebrospinal fluid CSF absorption by the arachnoid villi. Transient visual obscurations and papilledema with enlarged blind spots and loss of peripheral visual fields are additional manifestations. Most patients are young, female, and obese. They often have a history of exposure to provoking agents such as vitamin A and glucocorticoids. **Treatment of idiopathic intracranial hypertension is discussed in [Chap. 28](#).*

Cough A male-dominated (4:1) syndrome, cough headache is characterized by transient, severe head pain upon coughing, bending, lifting, sneezing, or stooping. Head pain persists for seconds to a few minutes. Many patients date the origins of the syndrome to a lower respiratory infection accompanied by severe coughing or to strenuous weight-lifting programs. Headache is usually diffuse but is lateralized in about

one-third of patients. The incidence of serious intracranial structural anomalies causing this condition is about 25%; the Chiari malformation ([Chap. 368](#)) is a common cause. Thus, [MRI](#) is indicated for most patients with cough headache. The benign disorder may persist for a few years; it responds dramatically to indomethacin at doses ranging from 50 to 200 mg daily. Approximately half of patients will also show a response to therapeutic lumbar puncture with removal of 40 mL of [CSF](#).

Many patients with migraine note that attacks of headache may be provoked by *sustained* physical exertion, such as during the third mile of a 5-mile run. Such headaches build up over hours, in contrast to cough headache. The term *effort migraine* has been used for this syndrome to avoid the ambiguous term *exertional headache*.

Lumbar Puncture Headache following lumbar puncture ([Chap. 356](#)) usually begins within 48 h but may be delayed for up to 12 days. Its incidence is between 10 and 30%. Head pain is dramatically positional; it begins when the patient sits or stands upright; there is relief upon reclining or with abdominal compression. The longer the patient is upright, the longer the latency before head pain subsides. It is worsened by head shaking and jugular vein compression. The pain is usually a dull ache but may be throbbing; its location is occipitofrontal. Nausea and stiff neck often accompany headache, and occasional patients report blurred vision, photophobia, tinnitus, and vertigo. The symptoms resolve over a few days but may on occasion persist for weeks to months.

Loss of [CSF](#) volume decreases the brain's supportive cushion, so that when a patient is upright there is probably dilation and tension placed on the brain's anchoring structures, the pain-sensitive dural sinuses, resulting in pain. Intracranial hypotension often occurs, but severe lumbar puncture headache may be present even in patients who have normal CSF pressure.

Treatment with intravenous caffeine sodium benzoate given over a few minutes as a 500-mg dose will promptly terminate headache in 75% of patients; a second dose given in 1 h brings the total success rate to 85%. An epidural blood patch accomplished by injection of 15 mL of autologous whole blood rarely fails for those who do not respond to caffeine. The mechanism for these treatment effects is not straightforward. The blood patch has an *immediate* effect, making it unlikely that sealing off a dural hole with blood clot is its mechanism of action.

Postconcussion Following seemingly trivial head injuries and particularly after rear-end motor vehicle collisions, many patients report varying combinations of headache, dizziness, vertigo, and impaired memory. Anxiety, irritability and difficulty with concentration are other hallmarks of this syndrome. Symptoms may remit after several weeks or persist for months and even years after the injury. Postconcussion headaches may occur whether or not a person was rendered unconscious by head trauma. Typically, the neurologic examination is normal with the exception of the behavioral abnormalities, and [CT](#) or [MRI](#) studies are unrevealing. Chronic subdural hematoma may on occasion mimic this disorder. Although the cause of postconcussive headache disorder is not known, it should not in general be viewed as a primary psychological disturbance. It often persists long after the settlement of pending lawsuits. The treatment is symptomatic support. Repeated encouragement that the syndrome

eventually remits is important.

Coital Headache This is another male-dominated (4:1) syndrome. Attacks occur periorgasmically, are very abrupt in onset, and subside in a few minutes if coitus is interrupted. These are nearly always benign events and usually occur sporadically; if they persist for hours or are accompanied by vomiting, subarachnoid hemorrhage must be excluded ([Chap. 361](#)).

PRINCIPAL CLINICAL VARIETIES OF RECURRENT HEADACHE

There is usually little difficulty in diagnosing the serious types of headaches listed above because of the clues provided by the associated symptoms and signs. It is when headache is chronic, recurrent, and unattended by other important signs of disease that the physician faces a challenging and unique medical problem. The following sections describe a variety of headache types, ranging from the most common (e.g., tension-type headache) to rare causes of recurrent headache.

TENSION-TYPE HEADACHE

The term *tension-type headache* is still commonly used to describe a chronic head pain syndrome characterized by bilateral tight, bandlike discomfort. Patients may report that the head feels as if it is in a vise or that the posterior neck muscles are tight. The pain typically builds slowly, fluctuates in severity, and may persist more or less continuously for many days. Exertion does not usually worsen the headache. The headache may be episodic or chronic (i.e., present more than 15 days per month). Tension-type headache is common in all age groups, and females tend to predominate. In some patients, anxiety or depression coexist with tension headache.

The pathophysiologic basis of tension-type headache remains unknown. Some investigators believe that periodic tension headache is biologically indistinguishable from migraine, whereas others believe that tension-type headache and migraine are two distinct clinical entities. Abnormalities of cervical and temporal muscle contraction are likely to exist, but the exact nature of the dysfunction has not yet been elucidated.

Relaxation almost always relieves tension-type headaches. Patients should be encouraged to find a means of relaxation, which, for a given individual, could include bed rest, massage, and/or formal biofeedback training. Pharmacologic treatment consists of either simple analgesics and/or muscle relaxants. Ibuprofen and naproxen sodium are useful treatments for most individuals. When simple over-the-counter analgesics such as acetaminophen, aspirin, ibuprofen, and/or other nonsteroidal anti-inflammatory drugs (NSAIDs) alone fail, the addition of butalbital and caffeine (in a combination compound such as Fiorinal, Fioricet) to these analgesics may be effective. A list of commonly used analgesics for tension-type headaches is presented in [Table 15-4](#). For chronic tension-type headache, prophylactic therapy is recommended. Low doses of amitriptyline (10 to 50 mg at bedtime) can provide effective prophylaxis.

MIGRAINE

Migraine, the most common cause of vascular headache, afflicts approximately 15% of

women and 6% of men. A useful definition of migraine is a benign and recurring syndrome of headache, nausea, vomiting, and/or other symptoms of neurologic dysfunction in varying admixtures ([Table 15-5](#)). Migraine can often be recognized by its activators (red wine, menses, hunger, lack of sleep, glare, estrogen, worry, perfumes, let-down periods) and its deactivators (sleep, pregnancy, exhilaration, sumatriptan). A classification of the many subtypes of migraine, as defined by the International Headache Society, is shown in [Table 15-1](#).

Severe headache attacks, regardless of cause, are more likely to be described as throbbing and associated with vomiting and scalp tenderness. Milder headaches tend to be nondescript -- tight, bandlike discomfort often involving the entire head -- the profile of tension-type headache.

Pathogenesis

Genetic Basis of Migraine Migraine has a definite genetic predisposition. Specific mutations leading to rare causes of vascular headache have been identified ([Table 15-6](#)). For example, the MELAS syndrome consists of a mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes and is caused by an A® G point mutation in the mitochondrial gene encoding for tRNA^{Leu(UUR)} at nucleotide position 3243. Episodic migraine-like headaches are another common clinical feature of this syndrome, especially early in the course of the disease. The genetic pattern of mitochondrial disorders is unique, since only mothers transmit mitochondrial DNA. Thus, all children of mothers with MELAS syndrome are affected with the disorder.

Familial hemiplegic migraine (FHM) is characterized by episodes of recurrent hemiparesis or hemiplegia during the aura phase of a migraine headache. Other associated symptoms may include hemianesthesia or paresthesia; hemianopic visual field disturbances; dysphasia; and variable degrees of drowsiness, confusion, and/or coma. In severe attacks, these symptoms can be quite prolonged and persist for days or weeks, but characteristically they last for only 30 to 60 min and are followed by a unilateral throbbing headache.

Approximately 50% of cases of [FHM](#) appear to be caused by mutations within the CACNL1A4 gene on chromosome 19, which encodes a P/Q type calcium channel subunit expressed only in the central nervous system. The gene is very large (>300 kb in length) and consists of 47 exons. Four distinct point mutations have been identified within the gene (in five different families) that cosegregate with the clinical diagnosis of FHM. Analysis of haplotypes in the two families with the same mutation suggest that each mutation arose independently rather than representing a founder effect. Thus, certain subtypes of FHM are caused by mutations in the CACNL1A4 gene. The function of the CACNL1A4 gene remains unknown, but it is likely to play a role in calcium-induced neurotransmitter release and/or contraction of smooth muscle. Different mutations within this gene are the cause of another neurogenetic disorder, episodic ataxia type 2 ([Chap. 364](#)).

In a genetic association study, a NcoI polymorphism in the gene encoding the D₂dopamine receptor (DRD2) was overrepresented in a population of patients with migraine with aura compared to a control group of nonmigraineurs, suggesting that

susceptibility to migraine with aura is modified by certain DRD2 alleles. In a Sardinian population, an association between different DRD2 alleles and migraine has also been demonstrated. Therefore, these initial studies suggest that variations in dopamine receptor regulation and/or function may alter susceptibility to migraine since molecular variations within the DRD2 gene have been associated with variations in dopaminergic function. However, since not all individuals with certain DRD2 genotypes suffer from migraine with aura, additional genes or factors must also be involved. Migraine is likely to be a complex disorder with polygenic inheritance and a strong environmental component.

The Vascular Theory of Migraine It was widely held for many years that the headache phase of migrainous attacks was caused by extracranial vasodilatation and that the neurologic symptoms were produced by intracranial vasoconstriction (i.e., the "vascular" hypothesis of migraine). Regional cerebral blood flow studies have shown that in patients with classic migraine there is, during attacks, a modest cortical hypoperfusion that begins in the visual cortex and spreads forward at a rate of 2 to 3 mm/min. The decrease in blood flow averages 25 to 30% (insufficient to explain symptoms on the basis of ischemia) and progresses anteriorly in a wavelike fashion independent of the topography of cerebral arteries. The wave of hypoperfusion persists for 4 to 6 h, appears to follow the convolutions of the cortex, and does not cross the central or lateral sulcus, progressing to the frontal lobe via the insula. Perfusion of subcortical structures is normal. Contralateral neurologic symptoms appear during temporoparietal hypoperfusion; at times, hypoperfusion persists in these regions after symptoms cease. More often, frontal spread continues as the headache phase begins. A few patients with classic migraine show no flow abnormalities; an occasional patient has developed focal ischemia sufficient to cause symptoms. However, focal ischemia does not appear to be *necessary* for focal symptoms to occur.

The ability of these changes to induce the symptoms of migraine has been questioned. Specifically, the decrease in blood flow that is observed does not appear to be significant enough to cause focal neurologic symptoms. Second, the increase in blood flow per se is not painful, and vasodilatation alone cannot account for the local edema and focal tenderness often observed in migraineurs. Moreover, in migraine without aura, no flow abnormalities are usually seen. Thus, it is unlikely that simple vasoconstriction and vasodilatation are the fundamental pathophysiologic abnormalities in migraine. However, it is clear that cerebral blood flow is altered during certain migraine attacks, and these changes may explain some, but clearly not all, of the clinical syndrome of migraine.

The Neuronal Theory of Migraine In 1941, the psychologist KS Lashley charted his own *fortification spectrum*, which is a migraine aura characterized by a slowly enlarging visual scotoma with luminous edges (see below). He was able to estimate that the evolution of his own scotoma proceeded across the occipital cortex at a rate of 3 mm/min. He speculated that a wavefront of intense excitation followed by a wave of complete inhibition of activity were propagated across the visual cortex. In 1944, the phenomenon that has come to be known as *spreading depression* was described by the Brazilian physiologist Leao in the cerebral cortex of laboratory animals. It is a slowly moving (2 to 3 mm/min), potassium-liberating depression of cortical activity, preceded by a wavefront of increased metabolic activity that can be produced by a variety of

experimental stimuli, including hypoxia, mechanical trauma, and the topical application of potassium. These observations suggest that neuronal abnormalities, most likely initiated in the brainstem, could be the cause of a migraine attack. More recently, both cortical and brainstem changes have been observed in positron emission tomography (PET) scan studies of migraine. Thus, the existence of a specific "brainstem generator" for migraine remains an intriguing possibility that might represent the pathophysiologic basis of migraine.

The Trigeminovascular System in Migraine Activation of cells in the trigeminal nucleus caudalis in the medulla (a pain-processing center for the head and face region) results in the release of vasoactive neuropeptides, including substance P and calcitonin gene-related peptide (CGRP), at vascular terminations of the trigeminal nerve. These peptide neurotransmitters have been proposed to induce a sterile inflammation that activates trigeminal nociceptive afferents originating on the vessel wall, further contributing to the production of pain. This mechanism also provides a potential mechanism for the soft tissue swelling and tenderness of blood vessels that attend migraine attacks. However, numerous pharmacologic agents that are effective in preventing or reducing inflammation in this animal model (e.g., selective 5-HT_{1D}agonists, NK-1 antagonists, endothelin antagonists) have failed to demonstrate any clinical efficacy in recent migraine trials.

5-Hydroxytryptamine in Migraine Pharmacologic and other data point to the involvement of the neurotransmitter 5-hydroxytryptamine (5-HT; also known as serotonin) in migraine. Approximately 40 years ago, methysergide was found to antagonize certain peripheral actions of 5-HT and was introduced as the first drug capable of preventing migraine attacks. Subsequently, it was found that platelet levels of 5-HT fall consistently at the onset of headache and that drugs that cause 5-HT to be released may trigger migrainous episodes. Such changes in circulating 5-HT levels proved to be pharmacologically trivial, however, and interest in the humoral role of 5-HT in migraine declined.

More recently, interest in the role of [5-HT](#) in migraine has been renewed due to the introduction of the triptan class of antimigraine drugs. The triptans are designed to stimulate selectively a particular subpopulation of 5-HT receptors. Molecular cloning studies have demonstrated that at least 14 specific 5-HT receptors exist in humans. The triptans (e.g., naratriptan, rizatriptan, sumatriptan, and zolmitriptan) are potent agonists of 5-HT_{1B}, 5-HT_{1D}, and 5-HT_{1F} receptors and are less potent at 5-HT_{1A} and 5-HT_{1E} receptors. A growing body of data indicates that the antimigraine efficacy of the triptans relates to their ability to stimulate 5-HT_{1B} receptors, which are located both on blood vessels and nerve terminals. Selective 5-HT_{1D} receptor agonists have, thus far, failed to demonstrate clinical efficacy in migraine. Triptans that are weak 5-HT_{1F} agonists are also effective in migraine; however, only 5-HT_{1B} efficacy is currently thought to be essential for antimigraine efficacy.

Physiologically, electrical stimulation near dorsal raphe neurons can result in migraine-like headaches. Blood flow in the pons and midbrain increases focally during migraine headache episodes; this alteration probably results from increased activity of cells in the dorsal raphe and locus caeruleus. There are projections from the dorsal raphe that terminate on cerebral arteries and alter cerebral blood flow. There are also

major projections from the dorsal raphe to important visual centers, including the lateral geniculate body, superior colliculus, retina, and visual cortex. These various serotonergic projections may represent the neural substrate for the circulatory and visual characteristics of migraine. The dorsal raphe cells stop firing during deep sleep, and sleep is known to ameliorate migraine; the antimigraine prophylactic drugs also inhibit activity of the dorsal raphe cells through a direct or indirect agonist effect.

Recent [PET](#) scan studies have demonstrated that midbrain structures near the dorsal raphe are differentially activated during a migraine attack. In one study of acute migraine, an injection of sumatriptan relieved the headache, but did not alter the brainstem changes noted on the PET scan. These data suggest that a "brainstem generator" may be the cause of migraine and that certain antimigraine medications may not interfere with the underlying pathologic process in migraine.

Dopamine in Migraine A growing body of biologic, pharmacologic, and genetic data support a role for dopamine in the pathophysiology of certain subtypes of migraine. Most migraine symptoms can be induced by dopaminergic stimulation. Moreover, there is dopamine receptor hypersensitivity in migraineurs, as demonstrated by the induction of yawning, nausea, vomiting, hypotension, and other symptoms of a migraine attack by dopaminergic agonists at doses that do not affect nonmigraineurs. Conversely, dopamine receptor antagonists are effective therapeutic agents in migraine, especially when given parenterally or concurrently with other antimigraine agents. As noted above, recent genetic data also suggest that molecular variations within dopamine receptor genes play a modifying role in the pathophysiology of migraine with aura. Therefore, modulation of dopaminergic neurotransmission should be considered in the therapeutic management of migraine.

The Sympathetic Nervous System in Migraine Biochemical changes occur within the sympathetic nervous system (SNS) of migraineurs before, during, and between migraine attacks. Factors that activate the SNS are all trigger factors for migraine. Specific examples include environmental changes (e.g., stress, sleep patterns, hormonal shifts, hypoglycemia) and agents that cause release and a secondary depletion of peripheral catecholamines [e.g., tyramine, phenylethylamine, fenfluramine, m-chlorophenylpiperazine (mCPP) and reserpine]. By contrast, effective therapeutic approaches to migraine share an ability to mimic and/or enhance the effects of norepinephrine in the peripheral SNS. For example, norepinephrine itself, sympathomimetics (e.g., isometheptene), monoamine oxidase inhibitors (MAOIs) and reuptake blockers alleviate migraine. Dopamine antagonists, prostaglandin synthesis inhibitors, and adenosine antagonists are pharmacologic agents effective in the acute treatment of migraine. These drugs block the negative feedback inhibition or norepinephrine release induced by endogenous dopamine, prostaglandins, and adenosine. Therefore, migraine susceptibility may relate to genetically based variations in the ability to maintain adequate concentrations of certain neurotransmitters within postganglionic sympathetic nerve terminals. This hypothesis has been called the *empty neuron theory* of migraine.

Clinical Features

Migraine without Aura (Common Migraine) In this syndrome no focal neurologic

disturbance precedes the recurrent headaches. Migraine without aura is by far the more frequent type of vascular headache. The International Headache Society criteria for migraine include moderate to severe head pain, pulsating quality, unilateral location, aggravation by walking stairs or similar routine activity, attendant nausea and/or vomiting, photophobia and phonophobia, and multiple attacks, each lasting 4 to 72 h.

Migraine with Aura (Classic Migraine) In this syndrome headache is associated with characteristic premonitory sensory, motor, or visual symptoms. Focal neurologic disturbances are more common during headache attacks than as prodromal symptoms. Focal neurologic disturbances without headache or vomiting have come to be known as *migraine equivalents* or *migraine accompaniments* and appear to occur more commonly in patients between the ages of 40 and 70 years. The term *complicated migraine* has generally been used to describe migraine with dramatic transient focal neurologic features or a migraine attack that leaves a persisting residual neurologic deficit.

The most common premonitory symptoms reported by migraineurs are visual, arising from dysfunction of occipital lobe neurons. Scotomas and/or hallucinations occur in about one-third of migraineurs and usually appear in the central portions of the visual fields. A highly characteristic syndrome occurs in about 10% of patients; it usually begins as a small paracentral scotoma, which slowly expands into a "C" shape. Luminous angles appear at the enlarging outer edge, becoming colored as the scintillating scotoma expands and moves toward the periphery of the involved half of the visual field, eventually disappearing over the horizon of peripheral vision. The entire process lasts 20 to 25 min. This phenomenon is pathognomonic for migraine, and has never been described in association with a cerebral structural anomaly. It is commonly referred to as a *fortification spectrum* because the serrated edges of the hallucinated "C" seemed to resemble a "fortified town with bastions all round it"; "spectrum" is used in the sense of an apparition or specter.

Basilar Migraine Symptoms referable to a disturbance in brainstem function, such as vertigo, dysarthria, or diplopia, occur as the only neurologic symptoms of the attack in about 25% of patients. A dramatic form of basilar migraine (Bickerstaff's migraine) occurs primarily in adolescent females. Episodes begin with total blindness accompanied or followed by admixtures of vertigo, ataxia, dysarthria, tinnitus, and distal and perioral paresthesia. In about one-quarter of patients, a confusional state supervenes. The neurologic symptoms usually persist for 20 to 30 min and are generally followed by a throbbing occipital headache. This basilar migraine syndrome is now known also to occur in children and in adults over age 50. An altered sensorium may persist for as long as 5 days and may take the form of confusional states superficially resembling psychotic reactions. Full recovery after the episode is the rule.

Carotidynia The carotidynia syndrome, sometimes called *lower-half headache* or *facial migraine*, is most common among older patients, with the incidence peaking in the fourth through sixth decades. Pain is usually located at the jaw or neck, although sometimes periorbital or maxillary pain occurs; it may be continuous, deep, dull, and aching, and it becomes pounding or throbbing episodically. There are often superimposed sharp, ice pick-like jabs. Attacks occur one to several times per week, each lasting several minutes to hours. Tenderness and prominent pulsations of the cervical carotid artery and soft tissue swelling overlying the carotid are usually present

ipsilateral to the pain; many patients also report throbbing ipsilateral headache concurrent with carotidynia attacks as well as between attacks. Dental trauma is a common precipitant of this syndrome. Carotid artery involvement also appears to be common in the more traditional forms of migraine; over 50% of patients with frequent migraine attacks are found to have carotid tenderness at several points on the side most often involved during hemicranial migraine attacks.

TREATMENT

Nonpharmacologic Approaches for All Migraineurs Migraine can often be managed to some degree by a variety of nonpharmacologic approaches ([Table 15-7](#)). The measures that apply to a given individual should be used routinely since they provide a simple, cost-effective approach to migraine management. Patients with migraine do not encounter more stress than headache-free individuals; overresponsiveness to stress appears to be the issue. Since the stresses of everyday living cannot be eliminated, lessening one's response to stress by various techniques is helpful for many patients. These include yoga, transcendental meditation, hypnosis, and conditioning techniques such as biofeedback. For most patients, this approach is, at best, an adjunct to pharmacotherapy. Avoidance of migraine trigger factors may also provide significant prophylactic benefits ([Table 15-7](#)). Unfortunately, these measures are unlikely to prevent all migraine attacks. When these measures fail to prevent an attack, then pharmacologic approaches are needed to abort an attack.

Pharmacologic Treatment of Acute Migraine The mainstay of pharmacologic therapy is the judicious use of one or more of the many drugs that are effective in migraine. The selection of the optimal regimen for a given patient depends on a number of factors, the most important of which is the severity of the attack ([Table 15-8](#)). Mild migraine attacks can usually be managed by oral agents; the average efficacy rate is 50-70%. Severe migraine attacks may require parenteral therapy. Most drugs effective in the treatment of migraine are members of one of three major pharmacologic classes: anti-inflammatory agents, 5-HT₁ agonists, and dopamine antagonists.

[Table 15-9](#) lists specific drugs effective in migraine. In general, an adequate dose of whichever agent is chosen should be used as soon as possible after the onset of an attack. If additional medication is required within 60 min because symptoms return or have not abated, the initial dose should be increased for subsequent attacks. Migraine therapy must be individualized for each patient; a standard approach for all patients is not possible. A therapeutic regimen may need to be constantly refined and personalized until one is identified that provides the patient with rapid, complete, and consistent relief with minimal side effects.

Nonsteroidal anti-inflammatory agents Both the severity and duration of a migraine attack can be reduced significantly by anti-inflammatory agents. Indeed, many undiagnosed migraineurs are self-treated with nonprescription anti-inflammatory agents ([Table 15-4](#)). A general consensus is that [NSAIDs](#) are most effective when taken early in the migraine attack. However, the effectiveness of anti-inflammatory agents in migraine is usually less than optimal in moderate or severe migraine attacks. The combination of acetaminophen, aspirin, and caffeine (Excedrin Migraine) has been approved for use by the U.S. Food and Drug Administration (FDA) for the treatment of mild to moderate

migraine. The combination of aspirin and metoclopramide has been shown to be equivalent to a single dose of sumatriptan. Major side effects of NSAIDs include dyspepsia and gastrointestinal irritation.

5-HT₁agonists

ORAL Stimulation of 5-HT₁ receptors can stop an acute migraine attack. Ergotamine and dihydroergotamine are nonselective receptor agonists, while the series of drugs known as triptans are selective 5-HT₁ receptor agonists. A variety of triptans (e.g., naratriptan, rizatriptan, sumatriptan, zolmitriptan) are now available for the treatment of migraine (Table 15-9).

Each of the triptan class of drugs has similar pharmacologic properties, but varies slightly in terms of clinical efficacy. Rizatriptan appears to be the fastest acting and most efficacious of the triptans currently available in the United States. Sumatriptan and zolmitriptan have similar rates of efficacy as well as time to onset, whereas naratriptan is the slowest acting and the least efficacious. Clinical efficacy appears to be related more to the $t_{\max}$ (time to peak plasma level) than to the potency, half-life, or bioavailability (Table 15-10). This observation is in keeping with a significant body of data indicating that faster-acting analgesics are more efficacious than slower-acting agents.

Unfortunately, monotherapy with a selective oral 5-HT₁ agonist does not result in rapid, consistent, and complete relief of migraine in all patients. Triptans are not effective in migraine with aura unless given after the aura is completed and the headache initiated. Side effects, although often mild and transient, occur in up to 89% of patients. Moreover, 5-HT₁ agonists are contraindicated in individuals with a history of cardiovascular disease. Recurrence of headache is a major limitation of triptan use, and occurs at least occasionally in 40 to 78% of patients.

Ergotamine preparations offer a nonselective means of stimulating 5-HT₁ receptors. A nonnauseating dose of ergotamine should be sought since a dose that provokes nausea is too high and may intensify head pain. Except for a sublingual formulation of ergotamine (Ergomar), oral formulations of ergotamine also contain 100 mg caffeine (theoretically to enhance ergotamine absorption and possibly to add additional vasoconstrictor activity). The average oral ergotamine dose for a migraine attack is 2 mg. Since the clinical studies demonstrating the efficacy of ergotamine in migraine predated the clinical trial methodologies used with the triptans, it is difficult to assess the clinical efficacy of ergotamine versus the triptans. In general, ergotamine appears to have a much higher incidence of nausea than triptans, but less headache recurrence.

NASAL The fastest acting nonparenteral antimigraine therapies that can be self-administered include nasal formulations of dihydroergotamine (Migranal) or sumatriptan (Imitrex Nasal). The nasal sprays result in substantial blood levels within 30 to 60 min. However, the nasal formulations suffer from inconsistent dosing, poor taste, and variable efficacy. Although in theory the nasal sprays might provide faster and more effective relief of a migraine attack than oral formulations, their reported efficacy is only approximately 50 to 60%.

PARENTERAL Parenteral administration of drugs such as dihydroergotamine (DHE-45

Injectable) and sumatriptan (Imitrex SC) is approved by the [FDA](#) for the rapid relief of a migraine attack. Peak plasma levels of dihydroergotamine are achieved 3 min after intravenous dosing, 30 min after intramuscular dosing, and 45 min after subcutaneous dosing. If an attack has not already peaked, subcutaneous or intramuscular administration of 1 mg dihydroergotamine suffices for about 80 to 90% of patients. Sumatriptan, 6 mg subcutaneously is effective in approximately 70 to 80% of patients.

Dopamine Antagonists

ORAL Oral dopamine antagonists should be considered as adjunctive therapy in migraine. Drug absorption is impaired during migrainous attacks because of reduced gastrointestinal motility. Delayed absorption occurs in the absence of nausea and is related to the severity of the attack and not its duration. Therefore, when oral [NSAIDs](#) and/or triptan agents fail, the addition of a dopamine antagonist such as metoclopramide, 10 mg, should be considered to enhance gastric absorption. In addition, dopamine antagonists decrease nausea/vomiting and restore normal gastric motility.

PARENTERAL Parenteral dopamine antagonists (e.g., chlorpromazine, prochlorperazine, metoclopramide) can also provide significant acute relief of migraine; they can be used in combination with parenteral [5-HT₁](#) agonists. A common intravenous protocol used for the treatment of severe migraine is the administration over 2 min of a mixture of 5 mg of prochlorperazine and 0.5 mg of dihydroergotamine.

Other Medications for Acute Migraine

ORAL The combination of acetaminophen, dichloralphenazone, and isometheptene (i.e., Midrin, Duradrin, generic), one to two capsules, has been classified by the [FDA](#) as "possibly" effective in the treatment of migraine. Since the clinical studies demonstrating the efficacy of this combination analgesic in migraine predated the clinical trial methodologies used with the triptans, it is difficult to assess the clinical efficacy of this sympathomimetic compound in comparison to other agents.

NASAL A nasal preparation of butorphanol is available for the treatment of acute pain. As with all narcotics, the use of nasal butorphanol should be limited to a select group of migraineurs, as described below.

PARENTERAL Narcotics are effective in the acute treatment of migraine. For example, intravenous meperidene (Demerol), 50 to 100 mg, is given frequently in the emergency room. This regimen "works" in the sense that the pain of migraine is eliminated. However, this regimen is clearly suboptimal in patients with recurrent headache for two major reasons. First, narcotics do not treat the underlying headache mechanism; rather, they act at the thalamic level to alter pain sensation. Second, the recurrent use of narcotics can lead to significant problems. In patients taking oral narcotics such as oxycodone (Percodan) or hydrocodone (Vicoden), narcotic addiction can greatly confuse the treatment of migraine. The headache that results from narcotic craving and/or withdrawal can be difficult to distinguish from chronic migraine. Therefore, it is recommended that narcotic use in migraine be limited to patients with severe, but infrequent, headaches that are unresponsive to other pharmacologic approaches.

Prophylactic Treatment of Migraine A substantial number of drugs are now available that have the capacity to stabilize migraine ([Table 15-11](#)). The decision of whether to use this approach depends on the frequency of attacks and on how well acute treatment is working. The occurrence of at least three attacks per month could be an indication for this approach. Drugs must be taken daily and there is usually a lag of at least 2 to 6 weeks before an effect is seen. The drugs that have been approved by the [FDA](#) for the prophylactic treatment of migraine include propranolol, timolol, sodium valproate, and methysergide. In addition, a number of other drugs appear to display prophylactic efficacy. This group of drugs includes amitriptyline, nortriptyline, verapamil, phenelzine, isocarbazid, and cyproheptadine. Phenelzine and methysergide are usually reserved for recalcitrant cases because of their serious potential side effects. Phenelzine is an [MAOI](#); therefore, tyramine-containing foods, decongestants, and meperidine are contraindicated. Methysergide may cause retroperitoneal or cardiac valvular fibrosis when it is used for more than 8 months, thus monitoring is required for patients using this drug; the risk of the fibrotic complication is about 1:1500 and is likely to reverse after the drug is stopped.

The probability of success with any one of the antimigraine drugs is 50 to 75%; thus, if one drug is assessed each month, there is a good chance that effective stabilization will be achieved within a few months. Many patients are managed adequately with low-dose amitriptyline, propranolol, or valproate. If these agents fail or lead to unacceptable side effects, then methysergide or phenelzine can be used. Once effective stabilization is achieved, the drug is continued for 5 to 6 months and then slowly tapered to assess the continued need. Many patients are able to discontinue medication and experience fewer and milder attacks for long periods, suggesting that these drugs may alter the natural history of migraine.

CLUSTER HEADACHE

A variety of names have been used for this condition, including *Raeder's syndrome*, *histamine cephalalgia*, and *sphenopalatine neuralgia*. *Cluster headache* is a distinctive and treatable vascular headache syndrome. The episodic type is most common and is characterized by one to three short-lived attacks of periorbital pain per day over a 4- to 8-week period, followed by a pain-free interval that averages 1 year. The chronic form, which may begin de novo or several years after an episodic pattern has become established, is characterized by the absence of sustained periods of remission. Each type may transform into the other. Men are affected seven to eight times more often than women; hereditary factors are usually absent. Although the onset is generally between ages 20 and 50, it may occur as early as the first decade of life. Propranolol and amitriptyline are largely ineffective. Lithium is beneficial for cluster headache and ineffective in migraine. The cluster syndrome is thus clinically, genetically, and therapeutically different from migraine. Nevertheless, mixed features of the two disorders are occasionally present, suggesting some common elements to their pathogenesis.

Clinical Features Periorbital or, less commonly, temporal pain begins without warning and reaches a crescendo within 5 min. It is often excruciating in intensity and is deep, nonfluctuating, and explosive in quality; only rarely is it pulsatile. Pain is strictly unilateral

and usually affects the same side in subsequent months. Attacks last from 30 min to 2 h; there are often associated symptoms of homolateral lacrimation, reddening of the eye, nasal stuffiness, lid ptosis, and nausea. Alcohol provokes attacks in about 70% of patients but ceases to be provocative when the bout remits; this on-off vulnerability to alcohol is pathognomonic of cluster headache. Only rarely do foods or emotional factors precipitate pain, in contrast to migraine.

There is a striking periodicity of attacks in at least 85% of patients. At least one of the daily attacks of pain recurs at about the same hour each day for the duration of a cluster bout. Onset is nocturnal in about 50% of the cases, and then the pain usually awakens the patient within 2 h of falling asleep.

Pathogenesis No consistent cerebral blood flow changes accompany attacks of pain. Perhaps the strongest evidence for a central mechanism is the periodicity of attacks; the existence of a central mechanism is also suggested by the observation that autonomic symptoms that accompany the pain are bilateral and are more severe on the painful side. The hypothalamus may be the site of activation in this disorder. The posterior hypothalamus contains cells that regulate autonomic functions, and the anterior hypothalamus contains cells (in the suprachiasmatic nuclei) that constitute the principal circadian pacemaker in mammals. Activation of both is necessary to explain the symptoms of cluster headache. The pacemaker is modulated via serotonergic dorsal raphe projections. It can be concluded tentatively that both migraine and cluster headache result from abnormal serotonergic neurotransmission, albeit at different loci.

TREATMENT

The most satisfactory treatment is the administration of drugs to prevent cluster attacks until the bout is over. Effective prophylactic drugs are prednisone, lithium, methysergide, ergotamine, sodium valproate, and verapamil. Lithium (600 to 900 mg daily) appears to be particularly useful for the chronic form of the disorder. A 10-day course of prednisone, beginning at 60 mg daily for 7 days followed by a rapid taper, may interrupt the pain bout for many patients. When ergotamine is used, it is most effective when given 1 to 2 h before an expected attack. Patients must be educated regarding the early symptoms of ergotism when ergotamine is used daily; a weekly limit of 14 mg should be adhered to.

For the attacks themselves, oxygen inhalation (9 L/min via a loose mask) is the most effective modality; 15 min of inhalation of 100% oxygen is often necessary. Sumatriptan, 6 mg subcutaneously, will usually shorten an attack to 10 to 15 min.

(Bibliography omitted in Palm version)

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16. BACK AND NECK PAIN - John W. Engstrom

The importance of back and neck pain in our society is underscored by the following: (1) the annual societal cost of back pain in the United States is estimated to be between \$20 and \$50 billion; (2) back symptoms are the most common cause of disability in patients under 45 years of age; (3) 50% of working adults, in one survey, admitted to having a back injury each year; and (4) approximately 1% of the U.S. population is chronically disabled because of back pain.

The enormous economic pressure to provide rational and efficient care of patients with back pain has resulted in clinical practice guidelines (CPGs) for these patients. CPGs are algorithms which guide evaluation or treatment at specific steps in patient care. CPGs for *acute low back pain* (ALBP) are based upon incomplete evidence (see algorithms, [Fig. 16-6](#)) but represent an attempt to standardize common medical practice. Major revisions in CPGs for back pain can be anticipated in the future. Management of patients with *chronic low back pain* (CLBP) is complex and not amenable to a simple algorithmic approach at this time.

ANATOMY OF THE SPINE

The anterior portion of the spine consists of cylindrical vertebral bodies separated by intervertebral disks and held together by the anterior and posterior longitudinal ligaments. The intervertebral disks are composed of a central gelatinous nucleus pulposus surrounded by a tough cartilaginous ring, the annulus fibrosis; disks are responsible for 25% of spinal column length ([Figs. 16-1](#) and [16-2](#)). The disks are largest in the cervical and lumbar regions where movements of the spine are greatest. The disks are elastic in youth and allow the bony vertebrae to move easily upon each other. Elasticity is lost with age. The function of the anterior spine is to absorb the shock of typical body movements such as walking and running.

The posterior portion of the spine consists of the vertebral arches and seven processes. Each arch consists of paired cylindrical pedicles anteriorly and paired laminae posteriorly ([Fig. 16-1](#)). The vertebral arch gives rise to two transverse processes laterally, one spinous process posteriorly, plus two superior and two inferior articular facets. The functions of the posterior spine are to protect the spinal cord and nerves within the spinal canal and to stabilize the spine by providing sites for the attachment of muscles and ligaments. The contraction of muscles attached to the spinous and transverse processes produces a system of pulleys and levers that results in flexion, extension, and lateral bending movements of the spine. Normal upright posture in humans places the center of gravity anterior to the spine. The graded contraction of well-developed paraspinal muscles attached to the laminae, transverse processes, and spinous processes is necessary to maintain normal upright posture.

The nerve roots exit at a level above their respective vertebral bodies in the cervical region (the C7 nerve root exits at the C6-C7 level) and below their respective vertebral bodies in the thoracic and lumbar regions (the T1 nerve root exits at the T1-T2 level). The spinal cord ends at the L1 or L2 level of the bony spine. Consequently, the lumbar nerve roots follow a long intraspinal course and can be injured anywhere from the upper lumbar spine to their exit at the intervertebral foramen. For example, it is common for

disk herniation at the L4-L5 level to produce compression of the S1 nerve root ([Fig. 16-3](#)). In contrast, cervical nerve roots follow a short intraspinal course and exit at the level of their respective spinal cord segments (upper cervical) or one segment below the corresponding levels (lower cervical cord). Cervical spine pathology can result in spinal cord compression, but lumbar spine pathology cannot.

Pain-sensitive structures in the spine include the vertebral body periosteum, dura, facet joints, annulus fibrosus of the intervertebral disk, epidural veins, and the posterior longitudinal ligament. Damage to these nonneural structures may cause pain. The nucleus pulposus of the intervertebral disk is not pain-sensitive under normal circumstances. Pain sensation is conveyed by the sinuvertebral nerve that arises from the spinal nerve at each spine segment and reenters the spinal canal through the intervertebral foramen at the same level. Disease of these diverse pain-sensitive spine structures may explain many cases of back pain without nerve root compression. The lumbar and cervical spine possess the greatest potential for movement and injury.

Approach to the Patient

Types of Back Pain An understanding of the nature of the pain as described by the patient is the essential first step in evaluation. Attention is also focused on identification of risk factors for serious underlying diseases that require specific evaluation.

Local pain is caused by stretching of pain-sensitive structures that compress or irritate sensory nerve endings. The site of the pain is near the affected part of the back.

Pain referred to the back may arise from abdominal or pelvic viscera. The pain is usually described as primarily abdominal or pelvic but is accompanied by back pain and usually unaffected by posture. The patient may occasionally complain of back pain only.

Pain of spine origin may be located in the back or referred to the buttocks or legs. Diseases affecting the upper lumbar spine tend to refer pain to the lumbar region, groin, or anterior thighs. Diseases affecting the lower lumbar spine tend to produce pain referred to the buttocks, posterior thighs, or rarely the calves or feet. Provocative injections into pain-sensitive structures of the spine (diskography) may produce leg pain that does not follow a dermatomal distribution. The exact pathogenesis of this "sclerotomal" pain is unclear, but it may explain many instances in which combined back and leg pain is unaccompanied by evidence of nerve root compression.

Radicular back pain is typically sharp and radiates from the spine to the leg within the territory of a nerve root (see "Lumbar Disk Disease," below). Coughing, sneezing, or voluntary contraction of abdominal muscles (lifting heavy objects or straining at stool) may elicit the radiating pain. The pain may increase in postures that stretch the nerves and nerve roots. Sitting stretches the sciatic nerve (L5 and S1 roots) because the nerve passes posterior to the hip. The femoral nerve (L2, L3, and L4 roots) passes anterior to the hip and is not stretched by sitting. The description of the pain alone often fails to distinguish clearly between sclerotomal pain and radiculopathy.

Pain associated with muscle spasm, although of obscure origin, is commonly associated with many spine disorders. The spasms are accompanied by abnormal posture, taut

paraspinal muscles, and dull pain.

Back pain at rest or unassociated with specific postures should raise the index of suspicion for an underlying serious cause (e.g., spine tumor, fracture, infection, or referred pain from visceral structures). Knowledge of the circumstances associated with the onset of back pain is important when weighing possible serious underlying causes for the pain. Some patients involved in accidents or work-related injuries may exaggerate their pain for the purpose of compensation or for psychological reasons.

Examination of the Back A physical examination that includes the abdomen and rectum is advisable. Back pain referred from visceral organs may be reproduced during palpation of the abdomen (pancreatitis, abdominal aortic aneurysm) or percussion over the costovertebral angles (pyelonephritis, adrenal disease, L1-L2 transverse process fracture).

The normal spine ([Fig. 16-2](#)) displays a thoracic kyphosis, lumbar lordosis, and cervical lordosis. Exaggeration of these normal alignments may result in hyperkyphosis (lameback) of the thoracic spine or hyperlordosis (swayback) of the lumbar spine. Spasm of lumbar paraspinal muscles results in flattening of the usual lumbar lordosis. Inspection may reveal lateral curvature of the spine (scoliosis) or an asymmetry in the appearance of the paraspinal muscles, suggesting muscle spasm. Taut paraspinal muscles limit the motion of the lumbar spine. Back pain of bony spine origin is often reproduced by palpation or percussion over the spinous process of the affected vertebrae.

Forward bending is frequently limited by paraspinal muscle spasm. Flexion of the hips is normal in patients with lumbar spine disease, but flexion of the lumbar spine is limited and sometimes painful. Lateral bending to the side opposite the injured spinal element may stretch the damaged tissues, worsen pain, and limit motion. Hyperextension of the spine (with the patient prone or standing) is limited when nerve root compression or bony spine disease is present.

Pain from hip disease may mimic the pain of lumbar spine disease. The first movement is typically internal rotation of the hip. Manual internal and external rotation at the hip with the knee and hip in flexion (Patrick sign) may reproduce the pain, as may percussion of the heel (of an outstretched leg) with the palm of the examiner's hand.

In the supine position passive flexion of the thigh on the abdomen while the knee is extended produces stretching of the L5 and S1 nerve roots and the sciatic nerve because the nerve passes posterior to the hip. Passive dorsiflexion of the foot during the maneuver adds to the stretch. While flexion to at least 80° is normally possible without causing pain, tight hamstrings commonly limit motion, may result in pain, and are readily identified by the patient. This *straight leg-raising (SLR) sign* is positive if the maneuver reproduces the patient's usual back or limb pain. Eliciting the SLR sign in the sitting position may help determine if the finding is reproducible. The patient may describe pain in the low back, buttocks, posterior thigh, or lower leg, but the key feature is reproduction of the patient's usual pain. The *crossed SLR* sign is positive when performance of the maneuver on one leg reproduces the patient's pain symptoms in the opposite leg or buttocks. The nerve or nerve root lesion is always on the side of the

pain. The *reverse* SLR sign is elicited by standing the patient next to the examination table and passively extending each leg while the patient continues to stand. This maneuver stretches the L2-L4 nerve roots and the femoral nerve because the nerves pass anterior to the hip. The reverse SLR test is positive if the maneuver reproduces the patient's usual back or limb pain.

The neurologic examination includes a search for weakness, muscle atrophy, focal reflex changes, diminished sensation in the legs, and signs of spinal cord injury. Findings with specific nerve root lesions are shown in [Table 16-1](#) and are discussed below.

Laboratory Studies Routine laboratory studies such as a complete blood count, erythrocyte sedimentation rate, chemistry panel, and urinalysis are rarely needed for the initial evaluation of acute (<3 months), nonspecific, low back pain. If risk factors for a serious underlying disease are present, then laboratory studies (guided by the history and examination) are indicated ([Fig. 16-6B](#)).

Plain films of the lumbar or cervical spine are helpful when risk factors for vertebral fracture (trauma, chronic steroid use) are present. *In the absence of risk factors, routine x-rays of the lumbar spine in the setting of acute, nonspecific, low back pain are expensive and rarely helpful.* Magnetic resonance imaging (MRI) and computed tomography (CT)-myelography have emerged as the radiologic tests of choice for evaluation of most serious diseases involving the spine. In general, the definition of soft tissue structures by MRI is superior, whereas CT-myelography provides optimal imaging of bony lesions in the region of the lateral recess and intervertebral foramen and is tolerated by claustrophobic patients. With rare exceptions, conventional myelography and bone scan are inferior to MRI and CT-myelography.

Electromyography (EMG) can be used to assess the functional integrity of the peripheral nervous system ([Chap. 357](#)) in the setting of back pain. Sensory nerve conduction studies are normal when focal sensory loss is due to nerve root damage because the nerve roots are proximal to the nerve cell bodies in the dorsal root ganglia. The diagnostic yield of needle EMG is higher than that of nerve conduction studies for radiculopathy. Denervation changes in a myotomal (segmental) distribution are detected by sampling multiple muscles supplied by different nerve roots and nerves; the pattern of muscle involvement indicates the nerve root(s) responsible for the injury. Needle EMG provides objective information about motor nerve fiber injury when the clinical evaluation of weakness is limited by pain or poor effort. EMG and nerve conduction studies will be normal when only limb pain or sensory nerve root injury or irritation is present. Mixed nerve somatosensory evoked potentials and F-wave studies are of uncertain value in the evaluation of radiculopathy.

CAUSES OF BACK PAIN

CONGENITAL ANOMALIES OF THE LUMBAR SPINE

Spondylolysis is a bony defect in the pars interarticularis (a segment near the junction of the pedicle with the lamina) of the vertebra; the etiology of the defect may be a stress fracture in a congenitally abnormal segment. The defect (usually bilateral) is best

visualized on oblique projections in plain x-rays or by [CT](#) scan and occurs in the setting of a single injury, repeated minor injuries, or growth.

Spondylolisthesis is the anterior slippage of the vertebral body, pedicles, and superior articular facets, leaving the posterior elements behind. Spondylolisthesis is associated with spondylolysis and degenerative spine disease and occurs more frequently in women. The slippage may be asymptomatic but may also cause low back pain, nerve root injury (the L5 root most frequently), or symptomatic spinal stenosis. Tenderness may be elicited near the segment that has "slipped" forward (most often L4 on L5 or occasionally L5 on S1). A "step" may be present on deep palpation of the posterior elements of the segment above the spondylolisthetic joint. The trunk may be shortened and the abdomen protuberant as a result of extreme forward displacement of L4 on L5 in severe degrees of spondylolisthesis. In these cases, cauda equina syndrome may occur ([Chap. 368](#)).

TRAUMA

Trauma is an important cause of acute low back pain. A patient complaining of back pain and inability to move the legs may have a spinal fracture or dislocation, and, with fractures above L1, spinal cord compression. In such cases care must be taken to avoid further damage to the spinal cord or nerve roots. The back should be immobilized pending results of plain x-rays.

Sprains and Strains The terms *low back sprain*, *strain*, or *mechanically induced muscle spasm* are used for minor, self-limited injuries associated with lifting a heavy object, a fall, or a sudden deceleration such as occurs in an automobile accident. These terms are used loosely and do not clearly describe a specific anatomic lesion. The pain is usually confined to the lower back, and there is no radiation to the buttocks or legs. Patients with low back pain and paraspinal muscle spasm often assume unusual postures.

Vertebral Fractures Most traumatic fractures of the lumbar vertebral bodies result from compression or flexion injuries producing anterior wedging or compression. With more severe trauma, the patient may sustain a fracture-dislocation or a "burst" fracture involving not only the vertebral body but posterior elements as well. Traumatic vertebral fractures are caused by falls from a height (a pars interarticularis fracture of the L5 vertebra is common), sudden deceleration in an automobile accident, or direct injury. Neurologic impairment is commonly associated with these injuries, and early surgical treatment is indicated ([Chap. 369](#)).

When fractures are atraumatic, the bone is presumed to be weakened by a pathologic process. The cause is usually postmenopausal (type 1) or senile (type 2) osteoporosis ([Chap. 342](#)). Underlying systemic disorders such as osteomalacia, hyperparathyroidism, hyperthyroidism, multiple myeloma, metastatic carcinoma, or glucocorticoid use may also weaken the vertebral body. The clinical context, neurologic signs, and x-ray appearance of the spine establish the diagnosis. Antiresorptive drugs including biphosphonates, alendronate, transdermal estrogen, and tamoxifen have been shown to reduce the risk of osteoporotic fractures.

LUMBAR DISK DISEASE

This disorder is a common cause of chronic or recurrent low back and leg pain. Disk disease is most likely to occur at the L4-L5 and L5-S1 levels, but upper lumbar levels are involved occasionally. The cause of the disk injury is often unknown; the risk is increased in overweight individuals. Degeneration of the nucleus pulposus and the annulus fibrosus increases with age and may be asymptomatic or painful. A sneeze, cough, or trivial movement may cause the nucleus pulposus to prolapse, pushing the frayed and weakened annulus posteriorly. In severe disk disease, the nucleus may protrude through the annulus (herniation) or become extruded to lie as a free fragment in the spinal canal.

The mechanism by which intervertebral disk injury causes back pain is controversial. The inner annulus fibrosus and nucleus pulposus are normally devoid of innervation. Inflammation and production of proinflammatory cytokines within the protruding or ruptured disk may trigger or perpetuate back pain. Ingrowth of nociceptive (pain) nerve fibers into inner portions of diseased intervertebral disk may be responsible for chronic "diskogenic" pain. Nerve root injury (*radiculopathy*) from disk herniation may be due to compression, inflammation, or both; pathologically, varying degrees of demyelination and axonal loss are usually present.

The symptoms of a ruptured intervertebral disk include back pain, abnormal posture, limitation of spine motion (particularly flexion), or radicular pain. A dermatomal pattern of sensory loss or a reduction in or loss of a deep tendon reflex is more suggestive of a specific root lesion than the pattern of pain. Motor findings (focal weakness, muscle atrophy, or fasciculations) occur less frequently than sensory or reflex changes, but a myotomal pattern of involvement can suggest specific nerve root injury. Lumbar disk disease is usually unilateral ([Fig. 16-4](#)), but bilateral involvement does occur with large central disk herniations that compress several nerve roots at the same level. Clinical manifestations of specific lumbosacral nerve root lesions are summarized in [Table 16-1](#). There is evidence to suggest that lumbar disk herniation with a nonprogressive nerve root deficit can be managed conservatively (i.e., nonsurgically) with a successful outcome. The size of the disk protrusion may naturally decrease over time.

Degeneration of the intervertebral disk without frank extrusion of disk tissue may give rise to low back pain only. There may be referred pain in the leg, buttock, or hip with little or no discomfort in the back and no signs of nerve root involvement. Lumbar disk syndromes are usually unilateral, but large central disk herniations can cause bilateral symptoms and signs and may produce a cauda equina syndrome.

Breakaway weakness describes a variable power of muscle contraction by a patient who is asked to provide maximal effort. The weakness may be due to pain or a combination of pain and underlying true weakness. Breakaway weakness without pain is due to lack of effort; patients who exhibit breakaway weakness should be asked if testing a specific muscle is painful. In uncertain cases, [EMG](#) can determine whether or not true weakness is present.

The differential diagnosis of lumbar disk disease includes a variety of serious and treatable conditions, including epidural abscess, hematoma, or tumor. Fever, constant

pain uninfluenced by position, sphincter abnormalities, or signs of spinal cord disease suggest an etiology other than lumbar disk disease. Bilateral absence of ankle reflexes can be a normal finding in old age or a sign of bilateral S1 radiculopathy. An absent deep tendon reflex or focal sensory loss may reflect injury to a nerve root, but other sites of injury along the nerve must also be considered. For example, an absent knee reflex may be due to a femoral neuropathy rather than an L4 nerve root injury. A focal decrease in sensation over the foot and distal lateral calf may result from a peroneal or lateral sciatic neuropathy rather than an L5 nerve root injury. Focal muscle atrophy may reflect loss of motor axons from a nerve root or peripheral nerve injury, an anterior horn cell disease, or disuse.

An [MRI](#) scan or [CT](#)-myelogram is necessary to establish the location and type of pathology. Simple MRI yields exquisite views of intraspinal and adjacent soft tissue anatomy and is more likely to establish a specific anatomic diagnosis than plain films or myelography. Bony lesions of the lateral recess or intervertebral foramen may be seen with optimal clarity on CT-myelographic studies.

The correlation of neuroradiologic findings to symptoms, particularly pain, is often problematic. As examples, contrast-enhancing tears in the annulus fibrosus or disk protrusions are widely accepted as common sources of back pain. However, one recent study found that over half of asymptomatic adults have annular tears on lumbar spine MR imaging, nearly all of which demonstrate contrast enhancement. Furthermore, asymptomatic disk protrusions are common in adults, and many of these abnormalities enhance with contrast. These observations strongly suggest that MRI findings of disk protrusion, tears in the annulus fibrosus, or contrast enhancement are common incidental findings that by themselves should not dictate management decisions for patients with back pain. The presence or absence of persistent disk herniation 10 years after surgical or conservative treatment has no bearing on a successful clinical outcome.

There are four indications for intervertebral disk surgery: (1) progressive motor weakness from nerve root injury demonstrated on clinical examination or [EMG](#), (2) bowel or bladder disturbance or other signs of spinal cord disease, (3) incapacitating nerve root pain despite conservative treatment for at least 4 weeks, and (4) recurrent incapacitating pain despite conservative treatment. The latter two criteria are more subjective and less well established than the others. Surgical treatment should also be considered if the pain and/or neurologic findings do not substantially improve over 4 to 12 weeks.

Surgery is preceded by MRI scan or CT-myelogram to define the location and type of pathology. The usual surgical procedure is a partial hemilaminectomy with excision of the involved and prolapsed intervertebral disk. Arthrodesis of the involved lumbar segments is considered only in the presence of significant spinal instability (i.e., degenerative spondylolisthesis or isthmic spondylolysis).

OTHER CAUSES OF LOW BACK PAIN

Spinal stenosis is an anatomic diagnosis reflecting a narrowed lumbar or cervical spinal canal. Classic *neurogenic claudication* occurs in the setting of moderate to severe spinal stenosis and typically consists of back and buttock or leg pain induced by walking or

standing. The pain is relieved by sitting. Symptoms in the legs are usually bilateral. Focal weakness, sensory loss, or reflex changes may occur when associated with radiculopathy. Unlike vascular claudication, the symptoms are often provoked by standing without walking. Unlike lumbar disk disease, the symptoms are usually relieved by sitting. Severe neurologic deficits, including paralysis and urinary incontinence, occur rarely. Spinal stenosis usually results from acquired (75%), congenital, or mixed acquired/congenital factors. Congenital forms (achondroplasia, idiopathic) are characterized by short, thick pedicles that produce both spinal canal and lateral recess stenosis. Acquired factors that may contribute to spinal stenosis include degenerative diseases (spondylosis, spondylolisthesis, scoliosis), trauma, spine surgery (postlaminectomy, fusion), metabolic or endocrine disorders (epidural lipomatosis, osteoporosis, acromegaly, renal osteodystrophy, hypoparathyroidism), and Paget's disease. [MRI](#) or [CT](#)-myelography provide the best definition of the abnormal anatomy ([Fig. 16-5](#)).

Conservative treatment includes nonsteroidal anti-inflammatory drugs (NSAIDs), exercise programs, and symptomatic treatment of acute pain exacerbations. Surgical therapy is considered when medical therapy does not relieve pain sufficiently to allow for activities of daily living or when significant focal neurologic signs are present. Between 65 and 80% of properly selected patients treated surgically experience >75% relief of back and leg pain. Up to 25% develop recurrent stenosis at the same spinal level or an adjacent level 5 years after the initial surgery; recurrent symptoms usually respond to a second surgical decompression.

Facet joint hypertrophy can produce unilateral radicular symptoms, due to bony compression, that are indistinguishable from disk-related radiculopathy. Patients may exhibit stretch signs, focal motor weakness, hyporeflexia, or sensory loss. Hypertrophic superior or inferior facets can often be visualized radiologically. Foraminotomy results in long-term relief of leg and back pain in 80 to 90% of patients.

Lumbar adhesive arachnoiditis with radiculopathy is the result of a fibrotic process following an inflammatory response to local tissue injury within the subarachnoid space. The fibrosis results in nerve root adhesions, producing back and leg pain associated with motor, sensory, and reflex changes. Myelography-induced arachnoiditis has become rare with the abandonment of oil-based contrast. Other causes of arachnoiditis include multiple lumbar operations, chronic spinal infections, spinal cord injury, intrathecal hemorrhage, intrathecal injection of steroids and anesthetics, and foreign bodies. The spine [MRI](#) appearance of arachnoiditis includes nerve roots clumping together centrally and adherent to the dura peripherally, or loculations of cerebrospinal fluid (CSF) within the thecal sac that obscure nerve root visualization. Treatment is often unsatisfactory. Microsurgical lysis of adhesions, dorsal rhizotomy, and dorsal root ganglionectomy have resulted in poor outcomes. Dorsal column stimulation for pain relief has produced varying results. Epidural steroid injections have been of limited value.

ARTHRITIS

Arthritis is a major cause of spine pain.

Spondylosis Osteoarthritic spine disease typically occurs in later life and primarily involves the cervical and lumbosacral spine. Patients often complain of back pain that is increased by motion and associated with stiffness or limitation of motion. The relationship between clinical symptoms and radiologic findings is usually not straightforward. Pain may be prominent when x-ray findings are minimal; alternatively, large osteophytes can be seen in asymptomatic patients in middle and later life. Hypertrophied facets and osteophytes may compress nerve roots in the lateral recess or intervertebral foramen. Osteophytes arising from the vertebral body may cause or contribute to central spinal canal stenosis. Loss of intervertebral disk height reduces the vertical dimensions of the intervertebral foramen; the descending pedicle may compress the nerve root exiting at that level. Osteoarthritic changes in the lumbar spine may rarely compress the cauda equina.

Ankylosing Spondylitis (See also [Chap. 315](#)) This distinctive arthritic spine disease typically presents with the insidious onset of low back and buttock pain. Patients are often males below age 40. Associated features include morning back stiffness, nocturnal pain, pain unrelieved by rest, an elevated sedimentation rate, and the histocompatibility antigen HLA-B27. The differential diagnosis includes tumor and infection. Onset at a young age and back pain characteristically improving with exercise suggest ankylosing spondylitis. Loss of the normal lumbar lordosis and exaggeration of thoracic kyphosis are seen as the disease progresses. Inflammation and erosion of the outer fibers of the annulus fibrosus at the point of contact with the vertebral body are followed by ossification and bone growth. Bony growth (syndesmophyte) bridges adjacent vertebral bodies and results in reduced spine mobility in all planes. The radiologic hallmarks of the disease are periarticular destructive changes, sclerosis of the sacroiliac joints, and bridging of vertebral bodies by bone to produce the fused "bamboo spine." Similar restricted movement may accompany Reiter's syndrome, psoriatic arthritis, and chronic inflammatory bowel disease. Stress fractures through the spontaneously ankylosed posterior bony elements of the rigid, osteoporotic spine may result in focal spine pain, spinal cord compression or cauda equina syndrome. Occasional atlantoaxial subluxation with spinal cord compression occurs. Bilateral ankylosis of the ribs to the spine and a decrease in the height of axial thoracic structures may cause marked impairment of respiratory function.

OTHER DESTRUCTIVE DISEASES

Neoplasm (See also [Chap. 370](#)) Back pain is the most common neurologic symptom among patients with systemic cancer. One-third of patients with undiagnosed back or neck pain and known systemic cancer have epidural extension or metastasis of tumor, and one-third have pain associated with vertebral metastases alone. About 11% have back pain unrelated to metastatic disease. Metastatic carcinoma (breast, lung, prostate, thyroid, kidney, gastrointestinal tract), multiple myeloma, and non-Hodgkin's and Hodgkin's lymphomas frequently involve the spine. Back pain may be the presenting symptom because the primary tumor site may be overlooked or asymptomatic. The pain tends to be constant, dull, unrelieved by rest, and worse at night. In contrast, mechanical low back pain is usually improved with rest. Plain x-rays usually, though not always, show destructive lesions in one or several vertebral bodies without disk space involvement. [MRI](#) or [CT](#)-myelography are the studies of choice in the setting of suspected spinal metastasis, but the trend of evidence favors the use of MRI. The procedure of

choice is the study most rapidly available because the patient may worsen during a diagnostic delay.

Infection *Vertebral osteomyelitis* is usually caused by staphylococci, but other bacteria or the tubercle bacillus (Pott's disease) may be the responsible organism. A primary source of infection, most often from the urinary tract, skin, or lungs, can be identified in 40% of patients. Intravenous drug use is a well-recognized risk factor. Back pain exacerbated by motion and unrelieved by rest, spine tenderness over the involved spine segment, and an elevated erythrocyte sedimentation rate are the most common findings. Fever or elevated white blood cell count are found in a minority of patients. Plain radiographs may show a narrowed disk space with erosion of adjacent vertebrae; these diagnostic changes may take weeks or months to appear. [MRI](#) and [CT](#) are sensitive and specific for osteomyelitis; MRI definition of soft tissue detail is exquisite. CT scan may be more readily available and better tolerated by some patients with severe back pain.

Spinal epidural abscess ([Chap. 368](#)) presents with back pain (aggravated by palpation or movement) and fever. The patient may exhibit nerve root injury or spinal cord compression accompanied by a sensory level, incontinence, or paraplegia. The abscess may track over multiple spinal levels and is best delineated by spine [MRI](#).

Osteoporosis and Osteosclerosis Considerable loss of bone may occur with or without symptoms in association with medical disorders, including hyperparathyroidism, chronic glucocorticoid use, or immobilization. Compression fractures occur in up to half of patients with severe osteoporosis. The risk of osteoporotic vertebral fracture is 4.5 times greater over 3 years among patients with a baseline fracture compared with osteoporotic controls. The sole manifestation of a compression fracture may be focal lumbar or thoracic aching (often after a trivial injury) that is exacerbated by movement. Other patients experience thoracic or upper lumbar radicular pain. Focal spine tenderness is common. When compression fractures are found, treatable risk factors should be sought. Compression fractures above the midthoracic region suggest malignancy.

Osteosclerosis is readily identifiable on routine x-ray studies (e.g., Paget's disease) and may or may not produce back pain. Spinal cord or nerve root compression may result from bony encroachment on the spinal canal or intervertebral foramina. Single dual-beam photon absorptiometry or quantitative [CT](#) can be used to detect small changes in bone mineral density. **For further discussion of these bone disorders, see [Chaps. 341 to 343](#).*

REFERRED PAIN FROM VISCERAL DISEASE

Diseases of the pelvis, abdomen, or thorax may produce referred pain to the posterior portion of the spinal segment that innervates the diseased organ. Occasionally, back pain may be the first and only sign. In general, pelvic diseases refer pain to the sacral region, lower abdominal diseases to the lumbar region (around the second to fourth lumbar vertebrae), and upper abdominal diseases to the lower thoracic or upper lumbar region (eighth thoracic to the first and second lumbar vertebrae). Local signs (pain with spine palpation, paraspinal muscle spasm) are absent, and minimal or no pain

accompanies normal spine movements.

Low Thoracic and Upper Lumbar Pain in Abdominal Disease Peptic ulcer or tumor of the posterior stomach or duodenum typically produces epigastric pain ([Chaps. 285 and 90](#)), but midline back or paraspinal pain may occur if retroperitoneal extension is present. Back pain due to peptic ulcer may be precipitated by ingestion of an orange, alcohol, or coffee and relieved by food or antacids. Fatty foods are more likely to induce back pain associated with biliary disease. Diseases of the pancreas may produce back pain to the right of the spine (head of the pancreas involved) or to the left (body or tail involved). Pathology in retroperitoneal structures (hemorrhage, tumors, pyelonephritis) may produce paraspinal pain with radiation to the lower abdomen, groin, or anterior thighs. A mass in the iliopsoas region often produces unilateral lumbar pain with radiation toward the groin, labia, or testicle. The sudden appearance of lumbar pain in a patient receiving anticoagulants suggests retroperitoneal hemorrhage.

Isolated low back pain occurs in 15 to 20% of patients with a contained rupture of an abdominal aortic aneurysm (AAA). The classic clinical triad of abdominal pain, shock, and back pain in an elderly man occurs in fewer than 20% of patients. Two of these three features are present in two-thirds of patients, and hypotension is present in half. Ruptured AAA has a high mortality rate; the typical patient is an elderly male smoker with back pain. The diagnosis is initially missed in at least one-third of patients because the symptoms and signs can be nonspecific. Common misdiagnoses include nonspecific back pain, diverticulitis, renal colic, sepsis, and myocardial infarction. A careful abdominal examination revealing a pulsatile mass (present in 50 to 75% of patients) is an important physical finding.

Lumbar Pain with Lower Abdominal Diseases Inflammatory bowel disorders (colitis, diverticulitis) or colonic neoplasms may produce lower abdominal pain, midlumbar back pain, or both. The pain may have a beltlike distribution around the body. A lesion in the transverse or initial descending colon may refer pain to the middle or left back at the L2-L3 level. Sigmoid colon disease may refer pain to the upper sacral or midline suprapubic regions or left lower quadrant of the abdomen.

Sacral Pain in Gynecologic and Urologic Disease Pelvic organs rarely cause low back pain, except for gynecologic disorders involving the uterosacral ligaments. The pain is referred to the sacral region. Endometriosis or uterine carcinoma may invade the uterosacral ligaments; malposition of the uterus may cause uterosacral ligament traction. The pain associated with endometriosis begins during the premenstrual phase and often continues until it merges with menstrual pain. Malposition of the uterus (retroversion, descensus, and prolapse) may lead to sacral pain after standing for several hours.

Menstrual pain may be felt in the sacral region. The poorly localized, cramping pain can radiate down the legs. Other pelvic sources of low back pain include neoplastic invasion of pelvic nerves, radiation necrosis, and pregnancy. Pain due to neoplastic infiltration of nerves is typically continuous, progressive in severity, and unrelieved by rest at night. Radiation therapy of pelvic tumors may produce sacral pain from late radiation necrosis of tissue or nerves. Low back pain with radiation into one or both thighs is common in the last weeks of pregnancy.

Urologic sources of lumbosacral back pain include chronic prostatitis, prostate carcinoma with spinal metastasis, and diseases of the kidney and ureter. Lesions of the bladder and testes do not usually produce back pain. The diagnosis of metastatic prostate carcinoma is established by rectal examination, spine imaging studies ([MRI](#) or [CT](#)), and measurement of prostate-specific antigen (PSA) ([Chap. 95](#)). Infectious, inflammatory, or neoplastic renal diseases may result in ipsilateral lumbosacral pain, as can renal artery or vein thrombosis. Ureteral obstruction due to renal stones may produce paraspinal lumbar pain.

Postural Back Pain There is a group of patients with chronic, nonspecific low back pain in whom no anatomic or pathologic lesion can be found despite exhaustive investigation. These individuals complain of vague, diffuse back pain with prolonged sitting or standing that is relieved by rest. The physical examination is unrevealing except for "poor posture." Imaging studies and laboratory evaluations are normal. Exercises to strengthen the paraspinal and abdominal muscles are sometimes therapeutic.

Psychiatric Disease Chronic low back pain ([CLBP](#)) may be encountered in patients with compensation hysteria, malingering, substance abuse, chronic anxiety states, or depression. Many patients with CLBP have a history of psychiatric illness (depression, anxiety, substance abuse) or childhood trauma (physical or sexual abuse) that antedates the onset of back pain. Preoperative psychological assessment has been used to exclude patients with marked psychological impairment who are at high risk for a poor surgical outcome. It is important to be certain that the back pain in these patients does not represent serious spine or visceral pathology in addition to the impaired psychological state.

Unidentified The cause of low back pain occasionally remains unclear. Some patients have had multiple operations for disk disease but have persistent pain and disability. The original indications for surgery may have been questionable with back pain only, no definite neurologic signs, or a minor disk bulge noted on [CT](#) or [MRI](#). Scoring systems based upon neurologic signs, psychological factors, physiologic studies, and imaging studies have been devised to minimize the likelihood of unsuccessful surgical explorations and to avoid selection of patients with psychological profiles that predict poor functional outcomes.

TREATMENT

Acute Low Back Pain A practical approach to the management of low back pain is to consider acute and chronic presentations separately. [ALBP](#) is defined as pain of less than 3 months' duration. Full recovery can be expected in 85% of adults with ALBP unaccompanied by leg pain. Most of these patients exhibit "mechanical" symptoms -- pain that is aggravated by motion and relieved by rest.

Observational, population-based studies have been used to justify a minimalist approach to individual patient care. These studies share a number of limitations: (1) a true placebo control group is often lacking; (2) patients who consult different provider groups (generalists, orthopedists, neurologists) are assumed to have similar etiologies

for their back pain; (3) no information is provided about the details of treatment within each provider group or between provider groups; and (4) no attempt to tabulate serious causes of [ALBP](#) is made. The appropriateness of specific diagnostic procedures or therapeutic interventions for low back pain cannot be assessed from these studies.

The proposed algorithms ([Fig. 16-6](#)) for management of [ALBP](#) in adults draw considerably from published guidelines. However, it must be emphasized that current [CPGs](#) for the treatment of low back pain are based on incomplete evidence -- for example, there is a paucity of well-designed studies documenting the natural history of disk lesions associated with a focal neurologic deficit. Guidelines should not substitute for sound clinical judgment.

The initial assessment excludes serious causes of spine pathology that require urgent intervention, including infection, cancer, and trauma. Risk factors for a possible serious underlying cause of back pain include: age > 50 years, prior diagnosis of cancer or other serious medical illness, bed rest without relief, duration of pain >1 month, urinary incontinence or recent nocturia, focal leg weakness or numbness, pain radiating into the leg(s) from the back, intravenous drug use, chronic infection (pulmonary or urinary), pain increasing with standing and relieved by sitting, history of spine trauma, and glucocorticoid use. Clinical signs associated with a possible serious etiology include unexplained fever, well-documented and unexplained weight loss, positive [SLR](#) sign or reverse SLR sign, crossed SLR sign, percussion tenderness over the spine or costovertebral angle, an abdominal mass (pulsatile or nonpulsatile), a rectal mass, focal sensory loss (saddle anesthesia or focal limb sensory loss), true leg weakness, spasticity, and asymmetric leg reflexes. Laboratory studies are unnecessary unless a serious underlying cause ([Fig. 16-6](#), Algorithms A and B) is suspected. Plain spine films are rarely indicated in the first month of symptoms unless a spine fracture is suspected.

The roles of bed rest, early exercise, and traction in the treatment of acute uncomplicated low back pain have been the subject of recent prospective studies. Clinical trials fail to demonstrate any benefit of prolonged (>2 days) bed rest for [ALBP](#). There is evidence that bed rest is also ineffective for patients with sciatica or for acute back pain with findings of nerve root injury. Theoretical advantages of early ambulation for ALBP include maintenance of cardiovascular conditioning, improved disk and cartilage nutrition, improved bone and muscle strength, and increased endorphin levels. A recent trial did not show benefit from an early vigorous exercise program, but the benefits of less vigorous exercise or other exercise programs remain unknown. The early resumption of normal physical activity (without heavy manual labor) is likely to be beneficial. Well-designed clinical studies of traction that include a sham traction group have failed to show a benefit of traction for ALBP. Despite this knowledge, one survey of physicians' perceptions of effective treatment identified strict bed rest for >3 days, trigger point injections (see below), and physical therapy (PT) as beneficial for more than 50% of patients with ALBP. In many instances, the behavior of treating physicians does not reflect the current medical literature.

Proof is lacking to support the treatment of acute back and neck pain with acupuncture, transcutaneous electrical nerve stimulation, massage, ultrasound, diathermy, or electrical stimulation. Cervical collars can be modestly helpful by limiting spontaneous and reflex neck movements that exacerbate pain. Evidence regarding the efficacy of ice

or heat is lacking, but these interventions are optional given the lack of negative evidence, low cost, and low risk. Biofeedback has not been studied rigorously. Facet joint, trigger point, and ligament injections are not recommended in the treatment of [ALBP](#).

A beneficial role for specific exercises or modification of posture has not been validated by rigorous clinical studies. As a practical matter, temporary suspension of activity known to increase mechanical stress on the spine (heavy lifting, prolonged sitting, bending or twisting, straining at stool) may be helpful.

Patient education is an important part of treatment. Studies reveal that patient satisfaction and the likelihood of follow-up increase when patients are educated about prognosis, treatment methods, activity modifications, and strategies to prevent future exacerbations. In one study, patients who felt they did not receive an adequate explanation for their symptoms wanted more diagnostic tests. Evidence for the efficacy of structured education programs ("back school") is inconclusive; in one controlled study, patients attending back school had a shorter duration of sick leave during the initial episode but not during subsequent episodes. Recent large, controlled, randomized studies of back school for primary prevention of low back injury and pain have failed to demonstrate a benefit.

Medications used in the treatment of [ALBP](#) include [NSAIDs](#), acetaminophen, muscle relaxants, and opioids. NSAIDs are superior to placebo for back pain relief. Acetaminophen is superior to placebo in the treatment of other types of pain but has not been compared against placebo for low back pain. Muscle relaxants provide short-term (4 to 7 days) benefit compared with placebo, but drowsiness often limits their daytime use. The efficacy of muscle relaxants compared to NSAIDs or in combination with NSAIDs is unclear. Opioid analgesics have not been shown to be more effective than NSAIDs or acetaminophen for relief of ALBP or likelihood of return to work. Short-term use of opioids in selected patients unresponsive to or intolerant of acetaminophen or NSAIDs may be helpful. There is no evidence to support the use of oral glucocorticoids or tricyclic antidepressants in treatment of ALBP.

The role of diagnostic and therapeutic nerve root blocks for patients with acute back or neck pain remains controversial. Equivocal data suggests that epidural steroids may occasionally produce short-term pain relief in patients with [ALBP](#) and radiculopathy, but proof is lacking for pain relief beyond 1 month. Epidural anesthetics, steroids, or opioids are not indicated as initial treatment for ALBP without radiculopathy. Diagnostic selective nerve root blocks have been advocated to determine if pain originates from a nerve root. However, these studies may be falsely positive due to a placebo effect, in patients with a painful lesion located distally along the peripheral nerve, or from anesthesia of the sinuvertebral nerve. Therapeutic selective nerve root blocks are an option after brief conservative measures fail, particularly when temporary relief of pain may be important for patient function. Needle position is confirmed under fluoroscopic guidance with nonionic contrast before injection of glucocorticoid and local anesthetic.

A short course of spinal manipulation or [PT](#) for symptomatic relief of uncomplicated [ALBP](#) is an option. A prospective, randomized study comparing PT, chiropractic manipulation, and education interventions for patients with ALBP found

modest trends toward benefit with both PT and chiropractic manipulation at 1 year. Costs per year were equivalent in the PT/chiropractic group and ~\$280 less for the group treated with the education booklet alone. The extent to which this modest improvement in symptoms and outcome is worth the cost must be determined for each patient. Extended duration of treatment or treatment of patients with radiculopathy is of unknown value and carries potential risk. The appropriate frequency or duration of spinal manipulation has not been addressed adequately.

Chronic Low Back Pain [CLBP](#) is defined as pain lasting longer than 12 weeks. Patients with CLBP account for 50% of back pain costs. Overweight individuals appear to be at particular risk. Other risk factors include: female gender, older age, prior history of back pain, restricted spinal mobility, pain radiating into a leg, high levels of psychological distress, poor self-rated health, minimal physical activity, smoking, job dissatisfaction, and widespread pain. Combinations of these premorbid factors have been used to predict which individuals with [ALBP](#) are likely to develop CLBP. The initial approach to these patients is similar to that for ALBP, and the differential diagnosis of CLBP includes most of the conditions described in this chapter. Treatment of this heterogeneous group of patients is directed toward the underlying cause when possible; the ultimate goal is to restore function to the greatest extent possible.

Many conditions that produce [CLBP](#) can be identified by the combination of neuroimaging and electrophysiologic studies. Spine [MRI](#) or [CT](#)-myelography are the techniques of choice but are generally not indicated within the first month after initial evaluation in the absence of risk factors for a serious underlying cause. Imaging studies should be performed only in circumstances where the results are likely to influence surgical or medical treatment.

Diskography is of questionable value in the evaluation of back pain. No additional anatomic information is provided beyond what is available by [MRI](#). Reproduction of the patient's typical pain with the injection is often used as evidence that a specific disk is the pain generator, but it is not known whether this information has any value in selecting candidates for surgery. There is no proven role for thermography in the assessment of radiculopathy.

The diagnosis of nerve root injury is most secure when the history, examination, results of imaging studies, and the [EMG](#) are concordant. The correlation between [CT](#) and EMG for localization of nerve root injury is between 65 and 73%. Up to one-third of asymptomatic adults have a disk protrusion detected by CT or [MRI](#) scans. Thus, surgical intervention based solely upon radiologic findings and pain increases the likelihood of an unsuccessful outcome.

[CLBP](#) can be treated with a variety of conservative measures. Acute and subacute exacerbations are managed with [NSAIDs](#) and comfort measures. There is no good evidence to suggest that one NSAID is more effective than another. Bed rest should not exceed 2 days. Activity tolerance is the primary goal, while pain relief is secondary. Exercise programs can reverse type II muscle fiber atrophy in paraspinal muscles and strengthen trunk extension. Supervised, intensive physical exercise or "work hardening" regimens (under the guidance of a physical therapist) have been effective in returning some patients to work, improving walking distances, and diminishing pain. The benefit

can be sustained with home exercise regimens; compliance with the exercise regimen strongly influences outcome. The role of manipulation, back school, or epidural steroid injections in the treatment of CLBP is unclear. Up to 30% of "blind" epidural steroid injections miss the epidural space even when performed by an experienced anesthesiologist. There is no strong evidence to support the use of acupuncture or traction in this setting. A reduction in sick leave days, long-term health care utilization, and pension expenditures may offset the initial expense of multidisciplinary treatment programs. In one study comparing 3 weeks of hydrotherapy versus routine ambulatory care, hydrotherapy resulted in diminished duration and intensity of back pain, reduced analgesic drug consumption, improved spine mobility, and improved functional score. Functional score returned to baseline at the 9-month follow-up, but all other beneficial effects were sustained. Percutaneous electrical nerve stimulation (PENS) has been shown to provide significant short-term relief of CLBP, but additional studies regarding long-term efficacy and cost are necessary.

PAIN IN THE NECK AND SHOULDER

Approach to the Patient

In one recent epidemiologic survey, the 6-month prevalence of disabling neck pain was 4.6% among adults. Neck pain commonly arises from diseases of the cervical spine and soft tissues of the neck. Neck pain arising from the cervical spine is typically precipitated by neck movements and may be accompanied by focal spine tenderness and limitation of motion. Pain arising from the brachial plexus, shoulder, or peripheral nerves can be confused with cervical spine disease, but the history and examination usually identify a more distal origin for the pain. Cervical spine trauma, disk disease, or spondylosis may be asymptomatic or painful and can produce a myelopathy, radiculopathy, or both. The nerve roots most commonly affected are C7 and C6.

TRAUMA TO THE CERVICAL SPINE

Unlike injury to the low back, trauma to the cervical spine (fractures, subluxation) places the spinal cord at risk for compression. Motor vehicle accidents, violent crimes, or falls account for 87% of spinal cord injuries, which can have devastating consequences ([Chap. 369](#)). Emergency immobilization of the neck prior to complete assessment is mandatory to minimize further spinal cord injury from movement of unstable cervical spine segments.

Whiplash injury is due to trauma (usually automobile accidents) causing cervical musculoligamentous sprain or strain due to hyperflexion or hyperextension. This diagnosis should not be applied to patients with fractures, disk herniation, head injury, or altered consciousness. One prospective study found that 18% of patients with whiplash injury had persistent injury-related symptoms 2 years after the car accident. Such patients were older, had a higher incidence of inclined or rotated head position at impact, greater intensity of initial neck and head pain, greater number of initial symptoms, and more osteoarthritic changes on cervical spine x-rays at baseline compared to patients who ultimately recovered. Objective data on the pathology of neck soft tissue injuries is lacking. Patients with severe initial injury are at increased risk for poor long-term outcome.

CERVICAL DISK DISEASE

Herniation of a lower cervical disk is a common cause of neck, shoulder, arm, or hand pain. Neck pain (worse with movement), stiffness, and limited range of neck motion are common. With nerve root compression, pain may radiate into a shoulder or arm. Extension and lateral rotation of the neck narrows the intervertebral foramen and may reproduce radicular symptoms (Spurling's sign). In young individuals, acute cervical nerve root compression from a ruptured disk is often due to trauma. Subacute radiculopathy is less likely to be related to a specific traumatic incident and may involve both disk disease and spondylosis. Cervical disk herniations are usually posterolateral near the lateral recess and intervertebral foramen. The usual patterns of reflex, sensory, and motor changes that accompany specific cervical nerve root lesions are listed in [Table 16-2](#). When evaluating patients with suspected cervical radiculopathy it is important to consider the following: (1) overlap in function between adjacent nerve roots is common, (2) the anatomic pattern of pain is the most variable of the clinical features, and (3) the distribution of symptoms and signs may be evident in only part of the injured nerve root territory.

Surgical management of cervical herniated disks usually consists of an anterior approach with discectomy followed by anterior interbody fusion. A simple posterior partial laminectomy with discectomy is an alternative approach. The risk of subsequent radiculopathy or myelopathy at cervical segments adjacent to the fusion is 3% per year and 26% at 10 years. Although the risk is sometimes portrayed as a late complication of cervical surgery, it may also reflect the natural history of degenerative cervical spine disease in this subpopulation of patients.

CERVICAL SPONDYLOSIS

Osteoarthritis of the cervical spine may produce neck pain that radiates into the back of the head, shoulders, or arms. Arthritic or other pathologic conditions of the upper cervical spine may be the source of headaches in the posterior occipital region (supplied by the C2-C4 nerve roots). Cervical spondylosis with osteophyte formation in the lateral recess or hypertrophic facet joints may produce a monoradiculopathy ([Fig. 16-7](#)). Narrowing of the spinal canal by osteophytes, ossification of the posterior longitudinal ligament, or a large central disk may compress the cervical spinal cord. In some patients, a combination of radiculopathy and myelopathy occur. An electrical sensation elicited by neck flexion and radiating down the spine from the neck (Lhermitte's symptom) usually indicates cervical or upper thoracic (T1-T2) spinal cord involvement. When little or no neck pain accompanies the cord compression, the diagnosis may be confused with amyotrophic lateral sclerosis ([Chap. 365](#)), multiple sclerosis ([Chap. 371](#)), spinal cord tumors ([Chap. 368](#)), or syringomyelia ([Chap. 368](#)). The possibility of this treatable cervical spinal cord disease must be considered even when the patient presents with leg complaints only. Furthermore, lumbar radiculopathy or polyneuropathy may mask an associated cervical myelopathy. [MRI](#) or [CT](#)-myelography can define the anatomic abnormalities, and [EMG](#) and nerve conduction studies can quantify the severity and localize the levels of motor nerve root injury.

OTHER CAUSES OF NECK PAIN

Rheumatoid arthritis (RA) ([Chap. 312](#)) of the cervical apophyseal joints results in neck pain, stiffness, and limitation of motion. In typical cases with symmetric inflammatory polyarthritis, the diagnosis of RA is straightforward. In advanced RA, synovitis of the atlantoaxial joint (C1-C2; [Fig. 16-2](#)) may damage the transverse ligament of the atlas, producing forward displacement of the atlas on the axis (atlantoaxial subluxation). Radiologic evidence of atlantoaxial subluxation occurs in 30% of patients with RA. Not surprisingly, the degree of subluxation correlates with the severity of erosive disease. When subluxation is present, careful neurologic assessment is important to identify early signs of myelopathy. Occasional patients develop high spinal cord compression leading to quadriplegia, respiratory insufficiency, and death. Although low back pain is common among RA patients, the frequency of facet disease, fracture, and spondylolisthesis is no greater than among age- and sex-matched controls with mechanical low back pain.

Ankylosing spondylitis can cause neck pain and on occasion atlantoaxial subluxation; when spinal cord compression is present or threatened, surgical intervention is indicated. Herpes zoster produces neck and posterior occipital pain in a C2-C3 distribution prior to the outbreak of vesicles. Neoplasms metastatic to the cervical spine, infections (osteomyelitis and epidural abscess), and metabolic bone diseases may also be the cause of neck pain. Neck pain may also be referred from the heart in the setting of coronary artery ischemia (cervical angina syndrome).

THORACIC OUTLET

The thoracic outlet is an anatomic region containing the first rib, the subclavian artery and vein, the brachial plexus, the clavicle, and the lung apex. Injury to these structures may result in posture or task-related pain around the shoulder and supraclavicular region. There are at least three subtypes of thoracic outlet syndrome (TOS). *True neurogenic TOS* results from compression of the lower trunk of the brachial plexus by an anomalous band of tissue connecting an elongate transverse process at C7 with the first rib. Neurologic deficits include weakness of intrinsic muscles of the hand and diminished sensation on the palmar aspect of the fourth and fifth digits. [EMG](#) and nerve conduction studies confirm the diagnosis. Definitive treatment consists of surgical division of the anomalous band compressing either the lower trunk of the brachial plexus or ventral rami of the C8 or T1 nerve roots. The weakness and wasting of intrinsic hand muscles typically does not improve, but surgery halts the insidious progression of weakness. The *arterial TOS* results from compression of the subclavian artery by a cervical rib; the compression results in poststenotic dilatation of the artery and thrombus formation. Blood pressure is reduced in the affected limb, and signs of emboli may be present in the hand; neurologic signs are absent. Noninvasive ultrasound techniques confirm the diagnosis. Treatment is with thrombolysis or anticoagulation (with or without embolectomy) and surgical excision of the cervical rib compressing the subclavian artery or vein. The *disputed TOS* includes a large number of patients with chronic arm and shoulder pain of unclear cause. The lack of sensitive and specific findings on physical examination or laboratory markers for this condition frequently results in diagnostic uncertainty. The role of surgery in disputed TOS is controversial; conservative approaches often include multidisciplinary pain management. Treatment is often unsuccessful.

BRACHIAL PLEXUS AND NERVES

Pain from injury to the brachial plexus or arm peripheral nerves can occasionally be confused with pain of cervical spine origin. Neoplastic infiltration of the lower trunk of the brachial plexus may produce shoulder pain radiating down the arm, numbness of the fourth and fifth fingers, and weakness of intrinsic hand muscles innervated by the ulnar and median nerves. Postradiation fibrosis (breast carcinoma is the most common setting) or a Pancoast tumor of the lung ([Chap. 88](#)) may produce similar findings. A Horner's syndrome is present in two-thirds of patients with a Pancoast tumor. Suprascapular neuropathy may produce severe shoulder pain, weakness, and wasting of the supraspinatus and infraspinatus muscles. *Acute brachial neuritis* is often confused with radiculopathy. It consists of the acute onset of severe shoulder or scapular pain followed over days to weeks by weakness of the proximal arm and shoulder girdle muscles innervated by the upper or middle trunks or cords of the brachial plexus. The onset is often preceded by an infection or immunization. Separation of this syndrome from cervical radiculopathy is important because slow, complete recovery of brachial neuritis occurs in 75% of patients after 2 years and in 89% after 3 years. Occasional cases of carpal tunnel syndrome produce pain and paresthesia extending into the forearm, arm, and shoulder resembling a C5 or C6 root lesion. Lesions of the radial or ulnar nerve can mimic a radiculopathy at C7 or C8, respectively. [EMG](#) and nerve conduction studies can accurately localize lesions to the nerve roots, brachial plexus, or nerves. **For further discussion of peripheral nerve disorders, see [Chap. 377](#).*

SHOULDER

Pain in the shoulder region can be difficult to separate clearly from neck pain. If the symptoms and signs of radiculopathy are absent, then the differential diagnosis includes mechanical shoulder pain (tendonitis, bursitis, rotator cuff tear, dislocation, adhesive capsulitis, and cuff impingement under the acromion) and referred pain (subdiaphragmatic irritation, angina, Pancoast tumor). Mechanical pain is often worse at night, associated with local shoulder tenderness, and aggravated by abduction, internal rotation, or extension of the arm. The pain of shoulder disease may at times radiate into the arm or hand, but the sensory, motor, and reflex changes that indicate disease of the nerve roots, plexus, or peripheral nerves are absent.

TREATMENT

A paucity of well-designed clinical trials exists for the treatment of neck pain. Symptomatic treatment of neck pain can include the use of analgesic medications and/or a soft cervical collar. Current indications for cervical disk surgery are similar to those for lumbar disk surgery; because of the risk of spinal cord injury with cervical spine disease, an aggressive approach is generally indicated whenever spinal cord injury is threatened. Surgical management of cervical herniated disks usually consists of an anterior approach with discectomy followed by anterior interbody fusion. A simple posterior partial laminectomy with discectomy is an acceptable alternative approach. The cumulative risk of subsequent radiculopathy or myelopathy at cervical segments adjacent to the fusion is approximately 3% per year and 26% per decade. Although this

risk is sometimes portrayed as a late complication of surgery, it may also reflect the natural history of degenerative cervical spine disease. Nonprogressive cervical radiculopathy (associated with a focal neurologic deficit) due to a herniated cervical disk may be treated conservatively with a high rate of success. Cervical spondylosis with bony, compressive cervical radiculopathy is generally treated with surgical decompression to interrupt the progression of neurologic signs. Cervical spondylotic myelopathy is typically managed with either anterior decompression and fusion or laminectomy. Outcomes in both surgical groups vary, but late functional deterioration occurs in 20 to 30% of patients; a prospective, controlled study comparing different surgical interventions is sorely needed.

(Bibliography omitted in Palm version)

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SECTION 2 -ALTERATIONS IN BODY TEMPERATURE

17. FEVER AND HYPERTHERMIA - Charles A. Dinarello, Jeffrey A. Gelfand

Body temperature is controlled by the hypothalamus. Neurons in both the preoptic anterior hypothalamus and the posterior hypothalamus receive two kinds of signals: one from peripheral nerves that reflect warmth/cold receptors and the other from the temperature of the blood bathing the region. These two types of signals are integrated by the thermoregulatory center of the hypothalamus to maintain normal temperature. In a neutral environment, the metabolic rate of humans consistently produces more heat than is necessary to maintain the core body temperature at 37°C. Therefore, the hypothalamus controls temperature by mechanisms of heat loss.

A normal body temperature is ordinarily maintained, despite environmental variations, because the hypothalamic thermoregulatory center balances the excess heat production derived from metabolic activity in muscle and the liver with heat dissipation from the skin and lungs. According to recent studies of healthy individuals 18 to 40 years of age, the mean oral temperature is $36.8^{\circ}\pm 0.4^{\circ}\text{C}$ ($98.2^{\circ}\pm 0.7^{\circ}\text{F}$), with low levels at 6 A.M. and higher levels at 4 to 6 P.M. The maximum normal oral temperature is 37.2°C (98.9°F) at 6 A.M. and 37.7°C (99.9°F) at 4 P.M.; these values define the 99th percentile for healthy individuals. In light of these studies, *an A.M. temperature of $>37.2^{\circ}\text{C}$ (98.9°F) or a P.M. temperature of $>37.7^{\circ}\text{C}$ (99.9°F) would define a fever.* The normal daily temperature variation is typically 0.5°C (0.9°F). However, in some individuals recovering from a febrile illness, this daily variation can be as great as 1.0°C . During a febrile illness, diurnal variations are usually maintained but at higher levels. Daily temperature swings do not occur in patients with hyperthermia (see below). Rectal temperatures are generally 0.4°C (0.7°F) higher than oral readings. The lower oral readings are probably attributable to mouth breathing, which is a particularly important factor in patients with respiratory infections and rapid breathing. Lower esophageal temperatures closely reflect core temperature. Tympanic membrane (TM) thermometers measure radiant heat energy from the tympanic membrane and nearby ear canal and display that absolute value (unadjusted mode) or a value automatically calculated from the absolute reading on the basis of nomograms relating the radiant temperature measured to actual core temperatures obtained in clinical studies (adjusted mode). These measurements, although convenient, may be more variable than directly determined oral or rectal values. Studies in adults show that readings are lower with unadjusted-mode than with adjusted-mode TM thermometers and that unadjusted-mode TM values are 0.8°C (1.6°F) lower than rectal temperatures.

In women who menstruate, the A.M. temperature is generally lower in the 2 weeks before ovulation; it then rises by about 0.6°C (1°F) with ovulation and remains at that level until menses occur. Seasonal variation in body temperature has been described but may reflect a metabolic change and is not common. Body temperature is elevated in the postprandial state, but this elevation does not represent fever. Pregnancy and endocrinologic dysfunction also affect body temperature. The daily temperature variation appears to be fixed in early childhood; in contrast, elderly individuals can exhibit a reduced ability to develop fever, with only a modest fever even in severe infections.

FEVER VERSUS HYPERTHERMIA

FEVER

Fever is an elevation of body temperature that exceeds the normal daily variation and occurs *in conjunction with an increase in the hypothalamic set point* -- for example, from 37°C to 39°C. This shift of the set point from "normothermic" to febrile levels very much resembles the resetting of the home thermostat to a higher level in order to raise the ambient temperature in a room. Once the hypothalamic set point is raised, neurons in the vasomotor center are activated and vasoconstriction commences. The individual first notices vasoconstriction in the hands and feet. Shunting of blood away from the periphery to the internal organs essentially decreases heat loss from the skin, and the person feels cold. For most fevers, body temperature increases by 1 to 2°C. Shivering, which increases heat production from the muscles, may begin at this time; however, shivering is not required if heat conservation mechanisms raise blood temperature sufficiently. Heat production from the liver also increases. In humans, behavioral instincts (e.g., putting on more clothing or bedding) lead to a reduction of exposed surfaces, which helps raise body temperature.

The processes of heat conservation (vasoconstriction) and heat production (shivering and increased metabolic activity) continue until the temperature of the blood bathing the hypothalamic neurons matches the new thermostat setting. Once that point is reached, the hypothalamus maintains the temperature at the febrile level by the same mechanisms of heat balance that are operative in the afebrile state. When the hypothalamic set point is again reset downward (due to either a reduction in the concentration of pyrogens or the use of antipyretics), the processes of heat loss through vasodilation and sweating are initiated. Behavioral changes triggered at this time include the removal of insulating clothing or bedding. Loss of heat by sweating and vasodilation continues until the blood temperature at the hypothalamic level matches the lower setting.

A fever of $>41.5^{\circ}\text{C}$ (106.7°F) is called *hyperpyrexia*. This extraordinarily high fever can develop in patients with severe infections but most commonly occurs in patients with central nervous system hemorrhages. In the preantibiotic era, fever due to a variety of infectious diseases rarely exceeded 106°F , and there has been speculation that this natural "thermal ceiling" is mediated by neuropeptides functioning as central antipyretics.

In some rare cases, the hypothalamic set point is elevated as a result of local trauma, hemorrhage, tumor, or intrinsic hypothalamic malfunction. The term *hypothalamic fever* is sometimes used to describe elevated temperature caused by abnormal hypothalamic function. However, most patients with hypothalamic damage have *subnormal*, not *supranormal*, body temperatures. These patients do not respond properly to mild environmental temperature changes. For example, when exposed to only mildly cold conditions, their core temperature falls quickly rather than over the normal period of a few hours. In the very few patients in whom elevated core temperature is suspected to be due to hypothalamic damage, diagnosis depends on the demonstration of other abnormalities in hypothalamic function, such as the production of hypothalamic releasing factors, abnormal response to cold, and absence of circadian temperature and

hormonal rhythms.

HYPERTHERMIA

Hyperthermia is characterized by *an unchanged (normothermic) setting of the thermoregulatory center* in conjunction with an uncontrolled increase in body temperature that exceeds the body's ability to lose heat. Exogenous heat exposure and endogenous heat production are two mechanisms by which hyperthermia can result in dangerously high internal temperatures. Excessive heat production can easily cause hyperthermia despite physiologic and behavioral control of body temperature. For example, over-insulating clothing can result in an elevated core temperature, and work or exercise in hot environments can produce heat faster than peripheral mechanisms can lose it.

Although most patients with elevated body temperature have fever, there are a few circumstances in which elevated temperature represents not fever but hyperthermia ([Table 17-1](#)). *Heat stroke*, caused by thermoregulatory failure in association with a warm environment, may be categorized as exertional or nonexertional. *Exertional heat stroke* typically occurs in younger individuals exercising at ambient temperatures and/or humidities that are higher than normal. Even in normal individuals, dehydration or the use of common medications (e.g., over-the-counter antihistamines with anticholinergic side effects) may help to precipitate exertional heat stroke. *Nonexertional* or *classic heat stroke* typically occurs in elderly individuals, particularly during heat waves. For example, in Chicago in July 1995, 465 deaths were certified as heat related. The elderly, the bedridden, persons taking anticholinergic or antiparkinsonian drugs or diuretics, and individuals confined to poorly ventilated and non-air-conditioned environments are most susceptible.

Drug-induced hyperthermia has become increasingly common as a result of the increased use of prescription psychotropic drugs and illicit drugs. Drug-induced hyperthermia may be caused by monoamine oxidase inhibitors, tricyclic antidepressants, and amphetamines and by the illicit use of phencyclidine, lysergic acid diethylamide (LSD), or cocaine.

Malignant hyperthermia occurs in individuals with an inherited abnormality of skeletal-muscle sarcoplasmic reticulum that causes a rapid increase in intracellular calcium levels in response to halothane and other inhalational anesthetics or to succinylcholine. Elevated temperature, increased muscle metabolism, rigidity, rhabdomyolysis, acidosis, and cardiovascular instability develop rapidly. This condition is often fatal. The *neuroleptic malignant syndrome* can occur with phenothiazines and other drugs such as haloperidol and is characterized by muscle rigidity, autonomic dysregulation, and hyperthermia. This disorder appears to be caused by the inhibition of central dopamine receptors in the hypothalamus, which results in increased heat generation and decreased heat dissipation. Thyrotoxicosis and pheochromocytoma can also cause increased thermogenesis.

It is important to distinguish between fever and hyperthermia since hyperthermia can be rapidly fatal and characteristically does not respond to antipyretics. However, there is no rapid way to make this distinction. Hyperthermia is often diagnosed on the basis of the

events immediately preceding the elevation of core temperature -- e.g., heat exposure or treatment with drugs that interfere with thermoregulation. However, in addition to the clinical history of the patient, the physical aspects of some forms of hyperthermia may alert the clinician. For example, in patients with heat stroke syndromes and in those taking drugs that block sweating, the skin is hot but dry. Moreover, antipyretics do not reduce the elevated temperature in hyperthermia, whereas in fever -- and even in hyperpyrexia -- adequate doses of either aspirin or acetaminophen usually result in some decrease in body temperature.

PYROGENS

The term *pyrogen* is used to describe any substance that causes fever. *Exogenous* pyrogens are derived from outside the patient; most are microbial products, microbial toxins, or whole microorganisms. The classic example of an exogenous pyrogen is the lipopolysaccharide endotoxin produced by all gram-negative bacteria. Endotoxins are potent not only as pyrogens but also as inducers of various pathologic changes in gram-negative infections. Another group of potent bacterial pyrogens is produced by gram-positive organisms and includes the enterotoxins of *Staphylococcus aureus* and the group A and B streptococcal toxins, also called *superantigens*. One staphylococcal toxin of clinical importance is the toxic shock syndrome toxin associated with isolates of *S. aureus* from patients with toxic shock syndrome. Like the endotoxins of gram-negative bacteria, the toxins produced by staphylococci and streptococci cause fever in experimental animals when injected intravenously at concentrations of <1 ug/kg of body weight. Endotoxin is a highly pyrogenic molecule in humans: a dose of 2 to 3 ng/kg produces fever and generalized symptoms of malaise in volunteers.

PYROGENIC CYTOKINES

Cytokines are small proteins (molecular mass, 10,000 to 20,000 Da) that regulate immune, inflammatory, and hematopoietic processes. For example, stimulation of lymphocyte proliferation during an immune response to vaccination is the result of the cytokines interleukin (IL) 2, IL-4, and IL-6. Another cytokine, granulocyte colony-stimulating factor, stimulates granulocytopoiesis in the bone marrow. Some cytokines cause fever and hence are called *pyrogenic cytokines*. From a historic point of view, the field of cytokine biology began in the 1940s with laboratory investigations into fever induction by products of activated leukocytes. These fever-producing molecules were called *endogenous pyrogens*. When endogenous pyrogens were purified from activated leukocytes, they were shown to possess various biologic activities, which are now recognized as the properties of the various cytokines.

The known pyrogenic cytokines include IL-1, IL-6, tumor necrosis factor (TNF), ciliary neurotropic factor (CNTF), and interferon (IFN) α . Others probably exist. Each cytokine is encoded by a separate gene, and each pyrogenic cytokine has been shown to cause fever in laboratory animals and in humans. When injected into humans, IL-1, IL-6, and TNF produce fever at low doses (10 to 100 ng/kg).

The synthesis and release of endogenous pyrogenic cytokines are induced by a wide spectrum of exogenous pyrogens, most of which have recognizable bacterial or fungal sources. Viruses also induce pyrogenic cytokines by infecting cells. However, in the

absence of microbial infection, inflammation, trauma, tissue necrosis, or antigen-antibody complexes can induce the production of [IL-1](#), [TNF](#), and/or IL-6, which -- individually or in combination -- trigger the hypothalamus to raise the set point to febrile levels. The cellular sources of pyrogenic cytokines are primarily monocytes, neutrophils, and lymphocytes, although many other types of cells can synthesize these molecules when stimulated.

ELEVATION OF THE HYPOTHALAMIC SET POINT BY CYTOKINES

During fever, levels of prostaglandin E₂ (PGE₂) are elevated in hypothalamic tissue and the third cerebral ventricle. The concentrations of PGE₂ are highest near the circumventricular vascular organs (organum vasculosum of lamina terminalis) -- networks of enlarged capillaries surrounding the hypothalamic regulatory centers. Destruction of these organs reduces the ability of pyrogens to produce fever. Most studies in animals have failed to show, however, that pyrogenic cytokines pass from the circulation into the brain itself. Thus, it appears that both exogenous and endogenous pyrogens interact with the endothelium of these capillaries and that this interaction is the first step in initiating fever -- i.e., in raising the set point to febrile levels.

The key events in the production of fever are illustrated in [Fig. 17-1](#). As has been mentioned, several cell types can produce pyrogenic cytokines. Pyrogenic cytokines such as [IL-1](#), IL-6, and [TNF](#) are released from the cells and enter the systemic circulation. Although the systemic effects of these circulating cytokines lead to fever by inducing the synthesis of [PGE₂](#), they also induce PGE₂ in peripheral tissues. The increase in PGE₂ in the periphery accounts for the nonspecific myalgias and arthralgias that often accompany fever. However, it is the induction of PGE₂ in the brain that starts the process of raising the hypothalamic set point for core temperature.

There are four receptors for [PGE₂](#), and each signals the cell in different ways. Of the four receptors, the third (EP-3) is essential for fever: when the gene for this receptor is deleted in mice, no fever follows the injection of [IL-1](#) or endotoxin. Deletion of the other PGE₂ receptor genes leaves the fever mechanism intact. Although PGE₂ is essential for fever, it is not a neurotransmitter. Rather, the release of PGE₂ from the brain side of the hypothalamic endothelium triggers the PGE₂ receptor on glial cells, and this stimulation results in the rapid release of cyclic adenosine 5'-monophosphate (cyclic AMP), which is a neurotransmitter. As shown in [Fig. 17-1](#), the release of cyclic AMP from the glial cells activates neuronal endings from the thermoregulatory center that extend into the area. The elevation of cyclic AMP is thought to account for changes in the hypothalamic set point either directly or indirectly by inducing the release of monoamine neurotransmitters. Since receptors for endotoxin are in many ways similar to IL-1 receptors, the activation of endotoxin receptors on the hypothalamic endothelium also results in PGE₂ production and fever.

PRODUCTION OF CYTOKINES IN THE CENTRAL NERVOUS SYSTEM

Several viral diseases produce active infection in the brain. Glial and possibly neuronal cells synthesize [IL-1](#), [TNF](#), and IL-6. [CNTF](#) is also synthesized by neural as well as neuronal cells. What role in the production of fever is played by these cytokines produced in the brain itself? In experimental animals, the concentrations of cytokine

required to cause fever are several orders of magnitude lower with direct injection into the brain than with intravenous injection. Therefore, central nervous system production of these cytokines apparently can raise the hypothalamic set point, bypassing the circumventricular organs involved in fever caused by circulating cytokines. Central nervous system cytokines may account for the hyperpyrexia of central nervous system hemorrhage, trauma, or infection.

Approach to the Patient

History It is in the diagnosis of a febrile illness that the science and art of medicine come together. In no other clinical situation is a meticulous history more important. Painstaking attention must be paid to the chronology of symptoms in relation to the use of prescription drugs (including drugs or herbs taken without a physician's supervision) or treatments such as surgical or dental procedures. The exact nature of any prosthetic materials and/or implanted devices should be ascertained. A careful occupational history should include exposures to animals; toxic fumes; potential infectious agents; possible antigens; or other febrile or infected individuals in the home, workplace, or school. A history of the geographic areas in which the patient has lived and a travel history should include locations during military service. Information on unusual hobbies, dietary proclivities (such as raw or poorly cooked meat, raw fish, and unpasteurized milk or cheeses), and household pets should be elicited, as should that on sexual orientation and practices, including precautions taken or omitted. Attention should be directed to the use of tobacco, marijuana, intravenous drugs, or alcohol; trauma; animal bites; tick or other insect bites; and prior transfusions, immunizations, drug allergies, or hypersensitivities. A careful family history should include information on family members with tuberculosis, other febrile or infectious diseases, arthritis or collagen vascular disease, or unusual familial symptomatology such as deafness, urticaria, fevers and polyserositis, bone pain, or anemia. Ethnic origin may be critical. For example, blacks are more likely than persons in other groups to have hemoglobinopathies. Turks, Arabs, Armenians, and Sephardic Jews are especially likely to have familial Mediterranean fever.

Physical Examination A meticulous physical examination should be repeated on a regular basis. All the vital signs are relevant. The temperature may be taken orally or rectally, but the site used should be consistent. Axillary temperatures are notoriously unreliable. Particular attention should be paid to daily (or sometimes more frequent) physical examination, which should continue until the diagnosis is certain and the anticipated response has been achieved. Special attention should be paid to the skin, lymph nodes, eyes, nail beds, cardiovascular system, chest, abdomen, musculoskeletal system, and nervous system. Rectal examination is imperative. The penis, prostate, scrotum, and testes should be examined carefully and the foreskin, if present, retracted. Pelvic examination must be part of every complete physical examination of a woman, with a search for such causes of fever as pelvic inflammatory disease and tubo-ovarian abscess.

Laboratory Tests Few signs and symptoms in medicine have as many diagnostic possibilities as fever. If the history, epidemiologic situation, or physical examination suggests more than a simple viral illness or streptococcal pharyngitis, then laboratory testing is indicated. The tempo and complexity of the workup will depend on the pace of

the illness, diagnostic considerations, and the immune status of the host. If findings are focal or if the history, epidemiologic setting, or physical examination suggests certain diagnoses, the laboratory examination can be focused. If fever is undifferentiated, the diagnostic nets must be cast farther, and certain guidelines are indicated, as follows.

CLINICAL PATHOLOGY The workup should include a complete blood count; a differential count should be performed manually or with an instrument sensitive to the identification of eosinophils, juvenile or band forms, toxic granulations, and Dohle bodies, the last three of which are suggestive of bacterial infection. Neutropenia may be present with some viral infections, particularly parvovirus B19 infection; drug reactions; systemic lupus erythematosus; typhoid; brucellosis; and infiltrative diseases of the bone marrow, including lymphoma, leukemia, tuberculosis, and histoplasmosis. Lymphocytosis may occur with typhoid, brucellosis, tuberculosis, and viral disease. Atypical lymphocytes are documented in many viral diseases, including infection with Epstein-Barr virus, cytomegalovirus, or HIV; dengue; rubella; varicella; measles; and viral hepatitis. This abnormality also occurs in serum sickness and toxoplasmosis. Monocytosis is a feature of typhoid, tuberculosis, brucellosis, and lymphoma. Eosinophilia may be associated with hypersensitivity drug reactions, Hodgkin's disease, adrenal insufficiency, and certain metazoan infections. If the febrile illness appears to be severe or is prolonged, the smear should be examined carefully for malarial or babesial pathogens (where appropriate) as well as for classic morphologic features, and the erythrocyte sedimentation rate should be determined. Urinalysis, with examination of urinary sediment, is indicated. It is axiomatic that any abnormal fluid accumulation (pleural, peritoneal, joint), even if previously sampled, merits reexamination in the presence of undiagnosed fever. Joint fluids should be examined for bacteria as well as crystals. Bone marrow biopsy (not simple aspiration) for histopathologic studies (as well as culture) is indicated when marrow infiltration by pathogens or tumor cells is possible. Stool should be inspected for occult blood; an inspection for fecal leukocytes, ova, or parasites also may be indicated.

CHEMISTRY Electrolyte, glucose, blood urea nitrogen, and creatinine levels should be measured. Liver function tests are usually indicated if efforts to identify the cause of fever do not point to the involvement of another organ. Additional assessments (e.g., measurement of creatine phosphokinase or amylase) can be added as the workup progresses.

MICROBIOLOGY Smears and cultures of specimens from the throat, urethra, anus, cervix, and vagina should be assessed when there are no localizing findings or when findings suggest the involvement of the pelvis or the gastrointestinal tract. If respiratory tract infection is suspected, sputum evaluation (Gram's staining, staining for acid-fast bacilli, culture) is indicated. Cultures of blood, abnormal fluid collections, and urine are indicated when fever is thought to reflect more than uncomplicated viral illness. Cerebrospinal fluid should be examined and cultured if meningismus, severe headache, or a change in mental status is noted.

RADIOLOGY A chest x-ray is usually part of the evaluation for any significant febrile illness.

Outcome of Diagnostic Efforts In most cases of fever, either the patient recovers

spontaneously or the history, physical examination, and initial screening laboratory studies lead to a diagnosis. When fever continues for 2 to 3 weeks, during which time repeat physical examinations and laboratory tests are unrevealing, the patient is provisionally diagnosed as having fever of unknown origin ([Chap. 125](#)).

TREATMENT

The Decision to Treat Fever Most fevers are associated with self-limited infections, most commonly of viral origin. In these cases, the general cause of the fever is easily identified. The routine use of antipyretics given automatically as "standing," "routine," or "prn" orders to treat low-grade fevers in adult patients on hospital wards is entirely unacceptable. This practice masks not only fever but also other important clinical indicators of a patient's course. The assumption underlying any decision to reduce fever with antipyretics is that there is no diagnostic benefit to be gained by allowing the fever to persist. However, there may be such a diagnostic benefit. For example, the daily highs and lows of normal temperature are exaggerated in most fevers, but the usual times of peak and trough temperatures may be reversed in typhoid fever and disseminated tuberculosis. Temperature-pulse dissociation (relative bradycardia) occurs in typhoid fever, brucellosis, leptospirosis, some drug-induced fevers, and factitious fever. In newborns, the elderly, patients with chronic renal failure, and patients taking glucocorticoids, fever may not be present despite infection, or core temperature may be hypothermic. Hypothermia is observed in patients with septic shock.

Some febrile diseases have characteristic patterns. With *relapsing* fevers, febrile episodes are separated by intervals of normal temperature; when paroxysms occur on the first and third days, the fever is called *tertian*. *Plasmodium vivax* causes tertian fevers. *Quartan* fevers are associated with paroxysms on the first and fourth days and are seen with *P. malariae*. Other relapsing fevers are related to *Borrelia* infections and rat-bite fever, which are both associated with days of fever followed by a several-day afebrile period and then a relapse of days of fever. Pel-Ebstein fever, with fevers lasting 3 to 10 days followed by afebrile periods of 3 to 10 days, is classic for Hodgkin's disease and other lymphomas. Another characteristic fever is that of cyclic neutropenia, in which fevers occur every 21 days and accompany the neutropenia. There is no periodicity of fever in patients with familial Mediterranean fever.

Mechanisms of Antipyretic Agents The synthesis of [PGE₂](#) depends on the constitutively expressed enzyme cyclooxygenase. The substrate for cyclooxygenase is arachidonic acid released from the cell membrane, and this release is the rate-limiting step in the synthesis of PGE₂. Inhibitors of cyclooxygenase are potent antipyretics. The antipyretic potency of various drugs is directly correlated with the inhibition of brain cyclooxygenase. Acetaminophen is a poor cyclooxygenase inhibitor in peripheral tissue and is without noteworthy anti-inflammatory activity; in the brain, however, acetaminophen is oxidized by the p450 cytochrome system, and the oxidized form inhibits cyclooxygenase activity.

Oral aspirin and acetaminophen are equally effective in reducing fever in humans. Nonsteroidal anti-inflammatory agents (NSAIDs) such as indomethacin and ibuprofen are also excellent antipyretics. Chronic high-dose therapy with antipyretics such as aspirin or the NSAIDs used in arthritis does not reduce normal core body temperature.

Thus, PGE₂ appears to play no role in normal thermoregulation.

As effective antipyretics, glucocorticoids act at two levels. First, similar to the cyclooxygenase inhibitors, glucocorticoids reduce PGE₂ synthesis by inhibiting the activity of phospholipase A₂, which is needed to release arachidonic acid from the cell membrane. Second, glucocorticoids block the transcription of the mRNA for the pyrogenic cytokines.

Drugs that interfere with vasoconstriction (phenothiazines, for example) can act as antipyretics, as can drugs that block muscle contractions. However, these agents are not true antipyretics since they can also reduce core temperature independently of hypothalamic control.

Indications and Regimens for the Treatment of Fever The objectives in treating fever are first to reduce the elevated hypothalamic set point and second to facilitate heat loss. There is no evidence that fever itself facilitates the recovery from infection or acts as an adjuvant to the immune system. In fact, peripheral PGE₂ production is a potent immunosuppressant. Hence, treating fever and its symptoms does no harm and does not slow the resolution of common viral and bacterial infections. Reducing fever with antipyretics also reduces systemic symptoms of headache, myalgias, and arthralgias.

Oral aspirin and NSAIDs effectively reduce fever but can adversely affect platelets and the gastrointestinal tract. Therefore, acetaminophen is preferred to all of these agents as an antipyretic. In children, acetaminophen must be used because aspirin increases the risk of Reye's syndrome. If the patient cannot take oral antipyretics, parenteral preparations of NSAIDs and rectal suppository preparations of various antipyretics can be used.

Treatment of fever in some patient groups is recommended. Fever increases the demand for oxygen (i.e., for every increase of 1°C over 37°C, there is a 13% increase in oxygen consumption) and can aggravate preexisting cardiac, cerebrovascular, or pulmonary insufficiency. Elevated temperature can induce mental changes in patients with organic brain disease. Children with a history of febrile or nonfebrile seizure should be aggressively treated to reduce fever, although it is unclear what triggers the febrile seizure and there is no correlation between absolute temperature elevation and onset of a febrile seizure in susceptible children.

In hyperpyrexia, the use of cooling blankets facilitates the reduction of temperature; however, cooling blankets should not be used without oral antipyretics. In hyperpyretic patients with central nervous system disease or trauma, reducing core temperature mitigates the ill effects of high temperature on the brain.

Treating Hyperthermia A high core temperature in a patient with an appropriate history (e.g., environmental heat exposure or treatment with anticholinergic or neuroleptic drugs, tricyclic antidepressants, succinylcholine, or halothane) along with appropriate clinical findings (dry skin, hallucinations, delirium, pupil dilation, muscle rigidity, and/or elevated levels of creatine phosphokinase) suggests hyperthermia. The attempt to lower the already normal hypothalamic set point is of little use. Physical cooling with sponging, fans, cooling blankets, and even ice baths should be initiated immediately in conjunction

with the administration of intravenous fluids and appropriate pharmacologic agents (see below). If insufficient cooling is achieved by external means, internal cooling can be achieved by gastric or peritoneal lavage with iced saline. In extreme circumstances, hemodialysis or even cardiopulmonary bypass with cooling of blood may be performed.

Malignant hyperthermia should be treated immediately with cessation of anesthesia and intravenous administration of dantrolene sodium. The recommended dose of dantrolene is 1 to 2.5 mg/kg of body weight given intravenously every 6 h for at least 24 to 48 h -- until oral dantrolene can be administered, if needed. Procainamide should also be administered to patients with malignant hyperthermia because of the likelihood of ventricular fibrillation in this syndrome. Dantrolene at similar doses is indicated in the neuroleptic malignant syndrome and in drug-induced hyperthermia and may even be useful in the hyperthermia of thyrotoxicosis. The neuroleptic malignant syndrome may also be treated with bromocriptine, levodopa, amantadine, or nifedipine or by induction of muscle paralysis with curare and pancuronium. Tricyclic antidepressant overdose may be treated with physostigmine.

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18. FEVER AND RASH - Elaine T. Kaye, Kenneth M. Kaye

The acutely ill patient with fever and rash ([Fig. 18-CD1](#)) often presents a diagnostic challenge for physicians. The distinctive appearance of an eruption in concert with a clinical syndrome may facilitate a prompt diagnosis and the institution of life-saving therapy or critical infection-control interventions.

Approach to the Patient

A thorough history of patients with fever and rash includes the following relevant information: immune status, medications taken within the previous month, specific travel history, immunization status, exposure to domestic pets and other animals, history of animal or arthropod bites, existence of cardiac abnormalities, presence of prosthetic material, recent exposure to ill individuals, and exposure to sexually transmitted diseases. The history should also include the site of onset of the rash and its direction and rate of spread.

A thorough physical examination entails close attention to the rash, with an assessment and precise definition of its salient features. First, it is critical to determine the *type* of lesions that make up the eruption. *Macules* are flat lesions defined by an area of changed color (i.e., a blanchable erythema). *Papules* are raised, solid lesions <5 mm in diameter; *plaques* are lesions >5 mm in diameter with a flat, plateau-like surface; and *nodules* are lesions >5 mm in diameter with a more rounded configuration. *Wheals* (urticaria, hives) are papules or plaques that are pale pink and may appear annular (ringlike) as they enlarge; classic (nonvasculitic) wheals are transient, lasting only 24 to 48 h in any defined area. *Vesicles* (<5 mm) and *bullae* (>5 mm) are circumscribed, elevated lesions containing fluid. *Pustules* are raised lesions containing purulent exudate; vesicular processes such as varicella or herpes simplex may evolve to pustules. *Nonpalpable purpura* is a flat lesion that is due to bleeding into the skin; if <3 mm in diameter, the purpuric lesions are termed *petechiae*; if >3 mm, they are termed *ecchymoses*. *Palpable purpura* is a raised lesion that is due to inflammation of the vessel wall (vasculitis) with subsequent hemorrhage. An *ulcer* is a defect in the skin extending at least into the upper layer of the dermis, and an *eschar* (tache noire) is a necrotic lesion covered with a black crust.

Other pertinent features of rashes include their *configuration* (i.e., annular or target), the *arrangement* of their lesions, and their *distribution* (i.e., central or peripheral). **For further discussion, see [Chaps. 55 and 57](#).*

CLASSIFICATION OF RASH

This chapter reviews rashes that reflect systemic disease but does not include localized skin eruptions (i.e., cellulitis, impetigo) that may also be associated with fever ([Chap. 128](#)). Rashes are classified herein on the basis of the morphology and distribution of lesions. For practical purposes, this classification system is based on the most typical disease presentations. However, morphology may vary as rashes evolve, and the presentation of diseases with rashes is subject to many variations ([Chap. 57](#)). For instance, the classic petechial rash of Rocky Mountain spotted fever (RMSF) may initially consist of blanchable erythematous macules distributed peripherally; at times,

the rash associated with RMSF may not be predominantly acral, or a rash may not develop at all.

Diseases with fever and rash may be classified by type of eruption: centrally distributed maculopapular, peripheral, confluent desquamative erythematous, vesiculobullous, urticarial, nodular, purpuric, ulcerated, or eschars ([Table 18-1](#)). For a more detailed discussion of each disease associated with a rash, the reader is referred to the chapter dealing with that specific disease. (Reference chapters and color plates are cited in the text and listed in [Table 18-1](#).)

Centrally Distributed Maculopapular Eruptions Centrally distributed rashes, in which lesions are primarily truncal, are the most common type of eruption. The rash of *measles* (rubeola) starts at the hairline 2 to 3 days into the illness and moves down the body, sparing the palms and soles ([Chap. 194](#)). It begins as discrete erythematous lesions, which become confluent as the rash spreads. Koplik's spots (1- to 2-mm white or bluish lesions with an erythematous halo on the buccal mucosa) are pathognomonic for measles and are generally seen during the first 2 days of symptoms. They should not be confused with Fordyce's spots (ectopic sebaceous glands), which have no erythematous halos and are found in the mouth of healthy individuals. Koplik's spots may briefly overlap with the measles exanthem.

German measles (rubella) also spreads from the hairline downward; unlike that of measles, however, the rash of rubella tends to clear from originally affected areas as it migrates and may be pruritic ([Chap. 195](#)). Forchheimer spots (palatal petechiae) may develop but are nonspecific since they also develop in mononucleosis ([Chap. 184](#)) and scarlet fever ([Chap. 140](#)). Postauricular and suboccipital adenopathy and arthritis are common among adults with German measles. Exposure of pregnant women to ill individuals should be avoided, as rubella causes severe congenital abnormalities. Numerous strains of enteroviruses ([Chap. 193](#)), primarily echoviruses and coxsackieviruses, cause nonspecific syndromes of fever and eruptions that may mimic rubella or measles. Patients with infectious mononucleosis caused by Epstein-Barr virus or with primary infection caused by HIV ([Chap. 309](#)) may exhibit pharyngitis, lymphadenopathy, and a nonspecific maculopapular exanthem.

The rash of *erythema infectiosum* (fifth disease), which is caused by human parvovirus B19, primarily affects children 3 to 12 years old; it develops after fever has resolved as a bright blanchable erythema on the cheeks ("slapped cheeks") with perioral pallor ([Chap. 187](#)). A more diffuse rash (often pruritic) appears the next day on the trunk and extremities and then rapidly develops into a lacy reticular eruption that may wax and wane (especially with temperature change) over 3 weeks. Adults with fifth disease often have arthritis, and fetal hydrops can develop in association with this condition in pregnant women.

Exanthem subitum (roseola, [Fig. 18-CD2](#)) is most common among children under 3 years of age ([Chap. 185](#)). As in erythema infectiosum, the rash usually appears after fever has subsided. It consists of 2- to 3-mm rose-pink macules and papules that rarely coalesce, occur initially on the trunk and sometimes on the extremities (sparing the face), and fade within 2 days.

Though drug reactions have many manifestations, including urticaria, exanthematous *drug-induced eruptions* ([Chap. 59](#)) are most common and are often difficult to distinguish from viral exanthems. Eruptions elicited by drugs are usually more intensely erythematous and pruritic than viral exanthems, but this distinction is not reliable. A history of new medications and an absence of prostration may help to distinguish a drug-related rash from an eruption of another etiology. Rashes may persist for up to 2 weeks after administration of the offending agent is discontinued. Certain populations are more prone than others to drug rashes. Of HIV-infected patients, 50 to 60% develop a rash in response to sulfa drugs; 50 to 100% of patients with mononucleosis due to Epstein-Barr virus develop a rash when given ampicillin.

Rickettsial illnesses ([Chap. 177](#)) should be considered in the evaluation of individuals with centrally distributed maculopapular eruptions. The usual setting for *epidemic typhus* is a site of war or natural disaster in which people are exposed to body lice. A diagnosis of recrudescent typhus should be considered in European immigrants to the United States. However, an indigenous form of typhus, presumably transmitted by flying squirrels, has been reported in the southeastern United States. *Endemic typhus* or *leptospirosis* (the latter caused by a spirochete; [Chap. 174](#)) may be seen in urban environments where rodents proliferate. Outside the United States, other rickettsial diseases cause a spotted-fever syndrome and should be considered in residents of or travelers to endemic areas. Similarly, *typhoid fever*, a nonrickettsial disease caused by *Salmonella typhi* ([Chap. 156](#)), is usually acquired during travel outside the United States.

Some centrally distributed maculopapular eruptions have distinctive features. Erythema chronicum migrans (ECM), the rash of Lyme disease ([Chap. 176](#)), typically manifests as singular or multiple annular plaques. Untreated ECM lesions usually fade within a month but may persist for more than a year. *Erythema marginatum*, the rash of acute rheumatic fever ([Chap. 235](#)), has a distinctive pattern of enlarging and shifting transient annular lesions.

Collagen vascular diseases may cause fever and rash. Patients with *systemic lupus erythematosus* ([Chap. 311](#)) typically develop a sharply defined, erythematous eruption in a butterfly distribution on the cheeks (malar rash) as well as many other skin manifestations. *Still's disease* ([Chap. 326](#)) manifests as an evanescent salmon-colored rash on the trunk and proximal extremities that coincides with fever spikes.

Peripheral Eruptions These rashes are alike in that they are most prominent peripherally or begin in peripheral (acral) areas before spreading centripetally. Early diagnosis and therapy are critical in *RMSF* ([Chap. 177](#)) because of its grave prognosis if untreated. Lesions evolve from macular to petechial, start on the wrists and ankles, spread centripetally, and appear on the palms and soles only later in the disease. The rash of *secondary syphilis* ([Chap. 172](#)), which may be diffuse but is prominent on the palms and soles, should be considered in the differential diagnosis of pityriasis rosea, especially in sexually active patients. *Atypical measles* ([Chap. 194](#)) is seen in individuals contracting measles who received the killed measles vaccine between 1963 and 1967 in the United States and who were not subsequently protected with the live vaccine. *Hand-foot-and-mouth disease* ([Chap. 193](#)) is distinguished by tender vesicles distributed peripherally and in the mouth; outbreaks commonly occur within families. The classic

target lesions of *erythema multiforme* appear symmetrically on the elbows, knees, palms, and soles. In relatively severe cases, these lesions may spread diffusely and involve mucosal surfaces. Lesions may develop on the hands and feet in *endocarditis* ([Chap. 126](#)).

Confluent Desquamative Erythemas These eruptions consist of diffuse erythema frequently followed by desquamation. The eruptions caused by group A *Streptococcus* or *Staphylococcus aureus* are toxin mediated. Certain disease features may provide diagnostic clues. *Scarlet fever* ([Chap. 140](#)) usually follows pharyngitis; patients have a facial flush, a "strawberry" tongue, and accentuated petechiae in body folds (Pastia's lines). *Kawasaki disease* ([Chaps. 57 and 317](#)) presents in the pediatric population as fissuring of the lips, a strawberry tongue, conjunctivitis, adenopathy, and sometimes cardiac abnormalities. *Streptococcal toxic shock syndrome* ([Chap. 140](#), [Fig. 18-CD3](#)) manifests with hypotension, multiorgan failure, and often a severe group A streptococcal infection (e.g., necrotizing fasciitis, [Fig. 18-CD4](#)). *Staphylococcal toxic shock syndrome* ([Chap. 139](#)) also presents with hypotension and multiorgan failure, but usually only *S. aureus* colonization -- not a severe *S. aureus* infection -- is documented. *Staphylococcal scalded-skin syndrome* ([Chap. 139](#)) is seen primarily in children and in immunocompromised adults. Generalized erythema is often evident during the prodrome of fever and malaise; profound tenderness of the skin is distinctive. In the exfoliative stage, the skin can be induced to form bullae with light lateral pressure (Nikolsky's sign). In a mild form, a scarlatiniform eruption mimics scarlet fever, but the patient does not exhibit a strawberry tongue or circumoral pallor. In contrast to the staphylococcal scalded-skin syndrome, in which the cleavage plane is superficial in the epidermis, *toxic epidermal necrolysis* ([Chap. 59](#)) ([Fig. 18-CD5](#)) involves sloughing of the entire epidermis, resulting in severe disease. *Exfoliative erythroderma syndrome* ([Chaps. 56 and 59](#)) is a serious reaction associated with systemic toxicity that is often due to eczema, psoriasis, mycosis fungoides, or a severe drug reaction.

Vesiculobullous Eruptions *Varicella* ([Chap. 183](#)) is highly contagious, often occurring in winter or spring. At a given time within a given region of the body, varicella lesions are in different stages of development. In immunocompromised hosts, varicella vesicles may lack the characteristic erythematous base or may appear hemorrhagic. *Rickettsialpox* ([Chap. 177](#)) is often documented in urban settings and is characterized by vesicles. It can be distinguished from varicella by an eschar at the site of the mouse-mite bite and the papule/plaque base of each vesicle. Disseminated *Vibrio vulnificus* infection ([Chap. 159](#)) or *ecthyma gangrenosum* due to *Pseudomonas aeruginosa* ([Chap. 155](#)) should be considered in immunosuppressed individuals with sepsis and hemorrhagic bullae.

Urticarial Eruptions Individuals with classic urticaria ("hives") usually have a hypersensitivity reaction without associated fever. In the presence of fever, urticarial eruptions are usually due to *urticarial vasculitis* ([Chap. 317](#)). Unlike individual lesions of classic urticaria, which last up to 48 h, these lesions may last up to 5 days. Etiologies include serum sickness (often induced by drugs such as penicillins, sulfas, salicylates, or barbiturates), connective-tissue disease (e.g., systemic lupus erythematosus or Sjogren's syndrome), and infection (e.g., with hepatitis B virus, coxsackievirus A9, or parasites). Malignancy may be associated with fever and chronic urticaria ([Chap. 57](#)).

Nodular Eruptions In immunocompromised hosts, nodular lesions often represent disseminated infection. Patients with disseminated *candidiasis* (often due to *Candida tropicalis*) may have a triad of fever, myalgias, and eruptive nodules ([Chap. 205](#)). Disseminated *cryptococcosis* lesions ([Chap. 204](#)) may resemble molluscum contagiosum. Necrosis of nodules should raise the suspicion of *aspergillosis* ([Chap. 206](#)) or *mucormycosis* ([Chap. 207](#)). *Erythema nodosum* presents with exquisitely tender nodules on the lower extremities. *Sweet's syndrome* ([Chap. 57](#)) should be considered in individuals with multiple nodules and plaques, often so edematous that they give the appearance of vesicles or bullae. Sweet's syndrome may affect either healthy individuals or persons with lymphoproliferative disease.

Purpuric Eruptions *Acute meningococcemia* ([Chap. 146](#)) classically presents in children as a petechial eruption, but initial lesions may appear as blanchable macules or urticaria. *RMSF* should be considered in the differential diagnosis of acute meningococcemia. *Echovirus 9 infection* ([Chap. 193](#)) may mimic acute meningococcemia; patients should be treated as if they have bacterial sepsis since prompt differentiation of these conditions may be impossible. Large ecchymotic areas of *purpura fulminans* ([Chaps. 124 and 146](#)) reflect severe underlying disseminated intravascular coagulation, which may be due to infectious or noninfectious causes. The lesions of *chronic meningococcemia* ([Chap. 146](#)) may have a variety of morphologies, including petechial. Purpuric nodules may develop on the legs and resemble erythema nodosum but lack its exquisite tenderness. Lesions of *disseminated gonococcemia* ([Chap. 147](#)) are distinctive, sparse, countable hemorrhagic pustules, usually located near joints. The lesions of chronic meningococcemia and those of gonococcemia may be indistinguishable in terms of appearance and distribution. *Viral hemorrhagic fever* ([Chaps. 198 and 199](#)) should be considered in patients with an appropriate travel history and a petechial rash. *Thrombotic thrombocytopenic purpura* ([Chaps. 57, 108, and 116](#)) is a noninfectious cause of fever and petechiae. *Cutaneous small-vessel vasculitis (leukocytoclastic vasculitis)* typically manifests as palpable purpura and has a wide variety of causes ([Chap. 57](#)).

Eruptions with Ulcers or Eschars The presence of an ulcer or eschar in the setting of a more widespread eruption can provide an important diagnostic clue. For example, the presence of an eschar may suggest the diagnosis of scrub typhus or rickettsialpox in the appropriate setting. In other illnesses (e.g., anthrax), an ulcer or eschar may be the only skin manifestation.

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19. APPROACH TO THE ACUTELY ILL INFECTED FEBRILE PATIENT - Tamar F. Barlam, Dennis L. Kasper

The physician treating the acutely ill febrile patient must be able to recognize infections that require emergent attention. If such infections are not adequately evaluated and treated at initial presentation, the opportunity to alter an adverse outcome may be lost. In this chapter, the clinical presentations of and approach to patients with relatively common infectious disease emergencies are discussed. These infectious processes are discussed in detail in other chapters. **Noninfectious causes of fever are not covered in this chapter; information on the approach to fever of unknown origin, including that eventually shown to be of noninfectious etiology, is presented in [Chap. 125](#).*

GENERAL CONSIDERATIONS

APPEARANCE

A physician must have a consistent approach to acutely ill patients. Even before the history is elicited and a physical examination performed, an immediate assessment of the patient's general appearance yields valuable information. The perceptive physician's subjective sense that a patient is septic or toxic often proves accurate. Visible agitation or anxiety in a febrile patient can be a harbinger of critical illness.

HISTORY

Presenting symptoms are frequently nonspecific. In addition to a general description of symptoms, it is important to obtain a sense of disease progression. Detailed questions should be asked about the onset and duration of symptoms and about changes in severity or rate of progression over time. Host factors and comorbid conditions may enhance the risk of infection with certain organisms or of a more fulminant course than is usually seen. Lack of splenic function, alcoholism with significant liver disease, intravenous drug use, HIV infection, diabetes, malignancy, and chemotherapy all predispose to specific infections and frequently to increased severity. The patient should be questioned about factors that might help identify a nidus for invasive infection, such as recent upper respiratory tract infections, influenza, or varicella; prior trauma; disruption of cutaneous barriers due to lacerations, burns, surgery, or decubiti; and the presence of foreign bodies, such as nasal packing after rhinoplasty, barrier contraceptives, tampons, arteriovenous fistulas, or prosthetic joints. Travel, contact with pets or other animals, or activities that might result in tick exposure can lead to diagnoses that would not otherwise be considered. Recent dietary intake, medication use, social contact with ill individuals, vaccination history, and menstrual history may be relevant. A review of systems should focus on any neurologic signs or sensorium alterations, rashes or skin lesions, and focal pain or tenderness and should also include a general review of respiratory, gastrointestinal, or genitourinary symptoms. It is especially important to determine the duration and progression of these symptoms in order to gain an appreciation of the pace and urgency of the process.

PHYSICAL EXAMINATION

A complete physical examination should be performed, with special attention to some

areas that are sometimes given short shrift in routine examinations. Assessment of the patient's general appearance and vital signs, skin and soft tissue examination, and the neurologic evaluation are of particular importance.

The patient may appear either anxious and agitated or lethargic and apathetic. Fever is usually present, although the elderly and compromised hosts, such as those who are uremic or cirrhotic and patients who are taking glucocorticoids or nonsteroidal anti-inflammatory agents, may be afebrile despite serious underlying infection. Measurement of blood pressure, heart rate, and respiratory rate helps determine the degree of hemodynamic and metabolic compromise. The patient's airway must be evaluated to rule out the risk of obstruction from an invasive oropharyngeal infection.

The etiologic diagnosis may become evident in the context of a thorough skin examination. Petechial rashes are typically seen with meningococcemia or Rocky Mountain spotted fever (RMSF); erythroderma is usual with toxic shock syndrome (TSS) and drug fever. The soft tissue and muscle examination is critical. Areas of erythema or duskiness, edema, and tenderness may indicate underlying necrotizing fasciitis, myositis, or myonecrosis. The neurologic examination must include a careful assessment of mental status for signs of early encephalopathy. Evidence of nuchal rigidity or focal neurologic findings should be sought. Focal findings, depressed mental status, or papilledema should be evaluated by brain imaging prior to lumbar puncture, which, in this setting, could initiate herniation.

SPECIFIC PRESENTATIONS

For most infections, there is time for careful evaluation, diagnostic testing, and consultation with other physicians. However, the infections considered below according to common clinical presentation can have rapidly catastrophic outcomes, and their immediate recognition can be life-saving. Recommended therapeutic regimens are presented in [Table 19-1](#).

SEPSIS WITHOUT AN OBVIOUS FOCUS OF PRIMARY INFECTION

These patients initially have a brief prodrome of nonspecific symptoms and signs that progresses quickly to hemodynamic instability with hypotension, tachycardia, tachypnea, or respiratory distress. A patient may display altered mental status. Disseminated intravascular coagulation (DIC) with clinical evidence of a hemorrhagic diathesis is a poor prognostic sign.

Septic Shock (See also [Chap. 124](#)) Patients with bacteremia leading to septic shock may have a primary site of infection (e.g., pneumonia, pyelonephritis, or cholangitis) that is not evident initially. Elderly patients with comorbid conditions, hosts compromised by malignancy and neutropenia, or patients who have recently undergone a surgical procedure or hospitalization are at increased risk for an adverse outcome. Gram-negative bacteremia with organisms such as *Pseudomonas aeruginosa*, *Aeromonas hydrophila*, or *Escherichia coli* and gram-positive infection with organisms such as *Staphylococcus aureus* or group A streptococci can present as intractable hypotension and multiorgan failure. Treatment can usually be initiated empirically on the basis of the presentation ([Table 124-3](#)).

Overwhelming Infection in Asplenic Patients (See also [Chap. 124](#)) Patients without splenic function are at risk for overwhelming bacterial sepsis. Asplenic patients succumb to sepsis at 600 times the rate of the general population; 50 to 70% of cases occur within the first 2 years after splenectomy, with a mortality rate of up to 80%. However, in the asplenic individual, an increased risk of overwhelming sepsis continues throughout life. In asplenia, encapsulated bacteria cause the majority of infections, and adults are at lower risk than children because they are more likely to have antibody to these organisms. *Streptococcus pneumoniae* infection is most common, but the risk of infection with *Haemophilus influenzae* or *Neisseria meningitidis* is also high. Severe clinical manifestations of infections due to *E. coli*, *S. aureus*, group B streptococci, *P. aeruginosa*, *Capnocytophaga*, *Babesia*, and *Plasmodium* have been described.

Babesiosis (See also [Chap. 214](#)) A history of recent travel to endemic areas should raise the possibility of infection with *Babesia*. Between 1 and 4 weeks after a tick bite, the patient experiences chills, fatigue, anorexia, myalgia, arthralgia, nausea, and headache; ecchymosis and/or petechiae are occasionally seen. The tick that most commonly transmits *Babesia*, *Ixodes scapularis*, also transmits *Borrelia burgdorferi* (the agent of Lyme disease) and *Ehrlichia*, and co-infection can occur, resulting in more severe disease. Infection with the European species *Babesia divergens* is more frequently fulminant than that due to the U.S. species *B. microti*, causing a febrile syndrome with hemolysis, jaundice, hemoglobinemia, and renal failure and a mortality rate of >50%. Severe babesiosis is especially common in asplenic hosts but does occur in hosts with normal splenic function.

Other Sepsis Syndromes Tularemia ([Chap. 161](#)) is seen throughout the United States, but primarily in Arkansas, Oklahoma, and Missouri, in association with wild rabbit, tick, and tabanid fly contact. The uncommon typhoidal form can be associated with gram-negative septic shock and a mortality rate of >30%. In the United States, plague ([Chap. 162](#)) is found primarily in New Mexico, Arizona, and Colorado after contact with ground squirrels, prairie dogs, or chipmunks. The septic form is particularly rare and is associated with shock, multiorgan failure, and a 30% mortality rate. These rare infections should be considered in the appropriate epidemiologic setting.

SEPSIS WITH SKIN MANIFESTATIONS (See also [Chap. 18](#))

Maculopapular rashes may reflect early meningococcal or rickettsial disease but are usually associated with nonemergent infections. Exanthems are usually viral.

Petechiae Petechial rashes caused by viruses are seldom associated with hypotension or a toxic appearance, although severe measles can be an exception. In other settings, petechial rashes require more urgent attention.

Meningococcemia (See also [Chap. 146](#)) Almost three-quarters of patients with bacteremic *N. meningitidis* infection have a rash. Meningococcemia most often affects young children (i.e., those 6 months to 5 years old, often in daycare). However, sporadic cases and outbreaks occur in schools (grade school through college) and army barracks. Between 10 and 20% of all cases have a fulminant course, with shock, [DIC](#), and multiorgan failure. Of these patients, 50 to 60% die, and survivors often require

extensive debridement or amputation of gangrenous extremities. Patients may exhibit fever, headache, nausea, vomiting, myalgias, change in mental status, and meningismus. However, the rapidly progressive form of disease is not usually associated with meningitis. The rash is initially pink, blanching, and maculopapular, appearing on the trunk and extremities, but then becomes hemorrhagic, forming petechiae. Petechiae are first seen at the ankles, wrists, axillae, mucosal surfaces, and palpebral and bulbar conjunctiva, with subsequent spread to the lower extremities and trunk. A cluster of petechiae may be seen at pressure points, e.g., where a blood pressure cuff has been inflated. In rapidly progressive meningococemia, the petechial rash quickly becomes purpuric ([Plate IID-44](#)) and patients develop DIC. Hypotension with petechiae for <12 h is associated with significant mortality. The mortality rate can exceed 90% in patients without meningitis who have rash, hypotension, and a normal or low white blood cell count and erythrocyte sedimentation rate. A better prognosis has been reported in cases where antibiotics are given before admission by the primary care provider. This observation suggests that early initiation of treatment may be life-saving.

Rocky Mountain spotted fever (See also [Chap. 177](#)) **RMSF** occurs throughout the United States. A history of tick bite is common; however, if such a history is lacking, a history of travel or outdoor activity (e.g., camping in tick-infested areas) can be ascertained. RMSF is caused by *Rickettsia rickettsii*. For the first 3 days, headache, fever, malaise, myalgias, nausea, vomiting, and anorexia are present. By day 3, half of patients have skin findings. Blanching macules develop initially on the wrists and ankles and then spread over the legs and trunk. The lesions become hemorrhagic and are frequently petechial. The rash spreads to palms and soles later in the course ([Plate IID-45](#)). The centripetal spread is a classic feature of RMSF. However, 10 to 15% of patients with RMSF never develop a rash. The patient can be hypotensive and develop noncardiogenic pulmonary edema, confusion, lethargy, and encephalitis progressing to coma. The cerebrospinal fluid (CSF) contains 10 to 100 cells/uL, usually with a predominance of mononuclear cells. The CSF glucose level is often normal; the protein concentration may be slightly elevated. Renal and hepatic injury and bleeding secondary to vascular damage are noted. Untreated infection has a mortality rate of 30%.

Purpura Fulminans (See also [Chaps. 124 and 146](#)) This is the cutaneous manifestation of **DIC** and presents as large ecchymotic areas and hemorrhagic bullae. Progression of petechiae to purpura and ecchymoses is associated with congestive heart failure, septic shock, acute renal failure, acidosis, hypoxia, hypotension, and death. Purpura fulminans has primarily been associated with *N. meningitidis* but, in the splenectomized patient, has been described in association with *S. pneumoniae* and *H. influenzae*.

Ecthyma Gangrenosum Septic shock caused by *P. aeruginosa* and *A. hydrophila* can be associated with ecthyma gangrenosum ([Plate IID-57C](#), [Fig. 19-CD1](#)): hemorrhagic vesicles surrounded by a rim of erythema with central necrosis and ulceration. These gram-negative bacteremias are most common among patients with neutropenia, extensive burns, and hypogammaglobulinemia.

Other Emergent Infections Associated with Rash *Vibrio vulnificus* and other noncholera *Vibrio* bacteremic infections ([Chap. 159](#)) can cause focal skin lesions and overwhelming sepsis in the host with liver disease. After ingestion of contaminated

shellfish, there is a sudden onset of malaise, chills, fever, and hypotension. The patient develops bullous or hemorrhagic skin lesions, usually on the lower extremities, and 75% of patients have leg pain. The mortality rate can be as high as 50%. *Capnocytophaga canimorsus* ([Chap. 127](#)) can cause septic shock in asplenic patients. Infection with this fastidious gram-negative rod typically presents after a dog bite as fever, chills, myalgia, vomiting, diarrhea, dyspnea, confusion, and headache. Findings can include an exanthem or erythema multiforme ([Plate IIE-67](#)), cyanotic mottling or peripheral cyanosis, petechiae, and ecchymosis. About 30% of patients with this fulminant form die of overwhelming sepsis and [DIC](#), and survivors may require amputation to treat gangrene.

Erythroderma TSS([Chaps. 139](#) and [140](#)) is usually associated with erythroderma ([Fig. 18-CD3](#)). The patient presents with fever, malaise, myalgias, nausea, vomiting, diarrhea, and confusion. There is a sunburn-type rash that may be subtle and patchy but is usually diffuse and is found on the face, trunk, and extremities. Erythroderma, which desquamates after 1 to 2 weeks, is more common in *Staphylococcus*-associated than in *Streptococcus*-associated TSS. Hypotension develops rapidly after onset of symptoms, often within hours. Multiorgan failure is seen. Often there is no indication of a primary focal infection. Colonization rather than overt infection of the vagina or a postoperative wound, for example, is typical with staphylococcal TSS, and the mucosal areas appear hyperemic but not infected. Early renal failure may distinguish this syndrome from other septic shock syndromes. Clinical evaluation constitutes the diagnosis because TSS is defined by the clinical criteria of fever, rash, hypotension, and multiorgan involvement. The mortality rate is 5% for menstruation-associated TSS, 10 to 15% for nonmenstrual TSS, and 30 to 70% for streptococcal TSS.

SEPSIS WITH A SOFT TISSUE/MUSCLE PRIMARY FOCUS (See also [Chap. 128](#))

Necrotizing Fasciitis This infection may arise at a site of minimal trauma or postoperative incision and may also be associated with recent varicella, childbirth, or muscle strain. The most common causes of necrotizing fasciitis are group A streptococci alone ([Chap. 140](#)) and a mixed facultative and anaerobic flora ([Chap. 128](#)). Diabetes mellitus, peripheral vascular disease, and intravenous drug use are associated risk factors. Use of nonsteroidal anti-inflammatory agents adversely affects granulocyte chemotaxis, phagocytosis, and bacterial killing, allowing progression of skin or soft tissue infections. The patient may have bacteremia and hypotension without other organ-system failure. Physical findings are minimal compared to the severity of pain and the degree of fever. The examination is often unremarkable except for soft tissue edema and erythema. The infected area is red, hot, shiny, swollen, and exquisitely tender. In untreated infection, the overlying skin develops blue-gray patches after 36 h, and cutaneous bullae and necrosis develop after 3 to 5 days. Necrotizing fasciitis due to a mixed flora, but not that due to group A streptococci, can be associated with gas production. Without treatment, pain decreases because of thrombosis of the small blood vessels and destruction of the peripheral nerves -- an ominous sign. The mortality rate is >30% overall, >70% in association with [TSS](#), and nearly 100% without surgical intervention. Life-threatening necrotizing fasciitis may also be due to *Clostridium perfringens* ([Chap. 145](#)); in this condition, the patient is extremely toxic and the mortality rate is high. Within 48 h, rapid tissue invasion and systemic toxicity associated with hemolysis and death ensue. The distinction between this entity and clostridial

myonecrosis is made by muscle biopsy.

Clostridial Myonecrosis (See also [Chap. 145](#)) Myonecrosis is often associated with trauma or surgery but can be spontaneous. The incubation period is usually 12 to 24 h long, and massive necrotizing gangrene develops within hours of onset. Systemic toxicity, shock, and death can occur within 12 h. The patient's pain and toxic appearance are out of proportion to physical findings. On examination, the patient is febrile, apathetic, tachycardic, and tachypneic and may express a feeling of impending doom. Hypotension and renal failure develop later, and hyperalertness is evident preterminally. The skin over the affected area is bronze-brown, mottled, and edematous. Bullous lesions with serosanguineous drainage and a mousy or sweet odor can be present. Crepitus can occur secondary to gas production in muscle tissue. The mortality rate is >65% with spontaneous myonecrosis, which is often associated with *C. septicum* and underlying malignancy. The mortality rates associated with trunk and limb infection are 63% and 12%, respectively, and any delay in surgical treatment increases the risk of death.

NEUROLOGIC INFECTIONS WITH OR WITHOUT SEPTIC SHOCK

Bacterial Meningitis (See also [Chap. 372](#)) Bacterial meningitis is one of the most common infectious emergencies involving the central nervous system. Although hosts with cell-mediated immune deficiency, including transplant recipients, diabetic patients, the elderly, and cancer patients treated with certain chemotherapeutic agents, are at particular risk for *Listeria monocytogenes* meningitis, most cases in adults are due to *S. pneumoniae* (30 to 50%) and *N. meningitidis* (10 to 35%). An early presentation of headache, meningismus, and fever is classic but is seen in only half of patients. The elderly can present without fever or meningeal signs despite lethargy and confusion. Cerebral dysfunction is evidenced by confusion, delirium, and lethargy that can progress to coma. The presentation is fulminant, with sepsis and brain edema, in some cases; papilledema at presentation is unusual and suggests another diagnosis (e.g., an intracranial lesion). Focal signs, including cranial nerve palsies (IV, VI, VII), can be seen in 10 to 20% of cases; 50 to 60% of patients have bacteremia. A poor neurologic outcome is associated with coma at any time during the course or with a CSF glucose level of <0.6 mmol/L (<10 mg/dL). Mortality is associated with coma, respiratory distress, shock, a CSF protein level of >2.5 g/L, a peripheral white blood cell count of <5000/uL, and a serum sodium level of <135 mmol/L.

Suppurative Intracranial Infections (See also [Chap. 372](#)) Other rare intracranial lesions that present with sepsis and hemodynamic instability are subdural empyema, septic cavernous sinus thrombosis, and septic superior sagittal sinus thrombosis. Rapid recognition of the toxic patient with central neurologic signs is crucial to improvement of the dismal prognosis of these entities.

Subdural Empyema This infection arises from the paranasal sinus in 60 to 70% of cases. Microaerophilic streptococci and staphylococci are the predominant etiologic organisms. The patient is toxic, with fever, headache, and nuchal rigidity. Of all patients, 75% have focal signs and 6 to 20% die.

Septic Cavernous Sinus Thrombosis This condition follows a facial or sphenoid sinus

infection; 70% of cases are due to staphylococci and the remainder to aerobic or anaerobic streptococci. A unilateral or retroorbital headache progresses to a toxic appearance and fever within days. Three-quarters of patients have unilateral periorbital edema that becomes bilateral and then progresses to ptosis, proptosis, ophthalmoplegia, and papilledema. The mortality rate is as high as 30%.

Septic Thrombosis of the Superior Sagittal Sinus This infection spreads from the ethmoid or maxillary sinuses. Its bacterial causes include *S. pneumoniae*, other streptococci, and staphylococci. The fulminant course is characterized by headache, nausea, vomiting, rapid progression to confusion and coma, nuchal rigidity, and brainstem signs. If the sinus is totally thrombosed, the mortality rate exceeds 80%.

Brain Abscess (See also [Chap. 372](#)) Brain abscess often occurs without systemic signs. Almost half of patients are afebrile, and presentations are more consistent with a space-occupying lesion in the brain; 70% have headache, 50% have focal neurologic signs, and 25% have papilledema. Abscesses can present as single or multiple lesions resulting from contiguous foci or hematogenous infection, such as unrecognized endocarditis. The infection progresses over several days from cerebritis to an abscess with a mature capsule. Abscesses arising hematogenously are especially apt to rupture into the ventricular space, causing a sudden and severe deterioration in clinical status and high mortality. Otherwise, mortality is low but morbidity is high (30 to 55%). Patients presenting with stroke and a parameningeal infectious focus, such as sinusitis or otitis, may have a brain abscess, and physicians must maintain a high level of suspicion. Prognosis worsens in patients with a fulminant course, delayed diagnosis, abscess rupture into the ventricles, multiple abscesses, or abnormal neurologic status at presentation.

Cerebral Malaria (See also [Chap. 214](#)) This entity should be urgently considered if patients who have recently traveled to areas endemic for malaria present with a febrile illness and lethargy or other neurologic signs. Fulminant malaria is caused by *Plasmodium falciparum* and is associated with temperatures of $>40^{\circ}\text{C}$ ($>104^{\circ}\text{F}$), hypotension, jaundice, adult respiratory distress syndrome, and bleeding. By definition, any patient with a change in mental status or repeated seizure in the setting of fulminant malaria has cerebral malaria. In adults this nonspecific febrile illness progresses to coma over several days; occasionally, coma occurs within hours and death within 24 h. Nuchal rigidity and photophobia are rare. On physical examination, symmetric encephalopathy is typical, and upper motor neuron dysfunction with decorticate and decerebrate posturing can be seen with advanced disease. Unrecognized infection results in a 30% mortality rate.

Spinal Epidural Abscesses (See also [Chap. 368](#)) Patients with spinal epidural abscesses often present with back pain and develop neurologic deficits late in their course. At-risk patients include those with diabetes mellitus; intravenous drug use; recent spinal trauma, surgery, or epidural anesthesia; and other comorbid conditions, such as HIV infection. The thoracic or lumbar spine is the most common location, and staphylococci are the most common etiologic agents; in HIV-infected intravenous drug users, therapy must cover gram-negative rods and methicillin-resistant *S. aureus*. If a patient gives a history of antecedent back pain and has new neurologic symptoms, this diagnosis must immediately be considered. Almost 60% of patients have fever and

almost 90% have back pain. Paresthesia, bowel and bladder dysfunction, radicular pain, and weakness are frequent neurologic complaints, and examination of the patient may reveal abnormal reflexes and motor and sensory deficits. Rapid recognition and treatment, including immediate drainage, can prevent or minimize permanent neurologic sequelae.

FOCAL SYNDROMES WITH A FULMINANT COURSE

Infection at virtually any primary focus (e.g., osteomyelitis, pneumonia, pyelonephritis, or cholangitis) can result in bacteremia and sepsis. [TSS](#) has been associated with focal infections such as septic arthritis, peritonitis, sinusitis, and wound infection. Death occurs secondary to septic shock or toxin production with hemodynamic instability and multiorgan failure. Rapid clinical deterioration and death can be associated with destruction of the primary site of infection, as is seen in endocarditis and in necrotizing infections of the oropharynx (in which edema suddenly compromises the airway).

Rhinocerebral Mucormycosis (See also [Chap. 207](#)) Patients with diabetes or malignancy are at risk for invasive rhinocerebral mucormycosis. Patients present with low-grade fever, dull sinus pain, diplopia, decreased mental status, decreased ocular motion, chemosis, proptosis, dusky or necrotic nasal turbinates, and necrotic hard-palate lesions that respect the midline. Without rapid recognition and intervention, the process continues an inexorable invasive course with high mortality.

Acute Bacterial Endocarditis (See also [Chap. 126](#)) This entity presents with a much more aggressive course than subacute endocarditis. Bacteria such as *S. aureus*, *S. pneumoniae*, *L. monocytogenes*, *Haemophilus* spp., and streptococci of groups A, B, and G attack native valves. Mortality rates range from 10 to 40%. The host may have comorbid conditions such as underlying malignancy, diabetes mellitus, intravenous drug use, or alcoholism. The patient presents with fever, fatigue, and malaise <2 weeks after onset of infection. On physical examination, a changing murmur and congestive heart failure may be noted. Hemorrhagic macules on palms or soles (*Janeway lesions*, [Fig. 19-CD2](#)) sometimes develop. Petechiae, Roth's spots ([Fig. 19-CD3](#)), splinter hemorrhages ([Fig. 19-CD4](#)), and splenomegaly are unusual. Rapid valvular destruction, particularly of the aortic valve, results in pulmonary edema and hypotension. Myocardial abscesses can form, eroding through the septum or into the conduction system and causing life-threatening arrhythmias or high-degree conduction block. Large friable vegetations can result in major arterial emboli, metastatic infection, or tissue infarction. Emboli can lead to stroke, change in mental status, visual disturbances, aphasia, ataxia, headache, meningismus, brain abscess, cerebritis, spinal cord infarct with paraplegia, arthralgia, osteomyelitis, splenic abscess, septic arthritis, and hematuria. Rapid intervention is crucial for a successful outcome.

DIAGNOSTIC WORKUP OF THE ACUTELY ILL PATIENT

After a quick clinical assessment, diagnostic material should be obtained rapidly and antibiotic and supportive treatment begun. In the sepsis syndromes, blood (for cultures; baseline complete blood count with differential; measurement of serum electrolytes, blood urea nitrogen, serum creatinine, and serum glucose; and liver function tests) can be obtained at the time an intravenous line is placed and before antibiotics are

administered. For patients with possible acute endocarditis, three sets of blood cultures should be performed. Asplenic patients should have a blood smear examined to confirm the presence of Howell-Jolly bodies (indicating the absence of splenic function) and a buffy coat examined for bacteria; these patients can have $>10^6$ organisms per milliliter of blood (compared to 10^4 /mL in patients with an intact spleen). Blood smears from patients with possible cerebral malaria or babesiosis must be examined for the diagnosis and quantitation of parasitemia. Blood smears may also be diagnostic in ehrlichiosis.

Patients with meningitis should have [CSF](#) obtained before the initiation of antibiotic therapy. *If focal neurologic signs, abnormal mental status, or papilledema mandates brain imaging before a lumbar puncture, antibiotics should be administered prior to imaging but after blood for cultures has been drawn.* If CSF cultures are negative, laboratory examination of CSF by latex agglutination or immunoprecipitation can be attempted to make an etiologic diagnosis. However, blood cultures will provide the diagnosis in 50 to 70% of cases.

Focal abscesses necessitate immediate computed tomography or magnetic resonance imaging as part of an evaluation for surgical intervention. Other diagnostic procedures, such as cultures of wounds or scraping of skin lesions, should not delay the initiation of treatment for more than minutes. Once emergent evaluation, diagnostic procedures, and (if appropriate) surgical consultation (see below) have been completed, other laboratory tests can be conducted. Appropriate radiography, computed axial tomography, magnetic resonance imaging, urinalysis, erythrocyte sedimentation rate determination, and transthoracic or transesophageal echocardiography may all prove important.

TREATMENT

[Table 19-1](#) lists first-line treatments for the infections considered in this chapter. (For a more detailed discussion of treatment, see specific chapters.) In addition to the initiation of parenteral antibiotic therapy, several of these infections require urgent surgical attention. General surgery for possible necrotizing fasciitis or myonecrosis, neurosurgical evaluation for subdural empyema or spinal epidural abscess, otolaryngologic surgery for possible mucormycosis, and cardiothoracic surgery for critically ill patients with acute endocarditis are as important as the rapid commencement of antibiotic therapy. For infections such as necrotizing fasciitis and clostridial myonecrosis, rapid surgical intervention supercedes other diagnostic or therapeutic maneuvers.

Acutely ill febrile patients require close observation, aggressive supportive measures, and -- in most cases -- admission to intensive care units. Adjunctive treatments, such as intravenous immunoglobulin administration for [TSS](#), can be considered after initial stabilization. The most important task of the physician is to recognize the acute infectious emergency and proceed with appropriate urgency.

(Bibliography omitted in Palm version)

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20. HYPOTHERMIA AND FROSTBITE - Daniel F. Danzl

HYPOTHERMIA

Accidental hypothermia occurs when there is an unintentional drop in the body's core temperature below 35°C (95°F). At this temperature, many of the compensatory physiologic mechanisms to conserve heat begin to fail. *Primary accidental hypothermia* is a result of the direct exposure of a previously healthy individual to the cold. The mortality rate is much higher for those patients who develop *secondary hypothermia* as a complication of a serious systemic disorder.

CAUSES

Primary accidental hypothermia is geographically and seasonally pervasive. Although most cases occur in the winter months and in colder climates, it is surprisingly common in warmer regions as well. In the United States, hypothermia accounts for more than 700 deaths each year, half of which occur in people age 65 or older.

Multiple variables make individuals at the extremes of age, the elderly and neonates, particularly vulnerable to hypothermia ([Table 20-1](#)). The elderly have diminished thermal proprioception and are more susceptible to immobility, malnutrition, and systemic illnesses that interfere with heat generation or conservation. Dementia, psychiatric illness, and socioeconomic factors often compound these problems by impeding adequate measures to prevent hypothermia. Neonates have high rates of heat loss because of their increased surface-to-mass ratio and their lack of effective shivering and adaptive behavioral responses. In addition, malnutrition can contribute to heat loss because of diminished subcutaneous fat and because of its association with depleted energy stores used for thermogenesis.

Individuals whose occupations or hobbies entail extensive exposure to cold weather are clearly at increased risk for hypothermia. Military history is replete with hypothermic tragedies. Hunters, sailors, skiers, and climbers also are at great risk of exposure, whether it involves injury, changes in weather, or lack of preparedness.

Ethanol causes vasodilatation (which increases heat loss), reduces thermogenesis and gluconeogenesis, and may impair judgment or lead to obtundation. Hypothermia is not an uncommon feature in Wernicke's encephalopathy and may mask its other manifestations. A number of medications are associated with altered thermal regulation. Phenothiazines, barbiturates, benzodiazepines, cyclic antidepressants, and many other medications reduce centrally-mediated vasoconstriction. Up to one-quarter of patients admitted to an intensive care unit because of drug overdose are hypothermic.

Anesthetics can block the shivering responses; their effects may be compounded when patients are not covered adequately in the operating or recovery rooms.

Several types of endocrine dysfunction can lead to hypothermia. Hypothyroidism -- particularly when extreme, as in myxedema coma -- reduces the metabolic rate and impairs thermogenesis and behavioral responses. Myxedema is more common in women than in men and may be occult. Adrenal insufficiency and hypopituitarism can also increase susceptibility to hypothermia. Hypoglycemia, most commonly caused by

insulin or oral hypoglycemic drugs, is associated with hypothermia, in part the result of neuroglycopenic effects on hypothalamic function. Increased osmolality and metabolic derangements associated with uremia, diabetic ketoacidosis, and lactic acidosis can lead to altered hypothalamic thermoregulation.

Neurologic injury from trauma, cerebrovascular accident, subarachnoid hemorrhage, or hypothalamic lesions increases susceptibility to hypothermia. Agenesis of the corpus callosum, or Shapiro syndrome, is one cause of episodic hypothermia, characterized by profuse perspiration followed by a rapid fall in temperature. Acute spinal cord injury disrupts the autonomic pathways that lead to shivering and prevents cold-induced reflex vasoconstrictive responses.

Hypothermia associated with sepsis is a poor prognostic sign. Hepatic failure causes decreased glycogen stores and gluconeogenesis, as well as a diminished shivering response. In acute myocardial infarction associated with low cardiac output, hypothermia may be reversed after adequate resuscitation. With extensive burns, psoriasis, erythrodermas, and other skin diseases, increased peripheral blood flow leads to excessive heat loss.

THERMOREGULATION

Heat loss occurs through five mechanisms: radiation (55 to 65% of heat loss), conduction (10 to 15% of heat loss, but much greater in cold water), convection (increase in the wind), respiration, and evaporation (which are affected by the ambient temperature and the relative humidity).

The preoptic anterior hypothalamus normally orchestrates thermoregulation ([Chap. 17](#)). The immediate defense of thermoneutrality is via the autonomic nervous system ([Chap. 72](#)), whereas delayed control is mediated by the endocrine system. Autonomic nervous system responses include the release of norepinephrine, increased muscle tone, and shivering, leading to thermogenesis and an increase in the basal metabolic rate. Cutaneous cold thermoreception causes direct reflex vasoconstriction to conserve heat. Prolonged exposure to cold also stimulates hypothalamic release of thyrotropin releasing hormone; this leads to increased levels of thyroid stimulating hormone (TSH), which stimulates the thyroid gland to produce thyroxine, a hormone that increases metabolic rate.

CLINICAL PRESENTATION

In most cases of hypothermia, the history of exposure to environmental factors, such as prolonged exposure to the outdoors without adequate clothing, makes the diagnosis straightforward. In urban settings, however, the presentation is often more subtle and the clinician may focus on other disease processes, toxin exposures, or psychiatric diagnoses.

After initial stimulation by hypothermia, there is progressive depression of all organ systems. The timing of the appearance of these clinical manifestations varies widely ([Table 20-2](#)). Without knowing the core temperature, it can be difficult to interpret other vital signs. For example, a tachycardia disproportionate to the core temperature

suggests secondary hypothermia resulting from hypoglycemia, hypovolemia, or a toxin overdose. Because carbon dioxide production declines progressively, the respiratory rate should be low; persistent hyperventilation suggests a central nervous system (CNS) lesion or one of the organic acidoses. A markedly depressed level of consciousness in a patient with mild hypothermia should raise suspicion of an overdose or CNS dysfunction due to infection or trauma.

Physical examination findings can also be altered by hypothermia. For instance, the assumption that areflexia is solely attributable to hypothermia can obscure and delay the diagnosis of a spinal cord injury. Patients with hypothermia may be confused or combative; these symptoms abate more rapidly with rewarming than with the use of restraints. A classic example of maladaptive behavior in patients with hypothermia is paradoxical undressing, which involves the inappropriate removal of clothing in response to a cold stress. The cold-induced ileus and abdominal rectus spasm can mimic, or mask, the presentation of an acute abdomen ([Chap. 14](#)).

When a patient in hypothermic cardiac arrest is first discovered, cardiopulmonary resuscitation is indicated, unless (1) a do-not-resuscitate status is verified, (2) obviously lethal injuries are identified, or (3) the depression of a frozen chest wall is not possible. As the resuscitation proceeds, the prognosis is grave if there is evidence of widespread cell lysis, as reflected by potassium levels exceeding 10 mEq/L. Other findings that may preclude continuing resuscitation include a core temperature $<12^{\circ}\text{C}$, a pH <6.5 , or evidence of intravascular thrombosis with a fibrinogen value <50 mg/dL. The decision to terminate resuscitation before rewarming the patient to 35°C is extremely difficult. There are no validated prognostic indicators for recovery from hypothermia. A history of asphyxia with secondary cooling is the most important negative predictor of survival.

DIAGNOSIS AND STABILIZATION

Hypothermia is confirmed by measuring the core temperature, preferably at two sites. Rectal probes should be placed to a depth of 15 cm and not adjacent to cold feces. A simultaneous esophageal measurement will be falsely high during heated inhalation therapy. The probe should be placed 24 cm below the larynx. The greatest discordance between the readings is usually during the transition phase before effective rewarming. Relying solely on infrared tympanic thermography is not advisable.

After a diagnosis of hypothermia is established, cardiac monitoring should be instituted, along with attempts to limit further heat loss. If the patient is in ventricular fibrillation, one sequence of 3 defibrillation attempts (2 J/kg) should be administered. If unsuccessful, active rewarming should be continued past 30° to 32°C . Supplemental oxygenation is always warranted, since tissue oxygenation is adversely affected by the leftward shift of the oxyhemoglobin dissociation curve. Pulse oximetry may be unreliable in patients with vasoconstriction. If protective airway reflexes are absent, gentle endotracheal intubation should be performed. Adequate pre-oxygenation will prevent ventricular arrhythmias.

Insertion of a gastric tube prevents dilatation secondary to decreased bowel motility. Indwelling bladder catheters facilitate monitoring of cold-induced diuresis. Dehydration is commonly encountered with chronic hypothermia, and most patients benefit from a bolus of crystalloid. Normal saline containing 5% dextrose is preferable to lactated

Ringer's solution, as the liver in hypothermic patients inefficiently metabolizes lactate. The placement of a pulmonary artery catheter, although of potential value, risks perforation of the less compliant pulmonary artery. The use of a central venous catheter should be avoided because of right atrial irritability.

Arterial blood gases should not be corrected for temperature ([Chap. 50](#)). This is termed the ectothermic or alpha-stat approach, which maximizes enzymatic function and maintains the normal distribution of charged metabolic intermediates. An uncorrected pH of 7.42 and a P_{CO_2} of 40 mmHg reflects appropriate alveolar ventilation and acid-base balance at any core temperature. Acid-base imbalances should be corrected gradually, since the bicarbonate buffering system is inefficient. When the P_{CO_2} increases 10 mmHg at 28°C, it doubles the pH decline of 0.08 that is normally induced at 37°C.

The severity of anemia may be underestimated because the hematocrit increases 2% for each 1°C drop in temperature. White blood cell sequestration and bone marrow suppression are common, potentially masking an infection. Although hypokalemia is more common in chronic hypothermia, hyperkalemia also occurs; the expected electrocardiographic changes can be obscured by hypothermia. Patients with renal insufficiency, metabolic acidoses, or rhabdomyolysis are most at risk for electrolyte disturbances.

Coagulopathies are common because cold inhibits the enzymatic reactions required for activation of the intrinsic cascade. In addition, the production of thromboxane B_2 by platelets is temperature-dependent, and platelet function is impaired. The administration of platelets and fresh frozen plasma is, therefore, not effective. The prothrombin or partial thromboplastin times reported by the laboratory appear deceptively normal and contrast with the observed coagulopathy. This contradiction appears because all coagulation tests are routinely performed at 37°C, and the enzymes are thus rewarmed.

REWARMING STRATEGIES

The key initial decision is whether to rewarm the patient passively or actively. *Passive external rewarming* simply involves covering and insulating the patient in a warm environment. With the head covered, the rate of rewarming is usually 0.5° to 2.0°C per hour. This technique is ideal for previously healthy patients who develop acute, mild primary accidental hypothermia. The patient must have sufficient fuel and glycogen to support endogenous thermogenesis.

There are reservations about the application of heat directly to the extremities of patients with chronic severe hypothermia. Extinguishing peripheral vasoconstriction in the dehydrated patient may precipitate core temperature "afterdrop" -- the continual decline in the core temperature after removal of the patient from the cold. This phenomenon results from conductive temperature equilibration and a circulatory convective mechanism. Rewarming frostbitten extremities before stabilization of the core temperature causes a significant core temperature afterdrop. In contrast, truncal heat application may minimize the risk of afterdrop.

Active rewarming is necessary under the following circumstances: core temperature < 32°C (poikilothermia), cardiovascular instability, age extremes, CNS

dysfunction, endocrine insufficiency, or any suspicion of secondary hypothermia. *Active external rewarming* is best accomplished with forced-air heating blankets. Other options include radiant heat sources and hot packs. Monitoring a patient with hypothermia in a heated tub is extremely difficult. Electric blankets should be avoided because vasoconstricted skin is easily burned. Widely available *active core rewarming* options include heated inhalation, heated infusion, and lavage (gastric, colonic, mediastinal, thoracic, pleural). The therapeutic options also include hemodialysis, venovenous, and continuous arteriovenous rewarming, in addition to formal cardiopulmonary bypass.

Arteriovenous anastomoses (AVA) rewarming provides exogenous heat by immersion of the hands, forearms, feet, and calves in 44° to 45°C water. Airway rewarming with heated humidified oxygen (40° to 45°C) is a convenient option via mask or endotracheal tube. Although airway rewarming provides less heat than some other forms of active core rewarming, it eliminates respiratory heat loss and adds 1° to 2°C to the overall rewarming rate. Crystalloids should be heated to 40° to 42°C. The quantity of heat provided is significant only during massive volume resuscitations. The most efficient method for heating and delivering fluid or blood is with a countercurrent in-line heat exchanger. Heated irrigation of the gastrointestinal tract or bladder transfers minimal heat because of the limited available surface area. These methods should be reserved for patients in cardiac arrest and then used in combination with all available active rewarming techniques. Closed thoracic lavage is far more efficient in severely hypothermic patients with cardiac arrest. The hemithoraces are irrigated through two large-bore thoracostomy tubes that are inserted into the left or both of the hemithoraces. Thoracostomy tubes should not be placed in the left chest of a spontaneously perfusing patient for purposes of rewarming. Peritoneal lavage with the dialysate at 40° to 45°C efficiently transfers heat when delivered through two catheters with outflow suction. Like peritoneal dialysis, standard hemodialysis is especially useful for patients with electrolyte abnormalities, rhabdomyolysis, or toxin ingestions.

With extracorporeal venovenous rewarming, the blood is removed from a central venous catheter, heated to 40°C, and returned through a second central or peripheral venous catheter. Continuous arteriovenous rewarming involves the use of percutaneously inserted femoral arterial and contralateral femoral venous 8.5 Fr catheters. The blood pressure must be at least 60 mmHg. Heparin-bonded tubing obviates the need for systemic anticoagulation. Full circulatory support with an oxygenator can only be provided through formal cardiopulmonary bypass (CPB). Femoral flow rates of 2 to 3 L/min elevate the core temperature 1° to 2°C every 3 to 5 min. CPB should be considered in nonperfusing patients without documented contraindications to resuscitation. Circulatory support may also be the only effective option in patients with completely frozen extremities, or those with significant tissue destruction coupled with rhabdomyolysis.

There is no evidence that extremely rapid rewarming improves survival in perfusing patients. The best strategy is usually a combination of passive, truncal active, and active core rewarming techniques.

DRUG THERAPY

When a patient is hypothermic, target organs and the cardiovascular system respond

minimally to most medications. Moreover, cumulative doses can cause toxicity during rewarming because of increased binding of drugs to proteins, and impaired metabolism and excretion. As an example, the administration of repeated doses of digoxin or insulin would be ineffective while the patient is hypothermic, and the residual drugs are potentially toxic during rewarming.

Any pharmacologic manipulation of the depressed and vasoconstricted cardiovascular system should generally be avoided. If the hypotension does not respond to crystalloid infusion and rewarming, low-dose dopamine (2 to 5 ug/kg per min) support should be considered. Atrial arrhythmias should initially be monitored without intervention, as the ventricular response will be slow, and most will convert spontaneously during rewarming. When indicated, bretylium tosylate is the class III ventricular antiarrhythmic of choice. During ventricular fibrillation, it should initially be administered at a dose of 10 mg/kg. Bretylium uniquely increases the ventricular arrhythmia threshold at low temperatures, although the wisdom of prophylaxis is unresolved.

Initiating empirical therapy for adrenal insufficiency is usually not warranted unless there is a history suggesting steroid dependence, hypoadrenalism, or a failure to rewarm with standard therapy. However, the administration of parenteral levothyroxine to euthyroid patients with hypothermia is potentially hazardous. Because laboratory results can be delayed and confounded by the presence of the sick euthyroid syndrome ([Chap. 330](#)), historical clues or physical findings suggestive of hypothyroidism should be sought. When myxedema is the cause of hypothermia, the relaxation phase of the Achilles reflex is prolonged more than the contraction phase.

Hypothermia obscures most of the symptoms and signs of infection, notably fever and leukocytosis. Shaking rigors from infection may be mistaken for shivering. Except in mild cases, extensive cultures and repeated physical examinations are essential. Unless an infectious source is identified, empirical antibiotic prophylaxis is most warranted in the elderly, neonates, and immunocompromised patients.

Preventive measures should be discussed with high-risk individuals, such as the elderly or people whose work frequently exposes them to extreme cold. The importance of layered clothing and headgear, adequate shelter, increased caloric intake, and the avoidance of ethanol should be emphasized, along with access to rescue services.

FROSTBITE

Peripheral cold injuries include both freezing and nonfreezing injuries to tissue. Frostbite occurs when the tissue temperature drops below 0°C. Ice crystal formation subsequently distorts and destroys the cellular architecture. Once the vascular endothelium is damaged, stasis progresses rapidly to microvascular thrombosis. Tissue freezes quickly when in contact with thermal conductors such as metal or volatile solutions. Other predisposing factors include constrictive clothing or boots, immobility, or vasoconstrictive medications.

Clinically, it is most practical to classify frostbite as superficial or deep. Superficial does not entail tissue loss. Classically, frostbite is retrospectively graded like a burn once the resultant pathology is demarcated over time. First-degree frostbite causes only

anesthesia and erythema. The appearance of superficial vesiculation surrounded by edema and erythema is considered second degree ([Plates IIA-18,IIA-19](#)). Hemorrhagic vesicles reflect a serious injury to the microvasculature, and indicate third-degree frostbite. Fourth-degree injuries damage subcuticular, muscular, and osseous tissues.

PATHOPHYSIOLOGY

Peripheral cold injury involves a cascade of events. Endothelial cells are very susceptible to cold injury. In the prefreeze phase, plasma leaks and there is the development of microvascular vasoconstriction. The radiation of heat from underlying tissues initially prevents crystallization. The freeze phase usually begins with extracellular fluid crystallization. Water exits the cell and causes intracellular dehydration, hyperosmolality, and ultimately cellular shrinkage and demise. Damaged tissue releases thromboxane A₂ and prostaglandin F_{2a}, which produce platelet aggregation, leukocyte immobilization, and vasoconstriction.

After the tissue thaws, the second phase of the cascade causes progressive dermal ischemia. The microvasculature begins to collapse, arteriovenous shunting increases tissue pressures, and there is progressive formation of edema. Finally, thrombosis, ischemia, and superficial necrosis appear. The development of mummification and demarcation may take weeks to months.

CLINICAL PRESENTATION

The initial presentation of frostbite can be deceptively benign. The symptoms always include a sensory deficiency affecting light touch, pain, and temperature perception. The acral areas and distal extremities are the most common insensate areas. Some patients complain of a clumsy or "chunk of wood" sensation in the extremity.

Deep frostbitten tissue can appear waxy, mottled, yellow, or violaceous-white. Favorable presenting signs include some warmth or sensation with normal color. The injury is often superficial if the subcutaneous tissue is pliable or if the dermis can be rolled over bony prominences.

The two most common nonfreezing peripheral cold injuries are *chilblain (pernio)* and *immersion (trench) foot*. Chilblain results from neuronal and endothelial damage induced by repetitive exposure to dry cold. Young females, particularly those with a history of Raynaud's phenomenon, are most at risk. Persistent vasospasticity and vasculitis can cause erythema, mild edema, and pruritus. Eventually plaques, blue nodules, and ulcerations develop. These lesions typically involve the dorsa of the hands and feet. In contrast, immersion (trench) foot results from repetitive exposure to wet cold above the freezing point. The feet initially appear cyanotic, cold, and edematous. The subsequent development of bullae is often indistinguishable from frostbite. This vesiculation rapidly progresses to ulceration and liquefaction gangrene. Patients with milder cases complain of hyperhidrosis, cold sensitivity, and painful ambulation for many years.

Various ancillary tests have been used in an attempt to diagnose the severity of peripheral cold injuries. None consistently predicts the extent of injury at presentation. For example, angiography and magnetic resonance imaging can demonstrate the

patency of large vessels but not the microvasculature. Ultrasonography and digital plethysmography are also insensitive. Thermography and technetium scintigraphy help evaluate perfusion several days after rewarming.

TREATMENT

Frozen tissue should be rapidly and completely thawed by immersion in circulating water at 37° to 40°C. Rapid rewarming often produces an initial hyperemia. The early formation of clear distal large blebs is more favorable than smaller proximal dark hemorrhagic blebs. A common error is the premature termination of thawing, since the reestablishment of perfusion is intensely painful. Parenteral narcotics will be necessary with deep frostbite. If cyanosis persists after rewarming, the tissue compartment pressures should be monitored carefully.

Numerous experimental antithrombotic and vasodilatory treatment regimens have been evaluated. There is no conclusive evidence that dextran, heparin, steroids, calcium channel blockers, or hyperbaric oxygen salvage tissue. A treatment protocol for frostbite is summarized in [Table 20-3](#).

Unless infection develops, any decision regarding debridement or amputation should be deferred until there is clear evidence of demarcation, mummification, and sloughing. The most common symptomatic sequelae reflect neuronal injury and the persistently abnormal sympathetic tone, including paresthesias, thermal misperception, and hyperhidrosis. Delayed findings include nail deformities, cutaneous carcinomas, and epiphyseal damage in children.

(Bibliography omitted in Palm version)

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SECTION 3 -NERVOUS SYSTEM DYSFUNCTION

21. FAINTNESS, SYNCOPE, DIZZINESS, AND VERTIGO - *Robert B. Daroff, Mark D. Carlson*

Syncope is defined as transient loss of consciousness due to reduced cerebral blood flow. Syncope is associated with postural collapse and spontaneous recovery. It may occur suddenly, without warning, or may be preceded by symptoms of varying duration. These include faintness or lightheadedness, "dizziness" without true vertigo, a feeling of warmth, diaphoresis, nausea, and visual blurring occasionally proceeding to blindness. These presyncopal symptoms may increase in severity until loss of consciousness occurs or may resolve prior to loss of consciousness if the cerebral ischemia is corrected. The differentiation of syncope from seizure is an important, sometimes difficult, diagnostic problem.

Syncope may be benign when it occurs as a result of normal cardiovascular reflex effects on heart rate and vascular tone or malignant when due to a life-threatening arrhythmia. Syncope may occur as a single event or may be recurrent. Recurrent, unexplained syncope, particularly in an individual with structural heart disease, is associated with a high risk for death (40% mortality within 2 years).

SYNCOPE

At the beginning of a syncopal attack, the patient is nearly always in the upright position, either sitting or standing. A cardiac etiology, such as an arrhythmia, is exceptional in this respect. The patient is warned of the impending faint by a sense of "feeling bad," of giddiness, and of movement or swaying of the floor or surrounding objects. The patient becomes confused and may yawn, visual spots and dimming may occur, and the ears may ring. Nausea and vomiting sometimes accompany these symptoms. There is a striking pallor or ashen gray color of the face, and generalized perspiration ensues. In some patients, a gradual onset with presyncopal symptoms may allow time for protection against injury; in others the syncope is sudden and without warning. The onset varies from instantaneous to 10 to 30 s, rarely longer.

The depth and duration of unconsciousness vary. Sometimes the patient remains partly aware of the surroundings, or there may be profound coma. The patient may remain in this state for seconds or minutes. Usually the patient lies motionless with skeletal muscles relaxed, but a few clonic jerks of the limbs and face may occur shortly after consciousness is lost. Sphincter control is usually maintained, in contrast to a seizure. The pulse is feeble or apparently absent, the blood pressure may be low or undetectable, and breathing may be almost imperceptible. Once the patient is in a horizontal position, gravity no longer hinders the flow of blood to the brain. The strength of the pulse may then improve, color begins to return to the face, breathing becomes quicker and deeper, and consciousness is regained. There is usually an immediate recovery of consciousness. Some patients, however, may be keenly aware of physical weakness, and rising too soon may precipitate another faint. In other patients, particularly those with tachyarrhythmias, there may be no residual symptoms following the initial syncope. Headache and drowsiness, which with mental confusion are the usual sequelae of a seizure, do not follow a syncopal attack.

PATHOPHYSIOLOGY

The more common types of faint are reducible to a few simple mechanisms. Syncope results from a sudden impairment of brain metabolism, usually brought about by hypotension with reduction of cerebral blood flow. Several mechanisms subserve circulatory adjustments to the upright posture. Approximately three-fourths of the systemic blood volume is contained in the venous bed, and any interference in venous return may lead to a reduction in cardiac output. Cerebral blood flow may still be maintained, as long as systemic arterial vasoconstriction occurs, but when this adjustment fails, serious hypotension with resultant cerebral underperfusion to less than half of normal results in syncope. Normally, the pooling of blood in the lower parts of the body is prevented by: (1) pressor reflexes that induce constriction of peripheral arterioles and venules, (2) reflex acceleration of the heart by means of aortic and carotid reflexes, and (3) improvement of venous return to the heart by activity of the muscles of the limbs. Placing a normal person on a tilt table to relax the muscles and tilting upright slightly diminishes cardiac output and allows the blood to accumulate in the legs to a slight degree; this may then be followed by a slight transitory fall in systolic arterial pressure and, in patients with defective vasomotor reflexes, may produce faints.

CAUSES OF SYNCOPE

Transiently decreased cerebral blood flow is usually due to one of three general mechanisms: disorders of vascular tone or blood volume, cardiovascular disorders including cardiac arrhythmias, or cerebrovascular disease ([Table 21-1](#)). Not infrequently, however, the cause of syncope is multifactorial.

Disorders of Vascular Tone or Blood Volume Disorders of autonomic control of the heart and circulation account for at least half of syncopal episodes. These disorders share common pathophysiologic mechanisms: a cardioinhibitory component (e.g., bradycardia due to increased efferent vagal activity), a vasodepressor component (e.g., inappropriate vasodilatation due to sympathetic withdrawal), or both.

Vasovagal (Vasodepressor, Neurocardiogenic) Syncope This form of syncope is the common faint that may be experienced by normal persons and accounts for approximately half of all episodes of syncope. It is frequently recurrent and commonly precipitated by a hot or crowded environment, alcohol, extreme fatigue, severe pain, hunger, prolonged standing, and emotional or stressful situations. Episodes are often preceded by a presyncopal prodrome lasting seconds to minutes. Vasovagal syncope rarely occurs in the supine position. The individual is usually sitting or standing and experiences weakness, nausea, diaphoresis, lightheadedness, blurred vision, and often a forceful heart beat with tachycardia followed by cardiac slowing prior to loss of consciousness. The individual appears pallid and has decreasing blood pressure prior to syncope. The duration of unconsciousness is rarely longer than a few minutes if the conditions that provoke the episode are reversed. Consciousness is usually regained shortly after assuming a recumbent posture, but unconsciousness may be prolonged if an individual remains upright. Although commonly benign, vasovagal syncope can be associated with prolonged asystole and hypotension, resulting in injury.

Vasovagal syncope occurs in the setting of increased sympathetic activity and venous pooling. Under these conditions, vigorous myocardial contraction of a relatively empty left ventricle activates ventricular mechanoreceptors and vagal afferent nerve fibers, inhibiting sympathetic efferent activity and increasing parasympathetic efferent activity. The resultant vasodilatation and bradycardia induce hypotension and syncope.

The central nervous system (CNS) mechanisms responsible for vasovagal syncope are not clear. Animal studies, not confirmed in humans, suggest that endogenous opiates (endorphins) may play a role. Serotonin (5-hydroxytryptamine) participates in blood pressure regulation and may also be involved in inhibition of sympathetic efferent activity (and arterial vasodilatation) associated with vasovagal syncope.

Although the reflex involving myocardial mechanoreceptors is the mechanism usually accepted as responsible for vasovagal syncope, other reflexes may also be operative. Patients with transplanted (denervated) hearts have experienced cardiovascular responses identical to those present during vasovagal syncope. This should not be possible if the response depends solely on the reflex mechanisms described above unless the transplanted heart has become reinnervated. Moreover, vasovagal syncope often occurs in response to stimuli (fear, emotional stress, or pain) that may not be associated with venous pooling in the lower extremities, which suggests a cognitive or cortical component to the reflex. Thus, a variety of afferent and efferent responses may cause vasovagal syncope.

Postural (Orthostatic) Hypotension This occurs in patients who have a chronic defect in, or variable instability of, vasomotor reflexes. Systemic arterial blood pressure falls on assumption of upright posture due to loss of vasoconstriction reflexes in resistance and capacitance vessels of the lower extremities. Although the syncopal attack differs little from vasodepressor syncope, the effect of posture is critical. Sudden rising from a recumbent position or standing quietly are precipitating circumstances. *Orthostatic hypotension may be the cause of syncope in up to 30% of the elderly; polypharmacy with antihypertensive or antidepressant drugs is often a contributor in these patients.*

Postural syncope may occur in otherwise normal persons with defective postural reflexes. Patients with *idiopathic postural hypotension* may be identified by a characteristic response to upright tilt on a table. Initially, the blood pressure diminishes slightly before stabilizing at a lower level. Shortly thereafter, the compensatory reflexes fail and the systemic arterial pressure falls precipitously. The condition is often familial.

Orthostatic hypotension, often accompanied by disturbances in sweating, impotence, and sphincter difficulties, is also a primary feature of the autonomic nervous system disorders discussed in [Chap. 366](#) and listed in [Table 366-1](#). The most common causes of neurogenic orthostatic hypotension are chronic diseases of the peripheral nervous system that involve postganglionic unmyelinated fibers (e.g., diabetic, nutritional, and amyloid polyneuropathy). Much less common are the multisystem atrophies, which are [CNS](#) disorders in which orthostatic hypotension is associated with (1) parkinsonism (Shy-Drager syndrome), (2) progressive cerebellar degeneration, or (3) a more variable parkinsonian and cerebellar syndrome (striatonigral degeneration). Very rarely, an acute postganglionic dysautonomia has been reported that appears to represent a variant of Guillain-Barre syndrome ([Chap. 378](#)).

There are several additional causes of postural syncope: (1) After physical deconditioning (such as after prolonged illness with recumbency, especially in elderly individuals with reduced muscle tone) or after prolonged weightlessness, as in space flight; (2) after sympathectomy that has abolished vasopressor reflexes; and (3) in patients receiving antihypertensive or vasodilator drugs and those who are hypovolemic because of diuretics, excessive sweating, diarrhea, vomiting, hemorrhage, or adrenal insufficiency.

Carotid Sinus Hypersensitivity Syncope due to carotid sinus hypersensitivity is precipitated by pressure on the carotid sinus baroreceptors, which are located just cephalad to the bifurcation of the common carotid artery. This typically occurs in the setting of shaving, a tight collar, or turning the head to one side. Carotid sinus hypersensitivity occurs predominantly in men, most of whom are 50 years of age or older. Activation of carotid sinus baroreceptors gives rise to impulses carried via the nerve of Hering, a branch of the glossopharyngeal nerve, to the medulla oblongata. These afferent impulses activate efferent sympathetic nerve fibers to the heart and blood vessels, cardiac vagal efferent nerve fibers, or both. In patients with carotid sinus hypersensitivity, these responses may cause sinus arrest or atrioventricular (AV) block (a cardioinhibitory response), vasodilatation (a vasodepressor response), or both (a mixed response). Although originally described in 1933, the mechanisms responsible for the syndrome are not clear, and validated diagnostic criteria do not exist; some authorities have questioned its very existence.

Situational Syncope A variety of activities, including cough, deglutition, micturition, and defecation, are associated with syncope in susceptible individuals. These syndromes are caused, at least in part, by abnormal autonomic control and may involve a cardioinhibitory response, a vasodepressor response, or both. Cough, micturition, and defecation are associated with maneuvers (such as Valsalva, straining, and coughing) that may contribute to hypotension and syncope by decreasing venous return. Increased intracranial pressure secondary to the increased intrathoracic pressure may also contribute by decreasing cerebral blood flow.

Cough syncope typically occurs in men with chronic bronchitis or chronic obstructive lung disease during or immediately after prolonged coughing fits. Micturition syncope occurs predominantly in middle-aged and older men, particularly those with prostatic hypertrophy and obstruction of the bladder neck; loss of consciousness usually occurs at night during or immediately after voiding. Deglutition and defecation syncope occur in men and women. Deglutition syncope may be associated with esophageal disorders, particularly esophageal spasm. In some individuals, particular foods and carbonated or cold beverages initiate episodes by activating esophageal sensory receptors that trigger reflex sinus bradycardia or AV block. Defecation syncope is probably secondary to a Valsalva maneuver in older individuals with constipation.

Glossopharyngeal Neuralgia Syncope due to glossopharyngeal neuralgia is preceded by pain in the oropharynx, tonsillar fossa, or tongue. Loss of consciousness is usually associated with asystole rather than vasodilatation. The mechanism is thought to involve activation of afferent impulses in the glossopharyngeal nerve which terminate in the nucleus solitarius of the medulla and, via collaterals, activate the dorsal motor

nucleus of the vagus nerve.

Cardiovascular Disorders Cardiac syncope results from a sudden reduction in cardiac output, caused most commonly by a cardiac arrhythmia. In normal individuals, heart rates between 30 and 180 beats per minutes (bpm) do not reduce cerebral blood flow, especially if the person is in the supine position. As the heart rate decreases, ventricular filling time and stroke volume increase to maintain normal cardiac output. At rates below 30 bpm, stroke volume can no longer increase to compensate adequately for the decreased heart rate. At rates greater than approximately 180 bpm, ventricular filling time is inadequate to maintain adequate stroke volume. In either case, cerebral hypoperfusion and syncope may occur. Upright posture, cerebrovascular disease, anemia, and coronary, myocardial, or valvular disease all reduce the tolerance to alterations in rate.

Bradyarrhythmias ([Chap. 229](#)) may occur as a result of an abnormality of impulse generation (e.g., sinoatrial arrest) or impulse conduction (e.g., [AV](#) block). Either may cause syncope if the escape pacemaker rate is insufficient to maintain cardiac output. Syncope due to bradyarrhythmias may occur abruptly, without presyncopal symptoms, and recur several times daily. Patients with *sick sinus syndrome* may have sinus pauses (>3 s), and those with syncope due to high-degree AV block (*Stokes-Adams-Morgagni syndrome*) may have evidence of conduction system disease (e.g., prolonged PR interval, bundle branch block). However, the arrhythmia is often transitory, and the surface electrocardiogram or continuous electrocardiographic monitor (Holter monitor) taken later may not reveal the abnormality. The *bradycardia-tachycardia syndrome* is a common form of sinus node dysfunction in which syncope generally occurs as a result of marked sinus pauses following termination of paroxysmal supraventricular tachycardia. Drugs are a common cause for bradyarrhythmias, particularly in patients with underlying structural heart disease. Digoxin, β -adrenergic receptor antagonists, calcium channel blockers, and many antiarrhythmic drugs may suppress sinoatrial node impulse generation or slow AV nodal conduction.

Syncope due to a *tachyarrhythmia* ([Chap. 230](#)) is usually preceded by palpitation or lightheadedness but may occur abruptly with no warning symptoms. *Supraventricular tachyarrhythmias* are unlikely to cause syncope in individuals with structurally normal hearts but may do so if they occur in patients with: (1) heart disease that also compromises cardiac output, (2) cerebrovascular disease, (3) a disorder of vascular tone or blood volume, or (4) a rapid ventricular rate. These tachycardias result most commonly from paroxysmal atrial flutter, atrial fibrillation, or reentry involving the [AV](#) node or accessory pathways that bypass part or all of the AV conduction system. Patients with the *Wolff-Parkinson-White syndrome* may experience syncope when a very rapid ventricular rate occurs due to reentry across an accessory AV connection.

In patients with structural heart disease, ventricular tachycardia, sometimes associated with ventricular fibrillation, is a common cause of syncope, particularly in patients with a prior myocardial infarction. Patients with aortic valvular stenosis and hypertrophic obstructive cardiomyopathy are also at risk for ventricular tachycardia. Individuals with abnormalities of ventricular repolarization (prolongation of the QT interval) are at risk to develop polymorphic ventricular tachycardia (*torsade de pointes*). Those with the inherited form of this syndrome often have a family history of sudden death in young

individuals. Genetic markers can identify some patients with familial long-QT syndrome but the clinical utility of these markers remains unproven. Drugs (i.e., certain antiarrhythmics and erythromycin) and electrolyte disorders (i.e., hypokalemia, hypocalcemia, hypomagnesemia) can prolong the QT interval and predispose to torsade de pointes. Antiarrhythmic medications may precipitate ventricular tachycardia, particularly in patients with structural heart disease.

In addition to arrhythmias, syncope may also occur with a variety of structural cardiovascular disorders. Episodes are usually precipitated when the cardiac output cannot increase to compensate adequately for peripheral vasodilatation. Peripheral vasodilatation may be appropriate, such as following exercise, or may occur due to inappropriate activation of left ventricular mechanoreceptor reflexes, as occurs in aortic outflow tract obstruction (aortic valvular stenosis or hypertrophic obstructive cardiomyopathy). Obstruction to forward flow is the most common reason that cardiac output cannot increase. Pericardial tamponade is a rare cause of syncope. Syncope occurs in up to 10% of patients with massive pulmonary embolism and may occur with exertion in patients with severe primary pulmonary hypertension. The cause is an inability of the right ventricle to provide appropriate cardiac output in the presence of obstruction or increased pulmonary vascular resistance. Loss of consciousness is usually accompanied by other symptoms such as chest pain and dyspnea. Atrial myxoma, a prosthetic valve thrombus, and, rarely, mitral stenosis may impair left ventricular filling, decrease cardiac output, and cause syncope.

Cerebrovascular Disease Cerebrovascular disease alone rarely causes syncope but may lower the threshold for syncope in patients with other causes. The vertebrobasilar arteries, which supply brainstem centers responsible for maintaining consciousness, are usually involved when cerebrovascular disease causes or contributes to syncope. An exception is the rare patient with tight bilateral carotid stenosis and recurrent syncope, often precipitated by standing or walking. Most patients who experience lightheadedness or syncope due to cerebrovascular disease also have symptoms of focal neurologic ischemia, such as arm or leg weakness, diplopia, ataxia, dysarthria, or sensory disturbances. Basilar artery migraine is a rare disorder that causes syncope in adolescents.

DIFFERENTIAL DIAGNOSIS

Anxiety Attacks and the Hyperventilation Syndrome Anxiety, such as occurs in panic attacks, is frequently interpreted as a feeling of faintness or dizziness resembling presyncope. The symptoms are not accompanied by facial pallor and are not relieved by recumbency. The diagnosis is made on the basis of the associated symptoms such as a feeling of impending doom, air hunger, palpitations, and tingling of the fingers and perioral region. Attacks can often be reproduced by hyperventilation, resulting in hypocapnia, alkalosis, increased cerebrovascular resistance, and decreased cerebral blood flow. The release of epinephrine in anxiety states also contributes to the symptoms.

Seizures A seizure may be heralded by an aura, which is caused by a focal seizure discharge and hence has localizing significance. The aura is usually followed by a rapid return to normal or by a loss of consciousness. Injury from falling is frequent in a seizure

and rare in syncope, because only in seizures are protective reflexes abolished instantaneously. Tonic-convulsive movements are characteristic of seizures and usually do not occur with syncope, although, as stated above, brief tonic-clonic seizure-like activity can accompany fainting episodes. The period of unconsciousness tends to be longer in seizures than in syncope. Urinary incontinence is frequent in seizures and rare in syncope. The return of consciousness is prompt in syncope, slow after a seizure. Mental confusion, headache, and drowsiness are common sequelae of seizures; physical weakness with a clear sensorium characterizes the postsyncopal state. Repeated spells of unconsciousness in a young person at a rate of several per day or month are more suggestive of epilepsy than syncope.

Hypoglycemia Severe hypoglycemia is usually due to a serious disease such as a tumor of the islets of Langerhans; advanced adrenal, pituitary, or hepatic disease; or to excessive administration of insulin.

Acute Hemorrhage Hemorrhage, usually within the gastrointestinal tract, is an occasional cause of syncope. In the absence of pain and hematemesis, the cause of the weakness, faintness, or even unconsciousness may remain obscure until the passage of a black stool.

Hysterical Fainting The attack is usually unattended by an outward display of anxiety. Lack of change in pulse and blood pressure or color of the skin and mucous membranes distinguish it from the vasodepressor faint.

Approach to the Patient

The diagnosis of syncope is often challenging. The cause may only be apparent at the time of the event, leaving few, if any, clues when the patient is seen later by the physician. In dealing with patients who have fainted, the physician should think first of those causes of fainting that constitute a therapeutic emergency. Among them are massive internal hemorrhage or myocardial infarction, which may be painless, and cardiac arrhythmias. In elderly persons, a sudden faint, without obvious cause, should arouse the suspicion of complete heart block or a tachyarrhythmia, even though all findings are negative when the patient is seen.

An algorithmic approach to syncope is presented in [Fig. 21-1](#). A careful history is the most important diagnostic tool, both to suggest the correct cause and to exclude other important potential causes ([Table 21-1](#)). Although no single element of the history is specific for a particular etiology of syncope, the nature of the events and their time course immediately prior to, during, and after an episode often provide valuable etiologic clues. Loss of consciousness in particular situations, such as during venipuncture, micturition, or in association with volume depletion, suggests an abnormality of vascular tone. The position of the patient at the time of the syncopal episode is very important; syncope in the supine position is unlikely to be vasovagal and suggests an arrhythmia or a seizure. Syncope due to carotid sinus syndrome may occur when the individual is wearing a shirt with a tight collar, turning the head (turning to look while driving in reverse), or manipulating the neck (as in shaving). The patient's medications must be noted, including nonprescription drugs or health store supplements, with particular attention to recent changes.

The physical examination should include evaluation of heart rate and blood pressure in the supine, sitting, and standing positions. In patients with unexplained recurrent syncope, an attempt to reproduce an attack may assist in diagnosis. Anxiety attacks induced by hyperventilation can be reproduced readily by having the patient breathe rapidly and deeply for 2 to 3 min. Cough syncope may be reproduced by inducing the Valsalva maneuver. Carotid sinus massage should generally be avoided, even in patients with suspected carotid sinus hypersensitivity; it is a risky procedure that can cause a transient ischemic attack (TIA) or stroke in susceptible individuals.

Diagnostic Tests The choice of diagnostic tests should be guided by the history and the physical examination. Measurements of serum electrolytes, glucose, and the hematocrit may help to establish the cause of syncope. Cardiac enzymes should be evaluated if myocardial ischemia is suspected. Blood and urine toxicology screens may reveal the presence of alcohol or other drugs. In patients with possible adrenocortical insufficiency, plasma aldosterone and mineralocorticoid levels should be obtained.

Although the surface electrocardiogram is unlikely to provide a definitive diagnosis, it may provide clues to the cause of syncope *and should be performed in almost all patients*. The presence of conduction abnormalities (PR prolongation and bundle branch block) suggests a bradyarrhythmia, whereas pathologic Q waves or prolongation of the QT interval suggests a ventricular tachyarrhythmia. Inpatients should undergo continuous electrocardiographic monitoring; outpatients should wear a Holter monitor for 24 to 48 h. Whenever possible, symptoms should be correlated with the occurrence of arrhythmias. Continuous electrocardiographic monitoring may establish the cause of syncope in as many as 15% of patients. Cardiac event monitors may be useful in patients with infrequent symptoms, particularly in patients with presyncope. The presence of a late potential on a signal-averaged electrocardiogram is associated with increased risk for ventricular tachyarrhythmias in patients with a prior myocardial infarction. Low-voltage (visually inapparent) T wave alternans is also associated with development of sustained ventricular arrhythmias.

Invasive cardiac electrophysiologic testing provides diagnostic and prognostic information regarding sinus node function, [AV](#) conduction, and supraventricular and ventricular arrhythmias. Abnormal findings include prolongation of the sinus node recovery time, prolongation of the histioventricle (HV) interval, induction of a supraventricular arrhythmia associated with hypotension, or induction of a supraventricular arrhythmia. Prolongation of the sinus node recovery time (>1500 ms) is a specific finding (85 to 100%) for diagnosis of sinus node dysfunction but has a low sensitivity; continuous electrocardiographic monitoring is usually more effective for diagnosing this abnormality. Prolongation of the HV interval and conduction block below the His bundle indicate that His-Purkinje disease may be responsible for syncope. Although an HV interval >100 ms is abnormal, this finding is not common in patients with syncope, and some patients with shorter intervals are also at risk for AV block. Programmed stimulation for ventricular arrhythmias is most useful in patients who have experienced a myocardial infarction; the sensitivity and specificity of this technique is lower in patients with normal hearts or those with heart disease other than coronary artery disease.

Upright tilt table testing is indicated for recurrent syncope, a single syncopal episode that caused injury, or a single syncopal event in a "high-risk" setting (pilot, commercial vehicle driver, etc.), whether or not there is a history of preexisting heart disease or prior vasovagal episodes. In susceptible patients, upright tilt at an angle between 60 and 80° for 30 to 60 min induces a vasovagal episode. The protocol can be shortened if upright tilt is combined with intravenous administration of drugs that cause venous pooling or increase adrenergic stimulation (isoproterenol, nitroglycerin, edrophonium, or adenosine). The sensitivity and specificity of tilt table testing is difficult to ascertain because of the lack of validated criteria. Moreover, the reflexes responsible for vasovagal syncope can be elicited in most, if not all, individuals given the appropriate stimulus. The reported accuracy of the test ranges from 30 to 80%, depending on the population studied and the techniques used. Whereas the reproducibility of a negative test is 85 to 100%, the reproducibility of a positive tilt table test is only between 62 and 88%.

A variety of other tests may be useful to determine the presence of structural heart disease that may cause syncope. The echocardiogram with Doppler examination detects valvular, myocardial, and pericardial abnormalities. The echocardiogram is the "gold standard" for the diagnosis of hypertrophic cardiomyopathy and atrial myxoma. Cardiac cine magnetic resonance (MR) imaging provides an alternative noninvasive modality that may be useful for patients in whom diagnostic-quality echocardiographic images cannot be obtained. This test is also indicated for patients suspected of having arrhythmogenic right ventricular dysplasia or right ventricular outflow tract ventricular tachycardia. Both are associated with right ventricular structural abnormalities that are better visualized on MR imaging than by echocardiogram. Exercise testing may detect ischemia or exercise-induced arrhythmias. In some patients, cardiac catheterization may be necessary to diagnose the presence or severity of coronary artery disease or valvular abnormalities. Ultrafast computed tomographic scan, ventilation-perfusion scan, or pulmonary angiography are indicated in patients in whom syncope may be due to pulmonary embolus.

In possible cases of cerebrovascular syncope, a variety of neuroimaging tests may be indicated, including Doppler ultrasound studies of the carotid and vertebrobasilar systems, MR imaging, MR angiography, and x-ray angiography of the cerebral vasculature ([Chaps. 358](#) and [361](#)). Electroencephalography is indicated if seizures are suspected.

TREATMENT

The treatment of syncope is directed toward the underlying cause. This discussion will focus on the treatment of disorders of autonomic control. **Arrhythmias are discussed in [Chaps. 229 and 230](#), valvular heart diseases in [Chap. 236](#), and cerebrovascular disorders in [Chap. 361](#).*

Certain precautions should be taken regardless of the cause of syncope. At the first sign of symptoms, patients should make every effort to avoid injury should they lose consciousness. Patients with frequent episodes, or those who have experienced syncope without warning symptoms should avoid situations in which sudden loss of consciousness might result in injury (e.g., climbing ladders, swimming alone, operating

heavy machinery, driving). Patients should lower their head to the extent possible, and preferably should lie down. Lowering the head by bending at the waist should be avoided because it may further compromise venous return to the heart. When appropriate, family members or other close contacts should be educated as to the problem. This will ensure appropriate therapy and may prevent delivery of inappropriate therapy (chest compressions associated with cardiopulmonary resuscitation) that may inflict trauma.

Patients who have lost consciousness should be placed in a position that maximizes cerebral blood flow, offers protection from trauma, and secures the airway. Whenever possible, the patient should be placed supine with the head turned to the side to prevent aspiration and the tongue from blocking the airway. Assessment of the pulse and direct cardiac auscultation may assist in determining if the episode is associated with a bradyarrhythmia or tachyarrhythmia. Clothing that fits tightly around the neck or waist should be loosened. Peripheral stimulation, such as by sprinkling cold water on the face, may be helpful. Patients should not be given anything by mouth or be permitted to rise until the sense of physical weakness has passed.

Patients with vasovagal syncope should be instructed to avoid situations or stimuli that have caused them to lose consciousness. Episodes associated with intravascular volume depletion may be prevented by salt and fluid loading prior to provocative events. β -Adrenoceptor antagonists, the most widely used agents, mitigate the increase in myocardial contractility that stimulates left ventricular mechanoreceptors and also block central serotonin receptors. Disopyramide, a vagolytic with negative inotropic properties, and another vagolytic, transdermal scopolamine, are used to treat vasovagal syncope. Paroxetine, a serotonin reuptake inhibitor used for depression, appears to be an effective treatment, as are theophylline and ephedrine. Midodrine, an α -agonist, has been a first-line agent for some patients. Permanent cardiac pacing is effective for patients with frequent episodes of vasovagal syncope and is indicated for those with prolonged asystole associated with vasovagal episodes.

Patients with orthostatic hypotension should be instructed to rise slowly and systematically (supine to seated, seated to standing) from the bed or a chair. Movement of the legs prior to rising facilitates venous return from the lower extremities. Whenever possible, medications that aggravate the problem (vasodilators, diuretics, etc.) should be discontinued. Elevation of the head of the bed [20 to 30 cm (8 to 12 in.)] and use of elastic stockings may help.

Therapeutic modalities include devices that prevent lower limb blood pooling, such as an antigravity or g suit or elastic stockings; salt loading; and a variety of pharmacologic agents including sympathomimetic amines, monamine oxidase inhibitors, beta blockers, and levodopa. **The treatment of orthostatic hypotension secondary to central or peripheral disorders of the autonomic nervous system is discussed in Chap. 366.*

Glossopharyngeal neuralgia is treated with carbamazepine, which is effective for the syncope as well as for the pain. Patients with carotid sinus syndrome should be instructed to avoid clothing and situations that stimulate carotid sinus baroreceptors. Patients should turn their entire body, rather than just their head, to look to one side. Those with intractable syncope due to the cardioinhibitory response to carotid sinus

stimulation should undergo permanent pacemaker implantation.

Patients with syncope should be hospitalized when the episode may have resulted from a life-threatening abnormality or if recurrence with significant injury seems likely. These individuals should be admitted to a bed with continuous electrocardiographic monitoring. Patients who are known to have a normal heart and for whom the history strongly suggests vasovagal or situational syncope may be treated as outpatients if the episodes are neither frequent nor severe.

DIZZINESS AND VERTIGO

Dizziness is a common and often vexing symptom. Patients use the term to encompass a variety of sensations, including those that seem semantically appropriate (e.g., lightheadedness, faintness, spinning, giddiness, etc.) and those that are misleadingly inappropriate, such as mental confusion, blurred vision, headache, or tingling. Moreover, some individuals with gait disorders complain of dizziness despite the absence of vertigo or other abnormal cephalic sensations. The causes include peripheral neuropathy, myelopathy, spasticity, parkinsonian rigidity, and cerebellar ataxia. In this context, the term *dizziness* is being used to describe disturbed mobility. There may be mild associated lightheadedness, particularly with impaired sensation from the feet or poor vision; this is known as *multiple-sensory-defect dizziness* and occurs in elderly individuals who complain of dizziness only during ambulation. Decreased position sense (secondary to neuropathy or myelopathy) and poor vision (from cataracts or retinal degeneration) create an overreliance on the aging vestibular apparatus. A less precise but sometimes comforting designation to patients is *benign dysequilibrium of aging*. Thus, a careful history is necessary to determine exactly what a patient who states, "Doctor, I'm dizzy," is experiencing. After eliminating the misleading symptoms or gait disturbance, "dizziness" usually means either *faintness* (presyncope) or *vertigo* (an illusory or hallucinatory sense of movement of the body or environment, most often a feeling of spinning). Operationally, dizziness is classified into three categories: (1) faintness, (2) vertigo, and (3) miscellaneous head sensations.

FAINTNESS

Prior to an actual faint (syncope), there are often prodromal presyncopal symptoms (faintness) reflecting ischemia to a degree insufficient to impair consciousness (see above).

VERTIGO

Vertigo is usually due to a disturbance in the vestibular system. The end organs of this system, situated in the bony labyrinths of the inner ears, consist of the three semicircular canals and the otolithic apparatus (utricle and saccule) on each side. The canals transduce angular acceleration, while the otoliths transduce linear acceleration and static gravitational forces, the latter providing a sense of head position in space. The neural output of the end organs is conveyed to the vestibular nuclei in the brainstem via the eighth cranial nerve. The principal projections from the vestibular nuclei are to the nuclei of cranial nerves III, IV, and VI, the spinal cord, the cerebral cortex, and the cerebellum. The vestibuloocular reflex (VOR) serves to maintain visual

stability during head movement and depends on direct projections from the vestibular nuclei to the sixth cranial nerve (abducens) nuclei in the pons and, via the medial longitudinal fasciculus, to the third (oculomotor) and fourth (trochlear) cranial nerve nuclei in the midbrain. These connections account for the nystagmus (to-and-fro oscillation of the eyes) that is an almost invariable accompaniment of vestibular dysfunction. The vestibular nerves and nuclei project to areas of the cerebellum (primarily the flocculus and nodulus) that modulate the VOR. The vestibulospinal pathways assist in the maintenance of postural stability. Projections to the cerebral cortex, via the thalamus, provide conscious awareness of head position and movement.

The vestibular system is one of three sensory systems subserving spatial orientation and posture; the other two are the visual system (retina to occipital cortex) and the somatosensory system that conveys peripheral information from skin, joint, and muscle receptors. The three stabilizing systems overlap sufficiently to compensate (partially or completely) for each other's deficiencies. Vertigo may represent either physiologic stimulation or pathologic dysfunction in any of the three systems.

Physiologic Vertigo This occurs when (1) the brain is confronted with a mismatch among the three stabilizing sensory systems; (2) the vestibular system is subjected to unfamiliar head movements to which it has never adapted, such as in seasickness; or (3) unusual head/neck positions, such as the extreme extension when painting a ceiling. Intersensory mismatch explains carsickness, height vertigo, and the visual vertigo most commonly experienced during motion picture chase scenes; in the latter, the visual sensation of environmental movement is unaccompanied by concomitant vestibular and somatosensory movement cues. *Space sickness*, a frequent transient effect of active head movement in the weightless zero-gravity environment, is another example of physiologic vertigo.

Pathologic Vertigo This results from lesions of the visual, somatosensory, or vestibular systems. Visual vertigo is caused by new or incorrect spectacles or by the sudden onset of an extraocular muscle paresis with diplopia; in either instance, [CNS](#) compensation rapidly counteracts the vertigo. Somatosensory vertigo, rare in isolation, is usually due to a peripheral neuropathy that reduces the sensory input necessary for central compensation when there is dysfunction of the vestibular or visual systems.

The most common cause of pathologic vertigo is vestibular dysfunction. The vertigo is frequently accompanied by nausea, jerk nystagmus, postural unsteadiness, and gait ataxia. Since vertigo increases with rapid head movements, patients tend to hold their heads still.

Labyrinthine Dysfunction This causes severe rotational or linear vertigo. When rotational, the hallucination of movement, whether of environment or self, is directed away from the side of the lesion. The fast phases of nystagmus beat away from the lesion side, and the tendency to fall is toward the side of the lesion.

When the head is straight and immobile, the vestibular end organs generate a tonic resting firing frequency that is equal from the two sides. With any rotational acceleration, the anatomic positions of the semicircular canals on each side necessitate an increased firing rate from one and a commensurate decrease from the other. This change in

neural activity is ultimately projected to the cerebral cortex, where it is summed with inputs from the visual and somatosensory systems to produce the appropriate conscious sense of rotational movement. After cessation of movement, the firing frequencies of the two end organs reverse; the side with the initially increased rate decreases, and the other side increases. A sense of rotation in the opposite direction is experienced; since there is no actual head movement, this hallucinatory sensation is *physiologic postrotational vertigo*.

Any disease state that changes the firing frequency of an end organ, producing unequal neural input to the brainstem and ultimately the cerebral cortex, causes vertigo. The symptom can be conceptualized as the cortex inappropriately interpreting the abnormal neural input from the brainstem as indicating actual head rotation. Transient abnormalities produce short-lived symptoms. With a fixed unilateral deficit, central compensatory mechanisms ultimately diminish the vertigo. Since compensation depends on the plasticity of connections between the vestibular nuclei and the cerebellum, patients with brainstem or cerebellar disease have diminished adaptive capacity, and symptoms may persist indefinitely. Compensation is always inadequate for severe fixed bilateral lesions despite normal cerebellar connections: these patients are permanently symptomatic.

Acute unilateral labyrinthine dysfunction is caused by infection, trauma, and ischemia. Often, no specific etiology is uncovered, and the nonspecific terms *acute labyrinthitis*, *acute peripheral vestibulopathy*, or *vestibular neuritis* are used to describe the event. The attacks are brief and leave the patient for some days with a mild positional vertigo. Infection with herpes simplex virus type 1 has been implicated. It is impossible to predict whether a patient recovering from the first bout of vertigo will have recurrent episodes.

Acute bilateral labyrinthine dysfunction is usually the result of toxins such as drugs or alcohol. The most common offending drugs are the aminoglycoside antibiotics which damage the fine hair cells of the vestibular end organs and may cause a permanent disorder of equilibrium.

Recurrent unilateral labyrinthine dysfunction, in association with signs and symptoms of cochlear disease (progressive hearing loss and tinnitus), is usually due to Meniere's disease ([Chap. 29](#)). When auditory manifestations are absent, the term *vestibular neuronitis* denotes recurrent monosymptomatic vertigo. [TIAs](#) of the posterior cerebral circulation (vertebrobasilar insufficiency) very infrequently cause recurrent vertigo without concomitant motor, sensory, visual, cranial nerve, or cerebellar signs.

Positional vertigo is precipitated by a recumbent head position, either to the right or to the left. Benign paroxysmal positional (or positioning) vertigo (BPPV) of the posterior semicircular canal is particularly common. Although the condition may be due to head trauma, usually no precipitating factors are identified. It generally abates spontaneously after weeks or months. The vertigo and accompanying nystagmus have a distinct pattern of latency, fatigability, and habituation that differs from the less common central positional vertigo ([Table 21-2](#)) due to lesions in and around the fourth ventricle. Moreover, the pattern of nystagmus in posterior canal BPPV is distinctive. The lower eye displays a large-amplitude torsional nystagmus, and the upper eye has a lesser degree of torsion combined with upbeating nystagmus. If the eyes are directed to the

upper ear, the vertical nystagmus in the upper eye increases in amplitude.

Vertigo of vestibular nerve origin may occur with diseases that involve the nerve in the petrous bone or the cerebellopontine angle. Except that it is less severe and less frequently paroxysmal, it has many of the characteristics of labyrinthine vertigo. The adjacent auditory division of the eighth cranial nerve also may be affected, which explains the frequent association of vertigo with tinnitus and deafness. The function of the eighth cranial nerve may be disturbed by tumors of the lateral recess (especially schwannomas), less frequently by meningeal inflammation in this region and, rarely, by an abnormal vessel that compresses the nerve.

Schwannomas involving the eighth cranial nerve (*acoustic neuroma*) grow slowly and produce such a gradual reduction of labyrinthine output that central compensatory mechanisms can prevent or minimize the vertigo; auditory symptoms of hearing loss and tinnitus are the most common manifestations. While lesions of the brainstem or cerebellum can cause acute vertigo, associated signs and symptoms usually permit distinction from a labyrinthine etiology ([Table 21-3](#)). However, labyrinthine ischemia, presumably due to occlusion of the labyrinthine branch of the internal auditory artery, may be the sole manifestation of vertebrobasilar insufficiency; patients with this syndrome present with the abrupt onset of severe vertigo, nausea and vomiting without tinnitus or hearing loss. Occasionally, an acute lesion of the vestibulocerebellum may present with monosymptomatic vertigo indistinguishable from a labyrinthopathy.

Vestibular epilepsy, vertigo secondary to temporal lobe epileptic activity, is rare and almost always intermixed with other epileptic manifestations.

Psychogenic vertigo, usually a concomitant of panic attacks or agoraphobia (fear of large open spaces, crowds, or leaving the safety of home), should be suspected in patients so "incapacitated" by their symptoms that they adopt a prolonged housebound status. Most patients with organic vertigo attempt to function despite their discomfort. Organic vertigo is accompanied by nystagmus; a psychogenic etiology is almost certain when nystagmus is absent during a vertiginous episode.

Miscellaneous Head Sensations This designation is used, primarily for purposes of initial classification, to describe dizziness that is neither faintness nor vertigo. Cephalic ischemia or vestibular dysfunction may be of such low intensity that the usual symptomatology is not clearly identified. For example, a small decrease in blood pressure or a slight vestibular imbalance may cause sensations different from distinct faintness or vertigo but that may be identified properly during provocative testing techniques. Other causes of dizziness in this category are hyperventilation syndrome, hypoglycemia, and the somatic symptoms of a clinical depression; these patients should have normal neurologic examinations and vestibular function tests.

Approach to the Patient

The most important diagnostic tool is a careful history focused on the meaning of "dizziness" to the patient. Is it faintness? Is there a sensation of spinning? If either of these is affirmed and the neurologic examination is normal, appropriate investigations for the multiple etiologies of cephalic ischemia or vestibular dysfunction are undertaken.

When the meaning of "dizziness" is uncertain, provocative tests may be helpful. These office procedures simulate either cephalic ischemia or vestibular dysfunction. Cephalic ischemia is obvious if the dizziness is duplicated during maneuvers that produce orthostatic hypotension. Further provocation involves the Valsalva maneuver, which decreases cerebral blood flow and should reproduce ischemic symptoms.

The simplest provocative test for vestibular dysfunction is rapid rotation and abrupt cessation of movement in a swivel chair. This always induces vertigo that the patients can compare with their symptomatic dizziness. The intense induced vertigo may be unlike the spontaneous symptoms, but shortly thereafter, when the vertigo has all but subsided, a lightheadedness supervenes that may be identified as "my dizziness." When this occurs, the dizzy patient, originally classified as suffering from "miscellaneous head sensations," is now properly diagnosed as having mild vertigo secondary to a vestibulopathy.

Patients with symptoms of positional vertigo should be appropriately tested ([Table 21-2](#)); positional testing is more sensitive with special spectacles that preclude visual fixation (Frenzel lenses).

A final provocative test, requiring the use of Frenzel lenses, is vigorous head shaking in the horizontal plane for about 10 s. If nystagmus develops after the shaking stops, even in the absence of vertigo, vestibular dysfunction is demonstrated. The maneuver can then be repeated in the vertical plane. If the provocative tests establish the dizziness as a vestibular symptom, an evaluation of vestibular vertigo is undertaken.

Evaluation of Patients with Pathologic Vestibular Vertigo The evaluation depends on whether a central etiology is suspected ([Table 21-3](#)). If so, [MR](#) imaging of the head is mandatory. Such an examination is rarely helpful in cases of recurrent monosymptomatic vertigo with a normal neurologic examination. Typical [BPPV](#) requires no investigation after the diagnosis is made ([Table 21-2](#)).

Vestibular function tests serve to (1) demonstrate an abnormality when the distinction between organic and psychogenic is uncertain, (2) establish the side of the abnormality, and (3) distinguish between peripheral and central etiologies. The standard test is electronystagmography (calorics), where warm and cold water (or air) are applied, in a prescribed fashion, to the tympanic membranes, and the slow-phase velocities of the resultant nystagmus from the right and left ears are compared. A velocity decrease from one side indicates hypofunction ("canal paresis"). An inability to induce nystagmus with ice water denotes a "dead labyrinth." Some institutions have the capability of quantitatively determining various aspects of the vestibuloocular reflex using computer-driven rotational chairs and precise oculographic recording of the eye movements.

Hyperventilation is the cause of dizziness in many anxious individuals; tingling of the hands and face may be absent. Forced hyperventilation for 1 min is indicated for patients with enigmatic dizziness and normal neurologic examinations. Similarly, depressive symptoms (which patients usually insist are "secondary" to the dizziness) must alert the examiner to a clinical depression as the *cause*, rather than the effect, of

the dizziness.

[CNS](#) disease can produce dizzy sensations of all types. Consequently, a neurologic examination is always required even if the history or provocative tests suggest a cardiac, peripheral vestibular, or psychogenic etiology. Any abnormality on the neurologic examination should prompt appropriate neurodiagnostic studies.

TREATMENT

Treatment of acute vertigo consists of bed rest and vestibular suppressant drugs such as antihistaminics (meclizine, dimenhydrinate, promethazine), or a tranquilizer with GABA-ergic effects (diazepam). If the vertigo persists beyond a few days, most authorities advise ambulation in an attempt to induce central compensatory mechanisms, despite the short-term discomfort to the patient. Chronic vertigo of labyrinthine origin may be treated with a systematized vestibular rehabilitation program to facilitate central compensation (see also [Table 21-4](#)).

[BPPV](#) is often self-limited but, when persistent, responds dramatically to specific repositioning exercise programs designed to empty particulate debris from the posterior semicircular canal. One of these exercises, the Epley procedure, is graphically demonstrated, in four languages, on a website for use in both physician's offices and self-treatment (<http://www.charite.de/ch/neuro/vertigo.html>).

Prophylactic measures to prevent recurrent vertigo are variably effective. Antihistamines are commonly utilized. Meniere's disease may respond to a diuretic or, more effectively, to a very low salt diet (1 g/day).

There are a variety of inner ear surgical procedures for refractory Meniere's disease, but these are only rarely necessary.

(Bibliography omitted in Palm version)

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22. WEAKNESS, MYALGIAS, DISORDERS OF MOVEMENT, AND IMBALANCE - *Richard K. Olney, Michael J. Aminoff*

Normal motor function requires integrated muscle activity with appropriate modulation by neuronal activity in the cerebral cortex, basal ganglia, cerebellum, and spinal cord. Symptoms and signs of motor system dysfunction may include weakness, fatigue, myalgias, spasms, cramps, dyskinetic movement, ataxia, imbalance, or disorders in the initiation or planning of movement.

WEAKNESS

Weakness is a reduction in normal power of one or more muscles. Patients may use the term differently; thus one or more specific examples of weakness should be elicited during the history. Increased fatigability or limitation in function due to pain is often confused with weakness by patients. *Increased fatigability* is the inability to sustain the performance of an activity that should be normal for a person of the same age, gender, and size.

Weakness is described commonly by severity and distribution. Paralysis and the suffix "-plegia" indicate weakness that is so severe that it is complete or nearly complete. "Paresis" refers to weakness that is mild or moderate. The prefix "hemi-" refers to one half of the body, "para-" to both legs, and "quadri-" to all four limbs.

Tone is the resistance of a muscle to passive stretch. Central nervous system (CNS) abnormalities that cause weakness generally produce *spasticity*, an increase in tone due to upper motor neuron disease. Spasticity is velocity-dependent, has a sudden release after reaching a maximum (the "clasp-knife" phenomenon), and predominantly affects antigravity muscles (i.e., upper limb flexors and lower limb extensors). Spasticity is distinct from rigidity and paratonia, two other types of increased tone. *Rigidity* is increased tone that is present throughout the range of motion (a "lead pipe" or "plastic" stiffness) and affects flexors and extensors equally. In some patients, rigidity has a cogwheel quality that is enhanced by voluntary movement of the contralateral limb (reinforcement). Rigidity occurs with certain extrapyramidal disorders. *Paratonia*, also referred to as *gegenhalten*, is increased tone that varies irregularly in a manner that may seem related to the degree of relaxation, is present throughout the range of motion, and affects flexors and extensors equally. Paratonia usually results from disease of the frontal lobes. Weakness with decreased tone (flaccidity) or normal tone occurs with disorders of the *motor unit*, that is, a single lower motor neuron and all of the muscle fibers it innervates.

Three basic patterns of weakness can usually be recognized based on the signs summarized in [Table 22-1](#). One results from upper motor neuron pathology, and the other two from disorders of the motor unit (lower motor neuron and myopathic weakness). Fasciculations and early atrophy help to distinguish lower motor neuron (neurogenic) weakness from myopathic weakness. A *fasciculation* is a visible or palpable twitch within a single muscle due to the spontaneous discharge of one motor unit. Neurogenic weakness also produces more prominent hypotonia and greater depression of tendon reflexes than myopathic weakness.

PATHOGENESIS

Upper Motor Neuron Weakness This pattern of weakness results from disorders that affect the upper motor neurons or their axons in the cerebral cortex, subcortical white matter, internal capsule, brainstem, or spinal cord ([Fig. 22-1](#)). Both the pyramidal and bulbospinal pathways contribute to normal strength, tone, coordination, and gait. Upper motor neuron lesions produce weakness through decreased activation of the lower motor neurons. In general, distal muscle groups are affected more severely than proximal ones, and axial movements are spared unless the lesion is severe and bilateral. With corticobulbar involvement, weakness is usually observed only in the lower face and tongue; extraocular, upper facial, pharyngeal, and jaw muscles are almost always spared. With bilateral corticobulbar lesions, *pseudobulbar palsy* often develops, in which dysarthria, dysphagia, dysphonia, and emotional lability accompany bilateral facial weakness. Spasticity accompanies upper motor neuron weakness but may not be present in the acute phase.

Upper motor neuron lesions also affect the ability to perform rapid repetitive movements. Such movements are slow and coarse, but normal rhythmicity is maintained. Finger-nose-finger and heel-knee-shin are performed slowly but adequately.

Lower Motor Neuron Weakness This pattern results from disorders of cell bodies of lower motor neurons in the brainstem motor nuclei and the anterior horn of the spinal cord, or from dysfunction of the axons of these neurons as they pass to skeletal muscle ([Fig. 22-2](#)).

Lower motor weakness is produced by a decrease in the number of motor units that can be activated, through a loss of the α motor neurons or disruption of their connections to muscle. With a decreased number of motor units, fewer muscle fibers are activated with full effort and maximum power is reduced. Loss of γ motor neurons does not cause weakness but decreases tension on the muscle spindles. Muscle tone and tendon reflexes depend on γ motor neurons, muscle spindles, spindle afferent fibers, and the α motor neurons. A tap on a tendon stretches muscle spindles and activates the primary spindle afferent fibers. These monosynaptically stimulate the α motor neurons in the spinal cord, producing a brief muscle contraction, which is the familiar tendon reflex.

When a motor unit becomes diseased, especially in anterior horn cell diseases, it may spontaneously discharge, producing a fasciculation. These isolated small twitches may be seen or felt clinically or recorded by electromyography (EMG) ([Chap. 357](#)). When α motor neurons or their axons degenerate, the denervated muscle fibers spontaneously discharge in a manner that cannot be seen or felt but can be recorded with EMG. These small single muscle fiber discharges are called *fibrillation potentials*. If significant lower motor neuron weakness is present, recruitment of motor units is delayed or reduced, with fewer than normal activated at a given discharge frequency. This contrasts with upper motor neuron weakness, in which a normal number of motor units are activated at a given frequency but in which the maximum discharge frequency is decreased.

Myopathic Weakness This pattern of weakness is produced by disorders within the motor unit that affect the muscle fibers or the neuromuscular junctions.

Two types of muscle fibers exist. Type I muscle fibers are rich in mitochondria and oxidative enzymes, produce relatively low force, but have low energy demands that can be supplied by ongoing aerobic metabolism. They produce sustained postural and nonforceful movements. Type II muscle fibers are rich in glycolytic enzymes, can produce relatively high force, but have high energy demands that cannot be supplied for long by ongoing aerobic metabolism. Thus, these units can be activated maximally for only brief periods of time to produce high-force movements.

For graded voluntary movements, type I muscle fibers are activated earlier in recruitment. For each muscle fiber, if the nerve terminal releases a normal number of acetylcholine molecules presynaptically and a sufficient number of postsynaptic acetylcholine receptors are opened, the end plate reaches threshold and thereby generates an action potential that spreads across the muscle fiber membrane and into the transverse tubular system. This electrical excitation activates intracellular events that produce an energy-dependent contraction of the muscle fiber (excitation-contraction coupling).

Myopathic weakness is produced by a decrease in the number or contractile force of muscle fibers activated within the motor unit. With muscular dystrophies, inflammatory myopathies, or myopathies with muscle fiber necrosis, decreased numbers of muscle fibers survive within many motor units. As demonstrated with [EMG](#), the size of each motor unit action potential is decreased so that motor units must be recruited more rapidly than normal to produce the power necessary for a certain movement. Neuromuscular junction diseases, such as myasthenia gravis, produce weakness in a similar manner, although the loss of muscle fibers within the motor unit is functional rather than actual. Furthermore, the number of muscle fibers activated can vary over time, depending on the state of rest of the neuromuscular junctions. Thus, fatigable weakness is suggestive of myasthenia gravis or another neuromuscular junction disease. Some myopathies produce weakness through loss of contractile force of muscle fibers or through relatively selective involvement of the type II muscle fibers. These may not affect the size of individual motor unit action potentials observed with EMG and are detected by a discrepancy between the electrical activity and force of a muscle.

Integrated Movements Most purposeful movements require the integrated coordination of many muscle groups. Consider a simple movement, such as grasping a ball. The primary movement is a flexion of the thumb and fingers of one hand, with opposition of the thumb and little finger. This requires the contraction of several muscles, including flexor digitorum superficialis, flexor digitorum profundus, flexor pollicis longus, flexor pollicis brevis, opponens pollicis, and opponens digiti minimi. These prime movers for this action are called *agonists*. In order for the grasping to be smooth and forceful, the thumb and finger extensors need to relax at the same rate as the flexors contract. The muscles that act in a directly opposing manner to the agonists are *antagonists*. A secondary action of the thumb and finger flexors is to flex the wrist; because wrist flexion tends to weaken finger flexion if both occur, activation of wrist extensors assists the grasping movement. Muscles that produce such complementary movements are *synergists*. Finally, the arm needs to be held in a stable position as the grasp occurs, so that the ball is not knocked away before it is secured. Muscles that stabilize the arm

position are *fixators*.

The coordination of activity by agonists, antagonists, synergists, and fixators is regulated by a three-level hierarchy of motor control. The lowest level of control is mediated through segmental reflexes in the spinal cord. These reflexes facilitate agonists and reciprocally inhibit the antagonists. Spinal segments also control rhythmic patterns of movement that involve more than a single pair of agonists and antagonists. For example, the lumbosacral spinal cord contains the basic programming for cyclical stepping movements that involve the synergistic activation of different muscle groups over time. The intermediate level of control is mediated through the descending bulbospinal pathways, which integrate visual, proprioceptive, and vestibular feedback into the execution of an action. For example, the locomotor center in the midbrain is required to modify the cyclical stepping movements in order that balance be maintained and forward movement occur. The highest level of control is mediated by the cerebral cortex. Superimposition of this highest level of control is necessary for activities such as walking to be goal-directed. Precise movements that are learned and improved through practice are also initiated and controlled by the motor cortex. Although only the agonists are directly activated, during the course of a complex sequence of actions such as playing the piano, the sequential activation of different groups of agonists for each note or chord is a part of the learned motor program. Further, the execution of these actions also involves input from the basal ganglia and cerebellar hemispheres to facilitate agonists, synergists, and fixators and to inhibit undesired antagonists.

Apraxia is a disorder of planning and initiating a skilled or learned movement ([Chap. 25](#)). Unilateral apraxia of the right hand may be due to a lesion of the left frontal lobe (especially anterior or inferior), the left temporoparietal region (especially the supramarginal gyrus), or their connections. Left body apraxia is produced by lesions of these regions in the right hemisphere or by lesions in the corpus callosum that disconnect the right temporoparietal or frontal regions from those on the left. Bilateral apraxia is often due to bilateral frontal lobe lesions or diffuse bilateral hemispheric disease.

Approach to the Patient

The mode of onset, distribution, and associated features of weakness should be carefully defined. When there is a discrepancy between the history and physical findings, it is usually because the patient complains of weakness, whereas symptoms are actually due to other causes, such as incoordination or pain limiting effort. Power may be examined in a variety of ways. The patient is asked to push or pull in a specified direction against resistance, and the strength in each muscle group is graded from 0 to 5 by the scale developed by the Medical Research Council ([Table 22-2](#)). A second method is indirect testing through observation of task performance such as holding the arms outstretched. This is especially useful in detecting mild, asymmetric upper motor neuron weakness through the observation of a downward drift with pronation of the forearm on one side. A third method is functional testing, which involves quantitation of activities. Common tests include counting the number of times a person can perform a deep-knee bend or step on a stool or chair, or timing the length of time the arms can be held abducted to 90 degrees. When performed serially, functional tests provide useful estimates of changes in the patient's status over time.

Other elements of the motor examination include appraisal of muscular bulk, inspection for fasciculations, and assessment of tone. Fasciculations are most easily determined by observing relaxed limbs that are illuminated from behind, but they can also be palpated as irregular low-amplitude twitches within the muscle. Tone is assessed by passive movement of each limb at its various joints and at several different speeds. In the clinical context of weakness, tone may be spastic or flaccid. The presence of cogwheel rigidity, lead-pipe rigidity, or paratonia suggests a disorder of integrated movements, rather than true weakness.

Hemiparesis Hemiparesis results from an upper motor neuron lesion above the midcervical spinal cord; most lesions that produce hemiparesis are located above the foramen magnum. The presence of language disorders, cortical sensory disturbances, cognitive abnormalities, disorders of visual-spatial integration, apraxia, or seizures indicates a cortical lesion. Homonymous visual field defects reflect either a cortical or a subcortical hemispheric lesion. A "pure motor" hemiparesis of the face, arm, and/or leg is due to a small, discrete lesion in the posterior limb of the internal capsule, cerebral peduncle, or upper pons. Some brainstem lesions produce the classic findings of ipsilateral cranial nerve signs and contralateral hemiparesis. These "crossed paralyses" are discussed further in [Chap. 361](#). The absence of cranial nerve signs or facial weakness suggests that a hemiparesis is due to a lesion in the high cervical spinal cord, especially if associated with ipsilateral loss of proprioception and contralateral loss of pain and temperature sense (the Brown-Sequard syndrome). However, most spinal cord lesions produce quadriparesis or paraparesis.

Acute or episodic hemiparesis usually has a vascular pathogenesis, either ischemia or a primary hemorrhage ([Chap. 361](#)). Less commonly, hemorrhage may occur into brain tumors ([Chap. 370](#)) or from rupture of normal vessels due to trauma ([Chap. 369](#)); the trauma may be trivial in patients who are anticoagulated or elderly. Less likely possibilities include a focal inflammatory lesion from multiple sclerosis ([Chap. 371](#)), abscess, or sarcoidosis ([Chap. 318](#)). Evaluation begins immediately with a computed tomography (CT) scan of the brain ([Fig. 22-3](#)). If CT is normal and an ischemic stroke is unlikely, magnetic resonance imaging (MRI) of the brain or cervical spine may be indicated.

Subacute hemiparesis that evolves over days or weeks has a long differential diagnosis. A common cause is subdural hematoma; this readily treatable condition must always be considered, especially in elderly or anticoagulated patients, even in the absence of a history of trauma ([Chap. 369](#)). Infectious possibilities include cerebral bacterial abscess ([Chap. 372](#)), fungal granuloma or meningitis ([Chap. 374](#)), and parasitic infection. Weakness from malignant primary and metastatic neoplasms may evolve over days to weeks ([Chap. 370](#)). AIDS ([Chap. 309](#)) may present with subacute hemiparesis due to toxoplasmosis or primary [CNS](#) lymphoma. Noninfectious inflammatory processes, such as multiple sclerosis ([Chap. 371](#)) or, less commonly, sarcoidosis, are further considerations. If the brain [MRI](#) is normal and if cortical and hemispheric signs are not present, MRI of the cervical spine may be required.

Chronic hemiparesis that evolves over months is usually due to a neoplasm ([Chap. 370](#)), an unruptured arteriovenous malformation ([Chap. 361](#)), a chronic subdural

hematoma ([Chap. 369](#)), or a degenerative disease ([Chaps. 363](#) to 366). The initial diagnostic test is often an [MRI](#) of the brain, especially if the clinical findings suggest brainstem pathology. If MRI of the brain is normal, the possibility of a foramen magnum or high cervical spinal cord lesion should be considered.

Paraparesis An intraspinal lesion at or below the upper thoracic spinal cord level is most commonly responsible. A sensory level over the trunk identifies the approximate level of the cord lesion. Paraparesis can also result from lesions at other locations that disturb upper motor neurons (especially parasagittal lesions and hydrocephalus) and lower motor neurons (anterior horn cell disorders, cauda equina syndromes, and occasionally peripheral neuropathies).

Acute or episodic paraparesis due to spinal cord disease may be difficult to distinguish from disorders affecting lower motor neurons or cerebral hemispheres. Recurrent episodes of paraparesis are often due to multiple sclerosis or to vascular malformations of the spinal cord. With acute spinal cord disease, the upper motor neuron deficit is usually associated with incontinence and a sensory disturbance of the lower limbs that extends rostrally to a level on the trunk; tone is typically flaccid, and tendon reflexes absent. In such cases, the diagnostic approach starts with an imaging study of the spinal cord ([Fig. 22-3](#)). Compressive lesions (particularly epidural tumor, abscess, or hematoma), spinal cord infarction (proprioception is usually spared), an arteriovenous fistula or other vascular anomaly, and transverse myelitis, among other causes may be responsible ([Chap. 368](#)). Diseases of the cerebral hemispheres that produce acute paraparesis include anterior cerebral artery ischemia (shoulder shrug also affected), superior sagittal sinus or cortical venous thrombosis, and acute hydrocephalus. If upper motor neuron signs are associated with drowsiness, confusion, seizures, or other hemispheric signs but not a sensory level over the trunk, the diagnostic approach starts with an [MRI](#) of the brain. Paraparesis is part of the cauda equina syndrome, which may result from trauma to the low back, a midline disk herniation, or intraspinal tumor; although sphincters are affected, hip flexion is often spared, as is sensation over the anterolateral thighs. Rarely, paraparesis is caused by a rapidly evolving peripheral neuropathy such as Guillain-Barre syndrome or by a myopathy. In such cases, electrophysiologic studies are diagnostically helpful and refocus the subsequent evaluation ([Chaps. 378](#) and [381](#)).

Subacute or chronic paraparesis with spasticity is caused by upper motor neuron disease. When paraparesis evolves over weeks or months with lower limb sensory loss and sphincter involvement, possible spinal cord disorders include multiple sclerosis, intraparenchymal tumor, chronic spinal cord compression from degenerative disease of the spine, subacute combined degeneration due to vitamin B₁₂ deficiency, viral infections (especially human T cell leukemia/lymphoma virus I), and hereditary or other degenerative diseases. Primary progressive multiple sclerosis usually presents in the fourth or fifth decade as progressive paraparesis ([Chap. 371](#)). Gliomas of the spinal cord typically produce a progressive myelopathy that is painful ([Chap. 370](#)). The clinical approach begins with an [MRI](#) of the spinal cord. If the imaging study is normal and spasticity is present, MRI of the brain may be indicated. If hemispheric signs are present, parasagittal meningioma or chronic hydrocephalus is likely and MRI of the brain is the initial test. Progression over months to years is typical of degenerative disorders such as primary lateral sclerosis ([Chap. 365](#)) and hereditary disorders such as

familial spastic paraparesis and adrenomyeloneuropathy ([Chap. 368](#)). In the rare situations when a chronic paraparesis is due to a lower motor neuron or myopathic etiology, the localization is usually suspected on clinical grounds by the absence of spasticity and confirmed by [EMG](#) and nerve conduction tests.

Quadriparesis or Generalized Weakness Generalized weakness may be due to disorders of the central nervous system or of the motor unit. Although the terms *quadriparesis* and *generalized weakness* are often used interchangeably, quadriparesis is more often chosen when an upper motor neuron cause is suspected and generalized weakness when a disease of the motor unit is likely. Weakness from [CNS](#) disorders is usually associated with changes in consciousness or cognition, with increased muscle tone and muscle stretch reflexes, and with alterations of sensation. Most neuromuscular causes of intermittent weakness are associated with normal mental function, diminished muscle tone, and hypoactive muscle stretch reflexes. Exceptions are some causes of acute quadriparesis due to upper motor neuron disorders in which transient hypotonia is present. The major causes of intermittent weakness are listed in [Table 22-3](#). A patient with generalized fatigability without objective weakness may have the *chronic fatigue syndrome* ([Chap. 384](#)).

Acute Quadriparesis Acute quadriparesis with onset over minutes may result from disorders of upper motor neurons (e.g., anoxia, hypotension, brainstem or cervical cord ischemia, trauma, and systemic metabolic abnormalities) or muscle (electrolyte disturbances, certain inborn errors of muscle energy metabolism, toxins, or periodic paralyses). Onset over hours to weeks may, in addition to the above, be due to lower motor neuron disorders. Guillain-Barre syndrome ([Chap. 378](#)) is the most common lower motor neuron weakness that progresses over days to several weeks; the finding of an elevated protein level in the cerebrospinal fluid is helpful but may be absent early in the course. If stupor or coma is present, the evaluation begins with a [CT](#) scan of the brain. If upper motor neuron signs are present but the patient is alert, the initial test is usually an [MRI](#) of the cervical cord. If weakness is lower motor neuron, myopathic, or uncertain in origin, the clinical approach starts with blood studies for muscle enzymes and electrolytes and an [EMG](#) and nerve conduction study.

Subacute or Chronic Quadriparesis When quadriparesis due to upper motor neuron disease develops over weeks, months, or years, the distinction between disorders of the cerebral hemispheres, brainstem, and cervical spinal cord is usually possible by clinical criteria alone. The diagnostic approach begins with an [MRI](#) of the clinically suspected site of pathology. Lower motor neuron disease usually presents with weakness that is most profound distally, whereas myopathic weakness is typically proximal; the evaluation then begins with [EMG](#) and nerve conduction studies.

Monoparesis This is usually due to lower motor neuron disease, with or without associated sensory involvement. Upper motor neuron weakness occasionally presents with a monoparesis of distal and nonantigravity muscles. Myopathic weakness is rarely limited to one limb.

Acute Monoparesis Distinguishing between upper and lower motor neuron disorders may be difficult clinically because tone and reflexes are frequently decreased in both at presentation. If the weakness is predominantly in distal and nonantigravity muscles and

not associated with sensory impairment or pain, focal cortical ischemia is likely ([Chap. 361](#)); in this setting, diagnostic possibilities are similar to those for acute hemiparesis. Sensory loss and pain usually accompany acute lower motor neuron weakness. The distribution of weakness is commonly localized to a single nerve root or peripheral nerve within one limb but occasionally reflects involvement of the brachial or lumbosacral plexus. If lower motor neuron weakness is suspected, or if the pattern of weakness is uncertain, the clinical approach begins with an [EMG](#) and nerve conduction study.

Subacute or Chronic Monoparesis Weakness with atrophy of one limb that develops over weeks or months is almost always lower motor neuron in origin. If the weakness is associated with numbness, a peripheral nerve or spinal root origin is likely; uncommonly, the brachial or lumbosacral plexus is affected. If numbness is absent, anterior horn cell disease is likely. In either case, an electrodiagnostic study is indicated. If upper rather than lower motor neuron signs are present, a tumor, vascular malformation, or other cortical lesion affecting the precentral gyrus may be responsible. Alternatively, if the leg is affected, a small thoracic cord lesion, often a tumor or multiple sclerosis, may be present. In these situations, the approach begins with an imaging study of the suspicious area.

Distal Weakness Involvement of two or four limbs distally suggests lower motor neuron or peripheral nerve disease. Acute distal lower limb weakness occurs occasionally from an acute toxic polyneuropathy or cauda equina syndrome. Distal symmetric weakness usually develops over weeks, months, or years and is due to metabolic, toxic, hereditary, degenerative, or inflammatory diseases of peripheral nerves ([Chap. 377](#)). With peripheral nerve disease, weakness is usually less severe than numbness. Anterior horn cell disease may begin distally but is typically asymmetric and is not associated with numbness ([Chap. 365](#)). Rarely, myopathies also present with distal weakness ([Chap. 381](#)). The first step in evaluation is an electrodiagnostic study ([Fig. 22-3](#)).

Proximal Weakness Proximal weakness of two or four limbs suggests a disorder of muscle or, less commonly, neuromuscular junction or anterior horn cell. Myopathy often produces symmetric weakness of the pelvic or shoulder girdle muscles ([Chap. 381](#)). Diseases of the neuromuscular junction (such as myasthenia gravis) may present with symmetric proximal weakness ([Chap. 380](#)), often associated with ptosis, diplopia, or bulbar weakness and fluctuating in severity during the day. Extreme fatigability present in some cases of myasthenia gravis may even suggest episodic weakness, but strength rarely returns fully to normal. The proximal weakness of anterior horn cell disease is most often asymmetric, but may be symmetric if familial ([Chap. 365](#)). Numbness does not occur with any of these diseases. The evaluation usually begins with determination of the serum creatine kinase level and electrophysiologic studies.

Weakness in a Restricted Distribution In some patients, weakness does not fit any of the above patterns. Examples include weakness limited to the extraocular, hemifacial, bulbar, or respiratory muscles. If unilateral, restricted weakness is usually due to lower motor neuron or peripheral nerve disease, such as in a facial palsy ([Chap. 367](#)) or an isolated superior oblique muscle paresis ([Chap. 28](#)). Relatively symmetric weakness of extraocular or bulbar muscles is usually due to a myopathy ([Chap. 381](#)) or neuromuscular junction disorder ([Chap. 380](#)). Bilateral facial palsy with areflexia

suggests Guillain-Barre syndrome ([Chap. 378](#)). Worsening of relatively symmetric weakness with fatigue is characteristic of neuromuscular junction disorders ([Chap. 380](#)). Asymmetric bulbar weakness is usually due to motor neuron disease. Weakness limited to respiratory muscles is uncommon and is usually due to motor neuron disease, myasthenia gravis, or polymyositis/dermatomyositis ([Chap. 382](#)).

MYALGIAS, SPASMS, AND CRAMPS

Spontaneous or exercise-related discomfort from muscles is usually benign and is rarely caused by a definable neuromuscular disease. However, a number of disorders of the motor system are characteristically painful. Some terms for muscular discomfort or involuntary contractions, such as myalgias, spasms, and cramps, are often used interchangeably by patients but have a more specific meaning to physicians. Other terms, such as aching, heaviness, and stiffness, are less specific. *Myalgias* are pains that are felt in muscle; the term does not imply an involuntary contraction. *Spasms* and *cramps* refer to episodes of involuntary contraction of one or more muscles. Cramps are usually painful, whereas spasms are not necessarily uncomfortable.

MYALGIAS

Proximal or generalized weakness associated with myalgias is usually due to an inflammatory, metabolic, endocrine, or toxic myopathy ([Chap. 381](#)). Spontaneous myalgias not accompanied by objective weakness are often without a clear cause unless associated with a well-defined systemic illness. Myalgias are a common manifestation of fever or infection, especially influenza. Muscle pains and stiffness with elevated serum creatine kinase concentration is common in hypothyroidism, even in patients without objective weakness. *Polymyalgia rheumatica* ([Chap. 317](#)) is characterized by diffuse myalgias and joint stiffness that predominantly affect the pelvic and shoulder girdles in a patient over 50 years of age who has anorexia, mild weight loss, and low-grade fever. Limitation of activity from the myalgias and joint stiffness also leads to disuse atrophy and may give the impression of weakness. However, [EMG](#), serum creatine kinase levels, and muscle biopsy are normal. The erythrocyte sedimentation rate is elevated in most patients, and features of giant-cell arteritis are present in 25%. Diffuse myalgias are common in many rheumatologic diseases, in which the diagnosis and treatment are based on other symptoms and signs. Myalgias are occasionally present in dermatomyositis/polymyositis, but most patients have weakness without significant pain. *Fibromyalgia* (fibrositis, fibromyositis) is associated with pain and tenderness of muscle and adjacent connective tissue ([Chap. 325](#)). Fatigue, insomnia, and depression are often present, but objective weakness, elevation of serum creatine kinase level, or elevation of the erythrocyte sedimentation rate does not occur. The diagnosis is dependent upon identifying characteristic focal "trigger points."

Focal Myalgias Focal muscle pain is often traumatic. Rupture of muscle tendons such as the biceps or gastrocnemius muscle may produce visible muscle shortening. Many such tears resolve without surgery but leave an abnormal appearance to the muscle belly. Nontraumatic focal muscle pain is often related to adjacent nonmuscular disorders (e.g., unilateral gastrocnemius pain due to deep venous thrombosis). Rarely, focal muscle pain may be caused by ischemic infarction or bacterial myositis, if acute, or by

neoplasm, parasitic infection, sarcoidosis, or other inflammation or infection, if subacute or chronic.

Exertional Myalgias Myalgias following unaccustomed, strenuous physical activity occur in normal individuals are often associated with laboratory evidence for muscle damage, such as an elevation of serum creatine kinase, edema of muscles on [MRI](#), necrosis of muscle fibers on biopsy, and rarely myoglobinuria. Similar symptoms and laboratory abnormalities characterize certain metabolic disorders of muscle, such as carnitine palmitoyl transferase and glycolytic pathway enzyme deficiencies. The association of objective weakness during an episode of myalgias suggests a metabolic muscle disease. The development of an acute contracture (the inability to relax a muscle due to energy depletion) with the myalgias suggests a metabolic muscle disease with a glycolytic enzyme deficiency ([Chap. 383](#)). Exertional myalgias with muscle fiber necrosis also occur in muscular dystrophy with partial deficiency of dystrophin, and certain mitochondrial cytopathies ([Chap. 383](#)). Exertional myalgias with elevated creatine kinase concentration but without weakness also occur in hypothyroidism, and when confined to the legs may be due to vascular or neurogenic intermittent claudication. Most patients with exertional myalgias and no weakness do not have a definable abnormality.

SPASMS AND CRAMPS

Involuntary contraction of muscle may occur with disorders of the [CNS](#), lower motor neuron, or muscle. Contractions that originate within the CNS and are associated with upper motor neuron signs are usually referred to as spasms and generally affect the flexors or extensors of one or more limbs. Those that originate within the CNS and are not associated with upper motor neuron signs include movement disorders discussed below, as well as the rare stiff-person syndrome and tetanus. Muscle rigidity from active muscle contraction can occur in the malignant hyperthermia syndrome, usually associated with general anesthesia. In the neuroleptic malignant syndrome, muscle rigidity arises from CNS overactivity and is present in muscle. Involuntary contractions that originate in the lower motor neurons are usually cramps, occasionally tetany, or rarely neuromyotonia. Spasms that originate in muscle or muscle membrane are usually a delayed relaxation after voluntary contraction, either myotonia or rarely a contracture. These conditions may be difficult to distinguish clinically but are often well characterized by [EMG](#) studies.

Stiff-Person Syndrome This rare syndrome is characterized by slowly progressive muscle stiffness and superimposed spasms. The stiffness commonly begins in the low back and spreads over months up the spine and into the limbs but not into the jaw. The gait becomes stiff, and there is hyperlordosis of the lumbar spine. Spasms are often produced by startle. Emotional stress tends to worsen the stiffness as well as the frequency and severity of spasms. The spontaneous motor activity disappears during sleep. The syndrome is often associated with diabetes mellitus and can be paraneoplastic, accompanying Hodgkin's lymphoma, small cell cancer of the lung, and breast cancer. Most patients have a serum antibody against glutamic acid decarboxylase, an enzyme responsible for synthesis of the inhibitory neurotransmitter g-aminobutyric acid (GABA). Stiffness results from loss of descending brainstem or segmental spinal inhibitory influences on the lower motor neurons. [EMG](#) studies reveal

continuous motor unit activity that is similar to voluntary effort with preservation of the silent period to muscle stretch. Stiffness and spasms typically respond partially to treatment with baclofen or benzodiazepines.

Tetanus This rare hyperexcitable state results from exposure to tetanus toxin in patients infected with *Clostridium tetani* ([Chap. 143](#)). Painful spasms typically begin with jaw closure (trismus) and soon become generalized. [EMG](#) studies reveal continuous motor unit activity that is similar to voluntary effort except for loss of the silent period to muscle stretch.

Cramps These are the most common type of involuntary muscle contraction. Cramps are a painful contraction of a single muscle that produces a palpable knot within the muscle for seconds to minutes and is relieved by passive stretch of the muscle or spontaneously. [EMG](#) studies reveal motor unit activity that has too high a discharge frequency to be voluntary. If cramps are associated with weakness, the weakness is almost always lower motor neuron in origin. When strength is normal, no definable condition is usually found, although dehydration, hypothyroidism ([Video 330-1](#)), or uremia is occasionally present. If prominent, membrane stabilizing drugs, such as carbamazepine, may provide symptomatic benefit.

Tetany Tetany is characterized by contraction of distal muscles of the hands (carpal spasm with extension of interphalangeal joints and adduction and flexion of the metacarpophalangeal joints) and feet (pedal spasm) and is associated with tingling around the mouth and distally in the limbs. Tetany with carpopedal spasms is a common manifestation of hypocalcemia or respiratory alkalosis (even from hyperventilation). [EMG](#) studies reveal single or more often grouped motor unit discharges at low discharge frequency.

Neuromyotonia (Isaac's Syndrome) Neuromyotonia is characterized by muscle stiffness at rest that persists during sleep and by delayed relaxation after voluntary effort. Distal limb muscles are usually affected most severely, but all skeletal muscle may be involved. Gait may be stiff, and close inspection of the muscle reveals undulation of the overlying skin due to continuous muscle fiber contractions (myokymia). The continuous muscle fiber activity generates heat, and excessive sweating is common. [EMG](#) studies commonly reveal myokymic discharges, especially in familial cases. Rarely, EMGs record high-frequency neuromyotonic discharges. Autoantibodies against voltage-gated potassium channels have been demonstrated in some cases, and plasma exchange may be effective.

Myotonia This is a nonpainful delay in the relaxation of muscle after voluntary activity. Delay in opening the hand after a forceful grip (grip myotonia) is common. These disorders are usually familial and worsen in cold weather. [EMG](#) demonstrates a waxing and waning discharge of individual muscle fibers.

Contracture A painful inability to relax a muscle after voluntary activity due to energy depletion characterizes certain metabolic disorders with failure of energy production, such as myophosphorylase deficiency (McArdle's disease). [EMG](#) studies reveal electrical silence.

MOVEMENT DISORDERS

Movement disorders are neurologic syndromes in which abnormal movements (or *dyskinesias*) occur due to a disturbance of fluency and speed of voluntary movement or the presence of unintended extra movements. Because they are so distinct from the pyramidal disorders that cause upper motor neuron weakness, movement disorders are often referred to as *extrapyramidal diseases*. *Hyperkinetic movement disorders* are those in which an excessive amount of spontaneous motor activity is seen or in which abnormal involuntary movements occur. *Hypokinetic movement disorders* are characterized by *akinesia* or *bradykinesia*, in which purposeful motor activity is absent or reduced. This is often described as "poverty of movement."

PATHOGENESIS

Movement disorders result from disease of the basal ganglia, paired subcortical gray matter structures consisting of the caudate and the putamen (which together are called the striatum), the internal and external segments of the globus pallidus, the subthalamic nucleus, and the substantia nigra. The major interconnections and neurotransmitters involved in basal ganglia circuits are illustrated in [Fig. 22-4A](#). An understanding of this circuitry can explain, in part, the perturbation that occurs in both the hypo- and hyperkinetic disorders.

Parkinson's disease ([Video 361-1](#)) ([Chap. 363](#)), the prototypic hypokinetic movement disorder, results from a loss of dopaminergic neurons in the substantia nigra pars compacta. This leads to less excitation of striatal neurons that express the D₁ type of dopamine receptors and less inhibition of D₂ striatal neurons, both contributing to reduced facilitation of cortically initiated movement ([Fig. 22-4B](#)). The resting tremor of Parkinson's disease is less readily explained by this model but may result from effects on cholinergic interneurons in the striatum. *Huntington's disease* ([Chap. 362](#)), a hyperkinetic movement disorder, may be explained by selective loss of D₂ striatal neurons, resulting in disinhibition of cortically initiated movements without normal feedback control. The pathogenesis of hemiballismus is similar -- a direct lesion of the glutamatergic neurons in the subthalamic nucleus (usually from a stroke) leads to disinhibition of thalamocortical projections.

Approach to the Patient

An algorithm for the interpretation of abnormal movements is illustrated in [Fig. 22-5](#). The initial step is to determine if the movement disorder is due to an excess or a poverty of movement (i.e., a hyperkinetic or a hypokinetic movement disorder).

Hyperkinetic Movement Disorders Abnormal involuntary movements are divided into those that are rhythmical and those that are irregular. Those that are rhythmical are termed *tremors*, with the uncommon exception of *palatal and segmental myoclonus*. Tremors are divided into three types: rest, postural, and intention tremor. A *rest tremor* is maximal at rest and becomes less prominent with activity. It is characteristic of parkinsonism, a hypokinetic movement disorder, and is therefore commonly associated with bradykinesia and cogwheel rigidity. A rest tremor that develops acutely is usually due to toxins [such as exposure to 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine

(MPTP)] or dopamine blocking drugs (such as phenothiazines). If insidious in onset, the diagnostic approach is the same as for Parkinson's disease ([Chap. 363](#)). A *postural tremor* is maximal while limb posture is actively maintained against gravity; it is lessened by rest and is not markedly enhanced during voluntary movement toward a target. A postural tremor that develops acutely is usually due to toxic or metabolic factors (for example, hyperthyroidism) or stress. The insidious onset of a postural tremor suggests a benign or familial essential tremor ([Chap. 363](#)). An *intention tremor* is most prominent during voluntary movement toward a target and is not present during postural maintenance or at rest. It is a sign of cerebellar disease ([Chap. 364](#)). *Asterixis*, which may superficially resemble a tremor, is an intermittent inhibition of muscle contraction that occurs with metabolic encephalopathy ([Chap. 376](#)). This leads, for example, to a momentary and repetitive partial flexion of the wrists during attempted sustained wrist extension.

Involuntary movements that are irregular are characterized further by their speed and site of occurrence and by whether they can be suppressed voluntarily. The slowest are athetosis and dystonia. *Athetosis* is a slow, writhing, sinuous movement that occurs nearly continuously in distal muscles. *Dystonia* is a slowly varying but nearly continuous deviation of posture about one or more joints; it may occur in a proximal or distal limb or in axial structures. Dystonia is a more sustained deviation of posture than athetosis, although these two phenomena overlap considerably. The further evaluation of athetosis and dystonia are discussed in [Chap. 363](#).

Among the rapid irregular movements, *tics* are controlled with voluntary effort, while the others are not. Tics often occur repetitively in a single location but are sometimes multifocal ([Chap. 363](#)).

Chorea, hemiballismus, and myoclonus are rapid, irregular jerks that cannot be consciously suppressed. *Hemiballismus* is the most distinctive among them. It is manifest as a sudden and often violent flinging movement of a proximal limb, usually an arm ([Chap. 363](#)). Hemiballismus usually develops acutely due to infarction of the contralateral subthalamic nucleus but occasionally develops subacutely or chronically due to other lesions of this nucleus.

Chorea is a rapid, jerky, irregular movement that tends to occur in the distal limbs or face but may also occur in proximal limb and axial structures. Acute or subacute onset is usually toxic due to excess levodopa or dopamine-agonist therapy or, less often, neuroleptics, birth control pills, pregnancy (chorea gravidarum), hyperthyroidism, or the antiphospholipid syndrome. In children, it may be associated with rheumatic fever and, in such cases, is referred to as *Sydenham's chorea*. The gradual onset of chorea is typical of degenerative neurologic diseases, such as Huntington's chorea ([Chap. 362](#)).

Myoclonus is a rapid, brief, irregular movement that is usually multifocal. Myoclonus can occur spontaneously at rest, in response to sensory stimuli, or with voluntary movements. It is a symptom that occurs in a wide variety of metabolic and neurologic disorders. Posthypoxic intention myoclonus is a special myoclonic syndrome that occurs as a sequel to transient cerebral anoxia. Myoclonus may result from lipid storage disease, encephalitis, Creutzfeldt-Jakob disease, or metabolic encephalopathies due to respiratory failure, chronic renal failure, hepatic failure, or electrolyte imbalance.

Myoclonus is also a feature of certain types of epilepsy, as discussed in [Chap. 360](#). *Palatal and segmental myoclonus* are uncommon rhythmic forms of myoclonus that may resemble tremor; they are caused by structural disease of the brainstem or spinal cord at the level of the abnormal movement.

Hypokinetic Movement Disorders These syndromes are manifest as bradykinesia, with a masked, expressionless facial appearance, loss of associated limb movements during walking, and rigid en bloc turning. If bradykinesia is associated only with a rest tremor, cogwheel rigidity, or impairment of postural reflexes (especially with a tendency to fall backwards), Parkinson's disease is likely ([Chap. 363](#)). If cognitive, language, upper motor neuron, sensory, or autonomic signs are also present, a *multisystem degenerative neurologic disease* is present. **These disorders are discussed in [Chaps. 363, 364, and 366](#).*

IMBALANCE AND DISORDERS OF GAIT

Imbalance is the impaired ability to maintain the intended orientation of the body in space. It is generally manifest as difficulty in maintaining an upright posture while standing or walking; a severe imbalance may also affect the ability to maintain posture while seated. Patients with imbalance commonly complain of a feeling of unsteadiness or dysequilibrium. Whereas imbalance and unsteadiness are synonymous, *dysequilibrium* implies the additional component of impaired spatial orientation even while lying down. Patients with dysequilibrium commonly also experience *vertigo*, defined as an hallucination of rotatory movement.

PATHOGENESIS

Imbalance and Limb Ataxia Imbalance results from disorders of the spinal cord (spinocerebellar) or vestibular sensory input, the integration of these inputs in the brainstem or midline cerebellum, or the motor output to the spinal neurons that control axial and proximal muscles. Limb ataxia results from disorders of the spinocerebellar and corticopontocerebellar inputs, the integration of these inputs in the intermediate and lateral cerebellum, or the output to the spinal neurons (via the red nucleus and rubrospinal tract) or to the cortex. These pathways ensure adequate speed, fluency, and integration of limb movements. The lateral cerebellar hemispheres coordinate a complex feedback circuit that modulates cortically initiated limb movement.

Sensory ataxia is caused by lesions that affect the peripheral sensory fibers, dorsal root ganglia cells, posterior columns of the spinal cord, lemniscal system in the brainstem, thalamus, or parietal cortex; relevant anatomy is discussed in [Chap. 23](#). Impairment of the proprioceptive sensory feedback to the cerebellum, basal ganglia, and cortex produces sensory ataxia. Sensory ataxia results in imbalance and disturbs the fluency and integration of movements that can be partially alleviated by visual feedback.

Disorders of Gait Walking is one of the most complicated motor activities. Essentially all structures discussed in this chapter participate in normal walking. Cyclical stepping movements produced by the lumbosacral spinal cord centers are modified by cortical, basal ganglionic, brainstem, and cerebellar influences based on proprioceptive, vestibular, and visual feedback.

Approach to the Patient

Examination of coordination, balance, and gait is typically performed at the same time. The finger-nose-finger and the heel-knee-shin maneuvers are observed for signs of incoordination in general and dysmetria in particular. *Dysmetria* consists of irregular errors in the amplitude and force of limb movements. This is accentuated near the target or point of intention and hence termed *intention tremor*. The patient is also asked to maintain the arms outstretched against a resistance that is suddenly removed; excessive *rebound* indicates cerebellar dysfunction. The ability of the patient to rapidly and repetitively tap the hands and feet is assessed for speed and rhythmicity. Errors in rhythm (irregular rate, velocity, or force) indicate *dysdiadochokinesia*. Slow, coarse, but rhythmical movements indicate upper motor neuron disorders. The patient is asked to demonstrate how to comb the hair or brush the teeth to assess the ability to initiate and execute a simple sequence of activity. Balance is examined by having the patient stand stationary with the feet together. If this position can be maintained, the eyes are closed for 5 to 10 s. Accentuation of sway or actual loss of balance is assessed. If balance is momentarily lost, several trials may be necessary to determine if the loss is consistently in the same direction. Walking along an uncrowded space, such as a hallway, is observed. Symmetry of arm swing and various phases of the gait cycle are observed. Walking is then performed for several steps on the heels, on the toes, and in tandem.

Imbalance An algorithm for interpretation of imbalance is presented in [Fig. 22-6](#).

Cerebellar ataxia results from disorders of the cerebellum or of its afferent inputs or efferent projections. Abnormalities of the midline cerebellar vermis or the flocculonodular lobe produce truncal ataxia which is usually revealed during the process of rising from a chair, assuming the upright stance with the feet together, or performing some other activity while standing. Once a desired position is reached, imbalance may be surprisingly mild. As walking begins, the imbalance recurs. Patients usually learn to lessen the imbalance by walking with the legs widely separated. The imbalance is usually not lateralized and may be accompanied by symmetric nystagmus.

Abnormalities of the intermediate and lateral portions of the cerebellum typically produce impaired limb movements rather than truncal ataxia. If involvement is asymmetric, lateralized imbalance is common and usually associated with asymmetric nystagmus. Clinical signs of cerebellar limb ataxia include dysmetria, intention tremor, dysdiadochokinesia, and abnormal rebound. Muscle tone is often modestly reduced; this contributes to the abnormal rebound due to decreased activation of segmental spinal cord reflexes and also to pendular reflexes, i.e., a tendency for a tendon reflex to produce multiple swings to and fro after a single tap. **For further discussion of cerebellar diseases, see [Chap. 364](#).*

Imbalance with vestibular dysfunction is characterized by a consistent tendency to fall to one side. The patient commonly complains of vertigo rather than imbalance, especially if the onset is acute. Acute vertigo associated with lateralized imbalance but no other neurologic signs is often due to disorders of the semicircular canal ([Chap. 21](#)); the presence of other neurologic signs suggests brainstem ischemia ([Chap. 361](#)) or multiple sclerosis ([Chap. 371](#)). When the vestibular dysfunction is peripheral, positional

nystagmus and vertigo tend to resolve if a provocative position is maintained (extinction) or repeated (habituation). Lateralized imbalance of gradual onset or persisting for more than 2 weeks, accompanied by nystagmus, may result from lesions of the semicircular canal or vestibular nerve, brainstem, or cerebellum.

Imbalance with sensory ataxia is characterized by marked worsening when visual feedback is removed. The patient can often assume the upright stance with feet together cautiously with eyes open. With eye closure, balance is rapidly lost (positive Romberg sign) in various directions at random. Sensory examination reveals impairment of proprioception at the toes and ankles, usually associated with an even more prominent abnormality of vibratory perception. Prompt evaluation for vitamin B₁₂ deficiency is important, as this disorder is reversible if recognized early ([Chap. 368](#)). Depression or absence of reflexes points to peripheral nerve disorders ([Chap. 377](#)). Spasticity with extensor plantar responses suggests posterior column and spinal cord disorders ([Chap. 368](#)). Rarely, sensory ataxia produces lateralized imbalance. In these cases, the disorder is usually in the parietal lobe or thalamus ([Chap. 23](#)), but may also be due to an asymmetric sensory neuropathy ([Chap. 377](#)) or posterior column disease ([Chap. 368](#)).

Sensory limb ataxia is similar to cerebellar limb ataxia but is markedly worse when the eyes are closed. Examination also reveals abnormal proprioception and vibratory perception. The approach focuses on localizing the proprioceptive impairment to the peripheral nerves ([Chap. 377](#)), the posterior columns of the spinal cord ([Chap. 368](#)), or rarely the parietal lobe.

Other forms of imbalance occur, but the fundamental problem is usually a primary disorder of strength, extrapyramidal function, or cortical initiation of movement.

Abnormal Gait Each of the disorders discussed in this chapter produces a characteristic gait disturbance. If the neurologic examination is normal except for an abnormal gait, diagnosis may be difficult even for the experienced clinician.

Hemiparetic gait characterizes spastic hemiparesis. In its most severe form, an abnormal posture of the limbs is produced by spasticity. The arm is adducted and internally rotated, with flexion of the elbow, wrist, and fingers and with extension of the hip, knee, and ankle. Forward swing of the spastic leg during walking requires abduction and circumduction at the hip, often with contralateral tilt of the trunk to prevent the toes catching on the floor as the leg is advanced. In its mildest form, the affected arm is held in a normal position, but swings less than the normal arm. The affected leg is flexed less than the normal leg during its forward swing and is more externally rotated. A hemiparetic gait is a common residual sign of a stroke ([Chap. 361](#)).

Paraparetic gait ([Video 361-3](#)) is a walking pattern in which both legs are moved in a slow, stiff manner with circumduction, similar to the leg movement in a hemiparetic gait. In many patients, the legs tend to cross with each forward swing ("scissoring"). A paraparetic gait is a common sign of spinal cord disease ([Chap. 368](#)) and also occurs in cerebral palsy.

Steppage gait is produced by weakness of ankle dorsiflexion. Because of the partial or

complete foot drop, the leg must be lifted higher than usual to avoid catching the toe on the floor during the forward swing of the leg. If unilateral, steppage gait is usually due to L5 radiculopathy, sciatic neuropathy, or peroneal neuropathy ([Chap. 377](#)). If bilateral, it is the common result of a distal polyneuropathy or lumbosacral polyradiculopathy ([Chap. 377](#)).

Waddling gait results from proximal lower limb weakness, most often from myopathy ([Chap. 381](#)) but occasionally from neuromuscular junction disease ([Chap. 380](#)) or a proximal symmetric spinal muscular atrophy ([Chap. 365](#)). With weakness of hip flexion, the trunk is tilted away from the leg that is being moved to lift the hip and provide extra distance between the foot and the floor, and the pelvis is rotated forward to assist with forward motion of the leg. Because pelvic girdle weakness is customarily bilateral, the pelvic lift and rotation alternates from side to side, giving the waddling appearance to the gait.

Parkinsonian gait ([Video 361-1](#)) is characterized by a forward stoop, with modest flexion at the hips and knees. The arms are flexed at the elbows and adducted at the shoulders, often with a 4- to 6-Hz resting pronation-supination tremor but little other movement, even during walking. Walking is initiated slowly by leaning forward and maintained with short rapid steps, during which the feet shuffle along the floor. The pace tends to accelerate (festination) as the upper body gradually leans further ahead of the feet, whether movement is forward (propulsion) or backward (retropulsion). The postural instability leads to falls ([Chap. 363](#)).

Apraxic gait ([Video 361-4](#)) results from bilateral frontal lobe disease with impaired ability to plan and execute sequential movements. This gait superficially resembles that of parkinsonism, in that the posture is stooped and any steps taken are short and shuffling. However, initiation and maintenance of walking are impaired in a different manner. Each movement that is required for walking can usually be performed, if tested in isolation while sitting or lying. However, when asked to step forward while standing, a long pause often occurs before any attempt is made to flex at the hip and advance, as if the patient is "glued to the ground." Once walking is initiated, it is not maintained, even in an abnormal festinating manner. Rather, after one or several steps are taken, walking is stopped for several seconds or longer. The process is then repeated. Dementia and incontinence may coexist.

Choreoathetotic gait is characterized by an intermittent, irregular movement that disrupts the smooth flow of a normal gait. Flexion or extension movements at the hip are common and unpredictable but readily observed as a pelvic lurch ([Chap. 363](#)).

Cerebellar ataxic gait ([Video 361-3](#)) is a broad-based gait disorder in which the speed and length of stride varies irregularly from step to step. With midline cerebellar disease, as in alcoholics, posture is erect but the feet are separated; lower limb ataxia is commonly present as well. Assumption of a particular stance or a change in position may cause instability, yet balance can usually be maintained well with the eyes open or closed. Walking may be rapid, but cadence is irregular. Although patients commonly lack confidence in the stability of their walking, only minimal support is often required for reassurance. With disease of the cerebellar hemispheres, limb ataxia and nystagmus are commonly present as well ([Chap. 364](#)).

Sensory ataxic gait may resemble a cerebellar gait, with its broad-based stance and difficulty with change in position. However, although balance may be maintained with the eyes open, loss of visual input through eye closure results in rapid loss of balance with a fall (positive Romberg sign), unless the physician assists the patient.

Vestibular gait is one in which the patient consistently tends to fall to one side, whether walking or standing. Cranial nerve examination demonstrates an obviously asymmetric nystagmus. The possibilities of unilateral sensory ataxia and hemiparesis are excluded by the findings of normal proprioception and strength ([Chap. 21](#)).

Astasia-abasia is a typical hysterical gait disorder. Although the patient usually has normal coordination of leg movements in bed or while sitting, the patient is unable to stand or walk without assistance. If distracted, stationary balance is sometimes maintained and several steps are taken normally, followed by a dramatic demonstration of imbalance with a lunge toward the examiner's arms or a nearby bed.

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23. NUMBNESS, TINGLING, AND SENSORY LOSS - Arthur K. Asbury

NORMAL SENSATION

Normal somatic sensation reflects a continuous day and night monitoring process that occupies considerable moment-to-moment nervous system capacity. Little of this activity reaches consciousness under ordinary conditions. In contrast, disordered sensation, particularly if experienced as painful, is alarming and dominates the sufferer's attention. Abnormalities of sensation, especially if painful, tend to make those suffering seek medical help. The physician must be able to recognize abnormal sensations by how they are described, know their type and likely site of origin, and understand their implications. **For a consideration of pain, see [Chap. 12](#).*

Positive and Negative Phenomena Abnormal sensory phenomena may be divided into two categories, positive and negative. The prototypical positive phenomenon is tingling (pins-and-needles), and the principal negative phenomenon is numbness. In addition to tingling, positive sensory phenomena include other altered sensations that are often described as pricking, bandlike, lightning-like shooting feelings (lancinations), aching, knifelike, twisting, drawing, pulling, tightening, burning, searing, electrical, or raw feelings. These descriptors are frequently the actual words used by patients. Such sensations may or may not be experienced as painful.

Positive phenomena usually result from trains of impulses generated at a site or sites of lowered threshold or heightened excitability along a sensory pathway, either peripheral or central. The nature and severity of an abnormal sensation depend on the number, rate, timing, and distribution of ectopic impulses and the type and function of nervous tissue in which they arise. Because positive phenomena represent excessive activity in sensory pathways, they are not necessarily associated with any sensory deficit (loss) upon examination.

Negative phenomena represent loss of sensory function and are characterized by diminished or absent feeling, often experienced as numbness. In contrast to positive phenomena, negative phenomena are accompanied by abnormal findings on sensory examination. In disorders affecting peripheral sensation, it is estimated that at least half the afferent axons innervating a given site are lost or functionless before sensory deficit can be demonstrated by clinical examination. This estimate probably varies according to how rapidly sensory nerve fibers have lost function. If the rate of loss is slow and chronic, lack of cutaneous feeling may be unnoticed by the patient and difficult to demonstrate on examination, even though few sensory fibers are functioning. Rapidly evolving sensory abnormality usually evokes both positive and negative phenomena and is readily recognized by patients. Subclinical degrees of sensory dysfunction not demonstrable on clinical sensory examination may be revealed by sensory nerve conduction studies or somatosensory cerebral evoked potentials ([Chap. 357](#)). Sensory symptoms may be either positive or negative, but sensory signs on examination are always a measure of negative phenomena.

Terminology Words used to characterize sensory disturbance are descriptive and have been arrived at mainly by convention. Paresthesia and dysesthesia are general terms used to denote sensory symptoms (positive phenomena) and are usually stated in the

plural form. *Paresthesias* usually refer to tingling or pins-and-needles sensations but may also include a wide variety of other abnormal sensations, excepting pain. Sometimes "paresthesias" carry the implication that the abnormal sensations are perceived without an apparent stimulus. *Dysesthesia* is a more general term used to subsume all types of abnormal sensations, even painful ones, whether a stimulus is evident or not.

While dysesthesias and paresthesias refer to sensations described by patients, another set of terms refers to sensory abnormalities found on examination. These include *hypesthesia* or *hypoesthesia* (reduction of cutaneous sensation to a specific type of testing such as pressure, light touch, and warm or cold stimuli); *anesthesia* (complete absence of skin sensation to the same stimuli plus pinprick); and *hypalgesia* (referring to reduced pain perception, i.e., nociception, such as the pricking quality elicited by a pin). *Hyperesthesia* means pain in response to touch. Similarly, *allodynia* describes the situation in which a nonpainful stimulus, once perceived, is experienced as painful, even excruciating. An example is elicitation of a painful sensation by application of a vibrating tuning fork. *Hyperalgesia* denotes severe pain in response to a mildly noxious stimulus, and *hyperpathia*, a broad term, encompasses all the phenomena described by hyperesthesia, allodynia, and hyperalgesia. With hyperpathia, the threshold for a sensory stimulus is increased and the perception is delayed but once felt, is unduly painful.

Disorders of deep sensation, arising from muscle spindles, tendons, and joints, affect proprioception (position sense). Manifestations include imbalance (particularly with eyes closed or in the dark), clumsiness of precision movements, and unsteadiness of gait, which are referred to collectively as *sensory ataxia* ([Chap. 22](#)). Other findings on examination usually, but not invariably, include reduced or absent joint position and vibratory sensibility and absent deep tendon reflexes in the affected limbs. Romberg's sign is positive, which means that the patient sways or topples when asked to stand with feet close together and eyes closed. In severe states of deafferentation involving deep sensation, the patient cannot walk or stand unaided or even sit unsupported. Continuous, sometimes wormlike involuntary movements, called *pseudoathetosis*, of the outstretched hands and fingers occur, particularly with eyes closed. Such patients are severely disabled.

Anatomy of Sensation Cutaneous afferent innervation is conveyed by a rich variety of receptors, both naked nerve endings (nociceptors and thermoreceptors) and encapsulated terminals (mechanoreceptors). Each type of receptor has its own set of sensitivities to specific stimuli, size and distinctness of receptive fields, and adaptational qualities. Much of the knowledge about these receptors has come from the development of techniques to study single intact nerve fibers intraneurally in awake unanesthetized human subjects. It is possible not only to record from single nerve fibers, large or small, but also to stimulate single fibers in isolation. A single impulse, whether elicited by a natural stimulus or evoked by electrical microstimulation, in a large myelinated afferent fiber may be both perceived and localized.

Afferent fibers of all sizes in peripheral nerve trunks traverse the dorsal roots and enter the dorsal horn of the spinal cord ([Fig. 23-1](#)). From there the smaller fibers take a different route to the parietal cortex than the larger fibers. The polysynaptic projections

of the smaller fibers (unmyelinated and small myelinated), which subserve mainly nociception, temperature sensibility, and touch, cross and ascend in the opposite anterior and lateral columns of the spinal cord, through the brainstem, to the ventral posterolateral (VPL) nucleus of the thalamus, and ultimately project to the postcentral gyrus of the parietal cortex ([Chap. 12](#)). This is referred to as the *spinothalamic pathway*, or *anterolateral system*. The larger fibers, which subserve tactile and position sense and kinesthesia, project rostrally in the posterior column on the same side of the spinal cord and make their first synapse in the gracile or cuneate nuclei of the lower medulla. The second-order neuron decussates and ascends in the medial lemniscus located medially in the medulla and in the tegmentum of the pons and midbrain and synapses in the VPL. The third-order neuron projects to parietal cortex; this large fiber system is referred to as the *posterior column-medial lemniscal pathway* (lemniscal, for short). Note that although the lemniscal and the anterolateral pathways both project up the spinal cord to the thalamus, it is the (crossed) anterolateral pathway that is referred to as the *spinothalamic tract*, by convention.

Although the fiber types and functions that make up the spinothalamic and lemniscal systems are relatively well known, it has been found that many other fibers, particularly those associated with touch, pressure, and position sense, ascend in a diffusely distributed pattern both ipsilaterally and contralaterally in the anterolateral quadrants of the spinal cord. This explains why an individual with a complete lesion of the posterior columns of the spinal cord may have little sensory deficit on examination.

EXAMINATION OF SENSATION

The main tasks of the sensory examination are tests of primary sensation. By convention these include the sense of pain, touch, vibration, joint position, and thermal sensation, both hot and cold ([Table 23-1](#)). Detailed descriptions of how to perform the various tests of the sensory examination can be found in standard texts (see "Bibliography").

Some general principles pertain. First, the examiner must depend on subjective patient response, particularly when using cutaneous stimuli (pin, touch, vibration, warm or cold). This factor may complicate the interpretation of the sensory examination. Second, with complaints of numbness, patients should be asked to outline on themselves the borders of numb areas. Third, some patients are only partially examinable. In a stuporous patient, sensory examination is reduced to observing the briskness of withdrawal in response to a pinch or other noxious stimulus. Comparison of response on one side of the body to the other is essential. In the alert but uncooperative patient, cutaneous sensation may be unexaminable. However, it is usually possible to get some idea of proprioceptive function by noting the patient's best performance of movements requiring balance and precision. Fourth, sensory examination of a patient who has no neurologic complaints should be abbreviated and may consist of pin, touch, and vibration testing in the hands and feet plus evaluation of stance and gait, including the Romberg maneuver. Evaluation of stance and gait also tests the integrity of motor and cerebellar systems.

Primary Sensation (See [Table 23-1](#)) The sense of pain is usually tested with a pin, asking the patient to focus on the pricking or unpleasant quality of the stimulus and not just the pressure or touch sensation elicited. Areas of hypalgesia should be mapped by

proceeding radially from the most hypalgesic site ([Figs. 23-2](#) and [23-3](#)).

Temperature sensation, to both hot and cold, is probably best tested with water flasks filled with water of the desired temperature, using a thermometer to verify the temperature. This is impractical in most settings. An alternative way to test cold sensation is to touch a metal object, such as a tuning fork at room temperature, to the skin. For testing warm temperatures, the tuning fork or other metal object may be held under warm water of the desired temperature and then used. Both cold and warm should be tested because different receptors respond to each.

Touch is usually tested with a wisp of cotton or a fine camelhair brush. In general, it is better to avoid testing touch on hairy skin because of the profusion of sensory endings that surround each hair follicle.

Joint position testing is a measure of proprioception, one of the most important functions of the sensory system. With the patient keeping eyes closed, joint position is tested in the great toe and in the fingers. If errors are made in recognizing the direction of passive movements of the toe or the finger, more proximal joints should be tested. A test of proximal joint position sense, primarily at the shoulder, is performed by asking the patient to bring the two index fingers together with the arms extended and the eyes closed. Normal individuals should be able to do this quite accurately, with errors of a centimeter or less.

The sense of vibration is tested with a tuning fork, preferably a large one that vibrates at 128 Hz. Vibration is usually tested at bony prominences, beginning distally at the malleoli of the ankles, and at the knuckles. If abnormalities are found, more proximal sites can be examined. Vibratory thresholds at the same site in the patient and the examiner can be compared for control purposes.

Quantitative Sensory Testing Effective sensory testing devices have been developed over the past two decades. Quantitative sensory testing is particularly useful for serial evaluation of cutaneous sensation in clinical trials. Threshold testing for touch and vibratory and thermal sensation is the most widely used application.

Cortical Sensation Cortical sensory testing includes two-point discrimination, touch localization, and bilateral simultaneous stimulation and tests for graphesthesia and stereognosis, to name the most commonly used methods. Abnormalities of these sensory tests, in the presence of normal primary sensation in an alert cooperative patient, signify a lesion of the parietal cortex or thalamocortical projections to the parietal lobe. If primary sensation is altered, these cortical discriminative functions will usually be abnormal, too. Comparisons should always be made between analogous sites on the two sides of the body because the deficit with a specific parietal lesion is likely to be hemilateral. Side-to-side comparisons hold true for all cortical sensory testing.

Two-point discrimination is tested by special calipers, the points of which may be set from 2 mm to several centimeters apart and then applied simultaneously to the site to be tested. The pulp of the fingertips is a common site to test; a normal individual can distinguish about 3-mm separation of points there.

Touch localization is usually carried out by light pressure with the examiner's fingertip, asking the patient, whose eyes are closed, to identify the site of touch. It is usual to ask the patient to touch the same site with a fingertip.

Bilateral simultaneous stimulation at analogous sites (e.g., the dorsa of both hands) can be carried out to determine whether the perception of touch is extinguished consistently on one side or the other. The phenomenon is referred to as *extinction* on bilateral simultaneous stimulation.

Graphesthesia means the capacity to recognize with eyes closed letters or numbers drawn by the examiner's fingertip on the palm of the hand. Once again, the comparison of one side with the other is of prime importance. Inability to recognize numbers or letters is termed *agraphesthesia*.

Stereognosis refers to the ability to identify common objects by palpation, recognizing their shape, texture, and size. Common standard objects are the best test objects, such as a marble, a paper clip, or coins. Patients with normal stereognosis should be able to distinguish a dime from a penny and a nickel from a quarter without looking. Patients should only be allowed to feel the object with one hand at a time. If they are unable to identify it in one hand, it should be placed in the other for comparison. Individuals unable to identify common objects and coins in one hand who can do so in the other are said to have *astereognosis* of the abnormal hand.

LOCALIZATION OF SENSORY ABNORMALITIES

Sensory symptoms and signs can result from lesions at almost any level of the nervous system, including parietal cortex, deep white matter, thalamus, brainstem, spinal cord, spinal root, peripheral nerve, and sensory receptor. Noting the distribution and nature of sensory symptoms and signs is the most important way to localize their source. The extent, configuration, symmetry, quality, and severity are the key observations.

Dysesthesias without sensory findings by examination can be difficult to interpret. To illustrate, tingling dysesthesias in an acral distribution (hands and feet) can have more than one interpretation. Distal dysesthesias can be systemic in origin, e.g., secondary to hyperventilation, or can be induced by a medication, such as the diuretic acetazolamide. Distal dysesthesias can also be an early event in an evolving polyneuropathy or can herald a myelopathy, such as with vitamin B₁₂ deficiency. Sometimes distal dysesthesias have no definable basis. In contrast, dysesthesias that correspond to a particular peripheral nerve territory denote a lesion of that nerve trunk. For instance, dysesthesias restricted to the fifth digit and the adjacent one-half of the fourth finger on one hand reliably point to disorder of the ulnar nerve, most commonly at the elbow.

Nerve and Root In focal nerve trunk lesions severe enough to cause a deficit, sensory abnormalities are readily mapped and generally have discrete boundaries ([Figs. 23-2](#) and [23-3](#)). Root lesions, referred to as radicular, are frequently accompanied by deep, aching pain along the course of the related nerve trunk. With compression of a fifth lumbar (L5) or first sacral (S1) root, as may occur with a ruptured intervertebral disc, sciatica is a frequent manifestation. With a lesion affecting a single root, sensory deficit

in the distribution of that root is often minimal or not demonstrable at all. This is because adjacent root territories overlap extensively.

Polyneuropathies are generally graded, distal, and symmetric in distribution of deficit ([Chap. 377](#)). Dyesthesias begin in the toes and ascend symmetrically, followed by numbness. When dyesthesias reach the knees, they have usually also appeared in the fingertips. The process appears to be nerve length-dependent, and the deficit is often described as "stocking-glove" in type. Although most polyneuropathies are pansenory and affect all modalities of sensation, selective sensory dysfunction according to nerve fiber size may occur. In polyneuropathies that affect small nerve fibers selectively, the hallmark is burning, painful dysesthesias with reduced pinprick and thermal sensation but with sparing of proprioception, motor function, and even deep tendon jerks. Touch is variably involved, but when spared, the sensory pattern is referred to as *sensory dissociation*. Sensory dissociation patterns can be seen with spinal cord lesions (see below) as well as with small fiber neuropathies. In contrast to small fiber polyneuropathies, large fiber polyneuropathies are characterized by position sense deficit, imbalance, absent tendon jerks, and variable motor dysfunction but preservation of most cutaneous sensation. Dyesthesias, if present at all, tend to be tingling or bandlike.

Spinal Cord (See [Chap. 368](#)) If the spinal cord is transected, all sensation is lost below the level of transection. Bladder and bowel function are also lost, as is motor function. Hemisection of the spinal cord produces the Brown-Sequard syndrome, which involves absent pain and temperature sensation on the opposite side below the lesion, and loss of proprioceptive sensation and loss of motor power on the same side below the lesion (see [Figs. 23-1](#) and [368-1](#)). Dissociated sensory deficit patterns (see above) are also a sign of spinothalamic tract involvement in the spinal cord, especially if the deficit is unilateral and has an upper level on the torso. Bilateral spinothalamic tract involvement occurs with lesions affecting the center of the spinal cord, such as happens with expansion of the central canal in syringomyelia. Sensory dissociation is characteristic of syringomyelia.

Brainstem Harlequin patterns of sensory disturbance, in which one side of the face and the opposite side of the body are affected, localize to the lateral medulla. Here a small lesion may damage both the ipsilateral descending trigeminal tract and ascending spinothalamic fibers subserving the opposite arm, leg, and hemitorso (see "Lateral medullary syndrome" in [Fig. 361-7](#)). In the tegmentum of the pons and midbrain, where the lemniscal and spinothalamic tracts merge, a lesion here causes pansenory loss on the contralateral body.

Thalamus Hemisensory disturbance with tingling numbness from head to foot is often thalamic in origin but can also be anterior parietal. If abrupt in onset, the lesion is likely to be due to a small stroke (lacunar infarction), particularly if localized to the thalamus. Occasionally, with lesions affecting the [VPL](#) or adjacent white matter, a syndrome of thalamic pain, also called *Dejerine-Roussy syndrome*, may ensue. This persistent unrelenting hemipainful state is often described in dramatic terms such as "like the flesh is being torn from my limbs" or "as though that side is bathed in acid" ([Chap. 12](#)).

Cortex With lesions of the parietal lobe, either of the cortex or of subjacent white matter,

the most prominent symptoms are contralateral hemineglect, hemi-inattention, and a tendency not to use the affected hand and arm. Tests of primary sensation may be normal or altered. Anterior parietal infarction may present as a pseudothalamic syndrome with crossed hemilateral loss of primary sensation. Dysesthesias or a sense of numbness may also occur, and rarely a painful state.

Focal Sensory Seizures These are generally due to lesions in or near the postcentral gyrus. Symptoms of focal sensory seizures are usually combinations of numbness and tingling, but frequently additional more complex sensations are present, such as a rushing feeling, a sense of warmth, a sense of movement without visible motion, or other unpleasant dysesthesias. Duration of seizures is variable; they may be transient, lasting only seconds, or they may persist for hours. Focal motor features (clonic jerking) may supervene, and seizures can become generalized with loss of consciousness. Likely sites of symptoms are unilaterally in the lips, face, digits, or foot, and symptoms may spread as in a Jacksonian march. On occasion, symptoms may occur in a symmetric bilateral fashion, for instance, in both hands; this results from involvement of the second sensory area (unilaterally) located in the rolandic area at and just above the Sylvian fissure.

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24. ACUTE CONFUSIONAL STATES AND COMA - Allan H. Ropper

Confusional states and coma are among the most common problems in general medicine. They account for a substantial portion of admissions to emergency wards and are a frequent cause of distress on all hospital services. Because clouding of consciousness and a diminished level of consciousness frequently coexist and result from many of the same diseases, they are presented together here, but from a medical perspective they have different clinical characteristics and physiologic explanations.

The basis of consciousness has long been a topic of great interest to psychologists and philosophers and is the subject of a vast literature. Physicians have been mainly concerned with impairments in the level of consciousness (coma, stupor, drowsiness) and with alterations of consciousness, meaning an inability to think coherently, i.e., with accustomed clarity and speed. The latter, broadly termed *confusion* relates to lessened awareness, perception, apperception (the interpretation of perceptions), thinking, expression in language and action, and all forms of intellection that are dependent on the continuous integration of mental processes. Normal awareness provides a background to our inner mental life, which flows from infancy to death like a "stream of thought," to use William James's metaphor. Self-awareness requires that a person experience these thoughts and be able to reflect and operate upon them.

Almost all instances of diminished alertness can be traced to widespread abnormalities of the cerebral hemispheres or to reduced activity of a special thalamocortical alerting system termed the *reticular activating system* (RAS). The proper functioning of this system, its ascending projections to the cortex, the cortex itself, and corticothalamic connections are required to maintain alertness and coherence of thought.

THE CONFUSIONAL STATE

Confusion is a mental and behavioral state of reduced comprehension, coherence, and capacity to reason. Inattention, as defined by the inability to sustain uninterrupted thought and actions, and disorientation are its earliest outward signs. As the state of confusion worsens, there are more global mental failings, including impairments of memory, perception, comprehension, problem solving, language, praxis, visuospatial function, and various aspects of emotional behavior that are each attributable to particular regions of the brain. In other instances an apparent confusional state may arise from an isolated deficit in mental function such as an impairment of language (*aphasia*), loss of memory (*amnesia*), or lack of appreciation of spatial relations of self and the external environment (*agnosia*), but the attributes of the problem are then quite different ([Chap. 25](#)). Confusion is also a feature of dementia, in which case the chronicity of the process, as in the instance of Alzheimer's disease, distinguishes it from an acute encephalopathy ([Chap. 26](#)).

The confused patient is usually subdued, not inclined to speak, and is physically inactive. A state of confusion that is accompanied by agitation, hallucinations, tremor, and illusions (misperceptions of environmental sight, sound, or touch) is termed *delirium*, as typified by delirium tremens from alcohol or drug withdrawal. In psychiatric circles, delirium often refers, albeit imprecisely, to all acute states of confusion with clouding of consciousness and incoherence of thought.

Approach to the Patient

Confusion and delirium always signify a disorder of the nervous system. They may be the major manifestation of a head injury; a seizure; drug toxicity (or drug withdrawal); a metabolic disorder resulting from hepatic, renal, pulmonary or cardiac failure; a systemic infection; meningitis or encephalitis; or a chronic dementing disease.

The search for these manifold causes begins with a careful history emphasizing the patient's condition before the onset of confusion. The clinical examination should focus on signs of diminished attentiveness, disorientation, and drowsiness and on the presence of localizing neurologic signs. From the clinical data the clinician is directed to the appropriate laboratory tests discussed further on. Often, even after all diagnostic tests are completed, one may still not know the cause of a confusional state. The proper approach is to observe the patient in the hospital for a number of days under stable conditions. New clues may appear or an obscure confusion perhaps related to a medication, may clear up, while other causes such as renal or hepatic failure may worsen and lead to coma.

Orientation and memory are tested by asking the patient in a forthright manner the date, inclusive of month, day, year, and day of week; the precise place; and some items of generally acknowledged and universally known information (the names of the President and Vice President, a recent national catastrophe, the state capital). Further probing may be necessary to reveal a defect -- why is the patient in the hospital; what is his or her address, zip code, telephone number, social security number? Problems of increasing complexity may be pursued, but they usually provide little additional information. Attention and coherence of thought can be gauged by the clarity of and speed of responses while the history is being given but are examined more explicitly by having the patient repeat strings of numbers (most adults easily retain seven digits forward and four backward), spell a word such as "world" backwards, and perform serial calculations -- tests of serial subtraction of 3 from 30 or 7 from 100 are useful. It is the inability to sustain coherent mental activity in performing tasks such as these that exposes the most subtle confusional states.

Other salient neurologic findings are the level of alertness, which fluctuates if there is drowsiness; indications of focal damage of the cerebrum such as hemiparesis, hemianopia, and aphasia; or adventitious movements of myoclonus or partial convulsions. The language of the confused patient may be disorganized and rambling, even to the extent of incorporating paraphasic words. These features, along with impaired comprehension that is due mainly to inattention, may be mistaken for aphasia.

One of the most specific signs of a metabolic encephalopathy is asterixis, which is an arrhythmic flapping tremor that is typically elicited by asking the patient to hold the arms outstretched with the wrists and hands fully extended. After a few seconds, there is a large jerking lapse in the posture of the hand and then a rapid return to the original position. The same movements can be appreciated in any tonically held posture, even of the tongue, and in extreme form the movements may intrude on voluntary limb motion. Bilateral asterixis always signifies a metabolic encephalopathy, e.g., from hepatic failure, hypercapnia, or from drug ingestion, especially with anticonvulsant

medications. Myoclonic jerking and tremor in an awake patient are typical of uremic encephalopathy or the use of antipsychotic drugs such as lithium, phenothiazines, or butyrophenones; myoclonus with coma may also signify anoxic cerebral damage.

Confusion in the postoperative period is common but at times so subtle as to escape attention. Cardiac and orthopedic procedures are particularly likely to produce disorientation or delirium in susceptible patients. Often a careful history will reveal that a mild but compensated dementia existed prior to the operation. Medications, particularly those with anticholinergic activity (including meperidine), inadvertent withdrawal from sleeping pills or alcohol, fever, and any of the endogenous metabolic derangements listed above may be responsible, or a stroke may have occurred.

Frequently confusion cannot be attributed to any single factor and it clears in several days. In many cases, particularly in the elderly, transient confusion and drowsiness arise with a febrile infection of the urinary tract, lungs, blood, or peritoneum. The term *septic encephalopathy* is currently used to describe this association, but the mechanism by which infection or inflammation leads to cerebral dysfunction is unknown. Fever can also alter brain function in a way that makes preexisting focal signs worse.

Distinguishing dementia from an acute confusional state is a great problem, especially in the elderly, since the two may coexist if a fever, other acute medical problem, or a poorly tolerated medication supervenes in a mildly demented patient, producing a so-called beclouded dementia. The memory loss of dementia brings about a confusional state that varies little in severity from hour to hour and day to day. Poor mental performance is derived mainly from incomplete recollection, inadequate access to names and ideas, and the inability to retain new information, thus affecting orientation and factual knowledge. In contrast to the acute confusional states, attention, alertness, and coherence are preserved until the most advanced stages. Eventually dementia produces a chronic confusion with breakdown of all types of mental performance, and the distinction from an acute encephalopathy depends mainly on the longstanding nature of the condition.

Treatment of the confusional state requires that all unnecessary medication be stopped, metabolic alterations be rectified, and infection be treated. Skilled nursing and a quiet room with a window are important. Careful explanations should be given at regular intervals to the family. In the elderly, regular reorientation and active measures to avoid risk factors (sleep deprivation, immobility, and vision and hearing impairments) reduce the number and severity of episodes of delirium in hospitalized patients.

COMA AND RELATED DISORDERS OF CONSCIOUSNESS

The unnatural situation of reduced alertness and responsiveness represents a continuum that in severest form is called *coma*, a deep sleeplike state from which the patient cannot be aroused. *Stupor* defines lesser degrees of unarousability in which the patient can be awakened only by vigorous stimuli, accompanied by motor behavior that leads to avoidance of uncomfortable or aggravating stimuli. *Drowsiness*, which is familiar to all persons, simulates light sleep and is characterized by easy arousal and the persistence of alertness for brief periods. Drowsiness and stupor are usually attended by some degree of confusion. In clinical practice these terms should be

supplemented by a narrative description of the level of arousal and of the type of responses evoked by various stimuli precisely as observed at the bedside. Such an account is preferable to ambiguous terms such as semicoma or obtundation, the definitions of which differ between physicians.

Several other neurologic conditions render patients apparently unresponsive and simulate coma, and certain other subsyndromes of coma must be considered separately because of their special significance. Among the latter, the *vegetative state* signifies an awake but unresponsive state. Most of these patients were earlier comatose and after a period of days or weeks emerge to an unresponsive state in which their eyelids are open, giving the appearance of wakefulness. Yawning, grunting, swallowing, as well as limb and head movements persist, but there are few, if any, meaningful responses to the external and internal environment -- in essence, an "awake coma." Although respiratory and autonomic functions are retained, the term "vegetative" is nonetheless unfortunate as it is subject to misinterpretation by lay persons. Always there are accompanying signs that indicate extensive damage in both cerebral hemispheres, e.g., decerebrate or decorticate limb posturing and absent responses to visual stimuli (see below). Cardiac arrest and head injuries are the most common causes of the vegetative state ([Chaps. 369](#) and [376](#)). The prognosis for regaining mental faculties once the vegetative state has supervened for several months is almost nil hence the term *persistent vegetative state*. Most instances of dramatic recovery, when investigated carefully, are found to yield to the usual rules for prognosis, but it must be acknowledged that rare instances of awakening to a condition of dementia and paralysis have been documented.

Certain other clinical states are prone to be misinterpreted as stupor or coma. *Akinetic mutism* refers to a partially or fully awake patient who is able to form impressions and think but remains immobile and mute, particularly when unstimulated. The condition may result from damage in the regions of the medial thalamic nuclei, the frontal lobes (particularly situated deeply or on the orbitofrontal surfaces), or from hydrocephalus. The term *abulia* is used to describe a mental and physical slowness and lack of impulse to activity that is in essence a mild form of akinetic mutism, with the same anatomic origins. *Catatonia* is a curious hypomobile and mute syndrome associated with a major psychosis. In the typical form patients appear awake with eyes open but make no voluntary or responsive movements, although they blink spontaneously, swallow, and may not appear distressed. As often, the eyes are half-open as if the patient is in a fog or light sleep. There are signs that indicate voluntary attempts to appear less than fully responsive, though it may take some ingenuity on the part of the examiner to demonstrate these. Eyelid elevation is actively resisted, blinking occurs in response to a visual threat, and the eyes move concomitantly with head rotation, all signs belying a brain lesion. It is characteristic but not invariable for the limbs to retain the posture, no matter how bizarre, in which they have been placed by the examiner ("waxy flexibility," or catalepsy.) Upon recovery, such patients have some memory of events that occurred during their catatonic stupor. The appearance is superficially similar to akinetic mutism, but clinical evidence of brain damage is lacking.

The *locked-in state* describes a pseudocoma in which an awake patient has no means of producing speech or volitional limb, face, and pharyngeal movements in order to indicate that he or she is awake, but vertical eye movements and lid elevation remain

unimpaired, thus allowing the patient to signal. Such individuals have written entire treatises using Morse code. Infarction or hemorrhage of the ventral pons, which transects all descending corticospinal and corticobulbar pathways, is the usual cause. A similar awake but deafferented state occurs as a result of total paralysis of the musculature in severe cases of Guillain-Barre syndrome ([Chap. 378](#)), critical illness neuropathy ([Chap. 376](#)), and pharmacologic neuromuscular blockade.

THE ANATOMY AND PHYSIOLOGY OF UNCONSCIOUSNESS

To the extent that all complex waking behaviors require the widespread participation of the cerebral cortex, consciousness cannot exist without the activity of these structures. A loosely grouped aggregation of neurons located in the upper brainstem and medial thalamus, the [RAS](#), maintains the cerebral cortex in a state of wakeful consciousness. It follows that the principal causes of coma are (1) lesions that damage a substantial portion of the RAS; (2) destruction of large portions of both cerebral hemispheres; and (3) suppression of thalamocerebral function by drugs, toxins, or by internal metabolic derangements such as hypoglycemia, anoxia, azotemia, or hepatic failure.

The classic animal experiments of Moruzzi and Magoun, published in 1949, and subsequent human clinicopathologic observations have established that the regions of the reticular formation that are critical to the maintenance of wakefulness extend from the caudal midbrain to the lower thalamus. A most important practical consideration derives from the anatomic proximity of the [RAS](#) to structures that are concerned with pupillary function and eye movements. Pupillary enlargement and loss of vertical and adduction movements of the globes suggest that upper brainstem damage may be the source of coma. Although circumscribed lesions confined to one or both cerebral hemispheres do not affect the brainstem RAS, a large mass on one side of the brain may cause coma by secondarily compressing the upper brainstem and consequently producing abnormalities of the pupils and eye movements (see discussion of transtentorial herniation below). This type of indirect effect is most typical of cerebral hemorrhages and of rapidly expanding tumors within a cerebral hemisphere. In all cases the degree of diminished alertness also relates to the rapidity of evolution and the extent of compression of the RAS.

The neurons of the [RAS](#) are thought to project rostrally to the cortex primarily via thalamic relay nuclei that in turn exert a tonic influence on the activity of the entire cerebral cortex. The behavioral arousal effected by somesthetic, auditory, and visual stimuli depends upon the rich reciprocal innervation that the RAS receives from these sensory systems. The relays between the RAS and the thalamic and cortical areas utilize a variety of neurotransmitters. Of these, the effect of arousal on acetylcholine and on the biogenic amines has been studied more extensively. Cholinergic fibers connect the midbrain to other areas of the upper brainstem, thalamus, and cortex. Serotonin and norepinephrine also subserve important functions in regulation of the sleep-wake cycle ([Chap. 27](#)). Their roles in arousal and coma have not been clearly established, although the alerting effects of amphetamines are likely to be mediated by catecholamine release.

Coma Due to Cerebral Mass Lesions and Herniations The cranial cavity is separated into compartments by infoldings of the dura -- the two cerebral hemispheres are

separated by the falx, and the anterior and posterior fossae by the tentorium. *Herniation* refers to displacement of brain tissue away from a mass and into a compartment that it normally does not occupy. Many of the signs associated with coma, and indeed coma itself, can be attributed to these tissue shifts. Herniation can be *transfalcial* (displacement of the cingulate gyrus under the falx and across the midline), *transtentorial* (displacement of the medial temporal lobe into the tentorial opening), and *foraminal* (downward forcing of the cerebellar tonsils into the foramen magnum; [Fig. 24-1](#)).

Uncal transtentorial herniation refers to impaction of the anterior medial temporal gyrus (the uncus) into the anterior portion of the tentorial opening. The displaced tissue compresses the third nerve as it traverses the subarachnoid space and results in enlargement of the ipsilateral pupil (putatively because the fibers subserving parasympathetic pupillary function are located peripherally in the nerve). The coma that follows may be due to lateral compression of the midbrain against the opposite tentorial edge by the displaced parahippocampal gyrus ([Fig. 24-2](#)). In some cases the lateral displacement causes compression of the opposite cerebral peduncle, producing a Babinski response and hemiparesis contralateral to the original hemiparesis (the Kernohan-Woltman sign). In addition to compressing the upper brainstem, tissue shifts, including herniations, may compress major blood vessels, particularly the anterior and posterior cerebral arteries as they pass over the tentorial reflections, thus producing brain infarctions. The distortions may also entrap portions of the ventricular system, resulting in regional hydrocephalus.

Central transtentorial herniation denotes a symmetric downward movement of the upper thalamic region through the tentorial opening. Miotic pupils and drowsiness are the heralding signs. Both temporal and central herniations are thought to cause progressive compression of the brainstem from above: first the midbrain, then the pons, and finally the medulla. The result is a sequential appearance of neurologic signs that corresponds to the affected level.

A direct relationship between the various configurations of transtentorial herniations and coma is, at best, tenuous. The orderly progression of signs from midbrain to medulla is often bypassed in catastrophic lesions where all brainstem functions are lost almost simultaneously. It is also clear that displacement of deep brain structures by a mass in any direction, with or without herniation, compresses the region of the [RAS](#) and results in coma. Furthermore, drowsiness and stupor typically occur with moderate lateral shifts at the level of the diencephalon (thalamus) well before transtentorial or other herniations are evident. Lateral shift is easily quantified on axial images of computed tomography (CT) and magnetic resonance imaging (MRI) scans ([Fig. 24-2](#)). In cases of *acutely appearing masses*, a fairly consistent and simple relationship exists between the degree of horizontal displacement of midline structures and the level consciousness. Specifically, horizontal displacement of the pineal calcification of 3 to 5 mm is generally associated with drowsiness, 6 to 8 mm with stupor, and >9 mm with coma. At the same time, intrusion of the medial temporal lobe into the tentorial opening may be apparent as an obliteration of the cisterns that surround the upper brainstem.

Coma and Confusional States Due to Metabolic Disorders A large variety of systemic metabolic abnormalities cause coma by interrupting the delivery of energy

substrates (hypoxia, ischemia, hypoglycemia) or by altering neuronal excitability (drug and alcohol intoxication, anesthesia, and epilepsy). The same metabolic abnormalities that produce coma may in milder form induce widespread cortical dysfunction and an acute confusional state. Thus, in metabolic encephalopathies, clouded consciousness and coma are a continuum. Neuropathologic changes in the various metabolic failures are variable -- very evident in hypoxia-ischemia, manifest as astrocytic changes in hepatic coma, and negligible in renal and other metabolic encephalopathies.

Cerebral neurons are fully dependent on cerebral blood flow (CBF) and the related delivery of oxygen and glucose. CBF approximates 75 mL per 100 g/min in gray matter and 30 mL per 100 g/min in white matter (mean = 55 mL per 100 g/min); oxygen consumption is 3.5 mL per 100 g/min, and glucose utilization is 5 mg per 100 g/min. Brain stores of glucose provide energy for approximately 2 min after blood flow is interrupted, and oxygen stores last 8 to 10 s after the cessation of blood flow. Simultaneous hypoxia and ischemia exhaust glucose more rapidly. The electroencephalogram (EEG) rhythm in these circumstances becomes diffusely slowed, typical of metabolic encephalopathies, and as conditions of substrate delivery worsen, eventually all recordable brain electrical activity ceases. In almost all instances of metabolic encephalopathy, the global metabolic activity of the brain is reduced in proportion to the degree of unconsciousness.

Conditions such as hyponatremia, hyperosmolality, hypercapnia, hypercalcemia, and hepatic and renal failure are associated with a variety of alterations in neurons and astrocytes. It should be stated at the outset that the reversible effects of these conditions on the brain are not understood, but they may in different circumstances impair energy supplies, change ion fluxes across neuronal membranes, and cause neurotransmitter abnormalities. For example, the high brain ammonia concentration that is associated with hepatic coma interferes with cerebral energy metabolism and with the Na⁺, K⁺-ATPase pump, increases the number and size of astrocytes, alters nerve cell function, and causes increased concentrations of potentially toxic products of ammonia metabolism; it may also result in abnormalities of neurotransmitters, including possible "false" neurotransmitters that may be active at receptor sites. Apart from hyperammonemia, which of these mechanisms is of critical importance is not clear. The mechanism of the encephalopathy of renal failure is also not known. Unlike ammonia, urea itself does not produce central nervous system (CNS) toxicity. A multifactorial causation has been proposed, including increased permeability of the blood-brain barrier to toxic substances such as organic acids and an increase in brain calcium or cerebrospinal fluid (CSF) phosphate content. Likewise, the basis of confusion and drowsiness that commonly accompanies the septic state has not been clarified.

Coma and seizures are a common accompaniment of any large shifts in sodium and water balance. These changes in osmolality may be the result of a number of systemic medical disorders including diabetic ketoacidosis, the nonketotic hyperosmolar state, and hyponatremia from any cause (e.g., water intoxication, excessive secretion of antidiuretic hormone or atrial natriuretic peptides). The volume of brain water correlates with the level of consciousness in these states, but other factors also play a role. Sodium levels below 125 mmol/L induce confusion, and below 115 mmol/L are associated with coma and convulsions. In hyperosmolar coma the serum osmolality generally exceeds 350 mosmol/L. *As in most other metabolic encephalopathies, the*

severity of neurologic change depends to a large degree on the rapidity with which the serum changes occur. Hypercapnia depresses the level of consciousness in proportion to the rise in CO₂ tension in the blood and depends very much on the rapidity of change. The pathophysiology of other metabolic encephalopathies such as hypercalcemia, hypothyroidism, vitamin B₁₂ deficiency, and hypothermia are incompletely understood but must also reflect derangements of [CNS](#) biochemistry and membrane function.

Epileptic Coma Although all metabolic derangements in some way alter neuronal electrophysiologic function, epilepsy is the only primary excitatory disturbance of brain electrical activity that is encountered in clinical practice. Continuous, generalized electrical discharges of the cortex (*seizures*) are associated with coma even in the absence of epileptic motor activity (*convulsions*). The self-limited coma that follows seizures, termed the *postictal state*, may be due to exhaustion of energy reserves or effects of locally toxic molecules that are the byproduct of seizures. The postictal state produces a pattern of continuous, generalized slowing of the background [EEG](#) activity similar to that of other metabolic encephalopathies.

Pharmacologic Coma This class of encephalopathy is in large measure reversible and leaves no residual damage providing hypoxia does not supervene. Many drugs and toxins are capable of depressing nervous system function. Some produce coma by affecting both the brainstem nuclei, including the [RAS](#), and the cerebral cortex. The combination of cortical and brainstem signs, which occurs in certain drug overdoses, may lead to an incorrect diagnosis of structural brainstem disease.

Approach to the Patient

The diagnosis and management of coma depend on knowledge of its main causes (see "Differential Diagnosis," below) and on interpretation of salient clinical signs, notably brainstem reflexes and motor function. Acute respiratory and cardiovascular problems should be attended to prior to neurologic assessment. A complete medical evaluation, except for the vital signs, funduscopy, and examination for nuchal rigidity, may be deferred until the neurologic evaluation has established the severity and nature of coma.

History In many cases, the cause of coma is immediately evident (e.g., trauma, cardiac arrest, or known drug ingestion). In the remainder, historic information about the onset of coma is often sparse, but certain historic points are especially useful: (1) the circumstances and rapidity with which neurologic symptoms developed; (2) the details of any immediately preceding medical and neurologic symptoms (confusion, weakness, headache, fever, seizures, dizziness, double vision, or vomiting); (3) the use of medications, illicit drugs, or alcohol; and (4) chronic liver, kidney, lung, heart, or other medical disease. Direct interrogation or telephone calls to family and observers on the scene are an important part of the initial evaluation. Ambulance technicians often provide the most useful information in an enigmatic case.

General Physical Examination The temperature, pulse, respiratory rate and pattern, and blood pressure should be measured quickly as the evaluation is getting under way. Fever suggests a systemic infection, bacterial meningitis, or encephalitis; only rarely is it attributable to a brain lesion that has disturbed temperature-regulating centers. A slight elevation in temperature may follow vigorous convulsions. High body temperature, 42 to

44°C, associated with dry skin should arouse the suspicion of heat stroke or anticholinergic drug intoxication. Hypothermia is observed with bodily exposure to lowered environmental temperature; alcoholic, barbiturate, sedative, or phenothiazine intoxication; hypoglycemia; peripheral circulatory failure; or hypothyroidism. Hypothermia itself causes coma only when the temperature is <31°C. Tachypnea may indicate acidosis or pneumonia. Aberrant respiratory patterns that may reflect brainstem disorders are discussed below. Marked hypertension, a sign of hypertensive encephalopathy or a rapid rise in intracranial pressure, may occur acutely after head injury. Hypotension is characteristic of coma from alcohol or barbiturate intoxication, internal hemorrhage, myocardial infarction, sepsis, profound hypothyroidism, or Addisonian crisis. The funduscopic examination is invaluable in detecting subarachnoid hemorrhage (subhyaloid hemorrhages), hypertensive encephalopathy (exudates, hemorrhages, vessel-crossing changes, papilledema), and increased intracranial pressure (papilledema). Generalized cutaneous petechiae suggest thrombotic thrombocytopenic purpura, meningococcemia, or a bleeding diathesis from which an intracerebral hemorrhage arises.

Neurologic Assessment The patient should be observed first without examiner intervention. Patients who toss about, reach up toward the face, cross their legs, yawn, swallow, cough, or moan are close to being awake. Lack of restless movements on one side or an outturned leg at rest suggests a hemiplegia. Intermittent twitching movements of a foot, finger, or facial muscle may be the only sign of seizures. Multifocal myoclonus almost always indicates a metabolic disorder, particularly azotemia, anoxia, or drug ingestion (lithium and haloperidol are particularly prone to cause this sign), or the rarer conditions of spongiform encephalopathy and Hashimoto disease. In a drowsy and confused patient bilateral asterixis is a certain sign of metabolic encephalopathy or drug ingestion.

The terms *decorticate rigidity* and *decerebrate rigidity*, or "posturing," describe stereotyped arm and leg movements occurring spontaneously or elicited by sensory stimulation. Flexion of the elbows and wrists and supination of the arm (decortication) suggests severe bilateral damage rostral to the midbrain, whereas extension of the elbows and wrists with pronation (decerebration) indicates damage to motor tracts in the midbrain or caudal diencephalon. The less frequent combination of arm extension with leg flexion or flaccid legs is associated with lesions in the pons. These concepts have been adapted from animal work and cannot be applied with the same precision to coma in humans. In fact, acute and widespread cerebral disorders of any type, regardless of location, frequently cause limb extension, and almost all such extensor posturing becomes predominantly flexor as time passes. Thus, posturing alone cannot be utilized for precise anatomic localization. Posturing may also be unilateral and may coexist with purposeful limb movements, usually reflecting incomplete damage to the motor system.

Level of Arousal and Elicited Movements If the patient is not aroused by a conversational volume of voice, a sequence of increasingly intense stimuli is used to determine the patient's threshold of arousal and the optimal motor response of each limb. It should be recognized that the results of this testing may vary from minute to minute and that serial examinations are most useful. Tickling the nostrils with a cotton wisp is a moderate stimulus to arousal -- all but deeply stuporous and comatose patients will move the head away and rouse to some degree. Using the hand to remove

an offending stimulus such as this one represents an even lesser degree of unresponsiveness.

Responses to noxious stimuli should be appraised critically. Stereotyped posturing indicates severe dysfunction of the corticospinal system. Abduction-avoidance movement of a limb is usually purposeful and denotes an intact corticospinal system extending from the contralateral cortex to the ipsilateral spinal cord. Pressure on the knuckles or bony prominences and pinprick are humane forms of noxious stimulus; pinching the skin causes unsightly ecchymoses and is generally not necessary but may be useful in eliciting abduction withdrawal movements of the limbs. Conversely, consistent (obligatory) adduction and flexion of stimulated limbs may be reflexive in origin and implies damage to the corticospinal system. Brief clonus or twitching may occur at the end of extensor posturing movements and should not be mistaken for convulsions.

Brainstem Reflexes Assessment of brainstem damage is essential to the localization of the lesion in coma ([Fig. 24-3](#)). The brainstem reflexes that are conveniently assessed are pupillary responses to light, spontaneous and elicited eye movements, corneal responses, and the respiratory pattern. As a rule, when these brainstem activities are preserved, particularly the pupil reactions and eye movements, coma must necessarily be ascribed to bilateral hemispherical disease. The converse, however, is not always true as a mass in the hemispheres may be the proximate cause of coma but nonetheless produce brainstem signs.

PUPILS Pupillary reactions are examined with a bright, diffuse light (not an ophthalmoscope); if the response is absent, this should be confirmed by observation through a magnifying lens. Reaction to light is often difficult to appreciate in pupils <2 mm in diameter, and bright room lighting mutes pupillary reactivity. Normally reactive and round pupils of midsize (2.5 to 5 mm) essentially exclude midbrain damage, either primary or secondary to compression. One unreactive and enlarged pupil (>6 mm) or one that is poorly reactive signifies a compression or stretching of the third nerve from the effects of a mass above. Enlargement of the pupil contralateral to a mass may occur but is infrequent. It may be found in cases of subdural hematoma or brain hemorrhage, possibly as a result of compression of the midbrain or third nerve against the opposite tentorial margin. An oval and slightly eccentric pupil is a transitional sign that accompanies early midbrain-third nerve compression. The most extreme pupillary sign, bilaterally dilated and unreactive pupils, indicates severe midbrain damage, usually from compression by a mass or from ingestion of drugs with anticholinergic activity. The use of mydriatic eye drops, by a previous examiner or self-administered by the patient, and direct ocular trauma are among the causes of misleading pupillary enlargement.

Unilateral miosis in coma has been attributed to dysfunction of sympathetic efferents originating in the posterior hypothalamus and descending in the tegmentum of the brainstem to the cervical cord. Reactive and bilaterally small (1 to 2.5 mm) but not pinpoint pupils are seen in metabolic encephalopathies or in deep bilateral hemispherical lesions such as hydrocephalus or thalamic hemorrhage. Very small but reactive pupils (<1 mm) characterize narcotic or barbiturate overdoses but also occur with extensive pontine hemorrhage. The response to naloxone and the presence of reflex eye movements (see below) distinguish these. The unilaterally small pupil of the Horner

syndrome is detected by failure of the pupil to enlarge in the dark. It is an occasional finding with a large cerebral hemorrhage that affects the thalamus.

OCULAR MOVEMENTS Eye movements are the second sign of importance in determining if the brainstem has been damaged. Abnormalities, implicate both midbrain and pontine functions, thus permitting the analysis of a large portion of the brainstem. The eyes are first observed by elevating the lids and noting the resting position and spontaneous movements of the globes. Lid tone, tested by lifting the eyelids and noting their resistance to opening and the speed of closure, is reduced progressively as coma deepens. Horizontal divergence of the eyes at rest is normal in drowsiness. As coma deepens, the ocular axes may become parallel again. An abducted eye indicates a medial rectus paresis due to third nerve dysfunction and has the same significance as pupillary enlargement. An adducted eye indicates lateral rectus paresis due to a sixth nerve lesion and, when bilateral, is often a sign of increased intracranial pressure. With few exceptions, vertical separation of the ocular axes (one eye lower than the other, i.e. skew deviation) results from pontine or cerebellar lesions but may also be a manifestation of a partial third nerve palsy.

Spontaneous eye movements in coma often take the form of conjugate horizontal roving. This finding alone exonerates the midbrain and pons and has the same meaning as normal reflex eye movements (see below). Cyclic vertical downward movements are seen in some circumstances. "Ocular bobbing" describes a brisk downward and slow upward movement of the eyes associated with loss of horizontal eye movements and is diagnostic of bilateral pontine damage, characteristically from thrombosis of the basilar artery. "Ocular dipping" is a slower, arrhythmic downward movement followed by a faster upward movement in patients with normal reflex horizontal gaze; it usually indicates diffuse cortical anoxic damage. The eyes may turn down and inward as a result of thalamic and upper midbrain lesions, typically with thalamic hemorrhage or dilatation of the third ventricle from hydrocephalus. Conjugate horizontal ocular deviation to one extreme at rest indicates damage to the pons on the side of the gaze paresis or a lesion in the frontal lobe on the opposite side. This phenomenon may be summarized by the following maxim: *The eyes look toward a hemispherical lesion and away from a brainstem lesion*. On rare occasions, the eyes may turn paradoxically away from the side of a deep hemispherical lesion ("wrong-way eyes"). Many other complex and interesting eye movements are known but do not have the same salience in coma as the ones already mentioned.

Oculocephalic reflexes are automatic movements of the eyes elicited by moving the head from side to side or vertically. As the activity of the hemispheres is subdued from whatever cause, eye movements are evoked in the direction opposite to the head movement ([Fig. 24-3](#)). These movements, called somewhat inappropriately "doll's eyes" (which more accurately refers to the reflex elevation of the eyelids with flexion of the neck) are suppressed by visual fixation, which requires the patient to be awake. Induced adduction of the globes tends to be less complete than abduction, hence subtle abnormalities in the doll's-eye maneuver should be interpreted with caution. Oculocephalic reflexes are generated by brainstem mechanisms originating in the labyrinths and in cervical proprioceptors and require the undiminished activity of the third nerve nucleus in the midbrain, the contralateral sixth nerve nucleus in the pons, and the medial longitudinal fasciculus (MLF) that runs virtually the length of the

brainstem and links the two. Preservation of reflex eye movements (particularly adduction) therefore informs the examiner that coma is probably not due to an upper brainstem lesion and by implication that the origin of unconsciousness lies in the cerebral diencephalic structures. However, the opposite -- the absence of eye movements -- may signify either damage within the brainstem or profound metabolic depression of all neuronal function including the brainstem nuclei. Metabolic causes of depressed neuronal function include overdoses of phenytoin, tricyclic antidepressants, barbiturates, alcohol, phenothiazines, diazepam, and neuromuscular blocking agents. The presence of normal pupillary size and light reaction will distinguish most drug-induced comas from structural brainstem damage.

Thermal, or "caloric," stimulation of the vestibular apparatus (oculovestibular response) provides a more intense stimulus that may be used to confirm the absence of the oculocephalic reflex but gives fundamentally the same information. The test is performed by irrigating the external auditory canal with cool water in order to induce convection currents in the labyrinths. After a brief latency, the result is tonic deviation of both eyes (lasting 30 to 120 s) to the side of cool-water irrigation. The integrity of the third and sixth nerve complexes and brainstem pathways from the labyrinths to the midbrain are thereby confirmed, thus excluding a brainstem lesion as the cause of coma. If the cerebral hemispheres are functioning, as in catatonic or hysterical pseudocoma, an obligate rapid corrective nystagmus is generated away from the side of tonic deviation. (The acronym "COWS" has been used to remind generations of medical students of the direction of compensatory nystagmus -- "cold water opposite, warm water same"). The absence of this nystagmus despite conjugate deviation of the globes signifies that the cerebral hemispheres are damaged or profoundly suppressed.

By touching the cornea with a wisp of cotton, a response consisting of brief bilateral lid closure is normally observed. Although the corneal reflexes are rarely useful alone, they may corroborate eye-movement abnormalities because they also depend on the integrity of pontine pathways. The response is lost if the reflex connections between the fifth (afferent) and both seventh (efferent) cranial nerves within the pons are damaged. [CNS](#)depressant drugs diminish or eliminate the corneal responses soon after reflex eye movements are paralyzed but before the pupils become unreactive to light. The corneal (and pharyngeal) response may be lost for a time on the side of an acute hemiplegia.

RESPIRATION Respiratory patterns have received much attention in coma diagnosis but are of less localizing value in comparison to other brainstem signs. Shallow, slow, but regular breathing suggests metabolic or drug depression. Cheyne-Stokes respiration in its classic cyclic form, ending with a brief apneic period, signifies bihemispherical damage or metabolic suppression and commonly accompanies light coma. Rapid, deep (Kussmaul) breathing usually implies metabolic acidosis but may also occur with pontomesencephalic lesions and, of course, severe pneumonia. Agonal gasps reflect bilateral lower brainstem damage and are well known as the terminal respiratory pattern of severe brain damage. A number of other cyclic breathing variations are of lesser significance for localization.

LABORATORY STUDIES AND IMAGING

The following studies are most useful in the diagnosis of confusional states and coma: chemical-toxicologic analysis of blood and urine, cranial [CT](#) or [MRI](#), [EEG](#), and [CSF](#) examination. Arterial blood-gas analysis is helpful in patients with lung disease and acid-base disorders. Chemical blood determinations are obtained routinely to disclose metabolic, toxic, or drug-induced encephalopathies. The metabolic aberrations commonly encountered in clinical practice require measurements of electrolytes, glucose, calcium, osmolality, and renal (blood urea nitrogen) and hepatic (NH_3) function. Toxicologic analysis is necessary in any case of coma where the diagnosis is not immediately clear. However, the presence of exogenous drugs or toxins, especially alcohol, does not exclude the possibility that other factors, particularly head trauma, are also contributing to the clinical state. An ethanol level of 43 mmol/L (200 mg/dL) in nonhabituated patients generally causes confusion and impaired mental activity and of >65 mmol/L (300 mg/dL) is associated with stupor. The development of tolerance may allow the chronic alcoholic to remain awake at levels >87 mmol/L (400 mg/dL).

The increased availability of [CT](#) and [MRI](#) has focused attention on causes of coma that are radiologically detectable (e.g., hemorrhages, tumors, or hydrocephalus). Resorting primarily to this approach, although at times expedient, is imprudent because most cases of coma (and confusion) are metabolic or toxic in origin. The notion that a normal CT scan excludes anatomic lesions as the cause of coma is also erroneous. Bilateral hemisphere infarction, small brainstem lesions, encephalitis, meningitis, mechanical shearing of axons as a result of closed head trauma, absent cerebral perfusion associated with brain death, sagittal sinus thrombosis, and subdural hematomas that are isodense to adjacent brain are some of the lesions that may not be visible. Nevertheless, if the source of coma remains unknown, a scan should be obtained.

The EEG is useful in metabolic or drug-induced confusional states but is rarely diagnostic, with the important exceptions of coma due to clinically unrecognized seizures, to herpesvirus encephalitis and Creutzfeldt-Jakob disease. The amount of background slowing of the EEG is a useful reflection of the severity of any diffuse encephalopathy. Predominant high-voltage slowing (d or triphasic waves) in the frontal regions is typical of metabolic coma, as from hepatic failure, and widespread fast (b) activity implicates sedative drugs (diazepines, barbiturates). A pattern of "a coma," defined by widespread, variable 8- to 12-Hz activity, superficially resembles the normal a rhythm of waking but is unresponsive to environmental stimuli. It results from pontine or diffuse cortical damage and has a poor prognosis. Most importantly, EEG recordings reveal coma that is due to persistent epileptic discharges that are not clinically manifested as convulsions. Normal activity on the EEG may also alert the clinician to the locked-in syndrome or to hysteria or catatonia.

Lumbar puncture is used more judiciously than in prior decades in cases of coma or confusion because neuroimaging scans effectively exclude intracerebral hemorrhage and most cases of subarachnoid hemorrhages. However, examination of the [CSF](#) is indispensable in the diagnosis of meningitis and encephalitis and in instances of suspected subarachnoid hemorrhage in which the scan is normal. Lumbar puncture should therefore not be deferred if meningitis is a possibility. Xanthochromia, indicating preexisting blood in the CSF, is documented by spinning the CSF in a large tube and comparing the supernatant to water. Measurement of the opening pressure within the subarachnoid space is of further help in interpreting abnormalities of the cell count and

protein content of the CSF.

DIFFERENTIAL DIAGNOSIS OF COMA ([Table 24-1](#))

In most instances confusion and coma are part of an obvious medical problem such as overt drug ingestion, hypoxia, stroke, trauma, or liver or kidney failure. Attention is then appropriately focused on the primary illness. Some general rules are helpful. Illnesses that cause sudden onset of coma are due to drug ingestion or to cerebral hemorrhage, trauma, cardiac arrest, epilepsy, or basilar artery embolism. Coma that appears subacutely is usually related to a preceding medical or neurologic problem, including the secondary brain swelling that surrounds a preexisting lesion such as a tumor or cerebral infarction.

The structural causes of coma can also be conceptualized in three broad categories: those without focal or lateralizing neurologic signs (e.g., metabolic encephalopathies); meningitis syndromes, characterized by stiff neck and an excess of cells in the spinal fluid (e.g., bacterial meningitis, subarachnoid hemorrhage); and those with prominent focal signs (e.g., stroke, cerebral hemorrhage). These are elaborated in [Table 24-1](#).

Cerebrovascular diseases cause the greatest difficulty in coma diagnosis. These are described in more detail in [Chap. 361](#) but may be summarized as follows: (1) basal ganglia and thalamic hemorrhage (acute but not instantaneous onset, vomiting, headache, hemiplegia, and characteristic eye signs); (2) pontine hemorrhage (sudden onset, pinpoint pupils, loss of reflex eye movements and corneal responses, ocular bobbing, posturing, hyperventilation, and excessive sweating); (3) cerebellar hemorrhage (occipital headache, vomiting, gaze paresis, and inability to stand); (4) basilar artery thrombosis (neurologic prodrome or warning spells, diplopia, dysarthria, vomiting, eye movement and corneal response abnormalities, and asymmetric limb paresis); and (5) subarachnoid hemorrhage (precipitous coma after headache and vomiting). The most common stroke, infarction in the territory of the middle cerebral artery, does not cause coma acutely but the surrounding edema may expand and act as a mass in a limited number of patients with large infarcts. The syndrome of acute hydrocephalus may accompany many intracranial diseases, particularly subarachnoid hemorrhage. Acute symmetric enlargement of both lateral ventricles causes headache and sometimes vomiting that may progress quickly to coma, with extensor posturing of the limbs, bilateral Babinski signs, small nonreactive pupils, and impaired vertical oculocephalic movements in the vertical direction.

If the history and examination do not suggest a large cerebral lesion or meningitic syndrome or a metabolic or drug cause, then information obtained from [CT](#) or [MRI](#) may be needed as outlined in [Table 24-1](#). As mentioned earlier, the majority of medical causes of coma can be established without a neuroimaging study.

BRAIN DEATH

This is a state in which there has been cessation of cerebral blood flow; as a result, global ischemia of the brain occurs while respiration is maintained by artificial means and the heart continues to function. It is the only type of brain damage that is unequivocally recognized as death. Many roughly equivalent criteria have been

advanced for the diagnosis of brain death, and it is essential to adhere to those endorsed as standards by the local medical community. Ideal criteria are simple, can be conducted at the bedside, and allow no chance of diagnostic error. They contain three essential elements: (1) widespread cortical destruction shown by deep coma -- unresponsiveness to all forms of stimulation; (2) global brainstem damage demonstrated by absent pupillary light reaction and the loss of oculovestibular and corneal reflexes; and (3) lower brainstem destruction indicated by complete apnea. The pulse rate is also invariant and unresponsive to atropine. Most patients have diabetes insipidus, but in some it develops only hours or days after the clinical signs of brain death. The pupils are often enlarged and may be mid-sized but should not be constricted. The absence of deep tendon reflexes is not required because the spinal cord may remain functional.

The proof that apnea is due to irreversible medullary damage requires that the P_{CO_2} be high enough to stimulate respiration during a test of spontaneous breathing (apnea test). This can be done safely in most patients by the use, prior to removing the ventilator, of diffusion oxygenation. This is accomplished by preoxygenation with 100% oxygen and then sustained during the test by a tracheal cannula connected to an oxygen supply. CO_2 tension increases approximately 0.3 to 0.4 kPa/min (2 to 3 mmHg/min) during apnea. At the end of the period of observation, typically several minutes in duration, arterial P_{CO_2} should be at least >6.6 to 8.0 kPa (50 to 60 mmHg) for the test to be valid.

The possibility of profound drug-induced or hypothermic depression of the nervous system should be excluded, and some period of observation, usually 6 to 24 h, is desirable during which this state is shown to be sustained. It is particularly advisable to delay clinical testing for up to 24 h if a cardiac arrest has caused brain death or if the inciting disease is not known. An isoelectric [EEG](#) may be used as a confirmatory test for total cerebral damage but is not absolutely necessary. Radionuclide brain scanning, cerebral angiography, or transcranial Doppler measurements may also be used to demonstrate the absence of cerebral blood flow, but with the exception of the latter, they are cumbersome and have not been correlated extensively with pathology.

There is no compelling reason to demonstrate brain death except when organ transplantation is involved. Although it is largely accepted in western society that the respirator can be disconnected from a brain-dead patient, problems frequently arise because of inadequate explanation and preparation of the family by the physician. Moreover, there is no proscription in reasonable medical practice to removing such support from patients who are not brain dead but whose condition is nonetheless hopeless and are likely to live for only a brief time.

TREATMENT

The immediate goal in acute coma is the prevention of further nervous system damage. Hypotension, hypoglycemia, hypercalcemia, hypoxia, hypercapnia, and hyperthermia should be corrected rapidly and assiduously. An oropharyngeal airway is adequate to keep the pharynx open in drowsy patients who are breathing normally. Tracheal intubation is indicated if there is apnea, upper airway obstruction, hypoventilation, or emesis, or if the patient is liable to aspirate because of coma. Mechanical ventilation is

required if there is hypoventilation or if there is an intracranial mass and a need to induce hypocapnia in order to lower intracranial pressure (ICP) as described below. Intravenous access is established and naloxone and dextrose are administered if narcotic overdose or hypoglycemia are even remote possibilities, and thiamine is given with glucose in order to avoid eliciting Wernicke disease in malnourished patients. In cases of suspected basilar thrombosis with brainstem ischemia, intravenous heparin or a thrombolytic agent is often utilized, keeping in mind that cerebellar and pontine hemorrhages resemble basilar artery occlusion. Physostigmine, when used by experienced physicians and with careful monitoring, may awaken patients with anticholinergic-type drug overdose, but many physicians believe that this is justified only to treat cardiac arrhythmias resulting from these overdoses. The use of benzodiazepine antagonists offers some prospect of improvement after overdoses of soporific drugs and has transient benefit in hepatic encephalopathy. Intravenous administration of hypotonic solutions should be monitored carefully in any serious acute brain illness because of the potential for exacerbating brain swelling. Cervical spine injuries must not be overlooked, particularly prior to attempting intubation or the evaluation of oculoccephalic responses. Headache accompanied by fever and meningismus indicates an urgent need for examination of the [CSF](#) to diagnose meningitis, and it is worth reemphasizing that lumbar puncture should not be delayed while awaiting a [aCT](#) scan. If the lumbar puncture in a case of suspected meningitis is delayed for any reason, an antibiotic such as a third-generation cephalosporin should be administered as soon as possible, preferably after obtaining blood cultures.

Enlargement of one pupil usually indicates secondary midbrain or third nerve compression by a hemispherical mass and requires that [ICP](#) be reduced ([Chap. 376](#)). Surgical evacuation of the mass may be appropriate in some cases (e.g., subdural and epidural hematoma.). Medical management to reduce ICP begins with the infusion of normal saline (safe because it is slightly hyperosmolar to serum). Therapeutic hyperventilation may be used to reduce ICP by inducing an arterial P_{CO_2} of 3.7 to 4.2 kPa (28 to 32 mmHg), but its effects are brief. Hyperosmolar therapy with mannitol or an equivalent agent is the mainstay of ICP reduction. It is used simultaneously with hyperventilation in critical cases. A ventricular puncture is necessary to decompress hydrocephalus if medical measures fail to improve alertness. The routine use of high-dose barbiturates and other neuronal-sparing agents soon after cardiac arrest or head trauma has not been shown in clinical studies to be beneficial, and glucocorticoids, although often still used, have no proven value except in cases of brain tumor with edema.

PROGNOSIS

The prediction of the outcome of coma must be considered in reference to long-term care and medical resources. One hopes to avoid the emotionally painful, hopeless outcomes associated with patients who are left severely disabled or vegetative. Several general rules pertain. The uniformly pessimistic outcome of the persistent vegetative state has already been mentioned. Children and young adults may have ominous early clinical findings such as abnormal brainstem reflexes and yet recover, so that temporization in offering a prognosis in this group of patients is wise. Metabolic comas have a far better prognosis than traumatic comas. All schemes for prognosis in adults should be taken as approximate indicators, and medical judgments must be tempered

by factors such as age, underlying systemic disease, and general medical condition. In an attempt to collect prognostic information from large numbers of patients with head injury, the Glasgow Coma Scale was devised; empirically it has predictive value in cases of brain trauma ([Chap. 369](#)). For anoxic and metabolic coma, clinical signs such as the pupillary and motor responses after 1 day, 3 days, and 1 week have been shown to have predictive value ([Chap. 376](#)). The absence of the cortical waves of the somatosensory evoked potentials has also proved a strong indicator of poor outcome in coma from any cause.

(Bibliography omitted in Palm version)

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25. APHASIAS AND OTHER FOCAL CEREBRAL DISORDERS - M.-Marsel Mesulam

The cerebral cortex of the human brain contains approximately 20 billion neurons spread over an area of 2 m². The *primary sensory* areas provide an obligatory portal for the entry of sensory information into cortical circuitry, whereas the *primary motor* areas provide a final common pathway for coordinating complex motor acts. The primary sensory and motor areas constitute <10% of the cerebral cortex. The rest is subsumed by unimodal, heteromodal, paralimbic, and limbic areas, collectively known as the *association cortex* (Fig. 25-1). The association cortex mediates the integrative processes that subserve cognition, emotion, and comportment. A systematic testing of these mental functions is necessary for the effective clinical assessment of the association cortex and its diseases.

According to current thinking, there are no centers for "hearing words," "perceiving space," or "storing memories." Cognitive and behavioral functions (domains) are coordinated by intersecting *large-scale neural networks* that contain interconnected cortical and subcortical components. The network approach to higher cerebral function has at least four implications of clinical relevance: (1) a single domain such as language or memory can be disrupted by damage to any one of several areas, as long as these areas belong to the same network; (2) damage confined to a single area can give rise to multiple deficits, involving the functions of all networks that intersect in that region; (3) damage to a network component may give rise to minimal or transient deficits if other parts of the network undergo compensatory reorganization; and (4) individual anatomic sites within a network display a relative (but not absolute) specialization for different behavioral aspects of the relevant function. Five anatomically defined large-scale networks are most relevant to clinical practice: a perisylvian network for language; a parietofrontal network for spatial cognition; an occipitotemporal network for face and object recognition; a limbic network for retentive memory; and a prefrontal network for attention and comportment.

THE LEFT PERISYLVIAN NETWORK FOR LANGUAGE: APHASIAS AND RELATED CONDITIONS

Language allows the communication and reshaping of thoughts and experiences by linking them to arbitrary symbols known as words. The neural substrate of language is composed of a distributed network centered in the perisylvian region of the *left* hemisphere. The posterior pole of this network is known as *Wernicke's area* and includes the posterior third of the superior temporal gyrus and a surrounding rim of the inferior parietal lobule. An essential function of Wernicke's area is to transform sensory inputs into their neural word representations so that these can establish the distributed associations that give the word its meaning. The anterior pole of the language network, known as *Broca's area*, includes the posterior part of the inferior frontal gyrus and a surrounding rim of prefrontal heteromodal cortex. An essential function of this area is to transform neural word representations into their articulatory sequences so that the words can be uttered in the form of spoken language. The sequencing function of Broca's area also appears to involve the ordering of words into sentences that contain a meaning-appropriate *syntax* (grammar). Wernicke's and Broca's areas are interconnected with each other and with additional perisylvian, temporal, prefrontal, and posterior parietal regions, making up a neural network subserving the various aspects of

language function. Damage to any one of these components or to their interconnections can give rise to language disturbances (*aphasia*). Aphasia should be diagnosed only when there are deficits in the formal aspects of language such as naming, word choice, comprehension, spelling, and syntax. Dysarthria and mutism do not, by themselves, lead to a diagnosis of aphasia. The language network shows a left hemisphere dominance pattern in the vast majority of the population. In approximately 90% of right handers and 60% of left handers, aphasia occurs only after lesions of the left hemisphere. In some individuals no hemispheric dominance for language can be discerned, and in some others (including a small minority of right handers) there is a right hemisphere dominance for language. A language disturbance occurring after a right hemisphere lesion in a right hander is called *crossed aphasia*.

Clinical Examination The clinical examination of language should include the assessment of naming, spontaneous speech, comprehension, repetition, reading, and writing. A deficit of naming (*anomia*) is the single most common finding in aphasic patients. When asked to name common objects (pencil or wristwatch) or their parts (eraser, lead, stem, band), the patient may fail to come up with the appropriate word, may provide a circumlocutious description of the object ("the thing for writing"), or may come up with the wrong word (*paraphasia*). If the patient offers an incorrect but legitimate word ("pen" for "pencil"), the naming error is known as a *semantic paraphasia*; if the word approximates the correct answer but is phonetically inaccurate ("plentil" for "pencil"), it is known as a *phonemic paraphasia*. Asking the patient to name body parts, geometric shapes, and component parts of objects (lapel of coat, cap of pen) can elicit mild forms of anomia in patients who can otherwise name common objects. In most anomias, the patient cannot retrieve the appropriate name when shown an object but can point to the appropriate object when the name is provided by the examiner. This is known as a one-way (or retrieval-based) naming deficit. A two-way naming deficit exists if the patient can neither provide nor recognize the correct name, indicating the presence of a language comprehension impairment. *Spontaneous speech* is described as "fluent" if it maintains appropriate output volume, phrase length, and melody or as "nonfluent" if it is sparse, halting, and average phrase length is below four words. The examiner should also note if the speech is paraphasic or circumlocutious; if it shows a relative paucity of substantive nouns and action verbs versus function words (prepositions, conjunctions); and if word order, tenses, suffixes, prefixes, plurals, and possessives are appropriate. *Comprehension* can be tested by assessing the patient's ability to follow conversation, by asking yes-no questions ("Can a dog fly?", "Does it snow in summer?") or asking the patient to point to appropriate objects ("Where is the source of illumination in this room?"). Statements with embedded clauses or passive voice construction ("If a tiger is eaten by a lion, which animal stays alive?") help to assess the ability to comprehend complex syntactic structure. Commands to close or open the eyes, stand up, sit down, or roll over should not be used to assess overall comprehension since appropriate responses aimed at such axial movements can be preserved in patients who otherwise have profound comprehension deficits.

Repetition is assessed by asking the patient to repeat single words, short sentences, or strings of words such as "No ifs, ands, or buts." The testing of repetition with tongue-twisters such as "hippopotamus" or "Irish constabulary" provides a better assessment of dysarthria and pallilalia than aphasia. Aphasic patients may have little difficulty with tongue-twisters but have a particularly hard time repeating a string of

function words. It is important to make sure that the number of words does not exceed the patient's attention span. Otherwise, the failure of repetition becomes a reflection of the narrowed attention span rather than an indication of an aphasic deficit. *Reading* should be assessed for deficits in reading aloud as well as comprehension. *Writing* is assessed for spelling errors, word order, and grammar. *Alexia* describes an inability to either read aloud or comprehend single words and simple sentences; *agraphia* (or dysgraphia) is used to describe an acquired deficit in the spelling or grammar of written language.

The correspondence between individual deficits of language function and lesion location does not display a rigid one-to-one relationship and should be conceptualized within the context of the distributed network model. Nonetheless, the classification of aphasic patients into specific clinical syndromes helps to determine the most likely anatomic distribution of the underlying neurologic disease and has implications for etiology and prognosis ([Table 25-1](#)). Aphasic syndromes can be divided into "central" syndromes, which result from damage to the two epicenters of the language network (Broca's and Wernicke's areas), and "disconnection" syndromes, which arise from lesions that interrupt the functional connectivity of these centers with each other and with the other components of the language network. The syndromes outlined below are idealizations; pure syndromes occur rarely.

Wernicke's Aphasia Comprehension is impaired for spoken and written language. Language output is fluent but is highly paraphasic and circumlocutious. The tendency for paraphasic errors may be so pronounced that it leads to strings of neologisms, which form the basis of what is known as "jargon aphasia." Speech contains large numbers of function words (e.g., prepositions, conjunctions) but few substantive nouns or verbs that refer to specific actions. The output is therefore voluminous but uninformative. For example, a patient attempts to describe how his wife accidentally threw away something important, perhaps his dentures: "We don't need it anymore, she says. And with it when that was downstairs was my teethtick...a...den...dentith...my dentist. And they happened to be in that bag...see? How could this have happened? How could a thing like this happen...So she says we won't need it anymore...I didn't think we'd use it. And now if I have any problems anybody coming a month from now, four months from now, or six months from now, I have a new dentist. Where my two...two little pieces of dentist that I use...that I...all gone. If she throws the whole thing away...visit some friends of hers and she can't throw them away."

Gestures and pantomime do not improve communication. The patient does not seem to realize that his or her language is incomprehensible and may appear angry and impatient when the examiner fails to decipher the meaning of a severely paraphasic statement. In some patients this type of aphasia can be associated with severe agitation and paranoid behaviors. One area of comprehension that may be preserved is the ability to follow commands aimed at axial musculature. The dissociation between the failure to understand simple questions ("What is your name") in a patient who rapidly closes his or her eyes, sits up, or rolls over when asked to do so is characteristic of Wernicke's aphasia and helps to differentiate it from deafness, psychiatric disease, or malingering. Patients with Wernicke's aphasia cannot express their thoughts in meaning-appropriate words and cannot decode the meaning of words in any modality of input. This aphasia therefore has expressive as well as receptive components.

Repetition, naming, reading, and writing are also impaired.

The lesion site most commonly associated with Wernicke's aphasia is the posterior portion of the language network and tends to involve at least parts of Wernicke's area. An embolus to the inferior division of the middle cerebral artery, and to the posterior temporal or angular branches in particular, is the most common etiology ([Chap. 361](#)). Intracerebral hemorrhage, severe head trauma, or neoplasm are other causes. A coexisting right hemi- or superior quadrantanopia is common, and mild right nasolabial flattening may be found, but otherwise the examination is often unrevealing. The paraphasic, neologistic speech in an agitated patient with an otherwise unremarkable neurologic examination may lead to the suspicion of a primary psychiatric disorder such as schizophrenia or mania, but the other components characteristic of acquired aphasia and the absence of prior psychiatric disease usually settle the issue. Some patients with Wernicke's aphasia due to intracerebral hemorrhage or head trauma may improve as the hemorrhage or the injury heals. In most other patients, prognosis for recovery is guarded.

Broca's Aphasia Speech is nonfluent, labored, interrupted by many word-finding pauses, and usually dysarthric. It is impoverished in function words but enriched in meaning-appropriate nouns and verbs. Abnormal word order and the inappropriate deployment of *bound morphemes* (word endings used to denote tenses, possessives, or plurals) lead to a characteristic agrammatism. Speech is telegraphic and pithy but quite informative. In the following passage, a patient with Broca's aphasia describes his medical history: "I see...the dotor, dotor sent me...Bosson. Go to hospital. Dotor...kept me beside. Two, tee days, doctor send me home."

Output may be reduced to a grunt or single word ("yes" or "no"), which is emitted with different intonations in an attempt to express approval or disapproval. In addition to fluency, naming and repetition are also impaired. Comprehension of spoken language is intact, except for syntactically difficult sentences with passive voice structure or embedded clauses. Reading comprehension is also preserved, with the occasional exception of a specific inability to read small grammatical words such as conjunctions and pronouns. The last two features indicate that Broca's aphasia is not just an "expressive" or "motor" disorder and that it may also involve a comprehension deficit for function words and syntax. Patients with Broca's aphasia can be tearful, easily frustrated, and profoundly depressed. Insight into their condition is preserved, in contrast to Wernicke's aphasia. Even when spontaneous speech is severely dysarthric, the patient may be able to display a relatively normal articulation of words when singing. This dissociation has been used to develop specific therapeutic approaches (melodic intonation therapy) for Broca's aphasia. Additional neurologic deficits usually include right facial weakness, hemiparesis or hemiplegia, and a buccofacial apraxia characterized by an inability to carry out motor commands involving oropharyngeal and facial musculature (e.g., patients are unable to demonstrate how to blow out a match or suck through a straw). Visual fields are intact. The cause is most often infarction of Broca's area (the inferior frontal convolution; [Fig. 25-1](#)) and surrounding anterior perisylvian and insular cortex, due to occlusion of the superior division of the middle cerebral artery ([Chap. 361](#)). Mass lesions including tumor, intracerebral hemorrhage, or abscess may also be responsible. Small lesions confined to the posterior part of Broca's area may lead to a nonaphasic and often reversible deficit of speech articulation, usually

accompanied by mild right facial weakness. When the cause of Broca's aphasia is stroke, recovery of language function generally peaks within 2 to 6 months, after which time further progress is limited.

Global Aphasia Speech output is nonfluent, and comprehension of spoken language is severely impaired. Naming, repetition, reading, and writing are also impaired. This syndrome represents the combined dysfunction of Broca's and Wernicke's areas and usually results from strokes that involve the entire middle cerebral artery distribution in the left hemisphere. Most patients are initially mute or say a few words, such as "hi" or "yes." Related signs include right hemiplegia, hemisensory loss, and homonymous hemianopia. Occasionally, a patient with a lesion in Wernicke's area will present with a global aphasia that soon resolves into Wernicke's aphasia.

Conduction Aphasia Speech output is fluent but paraphasic, comprehension of spoken language is intact, and repetition is severely impaired. Naming and writing are also impaired. Reading aloud is impaired, but reading comprehension is preserved. The lesion sites spare Broca's and Wernicke's areas but may induce a functional disconnection between the two so that neural word representations formed in Wernicke's area and adjacent regions cannot be conveyed to Broca's area for assembly into corresponding articulatory patterns. Occasionally, a Wernicke's area lesion gives rise to a transient Wernicke's aphasia that rapidly resolves into a conduction aphasia. The paraphasic output in conduction aphasia interferes with the ability to express meaning, but this deficit is not nearly as severe as the one displayed by patients with Wernicke's aphasia. Associated neurologic signs in conduction aphasia vary according to the primary lesion site.

Nonfluent Transcortical Aphasia (Transcortical Motor Aphasia) The features are similar to Broca's aphasia, but repetition is intact and agrammatism may be less pronounced. The neurologic examination may be otherwise intact, but a right hemiparesis can also exist. The lesion site disconnects the intact language network from prefrontal areas of the brain and usually involves the anterior watershed zone between anterior and middle cerebral artery territories or the supplementary motor cortex in the territory of the anterior cerebral artery.

Fluent Transcortical Aphasia (Transcortical Sensory Aphasia) Clinical features are similar to those of Wernicke's aphasia, but repetition is intact. The lesion site disconnects the intact core of the language network from other temporoparietal association areas. Associated neurologic findings may include hemianopia. Cerebrovascular lesions (e.g., infarctions in the posterior watershed zone) or neoplasms that involve the temporoparietal cortex posterior to Wernicke's area are the most common causes.

Isolation Aphasia This rare syndrome represents a combination of the two transcortical aphasias. Comprehension is severely impaired, and there is no purposeful speech output. The patient may parrot fragments of heard conversations (*echolalia*), indicating that the neural mechanisms for repetition are at least partially intact. This condition represents the pathologic function of the language network when it is isolated from other regions of the brain. Broca's and Wernicke's areas tend to be spared, but there is damage in surrounding frontal, parietal, and temporal cortex. Lesions are patchy and

can be associated with anoxia, carbon monoxide poisoning, or complete watershed zone infarctions.

Anomic Aphasia This form of aphasia may be considered the "minimal dysfunction" syndrome of the language network. Articulation, comprehension, and repetition are intact, but confrontation naming, word finding, and spelling are impaired. Speech is enriched in function words but impoverished in substantive nouns and verbs denoting specific actions. Language output is fluent but paraphasic, circumlocutious, and uninformative. The lesion sites can be anywhere within the left hemisphere language network, including the middle and inferior temporal gyri. *Anomic aphasia is the single most common language disturbance seen in head trauma, metabolic encephalopathy, and Alzheimer's disease.* The language impairment of Alzheimer's disease almost always leads to fluent aphasia (e.g., anomic, Wernicke's, conduction, or fluent transcortical aphasia). The insidious onset and relentless progression of nonfluent language disturbances (Broca's or nonfluent transcortical aphasia) can be seen in *primary progressive aphasia*, a degenerative syndrome most commonly associated with focal nonspecific neuronal loss or Pick's disease.

Pure Word Deafness This is not a true aphasic syndrome because the language deficit is modality-specific. The most common lesions are either bilateral or left-sided in the superior temporal gyrus. The net effect of the underlying lesion is to interrupt the flow of information from the unimodal auditory association cortex to Wernicke's area. Patients have no difficulty understanding written language and can express themselves well in spoken or written language. They have no difficulty interpreting and reacting to environmental sounds since primary auditory cortex and subcortical auditory relays are intact. Since auditory information cannot be conveyed to the language network, however, it cannot be decoded into neural word representations and the patient reacts to speech as if it were in an alien tongue that cannot be deciphered. Patients cannot repeat spoken language but have no difficulty naming objects. In time, patients with pure word deafness teach themselves lip reading and may appear to have improved. There may be no additional neurologic findings, but agitated paranoid reactions are frequent in the acute stages. Cerebrovascular lesions are the most frequent cause.

Pure Alexia Without Agraphia This is the visual equivalent of pure word deafness. The lesions (usually a combination of damage to the left occipital cortex and to a posterior sector of the corpus callosum -- the splenium) interrupt the flow of visual input into the language network. There is usually a right hemianopia, but the core language network remains unaffected. The patient can understand and produce spoken language, name objects in the left visual hemifield, repeat, and write. However, the patient acts as if illiterate when asked to read even the simplest sentence because the visual information from the written words (presented to the intact left visual hemifield) cannot reach the language network. Objects in the left hemifield may be named accurately because they activate nonvisual associations in the right hemisphere, which, in turn, can access the language network through transcallosal pathways anterior to the splenium. Patients with this syndrome may also lose the ability to name colors, although they can match colors. This is known as a *color anomia*. The most common etiology of pure alexia is a vascular lesion in the territory of the posterior cerebral artery or an infiltrating neoplasm in the left occipital cortex that involves the optic radiations as well as the crossing fibers of the splenium. Since the posterior cerebral artery also supplies medial temporal components

of the limbic system, the patient with pure alexia may also experience an amnesia, but this is usually transient because the limbic lesion is unilateral.

Aphemia There is an acute onset of severely impaired fluency (often mutism), which cannot be accounted by corticobulbar, cerebellar, or extrapyramidal dysfunction. Recovery is the rule and involves an intermediate stage of hoarse whispering. Writing, reading, and comprehension are intact, so this is not a true aphasic syndrome. Partial lesions of Broca's area or subcortical lesions that undercut its connections with other parts of the brain may be present. Occasionally, the lesion site is on the medial aspects of the frontal lobes and may involve the supplementary motor cortex of the left hemisphere.

Apraxia This generic term designates a complex motor deficit that cannot be attributed to pyramidal, extrapyramidal, cerebellar, or sensory dysfunction and that does not arise from the patient's failure to understand the nature of the task. The form that is most frequently encountered in clinical practice is known as *ideomotor apraxia*. Commands to perform a specific motor act ("cough," "blow out a match") or to pantomime the use of a common tool (a comb, hammer, straw, or toothbrush) in the absence of the real object cannot be followed. The patient's ability to comprehend the command is ascertained by demonstrating multiple movements and establishing that the correct one can be recognized. Some patients with this type of apraxia can imitate the appropriate movement (when it is demonstrated by the examiner) and show no impairment when handed the real object, indicating that the sensorimotor mechanisms necessary for the movement are intact. Some forms of ideomotor apraxia represent a disconnection of the language network from pyramidal motor systems: commands to execute complex movements are understood but cannot be conveyed to the appropriate motor areas, even though the relevant motor mechanisms are intact. *Buccofacial apraxia* involves apraxic deficits in movements of the face and mouth. *Limb apraxia* encompasses apraxic deficits in movements of the arms and legs. Ideomotor apraxia is almost always caused by lesions in the left hemisphere and is commonly associated with aphasic syndromes, especially Broca's aphasia and conduction aphasia. Its presence cannot be ascertained in patients with language comprehension deficits. The ability to follow commands aimed at axial musculature ("close the eyes," "stand up") is subserved by different pathways and may be intact in otherwise severely aphasic and apraxic patients. Patients with lesions of the anterior corpus callosum can display a special type of ideomotor apraxia confined to the left side of the body. Since the handling of real objects is not impaired, ideomotor apraxia, by itself, causes no limitation of daily living activities.

Ideational apraxia refers to a deficit in the execution of a goal-directed sequence of movements in patients who have no difficulty executing the individual components of the sequence. For example, when asked to pick up a pen and write, the sequence of uncapping the pen, placing the cap at the opposite end, turning the point towards the writing surface, and writing may be disrupted, and the patient may be seen trying to write with the wrong end of the pen or even with the removed cap. These motor sequencing problems are usually seen in the context of confusional states and dementias rather than focal lesions associated with aphasic conditions. *Limb-kinetic apraxia* involves a clumsiness in the actual use of tools that cannot be attributed to sensory, pyramidal, extrapyramidal, or cerebellar dysfunction. This condition can

emerge in the context of focal premotor cortex lesions or *corticobasal ganglionic degeneration*.

Gerstmann's Syndrome The combination of *acalculia* (impairment of simple arithmetic), *dysgraphia* (impaired writing), *finger anomia* (an inability to name individual fingers such as the index or thumb), and *right-left confusion* (an inability to tell whether a hand, foot, or arm of the patient or examiner is on the right or left side of the body) is known as Gerstmann's syndrome. In making this diagnosis it is important to establish that the finger and left-right naming deficits are not part of a more generalized anomia and that the patient is not otherwise aphasic. When Gerstmann's syndrome is seen in isolation, it is commonly associated with damage to the inferior parietal lobule (especially the angular gyrus) in the left hemisphere.

Aprosodia Variations of melodic stress and intonation influence the meaning and impact of spoken language. For example, the two statements "He *is* clever." and "He is clever?" contain an identical word choice and syntax but convey vastly different messages because of differences in the intonation and stress with which the statements are uttered. This aspect of language is known as *prosody*. Damage to perisylvian areas in the right hemisphere can interfere with speech prosody and can lead to syndromes of aprosodia. Ross has pointed out that damage to right hemisphere regions corresponding to Wernicke's area yields a greater impairment in the decoding of speech prosody, whereas damage to right hemisphere regions corresponding to Broca's area yields a greater impairment in the ability to introduce meaning-appropriate prosody into spoken language. The latter deficit is the most common type of aprosodia identified in clinical practice -- the patient produces grammatically correct language with accurate word choice but the statements are uttered in a monotone that interferes with the ability to convey the intended stress and affect. Patients with this type of aprosodia give the mistaken impression of being depressed or indifferent.

Subcortical Aphasias Damage to subcortical components of the language network (e.g., the striatum and thalamus of the left hemisphere) can also lead to aphasia. The resulting syndromes contain combinations of deficits in the various aspects of language but rarely fit the specific patterns described in [Table 25-1](#). An anomic aphasia accompanied by dysarthria or a fluent aphasia with hemiparesis should raise the suspicion of a subcortical lesion site.

THE PARIETOFRONTAL NETWORK FOR SPATIAL ORIENTATION: NEGLECT AND RELATED CONDITIONS

Hemispatial Neglect Adaptive orientation to significant events within the extrapersonal space is subserved by a large-scale network containing three major cortical components. The *cingulate cortex* provides access to a limbic-motivational mapping of the extrapersonal space, the *posterior parietal cortex* to a sensorimotor representation of salient extrapersonal events, and the *frontal eye fields* to motor strategies for attentional behaviors ([Fig. 25-2](#)). Subcortical components of this network include the striatum and the thalamus. Contralateral hemispatial neglect represents one outcome of damage to any of the cortical or subcortical components of this network. *The traditional view that hemispatial neglect always denotes a parietal lobe lesion is inaccurate.* In keeping with this anatomic organization, the clinical manifestations of

neglect display three behavioral components: sensory events (or their mental representations) within the neglected hemispace have a lesser impact on overall awareness; there is a paucity of exploratory and orienting acts directed toward the neglected hemispace; and the patient behaves as if the neglected hemispace was motivationally devalued.

According to one model of spatial cognition, the right hemisphere directs attention within the *entire* extrapersonal space, whereas the left hemisphere directs attention mostly within the contralateral right hemisphere. Consequently, unilateral left hemisphere lesions do not give rise to much contralesional neglect since the ipsilateral attentional mechanisms of the right hemisphere can compensate for the loss of the *contralaterally* directed attentional functions of the left hemisphere. Unilateral right hemisphere lesions, however, give rise to severe contralesional left hemispatial neglect because the unaffected left hemisphere does not contain ipsilateral attentional mechanisms. This model is consistent with clinical experience, which shows that contralesional neglect is more common, severe, and lasting after damage to the right hemisphere than after damage to the left hemisphere. Severe neglect for the right hemispace is rare, even in left handers with left hemisphere lesions.

Patients with severe neglect may fail to dress, shave, or groom the left side of the body; may fail to eat food placed on the left side of the tray; and may fail to read the left half of sentences. When the examiner draws a large circle [12 to 16 cm (5 to 6 in.) in diameter] and asks the patient to place the numbers 1 to 12 as if the circle represented the face of a clock, there is a tendency to crowd the numbers on the right side and leave the left side empty. When asked to copy a simple line drawing, the patient fails to copy detail on the left; and when asked to write, there is a tendency to leave an unusually wide margin on the left.

Two bedside tests that are useful in assessing neglect are *simultaneous bilateral stimulation* and *visual target cancellation*. In the former, the examiner provides either unilateral or simultaneous bilateral stimulation in the visual, auditory, and tactile modalities. Following right hemisphere injury, patients who have no difficulty detecting unilateral stimuli on either side experience the bilaterally presented stimulus as coming only from the right. This phenomenon is known as *extinction* and is a manifestation of the sensory-representational aspect of hemispatial neglect. In the target detection task, targets (e.g., A's) are interspersed with foils (e.g., other letters of the alphabet) on a 21.5 × 28.0 cm (8.5 × 11 in.) sheet of paper and the patient is asked to circle all the targets. A failure to detect targets on the left is a manifestation of the exploratory deficit in hemispatial neglect. Hemianopia, by itself, does not interfere with performance in this task since the patient is free to turn the head and eyes to the left. The normal tendency in target detection tasks is to start from the left upper quadrant and move systematically in horizontal or vertical sweeps. Some patients show a tendency to start the process from the right and proceed in a haphazard fashion. This represents a subtle manifestation of left neglect, even if the patient eventually manages to detect all the appropriate targets. Some patients with neglect may also deny the existence of hemiparesis and may even deny ownership of the paralyzed limb, a condition known as *anosognosia*.

Cerebrovascular lesions and neoplasms in the right hemisphere are the most common

causes of hemispatial neglect. Depending on the site of the lesion, the patient with neglect may also have hemiparesis, hemihypesthesia, and hemianopia on the left, but these are not invariant findings. The majority of patients display considerable improvement of hemispatial neglect, usually within the first several weeks.

Balint's Syndrome, Simultanagnosia, Dressing Apraxia, and Construction Apraxia

Bilateral involvement of the network for spatial attention, especially its parietal components, leads to a state of severe spatial disorientation known as *Balint's syndrome*. Balint's syndrome involves deficits in the orderly visuomotor scanning of the environment (*oculomotor apraxia*) and in accurate manual reaching toward visual targets (*optic ataxia*). The third and most dramatic component of Balint's syndrome is known as *simultanagnosia* and reflects an inability to integrate visual information in the center of gaze with more peripheral information. The patient gets stuck on the detail that falls in the center of gaze without attempting to scan the visual environment for additional information. The patient with simultanagnosia "misses the forest for the trees." Complex visual scenes cannot be grasped in their entirety, leading to severe limitations in the visual identification of objects and scenes. For example, a patient who is shown a table lamp and asked to name the object may look at its circular base and call it an ash tray. Some patients with simultanagnosia report that objects they look at may suddenly vanish, probably indicating an inability to look back at the original point of gaze after brief saccadic displacements. Movement and distracting stimuli greatly exacerbate the difficulties of visual perception. Simultanagnosia can sometimes occur without the other two components of Balint's syndrome.

A modification of the letter cancellation task described above can be used for the bedside diagnosis of simultanagnosia. In this modification, some of the targets (e.g., A's) are made to be much larger than the others [7.5 to 10 cm vs. 2.5 cm (3 to 4 in. vs. 1 in.) in height], and all targets are embedded among foils. Patients with simultanagnosia display a counterintuitive but characteristic tendency to miss the larger targets ([Fig. 25-3](#)). This occurs because the information needed for the identification of the larger targets cannot be confined to the immediate line of gaze and requires the integration of visual information across a more extensive field of view. The greater difficulty in the detection of the larger targets also indicates that poor acuity is not responsible for the impairment of visual function and that the problem is central rather than peripheral. Balint's syndrome results from bilateral dorsal parietal lesions; common settings include watershed infarction between the middle and posterior cerebral artery territories, hypoglycemia, sagittal sinus thrombosis, or atypical forms of Alzheimer's disease. In patients with Balint's syndrome due to stroke, bilateral visual field defects (usually inferior quadrantanopias) are common.

Another manifestation of bilateral (or right sided) dorsal parietal lobe lesions is *dressing apraxia*. The patient with this condition is unable to align the body axis with the axis of the garment and can be seen struggling as he or she holds a coat from its bottom or extends his or her arm into a fold of the garment rather than into its sleeve. Lesions that involve the posterior parietal cortex also lead to severe difficulties in copying simple line drawings. This is known as a *construction apraxia* and is much more severe if the lesion is in the right hemisphere. In some patients with right hemisphere lesions, the drawing difficulties are confined to the left side of the figure and represent a manifestation of hemispatial neglect; in others, there is a more universal deficit in reproducing contours

and three-dimensional perspective. Dressing apraxia and construction apraxia represent special instances of a more general disturbance in spatial orientation.

THE OCCIPITOTEMPORAL NETWORK FOR FACE AND OBJECT RECOGNITION: PROSOPAGNOSIA AND OBJECT AGNOSIA

Perceptual information about faces and objects is initially encoded in primary (striate) visual cortex and adjacent (upstream) peristriate visual association areas. This information is subsequently relayed first to the downstream visual association areas of occipitotemporal cortex and then to other heteromodal and paralimbic areas of the cerebral cortex. Bilateral lesions in the fusiform and lingual gyri of occipitotemporal cortex disrupt this process and interfere with the ability of otherwise-intact perceptual information to activate the distributed multimodal associations that lead to the recognition of faces and objects. The resultant face and object recognition deficits are known as *prosopagnosia* and *visual object agnosia*.

The patient with prosopagnosia cannot recognize familiar faces, including, sometimes, the reflection of his or her own face in the mirror. This is not a perceptual deficit since prosopagnosic patients can easily tell if two faces are identical or not. Furthermore, a prosopagnosic patient who cannot recognize a familiar face by visual inspection alone can use auditory cues to reach appropriate recognition if allowed to listen to the person's voice. The deficit in prosopagnosia is therefore modality-specific and reflects the existence of a lesion that prevents the activation of otherwise intact multimodal templates by relevant visual input. Damasio has pointed out that the deficit in prosopagnosia is not limited to the recognition of faces but that it can also extend to the recognition of individual members of larger generic object groups. For example, prosopagnosic patients characteristically have no difficulty with the generic identification of a face as a face or of a car as a car, but they cannot recognize the identity of an individual face or the make of an individual car. This reflects a visual recognition deficit for proprietary features that characterize individual members of an object class. When recognition problems become more generalized and extend to the generic identification of common objects, the condition is known as visual object agnosia. In contrast to prosopagnosic patients, those with object agnosia cannot recognize a face as a face or a car as a car. It is important to distinguish visual object agnosia from anomia. The patient with anomia cannot name the object but can describe its use. In contrast, the patient with visual agnosia is unable either to name a visually presented object or to describe its use. The characteristic lesions in prosopagnosia and visual object agnosia consist of bilateral infarctions in the territory of the posterior cerebral arteries. Associated deficits can include visual field defects (especially superior quadrantanopias) or a centrally based color blindness known as achromatopsia. Rarely, the responsible lesion is unilateral. In such cases, prosopagnosia is associated with lesions in the right hemisphere and object agnosia with lesions in the left.

THE LIMBIC NETWORK FOR MEMORY: AMNESIAS

Limbic and paralimbic areas (such as the hippocampus, amygdala, and entorhinal cortex), the anterior and medial nuclei of the thalamus, the medial and basal parts of the striatum, and the hypothalamus collectively constitute a distributed network known as the *limbic system*. The behavioral affiliations of this network include the coordination of

emotion, motivation, autonomic tone, and endocrine function. An additional area of specialization for the limbic network, and the one which is of most relevance to clinical practice, is that of declarative (conscious) memory for recent episodes and experiences. A disturbance in this function is known as *amnesic state*. In the absence of deficits in motivation, attention, language, or visuospatial function, the clinical diagnosis of a persistent global amnesic state is always associated with bilateral damage to the limbic network, usually within the hippocampo-entorhinal complex or the thalamus.

The memory disturbance in the amnesic state is multimodal and includes retrograde and anterograde components. The *retrograde amnesia* involves an inability to recall experiences that occurred before the onset of the amnesic state. Relatively recent events are more vulnerable to retrograde amnesia than more remote events. A patient who comes to the emergency room complaining that he cannot remember his identity but who can remember the events of the previous day is almost certainly not suffering from a neurologic cause of memory disturbance. The second and most important component of the amnesic state is the *anterograde amnesia*, which indicates an inability to store, retain, and recall new knowledge. Patients with amnesic states cannot remember what they ate a few minutes ago or the details of an important event they may have experienced a few hours ago. In the acute stages, there may also be a tendency to fill in memory gaps with inaccurate, fabricated, and often implausible information. This is known as *confabulation*. Patients with the amnesic syndrome forget that they forget and tend to deny the existence of a memory problem when questioned.

The patient with an amnesic state is almost always disoriented, especially to time. Accurate temporal orientation and accurate knowledge of current news rule out a major amnesic state. Memory can be tested with a list of four to five words read aloud by the examiner up to five times or until the patient can immediately repeat the entire list without intervening delay. In the next phase of testing, the patient is allowed to concentrate on the words and to rehearse them internally for 1 min before being asked to recall them. Accurate performance in this phase indicates that the patient is motivated and sufficiently attentive to hold the words on-line for at least 1 min. The final phase of the testing involves a retention period of 5 to 10 min, during which the patient is engaged in other tasks. Adequate recall at the end of this interval requires off-line storage, retention, and retrieval. Amnesic patients fail this phase of the task and may even forget that they were given a list of words to remember. Accurate recognition of the words by multiple choice in a patient who cannot recall them indicates a less severe memory disturbance that affects mostly the retrieval stage of memory.

Many neurologic diseases can give rise to an amnesic state. These include tumors (of the sphenoid wing, posterior corpus callosum, thalamus, or medial temporal lobe), infarctions (in the territories of the anterior or posterior cerebral arteries), head trauma, herpes simplex encephalitis, Wernicke-Korsakoff encephalopathy, paraneoplastic limbic encephalitis, and degenerative dementias such as Alzheimer's or Pick's disease. The one common denominator of all these diseases is that they lead to the bilateral lesions within one or more components in the limbic network, most commonly the hippocampus, entorhinal cortex, the mammillary bodies of the hypothalamus, and the limbic thalamus. Occasionally, unilateral left-sided lesions can give rise to an amnesic state, but the memory disorder tends to be transient. Depending on the nature and distribution of the underlying neurologic disease, the patient may also have visual field deficits, eye

movement limitations, or cerebellar findings. In many patients, such as those with *transient global amnesia* ([Chap. 26](#)), there are no associated neurologic findings; this sometimes leads incorrectly to the diagnosis of a psychiatric disorder.

Although the limbic network is the site of damage for amnestic states, it is almost certainly not the storage site for memories. Memories are stored in widely distributed form throughout the association cortex. The role attributed to the limbic network is to bind these distributed fragments into coherent events and experiences that can sustain conscious recall. Damage to the limbic network does not necessarily destroy memories but interferes with their conscious (declarative) recall in coherent form. The individual fragments of information remain preserved despite the limbic lesions and can sustain what is known as *implicit memory*. For example, patients with amnestic states can acquire new motor or perceptual skills, even though they may have no conscious knowledge of the experiences that led to the acquisition of these skills.

THE PREFRONTAL NETWORK FOR ATTENTION AND COMPORTMENT

Approximately one-third of all the cerebral cortex in the human brain is located in the frontal lobes. The frontal lobes can be subdivided into motor-premotor, dorsolateral prefrontal, medial prefrontal, and orbitofrontal components. The terms *frontal lobe syndrome* and *prefrontal cortex* refer only to the last three of these four components. These are the parts of the cerebral cortex that show the greatest phylogenetic expansion in primates and especially in humans. The dorsolateral prefrontal, medial prefrontal, and orbitofrontal areas, and the subcortical structures with which they are interconnected (i.e., the head of the caudate and the dorsomedial nucleus of the thalamus), collectively make up a large-scale network that coordinates exceedingly complex aspects of human cognition and comportment.

The prefrontal network plays an important role in behaviors that require an integration of thought with emotion and motivation. There is no simple formula for summarizing the diverse functional affiliations of the prefrontal network. Its integrity appears important for the simultaneous awareness of context, options, consequences, relevance, and emotional impact so as to allow the formulation of adaptive inferences, decisions, and actions. Damage to this part of the brain impairs mental flexibility, reasoning, hypothesis formation, abstract thinking, foresight, judgment, the on-line (attentive) holding of information, and the ability to inhibit inappropriate responses. Behaviors impaired by prefrontal cortex lesions, especially those related to the manipulation of mental content, are often referred to as "executive functions."

Even very large bilateral prefrontal lesions may leave all sensory, motor, and basic cognitive functions intact while leading to isolated but dramatic alterations of personality and comportment. The most common clinical manifestations of damage to the prefrontal network take the form of two relatively distinct syndromes. In the *frontal abulic syndrome*, the patient shows a loss of initiative, creativity, and curiosity and displays a pervasive emotional blandness and apathy. In the *frontal disinhibition syndrome*, the patient becomes socially disinhibited and shows severe impairments of judgment, insight, and foresight. The dissociation between intact intellectual function and a total lack of even rudimentary common sense is striking. Despite the preservation of all essential memory functions, the patient cannot learn from experience and continues to

display inappropriate behaviors without appearing to feel emotional pain, guilt, or regret when such behaviors repeatedly lead to disastrous consequences. The impairments may emerge only in real-life situations when behavior is under minimal external control and may not be apparent within the structured environment of the medical office. Testing judgment by asking patients what they would do if they detected a fire in a theater or found a stamped and addressed envelope on the road is not very informative since patients who answer these questions wisely in the office may still act very foolishly in the more complex real-life setting. The physician must therefore be prepared to make a diagnosis of frontal lobe disease on the basis of historic information alone even when the office examination of mental state may be quite intact.

The abulic syndrome tends to be associated with damage to the dorsolateral prefrontal cortex, and the disinhibition syndrome with the medial prefrontal or orbitofrontal cortex. These syndromes tend to arise almost exclusively after bilateral lesions, most frequently in the setting of head trauma, stroke, ruptured aneurysms, hydrocephalus, tumors (including metastases, glioblastoma, and falx or olfactory groove meningiomas), or focal degenerative diseases. Unilateral lesions confined to the prefrontal cortex may remain silent until the pathology spreads to the other side. The emergence of developmentally primitive reflexes such as grasping, rooting, and sucking are seen primarily in patients with large structural lesions that extend into the premotor components of the frontal lobes or in the context of metabolic encephalopathies. The vast majority of patients with prefrontal lesions and frontal lobe behavioral syndromes do not display these reflexes.

Damage to the frontal lobe disrupts a variety of attention-related functions including working memory (the transient on-line holding of information), concentration span, verbal fluency, the scanning and retrieval of stored information, the inhibition of immediate but inappropriate responses, and mental flexibility. The capacity for focusing on a trend of thought and the ability to voluntarily shift the focus of attention from one thought or stimulus to another can become impaired. Digit span (which should be seven forward and five reverse) is decreased; the recitation of the months of the year in reverse order (which should take less than 15 s) is slowed; and the number of words starting with a, f, or s that can be generated in 1 min (normally 12 or more per letter) is diminished even in nonaphasic patients. Characteristically, there is a progressive slowing of performance as the task proceeds; e.g., the patient asked to count backwards by 3s may say "100, 97, 94...91,...88," etc., and may not complete the task. In go-no go tasks (where the instruction is to raise the finger upon hearing one tap but to keep it still upon hearing two taps), the patient shows a characteristic inability to keep still in response to the "no go" stimulus; mental flexibility (tested by the ability to shift from one criterion to another in sorting or matching tasks) is impoverished; distractibility by irrelevant stimuli is increased; and there is a pronounced tendency for impersistence and perseveration.

These attentional deficits disrupt the orderly registration and retrieval of new information and lead to *secondary* memory deficits. Such memory deficits can be differentiated from the *primary* memory impairments of the amnesic state by showing that they improve when the attentional load of the task is decreased. Working memory (also known as immediate memory) is an attentional function based on the temporary on-line holding of information. It is closely associated with the integrity of the prefrontal network and the ascending reticular activating system. Retentive memory, on the other hand, depends

on the stable (off-line) storage of information and is associated with the integrity of the limbic network. The distinction of the underlying neural mechanisms is illustrated by the observation that severely amnesic patients who cannot remember events that occurred a few minutes ago may have intact if not superior working memory capacity as shown in tests of digit span.

Lesions in the caudate nucleus or in the dorsomedial nucleus of the thalamus (subcortical components of the prefrontal network) can also produce a frontal lobe syndrome. This is one reason why the mental state changes associated with degenerative basal ganglia diseases, such as Parkinson's or Huntington's disease, may take the form of a frontal lobe syndrome. Because of its widespread connections with other regions of association cortex, one essential computational role of the prefrontal network is to function as an integrator, or "orchestrator," for other networks. Bilateral multifocal lesions of the cerebral hemispheres, none of which are individually large enough to cause specific cognitive deficits such as aphasia or neglect, can collectively interfere with the connectivity and integrating function of prefrontal cortex. A frontal lobe syndrome is the single most common behavioral profile associated with a variety of bilateral multifocal brain diseases including metabolic encephalopathy, multiple sclerosis, vitamin B₁₂ deficiency, and others. In fact, the vast majority of patients with the clinical diagnosis of a frontal lobe syndrome tend to have lesions that do not involve prefrontal cortex but involve either the subcortical components of the prefrontal network or its connections with other parts of the brain. In order to avoid making a diagnosis of "frontal lobe syndrome" in a patient with no evidence of frontal cortex disease, it is advisable to use the diagnostic term *frontal network syndrome*, with the understanding that the responsible lesions can lie anywhere within this distributed network.

The patient with frontal lobe disease raises potential dilemmas in differential diagnosis: the abulia and blandness may be misinterpreted as depression, and the disinhibition as mania or acting-out. Appropriate intervention may be delayed while a treatable tumor keeps expanding. An informed approach to frontal lobe disease and its comportmental manifestations may help to avoid such errors.

CARING FOR THE PATIENT WITH DEFICITS OF HIGHER CEREBRAL FUNCTION

Some of the deficits described in this chapter are so complex that they may bewilder not only the patient and family but also the physician. It is imperative to carry out a systematic clinical evaluation in order to characterize the nature of the deficits and explain them in lay terms to the patient and family. Such an explanation can allay at least some of the anxieties, address the mistaken impression that the deficit (e.g., social disinhibition or inability to recognize family members) is psychologically motivated, and lead to practical suggestions for daily living activities. The consultation of a skilled neuropsychologist may aid in the formulation of diagnosis and management. Patients with simultanagnosia, for example, may benefit from the counterintuitive instruction to stand back when they cannot find an item so that a greater search area falls within the immediate field of gaze. In some patients, the history may be more important than the bedside examination. For example, patients with frontal lobe disease can be extremely irritable and abusive to spouses and yet display all the appropriate social graces during the visit to the medical office.

Reactive depression is common in patients with higher cerebral dysfunction and should be treated. These patients may be sensitive to the usual doses of antidepressants or anxiolytics and deserve a careful titration of dosage. Brain damage may cause a dissociation between feeling states and their expression, so that a patient who may superficially appear jocular could still be suffering from an underlying depression that deserves to be treated. In many cases, agitation may be controlled with reassurance. In other cases, treatment with benzodiazepines or sedating antidepressants may become necessary. The use of neuroleptics for the control of agitation should be reserved for refractory cases since extrapyramidal side effects are frequent in patients with coexisting brain damage.

Spontaneous improvement of cognitive deficits due to acute neurologic lesions is common. It is most rapid in the first few weeks but may continue for up to 2 years, especially in young individuals with single brain lesions. The mechanisms for this recovery are incompletely understood. Some of the initial deficits appear to arise from remote dysfunction (diaschisis) in parts of the brain that are interconnected with the site of initial injury. Improvement in these patients may reflect, at least in part, a normalization of the remote dysfunction. Other mechanisms may involve functional reorganization in surviving neurons adjacent to the injury or the compensatory use of homologous structures, e.g., the right superior temporal gyrus with recovery from Wernicke's aphasia. In some patients with large lesions involving Broca's and Wernicke's areas, only Wernicke's area may show contralateral compensatory reorganization (or bilateral functionality), giving rise to a situation where a lesion that should have caused a global aphasia becomes associated with a residual Broca aphasia. Prognosis for recovery from aphasia is best when Wernicke's area is spared. Cognitive rehabilitation procedures have been used in the treatment of higher cortical deficits. There are few controlled studies, but some do show a benefit of rehabilitation in the recovery from hemispatial neglect and aphasia. Some types of deficits may be more prone to recovery than others. For example, patients with nonfluent aphasias are more likely to benefit from speech therapy than patients with fluent aphasias and comprehension deficits. In general, lesions that lead to a denial of illness (e.g., anosognosia) are associated with cognitive deficits that are more resistant to rehabilitation. The recovery of higher cortical dysfunction is rarely complete. Periodic neuropsychological assessment is necessary for quantifying the pace of the improvement and for generating specific recommendations for cognitive rehabilitation, modifications in the home environment, and the timetable for returning to school or work.

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26. MEMORY LOSS AND DEMENTIA - *Thomas D. Bird*

DEFINITION

Dementia is a serious and common problem that affects more than 4 million Americans and costs society more than \$50 billion annually. Ten percent of persons over age 70 and 20 to 40% of individuals over age 85 have clinically identifiable memory loss. Dementia is a syndrome with many causes. A simple definition of dementia is a deterioration in cognitive abilities that impairs the previously successful performance of activities of daily living. Memory is the most common and most important cognitive ability that is lost. Other mental faculties may also be affected such as attention, judgment, comprehension, orientation, learning, calculation, problem solving, mood, and behavior. Agitation or withdrawal, hallucinations, delusions, insomnia, and loss of inhibitions are also common. Individuals with mental retardation and psychosis may become demented if a decline in intellectual function occurs. Many common forms of dementia are progressive, but some dementing illnesses are static and unchanging. Dementia is a chronic condition, whereas delirium is an acute confusional state associated with a change in level of consciousness (ranging from lethargy to agitation).

Memory is a complex function of the brain that has fascinated philosophers and scientists for centuries. Memory is currently viewed as a mental process that uses several storage buffers of differing capacity and duration ([Table 26-1](#)). Sensory memory lasts for about 250 ms in the visual mode (iconic memory) and 1 to 2 s in the auditory mode (echoic memory). Immediate (short-term or primary) memory has a duration of about half a minute and a limited capacity of approximately 5 to 10 items. Immediate memory is highly vulnerable to distraction, requiring attention and vigilance to maintain the content. It is often tested at the bedside by asking the patient to recall several digits forward and backward. Recent, or secondary, memory has been called both "short-term" and "long-term." It has a duration of minutes to weeks and exhibits a larger storage capacity than immediate memory. On entering this buffer, information undergoes a process of consolidation of variable duration. Recent memory is commonly tested in the clinical setting by asking a patient to recall three words after 3 to 5 min. Remote, or long-term, memory stores information lasting weeks to a lifetime and contains most of our personal experiences and knowledge. Some information appears to be stored accurately for an indefinite time, whereas other items fade or become distorted. Memory function includes registration (encoding or acquisition), retention (storage or consolidation), stabilization, and retrieval (decoding or recall). Registration and retrieval are conscious processes. Animal experiments have shown that long-term memory requires new protein synthesis, and the stabilization process probably involves physical changes at neuronal synapses.

Several additional classifications of memory are sometimes used by psychologists, particularly in reference to the content or use of the memory stores. Reference memory refers to a filing system that contains recent and remote information gained from previous experience. Working memory refers to an active process that is being updated continually by current experience. Episodic memory contains information about events occurring in a specific place and time. Semantic memory contains unchanging facts, principles, associations, and rules (for example, state capitals and the number of days in a week). Declarative (explicit) memory refers to facts about the world and past personal

events that must be consciously retrieved to be remembered. Procedural (implicit) memory, in contrast, is involved in learning and retaining a skill or procedure such as how to ride a bicycle, get dressed, or drive a car. Abilities stored in procedural memory become automatic and do not require conscious implementation.

Finally, the term *executive function* refers to mental activity involved in planning, initiating, and regulating behavior. It is considered the central organizing function of the brain that results in systematic, goal-directed activity. Executive functions are active in nonroutine situations where reflex or automatic behavior is not adequate. The anatomic and physiologic substrates of executive function are presumed to involve the frontal lobes ([Chap. 25](#)). Deficits in executive function occur frequently in patients with dementia.

FUNCTIONAL ANATOMY AND PATHOGENESIS

Dementia results from disorders of cerebral neuronal circuits and is a result of the total quantity of neuronal loss combined with the specific location of such loss ([Chap. 25](#)). The anatomic basis of memory was initially clarified from study of the alcohol/thiamine deficiency syndrome of Korsakoff and the consequences of temporal lobe surgery performed for the treatment of epilepsy. In Korsakoff's syndrome lesions in the hypothalamus, mammillary bodies, and dorsomedial nuclei of the thalamus showed that these areas were important for learning, recall, and recognition. Unilateral temporal lobe surgery for epilepsy produced mild to moderate amnesia for either verbal or nonverbal material. Bilateral medial temporal lobe excision involving the hippocampal formation, the parahippocampal gyrus, and part of the amygdala produced a severe anterograde learning disorder, i.e., an inability to store new memories, often with retained ability to recall old ones. The components of the medial temporal lobe memory system include the hippocampus and adjacent cortex, including the entorhinal, perirhinal, and parahippocampal regions ([Fig. 26-1](#)). This includes a circular pathway of neurons from the entorhinal cortex to the dentate gyrus, CA3 and CA1 neurons of the hippocampus to the subiculum, and back to the entorhinal cortex; this pathway is heavily damaged in Alzheimer's disease (AD). This system is fast, has limited capacity, and performs a crucial function at the time of learning and establishing declarative memory. Its role continues after learning during a lengthy period of reorganization and consolidation whereby memory stored in neocortex eventually becomes independent of the medial temporal lobe memory system. This process, by which the burden of long-term (permanent) memory storage is gradually assumed by neocortex, assures that the medial temporal lobe system is always available for the acquisition of new information. Recent functional brain imaging studies indicate that learning and memory involve many of the same regions of the cortex that process sensory information and control motor output. The forms of perceptual and motor learning that can occur without conscious recollections are mediated in part by contractions and expansions of representations in the sensory and motor cortex. One study, for example, has shown that the cortical representation of the fingers of the left hand of musical string players is larger than that in control individuals, suggesting that the representation of different parts of the body in the primary somatosensory cortex of humans depends on use and changes to conform to the current needs and experiences of the individual. Discrete cortical regions exist in which object knowledge (such as words related to color, animals, tools, or action) is organized as a distributed system in which the attributes of an object are stored close to

the regions of the cortex that mediate perception of those attributes ([Chap. 25](#)). That is, brain regions active during object identification are partly dependent on the intrinsic properties of the object. Procedural (implicit) memory appears to involve centers outside the hippocampus such as amygdala, cerebellum, and sensory cortex. Different frontal regions are activated for different kinds of memory storage. Functional magnetic resonance imaging (MRI) studies show that the magnitude of focal activation in left prefrontal-temporal regions or right prefrontal-bilateral parahippocampal regions predicts how well verbal or visual stimuli, respectively, will be remembered.

Biochemically, the cholinergic system plays an important role in memory. Anticholinergic agents such as atropine and scopolamine interfere with memory. Choline acetyl transferase (the enzyme catalyzing the formation of acetylcholine) and nicotinic cholinergic receptors are known to be deficient in the cortex of patients with [AD](#). The brains of patients with AD show severe neuronal loss in the nucleus basalis of Meynert, a major source of cholinergic input to the cerebral cortex. These findings form the basis for the use of cholinesterase inhibitors in the treatment of AD, with benefit presumably arising from increased available levels of acetylcholine. Behavior and mood are modulated by noradrenergic, serotonergic, and dopaminergic pathways; and norepinephrine has been shown to be reduced in the brainstem locus coeruleus in patients with AD. Neurotrophins are also postulated to play a role in memory in part by preserving cholinergic neurons.

Long-term potentiation (LTP), which refers to a long-lasting enhancement of synaptic transmission resulting from repetitive stimulation of excitatory synapses, is presumed to be involved in memory acquisition and storage. LTP occurs in the hippocampus and is mediated by *N*-methyl-D-aspartate (NMDA) receptors as well as cyclic AMP-responsive element binding protein (CREB). Gene knockout mouse models have been useful in the definition of secondary messenger systems that play a role in hippocampal LTP. For example, disruption of either calcium/calmodulin-dependent protein kinase or cytoplasmic tyrosine kinase (*fyn*) results in deficient hippocampal LTP and impaired spatial learning. In contrast, mice in which a neuronal glycoprotein *thy-1* has been inactivated show regionally selective impairment of LTP but intact spatial learning, suggesting that LTP in the entorhinal projection to the dentate gyrus of the hippocampus ([Fig. 26-1](#)) may not be necessary for some forms of spatial learning. Disruption of hippocampal levels of CREB impairs long-term memory in rats.

Most diseases causing dementia do not have highly restricted regions of pathology. Disorders such as [AD](#) appear to eventually represent relatively diffuse neuronal deterioration throughout the cerebral cortex, whereas multi-infarct dementia associated with recurrent strokes causes more focal damage in a random patchwork of cortical regions. Diffuse white matter damage may disrupt intracerebral connections and cause dementia syndromes such as those associated with leukodystrophies, multiple sclerosis, and Binswanger's disease. Subcortical structures such as the caudate, putamen, thalamus, and substantia nigra also modulate cognition and behavior in ways that are not yet well understood. Some investigators distinguish between cortical and subcortical types of dementia. A cortical dementia such as AD primarily presents as memory loss and is often associated with aphasia or other disturbance of language. Patients with subcortical dementia such as Huntington's disease (HD) are less likely to have memory and language problems and more likely to have difficulties with attention,

judgment, awareness, and behavior. Both the clinical and anatomic characteristics of the cortical and subcortical dementias show considerable overlap, and the conditions are often not distinct.

Lesions of some relatively specific cortical-subcortical pathways may have significant effects on behavior ([Chap. 25](#)). The dorsolateral prefrontal cortex has connections with the dorsolateral caudate, globus pallidus, and thalamus. Lesions of these pathways result in poor organization and planning, perseveration, and decreased cognitive flexibility with impaired judgment. The lateral orbital frontal cortex connects with the ventromedial caudate, globus pallidus, and thalamus. Lesions of these connections cause irritability, impulsiveness, and distractibility. The anterior cingulate cortex connects with the nucleus accumbens, globus pallidus, and thalamus. Interruption of these connections produces apathy and poverty of speech or even akinetic mutism.

The single strongest risk factor for dementia is increasing age. The prevalence of disabling memory loss increases with each decade over age 50 and is associated most often with the microscopic changes of [AD](#) at autopsy. Slow accumulation of mutations in neuronal mitochondria is also hypothesized to contribute to the increasing prevalence of dementia with age. Yet many centenarians have intact memory function and no evidence of clinically significant dementia. Whether dementia is an inevitable consequence of normal human aging remains controversial.

DIFFERENTIAL DIAGNOSIS

The many causes of dementia are listed in [Table 26-2](#). The frequency of each condition depends on the age group under study, the country of origin, and perhaps racial or ethnic variations. [AD](#) is the most common cause of dementia in western countries, affecting more than half of demented patients. Vascular disease is the second most common cause of dementia in the United States, affecting 10 to 20%; but it is more common than AD in some Asian countries. Dementia associated with chronic alcoholism and Parkinson's disease (PD) represent the next two most common categories. Chronic intoxications including those resulting from prescription drugs are an important, potentially treatable cause of dementia. Other disorders listed in the table are uncommon but important because many are reversible. The classification of dementing illnesses into two broad groups of reversible and irreversible disorders is a useful approach to the differential diagnosis of dementia.

Subtle cumulative memory loss is a natural part of aging. This frustrating experience, often the source of jokes and humor, is referred to as *benign forgetfulness of the elderly*. Benign means that it is not so progressive or serious that it impairs reasonably successful and productive daily functioning, although the distinction between benign and more significant memory loss can be difficult to make. A proportion of persons with benign memory loss progress to frank dementia, usually caused by [AD](#). It remains unclear why some individuals show progression and others do not. It was once assumed that a cumulative loss of hippocampal neurons with normal aging might underlie this forgetfulness, but recent quantitative neuronal counts indicate that this "natural" neuronal loss may not occur.

Alzheimer's disease is a slowly progressive dementing illness associated with diffuse

cortical atrophy and specific neuropathologic hallmarks of amyloid plaques and neurofibrillary tangles. Although quite common in the elderly, it remains a diagnosis of exclusion to be confirmed definitively only at autopsy. The clinical diagnosis of [AD](#) established by experienced neurologists proves to be correct at autopsy approximately 85 to 90% of the time. **This condition is described in greater detail in [Chap. 362](#).*

Two major types of vascular dementia can be identified ([Chap. 362](#)). The first, often called multi-infarct dementia, results from an accumulation of discrete cerebral strokes that produce disabling deficits of memory, behavior, and other cognitive abilities. Such patients usually give a history of sudden, separate stroke episodes with stepwise deterioration. On examination, focal neurologic deficits such as hemiparesis, unilateral Babinski reflex, aphasia, or visual field defect are common. Brain imaging shows multiple areas of stroke, which may have been ischemic or hemorrhagic. A second, more subtle and insidious type of vascular dementia, Binswanger's disease, is a dementing illness associated with diffuse, subcortical white matter damage often occurring in patients with chronic hypertension and/or severe atherosclerosis. The white matter changes are dramatically visualized by [MRI](#) and have also been called leukoencephalopathy. The pathogenesis of Binswanger's disease is unknown. Because [AD](#) and vascular dementia are common, occasional patients may have both conditions.

An inherited form of vascular dementia is CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy). It is caused by a mutation in the notch 3 gene and produces characteristic dense bodies in the media of arterioles in brain and skin. Affected persons have diffuse white matter deficits on brain [MRI](#) associated with migraine and recurrent stroke without hypertension.

Frontotemporal dementia (FTD) may represent 10 to 20% of persons with presenile dementia (onset before age 65). Initial symptoms are behavioral, such as disinhibition, apathy, or agitation with relatively intact memory. Brain imaging studies show focal lobar atrophy of the frontal and/or temporal lobes. Pick's disease is a form of FTD. Some cases are familial and associated with neuronal neurofibrillary tangle formation and mutations in the tau gene.

Dementia commonly accompanies chronic alcoholism ([Chap. 387](#)). This situation may be a result of associated malnutrition, especially of B vitamins and particularly thiamine. However, other as yet poorly defined aspects of chronic alcohol ingestion may also produce cerebral damage and atrophy. A rare idiopathic syndrome of dementia and seizures with degeneration of the corpus callosum has been reported primarily in male Italian drinkers of red wine (Marchiafava-Bignami disease).

Thiamine (vitamin B₁) deficiency causes Wernicke's encephalopathy. The clinical presentation is a malnourished individual (frequently but not necessarily alcoholic) with confusion, ataxia, and diplopia from ophthalmoplegia (Charcot's triad). Thiamine deficiency damages the thalamus, mammillary bodies, midline cerebellum, periaqueductal gray matter of the midbrain, and peripheral nerves. Damage to medial thalamic regions correlates most closely with memory loss. Prompt administration of parenteral thiamine (100 mg intravenously for 3 days followed by daily oral dosage) may reverse the disease if given in the first few days of symptom onset. However, prolonged

untreated thiamine deficiency can result in an irreversible dementia/amnestic syndrome (Korsakoff's psychosis) or even death.

In Korsakoff's syndrome, the patient is unable to recall new information despite normal immediate memory, attention span, and level of consciousness. Memory for new events is seriously impaired, whereas memory of knowledge prior to the illness is relatively intact. Patients are easily confused, disoriented, and incapable of recalling new information for more than a brief interval. Superficially, they may be conversant, entertaining, able to perform simple tasks, and follow immediate commands. Confabulation is common, although not always present, and may result in obviously erroneous statements and elaborations. There is no specific treatment because the previous thiamine deficiency has produced irreversible damage to the medial thalamic nuclei and mammillary bodies. Mammillary body atrophy may be visible on high-resolution [MRI](#).

Vitamin B₁₂ deficiency, as can occur in pernicious anemia, causes a macrocytic anemia and may also damage the nervous system ([Chaps. 107](#) and [368](#)). Neurologically it most commonly produces a spinal cord syndrome (myelopathy) affecting the posterior columns (loss of position and vibratory sense) and the lateral corticospinal tracts (hyperactive tendon reflexes and Babinski responses); it also damages peripheral nerves, resulting in sensory loss with depressed tendon reflexes. Damage to cerebral myelinated fibers may also cause dementia. The mechanism of neurologic damage is unclear but may be related to a deficiency of S-adenosylmethionine (required for methylation of myelin phospholipids) due to reduced methionine synthase activity or accumulation of methylmalonate and propionate, providing abnormal substrates for fatty acid synthesis in myelin. The neurologic signs of vitamin B₁₂ deficiency are usually associated with macrocytic anemia, but on occasion may occur in its absence. Treatment with parenteral vitamin B₁₂ (1000 ug intramuscularly daily for a week, weekly for a month, and monthly for life for pernicious anemia) stops progression of the disease if instituted promptly, but reversal of advanced nervous system damage will not occur.

Deficiency of nicotinic acid (pellagra) is associated with sun-exposed skin rash, glossitis, and angular stomatitis ([Chap. 75](#)). Severe dietary deficiency of nicotinic acid along with other B vitamins such as pyridoxine may result in spastic paraparesis, peripheral neuropathy, fatigue, irritability, and dementia. This syndrome has been seen in prisoner-of-war and concentration camps. Low serum folate levels appear to be a rough index of malnutrition, but isolated folate deficiency has not been proven to be a specific cause of dementia.

Approximately 20% of patients with [PD](#) ([Chap. 363](#)) eventually develop dementia. Treatment with L-dopa neither accelerates nor prevents this process. Some PD patients with dementia have cytoplasmic neuronal inclusions (Lewy bodies) or [AD](#) changes in the cerebral cortex, but others have no specific identifiable cortical pathology. Progressive supranuclear palsy is a dementing illness associated with parkinsonian features of rigidity, bradykinesia, and postural instability. Resting tremor is often absent, there is a vertical gaze palsy, and patients are resistant to treatment with L-dopa.

Infections of the central nervous system (CNS) usually cause delirium and other acute neurologic syndromes ([Chap. 24](#)). However, some chronic CNS infections such as

tuberculosis or cryptococcosis may produce a dementing illness ([Chap. 374](#)). Between 20 and 30% of patients in the advanced stages of infection with HIV become demented ([Chap. 309](#)). Cardinal features include psychomotor retardation, apathy, and impaired memory. This condition may result from secondary opportunistic infections but can also be caused by direct involvement of CNS neurons with HIV, where there is a multinucleated giant cell encephalitis and diffuse pallor of white matter. The neuronal toxicity may be mediated by cytokines or the direct neurotoxic effect of the gp 120 envelope glycoprotein. In the absence of CNS opportunistic infection, elevated $\delta 2$ -microglobulin in cerebrospinal fluid (CSF) is a useful marker for HIV dementia. Herpes simplex encephalitis ([Chap. 373](#)) has a predilection for the inferior temporal lobes and may present as subacute confusion and disorientation but more often as an acute syndrome rather than a chronic dementia. Computed tomography (CT), [MRI](#), and electroencephalogram (EEG) may all demonstrate the temporal lobe location of the lesions. CSF usually shows increased protein and a lymphocytic pleocytosis. CNS syphilis ([Chap. 172](#)) was a common cause of dementia in the preantibiotic era; it is uncommon now but can still be encountered in individuals with multiple sex partners. Characteristic CSF changes consist of pleocytosis, increased protein, and a positive Venereal Disease Research Laboratory (VDRL) test.

Prion disorders such as Creutzfeldt-Jakob disease (CJD) ([Chap. 375](#)) are rare conditions (approximately 1 per million population) that commonly produce dementia. CJD is typically a rapidly progressive disease associated with dementia, rigidity, and myoclonus, causing death in less than 1 to 2 years. These clinical characteristics may also rarely be seen in [AD](#), and the differential diagnosis usually depends on the slower progression of AD and the markedly abnormal periodic [EEG](#) discharges seen in CJD. Ataxia or cortical blindness may also accompany CJD. The transmissible agent, or prion, consists principally of an abnormal isoform of a host-encoded protein, the prion protein, which has undergone a physical conformational change and accumulates in affected brains. Bovine spongiform encephalopathy in the United Kingdom is thought to have resulted from cattle feed containing sheep tissues contaminated with infectious prions. Immunoassay for a 14-3-3 brain protein in [CSF](#) may be a useful marker for transmissible spongiform encephalopathies in patients with dementia.

Primary and metastatic neoplasms of the [CNS](#) ([Chap. 370](#)) usually produce focal neurologic findings and seizures rather than dementia. However, if tumor growth begins in the frontal or temporal lobes, the initial manifestations may be memory loss or behavioral changes. A rare paraneoplastic syndrome of dementia associated with occult carcinoma (usually small cell lung cancer) has been termed *limbic encephalitis* ([Chap. 101](#)). In this syndrome, confusion, agitation, seizures, poor memory, and frank dementia may occur in association with sensory neuropathy. The [CSF](#) often shows an increase in cells and protein. There is neuronal loss and perivascular lymphocytic infiltration in the hippocampus, amygdala, and cingulate and frontal cortex. Circulating antineuronal nuclear antibodies may be present. There is no specific treatment.

The syndrome of normal-pressure hydrocephalus ([Chap. 362](#)) is frequently discussed but difficult to diagnose. Clinically, a triad of memory loss, gait disturbance, and bladder incontinence is typical. The gait abnormality is often the initial symptom, and the dementia is usually mild. On imaging studies, the lateral ventricles are enlarged but there is minimal or no cortical atrophy. Lumbar puncture shows a normal or slightly

elevated opening pressure with normal [CSF](#). The condition may be idiopathic or the result of previous meningitis or subarachnoid blood from a ruptured aneurysm or head trauma. The pathogenetic mechanism is presumably a block of normal CSF flow over the convexity and delayed absorption into the venous system, with resulting stretch and distortion of white matter tracts within the corona radiata. Some individuals improve with ventricular shunting but many do not. The condition is difficult to distinguish from [AD](#) ([Chap. 362](#)).

A nonconvulsive seizure disorder may underlie a syndrome of confusion, clouding of consciousness, and garbled speech. Psychiatric disease is often suspected, but an [EEG](#) demonstrates the seizure discharges. If recurrent or persistent, the condition may be termed *complex partial status epilepticus*. The cognitive disturbance often responds to anticonvulsant therapy. The etiology may be previous small strokes or head trauma; some cases are idiopathic.

It is important to recognize systemic diseases that indirectly affect the brain and produce chronic confusion or dementia. Such conditions include dysthyroid states (especially hypothyroidism), vasculitis, and hepatic, renal, or pulmonary disease. Hepatic encephalopathy may begin with irritability and confusion and slowly progress to agitation, lethargy, and coma ([Chap. 376](#)).

Isolated angiitis of the [CNS](#) (CNS granulomatous angiitis) ([Chaps. 317](#) and [361](#)) occasionally causes a chronic encephalopathy associated with confusion, disorientation, and clouding of consciousness. Headache is common, and strokes and cranial neuropathies may occur. Brain imaging studies may be normal or nonspecifically abnormal. Studies of [CSF](#) reveal a mild pleocytosis or elevation in the protein level in half of the cases. Cerebral angiography often shows multifocal stenosis and narrowing of vessels. A few patients have only small-vessel disease that is not revealed on angiography. The angiographic appearance is not specific and may be mimicked by atherosclerosis, infection, or other causes of vascular disease. Brain or meningeal biopsy demonstrates abnormal arteries with endothelial cell proliferation and infiltrates of mononuclear cells. Autoantibodies and immune complexes are not present, and a cell-mediated process appears most likely. The prognosis is poor, but some patients respond to glucocorticoids or chemotherapy.

Chronic metal intoxications may also produce a dementing syndrome. The key to diagnosis is the elicitation of a history of exposure at work, home, or even as a consequence of a medical procedure such as dialysis. Lead poisoning has highly variable neurologic manifestations. Fatigue, depression, and confusion may be associated with episodic abdominal pain and peripheral neuropathy. Gray lead lines may appear in the gums. There is usually an associated anemia with basophilic stippling of red cells. The clinical presentation can resemble that of acute intermittent porphyria, including elevated levels of urine porphyrins as a result of the inhibition of δ -aminolevulinic acid dehydratase. Chronic lead poisoning from inadequately fired glazed pottery has been reported. The treatment is chelation therapy with agents such as ethylene diaminetetraacetic acid (EDTA). Chronic mercury poisoning may produce dementia, peripheral neuropathy, ataxia, and a fine tremulousness that may progress to a cerebellar intention tremor or choreoathetosis. The confusion and memory loss of chronic arsenic intoxication is also associated with nausea, weight loss, peripheral

neuropathy, pigmentation and scaling of the skin, and transverse white lines of the fingernails (Mee's lines). Treatment is chelation therapy with dimercaprol (BAL). Aluminum poisoning has been best documented with the dialysis dementia syndrome in which water used during renal dialysis was contaminated with excessive amounts of aluminum. This resulted in a progressive encephalopathy associated with confusion, memory loss, agitation, and, later, lethargy and stupor. Speech arrest and myoclonic jerking was common and associated with severe and generalized [EEG](#) changes. The condition was often fatal. There were no specific pathologic findings, but elevated brain aluminum content was documented. The condition has been eliminated by use of deionized water for dialysis. Although aluminum injected into experimental animals may produce neurofibrillary tangles, patients with dialysis dementia had neither tangles nor amyloid plaques, and there has been no direct association of aluminum poisoning with [AD](#).

Recurrent head trauma in professional boxers may lead to dementia, sometimes called the "punch drunk" syndrome or *dementia pugilistica*. The symptoms can be progressive and may begin late in a boxer's career or even long after retirement. The severity of the syndrome correlates with the length of the boxing career and the total number of bouts. Early in the condition there occurs a personality change associated with social instability and sometimes paranoia and delusions. Later, memory loss progresses to full dementia, often associated with parkinsonian signs and ataxia or intention tremor. At autopsy, the cerebral cortex may show changes similar to [AD](#), although neurofibrillary tangles are usually more predominant than amyloid plaques (which are usually diffuse rather than neuritic). There may also be loss of neurons in the substantia nigra. Chronic subdural hematoma is also occasionally associated with dementia, often in the context of underlying cortical atrophy from conditions such as [AD](#) or [HD](#). In these latter cases, evacuation of the subdural hematoma does not alter the underlying degenerative process.

Head injury ([Chap. 369](#)) may also be associated with temporary amnesia. The memory disturbance may include events that occurred both before the injury (retrograde amnesia) and during the postinjury period (posttraumatic or anterograde amnesia). Retrograde amnesia after severe head injury may extend back for hours or weeks before the injury; remote memory is usually intact. As patients recover, the extent of retrograde amnesia shrinks and may disappear. Often, retrograde amnesia causes permanent inability to recall the few minutes before the head injury, implying disruption of the immediate memory system and failure to register long-term memory. The length of posttraumatic amnesia generally corresponds to the length of the postconcussive confusional state, but posttraumatic amnesia may persist even in the presence of normal immediate memory and digit span. The duration of posttraumatic amnesia indicates the severity of head injury; the ability to learn new material is often the last cognitive deficit to recover. There are reports of recovery from retrograde amnesia occurring months or years after the initial brain insult; the recovery is sometimes stimulated by hypnosis, amobarbital interview, or electrical stimulation. One theory of such recovery envisions a resetting of distorted patterns of neuronal matrices subserving memory.

Transient global amnesia (TGA) is characterized by sudden onset of complete anterograde loss of memory and learning abilities, usually occurring in persons over age

50. Onset of memory loss may occur in the context of an emotional stimulus or physical exertion. During the attack the individual is alert and communicative, general cognition seems intact, and there are no other neurologic signs or symptoms. The patient may seem confused and repeatedly ask about present events. The ability to form new memories returns after a period of hours, and the individual returns to normal but has no recall for the period of the attack. Frequently no cause can be determined, but cerebrovascular disease, epilepsy (7% in one study), migraine, or cardiac arrhythmia sometimes may be implicated. A Mayo Clinic review of 277 patients with TGA found a past history of migraine in 14% and cerebrovascular disease in 11%, but these conditions were not temporally related to the TGA episodes. About one-fourth of the patients had recurrent attacks, but they were not at increased risk for subsequent stroke. Rare instances of permanent memory loss after sudden onset have been reported.

Psychogenic amnesia for personally important memories is common, although whether this amnesia results from deliberate avoidance of unpleasant memories or from unconscious repression may be impossible to establish. The event-specific amnesia is particularly common after violent crimes such as homicide of a close relative or friend or sexual abuse. It also may occur with severe drug or alcohol intoxication and sometimes with schizophrenia. More prolonged psychogenic amnesia occurs in fugue states that also commonly follow severe emotional stress. The patient with a fugue state suffers from a sudden loss of personal identity and may be found wandering far from home. In contrast to organic amnesia, fugue states are associated with amnesia for personal identity and events closely associated with the personal past. At the same time, memory for other recent events and the ability to learn and use new information are preserved. The episodes usually last hours or days and occasionally weeks or months while the patient takes on a new identity. On recovery, there is a residual amnesic gap for the period of the fugue.

Psychiatric diseases may mimic dementia. Severely depressed individuals may appear demented, a phenomenon called *pseudodementia*. Unlike cortical dementias, memory and language are usually intact when carefully tested in depressed persons. The patients may feel confused and are unable to accomplish routine tasks. Vegetative symptoms are common, such as insomnia, lack of energy, poor appetite, and concern with bowel function. The psychosocial milieu may suggest prominent reasons for depression. The patients respond to antidepressant treatment. Schizophrenia is usually not difficult to distinguish from dementia, but occasionally the distinction can be problematic. (Kraepelin's original term for schizophrenia was *dementia praecox*.) Schizophrenia usually has a much earlier age of onset (second and third decades) than most dementing illnesses. It is associated with intact memory, and the delusions and hallucinations of schizophrenia are usually more complex and bizarre than those of dementia. Some individuals with chronic schizophrenia develop an unexplained progressive dementia late in life that is not related to [AD](#). Memory loss may also be part of a conversion reaction. In this situation, patients commonly complain bitterly of memory loss, but careful cognitive testing either does not confirm the deficits or demonstrates inconsistent or unusual patterns of cognitive problems. The patients' behavior and "wrong" answers to questions often indicate that they both understand the question and know the answer.

Clouding of cognition by chronic drug or medication use, often prescribed by physicians, is an important cause of dementia. Sedatives, tranquilizers, and analgesics used to treat insomnia, pain, anxiety, or agitation may cause confusion, memory loss, and lethargy, especially in the elderly. Discontinuation of such medication often improves mentation.

Approach to the Patient

The approach to the patient with dementia should always keep two major questions in the forefront: What is the most accurate diagnosis, and is there a treatable or reversible condition? A broad overview of this approach is shown in [Table 26-3](#).

History The history should concentrate on the onset, duration, and tempo of the memory loss. Acute or subacute confusion may represent delirium and suggests intoxication, infection, or metabolic derangement. An elderly person with slowly progressive memory loss over several years is likely to have [AD](#). Initial symptoms often are difficulty with managing money, driving, shopping, following instructions, or finding one's way around town. A change in personality with disinhibition and intact memory may suggest [FTD](#). A history of sudden stroke with an irregular stepwise progression suggests multi-infarct dementia. Stroke is also commonly associated with a history of hypertension, atrial fibrillation, peripheral vascular disease, and diabetes. Rapid progression with rigidity and myoclonus suggests [CJD](#). Seizures may indicate stroke or neoplasm. Trouble in walking may suggest [PD](#) or normal-pressure hydrocephalus, especially the latter when associated with bladder incontinence. A history of multiple sex partners or intravenous drug use may indicate [CNS](#) infection, especially with HIV. A history of recurrent head trauma could indicate chronic subdural hematoma, dementia pugilistica, or normal-pressure hydrocephalus. Alcoholism may suggest malnutrition and thiamine deficiency. A remote history of gastric surgery resulting in loss of intrinsic factor might indicate vitamin B₁₂ deficiency. Certain occupations such as working in a battery or chemical factory might indicate heavy metal intoxication. Careful review of medication intake, especially of sedatives and tranquilizers, may raise the issue of chronic drug intoxication. A positive family history of dementia would be elicited in [HD](#), familial AD, and inherited FTD. The recent death of a loved one, insomnia, or poor appetite suggest depression.

Physical Examination A careful examination is essential to document the dementia, look for other signs of nervous system involvement, and search for clues suggesting other systemic disease. Cognitive function should be assessed in terms of orientation, recent and remote memory, and calculation. Many of the simple, commonly used bedside tests of cognitive function (such as serial 7s, digits forward and backward) are most useful when they are performed normally; this makes the diagnosis of dementia unlikely. Mistakes on these simple tests are more difficult to interpret and are of less diagnostic importance. Drawing a clock and the trail-making test are frequently used tests of immediate memory and visual-spatial abilities. The mini-mental status exam (MMSE) is an easily administered 30-points test of cognitive function ([Table 26-4](#)). It is used to quickly indicate a dementing process, provide a rough assessment of its severity, and follow progression of the illness. The MMSE is influenced by culture and education and is less useful in the early and late stages of dementia. Language function should be tested by the ability to read, write, comprehend, and name objects. Resting tremor, cogwheel rigidity, bradykinesia, and festinating gait indicate a parkinsonian

syndrome. Gait ataxia or apraxia (inability to initiate and coordinate steps in a sequential fashion) suggests normal-pressure hydrocephalus. Confusion, sixth cranial nerve paresis, and ataxia suggests thiamine deficiency. Myoclonic jerks are present in [CJD](#) but also occur in [AD](#). Hemiparesis or other focal neurologic deficits may occur in multi-infarct dementia or brain tumor. Bilateral hyperactive tendon reflexes, Babinski responses, and loss of vibration and position sensation suggest a myelopathy, such as occurs in vitamin B₁₂ deficiency. Stocking-glove sensory loss and diminished tendon reflexes suggest a peripheral neuropathy, which could indicate underlying diabetes, vitamin deficiency, or heavy metal intoxication. Dry cool skin, hair loss, and bradycardia suggest hypothyroidism. Confusion associated with repetitive stereotyped movements may indicate ongoing seizure activity. Hearing impairment or visual loss may produce confusion and disorientation misinterpreted as dementia. Such sensory deficits are common in the elderly.

Laboratory Tests The use of multiple laboratory tests in the evaluation of dementia is controversial. The physician does not want to miss a treatable cause, yet no single treatable cause stands out as common; thus a screen must employ multiple different tests, each of which has a low yield. Therefore, cost/benefit ratios are difficult to assess, and many laboratory screening algorithms for dementia discourage multiple tests. Nevertheless, even a test with only a 1 to 2% positive rate is probably worth undertaking if the alternative is missing a reversible or treatable cause of dementia. [Table 26-3](#) lists most screening tests for dementia. Neuroimaging studies ([CT](#) and [MRI](#)) are especially controversial because of their cost. However, they are clearly of value to identify primary and secondary neoplasms, locate areas of infarction, or suggest normal-pressure hydrocephalus or diffuse white matter disease. They also lend support to the diagnosis of [AD](#), especially if there is hippocampal atrophy in addition to diffuse cortical atrophy, and focal lobar atrophy may suggest [FTD](#). However, attempts to relate cognition to neuroimaging measures of atrophy and white matter changes have shown only modest correlations. A diagnosis of AD is reached primarily by exclusion of other causes of dementia. (The indications for apolipoprotein E testing for AD are discussed in [Chap. 362](#).) Serum levels of vitamin B₁₂ and TSH, complete blood count, electrolyte measurements, and a [VDRL](#) test are reasonable routine screening measures because they detect treatable conditions. Lumbar puncture need not be done routinely in the evaluation of dementia but is indicated if [CNS](#) infection is a serious consideration, for example, in patients with delirium, fever, or nuchal rigidity. [CSF](#) levels of tau protein are increased and those of Abamyloid are decreased in some patients with AD; however, the clinical usefulness of these changes is not yet clear. Formal psychometric testing is not necessary in every patient with dementia but can be used to document the severity of dementia, suggest psychogenic causes, and provide a semiquantitative method for following the disease course. [EEG](#) is rarely helpful except to suggest [CJD](#) (repetitive bursts of diffuse high voltage sharp waves) or an underlying nonconvulsive seizure disorder (epileptiform discharges). Brain biopsy (including meninges) is not commonly advised except to diagnose vasculitis, potentially treatable neoplasms, unusual infections (such as sarcoid), or in young persons where the diagnosis is in doubt. Angiography is not likely to be of use except when multiple strokes or cerebral vasculitis is a possible cause of the dementia.

TREATMENT

The two major goals of management are, first, to treat any correctable cause of the dementia and, second, to provide comfort and support to the patient and caregivers. Treatment of underlying causes might include thyroid replacement for hypothyroidism; vitamin therapy for thiamine and B₁₂ deficiency; antibiotics for opportunistic infections; ventricular shunting for normal-pressure hydrocephalus; and appropriate surgical, radiation, and/or chemotherapy for CNS neoplasms. Removal of sedating or cognition-impairing drugs and medications is often beneficial. If the patient is depressed rather than demented (pseudodementia), the depression should be vigorously treated. Patients with degenerative diseases such as AD and HD may also be depressed, and that portion of their condition may respond to antidepressant therapy. Antidepressants should be used with caution in demented patients because they may produce delirium. Antidepressants that have a low incidence of cognitive side effects, such as selective serotonin reuptake inhibitors, and tricyclic antidepressants with low anticholinergic activity such as desipramine and nortriptyline, are advisable. Anticonvulsants are used to control seizures. Agitation, hallucinations, delusions, and confusion are difficult to treat. These behavioral problems represent major causes for nursing home placement and institutionalization. Drugs such as phenothiazines, risperidone, haloperidol, and benzodiazepines may ameliorate the behavior problems but have untoward side effects such as sedation, rigidity, and dyskinesias. Medications that may calm agitation and insomnia without worsening dementia include low-dose haloperidol (0.5 to 2 mg), trazodone, buspirone, and propranolol. Olanzapine is increasingly used for patients with hallucinations. When patients do not respond, it is usually a mistake to advance to higher doses or to use anticholinergics or sedatives (such as barbiturates or benzodiazepines).

Cholinesterase inhibitors are being used to treat AD, and other drugs, such as estrogen, anti-inflammatory agents, and vitamin E are being investigated for the treatment or prevention of AD. These approaches are reviewed in [Chap. 362](#).

A proactive approach has been shown to reduce the occurrence of delirium in hospitalized patients. This scheme includes frequent orientation, cognitive activities, sleep enhancement measures, vision and hearing aids, and correction of dehydration.

Nondrug behavior therapy has an important place in the management of dementia. The primary goal is to make the life of the patient with dementia comfortable, uncomplicated, and safe. Preparing lists, schedules, calendars, and labels can be helpful. It is also useful to stress familiar routines, short-term tasks, brief walks, and simple physical exercises. For many patients with dementia, the memory for facts is worse than that for routine activities, and they still may be able to take part in remembered physical activities such as walking, bowling, dancing, and golf. Patients with dementia usually object to losing control over familiar tasks such as driving, cooking, and handling finances. Attempts to help or take over may be greeted with complaints, depression, or anger. Hostile responses on the part of the caretaker are useless and sometimes harmful. Explanation, reassurance, distraction, and calm statements are more productive responses in this setting. Eventually, tasks such as finances and driving must be assumed by others, and the patient will conform and adjust. Safety is an important issue that includes not only driving but the environment of the kitchen, bathroom, and sleeping area. These areas need to be monitored, supervised, and made as safe as possible. A move to a retirement home, assisted-living center, or nursing home can

initially increase confusion and agitation. Repeated reassurance, reorientation, and careful introduction to the new personnel will help to smooth the process. Provision of activities that are known to be enjoyable to the patient can be of considerable benefit. Attention should also be paid to frustration and depression in family members and caregivers. Caregiver guilt and burn-out are common. Family members often feel overwhelmed and helpless and may vent their frustrations on the patient, each other, and healthcare providers. Caregivers should be encouraged to take advantage of day-care facilities and respite breaks. Education and counseling about dementia are important. Local and national support groups can be of considerable help, such as the Alzheimer's Disease and Related Disorders Association.

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27. SLEEP DISORDERS - Charles A. Czeisler, John W. Winkelman, Gary S. Richardson

Disturbed sleep is among the most frequent health complaints physicians encounter. More than one-half of adults in the United States experience at least intermittent sleep disturbances. For most, it is an occasional night of poor sleep and/or daytime sleepiness. However, at least 15 to 20% of adults report chronic sleep disturbance or misalignment of circadian timing, which can lead to serious impairment of daytime functioning. In addition, such problems may contribute to or exacerbate medical or psychiatric conditions. Thirty years ago, many such complaints were treated with hypnotic medications without further diagnostic evaluation. Since then, a distinct class of sleep and arousal disorders has been identified, and the field of sleep disorders medicine is now an established clinical discipline. However, most physicians still only receive, on average, 1 h of education in sleep disorders in their medical school curriculum.

PHYSIOLOGY OF SLEEP AND WAKEFULNESS

Most adults sleep 7 to 8 h per night, although the timing, duration, and internal structure of sleep vary among healthy individuals and as a function of age. At the extremes, infants and the elderly have frequent interruptions of sleep. In the United States, adults of intermediate age tend to have one consolidated sleep episode per day, although in some cultures sleep may be divided into a midafternoon nap and a shortened night sleep. Two principal neurobiologic systems govern the sleep-wake cycle: one that actively generates sleep and sleep-related processes and another that times sleep within the 24-h day. Either intrinsic abnormalities in these systems or extrinsic disturbances (environmental, drug- or illness-related) can lead to sleep or circadian rhythm disorders.

STATES AND STAGES OF SLEEP

States and stages of human sleep are defined on the basis of characteristic patterns in the electroencephalogram (EEG), the electrooculogram (EOG -- a measure of eye-movement activity), and the surface electromyogram (EMG) measured on the chin and neck. The continuous recording of this array of electrophysiologic parameters to define sleep and wakefulness is termed *polysomnography*.

Polysomnographic profiles define two states of sleep: (1) rapid-eye-movement (REM) sleep, and (2) non-rapid-eye-movement (NREM) sleep. NREM sleep is in turn subdivided into four stages, characterized by increasing arousal threshold and slowing of the cortical [EEG](#). REM sleep is characterized by a low-amplitude, mixed-frequency EEG similar to that of NREM stage 1 sleep. The [EOG](#) shows bursts of REM similar to those seen during eyes-open wakefulness. Chin [EMG](#) activity is absent, reflecting the brainstem-mediated muscle atonia that is characteristic of that state.

ORGANIZATION OF HUMAN SLEEP

Normal nocturnal sleep in adults displays a consistent organization from night to night ([Fig. 27-1](#)). After sleep onset, sleep usually progresses through [NREM](#) stages 1 to 4

within 45 to 60 min. Slow-wave sleep predominates in the first third of the night and comprises 15 to 25% of total nocturnal sleep time in young adults. The percentage of slow-wave sleep is influenced by several factors, most notably age (see below). Prior sleep deprivation increases the rapidity of sleep onset and both the intensity and amount of slow-wave sleep.

The first [REM](#) sleep episode usually occurs in the second hour of sleep. More rapid onset of REM sleep in a young adult (particularly if less than 30 min) may suggest pathology such as endogenous depression, narcolepsy, circadian rhythm disorders, or drug withdrawal. [NREM](#) and REM alternate through the night with an average period of 90 to 110 min (the "ultradian" sleep cycle). Overall, REM sleep constitutes 20 to 25% of total sleep, and NREM stages 1 and 2 are 50 to 60% (increasing in elderly subjects).

Age has a profound impact on sleep state organization ([Fig. 27-1](#)). Slow-wave sleep is most intense and prominent during childhood, decreasing sharply at puberty and across the second and third decades of life. After age 30, there is a progressive, almost linear decline in the amount of slow-wave sleep, and the amplitude of delta [EEG](#) activity comprising slow-wave sleep is reduced. In the otherwise healthy older person, slow-wave sleep may be completely absent, particularly in males.

A different age profile exists for [REM](#) sleep. In infancy, REM sleep may comprise 50% of total sleep time, and the percentage is inversely proportional to developmental age. The amount of REM sleep falls off sharply over the first postnatal year as a mature REM-[NREM](#) cycle develops. During the rest of life into extreme old age, REM sleep occupies a relatively constant percentage of total sleep time.

NEUROANATOMY OF SLEEP

Lesion studies in animals and neurologic diseases in humans have suggested distinct neuroanatomic sites in the generation of normal sleep and wakefulness. Experimental studies in animals have variously implicated the medullary reticular formation, the thalamus, and the basal forebrain in the generation of sleep, while the brainstem reticular formation, the midbrain, the subthalamus, the thalamus, and the basal forebrain have all been suggested to play a role in the generation of wakefulness or [EEG](#) arousal ([Chap. 24](#)).

Current hypotheses suggest that the capacity for sleep and wakefulness generation is distributed along an axial "core" of neurons extending from the brainstem rostrally to the basal forebrain. Complex commingling of neuronal groups occurs at many points along this brainstem-forebrain axis. It was recently discovered that a cluster of g-aminobutyric acid (GABA) and galaninergic ventrolateral preoptic (VLPO) neurons, which innervate monoaminergic cell groups in the tuberomammillary nucleus that contribute to the ascending arousal system, are activated during sleep. This has led to the hypothesis that these hypothalamic VLPO neurons may play a key role in sleep regulation.

Moreover, the neuroanatomic correlates of [REM](#) sleep appear to be discretely localized. Specific regions in the pons are associated with the neurophysiologic correlates of REM sleep. Small lesions in the dorsal pons result in the loss of the descending muscle inhibition normally associated with REM sleep; microinjections of the cholinergic agonist

carbachol into the pontine reticular formation appear to produce a state with all of the features of REM sleep. These experimental manipulations are mimicked by pathologic conditions in humans and animals. In narcolepsy, for example, abrupt, complete, or partial paralysis (cataplexy) occurs in response to a variety of stimuli. In dogs with this condition, physostigmine, a central cholinesterase inhibitor, increases the frequency of cataplectic attacks, while atropine decreases their frequency. Conversely, in REM sleep behavior disorder (see below), patients suffer from incomplete motor inhibition during REM sleep, resulting in involuntary, occasionally violent movement during REM sleep.

NEUROCHEMISTRY OF SLEEP

Early experimental studies that focused on the raphe nuclei of the brainstem appeared to implicate serotonin as the primary sleep-promoting neurotransmitter, while catecholamines were considered to be responsible for wakefulness. Subsequent work has demonstrated that the raphe-serotonin system may facilitate sleep but is not necessary for its expression. Extensive pharmacologic studies of sleep and wakefulness suggest roles for other neurotransmitters as well. Cholinergic neurotransmission is known to play a role in [REM](#) sleep generation. The alerting influence of caffeine implicates adenosine, whereas the hypnotic effect of benzodiazepines and barbiturates suggests a role for endogenous ligands of the [GABA](#) receptor complex.

A variety of sleep-promoting substances have been identified, although it is not known whether or not they are involved in the endogenous sleep-wake regulatory process. These include prostaglandin D₂, delta sleep-inducing peptide, muramyl dipeptide, interleukin 1, fatty acid primary amides, and melatonin. The hypnotic effect of these substances is commonly limited to [NREM](#) or slow-wave sleep, although peptides that increase [REM](#) sleep have also been reported. Many putative "sleep factors," including interleukin 1 and prostaglandin D₂, are immunologically active as well, suggesting a link between immune function and sleep-wake states.

PHYSIOLOGY OF CIRCADIAN RHYTHMICITY

The sleep-wake cycle is the most evident of the many 24-h rhythms in humans. Prominent daily variations also occur in endocrine, thermoregulatory, cardiac, pulmonary, renal, gastrointestinal, and neurobehavioral functions. However, in evaluating a daily variation, it is important to distinguish between those rhythmic components passively evoked by periodic environmental or behavioral changes (e.g., the increase in blood pressure and heart rate upon assumption of the upright posture) and those actively driven by an endogenous oscillatory process (e.g., the circadian variation in plasma cortisol that persists under a variety of environmental and behavioral conditions).

The suprachiasmatic nuclei (SCN) of the hypothalamus act as the central neural pacemaker driving endogenous circadian rhythms in mammals. Bilateral destruction of these nuclei results in a loss of endogenous circadian rhythmicity that can only be restored by transplantation of the same structure from a donor animal. The genetically determined period of this endogenous neural oscillator, which averages ~24.2 h in humans, is normally synchronized to the 24-h period of the environmental light-dark cycle. Entrainment of mammalian circadian rhythms by the light-dark cycle is mediated

via the retinohypothalamic tract, a monosynaptic pathway that links the retina to the SCN. Humans are exquisitely sensitive to the resetting effects of light, even at low intensity.

The timing and internal architecture of sleep are directly coupled to the output of the endogenous pacemaker. Paradoxically, the endogenous circadian rhythms of sleep tendency, sleepiness, and [REM](#) sleep propensity all peak near the habitual wake time, just after the nadir of the endogenous circadian temperature cycle, whereas the circadian wake propensity rhythm peaks 1 to 3 h before the habitual bedtime. These rhythms are thus timed to oppose the homeostatic decline of sleep tendency during the habitual sleep episode and the rise of sleep tendency throughout the usual waking day, respectively. Misalignment of the output of the endogenous circadian pacemaker with the desired sleep-wake cycle can, therefore, induce insomnia, as well as decrements of alertness and neurobehavioral performance in night-shift workers and after jet lag.

BEHAVIORAL CORRELATES OF SLEEP STATES AND STAGES

Polysomnographic staging of sleep correlates with behavioral changes during specific states and stages. During the transitional state between wakefulness and sleep (stage 1 sleep), subjects may respond to faint auditory or visual signals without "awakening." Furthermore, memory incorporation is inhibited at the onset of [NREM](#) stage 1 sleep, and individuals aroused from that transitional sleep stage frequently deny having been asleep. Such transitions occur spontaneously after chronic partial sleep deprivation (e.g., 4 to 6 h of sleep per night) and acute total sleep deprivation (e.g., 24 h of wakefulness), notwithstanding attempts to remain continuously awake (see "Shift-Work Sleep Disorder," below).

Awakenings from [REM](#) sleep are associated with recall of vivid dream imagery more than 80% of the time. The reliability of dream recall increases with REM sleep episodes occurring later in the night. Imagery may also be reported after [NREM](#) sleep interruptions, though these typically lack the detail and vividness of REM sleep dreams. The incidence of NREM sleep dream recall can be increased by selective REM sleep deprivation, suggesting that REM sleep and dreaming per se are not inexorably linked.

PHYSIOLOGIC CORRELATES OF SLEEP STATES AND STAGES

All major physiologic systems are influenced by sleep. Changes in cardiovascular function include a decrease in blood pressure and heart rate during [NREM](#), and particularly during slow-wave sleep. During [REM](#) sleep, phasic activity (bursts of eye movements) is associated with variability in both blood pressure and heart rate mediated principally by the vagus. Cardiac dysrhythmias may occur selectively during REM sleep. Respiratory function also changes ([Chap. 263](#)). In comparison to relaxed wakefulness, respiratory rate becomes more regular during NREM sleep (especially slow-wave sleep) and tonic REM sleep and becomes very irregular during phasic REM sleep. Minute ventilation decreases in NREM sleep out of proportion to the decrease in metabolic rate at sleep onset, resulting in a higher P_{CO_2} .

Endocrine function also varies with sleep. The most prominent changes are apparent in neuroendocrine parameters. Slow-wave sleep is associated with secretion of growth

hormone, while sleep in general is associated with augmented secretion of prolactin. Sleep has a complex effect on the secretion of luteinizing hormone (LH): during puberty, sleep is associated with increased LH secretion, whereas sleep in the mature woman inhibits LH secretion in the early follicular phase of the menstrual cycle. Sleep onset (and probably slow-wave sleep) is associated with inhibition of thyroid-stimulating hormone and of the adrenocorticotrophic hormone-cortisol axis, an effect that is superimposed on the circadian rhythms in the two systems.

The pineal hormone melatonin is secreted predominantly at night in both day- and night-active species, reflecting the direct modulation of pineal activity by the circadian pacemaker through a circuitous neural pathway from the [SCN](#) to the pineal gland. Melatonin secretion is not dependent upon the occurrence of sleep, persisting in individuals kept awake at night. In addition, exogenous melatonin increases sleepiness and may potentiate sleep when administered to good sleepers attempting to sleep during daylight hours at a time when endogenous melatonin levels are low. However, there is little evidence to support the use of melatonin as a hypnotic in such individuals during nighttime hours, when endogenous melatonin levels are high and sleep is already consolidated. A large-scale, double-blind clinical trial is needed to evaluate the efficacy of melatonin as a sleep-promoting therapeutic for patients with insomnia.

Sleep is also associated with alterations of thermoregulatory function. [NREM](#) sleep is associated with an attenuation of thermoregulatory responses to either heat or cold stress, and animal studies of thermosensitive neurons in the hypothalamus document an NREM-sleep-dependent reduction of the thermoregulatory set-point. [REM](#) sleep is associated with complete absence of thermoregulatory responsiveness, effectively resulting in functional poikilothermy. However, the potential adverse impact of this failure of thermoregulation is blunted by inhibition of REM sleep by extreme ambient temperatures.

DISORDERS OF SLEEP AND WAKEFULNESS

Approach to the Patient

Patients may seek help from a physician because of one of several symptoms: (1) an acute or chronic inability to sleep adequately at night (insomnia); (2) chronic fatigue, sleepiness, or tiredness during the day; or (3) a behavioral manifestation associated with sleep itself. Complaints of insomnia or excessive daytime sleepiness should be viewed as symptoms (much like fever or pain) of underlying disorders. Knowledge of the differential diagnosis of these presenting complaints is essential to identify the underlying medical disorder. Only then can appropriate treatment, rather than nonspecific approaches (e.g., over-the-counter sleeping aids) be applied. Diagnoses of exclusion, such as primary insomnia, should be made only after other diagnoses have been ruled out. [Table 27-1](#) outlines the diagnostic and therapeutic approach to the patient with a complaint of excessive daytime sleepiness.

A careful history is essential in the evaluation of the patient with a sleep complaint. In particular, the duration, severity, and consistency of the complaint are important, along with the patient's estimate -- in the case of an insomnia complaint -- of the consequences of reported sleep loss on subsequent waking function. Information from a

friend or family member can be an invaluable aid in assessing the symptoms and the severity of the complaint for daytime functioning, as some patients may be unaware of, or will underreport, such potentially embarrassing symptoms as heavy snoring or falling asleep while driving.

Completion by the patient of a day-by-day sleep-work-drug log for at least 2 weeks can help the physician better understand the nature of the complaint. Work times and sleep times (including daytime naps and nocturnal awakenings) as well as drug and alcohol use, including caffeine and hypnotics, should be noted each day. The sleep times should be plotted to facilitate recognition of circadian rhythm sleep disorders such as delayed sleep phase syndrome (see below).

Polysomnography is necessary for the diagnosis of specific disorders such as narcolepsy and sleep apnea and may be of utility in other settings as well. In addition to the three electrophysiologic variables used to define sleep states and stages, the standard clinical polysomnogram includes measures of respiration (respiratory effort, air flow, and oxygen saturation), anterior tibialis [EMG](#), and electrocardiogram. Evaluation of penile tumescence during nocturnal sleep can also help determine whether the cause of erectile dysfunction in a patient is psychogenic or organic ([Chap. 51](#)).

INSOMNIA

Insomnia is the complaint of inadequate sleep; it can be classified according to the nature of sleep disruption and the duration of the complaint. The nature of the sleep disruption provides important information about the possible etiology of the insomnia and is also central to the selection of specific and appropriate treatment. Insomnia is subdivided into difficulty falling asleep (*sleep onset insomnia*), frequent or sustained awakenings (*sleep maintenance insomnia*), early morning awakenings (*sleep offset insomnia*), or persistent sleepiness despite sleep of adequate duration (*nonrestorative sleep*). Similarly, the duration of the symptom is an important determinant of the nature of appropriate treatment. An insomnia complaint lasting one to several nights (within a single episode) is termed *transient insomnia*. Transient insomnia is typically the result of situational stress or a change in sleep schedule or environment (e.g., jet lag). *Short-term insomnia* lasts from a few days to 3 weeks. Disruption of this duration is usually associated with more protracted stress, such as recovery from surgery or short-term illness. *Long-term insomnia*, or *chronic insomnia*, lasts for months or years and, in contrast with short-term insomnia, requires a thorough evaluation of underlying causes (see below). Chronic insomnia is often a waxing and waning disorder, with spontaneous or stressor-induced exacerbations.

While an occasional night of poor sleep, typically in the setting of stress or excitement about external events, is both common and without lasting consequences, persistent insomnia can have important adverse consequences in the form of impaired daytime function and increased risk of injury due to accidents. There is also clear evidence of increased risk of the development of major depression with insomnia of at least 1 year's duration. In addition, there is emerging evidence that individuals with chronic insomnia have increased utilization of health care resources, even after controlling for comorbid medical and psychiatric disorders.

Extrinsic Insomnia A number of sleep disorders are the result of extrinsic factors that interfere with sleep. *Transient situational insomnia* can occur after a change in the sleeping environment (e.g., in an unfamiliar hotel or hospital bed) or before or after a significant life event, such as a change of occupation, loss of a loved one, illness, or anxiety over a deadline or examination. Increased sleep latency, frequent awakenings from sleep, and early morning awakening can all occur. Recovery generally occurs rapidly, usually within a few weeks. Treatment is usually symptomatic, with intermittent use of hypnotics and resolution of the underlying stress. *Inadequate sleep hygiene* is characterized by a behavior pattern prior to sleep and/or a bedroom environment that is not conducive to sleep. Noise and/or light in the bedroom can interfere with sleep, as can a bed partner with periodic limb movements during sleep or one who snores loudly. Clocks can heighten the anxiety about the time it has taken to fall asleep. Drugs that act on the central nervous system, large meals, vigorous exercise, or hot showers just before sleep may interfere with sleep onset. Many individuals participate in stressful work-related activities in the evening, producing a state incompatible with sleep onset. In preference to hypnotic medications, patients should be counseled to avoid stressful activities before bed, develop a soporific bedtime ritual, and to prepare and reserve the bedroom environment for sleeping. Consistent, regular rising times should be maintained daily, including weekends.

Psychophysiologic Insomnia Persistent *psychophysiologic insomnia* is a behavioral disorder in which patients are preoccupied with a perceived inability to sleep adequately at night. The sleep disturbance is often triggered by an emotionally stressful event; however, the poor sleep habits and beliefs about sleep acquired during the stressful period persist long after the initial incident. Such patients become hyperaroused by their own persistent efforts to sleep and/or the sleep environment, and the insomnia is a conditioned or learned response. They may be able to fall asleep more easily at unscheduled times (when not trying) or outside the home environment. Polysomnographic recording in patients with psychophysiologic insomnia reveals an objective sleep disturbance, often with an abnormally long sleep latency; frequent nocturnal awakenings; and an increased amount of stage 1 transitional sleep. Rigorous attention should be paid to sleep hygiene and correction of counterproductive, arousing behaviors before bedtime. Behavioral therapies are the treatment modality of choice for psychophysiologic insomnia, with only intermittent use of medications. When patients are awake longer than 20 min, they should read or perform other relaxing activities to distract themselves from insomnia-related anxiety. In addition, bedtime and waketime should be scheduled to restrict time in bed to be equal to their perceived total sleep time. This will generally produce sleep deprivation, greater sleep drive, and, eventually, better sleep. Time in bed can then be gradually expanded.

Medication-, Drug-, or Alcohol-Dependent Insomnia Disturbed sleep can result from ingestion of a wide variety of agents. Caffeine is perhaps the most common pharmacologic cause of insomnia. It produces increased latency to sleep onset, more frequent arousals during sleep, and a reduction in total sleep time for up to 8 to 14 h after ingestion. As few as three to five cups of coffee can significantly disturb sleep in some patients; therefore, a 1- to 2-month trial without caffeine should be attempted in patients with these symptoms. Similarly, alcohol and nicotine can interfere with sleep, despite the fact that many patients use them to relax and promote sleep. Although alcohol can increase drowsiness and shorten sleep latency, even moderate amounts of

alcohol increase awakenings in the second half of the night. In addition, alcohol ingestion prior to sleep is contraindicated in patients with sleep apnea because of the inhibitory effects of alcohol on upper airway muscle tone. Acutely, amphetamines and cocaine suppress both [REM](#) sleep and total sleep time, which return to normal with chronic use. Withdrawal leads to a REM sleep rebound.

A number of prescribed medications can produce insomnia. Antidepressants, sympathomimetics, and glucocorticoids are common causes. In addition, severe rebound insomnia can result from the acute withdrawal of hypnotics, especially following the use of high doses of benzodiazepines with a short half-life. For this reason, hypnotic doses should be low to moderate, the total duration of hypnotic therapy should usually be limited to 2 to 3 weeks, and prolonged drug tapering is encouraged.

Altitude Insomnia Sleep disturbance is a common consequence of exposure to high altitude. Periodic breathing of the Cheyne-Stokes type occurs during [NREM](#) sleep about half the time at high altitude, with restoration of a regular breathing pattern during [REM](#) sleep. Both hypoxia and hypocapnia are thought to be involved in the development of periodic breathing. Frequent awakenings and poor quality sleep characterize altitude insomnia, which is generally worst on the first few nights at high altitude but may persist. Treatment with acetazolamide can decrease time spent in periodic breathing and substantially reduce hypoxia during sleep.

Restless Legs Syndrome (RLS) Patients with this sensory-motor disorder report a creeping or crawling dysesthesia deep within the calves or feet, or sometimes even in the upper extremities, that is associated with an irresistible urge to move the affected limbs. For most patients with RLS, the dysesthesias and restlessness are much worse in the evening or night compared to the daytime and frequently interfere with the ability to fall asleep. The disorder is exacerbated by inactivity and temporarily relieved by movement. In contrast, paresthesia secondary to peripheral neuropathy persists with activity. The severity of this chronic disorder may wax and wane with time and can be exacerbated by sleep deprivation, caffeine, and pregnancy. The prevalence is thought to be 5% of adults. Roughly one-third of patients will have multiple affected family members, possibly with an autosomal dominant pattern. Iron deficiency and renal failure may actually cause RLS, which is then considered secondary RLS. The symptoms of RLS are exquisitely sensitive to dopaminergic drugs (e.g., L-dopa or dopamine agonists). Narcotics, benzodiazepines, and certain anticonvulsants may also be of therapeutic value. Most patients with restless legs also experience periodic limb movement disorder during sleep, although the reverse is not the case.

Periodic Limb Movement Disorder *Periodic limb movement disorder*, previously known as *nocturnal myoclonus*, is the principal objective polysomnographic finding in 17% of patients with insomnia and 11% of those with excessive daytime somnolence ([Fig. 27-2](#)). It is often unclear whether it is an incidental finding or the cause of disturbed sleep. Stereotyped, 0.5- to 5.0-s extensions of the great toe and dorsiflexion of the foot recur every 20 to 40 s during [NREM](#) sleep, in episodes lasting from minutes to hours. Most such episodes occur during the first half of the night. The disorder occurs in a wide variety of sleep disorders (including narcolepsy, sleep apnea, [REM](#) sleep behavior disorder, and various forms of insomnia) and may be associated with frequent arousals and an increased number of sleep-stage transitions. The incidence increases with age:

44% of people over age 65 without a sleep complaint have >five periodic leg movements per hour of sleep. The pathophysiology is not well understood, though individuals with high spinal transections can exhibit periodic leg movements during sleep, suggesting the existence of a spinal generator. Polysomnography with bilateral surface [EMG](#) recording of the anterior tibialis is used to establish the diagnosis. Treatment options include dopaminergic medications or benzodiazepines.

Insomnia Associated with Mental Disorders Approximately 80% of patients with psychiatric disorders describe sleep complaints. There is considerable heterogeneity, however, in the nature of the sleep disturbance both between conditions and among patients with the same condition.

Depression can be associated with sleep onset insomnia, sleep maintenance insomnia, and/or early morning wakefulness. However, hypersomnia occurs in some depressed patients, especially adolescents and those with either bipolar or seasonal (fall/winter) depression ([Chap. 385](#)). Indeed, sleep disturbance is an important vegetative sign of depression and may commence before any mood changes are perceived by the patient. Consistent polysomnographic findings in depression include decreased [REM](#) sleep latency, lengthened first REM sleep episode, and shortened first [NREM](#) sleep episode; however, these findings are not specific for depression, and the extent of these changes varies with age and symptomatology. Depressed patients also show decreased slow-wave sleep and reduced sleep continuity.

In *mania* and *hypomania*, sleep latency is increased and total sleep time can be reduced. Patients with *anxiety disorders* tend not to show the changes in [REM](#) sleep and slow-wave sleep seen in endogenously depressed patients. Finally, *chronic alcoholics* lack slow-wave sleep, have decreased amounts of REM sleep (as an acute response to alcohol), and have frequent arousals throughout the night. This is associated with impaired daytime alertness. The sleep of chronic alcoholics may remain disturbed for years after discontinuance of alcohol usage. Sleep architecture and physiology are disturbed in *schizophrenia* (with a decreased amount of stage 4 sleep and a lack of augmentation of REM sleep following REM sleep deprivation); chronic schizophrenics often show day-night reversal, sleep fragmentation, and insomnia.

Insomnia Associated with Neurologic Disorders A variety of neurologic diseases result in sleep disruption through both indirect, nonspecific mechanisms (e.g., pain in cervical spondylosis or low back pain) or by impairment of central neural structures involved in the generation and control of sleep itself.

For example, *dementia* from any cause has long been associated with disturbances in the timing of the sleep-wake cycle, often characterized by nocturnal wandering and an exacerbation of symptomatology at night (so-called sundowning).

Epilepsy may rarely present as a sleep complaint ([Chap. 360](#)). Often the history is of abnormal behavior, at times with convulsive movements, during sleep, and the differential diagnosis includes [REM](#) sleep behavior disorder, sleep apnea syndrome, and periodic movements of sleep (see above). Diagnosis requires nocturnal [EEG](#) recording. Other neurologic diseases associated with abnormal movements, such as *Parkinson's disease*, *hemiballismus*, *Huntington's chorea*, and *Gilles de la Tourette syndrome*, are

also associated with disrupted sleep, presumably through secondary mechanisms. However, the abnormal movements themselves are greatly reduced during sleep. Headache syndromes may show sleep-associated exacerbations (*migraine* or *cluster headache*) ([Chap. 15](#)) by unknown mechanisms.

Fatal familial insomnia is a rare hereditary disorder caused by bilateral degeneration of anterior and dorsomedial nuclei of the thalamus. Insomnia is a prominent early symptom. Progressively, the syndrome produces autonomic dysfunction, dysarthria, myoclonus, coma, and death. The pathogenesis is a mutation in the prion protein ([Chap. 375](#)).

Insomnia Associated with Other Medical Disorders A number of medical conditions are associated with disruptions of sleep. The association is frequently nonspecific, e.g., that between sleep disruption and chronic pain from rheumatologic disorders. Attention to this association is important in that sleep-associated symptoms are the presenting complaint of many such patients. Treatment of the underlying medical disorder or symptom is the most useful approach to such patients. As noted above, sleep disruption can also result from the appropriate use of drugs such as glucocorticoids.

Among the most prominent associations is that between sleep disruption and *asthma*. In many asthmatics there is a prominent daily variation in airway resistance that results in marked increases in asthmatic symptoms at night, especially during sleep. In addition, treatment of asthma with theophylline-based compounds, adrenergic agonists, or glucocorticoids can independently disrupt sleep. When sleep disruption is a prominent side effect of asthma treatment, inhaled steroids (e.g., beclomethasone) that do not disrupt sleep may provide a useful alternative.

Cardiac ischemia may also be associated with sleep disruption. The ischemia itself may result from increases in sympathetic tone as a result of sleep apnea. Patients may present with complaints of nightmares or vivid, disturbing dreams, with or without awareness of the more classic symptoms of angina or of the sleep-disordered breathing. Treatment of the sleep apnea may substantially improve the angina and the nocturnal sleep quality. *Paroxysmal nocturnal dyspnea* can also occur as a consequence of sleep-associated cardiac ischemia that causes pulmonary congestion exacerbated by the recumbent posture.

Chronic obstructive pulmonary disease is also associated with sleep disruption, as is *cystic fibrosis*, *menopause*, *hyperthyroidism*, *gastroesophageal reflux*, *chronic renal failure*, and *liver failure*.

EVALUATION OF DAYTIME SLEEPINESS

Daytime impairment due to sleep loss may be difficult to quantify in the clinical setting for several reasons. First, sleepiness is not necessarily proportional to subjectively assessed sleep deprivation. In obstructive sleep apnea, for example, the repeated brief interruptions of sleep associated with resumption of respiration at the end of apneic episodes result in significant waking impairment, despite the fact that the patient may be unaware of the sleep fragmentation. Second, subjective descriptions of waking impairment vary from patient to patient. Patients may describe themselves as "sleepy,"

"fatigued," or "tired" and may have a clear sense of the meaning of those terms, while others may use the same terms to describe a completely different condition. Third, sleepiness, particularly when profound, may affect judgment in a manner analogous to ethanol, such that subjective awareness of the condition and the consequent cognitive and motor impairment is reduced. Finally, patients may be reluctant to admit that sleepiness is a problem, both because they are generally unaware of what constitutes normal alertness and because sleepiness is generally viewed pejoratively, ascribed more often to a deficit in motivation than to an inadequately addressed physiologic sleep need.

In assessing sleepiness in the clinical setting, specific questioning about the occurrence of sleep episodes during normal waking hours, both intentional and unintentional, can overcome the inconsistencies among subjective characterizations and help to interpret the adverse impact of sleepiness on daytime function. Specific areas to be addressed include the occurrence of inadvertent sleep episodes while driving or in other safety-related settings, sleepiness while at work or school (and the relationship of sleepiness to work and school performance), and the effect of sleepiness on social and family life. Evidence for significant daytime impairment [in association either with the diagnosis of a primary sleep disorder, such as narcolepsy or sleep apnea, or with imposed or self-selected sleep-wake schedules (see "Shift-Work Sleep Disorder," below)] raises the question of the physician's responsibility to notify motor vehicle licensing authorities of the increased risk of sleepiness-related vehicle accidents. As with epilepsy, legal requirements vary from state to state, and existing legal precedents do not provide a consistent interpretation of the balance between the physician's responsibility and the patient's right to privacy. At a minimum, physicians should document discussions with the patient regarding the increased risk of operating a vehicle, as well as a recommendation that driving be suspended until successful treatment or schedule modification can be instituted.

The distinction between fatigue and sleepiness can be useful in the differentiation of patients with complaints of fatigue or tiredness in the setting of disorders such as fibromyalgia, chronic fatigue syndrome ([Chap. 384](#)), or endocrine deficiencies such as hypothyroidism or Addison's disease. While patients with these disorders can typically distinguish their daytime symptoms from the sleepiness that occurs with sleep deprivation, substantial overlap can occur. This is particularly true when the primary disorder also results in chronic sleep disruption (e.g., sleep apnea in hypothyroidism) or in abnormal sleep (e.g., fibromyalgia).

While clinical evaluation of the complaint of excessive sleepiness is usually adequate, objective quantification is sometimes necessary for diagnostic purposes or for the evaluation of treatment response. Assessment of daytime functioning as an index of the adequacy of sleep can be made with the multiple sleep latency test (MSLT), which involves repeated measurement of sleep latency (time to onset of sleep) under standardized conditions during a day following quantified nocturnal sleep. The average latency across four to six tests (administered every 2 h across the waking day) is taken as an objective measure of daytime sleep tendency. Disorders of sleep that result in pathologic daytime somnolence can be reliably distinguished with the MSLT. In addition, the multiple measurements of sleep onset may identify direct transitions from wakefulness to [REM](#) sleep that are suggestive of specific pathologic conditions (e.g.,

narcolepsy).

NARCOLEPSY

Narcolepsy is both a disorder of the ability to sustain wakefulness voluntarily and a disorder of REM sleep regulation ([Table 27-2](#)). The classic "narcolepsy tetrad" consists of excessive daytime somnolence plus three specific symptoms related to an intrusion of REM sleep characteristics (e.g., muscle atonia, vivid dream imagery) into the transition between wakefulness and sleep: (1) sudden weakness or loss of muscle tone without loss of consciousness, often elicited by emotion (cataplexy); (2) hallucinations at sleep onset (hypnagogic hallucinations) or upon awakening (hypnopompic hallucinations); and (3) muscular paralysis upon awakening (sleep paralysis). The severity of cataplexy varies, as patients may have two to three attacks per day or per decade. The extent and duration of an attack may also vary, from a transient sagging of the jaw lasting a few seconds to rare cases of flaccid paralysis of the entire voluntary musculature for up to 20 to 30 min. Symptoms of narcolepsy typically begin in the second decade, although the onset ranges from ages 5 to 50. Once established, the disease is chronic without remissions. Secondary forms of narcolepsy have been described (e.g., after head trauma).

Narcolepsy affects about 1 in 4000 people in the United States and appears to have a genetic basis. Recently, two independent discoveries have revealed that hypothalamic neurons containing the neuropeptide orexin (hypocretin) may play an important role in the regulation of sleep/wakefulness: (1) a mutation in the orexin (hypocretin) receptor 2 gene has been associated with canine narcolepsy; and (2) orexin "knockout" mice that are genetically unable to produce this neuropeptide exhibit a phenotype, as assessed by behavioral and electrophysiologic criteria, that is similar to human narcolepsy. In addition, modafinil, a drug recently approved by the U.S. Food and Drug Administration (FDA) for the treatment of narcolepsy, activates orexin-containing neurons. However, the inheritance pattern of narcolepsy in humans is more complex than that of the canine model. A high rate of discordance in identical twins indicates that one or more nonheritable factors contributes to its development. First-degree relatives of narcoleptic patients nonetheless have about a 1% incidence of narcolepsy, much higher than the general population but much lower than is seen in the animal models. Of note, nearly all narcoleptics with cataplexy are positive for the human leukocyte antigen DQB*0106 (ordinarily found in 20 to 30% of the general population) ([Chap. 306](#)).

Diagnosis Definition of the essential and distinctive features of narcolepsy has continued to evolve, and the diagnostic criteria continue to be a matter of debate. Certainly, objective verification of excessive daytime somnolence, typically with [MSLT](#) mean sleep latencies <8 min, is an essential if nonspecific diagnostic feature. Other conditions that cause excessive sleepiness, such as sleep apnea or chronic sleep restriction, must be rigorously excluded. The other objective diagnostic feature of narcolepsy is the presence of [REM](#) sleep in at least two of the naps during the MSLT. This excessive REM "pressure" is also manifested by the appearance of REM sleep immediately or within minutes after sleep onset in 50% of narcoleptic patients, a rarity in unaffected individuals maintaining a conventional sleep-wake schedule. The REM-related symptoms of the classic narcolepsy tetrad are variably present. There is increasing evidence that narcoleptics with cataplexy (one-half to two-thirds of patients)

may represent a more homogeneous group than those without this symptom. However, a history of cataplexy can be difficult to establish reliably. Hypnagogic and hypnopompic hallucinations and sleep paralysis are often found in nonnarcoleptic individuals and may be present in only one-half of narcoleptics. Nocturnal sleep disruption is commonly observed in narcolepsy but is also a nonspecific symptom. Similarly, history of "automatic behavior" during wakefulness (a trance-like state during which simple motor behaviors persist) is not specific for narcolepsy and serves principally to corroborate the presence of daytime somnolence.

TREATMENT

The treatment of narcolepsy is symptomatic. Somnolence is treated with stimulants. Methylphenidate has long been considered the drug of choice by most; the usual initial dose is 10 mg bid, increasing as needed to a maximum of 20 mg qid. Pemoline, frequently used as an alternative due to its longer half-life, may be less effective and has recently been associated with fatal hepatic failure in several children. Dextroamphetamine, 10 mg bid, and methamphetamine are also frequently used alternatives. Recently, modafinil, a novel wake-promoting agent, has been approved by the [FDA](#) for treatment of the excessive daytime somnolence in narcolepsy; the dose is 200-400 mg/d given as a single dose. It is a long-acting agent that may cause fewer side effects than other medications.

Treatment of the [REM](#)-related phenomena cataplexy, hypnagogic hallucinations, and sleep paralysis requires the potent REM sleep suppression produced by antidepressant medications. The tricyclic antidepressants [e.g., protriptyline (10-40 mg/d) and clomipramine (25-50 mg/d)] and the selective serotonin reuptake inhibitors (SSRIs) [e.g., fluoxetine (10-20 mg/d)] are commonly used for this purpose in the United States. Efficacy of the antidepressants is limited largely by anticholinergic side effects (tricyclics) and by sleep disturbance and sexual dysfunction (SSRIs). Adequate nocturnal sleep time and planned daytime naps (when possible) are important preventative measures in narcolepsy.

SLEEP APNEA SYNDROMES

Respiratory dysfunction during sleep is a common, serious cause of excessive daytime somnolence as well as of disturbed nocturnal sleep. An estimated 2 to 5 million people in the United States have a reduction or cessation of breathing for 10 to 150 s, from thirty to several hundred times every night during sleep. These episodes may be due to either an occlusion of the airway (*obstructive sleep apnea*), absence of respiratory effort (*central sleep apnea*), or a combination of these factors (*mixed sleep apnea*) ([Fig. 27-2](#)). Failure to recognize and treat these conditions appropriately may lead to: significant, and often disabling, impairment of daytime alertness; increased risk of sleep-related motor vehicle accidents; hypertension and other serious cardiovascular complications; and increased mortality. Sleep apnea is particularly prevalent in overweight men and in the elderly, yet it is estimated to remain undiagnosed in 80 to 90% of affected individuals. This is unfortunate since effective treatments are available. **Readers are referred to Chap. 263 for a comprehensive review of the diagnosis and treatment of patients with these conditions.*

PARASOMNIAS

The term *parasomnia* refers to abnormal behaviors that arise from, or occur during, sleep. A continuum of parasomnias arise from [NREM](#) sleep, from brief confusional arousals to sleepwalking and night terrors. The presenting complaint is usually related to the behavior itself, but the parasomnias can disturb sleep continuity or lead to mild impairments in daytime alertness. Only one parasomnia is known to occur in [REM](#) sleep, i.e., REM sleep behavior disorder (RBD; see below).

Sleepwalking (Somnambulism) Patients affected by this disorder carry out automatic motor activities that range from simple to complex. Individuals may leave the bed, walk, urinate inappropriately, eat, or exit from the house while remaining only partially aware. Full arousal may be difficult, and some patients may respond to attempted awakening with agitation or even violence. Sleepwalking arises from stage 3 or 4 [NREM](#) sleep and is most common in children and adolescents, when these sleep stages are most robust. Episodes are usually isolated but may be recurrent in 1 to 6% of patients. The cause is unknown, though it has a familial basis in roughly one-third of cases.

Sleep Terrors This disorder, also called *pavor nocturnus*, occurs primarily in young children during the first several hours after sleep onset, in stages 3 and 4 of [NREM](#) sleep. The child suddenly screams, exhibiting autonomic arousal with sweating, tachycardia, and hyperventilation. The individual may be difficult to arouse and rarely recalls the episode on awakening in the morning. Recurrent attacks are rare, and treatment is usually by way of reassurance of parents. Both sleep terrors and sleepwalking represent abnormalities of arousal. In contrast, *nightmares* (dream anxiety attacks) occur during [REM](#) sleep and cause full arousal, with intact memory for the unpleasant episode.

REM Sleep Behavior Disorder [RBD](#) is a rare condition that is distinct from other parasomnias in that it occurs during [REM](#) sleep. It primarily afflicts men of middle age or older, many of whom have a history of prior neurologic disease. In fact, over one-third of patients will go on to develop Parkinson's disease within 10 to 20 years. Presenting symptoms are of agitated or violent behavior during sleep, reported by a bed partner. In contrast to typical somnambulism, injury to patient or bed partner is not uncommon, and, upon awakening, the patient reports vivid, often unpleasant, dream imagery. The principal differential diagnosis is that of nocturnal seizures, which can be excluded with polysomnography. In RBD, seizure activity is absent on the [EEG](#), and disinhibition of the usual motor atonia is observed in the [EMG](#) during REM sleep, at times associated with complex motor behaviors. The pathogenesis is unclear, but damage to brainstem areas mediating descending motor inhibition during REM sleep may be responsible. In support of this hypothesis are the remarkable similarities between RBD and the sleep of animals with bilateral lesions of the pontine tegmentum in areas controlling REM sleep motor inhibition. Treatment with clonazepam provides sustained improvement in almost all reported cases.

Sleep Bruxism Bruxism is an involuntary, forceful grinding of teeth during sleep that affects 10 to 20% of the population. The patient is usually unaware of the problem. The typical age of onset is 17 to 20 years, and spontaneous remission usually occurs by age 40. Sex distribution appears to be equal. Treatment is dictated by the risk of dental injury. In many cases, the diagnosis is made during dental examination, damage is

minor, and no treatment is indicated. In more severe cases, treatment with a rubber tooth guard is necessary to prevent disfiguring tooth injury. Stress management or, in some cases, biofeedback can be useful when bruxism is a manifestation of psychological stress. There are anecdotal reports of benefit using benzodiazepines.

Sleep Enuresis Bedwetting, like sleepwalking and night terrors, is another parasomnia that occurs during slow-wave sleep in the young. Before age 5 or 6, nocturnal enuresis should probably be considered a normal feature of development. The condition usually improves spontaneously at puberty, has a prevalence in late adolescence of 1 to 3%, and is rare in adulthood. The age threshold for initiation of treatment depends on parental and patient concern about the problem. Persistence of enuresis into adolescence or adulthood may reflect a variety of underlying conditions. In older patients with enuresis a distinction must be made between primary and secondary enuresis, the latter being defined as bedwetting in patients who have been fully continent for 6 to 12 months. Treatment of primary enuresis is reserved for patients of appropriate age (older than 5 or 6 years) and consists of bladder training exercises and behavioral therapy. Urologic abnormalities are more common in primary enuresis and must be assessed by urologic examination. Important causes of secondary enuresis include emotional disturbances, urinary tract infections or malformations, cauda equina lesions, epilepsy, sleep apnea, and certain medications. Symptomatic pharmacotherapy is usually accomplished with intranasal desmopressin, or oral oxybutynin chloride or imipramine.

Miscellaneous Parasomnias Other clinical entities fulfill the definition of a parasomnia in that they occur selectively during sleep and are associated with some degree of sleep disruption. Examples include *jactatio capitis nocturna* (nocturnal headbanging), sleep talking, nocturnal paroxysmal dystonia, and nocturnal leg cramps.

CIRCADIAN RHYTHM SLEEP DISORDERS

A subset of patients presenting with either insomnia or hypersomnia may have a disorder of sleep *timing* rather than sleep *generation*. Disorders of sleep timing can either be organic (i.e., due to an intrinsic defect in the circadian pacemaker or its input from entraining stimuli) or environmental (i.e., due to a disruption of exposure to entraining stimuli from the environment). Regardless of etiology, the symptoms reflect the influence of the underlying circadian pacemaker on sleep-wake function. Thus, effective therapeutic approaches should aim to entrain the oscillator at an appropriate phase.

RAPID TIME-ZONE CHANGE (JET LAG) SYNDROME

More than 60 million people experience transmeridian air travel annually, which is often associated with excessive daytime sleepiness, sleep onset insomnia, and frequent arousals from sleep, particularly in the latter half of the night. Gastrointestinal discomfort is common. The syndrome is transient, typically lasting 2 to 14 d depending on the number of time zones crossed, the direction of travel, and the traveler's age and phase-shifting capacity. Travelers who spend more time outdoors reportedly adapt more quickly than those who remain in hotel rooms, presumably due to bright (outdoor) light exposure.

SHIFT-WORK SLEEP DISORDER

More than 7 million workers in the United States regularly work at night, either on a permanent or rotating schedule. In addition, each week millions of Americans elect to remain awake at night to meet deadlines, drive long distances, or participate in recreational activities, leading to both sleep loss and misalignment of their circadian rhythms with respect to their sleep-wake cycle. Chronic shift workers have higher rates of cardiac, gastrointestinal, and reproductive disorders. Studies of regular night-shift workers indicate that the circadian timing system usually fails to adapt successfully to such inverted schedules. This leads to a misalignment between the desired work-rest schedule and the output of the pacemaker and in disturbed daytime sleep. Consequent sleep deprivation, increased length of time awake prior to work, and misalignment of circadian phase produce decreased alertness and performance, increased reaction time, and increased risk of performance lapses, thereby resulting in greater safety hazards among night workers and other sleep-deprived individuals.

Sleep onset is associated with marked attenuation in perception of both auditory and visual stimuli and lapses of consciousness. The sleepy individual may thus attempt to perform routine and familiar motor tasks during the transition state between wakefulness and sleep (stage 1 sleep) in the absence of adequate sensory input from the environment. Motor vehicle operators are especially vulnerable to sleep-related accidents since the sleep-deprived driver or operator often fails to heed the warning signs of fatigue. Such attempts to override the powerful biologic drive for sleep by the sheer force of will can yield a catastrophic outcome when sleep processes intrude involuntarily upon the waking brain. Such intrusions typically last only seconds but are known on occasion to persist for longer durations. These frequent brief intrusions of stage 1 sleep into behavioral wakefulness are a major component of the impaired psychomotor performance seen with sleepiness. Such intrusions and their associated performance lapses, which are preceded by a markedly increased subjective sense of sleepiness, will inevitably occur if the need for sleep is not satiated. There is a marked increase in the risk of sleep-related, fatal-to-the-driver highway crashes in the early morning and late afternoon hours, coincident with peaks in the daily rhythm of sleep tendency.

Safety programs should promote education about sleep and increase awareness of the hazards associated with night work and should be aimed at minimizing both circadian disruption and sleep deprivation. The work schedule should minimize: (1) exposure to night work, (2) the frequency of shift rotation so that shifts do not rotate more than once every 2 to 3 weeks, (3) the number of consecutive night shifts, and (4) the duration of night shifts. In fact, shift durations of greater than 18 h should be universally recognized as increasing the risk of sleep-related errors and performance lapses. Caffeine is undoubtedly the most widely used wake-promoting drug, but it cannot forestall sleep indefinitely and does not protect users from sleep-related performance lapses. Postural changes, exercise, and strategic placement of nap opportunities can sometimes temporarily reduce the risk of fatigue-related performance lapses. Properly timed exposure to bright light can facilitate rapid adaptation to night-shift work, where feasible. An adequate number of safe highway rest areas, shoulder rumble strips, and strict enforcement and compliance monitoring of hours-of-service policies are needed to

reduce the risk of sleep-related transportation crashes. Such steps can lead to improvements in performance and to reduced accident rates both at work and on the roadways.

DELAYED SLEEP PHASE SYNDROME

Delayed sleep phase syndrome is characterized by: (1) reported sleep onset and wake times intractably later than desired, (2) actual sleep times at nearly the same clock hours daily, and (3) essentially normal all-night polysomnography except for delayed sleep onset. Patients exhibit an abnormally delayed endogenous circadian phase, with the temperature minimum during the constant routine occurring later than normal. This delayed phase could be due to: (1) an abnormally long intrinsic period of the endogenous circadian pacemaker; (2) an abnormally reduced phase-advancing capacity of the pacemaker; or (3) an irregular prior sleep-wake schedule, characterized by frequent nights when the patient chooses to remain awake well past midnight (for social, school, or work reasons). In most cases, it is difficult to distinguish among these factors, since patients with an abnormally long intrinsic period are more likely to "choose" such late-night activities because they are unable to sleep at that time. Patients tend to be young adults. This self-perpetuating condition can persist for years and does not usually respond to attempts to reestablish normal bedtime hours.

Treatment methods involving bright-light phototherapy during the morning hours or melatonin administration in the evening hours show promise in these patients, although the relapse rate among such patients is very high.

ADVANCED SLEEP PHASE SYNDROME

Advanced sleep phase syndrome is the converse of the delayed sleep phase syndrome and tends to occur in the elderly. Patients with this condition report excessive daytime sleepiness during the evening hours, when they have great difficulty remaining awake, even in social settings. The patients awaken from 3 to 5 A.M. each day, often several hours before their desired wake times. Although such patients have not been studied extensively, familial inheritance of this condition has been reported. Some of these patients may benefit from bright-light phototherapy during the evening hours, designed to reset the circadian pacemaker to a later hour.

NON-24-H SLEEP-WAKE DISORDER

This condition can occur when the maximal phase-advancing capacity of the circadian pacemaker is not adequate to accommodate the difference between the 24-h geophysical day and the intrinsic period of the pacemaker in the patient. Alternatively, patients' self-selected exposure to artificial light may drive the circadian pacemaker to a longer than 24-h schedule. Affected patients are not able to maintain a stable phase relationship between the output of the pacemaker and the 24-h day. Such patients typically present with an incremental pattern of successive delays in sleep onsets and wake times, progressing in and out of phase with local time. When the patient's endogenous rhythms are out of phase with the local environment, insomnia coexists with excessive daytime sleepiness. Conversely, when the endogenous rhythms are in phase with the local environment, symptoms remit. The intervals between symptomatic

periods may last several weeks to several months. Blind individuals unable to perceive light are particularly susceptible to this disorder. Melatonin administration has been reported to improve sleep, and in some cases even to induce synchronization of the circadian pacemaker.

MEDICAL IMPLICATIONS OF CIRCADIAN RHYTHMICITY

Understanding the role of circadian rhythmicity in the pathophysiology of illness may lead to improvements in diagnosis and treatment. For example, prominent circadian variations have been reported in the incidence of *acute myocardial infarction*, *sudden cardiac death*, and *stroke*, the leading causes of death in the United States. Platelet aggregability is increased after arising in the early morning hours, coincident with the peak incidence of these cardiovascular events. A better understanding of the possible role of circadian rhythmicity in the acute destabilization of a chronic condition such as atherosclerotic disease could improve the understanding of the pathophysiology.

Diagnostic and therapeutic procedures may also be affected by the time of day at which data are collected. Examples include blood pressure, body temperature, the dexamethasone suppression test, and plasma cortisol levels. The timing of chemotherapy administration has been reported to have an effect on the outcome of treatment. Few physicians realize the extent to which routine measures are affected by the time (or sleep/wake state) when the measurement is made.

In addition, both the toxicity and effectiveness of drugs can vary during the day. For example, more than a fivefold difference has been observed in mortality rates following administration of toxic agents to experimental animals at different times of day. Anesthetic agents are particularly sensitive to time-of-day effects. Finally, the physician must be increasingly aware of the public health risks associated with the ever-increasing demands made by the duty-rest-recreation schedules in our round-the-clock society.

(Bibliography omitted in Palm version)

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SECTION 4 -DISORDERS OF EYES, EARS, NOSE, AND THROAT

28. DISORDERS OF THE EYE - Jonathan C. Horton

THE HUMAN VISUAL SYSTEM

The visual system provides a supremely efficient means for the rapid assimilation of information from the environment to aid in the guidance of behavior. The act of seeing begins with the capture of images focused by the cornea and lens upon a light-sensitive membrane in the back of the eye, called the *retina*. The retina is actually part of the brain, banished to the periphery to serve as a transducer for the conversion of patterns of light energy into neuronal signals. Light is absorbed by photopigment in two types of receptors: rods and cones. In the human retina there are 100 million rods and 5 million cones. The rods operate in dim (scotopic) illumination. The cones function under daylight (photopic) conditions. The cone system is specialized for color perception and high spatial resolution. The majority of cones are located within the macula, the portion of the retina serving the central 10° of vision. In the middle of the macula a small pit termed the *fovea*, packed exclusively with cones, provides best visual acuity.

Photoreceptors hyperpolarize in response to light, activating bipolar, amacrine, and horizontal cells in the inner nuclear layer. After processing of photoreceptor responses by this complex retinal circuit, the flow of sensory information ultimately converges upon a final common pathway: the ganglion cells. These cells translate the visual image impinging upon the retina into a continuously varying barrage of action potentials that propagates along the primary optic pathway to visual centers within the brain. There are a million ganglion cells in each retina, and hence a million fibers in each optic nerve.

Ganglion cell axons sweep along the inner surface of the retina in the nerve fiber layer, exit the eye at the optic disc, and travel through the optic nerve, optic chiasm, and optic tract to reach targets in the brain. The majority of fibers synapse upon cells in the lateral geniculate body, a thalamic relay station. Cells in the lateral geniculate body project in turn to the primary visual cortex. This massive afferent retinogeniculocortical sensory pathway provides the neural substrate for visual perception. Although the lateral geniculate body is the main target of the retina, separate classes of ganglion cells project to other subcortical visual nuclei involved in different functions. The pupillary reflex is mediated by input to the pretectal olivary nuclei in the midbrain. These pretectal nuclei send their output to the ipsilateral and contralateral Edinger-Westphal nuclei of the oculomotor nuclear complex. Cells in the Edinger-Westphal nuclei provide parasympathetic innervation to the iris sphincter via an interneuron in the ciliary ganglion. Circadian rhythms are timed by a retinal projection to the suprachiasmatic nucleus. Visual orientation and eye movements are served by retinal input to the superior colliculus. Gaze stabilization and optokinetic reflexes are governed by a group of small retinal targets known collectively as the *brainstem accessory optic system*. Finally, there is a sizeable retinal projection to the pulvinar, a large thalamic visual nucleus of obscure function.

The eyes must be rotated constantly within their orbits to place and maintain targets of visual interest upon the fovea. This activity, called *foveation*, or looking, is governed by an elaborate efferent motor system. Each eye is moved by six extraocular muscles,

supplied by cranial nerves from the oculomotor (III), trochlear (IV), and abducens (VI) nuclei. Activity in these ocular motor nuclei is coordinated by pontine and midbrain mechanisms for smooth pursuit, saccades, and gaze stabilization during head and body movements. Large regions of the frontal and parietooccipital cortex control these brainstem eye movement centers by providing descending supranuclear input.

Visual function can be disturbed in myriad ways. The eyes are mounted in a prominent position on the head, where they are vulnerable to trauma, exposure, and infection. Vision can be damaged by diseases intrinsic to the eye, such as glaucoma, cataract, or retinal detachment. Many neurologic diseases produce ocular symptoms, because extensive areas of the cortex, thalamus, cerebellum, and brainstem are devoted to visual perception or to the execution of eye movements. In genetic disorders, eye manifestations are common and often help the clinician to recognize a rare syndrome. Finally, the eyes are affected frequently by acquired systemic diseases.

The eye is a specialized organ, requiring unique optical instruments for proper examination. The slit lamp and ophthalmoscope proffer a beautiful, magnified view of the transparent anatomy of the eye and afford the only opportunity for direct inspection of blood vessels in a living subject. Some physicians do not acquire sufficient facility with these instruments to care for patients with eye problems. This is regrettable, for although it may be determined that a patient requires referral to an ophthalmologist, the initial evaluation of ocular symptoms lies within the purview of all physicians, and the assessment of visual acuity, pupils, eye movements, visual fields, and the fundi remain part of any general physical examination.

CLINICAL ASSESSMENT OF VISUAL FUNCTION

REFRACTIVE STATE

In approaching the patient with reduced vision, the first step is to decide whether refractive error is responsible. In *emmetropia*, parallel rays from infinity are focused perfectly upon the retina. Sadly, this condition is enjoyed by only a minority of the population. In *myopia*, the globe is too long, and light rays come to a focal point in front of the retina. Near objects can be seen clearly, but distant objects require a diverging lens in front of the eye. In *hyperopia*, the globe is too short, and hence a converging lens is used to supplement the refractive power of the eye. In *astigmatism*, the corneal surface is not perfectly spherical, necessitating a cylindrical corrective lens. In recent years it has become possible to correct refractive error with the excimer laser by performing either LASIK (laser in situ keratomileusis) or PRK (photorefractive keratectomy) to alter the curvature of the cornea.

With the onset of middle age, *presbyopia* develops as the lens within the eye becomes unable to increase its refractive power to accommodate upon near objects. To compensate for presbyopia, the emmetropic patient must use reading glasses. The patient already wearing glasses for distance correction usually switches to bifocals. The only exception is the myopic patient, who may achieve clear vision at near simply by removing glasses containing the distance prescription.

Refractive errors usually develop slowly and remain stable after adolescence, except in

unusual circumstances. For example, the acute onset of diabetes mellitus can produce sudden myopia because of fluid imbibition and swelling of the lens induced by hyperglycemia. Testing vision through a pinhole aperture is a useful way to screen quickly for refractive error. If the visual acuity is better through a pinhole than with the unaided eye, the patient needs a refraction to obtain best corrected visual acuity.

VISUAL ACUITY

The Snellen chart is used to test acuity at a distance of 6 m (20 ft). For convenience, a scale version of the Snellen chart, called the Rosenbaum card, is held at 36 cm (14 in) from the patient ([Fig. 28-1](#)). All subjects should be able to read the 6/6 m (20/20 ft) line with each eye using their refractive correction, if any. Patients who need reading glasses because of presbyopia must wear them for accurate testing with the Rosenbaum card. If 6/6 (20/20) acuity is not present in each eye, the deficiency in vision must be explained. For acuity worse than 6/240 (20/800), the ability to count fingers, see hand motions, or perceive a bright light should be recorded. Legal blindness is defined by the Internal Revenue Service as a best corrected acuity of 6/60 (20/200) or less in the better eye, or a binocular visual field subtending 20° or less. For driving the laws vary by state, but most require a corrected acuity of 6/12 (20/40) in at least one eye. Patients with a homonymous hemianopia should not drive.

PUPILS

The pupils should be tested individually in dim light with the patient fixating upon a distant target. If they respond briskly to light, there is no need to check the near response, because isolated loss of constriction (miosis) to accommodation does not occur. For this reason, the ubiquitous abbreviation PERRLA (pupils equal, round, and reactive to light and accommodation) implies a wasted effort with the last step. However, it is important to test the near response if the light response is poor or absent. Light-near dissociation occurs with neurosyphilis (Argyll Robertson pupil), lesions of the dorsal midbrain (obstructive hydrocephalus, pineal region tumors), and after aberrant regeneration (oculomotor nerve palsy, Adie's tonic pupil).

An eye with no light perception has no pupillary response to direct light stimulation. If the retina or optic nerve is only partially injured, the direct pupillary response will be weaker than the consensual pupillary response evoked by shining a light into the other eye. This *relative afferent pupillary defect* (Marcus Gunn pupil) can be elicited with the swinging flashlight test ([Fig. 28-2](#)). It is an extremely useful sign in retrobulbar optic neuritis and other optic nerve diseases, where it may be the sole objective evidence for disease.

Subtle inequality in pupil size, up to 0.5 mm, is a fairly common finding in normal persons. The diagnosis of essential or physiologic anisocoria is secure as long as the relative pupil asymmetry remains constant as ambient lighting varies. Anisocoria that increases in dim light indicates a sympathetic paresis of the iris dilator muscle. The triad of miosis with ipsilateral ptosis and anhidrosis constitutes Horner's syndrome, although anhidrosis is an inconstant feature. Brainstem stroke, carotid dissection, or neoplasm impinging upon the sympathetic chain are occasionally identified as the cause of Horner's syndrome, but most of cases are idiopathic.

Anisocoria that increases in bright light suggests a parasympathetic palsy. The first concern is an oculomotor nerve paresis. This possibility is excluded if the eye movements are full and the patient has no ptosis or diplopia. Acute pupillary dilation (mydriasis) can occur from damage to the ciliary ganglion in the orbit. Common mechanisms are infection (herpes zoster, influenza), trauma (blunt, penetrating, surgical), or ischemia (diabetes, temporal arteritis). After denervation of the iris sphincter the pupil does not respond well to light, but the response to near is often relatively intact. When the near stimulus is removed, the pupil redilates very slowly compared with the normal pupil, hence the term *tonic pupil*. In Adie's syndrome, a tonic pupil occurs in conjunction with weak or absent tendon reflexes in the lower extremities. This benign disorder, which occurs predominantly in healthy young women, is assumed to represent a mild dysautonomia. Tonic pupils are also associated with Shy-Drager syndrome, segmental hypohidrosis, diabetes, and amyloidosis. Occasionally, a tonic pupil is discovered incidentally in an otherwise completely normal, asymptomatic individual. The diagnosis is confirmed by placing a drop of dilute (0.125%) pilocarpine into each eye. Denervation hypersensitivity produces pupillary constriction in a tonic pupil, whereas the normal pupil shows no response. Pharmacologic dilation from accidental or deliberate instillation of anticholinergic agents (atropine, scopolamine drops) into the eye can also produce pupillary mydriasis. In this situation, normal strength (1%) pilocarpine causes no constriction.

Both pupils are affected equally by systemic medications. They are small with narcotic use (morphine, heroin) and large with anticholinergics (scopolamine). Parasympathetic agents (pilocarpine, demecarium bromide) used to treat glaucoma produce miosis. In any patient with an unexplained pupillary abnormality, a slit-lamp examination is helpful to exclude surgical trauma to the iris, an occult foreign body, perforating injury, intraocular inflammation, adhesions (synechia), angle-closure glaucoma, and iris sphincter rupture from blunt trauma.

EYE MOVEMENTS AND ALIGNMENT

Eye movements are tested by asking the patient with both eyes open to pursue a small target such as a penlight into the cardinal fields of gaze. Normal ocular versions are smooth, symmetric, full, and maintained in all directions without nystagmus. Saccades, or quick refixation eye movements, are assessed by having the patient look back and forth between two stationary targets. The eyes should move rapidly and accurately in a single jump to their target. Ocular alignment can be judged by holding a penlight directly in front of the patient at about 1 m. If the eyes are straight, the corneal light reflex will be centered in the middle of each pupil. To test eye alignment more precisely, the cover test is useful. The patient is instructed to gaze upon a small fixation target in the distance. One eye is covered suddenly while observing the second eye. If the second eye shifts to fixate upon the target, it was misaligned. If it does not move, the first eye is uncovered and the test is repeated on the second eye. If neither eye moves, the eyes are aligned orthotropically. If the eyes are orthotropic in primary gaze but the patient complains of diplopia, the cover test should be performed with the head tilted or turned in whatever direction elicits the patient's diplopia. With practice the examiner can detect an ocular deviation (heterotropia) as small as 1 to 2° with the cover test. Deviations can be measured by placing prisms in front of the misaligned eye to determine the power

required to neutralize the fixation shift evoked by covering the other eye.

STEREOPSIS

Stereoacuity is determined by presenting targets with retinal disparity separately to each eye using polarized images. The most popular office tests measure a range of thresholds from 800 to 40 seconds of arc. Normal stereoacuity is 40 seconds of arc. If a patient achieves this level of stereoacuity, one is assured that the eyes are aligned orthotropically and that vision is intact in each eye. Random dot stereograms have no monocular depth cues and provide an excellent screening test for strabismus and amblyopia in children.

COLOR VISION

The retina contains three classes of cones, with visual pigments of differing peak spectral sensitivity: red (560 nm), green (530 nm), and blue (430 nm). The red and green cone pigments are encoded on the X chromosome; the blue cone pigment on chromosome 7. Mutations of the blue cone pigment are exceedingly rare. Mutations of the red and green pigments cause congenital X-linked color blindness in 8% of males. Affected individuals are not truly color blind; rather, they differ from normal subjects in how they perceive color and how they combine primary monochromatic lights to match a given color. Anomalous trichromats have three cone types, but a mutation in one cone pigment (usually red or green) causes a shift in peak spectral sensitivity, altering the proportion of primary colors required to achieve a color match. Dichromats have only two cone types and will therefore accept a color match based upon only two primary colors. Anomalous trichromats and dichromats have 6/6 (20/20) visual acuity, but their hue discrimination is impaired. Ishihara color plates can be used to detect red-green color blindness. The test plates contain a hidden number, visible only to subjects with color confusion from red-green color blindness. Because color blindness is almost exclusively X-linked, it is worth screening only male children.

The Ishihara plates are often used to detect acquired defects in color vision, although they are intended as a screening test for congenital color blindness. Acquired defects in color vision frequently result from disease of the macula or optic nerve. For example, patients with a history of optic neuritis often complain of color desaturation long after their visual acuity has returned to normal. Color blindness can also occur from bilateral strokes involving the ventral portion of the occipital lobe (cerebral achromatopsia). Such patients can perceive only shades of gray and may also have difficulty recognizing faces (prosopagnosia). Infarcts of the dominant occipital lobe sometimes give rise to color anomia. Affected patients can discriminate colors, but they cannot name them.

VISUAL FIELDS

Vision can be impaired by damage to the visual system anywhere from the eyes to the occipital lobes. One can localize the site of the lesion with considerable accuracy by mapping the visual field deficit by finger confrontation and then correlating it with the topographic anatomy of the visual pathway ([Fig. 28-3](#)). More quantitative data can be obtained by formal perimetric examination of the visual fields. In kinetic perimetry, the patient faces a tangent screen or a hemispheric bowl (Goldmann perimeter) while the

examiner moves a small light target from the periphery towards the center. Such manual techniques have largely been supplanted by computer-driven perimeters (Humphrey, Octopus) that present a target of variable intensity at fixed positions in the visual field ([Fig. 28-3A](#)). By generating an automated printout of light thresholds, these static perimeters provide a sensitive means of detecting scotomas in the visual field. They are also useful for serial assessment of visual function in chronic diseases such as glaucoma or pseudotumor cerebri.

The crux of visual field analysis is to decide whether a lesion is before, at, or behind the optic chiasm. If a scotoma is confined to one eye, it must be due to a lesion anterior to the chiasm, involving either the optic nerve or retina. Retinal lesions produce scotomas that correspond optically to their location in the fundus. For example, a superior-nasal retinal detachment results in an inferior-temporal field cut. Damage to the macula causes a central scotoma ([Fig. 28-3B](#)).

Optic nerve disease produces characteristic patterns of visual field loss. Glaucoma selectively destroys axons that enter the superotemporal or inferotemporal poles of the optic disc, resulting in arcuate scotomas shaped like a Turkish scimitar, which emanate from the blind spot and curve around fixation to end flat against the horizontal meridian ([Fig. 28-3C](#)). This type of field defect mirrors the arrangement of the nerve fiber layer in the temporal retina. The superb acuity of humans is achieved by thrusting aside all retinal elements at the fovea except photoreceptors, to minimize absorption and scattering of light. To avoid passing over the fovea, axons from cells in the temporal retina must follow an indirect course arching around the fovea to reach the optic disc. Arcuate or nerve fiber layer scotomas also occur from optic neuritis, ischemic optic neuropathy, optic disc drusen, and branch retinal artery or vein occlusion.

Damage to the entire upper or lower pole of the optic disc causes an altitudinal field cut that follows the horizontal meridian ([Fig. 28-3D](#)). This pattern of visual field loss is typical of ischemic optic neuropathy but also occurs from retinal vascular occlusion, advanced glaucoma, and optic neuritis.

About half the fibers in the optic nerve originate from ganglion cells serving the macula. Damage to papillomacular fibers causes a cecocentral scotoma encompassing the blind spot and macula ([Fig. 28-3E](#)). If the damage is irreversible, pallor eventually appears in the temporal portion of the optic disc. Temporal pallor from a cecocentral scotoma may develop in optic neuritis, nutritional optic neuropathy, toxic optic neuropathy, Leber's hereditary optic neuropathy, and compressive optic neuropathy. It is worth mentioning that the temporal side of the optic disc is slightly more pale than the nasal side in most normal individuals. Therefore, it can sometimes be difficult to decide whether the temporal pallor visible on fundus examination represents a pathologic change. Pallor of the nasal rim of the optic disc is a less equivocal sign of optic atrophy.

At the optic chiasm, fibers from nasal ganglion cells decussate into the contralateral optic tract. Crossed fibers are damaged more by compression than uncrossed fibers. As a result, mass lesions of the sellar region cause a temporal hemianopia in each eye. Tumors anterior to the optic chiasm, such as meningiomas of the tuberculum sellae, produce a junctional scotoma characterized by an optic neuropathy in one eye and a superior temporal field cut in the other eye ([Fig. 28-3G](#)). More symmetric compression

of the optic chiasm by a pituitary adenoma ([Fig. 28-4](#)), meningioma, craniopharyngioma, glioma, or aneurysm results in a bitemporal hemianopia ([Fig. 28-3H](#)). The insidious development of a bitemporal hemianopia often goes unnoticed by the patient and will escape detection by the physician unless each eye is tested separately.

It is difficult to localize a postchiasmal lesion accurately, because injury anywhere in the optic tract, lateral geniculate body, optic radiations, or visual cortex can produce a homonymous hemianopia, i.e., a temporal hemifield defect in the contralateral eye and a matching nasal hemifield defect in the ipsilateral eye ([Fig. 28-3I](#)). A unilateral postchiasmal lesion leaves the visual acuity in each eye unaffected, although the patient may read the letters on only the left or right half of the eye chart. Lesions of the optic radiations tend to cause poorly matched or incongruous field defects in each eye. Damage to the optic radiations in the temporal lobe (Meyer's loop) produces a superior quadrantic homonymous hemianopia ([Fig. 28-3J](#)), whereas injury to the optic radiations in the parietal lobe results in an inferior quadrantic homonymous hemianopia ([Fig. 28-3K](#)). Lesions of the primary visual cortex give rise to dense, congruous hemianopic field defects. Occlusion of the posterior cerebral artery supplying the occipital lobe is a frequent cause of total homonymous hemianopia. Some patients with hemianopia after occipital stroke have macular sparing, because the macular representation at the tip of the occipital lobe is supplied by collaterals from the middle cerebral artery ([Fig. 28-3L](#)). Destruction of both occipital lobes produces cortical blindness. This condition can be distinguished from bilateral prechiasmal visual loss by noting that the pupil responses and optic fundi remain normal.

RED OR PAINFUL EYE

Corneal Abrasions These are seen best by placing a drop of fluorescein in the eye and looking with the slit lamp using a cobalt-blue light. A penlight with a blue filter will suffice if no slit lamp is available. Damage to the corneal epithelium is revealed by yellow fluorescence of the exposed basement membrane underlying the epithelium. It is important to check for foreign bodies. To search the conjunctival fornices, the lower lid should be pulled down and the upper lid everted. A foreign body can be removed with a moistened cotton-tipped applicator after placing a drop of topical anesthetic, such as proparacaine, in the eye. Alternatively, it may be possible to flush the foreign body from the eye by irrigating copiously with saline or artificial tears. If the corneal epithelium has been abraded, antibiotic ointment and a patch should be applied to the eye. A drop of an intermediate-acting cycloplegic, such as cyclopentolate hydrochloride 1%, helps to reduce pain by relaxing the ciliary body. The eye should be reexamined the next day. Minor abrasions may not require patching and cycloplegia.

Subconjunctival Hemorrhage This results from rupture of small vessels bridging the potential space between the episclera and conjunctiva. Blood dissecting into this space can produce a spectacular red eye, but vision is not affected and the hemorrhage resolves without treatment. Subconjunctival hemorrhage is usually spontaneous but can occur from blunt trauma, eye rubbing, or vigorous coughing. Occasionally it is a clue to an underlying bleeding disorder.

Pinguecula This is a small, raised conjunctival nodule at the temporal or nasal limbus. In adults such lesions are extremely common and have little significance, unless they

become inflamed (pingueculitis). A *pterygium* resembles a pinguecula but has crossed the limbus to encroach upon the corneal surface. Removal is justified when symptoms of irritation or blurring develop, but recurrence is a common problem.

Blepharitis This refers to inflammation of the eyelids. The most common form occurs in association with acne rosacea or seborrheic dermatitis. The eyelid margins are usually colonized heavily by staphylococcus. Upon close inspection, they appear greasy, ulcerated, and crusted with scaling debris that clings to the lashes. Treatment consists of warm compresses, strict eyelid hygiene, and topical antibiotics such as erythromycin. An external *hordeolum* (sty) is caused by staphylococcal infection of the superficial accessory glands of Zeis or Moll located in the eyelid margins. An internal hordeolum occurs after suppurative infection of the oil-secreting meibomian glands within the tarsal plate of the eyelid. Systemic antibiotics, usually tetracyclines, are sometimes necessary for treatment of meibomian gland inflammation (meibomitis) or chronic, severe blepharitis. A *chalazion* is a painless, granulomatous inflammation of a meibomian gland that produces a pealike nodule within the eyelid. It can be incised and drained, or injected with glucocorticoids. Basal cell, squamous cell, or meibomian gland carcinoma should be suspected for any nonhealing, ulcerative lesion of the eyelids.

Dacrocystitis An inflammation of the lacrimal drainage system, this can produce epiphora (tearing) and ocular injection. Gentle pressure over the lacrimal sac evokes pain and reflux of mucus or pus from the tear puncta. Dacrocystitis usually occurs after obstruction of the lacrimal system. It is treated with topical and systemic antibiotics, followed by probing or surgery to reestablish patency. *Entropion* (inversion of the eyelid) or *ectropion* (sagging or eversion of the eyelid) can also lead to epiphora and ocular irritation.

Conjunctivitis This is the most common cause of a red, irritated eye. Pain is minimal, and the visual acuity is reduced only slightly. The most common viral etiology is adenovirus infection. It causes a watery discharge, mild foreign-body sensation, and photophobia. Bacterial infection tends to produce a more mucopurulent exudate. Mild cases of infectious conjunctivitis are usually treated empirically with broad-spectrum topical ocular antibiotics, such as sulfacetamide 10%, polymixin-bacitracin-neomycin, or trimethoprim-polymixin combination. Smears and cultures are usually reserved for severe, resistant, or recurrent cases of conjunctivitis. To prevent contagion, patients should be admonished to wash their hands frequently, not to touch their eyes, and to avoid direct contact with others.

Allergic Conjunctivitis This condition is extremely common and often mistaken for infectious conjunctivitis. Three forms of allergic conjunctivitis are recognized, with closely overlapping manifestations. *Hay fever conjunctivitis* has a seasonal incidence, related to the release of airborne antigens into the air by plants. IgE-mediated activation of mast cells in the conjunctiva causes itching, redness, and edema. *Vernal conjunctivitis* is also seasonal, becoming worse during warm months. It affects exclusively children or adolescents and is more common in boys. The cause is unknown, but airborne antigens are thought to trigger symptoms. Itching, photophobia, epiphora, and mucous discharge are typical. The palpebral conjunctiva may become hypertrophic with giant excrescences called cobblestone papillae. Irritation from contact lenses or any chronic foreign body can also induce formation of cobblestone papillae.

Atopic conjunctivitis occurs in subjects with atopic dermatitis or asthma. Symptoms caused by allergic conjunctivitis can be alleviated with cold compresses, topical vasoconstrictors, antihistamines, and mast-cell stabilizers such as cromolyn sodium. Topical glucocorticoid solutions provide dramatic relief of immune-mediated forms of conjunctivitis, but their long-term use is ill-advised because of the complications of glaucoma, cataract, and secondary infection. Topical nonsteroidal anti-inflammatory agents (NSAIDs) such as ketorolac tromethamine are a better alternative.

Keratoconjunctivitis Sicca Also known as dry eye, it produces a burning, foreign-body sensation, injection, and photophobia. In mild cases the eye appears surprisingly normal, but tear production measured by wetting of a filter paper (Schirmer strip) is deficient. A variety of systemic drugs, including antihistaminic, anticholinergic, and psychotropic medications, result in dry eye by reducing lacrimal secretion. Disorders that involve the lacrimal gland directly, such as sarcoidosis or Sjogren's syndrome, also cause dry eye. Patients may develop dry eye after radiation therapy if the treatment field includes the orbits. Problems with ocular drying are also common after lesions affecting cranial nerves V or VII. Corneal anesthesia is particularly dangerous, because the absence of a normal blink reflex exposes the cornea to injury without pain to warn the patient. Dry eye is managed by frequent and liberal application of artificial tears and ocular lubricants. In severe cases the tear puncta can be plugged or cauterized to reduce lacrimal outflow.

Keratitis This is a threat to vision because of the risk of corneal clouding, scarring, and perforation. Worldwide, the two leading causes of blindness from keratitis are trachoma from chlamydial infection and vitamin A deficiency related to malnutrition. In the United States, contact lenses play a major role in corneal infection and ulceration. They should not be worn by anyone with an active eye infection. In evaluating the cornea, it is important to differentiate between a superficial infection (*keratoconjunctivitis*) and a deeper, more serious ulcerative process. The latter is accompanied by greater visual loss, pain, photophobia, redness, and discharge. Slit-lamp examination shows disruption of the corneal epithelium, a cloudy infiltrate or abscess in the stroma, and an inflammatory cellular reaction in the anterior chamber. In severe cases, pus settles at the bottom of the anterior chamber, giving rise to a hypopyon. Immediate empirical antibiotic therapy should be initiated after corneal scrapings are obtained for Gram's stain, Giemsa stain, and cultures. Fortified topical antibiotics are most effective, supplemented with subconjunctival antibiotics as required. The most frequent bacterial pathogens are *Staphylococcus*, *Streptococcus* (particularly *S. pneumoniae*), *Pseudomonas*, Enterobacteriaceae, *Haemophilus*, and *Neisseria*. For *Neisseria*, systemic antibiotics should be given in addition to topical antibiotics to eliminate systemic infection. A fungal etiology should always be considered in the patient with keratitis. Fungal infection is common in warm humid climates, especially after penetration of the cornea by plant or vegetable material.

Herpes Simplex The *herpes viruses* are a major cause of blindness from keratitis. Most adults in the United States have serum antibodies to herpes simplex, indicating prior viral infection ([Chap. 182](#)). Primary ocular infection is generally caused by herpes simplex type 1, rather than type 2. It manifests as a unilateral follicular blepharoconjunctivitis, easily confused with adenoviral conjunctivitis unless telltale vesicles appear on the periorcular skin or conjunctiva. A dendritic pattern of corneal

epithelial ulceration revealed by fluorescein staining is pathognomonic for herpes infection but is seen in only a minority of primary infections. Recurrent ocular infection arises from reactivation of the latent herpes virus. Viral eruption in the corneal epithelium may result in the characteristic herpes dendrite. Involvement of the corneal stroma produces edema, vascularization, and iridocyclitis. Herpes keratitis is treated with topical antiviral agents, cycloplegics, and oral acyclovir. Topical glucocorticoids are effective in mitigating corneal scarring but must be used with extreme caution because of the danger of corneal melting and perforation. Topical glucocorticoids also carry the risk of prolonging infection and inducing glaucoma.

Herpes Zoster Herpes zoster from reactivation of latent varicella (chickenpox) virus causes a dermatomal pattern of painful vesicular dermatitis. Ocular symptoms can occur after zoster eruption in any branch of the trigeminal nerve but are particularly common when vesicles form on the nose, reflecting nasociliary (V1) nerve involvement (Hutchinson's sign). Herpes zoster ophthalmicus produces corneal dendrites, which can be difficult to distinguish from those seen in herpes simplex. Stromal keratitis, anterior uveitis, raised intraocular pressure, ocular motor nerve palsies, acute retinal necrosis, and postherpetic scarring and neuralgia are other common sequelae. Herpes zoster ophthalmicus is treated with antiviral agents and cycloplegics. In severe cases, glucocorticoids may be added to prevent permanent visual loss from corneal scarring.

Episcleritis This is an inflammation of the episclera, a thin layer of connective tissue between the conjunctiva and sclera. Episcleritis resembles conjunctivitis but is a more localized process and discharge is absent. Most cases of episcleritis are idiopathic, but some occur in the setting of an autoimmune disease. *Scleritis* refers to a deeper, more severe inflammatory process, frequently associated with a connective tissue disease such as rheumatoid arthritis, lupus erythematosus, polyarteritis nodosa, Wegener's granulomatosis, or relapsing polychondritis. The inflammation and thickening of the sclera can be diffuse or nodular. In anterior forms of scleritis, the globe assumes a violet hue and the patient complains of severe ocular tenderness and pain. With posterior scleritis the pain and redness may be less marked, but there is often proptosis, choroidal effusion, reduced motility, and visual loss. Episcleritis and scleritis should be treated with [NSAIDs](#). If these agents fail, topical or even systemic glucocorticoid therapy may be necessary, especially if an underlying autoimmune process is active.

Uveitis Involving the anterior structures of the eye, this is called *iritis* or *iridocyclitis*. The diagnosis requires slit-lamp examination to identify inflammatory cells floating in the aqueous humor or deposited upon the corneal endothelium (keratic precipitates). Anterior uveitis develops in sarcoidosis, ankylosing spondylitis, juvenile rheumatoid arthritis, inflammatory bowel disease, psoriasis, Reiter's syndrome, and Behcet's disease. It is also associated with herpes infections, syphilis, Lyme disease, onchocerciasis, tuberculosis, and leprosy. Although anterior uveitis can occur in conjunction with many diseases, no cause is found to explain the majority of cases. For this reason, laboratory evaluation is usually reserved for patients with recurrent or severe anterior uveitis. Treatment is aimed at reducing inflammation and scarring by judicious use of topical glucocorticoids. Dilation of the pupil reduces pain and prevents the formation of synechiae.

Posterior Uveitis This is diagnosed by observing inflammation of the vitreous, retina, or

choroid on fundus examination. It is more likely than anterior uveitis to be associated with an identifiable systemic disease. Some patients have panuveitis, or inflammation of both the anterior and posterior segments of the eye. Posterior uveitis is a manifestation of autoimmune diseases such as sarcoidosis, Behcet's disease, Vogt-Koyanagi-Harada syndrome, and inflammatory bowel disease (see [Plate IV-1](#)). It also accompanies diseases such as toxoplasmosis, onchocerciasis, cysticercosis, coccidioidomycosis, toxocariasis, and histoplasmosis; infections caused by organisms such as *Candida*, *Pneumocystis carinii*, *Cryptococcus*, *Aspergillus*, herpes, and cytomegalovirus (see [Plate IV-2](#)); and other diseases such as syphilis, Lyme disease, tuberculosis, cat-scratch disease, Whipple's disease, and brucellosis. In multiple sclerosis, chronic inflammatory changes can develop in the extreme periphery of the retina (pars planitis or intermediate uveitis).

Acute Angle-Closure Glaucoma This is a rare and frequently misdiagnosed cause of a red, painful eye. Susceptible eyes have a shallow anterior chamber, either because the eye has a short axial length (hyperopia) or a lens enlarged by the gradual development of cataract. When the pupil becomes mid-dilated, the peripheral iris blocks aqueous outflow via the anterior chamber angle and the intraocular pressure rises abruptly, producing pain, injection, corneal edema, obscurations, and blurred vision. In some patients, ocular symptoms are overshadowed by nausea, vomiting, or headache, prompting a fruitless workup for abdominal or neurologic disease. The diagnosis is made by measuring the intraocular pressure during an acute attack or by performing gonioscopy to reveal the narrowed chamber angle by means of a specially mirrored contact lens. Acute angle closure is treated with oral or intravenous acetazolamide, topical beta blockers, apraclonidine, and pilocarpine to induce miosis. If these measures fail, a laser can be used to create a hole in the peripheral iris to relieve pupillary block. Many physicians are reluctant to dilate patients routinely for fundus examination because they fear precipitating an angle-closure glaucoma. The risk is actually remote and more than outweighed by the potential benefit to patients of discovering a hidden fundus lesion visible only through a fully dilated pupil. Moreover, a single attack of angle closure after pharmacologic dilation rarely causes any permanent damage to the eye and serves as an inadvertent provocative test to identify patients with narrow angles who would benefit from prophylactic laser iridectomy.

Endophthalmitis This occurs from bacterial, viral, fungal, or parasitic infection of the internal structures of the eye. It is usually acquired by hematogenous seeding from a remote site. Chronically ill, diabetic, or immunosuppressed patients, especially those with a history of indwelling intravenous catheters or positive blood cultures, are at greatest risk for endogenous endophthalmitis. Although most patients have ocular pain and injection, visual loss is sometimes the only symptom. Septic emboli, from a diseased heart valve or a dental abscess, that lodge in the retinal circulation can give rise to endophthalmitis. White-centered retinal hemorrhages (Roth's spots) are considered pathognomonic for subacute bacterial endocarditis, but they also appear in leukemia, diabetes, and many other conditions. Endophthalmitis also occurs as a complication of ocular surgery, occasionally months or even years after the operation. An occult penetrating foreign body or unrecognized trauma to the globe should be considered in any patient with unexplained intraocular infection or inflammation.

TRANSIENT OR SUDDEN VISUAL LOSS

Amaurosis Fugax This term refers to a transient ischemic attack of the retina. Because neural tissue has a high rate of metabolism, interruption of blood flow to the retina for more than a few seconds results in *transient monocular blindness*, a term used interchangeably with amaurosis fugax. Patients describe a rapid fading of vision like a curtain descending, sometimes affecting only a portion of the visual field. Amaurosis fugax usually occurs from an embolus that becomes stuck within a retinal arteriole (see [Plate IV-3](#)). If the embolus breaks up or passes, flow is restored and vision returns quickly to normal without permanent damage. With prolonged interruption of blood flow, the inner retina suffers infarction. Ophthalmoscopy reveals zones of whitened, edematous retina following the distribution of branch retinal arterioles. Complete occlusion of the central retinal artery produces arrest of blood flow and a milky retina with a cherry-red fovea (see [Plate IV-4](#)). Emboli are composed of either cholesterol (Hollenhorst plaque), calcium, or platelet-fibrin debris. The most common source is an atherosclerotic plaque in the carotid artery or aorta, although emboli can also arise from the heart, especially in patients with diseased valves, atrial fibrillation, or wall motion abnormalities.

In rare instances, amaurosis fugax occurs from low central retinal artery perfusion pressure in a patient with a critical stenosis of the ipsilateral carotid artery and poor collateral flow via the circle of Willis. In this situation, amaurosis fugax develops when there is a dip in systemic blood pressure or a slight worsening of the carotid stenosis. Sometimes there is contralateral motor or sensory loss, indicating concomitant hemispheric cerebral ischemia.

Retinal arterial occlusion also occurs rarely in association with retinal migraine, lupus erythematosus, anticardiolipin antibodies (see [Plate IV-4](#)), anticoagulant deficiency states (protein S, protein C, and antithrombin III deficiency), pregnancy, intravenous drug abuse, blood dyscrasias, dysproteinemias, and temporal arteritis.

Amaurosis fugax warns of a patient at high risk for stroke. The carotid arteries should be studied by ultrasound. Endarterectomy for a stenosis of $\geq 60\%$, even in asymptomatic patients, has been shown to reduce the subsequent rate of ipsilateral stroke ([Chap. 361](#)). Therapy with aspirin, warfarin, or other anticoagulants is appropriate in selected patients. If no carotid lesion is found, cardiac ultrasound should be performed. Ambulatory electrocardiographic monitoring may reveal that intermittent atrial fibrillation is giving rise to emboli.

Marked *systemic hypertension* causes sclerosis of retinal arterioles, splinter hemorrhages, focal infarcts of the nerve fiber layer (cotton-wool spots), and leakage of lipid and fluid (hard exudate) into the macula (see [Plate IV-5](#)). In hypertensive crisis, sudden visual loss can result from vasospasm of retinal arterioles and consequent retinal ischemia. In addition, acute hypertension may produce visual loss from ischemic swelling of the optic disc. Patients with acute hypertensive retinopathy should be treated by lowering the blood pressure. However, the blood pressure should not be reduced precipitously, because there is a danger of optic disc infarction from sudden hypoperfusion.

Impending *branch or central retinal vein occlusion* can produce prolonged visual

obscurations that resemble those described by patients with amaurosis fugax. The veins appear engorged and phlebitic, with numerous retinal hemorrhages (see [Plate IV-6](#)). In some patients, venous blood flow recovers spontaneously, while others evolve a frank obstruction with extensive retinal bleeding ("blood and thunder" appearance), infarction, and visual loss. Venous occlusion of the retina is often idiopathic, but hypertension, diabetes, and glaucoma are prominent risk factors. The benefit of treatment with anticoagulants is unproven and carries the risk of hemorrhage into the vitreous. Polycythemia, thrombocythemia, or other factors leading to an underlying hypercoagulable state should be corrected.

Anterior Ischemic Optic Neuropathy (AION) This is caused by insufficient blood flow through the posterior ciliary arteries supplying the optic disc. It produces sudden, painless, monocular visual loss, although patients occasionally report premonitory obscurations. The optic disc appears swollen and surrounded by nerve fiber layer splinter hemorrhages (see [Plate IV-7](#)). AION is divided into two forms: arteritic and nonarteritic. The nonarteritic form of AION is most common. No specific cause can be identified, although diabetes and hypertension are frequent risk factors. No treatment is available. About 5% of patients, especially those over age 60, develop the arteritic form of AION in conjunction with giant cell (temporal) arteritis ([Chap. 317](#)). It is urgent to recognize arteritic AION so that high doses of glucocorticoids can be instituted immediately to prevent blindness in the second eye. Symptoms of polymyalgia rheumatica may be present, and the sedimentation rate is usually elevated. In a patient with visual loss from suspected arteritic AION, temporal artery biopsy is helpful to confirm the diagnosis, but glucocorticoids should be started without waiting for the biopsy to be completed. The diagnosis of arteritic AION is difficult to sustain in the face of a normal sedimentation rate and a negative temporal artery biopsy, but such cases do occur rarely.

Posterior Ischemic Optic Neuropathy This is an infrequent cause of acute visual loss. It is induced by the combination of severe anemia and hypotension, causing infarction of the retrobulbar optic nerve. Cases have been reported after major blood loss during surgery, exsanguinating trauma, gastrointestinal bleeding, and renal dialysis. The fundus usually appears normal, although optic disc swelling develops if the process extends far enough anteriorly. Vision can be salvaged in some patients by prompt blood transfusion and reversal of hypotension.

Optic Neuritis This is a common inflammatory disease of the optic nerve. In the Optic Neuritis Treatment Trial (ONTT), the mean age of patients was 32 years, 77% were female, 92% had ocular pain (especially with eye movements), and 35% had optic disc swelling. In most patients, the demyelinating event was retrobulbar and the ocular fundus appeared normal on initial examination (see [Plate IV-8](#)), although optic disc pallor slowly developed over subsequent months.

Virtually all patients experience a gradual recovery of vision after a single episode of optic neuritis, even without treatment. This rule is so reliable that failure of vision to improve considerably after a first attack of optic neuritis casts doubt upon the original diagnosis. Treatment of optic neuritis is controversial because the favorable prognosis for visual recovery has made it difficult to demonstrate any benefit from glucocorticoids. The [ONTT](#) showed that patients treated with a conventional dose of oral glucocorticoids

(prednisone, 1 mg/kg per day for 14 days) did no better than patients treated with a placebo. A recent Danish trial of oral high-dose methylprednisolone (500 mg daily for 5 days, followed by a 10-day taper) reported a slight response at 1 and 3 weeks but none at 8 weeks. From these studies, it is apparent that oral glucocorticoids have little to offer in the treatment of optic neuritis. According to the ONTT, even high-dose intravenous methylprednisolone (250 mg every 6 h for 3 days) followed by oral prednisone (1 mg/kg per day for 11 days) makes no difference in final acuity (measured 6 months after the attack), although the recovery of visual function occurs more rapidly.

For some patients, optic neuritis remains an isolated event. However, the [ONTT](#) showed that the 5-year cumulative probability of developing clinically definite multiple sclerosis following optic neuritis is 30%. Remarkably, intravenous glucocorticoids were associated with a reduced rate of development of multiple sclerosis over a 2-year follow-up period, especially in the subgroup of patients with multiple foci of demyelination on their magnetic resonance (MR) scan. However, by the end of a 3-year follow-up period, patients treated with intravenous glucocorticoids versus placebo showed no difference in the rate of multiple sclerosis. Moreover, intravenous glucocorticoids did not reduce the likelihood of subsequent attacks of optic neuritis. To summarize, the organizers of the ONTT recommend an MR scan in patients with optic neuritis. If two or more foci of demyelination are found or visual loss is severe, they suggest treatment with intravenous glucocorticoids. The potential benefits of intravenous glucocorticoids are: (1) a slightly faster recovery of visual function, and (2) a potential reduction in the risk of subsequent neurologic events that would signify multiple sclerosis. Critics of the ONTT have questioned these recommendations, pointing out that: (1) visual outcome is the same in the long run, (2) evidence indicating a reduced risk of eventual multiple sclerosis with intravenous glucocorticoid treatment is based upon follow-up data in a rather small number of patients, and (3) the protection against multiple sclerosis is transient, and no longer apparent beyond 2 years of follow-up. In cases of unilateral optic neuritis, the decision whether to obtain an MR scan or to treat with intravenous glucocorticoids should be based upon clinical judgment and careful discussion with the patient. In cases of bilateral, simultaneous optic neuritis, the rationale for intravenous glucocorticoids is stronger.

Leber's Hereditary Optic Neuropathy This is a disease of young men, characterized by onset over a few weeks of painless, severe, central visual loss in one eye, followed weeks or months later by the same process in the other eye. Acutely, the optic disc appears mildly plethoric with surface capillary telangiectases, but no vascular leakage on fluorescein angiography. Eventually optic atrophy ensues. There is no treatment. Leber's optic neuropathy is caused by a point mutation at codon 11778 in the mitochondrial gene encoding nicotinamide adenine dinucleotide dehydrogenase (NADH) subunit 4. Subsequently, additional mutations responsible for the disease have been identified, most in mitochondrial genes encoding proteins involved in electron transport. Mitochondrial mutations causing Leber's neuropathy are inherited from the mother by all her children, but usually only sons develop symptoms. This curious male predilection is a mystery.

Toxic Optic Neuropathy This can result in acute visual loss with bilateral optic disc swelling and central or cecocentral scotomas. Such cases have been reported to result from exposure to ethambutol, methyl alcohol (moonshine), ethylene glycol (antifreeze),

or carbon monoxide. In toxic optic neuropathy, visual loss can also develop gradually and produce optic atrophy without a phase of acute optic disc edema (see [Plate IV-9](#)). Many agents have been implicated as a cause of toxic optic neuropathy, but the evidence supporting the association for many is weak. The following is a partial list of potential offending drugs or toxins: disulfiram, ethchlorvynol, chloramphenicol, amiodarone, monoclonal anti-CD3 antibody, ciprofloxacin, digitalis, streptomycin, lead, arsenic, thallium, D-penicillamine, isoniazid, emetine, and sulfonamides. Deficiency states, induced either by starvation, malabsorption, or alcoholism, can lead to insidious visual loss. Thiamine, vitamin B₁₂, and folate levels should be checked in any patient with unexplained, bilateral central scotomas and optic pallor.

Papilledema This connotes bilateral optic disc swelling from raised intracranial pressure (see [Plate IV-10](#)). Headache is a frequent, but not invariable, accompaniment. All other forms of optic disc swelling, e.g., from optic neuritis or ischemic optic neuropathy, should be called "optic disc edema." This convention is arbitrary but serves to avoid confusion. Often it is difficult to differentiate papilledema from other forms of optic disc edema by fundus examination alone. Transient visual obscurations are a classic symptom of papilledema. They can occur in only one eye or simultaneously in both eyes. They usually last seconds but can persist for minutes if the papilledema is fulminant. Obscurations follow abrupt shifts in posture or happen spontaneously. When obscurations are prolonged or spontaneous, the papilledema is more threatening. Visual acuity is not affected by papilledema unless the papilledema is severe, long-standing, or accompanied by macular edema and hemorrhage. Visual field testing shows enlarged blind spots and peripheral constriction ([Fig. 28-3F](#)). With unremitting papilledema, peripheral visual field loss progresses in an insidious fashion while the optic nerve develops atrophy. In this setting, reduction of optic disc swelling is an ominous sign of a dying nerve rather than an encouraging indication of resolving papilledema.

Evaluation of papilledema requires computed tomography (CT) or [MR](#) imaging to exclude an intracranial lesion. MR angiography is appropriate in selected cases to search for a dural venous sinus occlusion or an arteriovenous shunt. If neuroradiologic studies are negative, the subarachnoid opening pressure should be measured by lumbar puncture. An elevated pressure, with normal cerebrospinal fluid, points by exclusion to the diagnosis of *pseudotumor cerebri* (idiopathic intracranial hypertension). The majority of patients are young, female, and obese. Treatment with a carbonic anhydrase inhibitor such as acetazolamide lowers intracranial pressure by reducing the production of cerebrospinal fluid. Weight reduction is vital but often unsuccessful. If acetazolamide and weight loss fail, and visual field loss is progressive, lumboperitoneal shunting or optic nerve sheath fenestration should be undertaken without delay to prevent blindness. Occasionally, emergency surgery is required for sudden blindness caused by fulminant papilledema.

Optic Disc Drusen These are refractile deposits within the substance of the optic nerve head (see [Plate IV-11](#)). They are unrelated to drusen of the retina, which occur in age-related macular degeneration. Optic disc drusen are most common in people of northern European descent, with an incidence of 0.3 to 0.4%. Their diagnosis is obvious when they are visible as glittering particles upon the surface of the optic disc. However, in many patients they are hidden beneath the surface, producing an elevated optic disc with blurred margins that is easily mistaken for papilledema. It is important to recognize

pseudo-papilledema due to optic disc drusen to avoid an unnecessary evaluation for papilledema. Ultrasound or [CT](#) scanning are sensitive for detection of buried optic disc drusen because they contain calcium. In most patients, optic disc drusen are an incidental, innocuous finding, but they can produce visual obscurations. On perimetry they give rise to enlarged blind spots and arcuate scotomas from damage to the optic disc. With increasing age, drusen tend to become more exposed on the disc surface as optic atrophy develops. Hemorrhage, choroidal neovascular membrane, and [AION](#) are more likely to occur in patients with optic disc drusen. No treatment for drusen is available.

Vitreous Degeneration This occurs in all individuals with advancing age, leading to chronic and acute visual symptoms. Opacities develop in the vitreous, casting annoying shadows upon the retina. As the eye moves, these distracting "floaters" move synchronously, with a slight lag caused by inertia of the vitreous gel. Vitreous traction upon the retina causes mechanical stimulation, resulting in perception of flashing lights. This photopsia is brief and confined to one eye, in contrast to the bilateral, prolonged scintillations of cortical migraine. Contraction of the vitreous can result in sudden separation from the retina, heralded by an alarming shower of floaters and photopsia. This process, known as *vitreous detachment*, is a frequent involutional event in the elderly. It is not harmful unless it damages the retina. A careful examination of the dilated fundus is mandatory in any patient complaining of floaters or photopsia to search for peripheral tears or holes. If such a lesion is found, laser application or cryotherapy can forestall a retinal detachment. Occasionally a tear ruptures a retinal blood vessel, causing vitreous hemorrhage and sudden loss of vision. On attempted ophthalmoscopy the fundus is hidden by a dark red haze of blood. Ultrasound is required to examine the interior of the eye for a retinal tear or detachment. If the hemorrhage does not resolve spontaneously, the vitreous can be removed surgically. Vitreous hemorrhage also occurs from the fragile neovascular vessels that proliferate on the surface of the retina in diabetes, sickle cell anemia, and other ischemic ocular diseases.

Retinal Detachment This produces symptoms of floaters, flashing lights, and a scotoma in the peripheral visual field corresponding to the detachment (see [Plate IV-12](#)). If the detachment includes the fovea, there is an afferent pupil defect and the visual acuity is reduced. In most eyes, retinal detachment starts with a hole, flap, or tear in the peripheral retina (rhegmatogenous retinal detachment). Patients with peripheral retinal thinning (lattice degeneration) are particularly vulnerable to this process. Once a break has developed in the retina, liquified vitreous is free to enter the subretinal space, separating the retina from the pigment epithelium. The combination of vitreous traction upon the retinal surface and passage of fluid behind the retina leads inexorably to detachment. Patients with a history of myopia, trauma, or prior cataract extraction are at greatest risk for retinal detachment. The diagnosis is confirmed by ophthalmoscopic examination of the dilated eye.

Classic Migraine (See also [Chap. 15](#)) This usually occurs with a visual aura lasting about 20 min. In a typical attack, a small central disturbance in the field of vision marches toward the periphery, leaving a transient scotoma in its wake. The expanding border of migraine scotoma has a scintillating, dancing, or zig-zag edge, resembling the bastions of a fortified city, hence the term "fortification spectra." Patients' descriptions of fortification spectra vary widely and can be confused with amaurosis fugax. Migraine

patterns usually last longer and are perceived in both eyes, whereas amaurosis fugax is briefer and occurs in only one eye. Migraine phenomena also remain visible in the dark or with the eyes closed. Generally they are confined to either the right or left visual hemifield, but sometimes both fields are involved simultaneously. Patients often have a long history of stereotypic attacks. After the visual symptoms recede, headache develops in most patients.

Transient ischemic attacks from *vertebrobasilar insufficiency* result in acute homonymous visual symptoms. Many patients mistakenly describe symptoms in their left or right eye, when in fact they are occurring in the left or right hemifield of both eyes. Interruption of blood supply to the visual cortex causes a sudden fogging or graying of vision, occasionally with flashing lights or other positive phenomena that mimic migraine. Cortical ischemic attacks are briefer in duration than migraine, occur in older patients, and are not followed by headache. There may be associated signs of brainstem ischemia, such as diplopia, vertigo, numbness, weakness, or dysarthria.

Stroke This occurs when interruption of blood supply from the posterior cerebral artery to the visual cortex is prolonged. The only finding on examination is a homonymous visual field defect that stops abruptly at the vertical meridian. Occipital lobe stroke is usually due to thrombotic occlusion of the vertebrobasilar system, embolus, or dissection. Lobar hemorrhage, tumor, abscess, and arteriovenous malformation are other common causes of hemianopic cortical visual loss.

Factitious (Functional, Nonorganic) Visual Loss This is claimed by hysterics or malingerers. The latter comprise the vast majority, seeking sympathy, special treatment, or financial gain by feigning loss of sight. The diagnosis is suspected when the history is atypical, physical findings are lacking or contradictory, inconsistencies emerge on testing, and a secondary motive can be identified. In our litigious society, the fraudulent pursuit of recompense has spawned an epidemic of factitious visual loss.

CHRONIC VISUAL LOSS

Cataract This is a clouding of the lens sufficient to reduce vision. Most cataracts develop slowly as a result of aging, leading to gradual impairment of vision. The formation of cataract occurs more rapidly in patients with a history of ocular trauma, uveitis, or diabetes mellitus. Cataracts are acquired in a variety of genetic diseases, such as myotonic dystrophy, neurofibromatosis type 2, and galactosemia. Radiation therapy and glucocorticoid treatment can induce cataract as a side effect. The cataracts associated with radiation or glucocorticoids have a typical posterior subcapsular location. Cataract can be detected by noting an impaired red reflex when viewing light reflected from the fundus with an ophthalmoscope or by examining the dilated eye using the slit lamp.

The only treatment for cataract is surgical extraction of the opacified lens. Over a million cataract operations are performed each year in the United States. The operation is generally done under local anesthesia on an outpatient basis. Remarkable technical innovations have made it possible to aspirate the cataract while leaving the lens capsule intact (extracapsular cataract extraction), rather than removing the entire lens with its capsule (intracapsular cataract extraction). A plastic or silicone intraocular lens is then

placed within the empty lens capsule in the posterior chamber, substituting for the natural lens, and leading to rapid recovery of sight. More than 95% of patients who undergo cataract extraction can expect an improvement in vision. In many patients, the lens capsule remaining in the eye after cataract extraction eventually turns cloudy, causing a secondary loss of vision. A small opening is made in the lens capsule with a laser to restore clarity.

Glaucoma This is a slowly progressive, insidious optic neuropathy, usually associated with chronic elevation of intraocular pressure. In Americans of African descent it is the leading cause of blindness. The mechanism whereby raised intraocular pressure injures the optic nerve is not understood. Axons entering the inferotemporal and superotemporal aspects of the optic disc are damaged first, producing typical nerve fiber bundle or arcuate scotomas on perimetric testing. As fibers are destroyed, the neural rim of the optic disc shrinks and the physiologic cup within the optic disc enlarges (see [Plate IV-13](#)). This process is referred to colloquially as pathologic "cupping." The cup-to-disc diameter is expressed as a ratio, e.g., 0.2/1. The cup-to-disc ratio ranges widely in normal individuals, making it difficult to diagnose glaucoma reliably simply by observing an unusually large or deep optic cup. Careful documentation of serial prospective examinations is helpful. In the patient with physiologic cupping, the large cup remains stable, whereas in the patient with glaucoma it expands relentlessly over the years. Detection of visual field loss on formal perimetry also contributes to the diagnosis of glaucoma. Finally, most patients with glaucoma have raised intraocular pressure. However, a surprising number of patients with typical glaucomatous cupping and visual field loss have intraocular pressures that apparently never exceed the normal limit of 20 mmHg (so-called low-tension glaucoma).

In acute angle-closure glaucoma, the eye is red and painful due to abrupt, severe elevation of intraocular pressure. Such cases account for only a handful of patients with glaucoma. Most patients with glaucoma have open, nonoccludable anterior chamber angles. The cause of raised intraocular pressure in these patients is uncertain. Recent studies have implicated mutations in a gene encoding a glycoprotein expressed in the trabecular meshwork. This structure serves as a filter to drain aqueous from the eye. Because the elevation of intraocular pressure develops gradually and is less marked than in angle-closure glaucoma, there is no pain or ocular injection. The central visual field and foveal acuity are spared until end-stage disease is reached. For these reasons, severe and irreversible damage can occur before either the patient or physician recognizes the diagnosis. Screening of patients for glaucoma by noting the cup-to-disc ratio on ophthalmoscopy and by measuring intraocular pressure (using a Schiotz, Tonopen, air-puff, or Goldmann tonometer) is vital. Glaucoma is treated with topical adrenergic agonists (epinephrine, dipivefrin, apraclonidine, brimonidine), cholinergic agonists (pilocarpine), beta blockers (betaxolol, carteolol, levobunolol, metipranolol, and timolol), and prostaglandin analogues (latanaprost). Occasionally, systemic absorption of beta blocker from eye drops can be sufficient to cause side effects of bradycardia, hypotension, heart block, bronchospasm, impotence, or depression. Topical or oral carbonic anhydrase inhibitors are used to lower intraocular pressure by reducing aqueous production. Laser treatment of the trabecular meshwork in the anterior chamber angle improves aqueous outflow from the eye. If medical or laser treatments fail to halt optic nerve damage from glaucoma, a filter must be constructed surgically (trabeculectomy) to release aqueous from the eye in a controlled fashion.

Macular Degeneration This is a major cause of gradual, painless, bilateral central visual loss in the elderly. The old term, "senile macular degeneration," misinterpreted by many patients as an unflattering reference, has been replaced with "age-related macular degeneration." It occurs in a nonexudative (dry) form and an exudative (wet) form. The nonexudative process begins with the accumulation of extracellular deposits, called drusen, underneath the retinal pigment epithelium. On ophthalmoscopy, they are pleomorphic but generally appear as small discrete yellow lesions clustered in the macula (see [Plate IV-14](#)). With time they become larger, more numerous, and confluent. The retinal pigment epithelium becomes focally detached and atrophic, causing visual loss by interfering with photoreceptor function. There is currently no way to prevent the development of age-related macular degeneration. Concoctions of various vitamins (A, C, and E) and minerals (zinc, copper, and selenium) have been marketed, without good evidence that they retard the process of macular degeneration.

Exudative macular degeneration, which develops in only a minority of patients, occurs when neovascular vessels from the choroid grow through defects in Bruch's membrane into the potential space beneath the retinal pigment epithelium. Leakage from these vessels produces elevation of the retina and pigment epithelium, with distortion (metamorphopsia) and blurring of vision. Although onset of these symptoms is usually gradual, bleeding from subretinal choroidal neovascular membranes sometimes causes acute visual loss. The neovascular membranes can be difficult to see on fundus examination because they are beneath the retina. Fluorescein or indocyanine green angiography is extremely useful for their detection. In some patients, prompt laser ablation of choroidal neovascular membranes seen on fluorescein angiography can halt the exudative process. However, the neovascular membranes frequently recur, requiring constant vigilance and repeated photocoagulation.

Major or repeated hemorrhage under the retina from neovascular membranes results in fibrosis, development of a round (disciform) macular scar, and permanent loss of central vision. Surgical attempts to remove subretinal membranes in age-related macular degeneration have not improved vision in most patients. However, outcomes have been more encouraging for patients with choroidal neovascular membranes from ocular histoplasmosis syndrome.

Central Serous Chorioretinopathy This primarily affects males between the ages of 20 and 50. Leakage of serous fluid from the choroid causes small, localized detachment of the retinal pigment epithelium and the neurosensory retina. These detachments produce acute or chronic symptoms of metamorphopsia and blurred vision when the macula is involved. They are difficult to visualize with a direct ophthalmoscope because the detached retina is transparent and only slightly elevated. Diagnosis of central serous chorioretinopathy is made easily by fluorescein angiography, which shows dye streaming into the subretinal space. The cause of central serous chorioretinopathy is unknown. Symptoms may resolve spontaneously if the retina reattaches, but recurrent detachment is common. Laser photocoagulation has benefited some patients with this condition.

Diabetic Retinopathy A rare disease until 1921, when the discovery of insulin resulted in a dramatic improvement in life expectancy for patients with diabetes mellitus, it is now

a leading cause of blindness in the United States. The retinopathy of diabetes takes years to develop but eventually appears in nearly all cases. Regular surveillance of the dilated fundus is crucial for any patient with diabetes. In advanced diabetic retinopathy, the proliferation of neovascular vessels leads to blindness from vitreous hemorrhage, retinal detachment, and glaucoma (see [Plate IV-15](#)). These complications can be avoided in most patients by administration of panretinal laser photocoagulation at the appropriate point in the evolution of the disease. **For further discussion of the manifestations and management of diabetic retinopathy, see Chap. 333.*

Retinitis Pigmentosa This is a general term for a disparate group of rod and cone dystrophies characterized by progressive night blindness (nyctalopia), visual field constriction with a ring scotoma, loss of acuity, and an abnormal electroretinogram (ERG). It occurs sporadically or in an autosomal recessive, dominant, or X-linked pattern. Irregular black deposits of clumped pigment in the peripheral retina, called bone spicules because of their vague resemblance to the spicules of cancellous bone, give the disease its name (see [Plate IV-16](#)). The name is actually a misnomer because retinitis pigmentosa is not an inflammatory process. Most cases are due to a mutation in the gene for rhodopsin, the rod photopigment, or in the gene for peripherin, a glycoprotein located in photoreceptor outer segments. There is no effective treatment for retinitis pigmentosa. Vitamin A (15,000 IU/day) slightly retards the deterioration of the ERG but has no beneficial effect upon visual acuity or visual fields. Some forms of retinitis pigmentosa occur in association with rare, hereditary systemic diseases (olivopontocerebellar degeneration, Bassen-Kornzweig disease, Kearns-Sayre syndrome, Refsum's disease). Chronic treatment with chloroquine, hydroxychloroquine, and phenothiazines (especially thioridazine) can produce visual loss from a toxic retinopathy that resembles retinitis pigmentosa.

Epiretinal Membrane This is a fibrocellular tissue that grows across the inner surface of the retina, causing metamorphopsia and reduced visual acuity from distortion of the macula. With the ophthalmoscope one can see a crinkled, cellophane-like membrane on the retina. Epiretinal membrane is most common in patients over 50 years of age and is usually unilateral. Most cases are idiopathic, but some occur as a result of hypertensive retinopathy, diabetes, retinal detachment, or trauma. When visual acuity is reduced to the level of about 6/24 (20/80), vitrectomy and surgical peeling of the membrane to relieve macular puckering are recommended. Contraction of an epiretinal membrane sometimes gives rise to a *macular hole*. Most macular holes, however, are caused by local vitreous traction within the fovea. Vision is usually depressed to the level of 6/30 (20/100) or worse. Vitrectomy may improve visual acuity in some patients with macular hole. Fortunately, fewer than 10% of patients with a macular hole develop a hole in their other eye.

Melanoma and Other Tumors Melanoma is the most common primary tumor of the eye (see [Plate IV-17](#)). It causes photopsia, an enlarging scotoma, and loss of vision. A small melanoma is often difficult to differentiate from a benign choroidal nevus. Careful serial examinations are required to document a malignant pattern of growth. Treatment of melanoma is controversial. Options include enucleation, local resection, and irradiation. *Metastatic tumors* to the eye outnumber primary tumors of uveal origin. Breast and lung carcinoma have a special propensity to spread to the choroid or iris. Leukemia and lymphoma also commonly invade ocular tissues. Sometimes their only

sign on eye examination is cellular debris in the vitreous, which can masquerade as a chronic posterior uveitis. *Retrobulbar tumor* of the optic nerve (meningioma, glioma) or *chiasmal tumor* (pituitary adenoma, meningioma) produces gradual visual loss with few objective findings, except for optic disc pallor. Rarely, sudden expansion of a pituitary adenoma from infarction and bleeding (*pituitary apoplexy*) causes acute retrobulbar visual loss, with headache, nausea, and ocular motor nerve palsies. In any patient with visual field loss or optic atrophy, [CT](#) or [MR](#) scanning should be considered if the cause remains unknown after careful review of the history and thorough examination of the eye ([Fig. 28-4](#)).

PROPTOSIS

When the globes appear asymmetric, the clinician must first decide which eye is abnormal. Is one eye recessed within the orbit (*enophthalmos*) or is the other eye protuberant (*exophthalmos*, or *proptosis*)? A small globe or a Horner's syndrome can give the appearance of enophthalmos. True enophthalmos occurs commonly after trauma, from atrophy of retrobulbar fat, or fracture of the orbital floor. The position of the eyes within the orbits is measured using a Hertel exophthalmometer, a hand-held instrument that records the position of the anterior corneal surface relative to the lateral orbital rim. If this instrument is not available, relative eye position can be judged by bending the patient's head forward and looking down upon the orbits. A proptosis of only 2 mm in one eye is detectable from this perspective. The development of proptosis implies a space-occupying lesion in the orbit. [ACT](#) or [MR](#) scan should be obtained in any patient with proptosis, unless the diagnosis of Graves' ophthalmopathy is certain.

Graves' Ophthalmopathy This is the leading cause of proptosis in adults ([Chap. 330](#)). The proptosis is often asymmetric and can even appear to be unilateral. Orbital inflammation and engorgement of the extraocular muscles, particularly the medial rectus and the inferior rectus, account for the protrusion of the globe. Corneal exposure, lid retraction, conjunctival injection, restriction of gaze, diplopia, and visual loss from optic nerve compression are cardinal symptoms. Acute Graves' ophthalmopathy should be treated with oral prednisone (60 mg/day) for 1 month, followed by a taper over several months. Chronic manifestations can be managed by topical lubricants, eyelid surgery, eye muscle surgery, or radiation treatment. Optic nerve compression should be relieved promptly with glucocorticoids and orbital decompression to prevent permanent visual loss.

Orbital Pseudotumor This is an idiopathic, inflammatory orbital syndrome, frequently confused with Graves' ophthalmopathy. Symptoms are pain, limited eye movements, proptosis, and congestion. Evaluation for sarcoidosis, Wegener's granulomatosis, and other types of orbital vasculitis or collagen-vascular disease is negative. Imaging often shows swollen eye muscles (orbital myositis) with enlarged tendons. By contrast, in Graves' ophthalmopathy the tendons of the eye muscles are usually spared. The Tolosa-Hunt syndrome may be regarded as an extension of orbital pseudotumor through the superior orbital fissure into the cavernous sinus. The diagnosis of orbital pseudotumor is difficult. Biopsy of the orbit frequently yields nonspecific evidence of fat infiltration by lymphocytes, plasma cells, and eosinophils. A dramatic response to a therapeutic trial of systemic glucocorticoids indirectly provides the best confirmation of the diagnosis.

Orbital Cellulitis This causes pain, lid erythema, proptosis, conjunctival chemosis, restricted motility, decreased acuity, afferent pupillary defect, fever, and leukocytosis. It often arises from a paranasal sinus, especially by contiguous spread of infection from the ethmoid sinus through the thin lamina papyracea of the medial orbit. A history of recent upper respiratory tract infection, chronic sinusitis, thick mucous secretions, or dental disease is significant in any patient with suspected orbital cellulitis. Blood cultures should be obtained, but they are usually negative. Most patients respond to empiric therapy with broad-spectrum intravenous antibiotics. Occasionally, orbital cellulitis follows an overwhelming course, with massive proptosis, blindness, septic cavernous sinus thrombosis, and meningitis. To avert this disaster, orbital cellulitis should be managed aggressively in the early stages, with immediate antibiotic therapy and imaging of the orbits. Prompt surgical drainage of an orbital abscess or paranasal sinusitis is indicated if optic nerve function deteriorates despite antibiotics.

Tumors Tumors of the orbit cause painless, progressive proptosis. The most common primary tumors are hemangioma, lymphangioma, neurofibroma, dermoid cyst, adenoid cystic carcinoma, optic nerve glioma, optic nerve meningioma, and benign mixed tumor of the lacrimal gland. Metastatic tumor to the orbit occurs frequently in breast carcinoma, lung carcinoma, and lymphoma. Diagnosis by fine-needle aspiration followed by urgent radiation therapy can sometimes preserve vision.

Carotid Cavernous Fistulas With anterior drainage through the orbit these produce proptosis, diplopia, glaucoma, and tortuous, red conjunctival vessels. Direct fistulas usually result from trauma. They are easily diagnosed because of the dramatic signs produced by high-flow, high-pressure shunting. Indirect fistulas, or dural arteriovenous malformations, are more likely to occur spontaneously, especially in older women. The signs are more subtle and the diagnosis is frequently missed. The combination of slight proptosis, diplopia, enlarged muscles, and an injected eye is often mistaken for thyroid ophthalmopathy. A bruit heard upon auscultation of the head, or reported by the patient, is a valuable diagnostic clue. Imaging shows an enlarged superior ophthalmic vein in the orbits. Carotid cavernous shunts can be eliminated by intravascular embolization.

PTOSIS

Blepharoptosis This is an abnormal drooping of the eyelid. Unilateral or bilateral ptosis can be congenital, from dysgenesis of the levator palpebrae superioris, or from abnormal insertion of its aponeurosis into the eyelid. Acquired ptosis can develop so gradually that the patient is unaware of the problem. Inspection of old photographs is helpful in dating the onset. A history of prior trauma, eye surgery, contact lens use, diplopia, systemic symptoms (e.g., dysphagia or peripheral muscle weakness), or a family history of ptosis should be sought. Fluctuating ptosis that worsens late in the day is typical of myasthenia gravis. Examination should focus upon evidence for proptosis, eyelid masses or deformities, inflammation, pupil inequality, or limitation of motility. The width of the palpebral fissures is measured in primary gaze to quantitate the degree of ptosis. The ptosis will be underestimated if the patient is compensating by lifting the brow with the frontalis muscle.

Mechanical Ptosis This occurs in many elderly patients from stretching and

redundancy of eyelid skin and subcutaneous fat (dermatochalasis). The extra weight of these sagging tissues causes the lid to droop. Enlargement or deformation of the eyelid from infection, tumor, trauma, or inflammation also results in ptosis on a purely mechanical basis.

Aponeurotic Ptosis This is an acquired dehiscence or stretching of the aponeurotic tendon, which connects the levator muscle to the tarsal plate of the eyelid. It occurs commonly in older patients, presumably from loss of connective tissue elasticity. Aponeurotic ptosis is also a frequent sequela of eyelid swelling from infection or blunt trauma to the orbit, cataract surgery, or hard contact lens usage.

Myogenic Ptosis The causes of *myogenic ptosis* include myasthenia gravis ([Chap. 380](#)) and a number of rare myopathies that manifest with ptosis. The term *chronic progressive external ophthalmoplegia* refers to a spectrum of systemic diseases caused by mutations of mitochondrial DNA. As the name implies, the most prominent findings are symmetric, slowly progressive ptosis and limitation of eye movements. In general, diplopia is a late symptom because all eye movements are reduced equally. In the *Kearns-Sayre* variant, retinal pigmentary changes and abnormalities of cardiac conduction develop. Peripheral muscle biopsy shows characteristic "ragged-red fibers." *Oculopharyngeal dystrophy* is a distinct autosomal dominant disease with onset in middle age, characterized by ptosis, limited eye movements, and trouble swallowing. *Myotonic dystrophy*, another autosomal dominant disorder, causes ptosis, ophthalmoparesis, cataract, and pigmentary retinopathy. Patients have muscle wasting, myotonia, frontal balding, and cardiac abnormalities.

Neurogenic Ptosis This results from a lesion affecting the innervation to either of the two muscles that open the eyelid: Muller's muscle or the levator palpebrae superioris. Examination of the pupil helps to distinguish between these two possibilities. In Horner's syndrome, the eye with ptosis has a smaller pupil and the eye movements are full. In an oculomotor nerve palsy, the eye with the ptosis has a larger, or a normal, pupil. If the pupil is normal but there is limitation of adduction, elevation, and depression, a pupil-sparing oculomotor nerve palsy is likely (see next section). Rarely, a lesion affecting the small, central subnucleus of the oculomotor complex will cause bilateral ptosis with normal eye movements and pupils.

DOUBLE VISION

The first point to clarify is whether diplopia persists in either eye after covering the fellow eye. If it does, the diagnosis is monocular diplopia. The cause is usually intrinsic to the eye and therefore has no dire implications for the patient. Corneal aberrations (e.g., keratoconus, pterygium), uncorrected refractive error, cataract, or foveal traction may give rise to monocular diplopia. Occasionally it is a symptom of malingering or psychiatric disease. Diplopia alleviated by covering one eye is binocular diplopia and is caused by disruption of ocular alignment. Inquiry should be made into the nature of the double vision (purely side-by-side versus partial vertical displacement of images), mode of onset, duration, intermittency, diurnal variation, and associated neurologic or systemic symptoms. If the patient has diplopia while being examined, motility testing should reveal a deficiency corresponding to the patient's symptoms. However, subtle limitation of ocular excursions is often difficult to detect. For example, a patient with a

slight left abducens nerve paresis may appear to have full eye movements, despite a complaint of horizontal diplopia upon looking to the left. In this situation, the cover test provides a more sensitive method for demonstrating the ocular malalignment. It should be conducted in primary gaze, and then with the head turned and tilted in each direction. In the above example, a cover test with the head turned to the right will maximize the fixation shift evoked by the cover test.

Occasionally, a cover test performed in an asymptomatic patient during a routine examination will reveal an ocular deviation. If the eye movements are full and the ocular misalignment is equal in all directions of gaze (concomitant deviation), the diagnosis is strabismus. In this condition, which affects about 1% of the population, fusion is disrupted in infancy or early childhood. To avoid diplopia, vision is suppressed from the nonfixating eye. In some children, this leads to impaired vision (amblyopia, or "lazy" eye) in the deviated eye.

Binocular diplopia occurs from a wide range of processes: infectious, neoplastic, metabolic, degenerative, inflammatory, and vascular. One must decide if the diplopia is neurogenic in origin or due to restriction of globe rotation by local disease in the orbit. Orbital pseudotumor, myositis, infection, tumor, thyroid disease, and muscle entrapment (e.g., from a blowout fracture) cause restrictive diplopia. The diagnosis is confirmed by performing a forced duction test in the office. After applying topical anesthesia, the physician grasps the eye with forceps and pulls it toward the direction of deficient motion. If rotation of the globe is prevented by tethering, a restrictive process is at work. The utility of this test is limited by its unpopularity with patients; in practice, the diagnosis of restriction is made by recognizing other associated signs and symptoms of local orbital disease.

Myasthenia Gravis (See also [Chap. 380](#)) This is a major cause of diplopia. The diplopia is often intermittent, variable, and not confined to any single ocular motor nerve distribution. The pupils are always normal. Fluctuating ptosis may be present. Many patients have a purely ocular form of the disease, with no evidence of systemic muscular weakness. The diagnosis can be confirmed by an intravenous edrophonium injection or by an assay for antiacetylcholine receptor antibodies. Negative results from these tests do not exclude the diagnosis. *Botulism* from food or wound poisoning can mimic ocular myasthenia.

After restrictive orbital disease and myasthenia gravis are excluded, a lesion of a cranial nerve supplying innervation to the extraocular muscles is the most likely cause of binocular diplopia.

Oculomotor Nerve ([Video 28-1](#)) The third cranial nerve innervates the medial, inferior, and superior recti; inferior oblique; levator palpebrae superioris; and the iris sphincter. Total palsy of the oculomotor nerve causes ptosis, a dilated pupil, and leaves the eye "down and out" because of the unopposed action of the lateral rectus and superior oblique. This combination of findings is obvious. More challenging is the diagnosis of an early or partial oculomotor nerve palsy. In this setting, any combination of ptosis, pupil dilation, and weakness of the eye muscles supplied by the oculomotor nerve may be encountered. Frequent serial examinations during the evolving phase of the palsy and a high index of suspicion help ensure that the diagnosis is not missed. The advent of an

oculomotor nerve palsy with any degree of pupil involvement in an otherwise healthy patient, especially when accompanied by pain, raises the specter of a circle of Willis aneurysm. If an [MR](#) scan shows no compressive lesion, an arteriogram must be performed to rule out an aneurysm of either the posterior communicating artery or the basilar artery. If the pupil is entirely normal, with all other components of an oculomotor palsy present, aneurysm is so rare that an angiogram is seldom indicated.

A lesion of the oculomotor nucleus in the rostral midbrain produces signs that differ from those caused by a lesion of the nerve itself. There is bilateral ptosis because the levator muscle is innervated by a single central subnucleus. There is also weakness of the contralateral superior rectus, because it is supplied by the oculomotor nucleus on the other side. Occasionally both superior recti are weak. Isolated nuclear oculomotor palsy is quite rare. Usually neurologic examination reveals additional signs to suggest brainstem damage from infarction, hemorrhage, tumor, or infection.

Injury to structures surrounding fascicles of the oculomotor nerve descending through the midbrain has given rise to a number of classic eponymic designations. In *Nothnagel's syndrome*, injury to the superior cerebellar peduncle causes ipsilateral oculomotor palsy and contralateral cerebellar ataxia. In *Benedikt's syndrome*, injury to the red nucleus results in ipsilateral oculomotor palsy and contralateral tremor, chorea, and athetosis. *Claude's syndrome* incorporates features of both the aforementioned syndromes, by injury to both the red nucleus and the superior cerebellar peduncle. Finally, in *Weber's syndrome*, injury to the cerebral peduncle causes ipsilateral oculomotor palsy with contralateral hemiparesis.

In the subarachnoid space the oculomotor nerve is vulnerable to aneurysm, meningitis, tumor, infarction, and compression. In cerebral herniation the nerve becomes trapped between the edge of the tentorium and the uncus of the temporal lobe. Oculomotor palsy can also occur from midbrain torsion and hemorrhages during herniation. In the cavernous sinus, oculomotor palsy arises from carotid aneurysm, carotid cavernous fistula, cavernous sinus thrombosis, tumor (pituitary adenoma, meningioma, metastasis), herpes zoster infection, and the Tolosa-Hunt syndrome.

The etiology of an isolated, pupil-sparing oculomotor palsy often remains obscure, even after neuroimaging and extensive laboratory testing. Most cases are thought to result from microvascular infarction of the nerve, somewhere along its course from the brainstem to the orbit. Usually the patient complains of pain. Diabetes, hypertension, and vascular disease are major risk factors. Spontaneous recovery over a period of months is the rule. If this fails to occur, or if new findings develop, the diagnosis of microvascular oculomotor nerve palsy should be reconsidered. Aberrant regeneration is common when the oculomotor nerve is injured by trauma or compression (tumor, aneurysm). Miswiring of sprouting fibers to the levator muscle and the rectus muscles results in elevation of the eyelid upon downgaze or adduction. The pupil also constricts upon attempted adduction, elevation, or depression of the globe. Aberrant regeneration is not seen after oculomotor palsy from microvascular infarct and hence vitiates that diagnosis.

Trochlear Nerve The fourth cranial nerve originates in the midbrain, just caudal to the oculomotor nerve complex. Fibers exit the brainstem dorsally and cross to innervate the

contralateral superior oblique. The principal actions of this muscle are to depress and to intort the globe. A palsy therefore results in hypertropia and excyclotorsion. The cyclotorsion is seldom noticed by patients. Instead, they complain of vertical diplopia, especially upon reading or looking down. The vertical diplopia is also exacerbated by tilting the head toward the side with the muscle palsy, and alleviated by tilting it away. This "head tilt test" is a cardinal diagnostic feature.

Isolated trochlear nerve palsy occurs from all the causes listed above for the oculomotor nerve, except aneurysm. The trochlear nerve is particularly apt to suffer injury after closed head trauma. The mechanism is unknown, but the free edge of the tentorium may impinge upon the nerve during a concussive blow. Most isolated trochlear nerve palsies are idiopathic and hence diagnosed by exclusion as "microvascular." Spontaneous improvement occurs over a period of months in most patients. A base-down prism (conveniently applied to the patient's glasses as a stick-on Fresnel lens) may serve as a temporary measure to alleviate diplopia. If the palsy does not resolve, the eyes can be realigned by surgically adjusting other eye muscles.

Abducens Nerve (Video 28-2) The sixth cranial nerve innervates the lateral rectus muscle. A palsy produces horizontal diplopia, worse on gaze to the side of the lesion. A nuclear lesion has different consequences, because the abducens nucleus contains interneurons that project via the medial longitudinal fasciculus to the medial rectus subnucleus of the contralateral oculomotor complex. Therefore, an abducens nuclear lesion produces a complete lateral gaze palsy, from weakness of both the ipsilateral lateral rectus and the contralateral medial rectus. *Foville's syndrome* following dorsal pontine injury includes lateral gaze palsy, ipsilateral facial palsy, and contralateral hemiparesis incurred by damage to descending corticospinal fibers. *Millard-Gubler syndrome* from ventral pontine injury is similar, except for the eye findings. There is lateral rectus weakness only, instead of gaze palsy, because the abducens fascicle is injured rather than the nucleus. Infarct, tumor, hemorrhage, vascular malformation, and multiple sclerosis are the most common etiologies of brainstem abducens palsy.

After leaving the ventral pons, the abducens nerve runs forward along the clivus to pierce the dura at the petrous apex, where it enters the cavernous sinus. Along its subarachnoid course it is susceptible to meningitis, tumor (meningioma, chordoma, carcinomatous meningitis), subarachnoid hemorrhage, trauma, and compression by aneurysm or dolichoectatic vessels. At the petrous apex, mastoiditis can produce deafness, pain, and ipsilateral abducens palsy (*Gradenigo's syndrome*). In the cavernous sinus, the nerve can be affected by carotid aneurysm, carotid cavernous fistula, tumor (pituitary adenoma, meningioma, nasopharyngeal carcinoma), herpes infection, and Tolosa-Hunt syndrome.

Unilateral or bilateral abducens palsy is a classic sign of raised intracranial pressure. The diagnosis can be confirmed if papilledema is observed on fundus examination. The mechanism is still debated but is probably related to rostral-caudal displacement of the brainstem. The same phenomenon accounts for abducens palsy from low intracranial pressure (e.g., after lumbar puncture, spinal anesthesia, or spontaneous dural cerebrospinal fluid leak).

Treatment of abducens palsy is aimed at prompt correction of the underlying cause.

However, the cause remains obscure in many instances, despite diligent evaluation. As mentioned above for isolated trochlear or oculomotor palsy, most cases are assumed to represent microvascular infarcts because they often occur in the setting of diabetes or other vascular risk factors. Some cases may develop as a postinfectious mononeuritis (e.g., following a viral flu). Patching one eye or applying a temporary prism will provide relief of diplopia until the palsy resolves. If recovery is incomplete, eye muscle surgery can nearly always realign the eyes, at least in primary position. A patient with an abducens palsy that fails to improve should be reevaluated for an occult etiology (e.g., chordoma, carcinomatous meningitis, carotid cavernous fistula, myasthenia gravis).

Multiple Ocular Motor Nerve Palsies These should not be attributed to spontaneous microvascular events affecting more than one cranial nerve at a time. This remarkable coincidence does occur, especially in diabetic patients, but the diagnosis is made only in retrospect after exhausting all other diagnostic alternatives. Neuroimaging should focus on the cavernous sinus, superior orbital fissure, and orbital apex, where all three ocular motor nerves are in close proximity. In the diabetic or compromised host, fungal infection (*Aspergillus*, Mucorales, *Cryptococcus*) is a frequent cause of multiple nerve palsies. In the patient with systemic malignancy, carcinomatous meningitis is a likely diagnosis. Cytologic examination may be negative despite repeated sampling of the cerebrospinal fluid. The cancer-associated Lambert-Eaton myasthenic syndrome can also produce ophthalmoplegia. Giant cell (temporal) arteritis occasionally manifests as diplopia from ischemic palsies of extraocular muscles ([Figs. 28-CD1](#) and [28-CD2](#)). Fisher syndrome, an ocular variant of Guillain-Barre, can produce ophthalmoplegia with areflexia and ataxia. Often the ataxia is mild, and the areflexia is overlooked because the physician's attention is focused upon the eyes.

Supranuclear Disorders of Gaze These are often mistaken for multiple ocular motor nerve palsies. For example, Wernicke's encephalopathy can produce nystagmus and a partial deficit of horizontal and vertical gaze that mimics a combined abducens and oculomotor nerve palsy. The disorder occurs in malnourished or alcoholic patients and can be reversed by giving thiamine. Infarct, hemorrhage, tumor, multiple sclerosis, encephalitis, vasculitis, and Whipple's disease are other important causes of supranuclear gaze palsy.

The *frontal eye field* of the cerebral cortex is involved in generation of saccades to the contralateral side. After hemispheric stroke, the eyes usually deviate towards the lesioned side because of the unopposed action of the frontal eye field in the normal hemisphere. With time, this deficit resolves. Seizures generally have the opposite effect: the eyes deviate conjugately away from the irritative focus. *Parietal lesions* disrupt smooth pursuit of targets moving toward the side of the lesion. Bilateral parietal lesions produce *Balint's syndrome*, characterized by impaired eye-hand coordination (optic ataxia), difficulty initiating voluntary eye movements (ocular apraxia), and visuospatial disorientation (simultanagnosia).

Horizontal Gaze Descending cortical inputs mediating horizontal gaze ultimately converge at the level of the pons. Neurons in the paramedian pontine reticular formation are responsible for controlling conjugate gaze toward the same side. They project directly to the ipsilateral abducens nucleus. A lesion of either the paramedian pontine reticular formation or the abducens nucleus causes an ipsilateral conjugate gaze palsy.

Lesions at either locus produce nearly identical clinical syndromes, with the following exception: vestibular stimulation (oculocephalic maneuver or caloric) will succeed in driving the eyes conjugately to the side in a patient with a lesion of the paramedian pontine reticular formation, but not in a patient with a lesion of the abducens nucleus.

Internuclear Ophthalmoplegia This results from damage to the medial longitudinal fasciculus ascending from the abducens nucleus in the pons to the oculomotor nucleus in the midbrain (hence, "internuclear"). Damage to fibers carrying the conjugate signal from abducens interneurons to the contralateral medial rectus motoneurons results in a failure of adduction on attempted lateral gaze. For example, a patient with a left internuclear ophthalmoplegia will have slowed or absent adducting movements of the left eye. A patient with bilateral injury to the medial longitudinal fasciculus will have bilateral internuclear ophthalmoplegia. Multiple sclerosis is the most common cause, although tumor, stroke, trauma, or any brainstem process may be responsible. ***One-and-a half syndrome*** is due to a combined lesion of the medial longitudinal fasciculus and the abducens nucleus on the same side. The patient's only horizontal eye movement is abduction of the eye on the other side.

Vertical Gaze This is controlled at the level of the midbrain. The neuronal circuits affected in disorders of vertical gaze are not well elucidated, but lesions of the rostral interstitial nucleus of the medial longitudinal fasciculus and the interstitial nucleus of Cajal cause supranuclear paresis of upgaze, downgaze, or all vertical eye movements. Distal basilar artery ischemia is the most common etiology. ***Skew deviation*** refers to a vertical misalignment of the eyes, usually constant in all positions of gaze. The finding has poor localizing value because skew deviation has been reported after lesions in widespread regions of the brainstem and cerebellum.

Parinaud's Syndrome Also known as dorsal midbrain syndrome, this is a distinct supranuclear vertical gaze disorder from damage to the posterior commissure. It is a classic sign of hydrocephalus from aqueductal stenosis. Pineal region tumors (germinoma, pineoblastoma), cysticercosis, and stroke also cause Parinaud's syndrome. Features include loss of upgaze (and sometimes downgaze), convergence-retraction nystagmus on attempted upgaze, downwards ocular deviation ("setting sun" sign), lid retraction (Collier's sign), skew deviation, pseudoabducens palsy, and light-near dissociation of the pupils. Disorders of vertical gaze, especially downwards saccades, are an early feature of progressive supranuclear palsy. Smooth pursuit is affected later in the course of the disease. Parkinson's disease, Huntington's chorea, and olivopontocerebellar degeneration can also affect vertical gaze.

Nystagmus This is a rhythmical oscillation of the eyes, occurring physiologically from vestibular and optokinetic stimulation or pathologically in a wide variety of diseases. Abnormalities of the eyes or optic nerves, present at birth or acquired in childhood, can produce a complex, searching nystagmus with irregular pendular (sinusoidal) and jerk features. This nystagmus is commonly referred to as ***congenital sensory nystagmus***. It is a poor term, because even in children with congenital lesions, the nystagmus does not appear until several months of age. ***Congenital motor nystagmus***, which looks similar to congenital sensory nystagmus, develops in the absence of any abnormality of the sensory visual system. Visual acuity is also reduced in congenital motor nystagmus, probably by the nystagmus itself, but seldom below a level of 20/200.

Jerk Nystagmus This is characterized by a slow drift off the target, followed by a fast corrective saccade. By convention, the nystagmus is named after the quick phase. Jerk nystagmus can be downbeat, upbeat, horizontal (left or right), and torsional. The pattern of nystagmus may vary with gaze position. Some patients will be oblivious to their nystagmus. Others will complain of blurred vision, or a subjective, to-and-fro movement of the environment (oscillopsia) corresponding to their nystagmus. Fine nystagmus may be difficult to see upon gross examination of the eyes. Observation of nystagmoid movements of the optic disc on ophthalmoscopy is a sensitive way to detect subtle nystagmus. The slit lamp is also useful.

Gaze-Evoked Nystagmus This is the most common form of jerk nystagmus. When the eyes are held eccentrically in the orbits, they have a natural tendency to drift back to primary position. The subject compensates by making a corrective saccade to maintain the deviated eye position. Many normal patients have mild gaze-evoked nystagmus. Exaggerated gaze-evoked nystagmus can be induced by drugs (sedatives, anticonvulsants, alcohol); muscle paresis; myasthenia gravis; demyelinating disease; and cerebellopontine angle, brainstem, and cerebellar lesions.

Vestibular Nystagmus *Vestibular nystagmus* results from dysfunction of the labyrinth (Meniere's disease), vestibular nerve, or vestibular nucleus in the brainstem. Peripheral vestibular nystagmus often occurs in discrete attacks, with symptoms of nausea and vertigo. There may be associated tinnitus and hearing loss. Sudden shifts in head position may provoke or exacerbate symptoms.

Downbeat Nystagmus *Downbeat nystagmus* occurs from lesions near the craniocervical junction (Chiari malformation, basilar invagination). It has also been reported in brainstem or cerebellar stroke, lithium or anticonvulsant intoxication, alcoholism, and multiple sclerosis. *Upbeat nystagmus* is associated with damage to the pontine tegmentum, from stroke, demyelination, or tumor.

Opsoclonus This rare, dramatic disorder of eye movements consists of bursts of consecutive saccades (saccadomania). When the saccades are confined to the horizontal plane, the term *ocular flutter* is preferred. It can occur from viral encephalitis, trauma, or a paraneoplastic effect of neuroblastoma, breast carcinoma, and other malignancies. It has also been reported as a benign, transient phenomenon in otherwise healthy patients.

(Bibliography omitted in Palm version)

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29. DISORDERS OF SMELL, TASTE, AND HEARING - Anil K. Lalwani, James B. Snow, Jr.

SMELL

The sense of smell determines the flavor and palatability of food and drink. It serves, along with the trigeminal system, as a monitor of inhaled chemicals, including dangerous substances such as natural gas, smoke, and air pollutants. Loss of or decreased ability to smell affects approximately 1% of people under age 60 and more than half of the population beyond this age.

DEFINITIONS

Smell is the perception of odor by the nose. *Taste* is the perception of salty, sweet, sour, or bitter by the tongue. Related sensations during eating such as somatic sensations of coolness, warmth, and irritation are mediated through the trigeminal, glossopharyngeal, and vagal afferents in the nose, oral cavity, tongue, pharynx, and larynx. *Flavor* is the complex interaction of taste, smell, and somatic sensation.

Terms relating to disorders of smell include *anosmia*, an absence of the ability to smell; *hyposmia*, a decreased ability to smell; *hyperosmia* (an increased sensitivity to an odorant); *dysosmia* (distortion in the perception of an odor); *phantosmia*, perception of an odorant where none is present; and *agnosia*, inability to classify, contrast, or identify odor sensations verbally, even though the ability to distinguish between odorants or to recognize them may be normal. An odor stimulus is referred to as an *odorant*. Each category of smell dysfunction can be further subclassified as total (applying to all odorants) or partial (dysfunction of only select odorants).

PHYSIOLOGY OF SMELL

The *olfactory neuroepithelium* is located in the superior part of the nasal cavities. It contains an orderly arrangement of bipolar olfactory receptor cells, microvillar cells, sustentacular cells, and basal cells. The dendritic process of the bipolar cell has a bulb-shaped knob, or vesicle, that projects into the mucous layer and bears six to eight cilia. It is the cilia that contain the odorant receptors. The arrangement of cilia increases overall exposure to the environment and translates into each bipolar cell containing 56 cm² (9 in.²) of surface area to receive stimulus.

The microvillar cells are located adjacent to the receptor cells on the surface of the neuroepithelium. The sustentacular cells, unlike their counterparts in the respiratory epithelium, are not specialized to secrete mucus. Although they form a tight barrier separating the neurons from the outside environment, their complete function is unknown. The basal cells are progenitors of other cell types in the olfactory neuroepithelium, including the bipolar receptor cells. There is a regular turnover of the bipolar receptor cells, which function as the primary sensory neurons. In addition, with injury to the cell body or its axon, the receptor cell is replaced by a differentiated basal cell which reestablishes a central neural connection. Hence these primary sensory neurons are unique among sensory systems in that they are regularly replaced and regenerate after injury.

The unmyelinated axons of the receptor cells form the fila of the olfactory nerve, pass through the cribriform plate, and terminate within spherical masses of neuropil, termed *glomeruli*, in the olfactory bulb. The glomeruli are the focus of a high degree of convergence of information, since many more fibers enter than leave them. The main second-order neurons are the mitral cells. The primary dendrite of each mitral cell extends into a single glomerulus. Axons of the mitral cells project along with the axons of adjacent tufted cells to the limbic system, including the anterior olfactory nucleus, the prepiriform cortex, the periamygdaloid cortex, the olfactory tubercle, the nucleus of the lateral olfactory tract, and the corticomедial nucleus of the amygdala. Cognitive awareness of smell requires stimulation of the prepiriform cortex or the amygdaloid nuclei.

A secondary potential site of olfactory chemosensation is located in the epithelium of the vomeronasal organ, a tubular structure that opens on the ventral aspect of the nasal septum. Sensory neurons located in the vomeronasal organ detect pheromones, nonvolatile chemical signals that in lower mammals trigger innate and stereotyped reproductive and social behaviors, as well as neuroendocrine changes. The neurons from the organ project to the accessory olfactory bulbs and not the main olfactory bulb, as in the olfactory neuroepithelium. Whether humans use the vomeronasal organ to detect and respond to chemical signals from others remains controversial. Recent work in delineating the molecular pathology of Kallman syndrome suggests that development of the olfactory and vomeronasal system is required for normal sexual maturation ([Chap. 335](#)).

The sensation of smell begins with introduction of an odorant to the cilia of the bipolar neuron. Most odorants are hydrophobic; as they move from the air phase of the nasal cavity to the aqueous phase of the olfactory mucous, they are transported toward the cilia by small water-soluble proteins called *odorant-binding proteins* and reversibly bind to receptors on the cilia surface. Binding causes conformational changes in the receptor protein, which induces a chain of biochemical events and results in generation of action potentials in the primary neurons. Transduction depends on the activation of G protein-coupled second messengers. Intensity appears to be coded by the amount of firing in the afferent neurons.

Basic elements of the genetic coding involved in smell are now becoming understood. Olfactory receptor proteins belong to the large family of G protein-coupled receptors that also includes rhodopsins; α - and β -adrenergic receptors; muscarinic acetylcholine receptors; and neurotransmitter receptors for dopamine, serotonin, and substance P. Members of the G protein-coupled receptors are characterized by the presence of 7 putative α -helical transmembrane domains composed of 20 to 28 hydrophobic amino acid residues. In mammals, there are probably 300 to 1000 olfactory receptor genes belonging to 20 different families located on various chromosomes in clusters. The receptor genes are present at more than 25 different human chromosomal locations. The gene clusters have likely risen as a result of repeated duplication of individual genes or clusters of genes. Each olfactory neuron seems to express only one or, at most, a few receptor genes thus providing the molecular basis of odor discrimination. While the receptors are expressed in several tissues, including the olfactory neuroepithelium and mammalian germ cells, their primary role appears to be odorant

recognition and discrimination. Bipolar cells that express similar receptors appear to be scattered across discrete spatial zones. These similar cells converge on a select few number of glomeruli in the olfactory bulb. The result is a potential spatial map of how we receive odor stimulus, much like the tonotopic organization of how we perceive sound.

DISORDERS OF THE SENSE OF SMELL

Disorders of the sense of smell are caused by conditions that interfere with the access of the odorant to the olfactory neuroepithelium (transport loss), injure the receptor region (sensory loss), or damage central olfactory pathways (neural loss). At the present time, there are no clinical tests to differentiate these different types of olfactory losses. Fortunately, the history of the disease provides important clues to the cause. The leading causes of olfactory disorders are summarized in [Table 29-1](#); the most common etiologies are head trauma and viral infections. Head trauma is a frequent cause of anosmia in children and young adults, whereas viral etiologies predominate in older adults.

Cranial trauma is followed by unilateral or bilateral impairment of smell in up to 15% of cases; anosmia is more common than hyposmia. Olfactory dysfunction is more common when trauma is associated with loss of consciousness, moderately severe head injury (grades II to V), and skull fracture. Frontal injuries and fractures disrupt the cribriform plate and olfactory axons that perforate it. Sometimes there is an associated cerebrospinal fluid (CSF) rhinorrhea resulting from a tearing of the dura overlying the cribriform plate and paranasal sinuses. Anosmia may also follow blows to the occiput. Once traumatic anosmia develops, it is usually permanent; only 10% of patients ever improve or recover. Perversion of the sense of smell may occur as a transient phase in the recovery process.

Viral infections destroy the olfactory neuroepithelium, which is replaced by respiratory epithelium. Parainfluenza virus type 3 appears to be especially detrimental to human olfaction. HIV infection is associated with subjective distortion of taste and smell, which may become more severe as the disease progresses. The loss of taste and smell may play an important role in the development and progression of HIV-associated wasting. Congenital anosmias are rare but important. Kallmann syndrome is an X-linked disorder characterized by congenital anosmia and hypogonadotropic hypogonadism resulting from a failure of migration from the olfactory placode of olfactory receptor neurons and neurons synthesizing gonadotropin-releasing hormone ([Chap. 328](#)). The responsible gene (*KAL*) has been cloned. Anosmia can also occur in albinos. The receptor cells are present but are hypoplastic, lack cilia, and do not project above the surrounding supporting cells.

Meningioma of the inferior frontal region is the most frequent neoplastic cause of anosmia; loss of smell may be the only neurologic abnormality at presentation. Rarely, anosmia can occur with glioma of the frontal lobe. Occasionally, pituitary adenomas, craniopharyngiomas, suprasellar meningiomas, and aneurysms of the anterior part of the circle of Willis extend forward and damage olfactory structures. These tumors and hamartomas may also induce seizures with olfactory hallucinations, indicating involvement of the uncus of the temporal lobe.

Dysosmia, subjective distortions of olfactory perception, may occur with intranasal disease that partially impairs smell or may represent a phase in the recovery from a neurogenic anosmia. Most dysosmic disorders consist of disagreeable or foul odors, and they may be accompanied by distortions of taste. Dysosmia is associated with depression.

Approach to the Patient

The history of the onset and course of the disorder is often paramount in establishing an etiology. Unilateral anosmia is rarely a complaint and is only recognized by separate testing of smell in each nasal cavity. Bilateral anosmia, on the other hand, brings patients to medical attention. Anosmic patients usually complain of a loss of the sense of taste even though their taste thresholds may be within normal limits. In actuality, they are complaining of a loss of flavor detection, which is mainly an olfactory function. The physical examination should include a complete examination of the ears, upper respiratory tract, and head and neck. A neurologic examination emphasizing the cranial nerves and cerebellar and sensorimotor function is essential. The patient's general mood should be assessed, and any signs of depression should be noted.

The sensory evaluation of olfactory function is necessary to corroborate the patient's complaint, evaluate the efficacy of treatment, and assess the degree of permanent impairment. The degree to which qualitative sensations are present can be assessed by any of several methods. The Odor Stix test uses a commercially available odor-producing magic marker-like pen held approximately 8 to 15 cm (3 to 6 in.) from the patient's nose to check for gross perception of the odorant. Another gross perception of odorant test, the 30-cm alcohol test, uses a freshly opened isopropyl alcohol packet held approximately 30 cm (12 in.) from the patient's nose. There is a commercially available scratch-and-sniff card containing three odors available for testing olfaction grossly. A superior test is the University of Pennsylvania Smell Identification Test (UPSIT). This consists of a 40-item, forced choice, microencapsulated odor, scratch-and-sniff paradigm. For example, one of the items reads, "This odor smells most like (a) chocolate, (b) banana, (c) onion, or (d) fruit punch," and the patient is instructed to answer one of the alternatives. The test is highly reliable, is sensitive to age and sex differences, and provides an accurate quantitative determination of the olfactory deficit. Persons with a total loss of smell function score in the range of 7 to 19 out of 40. The average score for total anosmics is slightly higher than that expected on the basis of chance because of the inclusion of some odorants that act by trigeminal stimulation.

The second step is to establish a detection threshold for the odorant phenyl ethyl alcohol, using a graduated stimulus. Sensitivity for each side of the nose is determined with a detection threshold for phenyl ethyl methyl ethyl carbinol. Nasal resistance can also be measured with anterior rhinomanometry for each side of the nose.

Computed tomography (CT) or magnetic resonance imaging (MRI) of the head is required to rule out paranasal sinusitis, neoplasms of the anterior cranial fossa, nasal cavity, or paranasal sinuses and unsuspected fractures of the anterior cranial fossa. Bone abnormalities are best seen with CT. MRI is useful in evaluating olfactory bulbs, ventricles, and other soft tissue of the brain. Coronal CT is optimal for assessing

cribriform plate, anterior cranial fossa, and sinus anatomy.

Techniques have been developed to biopsy the olfactory neuroepithelium, but in view of the widespread degeneration of the olfactory neuroepithelium and intercalation of respiratory epithelium in the olfactory area of adults with no apparent olfactory dysfunction, biopsy material must be interpreted cautiously.

TREATMENT

Therapy for patients with transport olfactory losses due to allergic rhinitis, bacterial rhinitis and sinusitis, polyps, neoplasms, and structural abnormalities of the nasal cavities can be undertaken rationally and with a high likelihood for improvement. Allergy management, antibiotic therapy, topical and systemic glucocorticoid therapy, and surgery for nasal polyps, deviation of the nasal septum, and chronic hyperplastic sinusitis are frequently effective in restoring the sense of smell.

There is no treatment with demonstrated efficacy for sensorineural olfactory losses. Fortunately, spontaneous recovery often occurs. Zinc and vitamin therapy (especially with vitamin A) are advocated by some. Profound zinc deficiency can produce loss and distortion of the sense of smell but is not a clinically important problem except in very limited geographic areas ([Chap. 75](#)). The epithelial degeneration associated with vitamin A deficiency can cause anosmia, but in western societies the prevalence of vitamin A deficiency is low. Exposure to cigarette smoke and other airborne toxic chemicals can cause metaplasia of the olfactory epithelium. Spontaneous recovery can occur if the insult is discontinued. Patient counseling is therefore helpful in these cases.

As mentioned above, more than half of people over age 60 suffer from olfactory dysfunction. No effective treatment exists for presbyosmia, but patients are often reassured to learn that this problem is common in their age group. In addition, early recognition and counseling can help patients to compensate for the loss of smell. The incidence of natural gas-related accidents is disproportionately high in the elderly, perhaps due in part to the gradual loss of smell. Mercaptan, the pungent odor in natural gas, is an olfactory stimulant and does not activate taste receptors. Many elderly with olfactory dysfunction experience a decrease in flavor sensation and find it necessary to hyperflavor food, usually by increasing the amount of salt in their diet. The physician can assist patients in developing healthy strategies to deal with the decreased sense of smell.

TASTE

Compared with disorders of smell, gustatory disorders are uncommon and their pathogenesis poorly understood. Many patients with a loss of olfactory sensitivity also complain of a loss of the sense of taste. On testing, most of these patients have normal detection thresholds for taste.

DEFINITIONS

Disturbances of the sense of taste may be categorized as *total ageusia* -- total absence of gustatory function or inability to detect the qualities of sweet, salt, bitter, or sour;

partial ageusia -- ability to detect some of but not all the qualitative gustatory sensations; *specific ageusia* -- inability to detect the taste quality of certain substances; *total hypogeusia* -- decreased sensitivity to all tastants; *partial hypogeusia* -- decreased sensitivity to some tastants; and *dysgeusia* or *phantogeusia* -- distortion in the perception of a tastant, i.e., the perception of the wrong quality when a tastant is presented or the perception of a taste when there has been no tastant ingested. Confusions of sour and bitter are common and, at times, may be semantic misunderstandings. Frequently, however, they have physiologic or pathophysiologic bases. Other taste quality confusions occur between sour and salty and bitter. It may be possible to differentiate between the loss of flavor recognition in patients with olfactory losses who complain of a loss of taste as well as smell by asking if they are able to taste sweetness in sodas, saltiness in potato chips, etc.

PHYSIOLOGY OF TASTE

The taste receptor cells are located in the taste buds, spherical groups of cells arranged in a pattern resembling the segments of a citrus fruit. At the surface, the taste bud has a pore into which microvilli of the receptor cells project. Unlike the olfactory system, the receptor cell is not the primary neuron. Instead, gustatory afferent nerve fibers contact individual taste receptor cells. Transduction depends on activation of G protein-coupled second messengers but differs in details for each taste quality.

The sense of taste is mediated through the facial, glossopharyngeal, and vagal nerves. The gustatory system consists of at least five receptor populations. Taste buds are located in the papillae along the lateral margin and dorsum of the tongue at the junction of the dorsum and the base of the tongue, and in the palate, epiglottis, larynx, and esophagus. The chorda tympani branch of the facial nerve subserves taste from the anterior two-thirds of the tongue. The posterior third of the tongue is supplied by the lingual branch of the glossopharyngeal nerve. Afferents from the palate travel with the greater superficial petrosal nerve to the geniculate ganglion and then via the facial nerve to the brainstem. The internal branch of the superior laryngeal nerve of the vagus nerve contains the taste afferents from the larynx, including the epiglottis and esophagus.

The central connections of the nerves terminate in the brainstem in the nucleus of the tractus solitarius. The central pathway from the nucleus of the tractus solitarius projects to the ipsilateral parabrachial nuclei of the pons. Two divergent pathways project from the parabrachial nuclei. One ascends to the gustatory relay in the dorsal thalamus, synapses, and continues to the cortex of the insula. There is also evidence for a direct pathway from the parabrachial nuclei to the cortex. (Olfaction and gustation appear to be unique among sensory systems in that at least some fibers bypass the thalamus.) The other pathway from the parabrachial nuclei goes to the ventral forebrain, including the lateral hypothalamus, substantia innominata, central nucleus of the amygdala, and the stria terminalis.

Tastants gain access to the receptor cells through the taste pore. Four classes of taste are recognized: sweet, salt, sour, and bitter. Individual gustatory afferent fibers almost always respond to a number of different chemicals. Response patterns of gustatory afferent axons can be grouped into classes based on the stimulus chemical that produces the largest response. For example, for sucrose-best response neurons, the

second-best stimulus is almost always sodium chloride. The fact that individual gustatory afferent fibers respond to a large number of different chemicals led to the *across-fiber-pattern* theory of gustatory coding, while the best-stimulus analysis led to the concept of *labeled* afferents. It appears that labeled fibers are important for establishing gross quality, but the across-fiber pattern within a best-stimulus category, and perhaps among categories, is needed for discriminating chemicals within qualities. For example, sweetness may be carried by sucrose-best neurons, but the differentiation of sucrose and fructose may require a comparison of the relative activity among sucrose-best, salt-best, and quinine-best neurons. As with olfaction and other sensory systems, intensity appears to be encoded by the quantity of neural activity.

DISORDERS OF THE SENSE OF TASTE

Disorders of the sense of taste are caused by conditions that interfere with the access of the tastant to the receptor cells in the taste bud (transport loss), injure receptor cells (sensory loss), or damage gustatory afferent nerves and central gustatory pathways (neural loss) ([Table 29-2](#)). *Transport gustatory losses* result from xerostomia due to many causes, including Sjogren's syndrome, radiation therapy, heavy-metal intoxication, and bacterial colonization of the taste pore. *Sensory gustatory losses* are caused by inflammatory and degenerative diseases in the oral cavity; a vast number of drugs, particularly those that interfere with cell turnover such as antithyroid and antineoplastic agents; radiation therapy to the oral cavity and pharynx; viral infections; endocrine disorders; neoplasms; and aging. *Neural gustatory losses* occur with neoplasms, trauma, and surgical procedures in which the gustatory afferents are injured. Taste buds degenerate when their gustatory afferents are transected but remain when their somatosensory afferents are severed. Patients with renal disease have increased thresholds for sweet and sour tastes, which resolves with dialysis.

A side effect of medication is the single most common cause of taste dysfunction in clinical practice. The mechanism may be a change in the composition of saliva, an effect on receptor function or signal transduction, or disruption of the central processing of gustatory input. Unfortunately, the responsible mechanism is not well understood for most medications. Xerostomia, regardless of the etiology, can be associated with taste dysfunction. It is associated with poor oral clearance, poor dental hygiene, and can adversely affect the oral mucosa, all leading to dysgeusia. However, severe salivary gland failure does not necessarily lead to taste complaints. Xerostomia, along with the use of antibiotics or glucocorticoids, and compromised immune function can lead to overgrowth of *Candida*; overgrowth alone, without thrush or overt signs of infection can be associated with bad taste or hypogeusia. When taste dysfunction occurs in a patient at risk for fungal overgrowth, a trial of nystatin or other anti-fungal medication is warranted.

Upper respiratory infections and head trauma can lead to both smell and taste dysfunction; taste is more likely to improve than smell. The mechanism of taste disturbance in these situations is not well understood. Trauma to the chorda tympani branch of the facial nerve during middle ear surgery or third molar extractions is relatively common and can cause dysgeusia. Bilateral chorda tympani injuries are usually associated with hypogeusia, whereas unilateral lesions produce only limited symptoms, perhaps because responses from taste receptors are disinhibited by the

glossopharyngeal nerve.

Finally, aging itself may be associated with reduced taste sensitivity. The taste dysfunction may be limited to a single compound and may be mild. While many older patients may acknowledge loss of taste when asked, they are unlikely to seek medical attention for taste disturbance alone.

Approach to the Patient

Patients who complain of loss of taste should be evaluated for both gustatory and olfactory function. Clinical assessment of taste is not as well developed or standardized as that of smell. The first step is to perform suprathreshold whole-mouth taste testing for quality, intensity, and pleasantness perception of four taste qualities: sweet, salty, sour, and bitter. Most commonly used reagents for taste testing are sucrose, citric acid or hydrochloric acid, caffeine or quinine (sulfate or hydrochloride), and sodium chloride. The taste stimuli should be freshly prepared. For quantification, detection thresholds are obtained by applying graduated dilutions to the tongue quadrants or by whole-mouth sips. Electric taste testing (*electrogustometry*) is used clinically to identify taste deficits in specific quadrants of the tongue, following precise applications of stimuli. Regional gustatory testing may also be performed to assess for the possibility of loss localized to one or more receptor fields as a result of a peripheral or central lesion.

Once there is objective evidence of a disorder of taste, it is important to establish an anatomic diagnosis before proceeding to an etiologic diagnosis. The history of the disease often provides important clues to the cause. For example, absence of taste on the anterior two-thirds of the tongue associated with a facial paralysis indicates that the lesion is proximal to the juncture of the chorda tympani branch with the facial nerve in the mastoid.

TREATMENT

Therapy for gustatory loss is limited. Nonetheless, some etiologies of taste dysfunction are amenable to intervention. Taste disturbance related to drugs can often be resolved by changing the prescribed medication. Xerostomia can be treated with artificial saliva, providing some benefits to patients with a disturbed salivary milieu. Oral pilocarpine may be beneficial for a variety of forms of xerostomia. Appropriate treatment of bacterial and fungal infections of the oral cavity can be of great help in improving taste function. Taste dysfunction following trauma may resolve spontaneously without intervention and is more likely to do so than posttraumatic smell dysfunction. Altered taste due to surgical stretch injury of chorda tympani nerve usually improves within 3 to 4 months, while dysfunction is usually permanent with transection of the nerve. In most patients with idiopathic cases of altered taste sensitivity, the problem either remains stable or worsens. Zinc and vitamin therapy for gustatory losses is advocated by some but lacks demonstrated efficacy. No effective therapeutic strategies exist for the sensorineural disorders of taste.

HEARING

Hearing loss is one of the most common sensory disorders in humans. Nearly 10% of

the adult population has some hearing loss. For many, this impairment presents early in life. However, hearing loss can present at any age. Between 30 and 35% of individuals over the age of 65 have a hearing loss of sufficient magnitude to require a hearing aid.

PHYSIOLOGY OF HEARING ([Fig. 29-1](#))

Hearing occurs by air conduction and bone conduction. In air conduction, sound waves reach the ear by propagation in air, enter the external auditory canal, and set the tympanic membrane in motion, which in turn moves the malleus, incus, and stapes of the middle ear. Movement of the footplate of the stapes causes pressure changes in the fluid-filled inner ear eliciting a traveling wave in the basilar membrane of the cochlea. The tympanic membrane and the ossicular chain in the middle ear serve as an impedance-matching mechanism, improving the efficiency of energy transfer from air to the fluid-filled inner ear. Hearing by bone conduction occurs when the sounding source, in contact with the head, results in vibration of the bones of the skull, including the temporal bone, producing a traveling wave in the basilar membrane.

Stereocilia of the hair cells of the organ of Corti, which rests on the basilar membrane, are in contact with the tectorial membrane and are deformed by the traveling wave. A point of maximal displacement of the basilar membrane is determined by the frequency of the stimulating tone. High-frequency tones cause maximal displacement of the basilar membrane near the base of the cochlea. As the frequency of the stimulating tone decreases, the point of maximal displacement moves toward the apex of the cochlea.

The inner and outer hair cells of the organ of Corti have different innervation patterns, but both are mechanoreceptors. The afferent innervation relates principally to the inner hair cells, and the efferent innervation relates principally to outer hair cells. The motility of the outer hair cells alters the micromechanics of the inner hair cells creating a cochlear amplifier, which explains the exquisite sensitivity and frequency selectivity of the cochlea.

The current concept of cochlear transduction is that displacement of the tips of the stereocilia allows potassium to flow into the cell, resulting in its depolarization. The potassium influx opens calcium channels near the base of the cell, stimulating transmitter release. The neurotransmitter at the hair cell and cochlear nerve dendrite interface is thought to be glutamate. The action potential in the eighth nerve occurs 0.5 ms after the onset of the cochlear microphonic potential. Each of the cochlear nerve neurons can be activated at a frequency and intensity specific for that cell. This specificity is maintained at each point of the central auditory pathway: dorsal and ventral cochlear nuclei, trapezoid body, superior olivary complex, lateral lemniscus, inferior colliculus, medial geniculate body, and auditory cortex. At low frequencies, individual auditory nerve fibers can respond more or less synchronously with the stimulating tone. At higher frequencies, phase-locking occurs so that neurons alternate in response to particular phases of the cycle of the sound wave. Intensity is encoded by the amount of neural activity in individual neurons, the number of neurons that are active, and the specific neurons that are activated.

GENETIC CAUSES OF HEARING LOSS

More than half of childhood hearing impairment is thought to be hereditary; hereditary hearing impairment (HHI) can also manifest later in life. HHI may be classified as either nonsyndromic, when hearing loss is the only clinical abnormality, or syndromic, when hearing loss is associated with anomalies in other organ systems. Nearly two-thirds of HHIs are nonsyndromic and the remaining one-third are syndromic. Between 70 and 80% of nonsyndromic HHI is inherited in an autosomal recessive manner; another 15 to 20% is autosomal dominant. Less than 5% is X-linked or maternally inherited via the mitochondria.

Over 60 loci harboring genes for nonsyndromic [HHI](#) have been mapped, with equal numbers of dominant and recessive modes of inheritance; 14 different genes have been cloned ([Table 29-3](#)). The hearing genes fall into the categories of structural proteins (MYO7A, MYO15, TECTA, DIAPH1), transcription factors (POU3F4, POU4F3), ion channels (KCNQ4, PDS), and gap junction proteins (Cx26, Cx30, Cx31). Several of these genes, including connexin 26 (Cx26), TECTA, and MYO7A, cause both autosomal dominant and recessive forms of nonsyndromic HHI. In general, the hearing loss associated with dominant genes has its onset in adolescence or adulthood and varies in severity, whereas the hearing loss associated with recessive inheritance is congenital and profound. Connexin 26 is particularly important because it is associated with nearly 20% of cases of childhood deafness; in heterozygotes the onset of hearing loss may be in adolescence or adulthood. Two frame-shift mutations, 30delG and 167delT, account for >50% of the cases, making population screening feasible. The 167delT mutation is highly prevalent in Ashkenazi Jews; it is predicted that 1 in 1765 individuals in this population will be homozygous and affected. The hearing loss can also vary among the members of the same family, suggesting that other genes or factors likely influence the auditory phenotype.

The contribution of genetics to presbycusis (see below) is also becoming better understood. In addition to connexin 26, several other nonsyndromic genes are associated with hearing loss that progresses with age. It is likely that presbycusis has both environmental and genetic components.

Over 200 syndromes are associated with hearing loss. Common syndromic forms of hearing loss include Usher syndrome (retinitis pigmentosa and hearing loss), Waardenburg syndrome (pigmentary abnormality and hearing loss), Pendred syndrome (thyroid organification defect and hearing loss), Alport syndrome (renal disease and hearing loss), Jervell and Lange-Nielsen syndrome (prolonged QT interval and hearing loss), neurofibromatosis type 2 (bilateral acoustic schwannoma), and mitochondrial disorders [mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes (MELAS); myoclonic epilepsy and ragged red fibers (MERRF); progressive external ophthalmoplegia (PEO)] ([Table 29-4](#)).

Rapid progress in understanding the basis of these and related disorders has revealed a fascinating complexity, including evidence for genetic heterogeneity (different genes resulting in a similar clinical phenotype), allelic disorders (distinct phenotypes associated with different mutations in the same gene), and polygenic modifiers ([Chap. 65](#)).

DISORDERS OF THE SENSE OF HEARING

Hearing loss can result from disorders of the auricle, external auditory canal, middle ear, inner ear, or central auditory pathways ([Fig. 29-2](#)). *In general, lesions in the auricle, external auditory canal, or middle ear cause conductive hearing losses, whereas lesions in the inner ear or eighth nerve cause sensorineural hearing losses.*

Conductive Hearing Loss This results from obstruction of the external auditory canal by cerumen, debris, and foreign bodies; swelling of the lining of the canal; atresia of the ear canal; neoplasms of the canal; perforations of the tympanic membrane; disruption of the ossicular chain, as occurs with necrosis of the long process of the incus in trauma or infection; otosclerosis; or fluid, scarring, or neoplasms in the middle ear.

Cholesteatoma, i.e., the presence of stratified squamous epithelium in the middle ear or mastoid, occurs frequently in adults. A cholesteatoma is a benign, slowly growing lesion that destroys bone and normal ear tissue. Major theories of pathogenesis of acquired cholesteatoma include traumatic implantation and invasion, immigration and invasion through a perforation, and metaplasia following chronic infection and irritation. On examination, there is often a perforation filled with cheesy white squamous debris. A chronically draining ear that fails to respond to appropriate antibiotic therapy should raise the suspicion of a cholesteatoma. Conductive hearing loss secondary to ossicular erosion is common. Surgery is required to remove this insidiously growing and destructive disease process.

Conductive hearing loss in the presence of a normal ear canal and intact tympanic membrane is suggestive of ossicular pathology. Fixation of the stapes from *otosclerosis* is a common cause of low-frequency conductive hearing loss. It occurs with equal frequency in men and women and has a simple autosomal dominant inheritance with incomplete penetrance. Hearing impairment usually presents between the late teens to the forties. In women, the hearing loss is often first noticeable during pregnancy, as the otosclerotic process is accelerated during pregnancy. A hearing aid or a short outpatient surgical procedure (stapedectomy) can provide adequate auditory rehabilitation. Extension of otosclerosis beyond the stapes footplate to involve the cochlea (cochlear otosclerosis) can lead to mixed or sensorineural hearing loss. Fluoride therapy to prevent hearing loss associated with cochlear otosclerosis remains controversial.

Eustachian tube dysfunction is extremely common in adults and may predispose to acute otitis media (AOM) or serous otitis media (SOM). Trauma, AOM, or chronic otitis media are the usual factors responsible for tympanic membrane perforation. While small perforations often heal spontaneously, larger defects usually require surgical intervention. Tympanoplasty is highly effective (>90%) in the repair of tympanic membrane perforations. Otoscopy is usually sufficient to diagnose AOM, SOM, chronic otitis media, cerumen impaction, tympanic membrane perforation, and eustachian tube dysfunction.

Sensorineural Hearing Loss Damage to the hair cells of the organ of Corti may be caused by intense noise, viral infections, ototoxic drugs (e.g., salicylates, quinine and its synthetic analogues, aminoglycoside antibiotics, loop diuretics such as furosemide and ethacrynic acid, and cancer chemotherapeutic agents such as cisplatin), fractures of the temporal bone, meningitis, cochlear otosclerosis (see above), Meniere's disease, and

aging. Congenital malformations of the inner ear may be the cause of hearing loss in some adults. Genetic predisposition alone or in concert with environmental influences may also be responsible.

Presbycusis (age-associated hearing loss) is the most common cause of sensorineural hearing loss in adults. In the early stages, it is characterized by symmetric, gentle to sharply sloping high-frequency hearing loss. With progression, the hearing loss involves all frequencies. More importantly, the hearing impairment is associated with significant loss in clarity. There is a loss of discrimination for phonemes, recruitment (abnormal growth of loudness), and particular difficulty in understanding speech in noisy environments. Hearing aids may provide limited rehabilitation once the word recognition score deteriorates below 50%. Significant advancements and improvements in cochlear implants have made them the treatment of choice when hearing aids prove inadequate (<30% word recognition score with optimal amplification).

Meniere's disease is characterized by episodic vertigo, fluctuating sensorineural hearing loss, tinnitus, and aural fullness. Tinnitus and/or deafness may be absent during the initial attacks of vertigo, but they invariably appear as the disease progresses and are increased in severity during an acute attack. The annual incidence of Meniere's disease is 0.5 to 7.5 per 1000; onset is most frequently in the fifth decade of life but may also occur in young adults or the elderly. Histologically, there is distention of the endolymphatic system (endolymphatic hydrops) leading to degeneration of vestibular and cochlear hair cells. This may result from endolymphatic sac dysfunction secondary to infection, trauma, autoimmune disease, inflammatory causes, or tumor; an idiopathic etiology constitutes the largest category and is most accurately referred to as Meniere's disease. Although any pattern of hearing loss can be observed, typically, low-frequency, unilateral sensorineural hearing impairment is present. [MRI](#) should be obtained to exclude retrocochlear pathology such as cerebellopontine angle tumors or demyelinating disorders. Therapy is directed towards the control of vertigo. A low-salt diet is the mainstay of treatment of the control of rotatory vertigo. Diuretics, a short course of glucocorticoids, and intratympanic gentamicin may also be useful adjuncts in recalcitrant cases. Surgical therapy of vertigo is reserved for unresponsive cases and includes endolymphatic sac decompression, labyrinthectomy, and vestibular nerve section. Both labyrinthectomy and vestibular nerve section abolish rotatory vertigo in >90% of the cases. Unfortunately, there is no effective therapy for hearing loss, tinnitus, or aural fullness associated with Meniere's disease.

Sensorineural hearing loss may also result from any neoplastic, vascular, demyelinating, infectious, or degenerative disease or trauma affecting the central auditory pathways. Human immunodeficiency virus leads to both peripheral and central auditory system pathology and is associated with sensorineural hearing impairment.

An individual can have both conductive and sensory hearing loss termed *mixed hearing loss*. Mixed hearing losses are due to pathology that can affect the middle and inner ear simultaneously such as otosclerosis involving the ossicles and the cochlea, head trauma, chronic otitis media, cholesteatoma, middle ear tumors, and some inner ear malformations.

Trauma resulting in temporal bone fractures may be associated with conductive,

sensorineural, and mixed hearing loss. If the fracture spares the inner ear, there may simply be conductive hearing loss due to rupture of the tympanic membrane or disruption of the ossicular chain. These abnormalities are amenable to surgical correction. Profound hearing loss and severe vertigo are associated with temporal bone fractures involving the inner ear. A perilymphatic fistula associated with leakage of inner-ear fluid into the middle ear can occur and may require surgical repair. An associated facial nerve injury is not uncommon. [CT](#) is best suited to assess fracture of the traumatized temporal bone, evaluate the ear canal, and determine the integrity of the ossicular chain and the involvement of the inner ear. [CSF](#) leaks that accompany temporal bone fractures are usually self-limited; the use of prophylactic antibiotics is controversial.

Tinnitus is defined as the perception of a sound when there is no sound in the environment. It may have a buzzing, roaring, or ringing quality and may be pulsatile (synchronous with the heartbeat). Tinnitus is often associated with either a conductive or sensorineural hearing loss. The pathophysiology of tinnitus is not well understood. The cause of the tinnitus can usually be determined by finding the cause of the associated hearing loss. Tinnitus may be the first symptom of a serious condition such as a vestibular schwannoma. Pulsatile tinnitus requires evaluation of the vascular system of the head to exclude vascular tumors such as glomus jugulare tumors, aneurysms, and stenotic arterial lesions; it may also occur with [SOM](#).

Approach to the Patient

The goals in the evaluation of a patient with auditory complaints are to determine: (1) the nature of the hearing impairment (conductive vs. sensorineural), (2) the severity of the impairment (mild, moderate, severe, profound), (3) the anatomy of the impairment (external ear, middle ear, inner ear, or central auditory pathway pathology, and (4) the etiology. Initially, the history and the physical examination are critical in the identification of the underlying pathology leading to the auditory deficit. The history should elicit characteristics of the hearing loss, including the duration of deafness, nature of onset (sudden vs. insidious), rate of progression (rapid vs. slow), and involvement of the ear (unilateral vs. bilateral). The presence or absence of tinnitus, vertigo, imbalance, aural fullness, otorrhea, headache, facial nerve dysfunction, and head and neck paresthesias should be ascertained. Information regarding head trauma, exposure to ototoxins, occupational or recreational noise exposure, and family history of hearing impairment may also be important. A sudden onset of unilateral hearing loss, with or without tinnitus, may represent a viral infection of the inner ear or a vascular accident. Patients with unilateral hearing loss (sensory or conductive) usually complain of reduced hearing, poor sound localization, and difficulty hearing clearly with background noise. Gradual progression of a hearing deficit is common with otosclerosis, noise-induced hearing loss, vestibular schwannoma, or Meniere's disease. Small vestibular schwannomas typically present with asymmetric hearing impairment, tinnitus, and imbalance (rarely vertigo); cranial neuropathy, in particular of the trigeminal or facial nerve, may accompany larger tumors. In addition to hearing loss, Meniere's disease may be associated with episodic vertigo, tinnitus, and aural fullness. Hearing loss with otorrhea is most likely due to chronic otitis media or cholesteatoma.

Family history may be crucial in delineating a genetic basis of hearing impairment. The

history may also help identify environmental risk factors that lead to hearing impairment in a family. Sensitivity to aminoglycoside ototoxicity, maternally transmitted through a mitochondrial mutation, can be ascertained through a careful family history ([Chap. 67](#)). Susceptibility to noise-induced hearing loss or age-related hearing loss (presbycusis) may also be genetically determined.

The physical examination should evaluate the auricle, external ear canal, and tympanic membrane. The external ear canal of the elderly is often dry and fragile; it is preferable to clean cerumen with wall-mounted suction and cerumen loops and to avoid irrigation. In examining the eardrum, the topography of the tympanic membrane is more critical than the presence or absence of the highly touted light reflex. In addition to the pars tensa (the lower two-thirds of the eardrum), the pars flaccida above the short process of the malleus should also be examined for retraction pockets that may be evidence of chronic eustachian tube dysfunction or cholesteatoma. Insufflation of the ear canal is necessary to assess tympanic membrane mobility and compliance. Careful inspection of the nose, nasopharynx, and upper respiratory tract is indicated. Unilateral serous effusion in the adult should prompt a fiberoptic examination of the nasopharynx to exclude neoplasms. Cranial nerves should be carefully evaluated with special attention to facial and trigeminal nerves, which are commonly disturbed with tumors involving the cerebellopontine angle.

The Weber and Rinne tuning fork tests are used to differentiate conductive from sensorineural hearing losses and to confirm the findings of audiologic evaluation. Rinne's test compares the ability to hear by air conduction with the ability to hear by bone conduction. The tines of a vibrating tuning fork are held near the opening of the external auditory canal, and then the stem is placed on the mastoid process; for direct contact, it may be placed on teeth or dentures. The patient is asked to indicate whether the tone is louder by air conduction or bone conduction. Normally, and in the presence of sensorineural hearing loss, a tone is heard louder by air conduction than by bone conduction; however, with conductive hearing loss of ≥ 30 dB (see "Audiologic Assessment," below), the bone-conduction stimulus is perceived as louder than the air-conduction stimulus. The Rinne test is most sensitive in detecting mild conductive hearing losses if a 256-Hz tuning fork is used. The Weber test may be performed with a 256- or 512-Hz fork. The stem of a vibrating tuning fork is placed on the head in the midline and the patient asked whether the tone is heard in both ears or better in one ear than in the other. With a unilateral conductive hearing loss, the tone is perceived in the affected ear. With a unilateral sensorineural hearing loss, the tone is perceived in the unaffected ear. As a general rule, a 5-dB difference in hearing between the two ears is required for lateralization. The combined information from the Weber and Rinne tests permits a tentative conclusion as to whether a conductive or sensorineural hearing loss is present; however, these tests are associated with significant false-positive and -negative responses and therefore should be utilized only as screening tools.

LABORATORY ASSESSMENT OF HEARING

Audiologic Assessment The minimum audiologic assessment for hearing loss should include the measurement of pure tone air-conduction and bone-conduction thresholds, speech reception threshold, discrimination score, tympanometry, acoustic reflexes, and acoustic-reflex decay. This test battery provides a comprehensive screening evaluation

of the whole auditory system and allows one to determine whether further differentiation of a sensory (cochlear) from a neural (retrocochlear) hearing loss is indicated.

Pure tone audiometry assesses hearing acuity for pure tones. The test is administered by an audiologist and is performed in a sound-attenuated chamber. The pure tone stimulus is delivered with an audiometer, an electronic device that allows the presentation of specific frequencies (generally between 250 and 8000 Hz) at specific intensities. Air and bone conduction thresholds are established for each ear. Air conduction thresholds are established by presenting the stimulus in air with the use of headphones. Bone conduction thresholds are accomplished by placing the stem of a vibrating tuning fork or an oscillator of an audiometer in contact with the head. In the presence of a hearing loss, broad-spectrum noise is presented to the nontest ear for *masking* purposes so that responses are based on perception from the ear under test.

The responses are measured in decibels. An *audiogram* is a plot of intensity in decibels required to achieve threshold versus frequency. A decibel (dB) is equal to 20 times the logarithm of the ratio of the sound pressure required to achieve threshold in the patient to the sound pressure required to achieve threshold in a normal hearing person. Therefore, a change of 6 dB represents doubling of sound pressure, and a change of 20 dB represents a ten-fold change in sound pressure. Loudness, which depends on the frequency, intensity, and duration of a sound, doubles with approximately each 10-dB increase in sound pressure level. Pitch, on the other hand, does not directly correlate with frequency. The perception of pitch changes slowly in the low and high frequencies. In the middle tones, which are important for human speech, pitch varies more rapidly with changes in frequency.

Pure tone audiometry establishes the presence and severity of hearing impairment, unilateral vs. bilateral involvement, and the type of hearing loss. Conductive hearing losses with a large mass component, as is often seen in middle-ear effusions, produce elevation of thresholds that predominate in the higher frequencies. Conductive hearing losses with a large stiffness component, as in fixation of the footplate of the stapes in early otosclerosis, produce threshold elevations in the lower frequencies. Often, the conductive hearing loss involves all frequencies, suggesting involvement of both stiffness and mass. In general, sensorineural hearing losses such as presbycusis affect higher frequencies more than lower frequencies. An exception is Meniere's disease, which is characteristically associated with low-frequency sensorineural hearing loss. Noise-induced hearing loss has an unusual pattern of hearing impairment in which the loss at 4000 Hz is greater than at higher frequencies. Vestibular schwannomas characteristically affect the higher frequencies, but any pattern of hearing loss can be observed.

Speech recognition requires greater synchronous neural firing than is necessary for appreciation of pure tones. *Speech audiometry* tests the clarity with which one hears. The *speech reception threshold* (SRT) is defined as the intensity at which speech is recognized as a meaningful symbol and is obtained by presenting two-syllable words with an equal accent on each syllable. The intensity at which the patient can repeat 50% of the words correctly is the SRT. Once the SRT is determined, discrimination or word recognition ability is tested by presenting one-syllable words at 25 to 40 dB above the speech reception threshold. The words are phonetically balanced in that the phonemes

(speech sounds) occur in the list of words at the same frequency that they occur in ordinary conversational English. An individual with normal hearing or conductive hearing loss can repeat 88 to 100% of the phonetically balanced words correctly. Patients with a sensorineural hearing loss have variable loss of discrimination depending on the severity of hearing loss and the site of lesion. Further, as a general rule, neural lesions are associated with more deterioration in discrimination ability than are lesions in the inner ear. For example, in a patient with mild asymmetric sensorineural hearing loss, a clue to the diagnosis of vestibular schwannoma is the presence of greater than expected deterioration in discrimination ability. Deterioration in discrimination ability at higher intensities above the SRT also suggests a lesion in the eighth nerve or central auditory pathways.

Tympanometry measures the impedance of the middle ear to sound and is particularly useful in the identification and diagnosis of middle-ear effusions. A sounding source and microphone are introduced into the ear canal with an airtight seal. The amount of sound that is absorbed through the middle ear or reflected from the middle ear is measured at the microphone. In conductive hearing losses, more sound is reflected than in the normal middle ear. The pressure in the ear canal can be increased or decreased from atmospheric pressure. A *tympanogram* is the graphic representation of change in impedance or compliance as the pressure in the ear canal is changed. It provides information about the status of the tympanic membrane and the ossicular chain. Normally, the middle ear is most compliant at atmospheric pressure, and the compliance decreases as the pressure is increased or decreased; this pattern is seen with normal hearing or in the presence of sensorineural hearing loss. Compliance that does not change with change in pressure suggests middle-ear effusion. With a negative pressure in the middle ear, as with eustachian tube obstruction, the point of maximal compliance occurs with negative pressure in the ear canal. A tympanogram in which no point of maximal compliance can be obtained is most commonly seen with discontinuity of the ossicular chain. A reduction in the maximal compliance peak can be seen in otosclerosis.

During tympanometry, an intense tone (80 dB above the hearing threshold) elicits contraction of the stapedius muscle. The change in compliance of the middle ear with contraction of the stapedius muscle can be detected. The presence or absence of this *acoustic reflex* is important in the anatomic localization of facial nerve paralysis as well as hearing loss. Normal or elevated acoustic reflex thresholds in an individual with significant sensorineural hearing impairment suggests a cochlear hearing loss. Assessment of *acoustic reflex decay* helps differentiate sensory from neural hearing losses. In neural hearing loss, the reflex adapts or decays with time.

Otoacoustic emissions (OAE) can be measured with sensitive microphones inserted into the external auditory canal. The emissions may be spontaneous or evoked with sound stimulation. The presence of OAEs indicates that the outer hair cells of the organ of Corti are intact and can be used to assess auditory thresholds and to distinguish sensory from neural hearing losses.

Evoked Responses *Electrocochleography* measures the earliest evoked potentials generated in the cochlea and the auditory nerve. Receptor potentials recorded include the cochlear microphonic, generated by the outer hair cells of the organ of Corti, and the

summating potential, generated by the inner hair cells in response to sound. The whole nerve action potential representing the composite firing of the first-order neurons can also be recorded during electrocochleography. Clinically, the test is useful in the diagnosis of Meniere's disease where an elevation of the ratio of summating potential to action potential is seen.

Brainstem auditory evoked responses (BAERs) are useful in differentiating the site of sensorineural hearing loss ([Chap. 356](#)). In response to sound, five distinct electrical potentials arising from different stations along the peripheral and central auditory pathway can be recorded with computer averaging from scalp surface electrodes. BAERs are valuable in situations in which patients cannot or will not give reliable voluntary thresholds. They are also used to assess the integrity of the auditory nerve and brainstem in various clinical situations, including intraoperative monitoring and in determination of brain death.

Imaging Studies The choice of radiologic tests is largely determined by whether the goal is to evaluate the bony anatomy of the external, middle, and inner ear or to image the auditory nerve and brain. Axial and coronal [CT](#) of the temporal bone with fine 1-mm cuts is ideal for determining the caliber of the external auditory canal, integrity of the ossicular chain, and presence of middle-ear or mastoid disease; it can also detect inner-ear malformations. CT is also ideal for the detection of bone erosion often seen in the presence of chronic otitis media and cholesteatoma. [MRI](#) is superior to CT for imaging of retrocochlear pathology such as vestibular schwannoma, meningioma, other lesions of the cerebellopontine angle, demyelinating lesions of the brainstem, and brain tumors. Recent experience suggests that both CT and MRI are equally capable of identifying inner-ear malformations and assessing cochlear patency for preoperative evaluation of patients for cochlear implantation.

TREATMENT

In general, conductive hearing losses are amenable to surgical intervention and correction, while sensorineural hearing losses are permanent. The diagnosis of conductive hearing loss is usually straightforward, and the etiology of the conductive deficit is often apparent on physical examination. Atresia of the ear canal can be surgically repaired, often with significant improvement in hearing. Tympanic membrane perforations due to chronic otitis media or trauma can be repaired with an outpatient tympanoplasty. Likewise, conductive hearing loss associated with otosclerosis can be treated by stapedectomy, which is successful in 90 to 95% of cases. Tympanostomy tubes allow the prompt return of normal hearing in individuals with middle-ear effusions. Hearing aids are effective and well-tolerated in patients with conductive hearing losses.

Patients with mild, moderate, and severe sensorineural hearing losses are regularly rehabilitated with hearing aids of varying configuration and strength. Hearing aids have been improved to provide greater fidelity and have been miniaturized. The current generation of hearing aids can be placed entirely within the ear canal, thus reducing the stigma associated with their use. In general, the more severe the hearing impairment, the larger the hearing aid required for auditory rehabilitation. Digital hearing aids lend themselves to individual programming, and multiple and directional microphones at the ear level may be helpful in noisy surroundings. Since all hearing aids amplify noise as

well as speech, the only absolute solution to the problem found thus far is to place the microphone closer to the speaker than the noise source. This arrangement is not possible with a self-contained, cosmetically acceptable device. It is cumbersome and requires a user-friendly environment.

In many situations, including lectures and the theater, hearing-impaired persons benefit from assistive devices that are based on the principle of having the speaker closer to the microphone than any source of noise. Assistive devices include infrared and FM transmission as well as an electromagnetic loop around the room for transmission to the individual's hearing aid. Hearing aids with telecoils can also be used with properly equipped telephones in the same way.

In the event that the hearing aid provides inadequate rehabilitation, cochlear implants are appropriate. Criteria for implantation include severe to profound hearing loss with word recognition score $\leq 30\%$ under best aided conditions. Children with congenital and acquired profound hearing impairment are also appropriate candidates for cochlear implantation. Worldwide, more than 20,000 deaf individuals (including 4000 children) have received cochlear implants. Cochlear implants are neural prostheses that convert sound energy to electrical energy and can be used to stimulate the auditory division of the eighth nerve directly. In most cases of profound hearing impairment, the auditory hair cells are lost but the ganglionic cells of the auditory division of the eighth nerve are preserved. Cochlear implants consist of electrodes that are inserted into the cochlea through the round window, speech processors that extract acoustical elements of speech for conversion to electrical currents, and a means of transmitting the electrical energy through the skin. Patients with implants experience sound that helps with speech reading, allows open-set word recognition, and helps in modulating the person's own voice. Usually, within 3 months after implantation, adult patients can understand speech without visual cues. With the current generation of multichannel cochlear implants, nearly 75% of patients are able to converse on the telephone. It is anticipated that improvements in the electrode design and speech processors will permit further enhancement in understanding speech, especially in the presence of background noise.

For individuals who have had both eighth nerves destroyed by trauma or bilateral vestibular schwannomas (e.g., neurofibromatosis type 2), brainstem auditory implants placed near the cochlear nucleus may provide auditory rehabilitation. It is hoped that additional advances may provide benefits similar to those with the cochlear implant.

Tinnitus can often accompany hearing loss. The treatment of tinnitus is particularly problematic. Therapy is usually directed towards minimizing the appreciation of tinnitus. Relief of the tinnitus may be obtained by masking it with background music. Hearing aids are also helpful in tinnitus suppression, as are tinnitus maskers, devices that present a sound to the affected ear that is more pleasant to listen to than the tinnitus. The use of a tinnitus masker is often followed by several hours of inhibition of the tinnitus. Antidepressants have also shown beneficial effect in helping patients deal with tinnitus.

Tinnitus and background noise can significantly affect understanding of speech in individuals with hearing impairment. Hard-of-hearing individuals often benefit from a reduction in unnecessary noise (e.g., radio or television) to enhance the signal-to-noise

ratio. Speech comprehension is aided by lip reading; therefore, the impaired listener should be seated so that the face of the speaker is well illuminated and can be seen at all times. Speaking directly into the ear is occasionally helpful, but usually more is lost in communication than gained when the speaker's face cannot be seen. Speech should be slow enough to make each word distinct, but overly slow speech is distracting and loses contextual and speech-reading benefits. Although speech should be in a loud, clear voice, one should be aware that in sensorineural hearing losses in general and in elderly hard-of-hearing persons in particular, recruitment (the ability to hear loud sounds normally loud) may be troublesome. Above all, optimal communication cannot take place without both parties giving it their full and undivided attention.

PREVENTION

Conductive hearing losses may be prevented by prompt and appropriate antibiotic therapy of adequate duration for [AOM](#) and by ventilation of the middle ear with tympanostomy tubes in middle-ear effusions lasting 12 weeks or longer. Loss of vestibular function and deafness due to aminoglycoside antibiotics can largely be prevented by careful monitoring of serum peak and trough levels.

Some 10 million Americans have noise-induced hearing loss, and 20 million are exposed to hazardous noise in their employment. Noise-induced hearing loss can be prevented by avoidance of exposure to loud noise or by regular use of ear plugs or fluid-filled ear muffs to attenuate intense sound. Noise-induced hearing loss results from recreational as well as occupational activities and begins in adolescence. High-risk activities for noise-induced hearing loss include wood and metal working with electrical equipment and target practice and hunting with small firearms. All internal-combustion and electric engines, including snow and leaf blowers, snowmobiles, outboard motors, and chain saws, require protection of the user with hearing protectors. Virtually all noise-induced hearing loss is preventable through education, which should begin before the teenage years. Programs of industrial conservation of hearing are required when the exposure over an 8-h period averages 85 dB on the A scale. Workers in such noisy environments can be protected with preemployment audiologic assessment, the mandatory use of hearing protectors, and annual audiologic assessments.

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(Bibliography omitted in Palm version)

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30. INFECTIONS OF THE UPPER RESPIRATORY TRACT - *Marlene Durand, Michael Joseph*

Infections of the upper respiratory tract include some of the most common infectious diseases encountered by internists and other primary-care physicians. Pharyngitis, laryngitis, rhinitis, sinusitis, otitis externa, and otitis media account for millions of visits to physicians annually. Although these infections are usually mild enough to be treated on an outpatient basis, the primary-care physician must be able to recognize their serious complications, such as peritonsillar abscess from pharyngitis, subperiosteal abscess from frontal sinusitis, and temporal-bone osteomyelitis from invasive otitis externa. The physician must also identify potentially life-threatening infections of the head and neck, such as epiglottitis, Ludwig's angina, and rhinocerebral mucormycosis.

INFECTIONS OF THE NOSE AND FACE

Skin infections that commonly affect the nose and face include folliculitis, furunculosis, impetigo, and erysipelas. These are discussed in detail elsewhere ([Chap. 128](#), in particular [Table 128-1](#)).

Infection of the mucosal surface of the nose is most commonly due to respiratory viruses (e.g., rhinovirus) and presents as acute rhinitis. There are several rare, chronic intranasal infections. *Ozena*, or atrophic rhinitis, is characterized by atrophied mucosa overlaid by foul-smelling dry crusts (Greek *ozein*, "stench"). *Klebsiella ozaenae* is often isolated from nasal cultures, but whether it is a cause of illness or merely a colonizer is unclear. Intranasal irrigation with aminoglycosides (e.g., tobramycin ophthalmic solution) or oral administration of ciprofloxacin has resulted in clinical improvement in some cases. *Klebsiella rhinoscleromatis* causes *rhinoscleroma*, a chronic granulomatous disease of the upper respiratory tract mucosa that is seen in inhabitants of parts of Africa, Asia, and Latin America; it has been described in two patients positive for HIV. Mikulicz cells (foamy histiocytes) are seen in the submucosa of biopsy specimens. Rhinoscleroma can be treated with streptomycin, trimethoprim-sulfamethoxazole, a quinolone, or tetracycline for 2 months. *Pseudomonas mallei* causes *glanders*, a respiratory disease of horses. Infection is rare in humans; nasal inoculation may produce a purulent nasal discharge followed by granulomatous intranasal lesions that ulcerate. Treatment is with sulfadiazine.

Neonatal congenital syphilis may present as rhinitis (snuffles), and the generalized osteochondritis that follows may result in a "saddle-nose" deformity. In *leprosy*, *Mycobacterium leprae* infiltrates the nasal mucosa and may cause chronic nasal congestion and nosebleeds. Involvement of the nasal cartilage may also result in a saddle nose.

Rhinosporidium seeberi is a fungus-like organism, not yet cultured, that causes *rhinosporidiosis*. Pedunculated nasal masses that grow over months or years cause obstruction and a foul odor and must be surgically excised. *Blastomyces dermatitidis*, a fungus prevalent in the Mississippi and Ohio River valleys, usually causes pulmonary disease but may cause chronic ulcerative lesions of the skin and nasal mucosa. *Mucormycosis*, a life-threatening fungal illness that occurs primarily in diabetic patients, may present as black eschars in the nasal cavity (see "Fungal Sinusitis").

THE COMMON COLD

The common cold is a mild, self-limited viral infection of the upper respiratory tract. Adults average two to four colds per year and children six to eight. The most common causes are rhinovirus (40% of cases) and coronavirus (at least 10%), but parainfluenza virus, respiratory syncytial virus, influenza virus, and adenoviruses also account for some cases. Rhinovirus alone has more than 100 different immunotypes, and this diversity has hampered efforts to identify an effective therapy or vaccine. There is no specific treatment for the common cold, although antihistamines, decongestants, and ipratropium bromide nasal spray provide some relief of symptoms. One study found that zinc gluconate lozenges, taken every 2 h, may reduce the duration of symptoms but are associated with nausea in 20% of patients. The value of vitamin C in preventing colds has not been proved.

SINUSITIS

The paranasal sinuses are aerated cavities in the bones of the face that develop as outpouches of the nasal cavity and communicate with this cavity throughout life. The maxillary and ethmoid sinuses are present at birth; the frontal and sphenoid sinuses develop after ages 2 and 7, respectively. Like the nose, the sinuses are lined with respiratory epithelium that includes mucus-producing goblet cells and ciliated cells. The mucous blanket is carried toward the sinus openings (ostia) at a speed of up to 1 cm/min by the beating of the cilia. The ostia are small; the ethmoid sinus ostia, for example, are only 1 to 2 mm in diameter. Delay in the mucociliary transport time or -- more important -- obstruction of the ostia may lead to retained secretions and sinusitis.

Sinusitis is a common problem. In the United States, this infection accounts for millions of office visits annually. The most common type, maxillary sinusitis, is followed in frequency by ethmoid, frontal, and sphenoid sinusitis. A viral infection of the upper respiratory tract is the most common precursor of sinusitis, although only about 0.5% of such infections are complicated by clinically evident acute bacterial sinusitis. Sinusitis develops primarily through ostial obstruction due to mucosal edema. Viral upper respiratory infections also increase the amount of mucus produced and may damage ciliated cells, thereby delaying mucus transport time. Allergic rhinitis is another common cause of ostial obstruction, either by mucosal edema or by polyps. Nasotracheal or nasogastric intubation can result in obstruction of the ostia and is a major risk factor for nosocomial sinusitis in intensive care units. Dental infections may cause 5 to 10% of all cases of maxillary sinusitis; the roots of the upper back teeth (second bicuspid, first and second molars) abut the floor of the maxillary sinus. Other causes of sinusitis include barotrauma from deep-sea diving or airplane travel, mucus abnormalities (e.g., cystic fibrosis), and chemical irritants. Foreign bodies, tumors (e.g., midline granuloma, intranasal lymphoma, or squamous cell carcinoma), and granulomatous diseases (e.g., Wegener's granulomatosis or rhinoscleroma) may all cause sinusitis secondary to obstruction.

ACUTE BACTERIAL SINUSITIS

Manifestations Symptoms of acute sinusitis include purulent nasal or postnasal

drainage, nasal congestion, and sinus pain or pressure whose location depends on the sinus involved. Maxillary sinus pain is often perceived as being located in the cheek or upper teeth; ethmoid sinus pain, between the eyes or retroorbital; frontal sinus pain, above the eyebrow; and sphenoid sinus pain, in the upper half of the face or retroorbital with radiation to the occiput. Sinus pain is frequently worse when the patient bends over or is supine. Fever develops in about half of patients with acute maxillary sinusitis.

Diagnosis The diagnosis of bacterial sinusitis may be difficult, as symptoms may resemble those of the inciting viral upper respiratory infection. The persistence of cold symptoms for 7 to 10 days (or longer than usual for a particular patient) is the most consistent clinical feature of bacterial sinusitis, according to some authors. Four-view sinus x-rays are helpful in the diagnosis of acute sinusitis: radiologic opacity, an air-fluid level, or ≥ 4 mm of sinus mucosal thickening correlates well with active bacterial infection. Computed tomography (CT) of the sinus is much more sensitive than routine radiography, particularly for ethmoid and sphenoid disease. Its use should be reserved for complicated cases and for cases in hospitalized patients, however. In light of the finding that sinus CT often shows reversible acute changes in patients with common colds, it is apparent that routine early use of CT would lead to overdiagnosis of bacterial sinusitis.

Etiology The bacteriology of acute community-acquired maxillary sinusitis has been well defined by studies using direct sinus puncture and aspiration. In children and adults, *Streptococcus pneumoniae* and *Haemophilus influenzae* (not type b), the most common pathogens, cause about one-third and one-fourth of cases, respectively. In children, *Moraxella catarrhalis* is also important, accounting for 20% of cases. Rhinoviruses, influenza viruses, and parainfluenza viruses are found alone or with bacteria in one-fifth of adult cases.

TREATMENT

Empirical therapy for acute bacterial sinusitis should be directed against the common bacterial pathogens; sinus puncture is not indicated in routine cases, and cultures of nasal drainage are not very reliable. Amoxicillin (500 mg orally, three times daily for 10 to 14 days) or trimethoprim-sulfamethoxazole may be effective in the treatment of first-time cases.

¹The dosages and durations of antimicrobial therapy given in this chapter are appropriate for adults with normal renal function and should be adjusted for children, for patients with impaired renal function, and in light of the response to treatment. Other effective but more expensive antibiotics include amoxicillin/clavulanate, cefuroxime axetil, and clarithromycin. Treatment should be given for 1 to 2 weeks. Intravenous administration of antibiotics may be necessary for the treatment of patients with severe disease who appear toxic. In nosocomial sinusitis, *Staphylococcus aureus* and gram-negative bacilli are most common, and sinus cultures are indicated as an aid in tailoring therapy. Initial broad-spectrum intravenous therapy (e.g., with nafcillin and ceftriaxone) should be adjusted on the basis of culture results. Surgery to widen the ostia and drain thick secretions may be essential in severe acute sinusitis, particularly when ethmoid, frontal, or sphenoid disease fails to respond to initial intravenous therapy.

CHRONIC BACTERIAL SINUSITIS

Manifestations Chronic sinusitis is characterized by symptoms of sinus inflammation lasting ≥ 3 months. Most experts believe that this condition is caused by dysfunction of the mucociliary blanket, usually as a result of repeated past infections, rather than by the persistence of bacterial infection. Patients report constant sinus pressure, nasal congestion, and postnasal drainage, especially in the morning. A temperature of $\geq 38^{\circ}\text{C}$ ($\geq 100.5^{\circ}\text{F}$) is rare and may signify a superimposed acute bacterial infection. Many patients also note a change in nasal discharge (to thick and green) with acute exacerbations.

Diagnosis Sinus [CT](#) should be used in all cases of chronic sinusitis to define the extent of disease and to help exclude other diagnoses, such as an obstructing tumor. Patients should be evaluated for allergies and immunodeficiencies (e.g., hypogammaglobulinemia). Evaluation by an otolaryngologist is essential, as this specialist will be able to obtain additional information by an office nasal endoscopic examination. Surgery, now usually done endoscopically, may be necessary to correct blockage of the sinus ostia. This blockage occurs most often in the osteomeatal complex that drains the maxillary, frontal, and anterior ethmoid sinuses. Samples of sinus secretions obtained intraoperatively should be cultured for anaerobes, aerobes, and fungi. Fungal sinusitis may mimic chronic bacterial infection (see "Fungal Sinusitis").

Etiology The bacteriology of chronic sinusitis is not well defined. Nearly all patients with chronic disease, especially those who have had prior sinus surgery, have sinus cultures positive for bacteria. Such cultures may represent colonization rather than infection, however. Patients who have received multiple courses of antibiotics may be colonized by *S. aureus* or by *Pseudomonas* and other gram-negative bacilli. Anaerobes have been isolated from 100% of sinus specimens in some studies but from as few as 2% in others.

TREATMENT

The need for antibiotic therapy must be assessed on an individual basis, with antibiotics chosen in light of recent culture results.

COMPLICATIONS OF BACTERIAL SINUSITIS

Orbital complications of sinusitis, such as orbital cellulitis and orbital abscess, usually arise from ethmoid sinusitis, since the ethmoid is separated from the orbit by only a very thin bone (the lamina papyracea). Patients present with fever, unilateral periorbital edema and erythema, conjunctival injection and chemosis, and proptosis. Eye movement may be decreased; with orbital abscess, the eye is often fixed in the "down and out" position. [CT](#) or magnetic resonance imaging (MRI) should be used to rule out an orbital abscess. Treatment of orbital infections should include immediate drainage of any abscess, intravenous administration of broad-spectrum antibiotics -- e.g., nafcillin (1.5 to 2.0 g every 4 h) plus ceftriaxone (2 g/d) -- for at least 7 days, and a consideration of sinus drainage surgery.

Another extracranial complication of sinusitis is frontal subperiosteal abscess (Pott's puffy tumor) from frontal sinusitis. Patients present with a tender doughy swelling over the forehead. Treatment consists of surgical drainage of the abscess and the frontal sinus and 6 weeks of intravenous antibiotic therapy directed at the isolated organisms.

Intracranial complications such as epidural abscess, subdural empyema, meningitis, cerebral abscess, and dural-vein thrombophlebitis may result from sinusitis, particularly from frontal or sphenoid infections. Because the sphenoid sinus sits between the two cavernous sinuses, sphenoid sinusitis is a major cause of cavernous sinus thrombophlebitis.

FUNGAL SINUSITIS

Fungal sinusitis is categorized as noninvasive or invasive. *Noninvasive* disease is chronic and occurs in immunocompetent hosts. It has two forms that are analogous to the noninvasive pulmonary diseases of aspergilloma and allergic bronchopulmonary aspergillosis. A fungus ball (aspergilloma) inside a sinus may cause symptoms of obstruction without invading the mucosa. Typically, only one sinus (often maxillary) is affected, and patients have unilateral symptoms and opacification of only that sinus on [CT](#). Treatment is surgical only, unless special fungal stains show tissue invasion on histopathology. Allergic fungal sinusitis was first described in 1983 and is seen mainly in patients with a history of nasal polyposis and asthma. It is characterized by extremely thick sinus mucus ("allergic mucin") that, on histopathologic examination, is found to contain numerous Charcot-Leyden crystals, eosinophils, and rare fungal hyphae. There is no evidence of tissue invasion. Surgical removal of the inspissated mucus is often curative. Antifungal therapy is not indicated.

Invasive fungal sinusitis presents differently in immunocompetent and immunocompromised hosts. In immunocompromised individuals, fungal disease has an acute presentation. Rhinocerebral mucormycosis is a life-threatening infection due to fungi of the order Mucorales (*Rhizopus*, *Rhizomucor*, *Mucor*, *Absidia*, *Cunninghamella*). Mucormycosis usually involves diabetic patients (70% of cases), half of whom are in ketoacidosis at presentation. Other patients at risk include those with hematologic malignancies, transplant recipients, and patients receiving chronic glucocorticoid or iron chelation (deferoxamine) therapy. Mucormycosis in patients taking deferoxamine is generally due to *Cunninghamella* and is almost always fatal. Symptoms and signs of mucormycosis may be explained by the fungal predilection for blood vessels and nerves and for invasion into the orbital apex and cavernous sinus, with consequent compromise of cranial nerves II through VI. Patients most frequently present with unilateral ocular findings of 1 to 5 days' duration that may mimic bacterial orbital cellulitis: lid swelling and erythema (sometimes bluish in appearance), ptosis, proptosis, decreased extraocular movement, and impaired vision. Retroorbital or periorbital pain is a prominent complaint. There may be either increased or decreased sensation in the first division of the fifth cranial nerve on the involved side; facial palsy with involvement of cranial nerve VII has also been described. Patients may be afebrile and appear nontoxic. Individuals in whom mucormycosis is a consideration should undergo an immediate examination by an otolaryngologist, who will look for intranasal black eschars or necrotic turbinates. If found, these sites should be biopsied and frozen tissue sections examined by a pathologist. In rare cases, the nasal passage appears normal, but biopsy of the middle

turbinate reveals invasive fungi. The finding of tissue invasion by broad-based, nonseptate hyphae necessitates extensive surgical debridement and intravenous therapy with amphotericin B or liposomal amphotericin ([Chaps. 200](#) and [207](#)). *Aspergillus* and other filamentous fungi may also cause invasive sinus disease.

Immunocompetent hosts with invasive fungal sinusitis, in contrast, have slowly progressive disease. Fungi in ethmoid and sphenoid sinuses may invade the orbital apex, causing proptosis, ptosis, limitation of eye movement, and decreased vision. Patients may mistakenly be treated with glucocorticoids for presumed optic neuritis or orbital pseudotumor until sinus disease is recognized and biopsies are undertaken. Treatment consists of surgical debridement of the involved sinuses and prolonged intravenous therapy with amphotericin B. In all cases of invasive fungal sinusitis, follow-up [CT](#) and [MRI](#) should be conducted frequently to evaluate the progression of disease.

Mortality from invasive fungal sinusitis is high, even among immunocompetent hosts.

EAR AND MASTOID INFECTIONS

AURICULAR CELLULITIS AND PERICHONDritis

Auricular cellulitis usually presents as a swollen, erythematous, hot, tender ear. The lobule is especially swollen and red. There may be a history of minor trauma to the ear (e.g., involving earrings, cotton swabs, or scratching). Treatment consists of warm compresses and intravenous administration of antibiotics active against *S. aureus* and streptococci -- e.g., cefazolin (1 g every 8 h) or nafcillin.

Perichondritis, an infection of the perichondrium of the ear, is often accompanied by infection of the underlying cartilage of the pinna (chondritis). Associated interruption of the blood supply to the cartilage may lead to ear deformity. Patients present with a swollen, hot, red, and exquisitely tender pinna, usually with sparing of the lobule. The most common antecedents of the infection are burns, significant trauma to the ear (e.g., as a result of boxing), or ear piercing through the pinna. *Pseudomonas aeruginosa* and *S. aureus* are the most common pathogens. Perichondritis should be treated with antibiotics, such as intravenous ticarcillin/clavulanic acid (3.1 g every 4 h) or intravenous nafcillin plus oral ciprofloxacin, for several weeks. Incision and drainage may be helpful for culture and for resolution of infection, which is often slow. This infection must be distinguished from relapsing polychondritis, a rheumatologic condition ([Chap. 325](#)).

OTITIS EXTERNA

The external auditory canal is about 2.5 cm long and is lined by skin. Beneath this skin is cartilage in the lateral half of the canal, temporal bone in the medial half. The skin in the bony portion lacks a subcutaneous layer and is attached directly to the periosteum, an important feature in the pathogenesis of invasive otitis externa (see below). Cerumen, secreted by glands, acidifies the canal and suppresses bacterial growth. However, desquamated skin and retained moisture make the canal especially susceptible to the hydrophilic organism *P. aeruginosa*.

Acute otitis externa, or swimmer's ear, occurs mostly in the summer and may be due to a decrease in canal acidity and resulting bacterial overgrowth. The ear is pruritic and painful, and the canal appears swollen and red. The most common pathogens are *P. aeruginosa*, *S. aureus*, and streptococci. Treatment consists of cleansing of the ear with alcohol-acetic acid mixtures and the administration of topical antibiotic ear drops, such as polymyxin-neomycin (4 drops four times daily for 5 days). Herpes zoster in the external canal causes severe otalgia and is often accompanied by ipsilateral facial paralysis due to the involvement of the geniculate ganglion of cranial nerve VII (Ramsay Hunt syndrome). Reports suggest that treatment with intravenous acyclovir decreases the incidence of permanent facial-nerve palsy, but the results of relevant controlled trials have not yet been published.

Chronic otitis externa causes pruritus rather than ear pain and is often due to irritation from either repeated minor trauma to the canal (e.g., scratching or use of cotton swabs) or drainage of a chronic middle-ear infection. In the latter situation, treatment of chronic otitis media with oral antibiotics will also cover this condition.

Invasive ("malignant") otitis externa is a potentially life-threatening infection, almost always due to *P. aeruginosa*, that slowly invades from the external canal into adjacent soft tissues, mastoid, and temporal bone and eventually spreads across the base of the skull. It occurs primarily in diabetic patients whose diabetes, unlike that of patients with mucormycosis, is usually under control. There is a history of weeks to months of ear pain and drainage, often misdiagnosed as chronic otitis media (an entity that is rarely painful). Examination reveals an edematous canal, with granulation tissue in the posterior wall about halfway down the canal (the region of the cartilage-bone junction). Trismus or partial facial paralysis (cranial nerve VII) is evident in some instances. Cranial nerves IX, X, and XI are occasionally affected as well. Fever is rare in invasive otitis externa and, when it does develop, is usually low-grade.

Laboratory studies generally reveal a normal white blood cell count but a high erythrocyte sedimentation rate. [CT](#) and [MRI](#) studies are essential for defining the extent of bone and soft-tissue involvement. CT shows bony destruction of the skull base in advanced cases. For culture, biopsies of granulation tissue in the canal or of deeper tissues are preferable to swab specimens of ear drainage, which may be unreliable. In nearly all cases, antibiotics should be withheld until a deep-tissue specimen is obtained for culture and pathologic examination. Once this specimen has been collected, empirical therapy with intravenous antibiotics active against *Pseudomonas* -- e.g., ticarcillin (3 g every 4 h), piperacillin, or ceftazidime, plus an aminoglycoside) -- may be started intraoperatively. To avoid ototoxicity, ciprofloxacin should be substituted for the aminoglycoside if cultures grow a *Pseudomonas* strain that is sensitive to this drug. In more than 95% of cases, *P. aeruginosa* is the pathogen involved; in the remaining cases, the pathogens include *Staphylococcus epidermidis*, *Aspergillus*, *Fusobacterium*, and *Actinomyces*. Systemic antibiotic treatment should be continued for 6 to 8 weeks. In early cases due to sensitive *Pseudomonas* strains, oral ciprofloxacin alone (750 mg twice daily for 6 weeks) may follow the initial 2 weeks of combination intravenous therapy.

ACUTE OTITIS MEDIA

The middle ear is connected to the nasopharynx via the eustachian tube. When this tube is blocked, fluid collects in the middle-ear and mastoid cavities, providing a culture medium for any bacteria present. Acute otitis media (AOM), or middle-ear infection, may result. Viral upper respiratory infections, which can cause edema of the eustachian tube mucosa, often precede or accompany episodes of AOM. Otitis media, like upper respiratory tract infections, is most common in fall, winter, and spring. The incidence of AOM declines with age. More than two-thirds of children under age 3 have had at least one episode of AOM; the prevalence among adults is only 0.25%.

Symptoms include ear pain, fever, and decreased hearing acuity. On examination, the tympanic membrane moves poorly with insufflation and is usually red, opaque, bulging, or retracted. Spontaneous perforations of the tympanic membrane and otorrhea are occasionally documented.

The bacteriology of [AOM](#) has been delineated for pediatric disease: *S. pneumoniae* (35%), *H. influenzae* (25%), and *M. catarrhalis* (15%) are the most common organisms. Viruses, either alone or with bacteria, are found in one-quarter of pediatric cases. Small studies of AOM in adults have also found *S. pneumoniae* (21%) and *H. influenzae* (26%) to be the most common pathogens. More than 90% of *H. influenzae* infections are due to nontypable strains: those due to type b may be accompanied by bacteremia or meningitis.

TREATMENT

Treatment of otitis media is empirical, as diagnostic tympanocentesis is indicated only for patients who appear toxic, who are immunocompromised, or whose infection is refractory to initial therapy. Although about one-third of *H. influenzae* strains and at least three-quarters of *M. catarrhalis* strains are β -lactamase producers, most authorities still find amoxicillin therapy to be successful in routine cases. Other drugs effective against most β -lactamase-positive strains include amoxicillin/clavulanate (875 mg by mouth twice daily for 7 to 10 days), trimethoprim-sulfamethoxazole, erythromycin/sulfisoxazole, clarithromycin, and second-generation oral cephalosporins (e.g., loracarbef, cefpodoxime proxetil, and cefuroxime axetil). Penicillin resistance in pneumococci, now a major problem, is not mediated by β -lactamase ([Chap. 138](#)). Strains exhibiting intermediate resistance may respond to therapy with high-dose amoxicillin or to clindamycin, erythromycin, or trimethoprim-sulfamethoxazole. Quinolones such as levofloxacin, although not approved for use in children, may be effective in adults. Serious infections or those due to highly resistant strains require treatment with parenteral ceftriaxone or vancomycin. Adjunctive treatment of [AOM](#) with antihistamines is of no proven benefit.

Recurrent episodes of [AOM](#) in children are due to the same pathogens that cause primary AOM (*S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*). Most early recurrences (75%), however, are not relapses but are due to different organisms or to different strains of the organism that caused the initial episode. The pattern of recurrent AOM in adults is presumably similar but has not been well studied. Treatment for recurrent AOM should include drugs with activity against resistant strains. Patients with frequent recurrences (e.g., three episodes within 6 months) may benefit from antibiotic prophylaxis with once-daily amoxicillin or sulfisoxazole during the winter months,

although this benefit must be weighed against the risk of selecting more antibiotic-resistant strains of bacteria.

SEROUS OTITIS MEDIA

Otitis media with effusion, or serous otitis media, is characterized by the persistence of middle-ear fluid for several months without other signs of infection. This condition is associated with a 25-dB hearing loss in the affected ear. Cultures of middle-ear fluid are usually negative. Although some clinical trials have found that effusions resolve sooner in antibiotic-treated children than in controls, antibiotics are generally not recommended because the risk of increasing antibiotic resistance in the population is thought to outweigh the small benefit observed. Adenoidectomy, myringotomy, or tympanostomy tubes have been shown to decrease the duration of effusion in children.

CHRONIC SUPPURATIVE OTITIS MEDIA

In chronic suppurative otitis media, patients have painless hearing loss and intermittent purulent ear drainage. On examination, there is a central perforation in the tympanic membrane and purulent drainage from the middle ear. If a cholesteatoma is present, the perforation is peripheral. Culture of draining fluid reveals *P. aeruginosa* (40%), *S. aureus* (20%), *Klebsiella* (20%), and other enteric gram-negative bacilli. Anaerobes are found in 50% of cases, usually in mixed culture with aerobes. [CT](#) should be used to help evaluate a surgically treatable nidus of infection, such as a cholesteatoma or mastoid sequestrum. For therapeutic purposes, patients are divided into two groups: those with and without cholesteatoma. Those in the former group are cured with surgical excision of the cholesteatoma. Those without cholesteatoma require repeated courses of topical antibiotic drops for relapse of "active" (draining) disease, and true cures are rare. The role of systemic antibiotics is unclear. In one study, a course of intravenous antibiotics produced long-term success in 78% of children without cholesteatoma who had persistent otorrhea despite topical and oral antibiotic therapy.

Tuberculous otitis media is rare and is frequently misdiagnosed. It mimics nontuberculous chronic suppurative otitis media, but ear drainage fails to respond to routine antibiotics. On examination, the tympanic membrane often has multiple perforations, and "pearly" or "flabby" tissue is seen in the middle ear. Only 30% of patients have evidence of active tuberculosis on chest x-ray. Treatment is the same as for other types of extrapulmonary tuberculosis.

MASTOIDITIS

The mastoid is the portion of the temporal bone posterior to the ear that contains a honeycomb of air cells lined with respiratory epithelium. These air cells connect with the middle ear. Fluid in the middle ear, a prelude to otitis media, is almost always accompanied by fluid in the mastoid. True mastoiditis, however, has become rare in the antibiotic era, probably because of prompt treatment of otitis media.

Mastoiditis is characterized by erosion of the bony partitions between the mastoid air cells. Patients with acute mastoiditis present with pain, tenderness, and swelling over the mastoid. When there is an overlying subperiosteal abscess or cellulitis, the pinna is

pushed out and forward. [CT](#) may show bony destruction or a drainable mastoid abscess.

The reported bacteriology of mastoiditis has varied. Some cases involve organisms similar to those implicated in [AOM](#) (*S. pneumoniae*, *H. influenzae*); others are attributable to *S. aureus* and gram-negative bacilli, including *Pseudomonas*. Ideally, therapy should be guided by the results of cultures of middle-ear fluid obtained by tympanocentesis. Initial broad-spectrum therapy, such as that with intravenous ticarcillin/clavulanate plus gentamicin or ciprofloxacin, can later be narrowed.

COMPLICATIONS OF OTITIS MEDIA AND MASTOIDITIS

Extracranial complications include hearing loss, labyrinthitis and resulting vertigo, and facial-nerve palsy. Additional complications from mastoiditis develop when infection tracks under the periosteum of the temporal bone to cause a subperiosteal abscess or breaks through the mastoid tip to cause a neck abscess deep to the sternocleidomastoid muscle (Bezold's abscess). Intracranial complications include epidural abscess, dural venous thrombophlebitis (usually sigmoid sinus), meningitis, and brain abscess.

INFECTIONS OF THE ORAL CAVITY AND PHARYNX

ORAL CAVITY INFECTIONS

The oral cavity extends from the lips to the circumvallate papillae of the tongue and is heavily colonized with viridans streptococci and anaerobes. These organisms can cause several infections in this area. *Gingivitis* is an infection of the gums, the earliest form of periodontal disease. Anaerobes residing in the mouth, especially anaerobic gram-negative rods such as *Prevotella intermedia*, are the most common pathogens. Patients with *Vincent's angina*, also called *acute necrotizing ulcerative gingivitis* or *trench mouth*, have halitosis and ulcerations of the interdental papillae. Oral anaerobes are the cause, and therapy with oral penicillin plus metronidazole or with clindamycin alone is effective in both this condition and gingivitis.

Ludwig's angina is a rapidly spreading, life-threatening cellulitis of the sublingual and submandibular spaces that usually starts in an infected lower molar. Patients are febrile and may drool the secretions they cannot swallow. A brawny, boardlike edema in the sublingual area pushes the tongue up and back. Airway obstruction may result as the infection spreads to the supraglottic tissues. Treatment with intravenous antibiotics active against streptococci and oral anaerobes -- e.g., ampicillin/sulbactam (3 g every 6 h) or high-dose penicillin plus metronidazole -- should be followed by oral antibiotic therapy, with a total treatment duration of 14 days. Airway monitoring is also essential. Intubation or tracheostomy may be necessary.

Noma, or *cancrum oris*, is a fulminant gangrenous infection of the oral and facial tissues that occurs in severely malnourished and debilitated patients and is especially common among children. Beginning as a necrotic ulcer in the gingiva of the mandible, noma is caused by oral anaerobes, especially fusospirochetal organisms (e.g., *Fusobacterium nucleatum*). It is treated with high-dose penicillin, debridement, and correction of the underlying malnutrition.

Herpes simplex commonly causes cold sores of the lips but may also cause painful vesicles on the tongue and buccal mucosa. Primary infection may require intravenous hydration and should be treated with acyclovir. *Thrush*, or oropharyngeal candidiasis, is an infection caused by *Candida* spp. such as *C. albicans*. It occurs in neonates, patients who have received prolonged antibiotic therapy, and immunocompromised patients. More than 90% of patients with AIDS develop thrush. Patients with thrush report a "burning" tongue or "raw" throat and, on examination, have white plaques on the tongue and oral mucosa. Treatment consists of topical antifungal agents (clotrimazole, nystatin) or oral fluconazole. Therapy for fluconazole-resistant thrush in patients with AIDS may be difficult; itraconazole oral solution or amphotericin B oral suspension may be effective.

PHARYNGITIS

Most cases of pharyngitis are thought to be viral. Many occur as part of common colds caused by rhinovirus, coronavirus, or parainfluenza virus. Patients have a scratchy or sore throat as well as coryza and cough. The pharynx is inflamed and edematous, but no exudate is evident. Influenza virus and adenovirus may cause a particularly severe sore throat, along with fever and myalgias. In infection with either of the latter viruses, there is pharyngeal erythema and edema; however, adenovirus infection also commonly causes an exudate, thus mimicking streptococcal pharyngitis. *Infectious mononucleosis* due to Epstein-Barr virus often causes a severe sore throat. Exudative pharyngitis or tonsillitis is documented in half of mononucleosis cases and may also mimic streptococcal infection. *Herpangina*, caused by coxsackievirus, is characterized by fever, sore throat, myalgias, and a vesicular enanthem on the soft palate between the uvula and the tonsils. There are usually only two to six lesions, which begin as small papules that vesiculate and then ulcerate. Fever and nonexudative pharyngitis are common symptoms of the acute retroviral syndrome that develops several weeks after infection with [HIV](#).

The most important bacterial cause of pharyngitis is group A *Streptococcus* (*S. pyogenes*). This organism is responsible for about 15% of all cases of pharyngitis and can cause important complications, both suppurative (peritonsillar and retropharyngeal abscess) and nonsuppurative (scarlet fever, streptococcal toxic shock syndrome, rheumatic fever, acute poststreptococcal glomerulonephritis). Fever, severe sore throat, cervical adenopathy, and inflammation of the tonsils and pharynx (which are covered with exudate) are classic findings. However, many cases of streptococcal pharyngitis are mild, with minimal erythema and no exudate, and mimic the pharyngitis of the common cold. Although some patients may in fact have viral pharyngitis and may simply be colonized with group A streptococci, these individuals must nevertheless be treated for presumed streptococcal pharyngitis. Diagnosis is made by culture. Rapid antigen tests are now available. These tests are less sensitive than they are specific: a positive test may be considered equivalent to a positive culture, but a negative test requires culture confirmation. Either a single dose of intramuscular benzathine penicillin (1.2 million units) or a 10-day course of oral penicillin (250 mg four times daily) or erythromycin is necessary to eradicate the organism. Sensitivity to erythromycin should be verified if this agent is used, as an increase in erythromycin resistance has been noted, especially in Europe. Other antibiotics active against streptococci may be used

(e.g., amoxicillin, cefuroxime), and one trial showed that 4 days of cefuroxime therapy was as effective as 10 days of penicillin treatment in eradicating the organism. However, studies of the prevention of rheumatic fever are available only for penicillin ([Chap. 235](#)).

Other bacterial causes of pharyngitis include groups C and G *Streptococcus*, *Neisseria gonorrhoeae* ([Chap. 147](#)), *Arcanobacterium haemolyticum*, *Yersinia enterocolitica*, and -- very rarely -- *Corynebacterium diphtheriae* ([Chap. 141](#)). In addition, *Mycoplasma pneumoniae* ([Chap. 178](#)) and *Chlamydia pneumoniae* ([Chap. 179](#)) can cause pharyngitis.

A peritonsillar abscess (*quinsy*) may follow untreated streptococcal pharyngitis. Oral anaerobes also play a role in quinsy. Patients have a severe sore throat and speak with a "hot-potato" voice. Examination reveals pronounced unilateral peritonsillar swelling and erythema causing deviation of the uvula. Immediate aspiration by an otolaryngologist is required in conjunction with antibiotic therapy -- e.g., ampicillin/sulbactam (3 g intravenously every 6 h), penicillin plus metronidazole, or clindamycin.

LARYNGITIS, CROUP, AND EPIGLOTTITIS

LARYNGITIS

Laryngitis is characterized by hoarseness. Most cases of acute laryngitis are caused by viruses (rhinovirus, influenza virus, parainfluenza virus, coxsackievirus, adenovirus, or respiratory syncytial virus). Acute laryngitis may also be associated with group A *Streptococcus* and *M. catarrhalis*. Laryngitis must be differentiated from epiglottitis (see below). The goal of treatment is merely the relief of symptoms except when throat cultures are positive for group A *Streptococcus* (in which case penicillin should be used).

Chronic laryngitis due to infection is rare and must be distinguished from hoarseness of neoplastic etiology. *Tuberculous laryngitis* may be mistaken for laryngeal cancer when assessed by direct laryngoscopy. Laryngeal and supraglottic lesions include mucosal hyperemia and thickening, nodules, and ulcerations. In one study, a history of fever and night sweats was rare, and the most common chest radiographic finding was apical thickening and fibrosis. Biopsy reveals granulomas with acid-fast bacilli. Cultures should be performed to confirm the diagnosis and evaluate the sensitivities of the pathogen. Laryngeal tuberculosis is highly contagious and should be managed with the same precautions and therapy used for active pulmonary disease ([Chap. 169](#)). Fungal infections causing laryngitis include histoplasmosis ([Chap. 201](#)), blastomycosis ([Chap. 203](#)), and candidiasis ([Chap. 205](#)). *Histoplasma* and *Blastomyces* may cause nodules on the larynx, with or without ulcerations. *Candida* may cause laryngitis, along with thrush, in immunosuppressed patients or in patients with chronic mucocutaneous candidiasis.

CROUP

Croup, or acute laryngotracheobronchitis, is an infection of the upper and lower respiratory tract that causes marked subglottic edema. It mainly affects 2- and

3-year-old children and usually follows the onset of upper respiratory tract infection by 1 to 2 days. Symptoms include fever, hoarseness, a "seal's bark" cough, and inspiratory stridor. The most common etiology is parainfluenza virus, although croup may also be caused by other respiratory viruses (e.g., influenza or respiratory syncytial virus).

Croup must be differentiated from epiglottitis (see below). Epiglottitis usually progresses more rapidly and produces a more toxic appearance. Neck x-rays may be helpful but do not reliably exclude epiglottitis. In croup, the anterior-posterior neck x-ray shows subglottic edema (the "hourglass sign"); in epiglottitis, the lateral neck view shows a thick epiglottis.

Patients with severe croup should be hospitalized, monitored for hypoxemia through pulse oximetry, and watched for airway obstruction requiring intubation. Humidification is commonly prescribed, but few controlled trials have assessed its benefit. Nebulized racemic epinephrine provides temporary (2-h) improvement in patients with marked stridor, but such patients must be observed for rebound edema. Glucocorticoid therapy, either nebulized or parenteral, is clearly beneficial, and its effects are often evident within 1 h. One trial found that treatment with a single dose of either intramuscular dexamethasone or nebulized budesonide reduced the need for hospitalization of children with moderately severe croup by more than 50%.

EPIGLOTTITIS

Acute epiglottitis (supraglottitis) is a life-threatening, rapidly progressive cellulitis of the epiglottis that may cause complete airway obstruction. It begins as a cellulitis between the tongue base and the epiglottis that pushes the epiglottis posteriorly. The epiglottis itself then becomes swollen, threatening the airway. Before the introduction of *H. influenzae* type b (Hib) vaccine, epiglottitis was most common among children 2 to 4 years old. The disease is now rare in children, since the vaccine has reduced the incidence of invasive disease due to Hib by more than 95%. The incidence in adults has not changed.

The typical young child with epiglottitis has a several-hour history of fever, irritability, dysphonia, and dysphagia and presents sitting forward and drooling. Adolescents and adults usually have a less fulminant presentation, with symptoms (especially sore throat) of 1 or 2 days' duration. Adults may present with dyspnea (25%), drooling (15%), and stridor (10%). Epiglottitis constitutes a medical emergency, as airway occlusion may occur suddenly. Lateral neck films showing an enlarged epiglottis (the "thumb sign") are helpful if positive but may be falsely negative. The value of obtaining these films has also been questioned because doing so may cause a critical delay in securing the airway. Direct viewing of the pharynx by use of a tongue blade should not be attempted, as immediate laryngospasm and airway obstruction may result. Instead, a child with suspected epiglottitis should be transported -- while sitting up -- to the operating room for visualization of the epiglottis with a fiberoptic laryngoscope, with preparations made for immediate airway control. If the epiglottis is cherry-red, an uncuffed endotracheal tube should be placed. Diagnosis in adults is also made by direct viewing of the epiglottis with a flexible fiberoptic laryngoscope, again only after preparations are made to secure the airway.

All patients must be closely monitored in an intensive care unit and should be given antibiotics active against *H. influenzae*. Before [Hib](#) vaccine became available, this organism was responsible for nearly all pediatric cases and was isolated from the blood of almost 100% of the affected children. In adults, blood cultures are positive in about 25% of cases, all of which are due to *H. influenzae*. Other pathogens isolated from the pharynx of adults with epiglottitis include *H. parainfluenzae*, *S. pneumoniae*, group A *Streptococcus*, and (rarely) *S. aureus*; the correlation between throat and epiglottitis cultures is unclear, however. Children may be treated with intravenous cefuroxime, ceftriaxone, ampicillin/sulbactam, or trimethoprim-sulfamethoxazole. Adults may be treated for at least 7 days with cefuroxime, ampicillin/sulbactam (3 g intravenously every 6 h), or nafcillin plus ceftriaxone; those highly allergic to penicillin may be given clindamycin plus either trimethoprim-sulfamethoxazole or ciprofloxacin. If the patient with *H. influenzae* epiglottitis has household contacts that include an unvaccinated child under age 4, all members of the household and the patient should receive prophylactic rifampin to eradicate the carriage of *H. influenzae*.

DEEP NECK INFECTIONS

Deep neck infections may be life-threatening because of airway compromise, involvement of the carotid sheath, or spread into the mediastinum.

SUBMANDIBULAR SPACE INFECTIONS

See *Ludwig's angina* above (under "Oral Cavity Infections").

LATERAL PHARYNGEAL SPACE INFECTIONS

The lateral pharyngeal space, also called the parapharyngeal or pharyngomaxillary space, is in the superior lateral portion of the neck and extends from the hyoid bone to the base of the skull. It lies deep to the lateral wall of the pharynx and is lateral to the tonsil and carotid sheath and medial to the parotid gland. Infection in this space may follow tonsillitis, pharyngitis with adenoid involvement, parotitis, mastoiditis, or periodontal infection.

On presentation, most patients appear toxic and have fever, sore throat, pain on swallowing, and leukocytosis. Infection confined to the posterior (retrostyloid) portion of the lateral pharyngeal space causes swelling of the lateral pharyngeal wall, which may be missed because it is behind the palatopharyngeal arch. Involvement of the anterior portion of this space causes medial displacement of the tonsil, swelling over the parotid gland, and trismus. Rigidity of the neck or torticollis toward the opposite side may develop. Diagnosis is confirmed by [CT](#) with contrast.

Treatment includes securing of the airway, surgical drainage in the operating room, and administration for at least 10 days of intravenous antibiotics active against streptococci and oral anaerobes (e.g., ampicillin/sulbactam, 3 g every 6 h). Major complications result from involvement of the carotid sheath and the vessels it contains. These complications are frequently fatal and include jugular vein thrombophlebitis, erosion into the carotid artery, and mediastinitis. Jugular vein thrombophlebitis is characterized by high fevers, chills, and neck tenderness at the angle of the mandible. When it is caused

by *Fusobacterium necrophorum*, it may be accompanied by sepsis and septic pulmonary emboli (*Lemierre's syndrome*). Erosion into the carotid artery is usually heralded by repeated small bleeds into the mouth. The involvement of adjacent cranial nerves may result in ipsilateral Horner's syndrome, hoarseness, or unilateral tongue paresis. Extension of infection along the carotid sheath into the posterior mediastinum results in mediastinitis and a mortality of 50%. [MRI](#) is useful in delineating carotid and jugular involvement.

RETROPHARYNGEAL SPACE INFECTIONS

The retropharyngeal space lies between the pharynx and the prevertebral fascia and extends from the base of the skull into the mediastinum. Infection in this space may result from the spread of lateral pharyngeal space infection or from the lymphatic spread of infection in more cephalad sites (posterior sinuses, adenoids, nasopharynx) to the retropharyngeal lymph nodes. Retropharyngeal abscess is most common among infants and young children, probably because the retropharyngeal nodes later involute. Retropharyngeal abscess may also follow trauma to the posterior pharynx (e.g., endoscopy in adults, lollipop-stick perforation in children) or may result from anterior extension of infection from cervical osteomyelitis.

Symptoms include fever, marked difficulty and pain with swallowing, and a "hot-potato" voice. Physical examination may document drooling, nuchal rigidity, and bulging of the posterior pharyngeal wall. Advanced cases include dyspnea and stridor. Diagnosis may be confirmed by a lateral neck soft-tissue x-ray or [CT](#) scan. Treatment requires securing of the airway and emergency surgical drainage. Intravenous antibiotics should be given; the agents chosen should be active against streptococci, oral anaerobes, *S. aureus*, and *H. influenzae* (e.g., ampicillin/sulbactam alone or clindamycin plus ceftriaxone). Potential complications include airway obstruction, intraoral rupture of the abscess causing aspiration pneumonia, and mediastinitis.

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31. ORAL MANIFESTATIONS OF DISEASE - John S. Greenspan

A thorough oral examination, to include the oral and pharyngeal soft tissues as well as the teeth, is an important part of the physical examination. The common oral diseases are due to infection by bacteria, fungi, or viruses. The complex development of the orofacial structures leads to close interposition of diverse tissues, which are prone to developmental anomalies, growth disturbances, and neoplasia.

DISEASES OF THE TEETH

DENTAL CARIES, PULPAL AND PERIAPICAL DISEASE, AND COMPLICATIONS

Dental caries is a destructive disease of the hard tissues of the teeth due to infection with *Streptococcus mutans* and other bacteria. In the United States, fewer than half of those 17 years and younger now have carious lesions, although in many segments of the population and in developing countries the disease is more common. Artificial fluoridation of water to a level of 1 part per million, fluoride-containing toothpastes, and topical fluoride administration have reduced the incidence. Conversely, retention of teeth and the aging of the population have led to an increase in root caries. Increasing numbers of individuals surviving cancer therapy and other special populations (diabetic patients and those with xerostomia due to Sjogren's syndrome or to medications) may experience severe caries unless appropriate topical fluoride prophylaxis is used. Treatment of caries involves removal of the softened and infected hard tissues, sealing of the exposed dentine, and restoration of the lost tooth structure with silver amalgam, composite plastic, gold, or porcelain.

If the carious lesion progresses, infection of the dental pulp may occur, causing *acute pulpitis*. The tooth may become sensitive to hot or cold. When severe continuous throbbing pain ensues, pulp damage is irreversible, and root canal therapy becomes necessary. The contents of the pulp chamber and root canals are removed, followed by thorough cleaning, antisepsis, and filling with an inert material. Alternatively, extraction of the tooth may be indicated.

If the pulpitis is not treated successfully, infection may spread beyond the tooth apex into the periodontal ligament. Acute inflammation causes pain on chewing or on percussion, and a *periapical abscess* may form. Chronic inflammation may be painless or produce only slight pain, and a *periapical granuloma* may form within the alveolar bone. Proliferation of epithelial cell rests may convert the granuloma into a *periapical cyst*. Periapical radiolucencies may occur with the granuloma or the cyst but not with the abscess, unless it forms as a complication of one or two lesions. The pus in the periapical abscess may track through the alveolar bone into soft tissues, causing cellulitis and bacteremia, or may discharge into the oral cavity (*parulis* or *gumboil*), into the maxillary sinus, or through the skin of the face or submandibular area. A severe form of cellulitis, *Ludwig's angina*, originates from an infected mandibular molar, involves the submandibular space, and extends throughout the floor of the mouth, with elevation of the tongue, dysphagia, and difficulty breathing. Glottal edema may occur, necessitating tracheotomy.

EFFECT OF SYSTEMIC FACTORS ON TEETH

Enamel hypoplasia of the primary and/or permanent teeth, manifested by alterations ranging from white spots to gross defects in the surface structure of the crowns, may be caused by disturbances of calcium and phosphate metabolism such as are found in vitamin D-resistant rickets, hypoparathyroidism, gastroenteritis, and celiac disease. Premature birth or high fevers may also give rise to enamel hypoplasia. Tetracycline, when given during the second half of pregnancy, in infancy, and in childhood up to 8 years of age, causes both a permanent discoloration of the teeth and enamel hypoplasia. Daily ingestion of more than 1.5 mg fluoride can result in enamel discoloration (*mottling*). Larger teeth are associated with maternal diabetes, maternal hypothyroidism, and large birth size. Tooth size is reduced in Down's syndrome. Systemic disease may give rise to pain that simulates pulpal disease. Maxillary sinusitis is frequently manifested as pain in the maxillary teeth, including sensitivity to thermal changes and percussion. Angina pectoris may result in pain referred to the lower jaw, probably through the vagus nerve.

PERIODONTAL DISEASES

In adults, chronic destructive periodontal disease (*pyorrhea*) is responsible for more loss of teeth than caries, particularly in the aged. However, the prevalence and incidence of periodontal disease also appears to be declining in the United States. The most common form of periodontal disease starts as inflammation of the marginal gingiva (*gingivitis*), which is painless, although the gingiva may bleed on brushing. The disease spreads to involve the periodontal ligament, alveolar bone is slowly resorbed, and periodontal ligament attachment between tooth and bone is lost. The soft tissue separates from the tooth surface, causing "pocket" formation with bleeding on probing and during chewing. Acute inflammation may become superimposed on this chronic process, with the production of pus and the formation of a *periodontal abscess*. Ultimately, extreme bone loss, tooth mobility, and recurrent abscess formation lead to tooth exfoliation or may mandate tooth extraction.

Gingivitis and periodontitis are infections associated with the accumulation of *bacterial plaque*, which may become mineralized (*calculus*) and can be prevented by appropriate *oral hygiene* measures, including tooth brushing, flossing, antibacterial mouth rinses, and the removal of impacted food debris. Poorly fabricated or deteriorated restorations may contribute through overextended or inadequate margins. Therapy consists of removal of plaque and calculus, debridement of the pocket lining and superficial infected cementum, and elimination of other contributing factors.

Periodontal disease appears to be a group of conditions, including *adult periodontitis*, associated with *Porphyromonas gingivalis*, *Prevotella intermedia*, and other gram-negative organisms. *Localized juvenile periodontitis* (LJP) causes rapid, severe pocketing and bone loss and is associated with *Actinobacillus actinomycetemcomitans*, *Capnocytophaga*, *Eikenella corrodens*, and other anaerobes. *Acute necrotizing ulcerative gingivitis* (ANUG) involves sudden inflammation of the gingivae with necrosis, tissue loss, pain, bleeding, and halitosis and is associated with *P. intermedia* and spirochetes. ANUG and an aggressive and rapid form of periodontitis (*necrotizing ulcerative periodontitis*) are seen in association with HIV infection. Some of these cases progress to a destructive gangrene-like lesion of oral soft tissues and bone (*necrotizing*

stomatitis) resembling the *noma* seen in severely malnourished populations. Therapy involves local antibacterial measures, debridement, and, in severe cases, systemic antibiotics effective against anaerobes.

Host factors may be involved in the pathogenesis of periodontal disease in other populations as well. Severe periodontal disease may occur in persons with *Down's syndrome* and *diabetes mellitus*. During pregnancy there may be severe gingivitis and the formation of localized *pyogenic granulomas*. Certain drugs, notably the anticonvulsant *phenytoin* and the calcium channel blocker *nifedipine*, cause *fibrous hyperplasia* of the gingiva, which may cover the teeth, interfere with eating, and be unsightly. *Idiopathic familial gingival fibromatosis* may appear similar. Surgery may correct both conditions; change in medication may reverse the drug-induced form. The oral cavity is a significant reservoir for *Helicobacter pylori*. Uncontrolled diabetes mellitus leads to an exacerbation of oral infection, notably periodontal disease. In individuals genetically predisposed to diabetes, periodontal disease may also precipitate or exacerbate the diabetes. Oral infection has been proposed to contribute to coronary atherosclerosis as well as pregnancy outcomes such as premature labor and low birthweight.

Periapical and periodontal bacterial infections can cause transient bacteremia after tooth extraction and even routine dental prophylaxis. Antibiotic coverage is appropriate in patients with heart valves susceptible to infection or those with prosthetic joints.

DISEASES OF THE ORAL MUCOSA

INFECTIONS

Most oral mucosal diseases involve microorganisms ([Table 31-1](#)).

PIGMENTED LESIONS See [Table 31-2](#)

DERMATOLOGIC DISEASES See [Tables 31-1, 31-2](#), and [31-3](#) and [Chaps. 55, 56, 57, 58, 59](#), and [60](#)

DISEASES OF THE TONGUE See [Table 31-4](#)

HALITOSIS See [Table 31-5](#)

HIV DISEASE AND AIDS (See [Table 31-6](#) and also [Chaps. 191](#) and [309](#))

Immunosuppression induced by HIV infection predisposes to numerous oral infections, neoplasms, and autoimmune and idiopathic lesions. *Oral candidiasis* ([Plate IID-43](#)) and *hairy leukoplakia* ([Plate IID-42](#)) [a benign epithelial hyperplasia associated with Epstein-Barr virus (EBV)] are common features of HIV disease and often precede or accompany full-blown AIDS. Oral Kaposi's sarcoma and lymphoma are diagnostic of AIDS. Oral candidiasis is easily treated with topical or systemic antifungals: nystatin oral pastilles, clotrimazole oral troches, nystatin vaginal tablets used orally, fluconazole, and ketoconazole. While most oral lesions of HIV disease are also found in the general population. Necrotizing ulcerative periodontal disease and hairy leukoplakia are strongly

associated with HIV infection and are otherwise very rare.

HEMATOLOGIC AND NUTRITIONAL DISEASE

Gingival bleeding, necrotic ulcers, and enlargement due to malignant infiltrates are seen in all forms of leukemia, particularly *monocytic leukemia*. In *agranulocytosis* severe oral mucosal ulcers are seen, while in *thrombocytopenia* oral petechiae, ecchymoses, and gingival bleeding occur. In *Plummer-Vinson syndrome* ([Chap. 105](#)), atrophy of oral mucosa, particularly the tongue papillae, causes redness and soreness as well as dysphagia and is associated with increased susceptibility to oral cancer. A smooth tongue can also be seen in *pernicious anemia* ([Chap. 107](#)). Severe oral mucositis with ulcers, candidiasis, bacterial infections, and xerostomia complicate radiation therapy for head and neck cancers. Chemotherapy may also cause mucositis. Although now rarely seen in the United States, oral features of vitamin deficiency include oral mucositis and ulcers, glossitis, and burning sensations in the tongue (*B group vitamin deficiency*) and petechiae, gingival swelling, bleeding, and ulceration as well as loosening of teeth (*scurvy* of vitamin C deficiency).

DISEASES OF THE SALIVARY GLANDS

The major and minor salivary glands can be involved in mumps, sarcoidosis, tuberculosis, lymphoma, and Sjogren's syndrome ([Chap. 314](#)). The latter may cause dry eyes and dry mouth (*xerostomia*) and be associated with features of connective tissue diseases, including rheumatoid arthritis or systemic lupus erythematosus. Xerostomia may also be due to medications such as diuretics, antihistamines, or tricyclic antidepressants as well as radiation therapy for head and neck cancer. Without lysozyme-rich saliva, *cervical or incisal caries* and oral candidiasis may develop. Management includes fluoride mouth rinses and topical applications, saliva substitutes, salivary stimulation with sugarless candies, and the avoidance of sugar-containing drinks or food. Candidiasis is treated with nystatin or other antifungals. Salivary stones (*sialolithiasis*), usually in the duct of a major salivary gland, cause *sialoadenitis* with pain and swelling, often on eating, especially tart foods such as lemons.

The most common neoplasm of the salivary glands is the *pleomorphic adenoma*, which is benign but will recur unless fully resected; malignant tumors include *mucoepidermoid carcinoma*, *adenoid cystic carcinoma*, and *adenocarcinoma*. The pleomorphic adenoma causes a firm, slowly growing mass in the parotid, palate, or cheek, whereas malignant tumors grow faster and can cause ulceration and invade nerves, producing numbness or facial paralysis.

NEUROLOGIC DISTURBANCES AND OROFACIAL PAIN

The mouth and face may be the site of pain from a number of vascular, neurologic, muscle/connective tissue, or joint conditions. Interdisciplinary diagnosis and management programs involving neurologists, restorative dentists, oral surgeons, otorhinolaryngologists, and other specialists, together with new imaging techniques to diagnose or exclude organic lesions, have begun to clarify this complex field. *Temporal arteritis* causes pain in the face, jaws, and tongue and may mimic temporomandibular joint disease. Glucocorticoids may provide relief. *Myofascial pain* is a dull, constant ache

with local tenderness in the muscles of the jaws and difficulty in opening the mouth. Teeth clenching and grinding (*bruxism*) may play a role. *Arthralgia* of the temporomandibular joint causes local pain, which may extend to the face and head. Both myofascial pain and arthralgia can be relieved with heat, rest, and anti-inflammatory agents. Displacement of the meniscus or condyle may cause pain, clenching, or locking of the mandible in the open position. The joint may become involved in *osteoarthritis* with minimal symptoms, whereas *rheumatoid arthritis* causes pain and swelling in the joint and limitation of movement. *Ankylosis* may occur, necessitating condylectomy ([Chap. 312](#)).

Trigeminal neuralgia (tic douloureux) causes sudden, severe, unilateral lancinating pain initiated by touching a "trigger zone" or occurring spontaneously. Confusion with pulpal or periapical pain is common, leading to inappropriate endodontic or surgical therapy. Many cases respond to carbamazepine and phenytoin, but for a few, surgical intervention to decompress the trigeminal nerve is indicated. Similar symptoms in the distribution of the ninth cranial nerve (tongue, pharynx, soft palate) are due to *glossopharyngeal neuralgia*, which may be triggered by swallowing and may produce referred pain in the temporomandibular joint. *Postherpetic neuralgia* may follow trigeminal herpes zoster ([Chap. 367](#)) and cause burning, aching, and long-lasting pain. *Facial palsy* is usually unilateral and may be due to trauma, surgical intervention, tumor, or infection of the seventh cranial nerve. *Bell's palsy* is a form with acute onset and unknown cause, possibly viral infection such as herpes zoster. The corner of the mouth droops, and there may be difficulty in speech, eating, and in closing the eye. The symptoms usually disappear spontaneously, but residual facial immobility and lip drooping may persist. Abnormal or reduced *taste sensation* may be due to xerostomia, disturbances of the facial and glossopharyngeal nerves or their central connections, aging, or the wearing of dentures. Disease involving the hypoglossal nerve may cause atrophy of the tongue muscles with protrusion, if bilateral, or deviation toward the affected side, if unilateral. Numb chin (mental neuropathy) may be a sign of primary neural disease, but in the cancer patient it is often a harbinger of tumor relapse or progression.

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SECTION 5 -ALTERATIONS IN CIRCULATORY AND RESPIRATORY FUNCTIONS

32. DYSPNEA AND PULMONARY EDEMA - Roland H. Ingram, Jr., Eugene Braunwald

DYSPNEA

Breathing is controlled by central and peripheral mechanisms that adjust ventilation appropriate to increased metabolic demands during physical activity and increase ventilation in excess of metabolic demands in conditions such as anxiety and fear. A normal resting person is unaware of the act of breathing, and while he or she may become conscious of breathing during mild to moderate exertion, no discomfort is experienced. However, during and following exhausting exertion, an individual may become unpleasantly aware of breathing yet feel reasonably assured that the sensation will be transitory and is appropriate to the level of exercise. Therefore, as a cardinal symptom of diseases affecting the cardiorespiratory system, *dyspnea* is defined as an *abnormally uncomfortable awareness of breathing*.

Although dyspnea is not painful in the usual sense of the word, it is, like pain, involved with both the perception of a sensation and the reaction to that perception. Patients experience a number of uncomfortable sensations related to breathing and use an even larger number of verbal expressions to describe these sensations, such as "cannot get enough air," "air does not go all the way down," "smothering feeling or tightness or tiredness in the chest," and a "choking sensation." It may be necessary, therefore, to review the patient's history meticulously in order to ascertain whether the more abstruse descriptions do, in fact, represent dyspnea. Once it is established that a patient does have dyspnea, it is of paramount importance to define the circumstances in which it occurs and to assess associated symptoms. There are situations in which breathing appears labored but in which dyspnea does not occur. For example, the hyperventilation associated with metabolic acidemia is rarely accompanied by dyspnea. On the other hand, patients with apparently normal breathing patterns may complain of shortness of breath.

QUANTITATION OF DYSPNEA

The gradation of dyspnea is based on the amount of physical exertion required to produce the sensation. In assessing the severity of dyspnea, it is important to obtain a clear understanding of the patient's general physical condition, work history, and recreational habits. For example, the development of dyspnea in a trained runner upon running 2 mi may signify a much more serious disturbance than a similar degree of breathlessness in a sedentary person upon running a fraction of this distance. Interindividual variation in perception must also be considered. Some patients with severe disease may complain of only mild dyspnea; others with mild disease may experience more severe shortness of breath. Some patients with lung or heart disease may have such reduced capabilities due to other disease (e.g., peripheral vascular insufficiency or severe osteoarthritis of the hips or knees) that exertional dyspnea is precluded despite serious impairment of pulmonary or cardiac function.

Some patterns of dyspnea are not directly related to physical exertion. Sudden and

unexpected dyspneic episodes at rest can be associated with pulmonary emboli, spontaneous pneumothorax, hypercapnea secondary to breath holding, or anxiety. Nocturnal episodes of severe paroxysmal dyspnea are characteristic of left ventricular failure. Dyspnea upon assuming the supine posture, *orthopnea* (see below and [Chap. 232](#)), thought to be mainly characteristic of congestive heart failure, may also occur in some patients with asthma and chronic obstruction of the airways and is a regular finding in the rare occurrence of bilateral diaphragmatic paralysis. *Trepopnea* is used to describe the unusual circumstance in which dyspnea occurs only in a lateral decubitus position, most often in patients with heart disease, while *platypnea* is dyspnea that occurs only in the upright position. Positional alterations in ventilation-perfusion relationships ([Chap. 250](#)) have been invoked to explain these patterns.

MECHANISMS OF DYSPNEA (See [Fig. 32-1](#))

Dyspnea occurs whenever the work of breathing is excessive. Increased force generation is required of the respiratory muscles to produce a given volume change if the chest wall or lungs are less compliant or if resistance to airflow is increased. Increased work of breathing also occurs when the ventilation is excessive for the level of activity. Although an individual is more apt to become dyspneic when the work of breathing is increased, the work theory does not account for the perceptual difference between a deep breath with a normal mechanical load and a normal-sized breath with an increased mechanical load. The work might be the same with both breaths, but the normal one with the increased load will be associated with discomfort. In fact, with respiratory loading, such as adding a resistance at the mouth, there is an increase in respiratory center output that is disproportionate to the increase in the work of breathing. It has been postulated that whenever the force that muscles actually generate during breathing approaches some fraction of their maximal force-generating ability, which may vary among individuals, dyspnea ensues due to transduction of mechanical to neural stimuli.

In all likelihood, several different mechanisms operate to different degrees in the various clinical situations in which dyspnea occurs. In some circumstances, dyspnea is evoked by stimulation of receptors in the upper respiratory tract; in others it may originate from receptors in the lungs, airways, respiratory muscles, chest wall, or some combination of these structures. In any event, dyspnea is characterized by an excessive or abnormal activation of the respiratory centers in the brainstem. This activation comes about from stimuli transmitted from or through a variety of structures and pathways, including (1) intrathoracic receptors via the vagi; (2) afferent somatic nerves, particularly from the respiratory muscles and chest wall, but also from other skeletal muscles and joints; (3) chemoreceptors in the brain, aortic and carotid bodies, and elsewhere in the circulation; (4) higher (cortical) centers; and perhaps (5) afferent fibers in the phrenic nerves. In general, despite the interindividual variations described above, there is a reasonable correlation between the severity of dyspnea and the magnitude of disturbances of pulmonary or cardiac function that are responsible.

The mechanisms responsible for dyspnea may vary in different conditions ([Table 32-1](#)).

DIFFERENTIAL DIAGNOSIS

Obstructive Disease of Airways (See also [Chaps. 252](#) and [258](#)) Obstruction to airflow can be present anywhere from the extrathoracic airways out to the small airways in the periphery of the lung. Large extrathoracic airway obstruction can occur acutely, as with aspiration of food or a foreign body or with angioedema of the glottis. An allergic history together with a few scattered hives should raise the possibility of glottic edema. Acute upper airway obstruction is a medical emergency. More chronic forms can occur with tumors or with fibrotic stenosis following tracheostomy or prolonged endotracheal intubation. Whether acute or chronic, the cardinal symptom is dyspnea, and the characteristic signs are stridor and retraction of the supraclavicular fossae with *inspiration*.

Obstruction of intrathoracic airways can occur acutely and intermittently or can be present chronically with worsening during respiratory infections. Acute intermittent obstruction with wheezing is typical of *asthma* ([Chap. 252](#)). Chronic cough with expectoration is typical of *chronic bronchitis* ([Chap. 258](#)) and *bronchiectasis* ([Chap. 256](#)). Most often there are prolongation of expiration and coarse rhonchi that are generalized in chronic bronchitis and may be localized in the case of bronchiectasis. Intercurrent infection results in worsening of the cough, increased expectoration of purulent sputum, and more severe dyspnea. During such episodes, the patient may complain of nocturnal paroxysms of dyspnea with wheezing relieved by cough and expectoration of sputum. Despite the fact that severe limitation of expiratory flow and hyperinflation of the lung are characteristic of these diseases, the sensory experience is often that of an inability to take in a sufficiently deep breath rather than difficulty in exhaling.

The patient with predominant *emphysema* is characterized by many years of exertional dyspnea progressing to dyspnea at rest ([Chap. 258](#)). Although a parenchymal disease by definition, emphysema is invariably accompanied by obstruction of airways.

Diffuse Parenchymal Lung Diseases (See also [Chap. 259](#)) This category includes a large number of diseases ranging from acute pneumonia to chronic disorders such as sarcoidosis and the various forms of *pneumoconiosis* ([Chap. 254](#)). History, physical findings, and radiographic abnormalities often provide clues to the diagnosis. The patients are often tachypneic with arterial P_{CO_2} and P_{O_2} values below normal. Exertion often further reduces the arterial P_{O_2} . Lung volumes are decreased, and the lungs are stiffer, i.e., less compliant than normal.

Pulmonary Vascular Occlusive Diseases (See also [Chap. 261](#)) Repeated episodes of dyspnea at rest often occur with recurrent pulmonary emboli. Evidence of a source for emboli, such as phlebitis of a lower extremity or the pelvis, is quite helpful in leading the physician to suspect the diagnosis. Arterial blood gases are most often abnormal, but lung volumes are frequently normal or only minimally abnormal.

Diseases of the Chest Wall or Respiratory Muscles (See also [Chap. 263](#)) The physical examination establishes the presence of a chest wall disease such as severe kyphoscoliosis, pectus excavatum, or ankylosing spondylitis. Although all three of these deformities may be associated with dyspnea, only severe kyphoscoliosis regularly interferes with ventilation sufficiently to produce chronic cor pulmonale and respiratory failure.

Both weakness and paralysis of respiratory muscles can lead to respiratory failure and dyspnea ([Chap. 263](#)), but most often the signs and symptoms of the neurologic or muscular disorder are more prominently manifested in other systems.

Heart Disease In patients with cardiac disease, exertional dyspnea occurs most commonly as a consequence of an elevated pulmonary capillary pressure, which in turn may be due to left ventricular dysfunction ([Chaps. 231](#) and [232](#)), reduced left ventricular compliance, and mitral stenosis. The elevation of hydrostatic pressure in the pulmonary vascular bed tends to upset the Starling equilibrium (see "Pulmonary Edema," below) with resulting transudation of liquid into the interstitial space, reducing the compliance of the lungs and stimulating J (juxtacapillary) receptors in the alveolar interstitial space. When it is prolonged, pulmonary venous hypertension results in thickening of the walls of small pulmonary vessels and an increase in perivascular cells and fibrous tissue, causing a further reduction in compliance. The competition for space among vessels, airways, and increased liquid within the interstitial space compromises the lumina of small airways, increasing the airways' resistance. Diminution in compliance and an increase in the airways' resistance increase the work of breathing. In advanced congestive heart failure, usually involving elevation of both pulmonary and systemic venous pressures, hydrothorax may develop, interfering further with pulmonary function and intensifying dyspnea.

Orthopnea, i.e., dyspnea in the supine position, is the result of the alteration of gravitational forces when this position is assumed, which elevates pulmonary venous and capillary pressures. These, in turn, increase the pulmonary closing volume ([Chap. 250](#)) and reduce the vital capacity.

Paroxysmal (Nocturnal) Dyspnea Also known as *cardiac asthma*, this condition is characterized by attacks of severe shortness of breath that generally occur at night and usually awaken the patient from sleep. The attack is precipitated by stimuli that aggravate previously existing pulmonary congestion; frequently, the total blood volume is augmented at night because of the reabsorption of edema from dependent portions of the body during recumbency. A sleeping patient can tolerate relatively severe pulmonary engorgement and may awaken only when actual pulmonary edema and bronchospasm have developed, with the feeling of suffocation and with wheezing respirations.

Two other forms of nocturnal dyspnea must be distinguished from that due to heart failure. Chronic bronchitis is characterized by mucus hypersecretion and, after a few hours sleep, secretions can accumulate and produce dyspnea and wheezing, both of which are relieved by cough and expectoration of sputum. Asthma patients have circadian variations in their degree of airway obstruction. The obstruction becomes most severe between 2 A.M. and 4 A.M. and can be sufficiently severe that the patient awakens with a sense of suffocation, extreme dyspnea, and wheezing. Although there is a prominent inflammatory component to nocturnal asthma, inhaled bronchodilators usually improve symptoms quickly.

*Cheyne-Stokes respiration**[See Chap. 232](#)

Diagnosis The diagnosis of cardiac dyspnea depends on the recognition of heart disease on the basis of the clinical examination supplemented by noninvasive testing. There may be a history of antecedent myocardial infarction; third and fourth heart sounds may be audible; and/or there may be evidence of left ventricular enlargement, jugular neck vein distention, and/or peripheral edema. Often there are radiographic signs of heart failure, with evidence of interstitial edema, pulmonary vascular redistribution, and accumulation of liquid in the septal planes and pleural cavity. Transthoracic echocardiography is particularly useful in establishing the diagnosis of structural heart disease, which can be responsible for dyspnea. Specifically, left atrial and/or left ventricular dilatation, left ventricular hypertrophy, a reduced left ventricular ejection fraction, and disorders of left ventricular wall motion may be clues to the presence of a cardiac etiology of otherwise unexplained dyspnea.

DIFFERENTIATION BETWEEN CARDIAC AND PULMONARY DYSPNEA

In most patients with dyspnea there is obvious clinical evidence of disease of the heart and/or lungs. Like patients with cardiac dyspnea, patients with chronic obstructive lung disease may also waken at night with dyspnea, but, as pointed out above, this is usually associated with sputum production; the dyspnea is relieved after these patients rid themselves of secretions. The difficulty in the distinction between cardiac and pulmonary dyspnea may be compounded by the coexistence of diseases involving both organ systems.

In patients in whom the etiology of dyspnea is not clear, it is desirable to carry out pulmonary function testing, for these tests may be helpful in determining whether dyspnea is produced by heart disease, lung disease, abnormalities of the chest wall, or anxiety ([Chap. 250](#)). In addition to the usual means of assessing patients for heart disease, determination of the ejection fraction at rest and during exercise by echocardiography or radionuclide ventriculography is helpful in the differential diagnosis of dyspnea. The left ventricular ejection fraction is depressed in left ventricular failure, while the right ventricular ejection fraction may be low at rest or may decline during exercise in patients with severe lung disease. Both left and right ventricular ejection fractions are normal at rest and during exercise in dyspnea due to anxiety or malingering. Careful observation during the performance of an exercise treadmill test will often help in the identification of the patient who is malingering or whose dyspnea is secondary to anxiety. Under these circumstances, the patient usually complains of severe shortness of breath but appears to be breathing either effortlessly or totally irregularly. Cardiopulmonary testing, in which the patient's maximal functional exercise capacity is assessed while measurements of the electrocardiogram, blood pressure, oxygen consumption, arterial saturation (oximetry), and ventilation are carried out, is useful in the differentiation between cardiac and pulmonary dyspnea ([Table 32-2](#)).

ANXIETY NEUROSIS

Dyspnea experienced by a patient with an anxiety neurosis is difficult to evaluate. The signs and symptoms of acute and chronic hyperventilation do not serve to distinguish between anxiety neurosis and other processes, such as recurrent pulmonary emboli. Another potentially confusing situation is seen when chest pain and electrocardiographic changes accompany the hyperventilation syndrome. When present and attributable to

this condition, often referred to as *neurocirculatory asthenia* ([Chap. 13](#)), the chest pain is often sharp, fleeting, and in various loci, and the electrocardiographic changes are most often seen during repolarization. Frequent sighing respirations and an irregular breathing pattern point to a psychogenic origin of the dyspnea. Anxiety and depression in association with heart or lung disease can serve to intensify dyspnea symptoms beyond what would be expected for a given degree of dysfunction.

PULMONARY EDEMA (See [Table 32-3](#))

CARDIOGENIC PULMONARY EDEMA (See [Table 32-3](#), IA)

An increase in pulmonary venous pressure, which results initially in engorgement of the pulmonary vasculature, is common in most instances of dyspnea in association with congestive heart failure. The lungs become less compliant, the resistance of small airways increases, and there is an increase in lymphatic flow that apparently serves to maintain a constant pulmonary extravascular liquid volume. Mild tachypnea is present. If the increase in intravascular pressure is sufficient both in magnitude and duration, there is a net gain of liquid in the extravascular space, i.e., *interstitial* edema. At this point symptoms worsen, tachypnea increases, gas exchange deteriorates further, and radiographic changes, such as Kerley B lines and loss of distinct vascular margins, are seen. At this stage, the capillary endothelial intercellular junctions widen and allow passage of macromolecules into the interstices.

Further elevations in intravascular pressure disrupt the tight junctions between alveolar lining cells, and *alveolar* edema ensues, with outpouring of liquid that contains both red blood cells and macromolecules. With yet more severe disruption of the alveolar-capillary membrane, edematous liquid floods the alveoli and airways. At this point, full-blown clinical pulmonary edema with bilateral wet rales and rhonchi occurs, and the chest radiograph may show diffuse haziness of the lung fields with greater density in the more proximal hilar regions. Typically, the patient is anxious and perspires freely, and the sputum is frothy and blood-tinged. Gas exchange is more severely compromised with worsening hypoxia. Without effective treatment (described in [Chap. 232](#)), progressive acidemia, hypercapnia, and respiratory arrest ensue.

The earlier sequence of liquid accumulation described above follows the Starling law of capillary-interstitial liquid exchange:

The pressures tending to move liquid out of the vessel are P_c and p_{IF} , which are normally more than offset by pressures tending to move liquid back into the vasculature, i.e., the algebraic sum of P_{IF} and p_{pl} . Implicit in the preceding equation is that lymphatic flow can increase in the case of imbalance of forces and result in no net accumulation of interstitial liquid. Further elevations in P_c not only increase the outward movement of liquid in each capillary region but also recruit more of the capillary bed, which increases K . These two effects lead to liquid filtration that exceeds clearance capability by the lymphatics, and liquid accumulates in the loose interstitial spaces of the lung. Even greater increases in P_c open first the loose endothelial intercellular junctions and later the tight alveolar intercellular junctions with an increase in permeability to

macromolecules. This secondary disruption of both the function and structure of the alveolar-capillary membrane leads to alveolar flooding.

NONCARDIOGENIC PULMONARY EDEMA (See [Table 32-3](#), IB IC, II, III, and IV)

Several clinical conditions are associated with pulmonary edema based on an imbalance of Starling forces other than through primary elevations of pulmonary capillary pressure. Although diminished plasma oncotic pressure in hypoalbuminemic states (e.g., severe liver disease, nephrotic syndrome, protein-losing enteropathy) might be expected to lead to pulmonary edema, the balance of forces normally so strongly favors resorption that even in these conditions some elevation of capillary pressure is usually necessary before interstitial edema develops. Increased negativity of interstitial pressure has been implicated in the genesis of unilateral pulmonary edema following rapid evacuation of a large pneumothorax. In this situation, the findings may be apparent only by radiography, but occasionally the patient experiences dyspnea with physical findings localized to the edematous lung. It has been proposed that large negative intrapleural pressures during acute severe asthma may be associated with the development of interstitial edema. Lymphatic blockade secondary to fibrotic and inflammatory diseases or lymphangitic carcinomatosis may lead to interstitial edema. In such instances, both clinical and radiographic manifestations are dominated by the underlying disease process.

Other conditions characterized by increases in the interstitial liquid content of the lungs appear to be associated primarily with disruption of the alveolar-capillary membranes. Any number of spontaneously occurring or environmental toxic insults, including diffuse pulmonary infections, aspiration, and shock (particularly due to sepsis and hemorrhagic pancreatitis and following cardiopulmonary bypass), are associated with diffuse pulmonary edema that clearly does not have a hemodynamic origin. **These conditions, which may lead to the acute respiratory distress syndrome, are discussed in [Chap. 265](#).*

Other Forms of Pulmonary Edema There are three forms of pulmonary edema whose precise mechanism remains unexplained. *Narcotic overdose* is a well-recognized antecedent to pulmonary edema. Although illicit use of parenteral heroin is the most frequent cause, parenteral and oral overdoses of legitimate preparations of morphine, methadone, and dextropropoxyphene have also been associated with pulmonary edema. The earlier idea that injected impurities lead to the disorder is untenable. Available evidence suggests that there are alterations in the permeability of alveolar and capillary membranes rather than an elevation of pulmonary capillary pressure.

Exposure to high altitude in association with severe physical exertion is a well-recognized setting for pulmonary edema in unacclimatized yet otherwise healthy persons. Acclimatized high-altitude natives also develop this syndrome upon return to high altitude after a relatively brief sojourn at low altitudes. The syndrome is far more common in persons under the age of 25 years. The mechanism for high-altitude pulmonary edema (HAPE) remains obscure, and studies have been conflicting, some suggesting pulmonary venous constriction and others indicating pulmonary arteriolar constriction as the prime mechanisms. A role for hypoxia at high altitude is suggested by the fact that patients respond to the administration of oxygen and/or return to lower altitudes. Hypoxia per se does not alter permeability of the alveolar-capillary membrane.

Hence increased cardiac output and pulmonary arterial pressures with exercise combined with hypoxic pulmonary arteriolar constriction, which is more prominent in young persons, may combine to make this an example of prearteriolar, high-pressure pulmonary edema.

Neurogenic pulmonary edema has been described in patients with central nervous system disorders and without apparent preexisting left ventricular dysfunction. Although most experimental equivalents have implicated sympathetic nervous system activity, the mechanism whereby sympathetic efferent activity leads to pulmonary edema is a matter of speculation. It is known that a massive adrenergic nervous discharge leads to peripheral vasoconstriction with elevation of blood pressure and shifts of blood to the central circulation. In addition, it is probable that a reduction in left ventricular compliance also occurs, and both factors serve to increase left atrial pressures sufficiently to induce pulmonary edema on a hemodynamic basis. Some experimental evidence suggests that stimulation of adrenergic receptors increases capillary permeability directly, but this effect is relatively minor as compared with the imbalance of Starling forces.

TREATMENT OF PULMONARY EDEMA See [Chap. 232](#)

(Bibliography omitted in Palm version)

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33. COUGH AND HEMOPTYSIS - Steven E. Weinberger, Eugene Braunwald

COUGH

Cough is an explosive expiration that provides a normal protective mechanism for clearing the tracheobronchial tree of secretions and foreign material. When excessive or bothersome, it is also one of the most common symptoms for which medical attention is sought. Reasons for the latter include discomfort from the cough itself, interference with normal lifestyle, and concern for the cause of the cough, especially fear of cancer or AIDS.

MECHANISM

Coughing may be initiated either voluntarily or reflexively. As a defensive reflex it has both afferent and efferent pathways. The *afferent limb* includes receptors within the sensory distribution of the trigeminal, glossopharyngeal, superior laryngeal, and vagus nerves. The *efferent limb* includes the recurrent laryngeal nerve and the spinal nerves. The cough starts with a deep inspiration followed by glottic closure, relaxation of the diaphragm, and muscle contraction against a closed glottis. The resulting markedly positive intrathoracic pressure causes narrowing of the trachea. Once the glottis opens, the large pressure differential between the airways and the atmosphere coupled with tracheal narrowing produces rapid flow rates through the trachea. The shearing forces that develop aid in the elimination of mucus and foreign materials.

ETIOLOGY

Cough can be initiated by a variety of airway irritants, which enter the tracheobronchial tree by inhalation (smoke, dust, fumes) or by aspiration (upper airway secretions, gastric contents, foreign bodies). When cough is due to irritation by upper airway secretions (as with postnasal drip) or gastric contents (as with gastroesophageal reflux), the initiating factor may go unrecognized and the cough can be persistent. Additionally, prolonged exposure to such irritants may initiate airway inflammation, which can itself trigger cough and sensitize the airway to other irritants. Cough associated with gastroesophageal reflux is due only in part to aspiration of gastric contents, whereas vagally mediated reflex mechanisms appear to be responsible in many patients.

Any disorder resulting in inflammation, constriction, infiltration, or compression of airways can be associated with cough. Inflammation commonly results from airway infections, ranging from viral or bacterial bronchitis to bronchiectasis. In viral bronchitis, airway inflammation sometimes persists long after resolution of the typical acute symptoms, thereby producing a prolonged cough, lasting for weeks. Pertussis infection is also a possible cause of persistent cough in adults; however, diagnosis is generally made on clinical grounds ([Chap. 152](#)). Asthma is a common cause of cough. Although the clinical setting commonly suggests when a cough is secondary to asthma, some patients present with cough in the absence of wheezing or dyspnea, thus making the diagnosis more subtle ("cough variant asthma"). A neoplasm infiltrating the airway wall, such as bronchogenic carcinoma or a carcinoid tumor, is commonly associated with cough. Airway infiltration with granulomas may also trigger a cough, as seen with endobronchial sarcoidosis or tuberculosis. Compression of airways results from extrinsic

masses, including lymph nodes, mediastinal tumors, and aortic aneurysms.

Examples of parenchymal lung disease potentially producing cough include interstitial lung disease, pneumonia, and lung abscess. Congestive heart failure may be associated with cough, probably as a consequence of interstitial as well as peribronchial edema. A nonproductive cough complicates the use of angiotensin-converting enzyme (ACE) inhibitors in 5 to 20% of patients taking these agents. Onset is usually within 1 week of starting the drug but can be delayed up to 6 months. Although the mechanism is not known with certainty, it may relate to accumulation of bradykinin or substance P, both of which are degraded by ACE.

The most common causes of cough can be categorized according to the duration of the cough. Acute cough (<3 weeks) is most often due to upper respiratory infection (especially the common cold, acute bacterial sinusitis, and pertussis), but more serious disorders, such as pneumonia, pulmonary embolus, and congestive heart failure, can also present in this fashion. Chronic cough (>3 weeks) in a smoker raises the possibilities of chronic obstructive lung disease or bronchogenic carcinoma. In a nonsmoker who has a normal chest radiograph and is not taking an [ACE](#) inhibitor, the most common causes of chronic cough are postnasal drip, asthma, and gastroesophageal reflux.

Approach to the Patient

A detailed *history* frequently provides the most valuable clues for etiology of the cough. Particularly important questions include:

1. Is the cough acute or chronic?
2. At its onset, were there associated symptoms suggestive of a respiratory infection?
3. Is it seasonal or associated with wheezing?
4. Is it associated with symptoms suggestive of postnasal drip (nasal discharge, frequent throat clearing, a "tickle in the throat") or gastroesophageal reflux (heartburn or sensation of regurgitation)? (The absence of such suggestive symptoms does not exclude either of these diagnoses, particularly in the case of gastroesophageal reflux.)
5. Is it associated with fever or sputum? If sputum is present, what is its character?
6. Does the patient have any associated diseases or risk factors for disease (e.g., cigarette smoking, risk factors for infection with HIV, environmental exposures)?
7. Is the patient taking an [ACE](#) inhibitor?

The general *physical examination* may point to a nonpulmonary cause of cough, such as heart failure, primary nonpulmonary neoplasm, or AIDS. Examination of the oropharynx may provide suggestive evidence for postnasal drip, including oropharyngeal mucus or erythema, or a "cobblestone" appearance to the mucosa. Auscultation of the chest may demonstrate inspiratory stridor (indicative of upper airway

disease), rhonchi or expiratory wheezing (indicative of lower airway disease), or inspiratory crackles (suggestive of a process involving the pulmonary parenchyma, such as interstitial lung disease, pneumonia, or pulmonary edema).

Chest radiography may be particularly helpful in suggesting or confirming the cause of the cough. Important potential findings include the presence of an intrathoracic mass lesion, a localized pulmonary parenchymal infiltrate, or diffuse interstitial or alveolar disease. An area of honeycombing or cyst formation may suggest bronchiectasis, while symmetric bilateral hilar adenopathy may suggest sarcoidosis.

Pulmonary function testing ([Chap. 250](#)) is useful for assessing the functional abnormalities that accompany certain disorders producing cough. Measurement of forced expiratory flow rates can demonstrate reversible airflow obstruction characteristic of asthma. When asthma is considered but flow rates are normal, bronchoprovocation testing with methacholine or cold-air inhalation can demonstrate hyperreactivity of the airways to a bronchoconstrictive stimulus. Measurement of lung volumes and diffusing capacity is useful primarily for demonstration of a restrictive pattern, often seen with any of the diffuse interstitial lung diseases.

If *sputum* is produced, gross and microscopic examination may provide useful information. Purulent sputum suggests chronic bronchitis, bronchiectasis, pneumonia, or lung abscess. Blood in the sputum may be seen in the same disorders, but its presence also raises the question of an endobronchial tumor. Gram and acid-fast stains and cultures may demonstrate a particular infectious pathogen, while sputum cytology may provide a diagnosis of a pulmonary malignancy.

More specialized studies are helpful in specific circumstances. *Fiberoptic bronchoscopy* is the procedure of choice for visualizing an endobronchial tumor and collecting cytologic and histologic specimens. Inspection of the tracheobronchial mucosa can demonstrate endobronchial granulomas often seen in sarcoidosis, and endobronchial biopsy of such lesions or transbronchial biopsy of the lung interstitium can confirm the diagnosis. Inspection of the airway mucosa by bronchoscopy can also demonstrate the characteristic appearance of endobronchial Kaposi's sarcoma in patients with AIDS. *High-resolution computed tomography* (HRCT) can confirm the presence of interstitial disease and frequently suggests a diagnosis based on the pattern of disease. It is the procedure of choice for demonstrating dilated airways and confirming the diagnosis of bronchiectasis.

A diagnostic algorithm for evaluation of chronic cough is presented in [Fig. 33-1](#).

COMPLICATIONS

Common complications of coughing include chest and abdominal wall soreness, urinary incontinence, and exhaustion. On occasion, paroxysms of coughing may precipitate syncope (cough syncope; [Chap. 21](#)), consequent to markedly positive intrathoracic and alveolar pressures, diminished venous return, and decreased cardiac output. Although cough fractures of the ribs may occur in otherwise normal patients, their occurrence should at least raise the possibility of pathologic fractures, which are seen with multiple myeloma, osteoporosis, and osteolytic metastases.

TREATMENT

Definitive treatment of cough depends on determining the underlying cause and then initiating specific therapy. Elimination of an exogenous inciting agent (cigarette smoke, [ACE](#) inhibitors) or an endogenous trigger (postnasal drip, gastroesophageal reflux) is usually effective when such a precipitant can be identified. Other important management considerations are treatment of specific respiratory tract infections, bronchodilators for potentially reversible airflow obstruction, chest physiotherapy to enhance clearance of secretions in patients with bronchiectasis, and treatment of endobronchial tumors or interstitial lung disease when such therapy is available and appropriate.

Symptomatic or nonspecific therapy of cough should be considered when: (1) the cause of the cough is not known or specific treatment is not possible, and (2) the cough performs no useful function or causes marked discomfort. An irritative, nonproductive cough may be suppressed by an antitussive agent, which increases the latency or threshold of the cough center. Such agents include codeine (15 mg qid) or nonnarcotics such as dextromethorphan (15 mg qid). These drugs provide symptomatic relief by interrupting prolonged, self-perpetuating paroxysms. However, a cough productive of significant quantities of sputum should usually not be suppressed, since retention of sputum in the tracheobronchial tree may interfere with the distribution of ventilation, alveolar aeration, and the ability of the lung to resist infection.

Other agents working by a variety of mechanisms have also been used to control cough, but objective information assessing their benefit is meager. The inhaled anticholinergic agent, ipratropium bromide (2 to 4 puffs qid), has been used with the rationale of inhibiting the efferent limb of the cough reflex. Inhaled glucocorticoids, ideally administered with a spacer and dosed according to the particular agent, have been used for patients in whom airway inflammation is thought to be playing a role in the cough.

HEMOPTYSIS

Hemoptysis is defined as the expectoration of blood from the respiratory tract, a spectrum that varies from blood-streaking of sputum to coughing up large amounts of pure blood. *Massive hemoptysis* is variably defined as the expectoration of >100 to >600 mL over a 24-h period, although the patient's estimation of the amount of blood is notoriously unreliable. Expectoration of even relatively small amounts of blood is a frightening symptom and can be a marker for potentially serious disease, such as bronchogenic carcinoma. Massive hemoptysis, on the other hand, can represent an acutely life-threatening problem. Large amounts of blood can fill the airways and the alveolar spaces, not only seriously disturbing gas exchange but potentially causing the patient to suffocate.

ETIOLOGY

Because blood originating from the nasopharynx or the gastrointestinal tract can mimic blood coming from the lower respiratory tract, it is important to determine initially that the

blood is not coming from one of these alternative sites. Clues that the blood is originating from the gastrointestinal tract include a dark red appearance and an acidic pH, in contrast to the typical bright red appearance and alkaline pH of true hemoptysis.

The bronchial arteries, which are part of the high-pressure systemic circulation, originate either from the aorta or from intercostal arteries and are the source of bleeding in bronchitis or bronchiectasis or with endobronchial tumors.

An etiologic classification of hemoptysis can be based on the site of origin within the lungs ([Table 33-1](#)). The most common site of bleeding is the airways, i.e., the tracheobronchial tree, which can be affected by inflammation (acute or chronic bronchitis, bronchiectasis) or by neoplasm (bronchogenic carcinoma, endobronchial metastatic carcinoma, or bronchial carcinoid tumor). Blood originating from the pulmonary parenchyma can be either from a localized source, such as an infection (pneumonia, lung abscess, tuberculosis), or from a process diffusely affecting the parenchyma (as with a coagulopathy or with an autoimmune process such as Goodpasture's syndrome). Disorders primarily affecting the pulmonary vasculature include pulmonary embolic disease and those conditions associated with elevated pulmonary venous and capillary pressures, such as mitral stenosis or left ventricular failure.

Although the relative frequency of the different etiologies of hemoptysis varies from series to series, most recent studies indicate that bronchitis and bronchogenic carcinoma are the two most common causes. Despite the lower frequency of tuberculosis and bronchiectasis seen in recent compared to older series, these two disorders still represent the most common causes of massive hemoptysis in several series. Even after extensive evaluation, a sizable proportion of patients (up to 30% in some series) have no identifiable etiology for their hemoptysis. These patients are classified as having idiopathic or cryptogenic hemoptysis, and subtle airway or parenchymal disease is presumably responsible for the bleeding.

Approach to the Patient

The *history* is extremely valuable. Hemoptysis that is described as blood-streaking of mucopurulent or purulent sputum often suggests bronchitis. Chronic production of sputum with a recent change in quantity or appearance favors an acute exacerbation of chronic bronchitis. Fever or chills accompanying blood-streaked purulent sputum suggests pneumonia, whereas a putrid smell to the sputum raises the possibility of lung abscess. When sputum production has been chronic and copious, the diagnosis of bronchiectasis should be considered. Hemoptysis following the acute onset of pleuritic chest pain and dyspnea is suggestive of pulmonary embolism.

A history of previous or coexisting disorders should be sought, such as renal disease (seen with Goodpasture's syndrome or Wegener's granulomatosis), lupus erythematosus (with associated pulmonary hemorrhage from lupus pneumonitis), or a previous malignancy (either recurrent lung cancer or endobronchial metastasis from a nonpulmonary primary tumor). In a patient with AIDS, endobronchial or pulmonary parenchymal Kaposi's sarcoma should be considered. Risk factors for bronchogenic carcinoma, particularly smoking and asbestos exposure, should be sought. Patients

should be questioned about previous bleeding disorders, treatment with anticoagulants, or use of drugs that can be associated with thrombocytopenia.

The *physical examination* may also provide helpful clues to the diagnosis. For example, examination of the lungs may demonstrate a pleural friction rub (pulmonary embolism), localized or diffuse crackles (parenchymal bleeding or an underlying parenchymal process associated with bleeding), evidence of airflow obstruction (chronic bronchitis), or prominent rhonchi, with or without wheezing or crackles (bronchiectasis). Cardiac examination may demonstrate findings of pulmonary arterial hypertension, mitral stenosis, or heart failure. Skin examination may reveal Kaposi's sarcoma, arteriovenous malformations of Osler-Rendu-Weber disease, or lesions suggestive of systemic lupus erythematosus.

Diagnostic evaluation of hemoptysis starts with a chest radiograph to look for a mass lesion, findings suggestive of bronchiectasis ([Chap. 256](#)), or focal or diffuse parenchymal disease (representing either focal or diffuse bleeding or a focal area of pneumonitis). Additional initial screening evaluation often includes a complete blood count, a coagulation profile, and assessment for renal disease with a urinalysis and measurement of blood urea nitrogen and creatinine levels. When sputum is present, examination by Gram and acid-fast stains (along with the corresponding cultures) is indicated.

Fiberoptic bronchoscopy is particularly useful for localizing the site of bleeding and for visualization of endobronchial lesions. When bleeding is massive, rigid bronchoscopy is often preferable to fiberoptic bronchoscopy because of better airway control and greater suction capability. In patients with suspected bronchiectasis, [HRCT](#) is now the diagnostic procedure of choice, having replaced bronchography.

A diagnostic algorithm for evaluation of nonmassive hemoptysis is presented in [Fig. 33-2](#).

TREATMENT

The rapidity of bleeding and its effect on gas exchange determine the urgency of management. When the bleeding is confined to either blood-streaking of sputum or production of small amounts of pure blood, gas exchange is usually preserved; establishing a diagnosis is the first priority. When hemoptysis is massive, maintaining adequate gas exchange, preventing blood from spilling into unaffected areas of lung, and avoiding asphyxiation are the highest priorities. Keeping the patient at rest and partially suppressing cough may help the bleeding to subside. If the origin of the blood is known and is limited to one lung, the bleeding lung should be placed in the dependent position, so that blood is not aspirated into the unaffected lung.

With massive bleeding, the need to control the airway and maintain adequate gas exchange may necessitate endotracheal intubation and mechanical ventilation. In patients in danger of flooding the lung contralateral to the side of hemorrhage despite proper positioning, isolation of the right and left mainstem bronchi from each other can be achieved by selectively intubating the nonbleeding lung (often with bronchoscopic guidance) or by using specially designed double-lumen endotracheal tubes. Another

option involves inserting a balloon catheter through a bronchoscope by direct visualization and inflating the balloon to occlude the bronchus leading to the bleeding site. This technique not only prevents aspiration of blood into unaffected areas but also may promote tamponade of the bleeding site and cessation of bleeding.

Other available techniques for control of significant bleeding include laser phototherapy, electrocautery, embolotherapy, and surgical resection of the involved area of lung. With bleeding from an endobronchial tumor, the neodymium:yttrium-aluminum-garnet (Nd:YAG) laser can often achieve at least temporary hemostasis by coagulating the bleeding site. Electrocautery, which uses an electric current for thermal destruction of tissue, can be used similarly for management of bleeding from an endobronchial tumor. Embolotherapy involves an arteriographic procedure in which a vessel proximal to the bleeding site is cannulated, and a material such as Gelfoam is injected to occlude the bleeding vessel. Surgical resection is a therapeutic option either for the emergent therapy of life-threatening hemoptysis that fails to respond to other measures or for the elective but definitive management of localized disease subject to recurrent bleeding.

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34. APPROACH TO THE PATIENT WITH A HEART MURMUR - Patrick T. O'Gara, Eugene Braunwald

Auscultation of the heart constitutes the final step in the cardiovascular examination and, for many patients with established or suspected cardiac disease, represents a defining moment in the doctor-patient relationship. The examiner must bring to this exercise an integrated approach that incorporates pertinent information from several sources. The auscultatory findings must be interpreted in the context of the history and general physical examination and with the observations made regarding the venous wave forms and major arterial pulses. In this way, abnormalities of heart sounds, adventitious sounds, and murmurs can be placed in their proper perspective.

In many patients, a heart murmur is the only or the most conspicuous finding on physical examination. The recognition of a heart murmur usually leads to additional testing, such as electrocardiography, chest radiography, and echocardiography, and may result in referral to a cardiologist. The differential diagnosis of a heart murmur should begin with an unbiased and systematic evaluation of its major attributes: timing, duration, intensity, quality, frequency, configuration, location, radiation, and response to maneuvers (see [Table 225-1](#)). Laboratory testing can be pursued thereafter to clarify any remaining ambiguity and to provide additional anatomic and physiologic information that will impact on patient management.

Heart murmurs are defined in terms of their timing within the cardiac cycle. *Systolic murmurs* begin with or after the first heart sound (S_1) and terminate at or before the component (A_2 or P_2) of the second heart sound (S_2) that corresponds to their side of origin (left or right). *Diastolic murmurs* begin with or after the associated component of S_2 and end at or before the subsequent S_1 . *Continuous murmurs* are not confined to either phase of the cardiac cycle but rather begin in systole and proceed through S_2 into all or part of diastole.

The appropriate timing of heart murmurs is the first critical step in their identification. The distinction between S_1 and S_2 , and, therefore, systole and diastole, is usually a straightforward process but can be difficult in the setting of a tachyarrhythmia, in which case the heart sounds can be distinguished by simultaneous palpation of the carotid arterial pulse. The upstroke should closely follow S_1 . The principal causes of heart murmurs are shown in [Table 34-1](#), and the critical importance of the timing of heart murmurs in the differential diagnosis is shown in [Fig. 34-1](#).

SYSTOLIC HEART MURMURS

Systolic heart murmurs derive from the increased turbulence associated with (1) enhanced or accelerated flow across a normal semilunar valve, through a normal ventricular outflow tract, or into a dilated great vessel, (2) normal flow across a structurally abnormal semilunar valve or through a narrowed ventricular outflow tract, (3) flow across an incompetent atrioventricular valve, and (4) flow across the interventricular septum. One approach to their differential diagnosis further subdivides these murmurs according to their time of onset and duration within the systolic phase of the cardiac cycle.

EARLY SYSTOLIC MURMURS

Early systolic murmurs begin with S₁ and extend for a variable period of time, ending well before S₂. Their causes are relatively few in number. *Acute severe mitral regurgitation* into a normal-sized, relatively noncompliant left atrium results in an early and attenuated systolic murmur that is decrescendo in configuration and usually best heard at or just medial to the apical impulse ([Chap. 236](#)). These characteristics reflect the rapid rise in left atrial pressure caused by the sudden volume load into a nondilated chamber and contrast sharply with the auscultatory features of chronic mitral regurgitation. Clinical settings in which this occurs include: (1) papillary muscle rupture complicating acute myocardial infarction, (2) infective endocarditis, (3) rupture of chordae tendineae, and (4) blunt chest wall trauma.

Acute mitral regurgitation from papillary muscle rupture usually accompanies an inferior, posterior, or lateral infarction. The murmur is associated with a precordial thrill in approximately one-half of cases and is to be distinguished from that associated with postinfarction ventricular septal rupture. The latter is more commonly (90%) accompanied by a thrill at the left sternal edge, is holosystolic, and complicates anterior infarctions as often as inferior-posterior damage. The recognition of either of these mechanical defects mandates aggressive medical stabilization and emergent surgical intervention ([Chap. 243](#)).

The other potential causes of acute severe mitral regurgitation may be distinguished on the basis of associated findings. Spontaneous chordal rupture usually occurs on a substrate of myxomatous replacement, such as that underlying most forms of mitral valve prolapse ([Chap. 236](#)). This lesion may be part of a more generalized process, as can occur with the Marfan or Ehlers-Danlos syndromes, or it may be an isolated phenomenon. Infective endocarditis is associated with fever, peripheral embolic lesions, and positive blood cultures and most commonly occurs on a previously abnormal valvular apparatus ([Chap. 126](#)). Trauma is usually self-evident but may be disarmingly trivial ([Chap. 240](#)). It can result in papillary muscle contusion and rupture, chordal interruption, or leaflet avulsion or perforation.

Echocardiography should be performed in all cases of suspected acute severe mitral regurgitation to define the responsible mechanism, estimate the severity, and provide a preliminary assessment as to the feasibility of surgical repair (versus replacement).

Other causes of early systolic murmurs include congenital, small muscular ventricular septal defects. The duration of the murmur is attenuated by the closure of the defect during systolic contraction. The murmur is localized to the left sternal edge and is commonly of grade IV/VI or V/VI intensity. Signs of pulmonary hypertension or left ventricular volume overload are absent. Patients with anatomically large, uncorrected ventricular septal defects accompanied by pulmonary hypertension may also have murmurs confined to early systole. The elevated pulmonary vascular resistance attenuates the degree of shunting as pressures within the right and left ventricles equalize during the latter half of systole.

Tricuspid regurgitation with normal pulmonary artery pressures, such as that caused by infective endocarditis in injection drug users, may produce an early systolic murmur.

The murmur is soft, best heard at the lower left sternal edge, and may accentuate with inspiration (Carvallo's sign). Regurgitant c-v waves may be visible in the jugular venous pulse.

MIDSYSTOLIC MURMURS

Midsystolic murmurs begin at a short interval following S₁, end before S₂, and are usually crescendo-decrescendo in configuration ([Fig. 34-1C](#)). Semilunar valve stenosis is the classic prototype. With aortic valve stenosis, the murmur is usually loudest in the second right intercostal space (aortic area) and radiates along the carotid arteries ([Chap. 236](#)). The intensity of the murmur varies directly with the cardiac output; aortic valve stenosis with severe heart failure may produce a misleadingly soft systolic murmur. With a normal cardiac output, a systolic thrill is usually indicative of severe stenosis with a peak gradient in excess of 50 to 60 mmHg. An accompanying early systolic ejection click may be audible in younger patients with a bicuspid valve; its presence localizes the obstruction to the valvular (as opposed to the sub- or supra- valvular) level. The midsystolic murmur of aortic stenosis may be well transmitted to the apex, especially in older patients, where it becomes less harsh and slightly higher pitched (Gallavardin effect). The murmur of aortic stenosis should increase following a postpremature beat, whereas a mitral regurgitant murmur would not be expected to change in intensity.

Sclerosis of the aortic valve produces a murmur of similar location, radiation, and configuration, albeit without the usual signs of hemodynamic significance. The carotid upstroke is well preserved, the murmur peaks in midsystole and is not accompanied by a thrill, and only a modest gradient is estimated by Doppler echocardiography. Noncritical sclerodegenerative thickening of the aortic valve leaflets is perhaps the most common cause of a midystolic murmur in older adults. The similar midsystolic murmur of pulmonic valve stenosis, usually introduced by an ejection click, is best appreciated in the second and third left intercostal spaces (pulmonic area). The murmur lengthens and the intensity of P₂ diminishes with increasing degrees of stenosis. A midsystolic murmur in the aortic position can also be detected in hyperdynamic states (fever, thyrotoxicosis, pregnancy, anemia) and in the presence of isolated aortic regurgitation with the augmented flow into a dilated proximal aorta.

Crescendo-decrescendo midsystolic murmurs usually of grade II/VI intensity heard in the pulmonic area may be innocent if unaccompanied by any other signs of cardiac disease in children or young adults. They may also reflect enhanced flow into a normal pulmonary artery in hyperkinetic states or augmented flow into a dilated pulmonary artery. The latter may occur with an atrial septal defect, in which case splitting of S₂ is usually abnormal (fixed). Still's murmur is a vibratory, medium frequency, mid-systolic murmur heard best between the lower left sternal edge and the apex in normal children and young adults. It is generated by vibrations of the pulmonic valve leaflets at their attachments or by vibrations of a left ventricular false tendon.

The midsystolic murmur of hypertrophic cardiomyopathy ([Chap. 238](#)) is usually loudest between the left sternal edge and apex, of grade II/VI to III/VI intensity, and crescendo-decrescendo in configuration. In contrast to aortic valve stenosis, the murmur does *not* radiate into the neck and the carotid upstrokes are brisk and full and may even

be bifid. The intensity of the murmur associated with hypertrophic cardiomyopathy increases following maneuvers that decrease left ventricular volume (strain phase of the Valsalva maneuver, standing, amyl nitrite) or increase myocardial contractility (inotropic therapy). Conversely, the intensity of the systolic murmur decreases with maneuvers that increase ventricular volume (squatting, passive leg raising), impair contractility (beta-adrenoreceptor blockade), or raise preload and systemic afterload (squatting). Among these several maneuvers, auscultation in the standing and squatting positions, if possible, is perhaps the most sensitive technique to elicit a dynamic change in the intensity of the murmur associated with hypertrophic obstructive cardiomyopathy.

LATE SYSTOLIC MURMURS

A late systolic murmur begins well after the onset of ejection and is usually best heard at the left ventricular apex or between the apex and the left sternal edge. When introduced by a nonejection click, it is usually indicative of systolic prolapse of the mitral valve leaflet(s) into the left atrium. The click and murmur move closer to S₁ following maneuvers that decrease left ventricular volume (standing, Valsalva) and move oppositely upon increases in volume (leg raising, squatting). The intensity of the murmur augments with increases in systemic afterload (squatting, pressor agents) and decrease with vasodilation (amyl nitrite). Isometric exercise, which also delays the onset of the murmur, accentuates the intensity.

HOLOSYSTOLIC MURMURS

These murmurs, also termed *pansystolic murmurs*, begin with S₁ and continue through systole to S₂ ([Fig. 34-1B](#)). They are, with rare exception, indicative of atrioventricular valve regurgitation or of a ventricular septal defect; the differential diagnosis is shown in [Fig. 34-2](#). The murmur of mitral regurgitation is loudest at the left ventricular apex. Its radiation reflects the direction of the regurgitant jet. With a flail posterior mitral leaflet due to ruptured chordae tendineae, for example, the jet is directed anterosuperiorly, and the murmur radiates prominently to the base of the heart, where it might be confused with aortic valve stenosis unless the carotid upstrokes are carefully examined. Conversely, a flail anterior leaflet is associated with a posteriorly directed jet, which radiates into the axilla and the back. It may even strike the spine and be transmitted to the base of the neck. Severe mitral regurgitation is usually associated with a systolic thrill, a soft S₃, and a short diastolic rumbling murmur best appreciated in the left lateral decubitus position.

The holosystolic murmur of tricuspid regurgitation is generally softer (grades I to III/VI) than that of mitral regurgitation, is loudest at the left lower sternal edge, and increases in intensity upon inspiration. Associated signs include prominent "c-v" waves in the jugular venous pulse, systolic hepatic pulsations, and peripheral edema. Among the several causes of tricuspid regurgitation, annular dilatation from right ventricular enlargement in the setting of pulmonary artery hypertension is the most common.

Ventricular septal defect ([Chap. 234](#)) also produces a holosystolic murmur, the intensity of which varies inversely with the anatomic size of the defect. It is usually accompanied by a palpable thrill along the mid-left sternal border. The murmur of a ventricular septal defect is louder than that due to tricuspid regurgitation and does not share the latter's

inspiratory increase in intensity or associated peripheral signs.

DIASTOLIC HEART MURMURS

Like systolic murmurs, diastolic murmurs also can be subcategorized according to their time of onset.

EARLY DIASTOLIC MURMURS ([Fig. 34-1E](#))

Early diastolic murmurs result from semilunar valve incompetence and begin at the valve closure sound (A_2 or P_2), which reflects their site of origin. They are generally high pitched and decrescendo in configuration, especially in states of chronic regurgitation, in which their duration is a crude index of the severity of the lesion. The murmur of aortic regurgitation is generally, but not always, best heard in the second intercostal space at the left sternal edge. There is a tendency for the murmur associated with primary valvular pathology (e.g., rheumatic deformity, congenital bicuspid valve, endocarditis) to radiate more prominently along the *left* sternal border and to be well transmitted to the apex, while the murmur associated with primary aortic root pathology (e.g., annuloaortic ectasia, aortic dissection) radiates more often along the right sternal edge. It is occasionally necessary to examine the patient sitting forward in full expiration to appreciate the murmur, a maneuver that brings the aortic root closer to the anterior chest wall. Severe aortic regurgitation may be accompanied by a lower-pitched mid- to late-diastolic murmur at the apex (Austin Flint murmur), which is generally thought to reflect turbulence at the mitral inflow area from the mixing of the regurgitant (aortic) and forward (mitral) streams, and should be distinguished from mitral stenosis (see above). In the absence of significant heart failure, chronic severe aortic regurgitation is accompanied by several peripheral signs of significant diastolic runoff, including a wide systemic pulse pressure and water-hammer carotid upstrokes (Corrigan's pulse).

The murmur associated with *acute* aortic regurgitation is notably shorter in duration, lower pitched, and can be difficult to appreciate in the presence of tachycardia. Peripheral signs of significant diastolic runoff may be absent. These attributes reflect the abrupt rise in diastolic pressure within the noncompliant left ventricle, with a correspondingly rapid decline in the aortic diastolic-left ventricular pressure gradient.

The murmur of pulmonic valve regurgitation (Graham Steell murmur) begins with a loud (palpable) pulmonic closure sound (P_2) and is best heard in the pulmonic area with radiation along the left sternal border. Typically, it is high pitched, with a decrescendo quality, and is indicative of significant pulmonary artery hypertension with a diastolic pulmonary artery-right ventricular pressure gradient. Its increase in intensity upon inspiration is one means by which to distinguish it from aortic regurgitation. Signs of right ventricular pressure and volume overload are also usually present. With significant mitral stenosis, an early decrescendo diastolic murmur along the left sternal border is not uncommon and is almost always due to aortic rather than pulmonic regurgitation, despite the coexistence of pulmonary artery hypertension.

Pulmonic valve regurgitation in the absence of pulmonary artery hypertension can occur on a congenital basis and rarely with infective endocarditis. In these instances, the early diastolic murmur is softer and lower pitched than the classic Graham Steell murmur. It

begins at or even after P₂, which should be easily separable from A₂ and thus produce appreciation of an early diastolic pause.

MIDDIASTOLIC MURMURS

Middiastolic murmurs usually result from obstruction and/or augmented flow across the atrioventricular valves. The classic example is that of mitral stenosis due to rheumatic deformity ([Fig. 34-1F](#)). In the absence of extensive calcification, the first heart sound (S₁) is loud and the murmur begins after the opening snap; the time interval between S₂ and the opening snap is inversely related to the left atrial-left ventricular pressure gradient. The murmur is low pitched and best heard with the bell of the stethoscope over the apex, particularly in the left lateral decubitus position. While its intensity does not reflect the severity of the obstruction accurately, the duration of the murmur does provide some indication as to the magnitude of the obstruction. A longer murmur denotes persistence of a left atrioventricular pressure gradient over a greater proportion of the diastolic time interval. Presystolic accentuation of the murmur ([Fig. 34-1A](#)) is frequently appreciated in the presence of sinus rhythm and reflects a further increase in transmitral flow consequent to mechanical atrial systole.

The murmur associated with tricuspid stenosis shares many of these features, but it is best heard at the lower left sternal border and, like most right-sided events, increases in intensity upon inspiration. The observant examiner may discern a prolonged y descent in the jugular venous pulse. Signs of right heart failure may predominate.

There are several other causes of mid-diastolic murmurs that are important to distinguish from mitral stenosis. *Left atrial myxomas* ([Chap. 240](#)) may masquerade as mitral stenosis, but the diastolic murmur is not accompanied by an opening snap or pre-systolic accentuation. Augmented flow across the mitral valve in diastole, such as occurs with severe mitral regurgitation or with large left to right intra-cardiac (ventricular septal defect) or great vessel (patent ductus arteriosus) shunts may produce a short, low pitched mid-diastolic apical murmur. The murmur usually follows a soft S₃ that is lower pitched and later in timing than the opening snap ([Fig. 34-1G](#)). Severe tricuspid regurgitation can also result in enhanced diastolic tricuspid flow and produce a right-sided filling complex similar to that which accompanies severe mitral regurgitation. The Austin Flint murmur of severe aortic regurgitation has been previously described and occurs in the presence of chronic severe aortic regurgitation.

CONTINUOUS MURMURS

Continuous murmurs begin in systole, peak near S₂, and continue into all or part of diastole ([Fig. 34-1H](#)). Accordingly, they reflect the persistence of flow between two chambers during both phases of the cardiac cycle. The differential diagnosis of continuous murmurs is shown in [Table 34-1](#). Two innocent variants are the cervical venous hum and the mammary souffle. The former is audible in healthy children and young adults in the right supraclavicular fossa and can be abolished by compression over the internal jugular vein. Its diastolic component may be louder than its systolic counterpart. A mammary souffle represents augmented arterial flow through engorged breasts and becomes audible during the late third trimester of pregnancy or in the early postpartum period. Firm pressure with the diaphragm of the stethoscope can eliminate

the diastolic portion of the murmur. The murmur dissipates with time after delivery.

The classic continuous murmur is that due to a patent ductus arteriosus. It is best heard at or just above and to the left of the pulmonic area and may be audible in the back. Over time, a large uncorrected shunt may lead to elevation of the pulmonary vascular resistance, with resultant pulmonary artery hypertension and diminution or elimination of the diastolic component. A continuous murmur can also signify a ruptured congenital sinus of Valsalva aneurysm, which occurs either spontaneously or as a complication of infective endocarditis. Here, a high-pressure fistula is created between the aorta and a cardiac chamber, usually the right atrium or ventricle. The murmur is loudest along the right or left sternal border and is frequently accompanied by a thrill. Notably, the diastolic component is louder than the systolic component. It can be difficult to distinguish continuous murmurs from the temporally separate systolic and diastolic murmurs of mixed aortic valve disease or isolated severe aortic regurgitation. The emphasis is on the envelopment of S₂ by continuous murmurs and a gap between the to-and-fro murmurs of aortic valve disease.

A variety of other lesions can result in continuous murmurs. A coronary arteriovenous fistula sometimes produces a faint, continuous murmur with a louder diastolic component at the left sternal border or left ventricular apex. Severe atherosclerotic disease of a major systemic artery may produce a continuous bruit, the presence of which signifies very high-grade obstruction. Patients with peripheral pulmonary (branch) stenosis or with pulmonary atresia with extensive bronchial collaterals may also have continuous murmurs best heard in the back or along the lateral thoracic cage. Similar findings are present in patients with severe aortic coarctation, a lesion that should be identifiable on the basis of weak and delayed lower extremity pulses and upper extremity hypertension. The continuous murmurs emanate from the enlarged collateral (intercostal) arteries.

Approach to the Patient

It is widely recognized that, despite the importance placed on them by medical schools and training program directors, the auscultatory skills of medical students and residents have declined considerably since the advent of Doppler echocardiography. Recent surveys indicate that trainees fail to correctly identify up to 80% of adventitious sounds and murmurs. Diagnostic errors are as frequent among third-year medical residents as they are for first-year residents. Few training programs provide a dedicated educational curriculum for cardiac auscultation. In the aggregate, these deficiencies lead to an over-reliance on the use of echocardiography and increase the costs of evaluating patients with heart murmurs.

In many patients the cause of a heart murmur can be readily elucidated from careful assessment of the murmur itself, as described in this chapter, when considered in the light of the history, general physical examination, and other features of the cardiac examination, as described in [Chap. 225](#). When the diagnosis is in doubt, or when additional pathoanatomic and physiologic data are necessary in assessing the patient and planning treatment, transthoracic Doppler echocardiography is of great value in identifying not only the etiology of the murmur but also the severity of the responsible lesion ([Fig. 34-3](#)).

The majority of heart murmurs are midsystolic and soft (Grades I to II/VI). When such a murmur occurs in an asymptomatic child or young adult *without* other evidence of heart disease on clinical examination, it is usually benign and echocardiography is not generally required. On the other hand, echocardiographic examination is indicated in patients with loud systolic murmurs (³III/VI), especially those that are holosystolic or late systolic, in most patients with diastolic or continuous murmurs, and in patients with additional unexplained abnormal physical findings on cardiac examination.

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35. APPROACH TO THE PATIENT WITH HYPERTENSION - Gordon H. Williams

DEFINITION

Since there is no dividing line between normal and high blood pressure, arbitrary levels have been established to define persons who have an increased risk of developing a morbid cardiovascular event and/or will benefit from medical therapy. These definitions should take into account not only the level of diastolic pressure but also systolic pressure, age, sex, race, and concomitant diseases. For example, patients with a diastolic pressure >90 mmHg have a significant reduction in morbidity and mortality rate if they receive adequate therapy. These, then, are patients who have hypertension and who should be considered for treatment.

The level of *systolic* pressure is also important in assessing the influence of arterial pressure on cardiovascular morbidity. Some data suggest that it may be more important than diastolic pressure. For example, males with normal diastolic pressures (<82 mmHg) but elevated systolic pressures (>158 mmHg) have a cardiovascular mortality rate 2.5 times higher than individuals who have similar diastolic pressures but whose systolic pressures clearly are normal (<130 mmHg). A reduction in mortality and morbidity with treatment, specifically in the elderly, has been documented in these patients. This beneficial effect results mainly from a reduction in strokes and occurs in women as well. Other significant demographic factors that modify the influence of blood pressure on the frequency of morbid cardiovascular events are age, race, and sex, with young black males being most adversely affected by hypertension.

When hypertension is suspected, blood pressure should be measured at least twice during two separate examinations after the initial screening. In adults, a *diastolic* pressure below 85 mmHg is considered to be normal; one between 85 and 89 mmHg is high normal; one of 90 to 99 mmHg represents stage 1 or mild hypertension; one of 100 to 109 mmHg represents stage 2 or moderate hypertension; and one of ≥ 110 mmHg represents stage 3 or severe hypertension. A *systolic* pressure below 130 mmHg indicates normal blood pressure; one between 130 and 139 mmHg indicates high normal; one between 140 and 159 mmHg indicates stage 1 or mild hypertension; one between 160 and 179 mmHg indicates stage 2 or moderate hypertension; and one ≥ 180 mmHg indicates stage 3 or severe hypertension. Increasing use of 12- or 24-h blood pressure monitoring may provide additional useful information in patients who are difficult to classify. However, normal values for this procedure and its usefulness in relation to therapeutic outcomes are not currently known. A useful classification of hypertension derived from the Joint Committee on Detection, Evaluation, and Treatment of High Blood Pressure is shown in [Table 35-1](#).

Arterial pressure fluctuates in most persons, whether they are normotensive or hypertensive. Patients who are classified as having *labile hypertension* are those who sometimes, but not always, have arterial pressures in the hypertensive range. These patients are often considered to have borderline hypertension.

Sustained hypertension can become accelerated or enter a malignant phase, although that is unusual in treated patients. Though a patient with *malignant hypertension* often has a blood pressure above 200/140, the condition is defined by the presence of

papilledema, usually accompanied by retinal hemorrhages and exudates, rather than by the absolute pressure level. *Accelerated hypertension* is defined as a significant recent increase over previous hypertensive levels associated with evidence of vascular damage on fundusoscopic examination but without papilledema.

PATIENT EVALUATION

In evaluating patients with hypertension, the initial history, physical examination, and laboratory tests should be directed at (1) uncovering correctable secondary forms of hypertension ([Chap. 246](#)), (2) establishing a pretreatment baseline, (3) assessing factors that may influence the type of therapy or be changed adversely by therapy, (4) determining if target organ damage is present, and (5) determining whether other risk factors for the development of arteriosclerotic cardiovascular disease are present ([Chap. 241](#)). Ideally, this evaluation would also determine the underlying mechanism(s) in essential hypertension, particularly if such information leads to a more specific therapeutic program. Unfortunately, at present this aspect of the evaluation is limited by lack of knowledge of some of the underlying mechanisms, by uncertainty as to the correct treatment for a distinct subset even if the underlying mechanisms are known, or by the prohibitive cost of defining a subset of hypertensive patients even if specific therapy were available. However, with the accumulation of additional information, this sixth component of the evaluation of patients with hypertension may become increasingly important.

Symptoms and Signs Most patients with hypertension have no specific symptoms referable to their blood pressure elevation and are identified only in the course of a physical examination. When symptoms do bring the patient to the physician, they fall into three categories. They are related to (1) the elevated pressure itself, (2) the hypertensive vascular disease, and (3) the underlying disease, in the case of secondary hypertension. Though popularly considered a symptom of elevated arterial pressure, headache is characteristic only of severe hypertension; most commonly such headaches are localized to the occipital region and are present when the patient awakens in the morning but subside spontaneously after several hours. Other complaints that may be related to elevated blood pressure include dizziness, palpitations, easy fatigability, and impotence. Complaints referable to vascular disease include epistaxis, hematuria, blurring of vision owing to retinal changes, episodes of weakness or dizziness due to transient cerebral ischemia, angina pectoris, and dyspnea due to cardiac failure. Pain due to dissection of the aorta or to a leaking aneurysm is an occasional presenting symptom.

Examples of symptoms related to the underlying disease in secondary hypertension are polyuria, polydipsia, and muscle weakness secondary to hypokalemia in patients with primary aldosteronism or weight gain, and emotional lability in patients with Cushing's syndrome. The patient with a pheochromocytoma may present with episodic headaches, palpitations, diaphoresis, and postural dizziness.

History A strong family history of hypertension, along with the reported finding of intermittent pressure elevation in the past, favors the diagnosis of essential hypertension. Secondary hypertension often develops before the age of 35 or after 55. A history of use of adrenal steroids or estrogens is of obvious significance. A history of

repeated urinary tract infections suggests chronic pyelonephritis, although this condition may occur in the absence of symptoms; nocturia and polydipsia suggest renal or endocrine disease, while trauma to either flank or an episode of acute flank pain may be a clue to the presence of renal injury. A history of weight gain is compatible with Cushing's syndrome, and one of weight loss is compatible with pheochromocytoma. A number of aspects of the history aid in determining whether vascular disease has progressed to a dangerous stage. These include angina pectoris and symptoms of cerebrovascular insufficiency, congestive heart failure, and/or peripheral vascular insufficiency. Other risk factors that should be asked about include cigarette smoking, diabetes mellitus, lipid disorders, and a family history of early deaths due to cardiovascular disease. Finally, aspects of the patient's lifestyle that could contribute to the hypertension or affect its treatment should be assessed, including diet, physical activity, family status, work, and educational level.

Physical Examination The physical examination starts with the patient's general appearance. For instance, are the round face and truncal obesity of Cushing's syndrome present? Is muscular development in the upper extremities out of proportion to that in the lower extremities, suggesting coarctation of the aorta? The next step is to compare the blood pressures and pulses in the two upper extremities and in the supine and standing positions (for at least 2 min). A rise in diastolic pressure when the patient goes from the supine to the standing position is most compatible with essential hypertension; a fall, in the absence of antihypertensive medications, suggests secondary forms of hypertension. The patient's height and weight should be recorded. Detailed examination of the ocular fundi is mandatory, as funduscopic findings provide one of the best indications of the duration of hypertension and of prognosis. A useful guide is the Keith-Wagener-Barker classification of funduscopic changes ([Table 35-2](#)); the specific changes in each fundus should be recorded and a grade assigned. Palpation and auscultation of the carotid arteries for evidence of stenosis or occlusion are important; narrowing of a carotid artery may be a manifestation of hypertensive vascular disease, and it may also be a clue to the presence of a renal arterial lesion, since these two lesions may occur together. In examination of the heart and lungs, evidence of left ventricular hypertrophy and cardiac decompensation should be sought. Is there a left ventricular lift? Are third and fourth heart sounds present? Are there pulmonary rales? A third heart sound and pulmonary rales are unusual in uncomplicated hypertension. Their presence suggests ventricular dysfunction. Chest examination also includes a search for extracardiac murmurs and palpable collateral vessels that may result from coarctation of the aorta.

The most important part of the abdominal examination is auscultation for bruits originating in stenotic renal arteries. Bruits due to renal arterial narrowing nearly always have a diastolic component or may be continuous and are best heard just to the right or left of the midline above the umbilicus or in the flanks; they are present in many patients with renal artery stenosis due to fibrous dysplasia and in 40 to 50% of those with functionally significant stenosis due to arteriosclerosis. The abdomen should also be palpated for an abdominal aneurysm and for the enlarged kidneys of polycystic renal disease. The femoral pulses must be carefully felt, and, if they are decreased and/or delayed in comparison with the radial pulse, the blood pressure in the lower extremities must be measured. Even if the femoral pulse is normal to palpation, arterial pressure in the lower extremities should be recorded at least once in patients in whom hypertension

is discovered before the age of 30 years. Finally, examination of the extremities for edema and a search for evidence of a previous cerebrovascular accident and/or other intracranial pathology should be performed.

Laboratory Investigation There is controversy as to what laboratory studies should be performed in patients presenting with hypertension. In general, the disagreement centers on how extensively the patient should be evaluated for secondary forms of hypertension or subsets of essential hypertension. The *basic* laboratory studies that should be performed in all patients with sustained hypertension are described in ([Table 35-3](#)). **The secondary studies that should be added if (1) the initial evaluation indicates a form of secondary hypertension and/or (2) arterial pressure is not controlled after initial therapy are discussed in Chap. 246.*

Renal status is evaluated by assessing the presence of protein, blood, and glucose in the urine and measuring serum creatinine and/or blood urea nitrogen. Microscopic examination of the urine is also helpful. The serum potassium level should be measured both as a screen for mineralocorticoid-induced hypertension and to provide a baseline before diuretic therapy is begun. A blood glucose determination is helpful both because diabetes mellitus may be associated with accelerated arteriosclerosis, renal vascular disease, and diabetic nephropathy in patients with hypertension and because primary aldosteronism, Cushing's syndrome, and pheochromocytoma all may be associated with hyperglycemia. Furthermore, since antihypertensive therapy with diuretics, for example, can raise the blood glucose level, it is important to establish a baseline. The possibility of hypercalcemia may also be investigated. Serum cholesterol, high-density lipoprotein cholesterol, and triglyceride levels identify other factors that predispose to the development of arteriosclerosis.

An electrocardiogram should be obtained in all cases to permit assessment of cardiac status, particularly if left ventricular hypertrophy is present, and to provide a baseline. The echocardiogram is more sensitive than either the electrocardiogram or physical examination in determining whether cardiac hypertrophy is present. However, a complete, detailed echocardiographic study is expensive. Thus, in some circumstances, a cheaper, limited echocardiogram may be a useful addition to the *baseline* evaluation of a hypertensive patient, particularly as left ventricular hypertrophy is an independent cardiovascular risk factor and its presence suggests the need for vigorous antihypertensive therapy. Furthermore, while a substantial increase in arterial pressure usually correlates with the presence of left ventricular hypertrophy, a mild increase may not. Thus, one cannot use the blood pressure as a surrogate marker for the presence or absence of left ventricular hypertrophy. On the other hand, because of the cost of an echocardiogram and the uncertainty as to whether the resultant information would modify therapy, it is unclear that routine *follow-up* echocardiograms during therapy are justified. Furthermore, there are no data to suggest that reversal of left ventricular hypertrophy produces benefits beyond that conferred by blood pressure reduction. The chest roentgenogram may also be helpful by providing the opportunity to identify aortic dilation or elongation and the rib notching that occurs in coarctation of the aorta.

Certain clues from the history, physical examination, and basic laboratory studies may suggest an unusual cause for the hypertension and dictate the need for special studies as outlined in [Chap 246](#).

TREATMENT

See [Chap. 246](#).

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36. HYPOXIA AND CYANOSIS - Eugene Braunwald

HYPOXIA

The fundamental purpose of the cardiorespiratory system is to deliver O_2 (and substrates) to the cells and to remove CO_2 (and other metabolic products) from them. Proper maintenance of this function depends on intact cardiovascular and respiratory systems and a supply of inspired gas containing adequate O_2 . When hypoxia occurs consequent to respiratory failure, P_{aCO_2} usually rises ([Chap. 250](#)), and the hemoglobin-oxygen ($Hb-O_2$) dissociation curve (see [Fig. 106-2](#)) is displaced to the right. Under these conditions, the P_{aO_2} declines. Arterial hypoxemia, i.e., a reduction of O_2 saturation of arterial blood (S_{aO_2}), and consequent cyanosis are likely to be more marked when such depression of P_{aO_2} results from pulmonary disease than when the depression occurs as the result of a decline in the fraction of oxygen in inspired air (F_{IO_2}). In this situation P_{aCO_2} falls secondary to anoxia-induced hyperventilation and the $Hb-O_2$ dissociation curve is displaced to the left, limiting the decline in S_{aO_2} .

CAUSES OF HYPOXIA

Anemic Hypoxia Any reduction in the hemoglobin concentration of the blood is attended by a corresponding decline in the O_2 -carrying capacity of the blood. In anemic hypoxia, the P_{aO_2} is normal; but as a consequence of the reduction of the hemoglobin concentration, the absolute quantity of O_2 transported per unit volume of blood is diminished. As the anemic blood passes through the capillaries and the usual quantity of O_2 is removed from it, the P_{O_2} in the venous blood declines to a greater degree than would normally be the case.

Carbon Monoxide Intoxication (See also [Chap. 396](#)) Hemoglobin that is combined with carbon monoxide (carboxyhemoglobin, COHb) is unavailable for O_2 transport. In addition, the presence of COHb shifts the $Hb-O_2$ dissociation curve to the left (see [Fig. 106-2](#)) so that O_2 is unloaded only at lower tensions. By such formation of COHb, a given degree of reduction in O_2 -carrying power produces a far greater degree of tissue hypoxia than the equivalent reduction in hemoglobin due to simple anemia.

Respiratory Hypoxia Arterial unsaturation is a common finding in advanced pulmonary disease. The most common cause of respiratory hypoxia is ventilation-perfusion mismatch, which results from perfusion of poorly ventilated alveoli. As discussed in [Chap. 250](#), it may also be caused by hypoventilation, and it is then associated with an elevation of P_{aCO_2} . These two forms of respiratory hypoxia may be recognized because they are usually correctable by inspiring 100% O_2 for several minutes. A third cause is shunting of blood across the lung from right to left by perfusion of nonventilated portions of the lung, as in pulmonary atelectasis or through arteriovenous connections in the lung. The low P_{aO_2} in this situation is correctable only in part by an F_{IO_2} of 100%.

Hypoxia Secondary to High Altitude As one ascends rapidly to 3000 m (approximately 10,000 ft), the alveolar P_{O_2} declines to about 60 mmHg, and impaired memory and other cerebral symptoms of hypoxia may develop. At higher altitudes, arterial saturation declines rapidly and symptoms become more serious; and at 5000 m (approximately 15,000 ft) unacclimatized individuals usually cease to be able to function

normally.

Hypoxia Secondary to Right-to-Left Extrapulmonary Shunting From a physiologic viewpoint, this cause of hypoxia resembles intrapulmonary right-to-left shunting but is caused by congenital cardiac malformations such as tetralogy of Fallot, transposition of the great arteries, and Eisenmenger's syndrome ([Chap. 234](#)). As in pulmonary right-to-left shunting, the P_{aO_2} cannot be restored to normal with inspiration of 100% O_2 .

Circulatory Hypoxia As in anemic hypoxia, the P_{aO_2} is usually normal, but venous and tissue P_{O_2} values are reduced as a consequence of reduced tissue perfusion. Generalized circulatory hypoxia occurs in heart failure ([Chap. 232](#)) and in most forms of shock ([Chap. 38](#)).

Specific Organ Hypoxia Decreased perfusion of any organ resulting in localized circulatory hypoxia may occur secondary to organic arterial obstruction or as a consequence of vasoconstriction ([Chap. 248](#)). The latter is seen in the upper extremities in Raynaud's phenomenon. Ischemic hypoxia with accompanying pallor occurs in organic arterial obliterative disease. Localized hypoxia also may result from venous obstruction and the resultant congestion and reduced arterial inflow. Edema, which increases the distance through which O_2 diffuses before it reaches cells, also can cause localized hypoxia. In an attempt to maintain adequate perfusion to more vital organs, constriction may reduce perfusion in all limbs in patients with heart failure or hypovolemic shock.

Increased O_2 Requirements If the O_2 consumption of the tissues is elevated without a corresponding increase in perfusion, tissue hypoxia ensues and the P_{O_2} in venous blood becomes reduced. Ordinarily, the clinical picture of patients with hypoxia due to an elevated metabolic rate is quite different from that in other types of hypoxia; the skin is warm and flushed, owing to increased cutaneous blood flow that dissipates the excessive heat produced, and cyanosis is usually absent.

Exercise is a classic example of increased tissue O_2 requirements. These increased demands are normally met by several mechanisms operating simultaneously: (1) increasing the cardiac output and ventilation and thus O_2 delivery to the tissues; (2) preferentially directing the blood to the exercising muscles by changing vascular resistances in various circulatory beds, directly and/or reflexly; (3) increasing O_2 extraction from the delivered blood and widening the arteriovenous O_2 difference; and (4) reducing the pH of the tissues and capillary blood, thereby unloading more O_2 from hemoglobin. If the capacity of these mechanisms is exceeded, then hypoxia, especially of the exercising muscles, will result.

Improper Oxygen Utilization Cyanide ([Chap. 396](#)) and several other similarly acting poisons cause cellular hypoxia. The tissues are unable to utilize O_2 , and as a consequence, the venous blood tends to have a high O_2 tension. This condition has been termed *histotoxic hypoxia*.

EFFECTS OF HYPOXIA

Changes in the central nervous system, particularly the higher centers, are especially

important consequences of hypoxia. Acute hypoxia causes impaired judgment, motor incoordination, and a clinical picture closely resembling that of acute alcoholism. When hypoxia is long-standing, fatigue, drowsiness, apathy, inattentiveness, delayed reaction time, and reduced work capacity occur. As hypoxia becomes more severe, the centers of the brainstem are affected, and death usually results from respiratory failure. With the reduction of P_{aO_2} , cerebrovascular resistance decreases and cerebral blood flow increases, increasing O_2 delivery to the brain as a compensatory mechanism. However, when the reduction of P_{aO_2} is accompanied by hyperventilation and a reduction of P_{aCO_2} , cerebrovascular resistance rises, cerebral blood flow falls, and hypoxia is intensified. Hypoxia also causes pulmonary arterial constriction, which shunts blood away from poorly ventilated areas toward better-ventilated portions of the lung. However, it also increases pulmonary vascular resistance and right ventricular afterload.

Glucose is normally broken down to pyruvic acid. However, the further breakdown of pyruvate and the generation of adenosine triphosphate (ATP) consequent to it require O_2 ; and in the presence of hypoxia increasing proportions of pyruvate are reduced to lactic acid, which cannot be broken down further, causing metabolic acidosis. Under these circumstances, the total energy obtained from the breakdown of carbohydrate is greatly reduced, and the quantity of energy available for the production of ATP becomes inadequate.

An important component of the respiratory response to hypoxia originates in special chemosensitive cells in the carotid and aortic bodies, and in the respiratory center in the brainstem. The stimulation of these cells by hypoxia increases ventilation, with a loss of CO_2 , and leads to respiratory alkalosis. When combined with the metabolic acidosis resulting from the production of lactic acid, the serum bicarbonate level declines ([Chap. 50](#)).

Diminished P_{O_2} in any tissue results in local vasodilatation, and the diffuse vasodilatation that occurs in generalized hypoxia raises the cardiac output. In patients with underlying heart disease, the requirements of the peripheral tissues for an increase of cardiac output with hypoxia may precipitate congestive heart failure. In patients with ischemic heart disease, a reduced P_{aO_2} may intensify myocardial ischemia and further impair left ventricular function.

One of the important mechanisms of compensation for chronic hypoxia is an increase in the hemoglobin concentration and in the number of red blood cells in the circulating blood, i.e., the development of polycythemia secondary to erythropoietin production ([Chap. 110](#)).

CYANOSIS

Cyanosis refers to a bluish color of the skin and mucous membranes resulting from an increased quantity of reduced hemoglobin, or of hemoglobin derivatives, in the small blood vessels of those areas. It is usually most marked in the lips, nail beds, ears, and malar eminences. Cyanosis, especially if developed recently, is more commonly detected by a family member than the patient. The florid skin characteristic of polycythemia vera ([Chap. 110](#)) must be distinguished from the true cyanosis discussed here. A cherry-colored flush, rather than cyanosis, is caused by COHb ([Chap. 396](#)). The

degree of cyanosis is modified by the color of the cutaneous pigment and the thickness of the skin, as well as by the state of the cutaneous capillaries. The accurate clinical detection of the presence and degree of cyanosis is difficult, as proved by oximetric studies. In some instances, central cyanosis can be detected reliably when the SaO_2 has fallen to 85%; in others, particularly in dark-skinned persons, it may not be detected until it has declined to 75%. In the latter case, examination of the mucous membranes in the oral cavity and the conjunctivae rather than examination of the skin is more helpful in the detection of cyanosis.

The increase in the quantity of reduced hemoglobin in the mucocutaneous vessels that produces cyanosis may be brought about either by an increase in the quantity of venous blood as the result of dilatation of the venules and venous ends of the capillaries or by a reduction in the SaO_2 in the capillary blood. In general, cyanosis becomes apparent when the mean capillary concentration of reduced hemoglobin exceeds 40 g/L (4 g/dL). It is the *absolute* rather than the *relative* quantity of reduced hemoglobin that is important in producing cyanosis. Thus, in a patient with severe anemia, the relative amount of reduced hemoglobin in the venous blood may be very large when considered in relation to the total amount of hemoglobin in the blood. However, since the concentration of the latter is markedly reduced, the *absolute* quantity of reduced hemoglobin may still be small, and therefore patients with severe anemia and even *marked* arterial desaturation do not display cyanosis. Conversely, the higher the total hemoglobin content, the greater is the tendency toward cyanosis; thus, patients with marked polycythemia tend to be cyanotic at higher levels of SaO_2 than patients with normal hematocrit values. Likewise, local passive congestion, which causes an increase in the total amount of reduced hemoglobin in the vessels in a given area, may cause cyanosis. Cyanosis also is observed when nonfunctional hemoglobin such as methemoglobin or sulfhemoglobin ([Chap. 106](#)) is present in blood.

Cyanosis may be subdivided into central and peripheral types. In the *central* type, the SaO_2 is reduced or an abnormal hemoglobin derivative is present, and the mucous membranes and skin are both affected. *Peripheral* cyanosis is due to a slowing of blood flow and abnormally great extraction of O_2 from normally saturated arterial blood. It results from vasoconstriction and diminished peripheral blood flow, such as occurs in cold exposure, shock, congestive failure, and peripheral vascular disease. Often in these conditions the mucous membranes of the oral cavity or those beneath the tongue may be spared. Clinical differentiation between central and peripheral cyanosis may not always be simple, and in conditions such as cardiogenic shock with pulmonary edema there may be a mixture of both types.

DIFFERENTIAL DIAGNOSIS

Central Cyanosis ([Table 36-1](#)) Decreased SaO_2 results from a marked reduction in the PaO_2 . This reduction may be brought about by a decline in the FI_{O_2} without sufficient compensatory alveolar hyperventilation to maintain alveolar P_{O_2} . Cyanosis does not occur to a significant degree in an ascent to an altitude of 2500 m (8000 ft) but is marked in a further ascent to 5000 m (16,000 ft). The reason for this difference becomes clear on studying the S shape of the Hb- O_2 dissociation curve (see [Fig. 106-1](#)). At 2500 m (8000 ft) the FI_{O_2} is about 120 mmHg, the alveolar P_{O_2} is approximately 80 mmHg, and the SaO_2 is nearly normal. However, at 5000 m (16,000 ft) the FI_{O_2} and

alveolar P_{O_2} are about 85 and 50 mmHg, respectively, and the Sa_{O_2} is only about 75%. This leaves 25% of the hemoglobin in the arterial blood in the reduced form, an amount likely to be associated with cyanosis in the absence of anemia. Similarly, a mutant hemoglobin with a low affinity for O_2 (e.g., Hb Kansas) causes lowered Sa_{O_2} saturation and resultant central cyanosis ([Chap. 106](#)).

Seriously *impaired pulmonary function*, through perfusion of unventilated or poorly ventilated areas of the lung or alveolar hypoventilation, is a common cause of central cyanosis ([Chap. 250](#)). This condition may occur acutely, as in extensive pneumonia or pulmonary edema, or chronically with chronic pulmonary diseases (e.g., emphysema). In the last situation, secondary polycythemia is generally present, and clubbing of the fingers may occur. However, in many types of chronic pulmonary disease with fibrosis and obliteration of the capillary vascular bed, cyanosis does not occur because there is relatively little perfusion of underventilated areas.

Another cause of reduced Sa_{O_2} is *shunting of systemic venous blood into the arterial circuit*. Certain forms of congenital heart disease are associated with cyanosis ([Chap. 234](#)). Since blood flows from a higher-pressure to a lower-pressure region, for a cardiac defect to result in a right-to-left shunt, it must ordinarily be combined with an obstructive lesion distal to the defect or with elevated pulmonary vascular resistance. The most common congenital cardiac lesion associated with cyanosis in the adult is the combination of ventricular septal defect and pulmonary outflow tract obstruction (*tetralogy of Fallot*). The more severe the obstruction, the greater the degree of right-to-left shunting and resultant cyanosis. In patients with patent ductus arteriosus, pulmonary hypertension, and right-to-left shunt, *differential cyanosis* results; that is, cyanosis occurs in the lower but not in the upper extremities. *The mechanisms for the elevated pulmonary vascular resistance that may produce cyanosis in the presence of intra- and extracardiac communications without pulmonic stenosis (Eisenmenger syndrome) are discussed in Chap. 234.*

Pulmonary arteriovenous fistulae ([Chap. 57](#)) may be congenital or acquired, solitary or multiple, microscopic or massive. The severity of cyanosis produced by these fistulae depends on their size and number. They occur with some frequency in hereditary hemorrhagic telangiectasia. Sa_{O_2} reduction and cyanosis may also occur in some patients with cirrhosis, presumably as a consequence of pulmonary arteriovenous fistulas or portal vein-pulmonary vein anastomoses.

In patients with cardiac or pulmonary right-to-left shunts, the presence and severity of cyanosis depend on the size of the shunt relative to the systemic flow as well as on the Hb- O_2 saturation of the venous blood. With increased extraction of O_2 from the blood by the exercising muscles, the venous blood returning to the right side of the heart is more unsaturated than at rest, and shunting of this blood or its passage through lungs incapable of normal oxygenation intensifies the cyanosis. Also, since the systemic vascular resistance falls with exercise, the right-to-left shunt is augmented by exercise in patients with congenital heart disease and communications between the two sides of the heart. Secondary polycythemia occurs frequently in patients with arterial O_2 unsaturation and contributes to the cyanosis.

Cyanosis can be caused by small amounts of circulating methemoglobin and by even

smaller amounts of sulfhemoglobin ([Chap. 106](#)). Although they are uncommon causes of cyanosis, these abnormal hemoglobin pigments should be sought by spectroscopy when cyanosis is not readily explained by malfunction of the circulatory or respiratory systems. Generally, digital clubbing does not occur with them. The diagnosis of methemoglobinemia can be suspected if the patient's blood remains brown after being mixed in a test tube and exposed to air.

Peripheral Cyanosis Probably the most common cause of peripheral cyanosis is the normal vasoconstriction resulting from exposure to cold air or water. When cardiac output is low, as in severe congestive heart failure or shock, cutaneous vasoconstriction occurs as a compensatory mechanism so that blood is diverted from the skin to more vital areas such as the central nervous system and heart ([Chap. 232](#)), and intense cyanosis associated with cool extremities may result. Even though the arterial blood is normally saturated, the reduced volume flow through the skin and the reduced P_{O_2} at the venous end of the capillary result in cyanosis.

Arterial obstruction to an extremity, as with an embolus, or arteriolar constriction, as in cold-induced vasospasm (Raynaud's phenomenon, [Chap. 248](#)), generally results in pallor and coldness, but there may be associated cyanosis. Venous obstruction, as in thrombophlebitis, dilates the subpapillary venous plexuses and thereby intensifies cyanosis.

Approach to the Patient

Certain features are important in arriving at the cause of cyanosis:

1. The history, particularly the onset (cyanosis present since birth is usually due to congenital heart disease), and possible exposure to drugs or chemicals that may produce abnormal types of hemoglobin.
2. Clinical differentiation of central as opposed to peripheral cyanosis. Objective evidence by physical or radiographic examination of disorders of the respiratory or cardiovascular systems. Massage or gentle warming of a cyanotic extremity will increase peripheral blood flow and abolish peripheral but not central cyanosis.
3. The presence or absence of clubbing of the digits (see below). Clubbing without cyanosis is frequent in patients with infective endocarditis and ulcerative colitis; it may occasionally occur in healthy persons, and in some instances it may be occupational, e.g., in jackhammer operators. The combination of cyanosis and clubbing is frequent in patients with congenital heart disease and right-to-left shunting and is seen occasionally in persons with pulmonary disease such as lung abscess or pulmonary arteriovenous fistulae. In contrast, peripheral cyanosis or acutely developing central cyanosis is *not* associated with clubbed digits.
4. Determination of P_{aO_2} tension and Sa_{O_2} and spectroscopic and other examinations of the blood for abnormal types of hemoglobin (critical in the differential diagnosis of cyanosis).

CLUBBING

The selective bullous enlargement of the distal segments of the fingers and toes due to proliferation of connective tissue, particularly on the dorsal surface, is termed *clubbing*; there is increased sponginess of the soft tissue at the base of the nail. Clubbing may be hereditary, idiopathic, or acquired and associated with a variety of disorders, including cyanotic congenital heart disease, infective endocarditis, and a variety of pulmonary conditions (among them primary and metastatic lung cancer, bronchiectasis, lung abscess, cystic fibrosis, and mesothelioma), as well as with some gastrointestinal diseases (including regional enteritis, chronic ulcerative colitis, and hepatic cirrhosis).

Clubbing in patients with primary and metastatic lung cancer, mesothelioma, bronchiectasis, and hepatic cirrhosis may be associated with *hypertrophic osteoarthropathy*. In this condition, the subperiosteal formation of new bone in the distal diaphyses of the long bones of the extremities causes pain and symmetric arthritis-like changes in the shoulders, knees, ankles, wrists, and elbows. The diagnosis of hypertrophic osteoarthropathy may be confirmed by bone radiographs. Although the mechanism of clubbing is unclear, it appears to be secondary to a humoral substance that causes dilation of the vessels of the fingertip.

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37. EDEMA - Eugene Braunwald

Edema is defined as a clinically apparent increase in the interstitial fluid volume, which may expand by several liters before the abnormality is evident. Therefore, a weight gain of several kilograms usually precedes overt manifestations of edema, and a similar weight loss from diuresis can be induced in a slightly edematous patient before "dry weight" is achieved. *Ascites* ([Chap. 46](#)) and *hydrothorax* refer to accumulation of excess fluid in the peritoneal and pleural cavities, respectively, and are considered to be special forms of edema. *Anasarca* refers to gross, generalized edema.

Depending on its cause and mechanism, edema may be localized or have a generalized distribution; it is recognized in its generalized form by puffiness of the face, which is most readily apparent in the periorbital areas, and by the persistence of an indentation of the skin following pressure; this is known as "pitting" edema. In its more subtle form, it may be detected by noting that after the stethoscope is removed from the chest wall, the rim of the bell leaves an indentation on the skin of the chest for a few minutes. When the ring on a finger fits more snugly than in the past or when a patient complains of difficulty in putting on shoes, particularly in the evening, edema may be present.

PATHOGENESIS

About one-third of the total-body water is confined to the extracellular space. Approximately 25% of the latter, in turn, is composed of the plasma volume, and the remainder is interstitial fluid.

Starling Forces The forces that regulate the disposition of fluid between these two components of the extracellular compartment are frequently referred to as the *Starling forces* (see p. 202). The hydrostatic pressure within the vascular system and the colloid oncotic pressure in the interstitial fluid tend to promote movement of fluid from the vascular to the extravascular space. In contrast, the colloid oncotic pressure contributed by the plasma proteins and the hydrostatic pressure within the interstitial fluid, referred to as the *tissue tension*, promote the movement of fluid into the vascular compartment. Consequently there is a movement of water and diffusible solutes from the vascular space at the arteriolar end of the capillaries.

Fluid is returned from the interstitial space into the vascular system at the venous end of the capillary and by way of the lymphatics, and unless these channels are obstructed, lymph flow tends to increase with increases in net movement of fluid from the vascular compartment to the interstitium. These flows are usually balanced so that a steady state exists in the sizes of the intravascular and interstitial compartments, and yet a large exchange between them occurs. However, should any one of the hydrostatic or oncotic pressure gradients be altered significantly, a further net movement of fluid between the two components of the extracellular space will take place. The development of edema then depends on one or more alterations in the Starling forces so that there is increased flow of fluid from the vascular system into the interstitium or into a body cavity.

Edema due to increase in capillary pressure may result from an elevation of venous pressure due to obstruction in venous drainage. This increase in capillary pressure may be generalized, as occurs in congestive heart failure. The Starling forces may be

imbalanced when the colloid oncotic pressure of the plasma is reduced, owing to any factor that may induce hypoalbuminemia, such as saline expansion, malnutrition, liver disease, loss of protein into the urine or into the gastrointestinal tract, or a severe catabolic state.

Capillary Damage Edema may also result from damage to the capillary endothelium, which increases its permeability and permits the transfer of protein into the interstitial compartment. Injury to the capillary wall can result from drugs, viral or bacterial agents, and thermal or mechanical trauma. Increased capillary permeability may also be a consequence of a hypersensitivity reaction and is characteristic of immune injury. Damage to the capillary endothelium is presumably responsible for inflammatory edema, which is usually nonpitting, localized, and accompanied by other signs of inflammation -- redness, heat, and tenderness.

To formulate a hypothesis about the pathophysiology of an edematous state, it is important to discriminate between the *primary* events, such as localized or generalized venous or lymphatic obstruction, reduction of cardiac output, hypoalbuminemia, trapping of fluid in spaces such as the pleural or peritoneal cavities, or an increase in capillary permeability, and the predictable *secondary* consequences, which include the renal retention of salt and water in an attempt to restore the plasma volume when the latter has been reduced, as in venous obstruction (see below). Both the primary event and the secondary consequences contribute to the formation of edema. In some cases the primary event is the renal retention of salt and water. Examples are renal failure, nephrotic syndrome, glomerulonephritis, and early hepatic failure.

Reduction of Effective Arterial Volume In many forms of edema the *effective arterial blood volume*, an as yet poorly defined parameter of the filling of the arterial tree, is reduced, and as a consequence a series of physiologic responses designed to restore it to normal are set into motion. A key element of these responses is the retention of salt and therefore of water, principally by the renal proximal tubule ([Fig. 37-1](#)), and in many instances this repairs the deficit of the effective arterial blood volume; often this deficit is repaired without the development of overt edema. If, however, the retention of salt and water is insufficient to restore and maintain the effective arterial blood volume, the stimuli are not dissipated, the retention of salt and water continues, and edema may ultimately develop. This sequence of events is operative in dehydration and hemorrhage. Although in these conditions there is a reduction of effective arterial blood volume and activation of the entire sequence shown in the center of [Fig. 37-2](#), including the diminished excretion of salt and water, because the net sodium and water balance is negative rather than positive, edema does *not* occur. In most conditions that lead to edema, the mechanisms responsible for maintaining a normal effective osmolality in the body fluids operate efficiently so that sodium retention promotes thirst and secretion of the antidiuretic hormone. In edematous states, isotonic expansion of the extracellular fluid space may be massive, while the intracellular fluid volume is unchanged.

Reduced Cardiac Output A reduction of cardiac output, whatever the cause, is associated with a lowering of the effective arterial blood volume as well as of renal blood flow, constriction of the efferent renal arterioles, and an elevation of the filtration fraction, i.e., the ratio of glomerular filtration rate to renal plasma flow. In severe heart failure there is a reduction in the glomerular filtration rate. Activation of the sympathetic

nervous system and of the renin-angiotensin systems are responsible for renal vasoconstriction. The finding that α -adrenergic blocking agents and/or angiotensin-converting enzyme (ACE) inhibitors augment renal blood flow and induce diuresis supports the role of these two systems in elevating renal vascular resistance and salt and water retention.

Renal Factors Reduced cardiac output lowers effective arterial blood volume. There is increased tubular reabsorption of glomerular filtrate in both the proximal and distal tubules ([Fig. 37-1](#)). Alterations in intrarenal hemodynamics appear to play a significant role. Heart failure and other conditions, such as nephrotic syndrome and cirrhosis that reduce effective arterial blood volume, cause renal efferent arteriolar constriction. This, in turn, reduces the hydrostatic pressure while the increased filtration fraction raises the colloid osmotic pressure in the peritubular capillaries, thus enhancing salt and water reabsorption in the proximal tubule as well as in the ascending limb of the loop of Henle.

In addition, the diminished renal blood flow characteristic of states in which the effective arterial blood volume is reduced is translated by the renal juxtaglomerular cells into a signal for increased renin release ([Chap. 331](#)). The mechanisms responsible for this release include a baroreceptor response: reduced renal perfusion results in incomplete filling of the renal arterioles and diminished stretch of the juxtaglomerular cells, a signal that provides for the elaboration or release, or both, of renin. A second mechanism for renin release involves the macula densa; as a result of reduced glomerular filtration, the sodium chloride load reaching the distal renal tubules is reduced. This is sensed by the macula densa, which signals the neighboring juxtaglomerular cells to secrete renin. A third mechanism involves the sympathetic nervous system and circulating catecholamines. Activation of the α -adrenergic receptors in the juxtaglomerular cells stimulates renin release. These three mechanisms generally act in concert.

The Renin-Angiotensin-Aldosterone (RAA) System (See [Chap. 331](#)) Renin, an enzyme with a molecular weight of about 40,000, acts on its substrate, angiotensinogen, an α_2 globulin synthesized by the liver, to release angiotensin I, a decapeptide, which is broken down to angiotensin II (AII), an octapeptide. This has generalized vasoconstrictor properties; it is especially active on the efferent arterioles and independently increases Na^+ reabsorption in the proximal tubule. The RAA system has long been recognized as a hormone system. However, it also operates locally. Both circulating and intrarenally produced AII contribute to renal vasoconstriction and to salt and water retention. These renal effects of AII are mediated by activation of AII type 1 receptors, which can be blocked by specific antagonists such as losartan. AII also enters the circulation and stimulates the production of aldosterone by the zona glomerulosa of the adrenal cortex. In patients with heart failure, not only is aldosterone secretion elevated but the biologic half-life of aldosterone is prolonged, which further increases the plasma level of the hormone. A depression of hepatic blood flow, particularly during exercise, secondary to a reduction in cardiac output, is responsible for the reduced hepatic catabolism of aldosterone. Aldosterone, in turn, enhances Na^+ reabsorption (and K^+ excretion) by the collecting tubule. The activation of the RAA system is most striking in the early phase of acute, severe heart failure and is less intense in patients with chronic, stable, compensated heart failure.

Although increased quantities of aldosterone are secreted in heart failure and in other

edematous states and although blockade of the action of aldosterone by spironolactone (an aldosterone antagonist) or amiloride (a blocker of epithelial Na⁺ channels) often induces a moderate diuresis in edematous states, persistent augmented levels of aldosterone (or other mineralocorticoids) alone do not always promote accumulation of edema, as witnessed by the lack of striking fluid retention in most instances of primary aldosteronism ([Chap. 331](#)). Furthermore, although normal individuals retain some salt and water with the administration of potent mineralocorticoids, such as deoxycorticosterone acetate or fludrocortisone, this accumulation is self-limiting, despite continued exposure to the steroid, a phenomenon known as *mineralocorticoid escape*. The failure of normal individuals who receive large doses of mineralocorticoids to accumulate large quantities of extracellular fluid and to develop edema is probably a consequence of an increase in glomerular filtration rate (pressure natriuresis) and through the action of natriuretic substance(s) (see below). The continued secretion of aldosterone may be more important in the accumulation of fluid in edematous states because patients with edema secondary to heart failure, nephrotic syndrome, and cirrhosis are generally unable to repair the deficit in effective arterial blood volume. As a consequence they do not develop pressure natriuresis.

Blockade of the [RAA](#) system, by blocking [Ang](#) receptors or inhibiting ACE, reduces efferent arteriolar resistance and increases renal blood flow. This action (combined in patients with heart failure with a rise in cardiac output secondary to afterload reduction) as well as reduction in the secretion of aldosterone cause diuresis. However, in patients with moderate or severe impairment of renal function or with renal artery stenosis, interference with the RAA system can cause paradoxical sodium retention due to intensification of renal failure.

Arginine Vasopressin (AVP) and Endothelin (See also [Chap. 329](#)) The secretion of AVP occurs in response to increased intracellular osmolar concentration and by stimulating V₂ receptors increases the reabsorption of free water in the renal distal tubule and collecting duct, thereby increasing total-body water. Circulating AVP is elevated in many patients with heart failure secondary to a nonosmotic stimulus associated with decreased effective arterial volume. Such patients fail to show the normal reduction of AVP with a reduction of osmolality, contributing to hyponatremia and edema formation.

Endothelin This is a potent peptide vasoconstrictor released by endothelial cells; its concentration is elevated in heart failure and contributes to renal vasoconstriction, Na⁺ retention, and edema in heart failure.

Natriuretic Peptides Atrial distention and/or a sodium load cause release into the circulation of atrial natriuretic peptide (ANP), a polypeptide; a high-molecular-weight precursor of ANP is stored in secretory granules within atrial myocytes. Release of ANP causes (1) excretion of sodium and water by augmenting glomerular filtration rate, inhibiting sodium reabsorption in the proximal tubule, and inhibiting release of renin and aldosterone; and (2) arteriolar and venous dilatation by antagonizing the vasoconstrictor actions of [Ang](#), [AVP](#), and sympathetic stimulation. Thus, ANP has the capacity to oppose sodium retention and arterial pressure elevation in hypervolemic states.

The closely related brain natriuretic peptide (BNP) is stored primarily in cardiac ventricular myocardium and is released when ventricular diastolic pressure rises. Its

actions are similar to those of [ANP](#). Circulating levels of ANP and BNP are elevated in congestive heart failure but obviously not sufficient to prevent edema formation. In addition, in edematous states (particularly heart failure), there is abnormal resistance to the actions of natriuretic peptides.

CLINICAL CAUSES OF EDEMA

Obstruction of Venous (and Lymphatic) Drainage of a Limb In this condition the hydrostatic pressure in the capillary bed upstream to the obstruction increases so that an abnormal quantity of fluid is transferred from the vascular to the interstitial space. Since the alternative route (i.e., the lymphatic channels) may also be obstructed, an increased volume of interstitial fluid in the limb develops, i.e., there is a trapping of fluid in the extremity, causing local edema at the expense of the blood volume in the remainder of the body, thereby reducing effective arterial blood volume and leading to the consequences shown in [Fig. 37-2](#).

When venous and lymphatic drainage are obstructed in a limb, fluid accumulates in the interstitium at the expense of plasma volume. The latter stimulates the retention of salt and water until the deficit in plasma volume has been corrected. Tissue tension rises in the affected limb until it counterbalances the primary alterations in the Starling forces, at which time no further fluid accumulates. The net effect is a local increase in the volume of interstitial fluid. This same sequence occurs in ascites and hydrothorax, in which fluid is trapped or accumulates in the cavitary space, depleting the intravascular volume and leading to secondary salt and fluid retention, as already described.

Congestive Heart Failure (See also [Chap. 232](#)) In this disorder the defective systolic emptying of the chambers of the heart and/or the impairment of ventricular relaxation promotes an accumulation of blood in the heart and venous circulation at the expense of the effective arterial volume, and the aforementioned sequence of events ([Fig. 37-2](#)) is initiated. In mild heart failure, a small increment of total blood volume may repair the deficit of arterial volume and establish a new steady state. Through the operation of Starling's law of the heart, an increase in the volume of blood within the chambers of the heart promotes a more forceful contraction and may thereby increase the cardiac output (See [Fig. 232-1](#)). However, if the cardiac disorder is more severe, retention of fluid cannot repair the deficit in effective arterial blood volume. The increment in blood volume accumulates in the venous circulation, and the increase in capillary and lymphatic hydrostatic pressures promotes the formation of edema. In heart failure, a reduction occurs in baroreflex-mediated inhibition of the vasomotor center, which causes activation of renal vasoconstrictor nerves and the [RAA](#) system, causing sodium and water retention.

Incomplete ventricular emptying (systolic heart failure) and/or inadequate ventricular relaxation (diastolic heart failure) both lead to an elevation of ventricular diastolic pressure. If the impairment of cardiac function involves the right ventricle, pressures in the systemic veins and capillaries may rise, thereby augmenting the transudation of fluid into the interstitial space and enhancing the likelihood of peripheral edema in the presence of the accumulation of sodium and water, as described above. The elevated systemic venous pressure is transmitted to the thoracic duct with consequent reduction of lymph drainage, further increasing the accumulation of edema.

If the impairment of cardiac function (incomplete ventricular emptying and/or inadequate relaxation) involves the left ventricle primarily, then pulmonary venous and capillary pressures rise [leading in some instances to pulmonary edema ([Chap. 32](#))], as does pulmonary artery pressure; this in turn interferes with the emptying of the right ventricle, leading to an elevation of right ventricular diastolic and of central and systemic venous pressures, enhancing the likelihood of formation of peripheral edema. Pulmonary edema impairs gas exchange and may induce hypoxia, which embarrasses cardiac function still further, sometimes causing a vicious cycle.

Nephrotic Syndrome and Other Hypoalbuminemic States (See also [Chap. 274](#)) The primary alteration in this disorder is a diminished colloid oncotic pressure due to massive losses of protein into the urine. This promotes a net movement of fluid into the interstitium, causes hypovolemia, and initiates the edema-forming sequence of events described above, including activation of the [RAA](#) system. With severe hypoalbuminemia and the consequent reduced colloid osmotic pressure, the salt and water that are retained cannot be restrained within the vascular compartment, total and effective arterial blood volumes decline, and hence the stimuli to retain salt and water are not abated. A similar sequence of events occurs in other conditions that lead to severe hypoalbuminemia, including severe nutritional deficiency states, protein-losing enteropathy, congenital hypoalbuminemia, and severe, chronic liver disease. However, in the nephrotic syndrome, impaired renal Na^+ excretion contributes to edema, even in the absence of severe hypoalbuminemia.

Cirrhosis (See also [Chaps. 46](#) and [299](#)) This condition is characterized by hepatic venous outflow blockade, which in turn causes expansion of the splanchnic blood volume and increased hepatic lymph formation. Intrahepatic hypertension acts as a potent stimulus for renal Na^+ retention and perhaps systemic vasodilation and a reduction of effective arterial blood volume as well. These alterations are frequently complicated by hypoalbuminemia secondary to reduced hepatic synthesis and reduce the effective arterial blood volume even further, leading to activation of the [RAA](#) system, of renal sympathetic nerves, and other salt- and water-retaining mechanisms. The concentration of circulating aldosterone is elevated by the liver's failure to metabolize this hormone. Initially, the excess interstitial fluid is localized preferentially upstream to the congested portal venous system and obstructed hepatic lymphatics, i.e., in the peritoneal cavity. In later stages, particularly when there is severe hypoalbuminemia, peripheral edema may develop. The excess production of prostaglandins (PGE_2 and PGI_2) in cirrhosis attenuates renal Na^+ retention. When the synthesis of these substances is inhibited by nonsteroidal anti-inflammatory agents, renal function deteriorates and Na^+ retention increases.

Drug-Induced Edema A large number of widely used drugs can cause edema ([Table 37-1](#)). Mechanisms include renal vasoconstriction (nonsteroidal anti-inflammatory agents and cyclosporine), arteriolar dilatation (vasodilators), augmented renal sodium reabsorption (steroid hormones) and capillary damage (interleukin 2).

Idiopathic Edema This syndrome, which occurs almost exclusively in women, is characterized by periodic episodes of edema (unrelated to the menstrual cycle), frequently accompanied by abdominal distention. Diurnal alterations in weight occur with

orthostatic retention of sodium and water, so that the patient may weigh several pounds more after having been in the upright posture for several hours. Such large diurnal weight changes suggest an increase in capillary permeability that appears to fluctuate in severity and to be aggravated by hot weather. There is some evidence that a reduction in plasma volume occurs in this condition with secondary activation of the [RAA](#) system and impaired suppression of [AVP](#) release. Idiopathic edema should be distinguished from cyclical or premenstrual edema, in which the sodium and water retention may be secondary to excessive estrogen stimulation. There are also some cases in which the edema appears to be "diuretic-induced." It has been postulated that in these patients, chronic diuretic administration leads to mild blood volume depletion, which causes chronic hyperreninemia and juxtaglomerular hyperplasia. Salt-retaining mechanisms appear to overcompensate for the direct effects of the diuretics. *Acute* withdrawal of diuretics can then leave the sodium-retaining forces unopposed, leading to fluid retention and edema. Decreased dopaminergic activity and reduced urinary kallikrein and kinin excretion have been reported in this condition and may also be of pathogenetic importance.

TREATMENT

The treatment of idiopathic cyclic edema includes a reduction in salt intake, rest in the supine position for several hours each day, the wearing of elastic stockings (which are put on before arising in the morning), and an attempt to understand any underlying emotional problems. A variety of pharmacologic agents including [ACE](#) inhibitors, progesterone, the dopamine receptor agonist bromocriptine, and the sympathomimetic amine dextroamphetamine have all been reported to be useful when administered to patients who do not respond to simpler measures. Diuretics may be helpful initially but may lose their effectiveness with continuous administration; accordingly, they should be employed sparingly, if at all. Discontinuation of diuretics paradoxically leads to diuresis in "diuretic-induced" edema, described above.

DIFFERENTIAL DIAGNOSIS

The differences between the three major causes of generalized edema are shown in [Table 37-2](#).

Localized edema can usually be readily differentiated from generalized edema. The great majority of patients with generalized edema suffer from advanced cardiac, renal, hepatic, or nutritional disorders. Consequently, the differential diagnosis of generalized edema should be directed toward identifying or excluding these several conditions.

LOCALIZED EDEMA (See also [Chap. 248](#))

Edema originating from inflammation or hypersensitivity is usually readily identified. Localized edema due to venous or lymphatic obstruction may be caused by thrombophlebitis, chronic lymphangitis, resection of regional lymph nodes, filariasis, etc. Lymphedema is particularly intractable because restriction of lymphatic flow results in increased protein concentration in the interstitial fluid, a circumstance that aggravates retention of fluid.

EDEMA OF HEART FAILURE (See also [Chap. 232](#))

The presence of heart disease, as manifested by cardiac enlargement and gallop rhythm, together with evidence of cardiac failure, such as dyspnea, basilar rales, venous distention, and hepatomegaly, usually provides an indication on clinical examination that edema results from heart failure. Noninvasive tests such as echocardiography and radionuclide angiography may be helpful in establishing the diagnosis of heart failure.

EDEMA OF THE NEPHROTIC SYNDROME (See also [Chap. 274](#))

Marked proteinuria (>3.5 g/d), hypoalbuminemia (<35 g/L), and in some instances hypercholesterolemia are present. This syndrome may occur during the course of a variety of kidney diseases, which include glomerulonephritis, diabetic glomerulosclerosis, and hypersensitivity reactions. A history of previous renal disease may or may not be elicited.

EDEMA OF ACUTE GLOMERULONEPHRITIS AND OTHER FORMS OF RENAL FAILURE

The edema occurring during the acute phases of glomerulonephritis is characteristically associated with hematuria, proteinuria, and hypertension. Although some evidence supports the view that the fluid retention is due to increased capillary permeability, in most instances the edema in this disease results from primary retention of sodium and water by the kidneys owing to renal insufficiency. This state differs from congestive heart failure in that it is characterized by a normal (or sometimes even increased) cardiac output and a normal arterial-mixed venous oxygen difference. Patients with edema due to renal failure commonly have evidence of pulmonary congestion on chest roentgenograms before cardiac enlargement is significant, but they usually do not develop orthopnea. Patients with chronic impairment of renal function may also develop edema due to primary renal retention of sodium and water.

EDEMA OF CIRRHOSIS (See also [Chap. 299](#))

Ascites and biochemical and clinical evidence of hepatic disease (collateral venous channels, jaundice, and spider angiomas) characterize edema of hepatic origin. The ascites is frequently refractory to treatment because it collects as a result of a combination of obstruction of hepatic lymphatic drainage, portal hypertension, and hypoalbuminemia. Edema may also occur in other parts of the body in these patients as a result of hypoalbuminemia. Furthermore, the sizable accumulation of ascitic fluid may increase intraabdominal pressure and impede venous return from the lower extremities; hence, it tends to promote accumulation of edema in this region as well.

EDEMA OF NUTRITIONAL ORIGIN

A diet grossly deficient in protein over a prolonged period may produce hypoproteinemia and edema. The latter may be intensified by the development of beriberi heart disease, also of nutritional origin, in which multiple peripheral arteriovenous fistulas result in reduced effective systemic perfusion and effective arterial blood volume, thereby enhancing edema formation ([Chap. 75](#)). Edema may actually become intensified when

these famished subjects are first provided with an adequate diet. The ingestion of more food may increase the quantity of salt ingested, which is then retained along with water. So-called "refeeding edema" may also be linked to increased release of insulin, which directly increases tubular sodium reabsorption. In addition to hypoalbuminemia, hypokalemia and caloric deficits may be involved in the edema of starvation.

OTHER CAUSES OF EDEMA

These include hypothyroidism, in which the edema (myxedema) may be located typically in the pretibial region and which may also be associated with periorbital puffiness. Exogenous hyperadrenocortism, pregnancy, and administration of estrogens and vasodilators, particularly the calcium antagonist nifedipine, may also all cause edema.

DISTRIBUTION OF EDEMA

The distribution of edema is an important guide to the cause. Thus, edema limited to one leg or to one or both arms is usually the result of venous and/or lymphatic obstruction. Edema resulting from hypoproteinemia characteristically is generalized, but it is especially evident in the very soft tissues of the eyelids and face and tends to be most pronounced in the morning because of the recumbent posture assumed during the night. Less common causes of facial edema include trichinosis, allergic reactions, and myxedema. Edema associated with heart failure, on the other hand, tends to be more extensive in the legs and to be accentuated in the evening, a feature also determined largely by posture. When patients with heart failure have been confined to bed, edema may be most prominent in the presacral region. Unilateral edema occasionally results from lesions in the central nervous system affecting the vasomotor fibers on one side of the body; paralysis also reduces lymphatic and venous drainage on the affected side.

ADDITIONAL FACTORS IN DIAGNOSIS

The color, thickness, and sensitivity of the skin are significant. Local tenderness and increase in temperature suggest inflammation. Local cyanosis may signify a venous obstruction. In individuals who have had repeated episodes of prolonged edema, the skin over the involved areas may be thickened, indurated, and often red.

Measurement or estimation of the venous pressure is of importance in evaluating edema. Elevation in an isolated part of the body usually reflects localized venous obstruction. Generalized elevation of systemic venous pressure usually indicates the presence of congestive heart failure. Ordinarily, a significant generalized increase in venous pressure can be recognized by the level at which cervical veins collapse ([Chap. 225](#)). In patients with obstruction of the superior vena cava, edema is confined to the face, neck, and upper extremities, where the venous pressure is elevated compared with that in the lower extremities. Measurement of venous pressure in the upper extremities is also useful in patients with massive edema of the lower extremities and ascites; it is elevated in the upper extremities when the edema is on a cardiac basis (e.g., constrictive pericarditis or tricuspid stenosis) but is normal when it is secondary to cirrhosis. Severe heart failure may cause ascites that may be distinguished from the ascites caused by hepatic cirrhosis by the jugular venous pressure, which usually is

elevated in heart failure and normal in cirrhosis.

Determination of the concentration of serum albumin aids importantly in identifying those patients in whom edema is due, at least in part, to diminished intravascular colloid oncotic pressure. The presence of proteinuria also affords useful clues. The absence of proteinuria excludes nephrotic syndrome but cannot exclude nonproteinuric causes of renal failure. Slight to moderate proteinuria is the rule in patients with heart failure.

Approach to the Patient

An important first question is whether the edema is localized or generalized. If it is localized, those phenomena that may be responsible should be concentrated upon. Hydrothorax and ascites are forms of localized edema. Either may be a consequence of local venous or lymphatic obstruction, as in inflammatory or neoplastic disease.

If the edema is generalized, it should be determined, first, if there is serious hypoalbuminemia, e.g., serum albumin < 25 g/L. If so, the history, physical examination, urinalysis, and other laboratory data will help evaluate the question of cirrhosis, severe malnutrition, protein-losing gastroenteropathy, or the nephrotic syndrome as the underlying disorder. If hypoalbuminemia is not present, it should be determined if there is evidence of congestive heart failure of a severity to promote generalized edema. Finally, it should be determined whether the patient has an adequate urine output, or if there is significant oliguria or even anuria. **These abnormalities are discussed in [Chaps. 47, 269, and 270](#).*

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38. SHOCK - Ronald V. Maier

Shock is the clinical syndrome that results from inadequate tissue perfusion. Irrespective of cause, the hypoperfusion-induced imbalance between the delivery of and requirements for oxygen and substrate leads to cellular dysfunction. The cellular injury created by the inadequate delivery of oxygen and substrates also induces the production and release of inflammatory mediators that further compromise perfusion through functional and structural changes within the microvasculature. This leads to a vicious cycle in which impaired perfusion is responsible for cellular injury which causes maldistribution of blood flow, further compromising cellular perfusion; the latter causes multiple organ failure and, if the process is not interrupted, leads to the death of the patient. The clinical manifestations of shock are the result, in part, of sympathetic neuroendocrine responses to hypoperfusion as well as the breakdown in organ function induced by severe cellular dysfunction.

When very severe and/or persistent, inadequate oxygen delivery leads to irreversible cell injury, and only rapid restoration of oxygen delivery can reverse the progression of the shock state. The fundamental approach to management, therefore, is to recognize overt and impending shock in a timely fashion and to intervene emergently to restore perfusion. Except in cases of cardiogenic shock, this requires the expansion or reexpansion of blood volume. Control of any inciting pathologic process, e.g., continued hemorrhage, impairment of cardiac function, or infection, must occur simultaneously.

Clinical shock is usually accompanied by hypotension, i.e., a mean arterial pressure <60 mmHg in previously normotensive persons. Multiple classification schemes have been developed in an attempt to synthesize the seemingly dissimilar processes leading to shock. Strict adherence to a classification scheme may be difficult from a clinical standpoint because of the frequent combination of two or more causes of shock in any individual patient, but the classification shown in [Table 38-1](#) provides a useful reference point from which to discuss and further delineate the underlying processes. The individual classes are discussed below.

PATHOGENESIS AND ORGAN RESPONSE

MICROCIRCULATION

Normally when cardiac output falls, systemic vascular resistance rises to maintain a level of systemic pressure that is adequate to allow perfusion of the heart and brain at the expense of other tissues, especially muscle, skin, and the gastrointestinal tract. Systemic vascular resistance is determined primarily by the luminal diameter of arterioles. The metabolic rates of the heart and brain are high, and their stores of energy substrate are low. These organs are critically dependent on a continuous supply of oxygen and nutrients, and neither tolerates severe ischemia for more than brief periods. Autoregulation, i.e., the maintenance of blood flow over a wide range of perfusion pressures, is critical in sustaining cerebral and coronary perfusion despite significant hypotension. However, when mean arterial pressure drops to 60 mmHg, flow to these organs falls and their function deteriorates.

Arteriolar vascular smooth muscle has both α - and β -adrenergic receptors ([Chap. 72](#)).

The α_1 receptors mediate vasoconstriction, while the β_2 receptors mediate vasodilation. Efferent sympathetic fibers release norepinephrine, which acts primarily on α_1 receptors in one of the most fundamental compensatory responses to reduced perfusion pressure. Other constrictor substances that are increased in most forms of shock include angiotensin II, vasopressin, endothelin-1, and thromboxane A_2 . Both norepinephrine and epinephrine are released by the adrenal medulla, and the concentrations of these catecholamines in the blood stream rise. Circulating vasodilators in shock include prostacyclin (PGI_2), nitric oxide (NO), and, importantly, products of local metabolism such as adenosine that match flow to the metabolic needs of the tissue. The balance between these various vasoconstrictor and vasodilator influences acting upon the microcirculation determines local perfusion.

Transport to cells depends on microcirculatory flow; capillary permeability; the diffusion of oxygen, carbon dioxide, nutrients, and products of metabolism through the interstitium; and the exchange of these products across cell membranes. Impairment of the microcirculation, which is central to the pathophysiologic responses in the late stages of all forms of shock, results in the derangement of cellular metabolism, which is ultimately responsible for organ failure.

The normal response to mild or moderate hypovolemia is an attempt at restitution of intravascular volume through alterations in hydrostatic pressure and osmolarity. Constriction of arterioles leads to reductions in both the capillary hydrostatic pressure and the number of capillary beds perfused, thereby limiting the capillary surface area across which filtration occurs. When filtration is reduced while intravascular oncotic pressure remains constant or rises, there is net reabsorption of fluid into the vascular bed, in accord with Starling's law of capillary-interstitial liquid exchange ([Chap. 32](#)). Metabolic changes (including hyperglycemia and elevations in the products of glycolysis, lipolysis, and proteolysis) raise extracellular osmolarity, leading to an osmotic gradient between cells and interstitium that increases interstitial and intravascular volume at the expense of intracellular volume.

CELLULAR RESPONSES

Interstitial transport of nutrients is impaired, leading to a decline of intracellular high-energy phosphate stores. Mitochondrial dysfunction and uncoupling of oxidative phosphorylation are the most likely causes for decreased amounts of ATP. As a consequence, there is an accumulation of anaerobic metabolites including hydrogen ions, lactate, and other products of anaerobic metabolism. As shock progresses, these vasodilator metabolites override vasomotor tone, causing further hypotension and hypoperfusion. Dysfunction of cell membranes is thought to represent a common end-stage pathophysiologic pathway in the various forms of shock. Normal cellular transmembrane potential falls, and there is an associated increase in intracellular sodium and water, leading to cell swelling, which interferes further with microvascular perfusion.

NEUROENDOCRINE RESPONSE

Hypovolemia, hypotension, and hypoxia are sensed by baroreceptors and chemoreceptors, which contribute further to an autonomic response that attempts to

restore blood volume, maintain central perfusion, and mobilize metabolic substrates. Hypotension disinhibits the vasomotor center, resulting in increased adrenergic output and reduced vagal activity. Release of norepinephrine induces peripheral and splanchnic vasoconstriction, a major contributor to the maintenance of central organ perfusion, while reduced vagal activity increases the heart rate and cardiac output. The effects of circulating epinephrine released by the adrenal medulla in shock are largely metabolic, causing increased glycogenolysis and gluconeogenesis and reduced pancreatic insulin release.

Severe pain and other severe stress cause the hypothalamic release of adrenocorticotrophic hormone (ACTH). This stimulates cortisol secretion, which contributes to decreased peripheral uptake of glucose and amino acids, enhances lipolysis, and increases gluconeogenesis. Increased pancreatic secretion of glucagon during stress accelerates hepatic gluconeogenesis and further elevates blood glucose concentration. These hormonal actions act synergistically in the maintenance of blood volume. The importance of the cortisol response to stress is illustrated by the profound circulatory collapse that occurs in hypoadrenal patients (see below).

Renin release is increased in response to adrenergic discharge and reduced perfusion of the juxtaglomerular apparatus in the kidney. Renin induces the formation of angiotensin I, which is then converted to angiotensin II, an extremely potent vasoconstrictor and stimulator of aldosterone release by the adrenal cortex and of vasopressin by the posterior pituitary. Aldosterone contributes to the maintenance of intravascular volume by enhancing renal tubular reabsorption of sodium, resulting in the excretion of a low-volume, concentrated, sodium-free urine. Vasopressin has a direct action on vascular smooth muscle, contributing to vasoconstriction, and acts on the distal renal tubules to enhance water reabsorption.

CARDIOVASCULAR RESPONSE

Three variables -- ventricular filling (preload), the resistance to ventricular ejection (afterload), and myocardial contractility -- are paramount in controlling stroke volume ([Chap. 231](#)). Cardiac output, the major determinant of tissue perfusion, is the product of stroke volume and heart rate. Hypovolemia leads to decreased ventricular preload, which in turn reduces the stroke volume. An increase in heart rate is a useful but limited compensatory mechanism to maintain cardiac output. A shock-induced reduction in myocardial compliance is frequent, reducing ventricular end-diastolic volume and hence stroke volume at any given ventricular filling pressure. Restoration of intravascular volume then returns stroke volume to normal but only at elevated filling pressures. In addition, sepsis, ischemia, myocardial infarction, severe tissue trauma, hypothermia, general anesthesia, prolonged hypotension, and acidemia may all impair myocardial contractility and also reduce the stroke volume at any given ventricular end-diastolic volume. The resistance to ventricular ejection is influenced importantly by the systemic vascular resistance, which is elevated in most forms of shock. However, resistance is depressed in the early hyperdynamic stage of septic shock (see below), thereby allowing the cardiac output to be maintained.

The venous system contains nearly two-thirds of the total circulating blood volume, most in the small veins, and serves as a dynamic reservoir for autoinfusion of blood. Active

venoconstriction as a consequence of α -adrenergic activity is an important compensatory mechanism for the maintenance of venous return and therefore of ventricular filling during shock. On the other hand, venous dilatation, as occurs in neurogenic shock, reduces ventricular filling and hence stroke volume and cardiac output (see below).

PULMONARY RESPONSE

The response of the pulmonary vascular bed to shock parallels that of the systemic vascular bed, and the relative increase in pulmonary vascular resistance, particularly in septic shock, may exceed that of the systemic vascular resistance. Shock-induced tachypnea reduces tidal volume and increases both dead space and minute ventilation. Relative hypoxia and the subsequent tachypnea induce a respiratory alkalosis. Recumbency and involuntary restriction of ventilation secondary to pain reduce functional residual capacity and may lead to atelectasis. Shock is recognized as a major cause of acute lung injury and subsequent acute respiratory distress syndrome (ARDS; [Chap. 265](#)). These disorders are characterized by noncardiogenic pulmonary edema secondary to diffuse pulmonary capillary endothelial and alveolar epithelial injury, hypoxemia, and bilateral diffuse pulmonary infiltrates. Hypoxemia results from perfusion of underventilated and nonventilated alveoli. Loss of surfactant and lung volume in combination with increased interstitial and alveolar edema reduce lung compliance. The work of breathing and the oxygen requirements of respiratory muscles increase.

RENAL RESPONSE

Acute renal failure ([Chap. 269](#)), a serious complication of shock and hypoperfusion, occurs less frequently than heretofore because of early aggressive volume repletion. Acute tubular necrosis is now more frequently seen as a result of the interactions of shock, sepsis, the administration of nephrotoxic agents (such as aminoglycosides and angiographic contrast media), and rhabdomyolysis; the latter may be particularly severe in skeletal muscle trauma. The physiologic response of the kidney to hypoperfusion is to conserve salt and water. In addition to decreased renal blood flow, increased afferent arteriolar resistance accounts for diminished glomerular filtration rate, which together with increased ADH and aldosterone is responsible for reduced urine formation. Toxic injury causes necrosis of tubular epithelium and tubular obstruction by cellular debris with back-leak of filtrate. The depletion of renal ATP stores that occurs with prolonged renal hypoperfusion is related to subsequent impairment of renal function.

METABOLIC DERANGEMENTS

During shock, there is disruption of the normal cycles of carbohydrate, lipid, and protein metabolism. Through the citric acid cycle, alanine in conjunction with lactate (which is converted from pyruvate in the periphery in the presence of oxygen deprivation) enhances the hepatic production of glucose. With reduced availability of oxygen, the breakdown of glucose to pyruvate and ultimately lactate represents an inefficient cycling of substrate with minimal net energy production. An elevated plasma lactate/pyruvate ratio is consistent with anaerobic metabolism and reflects inadequate tissue perfusion. Decreased clearance of exogenous triglycerides coupled with increased hepatic

lipogenesis causes a significant rise in serum triglyceride concentrations. There is increased protein catabolism, a negative nitrogen balance, and, if the process is prolonged, severe muscle wasting.

INFLAMMATORY RESPONSES

Activation of an extensive network of proinflammatory mediator systems plays a significant role in the progression of shock and contributes importantly to the development of organ injury and failure.

Multiple humoral mediators are activated during shock and tissue injury. The complement cascade, activated through both the classic and alternate pathways, generates the anaphylatoxins C3a and C5a. Direct complement fixation to injured tissues can progress to the C5-C9 attack complex, causing further cell damage. Activation of the coagulation cascade causes microvascular thrombosis, with subsequent lysis leading to repeated episodes of ischemia and reperfusion. Components of the coagulation system, such as thrombin, are potent proinflammatory mediators that cause expression of adhesion molecules on endothelial cells and activation of neutrophils, leading to microvascular injury. Coagulation also activates the kallikrein-kininogen cascade, contributing to hypotension.

Eicosanoids are vasoactive and immunomodulatory products of arachidonic acid metabolism that include cyclooxygenase-derived prostaglandins and thromboxane A₂ as well as lipoxygenase-derived leukotrienes and lipoxins. Thromboxane A₂ is a potent vasoconstrictor that contributes to the pulmonary hypertension and acute tubular necrosis of shock. [PGI₂](#) and prostaglandin E₂ are potent vasodilators that enhance capillary permeability and edema formation. The cysteinyl leukotrienes LTC₄ and LTD₄ are pivotal mediators of the vascular sequelae of anaphylaxis, as well as of shock states resulting from sepsis or tissue injury. LTB₄ is a potent neutrophil chemoattractant and secretagogue that stimulates the formation of reactive oxygen species. Lipoxins are endogenous autocooids that inhibit leukotriene-mediated responses. Platelet-activating factor, an ether-linked, arachidonyl-containing phospholipid mediator, also carries potent bioactivities that include pulmonary vasoconstriction, bronchoconstriction, systemic vasodilation, increased capillary permeability, and the priming of macrophages and neutrophils to produce enhanced levels of inflammatory mediators.

Tumor necrosis factor (TNF) α , produced by activated macrophages, reproduces many components of the shock state including hypotension, lactic acidosis, and respiratory failure. Interleukin (IL) 1 is also produced by tissue-fixed macrophages and is critical to the inflammatory response occurring in hypoperfusion and septic states. Chemokines also participate in the systemic inflammatory response. For example, IL-8 is a potent neutrophil chemoattractant and activator that upregulates adhesion molecules on the neutrophil to enhance aggregation and adherence to the vascular endothelium. The endothelium normally produces nitric oxide (NO), a potent vasodilator. The inflammatory response stimulates the inducible isoform of NO synthase (iNOS), which is thought to overexpress toxic NO and oxygen-derived free radicals and contributes to the hyperdynamic cardiovascular response that occurs in sepsis.

Multiple inflammatory cells, including neutrophils, macrophages, and platelets, are a

major contributor to inflammation-induced injury. Margination of activated neutrophils in the microcirculation is a common pathologic finding in shock, causing secondary injury due to the release of potentially toxic oxygen radicals and proteases. Adhesion molecules are expressed on the surface of the endothelium and on cytokine-stimulated neutrophils. Tissue-fixed macrophages produce virtually all major components of the inflammatory response and orchestrate the progression and duration of the response.

Approach to the Patient

The underlying problem in all forms of shock is inadequate tissue perfusion and an imbalance between delivery and cellular needs of oxygen and metabolic substrate. It is important to recognize the onset of hypoperfusion at the earliest possible time in order to institute aggressive resuscitation and correction of the underlying etiology.

Monitoring Patients in shock require care in an intensive care unit. Careful and continuous assessment of the physiologic status is necessary. Arterial pressure through an indwelling line, pulse, and respiratory rate should be monitored continuously; a Foley catheter should be inserted to follow urine flow; and mental status assessed frequently.

Although there is ongoing debate as to the indications for using the flow-directed pulmonary artery catheter (PAC, Swan-Ganz catheter) in the management of patients in shock, most intensivists believe that the ability to predict the hemodynamic profiles of patients in shock accurately without a PAC is poor. The PAC is placed percutaneously via the subclavian or jugular vein through the central venous circulation and right heart into the pulmonary artery. There are ports both proximal in the right atrium and distal in the pulmonary artery to provide access for infusions and for cardiac output measurements. Right atrial and pulmonary artery pressures are measured, and the pulmonary capillary wedge pressure (PCWP) serves as an approximation of the left atrial pressure. Normal hemodynamic parameters are shown in [Table 228-3](#) and [Table 38-2](#).

Cardiac output is determined by the thermodilution technique, and high-resolution thermistors can also be used to determine right ventricular end-diastolic volume to monitor further the response of the right heart to fluid resuscitation. [APAC](#) with an oximeter port offers the additional advantage of on-line monitoring of the mixed venous oxygen saturation, an important index of tissue perfusion. Systemic and pulmonary vascular resistances are calculated as the ratio of the pressure drop across these vascular beds to the cardiac output ([Chap. 228](#)). Determinations of oxygen content in arterial and venous blood, together with cardiac output and hemoglobin concentration allow calculation of oxygen delivery, oxygen consumption, and oxygen-extraction ratio ([Table 38-3](#)). The hemodynamic patterns associated with the various form of shock are shown in [Table 38-4](#).

In resuscitation from shock, it is critical to restore tissue perfusion and optimize oxygen delivery, hemodynamics, and cardiac function rapidly. A goal of therapy is to achieve normal mixed venous oxygen saturation and arteriovenous oxygen-extraction ratio. To enhance oxygen delivery, red cell mass, arterial oxygen saturation, and cardiac output may be augmented singly or simultaneously. An increase in oxygen delivery not accompanied by an increase in oxygen consumption implies that oxygen availability is

adequate and that oxygen consumption is not flow-dependent. Conversely, an elevation of oxygen consumption with increased cardiac output implies that the oxygen supply is inadequate. A reduction in systemic vascular resistance accompanying an increase in cardiac output indicates that compensatory vasoconstriction is reversing due to improved tissue perfusion. The determination of stepwise expansion of blood volume on cardiac performance allows identification of the optimum preload.

SPECIFIC FORMS OF SHOCK

HYPOVOLEMIC SHOCK

This most common form of shock results either from the loss of red blood cell mass and plasma from hemorrhage or from the loss of plasma volume alone arising from extravascular fluid sequestration or gastrointestinal, urinary, and insensible losses. The signs and symptoms of nonhemorrhagic hypovolemic shock are the same as those of hemorrhagic shock, although they may have a more insidious onset. The normal physiologic response to hypovolemia is to maintain perfusion of the brain and heart while restoring an effective circulating blood volume. There is an increase in sympathetic activity, hyperventilation, collapse of venous capacitance vessels, release of stress hormones, and expansion of intravascular volume through the recruitment of interstitial and intracellular fluid and reduction of urine output.

Mild hypovolemia ($\leq 20\%$ of the blood volume) generates mild tachycardia but relatively few external signs, especially in a supine resting young patient ([Table 38-5](#)). With moderate hypovolemia (~ 20 to 40% of the blood volume) the patient becomes increasingly anxious and tachycardic; although normal blood pressure may be maintained in the supine position, there may be significant postural hypotension and tachycardia. If hypovolemia is severe ($\geq 40\%$ of the blood volume), the classic signs of shock appear; the blood pressure declines and becomes unstable even in the supine position, and the patient develops marked tachycardia, oliguria, and agitation or confusion. Perfusion of the central nervous system is well maintained until shock becomes severe. Hence, mental obtundation is an ominous clinical sign. The transition from mild to severe hypovolemic shock can be insidious or extremely rapid. If severe shock is not reversed rapidly, especially in elderly patients and those with comorbid illnesses, death is imminent. A very narrow time frame separates the derangements found in severe shock that can be reversed with aggressive resuscitation from those of progressive decompensation and irreversible cell injury.

Diagnosis Hypovolemic shock is readily diagnosed when there are signs of hemodynamic instability and the source of volume loss is obvious. The diagnosis is more difficult when the source of blood loss is occult, as into the gastrointestinal tract, or when plasma volume alone is depleted. After acute hemorrhage, hemoglobin and hematocrit values do not change until compensatory fluid shifts have occurred or exogenous fluid is administered. Thus, an initial normal hematocrit does not disprove the presence of significant blood loss. Plasma losses cause hemoconcentration, and free water loss leads to hypernatremia. These findings should suggest the presence of hypovolemia.

It is essential to distinguish between hypovolemic and cardiogenic shock (see below)

because definitive therapy differs significantly. Both forms are associated with a reduced cardiac output and a compensatory sympathetic mediated response characterized by tachycardia and elevated systemic vascular resistance. However, the findings in cardiogenic shock of jugular venous distention, rales, and an S₃ gallop distinguish it from hypovolemic shock and signify that volume expansion is undesirable.

TREATMENT

Initial resuscitation requires rapid reexpansion of the circulating blood volume along with interventions to control ongoing losses. In accordance with Starling's law ([Chap. 231](#)), stroke volume and cardiac output rise with the increase in preload. After resuscitation, the compliance of the ventricles may remain reduced due to increased interstitial fluid in the myocardium. Therefore, elevated filling pressures are required to maintain adequate ventricular performance.

Volume resuscitation is initiated with the rapid infusion of isotonic saline or a balanced salt solution such as Ringer's lactate through large-bore intravenous lines. No distinct benefit from the use of colloid has been demonstrated. The infusion of 2 to 3 L over 10 to 30 min should restore normal hemodynamic parameters. Continued hemodynamic instability implies that shock has not been reversed and/or that there are significant ongoing blood or volume losses. Continuing blood loss, with hemoglobin concentrations declining to 100 g/L (10 g/dL), should initiate blood transfusion, preferably as fully cross-matched blood. In extreme emergencies, type-specific or O-negative packed red cells may be transfused. In the presence of severe and/or prolonged hypovolemia, inotropic support with dopamine or dobutamine ([Chap. 72](#)) may be required to maintain adequate ventricular performance, after blood volume has been restored. Infusion of norepinephrine to increase arterial pressure by raising peripheral resistance is inappropriate, other than as a temporizing measure in severe shock while blood volume is reexpanded.

Successful resuscitation also requires support of respiratory function. Supplemental oxygen should be provided, and endotracheal intubation may be necessary to maintain arterial oxygenation. Following resuscitation from isolated hemorrhagic shock, end-organ damage is frequently less than following septic or traumatic shock. This may be due to the absence of the massive activation of inflammatory mediator response systems and the consequent nonspecific organ injury seen in the latter conditions.

TRAUMATIC SHOCK

Shock following trauma is, in large measure, due to hypovolemia. However, even when hemorrhage has been controlled, patients can continue to suffer loss of plasma volume into the interstitium of injured tissues. These fluid losses are compounded by injury-induced inflammatory responses, which contribute to the secondary microcirculatory injury. This causes secondary tissue injury and maldistribution of blood flow, intensifying tissue ischemia and leading to multiple organ system failure. Trauma to the heart, chest, or head can also contribute to the shock. For example, pericardial tamponade or tension pneumothorax impairs ventricular filling, while myocardial contusion depresses myocardial contractility.

TREATMENT

Inability to maintain a systolic blood pressure ≥ 90 mmHg after trauma-induced hypovolemia is associated with a mortality rate of ~50%. To prevent decompensation of homeostatic mechanisms, therapy must be promptly administered.

The initial management of the seriously injured patient requires attention to the "ABCs" of resuscitation: assurance of an airway (A), adequate ventilation (breathing, B), and establishment of an adequate blood volume to support the circulation (C). Control of hemorrhage requires immediate attention. Early stabilization of fractures, debridement of devitalized or contaminated tissues, and evacuation of hematomata all reduce the subsequent inflammatory response to the initial insult and minimize subsequent organ injury.

INTRINSIC CARDIOGENIC SHOCK

This form of shock is caused by failure, often sudden, of the heart as an effective pump. It occurs most commonly as a complication of acute myocardial infarction (AMI; [Chap. 243](#)), but it may also be seen in patients with severe brady- or tachyarrhythmias, valvular heart disease, or in the terminal stage of chronic heart failure of any cause, including ischemic heart disease and dilated cardiomyopathy. Cardiogenic shock is characterized by a low cardiac output, diminished peripheral perfusion, pulmonary congestion, and elevation of systemic vascular resistance and pulmonary vascular pressures. Acute right heart failure can arise as the result of right ventricular infarction or may complicate the acute respiratory distress syndrome and severe pulmonary hypertension of any etiology. As a consequence of right ventricular failure, left ventricular preload falls, and this, in turn, reduces systemic perfusion. In contrast to other forms of shock, absolute or relative hypovolemia is usually not present in cardiogenic shock.

The ineffective contractile activity of either the right or left side of the heart leads to the accumulation of blood in the venous circulation upstream to the failing ventricle. Cardiogenic shock with left-sided heart failure increases fluid in the lungs that can overwhelm the capacity of the pulmonary lymphatics and causes interstitial and sometimes alveolar edema. Interstitial lung edema usually occurs at pulmonary capillary pressures >18 mmHg, and overt pulmonary alveolar edema develops at pressures >24 mmHg ([Chap. 32](#)). Pulmonary edema impacts cardiac function further by impairing diffusion of oxygen, setting up a vicious cycle. The increase in interstitial and intraalveolar fluid causes a progressive reduction in lung compliance, thereby increasing the work of ventilation while increasing perfusion of poorly ventilated alveoli.

In establishing the diagnosis of cardiogenic shock, a history of cardiac disease or of [AMI](#) is of value. Associated physical findings include those of hemodynamic instability, peripheral vasoconstriction, and pulmonary and/or systemic venous congestion, as well as findings specific to the underlying cardiac abnormalities. An electrocardiogram may provide evidence of AMI or preexisting cardiac disease. The chest x-ray may show pulmonary edema and cardiomegaly. Transthoracic or transesophageal echocardiograms assist in the diagnosis of structural abnormalities and/or functional impairment of contractility. Serum cardiac markers will support the diagnosis of acute

cardiac injury. Hemodynamic monitoring is usually necessary. Placement of a [PAC](#) is helpful and will show a reduced cardiac output and an elevated [PCWP](#), and direct measurement of right atrial pressure allows calculation of systemic vascular resistance which is elevated.

TREATMENT

For all forms of cardiogenic shock, preload, afterload, and contractility should be modified using the information provided by the [PAC](#). [APCWP](#) of 15 to 20 mmHg should be the initial goal. If the PCWP is excessively elevated, inotropic agents may provide significant reduction. The goal is to increase contractility without significant increases in heart rate. Dopamine and norepinephrine exert both inotropic and vasoconstrictor actions ([Chap. 72](#)) that are useful in the presence of persistent hypotension.

Dobutamine, a positive inotropic agent with vasodilator properties, may be substituted when arterial pressure has been restored. Pulmonary congestion may be responsive to intravenous furosemide. Patients with an inadequate response to these measures can be supported by using intraaortic balloon counterpulsation to permit recovery of myocardial function. Additional measures to consider in cases of refractory cardiogenic shock include urgent myocardial revascularization in patients with [AMI](#) ([Chap. 243](#)), correction of anatomic cardiac defects such as rupture of the papillary muscles of the interventricular septum, the placement of ventricular assist devices, and even urgent cardiac transplantation.

COMPRESSIVE CARDIOGENIC SHOCK

With compression, the heart and surrounding structures are less compliant and, thus, normal filling pressures generate inadequate diastolic filling. Blood or fluid within the poorly distensible pericardial sac may cause tamponade ([Chap. 239](#)). Any cause of increased intrathoracic pressure, such as tension pneumothorax, herniation of abdominal viscera through a diaphragmatic hernia, or excessive positive pressure ventilation to support pulmonary function, can also cause compressive cardiogenic shock. Acute right heart failure with a sudden decline in cardiac output can be caused by pulmonary embolism obstructing right ventricular outflow and impairing left ventricular filling. Although initially responsive to increased filling pressures produced by volume expansion, as compression increases, cardiogenic shock occurs.

The diagnosis of compressive cardiogenic shock is most frequently based on clinical findings, the chest radiograph, and an echocardiogram. The diagnosis of compressive cardiac shock may be more difficult to establish in the setting of trauma when hypovolemia and cardiac compression are present simultaneously. The classic findings of pericardial tamponade include the triad of hypotension, neck vein distention, and muffled heart sounds ([Chap. 239](#)). Pulsus paradoxus, i.e., an inspiratory reduction in systolic pressure >10 mmHg, may also be noted. The diagnosis is confirmed by echocardiography, and treatment consists of immediate pericardiocentesis. A tension pneumothorax produces ipsilateral decreased breath sounds, tracheal deviation away from the affected thorax, and jugular venous distention. Radiographic findings include increased intrathoracic volume, depression of the diaphragm of the affected hemithorax, and shifting of the mediastinum to the contralateral side. Chest decompression must be carried out immediately. Release of air and restoration of normal cardiovascular

dynamics is both diagnostic and therapeutic.

SEPTIC SHOCK (See also [Chap. 124](#))

This form of shock is caused by the systemic response to a severe infection. It occurs most frequently in elderly or immunocompromised patients and in those who have undergone an invasive procedure in which bacterial contamination has occurred. Infections of the lung, abdomen, or urinary tract are most common, and approximately half of the patients have bacteremia. Gram-positive and -negative bacteria, viruses, fungi, rickettsiae, and protozoa have all been reported to produce the clinical picture of septic shock, and the overall response is generally independent of the specific type of invading organism. The clinical findings in septic shock are a consequence of the combination of metabolic and circulatory derangements driven by the systemic infection and the release of toxic components of the infectious organisms, e.g., the endotoxin of gram-negative bacteria or the exotoxins and enterotoxins of gram-positive bacteria. Organism toxins lead to the release of cytokines, including [IL](#)-1 and [TNF](#)- α , from tissue macrophages. Tissue factor expression and fibrin deposition are increased, and disseminated intravascular coagulation may develop. The inducible form of NO synthase is stimulated, and NO, a powerful vasodilator, is released. Hemodynamic changes in septic shock occur in two characteristic patterns: early, or hyperdynamic, and late, or hypodynamic, septic shock.

Hyperdynamic Response In hyperdynamic septic shock, tachycardia is present, the cardiac output is normal, and the systemic vascular resistance is reduced while the pulmonary vascular resistance is elevated. The extremities are usually warm. However, splanchnic vasoconstriction with decreased visceral flow is present. The venous capacitance is increased, which decreases venous return. With volume expansion cardiac output becomes supranormal. Myocardial contractility is depressed in septic shock by mediators including NO, [IL](#)-1, and/or [TNF](#)- α . Inflammatory mediator-induced processes include increased capillary permeability and continued loss of intravascular volume.

In septic shock, in contrast to other types of shock, total oxygen delivery may be increased while oxygen extraction is reduced due to maldistribution of microcirculatory perfusion and impaired utilization. In this setting the presence of a normal mixed venous oxygen saturation is not indicative of adequate peripheral perfusion, and even though the cardiac output may be elevated, it is still inadequate to meet the total metabolic needs. The toxicity of the infectious agents and their byproducts and the subsequent metabolic dysfunction drive the progressive deterioration of cellular and organ function. Acute respiratory distress syndrome, thrombocytopenia, and neutropenia are common complications.

Hypodynamic Response As sepsis progresses, vasoconstriction occurs and the cardiac output declines. The patient usually becomes markedly tachypneic, febrile, diaphoretic, and obtunded, with cool, mottled, and often cyanotic extremities. Oliguria, renal failure, and hypothermia develop; there may be striking increases in serum lactate.

TREATMENT

Aggressive volume expansion with a crystalloid solution to a [PCWP](#) of approximately 15 mmHg and the restoration of arterial oxygenation with inspired oxygen and frequently with mechanical ventilation are the highest priorities. In the presence of hypodynamic septic shock, augmentation of cardiac output may require inotropic support with dopamine or norepinephrine in the presence of hypotension or with dobutamine if arterial pressure is normal. Antibiotics should be administered, either appropriate for the results of cultures or empirical therapy based on the likely source of infection. Surgical debridement or drainage may also be necessary to control the infection.

NEUROGENIC SHOCK

Interruption of sympathetic vasomotor input after a high cervical spinal cord injury, inadvertent cephalad migration of spinal anesthesia, or severe head injury may result in neurogenic shock. In addition to arteriolar dilatation, venodilation causes pooling in the venous system, which decreases venous return and cardiac output. The extremities are often warm, in contrast to the usual vasoconstriction-induced coolness in hypovolemic or cardiogenic shock. Treatment involves a simultaneous approach to the relative hypovolemia and to the loss of vasomotor tone. Large volumes of fluid may be required to restore normal hemodynamics. Once hemorrhage has been ruled out, norepinephrine may be necessary to augment vascular resistance.

HYPOADRENAL SHOCK (See also [Chap. 331](#))

The normal host response to the stress of illness, operation, or trauma requires that the adrenal glands hypersecrete cortisol in excess of that normally required. Hypoadrenal shock occurs in settings in which unrecognized adrenal insufficiency complicates the host response to the stress induced by acute illness or major surgery. Adrenocortical insufficiency may occur as a consequence of the chronic administration of high doses of exogenous glucocorticoids. Recent studies have shown that prolonged stays in a critical state in an intensive care setting may also induce a relative hypoadrenal state. Other, less common causes include adrenal insufficiency secondary to idiopathic atrophy, tuberculosis, metastatic disease, bilateral hemorrhage, and amyloidosis. The shock produced by adrenal insufficiency is characterized by reductions in systemic vascular resistance, hypovolemia, and reduced cardiac output. The diagnosis of adrenal insufficiency may be established by means of an [ACTH](#) stimulation test ([Chap. 331](#)).

TREATMENT

In the hemodynamically unstable patient, dexamethasone sodium phosphate, 4 mg, should be given intravenously. This agent is preferred because unlike hydrocortisone it does not interfere with the [ACTH](#) stimulation test. If the diagnosis of adrenal insufficiency has been established, hydrocortisone, 100 mg every 6 to 8 h, can be given and tapered to a maintenance level as the patient achieves hemodynamic stability. Simultaneous volume resuscitation and pressor support is required.

ADJUNCTIVE THERAPIES

As described above, the sympathomimetic amines dobutamine, dopamine, and norepinephrine are widely used in the treatment of all forms of shock. The clinical

pharmacology of these agents is described in [Chap. 72](#).

POSITIONING

Positioning of the patient may be a valuable adjunct in the initial treatment of hypovolemic shock. Elevating the foot of the bed (i.e., placing it on "shock blocks") and assumption of the Trendelenburg position without flexion at the knees are effective but may increase work of breathing and risk for aspiration. Simply elevating both legs may be the optimal approach.

PNEUMATIC ANTISHOCK GARMENT (PASG)

The PASG and the military antishock trousers (MAST) are inflatable external compression devices that can be wrapped around the legs and abdomen and have been widely used in the prehospital setting as a means of providing temporary support of central hemodynamics in shock. They cause an increase in systemic vascular resistance and blood pressure by arterial compression, without causing a significant change in cardiac output. While the use of PASG has been recommended in noncardiogenic forms of shock, the most appropriate use appears to be as a means to tamponade bleeding and augment hemostasis. Inflation of the suit provides splinting of fractures of the pelvis and lower extremities and arrests hemorrhage from fractures.

REWARMING

Hypothermia is a potential adverse consequence of massive volume resuscitation. The infusion of large volumes of refrigerated blood products and room-temperature crystalloid solutions can rapidly drop core temperatures if fluid is not run through warming devices. Hypothermia may depress cardiac contractility and thereby further impair cardiac output and oxygen delivery. Hypothermia, particularly temperatures $<35^{\circ}\text{C}$, directly impairs the coagulation pathway, sometimes causing a significant coagulopathy. Rapid rewarming significantly decreases the requirement for blood products and an improvement in cardiac function. The most effective method for rewarming is extracorporeal countercurrent warmers through femoral artery and vein cannulation. This process does not require a pump and can rewarm from 30° to 36°C in <30 min.

(Bibliography omitted in Palm version)

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39. CARDIOVASCULAR COLLAPSE, CARDIAC ARREST, AND SUDDEN CARDIAC DEATH - Robert J. Myerburg, Agustin Castellanos

OVERVIEW AND DEFINITIONS

The vast majority of naturally occurring sudden deaths are caused by cardiac disorders. The magnitude of sudden *cardiac* death (SCD) as a public health problem is highlighted by estimates that more than 300,000 deaths occur each year in the United States by this mechanism, accounting for 50% of all cardiac deaths. SCD is a direct consequence of cardiac arrest, which is often reversible if responded to promptly. Since resuscitation techniques and emergency rescue systems are available to save patients who have out-of-hospital cardiac arrest, which was uniformly fatal in the past, understanding the SCD problem has practical importance.

[SCD](#) must be defined carefully. In the context of time, "sudden" is defined, for most clinical and epidemiologic purposes, as 1 h or less between the onset of the terminal clinical event, or an abrupt change in clinical status, and death. An exception is unwitnessed deaths in which pathologists may expand the definition of time to 24 h after the victim was last seen to be alive and stable.

Because of community-based interventions, victims may remain biologically alive for days or even weeks after a cardiac arrest that has resulted in irreversible central nervous system damage. Confusion in terms can be avoided by adhering strictly to definitions of death, cardiac arrest, and cardiovascular collapse ([Table 39-1](#)). Death is biologically, legally, and literally an absolute and irreversible event. Death may be delayed in a survivor of cardiac arrest, but "survival after sudden death" is an irrational term. Currently, the accepted definition of SCD is *natural death due to cardiac causes*, heralded by abrupt loss of consciousness within 1 h of the onset of acute symptoms, in an individual who may have known *preexisting* heart disease but in whom the *time* and *mode* of death are *unexpected*. When biologic death of the cardiac arrest victim is delayed because of interventions, the relevant pathophysiologic event remains the sudden and unexpected cardiac arrest that leads ultimately to death, even though delayed by artificial methods. The language used should reflect the fact that the index event was a cardiac arrest and that death was due to its delayed consequences.

ETIOLOGY, INITIATING EVENTS, AND CLINICAL EPIDEMIOLOGY

Clinical and epidemiologic studies have identified populations at high risk for [SCD](#). In addition, a large body of pathologic data provides information on the underlying *structural abnormalities* in victims of SCD, and studies of clinical physiology have begun to identify a group of *transient functional factors* that may convert a long-standing underlying structural abnormality from a stable to an unstable state ([Table 39-2](#)). This information is developing into an understanding of the causes and mechanisms of SCD.

Cardiac disorders constitute the most common causes of sudden *natural* death. After an initial peak incidence of sudden death between birth and 6 months of age (the sudden infant death syndrome), the incidence of sudden death declines sharply and remains low through childhood and adolescence. Among adolescents and young adults, the incidence of [SCD](#) is approximately 1 per 100,000 population per year. The incidence

begins to increase in adults over the age of 30 years, reaching a second peak in the age range of 45 to 75 years, when the incidence approximates 1 to 2 per 1000 per year among the unselected adult population. Increasing age within this range is a powerful risk factor for sudden *cardiac* death, and the proportion of cardiac causes among all sudden natural deaths increases dramatically with advancing years. From 1 to 13 years of age, only one of five sudden *natural* deaths is due to cardiac causes. Between 14 and 21 years of age, the proportion increases to 30%, and then to 88% in the middle-aged and elderly.

Young and middle-aged men and women have very different susceptibilities to [SCD](#), but the gender differences decrease with advancing age. In the 45- to 64-year-old age group, the male SCD excess is nearly 7:1. It falls to 2:1 or less in the 65- to 74-year-old age group. The difference in risk for SCD parallels the risks for other manifestations of coronary heart disease in men and women. As the gender gap for manifestations of coronary heart disease closes in the seventh and eighth decades of life, the excess risk of SCD in males also narrows. Despite the lower incidence among younger women, coronary risk factors such as cigarette smoking, diabetes, hyperlipidemia, and hypertension are highly influential, and SCD remains an important clinical and epidemiologic problem.

Hereditary factors contribute to the risk of [SCD](#), but largely in a nonspecific manner; they represent expressions of the hereditary predisposition to coronary heart disease. A few specific syndromes, such as congenital long QT interval syndromes ([Chap. 230](#)), right ventricular dysplasia, and the syndrome of right bundle branch block and non-ischemic ST-segment elevations (Brugada syndrome), are characterized by specific hereditary risk of SCD. There are also recent data suggesting a familial predisposition to SCD as a specific pattern of coronary heart disease expression.

The major categories of structural causes of, and functional factors contributing to, the [SCD](#) syndrome are listed in [Table 39-2](#). Worldwide, and especially in western cultures, coronary atherosclerotic heart disease is the most common structural abnormality associated with SCD. Up to 80% of all SCDs in the United States are due to the consequences of coronary atherosclerosis. The cardiomyopathies (dilated and hypertrophic, collectively; [Chap. 239](#)) account for another 10 to 15% of SCDs, and all the remaining diverse etiologies cause only 5 to 10% of these events. Transient ischemia in the previously scarred or hypertrophied heart, hemodynamic and fluid and electrolyte disturbances, fluctuations in autonomic nervous system activity, and transient electrophysiologic changes caused by drugs or other chemicals (e.g., proarrhythmia) have all been implicated as mechanisms responsible for transition from electrophysiologic stability to instability. In addition, reperfusion of ischemic myocardium may cause transient electrophysiologic instability and arrhythmias.

PATHOLOGY

Data from postmortem examinations of [SCD](#) victims parallel the clinical observations on the prevalence of coronary heart disease as the major structural etiologic factor. More than 80% of SCD victims have pathologic findings of coronary heart disease. The pathologic description often includes a combination of long-standing, extensive atherosclerosis of the epicardial coronary arteries and acute active coronary lesions,

which include a combination of fissured or ruptured plaques, platelet aggregates, hemorrhage, and thrombosis. In one study, chronic coronary atherosclerosis involving two or more major vessels with ³75% stenosis was observed in 75% of the victims. In another study, atherosclerotic plaque fissuring, platelet aggregates, and/or acute thrombosis were observed in 95 of 100 individuals who had pathologic studies after SCD.

As many as 70 to 75% of males who die suddenly have prior myocardial infarctions (MIs), but only 20 to 30% have recent acute MIs. A high incidence of left ventricular (LV) hypertrophy coexists with prior MIs.

CLINICAL DEFINITION OF FORMS OF CARDIOVASCULAR COLLAPSE ([Table 39-1](#))

Cardiovascular collapse is a general term connoting loss of effective blood flow due to acute dysfunction of the heart and/or peripheral vasculature. Cardiovascular collapse may be caused by vasodepressor syncope (vasovagal syncope, postural hypotension with syncope, neurocardiogenic syncope -- [Chap. 21](#)), a transient severe bradycardia, or cardiac arrest. The latter is distinguished from the transient forms of cardiovascular collapse in that it usually requires an intervention to achieve resuscitation. In contrast, vasodepressor syncope and many primary bradyarrhythmic syncopal events are transient and non-life-threatening, with spontaneous return of consciousness.

The most common electrical mechanism for true cardiac arrest is ventricular fibrillation (VF), which is responsible for 65 to 80% of cardiac arrests. Severe persistent bradyarrhythmias, asystole, and pulseless electrical activity (an organized electrical activity without mechanical response, formerly called electromechanical dissociation) cause another 20 to 30%. Sustained ventricular tachycardia (VT) with hypotension is a less common cause. Acute low cardiac output states, having precipitous onset, also may present clinically as a cardiac arrest. The causes include massive acute pulmonary emboli, internal blood loss from ruptured aortic aneurysm, intense anaphylaxis, cardiac rupture after myocardial infarction, and unexpected fatal arrhythmia due to electrolyte disturbances.

CLINICAL CHARACTERISTICS OF CARDIAC ARREST

PRODROME, ONSET, ARREST, DEATH

[SCD](#) may be presaged by days, weeks, or months of increasing angina, dyspnea, palpitations, easy fatigability, and other nonspecific complaints. However, these *prodromal complaints* are generally predictive of any major cardiac event; they are not specific for predicting SCD.

The *onset of the terminal event*, leading to cardiac arrest, is defined as an acute change in cardiovascular status preceding cardiac arrest by up to 1 h. When the onset is instantaneous or abrupt, the probability that the arrest is cardiac in origin is >95%. Continuous ECG recordings, fortuitously obtained at the onset of a cardiac arrest, commonly demonstrate changes in cardiac electrical activity in the minutes or hours before the event. There is a tendency for the heart rate to increase and for advanced

grades of premature ventricular contractions (PVCs) to evolve. Most cardiac arrests that are caused by [VF](#) begin with a run of sustained or nonsustained [VT](#), which then degenerates into VF.

Sudden unexpected loss of effective circulation may be separated into "arrhythmic events" and "circulatory failure." Arrhythmic events are characterized by a high likelihood of patients being awake and active immediately prior to the event, are dominated by [VF](#) as the electrical mechanism, and have a short duration of terminal illness (<1 h). In contrast, circulatory failure deaths occur in patients who are inactive or comatose, have a higher incidence of asystole than VF, have a tendency to a longer duration of terminal illness, and are dominated by noncardiac events preceding the terminal illness.

The onset of cardiac arrest may be characterized by typical symptoms of an acute cardiac event, such as prolonged angina or the pain of onset of [MI](#), acute dyspnea or orthopnea, or the sudden onset of palpitations, sustained tachycardia, or light-headedness. However, in many patients, the onset is precipitous, with minimal forewarning.

Cardiac arrest is, by definition, abrupt. Mentation may be impaired in patients with sustained [VT](#) during the onset of the terminal event. However, complete loss of consciousness is a *sine qua non* in cardiac arrest. Although rare spontaneous reversions occur, it is usual that cardiac arrest progresses to death within minutes (i.e., [SCD](#) has occurred) if active interventions are not undertaken promptly.

The probability of achieving successful resuscitation from cardiac arrest is related to the interval from onset to institution of resuscitative efforts, the setting in which the event occurs, the mechanism ([VF](#), [VT](#), pulseless electrical activity, asystole), and the clinical status of the patient prior to the cardiac arrest. Those settings in which it is possible to institute prompt cardiopulmonary resuscitation (CPR) provide a better chance of a successful outcome. However, the outcome in intensive care units and other in-hospital environments is heavily influenced by the patient's preceding clinical status. The immediate outcome is good for cardiac arrest occurring in the intensive care unit in the presence of an acute cardiac event or transient metabolic disturbance, but the outcome for patients with far-advanced chronic cardiac disease or advanced noncardiac diseases (e.g., renal failure, pneumonia, sepsis, diabetes, cancer) is not much more successful in hospital than in the out-of-hospital setting.

The success rate for initial resuscitation and survival to hospital discharge after an out-of-hospital cardiac arrest depends in part on the mechanism of the event. When the mechanism is [VT](#), the outcome is best; [VF](#) is the next most successful; and asystole and pulseless electrical activity generate dismal outcome statistics ([Fig. 39-1](#)). Advanced age also influences adversely the chances of successful resuscitation.

Progression to biologic death is a function of the mechanism of cardiac arrest and the length of the delay before interventions. [VF](#) or asystole without [CPR](#) within the first 4 to 6 min has a poor outcome, and there are few survivors among patients who had no life support activities for the first 8 min after onset. Outcome statistics are improved by lay bystander intervention (basic life support -- see below) prior to definitive interventions

(advanced life support -- defibrillation) and even more by early defibrillation. In regard to the latter, the notion that deployment of automatic external defibrillators in communities (e.g., police vehicles, large buildings, stadiums, etc.) will result in improved survival is currently being evaluated.

Death during the hospitalization after a successfully resuscitated cardiac arrest relates closely to the severity of central nervous system injury. Anoxic encephalopathy and infections subsequent to prolonged respirator dependence account for 60% of the deaths. Another 30% occur as a consequence of low cardiac output states that fail to respond to interventions. Recurrent arrhythmias are the least common cause of death, accounting for only 10% of in-hospital deaths.

In the setting of acute [MI](#), it is important to distinguish between primary and secondary cardiac arrests. *Primary* cardiac arrests refer to those that occur in the absence of hemodynamic instability, and *secondary* cardiac arrests are those that occur in patients in whom abnormal hemodynamics dominate the clinical picture before cardiac arrest. The success rate for immediate resuscitation in primary cardiac arrest during acute MI in a monitored setting should approach 100%. In contrast, as many as 70% of patients with secondary cardiac arrest succumb immediately or during the same hospitalization.

IDENTIFICATION OF PATIENTS AT RISK FOR SUDDEN CARDIAC DEATH

Primary prevention of cardiac arrest depends on the ability to identify individual patients at high risk. One must view the problem in the context of the total number of events and the population pools from which they are derived. The annual incidence of [SCD](#) among an unselected adult population is 1 to 2 per 1000 population ([Fig. 39-2A](#)), largely reflecting the prevalence of those coronary heart disease patients among whom SCD is the first clinically recognized manifestation (20 to 25% of first coronary events are SCD). The incidence (percent per year) increases progressively with the addition of identified coronary risk factors to populations free of prior coronary events. The most powerful factors are age, elevated blood pressure, [LV](#) hypertrophy, cigarette smoking, elevated serum cholesterol level, obesity, and nonspecific electrocardiographic abnormalities. These coronary risk factors are not specific for SCD but rather represent increasing risk for all coronary deaths. The proportion of coronary deaths that are sudden remains at approximately 50% in all risk categories. Despite the marked *relative* increased risk of SCD with addition of multiple risk factors (from 1 to 2 per 1000 population per year in an unselected population to as much as 50 to 60 per 1000 in subgroups having multiple risk factors for coronary artery disease), the *absolute* incidence remains relatively low when viewed as the relationship between the number of individuals who have a preventive intervention and the number of events that can be prevented. Specifically, a 50% reduction in annual SCD risk would be a huge *relative* decrease but would require an intervention in up to 200 unselected individuals to prevent one sudden death. These figures highlight the importance of primary prevention of coronary heart disease. Control of coronary risk factors may be the only practical method to prevent SCD in major segments of the population, because of the paradox that the majority of events occur in the large unselected subgroups rather than in the specific high-risk subgroups (compare "Events/Year" with "Percent/Year" in [Fig. 39-2A](#)). Under most conditions of higher level of risk, particularly those indexed to a recent major cardiovascular event (e.g., [MI](#), recent onset of heart failure, survival after out-of-hospital cardiac arrest), the highest risk of

sudden death occurs within the initial 6 to 18 months and then decreases toward baseline risk of the underlying disease ([Fig. 39-2B](#)). Accordingly, preventive interventions are most likely to be effective when initiated early.

For patients with acute or prior clinical manifestations of coronary heart disease, high-risk subgroups having a much higher ratio of [SCD](#) risk to population base can be identified. The acute, convalescent, and chronic phases of [MI](#) provide large population subsets with more highly focused risk ([Chap. 243](#)). The potential risk of cardiac arrest from the onset through the first 72 h after acute MI (the acute phase) may be as high as 15 to 20%. The highest risk of SCD in relation to MI is found in the subgroup that has experienced sustained [VT](#) or [VF](#) during the convalescent phase (3 days to 8 weeks) after MI. A greater than 50% mortality in 6 to 12 months has been observed among these patients, when managed with conservative medical therapy, and at least 50% of the deaths are sudden. Aggressive intervention techniques may reduce this incidence.

After the acute phase of [MI](#), long-term risk for total mortality and [SCD](#) are predicted by a number of factors. The most important for both SCD and non-SCD is the extent of myocardial damage sustained during the acute event. This is measured by the degree of reduction in the ejection fraction (EF), functional capacity, and/or the occurrence of heart failure. Increasing *frequency* of postinfarction [PVCs](#), with a plateau above the range of 10 to 30 PVCs per hour on 24-h ambulatory monitor recordings, also indicates increased risk, but advanced *forms* (salvos, nonsustained [VT](#)) may be more powerful predictors. PVCs interact strongly with decreased left ventricular EF. The combination of frequent PVCs, salvos or nonsustained VT, and an EF $\leq$ 35% identifies patients who have an annual risk of greater than 20%. The risk falls off sharply with decreasing PVC frequency and the absence of advanced forms, as well as with higher EF. Despite the risk implications of postinfarction PVCs, improved outcome as a result of PVC suppression has not been demonstrated ([Chap. 230](#)).

The extent of underlying disease due to any cause and/or prior clinical expression of risk of [SCD](#) (i.e., survival after out-of-hospital cardiac arrest not associated with acute [MI](#)) identify patients at very high risk for subsequent (recurrent) cardiac arrest. Survival after out-of-hospital cardiac arrest predicts up to a 30% 1-year recurrent cardiac arrest rate in the absence of specific interventions (see below).

A general rule is that the risk of [SCD](#) is approximately one-half the total cardiovascular mortality rate. As shown in [Fig. 39-2A](#), the very high risk subgroups provide more focused population fractions ("Percent/Year") for predicting cardiac arrest or SCD; but the impact on the overall population, indicated by the absolute number of preventable events ("Events/Year"), is considerably smaller. The requirements for achieving a major population impact are effective prevention of the underlying diseases and/or new epidemiologic probes that will allow better resolution of subgroups within large general populations.

TREATMENT

The individual who collapses suddenly is managed in four stages: (1) the initial response and basic life support; (2) advanced life support; (3) postresuscitation care; and (4) long-term management. The initial response and basic life support can be

carried out by physicians, nurses, paramedical personnel, and trained lay persons. There is a requirement for increasingly specialized skills as the patient moves through the stages of advanced life support, postresuscitation care, and long-term management.

Initial Response and Basic Life Support The initial response will confirm whether a sudden collapse is indeed due to a cardiac arrest. Observations for respiratory movements, skin color, and the presence or absence of pulses in the carotid or femoral arteries will promptly determine whether a life-threatening cardiac arrest has occurred. As soon as a cardiac arrest is suspected or confirmed, contacting an emergency rescue system (e.g., 911) should be the immediate priority.

Agonal respiratory movements may persist for a short time after the onset of cardiac arrest, but it is important to observe for severe stridor with a persistent pulse as a clue to aspiration of a foreign body or food. If this is suspected, a Heimlich maneuver (see below) may dislodge the obstructing body. A precordial blow, or "thump," delivered firmly by the clenched fist to the junction of the middle and lower third of the sternum may occasionally revert VT or VF, but there is concern about converting VT to VF. Therefore, it has been recommended to use precordial thumps as an advanced life support technique when monitoring and defibrillation are available. This conservative application of the technique remains controversial.

The third action during the initial response is to clear the airway. The head is tilted back and chin lifted so that the oropharynx can be explored to clear the airway. Dentures or foreign bodies are removed, and the Heimlich maneuver is performed if there is reason to suspect that a foreign body is lodged in the oropharynx. If respiratory arrest precipitating cardiac arrest is suspected, a second precordial thump is delivered after the airway is cleared.

Basic life support, more popularly known as CPR, is intended to maintain organ perfusion until definitive interventions can be instituted. The elements of CPR are the maintenance of ventilation of the lungs and compression of the chest. Mouth-to-mouth respiration may be used if no specific rescue equipment is immediately available (e.g., plastic oropharyngeal airways, esophageal obturators, masked Ambu bag). Conventional ventilation techniques during CPR require the lungs to be inflated 10 to 12 times per minute, i.e., once every fifth chest compression when two persons are performing the resuscitation and twice in succession every 15 chest compressions when one person is carrying out both ventilation and chest wall compression.

Chest compression is based on the assumption that cardiac compression allows the heart to maintain a pump function by sequential filling and emptying of its chambers, with competent valves maintaining forward direction of flow. The palm of one hand is placed over the lower sternum, with the heel of the other resting on the dorsum of the lower hand. The sternum is depressed, with the arms remaining straight, at a rate of approximately 80 to 100 per minute. Sufficient force is applied to depress the sternum 3 to 5 cm, and relaxation is abrupt.

Advanced Life Support Advanced life support is intended to achieve adequate ventilation, control cardiac arrhythmias, stabilize blood pressure and cardiac output, and restore organ perfusion. The activities carried out to achieve these goals include (1)

intubation with an endotracheal tube, (2) defibrillation/cardioversion and/or pacing, and (3) insertion of an intravenous line. Ventilation with O₂ (room air if O₂ is not immediately available) may promptly reverse hypoxemia and acidosis. The speed with which defibrillation/cardioversion is carried out is an important element for successful resuscitation. When possible, immediate defibrillation should precede intubation and insertion of an intravenous line; [CPR](#) should be carried out while the defibrillator is being charged. As soon as a diagnosis of [VT](#) or [VF](#) is obtained, a 200-J shock should be delivered. Additional shocks at higher energies, up to a maximum of 360 J, are tried if the initial shock does not successfully abolish VT or VF. Epinephrine, 1 mg intravenously, is given after failed defibrillation, and attempts to defibrillate are repeated. The dose of epinephrine may be repeated after intervals of 3 to 5 min (see [Fig. 39-3A](#)).

If the patient is less than fully conscious upon reversion, or if two or three attempts fail, prompt intubation, ventilation, and arterial blood gas analysis should be carried out. Intravenous NaHCO₃, which was formerly used in large quantities, is no longer considered routinely necessary and may be dangerous in larger quantities. However, the patient who is persistently acidotic after successful defibrillation and intubation should be given 1 meq/kg NaHCO₃ initially and an additional 50% of the dose repeated every 10 to 15 min.

After initial unsuccessful defibrillation attempts, or with persistent electrical instability, a bolus of 1 mg/kg lidocaine is given intravenously ([Chap. 243](#)), and the dose is repeated in 2 min in those patients who have persistent ventricular arrhythmias or remain in [VF](#). This is followed by a continuous infusion at a rate of 1 to 4 mg/min. If lidocaine fails to provide control, other antiarrhythmic therapies should be tried. For persistent, hemodynamically unstable ventricular arrhythmias, intravenous amiodarone has emerged as the treatment of choice (150 mg over 10 min, followed by 1 mg/min for up to 6 h, and 0.5 mg/min thereafter) ([Fig. 39-3A](#)). Intravenous procainamide (loading infusion of 100 mg/5 min to a total dose of 500 to 800 mg, followed by continuous infusion at 2 to 5 mg/min) may be tried for persisting, hemodynamically stable arrhythmias; or bretylium tosylate (loading dose 5 to 10 mg/kg in 5 min; maintenance dose 0.5 to 2 mg/min) may be tried as an alternative for unstable arrhythmias. Intravenous calcium gluconate is no longer considered safe or necessary for routine administration. It is used only in patients in whom acute hyperkalemia is known to be the triggering event for resistant VF, in the presence of known hypocalcemia, or in patients who have received toxic doses of calcium channel antagonists.

Cardiac arrest secondary to bradyarrhythmias or asystole is managed differently ([Fig. 39-3B](#)). The patient is promptly intubated, [CPR](#) is continued, and an attempt is made to control hypoxemia and acidosis. Epinephrine and/or atropine are given intravenously or by an intracardiac route. External pacing devices are now available to attempt to establish a regular rhythm, but the prognosis is generally very poor in this form of cardiac arrest, even with successful electrical pacing. Pulseless electrical activity (PEA) is treated similarly to bradyarrhythmias, but its outcome is also dismal. The one exception is bradyarrhythmic/asystolic cardiac arrest secondary to airway obstruction. This form of cardiac arrest may respond promptly to removal of foreign bodies by the Heimlich maneuver or, in hospitalized patients, by intubation and suctioning of obstructing secretions in the airway.

Postresuscitation Care This phase of management is determined by the clinical setting of the cardiac arrest. *Primary* VF in acute MI (Chap. 243) is generally very responsive to life-support techniques and easily controlled after the initial event. Patients are maintained on a lidocaine infusion at the rate of 2 to 4 mg/min for 24 to 72 h after the event. In the in-hospital setting, respirator support is usually not necessary or is needed for only a short time, and hemodynamics stabilize promptly after defibrillation or cardioversion. In *secondary* VF in acute MI (those events in which hemodynamic abnormalities predispose to the potentially fatal arrhythmia), resuscitative efforts are less often successful, and in those patients who are successfully resuscitated, the recurrence rate is high. The clinical picture and outcome are dominated by hemodynamic instability and the ability to control hemodynamic dysfunction. Bradyarrhythmias, asystole, and pulseless electrical activity are commonly secondary events in hemodynamically unstable patients.

The outcome after in-hospital cardiac arrest associated with *non-cardiac* diseases is poor, and in the few successfully resuscitated patients, the postresuscitation course is dominated by the nature of the underlying disease. Patients with cancer, renal failure, acute central nervous system disease, and uncontrolled infections, as a group, have a survival rate of less than 10% after in-hospital cardiac arrest. Some major exceptions are patients with transient airway obstruction, electrolyte disturbances, proarrhythmic effects of drugs, and severe metabolic abnormalities, most of whom may have an excellent chance of survival if they can be resuscitated promptly and maintained while the transient abnormalities are being corrected.

Long-Term Management after Survival of Out-of-Hospital Cardiac Arrest Patients who do not suffer irreversible injury of the central nervous system and who achieve hemodynamic stability should have extensive diagnostic and therapeutic testing to guide long-term management. This aggressive approach is driven by the fact that statistics from the 1970s indicated survival after out-of-hospital cardiac arrest was followed by a 30% recurrent cardiac arrest rate at 1 year, 45% at 2 years, and a total mortality rate of almost 60% at 2 years. Historical comparisons suggest that these dismal statistics may be significantly improved by newer interventions, but the magnitude of the improvement is unknown because of the lack of concurrently controlled intervention studies.

Among those patients in whom an acute transmural MI is the cause of out-of-hospital cardiac arrest, the management is the same as in any other patient who suffers cardiac arrest during the acute phase of a documented MI (Chap. 243). For almost all other categories of patients, however, extensive diagnostic studies are carried out to determine etiology, functional impairment, and electrophysiologic instability as guides to future management. In general, patients who have out-of-hospital cardiac arrest due to chronic ischemic heart disease, without an acute MI, are evaluated to determine whether transient ischemia or chronic electrophysiologic instability was the more likely cause of the event. If there is reason to suspect an ischemic mechanism, coronary revascularization by angioplasty or bypass surgery, plus drugs (most commonly beta blockers), are used to reduce ischemic burden.

Electrophysiologic instability has been identified by the use of programmed electrical stimulation to determine whether sustained VT or VF can be induced (Chap. 230). If so, this information can be used as a baseline against which to evaluate drug efficacy for

prevention of inducibility. The rationale for this approach is the assumption that suppression of inducibility predicts long-term benefit by the drug that achieves such suppression. For patients for whom successful drug therapy could not be identified by this technique, insertion of an implantable cardioverter-defibrillator (ICD), antiarrhythmic surgery (e.g., coronary bypass surgery, aneurysmectomy, cryoablation), or empiric amiodarone therapy have been recommended ([Chap. 230](#)). Primary surgical success, defined as surviving the procedure and reverting to a noninducible status without drug therapy, is better than 90% when patients are selected for ability to be mapped in the operating room. However, only a small fraction of patients meet the criteria. In addition, VT/VF *cannot* be induced in a number of survivors of cardiac arrest (30 to 50%), and inducible arrhythmias can be suppressed by drugs in no more than 20 to 30% of those whose arrhythmias can be induced. Because of these limitations of drug therapy and surgical approaches, ICD therapy has evolved into the most commonly used strategy for cardiac arrest survivors. ICDs have long been recognized to have very good success rates for sensing and reverting life-threatening arrhythmias, but improvement in long-term total survival outcomes remained lacking until a number of studies solidified the benefit of ICD therapy for specific subgroups. After empiric amiodarone therapy had been suggested to be as good as, or better than, conventional antiarrhythmic drug therapy for survivors of cardiac arrest, ICDs were demonstrated to be superior to amiodarone. Moreover, ICDs were also found to be superior for high risk patients with VT after myocardial infarction.

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SECTION 6 -ALTERATIONS IN GASTROINTESTINAL FUNCTION

40. DYSPHAGIA - Raj K. Goyal

Dysphagia is defined as a sensation of "sticking" or obstruction of the passage of food through the mouth, pharynx, or esophagus. It should be distinguished from other symptoms related to swallowing. *Aphagia* signifies complete esophageal obstruction, which is usually due to bolus impaction and represents a medical emergency. *Difficulty in initiating a swallow* occurs in disorders of the voluntary phase of swallowing. However, once initiated, swallowing is completed normally. *Odynophagia* means painful swallowing. Frequently, odynophagia and dysphagia occur together. *Globus pharyngeus* is the sensation of a lump lodged in the throat. However, no difficulty is encountered when swallowing is performed. *Misdirection of food*, resulting in nasal regurgitation and laryngeal and pulmonary aspiration of food during swallowing, is characteristic of oropharyngeal dysphagia. *Phagophobia*, meaning fear of swallowing, and *refusal to swallow* may occur in hysteria, rabies, tetanus, and pharyngeal paralysis due to fear of aspiration. Painful inflammatory lesions that cause odynophagia may also cause refusal to swallow. Some patients may feel the food as it goes down the esophagus. This esophageal sensitivity is not associated with either food sticking or obstruction, however. Similarly, the *feeling of fullness in the epigastrium* that occurs after a meal or after swallowing air should not be confused with dysphagia.

PHYSIOLOGY OF SWALLOWING

The process of swallowing begins with a voluntary (oral) phase during which a bolus of food is pushed into the pharynx by the contraction of the tongue. The bolus then activates oropharyngeal sensory receptors that initiate the involuntary (pharyngeal and esophageal) phase, or deglutition reflex. The deglutition reflex is a complex series of events and serves both to propel food through the pharynx and the esophagus and to prevent its entry into the airway. When the bolus is propelled backward by the tongue, the larynx moves forward and the upper esophageal sphincter opens. As the bolus moves into the pharynx, contraction of the superior pharyngeal constrictor against the contracted soft palate initiates a peristaltic contraction that proceeds rapidly downward to move the bolus through the pharynx and the esophagus. The lower esophageal sphincter opens as the food enters the esophagus and remains open until the peristaltic contraction has swept the bolus into the stomach. Peristaltic contraction in response to a swallow is called *primary peristalsis*. It involves inhibition followed by sequential contraction of muscles along the entire swallowing passage. The inhibition that precedes the peristaltic contraction is called *deglutitive inhibition*. Local distention of the esophagus from food activates intramural reflexes in the smooth muscle and results in *secondary peristalsis*, which is limited to the thoracic esophagus. *Tertiary contractions* are nonperistaltic because they occur simultaneously over a long segment of the esophagus. Tertiary contractions may occur in response to a swallow or esophageal distention, or they may occur spontaneously.

PATHOPHYSIOLOGY OF DYSPHAGIA

The normal transport of an ingested bolus through the swallowing passage depends on the size of the ingested bolus; the luminal diameter of the swallowing passage; the force

of peristaltic contraction; and deglutitive inhibition, including normal relaxation of upper and lower esophageal sphincters during swallowing. Dysphagia caused by a large bolus or luminal narrowing is called *mechanical dysphagia*, whereas dysphagia due to weakness of peristaltic contractions or to impaired deglutitive inhibition causing nonperistaltic contractions and impaired sphincter relaxation is called *motor dysphagia*.

Mechanical Dysphagia Mechanical dysphagia can be caused by a very large food bolus, intrinsic narrowing, or extrinsic compression of the lumen. In an adult, the esophageal lumen can distend up to 4 cm in diameter. When the esophagus cannot dilate beyond 2.5 cm in diameter, dysphagia to normal solid food can occur. Dysphagia is always present when the esophagus cannot distend beyond 1.3 cm. Circumferential lesions produce dysphagia more consistently than do lesions that involve only a portion of circumferences of the esophageal wall, as uninvolved segments retain their distensibility. The causes of mechanical dysphagia are listed in [Table 40-1](#). Common causes include carcinoma, peptic and other benign strictures, and lower esophageal ring.

Motor Dysphagia Motor dysphagia may result from difficulty in initiating a swallow or from abnormalities in peristalsis and deglutitive inhibition due to diseases of the esophageal striated or smooth muscle.

Diseases of the striated muscle involve the pharynx, upper esophageal sphincter, and cervical esophagus. The striated muscle is innervated by a somatic component of the vagus with cell bodies of the lower motor neurons located in the nucleus ambiguus. These neurons are cholinergic and excitatory and are the sole determinant of the muscle activity. Peristalsis in the striated muscle segment is due to sequential central activation of neurons innervating muscles at different levels along the esophagus. Motor dysphagia of the pharynx results from neuromuscular disorders causing muscle paralysis, simultaneous nonperistaltic contraction, or loss of opening of the upper esophageal sphincter. Loss of opening of the upper sphincter is caused by paralysis of geniohyoid and other suprahyoid muscles or loss of deglutitive inhibition of the cricopharyngeus muscle. Because each side of the pharynx is innervated by ipsilateral nerves, a unilateral lesion of motor neurons leads to unilateral pharyngeal paralysis. Although lesions of striated muscle also involve the cervical part of the esophagus, the clinical manifestations of pharyngeal dysfunction usually overshadow those due to esophageal involvement.

Diseases of the smooth-muscle segment involve the thoracic part of the esophagus and the lower esophageal sphincter. The smooth muscle is innervated by the parasympathetic component of the vagal preganglionic fibers and postganglionic neurons in the myenteric ganglia. The vagal pathway consists of parallel excitatory and inhibitory pathways that use acetylcholine and nitric oxide as neurotransmitters, respectively. The activation of inhibitory nerves causes inhibition that is followed by rebound contraction. These pathways are involved in the resting tone of the lower esophageal sphincter as well as swallow-induced lower esophageal sphincter opening and inhibition followed by peristaltic contractions in the esophageal body. Dysphagia results when the peristaltic contractions are weak or nonperistaltic or when the lower sphincter fails to relax normally. Loss of contractile power occurs due to muscle weakness, as in scleroderma. The nonperistaltic contractions and impaired relaxation of

the lower esophageal sphincter result from a defect in inhibitory vagal innervation and account for dysphagia in achalasia.

The causes of motor dysphagia are also listed in [Table 40-1](#). Important causes are pharyngeal paralysis, cricopharyngeal achalasia, scleroderma of the esophagus, achalasia, and diffuse esophageal spasm and related motor disorders.

Approach to the Patient

History The history can provide a presumptive diagnosis in over 80% of patients. The type of food causing dysphagia provides useful information. Difficulty only with solids implies mechanical dysphagia with a lumen that is not severely narrowed. In advanced obstruction, dysphagia occurs with liquids as well as solids. In contrast, motor dysphagia due to achalasia and diffuse esophageal spasm is equally affected by solids and liquids from the very onset. Patients with scleroderma have dysphagia to solids that is unrelated to posture and to liquids while recumbent but not upright. When peptic stricture develops in patients with scleroderma, dysphagia becomes more persistent.

The duration and course of dysphagia are helpful in diagnosis. Transient dysphagia may be due to an inflammatory process. Progressive dysphagia lasting a few weeks to a few months is suggestive of carcinoma of the esophagus. Episodic dysphagia to solids lasting several years indicates a benign disease characteristic of a lower esophageal ring.

The site of dysphagia described by the patient helps to determine the site of esophageal obstruction; the lesion is at or below the perceived location of dysphagia.

Associated symptoms provide important diagnostic clues. Nasal regurgitation and tracheobronchial aspiration with swallowing are hallmarks of pharyngeal paralysis or a tracheoesophageal fistula. Tracheobronchial aspiration unrelated to swallowing may be secondary to achalasia, Zenker's diverticulum, or gastroesophageal reflux.

Severe weight loss that is out of proportion to the degree of dysphagia is highly suggestive of carcinoma. When hoarseness precedes dysphagia, the primary lesion is usually in the larynx. Hoarseness following dysphagia may suggest involvement of the recurrent laryngeal nerve by extension of esophageal carcinoma. Sometimes hoarseness may be due to laryngitis secondary to gastroesophageal reflux. Association of laryngeal symptoms and dysphagia also occurs in various neuromuscular disorders. Hiccups may rarely occur with a lesion in the distal portion of the esophagus. Unilateral wheezing with dysphagia indicates a mediastinal mass involving the esophagus and a large bronchus.

Chest pain with dysphagia occurs in diffuse esophageal spasm and related motor disorders. Chest pain resembling diffuse esophageal spasms may occur in esophageal obstruction due to a large bolus. A prolonged history of heartburn and reflux preceding dysphagia indicates peptic stricture. A history of prolonged nasogastric intubation, ingestion of caustic agents, ingestion of pills without water, previous radiation therapy, or associated mucocutaneous diseases may provide the cause of esophageal stricture. If odynophagia is present, candidal or herpes esophagitis or pill-induced esophagitis

should be suspected.

In patients with AIDS or other immunodeficiency states, esophagitis due to opportunistic infections such as *Candida*, herpes simplex virus, or cytomegalovirus and tumors such as Kaposi's sarcoma and lymphoma should be suspected.

Physical Examination Physical examination is important in motor dysphagia due to skeletal muscle, neurologic, and oropharyngeal diseases. Signs of bulbar or pseudobulbar palsy, including dysarthria, dysphonia, ptosis, tongue atrophy, and hyperactive jaw jerk, in addition to evidence of generalized neuromuscular disease, should be sought. The neck should be examined for thyromegaly or a spinal abnormality. A careful inspection of the mouth and pharynx should disclose lesions that may interfere with passage of food because of pain or obstruction. Changes in the skin and extremities may suggest a diagnosis of scleroderma and other collagen-vascular diseases or mucocutaneous diseases such as pemphigoid or epidermolysis bullosa, which may involve the esophagus. Cancer spread to lymph nodes and liver may be evident. Pulmonary complications of acute aspiration pneumonia or chronic aspiration may be present.

Diagnostic Procedures Dysphagia is nearly always a symptom of organic disease rather than a functional complaint. If oropharyngeal dysphagia is suspected, videofluoroscopy of oropharyngeal swallowing should be obtained. If mechanical dysphagia is suspected on clinical history, barium swallow, esophagogastroscopy and endoscopic biopsies are the diagnostic procedures of choice. Barium swallow and esophageal motility studies are diagnostic tests for motor dysphagia. Esophagogastroscopy may be needed in patients with motor dysphagia to exclude an associated structural abnormality ([Chap. 284](#)).

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41. NAUSEA, VOMITING, AND INDIGESTION - William L. Hasler

Nausea is the subjective feeling of a need to vomit. Vomiting (emesis) is the oral expulsion of upper gastrointestinal contents resulting from contractions of gut and thoracoabdominal wall musculature. *Vomiting* is contrasted with regurgitation, the effortless passage of gastric contents into the mouth. *Rumination* is the repeated regurgitation of stomach contents, which are often rechewed and then reswallowed. In contrast to vomiting, these phenomena often exhibit some volitional control. *Indigestion* is a nonspecific term that encompasses a variety of upper abdominal complaints including nausea, vomiting, heartburn, regurgitation, and dyspepsia (upper abdominal discomfort or pain). Individuals with ulcer-like dyspepsia report epigastric burning or gnawing discomfort. Dysmotility-like dyspepsia is characterized by postprandial fullness, bloating, eructation (belching), anorexia (loss of appetite), and early satiety (an inability to complete a meal due to premature fullness).

NAUSEA AND VOMITING

MECHANISMS

Vomiting is coordinated by the brain stem and is effected by neuromuscular responses in the gut, pharynx, and thoracoabdominal wall. The mechanisms underlying nausea are poorly understood. Because nausea requires conscious perception, the sensation is probably mediated by the cerebral cortex. Electroencephalographic studies show activation of temporofrontal cortical regions with induction of nausea.

Coordination of Emesis Animal studies suggested that vomiting was coordinated by a single locus in the medullary reticular formation. However, further work has shown that no one "vomiting center" exists and that several brain stem nuclei initiate emesis, including the nucleus tractus solitarius; the dorsal vagal and phrenic nuclei; medullary nuclei that regulate respiration; and nuclei that control pharyngeal, facial, and tongue movements. The neurotransmitters involved in coordinating emesis are uncertain; however, neurokinin NK₁, serotonin, and vasopressin pathways are postulated.

Somatic and visceral muscles exhibit stereotypic responses during emesis. Inspiratory thoracic and abdominal wall muscles contract, producing high intrathoracic and intraabdominal pressures that facilitate expulsion of gastric contents. The gastric cardia herniates across the diaphragm, and the larynx moves upward to promote oral propulsion of the vomitus. Under normal conditions, distally migrating upper gut contractions are regulated by an electrical phenomenon, the slow wave, which cycles at 3 cycles per minute in the stomach and 11 cycles per minute in the duodenum. With emesis, slow waves are replaced by orally propagating spike activity, which induces retrograde contractions that assist in the oral expulsion of small intestinal contents.

Activators of Emesis Emetic stimuli act at several anatomic sites. Emesis provoked by noxious thoughts or smells originates in the cerebral cortex, whereas cranial nerves mediate vomiting after gag reflex activation. Motion sickness and inner ear disorders act on the labyrinthine apparatus, while gastric irritants and emetogenic anticancer agents such as cisplatin stimulate gastroduodenal vagal afferent nerves. Nongastric visceral afferents are activated by small intestinal and colonic obstruction and mesenteric

ischemia. The area postrema, a medullary nucleus, responds to bloodborne emetic stimuli and is termed the *chemoreceptor trigger zone*. Many emetic drugs act on the area postrema as do bacterial toxins and metabolic disorders such as uremia, hypoxia, and ketoacidosis.

Neurotransmitters that mediate induction of vomiting are selective for these anatomic sites. Labyrinthine disorders stimulate vestibular cholinergic muscarinic M₁ and histaminergic H₁ receptors, whereas gastroduodenal vagal afferent stimuli activate serotonin 5-HT₃ receptors. The area postrema is richly served by nerve fibers acting on diverse receptor subtypes including 5-HT₃, M₁, H₁, and dopamine D₂. Optimal pharmacologic management of the patient with vomiting requires an understanding of these pathways.

DIFFERENTIAL DIAGNOSIS

Nausea and vomiting are caused by conditions within and outside the gut as well as by drugs and circulating toxins ([Table 41-1](#)).

Intraperitoneal Disorders Visceral obstruction and inflammation of hollow and solid viscera may produce vomiting as the main symptom. Gastric obstruction results from ulcer disease and malignancy, whereas small bowel and colonic obstructions occur as a consequence of adhesions, benign or malignant tumors, volvulus, intussusception, or inflammatory diseases such as Crohn's disease. The superior mesenteric artery syndrome, occurring after weight loss or prolonged bed rest, results when the duodenum is compressed by the overlying superior mesenteric artery. Abdominal irradiation evokes emesis by impairing intestinal contractile function and by inducing strictures. Biliary colic causes nausea likely by action on visceral afferent nerves. Vomiting with pancreatitis, cholecystitis, and appendicitis is due to localized visceral irritation and induction of ileus. Enteric infections with viruses or bacteria such as *Staphylococcus aureus* and *Bacillus cereus* are among the most common causes of acute vomiting, especially in children. Opportunistic infections such as cytomegalovirus or herpes simplex induce emesis in immunocompromised individuals.

Disorders of gastrointestinal motor function also commonly cause nausea and vomiting. Gastroparesis is defined as a delay in emptying of food from the stomach and occurs after vagotomy for peptic ulcer, with pancreatic adenocarcinoma, or in systemic diseases such as diabetes, scleroderma, and amyloidosis. Idiopathic gastroparesis develops in the absence of systemic illness and may follow a viral prodrome suggesting an infectious etiology. Intestinal pseudoobstruction is characterized by disruption of intestinal and colonic motor activity and leads to intestinal retention of food residue and secretions, bacterial overgrowth, nutrient malabsorption, and development of nausea, vomiting, bloating, pain, and alteration of bowel pattern. Intestinal pseudoobstruction may be idiopathic, may be inherited as a familial visceral myopathy or neuropathy, or may result from systemic disease or be a paraneoplastic consequence of malignancy (especially small cell lung carcinoma).

Extraperitoneal Disorders Myocardial infarction and congestive heart failure are cardiac causes of nausea and vomiting. Nausea and vomiting occur after 25% of surgical operations, both within and outside the peritoneum. Postoperative emesis is

more common after laparotomy and orthopedic surgery than after laparoscopy and is more prevalent in women. Increased intracranial pressure from tumors, bleeding, abscess, or obstruction to cerebrospinal fluid outflow produces prominent vomiting with or without concurrent nausea. Motion sickness, labyrinthitis, and Meniere's disease evoke symptoms via labyrinthine pathways. Cyclic vomiting syndrome is a rare disorder of unknown etiology that produces episodes of intractable nausea and vomiting, usually in children. The syndrome shows a strong association with migraine headaches, suggesting that some cases may be migraine variants. Patients with psychiatric illnesses, including anorexia nervosa, bulimia nervosa, and depression, may report significant nausea. Psychogenic vomiting occurs most commonly in women with other emotional problems.

Medications and Metabolic Disorders Drugs are frequent causes of vomiting and may act on the stomach (analgesics, erythromycin) or area postrema (digoxin, opiates, anti-Parkinsonian drugs). Agents that cause emesis include antibiotics, antiarrhythmics, antihypertensives, oral hypoglycemics, and contraceptives. Cancer chemotherapy causes vomiting that is acute (within hours of administration), delayed (after 1 or more days), or anticipatory. Acute emesis resulting from highly emetogenic agents such as cisplatin is mediated by 5-HT₃ pathways, whereas delayed emesis is independent of 5-HT₃. Anticipatory nausea often responds better to anxiolytic therapy than to antiemetics.

Metabolic disorders are in the differential diagnosis in certain settings. Pregnancy is the most prevalent endocrinologic cause of nausea, occurring in 70% of women in the first trimester. Hyperemesis gravidarum is a severe form of nausea of pregnancy that can produce significant fluid loss and electrolyte disturbances. Uremia, ketoacidosis, adrenal insufficiency, as well as parathyroid and thyroid disease are other metabolic causes of emesis.

Circulating toxins evoke symptoms through effects on the area postrema. Endogenous toxins are generated in fulminant liver failure, whereas exogenous enterotoxins may be produced by enteric bacterial infection. Ethanol intoxication is a common toxic cause of nausea and vomiting.

Approach to the Patient

History and Physical Examination The history helps determine the etiology of unexplained nausea and vomiting. Drugs and toxins often cause acute symptoms, while established illnesses evoke chronic complaints. Vomiting within 1 h of eating characterizes pyloric obstruction, whereas emesis in the late postprandial period is reported with intestinal obstruction. Gastroparesis can produce nausea within minutes of food consumption but, in severe cases, leads to vomiting of meal residue ingested hours or days previously. Blood in the vomitus raises suspicion of an ulcer or malignancy; feculent emesis is noted with distal intestinal or colonic obstruction. Bilious vomiting excludes gastric obstruction, whereas emesis of undigested food is consistent with a pharyngoesophageal process such as Zenker's diverticulum or achalasia. Relief of abdominal pain by emesis characterizes small bowel obstruction, but vomiting has no effect on pancreatitis or cholecystitis pain. Pronounced weight loss raises concern about malignancy or obstruction. Fevers suggest inflammation, while an intracranial source is

considered if there are headaches or visual field changes. Vertigo or tinnitus indicate labyrinthine disease.

The physical examination complements the history. Abdominal auscultation may reveal absent bowel sounds with ileus. High-pitched rushes suggest bowel obstruction, while a succession splash on abrupt lateral movement of the patient is found with gastroparesis or pyloric obstruction. Tenderness or involuntary guarding raises suspicion of inflammation, whereas fecal blood suggests mucosal injury from ulcer, ischemia, or tumor. Neurologic etiologies present with papilledema, visual field loss, or focal neural abnormalities. Neoplasm is suggested by palpable masses or adenopathy.

The history and examination can characterize complications of emesis. Reports of lightheadedness with demonstration of orthostatic hypotension and reduced skin turgor indicate intravascular fluid loss. Hematemesis, especially with repeated vomiting, suggests a Mallory-Weiss tear of the gastroesophageal junction, while pulmonary abnormalities raise concern for aspiration of vomitus.

Diagnostic Testing With intractable symptoms or an elusive diagnosis, selected diagnostic tests can direct clinical management. Electrolyte replenishment is indicated if hypokalemia or metabolic alkalosis is found. Detection of iron-deficiency anemia mandates a search for mucosal injury. Pancreaticobiliary disease is suggested by abnormal pancreatic enzymes or liver biochemistries, whereas endocrinologic or rheumatologic etiologies are diagnosed by specific hormone or serologic testing. If luminal obstruction is considered, supine and upright abdominal radiographs may show intestinal air-fluid levels with reduced colonic air. Ileus is characterized by diffusely dilated air-filled bowel loops.

If initial testing is unrevealing, additional anatomic studies may be indicated. Upper endoscopy detects ulcer disease or gastroesophageal malignancy, and small bowel barium radiography diagnoses partial small bowel obstruction. Colonoscopy or barium enema can detect colonic obstruction. Ultrasound or computed tomography of the abdomen defines intraperitoneal inflammatory processes, while computed tomography or magnetic resonance imaging of the head can delineate intracranial sources of nausea and vomiting.

Gastrointestinal motility testing may detect a functional gastrointestinal disorder responsible for symptoms when investigation of anatomic abnormalities is negative. Gastroparesis is most commonly diagnosed with gastric scintigraphy, by which emptying of a radiolabeled meal is measured. A noninvasive means of quantitating gastric slow wave activity with cutaneous electrodes placed over the stomach, electrogastrography, has been proposed as an alternate means of diagnosing abnormal gastric emptying. With intestinal pseudoobstruction, small bowel barium radiography often suggests the diagnosis. Manometry of the small intestine may provide confirmation of the diagnosis as well as complementary information by characterizing the motor abnormality as neuropathic or myopathic based on contractile patterns. Such investigation can obviate the need for open biopsy of the intestine to evaluate for smooth muscle or neuronal degeneration.

TREATMENT

General Principles Therapy of vomiting is tailored to the underlying disease, with the medical or surgical correction of abnormalities if possible. Hospitalization is considered for severe dehydration, especially if oral fluid replenishment cannot be sustained. Once oral intake is tolerated, nutrients are restarted as liquids that are low in fat, as lipids delay gastric emptying. Foods high in indigestible residues are avoided because these also prolong gastric retention.

Antiemetic Medications Drugs that act on the central nervous system serve as antiemetic agents ([Table 41-2](#)). Antihistamines such as meclizine and dimenhydrinate and anticholinergic drugs such as scopolamine act on labyrinthine-activated pathways and are useful in the treatment of motion sickness and inner ear disorders. Phenothiazine and butyrophenone dopamine D₂antagonists are used to treat emesis evoked by area postrema stimuli and are effective for many medication, toxic, and metabolic etiologies. Dopamine antagonists freely cross the blood-brain barrier and may cause anxiety, dystonic reactions, hyperprolactinemic effects (galactorrhea and sexual dysfunction), and irreversible tardive dyskinesia.

Other drug classes have antiemetic properties. Serotonin 5-HT₃antagonists such as ondansetron and granisetron are useful in the treatment of postoperative vomiting and after radiation therapy but are mainly used to prevent cancer chemotherapy-induced emesis. The usefulness of 5-HT₃antagonists to control other causes of refractory emesis is less well established. Antidepressant drugs are established therapeutic options for patients with functional bowel disorders such as irritable bowel syndrome. Low-dose tricyclic antidepressants provide moderate symptomatic benefit in patients with unexplained nausea of a functional nature.

Gastrointestinal Motor Stimulants Drugs that stimulate gastric emptying are indicated for gastroparesis ([Table 41-2](#)). Cisapride, a serotonin 5-HT₄agonist that stimulates cholinergic nerves in the stomach, has become the preferred drug for outpatient management of gastroparesis. The drug is well tolerated but exhibits very rare drug interactions with selected antibiotics, antifungals, and other agents that predispose to fatal cardiac arrhythmias. Metoclopramide, a combined 5-HT₄agonist and D₂antagonist, is efficacious in the treatment of gastroparesis, but anti-dopaminergic side effects limit its use in 20% of patients. Erythromycin, a macrolide antibiotic, potently increases gastroduodenal motility by action on receptors for motilin, an endogenous stimulant of fasting motor activity. Erythromycin may be most useful when given intravenously to inpatients with refractory gastroparesis; however, oral forms of the drug also have some effect. Domperidone, a D₂antagonist not available in the United States, has prokinetic and antiemetic effects but does not cross into most other brain regions; thus, anxiety and dystonic reactions are rare. The main side effects of domperidone are induction of hyperprolactinemia through effects on pituitary regions served by a porous blood-brain barrier.

Patients with refractory upper gut motility disorders pose significant therapeutic challenges. Liquid suspensions of prokinetic drugs may be beneficial inasmuch as liquids empty from the stomach more rapidly than pills. Metoclopramide can be administered subcutaneously in patients who do not respond to oral drugs. Intestinal pseudoobstruction may respond to the somatostatin analogue octreotide, which induces

propagative small intestinal motor complexes. Placement of a feeding jejunostomy reduces hospitalizations and improves overall health in some patients with gastroparesis who do not respond to drug therapy. Surgical options are limited for refractory cases, but postvagotomy gastroparesis may improve with near-total resection of the stomach. Electrical pacing of the stomach may also be useful.

Selected Clinical Settings Cancer chemotherapeutic agents such as cisplatin are intensely emetogenic. Given prophylactically, 5-HT₃ antagonists prevent chemotherapy-induced acute vomiting in most cases ([Table 41-2](#)). Optimal antiemetic effects often are obtained with a 5-HT₃ antagonist in combination with a glucocorticoid. In high doses, metoclopramide is effective in controlling chemotherapy-evoked emesis, whereas benzodiazepines such as lorazepam are most useful in reducing anticipatory nausea and vomiting. In contrast, delayed emesis 1 to 5 days after chemotherapy is more refractory to treatment. Agents that act as neurokinin NK₁ antagonists in the brain stem may be potent antiemetic and antinausea drugs during both the acute and the delayed periods after chemotherapy. Cannabinoids such as tetrahydrocannabinol have been advocated for cancer-associated emesis, but these drugs produce significant side effects and are no more effective than antidopaminergic agents.

The clinician should exercise caution in the management of the patient with nausea of pregnancy. Studies of the teratogenic effects of available antiemetic agents have provided conflicting results. Few controlled trials have been performed in the nausea of pregnancy, although antihistamines such as meclizine and antidopaminergics such as prochlorperazine are more efficacious than placebo. As a consequence, alternative therapies such as pyridoxine or ginger have been recommended.

INDIGESTION

MECHANISMS

Most patients with indigestion have symptoms of a functional nature that result from gastroesophageal acid reflux or from gastric abnormalities including dysfunctional motor activity and afferent hypersensitivity; these symptoms comprise the syndrome functional dyspepsia. Some cases are a consequence of a more serious organic illness.

Gastroesophageal Acid Reflux Acid reflux results from selected physiologic defects. In scleroderma and pregnancy, lower esophageal sphincter (LES) tone is low, but most patients with acid reflux have normal LES pressures. Many individuals show frequent transient LES relaxations during which acid bathes the esophagus. The role of hiatal hernias is controversial -- although most reflux patients exhibit hiatal hernias, most individuals with hiatal hernias do not have excess heartburn.

Gastric Motor Dysfunction Disturbed gastric motility is purported to cause acid reflux in some patients with indigestion. Delayed gastric emptying also is found in 25 to 50% of individuals with functional dyspepsia. The relation of these defects to symptom induction is uncertain as many studies show poor correlation between symptom severity and the degree of motor dysfunction. Abnormal gastric fundic relaxation may cause dyspeptic symptoms such as bloating, fullness, nausea, and early satiety.

Visceral Afferent Hypersensitivity Disturbed gastric sensory function also may cause functional dyspepsia. Visceral afferent hypersensitivity was first demonstrated in patients with irritable bowel syndrome who had heightened perception of rectal balloon inflation without changes in rectal compliance. Patients with dyspepsia may experience discomfort with fundic distention to lower pressures than healthy control subjects.

Other Factors *Helicobacter pylori* has a clear etiologic role in peptic ulcer disease, but ulcers cause only a minority of cases of dyspepsia. The importance of *H. pylori* in the genesis of functional dyspepsia is controversial, but most investigators believe it is of minor importance. Analgesics cause dyspepsia; nitrates, calcium channel blockers, theophylline, and progesterone promote acid reflux. Other exogenous factors that induce acid reflux include ethanol, tobacco, and caffeine via [LES](#) relaxation. Finally, functional dyspepsia is exacerbated by stress, suggesting a pathogenic role for psychological factors.

DIFFERENTIAL DIAGNOSIS

Functional Causes Gastroesophageal reflux disease (GERD) is prevalent in Western society. Heartburn is reported once monthly by 40% of Americans and daily by 7%. Functional dyspepsia, defined as ≥ 3 months of dyspepsia without an organic cause, also is common. Nearly 25% of the populace has abdominal discomfort at least six times yearly, consistent with functional dyspepsia, but only 10 to 20% consult physicians. The clinician must distinguish these illnesses, which have a benign course, from conditions that have deleterious consequences.

Ulcer Disease In most cases of [GERD](#), the esophagus is not damaged. However, 5% of patients develop esophageal ulcers, and some form esophageal strictures. Functional dyspepsia is the cause of symptoms in 60% of individuals with dyspepsia. However, 15 to 25% of cases stem from ulcers of the stomach or duodenum. The most common causes of ulcer disease are gastric infection with *H. pylori* and use of nonsteroidal anti-inflammatory drugs. Other rare causes of gastroduodenal ulcer include Crohn's disease and Zollinger-Ellison syndrome, a condition resulting from gastrin overproduction by an endocrine tumor ([Chap. 285](#)).

Malignancy Patients with dyspepsia often seek care because of fear of cancer. However, <2% of cases result from gastroesophageal malignancy. Esophageal squamous cell carcinoma occurs most often in those patients with histories of tobacco or ethanol intake. Other risk factors include prior caustic ingestion, achalasia, and the hereditary disorder tylosis. Esophageal adenocarcinoma usually complicates long-standing acid reflux. Eight to 20% of patients with [GERD](#) exhibit glandular mucosal metaplasia of the squamous epithelium in the lower esophagus, termed *Barrett's metaplasia*. This condition predisposes to esophageal adenocarcinoma. Gastric malignancies include adenocarcinoma, which is more prevalent in certain Asian societies, and lymphoma (see [Chap. 90](#)).

Other Causes Alkaline reflux esophagitis produces [GERD](#)-like symptoms in patients who have had surgery for peptic ulcer disease. Opportunistic fungal or viral esophageal infections may produce heartburn or chest discomfort but more often cause painful swallowing. Although biliary colic is in the differential diagnosis of dyspepsia, most

patients with true biliary colic report discrete episodes of right upper quadrant or epigastric pain rather than chronic burning discomfort, nausea, and bloating. Lactose intolerance resulting from intestinal lactase deficiency produces gas, bloating, discomfort, and diarrhea. Lactase deficiency occurs in 15% of Caucasians of northern European descent but is more common in African Americans and Asians. Pancreatic disease (chronic pancreatitis and malignancy), hepatocellular carcinoma, celiac sprue, Menetrier's disease, infiltrative diseases (sarcoidosis and eosinophilic gastroenteritis), mesenteric ischemia, thyroid and parathyroid disease, and abdominal wall strain cause dyspepsia. Extraperitoneal etiologies of indigestion include congestive heart failure and tuberculosis.

Approach to the Patient

History and Physical Examination [GERD](#) classically produces heartburn, a substernal warmth beginning in the epigastrium that moves toward the neck. Heartburn often is exacerbated by meals and may awaken the patient. Associated symptoms include regurgitation of acid and water brash, the reflex release of salty salivary secretions into the mouth. Atypical symptoms include pharyngitis, asthma, cough, bronchitis, hoarseness, and chest pain that mimics angina. Some patients with acid reflux on esophageal pH testing do not report heartburn and instead note abdominal pain or other symptoms.

Individuals with ulcer-like dyspepsia have epigastric gnawing or burning that is relieved by meals or acid suppression. Dysmotility-like dyspepsia is a fullness or pain that is aggravated by eating and associated with nausea, bloating, eructation, and early satiety. There is overlap among the different dyspepsia subclasses and with other functional disorders such as irritable bowel syndrome.

The physical examination of individuals with functional causes of indigestion is usually normal. In atypical [GERD](#), pharyngeal erythema and wheezing over the lung fields may be present. Poor dentition may occur with prolonged acid regurgitation. Patients with functional dyspepsia may have epigastric tenderness or abdominal distension.

Discrimination between functional and organic causes of indigestion mandates exclusion of selected historical and examination features. Odynophagia suggests esophageal infection, while dysphagia promotes concern about a benign or malignant esophageal blockage. Other features that raise alarm include unexplained weight loss, recurrent vomiting with evidence of dehydration, occult or gross gastrointestinal bleeding, and a palpable mass or adenopathy.

Diagnostic Testing Because indigestion is prevalent in the community and because most cases result from functional illness, a general principle of diagnostic testing is to perform only limited and directed testing of selected individuals.

Once alarm factors are excluded, patients with typical [GERD](#) do not need further evaluation and are treated empirically. Upper endoscopy is indicated to exclude mucosal injury in patients with atypical symptoms, symptoms unresponsive to acid-suppressing drugs, or alarm factors. In patients with >5 years of heartburn, endoscopy is performed to screen for Barrett's metaplasia. Upper gastrointestinal

barium radiography has a slightly higher sensitivity for detecting strictures and rings than endoscopy; however, benign esophageal obstructions may be dilated with an endoscopic approach. Ambulatory esophageal pH testing is considered for atypical symptoms such as unexplained chest pain and for the symptoms that are unresponsive to appropriate medications. Esophageal manometry is most commonly ordered when surgical treatment of GERD is considered. A low [LES](#) pressure may predict failure with drug therapy and identify patients who may require surgery. The demonstration of disordered esophageal body peristalsis may affect the decision to operate or modify the type of operation chosen. Manometry with provocative testing may clarify the diagnosis in patients with atypical symptoms. Blinded perfusion of saline and then acid into the esophagus, known as the Bernstein test, can delineate whether unexplained chest discomfort results from acid reflux.

The approach to unexplained dyspepsia is dependent on the patient's age, symptom profile, and findings on examination. In individuals <45 years of age without alarm factors, blood serology for *H. pylori* is obtained to exclude the organism as a cause of ulcer disease. Upper endoscopy in this patient subset is reserved for those who fail to respond to treatment of *H. pylori*-positive or -negative dyspepsia. Upper endoscopy is performed as the initial diagnostic test in any individual with alarm factors or in patients >45 years of age because of the elevated risk of gastroesophageal malignancy with advancing age.

Further testing is indicated only if other factors are present. If there is blood loss, a blood count is obtained to exclude anemia. Thyroid chemistries or calcium levels screen for metabolic disease. With suspected pancreaticobiliary causes, blood is obtained for amylase, lipase, and liver chemistry determination. If biochemical abnormalities are found, abdominal ultrasound or computed tomography may give important information. Patients with dysmotility-like dyspepsia may selectively exhibit delayed gastric emptying; thus, gastric scintigraphy can be considered when drug treatment fails. Hydrogen breath testing after lactose ingestion may be performed for suspected lactase deficiency.

TREATMENT

General Principles In mild dyspepsia, reassurance that a careful evaluation revealed no serious organic disease may be the only intervention required. Drugs that cause acid reflux or dyspepsia should be stopped if possible. Patients with [GERD](#) should limit ethanol, caffeine, chocolate, and tobacco use because of their effects on the [LES](#). Other measures with efficacy in GERD include ingestion of a low-fat diet, avoidance of snacks before bedtime, and elevation of the head of the bed.

Specific therapy for organic diseases should be offered when possible. In disorders such as biliary colic, surgery is appropriate; whereas lactase deficiency and celiac sprue respond to special diets. Some illnesses such as peptic ulcer disease require specific medical regimens to effect cure. However, as most patients present with functional causes of indigestion, medications that reduce gastric acid, stimulate upper gut motility, or blunt gastric sensitivity are indicated.

Acid Suppressing or Neutralizing Medications Drugs that reduce or neutralize

gastric acid are the most prescribed agents for [GERD](#). Histamine H₂receptor antagonists such as cimetidine, ranitidine, famotidine, and nizatidine are useful in the treatment of mild to moderate GERD. For uncomplicated heartburn, H₂receptor antagonists are given for 4 weeks before endoscopy is considered. For severe symptoms or for many cases of erosive or ulcerative esophagitis, proton pump inhibitors such as omeprazole and lansoprazole are needed. These drugs, which inhibit gastric H⁺, K⁺-ATPase, are more potent than H₂receptor antagonists. Liquid antacids are useful for short-term control of mild GERD but are less effective for severe disease unless given at high doses that produce side effects (diarrhea with magnesium-containing agents and constipation with aluminum-containing agents). Sucralfate is a salt of aluminum hydroxide and sucrose octasulfate and buffers acid and binds pepsin and bile salts. Its efficacy in GERD and functional dyspepsia is unproven.

Acid suppressing drugs are advocated for first-line therapy of *H. pylori* negative dyspepsia, especially with ulcer-like symptoms. Ranitidine is of benefit in the treatment of functional dyspepsia versus placebo. In young patients without alarm symptoms, a 4-week trial of an H₂receptor antagonist or proton pump inhibitor is given. Endoscopy is performed only if symptoms do not improve.

***Helicobacter pylori* Eradication** Regimens to eradicate *H. pylori* are recommended for young patients with dyspepsia without alarm symptoms in whom the bacterium has been detected by serology. Several drug combinations show efficacy, but most include 10 to 14 days of a proton pump inhibitor or bismuth subsalicylate in concert with two antibiotics. If symptoms resolve, no further intervention is required. Most patients who respond to this "treatment-first" approach have underlying ulcer disease. The usefulness of *Helicobacter* eradication in patients with functional dyspepsia is unproven, but evidence suggests that <15% of cases relate to *H. pylori*. No evidence demonstrates that *H. pylori* eradication is useful in the treatment of [GERD](#).

Gastrointestinal Motor Stimulants Cisapride is superior to placebo in treating [GERD](#) and can be prescribed as sole therapy or as an adjunct to an acid-suppressing drug. Other motor stimulants such as metoclopramide, erythromycin, and domperidone are of limited use in the treatment of GERD.

Prokinetic agents are frequently used for treatment of functional dyspepsia. Cisapride and domperidone relieve symptoms more effectively than placebo. In general, these drugs are more potent than acid-reducing agents in the treatment of functional dyspepsia and may be given instead of acid suppressants as the initial empirical treatment of young patients with dyspepsia without alarm symptoms who are not infected with *H. pylori*. Patients with dysmotility-like dyspepsia may respond preferentially to motor-stimulating drugs.

Other Options In patients with [GERD](#) who do not respond to drug therapy, antireflux surgery may be offered. Operations include the Nissen fundoplication, in which the proximal stomach is wrapped completely around the [LES](#) to increase LES pressure, and the Belsey procedure, in which the wrap encircles 270° of the circumference of the LES. The latter is selected if esophageal peristalsis is suboptimal when a 360° wrap might cause dysphagia. Funduplications can be performed laparoscopically, thereby reducing the morbidity and the postoperative recuperation period.

Some patients with functional dyspepsia do not respond to acid suppressants or prokinetic drugs but may respond to low-dose tricyclic antidepressant therapy. The mechanism of action of these agents in functional dyspepsia is unknown but may involve blunting of visceral pain processing in the brain. Gas and bloating may be the most troubling symptoms in some patients with indigestion and can be difficult to treat. Successes with dietary exclusion of gas-producing foods such as legumes and use of the surface-active compound simethicone or the gas-absorptive agent activated charcoal have been reported. Psychological treatments have been proposed for functional dyspepsia; however, convincing data on their efficacy are lacking.

(Bibliography omitted in Palm version)

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42. DIARRHEA AND CONSTIPATION - David A. Ahlquist, Michael Camilleri

Diarrhea and constipation are exceedingly common and together exact an enormous toll in terms of morbidity, loss of work productivity, and consumption of medical resources. Worldwide, more than one billion people suffer one or more episodes of acute diarrhea each year. Among the 100 million persons affected annually by acute diarrhea in the United States, nearly half must restrict activities, 10% consult physicians, 250,000 require hospitalization, and roughly 3000 die (primarily the elderly). The economic burden to society is estimated at more than \$20 billion. Because of poor sanitation and more limited access to health care, acute infectious diarrhea remains one of the most common causes of mortality in developing countries, particularly among children, accounting for 5 to 8 million deaths per year. Population statistics on chronic diarrhea and constipation are more uncertain, perhaps due to variable definitions and reporting, but the frequency of these conditions is also high. Based on United States population surveys, prevalence rates for chronic diarrhea range from 2 to 7% and for chronic constipation from 3 to 17%. Diarrhea and constipation are among the most common patient complaints faced by internists and primary care physicians, and they account for nearly 50% of referrals to gastroenterologists.

Although diarrhea and constipation may present as mere nuisance symptoms at one extreme, they can be severe or life-threatening at the other. Even mild symptoms may signal a serious underlying gastrointestinal lesion, like colorectal cancer, or systemic disorder, like thyroid disease. Given the heterogeneous causes and potential severity of these common complaints, it is imperative for clinicians to appreciate the pathophysiology, etiologic classification, diagnostic strategies, and therapeutic principles of diarrhea and constipation so that rational and cost-effective care can be delivered.

NORMAL PHYSIOLOGY

The human small intestine and colon perform important functions including the secretion and absorption of water and electrolytes, the storage and subsequent transport of intraluminal contents aborally, and the salvage of some nutrients after bacterial metabolism of carbohydrate that are not absorbed in the small intestine. The main motor functions are summarized in [Table 42-1](#). Alterations in fluid and electrolyte handling contribute significantly to diarrhea. Alterations in motor and sensory functions of the human colon result in highly prevalent syndromes such as irritable bowel syndrome, chronic diarrhea, and chronic constipation.

NEURAL CONTROL

The small intestine and colon have intrinsic and extrinsic innervation. The *intrinsic innervation* also called the enteric nervous system, comprises myenteric, submucosal, and mucosal neuronal layers. The function of these layers is modulated by interneurons through the actions of neurotransmitter amines or peptides, including acetylcholine, opioids, norepinephrine, serotonin, ATP, and nitric oxide. The myenteric plexus regulates smooth muscle function, and the submucosal plexus affects secretion and absorption.

The *extrinsic innervations* of the small intestine and colon are part of the autonomic

nervous system and also modulate both motor and secretory functions. The parasympathetic nerve supply conveys both visceral sensory as well as excitatory pathways to the motor components of the colon. Parasympathetic fibers via the vagus nerve reach the small intestine and proximal colon along the branches of the superior mesenteric artery. The distal colon is supplied by sacral parasympathetic nerves (S₂₋₄) via the pelvic plexus; these fibers course through the wall of the colon as ascending intracolonic fibers as far as, and in some instances including, the proximal colon. The chief excitatory neurotransmitters controlling motor function are acetylcholine and the tachykinins, such as substance P. The sympathetic nerve supply modulates motor functions and reaches the small intestine and colon alongside the arterial arcades of the superior and inferior mesenteric vessels. Sympathetic input to the gut is generally excitatory to sphincters and inhibitory to nonsphincteric muscle. Visceral afferents convey sensation from the gut to the central nervous system; initially, they course along sympathetic fibers, but as they approach the spinal cord they separate, have cell bodies in the dorsal root ganglion, and enter the dorsal horn of the spinal cord. Afferent signals are conveyed to the brain along the lateral spinothalamic tract and the nociceptive dorsal column pathway and are then perceived. Other afferent fibers synapse in the prevertebral ganglia and reflexly modulate intestinal motility.

INTESTINAL FLUID ABSORPTION AND SECRETION

On an average day, 9 L of fluid enters the gastrointestinal tract; approximately 1 L of residual fluid reaches the colon; the stool excretion of fluid constitutes about 0.2 L/d. The colon has a large capacitance and functional reserve and may recover up to four times its usual volume of 0.8 L/d, provided the rate of flow permits reabsorption to occur. Thus, the colon can partially compensate for intestinal absorptive or secretory disorders.

In the colon, sodium absorption is predominantly electrogenic, and uptake takes place at the apical membrane; it is also compensated by the pumping out functions of the basolateral sodium pump. A variety of neural and non-neural mediators regulate colonic fluid and electrolyte balance, including cholinergic, adrenergic and serotonergic mediators. Angiotensin and aldosterone also influence colonic absorption, reflecting the common embryologic development of the distal colonic epithelium and the renal tubules.

ILEOCOLONIC STORAGE AND SALVAGE

The distal ileum acts as a reservoir, emptying intermittently by bolus movements. This action allows time for salvage of fluids, electrolytes, and nutrients. Segmentation by haustra compartmentalizes the colon and facilitates mixing, retention of residue, and formation of solid stools. In health, the ascending and transverse regions of colon function as reservoirs (average transit, 15 h), and the descending colon acts as a conduit (average transit, 3 h). The colon is efficient at conserving sodium and water, a function that is particularly important in sodium-depleted patients in whom the small intestine alone is unable to maintain sodium balance. Diarrhea or constipation may result from alteration in the reservoir function of the proximal colon, or the propulsive function of the left colon. Constipation may also result from disturbances of the rectal or sigmoid reservoir, typically as a result of dysfunction of the pelvic floor or the coordination of defecation.

SMALL INTESTINAL MOTILITY

During fasting, the motility of the small intestine is characterized by a cyclical event called the migrating motor complex (MMC), which serves to clear nondigestible residue from the small intestine. This organized, propagated series of contractions lasts on average 4 min, occurs every 60 to 90 min, and usually involves the entire small intestine. After food ingestion, the small intestine produces irregular, mixing contractions of relatively low amplitude, except in the distal ileum where more powerful contractions occur intermittently and empty the ileum by bolus transfers.

COLONIC MOTILITY AND TONE

The small intestinal [MMC](#) only rarely continues into the colon. However, short duration or phasic contractions mix colonic contents, and high amplitude propagated contractions (HAPCs) are sometimes associated with mass movements through the colon and occur approximately five times per day, usually on awakening in the morning and postprandially. Increased frequency of HAPCs may result in diarrhea. The predominant phasic contractions are irregular and nonpropagated and serve as a "mixing" function.

Colonic tone refers to the background contractility upon which phasic contractile activity (typically contractions lasting less than 15 s) is superimposed. It is an important cofactor in the colon's capacitance (volume accommodation) and sensation.

COLONIC MOTILITY AFTER MEAL INGESTION

After meal ingestion, colonic phasic and tonic contractility increase for a period of approximately 2 h. The initial phase (about 10 min) is mediated by the vagus nerve in response to mechanical distention of the stomach. The subsequent response of the colon requires caloric stimulation and is at least in part mediated by hormones, e.g., gastrin and serotonin.

DEFECATION

Tonic contraction of the puborectalis muscle, which forms a sling around the rectoanal junction, is important to maintain continence; during defecation, sacral parasympathetic nerves relax this muscle, facilitating the straightening of the rectoanal angle ([Fig. 42-1](#)). Distention of the rectum results in transient relaxation of the internal anal sphincter via intrinsic and reflex sympathetic innervation. As sigmoid and rectal contractions increase the pressure within the rectum, the rectosigmoid angle opens by more than 15°. Voluntary relaxation of the external anal sphincter (striated muscle innervated by the pudendal nerve) permits the evacuation of feces; this evacuation process can be augmented by an increase in intraabdominal pressure created by the Valsalva maneuver.

DIARRHEA

DEFINITION

Diarrhea is loosely defined as passage of abnormally liquid or unformed stools at an

increased frequency. For adults on a typical Western diet, stool weight exceeding 200 g/d can generally be considered diarrheal. Because of the fundamental importance of duration to diagnostic considerations, diarrhea may be further defined as *acute* if <2 weeks, *persistent* if 2 to 4 weeks, and *chronic* if >4 weeks in duration.

Two common conditions, usually associated with the passage of stool totaling <200 g/d, must be distinguished from diarrhea, as diagnostic and therapeutic algorithms differ. *Pseudodiarrhea*, or the frequent passage of small volumes of stool, often is associated with rectal urgency and accompanies the irritable bowel syndrome or anorectal disorders like proctitis. *Fecal incontinence* is the involuntary discharge of rectal contents and is most often caused by neuromuscular disorders or structural anorectal problems. Diarrhea and urgency, especially if severe, may aggravate or cause incontinence. Pseudodiarrhea and fecal incontinence occur at prevalence rates comparable to or higher than that of chronic diarrhea and should always be considered in patients complaining of "diarrhea." A careful history and physical examination generally allow these conditions to be discriminated from true diarrhea.

ACUTE DIARRHEA

More than 90% of cases of acute diarrhea are caused by infectious agents; these cases are often accompanied by vomiting, fever, and abdominal pain. The remaining 10% or so are caused by medications, toxic ingestions, ischemia, and other conditions.

Infectious Agents Most infectious diarrheas are acquired by fecal-oral transmission via direct personal contact or, more commonly, via ingestion of food or water contaminated with pathogens from human or animal feces. In the immunologically competent person, the resident fecal microflora, containing more than 500 taxonomically distinct species, are rarely the source of diarrhea and may actually play a role in suppressing the growth of ingested pathogens. Acute infection or injury occurs when the ingested agent overwhelms the host's mucosal immune and nonimmune (gastric acid, digestive enzymes, mucus secretion, peristalsis, and suppressive resident flora) defenses. Established clinical associations with specific enteropathogens may offer diagnostic clues.

In the United States, high risk groups are recognized:

1. *Travelers*. Nearly 40% of tourists to endemic regions of Latin America, Africa, and Asia develop so-called traveler's diarrhea, most commonly due to enterotoxigenic *Escherichia coli* as well as to *Campylobacter*, *Shigella*, and *Salmonella*. Visitors to Russia (especially St. Petersburg) may have increased risk of *Giardia*-associated diarrhea; visitors to Nepal may acquire *Cyclospora*. Campers, backpackers, and swimmers in wilderness areas may become infected with *Giardia*.
2. *Consumers of certain foods*. Diarrhea closely following food consumption at a picnic, banquet, or restaurant may suggest infection with *Salmonella*, *Campylobacter*, or *Shigella* from chicken; enterohemorrhagic *E. coli* (O157:H7) from undercooked hamburger; *Bacillus aureus* from fried rice; *Staphylococcus aureus* or *Salmonella* from mayonnaise or creams; *Salmonella* from eggs; and *Vibrio* species, *Salmonella*, or acute hepatitis A or B from seafood, especially if raw.

3. *Immunodeficient persons*. Individuals at risk for diarrhea include those with either primary immunodeficiency (e.g., IgA deficiency, common variable hypogammaglobulinemia, chronic granulomatous disease) or the much more common secondary immunodeficiency states (e.g., AIDS, senescence, pharmacologic suppression). Common enteropathogens often cause a more severe and protracted diarrheal illness; and, particularly in persons with AIDS, opportunistic infections, such as by *Mycobacterium* species, certain viruses (cytomegalovirus, adenovirus, and herpes simplex), and protozoa (*Cryptosporidium*, *Isospora belli*, Microsporidia, and *Blastocystis hominis*) may also play a role ([Chap. 309](#)). In patients with AIDS, agents transmitted venereally per rectum (e.g., *Neisseria gonorrhoeae*, *Treponema pallidum*, *Chlamydia*) may contribute to proctocolitis.

4. *Daycare participants and their family members*. Infections with *Shigella*, *Giardia*, *Cryptosporidium*, rotavirus, and other agents are very common and should be considered.

5. *Institutionalized persons*. Infectious diarrhea is one of the most frequent categories of nosocomial infections in many hospitals and long-term care facilities; the causes are a variety of microorganisms but most commonly *Clostridium difficile*.

The pathophysiology underlying acute diarrhea by infectious agents produces specific clinical features that may also be helpful in diagnosis ([Table 42-2](#)). Profuse watery diarrhea secondary to small bowel hypersecretion occurs with ingestion of preformed bacterial toxins, enterotoxin-producing bacteria, and enteroadherent pathogens. Diarrhea associated with marked vomiting and minimal or no fever may occur abruptly within a few hours after ingestion of the former two types; vomiting is usually less, and abdominal cramping or bloating is greater; fever is higher with the latter. Cytotoxin-producing and invasive microorganisms all cause high fever and abdominal pain. Invasive bacteria and *Entamoeba histolytica* often cause bloody diarrhea (referred to as *dysentery*). *Yersinia* invades the terminal ileal and proximal colon mucosa and may cause especially severe abdominal pain with tenderness mimicking acute appendicitis.

Finally, infectious diarrhea may be associated with systemic manifestations. Reiter's syndrome (arthritis, urethritis, and conjunctivitis) may accompany or follow infections by *Salmonella*, *Campylobacter*, *Shigella*, and *Yersinia*. Yersiniosis may also lead to an autoimmune-type thyroiditis, pericarditis, and glomerulonephritis. Both enterohemorrhagic *E. coli* (O157:H7) and *Shigella* can lead to the *hemolytic-uremic syndrome* with an attendant high mortality rate. Acute diarrhea can also be a major symptom of several systemic infections including *viral hepatitis*, *listeriosis*, *legionellosis*, and *toxic shock syndrome*.

Other Causes Side effects from medications are probably the most common noninfectious cause of acute diarrhea, and etiology may be suggested by a temporal association between use and symptom onset. Although innumerable medications may produce diarrhea, some of the more frequently incriminated include antibiotics, cardiac antidysrhythmics, antihypertensives, nonsteroidal anti-inflammatory drugs, certain antidepressants, chemotherapeutic agents, bronchodilators, antacids, and laxatives.

Occlusive or nonocclusive *ischemic colitis* typically occurs in persons older than 50 years of age, often presents as acute lower abdominal pain preceding watery, then bloody diarrhea, and generally results in acute inflammatory changes in the sigmoid or left colon while sparing the rectum. Acute diarrhea may accompany colonic *diverticulitis* and *graft-versus-host disease*. Acute diarrhea, often associated with systemic compromise, can follow ingestion of toxins including organophosphate insecticides, amanita and other mushrooms, arsenic, and preformed environmental toxins in seafoods, like ciguatera and scombroid. The conditions causing chronic diarrhea can also be confused with acute diarrhea early in their course. This confusion may occur with inflammatory bowel disease and some of the other inflammatory chronic diarrheas that may have an abrupt rather than insidious onset and exhibit features that mimic infection.

Approach to the Patient

The decision to evaluate acute diarrhea depends on its severity and duration and on various host factors ([Fig. 42-2](#)). Most episodes of acute diarrhea are mild and self-limited, and they do not justify the cost and potential morbidity of diagnostic or pharmacologic interventions. Indications for evaluation include profuse diarrhea with dehydration, grossly bloody stools, fever $\geq 38.5^{\circ}\text{C}$, duration $>48\text{ h}$ without improvement, new community outbreaks, associated severe abdominal pain in patients older than 50 years of age, and elderly (≥ 70 years) or immunocompromised patients. In some patients with moderately severe febrile diarrhea with fecal leukocytes (or increased fecal levels of the leukocyte proteins lactoferrin or calprotectin) present or with dysentery, a diagnostic evaluation might be eschewed in favor of an empiric antibiotic trial (see below).

The cornerstone of diagnosis in those suspected of severe acute infectious diarrhea is microbiologic analysis of the stool. Workup includes cultures for bacterial and viral pathogens, direct inspection for ova and parasites, and immunoassays for certain bacterial toxins (*C. difficile*), viral antigens (rotavirus), and protozoal antigens (*Giardia*, *E. histolytica*). The aforementioned clinical and epidemiologic associations may assist in focusing the evaluation. If a particular pathogen or set of possible pathogens is so implicated, then either the whole panel of routine studies may not be necessary or, in some instances, special cultures may be appropriate as for enterohemorrhagic and other types of *E. coli*, *Vibrio* species, and *Yersinia*. Molecular diagnosis of pathogens in stool can be made by identification of unique DNA sequences; and evolving microarray technologies could lead to a more rapid, sensitive, specific, and cost-effective diagnostic approach in the future.

Persistent diarrhea is commonly due to *Giardia*, but additional causative organisms that should be considered include *C. difficile* (especially if antibiotics had been administered), *E. histolytica*, *Cryptosporidium*, *Campylobacter*, and others. If stool studies are unrevealing, then flexible sigmoidoscopy with biopsies and upper endoscopy with duodenal aspirates and biopsies may be indicated.

Structural examination by sigmoidoscopy, colonoscopy, or abdominal CT scanning (or other imaging approaches) may be appropriate in patients with uncharacterized persistent diarrhea to exclude inflammatory bowel disease, or as an initial approach in

patients with suspected noninfectious acute diarrhea such as might be caused by ischemic colitis, diverticulitis, or partial bowel obstruction.

TREATMENT

Fluid and electrolyte replacement are of central importance to all forms of acute diarrhea. Fluid replacement alone may suffice for mild cases. Oral sugar-electrolyte solutions (sport drinks or designed formulations) should be instituted promptly with severe diarrhea to limit dehydration, which is the major cause of death. Profoundly dehydrated patients, especially infants and the elderly, require intravenous rehydration.

In moderately severe nonfebrile and nonbloody diarrhea, antimotility antisecretory agents like loperamide can be useful adjuncts to control symptoms. Such agents should be avoided with febrile dysentery, which may be exacerbated or prolonged by them. Bismuth subsalicylate may reduce symptoms of vomiting and diarrhea but should not be used to treat immunocompromised patients because of the risk of bismuth encephalopathy.

Judicious use of antibiotics is appropriate in selected instances of acute diarrhea and may reduce its severity and duration ([Fig. 42-2](#)). Many physicians treat moderately to severely ill patients with febrile dysentery empirically without diagnostic evaluation using a quinolone, such as ciprofloxacin (500 mg bid for 3 to 5 d). Empiric treatment can also be considered for suspected giardiasis with metronidazole (250 mg qid for 7d). Selection of antibiotics and dosage regimens is otherwise dictated by specific pathogens and conditions found ([Chaps. 131,153,156-162](#)). Antibiotic coverage is indicated whether or not a causative organism is discovered in patients that are immunocompromised, have mechanical heart valves or recent vascular grafts, or are elderly. Antibiotic prophylaxis is indicated for certain patients traveling to high-risk countries in whom the likelihood or seriousness of acquired diarrhea would be especially high, including those with immunocompromise, inflammatory bowel disease, or gastric achlorhydria. Use of trimethoprim/sulfamethoxazole or ciprofloxacin may reduce bacterial diarrhea in such travelers by 90%.

CHRONIC DIARRHEA

Diarrhea lasting more than 4 weeks warrants evaluation to exclude serious underlying pathology. In contrast to acute diarrhea, most of the many causes of chronic diarrhea are noninfectious. The classification of chronic diarrhea by pathophysiologic mechanism facilitates a rational approach to management ([Table 42-3](#)).

Secretory Causes Secretory diarrheas are due to derangements in fluid and electrolyte transport across the enterocolic mucosa. They are characterized clinically by watery, large-volume fecal outputs that are typically painless and persist with fasting. Because there is no malabsorbed solute, stool osmolality is accounted for by normal endogenous electrolytes with no fecal osmotic gap.

Medications Side effects from regular ingestion of drugs and toxins are the most common secretory causes of chronic diarrhea. Hundreds of prescription and over-the-counter medications (see "Other Causes of Acute Diarrhea," above) may

produce unwanted diarrhea. Surreptitious or habitual use of stimulant laxatives [e.g., senna, cascara, bisacodyl, ricinoleic acid (castor oil)] must also be considered. Chronic ethanol consumption may cause a secretory-type diarrhea due to enterocyte injury with impaired sodium and water absorption as well as to rapid transit and other alterations. Inadvertent ingestion of certain environmental toxins (e.g., arsenic) may lead to chronic rather than acute forms of diarrhea. Certain bacterial infections may occasionally persist and be associated with a secretory-type diarrhea.

Bowel resection, mucosal disease, or enterocolic fistula These conditions may result in a secretory-type diarrhea because of inadequate surface for resorption of secreted fluids and electrolytes. Unlike other secretory diarrheas, this subset of conditions tends to worsen with eating. With disease (e.g., Crohn's ileitis) or resection of <100 cm of terminal ileum, dihydroxy bile acids may escape absorption and stimulate colonic secretion (cholorrheic diarrhea). This mechanism may contribute to so-called *idiopathic secretory diarrhea*, in which bile acids are functionally malabsorbed from a normal-appearing terminal ileum. Partial bowel obstruction, ostomy stricture, or fecal impaction may paradoxically lead to increased fecal output due to hypersecretion.

Hormones Although uncommon, the classic examples of secretory diarrhea are those mediated by hormones. *Metastatic gastrointestinal carcinoid tumors* or, rarely, *primary bronchial carcinoids* may produce watery diarrhea alone or as part of the carcinoid syndrome that comprises episodic flushing, wheezing, dyspnea, and right-sided valvular heart disease. Diarrhea is due to the release into the circulation of potent intestinal secretagogues including serotonin, histamine, prostaglandins, and various kinins. Pellagra-like skin lesions may rarely occur as the result of serotonin overproduction with niacin depletion. *Gastrinoma*, one of the most common neuroendocrine tumors, most typically presents with refractory peptic ulcers, but diarrhea occurs in up to one-third of cases and may be the only clinical manifestation in 10%. While various secretagogues released with gastrin may play a role, the diarrhea most often results from fat maldigestion owing to pancreatic enzyme inactivation by low intraduodenal pH. The watery diarrhea hypokalemia achlorhydria (WDHA) syndrome, also called *pancreatic cholera*, is due to a non-b cell pancreatic adenoma, referred to as a VIPoma, that secretes vasoactive intestinal peptide (VIP) and a host of other peptide hormones including pancreatic polypeptide, secretin, gastrin, gastrin-inhibitory polypeptide, neurotensin, calcitonin, and prostaglandins. The secretory diarrhea is often massive with stool volumes >3 L/d; daily volumes as high as 20 L have been reported. Life-threatening dehydration, neuromuscular dysfunction from associated hypokalemia, hypomagnesemia, or hypercalcemia, flushing, and hyperglycemia may accompany vipoma. *Medullary carcinoma of the thyroid* may present with watery diarrhea caused by calcitonin, other secretory peptides, or prostaglandins. This tumor occurs sporadically or, in 25 to 50% of cases, as a feature of multiple endocrine neoplasia type IIa with pheochromocytomas and hyperparathyroidism. Prominent diarrhea is often associated with metastatic disease and poor prognosis. *Systemic mastocytosis*, which may be associated with the skin lesion urticaria pigmentosa, may cause diarrhea that is either secretory and mediated by histamine, or inflammatory and due to intestinal filtration by mast cells. Large *colorectal villous adenomas* may rarely be associated with a secretory diarrhea that may cause hypokalemia, can be inhibited by NSAIDs, and is apparently mediated by prostaglandins.

Congenital defects in ion absorption Rarely, these defects cause watery diarrhea from birth and include defective $\text{Cl}^-/\text{HCO}_3^-$ -exchange (*congenital chloridorrhea*) with alkalosis and defective Na^+/H^+ -exchange with acidosis. Some hormone deficiencies may be associated with watery diarrhea, such as occurs with adrenocortical insufficiency (Addison's disease) that may be accompanied by hyperpigmentation.

Osmotic Causes Osmotic diarrhea occurs when ingested, poorly absorbable, osmotically active solutes draw enough fluid lumenward to exceed the resorptive capacity of the colon. Fecal water output increases in proportion to such a solute load. Osmotic diarrhea characteristically ceases with fasting or with discontinued oral intake of the offending agent.

Osmotic laxatives Ingestion of magnesium-containing antacids, health supplements, or laxatives may induce osmotic diarrhea typified by a stool osmotic gap: $2([\text{Na}] + [\text{K}]) \ll 290 \text{ mosm/kg}$. Anionic laxatives containing sulfates or phosphates produce osmotic diarrhea without an osmotic gap, as sodium accompanies the anionic solutes; direct measurement of stool sulfates and phosphates may be necessary to confirm the cause of diarrhea.

Carbohydrate malabsorption Carbohydrate malabsorption due to acquired or congenital defects in brush-border disaccharidases and other enzymes leads to osmotic diarrhea with a low pH. One of the most common causes of chronic diarrhea in adults is *lactase deficiency*, which affects three-fourths of non-Caucasians worldwide and 5 to 30% of persons in the United States; most learn to avoid milk products without an intervention. Some sugars, such as sorbitol, are universally malabsorbed, and diarrhea ensues with ingestion of ample medications, gum, or candies sweetened with these nonabsorbable sugars. Lactulose, used to acidify stools in patients with hepatic failure, also causes diarrhea on this basis.

Steatorrheal Causes Fat malabsorption may lead to greasy, foul-smelling, difficult-to-flush diarrhea often associated with weight loss and nutritional deficiencies due to concomitant malabsorption of amino acids and vitamins. Increased fecal output is caused by the osmotic effects of fatty acids, especially after bacterial hydroxylation, and, to a lesser extent, by the burden of neutral fat. Quantitatively, steatorrhea is defined as stool fat exceeding the normal 7 g/d; daily fecal fat averages 15 to 25 g with small intestinal diseases and often exceeds 40 g with pancreatic exocrine insufficiency. Intraluminal maldigestion, mucosal malabsorption, or lymphatic obstruction may produce steatorrhea.

Intraluminal maldigestion This condition most commonly results from pancreatic exocrine insufficiency, which occurs when >90% of pancreatic secretory function is lost. *Chronic pancreatitis*, usually a sequela of ethanol abuse, most frequently causes pancreatic insufficiency. Other causes include *cystic fibrosis*, *pancreatic duct obstruction*, and rarely, *somatostatinoma*. Bacterial overgrowth in the small intestine may deconjugate bile acids and alter micelle formation that impair fat digestion; it occurs with stasis from a blind-loop, small bowel diverticulum, or dysmotility and is especially likely in the elderly. Finally, cirrhosis or biliary obstruction may lead to mild steatorrhea due to deficient intraluminal bile acid concentration.

Mucosal Malabsorption Mucosal malabsorption occurs from a variety of enteropathies, but most prototypically and perhaps most commonly from *celiac sprue*. This gluten-sensitive enteropathy characterized by villous atrophy and crypt hyperplasia in the proximal small bowel often presents with fatty diarrhea associated with multiple nutritional deficiencies of varying severity and affects all ages. *Tropical sprue* may produce a similar histologic and clinical syndrome, but it occurs in residents of or travelers to tropical climates; its often abrupt onset and response to antibiotics suggest an infectious etiology. *Whipple's disease*, due to the actinomycete *Treponema whipplei* and histiocytic infiltration of the small bowel mucosa, is a less common cause of steatorrhea that most typically occurs in young or middle-aged men; it is frequently associated with arthralgias, fever, lymphadenopathy, and extreme fatigue and may affect the central nervous system and endocardium. A similar clinical and histologic picture results from *Mycobacterium avium intracellulare* infection in patients with AIDS. *Abetalipoproteinemia* is a rare defect of chylomicron formation and fat malabsorption in children associated with acanthocytic erythrocytes, ataxia, and retinitis pigmentosa. Several other conditions may cause mucosal malabsorption including infections, especially with protozoa like *Giardia*, numerous medications (e.g., colchicine, cholestyramine, neomycin), and chronic ischemia.

Postmucosal lymphatic obstruction The pathophysiology of this condition, which is due to the rare *congenital intestinal lymphangiectasia* or to *acquired lymphatic obstruction* secondary to trauma, tumor, or infection, leads to the unique constellation of fat malabsorption with enteric losses of protein (often causing edema) and lymphocytes (with resultant lymphocytopenia) that enter the portal circulation directly. Carbohydrate and amino acid absorption are preserved.

Inflammatory Causes Inflammatory diarrheas are generally accompanied by pain, fever, bleeding, or other manifestations of inflammation. The mechanism of diarrhea may not only be exudation but, depending on lesion site, may include fat malabsorption, disrupted fluid/electrolyte absorption, and hypersecretion or hypermotility from release of cytokines and other inflammatory mediators. The unifying feature on stool analysis is the presence of leukocytes or leukocyte-derived proteins such as calprotectin. With severe inflammation, exudative protein loss can lead to anasarca (generalized edema). Any middle-aged or older person with chronic inflammatory-type diarrhea, especially with blood, should be carefully evaluated to exclude a colorectal or large enteric tumor.

Idiopathic inflammatory bowel disease The illnesses in this category, which include *Crohn's disease* and *chronic ulcerative colitis*, are among the most common organic causes of chronic diarrhea in adults and range in severity from mild to fulminant and life threatening. They may be associated with uveitis, polyarthralgias, cholestatic liver disease (primary sclerosing cholangitis), and various skin lesions (erythema nodosum, pyoderma gangrenosum). *Microscopic colitis*, including *collagenous colitis*, is an increasingly recognized cause of chronic watery diarrhea; biopsy of a normal appearing colorectum is required for histologic diagnosis.

Primary or secondary forms of immunodeficiency Immunodeficiency may lead to prolonged infectious diarrhea. With common, variable *hypogammaglobulinemia*, diarrhea is particularly prevalent and often the result of giardiasis.

Eosinophilic gastroenteritis Eosinophil infiltration of the mucosa, muscularis, or serosa at any level of the gastrointestinal tract may cause diarrhea, pain, vomiting, or ascites. Affected patients often have an atopic history, Charcot-Leyden crystals due to extruded eosinophil contents may be seen on microscopic inspection of stool, and peripheral eosinophilia is present in 50 to 75% of patients. While hypersensitivity to certain foods occurs in adults, true food allergy causing chronic diarrhea is rare.

Other Causes Chronic inflammatory diarrhea may be caused by *radiation enterocolitis*, *chronic graft-versus-host disease*, *Behcet's syndrome*, and *Cronkite-Canada syndrome*, among others.

Dysmotile Causes Rapid transit may accompany many diarrheas as a secondary or contributing phenomenon, but primary dysmotility is an unusual etiology of true diarrhea. Stool features often suggest a secretory diarrhea, but mild steatorrhea up to 14 g of fat per day can be produced by maldigestion from rapid transit alone. *Hyperthyroidism*, *carcinoid syndrome*, and certain drugs (e.g., prostaglandins, prokinetic agents) may produce hypermotility with resultant diarrhea. Primary visceral neuromyopathies or idiopathic acquired intestinal pseudo-obstruction may lead to stasis with secondary bacterial overgrowth causing diarrhea. *Diabetic diarrhea*, often accompanied by peripheral and generalized autonomic neuropathies, may occur in part because of intestinal dysmotility.

The exceedingly common *irritable bowel syndrome* (10% point prevalence, 1 to 2% per year incidence) is characterized by disturbed intestinal and colonic motor and sensory responses to various stimuli. Symptoms of stool frequency typically cease at night, alternate with periods of constipation, are accompanied by abdominal pain relieved with defecation, and rarely result in weight loss or true diarrhea.

Factitial Causes Factitial diarrhea accounts for up to 15% of unexplained diarrheas referred to tertiary care centers. Either as a form of *Munchausen syndrome* (deception or self-injury for secondary gain) or *bulimia*, some patients covertly self-administer laxatives alone or in combination with other medications (e.g., diuretics) or surreptitiously add water or urine to stool sent for analysis. Such patients are typically women, often with histories of psychiatric illness and disproportionately from careers in health care. Hypotension and hypokalemia are common co-presenting features. Such patients often deny this possibility when confronted, but they do benefit from psychiatric counseling when they acknowledge their behavior.

Approach to the Patient

The laboratory tools available to evaluate the very common problem of chronic diarrhea are extensive, and many are costly and invasive. As such, the diagnostic evaluation must be rationally directed by a careful history and physical examination, and simple triage tests are often warranted before complex investigations are launched ([Fig. 42-3](#)). The history, physical examination, and routine blood studies should attempt to characterize the mechanism of diarrhea, identify diagnostically helpful associations, and assess the patient's fluid/electrolyte and nutritional status. Patients should be questioned about the onset, duration, pattern, aggravants (especially diet), relieving factors, and stool characteristics of their diarrhea. The presence or absence of fecal

incontinence, fever, weight loss, pain, certain exposures (travel, medications, contacts with diarrhea), and common extraintestinal manifestations (skin changes, arthralgias, oral aphtha) should be noted. Physical findings may offer clues such as a thyroid mass, wheezing, heart murmurs, edema, hepatomegaly, abdominal masses, lymphadenopathy, mucocutaneous abnormalities, perianal fistulae, or anal sphincter laxity. Peripheral blood counts may reveal leukocytosis that suggests inflammation; anemia that reflects blood loss or nutritional deficiencies; or eosinophilia that may occur with parasitoses, neoplasia, collagen-vascular disease, allergy, or eosinophilic gastroenteritis. Blood chemistries may demonstrate electrolyte, hepatic, or other metabolic disturbances.

A therapeutic trial is often appropriate, definitive, and highly cost-effective when a specific diagnosis is suggested on the initial physician encounter. For example, chronic watery diarrhea, which ceases with fasting in an otherwise healthy young adult, may justify a trial of a lactose-restricted diet; bloating and diarrhea persisting since a mountain backpacking trip may warrant a trial of metronidazole for likely giardiasis; and postprandial diarrhea persisting since an ileal resection might be treated with cholestyramine before further evaluation. Persistent symptoms require additional investigation.

Certain diagnoses may be suggested on the initial encounter, e.g., idiopathic inflammatory bowel disease; however, additional focused evaluations may be necessary to confirm the diagnosis and characterize the severity or extent of disease so that treatment can be best guided. Patients suspected of having irritable bowel syndrome should be initially evaluated with proctosigmoidoscopy and mucosal biopsies; those with normal findings might be reassured and, as indicated, treated empirically with antispasmodics, antidiarrheals, bulk agents, anxiolytics, or antidepressants. Any patient who presents with chronic diarrhea and hematochezia should be evaluated with stool microbiologic studies and colonoscopy.

In an estimated two-thirds of cases, the cause for chronic diarrhea remains unclear after the initial encounter, and further testing is required. Quantitative stool collection and analyses can yield important objective data that may establish a diagnosis or characterize the type of diarrhea as a triage for focused additional studies ([Fig. 42-3](#)). If stool weight exceeds 200 g/d, additional stool analyses should be performed that might include electrolyte concentration, pH, occult blood testing, leukocyte inspection (or leukocyte protein assay), fat quantitation, and laxative screens.

For secretory diarrheas (watery, normal osmotic gap), possible medication-related side effects or surreptitious laxative use should be reconsidered. Microbiologic studies should be done including fecal bacterial cultures (including media for *Aeromonas* and *Plesiomonas*), inspection for ova and parasites, and *Giardia* antigen assay (the most sensitive test for giardiasis). Small bowel bacterial overgrowth can be excluded by intestinal aspirates with quantitative cultures or with glucose or xylose breath tests involving measurement of breath hydrogen or other metabolite (e.g., $^{14}\text{CO}_2$). However, interpretation of these breath tests may be confounded by disturbances of intestinal transit. When suggested by history or other findings, screens for peptide hormones should be pursued (e.g., serum gastrin, [VIP](#), calcitonin, and thyroid hormone/thyroid stimulating hormone, or urinary 5-hydroxyindolacetic acid and histamine). Upper

endoscopy and colonoscopy with biopsies and small bowel barium x-rays are helpful to rule out structural or occult inflammatory disease.

Further evaluation of osmotic diarrhea should include tests for lactose intolerance and magnesium ingestion, the two most common causes. Low fecal pH suggests carbohydrate malabsorption; lactose malabsorption can be confirmed by lactose breath testing or by a therapeutic trial with lactose exclusion and observation of the effect of lactose challenge (e.g., a quart of milk). Lactase determination on small bowel biopsy is generally not available. If fecal Mg^{2+} or laxative levels are elevated, then inadvertent or surreptitious ingestion should be considered and psychiatric help should be sought.

For those with proven fatty diarrhea, endoscopy with small bowel biopsy (including aspiration for *Giardia* and quantitative cultures) should be performed; if this procedure is unrevealing, a small bowel radiograph is often an appropriate next step. If small bowel studies are negative or if pancreatic disease is suspected, pancreatic exocrine insufficiency should be excluded with direct tests, such as the secretin-cholecystokinin stimulation test, or by indirect tests, such as assay of fecal chymotrypsin activity or a bentiromide test.

Chronic inflammatory-type diarrheas should be suspected by the presence of blood or leukocytes in the stool. Such findings warrant stool cultures, inspection for ova and parasites, *C. difficile* toxin assay, colonoscopy with biopsies, and if indicated, small bowel oral contrast studies.

TREATMENT

Treatment of chronic diarrhea depends on the specific etiology and may be curative, suppressive, or empiric. If the cause can be eradicated, treatment is curative as with resection of a colorectal cancer, antibiotic administration for Whipple's disease, or discontinuation of an offending drug. For many chronic conditions, diarrhea can be controlled by suppression of the underlying mechanism. Examples include elimination of dietary lactose for lactase deficiency or gluten for celiac sprue, use of glucocorticoids or other anti-inflammatory agents for idiopathic inflammatory bowel diseases, adsorptive agents such as cholestyramine for ileal bile acid malabsorption, proton pump inhibitors such as omeprazole for the gastric hypersecretion of gastrinomas, somatostatin analogues such as octreotide for malignant carcinoid, prostaglandin inhibitors such as indomethacin for medullary carcinoma of the thyroid, and pancreatic enzyme replacement for pancreatic insufficiency. When the specific cause or mechanism of chronic diarrhea evades diagnosis, empiric therapy may be beneficial. Mild opiates such as diphenoxylate or loperamide are often helpful in mild or moderate watery diarrhea. For those with more severe diarrhea, codeine or tincture of opium may be beneficial. Such antimotility agents should be avoided with inflammatory bowel disease, as toxic megacolon may be precipitated. Clonidine, an α_2 -adrenergic agonist, may allow control of diabetic diarrhea. For all patients with chronic diarrhea, fluid and electrolyte repletion is an important component of management (see "Acute Diarrhea," above). Replacement of fat-soluble vitamins may also be necessary in patients with chronic steatorrhea.

CONSTIPATION

DEFINITION

Constipation is a common complaint in clinical practice and usually refers to persistent, difficult, infrequent, or seemingly incomplete defecation. Because of the wide range of normal bowel habits, constipation is difficult to define precisely. Most persons have at least three bowel movements per week; however, stool frequency alone is not a sufficient criterion for the diagnosis of constipation because many constipated patients describe a normal frequency of defecation but subjective complaints of excessive straining, hard stools, lower abdominal fullness, and a sense of incomplete evacuation. The individual patient's symptoms must be analyzed in detail to ascertain what is meant by "constipation" or "difficulty" with defecation.

Stool form and consistency are well correlated with the time elapsed from the preceding defecation. Hard, pelleted stools occur with slow transit, while loose watery stools are associated with rapid transit. Small, pelleted stools are more difficult to expel than large ones.

The perception of hard stools or excessive straining is more difficult to assess objectively, and the need for enemas or digital disimpaction is a clinically useful way to corroborate the patient's perceptions of difficult defecation.

Psychosocial factors may also be important. A person whose parents attached great importance to daily defecation will become greatly concerned when he or she misses a daily bowel movement; some children withhold stool to gain attention; and some adults are simply too busy or too embarrassed to interrupt their work when the call to have a bowel movement is sensed.

CAUSES

Pathophysiologically, chronic constipation generally results from inadequate fiber intake or from disordered colonic transit or anorectal function as a result of a neurogastroenterologic disturbance, certain drugs, or in association with a large number of systemic diseases that affect the gastrointestinal tract ([Table 42-4](#)). Constipation of recent onset may be a symptom of significant organic disease such as tumor or stricture. In *idiopathic constipation*, a subset of patients exhibit delayed emptying of the ascending and transverse colon with prolongation of transit (often in the proximal colon) and a reduced frequency of propulsive colonic contractions ([HAPCs](#)). *Outlet obstruction to defecation* (also called *evacuation disorders*) may cause delayed colonic transit, which is usually corrected by biofeedback retraining of the disordered defecation. Constipation of any cause may be exacerbated by chronic illnesses that lead to physical or mental impairment and result in inactivity or physical immobility.

Approach to the Patient

A careful history should explore the patient's symptoms and confirm whether he or she is indeed constipated based on frequency (e.g., <3 bowel movements per week), consistency (lumpy/hard), excessive straining, prolonged defecation time, or need to support the perineum or digitate the anorectum. In the vast majority of cases (probably >90%), there is no underlying cause (e.g., cancer, depression, or hypothyroidism), and

constipation responds to ample hydration, exercise, and supplementation of dietary fiber (15 to 25 g/d). A good diet and medication history and attention to psychosocial issues are key. Physical examination and, particularly, a rectal examination should exclude most of the important diseases that present with constipation and possibly indicate features suggesting an evacuation disorder (e.g., high anal sphincter tone).

There is broad consensus on the selection of patients for further investigation. The presence of weight loss, rectal bleeding, or anemia with constipation mandates either sigmoidoscopy plus barium enema or colonoscopy alone, particularly in patients over 40 years of age, to exclude structural diseases such as cancer or strictures. Colonoscopy alone is most cost effective in this setting since it provides an opportunity to biopsy mucosal lesions, perform polypectomy, or dilate strictures. Barium enema has advantages over colonoscopy in the patient with isolated constipation, since it is less costly and identifies colonic dilatation and all significant mucosal lesions or strictures that are likely to present with constipation. Melanosis coli, or pigmentation of the colon mucosa, indicates the use of anthraquinone laxatives such as cascara or senna; however, this is usually apparent from a careful history. An unexpected disorder such as megacolon or cathartic colon may also be detected by colonic radiographs. Measurement of serum calcium and thyroid stimulating hormone levels will identify rare patients with metabolic disorders.

Patients with more troublesome constipation may not respond to fiber alone and may be helped by a bowel training regimen: taking an osmotic laxative and evacuating with enema or glycerine suppository as needed. After breakfast, a distraction-free 15 to 20 min on the toilet without straining is encouraged. Excessive straining may lead to development of hemorrhoids, and, if there is weakness of the pelvic floor or injury to the pudendal nerve, may result in obstructed defecation from descending perineum syndrome several years later. Those few who do not benefit from the simple measures delineated above or require long-term treatment with stimulant laxatives with the attendant risk of developing laxative abuse syndrome are assumed to have severe or intractable constipation and should have further investigation ([Fig. 42-4](#)).

INVESTIGATION OF SEVERE CONSTIPATION

A small minority (probably <5%) of all patients with constipation have cases that are considered severe or "intractable"; these are the patients most likely to be seen by gastroenterologists or in referral centers. Further observation of the patient may occasionally reveal a previously unrecognized cause, such as an evacuation disorder, laxative abuse, malingering, or psychiatric disorder. In these patients, recent studies suggest that evaluations of the physiologic function of the colon and pelvic floor and of psychological status aid in the rational choice of treatment. Even among these highly selected patients with severe constipation, a cause can be identified in only about 30% (see below).

Measurement of Colonic Transit Radiopaque marker transit tests are easy, repeatable, generally safe, inexpensive, reliable, and highly applicable in evaluating constipated patients in clinical practice. There are several validated methods that are very simple. For example, radiopaque markers are ingested, and an abdominal flat film taken 5 d later should indicate passage of 80% of the markers out of the colon. This test

does not provide useful information about the transit profile of the stomach and small bowel, and avoidance of laxatives or enemas during the testing period is essential.

Radioscintigraphy with a delayed-release capsule containing radiolabeled particles has been used to noninvasively characterize normal, accelerated, or delayed colonic function over 24 to 48 h with low radiation exposure. This approach simultaneously assesses gastric, small bowel, and colonic transit. The disadvantages are the greater cost and the need for specific materials prepared in a nuclear medicine laboratory.

Anorectal and Pelvic Floor Tests Pelvic floor dysfunction is suggested by the inability to evacuate the rectum, a feeling of persistent rectal fullness, rectal pain, the need to extract stool from the rectum digitally, application of pressure on the posterior wall of the vagina, support of the perineum during straining, and excessive straining. These significant symptoms should be contrasted with the sense of incomplete rectal evacuation, which is common in irritable bowel syndrome.

Patients with clinically suspected obstruction of defecation should also be evaluated by a psychologist to identify eating disorders or a "need to control," to provide stress management or relaxation training, and to identify depression.

A simple clinical test in the office to document a nonrelaxing puborectalis muscle is to have the patient strain to expel the index finger during a digital rectal exam. Motion of the puborectalis posteriorly during straining indicates proper coordination of the pelvic floor muscles.

Measurement of perineal descent is relatively easy to gauge clinically by placing the patient in the left decubitus position and watching the perineum to assess either paucity or lack of descent (<1.5 cm, a sign of pelvic floor dysfunction) or perineal ballooning during straining relative to bony landmarks (>4 cm, suggesting excessive perineal descent).

A useful overall test of evacuation is the balloon expulsion test. A urinary catheter is placed in the rectum, the balloon is inflated to 50 ml with water, and a determination is made about whether the patient can expel it while seated on a toilet or in the left lateral decubitus position. In the lateral position, the weight needed to facilitate expulsion of the balloon (normal, 0 to 200 g) is determined.

Anorectal manometry is not often contributory in the evaluation of patients presenting with severe constipation, except when an excessively high resting or squeeze anal sphincter tone suggests anismus (anal sphincter spasm). This test also identifies rare syndromes, such as adult Hirschsprung's disease, by the absence of the rectoanal inhibitory reflex or the presence of occult incontinence.

Defecography (a dynamic barium enema including lateral views obtained during barium expulsion) reveals "soft abnormalities" in many patients; the most relevant findings are the measured changes in rectoanal angle, anatomic defects of the rectum, and enteroceles or rectoceles. In a very small proportion of patients, significant anatomic defects associated with intractable constipation respond best to surgical treatment. These defects include severe intussusception with complete outlet obstruction due to

funnel-shaped plugging at the anal canal or an extremely large rectocele that is preferentially filled during attempts at defecation instead of expulsion of the barium through the anus. In summary, defecography requires an interested and experienced radiologist, and abnormalities are not pathognomonic for pelvic floor dysfunction. More commonly, outlet obstruction results from a nonrelaxing puborectalis muscle, which impedes rectal emptying, rather than from defects identified by defecography.

Dynamic imaging studies such as proctography during defecation or scintigraphic expulsion of artificial stool help measure perineal descent and the rectoanal angle during rest, squeezing and straining, and scintigraphic expulsion quantitates the amount of "artificial stool" emptied. Failure of the rectoanal angle to increase significantly ($\sim 15^\circ$) during straining confirms pelvic floor dysfunction.

Neurologic testing (EMG) is more helpful in the evaluation of patients with incontinence than of those with symptoms suggesting obstructed defecation. The absence of neurologic signs in the lower extremities suggests that any documented denervation of the puborectalis results from pelvic (e.g., obstetric) injury or from stretching of the pudendal nerve by chronic, long-standing straining.

Ultrasonography identifies sphincter or rectal wall defects and may help select patients for surgical correction. Spinal-evoked responses during electrical rectal stimulation or stimulation of external anal sphincter contraction by applying magnetic stimulation over the lumbosacral cord identify patients with limited sacral neuropathies with sufficient residual nerve conduction to attempt biofeedback training.

In summary, a balloon expulsion test is an important screening test for anorectal dysfunction. If positive, an anatomic evaluation of the rectum or anal sphincters and an assessment of pelvic floor relaxation are the tools for evaluating patients in whom obstructed defecation is suspected.

TREATMENT

After the cause of constipation is characterized, a treatment decision can be made. Slow transit constipation requires aggressive medical or surgical treatment; anismus or pelvic floor dysfunction usually responds to biofeedback management ([Fig. 42-4](#)). However, only about 30% of patients with severe constipation are found to have such a physiologic disorder.

Patients with slow transit constipation are treated with bulk, osmotic, and stimulant laxatives, including fiber, psyllium, milk of magnesia, lactulose, polyethylene glycol (colonic lavage solution), and bisacodyl. If a 2- to 3-month trial of medical therapy fails and patients continue to have documented slow transit constipation unassociated with obstructed defecation, colectomy with ileorectostomy is indicated. The decision to resort to surgery is facilitated in the presence of megacolon and megarectum. The complications after surgery include small bowel obstruction (11%) and fecal soiling, particularly at night during the first postoperative year.

Patients who have a combined disorder should pursue pelvic floor retraining (biofeedback and muscle relaxation), psychological counseling, and dietetic advice first,

followed by colectomy and ileorectostomy if colonic transit studies do not normalize with biofeedback alone. In patients with pelvic floor dysfunction alone, biofeedback training has a 70 to 80% success rate, measured by the acquisition of comfortable stool habits. Attempts to manage pelvic floor dysfunction with operations (internal anal sphincter or puborectalis muscle division) have achieved only mediocre success and have been largely abandoned.

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43. WEIGHT LOSS - Carol M. Reife

Significant unintentional weight loss in a previously healthy individual is often a harbinger of underlying systemic disease. During the routine medical history, therefore, inquiry should always be made about changes in weight; loss of 5% of body weight over 6 to 12 months should prompt further evaluation.

PHYSIOLOGY OF WEIGHT REGULATION

The normal individual maintains weight at a remarkably stable "set point," given the wide variation in daily caloric intake and level of activity. Because of the physiologic importance of maintaining energy stores, voluntary weight loss is difficult to achieve and sustain.

Appetite and metabolism are regulated by an intricate network of neural and hormonal factors. The hypothalamic feeding and satiety centers play a central role in these processes ([Chap. 77](#)). Neuropeptides, like corticotropin-releasing hormone (CRH), α -melanocyte stimulating hormone (α -MSH), and cocaine and amphetamine-related transcript (CART) induce anorexia by acting centrally on satiety centers. Epinephrine and norepinephrine cause a decrease in food intake and an increase in metabolic rate ([Chap. 72](#)). Amphetamines and related drugs used to suppress appetite act by releasing norepinephrine in the central nervous system. The gastrointestinal peptides glucagon, somatostatin, and particularly cholecystokinin induce a decrease in food intake by acting through a vagal mechanism to signal satiety. Hypoglycemia decreases levels of insulin which reduces glucose utilization and inhibits activity of the satiety center.

Leptin plays a central role in the long-term maintenance of weight homeostasis ([Chap. 77](#)). Leptin is produced by adipose tissue and acts on the hypothalamus to decrease food intake and increase energy expenditure. It suppresses expression of hypothalamic neuropeptide Y, a potent appetite stimulatory peptide. In parallel, leptin increases the expression of α -MSH, which decreases appetite by acting on the MC4R melanocortin receptor. Thus, leptin activates a series of downstream neural pathways that alter food-seeking behavior and metabolism. However, leptin deficiency, which occurs in conjunction with the loss of adipose tissue, stimulates appetite and induces other adaptive responses including inhibition of hypothalamic thyrotropin releasing hormone (TRH) and gonadotropin releasing hormone (GnRH).

A variety of cytokines, including tumor necrosis factor α (TNF- α), interleukin (IL) (IL-6), IL-1, interferon γ (IFN- γ), ciliary neurotrophic factor (CNTF), and leukemia inhibitory factor (LIF), can contribute to cachexia ([Chap. 17](#)). In addition to causing anorexia, these factors may induce fever, depress myocardial function, modulate immune and inflammatory responses, and induce a variety of specific metabolic alterations. TNF- α , for example, preferentially mobilizes fat but spares skeletal muscle. Levels of one or more of these cytokines may be increased in patients with cancer, sepsis, chronic inflammatory conditions, AIDS, and congestive heart failure.

Weight loss occurs when energy expenditure exceeds calories available for energy utilization. In most individuals, approximately half of food energy is utilized for basal

processes such as maintenance of body temperature. In a 70-kg person, basal activity consumes about 1800 kcal/d. About 40% of caloric intake is used for physical activity, although athletes may use more than 50% during vigorous exercise. About 10% of caloric intake is used for dietary thermogenesis, the energy expended for digestion, absorption, and metabolism of food.

Mechanisms of weight loss include decreased food intake, malabsorption, loss of calories, and increased energy requirements ([Fig. 43-1](#)). Changes in weight may reflect alterations in either tissue mass or body fluid content. A deficit of 3500 kcal generally correlates with the loss of 1 lb (0.45 kg) of body fat, but one must also consider water weight (2.2 lb/L) gained or lost. Weight loss that persists over weeks to months is almost invariably due to loss of tissue mass.

Food intake may be influenced by a wide variety of visual, olfactory, and gustatory stimuli as well by genetic, psychological, and social factors. Absorption may be impaired because of pancreatic insufficiency, cholestasis, celiac sprue, intestinal tumors, radiation injury, inflammatory bowel disease, infection, or medication effect. Manifestations of these disease processes may be suggested by changes in stool frequency and consistency. Calories also may be lost due to vomiting or diarrhea, glucosuria in diabetes mellitus, or fistulous drainage. Resting energy expenditure decreases with age and can be affected by thyroid status. Beginning at about age 60, body weight declines by an average of 0.5% per year. Body composition is also affected by aging; adipose tissue increases and lean muscle mass decreases with age.

SIGNIFICANCE OF WEIGHT LOSS

Unintentional weight loss, especially in the elderly, is not uncommon and is associated with increased morbidity and mortality rates, even after comorbid conditions have been taken into account. Prospective studies indicate that significant involuntary weight loss is associated with a mortality rate of 25% over the next 18 months. Retrospective studies of significant weight loss in the elderly document mortality rates of 9 to 38% over a 2- to 3-year period.

Cancer patients with weight loss have decreased performance status, response to chemotherapy, and median survival ([Chap. 79](#)). Marked degrees of weight loss also predispose to infection. Patients undergoing elective surgery, who have lost more than 10 lb (4.5 kg) in 6 months, have higher surgical mortality rates. Vitamin and nutrient deficiencies also can accompany significant weight loss ([Chap. 74](#)).

CAUSES OF WEIGHT LOSS

The list of possible causes of weight loss is extensive ([Table 43-1](#)). In the elderly, the most common causes of weight loss are depression, cancer, and benign gastrointestinal disease. Lung and gastrointestinal cancer are the most common malignancies in patients presenting with weight loss. In younger individuals, diabetes mellitus, hyperthyroidism, psychiatric disturbances including eating disorders, and infection, especially with HIV, should be considered.

The cause of involuntary weight loss is rarely occult. Careful history and physical

examination, in association with directed diagnostic testing, will identify the cause of weight loss in 75% of patients. The etiology of weight loss will not be found in the remaining patients, despite extensive testing. Patients with negative evaluations tend to have lower mortality rates than those found to have organic disease.

Patients with medical causes of weight loss usually have signs or symptoms that suggest involvement of a particular organ system. Gastrointestinal tumors, including those of the pancreas and liver, may affect food intake early in the course of illness, causing weight loss before other symptoms are apparent. Lung cancer may present with post-obstructive pneumonia, dyspnea, or cough and hemoptysis; however, it may be silent and should be considered even in those without a history of cigarette smoking. Depression and isolation can cause profound weight loss, especially in the elderly. Chronic pulmonary disease and congestive heart failure can produce anorexia and may also increase resting energy expenditure. Weight loss may be the presenting sign of infectious diseases such as HIV infection, tuberculosis, endocarditis, and fungal and parasitic infections. Hyperthyroidism or pheochromocytoma increase metabolism; elderly patients with apathetic hyperthyroidism may present with weight loss alone. New onset diabetes mellitus is often accompanied by weight loss, reflecting glucosuria and loss of the anabolic actions of insulin. Adrenal insufficiency may be suggested by increased pigmentation, hyponatremia, and hyperkalemia.

Approach to the Patient

Before extensive evaluation is undertaken, it is important to confirm that weight loss has occurred. Almost half of patients who claim significant weight loss have no actual change in weight when it is measured objectively. If weight loss is present, efforts should be made to determine the time interval over which it has occurred. In the absence of documentation, changes in belt notch size or the fit of clothing may help confirm loss of weight. Not infrequently, patients who have actually sustained significant weight loss are unaware that it has occurred. Routine documentation of weight during office visits is therefore important.

The review of systems should focus on signs or symptoms that are associated with disorders that commonly cause weight loss. These include fever, pain, shortness of breath or cough, palpitations, changes in pattern of urination, and evidence of neurologic disease. Gastrointestinal disturbances, including difficulty eating, dysphagia, anorexia, nausea, and change in bowel habits, should be sought. Use of cigarettes, alcohol, and all medications should be reviewed, and patients should be questioned about previous illness or surgery as well as diseases in family members. Risk factors for HIV infection should be assessed. Signs of depression, evidence of dementia, and social factors, including financial issues that might affect food intake, should be considered.

Physical examination should begin with weight determination and documentation of vital signs. The skin should be examined for pallor, jaundice, turgor, scars from prior surgery, and stigmata of systemic disease. The search for oral thrush or dental disease, thyroid gland enlargement, adenopathy, and respiratory or cardiac abnormalities and a detailed examination of the abdomen often lead to clues for further evaluation. Rectal examination, including prostate exam and testing of stool for occult blood, should be

performed in men; and all women should have a pelvic examination, even if they have had a hysterectomy. Neurologic examination should include mental status assessment and screening for depression.

Laboratory testing should confirm or exclude possible diagnoses elicited from the history and physical examination ([Table 43-2](#)). An initial phase of testing should include a complete blood count with differential, serum chemistry tests including glucose, electrolytes, renal and liver tests, calcium, thyroid stimulating hormone (TSH), urinalysis, and chest x-ray. Patients at risk for HIV infection should have HIV antibody testing. In all cases, recommended cancer screening tests appropriate for the gender and age group, such as mammograms and Pap smears, should be updated ([Chap. 80](#)). If gastrointestinal signs or symptoms are present, upper and/or lower endoscopy and abdominal imaging with either computed tomography (CT) or magnetic resonance imaging (MRI) have a relatively high yield, consistent with the high prevalence of gastrointestinal disorders in patients with weight loss. If an etiology of weight loss is not found, careful clinical follow-up, rather than persistent undirected testing, is reasonable.

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44. GASTROINTESTINAL BLEEDING - Loren Laine

Bleeding from the gastrointestinal (GI) tract may present in 5 ways. *Hematemesis* is vomitus of red blood or "coffee-grounds" material. *Melena* is black, tarry, foul-smelling stool. *Hematochezia* is the passage of bright red or maroon blood from the rectum. *Occult GI bleeding (GIB)* may be identified in the absence of overt bleeding by special examination of the stool (e.g., guaiac testing). Finally, patients may present only with *symptoms of blood loss or anemia* such as lightheadedness, syncope, angina, or dyspnea.

SOURCES OF GASTROINTESTINAL BLEEDING

UPPER GASTROINTESTINAL SOURCES OF BLEEDING ([Table 44-1](#))

The annual incidence of hospital admissions for upper GIB (UGIB) in the United States and Europe is approximately 0.1%, with a mortality rate of ~10%. Patients rarely die from exsanguination; rather, they die due to decompensation from other underlying illnesses. The mortality rate for patients under 60 years of age in the absence of malignancy or organ failure is <1%.

Peptic ulcers are the most common cause of [UGIB](#), accounting for about 50% of cases. Mallory-Weiss tears account for 5 to 15% of cases. The proportion of patients bleeding from varices varies widely from ~5 to 30%, depending on the population. Hemorrhagic or erosive gastropathy [e.g., due to nonsteroidal anti-inflammatory drugs (NSAIDs) or alcohol] and erosive esophagitis often cause mild UGIB, but major bleeding is rare.

Peptic Ulcers Clinical features that predict poorer outcome include hemodynamic instability, the number of units of blood transfused, red blood in the emesis and the stool, increasing age, and the presence of concurrent illness. Characteristics of the ulcer at endoscopy also provide important prognostic information. One-third of patients with active bleeding or a non-bleeding visible vessel have further bleeding that requires urgent surgery if they are treated conservatively. These patients clearly benefit from endoscopic therapy with bipolar electrocoagulation, heater probe, or injection therapy (e.g., absolute alcohol, 1:10,000 epinephrine), with reductions in bleeding, hospital stay, mortality rate, and costs. In contrast, patients with clean-based ulcers have rates of recurrent bleeding approaching zero. If there is no other reason for hospitalization, such patients may be discharged on the first hospital day, following stabilization. Patients without clean-based ulcers should usually remain in the hospital for 3 days, since most episodes of recurrent bleeding occur within 3 days.

Various pharmacologic agents have been assessed in the past for the treatment of ulcer bleeding without clearcut benefit. However, in recent controlled trials in Europe and Asia, high-dose intravenous omeprazole used to raise intragastric pH to 6 to 7 and enhance clot stability decreased further bleeding (but not mortality), even after the use of appropriate endoscopic therapy.

Approximately one-third of patients with a bleeding ulcer will rebleed within the next 1 to 2 years. Prevention of recurrent bleeding focuses on the three main factors in ulcer pathogenesis, *Helicobacter pylori*, [NSAIDs](#), and acid. Eradication of *H. pylori* in patients

with bleeding ulcers dramatically decreases rates of rebleeding to < 5%. If a bleeding ulcer develops in a patient taking NSAIDs, the NSAIDs should be discontinued if possible. If NSAIDs must be continued, initial treatment should be with a proton pump inhibitor, and subsequent prophylactic therapy with a proton pump inhibitor or misoprostol should be continued as long as the patient is taking NSAIDs. Changing from a standard NSAID to a COX-2-specific inhibitor should markedly lower the risk of recurrent [UGIB](#). Patients with bleeding ulcers unrelated to *H. pylori* or NSAIDs should remain on full-dose antisecretory therapy indefinitely. **Peptic ulcers are discussed in [Chap. 285](#).*

Mallory-Weiss Tears The classic history is vomiting, retching, or coughing preceding hematemesis, especially in an alcoholic patient. Bleeding from these tears, which are usually on the gastric side of the gastroesophageal junction, stops spontaneously in 80 to 90% of patients and recurs in only 0 to 5%. Endoscopic therapy is effective for actively bleeding Mallory-Weiss tears. Angiographic therapy with intra-arterial infusion of vasopressin or embolization also may be useful. Rarely, operative therapy with oversewing of the tear may be required. **Mallory-Weiss tears are discussed in [Chap. 284](#).*

Esophageal Varices Patients with [UGIB](#) and clinical evidence suggesting the possibility of liver disease should undergo early endoscopy to determine if varices are the sources of bleeding, because patients with variceal hemorrhage have poorer outcomes than patients with other sources of UGIB. Endoscopic therapy at this time decreases further bleeding, and repeated sessions with endoscopic therapy to eradicate esophageal varices significantly reduces rebleeding and mortality. Endoscopic ligation therapy is the endoscopic therapy of choice for esophageal varices because it has less rebleeding, a lower mortality rate, fewer local complications, and requires fewer treatment sessions to achieve variceal eradication as compared to sclerotherapy.

Acute treatment with octreotide (50 ug bolus and 50 ug/h intravenous infusion for 2 to 5 days) or somatostatin may help in the control of acute bleeding, and these agents have replaced vasopressin as the medical therapy of choice for acute variceal bleeding. Over the long term, treatment with nonselective beta blockers (e.g., propranolol) has also been shown to decrease recurrent bleeding from esophageal varices. These agents commonly are given along with chronic endoscopic therapy.

In patients who have persistent or recurrent bleeding despite endoscopic and medical therapy, more invasive therapy is warranted. Transjugular intrahepatic portosystemic shunt (TIPS) decreases rebleeding more effectively than endoscopic therapy, although hepatic encephalopathy is more common and the mortality rates are comparable. Most patients with TIPS have shunt stenosis within 1 to 2 years and require re-instrumentation. Therefore, TIPS is most appropriate for patients with more severe liver disease and those in whom transplant is anticipated. Patients with milder, well-compensated cirrhosis probably should undergo decompressive surgery (e.g., distal splenorenal shunt).

Portal hypertension is also responsible for bleeding from gastric varices, ectopic varices in the small and large intestine, and portal hypertensive gastropathy and enterocolopathy.

Hemorrhagic and Erosive Gastropathy ("Gastritis") Hemorrhagic and erosive gastropathy or gastritis refers to endoscopically visualized subepithelial hemorrhages and erosions. These are mucosal lesions and thus do not cause major bleeding. They develop in various clinical settings, the most important of which are ingestion of [NSAIDs](#), alcohol, and stress. Half of patients who chronically ingest NSAIDs have erosions (15 to 30% have ulcers), while up to 20% of actively drinking alcoholic patients with symptoms of [UGIB](#) have evidence of subepithelial hemorrhages or erosions.

Stress-related gastric mucosal injury occurs only in extremely sick patients: those who have experienced serious trauma, major surgery, burns covering more than one-third of the body surface area, major intracranial disease, and severe medical illness (ventilator dependency, coagulopathy). Significant bleeding probably does not develop unless ulceration occurs. The mortality rate in these patients is quite high because of their serious underlying illnesses.

The incidence of bleeding from stress-related gastric mucosal injury or ulceration has decreased dramatically in recent years, most likely due to better care of critically ill patients. Pharmacologic prophylaxis for bleeding may be considered in the high-risk patients mentioned above. The best clinical data suggest that intravenous H₂-receptor antagonist therapy is the treatment of choice, although sucralfate also is effective. Prophylactic therapy decreases bleeding, but it does not lower the mortality rate.

Other Causes Other, less frequent causes of [UGIB](#) include erosive duodenitis, neoplasms, aortoenteric fistulas, vascular lesions [including hereditary hemorrhagic telangiectasias (Osler-Weber-Rendu) and gastric antral vascular ectasia ("watermelon stomach")], Dieulafoy's lesion (in which an aberrant vessel in the mucosa bleeds from a pinpoint mucosal defect), prolapse gastropathy (prolapse of proximal stomach into esophagus with retching, especially in alcoholics), and hemobilia and hemosuccus pancreaticus (bleeding from the bile duct or pancreatic duct).

SMALL INTESTINAL SOURCES OF BLEEDING

Small intestinal sources of bleeding (bleeding from sites beyond the reach of the standard upper endoscope) are difficult to diagnose and are responsible for the majority of cases of obscure [GIB](#). Fortunately, small intestinal bleeding is uncommon. The most common causes are vascular ectasias and tumors (e.g., adenocarcinoma, leiomyoma, lymphoma, benign polyps, carcinoid, metastases, and lipoma). Other less common causes include Crohn's disease, infection, ischemia, vasculitis, small bowel varices, diverticula, Meckel's diverticula, duplication cysts, and intussusception. [NSAIDs](#) induce small intestinal erosions and ulcers and may be a relatively common cause of chronic, obscure GIB.

Meckel's diverticulum is the most common cause of significant lower GIB (LGIB) in children, decreasing in frequency as a cause of bleeding with age. In adults younger than 40 to 50 years, small bowel tumors often account for obscure [GIB](#), while in patients older than 50 to 60 years, vascular ectasias are usually responsible.

Vascular ectasias should be treated with endoscopic therapy if possible. Surgical

therapy can be used for vascular ectasias isolated to a segment of the small intestine when endoscopic therapy is unsuccessful; estrogen/progesterone compounds may also be tried. Isolated lesions, such as tumors, diverticula, or duplications, generally are treated with surgical resection.

COLONIC SOURCES OF BLEEDING

The incidence of hospitalizations for [LGIB](#) is about one-fifth that for [UGIB](#). Hemorrhoids are probably the most common cause of LGIB; anal fissures also cause minor bleeding and pain. If these local anal processes, which rarely require hospitalization, are excluded, the most common causes of LGIB in adults are diverticula, vascular ectasias (especially in the proximal colon of patients > 70 years), neoplasms (adenomatous polyps and adenocarcinoma), and colitis -- most commonly infectious or idiopathic inflammatory bowel disease, but occasionally ischemic or radiation-induced. Uncommon causes include post-polypectomy bleeding, solitary rectal ulcer syndrome, [NSAID](#)-induced ulcers or colitis, other neoplasms, trauma, ectopic varices (most commonly rectal), lymphoid nodular hyperplasia, vasculitis, and aorto-colic fistulas. In children and adolescents, the most common colonic causes of significant [GIB](#) are inflammatory bowel disease and juvenile polyps.

Diverticular bleeding is abrupt in onset, usually painless, sometimes massive, and often from the right colon; minor and occult bleeding is not characteristic. Clinical reports suggest that bleeding colonic diverticula stop bleeding spontaneously in approximately 80% of patients, and rebleed in 20 to 25% of patients. Intraarterial vasopressin may halt the bleeding, at least temporarily. If bleeding persists or recurs, segmental surgical resection is indicated.

Bleeding from right colonic vascular ectasias in the elderly may be overt or occult; it tends to be chronic and only occasionally is hemodynamically significant. Endoscopic hemostatic therapy may be useful in the treatment of vascular ectasias, as well as discrete bleeding ulcers and post-polypectomy bleeding, while endoscopic polypectomy, if possible, is used for bleeding colonic polyps. Surgical therapy is generally required for major, persistent, or recurrent bleeding from the wide variety of colonic sources of GIB that cannot be treated medically or endoscopically.

Approach to the Patient

Measurement of the heart rate and blood pressure is the best way to assess a patient with [GIB](#). Clinically significant bleeding leads to postural changes in heart rate or blood pressure, tachycardia, and, finally, recumbent hypotension. Patients also may have a vasovagal reaction with bradycardia during bleeding episodes.

In contrast, the hemoglobin does not fall immediately with acute [GIB](#), due to proportionate reductions in plasma and red cell volumes (i.e., "people bleed whole blood"). Thus, hemoglobin may be normal or only minimally decreased at the initial presentation of a severe bleeding episode. As extravascular fluid enters the vascular space to restore volume, the hemoglobin falls, but this process may take up to 72 h. Patients with slow, chronic GIB may have very low hemoglobin values despite normal blood pressure and heart rate. With the development of iron deficiency anemia, the

mean corpuscular volume will be low and red blood cell distribution width will be increased.

Differentiation of Upper from Lower GIB Hematemesis indicates an upper [GI](#) source of bleeding (above the ligament of Treitz). Melena indicates that blood has been present in the GI tract for at least 14 h. Thus, the more proximal the bleeding site, the more likely melena will occur. Hematochezia usually represents a lower GI source of bleeding, although an upper GI lesion may bleed so rapidly that blood does not remain in the bowel long enough for melena to develop. When hematochezia is the presenting symptom of [UGIB](#), it is associated with hemodynamic instability and dropping hemoglobin. Bleeding lesions of the small bowel may present as melena or hematochezia.

A non-bloody nasogastric aspirate may be seen in up to 16% of patients with [UGIB](#) -- usually from a duodenal source. Even a bile-stained appearance does not exclude a bleeding post-pyloric lesion since reports of bile in the aspirate are incorrect in about 50% of cases. Testing of aspirates that are not grossly bloody for occult blood is of no clinical value. Other clues to UGIB include hyperactive bowel sounds and an elevated BUN (due to volume depletion and absorbed blood proteins).

Diagnostic Evaluation of the Patient with [GIB](#)

UPPER GIB (Fig. 44-1) The history and physical exam seldom are diagnostic of the source of GIB. Upper endoscopy is the test of choice in patients with [UGIB](#), and should be performed urgently in patients with hemodynamic instability (hypotension, tachycardia, or postural changes in heart rate or blood pressure). Early routine endoscopy is also beneficial in cases of milder bleeding for management decisions. Patients with major bleeding and high risk endoscopic findings (varices, ulcers with active bleeding or a visible vessel) benefit from endoscopic hemostatic therapy, while patients with low-risk lesions (e.g., clean based ulcers, non-bleeding Mallory-Weiss tears, erosive or hemorrhagic gastropathy) who have stable vital signs and hemoglobin, and no other medical problems, can be discharged home.

LOWER [GIB](#) (Fig. 44-2) Patients with presumed [LGIB](#) may undergo early sigmoidoscopy for the detection of obvious, low-lying lesions. However, the procedure is difficult with brisk bleeding, and it often is impossible to identify the area of bleeding. Sigmoidoscopy is useful primarily in patients < 40 years with relatively minor bleeding. Patients with hematochezia and hemodynamic instability should have upper endoscopy to rule out an upper [GI](#) source before evaluation of the lower GI tract.

Colonoscopy after an oral lavage solution is the procedure of choice in patients with [LGIB](#) unless bleeding is too massive or unless sigmoidoscopy has disclosed an obvious actively bleeding lesion.^{99M}Tc-labeled red cell scan allows repeated imaging for up to 24 h and may identify the general location of bleeding. However, radionuclide scans should be interpreted with caution because results are highly variable. In active LGIB, angiography can detect the site of bleeding (extravasation of contrast into the gut) and permits treatment with intraarterial infusion of vasopressin or embolization. Even after bleeding has stopped, angiography may identify lesions with abnormal vasculature such as vascular ectasias or tumors.

GIB OF OBSCURE ORIGIN Obscure GIB is defined as recurrent acute or chronic bleeding for which no source has been identified by routine endoscopic and contrast studies. Push enteroscopy, with a specially designed enteroscope or a pediatric colonoscope to inspect the entire duodenum and part of the jejunum, is generally the next step. Push enteroscopy may identify probable bleeding sites in 20 to 40% of patients with obscure GIB. If enteroscopy is negative or unavailable, a specialized radiographic examination of the small bowel (e.g., enteroclysis) should be performed.

Patients with recurrent bleeding who require transfusions or repeated hospitalizations warrant further investigations.⁹⁹M Tc-labeled red blood cell scintigraphy should be employed. Angiography is useful even if bleeding has subsided, since it may disclose vascular anomalies or tumor vessels.⁹⁹M Tc-pertechnetate scintigraphy for diagnosis of Meckel's diverticulum should be done, especially in the evaluation of young patients with **LGIB**. When all tests are unrevealing, intraoperative endoscopy is indicated in patients with severe recurrent or persistent bleeding requiring repeated transfusions.

OCCULTGIB Occult GIB is manifested by either a positive test for fecal occult blood or iron deficiency anemia. Unless a patient has upper **GI** symptoms, evaluation of occult bleeding generally should begin with colonoscopy, particularly in patients older than 40 years. If evaluation of the colon is negative, some perform upper endoscopy only if iron deficiency anemia or upper GI symptoms are present, while others recommend upper endoscopy in all patients since up to 25 to 40% of these patients have some abnormality noted on upper endoscopy. If standard endoscopic tests are unrevealing, enteroscopy and/or enteroclysis may be considered in patients with iron-deficiency anemia.

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45. JAUNDICE - Daniel S. Pratt, Marshall M. Kaplan

Jaundice, or icterus, is a yellowish discoloration of tissue resulting from the deposition of bilirubin. Tissue deposition of bilirubin occurs only in the presence of serum hyperbilirubinemia and is a sign of either liver disease or, less often, a hemolytic disorder. The degree of serum bilirubin elevation can be estimated by physical examination. Slight increases in serum bilirubin are best detected by examining the sclerae which have a particular affinity for bilirubin due to their high elastin content. The presence of scleral icterus indicates a serum bilirubin of at least 3.0 mg/dL. The ability to detect scleral icterus is made more difficult if the examining room has fluorescent lighting. If the examiner suspects scleral icterus, a second place to examine is underneath the tongue. As serum bilirubin levels rise, the skin will eventually become yellow in light-skinned patients and even green if the process is longstanding; the green color is produced by oxidation of bilirubin to biliverdin.

The differential diagnosis for yellowing of the skin is limited. In addition to jaundice, it includes carotenoderma, the use of the drug quinacrine, and excessive exposure to phenols. Carotenoderma is the yellow color imparted to the skin by the presence of carotene; it occurs in healthy individuals who ingest excessive amounts of vegetables and fruits that contain carotene, such as carrots, leafy vegetables, squash, peaches, and oranges. Unlike jaundice, where the yellow coloration of the skin is uniformly distributed over the body, in carotenoderma the pigment is concentrated on the palms, soles, forehead, and nasolabial folds. Carotenoderma can be distinguished from jaundice by the sparing of the sclerae. Quinacrine causes a yellow discoloration of the skin in 4 to 37% of patients treated with it. Unlike carotene, quinacrine can cause discoloration of the sclerae.

Another sensitive indicator of increased serum bilirubin is darkening of the urine, which is due to the renal excretion of conjugated bilirubin. Patients often describe their urine as tea or cola colored. Bilirubinuria indicates an elevation of the direct serum bilirubin fraction and therefore the presence of liver disease.

Increased serum bilirubin levels occur when an imbalance exists between bilirubin production and clearance. A logical evaluation of the patient who is jaundiced requires an understanding of bilirubin production and metabolism.

PRODUCTION AND METABOLISM OF BILIRUBIN (See also [Chap. 294](#))

Bilirubin, a tetrapyrrole pigment, is a breakdown product of heme (ferroprotoporphyrin IX). About 70 to 80% of the 250 to 300 mg of bilirubin produced each day is derived from the breakdown of hemoglobin in senescent red blood cells. The remainder comes from prematurely destroyed erythroid cells in bone marrow and from the turnover of hemoproteins such as myoglobin and cytochromes found in tissues throughout the body.

The formation of bilirubin occurs in reticuloendothelial cells, primarily in the spleen and liver. The first reaction, catalyzed by the enzyme heme oxygenase, oxidatively cleaves the bridge of the porphyrin group and opens the heme ring. The end products of this reaction are biliverdin, carbon monoxide, and iron. The second reaction, catalyzed by

the cytosolic enzyme biliverdin reductase, reduces the central methylene bridge of biliverdin and converts it to bilirubin. Bilirubin formed in the reticuloendothelial cells is virtually insoluble in water. To be transported in blood, it must be solubilized. This is accomplished by its reversible, noncovalent binding to albumin. Unconjugated bilirubin bound to albumin is transported to the liver, where it, but not the albumin, is taken up by hepatocytes via a process that at least partly involves carrier-mediated membrane transport.

In the cytosol of the hepatocyte, unconjugated bilirubin is coupled predominantly to the protein ligandin (formerly called the Y protein). Ligandin was initially thought to be a transport protein facilitating the movement of bilirubin from the sinusoidal membrane to the endoplasmic reticulum. It is now thought to slow the cytosolic diffusion of bilirubin and to reduce its efflux back into serum. In the endoplasmic reticulum, bilirubin is solubilized by conjugation to glucuronic acid, forming bilirubin monoglucuronide and diglucuronide. The conjugation of glucuronic acid to bilirubin is catalyzed by bilirubin uridine-diphosphate (UDP) glucuronosyltransferase.

The now hydrophilic bilirubin conjugates diffuse from the endoplasmic reticulum to the canalicular membrane, where bilirubin monoglucuronide and diglucuronide are actively transported into canalicular bile by an energy-dependent mechanism involving the multiple organic ion transport protein/multiple drug resistance protein. The conjugated bilirubin excreted into bile drains into the duodenum and passes unchanged through the proximal small bowel. Conjugated bilirubin is not taken up by the intestinal mucosa. When the conjugated bilirubin reaches the distal ileum and colon, it is hydrolyzed to unconjugated bilirubin by bacterial β -glucuronidases. The unconjugated bilirubin is reduced by normal gut bacteria to form a group of colorless tetrapyrroles called urobilinogens. About 80 to 90% of these products are excreted in feces, either unchanged or oxidized to orange derivatives called urobilins. The remaining 10 to 20% of the urobilinogens are passively absorbed, enter the portal venous blood, and are reexcreted by the liver. A small fraction (usually less than 3 mg/dL) escapes hepatic uptake, filters across the renal glomerulus, and is excreted in urine.

MEASUREMENT OF SERUM BILIRUBIN

The terms direct- and indirect-reacting bilirubin are based on the original van den Bergh reaction. This assay, or a variation of it, is still used in most clinical chemistry laboratories to determine the serum bilirubin level. In this assay, bilirubin is exposed to diazotized sulfanilic acid, splitting into two relatively stable dipyrromethene azopigments that absorb maximally at 540 nm, allowing for photometric analysis. The direct fraction is that which reacts with diazotized sulfanilic acid in the absence of an accelerator substance such as alcohol. The direct fraction provides an approximate determination of the conjugated bilirubin in serum. The total serum bilirubin is the amount that reacts after the addition of alcohol. The indirect fraction is the difference between the total and the direct bilirubin and provides an estimate of the unconjugated bilirubin in serum.

With the van den Bergh method, the normal serum bilirubin concentration usually is <1 mg/dL (17 μ mol/L). Up to 30%, or 0.3 mg/dL (5.1 μ mol/L), of the total may be direct-reacting (conjugated) bilirubin. Total serum bilirubin concentrations are between 0.2 and 0.9 mg/dL in 95% of a normal population.

Several new techniques, although less convenient to perform, have added considerably to our understanding of bilirubin metabolism. First, they demonstrate that in normal people or those with Gilbert's syndrome, almost 100% of the serum bilirubin is unconjugated; less than 3% is monoconjugated bilirubin. Second, in jaundiced patients with hepatobiliary disease, the total serum bilirubin concentration measured by these new, more accurate methods is lower than the values found with diazo methods. This suggests that there are diazo-positive compounds distinct from bilirubin in the serum of patients with hepatobiliary disease. Third, these studies indicate that in jaundiced patients with hepatobiliary disease, monoglucuronides of bilirubin predominate over the diglucuronides. Fourth, part of the direct-reacting bilirubin fraction includes conjugated bilirubin that is covalently linked to albumin. This albumin-linked bilirubin fraction (*delta fraction* or *biliprotein*) represents an important fraction of total serum bilirubin in patients with cholestasis and hepatobiliary disorders. Albumin-bound conjugated bilirubin is formed in serum when hepatic excretion of bilirubin glucuronides is impaired and the glucuronides are present in serum in increasing amounts. By virtue of its tight binding to albumin, the clearance rate of albumin-bound bilirubin from serum approximates the half-life of albumin, 12 to 14 days, rather than the short half-life of bilirubin, about 4 h.

The prolonged half-life of albumin-bound conjugated bilirubin explains two previously unexplained enigmas in jaundiced patients with liver disease: (1) that some patients with conjugated hyperbilirubinemia do not exhibit bilirubinuria during the recovery phase of their disease because the bilirubin is bound to albumin and therefore not filtered by the renal glomeruli and (2) that the elevated serum bilirubin level declines more slowly than expected in some patients who otherwise appear to be recovering satisfactorily. Late in the recovery phase of hepatobiliary disorders, all the conjugated bilirubin may be in the albumin-linked form. Its value in serum falls slowly because of the long half-life of albumin.

MEASUREMENT OF URINE BILIRUBIN

Unconjugated bilirubin is always bound to albumin in the serum, is not filtered by the kidney, and is not found in the urine. Conjugated bilirubin is filtered at the glomerulus and the majority is reabsorbed by the proximal tubules; a small fraction is excreted in the urine. Any bilirubin found in the urine is conjugated bilirubin. The presence of bilirubinuria implies the presence of liver disease. A urine dipstick test (Ictotest) gives the same information as fractionation of the serum bilirubin. This test is very accurate. A false-negative test is possible in patients with prolonged cholestasis due to the predominance of conjugated bilirubin covalently bound to albumin.

THE EVALUATION OF JAUNDICE

The bilirubin present in serum represents a balance between input from production of bilirubin and hepatic/biliary removal of the pigment. Hyperbilirubinemia may result from (1) overproduction of bilirubin; (2) impaired uptake, conjugation, or excretion of bilirubin; or (3) regurgitation of unconjugated or conjugated bilirubin from damaged hepatocytes or bile ducts. An increase in unconjugated bilirubin in serum results from either overproduction, impairment of uptake, or conjugation of bilirubin. An increase in conjugated bilirubin is due to decreased excretion into the bile ductules or backward

leakage of the pigment. The initial steps in evaluating the patient with jaundice are to determine (1) whether the hyperbilirubinemia is predominantly conjugated or unconjugated in nature, and (2) whether other biochemical liver tests are abnormal. The thoughtful interpretation of limited data will allow for a rational evaluation of the patient ([Fig. 45-1](#)). This discussion will focus solely on the evaluation of the adult patient with jaundice.

ISOLATED ELEVATION OF SERUM BILIRUBIN

Unconjugated Hyperbilirubinemia The differential diagnosis of an isolated unconjugated hyperbilirubinemia is limited ([Table 45-1](#)). The critical determination is whether the patient is suffering from a hemolytic process resulting in an overproduction of bilirubin (hemolytic disorders and ineffective erythropoiesis) or from impaired hepatic uptake/conjugation of bilirubin (drug effect or genetic disorders).

Hemolytic disorders that cause excessive heme production may be either inherited or acquired. Inherited disorders include spherocytosis, sickle cell anemia, and deficiency of red cell enzymes such as pyruvate kinase and glucose-6-phosphate dehydrogenase. In these conditions, the serum bilirubin rarely exceeds 5 mg/dL. Higher levels may occur when there is coexistent renal or hepatocellular dysfunction, or in acute hemolysis such as a sickle cell crisis. In evaluating jaundice in patients with chronic hemolysis, it is important to remember the high incidence of pigmented (calcium bilirubinate) gallstones found in these patients, which increases the likelihood of choledocholithiasis as an alternative explanation for hyperbilirubinemia.

Acquired hemolytic disorders include microangiopathic hemolytic anemia (e.g., hemolytic-uremic syndrome), paroxysmal nocturnal hemoglobinuria, and immune hemolysis. Ineffective erythropoiesis occurs in cobalamin, folate, and iron deficiencies.

In the absence of hemolysis, the physician should consider a problem with the hepatic uptake or conjugation of bilirubin. Certain drugs, including rifampicin and probenecid, may cause unconjugated hyperbilirubinemia by diminishing hepatic uptake of bilirubin. Impaired bilirubin conjugation occurs in three genetic conditions: *Crigler-Najjar syndrome, types I and II*, and *Gilbert's syndrome*. *Crigler-Najjar type I* is an exceptionally rare condition found in neonates and characterized by severe jaundice (bilirubin > 20 mg/dL) and neurologic impairment due to kernicterus, frequently leading to death in infancy or childhood. These patients have a complete absence of bilirubin [UDP](#)glucuronosyltransferase activity, usually due to mutations in the critical 3' domain of the UDP glucuronosyltransferase gene, and are totally unable to conjugate, hence cannot excrete bilirubin. The only effective treatment is orthotopic liver transplantation. Use of gene therapy and allogeneic hepatocyte infusion are experimental approaches of future promise for this devastating disease.

Crigler-Najjar type II is somewhat more common than type I. Patients live into adulthood with serum bilirubin levels that range from 6 to 25 mg/dL. In these patients, mutations in the bilirubin [UDP](#)glucuronosyltransferase gene cause reduced but not completely absent activity of the enzyme. Bilirubin UDP glucuronosyltransferase activity can be induced by the administration of phenobarbital, which can reduce serum bilirubin levels in these patients. Despite marked jaundice, these patients usually survive into adulthood,

although they may be susceptible to kernicterus under the stress of intercurrent illness or surgery.

Gilbert's syndrome is also marked by the impaired conjugation of bilirubin due to reduced bilirubin UDPglucuronosyltransferase activity. Molecular analyses show that Gilbert's syndrome is due to reduced expression of UDP glucuronosyltransferase activity caused by lengthening of the TATAA box from A(TA)₆TAA to A(TA)₇TAA in the promoter element of the gene. This results in mild unconjugated hyperbilirubinemia with serum levels almost always less than 6 mg/dL. The serum levels may fluctuate and jaundice is often identified only during periods of fasting. Unlike both Crigler-Najjar syndromes, Gilbert's syndrome is very common. The reported incidence is 3 to 7% of the population with males predominating over females by a ratio of 2-7:1.

Conjugated Hyperbilirubinemia Elevated conjugated hyperbilirubinemia is found in two rare inherited conditions: *Dubin-Johnson syndrome* and *Rotor's syndrome* ([Table 45-1](#)). Patients with both conditions present with asymptomatic jaundice, typically in the second generation of life. The defect in Dubin-Johnson syndrome is a point mutation in the gene for the canalicular multispecific organic anion transporter. These patients have altered excretion of bilirubin into the bile ducts. Rotor's syndrome seems to be a problem with the hepatic storage of bilirubin. Differentiating between these syndromes is possible, but clinically unnecessary, due to their benign nature.

ELEVATION OF SERUM BILIRUBIN WITH OTHER LIVER TEST ABNORMALITIES

The remainder of this chapter will focus on the evaluation of the patient with a conjugated hyperbilirubinemia in the setting of other liver test abnormalities. This group of patients can be divided into those with a primary hepatocellular process and those with intra- or extrahepatic cholestasis. Being able to make this differentiation will guide the physician's evaluation ([Fig. 45-1](#)). This differentiation is made on the basis of the history and physical examination as well as the pattern of liver test abnormalities.

History A complete medical history is perhaps the single most important part of the evaluation of the patient with unexplained jaundice. Important considerations include the use of or exposure to any chemical or medication, either physician-prescribed or over-the-counter, such as herbal and vitamin preparations and other drugs such as anabolic steroids. The patient should be carefully questioned about possible parenteral exposures, including transfusions, intravenous and intranasal drug use, tattoos, and sexual activity. Other important questions include recent travel history, exposure to people with jaundice, exposure to possibly contaminated foods, occupational exposure to hepatotoxins, alcohol consumption, the duration of jaundice, and the presence of any accompanying symptoms such as arthralgias, myalgias, rash, anorexia, weight loss, abdominal pain, fever, pruritis, and changes in the urine and stool. While none of these latter symptoms are specific for any one condition, they can suggest a particular diagnosis. A history of arthralgias and myalgias predating jaundice suggests hepatitis, either viral or drug-related. Jaundice associated with the sudden onset of severe right upper quadrant pain and shaking chills suggests choledocholithiasis and ascending cholangitis.

Physical Examination The general assessment should include assessment of the

patient's nutritional status. Temporal and proximal muscle wasting suggests longstanding diseases such as pancreatic cancer or cirrhosis. Stigmata of chronic liver disease, including spider nevi, palmar erythema, gynecomastia, caput medusae, Dupuytren's contractures, parotid gland enlargement, and testicular atrophy are commonly seen in advanced alcoholic (Laennec's) cirrhosis and occasionally in other types of cirrhosis. An enlarged left supraclavicular node (Virchow's node) or periumbilical nodule (Sister Mary Joseph's nodule) suggest an abdominal malignancy. Jugular venous distention, a sign of right-sided heart failure, suggests hepatic congestion. Right pleural effusion, in the absence of clinically apparent ascites, may be seen in advanced cirrhosis.

The abdominal examination should focus on the size and consistency of the liver, whether the spleen is palpable and hence enlarged, and whether there is ascites present. Patients with cirrhosis may have an enlarged left lobe of the liver which is felt below the xiphoid and an enlarged spleen. A grossly enlarged nodular liver or an obvious abdominal mass suggests malignancy. An enlarged tender liver could be viral or alcoholic hepatitis or, less often, an acutely congested liver secondary to right-sided heart failure. Severe right upper quadrant tenderness with respiratory arrest on inspiration (Murphy's sign) suggests cholecystitis or, occasionally, ascending cholangitis. Ascites in the presence of jaundice suggests either cirrhosis or malignancy with peritoneal spread.

Laboratory Tests When the physician encounters a patient with unexplained jaundice, there are a battery of tests that are helpful in the initial evaluation. These include total and direct serum bilirubin with fractionation, aminotransferases, alkaline phosphatase, albumin, and prothrombin time tests. Enzyme tests [alanine aminotransferase (ALT), aspartate aminotransferase (AST), and alkaline phosphatase] are helpful in differentiating between a hepatocellular process and a cholestatic process (see [Table 293-1](#) and [Fig. 45-1](#)), a critical step in determining what additional workup is indicated. Patients with a hepatocellular process generally have a disproportionate rise in the aminotransferases compared to the alkaline phosphatase. Patients with a cholestatic process have a disproportionate rise in the alkaline phosphatase compared to the aminotransferases. The bilirubin can be prominently elevated in both hepatocellular and cholestatic conditions and therefore is not necessarily helpful in differentiating between the two.

In addition to the enzyme tests, all jaundiced patients should have additional blood tests, specifically an albumin and a prothrombin time, to assess liver function. A low albumin suggests a chronic process such as cirrhosis or cancer. A normal albumin is suggestive of a more acute process such as viral hepatitis or choledocholithiasis. An elevated prothrombin time indicates either vitamin K deficiency due to prolonged jaundice and malabsorption of vitamin K or significant hepatocellular dysfunction. The failure of the prothrombin time to correct with parenteral administration of vitamin K indicates severe hepatocellular injury.

The results of the bilirubin, enzyme, albumin, and prothrombin time tests will usually indicate whether a jaundiced patient has a hepatocellular or a cholestatic disease. The causes and evaluation of each of these is quite different.

Hepatocellular Conditions Hepatocellular diseases that can cause jaundice include viral hepatitis, drug or environmental toxicity, alcohol, and end-stage cirrhosis from any cause ([Table 45-2](#)). Wilson's disease should be considered in young adults. Autoimmune hepatitis is typically seen in young to middle-aged women, but may affect men and women of any age. Alcoholic hepatitis can be differentiated from viral and toxin-related hepatitis by the pattern of the aminotransferases. Patients with alcoholic hepatitis typically have an [AST:ALT](#) ratio of at least 2:1. The AST rarely exceeds 300 U/L. Patients with acute viral hepatitis and toxin-related injury severe enough to produce jaundice typically have aminotransferases greater than 500 U/L, with the ALT greater than or equal to the AST. The degree of aminotransferase elevation can occasionally help in differentiating between hepatocellular and cholestatic processes. While ALT and AST values less than 8 times normal may be seen in either hepatocellular or cholestatic liver disease, values 25 times normal or higher are seen primarily in acute hepatocellular diseases. Patients with jaundice from cirrhosis can have normal or only slight elevations of the aminotransferases.

When the physician determines that the patient has a hepatocellular disease, appropriate testing for acute viral hepatitis includes a hepatitis A IgM antibody, a hepatitis B surface antigen and core IgM antibody, and a hepatitis C viral RNA test. It can take many weeks for the hepatitis C antibody to become detectable, making it an unreliable test if acute hepatitis C is suspected. Depending on circumstances, studies for hepatitis D, E, Epstein-Barr virus (EBV), and cytomegalovirus (CMV) may be indicated. Ceruloplasmin is the initial screening test for Wilson's disease. Testing for autoimmune hepatitis usually includes an antinuclear antibody and measurement of specific immunoglobulins.

Drug-induced hepatocellular injury can be classified either as predictable or unpredictable. Predictable drug reactions are dose-dependent and affect all patients who ingest a toxic dose of the drug in question. The classic example is acetaminophen hepatotoxicity. Unpredictable or idiosyncratic drug reactions are not dose-dependent and occur in a minority of patients. A great number of drugs can cause idiosyncratic hepatic injury. Environmental toxins are also an important cause of hepatocellular injury. Examples include industrial chemicals such as vinyl chloride, herbal preparations containing pyrrolizidine alkaloids (Jamaica bush tea), and the mushrooms *Amanita phalloides* or *verna* containing highly hepatotoxic amatoxins.

Cholestatic Conditions When the pattern of the liver tests suggests a cholestatic disorder, the next step is to determine whether it is intra- or extrahepatic cholestasis ([Fig. 45-1](#)). Distinguishing intrahepatic from extrahepatic cholestasis may be difficult. History, physical examination, and laboratory tests are often not helpful. The next appropriate test is an ultrasound. The ultrasound is inexpensive, does not expose the patient to ionizing radiation, and can detect dilation of the intra- and extrahepatic biliary tree with a high degree of sensitivity and specificity. The absence of biliary dilatation suggests intrahepatic cholestasis, while the presence of biliary dilatation indicates extrahepatic cholestasis. False-negative results occur in patients with partial obstruction of the common bile duct or in patients with cirrhosis or primary sclerosing cholangitis (PSC) where scarring prevents the intrahepatic ducts from dilating.

Although ultrasonography may indicate extrahepatic cholestasis, it rarely identifies the

site or cause of obstruction. The distal common bile duct is a particularly difficult area to visualize by ultrasound because of overlying bowel gas. Appropriate next tests include computed tomography (CT) and endoscopic retrograde cholangiopancreatography (ERCP). CT scanning is better than ultrasonography for assessing the head of the pancreas and for identifying choledocholithiasis in the distal common bile duct, particularly when the ducts are not dilated. ERCP is the gold standard for identifying choledocholithiasis. It is performed by introducing a side-viewing endoscope perorally into the duodenum. The ampulla of Vater is visualized and a catheter is advanced through the ampulla. Injection of dye allows for the visualization of the common bile duct and the pancreatic duct. The success rate for cannulation of the common bile duct ranges from 80 to 95%, depending on the operator's experience. Beyond its diagnostic capabilities, ERCP allows for therapeutic interventions, including the removal of common bile duct stones and the placement of stents. In patients in whom ERCP is unsuccessful, transhepatic cholangiography can provide the same information. Magnetic resonance cholangiopancreatography (MRCP) is a rapidly developing, noninvasive technique for imaging the bile and pancreatic ducts; this may replace ERCP as the initial diagnostic test in cases where the need for intervention is felt to be small.

In patients with apparent *intrahepatic cholestasis*, the diagnosis is often made by serologic testing in combination with percutaneous liver biopsy. The list of possible causes of intrahepatic cholestasis is long and varied ([Table 45-3](#)). A number of conditions that typically cause a hepatocellular pattern of injury can also present as a cholestatic variant. Both hepatitis B and C can cause a cholestatic hepatitis (fibrosing cholestatic hepatitis) that has histologic features that mimic large duct obstruction. This disease variant has been reported in patients who have undergone solid organ transplantation. Hepatitis A, alcoholic hepatitis, [EBV](#), and [CMV](#) may also present as cholestatic liver disease.

Drugs may cause intrahepatic cholestasis, a variant of drug-induced hepatitis. Drug-induced cholestasis is usually reversible after eliminating the offending drug, although it may take many months for cholestasis to resolve. Drugs most commonly associated with cholestasis are the anabolic and contraceptive steroids. Cholestatic hepatitis has been reported with chlorpromazine, imipramine, tolbutamide, sulindac, cimetidine, and erythromycin estolate. It also occurs in patients taking trimethoprim, sulfamethoxazole, and penicillin-based antibiotics such as ampicillin, dicloxacillin, and clavulanic acid. Rarely, cholestasis may be chronic and associated with progressive fibrosis despite early discontinuation of the drug. Chronic cholestasis has been associated with chlorpromazine and prochlorperazine.

Primary biliary cirrhosis is a disease predominantly of middle-aged women in which there is a progressive destruction of interlobular bile ducts. The diagnosis is made by the presence of the antimitochondrial antibody that is found in 95% of patients. [Primary sclerosing cholangitis \(PSC\)](#) is characterized by the destruction and fibrosis of larger bile ducts. The disease may involve only the intrahepatic ducts and present as intrahepatic cholestasis. However, in 65% of patients with PSC, both intra- and extrahepatic ducts are involved. The diagnosis of PSC is made by [ERCP](#). The pathognomonic findings are multiple strictures of bile ducts with dilatations proximal to the strictures. Approximately 75% of patients with PSC have inflammatory bowel disease.

The *vanishing bile duct syndrome* and *adult bile ductopenia* are rare conditions in which there are a decreased number of bile ducts seen in liver biopsy specimens. The histologic picture is similar to that found in primary biliary cirrhosis. This picture is seen in patients who develop chronic rejection after liver transplantation and in those who develop graft-versus-host disease after bone marrow transplantation. Vanishing bile duct syndrome also occurs in rare cases of sarcoidosis, in patients taking certain drugs including chlorpromazine, and idiopathically. There are also familial forms of intrahepatic cholestasis, including the *familial intrahepatic cholestatic syndromes, I-III*. Benign recurrent cholestasis is an autosomal recessive disease that appears to be due to mutations in a P type ATPase, which probably acts as a bile acid transporter. The disease is marked by recurrent episodes of jaundice and pruritis; the episodes are self-limited but can be debilitating. *Cholestasis of pregnancy* occurs in the second and third trimesters and resolves after delivery. Its cause is unknown, but the condition is probably inherited and cholestasis can be triggered by estrogen administration.

Other causes of intrahepatic cholestasis include total parenteral nutrition (TPN), nonhepatobiliary sepsis, benign postoperative cholestasis, and a paraneoplastic syndrome associated with a number of different malignancies, including Hodgkin's disease, medullary thyroid cancer, hypernephroma, renal sarcoma, T cell lymphoma, prostate cancer, and several GI malignancies. In patients developing cholestasis in the intensive care unit, the major considerations should be sepsis, shock liver, and TPN jaundice. Jaundice occurring after bone marrow transplantation is most likely due to venoocclusive disease or graft-versus-host disease.

Causes of *extrahepatic cholestasis* can be split into malignant and benign ([Table 45-3](#)). Malignant causes include pancreatic, gallbladder, ampullary, and cholangiocarcinoma. The latter is most commonly associated with [PSC](#) and is exceptionally difficult to diagnose because its appearance is often identical to PSC. Pancreatic and gallbladder tumors, as well as cholangiocarcinoma, are rarely resectable and have poor prognoses. Ampullary carcinoma has the highest surgical cure rate of all the tumors that present as painless jaundice. Hilar lymphadenopathy due to metastases from other cancers may cause obstruction of the extrahepatic biliary tree.

Choledocholithiasis is the most common cause of extrahepatic cholestasis. The clinical presentation can range from mild right upper quadrant discomfort with only minimal elevations of the enzyme tests to ascending cholangitis with jaundice, sepsis, and circulatory collapse. [PSC](#) may occur with clinically important strictures limited to the extrahepatic biliary tree. In cases where there is a dominant stricture, patients can be effectively managed with serial endoscopic dilatations. Chronic pancreatitis rarely causes strictures of the distal common bile duct, where it passes through the head of the pancreas. AIDS cholangiopathy is a condition, usually due to infection of the bile duct epithelium with [CMV](#) or cryptosporidium, which has a cholangiographic appearance similar to PSC. These patients usually present with greatly elevated serum alkaline phosphatase levels, mean of 800 IU/L, but the bilirubin is often near normal. These patients do not typically present with jaundice.

SUMMARY

The goal of this chapter is not to provide an encyclopedic review of all of the conditions that can cause jaundice. Rather, it is intended to provide a framework that helps a physician to evaluate the patient with jaundice in a logical way ([Fig. 45-1](#)).

Simply stated, the initial step is to obtain appropriate blood tests to determine if the patient has an isolated elevation of serum bilirubin. If so, is the bilirubin elevation due to an increased unconjugated or conjugated fraction? If the hyperbilirubinemia is accompanied by other liver test abnormalities, is the disorder hepatocellular or cholestatic? If cholestatic, is it intra- or extrahepatic? All of these questions can be answered with a thoughtful history, physical examination, and interpretation of laboratory and radiologic tests and procedures.

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46. ABDOMINAL SWELLING AND ASCITES - Robert M. Glickman

ABDOMINAL SWELLING

Abdominal swelling or distention is a common problem in clinical medicine and may be the initial manifestation of a systemic disease or of otherwise unsuspected abdominal disease. *Subjective* abdominal enlargement, often described as a sensation of fullness or bloating, is usually transient and is often related to a functional gastrointestinal disorder when it is not accompanied by objective physical findings of increased abdominal girth or local swelling. *Obesity* and lumbar lordosis, which may be associated with prominence of the abdomen, may usually be distinguished from true increases in the volume of the peritoneal cavity by history and careful physical examination.

Clinical History Abdominal swelling may first be noticed by the patient because of a progressive increase in belt or clothing size, the appearance of abdominal or inguinal hernias, or the development of a localized swelling. Often, considerable abdominal enlargement has gone unnoticed for weeks or months, either because of coexistent obesity or because the ascites formation has been insidious, without pain or localizing symptoms. Progressive abdominal distention may be associated with a sensation of "pulling" or "stretching" of the flanks or groins and vague low back pain. Localized pain usually results from involvement of an abdominal organ (e.g., a passively congested liver, large spleen, or colonic tumor). Pain is uncommon in cirrhosis with ascites, and when it is present, pancreatitis, hepatocellular carcinoma, or peritonitis should be considered. Tense ascites or abdominal tumors may produce increased intraabdominal pressure, resulting in indigestion and heartburn due to gastroesophageal reflux or dyspnea, orthopnea, and tachypnea from elevation of the diaphragm. A coexistent pleural effusion, more commonly on the right, presumably due to leakage of ascitic fluid through lymphatic channels in the diaphragm, also may contribute to respiratory embarrassment. The patient with diffuse abdominal swelling should be questioned about increased alcohol intake, a prior episode of jaundice or hematuria, or a change in bowel habits. Such historic information may provide the clues that will lead one to suspect an occult cirrhosis, a colonic tumor with peritoneal seeding, congestive heart failure, or nephrosis.

Physical Examination A carefully executed general physical examination can yield valuable clues concerning the etiology of abdominal swelling. Thus palmar erythema and spider angiomas suggest an underlying cirrhosis, while supraclavicular adenopathy (Virchow's node) should raise the question of an underlying gastrointestinal malignancy.

Inspection of the abdomen is important. By noting the abdominal contour, one may be able to distinguish localized from generalized swelling. The tensely distended abdomen with tightly stretched skin, bulging flanks, and everted umbilicus is characteristic of ascites. A prominent abdominal venous pattern with the direction of flow away from the umbilicus often is a reflection of portal hypertension; venous collaterals with flow from the lower part of the abdomen toward the umbilicus suggest obstruction of the inferior vena cava; flow downward toward the umbilicus suggests superior vena cava obstruction. "Doming" of the abdomen with visible ridges from underlying intestinal loops is usually due to intestinal obstruction or distention. An epigastric mass, with evident peristalsis proceeding from left to right, usually indicates underlying pyloric obstruction.

A liver with metastatic deposits may be visible as a nodular right upper quadrant mass moving with respiration.

Auscultation may reveal the high-pitched, rushing sounds of early intestinal obstruction or a succussion sound due to increased fluid and gas in a dilated hollow viscus. Careful auscultation over an enlarged liver occasionally reveals the harsh bruit of a vascular tumor, especially a hepatocellular carcinoma, or the leathery friction rub of a surface nodule. A venous hum at the umbilicus may signify portal hypertension and an increased collateral blood flow around the liver. A fluid wave and flank dullness that shifts with change in position of the patient are important signs that indicate the presence of peritoneal fluid. In obese patients, small amounts of fluid may be difficult to demonstrate; on occasion, the fluid may be detected by abdominal percussion with patients on their hands and knees. Small amounts of ascites often can only be detected by ultrasound examination of the abdomen, which can detect as little as 100 mL of fluid. Careful percussion should serve to distinguish generalized abdominal enlargement from localized swelling due to an enlarged uterus, ovarian cyst, or distended bladder. Percussion also can outline an abnormally small or large liver. Loss of normal liver dullness may result from massive hepatic necrosis; it also may be a clue to free gas in the peritoneal cavity, as from perforation of a hollow viscus.

Palpation is often difficult with massive ascites, and ballottement of overlying fluid may be the only method of palpating the liver or spleen. A slightly enlarged spleen in association with ascites may be the only evidence of an occult cirrhosis. When there is evidence of portal hypertension, a soft liver suggests that obstruction to portal flow is extrahepatic; a firm liver suggests cirrhosis as the likely cause of the portal hypertension. A very hard or nodular liver is a clue that the liver is infiltrated with tumor, and when accompanied by ascites, it suggests that the latter is due to peritoneal seeding. The presence of a hard periumbilical nodule (Sister Mary Joseph's nodule) suggests metastatic disease from a pelvic or gastrointestinal primary tumor. A pulsatile liver and ascites may be found in tricuspid insufficiency.

An attempt should be made to determine whether a mass is solid or cystic, smooth or irregular, and whether it moves with respiration. The liver, spleen, and gallbladder should descend with respiration unless they are fixed by adhesions or extension of tumor beyond the organ. A fixed mass not descending with respiration may indicate that it is retroperitoneal. Tenderness, especially if localized, may indicate an inflammatory process such as an abscess; it also may be due to stretching of the visceral peritoneum or tumor necrosis. Rectal and pelvic examinations are mandatory; they may reveal otherwise undetected masses due to tumor or infection.

Radiographic and laboratory examinations are essential for confirming or extending the impressions gained on physical examination. Upright and recumbent films of the abdomen may demonstrate the dilated loops of intestine with fluid levels characteristic of intestinal obstruction or the diffuse abdominal haziness and loss of psoas margins suggestive of ascites. Ultrasonography is often of value in detecting ascites, determining the presence of a mass, or evaluating the size of the liver and spleen. Computed tomography (CT) scanning provides similar information. CT scanning is often necessary to visualize the retroperitoneum, pancreas, and lymph nodes. A plain film of the abdomen may reveal the distended colon of otherwise unsuspected ulcerative colitis

and give valuable information as to the size of the liver and spleen. An irregular and elevated right side of the diaphragm may be a clue to a liver abscess or hepatocellular carcinoma. Studies of the gastrointestinal tract with barium or other contrast media are usually necessary in the search for a primary tumor.

ASCITES

The evaluation of a patient with ascites requires that the cause of the ascites be established. In most cases ascites appears as part of a well-recognized illness, that is, cirrhosis, congestive heart failure, nephrosis, or disseminated carcinomatosis. In these situations, the physician should determine that the development of ascites is indeed a consequence of the basic underlying disease and not due to the presence of a separate or related disease process. This distinction is necessary even when the cause of ascites seems obvious. For example, when the patient with compensated cirrhosis and minimal ascites develops progressive ascites that is increasingly difficult to control with sodium restriction or diuretics, the temptation is to attribute the worsening of the clinical picture to progressive liver disease. However, an occult hepatocellular carcinoma, portal vein thrombosis, spontaneous bacterial peritonitis, or even tuberculosis may be responsible for the decompensation. The disappointingly low success in diagnosing tuberculous peritonitis or hepatocellular carcinoma in the patient with cirrhosis and ascites reflects the too-low index of suspicion for the development of such superimposed conditions. Similarly, the patient with congestive heart failure may develop ascites from a disseminated carcinoma with peritoneal seeding.

Diagnostic paracentesis (50 to 100 mL) should be part of the routine evaluation of the patient with ascites. The fluid should be examined for its gross appearance; protein content, cell count, and differential cell count should be determined; and Gram's and acid-fast stains and culture should be performed. Cytologic and cell-block examination may disclose an otherwise unsuspected carcinoma. [Table 46-1](#) presents some of the features of ascitic fluid typically found in various disease states. In some disorders, such as cirrhosis, the fluid has the characteristics of a transudate (<25 g protein per liter and a specific gravity of <1.016); in others, such as peritonitis, the features are those of an exudate. Rather than the total protein content of ascites, many authors prefer the use of a *serum-ascites albumin gradient (SAG)* to characterize ascites. The gradient correlates directly with portal pressure. A gradient >1.1 g/dL (high gradient) is characteristic of uncomplicated cirrhotic ascites and differentiates ascites due to portal hypertension from ascites not due to portal hypertension >95% of the time. A gradient <1.1 g/dL (low gradient) suggests that the ascites is not due to portal hypertension with >95% accuracy and mandates a search for other causes ([Table 46-1](#)). Although there is variability of the ascitic fluid in any given disease state, some features are sufficiently characteristic to suggest certain diagnostic possibilities. For example, blood-stained fluid with >25 g protein per liter is unusual in uncomplicated cirrhosis but is consistent with tuberculous peritonitis or neoplasm. Cloudy fluid with a predominance of polymorphonuclear cells and a positive Gram's stain are characteristic of bacterial peritonitis; if most cells are lymphocytes, tuberculosis should be suspected. The complete examination of each fluid is most important, for occasionally only one finding may be abnormal. For example, if the fluid is a typical transudate but contains >250 white blood cells per microliter, the finding should be recognized as atypical for cirrhosis and should warrant a search for tumor or infection. This is especially true in the evaluation of cirrhotic ascites where

occult peritoneal infection may be present with only minor elevations in the white blood cell count of the peritoneal fluid (300 to 500 cells per microliter). Since Gram's stain of the fluid may be negative in a high proportion of such cases, careful culture of the peritoneal fluid is mandatory. Bedside inoculation of blood culture flasks with ascitic fluid results in a dramatically increased incidence of positive cultures when bacterial infection is present (90 versus 40% positivity with conventional cultures done by the laboratory). Direct visualization of the peritoneum (laparoscopy) may disclose peritoneal deposits of tumor, tuberculosis, or metastatic disease of the liver. Biopsies are taken under direct vision, often adding to the diagnostic accuracy of the procedure.

Chylous ascites refers to a turbid, milky, or creamy peritoneal fluid due to the presence of thoracic or intestinal lymph. Such a fluid shows Sudan-staining fat globules microscopically and an increased triglyceride content by chemical examination. Opaque milky fluid usually has a triglyceride concentration of >1000 mg/dL. A turbid fluid due to leukocytes or tumor cells may be confused with chylous fluid (pseudochylous), and it is often helpful to carry out alkalization and ether extraction of the specimen. Alkali tend to dissolve cellular proteins and thereby reduce turbidity; ether extraction leads to clearing if the turbidity of the fluid is due to lipid. Chylous ascites is most often the result of lymphatic obstruction from trauma, tumor, tuberculosis, filariasis ([Chap. 221](#)), or congenital abnormalities. It also may be seen in the nephrotic syndrome.

Rarely, ascitic fluid may be *mucinous* in character, suggesting either pseudomyxoma peritonei ([Chap. 289](#)) or rarely a colloid carcinoma of the stomach or colon with peritoneal implants.

On occasion, ascites may develop as a seemingly isolated finding in the absence of a clinically evident underlying disease. Then, a careful analysis of ascitic fluid may indicate the direction the evaluation should take. A useful framework for the workup starts with an analysis of whether the fluid is classified as a high (transudate) or low (exudate) gradient fluid. *High gradient (transudative) ascites* of unclear etiology is most often due to occult cirrhosis, right-sided venous hypertension raising hepatic sinusoidal pressure, or hypoalbuminemic states such as nephrosis or protein-losing enteropathy. Cirrhosis with well-preserved liver function (normal albumin) resulting in ascites invariably is associated with significant portal hypertension ([Chap. 298](#)). Evaluation should include liver function tests, liver-spleen scan, or other hepatic imaging procedure (i.e., CT or ultrasound) to detect nodular changes in the liver or a colloid shift of isotope to suggest portal hypertension. On occasion, a wedged hepatic venous pressure can be useful to document portal hypertension. Finally, if clinically indicated, a liver biopsy will confirm the diagnosis of cirrhosis and perhaps suggest its etiology. Other etiologies may result in hepatic venous congestion and resultant ascites. Right-sided cardiac valvular disease and particularly constrictive pericarditis should raise a high index of suspicion and may require cardiac imaging and cardiac catheterization for definitive diagnosis. Hepatic vein thrombosis is evaluated by visualizing the hepatic veins with imaging techniques (Doppler ultrasound, angiography, CT scans, magnetic resonance imaging) to demonstrate obliteration, thrombosis, or obstruction by tumor. Uncommonly, transudative ascites may be associated with benign tumors of the ovary, particularly fibroma (Meigs' syndrome) with ascites and hydrothorax.

Low gradient (exudative) ascites should initiate an evaluation for primary peritoneal

processes, most importantly infection and tumor. Routine bacteriologic culture of ascitic fluid often yields a specific organism causing infectious peritonitis. Tuberculous peritonitis ([Table 46-1](#)) is best diagnosed by peritoneal biopsy, either percutaneously or via laparoscopy. Histologic examination invariably shows granulomata that may contain acid-fast bacilli. Since cultures of peritoneal fluid and biopsies for tuberculosis may require 6 weeks, characteristic histology with appropriate stains allows antituberculosis therapy to be started promptly. Similarly, the diagnosis of peritoneal seeding by tumor can usually be made by cytologic analysis of peritoneal fluid or by peritoneal biopsy if cytology is negative. Appropriate diagnostic studies can then be undertaken to determine the nature and site of the primary tumor. Pancreatic ascites ([Table 46-1](#)) is invariably associated with an extravasation of pancreatic fluid from the pancreatic ductal system, most commonly from a leaking pseudocyst. Ultrasound or CT examination of the pancreas followed by visualization of the pancreatic duct by direct cannulation [viz., endoscopic retrograde cholangiopancreatography (ERCP)] usually discloses the site of leakage and permits resective surgery to be carried out.

An analysis of the physiologic and metabolic factors involved in the production of ascites (detailed in [Chap. 298](#)), coupled with a complete evaluation of the nature of the ascitic fluid, invariably discloses the etiology of the ascites and permits appropriate therapy to be instituted.

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SECTION 7 -ALTERATIONS IN RENAL AND URINARY TRACT FUNCTION

47. AZOTEMIA AND URINARY ABNORMALITIES - *Bradley M. Denker, Barry M. Brenner*

Body homeostasis is maintained predominantly through the cellular processes that together comprise normal kidney function. Disturbances to any of these functions can lead to a constellation of abnormalities that may be detrimental to survival. The clinical manifestations of these diseases will depend upon the pathophysiology of the renal injury and will often be initially identified as a complex of symptoms, abnormal physical findings, and laboratory changes that will allow the identification of specific syndromes. These renal syndromes (summarized in [Table 47-1](#)) may arise as the consequence of a systemic illness or can occur as a primary renal disease. Nephrologic syndromes usually consist of several elements that reflect the underlying pathologic processes and the duration of the disease and typically include one or more of the following features: (1) disturbances in urine volume (oliguria, anuria, polyuria); (2) abnormalities of urine sediment [red blood cells (RBC); white blood cells, casts, and crystals]; (3) abnormal excretion of serum proteins (proteinuria); (4) reduction in glomerular filtration rate (GFR) (azotemia); (5) presence of hypertension and/or expanded total body volume (edema); (6) electrolyte abnormalities, or (7) in some syndromes, fever/pain. The combination of these findings should permit identification of one of the major nephrologic syndromes ([Table 47-1](#)) and will allow the differential diagnoses to be narrowed and the appropriate diagnostic evaluation and therapeutic course to be determined. Each of these syndromes and their associated diseases are discussed in more detail in subsequent chapters. This chapter will focus on several aspects of renal abnormalities that are critically important to distinguishing these processes: (1) reduction in GFR leading to azotemia, (2) alterations of the urinary sediment and/or protein excretion, and (3) abnormalities of urinary volume.

AZOTEMIA

ASSESSMENT OF GLOMERULAR FILTRATION RATE

Monitoring the [GFR](#) is important in both the hospital and outpatient settings, and several different methodologies are available (discussed below). In most acute clinical circumstances a measured GFR is not available, and it is necessary to estimate the GFR from the serum creatinine level in order to provide appropriate doses of drugs that are excreted into the urine. Serum creatinine is the most widely used marker for GFR and is related directly to the urine creatinine excretion and inversely to the serum creatinine (U_{Cr}/P_{Cr}). Based upon this relationship and some important caveats (discussed below), the GFR will fall proportionately with the increase in P_{Cr} . Failure to account for GFR reductions in drug dosing can lead to significant morbidity and mortality from drug toxicities (e.g., digoxin, aminoglycosides). In the outpatient setting, serial determinations of GFR are helpful for following the progression of chronic renal insufficiency, but again, the serum creatinine is often used as a surrogate for GFR (although much less accurate; see below). In patients with chronic progressive renal insufficiency there is an approximately linear relationship between $1/P_{Cr}$ and time. The slope of this line will remain constant for an individual patient, and when values are obtained that do not fall on this line, an investigation for a superimposed acute process

(e.g., volume depletion, drug reaction) should be initiated. It should be emphasized that the signs and symptoms of uremia will develop at significantly different levels of serum creatinine depending upon the patient (size, age, and sex), the underlying renal disease, existence of concurrent diseases, and true GFR. In general, patients do not develop symptomatic uremia until renal insufficiency is usually quite severe ($\text{GFR} < 15 \text{ mL/min}$) and in some patients it does not occur until the $\text{GFR} < 5 \text{ mL/min}$.

A reduced [GFR](#) leads to retention of nitrogenous waste products (azotemia) such as serum urea nitrogen and creatinine. Azotemia may result from reduced renal perfusion, intrinsic renal disease, or postrenal processes (ureteral obstruction; see below and [Fig. 47-1](#)). Precise determination of GFR is problematic as both commonly used markers (urea and creatinine) have characteristics that affect their accuracy as markers of clearance. Urea clearance is generally an underestimate of GFR because of tubule urea reabsorption and may be as low as one-half of GFR measured by other techniques.

Creatinine is a small, freely filtered solute that varies little from day to day (since it is derived from muscle metabolism of creatine). However, serum creatinine can increase acutely from dietary ingestion of cooked meat. Creatinine can be secreted by the proximal tubule through an organic cation pathway. There are many clinical settings where a creatinine clearance is not available, and decisions concerning drug dosing must be made based on the serum creatinine. A formula that allows an estimate of creatinine clearance in men that accounts for age-related decreases in [GFR](#), body weight, and sex has been derived by Cockcroft-Gault:

This value should be multiplied 0.85 for women, since a lower fraction of the body weight is composed of muscle. The gradual loss of muscle from chronic illness, chronic use of glucocorticoids, or malnutrition can mask significant changes in [GFR](#) with small or imperceptible changes in serum creatinine. More accurate determinations of GFR are available using inulin clearance or radionuclide-labeled markers such as ^{125}I -iothalamate or EDTA. These methods are highly accurate due to precise quantitation and the absence of any renal reabsorption/secretion and should be used to follow GFR in patients in whom creatinine is not likely to be a reliable indicator (patients with decreased muscle mass secondary to age, malnutrition, concurrent illnesses).

Approach to the Patient

Once it has been established that [GFR](#) is reduced, the physician must decide if this represents acute or chronic renal failure. The clinical situation, history, and laboratory data often make this an easy distinction. However, the laboratory abnormalities characteristic of chronic renal failure, including anemia, hypocalcemia, and hyperphosphatemia, are often also present in patients presenting with acute renal failure. Radiographic evidence of renal osteodystrophy ([Chap. 270](#)) would be seen only in chronic renal failure but is a very late finding, and these patients are usually on dialysis. The urinalysis and renal ultrasound can occasionally facilitate distinguishing acute from chronic renal failure. An approach to the evaluation of azotemic patients is shown in [Fig. 47-1](#). Patients with advanced chronic renal insufficiency often have some proteinuria, nonconcentrated urine (isosthenuria), and small kidneys on ultrasound

characterized by increased echogenicity and cortical thinning. Treatment should be directed toward slowing the progression of renal disease and providing symptomatic relief for edema, acidosis, anemia, and hyperphosphatemia, as discussed in [Chap. 270](#). Acute renal failure ([Chap. 269](#)) can result from processes affecting renal blood flow (prerenal azotemia), intrinsic renal diseases (affecting vessels, glomeruli, or tubules), or postrenal processes (obstruction to urine flow in ureters, bladder, or urethra) ([Chap. 281](#)).

Prerenal Failure Decreased renal perfusion accounts for 40 to 80% of acute renal failure and, if appropriately treated, is readily reversible. The etiologies of prerenal azotemia include any cause of decreased circulating blood volume including volume loss (gastrointestinal hemorrhage, burns, diarrhea, diuretics), volume sequestration (pancreatitis, peritonitis, rhabdomyolysis), or decreased effective circulating volume (cardiogenic shock, sepsis). Renal perfusion can also be affected by reductions in cardiac output from peripheral vasodilatation (sepsis, drugs) or profound renal vasoconstriction [severe heart failure, hepatorenal syndrome, drugs (such as nonsteroidal anti-inflammatory drugs (NSAIDs))]. True, or "effective," hypovolemia leads to a fall in mean arterial pressure, which in turn triggers a series of neural and humoral responses that include activation of the sympathetic nervous and renin-angiotensin-aldosterone systems and ADH release. [GFR](#) is maintained by prostaglandin-mediated relaxation of afferent arterioles and angiotensin II-mediated constriction of efferent arterioles. Once the mean arterial pressure falls below 80 mmHg, there is a steep decline in GFR.

Blockade of prostaglandin production by [NSAIDs](#) can result in severe vasoconstriction and acute renal failure under these circumstances. Angiotensin-converting enzyme (ACE) inhibitors decrease efferent arteriolar tone and can decrease glomerular capillary perfusion pressure. Patients on NSAIDs and/or ACE inhibitors are most susceptible to hemodynamically mediated acute renal failure when blood volume is reduced for any reason. Patients with renal artery stenosis are dependent upon efferent arteriolar vasoconstriction for maintenance of glomerular filtration pressure and are particularly susceptible to precipitous decline in [GFR](#) when given ACE inhibitors.

Prolonged renal hypoperfusion can lead to acute tubular necrosis (ATN; an intrinsic renal disease discussed below). The urinalysis and urinary electrolytes can be useful in distinguishing prerenal azotemia from ATN ([Table 47-2](#)). The urine of patients with prerenal azotemia can be predicted from the stimulatory actions of norepinephrine, angiotensin II, ADH, and low tubule fluid flow on salt and water reabsorption from the urine. In prerenal conditions the tubules are intact, leading to a concentrated urine (>500 mosm), avid Na retention (urine Na concentration <20 mM/L; fractional excretion of Na $<1\%$), and $U_{Cr}/P_{Cr} > 40$ ([Table 47-2](#)). The prerenal urine sediment is usually normal or has occasional hyaline and granular casts, while the sediment of ATN is usually filled with cellular debris and muddy brown granular casts.

Intrinsic Renal Disease When prerenal and postrenal azotemia have been excluded as etiologies of renal failure, an intrinsic parenchymal renal disease is present. Intrinsic renal disease can arise from processes involving large renal vessels, microvasculature and glomeruli, or tubulointerstitium. Ischemic and toxic [ATN](#) account for about 90% of acute intrinsic renal failure. As outlined in [Fig. 47-1](#), the clinical setting and urinalysis are

helpful in separating the possible etiologies of acute intrinsic renal failure. Prerenal azotemia and ATN are part of a spectrum of renal hypoperfusion; evidence of structural tubule injury is present in ATN, whereas prompt reversibility occurs with prerenal azotemia upon restoration of adequate renal perfusion. Thus, ATN can often be distinguished from prerenal azotemia by urinalysis and urine electrolyte composition ([Table 47-2](#) and [Fig. 47-1](#)). Ischemic ATN is observed most frequently in patients who have undergone major surgery, trauma, severe hypovolemia, overwhelming sepsis, or extensive burns. Nephrotoxic ATN complicates the administration of many common medications, usually by inducing a combination of intrarenal vasoconstriction, direct tubule toxicity, and/or tubular obstruction. The kidney is vulnerable to toxic injury by virtue of its rich blood supply (25% of cardiac output) and its ability to concentrate and metabolize toxins. A diligent search for hypotension and nephrotoxins will usually uncover the specific etiology of ATN. Discontinuation of nephrotoxins and stabilizing blood pressure will often suffice without the need for dialysis while the tubules recover. **An extensive list of potential drugs and toxins implicated in ATN can be found in [Chap. 269](#).*

Processes that involve the tubules and interstitium can lead to acute renal failure. These include drug-induced interstitial nephritis (especially antibiotics, [NSAIDs](#), and diuretics), severe infections (both bacterial and viral), systemic diseases (e.g., systemic lupus erythematosus), or infiltrative disorders (e.g., sarcoid, lymphoma, or leukemia). A list of drugs associated with allergic interstitial nephritis can be found in [Chap. 277](#). The urinalysis usually shows mild to moderate proteinuria, hematuria, and pyuria (approximately 75% of cases) and occasionally white blood cell casts. The finding of RBC casts in interstitial nephritis has been reported but should prompt a search for glomerular diseases. Occasionally renal biopsy will be needed to distinguish among these possibilities. The finding of eosinophils in the urine is suggestive of allergic interstitial nephritis and is optimally observed by using a Hansel stain. The absence of eosinophiluria, however, does not exclude the possibility of acute interstitial nephritis.

Occlusion of large renal vessels including arteries and veins is an uncommon cause of acute renal failure. A significant reduction in [GFR](#) by this mechanism suggests bilateral processes or a unilateral process in a patient with a single functioning kidney. Renal arteries can be occluded with atheroemboli, thromboemboli, in situ thrombosis, aortic dissection, or vasculitis. Atheroembolic renal failure can occur spontaneously but is most often associated with recent aortic instrumentation. The emboli are cholesterol-rich and lodge in medium and small renal arteries leading to an eosinophil-rich inflammatory reaction. Atheroembolic acute renal failure often has a normal urinalysis but may contain eosinophils and casts. The diagnosis can be confirmed by renal biopsy, but this is often unnecessary when other stigmata of atheroemboli are present (livedo reticularis, distal peripheral infarcts, eosinophilia). Renal artery thrombosis may lead to mild proteinuria and hematuria, whereas renal vein thrombosis typically induces heavy proteinuria and hematuria. **These vascular catastrophes often require angiography for confirmation and are discussed in [Chap. 278](#).*

Diseases of glomeruli (glomerulonephritis or vasculitis) and the renal microvasculature (hemolytic uremic syndromes, thrombotic thrombocytopenic purpura, or malignant hypertension) usually present with various combinations of glomerular injury: proteinuria, hematuria, reduced [GFR](#), and alterations of Na excretion leading to

hypertension, edema, and circulatory congestion (acute nephritic syndrome). These findings may occur as primary renal diseases or as renal manifestations of systemic diseases. The clinical setting and other laboratory data will help distinguish primary renal from systemic diseases. The finding of [RBC casts](#) in the urine is an indication for early renal biopsy ([Fig. 47-1](#)) as the pathologic pattern has important implications for diagnosis, prognosis, and treatment. Hematuria without RBC casts can also be an indication of glomerular disease, and this evaluation is summarized in [Fig. 47-2](#). **A detailed discussion of glomerulonephritis and diseases of the microvasculature can be found in Chap. 274.*

Postrenal Azotemia Urinary tract obstruction accounts for fewer than 5% of cases of acute renal failure, but it is usually reversible and must be ruled out early in the evaluation ([Fig. 47-1](#)). Since a single kidney is capable of adequate clearance, acute renal failure from obstruction requires obstruction at the urethra or bladder outlet, bilateral ureteral obstruction, or unilateral obstruction in a patient with a single functioning kidney. Obstruction is usually diagnosed by the presence of ureteral dilatation on renal ultrasound. However, early in the course of obstruction or if the ureters are unable to dilate (such as encasement by pelvic tumors), the ultrasound examination may be negative. **The specific urologic conditions that cause obstruction are discussed in Chap. 281.*

Oliguria and Anuria Oliguria refers to a 24-h urine output of <500 mL, and anuria is the complete absence of urine formation. Anuria can be caused by total urinary tract obstruction, total renal artery or vein occlusion, and shock (manifested by severe hypotension and intense renal vasoconstriction). Cortical necrosis, [ATN](#), and rapidly progressive glomerulonephritis can occasionally cause anuria. Oliguria can accompany any cause of acute renal failure and carries a more serious prognosis for renal recovery in all conditions except prerenal azotemia. *Nonoliguria* refers to urine output in excess of 500 mL/day in patients with acute or chronic azotemia. With nonoliguric ATN, disturbances of potassium and hydrogen balance are less severe than in oliguric patients and recovery to normal renal function is usually more rapid.

ABNORMALITIES OF THE URINE

PROTEINURIA

The evaluation of proteinuria is shown schematically in [Fig. 47-3](#) and is typically initiated after colorimetric detection of proteinuria by dipstick examination. Current methods for measuring proteinuria vary significantly. The dipstick measurement detects mostly albumin and gives false-positive results when pH > 7.0 and the urine is very concentrated or contaminated with blood. A very dilute urine may obscure significant proteinuria on dipstick examination, and proteinuria that is not predominantly albumin will be missed. This is particularly important for the detection of Bence Jones proteins in the urine of patients with multiple myeloma. Tests to measure total urine concentration accurately rely on precipitation with sulfosalicylic or trichloroacetic acids. Currently, ultrasensitive dipsticks are available to measure microalbuminuria (30 to 300 mg/d), an early marker of glomerular disease that has been shown to predict glomerular injury in early diabetic nephropathy ([Fig. 47-3](#)).

The magnitude of proteinuria and the protein composition in the urine depend upon the mechanism of renal injury leading to protein losses. Large amounts of plasma proteins normally course through the glomerular capillaries but do not enter the urinary space. Both charge and size selectivity prevent virtually all of albumin, globulin, and other large-molecular-weight proteins from crossing the glomerular wall. However, if this barrier is disrupted, there can be leakage of plasma proteins into the urine (glomerular proteinuria; [Fig. 47-3](#)). Smaller proteins (<20 kDa) are freely filtered but are readily reabsorbed by the proximal tubule. Normal individuals excrete less than 150 mg/d of total protein and only about 30 mg/d of albumin. The remainder of the protein in the urine is secreted by the tubules (Tamm-Horsfall, IgA, and urokinase) or represents small amounts of filtered β_2 -microglobulin, apoproteins, enzymes, and peptide hormones. Another mechanism of proteinuria occurs when there is excessive production of an abnormal protein that exceeds the capacity of the tubule for reabsorption. This most commonly occurs with plasma cell dyscrasias such as multiple myeloma and lymphomas that are associated with monoclonal production of immunoglobulin light chains.

The normal glomerular endothelial cell forms a barrier penetrated by pores of about 100 nm that holds back cells and other particles but offers little impediment to passage of most proteins. The glomerular basement membrane traps most large proteins (>100 kDa), while the foot processes of epithelial cells (podocytes) cover the urinary side of the glomerular basement membrane and produce a series of narrow channels (slit diaphragms) to allow molecular passage of small solutes and water ([Fig. 47-4](#)). The channels are coated with anionic glycoproteins that are rich in glutamate, aspartate, and sialic acid, which are negatively charged at physiologic pH. This negatively charged barrier impedes the passage of anionic molecules such as albumin. Some glomerular diseases, such as minimal change disease, cause fusion of glomerular epithelial cell foot processes, resulting in predominantly "selective" ([Fig. 47-3](#)) loss of albumin. Other glomerular diseases can present with disruption of the basement membrane and slit diaphragms (e.g., by immune complex deposition), resulting in large amounts of protein losses that include albumin and other plasma proteins. The fusion of foot processes causes increased pressure across the capillary basement membrane, resulting in areas with larger pore sizes. The combination of increased pressure and larger pores results in significant proteinuria ("nonselective"; [Fig. 47-3](#)).

When the total daily excretion of protein exceeds 3.5 g, there is often associated hypoalbuminemia, hyperlipidemia, and edema (nephrotic syndrome; [Table 47-1](#)). However, total daily urinary protein excretion greater than 3.5 g can occur without the other features of the nephrotic syndrome in a variety of other renal diseases ([Fig. 47-3](#)). Plasma cell dyscrasias (multiple myeloma) can be associated with large amounts of excreted light chains in the urine, which may not be detected by dipstick (which detects mostly albumin). The light chains produced from these disorders are filtered by the glomerulus and overwhelm the reabsorptive capacity of the proximal tubule. A sulfosalicylic acid precipitate that is out of proportion to the dipstick estimate is suggestive of light chains (Bence Jones protein), and light chains typically redissolve upon warming of the precipitate. Renal failure from these disorders occurs through a variety of mechanisms including tubule obstruction (cast nephropathy) and light chain deposition ([Chap. 275](#)).

Hypoalbuminemia in nephrotic syndrome occurs through excessive urinary losses, increased renal catabolism, and inadequate hepatic synthesis. The resulting decrease in plasma oncotic pressure contributes to edema formation by altering the Starling forces and favoring fluid movement from capillaries to interstitium. The resulting homeostatic mechanisms designed to correct the decrease in effective intravascular volume contribute to edema formation in some patients. These mechanisms include activation of the renin-angiotensin system, antidiuretic hormone, and the sympathetic nervous system, which contribute to excessive renal salt and water reabsorption and can contribute to unrelenting edema.

The severity of edema correlates with the degree of hypoalbuminemia and is modified by other factors such as heart disease or peripheral vascular disease. The diminished plasma oncotic pressure and urinary losses of regulatory proteins appear to stimulate hepatic lipoprotein synthesis. The resulting hyperlipidemia results in lipid bodies (fatty casts, oval fat bodies) in the urine. Other proteins are lost in the urine, leading to a variety of metabolic disturbances. These include thyroxine-binding globulin, cholecalciferol-binding protein, transferrin, and metal-binding proteins. A hypercoagulable state frequently accompanies severe nephrotic syndrome due to urinary losses of antithrombin III, reduced serum levels of proteins S and C, hyperfibrinogenemia, and enhanced platelet aggregation. Some patients develop severe IgG deficiency with resulting defects in immunity. Many diseases (some listed in [Fig. 47-3](#)) and drugs can cause the nephrotic syndrome, and a complete list can be found in [Chap. 274](#).

HEMATURIA, PYURIA, AND CASTS

Isolated hematuria without proteinuria, other cells, or casts is often indicative of bleeding from the urinary tract. Normal red blood cell excretion is up to 2 million [RBCs](#) per day. Hematuria is defined as two to five RBCs per high-power field (HPF) and can be detected by dipstick. Common causes of isolated hematuria include stones, neoplasms, tuberculosis, trauma, and prostatitis. Gross hematuria with blood clots is almost never indicative of glomerular bleeding; rather, it suggests a postrenal source in the urinary collecting system. Evaluation of patients presenting with microscopic hematuria is outlined in [Fig. 47-2](#). A single urinalysis with hematuria is common and can result from menstruation, viral illness, allergy, exercise, or mild trauma. Annual urinalysis of servicemen over a 10-year period showed an incidence of 38%. However, persistent or significant hematuria (>three RBCs/HPF on three urinalyses, or single urinalysis with >100 RBCs, or gross hematuria) identified significant renal or urologic lesions in 9.1% of over 1000 patients. Even patients who are chronically anticoagulated should be investigated as outlined in [Fig. 47-2](#). The suspicion for urogenital neoplasms in patients with isolated painless hematuria (nondysmorphic RBCs) increases with age. Neoplasms are rare in the pediatric population, and isolated hematuria is more likely to be "idiopathic" or associated with a congenital anomaly. Hematuria with pyuria and bacteriuria is typical of infection and should be treated with antibiotics after appropriate cultures. Acute cystitis or urethritis in women can cause gross hematuria. Hypercalciuria and hyperuricosuria are also risk factors for unexplained isolated hematuria in both children and adults. In some of these patients (50 to 60%), reducing calcium and uric acid excretion through dietary interventions can eliminate the microscopic hematuria.

Isolated microscopic hematuria can be a manifestation of glomerular diseases. The [RBCs](#) of glomerular origin are often dysmorphic when examined by phase-contrast microscopy. Irregular shapes of RBCs may also occur due to pH and osmolarity changes found in the distal tubule. There is, however, significant observer variability in detecting dysmorphic RBCs, especially if a phase-contrast microscope is not available. The most common etiologies of isolated glomerular hematuria are IgA nephropathy, hereditary nephritis, and thin basement membrane disease. IgA nephropathy and hereditary nephritis can have episodic gross hematuria. A family history of renal failure is often present in patients with hereditary nephritis, and patients with thin basement membrane disease often have other family members with microscopic hematuria. A renal biopsy is needed for the definitive diagnosis of these disorders, which are discussed in more detail in [Chap. 275](#). Hematuria with dysmorphic RBCs, RBC casts, and protein excretion >500 mg/d is virtually diagnostic of glomerulonephritis. RBC casts form as RBCs that enter the tubular fluid become trapped in a cylindrical mold of gelled Tamm-Horsfall protein. Even in the absence of azotemia, these patients should undergo serologic evaluation and renal biopsy as outlined in [Fig. 47-2](#).

Isolated pyuria is unusual since inflammatory reactions in the kidney or collecting system are also associated with hematuria. The presence of bacteria suggests infection, and white blood cell casts with bacteria are indicative of pyelonephritis. White blood cells and/or white blood cell casts may also be seen in tubulointerstitial processes such as interstitial nephritis, systemic lupus erythematosus, and transplant rejection. In chronic renal diseases, degenerated cellular casts called *waxy casts* can be seen in the urine. *Broad casts* are thought to arise in the dilated tubules of enlarged nephrons that have undergone compensatory hypertrophy in response to reduced renal mass (i.e., chronic renal failure). A mixture of broad casts typically seen with chronic renal failure together with cellular casts and [RBCs](#) may be seen in smoldering processes such as chronic glomerulonephritis with active glomerulitis.

ABNORMALITIES OF URINE VOLUME

The volume of urine produced varies depending upon the fluid intake, renal function, and physiologic demands of the individual. See "Azotemia," above, for discussion of decreased (oliguria) or absent urine production (anuria). **The physiology of water formation and renal water conservation are discussed in [Chap. 268](#).*

POLYURIA

By history, it is often difficult for patients to distinguish urinary frequency (often of small volumes) from polyuria, and a 24-h urine collection is needed for evaluation ([Fig. 47-5](#)). It is necessary to determine if the polyuria represents a solute or water diuresis and if the diuresis is appropriate for the clinical circumstances. The average person excretes between 600 and 800 mosmol of solutes per day, primarily as urea and electrolytes. The urine osmolality can help distinguish a solute from water diuresis. If the urine output is >3 L/d (arbitrarily defined as polyuria) and the urine is dilute (<250 mosmol/L), then total mosmol excretion is normal and a water diuresis is present. This circumstance could arise from polydipsia, inadequate secretion of vasopressin (central diabetes insipidus), or failure of renal tubules to respond to vasopressin (nephrogenic diabetes insipidus). If the urine volume is >3 L/d and urine osmolality is >300 mosmol/L, then a

solute diuresis is clearly present and a search for the responsible solute(s) is mandatory.

Excessive filtration of a poorly reabsorbed solute such as glucose, mannitol, or urea can depress reabsorption of NaCl and water in the proximal tubule and lead to enhanced excretion in the urine. Poorly controlled diabetes mellitus is the most common cause of a solute diuresis, leading to volume depletion and serum hypertonicity. Since the urine Na concentration is less than that of blood, more water than Na is lost, causing hyponatremia and hypertonicity. Common iatrogenic solute diuresis occurs from mannitol administration, radiocontrast media, and high-protein feedings (enterally or parenterally), leading to increased urea production and excretion. Less commonly, excessive Na loss may occur from cystic renal diseases, Bartter's syndrome, or during the course of a tubulointerstitial process (such as resolving [ATN](#)). In these so-called salt-wasting disorders, the tubule damage results in direct impairment of Na reabsorption and indirectly reduces the responsiveness of the tubule to aldosterone. Usually, the Na losses are mild, and the obligatory urine output is less than 2 L/d (resolving ATN and postobstructive diuresis are exceptions and may be associated with significant natriuresis and polyuria.)

Formation of large volumes of dilute urine represent polydipsic states or diabetes insipidus. Primary polydipsia can result from habit, psychiatric disorders, neurologic lesions, or medications. During deliberate polydipsia, extracellular fluid volume is normal or expanded and vasopressin levels are reduced because serum osmolality tends to be near the lower limits of normal.

Central diabetes insipidus may be idiopathic in origin or secondary to a variety of hypothalamic conditions including posthypophysectomy or trauma or neoplastic, inflammatory, vascular, or infectious hypothalamic diseases. Idiopathic central diabetes insipidus is associated with selective destruction of the vasopressin-secreting neurons in the supraoptic and paraventricular nuclei and can be inherited as an autosomal dominant trait or occur spontaneously. Nephrogenic diabetes insipidus can occur in a variety of clinical situations as summarized in [Fig. 47-5](#).

A plasma vasopressin level is recommended as the best method for distinguishing between central and nephrogenic diabetes insipidus. Alternatively, a water deprivation test plus exogenous vasopressin may also distinguish primary polydipsia from central and nephrogenic diabetes insipidus. **For a detailed discussion, see [Chap. 329](#).*

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48. INCONTINENCE AND LOWER URINARY TRACT SYMPTOMS - *Philippe E. Zimmern, John D. McConnell*

PHYSIOLOGY OF VOIDING

Normal bladder filling depends on unique elastic properties of the bladder wall that allow it to increase in volume at a pressure lower than that of the bladder neck and urethra (otherwise incontinence would occur). Despite provocative maneuvers such as coughing, voluntary bladder contractions do not occur. Emptying is dependent on the integrity of a complex neuromuscular network that causes relaxation of the urethral sphincter a few milliseconds before the onset of the detrusor (bladder muscle) contraction. With normal, sustained detrusor contraction, the bladder empties completely. A bladder that can fill and empty in this manner has a normal detrusor muscle and is described as *stable* according to conventional terminology.

Since the voluntary control of micturition depends on the neural connections between the cerebral cortex and the brainstem, disruption of these pathways (brain tumor, stroke, head trauma, Parkinson's disease) impairs the ability to suppress and control bladder contractions. A bladder contraction without voluntary effort characterizes an unstable bladder. Bladder or detrusor instability of neurologic origin is termed *detrusor hyperreflexia*. Conversely, the detrusor muscle that cannot contract during voiding is called *noncontractile*; underactivity of the detrusor due to a lesion of the sacral cord or pelvic nerves is termed *detrusor areflexia*.

Contrary to common belief, the center that controls normal micturition is not in the spinal cord but in the brainstem. Proper coordination (*synergia*) between the detrusor and urethral sphincters requires an intact neural (autonomic and somatic nervous systems) communication between bladder and urethra. Injury to the upper spinal cord, for example, can cause dyssynergia between bladder and urethra that results in urge incontinence, residual urine retention, bladder wall changes (trabeculation and fibrosis), and possibly renal insufficiency.

A simple way to classify voiding dysfunction is to determine whether it is primarily a *storage failure* or an *emptying failure* by asking two questions:

Is the voiding dysfunction due to the bladder or outlet (bladder neck or urethra) (failure to store)?

Is there neurologic dysfunction (failure to empty)?

Bladder storage and emptying problems may coexist in the same individual and can cause similar lower urinary tract symptoms (LUTS).

LOWER URINARY TRACT SYMPTOMS IN MEN

The most common cause of [LUTS](#) in men of middle age and older is prostatic hyperplasia, which causes obstruction to urine flow by encroachment on the urethral lumen ([Chap. 95](#)). Histologically, 50 to 80% of the prostatic volume is composed of stromal tissue (smooth muscle), while the remainder is glandular. The transitional zone,

which is responsible for benign prostatic growth, comprises 10 to 15% of the prostate at the end of puberty but increases in volume after age 40. However, prostatic enlargement is not always accompanied by symptoms because the direction of growth can be outward, so that little change may occur in urine flow. Alternatively, men with early histologic evidence of prostatic hyperplasia can experience significant voiding symptoms. In this circumstance, increased tone of the prostatic smooth muscle and enhanced prostatic tension within a nondistensible capsule can cause obstruction.

In response to obstruction, the bladder smooth-muscle cells hypertrophy to generate the higher pressures necessary for voiding, and the increase in bladder muscle mass leads to reduced elasticity, or compliance, and decreased bladder capacity. Detrusor dysfunction from bladder outlet obstruction can cause any combination of the [LUTS](#) described above. When the obstruction progresses, infiltration of extracellular matrix between the smooth-muscle bundles of the bladder wall can result in a hypocontractile or acontractile bladder (bladder failure).

Other complications such as urinary tract infections or bladder stones secondary to the large postvoid residuals (stasis) and upper tract damage (hydronephrosis, reflux) can develop during the course of the obstructive process. Although prostatic hyperplasia is the most common cause of bladder outlet obstruction in men, other sources of obstruction include prostate cancer, urethral stricture, and lack of proper sphincteric relaxation (neurologic cause). Nonobstructive causes of [LUTS](#) include diabetic neuropathy, which can affect the parasympathetic nerves of the bladder. Decreased sensation of bladder fullness leads to incomplete emptying and overdistention of the bladder and, in turn, to increased frequency and nocturia due to bladder overflow; these symptoms are frequently made worse by the polydipsia/polyuria of diabetes mellitus. At times, storage symptoms can be caused by other neurologic causes such as stroke, multiple sclerosis, or Parkinson's disease.

The International Prostate Symptom Score (IPSS) is used to assess the severity of [LUTS](#):

Decreased force of stream -- over the past month how often have you had a weak urinary stream?

Intermittency -- over the past month how often have you found you stopped and started again several times when urinating?

Incomplete emptying -- over the past month how often have you had a sensation of not emptying your bladder completely after finishing urination?

Straining -- over the past month how often have you had to push or strain to begin urination? The [IPSS](#) also assesses the impact of storage symptoms:

Frequency -- over the past month how often have you had to urinate again within 2 h after urinating?

Urgency -- over the past month how often have you found it difficult to postpone urination?

Nocturia -- over the past month how many times did you typically get up to urinate between going to bed and getting up in the morning? (Range: none to five or more times.) Except for nocturia, the answers range from 0 (not at all) to 5 (almost always). A total score of <8 indicates minimal voiding dysfunction; a total score of ≥ 13 is usually required to enroll patients in drug studies for the management of benign prostatic hyperplasia (BPH); and symptom scores >23 suggest significant bladder outlet obstruction. Because similar symptoms can result from neurologic causes, the [IPSS](#) questionnaire cannot be used to make the diagnosis of prostatic hyperplasia but is useful only as an index of severity and of the response to treatment.

LOWER URINARY TRACT SYMPTOMS IN WOMEN

Urethral obstruction is an uncommon cause of [LUTS](#) in women. A careful bimanual examination and passage of a urethral catheter are sufficient to exclude urethral stenosis, which is usually secondary to prior instrumentation or operative procedures, and urethral cancer. Urinary tract infection (cystitis) is more prevalent in women and must be excluded by urinalysis. Multiple sclerosis should be considered in middle-aged women presenting with frequency, urgency, or incontinence. In addition to many of the same disorders that produce voiding symptoms in men, estrogen deficiency, frequency-urgency syndrome, and interstitial cystitis (IC) with minimal pain must be considered. Cystocele and pelvic prolapse can cause urinary frequency secondary to impairment of bladder emptying.

EVALUATION

Men and women with [LUTS](#) and concomitant neurologic disease should undergo a complete urodynamic evaluation. In the absence of neurologic disease, men with LUTS most commonly have prostatic hyperplasia. However, it is necessary to exclude prostate cancer, especially if there is a positive family history, an abnormal prostate examination, or an elevated level of prostate-specific antigen (PSA). In both sexes bladder cancer can also cause storage symptoms and is suggested by microscopic hematuria and/or abnormal urine cytology. Usually, a detailed genitourinary history, a symptom assessment, a careful neurologic examination including rectal examination and assessment of the bulbocavernosus reflex, measurements of urine flow and postvoid residual urine volume (by bladder ultrasound), and limited laboratory evaluation (urinalysis, urine culture, PSA levels, urine cytology, urea/creatinine levels, as indicated) should be sufficient to direct therapy. More complex investigations of the lower urinary tract (cystoscopy, voiding cystography, urodynamics) and upper urinary tract (pyelogram or ultrasonography) are sometimes indicated. **For therapy of BPH, see [Chap. 95](#).*

INCONTINENCE

Incontinence is a condition where involuntary loss of urine is objectively demonstrated and is a social or hygienic problem. A common variant, *stress incontinence*, denotes involuntary loss of urine with physical exercise (coughing, sneezing, sports, sexual activity). *Urge incontinence* is an involuntary loss of urine associated with a strong desire to void, and *overflow incontinence* is an involuntary loss of urine when the

elevation of intravesical pressure with bladder overfilling or distention exceeds the maximal urethral pressure. Loss of urine through channels other than the urethra is rare (ectopic ureter, fistulae) but causes total or continuous incontinence.

INCONTINENCE IN WOMEN

Among noninstitutionalized women 60 years of age and older, 25 to 30% have urinary incontinence daily or weekly, and approximately half of institutionalized women are incontinent more than once a day. The annual cost of caring for incontinent persons is very high and, if not well managed, can be associated with complications such as decubitus ulcers.

Stress urinary incontinence (SUI) is secondary to urethral hypermobility or, less commonly (<10%), to intrinsic sphincteric deficiency (ISD). In the continent woman the bladder neck and proximal urethra are supported by the anterior vaginal wall and its lateral attachment to the levator muscles. Anterior vaginal wall relaxation causes urethral hypermobility, usually due to aging and/or estrogen deficiency or a prior traumatic delivery or pelvic surgery. Paradoxically, women can have clinical evidence of urethral hypermobility but no stress urinary incontinence.

Some women have an anatomically normal urethra and bladder neck but still have [SUI](#) due to damage to the internal sphincter (fixed, rigid, or "pipestem" urethra), due to prior anti-incontinence surgery, pelvic radiation or trauma, or neurologic disorders that cause denervation of the urethra. Urethral hypermobility and [ISD](#) can coexist in some patients and cause persistence (or rapid recurrence) of incontinence after a simple bladder neck suspension procedure that fixes the hypermobility but leaves the sphincter untreated.

Urge incontinence can be present alone or in association with [SUI](#) (mixed incontinence). The cause of the unsuppressible or uninhibited bladder contractions is usually idiopathic, but bacterial cystitis, bladder tumor, bladder outlet obstruction, and neurogenic bladder must be excluded. Overflow incontinence is due either to bladder outlet obstruction (rare in women), an acontractile bladder (diabetic neuropathy, multiple sclerosis), excessive smooth-muscle relaxation from drugs (anticholinergic medications), or psychogenic retention.

INCONTINENCE IN MEN

In men incontinence is less common than obstruction, but urgency and urge incontinence can occur as the result of bladder outlet obstruction (as from prostatic hyperplasia) that impairs detrusor smooth-muscle function and leads to detrusor instability. Men with neurogenic bladders (diabetic neuropathy, multiple sclerosis, Parkinson's disease, stroke) can develop urge incontinence. Other causes such as bacterial cystitis or bladder tumor must be excluded. [SUI](#) in men is usually the result of distal sphincteric damage, for example, as the result of radical prostatectomy for prostate cancer.

INCONTINENCE IN THE ELDERLY

Transient urinary incontinence is common in the elderly. A mnemonic devised by

Resnick delineates its numerous causes, namely *delirium*, *infection*, *atrophic urethritis*, *pharmacologic*, *psychological*, *excessive urine output* (hyperglycemia, congestive heart failure), *restricted mobility*, and *stool impaction* (DIAPPERS). Urge incontinence is the next most common disorder in this age group and is attributed to the progressive loss of the modulating influence of the frontal lobes of the cortex on the micturition center in the brainstem.

EVALUATION

The evaluation of urinary incontinence in women should include history and quality-of-life assessment, voiding diary, physical examination including pelvic examination, urinalysis and urine culture, and measurement of postvoid residual urine volume. For patients with an unclear history or after prior pelvic or anti-incontinence procedures, evaluation may include cystoscopy, urodynamic evaluation, and imaging studies (lower and/or upper urinary tract). The history should define the onset, duration, evolution, and triggering events of leakage. Prior treatments with medications, frequent voiding schedules, and exercise regimens should be noted. Severity of incontinence is denoted by recording the type and number of pads used per day or at night and how the incontinence affects daily activities (incontinence-impact questionnaire). The amount and type of fluid consumed, sexual history (hormonal status, deliveries, venereal diseases), gastrointestinal function (fecal incontinence, constipation), and past urologic history (bed-wetting, surgeries) must also be documented. The physical examination should place special emphasis on the abdominal, genital, pelvic (associated prolapses), and neurologic systems. [SUI](#) must be demonstrated by asking the patient to cough, strain, or even stand or squat. While leakage during a cough confirms SUI, leakage after a cough is due to bladder instability (stress-induced instability). SUI in the absence of urethral hypermobility raises the suspicion of a sphincteric defect. More complex testing is needed to determine whether the urethral anatomy is normal (evaluation of urethral mobility, lateral view of the urethra on the voiding cystourethrogram, cystoscopy), whether urethral function is normal with adequate closure (leak point pressure, urethral profilometry, videourodynamics), or whether bladder function is normal (bladder volume based on home diary, filling cystometrogram).

TREATMENT

Mild stress incontinence can be treated nonoperatively with medications, estrogen replacement, or biofeedback techniques. Modalities such as urethral plugs and anterior vaginal wall prostheses are under investigation. Moderate to severe stress incontinence responds to surgical procedures aimed at supporting the anterior vaginal wall (vaginal, laparoscopic, or abdominal operations) or enhancing urethral closure when stress incontinence is secondary to internal sphincter deficiency (periurethral injection of fat or collagen, autologous or cadaveric fascial sling, synthetic sling, or insertion of an artificial urinary sphincter).

Urge incontinence responds to the management of its cause. When it is due to neurogenic or idiopathic causes, anticholinergic agents are partially effective, although side effects such as mouth dryness, blurring of vision, or constipation can limit their usefulness. Better tolerated medications are now available including slow-release oxybutinin (Ditropan XL), which is administered as 5- to 10-mg tablets once daily, and

the more specific antimuscarinic agent, tolterodine (Detrol), which is usually given as 2 mg orally twice daily. Fluid restriction (which must be undertaken only with great caution) and bladder retraining with biofeedback may also be helpful. More aggressive intervention with bladder augmentation or urinary diversion are seldom necessary in the absence of neurologic disease.

BLADDER PAIN

Painful bladder disease is a general term for any bladder pathology that causes suprapubic, urethral, or pelvic pain. [IC](#) is the most common cause of bladder pain, but endometriosis, bacterial cystitis, and outlet obstruction that causes bladder instability can mimic the symptoms of IC.

INTERSTITIAL CYSTITIS

[IC](#) is a severe, chronic bladder disorder that causes frequency, nocturia, and suprapubic pain. The disorder usually affects women and is rare in blacks. Routine urine culture is uniformly negative, and the symptoms do not respond to antibiotic therapy. The etiology is probably multifactorial ([Table 48-1](#)). Current hypotheses as to etiology include autoimmune reaction against bladder antigens, deficiency in the glycosaminoglycan layer of the bladder surface allowing presumed toxins to penetrate the mucosa, mast cell infiltration and activation leading to the histamine release, and local bladder wall damage from bacteria.

The National Institutes of Health has established a series of criteria to define [IC](#) clinically ([Table 48-2](#)). The diagnosis is one of exclusion -- infection, radiation cystitis, urethral diverticula, herpes simplex, and malignancy must be excluded. Cystoscopy under anesthesia may be used for the following: (1) reveal glomerulations (submucosal vascular anomalies) or the infrequent Hunner's ulcer suggestive of IC; (2) make it possible to estimate bladder capacity (an important guide to treatment); (3) allow biopsy of the bladder wall when indicated; and (4) by bladder filling, sometimes provide therapeutic benefit with a reduction in pain level and urinary frequency up to 6 months or rarely longer.

EVALUATION

Chronic urinary frequency and bladder pain affect the quality of life to an extreme degree, though most patients experience a waxing and waning evolution; only 10% of patients have a consistent progression in symptoms. Evaluation should include a detailed history; physical examination designed to exclude neurologic and gynecologic pathology; voiding cystogram to exclude urethral defects; and urodynamic testing to eliminate a neurogenic bladder, bladder instability, or outlet obstruction and to document sensory instability. Referral to specialists may be indicated to exclude adnexal pathology, endometriosis, or bowel dysfunction or to utilize modern pain management techniques to prevent drug addiction.

TREATMENT

Empirical treatments that have been used include oral medications (amitriptyline,

hydroxyzine, pentosanpolysulfate) and intravesical agents (dimethyl sulfoxide, chlorpactin, heparin). These measures may improve the urinary symptoms and occasionally reduce pain but do not modify the long-term course. Surgical intervention (augmentation cystoplasty, urinary diversion) is indicated in fewer than 5% of cases because this is a non-life-threatening, chronic disease with occasional spontaneous remissions. "Last-resort" interventions such as removal of the bladder and urethra are not a guarantee of success because some patients continue to experience pelvic pain afterwards.

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49. FLUID AND ELECTROLYTE DISTURBANCES - Gary G. Singer, Barry M. Brenner

SODIUM AND WATER

COMPOSITION OF BODY FLUIDS

Water is the most abundant constituent in the body, comprising approximately 50% of body weight in women and 60% in men. This difference is attributable to differences in the relative proportions of adipose tissue in men and women. Total body water is distributed in two major compartments -- 55 to 75% is intracellular [intracellular fluid (ICF)], and 25 to 45% is extracellular [extracellular fluid (ECF)]. The [ECF](#) is further subdivided into intravascular (plasma water) and extravascular (interstitial) spaces in a ratio of 1:3.

The solute or particle concentration of a fluid is known as its *osmolality* and is expressed as milliosmoles per kilogram of water (mosmol/kg). Water crosses cell membranes to achieve osmotic equilibrium ([ECF](#) osmolality = [ICF](#) osmolality). The extracellular and intracellular solutes or osmoles are markedly different due to disparities in permeability and the presence of transporters and active pumps. The major ECF particles are Na^+ and its accompanying anions Cl^- and HCO_3^- , whereas K^+ and organic phosphate esters (ATP, creatine phosphate, and phospholipids) are the predominant ICF osmoles. Solutes that are restricted to the ECF or the ICF determine the *effective osmolality* (or *tonicity*) of that compartment. Since Na^+ is largely restricted to the extracellular compartment, total body Na^+ content is a reflection of ECF volume. Likewise, K^+ and its attendant anions are predominantly limited to the ICF and are necessary for normal cell function. Therefore, the number of intracellular particles is relatively constant, and a change in ICF osmolality is usually due to a change in ICF water content. However, in certain situations, brain cells can vary the number of intracellular solutes in order to defend against large water shifts. This process of *osmotic adaptation* is important in the defense of cell volume and occurs in chronic hyponatremia and hypernatremia. This response is mediated initially by transcellular shifts of K^+ and Na^+ , followed by synthesis, import, or export of organic solutes (so-called osmolytes) such as inositol, betaine, and glutamine. During chronic hyponatremia, brain cells lose solutes, thereby defending cell volume and diminishing neurologic symptoms. The converse occurs during chronic hypernatremia. Certain solutes, such as urea, do not contribute to water shift across cell membranes and are known as *ineffective osmoles*.

Fluid movement between the intravascular and interstitial spaces occurs across the capillary wall and is determined by the Starling forces -- capillary hydraulic pressure and colloid osmotic pressure. The transcapillary hydraulic pressure gradient exceeds the corresponding oncotic pressure gradient, thereby favoring the movement of plasma ultrafiltrate into the extravascular space. The return of fluid into the intravascular compartment occurs via lymphatic flow.

WATER BALANCE (See also [Chap. 268](#))

The normal plasma osmolality is 275 to 290 mosmol/kg and is kept within a narrow range by mechanisms capable of sensing a 1 to 2% change in tonicity. To maintain a

steady state, water intake must equal water excretion. Disorders of water homeostasis result in hypo- or hypernatremia. Normal individuals have an obligate water loss consisting of urine, stool, and evaporation from the skin and respiratory tract. Gastrointestinal excretion is usually a minor component of total water output, except in patients with vomiting, diarrhea, or high enterostomy output states. Evaporative or insensitive water losses are important in the regulation of core body temperature. Obligatory renal water loss is mandated by the minimum solute excretion required to maintain a steady state. Normally, about 600 mosmol must be excreted per day, and since the maximal urine osmolality is 1200 mosmol/kg a minimum urine output of 500 mL/d is required for neutral solute balance.

Water Intake The primary stimulus for water ingestion is *thirst*, mediated either by an increase in effective osmolality or a decrease in [ECF](#) volume or blood pressure. *Osmoreceptors*, located in the anterolateral hypothalamus, are stimulated by a rise in tonicity. Ineffective osmoles, such as urea and glucose, do not play a role in stimulating thirst. The average osmotic threshold for thirst is approximately 295 mosmol/kg and varies among individuals. Under normal circumstances, daily water intake exceeds physiologic requirements.

Water Excretion In contrast to the ingestion of water, its excretion is tightly regulated by physiologic factors. The principal determinant of renal water excretion is *arginine vasopressin* (AVP; formerly antidiuretic hormone), a polypeptide synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and secreted by the posterior pituitary gland. The binding of AVP to V₂ receptors on the basolateral membrane of principal cells in the collecting duct activates adenylyl cyclase and initiates a sequence of events that leads to the insertion of water channels into the luminal membrane. These water channels that are specifically activated by AVP are encoded by the *aquaporin-2* gene ([Chap. 329](#)). The net effect is passive water reabsorption along an osmotic gradient from the lumen of the collecting duct to the hypertonic medullary interstitium. The major stimulus for AVP secretion is hypertonicity. Since the major [ECF](#) solutes are Na⁺ salts, effective osmolality is primarily determined by the plasma Na⁺ concentration. An increase or decrease in tonicity is sensed by hypothalamic osmoreceptors as a decrease or increase in cell volume, respectively, leading to enhancement or suppression of AVP secretion. The osmotic threshold for AVP release is 280 to 290 mosmol/kg, and the system is sufficiently sensitive that plasma osmolality varies by no more than 1 to 2%.

Nonosmotic factors that regulate [AVP](#) secretion include *effective circulating* (arterial) *volume*, nausea, pain, stress, hypoglycemia, pregnancy, and numerous drugs. The hemodynamic response is mediated by baroreceptors in the carotid sinus. The sensitivity of these receptors is significantly lower than that of the osmoreceptors. In fact, depletion of blood volume sufficient to result in a decreased mean arterial pressure is necessary to stimulate AVP release, whereas small changes in effective circulating volume have little effect. In the setting of hypovolemia, the osmotic regulation of AVP remains intact. However, the osmotic threshold, or set point, for AVP release is decreased, and the sensitivity is increased.

To maintain homeostasis and a normal plasma Na⁺ concentration, the ingestion of solute-free water must eventually lead to the loss of the same volume of electrolyte-free

water. Three steps are required for the kidney to excrete a water load: (1) filtration and delivery of water (and electrolytes) to the diluting sites of the nephron; (2) active reabsorption of Na^+ and Cl^- without water in the thick ascending limb of the loop of Henle and, to a lesser extent, in the distal nephron; and (3) maintenance of a dilute urine due to impermeability of the collecting duct to water in the absence of [AVP](#). Abnormalities of any of these steps can result in impaired free water excretion, and eventual hyponatremia.

SODIUM BALANCE

Sodium is actively pumped out of cells by the Na^+, K^+ -ATPase pump. As a result, 85 to 90% of all Na^+ is extracellular, and the [ECF](#) volume is a reflection of total body Na^+ content. Normal volume regulatory mechanisms ensure that Na^+ loss balances Na^+ gain. If this does not occur, conditions of Na^+ excess or deficit ensue and are manifest as edematous or hypovolemic states, respectively. It is important to distinguish between disorders of osmoregulation and disorders of volume regulation since water and Na^+ balance are regulated independently. Changes in Na^+ concentration generally reflect disturbed water homeostasis, whereas alterations in Na^+ content are manifest as ECF volume contraction or expansion and imply abnormal Na^+ balance.

Sodium Intake Individuals eating a typical western diet consume approximately 150 mmol of NaCl daily. This normally exceeds basal requirements. As noted above, sodium is the principal extracellular cation. Therefore, dietary intake of Na^+ results in [ECF](#) volume expansion, which in turn promotes enhanced renal Na^+ excretion to maintain steady state Na^+ balance.

Sodium Excretion (See also [Chap. 268](#)) The regulation of Na^+ excretion is multifactorial and is the major determinant of Na^+ balance. A Na^+ deficit or excess is manifest as a decreased or increased effective circulating volume, respectively. Changes in effective circulating volume tend to lead to parallel changes in glomerular filtration rate (GFR). However, tubule Na^+ reabsorption, and not GFR, is the major regulatory mechanism controlling Na^+ excretion. Almost two-thirds of filtered Na^+ is reabsorbed in the proximal convoluted tubule -- this process is electroneutral and isoosmotic. Further reabsorption (25 to 30%) occurs in the thick ascending limb of the loop of Henle via the apical *$\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ cotransporter* -- this is an active process and is also electroneutral. Distal convoluted tubule reabsorption of Na^+ (5%) is mediated by the *thiazide-sensitive $\text{Na}^+ - \text{Cl}^-$ cotransporter*. Final Na^+ reabsorption occurs in the cortical and medullary collecting ducts, the amount excreted being reasonably equivalent to the amount ingested per day ([Chap. 268](#)).

HYPOVOLEMIA

ETIOLOGY

True volume depletion, or hypovolemia, generally refers to a state of combined salt and water loss exceeding intake, leading to [ECF](#) volume contraction. The loss of Na^+ may be renal or extrarenal ([Table 49-1](#)).

Renal Many conditions are associated with excessive urinary NaCl and water losses,

including diuretics. Pharmacologic diuretics inhibit specific pathways of Na⁺-reabsorption along the nephron with a consequent increase in urinary Na⁺-excretion. Enhanced filtration of non-reabsorbed solutes, such as glucose or urea, can also impair tubular reabsorption of Na⁺ and water, leading to an osmotic or solute diuresis. This often occurs in poorly controlled diabetes mellitus and in patients receiving high-protein hyperalimentation. Mannitol is a diuretic that produces an osmotic diuresis because the renal tubule is impermeable to mannitol. Many tubule and interstitial renal disorders are associated with Na⁺-wasting. Excessive renal losses of Na⁺ and water may also occur during the diuretic phase of acute tubular necrosis ([Chap. 269](#)) and following the relief of bilateral urinary tract obstruction. The natriuresis and water diuresis associated with these two conditions are often short-lived and an appropriate response to a state of [ECF](#) volume expansion that ensued as a result of prior oliguria. However, ongoing losses in the absence of adequate replacement fluids may eventually lead to a state of hypovolemia. Chronic renal insufficiency is associated with a diminished ability to regulate renal salt and water excretion appropriately ([Chap. 270](#)). Therefore, patients with a [GFR](#) of less than 25 mL/min have an obligatory renal Na⁺ loss that may result in progressive ECF volume depletion if Na⁺-intake is restricted. Finally, mineralocorticoid deficiency (hypoaldosteronism) causes salt wasting in the presence of normal intrinsic renal function.

Massive renal water excretion can also lead to hypovolemia. The [ECF](#) volume contraction is usually less severe since two-thirds of the volume lost is intracellular. Conditions associated with excessive urinary water loss include *central diabetes insipidus* (CDI) and *nephrogenic diabetes insipidus* (NDI). These two disorders are due to impaired secretion of and renal unresponsiveness to [AVP](#), respectively, and are discussed below.

Extrarenal Nonrenal causes of hypovolemia include fluid loss from the gastrointestinal tract, skin, and respiratory system and third space accumulations (burns, pancreatitis, peritonitis). Approximately 9 L of fluid enters the gastrointestinal tract daily, 2 L by ingestion and 7 L by secretion. Almost 98% of this volume is reabsorbed so that fecal fluid loss is only 100 to 200 mL/d. Impaired gastrointestinal reabsorption or enhanced secretion leads to volume depletion. Since gastric secretions have a low pH (high H⁺-concentration) and biliary, pancreatic, and intestinal secretions are alkaline (high HCO₃⁻-concentration), vomiting and diarrhea are often accompanied by metabolic alkalosis and acidosis, respectively.

Water evaporation from the skin and respiratory tract contributes to thermoregulation. These *insensible losses* amount to 500 mL/d. During febrile illnesses, prolonged heat exposure, or exercise, increased salt and water loss from skin, in the form of sweat, can be significant and lead to volume depletion. The Na⁺-concentration of sweat is normally 20 to 50 mmol/L and decreases with profuse sweating due to the action of aldosterone. Since sweat is hypotonic, the loss of water exceeds that of Na⁺. The water deficit is minimized by enhanced thirst. Nevertheless, ongoing Na⁺ loss is manifest as hypovolemia. Enhanced evaporative water loss from the respiratory tract may be associated with hyperventilation, especially in mechanically ventilated febrile patients.

Certain conditions lead to fluid sequestration in a *third space*. This compartment is extracellular but is not in equilibrium with either the [ECF](#) or the [ICF](#). The fluid is

effectively lost from the ECF and can result in hypovolemia. Examples include the bowel lumen in gastrointestinal obstruction, subcutaneous tissues in severe burns, retroperitoneal space in acute pancreatitis, and peritoneal cavity in peritonitis. Finally, severe hemorrhage from any source can result in volume depletion.

PATHOPHYSIOLOGY

[ECF](#) volume contraction is manifest as a decreased plasma volume and hypotension. Hypotension is due to decreased venous return (preload) and diminished cardiac output; it triggers baroreceptors in the carotid sinus and aortic arch and leads to activation of the sympathetic nervous system and the renin-angiotensin system. The net effect is to maintain mean arterial pressure and cerebral and coronary perfusion. In contrast to the cardiovascular response, the renal response is aimed at restoring the ECF volume by decreasing the [GFR](#) and filtered load of Na⁺ and, most importantly, by promoting tubular reabsorption of Na⁺. Increased sympathetic tone increases proximal tubular Na⁺ reabsorption and decreases GFR by causing preferential afferent arteriolar vasoconstriction. Sodium is also reabsorbed in the proximal convoluted tubule in response to increased angiotensin II and altered peritubular capillary hemodynamics (decreased hydraulic and increased oncotic pressure). Enhanced reabsorption of Na⁺ by the collecting duct is an important component of the renal adaptation to ECF volume contraction. This occurs in response to increased *aldosterone* and [AVP](#) secretion, and suppressed *atrial natriuretic peptide* secretion.

CLINICAL FEATURES

A careful history is often helpful in determining the etiology of [ECF](#) volume contraction (e.g., vomiting, diarrhea, polyuria, diaphoresis). Most symptoms are nonspecific and secondary to electrolyte imbalances and tissue hypoperfusion and include fatigue, weakness, muscle cramps, thirst, and postural dizziness. More severe degrees of volume contraction can lead to end-organ ischemia manifest as oliguria, cyanosis, abdominal and chest pain, and confusion or obtundation. Diminished skin turgor and dry oral mucous membranes are poor markers of decreased interstitial fluid. Signs of intravascular volume contraction include decreased jugular venous pressure, postural hypotension, and postural tachycardia. Larger and more acute fluid losses lead to hypovolemic shock, manifest as hypotension, tachycardia, peripheral vasoconstriction, and hypoperfusion -- cyanosis, cold and clammy extremities, oliguria, and altered mental status.

DIAGNOSIS

A thorough history and physical examination are generally sufficient to diagnose the etiology of hypovolemia. Laboratory data usually confirm and support the clinical diagnosis. The blood urea nitrogen (BUN) and plasma creatinine concentrations tend to be elevated, reflecting a decreased [GFR](#). Normally, the BUN:creatinine ratio is about 10:1. However, in *prerenal azotemia*, hypovolemia leads to increased urea reabsorption and a proportionately greater elevation in BUN than plasma creatinine, and a BUN:creatinine ratio of 20:1 or higher. An increased BUN (relative to creatinine) may also be due to increased urea production that occurs with hyperalimentation (high-protein), glucocorticoid therapy, and gastrointestinal bleeding.

Volume depletion may be associated with hyponatremia, hypernatremia, or a normal plasma Na⁺ concentration, depending on the tonicity of the fluid lost, the presence of thirst, and the access to water. Hypokalemia is common in settings of increased renal or gastrointestinal K⁺ loss, and hyperkalemia occurs in renal failure, adrenal insufficiency, and certain types of metabolic acidosis. Metabolic alkalosis occurs with diuretic-induced hypovolemia and in cases of vomiting or nasogastric suction. In contrast, metabolic acidosis is associated with renal failure, tubulointerstitial disorders, adrenal insufficiency, diarrhea, diabetic ketoacidosis, and lactic acidosis. Since albumin and erythrocytes are confined to the intravascular compartment, [ECF](#) volume contraction often leads to a relative elevation in hematocrit (hemoconcentration) and plasma albumin concentration.

The appropriate response to hypovolemia is enhanced renal Na⁺ and water reabsorption, which is reflected in the urine composition. Therefore, the urine Na⁺ concentration should usually be less than 20 mmol/L except in conditions associated with impaired Na⁺ reabsorption, as in acute tubular necrosis ([Chap. 269](#)). Another exception is hypovolemia due to vomiting, since the associated metabolic alkalosis and increased filtered HCO₃⁻ impair proximal Na⁺ reabsorption. In this case, the urine Cl⁻ is low (<20 mmol/L). The urine osmolality and specific gravity in hypovolemic subjects are generally greater than 450 mosmol/kg and 1.015, respectively, reflecting the presence of enhanced [AVP](#) secretion. However, in hypovolemia due to diabetes insipidus, urine osmolality and specific gravity are indicative of inappropriately dilute urine.

TREATMENT

The therapeutic goals are to restore normovolemia with fluid similar in composition to that lost and to replace ongoing losses. Symptoms and signs, including weight loss, can help estimate the degree of volume contraction and should also be monitored to assess response to treatment. Mild volume contraction can usually be corrected via the oral route. More severe hypovolemia requires intravenous therapy. Isotonic or normal saline (0.9% NaCl or 154 mmol/L Na⁺) is the solution of choice in normonatremic and mildly hyponatremic individuals and should be administered initially in patients with hypotension or shock. Severe hyponatremia may require hypertonic saline (3.0% NaCl or 513 mmol/L Na⁺). Hypernatremia reflects a proportionally greater deficit of water than Na⁺, and its correction will therefore require a hypotonic solution such as half-normal saline (0.45% NaCl or 77 mmol/L Na⁺) or 5% dextrose in water. Patients with significant hemorrhage, anemia, or intravascular volume depletion may require blood transfusion or colloid-containing solutions (albumin, dextran). Hypokalemia may be present initially or may ensue as a result of increased urinary K⁺ excretion; it should be corrected by adding appropriate amounts of KCl to replacement solutions.

HYPONATREMIA

ETIOLOGY

A plasma Na⁺ concentration less than 135 mmol/L usually reflects a hypotonic state. However, plasma osmolality may be normal or increased in some cases of hyponatremia, referred to as *pseudohyponatremia*. Plasma is 93% water, the remaining 7% consisting of plasma proteins and lipids. Since Na⁺ ions are dissolved in plasma

water, increasing the nonaqueous phase artificially lowers the Na⁺-concentration measured per liter of plasma (except when Na⁺-sensitive glass electrodes are used). The plasma osmolality and the Na⁺-concentration remain normal. This type of hyponatremia has little clinical significance, except to ascertain the cause of the hyperproteinemia or hyperlipidemia. Isotonic or slightly hypotonic hyponatremia may complicate transurethral resection of the prostate or bladder because large volumes of isoosmotic (mannitol) or hypoosmotic (sorbital or glycine) bladder irrigation solution can be absorbed and result in a dilutional hyponatremia. The metabolism of sorbitol and glycine to CO₂ and water may lead to hypotonicity if the accumulated fluid and solutes are not rapidly excreted. Hypertonic hyponatremia is usually due to hyperglycemia or, occasionally, intravenous administration of mannitol. Relative insulin deficiency causes myocytes to become impermeable to glucose. Therefore, during poorly controlled diabetes mellitus, glucose is an effective osmole and draws water from muscle cells, resulting in hyponatremia. Plasma Na⁺-concentration falls by 1.4 mmol/L for every 100 mg/dL rise in the plasma glucose concentration.

Most causes of hyponatremia are associated with a low plasma osmolality ([Table 49-2](#)). In general, hypotonic hyponatremia is due either to a primary water gain (and secondary Na⁺-loss) or a primary Na⁺ loss (and secondary water gain). In the absence of water intake or hypotonic fluid replacement, hyponatremia is usually associated with hypovolemic shock due to a profound sodium deficit and transcellular water shift. Contraction of the [ECF](#) volume stimulates thirst and [AVP](#) secretion. The increased water ingestion and impaired renal excretion result in hyponatremia. It is important to note that *diuretic-induced hyponatremia* is almost always due to thiazide diuretics. Loop diuretics decrease the tonicity of the medullary interstitium and impair maximal urinary concentrating capacity. This limits the ability of AVP to promote water retention. In contrast, thiazide diuretics lead to Na⁺ and K⁺-depletion, and AVP-mediated water retention. In the presence of a large K⁺-deficit, transcellular ion exchange (K⁺-exits and Na⁺-enters cells) may contribute to hyponatremia. Hyponatremia can also occur by a process of *desalination*. This occurs when the urine tonicity (the sum of the concentrations of Na⁺ and K⁺) exceeds that of administered intravenous fluids (including isotonic saline). This accounts for some cases of acute postoperative hyponatremia and cerebral salt wasting after neurosurgery.

Hyponatremia in the setting of [ECF](#) volume expansion is usually associated with edematous states, such as congestive heart failure, hepatic cirrhosis, and the nephrotic syndrome. These disorders all have in common a decreased effective circulating arterial volume, leading to increased thirst and increased [AVP](#) levels. Additional factors impairing the excretion of solute-free water include a reduced [GFR](#), decreased delivery of ultrafiltrate to the diluting site (due to increased proximal fractional reabsorption of Na⁺ and water), and diuretic therapy. The degree of hyponatremia often correlates with the severity of the underlying condition and is an important prognostic factor. Oliguric acute and chronic renal failure may be associated with hyponatremia if water intake exceeds the ability to excrete equivalent volumes.

Hyponatremia in the absence of [ECF](#) volume contraction, decreased effective circulating arterial volume, or renal insufficiency is usually due to increased [AVP](#) secretion resulting in impaired water excretion. Ingestion or administration of water is also required since high levels of AVP alone are usually insufficient to produce hyponatremia. This disorder,

commonly termed the *syndrome of inappropriate antidiuretic hormone secretion* (SIADH), is the most common cause of normovolemic hyponatremia and is due to the nonphysiologic release of AVP from the posterior pituitary or an ectopic source ([Chap. 329](#)). Renal free water excretion is impaired while the regulation of Na⁺ balance is unaffected. The most common causes of SIADH include neuropsychiatric and pulmonary diseases, malignant tumors, major surgery (postoperative pain), and pharmacologic agents. Severe pain and nausea are physiologic stimuli of AVP secretion; these stimuli are inappropriate in the absence of hypovolemia or hyperosmolality. A variety of central nervous system disorders may be associated with SIADH, such as meningitis, encephalitis, hemorrhage, stroke, psychosis, primary and metastatic tumors, and acute porphyria. Pneumonia, empyema, tuberculosis, and acute respiratory failure can be complicated by hyponatremia secondary to SIADH. Hypoxemia, hypercarbia, and positive-pressure ventilation are all nonosmotic stimuli for AVP release. Various tumors, notably oat cell carcinoma of the lung, have been demonstrated to secrete AVP ectopically. Many drugs either stimulate AVP release or potentiate its actions on the kidney. The pattern of AVP secretion can be used to classify SIADH into four subtypes: (1) erratic autonomous AVP secretion (ectopic production); (2) normal regulation of AVP release around a lower osmolality set point or *reset osmostat* (cachexia, malnutrition); (3) normal AVP response to hypertonicity with failure to suppress completely at low osmolality (incomplete pituitary stalk section); and (4) normal AVP secretion with increased sensitivity to its actions or secretion of some other antidiuretic factor (rare).

Hormonal excess or deficiency may cause hyponatremia. Adrenal insufficiency ([Chap. 331](#)) and hypothyroidism ([Chap. 330](#)) may present with hyponatremia and should not be confused with [SIADH](#). Although decreased mineralocorticoids may contribute to the hyponatremia of adrenal insufficiency, it is the cortisol deficiency that leads to hypersecretion of [AVP](#) both indirectly (secondary to volume depletion) and directly (cosecreted with corticotropin-releasing factor). The mechanisms by which hypothyroidism leads to hyponatremia include decreased cardiac output and [GFR](#) and increased AVP secretion in response to hemodynamic stimuli.

Finally, hyponatremia may occur in the absence of [AVP](#) or renal failure if the kidney is unable to excrete the dietary water load. In psychogenic or primary polydipsia, compulsive water consumption may overwhelm the normally large renal excretory capacity of 12 L/d ([Chap. 329](#)). These patients often have psychiatric illnesses and may be taking medications, such as phenothiazines, that enhance the sensation of thirst by causing a dry mouth. The maximal urine output is a function of the minimum urine osmolality achievable and the mandatory solute excretion. Metabolism of a normal diet generates about 600 mosmol/d, and the minimum urine osmolality in humans is 50 mosmol/kg. Therefore, the maximum daily urine output will be about 12 L ($600 \div 50 = 12$). A solute excretion rate of greater than ~750 mosmol/d is, by definition, an *osmotic diuresis*. A low-protein diet may yield as few as 250 mosmol/d, which translates into a maximal urine output of 5 L/d at a minimum urine tonicity of 50 mosmol/kg. Beer drinkers typically have a poor dietary intake of protein and electrolytes and consume large volumes (of beer), which may exceed the renal excretory capacity and result in hyponatremia. This phenomenon is referred to as *beer potomania*.

CLINICAL FEATURES

The clinical manifestations of hyponatremia are related to osmotic water shift leading to increased [ICF](#) volume, specifically brain cell swelling or cerebral edema. Therefore, the symptoms are primarily neurologic, and their severity is dependent on the rapidity of onset and absolute decrease in plasma Na^+ concentration. Patients may be asymptomatic or complain of nausea and malaise. As the plasma Na^+ concentration falls, the symptoms progress to include headache, lethargy, confusion, and obtundation. Stupor, seizures, and coma do not usually occur unless the plasma Na^+ concentration falls acutely below 120 mmol/L or decreases rapidly. As described above, adaptive mechanisms designed to protect cell volume occur in chronic hyponatremia. Loss of Na^+ and K^+ , followed by organic osmolytes, from brain cells decreases brain swelling due to secondary transcellular water shifts (from ICF to [ECF](#)). The net effect is to minimize cerebral edema and its symptoms. Hospitalized patients with hyponatremia have an increased mortality rate compared to normonatremic control subjects. However, the excess mortality is usually attributed to the underlying disorder rather than the electrolyte disturbance.

DIAGNOSIS

Hyponatremia is not a disease but a manifestation of a variety of disorders. The underlying cause can often be ascertained from an accurate history and physical examination, including an assessment of [ECF](#) volume status and effective circulating arterial volume. The differential diagnosis of hyponatremia, an expanded ECF volume, and decreased effective circulating volume includes congestive heart failure, hepatic cirrhosis, and the nephrotic syndrome. Hypothyroidism and adrenal insufficiency tend to present with a near-normal ECF volume and decreased effective circulating arterial volume. All of these diseases have characteristic signs and symptoms. Patients with [SIADH](#) are usually euvoletic.

Four laboratory findings often provide useful information and can narrow the differential diagnosis of hyponatremia: (1) the plasma osmolality, (2) the urine osmolality, (3) the urine Na^+ concentration, and (4) the urine K^+ concentration. Since [ECF](#) tonicity is determined primarily by the Na^+ concentration, most patients with hyponatremia have a decreased plasma osmolality. If the plasma osmolality is not low, pseudohyponatremia must be ruled out. The appropriate renal response to hypoosmolality is to excrete the maximum volume of dilute urine, i.e., urine osmolality and specific gravity of less than 100 mosmol/kg and 1.003, respectively. This occurs in patients with primary polydipsia. If this is not present, it suggests impaired free water excretion due to the action of [AVP](#) on the kidney. The secretion of AVP may be a physiologic response to hemodynamic stimuli or it may be inappropriate in the presence of hyponatremia and euvoletic. Since Na^+ is the major ECF cation and is largely restricted to this compartment, ECF volume contraction represents a deficit in total body Na^+ content. Therefore, volume depletion in patients with normal underlying renal function results in enhanced tubule Na^+ reabsorption and a urine Na^+ concentration less than 20 mmol/L. The finding of a urine Na^+ concentration greater than 20 mmol/L in hypovolemic hyponatremia implies a salt-wasting nephropathy, diuretic therapy, hypoaldosteronism, or occasionally vomiting. Both the urine osmolality and the urine Na^+ concentration can be followed serially when assessing response to therapy.

[SIADH](#) is characterized by hypoosmotic hyponatremia in the setting of an inappropriately concentrated urine (urine osmolality greater than 100 mosmol/kg). Patients are typically normovolemic and have normal Na⁺ balance. They tend to be mildly volume expanded secondary to water retention and have a urine Na⁺ excretion rate equal to intake (urine Na⁺ concentration usually greater than 40 mmol/L). By definition, they have normal renal, adrenal, and thyroid function and usually have normal K⁺ and acid-base balance. SIADH is often associated with hypouricemia due to the uricosuric state induced by volume expansion. In contrast, hypovolemic patients tend to be hyperuricemic secondary to increased proximal urate reabsorption.

CLINICAL APPROACH

See [Fig. 49-1](#).

TREATMENT

The goals of therapy are twofold: (1) to raise the plasma Na⁺ concentration by restricting water intake and promoting water loss; and (2) to correct the underlying disorder. Mild asymptomatic hyponatremia is generally of little clinical significance and requires no treatment. The management of asymptomatic hyponatremia associated with [ECF](#) volume contraction should include Na⁺ repletion, generally in the form of isotonic saline. The direct effect of the administered NaCl on the plasma Na⁺ concentration is trivial. However, restoration of euvolemia removes the hemodynamic stimulus for [AVP](#) release, allowing the excess free water to be excreted. The hyponatremia associated with edematous states tends to reflect the severity of the underlying disease and is usually asymptomatic. These patients have increased total body water that exceeds the increase in total body Na⁺ content. Treatment should include restriction of Na⁺ and water intake, correction of hypokalemia, and promotion of water loss in excess of Na⁺. The latter may require the use of loop diuretics with replacement of a proportion of the urinary Na⁺ loss to ensure net free water excretion. Dietary water restriction should be less than the urine output. Correction of the K⁺ deficit may raise the plasma Na⁺ concentration by favoring a shift of Na⁺ out of cells as K⁺ moves in. Water restriction is also a component of the therapeutic approach to hyponatremia associated with primary polydipsia, renal failure, and [SIADH](#) ([Chap. 329](#)).

The rate of correction of hyponatremia depends on the absence or presence of neurologic dysfunction. This, in turn, is related to the rapidity of onset and magnitude of the fall in plasma Na⁺ concentration. In asymptomatic patients, the plasma Na⁺ concentration should be raised by no more than 0.5 to 1.0 mmol/L per hour and by less than 10 to 12 mmol/L over the first 24 h. Acute or severe hyponatremia (plasma Na⁺ concentration <110 to 115 mmol/L) tends to present with altered mental status and/or seizures and requires more rapid correction. Severe symptomatic hyponatremia should be treated with hypertonic saline, and the plasma Na⁺ concentration should be raised by 1 to 2 mmol/L per hour for the first 3 to 4 h or until the seizures subside. Once again, the plasma Na⁺ concentration should probably be raised by no more than 12 mmol/L during the first 24 h. The quantity of Na⁺ required to increase the plasma Na⁺ concentration by a given amount can be estimated by multiplying the deficit in plasma Na⁺ concentration by the total body water. Under normal conditions, total body water is 50 or 60% of lean body weight in women or men, respectively. Therefore, to

raise the plasma Na⁺ concentration from 105 to 115 mmol/L in a 70-kg man requires 420 mmol $[(115 - 105) \times 70 \times 0.6]$ of Na⁺. The risk of correcting hyponatremia too rapidly is the development of the *osmotic demyelination syndrome* (ODS). This is a neurologic disorder characterized by flaccid paralysis, dysarthria, and dysphagia. The diagnosis is usually suspected clinically and can be confirmed by appropriate neuroimaging studies. There is no specific treatment for the disorder, which is associated with significant morbidity and mortality. Patients with chronic hyponatremia are most susceptible to the development of ODS, since their brain cell volume has returned to near normal as a result of the osmotic adaptive mechanisms described above. Therefore, administration of hypertonic saline to these individuals can cause sudden osmotic shrinkage of brain cells. In addition to rapid or overcorrection of hyponatremia, risk factors for ODS include prior cerebral anoxic injury, hypokalemia, and malnutrition, especially secondary to alcoholism. Water restriction in primary polydipsia and intravenous saline therapy in [ECF](#) volume-contracted patients may also lead to overly rapid correction of hyponatremia as a result of AVP suppression and a brisk water diuresis. This can be prevented by administration of water or use of an AVP analogue to slow down the rate of free water excretion. **For further discussion, see [Chap. 329](#).*

HYPERNATREMIA

ETIOLOGY

Hypernatremia is defined as a plasma Na⁺ concentration greater than 145 mmol/L. Since Na⁺ and its accompanying anions are the major effective [ECF](#) osmoles, hypernatremia is a state of hyperosmolality. As a result of the fixed number of [ICF](#) particles, maintenance of osmotic equilibrium in hypernatremia results in ICF volume contraction. Hypernatremia may be due to primary Na⁺ gain or water deficit. The two components of an appropriate response to hypernatremia are increased water intake stimulated by thirst and the excretion of the minimum volume of maximally concentrated urine reflecting [AVP](#) secretion in response to an osmotic stimulus.

In practice, the majority of cases of hypernatremia result from the loss of water. Since water is distributed between the [ICF](#) and the [ECF](#) in a 2:1 ratio, a given amount of solute-free water loss will result in a twofold greater reduction in the ICF compartment than the ECF compartment. For example, consider three scenarios: the loss of 1 L of water, isotonic NaCl, or half-isotonic NaCl. If 1 L of water is lost, the ICF volume will decrease by 667 mL, whereas the ECF volume will fall by only 333 mL. Due to the fact that Na⁺ is largely restricted to the ECF, this compartment will decrease by 1 L if the fluid lost is isoosmotic. One liter of half-isotonic NaCl is equivalent to 500 mL of water (one-third ECF, two-thirds ICF) plus 500 mL of isotonic saline (all ECF). Therefore, the loss of 1 L of half-isotonic saline decreases the ECF and ICF volumes by 667 mL and 333 mL, respectively.

The degree of hyperosmolality is typically mild unless the thirst mechanism is abnormal or access to water is limited. The latter occurs in infants, the physically handicapped, patients with impaired mental status, in the postoperative state, and in intubated patients in the intensive care unit. On rare occasions, impaired thirst may be due to *primary hypodipsia*. This usually occurs as a result of damage to the hypothalamic osmoreceptors that control thirst and tends to be associated with abnormal osmotic

regulation of [AVP](#) secretion. Primary hypodipsia may be due to a variety of pathologic changes including granulomatous disease, vascular occlusion, and tumors. A subset of hypodipsic hypernatremia, referred to as *essential hypernatremia*, does not respond to forced water intake. This appears to be due to a specific osmoreceptor defect resulting in nonosmotic regulation of AVP release. Thus, the hemodynamic effects of water loading lead to AVP suppression and excretion of dilute urine.

The source of free water loss is either renal or extrarenal. Nonrenal loss of water may be due to evaporation from the skin and respiratory tract (insensible losses) or loss from the gastrointestinal tract. Insensible losses are increased with fever, exercise, heat exposure, and severe burns and in mechanically ventilated patients. Furthermore, the Na^+ -concentration of sweat decreases with profuse perspiration, thereby increasing solute-free water loss. Diarrhea is the most common gastrointestinal cause of hypernatremia. Specifically, osmotic diarrheas (induced by lactulose, sorbitol, or malabsorption of carbohydrate) and viral gastroenteritides result in water loss exceeding that of Na^+ and K^+ . In contrast, secretory diarrheas (e.g., cholera, carcinoid, VIPoma) have a fecal osmolality (twice the sum of the concentrations of Na^+ and K^+) similar to that of plasma and present with [ECF](#) volume contraction and a normal plasma Na^+ -concentration or hyponatremia.

Renal water loss is the most common cause of hypernatremia and is due to drug-induced or osmotic diuresis or diabetes insipidus ([Chap. 329](#)). Loop diuretics interfere with the countercurrent mechanism and produce an isoosmotic solute diuresis. This results in a decreased medullary interstitial tonicity and impaired renal concentrating ability. The presence of non-reabsorbed organic solutes in the tubule lumen impairs the osmotic reabsorption of water. This leads to water loss in excess of Na^+ and K^+ , known as an osmotic diuresis. The most frequent cause of an osmotic diuresis is hyperglycemia and glucosuria in poorly controlled diabetes mellitus. Intravenous administration of mannitol and increased endogenous production of urea (high-protein diet) can also result in an osmotic diuresis. Hypernatremia secondary to nonosmotic urinary water loss is usually due to: (1) [CDI](#) or neurogenic diabetes insipidus characterized by impaired [AVP](#) secretion, or (2) [NDI](#) resulting from end-organ (renal) resistance to the actions of AVP. The most common cause of CDI is destruction of the neurohypophysis. This may occur as a result of trauma, neurosurgery, granulomatous disease, neoplasms, vascular accidents, or infection. In many cases, CDI is idiopathic and may occasionally be hereditary. The familial form of the disease is inherited in an autosomal dominant fashion and has been attributed to mutations in the propressophysin (AVP precursor) gene. NDI may be either inherited or acquired. Congenital NDI is an X-linked recessive trait due to mutations in the V_2 receptor gene. Mutations in the autosomal aquaporin-2 gene may also result in NDI. The aquaporin-2 gene encodes the water channel protein whose membrane insertion is stimulated by AVP. The causes of sporadic NDI are numerous and include drugs (especially lithium), hypercalcemia, hypokalemia, and conditions that impair medullary hypertonicity (e.g., papillary necrosis or osmotic diuresis). Pregnant women, in the second or third trimester, may develop NDI as a result of excessive elaboration of vasopressinase by the placenta.

Finally, although infrequent, a primary Na^+ gain may cause hypernatremia. For example, inadvertent administration of hypertonic NaCl or NaHCO_3 or replacing sugar with salt in

infant formula can produce this complication.

CLINICAL FEATURES

As a consequence of hypertonicity, water shifts out of cells, leading to a contracted [ICF](#) volume. A decreased brain cell volume is associated with an increased risk of subarachnoid or intracerebral hemorrhage. Hence, the major symptoms of hypernatremia are neurologic and include altered mental status, weakness, neuromuscular irritability, focal neurologic deficits, and occasionally coma or seizures. Patients may also complain of polyuria or thirst. For unknown reasons, patients with polydipsia from [CDI](#) tend to prefer ice-cold water. The signs and symptoms of volume depletion are often present in patients with a history of excessive sweating, diarrhea, or an osmotic diuresis. The mortality rate associated with a plasma Na^+ concentration greater than 180 mmol/L is very high. As with hyponatremia, the severity of the clinical manifestations is related to the acuity and magnitude of the rise in plasma Na^+ concentration. Chronic hypernatremia is generally less symptomatic as a result of adaptive mechanisms designed to defend cell volume. Brain cells initially take up Na^+ and K^+ salts, later followed by accumulation of organic osmolytes such as inositol. This serves to restore the brain ICF volume towards normal.

DIAGNOSIS

A complete history and physical examination will often provide clues as to the underlying cause of hypernatremia. Relevant symptoms and signs include the absence or presence of thirst, diaphoresis, diarrhea, polyuria, and the features of [ECF](#) volume contraction. The history should include a list of current and recent medications, and the physical examination is incomplete without a thorough mental status and neurologic assessment. Measurement of urine volume and osmolality are essential in the evaluation of hyperosmolality. The appropriate renal response to hypernatremia is the excretion of the minimum volume (500 mL/d) of maximally concentrated urine (urine osmolality >800 mosmol/kg). These findings suggest extrarenal or remote renal water loss or administration of hypertonic Na^+ salt solutions. The presence of a primary Na^+ excess can be confirmed by the presence of ECF volume expansion and natriuresis (urine Na^+ concentration usually >100 mmol/L). Many causes of hypernatremia are associated with polyuria and a submaximal urine osmolality. The product of the urine volume and osmolality, i.e., the solute excretion rate, is helpful in determining the basis of the polyuria (see above). To maintain a steady state, total solute excretion must equal solute production. As stated above, individuals eating a normal diet generate ~ 600 mosmol/d. Therefore, daily solute excretion in excess of 750 mosmol defines an osmotic diuresis. This can be confirmed by measuring the urine glucose and urea. In general, both [CDI](#) and [NDI](#) present with polyuria and hypotonic urine (urine osmolality <250 mosmol/kg). The degree of hypernatremia is usually mild unless there is an associated thirst abnormality. The clinical history, physical examination, and pertinent laboratory data can often rule out causes of acquired NDI. CDI and NDI can generally be distinguished by administering the [AVP](#) analogue desmopressin (10 μg intranasally) after careful water restriction. The urine osmolality should increase by at least 50% in CDI and will not change in NDI. Unfortunately, the diagnosis may sometimes be difficult due to partial defects in AVP secretion and action.

CLINICAL APPROACH

See [Fig. 49-2](#).

TREATMENT

The therapeutic goals are to stop ongoing water loss by treating the underlying cause and to correct the water deficit. The [ECF](#) volume should be restored in hypovolemic patients. The quantity of water required to correct the deficit can be calculated from the following equation:

In hypernatremia due to water loss, total body water is approximately 50 and 40% of lean body weight in men and women, respectively. For example, a 50-kg woman with a plasma Na^+ concentration of 160 mmol/L has an estimated free water deficit of 2.9 L $\text{icub}; [(160 - 140) \cdot 140] \cdot (0.4 \cdot 50) \text{rcub}$. As in hyponatremia, rapid correction of hypernatremia is potentially dangerous. In this case, a sudden decrease in osmolality could potentially cause a rapid shift of water into cells that have undergone osmotic adaptation. This would result in swollen brain cells and increase the risk of seizures or permanent neurologic damage. Therefore, the water deficit should be corrected slowly over at least 48 to 72 h. When calculating the rate of water replacement, ongoing losses should be taken into account, and the plasma Na^+ concentration should be lowered by 0.5 mmol/L per hour and by no more than 12 mmol/L over the first 24 h. The safest route of administration of water is by mouth or via a nasogastric tube (or other feeding tube). Alternatively, 5% dextrose in water or half-isotonic saline can be given intravenously. The appropriate treatment of [CDI](#) consists of administering desmopressin intranasally ([Chap. 329](#)). Other options for decreasing urine output include a low-salt diet in combination with low-dose thiazide diuretic therapy. In some patients with partial CDI, drugs that either stimulate [AVP](#) secretion or enhance its action on the kidney have been useful. These include chlorpropamide, clofibrate, carbamazepine, and nonsteroidal anti-inflammatory drugs (NSAIDs). The concentrating defect in [NDI](#) may be reversible by treating the underlying disorder or eliminating the offending drug. Symptomatic polyuria due to NDI can be treated with a low- Na^+ diet and thiazide diuretics as described above. This induces mild volume depletion, which leads to enhanced proximal reabsorption of salt and water and decreased delivery to the site of action of AVP, the collecting duct. By impairing renal prostaglandin synthesis, NSAIDs potentiate AVP action and thereby increase urine osmolality and decrease urine volume. Amiloride may be useful in patients with NDI who need to be on lithium. The nephrotoxicity of lithium requires the drug to be taken up into collecting duct cells via the amiloride-sensitive Na^+ channel.

POTASSIUM

POTASSIUM BALANCE

Potassium is the major intracellular cation. The normal plasma K^+ concentration is 3.5 to 5.0 mmol/L, whereas that inside cells is about 150 mmol/L. Therefore, the amount of K^+ in the [ECF](#) (30 to 70 mmol) constitutes less than 2% of the total body K^+ content (2500 to 4500 mmol). The ratio of [ICF](#) to ECF K^+ concentration (normally 38:1) is the principal

result of the resting membrane potential and is crucial for normal neuromuscular function. The basolateral Na⁺, K⁺-ATPase pump actively transports K⁺ in and Na⁺ out of the cell in a 2:3 ratio, and the passive outward diffusion of K⁺ is quantitatively the most important factor that generates the resting membrane potential. The activity of the electrogenic Na⁺, K⁺-ATPase pump may be stimulated as a result of an increased intracellular Na⁺ concentration and inhibited in the setting of digoxin toxicity or chronic illness such as heart failure or renal failure.

The distribution of K⁺ is also affected by several other factors, including hormones, acid-base balance, osmolality, and cell turnover. Insulin increases Na⁺, K⁺-ATPase activity indirectly and independent of its effect on glucose transport, leading to K⁺ shift into muscle and liver cells. Conversely, insulin deficiency results in K⁺ movement from the [ICF](#) to the [ECF](#) compartment. Catecholamines have variable effects on K⁺ distribution -- β_2 -adrenergic agonists promote whereas α -adrenergic agonists impair K⁺ uptake by cells. The Na⁺, K⁺-ATPase pump as well as insulin secretion are stimulated by β_2 -adrenergic agonists. In contrast, α -adrenergic agonists have the opposite effect. The major action of aldosterone is to increase K⁺ excretion (see below). The role of extracellular pH in K⁺ balance relates to the underlying acid-base disorder. In metabolic acidosis, 60% of the H⁺ load is buffered inside cells. To maintain electroneutrality, the H⁺ ion must either be accompanied by an anion or exchanged for intracellular K⁺ (leading to hyperkalemia). Organic acidoses are not usually associated with a pH-related K⁺ shift, since anions such as lactate and β -hydroxybutyrate can be readily taken up by the cell. The converse, movement of K⁺ into cells, may be seen with metabolic alkalosis. However, this is less important due to diminished intracellular buffering. Primary respiratory disturbances in acid-base balance result in minimal transcellular K⁺ shifts. In hyperosmolal states, K⁺ diffuses out of cells along with water due to *solvent drag*. The concentration gradient favoring K⁺ movement out of cells is also increased as a result of ICF water loss. Tissue destruction or breakdown results in the release of intracellular K⁺, whereas the production of new cells shifts K⁺ out of the ECF. Finally, moderate to severe exercise may be associated with K⁺ release from muscle, leading to glycogenolysis and local vasodilatation. This is usually transient but may affect the plasma K⁺ concentration if patients repeatedly clench and unclench their fist prior to venipuncture.

The K⁺ intake of individuals on an average western diet is 40 to 120 mmol/d or approximately 1 mmol/kg per day, 90% of which is absorbed by the gastrointestinal tract. Maintenance of the steady state necessitates matching K⁺ ingestion with excretion. Initially, extrarenal adaptive mechanisms, followed later by urinary excretion, prevent a doubling of the plasma K⁺ concentration that would occur if the dietary K⁺ load remained in the [ECF](#) compartment. Immediately following a meal, most of the absorbed K⁺ enters cells as a result of the initial elevation in the plasma K⁺ concentration and facilitated by insulin release and basal catecholamine levels. Eventually, however, the excess K⁺ is excreted in the urine (see below). The regulation of gastrointestinal K⁺ handling is not well understood. The amount of K⁺ lost in the stool can increase from 10 to 50 or 60% (of dietary intake) in chronic renal insufficiency. In addition, colonic secretion of K⁺ is stimulated in patients with large volumes of diarrhea, resulting in potentially severe K⁺ depletion.

POTASSIUM EXCRETION (See also [Chap. 268](#))

Renal excretion is the major route of elimination of dietary and other sources of excess K^+ . The filtered load of K^+ ($GFR \times \text{plasma } K^+ \text{ concentration} = 180 \text{ L/d} \times 4 \text{ mmol/L} = 720 \text{ mmol/d}$) is 10- to 20-fold greater than the $ECF K^+$ content. Some 90% of filtered K^+ is reabsorbed by the proximal convoluted tubule and loop of Henle. Proximally, K^+ is reabsorbed passively with Na^+ and water, whereas the luminal $Na^+-K^+-2Cl^-$ cotransporter mediates K^+ uptake in the thick ascending limb of the loop of Henle. Therefore, K^+ delivery to the distal nephron [distal convoluted tubule and cortical collecting duct (CCD)] approximates dietary intake. Net distal K^+ secretion or reabsorption occurs in the setting of K^+ excess or depletion, respectively. The cell responsible for K^+ secretion in the late distal convoluted tubule (or connecting tubule) and CCD is the principal cell. Virtually all regulation of renal K^+ excretion and total body K^+ balance occurs in the distal nephron. The driving force for K^+ secretion is a favorable electrochemical gradient across the luminal membrane of the principal cell. As a result of the action of the basolateral $Na^+, K^+-ATPase$ pump, the intracellular K^+ concentration far exceeds that of the fluid in the lumen of the CCD. The electrical gradient is created by electrogenic Na^+ reabsorption leading to a lumen-negative transepithelial potential difference (TEPD), favoring K^+ secretion. The generation of a lumen-negative TEPD depends on the relative rates of reabsorption of Na^+ and its accompanying anion (primarily Cl^-). Equimolar reabsorption of Na^+ and Cl^- at equivalent rates is electroneutral, whereas reabsorption of Na^+ in excess of Cl^- is electrogenic. The cellular uptake of Na^+ by the principal cell occurs via an apical Na^+ channel and is driven by a low intracellular Na^+ concentration relative to that in the lumen of the CCD. The mechanism and regulation of distal nephron Cl^- transport is less clear. Obviously, factors that impact on either Na^+ or Cl^- reabsorption by the principal cell will influence the TEPD. Potassium secretion is regulated by two physiologic stimuli -- aldosterone and hyperkalemia. Aldosterone is secreted by the zona glomerulosa cells of the adrenal cortex in response to high renin and angiotensin II or hyperkalemia. The actions of aldosterone on the principal cell include enhanced apical membrane Na^+ conductivity, stimulation of the basolateral $Na^+, K^+-ATPase$, and increased luminal K^+ channels. The plasma K^+ concentration, independent of aldosterone, can directly affect K^+ secretion. In addition to the K^+ concentration in the lumen of the CCD, renal K^+ loss depends on the urine flow rate, a function of daily solute excretion (see above). Since excretion is equal to the product of concentration and volume, increased distal flow rate can significantly enhance urinary K^+ output. Finally, in severe K^+ depletion, secretion of K^+ is reduced and reabsorption, via apical H^+ , $K^+-ATPase$ pumps in cortical and medullary collecting ducts, is upregulated.

HYPOKALEMIA

ETIOLOGY (See [Table 49-3](#))

Hypokalemia, defined as a plasma K^+ concentration $< 3.5 \text{ mmol/L}$, may result from one (or more) of the following: decreased net intake, shift into cells, or increased net loss. Diminished intake is seldom the sole cause of K^+ depletion since urinary excretion can be effectively decreased to less than 15 mmol/d as a result of net K^+ reabsorption in the distal nephron. With the exception of the urban poor and certain cultural groups, the amount of K^+ in the diet almost always exceeds that excreted in the urine. However, dietary K^+ restriction may exacerbate the hypokalemia secondary to increased gastrointestinal or renal loss. An unusual cause of decreased K^+ intake is ingestion of

clay (geophagia), which binds dietary K⁺ and iron. This custom was previously common among African Americans in the American South.

Redistribution into Cells Movement of K⁺ into cells may transiently decrease the plasma K⁺ concentration without altering total body K⁺ content. For any given cause, the magnitude of the change is relatively small, often less than 1 mmol/L. However, a combination of factors may lead to a significant fall in the plasma K⁺ concentration and may amplify the hypokalemia due to K⁺ wasting. Alkalosis, especially that due to a primary increase in plasma HCO₃⁻ (metabolic alkalosis), is often associated with hypokalemia. This occurs as a result of K⁺ redistribution as well as excessive renal K⁺ loss. Treatment of diabetic ketoacidosis with insulin may lead to hypokalemia due to stimulation of the Na⁺-H⁺ antiporter and (secondarily) the Na⁺, K⁺-ATPase pump. Furthermore, uncontrolled hyperglycemia often leads to K⁺ depletion from an osmotic diuresis (see below). Stress-induced catecholamine release and administration of β_2 -adrenergic agonists directly induce cellular uptake of K⁺ and promote insulin secretion by pancreatic islet β cells. *Hypokalemic periodic paralysis* is a rare condition characterized by recurrent episodic weakness or paralysis ([Chap. 381](#)). Since K⁺ is the major [ICF](#) cation, anabolic states can potentially result in hypokalemia due to a K⁺ shift into cells. This may occur following rapid cell growth seen in patients with pernicious anemia treated with vitamin B₁₂ or with neutropenia after treatment with granulocyte-macrophage colony stimulating factor. Massive transfusion with thawed washed red blood cells (RBCs) could cause hypokalemia since frozen RBCs lose up to half of their K⁺ during storage.

Nonrenal Loss of Potassium Excessive sweating may result in K⁺ depletion from increased integumentary and renal K⁺ loss. Hyperaldosteronism, secondary to [ECF](#) volume contraction, enhances K⁺ excretion in the urine ([Chap. 331](#)). Normally, K⁺ lost in the stool amounts to 5 to 10 mmol/d in a volume of 100 to 200 mL. Hypokalemia subsequent to increased gastrointestinal loss can occur in patients with profuse diarrhea (usually secretory), villous adenomas, VIPomas, or laxative abuse. However, the loss of gastric secretions does not account for the moderate to severe K⁺ depletion often associated with vomiting or nasogastric suction. Since the K⁺ concentration of gastric fluid is 5 to 10 mmol/L, it would take 30 to 80 L of vomitus to achieve a K⁺ deficit of 300 to 400 mmol typically seen in these patients. In fact, the hypokalemia is primarily due to increased renal K⁺ excretion. Loss of gastric contents results in volume depletion and metabolic alkalosis, both of which promote kaliuresis. Hypovolemia stimulates aldosterone release, which augments K⁺ secretion by the principal cells. In addition, the filtered load of HCO₃⁻ exceeds the reabsorptive capacity of the proximal convoluted tubule, thereby increasing distal delivery of NaHCO₃, which enhances the electrochemical gradient favoring K⁺ loss in the urine.

Renal Loss of Potassium In general, most cases of chronic hypokalemia are due to renal K⁺ wasting. This may be due to factors that increase the K⁺ concentration in the lumen of the [CCD](#) or augment distal flow rate. As described above, distal nephron K⁺ secretion is driven by a lumen-negative [TEPD](#), affected by aldosterone and the relative rates of reabsorption of Na⁺ and its accompanying anion(s). Mineralocorticoid excess commonly results in hypokalemia ([Chap. 331](#)). *Primary hyperaldosteronism* is due to dysregulated aldosterone secretion by an adrenal adenoma (Conn's syndrome) or carcinoma or to adrenocortical hyperplasia. In a rare subset of patients, the disorder

is familial (autosomal dominant) and aldosterone levels can be suppressed by administering low doses of exogenous glucocorticoid. The molecular defect responsible for *glucocorticoid-remediable hyperaldosteronism* is a rearranged gene (due to a chromosomal crossover), containing the 5'-regulatory region of the 11 β -hydroxylase gene and the coding sequence of the aldosterone synthase gene. Consequently, mineralocorticoid is synthesized in the zona fasciculata and regulated by corticotropin. A number of conditions associated with hyperreninemia result in secondary hyperaldosteronism and renal K⁺wasting. High renin levels are commonly seen in both renovascular and malignant hypertension. Renin-secreting tumors of the juxtaglomerular apparatus are a rare cause of hypokalemia. Other tumors that have been reported to produce renin include renal cell carcinoma, ovarian carcinoma, and Wilms' tumor. Hyperreninemia may also occur secondary to decreased effective circulating arterial volume.

In the absence of elevated renin or aldosterone levels, enhanced distal nephron secretion of K⁺ may result from increased production of non-aldosterone mineralocorticoids in *congenital adrenal hyperplasia* ([Chap. 331](#)).

Glucocorticoid-stimulated kaliuresis does not normally occur due to the conversion of cortisol to cortisone by 11 β -hydroxysteroid dehydrogenase (11 β -HSDH). Therefore, 11 β -HSDH deficiency or suppression allows cortisol to bind to the aldosterone receptor and leads to the *syndrome of apparent mineralocorticoid excess*. Drugs that inhibit the activity of 11 β -HSDH include glycyrrhetic acid, present in licorice, chewing tobacco, and carbenoxolone. The presentation of Cushing's syndrome may include hypokalemia if the capacity of 11 β -HSDH to inactivate cortisol is overwhelmed by persistently elevated glucocorticoid levels.

Liddle's syndrome is a rare familial (autosomal dominant) disease characterized by hypertension, hypokalemic metabolic alkalosis, renal K⁺wasting, and suppressed renin and aldosterone secretion ([Chap. 331](#)). Increased distal delivery of Na⁺ with a non-reabsorbable anion (not Cl⁻) enhances the lumen-negative [TEPD](#) and K⁺secretion. Classically, this is seen with *proximal (type 2) renal tubular acidosis* (RTA) and vomiting, associated with bicarbonaturia. Diabetic ketoacidosis and toluene abuse (glue-sniffing) can lead to increased delivery of β -hydroxybutyrate and hippurate, respectively, to the [CCD](#) and to renal K⁺ loss. High doses of penicillin derivatives administered to volume-depleted patients may likewise promote renal K⁺secretion as well as an osmotic diuresis. *Classic distal (type 1) RTA* is associated with hypokalemia due to increased renal K⁺ loss, the mechanism of which is uncertain. Amphotericin B causes hypokalemia due to increased distal nephron permeability to Na⁺ and K⁺ and to renal K⁺wasting.

Bartter's syndrome is a disorder characterized by hypokalemia, metabolic alkalosis, hyperreninemic hyperaldosteronism secondary to [ECF](#) volume contraction, and juxtaglomerular apparatus hyperplasia ([Chap. 331](#)). Finally, diuretic use and abuse are common causes of K⁺depletion. Carbonic anhydrase inhibitors, loop diuretics, and thiazides are all kaliuretic. The degree of hypokalemia tends to be greater with long-acting agents and is dose-dependent. Increased renal K⁺excretion is due primarily to increased distal solute delivery and secondary hyperaldosteronism (due to volume depletion).

CLINICAL FEATURES

The clinical manifestations of K⁺depletion vary greatly between individual patients, and their severity depends on the degree of hypokalemia. Symptoms seldom occur unless the plasma K⁺concentration is less than 3 mmol/L. Fatigue, myalgia, and muscular weakness of the lower extremities are common complaints and are due to a lower (more negative) resting membrane potential. More severe hypokalemia may lead to progressive weakness, hypoventilation (due to respiratory muscle involvement), and eventually complete paralysis. Impaired muscle metabolism and the blunted hyperemic response to exercise associated with profound K⁺depletion increase the risk of rhabdomyolysis. Smooth-muscle function may also be affected and manifest as paralytic ileus.

The electrocardiographic changes of hypokalemia ([Fig. 226-19](#)) are due to delayed ventricular repolarization and do not correlate well with the plasma K⁺concentration. Early changes include flattening or inversion of the T wave, a prominent U wave, ST-segment depression, and a prolonged QU interval. Severe K⁺depletion may result in a prolonged PR interval, decreased voltage and widening of the QRS complex, and an increased risk of ventricular arrhythmias, especially in patients with myocardial ischemia or left ventricular hypertrophy. Hypokalemia may also predispose to digitalis toxicity. Epidemiologic studies have linked a low-K⁺diet with an increased prevalence of hypertension, particularly among African Americans. Furthermore, in patients with essential hypertension, systemic blood pressure may be lowered by K⁺supplementation. The mechanism of the hypertensive effect of K⁺depletion is not certain but may relate to enhanced distal NaCl reabsorption.

Hypokalemia is often associated with acid-base disturbances related to the underlying disorder. In addition, K⁺depletion results in intracellular acidification and an increase in net acid excretion or new HCO₃⁻production. This is a consequence of enhanced proximal HCO₃⁻reabsorption, increased renal ammoniagenesis, and increased distal H⁺secretion. This contributes to the generation of metabolic alkalosis frequently present in hypokalemic patients. [NDI](#) (see above) is not uncommonly seen in K⁺depletion and is manifest as polydipsia and polyuria. Glucose intolerance may also occur with hypokalemia and has been attributed to either impaired insulin secretion or peripheral insulin resistance.

DIAGNOSIS

In most cases, the etiology of K⁺depletion can be determined by a careful history. Diuretic and laxative abuse as well as surreptitious vomiting may be difficult to identify but should be excluded. Rarely, patients with a marked leukocytosis (e.g., acute myeloid leukemia) and normokalemia may have a low measured plasma K⁺concentration due to white blood cell uptake of K⁺ at room temperature. This *pseudohypokalemia* can be avoided by storing the blood sample on ice or rapidly separating the plasma (or serum) from the cells. After eliminating decreased intake and intracellular shift as potential causes of hypokalemia, examination of the renal response can help to clarify the source of K⁺loss. The appropriate response to K⁺depletion is to excrete less than 15 mmol/d of K⁺ in the urine, due to increased reabsorption and decreased distal secretion. Hypokalemia with minimal renal K⁺excretion suggests that K⁺ was lost via the skin or gastrointestinal tract or that there is a remote history of vomiting or diuretic use. As

described above, renal K⁺ wasting may be due to factors that either increase the K⁺ concentration in the [CCD](#) or increase the distal flow rate (or both). The [ECF](#) volume status, blood pressure, and associated acid-base disorder may help to differentiate the causes of excessive renal K⁺ loss. A rapid and simple test designed to evaluate the driving force for net K⁺ secretion is the *transtubular K⁺ concentration gradient* (TTKG). The TTKG is the ratio of the K⁺ concentration in the lumen of the CCD ($[K^+]_{\text{CCD}}$) to that in peritubular capillaries or plasma ($[K^+]_{\text{P}}$). The validity of this measurement depends on three assumptions: (1) few solutes are reabsorbed in the medullary collecting duct (MCD), (2) K⁺ is neither secreted nor reabsorbed in the MCD, and (3) the osmolality of the fluid in the terminal CCD is known. Significant reabsorption or secretion of K⁺ in the MCD seldom occurs, except in profound K⁺ depletion or excess, respectively. When [AVP](#) is acting ($OSM_U \approx OSM_P$), the osmolality in the terminal CCD is the same as that of plasma, and the K⁺ concentration in the lumen of the distal nephron can be estimated by dividing the urine K⁺ concentration ($[K^+]_U$) by the ratio of the urine to plasma osmolality (OSM_U/OSM_P):

Hypokalemia with a [TTKG](#) greater than 4 suggests renal K⁺ loss due to increased distal K⁺ secretion. Plasma renin and aldosterone levels are often helpful in differentiating the various causes of hyperaldosteronism. Bicarbonaturia and the presence of other non-reabsorbed anions also increase the TTKG and lead to renal K⁺ wasting.

CLINICAL APPROACH

See [Fig. 49-3](#).

TREATMENT

The therapeutic goals are to correct the K⁺ deficit and to minimize ongoing losses. With the exception of periodic paralysis, hypokalemia resulting from transcellular shifts rarely requires intravenous K⁺ supplementation, which can lead to rebound hyperkalemia. It is generally safer to correct hypokalemia via the oral route. The degree of K⁺ depletion does not correlate well with the plasma K⁺ concentration. A decrement of 1 mmol/L in the plasma K⁺ concentration (from 4.0 to 3.0 mmol/L) may represent a total body K⁺ deficit of 200 to 400 mmol, and patients with plasma levels under 3.0 mmol/L often require in excess of 600 mmol of K⁺ to correct the deficit. Furthermore, factors promoting K⁺ shift out of cells (e.g., insulin deficiency in diabetic ketoacidosis) may result in underestimation of the K⁺ deficit. Therefore, the plasma K⁺ concentration should be monitored frequently when assessing the response to treatment. Potassium chloride is usually the preparation of choice and will promote more rapid correction of hypokalemia and metabolic alkalosis. Potassium bicarbonate and citrate (metabolized to HCO₃⁻) tend to alkalinize the patient and would be more appropriate for hypokalemia associated with chronic diarrhea or [RTA](#).

Patients with severe hypokalemia or those unable to take anything by mouth require intravenous replacement therapy with KCl. The maximum concentration of administered K⁺ should be no more than 40 mmol/L via a peripheral vein or 60 mmol/L via a central vein. The rate of infusion should not exceed 20 mmol/h unless paralysis or malignant

ventricular arrhythmias are present. Ideally, KCl should be mixed in normal saline since dextrose solutions may initially exacerbate hypokalemia due to insulin-mediated movement of K⁺ into cells. Rapid intravenous administration of K⁺ should be used judiciously and requires close observation of the clinical manifestations of hypokalemia (electrocardiogram and neuromuscular examination).

HYPERKALEMIA

ETIOLOGY

Hyperkalemia, defined as a plasma K⁺ concentration >5.0 mmol/L, occurs as a result of either K⁺ release from cells or decreased renal loss. Increased K⁺ intake is rarely the sole cause of hyperkalemia since the phenomenon of *potassium adaptation* ensures rapid K⁺ excretion in response to increases in dietary consumption. Iatrogenic hyperkalemia may result from overzealous parenteral K⁺ replacement or in patients with renal insufficiency. *Pseudohyperkalemia* represents an artificially elevated plasma K⁺ concentration due to K⁺ movement out of cells immediately prior to or following venipuncture. Contributing factors include prolonged use of a tourniquet with or without repeated fist clenching, hemolysis, and marked leukocytosis or thrombocytosis. The latter two result in an elevated serum K⁺ concentration due to release of intracellular K⁺ following clot formation. Pseudohyperkalemia should be suspected in an otherwise asymptomatic patient with no obvious underlying cause. If proper venipuncture technique is used and a plasma (not serum) K⁺ concentration is measured, it should be normal. Intravascular hemolysis, tumor lysis syndrome, and rhabdomyolysis all lead to K⁺ release from cells as a result of tissue breakdown. Metabolic acidoses, with the exception of those due to the accumulation of organic anions, can be associated with mild hyperkalemia resulting from intracellular buffering of H⁺ (see above). As previously described (p. 278), insulin deficiency and hypertonicity (e.g., hyperglycemia) promote K⁺ shift from the [ICF](#) to the [ECF](#). The severity of exercise-induced hyperkalemia is related to the degree of exertion. It is due to release of K⁺ from muscles and is usually rapidly reversible, often associated with rebound hypokalemia. Treatment with beta blockers rarely causes hyperkalemia but may contribute to the elevation in plasma K⁺ concentration seen with other conditions. *Hyperkalemic periodic paralysis* ([Chap. 381](#)) is a rare autosomal dominant disorder characterized by episodic weakness or paralysis, precipitated by stimuli that normally lead to mild hyperkalemia (e.g., exercise). The genetic defect appears to be a single amino acid substitution due to a mutation in the gene for the skeletal muscle Na⁺ channel. Hyperkalemia may occur with severe digitalis toxicity due to inhibition of the Na⁺, K⁺-ATPase pump. Depolarizing muscle relaxants such as succinylcholine can increase the plasma K⁺ concentration, especially in patients with massive trauma, burns, or neuromuscular disease.

Chronic hyperkalemia is virtually always associated with decreased renal K⁺ excretion due to either impaired secretion or diminished distal solute delivery ([Table 49-4](#)). The latter is seldom the only cause of impaired K⁺ excretion but may significantly contribute to hyperkalemia in protein-malnourished (low urea excretion) and [ECF](#) volume-contracted (decreased distal NaCl delivery) patients. Decreased K⁺ secretion by the principal cells results from either impaired Na⁺ reabsorption or increased Cl⁻ reabsorption, both of which give rise to a diminished (less lumen-negative) [TEPD](#) in the [CCD](#). *Hyporeninemic hypoaldosteronism* is a syndrome characterized by euvolemia or ECF volume

expansion and suppressed renin and aldosterone levels ([Chaps. 331](#) and [333](#)). This disorder is commonly seen in mild renal insufficiency, diabetic nephropathy, or chronic tubulointerstitial disease. Patients frequently have an impaired kaliuretic response to exogenous mineralocorticoid administration, suggesting that enhanced distal Cl⁻ reabsorption (electroneutral Na⁺ reabsorption) may account for many of the findings of hyporeninemic hypoaldosteronism. [NSAIDs](#) inhibit renin secretion and the synthesis of vasodilatory renal prostaglandins. The resultant decrease in [GFR](#) and K⁺ secretion is often manifest as hyperkalemia. As a rule, the degree of hyperkalemia due to hypoaldosteronism is mild in the absence of increased K⁺ intake or renal dysfunction. Angiotensin-converting enzyme (ACE) inhibitors block the conversion of angiotensin I to angiotensin II, resulting in impaired aldosterone release. Patients at increased risk of ACE inhibitor-induced hyperkalemia include those with diabetes mellitus, renal insufficiency, decreased effective circulating arterial volume, bilateral renal artery stenosis, or concurrent use of K⁺-sparing diuretics or NSAIDs.

Decreased aldosterone synthesis may be due to *primary adrenal insufficiency* (Addison's disease) or congenital adrenal enzyme deficiency ([Chap. 331](#)). Heparin (including low-molecular-weight heparin) inhibits production of aldosterone by the cells of the zona glomerulosa and can lead to severe hyperkalemia in a subset of patients with underlying renal disease; diabetes mellitus; or those receiving K⁺-sparing diuretics, [ACE](#) inhibitors, or [NSAIDs](#). *Pseudohypoaldosteronism* is a rare familial disorder characterized by hyperkalemia, metabolic acidosis, renal Na⁺ wasting, hypotension, high renin and aldosterone levels, and end-organ resistance to aldosterone. The gene encoding the mineralocorticoid receptor is normal in these patients, and the electrolyte abnormalities can be reversed with suprapharmacologic doses of an exogenous mineralocorticoid (e.g., 9α-fludrocortisone) or an inhibitor of [11β-HSDH](#) (e.g., carbenoxolone). The kaliuretic response to aldosterone is impaired by K⁺-sparing diuretics. Spironolactone is a competitive mineralocorticoid antagonist, whereas amiloride and triamterene block the apical Na⁺ channel of the principal cell. Two other drugs that impair K⁺ secretion by blocking distal nephron Na⁺ reabsorption are trimethoprim and pentamidine. These antimicrobial agents may contribute to the hyperkalemia often seen in patients infected with HIV who are being treated for *Pneumocystis carinii* pneumonia.

Hyperkalemia frequently complicates acute oliguric renal failure due to increased K⁺ release from cells (acidosis, catabolism) and decreased excretion. Increased distal flow rate and K⁺ secretion per nephron compensate for decreased renal mass in chronic renal insufficiency. However, these adaptive mechanisms eventually fail to maintain K⁺ balance when the [GFR](#) falls below 10 to 15 mL/min or oliguria ensues. Otherwise asymptomatic urinary tract obstruction is an often overlooked cause of hyperkalemia. Other nephropathies associated with impaired K⁺ excretion include drug-induced interstitial nephritis, lupus nephritis, sickle cell disease, and diabetic nephropathy.

Gordon's syndrome is a rare condition characterized by hyperkalemia, metabolic acidosis, and a normal [GFR](#). These patients are usually volume-expanded with suppressed renin and aldosterone levels as well as refractory to the kaliuretic effect of exogenous mineralocorticoids. It has been suggested that these findings could all be accounted for by increased distal Cl⁻ reabsorption (electroneutral Na⁺ reabsorption), also referred to as a *Cl⁻ shunt*. A similar mechanism may be partially responsible for the

hyperkalemia associated with cyclosporine nephrotoxicity. *Hyperkalemic distal (type 4) RTA* may be due to either hypoaldosteronism or a Cl⁻ shunt (aldosterone-resistant).

CLINICAL FEATURES

Since the resting membrane potential is related to the ratio of the $\frac{[K^+]_{ICF}}{[K^+]_{ECF}}$ concentration, hyperkalemia partially depolarizes the cell membrane. Prolonged depolarization impairs membrane excitability and is manifest as weakness, which may progress to flaccid paralysis and hypoventilation if the respiratory muscles are involved. Hyperkalemia also inhibits renal ammoniogenesis and reabsorption of NH₄⁺ in the thick ascending limb of the loop of Henle. Thus, net acid excretion is impaired and results in metabolic acidosis, which may further exacerbate the hyperkalemia due to K⁺ movement out of cells.

The most serious effect of hyperkalemia is cardiac toxicity, which does not correlate well with the plasma K⁺ concentration. The earliest electrocardiographic changes include increased T-wave amplitude, or peaked T waves. More severe degrees of hyperkalemia result in a prolonged PR interval and QRS duration, atrioventricular conduction delay, and loss of P waves. Progressive widening of the QRS complex and merging with the T wave produces a sinewave pattern. The terminal event is usually ventricular fibrillation or asystole.

DIAGNOSIS

With rare exceptions, chronic hyperkalemia is always due to impaired K⁺ excretion. If the etiology is not readily apparent and the patient is asymptomatic, pseudohyperkalemia should be excluded, as described above. Oliguric acute renal failure and severe chronic renal insufficiency should also be ruled out. The history should focus on medications that impair K⁺ handling and potential sources of K⁺ intake. Evaluation of the $\frac{[K^+]_{ICF}}{[K^+]_{ECF}}$ compartment, effective circulating volume, and urine output are essential components of the physical examination. The severity of hyperkalemia is determined by the symptoms, plasma K⁺ concentration, and electrocardiographic abnormalities.

The appropriate renal response to hyperkalemia is to excrete at least 200 mmol of K⁺ daily. In most cases, diminished renal K⁺ loss is due to impaired K⁺ secretion, which can be assessed by measuring the TTKG (see above). A TTKG <10 implies a decreased driving force for K⁺ secretion due to either hypoaldosteronism or resistance to the renal effects of mineralocorticoid. This can be determined by evaluating the kaliuretic response to administration of mineralocorticoid (e.g., 9 α -fludrocortisone). Primary adrenal insufficiency can be differentiated from hyporeninemic hypoaldosteronism by examining the renin-aldosterone axis. Renin and aldosterone levels should be measured in the supine and upright positions, following three days of Na⁺ restriction (Na⁺ intake <10 mmol/d) in combination with a loop diuretic to induce mild volume contraction. Aldosterone-resistant hyperkalemia can result from the various causes of impaired distal Na⁺ reabsorption or from a Cl⁻ shunt. The former leads to salt wasting, $\frac{[K^+]_{ICF}}{[K^+]_{ECF}}$ volume contraction, and high renin and aldosterone levels. In contrast, enhanced distal Cl⁻ reabsorption is associated with volume expansion and suppressed renin and aldosterone secretion. As mentioned above, hypoaldosteronism seldom causes severe hypokalemia in the absence of increased dietary K⁺ intake, renal

insufficiency, transcellular K⁺ shifts, or antikaliuretic drugs.

CLINICAL APPROACH

See [Fig. 49-4](#).

TREATMENT

The approach to therapy depends on the degree of hyperkalemia as determined by the plasma K⁺ concentration, associated muscular weakness, and changes on the electrocardiogram. Potentially fatal hyperkalemia rarely occurs unless the plasma K⁺ concentration exceeds 7.5 mmol/L and is usually associated with profound weakness and absent P waves, QRS widening, or ventricular arrhythmias on the electrocardiogram.

Severe hyperkalemia requires emergent treatment directed at minimizing membrane depolarization, shifting K⁺ into cells, and promoting K⁺ loss. In addition, exogenous K⁺ intake and antikaliuretic drugs should be discontinued. Administration of calcium gluconate decreases membrane excitability. The usual dose is 10 mL of a 10% solution infused over 2 to 3 min. The effect begins within minutes but is short-lived (30 to 60 min), and the dose can be repeated if no change in the electrocardiogram is seen after 5 to 10 min. Insulin causes K⁺ to shift into cells by mechanisms described previously and will temporarily lower the plasma K⁺ concentration. Although glucose alone will stimulate insulin release from normal pancreatic β cells, a more rapid response generally occurs when exogenous insulin is administered (with glucose to prevent hypoglycemia). A commonly recommended combination is 10 to 20 units of regular insulin and 25 to 50 g of glucose. Obviously, hyperglycemic patients should not be given glucose. If effective, the plasma K⁺ concentration will fall by 0.5 to 1.5 mmol/L in 15 to 30 min and the effect will last for several hours. Alkali therapy with intravenous NaHCO₃ can also shift K⁺ into cells. This is safest when administered as an isotonic solution of 3 ampules per liter (134 mmol/L NaHCO₃) and ideally should be reserved for severe hyperkalemia associated with metabolic acidosis. Patients with end-stage renal disease seldom respond to this intervention and may not tolerate the Na⁺ load and resultant volume expansion. When administered parenterally or in nebulized form, β_2 -adrenergic agonists promote cellular uptake of K⁺ (see above). The onset of action is 30 min, lowering the plasma K⁺ concentration by 0.5 to 1.5 mmol/L, and the effect lasts 2 to 4 h.

Removal of K⁺ can be achieved using diuretics, cation-exchange resin, or dialysis. Loop and thiazide diuretics, often in combination, may enhance K⁺ excretion if renal function is adequate. Sodium polystyrene sulfonate is a cation-exchange resin that promotes the exchange of Na⁺ for K⁺ in the gastrointestinal tract. Each gram binds 1 mmol of K⁺ and releases 2 to 3 mmol of Na⁺. When given by mouth, the usual dose is 25 to 50 g mixed with 100 mL of 20% sorbitol to prevent constipation. This will generally lower the plasma K⁺ concentration by 0.5 to 1.0 mmol/L within 1 to 2 h and last for 4 to 6 h. Sodium polystyrene sulfonate can also be administered as a retention enema consisting of 50 g of resin and 50 mL of 70% sorbitol mixed in 150 mL of tap water. The sorbitol should be omitted from the enema in postoperative patients due to the increased incidence of sorbitol-induced colonic necrosis, especially following renal transplantation. The most rapid and effective way of lowering the plasma K⁺ concentration is hemodialysis. This

should be reserved for patients with renal failure and those with severe life-threatening hyperkalemia unresponsive to more conservative measures. Peritoneal dialysis also removes K⁺ but is only 15 to 20% as effective as hemodialysis. Finally, the underlying cause of the hyperkalemia should be treated. This may involve dietary modification, correction of metabolic acidosis, cautious volume expansion, and administration of exogenous mineralocorticoid.

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50. ACIDOSIS AND ALKALOSIS - Thomas D. DuBose, Jr.

NORMAL ACID-BASE HOMEOSTASIS

Systemic arterial pH is maintained between 7.35 and 7.45 by extracellular and intracellular chemical buffering together with respiratory and renal regulatory mechanisms. The control of arterial CO₂ tension (P_aCO₂) by the central nervous system and respiratory systems and the control of the plasma bicarbonate by the kidneys stabilize the arterial pH by excretion or retention of acid or alkali. The metabolic and respiratory components that regulate systemic pH are described by the Henderson-Hasselbalch equation:

Under most circumstances, CO₂ production and excretion are matched, and the usual steady-state P_aCO₂ is maintained at 40 mmHg. Underexcretion of CO₂ produces hypercapnia, and overexcretion causes hypocapnia. Nevertheless, production and excretion are again matched at a new steady-state P_aCO₂. Therefore, the P_aCO₂ is regulated primarily by neural respiratory factors ([Chap. 263](#)) and is not subject to regulation by the rate of CO₂ production. Hypercapnia is usually the result of hypoventilation rather than of increased CO₂ production. Increases or decreases in P_aCO₂ represent derangements of neural respiratory control or are due to compensatory changes in response to a primary alteration in the plasma [HCO₃⁻].

Primary changes in P_aCO₂ can cause acidosis or alkalosis, depending on whether P_aCO₂ is above or below the normal value of 40 mmHg (respiratory acidosis or alkalosis, respectively). Primary alteration of P_aCO₂ evokes cellular buffering and renal adaptation, a slow process that becomes more efficient with time. A primary change in the plasma [HCO₃⁻] as a result of metabolic or renal factors results in compensatory changes in ventilation that blunt the changes in blood pH that would occur otherwise. Such respiratory alterations are referred to as *secondary*, or compensatory, changes, since they occur in response to primary metabolic changes.

The kidneys regulate plasma [HCO₃⁻] through three main processes: (1) "reabsorption" of filtered HCO₃⁻, (2) formation of titratable acid, and (3) excretion of NH₄⁺ in the urine. The kidney filters approximately 4000 mmol of HCO₃⁻ per day. To reabsorb the filtered load of HCO₃⁻, the renal tubules must therefore secrete 4000 mmol of hydrogen ions. Between 80 and 90% of HCO₃⁻ is reabsorbed in the proximal tubule. The distal nephron reabsorbs the remainder and secretes protons, as generated from metabolism, to defend systemic pH. While this quantity of protons, 40 to 60 mmol/d, is small, it must be secreted to prevent chronic positive H⁺ balance and metabolic acidosis. This quantity of secreted protons is represented in the urine as titratable acid and NH₄⁺. Metabolic acidosis in the face of normal renal function increases NH₄⁺ production and excretion. NH₄⁺ production and excretion are impaired in chronic renal failure, hyperkalemia, and renal tubular acidosis.

In sum, these regulatory responses, including chemical buffering, the regulation of P_aCO₂ by the respiratory system, and of [HCO₃⁻] by the kidneys, act in concert to maintain a systemic arterial pH between 7.35 and 7.45.

DIAGNOSIS OF GENERAL TYPES OF DISTURBANCES

The most common clinical disturbances are simple acid-base disorders, i.e., metabolic acidosis or alkalosis or respiratory acidosis or alkalosis. Since compensation is not complete, the pH is abnormal in simple disturbances. More complicated clinical situations can give rise to mixed acid-base disturbances.

SIMPLE ACID-BASE DISORDERS

Primary respiratory disturbances (primary changes in P_{aCO_2}) invoke compensatory metabolic responses (secondary changes in $[HCO_3^-]$), and primary metabolic disturbances elicit predictable compensatory respiratory responses. Physiologic compensation can be predicted from the relationships displayed in [Table 50-1](#). Primary changes in P_{aCO_2} or $[HCO_3^-]$ alter systemic pH and cause acidosis or alkalosis. To illustrate, metabolic acidosis due to an increase in endogenous acids (e.g., ketoacidosis) lowers extracellular fluid $[HCO_3^-]$ and decreases extracellular pH. This stimulates the medullary chemoreceptors to increase ventilation and to return the ratio of $[HCO_3^-]$ to P_{aCO_2} , and thus pH, toward normal, although not to normal. The degree of respiratory compensation expected in a simple form of metabolic acidosis can be predicted from the relationship: $P_{aCO_2} = (1.5 \times [HCO_3^-]) + 8$, i.e., the P_{aCO_2} is expected to decrease 1.25 mmHg for each mmol per liter decrease in $[HCO_3^-]$. Thus, a patient with metabolic acidosis and $[HCO_3^-]$ of 12 mmol/L would be expected to have a P_{aCO_2} between 24 and 28 mmHg. Values for P_{aCO_2} below 24 or greater than 28 mmHg define a mixed disturbance (metabolic acidosis and respiratory alkalosis or metabolic alkalosis and respiratory acidosis, respectively). Another way to judge the appropriateness of the response in $[HCO_3^-]$ or P_{aCO_2} is to use an acid-base nomogram ([Fig. 50-1](#)). While the shaded areas of the nomogram show the 95% confidence limits for normal compensation in simple disturbances, finding acid-base values within the shaded area does not necessarily rule out a mixed disturbance. Imposition of one disorder over another may result in values lying within the area of a third. Thus, the nomogram, while convenient, is not a substitute for the equations in [Table 50-1](#).

MIXED ACID-BASE DISORDERS

Mixed acid-base disorders -- defined as independently coexisting disorders, not merely compensatory responses -- are often seen in patients in critical care units and can lead to dangerous extremes of pH. A patient with diabetic ketoacidosis (metabolic acidosis) may develop an independent respiratory problem leading to respiratory acidosis or alkalosis. Patients with underlying pulmonary disease may not respond to metabolic acidosis with an appropriate ventilatory response because of insufficient respiratory reserve. Such imposition of respiratory acidosis on metabolic acidosis can lead to severe acidemia and a poor outcome. When metabolic acidosis and metabolic alkalosis coexist in the same patient, the pH may be normal or near normal. When the pH is normal, an elevated anion gap (see below) denotes the presence of a metabolic acidosis. A diabetic patient with ketoacidosis may have renal dysfunction resulting in simultaneous metabolic acidosis. Patients who have ingested an overdose of drug combinations such as sedatives and salicylates may have mixed disturbances as a result of the acid-base response to the individual drugs (metabolic acidosis mixed with

respiratory acidosis or respiratory alkalosis, respectively). Even more complex are triple acid-base disturbances. For example, patients with metabolic acidosis due to alcoholic ketoacidosis may develop metabolic alkalosis due to vomiting and superimposed respiratory alkalosis due to the hyperventilation of hepatic dysfunction or alcohol withdrawal.

DIAGNOSIS OF ACID-BASE DISORDERS

Care should be taken when measuring blood gases to obtain the arterial blood sample without using excessive heparin. In the determination of arterial blood gases by the clinical laboratory, both pH and P_{aCO_2} are measured, and the $[HCO_3^-]$ is calculated from the Henderson-Hasselbalch equation. This calculated value should be compared with the measured $[HCO_3^-]$ (total CO_2) on the electrolyte panel. These two values should agree within 2 mmol/L. If they do not, the values may not have been drawn simultaneously, a laboratory error may be present, or an error could have been made in calculating the $[HCO_3^-]$. After verifying the blood acid-base values, one can then identify the precise acid-base disorder.

The most common causes of acid-base disorders should be kept in mind while probing the history for clues about the etiology. For example, established chronic renal failure is expected to cause a metabolic acidosis, and chronic vomiting frequently causes metabolic alkalosis. Patients with pneumonia, sepsis, or cardiac failure frequently have respiratory alkalosis, and patients with chronic obstructive pulmonary disease or a sedative drug overdose often display a respiratory acidosis. The drug history is important since loop or thiazide diuretics may cause metabolic alkalosis, and the carbonic anhydrase inhibitor, acetazolamide, can result in metabolic acidosis.

Blood for electrolytes and arterial blood gases should be drawn simultaneously prior to therapy, since an increase in $[HCO_3^-]$ occurs with metabolic alkalosis and respiratory acidosis. Conversely, a decrease in $[HCO_3^-]$ occurs in metabolic acidosis and respiratory alkalosis.

Metabolic acidosis leads to hyperkalemia as a result of cellular shifts in which H^+ is exchanged for K^+ or Na^+ . For each decrease in blood pH of 0.10, the plasma $[K^+]$ should rise by 0.6 mmol/L. This relationship is not invariable. Diabetic ketoacidosis, lactic acidosis, diarrhea, and renal tubular acidosis (RTA) are often associated with potassium depletion because of urinary K^+ wasting.

Anion Gap All evaluations of acid-base disorders should include a simple calculation of the anion gap (AG); it represents those unmeasured anions in plasma (normally 10 to 12 mmol/L) and is calculated as follows: $AG = Na^+ - (Cl^- + HCO_3^-)$. The unmeasured anions include anionic proteins, phosphate, sulfate, and organic anions. When acid anions, such as acetoacetate and lactate, accumulate in extracellular fluid, the AG increases, causing a high-AG acidosis. An increase in the AG is most often due to an increase in unmeasured anions and less commonly is due to a decrease in unmeasured cations (calcium, magnesium, potassium). In addition, the AG may increase with an increase in anionic albumin, either because of increased albumin concentration or alkalosis, which alters albumin charge. A decrease in the AG can be due to: (1) an increase in unmeasured cations; (2) the addition to the blood of abnormal cations, such

as lithium (lithium intoxication) or cationic immunoglobulins (plasma cell dyscrasias); (3) a reduction in the major plasma anion albumin concentration (nephrotic syndrome); (4) a decrease in the effective anionic charge on albumin by acidosis; or (5) hyperviscosity and severe hyperlipidemia, which can lead to an underestimation of sodium and chloride concentrations.

In the face of a normal serum albumin, a high [AG](#) is usually due to non-chloride-containing acids that contain inorganic (phosphate, sulfate), organic (ketoacids, lactate, uremic organic anions), exogenous (salicylate or ingested toxins with organic acid production), or unidentified anions. By definition, therefore, a high-AG acidosis has two identifying features: a low $[\text{HCO}_3^-]$ and an elevated AG. The latter is present even if an additional acid-base disorder is superimposed to modify the $[\text{HCO}_3^-]$ independently. Simultaneous metabolic acidosis of the high-AG variety plus either chronic respiratory acidosis or metabolic alkalosis represents such a situation in which $[\text{HCO}_3^-]$ may be normal or even high. However, the AG is elevated, and $[\text{Cl}^-]$ is depressed.

Similarly, normal values for $[\text{HCO}_3^-]$, PaCO_2 , and pH do not ensure the absence of an acid-base disturbance. For instance, an alcoholic who has been vomiting may develop a metabolic alkalosis with a pH of 7.55, PaCO_2 of 48 mmHg, $[\text{HCO}_3^-]$ of 40 mmol/L, $[\text{Na}^+]$ of 135, $[\text{Cl}^-]$ of 80, and $[\text{K}^+]$ of 2.8. If such a patient were then to develop a superimposed alcoholic ketoacidosis with a β -hydroxybutyrate concentration of 15 mM, arterial pH would fall to 7.40, $[\text{HCO}_3^-]$ to 25 mmol/L, and the PaCO_2 to 40 mmHg. Although these blood gases are normal, the [AG](#) is elevated at 30 mmol/L, indicating a mixed metabolic alkalosis and metabolic acidosis.

METABOLIC ACIDOSIS

Metabolic acidosis can occur because of an increase in endogenous acid production (such as lactate and ketoacids), loss of bicarbonate (as in diarrhea), or accumulation of endogenous acids (as in renal failure). Metabolic acidosis has profound effects on the respiratory, cardiac, and nervous systems. The fall in blood pH is accompanied by a characteristic increase in ventilation, especially the tidal volume (Kussmaul respiration). Intrinsic cardiac contractility may be depressed, but inotropic function can be normal because of catecholamine release. Both peripheral arterial vasodilation and central venoconstriction can be present; the decrease in central and pulmonary vascular compliance predisposes to pulmonary edema with even minimal volume overload. Central nervous system function is depressed, with headache, lethargy, stupor, and, in some cases, even coma. Glucose intolerance may also occur.

There are two major categories of clinical metabolic acidosis: high-[AG](#) and normal-AG, or hyperchloremic acidosis ([Table 50-2](#) and [Table 50-3](#)).

TREATMENT

Treatment of metabolic acidosis with alkali should be reserved for severe acidemia except when the patient has no "potential HCO_3^- " in plasma. Potential $[\text{HCO}_3^-]$ can be estimated from the increment (D) in the anion gap ($\text{DAG} = \text{patient's AG} - 10$). It must be determined if the acid anion in plasma is metabolizable (i.e., β -hydroxybutyrate,

acetoacetate, and lactate) or nonmetabolizable (anions that accumulate in chronic renal failure and after toxin ingestion). The latter requires return of renal function to replenish the $[\text{HCO}_3^-]$ deficit, a slow and often unpredictable process. Consequently, patients with a normal AG acidosis (hyperchloremic acidosis), a slightly elevated AG (mixed hyperchloremic and AG acidosis), or an AG attributable to a nonmetabolizable anion in the face of renal failure should receive alkali therapy, either orally (NaHCO_3 or Shohl's solution) or intravenously (NaHCO_3), in an amount necessary to slowly increase the plasma $[\text{HCO}_3^-]$ into the 20 to 22 mmol/L range.

Controversy exists, however, in regard to the use of alkali in patients with a pure [AG](#) acidosis owing to accumulation of a metabolizable organic acid anion (ketoacidosis or lactic acidosis). In general, severe acidosis ($\text{pH} < 7.20$) warrants the intravenous administration of 50 to 100 meq of NaHCO_3 , over 30 to 45 min, during the initial 1 to 2 h of therapy. Provision of such modest quantities of alkali in this situation seems to provide an added measure of safety, but it is essential to monitor plasma electrolytes during the course of therapy, since the $[\text{K}^+]$ may decline as pH rises. The goal is to increase the $[\text{HCO}_3^-]$ to 10 meq/L and the pH to 7.25, not to increase these values to normal.

HIGH-ANION-GAP ACIDOSES

There are four principal causes of a high-[AG](#) acidosis: (1) lactic acidosis, (2) ketoacidosis, (3) ingested toxins ([Table 50-2](#)), and (4) acute and chronic renal failure. Initial screening to differentiate the high-AG acidoses should include: (1) a probe of the history for evidence of drug and toxin ingestion and measurement of arterial blood gas to detect coexistent respiratory alkalosis (salicylates); (2) determination of whether diabetes mellitus is present (diabetic ketoacidosis); (3) a search for evidence of alcoholism or increased levels of β -hydroxybutyrate (alcoholic ketoacidosis); (4) observation for clinical signs of uremia and determination of the blood urea nitrogen (BUN) and creatinine (uremic acidosis); (5) inspection of the urine for oxalate crystals (ethylene glycol); and (6) recognition of the numerous clinical settings in which lactate levels may be increased (hypotension, shock, cardiac failure, leukemia, cancer, and drug or toxin ingestion).

Lactic Acidosis An increase in plasma L-lactate may be secondary to poor tissue perfusion (type A) -- circulatory insufficiency (shock, circulatory failure), severe anemia, mitochondrial enzyme defects, and inhibitors (carbon monoxide, cyanide) -- or to aerobic disorders (type B) -- malignancies, diabetes mellitus, renal or hepatic failure, severe infections (cholera, malaria), seizures, AIDS, or drugs/toxins (biguanides, ethanol, methanol, isoniazid, AZT analogues, and fructose). Unrecognized bowel ischemia or infarction in a patient with severe atherosclerosis or cardiac decompensation receiving vasopressors is a common cause of lactic acidosis. D-Lactic acid acidosis, which may be associated with jejunoileal bypass or intestinal obstruction and is due to formation of D-lactate by gut bacteria, may cause both an increased [AG](#) and hyperchloremia.

TREATMENT

The underlying condition that disrupts lactate metabolism must first be corrected; tissue

perfusion must be restored when it is inadequate. Vasoconstrictors should be avoided, if possible, since they may worsen tissue perfusion. Alkali therapy is generally advocated for acute, severe acidemia ($\text{pH} < 7.1$) to improve cardiac function and lactate utilization. However, NaHCO_3 therapy may paradoxically depress cardiac performance and exacerbate acidosis by enhancing lactate production (HCO_3^- stimulates phosphofructokinase). While the use of alkali in moderate lactic acidosis is controversial, it is generally agreed that attempts to return the pH or $[\text{HCO}_3^-]$ to normal by administration of exogenous NaHCO_3 are deleterious. A reasonable approach is to infuse sufficient NaHCO_3 to raise the arterial pH to no more than 7.2 over 30 to 40 min.

NaHCO_3 therapy can cause fluid overload and hypertension because the amount required can be massive when accumulation of lactic acid is relentless. Fluid administration is poorly tolerated because of central venoconstriction, especially in the oliguric patient. If the underlying cause of the lactic acidosis can be remedied, blood lactate will be converted to HCO_3^- and may result in an overshoot alkalosis.

Ketoacidosis

Diabetic Ketoacidosis This condition is caused by increased fatty acid metabolism and the accumulation of ketoacids (acetoacetate and β -hydroxybutyrate). Diabetic ketoacidosis usually occurs in insulin-dependent diabetes mellitus in association with cessation of insulin or an intercurrent illness, such as an infection, gastroenteritis, pancreatitis, or myocardial infarction, which increases insulin requirements temporarily and acutely. The accumulation of ketoacids accounts for the increment in the [AG](#) and is accompanied most often by hyperglycemia [$\text{glucose} > 17 \text{ mmol/L}$ (300 mg/dL)]. It should be noted that since insulin prevents production of ketones, bicarbonate therapy is rarely needed except with extreme acidemia ($\text{pH} < 7.1$), and then in only limited amounts (see "Treatment" for lactic acidosis). **The management of this condition is described in [Chap. 333](#).*

Alcoholic Ketoacidosis Chronic alcoholics can develop ketoacidosis when alcohol consumption is abruptly curtailed; it is usually associated with binge drinking, vomiting, abdominal pain, starvation, and volume depletion. The glucose concentration is low or normal, and acidosis may be severe because of elevated ketones, predominantly β -hydroxybutyrate. Mild lactic acidosis may coexist because of alteration in the redox state. The nitroprusside ketone reaction (Acetest) can detect acetoacetic acid but not β -hydroxybutyrate, so that the degree of ketosis and ketonuria can be underestimated. Typically, insulin levels are low, and concentrations of triglyceride, cortisol, glucagon, and growth hormone are increased.

TREATMENT

Extracellular fluid deficits should be repleted by intravenous administration of saline and glucose (5% dextrose in 0.9% NaCl). Hypophosphatemia, hypokalemia, and hypomagnesemia may coexist and should be corrected. Hypophosphatemia usually emerges 12 to 24 h after admission, may be exacerbated by glucose infusion, and, if severe, may induce rhabdomyolysis. Upper gastrointestinal hemorrhage, pancreatitis, and pneumonia may accompany this disorder.

Drug- and Toxin-Induced Acidosis

Salicylates (See also [Chap. 396](#)) Salicylate intoxication in adults usually causes respiratory alkalosis, mixed metabolic acidosis-respiratory alkalosis, or a pure high-[AG](#) metabolic acidosis. In the latter example, which is less common, only a portion of the AG is due to the salicylates. Lactic acid production is also often increased.

TREATMENT

This should begin with vigorous gastric lavage with isotonic saline (not NaHCO_3) followed by administration of activated charcoal. In the acidotic patient, to facilitate removal of salicylate, intravenous NaHCO_3 is administered in amounts adequate to alkalinize the urine and to maintain urine output (urine pH > 7.5). While this form of therapy is straightforward in acidotic patients, a coexisting respiratory alkalosis may make this approach hazardous. Acetazolamide may be administered when an alkaline diuresis cannot be achieved, but this drug can cause systemic metabolic acidosis if HCO_3^- is not replaced. Hypokalemia may occur with an alkaline diuresis from NaHCO_3 and should be treated promptly and aggressively. Glucose-containing fluids should be administered because of the danger of hypoglycemia. Excessive insensible fluid losses may cause severe volume depletion and hypernatremia. If renal failure prevents rapid clearance of salicylate, hemodialysis can be performed against a bicarbonate dialysate.

Alcohols Under most physiologic conditions, sodium, urea, and glucose generate the osmotic pressure of blood. Plasma osmolality is calculated according to the following expression: $P_{\text{osm}} = 2\text{Na}^+ + \text{Glu} + \text{BUN}$ (all in mmol/L), or, using conventional laboratory values in which glucose and BUN are expressed in milligrams per deciliter: $P_{\text{osm}} = 2\text{Na}^+ + \text{Glu}/18 + \text{BUN}/2.8$. The calculated and determined osmolality should agree within 10 to 15 mmol/kg H_2O . When the measured osmolality exceeds the calculated osmolality by more than 15 to 20 mmol/kg H_2O , one of two circumstances prevails. Either the serum sodium is spuriously low, as with hyperlipidemia or hyperproteinemia (pseudohyponatremia), or osmolytes other than sodium salts, glucose, or urea have accumulated in plasma. Examples include mannitol, radiocontrast media, isopropyl alcohol, ethylene glycol, ethanol, methanol, and acetone. In this situation, the difference between the calculated osmolality and the measured osmolality (*osmolar gap*) is proportional to the concentration of the unmeasured solute. With an appropriate clinical history and index of suspicion, identification of an osmolar gap is helpful in identifying the presence of poison-associated [AG](#) acidosis.

Ethylene Glycol (See also [Chap. 396](#)) Ingestion of ethylene glycol (commonly used in antifreeze) leads to a metabolic acidosis and severe damage to the central nervous system, heart, lungs, and kidneys. The increased [AG](#) and osmolar gap are attributable to ethylene glycol and its metabolites, oxalic acid, glycolic acid, and other organic acids. Lactic acid production increases secondary to inhibition of the tricarboxylic acid cycle and altered intracellular redox state. Diagnosis is facilitated by recognizing oxalate crystals in the urine, the presence of an osmolar gap in serum, and a high-AG acidosis. Treatment should not be delayed while awaiting measurement of ethylene glycol levels in this setting.

TREATMENT

This includes the prompt institution of a saline or osmotic diuresis, thiamine and pyridoxine supplements, fomepizole or ethanol, and hemodialysis. The intravenous administration of the new alcohol dehydrogenase inhibitor, fomepizole (4-methylpyrazole; 7 mg/kg as a loading dose), or ethanol intravenously to achieve a level of 22 mmol/L (100 mg/dL) serves to lessen toxicity because they compete with ethylene glycol for metabolism by alcohol dehydrogenase. Fomepizole, although expensive, offers the advantages of a predictable decline in ethylene glycol levels without the adverse effects, such as excessive obtundation, associated with ethyl alcohol infusion.

Methanol (See also [Chap. 396](#)) The ingestion of methanol (wood alcohol) causes metabolic acidosis, and its metabolites formaldehyde and formic acid cause severe optic nerve and central nervous system damage. Lactic acid, ketoacids, and other unidentified organic acids may contribute to the acidosis. Due to its low molecular weight (32 Da), an osmolar gap is usually present.

TREATMENT

This is similar to that for ethylene glycol intoxication, including general supportive measures, fomepizole or ethanol administration, and hemodialysis.

Renal Failure (See also [Chaps. 269](#) and [270](#)) The hyperchloremic acidosis of moderate renal insufficiency is eventually converted to the high-[AG](#) acidosis of advanced renal failure. Poor filtration and reabsorption of organic anions contribute to the pathogenesis. As renal disease progresses, the number of functioning nephrons eventually becomes insufficient to keep pace with net acid production. Uremic acidosis is characterized, therefore, by a reduced rate of NH_4^+ production and excretion, primarily due to decreased renal mass. $[\text{HCO}_3^-]$ rarely falls below 15 mmol/L, and the AG rarely exceeds 20 mmol/L. The acid retained in chronic renal disease is buffered by alkaline salts from bone. Despite significant retention of acid (up to 20 mmol/d), the serum $[\text{HCO}_3^-]$ does not decrease further, indicating participation of buffers outside the extracellular compartment. Chronic metabolic acidosis results in significant loss of bone mass due to reduction in bone calcium carbonate. Chronic acidosis also increases urinary calcium excretion, proportional to cumulative acid retention.

TREATMENT

Both uremic acidosis and the hyperchloremic acidosis of renal failure require oral alkali replacement to maintain the $[\text{HCO}_3^-]$ between 20 and 24 mmol/L. This can be accomplished with relatively modest amounts of alkali (1.0 to 1.5 mmol/kg body weight per day). It is assumed that alkali replacement prevents the harmful effects of H^+ -balance on bone and prevents or retards muscle catabolism. Sodium citrate (Shohl's solution) or NaHCO_3 tablets are equally effective alkalinizing salts. Citrate enhances the absorption of aluminum from the gastrointestinal tract and should never be given together with aluminum-containing antacids because of the risk of aluminum intoxication. When hyperkalemia is present, furosemide (60 to 80 mg/d) should be added.

HYPERCHLOREMIC METABOLIC ACIDOSES

Alkali can be lost from the gastrointestinal tract in diarrhea or from the kidneys (renal tubular acidosis, [RTA](#)). In these disorders ([Table 50-3](#)), reciprocal changes in $[\text{Cl}^-]$ and $[\text{HCO}_3^-]$ result in a normal [AG](#). In pure hyperchloremic acidosis, therefore, the increase in $[\text{Cl}^-]$ above the normal value approximates the decrease in $[\text{HCO}_3^-]$. The absence of such a relationship suggests a mixed disturbance.

In diarrhea, stools contain a higher $[\text{HCO}_3^-]$ and decomposed HCO_3^- than plasma so that metabolic acidosis develops along with volume depletion. Instead of an acid urine pH (as anticipated with systemic acidosis), urine pH is usually around 6 because metabolic acidosis and hypokalemia increase renal synthesis and excretion of NH_4^+ , thus providing a urinary buffer that increases urine pH. Metabolic acidosis due to gastrointestinal losses with a high urine pH can be differentiated from [RTA](#) ([Chap. 276](#)) because urinary NH_4^+ excretion is typically low in RTA and high with diarrhea. Urinary NH_4^+ levels can be estimated by calculating the urine anion gap (UAG): $\text{UAG} = [\text{Na}^+ + \text{K}^+]_{\text{u}} - [\text{Cl}^-]_{\text{u}}$. When $[\text{Cl}^-]_{\text{u}} > [\text{Na}^+ + \text{K}^+]_{\text{u}}$, the urine ammonium level is appropriately increased, suggesting an extrarenal cause of the acidosis.

Loss of functioning renal parenchyma by progressive renal disease leads to hyperchloremic acidosis when the glomerular filtration rate (GFR) is between 20 and 50 mL/min and to uremic acidosis with a high [AG](#) when the GFR falls to <20 mL/min. Such a progression occurs commonly with tubulointerstitial forms of renal disease, but hyperchloremic metabolic acidosis can persist with advanced glomerular disease. In advanced renal failure, ammoniogenesis is reduced in proportion to the loss of functional renal mass, and ammonium accumulation and trapping in the outer medullary collecting tubule may also be impaired. Because of adaptive increases in K^+ secretion by the collecting duct and colon, the acidosis of chronic renal insufficiency is typically normokalemic.

Proximal [RTA](#) (type 2 RTA) is most often due to generalized proximal tubular dysfunction manifested by glycosuria, generalized aminoaciduria, and phosphaturia (Fanconi syndrome). With a low plasma $[\text{HCO}_3^-]$, the urine pH is acid ($\text{pH} < 5.5$). The fractional excretion of $[\text{HCO}_3^-]$ may exceed 10 to 15% when the serum $\text{HCO}_3^- > 20$ mmol/L. Since HCO_3^- is not reabsorbed normally in the proximal tubule, therapy with NaHCO_3 will enhance renal potassium wasting and hypokalemia.

The typical findings in classic distal [RTA](#) (type 1 RTA) ([Chap. 276](#)) include hypokalemia, hyperchloremic acidosis, low urinary NH_4^+ excretion (positive [UAG](#), low urine $[\text{NH}_4^+]$), and inappropriately high urine pH ($\text{pH} > 5.5$). Such patients are unable to acidify the urine below a pH of 5.5. Most patients have hypocitraturia and hypercalciuria, so that nephrolithiasis, nephrocalcinosis, and bone disease are common. In type 4 RTA, hyperkalemia is disproportionate to the reduction in [GFR](#) because of coexisting dysfunction of potassium and acid secretion. Urinary ammonium excretion is invariably depressed, and renal function may be compromised, for example, due to diabetic nephropathy, amyloidosis, or tubulointerstitial disease. **See [Chap. 276](#) for the pathophysiology, diagnosis, and treatment of RTA.*

Hyporeninemic Hypoaldosteronism (See also [Chap. 331](#)) This condition typically causes hyperchloremic metabolic acidosis, most commonly in older adults with diabetes mellitus or tubulointerstitial disease and renal insufficiency. Patients usually have mild to moderate renal insufficiency and acidosis, with elevation in serum $[K^+]$ (5.2 to 6.0 mmol/L), concurrent hypertension, and congestive heart failure. Both the metabolic acidosis and the hyperkalemia are out of proportion to impairment in [GFR](#). Nonsteroidal anti-inflammatory drugs -- trimethoprim, pentamidine, and ACE-inhibitors -- can also cause hyperkalemia with hyperchloremic metabolic acidosis in patients with renal insufficiency ([Table 50-3](#)).

METABOLIC ALKALOSIS

Metabolic alkalosis is manifested by an elevated arterial pH, an increase in the serum $[HCO_3^-]$, and an increase in P_{aCO_2} as a result of compensatory alveolar hypoventilation. It is often accompanied by hypochloremia and hypokalemia. The patient with a high $[HCO_3^-]$ and a low $[Cl^-]$ has either metabolic alkalosis or chronic respiratory acidosis. As shown in [Table 50-1](#), the P_{aCO_2} increases 6 mmHg for each 10-mmol/L increase in the $[HCO_3^-]$ above normal. Stated differently, in the range of $[HCO_3^-]$ from 10 to 40 mmol/L, the predicted P_{aCO_2} is approximately equal to the $[HCO_3^-] + 15$. The arterial pH establishes the diagnosis, since it is increased in metabolic alkalosis and decreased or normal in respiratory acidosis. Metabolic alkalosis frequently occurs in association with other disorders such as respiratory acidosis or alkalosis or metabolic acidosis.

PATHOGENESIS

Metabolic alkalosis occurs as a result of net gain of $[HCO_3^-]$ or loss of nonvolatile acid (usually HCl by vomiting) from the extracellular fluid. Since it is unusual for alkali to be added to the body, the disorder involves a generative stage, in which the loss of acid usually causes alkalosis, and a maintenance stage, in which the kidneys fail to compensate by excreting HCO_3^- because of volume contraction, a low [GFR](#), or depletion of Cl^- or K^+ .

Under normal circumstances, the kidneys have an impressive capacity to excrete HCO_3^- . Continuation of metabolic alkalosis represents a failure of the kidneys to eliminate HCO_3^- in the usual manner. For HCO_3^- to be added to the extracellular fluid, it must be administered exogenously or synthesized endogenously, in part or entirely by the kidneys. The kidneys will retain, rather than excrete, the excess alkali and maintain the alkalosis if (1) volume deficiency, chloride deficiency, and K^+ deficiency exist in combination with a reduced [GFR](#), which augments distal tubule H^+ secretion; or (2) hypokalemia exists because of autonomous hyperaldosteronism. In the first example, alkalosis is corrected by administration of NaCl and KCl, while in the latter it is necessary to repair the alkalosis by pharmacologic or surgical intervention, not with saline administration.

DIFFERENTIAL DIAGNOSIS

To establish the cause of metabolic alkalosis ([Table 50-4](#)), it is necessary to assess the status of the extracellular fluid volume (ECFV), the recumbent and upright blood pressure, the serum $[K^+]$, and the renin-aldosterone system. For example, the presence

of chronic hypertension and chronic hypokalemia in an alkalotic patient suggests either mineralocorticoid excess or that the hypertensive patient is receiving diuretics. Low plasma renin activity and normal urine $[\text{Na}^+]$ and $[\text{Cl}^-]$ in a patient who is not taking diuretics indicate a primary mineralocorticoid excess syndrome. The combination of hypokalemia and alkalosis in a normotensive, nonedematous patient can be due to Bartter's or Gitelman's syndrome, magnesium deficiency, vomiting, exogenous alkali, or diuretic ingestion. Determination of urine electrolytes (especially the urine $[\text{Cl}^-]$) and screening of the urine for diuretics may be helpful. If the urine is alkaline, with an elevated $[\text{Na}^+]$ and $[\text{K}^+]$ but low $[\text{Cl}^-]$, the diagnosis is usually either vomiting (overt or surreptitious) or alkali ingestion. If the urine is relatively acid and has low concentrations of Na^+ , K^+ , and Cl^- , the most likely possibilities are prior vomiting, the posthypercapnic state, or prior diuretic ingestion. If, on the other hand, neither the urine sodium, potassium, nor chloride concentrations are depressed, magnesium deficiency, Bartter's or Gitelman's syndrome, or current diuretic ingestion should be considered. Bartter's syndrome is distinguished from Gitelman's syndrome because of hypocalciuria and hypomagnesemia in the latter disorder. The genetic and molecular basis of these two disorders has been elucidated recently ([Chap. 276](#)).

Alkali Administration Chronic administration of alkali to individuals with normal renal function rarely, if ever causes alkalosis. However, in patients with coexistent hemodynamic disturbances, alkalosis can develop because the normal capacity to excrete HCO_3^- may be exceeded or there may be enhanced reabsorption of HCO_3^- . Such patients include those who receive oral or intravenous HCO_3^- , acetate loads (parenteral hyperalimentation solutions), citrate loads (transfusions), or antacids plus cation-exchange resins (aluminum hydroxide and sodium polystyrene sulfonate).

METABOLIC ALKALOSIS ASSOCIATED WITH [ECFV](#) CONTRACTION, K^+ DEPLETION, AND SECONDARY HYPERRENINEMIC HYPERALDOSTERONISM

Gastrointestinal Origin Gastrointestinal loss of H^+ from vomiting or gastric aspiration results in retention of HCO_3^- . The loss of fluid and NaCl in vomitus or nasogastric suction results in contraction of the [ECFV](#) and an increase in the secretion of renin and aldosterone. Volume contraction causes a reduction in [GFR](#) and an enhanced capacity of the renal tubule to reabsorb HCO_3^- . During active vomiting, there is continued addition of HCO_3^- to plasma in exchange for Cl^- , and the plasma $[\text{HCO}_3^-]$ exceeds the reabsorptive capacity of the proximal tubule. The excess NaHCO_3 reaches the distal tubule, where secretion is enhanced by an aldosterone and the delivery of the poorly reabsorbed anion, HCO_3^- . Because of contraction of the ECFV and hypochloremia, Cl^- is avidly conserved by the kidney. Correction of the contracted ECFV with NaCl and repair of K^+ deficits corrects the acid-base disorder.

Renal Origin

Diuretics (See also [Chap. 232](#)) Drugs that induce chloruresis, such as thiazides and loop diuretics (furosemide, bumetanide, torsemide, and ethracrynic acid), acutely diminish the [ECFV](#) without altering the total body bicarbonate content. The serum $[\text{HCO}_3^-]$ increases. The chronic administration of diuretics tends to generate an alkalosis by increasing distal salt delivery, so that K^+ and H^+ secretion are stimulated. The alkalosis is maintained by persistence of the contraction of the ECFV, secondary

hyperaldosteronism, K⁺ deficiency, and the direct effect of the diuretic (as long as diuretic administration continues). Repair of the alkalosis is achieved by providing isotonic saline to correct the ECFV deficit.

Bartter's Syndrome and Gitelman's Syndrome See [Chap. 276](#).

Nonreabsorbable Anions and Magnesium Deficiency Administration of large quantities of nonreabsorbable anions, such as penicillin or carbenicillin, can enhance distal acidification and K⁺ secretion by increasing the transepithelial potential difference (lumen negative). Mg²⁺ deficiency results in hypokalemic alkalosis by enhancing distal acidification through stimulation of renin and hence aldosterone secretion.

Potassium Depletion Chronic K⁺ depletion may cause metabolic alkalosis by increasing urinary acid excretion. Both NH₄⁺ production and absorption are enhanced and HCO₃⁻ reabsorption is stimulated. Chronic K⁺ deficiency upregulates the renal H⁺, K⁺-ATPase to increase K⁺ absorption at the expense of enhanced H⁺ secretion. Alkalosis associated with severe K⁺ depletion is resistant to salt administration, but repair of the K⁺ deficiency corrects the alkalosis.

After Treatment of Lactic Acidosis or Ketoacidosis When an underlying stimulus for the generation of lactic acid or ketoacid is removed rapidly, as with repair of circulatory insufficiency or with insulin therapy, the lactate or ketones are metabolized to yield an equivalent amount of HCO₃⁻. Other sources of new HCO₃⁻ are additive with the original amount generated by organic anion metabolism to create a surfeit of HCO₃⁻. Such sources include (1) new HCO₃⁻ added to the blood by the kidneys as a result of enhanced acid excretion during the preexisting period of acidosis, and (2) alkali therapy during the treatment phase of the acidosis. Acidosis-induced contraction of the [ECFV](#) and K⁺ deficiency act to sustain the alkalosis.

Posthypercapnia Prolonged CO₂ retention with chronic respiratory acidosis enhances renal HCO₃⁻ absorption and the generation of new HCO₃⁻ (increased net acid excretion). If the PaCO₂ is returned to normal, metabolic alkalosis results from the persistently elevated [HCO₃⁻]. Alkalosis develops if the elevated PaCO₂ is abruptly returned toward normal by a change in mechanically controlled ventilation. Associated [ECFV](#) contraction does not allow complete repair of the alkalosis by correction of the PaCO₂ alone, and alkalosis persists until Cl⁻ supplementation is provided.

METABOLIC ALKALOSIS ASSOCIATED WITH ECFV EXPANSION, HYPERTENSION, AND HYPERALDOSTERONISM

Mineralocorticoid administration or excess production [primary aldosteronism of Cushing's syndrome and adrenal cortical enzyme defects ([Chap. 331](#))] increases net acid excretion and may result in metabolic alkalosis, which may be worsened by associated K⁺ deficiency. [ECFV](#) expansion from salt retention causes hypertension and antagonizes the reduction in [GFR](#) and/or increases tubule acidification induced by aldosterone and by K⁺ deficiency. The kaliuresis persists and causes continued K⁺ depletion with polydipsia, inability to concentrate the urine, and polyuria. Increased aldosterone levels may be the result of autonomous primary adrenal overproduction or of secondary aldosterone release due to renal overproduction of renin. In both

situations, the normal feedback of ECFV on net aldosterone production is disrupted, and hypertension from volume retention can result.

Liddle's syndrome ([Chap. 276](#)) results from increased activity of collecting duct Na⁺-channel (ENaC) and is a rare inherited disorder associated with hypertension due to volume expansion manifested as hypokalemic alkalosis and normal aldosterone levels.

Symptoms With metabolic alkalosis, changes in central and peripheral nervous system function are similar to those of hypocalcemia ([Chap. 340](#)); symptoms include mental confusion, obtundation, and a predisposition to seizures, paresthesia, muscular cramping, tetany, aggravation of arrhythmias, and hypoxemia in chronic obstructive pulmonary disease. Related electrolyte abnormalities include hypokalemia and hypophosphatemia.

TREATMENT

This is primarily directed at correcting the underlying stimulus for HCO₃⁻-generation. If primary aldosteronism is present, correction of the underlying cause will reverse the alkalosis. [H⁺] loss by the stomach or kidneys can be mitigated by the use of H₂-receptor blockers, H⁺, K⁺-ATPase inhibitors, or the discontinuation of diuretics. The second aspect of treatment is to remove the factors that sustain HCO₃⁻-reabsorption, such as [ECFV](#) contraction or K⁺-deficiency. Although K⁺-deficits should be repaired, NaCl therapy is usually sufficient to reverse the alkalosis if ECFV contraction is present, as indicated by a low urine [Cl⁻].

If associated conditions preclude infusion of saline, renal HCO₃⁻-loss can be accelerated by administration of acetazolamide, a carbonic anhydrase inhibitor, which is usually effective in patients with adequate renal function but can worsen K⁺-losses. Dilute hydrochloric acid (0.1 N HCl) is also effective but can cause hemolysis. Alternatively, acidification can also be achieved with oral NH₄Cl, which should be avoided in the presence of liver disease. Hemodialysis against a dialysate low in [HCO₃⁻] and high in [Cl⁻] can be effective when renal function is impaired.

RESPIRATORY ACIDOSIS

Respiratory acidosis can be due to severe pulmonary disease, respiratory muscle fatigue, or abnormalities in ventilatory control and is recognized by an increase in PaCO₂ and decrease in pH ([Table 50-5](#)). In acute respiratory acidosis, there is an immediate compensatory elevation (due to cellular buffering mechanisms) in HCO₃⁻, which increases 1 mmol/L for every 10-mmHg increase in PaCO₂. In chronic respiratory acidosis (>24 h), renal adaptation increases the [HCO₃⁻] by 4 mmol/L for every 10-mmHg increase in PaCO₂. The serum HCO₃⁻ usually does not increase above 38 mmol/L.

The clinical features vary according to the severity and duration of the respiratory acidosis, the underlying disease, and whether there is accompanying hypoxemia. A rapid increase in PaCO₂ may cause anxiety, dyspnea, confusion, psychosis, and hallucinations and may progress to coma. Lesser degrees of dysfunction in chronic hypercapnia include sleep disturbances, loss of memory, daytime somnolence,

personality changes, impairment of coordination, and motor disturbances such as tremor, myoclonic jerks, and asterixis. Headaches and other signs that mimic raised intracranial pressure, such as papilledema, abnormal reflexes, and focal muscle weakness, are due to vasoconstriction secondary to loss of the vasodilator effects of CO_2 .

Depression of the respiratory center by a variety of drugs, injury, or disease can produce respiratory acidosis. This may occur acutely with general anesthetics, sedatives, and head trauma or chronically with sedatives, alcohol, intracranial tumors, and the syndromes of sleep-disordered breathing, including the primary alveolar and obesity-hypoventilation syndromes ([Chaps. 263](#) and [264](#)). Abnormalities or disease in the motor neurons, neuromuscular junction, and skeletal muscle can cause hypoventilation via respiratory muscle fatigue. Mechanical ventilation, when not properly adjusted and supervised, may result in respiratory acidosis, particularly if CO_2 production suddenly rises (because of fever, agitation, sepsis, or overfeeding) or alveolar ventilation falls because of worsening pulmonary function. High levels of positive end-expiratory pressure in the presence of reduced cardiac output may cause hypercapnia as a result of large increases in alveolar dead space ([Chap. 266](#)). Permissive hypercapnia is being used with increasing frequency because of studies suggesting lower mortality rates than with conventional mechanical ventilation, especially with severe central nervous system or heart disease. Although the potential beneficial effects of permissive hypercapnia may be mitigated by correction of the acidemia, it seems prudent, nevertheless, to keep the pH in the range of 7.2 to 7.3 by administration of NaHCO_3 .

Acute hypercapnia follows sudden occlusion of the upper airway or generalized bronchospasm as in severe asthma, anaphylaxis, inhalational burn, or toxin injury. Chronic hypercapnia and respiratory acidosis occur in end-stage obstructive lung disease. Restrictive disorders involving both the chest wall and the lungs can cause respiratory acidosis because the high metabolic cost of respiration causes ventilatory muscle fatigue. Advanced stages of intrapulmonary and extrapulmonary restrictive defects present as chronic respiratory acidosis.

The diagnosis of respiratory acidosis requires, by definition, the measurement of PaCO_2 and arterial pH. A detailed history and physical examination often indicate the cause. Pulmonary function studies ([Chap. 250](#)), including spirometry, diffusion capacity for carbon monoxide, lung volumes, and arterial PaCO_2 and O_2 saturation, usually make it possible to determine if respiratory acidosis is secondary to lung disease. The workup for nonpulmonary causes should include a detailed drug history, measurement of hematocrit, and assessment of upper airway, chest wall, pleura, and neuromuscular function.

TREATMENT

The management of respiratory acidosis depends on its severity and rate of onset. Acute respiratory acidosis can be life-threatening, and measures to reverse the underlying cause should be undertaken simultaneously with restoration of adequate alveolar ventilation. This may necessitate tracheal intubation and assisted mechanical ventilation. Oxygen administration should be titrated carefully in patients with severe

obstructive pulmonary disease and chronic CO₂ retention who are breathing spontaneously ([Chap. 258](#)). When oxygen is used in judiciously, these patients may experience progression of the respiratory acidosis. Aggressive and rapid correction of hypercapnia should be avoided, because the falling PaCO₂ may provoke the same complications noted with acute respiratory alkalosis (i.e., cardiac arrhythmias, reduced cerebral perfusion, and seizures). The PaCO₂ should be lowered gradually in chronic respiratory acidosis, aiming to restore the PaCO₂ to baseline levels and to provide sufficient Cl⁻ and K⁺ to enhance the renal excretion of HCO₃⁻.

Chronic respiratory acidosis is frequently difficult to correct, but measures aimed at improving lung function ([Chap. 258](#)) can help some patients and forestall further deterioration in most.

RESPIRATORY ALKALOSIS

Alveolar hyperventilation decreases PaCO₂ and increases the HCO₃⁻/PaCO₂ ratio, thus increasing pH ([Table 50-5](#)). Nonbicarbonate cellular buffers respond by consuming HCO₃⁻. Hypocapnia develops when a sufficiently strong ventilatory stimulus causes CO₂ output in the lungs to exceed its metabolic production by tissues. Plasma pH and [HCO₃⁻] appear to vary proportionately with PaCO₂ over a range from 40 to 15 mmHg. The relationship between arterial [H⁺] concentration and PaCO₂ is about 0.7 mmol/L per mmHg (or 0.01 pH unit/mmHg), and that for plasma [HCO₃⁻] is 0.2 mmol/L per mmHg. Hypocapnia sustained longer than 2 to 6 h is further compensated by a decrease in renal ammonium and titrable acid excretion and a reduction in filtered HCO₃⁻ reabsorption. Full renal adaptation to respiratory alkalosis may take several days and requires normal volume status and renal function. The kidneys appear to respond directly to the lowered PaCO₂ rather than to alkalosis per se. In chronic respiratory alkalosis a 1-mmHg fall in PaCO₂ causes a 0.4- to 0.5-mmol/L drop in [HCO₃⁻] and a 0.3-mmol/L fall (or 0.003 rise in pH) in [H⁺].

The effects of respiratory alkalosis vary according to duration and severity but are primarily those of the underlying disease. Reduced cerebral blood flow as a consequence of a rapid decline in PaCO₂ may cause dizziness, mental confusion, and seizures, even in the absence of hypoxemia. The cardiovascular effects of acute hypocapnia in the conscious human are generally minimal, but in the anesthetized or mechanically ventilated patient, cardiac output and blood pressure may fall because of the depressant effects of anesthesia and positive-pressure ventilation on heart rate, systemic resistance, and venous return. Cardiac arrhythmias may occur in patients with heart disease as a result of changes in oxygen unloading by blood from a left shift in the hemoglobin-oxygen dissociation curve (Bohr effect). Acute respiratory alkalosis causes intracellular shifts of Na⁺, K⁺, and PO₄⁻ and reduces free [Ca²⁺] by increasing the protein-bound fraction. Hypocapnia-induced hypokalemia is usually minor.

Chronic respiratory alkalosis is the most common acid-base disturbance in critically ill patients and, when severe, portends a poor prognosis. Many cardiopulmonary disorders manifest respiratory alkalosis in their early to intermediate stages, and the finding of normocapnia and hypoxemia in a patient with hyperventilation may herald the onset of rapid respiratory failure and should prompt an assessment to determine if the patient is becoming fatigued. Respiratory alkalosis is common during mechanical ventilation.

The hyperventilation syndrome may be disabling. Paresthesia, circumoral numbness, chest wall tightness or pain, dizziness, inability to take an adequate breath, and, rarely, tetany may themselves be sufficiently stressful to perpetuate the disorder. Arterial blood-gas analysis demonstrates an acute or chronic respiratory alkalosis, often with hypocapnia in the range of 15 to 30 mmHg and no hypoxemia. Central nervous system diseases or injury can produce several patterns of hyperventilation and sustained P_{aCO_2} levels of 20 to 30 mmHg. Hyperthyroidism, high caloric loads, and exercise raise the basal metabolic rate, but ventilation usually rises in proportion so that arterial blood gases are unchanged and respiratory alkalosis does not develop. Salicylates are the most common cause of drug-induced respiratory alkalosis as a result of direct stimulation of the medullary chemoreceptor ([Chap. 396](#)). The methylxanthines, theophylline, and aminophylline stimulate ventilation and increase the ventilatory response to CO_2 . Progesterone increases ventilation and lowers arterial P_{aCO_2} by as much as 5 to 10 mmHg. Therefore, chronic respiratory alkalosis is a common feature of pregnancy. Respiratory alkalosis is also prominent in liver failure, and the severity correlates with the degree of hepatic insufficiency. Respiratory alkalosis is often an early finding in gram-negative septicemia, before fever, hypoxemia, or hypotension develop.

The diagnosis of respiratory alkalosis depends on measurement of arterial pH and P_{aCO_2} . The plasma $[K^+]$ is often reduced and the $[Cl^-]$ increased. In the acute phase, respiratory alkalosis is not associated with increased renal HCO_3^- excretion, but within hours net acid excretion is reduced. In general, the HCO_3^- concentration falls by 2.0 mmol/L for each 10-mmHg decrease in P_{aCO_2} . Chronic hypocapnia reduces the serum $[HCO_3^-]$ by 5.0 mmol/L for each 10-mmHg decrease in P_{aCO_2} . It is unusual to observe a plasma $HCO_3^- < 12$ mmol/L as a result of a pure respiratory alkalosis.

When a diagnosis of respiratory alkalosis is made, its cause should be investigated. The diagnosis of hyperventilation syndrome is made by exclusion. In difficult cases, it may be important to rule out other conditions such as pulmonary embolism, coronary artery disease, and hyperthyroidism.

TREATMENT

The management of respiratory alkalosis is directed toward alleviation of the underlying disorder. If respiratory alkalosis complicates ventilator management, changes in dead space, tidal volume, and frequency can minimize the hypocapnia. Patients with the hyperventilation syndrome may benefit from reassurance, rebreathing from a paper bag during symptomatic attacks, and attention to underlying psychological stress. Antidepressants and sedatives are not recommended. β -Adrenergic blockers may ameliorate peripheral manifestations of the hyperadrenergic state.

(Bibliography omitted in Palm version)

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SECTION 8 -ALTERATIONS IN SEXUAL FUNCTION AND REPRODUCTION

51. ERECTILE DYSFUNCTION - Kevin T. McVary

Erectile dysfunction (ED) affects 10 to 25% of middle-aged and elderly men. Demographic changes, the popularity of newer treatments, and greater acceptance of ED by patients and society have led to increased diagnosis and associated health care expenditures for the management of this common disorder. Impairment of erectile function has a profound impact on the well-being of affected men. Because many patients are reluctant to initiate discussion of sexual function, the physician should address this topic directly to elicit a history of ED.

PHYSIOLOGIC CONTROL OF ERECTION AND MALE SEXUAL FUNCTION

Normal male sexual function requires (1) an intact libido, (2) the ability to achieve and maintain penile erection, (3) ejaculation, and (4) detumescence. *Libido* refers to sexual desire and is influenced by a variety of visual, olfactory, tactile, auditory, imaginative and hormonal stimuli. Sex steroids, particularly testosterone, act to increase libido. Libido can be diminished by hormonal or psychiatric disorders or by medications.

The major anatomic structures of the penis that are involved in erectile function include the three corpora, which consist of the paired cavernosa and a single spongiosum that encloses the urethra. A collagenous sheath, called the *tunica albuginea*, individually surrounds each corpora. The micro-architecture of the corpora is composed of a mass of smooth muscle (trabecula) which contains a network of endothelial-lined vessels (lacunar spaces).

Penile tumescence leading to erection depends on the increased flow of blood into the lacunar network after complete relaxation of the arteries and corporal smooth muscle. Subsequent compression of the trabecular smooth muscle against the fibroelastic tunica albuginea causes a passive closure of the emissary veins and accumulation of blood in the corpora. In the presence of a full erection and a competent valve mechanism, the corpora become noncompressible cylinders from which blood does not escape.

The central nervous system exerts an important influence by either stimulating or antagonizing spinal pathways that mediate erectile function and ejaculation. The erectile response is mediated by a combination of central (psychogenic) and peripheral (reflexogenic) innervation. Sensory nerves that originate from receptors in the penile skin and glans converge to form the dorsal nerve of the penis, which travels to the S2-S4 dorsal root ganglia via the pudendal nerve. Parasympathetic nerve fibers to the penis arise from neurons in the intermediolateral columns of S2-S4 sacral spinal segments. Sympathetic innervation originates from the T-11 to the L-2 spinal segments and descends through the hypogastric plexus.

Neural input to smooth muscle tone is crucial to the initiation and maintenance of an erection. There is also an intricate interaction between the corporal smooth muscle cell and its overlying endothelial cell lining ([Fig. 51-1A](#)). Nitric oxide, which induces vascular relaxation, promotes erection and is opposed by endothelin-1 (ET-1), which mediates vascular contraction. Nitric oxide is synthesized from L-arginine by nitric oxide synthase,

and is released from the nonadrenergic, noncholinergic (NANC) autonomic nerve supply to act postjunctionally on smooth muscle cells. Nitric oxide increases the production of cyclic 3',5'-guanosine monophosphate (cyclic GMP), which interacts with protein kinase G and decreases intracellular calcium, causing relaxation of the smooth muscle ([Fig. 51-1B](#)). Cyclic GMP is gradually broken down by phosphodiesterase type 5 (PDE-5). Inhibitors of PDE-5, such as the oral medication sildenafil, maintain erections by reducing the breakdown of cyclic GMP. However, if nitric oxide is not produced at some level, the addition of PDE-5 inhibitor is not effective, as the drug facilitates but does not initiate the initial enzyme cascade. In addition to nitric oxide, vasoactive prostaglandins (PGE₁, PGF_{2α}) are synthesized within the cavernosal tissue and increase cyclic AMP levels, also leading to relaxation of cavernosal smooth muscle cells.

Ejaculation is stimulated by the sympathetic nervous system, which results in contraction of the epididymis, vas deferens, seminal vesicles, and prostate, causing seminal fluid to enter the urethra. Seminal fluid emission is followed by rhythmic contractions of the bulbocavernosus and ischiocavernosus muscles, leading to ejaculation. *Premature ejaculation* is usually related to anxiety or a learned behavior and is amenable to behavioral therapy or treatment with medications such as selective serotonin reuptake inhibitors (SSRIs). *Retrograde ejaculation* results when the internal urethral sphincter does not close, and it may occur in men with diabetes or after surgery involving the bladder neck.

Detumescence is mediated by released norepinephrine from the sympathetic nerves, release of endothelin from the vascular surface, and contraction of smooth muscle induced by activation of postsynaptic α-adrenergic receptors. These events increase venous outflow and restore the flaccid state. Venous leak can cause premature detumescence and is thought to be caused by insufficient relaxation of the corporal smooth muscle rather than a specific anatomic defect. *Priapism* refers to a persistent and painful erection and may be associated with sickle cell anemia, hypercoagulable states, spinal cord injury, or injection of vasodilator agents into the penis.

ERECTILE DYSFUNCTION

EPIDEMIOLOGY

In the Massachusetts Male Aging Study (MMAS), a community-based survey of men between the ages of 40 and 70, 52% of responders reported some degree of [ED](#). Complete ED occurred in 10% of respondents, moderate ED occurred in 25%, and minimal ED in 17%. The incidence of moderate or severe ED more than doubled between the ages of 40 and 70. In the National Health and Social Life Survey (NHSLS), which was a nationally representative sample of men and women age 18 to 59 years, 10% of men reported being unable to maintain an erection (corresponding to the proportion of men in the MMAS reporting severe ED). Incidence was highest among men in the 50 to 59 age group (21%) and among men who were poor (14%), divorced (14%), and less educated (13%).

The incidence of [ED](#) is also higher among men with certain medical disorders. In the [MMAS](#), ED correlated with the presence of diabetes mellitus, heart disease, hypertension, and decreased HDL levels. Medications used to treat diabetes or

cardiovascular disease are additional risk factors (see below). There is a higher incidence of ED among men who have undergone radiation or surgery for cancer of the prostate and in those with a lower spinal cord injury. Psychological causes of ED include depression and anger. The [NHSLS](#) found a higher incidence of ED among men who reported fair-to-poor health or experienced stress from unemployment or other causes. ED is not considered a normal part of the aging process. Nonetheless, it is associated with certain physiologic and psychological changes related to age.

PATHOPHYSIOLOGY

[ED](#) may result from three basic mechanisms: (1) failure to initiate (psychogenic, endocrinologic, or neurogenic); (2) failure to fill (arteriogenic); or (3) failure to store (venoocclusive dysfunction) adequate blood volume within the lacunar network. The inability to initiate an erection may have psychogenic, endocrinologic, or neurogenic etiologies. These categories are not mutually exclusive, and multiple factors contribute to ED in many patients. For example, diminished filling pressure can lead secondarily to venous leak. Psychogenic factor frequently co-exist with other etiologic factors and should be considered in all cases. Diabetic, atherosclerotic, and drug-related causes account for >80% of cases of ED in older men.

Vasculogenic The most frequent organic cause of [ED](#) is a disturbance of blood flow to and from the penis. Atherosclerotic or traumatic arterial disease can decrease flow to the lacunar spaces, resulting in decreased rigidity and an increased time to full erection. Excessive outflow through the veins, despite adequate inflow, may also contribute to ED. In this case, the achieved perfusion pressures cannot compensate for the unrestricted outflow needed to ensure adequate erection. This situation may be due to insufficient relaxation of trabecular smooth muscle and may occur in anxious individuals with excessive adrenergic tone or in those with damaged parasympathetic outflow. Structural alterations to the fibroelastic components of the corpora may cause a loss of compliance and an inability to compress the tunical veins. This condition may result from aging, increased cross-leaking of collagen fibers induced by nonenzymatic glycosylation, hypoxia, or altered synthesis of collagen associated with hypercholesterolemia. Fibroelastic structures can also be damaged by surgery, radiation, or trauma to the penis.

Neurogenic Disorders that affect the sacral spinal cord or the autonomic fibers to the penis preclude nervous system relaxation of penile smooth muscle, thus leading to [ED](#). In patients with spinal cord injury, the degree of ED depends on the completeness and level of the lesion. Patients with incomplete lesions or injuries to the upper part of the spinal cord are more likely to retain erectile capabilities than those with complete lesions or injuries to the lower part. Although 75% of patients with spinal cord injuries have some erectile capability, only 25% have erections sufficient for penetration. Other neurologic disorders commonly associated with ED include multiple sclerosis and peripheral neuropathy. The latter is often due to either diabetes or alcoholism. Pelvic surgery may cause ED through disruption of the autonomic nerve supply.

Endocrinologic Androgens increase libido, but their exact role in erectile function remains unclear. Individuals with castrate levels of testosterone can achieve erections from visual or sexual stimuli. Nonetheless, normal levels of testosterone appear to be

important for erectile function, particularly in older males. Androgen replacement therapy can improve depressed erectile function when it is secondary to hypogonadism; it is not useful for ED when endogenous testosterone levels are normal. Increased prolactin may decrease libido by suppressing gonadotropin-releasing hormone (GnRH), and it also leads to decreased testosterone levels. Treatment of hyperprolactinemia with dopamine agonists can restore libido and testosterone.

Diabetic ED occurs in 35 to 75% of men with diabetes mellitus. Pathologic mechanisms are primarily related to diabetes-associated vascular and neurologic complications. Diabetic macrovascular complications are mainly related to age, whereas microvascular complications correlate with the duration of diabetes and the degree of glycemic control ([Chap. 333](#)). Individuals with diabetes also have reduced amounts of nitric oxide synthase in both endothelial and neural tissues.

Psychogenic Two mechanisms contribute to the inhibition of erections in psychogenic [ED](#). First, psychogenic stimuli to the sacral cord may inhibit reflexogenic responses, thereby blocking activation of vasodilator outflow to the penis. Second, excess sympathetic stimulation in an anxious man may increase penile smooth muscle tone. The most common causes of psychogenic ED are performance anxiety, depression, relationship conflict, loss of attraction, sexual inhibition, conflicts over sexual preference, sexual abuse in childhood, and fear of pregnancy or sexually transmitted disease. Almost all patients with ED, even when it has a clear-cut organic basis, develop a psychogenic component as a reaction to ED.

Medication-Related Medication-induced [ED](#) ([Table 51-1](#)) is estimated to occur in 25% of men seen in general medical outpatient clinics. Among the antihypertensive agents, the thiazide diuretics and beta blockers have been implicated most frequently. Calcium channel blockers and angiotensin-converting enzyme inhibitors are less frequently cited. These drugs may act directly at the corporal level (e.g., calcium channel blockers) or indirectly by reducing pelvic blood pressure, which is important in the development of penile rigidity. Alpha adrenergic blockers are less likely to cause ED. Estrogens, [GnRH](#) agonists, H_2 antagonists, and spironolactone cause ED by suppressing gonadotropin production or by blocking androgen action. Antidepressant and antipsychotic agents -- particularly neuroleptics, tricyclics, and [SSRIs](#) -- are associated with erectile, ejaculatory, orgasmic, and sexual desire difficulties. Digoxin induces ED via blockade of the Na^+, K^+ -ATPase pump, resulting in a net increase in intracellular calcium and increased corporal smooth muscle tone.

Although many medications can cause [ED](#), patients frequently have concomitant risk factors that confound the clinical picture. If there is a strong association between the institution of a drug and the onset of ED, alternative medications should be considered. Otherwise, it is often practical to treat the ED without attempting multiple changes in medications, as it may be difficult to establish a causal role for the drug.

CLINICAL EVALUATION

A good physician-patient relationship helps to unravel the possible causes of [ED](#), many of which require discussion of personal and sometimes embarrassing topics. For this reason, a primary care provider is often ideally suited to initiate the evaluation. A

complete medical and sexual history should be taken in an effort to assess whether the cause of ED is organic, psychogenic, or multifactorial ([Fig. 51-2](#)). Initial questions should focus on the onset of symptoms, the presence and duration of partial erections, and the progression of ED. A history of nocturnal or early morning erections is useful for distinguishing physiologic from psychogenic ED. Nocturnal erections occur during rapid eye movement (REM) sleep and require intact neurologic and circulatory systems. Organic causes of ED are generally characterized by a gradual and persistent change in rigidity or the inability to sustain nocturnal, coital, or self-stimulated erections. The patient should also be questioned about the presence of penile curvature or pain with coitus. It is also important to address libido, as decreased sexual drive and ED are sometimes the earliest signs of endocrine abnormalities (e.g., increased prolactin, decreased testosterone levels). It is useful to ask whether the problem is confined to coitus with one or other partners; ED arises not uncommonly in association with new or extramarital sexual relationships. Situational ED, as opposed to consistent ED, suggests psychogenic causes. Ejaculation is much less commonly affected than erection, but questions should be asked about whether ejaculation is normal, premature, delayed, or absent. Relevant risk factors should be identified, such as diabetes mellitus, coronary artery disease, lipid disorders, hypertension, peripheral vascular disease, smoking, alcoholism, and endocrine or neurologic disorders. The patient's surgical history should be explored with an emphasis on bowel, bladder, prostate, or vascular procedures. A complete drug history is also important, as medications constitute a major source of reversible ED. Social changes that may precipitate ED are also crucial to the evaluation, including health worries, spousal death, divorce, relationship difficulties, and financial concerns.

The physical examination is an essential element in the assessment of [ED](#). Signs of hypertension as well as evidence of thyroid, hepatic, hematologic, cardiovascular, or renal diseases should be sought. An assessment should be made of the endocrine and vascular systems, the external genitalia, and the prostate gland. The penis should be carefully palpated along the corpora to detect fibrotic plaques. Reduced testicular size and loss of secondary sexual characteristics are suggestive of hypogonadism. Neurologic examination should include assessment of anal sphincter tone, the bulbocavernosus reflex, and testing for peripheral neuropathy.

Selected laboratory testing is recommended in all cases. Although hyperprolactinemia is uncommon, a serum prolactin level should be measured, as decreased libido and/or erectile dysfunction may be the presenting symptoms of a prolactinoma or other mass lesions of the sella ([Chap. 328](#)). The serum testosterone level should be measured and, if low, gonadotropins should be measured to determine whether hypogonadism is primary (testicular) or secondary (hypothalamic-pituitary) in origin ([Chap. 335](#)). Serum chemistries, CBC, and lipid profiles may be of value, if not performed recently, as they can yield evidence of anemia, diabetes, hyperlipidemia, or other systemic diseases associated with [ED](#). Determination of serum PSA should be conducted according to recommended clinical guidelines ([Chap. 95](#)).

Additional diagnostic testing is rarely necessary in the evaluation of ED. However, in selected patients, specialized testing may provide insight into pathologic mechanisms of ED and aid in the selection of treatment options. Optional specialized testing includes: (1) studies of nocturnal penile tumescence and rigidity; (2) vascular testing (in-office

injection of vasoactive substances, penile Doppler ultrasound, penile angiography, dynamic infusion cavernosography/cavernosometry); (3) neurologic testing (biothesiometry-graded vibratory perception; somatosensory evoked potentials); and (4) psychological diagnostic tests. The information potentially gained from these procedures must be balanced against their invasiveness and cost.

TREATMENT

Patient Education Patient and partner education is essential in the treatment of [ED](#). In goal-directed therapy, education facilitates understanding of the disease, results of the tests, and selection of treatment. Discussion of treatment options helps to clarify how treatment is best offered, and to stratify first- and second-line therapies. Patients with high-risk lifestyle issues, such as smoking, alcohol abuse, or recreational drug use, should be counseled on the role these factors play in the development of ED.

Oral Agents Sildenafil is the only approved and effective oral agent for the treatment of [ED](#). Sildenafil has markedly improved the management of ED because it is effective for the treatment of a broad range of causes of ED, including psychogenic, diabetic, vasculogenic, post-radical prostatectomy (nerve-sparing procedures), and spinal cord injury. Sildenafil is a selective and potent inhibitor of PDE-5, the predominant phosphodiesterase isoform found in the penis. It is administered in doses of 25, 50, or 100 mg, and enhances erections after sexual stimulation. The onset of action is approximately 60 to 90 min. Reduced initial doses should be considered for patients who are elderly, have renal insufficiency, or are taking medications that inhibit the CYP3A4 metabolic pathway in the liver (e.g., erythromycin, cimetidine, ketoconazole, and, possibly, itraconazole and mibefradil), as they may increase the serum concentration of sildenafil. The drug does not affect ejaculation, orgasm, or sexual drive. Side effects associated with sildenafil include headaches (19%), facial flushing (9%), dyspepsia (6%) and nasal congestion (4%). Approximately 7% of men may experience transient altered color vision (blue halo effect). Sildenafil is contraindicated in men receiving nitrate therapy for cardiovascular disease, including agents delivered by oral, sublingual, transnasal, or topical routes. These agents can potentiate its hypotensive effect and may result in profound shock. Likewise, amyl/butyl nitrates (poppers) may have a fatal synergistic effect on blood pressure. Sildenafil should also be avoided in patients with congestive heart failure and cardiomyopathy because of the risk of vascular collapse. Because sexual activity leads to an increase in physiologic expenditure [5 to 6 metabolic equivalents (METs)], physicians have been advised to exercise caution in prescribing any drug for sexual activity to those with active coronary disease, heart failure, borderline hypotension, hypovolemia, and to those on complex antihypertensive regimens.

Androgen Therapy Testosterone replacement is used to treat both primary and secondary causes of hypogonadism ([Chap. 335](#)). Androgen supplementation in the setting of normal testosterone is rarely efficacious and is discouraged. Methods of androgen replacement include parenteral administration of long-acting testosterone esters (enanthate and cypionate), oral preparations (17 α -alkylated derivatives), and transdermal patches ([Chap. 335](#)). The long-acting 17 β -hydroxy esters of testosterone are the safest, most cost-effective, and practical preparations available. The administration of 200 to 300 mg intramuscularly every 2 to 3 weeks provides a practical

option but is far from an ideal physiologic replacement. Oral androgen preparations have the potential for hepatotoxicity and should be avoided. Transdermal delivery of testosterone more closely mimics physiologic testosterone levels, but it is unclear whether this translates into improved sexual function. Because testosterone gradually decreases into the hypogonadal range by 24 hours, patches need to be replaced daily. Testosterone therapy is contraindicated in men with androgen-sensitive cancers and may be inappropriate for men with bladder neck obstruction. It is generally advisable to measure PSA before giving androgen. Hepatic function should be tested before and during testosterone therapy.

Vacuum Constriction Devices Vacuum constriction devices (VCD) are a well-established, noninvasive therapy. They are a reasonable treatment alternative for select patients who cannot take sildenafil or do not desire other interventions. VCD draw venous blood into the penis and use a constriction ring to restrict venous return and maintain tumescence. Adverse events with VCD include pain, numbness, bruising, and altered ejaculation. Additionally, many patients complain that the devices are cumbersome and that the induced erections have a non-physiologic appearance.

Intraurethral Alprostadil If a patient fails to respond to oral agents, a reasonable next choice is intraurethral or self-injection of vasoactive substances. Intraurethral prostaglandin E₁ (alprostadil), in the form of a semisolid pellet (doses of 125 to 1000 µg), is delivered with an applicator. Approximately 65% of men receiving intraurethral alprostadil respond with an erection when tested in the office, but only 50% of those achieve successful coitus at home. Intraurethral insertion is associated with a markedly reduced incidence of priapism in comparison to intracavernosal injection.

Intracavernosal Self-Injection Injection of synthetic formulations of alprostadil is effective in 70 to 80% of patients with [ED](#), but discontinuation rates are high because of the invasive nature of administration. Doses range between 1 and 40 µg. Injection therapy is contraindicated in men with a history of hypersensitivity to the drug and in men at risk for priapism (hypercoagulable states, sickle cell disease). Side effects include local adverse events, prolonged erections, pain, and fibrosis with chronic use. Various combinations of alprostadil, phentolamine, and/or papaverine are sometimes used.

Surgery A less frequently used form of therapy for [ED](#) involves the surgical implantation of a semi-rigid or inflatable penile prosthesis. These surgical treatments are invasive, associated with potential complications, and generally reserved for treatment of refractory ED. Despite their high cost and invasiveness, penile prostheses are associated with high rates of patient satisfaction.

Sex Therapy A course of sex therapy may be useful for addressing specific interpersonal factors that may affect sexual functioning. Sex therapy generally consists of in-session discussion and at-home exercises specific to the person and the relationship. It is preferable if therapy includes both partners, provided the patient is involved in an ongoing relationship.

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52. DISTURBANCES OF MENSTRUATION AND OTHER COMMON GYNECOLOGIC COMPLAINTS IN WOMEN - *Bruce R. Carr, Karen D. Bradshaw*

Complaints related to the female reproductive tract can be categorized as disorders of menstruation, pelvic pain, disturbances in sexual function, or infertility. However, a single disorder, e.g., leiomyoma of the uterus, can present with symptoms referable to any one or more of these categories. Furthermore, sexual dysfunction can interdigitate with other problems in several ways. On the one hand, in women with complaints related to other reproductive tract functions, the underlying problem may actually be severe sexual dysfunction or marital conflict. Alternatively, women with severe organic disorders of the pelvis, e.g., pelvic inflammatory disease or endometriosis, may present with sexual dysfunction such as dyspareunia that in fact is only a minor manifestation of the underlying disease.

Since normal reproductive function depends on the integrated action of the central nervous system, the endocrine glands, and the reproductive organs, menstrual cycle abnormalities, sexual dysfunction, and infertility may be the result of systemic and psychological disorders as well as of primary defects in the endocrine and reproductive organs. The endocrine and physiologic control -- normal and abnormal -- of puberty, reproductive life, and menopause are discussed in [Chap. 336](#). The focus of this chapter is on the initial evaluation of women with disturbances of the reproductive tract.

DISTURBANCES IN MENSTRUATION

Disorders of menstruation can be divided into abnormal uterine bleeding and amenorrhea.

Abnormal Uterine Bleeding The menstrual cycle is defined as the interval between the onset of one bleeding episode and the onset of the next. In normal women the cycle averages 28 ± 3 days, the mean duration of menstrual flow is 4 ± 2 days, and the average blood loss is 35 to 80 mL. Between menarche and menopause most women experience one or more episodes of abnormal uterine bleeding, here defined as any bleeding pattern outside the normal ranges of frequency, duration, and/or amount of blood loss. The decision to evaluate a patient depends on the severity and frequency of the abnormal bleeding pattern.

When vaginal bleeding occurs, it should first be determined whether the blood is derived from the uterine endometrium. Rectal, bladder, cervical, and vaginal sources of bleeding must be excluded. Once the bleeding is established to be uterine in origin, a pregnancy-related disorder (such as threatened or incomplete abortion or ectopic pregnancy) must be ruled out by physical examination and appropriate laboratory tests. It should also be remembered that uterine bleeding may also be the initial or principal manifestation of a generalized bleeding diathesis. The remaining causes of abnormal uterine bleeding can be divided into those associated with ovulatory or anovulatory cycles.

Ovulatory Cycles Menstrual bleeding with ovulatory cycles is spontaneous, regular in onset, predictable in duration and amount of flow, and frequently associated with discomfort; it is the consequence of progesterone withdrawal at the end of the luteal

(postovulatory) phase and requires prior estrogen priming of the endometrium during the follicular (preovulatory) phase of the cycle. When deviations from an established pattern of menstrual flow occur but the cycles are still regular, the usual cause is disease of the outflow tract. For example, regular, prolonged, excessive bleeding episodes can result from abnormalities of the uterus such as submucous leiomyomas, adenomyosis, or endometrial polyps. On the other hand, cyclic, predictable menstruation characterized by spotting or light bleeding suggests obstruction of the outflow tract as with uterine synechiae or scarring of the cervix. Intermittent bleeding between cyclic ovulatory menses is often due to cervical or endometrial lesions.

Anovulatory Cycles Uterine bleeding that is irregular in occurrence, unpredictable as to amount and duration of flow, and usually painless is called *dysfunctional or anovulatory uterine bleeding*. This type of bleeding is the result of a failure of normal follicular maturation with consequent anovulation and may be either transient or chronic. Transient disruption of ovulatory cycles occurs most often in the early menarcheal years, during the perimenopausal period, or as the consequence of a variety of stresses and intercurrent illnesses. Persistent dysfunctional uterine bleeding during the reproductive years can occur in several organic diseases that affect ovarian function and is most often due to estrogen breakthrough bleeding. Estrogen breakthrough bleeding occurs when estrogen stimulation of the endometrium is continuous and is not interrupted by cyclic progesterone withdrawal, as can occur in polycystic ovarian disease.

Amenorrhea *Amenorrhea* is defined either as failure of menarche by age 16, regardless of the presence or absence of secondary sexual characteristics, or as the absence of menstruation for 6 months in a woman with previous periodic menses. Amenorrhea in a woman who has never menstruated is termed *primary amenorrhea*; cessation of menses is termed *secondary amenorrhea*. Because some disorders can cause both primary and secondary amenorrhea, we prefer a functional classification based on the nature of the underlying defect, namely, anatomic defects of the outflow tract (uterus, cervix, or vagina), ovarian failure, and chronic anovulation.

Anatomic defects of the outflow tract include congenital defects of the vagina, imperforate hymen, transverse vaginal septa, cervical stenosis, intrauterine adhesions (synechiae), absence of the vagina or uterus, and uterine maldevelopment. The diagnosis of an anatomic defect is usually made by physical examination but may be confirmed by demonstrating failure of bleeding following administration of estrogen plus a progestogen for 21 days. Pelvic ultrasonography, magnetic resonance imaging, hysterosalpingogram, or hysteroscopy may be helpful in defining the defect.

Causes of *ovarian failure* include gonadal dysgenesis, deficiency of 17 α -hydroxylase, resistant ovary syndrome, and premature ovarian failure. Ovarian failure encompasses disorders in which the ovary is deficient in germ cells and those in which the germ cells are resistant to follicle-stimulating hormone (FSH). The diagnosis of ovarian failure as the cause of amenorrhea is confirmed by an elevated plasma FSH level.

Women with *chronic anovulation* fail to ovulate spontaneously but have the capability of ovulating with appropriate therapy. In some women with chronic anovulation, total estrogen production is adequate, but it is not secreted in a cyclic fashion. In others,

estrogen production is deficient.

Women who have adequate estrogen production and demonstrate withdrawal bleeding after progestogen challenge often have polycystic ovarian disease (see [Fig. 336-8](#)). Other causes include hormone-secreting ovarian and adrenal tumors. Women with deficient or absent estrogen production, and therefore with absence of withdrawal bleeding after progestogen administration, usually have hypogonadotropic hypogonadism due to organic or functional disorders of the pituitary or central nervous system such as brain tumors, pituitary tumors (especially prolactin-secreting adenomas), primary hypopituitarism, or Sheehan's syndrome.

PELVIC PAIN

Pelvic pain may originate in the pelvis or be referred from another region of the body. A pelvic source is suggested by the history (e.g., dysmenorrhea and dyspareunia) and physical findings, but a high index of suspicion must be entertained for extrapelvic disorders that refer to the pelvis, such as appendicitis, diverticulitis, cholecystitis, intestinal obstruction, and urinary tract infections ([Chap. 14](#)).

"Physiologic" Pelvic Pain

Pain Associated with Ovulation ("Mittelschmerz") Many women experience low abdominal discomfort with ovulation, typically a dull aching pain at midcycle in one lower quadrant lasting from minutes to hours. It is rarely severe or incapacitating. The pain may result from peritoneal irritation by follicular fluid released into the peritoneal cavity at ovulation. The onset at midcycle and short duration of pain suggest this diagnosis.

Premenstrual or Menstrual Pain In normal ovulatory women, somatic symptoms during the few days prior to menses may be insignificant or disabling. Such symptoms include edema, breast engorgement, and abdominal bloating or discomfort. A symptom complex of cyclic irritability, depression, and lethargy is known as the *premenstrual syndrome* (PMS). PMS appears to be caused by changes in gonadal steroid levels. Although there is no consensus about therapy, randomized, controlled trials suggest significant improvement with the daily use of serotonin-reuptake inhibitors.

Severe or incapacitating uterine cramping during ovulatory menses and in the absence of demonstrable disorders of the pelvis is termed *primary dysmenorrhea*. Primary dysmenorrhea is caused by prostaglandin-induced uterine ischemia and is treated with prostaglandin synthetase inhibitors and/or oral contraceptive agents.

Pelvic Pain due to Organic Causes Severe dysmenorrhea associated with disease of the pelvis is termed *secondary dysmenorrhea*. Organic causes of pelvic pain can be classified as (1) uterine, (2) adnexal, (3) vulvar or vaginal, and (4) pregnancy-associated.

Uterine Pain Pain of uterine etiology is often chronic and continuous and increases in intensity during menstruation and intercourse. Causes include leiomyomas of the uterus (particularly submucous and degenerating leiomyomas), adenomyosis, and cervical stenosis. Infections of the uterus associated with intrauterine manipulation following

dilatation and curettage or with the insertion of intrauterine devices can also cause pelvic pain ([Chap. 336](#)). Pelvic pain due to endometrial or cervical cancer is usually a late manifestation ([Chap. 336](#)).

Adnexal Pain The most common cause of pain in the adnexae (fallopian tubes and ovaries) is infection ([Chap. 133](#)). Acute salpingo-oophoritis presents as low abdominal pain, fever, and chills; begins a few days after a menstrual period; and is usually due to chlamydial or gonococcal disease with or without a superimposed pyogenic infection. Chronic pelvic inflammatory disease results from either a single episode or multiple episodes of infection and may present as infertility associated with chronic pelvic pain that increases in intensity with menses and intercourse. On physical examination, cervical motion tenderness, adnexal tenderness, and adnexal thickening and/or masses may be present. Pelvic inflammatory disease may become a surgical emergency if peritonitis results from rupture of a tuboovarian abscess. Ovarian cysts or neoplasms may cause pelvic pain that becomes more severe with torsion or rupture of the mass, and ectopic pregnancy must be considered in the differential diagnosis (see below). Endometriosis involving fallopian tubes, ovaries, or peritoneum may cause both chronic low abdominal pain and infertility; the magnitude of tissue involvement does not always correlate with the severity of symptoms. Endometriosis pain typically increases with menstruation and, if the posterior ligaments of the uterus are involved, with intercourse.

Vulvar or Vaginal Pain Pain in these areas is most often due to infectious vaginitis caused by *Monilia*, *Trichomonas*, or bacteria and is characteristically associated with vaginal discharge and pruritus. Herpetic vulvitis, other dermatologic conditions of the vulva, condyloma acuminatum, and cysts or abscesses of Bartholin's glands may also cause vulvar pain.

Pregnancy-Associated Disorders Pregnancy must be considered in the differential diagnosis of pelvic pain during the reproductive years. Threatened abortion or incomplete abortion often presents with uterine cramping, bleeding, or passage of tissue following a period of amenorrhea. Ectopic pregnancy may be insidious in presentation or result in abrupt intraperitoneal hemorrhage and maternal death.

Evaluation of Pelvic Pain The evaluation of pelvic pain requires a careful history and pelvic examination. This often leads to the correct diagnosis and institution of appropriate treatment. If the pain is severe and the diagnosis is unclear, the workup should follow that outlined for the acute abdomen ([Chap. 14](#)). A culdocentesis may be indicated if a ruptured ectopic pregnancy is suspected. If there is a question of an adnexal mass or if the patient is so obese as to preclude a thorough pelvic examination, abdominal or vaginal sonography may be useful. Serial human chorionic gonadotropin (hCG) measurements may help in establishing a diagnosis of tubal pregnancy and are useful in determining if an intrauterine pregnancy is viable. Finally, diagnostic laparoscopy and laparotomy may be indicated with pain of undetermined etiology.

SEXUAL DYSFUNCTION

Some women with sexual dysfunction describe minor complaints related to the reproductive tract as a means of bringing sexual problems to the attention of the physician. Alternatively, sexual dysfunction may be thought to be the cause of low

abdominal discomfort or dyspareunia when the actual etiology is organic. However, more and more women seek medical advice because of sexual problems that interface in provenance between medicine, psychiatry, and sociology.

The normal sexual response begins with sexual arousal, which causes genital vasocongestion that results in vaginal lubrication in preparation for intromission. The lubrication is due to the formation of a transudate in the vagina and in conjunction with genital congestion produces the so-called orgasmic platform prior to orgasm. Sexual stimuli (visual, tactile, auditory, and olfactory) as well as healthy vaginal tissue are prerequisites for genital vasocongestion and vaginal lubrication. During the second stage of the sexual response, involuntary contractions of the muscles of the pelvis result in a pleasurable cortical sensory phenomenon known as orgasm. Direct or indirect stimulation of the clitoris is important in the production of the female orgasm. In simple terms, sexual dysfunction can be due to interference with the arousal or orgasmic phases of the sexual response. Either disorder can be due to an organic or functional cause or both.

Illnesses that impair neurologic function such as diabetes mellitus or multiple sclerosis can prevent normal sexual arousal. Local pelvic diseases such as vaginitis, endometriosis, and salpingo-oophoritis may preclude normal sexual response because of resulting dyspareunia. Debilitating systemic diseases such as cancer and cardiovascular diseases may inhibit normal sexual response indirectly.

More commonly, failure of a normal sexual response is due to psychological factors that impair sexual arousal. Such problems include misinformation, e.g., the perception of sexual satisfaction as bad, or feelings of guilt about previous psychologically traumatic events such as incest, rape, or unwanted pregnancy. In addition, women who have had previous hysterectomy or mastectomy may perceive themselves as "incomplete." Stresses such as anxiety, depression, fatigue, and marital or interpersonal conflicts may lead to failure of the vasocongestive response and prevent normal vaginal lubrication. Women with such experiences may be unable to achieve normal sexual response unless they receive professional counseling. Such problems are approached by attempting to identify and reduce the causative stresses.

Failure to achieve orgasm is a specific form of sexual dysfunction. In the absence of orgasm many women enjoy sexual encounters to variable degrees because of the pleasure derived from closeness in a cherished relationship, particularly with a loving partner. However, for other women sexual relations with rare or absent orgasms are frustrating and unsatisfying. In many instances, failure of orgasm is due to insufficient clitoral stimulation and may be rectified by appropriate counseling and patient education.

A specific entity, "vaginismus," painful, involuntary contractions of the musculature surrounding the entrance to the vagina, is a rare cause of dyspareunia. It is a conditioned response to a previous real or imagined frightening or traumatic sexual experience. Treatment is directed to elimination of the conditioned response by progressive vaginal dilation by the patient in conjunction with marital therapy.

REPRODUCTION

Infertility is discussed in detail in [Chap. 54](#). The approach to infertile couples always involves evaluation of both the man and woman. The history should address the frequency of intercourse, the sexual responses of both, the use of contraceptives or lubricants, prior pregnancies, interval to conception and outcome of pregnancy, previous or past medical illnesses, and all medications taken.

Male-associated factors account for a third of infertility problems. Therefore, one of the first procedures in the workup of infertile couples should be a semen analysis. The initial evaluation of the woman includes documentation of normal ovulatory cycles. A history of regular, cyclic, predictable, spontaneous menses usually indicates ovulatory cycles, which may be confirmed by basal body temperature graphs, properly timed endometrial biopsies, or plasma progesterone measurements during the luteal phase of the cycle. Also, the diagnosis of luteal-phase dysfunction (low progesterone secretion during the luteal phase) can be established by these methods. Transvaginal ultrasonography is useful for evaluating follicular development.

The most common cause of infertility in women is tubal disease, usually due to infection (pelvic inflammatory disease) or endometriosis. Tubal disease can be evaluated by obtaining a hysterosalpingogram or by diagnostic laparoscopy. Tubal diseases can usually be treated by laparoscopic tuboplasty and lysis of adhesions.

In many instances of infertility, it is now possible to use assisted reproductive technologies including in vitro fertilization and embryo transfer, gamete intrafallopian tube transfer, transfer of cryopreserved ova and embryos, donor oocytes or donor sperm, and ovarian hyperstimulation with clomiphene citrate or gonadotropins followed by intrauterine insemination.

The desire for contraception is also a frequent cause for women to seek medical treatment or evaluation. The most widely used methods for fertility control include (1) rhythm and withdrawal techniques, (2) barrier methods, (3) intrauterine devices, (4) oral steroid contraceptives, (5) sterilization, and (6) abortion. **These methods and their complications are discussed in [Chap. 54](#).*

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53. HIRSUTISM AND VIRILIZATION - David A. Ehrmann

Hirsutism, defined as excessive male-pattern hair growth, affects approximately 10% of women of reproductive age. Hirsutism may be mild, essentially representing a variation of normal hair growth, or rarely it may be the harbinger of a serious underlying condition. It is often idiopathic but may be caused by several conditions associated with androgen excess, such as polycystic ovarian syndrome (PCOS) or congenital adrenal hyperplasia (CAH) ([Table 53-1](#)). Cutaneous manifestations commonly associated with hirsutism include acne and male-pattern balding (androgenic alopecia). *Virilization*, on the other hand, refers to the state in which androgen levels are sufficiently high to cause additional signs and symptoms such as deepening of the voice, breast atrophy, increased muscle bulk, clitoromegaly, and increased libido; virilization is an ominous sign that suggests the possibility of an ovarian or adrenal neoplasm.

HAIR FOLLICLE GROWTH AND DIFFERENTIATION

Hair can be categorized as either *vellus* (fine, soft, and not pigmented) or *terminal* (long, coarse, and pigmented). The number of hair follicles does not change over an individual's lifetime, but the follicle size and type of hair can change in response to numerous factors, particularly androgens. Androgens are necessary for terminal hair and sebaceous gland development and mediate differentiation of pilosebaceous units (PSUs) into either a terminal hair follicle or a sebaceous gland. In the former case, androgens transform the vellus hair into a terminal hair; in the latter, the sebaceous component proliferates and the hair remains vellus.

There are three phases in the cycle of hair growth: (1) *anagen* (growth phase), (2) *catagen* (involution phase), and (3) *telogen* (rest phase). Depending on the body site, hormonal regulation may play an important role in the hair growth cycle. For example, the eyebrows, eyelashes, and vellus hairs are androgen-insensitive, whereas the axillary and pubic areas are sensitive to low doses of androgens. Hair growth on the face, chest, upper abdomen, and back requires greater levels of androgens and is therefore more characteristic of the pattern typically seen in males. Androgen excess in women leads to increased hair growth in most androgen-sensitive sites but will manifest with loss of hair in the scalp region, in part by reducing the time hairs spend in anagen phase.

Although androgen excess underlies most cases of hirsutism, there is only a modest correlation between androgen levels and the quantity of hair growth. This is due to the fact that hair growth from the follicle depends on local factors and variability in end-organ sensitivity, as well as circulating androgen concentrations. Genetic factors and ethnic background also influence hair growth. In general, dark-haired individuals tend to be more hirsute than blonde or fair individuals. Asians and Native Americans have relatively sparse hair in regions sensitive to high androgen levels, whereas people of Mediterranean descent are more hirsute. For these reasons, family history and ethnic background are important considerations when assessing the etiology and severity of hirsutism.

CLINICAL ASSESSMENT

Historic elements relevant to the assessment of hirsutism include the age of onset and rate of progression of hair growth and associated symptoms or signs (e.g., acne). Depending on the cause, excess hair growth is typically first noted during the second and third decades. The growth is usually slow but progressive. Sudden development and rapid progression of hirsutism suggests the possibility of an androgen-secreting neoplasm, in which case findings of virilization may also be present.

The age of onset of menstrual cycles (menarche) and the pattern of the menstrual cycle should be ascertained; irregular cycles from the time of menarche onward are more likely to result from ovarian rather than adrenal androgen excess. Associated symptoms such as galactorrhea should prompt evaluation for hyperprolactinemia ([Chap. 328](#)) and possibly hypothyroidism ([Chap. 330](#)). Hypertension, striae, easy bruising, centripetal weight gain, and weakness suggest hypercortisolism (Cushing's syndrome; [Chap. 331](#)). Rarely, patients with growth hormone excess (i.e., acromegaly) will present with hirsutism. Use of medications such as phenytoin, minoxidil, or cyclosporine may be associated with androgen-independent causes of excess hair growth (i.e., hypertrichosis). A family history of infertility and/or hirsutism may indicate disorders such as nonclassic congenital adrenal hyperplasia ([CAH](#)), a disorder particularly common in Ashkenazi Jews, among others ([Chap. 331](#)).

Physical examination should include measurement of height, weight, and calculation of body mass index (BMI). A BMI $>25 \text{ kg/m}^2$ is indicative of excess weight for height, and values $>30 \text{ kg/m}^2$ are often seen in association with hirsutism. Notation should be made of blood pressure. Cutaneous signs sometimes associated with androgen excess and insulin resistance include acanthosis nigricans and skin tags.

An objective clinical assessment of hair distribution and quantity is central to the evaluation in any woman presenting with hirsutism. This assessment permits the distinction between hirsutism and hypertrichosis and provides a baseline reference point to gauge the response to treatment. *Hypertrichosis* refers to the excessive growth of androgen-independent hair which is vellus, prominent in nonsexual areas, and most commonly familial or caused by metabolic disorders (e.g., thyroid disturbances, anorexia nervosa) or medications (e.g., phenytoin, minoxidil or cyclosporine).

A simple and commonly used method to grade hair growth is the modified scale of Ferriman and Gallwey ([Fig. 53-1](#)), where each of nine androgen-sensitive sites is graded from 0 to 4. Approximately 95% of Caucasian women have a score below 8 on this scale; thus, it is normal for most women to have some hair growth in androgen-sensitive sites. Scores above 8 suggest an excess of androgen-mediated hair growth, a finding that should be assessed further by hormonal evaluation (see below). In racial/ethnic groups that are less likely to manifest hirsutism (e.g., Asian women), additional cutaneous evidence of androgen excess should be sought, including pustular acne or thinning hair.

HORMONAL EVALUATION

Androgens are secreted by both the ovaries and adrenal glands in response to their respective tropic hormones, luteinizing hormone (LH) and adrenocorticotrophic hormone (ACTH). The principal circulating steroids involved in the etiology of hirsutism are

androstenedione, dehydroepiandrosterone (DHEA) and its sulfated form (DHEAS), and testosterone. The ovaries and adrenal glands normally contribute about equally to testosterone production. Further, approximately half of the total testosterone originates from direct glandular secretion, and the remainder is derived from the peripheral conversion of androstenedione and DHEA ([Chap. 335](#)).

Although it is the most important circulating androgen, testosterone is, in effect, the penultimate androgen in mediating hirsutism; it is converted to the more potent dihydrotestosterone (DHT) by the enzyme 5 α -reductase, which is located in the pilosebaceous unit. DHT has a higher affinity for, and slower dissociation from, the androgen receptor. The local production of DHT allows it to serve as the primary mediator of androgen action at the level of the pilosebaceous unit. There are two isoenzymes of 5 α -reductase: type 2 is found in the prostate gland and in hair follicles, whereas type 1 is primarily found in sebaceous glands.

One approach to testing for hyperandrogenemia is depicted in [Fig. 53-2](#). This involves measuring blood levels of testosterone and [DHEAS](#). It is also important to measure the level of free (or unbound) testosterone, because it is the fraction of testosterone that is not bound to its carrier protein, sex-hormone binding globulin (SHBG), that is biologically available. Hyperinsulinemia and/or androgen excess decrease hepatic production of SHBG, often resulting in levels of total testosterone within the high-normal range at a time when the free hormone is substantially elevated. Because adrenal androgens are readily suppressed by low doses of glucocorticoids, the dexamethasone androgen-suppression test may broadly distinguish ovarian from adrenal androgen overproduction. A blood sample is obtained before and after administering dexamethasone (0.5 mg orally every 6 h for 4 days). An adrenal source is suggested by suppression of plasma free testosterone into the normal range; incomplete suppression suggests ovarian androgen excess.

A baseline plasma total testosterone level >12 nmol/L (>3.5 ng/mL) usually indicates a virilizing tumor, whereas a level >7 nmol/L (>2 ng/mL) is suggestive. A basal [DHEAS](#) level >18.5 μ mol/L (>7000 μ g/L) suggests an adrenal tumor. Although DHEAS has been proposed as a "marker" of predominant adrenal androgen excess, it is not unusual to find modest elevations in DHEAS among women with [PCOS](#). Computed tomography (CT) or magnetic resonance imaging (MRI) should be used to localize an adrenal mass, and ultrasound will usually suffice to identify an ovarian mass, if clinical evaluation and hormonal levels suggest these possibilities.

[PCOS](#) is the most common cause of ovarian androgen excess ([Chap. 336](#)). However, the increased ratio of [LH](#) to follicle-stimulating hormone that is often seen in carefully studied patients with PCOS may not be exhibited in up to half of these women due to the pulsatility of gonadotropins. If performed, ultrasound shows enlarged ovaries and/or increased stroma in many women with PCOS. However, polycystic ovaries may also be found in women without clinical or laboratory features of PCOS. Therefore, polycystic ovaries are a relatively insensitive and nonspecific finding for the diagnosis of ovarian hyperandrogenism. Though it is not widely used, gonadotropin-releasing hormone agonist testing can be used to make a specific diagnosis of ovarian hyperandrogenism. A peak 17-hydroxyprogesterone level $\geq$ 7.8 nmol/L ($\geq$ 2.6 μ g/L), after the administration of 100 μ g nafarelin (or 10 μ g/kg leuprolide) subcutaneously, is virtually diagnostic of

ovarian hyperandrogenism.

Nonclassic [CAH](#) is most commonly due to 21-hydroxylase deficiency but can also be caused by autosomal recessive defects in other steroidogenic enzymes necessary for adrenal corticosteroid synthesis ([Chap. 331](#)). Because of the enzyme defect, the adrenal gland cannot secrete glucocorticoids efficiently (especially cortisol). This results in diminished negative feedback inhibition of [ACTH](#), leading to compensatory hyperplasia of the adrenal cortex and accumulation of steroid precursors proximal to the enzyme defect. These precursors are subsequently converted to androgen.

Deficiency of 21-hydroxylase can be reliably excluded by determining a morning 17-hydroxyprogesterone level <6 nmol/L (<2 ug/L) (drawn in the follicular phase). Alternatively, 21-hydroxylase deficiency can be diagnosed by measurement of 17-hydroxyprogesterone 1 h after administration of 250 ug of synthetic [ACTH](#) (cosyntropin) intravenously. Measurement after ACTH is slightly more cumbersome, though the results obtained in this manner are highly reproducible and can be compared to published nomograms.

TREATMENT

Treatment of hirsutism may be accomplished pharmacologically and by mechanical means of hair removal. Nonpharmacologic treatments should be considered in all patients, either as the only treatment or as an adjunct to drug therapy.

Nonpharmacologic treatments include (1) bleaching; (2) depilatory (removal from the skin surface) such as shaving and chemical treatments; or (3) epilatory (removal of the hair including the root) such as plucking, waxing, electrolysis, and laser therapy. Despite perceptions to the contrary, shaving does not increase the rate or density of hair growth. Chemical depilatory treatments may be useful for mild hirsutism that affects only limited skin areas, though they can cause skin irritation. Wax treatment removes hair temporarily but is uncomfortable. Electrolysis is effective for more permanent hair removal, particularly in the hands of a skilled electrologist. Laser phototherapy appears to be efficacious for hair removal. It delays hair regrowth and causes permanent hair removal in some patients. The long-term effects and complications associated with laser treatment are being evaluated.

Pharmacologic therapy for androgen excess is directed at interrupting one or more of the steps in the pathway leading to its expression: (1) suppression of adrenal and/or ovarian androgen production; (2) enhancement of androgen-binding to plasma-binding proteins, particularly [SHBG](#); (3) impairment of the peripheral conversion of androgen precursors to active androgen; and (4) inhibition of androgen action at the target tissue level. Attenuation of hair growth is typically not evident until 4 to 6 months after initiation of medical treatment and, in most cases, leads to a modest reduction in hair growth.

Combination estrogen-progestin therapy, in the form of an oral contraceptive, is usually the first-line endocrine treatment for hirsutism and acne, after cosmetic and dermatologic management. The estrogenic component of most oral contraceptives currently in use is either ethinyl estradiol or mestranol. The suppression of [LH](#) leads to reduced production of ovarian androgens. The reduced androgen levels also result in a

dose-related increase in [SHBG](#), thereby lowering the fraction of unbound plasma testosterone. Combination therapy has also been demonstrated to decrease [DHEAS](#), perhaps by reducing [ACTH](#) levels. Estrogens also have a direct, dose-dependent suppressive effect on sebaceous cell function.

The choice of a specific oral contraceptive should be predicated on the progestational component, as progestins vary in their suppressive effect on [SHBG](#) levels and in their androgenic potential. Ethynodiol diacetate has relatively low androgenic potential, whereas progestins such as norgestrel and levonorgestrel are particularly androgenic, as judged from their attenuation of the estrogen-induced increase in SHBG. Norgestimate exemplifies the newer generation of progestins that are virtually nonandrogenic. Oral contraceptives are contraindicated in women with a history of thromboembolic disease or in women with breast cancer or other estrogen-dependent cancers ([Chap. 336](#)). There is a relative contraindication to the use of oral contraceptives in smokers or in those with hypertension or a history of migraine headaches. In most trials, estrogen-progestin therapy alone improves the extent of acne by a maximum of 50 to 70%. In contrast, the effect on hair growth may not be evident for 6 months, and the maximum effect may require 9 to 12 months owing to the length of the hair growth cycle. Improvements in hirsutism are typically in the range of 20%, and often there is little more than arrest of further progression of hair growth.

Adrenal androgens are more sensitive than cortisol to the suppressive effects of glucocorticoids. Therefore, glucocorticoids are the mainstay of treatment in patients with [CAH](#). Although glucocorticoids have been reported to restore ovulatory function in some women with [PCOS](#), this effect is highly variable. Because of side effects from excessive glucocorticoids, low doses should be used. Dexamethasone (0.2 to 0.5 mg) or prednisone (5 to 10 mg) should be given at bedtime to achieve maximal suppression by inhibiting the nocturnal surge of [ACTH](#).

Cyproterone acetate is the prototypic antiandrogen. It acts mainly by competitive inhibition of the binding of testosterone and [DHT](#) to the androgen receptor. In addition, it may act to enhance the metabolic clearance of testosterone by inducing hepatic enzymes. Although not available for use in the United States, cyproterone acetate is widely used in Canada, Mexico, and Europe. Cyproterone (50 to 100 mg) is given on days 1 to 15 and ethinyl estradiol (50 ug) is given on days 5 to 26 of the menstrual cycle. Side effects of cyproterone acetate include irregular uterine bleeding, nausea, headache, fatigue, weight gain, and decreased libido.

Spirolactone, usually used as a mineralocorticoid antagonist, is also a weak antiandrogen. It is almost as effective as cyproterone acetate when used at high enough doses (100 to 200 mg daily). Patients should be monitored intermittently for hyperkalemia or hypotension, though these side effects are uncommon. Pregnancy should be avoided because of the risk of feminization of a male fetus. Spirolactone can also cause menstrual irregularity. It is often used in combination with an oral contraceptive, which helps in prevention of pregnancy and suppression of ovarian androgen production.

Flutamide is a potent nonsteroidal antiandrogen that is effective in treating hirsutism, but concerns about the induction of hepatocellular dysfunction have limited its use.

Finasteride is a competitive inhibitor of 5 α -reductase type 2. Beneficial effects on hirsutism have been reported, but the prominence of 5 α -reductase type 1 in the pilosebaceous unit appears to account for its limited efficacy. Finasteride would also be expected to impair sexual differentiation in a male fetus, and thus it should not be used in women who may become pregnant.

A prospective, randomized trial comparing low-dose flutamide, finasteride, and combination cyproterone acetate-ethinyl estradiol demonstrated relative superiority of flutamide and cyproterone acetate-ethinyl estradiol in the treatment of hirsutism. Ultimately, the choice of any specific agent(s) must be tailored to the unique needs of the patient being treated. As noted previously, pharmacologic treatments for hirsutism should be used in conjunction with nonpharmacologic approaches. Patients should be reminded about the relatively slow and usually modest responses to pharmacologic treatment. It is also helpful to review the pattern of female hair distribution in the normal population to dispel unrealistic expectations.

(Bibliography omitted in Palm version)

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54. INFERTILITY AND FERTILITY CONTROL - Janet E. Hall

The concept of reproductive choice is now firmly entrenched in developed countries and has dramatically altered reproductive behavior. The availability of effective contraceptive methods prevents unintended pregnancies and gives women the option of pursuing educational and career opportunities without interruption. Population control also has important economic and social implications. Infertility, on the other hand, can be accompanied by substantial stress and disappointment. Fortunately, the ability to diagnose and to treat various causes of infertility now provides an array of effective new approaches to this condition.

INFERTILITY

DEFINITION AND PREVALENCE

Infertility is defined as the inability to conceive after 12 months of unprotected sexual intercourse. In a study of 5574 English and American women who ultimately conceived, pregnancy occurred in 50% within 3 months, 72% within 6 months, and 85% within 12 months. These findings are consistent with predictions based on *fecundability*, the probability of achieving pregnancy in one menstrual cycle (approximately 20 to 25% in healthy young couples). Assuming a fecundability of 0.25, 98% of couples should conceive within 13 months. Based on this definition, the National Survey of Family Growth reports a 14% rate of infertility in the United States in married women aged 15 to 44. The infertility rate has remained relatively stable over the past 30 years, although the proportion of couples without children has risen, reflecting a trend to delay childbearing. This trend has important implications because of an age-related decrease in fecundability, which begins at age 35, and decreases markedly after age 40.

CAUSES OF INFERTILITY

There is a spectrum of infertility, ranging from reduced conception rates or the need for medical intervention to irreversible causes of infertility (*sterility*). Infertility can be attributed primarily to male factors in 25%, female factors in 58%, and is unexplained in about 17% of couples ([Fig. 54-1](#)). Not uncommonly, both male and female factors contribute to infertility.

Approach to the Patient

Initial Evaluation In all couples presenting with infertility, the initial evaluation includes discussion of the appropriate timing of intercourse and a description of the range of investigations that may be required. A brief description of infertility treatment options, including adoption, should be reviewed. Initial investigations are focused on determining whether the primary cause of the infertility is male, female, or both. These investigations include a semen analysis in the male, confirmation of ovulation in the female, and, in the majority of situations, documentation of tubal patency in the female. Although frequently used in the past, recent studies have not supported the efficacy of postcoital testing of sperm interaction with cervical mucus as a routine component of initial testing. Strategies for further evaluation are described below and in [Chaps. 335](#) and [336](#). In some cases, after an extensive workup excluding all male and female factors, a specific

cause cannot be identified and infertility may ultimately be classified as unexplained.

Psychological Aspects of Infertility Infertility is invariably associated with psychological stress related not only to the diagnostic and therapeutic procedures themselves but also to repeated cycles of hope and loss associated with each new procedure or cycle of treatment that does not result in the birth of a child. These feelings are often combined with a sense of isolation from friends and family. Counseling and stress-management techniques should be introduced early in the evaluation of infertility. In addition to the psychological benefits of stress management, it is possible that stress contributes to infertility in some couples (e.g., impaired ovulation). Importantly, infertility and its treatment do not appear to be associated with long-term psychological sequelae.

Female Causes Abnormalities in menstrual function constitute the most common cause of female infertility. These disorders, which include ovulatory dysfunction and abnormalities of the uterus or outflow tract, may present as amenorrhea (absence of menses) or as irregular or short menstrual cycles. A careful history and physical examination and a limited number of laboratory tests will help to determine whether the abnormality is: (1) hypothalamic or pituitary [low follicle-stimulating hormone (FSH), luteinizing hormone (LH), and estradiol with or without an increase in prolactin]; (2) polycystic ovarian syndrome (PCOS; irregular cycles and hyperandrogenism in the absence of other causes of androgen excess); (3) ovarian (low estradiol with increased FSH); or (4) uterine or outflow tract abnormality. The frequency of these diagnoses depends on whether the amenorrhea is primary or occurs after normal puberty and menarche ([Fig. 54-1](#)). **The approach to further evaluation of these disorders is described in detail in [Chap. 52](#).*

OVULATORY DYSFUNCTION In women with a history of regular menstrual cycles, *evidence of ovulation* should be sought by using urinary ovulation predictor kits (they reflect the preovulatory gonadotropin surge but do not confirm ovulation), basal body temperature charts, or a mid-luteal phase progesterone level. The mid-luteal phase progesterone increase (usually >3 ng/mL) confirms ovulation and corpus luteum function and is responsible for the rise in basal body temperature [$>0.3^{\circ}\text{C}$ ($>0.6^{\circ}\text{F}$) for 10 days]. An endometrial biopsy to exclude luteal phase insufficiency is no longer considered an essential part of the infertility workup for most patients. Even in the presence of ovulatory cycles, evaluation of *ovarian reserve* is recommended for women over 35 by measurement of [FSH](#) on day 3 of the cycle or in response to clomiphene, an estrogen antagonist (see below). An FSH level <10 IU/mL on cycle day 3 predicts adequate ovarian oocyte reserve. Inhibin B, an ovarian hormone that selectively suppresses FSH, is being investigated as an additional marker of ovarian reserve.

TUBAL DISEASE This may result from pelvic inflammatory disease (PID), appendicitis, endometriosis, pelvic adhesions, tubal surgery, and previous use of an intrauterine device (IUD). However, a cause is not identified in up to 50% of patients with documented tubal factor infertility. Because of the high prevalence of tubal disease, testing should occur early in the majority of couples with infertility. Subclinical infections with *Chlamydia trachomatis* may be an underdiagnosed cause of tubal infertility and requires the treatment of both partners. A hysterosalpingogram (HSG) is the most common screening test and will determine the presence of tubal patency and identify potential abnormalities of the uterine cavity.

ENDOMETRIOSIS *Endometriosis* is defined as the presence of endometrial glands or stroma outside the endometrial cavity and uterine musculature. Its presence is suggested by a history of dyspareunia (painful intercourse), worsening dysmenorrhea that often begins before menses, or by a thickened rectovaginal septum or deviation of the cervix on pelvic examination. The pathogenesis of the infertility associated with endometriosis is unclear but may involve indirect effects on the normal endometrium as well as the direct effects of adhesions in advanced disease. Endometriosis is often clinically silent, however, and can only be excluded definitively by laparoscopy.

Male Causes Known causes of male infertility include primary testicular disease, disorders of sperm transport, and hypothalamic-pituitary disease resulting in secondary hypogonadism. However, the etiology is not ascertained in up to half of men with suspected male factor infertility ([Fig. 54-1](#)). The key initial diagnostic test is a *semen analysis*. Although 95% confidence limits can be used to define normal semen parameters, data relating sperm counts to fecundability are more useful. Such studies suggest that sperm counts of <20 million/mL, with a motility of less than 40%, are associated with an increased risk of infertility. Analysis of sperm morphology is less well validated, but >40% normal forms are usually present in fertile men. Successful in vitro fertilization (IVF) can usually be accomplished with >14% normal forms (using strict Kruger criteria), whereas low fertilization is seen with <4% normal forms. Other tests such as the hamster egg penetration test and the zona-binding assay are not of proven value.

Testosterone levels should be measured if the sperm count is low on repeated examination or if there is clinical evidence of hypogonadism. A low testosterone level may result from *primary gonadal deficiency*; in this condition, levels of [LH](#) and [FSH](#) will be elevated. Less commonly, low testosterone and decreased spermatogenesis result from hypothalamic or pituitary disease, in which case the LH and FSH levels will be low ([Chap. 335](#)).

Abnormalities of spermatogenesis may have a genetic component. Y chromosome microdeletions and substitutions are increasingly recognized as a cause of *azoospermia* (absence of sperm) or *oligospermia* (low sperm count). Microdeletions (Yq6 region) have also been identified in a subset of men with elevated [FSH](#) levels or otherwise idiopathic infertility. Several candidate genes have been identified including *DAZ* (deleted in azoospermia) and *YRRM* (Y chromosome RNA recognition motif).

Acquired disorders of the testes are often associated with impaired spermatogenesis with relatively preserved Leydig cell function; thus, testosterone levels may be normal. Such abnormalities include viral orchitis (especially mumps) and other infectious causes such as tuberculosis or sexually transmitted diseases (STDs), chemotherapy (especially the alkylating agents cyclophosphamide and chlorambucil), ionizing radiation, and drugs that may impair fertility directly or through inhibition of testicular androgen production or action. Anabolic androgen abuse should be considered in a well-androgenized man with low gonadotropins and testosterone but a suppressed sperm count. Prolonged elevation of testicular temperature may impair spermatogenesis, e.g., after an acute febrile illness or in association with varicocele. A potential role for environmental toxins as a cause of impaired spermatogenesis has been suggested based on an apparent decrease in

sperm counts over the past several decades, but a direct cause-and-effect relationship has not been established.

SECONDARY HYPOGONADISM Low gonadotropin levels, associated with low testosterone, may signal the presence of a pituitary macroadenoma or hypothalamic tumor (in both cases prolactin levels may be elevated; [Chap. 328](#)) or may be the first presentation of hemochromatosis ([Chap. 345](#)) or other systemic illness. Recent studies have identified several genetic causes of gonadotropin-releasing hormone (GnRH) deficiency (*KAL* and *DAX-1*), as well as mutations that lead to isolated gonadotropin deficiency (GnRH receptor, [LHb](#), [FSHb](#) mutations) ([Chap. 328](#)).

DISORDERED SPERM TRANSPORT Patients with low sperm counts and normal hormonal levels may be found to have obstructive abnormalities of the vas deferens or epididymus. The most common causes of vas deferens obstruction are previous vasectomy or accidental ligation during inguinal surgery. Patency rates with microsurgical reversal techniques are high in the first 3 years after vasectomy but decrease markedly thereafter. Congenital absence of the vas deferens can be diagnosed by a deficiency of fructose in the ejaculate and is often associated with an abnormality of the cystic fibrosis transmembrane regulator (*CFTR*) gene. Young's syndrome, characterized by inspissated secretions, can also preclude normal sperm transport.

TREATMENT

The treatment of infertility should be tailored to the problems unique to each couple ([Table 54-1](#)). In many situations, including unexplained infertility, mild to moderate endometriosis, and/or borderline semen parameters, a stepwise approach to infertility is optimal, beginning with low-risk interventions and moving to more invasive, higher risk interventions only if necessary. After determination of all infertility factors and their correction, if possible, this approach might include, in increasing order of complexity: (1) expectant management, (2) clomiphene citrate (see below) with or without intrauterine insemination (IUI), (3) gonadotropins with or without IUI, and (4) [IVF](#). The time used to complete the evaluation, correction, and expectant management can be longer in women <30, but this process should be advanced rapidly in women >35. In some situations expectant management will not be appropriate.

Ovulatory Dysfunction Treatment of ovulatory dysfunction should first be directed at identification of the etiology of the disorder to allow specific management when possible. Dopamine agonists, for example, may be indicated in patients with hyperprolactinemia ([Chap. 328](#)); lifestyle modification may be successful in women with low body weight or a history of intensive exercise ([Chap. 78](#)).

Pulsatile GnRH is highly effective for restoring ovulation in patients with hypothalamic amenorrhea. When administered subcutaneously by an automated pump at a physiologic dose and frequency, pulsatile GnRH induces normal [LH](#) and [FSH](#) dynamics. Direct comparisons between pulsatile GnRH and gonadotropin treatment for ovulation induction indicate similar pregnancy rates; pulsatile GnRH is associated with lower rates of multiple gestation and virtually no risk of ovarian hyperstimulation.

Clomiphene citrate is a nonsteroidal estrogen antagonist that increases [FSH](#) and [LH](#) levels by blocking estrogen negative feedback at the hypothalamus. The efficacy of clomiphene for ovulation induction is highly dependent on patient selection. It induces ovulation in ~60% of women with [PCOS](#) and is the initial treatment of choice in these patients. The starting dose is 50 mg daily for 5 days beginning on day 5 of a spontaneous cycle or after a progestin-induced withdrawal bleed. The dose can be increased to 150 mg, if necessary, in subsequent cycles, and human chorionic gonadotropin (hCG) can be added as the ovulatory stimulus. In women with PCOS, the use of insulin-sensitizing agents, such as metformin appears to be particularly effective in combination with clomiphene.

Gonadotropins are highly effective for ovulation induction in women with hypogonadotropic hypogonadism and [PCOS](#). Gonadotropins are also used to induce multiple follicular recruitment in unexplained infertility and in older reproductive-aged women, particularly in conjunction with [IUI](#). Disadvantages include a significant risk of multiple gestation and the risk of ovarian hyperstimulation, a side effect that is more common in women with PCOS. However, careful monitoring and a conservative approach to ovarian stimulation reduce these risks; gonadotropin stimulation is an effective and safe treatment when applied by experienced practitioners. Currently available gonadotropins include urinary preparations of [LH](#) and [FSH](#), highly purified FSH, and recombinant FSH. Though FSH is the key component, there is growing data that the addition of some LH (or [hCG](#)) may improve results, particularly in hypogonadotropic patients.

None of these methods are effective in women with premature ovarian failure in whom donor oocyte or adoption are the methods of choice.

Tubal Disease If hysterosalpingography suggests a tubal or uterine cavity abnormality, or if a patient is ³⁵ at the time of initial evaluation, laparoscopy with tubal lavage is recommended, often with a hysteroscopy. Although tubal reconstruction may be attempted if tubal disease is identified, it is generally being replaced by the use of [IVF](#), as these patients are at increased risk of developing an ectopic pregnancy.

Endometriosis Though 60% of women with minimal or mild endometriosis may conceive within 1 year without treatment, laparoscopic resection or ablation appear to improve conception rates. Medical management of advanced stages of endometriosis is widely used for symptom control but has not been shown to enhance fertility ([Chap. 336](#)). In moderate to severe endometriosis, conservative surgery is associated with pregnancy rates of 50 and 39% respectively, compared with rates of 25 and 5% with expectant management alone. In some patients, [IVF](#) may be the treatment of choice.

Male Factor Infertility The treatment options for male factor infertility have expanded greatly in recent years. Secondary hypogonadism is highly amenable to treatment with pulsatile [GnRH](#) or gonadotropins ([Chap. 335](#)). In vitro techniques have provided new opportunities for patients with primary testicular failure and disorders of sperm transport. Choice of initial treatment options depends on sperm concentration and motility. Expectant management should be attempted initially in men with mild male factor infertility (sperm count of 15 to 20 × 10⁶/mL and normal motility). Moderate male factor infertility (10 to 15 × 10⁶/mL and 20 to 40% motility) should begin with [IUI](#) alone or in

combination with treatment of the female partner with clomiphene or gonadotropins, but it may require [IVF](#) with or without intracytoplasmic sperm injection (ICSI). For men with a severe defect (sperm count of $<10^6$ /mL, 10% motility), IVF with ICSI or donor sperm should be used.

Assisted Reproductive Technologies The development of assisted reproductive technologies (ART) has dramatically altered the treatment of male and female infertility. [IVF](#) is indicated for patients with many causes of infertility that have not been successfully managed with more conservative approaches. IVF or [ICSI](#) is often the treatment of choice in couples with a significant male factor or tubal disease, whereas IVF using donor oocytes is used in patients with premature ovarian failure and in women of advanced reproductive age. Success rates depend on the age of the woman and the cause of the infertility and are generally 18 to 24% per cycle when initiated in women <40 . In women >40 , there is a marked decrease in both the number of oocytes retrieved and their ability to be fertilized. Though often effective, IVF is expensive and requires careful monitoring of ovulation induction and invasive techniques including the aspiration of multiple follicles. IVF is associated with a significant risk of multiple gestation (29% twins, 7% triplets, and 0.6% higher order multiples). More recently developed blastocyst transfer protocols decrease the number of transfers but increase pregnancy rates.

CONTRACEPTION

Though various forms of contraception are widely available, approximately 30% of births in the United States are the result of unintended pregnancy. Teenage pregnancies continue to represent a serious public health problem in the United States, with >1 million unintended pregnancies each year -- a significantly greater incidence than in other industrialized nations ([Chap. 8](#)).

Contraceptive methods are widely used ([Table 54-2](#)). Only 15% of couples report having unprotected sexual intercourse in the past 3 months. A reversible form of contraception is used by $>50\%$ of couples. Sterilization (in either the male or female) has been employed as a permanent form of contraception by about 25% of couples. Pregnancy termination is relatively safe when directed by health care professionals but is rarely the option of choice.

No single contraceptive method is ideal, although all are safer than carrying a pregnancy to term. The effectiveness of a given method of contraception is dependent on the efficacy of the method itself, compliance, and appropriate use. Knowledge of the advantages and disadvantages of each contraceptive is essential for counseling an individual about the methods that are safest and most consistent with his or her lifestyle. Discrepancies between theoretical and actual effectiveness emphasize the importance of patient education and compliance when considering various forms of contraception ([Table 54-2](#)).

BARRIER METHODS

Barrier contraceptives, such as condoms, diaphragms, cervical caps, and spermicides, are easily available, reversible, and have fewer side effects than hormonal methods.

However, their effectiveness is highly dependent on compliance and proper use ([Table 54-2](#)). A major advantage of barrier contraceptives is the protection provided against [STDs](#) ([Chap. 132](#)). Consistent use is associated with a decreased risk of gonorrhea, nongonococcal urethritis, and genital herpes, probably due in part to the concomitant use of spermicides. Condom use also reduces the transmission of HIV infection. Natural membrane condoms may be less effective than latex condoms, and petroleum-based lubricants can degrade condoms and decrease their efficacy for preventing HIV infection. A highly effective female condom, which also provides protection against STDs, was approved in 1994 but has not achieved widespread use.

STERILIZATION

Sterilization is the method of birth control most frequently chosen by fertile men and multiparous women >30 ([Table 54-2](#)). Sterilization refers to a procedure that prevents fertilization by surgical interruption of the fallopian tubes in women or the vas deferens in men. Although tubal ligation and vasectomy are potentially reversible, these procedures should be considered permanent and should not be undertaken without careful patient counseling.

Several methods of *tubal ligation* have been developed, all of which are highly effective with a 10-year cumulative pregnancy rate of 1.85 per 100 women. However, when pregnancy does occur, the risk of ectopic pregnancy may be as high as 30%. The success rate of tubal reanastomosis depends on the method used -- the clip, silastic band, and modified Pomeroy procedures are easier to reverse than the Irving, Uchida, and electrocoagulation methods. Even after successful reversal, the risk of ectopic pregnancy remains great. In addition to prevention of pregnancy, tubal ligation reduces the risk of ovarian cancer, possibly by limiting the upward migration of potential carcinogens.

Vasectomy is an outpatient surgical procedure that has little risk and is highly effective. The development of azoospermia may be delayed for 2 to 6 months, and other forms of contraception must be used until two sperm-free ejaculations provide proof of sterility. Reanastomosis may restore fertility in 30 to 50% of men, but the success rate appears to decline with time after vasectomy and may be influenced by nonmechanical factors such as the development of anti-sperm antibodies.

INTRAUTERINE DEVICES

[IUDs](#) inhibit pregnancy primarily through a spermicidal effect caused by a sterile inflammatory reaction produced by the presence of a foreign body in the uterine cavity. There may also be effects on cervical mucus sperm transport through the oviduct. IUDs provide a high level of efficacy in the absence of systemic metabolic effects. An additional advantage is that ongoing motivation is not required to ensure efficacy once the device has been placed. However, only 1% of women in the United States use this method compared to a utilization rate of 15 to 30% in much of Europe and Canada. This relatively low utilization rate continues despite evidence that the newer devices are not associated with increased rates of pelvic infection and infertility, as occurred with earlier devices. Screening for [STD](#) should be performed prior to insertion, and an IUD should not be used in women at high risk for development of STD or in women at high risk for

bacterial endocarditis. In addition, the IUD may not be effective in women with uterine leiomyomas because they alter the size or shape of the uterine cavity. IUD use is associated with increased menstrual blood flow, although this is less pronounced with the progesterone-releasing IUD than the copper-containing device.

HORMONAL METHODS

No male hormonal contraceptive methods are currently approved in the United States. However, hormonal methods of male contraception, including [GnRH](#)-mediated suppression of the hypothalamic-pituitary-gonadal axis in combination with testosterone replacement, are under investigation.

Oral Contraceptive Pills Because of their ease of use and efficacy, oral contraceptive pills are the most widely used form of hormonal contraception. They act by suppressing ovulation, changing cervical mucus, and altering the endometrium. The current formulations are made from synthetic estrogens and progestins. The estrogen component of the pill consists of ethinyl estradiol or mestranol, which is metabolized to ethinyl estradiol. Multiple synthetic progestins are used. Norethindrone and its derivatives are used in many formulations. Low-dose norgestimate and third-generation progestins (desogestrel, gestodene) have a less androgenic profile; levonorgestrel appears to be the most androgenic of the progestins and should be avoided in patients with hyperandrogenic symptoms. The three major formulations of oral contraceptives include: (1) fixed-dose estrogen-progestin combination, (2) phasic estrogen-progestin combination, and (3) progestin only. Each of these formulations is administered daily for 3 weeks followed by a week of no medication during which menstrual bleeding generally occurs.

Current doses of ethinyl estradiol range from 20 to 50 ug. However, indications for the 50-ug dose are rare, and the majority of formulations contain 35 ug of ethinyl estradiol. The reduced estrogen and progesterone content in the second- and third-generation pills has decreased both side effects and risks associated with oral contraceptive use. At the currently used doses, patients must be cautioned not to miss pills due to the potential for ovulation. Side effects, including break-through bleeding, amenorrhea, and weight gain, are often responsive to a change in formulation. There is no evidence that low-dose oral contraceptives increase the risk of cardiovascular disease in women <30 or in nonsmoking women without additional risk factors. However, the risk of myocardial infarction and stroke in women who smoke is increased by the use of oral contraceptives. The risk of developing hypertension is increased somewhat, even with the low-dose preparations. An increased risk of venous thromboembolism occurs with all oral contraceptives and may be even greater with the third-generation preparations. The factor V Leiden mutation and other thrombophilic disorders ([Chap. 117](#)) are important risk factors for venous thrombosis during oral contraceptive therapy. However, biochemical or genetic screening for these disorders before starting oral contraceptives is not cost-effective at present. In most studies, oral contraceptive use has not been shown to increase the risk of breast cancer, but there is a slight increase in the risk of cervical cancer. Risks for endometrial and ovarian cancer are decreased in oral contraceptive users.

Previous thromboembolic events or stroke are absolute contraindications for the use of

oral contraceptive pills. A history of hormone-dependent tumors and liver disease are also contraindications. Oral contraceptive pills should not be given in pregnancy or in women with undiagnosed uterine bleeding or amenorrhea.

The microdose progestin-only minipill is less effective as a contraceptive, having a pregnancy rate of 2 to 7 per 100 women-years. However, it may be appropriate for women with cardiovascular disease or for women who cannot tolerate synthetic estrogens.

Injectable Contraceptives Depot medroxyprogesterone acetate (Depo-Provera) and Norplant ([Table 54-2](#)) act primarily by inhibiting ovulation and causing changes in the endometrium and cervical mucus that result in decreased implantation and sperm transport. Depo-Provera is effective for 3 months, but return of fertility after discontinuation may be delayed for up to 12 to 18 months. Norplant requires surgical insertion but is effective for up to 5 years after insertion; fertility is possible shortly after its removal. The U.S. Food and Drug Administration (FDA) has recently approved the use of covered rods in addition to the capsules. Amenorrhea, irregular bleeding, and weight gain are the most common adverse effects associated with both injectable forms of contraception. An injectable progestin/estrogen combination contraceptive will be available soon. It requires monthly injection, but irregular bleeding and weight gain are less common. A major advantage of the injectable progestin-based contraceptives is the apparent lack of increased arterial and venous thromboembolic events.

POSTCOITAL CONTRACEPTION

Postcoital contraceptive methods prevent implantation or cause regression of the corpus luteum and are highly efficacious if used appropriately. Although postcoital contraception is not specifically licensed for use in the United States, an FDA notice published in 1997 indicated that certain oral contraceptive pills could be used within 72 h of unprotected intercourse [Ovral (2 tablets 12 h apart) and Lo/Ovral (4 tablets 12 h apart)]. The Preven Emergency Contraceptive Kit contains four combination tablets (50 mg ethinyl estradiol and 0.25 mg levonorgestrel) and a pregnancy kit to rule out pregnancy before taking the pills. Side effects are common with these high doses of hormones and include nausea, vomiting, and breast soreness. Recent studies suggest that 600 mg mifepristone (RU486), a progesterone receptor antagonist, may be equally as effective or more effective than hormonal regimens, with fewer side effects. Mifepristone is not currently available in the United States.

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SECTION 9 -ALTERATIONS IN THE SKIN

55. APPROACH TO THE PATIENT WITH A SKIN DISORDER - *Thomas J. Lawley, Kim B. Yancey*

The challenge of examining the skin lies in distinguishing normal from abnormal, significant findings from trivial ones, and in integrating pertinent signs and symptoms into an appropriate differential diagnosis. The fact that the largest organ in the body is visible is both an advantage and a disadvantage to those who examine it. It is advantageous because no special instrumentation, other than a magnifying glass, is necessary and because the skin can be biopsied with little morbidity. However, the casual observer can be overwhelmed by a variety of stimuli and overlook important, subtle signs of skin or systemic disease. For instance, the sometimes minor differences in color and shape that distinguish a malignant melanoma (see [Plate IIC-30](#)) from a benign pigmented nevus (see [Plate IIC-28](#)) can be difficult to recognize. To aid in the interpretation of skin lesions, a variety of descriptive terms have been developed to characterize cutaneous lesions ([Tables 55-1](#) and [55-2](#) and [Fig. 55-1](#)) and to formulate a differential diagnosis ([Table 55-3](#)). For instance, the finding of large numbers of scaling papules, usually indicative of a primary skin disease, places the patient in a different diagnostic category than would hemorrhagic papules, which may indicate vasculitis or sepsis (see [Plates IIE-71](#) and [IID-44](#), respectively). It is important to differentiate primary skin lesions from secondary skin changes. If the examiner focuses on linear erosions overlying an area of erythema and scaling, he or she may incorrectly assume that the erosion is the primary lesion and the redness and scale are secondary, while the correct interpretation would be that the patient has a pruritic eczematous dermatitis and the erosions have been caused by scratching.

Approach to the Patient

In examining the skin it is usually advisable to assess the patient before taking a history. This way, the entire cutaneous surface is sure to be evaluated, and objective findings can be integrated with relevant historic data. Four basic features of any cutaneous lesion must be noted and considered in the examination of skin: the distribution of the eruption, the type(s) of primary lesion, the shape of individual lesions, and the arrangement of the lesions. In the initial examination it is important that the patient be disrobed as completely as possible. This will minimize chances of missing important individual skin lesions and make it possible to assess the distribution of the eruption accurately. The patient should first be viewed from a distance of about 1.5 to 2 m (4 to 6 ft) so that the general character of the skin and the distribution of lesions can be evaluated. Indeed, distribution of lesions often correlates highly with diagnosis ([Fig. 55-2](#)). For example, a hospitalized patient with a generalized erythematous exanthem is more likely to have a drug eruption than is a patient with a similar rash limited to the sun-exposed portions of the face. The presence or absence of lesions on mucosal surfaces should also be determined. Once the distribution of the lesions has been established, the nature of the primary lesion must be determined. Thus, when lesions are distributed on elbows, knees, and scalp, the most likely possibility based solely on distribution is psoriasis or dermatitis herpetiformis (see [Plates IIA-3](#) and [IIE-68](#), respectively). The primary lesion in psoriasis is a scaly papule that soon forms erythematous plaques covered with a white scale, whereas that of dermatitis

herpetiformis is an urticarial papule that quickly becomes a small vesicle. In this manner, identification of the primary lesion directs the examiner toward the proper diagnosis. Secondary changes in skin can also be quite helpful. For example, scale represents excessive epidermis, while crust is the result of an inadequate or discontinuous epithelial cell layer. Palpation of skin lesions can also yield insight into the character of an eruption. Thus red papules on the lower extremities that blanch with pressure can be a manifestation of many different diseases, but hemorrhagic red papules that do not blanch with pressure indicate palpable purpura characteristic of necrotizing vasculitis (see [Plate IIE-71](#)).

The shape of lesions is also an important feature. Flat, round, erythematous papules and plaques are common in many cutaneous diseases. However, target-shaped lesions that consist in part of erythematous plaques are specific for erythema multiforme (see [Plate IIE-67](#)). In the same way, the arrangement of individual lesions is important. Erythematous papules and vesicles can occur in many conditions, but their arrangement in a specific linear array suggests an external etiology such as allergic contact (see Plate IIA-8) or primary irritant dermatitis. In contrast, lesions with a generalized arrangement are common and suggest a systemic etiology.

As in other branches of medicine, a complete history should be obtained to emphasize the following features:

1. Evolution of lesions
 - a. Site of onset
 - b. Manner in which eruption progressed or spread
 - c. Duration
 - d. Periods of resolution or improvement in chronic eruptions
2. Symptoms associated with the eruption
 - a. Itching, burning, pain, numbness
 - b. What, if anything, has relieved symptoms
 - c. Time of day when symptoms are most severe
3. Current or recent medications (prescribed as well as over-the-counter)
4. Associated systemic symptoms (e.g., malaise, fever, arthralgias)
5. Ongoing or previous illnesses
6. History of allergies
7. Presence of photosensitivity

8. Review of systems

DIAGNOSTIC TECHNIQUES

Many skin diseases can be diagnosed on gross clinical appearance, but sometimes relatively simple diagnostic procedures can yield valuable information. In most instances, they can be performed at the bedside with a minimum of equipment.

Skin Biopsy A skin biopsy is a straightforward minor surgical procedure; however, it is important to biopsy the anatomic site most likely to yield diagnostic findings. This decision may require expertise in skin diseases and knowledge of superficial anatomic structures in selected areas of the body. In this procedure, a small area of skin is anesthetized with 1% lidocaine with or without epinephrine. The skin lesion in question can be excised with a scalpel or removed by punch biopsy. In the latter technique, a punch is pressed against the surface of the skin and rotated with downward pressure until it penetrates to the subcutaneous tissue. The circular biopsy is then lifted with forceps, and the bottom is cut with iris scissors. Biopsy sites may or may not need suture closure, depending on size and location.

KOH Preparation A potassium hydroxide (KOH) preparation is performed on scaling skin lesions when a fungal etiology is suspected. The edge of such a lesion is scraped gently with a scalpel blade, and the removed scale is collected on a glass microscope slide and treated with 1 to 2 drops of a solution of 10 to 20% KOH. KOH dissolves keratin and allows easier visualization of fungal elements. Brief heating of the slide accelerates dissolution of keratin. When the preparation is viewed under the microscope, the refractile hyphae will be seen more easily when the light intensity is reduced. This technique can be utilized to identify hyphae in dermatophyte infections (see [Plate IID-51](#)), pseudohyphae and budding yeast in *Candida* infections (see [Plate IID-43](#)), and fragmented hyphae and spores in tinea versicolor. The same sampling technique can be used to obtain scale for culture of selected pathogenic organisms.

Tzanck Smear A Tzanck smear is a cytologic technique most often used in the diagnosis of herpesvirus infections [simplex or varicella-zoster (see [Plates IID-36](#) and [IID-37](#))]. An early vesicle, not a pustule or crusted lesion, is unroofed, and the base of the lesion is scraped gently with a scalpel blade. The material is placed on a glass slide, air-dried, and stained with Giemsa or Wright's stain. Multinucleated giant cells suggest the presence of herpes, but culture or immunofluorescence testing must be performed to identify the specific virus.

Diascopy Diascopy is designed to assess whether a skin lesion will blanch with pressure as, for example, in determining whether a red lesion is hemorrhagic or simply blood-filled. For instance, a hemangioma (see [Plate IIA-17](#)) will blanch with pressure, whereas a purpuric lesion caused by necrotizing vasculitis (see [Plate IIE-71](#)) will not. Diascopy is performed by pressing a microscope slide or magnifying lens against a specified lesion and noting the amount of blanching that occurs. Granulomas often have an "apple jelly" appearance on diascopy.

Wood's Light A Wood's lamp generates 360-nm ultraviolet (or "black") light that can be

used to aid the evaluation of certain skin disorders. For example, a Wood's lamp will cause erythrasma (a superficial, intertriginous infection caused by *Corynebacterium minutissimum*) to show a characteristic coral red color, and wounds colonized by *Pseudomonas* to appear pale blue. Tinea capitis caused by certain dermatophytes such as *Microsporum canis* or *M. audouini* exhibits a yellow fluorescence. Pigmented lesions of the epidermis such as freckles are accentuated, while dermal pigment such as postinflammatory hyperpigmentation fades under a Wood's light. Vitiligo (see [Plate IIA-11](#)) appears totally white under a Wood's lamp, and previously unsuspected areas of involvement often become apparent. A Wood's lamp may also aid in the demonstration of tinea versicolor and in recognition of ash leaf spots in patients with tuberous sclerosis.

Patch Tests Patch testing is designed to document sensitivity to a specific antigen. In this procedure, a battery of suspected allergens is applied to the patient's back under occlusive dressings and allowed to remain in contact with the skin for 48 h. The dressings are removed, and the area is examined for evidence of delayed hypersensitivity reactions (e.g., erythema, edema, or papulovesicles). This test is best performed by physicians with special expertise in patch testing and is often helpful in the evaluation of patients with chronic dermatitis.

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56. ECZEMA, PSORIASIS, CUTANEOUS INFECTIONS, ACNE, AND OTHER COMMON SKIN DISORDERS - Robert A. Swerlick, Thomas J. Lawley

ECZEMA AND DERMATITIS

Eczema, or dermatitis, is a reaction pattern that presents with variable clinical and histologic findings and is the final common expression for a number of disorders, including atopic dermatitis, allergic contact and irritant contact dermatitis, dyshidrotic eczema, nummular eczema, lichen simplex chronicus, asteatotic eczema, and seborrheic dermatitis. Primary lesions may include papules, erythematous macules, and vesicles, which can coalesce to form patches and plaques. In severe eczema, secondary lesions from infection or excoriation, marked by weeping and crusting, may predominate. Long-standing dermatitis is often dry and is characterized by thickened, scaling skin (*lichenification*).

ATOPIC DERMATITIS

Atopic dermatitis (AD) is the cutaneous expression of the atopic state, characterized by a family history of asthma, hay fever, or dermatitis in up to 70% of patients. The criteria for the diagnosis of atopic eczema are shown in [Table 56-1](#). The prevalence of atopic dermatitis is increasing worldwide, with a point prevalence in Norwegian school children as high as 23%.

The etiology of [AD](#) is only partially defined. There is a clear genetic predisposition. When both parents are affected by AD, over 80% of their children manifest the disease. When only one parent is affected, the prevalence drops to slightly over 50%. A number of genes have been tentatively linked to AD including genes coding for IgE, the high-affinity IgE receptor, mast cell tryptase, and interleukin (IL) 4. Patients with AD may display a variety of immunoregulatory abnormalities including increased IgE synthesis; increased specific IgE to foods, aeroallergens, bacteria, and bacterial products; increased expression of CD23 (low-affinity IgE receptor) on monocytes and B cells; impaired delayed type hypersensitivity reactions; and increased type II and decreased type I cytokine responses.

The clinical presentation often varies with age. Half of patients with [AD](#) present within the first year of life, and 80% present by 5 years of age. Some 80% ultimately coexpress allergic rhinitis or asthma later in life. The infantile pattern is characterized by weeping inflammatory patches and crusted plaques that occur on the face, neck, extensor surfaces, and groin. The childhood and adolescent pattern is marked by dermatitis of flexural skin, particularly in the antecubital and popliteal fossae (see [Plate IIA-4](#)). AD may resolve spontaneously in adults, but the dermatitis will persist into adult life in over half of individuals affected as children. The distribution of lesions may be similar to those seen in childhood. However, adults affected with AD frequently have localized disease, manifesting as hand eczema or lichen simplex chronicus (see below).

Pruritus is a prominent characteristic of [AD](#), and many of the cutaneous findings in affected patients are secondary to rubbing and scratching. Other cutaneous stigmata of AD are perioral pallor, an extra fold of skin beneath the lower eyelid (Dennie's line), increased palmar markings, and increased incidence of cutaneous infections,

particularly with *Staphylococcus aureus*. Atopic individuals often have dry itchy skin, abnormalities in cutaneous vascular responses, and, in some instances, elevations in serum IgE.

Histologic examination of the skin affected by AD may demonstrate features of acute or chronic dermatitis. Immunopathology shows activated, memory T helper cells, which express the cutaneous lymphocyte antigen, the ligand for the inducible endothelial cell adhesion molecule E-selectin. AD skin lesions may also demonstrate IgE-bearing CD1a+ positive Langerhans cells, and these cells have been implicated in AD disease pathophysiology through mediation of hypersensitivity responses to environmental antigens.

TREATMENT

Therapy of AD should be based on avoidance of cutaneous irritants, adequate cutaneous hydration, judicious use of low- or midpotency topical glucocorticoids, and prompt treatment of secondarily infected skin lesions. Patients should be instructed to bathe using warm, but not hot, water and to limit their use of soap. Immediately after bathing while the skin is still moist, the skin should be lubricated with a low- or midpotency topical glucocorticoid in a cream or ointment base. Potent fluorinated topical glucocorticoids should not be used on the face or intertriginous areas. It takes a minimum of 30 g of glucocorticoid ointment to cover the entire body surface of an average adult.

Crusted and weeping skin lesions should be treated with systemic antibiotics with activity against *S. aureus* since secondary infection often exacerbates eczema. The frequency of macrolide-resistant organisms makes the use of penicillinase-resistant penicillins or cephalosporins preferable. Dicloxacillin or cephalexin (250 mg four times daily for 7 to 10 days) is generally adequate to decrease heavy colonization. As an adjunct, the use of triclosan-containing antibacterial washes and intermittent nasal mupirocin may be useful as prophylactic measures. The role of dietary allergens in atopic dermatitis is controversial, and there is little evidence that they play any role outside of infancy.

Control of pruritus is essential for treatment, since AD often represents "an itch that rashes." Antihistamines are useful to control the pruritus, but sedation may limit their usefulness. Unlike their effects in urticaria, nonsedating antihistamines are of little use since the effectiveness of antihistamines in the treatment of pruritus associated with AD is primarily related to their sedative effects as opposed to any specific action on histamine-mediated pathways.

Treatment with systemic glucocorticoids should be limited to severe exacerbations unresponsive to conservative topical therapy. In the patient with chronic AD, therapy with systemic glucocorticoids will generally clear the skin only briefly, but cessation of the systemic therapy will invariably be accompanied by return, if not worsening, of the dermatitis. Patients who do not respond to conventional therapies should be considered for patch testing to rule out allergic contact dermatitis. Immunotherapy with aeroallergens has not proven useful in AD, unlike its effect in allergic rhinitis and extrinsic asthma.

CONTACT DERMATITIS

Contact dermatitis is an inflammatory process in skin caused by an exogenous agent or agents that directly or indirectly injure the skin. This injury may be caused by an inherent characteristic of a compound -- irritant contact dermatitis (ICD). An example of ICD would be dermatitis induced by a concentrated acid or base. Agents that cause allergic contact dermatitis (ACD) induce an antigen-specific immune response. The clinical lesions of contact dermatitis may be acute (wet and edematous) or chronic (dry, thickened, and scaly), depending on the persistence of the insult (see Plate IIA-8). The most common presentation of contact dermatitis is hand eczema, and it is frequently related to occupational exposures. Occupation-related contact dermatitis represents a significant proportion of occupation-induced injury, affecting over 60,000 persons annually.

[ICD](#) is generally strictly demarcated and often localized to areas of thin skin (eyelids, intertriginous areas) or to areas where the irritant was occluded. Lesions may range from minimal skin erythema to areas of marked edema, vesicles, and ulcers. Chronic low-grade irritant dermatitis is the most common type of ICD and the most common area of involvement is the hands (see below). The most common irritants encountered are chronic wet work, soaps, and detergents. Treatment should be directed to avoidance of irritants and use of protective gloves or clothing.

[ACD](#) is a manifestation of delayed type hypersensitivity mediated by memory T lymphocytes in the skin. The most common cause of ACD is exposure to plants, specifically to members of the family Anacardiaceae; including the genera *Toxicodendron*, *Anacardium*, *Gluta*, *Mangifera*, and *Semecarpus*. Poison ivy, poison oak, and poison sumac are members of the genus *Toxicodendron* and cause an allergic reaction marked by erythema, vesiculation, and severe pruritus. The eruption is often linear, corresponding to areas where plants have touched the skin. However, other allergens may be more difficult to identify, especially if the exposure is chronic and the skin becomes thickened and scaly. The sensitizing antigen common to these plants is urushiol, an oleoresin containing the active ingredient pentadecylcatechol. The oleoresin may adhere to skin, clothing, tools, and pets, and contaminated articles may cause dermatitis even after prolonged storage. Blister fluid does not contain urushiol and is not capable of inducing skin eruption in exposed subjects.

TREATMENT

If [ACD](#) is suspected and an offending agent is identified and removed, the eruption will resolve. Usually, treatment with high-potency fluorinated topical glucocorticoids is enough to relieve symptoms while the ACD runs its course. For those patients who require systemic therapy, a tapering course over 2 to 3 weeks given as single morning doses is the preferred method.

Identification of a contact allergen can be a difficult and time-consuming task. Patients with dermatitis unresponsive to conventional therapy or with an unusual and patterned distribution should be suspected of having [ACD](#). They should be questioned carefully regarding occupational exposures, topical medicaments, and oral medications.

Common sensitizers include preservatives in topical preparations, nickel sulfate, potassium dichromate, thimerosal in ocular preparations, neomycin sulfate, fragrances, formaldehyde, and rubber-curing agents. Patch testing is helpful in identifying these agents, but should not be attempted on patients with widespread active dermatitis or on those taking systemic glucocorticoids.

HAND ECZEMA

Hand eczema is a very common, chronic skin disorder. It represents a large proportion of occupation-associated skin disease. It may be associated with other cutaneous disorders such as atopic dermatitis or may occur by itself. Similar to other forms of dermatitis, both exogenous and endogenous factors play important roles in the expression of hand dermatitis. Chronic, excessive exposure to water and detergents may initiate or aggravate this disorder. It may present with dryness and cracking of the skin of the hands as well as with variable amounts of erythema and edema. Often, the dermatitis will begin under rings where water and irritants are trapped. A variant of hand dermatitis, dyshidrotic eczema, presents with multiple, intensely pruritic, small papules and vesicles occurring on the thenar and hypothenar eminences and the sides of the fingers (see Plate IA-5). Lesions tend to occur in crops that slowly form crusts and heal.

The evaluation of a patient with hand eczema should include an assessment of potential occupation-associated exposures. Predominant involvement of the dorsal surface of the hands with sparing of the palmar surface suggests a possible contact dermatitis. The history should be directed to identifying possible irritant or allergen exposures. The use of rubber gloves to protect dermatitic skin is sometimes associated with the development of delayed type hypersensitivity reactions to agents used for cross-linking rubber. Such reactions can be detected by patch testing. Less commonly, patients may manifest hand dermatitis as a consequence of developing immediate type hypersensitivity reactions to latex. These are of particular concern since these patients are at risk for anaphylactic reactions. The most sensitive method of detection is the use of scratch testing with latex extract. However, this should be done with extreme caution only in a setting where an anaphylactic reaction can be treated. A latex radioallergosorbent test is available but is only about 60% sensitive.

TREATMENT

Therapy of hand dermatitis is directed toward avoidance of irritants, identification of possible contact allergens, treatment of coexistent infection, and application of topical glucocorticoids. Whenever possible, the hands should be protected by gloves, preferably vinyl. Most patients can be treated with cool moist compresses (dressings) to dry and debride acute inflammatory lesions and to decrease swelling, followed by application of a mid- to high-potency topical glucocorticoid in a cream or ointment base. As with atopic dermatitis, treatment of secondary infection by staphylococci or streptococci is essential for good control. Additionally, patients with hand dermatitis should be examined for dermatophyte infection by KOH preparation and culture (see below).

NUMMULAR ECZEMA

Nummular eczema is characterized by circular or oval "coinlike" lesions. Initially, this eruption consists of small edematous papules that become crusted and scaly. The most common locations are on the trunk or the extensor surfaces of the extremities, particularly on the pretibial areas or dorsum of the hands. It occurs more frequently in men and is most commonly seen in middle age. The etiology of nummular eczema is unknown. Whether nummular eczema represents a variant of atopic eczema is controversial. The treatment of nummular eczema is similar to that for other forms of dermatitis.

LICHEN SIMPLEX CHRONICUS

Lichen simplex chronicus may represent the end stage of a variety of pruritic and eczematous disorders. It consists of a well-circumscribed plaque or plaques with lichenified or thickened skin due to chronic scratching or rubbing. Common areas involved include the posterior nuchal region, dorsum of the feet, or ankles. Treatment of lichen simplex chronicus centers around breaking the cycle of chronic itching and scratching, which often occur during sleep. High-potency topical glucocorticoids are helpful in alleviating pruritus in most cases, but in recalcitrant cases, application of topical glucocorticoids under occlusion or intralesional injection of glucocorticoids may be required. Oral antihistamines such as hydroxyzine (10 to 50 mg every 6 h) or tricyclic antidepressants with antihistaminic activity such as doxepin (10 to 25 mg at bedtime) are useful as antipruritics primarily due to their sedating action, and are particularly useful at bedtime (see above). Patients need to be counseled regarding driving or operating heavy equipment after taking these medications due to their potentially potent sedative activity.

ASTEATOTIC ECZEMA

Asteatotic eczema, also known as xerotic eczema or "winter itch," is a mildly inflammatory variant of dermatitis that develops most commonly on the lower legs of elderly individuals during dry times of year. Fine cracks, with or without erythema, characteristically develop on the anterior surface of the lower extremities. Pruritus is variable. Asteatotic eczema responds well to avoidance of irritants, rehydration of the skin, and application of topical emollients.

STASIS DERMATITIS AND STASIS ULCERATION

Stasis dermatitis develops on the lower extremities secondary to venous incompetence and chronic edema. Early findings in stasis dermatitis consist of mild erythema and scaling associated with pruritus. The typical initial site of involvement is the medial aspect of the ankle, often over a distended vein (see [Plate IIA-7](#)). As the disorder progresses, the dermatitis becomes progressively pigmented, due to chronic erythrocyte extravasation leading to cutaneous hemosiderin deposition. As with other forms of dermatitis, stasis dermatitis may become acutely inflamed, with crusting and exudate. Chronic stasis dermatitis is often associated with dermal fibrosis that is recognized clinically as brawny edema of the skin. Stasis dermatitis is often complicated by secondary infection and contact dermatitis. Severe stasis dermatitis may precede the development of stasis ulcers.

TREATMENT

Avoidance of irritants and use of emollients and/or midpotency topical glucocorticoids are the cornerstones of therapy for stasis dermatitis. Control of chronic edema is important to prevent leg ulcers. Patients should be encouraged to elevate the affected extremity when sitting. A compression stocking with a gradient of at least 30 to 40 mmHg is most effective for edema control and is much more effective for preventing chronic edema than is antiembolism hose.

Stasis ulcers are difficult to treat, and resolution of these lesions is slow even under the best of circumstances. It is extremely important to elevate the affected limb as much as possible. The ulcer should be kept clear of necrotic material by gentle debridement and covered with a semipermeable dressing under pressure. Glucocorticoids should not be applied to ulcers, since they may retard healing. Secondarily infected lesions should be treated with appropriate oral antibiotics, but it should be noted that all ulcers will become colonized with bacteria, and the purpose of antibiotic therapy should not be to clear all bacterial growth. Some ulcers may take months to heal or require skin grafting.

SEBORRHEIC DERMATITIS

Seborrheic dermatitis is a common, chronic disorder, characterized by greasy scales overlying erythematous patches or plaques. The most common location is in the scalp where it may be recognized as severe dandruff. On the face, seborrheic dermatitis affects the eyebrows, eyelids, glabella, nasolabial fold, or ears (see [Plate IIA-6](#)). Scaling within the external ear is often mistaken for a chronic fungal infection (otomycosis), and postauricular dermatitis often becomes macerated and tender. Additionally, seborrheic dermatitis may develop in the central chest, axilla, groin, submammary folds, and gluteal cleft. Rarely, it may cause a widespread generalized dermatitis. Seborrheic dermatitis is usually symptomatic, with patients complaining of itching or burning.

Seborrheic dermatitis may be evident within the first few weeks of life, and within this context it occurs in the scalp ("cradle cap"), face, or groin. It is rarely seen in children beyond infancy but becomes evident again during adult life. Although it is frequently seen in patients with Parkinson's disease, in those who have had cerebrovascular accidents, and in those with human immunodeficiency virus (HIV) infection, the overwhelming majority of individuals with seborrheic dermatitis have no underlying disorder.

TREATMENT

Treatment with low-potency topical glucocorticoids in conjunction with shampoos containing coal tar and/or salicylic acid is generally sufficient to control activity of this disorder. High-potency topical glucocorticoid solutions (betamethasone or fluocinonide) are effective for control of scalp involvement. Fluorinated topical glucocorticoids should not be used on the face since this is often associated with the development of rebound worsening and steroid-induced rosacea or atrophy.

PAPULOSQUAMOUS DISORDERS ([Table 56-2](#))

PSORIASIS

Psoriasis is one of the most common dermatologic diseases, affecting up to 1 to 2% of the world's population. It is a chronic inflammatory skin disorder clinically characterized by erythematous, sharply demarcated papules and rounded plaques, covered by silvery micaceous scale. The skin lesions of psoriasis are variably pruritic. Traumatized areas often develop lesions of psoriasis (Koebner or isomorphic phenomenon). Additionally, other external factors may exacerbate psoriasis including infections, stress, and medications (lithium, beta blockers, and antimalarials).

The most common variety of psoriasis is called *plaque type*. Patients with plaque-type psoriasis will have stable, slowly growing plaques, which remain basically unchanged for long periods of time. The most common areas for plaque psoriasis to occur are the elbows, knees, gluteal cleft, and the scalp. Involvement tends to be symmetric. *Inverse psoriasis* affects the intertriginous regions including the axilla, groin, submammary region, and navel, it also tends to affect the scalp, palms, and soles. The individual lesions are sharply demarcated plaques (see [Plate IIA-3](#)) but may be moist due to their location. Plaque psoriasis generally develops slowly and runs an indolent course. It rarely remits spontaneously.

Eruptive psoriasis (guttate psoriasis) is most common in children and young adults. It develops acutely in individuals without psoriasis or in those with chronic plaque psoriasis. Patients present with many small erythematous, scaling papules, frequently after upper respiratory tract infection with *β*-hemolytic streptococci. The differential diagnosis should include pityriasis rosea and secondary syphilis. Patients with psoriasis may also develop pustular lesions. These may be localized to the palms and soles or may be generalized and associated with fever, malaise, diarrhea, and arthralgias.

About half of all patients with psoriasis have fingernail involvement, appearing as punctate pitting, nail thickening, or subungual hyperkeratosis. About 5 to 10% of patients with psoriasis have associated joint complaints, and these are most often found in patients with fingernail involvement. Although some have the coincident occurrence of classic rheumatoid arthritis ([Chap. 312](#)), many have joint disease that falls into one of three types associated with psoriasis: (1) asymmetric inflammatory arthritis most commonly involving the distal and proximal interphalangeal joints and less commonly the knees, hips, ankles, and wrists; (2) a seronegative rheumatoid arthritis-like disease; a significant portion of these patients go on to develop a severe destructive arthritis; or (3) disease limited to the spine (psoriatic spondylitis).

The etiology of psoriasis is still poorly understood. There is clearly a genetic component to psoriasis. Over 50% of patients with psoriasis report a positive family history, and a 65 to 72% concordance among monozygotic twins has been reported in twin studies. Psoriasis has been linked to HLA-Cw6 and, to a lesser extent, to HLA-DR7. Evidence has accumulated clearly indicating a role for T cells in the pathophysiology of psoriasis. Stimulation of immune function with cytokines such as [IL-2](#) has been associated with abrupt worsening of preexisting psoriasis, and bone marrow transplantation has resulted in clearance of disease. Psoriatic lesions are characterized by infiltration of skin with activated memory T cells, with CD8+ cells predominating in the epidermis. Agents that inhibit activated T cell function are often effective for the treatment of severe psoriasis.

Presumably, cytokines from activated T cells elaborate growth factors that stimulate keratinocyte hyperproliferation.

TREATMENT

Treatment of psoriasis depends on the type, location, and extent of disease. All patients should be instructed to avoid excess drying or irritation of their skin and to maintain adequate cutaneous hydration. Most patients with localized plaque-type psoriasis can be managed with midpotency topical glucocorticoids, although their long-term use is often accompanied by loss of effectiveness (tachyphylaxis). Crude coal tar (1 to 5% in an ointment base) is an old but useful method of treatment in conjunction with ultraviolet light therapy. A topical vitamin D analogue (calcipotriol) is also efficacious in the treatment of psoriasis.

Ultraviolet light is an effective therapy for patients with widespread psoriasis. The ultraviolet B (UV-B) spectrum is effective alone, or may be combined with coal tar (Goeckerman regimen) or anthralin (Ingram regimen). Natural sunlight or an artificial light source can be used. The combination of the ultraviolet A (UV-A) spectrum with either oral or topical psoralens (PUVA) is also extremely effective for the treatment of psoriasis, but long-term use may be associated with an increased incidence of squamous cell cancer and melanoma of the skin.

Various other agents can be used for widespread psoriatic disease. Methotrexate is an effective agent, especially in patients with associated psoriatic arthritis. Liver toxicity from long-term use limits its use to patients with widespread disease not responsive to less aggressive modalities. The synthetic retinoid, acetrein, has been shown to be effective in some patients with severe psoriasis but is a potent teratogen, thus limiting its use in women with childbearing potential. The evidence implicating psoriasis as a T cell-mediated disorder has created a new perspective relating to the treatment of psoriasis. Based on this presumed disease mechanism, immunomodulatory therapy utilizing cyclosporine has proven to be highly effective in selected patients with severe, crippling, and potentially life-threatening disease.

LICHEN PLANUS

Lichen planus (LP) is a papulosquamous disorder in which the primary lesions are pruritic, polygonal, flat-topped, violaceous papules. Close examination of the surface of these papules often reveals a network of gray lines (Wickham's striae). The skin lesions may occur anywhere but have a predilection for the wrists, shins, lower back, and genitalia (see [Plate IIA-9](#)). Involvement of the scalp may lead to hair loss. LP commonly involves mucous membranes, particularly the buccal mucosa, where it can present as a white netlike eruption. Its etiology is unknown, but cutaneous eruptions clinically resembling LP have been observed after administration of numerous drugs, including diuretics, gold, antimalarials, penicillamine, and phenothiazines, and in patients with skin lesions of chronic graft-versus-host disease. Additionally, LP associated with abnormal liver function has been correlated with viral hepatitis, particularly hepatitis C infection. The course of LP is variable, but most patients have spontaneous remissions 6 months to 2 years after the onset of disease. Topical glucocorticoids are the mainstay of therapy.

PITYRIASIS ROSEA

Pityriasis rosea (PR) is a papulosquamous eruption of unknown etiology that occurs more commonly in the spring and fall. Its first manifestation is the development of a 2- to 6-cm annular lesion (the herald patch). This is followed in a few days to a few weeks by the appearance of many smaller annular or papular lesions with a predilection to occur on the trunk (see [Plate IIA-13](#)). The lesions are generally oval, with their long axis parallel to the skin-fold lines. Individual lesions may range in color from red to brown and have a trailing scale. PR shares many clinical features with the eruption of secondary syphilis, but palm and sole lesions are extremely rare in PR and common in secondary syphilis. The eruption tends to be moderately pruritic and lasts 3 to 8 weeks. Treatment is generally directed at alleviating pruritus and consists of oral antihistamines, midpotency topical glucocorticoids, and, in some cases, the use of [UV-B](#) phototherapy.

CUTANEOUS INFECTIONS ([Table 56-3](#))

IMPETIGO AND ECTHYMA

Impetigo is a common superficial bacterial infection of skin caused by group Ab-hemolytic streptococci ([Chap. 140](#)) or *S. aureus* ([Chap. 139](#)). The primary lesion is a superficial pustule that ruptures and forms a characteristic yellow-brown honey-colored crust (see [Plate IID-38](#)). Lesions caused by staphylococci may be tense, clear bullae, and this less common form of the disease is called *bullous impetigo*. Lesions may occur on normal skin or in areas already affected by another skin disease. Ecthyma is a variant of impetigo that generally occurs on the lower extremities and causes punched-out ulcerative lesions. Treatment of both ecthyma and impetigo involves gentle debridement of adherent crusts, which is facilitated by the use of soaks and topical antibiotics, in conjunction with appropriate oral antibiotics.

ERYSIPELAS AND CELLULITIS

See [Chap. 128](#)

DERMATOPHYTOSIS

Dermatophytes are fungi that infect skin, hair, and nails and include members of the genera *Trichophyton*, *Microsporum*, and *Epidermophyton*. Infection of the foot (tinea pedis) is most common and is often chronic; it is characterized by variable erythema and edema, scaling, pruritus, and occasionally vesiculation. Involvement may be widespread or localized, but almost invariably the web space between the fourth and fifth toes is affected. Infection of the nails (tinea unguium) occurs in many patients with tinea pedis and is characterized by opacified, thickened nails and subungual debris. The groin is the next most commonly involved area (tinea cruris), with males affected much more often than females. It presents as a scaling erythematous eruption that spares the scrotum. Microscopic examination of either untreated tinea pedis or tinea cruris scale after digestion with KOH preparation will generally demonstrate hyphae.

Dermatophyte infection of the scalp (tinea capitis) has returned in epidemic proportions,

particularly affecting inner city children. The predominant organism is *T. tonsurans*. This organism can produce an inflammatory or relatively noninflammatory infection that may present with either well-defined or irregular, diffuse areas of mild scaling and hair loss. Tinea corporis, or infection on non-hair-bearing skin, may have a variable appearance, depending on the extent of the associated inflammatory reaction (see [Plate IID-51](#)). It may have the typical annular appearance of "ringworm" or appear as deep inflammatory nodules (on the scalp known as a *kerion*) or granulomas. KOH examination of scale or hair from patients with tinea capitis or inflammatory tinea corporis often does not reveal hyphae, and diagnosis may require culture or biopsy.

TREATMENT

Both topical and systemic therapies may be used to treat dermatophyte infection. Treatment depends on the site involved and the type of infection. Topical therapy is generally effective for uncomplicated tinea corporis, tinea cruris, and limited tinea pedis. It is not effective as monotherapy for tinea capitis or tinea unguium. Topical imidazoles (miconazole, ketoconazole, econazole, clotrimazole, oxiconazole, and sulconazole), triazoles (terconazole), and allylamines (terbinafine and naftifine) may all be effective topical therapies for dermatophyte infections. Haloprogin, undecylic acid, ciclopirox-olamine, and tolnaftate are also effective, but nystatin is not active against dermatophytes. Treatment should continue until the patient is clear of infection by clinical examination and culture. Tinea pedis often requires longer treatment courses and is associated with a high relapse rate.

Griseofulvin is the drug of choice for dermatophyte infections requiring systemic therapy. A daily dose of 500 mg of microsize or 350 mg of ultramicrosize griseofulvin administered with a fatty meal is an adequate dose for most dermatophyte infections. The duration of therapy may be as short as 2 weeks for uncomplicated tinea corporis but may be as long as 6 to 12 months for nail infections. The most common side effects of griseofulvin are gastrointestinal distress and headache. Dermatophyte infection of hair-bearing areas (e.g., tinea capitis) requires systemic antifungal therapy. The usual adult dose of griseofulvin is 1 g of microsize or 0.5 g of ultramicrosize given daily, and treatment should be continued for 6 to 8 weeks. Children should be treated with 15 to 20 mg/kg as a single daily dose given with a fatty meal. The adjunctive use of topical antifungal agents in addition to systemic therapy may be useful, but topical therapy alone is not adequate. Markedly inflammatory tinea capitis may result in scarring and hair loss, and systemic or topical glucocorticoids may be helpful in preventing this sequela. Recent studies in children have also suggested that both itraconazole (3 to 5 mg/kg for 6 to 10 weeks) and terbinafine (125 mg/d for 6 weeks) may be effective treatments for tinea capitis.

Until recently, griseofulvin was the recommended therapy for dermatophyte infection of the nails. However, despite prolonged treatment, cure rates were poor. Itraconazole given as either continuous daily therapy (200 mg/d for 3 months) or pulses (200 mg twice daily for 1 week per month for 3 consecutive months) has been shown to be a safe and effective therapy. Itraconazole has the potential for interactions with other drugs requiring the P450 enzyme system for metabolism. Similarly, terbinafine (250 mg/d for 3 months) has shown similar cure rates. Only limited data are available on the dosing and effectiveness of the newer antifungal agents in tinea corporis, tinea cruris,

and uncomplicated tinea pedis.

TINEA VERSICOLOR

Tinea versicolor is caused by a nondermatophyte dimorphic fungus that is a normal inhabitant of the skin. As the yeast form *Pityrosporum orbiculare*, it generally does not cause disease (except for folliculitis in certain individuals). However, in some individuals, it converts to the hyphal form and causes characteristic lesions. The expression of infection is promoted by heat and humidity. The typical lesions consist of oval scaly macules, papules, and patches concentrated on the chest, shoulders, and back but only rarely on the face or distal extremities. On dark skin, they often appear as hypopigmented areas, while on light skin, they are slightly hyperpigmented. In some darkly pigmented individuals, they may only appear as scaling patches. A KOH preparation from scaling lesions will demonstrate a confluence of short hyphae and round spores (so-called spaghetti and meatballs). Solutions containing sulfur, salicylic acid, or selenium sulfide will clear the infection if used daily for a week and then intermittently thereafter. Treatment with a single 400-mg dose of ketoconazole is also effective.

CANDIDIASIS

Candidiasis is a fungal infection caused by a related group of yeasts, whose manifestations may be localized to the skin, or rarely, may be systemic and life-threatening. The causative organism is usually *Candida albicans*, but may also be *C. tropicalis*, *C. parapsilosis*, or *C. krusei*. These organisms are normal saprophytic inhabitants of the gastrointestinal tract but may overgrow (usually due to broad-spectrum antibiotic therapy) and cause disease at a number of cutaneous sites. Other predisposing factors include diabetes mellitus, chronic intertrigo, oral contraceptive use, and cellular immune deficiency. Candidiasis is a very common infection in HIV-infected individuals ([Chap. 309](#)). The oral cavity is commonly involved. Lesions may occur on the tongue or buccal mucosa (thrush) and appear as white plaques (see [Plate IID-43](#)). Microscopic examination of scrapings demonstrate both pseudohyphae and yeast forms. Fissured, macerated lesions at the corners of the mouth (perleche) are often seen in individuals with poorly fitting dentures and may also be associated with candidal infection. Additionally, candidal infections have an affinity for sites that are chronically wet and macerated and may occur around nails (onycholysis and paronychia) and in intertriginous areas. Intertriginous lesions are characteristically edematous, erythematous, and scaly, with scattered "satellite pustules." In males, there is often involvement of the penis and scrotum as well as the inner aspect of the thighs. In contrast to dermatophyte infections, candidal infections are frequently accompanied by a marked inflammatory response. Diagnosis of candidal infection is based upon the clinical pattern and demonstration of yeast on KOH preparation, or culture.

TREATMENT

Treatment routinely involves removing any predisposing factors such as antibiotic therapy or chronic wetness and the use of appropriate topical or systemic antifungal therapy. Effective topical agents include nystatin or topical azoles (miconazole,

clotrimazole, econazole, or ketoconazole). These agents are generally effective in clearing mucous membrane or glabrous skin involvement in nonimmunosuppressed patients. The associated inflammatory response that often accompanies candidal infection on glabrous skin should be treated with a mild glucocorticoid lotion or cream (2.5% hydrocortisone). Systemic therapy is generally reserved for immunosuppressed patients or individuals with chronic or recurrent disease who fail to respond to or tolerate appropriate topical therapy. Vulvovaginal candidiasis may respond to treatment with a single dose of fluconazole (150 mg). Chronic recurrent oral or vaginal candidiasis may be treated with weekly to monthly oral fluconazole (150 to 200 mg) in conjunction with topical therapy.

WARTS

Warts are cutaneous neoplasms that are caused by papilloma viruses. Over 50 different human papilloma viruses (HPV) have been described, and this number will almost certainly continue to grow. Typical verruca vulgaris lesions are sessile, dome-shaped, usually about a centimeter in diameter, and their surface is made up of many small filamentous projections. The HPV that cause typical verruca vulgaris also cause typical plantar warts, flat warts (or verruca plana), and filiform warts in intertriginous areas. Plantar warts are endophytic and are covered by thick keratin. Paring of the wart will generally demonstrate a central core of keratinized debris and punctate bleeding points. Filiform warts are most commonly seen on the face, neck, and skin folds and present as papillomatous lesions on a narrow base. Flat warts are only slightly elevated and have a velvety, nonverrucous surface. They have a propensity for the face, arms, and legs and are often spread by shaving.

Multiple [HPV](#) types have been associated with genital tract lesions. They generally begin as small papillomas that may grow to form large fungating lesions. In women, they may involve either the labia, perineum, or perianal skin. Additionally, the mucosa of the vagina, urethra, and anus can be involved, as well as the cervical epithelium. In men, the lesions often occur initially in the coronal sulcus, but may be seen on the shaft of the penis, the scrotum, perianal skin, or in the urethra.

Within the past decade, appreciable evidence has accumulated that suggests [HPV](#) plays a role in the development of neoplasia of the uterine cervix and external genitalia ([Chap. 97](#)). HPV types 16 and 18 have been most intensely studied, while recent evidence also implicates other types. Lesions may initially appear as small, flat, velvety, hyperpigmented papules occurring on the genitalia or perianal skin. Histologic examination of biopsies from affected sites may reveal changes associated with typical warts and/or features typical of intraepidermal carcinoma (Bowen's disease). Squamous cell carcinomas associated with HPV infections have also been observed in extragenital skin ([Chap. 86](#)). This is most commonly seen in patients immunosuppressed after organ transplantation.

TREATMENT

There are many modalities available to treat warts, but no single therapy is universally effective. Factors that influence the choice of therapy include the location of the wart, extent of disease, the age and immunologic status of the patient, and the patient's

desire for therapy. Perhaps the most useful and convenient method for treating warts in almost any location is cryotherapy with liquid nitrogen. Equally effective, but requiring much more patient compliance, is the use of keratolytic agents such as salicylic acid plasters or combinations of lactic acid and salicylic acid. For genital warts, application of podophyllin solution is moderately effective but may be associated with marked local reactions in certain individuals. Dilute preparations of purified podophyllin permit physician-directed by patient-applied use, facilitating treatment of mucosal warts. Topical imiquimod, a potent inducer of local cytokine release, has also been approved for use in genital warts. Other topical agents that are used include trichloroacetic acid or cantharidin. Electrodesiccation and curettage or CO₂ laser excision are also effective therapies but require local anesthesia. Recurrence of warts appears to be common to all these modalities because viral genomic material is present in normal-appearing skin adjacent to the clinical lesions.

Treatment of warts should be tempered by the observation that an overwhelming majority of warts in normal individuals resolve spontaneously within 1 to 2 years. Also, only an extremely small proportion of warts is associated with neoplasia, and those are almost exclusively located on the genitalia or perianal skin.

HERPES SIMPLEX

[*See Chap. 182](#)

HERPES ZOSTER

[*See Chap. 183](#)

ACNE

ACNE VULGARIS

Acne vulgaris is a self-limited disorder primarily of teenagers and young adults, although perhaps 10 to 20% of adults may continue to experience some form of the disorder. The permissive factor for the expression of the disease in adolescence is the increase in sebum release by sebaceous glands after puberty. Small cysts, called *comedones*, form in hair follicles due to blockage of the follicular orifice by retention of sebum and keratinous material. The activity of lipophilic yeast (*Pityrosporum orbiculare*) and bacteria (*Propionibacterium acnes*) within the comedones releases free fatty acids from sebum, causes inflammation within the cyst, and results in rupture of the cyst wall. An inflammatory foreign-body reaction develops as a result of extrusion of oily and keratinous debris from the cyst.

The clinical hallmark of acne vulgaris is the comedone, which may be closed (whitehead) or open (blackhead). Closed comedones appear as 1- to 2-mm pebbly white papules, which are accentuated when the skin is stretched. They are the precursors of inflammatory lesions of acne vulgaris. The contents of closed comedones are not easily expressed. Open comedones, which rarely result in inflammatory acne lesions, have a large dilated follicular orifice and are filled with easily expressible oxidized, darkened, oily debris. Comedones are usually accompanied by inflammatory

lesions: papules, pustules, or nodules.

The earliest lesions seen in early adolescence are generally mildly inflamed or noninflammatory comedones on the forehead. Subsequently, more typical inflammatory lesions develop on the cheeks, nose, and chin (see [Plate IIA-1](#)). The most common location for acne is the face, but involvement of the chest and back is not uncommon. Most disease remains mild and does not lead to scarring. However, a small number of patients develop large inflammatory cysts and nodules, which may drain and result in significant scarring.

Exogenous and endogenous factors can alter the expression of acne vulgaris. Friction and trauma may rupture preexisting microcomedones and elicit inflammatory acne lesions. This is commonly seen with headbands or chin straps of athletic helmets. Application of comedogenic topical agents in cosmetics or hair preparations or chronic topical exposure to certain industrial compounds that are comedogenic may elicit or aggravate acne. Glucocorticoids, applied topically or administered systemically in high doses, may also elicit acne. Other systemic medications such as lithium, isoniazid, halogens, phenytoin, and phenobarbital may produce acneiform eruptions, or aggravate preexisting acne.

TREATMENT

Treatment of acne vulgaris is directed toward elimination of comedones by normalization of follicular keratinization, decreasing sebaceous gland activity, decreasing the population of lipophilic bacteria and yeast, and decreasing inflammation. Acne vulgaris may be treated with either local or systemic medications. Minimal to moderate, pauci-inflammatory disease may respond adequately to local therapy alone. Although areas affected with acne should be kept clean, there is little evidence to suggest that removal of surface oils plays an important role in therapy. Overly vigorous scrubbing may aggravate acne due to mechanical rupture of comedones. Topical agents such as retinoic acid, benzoyl peroxide, or salicylic acid may alter the pattern of epidermal desquamation, preventing the formation of comedones and aiding in the resolution of preexisting cysts. Topical antibacterial agents such as benzoyl peroxide, azelaic acid, topical erythromycin (with or without zinc), clindamycin, or tetracycline are also useful adjuncts to therapy.

Patients with moderate to severe acne with a prominent inflammatory component will benefit from the addition of systemic therapy. Oral tetracyclines or erythromycin in doses of 250 to 1000 mg/d will decrease follicular colonization with some of the lipophilic organisms. They also appear to have an anti-inflammatory effect independent of their antibacterial effect. Female patients who do not respond to oral antibiotics may benefit from hormonal therapy. Women placed on oral contraceptives containing ethinyl estradiol and norgestimate have demonstrated improvement in their acne when compared to a placebo control.

Severe nodulocystic acne not responsive to oral antibiotics, hormonal therapy, or topical therapy may be treated with the synthetic retinoid isotretinoin. It is used at doses of 0.5 to 2.0 mg/kg as a single daily dose for 15 to 20 weeks. The use of this drug is limited by its teratogenicity, and female patients must be screened for pregnancy prior to initiating

therapy, maintain a method of birth control during therapy, and be screened for pregnancy during treatment. Patients receiving this medication develop extremely dry skin and cheilitis and must be followed for development of hypertriglyceridemia.

ACNE ROSACEA

Acne rosacea is an inflammatory disorder predominantly affecting the central face. It is seen almost exclusively in adults, only rarely affecting patients under 30 years of age. Rosacea is seen more often in women, but those most severely affected are men. It is characterized by the presence of erythema, telangiectases, and superficial pustules (see [Plate IIA-2](#)), but is not associated with the presence of comedones. Rosacea only rarely involves the chest or back.

There is a relationship between the tendency for pronounced facial flushing and the subsequent development of acne rosacea. Often, individuals with rosacea initially demonstrate a pronounced flushing reaction. This may be in response to heat, emotional stimuli, alcohol, hot drinks, or spicy foods. As the disease progresses, the flush persists longer and longer and may eventually become permanent. Papules, pustules, and telangiectases can become superimposed on the persistent flush. Rosacea of very long standing may lead to connective tissue overgrowth, particularly of the nose (rhinophyma). Rosacea may also be complicated by various inflammatory disorders of the eye, including keratitis, blepharitis, iritis, and recurrent chalazion. These ocular problems are potentially sight-threatening and warrant ophthalmologic evaluation.

TREATMENT

Acne rosacea can generally be treated effectively with oral tetracycline in doses ranging from 250 to 1000 mg/d. Topical metronidazole or sodium sulfacetamide has also been shown to be effective. In addition, the use of low-potency, nonfluorinated topical glucocorticoids, particularly after cool soaks, is helpful in alleviating facial erythema. Fluorinated topical glucocorticoids should be avoided since chronic use of these preparations may actually elicit rosacea. Topical therapy is not effective treatment for ocular disease.

(Bibliography omitted in Palm version)

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57. SKIN MANIFESTATIONS OF INTERNAL DISEASE - Jean L. Bologna, Irwin M. Braverman

It is now a generally accepted concept in medicine that the skin can show signs of internal disease. Therefore, in textbooks of medicine one finds a chapter describing in detail the major systemic disorders that can be identified by cutaneous signs. The underlying assumption of such a chapter is that the clinician has been able to identify the disorder in the patient and needs only to read about it in the textbook. In reality, concise differential diagnoses and the identification of these disorders are actually difficult for the nondermatologist because he or she is not well versed in the recognition of cutaneous lesions or their spectrum of presentations. Therefore, the authors of this chapter have decided to cover this particular topic of cutaneous medicine not by discussing individual disorders but by describing and discussing the various presenting clinical signs and symptoms that indicate the presence of these disorders. Concise differential diagnoses will be generated in which the significant diseases will be briefly discussed and distinguished from the more common disorders that have no significance for internal diseases. The latter disorders are reviewed in table form and always need to be excluded when considering the former. For a detailed description of individual diseases, the reader should consult a dermatologic text.

PAPULOSQUAMOUS SKIN LESIONS ([Table 57-1](#))

When an eruption is characterized by elevated lesions, papules (<1 cm) or plaques (>1 cm), in association with scale, it is referred to as *papulosquamous*. The most common papulosquamous diseases -- *psoriasis*, *tinea*, *pityriasis rosea*, and *lichen planus* -- are primary cutaneous disorders ([Chap. 56](#)). When psoriatic lesions are accompanied by arthritis, the possibility of psoriatic arthritis or *Reiter's disease* should be considered. A history of oral ulcers, conjunctivitis, uveitis, and/or urethritis points to the latter diagnosis. In *guttate psoriasis* there is an acute onset of small, widely scattered, uniform lesions, often in association with a streptococcal infection. Lithium, beta blockers, HIV infection, and a rapid taper of systemic glucocorticoids are also known to exacerbate psoriasis.

Whenever the diagnosis of pityriasis rosea or lichen planus is made, it is important to review the patient's medications because the eruption can be treated by simply discontinuing the offending agent. Pityriasis rosea-like drug eruptions are seen most commonly with beta blockers, angiotensin-converting enzyme (ACE) inhibitors, gold, and metronidazole, while the drugs that can produce a lichenoid eruption include gold, antimalarials, thiazides, quinidine, phenothiazines, sulfonyleureas, and ACE inhibitors. Lichen planus-like lesions are also observed in chronic graft-versus-host disease.

Parapsoriasis is an intermediate disease, for it can remain solely as a primary cutaneous disease or it can progress to cutaneous T cell lymphoma (CTCL) after a latency period of as long as 40 years. There are several forms of parapsoriasis, including small plaque (0.5 to 5 cm), large plaque (>6 cm), and retiform. The lesions of both small plaque and large plaque parapsoriasis are thin and salmon-pink in color with fine white scale. In small plaque forms, the lesions are commonly on the trunk but can be widely scattered. In large plaque forms, the most common location is the "girdle" area, and fine wrinkling secondary to epidermal atrophy is often seen. Retiform

parapsoriasis forms a netlike pattern, and the individual papules are red-brown and flat-topped. The latter two forms of parapsoriasis, large plaque and retiform, can progress to CTCL.

A clue to the development of [CTCL](#) within lesions of large plaque or retiform parapsoriasis is an increase in the palpable component of the plaque (increased infiltration). In its early stages, CTCL may be confused with eczema or psoriasis, but it often fails to respond to the appropriate therapy for those inflammatory diseases. The diagnosis of CTCL is established by skin biopsy in which collections of atypical T lymphocytes are found in the epidermis and dermis. As the disease progresses, cutaneous tumors and lymph node involvement may appear.

In *secondary syphilis* there are scattered red-brown papules with thin scale. The eruption often involves the palms and soles and can resemble pityriasis rosea. Associated findings are helpful in making the diagnosis and include annular plaques on the face, nonscarring alopecia, condyloma lata (broad-based and moist), and mucous patches as well as lymphadenopathy, malaise, fever, headache, and myalgias. The interval between the primary chancre and the secondary stage is usually 4 to 8 weeks, and spontaneous resolution without appropriate therapy is seen.

ERYTHRODERMA ([Table 57-2](#))

Erythroderma is the term used when the majority of the skin surface is erythematous (red in color). There may be associated scale, erosions, or pustules as well as shedding of the hair and nails. Potential systemic manifestations include fever, chills, hypothermia, reactive lymphadenopathy, peripheral edema, hypoalbuminemia, and high-output cardiac failure. The major etiologies of erythroderma are (1) *cutaneous diseases* such as psoriasis and dermatitis ([Table 57-3](#)); (2) *drugs*; (3) *systemic diseases*, most commonly [CTCL](#); and (4) *idiopathic*. In the first three groups, the location and description of the initial lesions, prior to the development of the erythroderma, aid in the diagnosis. For example, a history of red scaly plaques on the elbows and knees would point to psoriasis. It is also important to examine the skin carefully for a migration of the erythema and associated secondary changes such as pustules or erosions. Migratory waves of erythema studded with superficial pustules are seen in *pustular psoriasis*.

Drug-induced erythroderma (exfoliative dermatitis) may begin as a morbilliform eruption ([Chap. 59](#)) or may arise as diffuse erythema. Fever and peripheral eosinophilia often accompany the eruption, and occasionally there is an associated allergic interstitial nephritis. A number of drugs can produce an erythroderma, including penicillins, sulfonamides, carbamazepine, phenytoin, gold, allopurinol, and captopril. While reactions to anticonvulsants can lead to a pseudolymphoma syndrome, reactions to allopurinol may be accompanied by hepatitis, gastrointestinal bleeding, and nephropathy.

The most common malignancy that is associated with erythroderma is [CTCL](#); in some series, up to 25% of the cases of erythroderma were due to CTCL. The patient may progress from isolated plaques and tumors, but more commonly the erythroderma is present throughout the course of the disease (Sezary syndrome). In the Sezary

syndrome, there are circulating atypical T lymphocytes, pruritus, and lymphadenopathy. In cases of erythroderma where there is no apparent cause (idiopathic), longitudinal follow-up is mandatory to monitor for the possible development of CTCL. There have been isolated case reports of erythroderma secondary to some solid tumors -- lung, liver, prostate, thyroid, and colon -- but it is usually in a late stage of the disease.

ALOPECIA ([Table 57-4](#))

The two major forms of alopecia are scarring and nonscarring. In *scarring alopecia* there is associated fibrosis, inflammation, and loss of hair follicles. A smooth scalp with a decreased number of follicular openings is usually observed clinically, but in some cases the changes are seen only in biopsy specimens from the affected areas. In *nonscarring alopecia* the hair shafts are gone, but the hair follicles are preserved, explaining the reversible nature of nonscarring alopecia.

Primary cutaneous disorders are the most common causes of nonscarring alopecia and they include *telogen effluvium*, *androgenetic alopecia*, *alopecia areata*, *tinea capitis*, and *traumatic alopecia* ([Table 57-5](#)). In women with androgenetic alopecia, an elevation in circulating levels of androgens may be seen as a result of ovarian or adrenal gland dysfunction. When there are signs of virilization, such as a deepened voice and enlarged clitoris, the possibility of an ovarian or adrenal gland tumor should be considered.

Exposure to various *drugs* can also cause diffuse hair loss, usually by inducing a telogen effluvium. An exception is the anagen effluvium observed with antimitotic agents such as daunorubicin. Alopecia is a side effect of the following drugs: warfarin, heparin, propylthiouracil, carbimazole, vitamin A, isotretinoin, acetrein, lithium, beta blockers, colchicine, and amphetamines. Fortunately, spontaneous regrowth usually follows discontinuation of the offending agent.

Less commonly, nonscarring alopecia is associated with *lupus erythematosus* and *secondary syphilis*. In systemic lupus there are two forms of alopecia -- one is scarring secondary to discoid lesions (see below) and the other is nonscarring. The latter form may be diffuse and involve the entire scalp, or it may be localized to the frontal scalp and result in multiple short hairs ("lupus hairs"). Scattered, poorly circumscribed patches of alopecia with a "moth-eaten" appearance are a manifestation of the secondary stage of syphilis. Diffuse thinning of the hair is also associated with hypothyroidism, hyperthyroidism, and HIV infection ([Table 57-4](#)).

Scarring alopecia is more frequently the result of a primary cutaneous disorder such as *lichen planus*, *folliculitis decalvans*, *cutaneous lupus*, or *linear scleroderma (morphea)* than it is a sign of systemic disease. Although the scarring lesions of *discoid lupus* can be seen in patients with systemic lupus, in the majority of cases the disease process is limited to the skin. Less common causes of scarring alopecia include *sarcoidosis* (see "Papulonodular Skin Lesions") and cutaneous *metastases*.

In the early phases of discoid lupus, lichen planus, and folliculitis decalvans, there are circumscribed areas of alopecia. Fibrosis and subsequent loss of follicles are observed primarily in the center of the individual lesions, while the inflammatory process is most

prominent at the periphery. The areas of active inflammation in discoid lupus are erythematous with scale, whereas the areas of previous inflammation are often hypopigmented with a rim of hyperpigmentation. In lichen planus the peripheral perifollicular macules are usually violet-colored. Complete examination of the skin and oral mucosa combined with a biopsy and direct immunofluorescence microscopy will aid in distinguishing these two entities. The peripheral active lesions in folliculitis decalvans are perifollicular pustules; these patients can develop a reactive arthritis.

FIGURATE SKIN LESIONS ([Table 57-6](#))

In *figurate* eruptions, the lesions form rings and arcs that are usually erythematous but can be flesh-colored to brown. Most commonly, they are due to primary cutaneous diseases such as *tinea*, *urticaria*, *erythema annulare centrifugum*, and *granuloma annulare* ([Chaps. 56](#) and [58](#)). An underlying systemic illness is found in a second, less common group of migratory annular erythemas. It includes *erythema gyratum repens*, *erythema migrans*, *erythema marginatum*, and *necrolytic migratory erythema*.

In *erythema gyratum repens*, one sees hundreds of mobile concentric arcs and wavefronts that resemble the grain in wood. A search for an underlying malignancy is mandatory in a patient with this eruption. *Erythema migrans* is the cutaneous manifestation of Lyme disease, which is caused by the spirochete *Borrelia burgdorferi*. In the initial stage (3 to 30 days after tick bite), a single annular lesion is usually seen, which can expand to 3–10 cm in diameter. Within several days, approximately half the patients develop multiple smaller erythematous lesions at sites distant from the bite. Associated symptoms include fever, headache, photophobia, myalgias, arthralgias, and malar rash. *Erythema marginatum* is seen in patients with rheumatic fever, primarily on the trunk. Lesions are pink-red in color, flat to mildly elevated, and transient.

There are additional cutaneous diseases that present as annular eruptions but lack an obvious migratory component. Examples include [CTCL](#), *annular cutaneous lupus*, also referred to as *subacute lupus*, *secondary syphilis*, and *sarcoidosis* (see "Papulonodular Skin Lesions").

ACNE ([Table 57-7](#))

Acne vulgaris and *acne rosacea* are the two major forms of acne ([Chap. 56](#)). Estrogens decrease sebaceous gland activity, whereas androgens enhance sebum production. Therefore, acne vulgaris in an adult, especially if it is of recent onset, may be a reflection of increased levels of circulating *androgens*. Dysfunction of the ovary or adrenal gland, e.g., polycystic ovary disease or Cushing's syndrome, can lead to the hormonal imbalance. Examination of the patient for signs such as hirsutism, androgenetic alopecia, hypertension, and redistribution of subcutaneous fat will aid in the diagnosis.

Exacerbations of acne vulgaris follow the ingestion of several *drugs*, such as anabolic steroids, glucocorticoids, lithium, and iodides as well as the application of oil-containing compounds. Acne-like lesions can be seen in patients with Behcet's disease (see "Ulcers"), and in immunocompromised hosts, disseminated fungal infections (e.g., cryptococcosis) may present as an acneiform eruption.

Patients with the carcinoid syndrome have episodes of flushing of the head, neck, and sometimes the trunk. Resultant skin changes of the face, in particular telangiectasias, may mimic the clinical appearance of acne rosacea.

PUSTULAR LESIONS

Acneiform eruptions (see "Acne") and *folliculitis* represent the most common pustular dermatoses. An important consideration in the evaluation of perifollicular pustules is a determination of the associated pathogen, e.g., normal flora, *Staphylococcus aureus*, *Pityrosporum*. Noninfectious forms of folliculitis include HIV-associated eosinophilic folliculitis and folliculitis secondary to drugs such as glucocorticoids and lithium. Administration of high-dose oral glucocorticoids can result in a widespread eruption of perifollicular pustules on the trunk, characterized by lesions in the same stage of development. With regard to underlying systemic diseases, pustules are a characteristic component of pustular psoriasis and can be seen in septic emboli of bacterial or fungal origin (see "Purpura").

TELANGIECTASIAS ([Table 57-8](#))

In order to distinguish the various types of telangiectasias, it is important to examine the shape and configuration of the dilated blood vessels. *Linear telangiectasias* are seen on the face of patients with *actinically damaged skin* and *acne rosacea* and they are found on the legs of patients with *venous hypertension* and *essential telangiectasia*. Patients with an unusual form of *mastocytosis* (telangiectasia macularis eruptiva perstans), the *carcinoid* syndrome (see "Acne"), and *ataxia-telangiectasia* also have linear telangiectasias. In ataxia-telangiectasia, linear telangiectasias appear on the bulbar conjunctiva during childhood. Eventually, there is involvement of the ears, eyelids, cheeks, and/or flexural areas such as the antecubital and popliteal fossae. Lastly, linear telangiectasias are found in areas of cutaneous inflammation. For example, lesions of discoid lupus frequently have telangiectasias within them.

Poikiloderma is a term used to describe a patch of skin with (1) reticulated hypo- and hyperpigmentation, (2) wrinkling secondary to epidermal atrophy, and (3) telangiectasias. Poikiloderma does not imply a single disease entity -- it is seen in skin damaged by *ionizing radiation*, in the disorders *poikiloderma vasculare atrophicans* (PVA) and *xeroderma pigmentosum*, as well as in patients with connective-tissue diseases, primarily *dermatomyositis* (DM). [PVA](#) is a precursor lesion of [CTCL](#), and the areas of poikiloderma usually begin in the flexural areas of the axillae and groin.

In *scleroderma*, the dilated blood vessels have a unique configuration and are known as *mat telangiectasias*. The lesions are broad macules that usually measure 2 to 7 mm in diameter but occasionally are larger. Mats have a polygonal or oval shape, and their erythematous color may be uniform or the result of delicate telangiectasias. The most common locations for mat telangiectasias are the face, oral mucosa, and hands -- peripheral sites that are prone to intermittent ischemia. The CREST (calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia) variant of scleroderma ([Chap. 313](#)) is associated with a chronic course and anticentromere antibodies. Mat telangiectasias are an important clue to the diagnosis of

the CREST syndrome as well as systemic scleroderma, for they may be the only cutaneous finding.

Periungual telangiectasias are pathognomonic signs of the three major connective tissue diseases -- *lupus erythematosus*, *scleroderma*, and [DM](#). They are easily visualized by the naked eye and occur in at least two-thirds of these patients. In both DM and lupus there is associated nailfold erythema, and in DM the erythema is often accompanied by "ragged" cuticles and fingertip tenderness. Under 10^x magnification, the blood vessels in the nailfolds of lupus patients are tortuous and resemble "glomeruli," whereas in scleroderma and DM there is a loss of capillary loops and those that remain are markedly dilated.

In *hereditary hemorrhagic telangiectasia* (Osler-Rendu-Weber disease), the lesions usually appear during adulthood and are most commonly seen on the mucous membranes, face, and distal extremities, including under the nails. They represent arteriovenous (AV) malformations of the dermal microvasculature, are dark red in color, and are usually slightly elevated. When the skin is stretched over an individual lesion, an eccentric punctum with radiating legs is seen. Although the degree of systemic involvement varies in this autosomal dominant disease (due to mutations in either the endoglin or activin receptor-like kinase gene), the major symptoms are recurrent epistaxis and gastrointestinal bleeding. The fact that these mucosal telangiectasias are actually AV communications helps to explain their tendency to bleed.

HYPOPIGMENTATION ([Table 57-9](#))

Disorders of hypopigmentation are classified as either diffuse or localized. The classic example of *diffuse hypopigmentation* is *oculocutaneous albinism* (OCA). The most common forms are due to mutations in the tyrosinase gene (type I) or the *P* gene (type II); patients with type IA OCA have a total lack of enzyme activity. At birth, different forms of OCA can appear similar -- white hair, gray-blue eyes, and pink-white skin. However, the patients with no tyrosinase activity maintain this phenotype, whereas those with decreased activity or *P* gene mutations will acquire some pigmentation of the eyes, hair, and skin as they age. The degree of pigment formation is also a function of racial background, and the pigmentary dilution is readily apparent when patients are compared to their first-degree relatives.

The ocular findings in [OCA](#) correlate with the degree of hypopigmentation and include decreased visual acuity, nystagmus, photophobia, and monocular vision. Generalized vitiligo, phenylketonuria, and homocystinuria are other unusual causes of diffuse pigmentary dilution. In generalized vitiligo, melanocytes are not found in affected skin, whereas in OCA they are present but have decreased activity. Appropriate laboratory tests exclude the other disorders of metabolism.

The differential diagnosis of *localized hypomelanosis* includes the following primary cutaneous disorders: *idiopathic guttate hypomelanosis*, *postinflammatory hypopigmentation*, *tinea (pityriasis) versicolor*, *vitiligo*, *chemical leukoderma*, *nevus depigmentosus* (see below), and *piebaldism* ([Table 57-9](#)). In this group of diseases, the areas of involvement are macules or patches with a decrease or absence of pigmentation. Patients with vitiligo also have an increased incidence of several

autoimmune disorders, including hypothyroidism, Graves' disease, pernicious anemia, Addison's disease, uveitis, alopecia areata, chronic mucocutaneous candidiasis, and the polyglandular autoimmune syndromes (types I and II). Diseases of the thyroid gland are the most frequently associated disorders, occurring in up to 30% of patients with vitiligo. Circulating autoantibodies are often found, and the most common ones are antithyroglobulin, antimicrosomal, and antiparietal cell antibodies.

There are three systemic diseases that should be considered in a patient with skin findings suggestive of vitiligo -- *Vogt-Koyanagi-Harada syndrome*, *scleroderma*, and *melanoma-associated leukoderma*. A history of aseptic meningitis, nontraumatic uveitis, tinnitus, hearing loss, and/or dysacusis points to the diagnosis of the Vogt-Koyanagi-Harada syndrome. In these patients, the face and scalp are the most common locations of pigment loss. The vitiligo-like leukoderma seen in patients with scleroderma has a clinical resemblance to idiopathic vitiligo that has begun to repigment as a result of treatment; that is, perifollicular macules of normal pigmentation are seen within areas of depigmentation. The basis of this leukoderma is unknown; there is no evidence of inflammation in areas of involvement, but it can resolve if the underlying connective tissue disease becomes inactive. In contrast to idiopathic vitiligo, melanoma-associated leukoderma often begins on the trunk, and its appearance should prompt a search for metastatic disease. The possibility exists that the destruction of normal melanocytes is the result of an immune response against malignant melanocytes.

There are two systemic disorders that may have the cutaneous findings of piebaldism ([Table 57-10](#)). They are *Hirschsprung's disease* and *Waardenburg's syndrome*. A possible explanation for both disorders is an abnormal embryonic migration or survival of two neural crest-derived elements, one of them being melanocytes and the other myenteric ganglion cells (Hirschsprung's disease) or auditory nerve cells (Waardenburg's syndrome). The latter syndrome is characterized by congenital sensorineural hearing loss, dystopia canthorum (lateral displacement of the inner canthi but normal interpupillary distance), heterochromic irises, and a broad nasal root, in addition to the piebaldism. Patients with Waardenburg's syndrome have been shown to have mutations in two genes that encode DNA-binding proteins, *PAX-3* and *MITF*, while patients with Hirschsprung's disease and white spotting have mutations in one of three genes -- endothelin 3, endothelin B receptor, and *SOX-10*.

In *tuberous sclerosis*, the earliest cutaneous sign is an ash leaf spot. These lesions are often present at birth and are usually multiple; however, detection may require Wood's lamp examination, especially in fair-skinned individuals. The pigment within them is reduced but not absent. The average size is 1 to 3 cm, and the common shapes are polygonal and lance-ovate. Examination of the patient for additional cutaneous signs such as adenoma sebaceum (multiple angiofibromas of the face), ungual and gingival fibromas, fibrous plaques of the forehead, and connective tissue nevi (shagreen patches) is recommended. It is important to remember that an ash leaf spot on the scalp will result in *poliosis*, which is a circumscribed patch of gray-white hair. Internal manifestations include seizures, mental retardation, central nervous system (CNS) and retinal hamartomas, renal angiomyolipomas, and cardiac rhabdomyomas. The latter can be detected in up to 60% of children (<18 years) with tuberous sclerosis by echocardiography.

Nevus depigmentosus is a stable, well-circumscribed hypomelanosis that is present at birth. There is usually a single circular or rectangular lesion, but occasionally the nevus has a dermatomal or whorled pattern. It is important to distinguish this more common entity from ash leaf spots especially when there are multiple lesions. In *hypomelanosis of Ito*, swirls and streaks of hypopigmentation run parallel to one another in a pattern that resembles a marble cake. Lesions may progress or regress with time, and in up to a third of patients, associated abnormalities are found including in the musculoskeletal system (asymmetry), the [CNS](#) (seizures and mental retardation), and the eyes (strabismus and hypertelorism). Chromosomal mosaicism has been detected in these patients; this lends support to the hypothesis that the pattern is the result of the migration of two clones of primordial melanocytes, each with a different pigment potential.

Localized areas of decreased pigmentation are commonly seen as a result of cutaneous inflammation ([Table 57-10](#)) and have been observed in the skin overlying active lesions of sarcoidosis (see "Papulonodular Skin Lesions") as well as in [CTCL](#). Cutaneous infections also present as disorders of hypopigmentation, and in *tuberculoid leprosy* there are a few asymmetric patches of hypomelanosis that have associated anesthesia, anhidrosis, and alopecia. Biopsy specimens of the palpable border show dermal granulomas that lack *Mycobacterium leprae* organisms.

HYPERPIGMENTATION ([Table 57-11](#))

Disorders of hyperpigmentation are also divided into two groups -- localized and diffuse. The *localized* forms are due to an epidermal alteration, a proliferation of melanocytes, or an increase in pigment production. Both seborrheic keratoses and acanthosis nigricans belong to the first group. *Seborrheic keratoses* are common lesions, but in one clinical setting they are a sign of systemic disease, and that setting is the sudden appearance of multiple lesions, often with an inflammatory base and in association with acrochordons (skin tags) and acanthosis nigricans. This is termed the *sign of Leser-Trelat* and signifies an internal malignancy. *Acanthosis nigricans* can also be a reflection of an internal malignancy, most commonly of the gastrointestinal tract, and it appears as velvety hyperpigmentation, primarily in flexural areas. In the majority of patients, acanthosis nigricans is associated with obesity, but it may be a reflection of an endocrinopathy such as acromegaly, Cushing's syndrome, the Stein-Leventhal syndrome, or insulin-resistant diabetes mellitus (type A, type B, and lipodystrophic forms).

A proliferation of melanocytes results in the following pigmented lesions: *lentigo*, *melanocytic nevus*, and *melanoma* ([Chap. 86](#)). In an adult, the majority of lentigines are related to sun exposure, which explains their distribution. However, in the Peutz-Jeghers and LEOPARD [lentigines; ECG abnormalities, primarily conduction defects, ocular hypertelorism; pulmonary stenosis and subaortic valvular stenosis; abnormal genitalia (cryptorchidism, hypospadias); retardation of growth; and deafness (sensorineural)] syndromes, lentigines do serve as a clue to systemic disease. In the multiple lentigines or *LEOPARD syndrome*, hundreds of lentigines develop during childhood and are scattered over the entire surface of the body. The lentigines in patients with *Peutz-Jeghers syndrome* are located primarily around the nose and mouth, on the hands and feet, and within the oral cavity. While the pigmented macules on the

face may fade with age, the oral lesions persist. However, similar intraoral lesions are also seen in Addison's disease and as a normal finding in darkly pigmented individuals. Patients with this autosomal dominant syndrome (due to mutations in a novel serine threonine kinase gene) have multiple benign polyps of the gastrointestinal tract, testicular tumors, and an increased risk of developing gastrointestinal (primarily colon), breast, and gynecologic cancers.

Lentigines are also seen in association with cardiac myxomas and have been described in two syndromes whose findings overlap: *LAMB* (lentigines, atrial myxomas, mucocutaneous myxomas, and blue nevi) syndrome and *NAME* [nevus, atrial myxoma, myxoid neurofibroma, and ephelides (freckles)] syndrome. These patients can also have evidence of endocrine overactivity in the form of Cushing's syndrome, acromegaly, or sexual precocity.

The third type of localized hyperpigmentation is due to a local increase in pigment production, and it includes *ephelides* and cafe au lait macules (CALM). The latter are most commonly associated with two disorders -- neurofibromatosis (NF) and McCune-Albright syndrome. CALM are flat, uniformly light brown in color, and can vary in size from 0.5 to 12 cm. Approximately 80% of adult patients with *type I NF* will have six or more CALM measuring 1.5 cm or greater in diameter. Additional findings are discussed in the section on neurofibromas (see "Papulonodular Skin Lesions"). In comparison with NF, the CALM in patients with *McCune-Albright syndrome* [polyostotic fibrous dysplasia with precocious puberty in females due to mosaicism for an activating mutation in a G protein (G_{sa}) gene] are usually larger, more irregular in outline, and tend to respect the midline. CALM have also been associated with pulmonary stenosis (Watson syndrome), tuberous sclerosis, the [LEOPARD](#) syndrome, and ataxia telangiectasia, but a few such lesions can be found in normal individuals.

In *incontinentia pigmenti*, dyskeratosis congenita, and bleomycin pigmentation, the areas of localized hyperpigmentation form a pattern -- swirled in the first, reticulated in the second, and flagellate in the third. Patients with the X-linked dominant disorder *incontinentia pigmenti* can have linear blisters and verrucous papules during infancy. During childhood, parallel swirls and streaks of hyperpigmentation appear on the trunk, and occasionally streaks of hypopigmentation appear on the extremities. Associated findings include seizures, mental retardation, retinal vascular abnormalities, and delayed or impaired dentition. Biopsy specimens of the streaks will show pigment within dermal macrophages ("incontinent pigment"). In *dyskeratosis congenita*, atrophic reticulated hyperpigmentation is seen on the neck, thighs, and trunk and is accompanied by nail dystrophy, pancytopenia, and leukoplakia of the oral and anal mucosa. The latter often develops into squamous cell carcinoma. In addition to the flagellate pigmentation (linear streaks) on the trunk, patients receiving bleomycin often have hyperpigmentation on the elbows, knees, and small joints of the hand.

Localized hyperpigmentation is seen as a side effect of several other *systemic medications*, including those that produce fixed drug reactions [phenolphthalein, nonsteroidal anti-inflammatory drugs (NSAIDs), sulfonamides, and barbiturates] and those that can complex with melanin (antimalarials). Fixed drug eruptions recur in the same location as circular areas of erythema that can become bullous and then resolve as brown macules. The eruption usually appears within hours of administration of the

offending agent, and common locations include the genitalia, extremities, and perioral region. Chloroquine and hydroxychloroquine produce gray-brown to blue-black discoloration of the shins, hard palate, and face, while blue macules can be seen on the lower extremities and in sites of inflammation with prolonged minocycline administration. Estrogen in oral contraceptives can induce melasma -- symmetric brown patches on the face, especially the cheeks, upper lip, and forehead. Similar changes are seen in pregnancy, in patients receiving hydantoin, and in the adult form of Gaucher's disease. In the latter group there is also hyperpigmentation of the distal lower extremities.

In the *diffuse* forms of hyperpigmentation, the darkening of the skin may be of equal intensity over the entire body or may be accentuated in sun-exposed areas. The causes of diffuse hyperpigmentation can be divided into four groups -- endocrine, metabolic, autoimmune, and drugs. The endocrinopathies that frequently have associated hyperpigmentation include *Addison's disease*, *Nelson syndrome*, and *ectopic ACTH syndrome*. In these diseases, the increased pigmentation is diffuse but is accentuated in the palmar creases, sites of friction, scars, and the oral mucosa. An overproduction of the pituitary hormones α -MSH (melanocyte-stimulating hormone) and ACTH can lead to an increase in melanocyte activity. These peptides are products of the proopiomelanocortin gene and exhibit homology; e.g., α -MSH and ACTH share 13 amino acids. A minority of the patients with Cushing's disease or hyperthyroidism have generalized hyperpigmentation.

The metabolic causes of hyperpigmentation include *porphyria cutanea tarda* (PCT), *hemochromatosis*, *vitamin B₁₂ deficiency*, *folic acid deficiency*, *pellagra*, *malabsorption*, and *Whipple's disease*. In patients with PCT (see "Vesicles/Bullae"), the skin darkening is seen in sun-exposed areas and is a reflection of the photoreactive properties of porphyrins. The increased level of iron in the skin of patients with hemochromatosis stimulates melanin pigment production and leads to the classic bronze color. Patients with pellagra have a brown discoloration of the skin, especially in sun-exposed areas, as a result of nicotinic acid (niacin) deficiency. In the areas of increased pigmentation, there is a thin varnish-like scale. These changes are also seen in patients who are vitamin B₆ deficient, have functioning carcinoid tumors (increased consumption of niacin), or take isoniazid. Approximately 50% of the patients with Whipple's disease have an associated generalized hyperpigmentation in association with diarrhea, weight loss, arthritis, and lymphadenopathy. A diffuse slate-blue color is seen in patients with melanosis secondary to metastatic melanoma. Although there is a debate as to whether the color is due to single-cell metastases in the dermis or to a widespread deposition of melanin resulting from the high concentration of circulating melanin precursors, there is more evidence to support the latter.

Of the autoimmune diseases associated with diffuse hyperpigmentation, *biliary cirrhosis* and *scleroderma* are the most common, and occasionally, both disorders are seen in the same patient. The skin is dark brown in color, especially in sun-exposed areas. In biliary cirrhosis the hyperpigmentation is accompanied by pruritus, jaundice, and xanthomas, whereas in scleroderma it is accompanied by sclerosis of the extremities, face, and, less commonly, the trunk. Additional clues to the diagnosis of scleroderma are telangiectasias, calcinosis cutis, Raynaud's phenomenon, and distal ulcerations (see "Telangiectasias"). The differential diagnosis of cutaneous sclerosis with hyperpigmentation includes the POEMS [polyneuropathy; organomegaly (liver, spleen,

lymph nodes); endocrinopathies (impotence, gynecomastia); *M*-protein; and skin changes] syndrome. The skin changes include hyperpigmentation, skin thickening, hypertrichosis, and angiomas.

In the late 1980s, an epidemic of the eosinophilia-myalgia syndrome was described that was presumably due to contaminated L-tryptophan preparations. In addition to maculopapular eruptions and alopecia, large areas of scleroderma-like induration were observed with overlying hyperpigmentation.

Diffuse hyperpigmentation that is due to *drugs* or *metals* can result from one of several mechanisms -- induction of melanin pigment formation, complexing of the drug or its metabolites to melanin, and deposits of the drug in the dermis. Busulfan, cyclophosphamide, long-term, high-dose ACTH, and inorganic arsenic induce pigment production. Complexes containing melanin or hemosiderin plus the drug or its metabolites are seen in patients receiving chlorpromazine and minocycline. The sun-exposed skin as well as the conjunctivae of patients on long-term, high-dose chlorpromazine can become blue-gray in color. Patients taking minocycline may develop a diffuse blue-gray, muddy appearance in sun-exposed areas in addition to pigmentation of the mucous membranes, teeth, nails, bones, and thyroid. Administration of amiodarone can result in both a phototoxic eruption (exaggerated sunburn) and/or a brown or blue-gray discoloration of sun-exposed skin. Biopsy specimens of the latter show yellow-brown granules in dermal macrophages, which represent intralysosomal accumulations of lipids, amiodarone, and its metabolites. Actual deposits of a particular drug or metal in the skin are seen with silver (argyria), where the skin appears blue-gray in color; gold (chrysiasis), where the skin has a brown to blue-gray color; and clofazimine, where the skin appears reddish brown. The associated hyperpigmentation is accentuated in sun-exposed areas, and discoloration of the eye is seen with gold (sclerae) and clofazimine (conjunctivae).

VESICLES/BULLAE ([Table 57-12](#))

Depending on their size, cutaneous blisters are referred to as *vesicles* (<0.5 cm) or *bullae* (>0.5 cm). The primary blistering disorders include *pemphigus vulgaris*, *pemphigus foliaceus*, *pemphigus erythematosus*, *paraneoplastic pemphigus*, *bullous pemphigoid*, *herpes gestationis*, *cicatricial pemphigoid*, *epidermolysis bullosa acquisita*, *linear IgA disease*, and *dermatitis herpetiformis* ([Chap. 58](#)).

Vesicles and bullae are also seen in *contact dermatitis*, both allergic and irritant forms ([Chap. 56](#)). When there is a linear arrangement of vesicular lesions, an exogenous cause should be suspected. Bullous disease secondary to the ingestion of drugs can take one of several forms, including phototoxic eruptions, isolated bullae, toxic epidermal necrolysis, and erythema multiforme ([Chap. 59](#)). Clinically, phototoxic eruptions resemble an exaggerated sunburn with diffuse erythema and bullae in sun-exposed areas. The most commonly associated drugs are thiazides, doxycycline, sulfonamides, [NSAIDs](#), and psoralens. The development of a phototoxic eruption is dependent on the doses of both the drug and UV-A irradiation.

Toxic epidermal necrolysis (TEN) is characterized by bullae that arise on widespread areas of erythema and then slough. This results in large areas of denuded skin. The

associated morbidity, such as sepsis, and mortality are relatively high and are a function of the extent of epidermal necrosis. In addition, these patients may also have involvement of the mucous membranes and intestinal tract. Drugs are the primary cause of TEN, and the most common offenders are phenytoin, barbiturates, sulfonamides, penicillins, and [NSAIDs](#). Severe acute graft-versus-host disease (grade 4) also can resemble TEN.

In *erythema multiforme* (EM), the primary lesions are pink-red macules and edematous papules, the centers of which may become vesicular. The clue to the diagnosis of EM, as opposed to a drug-induced morbilliform exanthem, is the development of a "dusky" violet color or petechiae in the center of the lesions. Target or iris lesions are also characteristic of EM and arise as a result of active centers and borders in combination with centrifugal spread. However, iris lesions need not be present to make the diagnosis of EM. Preferred sites of involvement include the distal extremities and mucous membranes (oral, nasal, ocular, and genital). Hemorrhagic crusts of the lips are characteristic of EM as well as herpes simplex, pemphigus vulgaris, and paraneoplastic pemphigus. Fever, malaise, myalgias, sore throat, and cough may precede or accompany the eruption. The lesions of EM usually resolve over 3 to 6 weeks but may be recurrent.

Drugs can induce [EM](#), in particular sulfonamides, phenytoin, barbiturates, penicillins, and carbamazepine, but they do not cause the majority of cases, especially in young adults. Infections with herpes simplex are the most common cause of EM in this age group, and the lesions appear 7 to 12 days after the viral eruption. Other infectious agents associated with EM include *Mycoplasma pneumoniae*, dimorphic fungi, and several viruses (echovirus, coxsackievirus, Epstein-Barr, and influenza). EM can also follow vaccinations with BCG, poliomyelitis, or vaccinia viruses; radiation therapy; and exposure to environmental toxins.

In addition to primary blistering disorders and hypersensitivity reactions, bacterial and viral infections can lead to vesicles and bullae. The most common infectious agents are herpes simplex ([Chap. 182](#)), herpes varicella-zoster ([Chap. 183](#)), and staphylococci ([Chap. 139](#)).

Staphylococcal scalded-skin syndrome (SSSS) and *bullous impetigo* are two blistering disorders associated with staphylococcal (phage group II) infection. In SSSS, the initial findings are redness and tenderness of the central face, neck, trunk, and intertriginous zones. This is followed by short-lived flaccid bullae and a slough or exfoliation of the superficial epidermis. Crusted areas then develop, characteristically around the mouth. SSSS is distinguished from [TEN](#) by the following features: younger age group, more superficial site of blister formation, no oral lesions, shorter course, less morbidity and mortality, and an association with staphylococcal exfoliative toxin ("exfoliatin"), not drugs. A rapid diagnosis of SSSS versus TEN can be made by a frozen section of the blister roof or exfoliative cytology of the blister contents. In SSSS the site of staphylococcal infection is usually extracutaneous (conjunctivitis, rhinorrhea, otitis media, pharyngitis, tonsillitis), and the cutaneous lesions are sterile, whereas in bullous impetigo the skin lesions are the site of infection. Impetigo is more localized than SSSS and usually presents with honey-colored crusts. Occasionally, superficial purulent blisters also form. *Cutaneous emboli* from gram-negative infections may present as

isolated bullae, but the base of the lesion is purpuric or necrotic, and it may develop into an ulcer (see "Purpura").

Several metabolic disorders are associated with blister formation, including diabetes mellitus, renal failure, and porphyria. Local hypoxia secondary to decreased cutaneous blood flow can also produce blisters, which explains the presence of bullae over pressure points in comatose patients (coma bullae). In *diabetes mellitus*, tense bullae with clear viscous fluid arise on normal skin. The lesions can be as large as 6 cm in diameter and are located on the distal extremities. There are several types of porphyria, but the most common form with cutaneous findings is [PCT](#). In sun-exposed areas (primarily the face and hands), the skin is very fragile, and trauma leads to erosions and tense vesicles. These lesions then heal with scarring and formation of milia; the latter are firm, 2- to 3-mm white or yellow papules that represent epidermoid inclusion cysts. Associated findings can include hypertrichosis of the lateral malar region (males) or face (females) and, in sun-exposed areas, hyperpigmentation and firm sclerotic plaques. An elevated level of urinary uroporphyrins confirms the diagnosis and is due to a decrease in uroporphyrinogen decarboxylase activity. Precipitating agents include alcohol, iron, chlorinated hydrocarbons, and hepatitis C infection.

The differential diagnosis of [PCT](#) includes (1) *porphyria variegata* -- the skin signs of PCT plus the systemic findings of acute intermittent porphyria; it has a diagnostic plasma porphyrin fluorescence emission at 626 nm; (2) *drug-induced bullous photosensitivity* (pseudoporphyria) -- the clinical and histologic findings are similar to PCT, but porphyrins are normal; etiologic agents include naproxen, furosemide, tetracycline, and nalidixic acid; (3) *bullous dermatosis of hemodialysis* -- the same appearance as PCT, but porphyrins are usually normal or occasionally borderline elevated; patients have chronic renal failure and are on hemodialysis; (4) PCT associated with hepatomas, hepatic carcinomas, and hemodialysis; and (5) *epidermolysis bullosa acquisita* ([Chap. 58](#)).

EXANTHEMS ([Table 57-13](#))

Exanthems are characterized by an acute generalized eruption. The two most common presentations are erythematous macules and papules (morbilliform) and confluent blanching erythema (scarlatiniform). *Morbilliform* eruptions are usually due to either *drugs* or *viral infections*. For example, up to 5% of the patients receiving penicillins, sulfonamides, phenytoin, or gold will develop a maculopapular eruption. Accompanying signs may include pruritus, fever, eosinophilia, and transient lymphadenopathy. Similar maculopapular eruptions are seen in the classic childhood viral exanthems, including (1) *rubeola* (measles) -- a prodrome of coryza, cough, and conjunctivitis followed by Koplik's spots on the buccal mucosa; the eruption begins behind the ears, at the hairline, and on the forehead and then spreads down the body, often becoming confluent; (2) *rubella* -- it begins on the forehead and face and then spreads down the body; it resolves in the same order and is associated with retroauricular and suboccipital lymphadenopathy; and (3) *erythema infectiosum* (fifth disease) -- erythema of the cheeks is followed by a reticulated pattern on extremities; it is secondary to a parvovirus B19 infection, and an associated arthritis is seen in adults.

Both measles and rubella are seen in unvaccinated young adults, and an atypical form

of measles is seen in adults immunized with either killed measles vaccine or killed vaccine followed in time by live vaccine. In contrast to classic measles, the eruption of atypical measles begins on the palms, soles, wrists, and knuckles, and the lesions may become purpuric. The patient with atypical measles can have pulmonary involvement and be quite ill. Rubelliform and roseoliform eruptions are also associated with *Epstein-Barr virus* (5 to 15% of patients), *echovirus*, *coxsackievirus*, and *adenovirus* infections. Detection of specific IgM antibodies or fourfold elevations in IgG antibodies allows the proper diagnosis. Occasionally, a maculopapular eruption is the result of a drug-viral interaction. For example, about 95% of the patients with infectious mononucleosis who are given ampicillin will develop a rash.

Of note, early in the course of infections with *Rickettsia* and *meningococcus*, prior to the development of purpura, the lesions may be erythematous macules and papules. This is also the case in chickenpox prior to the development of vesicles. Maculopapular eruptions are associated with early *HIV infection*, early secondary *syphilis*, *typhoid fever*, and *acute graft-versus-host disease*. In the last, lesions frequently begin on the palms and soles; the macular rose spots of typhoid fever involve primarily the anterior trunk.

The prototypic *scarlatiniform* eruption is seen in *scarlet fever* and is due to an erythrogenic toxin produced by group A β -hemolytic streptococcal infections, most commonly pharyngitis. This eruption is characterized by diffuse erythema, which begins on the neck and upper trunk, and red perifollicular puncta. Additional findings include a white strawberry tongue (white coating with red papillae) followed by a red strawberry tongue (red tongue with red papillae); petechiae of the palate; a facial flush with circumoral pallor; linear petechiae in the antecubital fossae; and desquamation of the involved skin, palms, and soles 5 to 20 days after onset of the eruption. A similar desquamation of the palms and soles is seen in toxic shock syndrome, Kawasaki's disease, and after severe febrile illnesses. Certain strains of staphylococci also produce an erythrogenic toxin that leads to the same clinical findings as in streptococcal scarlet fever, except that the antistreptolysin O titers are not elevated.

In *toxic shock syndrome* (TSS), staphylococcal (phage group I) infections produce an exotoxin (TSST-1) that causes the fever and rash, as well as enterotoxins. Initially, the majority of cases were reported in menstruating women who were using tampons. However, other sites of infection, including wounds and vaginitis, may produce TSS. The diagnosis of TSS is based on clinical criteria ([Chap. 139](#)), and three of these involve mucocutaneous sites. The clinical criteria are (1) fever; (2) diffuse erythema of the skin; (3) desquamation of the palms and soles 1 to 2 weeks after onset of illness; (4) hypotension; and (5) involvement of three or more organ systems, including the gastrointestinal tract, muscles, kidney, liver, [CNS](#), hematologic (thrombocytopenia), and mucous membranes. The latter is characterized as hyperemia of the vagina, oropharynx, or conjunctivae. Similar systemic findings have been described in *streptococcal toxic shock-like syndrome* ([Chap. 140](#)), and although an exanthem is seen less often than in TSS due to a staphylococcal infection, the underlying infection is often in the soft tissue.

The cutaneous eruption in *Kawasaki's disease* (mucocutaneous lymph node syndrome) ([Chap. 317](#)) is polymorphous, but the two most common forms are morbilliform and

scarlatiniform. The majority of cases are seen in children less than 5 years of age, but adult cases have been reported. The diagnosis is based on a fever lasting more than 5 days plus four of the five following criteria: (1) bilateral conjunctival injection; (2) exanthem; (3) cervical lymphadenopathy, usually unilateral; (4) erythema and edema of the hands and feet followed by desquamation; and (5) diffuse erythema of the oropharynx, red strawberry tongue, and erosions with crusting on the lips. This clinical picture can resemble [TSS](#) and scarlet fever, but clues to the diagnosis of Kawasaki's disease are the cervical lymphadenopathy, lip erosions, and increased platelets. The most serious associated systemic finding in this disease is coronary aneurysm secondary to arteritis. Aneurysms may lead to sudden death, primarily within the first 30 days of the illness. Scarlatiniform eruptions are also seen in the early phase of [SSSS](#) (see "Vesicles/Bullae") and as reactions to drugs.

URTICARIA ([Table 57-14](#))

Urticaria (hives) are transient lesions that are composed of a central wheal surrounded by an erythematous halo. Individual lesions are round, oval, or figurate and are often pruritic. *Acute* and *chronic* urticaria have a wide variety of allergic etiologies. Less common systemic causes of urticaria are mastocytosis (urticaria pigmentosa), hyperthyroidism, malignancy, and juvenile rheumatoid arthritis (JRA). In JRA, the lesions coincide with the fever spike and are transient but not migratory as in erythema marginatum.

The common *physical urticarias* include dermographism, solar urticaria, cold urticaria, and cholinergic urticaria. Patients with *dermographism* exhibit linear wheals following minor pressure or scratching of the skin. It is a common disorder, affecting approximately 5% of the population. *Solar urticaria* characteristically occur within minutes of sun exposure and are a skin sign of one systemic disease -- erythropoietic protoporphyria. In addition to the urticaria, these patients have subtle pitted scarring of the nose and hands. *Cold urticaria* are precipitated by exposure to the cold, and therefore exposed areas are usually affected. In some cases, the disease is associated with abnormal circulating proteins -- more commonly cryoglobulins and less commonly cryofibrinogens and cold agglutinins. Additional systemic symptoms include wheezing and syncope, thus explaining the need for these patients to avoid swimming in cold water. *Cholinergic urticaria* are precipitated by heat, exercise, or emotion and are characterized by small wheals with relatively large flares. They are occasionally associated with wheezing.

Whereas urticaria are the result of dermal edema, subcutaneous edema leads to the clinical picture of *angioedema*. Sites of involvement include the eyelids, lips, tongue, larynx, and gastrointestinal tract as well as the subcutaneous tissue. Angioedema occurs alone or in combination with urticaria, including urticarial vasculitis and the physical urticarias. Both acquired and hereditary (autosomal dominant) forms of angioedema occur ([Chap. 310](#)), and in the latter, urticaria is rarely seen.

Urticarial vasculitis is an immune complex disease that may be confused with simple urticaria. In contrast to simple urticaria, individual lesions tend to last longer than 24 h and usually develop central petechiae that can be observed even after the urticarial phase has resolved. The patient may also complain of burning rather than pruritus. On

biopsy, there is a leukocytoclastic vasculitis of the small blood vessels. Although many cases of urticarial vasculitis are idiopathic in origin, it can be a reflection of an underlying systemic illness such as lupus erythematosus, Sjogren's syndrome, or hereditary complement deficiency. There is a spectrum of urticarial vasculitis that ranges from purely cutaneous to multisystem involvement. The most common systemic signs and symptoms are arthralgias and/or arthritis, nephritis, and crampy abdominal pain, with asthma and chronic obstructive lung disease seen less often. Hypocomplementemia occurs in one- to two-thirds of patients, even in the idiopathic cases. Urticarial vasculitis can also be seen in patients with *hepatitis B* and *hepatitis C* infections, *serum sickness*, and *serum sickness-like illnesses*.

PAPULONODULAR SKIN LESIONS ([Table 57-15](#))

In the *papulonodular diseases*, the lesions are elevated above the surface of the skin and may coalesce to form plaques. The location, consistency, and color of the lesions are the keys to their diagnosis; this section is organized on the basis of color.

White Lesions In *calcinosis cutis* there are firm white to white-yellow papules with an irregular surface. When the contents are discharged, a chalky white material is seen. *Dystrophic* calcification is seen at sites of previous inflammation or damage to the skin. It develops in acne scars as well as on the distal extremities of patients with scleroderma and in the subcutaneous tissue and intermuscular fascial planes in [DM](#). The latter is more extensive and is more commonly seen in children. An elevated calcium phosphate product, as in secondary hyperparathyroidism, can lead to nodules of *metastatic* calcinosis cutis, which tend to be subcutaneous and periarticular. This form is often accompanied by calcification of muscular arteries and subsequent ischemic necrosis (calciophylaxis).

Skin-Colored Lesions There are several types of skin-colored lesions, including epidermoid inclusion cysts, lipomas, rheumatoid nodules, neurofibromas, angiofibromas, neuromas, and adnexal tumors such as tricholemmomas. Both *epidermoid inclusion cysts* and *lipomas* are very common mobile subcutaneous nodules -- the former are rubbery and compressible and drain cheeselike material (sebum and keratin) if incised. Lipomas are firm and somewhat lobulated on palpation. When extensive facial epidermoid inclusion cysts develop in childhood or there is a family history of such lesions, the patient should be examined for other signs of Gardner syndrome, including osteomas and desmoid tumors. *Rheumatoid nodules* are firm, 0.5- to 4-cm nodules that tend to localize around pressure points, especially the elbows. They are seen in approximately 20% of patients with rheumatoid arthritis and 6% of patients with Still's disease. Biopsies of the nodules show palisading granulomas. Similar lesions that are smaller and shorter-lived are seen in rheumatic fever.

Neurofibromas (benign Schwann cell tumors) are soft papules or nodules that exhibit the "button-hole" sign, that is, they invaginate into the skin with pressure in a manner similar to a hernia. Single lesions are seen in normal individuals, but multiple neurofibromas, usually in combination with six or more [CALM](#) measuring >1.5 cm (see "Hyperpigmentation") and multiple Lisch nodules, are seen in von Recklinghausen's disease ([NF](#) type I). Lisch nodules are 1-mm yellow-brown spots within the iris that are best observed with slit-lamp examination. Additional manifestations include axillary

freckling and peripheral and [CNS tumors \(Chap. 370\)](#). In some patients the neurofibromas are localized and unilateral, whereas in others they are limited to the CNS.

Angiofibromas are firm, pink to skin-colored papules that measure from 3 mm to several centimeters in diameter. When they are located on the central cheeks (adenoma sebaceum) or multiple fibromas are seen around the nails, the patient has tuberous sclerosis. It is an autosomal disorder due to mutations in two different genes, and the associated findings are discussed in the section on ash leaf spots as well as in [Chap. 370](#). Multiple facial angiofibromas have also been observed in patients with multiple endocrine neoplasia (MEN) syndrome, type 1.

Neuromas (benign proliferations of nerve fibers) are also firm, skin-colored papules. They are more commonly found at sites of amputation and as rudimentary supernumerary digits. However, when there are multiple neuromas on the eyelids, lips, distal tongue, and/or oral mucosa, the patient should be investigated for other signs of the [MEN syndrome, type 2b](#). Associated findings include marfanoid habitus, protuberant lips, intestinal ganglioneuromas, and medullary thyroid carcinoma (>75% of patients; [Chap. 339](#)).

Adnexal tumors are derived from pluripotential cells of the epidermis that can differentiate toward hair, sebaceous, apocrine, or eccrine glands or remain undifferentiated. *Basal cell epitheliomas* (BCEs) are examples of adnexal tumors that have little or no evidence of differentiation. Clinically, they are translucent papules with rolled borders, telangiectasias, and central erosion. BCEs commonly arise in sun-damaged skin of the head and neck. When a patient has multiple BCEs, especially prior to age 30, the possibility of the basal cell nevus syndrome should be raised. It is inherited as an autosomal dominant trait and is associated with jaw cysts, palmar and plantar pits, frontal bossing, medulloblastomas and calcification of the falx cerebri and diaphragma sellae. *Tricholemmomas* are also skin-colored adnexal tumors but differentiate toward hair follicles and can have a wartlike appearance. The presence of multiple tricholemmomas on the face and cobblestoning of the oral mucosa points to the diagnosis of Cowden's disease (multiple hamartoma syndrome) due to mutations in the *PTEN* gene. Internal organ involvement (in decreasing order of frequency) includes fibrocystic disease and carcinoma of the breast, adenomas and carcinomas of the thyroid, and gastrointestinal polyposis. Keratoses of the palms, soles, and dorsa of the hands are also seen.

Pink Lesions The cutaneous lesions associated with primary systemic *amyloidosis* are pink in color and translucent. Common locations are the face, especially the periorbital and perioral regions, and flexural areas. On biopsy, homogeneous deposits of amyloid are seen in the dermis and in the walls of blood vessels; the latter lead to an increase in vessel wall fragility. As a result, petechiae and purpura develop in clinically normal skin as well as in lesional skin following minor trauma, hence the term "pinch purpura." Amyloid deposits are also seen in the striated muscle of the tongue and result in macroglossia.

Even though specific mucocutaneous lesions are rarely seen in secondary amyloidosis and are present in only about 30% of the patients with primary amyloidosis, a rapid

diagnosis of systemic amyloidosis can be made by an examination of abdominal subcutaneous fat. By special staining, deposits are seen around blood vessels or individual fat cells in 40 to 50% of patients. There are also three forms of amyloidosis that are limited to the skin and that should not be construed as cutaneous lesions of systemic amyloidosis. They are macular amyloidosis (upper back), lichenoid amyloidosis (usually lower extremities), and nodular amyloidosis. In macular and lichenoid amyloidosis, the deposits are composed of altered epidermal keratin. Recently, macular and lichenoid amyloidosis have been associated with [MEN](#) syndrome, type 2a.

Patients with *multicentric reticulohistiocytosis* also have pink-colored papules and nodules on the face and mucous membranes as well as on the extensor surface of the hands and forearms. They have a polyarthritis that can mimic rheumatoid arthritis clinically. On histologic examination, the papules have characteristic giant cells that are not seen in biopsies of rheumatoid nodules. Pink to skin-colored papules that are firm, 2 to 5 mm in diameter, and often in a linear arrangement are seen in patients with *papular mucinosis*. This disease is also referred to as *lichen myxedematosus* or *scleromyxedema*. The latter name comes from the brawny induration of the face and extremities that may accompany the papular eruption. Biopsy specimens of the papules show localized mucin deposition, and serum protein electrophoresis demonstrates a monoclonal spike of IgG, usually with a λ light chain.

Yellow Lesions Several systemic disorders are characterized by yellow-colored cutaneous papules or plaques -- hyperlipidemia (xanthomas), gout (tophi), diabetes (necrobiosis lipoidica), pseudoxanthoma elasticum, and Torre syndrome (sebaceous tumors). Eruptive xanthomas are the most common form of *xanthomas*, and are associated with hypertriglyceridemia (types I, III, IV, and V). Crops of yellow papules with erythematous halos occur primarily on the extensor surfaces of the extremities and the buttocks, and they spontaneously involute with a fall in serum triglycerides. Increased β -lipoproteins (primarily types II and III) result in one or more of the following types of xanthoma: xanthelasma, tendon xanthomas, and plane xanthomas. Xanthelasma are found on the eyelids, whereas tendon xanthomas are frequently associated with the Achilles and extensor finger tendons; plane xanthomas are flat and favor the palmar creases, face, upper trunk, and scars. Tuberous xanthomas are frequently associated with hypertriglyceridemia, but they are also seen in patients with hypercholesterolemia (type II) and are found most frequently over the large joints or hand. Biopsy specimens of xanthomas show collections of lipid-containing macrophages (foam cells).

Patients with several disorders, including biliary cirrhosis, can have a secondary form of hyperlipidemia with associated tuberous and planar xanthomas. However, patients with myeloma have *normolipemic* flat xanthomas. This latter form of xanthoma may be 3-12 cm in diameter and is most frequently seen on the upper trunk or side of the neck. It is also important to note that the most common setting for eruptive xanthomas is uncontrolled diabetes mellitus. The least specific sign for hyperlipidemia is xanthelasma, because at least 50% of the patients with this finding have normal lipid profiles.

In *tophaceous gout* there are deposits of monosodium urate in the skin around the joints, particularly those of the hands and feet. Additional sites of *tophi* formation include

the helix of the ear and the olecranon and prepatellar bursae. The lesions are firm, yellow in color, and occasionally discharge a chalky material. Their size varies from 1 mm to 7 cm, and the diagnosis can be established by polarization of the aspirated contents of a lesion. Lesions of *necrobiosis lipoidica* are found primarily on the shins (90%), and patients can have diabetes mellitus or develop it subsequently. Characteristic findings include a central yellow color, atrophy (transparency), telangiectasias, and an erythematous border. Ulcerations can also develop within the plaques. Biopsy specimens show necrobiosis of collagen, granulomatous inflammation, and obliterative endarteritis.

In *pseudoxanthoma elasticum* (PXE) there is an abnormal deposition of calcium on the elastic fibers of the skin, eye, and blood vessels. In the skin, the flexural areas such as the neck, axillae, antecubital fossae, and inguinal area are the primary sites of involvement. Yellow papules coalesce to form reticulated plaques that have an appearance similar to that of plucked chicken skin. In severely affected skin, hanging, redundant folds develop. Some patients have a more subtle macular form of the disease, and careful inspection is required. Biopsy specimens of involved skin show swollen and irregularly clumped elastic fibers with deposits of calcium. In the eye, the calcium deposits in Bruch's membrane lead to angioid streaks and choroiditis; in the arteries of the heart, kidney, gastrointestinal tract, and extremities, the deposits lead to angina, hypertension, gastrointestinal bleeding, and claudication, respectively. Long-term administration of D-penicillamine can lead to PXE-like skin changes as well as elastic fiber alterations in internal organs.

Adnexal tumors that have differentiated toward sebaceous glands include sebaceous adenoma, sebaceous epithelioma, sebaceous carcinoma, and sebaceous hyperplasia. Except for sebaceous hyperplasia, which is commonly seen on the face, these tumors are fairly rare. Patients with Torre syndrome have *sebaceous adenomas*, and in the majority of cases there are multiple such tumors. These patients can also have sebaceous carcinomas and sebaceous hyperplasia as well as keratoacanthomas. The internal manifestations of Torre syndrome include *multiple* carcinomas of the gastrointestinal tract (primarily colon) as well as cancers of the larynx, genitourinary tract, and endometrium. Some patients also have a strong family history of cancer.

Red Lesions Cutaneous lesions that are red in color have a wide variety of etiologies; in an attempt to simplify their identification, they will be subdivided into papules, papules/plaques, and subcutaneous nodules. Common red papules include *arthropod bites* and *cherry hemangiomas*; the latter are small, bright-red, dome-shaped papules that represent benign proliferation of capillaries. In patients with AIDS, the development of multiple red hemangioma-like lesions points to bacillary angiomatosis, and biopsy specimens show clusters of bacilli that stain positive with the Warthin-Starry stain; the pathogens have been identified as *Bartonella henselae* and *B. quintana*. Disseminated visceral disease is seen primarily in immunocompromised hosts but can occur in immunocompetent individuals.

Multiple *angiokeratomas* are seen in Fabry's disease, an X-linked recessive lysosomal storage disease that is due to a deficiency of α -galactosidase A. The lesions are red to red-blue in color and can be quite small in size (1 to 3 mm), with the most common location being the lower trunk. Associated findings include chronic renal failure,

peripheral neuropathy, and corneal opacities (cornea verticillata). Electron photomicrographs of angiokeratomas and clinically normal skin demonstrate lamellar lipid deposits in fibroblasts, pericytes, and endothelial cells that are diagnostic of this disease. Widespread acute eruptions of erythematous papules are discussed in the section on exanthems.

There are several infectious diseases that present as erythematous papules or nodules in a sporotrichoid pattern, that is, in a linear arrangement along the lymphatic channels. The two most common etiologies are *Sporothrix schenckii* (sporotrichosis) and *M. marinum* (atypical mycobacteria). The organisms are introduced as a result of trauma, and a primary inoculation site is often seen in addition to the lymphatic nodules. Additional causes include *Nocardia*, *Leishmania*, and other dimorphic fungi; culture of lesional tissue will aid in the diagnosis.

The diseases that are characterized by erythematous plaques with scale are reviewed in the papulosquamous section, and the various forms of dermatitis are discussed in the section on erythroderma. Additional disorders in the differential diagnosis of red papules/plaques include *erysipelas*, *polymorphous light eruption* (PMLE), *lymphocytoma cutis*, *cutaneous lupus*, *lymphoma cutis*, and *leukemia cutis*. The first three diseases represent primary cutaneous disorders. PMLE is characterized by erythematous papules and plaques in a primarily sun-exposed distribution -- dorsum of the hand, extensor forearm, and face. Lesions follow exposure to UV-B and/or UV-A, and in northern latitudes PMLE is most severe in the late spring and early summer. A process referred to as "hardening" occurs with continued UV exposure, and the eruption fades, but in temperate climates it will recur in the spring. PMLE must be differentiated from cutaneous lupus, and this is accomplished by histologic examination and direct immunofluorescence of the lesions. Lymphocytoma cutis (pseudolymphoma) is a *benign* polyclonal proliferation of lymphocytes in the skin that presents as infiltrated pink-red to red-purple papules and plaques; it must be distinguished from lymphoma cutis.

Several types of red plaques are seen in patients with systemic *lupus*, including (1) erythematous urticarial plaques across the cheeks and nose in the classic butterfly rash; (2) erythematous discoid lesions with fine or "carpet-tack" scale, telangiectasias, central hypopigmentation, peripheral hyperpigmentation, follicular plugging, and atrophy located on the face, scalp, external ears, arms, and upper trunk; and (3) psoriasiform or annular lesions of subacute lupus with hypopigmented centers located on the face, extensor arms, and upper trunk. Additional cutaneous findings include (1) a violaceous flush on the face and V of the neck; (2) urticarial vasculitis (see "Urticaria"); (3) lupus panniculitis (see below); (4) diffuse alopecia; (5) alopecia secondary to discoid lesions; (6) periungual telangiectasias and erythema; (7) erythema multiforme-like lesions that may become bullous; and (8) distal ulcerations secondary to Raynaud's phenomenon, vasculitis, or livedoid vasculitis. Patients with only discoid lesions usually have the form of lupus that is limited to the skin. However, 2 to 10% of these patients eventually develop systemic lupus. Direct immunofluorescence of involved skin shows deposits of IgG or IgM and C3 in a granular distribution along the dermal-epidermal junction.

In *lymphoma cutis* there is a proliferation of malignant lymphocytes or histiocytes in the skin, and the clinical appearance resembles that of lymphocytoma cutis -- infiltrated pink-red to red-purple papules and plaques. Lymphoma cutis can occur anywhere on

the surface of the skin, whereas the sites of predilection for lymphocytomas include the malar ridge, tip of the nose, and earlobes. Patients with non-Hodgkin's lymphomas have specific cutaneous lesions more often than those with Hodgkin's disease, and occasionally, the skin nodules precede the development of extracutaneous non-Hodgkin's lymphoma or represent the only site of involvement. Arcuate lesions are sometimes seen in lymphoma and lymphocytoma cutis as well as in [CTCL](#). *Leukemia cutis* has the same appearance as lymphoma cutis, and specific lesions are seen more commonly in monocytic leukemias than in lymphocytic or granulocytic leukemias. Cutaneous chloromas (granulocytic sarcomas) may precede the appearance of circulating blasts in acute nonlymphocytic leukemia and, as such, represent a form of aleukemic leukemia cutis.

Common causes of erythematous subcutaneous nodules include inflamed epidermoid inclusion cysts, acne cysts, and furuncles. *Panniculitis*, an inflammation of the fat, also presents as subcutaneous nodules and is frequently a sign of systemic disease. There are several forms of panniculitis, including erythema nodosum, erythema induratum, lupus profundus, lipomembranous lipodermatosclerosis, α_1 -antitrypsin deficiency, factitial, and fat necrosis secondary to pancreatic disease. Except for erythema nodosum, these lesions may break down and ulcerate or heal with a scar. The shin is the most common location for the nodules of erythema nodosum, whereas the calf is the most common location for lesions of erythema induratum. In erythema nodosum the nodules are initially red but then develop a blue color as they resolve. Patients with erythema nodosum but no underlying systemic illness can still have fever, malaise, leukocytosis, arthralgias, and/or arthritis. However, the possibility of an underlying illness should be excluded, and the most common associations are streptococcal infections, upper respiratory infections, sarcoidosis, and inflammatory bowel disease. The less common associations include tuberculosis, histoplasmosis, coccidioidomycosis, psittacosis, drugs (oral contraceptives, sulfonamides, aspartame, bromides, iodides), cat-scratch fever, and infections with *Yersinia*, *Salmonella*, and *Chlamydia*.

In some patients, erythema induratum/nodular vasculitis is an idiopathic disease; however, in approximately 25 to 70% of patients, polymerase chain reaction (PCR) analysis will demonstrate *M. tuberculosis* complex DNA. The lesions of lupus profundus are found primarily on the upper arms and buttocks (sites of abundant fat) and are seen in both the cutaneous and systemic forms of lupus. The overlying skin may be normal, erythematous, or have the changes of discoid lupus. The subcutaneous fat necrosis that is associated with pancreatic disease is presumably secondary to circulating lipases and is seen in patients with pancreatic carcinoma as well as in patients with acute and chronic pancreatitis. In this disorder there may be an associated arthritis, fever, and inflammation of visceral fat. Histologic examination of deep incisional biopsy specimens will aid in the diagnosis of the particular type of panniculitis.

Subcutaneous erythematous nodules are also seen in *cutaneous polyarteritis nodosa* (PAN) and as a manifestation of *systemic vasculitis*, e.g., systemic PAN, allergic granulomatosis, or Wegener's granulomatosis ([Chap. 317](#)). Cutaneous PAN presents with painful subcutaneous nodules and ulcers within a red-purple, netlike pattern of livedo reticularis. The latter is due to slowed blood flow through the superficial horizontal venous plexus. The majority of lesions are found on the lower extremity, and while

arthralgias and myalgias may accompany cutaneous PAN, there is no evidence of systemic involvement. In both the cutaneous and systemic forms of vasculitis, skin biopsy specimens of the associated nodules will show the changes characteristic of a vasculitis; the size of the vessel involved will depend on the particular disease.

Red-Brown Lesions The cutaneous lesions in *sarcoidosis* ([Chap. 318](#)) are classically red to red-brown in color, and with diascopy (pressure with a glass slide) a yellow-brown residual color is observed that is secondary to the granulomatous infiltrate. The waxy papules and plaques may be found anywhere on the skin, but the face is the most common location. Usually there are no surface changes, but occasionally the lesions will have scale. Biopsy specimens of the papules show "naked" granulomas in the dermis, i.e., granulomas surrounded by a minimal number of lymphocytes. Other cutaneous findings in sarcoidosis include annular lesions with an atrophic or scaly center, papules within scars, hypopigmented macules and papules, alopecia, acquired ichthyosis, erythema nodosum, and lupus pernio (see below). Additional physical findings are peripheral lymphadenopathy and parotid and lacrimal gland enlargement. When there is cutaneous involvement of the hands, radiographs will often show lytic lesions in the underlying bone.

The differential diagnosis of sarcoidosis includes foreign-body granulomas produced by chemicals such as beryllium and zirconium, late secondary syphilis, and *lupus vulgaris*. Lupus vulgaris is a form of cutaneous tuberculosis that is seen in previously infected and sensitized individuals. There is often underlying active tuberculosis elsewhere, usually in the lungs or lymph nodes. At least 90% of the lesions occur in the head and neck area and are red-brown plaques with a yellow-brown color on diascopy. Secondary scarring and squamous cell carcinomas can develop within the plaques. Cultures or [PCR](#) analysis of the lesions should be done because it is rare for the acid-fast stain to show bacilli within the dermal granulomas.

Sweet's syndrome is characterized by red to red-brown plaques and nodules that are frequently painful and occur primarily on the head, neck, and upper extremities. The patients also have fever, neutrophilia, and a dense dermal infiltrate of neutrophils in the lesions. In approximately 10% of the patients there is an associated malignancy, most commonly acute nonlymphocytic leukemia. Sweet's syndrome has also been reported with lymphoma, chronic leukemia, myeloma, myelodysplastic syndromes, and solid tumors (primarily of the genitourinary tract). The differential diagnosis includes neutrophilic eccrine hidradenitis and atypical forms of pyoderma gangrenosum. Extracutaneous sites of involvement include joints, muscles, eye, kidney (proteinuria, occasionally glomerulonephritis), and lung (neutrophilic infiltrates). The idiopathic form of Sweet's syndrome is seen more often in women, following a respiratory tract infection.

A generalized distribution of red-brown macules and papules is seen in the form of mastocytosis known as *urticaria pigmentosa* ([Chap. 310](#)). Each lesion represents a collection of mast cells in the dermis, with hyperpigmentation of the overlying epidermis. Stimuli such as rubbing cause these mast cells to degranulate, and this leads to the formation of localized urticaria (Darier's sign). Additional symptoms can result from mast cell degranulation and include headache, flushing, diarrhea, and pruritus. Mast cells also infiltrate various organs such as the liver, spleen, and gastrointestinal tract in up to

30 to 50% of patients with urticaria pigmentosa, and accumulations of mast cells in the bones may produce either osteosclerotic or osteolytic shadows on radiographs. In the majority of these patients, however, the internal involvement remains fairly static. A subtype of chronic leukocytoclastic vasculitis, *erythema elevatum diutinum* (EED), also presents with papules that are red-brown in color. The papules coalesce into plaques on the extensor surfaces of knees, elbows, and the small joints of the hand. Flares of EED have been associated with streptococcal infections.

Blue Lesions Lesions that are blue in color are the result of either vascular ectasias and tumors or melanin pigment in the dermis. *Venous lakes* (ectasias) are compressible dark-blue lesions that are found commonly in the head and neck region. *Venous malformations* are also compressible blue papules and nodules that can occur anywhere on the body, including the oral mucosa. When they are multiple rather than single congenital lesions, the patient may have the blue rubber bleb syndrome or Mafucci's syndrome. Patients with the blue rubber bleb syndrome also have vascular anomalies of the gastrointestinal tract that may bleed, whereas patients with Mafucci's syndrome have associated dyschondroplasia and osteochondromas. *Blue nevi* (moles) are seen when there are collections of pigment-producing nevus cells in the dermis. These benign papular lesions are dome-shaped and occur most commonly on the dorsum of the hand or foot.

Violaceous Lesions Violaceous papules and plaques are seen in *lupus pernio*, *lymphoma cutis*, and *cutaneous lupus*. Lupus pernio is a particular type of sarcoidosis that involves the tip of the nose and the earlobes, with lesions that are violaceous in color rather than red-brown. This form of sarcoidosis is associated with involvement of the upper respiratory tract. The plaques of lymphoma cutis and cutaneous lupus may be red or violaceous in color and were discussed above.

Purple Lesions Purple-colored papules and plaques are seen in vascular tumors, such as *Kaposi's sarcoma* ([Chap. 309](#)) and *angiosarcoma*, and when there is extravasation of red blood cells into the skin in association with inflammation, as in *palpable purpura* (see "Purpura"). Patients with congenital or acquired [AV](#)fistulas and venous hypertension can develop purple papules on the lower extremities that can resemble Kaposi's sarcoma clinically and histologically; this condition is referred to as pseudo-Kaposi sarcoma (acral angiodermatitis). Angiosarcoma is found most commonly on the scalp and face of elderly patients or within areas of chronic lymphedema and presents as purple papules and plaques. In the head and neck region the tumor often extends beyond the clinically defined borders and may be accompanied by facial edema.

Brown and Black Lesions Brown- and black-colored papules are reviewed in "Hyperpigmentation."

Cutaneous Metastases These are discussed last because they can have a wide range of colors. Most commonly they present as either firm, skin-colored subcutaneous nodules or firm, red to red-brown papulonodules. The lesions of lymphoma cutis range from pink-red to plum in color, whereas metastatic melanoma can be pink, blue, or black in color. Cutaneous metastases develop from hematogenous or lymphatic spread and are most often due to the following primary carcinomas: in men, lung, colon, melanoma,

and oral cavity; and in women, breast, colon, and lung. These metastatic lesions may be the initial presentation of the carcinoma, especially when the primary site is the lung, kidney, or ovary.

PURPURA ([Table 57-16](#))

Purpura are seen when there is an extravasation of red blood cells into the dermis, and as a result, the lesions do not blanch with pressure. This is in contrast to those erythematous or violet-colored lesions that are due to localized vasodilatation -- they do blanch with pressure. Purpura (≥ 3 mm) and petechiae (≤ 2 mm) are divided into two major groups, palpable and nonpalpable. The most frequent causes of *nonpalpable* petechiae and purpura are primary cutaneous disorders such as *trauma*, *solar purpura*, and *capillaritis*. Less common causes are *steroid purpura* and *livedoid vasculitis* (see "Ulcers"). Solar purpura are seen primarily on the extensor forearms, while glucocorticoid purpura secondary to potent topical steroids or endogenous or exogenous Cushing's syndrome can be more widespread. In both cases there is alteration of the supporting connective tissue that surrounds the dermal blood vessels. In contrast, the petechiae that result from capillaritis are found primarily on the lower extremities. In capillaritis there is an extravasation of erythrocytes as a result of perivascular lymphocytic inflammation. The petechiae are bright red, 1 to 2 mm in size, and scattered within annular or coin-shaped yellow-brown macules. The yellow-brown color is caused by hemosiderin deposits within the dermis.

Systemic causes of nonpalpable purpura fall into several categories, and those secondary to clotting disturbances and vascular fragility will be discussed first. The former group includes *thrombocytopenia* ([Chap. 116](#)), *abnormal platelet function* as is seen in uremia, and *clotting factor defects*. The initial site of presentation for thrombocytopenia-induced petechiae is the distal lower extremity. Capillary fragility leads to nonpalpable purpura in patients with systemic *amyloidosis* (see "Papulonodular Skin Lesions"), disorders of collagen production such as *Ehlers-Danlos syndrome*, and *scurvy*. In scurvy there are flattened corkscrew hairs with surrounding hemorrhage on the lower extremities, in addition to gingivitis. Vitamin C is a cofactor for lysyl hydroxylase, an enzyme involved in the posttranslational modification of procollagen that is necessary for cross-link formation.

In contrast to the previous group of disorders, the purpura seen in the following group of diseases are associated with thrombi formation within vessels. It is important to note that these thrombi are demonstrable in skin biopsy specimens. This group of disorders includes disseminated intravascular coagulation (DIC), monoclonal cryoglobulinemia, thrombotic thrombocytopenic purpura, and reactions to warfarin. DIC is triggered by several types of infection (gram-negative, gram-positive, viral, and rickettsial) as well as by tissue injury and neoplasms. Widespread purpura and hemorrhagic infarcts of the distal extremities are seen. Similar lesions are found in purpura fulminans, which is a form of DIC associated with fever and hypotension that occurs more commonly in children following an infectious illness such as varicella, scarlet fever, or an upper respiratory tract infection. In both disorders, hemorrhagic bullae can develop in involved skin.

Monoclonal cryoglobulinemia is associated with multiple myeloma, Waldenstrom's

macroglobulinemia, lymphocytic leukemia, and lymphoma. Purpura, primarily of the lower extremities, and hemorrhagic infarcts of the fingers and toes are seen in these patients. Exacerbations of disease activity can follow cold exposure or an increase in serum viscosity. Biopsy specimens show precipitates of the cryoglobulin within dermal vessels. Similar deposits have been found in the lung, brain, and renal glomeruli. Patients with *thrombotic thrombocytopenic purpura* can also have hemorrhagic infarcts as a result of intravascular thromboses. Additional signs include thrombocytopenic purpura, fever, and microangiopathic hemolytic anemia ([Chap. 108](#)).

Administration of *warfarin* can result in painful areas of erythema that become purpuric and then necrotic with an adherent black eschar. This reaction is seen more often in women and in areas with abundant subcutaneous fat -- breasts, abdomen, buttocks, thighs, and calves. The erythema and purpura develop between the third and tenth day of therapy, most likely as a result of a transient imbalance in the levels of anticoagulant and procoagulant vitamin K-dependent factors. Continued therapy does not exacerbate preexisting lesions, and patients with an inherited or acquired deficiency of protein C are at increased risk for this particular reaction as well as for purpura fulminans.

Purpura secondary to *cholesterol emboli* are usually seen on the lower extremities of patients with atherosclerotic vascular disease. They often follow anticoagulant therapy or an invasive vascular procedure such as an arteriogram but also occur spontaneously from disintegration of atheromatous plaques. Associated findings include livedo reticularis, gangrene, cyanosis, subcutaneous nodules, and ischemic ulcerations. Multiple step sections of the biopsy specimen may be necessary to demonstrate the cholesterol clefts with the vessels. Petechiae are also an important sign of *fat embolism* and occur primarily on the upper body 2 to 3 days after a major injury. By using special fixatives, the emboli can be demonstrated in biopsy specimens of the petechiae. Emboli of tumor or thrombus are seen in patients with atrial myxomas and marantic endocarditis.

In the *Gardner-Diamond syndrome* (autoerythrocyte sensitivity), female patients develop large ecchymoses within areas of painful, warm erythema. An episode of significant trauma frequently precedes the onset of this syndrome. Intradermal injections of autologous erythrocytes or phosphatidyl serine derived from the red cell membrane can reproduce the lesions in some patients; however, there are instances where a reaction is seen at an injection site of the forearm but not in the midback region. The latter has led some observers to view Gardner-Diamond syndrome as a cutaneous manifestation of severe emotional stress. *Waldenstrom's hypergammaglobulinemic purpura* is a chronic disorder characterized by petechiae on the lower extremities. There are circulating complexes of IgG-anti-IgG molecules, and exacerbations are associated with prolonged standing or walking.

Palpable purpura are further subdivided into vasculitic and embolic. In the group of vasculitic disorders, *leukocytoclastic vasculitis* (LCV), also known as *allergic vasculitis*, is the one most commonly associated with palpable purpura ([Chap. 317](#)). *Henoch-Schonlein purpura* is a subtype of acute LCV that is seen primarily in children and adolescents following an upper respiratory infection. The majority of lesions are found on the lower extremities and buttocks. Systemic manifestations include fever, arthralgias (primarily of the knees and ankles), abdominal pain, gastrointestinal

bleeding, and nephritis. Direct immunofluorescence examination shows deposits of IgA within dermal blood vessel walls. In *polyarteritis nodosa*, specific cutaneous lesions result from a vasculitis of arterial vessels rather than postcapillary venules as in LCV. The arteritis leads to ischemia of the skin, and this explains the irregular outline of the purpura (see below).

Several types of infectious emboli can give rise to palpable purpura. These embolic lesions are usually *irregular* in outline as opposed to the lesions of leukocytoclastic vasculitis, which are *circular* in outline. The irregular outline is indicative of a cutaneous infarct, and the size corresponds to the area of skin that received its blood supply from that particular arteriole or artery. The palpable purpura in [LCV](#) are circular because the erythrocytes simply diffuse out evenly from the postcapillary venules as a result of inflammation. Infectious emboli are most commonly due to gram-negative cocci (meningococcus, gonococcus), gram-negative rods (Enterobacteriaceae), and gram-positive cocci (staphylococcus). Additional causes include *Rickettsia* and, in immunocompromised patients, *Candida* and opportunistic fungi.

The embolic lesions in *acute meningococcemia* are found primarily on the trunk, lower extremities, and sites of pressure, and a gunmetal-gray color often develops within them. Their size varies from 1 mm to several centimeters, and the organisms can be cultured from the lesions. Associated findings include a preceding upper respiratory tract infection, fever, meningitis, [DIC](#), and, in some patients, a deficiency of the terminal components of complement. In *disseminated gonococcal infection* (arthritis-dermatitis syndrome), a small number of papules and vesicopustules with central purpura or hemorrhagic necrosis are found over the joints of the distal extremities. Additional symptoms include arthralgias, tenosynovitis, and fever. To establish the diagnosis, a Gram stain of these lesions should be performed. *Rocky mountain spotted fever* is a tick-borne disease that is caused by *R. rickettsii*. A several-day history of fever, chills, severe headache, and photophobia precedes the onset of the cutaneous eruption. The initial lesions are erythematous macules and papules on the wrists, ankles, palms, and soles. With time, the lesions spread centripetally and become purpuric.

Lesions of *ecthyma gangrenosum* begin as edematous, erythematous papules or plaques and then develop central purpura and necrosis. Bullae formation also occurs in these lesions, and they are frequently found in the girdle region. The organism that is classically associated with ecthyma gangrenosum is *Pseudomonas aeruginosa*, but other gram-negative rods such as *Klebsiella*, *Escherichia coli*, and *Serratia* can produce similar lesions. In immunocompromised hosts, the list of potential pathogens is expanded to include *Candida* and opportunistic fungi.

ULCERS

The approach to the patient with a cutaneous ulcer, is outlined in [Table 57-17](#). **Peripheral vascular diseases of the extremities are reviewed in Chap. 248, as is Raynaud's phenomenon.*

Livedoid vasculitis (atrophie blanche) represents a combination of a vasculopathy with intravascular thrombosis. Purpuric lesions and livedo reticularis are found in association with painful ulcerations of the lower extremities. These ulcers are often slow to heal, but

when they do, irregularly shaped white scars are formed. The majority of cases are secondary to venous hypertension, but possible underlying illnesses include cryofibrinogenemia and disorders of hypercoagulability, e.g., the antiphospholipid syndrome ([Chap. 117](#)).

In *pyoderma gangrenosum*, the border of the ulcers has a characteristic appearance of an undermined necrotic bluish edge and a peripheral erythematous halo. The ulcers often begin as pustules that then expand rather rapidly to a size as large as 20 cm. Although these lesions are most commonly found on the lower extremities, they can arise anywhere on the surface of the body, including sites of trauma (pathergy). An estimated 30 to 50% of cases are idiopathic, and the most common associated disorders are ulcerative colitis and Crohn's disease. Less commonly, it is associated with chronic active hepatitis, seropositive rheumatoid arthritis, acute and chronic granulocytic leukemia, polycythemia vera, and myeloma. Additional findings in these patients, even those with idiopathic disease, are cutaneous anergy and a benign monoclonal gammopathy. Because the histology of pyoderma gangrenosum is nonspecific, the diagnosis is made clinically by excluding less common causes of similar-appearing ulcers such as necrotizing vasculitis, Meleney's ulcer (synergistic infection at a site of trauma or surgery), dimorphic fungi, cutaneous amebiasis, spider bites, and factitial. In the myeloproliferative disorders, the ulcers may be more superficial with a pustulobullous border, and these lesions provide a connection between classic pyoderma gangrenosum and acute febrile neutrophilic dermatosis (Sweet's syndrome).

FEVER AND RASH

The major considerations in a patient with a fever and a rash are inflammatory diseases versus infectious diseases. In the hospital setting, the most common scenario is a patient who has a drug rash plus a fever secondary to an underlying infection. However, it should be emphasized that a drug reaction can lead to both a cutaneous eruption and a fever ("drug fever"). Additional inflammatory diseases that are often associated with a fever include pustular psoriasis, erythroderma, and Sweet's syndrome. Lyme disease, secondary syphilis, and viral and bacterial exanthems (see "Exanthems") are examples of infectious diseases that produce a rash and a fever. Lastly, it is important to determine whether or not the cutaneous lesions represent septic emboli (see "Purpura"). Such lesions usually have evidence of ischemia in the form of purpura, necrosis, or impending necrosis (gunmetal-gray color). In the patient with thrombocytopenia, however, purpura can be seen in inflammatory reactions such as morbilliform drug eruptions and infectious lesions.

(Bibliography omitted in Palm version)

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58. IMMUNOLOGICALLY MEDIATED SKIN DISEASES - Kim B. Yancey, Thomas J. Lawley

A number of immunologically mediated skin diseases and cutaneous manifestations of immunologically mediated systemic disorders are now recognized as distinct entities with relatively consistent clinical, histologic, and immunopathologic findings. Many of these disorders are due to autoimmune mechanisms. Clinically, they are characterized by morbidity (pain, pruritus, disfigurement) and in some instances by mortality (largely due to loss of epidermal barrier function and/or secondary infection). The major features of the more common immunologically mediated skin diseases are summarized in this chapter ([Table 58-1](#)).

PEMPHIGUS VULGARIS

Pemphigus vulgaris (PV) is a blistering skin disease seen predominantly in elderly patients. Patients with PV have an increased incidence of the HLA-DR4 and -DRw6 serologically defined haplotypes. This disorder is characterized by the loss of cohesion between epidermal cells (a process termed *acantholysis*) with the resultant formation of intraepidermal blisters. Clinical lesions of PV typically consist of flaccid blisters on either normal-appearing or erythematous skin. These blisters rupture easily, leaving denuded areas that may crust and enlarge peripherally (Plate IIE-69). Substantial portions of the body surface may be denuded in severe cases. Manual pressure to the skin of these patients may elicit the separation of the epidermis (Nikolsky's sign). This finding, while characteristic of PV, is not specific to this disorder and is also seen in toxic epidermal necrolysis, Stevens-Johnson syndrome, and a few other skin diseases. Lesions in PV typically present on the oral mucosa, scalp, face, neck, axilla, and trunk. In half or more of patients, lesions begin in the mouth; approximately 90% of patients have oromucosal involvement at some time during the course of their disease. Involvement of other mucosal surfaces (e.g., pharyngeal, laryngeal, esophageal, conjunctival, vulval, or rectal) can occur in severe disease. Pruritus may be a feature of early pemphigus lesions; extensive denudation may be associated with severe pain. Lesions usually heal without scarring, except at sites complicated by secondary infection or mechanically induced dermal wounds. Nonetheless, postinflammatory hyperpigmentation is usually present at sites of healed lesions for some time.

Biopsies of early lesions demonstrate intraepidermal vesicle formation secondary to loss of cohesion between epidermal cells (i.e., acantholytic blisters). Blister cavities contain acantholytic epidermal cells, which appear as round homogeneous cells containing hyperchromatic nuclei. Basal keratinocytes remain attached to the epidermal basement membrane, hence blister formation is within the suprabasal portion of the epidermis. Lesional skin may contain focal collections of intraepidermal eosinophils within blister cavities; dermal alterations are slight, often limited to an eosinophil-predominant leukocytic infiltrate. Direct immunofluorescence microscopy of lesional or intact patient skin shows deposits of IgG on the surface of keratinocytes; in contrast, deposits of complement components are typically found in lesional but not uninvolved skin. Deposits of IgG on keratinocytes are derived from circulating autoantibodies directed against cell-surface antigens. Circulating autoantibodies can be demonstrated in 80 to 90% of [PV](#) patients by indirect immunofluorescence microscopy; monkey esophagus is the optimal substrate for demonstration of these autoantibodies. Patients with PV have

IgG autoantibodies directed against desmogleins (Dsgs), transmembrane desmosomal glycoproteins that belong to the cadherin supergene family of calcium-dependent adhesion molecules. While Dsg3 is specifically recognized by PV autoantibodies, approximately 50% of PV sera also contain IgG against Dsg1. Most patients with early PV and only mucosal involvement have only anti-Dsg3 autoantibodies, whereas most patients with advanced disease (i.e., involvement of skin and mucosa) have both anti-Dsg3 and anti-Dsg1 autoantibodies. Recent studies have shown that the anti-Dsg autoantibody profile in these patients' sera as well as the tissue distribution of Dsg3 and Dsg1 determine the site of blister formation in patients with pemphigus. Experimental studies have also shown that these autoantibodies are pathogenic (i.e., responsible for blister formation) and that their titer correlates with disease activity.

[PV](#) can be life-threatening. Prior to the availability of glucocorticoids, the mortality ranged from 60 to 90%; the current mortality is approximately 5%. Common causes of morbidity and mortality are infection and complications of treatment with glucocorticoids. Bad prognostic factors include advanced age, widespread involvement, and the requirement for high doses of glucocorticoids (with or without other immunosuppressive agents) for control of disease. The course of PV in individual patients is variable and difficult to predict. Some patients achieve remission (40% of patients in some series), but others may require long-term treatment or succumb to complications of their disease or its treatment. The mainstay of treatment is systemic glucocorticoids. Patients with moderate to severe disease are usually started on prednisone, 60 to 80 mg/d. If new lesions continue to appear after 1 to 2 weeks of treatment, the dose should be increased. Many regimens combine an immunosuppressive agent with systemic glucocorticoids for control of PV. The two most frequently used are either azathioprine (1 mg/kg per day) or cyclophosphamide (1 mg/kg per day). It is important to bring severe or progressive disease under control quickly to lessen the severity and/or duration of this disorder.

PEMPHIGUS FOLIACEUS

Pemphigus foliaceus (PF) is distinguished from [PV](#) by several features. In PF, acantholytic blisters are located high within the epidermis, usually just beneath the stratum corneum. Hence PF is a more superficial blistering disease than PV. The distribution of lesions in the two disorders is much the same, except that in PF mucous membrane lesions are very rare. Patients with PF rarely demonstrate intact blisters but rather exhibit shallow erosions associated with erythema, scale, and crust formation. Mild cases of PF resemble severe seborrheic dermatitis; severe PF may cause extensive exfoliation. Sun exposure (ultraviolet irradiation) may be an aggravating factor. A blistering skin disease endemic to south central Brazil known as *fogo selvagem*, or *Brazilian pemphigus*, is clinically, histologically, and immunopathologically indistinguishable from PF.

Patients with [PF](#) have immunopathologic features in common with [PV](#). Specifically, direct immunofluorescence microscopy of perilesional skin demonstrates IgG on the surface of keratinocytes. As in PV, patients with PF frequently have circulating IgG autoantibodies against keratinocyte cell surface antigens. Guinea pig esophagus is the optimal substrate for indirect immunofluorescence microscopy studies of sera from patients with PF. In PF, autoantibodies are directed against Dsg1, a 160-kDa desmosomal cadherin.

As noted for PV, the autoantibody profile in patients with PF (i.e., anti-Dsg1) and the normal tissue distribution of this autoantigen (i.e., low expression in oral mucosa) is thought to account for the distribution of lesions in this disease.

Although pemphigus has been associated with several autoimmune diseases, its association with thymoma and/or myasthenia gravis is particularly notable. To date, more than 30 cases of thymoma and/or myasthenia gravis have been reported in association with pemphigus, usually with [PF](#). Patients may also develop pemphigus as a consequence of drug exposure. The most frequently implicated agent is penicillamine; other offenders include captopril, rifampin, piroxicam, penicillin, and phenobarbital. Drug-induced pemphigus usually resembles PF rather than [PV](#); autoantibodies in these patients have the same antigenic specificity as they do in other pemphigus patients. In most patients, lesions resolve following discontinuation of the drug; however, some patients require treatment with systemic glucocorticoids and/or immunosuppressive agents.

[PF](#) is generally a far less severe disease than [PV](#) and carries a better prognosis. Localized disease can be treated conservatively with topical or intralesional glucocorticoids; more active cases can usually be controlled with systemic glucocorticoids.

PARANEOPLASTIC PEMPHIGUS

Paraneoplastic pemphigus (PNP) is an autoimmune acantholytic mucocutaneous disease associated with an occult or confirmed neoplasm. Patients with PNP typically show painful mucosal erosive lesions in association with pruritic papulosquamous eruptions that often progress to blisters. Palm and sole involvement is common in these patients and raises the possibility that prior reports of neoplasia-associated erythema multiforme actually may have represented unrecognized cases of PNP. Biopsies of lesional skin from these patients show varying combinations of acantholysis, keratinocyte necrosis, and vacuolar-interface dermatitis. Direct immunofluorescence microscopy of patient skin shows deposits of IgG and complement on the surface of keratinocytes and (variably) similar immunoreactants in the epidermal basement membrane zone. Patients with PNP have IgG autoantibodies against cytoplasmic proteins that are members of the plakin family (e.g., desmoplakins I and II, bullous pemphigoid antigen 1, envoplakin, periplakin, and plectin) and cell-surface proteins that are members of the cadherin family (e.g., Dsg3 and Dsg1). Because immunoadsorption of anti-Dsg3 IgG is sufficient to eliminate the ability of PNP sera to induce blisters in an experimental passive transfer animal model, these particular autoantibodies are thought to play a key pathogenic role in blister formation in these patients.

Although PNP is generally resistant to conventional therapies (i.e., those used to treat PV), patients may improve (or even remit) following resection of underlying neoplasms. The predominant neoplasms associated with this disorder are non-Hodgkin's lymphoma, chronic lymphocytic leukemia, Castleman's disease, thymoma, and spindle cell tumors.

BULLOUS PEMPHIGOID

Bullous pemphigoid (BP) is an autoimmune subepidermal blistering disease usually

seen in the elderly. Lesions typically consist of tense blisters on either normal-appearing or erythematous skin ([Plate IIE-72](#)). The lesions are usually distributed over the lower abdomen, groin, and flexor surface of the extremities; oral mucosal lesions are found in 10 to 40% of patients. Pruritus may be nonexistent or severe. As lesions evolve, tense blisters tend to rupture and be replaced by flaccid lesions or erosions with or without surmounting crust. Nontraumatized blisters heal without scarring. The major histocompatibility complex class II allele HLA-DQb1*0301 is prevalent in patients with BP. Despite isolated reports, several studies have shown that patients with BP do not have an increased incidence of malignancy in comparison with appropriately age- and sex-matched controls.

While biopsies of early lesional skin demonstrate subepidermal blisters, the histologic features depend on the character of the particular lesion. Lesions on normal-appearing skin generally show a sparse perivascular leukocytic infiltrate with some eosinophils; conversely, biopsies of inflammatory lesions typically show an eosinophil-rich infiltrate within the papillary dermis at sites of vesicle formation and in perivascular areas. In addition to eosinophils, cell-rich lesions also contain mononuclear cells and neutrophils. It is not always possible to distinguish [BP](#) from other subepidermal blistering diseases by routine histologic techniques.

Immunopathologic studies have broadened our understanding of this disease and aided its diagnosis. Direct immunofluorescence microscopy of normal-appearing perilesional skin shows linear deposits of IgG and/or C3 in the epidermal basement membrane. The sera of approximately 70% of these patients contain circulating IgG autoantibodies that bind the epidermal basement membrane of normal human skin in indirect immunofluorescence microscopy. An even higher percentage of patients shows reactivity to the epidermal side of 1 M NaCl split skin [an alternative immunofluorescence microscopy test substrate that is commonly used to distinguish circulating IgG anti-basement membrane autoantibodies in patients with [BP](#) from those in patients with similar, yet different, subepidermal blistering diseases (e.g., epidermolysis bullosa acquisita, see below)]. No correlation exists between the titer of these autoantibodies and disease activity. In BP, circulating autoantibodies recognize 230- and (in approximately 70% of BP patients) 180-kDa hemidesmosome-associated proteins in basal keratinocytes [i.e., bullous pemphigoid antigen (BPAG)1 and BPAG2, respectively]. Autoantibodies are thought to develop against these antigens (more specifically, initially against BPAG2), deposit in situ, and activate complement that subsequently produces dermal mast cell degranulation and granulocyte-rich infiltrates that cause tissue damage and blister formation.

[BP](#) may persist for months to years, with exacerbations or remissions. Although extensive involvement may result in widespread erosions and compromise cutaneous integrity, the mortality rate is low even in the absence of treatment. Nonetheless, deaths may occur in elderly and/or debilitated patients. The mainstay of treatment is systemic glucocorticoids. Patients with local or minimal disease can sometimes be controlled with topical glucocorticoids alone; patients with more extensive lesions generally respond to systemic glucocorticoids either alone or in combination with immunosuppressive agents. Patients will usually respond to prednisone, 40 to 60 mg/d. In some instances, azathioprine (1 mg/kg per day) or cyclophosphamide (1 mg/kg per day) are necessary adjuncts.

PEMPHIGOID GESTATIONIS

Pemphigoid gestationis (PG), also known as herpes gestationis, is a rare, nonviral, subepidermal blistering disease of pregnancy and the puerperium. PG may begin during any trimester of pregnancy or present shortly after delivery. Lesions are usually distributed over the abdomen, trunk, and extremities; mucous membrane lesions are rare. Skin lesions in these patients may be quite polymorphic and consist of erythematous urticarial papules and plaques, vesiculopapules, and/or frank bullae. Lesions are almost always very pruritic. Severe exacerbations of PG frequently occur after delivery, typically within 24 to 48 h. PG tends to recur in subsequent pregnancies, often beginning earlier during such gestations. Brief flare-ups of disease may occur with resumption of menses and may develop in patients later exposed to oral contraceptives. Occasionally, infants of affected mothers demonstrate transient skin lesions.

Biopsies of early lesional skin show teardrop-shaped subepidermal vesicles forming in dermal papillae in association with an eosinophil-rich leukocytic infiltrate. Differentiation of [PG](#) from other subepidermal bullous diseases by light microscopy is often difficult. However, direct immunofluorescence microscopy of perilesional skin from PG patients reveals the immunopathologic hallmark of this disorder -- linear deposits of C3 in the epidermal basement membrane zone. These deposits develop as a consequence of complement activation produced by low titer IgG anti-basement membrane zone autoantibodies. Recent studies have shown that the majority of PG sera contain autoantibodies that recognize [BPAG2](#), the same 180-kDa hemidesmosome-associated protein that is targeted by autoantibodies in roughly 70% of patients with [BP](#) -- a subepidermal bullous disease that resembles PG morphologically, histologically, and immunopathologically.

The goals of therapy in patients with [PG](#) are to prevent the development of new lesions, relieve intense pruritus, and care for erosions at sites of blister formation. Most patients require treatment with moderate doses of daily glucocorticoids (i.e., 20 to 40 mg of prednisone) at some point in their course. Mild cases (or brief flare-ups) may be controlled by vigorous use of potent topical glucocorticoids. Although PG was once thought to be associated with an increased risk of fetal morbidity and mortality, the best evidence now suggests that these infants may only be at increased risk of being slightly premature or "small for dates." Current evidence suggests that there is no difference in the incidence of uncomplicated live births in PG patients treated with systemic glucocorticoids and in those managed more conservatively. If systemic glucocorticoids are administered, newborns are at risk for development of reversible adrenal insufficiency.

DERMATITIS HERPETIFORMIS

Dermatitis herpetiformis (DH) is an intensely pruritic, papulovesicular skin disease characterized by lesions symmetrically distributed over extensor surfaces (i.e., elbows, knees, buttocks, back, scalp, and posterior neck) ([Plate IIE-68](#)). The primary lesion in this disorder is a papule, papulovesicle, or urticarial plaque. Because pruritus is prominent, patients may present with excoriations and crusted papules but no observable primary lesions. Patients sometimes report that their pruritus has a

distinctive burning or stinging component; the onset of such local symptoms reliably heralds the development of distinct clinical lesions 12 to 24 h later. Almost all DH patients have an associated, usually subclinical, gluten-sensitive enteropathy ([Chap. 286](#)), and more than 90% express the HLA-B8/DRw3 and HLA-DQw2 haplotypes. DH may present at any age, including childhood; onset in the second to fourth decades is most common. The disease is typically chronic.

Biopsy of early lesional skin reveals neutrophil-rich infiltrates within dermal papillae. Neutrophils, fibrin, edema, and microvesicle formation at these sites are characteristic of early disease. Older lesions may demonstrate nonspecific features of a subepidermal bulla or an excoriated papule. Because the clinical and histologic features of this disease can be variable and resemble other subepidermal blistering disorders, the diagnosis is confirmed by direct immunofluorescence microscopy of normal-appearing perilesional skin. Such studies demonstrate granular deposits of IgA (with or without complement components) in the papillary dermis and along the epidermal basement membrane zone. IgA deposits in the skin are unaffected by control of disease with medication; however, these immunoreactants may diminish in intensity or disappear in patients maintained for long periods on a strict gluten-free diet (see below). Patients with granular deposits of IgA in their epidermal basement membrane zone typically do not have circulating IgA anti-basement membrane autoantibodies and should be distinguished from individuals with linear IgA deposits at this site (see below).

Although most [DH](#) patients do not report overt gastrointestinal symptoms or laboratory evidence of malabsorption, biopsies of small bowel usually reveal blunting of intestinal villi and a lymphocytic infiltrate in the lamina propria. As is true for patients with celiac disease, this gastrointestinal abnormality can be reversed by a gluten-free diet. Moreover, if maintained, this diet alone may control the skin disease and eventuate in clearance of IgA deposits from these patients' epidermal basement membrane zone. Subsequent gluten exposure in such patients alters the morphology of their small bowel, elicits a flare-up of their skin disease, and is associated with the reappearance of IgA in their epidermal basement membrane zone. Additional evidence that DH develops as a consequence of dietary gluten exposure is the demonstration of IgA anti-endomysial antibodies in these patients' sera (as found in the sera of patients with ordinary gluten-sensitive enteropathy). Recent studies have shown that such autoantibodies are directed against tissue transglutaminase. Patients with DH also have an increased incidence of thyroid abnormalities, achlorhydria, atrophic gastritis, and antigastric parietal cell antibodies. These associations likely relate to the high frequency of the HLA-B8/DRw3 haplotype in these patients, since this marker is commonly linked to autoimmune disorders. The mainstay of treatment of DH is dapsone, a sulfone. Patients respond rapidly (24 to 48 h) to dapsone but require careful pretreatment evaluation and close follow-up to ensure that complications are avoided or controlled. All patients on more than 100 mg/d dapsone will have some hemolysis and methemoglobinemia. These are expected pharmacologic side effects of this agent. Gluten restriction can control DH and lessen dapsone requirements; this diet must rigidly exclude gluten to be of maximal benefit. Many months of dietary restriction may be necessary before a beneficial result is achieved. Good dietary counselling by a trained dietitian is essential.

LINEAR IGA DISEASE

Linear IgA disease, once considered a variant form of dermatitis herpetiformis, is actually a separate and distinct entity. Clinically, these patients may resemble patients with typical cases of [DH](#), [BP](#), or other subepidermal blistering diseases. Lesions typically consist of papulovesicles, bullae, and/or urticarial plaques, predominantly on extensor (as seen in "classic" DH), central, or flexural sites. Oral mucosal involvement occurs in some patients. Severe pruritus resembles that in patients with DH. Patients with linear IgA disease do not have an increased frequency of the HLA-B8/DRw3 haplotype or an associated enteropathy and hence are not candidates for a gluten-free diet.

The histologic alterations in early lesions may be virtually indistinguishable from those in [DH](#). However, direct immunofluorescence microscopy of normal-appearing perilesional skin reveals linear deposits of IgA (and often C3) in the epidermal basement membrane zone. Most patients with linear IgA disease demonstrate circulating IgA anti-basement membrane autoantibodies against epitopes in the extracellular domain of [BPAG2](#), a transmembrane protein found in hemidesmosomes of basal keratinocytes. These patients generally respond to treatment with dapsone, 50 to 150 mg/d.

EPIDERMOLYSIS BULLOSA ACQUISITA

[EBA](#) is a rare, noninherited, polymorphic, subepidermal blistering disease. (The inherited form is discussed in [Chap. 351](#).) Patients with classic or noninflammatory EBA have blisters on noninflamed skin, atrophic scars, milia, nail dystrophy, and oral lesions. Because lesions generally occur at sites exposed to minor trauma, classic EBA is considered to be a mechanobullous disease. Other patients with EBA have widespread inflammatory, scarring, bullous lesions and oromucosal involvement that resembles severe [BP](#). Some patients present with an inflammatory bullous disease that evolves into the classic noninflammatory form of this disorder. In general, EBA is chronic; associations with multiple myeloma, amyloidosis, inflammatory bowel disease, and diabetes mellitus have been reported. The HLA-DR2 haplotype is found with increased frequency in these patients.

The histology of lesional skin varies depending on the character of the lesion being studied. Noninflammatory bullae show subepidermal blisters with a sparse leukocytic infiltrate and resemble those in patients with porphyria cutanea tarda. Inflammatory lesions consist of a subepidermal blister and neutrophil-rich leukocytic infiltrates in the superficial dermis. [EBA](#) patients have continuous deposits of IgG (and frequently C3 as well as other complement components) in a linear pattern within the epidermal basement membrane zone. Ultrastructurally, these immunoreactants are found in the sublamina densa region in association with anchoring fibrils, wheat stack-like structures that extend from the lamina densa into the underlying papillary dermis. Approximately 25 to 50% of EBA patients have circulating IgG anti-basement membrane autoantibodies directed against type VII collagen -- the collagen species that comprises anchoring fibrils. Such IgG autoantibodies bind the dermal side of 1 M NaCl split skin (in contrast to IgG autoantibodies in patients with [BP](#) that bind either epidermal or both sides of this indirect immunofluorescence microscopy test substrate).

Treatment of [EBA](#) is generally unsatisfactory. Some patients with inflammatory EBA may respond to systemic glucocorticoids, either alone or in combination with immunosuppressive agents. Other patients (especially those with neutrophil-rich

inflammatory lesions) may respond to dapsone. The chronic, noninflammatory form of this disease is largely resistant to treatment, although some patients may respond to cyclosporine.

CICATRICAL PEMPHIGOID

Cicatricial pemphigoid (CP) is a rare, acquired, subepithelial blistering disease characterized by erosive lesions of mucous membranes and skin that result in scarring of at least some sites of involvement. Immunopathologically, perilesional mucosa and skin of patients with CP demonstrate in situ deposits of immunoreactants in epithelial basement membranes. Common sites of involvement include the oral mucosa (especially the gingiva) and conjunctiva; other sites that may be affected include the nasopharyngeal, laryngeal, esophageal, urogenital, and rectal mucosa. Skin lesions (present in about one-third of patients) tend to predominate on the scalp, face, and upper trunk and generally consist of a few scattered erosions or tense blisters on an erythematous or urticarial base. CP is typically a chronic and progressive disorder. Serious complications may arise as a consequence of ocular, laryngeal, esophageal, or urogenital lesions. Erosive conjunctivitis may result in shortened fornices, symblephara, ankyloblepharon, entropion, corneal opacities, and (in severe cases) blindness. Similarly, erosive lesions of the larynx may cause hoarseness, pain, and tissue loss that if unrecognized and untreated may eventuate in complete destruction of the airway. Esophageal lesions may result in stenosis and/or strictures that may place patients at risk for aspiration. Strictures may also complicate urogenital involvement.

Biopsies of lesional tissue generally demonstrate subepithelial vesiculobullae and a mononuclear leukocytic infiltrate. Neutrophils and eosinophils may be seen in biopsies of early lesions; older lesions may demonstrate a scant leukocytic infiltrate and fibrosis. Direct immunofluorescence microscopy of perilesional tissue typically demonstrates deposits of IgG, IgA, and/or C3 in these patients' epithelial basement membranes. Because many of these patients show no evidence of circulating anti-basement membrane autoantibodies, testing of perilesional skin is important diagnostically. Although [CP](#) was once thought to be a single nosologic entity, it is now largely regarded as a disease phenotype that may develop as a consequence of an autoimmune reaction against a variety of different molecules in epithelial basement membranes (e.g., [BPAG2](#), laminin 5, type VII collagen, and other antigens yet to be completely defined). Treatment of CP is largely dependent upon sites of involvement. Due to potentially severe complications, ocular, laryngeal, esophageal, and/or urogenital involvement require aggressive systemic treatment with dapsone, prednisone, or the latter in combination with another immunosuppressive agent (e.g., azathioprine or cyclophosphamide). Less threatening forms of the disease may be managed with topical or intralesional glucocorticoids.

AUTOIMMUNE SYSTEMIC DISEASES WITH PROMINENT CUTANEOUS FEATURES

DERMATOMYOSITIS

The cutaneous manifestations of dermatomyositis ([Chap. 382](#)) are often distinctive but at times may resemble those of systemic lupus erythematosus (SLE) ([Chap. 311](#)), scleroderma ([Chap. 313](#)), or other overlapping connective tissue diseases ([Chap. 313](#)).

The extent and severity of cutaneous disease may or may not correlate with the extent and severity of the myositis. Patients with severe muscle involvement may have relatively minor skin changes, whereas patients with marked skin involvement may have mild muscle disease. The cutaneous manifestations of dermatomyositis are similar whether the disease appears in childhood or old age, except that calcification of subcutaneous tissue is a common late sequela in childhood dermatomyositis.

The cutaneous signs of dermatomyositis may precede or follow the development of myositis by weeks to years. Cases lacking muscle involvement (i.e., dermatomyositis sine myositis) have also been reported. The most common manifestation is a purple-red discoloration of the upper eyelids, sometimes associated with scaling ("heliotrope" erythema; [Plate IIE-63](#)) and periorbital edema. Erythema on the cheeks and nose in a "butterfly" distribution may resemble the eruption in [SLE](#). Erythematous or violaceous scaling patches are common on the upper anterior chest, posterior neck, scalp, and the extensor surfaces of the arms, legs, and hands. Erythema and scaling may be particularly prominent over the elbows, knees, and the dorsal interphalangeal joints. Approximately one-third of patients have violaceous, flat-topped papules over the dorsal interphalangeal joints that are pathognomonic of dermatomyositis (Gottron's sign or Gottron's papules; [Plate IIE-65](#)). These lesions can be contrasted with the erythema and scaling on the dorsum of the fingers in some patients with SLE, which spares the skin over the interphalangeal joints. Periungual telangiectasia may be prominent, and a lacy or reticulated erythema may be associated with fine scaling on the extensor surfaces of the thighs and upper arms. Other patients, particularly those with long-standing disease, develop areas of hypopigmentation, hyperpigmentation, mild atrophy, and telangiectasia known as *poikiloderma vasculare atrophicans*. Poikiloderma is rare in both SLE and scleroderma and thus can serve as a clinical sign that distinguishes dermatomyositis from these two diseases. Cutaneous changes may be similar in scleroderma and dermatomyositis and may include thickening and binding down of the skin of the hands (sclerodactyly) as well as Raynaud's phenomenon. However, the presence of severe muscle disease, Gottron's papules, heliotrope erythema, and poikiloderma serve to distinguish patients with dermatomyositis. Skin biopsy of erythematous, scaling lesions of dermatomyositis may reveal only mild nonspecific inflammation but sometimes may show changes indistinguishable from those found in SLE, including epidermal atrophy, hydropic degeneration of basal keratinocytes, edema of the upper dermis, and a mild mononuclear cell infiltrate. Direct immunofluorescence microscopy of lesional skin is usually negative, although granular deposits of immunoglobulin(s) and complement in the epidermal basement membrane zone have been described in some patients. Treatment should be directed at the systemic disease. In the few instances where adjunctive cutaneous therapy is desirable, topical glucocorticoids are sometimes useful. These patients should avoid exposure to ultraviolet irradiation and use photoprotective measures such as sunscreens.

LUPUS ERYTHEMATOSUS

The cutaneous manifestations of lupus erythematosus (LE) ([Chap. 311](#)) can be divided into acute, subacute, and chronic (i.e., discoid LE) types. *Acute cutaneous LE* is characterized by erythema of the nose and malar eminences in a "butterfly" distribution ([Plate IIE-61](#)). The erythema is often sudden in onset, accompanied by edema and fine scale, and correlated with systemic involvement. Patients may have widespread

involvement of the face as well as erythema and scaling of the extensor surfaces of the extremities and upper chest. These acute lesions, while sometimes evanescent, usually last for days and are often associated with exacerbations of systemic disease. Skin biopsy of acute lesions may show only a sparse dermal infiltrate of mononuclear cells and dermal edema. In some instances, cellular infiltrates around blood vessels and hair follicles are notable, as is hydropic degeneration of basal cells of the epidermis. Direct immunofluorescence microscopy of lesional skin frequently reveals deposits of immunoglobulin(s) and complement in the epidermal basement membrane zone. Treatment is aimed at control of systemic disease; photoprotection in this, as well as in other forms of LE, is very important.

Subacute cutaneous lupus erythematosus (SCLE) is characterized by a widespread photosensitive, nonscarring eruption. About half of these patients have [SLE](#) in which severe renal and central nervous system involvement is uncommon. SCLE may present as a papulosquamous eruption that resembles psoriasis or annular lesions that resemble those seen in erythema multiforme. In the papulosquamous form, discrete erythematous papules arise on the back, chest, shoulders, extensor surfaces of the arms, and the dorsum of the hands; lesions are uncommon on the face, flexor surfaces of the arms, and below the waist. The slightly scaling papules tend to merge into large plaques, some with a reticulate appearance. The annular form involves the same areas and presents with erythematous papules that evolve into oval, circular, or polycyclic lesions. The lesions of SCLE are more widespread but have less tendency for scarring than do lesions of discoid [LE](#). Skin biopsy reveals a dense mononuclear cell infiltrate around hair follicles and blood vessels in the superficial dermis, combined with hydropic degeneration of basal cells in the epidermis. Direct immunofluorescence microscopy of lesional skin reveals deposits of immunoglobulin(s) in the epidermal basement membrane zone in about half these cases. A particulate pattern of IgG deposition around basal keratinocytes has recently been associated with SCLE. Most SCLE patients have anti-Ro antibodies. Local therapy is usually unsuccessful, and most patients require treatment with aminoquinoline antimalarials. Low-dose therapy with oral glucocorticoids is sometimes necessary; photoprotective measures against both ultraviolet B and A wavelengths are very important.

Discoid lupus erythematosus (DLE) is characterized by discrete lesions, most often on the face, scalp, or external ears. The lesions are erythematous papules or plaques with a thick, adherent scale that occludes hair follicles (follicular plugging). When the scale is removed, its underside will show small excrescences that correlate with the openings of hair follicles and is termed a "carpet tack" appearance. This finding is relatively specific for DLE. Long-standing lesions develop central atrophy, scarring, and hypopigmentation but frequently have erythematous, sometimes raised borders at the periphery ([Plate IIE-62](#)). These lesions persist for years and tend to expand slowly. Only 5 to 10% of patients with DLE meet the American Rheumatism Association criteria for [SLE](#). However, typical discoid lesions are frequently seen in patients with SLE. Biopsy of DLE lesions shows hyperkeratosis, follicular plugging, and atrophy of the epidermis. The dermal-epidermal junction reveals hydropic degeneration of basal keratinocytes, and a mononuclear cell infiltrate surrounding hair follicles and blood vessels. Direct immunofluorescence microscopy demonstrates immunoglobulin(s) and complement deposits at the basement membrane zone in about 90% of cases. Treatment is focused on control of local cutaneous disease and consists mainly of photoprotection and topical

or intralesional glucocorticoids. If local therapy is ineffective, use of aminoquinoline antimalarials may be indicated.

SCLERODERMA AND MORPHEA

The skin changes of scleroderma ([Chap. 313](#)) usually begin on the hands, feet, and face, with episodes of recurrent nonpitting edema. Sclerosis of the skin begins distally on the fingers (sclerodactyly) and spreads proximally, usually accompanied by resorption of bone of the fingertips, which may have punched out ulcers, stellate scars, or areas of hemorrhage ([Plate IIE-66](#)). The fingers may actually shrink in size and become sausage-shaped, and since the fingernails are usually unaffected, the nails may curve over the end of the fingertips. Periungual telangiectasias are usually present, but periungual erythema is rare. In advanced cases, the extremities show contractures and calcinosis cutis. Face involvement includes a smooth, unwrinkled brow, taut skin over the nose, shrinkage of tissue around the mouth, and perioral radial furrowing ([Plate IIE-64](#)). Matlike telangiectasias are often present, particularly on the face and hands. Involved skin feels indurated, smooth, and bound to underlying structures; hyperpigmentation and hypopigmentation are also often present. Raynaud's phenomenon, i.e., cold-induced blanching, cyanosis, and reactive hyperemia, is present in almost all patients and can precede development of scleroderma by many years. The combination of calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia has been termed the *CREST syndrome*. Anticentromere antibodies have been reported in a very high percentage of patients with the CREST syndrome but in only a small minority of patients with scleroderma. Skin biopsy reveals thickening of the dermis and homogenization of collagen bundles. Direct immunofluorescence microscopy of lesional skin is usually negative.

Morphea, which has been called *localized scleroderma*, is characterized by localized thickening and sclerosis of skin, usually affecting young adults or children. Morphea begins as erythematous or flesh-colored plaques that become sclerotic, develop central hypopigmentation, and demonstrate an erythematous border. In most cases, patients have one or a few lesions, and the disease is termed *localized morphea*. In some patients, widespread cutaneous lesions may occur, without systemic involvement. This form is called *generalized morphea*. Most patients with morphea do not have autoantibodies. Skin biopsy of morphea is indistinguishable from that of scleroderma. Linear scleroderma is a limited form of disease that presents in a linear, bandlike distribution and tends to involve deep as well as superficial layers of skin. Scleroderma and morphea are usually quite resistant to therapy. For this reason, physical therapy to prevent joint contractures and to maintain function is employed and is often helpful.

Diffuse fasciitis with eosinophilia is a clinical entity that can sometimes be confused with scleroderma. There is usually the sudden onset of swelling, induration, and erythema of the extremities frequently following significant physical exertion. The proximal portions of extremities (arms, forearms, thighs, legs) are more often involved than are the hands and feet. While the skin is indurated, it is usually not bound down as in scleroderma; contractures may occur early secondary to fascial involvement. The latter may also cause muscle groups to be separated (i.e., the "groove sign") and veins to appear depressed (i.e., sunken veins). These skin findings are accompanied by peripheral blood eosinophilia, increased erythrocyte sedimentation rate, and sometimes

hypergammaglobulinemia. Deep biopsy of affected areas of skin reveals inflammation and thickening of the deep fascia overlying muscle. An inflammatory infiltrate composed of eosinophils and mononuclear cells is usually found. Patients with eosinophilic fasciitis appear to be at increased risk to develop bone marrow failure or other hematologic abnormalities. While the ultimate course of eosinophilic fasciitis is uncertain, many patients respond favorably to treatment with prednisone in doses ranging from 40 to 60 mg/d.

The *eosinophilia-myalgia syndrome*, a disorder reported in epidemic numbers in 1989 and linked to ingestion of L-tryptophan manufactured by a single company in Japan, is a multisystem disorder characterized by debilitating myalgias and absolute eosinophilia in association with varying combinations of arthralgias, pulmonary symptoms, and peripheral edema. In a later phase (i.e., 3 to 6 months after initial symptoms), these patients often develop localized sclerodermatous skin changes, weight loss, and/or neuropathy ([Chap. 313](#)). The precise cause of this syndrome, which may resemble other sclerotic skin conditions, is unknown. However, the implicated lots of L-tryptophan contained the contaminant 1,1-ethylidene bis[tryptophan]. This contaminant may be pathogenic or a marker for another substance that provokes the disorder.

(Bibliography omitted in Palm version)

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59. CUTANEOUS DRUG REACTIONS - Robert S. Stern, Olivier M. Chosidow, Bruce U. Wintroub

Cutaneous reactions are among the most frequent adverse reactions to drugs. Prompt recognition of these reactions, drug withdrawal, and appropriate therapeutic interventions can minimize toxicity. This chapter focuses on adverse cutaneous reactions to drugs other than topical agents and reviews the incidence, patterns, and pathogenesis of cutaneous reactions to drugs and other therapeutic agents.

USE OF PRESCRIPTION DRUGS IN THE UNITED STATES

More than 1.5 billion prescriptions for 60,000 drug products, which include over 2000 different active agents, are dispensed each year in the United States. Hospital inpatients alone annually receive about 120 million courses of drug therapy, and half of adult Americans receive prescription drugs on a regular outpatient basis. Many additional patients use over-the-counter medicines that may cause adverse cutaneous reactions.

INCIDENCE OF CUTANEOUS REACTIONS

Although adverse drug reactions are common, it is difficult to ascertain their incidence, seriousness, and ultimate health effects. Available information comes from evaluations of hospitalized patients, epidemiologic surveys, premarketing studies, and voluntary reporting, most notably to the U.S. Food and Drug Administration's Medwatch System. None of these efforts provides comprehensive comparable data on the risk of cutaneous reactions associated with most medicines.

In one study about 2% of medical inpatients had skin reactions consisting of rash, urticaria, or pruritus during hospitalization. The overall reaction rate per course of drug therapy was about 3:1000. Among inpatients, penicillins, sulfonamides, and blood products accounted for two-thirds of cutaneous reactions. Among outpatients, reaction rates for many antibiotics were comparable to those observed in inpatients. Fluoroquinolones are notable causes of cutaneous reactions not observed in earlier studies. Reaction rates for selected commonly used antibiotics are summarized in [Table 59-1](#). Most cutaneous reactions occur within 2 weeks of exposure to a drug. The risk of allergic reactions does not vary greatly with age or sex. Among outpatients, the risk of a reaction to an antibiotic was comparable for first and subsequent courses of a given drug.

The distribution of morphologic patterns of drug eruptions cared for within a Finnish hospital dermatology department with a special interest in fixed drug eruptions included exanthematous reactions (32%), urticaria and/or angioedema (20%), fixed drug eruptions (34%), erythema multiforme (2%), Stevens-Johnson syndrome (SJS; 1%), exfoliative dermatitis (1%), and photosensitivity reactions (3%). Other studies suggest that about 80% of all cutaneous reactions are morbilliform or erythematous, 10 to 15% are urticaria or angioedema, and all other types of reactions are relatively rare.

The relative risk of [SJS](#) and toxic epidermal necrolysis (TEN), perhaps the most important severe cutaneous reactions, has been quantified in an international case control study and case series. Sulfonamide antibiotics, allopurinol, amine antiepileptic

drugs (phenytoin and carbamazepine), and lamotrigine (a new antiepileptic) are associated with the highest risk of these reactions.

PATHOGENESIS OF DRUG REACTIONS

Untoward cutaneous responses to drugs can arise as a result of immunologic or nonimmunologic mechanisms. Immunologic reactions require activation of host immunologic pathways and are designated *drug allergy*. Drug reactions occurring through nonimmunologic mechanisms may be due to activation of effector pathways, overdosage, cumulative toxicity, side effects, ecologic disturbance, interactions between drugs, metabolic alterations, exacerbation of preexisting dermatologic conditions, or inherited protein or enzyme deficiencies. It is often not possible to specify the responsible drug or pathogenic mechanism because the skin responds to a variety of stimuli through a limited number of reaction patterns. The mechanism of many drug reactions is unknown.

IMMUNOLOGIC DRUG REACTIONS

Drugs frequently elicit an immune response, but only a small number of individuals experience clinical hypersensitivity reactions. For example, most patients exposed to penicillin develop demonstrable antibodies to penicillin but do not manifest drug reactions when exposed to penicillin. Multiple factors determine the capacity of a drug to elicit an immune response, including the molecular characteristics of the drug and host effects.

Increases in *molecular* size and complexity are associated with increased immunogenicity, and macromolecular drugs such as protein or peptide hormones are highly antigenic. Most drugs are small organic molecules <1000 Da in size, and the capacity of such small molecules to elicit an immune response depends on their ability to act as haptens, i.e., to form stable, usually covalent, bonds with tissue macromolecules, an extremely rare event.

Route of administration of a drug or simple chemical can influence the nature of the *host* immune response. For example, topical application of antigens tends to induce delayed hypersensitivity, and exposure to antigens via oral or nasal cavities stimulates production of secretory immunoglobins, IgA and IgE, and occasionally IgM. Frequency of sensitization through intravenous administration of drugs varies, but anaphylaxis is a more likely consequence with this route of exposure than following oral administration.

The degree of drug exposure and individual variability in absorption and metabolism of a given agent may alter immunogenic load. The variable degree of *in vivo* acetylation of hydralazine provides a clinical example of this phenomenon. Hydralazine produces a lupus-like syndrome associated with antinuclear antibody formation more frequently in patients who acetylate the drug slowly. Frequent high-dose and interrupted courses of therapy are also important risk factors for development of drug allergy.

Pathogenesis of Allergic Drug Reactions> >

> >*IgE-Dependent Reactions* IgE-dependent drug reactions are usually manifest in the

skin and gastrointestinal, respiratory, and cardiovascular systems ([Chap. 310](#)). Primary symptoms and signs include pruritus, urticaria, nausea, vomiting, cramps, bronchospasm, and laryngeal edema and, on occasion, anaphylactic shock with hypotension and death. Immediate reactions may occur within minutes of drug exposure, and accelerated reactions occur hours or days after drug administration. Accelerated reactions are usually urticarial and may include laryngeal edema. Penicillin and related drugs are the most frequent causes of IgE-dependent reactions. Release of chemical mediators such as histamine, adenosine, leukotrienes, prostaglandins, platelet-activating factor, enzymes, and proteoglycans from sensitized tissue, mast cells, or circulating basophilic leukocytes results in vasodilation and edema. Release is triggered when polyvalent drug protein conjugates cross-link IgE molecules fixed to sensitized cells. The clinical manifestations are determined by interaction of the released chemical mediator with its target organ, i.e., skin, respiratory, gastrointestinal, and/or cardiovascular systems. Certain routes of administration favor different clinical patterns (i.e., oral route: gastrointestinal effects; intravenous route: circulatory effects).

Immune-Complex-Dependent Reactions Serum sickness is produced by circulating immune complexes and is characterized by fever, arthritis, nephritis, neuritis, edema, and an urticarial, papular, or purpuric rash ([Chap. 317](#)). The syndrome requires an antigen that remains in the circulation for prolonged periods so that when antibody is synthesized, circulating antigen-antibody complexes are formed. Serum sickness was first described following administration of foreign sera, but drugs are now the usual cause. Drugs that produce serum sickness include the penicillins, sulfonamides, thiouracils, cholecystographic dyes, phenytoin, aminosalicic acid, heparin, and antilymphocyte globulin. Cephalosporin administration in febrile children is associated with a high risk of a clinically similar reaction, but the mechanism of this reaction is unknown. In classic serum sickness, symptoms develop 6 days or more after exposure to a drug, the latent period representing the time needed to synthesize antibody. The antibodies responsible for immune-complex-dependent drug reactions are largely of the IgG or IgM class. Vasculitis, a relatively rare cutaneous complication of drugs, may also be a result of immune complex deposition ([Chap. 317](#)).

Cytotoxicity and Delayed Hypersensitivity Cytotoxicity and delayed hypersensitivity mechanisms may be important in the etiology of morbilliform exanthema, hypersensitivity syndrome, [SJS](#), or [TEN](#), but this is not proven. Systemic manifestations occur frequently. The nature of the antigen leading to cytotoxic reactions is unknown, but it is likely that different T lymphocyte populations are activated. T_H1 type cells will lead to the production of interleukin (IL)-2 and interferon (IFN)- γ and subsequent activation of cytotoxic T cells. In early lesions of morbilliform exanthema or TEN, histopathologic studies have shown expression of HLA-DR and intercellular adhesion molecule (ICAM-1) by keratinocytes, CD4 cells (in the dermis), and CD8 T cells (in the epidermis) and apoptosis of keratinocytes (facilitated by tumor necrosis factor α secretion and *fas*-ligand expression). T_H2 type cells produce cytokines such as IL-5, which may be involved in hypersensitivity syndrome (see below).

NONIMMUNOLOGIC DRUG REACTIONS

Nonimmunologic mechanisms are responsible for the majority of drug reactions; however, only the most important mechanisms will be discussed.

Nonimmunologic Activation of Effector Pathways Drug reactions may result from nonimmunologic activation of effector pathways by three mechanisms: First, drugs may release mediators directly from mast cells and basophils and present as anaphylaxis, urticaria, and/or angioedema. Urticarial anaphylactic reactions induced by opiates, polymyxin B, tubocurarine, radiocontrast media, and dextrans may occur by this mechanism. Second, drugs may activate complement in the absence of antibody. This is an additional mechanism through which radiocontrast media may act. Third, drugs such as aspirin and other nonsteroidal anti-inflammatory agents (NSAIDs) may alter pathways of arachidonic acid metabolism and induce urticaria.

Phototoxicity Phototoxic reactions may be drug-induced or may occur in metabolic disorders in which a photosensitizing chemical is overproduced. A phototoxic reaction occurs when enough chromophore (drug or metabolic product) absorbs sufficient radiation to cause a reaction or interaction with target tissue. Drug-induced phototoxic reactions can occur on first exposure. The incidence of phototoxicity is a direct function of the concentration of sensitizer and the amount of light of the appropriate wavelengths. At least three distinct photochemical mechanisms have been described: (1) the reaction between the excited state of a phototoxic molecule and a biologic target may cause formation of a covalent photoaddition product, (2) the phototoxic molecule may form stable photoproducts that are toxic to biologic substrates, and (3) radiation of a phototoxic molecule may result in transfer of energy to oxygen molecules and cause formation of toxic oxygen species, such as singlet oxygen superoxide anion, or hydroxyl radicals. Interaction of these reactive species with biologic targets produces photooxidized molecules. Phototoxic injury is usually manifest as a sunburn-like reaction.

Exacerbation of Preexisting Diseases A variety of agents can exacerbate preexisting diseases. For example, lithium can exacerbate acne and psoriasis in a dose-dependent manner. Beta-blocking agents and [IFN- \$\alpha\$](#) may induce psoriasis. Withdrawal of glucocorticoids can exacerbate psoriasis or atopic dermatitis.

Inherited Enzyme or Protein Deficiencies Specific genetically determined defects in the ability of an individual to detoxify toxic reactive drug metabolites may predispose such individuals to the development of severe drug reactions, especially hypersensitivity syndrome, and perhaps [TEN](#) associated with use of sulfonamides and anticonvulsants.

Alterations of Immunologic Status Alterations in patients' immunologic status may also modify the risk of cutaneous reactions. Bone marrow transplant patients, HIV-infected persons, and persons with Epstein-Barr virus infection are at higher risk of developing cutaneous reactions to drugs. Skin reactions to trimethoprim-sulfamethoxazole are seen in about a third of HIV-infected users of this drug, but desensitization can be accomplished. Dapsone, trimethoprim alone, and amoxicillin-clavulanate are also frequent causes of drug eruptions in HIV-infected patients. The advent of highly active antiretroviral therapy (HAART) may have decreased the risk of cutaneous reactions in HIV patients ([Chap. 309](#)).

[A CLINICAL CLASSIFICATION OF CUTANEOUS DRUG REACTIONS](#)

URTICARIA/ANGIOEDEMA

Urticaria is a skin reaction characterized by pruritic, red wheals. Lesions may vary from a small point to a large area. Individual lesions rarely last more than 24 h. When deep dermal and subcutaneous tissues are also swollen, this reaction is known as *angioedema*. Angioedema may involve mucous membranes and may be part of a life-threatening anaphylactic reaction. Urticarial lesions, along with pruritus and morbilliform (or maculopapular) eruptions, are among the most frequent types of cutaneous reactions to drugs.

Drug-induced urticaria may be caused by three mechanisms: an IgE-dependent mechanism, circulating immune complexes (serum sickness), and nonimmunologic activation of effector pathways. IgE-dependent urticarial reactions usually occur within 36 h but can occur within minutes. Reactions occurring within minutes to hours of drug exposure are termed *immediate reactions*, whereas those that occur 12 to 36 h after drug exposure are designated *accelerated reactions*. Immune-complex-induced urticaria associated with serum sickness usually occurs from 6 to 12 days after first exposure. In this syndrome, the urticarial eruption may be accompanied by fever, hematuria, arthralgias, hepatic dysfunction, and neurologic symptoms.

Certain drugs, such as [NSAIDs](#), angiotensin-converting enzyme (ACE) inhibitors, and radiographic dyes, may induce urticarial reactions, angioedema, and anaphylaxis in the absence of drug-specific antibody. Although ACE inhibitors, aspirin, penicillin, and blood products are the most frequent causes of urticarial eruptions, urticaria has been observed in association with nearly all drugs. Drugs also may cause chronic urticaria, which lasts more than 6 weeks. Aspirin frequently exacerbates this problem.

The treatment of urticaria or angioedema depends on the severity of the reaction and the rate at which it is evolving. In severe cases, especially with respiratory or cardiovascular compromise, epinephrine is the mainstay of therapy, but its effect is reduced in patients using beta blockers. For more seriously affected patients, treatment with systemic glucocorticoids, sometimes intravenously administered, are helpful. In addition to drug withdrawal, for patients with only cutaneous symptoms and without symptoms of angioedema or anaphylaxis, oral antihistamines are usually sufficient.

PHOTOSENSITIVITY ERUPTIONS

Photosensitivity eruptions are usually most marked in sun-exposed areas but may extend to sun-protected areas. Phototoxic reactions are more common with some drugs. Photoallergic reactions to systemically administered drugs are very rare. Phototoxic reactions usually resemble sunburn and can occur with the first exposure to a drug. Their severity depends on the tissue level of the drug, the extent of exposure to light, and the efficiency of the photosensitizer ([Chap. 60](#)).

Orally administered phototoxic drugs include many fluoroquinolones, chlorpromazine, tetracycline, thiazides, and at least two [NSAIDs](#) (benoxaprofen and piroxicam). The majority of the common phototoxic drugs have action spectrums in the long-wave ultraviolet A (UV-A) range. Phototoxic reactions abate with removal of either the drug or ultraviolet radiation. Because UV-A and visible light, which trigger these reactions, are

not easily absorbed by nonopaque sunscreens and are transmitted through window glass, these reactions may be difficult to block.

Photosensitivity reactions are treated by avoiding exposure to ultraviolet light (sunlight) and treating the reaction as one would a sunburn. Rarely, individuals develop persistent reactivity to light, necessitating long-term avoidance of sun exposure.

PIGMENTATION CHANGES

Drugs may cause a variety of pigmentary changes in the skin. Some drugs stimulate melanocytic activity and increase pigmentation. Drug deposition can also lead to pigmentation; this phenomenon occurs with heavy metals. Phenothiazines may be deposited in the skin and cause a slate-gray color. Antimalarial drugs may cause a slate-gray or yellow pigmentation. Long term minocycline use may cause slate-gray hyperpigmentation, especially in areas of chronic inflammation. Inorganic arsenic, once used to treat psoriasis, is associated with diffuse macular pigmentation. Other heavy metals that cause pigmentary changes include silver, gold, bismuth, and mercury. Long-term use of phenytoin can produce a chloasma-like pigmentation in women. Certain cytostatic agents can also cause pigmentary changes. Histologic examination is often diagnostic for drug deposition diseases.

Zidovudine (AZT) is a frequent cause of pigmentation, especially of the nails ([Chap. 309](#)). Nicotinic acid in large doses may cause brown pigmentation, and oral contraceptives may produce chloasma. In addition, amiodarone may cause violaceous hyperpigmentation that is increased in sun-exposed skin. Drugs such as heavy metals, copper antimalarial and arsenical agents, and ACTH also may discolor oral mucosa.

VASCULITIS

Cutaneous necrotizing vasculitis often presents as palpable purpuric lesions that may be generalized or limited to the lower extremities or other dependent areas ([Chap. 317](#)). Urticarial lesions, ulcers, and hemorrhagic blisters also occur. Vasculitis may involve other organs, including the liver, kidney, brain, and joints. Drugs are only one cause of vasculitis, with infection and collagen vascular disease responsible for the majority of cases.

Propylthiouracil induces a cutaneous vasculitis that is accompanied by leukopenia and splenomegaly. Direct immunofluorescent changes in these lesions suggest immune-complex deposition. Drugs implicated in vasculitic eruptions include allopurinol, thiazides, sulfonamides, penicillin, and some [NSAIDs](#).

HYPERSENSITIVITY SYNDROME

Initially described with phenytoin, hypersensitivity syndrome presents as an erythematous eruption that may become purpuric and is accompanied by many of the following features: fever, facial and periorbital edema, tender generalized lymphadenopathy, leukocytosis (often with atypical lymphocytes and eosinophils), hepatitis, and sometimes nephritis or pneumonitis. The cutaneous reaction usually begins 1 to 6 weeks after phenytoin is begun and usually resolves with drug cessation,

but symptoms, especially hepatitis, may persist. The eruption recurs with rechallenge, and cross-reactions among aromatic anticonvulsants, including phenytoin, carbamazepine, and barbiturates, are frequent. With phenytoin, an increased risk of this syndrome is associated with an inherited deficiency of epoxide hydrolase, an enzyme required for metabolism of a toxic intermediate arene oxide that is formed during metabolism of phenytoin by the cytochrome P450 system. Other drugs causing this syndrome include lamotrigine, dapsone, allopurinol, sulfonamides, minocycline, and sulfones. Systemic glucocorticoids (prednisone, 0.5 to 1.0 mg/kg) seem to reduce symptoms. Mortality as high as 10% has been reported.

WARFARIN NECROSIS OF THE SKIN

This rare reaction occurs usually between the third and tenth days of therapy with warfarin derivatives, usually in women. Lesions are sharply demarcated, erythematous, indurated, and purpuric and may resolve or progress to form large, irregular, hemorrhagic bullae with eventual necrosis and slow-healing eschar formation.

Development of the syndrome is unrelated to drug dose or underlying condition. Favored sites are breasts, thighs, and buttocks. The course is not altered by discontinuation of the drug after onset of the eruption. Similar reactions have been associated with heparin. Warfarin reactions are associated with protein C deficiency. Protein C is a vitamin K-dependent protein with a shorter half-life than other clotting proteins and is in part responsible for control of fibrinolysis. Since warfarin inhibits synthesis of vitamin K-dependent coagulation factors, warfarin anticoagulation in heterozygotes for protein C deficiency causes a precipitous fall in circulating levels of protein C, permitting hypercoagulability and thrombosis in the cutaneous microvasculature, with consequent areas of necrosis. Heparin-induced necrosis may have clinically similar features but is probably due to heparin-induced platelet aggregation with subsequent occlusion of blood vessels.

Warfarin-induced cutaneous necrosis is treated with vitamin K and heparin. Vitamin K reverses the effects of warfarin, and heparin acts as an anticoagulant. Treatment with protein C concentrates may also be helpful in individuals with deficiencies of protein C, the predisposing factor for development of these reactions.

MORBILLIFORM REACTIONS

Morbilliform or maculopapular eruptions are the most common of all drug-induced reactions, often start on the trunk or areas of pressure or trauma, and consist of erythematous macules and papules that are frequently symmetric and may become confluent. Involvement of mucous membranes, palms, and soles is variable; the eruption may be associated with moderate to severe pruritus and fever.

The pathogenesis is unclear. A hypersensitivity mechanism has been suggested, although these reactions do not always recur following drug rechallenge. Diagnosis is rarely assisted by laboratory or patch testing; differentiation from viral exanthem is the principal differential diagnostic consideration. Unless the suspect drug is essential it should be discontinued. Occasionally these eruptions may decrease or fade with continued use of the responsible drug.

Morbilloform reactions usually develop within 1 week of initiation of therapy and last 1 to 2 weeks; however, reactions to some drugs, especially penicillin and drugs with long half-lives, may begin more than 2 weeks after therapy has begun and last as long as 2 weeks after therapy has ceased.

Morbilloform eruptions are usually treated by discontinuing the suspect medications symptomatically. Oral antihistamines, emollients, and soothing baths are useful for treatment of pruritus. Short courses of potent topical glucocorticoids can reduce inflammation and symptoms and are probably helpful. The beneficial effect of systemic glucocorticoids relative to risk is less clear.

FIXED DRUG REACTIONS

These reactions are characterized by one or more sharply demarcated, erythematous lesions in which hyperpigmentation results after resolution of the acute inflammation; with rechallenge, the lesion recurs in the same (i.e., "fixed") location. Lesions often involve the lips, hands, legs, face, genitalia, and oral mucosa and cause burning. Most patients have multiple lesions. Patch testing is useful to establish the etiology. Fixed drug eruptions have been associated with phenolphthalein, sulfonamides, tetracyclines, phenylbutazone, [NSAIDs](#), and barbiturates. Although cross-sensitivity appears to occur between different tetracycline compounds, cross-sensitivity was not elicited when different sulfonamide compounds were administered to patients as part of provocation testing.

LICHENOID DRUG ERUPTIONS

A lichenoid cutaneous reaction, clinically and morphologically indistinguishable from lichen planus, is associated with a variety of drugs and chemicals. Eosinophils are more common when the reaction is drug-induced. Gold and antimalarials are most often associated with this eruption. Antihypertensive agents, including beta blockers and captopril, have also been reported to cause lichenoid reactions.

BULLOUS ERUPTIONS

Blisters accompany a wide variety of cutaneous reactions, including fixed drug eruptions, severe morbilliform eruptions in dependent areas of the body, and phototoxic reactions. [SJS](#) and [TEN](#) are the most serious and important bullous reactions to drugs. Nalidixic acid and furosemide cause blistering eruptions indistinguishable from the primary bullous diseases. A pemphigus foliaceus-like eruption is seen with penicillamine.

PUSTULAR ERUPTIONS

Acute generalized exanthematous pustulosis is often associated with exposure to drugs, most notably antibiotics. Usually beginning on the face or intertriginous areas, small nonfollicular pustules overlying erythematous and edematous skin may coalesce and lead to superficial ulceration. Fever is present and differentiating this eruption from [TEN](#) in its initial stages may be difficult. Acute generalized exanthematous pustulosis often

begins within a few days of initiating drug treatment.

ERYTHEMA MULTIFORME

Erythema multiforme is an acute, self-limited inflammatory disorder of skin and mucous membranes characterized by distinctive iris or target lesions, usually acraly distributed and often associated with sore throat, mucosal lesions, and malaise. Classic erythema multiforme usually has nondrug causes, most commonly herpes simplex infection, and must be differentiated from true [SJS](#), which is usually drug related.

STEVENS-JOHNSON SYNDROME

[SJS](#) is a blistering disorder that is usually more severe than erythema multiforme. Initial presentation is often a sore throat, malaise, and fever. Within a few days, in addition to erosions of multiple mucous membranes, small blisters developing on dusky or purpuric macules or atypical target lesions characterize this eruption. Total percent of body surface area blistering and eventual detachment is less than 10%. Overlap [SJS/TEN](#) shares characteristics of both SJS and TEN, with 10 to 30% of body surface area exhibiting epidermal detachment.

TOXIC EPIDERMAL NECROLYSIS

[TEN](#) is the most serious cutaneous drug reaction and may be fatal. Drugs are usually the cause of TEN. Onset is generally acute and is characterized by fever $>39^{\circ}\text{C}$ (102.2°F), blisters or ulcers of multiple mucous membranes, malaise, and epidermal necrosis involving $>30\%$ of body surface area. Intestinal and pulmonary involvement is associated with a poor prognosis, as is a greater extent of epidermal detachment and older age. About 30% of affected persons die. Many treatments affecting immune response or cytokines (thalidomide) or apoptosis (intravenous immunoglobulin) have been advocated, but none have been shown to be efficacious in well-controlled trials. In spite of its theoretical potential benefits, thalidomide therapy increases TEN-associated mortality. Supportive treatment in burn units is helpful in reducing morbidity and mortality.

DRUGS OF SPECIAL INTEREST

PENICILLIN

The incidence of cutaneous reactions to penicillin is about 1%. About 85% of cutaneous reactions to penicillin are morbilliform, and about 10% are urticaria or angioedema.

IgG, IgM, and IgE antibodies can be produced; IgG and IgM anti-penicillin antibodies play a role in the development of hemolytic anemia, whereas anaphylaxis and serum sickness appear to be due to IgE antibodies in serum.

In patients with suspected IgE-mediated reactions to penicillin for whom future treatment is anticipated, accurate tests for sensitization are available. Current practice is to perform skin testing with a commercially available penicilloyl determinant preparation (Pre-pen, Kremers-Urban) and with fresh penicillin and, if possible, with another source

of minor (nonpenicilloyl) determinants such as aged or base-treated penicillin. Antibodies to minor determinants are common in patients experiencing anaphylaxis, but testing with major determinants alone detects most patients at risk for anaphylaxis.

About one-fourth of patients with positive history of penicillin allergy have a positive skin test, while 6% (3 to 10%) with no history of penicillin sensitivity demonstrate a positive skin response to penicillin. Administering penicillin to those patients with a positive skin test produces reactions in a high proportion (50 to 100%); conversely, only a few patients (0.5%) with a negative skin test react to the drug, and reactions tend to be mild and to occur late. Since a false-negative skin test may occur during or just after an acute reaction, testing should be performed either prospectively or several months after a suspected reaction. As many as 80% of patients lose anaphylactic sensitivity and IgE antibody after several years. Radioallergosorbent tests and other in vitro tests offer no advantage over properly performed skin testing. Some cross-reactivity between penicillin and nonpenicillin b-lactam antibiotics (e.g., cephalosporins) occurs, but the majority of penicillin-allergic patients will tolerate cephalosporins. Persons who have negative skin tests to penicillin rarely develop reactions to cephalosporins.

In the face of a positive clinical history of penicillin reaction, another drug should be chosen. If this is not feasible or prudent (e.g., in a pregnant patient with syphilis or with enterococcal endocarditis), skin testing with penicillin is warranted. If skin tests are negative, cautious administration of penicillin is acceptable, although some recommend desensitization of such patients if the reaction was likely to be IgE-mediated. In those with positive skin tests, desensitization is mandatory if therapeutic use of b-lactam antibiotics is to be undertaken. Various protocols are available, including oral and parenteral approaches. Oral desensitization appears to have lower risk of serious anaphylactic reactions during desensitization. However, desensitization carries the risk of anaphylaxis regardless of how it is performed. After desensitization, many patients experience non-life-threatening IgE-mediated untoward reactions to penicillin during their course of therapy. Desensitization is not effective in those with exfoliative dermatitis or morbilliform reactions due to penicillin.

NONSTEROIDAL ANTI-INFLAMMATORY DRUGS

[NSAIDs](#), including aspirin and indomethacin (indometacin), cause two broad categories of allergic-like symptoms in susceptible individuals: (1) approximately 1% of persons experience urticaria or angioedema, and (2) about half as many (0.5%) experience rhinosinusitis and asthma; however, about 10% of adults with asthma and one-third of individuals with nasal polyposis and sinusitis may respond adversely to aspirin.

Urticaria/angioedema may be delayed up to 24 h and may occur at any age. The rhinosinusitis-asthma syndrome generally develops within 1 h of drug administration. In young patients, the reaction pattern often begins as watery rhinorrhea, which can be complicated by nasal and sinus infection, and polyposis, bloody discharge, and nasal eosinophilia. In many individuals with this syndrome, asthma that can be life-threatening eventually ensues whenever [NSAIDs](#) are subsequently ingested, and symptoms may persist despite avoidance of these drugs. Proof of the association of symptoms and NSAID use requires either clear-cut history of symptoms following drug ingestion or an oral challenge. For the latter to be performed with relative safety, (1) asthma must be

under good control, (2) the procedure must be conducted in a hospital setting by experienced personnel capable of recognizing and treating acute respiratory responses, and (3) the challenge should begin with very low doses (i.e., not >30 mg) of aspirin and increase every 1 to 2 h in doubling doses as tolerated to 650 mg.

While cross-reactivity between [NSAIDs](#) is common, it is not immunologic, and patients who are sensitive to NSAIDs cannot be identified by assessment of IgE antibody to aspirin, lymphocyte sensitization, or in vitro immunologic testing.

RADIOCONTRAST MEDIA

Large numbers of patients are exposed to radiocontrast agents. High-osmolality radiocontrast media are about five times more likely to induce urticaria (1%) or anaphylaxis than newer low-osmolality media. Severe reactions are rare with either type of contrast media. About one-third of those with mild reactions to previous exposure rereact on reexposure. In most cases, these reactions are probably not immunologic. Pretreatment with prednisone and diphenhydramine reduces reaction rates. Persons with a reaction to a high-osmolality contrast media should be given low-osmolality media if later contrast studies are required.

ANTICONSULSANTS

Of the anticonvulsants, the single orally administered agent with the highest risk of severe adverse cutaneous reactions is the antiseizure medicine lamotrigine. Older anticonvulsants, including phenytoin and carbamazepine, are also associated with many types of severe reactions and a high incidence of less severe reactions, particularly in children. In addition to [SJS](#), [TEN](#), and the hypersensitivity syndrome discussed above, the aromatic anticonvulsants can induce a pseudolymphoma syndrome and induce gingival hyperplasia.

SULFONAMIDES

Sulfonamides have perhaps the highest risk of causing cutaneous eruptions and are the drugs most frequently implicated in [SJS](#) and [TEN](#). The combination of sulfamethoxazole and trimethoprim frequently induces adverse cutaneous reactions in patients with AIDS ([Chap. 309](#)). Desensitization is often successful in AIDS patients with morbilliform eruptions but is a high-risk procedure in AIDS patients who manifest erythroderma, fever, or a bullous reaction in response to their earlier sulfonamide exposure.

AGENTS USED IN CANCER CHEMOTHERAPY

Since many agents used in cancer chemotherapy inhibit cell division, rapidly proliferating elements of the skin, including hair, mucous membranes, and appendages, are sensitive to their effects; as a result, stomatitis and alopecia are among the most frequent dose-dependent side effects of chemotherapy. Onychodystrophy (dystrophic changes in nails) is also seen with bleomycin, hydroxyurea (hydroxycarbamide), and 5-fluorouracil. Sterile cellulitis and phlebitis and ulceration of pressure areas occur with many of these agents. Urticaria, angioedema, exfoliative dermatitis, and erythema of the palms and soles have also been seen, as has local and diffuse hyperpigmentation.

GLUCOCORTICOIDS

Both systemic and topical glucocorticoids cause a variety of skin changes, including acneiform eruptions, atrophy, striae, and other stigmata of Cushing's syndrome, and in sufficiently high doses can retard wound healing. Patients using glucocorticoids are at higher risk for bacterial, yeast, and fungal skin infections that may be misinterpreted as drug eruptions but are instead drug side effects.

CYTOKINE THERAPY

Alopecia is a common complication of [IFN-a](#). Induction or exacerbation of various immune-mediated disorders (psoriasis, lichen planus, lupus erythematosus) has been also reported with this agent. IFN- β injection has been associated with local necrosis of the skin. Granulocyte colony stimulating factor may induce various neutrophilic dermatosis, including Sweet's syndrome, pyoderma gangrenosum, neutrophilic eccrine hidradenitis, and vasculitis, and can exacerbate psoriasis.

[IL-2](#) is associated with frequent cutaneous reactions including exanthema, facial edema, xerosis, and pruritus. Cases of pemphigus vulgaris, linear IgA disease, psoriasis, and vitiligo have also been described in association with this drug.

ANTIMALARIAL AGENTS

Antimalarial agents are used as therapy for several skin diseases, including the skin manifestations of lupus and polymorphous light eruption, but they can also induce cutaneous reactions. Although also used to treat porphyria cutanea tarda at low doses, in patients with asymptomatic porphyria cutanea tarda, higher doses of chloroquine increase porphyrin levels to such an extent that they may exacerbate the disease.

Pigmentation disturbances, including black pigmentation of the face, mucous membranes, and pretibial and subungual areas, occur with antimalarials. Quinacrine (mepacrine) causes generalized, cutaneous yellow discoloration.

GOLD

Chrysotherapy has been associated with a variety of dose-related dermatologic reactions (including maculopapular eruptions), which can develop as long as 2 years after initiation of therapy and require months to resolve. Erythema nodosum, psoriasiform dermatitis, vaginal pruritus, eruptions similar to those of pityriasis rosea, hyperpigmentation, and lichenoid eruptions resembling those seen with antimalarial agents have been reported. After a cutaneous reaction, it is sometimes possible to reinstitute gold therapy at lower doses without recurrence of the dermatitis.

DIAGNOSIS OF DRUG REACTIONS

Possible causes of an adverse reaction can be assessed as definite, probable, possible, or unlikely based on six variables: (1) previous experience with the drug in the general population, (2) alternative etiologic candidates, (3) timing of events, (4) drug levels or

evidence of overdose, (5) patient reaction to drug discontinuation, and (6) patient reaction to rechallenge.

PREVIOUS EXPERIENCE

Tables of relative reaction rates are available and are useful to assess the likelihood that a given drug is responsible for a given cutaneous reaction. The specific morphologic pattern of a drug reaction, however, may modify these reaction rates by increasing or decreasing the likelihood that a given drug is responsible for a given reaction. For example, since fixed eruptions due to drugs are more often seen with barbiturates than with penicillin, a fixed drug reaction in a patient taking both types of agents is more likely to be due to the barbiturate, even though penicillins have a higher overall drug reaction rate.

ALTERNATIVE ETIOLOGIC CANDIDATES

A cutaneous eruption may be due to exacerbation of preexisting disease or to development of new disease unrelated to drugs. For example, a patient with psoriasis may have a flare-up of disease coincidental with administration of penicillin for streptococcal infection; in this case, infection is a more likely cause for the flare-up than drug reaction.

TIMING OF EVENTS

Most drug reactions of the skin occur within 1 to 2 weeks of initiation of therapy. Hypersensitivity syndrome may occur later (up to 8 weeks) after initiating drug therapy. Fixed drug reactions and generalized exanthematous pustulosis often occur earlier (within 48 h), as do reactions of all types in persons with prior sensitization to that drug or a cross-sensitizing agent.

DRUG LEVELS

Some cutaneous reactions are dependent on dosage or cumulative toxicity. For example, lichenoid dermatoses due to gold administration appear more often in patients taking high doses.

DISCONTINUATION

Most adverse cutaneous reactions to drugs remit with discontinuation of the suspected agent. A reaction is considered unlikely to be drug-related if improvement occurs while the drug is continued or if a patient fails to improve after stopping the drug and appropriate therapy.

RECHALLENGE

Rechallenge provides the most definitive information concerning adverse cutaneous reactions to drugs, since a reaction failing to recur on rechallenge with a drug is unlikely to be due to that agent. Rechallenge is usually impractical, however, because the need to ensure patient safety and comfort outweighs the value of the possible information

derived from rechallenge.

Of special importance is the rapid recognition of reactions that may become serious or life-threatening. [Table 59-2](#) lists clinical and laboratory features that, if present, suggest the reaction may be serious. [Table 59-3](#) provides key features of the most serious adverse cutaneous reactions.

DIAGNOSIS OF DRUG ALLERGY

Tests for IgE responses include in vivo and in vitro methods, but such tests are available for only a limited number of drugs, including penicillins and cephalosporins, some peptide and protein drugs (insulin, xenogeneic sera), and some agents used for general anesthesia. In vivo testing is accomplished by prick puncture and/or by intradermal skin testing. A wheal-and-flare response 2 × 2 mm greater than that seen with a saline control within 20 min is considered indicative of IgE-mediated mast cell degranulation, provided (1) the patient is not dermographic, (2) the drug does not nonspecifically degranulate mast cells, (3) the drug concentration is not high enough to be irritating, and (4) the buffer itself does not cause wheal-and-flare responses.

Skin testing with major and minor determinants of penicillins or cephalosporins has proved useful for identifying patients at risk of anaphylactic reactions to these agents. However, skin tests themselves carry a small risk of anaphylaxis. Negative skin tests do not rule out IgE-mediated reactivity, and the risk of anaphylaxis in response to penicillin administration in patients with negative skin tests is about 1%; about two-thirds of patients with a positive skin test and history of a previous adverse reaction to penicillin experience an allergic response on rechallenge. Skin tests may be negative in allergic patients receiving antihistamines or in those whose allergy is to determinants not present in the test reagent. Although less well studied, similar techniques can identify patients who are sensitive to protein drugs and to agents such as gallamine and succinylcholine. Most other drugs are small molecules, and skin testing with them is unreliable.

There are no generally available and reliable tests for assessing causality of non-IgE-mediated reactions, except possibly patch tests for assessment of fixed drug reactions. Therefore, diagnosis usually relies on clinical factors rather than test results.

(Bibliography omitted in Palm version)

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60. PHOTSENSITIVITY AND OTHER REACTIONS TO LIGHT - *David R. Bickers*

SOLAR RADIATION

Sunlight is the most visible and obvious source of comfort in the environment. This natural proclivity for the sun has the beneficial results of warmth and vitamin D synthesis but also can produce pathologic consequences. Few effects of sun exposure beyond those affecting the skin have been identified, but cutaneous exposure to sunlight can evoke immunosuppressive responses and genetic changes that may be relevant to the pathogenesis of nonmelanoma skin cancer and perhaps infections such as herpes simplex.

The sun's energy encompasses a broad range from ultrashort highly energetic ionizing radiation (10⁻²μm) to ultralong radiowaves of very low photon energy (10⁷μm). Thus, the emission spectrum ranges over nine orders of magnitude, but that reaching the earth's surface is narrow and is limited to components of the ultraviolet (UV), visible light, and portions of the infrared. The cutoff at the short end of the UV is at approximately 290 nm, because stratospheric ozone is formed by ionizing radiation of wavelengths less than 100 nm and absorbs solar energy between 120 and 310 nm, thereby preventing penetration to the earth's surface of the shorter, more energetic, potentially more harmful wavelengths of solar radiation. Indeed, concern about destruction of the ozone layer by chlorofluorocarbons released into the atmosphere has led to international agreements to reduce production of these chemicals.

Measurements of solar flux indicate that there is a twentyfold regional variation in the amount of energy at 300 nm that reaches the earth's surface. This variability relates to seasonal effects, the path of sunlight transmission through ozone and air, the altitude (4% increase for each 300 m of elevation), the latitude (increasing intensity with decreasing latitude), and the amount of cloud cover, fog, and pollution.

The major components of the photobiologic action spectrum include the [UV](#) and visible wavelengths between 290 and 700 nm. In addition, the wavelengths beyond 700 nm in the infrared primarily evoke heat, but warming of the skin may enhance biologic responses to wavelengths in the UV and visible spectrum.

The [UV](#) spectrum is arbitrarily divided into three major segments: C, B, and A. This includes the wavelengths between 10 and 400 nm. Ultraviolet C (UV-C) consists of wavelengths between 10 and 290 nm and does not reach the earth because of its absorption by stratospheric ozone. These wavelengths are not a cause of photosensitivity except in occupational settings where artificial sources of this energy are employed -- e.g., for germicidal effects. Ultraviolet B (UV-B) consists of wavelengths between 290 and 320 nm. This portion of the photobiologic action spectrum is the most efficient in producing redness or erythema in human skin and hence is sometimes known as the "sunburn spectrum." Ultraviolet A (UV-A) represents those wavelengths between 320 and 400 nm and is approximately 1000-fold less efficient in producing skin hyperemia than is UV-B. The UV-A has also been divided into two parts known as UV-A 1 (340 to 400 nm) and UV-A 2 (320 to 340 nm).

The visible wavelengths between 400 and 700 nm include the familiar white light which

when directed through a prism can be shown to consist of various colors including violet, indigo, blue, green, yellow, orange, and red. The energy possessed by photons in the visible spectrum is not capable of damaging human skin in the absence of a photosensitizing chemical. The absorption of energy is critical to the development of photosensitivity. Thus the *absorption spectrum* of a molecule is defined as the range of wavelengths absorbed by it, whereas the *action spectrum* for an effect of incident radiation is defined as the range of wavelengths that evoke the response.

Photosensitivity occurs when a photon-absorbing chemical (chromophore) present in the skin absorbs incident energy, becomes excited, and transfers the absorbed energy to various structures or to oxygen. The absorbed energy must be dissipated by processes including heat, fluorescence, and phosphorescence. It is important to emphasize that absorption spectra and action spectra need not be superimposable, but there must be overlap at some point to produce photosensitization.

STRUCTURE AND FUNCTION OF SKIN

The skin's exposure to sunlight permits the absorption of some wavelengths and the transmission of others. Essentially, human skin is a sandwich of two distinctive compartments, the epidermis and dermis, separated by a basement membrane. The outer epidermis is a stratified squamous epithelium comprising the surface stratum corneum (a protein- and lipid-rich compact membrane), the stratum granulosum, stratum spinosum, and the basal cell layer. The basal cell layer contains a heterogeneous population of cells, a subset of which migrate upward in the process of terminal differentiation that results in the expression of specific keratin genes and the formation of the stratum corneum. Epidermal cells include resident keratinocytes and melanocytes and immigrant cells, including the immunologically active Langerhans cells, lymphocytes, polymorphonuclear leukocytes, monocytes, and macrophages, making the epidermis a major component of the immune system. Branches of sensory nerve endings also reach into this compartment.

The second major component of skin is the dermis, which is relatively large and less densely populated with cells that include fibroblasts, endothelial cells within dermal vessels, and mast cells. Tissue macrophages and sparsely distributed inflammatory cells are also present. All these cells exist within an extracellular matrix of collagen, elastin, and glycosaminoglycans. In contrast to the epidermis, rich vascularization of the dermis allows it to play an important role in temperature regulation and in inflammatory responses to skin injury.

UV RADIATION (UVR) AND SKIN

The epidermis and the dermis contain several chromophores capable of interacting with incident solar energy. These interactions include reflection, refraction, absorption, and transmission. The stratum corneum is a major impediment to the transmission of [UV-B](#), and less than 10% of incident wavelengths in this region penetrate the basement membrane. Approximately 3% of radiation below 300 nm, 20% of radiation below 360 nm, and 33% of short visible radiation reaches the basal cell layer in untanned human skin. Proteins and nucleic acids absorb intensely in the short UV-B. In contrast, UV-A 1 and 2 penetrate the epidermis efficiently to reach the dermis, where they likely produce

changes in structural and matrix proteins that contribute to the aged appearance of chronically sun-exposed skin, particularly in individuals of light complexion.

One of the consequences of [UV-B](#) absorption by DNA is the production of pyrimidine dimers. These structural changes can be repaired by mechanisms that result in their recognition and excision, and the reestablishment of normal base sequences. The efficient repair of these structural aberrations is crucial, since individuals with defective DNA repair are at high risk for the development of cutaneous cancer. For example, patients with xeroderma pigmentosum, an autosomal recessive disorder, are characterized by variably decreased repair of UV-induced photoproducts, and their skin may develop the xerotic appearance of photoaging as well as basal cell and squamous cell carcinomas and melanoma in the first two decades of life. Studies in mice using knockout gene technology have verified the importance of genes regulating these repair pathways in preventing the development of UV-induced cancer.

Cutaneous Optics and Chromophores Chromophores are endogenous or exogenous chemical components that can absorb physical energy. Endogenous chromophores of skin are of two types: (1) chemicals that are normally present, including nucleic acids, proteins, lipids, and 7-dehydrocholesterol, the precursor of vitamin D; and (2) chemicals, such as porphyrins, synthesized elsewhere in the body that circulate in the bloodstream and diffuse into the skin. Normally, only trace amounts of porphyrins are present in the skin, but in the diseases known as the porphyrias, increased amounts are released into the circulation and are transported to the skin, where they absorb incident energy both in the Soret band around 400 nm (short visible) and to a lesser extent in the red portion of the visible spectrum (580 to 660 nm). This results in structural damage to the skin that may be manifest as erythema, edema, urticaria, or blister formation ([Chap. 346](#)).

Acute Effects of Sun Exposure The immediate cutaneous consequences of sun exposure include sunburn and vitamin D synthesis.

Sunburn This very common affliction of human skin is caused by exposure to [UVR](#). Generally speaking, the individual's ability to tolerate sunlight is inversely proportional to his or her melanin pigmentation. Melanin is a complex polymer of tyrosine that functions as an efficient neutral-density filter with broad absorbance within the [UV](#) portion of the solar spectrum. Melanin is synthesized in specialized epidermal dendritic cells termed *melanocytes* and is packaged into *melanosomes* that are transferred via dendritic processes into *keratinocytes*, where they provide photoprotection. Sun-induced melanogenesis is a consequence of increased tyrosinase activity in melanocytes that in turn may be due to a combination of eicosanoid and endothelin-1 release. Tolerance of sun exposure is a function of the efficiency of the epidermal-melanin unit and can usually be ascertained by asking an individual two questions: (1) Do you burn after sun exposure? and (2) Do you tan after sun exposure? By the answers to these questions, it is usually possible to divide the population into six skin types varying from type I (always burn, never tan) to type VI (never burn, always tan) ([Table 60-1](#)).

There are two general theories about the pathogenesis of the sunburn response. First, the lag phase in time between skin exposure and the development of visible redness (usually 4 to 12 h) suggests an epidermal chromophore that causes delayed production and/or release of vasoactive mediator(s), or cytokines, that diffuse to the dermal

vasculature to evoke vasodilatation. Indeed, [UVR](#) stimulates the release of numerous proinflammatory cytokines and nitric oxide by keratinocytes. Second, it is possible that the small amount of incident [UV-B](#) radiation (10% or less) that penetrates to the dermis can be absorbed directly by endothelial cells in the vasculature, thereby resulting in vasodilatation. The issue remains unresolved.

The action spectrum for sunburn erythema includes the [UV-B](#) and UV-A regions. Photons in the shorter UV-B are at least 1000-fold more efficient than photons in the longer UV-B and the UV-A in evoking the response. However, UV-A may contribute to sunburn erythema at midday when much more UV-A than UV-B is present.

The mechanism of injury remains poorly defined, but the action spectrum for [UV-B](#) erythema closely resembles the absorption spectrum for DNA after adjusting for the absorbance of incident energy by the stratum corneum. Apoptotic keratinocytes (so-called sunburn cells) are visible histologically within an hour of exposure and are maximal within 24 h. UV-A is less effective than UV-B in producing sunburn cells. Mast cells may release inflammatory mediators after exposure to UV-B and UV-A. For example, erythema doses of both UV-B and UV-A increase histamine levels in experimentally induced suction blisters of human skin that return to normal after 24 h (before visible erythema has subsided). Prostaglandin E₂ increases to approximately 150% of control levels after 24 h and then diminishes. Since prostaglandins evoke both pain and redness when injected intradermally, their presence in suction blisters after UV-B exposure suggests a role in UV-B erythema. Age-related declines occur in the amount of inflammatory mediators detectable in human skin after UV-B irradiation. UV-A erythema results in few epidermal sunburn cells, but vascular endothelial injury is greater than with UV-B. In addition, there are increased levels of arachidonic acid and of prostaglandins D₂, E₂, and I₂ that peak within 5 to 9 h and then subside before peak redness occurs. Despite evidence for the role of prostaglandins in both UV-B- and UV-A-irradiated skin, administration of nonsteroidal anti-inflammatory drugs is more effective in reducing erythema evoked by UV-B than by UV-A. UV-B also induces cutaneous matrix-degrading metalloproteinases within hours of exposure.

Vitamin D Photochemistry Cutaneous exposure to [UV-B](#) causes photolysis of epidermal previtamin D₃ (7-dehydrocholesterol) to previtamin D₃, which then undergoes a temperature-dependent isomerization to form the stable hormone vitamin D₃. This compound then diffuses to the dermal vasculature and circulates systemically where it is converted to the functional hormone 1,25-dihydroxy vitamin D₃ [1,25(OH)₂D₃]. Vitamin D metabolites from the circulation or those produced in the skin itself can augment epidermal differentiation signaling. Aging substantially decreases the ability of human skin to produce vitamin D₃. This, coupled with the widespread use of sunscreens that filter out [UV-B](#), has led to concern that vitamin D deficiency may become a significant clinical problem in the elderly. Indeed, studies have shown that the use of sunscreens can diminish the production of vitamin D₃ in human skin.

Chronic Effects of Sun Exposure: Nonmalignant The clinical features of photodamaged sun-exposed skin consist of wrinkling, blotchiness, telangiectasia, and a roughened, irregular, "weather-beaten" appearance. Whether these changes, which some refer to as *photoaging* or *dermatoheliosis*, represent accelerated chronologic aging or a separate and distinct process is not clear.

Within chronically sun-exposed epidermis, there is thickening (acanthosis) and morphologic heterogeneity within the basal cell layer. Higher but irregular melanosome content may be present in some keratinocytes, indicating prolonged residence of the cells in the basal cell layer. These structural changes may help to explain the leathery texture and the blotchy discoloration of sun-damaged skin.

The dermis is the major site for sun-associated chronic damage, manifest as a massive increase in thickened irregular masses of tangled elastic fibers resulting from enhanced expression of elastin genes. Collagen fibers are also abnormally clumped in the deeper dermis. Fibroblasts are increased in number and show morphologic signs suggesting activation. Degraded mast cells may be present in the dermis, the relevance of which remains unclear.

These morphologic changes, both gross and microscopic, are features of chronically sun-exposed skin. The chromophore(s), the action spectra, and the specific biochemical events orchestrating these changes are unknown.

Chronic Effects of Sun Exposure: Malignant One of the major known consequences of chronic skin exposure to sunlight is nonmelanoma skin cancer. The two types of nonmelanoma skin cancer are basal cell and squamous cell carcinoma ([Chap. 86](#)). There are three major steps for cancer induction: initiation, promotion, and progression. Chronic exposure of animal skin to artificial light sources that mimic solar [UVR](#) results in *initiation*, a step whereby structural (mutagenic) changes in DNA evoke an irreversible alteration in the target cell (keratinocyte) that begins the tumorigenic process. Exposure to a tumor initiator is believed to be a necessary but not sufficient step in the malignant process, since initiated skin cells not exposed to tumor promoters do not generally develop tumors. The second stage in tumor development is *promotion*, a multistep process whereby initiated cells are exposed to chemical and physical agents that evoke epigenetic changes that culminate in the clonal expansion of initiated cells and cause the development, over a period of weeks to months, of benign growths known as *papillomas*. Again, using transgenic animals, the importance of [UV](#) effects on the expression of additional oncogenes such as *fos* and *jun* in developing papillomas has been demonstrated. UV-B is a *complete carcinogen*, meaning that it can function as both an initiator and a promoter, leading to tumor induction. *Incomplete carcinogens* can initiate tumorigenesis but require additional skin exposure to tumor promoters to elicit tumors. The prototype tumor promoter is the phorbol ester 12-*O*-tetradecanoyl phorbol-13-acetate. Tumor promotion usually requires multiple exposures over time to evoke a neoplasm.

The final step in the malignant process is the conversion of benign precursors into malignant lesions, a process thought to require additional genetic alterations in already transformed cells. Indeed, *ras* gene mutations have been detected in a minority of human nonmelanoma skin cancers. Mutations of the tumor suppressor gene p53 also occur in sun-damaged human skin.

Sun exposure causes nonmelanoma and melanoma cancers of the skin, although the evidence is far more direct for its role in nonmelanoma (basal cell and squamous cell carcinoma) than in melanoma. Approximately 80% of nonmelanoma skin cancers

develop on exposed body area, including the face, the neck, and the hands. Men of fair complexion who work outdoors are twice as likely as women to develop these types of cancers. Whites of darker complexions (e.g., Hispanics) have one-tenth the risk of developing such cancers as do light-skinned individuals. Blacks are at lowest risk for all forms of skin cancer. Between 600,000 and 800,000 individuals in the United States develop nonmelanoma skin cancer annually, and the lifetime risk for a white individual to develop such a neoplasm is estimated at approximately 15%. A consensus exists that the incidence of nonmelanoma skin cancer in the population is rising, for reasons that are unclear.

The relationship of sun exposure to melanoma is less clear-cut, but suggestive evidence supports an association. Melanomas occasionally develop by the teenage years, indicating that the latent period for tumor growth is less than that of nonmelanoma skin cancer. Melanomas are among the most rapidly increasing of all human malignancies ([Chap. 86](#)). Epidemiologic studies of immigrants of similar ethnic stock indicate that individuals born in one area or who migrated to the same locale before age 10 have higher age-specific melanoma rates than individuals arriving later. It is thus reasonable to conclude that life in a sunny climate from birth or early childhood increases the risk of melanoma. In general, risk does not correlate with cumulative sun exposure but may relate to sequelae of sun exposure in childhood. Thus, a blistering sunburn is associated with a doubling of melanoma risk at the site of the reaction.

Immunologic Effects Exposure to solar radiation influences both local and systemic immune responses. [UV-B](#) appears to be most efficient in altering immune responses, likely related to the capacity of such energy to affect antigen presentation in skin by interacting with epidermal Langerhans cells. These bone marrow-derived dendritic cells possess surface markers characteristic of monocytes and macrophages. Following skin exposure to erythema doses of UV-B, Langerhans cells undergo both morphologic and functional changes that result in decreased contact allergic responses when haptens are applied to the irradiated site. This diminished capacity for sensitization is due to the induction of antigen-specific suppressor T lymphocytes. Indeed, while the immunosuppressive effect of irradiation is limited to haptens applied to the irradiated site, the net result is systemic immune suppression to that antigen because of the induction of suppressor T cells.

Higher doses of radiation evoke diminished immunologic responses to antigens introduced either epicutaneously or intracutaneously at sites distant from the irradiated site. These suppressed responses are also associated with the induction of antigen-specific suppressor T lymphocytes and may be mediated by as yet undefined factors that are released from epidermal cells at the irradiated site. The implications of this generalized immune suppression in terms of altered susceptibility to cutaneous cancer or to infection remain to be defined.

It is known that [UV](#)-induced tumors in murine skin are antigenic and are rapidly rejected when transplanted into normal syngeneic animals. If the tumors are transplanted into animals previously exposed to subcarcinogenic doses of UV-B, they are not rejected and instead grow progressively in the recipients. This failure of irradiated animals to reject the transplanted tumors is due to the development of T suppressor cells that prevent the rejection response. While the mechanism of suppression of tumor rejection

is unknown, such a response might be a critical determinant of cancer risk in human skin.

PHOTOSENSITIVITY DISEASES

The diagnosis of photosensitivity requires a careful history to define the duration of the signs and symptoms, the length of time between exposure to sunlight and the development of subjective complaints, and visible changes in the skin. The age of onset also can be a helpful clue; for example, the acute photosensitivity of erythropoietic protoporphyria almost always begins in childhood, whereas the chronic photosensitivity of porphyria cutanea tarda typically begins in the fourth and fifth decades. A history of exposure to topical and systemic drugs and chemicals may provide important information. Many classes of drugs can cause photosensitivity on the basis of either phototoxicity or photoallergy. Fragrances such as musk ambrette that were previously present in numerous cosmetic products are also potent photosensitizers.

Examination of the skin may also offer important clues. Anatomic areas that are naturally protected from direct sunlight such as the hairy scalp, the upper eyelids, the retroauricular areas, and the infranasal and submental regions may be spared, whereas exposed areas show characteristic features of the pathologic process. These anatomic localization patterns are often helpful but not infallible in making the diagnosis. For example, airborne contact sensitizers that are blown onto the skin may produce dermatitis that can be difficult to distinguish from photosensitivity, despite the fact that such material may trigger skin reactivity in areas shielded from direct sunlight.

Many dermatologic conditions may be caused or aggravated by light ([Table 60-2](#)). The role of light in evoking these responses may be dependent on genetic abnormalities ranging from well-described defects in DNA repair that occur in xeroderma pigmentosum to the inherited abnormalities in heme synthesis that characterize the porphyrias. In certain photosensitivity diseases, the chromophore has been identified, whereas in the majority, the energy-absorbing agent is unknown.

Polymorphous Light Eruption After sunburn, the most common type of photosensitivity disease is *polymorphous light eruption*, the mechanism of which is unknown. Many affected individuals never seek medical attention because the condition is often transient, becoming manifest each spring with initial sun exposure but then subsiding spontaneously with continuing exposure, a phenomenon known as "hardening." The major manifestations of polymorphous light eruption include pruritic (often intensely so) erythematous papules that may coalesce into plaques on exposed areas of the face and arms or other areas as well, making the distribution spotty and uneven.

The diagnosis can be confirmed by skin biopsy and by performing phototest procedures in which skin is exposed to multiple erythema doses of [UV](#)-A and UV-B. The action spectrum for polymorphous light eruption is usually within these portions of the solar spectrum.

Treatment of this disease includes the induction of hardening by the cautious administration of [UV](#)light, either alone or in combination with photosensitizers such as

the psoralens (see below).

Phototoxicity and Photoallergy These photosensitivity disorders are related to the topical or systemic administration of drugs and other chemicals. Both reactions require the absorption of energy by a drug or chemical resulting in the production of an excited-state photosensitizer that can transfer its absorbed energy to a bystander molecule or to molecular oxygen, thereby generating tissue-destructive chemical species.

Phototoxicity is a nonimmunologic reaction caused by drugs and chemicals, a few of which are listed in [Table 60-3](#). The usual clinical manifestations include erythema resembling a sunburn that quickly desquamates or "peels" within several days. In addition, edema, vesicles, and bullae may occur.

Photoallergy is distinct in that the immune system participates in the pathologic process. The excited-state photosensitizer may create highly unstable haptenic free radicals that bind covalently to macromolecules to form a functional antigen capable of evoking a delayed hypersensitivity response. Some of the drugs and chemicals that produce photoallergy are listed in [Table 60-4](#). The clinical manifestations typically differ from those of phototoxicity in that an intensely pruritic eczematous dermatitis tends to predominate and evolves into lichenified, thickened, "leathery" changes in sun-exposed areas. A small subset (perhaps 5 to 10%) of patients with photoallergy may develop a persistent exquisite hypersensitivity to light even when the offending drug or chemical is identified and eliminated. Known as *persistent light reaction*, this may be incapacitating for years. Some have used the term *chronic actinic dermatitis* to encompass these chronic hyperresponsive states.

Diagnostic confirmation of phototoxicity and photoallergy often can be obtained using phototest procedures. In patients with suspected phototoxicity, determination of the minimal erythema dose (MED) while the patient is exposed to a suspected agent and then repeating the MED after discontinuation of the agent may provide a clue to the causative drug or chemical. Photopatch testing can be performed to confirm the diagnosis of photoallergy. This is a simple variant of ordinary patch testing in which a series of known photoallergens is applied to the skin in duplicate and one set is irradiated with a suberythema dose of UV-A. Development of eczematous changes at sites exposed to sensitizer and light is a positive result. The characteristic abnormality in patients with persistent light reaction is a diminished threshold to erythema evoked by UV-B. Patients with chronic actinic dermatitis may have a broad spectrum of UV hyperresponsiveness.

The management of drug photosensitivity is first and foremost to eliminate exposure to the chemical agents responsible for the reaction and to minimize sun exposure. The acute symptoms of phototoxicity may be ameliorated by cool, moist compresses, topical glucocorticoids, and systemically administered nonsteroidal antiinflammatory agents. In severely affected individuals, a rapidly tapered course of systemic glucocorticoids may be useful. Judicious use of analgesics may be necessary.

Photoallergic reactions require a similar management approach. Furthermore, individuals suffering from persistent light reactivity must be meticulously protected

against light exposure. In selected patients in whom chronic systemic high-dose glucocorticoids pose unacceptable risks, it may be necessary to employ cytotoxic agents such as azathioprine or cyclophosphamide.

Porphyria The porphyrias ([Chap. 346](#)) are a group of diseases that have in common various derangements in the synthesis of heme. Heme is an iron-chelated tetrapyrrole or porphyrin, and the nonmetal chelated porphyrins are potent photosensitizers that absorb light intensely in both the short (400 to 410 nm) and the long (580 to 650 nm) portions of the visible spectrum.

Heme cannot be reutilized and must be continuously synthesized, and the two body compartments with the largest capacity for its production are the bone marrow and the liver. Accordingly, the porphyrias originate in one or the other of these organs, with the end result of excessive endogenous production of potent photosensitizing porphyrins. The porphyrins circulate in the bloodstream and diffuse into the skin, where they absorb solar energy, become photoexcited, and evoke cutaneous photosensitivity. The mechanism of porphyrin photosensitization is known to be photodynamic or oxygen-dependent and is mediated by reactive oxygen species such as superoxide anions.

Porphyria cutanea tarda is the most common type of human porphyria and is associated with decreased activity of the enzyme uroporphyrinogen decarboxylase associated with a number of gene mutations. There are two basic types of porphyria cutanea tarda: the sporadic or acquired type, generally seen in individuals ingesting ethanol or receiving estrogens; and the inherited type, in which there is autosomal dominant transmission of deficient enzyme activity. Both forms are associated with increased hepatic iron stores.

In both types of porphyria cutanea tarda, the predominant feature is a chronic photosensitivity characterized by increased fragility of sun-exposed skin, particularly areas subject to repeated trauma such as the dorsa of the hands, the forearms, the face, and the ears. The predominant skin lesions are vesicles and bullae that rupture, producing moist erosions, often with a hemorrhagic base, that heal slowly with crusting and purplish discoloration of the affected skin. Hypertrichosis, mottled pigmentary change, and scleroderma-like induration are associated features. Biochemical confirmation of the diagnosis can be obtained by measurement of urinary porphyrin excretion, plasma porphyrin assay, and by assay of erythrocyte and/or hepatic uroporphyrinogen decarboxylase. Multiple mutations of the uroporphyrinogen decarboxylase gene have been identified in human populations, including exon skipping and base substitutions.

Treatment consists of repeated phlebotomies to diminish the excessive hepatic iron stores and/or intermittent low doses of the antimalarial drugs chloroquine and hydroxychloroquine. Long-term remission of the disease can be achieved if the patient eliminates exposure to porphyrinogenic agents.

Erythropoietic protoporphyria originates in the bone marrow and is due to a decrease in the mitochondrial enzyme ferrochelatase secondary to numerous gene mutations. The major clinical features include an acute photosensitivity characterized by subjective burning and stinging of exposed skin that often develops during or just after exposure.

There may be associated skin swelling and, after repeated episodes, a waxlike scarring.

The diagnosis is confirmed by demonstration of measurement of free elevated erythrocyte protoporphyrin. Detection of increased plasma protoporphyrin helps to differentiate lead poisoning and iron-deficiency anemia, in both of which elevated erythrocyte protoporphyrin occurs in the absence of cutaneous photosensitivity and of elevated plasma protoporphyrin.

Treatment consists of reducing sun exposure and the oral administration of the carotenoid β -carotene, which is an effective scavenger of free radicals. This drug increases tolerance to sun exposure in many affected individuals, although it has no effect on deficient ferrochelatase.

An algorithm for the approach to a patient with photosensitivity is illustrated in [Fig. 60-1](#).

PHOTOPROTECTION

Since photosensitivity of the skin results from exposure to sunlight, it follows that avoidance of the sun would eliminate these disorders. Unfortunately, social pressures make this an impractical alternative for most individuals, and this has led to a search for better approaches to photoprotection.

Natural photoprotection is provided by structural proteins in the epidermis, particularly keratins and melanin. The amount of melanin and its distribution in cells is genetically regulated, and individuals of darker complexion (skin types IV to VI) are at decreased risk for the development of cutaneous malignancy.

Other forms of photoprotection include clothing and sunscreens. Clothing constructed of tightly woven sun-protective fabrics, irrespective of color, affords substantial protection. Wide-brimmed hats, long sleeves, and trousers all reduce direct exposure. Sunscreens are of two major types -- chemical and physical. Chemical sunscreens are chromophores that absorb energy in the [UV-B](#) and/or [UV-A](#) regions, thereby diminishing photon absorption by the skin ([Table 60-5](#)). Sunscreens are rated for their photoprotective effect by their *sun protective factor* (SPF). The SPF is simply a ratio of the time required to produce sunburn erythema with and without sunscreen application. SPF ratings of 15 or higher provide effective protection against [UV-B](#) and, to a lesser extent, [UV-A](#). The major categories of chemical sunscreens include *p*-aminobenzoic acid and its esters, benzophenones, anthranilates, cinnamates, and salicylates. Physical sunscreens are light-opaque mixtures containing metal particles such as titanium oxide and zinc oxide that scatter light, thereby reducing photon absorption by the skin.

In addition to light absorption, a critical determinant of the photoprotective effect of sunscreens is their ability to remain on the skin, a property known as *substantivity*. In general, the *p*-aminobenzoic acid esters formulated in moisturizing vehicles provide the greatest substantivity.

Photoprotection can also be achieved by limiting the time of exposure during the day. Since the majority of an individual's total lifetime sun exposure may occur by the age of

18, it is important to educate parents and young children about the hazards of sunlight. Simply eliminating exposure at midday will substantially reduce lifetime [UV](#)-B exposure.

PHOTOTHERAPY AND PHOTOCHEMOTHERAPY

[UVR](#) can also be used therapeutically. The administration of [UV](#)-B alone or in combination with topically applied agents can induce remissions of psoriasis and atopic dermatitis.

Photochemotherapy in which topically applied or systemically administered *psoralens* are combined with [UV](#)-A (PUVA) is also effective in treating psoriasis and in the early stages of cutaneous T cell lymphoma and vitiligo. Psoralens are tricyclic furocoumarins that, when intercalated into DNA and exposed to UV-A, form adducts with pyrimidine bases and eventually form DNA cross-links. These structural changes are thought to decrease DNA synthesis and relate to improvement that occurs in psoriasis. The reason that PUVA photochemotherapy is effective in cutaneous T cell lymphoma is not clear.

In addition to its effects on DNA, PUVA photochemotherapy also stimulates melanin synthesis, and this provides the rationale for its use in the depigmenting disease vitiligo. Oral 8-methoxypsoralen and [UV](#)-A appear to be most effective in this regard, but as many as 100 treatments extending over 12 to 18 months may be required to promote satisfactory repigmentation.

The major side effects of [UV](#)-B phototherapy and PUVA photochemotherapy are due to the cumulative effects of photon absorption and include skin dryness, actinic keratoses, and an increased risk of nonmelanoma skin cancer. Despite these risks, the therapeutic index of these modalities is quite acceptable.

(Bibliography omitted in Palm version)

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SECTION 10 -HEMATOLOGIC ALTERATIONS

61. ANEMIA AND POLYCYTHEMIA - John W. Adamson, Dan L. Longo

HEMATOPOIESIS AND THE PHYSIOLOGIC BASIS OF RED CELL PRODUCTION

Hematopoiesis is the process by which the formed elements of the blood are produced. The process is regulated through a series of steps beginning with the pluripotent hematopoietic stem cell. Stem cells are capable of producing red cells, all classes of granulocytes, monocytes, platelets, and the cells of the immune system. Commitment of the stem cell to the specific cell lineages appears not to be regulated by known exogenous growth factors or cytokines. Rather, stem cells develop into differentiated cell types through incompletely defined molecular events that are intrinsic to the stem cell itself ([Chap. 104](#)). Following lineage commitment (or differentiation), hematopoietic progenitor and precursor cells come increasingly under the regulatory influence of growth factors and hormones, such as erythropoietin (EPO) for red cell production. EPO is required for the maintenance of committed erythroid progenitor cells which, in the absence of the hormone, undergo programmed cell death (*apoptosis*). The regulated process of red cell production is *erythropoiesis*, and its key elements are illustrated in [Fig. 61-1](#).

In the bone marrow, the first morphologically recognizable erythroid precursor is the pronormoblast. This cell can undergo 4 to 5 cell divisions that result in the production of 16 to 32 mature red cells. With increased [EPO](#) production, or the administration of EPO as a drug, early progenitor cell numbers are amplified and, in turn, give rise to increased numbers of erythrocytes. The regulation of EPO production itself is linked to O₂ transport.

In mammals, O₂ is transported to tissues bound to the hemoglobin contained within circulating red cells. The mature red cell is 8µm in diameter, anucleate, discoid in shape, and extremely pliable in order for it to traverse the microcirculation successfully; its membrane integrity is maintained by the intracellular generation of ATP. Normal red cell production results in the daily replacement of 0.8 to 1% of all circulating red cells in the body. The average red cell lives 100 to 120 days. The machinery responsible for red cell production is called the *erythron*. The erythron is a dynamic organ made up of a rapidly proliferating pool of marrow erythroid precursor cells and a large mass of mature circulating red blood cells. The size of the red cell mass reflects the balance of red cell production and destruction. The physiologic basis of red cell production and destruction provides an understanding of the mechanisms that can lead to anemia.

The physiologic regulator of red cell production, the glycoprotein hormone [EPO](#), is produced and released by peritubular capillary lining cells within the kidney. These cells are highly specialized epithelial-like cells. A small amount of EPO is produced by hepatocytes. The fundamental stimulus for EPO production is the availability of O₂ for tissue metabolic needs. Impaired O₂ delivery to the kidney can result from a decreased red cell mass (*anemia*), impaired O₂ loading of the hemoglobin molecule (*hypoxemia*), or, rarely, impaired blood flow to the kidney (renal artery stenosis). EPO governs the day-to-day production of red cells, and ambient levels of the hormone can be measured in the plasma by sensitive immunoassays -- the normal level being 10 to 25 U/L. When

the hemoglobin concentration falls below 100 to 120 g/L (10 to 12 g/dL), plasma EPO levels increase logarithmically in inverse proportion to the severity of the anemia. In circulation, EPO has a half-clearance time of 6 to 9 h. EPO acts by binding to specific receptors on the surface of marrow erythroid precursors, inducing them to proliferate and to mature. Under the stimulus of EPO, red blood cell production can increase four- to fivefold within a 1- to 2-week period but only in the presence of adequate nutrients, especially iron. The functional capacity of the erythron, therefore, requires normal renal production of EPO, a functioning erythroid marrow, and an adequate supply of substrates for hemoglobin synthesis. A defect in any of these key components can lead to anemia. Generally, anemia is recognized in the laboratory when a patient's hemoglobin level or hematocrit is reduced below an expected value (the normal range). The likelihood and severity of anemia are defined based on the deviation of the patient's hemoglobin/hematocrit from values expected for age- and sex-matched normal subjects. The lower ranges of distribution of hemoglobin/hematocrit values for adult males and females are shown in [Fig. 61-2](#). The hemoglobin concentration in adults has a Gaussian distribution. The mean hematocrit value for adult males is 47% ($\pm$ SD 7) and that for adult females is 42% ($\pm$ 5). Any individual hematocrit or hemoglobin value carries with it a likelihood of associated anemia. Thus, a hematocrit of $\leq$ 39% in an adult male or $<$ 35% in an adult female has only about a 25% chance of being normal. Suspected low hemoglobin or hematocrit values are more easily interpreted if there are historic values for the same patient for comparison.

The critical elements of erythropoiesis -- [EPO](#) production, iron availability, the proliferative capacity of the bone marrow, and effective maturation of red cell precursors -- are used for the initial classification of anemia (see "Definition and Classification, below).

ANEMIA

CLINICAL PRESENTATION OF ANEMIA

Signs and Symptoms Anemia is most often recognized by abnormal screening laboratory tests. Patients only occasionally present with advanced anemia and its attendant signs and symptoms. Acute anemia is nearly always due to blood loss or hemolysis. In fact, with acute blood loss, hypovolemia dominates the clinical picture and the hematocrit and hemoglobin levels do not reflect the volume of blood lost. Signs of vascular instability dominate with acute losses of 10 to 15% of the total blood volume. In such patients, the issue is not anemia but hypotension and decreased organ perfusion. When $>$ 30% of the blood volume is lost suddenly, patients are unable to compensate with the usual mechanisms of vascular contraction and changes in regional blood flow. The patient prefers to remain supine and will show postural hypotension and tachycardia if upright. If the volume of blood lost is $>$ 40% (i.e., $>$ 2 L in the average-sized adult), signs of hypovolemic shock including confusion, air hunger, diaphoresis, hypotension, and tachycardia appear ([Chap. 108](#)). Such patients have significant deficits in vital organ perfusion and require immediate volume replacement. With mild blood loss, enhanced O_2 delivery is achieved through changes in the O_2 -hemoglobin dissociation curve mediated by a decreased pH or increased CO_2 (*Bohr effect*).

With acute hemolytic disease, the signs and symptoms depend on the mechanism that

leads to red cell destruction. Intravascular hemolysis with release of free hemoglobin may be associated with acute back pain, free hemoglobin in the plasma and urine, and renal failure. Symptoms associated with more chronic or progressive anemia depend on the age of the patient and the adequacy of blood supply to critical organs. Symptoms associated with moderate anemia include fatigue, loss of stamina, breathlessness, and tachycardia (particularly with physical exertion). However, because of the intrinsic compensatory mechanisms that govern the O₂-hemoglobin dissociation curve, the gradual onset of anemia -- particularly in young patients -- may not be associated with signs or symptoms until the anemia is severe [hemoglobin <70 to 80 g/L (7 to 8 g/dL)]. When anemia develops over a period of days or weeks, the total blood volume is normal to slightly increased and changes in cardiac output and regional blood flow help compensate for the overall loss in O₂-carrying capacity. Changes in the position of the O₂-hemoglobin dissociation curve account for some of the compensatory response to anemia. With chronic anemia, intracellular levels of 2,3-bisphosphoglycerate (BPG) rise, shifting the dissociation curve to the right and facilitating O₂ unloading. This compensatory mechanism can only maintain normal tissue O₂ delivery in the face of a 20 to 30 g/L (2 to 3 g/dL) deficit in hemoglobin concentration. Finally, further protection of O₂ delivery to vital organs is achieved by the shunting of blood away from organs that are relatively rich in blood supply, particularly the kidney, gut, and skin.

Certain disorders are commonly associated with anemia. Chronic inflammatory states (e.g., infection, rheumatoid arthritis) are associated with mild to moderate anemia, whereas lymphoproliferative disorders, such as chronic lymphocytic leukemia and certain other B cell neoplasms, may be associated with autoimmune hemolysis.

Approach to the Patient

The evaluation of the patient with anemia requires a careful history and physical examination. Historic information that may be useful includes exposure to certain toxic agents or drugs and symptoms related to other disorders commonly associated with anemia. These include symptoms and signs such as bleeding, fatigue, malaise, fever, weight loss, night sweats, and other systemic symptoms. Clues to the mechanisms of anemia may be provided on physical examination by findings of infection, blood in the stool, lymphadenopathy, splenomegaly, or petechiae. Splenomegaly and lymphadenopathy suggest an underlying lymphoproliferative disease, while petechiae suggest platelet dysfunction. If it is uncertain whether a mild anemia represents an extreme normal value or an abnormal finding, past laboratory measurements may be helpful. Nutritional history related to drugs or alcohol intake and family history of anemia should always be assessed. Certain geographic backgrounds and ethnic origins are associated with an increased likelihood of an inherited disorder of the hemoglobin molecule or intermediary metabolism. Glucose-6-phosphate dehydrogenase deficiency and certain hemoglobinopathies are seen more commonly in those of middle-Eastern or African origin.

In the anemic patient, physical examination may demonstrate a forceful heartbeat, strong peripheral pulses, and a systolic "flow" murmur. The skin and mucous membranes may be pale if the hemoglobin is <80 to 100 g/L (8 to 10 g/dL). This part of the physical examination should focus on areas where vessels are close to the surface such as the mucous membranes, nail beds, and palmar creases. If the palmar creases

are lighter in color than the surrounding skin when the hand is hyperextended, the hemoglobin level is usually <80 g/L (8 g/dL).

Laboratory Evaluation [Table 61-1](#) lists the tests used in the initial workup of anemia. A routine complete blood count (CBC) is required as part of the evaluation and includes the hemoglobin, hematocrit, and red cell indices: the mean cell volume (MCV) in femtoliters, mean cell hemoglobin (MCH) in picograms per cell, and mean concentration of hemoglobin per volume of red cells (MCHC) in grams per liter (non-SI: grams per deciliter). The red cell indices are calculated as shown in [Table 61-2](#), and the normal variations in the CBC with age are shown in [Table A-7](#). A number of physiologic factors affect the normal CBC values including age, gender, pregnancy, smoking, and altitude. High-normal hemoglobin values may be seen in men and women who live at altitude or smoke heavily. The elevations in smokers reflect normal compensation due to the displacement of O₂ by CO in hemoglobin binding. Other important information is provided by the reticulocyte count and measurements of iron supply including the *serum iron*, the *total iron-binding capacity* (TIBC; an indirect measure of the transferrin level), and *serum ferritin*. Marked alterations in the red cell indices usually reflect disorders of maturation or iron deficiency. Clinical laboratories also provide a description of both the red and white cells, a white cell differential count, and the platelet count. In patients with severe anemia and abnormalities in red blood cell morphology, a bone marrow aspirate or biopsy may be important to assist in the diagnosis. Other tests of value in the diagnosis of specific anemias are discussed in chapters on specific disease states.

The components of the [CBC](#) also help in the classification of anemia. *Microcytosis* is reflected by a lower than normal [MCV](#) (<80), whereas high values (>100) reflect *macrocytosis*. The [MCH](#) and [MCHC](#) reflect defects in hemoglobin synthesis (*hypochromia*). Automated cell counters describe the red cell volume distribution width (RDW). The MCV (representing the peak of the distribution curve) is insensitive to the appearance of small populations of macrocytes or microcytes. An experienced laboratory technician will be able to identify minor populations of large or small cells or hypochromic cells before the red cell indices change.

PERIPHERAL BLOOD SMEAR The peripheral blood smear provides important information about defects in red cell production. As a complement to the red cell indices, the blood smear also reveals variations in cell size (*anisocytosis*) and shape (*poikilocytosis*). The degree of anisocytosis usually correlates with increases in the [RDW](#) or the range of cell sizes. Poikilocytosis suggests a defect in the maturation of red cell precursors in the bone marrow or fragmentation of circulating red cells. The blood smear may also reveal *polychromasia* -- red cells that are slightly larger than normal and grayish blue in color on the Wright-Giemsa stain. These cells are reticulocytes that have been prematurely released from the bone marrow, and their color represents residual amounts of ribosomal RNA. These cells appear in circulation in response to [EPO](#) stimulation or to architectural damage of the bone marrow (fibrosis, infiltration of the marrow by malignant cells, etc.) that results in their disordered release from the marrow. The appearance of nucleated red cells, Howell-Jolly bodies, target cells, sickle cells, and others may provide clues to specific disorders (see [Plates V-2, V-3, V-8, V-9, V-16, V-21, V-24, V-26, V-27, V-28, and V-39](#)).

RETICULOCYTE COUNT An accurate reticulocyte count is key to the initial

classification of anemia. Normally, reticulocytes are red cells that have been recently released from the bone marrow. They are identified by staining with a supravital dye that precipitates the residual ribosomal RNA. These precipitates appear as blue or black punctate spots. This residual RNA is metabolized over the first 24 to 36 h of the reticulocyte's lifespan in circulation. Normally, the reticulocyte count ranges from 1 to 2% and reflects the daily replacement of 0.8 to 1.0% of the circulating red cell population. A correctly interpreted reticulocyte count provides a reliable measure of red cell production.

In the initial classification of anemia, the patient's reticulocyte count is compared with the expected reticulocyte response. In general, if the [EPO](#) and erythroid marrow responses to moderate anemia [hemoglobin < 100 g/L (10 g/dL)] are intact, the red cell production rate increases to two to three times normal within 10 days following the onset of anemia. In the face of established anemia, a reticulocyte response less than two to three times normal indicates an inadequate marrow response.

In order to use the reticulocyte count to estimate marrow response, two corrections are necessary. The first correction adjusts the reticulocyte count based on the reduced number of circulating red cells. With anemia, the percentage of reticulocytes may be increased while the absolute number is unchanged. To correct for this effect, the reticulocyte percentage is multiplied by the ratio of the patient's hemoglobin or hematocrit to the expected hemoglobin/hematocrit for the age and gender of the patient ([Table 61-3](#)). This provides an estimate of the absolute reticulocyte count. In order to convert the corrected reticulocyte count to an index of marrow production, a further correction is required, depending on whether some of the reticulocytes in circulation have been released from the marrow prematurely. For this second correction, the peripheral blood smear is examined to see if there are polychromatophilic macrocytes present. These cells, representing prematurely released reticulocytes, are referred to as "shift" cells, and the relationship between the degree of shift (and the necessary shift correction factor) is shown in [Fig. 61-3](#). The correction is necessary because these prematurely released cells survive as reticulocytes in circulation for >1 day, thereby providing a falsely high estimate of daily red cell production. If polychromasia is increased, the reticulocyte count, already corrected for anemia, should be divided again by a factor of 2 to account for the prolonged reticulocyte maturation time. The second correction factor varies from 1 to 3 depending upon the severity of anemia. In general, a correction of 2 is commonly used. An appropriate correction is shown in [Table 61-3](#). If polychromatophilic cells are not seen on the blood smear, the second correction is not required. The now doubly corrected reticulocyte count is the *reticulocyte production index*, and it provides an estimate of marrow production relative to normal.

Premature release of reticulocytes is normally due to increased [EPO](#) stimulation. However, if the integrity of the bone marrow release process is lost through tumor infiltration, fibrosis, or other disorders, the appearance of nucleated red cells or polychromatophilic macrocytes should still invoke the second reticulocyte correction. The shift correction should always be applied to a patient with anemia and a very high reticulocyte count to provide a true index of effective red cell production. Patients with severe chronic hemolytic anemia may increase red cell production as much as six- to sevenfold. This measure alone, therefore, confirms the fact that the patient has an appropriate EPO response, a normally functioning bone marrow, and sufficient iron

available to meet the demands for new red cell formation. If the reticulocyte production index is <2 in the face of established anemia, a defect in erythroid marrow proliferation or maturation must be present.

TESTS OF IRON SUPPLY AND STORAGE The laboratory measurements that reflect the availability of iron for hemoglobin synthesis include the serum iron, the [TIBC](#), and the percent transferrin saturation. The percent transferrin saturation is derived by dividing the serum iron level ($\times 100$) by the TIBC. The normal serum iron ranges from 9 to 27 $\mu\text{mol/L}$ (50 to 150 $\mu\text{g/dL}$), while the normal TIBC is 54 to 64 $\mu\text{mol/L}$ (300 to 360 $\mu\text{g/dL}$); the transferrin saturation ranges from 25 to 50%. A diurnal variation in the serum iron leads to a variation in the percent transferrin saturation. The serum ferritin is used to evaluate total-body iron stores. Adult males have serum ferritin levels that average about 100 $\mu\text{g/L}$, corresponding to iron stores of about 1 g. Adult females have lower serum ferritin levels averaging 30 $\mu\text{g/L}$, reflecting lower iron stores. A serum ferritin level of 10 to 15 $\mu\text{g/L}$ represents depletion of body iron stores. However, ferritin is also an acute-phase reactant and, in the presence of acute or chronic inflammation, may rise severalfold above baseline levels. As a rule, a serum ferritin $>200 \mu\text{g/L}$ means there is at least some iron in tissue stores.

BONE MARROW EXAMINATION A bone marrow aspirate and smear or a needle biopsy may be useful in the diagnosis of a marrow disorder such as myelofibrosis, a red cell maturation defect, or an infiltrative disease ([Plates V-5, V-13, V-14, V-15, V-19, V-29, and V-33](#)). The increase or decrease of one cell lineage (myeloid vs. erythroid) compared to another is obtained by a differential count of nucleated cells in a bone marrow smear [the erythroid/granulocytic (E/G) ratio]. A patient with a hypoproliferative anemia (see below) and a reticulocyte production index <2 will demonstrate an E/G ratio of 1:2 or 1:3. In contrast, patients with hemolytic disease and a production index >3 will have an E/G ratio of at least 1:1. Maturation disorders are identified from the discrepancy between a high E/G ratio and a low reticulocyte production index (see below). Either the marrow smear or biopsy can be stained for the presence of iron stores or iron in developing red cells. The storage iron is in the form of *ferritin* or *hemosiderin*. On carefully prepared bone marrow smears, small ferritin granules can normally be seen in 10 to 20% of developing erythroblasts. Such cells are called *sideroblasts*.

OTHER LABORATORY MEASUREMENTS

Additional laboratory tests may be of value in confirming specific diagnoses. **For details of these tests and how they are applied in individual disorders, see [Chaps. 105 to 109](#).*

DEFINITION AND CLASSIFICATION OF ANEMIA

Initial Classification of Anemia Classifying an anemia according to the functional defect in red cell production helps organize the subsequent use of laboratory studies. The three major classes of anemia are: (1) marrow production defects (*hypoproliferation*), (2) red cell maturation defects (*ineffective erythropoiesis*), and (3) decreased red cell survival (*blood loss/hemolysis*). This functional classification of anemia then guides the selection of specific clinical and laboratory studies designed to complete the differential diagnosis and to plan appropriate therapy. The classification is

shown in [Fig. 61-4](#). A hypoproliferative anemia is typically seen with a low reticulocyte production index together with little or no change in red cell morphology (a normocytic, normochromic anemia) ([Chap. 105](#)). Maturation disorders typically have a slight to moderately elevated reticulocyte production index that is accompanied by either macrocytic ([Chap. 107](#)) or microcytic ([Chaps. 105,106](#)) red cell indices. Increased red blood cell destruction secondary to hemolysis results in an increase in the reticulocyte production index to at least three times normal ([Chap. 108](#)), provided sufficient iron is available for hemoglobin synthesis. Hemorrhagic anemia does not typically result in production indices of more than 2.5 times normal because of the limitations placed on expansion of the erythroid marrow by iron availability.

In the first branch point of the classification of anemia, a reticulocyte production index >2.5 indicates that hemolysis is most likely. A reticulocyte production index of <2 indicates either a hypoproliferative anemia or maturation disorder. The latter two possibilities can often be distinguished by the red cell indices, by examination of the peripheral blood smear, or by a marrow examination. If the red cell indices are normal, the anemia is almost certainly hypoproliferative in nature. Maturation disorders are characterized by ineffective red cell production and a low reticulocyte production index with bizarre red cell shapes -- macrocytes or hypochromic microcytes on the peripheral blood smear. With a hypoproliferative anemia, no erythroid hyperplasia is noted in the marrow, whereas patients with ineffective red cell production have erythroid hyperplasia and an E/G ratio³1:1.

Hypoproliferative Anemias At least 75% of all cases of anemia are hypoproliferative in nature. A hypoproliferative anemia reflects absolute or relative marrow failure in which the erythroid marrow has not proliferated appropriately for the degree of anemia. The majority of hypoproliferative anemias are due to mild to moderate iron deficiency or inflammation. A hypoproliferative anemia can result from marrow damage, iron deficiency, or inadequate [EPO](#) stimulation. The last may reflect impaired renal function, suppression of EPO production by inflammatory cytokines such as interleukin 1, or reduced tissue needs for O₂ from metabolic disease such as hypothyroidism. Only occasionally is the marrow unable to produce red cells at a normal rate, and this is most prevalent in patients with renal failure. In general, hypoproliferative anemias are characterized by normocytic, normochromic red cells, although microcytic, hypochromic cells may be observed with mild iron deficiency or long-standing chronic inflammatory disease. The key laboratory tests in distinguishing between the various forms of hypoproliferative anemia include the serum iron and iron-binding capacity, evaluation of renal and thyroid function, a marrow biopsy or aspirate to detect marrow damage or infiltrative disease, and serum ferritin to assess iron stores. Occasionally, an iron stain of the marrow will be needed to determine the pattern of iron distribution. Patients with the anemia of acute or chronic inflammation show a distinctive pattern of serum iron (low), [TIBC](#) (normal or low), percent transferrin saturation (low), and serum ferritin (normal or high). A distinct pattern of results is noted in mild to moderate iron deficiency (low serum iron, high TIBC, low percent transferrin saturation, low serum ferritin) ([Chap. 105](#)). Marrow damage by a drug, infiltrative disease such as leukemia or lymphoma, or marrow aplasia can usually be diagnosed from the peripheral blood and bone marrow morphology. With infiltrative disease or fibrosis, a marrow biopsy will likely be required.

Maturation Disorders The presence of anemia with an inappropriately low reticulocyte

production index, macro- or microcytosis on smear, and abnormal red cell indices suggests a maturation disorder. Maturation disorders are divided into two categories: nuclear maturation defects, associated with macrocytosis and abnormal marrow development, and cytoplasmic maturation defects, associated with microcytosis and hypochromia usually from defects in hemoglobin production. The low reticulocyte production index is a reflection of the ineffective erythropoiesis that results from the destruction within the marrow of developing erythroblasts. Marrow morphology shows an E/G ratio of $\approx 1:1$, diagnostic of erythroid hyperplasia.

Nuclear maturation defects result from vitamin B₁₂ or folic acid deficiency, drug damage, or myelodysplasia. Drugs that interfere with cellular DNA metabolism, such as methotrexate or alkylating agents, can produce a nuclear maturation defect. Alcohol, alone, is also capable of producing macrocytosis and a variable degree of anemia, but this is usually associated with coincident folic acid deficiency. Measurements of folic acid and vitamin B₁₂ are key not only in identifying the specific vitamin deficiency but also because they reflect different pathogenetic mechanisms.

Cytoplasmic maturation defects result from severe iron deficiency or abnormalities in globin or heme synthesis. Iron deficiency occupies an unusual position in the classification of anemia. If the iron-deficiency anemia is mild to moderate, erythroid marrow proliferation is decreased and the anemia is classified as hypoproliferative. However, if the anemia is severe and prolonged, the erythroid marrow will become hyperplastic despite the inadequate iron supply, and the anemia will be classified as ineffective erythropoiesis with a cytoplasmic maturation defect. In either case, a reduced reticulocyte production index, microcytosis, and a classic pattern of iron values make the diagnosis clear and easily distinguish iron deficiency from other cytoplasmic maturation defects such as the thalassemias. Defects in heme synthesis, in contrast to globin synthesis, are less common and may be acquired or inherited ([Chap. 346](#)). Acquired abnormalities are usually associated with myelodysplasia, may lead to either a macro- or microcytic anemia, and are frequently associated with mitochondrial iron loading. In these cases, iron is taken up by the mitochondria of the developing erythroid cell but not incorporated into heme. The iron-encrusted mitochondria surround the nucleus of the erythroid cell, forming a ring. Based on the distinctive finding of so-called ringed sideroblasts on the marrow iron stain (see [Plate V-37](#)), patients are diagnosed as having a sideroblastic anemia -- almost always reflecting myelodysplasia. Again, studies of iron parameters are helpful in the differential diagnosis and management of these patients.

Blood Loss/Hemolytic Anemia In contrast to anemias associated with an inappropriately low reticulocyte production index, blood loss or hemolysis is associated with red cell production indices of ≥ 2.5 times normal. The stimulated erythropoiesis is reflected in the blood smear by the appearance of increased numbers of polychromatophilic macrocytes. A marrow examination is rarely indicated if the reticulocyte production index is increased appropriately. The red cell indices are typically normocytic or slightly macrocytic, reflecting the increased number of reticulocytes. Acute blood loss is not associated with an increased reticulocyte production index because of the time required to increase [EPO](#) production and, subsequently, marrow proliferation. Subacute blood loss may be associated with modest reticulocytosis because iron is lost along with the red cells. Anemia from chronic

blood loss more often presents as iron deficiency than with the picture of increased red cell production.

The evaluation of blood loss anemia is usually not difficult. Most problems arise when a patient presents with an increased red cell production index from an episode of acute blood loss that went unrecognized. The cause of the anemia and increased red cell production may not be obvious. The confirmation of a recovering state may require observations over a period of 2 to 3 weeks, during which the hemoglobin concentration will be seen to rise and the reticulocyte production index fall.

Hemolytic disease, while dramatic, is among the least common forms of anemia. The ability to sustain a high reticulocyte production index reflects the ability of the erythroid marrow to compensate for hemolysis and the efficient recycling of iron from the destroyed red cells to support new hemoglobin synthesis. The level of response will depend on the severity of the anemia and the nature of the underlying disease process.

Hemolytic anemias present in different ways. Some appear suddenly as an acute, self-limited episode of intravascular or extravascular hemolysis, a presentation pattern often seen in patients with autoimmune hemolysis or with inherited defects of the Embden-Myerhof pathway or the glutathione reductase pathway. Patients with inherited disorders of the hemoglobin molecule or red cell membrane generally have a lifelong clinical history typical of the disease process. Those with chronic hemolytic disease, such as hereditary spherocytosis, may actually present not with anemia but with a complication stemming from the prolonged increase in red cell destruction such as aplastic crisis, symptomatic bilirubin gallstones, or splenomegaly.

The differential diagnosis of an acute or chronic hemolytic event requires the careful integration of family history, pattern of clinical presentation, and a number of highly specific laboratory studies ([Chap. 108](#)). Some of the more common congenital hemolytic anemias may be identified from the red cell morphology, a routine laboratory test such as hemoglobin electrophoresis, or a screen for red cell enzymes. Acquired defects in red cell survival are often immunologically mediated and require the immunoglobulin test or a cold agglutinin titer to detect the presence of hemolytic antibodies or complement-mediated red cell destruction.

TREATMENT

An overriding principle is to not initiate treatment of mild to moderate anemia without a specific diagnosis. Rarely, in the acute setting, anemia may be so severe that red cell transfusions are required before a specific diagnosis is made. Whether the anemia is of acute or gradual onset, the selection of the appropriate treatment is determined by the documented cause(s) of the anemia. Often, the cause of the anemia may be multifactorial. For example, a patient with severe rheumatoid arthritis who has been taking anti-inflammatory drugs may have a hypoproliferative anemia associated with chronic inflammation as well as chronic blood loss associated with intermittent gastrointestinal bleeding. In every circumstance, it is important to evaluate the patient's iron status fully before and during the treatment of any anemia. **Transfusion is discussed in [Chap. 114](#); iron therapy is discussed in [Chap. 105](#); treatment of megaloblastic anemia is discussed in [Chap. 107](#); treatment of other entities is discussed*

in their respective chapters (sickle cell anemia, [Chap. 106](#); hemolytic anemias, [Chap. 108](#); aplastic anemia and myelodysplasia, [Chap. 109](#)).

Therapeutic options for the treatment of anemias have expanded dramatically during the past 25 years. Blood component therapy is available and safe. Recombinant [EPO](#) as an adjunct to anemia management has transformed the lives of patients with chronic renal failure on dialysis. Improvements in the management of sickle cell crises and sickle cell anemia have also occurred. Eventually, patients with inherited disorders of globin synthesis or mutations in the globin gene, such as sickle cell disease, may benefit from the successful introduction of targeted genetic therapy ([Chap. 69](#)).

POLYCYTHEMIA

Polycythemia is defined as an increase in circulating red blood cells above normal. This increase may be real or only apparent (spurious or relative) because of a decrease in plasma volume. The term *erythrocytosis* may be used interchangeably with polycythemia, but some draw a distinction between them; erythrocytosis implies documentation of increased red cell mass, whereas polycythemia refers to any increase in red cells. Often patients with polycythemia are detected through an incidental finding of elevated hemoglobin or hematocrit levels. Concern that the hemoglobin level may be abnormally high is usually triggered at 170 g/L (17 g/dL) for men and 150 g/L (15 g/dL) for women. Hematocrit levels >50% in men or >45% in women may be abnormal. Hematocrits >60% in men and >55% in women are almost invariably associated with increased red cell mass.

Historic features useful in the differential diagnosis include smoking history; living at high altitude; or a history of congenital heart disease, peptic ulcer disease, sleep-apnea, chronic lung disease, or renal disease.

Patients with polycythemia may be asymptomatic or experience symptoms related to the increased red cell mass or an underlying disease process that leads to increased red cell production. The dominant symptoms from increased red cell mass are thrombotic (both venous and arterial), because the blood viscosity increases logarithmically at hematocrits >55%. Manifestations range from digital ischemia to Budd-Chiari syndrome with hepatic vein thrombosis. Abdominal thromboses are particularly common. Neurologic symptoms such as vertigo, tinnitus, headache, and visual disturbances may occur. Hypertension is often present. Patients with *polycythemia vera* may have aquagenic pruritus and symptoms related to hepatosplenomegaly. Patients may have easy bruising, epistaxis, or bleeding from the gastrointestinal tract. Patients with hypoxemia may develop cyanosis on minimal exertion or have headache, impaired mental acuity, and fatigue.

The physical examination usually reveals a ruddy complexion. Splenomegaly favors polycythemia vera as the diagnosis ([Chap. 110](#)). The presence of cyanosis or evidence of a right-to-left shunt suggests congenital heart disease presenting in the adult, particularly tetralogy of Fallot or Eisenmenger syndrome ([Chap. 234](#)). Increased blood viscosity raises pulmonary artery pressure; hypoxemia can lead to increased pulmonary vascular resistance. Together these factors can produce cor pulmonale.

Polycythemia can be spurious (related to a decrease in plasma volume; Gaisbock's syndrome), primary, or secondary in origin. The secondary causes are all associated with increases in [EPO](#) levels: either a physiologically adapted appropriate elevation based upon tissue hypoxia (lung disease, high altitude, CO poisoning, high-affinity hemoglobinopathy) or an abnormal overproduction (renal disease, tumors with ectopic EPO production). A rare familial form of polycythemia is associated with normal EPO levels but mutations producing hyperresponsive EPO receptors.

Approach to the Patient

As shown in [Fig. 61-5](#), the first step is to document the presence of an increased red cell mass using the principle of isotope dilution by administering ^{51}Cr -labeled autologous red blood cells to the patient and sampling blood radioactivity over a 2-h period. If the red cell mass is normal (<36 mL/kg in men, <32 mL/kg in women), the patient has spurious polycythemia. If the red cell mass is increased (>36 mL/kg in men, >32 mL/kg in women), serum [EPO](#) levels should be measured. If EPO levels are low or absent, the patient most likely has polycythemia vera. Ancillary tests that support this diagnosis include elevated white blood cell count, increased absolute basophil count, thrombocytosis, elevated leukocyte alkaline phosphatase levels, and elevated serum vitamin B₁₂ and vitamin B₁₂-binding protein levels.

If serum [EPO](#) levels are elevated, one attempts to distinguish whether the elevation is a physiologic response to hypoxia or is related to autonomous production. Patients with low arterial O₂ saturation ($<92\%$) should be further evaluated for the presence of heart or lung disease, if they are not living at high altitude. Patients with normal O₂ saturation who are smokers may have elevated EPO levels because of CO displacement of O₂. If carboxyhemoglobin (COHb) levels are high, the diagnosis is smoker's polycythemia. Such patients should be urged to stop smoking. Those who cannot stop smoking require phlebotomy to control their polycythemia. Patients with normal O₂ saturation who do not smoke either have an abnormal hemoglobin that does not deliver O₂ to the tissues (evaluated by finding elevated O₂-hemoglobin affinity) or have a source of EPO production that is not responding to the normal feedback inhibition. Further workup is dictated by the differential diagnosis of EPO-producing neoplasms. Hepatoma, uterine leiomyoma, and renal disease or cysts are all detectable with abdominopelvic computed tomography scans. Cerebellar hemangiomas may produce EPO, but they nearly always present with localizing neurologic signs and symptoms rather than polycythemia-related symptoms.

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Dr. Robert S. Hillman wrote this chapter in the 14th edition, and elements of his chapter were retained here.

(Bibliography omitted in Palm version)

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62. BLEEDING AND THROMBOSIS - Robert I. Handin

Hemorrhage, intravascular thrombosis, and embolism are common clinical manifestations of many diseases. The normal hemostatic system limits blood loss by precisely regulated interactions between components of the vessel wall, blood platelets, and plasma proteins. However, when disease or trauma damages large arteries and veins, excessive bleeding may occur, despite a normal hemostatic system. Less frequently, hemorrhage is caused by an inherited or acquired disorder of the hemostatic machinery itself. A large number of such bleeding disorders have been identified.

In addition, unregulated activation of the hemostatic system may cause thrombosis and embolism, which can reduce blood flow to critical organs such as the brain and myocardium. Although we understand less about the pathophysiology of thrombosis than of hemostatic failure, certain patient groups have been identified that are particularly prone to thrombosis and embolism. These include patients who (1) are immobilized after surgery, (2) have chronic congestive heart failure, (3) have atherosclerotic vascular disease, (4) have a malignancy, or (5) are pregnant. Most of these "thrombosis-prone" patients have inherited or acquired "hypercoagulable" or "prethrombotic" disorders.

Certain information in the patient's history, such as the mode of onset and sites of bleeding, a family bleeding tendency, and a record of drug ingestion, helps establish the correct diagnosis. Physical examination can identify bleeding in the skin or joint deformities due to previous hemarthroses. Ultimately, however, bleeding disorders are diagnosed by laboratory tests. General screening tests are used first, to document a systemic disorder, and are then supplemented by specific tests of coagulation protein or platelet function to arrive at an accurate diagnosis.

The hypercoagulable or prethrombotic patient can also be identified by a careful history. Three important clues to this diagnosis are: (1) repeated episodes of thromboembolism without an obvious predisposing condition, (2) a family history of thrombosis, and (3) well-documented thromboembolism in adolescents and young adults. All of the known inherited prethrombotic disorders can be diagnosed with specific immunologic, functional, and, in some cases, genetic tests.

NORMAL HEMOSTASIS

Accurate diagnosis and treatment of patients with either bleeding or thrombosis require knowledge of the pathophysiology of hemostasis. The process can be divided into primary and secondary components and is initiated when trauma, surgery, or disease disrupts the vascular endothelial lining and blood is exposed to subendothelial connective tissue. *Primary hemostasis* is the name given to the process of platelet plug formation at sites of injury. It occurs within seconds of injury and is of prime importance in stopping blood loss from capillaries, small arterioles, and venules ([Fig. 62-1](#)).

Secondary hemostasis consists of the reactions of the plasma coagulation system that result in fibrin formation. It requires several minutes for completion. The fibrin strands that are produced strengthen the primary hemostatic plug. This reaction is particularly important in larger vessels and prevents bleeding from recurring hours or days after the injury. Although presented here as separate events, primary and secondary hemostasis

are closely linked. For example, activated platelets accelerate plasma coagulation, and products of the plasma coagulation reaction, such as thrombin, induce platelet activation.

Effective primary hemostasis requires three critical events -- platelet adhesion, granule release, and platelet aggregation. Within a few seconds of injury, platelets adhere to collagen fibrils in vascular subendothelium via a specific platelet collagen receptor, glycoprotein Ia/IIa, which is a member of the integrin family. As shown in [Fig. 62-2](#), this interaction is stabilized by the von Willebrand factor, an adhesive glycoprotein that allows platelets to remain attached to the vessel wall despite the high shear forces generated within the vascular lumen. The von Willebrand factor accomplishes this task by forming a link between a platelet receptor site on glycoprotein Ib/IX and subendothelial collagen fibrils. The adherent platelets then release preformed granule constituents and generate de novo mediators like those depicted in [Fig. 62-1](#).

As in other cells, platelet activation and secretion are regulated by changes in the level of cyclic nucleotides, the influx of calcium, hydrolysis of membrane phospholipids, and phosphorylation of critical intracellular proteins. The relevant pathways are depicted in [Figs. 62-3](#) and [62-4](#). The binding of agonists such as epinephrine, collagen, or thrombin to platelet surface receptors activates two membrane enzymes -- phospholipase C and phospholipase A₂. These enzymes catalyze the release of arachidonic acid from two of the major membrane phospholipids, phosphatidylinositol and phosphatidylcholine. Initially, a small quantity of the released arachidonic acid is converted to thromboxane A₂ (TXA₂), which, in turn, can activate phospholipase C. The formation of TXA₂ from arachidonic acid is mediated by the enzyme cyclooxygenase ([Fig. 62-3](#)). This enzyme is inhibited by aspirin and nonsteroidal anti-inflammatory drugs. Inhibition of TXA₂ synthesis is a cause of mild bleeding in some patients and is the same way some antithrombotic drugs work.

Hydrolysis of the membrane phospholipid phosphatidylinositol 4,5-bisphosphate produces diacylglycerol (DAG) and inositol triphosphate (IP₃), both of which play critical roles in platelet metabolism. IP₃ mediates the movement of calcium into the platelet cytosol and stimulates the phosphorylation of myosin light chains. The latter interact with actin to facilitate granule movement and platelet shape change. DAG activates protein kinase C, which, in turn, phosphorylates several substrates, including myosin light chain kinase and a 47-kDa protein (plekstrin). Phosphorylation of these or other proteins may regulate platelet granule secretion.

A finely balanced mechanism controls the rate and extent of platelet activation ([Fig. 62-3](#)). TXA₂, a platelet product of arachidonic acid, stimulates platelet activation and secretion. In contrast, prostacyclin, an endothelial cell product of arachidonic acid metabolism, inhibits platelet activation by raising intraplatelet levels of cyclic adenosine monophosphate.

Following activation, platelets secrete their granule contents into plasma. Endoglycosidases and a heparin-cleaving enzyme are released from lysosomes; calcium, serotonin, and adenosine diphosphate (ADP) are released from the dense granules; and several proteins, including the von Willebrand factor, fibronectin, thrombospondin, the platelet-derived growth factor (PDGF), and a heparin-neutralizing

protein (platelet factor 4), are released from granules. Released ADP binds to purinergic receptors, which, when activated, change the conformation of the glycoprotein IIb/IIIa complex so that it binds fibrinogen, linking adjacent platelets into a hemostatic plug ([Fig. 62-2](#)). Released PDGF stimulates the growth and migration of fibroblasts and smooth-muscle cells within the vessel wall, an important part of the repair process.

As the primary hemostatic plug is being formed, plasma coagulation proteins are activated to initiate secondary hemostasis. An overall picture of the coagulation scheme, including the role of various inhibitors, is shown in [Fig. 62-5](#). The coagulation pathway can be broken down into a series of reactions ([Fig. 62-6](#)) that culminate in the production of enough thrombin to convert a small amount of plasma fibrinogen to fibrin. Each of the reactions requires the formation of a surface-bound complex and the conversion of inactive precursor proteins into active proteases by limited proteolysis, and each is regulated by both plasma and cellular cofactors and calcium.

In *reaction 1*, the intrinsic or contact phase of coagulation, three plasma proteins, Hageman factor (factor XII), high-molecular-weight kininogen (HMWK), and prekallikrein (PK), form a complex on vascular subendothelial collagen. After binding to HMWK, factor XII is slowly converted to an active protease (XIIa), which then activates PK to kallikrein and factor XI to XIa. Kallikrein accelerates the conversion of XII to XIIa, while XIa participates in subsequent coagulation reactions. An alternative mechanism for the activation of factor XI may exist, as patients who are deficient in either factor XII, HMWK, or PK have apparently normal hemostasis and no clinical bleeding.

Reaction 2 provides a second pathway to initiate coagulation by activating factor VII to a protease. In this extrinsic or tissue-factor-dependent pathway, a complex is formed between factor VII, calcium, and tissue factor, a ubiquitous lipoprotein present in cellular membranes and exposed by cellular injury. The tissue factor-VII pathway is continuously active and makes a major contribution to basal coagulation. Factor VII and three other coagulation proteins -- factors II (prothrombin), IX, and X -- require calcium and vitamin K for biologic activity. These proteins are synthesized in the liver, where a vitamin K-dependent carboxylase catalyzes a unique posttranslational modification that adds a second carboxyl group to certain glutamic acid residues. Pairs of these di-g-carboxyglutamic acid (Gla) residues bind calcium, which alters protein conformation for binding to phospholipid surfaces and confers biologic activity. Inhibition of this modification by vitamin K antagonists (e.g., warfarin) is the basis of one of the most common forms of anticoagulant therapy.

In *reaction 3*, factor X is activated by the proteases generated in the two previous reactions. In one reaction, a calcium- and lipid-dependent complex is formed between factors VIII, IX, and X. Within this complex, factor IX is first converted to IXa by factor XIa that was generated within the intrinsic pathway (*reaction 1*). Factor X is then activated by factor IXa in concert with factor VIII. Alternatively, both factors IX and X can be activated more directly by factor VIIa, generated via the extrinsic pathway (*reaction 2*). Activation of factors IX and X provides a link between the intrinsic and extrinsic coagulation pathways ([Fig. 62-5](#)).

Reaction 4, the final step, converts prothrombin to thrombin in the presence of factor V,

calcium, and phospholipid. Although prothrombin conversion can take place on various natural and artificial phospholipid-rich surfaces, it proceeds several thousand times faster on the surface of activated platelets or endothelial cells. Thrombin has multiple functions in hemostasis. Although its principal role in hemostasis is the conversion of fibrinogen to fibrin, it also activates factors V, VIII, and XIII and stimulates platelet aggregation and secretion. Following the release of fibrinopeptides A and B from the α and β chains of fibrinogen, the modified molecule, now called *fibrin monomer*, polymerizes into an insoluble gel. The fibrin polymer is then stabilized by the cross-linking of individual chains by factor XIIIa, a plasma transglutaminase ([Fig. 62-5](#)).

Although the classic view of coagulation had clinical utility, it left several important questions unanswered: (1) Why does factor XII deficiency dramatically prolong partial thromboplastin time (PTT) but not cause bleeding? (2) Why is there heterogeneity in the bleeding symptoms of patients with factor XI deficiency? (3) Why do deficiencies in factors VIII or IX produce such dramatic bleeding even though the "extrinsic" pathway remains intact? Activation of factors IX and X by the tissue factor-VIIa complex is thought to play a major role in the initiation of hemostasis. Once coagulation is initiated by this interaction, the tissue factor pathway inhibitor (TFPI) blocks the pathway, and elements of the intrinsic pathway, particularly factors VIII and IX, become the dominant regulators of thrombin generation. This step in the pathway explains why factor XII-deficient patients are asymptomatic and why factor XI-deficient patients have only a mild to moderate bleeding diathesis ([Fig. 62-7](#)).

Clot lysis and vessel repair begin immediately after the formation of the definitive hemostatic plug. Three potential activators of the fibrinolytic system are: Hageman factor fragments, urinary plasminogen activator (uPA) or urokinase, and tissue plasminogen activator (tPA). The principal physiologic activators, tPA and uPA, diffuse from endothelial cells and convert plasminogen, adsorbed to the fibrin clot, into plasmin ([Fig. 62-8](#)). Plasmin then degrades fibrin polymer into small fragments, which are cleared by the monocyte-macrophage scavenger system. Although plasmin can also degrade fibrinogen, the reaction remains localized because (1) tPA and some forms of uPA activate plasminogen more effectively when it is adsorbed to fibrin clots; (2) any plasmin that enters the circulation is rapidly bound and neutralized by the α_2 plasmin inhibitor (patients who lack this factor have unchecked fibrinolysis and bleed); and (3) endothelial cells release a plasminogen activator inhibitor (PAI-1), which blocks the action of tPA.

Only a small quantity of each coagulation enzyme is converted to its active form. As a consequence, the hemostatic plug does not propagate beyond the site of injury. Precise regulation is important, since each milliliter of blood contains enough clotting potential to clot all the fibrinogen in the body in 10 to 15 s. Blood fluidity is maintained by the flow of blood, the adsorption of coagulation factors to surfaces and their trapping in the emerging clot, and by multiple inhibitors in plasma. Antithrombin, proteins C and S, and [TFPI](#) are important inhibitors that maintain blood fluidity.

These inhibitors have distinct modes of action. Antithrombin forms complexes with all serine protease coagulation factors except factor VII ([Fig. 62-5](#)). Rates of complex formation are accelerated by heparin and heparin-like molecules on the surface of the endothelial cells. Heparin's ability to accelerate antithrombin activity is the basis for its

anticoagulant action. Protein C is converted to an active protease by thrombin after it is bound to an endothelial cell protein called *thrombomodulin*. Activated protein C then inactivates the two plasma cofactors V and VIII by limited proteolysis, which slows down two critical coagulation reactions. Protein C may also stimulate the release of [tPA](#) from endothelial cells. The inhibitory function of protein C is enhanced by protein S. Reduced levels of antithrombin or proteins C and S, or dysfunctional forms of these molecules, result in a hypercoagulable or prethrombotic state. In addition, a particularly common heritable defect associated with a hypercoagulable state is the presence of a form of factor V (factor V Leiden) that is resistant to protein C inhibition. Between 20 and 50% of patients with unexplained venous thromboembolism have this defect.

Blood coagulation is not uniform throughout the body. The composition of the blood clot varies with the site of injury. Hemostatic plugs or thrombi that form in veins where blood flow is slow are rich in fibrin and trapped red blood cells and contain relatively few platelets. They are often called *red thrombi* because of their appearance in surgical and pathologic specimens. The friable ends of these red thrombi, which most often form in leg veins, can break off and embolize to the pulmonary circulation. Conversely, clots that form in arteries under conditions of high flow are predominantly composed of platelets and have little fibrin. These *white thrombi* may readily dislodge from the arterial wall and embolize to distant sites, causing temporary or permanent ischemia. These clots are a particularly common cause of embolism in the cerebral and retinal circulation, where they may lead to transient neurologic dysfunction (transient ischemic attacks), including temporary monocular blindness (amaurosis fugax), or to strokes. In addition, most episodes of myocardial infarction are due to thrombi that form after the rupture of atherosclerotic plaques within diseased coronary arteries. Hemostatic plugs, which are a physiologic response to injury, are very similar to pathologic thrombi. Thrombosis has been described as coagulation occurring in the wrong place or at the wrong time.

CLINICAL EVALUATION

HISTORY

Certain elements of the history are particularly useful in determining whether bleeding is caused by an underlying hemostatic disorder or by a local anatomic defect. One clue is a history of bleeding following common hemostatic stresses such as dental extraction, childbirth, or minor surgery. Bleeding that is sufficiently severe to require a blood transfusion merits special attention. A family history of bleeding and bleeding from multiple sites that cannot be linked to trauma or surgery also suggest a systemic disorder. Since bleeding can be mild, lack of a family history of bleeding does not exclude an inherited hemostatic disorder.

Bleeding from a platelet disorder is usually localized to superficial sites such as the skin and mucous membranes, comes on immediately after trauma or surgery, and is readily controlled by local measures ([Table 62-1](#)). In contrast, bleeding from secondary hemostatic or plasma coagulation defects occurs hours or days after injury and is unaffected by local therapy. Such bleeding most often occurs in deep subcutaneous tissues, muscles, joints, or body cavities. A careful and thorough history may establish the presence of a hemostatic disorder and guide initial laboratory testing.

PHYSICAL EXAMINATION

The most common site to observe bleeding is in the skin and mucous membranes. Collections of blood in the skin are called *purpura* and may be subdivided on the basis of the site of bleeding in the skin. Small pinpoint hemorrhages into the dermis due to the leakage of red cells through capillaries are called *petechiae* and are characteristic of platelet disorders -- in particular, severe thrombocytopenia. Larger subcutaneous collections of blood due to leakage of blood from small arterioles and venules are called *ecchymoses* (common bruises) or, if somewhat deeper and palpable, *hematomas*. They are also common in patients with platelet defects and result from minor trauma. Dilated capillaries, or *telangiectasia*, may cause bleeding without any hemostatic defect. In addition, the loss of connective tissue support for capillaries and small veins that accompanies aging increases the fragility of superficial vessels, such as those on the dorsum of the hand, leading to extravasation of blood into subcutaneous tissue -- *senile purpura*. Menorrhagia is sometimes a serious problem in women with severe thrombocytopenia or platelet dysfunction. Some patients with primary hemostatic defects, especially von Willebrand's disease, may have recurrent gastrointestinal hemorrhage, often associated with angiodysplasia, a common vascular malformation in the gastrointestinal tract.

Bleeding into body cavities, the retroperitoneum, or joints is a common manifestation of plasma coagulation defects. Repeated joint bleeding may cause synovial thickening, chronic inflammation, and fluid collections and may erode articular cartilage and lead to chronic joint deformity and limited mobility. Such deformities are particularly common in deficiencies of factors VIII and IX, the two sex-linked coagulation disorders referred to as the *hemophilias*. For unclear reasons, hemarthroses are much less common in patients with other plasma coagulation defects. Blood collections in various body cavities or soft tissues can cause secondary necrosis of tissues or nerve compression. Retroperitoneal hematomas can cause femoral nerve compression, and large collections of poorly coagulated blood in soft tissues occasionally mimic malignant growths -- the pseudotumor syndrome. Two of the most life-threatening sites of bleeding are in the oropharynx, where bleeding can compromise the airway, and in the central nervous system. Intracerebral hemorrhage is one of the leading causes of death in patients with severe coagulation disorders. Because of their need for plasma and factor concentrates derived from multiple donors, many patients with hemophilia were infected with HIV before effective testing of donors was in place.

LABORATORY TESTS

The most important screening tests of the primary hemostatic system are (1) a *bleeding time* (a sensitive measure of platelet function), and (2) a *platelet count*. The latter correlates well with the propensity to bleed. The normal platelet count is 150,000 to 450,000/uL of blood. As long as the count is >100,000/uL, patients are usually not symptomatic and the bleeding time remains normal. Platelet counts of 50,000 to 100,000/uL cause mild prolongation of the bleeding time; bleeding occurs only from severe trauma or other stress. Patients with platelet counts <50,000/uL have easy bruising, manifested by skin purpura after minor trauma and bleeding after mucous membrane surgery. Patients with a platelet count <20,000/uL have an appreciable

incidence of spontaneous bleeding, usually have petechiae, and may have intracranial or other spontaneous internal bleeding. The major causes of thrombocytopenia are outlined in [Table 62-2](#).

Patients with qualitative platelet abnormalities have a normal platelet count and a prolonged bleeding time ([Table 62-3](#)). The bleeding time is ascertained by making a small, superficial skin incision and timing the duration of blood flow from the wounded area. By careful standardization, bleeding time is a reliable and sensitive test of platelet function. A template or an automated scalpel controls the length and depth of the incision (usually 1 mm deep by 9 mm long), and a sphygmomanometer inflated to 40 mmHg distends the capillary bed of the forearm uniformly. The bleeding time test must be performed by an experienced technician, as small differences in technique have a big effect on outcome. Any patient with a bleeding time >10 min has an increased risk of bleeding, but the risk does not become great until the bleeding time >15 or 20 min. As shown in [Fig. 62-9](#), the relationship between the platelet count and the bleeding time is roughly linear. When a defect in primary hemostasis is uncovered, specialized testing is needed to determine the cause of the platelet dysfunction ([Table 62-3](#)). A precise diagnosis is important in determining the proper treatment. Occasional patients with a strong history of bleeding, particularly those with mild von Willebrand's disease, may have a normal bleeding time when initially tested, owing to cyclical variations in the level of the von Willebrand factor. Repeated testing may be necessary to establish an accurate diagnosis. Bleeding time is not an effective screening test for preoperative patients.

Plasma coagulation function is readily assessed with the [PTT](#), prothrombin time (PT), thrombin time (TT), and quantitative fibrinogen determination ([Fig. 62-5, Table 62-4](#)). The PTT screens the intrinsic limb of the coagulation system and tests for the adequacy of factors XII, [HMWK, PK](#), XI, IX, and VIII. The PT screens the extrinsic or tissue factor-dependent pathway. Both tests also evaluate the common coagulation pathway involving all the reactions that occur after the activation of factor X. Prolongation of the PT and PTT that does not resolve after the addition of normal plasma suggests a coagulation inhibitor. A specific test for the conversion of fibrinogen to fibrin is needed when both the PTT and PT are prolonged -- either a TT or a clottable fibrinogen level can be employed. When abnormalities are noted in any of the screening tests, more specific coagulation factor assays can be ordered to determine the nature of the defect.

Several rare coagulation abnormalities that may be missed as they do not affect these screening tests: factor XIII deficiency, α_2 plasmin inhibitor deficiency, [PAI-1](#) deficiency (PAI-1 is the major inhibitor of plasminogen activators), and Scott's syndrome, a platelet coagulant defect. A test for factor XIII-dependent fibrin cross-linking, such as clot solubility in 5 M urea, should be ordered when the [PT](#) and [PTT](#) are both normal but the history of bleeding is strong. The fibrinolytic system can be assessed by measuring the rate of clot lysis with the euglobulin lysis or whole blood clot lysis tests and by measuring the levels of α_2 plasmin inhibitor and PAI-1. Scott's syndrome can be detected by measuring the serum PT, which assesses the amount of residual prothrombin.

Conditions associated with thrombosis are listed in [Table 62-5](#). Patients suspected of having a hypercoagulable or prethrombotic disorder on the basis of clinical information should be tested with specific assays to screen for the known defects. Currently

available tests can identify 50 to 60% of the cases of familial or recurrent venous thrombosis.

Inhibitor syndromes or circulating anticoagulants are usually due to antibodies that impair coagulation factor activity. They are an infrequent cause of bleeding and require specialized diagnostic testing. Inhibitors are likely when screening test abnormalities cannot be reversed by adding normal plasma to patient plasma. Antibodies against specific coagulation factors may develop in (1) postpartum women, (2) patients with autoimmune disorders such as systemic lupus erythematosus, (3) patients taking drugs such as penicillin and streptomycin, and (4) otherwise healthy elderly individuals. In addition, between 10 to 20% of patients with severe hemophilia who have received multiple plasma infusions develop inhibitory antibodies. Some patients, especially those with systemic lupus erythematosus, may also have a nonspecific form of anticoagulant antibody that interferes with phospholipid binding of coagulation factors and prolongs the [PT](#) and [PTT](#) but does not cause clinical bleeding. The presence of the lupus anticoagulant may increase the risk of thromboembolism and may cause placental infarction, recurrent midtrimester abortion, and venous and arterial thrombosis. The lupus-like anticoagulant is one manifestation of the anticardiolipin antibody syndrome. Patients may have anticardiolipin antibodies that do not prolong the PTT, but patients are still at risk from thrombosis. Occasionally, patients develop inhibitors that are not antibodies. For example, several patients with clinical bleeding have been found to have circulating mucopolysaccharides that have heparin-like activity.

(Bibliography omitted in Palm version)

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63. ENLARGEMENT OF LYMPH NODES AND SPLEEN - Patrick H. Henry, Dan L. Longo

This chapter is intended to serve as a guide to the evaluation of patients who present with enlargement of the lymph nodes (*lymphadenopathy*) or the spleen (*splenomegaly*). Lymphadenopathy is a rather common clinical finding in primary care settings, whereas palpable splenomegaly is less so.

LYMPHADENOPATHY

Lymphadenopathy may be an incidental finding in patients being examined for various reasons or it may be a presenting sign or symptom of the patient's illness. The physician must eventually decide whether the lymphadenopathy is a normal finding or one that requires further study, up to and including biopsy. Soft, flat, submandibular nodes (<1 cm) are often palpable in healthy children and young adults, and healthy adults may have palpable inguinal nodes of up to 2 cm, which are considered normal. Further evaluation of these normal nodes is not warranted. In contrast, if the physician believes the node(s) to be abnormal, then pursuit of a more precise diagnosis is needed.

Approach to the Patient

Lymphadenopathy may be a primary or secondary manifestation of numerous disorders, as shown in [Table 63-1](#). Many of these disorders are infrequent causes of lymphadenopathy. Analysis of lymphadenopathy in primary care practice has shown that more than two-thirds of patients have nonspecific causes or upper respiratory illnesses (viral or bacterial), and fewer than 1% have a malignancy. In one study, researchers reported that 186 of 220 patients (84%) referred for evaluation of lymphadenopathy had a "benign" diagnosis. The remaining 34 patients (16%) had a malignancy (lymphoma or metastatic adenocarcinoma). Sixty-three percent (112) of the 186 patients with benign lymphadenopathy had a nonspecific or reactive etiology (no causative agent found), and the remainder had a specific cause demonstrated, most commonly infectious mononucleosis, toxoplasmosis, or tuberculosis. Thus, the vast majority of patients with lymphadenopathy will have a nonspecific etiology requiring few diagnostic tests.

Clinical Assessment The physician will be aided in the pursuit of an explanation for the lymphadenopathy by a careful medical history, physical examination, selected laboratory tests, and perhaps an excisional lymph node biopsy.

The *medical history* should reveal the setting in which lymphadenopathy is occurring. Symptoms such as sore throat, cough, fever, night sweats, fatigue, weight loss, or pain in the nodes should be sought. The patient's age, sex, occupation, exposure to pets, sexual behavior, and use of drugs such as diphenylhydantoin are other important historic points. For example, children and young adults usually have benign (i.e., nonmalignant) disorders, such as viral or bacterial upper respiratory infections, infectious mononucleosis, toxoplasmosis, and, in some countries, tuberculosis, which account for the observed lymphadenopathy. In contrast, after age 50 the incidence of malignant disorders increases and that of benign disorders decreases.

The *physical examination* can provide useful clues such as the extent of lymphadenopathy (localized or generalized), size of nodes, texture, presence or absence of nodal tenderness, signs of inflammation over the node, skin lesions, and splenomegaly. A thorough ear, nose, and throat (ENT) examination is indicated in adult patients with cervical adenopathy and a history of tobacco use. Localized or regional adenopathy implies involvement of a single anatomic area. Generalized adenopathy has been defined as involvement of three or more noncontiguous lymph node areas. Many of the causes of lymphadenopathy ([Table 63-1](#)) can produce localized or generalized adenopathy, so this distinction is of limited utility in the differential diagnosis. Nevertheless, generalized lymphadenopathy is frequently associated with nonmalignant disorders such as infectious mononucleosis [Epstein-Barr virus (EBV) or cytomegalovirus (CMV)], toxoplasmosis, AIDS, other viral infections, systemic lupus erythematosus (SLE), and mixed connective tissue disease. Acute and chronic lymphocytic leukemias and malignant lymphomas also produce generalized adenopathy in adults.

The site of localized or regional adenopathy may provide a useful clue about the cause. Occipital adenopathy often reflects an infection of the scalp, and preauricular adenopathy accompanies conjunctival infections and cat-scratch disease. The most frequent site of regional adenopathy is the neck, and most of the causes are benign -- upper respiratory infections, oral and dental lesions, infectious mononucleosis, other viral illnesses. The chief malignant causes include metastatic cancer from head and neck, breast, lung, and thyroid primaries. Enlargement of supraclavicular and scalene nodes is always abnormal. Because these nodes drain regions of the lung and retroperitoneal space, they can reflect either lymphomas, other cancers, or infectious processes arising in these areas. Virchow's node is an enlarged left supraclavicular node infiltrated with metastatic cancer from a gastrointestinal primary. Metastases to supraclavicular nodes also occur from lung, breast, testis, or ovarian cancers. Tuberculosis, sarcoidosis, and toxoplasmosis are nonneoplastic causes of supraclavicular adenopathy. Axillary adenopathy is usually due to injuries or localized infections of the ipsilateral upper extremity. Malignant causes include melanoma or lymphoma and, in women, breast cancer. Inguinal lymphadenopathy is usually secondary to infections or trauma of the lower extremities and may accompany sexually transmitted diseases such as lymphogranuloma venereum, primary syphilis, genital herpes, or chancroid. These nodes may also be involved by lymphomas and metastatic cancer from primary lesions of the rectum, genitalia, or lower extremities (melanoma).

The size and texture of the lymph node(s) and the presence of pain are useful parameters in evaluating a patient with lymphadenopathy. Nodes $<1.0 \text{ cm}^2$ in area ($1.0 \times 1.0 \text{ cm}$ or less) are almost always secondary to benign, nonspecific reactive causes. In one retrospective analysis of younger patients (9 to 25 years) who had a lymph node biopsy, a maximum diameter of $>2 \text{ cm}$ served as one discriminant for predicting that the biopsy would reveal malignant or granulomatous disease. Another study showed that a lymph node size of 2.25 cm^2 ($1.5 \text{ cm} \times 1.5 \text{ cm}$) was the best discriminating limit for distinguishing malignant or granulomatous lymphadenopathy from other causes of lymphadenopathy. Patients with node(s) $\geq 1.0 \text{ cm}^2$ should be observed after excluding infectious mononucleosis and/or toxoplasmosis unless there are symptoms and signs of an underlying systemic illness.

The texture of lymph nodes may be described as soft, firm, rubbery, hard, discrete, matted, tender, movable, or fixed. Tenderness is found when the capsule is stretched during rapid enlargement, usually secondary to an inflammatory process. Some malignant diseases such as acute leukemia may produce rapid enlargement and pain in the nodes. Nodes involved by lymphoma tend to be large, discrete, symmetric, rubbery, firm, mobile, and nontender. Nodes containing metastatic cancer are often hard, nontender, and nonmovable because of fixation to surrounding tissues. The coexistence of splenomegaly in the patient with lymphadenopathy implies a systemic illness such as infectious mononucleosis, lymphoma, acute or chronic leukemia, [SLE](#), sarcoidosis, toxoplasmosis, cat-scratch disease, or other less common hematologic disorders. The patient's story should provide helpful clues about the underlying systemic illness.

Nonsuperficial presentations (thoracic or abdominal) of adenopathy are usually detected as the result of a symptom-directed diagnostic workup. Thoracic adenopathy may be detected by routine chest roentgenography or during the workup for superficial adenopathy. It may also be found because the patient complains of a cough or wheezing from airway compression; hoarseness from recurrent laryngeal nerve involvement; dysphagia from esophageal compression; or swelling of the neck, face, or arms secondary to compression of the superior vena cava or subclavian vein. The differential diagnosis of mediastinal and hilar adenopathy includes primary lung disorders and systemic illnesses that characteristically involve mediastinal or hilar nodes. In the young, mediastinal adenopathy is associated with infectious mononucleosis and sarcoidosis. In endemic regions, histoplasmosis can cause unilateral paratracheal lymph node involvement that mimics lymphoma. Tuberculosis can also cause unilateral adenopathy. In older patients, the differential diagnosis includes primary lung cancer (especially among smokers), lymphomas, metastatic carcinoma (usually lung), tuberculosis, fungal infection, and sarcoidosis.

Enlarged intraabdominal or retroperitoneal nodes are usually malignant. Although tuberculosis may present as mesenteric lymphadenitis, these masses usually contain lymphomas or, in young men, germ cell tumors.

Laboratory Investigation The laboratory investigation of patients with lymphadenopathy must be tailored to elucidate the etiology suspected from the patient's history and physical findings. One study from a family practice clinic evaluated 249 younger patients with "enlarged lymph nodes, not infected" or "lymphadenitis." No laboratory studies were obtained in 51%. When studies were performed, the most common were a complete blood count (33%), throat culture (16%), chest x-ray (12%), or monospot test (10%). Only eight patients (3%) had a node biopsy, and half of those were normal or reactive. The complete blood count can provide useful data for the diagnosis of acute or chronic leukemias, [EBV](#) or [CMV](#) mononucleosis, lymphoma with a leukemic component, pyogenic infections, or immune cytopenias in illnesses such as [SLE](#). Serologic studies may demonstrate antibodies specific to components of EBV, CMV, HIV, and other viruses; *Toxoplasma gondii*; *Brucella*; etc. If SLE is suspected, then antinuclear and anti-DNA antibody studies are warranted.

The chest x-ray is usually negative, but the presence of a pulmonary infiltrate or mediastinal lymphadenopathy would suggest tuberculosis, histoplasmosis, sarcoidosis, lymphoma, primary lung cancer, or metastatic cancer and demands further

investigation.

A variety of imaging techniques [computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, color Doppler ultrasonography] have been employed to differentiate benign from malignant lymph nodes, especially in patients with head and neck cancer. CT and MRI are comparably accurate (65 to 90%) in the diagnosis of metastases to cervical lymph nodes. Ultrasonography has been used to determine the long (L) axis, short (S) axis, and a ratio of long to short axis in cervical nodes. An L/S ratio of <2.0 has a sensitivity and a specificity of 95% for distinguishing benign and malignant nodes in patients with head and neck cancer. This ratio has greater specificity and sensitivity than palpation or measurement of either the long or the short axis alone.

The indications for lymph node biopsy are imprecise, yet it is a valuable diagnostic tool. The decision to biopsy may be made early in a patient's evaluation or delayed for up to 2 weeks. Prompt biopsy should occur if the patient's history and physical findings suggest a malignancy; examples include a solitary, hard, nontender cervical node in an older patient who is a chronic user of tobacco; supraclavicular adenopathy; and solitary or generalized adenopathy that is firm, movable, and suggestive of lymphoma. If a primary head and neck cancer is suspected as the basis of a solitary, hard cervical node, then a careful [ENT](#) examination should be performed. Any mucosal lesion that is suspicious for a primary neoplastic process should be biopsied first. If no mucosal lesion is detected, an excisional biopsy of the largest node should be performed. Fine-needle aspiration should not be performed as the first diagnostic procedure. Most diagnoses require more tissue than such aspiration can provide and it often delays a definitive diagnosis. Fine-needle aspiration should be reserved for thyroid nodules and for confirmation of relapse in patients whose primary diagnosis is known. If the primary physician is uncertain about whether to proceed to biopsy, consultation with a hematologist or medical oncologist should be helpful. In primary care practices, fewer than 5% of lymphadenopathy patients will require a biopsy. That percentage will be considerably larger in referral practices, i.e., hematology, oncology, or otolaryngology (ENT).

Two groups have reported algorithms that they claim will identify more precisely those lymphadenopathy patients who should have a biopsy. Both reports were retrospective analyses in referral practices. The first study involved patients 9 to 25 years of age who had a node biopsy performed. Three variables were identified that predicted those young patients with peripheral lymphadenopathy who should undergo biopsy; lymph node size >2 cm in diameter and abnormal chest x-ray had positive predictive value, whereas recent [ENT](#) symptoms had negative predictive values. The second study evaluated 220 lymphadenopathy patients in a hematology unit and identified five variables [lymph node size, location (supraclavicular or non-supraclavicular), age (>40 years or <40 years), texture (nonhard or hard), and tenderness] that were utilized in a mathematical model to identify those patients requiring a biopsy. Positive predictive value was found for age >40 years, supraclavicular location, node size >2.25 cm², hard texture, and lack of pain or tenderness. Negative predictive value was evident for age <40 years, node size <1.0 cm², nonhard texture, and tender or painful nodes. Ninety-one percent of those who required biopsy were correctly classified by this model. Since both of these studies were retrospective analyses and one was limited to young patients, it is not known how useful these models would be if applied prospectively in a primary care

setting.

Most lymphadenopathy patients do not require a biopsy, and at least half require no laboratory studies. If the patient's history and physical findings point to a benign cause for lymphadenopathy, then careful follow-up at a 2- to 4-week interval can be employed. The patient should be instructed to return for reevaluation if the node(s) increase in size. Antibiotics are not indicated for lymphadenopathy unless there is strong evidence of a bacterial infection. Glucocorticoids should not be used to treat lymphadenopathy because their lympholytic effect obscures some diagnoses (lymphoma, leukemia, Castleman's disease) and they contribute to delayed healing or activation of underlying infections. An exception to this statement is the life-threatening pharyngeal obstruction by enlarged lymphoid tissue in Waldeyer's ring that is occasionally seen in infectious mononucleosis.

SPLENOMEGALY

STRUCTURE AND FUNCTION OF THE SPLEEN

The spleen is a reticuloendothelial organ that has its embryologic origin in the dorsal mesogastrium at about 5 weeks' gestation. It arises in a series of hillocks, migrates to its normal adult location in the left upper quadrant (LUQ), and is attached to the stomach via the gastrosplenic ligament and to the kidney via the lienorenal ligament. When the hillocks fail to unify into a single tissue mass, accessory spleens may develop in around 20% of persons. The function of the spleen has been elusive. Galen believed it was the source of "black bile" or melancholia, and the word *hypochondria* (literally, beneath the ribs) and the idiom "to vent one's spleen" attest to the beliefs that the spleen had an important influence on the psyche and emotions. In humans, its normal physiologic roles seem to be the following:

1. Maintenance of quality control over erythrocytes in the red pulp by removal of senescent and defective red blood cells. The spleen accomplishes this function through a unique organization of its parenchyma and vasculature ([Fig. 63-1](#)).
2. Synthesis of antibodies in the white pulp.
3. The removal of antibody-coated bacteria and antibody-coated blood cells from the circulation.

An increase in these normal functions may result in splenomegaly.

The spleen is composed of red pulp and white pulp, which are Malpighi's terms for the red blood-filled sinuses and reticuloendothelial cell-lined cords and the white lymphoid follicles arrayed within the red pulp matrix. The spleen is in the portal circulation. The reason for this is unknown but may relate to the fact that lower blood pressure allows less rapid flow and minimizes damage to normal erythrocytes. Blood flows into the spleen at a rate of about 150 mL/min through the splenic artery, which ultimately ramifies into central arterioles. Some blood goes from the arterioles to capillaries and then to splenic veins and out of the spleen, but the majority of blood from central arterioles flows into the macrophage-lined sinuses and cords. The blood entering the

sinuses reenters the circulation through the splenic venules, but the blood entering the cords is subjected to an inspection of sorts. In order to return to the circulation, the blood cells in the cords must squeeze through slits in the cord lining to enter the sinuses that lead to the venules. Old and damaged erythrocytes are less deformable and are retained in the cords, where they are destroyed and their components recycled. Red cell inclusion bodies such as parasites, nuclear residua (Howell-Jolly bodies), or denatured hemoglobin (Heinz bodies) are pinched off in the process of passing through the slits, a process called *pitting*. The culling of dead and damaged cells and the pitting of cells with inclusions appear to occur without significant delay since the blood transit time through the spleen is only slightly slower than in other organs.

The spleen is also capable of assisting the host in adapting to its hostile environment. It has at least three adaptational functions: (1) clearance of bacteria and particulates from the blood, (2) the generation of immune responses to certain invading pathogens, and (3) the generation of cellular components of the blood under circumstances in which the marrow is unable to meet the needs (i.e., extramedullary hematopoiesis). The latter adaptation is a recapitulation of the blood-forming function the spleen plays during gestation. In some animals, the spleen also serves a role in the vascular adaptation to stress because it stores red blood cells (often hemoconcentrated to higher hematocrits than normal) under normal circumstances and contracts under the influence of α -adrenergic stimulation to provide the animal with an autotransfusion and improved oxygen-carrying capacity. However, the normal human spleen does not sequester or store red blood cells and does not contract in response to sympathetic stimuli. The normal human spleen contains approximately one-third of the total body platelets and a significant number of marginated neutrophils. These sequestered cells are available when needed to respond to bleeding or infection.

Approach to the Patient

Clinical Assessment The most common *symptoms* produced by diseases involving the spleen are pain and a heavy sensation in the [LUQ](#). Massive splenomegaly may cause early satiety. Pain may result from acute swelling of the spleen with stretching of the capsule, infarction, or inflammation of the capsule. For many years, it was believed that splenic infarction was clinically silent, which at times is true. However, Soma Weiss, in his classic 1942 report of the self-observations by a Harvard medical student on the clinical course of subacute bacterial endocarditis, documented that severe LUQ and pleuritic chest pain may accompany thromboembolic occlusion of splenic blood flow. Vascular occlusion, with infarction and pain, is commonly seen in children with sickle cell crises. Rupture of the spleen, either from trauma or infiltrative disease that breaks the capsule, may result in intraperitoneal bleeding, shock, and death. The rupture itself may be painless.

A palpable spleen is the major *physical sign* produced by diseases affecting the spleen and suggests enlargement of the organ. The normal spleen is said to weigh less than 250 g, decreases in size with age, normally lies entirely within the rib cage, has a maximum cephalocaudad diameter of 13 cm by ultrasonography or maximum length of 12 cm and/or width of 7 cm by radionuclide scan, and is usually not palpable. However, a palpable spleen was found in 3% of 2200 asymptomatic, male, freshman college students. Follow-up at 3 years revealed that 30% of those students still had a palpable

spleen without any increase in disease prevalence. Ten-year follow-up found no evidence for lymphoid malignancies. Furthermore, in some tropical countries (e.g., New Guinea) the incidence of splenomegaly may reach 60%. Thus, the presence of a palpable spleen does not always equate with presence of disease. Even when disease is present, splenomegaly may not reflect the primary disease, but rather a reaction to it. For example, in patients with Hodgkin's disease, only two-thirds of the palpable spleens show involvement by the cancer.

Physical examination of the spleen utilizes primarily the techniques of palpation and percussion. Inspection may reveal a fullness in the [LUQ](#) that descends on inspiration, a finding associated with a massively enlarged spleen. Auscultation may reveal a venous hum or a friction rub.

Palpation can be accomplished by bimanual palpation, ballotment, and palpation from above (Middleton maneuver). For bimanual palpation, which is at least as reliable as the other techniques, the patient is supine with flexed knees. The examiner's left hand is placed on the lower rib cage and pulls the skin toward the costal margin, allowing the fingertips of the right hand to feel the tip of the spleen as it descends while the patient inspires slowly, smoothly, and deeply. Palpation is begun with the right hand in the left lower quadrant with gradual movement toward the left costal margin, thereby identifying the lower edge of a massively enlarged spleen. When the spleen tip is felt, the finding is recorded as centimeters below the left costal margin at some arbitrary point, i.e., 10 to 15 cm, from the midpoint of the umbilicus or the xiphisternal junction. This allows other examiners to compare findings or the initial examiner to determine changes in size over time. Bimanual palpation in the right lateral decubitus position adds nothing to the supine examination.

Percussion for splenic dullness is accomplished with any of three techniques described by Nixon, Castell, or Barkun:

1. *Nixon's method*: The patient is placed on the right side so that the spleen lies above the colon and stomach. Percussion begins at the lower level of pulmonary resonance in the posterior axillary line and proceeds diagonally along a perpendicular line toward the lower midanterior costal margin. The upper border of dullness is normally 6 to 8 cm above the costal margin. Dullness greater than 8 cm in an adult is presumed to indicate splenic enlargement.

2. *Castell's method*: With the patient supine, percussion in the lowest intercostal space in the anterior axillary line (8th or 9th) produces a resonant note if the spleen is normal in size. This is true during expiration or full inspiration. A dull percussion note on full inspiration suggests splenomegaly.

3. *Percussion of Traube's semilunar space*: The borders of Traube's space are the sixth rib superiorly, the left midaxillary line laterally, and the left costal margin inferiorly. The patient is supine with the left arm slightly abducted. During normal breathing, this space is percussed from medial to lateral margins, yielding a normal resonant sound. A dull percussion note suggests splenomegaly.

Studies comparing methods of percussion and palpation with a standard of

ultrasonography or scintigraphy have revealed sensitivity of 56 to 71% for palpation and 59 to 82% for percussion. Reproducibility among examiners is better for palpation than percussion. Both techniques are less reliable in obese patients or patients who have just eaten. Thus, the physical examination techniques of palpation and percussion are imprecise at best. It has been suggested that the examiner perform percussion first and, if positive, proceed to palpation; if the spleen is palpable, then one can be reasonably confident that splenomegaly exists. However, not all [LUQ](#) masses are enlarged spleens; gastric or colon tumors and pancreatic or renal cysts or tumors can mimic splenomegaly.

The presence of an enlarged spleen can be more precisely determined, if necessary, by liver-spleen radionuclide scan, [CT](#), [MRI](#), or ultrasonography. The latter technique is the current procedure of choice for routine assessment of spleen size (normal = a maximum cephalocaudad diameter of 13 cm) because it has high sensitivity and specificity and is safe, noninvasive, quick, mobile, and less costly. Nuclear medicine scans are accurate, sensitive, and reliable but are costly, require greater time to generate data, and utilize immobile equipment. They have the advantage of demonstrating accessory splenic tissue. CT and MRI provide accurate determination of spleen size, but the equipment is immobile and the procedures are expensive. MRI appears to offer no advantage over CT. Changes in spleen structure such as mass lesions, infarcts, inhomogeneous infiltrates, and cysts are more readily assessed by CT, MRI, or ultrasonography. None of these techniques is very reliable in the detection of patchy infiltration (e.g., Hodgkin's disease).

Differential Diagnosis Many of the diseases associated with splenomegaly are listed in [Table 63-2](#). They are grouped according to the presumed basic mechanisms responsible for organ enlargement:

1. Hyperplasia or hypertrophy related to a particular splenic function such as reticuloendothelial hyperplasia (work hypertrophy) in diseases such as hereditary spherocytosis or thalassemia syndromes that require removal of large numbers of defective red blood cells; immune hyperplasia in response to systemic infection (infectious mononucleosis, subacute bacterial endocarditis) or to immunologic diseases (immune thrombocytopenia, [SLE](#), Felty's syndrome).
2. Passive congestion due to decreased blood flow from the spleen in conditions that produce portal hypertension (cirrhosis, Budd-Chiari syndrome, congestive heart failure).
3. Infiltrative diseases of the spleen (lymphomas, metastatic cancer, amyloidosis, Gaucher's disease, myeloproliferative disorders with extramedullary hematopoiesis).

The differential diagnostic possibilities are much fewer when the spleen is "massively enlarged," that is, it is palpable more than 8 cm below the left costal margin or its drained weight is ≥ 1000 g ([Table 63-3](#)). The vast majority of such patients will have non-Hodgkin's lymphoma, chronic lymphocytic leukemia, hairy cell leukemia, chronic myelogenous leukemia, myelofibrosis with myeloid metaplasia, or polycythemia vera.

Laboratory Assessment The major laboratory abnormalities accompanying splenomegaly are determined by the underlying systemic illness. Erythrocyte counts

may be normal, decreased (thalassemia major syndromes, [SLE](#), cirrhosis with portal hypertension), or increased (polycythemia vera). Granulocyte counts may be normal, decreased (Feltz's syndrome, congestive splenomegaly, leukemias), or increased (infections or inflammatory disease, myeloproliferative disorders). Similarly, the platelet count may be normal, decreased when there is enhanced sequestration or destruction of platelets in an enlarged spleen (congestive splenomegaly, Gaucher's disease, immune thrombocytopenia), or increased in the myeloproliferative disorders such as polycythemia vera.

The complete blood count may reveal cytopenia of one or more blood cell types, which should suggest *hypersplenism*. This condition is characterized by splenomegaly, cytopenia(s), normal or hyperplastic bone marrow, and a response to splenectomy. The latter characteristic is less precise because reversal of cytopenia, particularly granulocytopenia, is sometimes not sustained after splenectomy. The cytopenias result from increased destruction of the cellular elements secondary to reduced flow of blood through enlarged and congested cords (congestive splenomegaly) or to immune-mediated mechanisms. In hypersplenism, various cell types usually have normal morphology on the peripheral blood smear, although the red cells may be spherocytic due to loss of surface area during their longer transit through the enlarged spleen. The increased marrow production of red cells should be reflected as an increased reticulocyte production index, although the value may be less than expected due to increased sequestration of reticulocytes in the spleen.

The need for additional laboratory studies is dictated by the differential diagnosis of the underlying illness of which splenomegaly is a manifestation.

SPLENECTOMY

Splenectomy is infrequently performed for diagnostic purposes, especially in the absence of clinical illness or other diagnostic tests that suggest underlying disease. More often splenectomy is performed for staging the extent of disease in patients with Hodgkin's disease, for symptom control in patients with massive splenomegaly, for disease control in patients with traumatic splenic rupture, or for correction of cytopenias in patients with hypersplenism or immune-mediated destruction of one or more cellular blood elements. Splenectomy is necessary for routine staging of patients with Hodgkin's disease only in those with clinical stage I or II disease in whom radiation therapy alone is contemplated as the treatment. Noninvasive staging of the spleen in Hodgkin's disease is not a sufficiently reliable basis for treatment decisions because one-third of normal-sized spleens will be involved with Hodgkin's disease and one-third of enlarged spleens will be tumor-free. Although splenectomy in chronic myelogenous leukemia does not affect the natural history of disease, removal of the massive spleen usually makes patients significantly more comfortable and simplifies their management by significantly reducing transfusion requirements. Splenectomy is an effective secondary or tertiary treatment for two chronic B cell leukemias, hairy cell leukemia and prolymphocytic leukemia, and for the very rare splenic mantle cell or marginal zone lymphoma. Splenectomy in these diseases may be associated with significant tumor regression in bone marrow and other sites of disease. Similar regressions of systemic disease have been noted after splenic irradiation in some types of lymphoproliferative disease, especially chronic lymphocytic leukemia and prolymphocytic leukemia. This

has been termed the *abscopal effect*. Such systemic tumor responses to local therapy directed at the spleen suggest that there may be some hormone or growth factor produced by the spleen that affects tumor cell proliferation, but this conjecture is not yet substantiated. The most common indication for splenectomy is traumatic or iatrogenic splenic rupture. In a fraction of patients with splenic rupture, peritoneal seeding of splenic fragments can lead to splenosis -- the presence of multiple rests of spleen tissue not connected to the portal circulation. This ectopic spleen tissue may cause pain or gastrointestinal obstruction, as in endometriosis. A large number of hematologic, immunologic, and congestive causes of splenomegaly can lead to destruction of one or more cellular blood elements. In the majority of such cases, splenectomy can correct the cytopenias, particularly anemia and thrombocytopenia. Perhaps the only contraindication to splenectomy is the presence of marrow failure, in which the enlarged spleen is the only source of hematopoietic tissue.

The absence of the spleen has minimal long-term effects on the hematologic profile. In the immediate postsplenectomy period, there may be some leukocytosis (up to 25,000/uL) and thrombocytosis (up to 1×10^6 /uL), but within 2 to 3 weeks, blood cell counts and survival of each cell lineage are usually normal. The chronic manifestations of splenectomy are marked variation in size and shape of erythrocytes (anisocytosis, poikilocytosis) and the presence of Howell-Jolly bodies (nuclear remnants), Heinz bodies (denatured hemoglobin), basophilic stippling, and an occasional nucleated erythrocyte in the peripheral blood. When such erythrocyte abnormalities appear in a patient whose spleen has not been removed, one should suspect splenic infiltration by tumor that has interfered with its normal culling and pitting function.

The most serious consequence of splenectomy is increased susceptibility to bacterial infections, particularly those with capsules such as *Streptococcus pneumoniae*, *Haemophilus influenzae*, and some gram-negative enteric organisms. Patients under age 20 years are particularly susceptible to overwhelming sepsis with *S. pneumoniae*, and the overall actuarial risk of sepsis in patients who have had their spleens removed is about 7% in 10 years. The case-fatality rate for pneumococcal sepsis in splenectomized patients is 50 to 80%. About 25% of patients without spleens will develop a serious infection at some time in their life. The frequency is highest within the first 3 years after splenectomy. About 15% of the infections are polymicrobial, and lung, skin, and blood are the most common sites. No increased risk of viral infection has been noted in patients who have no spleen. The susceptibility to bacterial infections relates to the inability to remove opsonized bacteria from the bloodstream and a defect in making antibodies to T cell-independent antigens such as the polysaccharide components of bacterial capsules. Pneumococcal vaccine (23-valent polysaccharide vaccine) should be administered to all patients 2 weeks before elective splenectomy. The Advisory Committee on Immunization Practices recommends that even splenectomized patients receive pneumococcal vaccine with a repeat vaccination 5 years later. Efficacy has not been proven in this setting, and the recommendation discounts the possibility that administration of the vaccine may actually lower the titer of specific pneumococcal antibodies. A more effective pneumococcal vaccine that involves T cells in the response is in development. The vaccine to *H. influenzae* should also be given to patients in whom elective splenectomy is planned. No other vaccines are routinely recommended in this setting.

Splenectomized patients should be educated to consider any unexplained fever as a medical emergency. Prompt medical attention with evaluation and treatment of suspected bacteremia may be life-saving. Routine chemoprophylaxis with oral penicillin can result in the emergence of drug-resistant strains and is not recommended.

In addition to an increased susceptibility to bacterial infections, splenectomized patients are also more susceptible to the parasitic disease babesiosis. The splenectomized patient should avoid areas where the parasite *Babesia* is endemic (e.g., Cape Cod, MA).

Surgical removal of the spleen is an obvious cause of *hyposplenism*. Patients with sickle cell disease often suffer from autosplenectomy as a result of splenic destruction by the numerous infarcts associated with sickle cell crises during childhood. Indeed, the presence of a palpable spleen in a patient with sickle cell disease after age 5 suggests a coexisting hemoglobinopathy, e.g., thalassemia or hemoglobin C. In addition, patients who receive splenic irradiation for a neoplastic or autoimmune disease are also functionally hyposplenic. The term *hyposplenism* is preferred to *asplenism* in referring to the physiologic consequences of splenectomy because asplenia is a rare, specific, and fatal congenital abnormality in which there is a failure of the left side of the coelomic cavity (which includes the splenic anlagen) to develop normally. Infants with asplenia have no spleens, but that is the least of their problems. The right side of the developing embryo is duplicated on the left so there is liver where the spleen should be, there are two right lungs, and the heart comprises two right atria and two right ventricles.

(Bibliography omitted in Palm version)

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64. DISORDERS OF GRANULOCYTES AND MONOCYTES - Steven M. Holland, John I. Gallin

Leukocytes are the major cells comprising inflammatory and immune responses and include neutrophils, T and B lymphocytes, natural killer (NK) cells, monocytes, eosinophils, and basophils. These cells have specific functions, such as antibody production by B lymphocytes or destruction of bacteria by neutrophils, but in no single infectious disease is the exact role of the cell types completely established. Thus, whereas neutrophils are classically thought to be critical to host defense against bacteria, they may also play important roles in defense against viral infections.

The blood delivers leukocytes to the various tissues from the bone marrow, where they are produced. Normal blood leukocyte counts are given in the Appendix (Tables A-7 and A-8). The various leukocytes are derived from a common stem cell in the bone marrow. Three-fourths of the nucleated cells of bone marrow are committed to the production of leukocytes. Leukocyte maturation in the marrow is under the regulatory control of a number of different factors, known as colony stimulating factors and interleukins ([Chap. 104](#)). Because an alteration in the number and type of leukocytes is often associated with disease processes, total white blood count (WBC) (cells per microliter) and differential counts are informative. The lymphocytes and basophils are discussed in [Chaps. 305](#) and [310](#), respectively. This chapter focuses on the neutrophils, monocytes, and eosinophils.

NEUTROPHILS

MATURATION

Important events in neutrophil life are summarized in [Fig. 64-1](#). In normal humans, neutrophils are produced only in the bone marrow. The minimum number of stem cells necessary to support hematopoiesis is estimated to be 400 to 500. Human blood monocytes, tissue macrophages, and stromal cells produce colony stimulating factors, hormones required for the growth of monocytes and neutrophils in the bone marrow. The hematopoietic system not only produces enough neutrophils ($\sim 1.3 \times 10^{11}$ cells per 80-kg person per day) to carry out physiologic functions but also has a large reserve stored in the marrow which can be mobilized in response to inflammation or infection. An increase in the number of blood neutrophils is called neutrophilia, and the presence of immature cells is termed a shift to the left. A decrease in the number of blood neutrophils is called neutropenia.

Neutrophils and monocytes evolve from pluripotent stem cells under the influence of cytokines and colony stimulating factors ([Fig. 64-2](#)). The proliferation phase through the metamyelocyte takes about 1 week, while the maturation phase from metamyelocyte to mature neutrophil takes another week. The myeloblast is the first recognizable precursor cell and is followed by the *promyelocyte* ([Plate V-23](#)). The promyelocyte evolves when the classic lysosomal granules, called the *primary* or *azurophil granules*, are produced. The primary granules contain hydrolases, elastase, myeloperoxidase, cationic proteins, and bactericidal/permeability-increasing protein, which is important for killing gram-negative bacteria. Azurophil granules also contain *defensins*, a family of cysteine-rich polypeptides with broad antimicrobial activity against bacteria, fungi, and

certain enveloped viruses. The promyelocyte divides to produce the *myelocyte*, a cell responsible for the synthesis of the *specific* or *secondary granules* which contain unique (specific) constituents such as lactoferrin, vitamin B₁₂-binding proteins, membrane components of the nicotinamide-adenine dinucleotide phosphate (NADPH) oxidase required for hydrogen peroxide production, histaminase, and receptors for certain chemoattractants and adherence-promoting factors (CR3) as well as receptors for the basement membrane component, laminin. The secondary granules do not contain acid hydrolases and therefore are not classic lysosomes. Packaging of secondary granule contents during myelopoiesis is controlled by CCAAT/enhancer binding protein-ε. Secondary granule contents are readily released extracellularly, and their mobilization is important in modulating inflammation. During the final stages of maturation no cell division occurs, and the cell passes through the *metamyelocyte* stage and then to the *band* neutrophil with a sausage-shaped nucleus ([Plate V-35](#)). As the band cell matures, the nucleus assumes a lobulated configuration. The nucleus of neutrophils normally contains up to four segments. Excessive segmentation (more than five nuclear lobes) may be a manifestation of folate or vitamin B₁₂ deficiency ([Plate V-38](#)). The Pelger-Huet anomaly ([Plate V-34B](#)), an infrequent dominant benign inherited trait, results in neutrophils with distinctive bilobed nuclei that must be distinguished from band forms. The physiologic role of the multilobed nucleus of neutrophils is unknown, but it may allow great deformation of neutrophils during migration into tissues at sites of inflammation.

In severe acute bacterial infection, prominent neutrophil cytoplasmic granules called *toxic granulations* are occasionally seen ([Plate V-11](#)). Toxic granulations are immature or abnormally staining azurophil granules. Cytoplasmic inclusions, also called *Dohle bodies* ([Plate V-35](#)), can be seen during infection and are fragments of ribosome-rich endoplasmic reticulum. Large neutrophil vacuoles are often present in acute bacterial infection and probably represent pinocytosed (internalized) membrane.

Neutrophils are heterogeneous in function. Monoclonal antibodies have been developed that recognize only a subset of mature neutrophils. The meaning of neutrophil heterogeneity is not known.

MARROW RELEASE AND CIRCULATING COMPARTMENTS

Specific signals, including interleukin (IL) 1, tumor necrosis factor-α (TNF-α), the colony stimulating factors, complement fragment C3e, and perhaps other cytokines mobilize leukocytes from the bone marrow and deliver them to the blood in an unstimulated state. Under normal conditions, about 90% of the neutrophil pool is in the bone marrow, 2 to 3% in the circulation, and the remainder in the tissues ([Fig. 64-3](#)).

The circulating pool exists in two dynamic compartments: one freely flowing and one marginated. The freely flowing pool is about one-half the neutrophils in the basal state and is composed of those cells that are in the blood and not in contact with the endothelium. Marginated leukocytes are those that are in close physical contact with the endothelium ([Fig. 64-4](#)). In the pulmonary circulation, where an extensive capillary bed (~1000 capillaries per alveolus) exists, margination occurs because the capillaries are about the same size as a mature neutrophil. Therefore, neutrophil fluidity and deformability are necessary to make the transit through the pulmonary bed. Increased

neutrophil rigidity and decreased deformability lead to augmented neutrophil trapping and margination in the lung. In contrast, in the systemic postcapillary venules, margination is mediated by the interaction of specific cell-surface molecules. *Selectins* are glycoproteins expressed on neutrophils and endothelial cells, among others, that cause a low-affinity interaction, resulting in "rolling" of the neutrophil along the endothelial surface. On neutrophils, the molecule L-selectin [cluster determinant (CD) 62L] binds to glycosylated proteins on endothelial cells [e.g., glycosylation-dependent cell adhesion molecule (GlyCAM1) and CD34]. Glycoproteins on neutrophils, most importantly sialyl-Lewis_x (SLe_x, CD15s), are targets for binding of selectins expressed on endothelial cells [E-selectin (CD62E) and P-selectin (CD62P)] and other leukocytes. In response to chemotactic stimuli from injured tissues (e.g., complement product C5a, leukotriene B₄, [IL-8](#)) or bacterial products [e.g., *N*-formylmethionylleucylphenylalanine (f-metleuphe)], neutrophil adhesiveness increases, and the cells "stick" to the endothelium through *integrins*. The integrins are leukocyte glycoproteins that exist as complexes of a common CD18 b-chain with CD11a (LFA-1), CD11b (also called either Mac-1, CR3, or the C3b receptor), and CD11c (p150,95). CD11a/CD18 and CD11b/CD18 bind to specific endothelial receptors [intercellular adhesion molecules (ICAM) 1 and 2].

On cell stimulation, L-selectin is shed; receptors for chemoattractants and opsonins are mobilized; the phagocytes orient toward the chemoattractant source in the extravascular space, increase their motile activity (chemokinesis), and migrate directionally (chemotaxis) into tissues. The process of migration into tissues is called *diapedesis* and involves the crawling of neutrophils between postcapillary endothelial cells that open junctions between adjacent cells to permit leukocyte passage. Diapedesis involves platelet/endothelial cell adhesion molecule (PECAM) 1 (CD31), which is expressed on both the emigrating leukocyte and the endothelial cells. The endothelial responses (increased blood flow from increased vasodilation and permeability) are mediated by anaphylatoxins (e.g., C3a and C5a) as well as vasodilators such as histamine, bradykinin, serotonin, nitric oxide, vascular endothelial growth factor (VEGF), and prostaglandins E and I. Cytokines regulate some of these processes [e.g., [TNF-α](#) induction of VEGF, interferon (IFN) γ inhibition of prostaglandin E].

In the healthy adult, most neutrophils leave the body by migration through the mucous membrane of the gastrointestinal tract. Normally, neutrophils spend a short time in the circulation (half-life, 6 to 7 h). Senescent neutrophils are cleared from the circulation by macrophages in the lung and spleen. Once in the tissues, neutrophils release enzymes, such as collagenase and elastase, that help establish abscess cavities. Neutrophils ingest pathogenic materials that have been opsonized by IgG and C3b. Fibronectin and the tetrapeptide tuftsin facilitate phagocytosis.

With phagocytosis comes a burst of oxygen consumption and activation of the hexose-monophosphate shunt. A membrane-associated [NADPH](#) oxidase, consisting of membrane and cytosolic components, is assembled and catalyzes the reduction of oxygen to superoxide anion, which is then converted to hydrogen peroxide and other toxic oxygen products (e.g., hydroxyl radical). Hydrogen peroxide + chloride + neutrophil myeloperoxidase generates hypochlorous acid (bleach), hypochlorite, and chlorine. These products oxidize and halogenate microorganisms and tumor cells and, when uncontrolled, can damage host tissue. Strongly cationic proteins, defensins, and

probably nitric oxide also participate in microbial killing. Other enzymes, such as lysozyme and acid proteases, help digest microbial debris. After 1 to 4 days in tissues neutrophils die. The apoptosis of neutrophils is also cytokine regulated; granulocyte colony stimulating factor (G-CSF) and [IFN- \$\gamma\$](#) prevent their death. Under certain conditions, such as in delayed-type hypersensitivity, monocyte accumulation occurs within 6 to 12 h of initiation of inflammation. Neutrophils, monocytes, microorganisms in various states of digestion, and altered local tissue cells make up the inflammatory exudate, pus. Myeloperoxidase confers the characteristic green color to pus and may participate in turning off the inflammatory process by inactivating chemoattractants and immobilizing phagocytic cells.

Neutrophils respond to certain cytokines [[IFN- \$\gamma\$](#) , granulocyte-macrophage colony stimulating factor (GM-CSF), [IL-8](#)] and produce cytokines and chemotactic signals [[TNF- \$\alpha\$](#) , IL-8, macrophage inflammatory protein (MIP) 1] that modulate the inflammatory response. In the presence of fibrinogen, f-metleuphe or leukotriene B₄ induces IL-8 production by neutrophils, providing autocrine amplification of inflammation. *Chemokines (chemoattractant cytokines)* are small proteins produced by many different cell types, including endothelial cells, fibroblasts, epithelial cells, neutrophils, and monocytes, that regulate neutrophil and monocyte recruitment and activation. The chemokines transduce their signals through heterotrimeric G protein-linked receptors that have seven cell membrane-spanning domains, the same type of cell-surface receptor that mediates the response to the classical chemoattractants *N*-f-metleuphe and C5a. Four major groups of chemokines are recognized based on the cysteine structure near the N terminus: C, CC, CXC, and CXXXC. The CXC cytokines like IL-8 mainly attract neutrophils; CC chemokines like [MIP-1 \$\alpha\$](#) attract lymphocytes, monocytes, eosinophils, and basophils; the C chemokine lymphotactin is T cell tropic; the CXXXC chemokine fractalkine attracts neutrophils, monocytes, and T cells. These molecules and their receptors not only regulate the trafficking and activation of inflammatory cells, but chemokine receptors serve as co-receptors for HIV infection ([Chap. 309](#)).

NEUTROPHIL ABNORMALITIES

A defect in the neutrophil life cycle can lead to dysfunction and compromised host defenses. Inflammation is often depressed, and the clinical result is often recurrent and severe bacterial and fungal infections. Aphthous ulcers of mucous membranes (gray ulcers without pus) and gingivitis and periodontal disease suggest a phagocytic cell disorder. Patients with congenital phagocyte defects can have infections within the first few days of life. Skin, ear, upper and lower respiratory tract, and bone infections are common. Sepsis and meningitis are rare. In some disorders the frequency of infection is variable, and patients can go for months or even years without major infection. Aggressive management of these congenital diseases has extended the life span of patients beyond 30 years.

Neutropenia The consequences of absent neutrophils are dramatic. Susceptibility to infectious diseases increases sharply when neutrophil counts fall below 1000 cells/uL. When the absolute neutrophil count (ANC; band forms and mature neutrophils combined) falls below 500 cells/uL, control of endogenous microbial flora (e.g., mouth, gut) is impaired; when the ANC is < 200/uL, the inflammatory process is absent. Neutropenia can be due to depressed production, increased peripheral destruction, or

excessive peripheral pooling. A falling neutrophil count or a significant decrease in the number of neutrophils below steady state levels, together with a failure to increase neutrophil counts in the setting of infection or other challenge, requires investigation. Acute neutropenia, such as that caused by cancer chemotherapy, is more likely to be associated with increased risk of infection than neutropenia of long duration (months to years) that reverses in response to infection or carefully controlled administration of endotoxin (see "Laboratory Diagnosis," below).

Some causes of inherited and acquired neutropenia are listed in [Table 64-1](#). The most common neutropenias are iatrogenic, resulting from the use of cytotoxic or immunosuppressive therapies for malignancy or control of autoimmune disorders. These drugs cause neutropenia because they result in decreased production of rapidly growing progenitor (stem) cells of the marrow. Certain antibiotics such as chloramphenicol, trimethoprim-sulfamethoxazole, flucytosine, vidarabine, and the antiretroviral drug zidovudine may cause neutropenia by inhibiting proliferation of myeloid precursors. The marrow suppression is generally dose-related and dependent on continued administration of the drug. Recombinant human [G-CSF](#) reverses this form of neutropenia.

Another important mechanism for iatrogenic neutropenia is the effect of drugs that serve as immune haptens and sensitize neutrophils or neutrophil precursors to immune-mediated peripheral destruction. This form of drug-induced neutropenia can be seen within 7 days of exposure to the drug; with previous drug exposure, resulting in preexisting antibodies, neutropenia may occur a few hours after administration of the drug. Although any drug can cause this form of neutropenia, the most frequent causes are commonly used antibiotics, such as sulfa-containing compounds, penicillins, and cephalosporins. Fever and eosinophilia also may be associated drug reactions, but often these signs are not present. Drug-induced neutropenia can be severe, but discontinuation of the sensitizing drug is sufficient for recovery, which is usually seen within 5 to 7 days and is complete by 10 days. Readministration of the sensitizing drug should be avoided, since abrupt neutropenia often will result. For this reason, diagnostic challenge should be avoided.

Autoimmune neutropenias caused by circulating antineutrophil antibodies are another form of acquired neutropenia that results in increased destruction of neutrophils. Acquired neutropenia also may be seen with viral infections, including infection with HIV. Acquired neutropenia may be cyclic in nature, occurring at intervals of several weeks. Acquired cyclic or stable neutropenia may be associated with an expansion of large granular lymphocytes (LGL), which may be T cells, [NK](#) cells, or NK-like cells. Patients with LGL lymphocytosis may have moderate blood and bone marrow lymphocytosis, neutropenia, polyclonal hypergammaglobulinemia, splenomegaly, rheumatoid arthritis, and absence of lymphadenopathy. Such patients may have a chronic and relatively stable course. Recurrent bacterial infections are frequent. Benign and malignant forms of this syndrome occur. In some patients, a spontaneous regression has occurred even after 11 years, suggesting an immunoregulatory defect as the basis for at least one form of the disorder. Glucocorticoids, cyclosporine, [IFN- \$\alpha\$](#) , and nucleosides such as 2-chlorodeoxyadenosine each have induced remission.

Hereditary Neutropenias Hereditary neutropenias are rare and may manifest in early

childhood as a profound constant neutropenia or agranulocytosis. Congenital forms of neutropenia include Kostmann's syndrome (neutrophil count <100/uL), which is often fatal; more benign chronic idiopathic neutropenia (neutrophil count of 300 to 1500/uL); the cartilage-hair hypoplasia syndrome; Shwachman's syndrome associated with pancreatic insufficiency; myelokathexis, a congenital disorder characterized by neutrophil degeneration, hypersegmentation, and myeloid hyperplasia in the marrow associated with decreased expression of bcl-X_L in myeloid precursors and accelerated apoptosis; and neutropenias associated with other immune defects (X-linked agammaglobulinemia, ataxia telangiectasia, IgA deficiency). Mutations in the [G-CSF](#) receptor on chromosome 1 associated with poor response to G-CSF can occur with severe congenital neutropenia and predispose to myeloid malignancy. Hereditary cyclic neutropenia, an autosomal dominant trait, may occur in infancy and is characterized by a remarkably regular 3-week cycle. Hereditary cyclic neutropenia actually is cyclic hematopoiesis, due to mutations in the neutrophil elastase gene. Glucocorticoids and G-CSF blunt the cycling in some patients.

Maternal factors can be associated with neutropenia in the newborn. Transplacental transfer of IgG directed against antigens on fetal neutrophils can result in peripheral destruction. Drugs (e.g., thiazides) ingested during pregnancy can cause neutropenia in the newborn by either depressed production or peripheral destruction.

The presence of immunoglobulin directed toward neutrophils is seen in Felty's syndrome -- a triad of rheumatoid arthritis, splenomegaly, and neutropenia ([Chap. 312](#)). Patients with Felty's syndrome who respond to splenectomy with an increase in their neutrophil count also have lower postoperative serum neutrophil-binding IgG. Some of these patients have neutropenia associated with an increased number of [LGL](#). Splenomegaly with peripheral trapping and destruction of neutrophils is also seen in lysosomal storage diseases and in portal hypertension.

Neutrophilia Neutrophilia results from increased neutrophil production, increased marrow release, or defective margination ([Table 64-2](#)). The most important acute cause of neutrophilia is infection. Neutrophilia from acute infection represents both increased production and increased marrow release. Increased production is also associated with chronic inflammation and certain myeloproliferative diseases. Increased marrow release and mobilization of the marginated leukocyte pool are induced by glucocorticoids. Release of epinephrine, as with vigorous exercise, excitement, or stress, will demarginate neutrophils in the spleen and lungs and double the neutrophil count in minutes. Leukocytosis with cell counts of 10,000 to 25,000/uL occurs in response to infection and other forms of acute inflammation and results from both release of the marginated pool and mobilization of marrow reserves. Persistent neutrophilia with cell counts of 30,000 to 50,000/uL or higher is called a *leukemoid reaction*, a term often used to distinguish this degree of neutrophilia from leukemia. In a leukemoid reaction, the circulating neutrophils are usually mature and not clonally derived.

Abnormal Neutrophil Function Inherited and acquired abnormalities of phagocyte function are listed in [Table 64-3](#). The resulting diseases are best considered in terms of the functional defects of adherence, chemotaxis, and microbicidal activity. The distinguishing features of the important inherited disorders of phagocyte function are shown in [Table 64-4](#).

Disorders of Adhesion Two types of leukocyte adhesion deficiency (LAD) have been described. Both are autosomal recessive traits and result in the inability of neutrophils to exit the circulation to sites of infection, leading to leukocytosis and increased susceptibility to infection ([Fig. 64-4](#)). Patients with LAD 1 have mutations in CD18, the common component of the integrins LFA-1, Mac-1, and p150,95, leading to a defect in tight adhesion between neutrophils and the endothelium. The heterodimer formed by CD18/CD11b (Mac-1) is also the receptor for the complement-derived opsonin C3bi (CR3). The CD18 gene is located on distal chromosome 21q. Variable expression of the defect determines the severity of clinical disease. Complete lack of expression of the leukocyte adhesion proteins results in the severe phenotype in which inflammatory cytokines do not increase the expression of leukocyte adhesion proteins on neutrophils or activated T and B cells. Neutrophils (and monocytes) from patients with LAD 1 adhere poorly to endothelial cells and protein-coated surfaces and exhibit defective spreading, aggregation, and chemotaxis. Patients with LAD 1 have recurrent bacterial and fungal infections involving skin, oral and genital mucosa, and respiratory and intestinal tracts; persistent leukocytosis (neutrophil counts of 15,000 to 20,000/uL) because cells do not marginate; and, in severe cases, a history of delayed separation of the umbilical stump. Infections, especially of the skin, may become necrotic with progressively enlarging borders, slow healing, and development of dysplastic scars. The most common bacteria are *Staphylococcus aureus* and enteric gram-negative bacteria. LAD 2 is caused by an abnormality of SLe^x(CD15s), the ligand on neutrophils that interacts with selectins on endothelial cells.

Disorders of Neutrophil Granules The most common neutrophil defect is *myeloperoxidase deficiency*, a primary granule defect inherited as an autosomal recessive trait; the incidence is ~1 in 2000 persons. Isolated myeloperoxidase deficiency is not associated with clinically compromised defenses, because other defense systems such as hydrogen peroxide generation are amplified. Microbicidal activity of neutrophils is delayed but not absent. Myeloperoxidase deficiency may make other acquired host defense defects more serious. An acquired form of myeloperoxidase deficiency occurs in myelomonocytic leukemia and acute myeloid leukemia.

Chediak-Higashi syndrome (CHS) is a rare disease with autosomal recessive inheritance due to defects in the lysosomal transport protein LYST, encoded by the gene *CHS1* at 1q42. This protein is required for normal packaging and disbursement of granules. Neutrophils (and all cells containing lysosomes) from patients with CHS characteristically have large granules ([Plate V-34A](#)). Patients with CHS have an increased number of infections resulting from many agents. CHS neutrophils and monocytes have impaired chemotaxis and abnormal rates of microbial killing due to slow rates of fusion of the lysosomal granules with phagosomes. [NK](#) cell function is also impaired.

Specific granule deficiency is a rare autosomal recessive disease in which the production of secondary granules and their contents, as well as primary granule component defensins, is defective. The defect in bacterial killing leads to severe bacterial infections. One type of specific granule deficiency is due to a mutation in the CCAAT/enhancer binding protein-ε, a regulator of expression of granule components.

Chronic granulomatous disease Chronic granulomatous disease (CGD) is a group of disorders of granulocyte and monocyte oxidative metabolism. Although CGD is rare, with an incidence of 1 in 200,000 individuals, it is an important model of defective neutrophil oxidative metabolism. Most often CGD is inherited as an X-linked recessive trait; 30% of patients inherit the disease in an autosomal recessive pattern. Mutations in the genes for the four proteins that assemble at the plasma membrane account for all patients with CGD. Two proteins (a 91-kDa protein, abnormal in X-linked CGD, and a 22-kDa protein, absent in one form of autosomal recessive CGD) form the heterodimer cytochrome b-558 in the plasma membrane. Two other proteins (47 and 67 kDa, abnormal in the other autosomal recessive forms of CGD) are cytoplasmic in origin and interact with the cytochrome after cell activation to form [NADPH](#)oxidase, required for hydrogen peroxide production. Leukocytes from patients with CGD have severely diminished hydrogen peroxide production. The genes involved in each of the defects have been cloned and sequenced and the chromosome locations identified. Patients with CGD characteristically have increased numbers of infections due to catalase-positive microorganisms (organisms that destroy their own hydrogen peroxide). When patients with CGD become infected, they often have extensive inflammatory reactions, and lymph node suppuration is common despite the administration of appropriate antibiotics. Aphthous ulcers and chronic inflammation of the nares are often present. Granulomas are frequent and can obstruct the gastrointestinal or genitourinary tracts. The excessive inflammation probably reflects failure to degrade chemoattractants and antigens, leading to persistent neutrophil accumulation. Impaired killing of intracellular microorganisms by macrophages may lead to persistent cell-mediated immunity and granuloma formation.

MONONUCLEAR PHAGOCYTES

The mononuclear phagocyte system is composed of monoblasts, promonocytes, and monocytes in addition to the structurally diverse tissue macrophages that make up what was previously referred to as the reticuloendothelial system. Macrophages are long-lived phagocytic cells capable of many of the functions of neutrophils. They are also secretory cells that participate in many immunologic and inflammatory processes distinct from neutrophils. Monocytes leave the circulation by diapedesis more slowly than neutrophils and have a half-life in the blood of 12 to 24 h.

After blood monocytes arrive in the tissues, they differentiate into macrophages ("big eaters") with specialized functions suited for specific anatomic locations. Macrophages are particularly abundant in capillary walls of the lung, spleen, liver, and bone marrow, where they function to remove microorganisms and other noxious elements from the blood. Alveolar macrophages, liver Kupffer cells, splenic macrophages, peritoneal macrophages, bone marrow macrophages, lymphatic macrophages, brain microglial cells, and dendritic macrophages all have specialized functions. Macrophage-secreted products include lysozyme, neutral proteases, acid hydrolases, arginase, complement components, enzyme inhibitors (plasmin, α_2 -macroglobulin), binding proteins (transferrin, fibronectin, transcobalamin II), nucleosides, and cytokines ([TNF- \$\alpha\$](#) ; [IL-1](#), -8, -12, and -18). [IL-1](#) ([Chaps. 17](#) and [305](#)) has many functions, including initiating fever in the hypothalamus, mobilizing leukocytes from the bone marrow, activating lymphocytes and neutrophils. [TNF- \$\alpha\$](#) is a pyrogen that duplicates many of the actions of [IL-1](#) and plays an important role in the pathogenesis of gram-negative shock ([Chap. 124](#)). [TNF- \$\alpha\$](#)

stimulates production of hydrogen peroxide and related toxic oxygen species by macrophages and neutrophils. In addition, TNF- α induces catabolic changes that contribute to the profound wasting (cachexia) associated with many chronic diseases.

Other macrophage-secreted products include reactive oxygen and nitrogen metabolites, bioactive lipids (arachidonic acid metabolites and platelet-activating factors), chemokines, colony stimulating factors, and factors stimulating fibroblast and vessel proliferation. Macrophages help regulate the replication of lymphocytes and participate in the killing of tumors, viruses, and certain bacteria (*Mycobacterium tuberculosis* and *Listeria monocytogenes*). Macrophages are key effector cells in the elimination of intracellular microorganisms. Their ability to fuse to form giant cells that coalesce into granulomas in response to some inflammatory stimuli is important in the elimination of intracellular microbes and is under the control of IFN- γ . Nitric oxide induced by IFN- γ is an important effector against intracellular parasites including tuberculosis and *Leishmania*.

Macrophages play an important role in the immune response ([Chap. 305](#)). They process and present antigen to lymphocytes and secrete cytokines that modulate and direct lymphocyte development and function. Macrophages participate in autoimmune phenomena by removing immune complexes and other substances from the circulation. Polymorphisms in macrophage receptors for immunoglobulin (Fc γ RII) determine susceptibility to some infections and autoimmune diseases. In wound healing, they dispose of senescent cells, and they contribute to atheroma development. Macrophage elastase mediates development of emphysema from cigarette smoking.

DISORDERS OF THE MONONUCLEAR PHAGOCYTE SYSTEM

Many disorders of neutrophils extend to mononuclear phagocytes. Thus, drugs that suppress neutrophil production in the bone marrow can cause monocytopenia. Transient monocytopenia occurs after stress or glucocorticoid administration. Monocytosis is associated with tuberculosis, brucellosis, subacute bacterial endocarditis, Rocky Mountain spotted fever, malaria, and visceral leishmaniasis (kala azar). Monocytosis also occurs with malignancies, leukemias, myeloproliferative syndromes, hemolytic anemias, chronic idiopathic neutropenias, and granulomatous diseases such as sarcoidosis, regional enteritis, and some collagen vascular diseases. Patients with [LAD](#), hyperimmunoglobulin E-recurrent infection (Job's) syndrome, [CHS](#), and [CGD](#) all have defects in the mononuclear phagocyte system.

Monocyte cytokine production is impaired in some patients with disseminated nontuberculous mycobacterial infection who are not infected with HIV. Genetic defects in IFN- γ receptors 1 and 2 impair monocyte killing of intracellular parasites, as do lesions in the potent IFN- γ inducer, IL-12 and its receptor ([Fig. 64-5](#)).

Certain viral infections impair mononuclear phagocyte function. For example, influenza virus infection causes abnormal monocyte chemotaxis. Mononuclear phagocytes can be infected by HIV using CCR5, the chemokine receptor that acts as a coreceptor with CD4 for HIV. T lymphocytes produce IFN- γ , which induces FcR expression and phagocytosis and stimulates hydrogen peroxide production by mononuclear phagocytes and neutrophils. In certain diseases, such as AIDS, IFN- γ production may be deficient, while

in other diseases, such as T cell lymphomas, excessive release of IFN-g may be associated with erythrophagocytosis by splenic macrophages.

Monocytopenia occurs with acute infections, with stress, and after treatment with glucocorticoids. Monocytopenia also occurs in aplastic anemia, hairy cell leukemia, acute myeloid leukemia, and as a direct result of myelotoxic drugs.

EOSINOPHILS

Eosinophils and neutrophils share similar morphology, many lysosomal constituents, phagocytic capacity, and oxidative metabolism. Eosinophils express a specific chemoattractant receptor and respond to a specific chemokine, eotaxin. Little is known about the role of eosinophils. Eosinophils are much longer lived than neutrophils, and unlike neutrophils, tissue eosinophils can recirculate. During most infections, eosinophils are not important. However, in invasive helminthic infections, such as hookworm, schistosomiasis, strongyloidiasis, toxocariasis, trichinosis, filariasis, echinococcosis, and cysticercosis, the eosinophil plays a central role in host defense. Eosinophils are associated with bronchial asthma, cutaneous allergic reactions, and other hypersensitivity states.

The distinctive feature of the red-staining (Wright's stain) eosinophil granules is its crystalline core consisting of an arginine-rich protein (major basic protein) with histaminase activity, important in host defense against parasites. Eosinophil granules also contain a unique eosinophil peroxidase that catalyzes the oxidation of many substances by hydrogen peroxide and may facilitate killing of microorganisms.

Eosinophil peroxidase, in the presence of hydrogen peroxide and halide, initiates mast cell secretion in vitro and thereby promotes inflammation. Eosinophils contain cationic proteins, some of which bind to heparin and reduce its anticoagulant activity. Eosinophil-derived neurotoxin and eosinophil cationic protein are ribonucleases that can kill respiratory syncytial virus. Eosinophil cytoplasm contains Charcot-Leyden crystal protein, a hexagonal bipyramidal crystal first observed in a patient with leukemia and then in sputum of patients with asthma; this protein is lysophospholipase and may function to detoxify certain lysophospholipids.

Several factors enhance the eosinophil's function in host defense. T cell-derived factors enhance the ability of eosinophils to kill parasites. Mast cell-derived eosinophil chemotactic factor of anaphylaxis (ECFa) increases the number of eosinophil complement receptors and enhances eosinophil killing of parasites. Eosinophil colony stimulating factors (e.g., [IL-5](#)) produced by macrophages increase eosinophil production in the bone marrow and activate eosinophils to kill parasites.

EOSINOPHILIA

Eosinophilia is the presence of >500 eosinophils per microliter of blood and is common in many settings besides parasite infection. Significant tissue eosinophilia can occur without an elevated blood count. The most common cause of eosinophilia is allergic reactions to drugs (iodides, aspirin, sulfonamides, nitrofurantoin, penicillins, and cephalosporins). Allergies such as hay fever, asthma, eczema, serum sickness, allergic

vasculitis, and pemphigus are associated with eosinophilia. Eosinophilia also occurs in collagen vascular diseases (e.g., rheumatoid arthritis, eosinophilic fasciitis, allergic angitis, and periarteritis nodosa) and malignancies (e.g., Hodgkin's disease; mycosis fungoides; chronic myelogenous leukemia; and cancer of the lung, stomach, pancreas, ovary, or uterus), as well as in Job's syndrome and [CGD](#). Eosinophilia commonly is present in the helminthic infections. [IL-5](#) is the dominant eosinophil growth factor. Therapeutic administration of the cytokines IL-2 and [GM-CSF](#) frequently leads to transient eosinophilia. The most dramatic hypereosinophilic syndromes are Loeffler's syndrome, tropical pulmonary eosinophilia, Loeffler's endocarditis, eosinophilic leukemia, and idiopathic hypereosinophilic syndrome (50,000 to 100,000/uL).

The idiopathic hypereosinophilic syndrome represents a heterogeneous group of disorders with the common feature of prolonged eosinophilia of unknown cause and organ system dysfunction, including the heart, central nervous system, kidneys, lungs, gastrointestinal tract, and skin. The bone marrow is involved in all affected individuals, but the most severe complications involve the heart and central nervous system. Clinical manifestations and organ dysfunction are highly variable. Eosinophils are found in the involved tissues and likely cause tissue damage by local deposition of toxic eosinophil proteins such as eosinophil cationic protein and major basic protein. In the heart, the pathologic changes lead to thrombosis, endocardial fibrosis, and restrictive endomyocardialopathy. The damage to tissues in other organ systems is similar. The mechanism for the hypereosinophilia is not known. Glucocorticoids usually induce remission. In patients who do not respond to glucocorticoids, a cytotoxic agent such as hydroxyurea has been used successfully to lower the peripheral blood eosinophil counts and to improve markedly the prognosis. [IFN- \$\alpha\$](#) is also effective in some patients, including those unresponsive to hydroxyurea. Aggressive medical and surgical approaches are used to manage patients with cardiovascular complications.

The *eosinophilia-myalgia syndrome* is a multisystem disease with prominent cutaneous, hematologic, and visceral manifestations that frequently evolves into a chronic course and can occasionally be fatal. The syndrome is characterized by eosinophilia (eosinophil count >1000/uL) and generalized disabling myalgias without other recognized causes. Eosinophil fasciitis, pneumonitis, and myocarditis; neuropathy culminating in respiratory failure; and encephalopathy may occur. The disease is caused by ingesting contaminants in L-tryptophan-containing products. Eosinophils, lymphocytes, macrophages, and fibroblasts accumulate in the affected tissues, but their role in pathogenesis is unclear. Activation of eosinophils and fibroblasts and the deposition of eosinophil-derived toxic proteins in affected tissues may contribute. [IL-5](#) and transforming growth factor β have been implicated as potential mediators. Treatment is withdrawal of L-tryptophan-containing products and the administration of glucocorticoids. Most patients recover fully, remain stable, or show slow recovery; but the disease can be fatal in up to 5% of patients.

EOSINOPENIA

Eosinopenia occurs with stress, such as acute bacterial infection, and after treatment with glucocorticoids. The mechanism of eosinopenia of acute bacterial infection is unknown but is independent of endogenous glucocorticoids, since it occurs in animals after total adrenalectomy. There is no known adverse effect of eosinopenia.

HYPERIMMUNOGLOBULIN E-RECURRENT INFECTION SYNDROME

The hyperimmunoglobulin E-recurrent infection (HIE) syndrome or *Job's syndrome* is a rare multisystem disease in which the immune system, bone, teeth, lung and skin are affected. Abnormal chemotaxis is a variable feature. The molecular basis for this syndrome is not known, but some cases show autosomal dominant transmission with linkage to 4q. Patients with this syndrome have characteristic facies with broad nose, kyphoscoliosis and osteoporosis, and eczema. The primary teeth erupt normally but do not deciduate, often requiring extraction. Patients develop recurrent sinopulmonary and cutaneous infections that tend to be much less inflamed than appropriate for the degree of infection and have been referred to as "cold abscesses." A high degree of suspicion is required to diagnose infections in these patients, who may appear well despite extensive disease. The cold abscesses have been considered a reflection of impaired chemotaxis with too few phagocytes arriving too late, perhaps due to a lymphocyte factor inhibiting chemotaxis. However, the chemotactic defect in these patients is variable, and the fundamental basis for the impaired defenses is complex and poorly defined.

LABORATORY DIAGNOSIS AND MANAGEMENT

Initial studies of [WBC](#) and differential and often a bone marrow examination are followed by assessment of bone marrow reserves (steroid challenge test), marginated circulating pool of cells (epinephrine challenge test), and marginating ability (endotoxin challenge test) ([Fig. 64-3](#)). In vivo assessment of inflammation is possible with a Rebuck skin window test or an in vivo blister assay, which measures the ability of leukocytes and inflammatory mediators to accumulate locally in the skin. In vitro tests of phagocyte aggregation, adherence, chemotaxis, phagocytosis, degranulation, and microbicidal activity (for *S. aureus*) may help pinpoint cellular or humoral lesions. Deficiencies of oxidative metabolism are detected with the nitroblue tetrazolium (NBT) dye test, which is based on the ability of products of oxidative metabolism to reduce yellow, soluble NBT to blue-black formazan, an insoluble material that can be seen microscopically. Studies of superoxide and hydrogen peroxide production may further define neutrophil oxidative function.

Patients with leukopenias or leukocyte dysfunction often have delayed inflammatory responses. Therefore, clinical manifestations may be minimal despite overwhelming infection, and unusual infections must always be suspected. Early signs of infection demand prompt, aggressive culturing for microorganisms, use of antibiotics, and surgical drainage of abscesses. Prolonged antibiotics are often required. In patients with [CGD](#), prophylactic antibiotics (trimethoprim-sulfamethoxazole) diminish the frequency of life-threatening infections. Short courses of glucocorticoids may relieve gastrointestinal or genitourinary tract obstruction by granulomas in patients with CGD. Recombinant human [IFN-g](#), which nonspecifically stimulates phagocytic cell function, reduces the frequency of infections in patients with CGD by 70% and reduces the severity of infection. This effect of IFN-g in CGD is additive to the effect of prophylactic antibiotics. The recommended dose is 50 ug/m² subcutaneously three times weekly. IFN-g also has been used successfully in the treatment of leprosy, nontuberculous mycobacteria, and visceral leishmaniasis.

Rigorous oral hygiene reduces but does not eliminate the discomfort of gingivitis, periodontal disease, and aphthous ulcers; chlorhexidine mouthwash and tooth brushing with a hydrogen peroxide-sodium bicarbonate paste helps many patients. Oral antifungal agents (fluconazole) have reduced mucocutaneous candidiasis in patients with Job's syndrome. Androgens, glucocorticoids, lithium, and immunosuppressive therapy have been used to restore myelopoiesis in patients with neutropenia due to impaired production. Recombinant [G-CSF](#) is useful in the management of certain forms of neutropenia due to depressed neutrophil production, especially that related to cancer chemotherapy. Patients with chronic neutropenia with evidence of a good bone marrow reserve need not receive prophylactic antibiotics.

Patients with constant or cyclic neutrophil counts <500/uL may benefit from prophylactic antibiotics and [G-CSF](#) during periods of neutropenia. Oral trimethoprim-sulfamethoxazole (160/800 mg) twice daily can prevent infection. Increased numbers of fungal infections are not seen in patients with [CGD](#) on this regimen. Oral quinolones such as norfloxacin and ciprofloxacin are alternatives.

In the setting of cytotoxic chemotherapy with severe, persistent neutropenia, trimethoprim-sulfamethoxazole prevents *Pneumocystis carinii* pneumonia. These patients, and patients with phagocytic cell dysfunction, should avoid heavy exposure to airborne soil, dust, or decaying matter (mulch, manure), which are often rich in spores of *Aspergillus* or other fungi. Restriction of activities or social contact has no proven role in reducing risk of infection.

Cure of some congenital phagocyte defects is possible by bone marrow transplantation ([Chap. 115](#)). However, complications of bone marrow transplantation are still serious, and with rigorous medical care many patients with phagocytic disorders can go for years without a life-threatening infection. The identification of specific gene defects in patients with [LAD 1](#) and [CGD](#) has led to gene therapy trials in a number of genetic white cell disorders.

(Bibliography omitted in Palm version)

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PART FOUR -CLINICAL PHARMACOLOGY

70. PRINCIPLES OF DRUG THERAPY - John A. Oates, Dan M. Roden, Grant R. Wilkinson

Safe and effective drug therapy requires that drugs be delivered to their molecular targets in tissues at concentrations within the range that yields efficacy without toxicity. Variability in drug effects between individual patients can thus be attributed, in part, to differences in the processes that determine these concentrations, i.e., drug disposition involving absorption, distribution, and elimination. *Pharmacokinetics* provides a quantitative description of these processes, i.e., what the body does to the drug. In addition, variability in response may also reflect differences in the drug-target interaction that affect the concentration-related effects of the drug on the body, termed *pharmacodynamics*. This chapter reviews the principles of pharmacokinetics and pharmacodynamics and their application to optimizing therapeutic regimens.

CLINICAL PHARMACOKINETICS

PLASMA LEVELS AFTER A SINGLE DOSE

The levels of lidocaine in plasma following intravenous administration decline in two phases, as illustrated in [Fig. 70-1](#); such a biphasic decline is typical for many drugs. Immediately after rapid injection, essentially all of the drug is in the plasma or central compartment, and the high initial plasma level reflects the confinement of the drug to this small volume. The drug is then transferred into an extravascular or peripheral compartment during a period called the *distribution phase*. For lidocaine, the distribution phase is virtually complete within 30 min. A phase of slower decline, the *equilibrium phase*, then occurs. During this phase, the drug levels in the plasma and in tissues change in parallel. Although this disposition profile is common to many drugs given intravenously, the characteristic parameters vary among drugs.

Distribution Phase The pharmacologic effects during the distribution phase depend on whether the concentration of drug at the receptor site is similar to that in the plasma. If that is the case, the pharmacologic effects may be intense during this period because of the high initial levels in the plasma. For example, after a small bolus dose (50 mg) of lidocaine, antiarrhythmic effects may be evident during the early distribution phase but will disappear as levels fall below those that are minimally effective, even before equilibrium between plasma and tissue is reached. Thus, a larger single dose or multiple small doses must be administered to achieve an effect that is sustained into the equilibrium phase. Toxicity resulting from high levels of some drugs during the distribution phase precludes administration of a single intravenous loading dose that will yield therapeutic levels during the equilibrium phase. For example, the administration of the entire loading dose of phenytoin as a single intravenous bolus can cause cardiovascular collapse due to the high levels during the distribution phase. If a loading dose of phenytoin is administered intravenously, it must be given slowly, e.g., at infusion rates of 25 to 50 mg/min. For similar reasons, the intravenous loading dose of many potent drugs that equilibrate rapidly with their receptors is either divided into fractional doses given at intervals or administered by infusion over a similar period.

A dose given orally results in lower plasma levels during the initial period than an intravenous bolus dose that delivers the same amount of drug to the systemic circulation. Because the drug is not absorbed instantly after oral administration and is delivered into the systemic circulation more slowly, much of it has been distributed by the time absorption is complete. Thus, procainamide, which is almost totally absorbed after oral administration, can be given orally as a single 750-mg loading dose with little risk of hypotension; in contrast, loading of this drug by the intravenous route is more safely accomplished by giving the dose in fractions of about 100 mg at 5-min intervals or by slow infusion to avoid hypotension during the distribution phase.

Some drugs are so predictably lethal when infused too rapidly that special precautions should be taken to prevent even the inadvertent occurrence. For example, solutions of potassium for intravenous administration in excess of 20 meq/L should be avoided in all but the most exceptional and carefully monitored circumstances. This minimizes the possibility of cardiac arrests, which can occur as a result of accidental increases in infusion rates of more concentrated solutions.

As these examples illustrate, excessively rapid administration of many drugs can lead to catastrophic consequences that result from high concentrations in the blood during the distribution phase.

In contrast, for some centrally active drugs, the higher concentration of drug during the distribution phase after intravenous administration is used to advantage. The use of midazolam for "IV sedation," for example, depends upon its rapid uptake by the brain during the distribution phase to produce sedation quickly, with subsequent egress from the brain during the redistribution of the drug as equilibrium is achieved.

Some drugs are distributed slowly to their sites of action during the distribution phase, i.e., concentration at the relevant receptor does not parallel that in plasma early after drug administration. For example, the level of digoxin at the receptor site (and the drug's pharmacologic effect) do not reflect plasma levels during the distribution phase. Digoxin is transported (or bound) to its cardiac receptors slowly by a process that proceeds throughout distribution. Thus, over the distribution phase of several hours, plasma levels fall while the level at the site of action and the pharmacologic effect increase. Only at the end of the distribution phase, when the drug has reached equilibrium with the receptor, does the concentration of digoxin in plasma reflect pharmacologic effect. For this reason, there should be a 6- to 8-h wait after administration before plasma levels of digoxin are measured as a guide to therapy.

Equilibrium Phase After the concentration of drug in plasma has reached a dynamic equilibrium with that in the tissues, the levels in plasma and tissues fall in parallel as the drug is eliminated from the body. Thus, the equilibrium phase is also called the *elimination phase*. During this phase, drug concentrations measured in plasma can provide a useful index of drug levels in tissues.

Most drugs are eliminated as a first-order process. This means that the time required for the plasma level of the drug to fall to one-half the original value (the half-life, $t_{1/2}$) is the same regardless of the point on the plasma level curve at which measurement begins. Another characteristic of the first-order process is that a plot of the logarithm of the

plasma concentration versus time is linear. From such a plot ([Fig. 70-1](#)), it can be seen that the half-life of lidocaine is 108 min. If the half-life is known, the amount of a dose remaining in the body at any time following administration of a single dose can be calculated. [Table 70-1](#) shows how this amount changes over five successive half-lives.

From a clinical standpoint, elimination is essentially complete when it has reached about 90%. Therefore, for practical purposes, *a first-order elimination process reaches completion after three to four half-lives.*

DRUG ACCUMULATION -- LOADING AND MAINTENANCE DOSES

If a drug is given repeatedly at intervals shorter than the time required to eliminate a dose, both the amount of drug in the body and its pharmacologic effect increase with successive doses until they reach a plateau. [Figure 70-2](#) shows the accumulation of digoxin administered in repeated maintenance doses (without a loading dose). Since the half-life of digoxin is about 1.6 days in a patient with normal renal function, 65% of a digoxin dose remains in the body at the end of 1 day. Thus, the second dose will raise the amount of digoxin in the body (and the average plasma level) to 165% of the level produced by the first dose. Each subsequent dose causes a further increase until a *steady state* is achieved. At that point, the drug dosing rate (bioavailable dose/dosage interval) is equal to the rate of elimination, with the fluctuation between peak and trough plasma levels remaining constant. If the rate of drug delivery is then altered, a new steady state will be attained. Continuous infusion of a drug at constant rate also results in progressive accumulation to a predictable steady state ([Fig. 70-2](#)). In this case, the steady-state plasma level (C_{ss}) is equivalent to the average between the peak and trough levels ($C_{avg,ss}$) produced by intermittent administration of the same amount of drug over the same period. For *all* drugs with first-order kinetics, the time required to achieve steady-state levels can be predicted from the half-life, because accumulation is a first-order process with a half-life identical to that for elimination. Thus, accumulation reaches 90% of steady-state levels at the end of three to four half-lives. This is true for either intermittent or continuous dosing ([Fig. 70-2](#)).

When a therapeutic effect is required urgently, simply administering the maintenance dose of a drug with a long half-life results in an unacceptable delay in reaching steady-state levels of the drug and its intended effect. The time required to achieve the desired pharmacologic effect may be shortened by the administration of a *loading dose* -- the amount of drug that will bring the plasma concentration rapidly to the steady-state level. Therefore, if treatment with lidocaine ($t_{1/2} \sim 108$ min) were initiated by infusion at only the maintenance dose level, it would take about 4 to 8 h before the drug's maximal effect was achieved. Because ventricular arrhythmias may be life-threatening, it is not reasonable to wait that long to achieve an effective steady-state drug level. Accordingly, it would be appropriate to administer one or more loading doses at the onset of therapy, together with infusion at the rate of the maintenance dose.

Loading may be accomplished by the administration of a single loading dose or, if that would create a risk of toxicity, by the administration of the loading dose in a series of fractions of that dose over time. The latter approach is particularly appropriate for most drugs that have a low therapeutic index. A divided loading dose strategy or the administration of the loading dose in a slow intravenous infusion are particularly

advisable when the drug is given intravenously. Thus, in the case of lidocaine, a common regimen is to administer an initial intravenous bolus of 1 mg/kg followed by up to three additional bolus injections of 0.5 mg/kg every 8 to 10 min as necessary, and a maintenance infusion of 2 mg/min.

Regardless of the size of the loading dose, *after maintenance therapy has been given for three to four half-lives, the amount of drug in the body is determined by the maintenance dose.* The independence of the steady state plasma levels from the loading dose is illustrated in [Fig. 70-2](#), which shows that the elimination of the loading dose would be practically complete after three to four half-lives.

DETERMINANTS OF DRUG DISPOSITION

A number of physiologic and pathophysiologic factors determine the disposition of a drug and hence its pharmacokinetics in an individual patient. The three most important are *clearance*, a measure of the body's ability to eliminate drug; *volume of distribution*, an indication of the extent to which the drug is distributed outside of the blood compartment; and *bioavailability*, the fraction of the administered dose that reaches the systemic circulation. The elimination half-life ($t_{1/2}$), which measures the rate of drug removal from the body, is determined by the relationship between the physiologically determined clearance and volume of distribution.

Clearance The majority of drugs are given over a prolonged period of time according to a multiple dosage regimen, e.g., x mg every y h. The clinical goal is to maintain the drug's steady-state concentration within the therapeutic range for the individual patient; if the level is too low, reduced efficacy results, whereas if it is too high, the likelihood of adverse effects increases.

Steady state is achieved when the rate of drug elimination equals the rate of drug delivery into the systemic circulation, which, if bioavailability is complete, corresponds to the rate at which the drug dose is administered:

where Cl is clearance and C_{ss} is the steady-state concentration in plasma. Therefore, *at a given dosing rate the concentration of drug in plasma is completely dependent on its clearance.* If the drug is infused, the concentration remains constant so long as drug delivery continues; however, when the drug is given intermittently, the above relationship is expressed as:

where $C_{avg,ss}$ is the *average* value during the dosage interval and is equal to C_{ss} although the *actual* concentrations will be higher or lower at various points during this period. Thus, clearance determines the rate at which a drug should be administered in order to obtain a desired steady-state concentration; stated in a different fashion, steady-state drug levels can be modified either up or down by changing the dosage rate. *The clearance of the vast majority of drugs is constant over the therapeutic range of concentrations.*

Clearance is a measure of the rate at which the organs that eliminate drug from the body remove drug from the blood.

Accordingly, clearance reflects the volume of blood (or plasma) from which the drug would have to be removed per unit time to account for the elimination; it can be related to either total or unbound drug.

Drug elimination generally occurs as a result of metabolism and/or excretion in the liver, kidney, and possibly other organs. Clearance of drug from the body, therefore, reflects the overall contribution of each of these organs, as indicated by their individual rates of elimination normalized to the concentration of drug, and is additive.

Clearance of a drug is usually estimated following administration of an intravenous dose (dose_{iv}) and measurement of the resulting total area under the curve (AUC) for the blood or plasma concentration-time curve (AUC_{iv}) from zero to infinite time.

[Table 70-2](#) indicates the marked differences in plasma clearance for some commonly used drugs. Some drugs such as phenobarbital and valproic acid have relatively low values (<10 mL/min), whereas others such as procainamide and lidocaine have much larger clearances (>500 mL/min). Such differences mainly reflect different rate-limiting determinants such as blood flow through the organ(s) of clearance, the extent of binding of the drug to plasma proteins, and the efficiency of the clearance process to remove drug from tissue water by metabolism and/or excretion. The data in [Table 70-2](#) also demonstrate that the relative contributions of the two major routes of elimination, i.e., renal and nonrenal, also vary according to the individual drug. In some cases, such as amikacin, digoxin, gentamicin, lithium, and tobramycin, excretion by the kidneys is predominant. However, with many other drugs (e.g., carbamazepine, lidocaine), nonrenal elimination, which usually reflects metabolism in the liver, is more important.

Volume of Distribution The relationship between the amount of drug in the body and the concentration of drug in the plasma provides a measure of the apparent volume of distribution:

This volume does not, except in a limited number of special cases, reflect an identifiable physiologic volume but corresponds to the virtual volume of fluid that would be required to contain all of the drug in the body at the same concentration as in the plasma. In a typical 70-kg human, plasma volume is about 3 L, blood volume around 5.5 L, and extracellular water outside the vasculature is approximately 42 L. The volume of distribution of drugs that are extensively bound to plasma proteins but are not bound to tissue components approaches plasma volume. However, for most drugs, the volume of

distribution is far greater than any physiologic space. For example, the volume of distribution of digoxin is about 700 L, which obviously exceeds total body volume. This simply indicates that digoxin is largely distributed outside the vascular system and hence the proportion of the drug present in the plasma compartment is low.

The volume of distribution may be estimated by back-extrapolation of the plasma concentration-time curve to zero time (C_0) ([Fig. 70-1](#)) and dividing this into the dose of drug administered intravenously.

When distribution of the drug is not instantaneous, a more useful volume estimate is based on the area under the plasma-concentration time curve (AUC) and the terminal elimination half-life of the drug ($t_{1/2}$).

Extent and Rate of Bioavailability After intravenous administration, all of the administered dose reaches the systemic circulation. In contrast, with all other routes of administration, such as oral, intramuscular, and subcutaneous, there is the potential for only a part of the dose to be absorbed; this fraction (F) is termed the drug's *bioavailability*. Lack of complete bioavailability may reflect the inability of the drug to be completely released from the dosage form or vehicle, chemical destruction at the site of administration, incomplete absorption into the vascular system, and metabolism and/or excretion during translocation of the drug from its site of administration to the systemic circulation. In the case of oral dosing, this would include the intestinal epithelium and the liver and lungs. Metabolism and/or excretion by the gastrointestinal tract (F_G) and liver (F_L) are collectively referred to as the *first-pass effect*, since the resulting drug elimination only occurs during drug delivery to the systemic circulation. Loss of drug because of a first-pass effect thus requires that the dosing rate appropriately take into account bioavailability. For example, after oral drug administration:

Bioavailability and the first-pass effect are important with respect to possible differences in drug responsiveness dependent on the route of administration. For example, glyceryl trinitrate has such a large oral first-pass effect (>99%) that systemic concentrations after an oral dose are negligible and no antianginal effect is present. Giving the drug by the sublingual or transdermal routes bypasses the splanchnic organs and allows essentially all of the drug to reach the systemic circulation. For drugs that are efficiently metabolized by the intestinal epithelium and/or liver, i.e., drugs with high extraction ratios in either of these organs, differences in the extent of the first-pass effect between individuals frequently explain variability in drug response. With propranolol, for example, the 15-fold variability in plasma concentrations after the same oral dose results from differences in the individual hepatic extraction ratios reflective of different levels of drug metabolizing activity.

Drug administration by nonintravenous routes involves an absorption process characterized by the plasma level increasing to a maximum value at some time after

administration and then declining as the rate of drug elimination exceeds the rate of absorption. Thus, the peak concentration is lower and occurs later than after the same dose given by rapid intravenous injection. The rate of absorption can be an important consideration during the initial period after drug administration, especially for drugs with a narrow *therapeutic index* -- the ratio of the toxic dose to the therapeutic dose. If absorption is too rapid, then the resulting high concentration may cause adverse effects not observed with a more slowly available formulation. At the other extreme, slow absorption is deliberately designed into "slow-release" or "sustained-release" drug formulations in order to maintain plasma concentrations essentially constant during the dosage interval, because the drug's rate of elimination is offset by an equivalent rate of absorption controlled by formulation factors.

Half-Life The organs of elimination can only clear drug from the blood. Thus, the rate at which drug is eliminated from the body is a function of both clearance and the extent to which drug is distributed outside of the vascular compartment. The fraction of total drug in the body that is eliminated in a given time is designated the *fractional elimination constant* (k).

For example, if the volume of distribution is 10 L and clearance is 1 L/min, then one-tenth of the drug is eliminated per minute. If k is multiplied by the total amount of drug in the body, the actual rate of elimination at any given time can be determined:

This relationship, indicating that the rate of drug elimination is proportional to the drug concentration, describes a first-order, or monoexponential, process. With a few notable exceptions, the elimination of drugs used clinically is first-order.

Half-life ($t_{1/2}$) is the time that it takes for the plasma concentration or amount of drug in the body to decline by 50%. This parameter is related to k as follows:

where 0.693 is the natural logarithm of $2(C_0/0.5 C_0)$.

Because

then

This is an important relationship since it indicates that the rate of drug elimination, reflected by $t_{1/2}$, is dependent on both the efficiency of drug removal (Cl) and the drug's volume of distribution (V). When V remains constant, $t_{1/2}$ is a reflection of clearance. Thus, $t_{1/2}$ is shortened when rifampin induces the enzymes responsible for a drug's

hepatic clearance and is lengthened when a drug's renal clearance is impaired in renal failure. However, when there are concomitant alterations in V , as occurs for some drugs in cardiac failure, $t_{1/2}$ is not an accurate measure of Cl or drug dose.

DESIGNING DOSAGE REGIMENS

Most drugs are administered as part of long-term therapy involving multiple dosing, and it is critical that the dosage regimen be optimized to the individual patient. With some drugs, the desired response, e.g., coagulation or blood pressure, is readily measurable and an individualized dosage regimen can be developed with dosage titration. However, dosage changes should be conservative (<50% for drugs with a low therapeutic index) and not more frequent than every three to four half-lives. Other drugs have little dose-related toxicity so the therapeutic window is large, e.g., penicillins and β -adrenoceptor antagonists. In these situations, effective and prolonged drug effects may be obtained by a "maximal dose" strategy. It is also possible to use this strategy to extend the duration of action of a drug, especially one that is eliminated rapidly from the body. Thus, 75 mg of captopril will result in reduced blood pressure for up to 12 h, even though the elimination half-life of this angiotensin-converting enzyme (ACE) inhibitor is about 2 h; this is because the dose raises the concentration of drug in plasma many times higher than the threshold for its pharmacologic effect.

Determination of the Maintenance Dose The relationship between the maintenance dose and the final steady-state concentration is

Thus, steady-state concentrations can be predictably increased or decreased by appropriate modification of the maintenance dosing rate to achieve a desired target value:

In most cases, this is best achieved by changing the drug dose but not the dosing interval, e.g., by giving 250 mg every 8 h instead of 200 mg every 8 h. However, this approach is acceptable only if the resulting maximum concentration is not toxic and the trough value does not fall below the minimum effective concentration for an undesirable period of time. Alternatively, the steady state may be changed by altering the frequency of intermittent dosing but not the size of each dose. In this case, the magnitude of the fluctuations around the average steady-state level will change -- the shorter the dosing interval, the smaller the difference between peak and trough levels ([Fig. 70-3](#)).

The extent of fluctuation is determined by the relationship between the dosing interval and the drug's half-life. For example, if the dosing interval is equal to the drug's half-life, then the fluctuation would be twofold, which is usually a tolerable variation. If a longer dosing interval is used, then the difference between the maximum and minimum plasma levels will be greater ([Fig. 70-3](#)). Marked fluctuations increase the likelihood of increased concentration-dependent drug effects early during the dosing interval and possible ineffectiveness at the end of the period, even though the average steady-state drug concentration is the same as that following administration at the same dosing rate but at

shorter intervals.

Determination of the Loading Dose The loading dose can be estimated if both the desired plasma level (C) and the apparent volume of distribution (V) are known:

The loading amount required to achieve steady-state plasma levels can also be determined from the fraction of drug eliminated during the dosing interval and the maintenance dose. For example, if the fraction of digoxin eliminated daily is 35% and the planned maintenance dose is 0.25 mg daily, then the loading dose to achieve steady-state levels would be $(100/35)$ times the maintenance dose, or approximately 0.75 mg. Thus,

NONLINEAR DRUG ELIMINATION

The elimination of some drugs (e.g., phenytoin, salicylate, propafenone, and theophylline) does not follow first-order kinetics because the clearance of these drugs changes as levels in the body fall during elimination or change after alterations in dose. Such elimination is called *concentration-dependent* or *dose-dependent*. Accordingly, the time for the concentration to fall to one-half becomes less as plasma levels fall. (This halving time is not truly a half-life, because the term *half-life* applies to first-order kinetics and is a constant.) When a drug is eliminated by first-order kinetics, the plasma level at steady state is directly related to the amount of the maintenance dose, and a doubling of the dose should lead to doubling of the steady-state plasma level. However, for drugs with dose-dependent kinetics, an increase in the dose may be accompanied by a disproportionate increase in the plasma level. For example, a threefold increase in the dose of propafenone (from 300 mg to 900 mg daily) leads to a tenfold increase in the concentration of propafenone in plasma. Changes in dosage regimens for drugs with dose-dependent kinetics should always be accompanied by surveillance for adverse effects and by measurement of the concentration of the drug in plasma during the time of transition to the new steady-state, if this is feasible.

INDIVIDUALIZATION OF DRUG THERAPY

EFFECTS OF RENAL DISEASE

Whether a drug's dosing rate needs to be modified in patients with renal dysfunction depends on whether the drug is primarily excreted through the kidneys and whether increased drug levels, secondary to impaired renal clearance, will be associated with adverse effects. If both of these factors are present, it is likely that with decreased renal clearance the drug will accumulate to a greater extent than in patients with normal renal function and toxicity will result. This is especially true for drugs with long half-lives and narrow therapeutic indexes (e.g., digoxin). In general, over 60 to 70% of the drug must be renally excreted for dosage modification to be necessary and then only when renal function is less than about 30 to 50% of normal.

The goal of any dosing rate adjustment is to modify the dosing schedule so that the drug's plasma concentration-time profile is as similar to the desired one as possible and that the steady state is reached in about the same time as in a patient with normal renal function. To obtain the desired profile, a modification may be made by decreasing the dose while maintaining the dosage interval, keeping the dose the same but increasing the dosing interval, or a combination of these two approaches.

A drug's renal clearance is proportional to creatinine's clearance (Cl_{CR}), which may be measured directly or estimated from the serum creatinine level (C_{CR}). In men:

For women, the estimate by the above equation should be multiplied by 0.85 to reflect their smaller muscle mass. It should also be noted that this equation is not valid for patients with severe renal insufficiency ($C_{CR} < 5$ mg/dL) or when renal function is changing rapidly. For simplicity, normal creatinine clearance is conveniently considered to be 100 mL/min. Thus, if the relative contributions of renal and nonrenal elimination to systemic clearance are known, an appropriate modification of the dose in a patient with a given level of insufficiency can be estimated. For example, if the fraction of drug excreted unchanged is 0.9 and creatinine clearance is reduced to 10% of normal, the dosing rate should be reduced to 19% of normal. This modification, which in practice would be rounded to 20%, is based on the fact that nonrenal clearance is unchanged (10% of normal clearance); renal clearance is reduced from 90% to 9% of normal Cl ; thus systemic clearance is reduced to $10 + 9 = 19\%$ of normal Cl .

In clinical practice today, most decisions involving dosing adjustment in patients with renal failure use published tables of recommended dosage reduction or dosing interval lengthening based on the level of renal function indicated by Cl_{CR} , or similar information provided in the drug "label" ([Table 70-3](#)). Such modifications are, however, rigorously based on pharmacokinetic principles and are best used when resulting plasma concentration data and clinical observation are used, as necessary, to further optimize therapy for the individual patient.

Often metabolites of the drug are pharmacologically active or cause toxicity, and renal insufficiency may result in their unanticipated accumulation. Meperidine, for example, is extensively metabolized, and renal failure has little effect on its plasma concentration; however, its metabolite, normeperidine, accumulates above its usual level when renal function is impaired. Because normeperidine has greater convulsant activity than meperidine, this accumulation probably accounts for the signs of central nervous system (CNS) excitation, such as irritability, twitching, and seizures, that appear when multiple doses of meperidine are administered to patients with renal disease.

EFFECTS OF LIVER DISEASE

In contrast to the predictable decline in renal clearance of drugs in renal insufficiency, it is not possible to make a general prediction of the effect of liver disease on hepatic biotransformation of drugs ([Chap. 292](#)). Rather, the possible effects of hepatitis or cirrhosis range from impaired to increased drug clearance. Even in advanced hepatocellular disease, drug clearance is usually impaired only about two- to fivefold.

The extent of such changes, however, cannot be predicted by the common tests of liver function. Consequently, even when it is suspected that drug elimination is altered in liver disease, there is no quantitative basis on which to adjust the dosage regimen other than assessment of clinical response and the concentration of the drug in plasma.

A drug's oral bioavailability may markedly increase in patients with liver disease. This is particularly the case for those drugs that normally are very well extracted by the liver and thus have a high first-pass effect. In addition, the presence of portacaval shunts may further reduce first-pass elimination and lead to higher drug concentrations reaching the systemic circulation, with the increased risk of adverse effects. For example, the oral availability for high first-pass drugs such as morphine, meperidine, midazolam, and nifedipine is almost doubled in patients with cirrhosis, compared to those with normal liver function. The size of the oral dose of such drugs should, therefore, be reduced in such patients.

EFFECTS OF CIRCULATORY INSUFFICIENCY -- CARDIAC FAILURE AND SHOCK

Under conditions of decreased tissue perfusion, the cardiac output is redistributed to preserve blood flow to the heart and brain at the expense of other tissues ([Chap. 38](#)). As a result, the drug may be distributed into a smaller volume of distribution, higher drug concentrations will be present in the plasma, and the tissues that are best perfused will be exposed to these higher concentrations. If either the brain or heart is sensitive to the drug, an alteration in response will occur.

Furthermore, the decreased perfusion of the kidney and liver may impair drug clearance by these organs, directly or indirectly. Thus, in severe congestive heart failure, in hemorrhagic shock, and in cardiogenic shock, the response to the usual dose of drug may be excessive, and dosage modification may be necessary. For example, the clearance of lidocaine is reduced by about 50% in cardiac failure, and therapeutic plasma levels are achieved at infusion rates only about half those usually required. The volume of distribution of lidocaine is also reduced, meaning that the correct loading dose will be smaller than usual. Similar situations are thought to exist for procainamide, theophylline, and possibly quinidine. Unfortunately, predictors of these types of pharmacokinetic alterations are unavailable. Therefore, loading doses should be conservative, and continued therapy should be monitored closely, following clinical indicators of toxicity and plasma levels.

DISEASE-INDUCED CHANGES IN PLASMA BINDING

Many drugs circulate in the plasma partly bound to plasma proteins. Since only the unbound (free) drug can distribute to the site of pharmacologic action, the therapeutic response should be related to the free rather than the total circulating plasma drug concentration. In most cases, the degree of binding is fairly constant across the therapeutic concentration range, so that the total drug levels in plasma can be used as a basis for adjusting dosage without resulting in significant error. However, conditions such as hypoalbuminemia, liver disease, and renal disease can decrease the extent of drug binding, particularly of acidic and neutral drugs so that at any total plasma level there is a greater concentration of free drug than usual and thus a risk of increased response and toxicity. By contrast, conditions that lead to an increased plasma

concentration of the acute-phase reactant α_1 -acid glycoprotein -- such as myocardial infarction, surgery, neoplastic disease, rheumatoid arthritis, and burns -- cause an increase in drug binding for the basic drugs, e.g., lidocaine and quinidine, that bind to this macromolecule, resulting in an opposite set of effects. The drugs for which changes in binding are important are those that are normally highly bound to plasma proteins (>90%), because a small alteration in the extent of binding produces a large change in the amount of unbound drug.

For many drugs, elimination and distribution are restricted largely to the unbound fraction, and so a decrease in binding leads to an increase in the clearance and distribution of the drug. The relative magnitudes of these changes are such that the net effect is a shortened half-life.

GENETIC DETERMINANTS OF THE RESPONSE TO DRUGS

Knowledge of the enzyme that catalyzes the predominant pathway of metabolism of a drug provides a basis for understanding the therapeutic consequences of variations in the genotype of that enzyme. For a number of the enzymes that metabolize drugs, there are differences (polymorphisms) in catalytic function that are genetically determined ([Chap. 65](#)). *A phenotypic trait or its corresponding gene is said to be polymorphic if there is more than one form of the trait or gene in the population.* Polymorphisms in the function of an enzyme are determined by allelic variants in its gene. Increasingly it is possible to individualize treatment based on analysis of the phenotype and/or genotype of the relevant drug-metabolizing enzyme.

Similarly, polymorphism in the receptor for a drug can determine variability in its pharmacologic effect. Genotyping those drug receptors for which polymorphisms influence response may also assist in individualizing drug therapy.

THIOPURINE S-METHYLTRANSFERASE (TPMT)

The metabolism of azathioprine provides an example of the importance of genetic polymorphisms of enzymes. Azathioprine exerts its immunosuppressive action via an active metabolite, 6-mercaptopurine. Within target cells, the major pathway of inactivation of 6-mercaptopurine is by TPMT. Genetic polymorphisms in this enzyme lead to differences in inactivation of 6-mercaptopurine, with corresponding vast differences in the sensitivity of patients to the toxic and therapeutic effects of azathioprine. Homozygotes for alleles encoding inactive TPMT (0.3 to 1% of the population) predictably exhibit severe pancytopenia on standard doses of azathioprine. Heterozygotes for alleles encoding enzymes with deficient TPMT activity also experience more bone marrow suppression on "usual" doses. From a therapeutic standpoint it is likely that the bone marrow suppression in the heterozygotes has influenced the empiric determination of the "usual" dose range of azathioprine and that this results in undertreatment of some of the patients homozygous for the allele encoding a TPMT with full catalytic activity. To detect the TPMT deficient phenotype in patients anticipating therapy with azathioprine, the catalytic function of TPMT may be measured in red blood cells (if no blood or red cell transfusions have been given within 2 months). A high concordance of genotype with phenotype (~95%) suggests that analysis of genotype may be used to individualize dosing and therefore improve

treatment with azathioprine (and 6-mercaptopurine) in the future.

ACETYLATION

Isoniazid, hydralazine, sulfonamides, procainamide, and a number of other drugs are metabolized by acetylation of a hydrazino or amino group. This reaction is catalyzed by *N*-acetyl transferase-2 (NAT-2), an enzyme in the liver cytosol that transfers an acetyl group from acetyl coenzyme A to the drug. Individuals differ markedly in the rate at which drugs are acetylated, because of polymorphisms in the NAT-2 gene, resulting in a bimodal distribution of the population into "rapid acetylators" and "slow acetylators."

Acetylation phenotype can be determined by measuring the ratio of acetylated to nonacetylated forms of the probe drugs, dapsone, caffeine, or sulfamethazine, in plasma or urine following administration of a test dose of these acetylation substrates. Slow, intermediate, and rapid acetylators may be identified by these methods for phenotyping. It is also possible to identify slow acetylators by genotyping, using genomic DNA obtained from blood leukocytes.

METABOLISM BY CYTOCHROME P450 MONOOXYGENASES

In healthy individuals taking no other medications, the major determinant of the rate of metabolism of drugs by the cytochrome P450 monooxygenases is genetic. Hepatic endoplasmic reticulum contains a family of cytochrome P450 (CYP) isoforms with different substrate specificities. Many drugs undergo oxidative metabolism by more than one isoform, and the steady-state concentrations of such drugs in the plasma is a function of the sum of the activities of these and other metabolizing enzymes. When a drug is metabolized by multiple pathways, the catalytic activities of the participating enzymes are regulated by a number of genes, so that the clearance rates and steady-state concentrations of the drug tend to distribute unimodally within the population. The range of activity may differ markedly (3tenfold) between different individuals, as is the case for chlorpromazine, and there is no way to predict the rate before beginning therapy.

Certain metabolic pathways show a bimodal or trimodal distribution of activity, suggesting control by a single gene, and polymorphisms in these genes have been identified. Most individuals are extensive metabolizers (EM phenotype); a smaller group have a lower ability to metabolize the drug (or no ability at all) and are called poor metabolizers (PM phenotype). Heterozygotes for genes encoding the enzymes lacking catalytic activity may be intermediate metabolizers (IM phenotype). And patients with duplicate or multiple copies of the gene may exhibit ultrarapid metabolism. These polymorphisms are of greatest clinical relevance during administration of substrate drugs for which there are no major alternative routes of elimination. The clinical consequences of the PM phenotype will then depend on the resultant accumulation of the drug or occasionally on the absence of generation of active metabolites.

The cytochrome P450 isoform CYP2D6 is polymorphically distributed in the population, and about 8 to 10% of Caucasians are deficient in this enzyme. CYP2D6 represents the main metabolic pathway for a number of drugs, including antiarrhythmic agents (propafenone, flecainide), β -adrenoceptor blockers (timolol, metoprolol, and alprenolol),

tricyclic antidepressants (nortriptyline, desipramine, imipramine, clomipramine), neuroleptic drugs (perphenazine, thioridazine, and possibly haloperidol), selective serotonin reuptake inhibitors (fluoxetine and paroxetine), and certain opiates, such as codeine and dextromethorphan. Thus, codeine has a much lower analgesic effect in [PM](#) patients because of impaired production of the active metabolite, morphine. Conversely, a patient with duplicate or multiple copies of CYP2D6 will exhibit an exaggerated response to codeine. Patients with the PM phenotype experience more pronounced systemic β -adrenoceptor blockade after the administration of timolol ophthalmic solution. The catalytic activity of CYP2D6 in humans may be assessed by using a test drug, debrisoquin, which is eliminated almost entirely via hydroxylation by CYP2D6. Individuals with the PM phenotype can be identified by genotyping for the alleles that encode proteins with loss of catalytic function. There are ethnic variations in the frequency of the PM phenotype, which occurs in 5 to 10% of Caucasians but with a lesser frequency in Asians (1 to 2%).

The isoform CYP2C19 also exhibits polymorphism; it was initially detected with the hydroxylation of mephenytoin, which is used as a probe drug for the function of this P450 isoform. This enzyme catalyzes the major metabolic pathway of omeprazole, proguanil, diazepam, and citalopram. The impact of the polymorphism in CYP2C19 on treatment outcome is clearly illustrated by omeprazole. The efficacy of omeprazole (20 mg in combination with amoxicillin) in eradicating *Helicobacter pylori* is markedly reduced in persons with the homozygous [EM](#) genotype (29% cured) as compared with 100% cure in those with homozygous [PM](#) genotype. This reflects the relative lack of effect of the recommended dose of omeprazole (20 mg) on gastric acid secretion and ulcer healing in patients with the CYP2C19 EM genotype. Certainly knowledge of a patient's CYP2C19 genotype would improve therapy with this proton pump inhibitor. Impaired hydroxylation of mephenytoin is present in only 3 to 5 percent of Caucasians, but the incidence is about 20 percent in individuals of Japanese and Chinese descent.

CYP2C9 catalyzes the major pathways of metabolism of warfarin and phenytoin. There are allelic variants of the gene for this enzyme that encode proteins with loss of catalytic function. These variant alleles are associated with requirement for a very low dose of warfarin, difficulties in initiating warfarin therapy, and an increased risk of bleeding complications. Similarly, high concentrations of phenytoin in plasma and resulting adverse effects of phenytoin occur in patients with loss of function alleles for CYP2C9.

Polymorphisms in drug-metabolizing ability may be associated with large differences in the disposition of a drug among individuals, especially when the involved pathway makes a major contribution to the elimination of the drug. For example, the clearance of mephenytoin given orally differs 100- to 200-fold between individuals of the [EM](#) and [PM](#) phenotypes. As a result, the peak plasma concentrations and bioavailability after oral administration are much higher, and the rate of drug elimination much lower, in PM than in EM individuals. In PM individuals, the result is excessive drug accumulation and exaggerated pharmacologic responses, including toxicity, when usual drug dosages are administered. Individualization of drug therapy is especially critical for drugs that exhibit polymorphic drug metabolism. The increasing availability of laboratory methods to identify the PM phenotype for NAT-2, CYP2D6, and CYP2C19 by genotyping should be useful for this purpose.

INTERINDIVIDUAL VARIABILITY IN THE MOLECULAR TARGETS WITH WHICH DRUGS INTERACT

The increasing emphasis on identifying molecular mechanisms of disease ([Chap. 65](#)) has important consequences for further understanding a genetic basis for individual variability in drug actions. As molecular approaches identify the role of specific gene products in human physiology, polymorphisms that alter expression or function of those gene products are being recognized; it is estimated that such polymorphisms occur in 1 in 1000 bp in the human genome. These genes in turn, are now being recognized as the molecular targets with which available and new drugs interact to produce beneficial and adverse effects.

Genome-wide searches in families with premature Alzheimer's disease identified the *APOE* locus as linked to the disease ([Chap. 362](#)). Specifically, the *E4* allele of the *APOE* gene appears associated with a worse prognosis, and this is thought to relate to reduced expression of choline acetyl transferase. Further, a therapeutic response to the choline acetyl transferase inhibitor, tacrine, appears to be more common with the prognostically more benign *APOE2* or *APOE3* alleles. Multiple polymorphisms identified in the β_2 -adrenergic receptor appear to be linked to specific phenotypes in asthma and congestive heart failure, diseases in which β_2 -receptor function might be expected to determine prognosis. It has been suggested that polymorphisms in the β_2 -receptor may be a determinant of response to inhaled β_2 -receptor agonists.

The development of marked QT prolongation and the polymorphic ventricular tachycardia, *torsade de pointes* ([Chap. 230](#)), in response to certain action potential-prolonging drugs such as quinidine used to be characterized as an "idiosyncratic" response. Advances in understanding the molecular basis of normal cardiac repolarization have resulted in identification of genes encoding ion channel proteins, the molecules whose normal function results in physiologic cardiac repolarization. Mutations in these genes cause congenital arrhythmia syndromes, such as the long QT syndrome, and block of ion channels is a common mechanism whereby drugs prolong QT intervals. Patients with mutations in these genes that remain subclinical until challenge with drugs are now recognized. In summary, continuing efforts to unravel the molecular basis of disease are likely also to provide insights into determinants of the response to drug therapy.

DRUG USE IN THE ELDERLY (See also [Chap. 9](#))

Aging results in changes in organ function, especially of the organs involved in drug disposition, as well as alterations in body size and composition. Not surprisingly, therefore, pharmacokinetics are often different in elderly individuals than in younger adults. Also, elderly patients often have multiple diseases and may therefore be taking a large number of drugs. Consequently, drug interactions, as well as an increased vulnerability to morbidity and mortality, contribute to the higher incidence of adverse drug reactions in elderly patients. Increased sensitivity of target organs and impairment of physiologic control systems, such as those involved in the regulation of the circulation, may also be a factor. Accordingly, optimization of drug therapy in the elderly, particularly in frail patients, is often difficult, as a variety of factors (often poorly defined) accentuate the usual interindividual variability in drug response.

Although many individuals preserve good renal function into old age, elderly patients as a group have an increased likelihood of impaired renal excretion of drugs. Even in the absence of kidney disease, renal clearance is generally reduced by about 35 to 50% in elderly patients. Dosage adjustments analogous to those in patients with renal dysfunction (see above) are therefore necessary for drugs that are eliminated mainly by the kidneys, such as digoxin, aminoglycosides, lithium, and other drugs listed in [Table 70-3](#). In this regard, it is important to recognize that the reduced muscle mass of older individuals results in a reduced rate of creatinine production; thus, a normal serum creatinine concentration can be present even though creatinine clearance is impaired.

Aging also results in a decrease in the size of and blood flow to the liver and possibly in the activity of hepatic drug-metabolizing enzymes; accordingly, the hepatic clearance of some drugs is impaired in the elderly. Unfortunately, no consistent pattern of clinical application appears to be present. Moreover, the changes are often modest relative to other causes for interindividual variability in these patients. However, even a small reduction in hepatic extraction may significantly increase the oral bioavailability of drugs with a high first-pass effect, such as propranolol and labetalol.

Impaired clearance and/or increased distribution may cause the elimination half-life of a drug to increase with aging. Thus, if a dosage modification in an elderly patient is required, it is often possible to accomplish it by decreasing the frequency of drug administration, possibly along with a reduction in dose.

Even if the pharmacokinetics of a drug are not altered, an elderly patient may require a smaller dosage because of an increase in pharmacodynamic sensitivity. Examples include increased analgesic effects of opioids, increased sedation from benzodiazepines and other [CNS](#) depressants, and increased risk of bleeding while receiving anticoagulant therapy, even when clotting parameters are well controlled. Exaggerated responses to cardiovascular drugs are also common because of the impaired responsiveness of normal homeostatic mechanisms. Such age-related changes require close monitoring of the patient's clinical response and appropriate dosage titration. Accordingly, in the elderly, initial doses should be less than the usual adult dosage and should be increased slowly. The final therapeutic regimen should be as simple as possible, and the number of different drugs used should be kept as low as possible. Also, because interindividual variability in drug responsiveness is greater in geriatric patients than in younger adults, individualization of therapy is even more critical.

INTERACTIONS BETWEEN DRUGS

The effect of some drugs can be altered markedly by the administration of other agents. Such interactions can complicate therapy by adversely increasing or decreasing the action of a drug. Drug interactions must be considered in the differential diagnosis of unexpected responses to drugs, and it should be recognized that patients often come to the physician with a legacy of drugs acquired during previous medical experiences. A meticulous drug history will minimize such unknown elements. It should include examination of the patient's medications and, if necessary, calls to the pharmacist to identify prescriptions. It should also address the use agents not often volunteered on

initial questioning, such as over-the-counter drugs, health food supplements, and topical agents such as eye drops.

There are two principal types of interactions between drugs. In *pharmacokinetic interactions*, the delivery of a drug to its site of action is altered, whereas in *pharmacodynamic interactions*, the responsiveness of the target organ or system is modified.

An index of the drug interactions discussed in this chapter is provided in [Table 70-4](#). The table includes interactions that have verified significance in patients, plus a few that are so potentially dangerous that cognizance should be taken of the experimental data or case reports suggesting they occur.

I. PHARMACOKINETIC INTERACTIONS CAUSING DIMINISHED DRUG DELIVERY

A. Impaired Gastrointestinal Absorption Examples include aluminum ions, present in antacids, which form insoluble chelates with the tetracyclines, preventing absorption of these drugs. Ferrous ions similarly block tetracycline absorption. Kaolin-pectin suspensions bind digoxin, and when these substances are administered together, digoxin absorption is reduced by about one-half. However, when kaolin-pectin is administered 2 h after digoxin, digoxin absorption is unaffected.

Ketoconazole is a weak base that dissolves well only at acidic pH. Histamine H₂receptor antagonists, such as ranitidine and cimetidine, reduce gastric acidity and thus impair the dissolution and absorption of ketoconazole. By contrast, the absorption of fluconazole is not impaired by an increase in gastric pH.

B. Induction of Hepatic Drug-Metabolizing Enzymes When a drug is eliminated largely by metabolism, an increase in the rate of its metabolism reduces its availability to sites of action. Most drugs are metabolized largely in the liver because of this organ's large mass, high blood flow, and high concentration of drug-metabolizing enzymes. The first step in the metabolism of many drugs is catalyzed by a group of cytochrome P450 mixed-function oxidases located in the endoplasmic reticulum (see "Metabolism by Cytochrome P450 Monooxygenases," above). These enzyme systems oxidize drug molecules by a variety of reactions, including aromatic hydroxylations, *N*-demethylations, *O*-demethylations, and sulfoxidations. The products of these reactions are usually more polar than the parent compound (and more readily excreted by the kidney).

The expression of some of the mixed-function oxidase (CYP) isoforms is regulated, and their content in the liver can be increased, by a number of drugs. Phenobarbital is the prototype of these inducers, and all the barbiturates in clinical use increase CYP enzyme activity. Induction with phenobarbital can occur with doses of as little as 60 mg daily. Mixed-function oxidases are also induced by rifampin, carbamazepine, phenytoin, and glutethimide and by smoking, exposure to chlorinated insecticides such as DDT, and chronic alcohol ingestion.

Phenobarbital, rifampin, and other inducers lower plasma levels of many drugs, including warfarin, quinidine, mexiletine, verapamil, ketoconazole, itraconazole,

cyclosporine, dexamethasone, methylprednisolone, prednisolone (the active metabolite of prednisone), oral contraceptive steroids, methadone, metronidazole, and metyrapone. These interactions all have obvious clinical significance. In the case of the coumarin anticoagulants, the patient is placed at major risk if the appropriate level of anticoagulation is achieved when an inducer is also being administered and the inducer is later discontinued (for example, at discharge from the hospital). The plasma levels of the coumarin anticoagulant will rise as the induction effect wears off, leading to excessive anticoagulation. There is considerable variation among individuals in the extent to which drug metabolism can be induced.

C. Inhibition of Cellular Uptake or Binding The guanidinium antihypertensive agents guanethidine and guanadrel are transported to their site of action in adrenergic neurons by an energy-requiring membrane transport system for biogenic monoamines; the physiologic function of this system is reuptake of the adrenergic neurotransmitter. Inhibitors of norepinephrine uptake prevent the uptake of the guanidinium antihypertensive agents into adrenergic neurons and thereby block their pharmacologic effects. The tricyclic antidepressants are potent inhibitors of norepinephrine uptake. Consequently, concomitant administration of clinical doses of tricyclic antidepressants, including desipramine, protriptyline, nortriptyline, and amitriptyline, almost totally abolishes the antihypertensive effects of guanethidine and guanadrel. Although they are less potent inhibitors of norepinephrine uptake, doxepin and chlorpromazine produce dose-related antagonism of the action of the guanidinium antihypertensives.

The antihypertensive effect of clonidine is partially antagonized by tricyclic antidepressants. Clonidine lowers arterial pressure by reducing sympathetic outflow from the blood pressure-regulating centers in the hindbrain ([Chap. 246](#)). This central hypotensive action is antagonized by the tricyclic antidepressants.

II. PHARMACOKINETIC INTERACTIONS CAUSING INCREASED DRUG DELIVERY

A. Inhibition of Drug Metabolism If the active form of a drug is eliminated largely by biotransformation, inhibition of its metabolism leads to reduced clearance, prolonged half-life, and accumulation of the drug during maintenance therapy. Excessive accumulation due to inhibited metabolism can lead to adverse effects.

Cimetidine is a potent inhibitor of the oxidative metabolism of many drugs, including warfarin, quinidine, nifedipine, lidocaine, theophylline, and phenytoin. Adverse reactions, many of them severe, have resulted from the administration of these drugs in conjunction with cimetidine. Cimetidine is a more potent inhibitor of mixed-function oxidases than ranitidine, whereas ranitidine is more potent as a histamine H₂receptor antagonist. Famotidine and nizatidine are not known to produce clinically appreciable inhibition of drug metabolism.

Knowledge of the CYP isoforms that catalyze the main pathway of metabolism of a drug provides a basis for predicting and understanding drug interactions. For example, the CYP3A subfamily of isoforms catalyzes the metabolism of many drugs for which blockage of metabolism results in toxicity. Drugs that depend on CYP3A as a major route of metabolism include cyclosporine, quinidine, lovastatin, simvastatin, atorvastatin, nifedipine, lidocaine, cisapride, erythromycin, methylprednisolone, carbamazepine,

midazolam, and triazolam.

The antifungal agents ketoconazole and itraconazole are potent inhibitors of enzymes in the CYP3A family. When fluconazole levels are elevated as a result of higher doses and/or renal insufficiency, this drug can also inhibit CYP3A. The macrolide antibiotics erythromycin and clarithromycin inhibit CYP3A4 to a clinically significant extent, but azithromycin does not inhibit this enzyme. Some of the calcium antagonists, diltiazem, nicardipine, and verapamil can also inhibit CYP3A, as can some of its other substrates, such as cyclosporine.

Cyclosporine can cause serious toxicity when its metabolism is inhibited by erythromycin, ketoconazole, diltiazem, nicardipine, or verapamil. A serious complication of HMG-CoA reductase inhibitors is myopathy. Fortunately, this is infrequent except in the context of interactions of a subset of the HMG-CoA reductase inhibitors with other drugs, particularly those that inhibit CYP3A4. The disposition of lovastatin is reduced markedly by drugs that inhibit CYP3A4, causing increases in plasma levels by more than tenfold. As a consequence, lovastatin has produced severe myopathy with rhabdomyolysis when administered together with erythromycin or cyclosporine. Not all of the HMG-CoA reductase inhibitors are as dependent on CYP3A4 for disposition as is lovastatin. Blocking CYP3A4 causes moderate elevations of plasma levels of simvastatin and atorvastatin (increases of severalfold), whereas elevations of the levels of fluvastatin and cerivastatin are only slight. Pravastatin disposition and plasma levels are not altered by inhibitors of CYP3A4. Cisapride can cause polymorphic ventricular tachycardia (torsade de pointes) when its metabolism is blocked by inhibitors of CYP3A, such as ketoconazole, itraconazole, clarithromycin, and erythromycin.

Whenever an inhibitor of CYP3A4 is administered to a patient, the physician should be alert to the possibility of serious interactions with drugs that are metabolized by CYP3A.

The CYP2D6 isoform that catalyzes the polymorphic metabolism of debrisoquin is markedly inhibited by quinidine and is also blocked by a number of neuroleptic drugs, such as chlorpromazine and haloperidol, and by fluoxetine. The analgesic effect of codeine depends on its metabolism to morphine via CYP2D6 in individuals with the [EM](#) phenotype. Thus, quinidine reduces the analgesic efficacy of codeine in EMs. Since desipramine is cleared largely by metabolism via CYP2D6 in EMs, its levels are increased substantially by concurrent administration of quinidine, fluoxetine, or the neuroleptic drugs that inhibit CYP2D6.

Some drugs are inactivated by mechanisms other than the hepatic drug-metabolizing enzymes. Azathioprine is converted in the body to an active metabolite, 6-mercaptopurine, which in turn is oxidized by xanthine oxidase to 6-thiouric acid. When allopurinol, a potent inhibitor of xanthine oxidase, is administered concurrently with standard doses of azathioprine or 6-mercaptopurine, life-threatening toxicity (bone marrow suppression) can result.

Other drugs that inhibit biotransformation of pharmacologic compounds (with examples of drugs whose metabolism is blocked by the inhibitor listed in parenthesis) include:

- Amiodarone (warfarin, quinidine)

- Clofibrate (phenytoin, tolbutamide)
- Excessive ingestion of ethanol (warfarin)
- Isoniazid (phenytoin)
- Metronidazole, cotrimoxazole (warfarin)
- Phenylbutazone (warfarin, phenytoin, tolbutamide)

B. Inhibition of Drug Transport Specific molecules that transport drugs into and out of cells are increasingly recognized, and inhibition of their function can be a major cause of clinically important drug interactions. The best studied to date is P-glycoprotein, originally isolated from tumor cells displaying resistance to multiple, structurally unrelated anticancer agents. The mechanism underlying this "multidrug resistance" phenomenon is P-glycoprotein-mediated pumping of anticancer agents out of cells, thereby inhibiting their anticancer effects. P-glycoprotein is also expressed in normal tissues (the luminal aspect of intestinal and renal tubular cells, the canalicular aspect of hepatocytes, the capillary endothelium of the blood-brain barrier), where it is responsible for efflux of not only antineoplastics but also digoxin and HIV protease inhibitors. Quinidine inhibits P-glycoprotein function in vitro, and it now seems apparent that the widely recognized doubling of plasma digoxin when quinidine is coadministered reflects this action in vivo, particularly since the effects of quinidine (increased digoxin bioavailability and reduced renal and hepatic secretion) occur at the sites of P-glycoprotein expression. Many other drugs also elevate digoxin concentrations (e.g., amiodarone, verapamil, cyclosporine, itraconazole, and erythromycin), and a similar mechanism seems likely. Reduced [CNS](#) penetration of multiple HIV protease inhibitors (with the attendant risk of facilitating a sanctuary site for the virus) appears attributable to P-glycoprotein-mediated exclusion of the drug from the CNS.

A number of drugs are secreted by the renal tubular transport systems for organic anions. Inhibition of this tubular transport system can cause excessive accumulation of a drug. Phenylbutazone, probenecid, and salicylates competitively inhibit this transport system. Salicylate, for example, reduces the renal clearance of methotrexate, an interaction that may lead to methotrexate toxicity. Renal tubular secretion contributes substantially to the elimination of penicillin, which can be inhibited by probenecid.

Inhibition of the tubular cation transport system by cimetidine impedes the renal clearance of procainamide and its active metabolite *N*-acetylprocainamide.

III. PHARMACODYNAMIC AND OTHER INTERACTIONS BETWEEN DRUGS

Therapeutically useful interactions occur in which the effect of two drugs in combination is greater than the sum of their effects when used individually. Favorable drug combinations are described in specific therapeutic sections in this text, and this section focuses on interactions that create unwanted effects. Two drugs may act on separate components of a common process to yield effects greater than either has alone. For example, although small doses of aspirin (<1 g daily) do not alter the prothrombin time

appreciably in patients who are receiving warfarin therapy, aspirin nevertheless increases the risk of bleeding in these patients because it inhibits platelet aggregation. Thus the combination of impaired functions of platelets and the clotting system, while useful for some therapeutic purposes, also increases the potential for hemorrhagic complications in patients receiving warfarin therapy.

Nonsteroidal antiinflammatory drugs (NSAIDs) cause gastric and duodenal ulcers, and, in patients treated with warfarin, the risk of bleeding from a peptic ulcer is increased almost threefold by concomitant use of a NSAID. This clearly is a serious drug interaction.

Indomethacin, piroxicam, and probably other NSAIDs antagonize the antihypertensive effects of β -adrenergic receptor blockers, diuretics, ACE inhibitors, and other drugs. The resulting elevation in blood pressure ranges from trivial to severe. Aspirin and sulindac, however, do not elevate the blood pressure in treated hypertensive patients.

Polymorphic ventricular tachycardia (torsade de pointes) during quinidine administration occurs much more frequently in patients receiving diuretics, probably owing to potassium and/or magnesium depletion.

The administration of supplemental potassium leads to more frequent and more severe hyperkalemia when potassium elimination is reduced by concurrent treatment with ACE inhibitors, spironolactone, amiloride, or triamterene.

The pharmacologic effects of sildenafil result from inhibition of the phosphodiesterase type 5 isoform that inactivates cyclic GMP in the vasculature. Nitroglycerin and related nitrates produce vasodilation by elevating cyclic GMP. Thus, coadministration of these nitrates with sildenafil will cause profound and potentially catastrophic hypotension.

CONCENTRATION OF DRUGS IN PLASMA AS A GUIDE TO THERAPY

In many cases, the plasma concentration of a drug is measured as a guide in the individualization of therapy. Genetic variation in elimination rates, interactions with other drugs, disease-induced alterations in elimination and distribution, and other factors combine to yield a wide range of plasma levels in patients given the same dose. Furthermore, the problem of noncompliance with prescribed regimens during continuing therapy is an endemic and elusive cause of therapeutic failure (see below). Clinical indicators assist in the titration of some drugs into the desired range, but no chemical determination is a substitute for careful observations of the response to treatment. However, the therapeutic and adverse effects are not precisely quantifiable for all drugs, and, in complex clinical situations, estimates of the action of a drug may be misleading. For example, previously existing neurologic disease may obscure the neurologic consequences of intoxication with phenytoin. Because clearance, half-life, accumulation, and steady-state plasma levels are difficult to predict, the measurement of plasma levels is often useful as a guide to the optimal dose. This is particularly true when there is a narrow range between the plasma levels yielding therapeutic and adverse effects. For drugs having such characteristics -- e.g., digoxin, theophylline, lidocaine, aminoglycosides, cyclosporine, and anticonvulsants -- dose optimization should involve modification of the standard dose on the basis of the pharmacokinetic

principles described above. In certain instances, predictive nomograms and algorithms have been developed to facilitate the necessary modifications. However, the most flexible and accurate method for individualizing drug dosage appears to be a feedback approach using a small number of previously obtained plasma levels and Bayesian forecasting. In controlled studies, this type of computer-assisted dosing has been shown to improve patient care. However, the overall cost/benefit ratio of such methods in routine management still remains to be conclusively demonstrated.

For drugs with a narrow therapeutic window that exhibit first-order elimination, then, dosage adjustments may be made on the assumption that the average, maximum, and minimum steady-state concentrations are related linearly to the dosing rate. Accordingly, the dose may be adjusted on the basis of the ratio between the desired and measured concentrations:

For drugs that have dose-dependent kinetics (e.g., phenytoin and theophylline), plasma concentrations change disproportionately more than the alteration in the dosing rate. Not only should changes in dose be small to minimize the degree of unpredictability, but plasma concentration monitoring is also critical to ensure appropriate modification.

The variability among individual responses to given plasma levels must be recognized. This is illustrated by a hypothetical population concentration-response curve ([Fig. 70-4](#)) and its relationship to the therapeutic range or therapeutic window of desired plasma levels. The defined therapeutic window should include the levels at which the intended pharmacologic effect is achieved in most patients. However, a few persons, who are sensitive to the therapeutic effects, respond to lower levels, whereas others are refractory enough to require levels that may cause adverse effects. For example, a few patients with strong seizure foci require plasma levels of phenytoin exceeding 20 ug/mL to control seizures. Dosages to achieve this effect may be appropriate if tolerated.

As also illustrated in [Fig. 70-4](#), some patients are prone to adverse effects at levels that are tolerated by most of the population. Therefore, raising the plasma concentration of a drug to a level that has a high probability of being therapeutically effective may bring on unwanted actions in an occasional patient. [Table 70-2](#) presents for a number of drugs the plasma concentrations that are associated with adverse and therapeutic effects in most patients. Use of this information according to the guidelines discussed should permit more effective and safer therapy for those patients who are not "average."

EFFECTIVE PARTICIPATION OF THE PATIENT IN THERAPY

Measurement of the concentration of a drug in plasma is the most effective way to detect failure to take a drug. Such "noncompliance" is a frequent problem in the long-term treatment of diseases such as hypertension and epilepsy, occurring in 25% or more of patients in therapeutic environments in which no special effort is made to involve patients in the responsibility for their own health. Occasionally, noncompliance can be uncovered by sympathetic, nonincriminating questioning, but more often it is recognized only after determining that the concentration of drug in plasma is nil or is recurrently low. Because other factors can cause plasma levels to be lower than

expected, comparison with levels obtained during inpatient treatment may be required to confirm that noncompliance has occurred. Once the physician is certain of noncompliance, a nonaccusatory discussion of the problem with the patient may clarify the reason for the noncompliance and serve as a basis for more effective cooperation on the part of the patient. Many approaches have been tried to help patients exercise more responsibility for their own treatment, most based on better communication regarding the nature of the disease and the chances of success or failure of the treatment. The patient is given a chance to discuss problems associated with treatment. The process may be improved by the involvement of nurses and other paramedical personnel. Minimizing the complexity of the regimen is helpful in terms of both the number of drugs and the frequency of administration. Educating patients to assume the principal role in their own health care requires a blend of the art and science of medicine.

(Bibliography omitted in Palm version)

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71. ADVERSE REACTIONS TO DRUGS - *Alastair J. J. Wood*

The beneficial effects of drugs are coupled with the inescapable risk of untoward effects. The morbidity and mortality from these untoward effects often present diagnostic problems because they can involve every organ and system of the body and are frequently mistaken for signs of underlying disease. Major advances in the investigation, development, and regulation of drugs ensure in most instances that they are uniform, effective, and relatively safe and that their recognized hazards are publicized. However, prior to regulatory approval and marketing, new drugs are tested in relatively few patients who tend to be less sick and to have fewer concomitant diseases than those patients who subsequently receive the drug therapeutically. Because of the relatively small number of patients studied in clinical trials, and the selected nature of these patients, rare adverse effects may not be detected prior to a drug's approval, and physicians therefore need to be cautious in the prescription of new drugs and alert for the appearance of previously unrecognized adverse events.

The large number and variety of drugs available over the counter (OTC), herbal preparations, and by prescription, make it impossible for patient or physician to obtain or retain the knowledge necessary to use all drugs well. It is understandable, therefore, that many OTC drugs are used unwisely by the public and that restricted drugs may be prescribed incorrectly by physicians.

Most physicians use no more than 50 drug products in their practice, gaining familiarity with their effectiveness and safety. Most patients probably use only a limited number of [OTC](#) drugs. Nevertheless, many patients receive care and drug prescriptions from more than one physician, and in any 30-day period, many patients consume more than three different OTC drug products containing nine or more different chemical agents.

Some 25 to 50% of patients make errors in self-administration of prescribed medicines, and these errors can be responsible for adverse drug effects. Elderly patients are the group most likely to commit such errors, perhaps in part because they consume more medicines. One-third or more of patients also may not take their prescribed medications. Similarly, patients commit errors in taking [OTC](#) drugs by not reading or following the directions on the containers. Physicians must recognize that providing directions with prescriptions does not always guarantee compliance.

Every drug can produce untoward consequences, even when used according to standard or recommended methods of administration. When used incorrectly, the effectiveness may be reduced, and adverse reactions can be expected to occur more frequently. Also, the administration of several drugs concurrently may result in adverse drug interactions ([Chap. 70](#)). In the hospital, all drugs a patient is given should be under the control of a physician, and patient compliance is, in general, ensured. Errors may occur nevertheless -- the wrong drug or dose may be given, or the drug may be given to the wrong patient -- although improved drug distribution and administration systems have reduced this problem. On the other hand, there are no easy means for controlling how ambulatory patients take prescription or [OTC](#) drugs.

EPIDEMIOLOGY

Epidemiologic studies of adverse drug reactions have been helpful in evaluating the magnitude of the overall problem, in calculating the rate of reactions to individual drugs, and in characterizing some of the determinants of adverse drug effects.

Patients receive, on average, 10 different drugs during each hospitalization. The sicker the patient, the more drugs are given, and there is a corresponding increase in the likelihood of adverse drug reactions. When fewer than 6 different drugs are given to hospitalized patients, the probability of an adverse reaction is about 5%, but if more than 15 drugs are given, the probability is over 40%. Retrospective analyses of ambulatory patients have revealed adverse drug effects in 20%.

Thus, the magnitude of drug-induced disease is large. Of patients admitted to the medical and pediatric services of general hospitals, 2 to 5% are admitted because of illnesses attributed to drugs. The case/ fatality ratio from drug-induced disease in hospitalized patients varies from 2 to 12%. Furthermore, some fetal or neonatal abnormalities are due to medicines taken by the mother during pregnancy or parturition.

A small group of widely used drugs accounts for a disproportionate number of reactions. Aspirin and other nonsteroidal anti-inflammatory drugs, analgesics, digoxin, anticoagulants, diuretics, antimicrobials, glucocorticoids, antineoplastics, and hypoglycemic agents account for 90% of reactions, although the drugs involved differ between ambulatory and hospitalized patients. Estimates of the cost of drug-related morbidity and mortality in the ambulatory setting range from \$30 billion to \$130 billion.

ADVERSE DRUG REACTIONS IN THE ELDERLY (See also [Chap. 9](#))

The elderly as a group have a greater burden of disease and receive a greater number of medications than other persons. Thus, it is not surprising that adverse drug reactions occur frequently in elderly patients. The issue of whether an elderly individual is more likely to develop an adverse drug reaction than a young person with a similar number of concurrent diseases and taking the same number of drugs has not been answered unequivocally. However, in population surveys of the noninstitutionalized elderly, as many as 10% report having had at least one adverse drug reaction in the last year. The incidence appears to be even greater in hospitalized elderly patients. Although it is widely believed that the elderly are more sensitive to drugs than the young, that is not true for all drugs. For example, a consistent decrease in sensitivity to drugs acting at the α -adrenergic receptor has been demonstrated in the elderly. The consequences of adverse drug effects may differ in the elderly because of their greater likelihood of other disease. For example, use of long-half-life benzodiazepines is linked to the occurrence of hip fractures in elderly patients, perhaps reflecting both a risk of falls from these drugs and the increased incidence of osteoporosis in elderly patients. Even when a drug impairs function similarly in patients of different age groups, the poorer baseline function in elderly persons may put them at greater risk for an adverse drug reaction. When prescribing for an elderly patient, the possibility that hepatic or renal mechanisms of drug excretion may be impaired should be taken into account. Adverse drug effects in the elderly may be subtle and, as in all populations, the physician must be alert to the possibility that a patient's signs and symptoms reflect an adverse effect of medication.

ETIOLOGY

Most adverse drug reactions are preventable, and recent studies using a systems analysis approach suggest that the most common system failure associated with an adverse drug reaction is the failure to disseminate knowledge about drugs to individuals involved in prescribing and administering them. Most adverse reactions can be classified into two groups. The most frequent ones result from an exaggeration of a predicted pharmacologic action of the drug. Other adverse reactions ensue from toxic effects unrelated to the intended pharmacologic actions. The latter effects are often unpredictable, are frequently severe, and result from recognized as well as undiscovered mechanisms. Some mechanisms unrelated to the drug's primary pharmacologic activity may include direct cytotoxicity, initiation of abnormal immune responses, and perturbation of metabolic processes in individuals with genetic enzymatic defects. Further understanding of interindividual differences in the expression of the enzymes responsible for drug metabolism has contributed to the understanding of adverse drug reactions that previously were thought to be idiosyncratic (see below). Prior consideration of the factors known to modify drug action often make it possible to prevent adverse reactions of this type.

Genetic Variations in Drug Oxidation by Cytochromes There is considerable interindividual variability in drug metabolism, resulting in variability in drug concentrations ([Chap. 70](#)). The majority of drugs are oxidized by cytochrome P450s (CYP) in the liver and gut. Some of these enzymes exhibit genetic polymorphisms resulting in enzymes with absent or reduced drug metabolizing activity, which may result in concentration-dependent toxicity. Conversely, where toxicity or pharmacologic effect is produced by a metabolite, individuals with low enzyme activity may have reduced drug effect whereas those with genetically determined increased enzyme activity will have increased drug effect. Examples of such polymorphically distributed oxidation enzymes include CYP2D6, CYP2C9, and CYP2C19.

The clinical consequences of the poor metabolizer phenotype are now becoming clearer and depend on the specific consequences of excessive drug concentrations. For example, the more potent (S) isomer of warfarin is metabolized by the polymorphically distributed enzyme [CYP2C9](#), resulting in lower (S) warfarin clearance and higher concentrations in both heterozygotes and homozygotes for the allelic variants associated with reduced enzyme activity. Recently, the clinical consequences of this polymorphism have been demonstrated in patients followed in an anticoagulant clinic. Patients who were stabilized on warfarin doses of 1.5 mg/d or less had an increased frequency of the genotypes associated with low warfarin metabolism compared to either community controls or patients requiring higher doses of warfarin ([Table 71-1](#)). In addition, the group stabilized on low-dose warfarin had a greater incidence of initial over-anticoagulation and hemorrhage than the group on the higher dose. This serves as an example of how genetic variations of a cytochrome P450 enzyme alter the response to a drug.

The oral hypoglycemic glipizide is also metabolized by [CYP2C9](#), and excessively low blood glucose concentrations occur after usual doses in genetic CYP2C9 poor metabolizers. Another oral hypoglycemic, phenformin, produced lactic acidosis in some patients. It is now recognized that phenformin is metabolized by another polymorphically distributed oxidative enzyme, CYP2D6. Patients who have genetically determined low

activity of CYP2D6 may be at particular risk from phenformin-induced lactic acidosis.

Genotypically determined variability in drug toxicity may also occur with drugs metabolized by enzymes other than cytochrome P450s. Such toxicity can be severe with the clinical use of the antimetabolites azathioprine and 6-mercaptopurine (to which it is converted *in vivo*). The cytotoxic thioguanine nucleotides produced *in vivo* are detoxified by further metabolism by xanthine oxidase and thiopurine methyltransferase (TPMT). The latter enzyme shows a trimodal distribution ([Fig. 71-1](#)). In children receiving mercaptopurine for treatment of leukemia, low activity of this enzyme is associated with excessive myelosuppression, whereas children with high TPMT levels have a poor antileukemic response. Azathioprine is currently used as a disease-modifying agent in the treatment of rheumatoid arthritis. Individuals with mutant TPMT alleles that result in impaired metabolism developed toxicity rapidly after beginning azathioprine and were uniformly unable to take the drug chronically. Thus the genotypic basis for drug toxicity is beginning to emerge.

Pharmacokinetic Bases for Adverse Reactions *An abnormally high drug concentration at the receptor site* (site of action) owing to pharmacokinetic variability is the usual cause of these reactions ([Chap. 70](#)). For example, a reduction in the volume of distribution, in the rate of metabolism, or in the rate of excretion all result in higher than expected concentration of drug at the receptor site, with a consequent increase in the pharmacologic effect.

Alteration in the dose-response curve due to increased receptor sensitivity results in an increase in drug effect at a given drug concentration. An example is the excessive response of elderly persons to the anticoagulant warfarin at normal or lower than normal blood levels. Such alterations in the dose-response curve may reflect altered drug sensitivity due to receptor polymorphisms, which are now being recognized. One such example is the prolonged QT syndrome, which has both a genetic basis, in individuals with abnormal potassium channels involved in cardiac repolarization, and a pharmacologic basis in individuals who receive drugs known to prolong the QT interval. Such individuals may develop torsade de pointes ([Chap. 230](#)). A large number of drugs have now been identified that can produce this potentially lethal effect ([Table 71-2](#); also <http://www.dml.georgetown.edu/depts/pharmacology/torsades.html>).

The shape of the dose-response curve also determines the likelihood of adverse drug reactions. Drugs with a steep dose-response curve or a narrow therapeutic index ([Chap. 70](#)) are more likely to cause dose-related toxicity because a small increase in dose produces a large change in pharmacologic effect. An increase in the dose of drugs that exhibit nonlinear kinetics, such as phenytoin ([Chap. 70](#)), may produce a proportionately greater increase in the blood level, resulting in toxicity.

Concurrent administration of other drugs may affect pharmacokinetics or pharmacodynamics. Pharmacokinetics may be affected by alterations in bioavailability, protein binding, or the rate of metabolism or excretion. Pharmacodynamics may be altered by another drug that competes for the same receptor sites, that prevents the drug from reaching its site of action, or that antagonizes or enhances the drug's pharmacologic effect. Inhibition of the metabolism of one drug by another may occur when both drugs bind to the same [CYP](#). Therefore, as the specific CYPs responsible for

the metabolism of individual drugs become known, prediction of drug interactions is put on a more rational scientific basis. An important example of such a mechanism is the inhibition of terfenadine's metabolism by inhibitors of CYP3A, such as erythromycin and systemic antimycotics. Such inhibition has resulted in torsade de pointes and lethal cardiac arrhythmia ([Chap. 70](#)).

TOXICITY UNRELATED TO A DRUG'S PRIMARY PHARMACOLOGIC ACTIVITY

Cytotoxic Reactions The understanding of so-called idiosyncratic reactions has greatly improved with the recognition that many of them are due to irreversible binding of a drug or its metabolites to tissue macromolecules by covalent bonds. Some chemical carcinogens, such as the alkylating agents, combine directly with DNA. Usually, it is only after metabolic activation of a drug to reactive metabolites that covalent binding occurs. This activation usually occurs in the microsomal mixed-function oxidase system, the hepatic enzyme system responsible for the metabolism of many drugs ([Chap. 70](#)). During the course of drug metabolism, reactive metabolites may covalently bind to tissue macromolecules, causing tissue damage. Because of the reactive nature of these metabolites, covalent binding often occurs close to the site of production. Typically that is the liver, but the mixed-function oxidase system is found in other tissues as well.

An example of this type of adverse drug reaction is the hepatotoxicity associated with isoniazid. This drug is metabolized principally by acetylation to acetylisoniazid, which is then hydrolyzed to acetylhydrazine. The further metabolism of acetylhydrazine by the mixed-function oxidase system liberates reactive metabolites that covalently bind to hepatic macromolecules, causing hepatic necrosis. The administration of drugs known to increase the activity of the mixed-function oxidase system, such as phenobarbital or rifampin, together with isoniazid results in the production of increased amounts of reactive metabolites, increased covalent binding, and a greater risk of hepatic damage.

The hepatic necrosis produced by overdosage of acetaminophen is also caused by reactive metabolites. Normally these metabolites are detoxified by combining with hepatic glutathione. When glutathione becomes exhausted, the metabolites bind instead to hepatic protein, with resultant hepatocyte damage. The hepatic necrosis produced by the ingestion of acetaminophen can be prevented, or at least attenuated, by the administration of substances such as *N*-acetylcysteine that reduce the binding of electrophilic metabolites to hepatic proteins. The risk of hepatic necrosis is increased in patients receiving drugs such as phenobarbital that increase the rate of drug metabolism and the rate of production of toxic metabolite(s).

It is likely, though as yet not proved, that other idiosyncratic reactions are caused by the covalent binding of reactive metabolites to tissue macromolecules, resulting either in direct cytotoxicity or in the initiation of an immune response.

Immunologic Mechanisms Most pharmacologic agents are poor immunogens because they are small molecules with molecular weights of less than 2000. Stimulation of antibody synthesis or sensitization of lymphocytes by a drug or one of its metabolites usually requires in vivo activation and covalent linkage to protein, carbohydrate, or nucleic acid.

Drug stimulation of antibody production may mediate tissue injury by one of several mechanisms. The antibody may attack the drug when the drug is covalently attached to a cell, and thereby destroy the cell. This mechanism occurs in penicillin-induced hemolytic anemia. Antibody-drug-antigen complexes may be passively adsorbed by a bystander cell, which is then destroyed by activation of complement; this occurs in quinine- and quinidine-induced thrombocytopenia. Drugs or their reactive metabolites may alter a host tissue, rendering it antigenic and eliciting autoantibodies. For example, hydralazine and procainamide can chemically alter nuclear material, stimulating the formation of anti-nuclear antibodies and occasionally causing lupus erythematosus. Autoantibodies can be elicited by drugs that neither interact with the host antigen nor have any chemical similarity to the host tissue; for example, α-methyldopa frequently stimulates the formation of antibodies to host erythrocytes, yet the drug neither attaches to the erythrocyte nor shares any chemical similarities with the antigenic determinants on the erythrocyte.

Drug-induced pure red cell aplasia ([Chap. 109](#)) is due to an immune-based drug reaction. Red cell formation in bone marrow cultures can be inhibited by phenytoin and purified IgG obtained from a patient with pure red cell aplasia associated with phenytoin.

Serum sickness ([Chap. 310](#)) results from the deposition of circulating drug-antibody complexes on endothelial surfaces. Complement activation occurs, chemotactic factors are generated locally, and an inflammatory response develops at the site of complex entrapment. Arthralgias, urticaria, lymphadenopathy, glomerulonephritis, or cerebritis may result. Penicillin is the most common cause of serum sickness today. Many drugs, particularly antimicrobial agents, induce production of IgE, which binds to mast cell membranes. Contact with a drug antigen initiates a series of biochemical events in the mast cell and results in the release of mediators that can produce the urticaria, wheezing, flushing, rhinorrhea, and (occasionally) hypotension characteristic of anaphylaxis.

Drugs may also excite cell-mediated immune responses. Topically administered substances may interact with sulfhydryl or amino groups in the skin and react with sensitized lymphocytes to produce the rash characteristic of contact dermatitis. Other types of rashes may also result from the interaction of serum factors, drugs, and sensitized lymphocytes. The role of drug-activated lymphocytes in the immune mechanisms governing destruction of visceral tissue is unknown.

Toxicity Associated with Genetically Determined Enzymatic Defects In the porphyrias, drugs that increase the activity of enzymes proximal to the deficient enzyme in the biosynthetic pathway of porphyrins can increase the quantity of porphyrin precursors that accumulate proximal to the deficient enzyme ([Chap. 346](#)). These drugs are listed in [Table 71-1](#).

Patients with a deficiency of glucose-6-phosphate dehydrogenase (G6PD) develop hemolytic anemia in response to primaquine and a number of other drugs ([Table 71-1](#)) that do not cause hemolysis in patients with adequate quantities of this enzyme ([Chap. 108](#)).

DIAGNOSIS

The manifestations of drug-induced diseases frequently resemble those of other diseases, and a given set of manifestations may be produced by different and dissimilar drugs. Recognition of the role of a drug or drugs in an illness depends on appreciation of the possible adverse reactions to drugs in any disease, on identification of the temporal relationship between drug administration and development of the illness, and on familiarity with the common manifestations of the drugs. Many associations between particular drugs and specific reactions have been described, but there is always a "first time" for a novel association, and any drug should be suspected of causing an adverse effect if the clinical setting is appropriate.

Illness related to a drug's pharmacologic action is often more easily recognized than illness attributable to immune or other mechanisms. For example, side effects such as cardiac arrhythmias in patients receiving digitalis, hypoglycemia in patients given insulin, and bleeding in patients receiving anticoagulants are more readily related to a specific drug than are symptoms such as fever or rash, which may be caused by many drugs or by other factors.

Once an adverse reaction is suspected, discontinuance of the suspected drug followed by disappearance of the reaction is presumptive evidence of a drug-induced illness. Confirming evidence may be sought by cautiously reintroducing the drug and seeing if the reaction reappears. However, that should be done only if confirmation would be useful in the future management of the patient and if the attempt would not entail undue risk. With concentration-dependent adverse reactions, lowering the dosage may cause the reaction to disappear, and raising it may cause the reaction to reappear. When the reaction is thought to be allergic, however, readministration of the drug may be hazardous, since anaphylaxis may develop. Readministration is unwise under these conditions unless no alternative drugs are available and treatment is necessary.

If the patient is receiving many drugs when an adverse reaction is suspected, the drugs likeliest to be responsible can usually be identified. All drugs may be discontinued at once, or, if that is not practical, they should be discontinued one at a time, starting with the one that is most suspect, and the patient observed for signs of improvement. The time needed for a concentration-dependent adverse effect to disappear depends on the time required for the concentration to fall below the range associated with the adverse effect, and that, in turn, depends on the initial blood level and on the rate of elimination or metabolism of the drug. Adverse effects of drugs with long half-lives, such as phenobarbital, take a considerable time to disappear.

Drugs recognized as producing a number of reactions are listed in [Table 71-1](#). This table includes both well-documented and some less well-documented reactions, focusing on those that are sufficiently important to require consideration. This information should be used to suggest the drug likely to be causing a reaction; the absence of a drug from the table does not mean that it cannot be responsible for the reaction, however.

Serum antibody has been demonstrated in some persons with drug allergies involving cellular blood elements, as in agranulocytosis, hemolytic anemia, and thrombocytopenia. For example, both quinine and quinidine can produce platelet agglutination in vitro in the presence of complement and the serum from a patient who

has developed thrombocytopenia following use of this drug.

Eliciting a drug history from patients is important for diagnosis. Attention must be directed to [OTC](#) drugs and herbal preparations as well as to prescription drugs. Each type can be responsible for adverse drug effects, and adverse interactions may occur between OTC drugs and prescribed drugs. In addition, it is common for patients to be cared for by several physicians, and duplicative, additive, counteractive, or synergistic drug combinations may therefore be administered if the physicians are not aware of the patients' drug histories. Every physician should determine what drugs a patient has been taking, at least during the preceding 30 days, before prescribing any medications. A frequently overlooked source of additional drug exposure is topical therapy; for example, a patient complaining of bronchospasm may not mention that an ophthalmic beta blocker is being used unless specifically asked. A history of previous adverse drug effects in patients is common. Since these patients have shown a predisposition to drug-induced illnesses, such a history should dictate added caution in prescribing drugs.

Patients with biochemical abnormalities such as erythrocyte [G6PD](#) deficiency can be identified. Most patients with the G6PD defect are of African or Mediterranean descent. Drug-induced hemolytic crisis can be avoided by testing for the enzyme defect before administering drugs that could cause the reaction. Similarly, persons with an abnormal serum pseudocholinesterase level may have abnormally prolonged apnea when given succinylcholine.

GENERAL COMMENTS

No drug is completely without side effects, and a side effect in one patient may be the desired pharmacologic effect in another. Current drug regulations allow physicians to have considerable confidence in the purity, bioavailability, and effectiveness of the drugs they prescribe. However, physicians have to weigh potential toxicity against possible benefits. Toxicity that would be acceptable for an effective antineoplastic agent would not be permitted in an oral contraceptive, for example. Because of the necessarily small number of patients treated in premarketing studies, rare adverse reactions may not be identified, so the first responsibility for identifying and reporting these effects must rest with the practicing clinician through the use of the various national adverse reaction reporting systems, such as those operated by the Food and Drug Administration in the United States and the Committee on Safety of Medicines in Great Britain. The publication of a newly recognized adverse reaction can in a short time stimulate many similar such reports of reactions that previously had gone unrecognized.

The prevention of adverse drug reactions first involves a high index of suspicion that the development of a new symptom or sign may be drug-related. Reduction of the dose or discontinuation of the suspected agent usually clarifies the issue in concentration-dependent toxic reactions. Physicians should be familiar with the common adverse effects of the drugs they use and, when in doubt, should consult the literature.

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72. PHYSIOLOGY AND PHARMACOLOGY OF THE AUTONOMIC NERVOUS SYSTEM - Lewis Landsberg, James B. Young

FUNCTIONAL ORGANIZATION OF THE AUTONOMIC NERVOUS SYSTEM

The autonomic nervous system innervates vascular and visceral smooth muscle, exocrine and endocrine glands, and parenchymal cells throughout the various organ systems. Functioning below the conscious level, the autonomic nervous system responds rapidly and continuously to perturbations that threaten the constancy of the internal environment. The many functions governed by this system include the distribution of blood flow and the maintenance of tissue perfusion, the regulation of blood pressure, the regulation of the volume and composition of the extracellular fluid, the expenditure of metabolic energy and supply of substrate, and the control of visceral smooth muscle and glands.

ANATOMIC ORGANIZATION

The autonomic neurons, located in ganglia outside the central nervous system (CNS), give rise to the postganglionic autonomic nerves that innervate organs and tissues throughout the body ([Fig. 72-1](#)). The activity of autonomic nerves is regulated by central neurons responsive to diverse afferent inputs. After central integration of afferent information, autonomic outflow is adjusted to permit the functioning of the major organ systems in accordance with the needs of the organism as a whole. Connections between the cerebral cortex and the autonomic centers in the brainstem coordinate autonomic outflow with higher mental functions.

The Sympathetic and Parasympathetic Divisions The preganglionic neurons of the parasympathetic nervous system leave the [CNS](#) in the third, seventh, ninth, and tenth cranial nerves and in the second and third sacral nerves, while the preganglionic neurons of the sympathetic nervous system exit the spinal cord between the first thoracic and the second lumbar segments ([Fig. 72-1](#)). Responses to sympathetic and parasympathetic stimulation are frequently antagonistic, as exemplified by their opposing effects on heart rate and gut motility. This antagonism reflects highly coordinated interactions within the CNS; the resultant changes in parasympathetic and sympathetic activity, often reciprocal, provide more precise control of autonomic responses than could be achieved by the modulation of a single system. Moreover, both sympathetic and parasympathetic portions of the autonomic nervous system are composed of multiple function-specific subdivisions. Neurons with the various subdivisions differ neurochemically and neurophysiologically and are controlled by distinct regions within the CNS. This specialization within sympathetic and parasympathetic divisions contributes to the precision and specificity of autonomic regulation.

Neurotransmitters *Acetylcholine* (ACh) is the preganglionic neurotransmitter for both divisions of the autonomic nervous system as well as the postganglionic neurotransmitter of the parasympathetic neurons. Nerves that release ACh are said to be cholinergic. *Norepinephrine* (NE) is the neurotransmitter of the postganglionic sympathetic neurons; these nerves are said to be adrenergic. Within the sympathetic outflow, postganglionic neurons innervating the eccrine sweat glands (and perhaps

some blood vessels supplying skeletal muscle) are of the cholinergic type.

THE SYMPATHETIC NERVOUS SYSTEM AND ADRENAL MEDULLA

CATECHOLAMINES

All three of the naturally occurring catecholamines, [NE](#), *epinephrine* (E), and *dopamine*, function as neurotransmitters within the [CNS](#). NE, the neurotransmitter of postganglionic sympathetic nerve endings, exerts its effects locally, in the immediate vicinity of its release. Epinephrine, the circulating hormone of the adrenal medulla, influences processes throughout the body. A peripheral dopaminergic system also exists but has not been characterized in detail.

Biosynthesis (Fig. 72-2) Catecholamines are synthesized from the amino acid tyrosine, which is sequentially hydroxylated to form dihydroxyphenylalanine (dopa), decarboxylated to form dopamine, and hydroxylated on the β position of the side chain to form [NE](#). The initial step, the hydroxylation of tyrosine, is rate-limiting and is regulated so that synthesis of dopa is coupled to NE release. This regulation is achieved by alterations in both the activity and the amount of tyrosine hydroxylase. In the adrenal medulla and in those central neurons utilizing epinephrine as neurotransmitter, NE is *N*-methylated to epinephrine by the enzyme phenylethanolamine-*N*-methyltransferase (PNMT).

Catecholamine Metabolism The major metabolic transformations of catecholamines involve *O*-methylation at the meta-hydroxyl group and oxidative deamination. *O*-Methylation is catalyzed by the enzyme catechol-*O*-methyltransferase (COMT), and oxidative deamination is promoted by monoamine oxidase (MAO). COMT in liver and kidney is important in the metabolism of circulating catecholamines. MAO, a mitochondrial enzyme present in most tissues, including nerve endings, has a lesser role in the metabolism of circulating catecholamines but is important in regulating the catecholamine stores within the peripheral sympathetic nerve endings. The metanephrines and 3-methoxy-4-hydroxymandelic acid (vanilmandelic acid, VMA) are the major end products of E and [NE](#) metabolism. Homovanillic acid (HVA) is the end product of dopamine metabolism.

STORAGE AND RELEASE OF CATECHOLAMINES

In both the adrenal medulla and sympathetic nerve endings catecholamines are stored in subcellular vesicles and released by exocytosis. The large stores of catecholamines in these tissues provide an important physiologic reserve that maintains an adequate supply of catecholamines in the face of intense stimulation. A variety of substances may be stored along with catecholamines in sympathetic nerve endings and adrenal medulla and released with catecholamines during exocytosis. These substances, which may function as cotransmitters or neuromodulators, include peptides such as neuropeptide Y, substance P, and enkephalins; purines such as ATP and adenosine; and other amines such as serotonin. At the neuroeffector junction, coreleased neuromodulators modify the response to [NE](#), while cotransmitters exert physiologic effects independent of those induced by NE.

Adrenal Medulla The adrenal medullary chromaffin tissue in a pair of normal human adrenal glands weighs about 1 g and contains approximately 6 mg catecholamines, 85% of which is epinephrine.

Catecholamine secretion, stimulated by [ACh](#) from the preganglionic sympathetic nerves, occurs after calcium influx triggers fusion of the chromaffin granule membrane and cell membrane; obliteration of the cell membrane at the point of fusion and extrusion of the entire soluble contents of the granule into the extracellular space complete the process of exocytosis ([Fig. 72-2](#)). Although the molecular mechanisms involved in the exocytotic process are only partially understood, evidence has accumulated that specific calcium-binding proteins are involved. Once bound, calcium induces a conformational change in these proteins that induces fusion of granules and docking of granules at the cell membrane.

Peripheral Sympathetic Nerve Endings The peripheral sympathetic nerve endings form a reticulum or ground plexus that brings the terminal fibers into close contact with effector cells. All the [NE](#) in peripheral tissues is in the sympathetic nerve endings, and heavily innervated tissues contain as much as 1 to 2 ug/g of tissue. NE stored in the nerve endings is in discrete subcellular particles analogous to the adrenal medullary chromaffin granules. [MAO](#) in the mitochondria of the nerve endings plays an important role in regulating the local concentration of NE ([Fig. 72-2](#)). Amines in storage vesicles are protected from oxidative deamination; amines within the cytoplasm, however, are deaminated to inactive metabolites. Release from the nerve ending occurs in response to action potentials propagated in terminal sympathetic fibers.

THE PERIPHERAL ADRENERGIC NEUROEFFECTOR JUNCTION

The peripheral sympathetic nerve endings possess an amine transport system that actively takes up amines from the extracellular fluid. Neuronal uptake or recapture of locally released [NE](#) terminates the action of the transmitter and contributes to the constancy of the NE stores.

A variety of factors alter the relationship between neuronal impulse traffic and [NE](#) release. Diminished temperature and acidosis, for example, both decrease the amount of NE released in response to sympathetic impulses. Several chemical mediators operate at the peripheral sympathetic nerve ending (referred to as *prejunctional* or *presynaptic sites*) to modify sympathetic neurotransmission by influencing the amount of NE released in response to nerve impulses. Prejunctional modulation may be either inhibitory or facilitatory. Certain modulators, such as catecholamines and [ACh](#), may either inhibit or facilitate NE release, antagonistic effects that are mediated by different adrenergic or cholinergic receptors, respectively. Those compounds exerting an *inhibitory* effect on NE release at the prejunctional nerve ending include the following: catecholamines (α_2 receptor), ACh (muscarinic receptor), dopamine (D_2 receptor), histamine (H_2 receptor), serotonin, adenosine, enkephalins, and prostaglandins.

Catecholamines reduce [NE](#) release via prejunctional α_2 receptors in a classic negative-feedback system. Feedback regulation is complicated by the fact that β -receptor activation facilitates NE release.

Though both inhibitory and facilitatory effects of [ACh](#) on [NE](#) release have been described, the inhibitory effect of ACh, mediated by the muscarinic cholinergic receptor, occurs at lower ACh concentrations and is probably of greater physiologic significance.

CENTRAL REGULATION OF SYMPATHOADRENAL OUTFLOW

Brainstem Sympathetic Centers Sympathetic outflow is initiated from the reticular formation of the medulla oblongata and pons and from centers in the hypothalamus. The rostral ventral portion of the medulla, particularly the area designated the rostral ventrolateral medulla (RVLM), appears to contain especially important sympathoexcitatory areas. Descending fibers originating from all these centers synapse in the intermediolateral cell column of the spinal cord with the preganglionic sympathetic neurons. Changes in the physical and chemical properties of the extracellular fluid, including the circulating levels of hormones and substrates, also affect sympathetic nervous system outflow. The area postrema, in the floor of the fourth ventricle, along with other circumventricular organs lie outside the blood-brain barrier and may play an important role in this regard. Although the hallmark of intense sympathoadrenal stimulation is a global response (the fight-or-flight reaction of Cannon), discrete changes in sympathetic outflow to different organ systems continuously regulate many autonomic functions.

Relationship Between the Sympathetic Nervous System and the Adrenal Medulla

Sympathetic nervous system activity and adrenal medullary secretion are coordinated but not always congruent. During periods of intense sympathetic stimulation, such as cold exposure and exhaustive exercise, the adrenal medulla is progressively recruited, and circulating epinephrine reinforces the physiologic effects of sympathetic stimulation. In other situations, the sympathetic nervous system and the adrenal medulla are stimulated independently. The response to upright posture, for example, involves predominantly the sympathetic nervous system, while hypoglycemia stimulates only the adrenal medulla.

Sympathetic Regulation of the Cardiovascular System Stretch receptors in the systemic and pulmonary arteries and veins continuously monitor intravascular pressures; the resulting afferent impulses, after relay and integration in the brainstem, alter sympathetic activity in defense of blood pressure and blood flow to critical areas ([Fig. 72-3](#)).

Arterial Baroreceptors An increase in blood pressure stimulates receptors in the carotid sinus and aortic arch. The ensuing afferent impulses, after relay within the nucleus of the solitary tract (NTS) in the brainstem, suppress the brainstem sympathetic centers ([Fig. 72-3](#)). This baroreceptor reflex arc forms a negative-feedback loop in which a rise in arterial pressure results in the inhibition of central sympathetic outflow. A brainstem noradrenergic pathway interacts with the NTS to participate in suppression of sympathetic outflow. This noradrenergic inhibitory pathway is stimulated by centrally acting adrenergic agonists and may be involved in the action of certain antihypertensive drugs, such as clonidine, that potentiate the baroreceptor-mediated vasodepressor response ([Chap. 246](#)). In the opposite manner, when the blood pressure falls, decreased afferent impulses diminish central inhibition, resulting in an increase in

sympathetic outflow and a rise in arterial pressure.

Central Venous Pressure Receptors in the walls of the great veins and within the atria are also involved in the regulation of sympathetic outflow. Stimulation of these receptors by high venous pressure suppresses the brainstem sympathetic centers; when central venous pressure is low, sympathetic outflow increases. The central connections are poorly understood, but the afferent impulses are carried in the vagus ([Fig. 72-3](#)).

ASSESSMENT OF SYMPATHOADRENAL ACTIVITY

The clinical assessment of sympathoadrenal activity involves the measurement of catecholamines in plasma and of catecholamines and catecholamine metabolites in urine, and the assessment of sympathetic nerve impulse traffic by microneurography. Microneurography, utilizing microelectrodes implanted in nerves supplying skeletal muscle (such as the peroneal nerve), is primarily a research tool. Quantitation of urinary catecholamines and metabolites is useful in the diagnosis of pheochromocytoma ([Chap. 332](#)).

Plasma Catecholamines Catecholamines in human plasma may be measured by radioenzymatic isotope derivative techniques or by high-performance liquid chromatography in conjunction with electrochemical detection. Plasma catecholamine measurements provide an index of sympathetic nervous system and adrenal medullary activity and have been widely used to assess sympathoadrenal activity in clinical investigation in human subjects. The usefulness of plasma catecholamine measurements, however, is compromised by factors that alter the relationship between the plasma concentration of catecholamines and the functional state of the sympathoadrenal system, and also by important regional differences in sympathetic outflow. Techniques utilizing tracer infusions of tritiated [NE](#), which correct for changes in NE clearance when applied across a particular anatomic region, estimate regional sympathetic outflow with some precision and have helped to define differentiated sympathetic nervous system activity in the investigational setting. The clinical usefulness of plasma catecholamine levels remains limited to the evaluation of patients with autonomic insufficiency and, on occasion, patients with suspected pheochromocytoma ([Chap. 332](#)).

Basal plasma [NE](#) concentrations are in the range of 0.09 to 1.8 nmol/L (150 to 350 pg/mL); basal E levels are about 135 to 270 pmol/L (25 to 50 pg/mL). The half-time of disappearance of NE from the circulation is approximately 2 min. The plasma NE level is markedly affected by a variety of factors, including posture; accordingly, the conditions under which blood is obtained for assay must be controlled. By convention, basal plasma NE levels are those obtained through an indwelling intravenous line after the patient has rested supine in a relaxed environment for 30 min.

Plasma NE response to upright posture The predictable increase in circulating NE concentration during upright posture provides a convenient test of sympathetic nervous system function. Five minutes of quiet standing results in a two- to threefold increase in plasma NE level. A normal response requires an intact afferent system, appropriate [CNS](#) relays, and an intact peripheral sympathetic nervous system; a defect of any of these components reduces the increment in circulating NE.

Plasma E levels are also dependent on the physical and mental state of the subject. Change in plasma E with upright posture is usually small. Hypoglycemia, strenuous exercise, and various types of mental stress, however, can cause large increases in the plasma E level.

PERIPHERAL DOPAMINERGIC SYSTEM

In addition to its role as neurotransmitter in the [CNS](#), dopamine functions as an inhibitory transmitter in the carotid body and the sympathetic ganglia. A distinct peripheral dopaminergic system is also believed to exist. Dopamine elicits a variety of responses not attributable to stimulation of classic adrenergic receptors; it relaxes the lower esophageal sphincter, delays gastric emptying, causes vasodilation in the renal and mesenteric arterial circulation, suppresses aldosterone secretion, directly stimulates renal sodium excretion, and suppresses [NE](#) release at sympathetic nerve terminals by a presynaptic inhibitory mechanism. The mediation of these dopaminergic effects in vivo is poorly understood. Dopamine does not appear to be a circulating hormone.

ADRENERGIC RECEPTORS

Catecholamines influence effector cells by interacting with specific surface *receptors* coupled to G proteins. Two major categories of response to catecholamines reflect the activation of two populations of adrenergic receptors, designated α and β . Both α and β receptors have been further divided into subtypes that serve different functions and are susceptible to differential stimulation and blockade.

α -ADRENERGIC RECEPTORS

α -Adrenergic receptors mediate vasoconstriction, intestinal relaxation, and pupillary dilatation. Epinephrine and [NE](#) are approximately equipotent as α -receptor agonists. Distinct α_1 - and α_2 -receptor subtypes are also recognized. Originally the postsynaptic or postjunctional α -adrenergic receptors on effector cells were designated α_1 , while the presynaptic α -adrenergic receptors on the sympathetic nerve endings were designated α_2 . It is now recognized that nonneuronal (postsynaptic) processes are mediated by the α_2 receptor as well. The α_1 receptor mediates the classic α effects, including vasoconstriction; phenylephrine and methoxamine are selective α_1 agonists, and prazosin is a selective α_1 antagonist. The α_2 receptor mediates presynaptic inhibition of NE release from adrenergic nerves and other responses, including inhibition of [ACh](#) release from cholinergic nerves, inhibition of lipolysis in adipocytes, inhibition of insulin secretion, stimulation of platelet aggregation, and vasoconstriction in some vascular beds. Specific α_2 agonists include clonidine and α -methylnorepinephrine; these agents, the latter derived from α -methyldopa in vivo, exert an antihypertensive effect by interacting with α_2 receptors within the brainstem sympathetic centers that regulate blood pressure. Yohimbine is a specific α_2 antagonist.

β -ADRENERGIC RECEPTORS

Physiologic events associated with β -adrenergic receptor responses include stimulation of heart rate and contractility, vasodilation, bronchodilation, and lipolysis. β -Receptor

responses can also be divided into two types. The β_1 receptor responds equally to E and NE and mediates cardiac stimulation and lipolysis. The β_2 receptor is more responsive to E than to NE and mediates responses such as vasodilation and bronchodilation. Isoproterenol stimulates and propranolol blocks both β_1 and β_2 receptors. Other agonists and antagonists that have partial selectivity for the β_1 or β_2 receptors have been used therapeutically where the desired response involves predominantly one of the two subtypes.

Both pharmacologic and molecular genetic studies have demonstrated an additional distinct β_3 -adrenergic receptor that subserves lipolysis in white and brown adipose tissue as well as the stimulation of heat production in brown adipose tissue. The human β_3 -adrenergic receptor has been cloned, and a distinct polymorphism noted that may, in some populations, be associated with weight gain, insulin resistance, and type 2 diabetes mellitus. The β_3 -adrenergic receptor has a much greater affinity for NE than E and, unlike the β_1 and β_2 receptors, does not undergo desensitization. Synthetic agonists for the β_3 receptor, currently under development, have a potential role in the treatment of obesity by increasing metabolic rate.

DOPAMINERGIC RECEPTORS

Specific dopaminergic receptors, distinct from the classic α - and β -adrenergic receptors, are present in the CNS and peripheral nervous system and in several nonneural tissues. Two types of dopaminergic receptors serve different functions and have different second messengers. Dopamine is a potent agonist of both types of receptors; the action of dopamine is antagonized by phenothiazines and thioxanthenes. The D_1 receptor mediates vasodilation in the renal, mesenteric, coronary, and cerebral vascular beds. Fenoldopam is an agonist selective for the D_1 receptor. The D_2 receptor inhibits transmission in the sympathetic ganglia, inhibits NE release from sympathetic nerve endings by an effect on the presynaptic membrane (Fig. 72-2), inhibits prolactin release from the pituitary, and causes vomiting. Selective agonists of the D_2 receptor include bromocriptine, cabergoline, and apomorphine, while butyrophenones such as haloperidol (active within the CNS), domperidone (does not cross blood-brain barrier readily), and the benzamide sulpiride are relatively selective D_2 antagonists.

STRUCTURE AND FUNCTION OF ADRENERGIC RECEPTORS

Adrenergic receptors belong to a superfamily of related membrane proteins, including the visual protein rhodopsin and the muscarinic acetylcholine receptors, that interacts with G proteins. These proteins share significant sequence homologies and, as deduced from the properties of the constituent amino acids, a similar topographic structure in the cell membrane. The postulated structure of this family of receptor proteins is shown schematically in Fig. 72-4. The characteristic features include seven membrane-spanning hydrophobic domains containing 20 to 28 amino acids each. The membrane-spanning domains, particularly M-7 (Fig. 72-4), appear to be important in determining the characteristic agonist binding.

Coupling of Receptor Occupancy with Cellular Response The major mediators of adrenergic (as well as many other) cellular responses are a family of regulatory proteins termed *G proteins* that, when activated, bind the nucleotide guanosine triphosphate

(GTP). The best-characterized G proteins are those that stimulate or inhibit adenylyl cyclase, designated G_s or G_i , respectively (Fig. 72-5). The β_1 , β_2 , and D_1 receptors are coupled to G_s ; receptor occupancy is therefore associated with stimulation of adenylyl cyclase and results in an increase in intracellular cyclic adenosine monophosphate (AMP), which in turn results in activation of protein kinase A and other cyclic AMP-dependent protein kinases. The resultant protein phosphorylation alters the activity of enzymes and the function of other proteins, culminating in a cellular response that is characteristic of the tissue being stimulated. The α_2 , M_2 subtype of the muscarinic acetylcholine receptor and the D_2 receptor are coupled to G_i , resulting in diminished adenylyl cyclase activity and a fall in cyclic AMP. The subsequent alterations in enzyme activity and function of other proteins produce an alternate, frequently opposite, series of cellular responses. Although many α_2 responses can be explained by inhibition of adenylyl cyclase, other mechanisms may be involved as well.

The α_1 -adrenergic receptor (as well as the M_1 subtype of the acetylcholine receptor) is coupled to a different G protein that activates phospholipase C; this G protein has not been as well characterized but is sometimes designated G_q . Receptor occupancy in this system stimulates phospholipase C, which catalyzes the breakdown of membrane-bound phospholipids, particularly phosphatidylinositol-4,5-bisphosphate (PIP_2) with the production of inositol-1,4,5-trisphosphate (IP_3) and 1,2-diacylglycerol (DAG), both of which act as second messengers (Fig. 72-5). IP_3 rapidly mobilizes calcium from intracellular stores within the endoplasmic reticulum, producing an increase in free cytoplasmic calcium which by itself and via calcium-calmodulin-dependent protein kinases influences cellular processes appropriate to the stimulated cell. The transient rise in calcium induced by IP_3 from the intracellular stores is reinforced in the presence of continued agonist stimulation by alterations in membrane calcium flux that result eventually in net calcium uptake from the extracellular fluid by mechanisms that have been incompletely defined.

DAG, the other second messenger produced by the action of phospholipase C on PIP_2 (as well as other membrane phospholipids), remains associated with the cell membrane and activates protein kinase C, which has different substrates than the calcium-calmodulin kinases stimulated by IP_3 . Protein phosphorylation stimulated by protein kinase C contributes to the tissue-specific response in ways that remain poorly understood. Increases in intracellular calcium also potentiate the activation of protein kinase C (Fig. 72-5).

REGULATION OF ADRENERGIC RECEPTORS

Prolonged exposure to α - or β -adrenergic agonists decreases the number of corresponding adrenergic receptors on effector cells. Although the biochemical mechanisms involved are obscure, internalization of the β -adrenergic receptor within the cell occurs during agonist exposure in some systems, suggesting that internal translocation contributes to the decrease in receptor number under these circumstances.

Alteration in agonist concentration may also affect the affinity of the receptor for the agonist. Adrenergic receptors that utilize adenylyl cyclase for the second messenger (β receptors, α_2 receptors) exist in high- and low-affinity states; exposure to agonist

diminishes the proportion of receptors in the high-affinity state. Such alterations in adrenergic receptors induced by adrenergic agonists are termed *homologous regulation*. Agonist-induced alterations in adrenergic-receptor density and affinity are believed to contribute to the diminished physiologic response that occurs after prolonged exposure of an effector tissue to adrenergic agonist, a phenomenon known as *tachyphylaxis* or *desensitization*.

Adrenergic receptors are also influenced by factors other than adrenergic agonists, so-called *heterologous regulation*. Enhanced α -adrenergic-receptor affinity, for example, may underlie the potentiation of α -adrenergic responses that occur in response to lowered environmental temperatures. Thyroid hormones potentiate β -receptor responses by alterations in β -receptor number and in the efficiency of coupling receptor occupancy with physiologic response. Estrogen and progesterone alter the sensitivity of the myometrium to catecholamines by effects on α -adrenergic receptors. Glucocorticoids may influence adrenergic function by antagonizing agonist-induced decreases in adrenergic receptors, thereby counteracting tachyphylaxis in response to intense adrenergic stimulation.

Alterations in sensitivity to catecholamines also occur as a consequence of postreceptor changes, although the latter remain poorly characterized.

PHYSIOLOGY OF THE SYMPATHOADRENAL SYSTEM

Catecholamines influence all of the major organ systems. The effects take place in seconds and may occur in anticipation of physiologic requirement. An increase in sympathoadrenal activity prior to strenuous exercise, for example, lessens the impact of exercise on the internal environment.

DIRECT EFFECTS OF CATECHOLAMINES

Cardiovascular System Catecholamines stimulate vasoconstriction in the subcutaneous, mucosal, splanchnic, and renal vascular beds by α -receptor-mediated mechanisms. Although vasoconstriction was originally considered an α_1 -receptor response, vascular tone appears to be more complexly regulated and, in many areas, involves α_2 -mediated responses as well. The venous portion of the circulation, in particular, is endowed with α_2 receptors. Differential regulation of the two types of α receptors, under certain circumstances, contributes to an integrated physiologic response. Since vasoconstriction in the coronary and cerebral circulations is minimal, flow to these areas is maintained during sympathetic stimulation. Skeletal muscle vasculature contains β receptors sensitive to low circulating levels of epinephrine so that skeletal muscle blood flow is augmented during adrenal medullary activation.

The effects of catecholamines on the heart are mediated by β_1 receptors and include increase in heart rate, enhancement of cardiac contractility, and increase in conduction velocity. The increase in myocardial contractility is illustrated by a leftward and upward shift of the ventricular function curve (see [Fig. 231-6](#)) that relates cardiac work to ventricular diastolic fiber length; at any initial fiber length, catecholamines increase cardiac work. Catecholamines also enhance cardiac output by stimulating venoconstriction, enhancing venous return, and increasing the force of atrial contraction,

thereby augmenting diastolic volume and hence fiber length. The acceleration of conduction in the junctional tissues results in a more synchronous, and hence more effective, ventricular contraction. Cardiac stimulation increases myocardial oxygen consumption, a major factor in the pathogenesis and treatment of myocardial ischemia.

Metabolism Catecholamines increase metabolic rate. In small mammals, mitochondrial respiration in brown adipose tissue is functionally uncoupled by [NE](#). In a reaction unique to brown adipose tissue, NE stimulates the β_3 -adrenergic receptor that activates a specific mitochondrial uncoupling protein that dissipates the proton gradient between the inner mitochondrial matrix and the cytoplasm, thereby uncoupling substrate utilization and ATP synthesis. In humans, a functional role for brown adipose tissue has not been established with certainty.

Substrate Mobilization In a variety of tissues, catecholamines stimulate the breakdown of stored fuel with the production of substrate for local consumption; glycogenolysis in the heart, for example, provides substrate for immediate metabolism by the myocardium. Catecholamines also accelerate fuel mobilization in liver, adipose tissue, and skeletal muscle, liberating substrates (glucose, free fatty acids, lactate) into the circulation for use throughout the body.

Fluids and Electrolytes By a direct action on the renal tubule, [NE](#) stimulates sodium reabsorption, thereby defending extracellular fluid volume. Dopamine, in contrast, promotes sodium excretion. NE and E also promote cellular uptake of potassium.

Viscera Catecholamines affect visceral function by actions on smooth muscle and glandular epithelium. Urinary bladder and intestinal smooth muscle are relaxed while the corresponding sphincters are stimulated. Gallbladder emptying also involves sympathetic mechanisms. Catecholamine-mediated smooth-muscle contraction in the female aids ovulation and ovum transport along the fallopian tubes, and in the male provides propulsive force for the seminal fluid during ejaculation. Inhibitory α_2 -receptors on cholinergic neurons within the gut contribute to intestinal relaxation. Catecholamines induce bronchodilation by β_2 -receptor mechanism.

INDIRECT EFFECTS OF CATECHOLAMINES

The ultimate physiologic response induced by catecholamines involves changes in hormone secretion and in blood flow distribution, both of which support and amplify the direct effects of catecholamines.

Endocrine System Catecholamines influence the secretion of renin, insulin, glucagon, calcitonin, parathormone, thyroxine, gastrin, erythropoietin, progesterone, and, possibly, testosterone. Secretion of each of these hormones is governed by complex feedback loops. With the exception of thyroxine and the gonadal steroids, each is a polypeptide not under the direct control of the pituitary gland. Sympathoadrenal input into the secretion of these hormones provides a mechanism for regulation by the [CNS](#) and ensures a coordinated hormonal response in accord with the homeostatic needs of the organism.

Renin (See also [Chap. 246](#)) Sympathetic stimulation increases renin release by a direct

b-receptor effect independent of vascular changes within the kidney. The renin response to volume depletion is sympathetically mediated and is initiated by a fall in central venous pressure. Since renin secretion activates the angiotensin-aldosterone system, angiotensin-induced vasoconstriction supports the direct effects of catecholamines on blood vessels, while aldosterone-mediated sodium reabsorption complements the direct increase in sodium reabsorption induced by sympathetic stimulation. b-receptor blocking agents suppress renin secretion.

Insulin and Glucagon Stimulation of pancreatic sympathetic nerves or an elevation in circulating catecholamines suppresses insulin and increases glucagon release. Inhibition of insulin secretion is mediated by the α_2 receptor, and stimulation of glucagon is mediated by the β receptor. This combination of effects supports substrate mobilization, reinforcing the direct effects of catecholamines on hepatic glucose output and lipolysis. Although α -receptor-mediated suppression of insulin release usually predominates, a b-receptor mechanism may augment insulin secretion under some circumstances.

SYMPATHOADRENAL FUNCTION IN SELECTED PHYSIOLOGIC AND PATHOPHYSIOLOGIC STATES

Support of the Circulation The sympathetic nervous system functions to maintain an adequate circulation. During upright posture and volume depletion, reduction of afferent venous and arterial baroreceptor impulse traffic diminishes an inhibitory input to the vasomotor center, thereby increasing sympathetic activity ([Fig. 72-3](#)) and reducing efferent vagal tone. As a result, heart rate is increased, and cardiac output is diverted from the skin, subcutaneous tissues, mucosa, and viscera. Sympathetic stimulation of the kidney increases sodium reabsorption, and sympathetically mediated venoconstriction enhances venous return ([Fig. 72-6](#)). With pronounced hypotension, the adrenal medulla is recruited and epinephrine reinforces the effects of the sympathetic nervous system.

The intense sympathoadrenal stimulation that accompanies severe volume depletion may contribute to the development of ketoacidosis in alcoholics as well as to the ketoacidosis sometimes seen in association with hyperemesis gravidarum. Catecholamine-mediated suppression of insulin and stimulation of glucagon markedly potentiate ketogenesis in these disease states. Volume resuscitation and provision of adequate glucose promptly reverse the ketoacidosis in most cases.

Congestive Heart Failure The sympathetic nervous system also provides circulatory support during congestive heart failure ([Chap. 232](#)). Venoconstriction and sympathetic stimulation of the heart increase cardiac output while peripheral vasoconstriction directs blood flow to the heart and brain. The afferent signals are less clear than in simple volume depletion because the venous pressure is usually elevated. In severe heart failure, depletion of cardiac [NE](#) may impair the effectiveness of sympathetic circulatory support. On the other hand, the possibility has been raised that intense sympathetic stimulation may further impair cardiac function, suggesting possible benefit from β -adrenergic blockade. The use of beta blockers in the treatment of congestive heart failure, in fact, has increased in recent years.

Trauma and Shock In acute traumatic injury or shock, adrenal catecholamines support the circulation and mobilize substrates. In the chronic, reparative phase following injury, catecholamines also contribute to substrate mobilization and to the elevation in metabolic rate.

Exercise Sympathetic activation during exercise increases cardiac output and ensures sufficient substrate to meet the increased metabolic needs. Central neural factors, such as anticipation, and circulatory factors, such as fall in venous pressure, trigger the sympathetic response. Mild degrees of exercise stimulate the sympathetic nervous system alone; during more severe exertion the adrenal medulla is activated as well. Cardiovascular conditioning is associated with a decrease in sympathetic nervous system activity both at rest and during exercise, in comparison with the untrained state.

Hypoglycemia (See also [Chap. 334](#)) Hypoglycemia causes a marked increase in adrenal medullary epinephrine secretion. When glucose concentrations fall below overnight fasting levels, regulatory glucose-sensitive neurons in the [CNS](#) initiate a prompt increase in adrenal medullary secretion. The increase is especially intense at plasma glucose levels below 2.8 mmol/L (50 mg/dL), when plasma E levels increase 25 to 50 times above baseline, thereby increasing hepatic glucose output, providing alternative substrate in the form of free fatty acids, suppressing endogenous insulin release, and inhibiting insulin-mediated glucose utilization in muscle. Many clinical manifestations of hypoglycemia, such as tachycardia, palpitations, nervousness, tremor, and widened pulse pressure, are secondary to increased E secretion, and these manifestations constitute an "early warning" system in insulin-requiring diabetics. In patients with long-standing diabetes mellitus, however, the E response to hypoglycemia may be diminished or absent, leaving affected patients at greater risk to develop severe hypoglycemia.

Cold Exposure The sympathetic nervous system plays a critical role in the maintenance of normal body temperature during exposure to a cold environment. Receptors in the skin and [CNS](#) respond to a fall in temperature by activating hypothalamic and brainstem centers that increase sympathetic activity. Sympathetic stimulation leads to vasoconstriction in the superficial vascular beds, thereby diminishing heat loss. The sympathetic response involves a complex interaction between lowered environmental temperatures and α_2 -adrenergic receptors. Acclimatization during chronic cold exposure increases the capacity for metabolic heat production in response to sympathetic stimulation.

Dietary Intake Fasting suppresses and overfeeding stimulates the sympathetic nervous system. The reduction in sympathetic activity during fasting or starvation contributes to the decrease in metabolic rate, bradycardia, and hypotension in these states. Enhanced sympathetic activity during periods of increased caloric intake contributes to the elevation in metabolic rate associated with a chronic increase in dietary intake.

Hypoxia Chronic hypoxia is associated with stimulation of the sympathoadrenal system, and some of the cardiovascular changes attendant to hypoxia are dependent on catecholamines.

Orthostatic Hypotension The maintenance of arterial pressure during upright posture

depends on an adequate blood volume, an unimpaired venous return, and an intact sympathetic nervous system. Significant postural hypotension, therefore, often reflects extracellular fluid volume depletion or dysfunction of the circulatory reflexes. Diseases of the nervous system, such as tabes dorsalis, syringomyelia, or diabetes mellitus, may disrupt these sympathetic reflexes with resultant orthostatic hypotension. Although any antiadrenergic agent may impair the postural sympathetic response, orthostatic hypotension is most prominent with drugs that block neurotransmission within the ganglia or adrenergic neurons.

The term *idiopathic orthostatic hypotension* refers to a group of degenerative diseases involving either the pre- or postganglionic sympathetic neurons ([Chaps. 21](#) and [365](#)).

Treatment of orthostatic hypotension is usually unsatisfactory except in the mildest cases. There is no way of reestablishing the normal relationship between fall in venous return and sympathetic neuronal activation. Volume expansion with fludrocortisone and a liberal salt diet in conjunction with fitted stockings to the waist, as well as elevation of the head of the bed to avoid recumbency, will maintain plasma volume and venous return and frequently provide symptomatic improvement.

PHARMACOLOGY OF THE SYMPATHOADRENAL SYSTEM

A variety of therapeutic agents affect sympathetic nervous system function or interact with adrenergic receptors, making it possible to stimulate or suppress effects mediated by catecholamines with some degree of specificity ([Table 72-1](#)).

SYMPATHOMIMETIC AMINES

Sympathomimetic amines may directly activate adrenergic receptors (direct acting) or release [NE](#) from the sympathetic nerve endings (indirect acting). Many agents have both direct and indirect effects.

Epinephrine and Norepinephrine The naturally occurring catecholamines act predominantly by the direct stimulation of adrenergic receptors. [NE](#) is employed to support the circulation and elevate the blood pressure in hypotensive states ([Chap. 38](#)). Peripheral vasoconstriction is the major effect, although cardiac stimulation occurs as well. Epinephrine, also employed as a pressor, has special usefulness in the treatment of allergic reactions, especially those associated with anaphylaxis. Epinephrine antagonizes the effects of histamine and other mediators on vascular and visceral smooth muscle and is useful in the treatment of bronchospasm.

Dopamine *Dopamine* is used in treating hypotension, shock ([Chap. 38](#)), and certain forms of heart failure ([Chap. 232](#)). At low infusion rates it exerts a positive inotropic effect both by a direct action on the cardiac β_1 receptors and by the indirect release of [NE](#) from sympathetic nerve endings in the heart. At low doses direct stimulation of dopaminergic receptors in the renal and mesenteric vasculature also results in vasodilation in the gut and kidney and facilitates sodium excretion. At higher infusion rates interaction with α -adrenergic receptors results in vasoconstriction, an increase in peripheral resistance, and an elevation of blood pressure.

b-Receptor Agonists *Isoproterenol*, a direct-acting b-receptor agonist, stimulates the heart, decreases peripheral resistance, and relaxes bronchial smooth muscle. It raises the cardiac output and accelerates atrioventricular conduction while increasing the automaticity of ventricular pacemakers. *Isoproterenol* was formerly used in the treatment of heart block and bronchoconstriction. *Dobutamine*, a congener of dopamine with relative selectivity for the β_1 receptor and with a greater effect on myocardial contractility than on heart rate, is also used in the treatment of congestive heart failure, often in combination with vasodilators ([Chap. 232](#)). In conjunction with radionuclide imaging or echocardiographic assessment of wall motion, *dobutamine*, as well as other investigational congeners that have a relatively greater effect on heart rate, is used in the diagnosis of demand-induced myocardial ischemia.

Selective β_2 -Receptor Agonists The cardiac stimulation caused by nonselective b agonists, such as *isoproterenol* or *epinephrine*, is occasionally dangerous when these agents are used in the treatment of bronchoconstriction ([Chap. 252](#)). Selective β_2 agonists, administered by inhalation for bronchoconstriction, include agents with an intermediate duration of action (*metaproterenol*, *albuterol*, *terbutaline*, *pirbuterol*, *isoetharine*, and *bitolterol*) and the newer long-acting agents (*salmeterol* and *formoterol*); these drugs improve the therapeutic ratio by achieving bronchial dilatation with less activation of the cardiovascular system ([Chaps. 252](#) and [258](#)). Although selectivity is relative and cardiac stimulation may occur with these agents at higher dose levels, inhaled agonists at the usual doses result in relatively little cardiac stimulation. Oral administration, which is no longer preferred, is associated with more systemic b-agonist effects. *Ritodrine*, another selective β_2 agonist, is used as a tocolytic agent (as is *terbutaline*) to relax the uterus and antagonize premature labor.

a-Adrenergic Agonists *Phenylephrine* and *methoxamine* are direct-acting a agonists that elevate blood pressure by increasing peripheral vasoconstriction. They are used primarily in the treatment of hypotension and paroxysmal supraventricular tachycardia ([Chap. 230](#)), in the latter case by increasing cardiac vagal tone through reflex baroreceptor stimulation. *Phenylephrine* and a related proprietary compound, *phenylpropanolamine*, are common constituents of decongestant medications (often combined with antihistamines) for the treatment of allergic rhinitis and upper respiratory infections.

Miscellaneous Sympathomimetic Amines with Mixed Actions *Ephedrine* has both direct b-receptor agonist properties and an indirect effect on sympathetic nerve endings, from which it releases [NE](#), and is used primarily as a bronchodilator. *Sudaphedrine*, a congener of *ephedrine*, is less potent at dilating bronchi and serves as a nasal decongestant. *Metaraminol* has both direct and indirect effects on sympathetic nerve endings and is employed in the treatment of hypotensive states.

Dopaminergic Agonists The D_2 -receptor agonists, *bromocriptine* and *cabergoline*, are used to suppress prolactin secretion ([Chap. 328](#)). *Apomorphine*, another D_2 -receptor agonist, is used to induce emesis. The D_1 -receptor agonist, *fenoldopam*, has recently been approved for the short-term in-hospital treatment of severe hypertension.

ANTIADRENERGIC OR SYMPATHOLYTIC AGENTS (See also [Chap. 246](#))

Agents Inhibiting Central Sympathetic Outflow The antihypertensive agents *methyldopa*, *clonidine*, *guanabenz*, and *guanfacine* diminish central sympathetic outflow by stimulating a central adrenergic pathway (α_2 receptor) that diminishes vasomotor outflow. CNS side effects such as sedation are common. When administration of clonidine is stopped abruptly, a withdrawal syndrome characterized by rebound hyperactivity of the sympathetic nervous system can produce a state resembling the crises of patients with pheochromocytoma. *Opiates* also may exert a central sympatholytic effect; the sympathetic excitation of morphine withdrawal responds to clonidine and vice versa. *Propranolol* and *reserpine* may exert some sympatholytic effects at the level of the CNS.

Ganglionic Blocking Agents Ganglionic transmission may be antagonized by drugs that block the (nicotinic) cholinergic synapse between the pre- and postganglionic autonomic nerves. These agents inhibit the parasympathetic as well as the sympathetic nervous system. Only *trimethaphan* is in general clinical use; its major application is in the treatment of hypertensive crises, particularly aortic dissection, when controlled hypotension and decreased myocardial contractility are desirable ([Chap. 246](#)).

Agents Acting at the Peripheral Sympathetic Nerve Endings Adrenergic neuron-blocking agents depress the function of the peripheral sympathetic nerves by decreasing the amount of neurotransmitter released. *Guanethidine*, the prototype of this class of drugs, is concentrated in the sympathetic nerve endings by the amine-uptake mechanism. Within the terminal it blocks the release of NE in response to nerve impulses and eventually depletes the nerve of NE by displacing it from the intraneuronal storage granules. The drug was formerly useful in the management of severe hypertension, although orthostatic hypotension was a limiting side effect. *Bretylium*, an agent whose effects are similar to those of guanethidine, is employed in the treatment of ventricular fibrillation ([Chap. 230](#)). Both guanethidine and bretylium are antagonized by agents that affect the amine-uptake transport process such as sympathomimetic amines, tricyclic antidepressants, phenoxybenzamine, and phenothiazines. The antihypertensive action of guanethidine may be rapidly reversed by these drugs.

Reserpine depletes catecholamines from the peripheral sympathetic nerve endings, the brain, and the adrenal medulla. Its antihypertensive effect in humans is usually attributed to depletion of peripheral NE stores within sympathetic nerve endings. The sedation and occasionally morbid depression attending its use result from NE depletion within the CNS.

Adrenergic-Receptor Blocking Agents Adrenergic blocking agents antagonize the effects of catecholamines at the level of the peripheral tissue.

α -Adrenergic-receptor blocking agents *Phenoxybenzamine* and *phentolamine* are utilized principally in treating pheochromocytoma ([Chap. 332](#)). Phenoxybenzamine produces prolonged, noncompetitive α blockade, while phentolamine leads to reversible, competitive blockade. Because of its rapid action and short duration, phentolamine is commonly used in the treatment of acute hypertensive paroxysms secondary to catecholamine excess, such as occur with pheochromocytoma. Both phentolamine and phenoxybenzamine antagonize α_1 and α_2 receptors, although phenoxybenzamine is more potent at the α_1 -receptor site. *Prazosin*, an α -adrenergic

blocking agent with selectivity for the α_1 -receptor, possesses properties that resemble those of primary vasodilators and has been used in the treatment of essential hypertension, as an afterload-reducing agent in congestive heart failure, and as an adjunct in the treatment of pheochromocytoma ([Chap. 332](#)). *Doxazosin* and *terazosin*, long-acting selective α_1 -blockers, are more useful in the treatment of essential hypertension because of better dosing schedule and less orthostatic hypotension. These agents also lower triglyceride levels and raise high-density lipoprotein (HDL) cholesterol levels. These selective α_1 -blockers, along with *tamsulosin* are useful in the symptomatic treatment of urinary outflow track obstruction and prostatism because they antagonize contraction of the sphincter at the bladder trigone and the prostate smooth muscle.

b-Adrenergic-receptor blocking agents These drugs antagonize the effects of catecholamines on the heart and are useful in the treatment of angina pectoris, hypertension, and cardiac arrhythmias. The benefit of beta blockade in angina derives from the decrease in myocardial oxygen consumption following reduction in heart rate and myocardial contractility ([Chap. 244](#)). The hypotensive effect of beta blockade is not clearly understood ([Chap. 246](#)). Diminished cardiac output, decreased NE release at postganglionic sympathetic nerve endings, reduced renin secretion, and suppressed central sympathetic outflow are possible mechanisms. The efficacy of b-blocking agents in the treatment of arrhythmias depends on reduction of the rate of spontaneous depolarization of pacemaker cells in the sinus node and junctional pacemakers and on slowing conduction within the atria and atrioventricular node. Beta blockade is also effective in the symptomatic management of hyperthyroidism and the control of tachycardia and arrhythmias in patients with pheochromocytoma. b-adrenergic blocking agents are also useful in the treatment of migraine, essential tremor, idiopathic hypertrophic subaortic stenosis, and aortic dissection. Several trials have demonstrated that b-receptor blocking agents, administered long-term, diminish mortality following acute myocardial infarction. The mechanism of this cardioprotective effect may involve antiarrhythmic action, prevention of reinfarction, and reduction in infarct size ([Chap. 243](#)).

Pharmacologic Properties of b-Receptor Blocking Agents Fourteen beta-blocking agents (atenolol, acebutolol, betaxolol, bisoprolol, carvedilol, carteolol, esmolol, metoprolol, nadolol, pindolol, penbutolol, propranolol, sotalol, and timolol) are available for use in the United States. Other agents (alprenolol, bevantolol, dilevalol, oxprenolol, etc.) are in use in other countries and investigational within the United States. The utility of these agents is derived predominantly from blockade of b-adrenergic receptors.

Although much has been written about other pharmacologic properties, including cardioselectivity, membrane-stabilizing (local anesthetic) effects, intrinsic sympathomimetic (partial-agonist) activity, and lipid solubility, the clinical significance of these additional properties is small. Local anesthetic properties are most prominent with propranolol; however, membrane stabilization probably does not contribute substantially to the clinical utility. The various beta blockers do differ in their water and lipid solubility. The lipophilic agents (propranolol, metoprolol, oxprenolol, bisoprolol, carvedilol) are readily absorbed from the gastrointestinal tract, metabolized by the liver, have large volumes of distribution, and penetrate the CNS well; the hydrophilic agents (acebutolol, atenolol, betaxolol, carteolol, nadolol, sotalol) are less readily absorbed, not extensively

metabolized, and have relatively long plasma half-lives. As a consequence, the hydrophilic agents may be administered once per day. Hepatic failure may prolong the plasma half-life of the lipophilic agents, whereas renal failure may prolong the action of the hydrophilic group. The degree of lipid solubility, therefore, provides a basis for choice of a particular agent in patients with hepatic or renal insufficiency. Although the hydrophilic agents penetrate the CNS less well, CNS side effects (sedation, depression, hallucinations) are well described with the hydrophilic as well as with the lipophilic agents.

Some β -adrenergic blocking agents possess β -agonist activity. This has been referred to as "intrinsic sympathomimetic activity" (ISA). Agents with partial agonist activity (pindolol, alprenolol, acebutolol, carteolol, dilevalol, oxprenolol) cause little or no depression of resting heart rate (partial agonist effect) while blocking the increase in heart rate that occurs in response to exercise or the administration of a β agonist such as isoproterenol. The presence of partial agonist activity may be useful when bradycardia limits treatment in patients with slow resting heart rates. Pindolol also produces mild vasodilation, perhaps in part related to peripheral α_2 stimulation. Agents with partial agonist activity cause less change in blood lipid levels than agents without agonist properties. On theoretical grounds, intrinsic sympathomimetic activity would be undesirable in the treatment of thyrotoxicosis, idiopathic hypertrophic subaortic stenosis, aortic dissection, and tachyarrhythmias.

Cardioselective (β_1) Adrenergic-Receptor Blocking Agents Propranolol, the prototype of the nonselective β -adrenergic blocking agent, induces a competitive blockade of both β_1 and β_2 receptors. Other nonselective β -blocking agents include alprenolol, carteolol, dilevalol, nadolol, oxprenolol, penbutolol, pindolol, sotalol, timolol, and carvedilol. Metoprolol, esmolol, acebutolol, atenolol, and betaxolol possess relative selectivity for the β_1 receptor. Although β_1 -(cardio-) selective agents have the theoretical advantage of producing less bronchoconstriction and less peripheral vasoconstriction, a clear-cut clinical advantage of the cardioselective agents has not been decisively demonstrated, since the β_1 selectivity is only relative. Bronchoconstriction may occur when β_1 -selective agents are administered in full therapeutic doses.

Adverse Effects of β -Receptor Blocking Agents Aside from the effects on the [CNS](#), most adverse reactions to β -blocking agents are consequences of β -adrenergic blockade. These include the precipitation of heart failure in patients in whom cardiac compensation depends on enhanced sympathetic drive; the aggravation of bronchospasm in patients with asthma; predisposition to the development of hypoglycemia in insulin-requiring diabetics (blockade of catecholamine-mediated counterregulation and antagonism of the adrenergic warning signs of hypoglycemia); the development of hyperkalemia in diabetic or uremic patients with impaired potassium tolerance; the enhancement of coronary or peripheral arterial vasospasm; and elevation in triglycerides and depression of [HDL](#) levels. The lipid (and perhaps the peripheral vascular) effects are less (or absent) in agents with partial (β_2) agonist activity or α -blocking properties (carvedilol).

Miscellaneous Adrenergic Blocking Agents Labetalol, approved for use in the United States as an antihypertensive agent, is a competitive antagonist of both α - and β -adrenergic receptors. Although labetalol induces relatively more β - than α -receptor

blockade, fall in peripheral resistance may be marked following acute administration of the drug. Vasodilation may be mediated in part by a partial agonist effect on the β_2 -adrenergic receptor; labetalol does not possess partial agonist activity for the β_1 (cardiac) receptor.

Metoclopramide is a dopaminergic antagonist with cholinergic agonist properties. It enhances gastric emptying, increases the tone of the lower esophageal sphincter, increases prolactin and aldosterone secretion, and antagonizes emesis induced by apomorphine. It is useful clinically in enhancing gastric emptying (in the absence of organic obstruction such as in diabetic gastroparesis), in antagonizing gastroesophageal reflux, and as an antiemetic during cancer chemotherapy.

THE PARASYMPATHETIC NERVOUS SYSTEM

ACETYLCHOLINE

[ACh](#) serves as the neurotransmitter at all autonomic ganglia, at the postganglionic parasympathetic nerve endings, at the postganglionic sympathetic nerve endings innervating the eccrine sweat glands, and at the skeletal muscle end plate (neuromuscular junction). The enzyme choline acetyltransferase catalyzes the synthesis of ACh from acetyl coenzyme A (CoA) produced within the nerve ending and from choline, actively taken up from the extracellular fluid. Within the cholinergic nerve endings, ACh is stored in discrete synaptic vesicles and released in response to nerve impulses that depolarize the nerve terminals and increase calcium influx.

Cholinergic Receptors Different receptors for [ACh](#) exist on the postganglionic neurons within the autonomic ganglia and at the postjunctional autonomic effector sites. Those within the autonomic ganglia and adrenal medulla are stimulated predominantly by nicotine (*nicotinic receptors*) and those on autonomic effector cells by the alkaloid muscarine (*muscarinic receptors*). Ganglionic blocking agents antagonize the nicotinic receptors, while atropine blocks the muscarinic receptors. The muscarinic (M) receptor, furthermore, has been recently subdivided into additional types. The M_1 receptor is localized to the [CNS](#) and perhaps parasympathetic ganglia; the M_2 receptor is the nonneuronal muscarinic receptor on smooth muscle, cardiac muscle, and glandular epithelium. Bethanechol is a selective agonist of the M_3 receptor; pirenzepine, an investigational agent, is a selective antagonist of the M_1 receptor that markedly reduces gastric acid secretion. The M_2 receptor inhibits adenylyl cyclase and utilizes the regulatory G protein; the M_1 receptor interacts with G_q and stimulates phospholipase C ([Fig. 72-5](#)). The M_3 receptor, present on smooth muscle and secretory glands, is antagonized by atropine and utilizes phospholipase C, [IP₃](#), and [DAG](#) as second messengers. Other subtypes have been identified by molecular biologic techniques but have not yet been fully characterized.

Acetylcholinesterase Hydrolysis of [ACh](#) by acetylcholinesterase inactivates the neurotransmitter at cholinergic synapses. This enzyme (also known as specific or true cholinesterase) is present within neurons and is distinct from butyrylcholinesterase (serum cholinesterase or pseudocholinesterase). The latter enzyme is present in plasma and nonneuronal tissues and is not primarily involved in the termination of the effects of ACh at autonomic effector sites. The pharmacologic effects of

anticholinesterase agents are due to inhibition of neuronal (true) acetylcholinesterase.

PHYSIOLOGY OF THE PARASYMPATHETIC NERVOUS SYSTEM

The parasympathetic nervous system participates in the regulation of the cardiovascular system, the gastrointestinal tract, and the genitourinary system. Tissues such as liver, kidney, pancreas, and thyroid also receive parasympathetic innervation, suggesting a role for the parasympathetic nervous system in metabolic regulation as well, although cholinergic effects on metabolism are not well characterized.

Cardiovascular System Parasympathetic effects on the heart are mediated by the vagus nerve. [ACh](#) reduces the rate of spontaneous depolarization of the sinoatrial node and decreases heart rate. ACh also delays impulse conduction within the atrial musculature while shortening the effective refractory period, a combination of factors that may initiate or perpetuate atrial arrhythmias. At the atrioventricular node, ACh reduces conduction velocity, increases the effective refractory period, and thus diminishes the ventricular response during atrial flutter or fibrillation ([Chap. 230](#)). The decrease in inotropy induced by ACh is related to a prejunctional inhibitory effect on sympathetic nerve endings as well as to a direct inhibitory effect on the atrial myocardium. The ventricular myocardium is not much affected since innervation by cholinergic fibers is minimal. A direct cholinergic contribution to the regulation of peripheral resistance appears unlikely since parasympathetic innervation of the vasculature is not extensive. The parasympathetic nervous system, however, may influence peripheral resistance indirectly by inhibiting [NE](#) release from sympathetic nerves.

Gastrointestinal Tract Parasympathetic innervation of the gut is via the vagus nerve and the pelvic sacral nerves. The parasympathetic nervous system increases the tone of gastrointestinal smooth muscle, enhances peristaltic activity, and relaxes the gastrointestinal sphincters. [ACh](#) stimulates exocrine secretion from the glandular epithelium and enhances the secretion of gastrin, secretin, and insulin.

Genitourinary and Respiratory Systems Sacral parasympathetic nerves supply the urinary bladder and genitalia. [ACh](#) increases ureteral peristalsis, contracts the urinary detrusor muscle, and relaxes the trigone and sphincter, thereby playing a critical role in the coordination of urination. The respiratory tract is innervated with parasympathetic fibers derived from the vagus nerve. ACh increases tracheobronchial secretions and stimulates bronchial constriction.

PHARMACOLOGY OF THE PARASYMPATHETIC NERVOUS SYSTEM

Cholinergic Agonists [ACh](#) itself has no therapeutic role because of its widespread effects and short duration of action. Congeners of ACh are less susceptible to hydrolysis by cholinesterase and have a narrower range of physiologic effects. Bethanechol, the only systemic cholinergic agonist in general use, stimulates gastrointestinal and genitourinary smooth muscle with minimal effect on the cardiovascular system. It is used in the treatment of urinary retention in the absence of outflow tract obstruction and, less commonly, in gastrointestinal disorders such as postvagotomy gastric atony. Pilocarpine and carbachol are topical cholinergic agonists used in the treatment of

glaucoma.

Acetylcholinesterase Inhibitors Cholinesterase inhibitors enhance the effects of parasympathetic stimulation by diminishing the inactivation of [ACh](#). The therapeutic application of reversible cholinesterase inhibitors depends on the role of ACh as neurotransmitter at the skeletal muscle neuroeffector junction and within the [CNS](#) and includes the treatment of myasthenia gravis ([Chap. 380](#)), the termination of neuromuscular blockade following general anesthesia, and the reversal of intoxication by agents with a central anticholinergic action. Physostigmine, a tertiary amine, penetrates the CNS well, while related quaternary amines (neostigmine, pyridostigmine, ambenonium, and edrophonium) do not. Organophosphorous cholinesterase inhibitors produce irreversible cholinesterase blockade; these agents are used principally as insecticides and are primarily of toxicologic interest. With regard to the autonomic nervous system, cholinesterase inhibitors are of limited use in the treatment of intestinal and bladder smooth-muscle dysfunction such as occurs in paralytic ileus and atonic urinary bladder. Cholinesterase inhibitors induce a vagotonic response in the heart and may be useful in terminating attacks of paroxysmal supraventricular tachycardia ([Chap. 230](#)).

Cholinergic-Receptor Blocking Agents *Atropine* blocks muscarinic cholinergic receptors, with little effect on cholinergic transmission at the autonomic ganglia and the neuromuscular junctions. Many of the [CNS](#) actions of atropine and atropine-like drugs are attributable to blockade of central muscarinic synapses. The related alkaloid, *scopolamine*, is similar to atropine but causes drowsiness, euphoria, and amnesia, effects that make it suitable as a preanesthetic medication.

Atropine increases heart rate and enhances atrioventricular conduction, actions that may be useful in combating the bradycardia or heart block associated with heightened vagal tone. In addition, atropine reverses cholinergically mediated bronchoconstriction and diminishes respiratory tract secretions. These effects contribute to its utility as a preanesthetic medication.

Atropine also decreases gastrointestinal tract motility and secretion. Although various derivatives and congeners of atropine (such as *propantheline*, *isopropamide*, and *glycopyrrolate*) have been advocated in patients with peptic ulcer or with diarrheal syndromes, the chronic use of such agents is limited by other manifestations of parasympathetic inhibition such as dry mouth and urinary retention. The investigational selective M₁ inhibitor pirenzepine inhibits gastric secretion at doses that have minimal anticholinergic effects at other sites; this agent may be useful in the treatment of peptic ulcer. Atropine and its congener *ipratropium*, when given by inhalation, cause bronchodilation and have been used experimentally in the treatment of asthma.

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PART FIVE -NUTRITION

73. NUTRITIONAL REQUIREMENTS AND DIETARY ASSESSMENT - *Johanna Dwyer*

Nutrients are substances that are not synthesized in the body in sufficient amounts and therefore must be supplied by the diet. Nutrient requirements for groups of healthy persons have been thoroughly defined on the basis of experimental evidence. For good health we require energy-providing nutrients (protein, fat, and carbohydrate), vitamins, minerals, and water. Specific nutrient requirements include 9 essential amino acids, several fatty acids, 4 fat-soluble vitamins, 10 water-soluble vitamins, and choline. Several inorganic substances, including four minerals, seven trace minerals, three electrolytes, and the ultratrace elements, must also be supplied in the diet ([Chap. 75](#)).

The required amounts of the essential nutrients differ by age and physiologic state. Conditionally essential nutrients are not required in the diet but must be supplied to individuals who do not synthesize them in adequate amounts, such as those with genetic defects, those having pathologic states with nutritional implications, and developmentally immature infants ([Chap. 74](#)). Many organic phytochemicals and zoochemicals present in foods have various health effects. For example, dietary fiber has been shown to have beneficial effects on gastrointestinal function.

ESSENTIAL NUTRIENT REQUIREMENTS

Energy For weight to remain stable, energy intake must match energy output ([Chap. 77](#)). The major categories of energy output are resting energy expenditure (REE) and physical activity; minor sources include the energy cost of metabolizing food (thermic effect of food or specific dynamic action) and shivering thermogenesis (e.g., cold-induced thermogenesis). The average energy intake is about 2800 kcal/d for American men and about 1800 kcal/d for American women, though these estimates vary with body size and activity level. Formulas for estimating REE are useful for assessing the energy needs of an individual whose weight is stable. Thus, for males, $REE = 900 + 10w$, and for females, $REE = 700 + 7w$, where w is weight in kg. The calculated REE is then adjusted for physical activity level by multiplying by 1.2 for sedentary, 1.4 for moderately active, or 1.8 for very active individuals. The final figure provides an estimate of total caloric needs in a state of energy balance.

Illness often alters energy needs. Unstressed hospitalized patients at bed rest usually require 1.2 times their [REE](#), whereas those who are stressed, febrile, and catabolic require 1.5 to 2 times their REE ([Chap. 74](#)). Intestinal malabsorption may decrease net utilizable energy to as little as 25% of ingested energy and may necessitate feeding by parenteral routes ([Chap. 76](#)). Fever increases energy expenditure by 10 to 13% per degree Celsius above normal. Other diseases increase energy needs by varying amounts, such as burns (40 to 100%), trauma (40 to 100%), and hyperthyroidism (10 to 100%). Hypothyroidism and adrenal insufficiency decrease resting energy needs, but these alterations are corrected after adequate hormone replacement. In obese patients, weight reduction can be accomplished by reducing energy intakes by approximately 500 kcal/d to achieve a loss of 0.5 kg of fat per week, or 1000 kcal/d to lose 1 kg per week ([Chap. 77](#)).

Protein Dietary protein consists of both essential and nonessential amino acids that are required for protein synthesis, whereas certain amino acids can also be used for energy and gluconeogenesis ([Chap. 334](#)). The nine essential amino acids are histidine, isoleucine, leucine, lysine, methionine/cystine, phenylalanine/tyrosine, threonine, tryptophan, and valine. When energy intake is inadequate, protein intake must be increased, since ingested amino acids are diverted into pathways of glucose synthesis and oxidation. In extreme energy deprivation, protein-calorie malnutrition may ensue ([Chaps. 74](#) and [76](#)).

For adults, the recommended dietary allowance (RDA) for protein is about 0.6 g/kg desirable body weight per day, assuming that energy needs are met and that the protein is of relatively high biologic value. Current recommendations for a healthy diet call for at least 10 to 14% of calories from protein. Biologic value tends to be highest for animal proteins, followed by proteins from legumes (beans), cereals (rice, wheat, corn), and roots. Combinations of plant proteins that complement one another in biologic value or combinations of animal and plant proteins can increase biologic value and lower total protein requirements.

Protein needs increase during growth, pregnancy, lactation, and rehabilitation during treatment of malnutrition. The tolerance of dietary protein is decreased in renal insufficiency and liver failure. Normal protein intake can precipitate encephalopathy in patients with cirrhosis of the liver ([Chap. 299](#)) or worsen uremia in those with renal failure ([Chap. 270](#)).

Fat and Carbohydrate Fats are a concentrated source of energy and constitute on average 34% of calories in U.S. diets. However, for optimal health, fat intake should total no more than 30% of calories. Saturated fat and trans-fat should be limited to <10% of calories, and polyunsaturated fats to <10% of calories, with monounsaturated fats comprising the remainder of fat intake. At least 55% of total calories should be derived from carbohydrates. The brain requires about 100 g/d of glucose for fuel; other tissues use about 50 g/d. Over time, adaptations in carbohydrate needs are possible in hypocaloric states. For example, reduced insulin levels lead to adipose tissue breakdown and cause the body to burn more fatty acids. However, some tissues (e.g., brain and red blood cells) rely on glucose supplied either exogenously or from muscle proteolysis ([Chap. 334](#)).

Water For adults, 1 to 1.5 mL water per kcal of energy expenditure is sufficient under usual conditions to allow for normal variations in physical activity levels, sweating, and solute load of the diet. Water losses include 50 to 100 mL/d in the feces, 500 to 1000 mL/d by evaporation or exhalation, and, depending on the renal solute load, ³1000 mL/d in the urine. If external losses increase, intakes must increase accordingly to avoid underhydration. Fever increases water losses by approximately 200 mL/d per°C; diarrheal losses vary but may be as great as 5 L/d in severe diarrhea. Heavy sweating and vomiting also increase water losses. When renal function is normal and solute intakes are adequate, the kidneys can adjust to increased water intake by excreting up to 18 L/d of excess water ([Chap. 329](#)). However, obligatory urine outputs can compromise hydration status when there is inadequate intake or when losses increase in disease or kidney damage.

Infants have high requirements for water because of their large ratio of surface area to volume, the limited capacity of the immature kidney to handle high renal solute loads, and their inability to communicate their thirst. Increased water needs during pregnancy are low, perhaps an additional 30 mL/d; but during lactation, milk production increases water requirements so that approximately 1000 mL/d of additional water is needed, or 1 mL for each mL of milk produced. Special attention must be paid to the water needs of the elderly, who have reduced total body water and blunted thirst sensation and may be taking diuretics.

Other Nutrients The vitamins and minerals required for health and the clinical disorders caused by vitamin deficiency or excess are discussed in [Chap. 75](#).

DIETARY REFERENCE INTAKES, RECOMMENDED ALLOWANCES, AND TOLERANCES

Fortunately, human life and well-being can be maintained within a fairly wide range for most nutrients. However, the capacity for adaptation is not infinite -- too much, as well as too little, intake of a nutrient may have adverse effects or alter the health benefits conferred by another nutrient ([Chap. 75](#)). Therefore, benchmark recommendations on nutrient intakes have been developed to guide clinical practice. These quantitative estimates of nutrient intakes are collectively referred to as the *dietary reference intakes* (DRIs). The DRIs supplant the [RDAs](#), the single reference values used in the United States since 1989. DRIs include the estimated average requirement (EAR) for nutrients, as well as three other reference values used for dietary planning for individuals: the RDAs, the adequate intake (AI), and the safe upper level (UL). The current RDAs and AIs are provided in [Tables 73-1](#) and [73-2](#), respectively.

Estimated Average Requirement When florid dietary deficiency diseases such as rickets, scurvy, xerophthalmia, and protein-calorie malnutrition were common, nutrient adequacy was assumed by the absence of clinical signs of a dietary deficiency disease. Later, it was determined that biochemical and other changes were evident long before the clinical deficiency became apparent. Consequently, criteria of adequacy are chosen using such biologic markers when they are available. Current efforts focus on the amount of a nutrient that reduces the risk of chronic degenerative diseases. Priority is given to sensitive biochemical, physiologic, or behavioral tests that reflect early changes in regulatory processes or maintenance of body stores of nutrients.

The [EAR](#) is the amount of a nutrient estimated to be adequate for half of the healthy individuals of a specific age and sex. The types of evidence and criteria used to establish nutrient requirements vary by nutrient, age, and physiologic group. The EAR is not an effective estimate of nutrient adequacy in individuals because it is a median requirement for a group, and the variation around this number is considerable. As the EAR specifies, 50% of individuals in a group fall below the requirement and 50% fall above it. Thus, a person with a usual intake at the EAR has a 50% risk of an inadequate intake during the reporting period. For these reasons, other standards, described below, are more useful for clinical purposes.

Recommended Dietary Allowances The [RDA](#) is the average daily dietary intake level

that meets the nutrient requirements of nearly all healthy persons of a specific sex, age, life stage, or physiologic condition (such as pregnancy or lactation). The RDA is commonly used as a nutrient-intake goal for planning diets of individuals.

The [RDA](#) is defined statistically as 2 standard deviations (SD) above the [EAR](#) to ensure that the needs of any given individual are met. The RDAs are used to formulate food guides such as the U.S. Department of Agriculture (USDA) Food Guide Pyramid for individuals, food exchange lists for therapeutic diet planning, and as a standard for describing the nutritional content of processed foods and nutrient supplements. The nutrient content in a food is stated by weight or as a percent of the daily value (DV), a variant of the RDA which, for an adult, represents the highest RDA for an adult consuming 2000 kcal/d.

The risk of dietary inadequacy increases as intakes fall further below the [RDA](#). However, the RDA is an overly generous criterion for evaluating nutrient adequacy. For example, by definition the RDA exceeds the actual requirements of all but about 2 to 3% of the population. Therefore, many people whose intakes fall below the RDA may still be getting enough of the nutrient.

Adequate Intake It is not possible to set an [RDA](#) for some nutrients that do not have an established [EAR](#). In this circumstance, the [AI](#) is based on observed, or experimentally determined, approximations of nutrient intakes in healthy people. In the [DRIs](#) established to date, AIs rather than RDAs are proposed for infants up to age 1, as well as for calcium, vitamin D, fluoride, pantothenic acid, biotin, and choline for persons of all ages.

Tolerable Upper Levels of Nutrient Intake Excessive nutrient intake can disturb body functions and cause acute, progressive, or permanent disabilities ([Chap. 75](#)). Some diseases of nutritional excess include fluorosis, hypervitaminosis A, hypervitaminosis D, and obesity. The tolerable [UL](#) is the highest level of chronic nutrient intake (usually daily) that is unlikely to pose a risk of adverse health effects for most of the population. An uncertainty factor is applied to ensure that even very sensitive persons would not experience adverse effects at the UL dose chosen. For many nutrients, data on the adverse effects of large amounts of the nutrient are unavailable or too limited to establish a UL. Therefore, the lack of a UL does *not* mean that the risk of adverse effects from high intakes is nonexistent; caution is warranted in those who consume large amounts of such nutrients. Healthy individuals derive no established benefit from consuming nutrient levels above the [RDA](#) or [AI](#). Individual nutrients in foods that most people eat rarely reach levels that exceed the UL. However, nutritional supplements provide more concentrated amounts of nutrients per dose and, as a result, pose a potential risk of toxicity. Nutrient supplements are labeled with "supplement facts" that express the amount of nutrient in absolute units or as the percent of the [DV](#) provided per recommended serving size. Those who use supplements should be advised that total nutrient consumption, including both food and supplements, should not exceed RDA levels.

FACTORS ALTERING NUTRIENT NEEDS

The [DRIs](#) are affected by age, sex, rate of growth, pregnancy, lactation, physical activity, composition of diet, concomitant diseases, and drugs. When only slight differences exist

between the requirements for nutrient sufficiency and excess, dietary planning becomes more difficult. Renal insufficiency provides one example in which protein intakes must be sufficient to maintain protein nutritional status, while avoiding exacerbation of uremic symptoms because of protein excess.

Physiologic Factors Growth, strenuous physical activity, pregnancy, and lactation increase needs for energy and several essential nutrients. Energy needs rise during pregnancy, due to the demands of fetal growth, and during lactation, because of the increased energy required for milk production. Energy needs decrease with loss of lean body mass, the major determinant of [REE](#). Because both health and physical activity tend to decline with age, energy needs in older persons, especially those over 70, tend to be less than those of younger persons.

Dietary Composition Dietary composition affects the biologic availability and utilization of nutrients. For example, the absorption of iron may be impaired by high amounts of calcium or lead; non-heme iron uptake may be impaired by the lack of ascorbic acid and amino acids in the meal. The absorption of calcium and magnesium is decreased by large amounts of phytates in the diet. Protein utilization by the body may be decreased when essential amino acids are not present in sufficient amounts. Animal foods, such as milk, eggs, and meat, have high biologic values with most of the needed amino acids present in adequate amounts. Plant proteins in corn (maize), soy, and wheat have lower biologic values and must be combined with other plant or animal proteins to achieve optimal utilization by the body.

Route of Administration The [RDAs](#) apply only to oral intakes. When nutrients are administered parenterally, similar values can sometimes be used for amino acids, carbohydrates, fats, sodium, chloride, potassium, and most of the vitamins, since their intestinal absorption is nearly 100% ([Chap. 75](#)). However, the oral bioavailability of most mineral elements may be only half that obtained by parenteral administration. For some nutrients that are not readily stored in the body, or cannot be stored in large amounts, timing of administration may also be important. For example, amino acids cannot be used for protein synthesis if they are not supplied together; instead they will be used for energy production.

Disease Specific dietary deficiency diseases include protein-calorie malnutrition; iron, iodine, and vitamin A deficiency; megaloblastic anemia due to vitamin B₁₂ or folic acid deficiency; vitamin D deficiency rickets; and scurvy, beriberi, and pellagra ([Chaps. 74 and 75](#)). Each deficiency disease is characterized by imbalances at the cellular level between the supply of nutrients or energy and the body's nutritional needs for growth, maintenance, and other functions. Imbalances in nutrient intakes are recognized as risk factors for certain chronic degenerative diseases, such as saturated fat and cholesterol in coronary artery disease; sodium in hypertension; obesity in hormone-dependent endometrial, breast, and prostate cancers; and ethanol in alcoholism. Since the etiology and pathogenesis of these disorders are multifactorial, diet is only one of many risk factors. Osteoporosis, for example, is associated with calcium deficiency, as well as risk factors related to environment (e.g., smoking, sedentary lifestyle), physiology (e.g., estrogen deficiency), genetic determinants (e.g., defects in collagen metabolism), and drug use (chronic steroids) ([Chap. 342](#)).

DIETARY ASSESSMENT

In clinical situations, nutritional assessment is an iterative process that involves: (1) screening for malnutrition, (2) assessing the diet and other data to establish either the absence or presence of malnutrition and its possible causes, and (3) planning for the most appropriate nutritional therapy. Some disease states affect the bioavailability, requirements, utilization, or excretion of specific nutrients. In these circumstances, specific measurements of various nutrients may be required to assure adequate replacement ([Chap. 75](#)).

Most health care facilities have a nutrition screening process in place for identifying possible malnutrition after hospital admission. Nutritional screening is required by the Joint Commission on Accreditation of Healthcare Organizations (JCAHO), but there are no universally recognized or validated standards, so techniques vary. The factors that are usually assessed include: abnormal weight for height or body mass index (e.g., BMI <19 or >25); reported weight change (involuntary loss or gain of >5 kg in past 6 months) ([Chap. 43](#)); diagnoses with known nutritional implications (metabolic disease, any disease affecting the gastrointestinal tract, alcoholism, and others); present therapeutic dietary prescription; chronic poor appetite; presence of chewing and swallowing problems or major food intolerances; need for assistance with preparing or shopping for food, eating, or other aspects of self care; and social isolation. Reassessment of nutrition status should occur periodically in hospitalized patients -- at least once every week.

A more complete dietary assessment is indicated for patients who exhibit a high risk of malnutrition on nutrition screening. The type of assessment varies based on the clinical setting, severity of the patient's illness, and stability of his or her condition.

Acute Care Settings In acute care settings, anorexia, various diseases, test procedures, and medications can compromise dietary intake. Under such circumstances, the goal is to identify and avoid inadequate intake and assure appropriate alimentation. Dietary assessment in acute care situations focuses on what patients are currently eating, whether they are able and willing to eat, and whether they experience any problems with eating. Dietary intake assessment is based on information from observed intakes; medical record; history; clinical examination; and anthropometric, biochemical, and functional status. The objective is to gather enough information to establish the likelihood of malnutrition due to poor dietary intake or other causes in order to determine whether nutritional therapy is indicated.

Simple observations may suffice to suggest inadequate oral intake. These include dietitians' and nurses' notes, the amount of food eaten on trays, frequent tests and procedures that are likely to cause meals to be skipped, nutritionally inadequate diet orders such as clear liquids or full liquids for more than a few days, fever, gastrointestinal distress, vomiting, diarrhea, or a comatose state. Patients with diseases or treatments that involve any part of the alimentary tract are at high nutritional risk. Acutely ill patients with diet-related diseases such as diabetes need assessment because an inappropriate diet may exacerbate these conditions and adversely affect other therapies. Abnormal biochemical values [serum albumin levels <35 g/L (<3.5 mg/dL); serum cholesterol levels <3.9 mmol/L (<150 mg/dL)] are nonspecific but may

also indicate a need for further nutritional assessment.

Most therapeutic diets offered in hospitals are calculated to meet individual nutrient requirements and the [RDA](#). Exceptions include clear liquids, some full liquid diets, and test diets, which are inadequate for several nutrients and should not be used, if possible, for more than 24 h. As much as half of the food served to hospitalized patients is not eaten, and so it cannot be assumed that the intakes of hospitalized patients are adequate. The dietary assessment should therefore compare how much and what food the patient has consumed with the diet that has been provided in the hospital. Major deviations in intakes of energy, protein, fluids, or other nutrients of special concern for the patient's illness should be noted and corrected.

Nutritional monitoring is especially important for patients who are very ill and who have extended lengths of stay. Patients who are fed by special enteral and parenteral routes also require special nutritional assessment and monitoring by physicians with training in nutrition support and/or dietitians with certification in nutrition support ([Chap. 76](#)).

Ambulatory Settings The aim of dietary assessment in the outpatient setting is to determine whether the patient's usual diet is a health risk in itself or if it contributes to existing chronic disease-related problems. It also provides the basis for planning a diet that fulfills therapeutic goals while ensuring patient compliance. The outpatient dietary assessment should review the adequacy of present and usual food intakes, including vitamin and mineral supplements, medications, and alcohol, as all of these may affect the patient's nutritional status. The dietary assessment should focus on the dietary constituents that are most likely to be involved or compromised by a specific diagnosis, as well as any comorbidities that are present. More than one day's intake should be reviewed to provide a better representation of the usual diet.

There are many ways to assess the adequacy of the patient's habitual diet. These include a food guide, a food exchange list, a diet history, or a food frequency questionnaire. A commonly used food guide for healthy persons is the [USDA's](#) food pyramid, which is useful as a basis for identifying inadequate intakes of essential nutrients, as well as likely excesses in fat, saturated fat, sodium, sugar, and alcohol ([Table 73-3](#)). The guide is calculated to provide approximately 1600 kcal for sedentary women and some older adults; 2200 kcal for most children, teenage girls, active women, and many sedentary men (women who are pregnant or breastfeeding may need somewhat more); and 2800 kcal for teenage boys, most active men, and some very active women. Results provide a rough guide to food groups that may be eaten either in excess of recommendations or in insufficient quantities. Respondents who follow ethnic or unusual dietary patterns may need extra instruction on how foods should be categorized, as well as the appropriate portion sizes that constitute a serving. The process of reviewing the guide with patients helps them transition to healthier dietary patterns. For those on therapeutic diets, assessment against food exchange lists may be useful. These include, for example, the American Diabetes Association food exchange lists for diabetes, or the American Dietetic Association food exchange lists for renal disease.

Nutritional Status Assessment Full nutritional status assessment is a complex, time-consuming, and expensive process that requires considerable expertise.

Candidates include seriously ill patients and those at very high nutritional risk when the cause of malnutrition is still uncertain after initial clinical evaluation and dietary assessment. Full nutritional status assessment involves multiple dimensions, including documentation of dietary intake, anthropometric measurements, biochemical measurements of blood and urine, clinical examination, health history, and functional. **For further discussion of Nutritional Assessment, see [Chap. 74](#).*

(Bibliography omitted in Palm version)

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74. MALNUTRITION AND NUTRITIONAL ASSESSMENT - Charles H. Halsted

Malnutrition is a frequent and integral component of acute and chronic illness. When recognized by appropriate clinical assessment, malnutrition is found in >50% of all hospitalized adults. It contributes to increased in-hospital morbidity and mortality in both medical and surgical patients, and leads to more frequent hospital admissions among the elderly. Malnutrition results from various combinations of starvation, including inadequate intake or abnormal gastrointestinal assimilation of the diet, the stress response to acute injury or chronic inflammation, and abnormal nutrient metabolism. Nutritional assessment should be considered an integral part of the clinical evaluation and be used as a basis for nutritional support in the overall therapeutic plan.

DEFINITIONS OF MALNUTRITION

In the strict sense, the term *malnutrition* includes extremes of underweight and overweight. The current chapter, however, focuses on the evaluation of the undernourished patient who presents with diminished body protein and energy stores and micronutrient deficiencies.

To the practicing physician, both outpatients and inpatients should be considered at risk for malnutrition if they meet one or more of the following criteria: (1) unintentional loss of >10% of usual body weight in the preceding 3 months, (2) body weight <90% of ideal for height, or (3) body mass index (BMI; the weight in kilograms divided by the height in square meters) <18.5. With regard to varying levels of severity, body weight <90% of ideal for height represents risk of malnutrition, body weight <85% of ideal constitutes *malnutrition*, <70% of ideal represents *severe malnutrition*, and <60% of ideal is usually incompatible with survival.

Malnutrition may be endemic in regions of famine, and two forms of severe malnutrition are recognized under conditions of inadequate food supply or distribution: *marasmus* refers to generalized starvation with loss of body fat and protein, whereas *kwashiorkor* refers to selective protein malnutrition with edema and fatty liver. The latter form occurs following restriction of dietary protein among children in settings of recurrent diarrheal illness. These distinctions, however, seldom apply to malnourished patients in more developed societies. In this setting, features of combined protein-calorie malnutrition (PCM) are more commonly seen in the context of a wide variety of acute and chronic illnesses that lead to depletion of body fat, muscle wasting, multiple signs of micronutrient deficiencies, decubitus ulcers, and life-threatening infections. An overview of the evaluation of malnutrition in the sick adult is depicted in [Fig. 74-1](#).

PATHOPHYSIOLOGY AND ETIOLOGIES OF MALNUTRITION

In simple terms, patients lose weight when: (1) the intake or gastrointestinal assimilation of dietary calories is insufficient to meet normal energy expenditure; (2) the expenditure of body energy stores is greater than energy normally consumed and assimilated by the body; or (3) the metabolism of energy supplies, protein, and other nutrients is significantly impaired by the intrinsic disease process.

Body Composition As depicted in [Fig. 74-2](#), the human body stores between 15 and

25% of its energy as fat (greater in women than men), which is available for the metabolism of endogenous fatty acids during starvation. The remaining fat-free mass (FFM) is composed of extracellular and intracellular water, the bony skeleton, glycogen, and skeletal and visceral protein. Aside from body fat, energy reserves are also provided by intracellular glycogen and protein, which, together with intracellular water, constitute the body cell mass (BCM). Thus, in addition to the enzymes that support the normal metabolic machinery of the body, the BCM provides reserve protein for energy production by gluconeogenesis during the stress response.

The Metabolic Response to Starvation and Stress The expenditure of body stores of energy (as fat, glycogen, and protein) is different during *starvation* (due to decreased intake and/or assimilation of the diet) and *stress* (due to excessive expenditure of energy and body protein). Consequently, these events affect body compartments differently. Starvation decreases the size of all body compartments, whereas stress reduces [BCM](#), increases extracellular water, and has variable effects on body fat.

A normal 70-kg man stores fuel at about 15 kg as fat, 6 kg as protein, and 0.4 kg as glycogen. During a 24-h fast, energy needs are met by the consumption of liver glycogen stores and the conversion of up to 75 g of body protein to glucose (by gluconeogenesis). During prolonged starvation, metabolism is supported by stores of body fat (about 150 g/d), which provides fatty acid-derived ketones, and muscle protein (about 20 g/d), which is used for gluconeogenesis. Under these conditions, total energy expenditure is decreased in order to conserve energy. While normal-weight individuals can sustain total fasting for about 2 months, obese individuals can fast for periods >12 months, depending on their fat stores.

The metabolic responses to the stress of acute critical illness (e.g., following accidental or surgical trauma or sepsis) significantly modify this sequence of events. In contrast to the hypometabolism, protein conservation, and reliance on body fat stores for energy needs during starvation, the acute stress response is characterized by hypermetabolism, in which the demands of accelerated energy expenditure are met by skeletal and visceral proteolysis to provide amino acid substrate for gluconeogenesis. Muscle proteolysis and gluconeogenesis are promoted by high levels of circulating catecholamines, glucagon, cortisol, and cytokines, including tumor necrosis factor (TNF) α and interleukins 1 and 6, in the setting of insulin resistance. When untreated, body protein catabolism is accelerated to 240 g/d, which is sufficient to deplete 50% of body protein stores within 3 weeks.

A more common clinical situation is the malnourished patient with chronic illness in whom acute trauma or sepsis superimposes cytokine-mediated proteolysis with increased metabolic demands. If unchecked by appropriate therapy, the process of progressive [PCM](#) in such patients is associated with decreased cardiac and renal function, fluid retention, intestinal mucosal atrophy, loss of intracellular minerals (zinc, magnesium, and phosphorus), diminished cell-mediated immune functions, increased risk of infection, and eventual death ([Fig. 74-3](#)).

Etiologies of Malnutrition The causes of decreased dietary intake are diverse and include social and economic conditions, psychiatric diseases, neurodegenerative dementias, cytokine-mediated appetite suppression in chronic infections such as AIDS

or in disseminated cancer, and self-limited food intake in abdominal pain syndromes ([Table 74-1;Chap. 43](#)). Given the central role of the gastrointestinal tract in the assimilation of nutrients,[PCM](#) is a predictable component of many chronic gastrointestinal diseases. These diseases promote starvation through decreased assimilation of the diet by: (1) blocking the transit of dietary constituents to the intestinal absorbing surface, (2) impairing normal processes of pancreatic or biliary digestion, or (3) preventing the intestinal mucosal transport of dietary constituents. Diseases that are characterized by increased catabolism of stored energy and protein include acute surgical or medical critical illness and acute or chronic inflammatory or infectious disorders affecting diverse organ systems. Other chronic diseases promote malnutrition through mixed mechanisms that contribute to abnormal nutrient metabolism. Both AIDS and disseminated malignancy, for example, cause progressive malnutrition through combinations of anorexia and futile cycles of fatty acid and glucose metabolism. Chronic obstructive pulmonary disease increases risk of malnutrition through the increased energy expenditure of labored respiration, chronic indolent bronchial infection, and the anorexic side effects of many bronchodilating drugs. Chronic liver disease is often associated with PCM caused by the cumulative effects of anorexia; decreased biliary circulation; and abnormal lipid, carbohydrate, and protein metabolism. The chronic intestinal inflammation of Crohn's disease or ulcerative colitis accelerates fecal losses of protein, electrolytes, and zinc.

CLINICAL EVALUATION OF THE MALNOURISHED PATIENT

THE PATIENT HISTORY

The clinical nutritional history should include diet and weight change, socioeconomic conditions, and symptoms unique to each clinical setting ([Table 74-2](#)). Social and economic conditions that may lead to poverty include inadequate income, homelessness, and activities that restrict real income and promote involuntary diet restriction, such as drug abuse or chronic alcoholism. Anorexia, or loss of appetite, is a feature of psychiatric disorders, such as anorexia nervosa and neurodegenerative dementia in the elderly. Many self-selected, inadequate diets may promote malnutrition. During binge drinking, chronic alcoholics typically substitute more than half their daily food calories with excessive amounts of ethanol, the metabolism of which consumes energy and promotes unbalanced metabolism of fat and carbohydrates. Other inadequate diets include unbalanced and commercially promoted formulas for rapid weight loss and strict vegetarianism, which may lead to selective deficiencies of specific nutrients such as vitamin B₁₂ and iron.

Digestive diseases are major causes of malnutrition, both in the inpatient and outpatient settings. The malnourished patient with digestive disease may present with symptoms of: (1) dysphagia or recurrent vomiting due to benign or malignant esophageal or gastrointestinal obstruction; (2) chronic diarrhea due to abnormal pancreatic or biliary digestion, intestinal mucosal malabsorption, or protein-losing enteropathy; or (3) recurrent abdominal pain exacerbated by eating, as occurs in patients with chronic pancreatitis, inflammatory bowel disease, or intestinal ischemia.

On the general medical service,[PCM](#) is prevalent in patients with multiple chronic illnesses that are associated with anorexia, recurrent stress, and abnormal nutrient

metabolism. In addition, PCM is comorbid with chronic recurrent pancreatitis, renal failure, chronic liver disease, chronic obstructive pulmonary disease, disseminated cancer, and chronic infections such as AIDS and tuberculosis. Depending on the severity of injury or illness, critically ill surgical and medical patients predictably develop stress-related PCM if increased nutritional needs are not met after 5 to 10 days.

THE PHYSICAL EXAMINATION

A careful physical examination can both characterize and define the extent of malnutrition. Measurements of unclothed weight and height are essential for establishing the severity of malnutrition in all patients but may be confounded by the effects of fluid overload as a result of edema and ascites. The normal values for weight (in kg) and height (in cm) in men and women are provided in [Table 74-3](#). These values can be adjusted by $\pm 10\%$ to account for variability in body frame.

Anthropometry Measurements of subcutaneous fat and skeletal muscle are important to determine the severity of [PCM](#). Using specialized calipers and a tape measure, anthropometry estimates body fat from the thickness of the skin-fold of the posterior mid-upper arm. Anthropometric measurements in healthy and malnourished adults are shown in [Table 74-4](#). Mid-arm muscle circumference is estimated from the equation:

The use of anthropometry is limited by the requirement for specialized calipers, the experience of the observer, and potential confounding effects of edema or dehydration.

Specific Physical Findings of Malnutrition During the conventional physical examination, the observant and experienced clinician can identify multiple and specific findings of [PCM](#) and its associated micronutrient deficiencies ([Chap. 75](#)). A variety of nutritional deficiencies can be identified by examination of the patient's general appearance, including skin, hair, nails, mucus membranes, and neurologic system ([Table 74-5](#)). Initially, a pinch of the posterior upper arm may reveal loss of subcutaneous fat in the malnourished patient. Hollowing of the temporal muscles, wasting of upper arms and thigh muscles, easily plucked hair, and peripheral edema are all consistent with protein deficiency. Examination of the skin may reveal the papular keratitis ("goose bump rash") of vitamin A deficiency, perifollicular hemorrhages of vitamin C deficiency, ecchymoses of vitamin K deficiency, the "flaky paint" lower extremity rash of zinc deficiency, hyperpigmentation of skin-exposed areas from niacin deficiency, seborrhea of essential fatty acid deficiency, spooning of nails in iron deficiency, and transverse nail pigmentation in protein deficiency. The eye examination yields conjunctival pallor of anemia, pericorneal and corneal opacities of severe vitamin A deficiency ("Bitot spots"), and nystagmus and isolated ocular muscle paresis of thiamine deficiency. The oral examination may reveal angular stomatitis and cheilosis of either riboflavin or niacin deficiency; glossitis with smooth and red tongue of riboflavin, niacin, vitamin B₁₂, or pyridoxine deficiency; and hypertrophied bleeding gums of vitamin C deficiency. Examination of the neurologic system, particularly in the setting of chronic alcohol abuse, may detect memory loss with confabulation, a wide-based gait, and past pointing, which, together with ophthalmoplegia and peripheral neuropathy, constitute the Wernicke-Korsakoff syndrome of thiamine deficiency. Other neurologic causes of

dementia include pellagra due to niacin and/or tryptophan deficiency. Additional causes of peripheral neuropathy include deficiencies of pyridoxine or vitamin E; loss of distal vibratory and position sense is characteristic of the subacute combined degeneration of vitamin B₁₂ deficiency.

LABORATORY ASSESSMENT

Selected use of laboratory tests, most of which are widely available, is essential for characterizing and quantifying malnutrition. Laboratory findings that are often attributed to chronic disease may, in actuality, reflect the response to [PCM](#) or selected micronutrient deficiencies in the setting of chronic illness.

Serum Visceral Proteins Serum albumin, which has a 2- to 3-week half-life, is a highly sensitive but nonspecific measure of [PCM](#). A normal serum albumin level in a well-hydrated patient is inconsistent with PCM. On the other hand, a low serum albumin level must be interpreted in its clinical context, since the concentration of albumin is decreased in the setting of increased plasma volume (as seen in acute trauma or sepsis and in chronic liver, renal, or cardiopulmonary failure). The acute stress of surgery, sepsis, or other acute inflammatory illness lowers the serum albumin level because of a combination of increased circulating extracellular volume and [TNF](#)- α -mediated inhibition of albumin synthesis. Hepatic albumin synthesis is inhibited in the setting of liver cirrhosis, AIDS, and disseminated cancer, whereas albumin loss from the body is accelerated in inflammatory bowel diseases, including ulcerative colitis, Crohn's disease, and radiation enteritis. Several shorter-lived visceral proteins can also be measured for estimation of the severity of PCM. These include transferrin (1-week half-life), prealbumin or retinol-binding protein complex (2-day half-life), and fibronectin (1-day half-life). However, like the serum albumin level, the circulating level of each of these proteins is affected by the changes in extracellular volume that occur in acute and chronic illnesses.

Vitamin and Mineral Assays Specific micronutrient deficiencies can be measured by a variety of serum and red blood cell assays, often utilizing high-performance liquid chromatography or enzyme or microbiologic assays ([Chap. 75](#)). Commonly available assays and their interpretations are listed in [Table 74-6](#). [PCM](#) is typically associated with low serum levels of vitamin A, zinc, and magnesium. Abnormal digestion and absorption of dietary fat are associated with deficiencies of fat-soluble vitamins A, D, and E, whereas intestinal mucosal malabsorption (as in celiac disease) is commonly associated with additional deficiencies of iron and folic acid. Chronic alcoholism is frequently associated with thiamine, folate, vitamin A, and zinc deficiencies. Vitamin B₁₂ deficiency due to achlorhydria occurs in up to 15% of elderly individuals as well as in those with pernicious anemia or with diseases involving the terminal ileum. As described in [Chap. 105](#), both folate and vitamin B₁₂ deficiencies are associated with elevations in plasma homocysteine; vitamin B₁₂ deficiency can also elevate the plasma level of methylmalonic acid.

Assessment of Immune Function [PCM](#) is associated with atrophy of thymic-dependent lymphoid structures and reduced T cell-mediated immunity. Conversely, B cell-mediated production of immunoglobulins is usually unaffected. Total lymphocyte count (total white cell count $\times$ fraction as lymphocytes) is often $<1000/\mu\text{L}$ in

PCM and may be accompanied by anergy to common skin test antigens. While sensitive for PCM, these measures of cell-mediated immunity are nonspecific and can be affected by other disorders such as acute or chronic infections, uremia, or immunosuppressive therapy.

SPECIALIZED PROCEDURES FOR NUTRITIONAL ASSESSMENT

Several specialized procedures are used to assess energy and protein stores and energy expenditure in malnourished patients. These procedures may be employed during the initial nutritional assessment or may serve as an index of the efficacy of nutritional support during the treatment of malnourished patients.

Bioelectric Impedance Analysis Bioelectric impedance analysis (BIA) is a simplified and portable method for measurement of body fat, [FFM](#), and total-body water. BIA is based on differences in the electric conductivity of a weak current between electrodes placed on the dorsal surfaces of the hands and feet. The measurement reflects differences in the impedance to electric current, which is greatest through fat and least through water. Lean body mass can be calculated as the difference between fat mass and body weight or as total-body water divided by 0.73.

Overall, [BIA](#) is most useful in assessing body fat and [FFM](#) in stable patients and in those who suffer from conditions leading to relative starvation. However, BIA can also be used to assess critically ill patients with decreased intracellular water space and [BCM](#) and expanded extracellular compartment size. Reduced BCM correlates inversely with increased metabolic rate. BIA may be confounded in AIDS patients receiving protease-inhibitor therapy, if they exhibit lipodystrophy with associated redistribution of interscapular, abdominal, and breast fat ([Chap. 309](#)).

Energy Expenditure Body weight and energy balance are sustained in health by the consumption of dietary calories in an amount equal to the daily expenditure of energy. Therefore, caloric needs can be determined from the estimated daily total energy expenditure (TEE), which is composed of basal or resting energy expenditure (REE, about 75% of total), the thermic expenditures of digestion (about 10% of total), and modest physical activity (about 15% of total). The REE is directly proportional to both the [FFM](#) and [BCM](#) and can be estimated in healthy people using the Harris and Benedict formula on the basis of weight in kg (W), height in cm (H), and age in years (A):

A simplified bedside estimation for [TEE](#) in sick patients is 25 kcal/kg of body weight, to which is added 10% for digestion or metabolism of intravenous or enteral nutrition. In the acutely ill patient, one should include an additional 12.5% for each degree of fever over 37°C, as well as an additional multiplier commensurate with the severity of illness (e.g., 25% for general surgery, 50% for sepsis, and 100% for extensive third-degree burns).

While [REE](#) can be predicted by the Harris-Benedict equations in healthy persons, it is

decreased in starvation because of hypometabolism. In contrast, REE is increased in the hypermetabolic stress that accompanies critical illness. REE and caloric requirements cannot be predicted in certain clinical conditions. These include the relatively starved, chronically ill patient admitted with a critical illness, the obese patient who develops a critical illness on the background of both increased body fat and [FFM](#), or the patient with chronic liver disease accompanied by combinations of anorexia and ongoing hepatic inflammation. In these situations, REE can be measured accurately by the gas-exchange method of indirect calorimetry. In practice, indirect calorimetry is performed at the bedside using a mobile metabolic cart. This procedure is applicable to ventilator-independent and -dependent patients whose fractional intake of oxygen is less than 0.45. Because the goal is to reach an accurate approximation of the 24-h energy requirement, measurements must be taken at intervals during the day and must account for several variables, including food intake and activity. To calculate the energy cost of metabolism by indirect calorimetry, the volumes (V) of oxygen consumed and carbon dioxide produced are measured over a given period of time, according to the modified Weir equation where

Indirect calorimetry also provides the respiratory quotient (RQ), which is the ratio of carbon dioxide produced to oxygen consumed during the process of gas collection. The RQ decreases when fat is the predominant substrate for metabolism (as in starvation) and increases when the contribution of carbohydrate increases (as during stress with gluconeogenesis). In healthy individuals, the RQ usually falls between 0.80 and 0.90. A RQ <0.7 is consistent with active ketogenesis from endogenous fatty acid metabolism with limited generation of carbon dioxide. An RQ >1.0 indicates net lipogenesis, or the conversion of substrate carbohydrate to fat, a situation that occurs with overfeeding. Values that fall outside the range of 0.65 to 1.25 suggest an error in measurement technique.

Creatinine Excretion in the 24-h Urine Creatinine, the metabolic product of skeletal muscle creatine, is produced at a constant rate and in an amount directly proportional to skeletal muscle mass. With steady-state day-to-day renal function, each gram of creatinine in the 24-h urine collection represents 18.5 g of fat-free skeletal muscle. Since skeletal muscle is the major component of [FFM](#) and [BCM](#), measurement of creatinine in the 24-h urine collection can be used as a relative measure of these body compartments during the initial assessment and/or during the course of nutritional support. The *creatinine coefficient* represents the amount of creatinine excreted per kilogram of body weight; it is equal to 23 mg/kg of ideal body weight in men and 18 mg/kg of ideal body weight in women. The *creatinine-height index* represents the ratio of the measured 24-h urine creatinine excretion to the value predicted by the creatinine coefficient for the patient's ideal body weight. These values can be calculated from estimation of the patient's ideal body weight ([Table 74-3](#)) or from tables that relate creatinine excretion to height in men and women ([Table 74-7](#)). In practice, the accuracy of the 24-h urine creatinine depends primarily on completeness of the urine collection. Together with variations due to fever and fluctuations in dietary intake, inaccuracies of urine collections may result in as much as 10% error in the quantitative 24-h urine creatinine measurement. The constancy of creatinine excretion depends on steady-state renal function, and unpredictable creatinine excretion may occur through

feces or skin in patients with serum creatinine levels $>530 \text{ } \mu\text{mol/L}$ ($>6 \text{ mg/dL}$). The presence of ascites, however, apparently does not compromise the accuracy of the 24-h urine creatinine as a reflection of FFM or BCM in patients with chronic liver disease.

Urine Nitrogen Excretion and Nitrogen Balance Nitrogen balance provides an index of protein gain or loss: 1 g nitrogen is equivalent to 6.25 g protein. Nitrogen balance can be assessed by measuring the difference between nitrogen consumed through the mouth, enteral tube, or intravenous sources and nitrogen excreted in the urine, feces, and other intestinal sources. Protein requirements to achieve zero or positive balance are less in starvation states, where daily protein losses are minimized because of hypometabolism, than in clinical states of stress, where the catabolism of skeletal muscle is accelerated for gluconeogenesis. Accurate measurement of nitrogen balance requires complete measurement of nitrogen losses from all possible excretory routes. In most cases, total urine nitrogen can be calculated by dividing 24-h urinary urea nitrogen by 0.85 and assuming approximately 2 g/d for nitrogen losses in feces and sweat. On the other hand, when the clinical condition includes extensive diarrhea and/or protein losses from pancreatic or enterocutaneous fistulas, the accuracy of nitrogen balance requires measurement of total nitrogen by the modified Kjeldahl technique in both urine and enteric sources. Total nitrogen measurements are also advisable in patients with liver failure, where urinary ammonia becomes a major and alternative source of nitrogen.

INTEGRATED BEDSIDE NUTRITIONAL ASSESSMENT

Several different approaches have been developed in order to simplify the process of nutritional assessment by using selective measurements that relate malnutrition to the specific medical condition and the severity of the underlying disease process.

Subjective Global Assessment This approach incorporates historic and physical findings as a basis for nutrition assessment by the trained physician. Major components in the history include evaluation of the extent of recent weight loss, changes in dietary intake, presence of significant gastrointestinal symptoms persisting more than 2 weeks, alterations in functional status, and the metabolic demand of the patient's underlying disease. Emphasis in the physical examination is placed on findings of depletion of subcutaneous body fat; skeletal muscle wasting; typical changes in skin, mucus membranes, and neurologic examination; as well as the presence of edema. Integration of the historic and physical data permits ranking of patients according to the following categories: adequate nutrition, moderate malnutrition, or severe malnutrition. Though the developers of the subjective global assessment have reported good sensitivity and specificity, the approach is still quite dependent on the training and experience of the clinician.

Prognostic Nutritional Assessment Several paradigms have been developed to link different parameters of nutritional assessment with clinical prognosis. Each approach links specific features of malnutrition with certain measurements of cell-mediated immunity, since abnormal immune function is a common pathway for increased risk in the malnourished patient ([Fig. 74-3](#)). A surgical prognostic nutritional index predicts morbidity based on preoperative measurements of serum albumin, transferrin, triceps skin-fold thickness, and delayed hypersensitivity to skin-test antigens.

Another [PCM](#) score was developed to link survival in alcoholic liver disease to both skin-fold and mid-arm muscle measurements; the creatinine-height index; values for serum albumin, transferrin, prealbumin, and retinol-binding protein; the total lymphocyte count; and the skin-test response to a series of antigens. The Maastricht index predicts survival in patients with serious gastrointestinal diseases on the basis of factors related to serum albumin, retinol-binding protein, lymphocyte count, and deviation from the patient's ideal body weight.

(Bibliography omitted in Palm version)

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75. VITAMIN AND TRACE MINERAL DEFICIENCY AND EXCESS- *Robert M. Russell*

Vitamins and trace minerals are required constituents of the human diet since they are either inadequately synthesized or not synthesized in the human body. Only small amounts of these substances are needed for carrying out essential biochemical reactions (e.g., acting as coenzymes or prosthetic groups). Overt vitamin or trace mineral deficiencies are rare in western countries due to a plentiful, varied, and inexpensive food supply; however, multiple nutrient deficiencies may appear together in persons who are ill or alcoholic. Moreover, subclinical vitamin and trace mineral deficiencies, as diagnosed by laboratory testing, are quite common in the normal population -- especially the geriatric population.

Body stores of vitamins and minerals vary tremendously. For example, vitamin B₁₂ and vitamin A stores are large, and an adult may not become deficient for 1 or more years after being on a depleted diet. However, folate and thiamine may become depleted within weeks when eating a deficient diet. Therapeutic modalities can deplete essential nutrients from the body; for example, hemodialysis removes water-soluble vitamins, which must be replaced by supplementation.

There are several roles for vitamins and trace minerals in diseases: (1) deficiencies of vitamins and minerals may be caused by disease states such as malabsorption; (2) both deficiency and excess of vitamins and minerals can cause disease in and of themselves (e.g., vitamin A intoxication and liver disease); and (3) vitamins and minerals in high doses may be used as drugs (e.g., niacin for hypercholesterolemia). The hematologic-related vitamins and minerals ([Chaps. 105,107](#)) are not considered in this chapter, nor are the bone-related vitamins and minerals (vitamin D, calcium, phosphorus; [Chap. 340](#)), as they are covered elsewhere.

VITAMINS

THIAMINE (VITAMIN B₁)

Thiamine was the first B vitamin to be identified and is therefore also referred to as vitamin B₁. Thiamine pyrophosphate, the coenzyme form of thiamine, is required for branched-chain amino acid metabolism and carbohydrate metabolism ([Fig. 75-1](#)).

Thiamine functions in the decarboxylation of α -ketoacids, such as pyruvate and α -ketoglutarate, and branched-chain amino acids and thus is a source of energy generation. In addition, thiamine pyrophosphate acts as a coenzyme for a transketolase reaction that mediates the conversion of hexose and pentose phosphates. It has also been postulated that thiamine plays a role in peripheral nerve conduction, although the exact chemical reactions underlying this function are unknown.

Absorption and Requirements At high doses, thiamine is absorbed by a passive mechanism; at low doses, it is absorbed by a carrier-mediated, active transport system and becomes phosphorylated in the process. Once absorbed, thiamine circulates bound to plasma proteins (mainly albumin) and erythrocytes. Storage sites for thiamine include muscle, heart, liver, kidney, and brain, although muscle is the principal storage site. The total body store of thiamine, mainly in the form of thiamine pyrophosphate, is approximately 30 mg, and its biologic half-life ranges between 9 and 18 days.

Given the fact that thiamine is involved in carbohydrate metabolism and energy generation, the Recommended Dietary Allowance (RDA) for males has been adjusted upward to account for increased energy utilization. Experiments have shown that heavy athletic training also increases thiamine utilization slightly. The RDA for thiamine is 1.2 mg/d for males and 1.1 mg/d for females. The median intake of thiamine in the United States from food alone is 2 mg/d. There is a 10% increase in the need for thiamine in pregnancy, and a small further increase in lactating females.

Primary food sources for thiamine include yeast, pork, legumes, beef, whole grains, and nuts. Milled and polished rice contain little, if any, thiamine. Thiamine deficiency is therefore more common in cultures that rely heavily on a rice-based diet. The molecule is heat-sensitive and is destroyed at pH > 8. Tea, coffee (caffeinated and decaffeinated), raw fish, and shellfish contain thiamineases, which can destroy the vitamin. Thus, drinking large amounts of tea or coffee can theoretically lower thiamine body stores.

Deficiency Most dietary deficiency of thiamine worldwide is the result of poor dietary intake. In western countries, the primary causes of thiamine deficiency are alcoholism and chronic illness, such as cancer. Alcohol is known to interfere directly with the absorption of thiamine and with the synthesis of thiamine pyrophosphate. Malnourished individuals with alcoholic liver disease are also at increased risk of thiamine deficiency because of diminished storage sites in liver and muscle. Thiamine should always be replenished when refeeding a patient with alcoholism, as carbohydrate repletion without adequate thiamine can precipitate acute thiamine deficiency.

Thiamine deficiency in its early stage induces anorexia, irritability, apathy, and generalized weakness. Prolonged thiamine deficiency causes beriberi, which is classically categorized as wet or dry, although there is considerable overlap. In either form of beriberi, patients may complain of pain and paresthesia. *Wet beriberi* presents primarily with cardiovascular symptoms, due to impaired myocardial energy metabolism and dysautonomia, and can occur after 3 months of a thiamine-deficient diet. Patients present with an enlarged heart, tachycardia, high-output congestive heart failure, peripheral edema, and peripheral neuritis. Patients with *dry beriberi* present with a symmetric peripheral neuropathy of the motor and sensory systems with diminished reflexes. The neuropathy affects the legs most markedly, and patients have difficulty rising from a squatting position.

Alcoholic patients with chronic thiamine deficiency may also have central nervous system manifestations known as *Wernicke's encephalopathy*, consisting of horizontal nystagmus, ophthalmoplegia (due to weakness of one or more extraocular muscles), cerebellar ataxia, and mental impairment ([Chap. 387](#)). When there is an additional loss of memory and a confabulatory psychosis, the syndrome is known as *Wernicke-Korsakoff syndrome*. Although this syndrome is generally described in alcoholic patients, there may be a genetic predisposition to Wernicke-Korsakoff that involves a variant transketolase isozyme.

In severely malnourished infants 2 to 3 months old, thiamine deficiency may occur precipitously with sudden cardiovascular failure and collapse, resulting in death within

hours. In addition, infants with thiamine deficiency may present with features suggesting meningitis, including vomiting, nystagmus, and convulsions. An aphonic presentation has also been described in which there is extreme irritability and either a very hoarse cry or total inability to emit any noise whatsoever (a silent scream).

The laboratory diagnosis of thiamine deficiency is usually made by a functional enzymatic assay of transketolase activity measured before and after the addition of thiamine pyrophosphate. A >25% stimulation by the addition of thiamine pyrophosphate (an activity coefficient of 1.25) is taken as abnormal. Thiamine or the phosphorylated esters of thiamine in serum or blood can also be measured by high-performance liquid chromatography (HPLC) to detect deficiency. Moreover, a urinary level of thiamine <27 ug per gram of creatinine per day is abnormal. In measuring urinary excretion of thiamine, one should make sure the patient is not taking diuretics, which increase thiamine excretion.

TREATMENT

In acute thiamine deficiency with either cardiovascular or neurologic signs, 100 mg/d of thiamine should be given parenterally for 7 days, followed by 10 mg/d orally until there is complete recovery. Cardiovascular improvement occurs in $\leq$ 12 h, and ophthalmoplegic improvement occurs within 24 h. Other manifestations gradually clear, although psychosis in the Wernicke-Korsakoff syndrome may be permanent or persist for several months. Consistent with this, pathologic changes occur in the cortex, cerebellum, and mammillary bodies of the thalamus. Parenteral thiamine should be given prophylactically to all chronic alcoholic patients in the emergency room, or as soon as they are admitted, to prevent precipitation of thiamine deficiency after the provision of glucose-containing solutions.

Thiamine-responsive conditions requiring pharmacologic doses of thiamine include branched-chain ketoaciduria (maple sugar urine disease), subacute necrotizing encephalopathy due to thiamine triphosphate deficiency in the brain (Leigh syndrome), thiamine-responsive lactic acidosis, and thiamine-responsive megaloblastic anemia associated with diabetes mellitus and deafness ([Chap. 353](#)). The gene for this recessive disorder, *SLC19A2*, encodes a thiamine transporter.

Toxicity Although anaphylaxis has been reported after high doses of thiamine, no adverse effects have been recorded from either food or supplements at high doses. Thiamine supplements may be bought over the counter in doses of up to 50 mg/d.

RIBOFLAVIN (VITAMIN B₂)

Riboflavin is important for the metabolism of fat, carbohydrate, and protein, reflecting its role as a respiratory coenzyme and an electron donor. Riboflavin is esterified with phosphoric acid in the body to form two coenzymes, flavin-mononucleotide (FMN) and flavin adenine dinucleotide (FAD) which are involved in a variety of cellular oxidation-reduction processes. Enzymes that contain FAD or FMN as prosthetic groups are known as *flavoenzymes* (e.g., succinic acid dehydrogenase, monoamine oxidase, glutathione reductase). Riboflavin plays an important role in niacin metabolism, since flavoenzymes act as intermediaries in the oxidation of the reduced forms of

nicotinamide adenine dinucleotide (NAD) and NAD phosphate (NADP).

Riboflavin is normally absorbed by active and carrier-mediated saturable mechanisms, whereas diffusion is the principal mechanism of absorption at high concentrations. Riboflavin phosphorylation takes place mainly in the wall of the small intestine. Both [FMN](#) and [FAD](#) are bound to immunoglobulins and albumin in the circulation, and both forms are stored to some degree in liver and muscle.

Although much is known about the chemical and enzymatic reactions of riboflavin, the clinical manifestations of riboflavin deficiency are nonspecific and similar to those of other B vitamin deficiencies. Further, riboflavin deficiency usually occurs in combination with other water-soluble vitamin deficiencies ([Chap. 74](#)). Riboflavin deficiency is manifested principally by lesions of the mucocutaneous surfaces of the mouth (angular stomatitis, cheilosis, atrophic glossitis, magenta tongue, pharyngitis) and skin (seborrhea, genital dermatitis). In addition to the mucocutaneous lesions, corneal vascularization, anemia, and personality changes have been described with riboflavin deficiency.

Deficiency and Excess Riboflavin deficiency is almost always due to dietary deficiency. The requirement for riboflavin is increased during pregnancy and lactation and possibly by heavy exercise. The use of phenothiazines and antibiotics also appears to increase the need for riboflavin. Milk, other dairy products, and enriched breads and cereals are the most important dietary sources of riboflavin in the United States, although lean meat, fish, eggs, broccoli, and legumes are also good sources. Riboflavin is extremely sensitive to light, and milk should be stored in containers that protect against photodegradation. In non-milk-drinking societies (e.g., Central America), the laboratory diagnosis of riboflavin deficiency is common. Laboratory diagnosis of riboflavin deficiency can be made by measurement of red blood cell or urinary riboflavin concentrations or by measurement of erythrocyte glutathione reductase activity, with and without added [FAD](#). A stimulation (activity coefficient) of >1.4 is diagnostic of a deficient state. The [RDA](#) for riboflavin is 1.1 to 1.3 mg/d in adults, with slightly higher recommendations for lactating and pregnant women. Rare genetic defects of flavoprotein synthesis may require pharmacologic doses of riboflavin for treatment. Because the capacity of the gastrointestinal tract to absorb riboflavin is limited (~20 mg if given in one oral dose), riboflavin toxicity has not been described. Thus, the most recent revision of the RDAs did not set an upper limit for this nutrient.

NIACIN (VITAMIN B₃)

The term *niacin* refers to nicotinic acid and nicotinamide and their biologically active derivatives. Nicotinic acid and nicotinamide serve as precursors of two coenzymes, [NAD](#) and [NADP](#). These coenzymes are important in numerous oxidation and reduction reactions in the body. NAD and NADP serve as cofactors for dehydrogenases and are involved in the transfer of the hydride ion in many redox reactions. Thus, niacin is important in pentose, steroid, and fatty acid biosynthesis; glycolysis; protein metabolism; and the oxidation of fuels such as lactate, pyruvate, and alcohol. In addition, NAD and NADP are active in adenine diphosphate-ribose transfer reactions involved in DNA repair and calcium mobilization.

Absorption, Metabolism, and Requirements Nicotinic acid and nicotinamide are absorbed well from the stomach and small intestine. Both forms of niacin are absorbed by a sodium-dependent, facilitated diffusion mechanism at low doses, whereas passive diffusion occurs at high doses. Some storage of [NAD](#) takes place in the liver. The amino acid tryptophan can be converted to niacin with an efficiency of 60:1 by weight. Thus, the [RDA](#) for niacin is expressed in niacin equivalents. Greater conversion of tryptophan to niacin occurs in niacin-deficient states, pregnancy, and in women using oral contraceptives. However, a lower conversion efficiency occurs if a patient is vitamin B₆- or riboflavin-deficient. The drug isoniazid inhibits the conversion of tryptophan to niacin. The urinary excretion products of niacin include nicotinic acid and niacin oxide; however, the major urinary metabolites are 2-pyridone and 2-methyl nicotinamide, measurements of which are used in diagnosis of niacin deficiency.

The [RDA](#) for niacin is 16 niacin equivalents per day for men and 14 niacin equivalents for women. Median intakes of niacin in the United States considerably exceed these values. Diets that are corn-based can predispose to niacin deficiency due to the low tryptophan and niacin content. Niacin bioavailability is high from beans, milk, meat, and eggs; bioavailability from cereal grains is lower. Since flour is enriched with the "free" niacin (i.e., non-coenzyme form), bioavailability is excellent.

Deficiency Niacin deficiency causes *pellagra*, which is mostly found among people eating corn-based diets in parts of China, Africa, and India. Pellagra in North America is found mainly among alcoholics; in patients with congenital defects of intestinal and kidney absorption of tryptophan (Hartnup's disease; [Chap. 352](#)); and in patients with carcinoid syndrome ([Chap. 93](#)), in which there is increased conversion of tryptophan to serotonin. The early symptoms of pellagra include loss of appetite, generalized weakness and irritability, abdominal pain, and vomiting. Epithelial cell changes then ensue with stomatitis and bright-red glossitis, followed by a characteristic skin rash that is pigmented and scaling, particularly in skin areas exposed to sunlight. This rash is known as "Casal's necklace," when it rings the neck, and is seen in advanced cases. Vaginitis and esophagitis may also occur. Diarrhea (in part due to proctitis and in part due to malabsorption), depression, seizures, and dementia are also part of the pellagra syndrome -- the four D's: dermatitis, diarrhea, and dementia leading to death.

The diagnosis of niacin deficiency is based on low levels of the urinary metabolites 2-methyl nicotinamide and 2-pyridone. Treatment of pellagra consists of oral supplementation of 100 to 200 mg of nicotinamide or nicotinic acid three times daily for 5 days. High doses of nicotinic acid (3 g nicotinic acid per day) are used for the treatment of elevated cholesterol levels and in the treatment of types 2, 4, and 5 hyperlipidemias ([Chap. 344](#)).

Toxicity Prostaglandin-mediated flushing has been observed at daily doses as low as 50 mg of niacin when taken as a supplement or as therapy for hypertriglyceridemia. No toxicity has been seen from niacin derived from food sources. Flushing may be accompanied by skin dryness, itching, and headache. Premedication with aspirin may alleviate these symptoms. Nausea, vomiting, and abdominal pain also occur at similar doses of niacin. Hepatic toxicity is the most serious toxic reaction due to niacin and may present as jaundice with elevated AST and ALT levels. A few cases of fulminant hepatitis requiring liver transplantation have been reported at doses of 3 to 9 g/d. Other

toxic reactions include glucose intolerance, macular edema, and macular cysts. It is not clear whether sustained-release forms of nicotinic acid are more toxic than regular forms. The upper limit for daily niacin intake has been set at 35 mg. However, this upper limit does not pertain to the therapeutic use of niacin.

PYRIDOXINE (VITAMIN B₆)

Vitamin B₆ refers to a family of compounds including pyridoxine, pyridoxal, pyridoxamine, and their 5¢-phosphate derivatives. 5¢-Pyridoxal phosphate (PLP) is a cofactor for more than 100 enzymes involved in amino acid metabolism (e.g., 5¢-PLP is a cofactor for the transulfuration enzymes involved in the conversion of homocysteine to cystathionine; [Chap. 352](#)). Vitamin B₆ is also involved in heme and neurotransmitter synthesis and in the metabolism of glycogen, lipids, steroids, sphingoid bases, and several vitamins, including the conversion of tryptophan to niacin.

Absorption and Metabolism Approximately 75% of vitamin B₆ is absorbed from a mixed diet by a nonsaturable, passive process. Much of dietary vitamin B₆ is in the phosphorylated form, and the phosphate must be removed by intestinal alkaline phosphatase before absorption takes place. Once absorbed, the vitamin becomes rephosphorylated in the liver, where the various forms can be interconverted. In the liver, [PLP](#) binds avidly to cellular proteins and albumin. Since these binding proteins protect it from phosphatase activity, tissue levels of PLP can become quite high with continuous supplementation. Sixty mg of vitamin B₆ is stored in the body, and much is in the form of PLP bound to phosphorylase A in muscle. The biologic half-life of vitamin B₆ is 25 days.

Dietary Sources Plants contain vitamin B₆ in the form of pyridoxine, whereas animal tissues contain [PLP](#) and pyridoxamine phosphate. The vitamin B₆ contained in plants is less bioavailable than that from animal tissues. All forms of vitamin B₆ are labile in alkaline conditions. Rich food sources of vitamin B₆ include legumes, nuts, wheat bran, and meat, although the vitamin is present in all food groups. The [RDA](#) for young adults (both males and females) has been set at 1.3 mg/d. For older adults, the RDA is slightly higher (1.5 mg/d for women, 1.7 mg/d for men).

Deficiency Symptoms of vitamin B₆ deficiency include seborrheic dermatitis, glossitis, stomatitis, and cheilosis, as frequently seen with other B vitamin deficiencies ([Chap. 74](#)). In addition, severe vitamin B₆ deficiency can lead to generalized weakness, irritability, peripheral neuropathy, abnormal electroencephalograms, and personality changes including depression and confusion. In infants, diarrhea, seizures, and anemia have been reported. Microcytic, hypochromic anemia is due to diminished hemoglobin synthesis, since the first enzyme involved in heme biosynthesis (amino-levulinate synthase) requires [PLP](#) as a cofactor ([Chap. 104](#)). In some case reports, platelet dysfunction has also been reported. Since vitamin B₆ is necessary for the conversion of homocysteine to cystathionine, it is possible that chronic low-grade vitamin B₆ deficiency may result in hyperhomocystinemia and increased risk of cardiovascular disease ([Chaps. 242](#) and [352](#)).

Certain medications such as isoniazid, L-dopa, penicillamine, and cycloserine interact with [PLP](#) due to a reaction with carbonyl groups. Oral contraceptives have been reported

to decrease vitamin B₆ status indicators, although the mechanism for this is uncertain. Alcoholism also decreases vitamin B₆ status due to poor diet, liver disease, and the fact that acetaldehyde can compete with PLP for protein binding, leading to increased degradation and excretion. The increased ratio of aspartate aminotransferase (AST or SGOT) to alanine aminotransferase (ALT or SGPT) seen in alcoholic liver disease reflects the relative vitamin B₆ dependence of ALT. Vitamin B₆ requirements are higher in preeclampsia, eclampsia, and hemodialysis. Vitamin B₆ dependency syndromes that require pharmacologic doses of vitamin B₆ are rare, but include cystathionine b-synthase deficiency, pyridoxine-responsive (primarily sideroblastic) anemias, and gyrate atrophy with chorioretinal degeneration due to decreased activity of the mitochondrial enzyme ornithine aminotransferase. In these situations, 100 to 200 mg/d of oral vitamin B₆ are required for treatment.

High doses of vitamin B₆ have been used to treat carpal tunnel syndrome, premenstrual tension, schizophrenia, autism, and diabetic neuropathy but have not been found to be effective.

The laboratory diagnosis of vitamin B₆ deficiency is generally made on the basis of low plasma [PLP](#) values (<20 nmol/L). Other measures of vitamin B₆ deficiency include low erythrocyte levels of PLP, low plasma pyridoxal, and low urinary levels of 4-pyridoxic acid. Treatment of vitamin B₆ deficiency is 50 mg/d; higher doses of 100 to 200 mg/d are given if vitamin B₆ deficiency is related to medication use. Vitamin B₆ should not be given with L-dopa, since the vitamin interferes with the action of this drug.

Toxicity The safe upper limit for vitamin B₆ has been set at 100 mg/d, although the lowest dose at which toxicity (sensory neuropathy) has been seen is 500 mg/d. No adverse effects have been associated with high intakes of vitamin B₆ from food sources only. When toxicity occurs, it causes a severe sensory neuropathy, leaving patients unable to walk. Some cases of photosensitivity and dermatitis have also been reported.

VITAMIN C

Both ascorbic acid and its oxidized product dehydroascorbic acid are biologically active. Vitamin C participates in oxidation-reduction reactions and hydrogen ion transfer reactions. As an antioxidant, vitamin C donates electrons to quench reactive free radical and oxygen species. It also acts to regenerate other antioxidants such as vitamin E, flavonoids, and glutathione. Other actions of vitamin C include promotion of nonheme iron absorption, carnitine biosynthesis, and the conversion of dopamine to norepinephrine. Vitamin C is also important for connective tissue metabolism and cross-linking and is a component of many drug-metabolizing enzyme systems, particularly the mixed-function oxidase systems. As such, the vitamin participates in the synthesis of corticosteroids, aldosterone, and the metabolism of cholesterol. Vitamin C also participates in enzymatic reactions requiring a reduced metal, although the exact molecular basis for this role has not been delineated.

Absorption and Physiology Vitamin C is absorbed by an energy-dependent, saturable transport system, and a progressively smaller proportion of the vitamin is absorbed with increasing dose. Almost complete absorption of the vitamin occurs if <100 mg is administered in a single dose; however, only 50% or less is absorbed at doses >1 g.

Enhanced degradation and fecal and urinary excretion of vitamin C occur at higher intake levels. High levels of the reduced form of vitamin C are contained in white blood cells, lens tissue, and brain. The maximum body pool in adult males is approximately 1500 mg, and 3% of this body pool is turned over each day, resulting in a half-life of approximately 18 days.

Dietary Sources and Requirements Good dietary sources of vitamin C include citrus fruits, green vegetables (especially broccoli), tomatoes, and potatoes. Appreciable amounts of vitamin C may be consumed as an antioxidant food additive, and the consumption of five servings of fruits and vegetables a day provides vitamin C in excess of the [RDA](#) of 60 mg/d for males and females. Moreover, approximately 40% of the U.S. population takes vitamin C as a dietary supplement. Vitamin C requirements are increased slightly to 70 mg in pregnancy and are increased further to 90 to 95 mg/d during lactation. Smoking, hemodialysis, and stress (e.g., infection, trauma) appear to increase vitamin C requirements. "Natural forms" of vitamin C are no more bioavailable than synthetic forms.

Deficiency Vitamin C deficiency causes scurvy; in the United States, this is seen primarily among poor and elderly people and alcoholics who consume <10 mg/d of vitamin C. Vitamin C deficiency has also been described among individuals consuming macrobiotic diets. Scurvy occurs when the body pool for vitamin C drops to <300 mg/d and plasma levels drop to <11 $\mu\text{mol/L}$. Symptoms of scurvy primarily reflect impaired formation of mature connective tissue and include bleeding into skin (petechiae, ecchymoses, perifollicular hemorrhages); inflamed and bleeding gums; and manifestations of bleeding into joints, the peritoneal cavity, pericardium, and the adrenal glands. Other generalized symptoms include weakness, fatigue, and depression. In children, vitamin C deficiency may cause impaired bone growth. Laboratory diagnosis of vitamin C deficiency is made on the basis of low plasma or leukocyte levels.

Administration of vitamin C (200 mg/d) results in marked improvement in the symptoms of scurvy in a matter of several days. High-dose vitamin C supplementation (e.g., 1 to 2 g/d) has been shown to slightly decrease the symptoms and duration of upper respiratory tract infections and to improve glycemic control. Vitamin C supplementation has also been reported to be useful in Chediak-Higashi syndrome ([Chap. 64](#)) and osteogenesis imperfecta ([Chap. 351](#)). It has been claimed that foods high in vitamin C may lower the incidence of certain cancers, particularly esophageal and gastric cancers. If proven, this effect may be due to the fact that vitamin C can prevent the conversion of nitrites and secondary amines to carcinogenic nitrosamines. However, one intervention study from China did not show vitamin C to be protective. Other chronic diseases for which diets high in vitamin C have been reported to be protective include cardiovascular disease, stroke, and cataracts. However, these studies are correlational, and no large-scale intervention studies have been reported.

Toxicity Taking >2 g of vitamin C in a single dose may result in abdominal pain, diarrhea, and nausea; doses >3 g have been reported to elevate blood levels of alanine aminotransferase, lactic acid dehydrogenase, and uric acid. Since vitamin C may be metabolized to oxalate, it has been feared that chronic, high-dose vitamin C supplementation could result in an increased prevalence of kidney stones. However, this has not been borne out in several trials, except in individual patients with preexisting

renal disease. Thus, it is reasonable to advise patients with a past history of kidney stones not to take large doses of vitamin C. There is also an unproven, but possible risk that chronic high doses of vitamin C could promote iron overload in patients taking supplemental iron. High doses of vitamin C can induce hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency, and doses >1 g/d can cause false-negative guaiac reactions as well as interfering with tests for urinary glucose.

BIOTIN

Biotin is a water-soluble vitamin with a bicyclic structure. The vitamin plays an important role in gluconeogenesis and fatty acid synthesis and serves as a CO₂carrier on the surface of both cytosolic and mitochondrial carboxylase enzymes. The vitamin also functions in the catabolism of specific amino acids (e.g., leucine).

Biotin in food sources is bound to protein from which it must be cleaved in order to be absorbed. The enzyme biotinidase dissociates the vitamin and facilitates its subsequent transport. Excellent food sources of biotin include liver, soy, beans, yeast, and egg yolks, although egg white contains the protein avidin that strongly binds the vitamin and reduces its bioavailability. Biotin is contained in moderate amounts in legumes, nuts, mushrooms, cauliflower, and certain cereals. Although biotin is synthesized by intestinal bacteria, the relative importance of this source in humans is uncertain. The recommended intake of biotin for adults is 30 ug/d and 35 ug/d in lactating women.

Biotin deficiency has been induced by experimental feeding of egg white diets and in patients with short bowels who received biotin-free parenteral nutrition. In the adult, biotin deficiency results in mental changes (depression, hallucinations), paresthesia, anorexia, and nausea. A scaling, seborrheic, and erythematous rash may occur around the eyes, nose, and mouth as well as on the extremities. In infants, biotin deficiency presents as hypotonia, lethargy, and apathy. In addition, the infant may develop alopecia and a characteristic rash that includes the ears. Two types of inherited infantile biotin deficiency states have been described. Multiple carboxylase deficiency syndrome is an autosomal recessive disorder that is expressed during the first week of life and is characterized by severe metabolic ketoacidosis and dermatitis. Treatment requires pharmacologic doses of biotin, using up to 10 mg/d. Late-onset infantile biotin deficiency due to absorptive and transport defects occurs between 3 and 6 months with dermatitis, seizures, ataxia, hypotonia, and variable metabolic acidosis. The laboratory diagnosis of biotin deficiency can be established based on a decreased urinary concentration.

PANTOTHENIC ACID

Pantothenic acid is a component of coenzyme A and phosphopantetheine, which are involved in fatty acid metabolism and the synthesis of cholesterol, steroid hormones, and all compounds formed from isoprenoid units. In addition, pantothenic acid is involved in the acetylation of proteins. Pantothenic acid is actively transported when given at low doses, but it is passively absorbed when given at high doses. The vitamin is excreted in the urine, and the laboratory diagnosis of deficiency is made on the basis of low urinary vitamin levels.

The vitamin is ubiquitous in the food supply. Liver, yeast, egg yolks, and vegetables are

particularly good sources. The recommended adequate intake for adults is 5 mg/d. Human pantothenic acid deficiency has only been demonstrated in experimental feeding of diets low in pantothenic acid or by giving a specific pantothenic acid antagonist. The symptoms of pantothenic acid deficiency are nonspecific and include gastrointestinal disturbance, depression, muscle cramps, paresthesia, ataxia, and hypoglycemia. Pantothenic acid deficiency was thought to cause the burning feet syndrome seen in prisoners of war during World War II. No toxicity of this vitamin has been reported.

CHOLINE

Choline is a precursor for acetylcholine, phospholipids, and betaine. Choline is necessary for the structural integrity of cell membranes, cholinergic neurotransmission, lipid and cholesterol metabolism, and transmembrane signaling. Recently, a recommended adequate intake was set at 550 mg/d for adult males and 425 mg/d for adult females. Choline is thought to be a "conditionally essential" nutrient, in that de novo synthesis occurs in the liver and is less than the vitamin's utilization only under certain stress conditions. Choline deficiency has occurred in patients receiving parenteral nutrition devoid of choline. Deficiency results in fatty liver and elevated transaminase levels. The diagnosis of choline deficiency is made on the basis of low plasma levels.

Toxicity from choline results in hypotension, cholinergic sweating, diarrhea, salivation, and a fishy body odor. The upper limit for choline has been set at 3.5 g/d. Therapeutically, choline has been suggested for patients with dementia and for patients at high risk of cardiovascular disease, due to its ability to lower cholesterol and homocysteine levels. However, such benefits have yet to be documented.

VITAMIN A

Vitamin A, in the strictest sense, refers to retinol. However, the oxidized metabolites, retinaldehyde and retinoic acid, are also biologically active compounds. The term *retinoids* includes synthetic molecules that are chemically related to retinol. Retinaldehyde is the essential form of vitamin A that is required for normal vision, whereas retinoic acid is necessary for normal morphogenesis, growth, and cell differentiation. Retinoic acid does not function in vision and, in contrast to retinol, is not involved in reproduction. Vitamin A also plays a role in iron utilization, humoral immunity, T cell-mediated immunity, natural killer cell activity, and phagocytosis. Vitamin A is commercially available in esterified forms (e.g., acetate, palmitate) since it is more stable as an ester.

There are over 600 carotenoids in nature, and approximately 50 of these can be metabolized to vitamin A. β -Carotene is the most prevalent carotenoid in the food supply that has provitamin A activity. Although the breakdown of β -carotene should theoretically yield two molecules of vitamin A, the conversion of carotenoids to vitamin A, in fact, is much less efficient. It is estimated that 6 μ g or greater of dietary β -carotene is equivalent to 1 μ g of retinol, whereas 12 μ g or greater of other dietary provitamin A carotenoids (e.g., cryptoxanthin, α -carotene) is equivalent to 1 μ g of retinol.

Absorption and Metabolism Approximately 80% of preformed vitamin A is absorbed

from food, and absorption is via a carrier-mediated mechanism at low concentrations and passive diffusion at high concentrations. Approximately 15 to 30% of provitamin A carotenoids are absorbed passively from the diet, and the absorption becomes much less efficient at high dosage. The absorption of both vitamin A and carotenoids are partially dependent on an adequate bile concentration within the intestinal lumen for the formation of micelles. Once a provitamin A carotene is absorbed into the epithelial cell, a small proportion of it is split to form vitamin A. At higher doses of *b*-carotene, the conversion to vitamin A is less efficient, thereby preventing vitamin A toxicity. The absorption of both vitamin A and intact *b*-carotene is via the lymphatics after chylomicron formation.

Hepatic clearance of vitamin A in chylomicrons is efficient, and the liver contains approximately 90% of the vitamin A reserves. Approximately 10 to 40% of a vitamin A dose is oxidized or conjugated in the liver and excreted in urine or bile. Of a given dose of vitamin A, approximately 50% enters the liver storage pool. Storage of vitamin A takes place in the lipid storage (Ito) cell of the liver, which is also a collagen-producing cell. The liver secretes vitamin A in the form of retinol, which is bound to retinol-binding protein. Once this has occurred, the retinol-binding protein complex interacts with a second protein, transthyretin. This trimolecular complex functions to prevent vitamin A from being filtered by the kidney glomerulus, to protect the body against the toxicity of retinol and to allow retinol to be taken up by specific cell-surface receptors that recognize retinol-binding protein. A certain amount of vitamin A enters peripheral cells even if it is not bound to retinol-binding protein. After retinol is internalized by the cell, it becomes bound to a series of cellular retinol-binding proteins, which function as sequestering and transporting agents as well as cofactors for enzymatic reactions. Certain cells also contain retinoic acid-binding proteins, which have the same sequestering functions as well as enabling retinoic acid metabolism.

11-*cis*-Retinaldehyde functions as a visual pigment chromophore to capture light. Rhodopsin is composed of the protein opsin and retinaldehyde and is contained in the rod cells, whereas iodopsin is contained in cones. When the dark-adapted retina is exposed to light, the 11-*cis*-retinaldehyde contained in rhodopsin isomerizes to an all-*trans* form. This conformational change causes dissociation from the opsin, resulting in a nerve impulse and a visual response. Once the retina returns to dim light conditions, rhodopsin is regenerated.

Retinoic acid is a ligand for certain nuclear receptors that act as transcription factors. Two families of receptors (RAR and RXR receptors) are active in retinoid-mediated gene transcription. Retinoid receptors regulate transcription by binding as dimeric complexes to specific DNA sites, the retinoic acid response elements, in target genes ([Chap. 327](#)). The receptors can either stimulate or repress gene expression in response to their ligands. RAR binds all-*trans* retinoic acid and 9-*cis* retinoic acid, whereas RXR binds only 9-*cis* retinoic acid.

The retinoid receptors play an important role in controlling cell proliferation and differentiation. Retinoic acid is useful in the treatment of promyelocytic leukemia ([Chap. 111](#)). In this case, a gene rearrangement fuses the RAR to one of several other genes [e.g., t(15;17)], causing an apparent block in cell differentiation. Treatment with retinoic acid activates the RAR, dissociating repressor complexes and leading to cell

differentiation and more normal cell turnover. Retinoic acid is also used in the treatment of cystic acne because it inhibits keratinization, decreases sebum secretion, and possibly alters the inflammatory reaction ([Chap. 56](#)). RXRs dimerize with other nuclear receptors to function as coregulators of genes responsive to retinoids, thyroid hormone, and calcitriol. RXR agonists induce insulin sensitivity experimentally, perhaps because RXR is a cofactor for the peroxisome-proliferator-activated receptors (PPARs), which are targets for the thiazolidinedione drugs such as rosiglitazone and troglitazone ([Chap. 333](#)).

Dietary Sources The retinol equivalent (RE) is used to express the vitamin A value of food. One RE is defined as 1 μg of retinol (0.003491 mmol). In the past, 1 RE was considered to be equal to 6 μg of β -carotene, but additional studies indicate that 1 RE may, in fact, be equal to 12 to 20 μg of β -carotene from a dietary source. In older literature, vitamin A was often expressed in international units (IU), with 1 RE being equal to 3.33 IU of retinol and 12 IU of β -carotene, but these units are no longer in current medical or scientific use. The RDA for vitamin A is set at 1000 RE for adult males and 800 RE for adult females.

Liver and fish are excellent food sources for preformed vitamin A; vegetable sources of provitamin A carotenoids include dark-green and -colored fruits and vegetables. Diets consisting mainly of rice, wheat, maize, and tubers can produce vitamin A deficiency, as few carotenoids are contained in these foods. In areas where these foods are staples, children are particularly susceptible to vitamin A deficiency because neither breast nor cow's milk supplies enough vitamin A to prevent deficiency. Areas of the world where vitamin A deficiency is particularly prevalent include parts of Africa, South America, and Southeast Asia. Vitamin A deficiency occurs in more than 250,000 children each year, resulting in blindness and a 50% mortality rate within the year. In western countries, vitamin A deficiency is seen primarily among patients with diseases associated with fat malabsorption (e.g., celiac sprue, short-bowel syndrome). Concurrent zinc deficiency can interfere with the mobilization of vitamin A from liver stores as well as the synthesis of rhodopsin in the eye; thus vitamin A deficiency is exacerbated by concurrent zinc deficiency. Alcohol also interferes with the conversion of retinol to retinaldehyde in the eye by competing for alcohol (retinol) dehydrogenase. Drugs that interfere with the absorption of vitamin A include mineral oil, neomycin, and cholestyramine.

Deficiency Symptoms of vitamin A deficiency include hyperkeratotic skin lesions, night blindness (inability to see in dim light), dryness of the eyes, xerosis, and Bitot spots, which are white patches of keratinized epithelium appearing on the sclera ([Fig. 75-CD1](#)). Aggressive xerophthalmia can result in corneal ulceration. If untreated, proteolytic destruction and rupture of the cornea ensues with permanent blindness, although vitamin A treatment of patients with corneal ulcers can also result in blindness due to permanent corneal scarring. Children with vitamin A deficiency have increased mortality, primarily from infectious diseases, measles, respiratory diseases, and diarrhea. Extremely low birth weight infants (<1000 g) should be treated parenterally with 5000 IU (1500 μg or RE) of vitamin A three times a week for 4 weeks.

There are no specific deficiency signs or symptoms that result from carotenoid deficiency. However, dietary carotenoids have been suggested to protect against cataract formation, low-density lipoprotein (LDL) oxidation, and certain cancers. It was

hoped that β -carotene would be an effective chemopreventive for cancer because numerous epidemiologic studies had shown that diets high in β -carotene were associated with lower incidences of cancers of the respiratory and digestive system. However, intervention studies using high doses of β -carotene actually resulted in more lung cancers than in placebo-treated groups. Non-provitamin A carotenoids, such as lutein and zeaxanthin, have been suggested to protect against macular degeneration. The non-provitamin A carotenoid lycopene has been suggested to protect against prostate cancer. However, the effectiveness of these agents has not been proven by intervention studies, and the mechanisms underlying these purported biologic actions are unknown.

The diagnosis of vitamin A deficiency is made by measurement of serum retinol (normal range, 30 to 65 $\mu\text{g/dL}$), tests of dark adaptation, impression cytology of the conjunctiva (decreased numbers of mucous-secreting cells), or measurement of body storage pools, either directly by liver biopsy or by isotopic dilution after administering a stable isotope of vitamin A.

Vitamin A deficiency with ocular changes should be treated by administering 100,000 IU (30 mg) of vitamin A intramuscularly, or 200,000 IU (60 mg) orally. In areas of endemic vitamin A deficiency, this is followed by vitamin A capsules of 200,000 IU at 6-month intervals. Vitamin A deficiency in patients with malabsorptive diseases, who have abnormal dark adaptation or symptoms of night blindness without ocular changes, should be treated for 1 month with 50,000 IU/d (15 mg/d) orally of a water micelle preparation of vitamin A. This is followed by lower maintenance doses with the exact amount determined by monitoring serum retinol.

Toxicity Acute toxicity of vitamin A was first noted in Arctic explorers after eating polar bear liver and has been seen after administration of 150 mg in adults or 100 mg in children. Acute toxicity is manifest by increased intracranial pressure, vertigo, diplopia, bulging fontanels in children, seizures, and exfoliative dermatitis; it may result in death. Chronic vitamin A intoxication has been seen in normal adults who ingest 50,000 IU/d (15 mg/d) of vitamin A for a period of several months and in children who ingest 20,000 IU/d (6 mg/d). Manifestations include dry skin, cheilosis, glossitis, vomiting, alopecia, bone pain, hypercalcemia, lymph node enlargement, hyperlipidemia, amenorrhea, and features of pseudotumor cerebri with increased intracranial pressure and papilledema. Liver fibrosis with portal hypertension and bone demineralization may also result from chronic vitamin A intoxication. When vitamin A is provided in excess of pregnant women, congenital malformations have included spontaneous abortions, craniofacial abnormalities, and valvular heart disease. In pregnancy, the daily dose of vitamin A should not exceed 10,000 IU (3 mg). Elderly individuals appear to be more prone to vitamin A intoxication, as are alcoholics and patients with liver disease. In fact acute hepatitis may precipitate vitamin A intoxication in patients who have extremely high vitamin A stores in the liver. It should be noted that the commercially available retinoid derivatives are also toxic, including 13-*cis*-retinoic acid, which has been associated with birth defects. As a result, contraception should be continued for at least 1 year, and possibly longer, in women who have taken 13-*cis* retinoic acid.

High doses of carotenoids do not result in toxic symptoms. However, carotenemia, which is characterized by a yellowing of the skin (creases of the palms and soles) but

not the sclerae, may be seen after ingestion of >30 mg of β -carotene on a daily basis. Hypothyroid patients are particularly susceptible to the development of carotenemia due to impaired breakdown of carotene to vitamin A. Reduction of carotenes from the diet results in the disappearance of skin yellowing and carotenemia over a period of 30 to 60 days.

VITAMIN D (See [Chap. 340](#)).

VITAMIN E

Vitamin E is a collective name for a group of tocopherols and tocotrienols, the latter having an unsaturated sidechain. There are eight naturally occurring plant compounds with vitamin E activity. RRR- α tocopherol is the most active, while synthetic stereoisomers of vitamin E are less biologically active. Vitamin E acts as a chain-breaking antioxidant and is an efficient peroxyl radical scavenger, which protects [LDLs](#) and polyunsaturated fats in membranes from oxidation. A network of other antioxidants (e.g., vitamin C, glutathione) and enzymes maintains vitamin E in a reduced state. Vitamin E also inhibits prostaglandin synthesis and the activities of protein kinase C and phospholipase A₂.

Absorption and Metabolism Vitamin E is a fat-soluble vitamin and requires all the processes needed for micelle formation to be absorbed. About 15 to 40% is absorbed passively from a single physiologic dose, and there is less efficient absorption at high doses. Polyunsaturated fat may inhibit absorption. Vitamin E is taken up from chylomicrons by the liver, and an hepatic tocopherol transport protein is involved in intracellular vitamin E transport and incorporation into very low density lipoprotein (VLDL). The transport protein has particular affinity for the RRR isomeric form of a tocopherol; thus this natural isomer has the most biologic activity. In the circulation, vitamin E is bound to all lipoprotein classes and becomes widely distributed in tissues, with fat and muscle being the most important storage depots. Vitamin E metabolites are mainly excreted in feces, although some are also excreted in urine.

Requirement The [RDA](#) for vitamin E is currently 10 mg for adults. Additional vitamin E is recommended during pregnancy (12 mg/d) and lactation (14 mg/d). Vitamin E is widely distributed in the food supply. The RRR- α isomers are particularly high in sunflower oil, safflower oil, and wheat germ oil; γ tocotrienols are notably present in soybean and corn oils. Vitamin E is also found in meats, nuts, and cereal grains, and small amounts are present in fruits and vegetables. Vitamin E pills containing doses of 50 to 1000 mg are ingested by a large fraction of the U.S. population. In the older literature, 1 IU of vitamin E is equal to 1 mg *all-racemic* tocopherol acetate. Diets high in polyunsaturated fats may necessitate a slightly higher requirement for vitamin E.

Dietary deficiency of vitamin E does not exist. Vitamin E deficiency is seen only in severe and prolonged malabsorptive diseases, such as celiac disease, or after small-intestinal resection, leading to short-bowel syndrome. Children with cystic fibrosis or prolonged cholestasis may develop vitamin E deficiency characterized by areflexia and hemolytic anemia. Children with abetalipoproteinemia cannot absorb or transport vitamin E and become deficient quite rapidly. A familial form of isolated vitamin E deficiency also exists, which is due to a defect in the tocopherol transport protein.

Vitamin E deficiency causes axonal degeneration of the large myelinated axons and results in posterior column and spinocerebellar symptoms. Peripheral neuropathy is initially characterized by areflexia, with progression to an ataxic gait, and by decreased vibration and position sensations. Ophthalmoplegia, skeletal myopathy, and pigmented retinopathy may also be features of vitamin E deficiency. The laboratory diagnosis of vitamin E deficiency is made on the basis of low blood levels of a tocopherol (<5 ug/mL, or <0.8 mg of a tocopherol per gram of total lipids).

TREATMENT

Symptomatic vitamin E deficiency should be treated with 800 to 1200 mg of a tocopherol per day. Patients with abetalipoproteinemia may need as much as 5000 to 7000 mg/d. Children with symptomatic vitamin E deficiency should be treated with 400 mg/d orally of water-soluble esters; alternatively, 2 mg/kg per day may be administered intramuscularly. Vitamin E in high doses may protect against oxygen-induced retrolental fibroplasia and bronchopulmonary dysplasia in prematurity, as well as intraventricular hemorrhage of prematurity. Vitamin E has been suggested to increase sexual performance, to treat intermittent claudication, and to slow the aging process, but evidence for these properties is lacking. High doses (60 to 800 mg/d) of vitamin E have been shown in controlled trials to improve parameters of immune function, and there are two intervention studies showing that vitamin E at 400 to 800 mg/d may be protective against cardiovascular disease, possibly by inhibiting [LDL](#) oxidation. Also, supplemental intake of vitamin E (100 to 200 mg/d) has been associated with a decreased risk of cataracts.

Toxicity High doses of vitamin E (>800 mg/d) may reduce platelet aggregation and interfere with vitamin K metabolism and are therefore contraindicated in patients taking coumadin. Nausea, flatulence, and diarrhea have been reported at doses >1 g/d.

VITAMIN K

There are two natural forms of vitamin K: vitamin K I, also known as *phylloquinone*, from vegetable and animal sources, and vitamin K II, or *menaquinone*, which is synthesized by bacterial flora and found in hepatic tissue. *Menadione*, or vitamin K III, is a chemically synthesized pro-vitamin that can be converted to menaquinone by the liver.

Phylloquinone and menaquinones differ only in their lipophilic sidechains, and both are destroyed in an alkaline pH and by ultraviolet light.

Absorption and Physiology As with other fat-soluble vitamins, vitamin K absorption is dependent on normal pancreatic function and the presence of bile salts. Phylloquinones are absorbed by a saturable energy-dependent mechanism in the proximal small intestine, whereas menaquinones are absorbed by passive diffusion in the small intestine and colon. Approximately 100 ug of vitamin K is stored in the liver as well as in lung, bone marrow, kidneys, and adrenal glands. Most vitamin K circulates bound to [VLDL](#), although it is also carried by [LDL](#) and high-density lipoprotein (HDL). The half-life of vitamin K is only 1½ days, despite the presence of a vitamin K regeneration cycle.

Vitamin K is necessary for the posttranslational carboxylation of glutamic acid, which is

necessary for calcium binding to γ -carboxylated proteins such as prothrombin (factor II); factors VII, IX, and X; protein C; protein S; and proteins found in bone (bone gla, matrix gla protein, and osteocalcin). The importance of vitamin K for bone mineralization is not known. Warfarin-type drugs inhibit γ -carboxylation by preventing the conversion of vitamin K to its active hydroquinone form. Vitamin E, at high doses, may act as a vitamin K antagonist.

Dietary Sources Vitamin K is found in green leafy vegetables such as kale and spinach, but appreciable amounts are also present in butter, margarine, liver, milk, ground beef, coffee, and pears. Vitamin K is present in vegetable oils and is particularly rich in olive oil and soybean oil. The recommended intake of vitamin K is 70 $\mu\text{g}/\text{d}$ in adults. The average daily intake by Americans is estimated to be approximately 100 $\mu\text{g}/\text{d}$.

Deficiency The symptoms of vitamin K deficiency are due to hemorrhage, and newborns are particularly susceptible because of low fat stores, low breast milk levels of vitamin K, sterility of the infantile intestinal tract, liver immaturity, and poor placental transport. Intracranial bleeding, as well as gastrointestinal and skin bleeding, can be seen in vitamin K-deficient infants 1 to 7 days after birth. Thus, vitamin K (1 mg intramuscularly) is given prophylactically at the time of delivery.

Vitamin K deficiency in adults may be seen in patients with chronic small-intestinal disease (e.g., celiac disease, Crohn's disease), obstructed biliary tracts, or after small-bowel resection. Broad-spectrum antibiotic treatment can precipitate vitamin K deficiency by reducing gut bacteria, which synthesize menaquinones, as well as by inhibiting the metabolism of vitamin K. The diagnosis of vitamin K deficiency is usually made on the basis of an elevated prothrombin time or reduced clotting factors. Vitamin K may also be measured directly by [HPLC](#). In addition, undercarboxylated prothrombin and low gla levels in urine are indicative of vitamin K deficiency. Vitamin K deficiency is treated using a parenteral dose of 10 mg. For patients with chronic malabsorption, 1 to 2 mg/d of vitamin K may be given orally, or 1 to 2 mg/week can be taken parenterally. Patients with liver disease may have an elevated prothrombin time because of liver cell destruction as well as vitamin K deficiency. If an elevated prothrombin time does not improve on vitamin K therapy, it can be assumed that it is not the result of vitamin K deficiency.

Toxicity Parenteral doses of the water-soluble vitamin K derivative (menadione) have been reported to cause hemolytic anemia and hypobilirubinemia in infants. Toxicity from dietary phyloquinones and menaquinones has not been described. High doses of vitamin K can impair the actions of oral anticoagulants.

TRACE MINERALS (See [Table 75-1](#))

ZINC

Zinc is an integral component of many metalloenzymes in the body; it is involved in the synthesis and stabilization of proteins, DNA, and RNA and plays a structural role in ribosomes and membranes. Zinc is necessary for the binding of steroid hormone receptors and several other transcription factors to DNA and thereby plays an important

role in the regulation of gene transcription. Zinc is absolutely required for normal spermatogenesis, fetal growth, and embryonic development.

Absorption and Physiology Zinc is absorbed in the small intestine by a carrier-mediated mechanism. The absorption of zinc from the diet is inhibited by dietary phytate, fiber, oxalate, iron, and copper, as well as by certain drugs including penicillamine, sodium valproate, and ethambutol. The [RDA](#) for zinc is 15 mg in males and 12 mg in females, with an additional 3 mg in pregnancy and 4 to 7 mg during lactation. Supplemental zinc is recommended for women taking ³60 mg/d of iron during pregnancy.

Meat, shellfish, nuts, and legumes are good sources of bioavailable zinc, whereas zinc in grains is less available for absorption. Zinc is excreted mainly in the feces but also in urine and sweat. The body contains approximately 2 g of zinc, and high concentrations are found in liver, prostate, pancreas, bone, and brain (hippocampus and cerebral cortex), where the metal may function in neural transmission.

Deficiency Mild zinc deficiency has been described in many diseases including diabetes mellitus, AIDS, cirrhosis, alcoholism, inflammatory bowel disease, malabsorption syndromes, and sickle cell anemia ([Figs. 75-CD2](#) and [75-CD3](#)). In these diseases, mild chronic zinc deficiency can cause stunted growth in children, decreased taste sensation (hypogusia), impaired immune function, and night blindness due to impaired conversion of retinol to retinaldehyde. Severe chronic zinc deficiency has been described as a cause of hypogonadism and dwarfism in several Middle Eastern countries. In these children, hypopigmented hair is also part of the syndrome. Acrodermatitis enteropathica is a rare autosomal recessive disorder characterized by abnormalities in zinc absorption. Clinical manifestations include diarrhea, alopecia, muscle wasting, depression, irritability, and a rash involving the extremities, face, and perineum. The rash is characterized by vesicular and pustular crusting with scaling and erythema. In addition, hypopigmentation and corneal edema have been described in these patients. Occasional patients with Wilson's disease have developed zinc deficiency as a consequence of penicillamine therapy. Patients on long-term parenteral nutrition have developed deficiency when zinc has been omitted from the total parenteral nutrition (TPN) solution.

The diagnosis of zinc deficiency is usually made by a serum zinc level of <12 $\mu\text{mol/L}$ (<70 $\mu\text{g/dL}$). Pregnancy and birth control pills may cause a slight depression in serum zinc levels, and hypoalbuminemia from any cause can result in hypozincemia. In acute stress situations, zinc may be redistributed from serum into tissues. Zinc deficiency may be treated with 60 mg elemental zinc, given orally twice a day. Zinc gluconate lozenges (13 mg elemental Zn every 2 h while awake) have been reported to reduce the duration and symptoms of the common cold in adults, but these studies are conflicting.

Toxicity Acute zinc toxicity after oral ingestion causes nausea and vomiting, fever, and respiratory distress. Zinc fumes from welding may also be toxic and cause fever, chills, excessive salivation, sweating, and headache. Chronic large doses of zinc may depress immune function and cause hypochromic anemia as a result of copper deficiency.

COPPER

Copper is an integral part of numerous enzyme systems including amine oxidases, ferroxidase (ceruloplasmin), cytochrome-c oxidase, superoxide dismutase, and dopamine hydroxylase. As such, copper plays a role in iron metabolism, melanin synthesis, and central nervous system function; the synthesis and cross-linking of elastin and collagen; and the scavenging of superoxide radicals.

Copper is absorbed in the proximal small intestine, and 90% of circulating copper is bound to ceruloplasmin. The body contains 50 to 120 mg of copper, and high concentrations are found in liver, brain, heart, spleen, kidney, and blood. The U.S. [RDA](#) is 1.5 to 3 mg of copper intake per day, although World Health Organization recommendations are somewhat lower. Dietary sources of copper include shellfish, liver, nuts, legumes, bran, and organ meats, whereas milk is a very poor source. Copper is primarily excreted in the feces, and small amounts are also excreted in urine.

Deficiency Dietary copper deficiency is relatively rare, although it has been described in premature infants fed milk diets and in infants with malabsorption. Signs and symptoms of copper deficiency include a hypochromic-normocytic anemia, osteopenia, depigmentation, mental retardation, and psychomotor abnormalities. Copper deficiency anemia has been reported in patients with malabsorptive diseases and nephrotic syndrome and in patients treated for Wilson's disease with chronic high doses of oral zinc, which can interfere with copper absorption. Menkes kinky hair syndrome is an X-linked metabolic disturbance of copper metabolism characterized by mental retardation, hypocupremia, and decreased circulating ceruloplasmin ([Chap. 351](#)). It is caused by mutations in a copper-transporting *ATP7A* gene. Children with this disease often die within 5 years due to dissecting aneurysms or cardiac rupture.

The diagnosis of copper deficiency is usually made on the basis of low serum levels of copper (<65 ug/dL) and low ceruloplasmin levels (<18 mg/dL). Serum levels of copper may be elevated in pregnancy or stress conditions since ceruloplasmin is an acute-phase reactant.

Toxicity Toxicity due to copper is usually accidental and may include nausea, vomiting, diarrhea, and hemolytic anemia. In severe cases, kidney failure, liver failure, and coma may ensue. In Wilson's disease, mutations in the copper-transporting *ATP7B* gene lead to accumulation of copper in the liver and brain, with low blood levels due to decreased ceruloplasmin ([Chap. 348](#)). Indian childhood cirrhosis is another hereditary disease characterized by extremely high copper levels in the liver. The World Health Organization recommends that adult females should not ingest >10 mg/d and males should not take in >12 mg/d of copper.

SELENIUM

Selenium, in the form of selenocysteine, is a component of the enzyme glutathione peroxidase, which serves to protect proteins, cell membranes, lipids, and nucleic acids from oxidant molecules. Selenocysteine is also found in the deiodinase enzymes, which mediate the deiodination of thyroxine to the more active triiodothyronine ([Chap. 330](#)). Rich sources of selenium include seafood, muscle meat, and cereals, although the selenium content of cereal is determined by the soil concentration. Countries with low

soil concentrations include parts of Scandinavia, China, and New Zealand. *Keshan disease* is an endemic cardiomyopathy found in children and young women residing in regions of China where dietary intake of selenium is low (<20 ug/d). Concomitant deficiencies of iodine and selenium may worsen the clinical manifestations of cretinism. The adult [RDAs](#) for selenium in the United States are 55 and 70 ug/d for females and males, respectively. Low blood levels of selenium in various populations have been correlated with an increase in coronary artery disease and certain cancers, although the data are not consistent. Selenosis occurs at intakes of ³400 ug/d and can result in nausea, vomiting, loss of hair, nail changes, peripheral neuropathy, and fatigue.

CHROMIUM

Chromium potentiates the action of insulin in patients with impaired glucose tolerance, presumably by increasing insulin receptor-mediated signaling. In addition, in some patients, improvement in blood lipid profiles has been reported. The usefulness of chromium supplements in muscle building are not substantiated. Rich food sources of chromium include yeast, meat, and grain products. Chromium deficiency has been reported to cause glucose intolerance, peripheral neuropathy, and confusion. The suggested intake of chromium for adults is 50 to 200 ug/d. Chromium in the trivalent state is found in supplements and is largely nontoxic; however, chromium-6 is a product of stainless steel welding and is a known pulmonary carcinogen, as well as causing liver, kidney, and central nervous system.

MAGNESIUM See [Chap. 340](#)

FLUORIDE, MANGANESE, AND ULTRATRACE ELEMENTS

An essential function for *fluoride* in humans has not been described, although it is useful for the maintenance of structure in teeth and bone. An adequate intake for fluoride (on the basis of protection against dental caries) has been set at 3.1 and 3.8 mg/d in adult females and males, respectively. Adult fluorosis can occur at an intake of 10 mg/d for prolonged periods and results in mottled and pitted defects in tooth enamel as well as brittle bone (skeletal fluorosis). Much lower doses of fluoride (0.7 to 2 mg) can cause dental fluorosis or mottled enamel in infants and children.

Manganese and molybdenum deficiencies have been reported in patients with rare genetic abnormalities as well as in a few patients receiving prolonged [TPN](#). Several manganese-specific enzymes have been identified (e.g., manganese superoxide dismutase). The estimated adequate daily dietary *manganese* intake for adults is 2 to 3 mg/d. Deficiencies of manganese have been reported to result in bone demineralization, poor growth, ataxia, and convulsions.

Ultratrace elements are those for which the need is <1 mg/d. Essentiality has not been established for most ultratrace elements, although *iodine* is clearly essential ([Chap. 330](#)). *Molybdenum* is necessary for the activity of sulfite and xanthine oxidase, and molybdenum deficiency may result in skeletal and brain lesions. The minimum required daily molybdenum intake is estimated to be ~25 ug/d. There is circumstantial evidence to suggest that *arsenic* (impaired growth, infertility), *boron* (impaired energy metabolism, impaired brain function), *nickel* (impaired-growth and reproduction), *silicon* (impaired

growth) and *vanadium* (impaired skeletal formation) might also be essential.

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76. ENTERAL AND PARENTERAL NUTRITION THERAPY - Lyn Howard

Parenteral and enteral nutrition provide life-sustaining therapy for patients who cannot take adequate food by mouth and who consequently are at risk for malnutrition and its effects, including susceptibility to infection, weakness and immobility; these features predispose the patient to aspiration pneumonia, pulmonary embolism, and pressure sores, all of which delay recovery from illness and increase mortality.

The term *enteral* refers to feeding via the gut and hence includes normal eating, but in the present context implies the infusion of formulas via a tube into the upper gastrointestinal tract. *Parenteral* refers to the infusion of nutrient solutions into the bloodstream. While these are different approaches to nutritional support, their goals are the same. Where feasible, enteral nutrition is the preferred route because it sustains the digestive, absorptive, and immunologic barrier functions of the gastrointestinal tract. The cost of enteral tube feeding is about one-tenth the cost of parenteral feeding.

Several developments have made tube feeding easier and more acceptable to patients. Small-bore pliable tubes have largely replaced large-bore rubber tubes, and double-lumen tubes are now available for simultaneous gastric suction and jejunal feeding when there is concern about gastric retention and aspiration. Enteral tubes can be inserted into the stomach or jejunum through the nose or, for long-term use, directly through the abdominal wall, using endoscopic, radiologic, or surgical techniques. Once the enterocutaneous tract is established, the protruding tube can be replaced by a "button" entry port, flush with the abdominal wall.

Complete nutrition by vein with sufficient calories, amino acids, minerals, and vitamins to permit wound healing, restoration of normal body composition of a cachectic patient, or growth in children became feasible in the 1960s with the development of high-flow central vein catheters. Parental nutrition is now available in all large hospitals and for some patients at home. Adequate calories and other nutrients can be delivered in the form of high-energy, isotonic intravenous fat solutions via a peripheral vein. However, peripheral veins usually cannot sustain such infusions indefinitely, and long-term support requires central venous access.

THE DECISION PROCESS FOR USING PARENTERAL OR ENTERAL NUTRITION

The decision to use specialized nutrition support should be based on the likelihood that averting or redressing malnutrition will improve the quality of life or the ability to recover from a serious illness.

Approximately 15 to 20% of hospitalized patients have evidence of malnutrition. Some malnourished patients benefit from specialized nutrition support; for others, wasting is an inevitable component of a terminal disease. Selecting the appropriate form of nutritional support for the patient requires knowledge of the potential benefits and risks of nutritional support, and the physician must inform the patient and family of these issues. A flow diagram of the steps involved in deciding whether specialized nutrition support should be undertaken and, if so, how, is depicted in [Fig. 76-1](#). Like all life-support measures, these therapies are difficult to withdraw once started.

The first step requires consideration of the nutritional implications of the disease process. Is the condition or its treatment likely to impair appetite or food ingestion and absorption for a prolonged period of time? Because prevention of malnutrition is easier than repleting a cachectic patient, this issue must be considered in the initial evaluation ([Chap. 74](#)). The second step is to determine whether the patient is already sufficiently malnourished that lean body mass is decreased and critical functions such as healing and ventilation are impaired. The presence or absence of metabolic stress should be noted, since injury or infection can evoke the secretion of hormonal and cytokine factors that reduce the efficiency of nutrition repletion.

Weight loss without physiologic impairment is probably of no consequence. Physiologic impairment usually develops when more than 20% of body protein is lost and is more likely if key organ systems, such as the gut or liver, are directly affected by disease. Once it is recognized that the patient is malnourished or at major risk, the next question is whether specialized nutritional support will impact positively on the patient's response to the disease, improving the quality of life. While the provision of food and water is part of basic medical care, nutrition support by enteral or parenteral means is associated with risk and discomfort and should be recommended only when potential benefit exceeds risk and undertaken only with the consent of the patient.

If it is decided that preventing or treating malnutrition with specialized nutrition support would improve the prognosis and quality of life, the nutritional requirements must be determined and the route of nutrient delivery must be selected.

RISKS AND BENEFITS OF NUTRITION SUPPORT

The risks are determined primarily by the route required to deliver nutrition support. Providing nutritional requirements by special attention to oral intake of food, or by adding oral liquid supplements and monitoring food intake with frequent calorie counts, is the safest and least costly approach. It is also the most metabolically efficient since normal eating initiates the cephalic phase of digestion. Tube-fed infants grow better if the cephalic phase is stimulated by having the infant suck on a pacifier.

Anorexia, impairment of swallowing, or bowel disease may limit oral intake or the absorption of oral nutrients, in which case tube enteral nutrition is the next consideration. The bowel and its associated digestive organs derive 70% of their required nutrients directly from food in the lumen. In addition, glutamine, short-chain fatty acids, and nucleotides may have particular importance in maintaining gut integrity. Enteral feeding also supports gut function by stimulating splanchnic blood flow, neuronal activity, IgA antibody release, and secretion of gastrointestinal hormones such as epidermal growth factor that stimulate gut trophic activity. All these factors support the gut as an immunologic barrier against enteric pathogens, reducing the likelihood of bacterial overgrowth. For these reasons, some enteral nutrition should always be provided if possible, even when parenteral nutrition is required to provide most of the support. In the past, bowel rest through parenteral nutrition was thought to be the cornerstone of treatment of many severe gastrointestinal disorders, but the value of some enteral nutrition is now widely accepted, and strict bowel rest is rarely appropriate. Parenteral nutrition alone is necessary in severe hemorrhagic pancreatitis, necrotizing enterocolitis, prolonged ileus, and distal bowel obstruction.

Specialized nutrition support is expensive, accounting for >1% of all health care dollars. Consequently, hard clinical endpoints such as mortality rate, incidence of major complications, and duration of hospital stay are required of risk-benefit studies. Better nitrogen balance, increased levels of serum albumin, and improved delayed hypersensitivity are softer endpoints. [Table 76-1](#) summarizes clinical trials that evaluate the use of specialized nutrition support in different disease states.

Perioperative Nutrition There is a clear-cut association between preoperative malnutrition and poor surgical outcome, but it has been difficult to demonstrate the benefit of preoperative parenteral nutrition on the outcome of surgery in malnourished patients. However, a meta-analysis of several small studies and a large cooperative Veterans Administration study indicates that preoperative parenteral nutrition does improve the outcome of severely malnourished surgical patients. In treated patients, noninfectious complications (e.g., pulmonary emboli and delayed wound healing) are reduced in the postoperative period. Effective preoperative restoration of nutrition by the parenteral route requires at least 7 to 14 days. If feasible, a safer and less costly approach is preoperative enteral nutrition, especially if provided at home.

Immediate postoperative nutritional support is appropriate for patients who received preoperative support and for patients unlikely to resume oral feeding within 10 days. The parenteral route is commonly used because of postoperative ileus or concern about disrupting a new bowel anastomosis. However, cautious jejunal feeding is often tolerated. Specialized enteral formulas supplemented with conditionally essential nutrients may be particularly beneficial in debilitated and immunosuppressed postoperative patients ([Table 76-2](#)).

Critical Illness Very early nutrition support (within the first 48 h) improves survival and reduces infections and length of hospital stay in patients with severe head injuries, burns, and major abdominal trauma. Enteral therapy, where feasible, is superior to parenteral therapy in several randomized trials. Enteral nutrition equally benefits malnourished and well-nourished injured patients. Animal studies show that enteral feeding reduces translocation of gut bacteria and the systemic catabolic response; however, these phenomena have not been substantiated in humans. Early enteral feeding may prevent bacterial overgrowth and decrease aspiration pneumonia.

The practical issue is obtaining jejunal access in a critically ill patient, who is not easily transferred out of the intensive care unit for endoscopic or radiologic tube placement. Sometimes a nasal or percutaneous combined gastric suction and jejunal feeding tube can be inserted at the bedside. If a surgical laparotomy is indicated, a feeding tube can be placed simultaneously.

Most studies of enteral feeding in critically ill patients used either a general polymeric formula or one with hydrolyzed protein. Formulas supplemented with conditionally essential nutrients reduce infections and length of hospital stay. Parenteral formulas enriched with large amounts of branched-chain amino acids (BCAA) improve nitrogen balance but do not appear to affect clinical outcome.

Cancer Cachexia Early nonrandomized studies suggested that patients with cancer

cachexia benefitted from parenteral nutrition, but randomized trials demonstrated more risk than benefit for patients receiving chemotherapy or radiation. Severely malnourished cancer patients undergoing surgery benefit from preoperative parenteral nutrition, as do other malnourished patients.

In two randomized, prospective trials, patients undergoing bone marrow transplantation had better long-term survival after parenteral or enteral nutrition in the cytoreduction phase; nutrition support did not influence the initial infection rate or the frequency of graft rejection or graft-versus-host disease. Immediate morbidity is reduced if glutamine supplements are added to the parenteral nutrition solution. Parenteral nutrition continued at home delays return to oral feeding. For cancer patients with unresectable upper gastrointestinal cancer, enteral feeding is usually justified if it is desired by the patient and family. Parenteral feeding should be provided only if clinical improvement can be expected and when quality survival at home for several months is predicted.

Liver Failure Malnutrition is common in advanced liver disease. Patients with acute or chronic liver failure have decreased levels of [BCAA](#) and elevated levels of aromatic amino acids (AAA) in plasma and cerebrospinal fluid. Randomized, prospective trials with parenteral and enteral formulas high in BCAA and low in AAA have demonstrated better nitrogen balance and less risk of encephalopathy. One large multicenter study also reported improved survival. The BCAA-enriched formulas are expensive and should be used only in patients who have encephalopathy or who develop encephalopathy when fed a standard protein formula providing 0.8 g protein/kg per day.

Renal Failure Since renal failure is associated with impaired nitrogen excretion, it is rational to assume that protein restriction might benefit patients with both acute and chronic renal failure. Patients with acute renal failure given parenteral calories and amino acids have fewer infectious complications and a better chance of leaving the hospital than similar patients given only calories. An early randomized study showed benefit when essential amino acids were the sole source of nitrogen, but in other studies, standard solutions supplying both essential and nonessential amino acids provide similar advantage. Thus, the benefit of using expensive formulas containing only essential amino acids or their keto analogues is not established. A large national study failed to show any benefit of a low-protein diet on slowing progression of renal impairment in patients on chronic dialysis. Some 15 to 20% of patients on chronic dialysis have significant nutritional impairment, usually due to profound anorexia. The anorexia may improve with stepped-up dialysis or treatment of gastritis but usually persists. The resulting growth impairment in younger patients has been treated with supplemental tube enteral nutrition. This approach has not been widely used in adult patients. Limited parenteral calories and amino acids can be provided in the last 90 min of hemodialysis treatment (intradialytic parenteral nutrition). This may improve appetite, serum protein levels, and body weight. No randomized studies have documented better survival, so the appropriateness of this regimen is not established. Standard dialysis uses glucose to provide an osmotic load, and some glucose calories are absorbed. During continuous ambulatory peritoneal dialysis, amino acids can be substituted for glucose and are also partly absorbed, offsetting the loss of endogenous amino acids into the dialysate. This approach to nutrition repletion is expensive and also awaits a randomized study.

Pancreatitis Parenteral nutrition does not improve the outcome of patients with mild or moderate pancreatitis. However, in severe pancreatitis, survival decreases as malnutrition becomes more severe. When parenteral nutrition support was delayed beyond 72 h, patients with severe pancreatitis had a threefold higher complication and mortality rate, compared to similar patients treated earlier. In the absence of severe hyperlipidemia or thrombocytopenia, intravenous lipids appear safe and are especially useful if glucose intolerance is present. Several studies report successful enteral jejunal feeding in acute pancreatitis and, compared to parenteral nutrition, the inflammatory response and infectious complications are less.

Inflammatory Bowel Disease Evidence of nutritional deficiencies such as weight loss, growth failure, anemia, and hypoalbuminemia are common in inflammatory bowel disease (IBD), more so in Crohn's disease than in ulcerative colitis ([Chap. 287](#)). Nutrition support plays a role in correcting these nutritional deficiencies, particularly prior to elective surgery. Since IBD often improves with diversion of the fecal stream, the question is whether bowel rest and parenteral nutrition have a role as primary treatment. However, randomized, prospective studies have shown no special benefit from bowel rest. Elemental diets are not quite as effective as glucocorticoids for inducing remission in acute Crohn's disease but may be preferable in children to avoid growth impairment. Relapse is common when a regular diet is resumed. In controlled studies, remissions are prolonged if the Crohn's patient does not return to a regular diet but instead eliminates from the diet those foods that induce gastrointestinal symptoms. For the majority of Crohn's patients this leads to avoidance of cereals, yeast, green vegetables, and, early on, dairy products. Because of the possibility that diets high in omega-3 fatty acids have a beneficial effect in immune disorders by altering prostaglandin synthesis, their value in IBD is under investigation. Some studies suggest that high-fiber diets benefit IBD patients, but fiber can also cause obstruction in patients with bowel strictures.

Short Bowel Syndrome Before the advent of parenteral nutrition, patients with acute short bowel syndrome from mesenteric vascular infarction or massive small bowel surgical resection seldom survived. Parenteral nutrition has allowed many patients to survive indefinitely with only a foot or two of small intestine. In some, the remaining bowel eventually adapts and allows the absorption of adequate calories and protein. This is especially true of patients who retain their ileocecal valve and colon. However, fluid and electrolyte imbalance may persist, necessitating some parenteral fluid and electrolyte support. A gradual switch to overnight tube enteral hydration or constant sipping of an electrolyte solution may allow discontinuation of all parenteral support.

Pulmonary Disease Weight loss in patients with advanced pulmonary disease is due to increased work of breathing and poor food intake. Patients with chronic pulmonary disease who are <90% of their ideal weight have a higher 5-year mortality, independent of pulmonary status. The recommended energy intake for these patients is 1.7 times their resting energy expenditure. The use of a low-carbohydrate formula is beneficial in the weaning of patients from ventilators, but the superiority of such formulas in ambulatory patients with chronic lung disease is not established. In cystic fibrosis, malnutrition may hasten pulmonary deterioration, and enteral tube feeding enhances growth and stabilizes or improves pulmonary function, particularly in young children. Tube feeding is safest when delivered into the jejunum. Postpyloric feeding is no safer

than gastric feeding.

HIV Disease Specialized nutrition support can replete body cell mass if the weight loss is due to inadequate oral intake caused by oral or esophageal disease or to inadequate intestinal absorption, which is common in HIV patients with cryptosporidiosis or microsporidiosis infections ([Chap. 309](#)). The route of nutrition support has usually been parenteral, but patients respond equally well with an isocaloric semi-elemental oral diet. Patients using the oral supplement experience a better quality of life, and their medical costs are significantly lower. Wasting due to systemic infection and increased cytokine secretion is not redressed by specialized nutrition support. Survival, CD4+ counts, and intestinal function also are not improved by specialized nutrition support.

Pregnancy Severe hyperemesis gravidarum can make any oral or tube enteral nutrition impossible, and profound weight loss and ketosis may harm the developing fetus. The underlying mechanism of the disorder is not understood, but it is cured by abortion or delivery. Temporary parenteral nutrition usually results in a successful outcome, but nausea and vomiting tend to persist, despite bowel rest.

Home Parenteral and Enteral Nutrition Some patients require long-term nutrition support, and for many this can be administered at home. Clinical outcomes of patients with severe intestinal disorders that used either parenteral or enteral nutrition are summarized in [Table 76-3](#). Nutrition support is not usually appropriate in terminally ill patients but is an option if the patient is expected to survive for several months. Such therapy must make sense to the patient, and sufficient help must be available so the treatment can be given at home without undue hazard. Both home therapies are relatively safe, with <5% therapy-related mortality.

THE DESIGN OF INDIVIDUAL REGIMENS

Fluid Requirements These can be estimated by adding the normal daily requirement (120 mL/kg of body weight for infants, 35 mL/kg of body weight for adults) to any abnormal loss. If the patient is on parenteral therapy, any enteral intake should be subtracted from the estimate ([Table 76-4](#)). Since abnormal loss of enteric fluid implies significant mineral losses, extra amounts of these nutrients, as well as fluid ([Table 76-5](#)), must be added to the parenteral formula.

Energy Requirements These can be determined as outlined in [Chaps. 73](#) and [74](#). In the long run, energy expenditure dictates energy requirements, but in the early phase of nutrition repletion, requirements may not reflect expenditure. For example, malnourished patients are hypometabolic and may expend only 85 kJ/kg (20 kcal/kg) per day, but more calories are needed both for tissue repletion and because the metabolic rate increases with refeeding. Conversely, a highly stressed patient (sepsis, trauma) may expend 165 kJ/kg (40 kcal/kg) per day with a significant proportion of the calories coming from protein breakdown and gluconeogenesis and from catecholamine-induced lipolysis. Oxidation of exogenous glucose plateaus at 100 kJ/kg (25 kcal/kg) per day, and administering additional glucose induces hepatic steatosis. Providing such patients with additional calories as exogenous fat does not suppress endogenous lipolysis. Furthermore, lipid solutions are made from vegetable oil and egg phospholipid and lack apoproteins, which they acquire from endogenous lipoproteins.

Initially, the artificial chylomicron may be taken up by the reticuloendothelial system enhancing its blockade. For all these reasons, modest hypocaloric glucose feeding with minimal parenteral fat is safer in the acutely stressed subject.

Parenteral lipid solutions are available as 10 or 20% isotonic solutions and are infused separately from amino acids and glucose or as a combined "three-in-one solution," obviating the need for an extra pump. Three-in-one solutions are less stable than the glucose and amino acid mix, and destabilized fat particles have the potential to coalesce into larger droplets, becoming fat emboli. For this reason, three-in-one solutions have a shorter storage life and must be mixed by a pharmacist knowledgeable about the correct mixing sequence and safe levels of electrolytes and trace elements. Iron, for example, cannot be added to this solution.

Polyunsaturated vegetable oils are used in most enteral formulas because they are absorbed better than animal fat by a diseased gastrointestinal tract. Fat must supply the essential fatty acid requirement (1 to 4% of energy from linoleic and linolenic acid) ([Table 76-6](#)). Larger amounts (30% of energy) are safe in relatively stable patients and avoid the problems of providing large amounts of glucose (e.g., hyperglycemia and hepatic steatosis). Substituting omega-3 polyunsaturated fish oils for polyunsaturated vegetable fats may reduce the catabolic response to burn injury, trauma, and radiation by reducing the synthesis of prostaglandins that enhance the inflammatory response ([Table 76-2](#)). Such fats are available in enteral formulas and are currently being tested in parenteral formulas.

Protein or Amino Acid Requirements The recommended dietary protein allowance of 0.8 g/kg per day is adequate for nonstressed patients, such as a starved patient with a high-grade esophageal stricture. Catabolic patients, in contrast, may require up to 1.5 g/kg per day of protein to induce positive nitrogen balance and reconstitute normal body mass. Early studies showed that recombinant human growth hormone (rHGH) increases lean body mass. However, subsequent trials have shown that it is associated with increased mortality in critically ill patients, and it should not be used in this setting.

In a stable patient the adequacy of protein support can be assessed by analyzing protein balance:

where protein loss = $[(24\text{-h urine urea nitrogen (g)} + 4) \times 6.25]$. Over a long period, protein balance is assessed by documenting wound healing, restoration of normal body composition, or resumption of longitudinal growth. In states of disturbed protein utilization (e.g., renal and hepatic failure), azotemia and abnormal plasma amino acid patterns develop. The benefit of special enteral and parenteral solutions that correct these aberrations is only established in hepatic encephalopathy (see "Risks and Benefits of Nutrition Support").

Certain nutrients that can normally be synthesized endogenously become essential in severely ill patients when endogenous production or salvage pathways are impaired. This is true of glutamine, nucleotides, and the products of methionine metabolism ([Table 76-2](#)). Glutamine, an important fuel for the enterocyte and lymphocyte, is fairly insoluble

and is absent from standard parenteral formulas and present in low concentrations in most enteral formulas. Soluble glutamine-containing dipeptides are under investigation.

Nucleotides and their related metabolic products have beneficial effects on the immune system, growth of the small intestine, lipid metabolism, and hepatic function. Nucleotides can be synthesized de novo in all cells only in small amounts, and the body therefore depends on dietary nucleotides or on salvage pathways that recycle nucleotides from purine and pyrimidine turnover. Nutritionally depleted patients benefit from formulas enriched in nucleotides.

When amino acids are infused systemically, rather than via the more physiologic portal vein, methionine, the only sulfur-containing amino acid in most parenteral solutions, is transaminated in peripheral tissues rather than transsulfurated in the liver. As a result, downstream sulfur products such as carnitine, taurine, and glutathione become relatively deficient ([Chap. 352](#)). Preliminary studies suggest that the addition of an intermediate compound, S-adenosyl methionine, to parenteral solutions results in less cholestasis.

Mineral and Vitamin Requirements Parenteral and enteral mineral and vitamin requirements are summarized in [Table 76-6](#). Electrolyte modifications are necessary if the patient has significant gastrointestinal losses ([Table 76-5](#)) or renal failure. Requirements of some minerals and vitamins are higher when administered parenterally for several reasons: (1) many micronutrients delivered into the systemic rather than the portal circulation are not captured by the liver and instead pass directly into the urine; (2) patients with bowel disease may have enteric loss of sodium, potassium, chloride, and bicarbonate and malabsorption of divalent cations, fat-soluble vitamins, and vitamin B₁₂; and (3) nutrients may adhere to the tubing and delivery bags, and exposure to oxygen and light may destroy vitamins (particularly vitamin A).

PARENTERAL NUTRITION

Infusion Technique and Patient Monitoring Partial and short-term total parenteral nutrition can be provided via a peripheral vein if the majority of the energy is supplied by isotonic fat solutions; long-term total parenteral nutrition using glucose as the chief energy source requires administration via a central vein catheter so the hypertonic solution can be rapidly diluted in a high-flow system. The preferred site for central vein infusion is the superior vena cava. Access sites and catheter choices are summarized in [Table 76-7](#). Peripherally inserted central catheters are the most economical option for short-term parenteral nutrition. In one randomized study, the number of catheter-related infections was the same with peripherally and centrally inserted catheters. Tunneled catheters and implanted subcutaneous ports require operating room insertion and are more stable for long-term use. Central catheters should be changed when clinically indicated; routine changes are costly and hazardous. Chlorhexidine solution is a more effective local antiseptic than iodophor or alcohol. Although transparent dressings are helpful in stabilizing catheters and allow easy observation of the skin site, the incidence of catheter-related sepsis is higher than with traditional dry gauze dressings; newer transparent dressings that trap less moisture are under investigation. Catheters made from Silastic material or polyurethane are associated with lower complications than polyvinylchloride catheters. Several types of needleless systems use hub valves, and

contamination rates are higher with these devices when used for long-term parenteral nutrition. Appropriate clinical and laboratory monitoring for patients on parenteral nutrition are summarized in [Table 76-8](#).

Complications (See [Table 76-9](#))

Mechanical The insertion of a central venous catheter should be done only by trained personnel under aseptic techniques. Major mechanical complications include pneumothorax; hemothorax from laceration of the subclavian artery or vein; brachial plexus injury; and malpositioning of the catheter in a cerebral vein, the azygos vein, or the right ventricle. The correct catheter position must be confirmed by x-ray before hypertonic nutrient solution is infused. Catheters can subsequently dislocate, develop leaks, or become detached from the hub and embolize into the heart or pulmonary artery. Catheter thrombosis may occur, especially if the catheter is used for withdrawing blood samples, and extension of the thrombosis to the central vein is frequently coincident with infection. Thrombosed catheters can sometimes be unblocked by urokinase treatment. The addition of low-dose heparin (1000 U/L) to limit thrombosis in parenteral catheters is controversial; no randomized, controlled studies demonstrate benefit, and heparin can contribute to loss of bone mineral, which is already a problem with long-term parenteral nutrition.

Metabolic Fluid overload can cause congestive heart failure, particularly in elderly and debilitated patients. Glucose overload can cause an osmotic diuresis or stimulate insulin secretion, which in turn promotes extracellular to intracellular shifts of potassium and phosphorous. Such shifts are most dangerous in cachectic patients with depletion of potassium and phosphorus stores and can cause arrhythmias, cardiopulmonary dysfunction, and neurologic symptoms. To avoid these problems, parenteral nutrition should be started slowly and monitored carefully. Glucose content is increased gradually as the patient demonstrates tolerance of the high glucose load. Late metabolic complications include cholestatic liver disease with bile sludging and gallstone formation. The exact cause of the liver disease is not understood, but lack of enteral stimulation to bile flow and defective sulfur amino acid metabolism and cholesterol solubilization appear to play a role. Cholestasis is less likely to occur if some enteral feeding is maintained. Parenteral nutrition induces hypercalciuria, which can result in negative calcium balance and osteopenia. Hypercalciuria may have several causes, including the high fixed-acid load of infused amino acids and the bisulfite preservative in parenteral solutions. Earlier, protein hydrolysates were used as an amino acid source and were contaminated with aluminum, which blocked bone mineralization. Aluminum is still a contaminant of some additives such as calcium gluconate. Once patients on long-term parenteral nutrition change from catabolic breakdown to sustained anabolism, deficiencies of micronutrients such as essential fatty acids, trace minerals, and vitamins may develop unless they are supplied in adequate amounts ([Table 76-6](#)).

Infectious Infection of the access line rarely occurs in the first 72 h, and fever during this period is usually due to infection elsewhere or some other cause. Infection of the access line is likely if the fever defervesces when the infusion of the parenteral formula is tapered.

Positive central line cultures suggest catheter sepsis, especially if no other infectious

source is identified and if the organism is *Staphylococcus* or *Candida*. Although removal of the central catheter may allow fungemia to clear spontaneously, antibiotic therapy is recommended for bacterial infections and the more invasive fungi. Catheter sepsis rates are similar in single lumen central lines dedicated to parenteral nutrition whether inserted peripherally via the subclavian vein or tunneled; multiple-lumen catheters are associated with a greater incidence of sepsis. While there is no evidence to support the use of prophylactic antibiotics, recurrent catheter sepsis may be avoided if cuffs are used around the catheter exit site or small amounts of an antibiotic solution are left in the line along with a heparin lock.

ENTERAL NUTRITION

Tube Placement and Patient Monitoring The types of enteral feeding tubes, methods of insertion, their clinical uses, and potential complications are outlined in [Table 76-10](#). The different types of enteral formulas are listed in [Table 76-11](#). Patients on enteral feeding are at risk for many of the same metabolic complications as those receiving parenteral nutrition and should be monitored in the same way ([Table 76-8](#)). Since small-bore tubes are easily displaced, tube position should be checked at intervals by aspirating and measuring the pH of the gut fluid (<4 in stomach, >6 in jejunum).

Complications

Aspiration The debilitated patient with poor gastric emptying and impairment of swallowing and cough mechanisms is at risk for aspiration; this is particularly so for those on respirators. Tracheal suctioning induces coughing and gastric regurgitation, and cuffs on endotracheal or tracheostomy tubes seldom provide protection against aspiration. Under these circumstances, it may be safer to use a large-bore feeding tube to allow for temporary removal of gastric contents during tracheal suction or to use jejunal feeding. Constant gastric infusion of an enteral formula is better tolerated in sick patients than intermittent bolus feeding. A continuous infusion is best achieved with a pump, especially when using fine-bore tubes that have a greater potential to clog. If long-term feeding is anticipated, endoscopic, radiologic, or surgical placement of a gastric tube is preferred by most patients. For long-term ambulatory patients, a gastrostomy tube can be converted to a gastric "button," an access device that is flush with the skin. A nasojejunal tube reduces the risk of aspiration. However, fluoroscopically guided placement of fine-bore tubes through the pylorus is time-consuming, and such tubes frequently pull back into the stomach. A percutaneous combined gastric-suction and jejunal-feeding tube is more reliable. This can be placed radiologically, endoscopically, or surgically.

Diarrhea Enteral feeding often causes diarrhea, especially if bowel function is compromised by bowel disease or drugs. The diarrhea may be controlled by the use of continuous feeding of fiber-containing formulas or by adding an anticholinergic medication to the formula. Diarrhea associated with enteral feeding does not necessarily imply inadequate absorption of nutrients, other than water and electrolytes. Furthermore, since luminal nutrients exert trophic effects on the gut mucosa and enhance the enteric immunologic barrier, it is often appropriate to persist with tube feeding, despite the diarrhea, even when this necessitates supplemental parenteral fluid support.

THE SCOPE AND COST OF NUTRITION SUPPORT

As many as 25% of patients entering tertiary care hospitals have central catheters placed, and 20 to 30% of these catheters are used for parenteral nutrition. The incidence of catheter-related infection reflects the severity of the underlying medical condition and varies from 2 to 30 per thousand catheter days, depending on the type of patients involved. In critically ill patients, catheter sepsis is associated with a 35% mortality rate and a high cost per survivor. Most catheter-related complications derive from faulty insertion and management of the catheter rather than defects in the device. In large tertiary care hospitals, the insertion and management of these lines by specially trained teams can reduce complications by 80%, impacting significantly on outcome and costs. A growing shift from parenteral to enteral nutrition also promises significant cost savings. Home parenteral nutrition costs approximately half as much as similar treatment in the hospital, and home enteral nutrition costs much less.

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77. OBESITY - Jeffrey S. Flier

In a world where food supplies are intermittent, the ability to store energy in excess of what is required for immediate use is essential for survival. Fat cells, residing within widely distributed adipose tissue depots, are adapted to store excess energy efficiently as triglyceride and, when needed, to release stored energy as free fatty acids for use at other sites. This physiologic system, orchestrated through endocrine and neural pathways, permits humans to survive starvation for as long as several months. However, in the presence of nutritional abundance and a sedentary lifestyle, and influenced importantly by genetic endowment, this system increases adipose energy stores and produces adverse health consequences.

DEFINITION AND MEASUREMENT

Obesity is a state of excess adipose tissue mass. Although often viewed as equivalent to increased body weight, this need not be the case -- lean but very muscular individuals may be overweight by arbitrary standards without having increased adiposity. Body weights are distributed continuously in populations, so that a medically meaningful distinction between lean and obese is somewhat arbitrary. Obesity is therefore more effectively defined by assessing its linkage to morbidity or mortality.

Although not a direct measure of adiposity, the most widely used method to gauge obesity is the *body mass index* (BMI), which is equal to weight/height^2 (in kg/m^2) ([Fig. 77-1](#)). Other approaches to quantifying obesity include anthropometry (skin-fold thickness), densitometry (underwater weighing), computed tomography (CT) or magnetic resonance imaging (MRI), and electrical impedance. Using data from the Metropolitan Life Tables, BMIs for the midpoint of all heights and frames among both men and women range from 19 to 26 kg/m^2 ; at a similar BMI, women have more body fat than men. Based on unequivocal data of substantial morbidity, a BMI of 30 is most commonly used as a threshold for obesity in both men and women. Large-scale epidemiologic studies suggest that all-cause, metabolic, and cardiovascular morbidity begin to rise (albeit at a slow rate) when BMIs are ³ 25, suggesting that the cut-off for obesity should be lowered. Some authorities use the term *overweight* (rather than obese) to describe individuals with BMIs between 25 or 27 and 30. A BMI between 25 and 30 should be viewed as medically significant and worthy of therapeutic intervention, especially in the presence of risk factors that are influenced by adiposity, such as hypertension and glucose intolerance.

The distribution of adipose tissue in different anatomic depots also has substantial implications for morbidity. Specifically, intraabdominal and abdominal subcutaneous fat have more significance than subcutaneous fat present in the buttocks and lower extremities. This distinction is most easily made by determining the waist-to-hip ratio, with a ratio >0.9 in women and >1.0 in men being abnormal. Many of the most important complications of obesity, such as insulin resistance, diabetes, hypertension, and hyperlipidemia, and hyperandrogenism in women, are linked more strongly to intraabdominal and/or upper body fat than to overall adiposity. The mechanism underlying this association is unknown but may relate to the fact that intraabdominal adipocytes are more lipolytically active than those from other depots. Release of free fatty acids into the portal circulation has adverse metabolic actions, especially on the

liver.

PREVALENCE

Recent data from the National Health and Nutrition Examination Surveys (NHANES) show that the percent of the American adult population with obesity ([BMI](#) > 30) has increased from 14.5% (between 1976 and 1980) to 22.5% (between 1998 and 1994). As many as 50% of U.S. adults 20 years of age were overweight (defined as BMI > 25) between the years of 1998 and 1991. Because substantial health risks exist in many individuals with BMI between 25 and 30, the increasing prevalence of medically significant obesity raises great concern. Obesity is more common among women and in the poor; the prevalence in children is also rising at a worrisome rate.

PHYSIOLOGIC REGULATION OF ENERGY BALANCE

Substantial evidence suggests that body weight is regulated by both endocrine and neural components that ultimately influence the effector arms of energy intake and expenditure. This complex regulatory system is necessary because even small imbalances between energy intake and expenditure will ultimately have large effects on body weight. For example, a 0.3% positive imbalance over 30 years would result in a 9-kg (20-lb) weight gain. Alterations in stable weight by forced overfeeding or food deprivation induce physiologic changes that resist these perturbations: with weight loss, appetite increases and energy expenditure falls; with overfeeding, appetite falls and energy expenditure increases. This latter compensatory mechanism frequently fails, however, permitting obesity to develop when food is abundant and physical activity is limited. A major regulator of these adaptative responses is the adipocyte-derived hormone leptin, which acts through brain circuits (predominantly in the hypothalamus) to influence appetite, energy expenditure, and neuroendocrine function (see below).

Appetite is influenced by many factors that are integrated by the brain, most importantly within the hypothalamus ([Fig. 77-2](#)). Signals that impinge on the hypothalamic center include neural afferents, hormones, and metabolites. Vagal inputs are particularly important, bringing information from viscera, such as gut distention. Hormonal signals include leptin, insulin, cortisol, and gut peptides such as cholecystokinin, which signals to the brain through the vagus nerve. Metabolites, including glucose, can influence appetite, as seen by the effect of hypoglycemia to induce hunger; however, glucose is not normally a major regulator of appetite. These diverse hormonal, metabolic, and neural signals act by influencing the expression and release of various hypothalamic peptides [e.g., neuropeptide Y (NPY), Agouti-related peptide (AgRP), melanocyte-stimulating hormone (MSH), and melanin concentrating hormone (MCH)] that are integrated with serotonergic, catecholaminergic, and opioid signaling pathways (see below). Psychological and cultural factors also appear to play a role in the final expression of appetite. Apart from rare syndromes involving leptin, its receptor, and the melanocortin system (see below), the defects in this complex appetite control network that account for common causes of obesity are not well understood.

Energy expenditure includes the following components: (1) resting or basal metabolic rate; (2) the energy cost of metabolizing and storing food; (3) the thermic effect of exercise; and (4) adaptive thermogenesis, which varies in response to chronic caloric

intake (rising with increased intake). Basal metabolic rate accounts for about 70% of daily energy expenditure, whereas active physical activity contributes 5 to 10%. Thus, a significant component of daily energy consumption is fixed.

Adaptive thermogenesis occurs in *brown adipose tissue* (BAT), which plays an important role in energy metabolism in many mammals. In contrast to white adipose tissue, which is used to store energy in the form of lipids, BAT expends stored energy as heat. A mitochondrial *uncoupling protein* (UCP-1) in BAT dissipates the hydrogen ion gradient in the oxidative respiration chain and releases energy as heat. The metabolic activity of BAT is increased by a central action of leptin, acting through the sympathetic nervous system, which heavily innervates this tissue. In rodents, BAT deficiency causes obesity and diabetes; stimulation of BAT with a specific adrenergic agonist (β_3 agonist) protects against diabetes and obesity. Although BAT exists in humans (especially neonates), its physiologic role is not yet established. Homologues of UCP-1 may mediate uncoupled mitochondrial respiration in other tissues.

THE ADIPOCYTE AND ADIPOSE TISSUE

Adipose tissue is composed of the lipid-storing adipose cell and a stromal/vascular compartment in which preadipocytes reside. Adipose mass increases by enlargement of adipose cells through lipid deposition, as well as by an increase in the number of adipocytes. The process by which adipose cells are derived from a mesenchymal preadipocyte involves an orchestrated series of differentiation steps mediated by a cascade of specific transcription factors. One of the key transcription factors is *peroxisome proliferator-activated receptor γ* (PPAR γ), a nuclear receptor that binds the thiazolidinedione class of insulin-sensitizing drugs used in the treatment of type 2 diabetes ([Chap. 333](#)).

Although the adipocyte has generally been regarded as a storage depot for fat, it is also an endocrine cell that releases numerous molecules in a regulated fashion ([Fig. 77-3](#)). These include the energy balance-regulating hormone leptin, cytokines such as tumor necrosis factor (TNF) α , complement factors such as factor D (also known as adipsin), prothrombotic agents such as plasminogen activator inhibitor 1, and a component of the blood pressure regulating system, angiotensinogen. These factors, and others not yet identified, play a role in the physiology of lipid homeostasis, insulin sensitivity, blood pressure control, and coagulation and are likely to contribute to obesity-related pathologies.

ETIOLOGY OF OBESITY

Though the molecular pathways regulating energy balance are beginning to be illuminated, the causes of obesity remain elusive. In part, this reflects the fact that obesity is a heterogeneous group of disorders. At one level, the pathophysiology of obesity seems simple: a chronic excess of nutrient intake relative to the level of energy expenditure. However, due to the complexity of the neuroendocrine and metabolic systems that regulate energy intake, storage, and expenditure, it has been difficult to quantitate all the relevant parameters (e.g., food intake and energy expenditure) over time in human subjects.

Role of Genes vs. Environment Obesity is commonly seen in families. Inheritance is usually not Mendelian, however, and it is difficult to distinguish the role of genes and environmental factors. Adoptees usually resemble their biologic rather than adoptive parents with respect to obesity, providing strong support for genetic influences. Likewise, identical twins have very similar [BMIs](#) whether reared together or apart, and their BMIs are much more strongly correlated than those of dizygotic twins. These genetic effects appear to relate to both energy intake and expenditure.

Whatever the role of genes, it is clear that the environment plays a key role in obesity, as evidenced by the fact that famine prevents obesity in even the most obesity-prone individual. In addition, the recent increase in the prevalence of obesity in the United States is too rapid to be due to changes in the gene pool. Cultural factors are also important -- these relate to both availability and composition of the diet and to changes in the level of physical activity. In industrial societies, obesity is more common among poor women, whereas in underdeveloped countries, wealthier women are more often obese. In children, obesity correlates to some degree with time spent watching television. High-fat diets may promote obesity, as may diets rich in simple (as opposed to complex) carbohydrates.

Specific Genetic Syndromes Obesity in rodents has been known for many years to be caused by a number of distinct mutations distributed through the genome. Most of these single-gene mutations cause both hyperphagia and diminished energy expenditure, suggesting a link between these two parameters of energy homeostasis. Identification of the *ob* gene mutation in genetically obese (*ob/ob*) mice represents a major breakthrough in the field. The *ob/ob* mouse develops severe obesity, insulin resistance, and hyperphagia, as well as efficient metabolism (e.g., it gets fat even when given the same number of calories as lean littermates). The product of the *ob* gene is the peptide leptin, a name derived from the Greek root *leptos*, meaning thin. Leptin is secreted by adipose cells and acts through the hypothalamus. Its level of production provides an index of adipose energy stores ([Fig. 77-4](#)). High leptin levels decrease food intake and increase energy expenditure. Another mouse mutant, *db/db*, which is resistant to leptin, has a mutation in the leptin receptor and develops a similar syndrome. The *ob* gene is present in humans and expressed in fat. Several families with morbid, early-onset obesity due to inactivating mutations in either leptin or the leptin receptor have been described, thus demonstrating the biologic relevance of leptin in humans. The obesity in these individuals begins shortly after birth, is severe, and is accompanied by neuroendocrine abnormalities. The most prominent of these is hypogonadotropic hypogonadism, which is reversed by leptin replacement. Central hypothyroidism and growth retardation are seen in the mouse model, but their occurrence in leptin-deficient humans is less clear. To date, there is no evidence to suggest that mutations or polymorphisms in the leptin or leptin receptor genes play a prominent role in common forms of obesity.

Mutations in several other genes cause severe obesity in humans ([Table 77-1](#)), each of these syndromes is rare. Mutations in the gene encoding proopiomelanocortin (POMC) cause severe obesity through failure to synthesize α -[MSH](#), a key neuropeptide that inhibits appetite in the hypothalamus. The absence of POMC also causes secondary adrenal insufficiency due to absence of adrenocorticotrophic hormone (ACTH), as well as pale skin and red hair due to absence of MSH. Proenzyme convertase 1 (PC-1)

mutations are thought to cause obesity by preventing synthesis of α -MSH from its precursor peptide, POMC. α -MSH binds to the type 4 melanocortin receptor (MC4R), a key hypothalamic receptor that inhibits eating; mutations of this receptor also cause obesity. These three genetic defects, although rare, define a pathway through which leptin (by stimulating POMC and increasing MSH) restricts food intake and limits weight ([Fig. 77-5](#)).

In addition to these human obesity genes, studies in rodents reveal several other molecular candidates for hypothalamic mediators of human obesity or leanness. The *tub* gene encodes a hypothalamic peptide of unknown function; mutation of this gene causes late-onset obesity. The *fat* gene encodes carboxypeptidase E, a peptide-processing enzyme; mutation of this gene is thought to cause obesity by disrupting production of one or more neuropeptides. [AgRP](#) is coexpressed with [NPY](#) in arcuate nucleus neurons. AgRP antagonizes [MSH](#) action at MC4 receptors, and its overexpression induces obesity. A putative activating mutation in the gene encoding PPAR γ , the adipocyte transcription factor required for adipogenesis, has been linked to obesity in a group of German subjects.

A number of complex human syndromes with defined inheritance are associated with obesity ([Table 77-2](#)). Although specific genes are undefined at present, their identification will likely enhance our understanding of more common forms of human obesity. In the Prader-Willi syndrome, obesity coexists with short stature, mental retardation, hypogonadotropic hypogonadism, hypotonia, small hands and feet, fish-shaped mouth, and hyperphagia. Most patients have a chromosome 15 deletion ([Chap. 66](#)). Laurence-Moon-Biedl syndrome involves obesity, mental retardation, retinitis pigmentosa, polydactyly, and hypogonadotropic hypogonadism.

Other Specific Syndromes Associated with Obesity

Cushing's Syndrome Although obese patients commonly have central obesity, hypertension, and glucose intolerance, they lack other specific stigmata of Cushing's syndrome ([Chap. 331](#)). Nonetheless, a potential diagnosis of Cushing's syndrome is often entertained. Cortisol production and urinary metabolites (17OH steroids) may be increased in simple obesity. Unlike in Cushing's syndrome, however, cortisol levels in blood and urine in the basal state and in response to CRH or [ACTH](#) are normal; the overnight 1-mg dexamethasone suppression test is normal in 90%, with the remainder being normal on a standard 2-day low-dose dexamethasone suppression test.

Hypothyroidism The possibility of hypothyroidism should be considered when evaluating obesity, but it is an uncommon cause of obesity; hypothyroidism is easily ruled out by measuring thyroid stimulating hormone (TSH). Much of the weight gain that occurs in hypothyroidism is due to myxedema ([Chap. 330](#)).

Insulinoma Patients with insulinoma often gain weight as a result of overeating to avoid hypoglycemia symptoms ([Chap. 334](#)). The increased substrate plus high insulin levels promotes energy storage in fat. This can be marked in some individuals but is modest in most.

Craniopharyngioma and Other Disorders Involving the Hypothalamus Whether through

tumors, trauma, or inflammation, hypothalamic dysfunction of systems controlling satiety, hunger, and energy expenditure can cause varying degrees of obesity ([Chap. 328](#)). It is uncommon to identify a discrete anatomic basis for these disorders. Subtle hypothalamic dysfunction is probably a more common cause of obesity than can be documented using currently available techniques. Growth hormone (GH), which exerts lipolytic activity, is diminished in obesity and increases with weight loss. Despite low growth hormone levels, insulin-like growth factor (IGF) I (somatomedin) production is normal, suggesting that GH suppression is a compensatory response to increased nutritional supply.

Pathogenesis of Common Obesity Obesity can result from increased energy intake, decreased energy expenditure, or a combination of the two. Thus, identifying the etiology of obesity should involve measurements of both parameters. However, it is nearly impossible to perform direct and accurate measurements of energy intake in free-living individuals. Obese people, in particular, appear to underreport intake. Measurements of chronic energy expenditure have only recently become available using doubly-labeled water or metabolic chamber/rooms. In subjects at stable weight and body composition, energy intake equals expenditure. Consequently, these techniques allow determination of energy intake in free-living individuals. The level of energy expenditure differs in established obesity, during periods of weight gain or loss, and in the pre- or postobese state. Studies that fail to take note of this phenomenon are not easily interpreted.

There is increased interest in the concept of a body weight "set point." This idea is supported by physiologic mechanisms centered around a sensing system in adipose tissue that reflects fat stores, and a receptor, or "adipostat," that is in the hypothalamic centers. When fat stores are depleted, the adipostat signal is low, and the hypothalamus responds by stimulating hunger and decreasing energy expenditure to conserve energy. Conversely, when fat stores are abundant, the signal is increased, and the hypothalamus responds by decreasing hunger and increasing energy expenditure. The recent discovery of the *ob* gene, and its product leptin, provides a molecular basis for this physiologic concept (see above).

What Is the Status of Food Intake in Obesity (Do the Obese Eat More Than the Lean?) This question has stimulated much debate, due in part to the methodologic difficulties inherent in determining food intake. Many obese individuals believe that they eat small quantities of food, and this claim has often been supported by the results of food intake questionnaires. However, it is now established that average energy expenditure increases as people get more obese, due primarily to the fact that metabolically active lean tissue mass increases with obesity. Given the laws of thermodynamics, the obese person must therefore eat more than the average lean person to maintain their increased weight. It may be the case, however, that a subset of individuals who are predisposed to obesity have the capacity to become obese initially without an absolute increase in caloric consumption.

What Is the State of Energy Expenditure in Obesity? The average total daily energy expenditure is higher in obese than lean individuals when measured at stable weight. However, energy expenditure falls as weight is lost, due in part to loss of lean body mass and to decreased sympathetic nerve activity. When reduced to near-normal

weight and maintained there for a while, (some) obese individuals have lower energy expenditure than (some) lean individuals. There is also a tendency for those who develop obesity as infants or children to have lower resting energy expenditure rates than those who remain lean.

The physiologic basis for variable rates of energy expenditure (at a given body weight and level of energy intake) is essentially unknown. A mutation in the human β_3 adrenergic receptor may be associated with increased risk of obesity and/or insulin resistance in certain (but not all) populations. Homologues of the [BAT](#) uncoupling protein, named [UCP](#)-2 and UCP-3, have been identified in both rodents and humans. UCP-2 is expressed widely, whereas UCP-3 is primarily expressed in skeletal muscle. These proteins may play a role in disordered energy balance.

One newly described component of thermogenesis, called *nonexercise activity thermogenesis* (NEAT), has been linked to obesity. It is the thermogenesis that accompanies physical activities other than volitional exercise, such as the activities of daily living, fidgeting, spontaneous muscle contraction, and maintaining posture. NEAT accounts for about two-thirds of the increased daily energy expenditure induced by overfeeding. The wide variation in fat storage seen in overfed individuals is predicted by the degree to which NEAT is induced. The molecular basis for NEAT and its regulation are unknown.

Leptin in Typical Obesity The vast majority of obese people have increased leptin levels but do not have mutations of either leptin or its receptor. They appear, therefore, to have a form of functional "leptin resistance." Data suggesting that some individuals produce less leptin per unit fat mass than others or have a form of relative leptin deficiency that predisposes to obesity are at present contradictory and unsettled. The mechanism for leptin resistance, and whether it can be overcome by raising leptin levels, is not yet established. Some data suggest that leptin may not effectively cross the blood-brain barrier as levels rise. It is also possible that leptin signaling inhibitors are involved in the leptin-resistant state.

PATHOLOGIC CONSEQUENCES OF OBESITY

Obesity has major adverse effects on health. Morbidly obese individuals (>200% ideal body weight) have as much as a twelvefold increase in mortality. Mortality rates rise as obesity increases, particularly when obesity is associated with increased intraabdominal fat (see above). It is also apparent that the degree to which obesity affects particular organ systems is influenced by susceptibility genes that vary in the population.

Insulin Resistance and Type 2 Diabetes Mellitus Hyperinsulinemia and insulin resistance are pervasive features of obesity, increasing with weight gain and diminishing with weight loss. Insulin resistance is more strongly linked to intraabdominal fat than to fat in other depots. The molecular link between obesity and insulin resistance has been sought for many years, with the major factors under investigation being: (1) insulin itself, by inducing receptor downregulation; (2) free fatty acids, known to be increased and capable of impairing insulin action; and (3) the cytokine [TNF](#)- α , which is produced by adipocytes, overexpressed in obese adipocytes, and capable of inhibiting insulin action. Despite insulin resistance, most obese individuals do not develop diabetes, suggesting

that the onset of diabetes requires an interaction between obesity-induced insulin resistance and other factors that predispose to diabetes, such as impaired insulin secretion ([Chap. 333](#)). Obesity, however, is a major risk factor for diabetes, and as many as 80% of patients with type 2 diabetes mellitus are obese. Weight loss, even of modest degree, is associated with increased insulin sensitivity and often improves glucose control in diabetes.

Reproductive Disorders Disorders that affect the reproductive axis are associated with obesity in both men and women. Male hypogonadism is associated with increased adipose tissue, often distributed in a pattern more typical of females. In men >160% ideal body weight, plasma testosterone and sex hormone-binding globulin (SHBG) are often reduced, and estrogen levels (derived from conversion of adrenal androgens in adipose tissue) are increased ([Chap. 335](#)). Gynecomastia may be seen. However, masculinization, libido, potency, and spermatogenesis are preserved in most of these individuals. Free testosterone may be decreased in morbidly obese men whose weight exceeds 200% ideal body weight.

Obesity has long been associated with menstrual abnormalities in women, particularly in women with upper body obesity ([Chaps. 52 and 336](#)). Common findings are increased androgen production, decreased SHBG, and increased peripheral conversion of androgen to estrogen. Most obese women with oligomenorrhea have the polycystic ovarian syndrome (PCOS), with its associated anovulation and ovarian hyperandrogenism; 40% of women with PCOS are obese. Interestingly, most nonobese women with PCOS are also insulin-resistant, suggesting that insulin resistance, hyperinsulinemia, or the combination of the two are causative or contribute to the ovarian pathophysiology in PCOS in both obese and lean individuals. In obese women with PCOS, weight loss or treatment with insulin-sensitizing drugs often restores normal menses, along with a fall in estrone levels and normalized gonadotropin secretion. The increased conversion of androstenedione to estrogen, which occurs to a greater degree in women with lower body obesity, may contribute to the increased incidence of uterine cancer in postmenopausal women with obesity.

Cardiovascular Disease The Framingham Study revealed that obesity was an independent risk factor for the 26-year incidence of cardiovascular disease in men and women [including coronary disease, stroke, and congestive heart failure (CHF)]. The waist/hip ratio may be the best predictor of these risks. When the additional effects of hypertension and glucose intolerance associated with obesity are included, the adverse impact of obesity is even more evident. The effect of obesity on cardiovascular mortality in women may be seen at BMIs as low as 25. Obesity, especially abdominal obesity, is associated with an atherogenic lipid profile, with increased low-density lipoprotein (LDL) cholesterol, very low density lipoprotein and triglyceride, and decreased high-density lipoprotein cholesterol ([Chap. 344](#)). Obesity is also associated with hypertension. Measurement of blood pressure in the obese requires use of a larger cuff size to avoid artifactual increases. Obesity-induced hypertension is associated with increased peripheral resistance and cardiac output, increased sympathetic nervous system tone, increased salt sensitivity, and insulin-mediated salt retention; it is often responsive to modest weight loss.

Pulmonary Disease Obesity may be associated with a number of pulmonary

abnormalities. These include reduced chest wall compliance, increased work of breathing, increased minute ventilation due to increased metabolic rate, and decreased total lung capacity and functional residual capacity ([Chap. 250](#)). Severe obesity may be associated with obstructive sleep apnea and the "obesity hypoventilation syndrome" ([Chap. 263](#)). Sleep apnea can be obstructive (most common), central, or mixed. Weight loss (10 to 20 kg) can bring substantial improvement, as can major weight loss following gastric bypass or restrictive surgery. Continuous positive airway pressure has been used with some success.

Gallstones Obesity is associated with enhanced biliary secretion of cholesterol, supersaturation of bile, and a higher incidence of gallstones, particularly cholesterol gallstones ([Chap. 302](#)). A person 50% above ideal body weight has about a sixfold increased incidence of symptomatic gallstones. Paradoxically, fasting increases supersaturation of bile by decreasing the phospholipid component. Fasting-induced cholecystitis is a complication of extreme diets.

Cancer Obesity in males is associated with higher mortality from cancer of the colon, rectum, and prostate; obesity in females is associated with higher mortality from cancer of the gallbladder, bile ducts, breasts, endometrium, cervix, and ovaries. Some of the latter may be due to increased rates of conversion of androstenedione to estrone in adipose tissue of obese individuals.

Bone, Joint, and Cutaneous Disease Obesity is associated with an increased risk of osteoarthritis, no doubt partly due to the trauma of added weight bearing. The prevalence of gout may also be increased ([Chap. 322](#)). Among the skin problems associated with obesity is acanthosis nigricans, manifested by darkening and thickening of the skin folds on the neck, elbows, and dorsal interphalangeal spaces. Acanthosis reflects the severity of underlying insulin resistance and diminishes with weight loss. Friability of skin may be increased, especially in skin folds, enhancing the risk of fungal and yeast infections. Finally, venous stasis is increased in the obese.

TREATMENT

Obesity is a chronic medical condition. Successful treatment, defined as the sustained attainment of normal body weight without producing unacceptable treatment-induced morbidity, is rarely achieved in clinical practice. Many approaches produce short-term weight loss, and this has clear benefits for associated morbidities such as hypertension and diabetes. Despite the fact that sustained weight loss is uncommon, enormous resources are expended in pursuit of this goal.

Treatment goals should be guided by the health risks of obesity in any given individual ([Fig. 77-6](#)). The clinician should always consider the possibility that an individual has an identified cause of obesity, such as hypothyroidism, hypercortisolism, male hypogonadism, insulinoma, or central nervous system disease that affects hypothalamic function. Although they are infrequent causes of obesity, specific therapy may be available.

Behavior Modification The principles of behavior modification provide the underpinnings for many current programs of weight reduction. Typically, the patient is

requested to monitor and record the circumstances related to eating, and rewards are designed to modify maladaptive behaviors. Patients may benefit from counseling offered in a stable group setting for extended periods of time, including after weight loss.

Diet Reduced caloric intake is the cornerstone of obesity treatment. The fundamental goal is the sustained reduction of energy intake below that of energy expenditure. The difficulty in achieving this goal has led to a wide array of suggested diets that vary in recommended calorie content (from total fasting to mild reductions), as well as specific food content and form (e.g., liquid vs. solid). There is no scientific evidence to validate the utility of specific "fad diets." The main diet regimens in use follow several general facts relevant to food intake and weight loss. First, a deficit of 7500 kcal will produce a weight loss of approximately 1 kg. Therefore, eating 100 kcal/d less for a year should cause a 5-kg weight loss, and a deficit of 1000 kcal/d should cause a loss of approximately 1 kg per week. The rate of weight loss on a given caloric intake is related to the rate of energy expenditure. Because obese individuals have a higher metabolic rate than lean individuals, and because men have a higher metabolic rate than women (due to their greater lean body mass), the rate of weight loss is greater among the more obese and among men (relative to women). With chronic caloric restriction, metabolic rate diminishes, but because of reduced lean body mass (along with much greater loss in fat mass) and possibly because of other adaptations. This fall in metabolic rate with food restriction slows the rate of weight loss on a constant diet. With total starvation or diets restricted to <600 kcal/d, initial weight loss over the first week results predominantly from natriuresis and the loss of fluids.

Very low energy diets (e.g., 400 to 600 kcal/d) are widely used. The liquid protein diets popularized in the 1970s were proved to be unsafe, causing >60 deaths. Life-threatening arrhythmias were documented in the clinical research setting, a consequence of both low-quality protein and deficiencies of vitamins, minerals, and trace elements. These types of diets have now been substantially modified. A very low energy diet consisting of 45 to 70 g high-quality protein, 30 to 50 g carbohydrate, and approximately 2 g fat per day, as well as supplements of vitamins, minerals, and trace elements, appears to be safe in selected patients under medical supervision. Patients should not be started on such diets unless they are >130% of their ideal body weight. Contraindications include pregnancy, cancer, recent myocardial infarction, cerebrovascular disease, hepatic disease, or untreated psychiatric disease. When used in patients with diabetes who are receiving insulin or oral agent therapy, close supervision is required and diabetic treatment may need to be adjusted. Whenever possible, exercise regimens and behavioral modification approaches should be used in conjunction with the diet.

Advantages of very low calorie diets are the greater rate of weight loss compared to less restrictive diets, as well as the possible beneficial effect of hunger suppression brought about by the production of ketones. In patients on such diets, blood pressure, blood glucose, cholesterol, and triglyceride levels fall, and pulmonary function and exercise tolerance improve. Sleep apnea may improve within a few weeks. Complications of these very low energy diets are usually minor and include fatigue, constipation or diarrhea, dry skin, hair loss, menstrual irregularities, orthostatic dizziness, and difficulty concentrating. Cholelithiasis and pancreatitis may occur when such diets are interrupted by binge eating; gallstones have been shown to develop in as many as 25% of patients

while on the diet.

Low-calorie diets, >800 kcal/d, are applicable to most patients and have fewer restrictions than the very low calorie diets. Considerable controversy has attended the question of which diet composition is most appropriate for promoting weight loss. Though commonly recommended, benefits resulting from very low fat diets are modest at best. Nonetheless, the health effects of low-fat diets -- apart from curbing obesity -- may be important. A diet rich in fruits, vegetables, and whole grains may promote weight loss and is preferable to low-fat diets in which large amounts of simple carbohydrates are substituted for fats. The latter may actually promote obesity. Some have advocated diets with protein replacement of simple carbohydrates in an effort to minimize insulin production. The efficacy of this strategy, aside from overall calorie reduction, is unknown.

Exercise Exercise is an important component of the overall approach to treating obesity. Increased energy expenditure is the most obvious mechanism for an effect of exercise. The impact of an exercise regimen as a sole therapy of obesity has been difficult to document. On the other hand, exercise appears to be a valuable means to sustain diet therapy ([Fig. 77-7](#)). Even if exercise had no such salutary effect, it would be valuable in the obese individual for its effects on cardiovascular tone and blood pressure. Because many obese individuals have not engaged in exercise on a regular basis and may have cardiovascular risk factors, it should be introduced gradually and under medical supervision, especially in the most obese individuals.

Drugs Unfortunately, drug treatment of obesity is rarely efficacious. Despite short-term benefits, medication-induced weight loss is often associated with rebound weight gain after the cessation of drug use, side effects from the medications, and the potential for drug abuse. Given the need for effective therapies, many possible compounds have been evaluated. On the basis of placebo-controlled trials, the U.S. Food and Drug Administration (FDA) approved several amphetamine-like agents for short-term use. Phentermine is an amphetamine-like drug with low addictive potential that has shown modest efficacy (10 vs. 4.4 kg of weight loss over a 24-week period in well-controlled study). This class of drugs is thought to act centrally by reducing appetite. Effects on energy expenditure are less clear. Over-the-counter drugs, such as phenylpropanolamine HCl, have similar efficacy to prescription appetite suppressants in short-term studies. Drugs that promote serotonin release or inhibit serotonin reuptake, such as fenfluramine, also have modest efficacy. When fenfluramine was administered together with phentermine, as "fen-phen," the combination was widely used based on controlled trials that demonstrated modest but definite efficacy. However, the risk of primary pulmonary hypertension was increased up to 20-fold in association with this treatment. The FDA withdrew approval of the fen-phen combination in 1997 when reports suggested an association with right- and left-sided valvular heart disease. The histopathologic features of the valvular disease are similar to those seen in carcinoid syndrome and are thought to result from fenfluramine. Though the true incidence and long-term effects of these valvular lesions are currently unknown, the occurrence of this complication has been verified in multiple studies.

Sibutramine is a novel central reuptake inhibitor of both norepinephrine and serotonin. Using a once-daily dose over 24 weeks, it produced a 7% weight loss in a double-blind,

placebo-controlled trial. It lowered cholesterol and triglyceride levels and exhibited similar clinical efficacy to fenfluramine. Sibutramine increases pulse and blood pressure in some patients, and long-term safety is not established. Orlistat is an inhibitor of intestinal lipase that causes modest weight loss due to drug-induced fat malabsorption. A randomized, double-blind trial over 2 years revealed modest weight loss (8.7 kg for 120 mg orlistat versus 5.8 kg from diet alone) during the first year and better maintenance of weight loss in a second year compared to the placebo-treated group (3.2 kg regained versus 5.6 kg regained for placebo). [LDL](#) cholesterol and insulin levels were also reduced. In patients with obesity and type 2 diabetes mellitus, the antidiabetic medication metformin tends to decrease body weight. The mechanism appears to involve inhibition of appetite. Thyroid hormone has little place in the treatment of obesity, as the vast majority of obese individuals are euthyroid. It promotes loss of lean body mass and raises the risk of complications from the hyperthyroid state.

β_3 -Adrenergic receptor agonists may provide a new treatment approach for obesity. Drugs of this class are in clinical trials. In animals, β_3 agonists promote leanness by stimulating thermogenesis in [BAT](#); they also stimulate lipolysis in white adipose tissue. These drugs also reduce insulin resistance and lower blood glucose in animal models by a mechanism that is not yet defined. Recombinant human leptin is also in clinical trials. In the rare cases of leptin deficiency caused by mutations of the leptin gene, the administration of recombinant leptin is highly effective. Preliminary reports suggest that the response to leptin is limited or absent in common causes of obesity (which are associated with hyperleptinemia and leptin resistance). New drugs are also being developed based on insights into central pathways that regulate body weight. These include antagonists for [NPY](#) receptors (subtypes Y1, Y5) and agonists for melanocortin 4 receptors.

Surgery Morbid obesity, commonly defined as either 45 kg (100 lb) or 100% above ideal body weight, is estimated to increase mortality by as much as twelvefold in men between 25 and 34 years of age and sixfold between 35 and 45 years of age. Deaths from cardiovascular disease, diabetes, and accidents have been documented. In response to ineffective treatment using diet, exercise, and available drugs, surgical approaches have been tried. The potential benefits of surgery include major weight loss and improvement in hypertension, diabetes, sleep apnea, [CHF](#), angina, hyperlipidemia, and venous disease. Many different approaches have been used, often without adequate assessment of efficacy and complications. Jejunoileal bypass surgery has largely been abandoned because of complications, which have included electrolyte disturbances, nephrolithiasis, gallstones, gastric ulcers, arthritis, and hepatic dysfunction, with cirrhosis occurring in as many as 7% of patients. Two procedures in common use today are the vertical-banded gastroplasty and the Roux-en-Y gastric bypass ([Fig. 77-8](#)).

Following the National Institutes of Health Consensus Conference on Gastrointestinal Surgery for Severe Obesity in 1991, it was recommended that suitable patients be selected using the following criteria: (1) the presence of 45 kg (100 lb) or 100% above ideal body weight, or one or more severe medical conditions related to refractory obesity; (2) repeated failures of other therapeutic approaches; (3) at eligible weight for 3 to 5 years; (4) capability of tolerating surgery; (5) absence of alcoholism, other addictions, or major psychopathology; and (6) prior clearance by a psychiatrist. It is

recommended that an appropriately experienced surgeon work together with nutritionists and other support personnel; evaluation and follow-up programs should be monitored closely.

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(Bibliography omitted in Palm version)

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78. EATING DISORDERS - B. Timothy Walsh

Anorexia nervosa and bulimia nervosa are characterized by severe disturbances of eating behavior. The salient feature of *anorexia nervosa* is a refusal to maintain a minimally normal body weight. *Bulimia nervosa* is characterized by recurrent episodes of binge eating followed by abnormal compensatory behaviors, such as self-induced vomiting. Anorexia nervosa and bulimia nervosa are closely related. Both occur primarily among previously healthy young women who become overly concerned with body shape and weight. Many patients with bulimia nervosa have past histories of anorexia nervosa, and many patients with anorexia nervosa engage in binge eating and purging behavior. In the current diagnostic system, the critical distinction between anorexia nervosa and bulimia nervosa depends on body weight: patients with anorexia nervosa are, by definition, significantly underweight, whereas the weights of patients with bulimia nervosa are in the normal range or above.

Another syndrome of disturbed eating behavior has been described recently: *Binge eating disorder* is characterized by repeated episodes of binge eating, similar to those of bulimia nervosa, in the absence of inappropriate compensatory behavior. Patients with binge eating disorder are typically middle-aged men or women with significant obesity. They have an increased frequency of anxiety and depression compared to similarly obese patients without binge eating disorder. It is not known whether patients with binge eating disorder are at increased risk for medical complications or what treatments are most useful.

EPIDEMIOLOGY

In women, the full syndrome of anorexia nervosa occurs with a lifetime prevalence of approximately 0.5%; bulimia nervosa occurs with a lifetime prevalence of 1 to 3%. Variants of these eating disorders with only some features of anorexia nervosa or bulimia nervosa are much more common and occur in 5 to 10% of young women. Both anorexia nervosa and bulimia nervosa also occur in males but at frequencies approximately one-tenth of those in females.

Anorexia nervosa and bulimia nervosa are more prevalent in cultures where food is plentiful and in which being thin is associated with attractiveness. These disorders are more frequent among young women who place a premium on thinness, such as ballet dancers and models. The incidence of anorexia nervosa appears to have increased in recent decades. The frequency of bulimia nervosa increased dramatically in the early 1970s and 1980s but may have declined somewhat in recent years.

DIAGNOSIS

The diagnosis of eating disorders is based on the presence of characteristic behavioral, psychological, and physical attributes ([Tables 78-1](#) and [78-2](#)). Widely accepted diagnostic criteria are provided by the American Psychiatric Association's *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV).

ANOREXIA NERVOSA

For anorexia nervosa, these criteria include weight <85% of the expected weight for age and height, which is roughly equivalent to a body mass index (BMI) of 18.5 kg/m² for adult women. This weight criterion is somewhat arbitrary, so that a patient who meets all other diagnostic criteria but weighs between 85 and 90% of expected would still merit the diagnosis of anorexia nervosa. Despite being underweight, patients with anorexia nervosa are irrationally afraid of gaining weight, often out of a concern that weight gain will get "out of control." They also exhibit a distortion of body image (criterion 3, [Table 78-1](#)), which may express itself in several ways. For example, despite being emaciated, patients with anorexia nervosa may believe that their body as a whole, or some part of their body, is too fat and experience additional weight loss as a highly rewarding achievement. The current diagnostic criteria require that women with anorexia nervosa not have spontaneous menses, but occasional patients with the characteristics and complications of anorexia nervosa describe regular menses. Two mutually exclusive subtypes of anorexia nervosa are specified in [DSM-IV](#). Patients whose weight loss is maintained primarily by caloric restriction, perhaps augmented by excessive exercise, are considered to have the "restricting" subtype of anorexia nervosa. The "binge eating/purging" subtype is characterized by self-induced vomiting or laxative abuse. Patients with the binge/purge subtype are more prone to develop electrolyte imbalances, are more emotionally labile, and are more likely to have other problems with impulse control, such as drug abuse.

The diagnosis of anorexia nervosa can usually be made confidently on the basis of history when significant weight loss is accomplished by restrictive dieting and excessive exercise and is accompanied by a marked reluctance to gain weight. Patients with anorexia nervosa often deny that they have a serious problem and may be brought to medical attention by concerned family or friends. In atypical presentations, other causes of significant weight loss in previously healthy young people should be considered, including inflammatory bowel disease, gastric outlet obstruction, central nervous system (CNS) tumors, and neoplasm ([Chap. 43](#)).

BULIMIA NERVOSA

The critical diagnostic features of bulimia nervosa are repeated episodes of binge eating followed by inappropriate and abnormal behaviors aimed at avoiding weight gain. During binges, patients with this disorder tend to consume large amounts of sweet foods with a high fat content, such as dessert items. The most frequent compensatory behaviors are self-induced vomiting and laxative abuse, but a wide variety of techniques have been described, including the omission of insulin injections by diabetics. Typically, patients with bulimia nervosa are ashamed of their behavior and endeavor to keep their disorder hidden from family and friends. Like patients with anorexia nervosa, those with bulimia nervosa place an unusual emphasis on weight and shape as a basis for their self-esteem.

As in anorexia nervosa, there are two mutually exclusive subtypes of bulimia nervosa. Patients with the "purging" subtype utilize compensatory behaviors that directly rid the body of calories or fluids (e.g., self-induced vomiting, laxative or diuretic abuse), whereas those with the "nonpurging" subtype attempt to compensate for binges by fasting or by excessive exercise. Patients with the nonpurging subtype tend to be heavier and are less prone to fluid and electrolyte disturbances.

The diagnosis of bulimia nervosa requires a candid history from the patient detailing recurrent, large eating binges followed by the purposeful use of inappropriate mechanisms to avoid weight gain. Most patients with bulimia nervosa who present for treatment are distressed by their inability to control their eating behavior but are able to provide such details if queried in a supportive and nonjudgmental fashion.

ETIOLOGY

The fundamental etiology of the eating disorders is unknown but is believed to involve a combination of psychological, biologic, and cultural risk factors. Many of these risk factors, such as sexual or physical abuse and a family history of mood disturbance or substance abuse, are best viewed as nonspecific risk factors that increase vulnerability to a range of psychiatric disorders. Other factors appear to be more specific to the development of an eating disorder.

Patients who develop anorexia nervosa are inclined to be more obsessional and perfectionist than their peers. The disorder often begins as a diet not distinguishable at the outset from those undertaken by many adolescents and young women. As weight loss progresses, the fear of gaining weight grows; dieting becomes stricter; and psychological, behavioral, and medical aberrations increase. The fact that most cases are reported from countries where food is plentiful and where thinness, especially among women, is highly valued suggests that cultural factors play a significant role in the development of anorexia nervosa. However, it is notable that the clinical syndrome was well described over a century ago, when the cultural pressures were quite different.

Numerous physiologic disturbances, including abnormalities in a variety of neurotransmitter systems, have been described in anorexia nervosa (see below). It is difficult to distinguish neurochemical, metabolic, and hormonal changes that may have a role in the initiation or perpetuation of the syndrome from those that are secondary to the disorder. The resolution of most of these abnormalities with weight restoration argues against their having a critical etiologic role.

Bulimia nervosa typically begins during or following an episode of dieting, often in association with depressed mood. Patients who develop bulimia nervosa describe a higher-than-expected prevalence of childhood and parental obesity, suggesting that a predisposition towards obesity may increase vulnerability to this eating disorder. The marked increase in the number of cases of bulimia nervosa during the past 25 years and the rarity of bulimia nervosa in underdeveloped countries suggest that cultural factors are important. Several biologic abnormalities in patients with bulimia nervosa may perpetuate this disorder once it has begun. These include abnormalities of [CNS](#) serotonergic function, which is involved in the regulation of eating behavior, and disruption of peripheral satiety mechanisms, including the release of cholecystokinin (CCK) from the small intestine.

Genetic factors probably contribute to the risk of development of eating disorders, as the incidence of these disorders is greater in families with one affected member and the concordance in monozygotic twins is greater than in dizygotic twins. However, specific genes have not been identified, and the range of the estimates of heritability is large.

CLINICAL FEATURES ([Table 78-3](#))

ANOREXIA NERVOSA

Anorexia nervosa typically begins in mid to late adolescence, sometimes in association with a stressful life event such as leaving home for school. The disorder occasionally develops in early puberty, before menarche, but seldom begins after age 40. Despite being underweight, patients with anorexia nervosa rarely complain of hunger or fatigue and often exercise extensively. Further weight loss is viewed by the patient as a fulfilling accomplishment, while weight gain is seen as a personal failure. Patients tend to become socially withdrawn and increasingly committed to work or study, dieting, and exercise. As weight loss progresses, thoughts of food dominate mental life and idiosyncratic rules develop around eating. Patients with anorexia nervosa may obsessively collect cookbooks and recipes and be drawn to food-related occupations. Despite the denial of hunger, one-quarter to one-half of patients with anorexia nervosa engage in eating binges.

Physical Features Patients with anorexia nervosa typically have few physical complaints but may note cold intolerance and constipation. Some women who develop anorexia nervosa after menarche report that their menses ceased before significant weight loss occurred. Weight and height should be measured to allow calculation of [BMI](#) (kg/m^2). Vital signs may reveal bradycardia, hypotension, and hypothermia. Soft, downy hair growth (lanugo) sometimes occurs, especially on the back, and alopecia may be seen. Salivary gland enlargement, which is associated with starvation as well as with binge eating and vomiting, may make the face appear surprisingly full in contrast to the marked general wasting. Acrocyanosis of the digits is common, and peripheral edema can be seen in the absence of hypoalbuminemia, particularly when the patient begins to regain weight. Some patients who consume large amounts of vegetables containing vitamin A develop a yellow tint to the skin (*hypercarotenemia*), which is especially notable on the palms.

Laboratory Abnormalities Mild normochromic, normocytic anemia is frequent, as is mild to moderate leukopenia, with a disproportionate reduction of polymorphonuclear leukocytes. Dehydration may result in slightly increased levels of blood urea nitrogen and creatinine. Serum liver enzyme levels may increase, especially during the early phases of refeeding. The level of serum proteins is usually normal. Blood sugar is often low and serum cholesterol may be moderately elevated. Gastrointestinal motility is diminished, leading to reduced gastric emptying and constipation. A range of electrolyte disturbances may develop, reflecting the degree to which the patient restricts or overconsumes fluids and whether the patient engages in purging behavior. Hypokalemic alkalosis suggests self-induced vomiting or the use of diuretics. Hyponatremia is common and may result from excess fluid intake and disturbances in the secretion of antidiuretic hormone.

Endocrine Abnormalities The regulation of virtually every endocrine system is altered in anorexia nervosa, but the most striking changes occur in the reproductive system. Amenorrhea is hypothalamic in origin and reflects diminished production of gonadotropin-releasing hormone (GnRH). When exogenous GnRH is administered in a

physiologic pulsatile manner, pituitary responses of luteinizing hormone (LH) and follicle stimulating hormone (FSH) are normalized, indicating the absence of a primary pituitary abnormality. The resulting gonadotropin deficiency causes low plasma estrogen in women and reduced testosterone in men. The hypothalamic GnRH pulse generator is exquisitely sensitive, particularly in women, to body weight, stress, and exercise, each of which may contribute to *hypothalamic amenorrhea* in anorexia nervosa ([Chap. 336](#)). Although the mechanisms underlying these effects are unknown, the decreased adipose tissue associated with weight loss leads to a marked reduction in leptin, a hormone that plays a permissive role in GnRH production ([Chap. 77](#)). In many patients, weight gain to a specific threshold triggers restoration of the GnRH pulse generator, initially recapitulating the pubertal pattern of nocturnal gonadotropin secretion before returning to the normal adult pattern.

Serum cortisol and 24-h urine free cortisol levels are generally elevated but without characteristic clinical signs of cortisol excess. Thyroid function tests resemble the pattern seen in euthyroid sick syndrome ([Chap. 330](#)). Thyroxine (T₄) and free T₄ levels are usually in the low-normal range, triiodothyronine (T₃) levels are reduced, and reverse T₃(rT₃) is elevated. The level of thyroid stimulating hormone (TSH) is normally or partially suppressed. Growth hormone is increased, but insulin-like growth factor 1 (IGF-1), which is produced mainly by the liver, is reduced, as it is in other conditions of starvation. Diminished bone density is routinely observed in anorexia nervosa and reflects the effects of multiple nutritional deficiencies, reduced gonadal steroids, and increased cortisol. The degree of bone density reduction is proportional to the length of the illness, and patients are at risk for the development of symptomatic fractures. The occurrence of anorexia nervosa during adolescence may lead to the premature cessation of linear bone growth and a failure to achieve expected adult height.

Cardiac Abnormalities Cardiac output is reduced, and congestive heart failure occasionally occurs during rapid refeeding. The electrocardiogram usually shows sinus bradycardia, reduced QRS voltage, and nonspecific ST-T-wave abnormalities. Some patients develop a prolonged QT_c interval, which may predispose to serious arrhythmias.

BULIMIA NERVOSA

The typical patient presenting for treatment of bulimia nervosa is a woman of normal weight in her mid-twenties who reports binge eating and purging 5 to 10 times a week for 5 to 10 years. The disorder usually begins in late adolescence or early adulthood during or following a diet. The self-imposed caloric restriction leads to increased hunger and to overeating. In an attempt to avoid weight gain, the patient induces vomiting, takes laxatives or diuretics, or engages in some other form of compensatory behavior. Initially, patients may experience a sense of satisfaction that appealing food can be eaten without weight gain. However, as the disorder progresses, patients perceive diminished control over eating. Binges increase in size and frequency and are provoked by a variety of stimuli, such as transient depression, anxiety, or a sense that too much food has been consumed in a normal meal. Between binges, patients attempt to restrict caloric intake, which increases hunger and sets the stage for the next binge.

Although vomiting may be triggered initially by manual stimulation of the gag reflex, most patients with bulimia nervosa develop the ability to induce vomiting at will. Rarely,

patients resort to the regular use of syrup of ipecac. Laxatives and diuretics are frequently taken in impressive quantities, such as 30 or 60 laxative pills on a single occasion. The resulting fluid loss produces dehydration and a feeling of emptiness but has little impact on caloric balance.

The physical abnormalities associated with bulimia nervosa primarily result from the purging behavior. Painless bilateral salivary gland hypertrophy (sialadenosis) may be noted. A scar or callus on the dorsum of the hand may develop due to repeated trauma from the teeth among patients who manually stimulate the gag reflex. Recurrent vomiting and the exposure of the lingual surfaces of the teeth to stomach acid leads to loss of dental enamel and eventually to chipping and erosion of the front teeth. Laboratory abnormalities are surprisingly infrequent, but hypokalemia, hypochloremia, and hyponatremia are observed occasionally. Repeated vomiting may lead to alkalosis, whereas repeated laxative abuse may produce a mild metabolic acidosis. Serum amylase may be mildly elevated due to an increase in the salivary isoenzyme.

Serious physical complications resulting from bulimia nervosa are rare. Oligomenorrhea and amenorrhea are more frequent than in women without eating disorders. Arrhythmias occasionally occur secondary to electrolyte disturbances. Tearing of the esophagus and rupture of the stomach, which constitute life-threatening events, have been reported. Some patients who have chronically abused laxatives or diuretics develop transient peripheral edema when this behavior ceases, presumably due to high levels of aldosterone resulting from persistent fluid and electrolyte depletion.

PROGNOSIS

The course and outcome of anorexia nervosa are highly variable. One-quarter to one-half of patients eventually recover fully, with few psychological or physical sequelae. However, many patients have persistent difficulties with weight maintenance, depression, and eating disturbances, including bulimia nervosa. The development of obesity following anorexia nervosa is rare. The long-term mortality of anorexia nervosa is among the highest associated with any psychiatric disorder. Approximately 5% of patients die per decade of follow-up, primarily due to the physical effects of chronic starvation or by suicide.

Virtually all of the physiologic abnormalities associated with anorexia nervosa are observed in other forms of starvation and markedly improve or disappear with weight gain. A worrisome exception is the reduction in bone mass, which may not recover fully, particularly when anorexia nervosa occurs during adolescence when peak bone mass is normally achieved.

The prognosis of bulimia nervosa is much more favorable. Mortality is low, and full recovery occurs in approximately 50% of patients within 10 years. Approximately 25% of patients have persistent symptoms of bulimia nervosa over many years. Few patients progress from bulimia nervosa to anorexia nervosa.

TREATMENT

Anorexia Nervosa Because of the profound physiological and psychological effects of

starvation, there is a broad consensus that weight restoration to 90% of predicted weight is the primary goal in the treatment of anorexia nervosa. Unfortunately, because most patients resist this goal, its accomplishment is often accompanied by frustration for the patient, the family, and the physician. In attempting to engage the patient in treatment, it may be useful for the physician to elicit the patient's physical concerns (e.g., about osteoporosis, weakness, or fertility) and, if possible, educate the patient regarding the importance of normalizing nutritional status in order to address those concerns. The physician should attempt to reassure the patient that weight gain will not be permitted to get "out of control" but simultaneously emphasize that weight restoration is medically and psychologically imperative.

The intensity of the initial treatment, including the need for hospitalization, is determined by the patient's current weight, the rapidity of recent weight loss, and the severity of medical and psychological complications ([Fig. 78-1](#)). Hospitalization should be strongly considered for patients weighing <75% of expected, even if the results of routine blood studies are within normal limits. Acute medical problems, such as severe electrolyte imbalances, should be identified and addressed. Nutritional restoration can almost always be successfully accomplished by oral feeding, and parenteral methods are rarely required. For severely underweight patients, sufficient calories (approximately 1500 to 1800 kcal/d) should be provided initially in divided meals as food or liquid supplements to maintain weight and to permit stabilization of fluid and electrolyte balance. Calories can then be gradually increased to achieve a weight gain of 1 to 2 kg (2 to 4 lb) per week, typically requiring an intake of 3000 to 4000 kcal/d. Meals must be supervised, ideally by personnel who are firm regarding the necessity of food consumption, empathic regarding the challenges entailed, and reassuring regarding the patient's eventual recovery. Patients have great psychological difficulty complying with the need for increased caloric consumption, and the assistance of psychiatrists or psychologists experienced in the treatment of anorexia nervosa is usually necessary.

Psychiatric treatment focuses primarily on two issues. First, patients require much emotional support during the period of weight gain. Second, patients must learn to base their self-esteem, not on the achievement of an inappropriately low weight, but on the development of satisfying personal relationships and the attainment of reasonable academic and occupational goals. For younger patients, the active involvement of the family in treatment is crucial.

Less severely affected patients may be treated in a partial hospitalization program where medical and psychiatric supervision is available and several meals can be monitored each day. Outpatient treatment may suffice for mildly ill patients. Weight must be monitored at frequent intervals, and explicit goals agreed on for weight gain, with the understanding that more intensive treatment will be required if the level of care initially employed is not successful.

Medical complications occasionally occur during refeeding. Most patients transiently retain excess fluid, occasionally resulting in peripheral edema. Fluid retention occurs during recovery from other forms of malnutrition and generally does not require specific treatment in the absence of cardiac, renal, or hepatic dysfunction. Congestive heart failure and acute gastric dilatation have been described when refeeding has been rapid. Transient modest elevations in serum levels of liver enzymes occasionally occur. Low

levels of magnesium and phosphate should be repleted. Multivitamins should be given, and it is important to ensure adequate intake of vitamin D (400 IU/d) and calcium (1500 mg/d) to minimize bone loss.

No psychotropic medications are of established value in the treatment of anorexia nervosa; tricyclic antidepressants are contraindicated when there is prolongation of the QTc interval. The alterations of cortisol and thyroid hormone metabolism do not require specific treatment and are corrected by weight gain. Estrogen treatment appears to have minimal impact on bone density in underweight patients but may be helpful to relieve symptoms of estrogen deficiency.

Bulimia Nervosa Bulimia nervosa can usually be treated on an outpatient basis. Cognitive behavioral therapy (CBT) is a short-term (4 to 6 months) psychological treatment that focuses on the intense concern with shape and weight, the persistent dieting, and the binge eating and purging that characterize this disorder. Patients are directed to monitor the circumstances, thoughts, and emotions associated with binge/purge episodes, to eat regularly, and to challenge their assumptions linking weight to self-esteem. CBT produces symptomatic remission in 25 to 50% of patients.

Numerous double-blind, placebo-controlled trials have documented that antidepressant medications are useful in the treatment of bulimia nervosa but are probably somewhat less effective than CBT. Although efficacy has been established for virtually all chemical classes of antidepressants, only the selective serotonin reuptake inhibitor fluoxetine (Prozac) has been approved for use in bulimia nervosa by the U.S. Food and Drug Administration. Antidepressant medications are helpful even for patients with bulimia nervosa who are not depressed, and the dose of fluoxetine recommended for bulimia nervosa (60 mg/d) is higher than that typically used to treat depression. These observations suggest that different mechanisms may underlie the utility of these medications in bulimia nervosa and in depression.

A substantial minority of patients with bulimia nervosa do not respond adequately to CBT, antidepressant medication, or their combination. More intensive forms of treatment, including hospitalization, may be required for such patients.

(Bibliography omitted in Palm version)

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PART SEVEN -INFECTIOUS DISEASES

SECTION 1 -BASIC CONSIDERATIONS IN INFECTIOUS DISEASES

119. INTRODUCTION TO INFECTIOUS DISEASES: HOST-PARASITE INTERACTIONS - Lawrence C. Madoff, Dennis L. Kasper

Despite decades of dramatic progress in their treatment and prevention, infectious diseases remain a major cause of death and debility and are responsible for worsening the living conditions of many millions of people around the world. Infections frequently challenge the physician's diagnostic skill and must be considered in the differential diagnoses of syndromes affecting every organ system.

CHANGING EPIDEMIOLOGY OF INFECTIOUS DISEASES

With the advent of antimicrobial agents, some medical leaders believed that infectious diseases would soon be eliminated and become of historic interest only. Indeed, the hundreds of chemotherapeutic agents developed since World War II, most of which are potent and safe, include drugs effective not only against bacteria but also against viruses, fungi, and parasites. Nevertheless, we now realize that as we developed antimicrobial agents, microbes developed the ability to elude our best weapons and to counterattack with new survival strategies. Antibiotic resistance occurs at an alarming rate among all classes of mammalian pathogens. Pneumococci resistant to penicillin and enterococci resistant to vancomycin have become commonplace. Even *Staphylococcus aureus* with reduced susceptibility to vancomycin has appeared. Such pathogens present real clinical problems in managing infections that were easily treatable just a few years ago. Diseases once thought to have been nearly eradicated from the developed world -- tuberculosis, cholera, and rheumatic fever, for example -- have rebounded with renewed ferocity. Newly discovered and emerging infectious agents appear to have been brought into contact with humans by changes in the environment and by movements of human and animal populations. An example of the propensity for pathogens to escape from their usual niche is the alarming 1999 outbreak in New York of encephalitis due to a flavivirus similar or identical to West Nile virus, which had never previously been isolated in the Americas.

Many infectious agents have been discovered only in recent decades. Ebola virus, hantavirus, the agent of human granulocytotropic ehrlichiosis, and retroviruses such as HIV humble us despite our deepening understanding of pathogenesis at the most basic molecular level. Even in developed countries, infectious diseases have made a resurgence. Between 1980 and 1996, mortality from infectious diseases in the United States increased by 64% to levels not seen since the 1940s.

The role of infectious agents in the etiology of diseases once believed to be noninfectious is being increasingly recognized. For example, it is now widely accepted that *Helicobacter pylori* is the causative agent of peptic ulcer disease and perhaps of gastric malignancy. Human papillomavirus is likely to be the most important cause of invasive cervical cancer. A new human herpesvirus (HHV-8) is believed to be the cause of most cases of Kaposi's sarcoma. Epstein-Barr virus is a cause of certain lymphomas and may play a role in the genesis of Hodgkin's disease. The possibility certainly exists

that other diseases of unknown cause, such as rheumatoid arthritis, sarcoidosis, or inflammatory bowel disease, have infectious etiologies. There is even evidence that atherosclerosis may have an infectious component.

Medical advances over infectious diseases have been hindered by changes in the patient population. Immunocompromised hosts now constitute a significant proportion of the seriously infected population. Physicians immunosuppress their patients to prevent the rejection of transplants and to treat neoplastic and inflammatory diseases. Some infections, most notably that caused by HIV, immunocompromise the host in and of themselves. Lesser degrees of immunosuppression are associated with other infections, such as influenza and syphilis. Infectious agents that coexist peacefully with immunocompetent hosts wreak havoc in those who lack a complete immune system. AIDS has brought to prominence once-obscure organisms such as *Pneumocystis carinii*, *Cryptosporidium parvum*, and *Mycobacterium avium*.

BIOTERRORISM

In recent years, the efforts of some governments and terrorist organizations to use biological weaponry have refocused public concern on the topic. To date, there is little evidence that biological weapons have ever been effectively used; indeed, their ease of use may be overstated. However, the ability of infectious agents to inflict widespread illness and thus to cause societal disruption and panic, together with the relatively low cost of these agents, has led to their being called a "poor man's nuclear arsenal."

Several pathogens have been considered likely candidates for biological warfare. *Bacillus anthracis*, which causes the zoonosis anthrax, is widely viewed as the leading contender. The hardy spores of the bacillus can be distributed by bombardment or other dispersal mechanisms. Inhalation of this pathogen results in severe pneumonia with a mortality rate of 95% in untreated persons. A World Health Organization report estimated that 50 kg of *B. anthracis* released upwind of a city with a population of 500,000 would result in up to 95,000 fatalities, with an additional 125,000 persons incapacitated. Moreover, the attack might go undetected until large numbers of seriously ill individuals presented with overt disease. In 1979, an accidental release from a military microbiology facility near Sverdlovsk in the former Soviet Union caused at least 66 deaths from inhalational anthrax along a 4-km-wide path downwind of the facility. The vast scope of the Soviet Union's biological warfare program, employing 60,000 people at its height, has only recently come to light. This endeavor is thought to be echoed by efforts in many other countries in the world today. In response to the perceived threat of anthrax as a biological weapon, the U.S. military recently decided to vaccinate more than 2 million of its members against this infection.

Smallpox, an ancient scourge caused by variola virus, has also been considered as a bioweapon owing to its contagiousness and high mortality rate and to the declining population of immunized persons. Indeed, one of the earliest accounts of biological warfare involved the distribution of smallpox-infected blankets to Native American tribes by British troops. Debate continues about whether to eradicate the two known existing stocks of the virus in U.S. and Russian government laboratories. Many investigators believe that additional undocumented stockpiles of the virus exist around the world.

Other infectious organisms that combine the virulence and stability necessary for biological weapons include *Yersinia pestis*, the agent of plague, and *Francisella tularensis*, the agent of tularemia. Viral hemorrhagic fever agents such as the Ebola and Marburg viruses as well as toxins such as that from *Clostridium botulinum* have also been considered as biological weapons. While some of the diseases caused by these agents can be effectively treated or prevented if sufficient resources exist to do so, it may also be possible for an aggressor to render organisms resistant to antibiotics or even to vaccines through genetic manipulation or other means.

HOST FACTORS IN INFECTION

For any infectious process to occur, the parasite and the host must first encounter each other. Factors such as geography, environment, and behavior thus influence the likelihood of infection. Though the initial encounter between a susceptible host and a virulent organism frequently results in disease, some organisms can be harbored in the host for years before disease becomes clinically evident. For a complete view, individual patients must be considered in the context of the population to which they belong. Infectious diseases do not often occur in isolation; rather, they spread through a group exposed from a point source (e.g., a contaminated water supply) or from individual to individual (e.g., via respiratory droplets). Thus, the clinician must be alert to infections prevalent in the community as a whole. A detailed history, including information on travel, behavioral factors, exposures to animals or potentially contaminated environments, and living and occupational conditions, must be elicited. For example, the likelihood of infection by *Plasmodium falciparum* can be significantly affected by altitude, climate, terrain, season, and even time of day. Antibiotic-resistant strains are localized to specific geographic regions, and a seemingly minor alteration in a travel itinerary can dramatically influence the likelihood of acquiring chloroquine-resistant malaria. If such important details in the history are overlooked, inappropriate treatment may result in the death of the patient. Likewise, the chance of acquiring a sexually transmitted disease can be greatly affected by a relatively minor variation in sexual practices, such as the method used for birth control. Knowledge of the relationship between specific risk factors and disease allows the physician to influence a patient's health even before the development of infection by modification of these risk factors and -- when a vaccine is available -- by immunization.

Many specific host factors influence the likelihood of acquiring an infectious disease. Age, immunization history, prior illnesses, level of nutrition, pregnancy, coexisting illness, and perhaps emotional state all have some impact on the risk of infection after exposure to a potential pathogen. The importance of individual host defense mechanisms, either specific or nonspecific, becomes apparent in their absence, and our understanding of these immune mechanisms is enhanced by studies of clinical syndromes developing in immunodeficient patients ([Table 119-1](#)). For example, the frequent occurrence of meningococcal disease in people with deficiencies in specific complement proteins of the "membrane attack complex" underscores the importance of an intact complement system in the prevention of meningococcal infection.

Medical care itself increases the patient's risk of acquiring an infection in several ways: (1) through contact with pathogens during hospitalization, (2) through breaching of the skin (with intravenous devices or surgical incisions) or mucosal surfaces (with

endotracheal tubes or bladder catheters), (3) through introduction of foreign bodies, (4) through alteration of the natural flora with antibiotics, and (5) through treatment with immunosuppressive drugs.

THE IMMUNE RESPONSE

Infection involves complicated interactions of parasite and host and inevitably affects both. In most cases, a pathogenic process consisting of several steps is required for the development of infections. Since the competent host has a complex series of barricades in place to prevent infection, the successful parasite must utilize specific strategies at each of these steps. The specific strategies used by bacteria, viruses, and parasites ([Chaps. 120](#) and [180](#)) have some remarkable conceptual similarities, but the strategic details are unique not only for each class of organism but also for individual species within a class.

Once in the bloodstream or a normally sterile body site, the microorganism faces the host's tightly integrated cellular and humoral immune systems. Cellular immunity ([Chap. 305](#)), comprising T lymphocytes, macrophages, and natural killer cells, primarily recognizes and combats pathogens that proliferate intracellularly. Cellular immune mechanisms are important in immunity to all classes of infectious agents, including most viruses and many bacteria (e.g., *Mycoplasma*, *Chlamydia*, *Listeria*, *Salmonella*, and *Mycobacterium*), parasites (e.g., *Trypanosoma*, *Toxoplasma*, and *Leishmania*), and fungi (e.g., *Histoplasma*, *Cryptococcus*, and *Coccidioides*). Usually, T lymphocytes are activated by macrophages and B lymphocytes, which present foreign antigens along with the host's own major histocompatibility complex antigen. Activated T cells may then act in several ways to fight infection. Cytotoxic T cells may directly attack and lyse host cells that express foreign antigens. Helper T cells stimulate the proliferation of B cells and the production of immunoglobulins. B cells and T cells communicate with each other via a variety of signals, and often more than one signal is employed simultaneously. For example, costimulation through the CD40-CD40 ligand increases B cell responses, and costimulation via the B7-CD28 axis is required for activation of the CD4⁺ helper T cell. T cells elaborate cytokines (e.g., interferon), which directly inhibit the growth of pathogens or stimulate killing by host macrophages and cytotoxic cells. Cytokines also augment the host's immunity by stimulating the inflammatory response (fever, the production of acute-phase serum components, and the proliferation of leukocytes). Cytokine stimulation does not always result in a favorable response in the host; septic shock ([Chap. 124](#)) and toxic shock syndrome ([Chaps. 139](#) and [140](#)) are among the conditions that are mediated by these inflammatory substances.

The reticuloendothelial system comprises monocyte-derived phagocytic cells that are located in the liver (Kupffer cells), lung (alveolar macrophages), spleen, kidney (mesangial cells), brain (microglia), and lymph nodes and that clear circulating microorganisms. Although these tissue macrophages and polymorphonuclear leukocytes (PMNs) are capable of killing microorganisms without help, they function much more efficiently when pathogens are first *opsonized* (Greek, "to prepare for eating") by components of the complement system such as C3b and/or by antibodies.

Extracellular pathogens, including most encapsulated bacteria, are attacked by the humoral immune system, which includes antibodies, the complement cascade, and

phagocytic cells. Antibodies are complex glycoproteins (also called immunoglobulins) that are produced by mature B lymphocytes, circulate in body fluids, and are secreted on mucosal surfaces. Antibodies specifically recognize and bind to foreign antigens. One of the most impressive features of the immune system is the ability to generate an incredible diversity of antibodies capable of recognizing virtually every foreign antigen yet not reacting with self. In addition to being exquisitely specific for antigens, antibodies come in different structural and functional classes: IgG predominates in the circulation and persists for many years after exposure; IgM is the earliest specific antibody to appear in response to infection; secretory IgA is important in immunity at mucosal surfaces, while monomeric IgA appears in the serum; and IgE is important in allergic and parasitic diseases. Antibodies may directly impede the function of an invading organism, neutralize secreted toxins and enzymes, or facilitate the removal of the antigen (invading organism) by phagocytic cells. Immunoglobulins participate in cell-mediated immunity by promoting the antibody-dependent cellular cytotoxicity functions of certain T lymphocytes. Antibodies also promote the deposition of complement components on the surface of the invader.

The complement system ([Chap. 305](#)) consists of a group of serum proteins functioning as a cooperative, self-regulating cascade of enzymes that adhere to -- and in some cases disrupt -- the surface of invading organisms. Some of these surface-adherent proteins (e.g., C3b) can then act as opsonins for destruction of microbes by phagocytes. The later, "terminal" components (C7, C8, and C9) can directly kill some bacterial invaders (notably, many of the neisseriae) by forming a "membrane attack complex" and disrupting the integrity of the bacterial membrane, thus causing bacteriolysis. Other complement components, such as C5a, act as chemoattractants for [PMNs](#). Complement activation and deposition occur by either or both of two pathways: the classic pathway is activated primarily by immune complexes (i.e., antibody bound to antigen), and the alternative pathway is activated by microbial components, frequently in the absence of antibody. PMNs have receptors for both antibody and C3b, and antibody and complement function together to aid in the clearance of infectious agents.

[PMNs](#), short-lived white blood cells that engulf and kill invading microbes, are first attracted to inflammatory sites by chemoattractants such as C5a, which is a product of complement activation at the site of infection. PMNs localize to the site of infection by adhering to cellular adhesion molecules expressed by endothelial cells. Endothelial cells express these receptors, called *selectins* (CD-62, ELAM-1), in response to inflammatory cytokines such as tumor necrosis factor (TNF) α and interleukin 1. The binding of these selectin molecules to specific receptors on PMNs results in the adherence of the PMNs to the endothelium. Cytokine-mediated upregulation and expression of intercellular adhesion molecule 1 (ICAM 1) on endothelial cells then take place, and this latter receptor binds to β_2 integrins on PMNs, thereby facilitating diapedesis into the extravascular compartment. Once the PMNs are in the extravascular compartment, various molecules such as arachidonic acids further enhance the inflammatory process.

Approach to the Patient

The clinical manifestations of infectious diseases at presentation are myriad, varying from fulminant life-threatening processes to brief and self-limited conditions to indolent chronic maladies. The clinician must use all the skills of medicine to diagnose the

infection and prescribe appropriate treatment. First, a careful history is essential and must include details on underlying chronic diseases; medications; occupation; travel; and risk factors for exposure to certain types of pathogens, such as those associated with sexual contacts, family illnesses, illicit drug use, particular animals, blood transfusions, ingestion of contaminated liquids or foods, or bites of insect vectors. Since infectious diseases may involve many organ systems, a careful review of systems may elicit important clues as to the disease process. The physical examination must be thorough, and attention must be paid to seemingly minor details: a soft heart murmur that might indicate bacterial endocarditis; an evanescent skin rash that suggests rheumatic fever; or a retinal lesion that suggests disseminated candidiasis or cytomegalovirus (CMV) infection.

LABORATORY INVESTIGATIONS

Laboratory studies must be carefully considered and directed toward establishing an etiologic diagnosis in the shortest possible time, at the lowest possible cost, and with the least possible discomfort to the patient. Cultures must be performed in a manner that minimizes the likelihood of contamination with normal flora while maximizing the yield. A sputum sample is far more likely to be valuable when elicited with careful coaching by the clinician than when collected in a container simply left at the bedside with cursory instructions. Gram's stains of specimens should be interpreted carefully and the quality of the specimen assessed. The findings on Gram's staining should correspond to the results of culture; a discrepancy may suggest diagnostic possibilities such as infection due to fastidious or anaerobic bacteria.

The microbiology laboratory must be an ally in the diagnostic endeavor ([Chap. 121](#)). Astute laboratory personnel will suggest optimal culture and transport conditions or alternative tests to facilitate diagnosis. If informed about specific potential pathogens, an alert laboratory staff will allow sufficient time for these organisms to become evident in culture, even when present in small numbers or when slow-growing. The parasitology technician who is attuned to the specific diagnostic considerations relevant to a particular case may be able to detect the rare, otherwise-elusive egg or cyst in a stool specimen. In cases where a diagnosis appears difficult, serum should be stored during the early acute phase of the illness so that a diagnostic rise in titer of antibody to a specific pathogen can be detected later. Bacterial and fungal antigens can sometimes be detected in body fluids, even when cultures are negative or are rendered sterile by antibiotic therapy. Techniques such as the polymerase chain reaction allow the amplification of specific DNA sequences so that minute quantities of foreign nucleic acids can be recognized in host specimens.

TREATMENT

Optimal therapy for infectious diseases requires a broad knowledge of medicine and careful clinical judgment. Life-threatening infections such as bacterial meningitis or sepsis, viral encephalitis, or falciparum malaria must be treated immediately, often before a specific causative organism is identified. Antimicrobial agents must be chosen empirically and must be active against the range of potential infectious agents consistent with the clinical scenario. In contrast, good clinical judgment sometimes dictates withholding of antimicrobials in a self-limited process or until a specific

diagnosis is made. The dictum *primum non nocere* should be adhered to, and it should be remembered that all antimicrobials carry a risk (and a cost) to the patient. Direct toxicity may be encountered -- e.g., ototoxicity due to aminoglycosides, lipodystrophy due to HIV protease inhibitors, and hepatotoxicity due to antituberculous agents such as isoniazid and rifampin. Allergic reactions are common and can be serious. Since superinfection sometimes follows the eradication of the normal flora and colonization by a resistant organism, one invariable principle is that infectious disease therapy should be directed toward as narrow a spectrum of infectious agents as possible. Treatment specific for the pathogen should result in as little perturbation as possible of the host's microflora. With few exceptions, abscesses require surgical or percutaneous drainage for cure. Foreign bodies, including medical devices, must generally be removed in order to eliminate an infection of the device or of the adjacent tissue. Other infections, such as necrotizing fasciitis, peritonitis due to a perforated organ, gas gangrene, and chronic osteomyelitis, require surgery as the primary means of cure; in these conditions, antibiotics play only an adjunctive role.

The role of immunomodulators in the management of infectious diseases has received increasing attention. Glucocorticoids have been shown to be of benefit in the treatment of *Haemophilus influenzae* meningitis in children and in therapy for *P. carinii* pneumonia in patients with AIDS. The use of these agents in other infectious processes remains less clear and in some cases (in cerebral malaria and septic shock, for example) is detrimental. Other agents that modulate the immune response include prostaglandin inhibitors, specific lymphokines, and [TNF](#) inhibitors. Specific antibody therapy plays a role in the treatment and prevention of many diseases. Specific immunoglobulins have long been known to prevent the development of symptomatic rabies and tetanus. More recently, [CMV](#) immune globulin has been recognized as important not only in preventing the transmission of the virus during organ transplantation but also in treating CMV pneumonia in bone marrow transplant recipients. There is a strong need for well-designed clinical trials to evaluate each new interventional modality.

PERSPECTIVE

The genetic simplicity of many infectious agents allows them to undergo rapid evolution and to develop selective advantages that result in constant variation in the clinical manifestations of infection. Moreover, changes in the environment and the host can predispose new populations to a particular infection. An epidemic of lethal respiratory failure -- later identified as hantavirus pulmonary syndrome -- on a Navajo reservation in the southwestern United States in 1993 caused nationwide alarm, exemplifying the fear that new plagues induce in the human psyche.

The potential for infectious agents to emerge in novel and unexpected ways requires that physicians and public health officials be knowledgeable, vigilant, and open-minded in their approach to unexplained illness. The emergence of antimicrobial-resistant pathogens (e.g., enterococci that are resistant to all known antimicrobial agents and cause infections that are essentially untreatable) has led some to conclude that we are entering the "postantibiotic era." Others have held to the perception that infectious diseases no longer represent as serious a concern to world health as they once did. The progress that science, medicine, and society as a whole have made in combating these maladies is impressive, and it is ironic that, as we stand on the threshold of an

understanding of the most basic biology of the microbe, infectious diseases are posing renewed problems. We are threatened by the appearance of new diseases such as AIDS, hepatitis C, and Ebola virus infection and by the reemergence of old foes such as tuberculosis, cholera, plague, and *Streptococcus pyogenes* infection. True students of infectious diseases were perhaps less surprised than anyone else by these developments. Those who know pathogens are aware of their incredible adaptability and diversity. As ingenious and successful as therapeutic approaches may be, our ability to develop methods to counter infectious agents so far has not matched the myriad strategies employed by the sea of microbes that surrounds us. Their sheer numbers and the rate at which they can evolve are daunting. Moreover, environmental changes, rapid global travel, population movements, and medicine itself -- through its use of antibiotics and immunosuppressive agents -- all increase the impact of infectious diseases. Although new vaccines, new antibiotics, improved global communication, and new modalities for treating and preventing infection will be developed, pathogenic microbes will continue to develop new strategies of their own, presenting us with an unending and dynamic challenge.

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120. MOLECULAR MECHANISMS OF MICROBIAL PATHOGENESIS- Gerald B. Pier

Over the past two decades, molecular studies of microbial pathogenesis have yielded an explosion of information about the various microbial and host molecules that contribute to the processes of infection and disease. These processes can be classified into several stages: microbial encounter with and entry into the host; microbial growth after entry; avoidance of innate host defenses; tissue invasion and tropism; tissue damage; and transmission to new hosts. Virulence is the measure of an organism's capacity to cause disease and is a function of the pathogenic factors elaborated by microbes. These factors promote colonization (the simple presence of potentially pathogenic microbes in or on a host), infection (attachment and growth of pathogens and avoidance of host defenses), and disease (often, but not always, the activities of secreted toxins or toxic metabolites). In addition, the host's inflammatory response to infection greatly contributes to disease and its attendant clinical signs and symptoms. Knowledge of the molecular structures of the microbial surface, their interactions with the host, and the host response is critical to an understanding of the basic processes of infection and disease.

MICROBIAL ENTRY AND ADHERENCE

ENTRY SITES

A microbial pathogen can potentially enter any part of a host organism. In general, the type of disease produced by a particular microbe is often a direct consequence of its route of entry into the body. The most common sites of entry are body parts in contact with the external environment, including mucosal surfaces (particularly those of the respiratory, alimentary, and urogenital tracts) and the skin. Ingestion, inhalation, and sexual contact are typical routes of microbial entry. Other portals of entry include injuries to the skin (cuts, bites, burns, trauma) along with injection via natural (i.e., vector-borne) or artificial (i.e., needle-stick injury) routes. A few pathogens, such as *Schistosoma* spp., can penetrate unbroken skin. The conjunctiva can serve as an entry point for pathogens of the eye.

Microbial entry usually relies on the organism's biologic characteristics and reflects the presence of specific microbial factors needed for persistence and growth in a tissue. Fecal-oral spread via the alimentary tract requires a biology consistent with survival in the varied environments of the gastrointestinal tract (including the low pH of the stomach and the high bile content of the intestine) as well as in contaminated food or water outside the host. Organisms that gain entry via the respiratory tract are most often those that survive well in small moist droplets produced during sneezing and coughing; most such pathogens do not survive well once they dry out. Pathogens that enter by venereal routes often survive best on the warm moist environment of the urogenital mucosa. Many sexually transmitted human pathogens have restricted host ranges and do not infect other animals (e.g., *Neisseria gonorrhoeae*, *Treponema pallidum*, and HIV).

The biology of microbes entering through the skin is highly varied. Some of these organisms can survive in a broad range of environments, such as the salivary glands or alimentary tracts of arthropod vectors, the mouths of larger animals, soil, and water. A

complex biology allows protozoan parasites such as *Plasmodium*, *Leishmania*, and *Trypanosoma* spp. to undergo morphogenic changes that allow the organism to be transmitted to mammalian hosts during insect feeding for blood meals. Plasmodia are injected as infective sporozoites from the salivary glands during mosquito feeding. *Leishmania* parasites are regurgitated as promastigotes from the alimentary tract of sandflies and injected by bite into a susceptible host. Trypanosomes are first ingested from infected hosts by reduviid bugs; the pathogens then multiply in the gastrointestinal tract of the insects and are released in feces onto the host's skin during subsequent feedings. Most microbes that land directly on intact skin are destined to die, as survival on the skin or in hair follicles requires resistance to fatty acids, low pH, and other antimicrobial factors on skin. Once it is damaged (and particularly if it becomes necrotic), the skin can be a major portal of growth and entry for pathogens or their toxic products. Tetanus and burn wound infections are clear examples. After animal bites, pathogens resident in the animal's saliva gain access through the skin to the victim's tissues. Rabies is the paradigm for this pathogenic process; rabies virus grows in striated muscle cells at the site of inoculation.

MICROBIAL ADHERENCE

Once in or on a host, most microbes must anchor themselves to a tissue or tissue factor; the possible exceptions are organisms that directly enter the bloodstream and multiply there. Specific microbial ligands or adhesins for host receptors constitute a major area of study in the field of microbial pathogenesis. Adhesins comprise a wide range of surface structures, not only anchoring the microbe to a tissue and promoting cellular entry where appropriate but also eliciting host responses critical to the pathogenic process ([Table 120-1](#)). Most microbes produce multiple adhesins specific for multiple host receptors. These adhesins are often redundant, are serologically variable, and act additively or synergistically with other microbial factors to promote microbial sticking to host tissues. In addition, some microbes (such as *Mycobacterium tuberculosis* and *Legionella pneumophila*) adsorb host proteins (such as complement components) onto their surface and utilize the natural host protein receptor for microbial binding and entry into target cells.

All viral pathogens must bind to host cells, enter them, and replicate within them. Viral coat proteins serve as the ligands for cellular entry, and more than one ligand-receptor interaction may be needed; for example, HIV utilizes its envelope glycoprotein (gp) 120 to enter host cells by binding to both CD4 and one of several receptors for chemokines. Similarly, the measles virus H glycoprotein binds to both CD46 and the membrane-organizing protein moesin. The gC protein on herpes simplex virus binds to heparin sulfate; this step is followed by attachment to cells mediated by the viral gD (and possibly gH) protein. CD46 has now been shown to be the cellular receptor for human herpesvirus type 6. Eukaryotic parasites use complicated surface glycoproteins as adhesins, some of which are lectins with specificity for carbohydrates on host cells.

Among the microbial adhesins studied in greatest detail are bacterial pili and flagella. *Pili* or *fimbriae* are commonly used by gram-negative bacteria for attachment to host cells and tissues. In electron micrographs, these hairlike projections (up to several hundred per cell) may be confined to one end of the organism (polar pili) or distributed more evenly over the surface. An individual cell may have pili with a variety of functions.

Most pili are made up of a major pilin protein subunit (molecular weight, 17,000 to 30,000) that polymerizes to form the pilus. Many strains of *Escherichia coli* express mucus-binding type 1 pili, whose binding to host tissues is inhibited by D-mannose. Other strains produce the Pap (pyelonephritis-associated) or P pilus adhesin that mediates binding to digalactose (gal-gal) residues on globosides of the human P blood groups. These pili have proteins located at the tips of the main pilus unit that are critical to the binding specificity of the whole pilus unit. Immunization with the mannose-binding FimH tip protein of type 1 pili prevents experimental *E. coli* bladder infections in mice and monkeys. *E. coli* cells causing diarrheal disease express pilus-like receptors for enterocytes on the small bowel, along with other receptors termed *colonization factors*.

A common type of pilus found in *Neisseria* spp., *Moraxella* spp., *Vibrio cholerae*, and *Pseudomonas aeruginosa* mediates adherence of these organisms to target surfaces. These pili tend to have a relatively conserved amino-terminal region and a more variable carboxyl-terminal region. For some species such as *N. gonorrhoeae* and *Neisseria meningitidis*, the pili are critical for attachment to mucosal epithelial cells. For others, such as *P. aeruginosa*, the pili only partially mediate the cells' adherence to host tissues. *V. cholerae* cells appear to use two different types of pili for intestinal colonization. Whereas interference with this stage of colonization would appear to be an effective antibacterial strategy, attempts to develop pilus-based vaccines for human diseases have not been highly successful to date.

Flagella are long appendages attached at either one or both ends of the bacterial cell (polar flagella) or distributed over the entire cell surface (peritrichous flagella). Flagella, like pili, are composed of a polymerized or aggregated basic protein. In flagella, the protein subunits form a tight helical structure and vary serologically with the species. Spirochetes such as *T. pallidum* and *Borrelia burgdorferi* have axial filaments similar to flagella running down the long axis of the center of the cell, and they "swim" by rotation around these filaments. Some bacteria can glide over a surface in the absence of obvious motility structures.

Other bacterial structures involved in adherence to host tissues include specific staphylococcal and streptococcal proteins that bind to human extracellular matrix proteins such as fibrin, fibronectin, laminin, and collagen. Fibronectin appears to be a commonly used receptor for various pathogens; a particular sequence, Arg-Gly-Asp or RGD, is critical for bacterial binding. Surface lipoteichoic acids may also promote streptococcal adherence to mucosal surfaces. The surface lipopolysaccharide (LPS) of *P. aeruginosa* mediates binding to the cystic fibrosis transmembrane conductance regulator (CFTR) on airway epithelial cells. Coagulase-negative staphylococci readily colonize prosthetic devices and catheters commonly used in medical care; the surface capsular polysaccharide of these organisms promotes binding to the prosthetic material. It has been reported that *Staphylococcus aureus* produces the same capsular polysaccharide and may also use this material to colonize prosthetic devices.

FUNGAL ADHESINS

Several fungal adhesins have been described that mediate colonization of epithelial surfaces, particularly adherence to structures like fibronectin, laminin, and collagen. The product of the *Candida albicans* *INT1* gene, int1p, bears similarity to mammalian

integrins that bind to extracellular matrix proteins. Transformation of normally nonadherent *Saccharomyces cerevisiae* with this gene allows these yeast cells to adhere to human epithelial cells. Disruption of *INT1* in *C. albicans* diminishes but does not eliminate epithelial cell adhesion; this result indicates that both int1p and other adhesins mediate binding of *C. albicans* to epithelial cells. Moreover, int1p is needed for filamentous growth of *C. albicans* -- a phenotype linked to virulence, and particularly to the ability to penetrate keratinized epithelium. *INT1*-deficient *C. albicans* exhibits markedly reduced virulence in a mouse model of infection.

For several fungal pathogens that initiate infections after inhalation of infectious material, the inoculum is ingested by alveolar macrophages, in which the cells transform to pathogenic phenotypes. Like *C. albicans*, *Blastomyces dermatitidis* binds to CD11b/CD18 integrins as well as to CD14 on macrophages. *B. dermatitidis* produces a 120-kDa surface protein, designated WI-1, that mediates this adherence. The binding domain of WI-1 is homologous to the invasin protein of *Yersinia* that binds to the same type of host cell receptor. An unidentified factor on *Histoplasma capsulatum* also mediates binding of this fungal pathogen to the integrin surface proteins.

HOST RECEPTORS

Host receptors are found both on target cells (such as epithelial cells lining mucosal surfaces) and within the mucus layer covering these cells. Microbial pathogens bind to a wide range of host receptors to establish infection ([Table 120-1](#)). Selective loss of host receptors for a pathogen may confer natural resistance to an otherwise susceptible population. For example, *Plasmodium vivax*, one of four *Plasmodium* species causing malaria, binds to the Duffy blood group antigen, Fy, on erythrocytes. In West Africa, 70% of individuals lack Fy antigens and are resistant to *P. vivax* infection. *Salmonella typhi*, the etiologic agent of typhoid fever, uses [CFTR](#) to enter the gastrointestinal submucosa after being ingested. As homozygous mutations in *CFTR* are the cause of the life-shortening disease cystic fibrosis, heterozygote carriers (e.g., 4 to 5% of individuals of European ancestry) may have had a selective advantage due to decreased susceptibility to *S. typhi* infection.

Numerous virus-target cell interactions have been described, and it is now clear that different viruses can use similar host cell receptors for entry. The list of certain and likely host receptors for viral pathogens is long. Among the host membrane components that can serve as receptors for viruses are sialic acids, gangliosides, glycosaminoglycans, integrins and other members of the immunoglobulin superfamily, histocompatibility antigens, and regulators and receptors for complement components.

MICROBIAL GROWTH AFTER ENTRY

Once established on a mucosal or skin site, pathogenic microbes must replicate before causing full-blown infection and disease. Within cells, viral particles release their nucleic acids, which may be directly translated into viral proteins (positive-strand RNA viruses), transcribed from a negative strand of RNA into a complementary mRNA (negative-strand RNA viruses), or transcribed into a complementary strand of DNA (retroviruses); for DNA viruses, mRNA may be transcribed directly from viral DNA, either in the cell nucleus or in the cytoplasm. To grow, bacteria must acquire specific nutrients

or synthesize them from precursors in host tissues. For example, since aromatic amino acids are not available as nutrients in host tissues, pathogenic bacteria must synthesize them from precursors. Many infectious processes are usually confined to specific epithelial surfaces -- influenza to the respiratory mucosa, gonorrhea to the urogenital epithelium, shigellosis to the gastrointestinal epithelium. While there are multiple reasons for this specificity, one important consideration is probably the ability of these pathogens to obtain from these specific environments the nutrients needed for growth and survival.

Temperature restrictions also play a role in limiting certain pathogens to specific tissues. Rhinoviruses, a cause of the common cold, grow best at 33°C and replicate in cooler nasal tissues but not in the lung. Leprosy lesions due to *Mycobacterium leprae* are found in and on relatively cool body sites. Fungal pathogens that infect the skin, hair follicles, and nails (dermatophyte infections) remain confined to the cooler, exterior, keratinous layer of the epithelium.

A topic of major interest is the ability of many bacterial, fungal, and protozoal species to grow in multicellular masses referred to as *biofilms*. These masses are biochemically and morphologically quite distinct from the free-living individual cells referred to as *planktonic cells*. Growth in biofilms leads to altered microbial metabolism, production of extracellular virulence factors, and decreased susceptibility to biocides, antimicrobial agents, and host defense molecules and cells. Regulation of biofilm morphogenesis is controlled by bacterial quorum-sensing systems. Quorum sensing for some organisms involves the production of homoserine lactone molecules such as *N*-(3-oxododecanoyl)homoserine lactone and *N*-butyrylhomoserine lactone, which combine with transcriptional activators to control gene expression. *P. aeruginosa* growing on the bronchial mucosa during chronic infection, staphylococci and other pathogens growing on implanted medical devices, and dental pathogens growing on tooth surfaces to form plaques represent several examples of microbial biofilm growth associated with human disease. Many other pathogens can form biofilms during in vitro growth, including *Helicobacter pylori*, the cause of stomach ulcers; *E. coli* O157, one cause of the hemolytic-uremic syndrome; and *Gardnerella vaginalis*, an organism associated with bacterial vaginosis.

AVOIDANCE OF INNATE HOST DEFENSES

As microbes have probably interacted with mucosal/epithelial surfaces since the emergence of multicellular organisms, it is not surprising that multicellular hosts have a variety of innate surface defense mechanisms that can sense when pathogens are present and contribute to their elimination. The skin, a formidable physical barrier to microbial entry, is both acidic and bathed with fatty acids toxic to many microbes. Successful skin pathogens such as staphylococci must tolerate these adverse conditions. Mucosal surfaces themselves present a barrier composed of a thick mucus layer that entraps microbes and facilitates their transport out of the body by such processes as mucociliary clearance, coughing, and urination. Mucous secretions, saliva, and tears contain antibacterial factors such as lysozyme and antiviral factors such as interferons. Gastric acidity is inimical to the survival of many ingested pathogens, and many mucosal surfaces -- particularly the nasopharynx, the vaginal tract, and the gastrointestinal tract -- contain a resident flora of commensal microbes that interfere

with the ability of pathogens to colonize and infect a host.

Pathogens that survive these factors must still contend with host endocytic, phagocytic, and inflammatory responses as well as with host genetic factors that determine the degree to which a pathogen can survive and grow. The growth of viral pathogens entering skin or mucosal epithelial cells can be limited by a variety of host genetic factors, including production of interferons, modulation of receptors for viral entry, and age- and hormone-related susceptibility factors; by nutritional status; and even by personal habits such as smoking and exercise.

ENCOUNTERS WITH EPITHELIAL CELLS

Over the past decade, many bacterial pathogens have been shown to enter epithelial cells ([Fig. 120-1](#)), often using specialized surface structures that bind to receptors for internalization. However, the exact role and the importance of this process in infection and disease are not well defined for most of these pathogens. Bacterial entry into host epithelial cells is seen as a means for dissemination to adjacent or deeper tissues or as a route to sanctuary to avoid ingestion and killing by professional phagocytes. Epithelial cell entry appears, for instance, to be a critical aspect of dysentery induction by *Shigella*.

Curiously, the less virulent strains of many bacterial pathogens are more adept at entering epithelial cells than are more virulent strains; examples include pathogens that lack the surface polysaccharide capsule needed to cause serious disease. Thus, for *Haemophilus influenzae*, *Streptococcus pneumoniae*, *S. agalactiae* (group B *Streptococcus*), and *S. pyogenes*, isogenic mutants or variants lacking capsules enter epithelial cells better than the wild-type, encapsulated parental forms that cause disseminated disease. These observations have led to the proposal that epithelial cell entry may be a manifestation of host defense, resulting in bacterial clearance by both shedding of epithelial cells containing internalized bacteria and initiation of a subclinical inflammatory response. However, a consequence of this process would be the opening of a hole in the epithelium, potentially allowing uningested organisms to enter the submucosa. This scenario has been documented in murine *Salmonella typhimurium* infections and in experimental bladder infections with uropathogenic *E. coli*. In the latter system, bacterial pili mediate cell attachment to integral membrane glycoproteins called uroplakins that coat the host cells, resulting in exfoliation of the cells with attached bacteria. Subsequently, infection is produced by residual bacterial cells that invade the denuded epithelium. Perhaps at low bacterial inocula epithelial cell ingestion and subclinical inflammation are efficient means to eliminate pathogens, while at higher inocula a proportion of surviving bacterial cells enter the host tissue through the damaged mucosal surface and multiply, producing disease. Alternatively, failure of the appropriate epithelial cell response to a pathogen may allow the organism to survive on a mucosal surface where, if it avoids other host defenses, it can grow and cause a local infection. Along these lines, as noted above, *P. aeruginosa* is taken into epithelial cells by [CFTR](#), a protein missing or nonfunctional in most severe cases of cystic fibrosis. The major clinical consequence of this disease is chronic airway-surface infection with *P. aeruginosa* in 80 to 90% of patients with cystic fibrosis. The failure of airway epithelial cells to ingest and promote the removal of *P. aeruginosa* has been proposed as a key component of the hypersusceptibility of these patients to chronic airway infection.

ENCOUNTERS WITH PHAGOCYTES

Phagocytosis of microbes is a major innate host defense that limits the growth and spread of pathogens. Phagocytes appear rapidly at sites of infection in conjunction with the initiation of inflammation. Ingestion of microbes by both tissue-fixed macrophages and migrating phagocytes probably accounts for the limited ability of most microbial agents to cause disease. A family of related molecules called *collectins*, *soluble defense collagens*, or *pattern recognition molecules* are found in blood (mannose-binding lectin), in lung (surfactant proteins A and D), and most likely in other tissues as well and bind to carbohydrates on microbial surfaces to promote phagocyte clearance. Bacterial pathogens seem to be ingested principally by polymorphonuclear neutrophils (PMNs), while eosinophils are frequently found at sites of infection by protozoan or multicellular parasites. Successful pathogens, by definition, must avoid being cleared by professional phagocytes. One of several antiphagocytic strategies employed by bacteria and by the fungal pathogen *Cryptococcus neoformans* is to elaborate large-molecular-weight surface polysaccharide antigens, often in the form of a capsule that coats the cell surface. Most pathogenic bacteria produce such antiphagocytic capsules.

As activation of local phagocytes in tissues is a key step in initiating inflammation and migration of additional phagocytes into infected sites, much attention has been paid to microbial factors that initiate inflammation. Encounters with phagocytes are governed largely by the structure of the microbial constituents that elicit inflammation, and detailed knowledge of these structures for bacterial pathogens has contributed greatly to our understanding of molecular mechanisms of microbial pathogenesis ([Fig. 120-2](#)). The best-studied system involves the interaction of [LPS](#) from gram-negative bacteria and the glycosylphosphatidylinositol (GPI)-anchored membrane protein CD14 found on the surface of professional phagocytes, including migrating and tissue-fixed macrophages and [PMNs](#). A soluble form of CD14 is also found in plasma and on mucosal surfaces. A plasma protein, LPS-binding protein (LBP), transfers LPS to membrane-bound CD14 on myeloid cells and promotes binding of LPS to soluble CD14. Soluble CD14/LPS/LBP complexes bind to many cell types and may be internalized to initiate cellular responses to microbial pathogens. It has been shown that peptidoglycan and lipoteichoic acid from gram-positive bacteria and cell-surface products of mycobacteria and spirochetes can interact with CD14 ([Fig. 120-2](#)).

[GPI](#)-anchored receptors do not have intracellular signaling domains, and mammalian Toll-like receptors (TLRs) transduce signals for cellular activation due to [LPS](#) binding. TLRs initiate cellular activation through a series of signal-transducing molecules ([Fig. 120-2](#)) that lead to nuclear translocation of the transcription factor NF- κ B, a master-switch for production of important inflammatory cytokines such as tumor necrosis factor (TNF- α) and interleukin (IL) 1.

The initiation of inflammation can occur not only with LPS and peptidoglycan but also with viral particles and other microbial products such as polysaccharides, enzymes, and toxins.

Bacterial Cell Wall Structure Gram-positive bacteria have a rigid cell wall that gives the organisms their characteristic shape, differentiates them from eukaryotic cells, and allows them to survive in osmotically unfavorable environments. The cell wall is

composed mainly of peptidoglycan ([Fig. 120-3](#), panel C; a polymer of *N*-acetylglucosamine and its lactyl ether, *N*-acetylmuramic acid), with peptide side chains covalently bound to the lactyl group ([Fig. 120-3](#), panel A). The peptide chains consist of alternating D and L amino acids and are usually linked to each other by a pentaglycine bridge binding a terminal D-alanine on one peptide substituent to the penultimate L-lysine on a neighboring peptide. Variations in this basic structure have been described for a number of bacterial genera. In addition, the cell walls of gram-positive bacteria contain teichoic acids ([Fig. 120-3](#), panel D), phosphate-linked polymers of ribitol or glycerol that can have additional compounds linked to available side groups. Lipid tails anchor these acids to the cytoplasmic membrane, giving rise to lipoteichoic acids.

Gram-negative bacteria possess a cytoplasmic membrane and a peptidoglycan layer similar to but reduced from that found in gram-positive organisms, but these organisms also produce an outer membrane that is covalently linked to the tetrapeptides of the peptidoglycan layer by a lipoprotein ([Fig. 120-3](#), panel B). Embedded in the outer membrane are special proteins with important functions, including maintaining the outer membrane's integrity, acting as a selective barrier for diffusion of molecules into the cell, serving as receptors for bacteriophages, and binding siderophores that scavenge iron for transport into the bacterial cell. The exterior layer of the outer membrane contains the major surface glycolipid, which can be either a classical bacterial [LPS](#) or a lipooligosaccharide (LOS); pathogens such as *Neisseria* and *Haemophilus* spp. express LOS, which contains smaller polysaccharide constituents. Although LPS/LOS was thought to be essential to the viability of gram-negative bacteria, a viable strain of *N. meningitidis* lacking LOS has now been made.

Exterior to the [LPS](#) for many, but not all, gram-negative pathogens is a capsular polysaccharide, which (along with LPS for some pathogenic species) confers resistance to phagocytosis by preventing innate host opsonins, such as the complement proteins C3 and C4, from coating the organisms -- a process that promotes their uptake by phagocytes. Capsular polysaccharides are also important extracellular components of gram-positive bacteria, serving as critical factors in bacterial resistance to opsonophagocytosis and phagocytic killing. Variation in the expression of the capsule in *S. pneumoniae* accounts for the different morphologies of colonies of this pathogen on agar plates (smooth and rough phenotypes); this property was exploited in studies proving that DNA carries genetic information in a cell.

Lipopolysaccharide Most of the important biologic properties associated with [LPS](#) (endotoxin) are due to the lipid A portion ([Fig. 120-3](#), panel E), a relatively conserved, highly acylated di-*N*-acetylglucosamine backbone linked at 1 and 6 and containing phosphate groups on the reducing 1 and nonreducing 4 carbons. Attached to carbon 6 is the inner polysaccharide core, which is usually, but not always, composed of a di- or trisaccharide of 2-keto-3-deoxyoctonate (KDO). Additional sugar substituents are linked to the inner core, forming a complete core. Attached to the complete core are either short polysaccharide side chains (forming [LOS](#)) or longer O polysaccharide side chains (forming complete LPS) composed of a variety of monosaccharides, substituted with a variety of components, such as formyl, acetyl, and hydroxy-buteryl side chains; amino acids or peptides; and phosphate groups. Further biologic functions of LOS/LPS that are important to the survival of microbes after entry

into a host include resistance to the bacteriolytic effects of complement and protection against antimicrobial factors such as defensins and bactericidal permeability-increasing protein, a molecule closely related to [LBP](#) in structure and function. Defensins are found in high concentrations in granules of myeloid cells, including platelets, and are usually highly cationic peptides capable of insertion into bacterial cells and killing of these cells.

Additional Interactions of Microbial Pathogens and Phagocytes Other ways that microbial pathogens avoid destruction by phagocytes include production of factors that are toxic to the phagocytes or that interfere with the chemotactic and ingestion function of phagocytes. Hemolysins, leukocidins, and the like are microbial proteins that can kill phagocytes that are attempting to ingest organisms elaborating these substances. For example, staphylococcal hemolysins inhibit macrophage chemotaxis and kill these phagocytes. Streptolysin O made by *S. pyogenes* binds to cholesterol in phagocyte membranes and initiates a process of internal degranulation, with the release of normally granule-sequestered toxic components into the phagocyte's cytoplasm. *Entamoeba histolytica*, an intestinal protozoan that causes amebic dysentery, can disrupt phagocyte membranes after direct contact via the release of protozoal phospholipase A and pore-forming peptides.

Microbial Survival Inside Phagocytes Many important microbial pathogens use a variety of strategies to survive inside phagocytes (particularly macrophages) after ingestion. Inhibition of fusion of the phagocytic vacuole (the phagosome) containing the ingested microbe with the lysosomal granules containing antimicrobial substances (the lysosome) allows *M. tuberculosis*, *S. typhi*, and *Toxoplasma gondii* to survive inside macrophages. Some organisms, such as *Listeria monocytogenes*, escape into the phagocyte's cytoplasm to grow and eventually spread to other cells. Resistance to killing within the macrophage and subsequent growth are critical to successful infection by herpes-type viruses, measles virus, poxviruses, *Salmonella*, *Yersinia*, *Legionella*, *Mycobacterium*, *Trypanosoma*, *Nocardia*, *Histoplasma*, *Toxoplasma*, and *Rickettsia*. *Salmonella* spp. use a master regulatory system, in which the *PhoP/PhoQ* genes control other genes, to enter and survive within cells, with intracellular survival entailing structural changes in the cell envelope [LPS](#).

TISSUE INVASION AND TISSUE TROPISM

TISSUE INVASION

Most viral pathogens cause disease by growth at skin or mucosal entry sites, but some pathogens spread from the initial site to deeper tissues. Virus can spread via the nerves (rabies virus) or plasma (picornaviruses) or within migratory blood cells (poliovirus, Epstein-Barr virus, and many others). Specific viral genes determine where and how individual viral strains can spread.

Bacteria may invade deeper layers of mucosal tissue via intracellular uptake by epithelial cells, traversal of epithelial cell junctions, or penetration through denuded epithelial surfaces. Among virulent *Shigella* strains and invasive *E. coli*, outer-membrane proteins are critical to epithelial cell invasion and bacterial multiplication. *Neisseria* and *Haemophilus* spp. penetrate mucosal cells by poorly understood mechanisms before dissemination into the bloodstream. Staphylococci and

streptococci elaborate a variety of extracellular enzymes, such as hyaluronidase, lipases, nucleases, and hemolysins, that are probably important in breaking down cellular and matrix structures and allowing the bacteria access to deeper tissues and blood. Organisms that colonize the gastrointestinal tract can often translocate through the mucosa into the blood and, under circumstances in which host defenses are inadequate, cause bacteremia. *Yersinia enterocolitica* can invade the mucosa through the activity of the invasin protein. Some bacteria (e.g., *Brucella*) can be carried from a mucosal site to a distant site by phagocytic cells (e.g., [PMNs](#)) that ingest but fail to kill the bacteria.

Fungal pathogens almost always take advantage of host immunocompromise to spread hematogenously to deeper tissues. The AIDS epidemic has resoundingly illustrated this principle: the immunodeficiency of many HIV-infected patients permits the development of life-threatening fungal infections of the lung, blood, and brain. Other than the capsule of *C. neoformans*, specific fungal antigens involved in tissue invasion are not well characterized. Both fungal pathogens and protozoal pathogens (e.g., *Plasmodium* spp. and *E. histolytica*) undergo morphologic changes to spread within a host. Malarial parasites grow in liver cells as merozoites and are released into the blood to invade erythrocytes and become trophozoites. *E. histolytica* is found as both a cyst and a trophozoite in the intestinal lumen, through which this pathogen enters the host, but only the trophozoite form can spread systemically to cause amebic liver abscesses. Other protozoal pathogens, such as *T. gondii*, *Giardia lamblia*, and *Cryptosporidium*, also undergo extensive morphologic changes after initial infection to spread to other tissues.

TISSUE TROPISM

The propensity of certain microbes to cause disease by infecting specific tissues has been known since the early days of bacteriology, yet the molecular basis for this propensity is understood somewhat better for viral pathogens than for other agents of infectious disease. Specific receptor-ligand interactions clearly underlie the ability of certain viruses to enter cells within tissues and disrupt normal tissue function, but the mere presence of a receptor for a virus on a target tissue is not sufficient for tissue tropism. Factors in the cell, route of viral entry, viral capacity to penetrate into cells, viral genetic elements that regulate gene expression, and pathways of viral spread in a tissue all affect tissue tropism. Some viral genes are best transcribed in specific target cells, such as hepatitis B genes in liver cells and Epstein-Barr virus genes in B lymphocytes. The route of inoculation of poliovirus determines its neurotropism, although the molecular basis for this circumstance is not understood.

The lesser understanding of the tissue tropism of bacterial and parasitic infections is exemplified by *Neisseria* spp. There is no well-accepted explanation of why *N. gonorrhoeae* colonizes and infects the human genital tract while the closely related species *N. meningitidis* principally colonizes the human oropharynx. *N. meningitidis* expresses a capsular polysaccharide, while *N. gonorrhoeae* does not; however, there is no indication that this property plays a role in the different tissue tropisms displayed by these two bacterial species. *N. gonorrhoeae* can use cytidine monophosphate *N*-acetylneuraminic acid from host tissues to add *N*-acetylneuraminic acid (sialic acid) to its [LOS](#) O side chain, and this alteration appears to make the organism resistant to host defenses. Lactate, present at high levels on genital mucosal surfaces, stimulates

sialylation of gonococcal LOS. Bacteria with sialic acid sugars in their capsules, such as *N. meningitidis*, *E. coli* K-1, and group B streptococci, have a propensity to cause meningitis, but this generalization has many exceptions. For example, all recognized serotypes of group B streptococci contain sialic acid in their capsules, but only one serotype (III) is responsible for most cases of group B streptococcal meningitis. Moreover, both *H. influenzae* and *S. pneumoniae* can readily cause meningitis, but these organisms do not have sialic acid in their capsules.

TISSUE DAMAGE AND DISEASE

Disease is a complex phenomenon resulting from tissue invasion and destruction, toxin elaboration, and host response. Viruses cause much of their damage by exerting a cytopathic effect on host cells and inhibiting host defenses. The growth of bacterial, fungal, and protozoal parasites in tissue, which may or may not be accompanied by toxin elaboration, can also compromise tissue function and lead to disease. For some bacterial and possibly some fungal pathogens, toxin production is one of the best-characterized molecular mechanisms of pathogenesis, while host factors such as [IL-1](#), [TNF- \$\alpha\$](#) , kinins, inflammatory proteins, products of complement activation, and mediators derived from arachidonic acid metabolites (leukotrienes) and cellular degranulation (histamines) readily contribute to the severity of disease.

VIRAL DISEASE

Viral pathogens are well known to inhibit host immune responses by a variety of mechanisms. Immune responses can be affected by down-regulating production of most major histocompatibility complex (MHC) molecules (adenovirus E3 protein), by diminishing cytotoxic T cell recognition of virus-infected cells (Epstein-Barr virus EBNA1 antigen and cytomegalovirus IE protein), by producing virus-encoded complement receptor proteins (herpesvirus and vaccinia virus) that protect infected cells from complement-mediated lysis, by making proteins that interfere with the action of interferon (influenza virus and poxvirus), and by elaborating superantigen-like proteins (mouse mammary tumor virus and related retroviruses, rabies nucleocapsid, and possibly the Nef protein of HIV). Superantigens activate large populations of T cells that express particular subsets of the T cell receptorb protein, causing massive cytokine release and subsequent host reactions. Another molecular mechanism of viral virulence involves the production of peptide growth factors for host cells, which disrupt normal cellular growth, proliferation, and differentiation. In addition, viral factors can bind to and interfere with the function of host receptors for signaling molecules. Modulation of cytokine production during viral infection can stimulate viral growth inside cells with receptors for the cytokine, and virus-encoded cytokine homologues (e.g., the Epstein-Barr virus BCRF1 protein, which is highly homologous to the immunoinhibitory [IL-10](#) molecule) can potentially prevent immune-mediated clearance of viral particles. Viruses can cause disease in neural cells by interfering with levels of neurotransmitters without necessarily destroying the cells, or they may induce either programmed cell death (apoptosis) to destroy tissues or inhibitors of apoptosis to allow for prolonged viral infection of cells. Overall, any disruption of normal cellular and tissue function due to viral infection can underlie the resultant clinical disease.

BACTERIAL TOXINS

Among the first infectious diseases to be understood were those due to toxin-elaborating bacteria. Diphtheria, botulism, and tetanus toxins are responsible for the diseases associated with local infections due to *Corynebacterium diphtheriae*, *Clostridium botulinum*, and *Clostridium tetani*, respectively. Enterotoxins produced by *E. coli*, *Salmonella*, *Shigella*, *Staphylococcus*, and *V. cholerae* contribute to diarrheal disease caused by these organisms. Staphylococci, streptococci, *P. aeruginosa*, and *Bordetella* elaborate various toxins that cause or contribute to disease, including toxic shock syndrome toxin 1 (TSST-1); erythrogenic toxin; exotoxins A, S, and U; and pertussis toxin. A number of these toxins (e.g., cholera toxin, diphtheria toxin, pertussis toxin, *E. coli* heat-labile toxin, and *P. aeruginosa* exotoxin) have adenosine diphosphate (ADP)-ribosyltransferase activity; i.e., the toxins enzymatically catalyze the transfer of the ADP-ribosyl portion of nicotinamide adenine diphosphate to target proteins and inactivate them. The staphylococcal enterotoxins, TSST-1, and the streptococcal pyogenic exotoxins behave as superantigens, stimulating certain T cells to proliferate without processing of the protein toxin by antigen-presenting cells. Part of this process involves stimulation of the antigen-presenting cells to produce [IL-1](#) and [TNF- \$\alpha\$](#) , which have been implicated in many of the clinical features of diseases like toxic shock syndrome and scarlet fever. A number of gram-negative pathogens (*Salmonella*, *Yersinia*, and *P. aeruginosa*) possess the ability to inject toxins directly into host target cells by means of a complex set of proteins referred to as the type III secretion system.

ENDOTOXIN

The lipid A portion of gram-negative [LPS](#) has potent biologic activities that cause many of the clinical manifestations of gram-negative bacterial sepsis, including fever, muscle proteolysis, uncontrolled intravascular coagulation, and shock. The effects of lipid A appear to be mediated by the production of potent cytokines due to LPS binding to CD14 and signal transduction via [TLRs](#), particularly TLR4. Cytokines exhibit potent hypothermic activity through effects on the hypothalamus; they also increase vascular permeability, alter the activity of endothelial cells, and induce endothelial-cell procoagulant activity. Numerous therapeutic strategies aimed at neutralizing the effects of endotoxin are under investigation, but so far the results have been disappointing.

INVASION

Many diseases are caused primarily by pathogens growing in tissue sites that are normally sterile. Pneumococcal pneumonia is mostly attributable to the growth of *S. pneumoniae* in the lung and the attendant host inflammatory response, although specific factors that enhance this process (e.g., pneumolysin) may be responsible for some of the pathogenic potential of the pneumococcus. Disease that follows bacteremia and invasion of the meninges by meningitis-producing bacteria such as *N. meningitidis*, *H. influenzae*, *E. coli* K1, and group B streptococci appears to be due solely to the ability of these organisms to gain access to these tissues, multiply in them, and provoke cytokine production leading to tissue-damaging host inflammation.

Specific molecular mechanisms accounting for tissue invasion by fungal and protozoal pathogens are less well described. Except for studies pointing to factors like capsule and melanin production by *C. neoformans* and possibly levels of cell wall glucans in

some pathogenic fungi, the molecular basis for fungal invasiveness is not well defined. Melanin has been shown to protect the fungal cell against death caused by phagocyte factors such as nitric oxide, superoxide, and hypochlorite. Morphogenic variation and production of proteases (e.g., the *Candida* aspartyl proteinase) have been implicated in fungal invasion of host tissues.

If pathogens are effectively to invade host tissues (particularly the blood), they must avoid the major host defenses represented by complement and phagocytic cells. Bacteria most often avoid these defenses through their cell surface polysaccharides -- either capsular polysaccharides or long O-side-chain antigens characteristic of the smooth [LPS](#) of gram-negative bacteria. These molecules can prevent the activation and/or deposition of complement opsonins or limit the access of phagocytic cells with receptors for complement opsonins to these molecules when they are deposited on the bacterial surface below the capsular layer. Another potential mechanism of microbial virulence is the ability of some organisms to present the capsule as an apparent self antigen through molecular mimicry. For example, the polysialic acid capsule of group B *N. meningitidis* is chemically identical to an oligosaccharide found on human brain cells.

Immunochemical studies of capsular polysaccharides have led to an appreciation of the tremendous chemical diversity that can result from the linking of a few monosaccharides. For example, three hexoses can link up in more than 300 different, potentially serologically distinct ways, while three amino acids have only six possible peptide combinations. Capsular polysaccharides have been used as effective vaccines against meningococcal meningitis as well as against pneumococcal and *H. influenzae* infections and may prove to be of value as vaccines against any organisms that express a nontoxic, immunogenic capsular polysaccharide. In addition, most encapsulated pathogens become virtually avirulent when capsule production is interrupted by genetic manipulation; this observation emphasizes the importance of this structure in pathogenesis.

HOST RESPONSE

The inflammatory response of the host is critical for interruption and resolution of the infectious process but also is often responsible for the signs and symptoms of disease. Infection promotes a complex series of host responses involving the complement, kinin, and coagulation pathways. The production of cytokines such as [IL-1](#), [TNF- \$\alpha\$](#) , and other factors regulated in part by the NF- κ B transcription factor leads to fever, muscle proteolysis, and other effects, as noted above. An inability to kill or contain the microbe usually results in further damage due to the progression of inflammation and infection. For example, in many chronic infections, degranulation of host inflammatory cells can lead to release of host proteases, elastases, histamines, and other toxic substances that can degrade host tissues. Chronic inflammation in any tissue can lead to the destruction of that tissue and to clinical disease associated with loss of organ function, such as sterility from pelvic inflammatory disease caused by chronic infection with *N. gonorrhoeae*.

The nature of the host response elicited by the pathogen often determines the pathology of a particular infection. Local inflammation produces local tissue damage, while systemic inflammation, such as that seen during sepsis, can result in the signs and

symptoms of septic shock. The severity of septic shock is associated with the degree of production of host effectors. Disease due to intracellular parasitism results from the formation of granulomas, wherein the host attempts to wall off the parasite inside a fibrotic lesion surrounded by fused epithelial cells that make up so-called multinucleated giant cells. A number of pathogens, particularly anaerobic bacteria, staphylococci, and streptococci, provoke the formation of an abscess, probably because of the presence of zwitterionic surface polysaccharides such as the capsular polysaccharide of *Bacteroides fragilis*. The outcome of an infection depends on the balance between an effective host response that eliminates a pathogen and an excessive inflammatory response that is associated with an inability to eliminate a pathogen and with the resultant tissue damage that leads to disease.

TRANSMISSION TO NEW HOSTS

As part of the pathogenic process, most microbes are shed from the host, often in a form infectious for susceptible individuals. However, the rate of transmissibility may not necessarily be high, even if the disease is severe in the infected individual, as these traits are not linked. Most pathogens exit via the same route by which they entered: respiratory pathogens by aerosols from sneezing or coughing or through salivary spread, gastrointestinal pathogens by fecal-oral spread, sexually transmitted diseases by venereal spread, and vector-borne organisms by either direct contact with the vector through a blood meal or indirect contact with organisms shed into environmental sources such as water. Microbial factors that specifically promote transmission are not well characterized. Respiratory shedding is facilitated by overproduction of mucous secretions, with consequently enhanced sneezing and coughing. Diarrheal toxins such as cholera toxin, *E. coli* heat-labile toxins, and *Shigella* toxins probably facilitate fecal-oral spread of microbial cells in the high volumes of diarrheal fluid produced during infection. The ability to produce phenotypic variants that resist hostile environmental factors (e.g., the highly resistant cysts of *E. histolytica* shed in feces) represents another mechanism of pathogenesis relevant to transmission. Blood parasites such as *Plasmodium* spp. change phenotype after ingestion by a mosquito -- a prerequisite for the continued transmission of this pathogen. Venereally transmitted pathogens may undergo phenotypic variation due to the production of specific factors to facilitate transmission, but shedding of these pathogens into the environment does not result in the formation of infectious foci.

In summary, the molecular mechanisms used by pathogens to colonize, invade, infect, and disrupt the host are numerous and diverse. Each phase of the infectious process involves a variety of microbial and host factors interacting in a manner that can result in disease. Recognition of the coordinated genetic regulation of virulence factor elaboration when organisms move from their natural environment into the mammalian host emphasizes the complex nature of the host-parasite interaction. Fortunately, the need for diverse factors in successful infection and disease implies that a variety of therapeutic strategies may be developed to interrupt this process and thereby prevent and treat microbial infections.

(Bibliography omitted in Palm version)

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121. LABORATORY DIAGNOSIS OF INFECTIOUS DISEASES - Andrew B. Onderdonk

The laboratory diagnosis of infection requires the demonstration, either direct or indirect, of viral, bacterial, fungal, or parasitic agents in tissues, fluids, or excreta of the host. Clinical microbiology laboratories are responsible for processing these specimens and also for determining the antibiotic susceptibility of bacterial pathogens. Traditionally, detection of pathogenic agents has relied largely on either the microscopic visualization of pathogens in clinical material or the growth of microorganisms in the laboratory. Identification is generally based on phenotypic characteristics, such as fermentation profiles for bacteria, cytopathic effects in tissue culture for viral agents, and microscopic morphology for fungi and parasites. These techniques are reliable but are often time-consuming. Increasingly, the use of nucleic acid probes is becoming a standard detection and/or identification method in the clinical microbiology laboratory, gradually replacing phenotypic characterization and microscopic visualization methods.

DETECTION METHODS

Reappraisal of the methods employed in the clinical microbiology laboratory has led to the development of strategies for detection of pathogenic agents through nonvisual biologic signal detection systems. Much of this methodology is based on either computerization of detection systems with relatively inexpensive but sophisticated computers or the use of nucleic acid probes directed at specific DNA or RNA targets. This chapter discusses both the methods that are currently available and those that are being developed.

BIOLOGIC SIGNALS

A *biologic signal* is a material that can be reproducibly differentiated from other substances present in the same physical environment. Key issues in the use of a biologic (or electronic) signal are distinguishing it from background noise and translating it into meaningful information. Examples of biologic signals applicable to clinical microbiology include structural components of bacteria, fungi, and viruses; specific antigens; metabolic end products; unique DNA or RNA base sequences; enzymes; toxins or other proteins; and surface polysaccharides.

DETECTION SYSTEMS

A detector is used to sense a signal and to discriminate between the signal and background noise. Detection systems range from the trained eyes of a technologist assessing morphologic variations to sensitive electronic instruments, such as gas-liquid chromatographs coupled to computer systems for signal analysis. The sensitivity with which signals can be detected varies widely. It is essential to use a detection system that discerns small amounts of signal even when biologic background noise is present -- i.e., that is both sensitive and specific. Common detection systems include immunofluorescence; chemiluminescence for DNA/RNA probes; flame ionization detection of short- or long-chain fatty acids; and detection of substrate utilization or end-product formation as color changes, of enzyme activity as a change in light absorbance, of turbidity changes, of cytopathic effects in cell lines, and of particle

agglutination.

AMPLIFICATION

Amplification enhances the sensitivity with which weak signals can be detected. The most common microbiologic amplification technique is growth of a single bacterium into a discrete colony on an agar plate or into a suspension containing many identical organisms. The advantage of growth as an amplification method is that it requires only an appropriate growth medium; the disadvantage is the amount of time required for amplification. More rapid specific amplification of biologic signals can be achieved with techniques such as polymerase (ligase) chain reactions (PCRs, for DNA/RNA), enzyme immunoassays (EIAs, for antigens and antibodies), electronic amplification (for gas-liquid chromatography assays), antibody capture methods (for concentration and/or separation), and selective filtration or centrifugation. Although a variety of methods are available for the amplification and detection of biologic signals in research, thorough testing is required before they are validated as diagnostic assays.

DIRECT DETECTION

MICROSCOPY

The field of microbiology has been defined largely by the development and use of the microscope. The examination of specimens by microscopic methods rapidly provides useful diagnostic information. Staining techniques permit organisms to be seen more clearly.

The simplest method for microscopic evaluation is the wet mount, which is used, for example, to examine cerebrospinal fluid (CSF) for the presence of *Cryptococcus neoformans*, with India ink as a background against which to visualize large-capsuled yeast cells. Wet mounts with dark-field illumination are also used to detect spirochetes from genital lesions and to reveal *Borrelia* or *Leptospira* in blood. Skin scrapings and hair samples can be examined with use of either 10% KOH wet-mount preparations or the calcofluor white method and ultraviolet illumination to detect fungal elements as fluorescing structures. Staining of wet mounts -- for example, with lactophenol cotton blue stain for fungal elements -- is often used for morphologic identification. These techniques enhance signal detection and decrease the background, making it easier to identify specific fungal structures.

STAINING

Gram's Stain Without staining, bacteria are difficult to see at the magnifications (400 to 1000 $\times$) used for their detection. Although simple one-step stains can be used, differential stains are more common. Gram's stain differentiates between organisms with thick peptidoglycan cell walls (gram-positive) and those with outer membranes that can be dissolved with alcohol or acetone (gram-negative).

Gram's stain is particularly useful for examining sputum for polymorphonuclear leukocytes (PMNs) and bacteria. Sputum specimens with 25 or more PMNs and fewer than 10 epithelial cells per low-power field often provide clinically useful information.

However, the presence in "sputum" samples of more than 10 epithelial cells per low-power field and of multiple bacterial types suggests contamination with oral microflora. Despite the difficulty of discriminating between normal microflora and pathogens, Gram's stain may prove useful for specimens from areas with a large resident microflora if a useful biologic marker (signal) is available. Gram's staining of vaginal swab specimens can be used to detect epithelial cells covered with gram-positive bacteria in the absence of lactobacilli and the presence of gram-negative rods -- a scenario regarded as a sign of bacterial vaginosis. Similarly, examination of stained stool specimens for leukocytes is useful as a screening procedure before testing for *Clostridium difficile* toxin or other enteric pathogens.

The examination of CSF and joint, pleural, or peritoneal fluid with Gram's stain is useful for determining whether bacteria and/or PMNs are present. The sensitivity is such that >10⁴ bacteria per milliliter should be detected. Centrifugation is often performed before staining to concentrate specimens thought to contain low numbers of organisms. The pellet is examined after staining. This simple method is particularly useful for examination of CSF for bacteria and white blood cells or of sputum for acid-fast bacilli (AFB).

Acid-Fast Stain The acid-fast stain identifies organisms that retain carbol fuchsin dye after acid/organic solvent disruption (e.g., *Mycobacterium* spp.). Modifications of this procedure allow the differentiation of *Actinomyces* from *Nocardia* or other weakly acid-fast organisms. The acid-fast stain is applied to sputum, other fluids, and tissue samples when AFB (e.g., *Mycobacterium* spp.) are suspected. The identification of the pink/red AFB against the blue background of the counterstain requires a trained eye, since few AFB may be detected in an entire smear, even when the specimen has been concentrated by centrifugation. An alternative method is the auramine-rhodamine combination fluorescent dye technique.

Fluorochrome Stains Fluorochrome stains, such as acridine orange, are used to identify white blood cells, yeasts, and bacteria in body fluids. Other specialized stains, such as Dappe's stain, may be used for the detection of *Mycoplasma* in cell cultures. Capsular, flagellar, and spore stains are used for identification or demonstration of characteristic structures.

Immunofluorescent Stains The direct immunofluorescent antibody technique uses antibody coupled to a fluorescing compound, such as fluorescein, and directed at a specific antigenic target to visualize organisms or subcellular structures. When samples are examined under appropriate conditions, the fluorescing compound absorbs ultraviolet light and reemits light at a higher (visible) wavelength detectable by the human eye. In the indirect immunofluorescent antibody technique, an unlabeled (target) antibody binds a specific antigen. The specimen is then stained with fluorescein-labeled polyclonal antibody directed at the target antibody. Because each unlabeled target antibody attached to the appropriate antigen has multiple sites for attachment of the second antibody, the visual signal can be intensified (i.e., amplified). This form of staining is called *indirect* because a two-antibody system is used to generate the signal for detection of the antigen. Both direct and indirect methods detect viral inclusions (e.g., cytomegalovirus and herpes simplex virus) within cultured cells as well as many difficult-to-grow bacterial agents (e.g., *Legionella pneumophila*) directly in clinical

specimens.

MACROSCOPIC ANTIGEN DETECTION

Latex agglutination assays and [EIAs](#) are rapid and inexpensive methods for identifying organisms, extracellular toxins, and viral agents by means of protein and polysaccharide antigens. Such assays may be performed directly on clinical samples or after growth of organisms on agar plates or in viral cell cultures. The biologic signal in each case is the antigen to be detected. Monoclonal or polyclonal antibodies coupled to a reporter (such as latex particles or an enzyme) are used for detection of antibody-antigen binding reactions.

Techniques such as direct agglutination of bacterial cells with specific antibody are simple but relatively insensitive, while latex agglutination and [EIAs](#) are more sensitive. Some cell-associated antigens, such as capsular polysaccharides and lipopolysaccharides, can be detected by agglutination of a suspension of bacterial cells when antibody is added; this method is useful for typing of the somatic antigens of *Shigella* and *Salmonella*. In systems such as EIAs, which employ antibodies coupled to an enzyme, an antigen-antibody reaction results in the conversion of a colorless substrate to a colored product. Because the coupling of an enzyme to the antibody can amplify a weak biologic signal, the sensitivity of such assays is often high. In each instance, the basis for antigen detection is antigen-antibody binding, with the detection system changed to accommodate the biologic signal. Most such assays provide information as to whether antigen is present but do not quantify the antigen. EIAs are also useful for detecting bacterial toxins -- e.g., *C. difficile* toxins A and B in stool.

DETECTION OF PATHOGENIC AGENTS BY CULTURE

SPECIMEN COLLECTION AND TRANSPORT

To culture bacterial, mycotic, or viral pathogens, an appropriate sample must be placed into the proper medium for growth (amplification). The success of efforts to identify a specific pathogen often depends on the collection and transport process coupled to a laboratory-processing algorithm suitable for the specific sample/agent. In some instances, it is better for specimens to be plated at the time of collection rather than first being transported to the laboratory (e.g., urethral swabs being cultured for *Neisseria gonorrhoeae* or sputum specimens for pneumococci). In general, the more rapidly a specimen is plated onto appropriate media, the better the chance for isolating bacterial pathogens. Appendix B lists procedures for collection and transport of common specimens. Because there are many pathogen-specific paradigms for these procedures, it is important to seek advice from the microbiology laboratory when in doubt about a particular situation.

ISOLATION OF BACTERIAL PATHOGENS

Isolation of suspect pathogen(s) from clinical material relies on the use of artificial media that support bacterial growth in vitro. Such media are composed of agar, which is not metabolized by bacteria; nutrients to support the growth of the species of interest; and sometimes substances to inhibit the growth of other bacteria. Broth is employed for

growth (amplification) of organisms from specimens with few bacteria, such as peritoneal dialysis fluid, [CSF](#), or samples in which anaerobes or other fastidious organisms may be present. The general use of liquid medium for all specimens is not worthwhile.

Two basic strategies are used to isolate pathogenic bacteria. The first is to employ enriched media that support the growth of any bacteria that may be present in a sample such as blood or [CSF](#), which contain no bacteria under normal conditions. Broths that allow the growth of small numbers of organisms may be subcultured to solid media when growth is detected. The second strategy is to isolate (amplify) specific bacterial species from stool, genital tract secretions, or sputum -- sites that contain many bacteria under normal conditions. Antimicrobial agents or other inhibitory substances are incorporated into the agar medium to inhibit growth of all but the bacteria of interest. After incubation, organisms that grow on such media are further characterized to determine whether they are pathogens. Selection for organisms that may be pathogens from the normal microflora shortens the time required for diagnosis ([Fig. 121-1](#)).

ISOLATION OF VIRAL AGENTS (See also [Chap. 180](#))

Pathogenic viral agents often are cultured when the presence of serum antibody is not a criterion for active infection or when an increase in serum antibody may not be detected during infection. The biologic signal -- virus -- is amplified to a detectable level. Although a number of techniques are available, an essential element is a monolayer of cultured mammalian cells sensitive to infection with the suspected virus. These cells serve as the amplification system by allowing the proliferation of viral particles. Virus may be detected by direct observation of the cultured cells for cytopathic effects or by immunofluorescent detection of viral antigens following incubation. Culture methods are particularly useful for detection of rapidly propagated agents, such as cytomegalovirus or herpes simplex virus.

AUTOMATION OF MICROBIAL DETECTION IN BLOOD

The detection of microbial pathogens in blood is difficult because the number of organisms present in the sample is often low and the organisms' integrity and ability to replicate may be damaged by humoral defense mechanisms or antimicrobial agents. Over the years, systems that rely on the detection of CO₂ produced by bacteria and yeasts in blood culture medium have allowed the automation of the detection procedure. The most common systems involve either the insertion of a sampling device into each culture bottle at periodic intervals, with drawing off of the head-space gas for analysis by an infrared monitor, or the use of reflectance optics, with a light-emitting diode and photodiode employed to detect a color change in a CO₂-sensitive indicator built into the bottom of the culture bottle. These systems measure CO₂ concentration as indicative of microbial growth. Sophisticated algorithms are used to evaluate the rate at which CO₂ is being produced and then to determine whether the rate of change is consistent with microbial growth. Such methods are no more sensitive than the human eye in detecting a positive culture; however, because the bottles in an automated system are monitored more frequently, a positive culture is often detected more rapidly than by manual techniques, and important information, including the result of Gram's stain and preliminary susceptibility assays, can be obtained sooner. One advantage of reflectance

optic systems is that the bottles are scanned continuously in a noninvasive monitoring procedure, and thus the likelihood of laboratory contamination is decreased.

Automated systems also have been applied to the detection of microbial growth from specimens other than blood, such as peritoneal and other normally sterile fluids. *Mycobacterium* spp. can be detected in certain automated systems if appropriate liquid media are used for culture.

DETECTION OF PATHOGENIC AGENTS BY SEROLOGIC METHODS

Measurement of serum antibody provides an indirect marker for past or current infection with a specific viral agent or other pathogens, including *Brucella*, *Legionella*, *Rickettsia*, and *Helicobacter pylori*. The biologic signal is usually either IgM or IgG antibody directed at surface-expressed antigen(s). The detection systems include those used for bacterial antigens (agglutination reactions, immunofluorescence, and EIA) and unique systems such as hemolysis inhibition and complement fixation. Serologic methods generally fall into two categories: those that determine protective antibody levels and those that measure changing antibody titers during infection. Determination of an antibody response as a measure of current immunity is important in the case of viral agents for which there are vaccines, such as rubella virus or varicella-zoster virus; assays for this purpose normally use one or two dilutions of serum for a qualitative determination of protective antibody levels. Quantitative serologic assays to detect increases in antibody titers most often employ paired serum samples obtained 10 to 14 days apart (i.e., acute- and convalescent-phase samples). Since the incubation period before symptoms are noted may be long enough for an antibody response to occur, the demonstration of acute-phase antibody alone is often insufficient to establish the diagnosis of active infection as opposed to past exposure. In such circumstances, IgM may be useful as a measure of an early, acute-phase antibody response. A fourfold increase in total antibody titer or in EIA activity between the acute- and convalescent-phase samples is also regarded as evidence for active infection.

For certain viral agents, such as Epstein-Barr virus, the antibodies produced may be directed at different antigens during different phases of the infection. For this reason, most laboratories test for antibody directed at both viral capsid antigens and antigens associated with recently infected host cells to determine the stage of infection.

IDENTIFICATION METHODS

Once bacteria are isolated, traditional methods of phenotypic characterization are often used to identify specific isolates. An organism's phenotypic characteristics include traits that are readily detectable after growth on agar media (colony size, color, hemolytic reactions, odor), use of specific substrates and carbon sources (such as carbohydrates), formation of specific end products during growth, and microscopic appearance. Broth tubes containing specific substrates are commonly employed for phenotypic characterization. While such methods have been used since the time of Pasteur, their simplicity and low cost continue to make them appealing today.

CLASSIC PHENOTYPING

Automated systems allow rapid phenotypic identification of bacterial pathogens. Most such systems are based on biotyping techniques, in which isolates are grown on multiple substrates and the reaction pattern is compared with known patterns for various bacterial species. This procedure is relatively fast, and commercially available systems include miniaturized fermentation, coding to simplify recording of results, and probability calculations for the most likely pathogens. If the biotyping approach is automated and the reading process is coupled to computer-based data analysis, rapidly growing organisms, such as Enterobacteriaceae, can be identified within hours of detection on agar plates.

Several systems use preformed enzymes for even speedier identification (within 2 to 3 h). Such systems do not rely on bacterial growth per se to determine whether a substrate has been used or not. They employ a heavy inoculum in which specific bacterial enzymes are present in sufficient quantity to convert substrate to product rapidly. In addition, some systems use fluorogenic substrate/end-product detection methods to increase sensitivity (through signal amplification).

GAS-LIQUID CHROMATOGRAPHY

Gas-liquid chromatography is often used to detect metabolic end products of bacterial fermentations. One common application is identification of short-chain fatty acids produced by obligate anaerobes during glucose fermentation. Because the types and relative concentrations of volatile acids differ among the various genera and species that make up this group of organisms, such information serves as a metabolic "fingerprint" for a particular isolate.

Gas-liquid chromatography can be coupled to a sophisticated signal-analysis software system for identification and quantitation of long-chain fatty acids (LCFAs) in the outer membranes and cell walls of bacteria and fungi. For any given species, the types and relative concentrations of LCFAs are distinctive enough to allow identification of even closely related species. An organism may be identified definitively within a few hours after detection of growth on appropriate media. LCFA analysis is one of the most advanced procedures currently available for phenotypic characterization.

NUCLEIC ACID PROBES

Techniques for the detection and quantitation of specific DNA and RNA base sequences in clinical specimens have become powerful tools for the diagnosis of bacterial, viral, parasitic, and fungal infections. The basic strategy is to detect a relatively short sequence of bases specific for a particular pathogen on single-stranded DNA or RNA by hybridization of a complementary sequence of bases (probe) coupled to a "reporter" system that serves as the signal for detection. Detection of an organism by nucleic acid probes offers a decided advantage over culture methods for difficult-to-grow organisms. Current technology encompasses a wide array of methods for amplification and signal detection, some of which have been approved by the U.S. Food and Drug Administration (FDA) for clinical diagnosis.

Use of nucleic acid probes generally involves lysis of intact cells and denaturation of the DNA or RNA to render it single-stranded. The probe may be hybridized to the target

sequence in a solution or on a solid support, depending on the system employed. In situ hybridization of a probe to a target is also possible and allows the use of probes with agents present in tissue specimens. Once the probe has been hybridized to the target (biologic signal), a variety of strategies may be employed to amplify and/or quantify the target-probe complex ([Fig. 121-2](#)).

Probes for Direct Detection of Pathogens in Clinical Specimens Nucleic acid probes are available commercially for direct detection of various bacterial and parasitic pathogens, including *L. pneumophila*, *Chlamydia trachomatis*, *N. gonorrhoeae*, group A *Streptococcus*, *Gardnerella vaginalis*, *Mycoplasma hominis*, and *Giardia lamblia*. In addition, probes for direct detection of human papillomavirus, *Candida* spp., and *Trichomonas vaginalis* have been approved. An assortment of probes for confirming the identity of cultured pathogens, such as *Mycobacterium* and *Salmonella* spp., are also available. Probes for the direct detection of bacterial pathogens are often aimed at highly conserved 16S ribosomal RNA sequences, of which there are many more copies than there are of any single genomic DNA sequence in a bacterial cell. The sensitivity and specificity of probe assays for direct detection are comparable to those of more traditional assays, including [EIA](#) and culture. Many laboratories have developed their own probes for pathogens; however, unless a method-validation protocol for diagnostic testing has been performed, the use of such probes is restricted to research by federal law in the United States.

Nucleic Acid Probe Target-Amplification Strategies In theory, a single target nucleic acid sequence can be amplified to detectable levels. There are several strategies for target and/or probe amplification, including [PCR](#), ligase chain reaction, strand displacement amplification, and self-sustaining sequence replication. In each case, a target sequence or hybridized probe is amplified exponentially to obtain sufficient signal for detection, usually by the attachment of chemiluminescent reporter groups to the amplified product. The PCR strategy requires repeated heating of the DNA or RNA to separate the two complementary strands of the double helix, hybridization of a primer sequence to the appropriate target sequence, target amplification using the PCR for complementary strand extension, and signal detection via a labeled probe. The sensitivity of such assays is far greater than that of traditional assay methods such as culture. However, the care with which the assays are performed is important, because cross-contamination of clinical material with DNA or RNA from other sources (even at low levels) can cause false-positive results. An alternative method employs transcription-mediated amplification, in which an RNA target sequence is converted to DNA, which is then exponentially transcribed into RNA target. The advantage of this method is that only a single heating/annealing step is required for amplification. At present, amplification assays for *Mycobacterium tuberculosis*, *N. gonorrhoeae*, *C. trachomatis*, and *M. hominis* are on the market. Again, many laboratories have used commercially available *taq* polymerase, probe sequences, and reagents to develop "in-house" assays for diagnostic use. Issues related to quality control, interpretation of results, sample processing, and regulatory requirements have slowed the commercial development of diagnostic assay kits.

Signal Amplification Strategies Alternative systems for signal amplification have great appeal, particularly for quantitative determination of the amount of target present in a given specimen. With the advent of newer therapeutic regimens for HIV-associated

disease, cytomegalovirus infection, and hepatitis C virus infection, the response to therapy has been monitored by determining both genotype and "viral load" at various times after treatment initiation. Target amplification ([PCR](#), transcription-mediated amplification) is difficult to control in a manner that allows accurate determination of the original target (genome) concentration. In other systems, probes attached to complementary target sequences are amplified by the attachment of a second probe and an amplification multimer to the original probe. In one such system, branched-chain DNA (bDNA)-based amplification, bDNA is attached to a site different from the target-binding sequence of the original probe. Chemiluminescent-labeled oligonucleotides can then bind to multiple repeating sequences on the bDNA. The amplified bDNA signal is detected by chemiluminescence. Alternatively, a DNA probe may be attached to an RNA target and the resulting DNA/RNA hybrid captured on a solid support by antibody specific for DNA/RNA hybrids (concentration/amplification) and detected by chemiluminescent-labeled antibody specific for DNA/RNA hybrids. Both methods can be used to determine the approximate number of target copies (virus) in the starting material. The advantage of these systems over PCR is that only a single heating/annealing step is required to hybridize the target-binding probe to the target sequence for amplification.

Application of Nucleic Acid Probe Technology Nucleic acid probe technology is being used to identify difficult-to-grow or noncultivable bacterial pathogens, such as *Mycobacterium*, *Legionella*, *Ehrlichia*, *Rickettsia*, *Babesia*, *Borrelia*, and *Tropheryma whippelii*. Amplification methods are also being used to detect chronic viral infections, such as herpes simplex encephalitis, cytomegalovirus infection, and hepatitis C. The monitoring of therapy with quantitative viral-load testing is a significant new application of nucleic acid technology. Further applications will likely include the replacement of culture for identification of many pathogens with solid-state DNA/RNA chip technology, in which thousands of unique nucleic acid sequences can be detected on a single computer chip. Probe technology also has the potential to detect viral pathogens faster than is possible with current culture techniques. However, if laboratories are to take full advantage of probe technology, the cost of reagents and assay automation must be competitive with the cost of existing methodology. At present, the detection of agents such as *C. trachomatis* or *N. gonorrhoeae* by probe technology is more expensive for most laboratories than detection by traditional culture or [EIA](#). Moreover, because automated processing equipment is just beginning to find its way into the laboratory for these assays, nucleic acid amplification methods are both more labor-intensive and more expensive than other detection systems. In the absence of clear documentation of clinical utility, many laboratories continue to wait for [FDA](#) approval of commercially available DNA/RNA probe assays rather than validating in-house assays.

SUSCEPTIBILITY TESTING

A principal responsibility of the clinical microbiology laboratory is to determine which antimicrobial agents inhibit a specific bacterial isolate. Such testing is used to screen for infection control problems, such as methicillin-resistant *Staphylococcus aureus*, vancomycin-resistant *Enterococcus faecium*, or extended-spectrum b-lactamase-producing organisms. Two approaches are useful. The first is a qualitative assessment of susceptibility, with responses categorized as susceptible, resistant, or intermediate. This approach can involve either the placement of paper disks containing

antibiotics on an agar surface inoculated with the bacterial strain to be tested (Kirby-Bauer or disk/agar diffusion method), with measurement of the zones of growth inhibition following incubation, or the use of broth tubes containing a set concentration of antibiotic (breakpoint method). These methods have been carefully calibrated against quantitative methods and clinical experience with each antibiotic, and zones of inhibition and breakpoints have been calculated on a species-by-species basis.

The second approach is to inoculate the test strain of bacteria into a series of broth tubes (or agar plates) with increasing concentrations of antibiotic. The lowest concentration of antibiotic that inhibits microbial growth in this test system is known as the *minimum inhibitory concentration* (MIC). If tubes in which no growth occurs are subcultured, the minimum concentration of antibiotic required to kill the starting inoculum can also be determined (*minimum bactericidal concentration*, or MBC). Quantitative susceptibility testing by the microbroth dilution technique, a miniaturized version of the broth dilution technique using microwell plates, lends itself to automation and is commonly used in larger clinical laboratories.

A novel version of the disk/agar diffusion method employs a quantitative diffusion gradient, or epsilometer (E-test), and uses an absorbent strip with a known gradient of antibiotic concentrations along its length. When the strip is placed on the surface of an agar plate seeded with a bacterial strain to be tested, antibiotic diffuses into the medium, and bacterial growth is inhibited. The [MIC](#) is estimated as the lowest concentration that inhibits growth.

For some organisms, such as obligate anaerobes, routine susceptibility testing generally is not performed because of the difficulty of growing the organisms and the predictable sensitivity of most isolates to specific antibiotics.

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122. IMMUNIZATION PRINCIPLES AND VACCINE USE - Gerald T. Keusch, Kenneth J. Bart

Most humans live their lives ignoring the certainty of their own mortality. Perhaps this fact explains why the adage "an ounce of prevention is worth a pound of cure" has so little effect on their everyday behavior. Even when it comes to acting to protect their young, parents are capable of ignoring the potential for mortality among their children (in the developed world) and of accepting the certainty of childhood deaths (in the developing world). In both settings, parents all too often fail to seek out and demand the best preventive measures available. Unless mandated by the law in the former setting or provided by benevolent organizations or governments in the latter, universal immunization has invariably remained an unattained goal. Compulsion and benevolence, it seems, are two essential components of immunization.

However, the integration of immunization practices (a major component of primary disease prevention) into routine health care services has provided caregivers with control over a substantial proportion of the disease and mortality that plagued the United States during the first half of the twentieth century ([Table 122-1](#)). For society today, immunization represents one of the most cost-effective means of preventing infectious disease. For every dollar spent, diphtheria/tetanus/pertussis (DTP) vaccine saves \$29, measles/mumps/rubella (MMR) vaccine saves \$21, trivalent oral poliovirus vaccine (OPV) saves \$6, varicella vaccine saves \$5, and *Haemophilus influenzae* type b vaccine saves \$2. At present, >50 biologic products are licensed in the United States, and 6 vaccines (12 antigens) are used for routine immunization in the young, including diphtheria/tetanus/acellular pertussis vaccine (DTaP), inactivated poliovirus vaccine (IPV), MMR, *H. influenzae* type b (Hib) vaccine, hepatitis B virus (HBV) vaccine, and varicella vaccine. Five vaccines are designed for routine use in adults: tetanus/diphtheria (Td) toxoids, adsorbed, for adult use; HBV vaccine; influenza virus vaccine; polyvalent pneumococcal polysaccharide vaccine; and varicella vaccine. Some preparations are designated as special-use vaccines (e.g., hepatitis A vaccine for travelers). Unfortunately, vaccines for eukaryotic pathogens (protozoa and helminths), which affect a large proportion of the world's population, have been difficult to develop and remain only a hope for the future.

IMPACT OF IMMUNIZATION

The epidemiologically appropriate use of vaccines has resulted in the global eradication of smallpox and in the potential eradication of poliomyelitis in the next few years and of measles by 2020. Already achieved are the virtual elimination of congenital rubella syndrome, tetanus, and diphtheria as well as a dramatic reduction in pertussis, rubella, measles, and mumps in the United States. The introduction of [Hib](#) conjugate vaccines for immunization of infants has all but eliminated invasive *Haemophilus* infections (including meningitis and pneumonia), presumably because these vaccines also reduce nasopharyngeal carriage of Hib and induce protection before the period of greatest vulnerability in infancy. The recently licensed polyvalent pneumococcal polysaccharide conjugate vaccine promises to have the same impact on invasive pneumococcal disease, including otitis media.

DEFINITIONS

Vaccination and *immunization* are often used as interchangeable terms. However, the former denotes only the administration of a vaccine or toxoid, whereas the latter describes the process of inducing or providing immunity by any means, whether active or passive. Thus, vaccination does not guarantee immunization. *Active immunization* refers to the induction of immune defenses by the administration of antigens in appropriate forms, whereas *passive immunization* involves the provision of temporary protection by the administration of exogenously produced immune substances. Immunizing agents thus include vaccines, toxoids, and antibody-containing immunoglobulin preparations from human or animal donors ([Table 122-2](#)).

PRINCIPLES OF IMMUNIZATION

Artificial induction of immunity closely follows two well-tested principles of nature. The first, active immunization, can be traced at least as far back as Thucydides, who noted that people surviving epidemics of plague in Athens were spared during later outbreaks of the same disease. The second, passive immunization, is a natural process as well and is exemplified by the transplacental transmission of maternal antibodies to the fetus to provide protection against several diseases during the first months of life. Use of the two measures together may produce a complementary effect (as with [HBV](#) vaccine plus hepatitis B immune globulin) or may actually interfere with the development of immunity (as when measles vaccine is administered within 6 weeks of immunoglobulin). Depending on whether there are multiple species or serotypes of an organism and -- if so -- whether there are common, cross-reactive, protective antigens, a specific vaccine may induce protection against all representative forms of an infectious agent or against the immunizing strain only. One of the intrinsic virtues of whole-organism vaccines is that they potentially contain all protective antigens of the organism. However, this virtue is counterbalanced by an inherent problem with such vaccines: the possibility of adverse responses to reactive but nonprotective antigens present in the mix. Because the immune response to specific antigens is controlled genetically, all individuals cannot be expected to respond identically to the same vaccine.

APPROACHES TO ACTIVE IMMUNIZATION

The two standard approaches to active immunization are (1) the use of live, generally attenuated, infectious agents (e.g., measles virus); and (2) the use of inactivated agents or their constituents or products obtained by genetic recombination (e.g., acellular pertussis vaccines). For many diseases (e.g., poliomyelitis, influenza), both approaches have been employed. Live attenuated vaccines are believed to induce an immunologic response more nearly like that resulting from natural infection than the response induced by killed vaccines. Currently available inactivated or killed vaccines consist of inactivated whole organisms (e.g., plague vaccine); detoxified protein exotoxins (e.g., tetanus toxoid); recombinant protein antigens (e.g., [HBV](#) vaccine); or carbohydrate antigens, either present as soluble purified capsular material (e.g., *Streptococcus pneumoniae* polysaccharides) or conjugated to a protein carrier (e.g., [Hib](#) polysaccharide conjugated to diphtheria or tetanus toxoids).

APPROACHES TO PASSIVE IMMUNIZATION

Passive immunization is generally used to provide temporary immunity in an unimmunized subject exposed to an infectious disease when active immunization either is unavailable (e.g., for cytomegalovirus infection) or has not been implemented before exposure (e.g., for rabies). Passive immunization is used in the treatment of certain disorders associated with toxins (e.g., diphtheria), in certain bites (those of snakes and spiders), and as a specific or nonspecific immunosuppressant [Rho(D) immune globulin and antilymphocyte globulin, respectively].

Three types of preparations are used in passive immunization: (1) standard human immune serum globulin for general use (e.g., gamma globulin), administered intramuscularly or intravenously; (2) special immune serum globulins with a known content of antibody for specific agents (e.g., [HBV](#) or varicella-zoster immune globulin); and (3) animal sera and antitoxins.

ROUTE OF ADMINISTRATION

The route of administration in part determines the rapidity and nature of the immune responses to vaccines. Vaccines can be administered orally, intranasally, intradermally, subcutaneously, or intramuscularly. Parenterally administered vaccine may not induce mucosal secretory IgA, and mucosal immunization may not induce good systemic responses. Vaccines must be administered by the licensed route to ensure immunogenicity and safety. For example, administration of [HBV](#) vaccine into the gluteal rather than the deltoid muscle often fails to induce an adequate immune response, while subcutaneous rather than intramuscular administration of [DTP](#) increases the risk of reactions.

AGE

Because age influences the response to vaccines, schedules for immunization are based on age-dependent responses determined empirically from clinical trials. The presence of high levels of maternal antibody and/or the immaturity of the immune system in the early months of life impairs the initial immune response to some vaccines (e.g., measles or [Hib](#) polysaccharide but not [HBV](#)). In the elderly, vaccine responses may be diminished because of natural waning of the immune system. Hence, larger amounts of an antigen may be required to produce the desired response (e.g., in vaccination against influenza).

ADJUVANT POTENTIATION

The immune response to some antigens is potentiated by the addition of adjuvants such as aluminum salts or, in the case of polysaccharides (e.g., the polyribose phosphate oligosaccharide of [Hib](#)), by conjugation to a carrier protein. Adjuvants, nonspecific boosters of immune responses, are used with inactivated products such as diphtheria and tetanus toxoids, acellular pertussis (aP) vaccine, and [HBV](#) vaccine. The mechanism for adjuvant enhancement of immunogenicity is not well defined but relates in part to the rendering of soluble antigens into a particulate form, the mobilization of phagocytes to the site of antigen deposition, and the slowing down of the release of antigens, which prolongs stimulation of the immune response.

THE IMMUNE RESPONSE

While many constituents of infectious microorganisms and their products, such as exotoxins, are or can be made to be immunogenic, only a limited number stimulate a protective immune response. The immune system is complex, and antigen composition and presentation are critical for stimulation of the desired immune responses.

The Primary Response In the primary response to a vaccine antigen, an apparent latent period of several days precedes the detection of humoral and cell-mediated immunity. Although the immune response is turned on by contact with the antigen and the immune system, measurable circulating antibodies do not appear for 7 to 10 days. The immunoglobulin class of the response also changes over time. Early-appearing IgM antibodies generally exhibit only low affinity for the antigen, whereas later-appearing IgG antibodies display high affinity. For "thymus-dependent" antigens, CD4⁺ T helper lymphocytes control the switch from IgM to IgG. Some individuals do not respond, even when presented repeatedly with a vaccine antigen, often because they lack the major histocompatibility complex determinants required to recognize the antigen. This situation is known as *primary vaccine failure*.

The Secondary Response Heightened humoral or cell-mediated responses are elicited by a second exposure to the same antigen. These secondary responses occur rapidly, usually within 4 or 5 days, and result, for example, in increased titers of IgG antibody. The secondary response depends on immunologic memory after the first exposure and is characterized by a marked proliferation of antibody-producing B lymphocytes and/or effector T cells. Polysaccharide vaccines, such as that for *S. pneumoniae*, evoke immune responses that are independent of T cells and are not enhanced by repeated administration. Covalent linking of polysaccharides to proteins converts the former to T cell-dependent antigens that induce immunologic memory and secondary responses to revaccination. Although levels of vaccine-induced antibodies may decline over time (*secondary vaccine failure*), revaccination or exposure to the organism may elicit a rapid protective secondary response consisting of IgG antibodies with little or no detectable IgM. This *anamnestic response* indicates that immunity has persisted. The lack of measurable antibody does not necessarily mean that the individual is unprotected. Furthermore, the mere presence of detectable antibodies after the administration of some vaccines and toxoids does not ensure clinical protection. A minimal circulating level of antibody is known to be required for protection from some diseases (e.g., 0.01 IU/mL for tetanus antitoxin).

Hypersensitivity Reactions Independent of antibody production, the stimulation of the immune system by vaccination may elicit unanticipated responses, especially hypersensitivity reactions. In the past, killed measles vaccine induced incomplete humoral immunity and cell-mediated hypersensitivity, resulting in the development of a syndrome of atypical measles in some children after subsequent exposure; thus this type of vaccine is no longer in use.

Mucosal Immunity Some pathogens are confined to and replicate only at mucosal surfaces (e.g., *Vibrio cholerae*), while others are able to penetrate the mucosa and replicate (e.g., poliovirus, rubella virus, and influenza virus). At the mucosal site, these organisms induce secretory IgA. The induction of secretory IgA by vaccines may be an

efficient way to block the essential first steps in pathogenesis, whether the organism is restricted to mucosal surfaces or systemically invades the host across mucosal surfaces.

Measurement of the Immune Response Immune responses to vaccines are often gauged by the concentration of specific antibody in serum. While seroconversion serves as a dependable indicator of an immune response, it measures only one immunologic parameter and does not necessarily indicate protection. The development of circulating antibodies after immunization often correlates directly with clinical protection (e.g., against measles or rubella). Some responses may not in themselves confer immunity but may be sufficiently associated with protection that they remain useful proxy measures of protective immunity (e.g., vibriocidal serum antibodies in cholera).

HERD IMMUNITY

It is not necessary to immunize every person in order to stop transmission of an infectious agent through a population. For those organisms dependent on person-to-person transmission, there may be a definable prevalence of immunity in the population above which it becomes difficult for the organism to circulate and reach new susceptibles. This prevalence is called *herd immunity*. When herd immunity is operative, the goals of immunization are converted from the immunization of every person in the community to the immunization of a specified minimum percentage of persons at risk. Herd immunity may wane if immunization capacity fails (as in diphtheria in the new independent states of the former Soviet Union) or if a sufficient percentage of individuals refuse to be immunized (as in pertussis in the United Kingdom and Japan in the 1970s because concern about infrequent -- albeit severe -- vaccine reactions came to exceed the fear of the disease itself). In both situations, loss of herd immunity led to renewed circulation of the organism and increased susceptibility to infection, with subsequent large outbreaks.

TARGET POPULATIONS AND TIMING OF IMMUNIZATION

For common and highly communicable childhood diseases like measles, the target population is the universe of susceptible individuals, and the time to immunize is as early in life as is feasible. Epidemiologic differences in measles in different settings, however, dictate different strategies of immunization. In the industrialized world, immunization with live-virus vaccine at 12 to 15 months of age has been the norm because the vaccine protects >95% of those immunized at this age and there is little measles morbidity/mortality among very young infants. In contrast, in the developing world, measles accounts for a significant proportion of deaths of young infants. Thus it is desirable to immunize children during the first few months of life in order to narrow the window of vulnerability between the rapid decline of maternal antibody after 4 to 6 months and the development of vaccine-induced active immunity.

[Hib](#) causes meningitis, epiglottitis, and pneumonia in early childhood, with rates rising sharply after the disappearance of maternally derived antibody. The first Hib vaccines often failed when administered during infancy; this failure was due mainly to an age-related inability to respond to polysaccharide antigens. To overcome this problem, the protective polysaccharide was coupled to protein and converted to a T

cell-dependent antigen to which young infants could respond.

In contrast to measles and [Hib](#) infection, rubella is primarily a threat to the fetus; young infants and children are not at risk of serious illness. Given the susceptibility of the fetus, immunization of all women of reproductive age before pregnancy would be an ideal strategy. However, it is difficult to systematically vaccinate adolescent and young-adult females. Thus, to assure the protection of as many women as possible, the rubella component is included in a combination vaccine with mumps and measles ([MMR](#)) that is administered during infancy.

Some vaccines are now used primarily for adults. For example, influenza virus and polyvalent pneumococcal polysaccharide vaccines are used to prevent pneumonia deaths in the elderly. Unfortunately, these vaccines are underutilized, in part because physicians and otherwise healthy individuals in the target group ignore the indications and in part because there is still a tendency to think about disease prevention with vaccines as a strategy for children. Pneumococcal polysaccharide vaccine is also recommended for children >2 years old who are at risk of severe or even life-threatening pneumococcal infection, such as those with sickle cell disease, asplenia (whether functional or anatomic), renal failure with nephrotic syndrome, cerebrospinal fluid leak, and HIV infection or other immunosuppressive disease states.

THE DEVELOPMENT OF VACCINES

BIOLOGIC IMPEDIMENTS

There are often major technical problems to overcome in vaccine development. Although just one major antigenic type of influenza virus is typically in circulation at any one time, the virus is characterized biologically by its antigenic drift. Thus, a new antigenic version capable of causing a global pandemic emerges periodically, and a new vaccine must be rapidly devised, produced, and distributed. In contrast, many prevalent pneumococcal polysaccharide serotypes circulate at all times. Because immunity to the pneumococcus is serotype specific, an individual is susceptible to all serotypes against which he or she lacks antibody. Serotype-specific protection has made it more difficult to develop an effective pneumococcal vaccine than it was to develop a vaccine against *H. influenzae*, of which one capsular serotype (type b) is associated with nearly all cases of severe disease. To overcome this problem, pneumococcal vaccine currently includes 23 polysaccharides that represent ~80% of the virulent serotypes commonly encountered in the United States. Unfortunately, some serotypes are poorly immunogenic, and immunized individuals remain susceptible to the serotypes not included in the vaccine.

STRATEGY FOR VACCINE DEVELOPMENT

Vaccine development depends on the systematic application of a four-phase strategy: (1) studies in animals to identify protective antigen, (2) determination of how to present this antigen effectively to the immune system, (3) assessment of the safety and immunogenicity of the preparation in small and then in large human populations at various ages, and (4) evaluation of safety and efficacy in the target population. Each of these steps is simple in concept but difficult in execution, not least because of the

clinical trials necessary to assess safety and efficacy; failure at any level stops the process. Thus, in 1995, >190 candidate vaccines were under investigation, but just 5 new products were licensed in the United States. Progress in immunology has taught us much about the organization and function of the immune system ([Chap. 305](#)); it has also taught us that the immune system is complicated and that details of antigen composition and presentation are critical for stimulating desired immune responses.

Ultimately, vaccines for humans must be tested in humans. After initial animal studies and small phase 1 and 2 human studies to assess immune responses, optimal dosage, and safety, clinical trials of vaccine efficacy are performed, sometimes with informed volunteers who are challenged with a virulent strain. Larger clinical effectiveness trials in the community, typically involving 1000 to 10,000 vaccinees, may lead to application for licensure. Because of their limited size, however, these trials cannot be expected to detect rare adverse effects. Thus, licensing does not guarantee that a new vaccine is completely safe, and postlicensing monitoring is needed to ensure effectiveness and to document the occurrence of adverse events of low frequency. In 1999, the recently licensed rhesus rotavirus vaccine was withdrawn because postmarketing surveillance uncovered an association with a rare event in infants, intussusception of the bowel.

The development of vaccines goes beyond technology and proof of principle to issues such as development costs, manufacturers' liability and indemnity, perceived public health needs, and the likelihood that a product will be used or sold. Given the complex science required, the costs of vaccine development are high and success is uncertain, adding risk to the development decision. It is unfortunate that the one sure implication of uncertainty in vaccine development is increased cost. In addition, a rational assignment of costs for development between the public and private sectors in the United States has never been achieved.

VACCINE FORMULATIONS

Studies of clinical immunology have shown that living and dead antigens do not necessarily induce the same immune responses and that the requirements for the development of protective immunity differ with the organism. These insights, together with the refinement of epidemiologic concepts surrounding immunization, have changed the strategy of vaccine development. The goal is not only to select the correct antigens but also to ensure that the vaccines will result in the type of immune response needed for protection, whether the T cell-mediated activation of macrophages or the generation of cytotoxic T cells, B cell-mediated secretory IgA, or a particular IgG subtype response to a specific polysaccharide epitope.

Live vaccines consist of selected or genetically altered organisms that are avirulent or dramatically attenuated yet remain immunogenic. These agents are expected to cause a subclinical illness that mimics natural infection except for the lack of clinically significant disease. They offer the advantage of replication *in vivo*, which increases the antigenic load presented to the host's immune system; they may confer lifelong protection with one dose; they present all expressed antigens, thus overcoming immunogenetic restrictions in some hosts; they may reach the local sites most relevant to the induction of protective immunity; and they may produce important protective antigens *in vivo* that are not efficiently expressed *in vitro*.

Nonviable vaccines may fail to elicit mucosal IgA-mediated immunity, as they lack a delivery system that will effectively transport them to local antigen-processing cells. Moreover, except for pure polysaccharide antigens, these preparations must almost always be given in multiple doses to induce effective responses. However, killed vaccines can be extremely effective. For example, the nonviable hepatitis A vaccine formulation appears to be close to 100% effective in inducing protective immunity. Methods are under development to incorporate vaccine antigens into degradable polymers that may release antigen at predictable times after a single inoculation and simulate multiple injections over time of the same vaccine.

In spite of their advantages, live vaccines are not always to be preferred. For example, live [OPV](#) is contraindicated for use in children with immune-deficiency diseases and in their adult contacts. In addition, even though killed poliovirus vaccine does not completely immunize the gut and can neither reduce the circulation of wild-type poliovirus nor immunize contacts of vaccine recipients, the United States has now switched to a four-dose schedule of this vaccine because of the risk of vaccine-associated polio posed by live OPV.

To create a deliverable vaccine, constituents other than the antigens are required ([Table 122-3](#)). These constituents can affect the immunogenicity, efficacy, and safety of a vaccine and can render one formulation superior to another.

NEW VACCINE APPROACHES

The first generation of vaccines included whole killed -- or, more recently, live -- attenuated microorganisms or partially purified microbial products, such as tetanus toxoid, that induced protective antibodies. The second generation of vaccines has taken advantage of molecular genetics and protein chemistry to isolate and manipulate purified proteins or components or subunits of organisms or to generate genetically engineered and attenuated live native organisms or cloned antigens expressed by harmless vector organisms. One conceptual leap is the production of transgenic plants expressing protective vaccine antigens (cloned, for example, in potatoes or bananas) that, when ingested orally, induce mucosal and systemic immune responses to homologous infectious challenges. While the practical use of this technique awaits further refinement, the concept that protective immunity can be induced in this manner has been proven in both animals and humans. Ease of production, stability, ease of administration without equipment, and low cost are the obvious advantages.

Another conceptual leap has led to a third generation of vaccines, in which nucleic acids (either DNA or RNA) are used to induce immunity. Development of DNA vaccines is at a more advanced stage. The principle is simple. First, a DNA plasmid containing the gene sequence for the immunogenic protein or fragment of interest is assembled, and the gene is placed under the control of a strong promoter and an appropriate transcription termination sequence. A single immunization with the plasmid (via intramuscular or intradermal injection, helium-accelerated gene gun injection of DNA-coated gold particles, compressed-air pneumatic jet injection of soluble DNA, direct skin application after suitable preparation, or even insertion of biodegradable stents loaded with the DNA of interest) results in DNA uptake into cells where the gene is expressed and

processed normally; thus the product stimulates an immune response. Alteration of the DNA construct or of the mode of administration or the coadministration of cytokine genes can determine whether the immune response is humoral or cellular or whether it involves primarily Th1, Th2, or cytotoxic T cells. Such decisions can be used to optimize the protective immunity induced.

This form of immunization offers real advantages and only theoretical and remote disadvantages ([Table 122-4](#)). Moreover, DNA vaccines may be useful in inducing tumor immunity, treating allergy (by suppressing IgE production), or even administering genes for gene therapy. RNA vaccines would avoid some of the potential concerns raised by DNA vaccines because RNA is less stable and does not persist or integrate into the chromosome or cause insertional mutagenesis. However, this lack of stability and the likely need for multiple doses, along with the increased cost of producing, storing, and transporting RNA, are significant disadvantages that remain to be overcome. The concept of nucleic acid vaccines -- whether based on DNA or RNA -- has been validated experimentally, and early human trials have begun. There is great optimism for the future but much to be learned if we are to apply this powerful new immunization technique successfully.

PRODUCTION OF VACCINES

As products to be given to healthy individuals to prevent disease, vaccines must not only be efficacious but also cause no harm. In the United States, quality assurance is the responsibility of vaccine manufacturers. Standards of manufacture of biologics [known as good manufacturing practices (GMPs)] are regulated and supervised by the U.S. Food and Drug Administration (FDA). Proof of the safety, efficacy, sterility, and purity of products is required before licensure, and sterility and purity are continually monitored for all lots of vaccine after licensure. Postmarketing studies of safety (phase IV studies) are part of routine regulatory control. On rare occasions, either GMP or quality assurance is inadequate; for example, the release of incompletely killed Salk polio vaccine in 1955 caused an outbreak of poliomyelitis in nearly 200 vaccine recipients and their contacts. Unregulated and uncontrolled manufacture of vaccines in developing countries has sometimes led to immunization with inactive products that fail to provide the expected protective immunity.

Another problem in the production of vaccines has unexpectedly cropped up in the past decade. For various reasons, including the high costs of vaccine development and the prospect of much higher profitability from investments in other products, the number of vaccine manufacturers in the United States has declined and the cost of some basic childhood vaccines has increased. Concern therefore exists about the future availability of these essential biologics for national use. Furthermore, pricing decisions made within the private-sector pharmaceutical industry can have a major impact on vaccine use. This situation has stimulated an initiative toward increased public involvement in supplying vaccine to individuals for whom price is an issue as well as in oversight of the vaccine supply and of price negotiations with the industry.

ADMINISTRATION OF VACCINES

Health care workers administering vaccines must take the precautions necessary to

minimize the risk of spreading disease -- for example, hand washing between immunizations. They should be immunized against hepatitis B, measles, rubella, influenza, and varicella. Different vaccines should not be mixed in the same syringe unless such a practice is specifically endorsed by licensure. Disposable needles and syringes should be discarded in labeled, puncture-proof containers to prevent inadvertent needlestick injury or reuse.

The addition of new, individually injectable vaccines to the immunization schedule has heightened parental concerns about the administration of up to four injections at a single clinic visit. The development and use of combinations of vaccines are intended to mitigate these concerns. Even when multiple injections are required, providers must make every effort to administer all indicated vaccines at each visit.

Wherever effective primary health care systems ensure access to medical services for the majority and the population is educated about the need for and efficacy of vaccines, coverage rates for basic immunization are usually high, regardless of the route of vaccine administration or the number of doses necessary. However, without systematic attention to the completion of multiple-dose vaccine schedules, coverage rates for second, third, and booster doses may drop off significantly.

USE OF VACCINES

Recommendations for vaccine use in the United States are developed by several different groups. These recommendations are the result of a collaborative process among the recommending groups, the pharmaceutical industry, and the [FDA](#).

Vaccines recommended in 1999 for routine administration to infants, children, and adults are shown in [Table 122-5](#); vaccines recommended for special use are shown in [Table 122-6](#); and schedules for immunization of children and adults are shown in [Fig. 122-1](#) and [Table 122-7](#), respectively. The recommendations on route, site, and dosages for vaccination are derived from theoretical considerations, experimental trials, and clinical experience; deviation from these recommendations can result in inadequate protection. The administration of doses at intervals longer than those recommended does not diminish the ultimate protective response but merely delays it. It is not necessary to restart an interrupted series from the beginning or to add an extra dose. In contrast, giving vaccines at shorter-than-recommended intervals may result in poor responses.

RECORDING AND REPORTING REQUIREMENTS

Certain aspects of vaccine use are regulated by the National Childhood Vaccine Injury Act (NCVIA) of 1986 (modified in 1995). The act requires that all mandated childhood vaccinations be recorded by health care providers in the child's permanent medical record, including date of administration, manufacturer and lot number, and name of the provider administering the vaccine. State-based immunization information systems and registries are being developed to help public and private providers manage their immunization activities and particularly to address the problem of assessing immunization coverage when an individual's records are divided among multiple medical facilities.

Parents must be informed about the benefits and risks of immunization and should maintain an up-to-date immunization record on their children. Educational materials providing the required information (Vaccine Information Statements, VISs) are available from the American Academy of Pediatrics (AAP) or the Centers for Disease Control and Prevention (CDC).

VACCINES FOR ROUTINE USE

Infants and Children Recommended routine-use vaccines and schedules for their administration to infants and children are shown in [Table 122-5](#) and [Fig. 122-1](#), respectively. It is current practice for all children in the United States to receive [DTaP](#), poliovirus, [MMR](#), [Hib](#), [HBV](#), and varicella vaccines unless there are specific contraindications. Hepatitis A vaccine is currently recommended when there is a special risk of exposure to infection due to residence in communities with elevated rates of hepatitis A or travel to highly endemic countries.

Adults (See [Table 122-7](#)) All adults should be immune to diphtheria and tetanus. If not previously immunized, adults require a primary immunizing course of [Td](#). Many individuals remain immune to tetanus into adulthood because they have received tetanus toxoid rather than Td after injuries, but they are commonly at risk of diphtheria because of the decline in titer of diphtheria antitoxin and the lack of boosting against diphtheria. The development of acellular pertussis vaccines that appear to be safe in adults may lead to a recommendation for booster immunization of adults if clinical trials confirm safety and efficacy. Routine immunization against polio is not recommended for adults unless they are at particular risk of exposure (e.g., through travel to endemic regions, as discussed below) or are the parents or guardians of a child with an immunodeficiency disorder. Adults should be protected from measles, mumps, and rubella; they should be vaccinated unless they are known to have received vaccine on or after their first birthday or to have had physician-diagnosed disease. Rubella vaccine should be given to all women of childbearing age unless they have documentary proof of immunization after their first birthday or laboratory evidence of immunity. An unsupported history of rubella disease is unreliable and should not be accepted. Adults without a clear history of chickenpox should receive varicella vaccine. College students, particularly freshmen living in a dormitory, are at increased risk of meningococcal meningitis. They should be made aware of the polysaccharide vaccine for serogroups A, C, Y, and W-135 and should be offered the option of immunization.

Current recommendations also include influenza vaccine for routine annual administration to adults³⁵ years of age and to individuals with chronic illness at any age. Polyvalent pneumococcal polysaccharide vaccine is similarly recommended for the elderly or chronically ill. [HBV](#) vaccine is recommended for individuals at high risk of exposure, including health care workers exposed to potentially infected blood or blood products, homosexuals, injection drug users, individuals living and working in institutions for the mentally retarded, and household contacts of known carriers of hepatitis B surface antigen (HBsAg). A new recombinant outer-surface protein A (rOspA) is licensed for persons 15 to 70 years of age for Lyme disease (LYMErix, SmithKline Beecham Pharmaceuticals), with use based on individual risk (geography and risk of exposure to ticks).

Adverse Events Modern vaccines, while safe and effective, are associated with adverse effects that range from infrequent and very mild to rare and life-threatening. The decision to use a vaccine involves an assessment of the risks of disease, the benefits of vaccination, and the risks associated with vaccination. Because these factors may change over time, continued assessment is essential. [Table 122-8](#) lists valid and invalid contraindications to immunization and describes appropriate precautions in the use of specific vaccines. Antivaccine advocacy groups actively encourage avoidance of immunization because of their unproven belief that vaccines may cause certain disorders (for example, autism). This situation presents a challenge to the physician in educating parents about vaccine benefits and risks.

Vaccine components, including protective antigens, animal proteins introduced during vaccine production, and antibiotics or other preservatives or stabilizers, can cause allergic reactions in some recipients. These reactions may be local or systemic and may include urticaria and serious anaphylaxis. The most common extraneous allergen is egg protein introduced when vaccines such as those for measles, mumps, influenza, and yellow fever are prepared in embryonated eggs. Local or systemic reactions can result from too-frequent administration of vaccines such as [Td](#), diphtheria/tetanus (DT), or rabies; these reactions are probably due to antigen-antibody complexes. In addition, live-virus vaccines can interfere with tuberculin test responses. When a tuberculin skin test is indicated, it should be done either on the day of immunization or 6 weeks later. When influenza vaccine is given to children <13 years old, only "split-virus" preparations should be used since whole-virus vaccines are associated with higher rates of adverse reactions in young children. Cumulative exposure to mercury in thimerosal-preserved vaccines is a concern, and plans are under way to replace current vaccines with thimerosal-free products. In the interim, infants born to HBsAg-negative mothers should not receive the initial dose of HBV vaccine at birth.

All detected adverse events temporally related to vaccination are expected to be reported to both the local health department and the vaccine manufacturer. The [NCVIA](#) requires health care providers to report certain suspected adverse events following the receipt of a mandated vaccine to the [FDA](#)'s Vaccine Adverse Events Reporting System ([Table 122-9](#)). Although a temporal relationship does not establish cause and effect, this surveillance system remains the only mechanism for collecting the data needed to form conclusions and make decisions.

USE OF VACCINES IN SPECIAL CIRCUMSTANCES

Pregnancy Because of theoretical risk to the fetus and real risk of litigation to the practitioner, routine immunization of pregnant women is best avoided. However, wherever hygienic conditions during delivery cannot be guaranteed, it is essential to ensure that pregnant women are immune to tetanus: the transfer of maternal antitoxin is an important means of preventing neonatal tetanus, and pregnant women can safely receive tetanus as well as diphtheria toxoids. Although live-virus vaccines in general should be withheld during pregnancy, polio and yellow fever vaccines are exceptions and may be administered if the risk of exposure to disease is great. If indicated, some inactivated vaccines (e.g., [HBV](#), influenza, and pneumococcal vaccines) are safe for pregnant women. Known pregnancy is considered a contraindication to the receipt of

rubella, measles, mumps, and varicella vaccines. Although of theoretical concern, no cases of congenital rubella syndrome or abnormalities attributable to rubella vaccine virus have been observed in infants born to susceptible mothers who received rubella vaccine during pregnancy.

Breast Feeding Neither killed nor live vaccine affects the safety of breast feeding for either mother or infant. Breast-fed infants can be immunized on a normal schedule.

Occupational Exposure Immunization recommendations for most occupational groups remain to be developed. Specific practices are now mandated by the Occupational Safety and Health Administration for the immunization of health care workers against hepatitis B in the United States. Rubella is transmitted to and from health care workers in medical facilities, particularly in pediatric practice. Health care workers who might transmit rubella to pregnant patients should be immune to rubella; it is prudent to screen these employees for antibodies to rubella virus and to immunize susceptible individuals. Persons providing health care are also at greater risk from measles and varicella than the general public, and those who are likely to come into contact with measles- and varicella-infected patients should be immune. Persons employed in caring for patients with chronic diseases can transmit influenza; such workers should be vaccinated annually. Unfortunately, these recommendations often are not fully implemented, even in academic institutions.

HIV Infection and Other Immunocompromised States Limited studies in HIV-infected individuals have found no increase in the risk of adverse events from live or inactivated vaccines. However, immune responses may not be as vigorous in immunocompromised individuals as in those with a normal immune system. Persons known to be infected with HIV should be immunized with recommended vaccines in the same manner as individuals with a normal immune system and as early in the course of their disease as possible, before immune function becomes significantly impaired. Live attenuated [MMR](#) vaccine can be administered to this group, but [OPV](#) cannot ([Table 122-10](#)). [IPV](#) should be used when vaccination against polio is indicated. Household contacts of immunocompromised individuals should be immune to polio; when vaccinated, they should receive [IPV](#). In practice, it is not necessary to test for HIV before making decisions about the immunization of asymptomatic individuals from known HIV risk groups.

Live attenuated vaccines are normally contraindicated in immunocompromised patients, including those with congenital immunodeficiency syndromes and those receiving immunosuppressive therapy. Passive immunization with immunoglobulin preparations or antitoxins can be considered in individual cases, either as postexposure prophylaxis or as part of the treatment of established infection.

Postexposure Immunization For certain infections, active or passive immunization soon after exposure prevents or attenuates disease expression. Recommended postexposure immunization regimens are compiled in [Table 122-11](#). Measles immune globulin given within 6 days of exposure may prevent or modify infection, and measles vaccine given within the first few days after exposure may prevent symptomatic infection. Although clinical manifestations of rubella in pregnant women are minimized by postexposure passive immunization, this approach may not prevent maternal

viremia, fetal infection, and congenital rubella syndrome. Therefore, the administration of immune globulin is recommended only for women developing rubella during pregnancy who will not consider abortion under any circumstances. Tetanus immune globulin can be used in patients with tetanus. Survivors with no history of tetanus immunization should receive a primary series of toxoid injections since disease does not result in the development of protective levels of antitoxin. Administration of rabies immune globulin plus rabies vaccine in the immediate postexposure period is highly effective in preventing disease. Similarly, for persons who have not been actively immunized, the use of immune globulin within 2 weeks of exposure to hepatitis A is likely to prevent clinical illness. Good data indicate the efficacy of human hepatitis B immune globulin in preventing disease after exposure. While no high-titer preparation is available for postexposure protection against non-A, non-B hepatitis, standard human immune serum globulin is efficacious.

Simultaneous Administration of Multiple Vaccines The simultaneous administration of the most widely used live and inactivated vaccines has not resulted in impaired antibody responses or in increased rates of adverse reactions. Simultaneous administration of vaccines is advantageous in that it increases the probability that a child will ultimately be fully immunized; it is also useful in any age group when the potential exists for exposure to multiple infectious diseases during travel to endemic countries. However, combination [DTaP/Hib](#) vaccines should not be used for primary immunization of infants because the response to Hib is blunted and suboptimal. Live-virus vaccines not given together on the same day should generally be administered at least 30 days apart.

High doses of immune globulin may inhibit the efficacy of measles and rubella vaccines, and an interval of at least 3 months is recommended between the administration of immune globulin and that of [MMR](#) vaccine or its components. Postpartum vaccination of rubella-susceptible women should not be delayed because of the administration of anti-Rho(D) immune globulin or any other blood product during the last trimester or at delivery. Should administration of an immune globulin preparation become necessary after vaccination, it should be postponed, if possible, for at least 14 days to allow time for vaccine-virus replication and development of immunity. In general, there is little interaction of immune globulin with inactivated vaccines, and postexposure passive prophylaxis can be given together with [HBV](#) vaccine or tetanus toxoid, resulting in both immediate and long-lasting protection.

Travel The International Sanitary Regulations allow countries to impose requirements for yellow fever and killed cholera vaccines as a condition for admission, even though the latter is not an effective public health tool. Travelers should know whether these vaccines are required for entry into the countries on their itinerary to avoid being turned back or immunized on the spot. Infants, children, and adults should have all routine immunizations updated before traveling, with particular attention to polio, measles, and [DTP/DTaP](#) or [Td](#) vaccines. The use of hepatitis A vaccine may be advisable for travelers to some locales. Special-use vaccines ([Table 122-6](#)), including rabies, meningococcal polysaccharide, typhoid (oral live or Vi polysaccharide), Japanese encephalitis, and plague vaccines, should be considered for those individuals who expect to go beyond the usual tourist routes or to spend extended periods in rural areas in disease-endemic regions. Most U.S. cities have at least one travel clinic that

maintains up-to-date epidemiologic information and can provide the appropriate vaccines.

DELIVERY OF VACCINES

Over the past 25 years, considerable progress has been made to ensure that every child in the United States is fully immunized by the time of school entry. All 50 states now require immunization for school entry, and most have laws addressing attendance at preschools and day-care centers. The impact of immunization and of other improvements in the health care provided to the American population on the incidence of vaccine-preventable illness is shown in [Table 122-1](#). Nonetheless, many children are not fully immunized, especially in poor and underserved communities. The failure to vaccinate preschool children was largely responsible for the resurgence of measles between 1989 and 1991, with >55,000 cases and >130 measles-related deaths. Outbreaks of pertussis, mumps, and congenital rubella syndrome have occurred for the same reason: low immunization rates among preschool children.

ACCESS TO IMMUNIZATION

Four major barriers to infant and childhood immunization have been identified within the health care system: (1) low public awareness and lack of public demand for immunization, (2) inadequate access to immunization services, (3) missed opportunities to administer vaccines, and (4) inadequate resources for public health and preventive programs. These problems are sources of public concern, and their solution is a priority for national health policy in the United States. In response, the Children's Immunization Initiative was begun in 1990. At the national level, this program includes outreach and educational campaigns to promote parental awareness of the value of vaccination and to encourage health care providers to use every opportunity to vaccinate the children in their care. At the state and local levels, community and business groups, religious and service groups, schools, and the media have joined together in community-based networks. A National Immunization Week each April has been established to focus attention on the vaccination needs of infants and children. To improve the quality and quantity of vaccination services, expanded immunization-clinic hours and computerization of immunization records are being implemented as well.

There has been only modest progress towards the goals for adult immunization in the United States. Adult-immunization goals are important: As many as 60,000 adults die each year of vaccine-preventable diseases for which effective vaccines are not being optimally used. Most persons ³65 years of age do not receive influenza vaccine each year, and even fewer have ever received pneumococcal vaccine. Health care providers more often miss vaccination opportunities with adults than with infants and children. From 60 to 90% of adults hospitalized for or dying of influenza-associated respiratory disease have received medical care during the previous year and could have been immunized at that time. Medicare reimbursement for excess hospitalization during influenza epidemics ranges from \$750 million to \$1 billion. Additional efforts are required to ensure that adults receive pneumococcal, [Td](#), and [HBV](#) vaccines as well.

A special setting for adult immunization is the administration of certain vaccines to pregnant women to enhance passive immunity in their offspring (e.g., tetanus toxoid). In

most cases the mother herself derives important benefits as well. Immunization of the mother should be undertaken at least 6 weeks before delivery to allow for efficient transplacental transfer of antibody to the fetus.

STANDARDS FOR IMMUNIZATION PRACTICES

National standards of immunization for adult and pediatric practice have been established to define common policies and practices for public health clinics and in physicians' private offices ([Table 122-12](#)). These guidelines highlight the need to distinguish between valid contraindications and conditions that are often considered but are not in fact contraindications ([Table 122-8](#)). Among the valid contraindications applicable to all vaccines are a history of anaphylaxis or other serious allergic reactions to a vaccine or vaccine component and the presence of a moderate or severe illness, with or without fever. Infants who develop encephalopathy within 72 h of a dose of [DTP](#) or [DTaP](#) should not receive further doses; those who develop a "precaution" ([Table 122-8](#)) should not normally receive further doses. Because of theoretical risks to the fetus, pregnant women should not receive [MMR](#) or varicella vaccine. Diarrhea, minor respiratory illness with or without fever, mild to moderate local reactions to a previous dose of vaccine, the concurrent or recent use of antimicrobial agents, mild to moderate malnutrition, or the convalescent phase of an acute illness are not valid contraindications to routine immunization. Failure to vaccinate children because of these conditions is increasingly viewed as a missed opportunity for immunization.

BIOTERRORISM

The end of the twentieth century witnessed a rise in the risk of bioterrorism. While smallpox has been eradicated, known stocks of smallpox virus still exist in the United States and Russia, and unknown stocks probably exist in other countries considered likely to engage in terrorism. Global supplies of smallpox vaccine for use in case of deliberate release of smallpox virus are inadequate, and millions of people are likely to become infected and die in the event of such a release. Steps are only now being taken to ensure sufficient stockpiles of vaccine for this eventuality, and it will be several more years before these reach a critical size.

THE NATIONAL VACCINE INJURY COMPENSATION PROGRAM

The use of mandated vaccines benefits society as a whole by reducing morbidity and the cost of care for preventable diseases and by reducing childhood mortality. Thus, in the United States, society has assumed the obligation to care for those injured by the administration of mandated vaccines. The [NCVIA](#) of 1986 (modified in 1995) is the instrument in use to ensure fairness to injured persons as well as protection for federal, state, and local immunization programs; private immunization providers; and vaccine manufacturers. The act was designed to implement two vital public policies: (1) to provide prompt and fair compensation to the families of children who have died or have been injured as a result of routine mandated immunization; and (2) to reduce the adverse impact of the tort system on vaccine supply, cost, and innovation/development. The success of immunization programs in the United States depends upon the continued viability of the National Vaccine Injury Compensation Program.

CONTROL OF VACCINE-PREVENTABLE DISEASE

A continuing task of public health practice is to maintain individual and herd immunity. The job is not over once a population is fully vaccinated; rather, it is imperative to immunize each subsequent generation as long as the threat of the disease persists. Ongoing surveillance and prompt reporting of disease to local or state health departments are essential to this goal, ensuring a continuing awareness of the possibility of vaccine-preventable illness. Nearly all vaccine-preventable diseases are now notifiable, and individual case data are routinely forwarded to the [CDC](#). These data are used to detect outbreaks or other unusual events that require investigation and to evaluate prevention and control policies, practices, and strategies.

As a direct consequence of successes in immunization, vaccine-preventable diseases have become less visible; ironically, this situation may foster complacency among parents and health care providers about routine immunization of children. Even among the affluent and educated, immunization levels may be low, reflecting a misunderstanding of the continuing threat of disease with which parents and health care providers have limited experience or perhaps an unjustifiably greater fear of adverse reactions to vaccination than of the potential for illness and death due to vaccine-preventable diseases. Health care workers play an essential role in influencing the attitudes of patients regarding appropriate immunization; therefore, it is essential that these professionals continually update their own knowledge about vaccines and about the epidemiology of vaccine-preventable illnesses.

RESEARCH ON VACCINES AND IMMUNIZATION

The potential to eradicate selected diseases and to build sustainable immunization programs that reach every child is not being fulfilled with existing vaccines and delivery technology. New vaccines or new formulations that will not only improve protective responses but also simplify the immunization schedule are needed. The ideal would be vaccines that can be administered orally early in life, that provide lifelong protection against multiple infections, that can be given as one or only a few doses, and that are less reactive and more heat stable than current vaccines. To attain these ambitious goals may take decades, but progress is already being made in the development of new combinations of current vaccines to facilitate complete immunization. The results will be applicable to immunization programs in both developed and developing countries.

REEMERGENCE OF CONTROLLED DISEASE AND EMERGENCE OF NEW DISEASE

The emergence of new pathogens is fostered by the genetic potential of microbes to evolve as well as by rapid changes in human demographics and behavior and in a global ecology that creates new or more favorable hosts. Proof of the need for continuing vaccine research is found in the emergence of new infectious diseases such as HIV infection, Lyme borreliosis, hantavirus pulmonary syndrome, and hepatitis C; the appearance of a new epidemic cholera strain (serotype O139 Bengal) that exhibits no cross-immunity with the traditional O1 serotype; and the increase in global incidence and in drug resistance of familiar diseases that were once considered under control, such as tuberculosis and malaria. In addition, some common illnesses without a

previously known etiology, such as peptic ulcer disease and cervical and nasopharyngeal cancer, have now been epidemiologically linked to specific infectious agents and have thus become potentially vaccine-preventable conditions.

DEVELOPMENT OF NEW VACCINES

For many serious or even life-threatening infectious diseases, no effective vaccines are available. Although many new vaccines are undergoing human trials, the task of developing vaccines is proving very complex. Priorities for the United States currently include research on the following vaccines: HIV, pneumococcus (conjugate), group B *Streptococcus*, respiratory syncytial virus, rotavirus, *Mycobacterium tuberculosis*, herpes simplex virus, influenza A and B viruses, and hepatitis C virus. Also of high priority are vaccines for two virus-associated tumors: cervical cancer (human papillomavirus) and nasopharyngeal cancer (Epstein-Barr virus).

INTERNATIONAL CONSIDERATIONS

Since the establishment of the World Health Organization's Expanded Programme on Immunization in 1981, levels of coverage for the recommended basic children's vaccines (bacille Calmette-Guerin, polio, [DTP/DTaP](#), measles, and [HBV](#)) have risen from 5% to ~80% worldwide. Each year, at least 2.7 million deaths from measles, neonatal tetanus, and pertussis and 200,000 cases of paralysis due to polio are prevented by immunization. Despite the successes of this program, many vaccine-preventable diseases remain prevalent in the developing world. Measles, for example, continues to kill an estimated 1.5 million children each year, and cases of diphtheria, whooping cough, polio, and neonatal tetanus still occur at unacceptably high rates. It is estimated that between 20 and 35% of all deaths of children under the age of 5 years are still associated with vaccine-preventable diseases.

In addition to the antigens included in the Expanded Programme for routine use in the developing world, others ([Hib](#), Japanese B encephalitis, yellow fever, group A meningococcus, mumps, and rubella) are used regionally, depending on disease epidemiology and resources. Polio has been targeted for eradication; this disease has already been eliminated in the Americas and Europe and is close to elimination in the Western Pacific.

Because infectious diseases know no geographic or political boundaries, uncontrolled disease anywhere in the world poses a threat to the United States. Vaccines offer the opportunity to control and even eradicate some diseases, and eradication means that vaccines are no longer needed. Vaccines represent the best hope for stopping the pandemic of HIV infection throughout the world. The experience with smallpox has shown that the eradication of disease is a remarkably good economic investment. The entire sum that the United States spent for the global smallpox eradication campaign has been recouped, in 1968 dollars, every 2.5 months since 1971. The global eradication of polio will save the United States over \$300 million a year in vaccine and associated delivery costs and will save over \$1.5 billion a year worldwide.

Issues of cost, liability, risk, and profitability limit the interest of the pharmaceutical industry in the development of vaccines (e.g., for malaria) that will be used primarily in

poor developing countries. Efforts have been made to create partnerships in public research and privately funded development. Activities of established international organizations (such as the World Health Organization) and some new organizations (such as the Global Alliance for Vaccines and Immunization, the International AIDS Vaccine Initiative, and the Bill and Melinda Gates Foundation) have helped to move the process forward with strategy development and implementation or new funding. New international schemes are being considered by wealthy industrial nations; for example, advance-purchase schemes are being proposed in which the purchase of effective vaccines is guaranteed, ensuring the profitability that the marketplace has provided for industry in the wealthy countries. The effectiveness of such approaches remains to be seen, but they offer much-needed hope for at-risk populations around the world.

SOURCES OF INFORMATION ON IMMUNIZATION

- Official vaccine package circulars and Vaccine Administration Statements from the Centers for Disease Control and Prevention
- Report of the Committee on Infectious Diseases of the American Academy of Pediatrics ("Red Book")
- Recommendations of the Advisory Committee on Immunization Practices, Centers for Disease Control and Prevention
- Guide for Adult Immunization, American College of Physicians
- Health Information for International Travel (published yearly) and Advisory Memoranda on Travel (published periodically), Centers for Disease Control and Prevention
- Control of Communicable Diseases in Man, American Public Health Association
- Technical Bulletin of the College of Obstetrics and Gynecology
- National Network for Immunization Information, Infectious Diseases Society of America/Pediatric Infectious Diseases Society/American Academy of Pediatrics/American Nurses Association

(Bibliography omitted in Palm version)

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123. HEALTH ADVICE FOR INTERNATIONAL TRAVEL - J.S. Keystone, P.E. Kozarsky

In 1991, the World Health Organization estimated that more than 30 million persons traveled from industrialized countries to the developing world. Studies show that between 50 and 75% of short-term travelers to the tropics or subtropics report some health impairment. Most of these health problems are minor, with only 5% requiring medical attention and fewer than 1% requiring hospitalization.

Although infectious agents contribute substantially to morbidity in travelers, these pathogens account for only about 1% of deaths among this population. Cardiovascular disease and injuries are the most frequent causes of death among travelers from the United States, accounting for 49 and 22% of deaths, respectively. Age-specific rates of mortality due to cardiovascular disease are similar to those among nontravelers. In contrast, rates of death due to injury -- the majority from motor vehicle, drowning, or aircraft accidents -- are several times higher among travelers. [Figure 123-1](#) summarizes the monthly incidence of health problems during travel in developing countries.

GENERAL ADVICE

Staying healthy during travel requires familiarity with the health risks that may be encountered at a given destination. However, health maintenance recommendations are based not only on the traveler's destination but also on risk assessment, which is determined by health status, specific itinerary, and lifestyle during travel. Detailed information regarding country-specific risks and recommendations may be obtained from many sources, including those listed under "Sources of Information on Travel Medicine."

IMMUNIZATIONS FOR TRAVEL

Immunizations for travel are generally divided into three categories: routine (childhood/adult boosters that are necessary regardless of travel), required (immunizations that are mandated by international regulations for entry into certain areas or for border crossings), and recommended (immunizations that are desirable because they confer protection against a variety of illnesses for which travel increases the risk). Vaccines commonly given to travelers are listed in [Table 123-1](#).

Routine Immunizations

Diphtheria, Tetanus, and Polio Diphtheria continues to be a problem worldwide, with large outbreaks in the independent states formerly encompassed by the Soviet Union. Serosurveys show that tetanus antitoxin is lacking in many North Americans, especially those over the age of 50. Although the risk of polio to the international traveler is extremely low and although polio has been eradicated from the western hemisphere, studies in the United States have found varying levels of immunity in the general population; data indicate that 12% of adult American travelers are unprotected against at least one poliovirus serogroup. Foreign travel offers an ideal opportunity to have these immunizations updated.

Measles Measles (rubeola) continues to be a major cause of morbidity and mortality in the developing world ([Chap. 194](#)). Several outbreaks of measles in the United States have been linked to imported cases. The group at highest risk consists of persons born after 1956 and vaccinated before 1980, in many of whom primary vaccination failed. Travelers in this group should be reimmunized.

Influenza Influenza occurs year-round in the tropics and during the summer months in the southern hemisphere (which coincide with the winter months in the northern hemisphere). Vaccination should be considered for all travelers to these regions, particularly those who are elderly or chronically ill. The largest outbreak of travel-related influenza occurred in the summer of 1998 in Alaska and the Northwest Territories of Canada among cruise ship passengers and staff ([Chap. 190](#)).

Pneumococcal Infection Pneumococcal vaccine should be administered routinely to persons at high risk of serious pneumococcal infection, such as individuals with chronic heart, lung, or renal disease and those who have been splenectomized or have sickle cell disease.

Required Immunizations

Yellow Fever Documentation of yellow fever vaccination may be required for entry into countries of sub-Saharan Africa and equatorial South America, where the disease is endemic or epidemic, or into countries that are at risk of having the infection introduced. This vaccine is given only by state-authorized yellow fever centers, and its administration must be documented on an official International Certificate of Vaccination. The incidence of yellow fever among travelers is extremely low, probably because the vaccine is highly efficacious.

Cholera According to the World Health Organization, cholera vaccine should no longer be required for entry into any country. (However, see *recommended* use below.)

Meningococcal Meningitis Meningococcal vaccination is required for entry into Saudi Arabia during the Hajj. (For information on *recommended* use, see below.)

Recommended Immunizations

Hepatitis A and B Hepatitis A is the most frequent vaccine-preventable infection of travelers; the incidence of symptomatic infection during a 1-month stay in a developing country ranges from 3 to 6 cases per 1000. The risk is six times greater for those who stray from the usual tourist routes. The mortality rate for hepatitis A increases with age, reaching almost 3% among symptomatic individuals over age 50. Several vaccines are available in North America, each of which has an efficacy rate of >95%. The monthly incidence of hepatitis B infection, both symptomatic and asymptomatic, is 80 to 240 cases per 100,000. For reasons that are not entirely clear, long-stay overseas workers are at considerable risk for hepatitis B infection. In the near future, a combined hepatitis A and B vaccine should become available in the United States.

Typhoid Fever The attack rate for typhoid fever is 1 case per 30,000 per month of travel to the developing world ([Chap. 156](#)). However, the rates in India, Senegal, and North

Africa are tenfold higher, and, within these areas, rates are especially high among travelers to relatively remote destinations and among persons who are returning to their homelands to stay with relatives or friends. Each of the three available vaccines -- one oral and two injectable -- has an efficacy rate of approximately 70%.

Meningococcal Meningitis Although the risk of meningococcal disease among travelers has not been quantified, it is likely to be higher among those who live with poor indigenous populations in overcrowded conditions. The vaccine is recommended for persons traveling to sub-Saharan Africa during the dry season or to areas of the world where there are epidemics. Meningococcal vaccine, which protects against serogroups A/C/Y/W-135, has an efficacy rate of >90%.

Japanese Encephalitis The risk of Japanese encephalitis, an infection transmitted by mosquitoes in rural Asia and Southeast Asia, is approximately 1 case per 5000 per month of stay in an endemic area. Most symptomatic infections in U.S. residents have involved military personnel or their families. The vaccine efficacy rate is >80%. Serious allergic reactions sometimes occur; these reactions may be delayed in onset, developing up to 10 days after immunization. The vaccine is recommended for persons staying >1 month in endemic areas.

Cholera The risk of cholera is extremely low, with approximately 1 case per 500,000 journeys to endemic areas. Cholera vaccine is rarely recommended but should be considered for aid workers in refugee camps or in disaster/war-torn areas. The injectable vaccine available in the United States is only 30 to 50% effective, and its protective effect persists for only a short period. A more effective oral cholera vaccine is available in other countries.

Rabies Many cases of rabies have been reported in travelers, but there are no data on the risk of infection. Domestic animals are the major transmitters of rabies in developing countries ([Chap. 197](#)). Countries where canine rabies is highly endemic include Mexico, the Philippines, Sri Lanka, India, Thailand, and Vietnam. Each of the three vaccines available in the United States provides >90% protection. Rabies vaccine is recommended for long-stay travelers, particularly children, and persons who may be occupationally exposed in endemic areas.

PREVENTION OF MALARIA AND OTHER INSECT-BORNE DISEASES

It is estimated that more than 30,000 American and European travelers develop malaria each year ([Chap. 214](#)). Nevertheless, several studies indicate that fewer than 50% of U.S. travelers to malaria-endemic regions adhere to basic recommendations for malaria prevention.

The risk of malaria is highest in sub-Saharan Africa and Oceania (1:50 to 1:1000) and during the past decade has increased by more than fivefold for travelers to Kenya. The risk is intermediate (1:1000 to 1:12,000) for travelers to Haiti and the Indian subcontinent and is low (<1:50,000) for travelers to Asia and to Central and South America. Of the 1000 cases of malaria reported annually in the United States, 90% of those due to *Plasmodium falciparum* occur in travelers returning or immigrating from Africa and Oceania. With the worldwide increase in chloroquine- and multidrug-resistant

falciparum malaria, decisions about chemoprophylaxis have become more difficult. Moreover, the spread of malaria due to primaquine- and chloroquine-resistant strains of *P. vivax* has added to the complexity of treatment. The case-fatality rate of falciparum malaria in the United States is 4%; however, in only one-third of patients who die is the diagnosis of malaria considered before death.

Compliance with chemoprophylaxis regimens and use of personal protection measures are keys to the prevention of malaria. Personal protection measures are aimed at preventing mosquito bites, especially between dusk and dawn, and include the use of DEET-containing insect repellents, permethrin-impregnated bed nets and clothing, screened sleeping accommodations, and protective clothing. These practices also help prevent other insect-transmitted illnesses, such as dengue fever. Over the past decade, the incidence of dengue has increased, particularly in the Caribbean region, Latin America, and Southeast Asia. Since dengue fever is transmitted by a day-biting, urban-dwelling mosquito, attention to personal protection measures is required around the clock in most tropical areas.

The decision about whether or not to use malaria prophylaxis is based on the traveler's destination; the particular medication prescribed is determined not only by destination but also by the traveler's preference and medical history. [Table 123-2](#) lists the currently recommended drugs of choice for prophylaxis of malaria by destination. Alternative medications for prophylaxis that are in use by some physicians include primaquine, atovaquone/proguanil, and chloroquine/proguanil.

PREVENTION OF GASTROINTESTINAL ILLNESS

Diarrhea is the leading cause of illness in travelers ([Chap. 131](#)) and is usually a short-lived, self-limited condition; however, 40% of affected individuals must alter their scheduled activities, and another 20% are confined to bed. The most important determinant of risk is the traveler's destination. Incidence rates per 2-week stay have been reported to be as low as 8% in industrialized countries and as high as 55% in parts of Africa, Central and South America, and Southeast Asia. Infants and young adults are at particularly high risk. The incidence of diarrhea is proportional to the number of dietary indiscretions. Studies of U.S. students in Mexico showed that eating meals in restaurants and cafeterias or consuming food from street vendors was associated with increased risk.

The most frequently identified pathogen causing traveler's diarrhea is toxigenic *Escherichia coli*, although in some parts of the world (notably northern Africa and Southeast Asia) *Campylobacter* infections appear to predominate. Other common causative organisms include *Salmonella*, *Shigella*, rotavirus, and the Norwalk agent. Except for giardiasis, parasitic infections are uncommon causes of traveler's diarrhea. A growing problem for travelers is the development of antibiotic resistance in many bacterial pathogens, including strains of *Campylobacter* and *Salmonella* resistant to quinolones and strains of *E. coli*, *Shigella*, and *Salmonella* resistant to trimethoprim-sulfamethoxazole.

Although the mainstay of prevention of traveler's diarrhea involves food and water precautions, the literature has repeatedly documented dietary indiscretions in 98% of

travelers within the first 48 h after arrival at their destination. The old maxim "Boil it, cook it, peel it, or forget it!" is easy to remember but appears to be difficult to adhere to. General food and water precautions include eating foods piping hot; avoiding foods that are raw, poorly cooked, or sold by street vendors; and drinking boiled or commercially bottled beverages, particularly those that are carbonated. Heating kills diarrhea-causing organisms, whereas freezing does not; therefore, ice cubes made from unpurified water should be avoided.

As traveler's diarrhea can occur despite rigorous food and water precautions, travelers should carry medications for self-treatment. For mild to moderate diarrhea, loperamide and fluid replacement may be sufficient. An antibiotic is useful in reducing the frequency of bowel movements and duration of illness in moderate to severe diarrhea. The standard regimen is a 3-day course of a quinolone taken twice daily or (in the case of some of the newer agents) once daily. However, studies have shown that a single large dose of a quinolone may be as effective as the 3-day regimen, particularly if infection with a multidrug-resistant organism is not suspected. For diarrhea acquired in areas such as Thailand, where >70% of *Campylobacter* infections are quinolone resistant, azithromycin may be a better choice.

Prophylaxis of traveler's diarrhea with bismuth subsalicylate is widely used but only about 60% effective. For certain individuals (e.g., athletes who must be in peak physical condition to perform, persons with a repeated history of traveler's diarrhea, and some persons with chronic diseases), a single daily dose of a quinolone antibiotic during travel of <1 month's duration is highly effective.

PREVENTION OF OTHER TRAVEL-RELATED PROBLEMS

Travelers are at high risk for *sexually transmitted diseases*. Surveys have shown that large numbers engage in casual sex, and there is a reluctance to use condoms consistently. An increasing number of travelers are being diagnosed with *schistosomiasis*. Travelers should be cautioned to avoid bathing, swimming, or wading in freshwater lakes, streams, or rivers in the parts of tropical South America, the Caribbean, Africa, and Southeast Asia where this infection can be acquired. Travelers are cautioned to avoid walking barefoot because of the risk of *hookworm* and *strongyloidiasis* and, at night, *snakebites*. Prevention of *travel-associated injury* depends mostly on common-sense precautions. Riding on motorcycles and overcrowded public vehicles is not recommended; in particular, individuals should not travel by road after dark in rural areas. In addition to its association with motor vehicle accidents, excessive alcohol use has been a significant factor in drownings, assaults, and injuries.

THE TRAVELER'S MEDICAL KIT

A traveler's medical kit is strongly advisable, particularly for long-stay travelers. The contents may vary widely, depending on the itinerary, duration of stay, style of travel, and local medical facilities. While many medications are available abroad, often over the counter, directions for their use may be nonexistent or in a foreign language, and, more important, a product may be outdated or counterfeit. Therefore, if possible, a complete supply of medications should accompany the traveler. In the kit, the short-term traveler should consider carrying an analgesic, an antidiarrheal agent, antihistamines, a laxative,

oral rehydration salts, sunscreen with a skin protection factor of at least 30, insect repellents (DEET) for the skin, an insecticide for clothing (permethrin), and (if necessary) an antimalarial. To these medications the long-stay traveler might add a broad-spectrum general-purpose antibiotic, an antibacterial eye and skin ointment, and a topical antifungal cream. Regardless of the duration of travel, a first-aid kit containing items such as scissors, tweezers, and bandages should be considered.

TRAVEL AND SPECIAL HOSTS

PREGNANCY AND TRAVEL

A woman's medical history and itinerary, the quality of medical care at her destinations, and her degree of flexibility determine whether travel is wise during pregnancy. According to the American College of Obstetrics and Gynecology, the safest part of pregnancy in which to travel is the second trimester (between 18 and 24 weeks), when there is the least danger of spontaneous abortion or premature labor. Some obstetricians prefer that women stay within a few hundred miles of home after the 28th week of pregnancy in case problems arise; in general, however, healthy women may be advised that it is acceptable to travel.

Despite this general recommendation, there are some relative contraindications to international travel during pregnancy, including certain obstetric risk factors: a history of miscarriage, premature labor, incompetent cervix, or toxemia. General medical problems such as diabetes, heart failure, severe anemia, or a history of thromboembolic disease should also prompt the pregnant woman to postpone her travels. Finally, regions in which the pregnant woman and her fetus may be at excessive risk (e.g., those at high altitudes and those where live-virus vaccines are required or where multidrug-resistant malaria is endemic) are not ideal destinations during any trimester.

Malaria Malaria during pregnancy carries a significant risk of morbidity and death. Levels of parasitemia are highest and failure to clear the parasites after chloroquine treatment are most frequent among primigravidae. Severe disease, with complications such as cerebral malaria, massive hemolysis, and renal failure, is especially likely in pregnancy. Fetal sequelae include spontaneous abortion, stillbirth, preterm delivery, and congenital infection.

Traveler's Diarrhea Because dehydration due to traveler's diarrhea can lead to inadequate placental blood flow, pregnant travelers must be extremely cautious regarding their food and beverage intake. The exclusive consumption of bottled (carbonated) or boiled drinks without ice, the eating of well-cooked meats and pasteurized dairy products, and the avoidance of pre-prepared salad items should help protect against traveler's diarrhea due to the usual causes as well as against infections such as toxoplasmosis, hepatitis E, and listeriosis, which can have serious sequelae in pregnancy.

The mainstay of therapy for traveler's diarrhea is rehydration. Kaolin-pectin combinations and loperamide may be used if necessary, but many of the usual antibiotics (e.g., quinolones) are contraindicated during pregnancy. Ampicillin alone or with clavulanic acid may be used, but many strains of *E. coli* and other organisms

implicated in traveler's diarrhea are resistant. Azithromycin or an oral third-generation cephalosporin may be the best option.

Because of the major problems encountered when infants are given local foods and beverages, women are strongly encouraged to breast-feed when traveling with a neonate. A nursing mother with traveler's diarrhea should not stop breast-feeding but should increase her fluid intake.

Air Travel and High-Altitude Destinations Commercial air travel is not a risk to the healthy pregnant woman or to the fetus. Fetal oxygenation is not adversely affected by the decreased cabin pressures because of the fetal hemoglobin dissociation curve; the higher radiation levels reported at altitudes >10,500 m (35,000 ft) should pose no problem to the healthy pregnant traveler. Since each airline has a policy regarding pregnancy and flying, it is best to check with the specific carrier when booking reservations. Domestic air travel is usually permitted until the 36th week, whereas international air travel is generally curtailed after the 32nd week.

There are no known risks for pregnant women who travel to high-altitude destinations and stay for short periods. However, there are likewise no data on the safety of pregnant women at altitudes >4500 m (15,000 ft). Because of the harsh conditions usually associated with such trips, they are generally contraindicated for other reasons.

THE HIV-INFECTED TRAVELER

The traveler infected with HIV is at special risk of serious infections due to a number of pathogens that may be more prevalent at travel destinations than at home. However, the degree of risk depends primarily on the state of the immune system at the time of travel. For persons whose CD4⁺ cell counts are normal or >500/uL, no data suggest a greater risk during travel than for persons without HIV infection. Individuals with AIDS (CD4⁺ counts <200/uL) and others who are symptomatic need special counseling and should visit a travel medicine practitioner before departure, especially when traveling to the developing world.

Several countries now routinely deny entry to HIV-positive individuals, even though no data show that these restrictions decrease rates of transmission of the virus. In general, HIV testing is required of those individuals who wish to stay abroad longer than 3 months or who intend to work or study abroad. Some countries will accept an HIV serologic test done within 6 months of departure, whereas others will not accept a blood test done at any time in the traveler's home country. In addition, border officials often have the authority to make inquiries of individuals entering a country and to check the medications they are carrying. If a drug such as zidovudine (AZT) is identified, the person may be barred from entering the country. Information on testing requirements for specific countries is available from consular offices but is subject to frequent change.

Health insurance policies should be checked to make sure they are valid for care in other countries. The HIV-positive traveler should strongly consider obtaining trip cancellation insurance and evacuation insurance in case of illness. It is ideal to have the name of a physician at the travel destination who is familiar with the treatment of patients with AIDS, as the clinical findings associated with infection may be atypical in a

patient with AIDS, and several infections may exist simultaneously. The traveler should be encouraged to visit the physician promptly if problems arise.

Immunizations All of the HIV-infected traveler's routine immunizations should be up to date ([Chap. 122](#)). The response to immunization may be impaired at CD4+ cell counts of <200/uL (and in some cases at even higher counts). However, when the risk of illness is high or the sequelae of illness are serious, immunization is recommended. In certain circumstances, it may be prudent to check the adequacy of the serum antibody response before departure (e.g., yellow fever neutralization inhibition if exposure is unavoidable).

Because of the increased risk of infections due to *Streptococcus pneumoniae* and other bacterial pathogens that cause pneumonia following influenza, pneumococcal polysaccharide and influenza vaccines should be administered. The estimated rates of response to influenza vaccine are >80% among persons with asymptomatic HIV infection and <50% among those with AIDS.

In general, live attenuated vaccines are contraindicated for persons with immune dysfunction. Live oral polio vaccine should not be given to HIV-infected patients or to members of their households. Instead, inactivated polio vaccine (eIPV) should be used; most HIV-infected individuals without AIDS will develop protective antibody levels in response to this vaccine.

Because measles (rubeola) can be a severe and lethal infection in HIV-positive patients, the measles vaccine (or the combination measles-mumps-rubella vaccine) should be given to these individuals. Although this is a live vaccine, there have been no reports of serious complications in this population. Between 18 and 58% of symptomatic HIV-infected vaccinees develop adequate antibody titers, and between 50 and 100% of those who are infected but asymptomatic seroconvert.

The decision of whether or not to administer any of the special vaccines to an HIV-infected traveler should be based on the individual's risk. Inactivated vaccines can be administered without concern for safety but with concern about adequate protection. For example, data suggest that HIV-infected persons do not have as strong an antibody response to the meningococcal meningitis vaccine as do uninfected persons. Moreover, few data are available on the efficacy of many of the other vaccines (e.g., those for hepatitis A, typhoid, and cholera).

It is recommended that the live yellow fever vaccine not be given to HIV-infected travelers. Nevertheless, when inadvertently administered to HIV-positive military personnel, this vaccine elicited no adverse reactions. Therefore, if the traveler's CD4+ count is >200/uL and travel in an endemic area is absolutely necessary, the vaccine can probably be administered safely. HIV-infected persons whose CD4+ count is <200/uL should be discouraged from traveling to endemic regions. If the traveler is passing through or traveling to an area where the vaccine is required but the disease risk is low, a physician's waiver should be issued. Bacille Calmette-Guerin vaccine should not be given because of reports of disseminated infection in HIV-infected persons.

A transient (days to weeks) burst of viremia has been demonstrated in HIV-infected

individuals following immunization with vaccines for such diseases as influenza, pneumococcal infection, and tetanus ([Chap. 309](#)). However, at this point, there is no evidence that this transient increase in viremia is detrimental over time. Furthermore, it is likely that immune activation associated with infection with the live organisms in question would result in increases in viremia of greater magnitude and duration than those associated with vaccination. Therefore, the vaccination recommendations discussed above need not be modified at this time.

Gastrointestinal Illness Decreased levels of gastric acid, abnormal gastrointestinal mucosal immunity, other complications of HIV infection, and medications taken by HIV-infected patients make traveler's diarrhea especially problematic in these individuals. Traveler's diarrhea is likely to occur more frequently, be more severe, and be more difficult to treat in association with HIV infection. *Salmonella*, *Shigella*, and *Campylobacter* infections are also more protracted and more often accompanied by bacteremia in HIV-infected persons.

Cryptosporidium ([Chap. 218](#)), a common cause of diarrhea in tropical countries, produces severe chronic diarrhea and cholecystitis with increased mortality among patients with AIDS. *Isospora belli* causes infections at high rates among AIDS patients in the developing world; this infection is associated with malabsorption, weight loss, and relapses after treatment. Persistent diarrhea due to microsporidiosis has been reported.

Because of these potential problems, the HIV-infected traveler must be careful to consume only appropriately prepared foods and beverages. In addition, this group of individuals may benefit from prophylaxis for traveler's diarrhea, using bismuth subsalicylate or a daily antibiotic (ideally a quinolone derivative) for short-term travel to the developing world. If the traveler is already taking a sulfonamide preparation for prophylaxis of *Pneumocystis* pneumonia, a regimen of self-treatment with a quinolone would be appropriate.

Other Travel-Related Infections Data are lacking on the severity of vector-borne diseases in HIV-infected individuals. Malaria is especially severe in asplenic and certain immunocompromised hosts, although increased severity has not been demonstrated in AIDS. *Babesia* infection is known to cause serious illness and to recur in HIV-infected patients; this tick-transmitted illness occurs in parts of the United States but is not known to be a widespread problem.

Visceral leishmaniasis ([Chap. 215](#)) has been reported in numerous HIV-infected travelers. Because the usual signs -- splenomegaly and hyperglobulinemia -- are nonspecific and may even be lacking, the diagnosis is difficult to make. In addition, serologic results are often negative. This infection is difficult to treat, and its associated mortality is high. Even short-term travelers to southern Europe have developed the illness; thus, the avoidance of sandfly bites is critical.

Certain respiratory illnesses, such as histoplasmosis and coccidioidomycosis, cause greater morbidity and mortality among patients with AIDS than in the general population. Though tuberculosis is common among HIV-infected persons (especially in developing countries), the acquisition of this infection by the short-term traveler is not a major concern. The possibility of acquiring *Legionella* infections from spas should be

considered, although no data confirm an increase in the severity of such infections in AIDS.

Finally, the HIV-infected traveler should always be cautioned about safe sexual practices, which may help prevent both the transmission of HIV to others and the acquisition by the traveler of other sexually transmitted diseases that may be drug resistant or may result in serious sequelae (e.g., syphilis).

Medications Adverse events due to medications and drug interactions are common and raise complex issues for HIV-infected persons. In addition, rates of cutaneous reaction are unusually high among patients with AIDS. Physicians advising these travelers need to consider the problems that may arise from the use of agents such as antimalarial drugs, medications for altitude acclimatization, or antidiarrheal compounds; one example is increased cutaneous sensitivity to sulfonamides. Since zidovudine is metabolized by hepatic glucuronidation, inhibitors of this process may elevate serum levels of the drug. Though quinine does not affect levels of zidovudine, there are no relevant data on chloroquine, primaquine, or mefloquine. Furthermore, it is not known whether the antagonistic effect of zidovudine on pyrimethamine has clinical relevance in the treatment or prevention of plasmodial infections.

CHRONIC ILLNESS, DISABILITY, AND TRAVEL

Evaluating fitness for travel is a growing issue in view of the increased number of elderly and chronically ill individuals journeying to exotic destinations. Conditions encountered during flight are of particular concern in these cases. Since most commercial aircraft are pressurized to 2500 m (8000 ft) above sea level, corresponding to a P_{aO_2} of about 55 mmHg, individuals with serious cardiopulmonary problems should be evaluated before travel. In addition, those who have recently had surgery, a myocardial infarction, a cerebrovascular accident, or another medical crisis may be at high risk for adverse events in flight. A summary of current recommendations regarding fitness to fly has been published by the Aerospace Medical Association Air Transport Medical Committee. Chronic health problems should not prevent travel, but special measures can make the journey safer and more comfortable.

Heart Disease Cardiovascular events are the main cause of deaths among travelers and of in-flight emergencies on commercial aircraft. Persons with underlying heart disease should review their itineraries with a physician prior to departure; travel in harsh environments or to remote destinations is not wise. Extra supplies of all medications should be kept in carry-on luggage, along with a recent copy of an electrocardiogram and the name and telephone number of the traveler's physician at home. Pacemakers are not affected by airport security devices, but electronic telephone checks of pacemaker function cannot be transmitted by international satellites. The traveler may benefit from supplemental oxygen, which should be ordered by a physician (since oxygen delivery systems are not standard) 48 to 72 h before flight time. Personal oxygen tanks are not permitted on aircraft. Travelers should request aisle seating and should walk, perform stretching and flexing exercises, and remain hydrated during the flight to prevent venous thrombosis and pulmonary embolism.

Chronic Lung Disease Chronic obstructive pulmonary disease (COPD) is one of the

most common diagnoses in patients who require emergency-room evaluation for symptoms occurring during airline flights. Patients with COPD experience dyspnea, edema, wheezing, cyanosis, and chest pain. The best predictor of the development of these symptoms is the sea level PaO_2 . A PaO_2 of at least 72 mmHg corresponds to an in-flight PaO_2 of 55 mmHg when the cabin is pressurized to 2500 m (8000 ft). Therefore, if the traveler's baseline PaO_2 is <72 mmHg, the provision of supplemental oxygen during the flight should be considered. Pulmonary function is also maximized by continuing bronchodilator treatment and the use of glucocorticoids as prescribed. Contraindications to flight include active bronchospasm, lower respiratory infection, phlebitis, pulmonary hypertension, and recent thoracic surgery (within the preceding 3 weeks) or pneumothorax. Consideration should be given to decreasing the amount of outdoor activity at the destination if there is excessive air pollution.

Diabetes Mellitus Alterations in glucose control and changes in insulin requirements are common problems when diabetic patients travel. Changes in time zone, in the amount and timing of food intake, and in physical activity demand more vigilant assessment of metabolic control. The diabetic traveler should pack medication (including a bottle of regular insulin for emergencies), insulin syringes and needles, equipment and supplies for glucose monitoring, and snacks in carry-on luggage. Insulin is stable for about 3 months at room temperature but should be kept as cool as possible. The name and telephone number of the home physician and a card and necklace listing the medical problems and the type and dose of insulin used should accompany the traveler. When six or more time zones are crossed, insulin requirements may be temporarily altered, depending on food intake and physical activity. In traveling eastward (e.g., from the United States to Europe), the morning insulin dose on arrival may need to be decreased. The blood glucose can then be checked during the day to determine whether additional insulin is required. For flights westward, with lengthening of the day, an additional dose of regular insulin may be required. Comfortable footwear is essential for the diabetic traveler.

Other Special Groups Other groups for whom special travel measures are now being encouraged include patients undergoing dialysis, those with transplants, and those with other disabilities. Up to 13% of travelers have some disability, but few advocacy groups and tour companies dedicate themselves to this growing population. The key to safe travel in each case is adequate research ahead of time. Patients undergoing chronic ambulatory peritoneal dialysis may ship their dialysis solutions to their destinations before traveling. They should carry essential medical records as well as antibiotics for self-treatment of presumed peritonitis. Hemodialysis patients need to reserve appointments at dialysis centers prior to their departure from home. Travel by transplant recipients to distant destinations should ideally be scheduled at least 1 year after surgery, as most rejection episodes occur early. Medication interactions are a source of serious concern for these travelers, and appropriate medical information should be carried, along with the home physician's name and telephone number. Some travelers taking glucocorticoids carry stress doses in case they become ill. Immunization of these immunocompromised travelers may result in less than adequate protection against certain diseases. Thus, the traveler and physician must carefully consider which destinations are appropriate.

PROBLEMS AFTER RETURN

The most frequent medical problems encountered by travelers after their return home are diarrhea, fever, respiratory illnesses, and skin diseases. Frequently ignored problems are fatigue and emotional stress, especially in long-stay travelers. The approach to diagnosis requires some knowledge of geographic medicine, in particular the epidemiology and clinical presentation of infectious disorders. A geographic history should focus on the traveler's exact itinerary, including dates of arrival and departure; exposure history (food indiscretions, drinking-water sources, freshwater contact, sexual activity, animal contact, insect bites); location and style of travel (urban vs. rural, first-class hotel accommodation vs. camping); immunization history; and use of antimalarial chemosuppression.

DIARRHEA

Although extremely common, acute traveler's diarrhea is usually self-limited or amenable to antibiotic therapy. Bowel symptoms that persist after the traveler's return home have a less well-defined etiology and may require medical attention from a specialist. Infectious agents appear to be responsible for only a small proportion of cases with persistent bowel symptoms. Of the pathogens detected in these instances, *Giardia lamblia* ([Chap. 218](#)) is by far the most common; *Cyclospora cayetanensis*, *Cryptosporidium* spp., and *Entamoeba histolytica* are rare isolates. The most frequent causes of persistent diarrhea after travel are postinfectious sequelae, such as lactose intolerance or an irritable bowel syndrome. When no infectious etiology can be identified, a trial of metronidazole therapy for presumed giardiasis, a strict lactose-free diet for 1 week, or a several-week trial of high-dose hydrophilic mucilloid relieves the symptoms of many patients.

FEVER

Fever in a traveler who has returned from a malarious area should be considered a medical emergency because death from *P. falciparum* malaria can follow an illness of only several days' duration. Although "fever from the tropics" does not always have a tropical cause, malaria should be the first diagnosis considered. The risk of *P. falciparum* malaria is highest among travelers returning from Africa or Oceania and among those who become symptomatic within the first 2 months after return. Other important causes of fever after travel include viral hepatitis (hepatitis A and E), typhoid fever, bacterial enteritis, arbovirus infections (e.g., dengue fever), rickettsial infections (including tick and scrub typhus or Q fever) and -- in rare instances -- leptospirosis, acute HIV infection, and amebic liver abscess. In at least 25% of cases, no etiology can be found, and the illness resolves spontaneously. Clinicians should keep in mind that no present-day antimalarial agent guarantees protection from malaria and that some immunizations -- notably, those against typhoid and cholera -- are only partially protective.

As noted above, the approach to the febrile returned traveler begins with a detailed medical and geographic history. Knowing exact dates of arrival and departure from tropical areas enables the physician to ascertain the shortest and longest possible incubation periods for illnesses in the differential diagnosis. For example, a traveler who develops fever <1 week after arrival in a malarious area cannot have malaria because

the incubation period is too short, whereas a fever whose onset comes >2 weeks after departure from an endemic area cannot be dengue fever because the incubation period is too long. In the physical examination, particular attention should be given to the skin so as not to miss a subtle rash or eschar.

When no specific diagnosis is forthcoming, the following investigations may be helpful: complete blood count, liver function tests, thick/thin blood films for malaria (repeated twice if necessary), urinalysis, blood cultures (repeated once if necessary), and collection of an acute-phase serum sample to be held for later examination along with a paired convalescent-phase serum sample.

SKIN DISEASES

Pyodermas, sunburn, insect bites, skin ulcers, and cutaneous larva migrans are the most common skin conditions encountered in travelers after their return home. In those with persistent skin ulcers, the diagnoses of cutaneous leishmaniasis, mycobacterial infection, or fungal infection should be considered. Careful, complete inspection of the skin is important in detecting the rickettsial eschar in a febrile patient or the central breathing hole in a "boil" due to myiasis.

EMERGING INFECTIOUS DISEASES

In recent years, travel and commerce have fostered the worldwide spread of HIV infection, led to the reemergence of cholera as a global health threat, and created considerable fear about the possible spread of Ebola virus infection and plague. For travelers, there are more realistic concerns. One of the largest outbreaks of dengue fever ever documented is now raging in Latin America; schistosomiasis is being described in previously unaffected lakes in Africa; and antibiotic-resistant strains of sexually transmitted and enteric pathogens are emerging at an alarming rate in the developing world. As Nobel Laureate Dr. Joshua Lederberg pointed out, "The microbe that felled one child in a distant continent yesterday can reach yours today and seed a global pandemic tomorrow." The vigilant clinician understands that the importance of a thorough travel history cannot be overemphasized.

SOURCES OF INFORMATION ON TRAVEL MEDICINE

- CDC publication *Health Information for International Travel*
- CDC home page: <http://www.cdc.gov>
- CDC travel information: <http://www.cdc.gov/travel/travel.html>
- Health Canada: <http://www.hwc.ca/hpb/lcdc>
- International Society of Travel Medicine: <http://www.istm.org>

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SECTION 2 -CLINICAL SYNDROMES: COMMUNITY-ACQUIRED INFECTIONS

124. SEPSIS AND SEPTIC SHOCK - Robert S. Munford

DEFINITIONS (See [Table 124-1](#))

The host's reaction to invading microbes involves a rapidly amplifying polyphony of signals and responses that may spread beyond the invaded tissue. Fever or hypothermia, tachypnea, and tachycardia often herald the onset of *sepsis*, the systemic response to microbial invasion. When counterregulatory control mechanisms are overwhelmed, homeostasis may fail, and dysfunction of major organs may supervene (*severe sepsis*). Further regulatory imbalance leads to *septic shock*, which is characterized by hypotension as well as organ dysfunction. As sepsis progresses to septic shock, the risk of dying increases substantially. Sepsis is usually reversible, whereas patients with septic shock often succumb despite aggressive therapy.

The *systemic inflammatory response syndrome* (SIRS), as defined in the early 1990s by critical care specialists, may have an infectious or a noninfectious etiology. If infection is suspected or proven, a patient with SIRS is said to have sepsis.

ETIOLOGY

Sepsis can be a response to any class of microorganism. Microbial invasion of the bloodstream is not essential for the development of sepsis, since local or systemic spread of microbial signal molecules or toxins can also elicit the response. Blood cultures yield bacteria or fungi in ~20 to 40% of cases of severe sepsis and 40 to 70% of cases of septic shock. Individual gram-negative or gram-positive bacteria account for ~70% of these isolates; the remainder are fungi or a mixture of microorganisms ([Table 124-2](#)). In patients whose blood cultures are negative, the etiologic agent is often established by culture or microscopic examination of infected material from a local site. In some case series, a majority of patients with a clinical picture of severe sepsis or septic shock have had negative microbiologic data.

Factors that predispose to gram-negative bacillary bacteremia include diabetes mellitus, lymphoproliferative diseases, cirrhosis of the liver, burns, invasive procedures or devices, and treatment with drugs that cause neutropenia. Major risk factors for gram-positive bacteremia include vascular catheterization, the presence of indwelling mechanical devices, burns, and intravenous drug use. Fungemia occurs most often in immunosuppressed patients with neutropenia, often after broad-spectrum antimicrobial therapy. In patients who experience bacteremia, factors that increase the risk of developing severe sepsis include age (>50 years) and a primary pulmonary, abdominal, or neuromeningeal site of infection. Bacteremia that arises from an intravascular catheter or the urinary tract is less likely to induce severe sepsis.

EPIDEMIOLOGY

The septic response is now a contributing factor in >100,000 deaths per year in the United States. The incidence of severe sepsis and septic shock has increased over the last 15 years and is now probably between 300,000 and 500,000 cases per year.

Approximately two-thirds of cases occur in patients hospitalized for other illnesses. The increasing incidence of severe sepsis in the United States is attributable to the aging of the population, the increasing longevity of patients with chronic diseases, and the relatively high frequency with which sepsis develops in patients with AIDS. The widespread use of antimicrobial agents, glucocorticoids, indwelling catheters and mechanical devices, and mechanical ventilation also plays a role.

PATHOPHYSIOLOGY

The septic response is often triggered when microorganisms spread from the gastrointestinal tract or skin into contiguous tissues. Localized tissue infection may then lead to bacteremia or fungemia. Alternatively, microorganisms may be introduced directly into the bloodstream (for example, via intravenous catheters). In general, the septic response occurs when immune defenses fail to contain an invading microbe. Since most cases are triggered by microbes that do not ordinarily cause systemic disease in normal hosts ([Table 124-2](#)), deficiencies in nonadaptive host factors may be most important. The septic response may also be induced by microbial exotoxins that act as superantigens (e.g., toxic shock syndrome toxin 1; [Chap. 139](#)).

Microbial Signals Animals recognize certain microbial molecules as signals that microorganisms have invaded. Lipopolysaccharide (LPS, also called *endotoxin*) is the most potent and best-studied gram-negative bacterial signal molecule. A plasma protein (LPS-binding protein, or LBP) transfers LPS to CD14 on the surfaces of monocytes, macrophages, and neutrophils. This interaction rapidly triggers the production and release of mediators, such as tumor necrosis factor (TNF) α (see below), that amplify the LPS signal and transmit it to other cells and tissues. Soluble CD14 may also bind LPS in plasma and transfer it to cells that lack cell-surface CD14. The peptidoglycan and lipoteichoic acids of gram-positive bacteria, certain polysaccharides, extracellular enzymes, and toxins elicit responses in animals that are similar to those induced by LPS; some of these molecules may also bind CD14. CD14 thus attracts numerous non-self molecules to the surfaces of myeloid cells, greatly increasing the sensitivity with which these molecules can be recognized by the host. Evidence suggests that signal specificity occurs in or on the plasma membrane and is conferred, at least in part, by members of the toll-like receptor family of transmembrane proteins. Other innate immune mechanisms for recognizing microbial molecules include complement (principally the alternative pathway), mannose-binding protein, and C-reactive protein.

Host Responses The septic response involves complex interactions among microbial signal molecules, leukocytes, humoral mediators, and the vascular endothelium.

Cytokines Inflammatory cytokines amplify and diversify the response. These proteins can exert endocrine, paracrine, and autocrine effects ([Chap. 305](#)). **TNF** stimulates leukocytes and vascular endothelial cells to release other cytokines (as well as additional TNF- α), to express cell-surface adhesion molecules, and to increase arachidonic acid turnover. Blood levels of TNF- α are high in most patients with severe sepsis or septic shock. Moreover, intravenous infusion of TNF- α can elicit many of the characteristic abnormalities of sepsis, including fever, tachycardia, tachypnea, leukocytosis, myalgias, and somnolence. In animals, larger doses of TNF- α induce shock, disseminated intravascular coagulation (DIC), and death. Specific TNF- α

antagonists can abrogate the septic response and prevent the deaths of experimental animals challenged with endotoxin.

Although [TNF- \$\alpha\$](#) is a central mediator, it is only one of many cytokines that contribute to the septic process. Interleukin (IL) 1 β , for example, which exhibits many of the same activities as TNF- α , seems to play an increasingly significant role as sepsis intensifies. TNF- α , IL-1 β , interferon- γ , IL-8, and other cytokines probably interact synergistically with each other and with additional mediators. Moreover, some mediators (such as IL-1 β and TNF- α) may enhance their own rates of synthesis by positive feedback. As sepsis progresses, the mixture of cytokines and other molecules becomes very complex: elevated blood levels of >50 molecules have been found in patients with septic shock. In animal models, the septic response can be interrupted by early interventions that neutralize one or another of its many components; this observation testifies to the importance of mediator interactions for the overall outcome. It has been much more difficult to rescue animals from severe sepsis and septic shock.

Phospholipid-Derived Mediators Arachidonic acid, released from membrane phospholipids by phospholipase A₂, is converted by the cyclooxygenase pathway into prostaglandins and thromboxanes. Prostaglandin E₂ and prostacyclin cause peripheral vasodilatation, whereas thromboxane is a vasoconstrictor and promotes platelet aggregation. Administration of the cyclooxygenase inhibitor ibuprofen for 48 h to patients with severe sepsis suppressed production of these metabolites and decreased body temperature, heart rate, and metabolic acidosis without reducing mortality. Leukotrienes are also potent mediators of ischemia and shock; the fact that the reaction to endotoxin challenge is normal in mice that lack the 5-lipoxygenase gene, however, casts doubt on the role of leukotrienes in the septic response.

Another important phospholipid-derived mediator is platelet-activating factor (PAF; 1-O-alkyl-2-acetyl-*sn*-glycero-3-phosphorylcholine). PAF potently stimulates neutrophil aggregation and degranulation, promotes platelet aggregation, and may contribute to tissue injury.

Coagulation Factors Intravascular fibrin deposition, thrombosis, and [DIC](#) are important features of the septic response. [IL-6](#) and other mediators promote intravascular coagulation initially by inducing blood monocytes to express tissue factor ([Chap. 62](#)). When tissue factor is expressed on monocytes, it binds to factor VIIa to form an active complex that can convert factors X and IX to enzymatically active forms. The result is activation of both extrinsic and intrinsic clotting pathways, culminating in the generation of fibrin. Clotting is also favored by impaired function of the protein C-protein S inhibitory pathway and depletion of antithrombin, while fibrinolysis is prevented by increased plasma levels of plasminogen activator inhibitor 1. Thus, there may be a striking propensity to intravascular fibrin deposition, thrombosis, and bleeding ([Chap. 146](#)). Contact-system activation occurs during sepsis but contributes more to the development of hypotension than to DIC.

Complement C5a and other products of complement activation may promote neutrophil reactions such as chemotaxis, aggregation, degranulation, and oxygen-radical production. When administered to animals, C5a induces hypotension, pulmonary vasoconstriction, neutropenia, and vascular leakiness due in part to endothelial

damage.

Activation of the Vascular Endothelium Many tissues may be damaged by the septic response. The probable underlying mechanism is widespread vascular endothelial injury, with fluid extravasation and microthrombosis that decrease oxygen and substrate utilization by the affected tissues. Leukocyte-derived mediators and platelet-leukocyte-fibrin thrombi contribute to this injury, but the vascular endothelium itself seems to play an active role. Stimuli such as [TNF- \$\alpha\$](#) induce vascular endothelial cells to produce and release cytokines, procoagulant molecules, [PAF](#), endothelium-derived relaxing factor (nitric oxide), and other mediators. In addition, regulated cell-adhesion molecules promote the adherence of neutrophils to endothelial cells. While these responses may attract phagocytes to infected sites and activate their antimicrobial arsenals, endothelial cell activation can also promote increased vascular permeability, microvascular thrombosis, [DIC](#), and hypotension. Moreover, vascular integrity may be damaged by neutrophil enzymes (such as elastase) and toxic oxygen metabolites so that local hemorrhage ensues. Blocking the adhesion of leukocytes to endothelial cell surfaces, as with monoclonal antibodies to intercellular adhesion molecule 1, can prevent tissue necrosis in response to endotoxin administration in animals.

Septic Shock Much evidence now implicates nitric oxide, produced by inducible nitric oxide synthase (iNOS), as a mediator of septic shock in experimental animals and probably in humans. Mice that lack the *iNOS* gene may not be resistant to endotoxic shock, however. Other prominent hypotensive molecules are β -endorphin, bradykinin, [PAF](#), and prostacyclin. Agents that inhibit the synthesis or action of each of these mediators can prevent or reverse endotoxic shock in animals. However, in clinical trials, neither a PAF receptor antagonist nor a bradykinin antagonist improved the survival rate of patients with septic shock, and a NOS inhibitor, L- N^G -methylarginine HCl, actually increased the mortality rate.

Control Mechanisms Elaborate host mechanisms regulate both microbial signals and the inflammatory response. While plasma LBP promotes the inflammatory response by facilitating the interaction of [LPS](#) with monocyte cell-surface CD14, LBP and other plasma proteins (such as phospholipid transfer protein) also can prevent LPS signaling by transferring LPS molecules into plasma lipoprotein particles. The relative plasma concentrations of LPS, LBP, CD14, and lipoproteins may therefore govern the intensity with which LPS -- and probably other microbial molecules -- can trigger host responses. The mechanisms that control the inflammatory response are also complex, overlapping, and poorly understood. Glucocorticoids inhibit cytokine synthesis by monocytes in vitro and, when administered with or shortly after an inflammatory stimulus, may protect animals from septic shock. The increase in blood cortisol levels early in the septic response presumably plays a similar inhibitory role. In addition, certain cytokine antagonists may contribute. Blood levels of [IL-1 receptor antagonist \(IL-1Ra\)](#) often greatly exceed those of circulating IL-1 β , and this excess may result in inhibition of the binding of IL-1 β to its receptors. Transforming growth factor β (TGF- β) and IL-10 can also inhibit LPS-induced responses by human monocytes in vitro and prevent endotoxic death in animals. Blood and tissue levels of prostaglandin E_2 , TGF- β , α -melanocyte-stimulating hormone, cortisol, IL-1Ra, soluble TNF receptors, and IL-10 increase during the septic response, and these molecules probably act in concert

to diminish its intensity. In fact, very high concentrations of many anti-inflammatory molecules are found in the blood of patients with severe sepsis or septic shock, so that the net mediator balance in the blood of these extremely sick patients may actually be anti-inflammatory. In addition, blood leukocytes from patients with severe sepsis are often hyporesponsive to agonists such as LPS. In patients with severe sepsis, persistence of leukocyte hyporesponsiveness has been associated with an increased risk of dying. Research is needed to clarify the role of the anti-inflammatory response in the septic process.

CLINICAL MANIFESTATIONS

The manifestations of the septic response are usually superimposed on the symptoms and signs of the patient's underlying illness and primary infection. The systemic response to infection often intensifies over time from mild (sepsis) to extremely severe (septic shock). The rate at which the response increases may differ from patient to patient, and there are striking individual variations in its manifestations. For example, some patients with sepsis are normo- or hypothermic; the absence of fever is most common in neonates, in elderly patients, and in persons with uremia or alcoholism.

Hyperventilation is often an early sign. Disorientation, confusion, and other manifestations of encephalopathy may also develop early in the septic response, particularly in the elderly and in individuals with preexisting neurologic impairment. Focal neurologic signs are uncommon, although preexisting focal deficits may become more prominent.

Hypotension and DIC predispose to acrocyanosis and ischemic necrosis of peripheral tissues, most commonly the digits ([Fig. 124-CD1](#)). Cellulitis, pustules, bullae, or hemorrhagic lesions may develop when hematogenous bacteria or fungi seed the skin or underlying soft tissue ([Fig. 124-CD2](#)). Bacterial toxins may also be distributed hematogenously to elicit diffuse cutaneous reactions. On occasion, skin lesions may suggest specific pathogens. When sepsis is accompanied by cutaneous petechiae or purpura, infection with *Neisseria meningitidis* (or, less commonly, *Haemophilus influenzae*) should be suspected ([Plate IID-44](#)); in a patient who has been bitten by a tick while in an endemic area, petechial lesions also suggest Rocky Mountain spotted fever ([Plate IID-45](#)). A cutaneous lesion seen almost exclusively in neutropenic patients is ecthyma gangrenosum ([Fig. 19-CD1](#)), usually caused by *Pseudomonas aeruginosa*. It is a bullous lesion, surrounded by edema, that undergoes central hemorrhage and necrosis ([Plate IID-57C](#)). Histopathologic examination shows bacteria in and around the wall of a small vessel, with little or no neutrophilic response. Hemorrhagic or bullous lesions in a septic patient who has recently eaten raw oysters suggest *Vibrio vulnificus* bacteremia, while such lesions in a patient who has recently suffered a dog bite may indicate bloodstream infection due to *Capnocytophaga canimorsus* or *C. cynodegmi*. Generalized erythroderma in a septic patient suggests the toxic shock syndrome due to *Staphylococcus aureus* or *Streptococcus pyogenes*.

Gastrointestinal manifestations such as nausea, vomiting, diarrhea, and ileus may suggest acute gastroenteritis. Stress ulceration can lead to upper gastrointestinal bleeding. Cholestatic jaundice, with elevated levels of serum bilirubin (mostly conjugated) and alkaline phosphatase, may precede other signs of sepsis.

Hepatocellular or canalicular dysfunction appears to underlie most cases, and the results of hepatic function tests return to normal with resolution of the infection. Prolonged or severe hypotension may induce acute hepatic injury or ischemic bowel necrosis.

Many tissues may be unable to extract oxygen normally from the blood, so that anaerobic metabolism occurs despite near-normal mixed venous oxygen saturation. Blood lactate levels rise early, in part because of increased glycolysis with impaired clearance of the resulting lactate and pyruvate by the liver and kidneys. As hypoperfusion develops, tissue hypoxia generates more lactic acid, contributing to metabolic acidosis. The blood glucose concentration often increases, particularly in patients with diabetes, although impaired gluconeogenesis and excessive insulin release on occasion produce hypoglycemia. The cytokine-driven acute-phase response inhibits the synthesis of albumin and transthyretin while enhancing the production of C-reactive protein, [LBP](#), fibrinogen, and complement components. Protein catabolism is often markedly accelerated.

MAJOR COMPLICATIONS

Cardiopulmonary Complications Ventilation-perfusion mismatching produces a fall in arterial P_{O_2} early in the course. Increasing alveolar capillary permeability results in an increased pulmonary water content, which decreases pulmonary compliance and interferes with oxygen exchange. Progressive diffuse pulmonary infiltrates and arterial hypoxemia ($P_{aO_2}/F_{iO_2} < 200$ mmHg) indicate the development of the acute respiratory distress syndrome (ARDS). ARDS develops in ~50% of patients with severe sepsis or septic shock. The failure of the respiratory muscles can exacerbate hypoxemia and hypercapnia. An elevated pulmonary capillary wedge pressure (>18 mmHg) suggests fluid volume overload or cardiac failure rather than ARDS. Pneumonia caused by viruses or by *Pneumocystis carinii* may be clinically indistinguishable from ARDS.

Sepsis-induced hypotension usually results from a generalized maldistribution of blood flow and blood volume and from hypovolemia that is due, at least in part, to diffuse capillary leakage of intravascular fluid. Other factors that may decrease effective intravascular volume include dehydration from antecedent disease or insensible fluid losses, vomiting or diarrhea, and polyuria. During early septic shock, systemic vascular resistance is usually elevated and cardiac output may be low. After fluid repletion, in contrast, cardiac output typically increases and systemic vascular resistance falls. Indeed, normal or increased cardiac output and decreased systemic vascular resistance distinguish septic shock from cardiogenic, extracardiac obstructive, and hypovolemic shock; other processes that can produce this combination include anaphylaxis, beriberi, cirrhosis, and overdoses of nitroprusside or narcotics ([Chap. 38](#)).

Depression of myocardial function, manifested as increased end-diastolic and systolic ventricular volumes with a decreased ejection fraction, develops within 24 h in most patients with severe sepsis. Cardiac output is maintained despite the low ejection fraction because ventricular dilatation permits a normal stroke volume. In survivors, myocardial function returns to normal over several days. Although myocardial dysfunction may contribute to hypotension, refractory hypotension is usually due to a low systemic vascular resistance, and death results from refractory shock or the failure

of multiple organs rather than from cardiac dysfunction per se.

Renal Complications Oliguria, azotemia, proteinuria, and nonspecific urinary casts are frequently found. Many patients are inappropriately polyuric; hyperglycemia may exacerbate this tendency. Most renal failure is due to acute tubular necrosis induced by hypotension or capillary injury, although some patients also have glomerulonephritis, renal cortical necrosis, or interstitial nephritis. Drug-induced renal damage may complicate therapy, particularly when hypotensive patients are given aminoglycoside antibiotics.

Coagulation Although thrombocytopenia occurs in 10 to 30% of patients, the underlying mechanism(s) are not understood. Platelet counts are usually very low ($<50,000/\mu\text{L}$) in patients with [DIC](#); these low counts typically reflect diffuse endothelial injury or microvascular thrombosis.

Neurologic Complications When the septic illness lasts for weeks to months, "critical-illness" polyneuropathy may prevent weaning from ventilatory support and produce distal motor weakness. Electrophysiologic studies are diagnostic. Guillain-Barre syndrome, metabolic disturbances, and toxin activity must be ruled out.

LABORATORY FINDINGS

Abnormalities that occur early in the septic response may include leukocytosis with a left shift, thrombocytopenia, hyperbilirubinemia, and proteinuria. Leukopenia may develop. The neutrophils may contain toxic granulations ([Fig. 124-CD3](#)), Dohle bodies, or cytoplasmic vacuoles. As the septic response becomes more severe, thrombocytopenia worsens (often with prolongation of the thrombin time, decreased fibrinogen, and the presence of D-dimers, suggesting [DIC](#)), azotemia and hyperbilirubinemia become more prominent, and levels of aminotransferases rise. Active hemolysis suggests clostridial bacteremia, malaria, a drug reaction, or DIC; in the case of DIC, microangiopathic changes may be seen on a blood smear.

During early sepsis, hyperventilation induces respiratory alkalosis. With respiratory muscle fatigue and the accumulation of lactate, metabolic acidosis (with increased anion gap) typically supervenes. Evaluation of arterial blood gases reveals hypoxemia, which is initially correctable with supplemental oxygen but whose later refractoriness to 100% oxygen inhalation indicates right-to-left shunting. The chest radiograph may be normal or may show evidence of underlying pneumonia, volume overload, or the diffuse infiltrates of [ARDS](#). The electrocardiogram may show only sinus tachycardia or nonspecific ST-T wave abnormalities.

Most diabetic patients with sepsis develop hyperglycemia. Severe infection may precipitate diabetic ketoacidosis, which may exacerbate hypotension ([Chap. 333](#)). Hypoglycemia occurs rarely. The serum albumin level, initially within the normal range, declines as sepsis continues. Serum lipid concentrations are often elevated. Hypocalcemia is rare.

DIAGNOSIS

There is no specific test. Diagnostically sensitive findings in a patient with suspected or proven infection include fever or hypothermia, tachypnea, tachycardia, and leukocytosis or leukopenia ([Table 124-1](#)); acutely altered mental status, thrombocytopenia, or hypotension also suggests the diagnosis. The septic response can be quite variable, however. In one study, 36% of patients with severe sepsis had a normal temperature, 40% had a normal respiratory rate, 10% had a normal pulse rate, and 33% had normal white blood cell counts. Moreover, the systemic responses of uninfected patients with other conditions may be similar to those characteristic of sepsis. Noninfectious etiologies of [SIRS](#) ([Table 124-1](#)) include pancreatitis, burns, trauma, adrenal insufficiency, pulmonary embolism, dissecting or ruptured aortic aneurysm, myocardial infarction, occult hemorrhage, cardiac tamponade, post-cardiopulmonary bypass syndrome, anaphylaxis, and drug overdose.

Definitive etiologic diagnosis requires isolation of the microorganism from blood or a local site of infection. At least two blood samples (10 mL each) should be obtained (from different venipuncture sites) for culture. Because gram-negative bacteremia is typically low-grade (<10 organisms per milliliter of blood), multiple blood cultures or prolonged incubation of cultures may be necessary; *S. aureus* grows more readily and is detectable in blood cultures within 48 h in most instances. In many cases, blood cultures are negative; this result can reflect prior antibiotic administration, the presence of slow-growing or fastidious organisms, or the absence of microbial invasion of the bloodstream. In these cases, Gram's staining and culture of material from the primary site of infection or of infected cutaneous lesions may help establish the microbial etiology. The skin and mucosae should be examined carefully and repeatedly for lesions that might yield diagnostic information. With overwhelming bacteremia (e.g., pneumococcal sepsis in splenectomized individuals or fulminant meningococcemia), microorganisms are sometimes visible on buffy coat smears of peripheral blood.

Detection of endotoxin in blood by the limulus lysate test may portend a poor outcome, but this assay is not useful for diagnosing gram-negative bacterial infections, including gram-negative bacteremia. Although blood levels of [IL-6](#) also may correlate with prognosis, cytokine assays are poorly standardized and currently have limited clinical value.

TREATMENT

Patients in whom sepsis is suspected must be managed expeditiously. This task is best accomplished in an intensive care unit by personnel who are experienced in the care of the critically ill. Successful management requires urgent measures to treat the local site of infection, to provide hemodynamic and respiratory support, and to eliminate the offending microorganism. The outcome is also influenced by the patient's underlying disease, which should be managed aggressively.

Antimicrobial Agents Antimicrobial chemotherapy should be initiated as soon as samples of blood and other relevant sites have been cultured. The choice of initial therapy is based on knowledge of the likely pathogens at specific sites of local infection. Available information about patterns of antimicrobial susceptibility among bacterial isolates from the community, the hospital, and the patient also should be taken into account. It is important, pending culture results, to initiate empirical antimicrobial therapy

that is effective against both gram-positive and gram-negative bacteria ([Table 124-3](#)). Maximal recommended doses of antimicrobial drugs should be given intravenously, with adjustment for impaired renal function when necessary. When culture results become available, the regimen can often be simplified, as a single antimicrobial agent is frequently adequate for the treatment of a known pathogen. Most patients require antimicrobial therapy for at least 1 week; the duration of treatment is typically influenced by factors such as the site of tissue infection, the adequacy of surgical drainage, the patient's underlying disease, and the antimicrobial susceptibility of the bacterial isolate(s).

Removal of the Source of Infection Removal or drainage of a focal source of infection is essential. Sites of occult infection should be sought carefully. Indwelling intravenous catheters should be removed, the tip rolled over a blood agar plate for quantitative culture, and a new catheter inserted at a different site. Foley and drainage catheters should be replaced. The possibility of paranasal sinusitis (often caused by gram-negative bacteria) should be considered if the patient has undergone nasal intubation. In the neutropenic patient, cutaneous sites of tenderness and erythema, particularly in the perianal region, must be carefully sought. In patients with sacral or ischial decubitus ulcers, it is important to exclude pelvic or other soft-tissue pus collections (by computed tomography or magnetic resonance imaging, if necessary). In patients with severe sepsis arising from the urinary tract, sonography or computed tomography should be used to rule out ureteral obstruction, perinephric abscess, and renal abscess. These studies are not so urgent in patients with less severe urosepsis, provided that a clinical response is evident within 48 to 72 h.

Hemodynamic, Respiratory, and Metabolic Support (See also [Chap. 38](#)) The primary goal is to restore adequate oxygen and substrate delivery to the tissues. Adequate organ perfusion is essential. Effective intravascular volume depletion is common in patients with sepsis, and initial management of hypotension should include the administration of intravenous fluids, typically 1 to 2 L of normal saline over 1 to 2 h. The pulmonary capillary wedge pressure or the central venous pressure must be monitored in patients with refractory shock or underlying cardiac or renal disease. To avoid pulmonary edema, the pulmonary capillary wedge pressure should be maintained between 12 and 16 mmHg or the central venous pressure between 10 and 12 cmH₂O. The urine output rate should be kept above 30 mL/h by continuing fluid administration; a diuretic such as furosemide may be used if needed. In about one-third of patients, hypotension and organ hypoperfusion respond to fluid resuscitation; a reasonable goal is to maintain a mean arterial blood pressure of >60 mmHg (systolic pressure, >90 mmHg) and a cardiac index of ≥ 4 (L/min)/m². If these guidelines cannot be met by volume infusion, inotropic and vasopressor therapy is indicated ([Chap. 38](#)). Circulatory adequacy is also assessed by clinical parameters (mentation, urine output, skin perfusion) and, when possible, by measurements of oxygen delivery and consumption.

Adrenal insufficiency should be considered in septic patients with refractory hypotension, fulminant *N. meningitidis* bacteremia, prior glucocorticoid use, disseminated tuberculosis, or AIDS. The cosyntropin (α_{1-24} -ACTH) stimulation test ([Chap. 331](#)) may suggest absolute or partial adrenal insufficiency. Supplemental hydrocortisone (50 mg intravenously every 6 h) may be given while the results of the cosyntropin test are awaited.

Ventilator therapy is indicated for progressive hypoxemia, hypercapnia, neurologic deterioration, or respiratory muscle failure. Sustained tachypnea (respiratory rate, >30 breaths/min) is frequently a harbinger of impending respiratory collapse; mechanical ventilation is often initiated to ensure adequate oxygenation, divert blood from the muscles of respiration, prevent aspiration of oropharyngeal contents, and reduce the cardiac afterload. Blood or erythrocyte transfusion is indicated if oxygen delivery is compromised by a low hemoglobin concentration (<8 to 10 g/dL).

Bicarbonate is sometimes administered for severe metabolic acidosis (arterial pH < 7.2). [DIC](#), if complicated by major bleeding, should be treated with transfusion of fresh-frozen plasma and platelets. Successful treatment of the underlying infection is essential to reverse both acidosis and DIC.

These are consensus recommendations; none of these generally accepted components of resuscitative care has been validated in randomized clinical trials.

General Support In patients with prolonged severe sepsis (i.e., that lasting more than 2 or 3 days), nutritional supplementation may reduce the impact of protein hypercatabolism; available evidence favors the enteral delivery route. Recovery is also assisted by preventing skin breakdown, deep venous thrombosis, nosocomial infections, and stress ulcers.

Other Measures Despite aggressive management, many patients with severe sepsis or septic shock die. Two kinds of agents that may help prevent these deaths are being investigated: (1) drugs that neutralize bacterial endotoxin, thereby potentially benefiting the fraction (approximately half) of septic patients who have gram-negative bacterial infection, and (2) drugs that interfere with one or more mediators of the inflammatory response and may benefit all patients with sepsis.

Antiendotoxin Agents Lipid A, the toxic moiety of endotoxin, is conserved in the [LPS](#) of gram-negative bacteria. Despite much effort to develop drugs that bind lipid A and neutralize endotoxin in vivo, the potential of endotoxin as a target for therapeutic intervention remains controversial. In placebo-controlled clinical trials, two monoclonal antibodies to endotoxin did not prevent the death of patients with severe gram-negative bacterial sepsis. In retrospective studies, these antibodies did not bind to LPS with high affinity, and one was reported to be a polyreactive autoantibody. A theoretically more promising agent is bactericidal permeability-increasing protein, a human neutrophil protein that neutralizes the toxicity of lipid A and may be bactericidal to many gram-negative bacteria. In one clinical trial, this protein decreased morbidity and mortality among children with fulminant meningococemia. Other investigational drugs include nontoxic lipid A analogs that reduce host responses to endotoxins and lipoproteins (such as high-density lipoprotein) that bind and neutralize endotoxin in the circulation and can remove it from the surfaces of myeloid cells.

Antimediator Agents Other adjunctive therapies are intended to control the inflammatory response, regardless of the microbial stimulus. However, numerous agents that directly or indirectly interfere with the actions of inflammatory mediators have not prevented the death of patients with severe sepsis or septic shock. Many

factors have probably contributed to the unsuccessful outcomes of these trials, including problems with study design (inappropriate end points, inadequate sample size, population heterogeneity, multiple covariates) and drug administration (wrong dose, time, or duration of administration). Anti-inflammatory drugs tested in clinical trials include methylprednisolone, ibuprofen, recombinant IL-1Ra, genetically engineered soluble receptors for [TNF-α](#), and monoclonal antibodies to TNF-α. Because TNF-α and [IL-1β](#) doubtless play key roles in antimicrobial host defense, neutralizing these cytokines could be detrimental in some cases. In addition, studies suggest that many anti-inflammatory molecules, including soluble TNF receptors and IL-1Ra, may already be present at high concentrations in the plasma of patients with septic shock. Identifying beneficial regimens of treatment with drugs that neutralize TNF-α and IL-1β may therefore be very difficult. Clinical trials are testing drugs (antithrombin, activated protein C, tissue factor pathway inhibitor) intended to prevent or reverse microthrombosis and evaluating regimens in which low doses of glucocorticoids are administered for prolonged periods.

All of the more recent clinical trials have enrolled patients with severe sepsis or septic shock. Neither the ability of adjunctive agents to prevent severe sepsis or septic shock in high-risk patients nor the value of combination therapy with two or more adjunctive drugs has been tested.

PROGNOSIS

Approximately 20 to 35% of patients with severe sepsis and 40 to 60% of patients with septic shock die within 30 days. Others die within the ensuing 6 months. Late deaths often result from poorly controlled infection, complications of intensive care, failure of multiple organs, or the patient's underlying disease.

Several prognostic stratification systems indicate that factoring in the patient's age, underlying condition, and various physiologic variables can yield estimates of the risk of dying of severe sepsis. Of the individual covariates, the severity of underlying disease most strongly influences the risk of dying. Septic shock is also a strong predictor of short- and long-term mortality. Case-fatality rates are similar for culture-positive and culture-negative severe sepsis.

PREVENTION

Prevention offers the best opportunity to reduce morbidity and mortality. Most episodes of severe sepsis and septic shock are nosocomial. These cases might be prevented by reducing the number of invasive procedures undertaken, by limiting the use (and duration of use) of indwelling vascular and bladder catheters, by reducing the incidence and duration of profound neutropenia (<500 neutrophils/uL), and by more aggressively treating localized nosocomial infections. Indiscriminate use of antimicrobial agents and glucocorticoids should be avoided, and optimal infection-control measures ([Chap. 134](#)) should be used. In addition, prompt and aggressive management of patients with sepsis is imperative. Studies indicate that 50 to 70% of patients who develop nosocomial severe sepsis or septic shock have experienced a less severe stage of the septic response (e.g., [SIRS](#), sepsis) on at least one previous day in the hospital. Research is needed to identify patients at high risk for severe sepsis and to develop adjunctive

agents that can damp the septic response before organ dysfunction or hypotension occurs.

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125. FEVER OF UNKNOWN ORIGIN - Jeffrey A. Gelfand

DEFINITION AND CLASSIFICATION

Fever of unknown origin (FUO) was defined by Petersdorf and Beeson in 1961 as (1) temperatures $>38.3^{\circ}\text{C}$ (101°F) on several occasions; (2) a duration of fever of >3 weeks; and (3) failure to reach a diagnosis despite 1 week of inpatient investigation. While this classification has stood for more than 30 years, Durack and Street have proposed a new system for classification of FUO: (1) classic FUO; (2) nosocomial FUO; (3) neutropenic FUO; and (4) FUO associated with HIV infection ([Table 125-1](#)).

Classic FUO Classic [FUO](#) corresponds closely to the earlier definition of FUO, differing only with regard to the prior requirement for 1 week's study in the hospital. The new definition is broader, stipulating three outpatient visits or 3 days in the hospital without elucidation of a cause or 1 week of "intelligent and invasive" ambulatory investigation.

Nosocomial FUO In nosocomial [FUO](#), a temperature of $\geq 38.3^{\circ}\text{C}$ (101°F) develops on several occasions in a hospitalized patient who is receiving acute care and in whom infection was not manifest or incubating on admission. Three days of investigation, including at least 2 days' incubation of cultures, is the minimum requirement for this diagnosis.

Neutropenic FUO Neutropenic [FUO](#) is defined as a temperature of $\geq 38.3^{\circ}\text{C}$ (101°F) on several occasions in a patient whose neutrophil count is $<500/\mu\text{L}$ or is expected to fall to that level in 1 to 2 days. The diagnosis of neutropenic FUO is invoked if a specific cause is not identified after 3 days of investigation, including at least 2 days' incubation of cultures.

HIV-Associated FUO [FUO](#) associated with HIV infection is defined by a temperature of $\geq 38.3^{\circ}\text{C}$ (101°F) on several occasions over a period of >4 weeks for outpatients or >3 days for hospitalized patients with HIV infection. This diagnosis is invoked if appropriate investigation over 3 days, including 2 days' incubation of cultures, reveals no source.

Adoption of these categories of [FUO](#) on a wide scale in the literature would allow a more rational compilation of data regarding these disparate groups. In the remainder of this chapter, the discussion will focus on classic FUO unless otherwise specified.

CAUSES OF CLASSIC FUO

[Table 125-2](#) summarizes the findings of several large studies of [FUO](#) carried out since the advent of the antibiotic era, including a prospective study of 167 adult patients with FUO encompassing all 8 university hospitals in the Netherlands and using a standardized protocol in which the first author reviewed every patient. Coincident with the widespread use of antibiotics, increasingly useful diagnostic technologies -- both noninvasive and invasive -- have been developed. Newer studies reflect not only changing patterns of disease but also the impact of diagnostic techniques that make it possible to eliminate many patients with specific illness from the FUO category. The ubiquitous use of microbiologic cultures and the widespread use of potent broad-spectrum antibiotics may have decreased the number of infections causing FUO.

The wide availability of ultrasonography, computed tomography (CT), and magnetic resonance imaging (MRI) has enhanced the detection of occult neoplasms and lymphomas in patients previously thought to have FUO. Likewise, the widespread availability of highly specific and sensitive immunologic testing has reduced the number of undetected cases of systemic lupus erythematosus and other autoimmune diseases.

Several generalizations can be made. Infections, especially extrapulmonary tuberculosis, remain the leading diagnosable cause of [FUO](#). Prolonged mononucleosis syndromes caused by Epstein-Barr virus, cytomegalovirus (CMV), or HIV are conditions whose consideration as a cause of FUO is sometimes confounded by delayed antibody responses. Intraabdominal abscesses (sometimes poorly localized) and renal, retroperitoneal, and paraspinal abscesses continue to be difficult to diagnose. Renal malacoplakia, with submucosal plaques or nodules involving the urinary tract, may cause FUO and is often fatal if untreated. It is associated with coliform infection, is seen most often in patients with defects of intracellular bacterial killing, and is treated with fluoroquinolones or trimethoprim-sulfamethoxazole. Occasionally, other organs may be involved. Osteomyelitis, especially where prosthetic devices have been implanted, and infective endocarditis must be considered. Although true culture-negative infective endocarditis is rare, one may be misled by slow-growing organisms of the HACEK group (*Haemophilus aphrophilus*, *Actinobacillus actinomycetemcomitans*, *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella kingae*; [Chap. 150](#)), *Bartonella* spp. (previously *Rochalimaea*), *Legionella* spp., *Coxiella burnetii*, *Chlamydia psittaci*, and fungi. Prostatitis, dental abscesses, sinusitis, and cholangitis continue to be sources of occult fever.

Fungal disease, most notably histoplasmosis involving the reticuloendothelial system, may cause [FUO](#). FUO with headache should prompt examination of spinal fluid for *Cryptococcus neoformans*. Malaria (which may result from transfusion, the failure to take a prescribed prophylactic agent, or infection with a drug-resistant strain) continues to be a cause, particularly of nonsynchronized FUO. A related protozoan species, *Babesia*, may cause FUO and is increasing in incidence.

In most earlier series, neoplasms were the next most common cause of [FUO](#) after infections ([Table 125-3](#)). In the two most recent series, a decrease in the percentage of FUO cases due to malignancy was attributed to improvement in diagnostic technologies. This observation does not diminish the importance of considering neoplasia in the initial diagnostic evaluation of a patient with fever. A number of patients in these series had temporal arteritis, adult Still's disease, drug-related fever, and factitious fever. In recent series, approximately 25 to 30% of cases of FUO have remained undiagnosed. The general term *noninfectious inflammatory diseases* applies to systemic rheumatologic or vasculitic diseases such as polymyalgia rheumatica, lupus, and adult Still's disease as well as to granulomatous diseases such as sarcoidosis and Crohn's and granulomatous hepatitis.

In the elderly, multisystem disease is the most frequent cause of [FUO](#), giant cell arteritis being the leading etiologic entity in this category. Tuberculosis is the most common infection causing FUO in the elderly, and colon cancer is an important cause of FUO with malignancy.

Many diseases have been grouped in the various studies as "miscellaneous." On this list are drug fever, pulmonary embolism, factitious fever, familial Mediterranean fever, and Fabry's disease.

A drug-related etiology must be considered in any case of prolonged fever. Any febrile pattern may be elicited by a drug, and both relative bradycardia and hypotension are uncommon. Eosinophilia and/or rash is found in only one-fifth of patients with drug fever, which usually begins 1 to 3 weeks after the start of therapy and remits 2 to 3 days after therapy is stopped. Virtually all classes of drugs cause fever, but antimicrobials (especially b-lactam antibiotics), cardiovascular drugs (e.g., quinidine), antineoplastic drugs, and drugs acting on the central nervous system (e.g., phenytoin) are particularly common causes.

It is axiomatic that, as the duration of fever increases, the likelihood of an infectious cause decreases ([Table 125-4](#)). In a series of 347 patients referred to the National Institutes of Health from 1961 to 1977, only 6% had an infection. A significant proportion (9%) had factitious fevers -- i.e., fevers due either to false elevations of temperature or to self-induced disease. A substantial number of these factitious cases were in young women in the health professions. It is worth noting that 8% of the patients with prolonged fevers (some of whom had completely normal liver function studies) had granulomatous hepatitis, and 6% had adult Still's disease. After prolonged investigation, 19% of cases still had no specific diagnosis. A total of 27% of patients either had no actual fever during the weeks of inpatient observation or had an exaggerated circadian temperature rhythm without chills, elevated pulse, or other abnormalities.

The conditions that may be considered in a differential diagnosis of classic [FUO](#) in adults are listed in [Table 125-5](#). This list applies strictly to the United States; the frequency of global travel underscores the need for a detailed travel history, and the continuing emergence of new infectious diseases makes this listing potentially incomplete.

SPECIALIZED DIAGNOSTIC STUDIES

Classic FUO Certain specific diagnostic maneuvers become critical in dealing with prolonged fevers. If factitious fever is suspected, electronic thermometers should be used, temperature-taking should be supervised, and simultaneous urine and body temperatures should be measured. Any tissue removed during prior relevant surgery should be reexamined; slides should be requested, and, if need be, paraffin blocks of fixed pathologic material should be reexamined and additional special studies performed. Relevant x-rays should be reexamined; reviewing of prior radiologic reports may be insufficient. Serum should be set aside in the laboratory as soon as possible and retained for future examination for rising antibody titers. *Febrile agglutinins* is a vague term that in most laboratories refers to serologic studies for salmonellosis, brucellosis, and rickettsial diseases. These studies are seldom useful, having low sensitivity and variable specificity. Rising titers of antibody to *Brucella* ([Chap. 160](#)) are usually diagnostic, but false-positive results may be obtained in typhoid fever, tularemia, and yersinial infections. Infection with *Brucella canis* may be missed with standard antibody tests for *Brucella*. *Salmonella* infection ([Chap. 156](#)) elevates antibody titers to the H and O antigens. High titers of antibody to the H antigen persist for years and may reflect previous infection or immunization. Serology for *Yersinia enterocolitica* may be

useful. The measurement of specific antirickettsial titers should be requested for the diagnosis of Rocky Mountain spotted fever and Q fever. Multiple blood samples (no fewer than three, rarely more than six), including samples for anaerobic culture, should be cultured in the laboratory for at least 2 weeks to ensure that any HACEK-group organisms that may be present have ample time to grow ([Chap. 150](#)).

Lysis-centrifugation blood culture techniques should be employed in cases where prior antimicrobial therapy or fungal or atypical mycobacterial infection is suspected. Blood culture media should be supplemented with L-cysteine or pyridoxal to assist in the isolation of nutritionally variant streptococci. It should be noted that sequential cultures positive for multiple organisms may reflect self-injection of contaminated substances. Urine cultures, including cultures for mycobacteria, fungi, and [CMV](#), are indicated. Blood, urine, or cerebrospinal fluid (CSF) can now be tested for a variety of pathogens such as CMV or hepatitis C virus by using the polymerase chain reaction (PCR) to amplify and hence detect viral nucleic acid ([Chap. 121](#)). Liver biopsy, even when the results of liver function studies are normal, should be considered and pursued if the diagnosis remains elusive. Specimens should be cultured for mycobacteria and fungi. Likewise, bone marrow biopsy (not simple aspiration) should be used to obtain specimens for histology and culture. The blood smear should be examined for *Plasmodium*, *Babesia*, *Trypanosoma*, *Leishmania*, and *Borrelia*.

In an [FUO](#) workup, the erythrocyte sedimentation rate (ESR) should be determined. Striking elevation of the ESR and anemia of chronic disease are frequently seen in association with giant cell arteritis or polymyalgia rheumatica, common causes of FUO in patients over 50 years of age. Still's disease is also suggested by elevations of ESR, leukocytosis, and anemia and is often accompanied by arthralgias, polyserositis (pleuritis, pericarditis), lymphadenopathy, splenomegaly, and rash. Antinuclear antibody, antineutrophil cytoplasmic antibody, rheumatoid factor, and serum cryoglobulins should be measured to rule out other collagen vascular diseases and vasculitis. Another cause of an extremely high ESR may be a false-positive value attributable to a cold agglutinin with a broad thermal amplitude. The ESR test is nonspecific, yielding values that depend on certain serum proteins (most notably fibrinogen) known to interfere with the zeta-potential that keeps erythrocytes from clumping. When fibrinogen levels go up, the zeta-potential is inhibited, erythrocytes clump, and the ESR is high. A cold agglutinin, by binding to erythrocytes, can produce a false-positive agglutinin that mimics an acute-phase response; cold agglutinins may be seen in *Mycoplasma* and Epstein-Barr virus infections and in lymphomas.

With rare exceptions, the intermediate-strength purified protein derivative (PPD) skin test should be used to screen for tuberculosis in patients with classic [FUO](#). Concurrent control tests, such as the CMI test (Connaught Labs, Swiftwater, PA), which is especially effective, should be employed. It should be kept in mind that both the PPD skin test and control tests may yield negative results in miliary tuberculosis, sarcoidosis, Hodgkin's disease, malnutrition, or AIDS. Noninvasive procedures should include an upper gastrointestinal contrast study with small-bowel follow-through and barium enema to include the terminal ileum and cecum. Chest x-rays should be repeated if new symptoms arise. In some cases, pulmonary function studies may be necessary. A diminished carbon monoxide diffusing capacity may indicate a restrictive lung disease such as sarcoidosis, even with a normal chest x-ray. In such cases, transbronchial biopsy may prove diagnostic. Flexible colonoscopy may be advisable, since colon

carcinoma is a cause of FUO and easily escapes detection by ultrasound and [CT](#).

[CT](#) of the chest and abdomen should be performed. If a spinal or paraspinal lesion is suspected, however, [MRI](#) is preferred. MRI may be superior to CT in demonstrating intraabdominal abscesses and aortic dissection, but the relative utility of MRI and CT in the diagnosis of [FUO](#) is unknown. At present, it appears that abdominal CT, with oral and intravenous contrast, should be used unless MRI is specifically indicated.

Arteriography may be useful for patients in whom systemic necrotizing vasculitis is suspected. Saccular aneurysms may be seen, most commonly in renal or hepatic vessels, and may permit diagnosis of arteritis when biopsy is difficult. [Figure 125-1](#) shows a renal angiogram of a patient with polyarteritis nodosa. Ultrasonography of the abdomen is useful for the investigation of the hepatobiliary tract, kidneys, spleen, and pelvis. Echocardiography may be helpful in an evaluation for bacterial endocarditis, pericarditis, nonbacterial thrombotic endocarditis, and atrial myxomas. Transesophageal echocardiography is especially sensitive for these lesions.

Radionuclide scanning procedures using technetium (Tc) 99m sulfur colloid, gallium (Ga) 67 citrate, or indium (In) 111-labeled leukocytes or immunoglobulin may be useful in identifying and/or localizing inflammatory processes. In a recent study, Ga scintigraphy yielded useful diagnostic information in almost one-third of cases, and it was suggested that this procedure might actually be used before other imaging techniques if no specific organ is suspected of being abnormal. Tc bone scan should be undertaken to look for osteomyelitis or bony metastases; ⁶⁷Ga scan may be used to identify sarcoidosis ([Chap. 318](#)) or *Pneumocystis carinii* ([Chap. 209](#)) in the lungs or Crohn's disease ([Chap. 287](#)) in the abdomen. ¹¹¹In-labeled white blood cell (WBC) scan may be used to locate abscesses; ¹¹¹In-labeled immunoglobulin scan also shows promise in this regard. With ⁶⁷Ga, ¹¹¹In-WBC, and ¹¹¹In-immunoglobulin scans, false-positive and false-negative findings are common.

Biopsy of the liver and bone marrow should be considered routine in the workup of [FUO](#) if the studies mentioned above are unrevealing or if fever is prolonged. It goes without saying that areas of suspected abnormality should be sampled for pathologic examination whenever practical. When possible, a section of the tissue block should be retained for further sections or stains. [PCR](#) technology makes it possible to identify and speciate mycobacterial DNA in paraffin-embedded, fixed tissues. Thus, in some cases, it is possible to make a retrospective diagnosis based on studies of long-fixed pathologic tissues. In a patient over age 50 (or occasionally in a younger patient) with the appropriate symptoms and laboratory findings, "blind biopsy" of one or both temporal arteries may yield a diagnosis of arteritis. If noted, tenderness or decreased pulsation should guide the selection of a site for biopsy. Lymph node biopsy may be helpful if nodes are enlarged, but inguinal nodes are often palpable and are seldom diagnostically useful.

Exploratory laparotomy has been performed when all other diagnostic procedures fail but has largely been replaced by modern imaging and guided-biopsy techniques. Laparoscopic biopsy may provide more adequate guided sampling of lymph nodes or liver.

Nosocomial FUO The primary considerations in diagnosing nosocomial [FUO](#) are the

underlying susceptibility of the patient coupled with the potential complications of hospitalization. The original surgical or procedural field is the place to begin a directed physical and laboratory examination for abscesses, hematomas, or infected foreign bodies. More than 50% of patients with nosocomial FEO are infected, and intravascular lines, septic phlebitis, and prostheses are all suspect. In this setting, the approach is to focus on sites where occult infections may be sequestered, such as the sinuses of intubated patients or a prostatic abscess in a man with a urinary catheter. *Clostridium difficile* colitis may be associated with fever and leukocytosis before the onset of diarrhea. In approximately 25% of patients with nosocomial FEO, the fever has a noninfectious cause. Among these causes are acalculous cholecystitis, deep vein thrombophlebitis, and pulmonary embolism. Drug fever, transfusion reactions, alcohol/drug withdrawal, adrenal insufficiency, thyroiditis, pancreatitis, gout, and pseudogout are among the many possible causes to consider. As in classic FEO, repeated meticulous physical examinations, coupled with focused diagnostic techniques, are imperative. Multiple blood, wound, and fluid cultures are mandatory. The pace of diagnostic tests is accelerated, and the threshold for procedures -- [CT](#) scans, ultrasonography,¹¹¹In-WBC scans, noninvasive venous studies -- is low. Even so, 20% of cases of nosocomial FEO may go undiagnosed.

Like diagnostic measures, therapeutic maneuvers must be swift and decisive, as many patients are already critically ill. Intravenous lines must be changed (and cultured), drugs stopped for 72 h, and empirical therapy started if bacteremia is a threat. In many hospital settings, empirical antibiotic coverage for nosocomial FEO now includes vancomycin for methicillin-resistant *Staphylococcus aureus* as well as broad-spectrum gram-negative coverage with piperacillin/tazobactam, ticarcillin/clavulanate, imipenem, or meropenem. Practice guidelines covering many of these issues have been published jointly by the Infectious Diseases Society of America (IDSA) and the Society for Critical Care Medicine and can be accessed on the IDSA website (<http://www.idsociety.org/practice/index.html>).

Neutropenic FEO (See also [Chap. 85](#)) Neutropenic patients are susceptible to focal bacterial and fungal infections, to bacteremic infections, to infections involving catheters (including septic thrombophlebitis), and to perianal infections. *Candida* and *Aspergillus* infections are common. Infections due to herpes simplex virus or [CMV](#) are sometimes causes of [FEO](#) in this group. While the duration of illness may be short in these patients, the consequences of untreated infection may be catastrophic, with 50 to 60% infected, and 20% bacteremic. The [IDSA](#) has published extensive practice guidelines covering these critically ill neutropenic patients; these guidelines appear on the website cited in the previous section. In these patients, severe mucositis, quinolone prophylaxis, colonization with methicillin-resistant *S. aureus*, obvious catheter-related infection, or hypotension would dictate the use of vancomycin plus ceftazidime or imipenem to provide empirical coverage for bacterial sepsis.

HIV-Associated FEO HIV infection alone may be a cause of fever. Infection due to *Mycobacterium avium* or *Mycobacterium intracellulare*, tuberculosis, toxoplasmosis, [CMV](#) infection, *P. carinii* infection, salmonellosis, cryptococcosis, histoplasmosis, non-Hodgkin's lymphoma, and (of particular importance) drug fever are all possible causes of [FEO](#). Mycobacterial infection can be diagnosed by blood cultures and by liver, bone marrow, and lymph node biopsies. Chest [CT](#) should be performed to

identify enlarged mediastinal nodes. Serologic studies may reveal cryptococcal antigen, and ^{67}Ga scan may help identify *P. carinii* pulmonary infection. More than 80% of HIV patients with FUO are infected, but drug fever and lymphoma remain important considerations. **Treatment of HIV-associated FUO depends on many factors and is discussed in [Chap. 309](#).*

TREATMENT

The emphasis in patients with classic [FUO](#) is on continued observation and examination, with the avoidance of "shotgun" empirical therapy. Empirical treatment for endocarditis, for example, should be avoided unless there are specific reasons beyond fever to invoke this diagnosis. Every patient with FUO should undergo an exhaustive examination for tuberculosis. If the [PPD](#) skin test is positive or if granulomatous hepatitis or other granulomatous disease is present with anergy (and sarcoid seems unlikely), then a therapeutic trial with isoniazid and rifampin (and possibly a third drug) should be undertaken, with treatment usually continued for up to 6 weeks. A failure of the fever to respond over this period suggests an alternative diagnosis.

The response of rheumatic fever and Still's disease to aspirin and nonsteroidal anti-inflammatory agents (NSAIDs) may be dramatic. The effects of glucocorticoids on temporal arteritis, polymyalgia rheumatica, and granulomatous hepatitis are equally dramatic. Colchicine is highly effective in preventing attacks of familial Mediterranean fever but is of little use once an attack is well under way. The ability of glucocorticoids and NSAIDs to mask fever while permitting the spread of infection dictates that their use be avoided unless infection has been largely ruled out and unless inflammatory disease is both probable and debilitating or threatening.

When no underlying source of [FUO](#) is identified after prolonged observation (>6 months), the prognosis is generally good, however vexing the fever may be to the patient. Under such circumstances, debilitating symptoms are treated with [NSAIDs](#), and glucocorticoids are the last resort. The initiation of empirical therapy does not mark the end of the diagnostic workup; rather, it commits the physician to continued thoughtful reexamination and evaluation. Patience, compassion, equanimity, and intellectual flexibility are indispensable attributes for the clinician in dealing successfully with FUO.

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Sheldon M. Wolff, MD, now deceased, was an author of a previous version of this chapter. It is to his memory that the chapter is dedicated.

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126. INFECTIVE ENDOCARDITIS - Adolf W. Karchmer

INTRODUCTION

The proliferation of microorganisms on the endothelium of the heart results in infective endocarditis. The prototypic lesion at the site of infection, the *vegetation* (see [Plate IID-59, Fig. 126-CD1](#)), is a mass of platelets, fibrin, microcolonies of microorganisms, and scant inflammatory cells. Infection most commonly involves heart valves (either native or prosthetic) but may also occur on the low-pressure side of the ventricular septum at the site of a defect, on the mural endocardium where it is damaged by aberrant jets of blood or foreign bodies, or on intracardiac devices themselves. The analogous process involving arteriovenous shunts, arterioarterial shunts (patent ductus arteriosus), or a coarctation of the aorta is called *infective endarteritis*.

Endocarditis may be classified according to the temporal evolution of disease, the site of infection, the cause of infection, or a predisposing risk factor such as injection drug use. While each classification criterion provides therapeutic and prognostic insight, the methods overlap and none is sufficient alone. The classification of endocarditis as acute and subacute was initially used to describe the illness and the time elapsed until death; presently it is applied to the features and progression of infection until diagnosis. *Acute endocarditis* is a hectically febrile illness, rapidly damages cardiac structures, hematogenously seeds extracardiac sites, and, if untreated, progresses to death within weeks. *Subacute endocarditis* follows an indolent course; causes structural cardiac damage only slowly, if at all; rarely causes metastatic infection; and is gradually progressive unless complicated by a major embolic event or ruptured mycotic aneurysm.

In developed countries, the incidence of endocarditis ranges from 1.5 to 6.2 cases per 100,000 population per year. In the late 1980s in a metropolitan area of the United States (Philadelphia), endocarditis occurred in 9.3 persons per 100,000 population per year. However, half of these cases arose as a consequence of injection drug use. The incidence of endocarditis is notably increased among the elderly. The cumulative rate of prosthetic valve endocarditis is 1.5 to 3.0% at 1 year after valve replacement and 3 to 6% at 5 years; the risk is greatest during the first 6 months after valve replacement.

ETIOLOGY

A vast array of microorganisms, including many species of bacteria and fungi, have been reported to cause sporadic episodes of endocarditis. Nevertheless, a small number of bacterial species cause the majority of cases ([Table 126-1](#)). The causative microorganisms vary somewhat among the major clinical types of endocarditis, in part because of the different portals of entry. The oral cavity, skin, and upper respiratory tract are the respective primary portals for the viridans streptococci, staphylococci, and HACEK organisms (*Haemophilus*, *Actinobacillus*, *Cardiobacterium*, *Eikenella*, and *Kingella*) causing community-acquired native valve endocarditis. *Streptococcus bovis* originates from the gastrointestinal tract, where it is associated with polyps and colonic tumors, and enterococci enter the bloodstream from the genitourinary tract. Nosocomial native valve endocarditis is largely the consequence of bacteremia arising from intravascular catheters and less commonly from nosocomial wound and urinary tract

infection. Endocarditis complicates 6 to 25% of episodes of catheter-associated *Staphylococcus aureus* bacteremia; higher rates are detected by careful transesophageal echocardiography (TEE) screening (see "Echocardiography," below).

Prosthetic valve endocarditis arising within 2 months of valve surgery is generally the result of intraoperative contamination of the prosthesis or a bacteremic postoperative complication. The nosocomial nature of these infections is reflected in their primary microbial causes: coagulase-negative staphylococci, *S. aureus*, facultative gram-negative bacilli, diphtheroids, and fungi. The portals of entry and organisms causing cases beginning >12 months after surgery are similar to those in community-acquired native valve endocarditis. Epidemiologic evidence suggests that prosthetic valve endocarditis due to coagulase-negative staphylococci that presents between 2 and 12 months after surgery is often nosocomial in origin but with a delayed onset. At least 85% of coagulase-negative staphylococci that cause prosthetic valve endocarditis within 12 months of surgery are methicillin-resistant; the rate of methicillin resistance decreases to 25% among coagulase-negative staphylococci causing prosthetic endocarditis that presents >1 year after valve surgery.

Transvenous pacemaker lead and/or implanted defibrillator-associated endocarditis is usually a nosocomial infection. The majority of episodes occur within weeks of implantation or generator change and are caused by *S. aureus* or coagulase-negative staphylococci.

Endocarditis occurring among injection drug users, especially when infection involves the tricuspid valve, is commonly caused by *S. aureus* strains, many of which are methicillin-resistant. The causes of left-sided valve infection in addicts are more varied, and the involved valves have often been damaged by prior episodes of endocarditis. A number of these cases are caused by *Pseudomonas aeruginosa* and *Candida* species, and sporadic cases are due to unusual organisms such as *Bacillus*, *Lactobacillus*, and *Corynebacterium* species. Polymicrobial endocarditis occurs more frequently in injection drug users than in patients who do not inject drugs. The presence of HIV in this population does not significantly impact the causes of endocarditis.

From 5 to 15% of patients with endocarditis have negative blood cultures; in one-third to one-half of these cases, cultures are negative because of prior antibiotic exposure. The remainder of these patients are infected by fastidious organisms, such as pyridoxal-requiring streptococci (now designated *Abiotrophia* species), the gram-negative coccobacillary HACEK organisms, *Bartonella henselae*, or *Bartonella quintana*. Some fastidious organisms that cause endocarditis have characteristic epidemiologic settings (e.g., *Coxiella burnetii* in Europe, *Brucella* species in the Middle East). *Tropheryma whippelii* causes an indolent, culture-negative, afebrile form of endocarditis.

PATHOGENESIS

Unless it is injured, the normal endothelium is resistant to infection by most bacteria and to thrombus formation. Endothelial injury (e.g., at the site of impact of high-velocity jets or on the low-pressure side of a cardiac structural lesion) causes aberrant flow and allows either direct infection by virulent organisms or the development of an uninfected

platelet-fibrin thrombus -- a condition called *nonbacterial thrombotic endocarditis* (NBTE). The thrombus subsequently serves as a site of bacterial attachment during transient bacteremia. The cardiac lesions most commonly resulting in NBTE are mitral regurgitation, aortic stenosis, aortic regurgitation, ventricular septal defects, and complex congenital heart disease. These lesions result from rheumatic heart disease (particularly in the developing world, where rheumatic fever remains prevalent), mitral valve prolapse, degenerative heart disease, and congenital malformations. NBTE also arises as a result of a hypercoagulable state; this phenomenon gives rise to the clinical entity of *marantic endocarditis* (seen in patients with malignancy) and to bland vegetations complicating systemic lupus erythematosus and the antiphospholipid antibody syndrome.

Organisms that cause endocarditis generally enter the bloodstream from mucosal surfaces, the skin, or sites of focal infection. Except for more virulent bacteria (e.g., *S. aureus*) that can adhere directly to intact endothelium or exposed subendothelial tissue, microorganisms in the blood adhere to thrombi. If resistant to the bactericidal activity of serum and the microbicidal peptides released by platelets, the organisms proliferate and either stimulate tissues to produce a procoagulant or themselves exert procoagulant activity leading to further platelet-fibrin deposition and vegetation formation. Although an enormous variety of microorganisms circulate transiently in the bloodstream, only a limited number commonly cause endocarditis. The etiologic organisms of endocarditis bear surface components that facilitate adherence to injured endothelium and host proteins or to thrombi. Experiments suggest that fibronectin receptors present on many gram-positive bacteria, clumping factor (a fibrinogen-binding surface protein) on *S. aureus*, and dextrans on streptococci facilitate adherence. Organisms become enmeshed in the growing platelet-fibrin vegetation and, in the absence of host defenses, proliferate to form dense microcolonies. More than 90% of the organisms in vegetations are metabolically inactive (nongrowing) and thus are relatively resistant to killing by antimicrobial agents. Proliferating surface organisms are shed into the bloodstream continuously, whereupon some are cleared by the reticuloendothelial system and others are redeposited on the vegetation and stimulate further vegetation growth.

The pathophysiologic consequences and clinical manifestations of endocarditis -- other than constitutional symptoms, which are probably a result of cytokine production -- arise from damage to intracardiac structures; embolization of vegetation fragments, leading to infection or infarction of remote tissues; hematogenous infection of sites during bacteremia; and tissue injury due to the deposition of circulating immune complexes or immune responses to deposited bacterial antigens.

CLINICAL MANIFESTATIONS

The clinical syndrome of infective endocarditis is highly variable, may involve multiple organs, and spans a continuum between acute and subacute presentations. Native valve endocarditis (whether acquired in the community or nosocomially), prosthetic valve endocarditis, and endocarditis due to injection drug use share clinical and laboratory manifestations ([Table 126-2](#)). Although the relationship is not absolute, the causative microorganism is primarily responsible for the temporal course of endocarditis. β -Hemolytic streptococci, *S. aureus*, and pneumococci typically result in an acute course, although *S. aureus* occasionally causes subacute disease.

Endocarditis caused by *Staphylococcus lugdunensis* (a coagulase-negative species) or by enterococci may present acutely. Subacute endocarditis is typically caused by viridans streptococci, enterococci, coagulase-negative staphylococci, and the HACEK group. Endocarditis caused by *Bartonella* species and the agent of Q fever, *C. burnetii*, is exceptionally indolent.

The clinical features of endocarditis are nonspecific. However, these symptoms in a febrile patient with valvular abnormalities or a behavior pattern (injection drug use) that predisposes to endocarditis suggest the diagnosis, as do bacteremia with organisms that frequently cause endocarditis, otherwise unexplained arterial emboli, and progressive cardiac valvular incompetence. In patients with subacute presentations, fever is typically low-grade and rarely exceeds 39.4°C (103°F); in contrast, temperatures between 39.4 and 40°C (103 and 104°F) are often noted in acute endocarditis. Fever may be blunted or absent in patients who are elderly or severely debilitated or who have marked cardiac or renal failure.

Cardiac Manifestations Although heart murmurs are usually indicative of the predisposing cardiac pathology rather than of endocarditis, valvular damage and ruptured chordae may result in new regurgitant murmurs. In acute endocarditis involving a normal valve, murmurs are heard on presentation in only 30 to 45% of patients but ultimately are detected in 85%. Congestive heart failure develops in 30 to 40% of patients; it is usually a consequence of valvular dysfunction but occasionally is due to endocarditis-associated myocarditis or an intracardiac fistula. The temporal progression of heart failure is variable and depends upon the severity of valvular dysfunction; failure due to aortic valve dysfunction progresses more rapidly than that due to mitral valve dysfunction. Extension of infection beyond valve leaflets into adjacent annular or myocardial tissue results in perivalvular abscesses, which in turn may cause fistulae (from the root of the aorta into cardiac chambers or between cardiac chambers) with new murmurs. Abscesses may burrow from the aortic valve annulus through the epicardium, causing pericarditis. Extension of infection into paravalvular tissue adjacent to either the right or the noncoronary cusp of the aortic valve may interrupt the conduction system in the upper interventricular septum, leading to varying degrees of heart block. Although perivalvular abscesses arising from the mitral valve may potentially interrupt conduction pathways near the atrioventricular node or in the proximal bundle of His, such interruption occurs infrequently. Emboli to a coronary artery may result in myocardial infarction; nevertheless, embolic transmural infarcts are rare.

Noncardiac Manifestations ([Figs. 19-CD2, 124-CD2, and 126-CD2](#)) The classic nonsuppurative peripheral manifestations of subacute endocarditis are related to the duration of infection and, with early diagnosis and treatment, have become infrequent. In contrast, septic embolization mimicking some of these lesions (subungual hemorrhage, [Fig. 19-CD4](#)). Osler's nodes, [Fig. 126-CD3](#)) is common in patients with acute *S. aureus* endocarditis (see [Plate IID-58](#)). Musculoskeletal symptoms, including nonspecific inflammatory arthritis and back pain, usually remit promptly with treatment but must be distinguished from the symptoms of focal metastatic infection. Hematogenously seeded focal infection may involve any organ but most often is clinically evident in the skin, spleen, kidneys, skeletal system, and meninges. Arterial emboli, which may be asymptomatic and discovered only at autopsy, are clinically

apparent in up to 50% of patients. Vegetations >10 mm in diameter (as measured by echocardiography) and those located on the mitral valve are more likely to embolize than are smaller or nonmitral vegetations. Embolic events -- often with infarction -- involving the extremities, spleen, kidneys ([Fig. 126-1](#)), bowel, or brain are often noted at presentation. With antibiotic treatment, the frequency of embolic events decreases from 13 per 1000 patient-days during the initial week to 1.2 per 1000 patient-days after the third week. Emboli occurring late during or after effective therapy do not in themselves constitute evidence of failed antimicrobial treatment. Neurologic symptoms, most often resulting from embolic strokes, occur in up to 40% of patients. Other neurologic complications include aseptic or purulent meningitis, intracranial hemorrhage due to hemorrhagic infarcts or ruptured mycotic aneurysms, seizures, and encephalopathy. Microabscesses in brain and meninges occur commonly in *S. aureus* endocarditis; surgically drainable abscesses are infrequent.

Immune complex deposition on the glomerular basement membrane causes diffuse hypocomplementemic glomerulonephritis and renal dysfunction, which typically improve with effective antimicrobial therapy. Embolic renal infarcts cause flank pain and hematuria but rarely cause renal dysfunction.

Manifestations of Specific Predisposing Conditions Among injection drug users, infection involving valves on the left side of the heart presents with the typical clinical features of endocarditis. In almost 50% of patients with endocarditis associated with injection drug use, infection is limited to the tricuspid valve. These patients present with fever, faint or no murmur, and (in 75% of cases) prominent pulmonary findings, including cough, pleuritic chest pain, nodular pulmonary infiltrates, and occasionally pyopneumothorax.

Nosocomial endocarditis (defined as that which results from hospital care within the prior month and most commonly presenting as intravascular catheter-associated bacteremia), if not associated with a retained intracardiac device, has typical manifestations. Endocarditis associated with flow-directed pulmonary artery catheters is often cryptic, with symptoms masked by comorbid critical illness, and is commonly diagnosed at autopsy. Transvenous pacemaker lead and/or implanted defibrillator-associated endocarditis commonly follows initial implantation or a generator unit change; may be associated with obvious or cryptic generator pocket infection; and results in fever, minimal murmur, and pulmonary symptoms similar to those encountered in addicts with tricuspid endocarditis.

Prosthetic valve endocarditis presents with typical clinical features. Cases arising within 60 days of valve surgery (early onset) lack peripheral vascular manifestations and may be obscured by comorbidity associated with recent surgery. In both early-onset and more delayed presentations, paravalvular infection is common and often results in partial valve dehiscence, regurgitant murmurs, congestive heart failure, or disruption of the conduction system.

DIAGNOSIS

The Duke Criteria The diagnosis of infective endocarditis is established with certainty only when vegetations obtained at cardiac surgery, at autopsy, or from an artery (an

embolus) are examined histologically and microbiologically. Nevertheless, a highly sensitive and specific diagnostic schema -- known as the *Duke criteria* -- has been developed on the basis of clinical, laboratory, and echocardiographic findings ([Table 126-3](#)). Documentation of two major criteria, of one major and three minor criteria, or of five minor criteria allows a clinical diagnosis of definite endocarditis. The diagnosis of endocarditis is rejected if an alternative diagnosis is established, if symptoms resolve and do not recur with £4 days of antibiotic therapy, or if surgery or autopsy after £4 days of antimicrobial therapy yields no histologic evidence of endocarditis. Illnesses not classified as definite endocarditis or rejected are considered cases of possible infective endocarditis. When pathologically confirmed cases have been scored retrospectively by these criteria, 90% fulfill the definition of definite or possible endocarditis; 10% are rejected (primarily because of an incomplete echocardiographic evaluation). In comparison with expert opinion, the Duke criteria identify cases considered to be endocarditis but also accept a small percentage of cases rejected by the experts. This potential for a false-positive diagnosis is the major deficiency in this schema when used clinically. If all patients with a diagnosis of definite or possible endocarditis are fully treated for endocarditis, this reduced specificity results in excess treatment for some patients. A modification of the schema has been proposed in order to increase its specificity without significantly reducing its sensitivity. This modification would require documentation of at least one major or three minor criteria for cases to be categorized as possible endocarditis.

The roles of bacteremia and echocardiographic findings in the diagnosis of endocarditis are appropriately emphasized in the Duke criteria. That multiple blood cultures obtained over time are positive is consistent with the known continuous low-density nature of bacteremia in patients with endocarditis (£100 organisms per milliliter). Among untreated endocarditis patients who ultimately have a positive blood culture, 95% of all blood cultures are positive, and in 98% of cases one of the initial two sets of cultures yields the microorganism. The diagnostic criteria attach significance to the species of organism isolated from blood cultures. To fulfill a major criterion, the isolation of an organism that causes both endocarditis and bacteremia in the absence of endocarditis (e.g., *S. aureus*, enterococci) must take place repeatedly (i.e., persistent bacteremia) and in the absence of a primary focus of infection. Organisms that rarely cause endocarditis but commonly contaminate blood cultures (e.g., diphtheroids, coagulase-negative species) must be isolated repeatedly if their isolation is to serve as a major criterion.

Blood Cultures Isolation of the causative microorganism from blood cultures is critical not only for diagnosis but also for determination of antimicrobial susceptibility and planning of treatment. In the absence of prior antibiotic therapy, a total of three blood culture sets, ideally with the first separated from the last by at least 1 h, should be obtained from different venipuncture sites over 24 h. If the cultures remain negative after 48 to 72 h, two or three additional blood cultures, including a lysis-centrifugation culture, should be obtained, and the laboratory should be asked to pursue fastidious microorganisms by prolonging incubation time and performing special subcultures. Empirical antimicrobial therapy should not be administered initially to hemodynamically stable patients with subacute endocarditis, especially those who have received antibiotics within the preceding 2 weeks; thus, if necessary, additional blood cultures can be obtained without the confounding effect of empirical treatment. Patients with

acute endocarditis or with deteriorating hemodynamics that may require urgent surgery should be treated empirically immediately after obtaining the initial three sets of blood cultures.

Non-Blood-Culture Tests for the Etiologic Agent Serologic tests can be used to identify some organisms causing endocarditis that are difficult to recover by blood culture: *Brucella*, *Bartonella*, *Legionella*, and *C. burnetii*. Pathogens can also be identified in vegetations by culture, by microscopic examination with special stains, and by use of polymerase chain reaction to recover unique microbial DNA or 16S rRNA.

Echocardiography Cardiac imaging with echocardiography allows anatomic confirmation of infective endocarditis, sizing of vegetations, detection of intracardiac complications, and assessment of cardiac function. A two-dimensional study with color flow and continuous as well as pulsed Doppler is optimal. Transthoracic echocardiography (TTE) is noninvasive and exceptionally specific; however, it cannot image vegetations <2 mm in diameter, and in 20% of patients it is technically inadequate because of emphysema or body habitus. Thus, TTE detects vegetations in only 65% of patients with definite clinical endocarditis (i.e., it has a sensitivity of 65%). Moreover, TTE is not adequate for evaluating prosthetic valves or detecting intracardiac complications. Transesophageal echocardiography (TEE) is safe and significantly more sensitive than TTE. It detects vegetations in >90% of patients with definite endocarditis; nevertheless, false-negative studies are noted in 6 to 18% of endocarditis patients. TEE is the optimal method for the diagnosis of prosthetic endocarditis or the detection of myocardial abscess, valve perforation, or intracardiac fistulae.

Experts favor echocardiographic evaluation of all patients with a clinical diagnosis of endocarditis; however, the test should not be used to screen patients with otherwise explained positive blood cultures or patients with unexplained fever. In patients with a low pretest likelihood of endocarditis (<5%), a high-quality TTE that is negative is sufficient to exclude endocarditis. For patients whose habitus makes them difficult to study with TTE and for those who may have prosthetic valve endocarditis or who are at high risk of intracardiac complications, TEE is the preferred imaging modality. For patients with a pretest probability of endocarditis ranging from 5 to 50%, initial evaluation by TEE -- in lieu of a sequential strategy of TTE, which, if negative, will be followed by TEE -- is cost-effective. A negative TEE when endocarditis is likely does not exclude the diagnosis but rather warrants repetition of the study in 7 to 10 days with optimal multiplanar technique.

Other Studies Many laboratory studies that do not aid in diagnostic evaluation are nevertheless important in the management of patients with endocarditis; these studies include complete blood counts, creatinine measurement, chest radiography, and electrocardiography. The erythrocyte sedimentation rate, C-reactive protein level, circulating immune complex titer, and rheumatoid factor concentration are commonly increased in endocarditis (Table 126-2). Cardiac catheterization is useful only to assess coronary artery patency in older individuals who are to undergo surgery for endocarditis.

TREATMENT

Antimicrobial Therapy It is difficult to eradicate bacteria from the avascular vegetation

in infective endocarditis because this site is relatively inaccessible to host defenses and because the bacteria are nongrowing and metabolically inactive. Since all bacteria in the vegetation must be killed, therapy for endocarditis must be bactericidal and must be given for prolonged periods. Antibiotics are generally given parenterally and must reach high serum concentrations that will, through passive diffusion, lead to effective concentrations in the depths of the vegetation. The choice of effective therapy requires precise knowledge of the susceptibility of the causative microorganisms. The initiation of treatment before a cause is defined must balance the need to establish a microbiologic diagnosis against the potential progression of disease or the need for urgent surgery (see "Blood Cultures" above). The individual vulnerabilities of the patient should be weighed in the selection of therapy -- e.g., allergies, end-organ dysfunction, interactions with concomitant medications, and risks of adverse events.

Although given for several weeks longer, the regimens recommended for the treatment of endocarditis involving prosthetic valves (except for staphylococcal infections) are similar to those used to treat native valve infection ([Table 126-4](#)). Recommended doses and duration of therapy should be adhered to unless alterations are required by adverse events.

Organism-Specific Therapies

STREPTOCOCCI Although most strains of viridans streptococci and *S. bovis* that cause endocarditis are susceptible to penicillin [minimum inhibitory concentration (MIC) ≤ 0.1 $\mu\text{g/mL}$], recent reports indicate increasing penicillin resistance among viridans streptococci recovered from blood cultures. In the selection of optimal therapy, the penicillin MIC must be determined ([Table 126-4](#)). The 2-week penicillin/gentamicin regimen should not be used to treat complicated native valve infection or prosthetic valve endocarditis. Although small studies have suggested that a 2-week regimen of single daily doses of ceftriaxone (2 g IV) plus gentamicin (3 mg/kg) or netilmicin (4 mg/kg) is effective for penicillin-susceptible streptococcal endocarditis, the data are not sufficient to support routine use of this regimen. Penicillin/gentamicin is recommended for the treatment of endocarditis caused by group B streptococci.

ENTEROCOCCI Enterococci are resistant to oxacillin, nafcillin, and the cephalosporins and are inhibited only by penicillin, ampicillin, teicoplanin (not available in the United States), and vancomycin. To kill enterococci requires the synergistic interaction of a cell wall-active antibiotic (penicillin, ampicillin, vancomycin, or teicoplanin) that is effective at achievable serum concentrations and an aminoglycoside (gentamicin or streptomycin) to which the isolate does not exhibit high-level resistance. An isolate's resistance to cell wall-active agents or ability to replicate in the presence of gentamicin at ≥ 500 $\mu\text{g/mL}$ or streptomycin at 2000 $\mu\text{g/mL}$ -- a phenomenon called *high-level aminoglycoside resistance* -- indicates that the ineffective antimicrobial cannot participate in the interaction to produce killing. High-level resistance to gentamicin predicts that tobramycin, netilmicin, amikacin, and kanamycin will also be ineffective. In fact, even when enterococci are not highly resistant to gentamicin, it is difficult to predict the ability of these other aminoglycosides to participate in synergistic killing; consequently, they should not in general be used to treat enterococcal endocarditis.

Clearly, enterococci causing endocarditis must be tested for high-level resistance to

streptomycin and gentamicin, b-lactamase production, and susceptibility to penicillin and ampicillin (MIC, £16 ug/mL) and to vancomycin (MIC, £8 ug/mL). If the isolate produces b-lactamase, ampicillin/sulbactam or vancomycin can be used as the cell wall-active component; if the penicillin/ampicillin MIC is >16 ug/mL, vancomycin can be considered; and if the vancomycin MIC is >8 ug/mL, penicillin or ampicillin may be considered. Based on the absence of high-level resistance, gentamicin or streptomycin should be used as the aminoglycoside. If there is high-level resistance to both these drugs, no aminoglycoside should be given; instead, an 8- to 12-week course of a single cell wall-active agent is suggested. If single-drug therapy fails or the isolate is resistant to all of the commonly used agents, surgical treatment is advised. The role of newer agents potentially active against multidrug-resistant enterococci (quinupristin/dalfopristin, linezolid, and daptomycin) in the treatment of endocarditis has not been established.

STAPHYLOCOCCI The regimens used to treat staphylococcal endocarditis are not based upon coagulase production but rather upon the presence or absence of a prosthetic valve or foreign device, the native valve(s) involved, and the resistance of the isolate to penicillin and methicillin. Penicillinase is produced by 95% of staphylococci; thus, all isolates should be considered penicillin-resistant until shown not to produce this enzyme. The addition of gentamicin (if the isolate is susceptible) to a b-lactam antibiotic to enhance therapy for native mitral or aortic valve endocarditis is optional. Its addition hastens eradication of bacteremia but does not improve survival rates. If added, gentamicin should be limited to the initial 3 to 5 days of therapy to avoid nephrotoxicity. Gentamicin generally is not added to the vancomycin regimen in this setting.

Methicillin-susceptible *S. aureus* endocarditis that is uncomplicated and limited to the tricuspid or pulmonic valve -- a condition occurring almost exclusively in injection drug users -- can often be treated with a 2-week course that combines oxacillin or nafcillin (but not vancomycin) with gentamicin. Prolonged fevers (>5 days) during therapy suggest that these patients should receive standard therapy.

Staphylococcal prosthetic valve endocarditis is treated for 6 to 8 weeks with a multidrug regimen. Rifampin is an essential component because it kills staphylococci that are adherent to foreign material. Two other agents (selected on the basis of susceptibility testing) are combined with rifampin to prevent in vivo emergence of resistance. Because many staphylococci, particularly methicillin-resistant *S. aureus* and *S. epidermidis*, are resistant to gentamicin, the utility of gentamicin should be established before rifampin treatment is begun. If the isolate is resistant to gentamicin, another aminoglycoside or a fluoroquinolone (chosen in light of susceptibility results) should be substituted.

OTHER ORGANISMS Although penicillin is the therapy of choice for endocarditis caused by *S. pneumoniae*, therapy should be initiated with ceftriaxone and vancomycin until susceptibility to penicillin is established. *P. aeruginosa* endocarditis is treated with an antipseudomonal penicillin (ticarcillin or piperacillin) and high doses of tobramycin (8 mg/kg per day in three divided doses). Endocarditis caused by Enterobacteriaceae is treated with a potent b-lactam antibiotic plus an aminoglycoside. Corynebacterial endocarditis is treated with penicillin plus an aminoglycoside (if the organism is susceptible to the aminoglycoside) or with vancomycin, which is highly bactericidal for most strains. Therapy for *Candida* endocarditis consists of amphotericin B plus

flucytosine and early surgery; long-term (if not indefinite) suppression with fluconazole is used increasingly.

Empirical Therapy In designing and executing therapy without culture data (i.e., before culture results are known or when cultures are negative), clinical and epidemiologic clues to etiology must be weighed, and both the pathogens associated with the specific endocarditis syndrome and the hazards of suboptimal therapy must be considered. Thus, empirical therapy for acute endocarditis in an injection drug user should cover methicillin-resistant *S. aureus* and gram-negative bacilli. The initiation of treatment with vancomycin plus gentamicin immediately after blood is obtained for cultures covers these as well as many other potential causes. In treating culture-negative episodes, marantic endocarditis must be excluded and fastidious organisms sought serologically. In the absence of confounding prior antibiotic therapy, it is unlikely that *S. aureus* or enterococcal infection will present with negative blood cultures. Thus, in this situation, these organisms are not the determinants of therapy for subacute endocarditis. Blood culture-negative native valve endocarditis is treated with ceftriaxone (or ampicillin) plus gentamicin; these two antimicrobials plus vancomycin should be used if prosthetic valves are involved.

Outpatient Antimicrobial Therapy Fully compliant patients who have sterile blood cultures, are afebrile during therapy, and have no clinical or echocardiographic findings that suggest an impending complication may complete therapy as outpatients. Careful follow-up and a stable home setting are necessary, as are predictable intravenous access and selection of antimicrobials that are stable in solution.

Monitoring Antimicrobial Therapy The serum bactericidal titer -- the highest dilution of the patient's serum during therapy that kills 99.9% of the standard inoculum of the infecting organism -- is no longer recommended for assessment of patients receiving standard regimens. However, in the treatment of endocarditis caused by unusual organisms, this measurement, although not standardized and difficult to interpret, may provide a patient-specific assessment of in vivo antibiotic effect. Serum concentrations of aminoglycosides and vancomycin should be monitored.

Antibiotic toxicities, including allergic reactions, occur in 25 to 40% of patients and commonly arise during the third week of therapy. Blood tests to detect antibiotic-specific potential end-organ toxicity should be performed periodically.

In most patients effective antibiotic therapy results in subjective improvement and resolution of fever within 5 to 7 days. Blood cultures should be repeated daily until sterile, rechecked if there is recrudescence fever, and performed again 4 to 6 weeks after therapy to document cure. Blood cultures become sterile within 2 days after the start of appropriate therapy when infection is caused by viridans streptococci, enterococci, or HACEK organisms. In *S. aureus* endocarditis, b-lactam therapy results in sterile cultures in 3 to 5 days, whereas positive cultures may persist for 7 to 9 days with vancomycin treatment. When fever persists for 7 days in spite of appropriate antibiotic therapy, patients should be evaluated for paravalvular abscess and for extracardiac abscesses (spleen, kidney) or complications (embolic events). Recrudescence fever raises the question of these complications but also of drug reactions or complications of hospitalization. Serologic abnormalities (e.g., erythrocyte sedimentation rate,

rheumatoid factor) resolve slowly and do not reflect response to treatment. Vegetations become smaller with effective therapy, but at 3 months after cure half are unchanged and 25% are slightly larger.

Surgical Treatment Intracardiac and central nervous system complications of endocarditis are important causes of the morbidity and mortality associated with this infection. In some cases, effective treatment for these complications requires surgery. Most of the clinical indications for surgical treatment of endocarditis are not absolute ([Table 126-5](#)). The risks and benefits as well as the timing of surgical treatment must therefore be individualized.

Intracardiac Surgical Indications Most surgical interventions are clearly warranted by intracardiac findings, often detected by echocardiography. Because of the highly invasive nature of prosthetic valve endocarditis, as many as 40% of affected patients merit surgical treatment. In many patients, coincident rather than single intracardiac events necessitate surgery.

CONGESTIVE HEART FAILURE Moderate to severe refractory congestive heart failure caused by new or worsening valve dysfunction is the major indication for cardiac surgical treatment of endocarditis. Of patients with moderate to severe heart failure due to valve dysfunction who are treated medically, 60 to 90% die within 6 months. In the setting of similar hemodynamic dysfunction, surgical treatment is associated with mortality rates of 20 to 40% with native valve endocarditis and 35 to 55% with prosthetic valve infection. Surgery may be required to relieve functional stenosis due to large vegetations or to restore competence to damaged regurgitant valves.

PERIVALVULAR INFECTION This complication, which occurs in 10 to 15% of native valve and 45 to 60% of prosthetic valve infections, is suggested by persistent unexplained fever during appropriate therapy, new electrocardiographic conduction disturbances, and pericarditis. Extension can occur from any valve but is most common with aortic valve infection. [TEE](#) with color Doppler is the test of choice to detect perivalvular abscesses (sensitivity, 85%). Although occasional perivalvular infections are cured medically, surgery is warranted when fever persists, fistulae develop, prostheses are dehiscent and unstable, and infection relapses after appropriate treatment. Cardiac rhythm must be monitored since high-grade heart block may require insertion of a pacemaker.

UNCONTROLLED INFECTION Continued positive blood cultures or otherwise unexplained persistent fevers (in patients with either blood culture-positive or -negative endocarditis) despite optimal antibiotic therapy may reflect uncontrolled infection and warrant surgery. Surgical treatment is also advised for endocarditis caused by those organisms against which clinical experience indicates that effective antimicrobial therapy is lacking. This category includes infections caused by yeasts, fungi, *P. aeruginosa*, other highly resistant gram-negative bacilli, *Brucella* species, and probably *C. burnetii*.

S. AUREUS ENDOCARDITIS Mortality rates for *S. aureus* prosthetic valve endocarditis exceed 70% with medical treatment but are reduced to 25% with surgical treatment. In patients with intracardiac complications associated with *S. aureus* prosthetic valve

infection, surgical treatment reduces mortality by twentyfold. Surgical treatment should be considered for patients with *S. aureus* native aortic or mitral valve infection who have [TTE](#)-demonstrable vegetations and remain septic during the initial week of therapy. Isolated tricuspid valve endocarditis, even with persistent fever, rarely requires surgery.

PREVENTION OF SYSTEMIC EMBOLI Mortality and persisting morbidity due to emboli are largely limited to patients suffering occlusion of cerebral or coronary arteries. Echocardiographic determination of vegetation size and anatomy, although predictive of patients at high risk of systemic emboli, does not identify those patients in whom the benefits of surgery to prevent emboli clearly exceed the risks of the surgical procedure. Net benefits favoring surgery are most likely when the risk of embolism is high and other surgical benefits can be achieved simultaneously -- e.g., repair of a moderately dysfunctional valve or debridement of a paravalvular abscess.

Timing of Cardiac Surgery Surgery to correct valvular dysfunction and progressive congestive heart failure should not be delayed simply to permit additional antibiotic therapy, since this course of action increases the risk of mortality. Similarly, surgery should not be delayed when the indication is uncontrolled or perivalvular infection. Delay is justified only when infection is controlled and congestive heart failure is fully compensated with medical therapy. Recrudescence of endocarditis involving a prosthetic valve follows surgery in 2% of patients with culture-positive native valve endocarditis and 15% of patients with active prosthetic valve endocarditis. These risks are more acceptable than the high mortality rates that result when surgery is inappropriately delayed or not performed.

Among patients who have experienced a neurologic complication of endocarditis, further neurologic deterioration can occur as a consequence of cardiac surgery. The risk of significant neurologic exacerbation is related to the interval between the complication and surgery. Where feasible, cardiac surgery should be delayed for 2 to 3 weeks after a nonhemorrhagic embolic stroke and for 4 weeks after a hemorrhagic embolic stroke. A ruptured mycotic aneurysm should be clipped and cerebral edema allowed to resolve prior to cardiac surgery.

Extracardiac Complications Splenic abscess develops in 3 to 5% of patients with endocarditis. Effective therapy requires either computed tomography-guided percutaneous drainage or splenectomy. Mycotic aneurysms occur in 2 to 15% of endocarditis patients; half of these cases involve the cerebral arteries and present as headaches, focal neurologic symptoms, or hemorrhage. Cerebral aneurysms should be monitored by angiography. Some will resolve, but those that persist, enlarge, or leak should be treated surgically if possible. Extracerebral aneurysms present as local pain, a mass, local ischemia, or bleeding; generally these aneurysms are treated by resection.

OUTCOME

The outcome of infective endocarditis is affected by a variety of factors, some of which are interrelated. Factors with an adverse impact include older age, severe comorbid conditions, delayed diagnosis, involvement of prosthetic valves or the aortic valve, an invasive (*S. aureus*) or antibiotic-resistant (*P. aeruginosa*, yeast) pathogen, intracardiac

complications, and major neurologic complications. Death and poor outcome often are related not to failure of antibiotic therapy but rather to the interactions of comorbidities and endocarditis-related end-organ complications. The overall survival rate for patients with native valve endocarditis caused by viridans streptococci, HACEK organisms, or enterococci (susceptible to synergistic therapy) ranges from 85 to 90%. For *S. aureus* native valve endocarditis in patients who do not inject drugs, survival rates are 55 to 70%, whereas 85 to 90% of injection drug users survive this infection. Prosthetic valve endocarditis beginning within 2 months of valve replacement results in mortality rates of 40 to 50%, whereas rates are only 10 to 20% in later-onset cases.

PREVENTION

Antibiotics have been administered in conjunction with selected procedures considered to entail a risk for bacteremia and endocarditis. The benefits of antibiotic prophylaxis are not established and in fact may be modest: only 50% of patients with native valve endocarditis knew that they had a valve lesion predisposing to infection, most endocarditis cases do not follow a procedure, and 35% of cases are caused by organisms not targeted by prophylaxis. Dental treatments, the procedures most widely accepted as predisposing to endocarditis, are no more frequent during the 3 months preceding this diagnosis than in uninfected matched controls. Nevertheless, an expert committee of the American Heart Association, along with similar advisory groups in other developed countries, has identified procedures that may precipitate bacteremia with organisms that cause endocarditis ([Table 126-6](#)), patients who should receive prophylaxis based on the relative risk for developing endocarditis and the severity of subsequent infection ([Table 126-7](#)), and regimens that may be used for prophylaxis ([Table 126-8](#)). Except for an isolated secundum atrial septal defect and a totally corrected patent ductus arteriosus, ventricular septal defect, or pulmonary stenosis, patients with congenital heart defects continue to experience high rates of endocarditis despite total surgical correction of the defect. In vulnerable patients, maintaining good dental hygiene and aggressively treating local infections may reduce the risk of endocarditis.

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127. INFECTIOUS COMPLICATIONS OF BITES AND BURNS - Lawrence C. Madoff

The skin is an essential component of the nonspecific immune system, protecting the host from potential pathogens in the environment. Breaches in this protective barrier thus represent a form of immunocompromise that predisposes the patient to infection. Bites and scratches from animals and humans allow the inoculation of microorganisms past the skin's protective barrier into deeper, susceptible host tissues. Thermal burns may cause massive destruction of the integument as well as derangements in humoral and cellular immunity, enabling environmental opportunists and components of the host's own skin flora to cause infection.

ANIMAL BITES AND SCRATCHES

Each year in the United States, between 1 and 2 million animal-bite wounds are sustained; the vast majority are inflicted by pet dogs and cats, which number more than 100 million. Other bite wounds are a consequence of encounters with animals in the wild or in occupational settings. While many of these wounds require minimal or no therapy, a significant number result in infection, which may be life-threatening. The microbiology of bite-wound infections in general reflects the oropharyngeal flora of the biting animal, although organisms from the soil, the skin of the animal and victim, and the animal's feces may also be involved.

Dog Bites Dogs are responsible for approximately 80% of bite wounds, an estimated 15 to 20% of which become infected. A study for the period 1992 through 1994 found that dog bites resulted in more than 900 emergency department visits each day in the United States. Most dog bites are provoked and are inflicted by the victim's pet or by a dog known to the victim. These bites frequently occur during efforts to break up a dogfight. Victims tend to be male, and bites most often involve a lower extremity. Infection typically manifests 8 to 24 h after the bite as pain at the site of injury with cellulitis accompanied by purulent, sometimes foul-smelling discharge. Septic arthritis and osteomyelitis may develop if the canine tooth penetrates synovium or bone. Systemic manifestations such as fever, lymphadenopathy, and lymphangitis may also occur. The microbiology of dog-bite wound infections is usually mixed and includes a hemolytic streptococci, *Pasteurella* spp., *Staphylococcus* spp., *Eikenella corrodens*, and *Capnocytophaga canimorsus* (formerly designated DF-2). Many wounds also include anaerobic bacteria such as *Actinomyces*, *Fusobacterium*, *Prevotella*, and *Porphyromonas* spp.

While most infections resulting from dog-bite injuries are localized to the area of injury, many of the microorganisms involved are capable of causing systemic infection, including bacteremia, meningitis, brain abscess, endocarditis, and chorioamnionitis. These infections are particularly likely in hosts with edema or compromised lymphatic drainage in the involved extremity (e.g., following a bite on the arm after radical or modified radical mastectomy) and in patients who are immunocompromised by medication or disease (e.g., glucocorticoid use, systemic lupus erythematosus, acute leukemia, or hepatic cirrhosis). In addition, dog bites and scratches may result in systemic illnesses such as rabies ([Chap. 197](#)) and tetanus ([Chap. 143](#)).

Infection with *C. canimorsus* following dog-bite wounds may result in fulminant sepsis,

disseminated intravascular coagulation, and renal failure, particularly in hosts who have impaired hepatic function, who have undergone splenectomy, or who are immunosuppressed. This organism is a thin gram-negative rod that is difficult to culture on most solid media but grows in a variety of liquid media. The bacteria are occasionally seen within polymorphonuclear leukocytes on Wright-stained smears of peripheral blood from septic patients.

Cat Bites Although less common than dog bites, cat bites and scratches result in infection in more than half of all cases. Because the narrow, sharp feline incisors penetrate deeply into tissue, cat bites are more likely than dog bites to cause septic arthritis and osteomyelitis; the development of these conditions is particularly likely when punctures are located over or near a joint, especially in the hand. Women sustain cat bites more frequently than do men. These bites most often involve the hands and arms. Both bites and scratches from cats are prone to infection from organisms in the cat's oropharynx. *Pasteurella multocida*, a normal component of the feline oral flora, is a small gram-negative coccobacillus implicated in the majority of cat-bite wound infections. Like that of dog-bite wound infections, however, the microflora of cat-bite wound infections is usually mixed. Other microorganisms causing infection after cat bites are similar to those causing dog-bite wound infections.

The same risk factors for systemic infection following dog-bite wounds apply to cat-bite wounds. *Pasteurella* infections tend to advance rapidly, often within hours, causing severe inflammation accompanied by purulent drainage; *Pasteurella* may also be spread by respiratory droplets from animals, resulting in pneumonia or bacteremia. Like dog-bite wounds, cat-bite wounds may result in the transmission of rabies or in the development of tetanus. Infection with *Bartonella henselae* causes cat-scratch disease ([Chap. 163](#)) and is an important late consequence of cat bites and scratches. Tularemia ([Chap. 161](#)) has also been reported to follow cat bites.

Other Animal Bites Infections have been attributed to bites from many animal species, often as a consequence of occupational exposure (farmers, laboratory workers, veterinarians) or recreational exposure (hunters and trappers, wilderness campers, owners of exotic pets). Generally, the microflora of bite wounds reflects the oral flora of the biting animal. Most members of the cat family, including feral cats, harbor *P. multocida*. Bite wounds from aquatic animals such as alligators or piranhas may contain *Aeromonas hydrophila*. Venomous snakebites ([Chap. 397](#)) result in severe inflammatory responses and tissue necrosis -- conditions that render these injuries prone to infection. The snake's oral flora includes many species of aerobes and anaerobes, such as *Pseudomonas aeruginosa*, *Proteus* spp., *Staphylococcus epidermidis*, *Bacteroides fragilis*, and *Clostridium* spp. Bites from nonhuman primates are highly susceptible to infection with pathogens similar to those isolated from human bites (which are discussed later in this chapter). Bites from Old World monkeys (*Macaca*) may also result in the transmission of B virus (*Herpesvirus simiae*, cercopithecine herpesvirus), a cause of serious infection of the human central nervous system. Bites of seals, walruses, and polar bears may cause a chronic suppurative infection known as *seal finger*, which is probably due to one or more species of *Mycoplasma* colonizing these animals.

Small rodents, including rats, mice, and gerbils, as well as animals that prey on rodents may transmit *Streptobacillus moniliformis* (a microaerophilic, pleomorphic gram-negative

rod) or *Spirillum minor* (a spirochete), which cause a clinical illness known as *rat-bite fever*. The vast majority of cases in the United States are streptobacillary, whereas *Spirillum* infection occurs mainly in Asia.

In the United States, the risk of rodent bite mainly affects laboratory workers or inhabitants of rodent-infested dwellings (particularly children). Rat-bite fever is distinguished from acute bite-wound infection by its typical manifestation after the initial wound has healed. Streptobacillary disease follows an incubation period of 3 to 10 days. Fever, chills, myalgias, headache, and severe migratory arthralgias are usually followed by a maculopapular rash, which characteristically involves the palms and soles and may become confluent or purpuric. Complications include endocarditis, myocarditis, meningitis, pneumonia, and abscesses in many organs. *Haverhill fever* is an *S. moniliformis* infection acquired from contaminated milk or drinking water and has similar manifestations. Streptobacillary rat-bite fever was frequently fatal in the preantibiotic era. The differential diagnosis includes Rocky Mountain spotted fever, Lyme disease, leptospirosis, and secondary syphilis. The diagnosis is made by direct observation of the causative organisms in tissue or blood, by culture on enriched media, or by serologic testing with specific agglutinins.

Spirillum infection (referred to in Japan as *sodoku*) causes pain and purple swelling at the site of the initial bite, with associated lymphangitis and regional lymphadenopathy, after an incubation period of 1 to 4 weeks. The systemic illness includes fever, chills, and headache. The original lesion may eventually progress to an eschar. The infection is diagnosed by direct visualization of the spirochetes in blood or tissue or by animal inoculation.

Human Bites Human bites may be self-inflicted; may be sustained by medical personnel caring for patients; or may take place during fights, domestic abuse, or sexual activity. Human bites more frequently become infected than do bites inflicted by other animals. These infections reflect the diverse oral microflora of humans, which includes multiple species of aerobic and anaerobic bacteria. Common aerobic isolates include viridans streptococci, *Staphylococcus aureus*, *E. corrodens* (which is particularly common in clenched-fist injury; see below), and *Haemophilus influenzae*. Anaerobic species, including *Fusobacterium nucleatum* and *Prevotella*, *Porphyromonas*, and *Peptostreptococcus* spp., are isolated from 50% of human-bite wound infections; many of these isolates produce β -lactamases. The oral flora of hospitalized and debilitated patients often includes Enterobacteriaceae in addition to the usual organisms. Both HIV and hepatitis B virus have been reported to be transmitted by human bite, but these instances appear to be quite rare.

Human bites are categorized as "occlusional" injuries, which are inflicted by actual biting, and "clenched-fist" injuries, which are sustained when the fist of one individual strikes the teeth of another, causing traumatic laceration of the hand. For several reasons, clenched-fist injuries result in particularly serious infections. The deep spaces of the hand, including the bone, joint, and tendons, are frequently inoculated with organisms in the course of such injuries. The clenched position of the fist during injury, followed by extension of the hand, may further promote the introduction of bacteria as contaminated tendons retract beneath the skin's surface. Moreover, medical attention is often sought only after frank infection develops.

TREATMENT

Initial Assessment A careful history should be elicited, including the type of biting animal, the type of attack (provoked or unprovoked), and the amount of time elapsed since injury. Local and regional authorities should be contacted to determine whether an individual species could be rabid and/or to locate and observe the biting animal when rabies prophylaxis may be indicated ([Chap. 197](#)). Suspicious human-bite wounds should provoke careful questioning regarding domestic or child abuse. Details on antibiotic allergies, immunosuppression, splenectomy, liver disease, mastectomy, and immunization history should be obtained. The wound should be inspected carefully for evidence of infection, including redness, exudate, and foul odor. The type of wound (puncture, laceration, or scratch); the depth of penetration; and the possible involvement of joints, tendons, nerves, and bone should be assessed. It is often useful to include a diagram or photograph of the wound in the medical record. In addition, a general physical examination should be conducted and should include an assessment of vital signs as well as an evaluation for evidence of lymphangitis, lymphadenopathy, dermatologic lesions, and functional limitations. Injuries to the hand warrant consultation with a hand surgeon for the assessment of tendon, nerve, and muscular damage. Radiographs should be obtained when the bone may have been penetrated or a tooth fragment may be present. Culture and Gram's staining of all infected wounds are essential; anaerobic cultures should be undertaken if abscesses, devitalized tissue, or foul-smelling exudate is present. A small-tipped swab may be used to culture deep punctures or small lacerations. It is also reasonable to culture samples from uninfected wounds due to bites inflicted by animals other than dogs and cats, since the microorganisms causing disease are less predictable in these cases. A white blood cell count should be determined and blood cultured if systemic infection is suspected.

Wound Management Wound closure is controversial in bite injuries. Many authorities prefer not to attempt primary closure of wounds that are or may become infected, preferring to irrigate these wounds copiously, debride devitalized tissue, remove foreign bodies, and approximate the wound edges. Delayed primary closure may be undertaken after the risk of infection is over. Small uninfected wounds may be allowed to close by secondary intention. Puncture wounds due to cat bites should be left unsutured because of the high rate at which they become infected. Facial wounds are usually sutured after thorough cleaning and irrigation because of the importance of a good cosmetic result in this area and because anatomic factors such as an excellent blood supply and the absence of dependent edema lessen the risk of infection.

Antibiotic Therapy

Established Infection Antibiotics should be administered in all established bite-wound infections and should be chosen in light of the most likely potential pathogens, as indicated by the biting species and by Gram's stain and culture results ([Table 127-1](#)). For dog and cat bites, antibiotics should be effective against *S. aureus*, *Pasteurella* spp., *C. canimorsus*, streptococci, and oral anaerobes. For human bites, agents with activity against *S. aureus*, *H. influenzae*, and β -lactamase-positive oral anaerobes should be used. The combination of an extended-spectrum penicillin with a β -lactamase inhibitor (amoxicillin/clavulanic acid, ticarcillin/clavulanic acid, ampicillin/sulbactam) appears to

offer the most reliable coverage for these pathogens. Second-generation cephalosporins (cefuroxime, cefoxitin) also offer substantial coverage. The choice of antibiotics in penicillin-allergic patients (particularly those in whom immediate-type hypersensitivity makes the use of cephalosporins hazardous) is more difficult and is based primarily on in vitro sensitivity since data on clinical efficacy are inadequate. The combination of an antibiotic active against gram-positive cocci and anaerobes (such as clindamycin) with trimethoprim-sulfamethoxazole or a fluoroquinolone, which is active against many of the other potential pathogens, would appear reasonable. In vitro data suggest that either trovafloxacin or azithromycin alone provides coverage against most commonly isolated bite-wound pathogens.

Antibiotics are normally given for 10 to 14 days, but the response to therapy must be carefully monitored. Failure to respond should prompt a consideration of diagnostic alternatives and surgical evaluation for possible drainage or debridement. Complications such as osteomyelitis or septic arthritis mandate a longer duration of therapy.

Management of *C. canimorsus* sepsis requires a 2-week course of intravenous penicillin G (2 million units intravenously every 4 h) and supportive measures. Alternative agents for the treatment of *C. canimorsus* infection include cephalosporins and fluoroquinolones. Serious infection with *P. multocida* (e.g., pneumonia, sepsis, or meningitis) should also be treated with intravenous penicillin G. Alternative agents include second- or third-generation cephalosporins or ciprofloxacin.

Bites by venomous snakes may not require antibiotic treatment, but it is often difficult to distinguish signs of infection from tissue damage caused by the envenomation. Thus many authorities continue to recommend treatment directed against the snake's oral flora -- i.e., the administration of broadly active agents such as ceftriaxone (1 to 2 g intravenously every 12 to 24 h) or ampicillin/sulbactam (1.5 to 3.0 g intravenously every 6 h).

Seal finger appears to respond to doxycycline (100 mg twice daily for an interval guided by the response to therapy).

Presumptive or Prophylactic Therapy The use of antibiotics in patients presenting early after bite injury (within 8 h) is controversial. Although symptomatic infection will not yet be manifest in many of these wounds at this point, many early wounds will harbor pathogens, and many will become infected. Studies of the use of prophylactic antibiotics in wound infections are limited and have often included small numbers of cases in which various types of wounds have been managed according to various protocols. A recent meta-analysis of eight randomized trials of prophylactic antibiotics in patients with dog-bite wounds demonstrated a reduction of the rate of infection by approximately 50% with prophylaxis. However, in the absence of sound clinical trials, many clinicians base the decision to treat bite wounds with empirical antibiotics on the species of the biting animal; the location, severity, and extent of the bite wound; and the existence of comorbid conditions in the host. All human- and monkey-bite wounds should be treated presumptively because of the high rate of infection. Most cat-bite wounds, particularly those involving the hand, should be treated. Other factors favoring treatment for bite wounds include severe injury, as in crush wounds; potential bone or joint involvement; involvement of the hands or genital region; host immunocompromise, including that due

to liver disease or splenectomy; and prior mastectomy on the side of an involved upper extremity. When prophylactic antibiotics are administered, they are usually given for 3 to 5 days.

Rabies and Tetanus Prophylaxis Rabies prophylaxis, consisting of both passive administration of rabies immune globulin (with as much of the dose as possible infiltrated in and around the wound) and active immunization with rabies vaccine, should be given in consultation with local and regional public health authorities for many wild-animal (and some domestic-animal) bites and scratches as well as for certain nonbite exposures ([Chap. 197](#)). Rabies is endemic in a variety of animals, including dogs and cats in many areas of the world. Many local health authorities require the reporting of all animal bites. A tetanus booster immunization should be given if the patient has undergone primary immunization but has not received a booster dose in the past 5 years. Patients who have not previously completed primary immunization should be immunized and should also receive tetanus immune globulin. Elevation of the site of injury is an important adjunct to antimicrobial therapy. Immobilization of the infected area, especially the hand, is also beneficial.

BURNS

Epidemiology More than 2 million burn injuries are brought to medical attention in the United States each year. While many burn injuries are minor and require little or no intervention, approximately 70,000 persons are hospitalized for these injuries, and 20,000 of this number are burned severely enough to require admission to a specialized burn unit. Scalds, structural fires, and flammable liquids and gases are the major causes of burns, but electrical, chemical, and smoking-related sources are also important. Burns predispose to infection by damaging the protective barrier function of the skin, thus facilitating the entry of pathogenic microorganisms, and by inducing systemic immunosuppression. It is therefore not surprising that infectious complications are the major cause of morbidity and mortality in serious burn injury and that as many as 10,000 patients in the United States die of burn-related infections each year.

Pathophysiology Loss of the cutaneous barrier facilitates entry of the patient's own flora and of organisms from the hospital environment into the burn wound. The wound often contains devitalized or frankly necrotic tissue that quickly becomes contaminated with bacteria. Invasive infection -- localized and/or systemic -- occurs when bacteria penetrate viable tissue, usually below the eschar. Streptococci and staphylococci were the predominant causes of burn-wound infection in the preantibiotic era and remain important pathogens at present. With the advent of antimicrobial agents, *P. aeruginosa* became a major problem in burn-wound management. As antibiotics more effective against *Pseudomonas* have become available, fungi (particularly *Candida albicans*, *Aspergillus* spp., and the agents of mucormycosis) have emerged as increasingly important pathogens in burn-wound patients. Herpes simplex virus infection has also been found in burn wounds, especially on the face.

The frequency of infection parallels the extent and severity of burn injury. Severe burns cause defects in both cellular and humoral immunity that have a major impact on infection. For example, decreases in the number and activity of circulating helper T cells, increases in suppressor T cells, and diminution in levels of immunoglobulin follow

major burns. Neutrophil function has also been shown to be impaired after burns. The increased levels of multiple cytokines detected in burn patients are compatible with the widely held belief that the inflammatory response becomes dysregulated in these individuals. Increased permeability of the gut wall to bacteria and their components, such as endotoxin, also contributes to immune dysregulation and sepsis. Thus, the burn patient is predisposed to infection at remote sites (see below) as well as at the sites of burn injury.

Clinical Manifestations Since clinical indications of wound infection are difficult to interpret, wounds must be monitored carefully for changes that may reflect infection. A margin of erythema frequently surrounds the sites of burns and by itself is not usually indicative of infection. Signs of infection include the conversion of a partial-thickness to a full-thickness burn, color changes (e.g., the appearance of a dark brown or black discoloration of the wound), the new appearance of erythema or violaceous edema in normal tissue at the wound margins, the sudden separation of the eschar from subcutaneous tissues, and the degeneration of the wound with the appearance of a new eschar. The appearance of a green discoloration of the wound or subcutaneous fat or the development of ecthyma gangrenosum at a remote site points to a diagnosis of invasive *P. aeruginosa* infection. Changes in body temperature, hypotension, tachycardia, altered mentation, neutropenia or neutrophilia, thrombocytopenia, and renal failure may result from invasive burn wounds and sepsis. However, because profound alterations in homeostasis occur as a consequence of burns per se and because inflammation without infection is a normal component of these injuries, the assessment of these changes is complicated. Alterations in body temperature, for example, are attributable to thermoregulatory dysfunction; tachycardia and hyperventilation accompany the metabolic changes induced by extensive burn injury and are not necessarily indicative of bacterial sepsis.

Given the difficulty of evaluating burn wounds solely on the basis of clinical observation and laboratory data, wound biopsies are necessary for definitive diagnosis of infection. The timing of these biopsies can be guided by clinical changes, but in some centers burn wounds are routinely biopsied at regular intervals. The biopsy specimen is examined for histologic evidence of bacterial invasion, and quantitative microbiologic cultures are performed. The presence of $>10^5$ viable bacteria per gram of tissue is highly suggestive of invasive infection and of a dramatically increased risk of sepsis. Histopathologic evidence of invasion of viable tissue by microorganisms is a more definitive indicator of infection. A blood culture positive for the same organism seen in large quantities in biopsied tissue is a reliable indicator of burn sepsis. Surface cultures may provide some indication of the microorganisms present in the hospital environment but are not indicative of the etiology of infection.

In addition to infection of the burn wound itself, a number of other infections due to the immunosuppression caused by extensive burns and the manipulations necessary for clinical care put burn patients at risk. Pneumonia, now the most common infectious complication among hospitalized burn patients, is most often nosocomially acquired via the respiratory route; septic pulmonary emboli may also occur. Suppurative thrombophlebitis may complicate the vascular catheterization necessary for fluid and nutritional support in burns. Endocarditis, urinary tract infection, bacterial chondritis (particularly in patients with burned ears), and intraabdominal infection also complicate

serious burn injury.

TREATMENT

The ultimate goal of burn-wound management is closure and healing of the wound. Early surgical excision of burned tissue, with extensive debridement of necrotic tissue and grafting of skin or skin substitutes, greatly decreases the mortality associated with severe burns. In addition, the three widely used topical antimicrobial agents -- silver sulfadiazine cream, mafenide acetate cream, and silver nitrate -- dramatically decrease the bacterial burden of burn wounds and reduce the incidence of burn-wound infection; they are routinely applied to partial- and full-thickness burns. All three agents are broadly active against many bacteria and against some fungi and are useful before bacterial colonization is established. Silver sulfadiazine is often used initially, but its value can be limited by bacterial resistance. Mafenide acetate has broader activity; the cream penetrates eschars and thus can prevent or treat infection beneath the eschars. The foremost disadvantages of this agent are that it can inhibit carbonic anhydrase, resulting in metabolic acidosis, and that it elicits hypersensitivity reactions in up to 7% of patients. This agent is most often used when gram-negative bacteria invade the burn wound and when treatment with silver sulfadiazine fails.

When invasive wound infection is diagnosed, topical therapy should be changed to mafenide acetate. Subeschar clysis (the direct instillation of an antibiotic, often piperacillin, under the eschar into wound tissues) is a useful adjunct to surgical and systemic antimicrobial therapy. Systemic treatment with antibiotics active against the pathogens present in the wound should be instituted. In the absence of culture data, treatment is broad and should cover organisms commonly encountered in the particular burn unit. Usually such coverage is achieved with an antibiotic active against gram-positive pathogens, such as oxacillin (2 g intravenously every 4 h), and with antibiotics active against *P. aeruginosa* and other gram-negative rods, such as mezlocillin (3 g intravenously every 4 h) and gentamicin (5 mg/kg intravenously per day). In the penicillin-allergic patient, vancomycin (1 g intravenously every 12 h) may be substituted for oxacillin (and is efficacious when methicillin-resistant *S. aureus* is present), and ciprofloxacin (400 mg intravenously every 12 h) may be substituted for mezlocillin. Patients with burn wounds frequently have alterations in metabolism and renal clearance mechanisms that mandate the monitoring of serum antibiotic levels; the levels achieved with standard doses are often subtherapeutic.

In general, prophylactic systemic antibiotics have no role in the management of burn wounds (except for minor burns in outpatients) and can in fact lead to colonization with resistant microorganisms. An exception involves cases requiring burn-wound manipulation. Since procedures such as debridement, excision, or grafting frequently result in bacteremia, prophylactic systemic antibiotics are administered at the time of burn-wound manipulation; the particular agents used should be chosen on the basis of data obtained by wound culture or data on the hospital's resident flora. All burn-injury patients should undergo tetanus booster immunization if they have completed primary immunization but have not received a booster dose in the past 5 years. Patients without prior immunization should receive tetanus immune globulin and undergo primary immunization. Infection control measures play a major role in preventing burn-wound infection and limiting the spread of antibiotic-resistant nosocomial pathogens.

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128. INFECTIONS OF THE SKIN, MUSCLE, AND SOFT TISSUES - Dennis L. Stevens

ANATOMIC RELATIONSHIPS: CLUES TO THE DIAGNOSIS OF SOFT TISSUE INFECTIONS

Protection against infection of the epidermis is dependent on the mechanical barrier afforded by the stratum corneum, since the epidermis itself is devoid of blood vessels ([Fig. 128-1](#)). Disruption of this layer by burns or bites ([Chap. 127](#)), abrasions, foreign bodies, primary dermatologic disorders (e.g., herpes simplex, varicella, and ecthyma gangrenosum), surgery, or vascular or pressure ulcer allows penetration of bacteria to the deeper structures. Similarly, the hair follicle can serve as a portal either for components of the normal flora (e.g., *Staphylococcus*) or for extrinsic bacteria (e.g., *Pseudomonas* in hot-tub folliculitis). Intracellular infection of the squamous epithelium with vesicle formation may arise from cutaneous inoculation, as in infection with herpes simplex virus (HSV) type 1; from the dermal capillary plexus, as in varicella and infections due to other viruses associated with viremia; or from cutaneous nerve roots, as in herpes zoster. Bacteria infecting the epidermis, such as *Streptococcus pyogenes*, may be translocated laterally to deeper structures via lymphatics, an event that results in the rapid superficial spread of erysipelas. Later, engorgement or obstruction of lymphatics causes flaccid edema of the epidermis, another characteristic of erysipelas.

The rich plexus of capillaries beneath the dermal papillae provides nutrition to the stratum germinativum, and physiologic responses of this plexus produce important clinical signs and symptoms. For example, infective vasculitis of the plexus results in petechiae, Osler's nodes ([Fig. 126-CD3](#)), Janeway lesions ([Fig. 19-CD2](#)), and palpable purpura, which are important clues to the existence of endocarditis ([Chap. 126](#)). In addition, metastatic infection within this plexus can result in cutaneous manifestations of disseminated fungal infection ([Chap. 205](#)), gonococcal infection ([Chap. 147](#)), *Salmonella* infection ([Chap. 156](#)), *Pseudomonas* infection (i.e., ecthyma gangrenosum; [Chap. 155](#)), meningococemia ([Chap. 146](#)), and staphylococcal infection ([Chap. 139](#)). The plexus also provides access for bacteria to the circulation, thereby facilitating local spread or bacteremia. The postcapillary venules of this plexus are a major site of polymorphonuclear leukocyte sequestration, diapedesis, and chemotaxis to the site of cutaneous infection.

Exaggeration of these physiologic mechanisms by excessive levels of cytokines or bacterial toxins causes leukostasis, venous occlusion, and pitting edema. Edema with purple bullae, ecchymosis, and cutaneous anesthesia suggests loss of vascular integrity and necessitates exploration of the deeper structures for evidence of necrotizing fasciitis or myonecrosis. An early diagnosis requires a high level of suspicion in instances of unexplained fever and of pain and tenderness in the soft tissue, even in the absence of acute cutaneous inflammation.

INFECTIONS ASSOCIATED WITH VESICLES ([Table 128-1](#))

Vesicle formation due to infection is caused by viral proliferation within the epidermis. In varicella and variola, viremia precedes the onset of a diffuse centripetal rash that progresses from macules to vesicles, then to pustules, and finally to scabs over the

course of 1 to 2 weeks. Vesicles of varicella have a "dewdrop" appearance and develop in crops randomly about the trunk, extremities, and face over 3 to 4 days. Herpes zoster occurs in a single dermatome; the appearance of vesicles is preceded by pain for several days. Zoster may occur in persons of any age but is most common among immunosuppressed individuals and elderly patients, whereas most cases of varicella occur in young children. Vesicles due to [HSV](#) are found on or around the lips (HSV-1) or genitals (HSV-2) but may appear on the head and neck of young wrestlers (herpes gladiatorum) or on the digits of health care workers (herpetic whitlow). Coxsackievirus A16 characteristically causes vesicles on the hands, feet, and mouth of children. Orf is caused by a DNA virus related to smallpox virus and infects the fingers of individuals who work around goats and sheep. Molluscum contagiosum virus induces flaccid vesicles on the skin of healthy and immunocompromised individuals.

Rickettsialpox begins following mite-bite inoculation of *Rickettsia akari* into the skin. A papule with a central vesicle evolves to form a 1- to 2.5-cm painless crusted black eschar with an erythematous halo and proximal adenopathy. While more common in the northeastern United States and the Ukraine in 1940-1950, rickettsialpox has recently been described in Ohio, Arizona, and Utah. Blistering dactylitis is a painful, vesicular, localized *Staphylococcus aureus* or group A streptococcal infection of the pulps of the distal digits of the hands.

INFECTIONS ASSOCIATED WITH BULLAE ([Table 128-1](#))

Staphylococcal scalded-skin syndrome (SSSS) ([Fig. 18-CD5](#)) in neonates is caused by a toxin (exfoliatin) from phage group II *S. aureus*. SSSS must be distinguished from toxic epidermal necrolysis (TEN), which occurs primarily in adults, is drug-induced, and has a higher mortality. Punch biopsy with frozen section is useful in making this distinction since the cleavage plane is the stratum corneum in SSSS ([Fig. 128-1](#)) and the stratum germinativum in TEN. Intravenous g-globulin is a promising treatment for TEN. Necrotizing fasciitis and gas gangrene also induce bulla formation (see "Necrotizing Fasciitis," below). Halophilic vibrio infection can be as aggressive and fulminant as necrotizing fasciitis; a helpful clue in its diagnosis is a history of exposure to waters of the Gulf of Mexico or the Atlantic seaboard or (in a patient with cirrhosis) the ingestion of raw seafood. The etiologic organism (*Vibrio vulnificus*) is highly susceptible to tetracycline.

INFECTIONS ASSOCIATED WITH CRUSTED LESIONS ([Table 128-1](#))

Impetigo contagiosa is caused by *S. pyogenes*, and bullous impetigo is due to *S. aureus* ([Fig. 128-CD1](#)). Both skin lesions may have an early bullous stage but then appear as thick crusts with a golden-brown color ([Fig. 128-CD2](#)). Streptococcal lesions are most common among children 2 to 5 years of age, and epidemics may occur in settings of poor hygiene, particularly among children of lower socioeconomic status in tropical climates. It is important to recognize impetigo contagiosa because of its relationship to poststreptococcal glomerulonephritis. Superficial dermatophyte infection (ringworm) can occur on any skin surface, and skin scrapings with KOH staining are diagnostic. Primary infections with dimorphic fungi such as *Blastomyces dermatitidis* and *Sporothrix schenckii* can initially present as crusted skin lesions resembling ringworm. Disseminated infection with *Coccidioides immitis* can also involve the skin, and biopsy

and culture should be performed on crusted lesions in patients from endemic areas. Crusted nodular lesions caused by *Mycobacterium chelonae* have been described in HIV-seropositive patients. Treatment with clarithromycin looks promising.

FOLLICULITIS ([Table 128-1](#))

Hair follicles serve as a portal for a number of bacteria, though *S. aureus* is the most common cause of localized folliculitis ([Fig. 128-CD3](#)). Sebaceous glands empty into hair follicles and ducts and, if blocked, form sebaceous cysts, which may resemble staphylococcal abscesses or may become secondarily infected. Infection of sweat glands (hidradenitis suppurativa) can also mimic infection of hair follicles, particularly in the axillae. Chronic folliculitis is uncommon except in acne vulgaris, where constituents of the normal flora (e.g., *Propionibacterium acnes*) may play a role.

Diffuse folliculitis occurs in two settings. "Hot-tub folliculitis" ([Fig. 128-CD4](#)) is caused by *Pseudomonas aeruginosa* in waters that are insufficiently chlorinated and maintained at temperatures between 37 and 40°C. Infection is usually self-limited, though bacteremia and shock have been reported. Swimmer's itch occurs when a skin surface is exposed to water infested with freshwater avian schistosomes. Warm water temperatures and alkaline pH are suitable for mollusks that serve as intermediate hosts between birds and humans. Free-swimming schistosomal cercariae readily penetrate human hair follicles or pores but quickly die and elicit a brisk allergic reaction causing intense itching and erythema.

PAPULAR AND NODULAR LESIONS ([Table 128-1](#))

Raised lesions of the skin occur in many different forms. *Mycobacterium marinum* infections of the skin may present as cellulitis or as raised erythematous nodules. Erythematous papules are early manifestations of cat-scratch disease (primary site of inoculation) and bacillary angiomatosis (*Bartonella henselae*). Raised serpiginous or linear eruptions are characteristic of cutaneous larva migrans, which is caused by burrowing larvae of dog or cat hookworms (*Ancylostoma braziliense*) and which humans acquire through contact with soil that has been contaminated with dog or cat feces. Similar burrowing raised lesions are present in dracunculiasis caused by migration of the adult female nematode *Dracunculus medinensis*. Nodules caused by *Onchocerca volvulus* may range from 1 to 10 cm in diameter and occur largely in persons bitten by *Simulium* flies in Africa. The nodules contain the adult worm encased in fibrous tissue. Migration of microfilariae into the eyes may result in blindness. Verruca peruana is caused by *Bartonella bacilliformis*, which is transmitted to humans by the sandfly *Phlebotomus*. This condition can take the form of single gigantic lesions (several centimeters in diameter) or multiple small lesions (several millimeters in diameter). Numerous subcutaneous nodules may also be present in cysticercosis caused by larvae of *Taenia solium*. Multiple erythematous papules develop in schistosomiasis; each represents a cercarial invasion site. Skin nodules as well as thickened subcutaneous tissue are prominent features of lepromatous leprosy. Large nodules or gummas are features of tertiary syphilis ([Fig. 128-CD5](#)), whereas flat papulosquamous lesions are characteristic of secondary syphilis. Human papillomavirus may cause singular warts (verruca vulgaris; [Fig. 128-CD6](#)) or multiple warts in the anogenital area (condylomata acuminata).

ULCERS WITH OR WITHOUT ESCHARS ([Table 128-1](#))

Cutaneous anthrax begins as a pruritic papule, which develops within days into an ulcer with surrounding vesicles and edema and then into an enlarging ulcer with a black eschar. Cutaneous anthrax may cause chronic nonhealing ulcers with an overlying dirty-gray membrane, though lesions may also mimic psoriasis, eczema, or impetigo. Ulceroglandular tularemia may have associated ulcerated skin lesions with painful regional adenopathy. Although buboes are the major cutaneous manifestation of plague, in 25% of cases ulcers with eschars, papules, or pustules are also present.

Mycobacterium ulcerans typically causes chronic skin ulcers on the extremities of individuals living in the tropics. *Mycobacterium leprae* may be associated with cutaneous ulcerations in patients with lepromatous leprosy related to Lucio's phenomenon or during reversal reactions. *Mycobacterium tuberculosis* may also cause ulcerations, papules, or erythematous macular lesions of the skin in both normal and immunocompromised patients.

Decubitus ulcers are due to tissue hypoxia secondary to pressure-induced vascular insufficiency and may become secondarily infected with components of the skin and gastrointestinal flora, including anaerobes. Ulcerative lesions on the anterior shins may be due to pyoderma gangrenosum, which must be distinguished from similar lesions of infectious etiology by histologic evaluation of biopsy sites. Ulcerated lesions on the genitals may be either painful (chancroid) or painless (primary syphilis).

ERYSIPELAS ([Table 128-1](#))

Erysipelas is due to *S. pyogenes* and is characterized by an abrupt onset of fiery-red swelling of the face or extremities. The distinctive features of erysipelas are well-defined indurated margins, particularly along the nasolabial fold; rapid progression; and intense pain. Flaccid bullae may develop during the second or third day of illness, but extension to deeper soft tissues is rare. Treatment with penicillin is effective; swelling may progress despite appropriate treatment, though fever, pain, and the intense red color diminish. Desquamation of the involved skin occurs 5 to 10 days into the illness. Infants and elderly adults are most commonly afflicted, and the severity of systemic toxicity varies.

CELLULITIS ([Table 128-1](#))

Cellulitis is an acute inflammatory condition of the skin that is characterized by localized pain, erythema, swelling, and heat. Cellulitis may be caused by indigenous flora colonizing the skin and appendages (e.g., *S. aureus* and *S. pyogenes*) or by a wide variety of exogenous bacteria. Because the exogenous bacteria involved in cellulitis occupy unique niches in nature, a thorough history including epidemiologic data provides important clues to etiology. When there is drainage, an open wound, or an obvious portal of entry, Gram's stain and culture provide a definitive diagnosis. In the absence of these findings, the bacterial etiology of cellulitis is difficult to establish, and in some cases staphylococcal and streptococcal cellulitis may have similar features. Even with needle aspiration of the leading edge or a punch biopsy of the cellulitis tissue itself,

cultures are positive in only 20% of cases. This observation suggests that relatively low numbers of bacteria may cause cellulitis and that the expanding area of erythema within the skin may be a direct effect of extracellular toxins or of the soluble mediators of inflammation elicited by the host.

Bacteria may gain access to the epidermis through cracks in the skin, abrasions, cuts, burns, insect bites, surgical incisions, and intravenous catheters. Cellulitis caused by *S. aureus* spreads from a central localized infection, such as an abscess, folliculitis, or an infected foreign body (e.g., a splinter, a prosthetic device, or an intravenous catheter). In contrast, cellulitis due to *Staphylococcus pyogenes* is a more rapidly spreading, diffuse process frequently associated with lymphangitis and fever. Recurrent streptococcal cellulitis of the lower extremities may be caused by organisms of group A, C, or G in association with chronic venous stasis or with saphenous venectomy for coronary artery bypass surgery. Streptococci also cause recurrent cellulitis among patients with chronic lymphedema resulting from elephantiasis, lymph node dissection, or Milroy's disease. Recurrent staphylococcal cutaneous infections are more common among individuals who have eosinophilia and elevated serum levels of IgE (Job's syndrome) and among nasal carriers of staphylococci. Cellulitis caused by *S. agalactiae* (group B streptococci) occurs primarily in elderly patients and those with diabetes mellitus or peripheral vascular disease. *Haemophilus influenzae* typically causes periorbital cellulitis in children in association with sinusitis, otitis media, or epiglottitis. It is unclear whether this form of cellulitis will (like meningitis) become less common as a result of the impressive efficacy of the *H. influenzae* type b vaccine.

Many other bacteria also cause cellulitis. Fortunately, these organisms occur in such characteristic settings that a good history provides useful clues to the diagnosis. Cellulitis associated with cat bites and, to a lesser degree, with dog bites is commonly caused by *Pasteurella multocida*, though in the latter case *Staphylococcus intermedius* and *Capnocytophaga canimorsus* (formerly DF-2) must also be considered. Sites of cellulitis and abscesses associated with dog bites and human bites also contain a variety of anaerobic organisms, including *Fusobacterium*, *Bacteroides*, aerobic and anaerobic streptococci, and *Eikenella corrodens*. *Pasteurella* is notoriously resistant to dicloxacillin and nafcillin but is sensitive to all other b-lactam antimicrobials as well as to quinolones, tetracycline, and erythromycin. Ampicillin/clavulanate, ampicillin/sulbactam, and cefoxitin are good choices for the treatment of animal or human bite infections. *Aeromonas hydrophila* causes aggressive cellulitis in tissues surrounding lacerations sustained in fresh water (lakes, rivers, and streams). This organism remains sensitive to aminoglycosides, fluoroquinolones, chloramphenicol, trimethoprim-sulfamethoxazole, and third-generation cephalosporins; it is resistant to ampicillin, however.

P. aeruginosa causes three types of soft tissue infection: ecthyma gangrenosum in neutropenic patients, hot-tub folliculitis, and cellulitis following penetrating injury. Most commonly, *P. aeruginosa* is introduced into the deep tissues when a person steps on a nail. Treatment includes surgical inspection and drainage, particularly if the injury also involves bone or joint capsule. Choices for empirical treatment while antimicrobial susceptibility data are awaited include an aminoglycoside, a third-generation cephalosporin (ceftazidime, cefoperazone, or cefotaxime), a semisynthetic penicillin (ticarcillin, mezlocillin, or piperacillin), or a fluoroquinolone (though drugs of the last class are not indicated for the treatment of children <13 years old).

Gram-negative bacillary cellulitis, including that due to *P. aeruginosa*, is most common among hospitalized, immunocompromised hosts. Cultures and sensitivity tests are critically important in this setting because of multidrug resistance ([Chap. 155](#)).

The gram-positive aerobic rod *Erysipelothrix rhusiopathiae* is most often associated with fish and domestic swine and causes cellulitis primarily in bone renderers and fishmongers ([Fig. 128-CD7](#)). *E. rhusiopathiae* remains susceptible to most β -lactam antibiotics (including penicillin), erythromycin, clindamycin, tetracycline, and cephalosporins but is resistant to sulfonamides, chloramphenicol, and vancomycin. Its resistance to vancomycin, which is unusual among gram-positive bacteria, is of potential clinical significance since this agent is sometimes used in empirical therapy for skin infection. Fish food containing the water flea *Daphnia* is sometimes contaminated with *M. marinum*, which can cause cellulitis or granulomas on skin surfaces exposed to the water in aquariums or injured in swimming pools. Rifampin plus ethambutol has been an effective therapeutic combination in some cases, though no comprehensive studies have been undertaken. In addition, some strains of *M. marinum* are susceptible to tetracycline or to trimethoprim-sulfamethoxazole.

NECROTIZING FASCIITIS ([Table 128-1](#), [Fig. 18-CD4](#))

Necrotizing fasciitis, formerly called streptococcal gangrene, may be associated with group A *Streptococcus* or mixed aerobic-anaerobic bacteria or may occur as part of gas gangrene caused by *Clostridium perfringens*. Early diagnosis may be difficult when pain or unexplained fever is the only presenting manifestation. Swelling then develops and is followed by brawny edema and tenderness. With progression, dark red induration of the epidermis appears along with bullae filled with blue or purple fluid. Later the skin becomes friable and takes on a bluish, maroon, or black color. By this stage, thrombosis of blood vessels in the dermal papillae ([Fig. 128-1](#)) is extensive. Extension of infection to the level of the deep fascia causes this tissue to take on a brownish-gray appearance. Rapid spread occurs along fascial planes, through venous channels and lymphatics. Patients in the later stages are toxic and frequently manifest shock and multiorgan failure.

Necrotizing fasciitis caused by mixed aerobic-anaerobic bacteria begins with a breach in the integrity of a mucous membrane barrier, such as the mucosa of the gastrointestinal or genitourinary tract. The portal can be a malignancy, diverticulum, hemorrhoid, anal fissure, or urethral tear. Other predisposing factors include peripheral vascular disease, diabetes mellitus, surgery, and penetrating injury to the abdomen. Leakage into the perineal area results in a syndrome called *Fournier's gangrene*, characterized by massive swelling of the scrotum and penis with extension into the perineum or the abdominal wall and legs.

Necrotizing fasciitis caused by *S. pyogenes* has increased in frequency and severity since 1985. It frequently begins deep at the site of a nonpenetrating minor trauma such as a bruise or a muscle strain. Seeding of the site via transient bacteremia is likely, though most patients deny antecedent streptococcal infection. Alternatively, *S. pyogenes* may reach the deep fascia from a site of cutaneous infection or penetrating trauma. Toxicity is severe, and renal impairment may precede the development of

shock. In 20 to 40% of cases, myositis occurs concomitantly, and, as in gas gangrene (see below), serum creatinine phosphokinase values may be markedly elevated. Necrotizing fasciitis due to mixed aerobic-anaerobic bacteria may be associated with gas in the deep tissue, but gas is not usually present when the cause is *S. pyogenes*. Prompt surgical exploration down to the deep fascia and muscle is essential. Necrotic tissue must be surgically removed, and Gram's staining and culture of excised tissue are useful in establishing whether group A streptococci, mixed aerobic-anaerobic bacteria, or *Clostridium* spp. are present (see "Treatment" below).

MYOSITIS/MYONECROSIS ([Table 128-1](#))

Muscle involvement can occur with virus infection (e.g., influenza virus, dengue virus, or coxsackievirus B) or parasitic invasion (e.g., trichinosis, cysticercosis, or toxoplasmosis). Although myalgia can occur in most of these infections, severe muscle pain is the hallmark of pleurodynia (coxsackie virus B), trichinosis, and bacterial infection. Acute rhabdomyolysis predictably occurs with clostridial and streptococcal myositis but may also be associated with influenza virus, echovirus, coxsackievirus, Epstein-Barr virus, and *Legionella* infection.

Pyomyositis is usually due to *S. aureus*, is common in tropical areas, and generally has no known portal of entry. Infection remains localized, and shock does not develop unless organisms produce toxic shock syndrome toxin 1 or certain enterotoxins and the patient lacks antibodies to the toxin produced by the infecting organisms. In contrast, *S. pyogenes* may induce primary myositis referred to as *streptococcal necrotizing myositis*, which is associated with severe systemic toxicity. Myonecrosis occurs concomitantly with necrotizing fasciitis in about 50% of cases. Both are part of the streptococcal toxic shock syndrome.

Gas gangrene usually follows severe penetrating injuries that result in interruption of the blood supply and introduction of soil into wounds. Such cases of traumatic gangrene are usually caused by *C. perfringens*, *C. septicum*, or *C. histolyticum*. Rarely, latent or recurrent gangrene can occur years after penetrating trauma, most likely owing to dormant spores that reside at the site of previous injury. Spontaneous nontraumatic gangrene among patients with neutropenia, gastrointestinal malignancy, diverticulosis, or recent radiation therapy to the abdomen is caused by several clostridial species, although *C. septicum* is most common. The tolerance of this anaerobe to oxygen probably explains why it can initiate infection spontaneously in normal tissue anywhere in the body.

Synergistic nonclostridial anaerobic myonecrosis, also known as necrotizing cutaneous myositis and synergistic necrotizing cellulitis, is a variant of necrotizing fasciitis caused by mixed aerobic and anaerobic bacteria with the exclusion of clostridial organisms (see "Necrotizing Fasciitis," above).

DIAGNOSIS

This chapter has emphasized the physical appearance and location of lesions within the soft tissues as important diagnostic clues. The temporal progression of the lesions as well as the patient's travel history, animal exposure or bite history, age, underlying

disease status, and lifestyle are also crucial considerations in the formulation of a narrowed differential diagnosis. However, even the astute clinician may find it challenging to diagnose all infections of the soft tissues by history and inspection alone. Soft tissue radiography, computed tomography, and magnetic resonance imaging may be useful in determining the depth of infection and should be performed in patients with rapidly progressing lesions or in those with evidence of systemic inflammatory response syndrome. These tests are particularly valuable for defining a localized abscess or detecting gas in tissue. Unfortunately, they may reveal only soft tissue swelling and thus are not specific for fulminant infections such as necrotizing fasciitis or myonecrosis caused by group A *Streptococcus*, where gas is not found in lesions.

Aspiration of the leading edge or punch biopsy with frozen section may be helpful if the results are positive, but false-negative results occur in approximately 80% of cases. There is some evidence that aspiration alone may be superior to injection and aspiration using normal saline. Frozen sections are especially useful in distinguishing [SSSS](#) from TEN and are quite valuable in cases of necrotizing fasciitis. Open surgical inspection with debridement as indicated is clearly the best way to determine the extent and severity of infection and to obtain material for Gram's staining and culture. Such an aggressive approach is important and may be lifesaving if undertaken early in the course of fulminant infections where there is evidence of systemic toxicity.

TREATMENT

A full description of the treatment of all the clinical entities described herein is beyond the scope of this chapter. As a guide to the clinician in selecting appropriate treatment, the antimicrobial agents useful in the most common and the most fulminant cutaneous infections are listed in [Table 128-2](#).

Early and aggressive surgical exploration is essential in patients with suspected necrotizing fasciitis, myositis, or gangrene in order to (1) visualize the deep structures, (2) remove necrotic tissue, (3) reduce compartment pressure, and (4) obtain suitable material for Gram's staining and for aerobic and anaerobic cultures. Appropriate empirical antibiotic treatment for mixed aerobic-anaerobic infections could consist of ampicillin/sulbactam, cefoxitin, or the following combination: (1) clindamycin (600 to 900 mg intravenously every 8 h) or metronidazole (750 mg every 6 h) plus (2) ampicillin or ampicillin/sulbactam (2 to 3 g intravenously every 6 h) plus (3) gentamicin (1.0 to 1.5 mg/kg every 8 h). Group A streptococcal and clostridial infection of the fascia and/or muscle carries a mortality rate of 20 to 50% with penicillin treatment. In experimental models of streptococcal and clostridial necrotizing fasciitis/myositis, clindamycin has exhibited markedly superior efficacy, but no comparative trials have been performed in humans. Hyperbaric oxygen treatment may also be useful in gas gangrene due to clostridial species. Antibiotic treatment should be continued until all signs of systemic toxicity have resolved, all devitalized tissue has been removed, and granulation tissue has developed ([Chaps. 140, 145, and 167](#)).

In summary, infections of the skin and soft tissues are diverse in presentation and severity and offer a great challenge to the clinician. This chapter provides an approach to diagnosis and understanding of the pathophysiologic mechanisms involved in these infections. More in-depth information is found in chapters on specific infections.

(Bibliography omitted in Palm version)

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129. OSTEOMYELITIS - James H. Maguire

Osteomyelitis, an infection of bone, is caused most commonly by pyogenic bacteria and mycobacteria. Classification of cases on the basis of the causative agent; the route, duration, and anatomic location of infection; and local and systemic host factors provides a useful framework for evaluating the patient and planning treatment.

PATHOGENESIS AND PATHOLOGY

Microorganisms enter bone by the hematogenous route, by direct introduction from a contiguous focus of infection, or by a penetrating wound. Trauma, ischemia, and foreign bodies enhance the susceptibility of bone to microbial invasion by exposing sites to which bacteria can bind. Phagocytes attempt to contain the infections and, in the process, release enzymes that lyse bone. Pus spreads into vascular channels, raising intraosseous pressure and impairing the flow of blood; as the untreated infection becomes chronic, ischemic necrosis of bone results in the separation of large devascularized fragments (*sequestra*). When pus breaks through the cortex, subperiosteal or soft tissue abscesses form, and the elevated periosteum deposits new bone (the *involucrum*) around the sequestrum. Bacteria escape host defenses by adhering tightly to damaged bone, by entering and persisting within osteoblasts, and by coating themselves and underlying surfaces with a protective polysaccharide-rich biofilm.

Microorganisms, infiltrates of neutrophils, and congested or thrombosed blood vessels are the principal histologic findings of acute osteomyelitis. The distinguishing feature of chronic osteomyelitis is necrotic bone, which is characterized by the absence of living osteocytes. Mononuclear cells predominate in chronic infections, and granulation and fibrous tissues replace bone that has been resorbed by osteoclasts. In the chronic stage, organisms may be too few to be seen.

HEMATOGENOUS OSTEOMYELITIS

Hematogenous infection accounts for ~20% of cases of osteomyelitis and primarily affects children, in whom the long bones are infected, and older adults and intravenous drug users, in whom the spine is the usual site of infection.

Acute Hematogenous Osteomyelitis Infection usually involves a single bone, most commonly the tibia, femur, or humerus. Bacteria settle in the well-perfused metaphysis, where functioning phagocytes are scarce, a network of venous sinusoids slows the flow of blood, and fenestrations in capillaries allow organisms to escape into the extravascular space. Because vascular anatomy changes with age, hematogenous infection of long bones is uncommon during adulthood and, when it occurs, usually involves the diaphysis.

In children, the source of bacteremia is often inapparent, although there may have been recent blunt trauma to the extremity leading to a small intraosseous hematoma or vascular obstruction. On presentation, the child usually appears acutely ill, with high fever, chills, localized pain and tenderness, and leukocytosis. Cutaneous erythema and swelling indicate extension of pus through the cortex. During infancy and after puberty,

infection may spread through the epiphysis into the joint space. In children of other ages (i.e., between infancy and puberty), extension of infection through the cortex results in involvement of joints if the metaphysis is intracapsular. Thus, septic arthritis of the elbow, shoulder, and hip may complicate osteomyelitis of the proximal radius, humerus, and femur, respectively.

Plain radiographs initially show soft tissue swelling, but the first change in bone -- a periosteal reaction -- is not evident until at least 10 days after the onset of infection. Lytic changes can be detected after 2 to 6 weeks, when 50 to 75% of bone density has been lost. Rarely, a well-circumscribed lytic lesion, or *Brodie's abscess*, is seen in a child who has been in pain for several months but has had no fever.

Chronic Hematogenous Osteomyelitis With prompt treatment, <5% of cases of acute hematogenous osteomyelitis progress to chronic osteomyelitis. On average, 10 days are required for the formation of necrotic bone, but plain radiographs are unable to detect sequestra or sclerotic new bone for many weeks.

A protracted clinical course, long periods of quiescence, and recurrent exacerbations are characteristic of chronic osteomyelitis. Sinus tracts between bone and skin may drain purulent material and occasionally pieces of necrotic bone. An increase in drainage, pain, or the erythrocyte sedimentation rate (ESR) signals an exacerbation. Fever is unusual except when obstruction of a sinus tract leads to soft tissue infection. Rare late complications include pathologic fractures, squamous cell carcinoma of the sinus tract, and amyloidosis.

Vertebral Osteomyelitis Organisms reach the well-perfused vertebral body of adults via spinal arteries and quickly spread from the end plate into the disk space and then to the adjacent vertebral body. The infection may originate in the urinary tract, and it does so particularly often among elderly men. Other sources of bacteremia include endocarditis, soft tissue infection, and a contaminated intravenous line; these sources are usually obvious. Diabetes mellitus, hemodialysis, and intravenous drug use carry an increased risk of spinal infection. Penetrating injuries and surgical procedures to the spine may cause nonhematogenous vertebral osteomyelitis or infection localized to the disk.

Most patients with vertebral osteomyelitis report neck or back pain; 15% describe atypical pain in the chest, the abdomen, or an extremity that is due to irritation of nerve roots. Symptoms are localized to the lumbar spine more often than to the thoracic spine (>50% vs. 35% of cases) or the cervical spine in pyogenic infections, but the thoracic spine is involved most commonly in tuberculous spondylitis (Pott's disease). Percussion over the involved vertebra elicits tenderness, and physical examination may reveal spasm of the paraspinal muscles and a limitation of motion. More than 50% of patients experience a subacute illness in which a vague, dull pain gradually intensifies over 2 to 3 months; fever is low grade or absent, and the white blood cell count is normal. An acute presentation with high fever and toxicity is less common and suggests ongoing bacteremia.

Usually, by the time the patient seeks medical attention, the [ESR](#) is elevated, and plain radiographs show irregular erosions in the end plates of adjacent vertebral bodies and

narrowing of the intervening disk space. This radiographic pattern is virtually diagnostic of bacterial infection because tumors and other diseases of the spine rarely cross the disk space. Computed tomography (CT) or magnetic resonance imaging (MRI) may demonstrate epidural, paraspinal, retropharyngeal, mediastinal, retroperitoneal, or psoas abscesses that originate in the spine. An epidural abscess may evolve suddenly or over several weeks; irreversible paralysis may result from failure to recognize the classic clinical presentation of a spinal epidural abscess: spinal pain progressing to radicular pain and weakness.

Microbiology More than 95% of cases of hematogenous osteomyelitis are caused by a single organism. *Staphylococcus aureus* accounts for 50% of isolates. Other common pathogens include group B streptococci and *Escherichia coli* during the newborn period and group A streptococci and *Haemophilus influenzae* in early childhood. Vertebral osteomyelitis is due to *E. coli* and other enteric bacilli in ~25% of cases. *S. aureus*, *Pseudomonas aeruginosa*, and *Serratia* infections are associated with intravenous drug use in some parts of the United States and may involve the sacroiliac, sternoclavicular, or pubic joints as well as the spine. *Salmonella* spp. and *S. aureus* are the major causes of long-bone osteomyelitis complicating sickle cell anemia and other hemoglobinopathies. Tuberculosis and brucellosis affect the spine more often than other bones. Other common sites of tuberculous osteomyelitis include the small bones of the hands and feet, the metaphyses of long bones, the ribs, and the sternum.

Unusual causes of hematogenous osteomyelitis include disseminated histoplasmosis, coccidioidomycosis, and blastomycosis in endemic areas. Immunocompromised persons on rare occasions develop osteomyelitis due to atypical mycobacteria, *Bartonella henselae*, or *Pneumocystis carinii* or to species of *Candida*, *Cryptococcus*, or *Aspergillus*. Syphilis, yaws, varicella, and vaccinia may involve bone. The etiology of chronic relapsing multifocal osteomyelitis, an inflammatory condition of children that is characterized by recurrent episodes of painful lytic lesions in multiple bones, has not yet been identified.

OSTEOMYELITIS SECONDARY TO A CONTIGUOUS FOCUS OF INFECTION

Clinical Features This broad category of osteomyelitis includes infections introduced by penetrating injuries and surgical procedures and by direct extension of infection from adjacent soft tissues. It accounts for the greatest number of cases of osteomyelitis and occurs most commonly in adults.

Frequently, the diagnosis is not made until the infection has already become chronic. The pain, fever, and inflammatory signs due to acute osteomyelitis may be attributed to the original injury or soft tissue infection. An indolent infection may become apparent only weeks or months later, when a sinus tract develops, a surgical wound breaks down, or a fracture fails to heal. It may be impossible to distinguish radiographic abnormalities due to osteomyelitis from those due to the precipitating condition.

A special type of contiguous-focus osteomyelitis occurs in the setting of peripheral vascular disease and nearly always involves the small bones of the feet of adult diabetic patients. Diabetic neuropathy exposes the foot to frequent trauma and pressure sores, and the patient may be unaware of infection as it spreads into bone. Poor tissue

perfusion impairs normal inflammatory responses and wound healing and creates a milieu that is conducive to anaerobic infections. It is often during the evaluation of a nonhealing ulcer, a swollen toe, or acute cellulitis that a radiograph provides the first evidence of osteomyelitis. If bone is palpable during examination of the base of an ulcer with a blunt surgical probe, osteomyelitis is likely.

Microbiology *S. aureus* is a pathogen in more than half of cases of contiguous-focus osteomyelitis. However, in contrast to hematogenous osteomyelitis, these infections are often polymicrobial and are more likely to involve gram-negative and anaerobic bacteria. Hence a mixture of staphylococci, streptococci, enteric organisms, and anaerobic bacteria may be isolated from a diabetic foot infection or pelvic osteomyelitis underlying a decubitus ulcer. Aerobic and anaerobic bacteria cause osteomyelitis following surgery or soft tissue infection of the oropharynx, paranasal sinuses, gastrointestinal tract, or female genital tract. *S. aureus* is the principal cause of postoperative infections; coagulase-negative staphylococci are common pathogens after implantation of orthopedic appliances; and these organisms as well as gram-negative enteric bacilli, atypical mycobacteria, and *Mycoplasma* may cause sternal osteomyelitis after cardiac surgery. Infection with *P. aeruginosa* is frequently associated with puncture wounds of the foot or with thermal burns, and *Pasteurella multocida* infection commonly follows cat bites ([Chap. 127](#)).

DIAGNOSIS

Early diagnosis of acute osteomyelitis is critical because prompt antibiotic therapy may prevent the necrosis of bone. The evaluation usually begins with plain radiographs because of their ready availability, although they frequently show no abnormalities during early infection. The [ESR](#) and C-reactive protein levels are elevated in most cases of active osteomyelitis, including those in which constitutional symptoms and leukocytosis are lacking. These findings are not specific to osteomyelitis, however, and the ESR is occasionally normal in early infections. In 95% of cases, the technetium radionuclide scan using ^{99m}Tc diphosphonate is positive within 24 h of the onset of symptoms. Falsely negative scans usually indicate obstruction of blood flow to the bone. Because the uptake of technetium reflects osteoblastic activity and skeletal vascularity, the bone scan cannot differentiate osteomyelitis from fractures, tumors, infarction, or neuropathic osteopathy. ^{67}Ga citrate- and ^{111}In -labeled leukocyte or immunoglobulin scans, which have greater specificity for inflammation, may help distinguish infectious from noninfectious processes and indicate inflammatory changes within bones that for other reasons are already abnormal on radiography and technetium scanning. Ultrasound can be used to diagnose osteomyelitis by the detection of subperiosteal fluid collections, soft tissue abscesses adjacent to bone, and periosteal thickening and elevation.

[MRI](#) is as sensitive as the bone scan for the diagnosis of acute osteomyelitis because it is able to detect changes in the water content of marrow. MRI yields better anatomic resolution of epidural abscesses and other soft tissue processes than [CT](#) and is currently the imaging technique of choice for vertebral osteomyelitis ([Fig. 129-1](#)).

The role of diagnostic imaging in chronic osteomyelitis is to determine the presence of active infection and delineate the extent of debridement necessary to remove necrotic

bone and abnormal soft tissues. Although plain films accurately reflect chronic changes, the [CT](#) scan is more sensitive for the detection of sequestra, sinus tracts, and soft tissue abscesses. Both CT and ultrasound are useful for guiding percutaneous aspiration of subperiosteal and soft tissue fluid collections. Sequential technetium and gallium or indium scans may help determine whether infection is active and may distinguish infection from noninflammatory bone changes; these methods do not, however, provide good anatomic detail. [MRI](#) provides detailed information about the activity and the anatomic extent of infection but does not always distinguish osteomyelitis from healing fractures and tumors. MRI is particularly useful in distinguishing cellulitis from osteomyelitis in the diabetic foot; however, no imaging modality consistently distinguishes infection from neuropathic osteopathy.

Appropriate samples for microbiologic studies should be obtained in all cases of suspected osteomyelitis before the initiation of antimicrobial therapy. Blood cultures are indicated in acute cases and are positive in more than one-third of cases of hematogenous osteomyelitis in children and in 25% of cases of vertebral osteomyelitis in adults. If the clinical picture demands immediate antibiotic therapy or if blood cultures are negative, samples from needle aspiration of pus in bone or soft tissues or from a bone biopsy should be obtained for culture.

The results of culture of specimens obtained by swabbing of a sinus tract or the base of an ulcer correlate poorly with the organisms infecting the bone. For this reason, in cases of chronic osteomyelitis and contiguous-focus osteomyelitis, samples for aerobic and anaerobic culture should be obtained from several sites by percutaneous needle aspiration, percutaneous biopsy, or intraoperative biopsy at the time of debridement. Isolates of coagulase-negative staphylococci and other organisms of low virulence should not automatically be disregarded as contaminants, especially in the presence of prosthetic materials. Special culture media may be necessary for the isolation of mycobacteria, fungi, and less common pathogens. In some cases, histopathologic examination of biopsy specimens may be the only way to make a diagnosis.

TREATMENT

Antibiotic Therapy Antibiotics are administered only after appropriate specimens have been obtained for culture. The antibiotics selected should be bactericidal and, at least initially, should be given intravenously. When necessary, empirical therapy is guided by findings on Gram's staining of a specimen from the bone or abscess or is chosen to cover the most likely pathogens. Empirical therapy in most cases should include high doses of an agent active against *S. aureus* (such as oxacillin, nafcillin, a cephalosporin, or vancomycin) and, if gram-negative organisms are likely to be involved, a third-generation cephalosporin, an aminoglycoside, or a fluoroquinolone.

Specific intravenous therapy is based on the in vitro susceptibility of the organism(s) isolated from bone or blood. Penicillin G (3 to 4 million units every 4 h) is the drug of choice for the treatment of infections due to penicillin-sensitive staphylococci and streptococci; nafcillin or oxacillin (2 g every 4 h) is preferred for penicillin-resistant, methicillin-sensitive staphylococci. Cefazolin (1 to 2 g every 8 h) or vancomycin [15 mg/kg (up to 1 g) every 12 h] is an alternative for persons allergic to penicillins. Infections due to methicillin-resistant staphylococci are treated with vancomycin.

Regimens for infections due to susceptible gram-negative rods include ampicillin (2 g every 4 h), cefazolin, a second-generation cephalosporin such as cefuroxime (1.5 g every 8 h), or a fluoroquinolone such as ciprofloxacin (400 mg every 12 h) or levofloxacin (500 mg every 24 h). Initial therapy for osteomyelitis due to *P. aeruginosa* or *Enterobacter* spp. should not consist of a b-lactam antibiotic alone because of the potential for these organisms to develop resistance during therapy. Appropriate intravenous therapies for *P. aeruginosa* infections include tobramycin (1.7 mg/kg every 8 h, or 5 to 7 mg/kg every 24 h) and a broad-spectrum b-lactam compound such as ticarcillin (3 g every 4 h), ceftazidime (1 to 2 g every 8 h), or aztreonam (1 to 2 g every 8 h); a fluoroquinolone may be substituted for one of the latter. *Enterobacter* infections can be treated with a fluoroquinolone alone or with combinations of a broad-spectrum b-lactam antibiotic and gentamicin in the same doses as tobramycin. Serum levels of aminoglycosides should be monitored closely to avoid toxicity.

The duration of therapy is typically 4 to 6 weeks; at-home intravenous administration of antibiotics or oral therapy is appropriate for motivated and medically stable patients. Antibiotics that require infrequent dosing, such as ceftriaxone and vancomycin, facilitate home therapy. Children with acute hematogenous osteomyelitis routinely receive oral antibiotics after 5 to 10 days of parenteral therapy if signs of active infection have resolved; such treatment has been as successful as standard parenteral therapy. The doses of oral penicillins or cephalosporins required for the treatment of osteomyelitis are several times higher than the doses of these drugs given for common infections. Adults may not tolerate these high doses as well as children, and, except in the case of the fluoroquinolones, few data support the use of oral antibiotics by adults. For treatment of osteomyelitis due to Enterobacteriaceae, oral administration of an agent such as ciprofloxacin (750 mg every 12 h) or levofloxacin (500 mg every 24 h) has been as successful as intravenous administration of b-lactam antibiotics. Caution should be exercised in the use of fluoroquinolones as the sole agents for treatment of infection due to *S. aureus* or *P. aeruginosa* because resistance may develop during therapy. Addition of rifampin to a quinolone has yielded encouraging results in infections due to *S. aureus*, but further studies are necessary to confirm these findings. Oral administration of clindamycin (300 to 450 mg every 6 h) or metronidazole (500 mg every 8 h) results in high drug levels in serum and can take the place of intravenous regimens for the treatment of *Bacteroides* infections. Oral clindamycin has produced good results in therapy for osteomyelitis due to *S. aureus*, especially in children. There are few data to support the routine use of the serum minimal bactericidal concentration (MBC) other than to document adherence to treatment.

Acute Osteomyelitis Early treatment of acute hematogenous osteomyelitis of childhood with 4 to 6 weeks of an appropriate antibiotic is usually successful; treatment for <3 weeks has resulted in a 10-fold greater rate of failure. Surgical intervention in childhood cases is indicated for intraosseous or subperiosteal abscesses, concomitant septic arthritis, and failure of the acute signs of infection to improve in 24 to 48 h. Acute hematogenous osteomyelitis of bones other than the spine in adults often requires surgical debridement.

Vertebral Osteomyelitis A 4- to 6-week course of treatment with an appropriate antibiotic is usually sufficient to cure vertebral osteomyelitis. Failure of the [ESR](#) to drop by two-thirds or more of its pretreatment level is an indication for longer treatment.

Surgery is seldom necessary, even in cases of many months' duration, except in instances of spinal instability, new or progressive neurologic deficits, large soft tissue abscesses that cannot be drained percutaneously, or a failure of medical treatment. Patients should maintain bed rest until back pain has declined to the point at which ambulation is possible. Body casts are no longer used. Spontaneous fusion of involved vertebrae occurs in the majority of cases after successful treatment.

Contiguous-Focus Osteomyelitis Even when diagnosed early, contiguous-focus osteomyelitis usually requires surgery in addition to 4 to 6 weeks of appropriate antibiotic therapy because of underlying soft tissue infection or damage to bone from an injury or surgery. A 2-week course of antibiotics following thorough debridement and soft tissue coverage has yielded excellent results in treatment of superficial osteomyelitis involving only the outer cortex of bone.

Chronic Osteomyelitis The risks and benefits of aggressive therapy for chronic osteomyelitis should be weighed before any attempt is made to eradicate the infection. Some patients with extensive disease prefer to live with their infections rather than undergo multiple surgical procedures, take prolonged courses of antimicrobial therapy, and face the risk of loss of an extremity. Such persons often benefit from intermittent courses of oral antibiotics to suppress acute exacerbations.

Once the decision has been made to treat chronic osteomyelitis aggressively, the patient's nutritional and metabolic status should be optimized to expedite healing of soft tissues and bone. Antibiotic administration should be started several days before surgery to reduce inflammation if the etiology of the infection is known preoperatively. If not, antibiotic therapy should be withheld until surgical debridement. An empirical antibiotic regimen is started intraoperatively after culture specimens are obtained. A 4- to 6-week course of appropriate antibiotic therapy is given postoperatively on the basis of the susceptibility pattern of organisms isolated from the bone. The benefit of prolonged oral antibiotic therapy after 4 to 6 weeks of parenteral therapy remains unproven. There is insufficient information to recommend the routine use of hyperbaric oxygen to enhance the killing of microorganisms by phagocytes or of instillation pumps and antibiotic-impregnated methacrylate beads to deliver high levels of antibiotics to the bone.

The success of therapy for chronic osteomyelitis rests largely on the complete surgical removal of necrotic bone and abnormal soft tissues. Modern imaging techniques allow accurate preoperative delineation of tissues to be debrided, but it remains difficult for the surgeon to determine intraoperatively whether all necrotic and infected tissue has been removed. In the past, the inability to repair large defects in bone and soft tissue limited the extent of debridement. Muscle flaps and skin grafts are now used routinely to cover large soft tissue defects and fill dead space, and bone grafts and vascularized bone transfer may restore a seriously compromised bone to a functional state.

In infections of recent fractures, internal fixators are often left in place, and the infection is controlled by limited debridement and suppressive antibiotic therapy. Definitive surgical/antimicrobial therapy is delayed until after bony union of the fracture is achieved. If there is nonunion of the fracture or loosening of the fixator, the appliance should be removed, the bone debrided, and an external fixator or a new internal fixator

applied.

Osteomyelitis of the small bones of the feet in persons with vascular disease also requires surgical treatment. The effectiveness of the surgery is limited by the blood supply to the site and the body's ability to heal the wound. Revascularization of the extremity is indicated if the vascular disease involves large arteries. In cases of decreased perfusion due to small-vessel disease, foot-sparing surgery may fail, and the best option is suppressive therapy or amputation. The duration of antibiotic therapy depends on the surgical procedure performed. When the infected bone is removed entirely but residual infection of soft tissues remains, antibiotic therapy should be given for 2 weeks; if amputation eliminates infected bone and soft tissue, standard surgical prophylaxis is given; otherwise, postoperative antibiotics must be given for 4 to 6 weeks.

(Bibliography omitted in Palm version)

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130. INTRAABDOMINAL INFECTIONS AND ABSCESES - Dori F. Zaleznik, Dennis L. Kasper

Intraperitoneal infections generally arise because a normal anatomic barrier is disrupted. This disruption may occur when the appendix, a diverticulum, or an ulcer ruptures; when the bowel wall is weakened by ischemia, tumor, or inflammation (e.g., in inflammatory bowel disease); or with adjacent inflammatory processes, such as pancreatitis or pelvic inflammatory disease, in which enzymes (in the former case) or organisms (in the latter) may leak into the peritoneal cavity. Whatever the inciting event, once inflammation develops and organisms usually contained within the bowel or another organ enter the normally sterile peritoneal space, a predictable series of events takes place. Intraabdominal infections occur in two stages: peritonitis and -- if it goes untreated -- abscess formation. The types of microorganisms predominating in each stage of infection are responsible for the pathogenesis of disease.

PERITONITIS

The peritoneal cavity is large but is divided into compartments. The upper and lower peritoneal cavities are divided by the transverse mesocolon; the greater omentum extends from the transverse mesocolon and from the lower pole of the stomach to line the lower peritoneal cavity. The pancreas, duodenum, and ascending and descending colon are located in the anterior retroperitoneal space; the kidneys, ureters, and adrenals are found in the posterior retroperitoneal space. The other organs, including liver, stomach, gallbladder, spleen, jejunum, ileum, transverse and sigmoid colon, cecum, and appendix, are found within the peritoneal cavity itself. Normally the cavity is lined with a serous membrane that can serve as a conduit for fluids -- a property utilized in peritoneal dialysis. A small amount of fluid, sufficient to allow movement of organs, is normally present in the peritoneal space. This fluid is serous, with a protein content (consisting mainly of albumin) of <30 g/L and fewer than 300 white blood cells (WBCs, generally mononuclear cells) per microliter. In the presence of infection, some of these compartments collect fluid or pus more often than others. These compartments include the pelvis (the lowest portion), the subphrenic spaces on the right and left sides, and Morrison's pouch, which is a posterosuperior extension of the subhepatic spaces and is the lowest part of the paravertebral groove when a patient is recumbent. The falciform ligament separating the right and left subphrenic spaces appears to act as a barrier to the spread of infection; consequently, it is unusual to find bilateral subphrenic collections.

SPONTANEOUS BACTERIAL PERITONITIS

Peritonitis is either primary (without an apparent source of contamination) or secondary. The types of organisms found and the clinical presentation of these two processes are different. In adults, primary or spontaneous bacterial peritonitis (SBP) occurs most commonly in conjunction with cirrhosis of the liver (frequently the result of alcoholism). It virtually always develops in patients with ascites. Nevertheless, it is not a common event, occurring in ~10% of cirrhotic patients. The cause of SBP has not been established definitively but is believed to involve hematogenous spread of organisms in a patient in whom a diseased liver and altered portal circulation result in a defect in the usual filtration function. Organisms are able to multiply in ascites, a good medium for

growth. The proteins of the complement cascade have been found in peritoneal fluid, with lower levels in cirrhotic patients than in patients with ascites of other etiologies. The opsonic and phagocytic properties of neutrophils are decreased in patients with advanced liver disease.

The presentation of [SBP](#) differs from that of secondary peritonitis. The most common manifestation is fever, which is reported in as many as 80% of patients. Ascites is found but virtually always predates infection. Abdominal pain, an acute onset of symptoms, and peritoneal irritation detected during physical examination can be helpful diagnostically, but the absence of any of these findings does not exclude this often-subtle diagnosis. It is vital to sample the peritoneal fluid of any cirrhotic patient with ascites and fever. The finding of >300 polymorphonuclear leukocytes (PMNs) per microliter is diagnostic for SBP, according to Conn. The microbiology of SBP is also distinctive. While enteric gram-negative bacilli such as *Escherichia coli* are most commonly encountered, gram-positive organisms such as streptococci, enterococci, or even pneumococci are sometimes found. In SBP, a single organism is typically isolated; anaerobes are found less frequently in SBP than in secondary peritonitis, in which a mixed flora including anaerobes is the rule. In fact, if SBP is suspected and multiple organisms including anaerobes are recovered from the peritoneal fluid, the diagnosis must be reconsidered and a source of secondary peritonitis sought.

The diagnosis of [SBP](#) is not easy. It depends on the exclusion of a primary intraabdominal source of infection. Contrast-enhanced computed tomography (CT) is very useful in identifying an intraabdominal source for infection. It may be difficult to recover organisms from cultures of peritoneal fluid, presumably because the burden of organisms is low. However, the yield can be improved if 10 mL of peritoneal fluid is placed directly into a blood culture bottle. Bacteremia frequently accompanies SBP; therefore, blood should be cultured simultaneously. No specific radiographic studies are helpful in the diagnosis of SBP. A plain film of the abdomen would be expected to show ascites. Chest and abdominal radiography should be performed in patients with abdominal pain to exclude free air, which signals a perforation.

TREATMENT

Treatment for [SBP](#) is directed at the isolate from blood or peritoneal fluid. Gram's staining of peritoneal fluid often gives negative results in primary peritonitis; therefore, until culture results become available, empirical therapy should cover gram-negative aerobic bacilli and gram-positive cocci. Ampicillin plus gentamicin is a reasonable initial regimen. Third-generation cephalosporins, carbapenems, or broad-spectrum penicillin/b-lactamase inhibitor combinations are also options. Empirical coverage for anaerobes is not necessary. After the infecting organism is identified, therapy should be narrowed to target that specific pathogen. Patients with SBP usually respond within 72 h to appropriate antibiotic therapy.

SECONDARY PERITONITIS

Secondary peritonitis develops when bacteria contaminate the peritoneum as a result of spillage from an intraabdominal viscus. The organisms found almost always constitute a mixed flora in which facultative gram-negative bacilli and anaerobes predominate,

especially when the contaminating source is colonic. Early in the course of infection, when the host response is directed toward containment of the infection, exudate containing fibrin and [PMNs](#) is found. Early death in this setting is attributable to gram-negative bacillary sepsis and to potent endotoxins circulating in the bloodstream ([Chap. 124](#)). Gram-negative bacilli, particularly *E. coli*, are common bloodstream isolates, but *Bacteroides fragilis* bacteremia occurs as well. The severity of abdominal pain and the clinical course depend on the inciting process. The species of organisms isolated from the peritoneum also vary with the source of the initial process and the normal flora present at that site. Peritonitis can result primarily from chemical irritation or bacterial contamination. For example, as long as the patient is not achlorhydric, a ruptured gastric ulcer will release low-pH gastric contents that will serve as a chemical irritant. The normal flora of the stomach comprises the same organisms found in the oropharynx ([Chap. 167](#)) but in lower numbers. The surfaces of teeth contain $\sim 10^7$ aerobic and 10^7 anaerobic organisms per milliliter of saliva; the normally acidic stomach contains an equal ratio of aerobic and anaerobic species, but in concentrations more in the range of 10^5 /mL. After meals, when gastric acidity is highest, this number may fall to 10^3 /mL. Thus, the bacterial burden in a ruptured gastric ulcer -- or even a duodenal ulcer -- is negligible compared with that in a ruptured appendix. The normal flora of the colon below the ligament of Treitz contains $\sim 10^{11}$ anaerobic organisms per gram of feces but only 10^8 aerobes per gram; therefore, anaerobic species account for 99% of the bacteria. Leakage of colonic contents (pH 7 to 8) does not cause significant chemical peritonitis, but infection is intense because of the heavy bacterial load.

Depending on the inciting event, local symptoms may initially be found in secondary peritonitis -- for example, epigastric pain from a ruptured gastric ulcer. In appendicitis ([Chap. 291](#)), the initial presenting symptoms are often vague, with periumbilical discomfort and nausea followed in a number of hours by pain more localized to the right lower quadrant. Unusual locations of the appendix (including a retrocecal position) can complicate this presentation further. Once infection has spread to the peritoneal cavity, however, pain increases, particularly with infection involving the parietal peritoneum, which is innervated extensively. Patients usually lie motionless, often with knees drawn up to avoid stretching the nerve fibers of the peritoneal cavity. Coughing and sneezing, which increase pressure within the peritoneal cavity, are associated with sharp pain. There may or may not be pain localized to the infected or diseased organ from which secondary peritonitis has arisen. Patients with secondary peritonitis generally have abnormal findings on abdominal examination, with marked voluntary and involuntary guarding of the anterior abdominal musculature. Later findings include tenderness, especially rebound tenderness. In addition, there may be localized findings in the area of the inciting event. In general, patients are febrile, with marked leukocytosis and a left shift of the [WBCs](#) to earlier granulocyte forms.

While recovery of organisms from peritoneal fluid is easier in secondary than in primary peritonitis, a tap of the abdomen is rarely the procedure of choice in secondary peritonitis. An exception is in cases involving trauma, where the possibility of a hemoperitoneum may need to be excluded early.

TREATMENT

Treatment for secondary peritonitis includes early administration of antibiotics aimed

particularly at aerobic gram-negative bacilli and anaerobes (see below) as well as etiologic studies. Secondary peritonitis usually requires both surgical intervention to address the inciting process and antibiotic administration to treat early bacteremia, to decrease the incidence of abscess formation and wound infection, and to prevent more distant spread of infection. In [SBP](#) in adults, surgery is rarely indicated. In secondary peritonitis, surgery may be life-saving.

PERITONITIS IN PATIENTS UNDERGOING CAPD

A third type of peritonitis arises in patients who are undergoing continuous ambulatory peritoneal dialysis (CAPD). Unlike primary and secondary peritonitis, which are caused by endogenous bacteria, peritonitis in CAPD patients usually involves skin organisms. The pathogenesis of infection is similar to that of intravascular-device infection, in which skin organisms migrate along the catheter, which both serves as an entry point and exerts the effects of a foreign body. Exit-site or tunnel infection may or may not accompany CAPD peritonitis. Like primary peritonitis, CAPD peritonitis is usually caused by a single organism. Peritonitis is, in fact, the most common reason for discontinuation of CAPD. Improvements in equipment design, especially that of the Y-set connector, have resulted in a decrease from one case of peritonitis per 9 months of CAPD to one case per 15 months.

The clinical presentation of [CAPD](#) peritonitis resembles that of secondary peritonitis in that diffuse pain and peritoneal signs are common. The dialysate is usually cloudy and contains >100 [WBCs](#) per microliter, $>50\%$ of which are neutrophils. The most common etiologic organism is coagulase-negative *Staphylococcus*, which accounts for $\sim 30\%$ of cases. *S. aureus* causes $\sim 10\%$ of cases, is more commonly identified among patients who are nasal carriers of the organism, and is the most frequent pathogen in those with an overt exit-site infection. Gram-negative bacilli and fungi such as *Candida* species are also found. Vancomycin-resistant enterococci (VRE) and vancomycin-intermediate *S. aureus* (VISA) have been reported to produce peritonitis in CAPD patients. The finding of more than one organism in dialysate culture should prompt a search for a cause of secondary peritonitis. As with primary peritonitis, culture of dialysate fluid in blood culture bottles improves the yield.

TREATMENT

Empirical therapy for [CAPD](#) peritonitis should be directed at coagulase-negative *Staphylococcus*, *S. aureus*, and gram-negative bacilli until the results of cultures are available. Since the advent of [VRE](#) and [VISA](#), recommended treatment has changed from vancomycin and an aminoglycoside to a first-generation cephalosporin such as cefazolin and an aminoglycoside, which can be administered together in the same bag. A loading dose of cefazolin (500 mg/L) is administered intraperitoneally, with a maintenance dose of 125 mg/L in each bag. Ototoxicity is a significant concern in patients receiving aminoglycosides; some data suggest that administration of a single daily dose lessens this risk. Thus, gentamicin is usually administered at a dose of 20 mg/L once a day. If methicillin-resistant *S. aureus* is a relatively common isolate in a community, vancomycin may still be a reasonable first choice for empirical therapy, especially in a toxic-appearing patient or a patient with an overt exit-site infection. The dose (2 g) is allowed to remain in the peritoneal cavity for 6 h. The clinical response to

an empirical treatment regimen should be rapid; if the patient has not responded after 48 h of treatment, catheter removal should be considered.

INTRAPERITONEAL ABSCESES

Abscess formation is common in untreated peritonitis if overt gram-negative sepsis either does not develop or develops but is not fatal. In experimental models of abscess formation, mixed aerobic and anaerobic organisms have been implanted intraperitoneally. Without therapy directed at anaerobes, animals develop intraabdominal abscesses. As in humans, these experimental abscesses may stud the peritoneal cavity, lie within the omentum or mesentery, or even develop on the surface of or within viscera such as the liver.

PATHOGENESIS AND IMMUNITY

There is often disagreement about whether an abscess represents a disease state or a host response. In a sense, it represents both: While an abscess is an infection in which viable infecting organisms and [PMNs](#) are contained in a fibrous capsule, it is also a process by which the host confines microbes to a limited space, thereby preventing further spread of infection. Experimental work has helped to define both the host cells and the bacterial virulence factors responsible -- most notably, in the case of *B. fragilis*. This organism, although accounting for only 0.5% of the normal colonic flora, is the anaerobe most frequently isolated from intraabdominal infections and is the most common anaerobic bloodstream isolate. On clinical grounds, therefore, *B. fragilis* appears to be uniquely virulent. Moreover, *B. fragilis* causes abscesses in animal models of intraabdominal infection, whereas most other *Bacteroides* species must act synergistically with a facultative organism to induce abscess formation.

Of the several virulence factors identified in *B. fragilis*, one is critical -- the capsular polysaccharide complex (CPC) found on the bacterial surface. The CPC comprises several distinct surface polysaccharides. Structural analysis of the polysaccharides in the CPC has shown an unusual motif of oppositely charged sugars. Polysaccharides having these *zwitterionic* characteristics evoke a host response in the peritoneal cavity that localizes bacteria into abscesses. *B. fragilis* and the CPC have been found to adhere to primary mesothelial cells in vitro; this adherence, in turn, stimulates the production of tumor necrosis factor (TNF- α) and intercellular adhesion molecule 1 (ICAM-1) by peritoneal macrophages. Mice treated with antibodies to TNF- α or ICAM-1 did not develop abscesses in the mouse peritonitis model. Although abscesses characteristically contain [PMNs](#), the process of abscess induction depends on the stimulation of T lymphocytes by these unique polysaccharides. Experimentally, the essential role of T cells in initiating abscess formation has been proven following blockage of the CD28-B7 costimulatory pathway. The alternative pathways of complement and fibrinogen also participate in abscess formation.

While antibodies to the [CPC](#) are not critical in immunity to abscesses, they enhance bloodstream clearance of *B. fragilis*. When administered subcutaneously, *B. fragilis* surface polysaccharide A (PS A) has immunomodulatory characteristics and stimulates T cells via an interleukin-2-dependent mechanism to inhibit the host response of abscess formation to intraperitoneal challenge with *B. fragilis*. Treatment of

experimental animals with PS A or other zwitterionic molecules reduces abscess development and can be administered after bacterial contamination of the peritoneal cavity.

CLINICAL PRESENTATION

Most intraperitoneal abscesses result from fecal spillage from a colonic source, such as an inflamed appendix. Of all intraabdominal abscesses, 74% are intraperitoneal or retroperitoneal and are not visceral. Abscesses can also arise from a number of other processes. They usually form within weeks of the development of peritonitis and may be found in a variety of locations, from omentum to mesentery, pelvis to psoas muscles, and subphrenic space to a visceral organ such as the liver, where they may develop either on the surface of the organ or within it. Periappendiceal and diverticular abscesses have traditionally been frequent. Diverticular abscesses are least likely to rupture. Infections of the female genital tract and pancreatitis are also among the more common causative events. When abscesses occur in the female genital tract -- either as a primary infection (e.g., tuboovarian abscess) or as an infection extending into the pelvic cavity or peritoneum -- *B. fragilis* figures prominently among the organisms isolated. *B. fragilis* is not found in large numbers in the normal vaginal flora. It is encountered less commonly in pelvic inflammatory disease and endometritis, for example, without an associated abscess. In pancreatitis with leakage of damaging pancreatic enzymes, inflammation is prominent. Therefore, clinical findings such as fever, leukocytosis, and even abdominal pain do not distinguish pancreatitis itself from complications such as pancreatic pseudocyst, pancreatic abscess ([Chap. 304](#)), or intraabdominal collections of pus. Especially in cases of necrotizing pancreatitis, in which the incidence of local pancreatic infection may be as high as 30%, needle aspiration under [CT](#) guidance is performed as often as once a week to sample fluid for culture. Many centers prescribe prophylactic antibiotics to prevent infection in patients with necrotizing pancreatitis. Imipenem is the most frequently used drug for this purpose since it achieves high tissue levels in the pancreas, although it is not unique in this regard. If infected fluid is removed during needle aspiration, most experts agree that surgery is superior to percutaneous drainage.

The psoas muscle of the anterior back is another location in which abscesses are encountered. These abscesses may arise from a presumed hematogenous source, by contiguous spread from an intraabdominal or pelvic process, or by contiguous spread from nearby bony structures such as vertebral bodies. Associated osteomyelitis due to spread from bone to muscle or from muscle to bone is common in psoas abscesses. When Pott's disease was common, *Mycobacterium tuberculosis* was a frequent cause of psoas abscess. Currently in the United States, the usual isolates from psoas abscesses are either *S. aureus* or a mixture of enteric organisms including aerobic gram-negative bacilli. *S. aureus* is most likely to be isolated when a psoas abscess arises from hematogenous spread or a contiguous focus of osteomyelitis; a mixed enteric flora is most likely when the abscess has an intraabdominal or pelvic source.

DIAGNOSIS

A variety of scanning procedures have considerably facilitated the diagnosis of intraabdominal abscesses. Abdominal [CT](#) probably has the highest yield, although

ultrasonography is particularly useful for the right upper quadrant, kidneys, and pelvis. Both indium-labeled [WBCs](#) and gallium tend to localize in abscesses and may be useful in finding a collection. Since gallium is taken up in the bowel, indium-labeled WBCs may have a slightly greater yield for abscesses near the bowel. Neither indium-labeled WBC nor gallium scans serve as a basis for a definitive diagnosis, however; both need to be followed by other, more specific studies, such as CT, if an area of possible abnormality is identified. Abscesses contiguous with or contained within outpouchings of bowel are particularly difficult to diagnose with scanning procedures. Occasionally, a barium enema may detect a diverticular abscess not diagnosed by other procedures, although barium should not be injected if a free perforation is suspected. If one study is negative, a second study sometimes reveals a collection. On occasion, exploratory laparotomy still must be undertaken if an abscess is strongly suspected on clinical grounds, although this procedure has been less commonly used since the advent of CT.

TREATMENT

An algorithm for the management of patients with intraabdominal abscesses is presented in [Fig. 130-1](#). The treatment of intraabdominal infections involves the determination of the initial focus of infection, the administration of broad-spectrum antibiotics targeted at organisms involved in the associated infection, and the performance of a drainage procedure if one or more definitive abscesses have formed already. It cannot be overemphasized that antimicrobial therapy, in general, is adjunctive to drainage and/or surgical correction of an underlying lesion or process in intraabdominal abscesses ([Fig. 130-CD1](#)). Unlike the intraabdominal abscesses precipitated by most infections, for which drainage of some kind is generally required, abscesses associated with diverticulitis usually wall off locally after rupture of a diverticulum, so that surgical intervention is not routinely required.

A number of antimicrobial agents exhibit excellent activity against aerobic gram-negative bacilli. Since mortality in intraabdominal sepsis is linked to gram-negative bacteremia, empirical therapy for intraabdominal infection always needs to include adequate coverage of gram-negative aerobic and facultative organisms. Aminoglycosides and second- and third-generation cephalosporins are the agents most widely tested and used in intraabdominal processes. Newer antibiotics, such as aztreonam, imipenem, ticarcillin/clavulanic acid, piperacillin/tazobactam, and quinolones (e.g., ciprofloxacin), cover these organisms as well, although at a higher cost. Second-generation cephalosporins, such as cefoxitin or cefotetan, are not as uniformly active as the other agents against all of the aerobic gram-negative species. Aztreonam, ciprofloxacin, aminoglycosides, and most of the third-generation cephalosporins are not active against anaerobes; for the treatment of intraabdominal infections, these drugs need to be used in combination with another antibiotic. Since a number of antibiotics highly effective against anaerobes are available, third-generation cephalosporins generally should not be considered for use against the anaerobic bacteria involved in intraabdominal sepsis.

The most active and cost-effective antibiotic for anaerobic coverage currently is metronidazole ([Chap. 167](#)). Only rare isolates of *B. fragilis* have been reported to be resistant to this drug. In a study of 3177 anaerobic isolates from eight centers in the United States over a 5-year period, no strains resistant to metronidazole were

documented among *B. fragilis* isolates, and resistance to imipenem, ampicillin/sulbactam, and piperacillin/tazobactam was exceedingly rare. In contrast, resistance to cefotetan, ceftizoxime, and clindamycin increased during this interval, and resistance to cefoxitin was measurable but unchanged during the study. Despite increasing reports of in vitro resistance of *B. fragilis* to a number of agents, clinical failures are still limited to case reports; therefore, the clinical significance of antimicrobial resistance in anaerobes is uncertain. One report describes a bloodstream isolate of *B. fragilis* with resistance to metronidazole and with reduced susceptibility to imipenem and amoxicillin/clavulanic acid that became resistant to both of the latter two drugs after treatment of the patient with imipenem. Among newer agents, imipenem, ticarcillin/clavulanic acid, piperacillin/tazobactam, meropenem, and ampicillin/sulbactam are highly active against anaerobes. Chloramphenicol, which exhibits strong activity against *B. fragilis* in vitro, nevertheless should probably not be considered a first-line drug for use against anaerobes, since failures of treatment have been documented in both experimental and clinical intraabdominal infections. Neither metronidazole nor clindamycin covers aerobic gram-negative bacilli; thus, these drugs must be combined with other agents for use in this setting. Metronidazole is also less active against gram-positive than against gram-negative anaerobic species.

VISCERAL ABSCESSSES

Liver Abscesses The liver is the organ most subject to the development of abscesses. Altemeier and associates studied 540 intraabdominal abscesses over a 12-year period. Of these abscesses 26% were visceral. Liver abscesses made up 13% of the total number of abscesses, or 48% of all visceral abscesses. Liver abscesses may be solitary or multiple ([Fig. 130-CD2](#)); they may arise from hematogenous spread of bacteria or from local spread from contiguous sites of infection within the peritoneal cavity. In the past, appendicitis with rupture and subsequent spread of infection was the most common route for the development of a liver abscess. Currently, associated disease of the biliary tract is most often the etiology. Suppurative pylephlebitis (suppurative thrombosis of the portal vein), usually arising from infection in the pelvis but sometimes from infection elsewhere in the peritoneal cavity, is another common source for bacterial seeding of the liver.

Fever is the most common presenting sign of liver abscess. Some patients, particularly those with active associated disease of the biliary tract, have symptoms and signs localized to the right upper quadrant, including pain, guarding, punch tenderness, and even rebound tenderness. Nonspecific symptoms, such as chills, anorexia, weight loss, nausea, and vomiting, may also develop. Only 50% of patients with liver abscesses, however, have hepatomegaly, right-upper-quadrant tenderness, or jaundice; thus, half of patients have no symptoms or signs that would direct attention to the liver. Fever of unknown origin (FUO) may be the only presenting manifestation of liver abscess, especially in the elderly. Diagnostic studies of the abdomen, especially the right upper quadrant, should be a part of any FUO workup. The single most reliable laboratory finding is an elevated serum concentration of alkaline phosphatase, which is documented in 70% of patients with liver abscesses. Other tests of liver function may yield normal results, but 50% of patients have elevated serum levels of bilirubin, and 48% have elevated concentrations of aspartate aminotransferase. Other associated laboratory findings include leukocytosis in 77% of patients, anemia (usually

normochromic, normocytic) in 50%, and hypoalbuminemia in 33%. Concomitant bacteremia is found in one-third of patients. A liver abscess is sometimes suggested by chest radiography, especially if a new elevation of the right hemidiaphragm is seen; other suggestive findings include a right basilar infiltrate and a right pleural effusion.

Imaging studies are the most reliable methods for diagnosing liver abscesses. These studies include ultrasonography, [CT](#), indium-labeled [WBC](#) or gallium scans, and even magnetic resonance imaging. In an occasional case, more than one such study may be required. Organisms recovered from liver abscesses vary with the etiology. In liver infection arising from the biliary tree, enteric gram-negative aerobic bacilli and enterococci are common isolates. Unless previous surgery has been performed, anaerobes are not generally involved in liver abscesses arising from biliary infections. In contrast, in liver abscesses arising from pelvic and other intraperitoneal sources, a mixed flora including aerobic and anaerobic species (especially *B. fragilis*) is common. With hematogenous spread of infection, usually only a single organism is encountered; this species may be *S. aureus* or a streptococcal species such as *S. milleri*.

Liver abscesses may also be caused by *Candida* species; such abscesses usually follow fungemia in patients receiving chemotherapy for cancer and often present when neutrophils return after a period of neutropenia. Amebic liver abscesses are not an uncommon problem ([Chap. 213](#)). Amebic serologic testing gives positive results in >95% of cases; thus, a negative result helps to exclude this diagnosis.

TREATMENT

While drainage -- either percutaneous (with a pigtail catheter kept in place) or surgical -- remains the mainstay of therapy for intraabdominal abscesses (including liver abscesses), there is growing interest in medical management alone for pyogenic liver abscesses. The drugs used in empirical broad-spectrum antibiotic therapy include the same ones used in intraabdominal sepsis. Usually, a diagnostic aspirate of abscess contents should be obtained before the initiation of empirical therapy, with antibiotic choices adjusted when the results of Gram's staining and culture become available. Cases treated without definitive drainage generally require longer courses of antibiotic therapy. When percutaneous drainage was compared with open surgical drainage, the average length of hospital stay for the former was almost twice that for the latter, although both the time required for fever to resolve and mortality were the same for the two procedures. Mortality was appreciable despite treatment, averaging 15%. Several factors may predict the failure of percutaneous drainage and therefore may favor primary surgical intervention. These factors include the presence of multiple, sizable abscesses; viscous abscess contents that tend to plug the catheter; associated disease (e.g., disease of the biliary tract) that requires surgery; or the lack of a clinical response to percutaneous drainage in 4 to 7 days.

Treatment of candidal liver abscesses usually entails lengthy administration of amphotericin B, although reports have described successful maintenance therapy with fluconazole after an initial course of amphotericin ([Chap. 205](#)).

Splenic Abscesses Splenic abscesses are much less common than liver abscesses. In fact, no splenic abscesses were observed in Altemeier's series of 540 intraabdominal

abscesses. The incidence of splenic abscesses has ranged from 0.14 to 0.7% in various autopsy series. The clinical setting and the organisms isolated usually differ from those for liver abscesses. The degree of clinical suspicion for splenic abscess needs to be high, as this condition is frequently fatal if left untreated. Even in the most recently published series, diagnosis was made only at autopsy in 37% of cases. While splenic abscesses may arise occasionally from contiguous spread of infection or from direct trauma to the spleen, hematogenous spread of infection is the usual mode of development. Bacterial endocarditis is the most common associated infection. Splenic abscesses can develop in patients who have received extensive immunosuppressive therapy (particularly those with malignancy involving the spleen) and in patients with hemoglobinopathies or other hematologic disorders (especially sickle cell anemia).

While ~50% of patients with splenic abscesses have abdominal pain, the pain is localized to the left upper quadrant in only half of these cases. Splenomegaly is found in ~50% of cases. Fever and leukocytosis are generally present; the development of fever preceded diagnosis by an average of 20 days in one series. Left-sided chest findings may include abnormalities to auscultation, and chest radiographic findings may include an infiltrate or a left-sided pleural effusion. When splenic abscesses are being considered in a differential diagnosis, [CT](#) scan of the abdomen has been the most sensitive diagnostic tool. Ultrasonography can yield the diagnosis, but cases have been missed with this modality. Liver-spleen scan or gallium scan may also be useful. Streptococcal species are the most common bacterial isolates from splenic abscesses, and *S. aureus* is the next most common; presumably these prevalences reflect the bacterial cause of the associated endocarditis. An increase in the frequency of isolation of gram-negative aerobic organisms from splenic abscesses has been reported; these organisms often derive from a urinary tract focus, with associated bacteremia, or from another intraabdominal source. *Salmonella* species are seen fairly commonly, especially in patients with sickle cell hemoglobinopathy. Anaerobic species accounted for only 5% of isolates in the largest collected series, but the reporting of a number of "sterile abscesses" may indicate that optimal techniques for the isolation of anaerobes were not employed.

TREATMENT

Because of the high mortality figures reported for splenic abscesses, the treatment of choice is splenectomy with adjunctive antibiotics. However, percutaneous drainage has been successful. The most important factor in successful treatment of splenic abscesses is early consideration of the diagnosis.

Perinephric and Renal Abscesses Perinephric and renal abscesses are not common: The former accounted for only ~0.02% of hospital admissions and the latter for ~0.2% in Altemeier's series of 540 intraabdominal abscesses. While liver abscesses generally arise from contiguous foci of infection or track from other intraabdominal sources and splenic abscesses usually arise from hematogenous spread (e.g., spread from bacterial endocarditis), perinephric and renal abscesses have a different pathogenesis. Before antibiotics became available, most renal and perinephric abscesses were hematogenous in origin, with *S. aureus* most commonly recovered. Now, in contrast, >75% of perinephric and renal abscesses arise from an initial urinary tract infection. Infection ascends from the bladder to the kidney, with pyelonephritis occurring

first. Bacteria may directly invade the renal parenchyma from medulla to cortex. Local vascular channels within the kidney may also facilitate the transport of organisms. Areas of abscess developing within the parenchyma may rupture into the perinephric space. The kidneys and adrenal glands are surrounded by a layer of perirenal fat that, in turn, is surrounded by Gerota's fascia, which extends superiorly to the diaphragm and inferiorly to the pelvic fat. When abscesses extend into the perinephric space, tracking may occur through Gerota's fascia into the psoas or transversalis muscles, into the anterior peritoneal cavity, superiorly to the subdiaphragmatic space, or inferiorly to the pelvis. Of the several risk factors that have been associated with the development of perinephric abscesses, the most important is the presence of concomitant nephrolithiasis producing local obstruction to urinary flow. Of patients with perinephric abscess, 20 to 60% have renal stones. In addition, other structural abnormalities of the urinary tract, a history of urologic surgery, trauma, and diabetes mellitus have all been identified as risk factors.

The organisms most frequently encountered in perinephric and renal abscesses are *E. coli*, *Proteus* species, and *Klebsiella* species. *E. coli*, the aerobic species most commonly found in colonic flora, seems to have unique virulence properties in the urinary tract, including factors promoting adherence to uroepithelial cells. The urease of *Proteus* species splits urea, thereby creating a more alkaline and hospitable environment for bacterial proliferation. *Proteus* species are frequently found in association with large struvite stones caused by the precipitation of magnesium ammonium sulfate in an alkaline environment. These stones serve as a nidus for recurrent urinary tract infection. While a single bacterial species is usually recovered from a perinephric or renal abscess, multiple species may also be found. If a urine culture is not contaminated with periurethral flora and is found to contain more than one organism, a perinephric abscess or renal abscess should be considered in the differential diagnosis. Urine cultures may also be polymicrobial in cases of bladder diverticulum.

Candida species should be considered in the etiology of renal abscesses. This fungus may spread to the kidney via the hematogenous route or by ascension from the bladder. The hallmark of the latter route of infection is ureteral obstruction with large fungal balls.

The presentation of perinephric and renal abscesses is quite nonspecific. Flank pain and abdominal pain are common. At least 50% of patients are febrile. Pain may be referred to the groin or leg, particularly with extension of infection. The diagnosis of perinephric abscess, like that of splenic abscess, is frequently delayed, and mortality in some series is appreciable, although lower than in the past. Perinephric or renal abscess should be most seriously considered when a patient presents with symptoms and signs of pyelonephritis and remains febrile after 4 or 5 days, by which time the fever should have resolved. Moreover, when a urine culture yields a polymicrobial flora, when a patient is known to have renal stone disease, or when fever and pyuria coexist with a sterile urine culture, the diagnosis of perinephric or renal abscess should be entertained.

Renal ultrasonography and abdominal [CT](#) are the most useful diagnostic modalities. If a renal abscess or perinephric abscess is diagnosed, nephrolithiasis should be excluded, especially when a high urinary pH suggests the presence of a urea-splitting organism.

TREATMENT

Treatment for perinephric or renal abscesses, like that for other intraabdominal abscesses, includes drainage of pus and antibiotic therapy directed at the organism(s) recovered. For perinephric abscesses, percutaneous drainage is usually successful.

Pancreatic Abscesses*[See Chap. 304.](#)

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131. ACUTE INFECTIOUS DIARRHEAL DISEASES AND BACTERIAL FOOD POISONING - Joan R. Butterton, Stephen B. Calderwood

Ranging from mild annoyances during vacations to devastating dehydrating illnesses that can kill within hours, acute gastrointestinal illnesses rank second only to acute upper respiratory illnesses as the most common diseases worldwide. In children <5 years old, attack rates range from 2 to 3 illnesses per child per year in developed countries to as high as 10 to 18 illnesses per child per year in developing countries. In Asia, Africa, and Latin America, acute diarrheal illnesses are not only a leading cause of morbidity in children -- with an estimated 1 billion cases per year -- but also the major cause of mortality, being responsible for 4 to 6 million deaths per year, or a sobering total of 12,600 deaths per day. In some areas, more than 50% of childhood deaths are directly attributable to acute diarrheal illnesses. In addition, by contributing to malnutrition and thereby reducing resistance to other infectious agents, gastrointestinal illnesses may be indirect factors in a far greater burden of disease.

The wide range of clinical manifestations of acute gastrointestinal illnesses is matched by the wide variety of infectious agents involved, including viruses, bacteria, and parasitic pathogens ([Table 131-1](#)). This chapter will discuss factors that enable gastrointestinal pathogens to cause disease, will review host defense mechanisms, and will delineate an approach to the evaluation and treatment of patients presenting with acute diarrhea. Individual organisms causing acute gastrointestinal illnesses are discussed in detail in subsequent chapters.

PATHOGENIC MECHANISMS

Enteric pathogens have developed a variety of tactics to overcome host defenses. Understanding the virulence factors employed by these organisms is important in the diagnosis and treatment of clinical disease.

Inoculum Size The number of microorganisms that must be ingested to cause disease varies considerably from species to species. For *Shigella*, enterohemorrhagic *Escherichia coli*, *Giardia lamblia*, or *Entamoeba*, as few as 10 to 100 bacteria or cysts can produce infection, while 10⁵ to 10⁸ *Vibrio cholerae* organisms must be ingested orally to cause disease. The infective dose of *Salmonella* varies widely, depending on the species, host, and food vehicle. The ability of organisms to overcome host defenses has important implications for transmission; *Shigella*, enterohemorrhagic *E. coli*, *Entamoeba*, and *Giardia* can spread by person-to-person contact, whereas under some circumstances *Salmonella* may have to grow in food for several hours before reaching an effective infectious dose.

Adherence Many organisms must adhere to the gastrointestinal mucosa as an initial step in the pathogenic process; thus, organisms that can compete with the normal bowel flora and colonize the mucosa have an important advantage in causing disease. Specific cell-surface proteins involved in attachment of bacteria to intestinal cells are important virulence determinants. *V. cholerae*, for example, adheres to the brush border of small-intestinal enterocytes via specific surface adhesins, including the toxin-coregulated pilus and other accessory colonization factors. Enterotoxigenic *E. coli* produces an adherence protein called *colonization factor antigen* that is necessary for

colonization of the upper small intestine by the organism prior to the production of enterotoxin. Enteropathogenic and enterohemorrhagic strains of *E. coli* produce virulence determinants that allow these organisms to attach to and efface the brush border of the intestinal epithelium.

Toxin Production The production of one or more exotoxins is important in the pathogenesis of numerous enteric organisms. Such toxins include *enterotoxins*, which cause watery diarrhea by acting directly on secretory mechanisms in the intestinal mucosa; *cytotoxins*, which cause destruction of mucosal cells and associated inflammatory diarrhea; and *neurotoxins*, which act directly on the central or peripheral nervous system. Some exotoxins act by more than one mechanism; *Shigella dysenteriae* type 1, for example, produces an exotoxin that has both enterotoxic and cytotoxic activities.

The prototypical enterotoxin is cholera toxin, a heterodimeric protein composed of one A and five B subunits. The A subunit contains the enzymatic activity of the toxin, while the B subunit pentamer binds holotoxin to the enterocyte surface receptor, the ganglioside GM₁. After the binding of holotoxin, a fragment of the A subunit is translocated across the eukaryotic cell membrane into the cytoplasm, where it catalyzes the ADP-ribosylation of a GTP-binding protein and causes persistent activation of adenylate cyclase. The end result is an increase of cyclic AMP in the intestinal mucosa, which increases Cl⁻ secretion and decreases Na⁺ absorption, leading to loss of fluid and the production of diarrhea.

Enterotoxigenic strains of *E. coli* may produce a protein called *heat-labile enterotoxin* (LT) that is similar to cholera toxin and causes secretory diarrhea by the same mechanism. Alternatively, enterotoxigenic strains of *E. coli* may produce *heat-stable enterotoxin* (ST), one form of which causes diarrhea by activation of guanylate cyclase and elevation of intracellular cyclic GMP. Some enterotoxigenic strains produce both LT and ST.

Bacterial cytotoxins, in contrast, destroy intestinal mucosal cells and produce the syndrome of dysentery, with bloody stools containing inflammatory cells. Enteric pathogens that produce such cytotoxins include *S. dysenteriae* type 1, *Vibrio parahaemolyticus*, and *Clostridium difficile*. Shiga toxin-producing strains of *E. coli*, most commonly serotype O157:H7 in the United States, produce potent cytotoxins that are highly related to the Shiga toxin of *S. dysenteriae* type 1. Such strains of *E. coli* have been associated with outbreaks of hemorrhagic colitis and hemolytic-uremic syndrome.

Neurotoxins usually are produced by the responsible organism outside the host and therefore cause symptoms soon after ingestion. Included are the staphylococcal and *Bacillus cereus* toxins, which act on the central nervous system to produce vomiting.

Invasion Dysentery may result not only from the production of cytotoxins but also from bacterial invasion and destruction of intestinal mucosal cells. Infections due to *Shigella* and enteroinvasive *E. coli*, for example, are characterized by the organisms' invasion of mucosal epithelial cells, intraepithelial multiplication, and subsequent spread to adjacent cells. *Salmonella*, on the other hand, causes inflammatory diarrhea by invasion of the bowel mucosa but generally is not associated with the destruction of enterocytes or the

full clinical syndrome of dysentery. *Salmonella typhi* and *Yersinia enterocolitica* can penetrate intact intestinal mucosa, multiply intracellularly in Peyer's patches and intestinal lymph nodes, and then disseminate through the bloodstream to cause enteric fever, a syndrome characterized by fever, headache, relative bradycardia, abdominal pain, splenomegaly, and leukopenia.

HOST DEFENSES

Given the enormous number of microorganisms ingested with every meal, the normal host must possess effective defense mechanisms to combat a constant influx of potential enteric pathogens. Studies of infections in patients with alterations in these defenses have led to a greater understanding of the variety of ways in which the normal host can protect itself against disease.

Normal Flora The large numbers of bacteria that normally inhabit the intestine act as an important host defense by preventing colonization by potential enteric pathogens. Persons with fewer intestinal bacteria, such as infants who have not yet developed normal enteric colonization or patients receiving antibiotics, are at significantly greater risk of developing infections with enteric pathogens. The composition of the intestinal flora is as important as the number of organisms present. More than 99% of the normal colonic flora is made up of anaerobic bacteria, and the acidic pH and volatile fatty acids produced by these organisms appear to be critical elements in resistance to colonization.

Gastric Acid The acidic pH of the stomach is an important barrier to enteric pathogens, and an increased frequency of infections due to *Salmonella*, *G. lamblia*, and a variety of helminths has been reported among patients who have undergone gastric surgery or are achlorhydric for some other reason. Neutralization of gastric acid with antacids or H₂blockers -- common among hospitalized patients -- similarly increases the risk of enteric colonization. Some microorganisms, however, can survive the extreme acidity of the gastric environment; rotavirus, for example, is highly stable to acidity.

Intestinal Motility Normal peristalsis is the major mechanism for clearance of bacteria from the proximal small intestine, although gastric acidity and secreted immunoglobulins also play a role in limiting the number of organisms present. When intestinal motility is impaired -- for example, by treatment with opiates or other antimotility drugs, anatomic abnormalities (diverticula, fistulas, or afferent-loop stasis following surgery), or hypomotility states (as in diabetes mellitus or scleroderma) -- the frequency of bacterial overgrowth and infection of the small bowel with enteric pathogens is much increased. Some patients in whom *Shigella* infection is treated with diphenoxylate hydrochloride with atropine (Lomotil) experience prolonged fever and shedding of organisms, while patients treated with opiates for mild *Salmonella* gastroenteritis have a higher frequency of bacteremia than those not treated with opiates.

Immunity Both cellular immune responses and antibody production play important roles in protecting susceptible hosts from enteric infections. The wide spectrum of viral, bacterial, parasitic, and fungal gastrointestinal infections in patients with AIDS highlights the significance of cell-mediated immunity in protecting the normal host from these pathogens. Humoral immunity is also important and consists of systemic IgG and IgM

as well as secretory IgA. Growing evidence supports the concept of a mucosal immune system for secretory IgA in which binding of bacterial antigens to the luminal surface of M cells in the distal small bowel and subsequent presentation of the antigens to subepithelial lymphoid tissue lead to the proliferation of sensitized lymphocytes. These lymphocytes circulate and populate all of the mucosal tissues of the body as IgA-secreting plasma cells.

Approach to the Patient

The approach to the patient with possible infectious diarrhea or bacterial food poisoning is shown in [Fig. 131-1](#).

History The answers to questions with high discriminating value can quickly narrow the range of potential causes of diarrhea and help determine whether treatment is needed. Important elements of the narrative history are detailed in [Fig. 131-1](#).

Physical Examination The examination of patients for signs of dehydration provides essential information about the severity of the diarrheal illness and the need for rapid therapy. Mild dehydration is indicated by thirst, dry mouth, decreased axillary sweat, decreased urine output, and slight weight loss. Signs of moderate dehydration include an orthostatic fall in blood pressure, skin tenting, and sunken eyes (or, in infants, a sunken fontanelle). Signs of severe dehydration range from hypotension and tachycardia to confusion and frank shock.

Diagnostic Approach After the severity of illness is assessed, the most important distinction that the clinician must make is between *inflammatory* and *noninflammatory* disease. Using the history and epidemiologic features of the case as guides in making this distinction, the clinician can rapidly evaluate the need for further efforts to define a specific etiology and for therapeutic intervention. Examination of a stool sample is an important supplement to the narrative history ([Fig. 131-CD1](#)). Grossly bloody or mucoid stool suggests an inflammatory process, but all stools should be examined for fecal leukocytes; the latter task is accomplished by the preparation of a thin smear of the stool on a glass slide, the addition of a drop of methylene blue, and examination of the wet mount. Causes of acute infectious diarrhea, categorized as inflammatory and noninflammatory, are listed in [Table 131-1](#).

EPIDEMIOLOGY

Travel History Of the 12 to 20 million people who travel from temperate industrialized countries to tropical regions of Asia, Africa, and Central and South America each year, 20 to 50% experience a sudden onset of abdominal cramps, anorexia, and watery diarrhea; thus *traveler's diarrhea* is the most common travel-related illness ([Chap. 123](#)). The time of onset is usually 3 days to 2 weeks after the traveler's arrival in a tropical area; most cases begin within the first 3 to 5 days. The illness is generally self-limited, lasting 1 to 5 days. The high rate of diarrhea among travelers to underdeveloped areas is related to the ingestion of contaminated food or water.

The organisms that cause traveler's diarrhea vary considerably with location. In all areas, enterotoxigenic *E. coli* is the most common isolate from persons with the classic

secretory traveler's diarrhea syndrome; the proportion of cases accounted for by this organism ranges from a high of approximately 50% in Latin America to a low of 15% in Asia. *Shigella*, *Salmonella*, and *Campylobacter* spp. are classically considered to cause more invasive dysenteric disease than enterotoxigenic *E. coli*, but clinical differentiation of infections attributable to these organisms can be difficult. *Shigella*, *Salmonella*, and *Campylobacter* are isolated in 1 to 15% of cases, with different organisms being more common in different locations. *Vibrio* spp. are most common in Asia, although *V. cholerae* O1 reached epidemic proportions in parts of Central and South America in 1991 and remains a significant source of concern to travelers to these regions. Epidemic *V. cholerae* O139 Bengal has spread throughout India and Southeast Asia since 1992. Less frequently isolated bacteria are *Aeromonas hydrophila* and *Plesiomonas shigelloides*, which are more common among travelers to Thailand. Parasitic causes of traveler's diarrhea include *Entamoeba histolytica*, which is responsible for up to 5% of cases in Mexico and Thailand, and *G. lamblia*, which has been associated with contaminated freshwater supplies in many areas of the world. *Giardia* is found in association with zoonotic reservoirs in the northern United States and poses a risk to hikers and campers who drink from freshwater streams. A striking association of *Giardia* with contaminated water supplies has likewise been noted in Russia. *Cryptosporidium* has been recognized as a problem in travelers to the former Soviet Republics, Mexico, and Africa and has caused large-scale urban outbreaks of infection in the United States. Disease due to *Cyclospora* and microsporidia has recently been recognized. Viruses such as rotavirus and Norwalk-like viruses have been isolated from as many as 10 to 40% of visitors to areas of Latin America, Asia, and Africa who develop traveler's diarrhea.

Location Day-care centers are sites of particularly high attack rates of enteric infections. Rotavirus is most common among children <2 years old, with attack rates of 75 to 100% among those exposed. *G. lamblia* is more common among older children, with somewhat lower attack rates. Other common organisms, often spread by fecal-oral contact, are *Shigella*, *Campylobacter jejuni*, and *Cryptosporidium*. A characteristic feature of infection in day-care centers is the high rate of secondary cases among family members.

Similarly, hospitals are sites for concentrations of enteric infections. In medical intensive-care units and pediatric wards, diarrhea is among the most common nosocomial infections. *C. difficile* is the predominant cause of nosocomial diarrhea among adults in the United States; viral pathogens, especially rotavirus, can spread rapidly in pediatric wards. Enteropathogenic *E. coli* has been associated with outbreaks of diarrhea in nurseries for newborns. One-third of elderly patients in chronic-care institutions develop a significant diarrheal illness each year. Surveillance stool cultures suggest that 25% of the residents of these institutions harbor cytotoxin-producing *C. difficile*, which causes more than half of all cases of diarrhea in this population. Antimicrobial therapy can predispose to pseudomembranous colitis by altering the normal colonic flora and allowing the multiplication of *C. difficile*.

Age Most of the morbidity and mortality from enteric pathogens involves children <5 years of age. Breast-fed infants are protected from contaminated food and water and derive some protection from maternal antibodies, but their risk of infection rises dramatically when they begin to eat solid foods. Infants and younger children are more

likely than adults to develop rotaviral disease, while older children and adults are more commonly infected with Norwalk-like viruses. Other organisms with higher attack rates among children than among adults include enterotoxigenic, enteropathogenic, and enterohemorrhagic *E. coli*; *C. jejuni*; and *G. lamblia*. In children, the incidence of *Salmonella* infections is highest among infants <1 year of age, while the attack rate for *Shigella* infections is greatest among children aged 6 months to 4 years.

Bacterial Food Poisoning If the history and the stool examination indicate a noninflammatory etiology of diarrhea and there is evidence of a common-source outbreak, questions concerning the ingestion of specific foods and the time of onset of the diarrhea after a meal can provide clues to the bacterial cause of the illness. Potential causes of bacterial food poisoning are shown in [Table 131-2](#).

Bacterial disease caused by an enterotoxin elaborated outside the host, such as that due to *Staphylococcus aureus* or *B. cereus*, has the shortest incubation period (1 to 6 h) and generally lasts <12 h. Most cases of staphylococcal food poisoning are caused by contamination from infected human carriers. Staphylococci can multiply at a wide range of temperatures; thus, if food is left to cool slowly and remains at room temperature after cooking, the organisms will have the opportunity to form enterotoxin. Outbreaks following picnics where potato salad, mayonnaise, and cream pastries have been served offer classic examples of staphylococcal food poisoning. Diarrhea, nausea, vomiting, and abdominal cramping are common, while fever is less so.

B. cereus can produce either a syndrome with a short incubation period -- the *emetic* form, mediated by a staphylococcal type of enterotoxin -- or one with a longer incubation period (8 to 16 h) -- the *diarrheal* form, caused by an enterotoxin resembling *E. coli* [LT](#), in which diarrhea and abdominal cramps are characteristic but vomiting is uncommon. The emetic form of *B. cereus* food poisoning is associated with contaminated fried rice; the organism is common in uncooked rice, and its heat-resistant spores survive boiling. If cooked rice is not refrigerated, the spores can germinate and produce toxin. Frying before serving may not destroy the preformed, heat-stable toxin.

Food poisoning due to *C. perfringens* also has a slightly longer incubation period (8 to 14 h) and results from the survival of heat-resistant spores in inadequately cooked meat, poultry, or legumes. After ingestion, toxin is produced in the intestinal tract, causing moderately severe abdominal cramps and diarrhea; vomiting is rare, as is fever. The illness is self-limited, rarely lasting for more than 24 h.

Not all food poisoning has a bacterial cause. Diagnostic confusion can result from diarrhea caused by nonbacterial agents of short-incubation food poisoning, including capsaicin, which is found in hot peppers, and a variety of toxins found in fish and shellfish ([Chap. 397](#)).

LABORATORY EVALUATION

Many cases of noninflammatory diarrhea are self-limited or can be treated empirically, and in these instances the clinician may not need to determine a specific etiology. Potentially pathogenic *E. coli* cannot be distinguished from normal fecal flora by routine culture. Special tests to detect [LT](#) and [ST](#) are not available in most clinical laboratories. In

situations in which cholera is a concern, stool should be cultured on thiosulfate-citrate-bile salts-sucrose (TCBS) agar. A latex agglutination test has made the rapid detection of rotavirus in stool practical for many laboratories, while reverse transcriptase polymerase chain reaction and specific antigen enzyme immunoassays have been developed for the identification of Norwalk-like viruses. At least three stool specimens should be examined for *Giardia* cysts or stained for *Cryptosporidium* if the level of clinical suspicion regarding the involvement of these organisms is high.

All patients with fever and evidence of inflammatory disease acquired outside the hospital should have stool cultured for *Salmonella*, *Shigella*, and *Campylobacter*. *Salmonella* and *Shigella* can be selected on MacConkey's agar as non-lactose-fermenting (colorless) colonies or can be grown on *Salmonella-Shigella* agar or in selenite enrichment broth, both of which inhibit most organisms except these pathogens. Evaluation of nosocomial diarrhea should initially focus on *C. difficile*; stool culture for other pathogens in this setting has an extremely low yield and is not cost-effective. Pathogenic strains of *C. difficile* generally produce two toxins, A and B. Toxin B can be detected with a cytotoxin assay; if the toxin is present, a monolayer culture of fibroblasts will show cytopathic effects within 6 to 24 h. Rapid enzyme immunoassays and latex agglutination tests for both toxin A and toxin B have recently been developed ([Chap. 145](#)). Isolation of *C. jejuni* requires inoculation of fresh stool onto selective growth medium and incubation at 42°C in a microaerophilic atmosphere. In many laboratories in the United States, *E. coli* O157:H7 is among the most common pathogens isolated from visible bloody stools. Strains of this enterohemorrhagic serotype can be identified in specialized laboratories by serotyping but also can be identified presumptively as lactose-fermenting, indole-positive colonies of sorbitol nonfermenters (white colonies) on sorbitol MacConkey plates. Fresh stools should be examined for amebic cysts and trophozoites.

TREATMENT

In many cases, a specific diagnosis is not necessary or not available to guide treatment. The clinician can proceed with the information obtained from the history, stool examination, and evaluation of the severity of dehydration. Empirical regimens for the treatment of traveler's diarrhea are listed in [Table 131-3](#).

The mainstay of treatment is adequate rehydration. The treatment of cholera and other dehydrating diarrheal diseases was revolutionized by the promotion of oral rehydration solutions, the efficacy of which depends on the fact that glucose-facilitated absorption of sodium and water in the small intestine remains intact in the presence of cholera toxin. The use of oral rehydration solutions has reduced mortality due to cholera from >50% (in untreated cases) to <1%. The World Health Organization recommends a solution containing 3.5 g sodium chloride, 2.5 g sodium bicarbonate, 1.5 g potassium chloride, and 20 g glucose (or 40 g sucrose) per liter of water. Patients who are severely dehydrated or in whom vomiting precludes the use of oral therapy should receive intravenous solutions such as Ringer's lactate.

Although most secretory forms of traveler's diarrhea -- usually due to enterotoxigenic *E. coli* -- can be treated effectively with rehydration, bismuth subsalicylate, or antiperistaltic agents, antimicrobial agents can shorten the duration of illness from between 3 and 4

days to between 24 and 36 h.

PROPHYLAXIS

Improvements in hygiene to limit fecal-oral spread of enteric pathogens will be necessary if the prevalence of diarrheal diseases is to be significantly reduced in developing countries. Travelers can reduce their risk of diarrhea by eating only hot, freshly cooked food; by avoiding raw vegetables, salads, and unpeeled fruit; and by drinking only boiled or treated water and avoiding ice. In one cross-sectional epidemiologic survey, fewer than 3% of all European and North American travelers to Jamaica adhered to prescribed dietary restrictions, and travel health advice had no impact on the incidence of traveler's diarrhea; overall, the diarrhea attack rate among these travelers was 23.6%, with classic traveler's diarrhea in 11.7%.

Bismuth subsalicylate is an inexpensive agent for the prophylaxis of traveler's diarrhea; it is taken at a dosage of 2 tablets (525 mg) four times a day. Treatment appears to be effective and safe for up to 3 weeks. Prophylactic antimicrobial agents, although effective, are not generally recommended for the prevention of traveler's diarrhea, except when travelers are immunosuppressed or have other underlying illnesses that place them at high risk for morbidity from gastrointestinal infection. The risk of side effects and the possibility of developing an infection with a drug-resistant organism or with more harmful, invasive bacteria make it more reasonable to institute a short course of treatment once symptoms have developed.

The possibility of exerting a major impact on the worldwide morbidity and mortality associated with diarrheal diseases has led to intense efforts to develop effective vaccines against the common bacterial and viral enteric pathogens. Recent research has shown promising advances in the development of vaccines against rotavirus, *Shigella*, *V. cholerae*, *S. typhi*, and enterotoxigenic *E. coli*.

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132. SEXUALLY TRANSMITTED DISEASES: OVERVIEW AND CLINICAL APPROACH - King K. Holmes

In all societies, sexually transmitted diseases (STDs) rank among the most common of all infectious diseases, with over 30 infections now classified as predominantly sexually transmitted or as frequently sexually transmissible ([Table 132-1](#)). The many new sexually transmitted pathogens recognized and characterized since 1980 include HIV types 1 and 2, human T cell lymphotropic virus (HTLV) types I and II, many genital types of human papillomavirus (HPV), *Mycoplasma genitalium*, two species of *Mobiluncus* (associated with bacterial vaginosis), two species of *Helicobacter* (initially associated with proctocolitis in homosexual men and later, during the AIDS era, with bacteremia, dermatitis, and fever among immunosuppressed individuals), and the Kaposi's sarcoma-associated herpesvirus (human herpesvirus type 8, or HHV-8).

In developing countries, with three-quarters of the world's population and 90% of the world's [STDs](#), such factors as population growth (especially in adolescent and young-adult age groups), rural-to-urban migration, wars, and poverty create exceptional vulnerability to disease resulting from risky sexual behaviors. This situation leads to the spread of STD, with the emergence of new pathogens and new variants of old pathogens. During the 1990s, in China, Russia, the states of the former Soviet Union, and South Africa, internal social structures changed rapidly as borders opened to the West, unleashing enormous new epidemics of HIV infection and other STDs. HIV has become the leading cause of death in some developing countries, and [HPV](#) and hepatitis B virus (HBV) remain important causes of cervical and hepatocellular carcinoma, respectively -- two of the commonest malignancies in the developing world. Sexually transmitted herpes simplex virus (HSV) infections now cause most genital ulcer disease throughout the world and an increasing proportion of cases of genital herpes in developing countries with generalized HIV epidemics, where the positive feedback loop between HSV and HIV transmission is a growing, intractable problem. Globally, the agents of curable STDs -- gonorrhea, chlamydial infections, syphilis, chancroid, and trichomoniasis -- caused ~350 million new infections annually in the mid-1990s. Bacterial vaginosis (arguably acquired sexually) occurs in up to 50% of women of reproductive age in developing countries. Thus, there are probably close to 1 billion cases of these curable infections annually, all six of which are associated with increased risk of HIV transmission or acquisition.

In the industrialized countries, fear of HIV infection since the mid- 1980s, coupled with widespread behavioral interventions and better- organized systems of care for the curable [STDs](#), have helped curb the transmission of the latter diseases. Nonetheless, viral STDs, such as genital herpes and [HPV](#) infection, had not obviously decreased in incidence at the turn of the millennium, and infection with HIV remains a leading cause of death in persons 25 to 44 years of age in the United States, as in developing countries, despite the advent of potent antiretroviral therapy.

Although rates of the bacterial [STDs](#) have fallen in all industrialized countries over the past 20 years, foci of hyperendemic transmission persist in the southeastern United States and in most large U.S. cities. Rates of gonorrhea and syphilis remain higher in the United States than in any other Western industrialized country. The reemergence of syphilis and gonorrhea epidemics in Russia and the former Soviet states has created

similar foci for reintroduction of these STDs into western Europe. The remarkable return of high rates of gonorrhea and syphilis among homosexual and bisexual men in many parts of the United States reflects increased risk-taking since the advent of potent antiretroviral therapy and may portend resurgent transmission of HIV in this group.

CLASSIFICATION AND EPIDEMIOLOGY

Sexually transmitted infection (STI) may or may not result in disease ([STD](#)). Some prefer the term *reproductive tract infection* to destigmatize the diagnosis and treatment of STIs and to encompass conditions such as bacterial vaginosis, whose designation as an STD is debated.

Certain [STDs](#) (such as syphilis, gonorrhea, HIV infection, hepatitis B, and chancroid) are most concentrated within "core populations" having high rates of partner change, concurrent partners, or "dense" sexual networks -- for example, prostitutes and their clients and persons involved in the use of illicit drugs, particularly crack cocaine. Poor access to or low motivation for obtaining early treatment also fosters spread of the curable STIs. In most of the United States, groups most vulnerable to STDs, including HIV infection, consist predominantly of young unmarried individuals of low socioeconomic status who often reside within crowded urban neighborhoods, although some rural areas (e.g., in the southeastern United States) also have high rates of STIs. Other STDs are distributed more evenly in society. For example, chlamydial infections can persist for many months, often asymptotically, and can propagate widely in populations that do not share all of the characteristics of core groups described above. Similarly, genital [HPV](#) infections and genital herpes persist and spread efficiently in relatively low-risk populations.

In general, the product of three factors determines the initial rate of spread of any [STI](#) within a population: rate of exposure, efficiency of transmission per exposure, and duration of infectivity of those infected. Efforts to prevent and control STIs attempt to decrease the duration of infectivity (through early diagnosis and curative or suppressive treatment), to decrease the efficiency of transmission (e.g., through promotion of condom use and safer sexual practices), and to decrease the rate of exposure of susceptibles to infected persons (e.g., through provision of information, health education, and counseling and efforts to change the norms of sexual behavior).

MANAGEMENT OF COMMON STD SYNDROMES

Although other chapters discuss management of specific [STIs](#), delineating treatment based on diagnosis of a specific infection, most patients are actually managed (at least initially) on the basis of presenting symptoms and signs and associated risk factors, even in industrialized countries. [Table 132-2](#) lists some of the most common clinical [STD](#) syndromes and their microbial etiologies. Strategies for their management are outlined below. [Chaps. 191](#) and [309](#) address the management of infections with human retroviruses.

RISK ASSESSMENT

Routine patient care begins with risk assessment (e.g., for heart disease, cancer). An

overall risk assessment interview should include STD/HIV in primary care, urgent care, and emergency care settings as well as in specialty clinics providing adolescent, prenatal, and family planning services. STD/HIV risk assessment guides interpretation of symptoms that could reflect an STD; decisions on screening or prophylactic/preventive treatment; risk reduction counseling and intervention (e.g., hepatitis B vaccination); and notification of partners of patients with known infections. Consideration of routine demographic data (e.g., gender, age, marital status, area of residence) is a simple first step in STD/HIV risk assessment. For example, national guidelines recommend routine screening of sexually active females ≥ 25 years of age for *C. trachomatis* infection. [Table 132-3](#) provides a set of 10 STD/HIV risk assessment questions that clinicians can pose verbally or that health care systems can adapt (with yes/no responses) into a routine self-administered questionnaire for use in clinics. The initial framing statement gives permission to discuss taboo topics.

Risk assessment is followed by clinical assessment (elicitation of information on specific current symptoms and signs of STDs). Confirmatory diagnostic tests (for those with symptoms or signs) or screening tests (for those without symptoms or signs) may involve microscopic examination, culture, antigen detection tests, genetic probe or amplification tests, or serology. Initial syndrome-based treatment should cover the most likely causes. For certain syndromes, results of rapid tests can narrow the spectrum of this initial therapy (e.g., wet mount of vaginal fluid for women with vaginal discharge, Gram's stain of urethral discharge for men with urethral discharge, rapid plasma reagin test for genital ulcer). After the institution of treatment, STD management proceeds to the "4 C's" of prevention and control: contact tracing (see "Prevention and Control of STDs," below), ensuring compliance with therapy, and counseling on risk reduction, including condom promotion and provision.

URETHRITIS IN MEN

The incidence of reported gonococcal urethritis ([Fig. 145-CD1](#)) in the United States has fallen to the lowest level since reporting began, while that of nongonococcal urethritis (NGU) remains high -- a pattern typical of all industrialized countries. Until recently, *Chlamydia trachomatis* caused ~30 to 40% of NGU cases, but the proportion of cases due to this organism may have declined in some populations served by effective chlamydial control programs. [HSV](#) and *Trichomonas vaginalis* each cause a small proportion of NGU cases in the United States. Case-control studies have also implicated *Ureaplasma urealyticum* and *M. genitalium* as probable causes of many *Chlamydia*-negative cases, and coliform bacteria can cause urethritis in men who practice insertive anal intercourse. The initial diagnosis of urethritis in men currently includes specific tests only for *Neisseria gonorrhoeae* and *C. trachomatis*. [Table 132-4](#) summarizes the steps in management of sexually active men with symptoms of urethral discharge and/or dysuria.

1. *Establish the presence of urethritis.* If proximal-to-distal "milking" of the urethra does not express a purulent or mucopurulent discharge, even after the patient has not voided for several hours or preferably overnight, the centrifuged sediment of the first 20 to 30 mL of voided urine can be examined for inflammatory cells, either by microscopy or by the leukocyte esterase test. In urethral gonococcal or chlamydial infection, a Gram's-stained smear of overt discharge or of an anterior urethral specimen obtained

by passage of a small urethrogenital swab 2 to 3 cm into the urethra usually reveals ³⁵ neutrophils per 1000 field in areas containing cells; in gonococcal infection, such a smear also usually reveals gram-negative intracellular diplococci. Patients with symptoms who lack objective evidence of urethritis may have functional rather than organic problems and generally do not benefit from repeated courses of antibiotics.

2. *Evaluate for complications or alternative diagnoses.* A brief history and examination will exclude epididymitis and systemic complications, such as disseminated gonococcal infection and Reiter's syndrome. Although digital examination of the prostate gland seldom contributes to the evaluation of sexually active young men with urethritis, men with dysuria who lack evidence of urethritis as well as sexually inactive men with urethritis should undergo prostate palpation, urinalysis, and urine culture to exclude bacterial prostatitis and cystitis.

3. *Evaluate for gonococcal and chlamydial infection.* An absence of typical gram-negative diplococci on Gram's-stained smear of urethral exudate containing inflammatory cells warrants a preliminary diagnosis of [NGU](#) and should lead to testing of the urethral specimen for *C. trachomatis*. Culture or DNA detection tests for *N. gonorrhoeae* may be positive when Gram's staining is negative; certain strains of *N. gonorrhoeae* reportedly can result in negative urethral Gram's stains in up to 30% of cases of urethritis. Results of tests for gonococcal and chlamydial infection predict the patient's prognosis (with greater risk for recurrent NGU if neither chlamydiae nor gonococci are found than if either is detected) and can guide both the counseling given to the patient and the management of the patient's sexual partner(s).

4. *Treat urethritis.*

TREATMENT

In practice, if Gram's stain does not reveal gonococci, urethritis is treated with a regimen effective for [NGU](#), such as azithromycin (1.0 g orally in a single dose). If gonococci are demonstrated by Gram's stain or if no diagnostic tests are performed to definitively exclude gonorrhea, treatment should also include a single-dose regimen for gonorrhea ([Chap. 147](#)). Sexual partners should be tested for gonorrhea and chlamydial infection and should receive the same regimen given to the male index case.

EPIDIDYMITIS

Acute epididymitis, almost always unilateral, must be differentiated from testicular torsion, tumor, and trauma. Torsion, a surgical emergency, usually occurs in the second or third decade of life and produces a sudden onset of pain, elevation of the testicle within the scrotal sac, rotation of the epididymis from a posterior to an anterior position, and absence of blood flow on Doppler examination or ^{99m}Tc scan. Persistence of symptoms after a course of therapy for epididymitis suggests the possibility of testicular tumor. In sexually active men under age 35, acute epididymitis is caused most frequently by *C. trachomatis* and less commonly by *N. gonorrhoeae* and is usually associated with overt or subclinical urethritis. Acute epididymitis in older men or following urinary tract instrumentation is usually caused by urinary pathogens. Similarly, epididymitis in men who have practiced insertive rectal intercourse is often caused by

Enterobacteriaceae. These men usually have no urethritis but do have bacteriuria.

TREATMENT

Ofloxacin (300 mg orally bid for 10 days) is an optimal agent for syndrome-based treatment of epididymitis because of its effectiveness against *N. gonorrhoeae*, *C. trachomatis*, and Enterobacteriaceae. Alternatively, ceftriaxone (250 mg intramuscularly) followed by doxycycline (100 mg orally bid for 10 days) is effective for epididymitis caused by *N. gonorrhoeae* or *C. trachomatis*.

URETHRITIS AND THE URETHRAL SYNDROME IN WOMEN

C. trachomatis, *N. gonorrhoeae*, and occasionally [HSV](#) cause symptomatic urethritis -- known as the urethral syndrome in women -- characterized by "internal" dysuria (usually without urinary urgency or frequency) and pyuria, with *Escherichia coli* or other uropathogens not present in urine at counts of $\geq 10^2$ /mL. In contrast, the dysuria associated with vulvar herpes or vulvovaginal candidiasis (and perhaps with trichomoniasis) is often described as "external," being caused by painful contact of urine with the inflamed or ulcerated labia or introitus. Acute onset, association with urinary urgency or frequency, hematuria, or suprapubic bladder tenderness suggests bacterial cystitis. Among women with symptoms of acute bacterial cystitis, costovertebral pain and tenderness or fever suggests acute pyelonephritis. The management of bacterial urinary tract infection (UTI) is discussed in [Chap. 280](#).

Signs of vulvovaginitis, coupled with symptoms of external dysuria, suggest vulvar infection (e.g., with [HSV](#) or *Candida albicans*). Among dysuric women without signs of vulvovaginitis, bacterial [UTI](#) must be differentiated from the urethral syndrome by assessment of risk, evaluation of the pattern of symptoms and signs, and specific microbiologic testing. An [STD](#) etiology of the urethral syndrome is suggested by young age, more than one current sexual partner or a new partner within the past month, or coexisting mucopurulent cervicitis (MPC; see below). The finding of a single urinary pathogen, such as *E. coli* or *Staphylococcus saprophyticus*, at a concentration of $\geq 10^2$ /mL in a properly collected specimen of midstream urine from a dysuric woman with pyuria indicates probable bacterial UTI, whereas pyuria with $< 10^2$ conventional uropathogens per milliliter of urine ("sterile" pyuria) suggests acute urethral syndrome due to *C. trachomatis* or *N. gonorrhoeae*. Gonorrhea and chlamydial infection should be sought by specific tests. Among women with sterile pyuria caused by chlamydial infection, treatment with doxycycline (100 mg bid for 7 days) alleviates dysuria.

VULVOVAGINAL INFECTIONS

Abnormal Vaginal Discharge If directly questioned during routine health checkups, many women acknowledge having nonspecific symptoms of vaginal discharge that do not correlate with objective signs of inflammation or with actual infection. However, unsolicited reporting of abnormal vaginal discharge does suggest bacterial vaginosis or trichomoniasis. Specifically, an abnormally increased amount, an abnormal odor, and an abnormal yellow color of the discharge are associated with one or both of these conditions. Cervical infection with *N. gonorrhoeae* or *C. trachomatis* does not appear to cause an increased amount or abnormal odor of discharge, but cervicitis, like

trichomoniasis, can include the production of an increased number of neutrophils in vaginal fluid, resulting in a yellow color. Vulvar conditions such as genital herpes or vulvovaginal candidiasis can cause vulvar pruritus, burning, irritation, or lesions as well as external dysuria (as urine passes over the inflamed vulva) or vulvar dyspareunia.

Certain vulvovaginal infections may have serious sequelae. Trichomoniasis, bacterial vaginosis, and vulvovaginal candidiasis have all been associated with increased risk of acquisition of HIV infection. Vaginal trichomoniasis and bacterial vaginosis early in pregnancy independently predict premature onset of labor. Bacterial vaginosis can also lead to anaerobic bacterial infection of the endometrium and salpinges. Vaginitis may be an early and prominent feature of toxic shock syndrome, and recurrent or chronic vulvovaginal candidiasis develops with increased frequency among women with systemic illnesses, such as diabetes mellitus or HIV-related immunosuppression (although only a very small proportion of women with recurrent vulvovaginal candidiasis in the United States actually have a serious predisposing illness).

Thus vulvovaginal symptoms or signs warrant careful evaluation, including pelvic examination, simple rapid diagnostic tests, and appropriate therapy specific for the anatomic site and type of infection. Unfortunately, a recent survey in the United States indicated that clinicians seldom perform the tests required to establish the cause of such symptoms. The diagnosis and treatment of the three most common types of vaginal infection are summarized in [Table 132-5](#).

Inspection of the vulva and perineum may reveal tender genital ulcerations (typically due to [HSV](#) infection, occasionally to chancroid) or fissures (typically due to vulvovaginal candidiasis) or discharge visible at the introitus before insertion of a speculum (suggestive of bacterial vaginosis or trichomoniasis). Speculum examination permits the clinician to discern whether the discharge in fact looks abnormal and whether any abnormal discharge in the vagina emanates from the cervical os (mucoid and, if abnormal, yellow) or from the vagina (not mucoid, since the vaginal epithelium does not produce mucus). Symptoms or signs of abnormal vaginal discharge should prompt testing of vaginal fluid for pH, odor when mixed with 10% KOH, and microscopic features when mixed with saline and with 10% KOH. Additional objective laboratory tests useful for establishing the cause of abnormal vaginal discharge include Gram's staining to detect alterations in the vaginal flora; new card and dipstick tests for bacterial vaginosis, as described below; and a new DNA probe test purported to detect *T. vaginalis* and *C. albicans* as well as the increased concentrations of *Gardnerella vaginalis* associated with bacterial vaginosis.

TREATMENT

Patterns of treatment for vaginal discharge vary widely. In developing countries, where clinics or pharmacies often dispense treatment based on symptoms alone without examination or testing, oral treatment with metronidazole, either as a 2-g single dose or as a 7-day regimen, provides reasonable coverage against both trichomoniasis and bacterial vaginosis, the usual causes of symptoms of vaginal discharge; metronidazole treatment of sex partners would prevent reinfection of women with trichomoniasis even if it does not help prevent the recurrence of bacterial vaginosis. Guidelines promulgated during the 1990s by the World Health Organization suggested treatment for cervical

infection and for vulvovaginal candidiasis in women with symptoms of abnormal vaginal discharge; in retrospect, these recommendations were faulty, since these conditions seldom produce such symptoms.

In industrialized countries, clinicians treating symptoms and signs of abnormal vaginal discharge should at least differentiate between bacterial vaginosis and trichomoniasis, because optimal management of patients and partners differs for these two conditions, as discussed briefly below.

Vaginal Trichomoniasis (See also [Chap. 218](#)) Symptomatic trichomoniasis characteristically produces a profuse, yellow, purulent, homogeneous vaginal discharge and vulvar irritation, often with visible inflammation of the vaginal and vulvar epithelium and petechial lesions on the cervix (the so-called strawberry cervix, usually evident only by colposcopy). The pH of vaginal fluid usually rises to ≥ 5.0 . In women with typical symptoms and signs of trichomoniasis, microscopic examination of vaginal discharge mixed with saline reveals motile trichomonads in at least 80% of culture-positive cases. However, in the absence of symptoms or signs, culture is often required for detection of the organism. Treatment of asymptomatic as well as symptomatic cases reduces rates of transmission and prevents later development of symptoms.

TREATMENT

Only nitroimidazoles consistently cure trichomoniasis. Tinidazole and ornidazole have longer half-lives than metronidazole but do not give better results than a single 2-g oral dose of metronidazole, the treatment of choice. Treatment of male sexual partners -- often facilitated by dispensing metronidazole to the female patient to give to her partner(s), with a warning about avoiding the concurrent use of alcohol -- reduces both the risk of reinfection and the reservoir of infection. Treatment with 0.75% metronidazole gel intravaginally, although effective for bacterial vaginosis, is not reliable for vaginal trichomoniasis. Systemic use of metronidazole is not recommended during the first trimester of pregnancy but is considered safe thereafter.

Bacterial Vaginosis This syndrome (formerly termed *nonspecific vaginitis*, *Haemophilus vaginitis*, *anaerobic vaginitis*, or *Gardnerella-associated vaginal discharge*) is characterized by symptoms of vaginal malodor and a slightly to moderately increased white discharge, which appears homogeneous, is low in viscosity, and smoothly coats the vaginal mucosa. Risk factors include multiple sexual partners and recent intercourse with a new partner, but antibiotic treatment of male partners has not reduced the rate of recurrence among affected women.

The vaginal fluid of women with bacterial vaginosis is characterized by markedly increased prevalences and concentrations of *G. vaginalis*, *Mycoplasma hominis*, and several anaerobic bacteria [e.g., *Mobiluncus* spp., *Prevotella* spp. (formerly *Bacteroides* spp.), and some *Peptostreptococcus* spp.]. The vaginal fluid usually lacks hydrogen peroxide-producing *Lactobacillus* spp., which constitute most of the normal vaginal flora and perhaps help protect against certain cervical and vaginal infections. Vaginal douching, use of intravaginal nonoxynol-9 spermicide, and new sexual partners can all result in loss of vaginal colonization by hydrogen peroxide-producing lactobacilli.

Bacterial vaginosis is conventionally diagnosed clinically with the *Amsel criteria*, which include any three of the following four clinical abnormalities: (1) objective signs of increased white homogeneous vaginal discharge; (2) a vaginal discharge pH of >4.5; (3) liberation of a distinct fishy odor (attributable to volatile amines such as trimethylamine) immediately after vaginal secretions are mixed with a 10% solution of KOH; and (4) microscopic demonstration of "clue cells" (vaginal epithelial cells coated with coccobacillary organisms giving them a granular appearance and indistinct borders; [Fig. 132-1](#)) on a wet mount prepared by mixing vaginal secretions with normal saline in a ratio of ~1:1. A new card test now facilitates screening of vaginal fluid for pH >4.5 and trimethylamine, and a new dipstick test detects proline aminopeptidase, an enzyme associated with this syndrome.

Alternatively, the microbiology laboratory can determine the *Nugent score* by examining a Gram-stained smear of vaginal discharge. A score of 7 to 10, based on reduced numbers or the absence of large gram-positive rods (lactobacilli) and the presence of small gram-negative or variable rods (*Gardnerella* and anaerobic rods) and of curved gram-negative or variable rods (*Mobiluncus*), has high sensitivity and specificity for the diagnosis of bacterial vaginosis. Attempts to isolate *G. vaginalis*, genital mycoplasmas, or anaerobic bacteria do not aid in the diagnosis of bacterial vaginosis because these organisms occur (albeit in much lower concentrations) in the vaginal flora of many women without the syndrome.

TREATMENT

The standard dosage of metronidazole for the treatment of bacterial vaginosis is 500 mg orally bid for 7 days. Alternatively, the single 2-g oral dose of metronidazole recommended for trichomoniasis produces short-term rates of recurrence of bacterial vaginosis somewhat higher than those obtained with the 7-day regimen. Intravaginal treatment with 2% clindamycin cream [one full applicator (5 g containing 100 mg of clindamycin phosphate) each night for 7 nights] or with 0.75% metronidazole gel [one full applicator (5 g containing 37.5 mg of metronidazole) twice daily for 5 days] is also effective and does not elicit systemic adverse reactions. Nonetheless, long-term recurrence (i.e., after several months) is distressingly common after either oral or intravaginal treatment. Treatment of male partners with metronidazole does not prevent recurrence of bacterial vaginosis, even though new sexual partners have been implicated as a risk factor for recurrence.

No controlled data support the use of currently available vaginal or oral preparations of lactobacilli for the treatment or prevention of recurrence of bacterial vaginosis. Clinical trials are evaluating prevention of recurrence by repeated intravaginal inoculation of a vaginal *Lactobacillus* species that produces hydrogen peroxide and adheres to vaginal epithelium.

Vulvovaginal Pruritus, Burning, or Irritation Vulvovaginal candidiasis produces vulvar pruritus, burning, or irritation, generally without symptoms of increased vaginal discharge or malodor. Genital herpes can produce similar symptoms, with lesions sometimes difficult to distinguish from the fissures caused by candidiasis. Signs of vulvovaginal candidiasis include vulvar erythema, edema, fissures, and tenderness. With candidiasis, a white scanty vaginal discharge sometimes takes the form of white

thrushlike plaques or cottage cheese- like curds adhering loosely to the vaginal mucosa. *C. albicans* accounts for nearly all cases of symptomatic vulvovaginal candidiasis, which probably arise from endogenous strains of *C. albicans* that have colonized the vagina or the intestinal tract.

The diagnosis of vulvovaginal candidiasis usually involves the demonstration of pseudohyphae or hyphae by microscopic examination of vaginal fluid mixed with saline or 10% KOH or subjected to Gram's staining. Microscopic examination is less sensitive than culture but correlates better with symptoms.

TREATMENT

Symptoms and signs of vulvovaginal candidiasis warrant treatment, usually intravaginal administration of any of several imidazole antibiotics (e.g., miconazole or clotrimazole) for 3 to 7 days. Over-the-counter marketing of such preparations has reduced the cost of care and made treatment more convenient for many women with recurrent yeast vulvovaginitis. However, most women who purchase these preparations do not have vulvovaginal candidiasis, while many do have other vaginal infections that require different treatment. Therefore, only women with classic symptoms of vulvar pruritus and a history of previous episodes of yeast vulvovaginitis documented by an experienced clinician should self-treat. Single-dose oral treatment with fluconazole (150 mg) is also effective and is preferred by many patients. Prolonged or periodic oral therapy may benefit women with severe or frequently recurrent vulvovaginal candidiasis and those who do not respond to intravaginal or single-dose oral therapy. Such patients probably should be evaluated for diabetes and HIV infection, although such systemic illnesses seldom explain recurrent vulvovaginal candidiasis. Treatment of sexual partners is not routinely indicated.

Other Causes of Vaginal Discharge or Vaginitis In the ulcerative vaginitis associated with staphylococcal toxic shock syndrome, *Staphylococcus aureus* should be promptly identified in vaginal fluid by Gram's stain and by culture. In desquamative inflammatory vaginitis, smears of vaginal fluid reveal neutrophils, massive vaginal epithelial cell exfoliation with increased numbers of parabasal cells, and gram-positive cocci; this syndrome responds to treatment with 2% clindamycin cream. Additional causes of vaginitis and vulvovaginal symptoms in women include retained foreign bodies (e.g., tampons), cervical caps, vaginal spermicides, vaginal antiseptic preparations or douches, vaginal epithelial atrophy in postmenopausal women or in the postpartum period during prolonged breast-feeding, allergic reactions to latex condoms, vaginal aphthae associated with HIV infection or Behcet's syndrome, and vestibulitis (a poorly understood syndrome).

MUCOPURULENT CERVICITIS

[MPC](#) refers to inflammation of the columnar epithelium and subepithelium of the endocervix and of any contiguous columnar epithelium that lies exposed in an ectopic position on the exocervix. MPC in women represents the "silent partner" of urethritis in men, being equally common and often caused by the same agents (*N. gonorrhoeae* or *C. trachomatis*) but more difficult to recognize. As the most common manifestation of these serious bacterial infections in women, MPC can be a harbinger or sign of pelvic

inflammatory disease (PID) and -- in pregnant women -- can lead to obstetric complications. More than half of all cases of this syndrome in the United States today remain idiopathic.

The diagnosis of [MPC](#) rests on the detection of yellow mucopurulent discharge from the cervical os or of increased numbers of polymorphonuclear leukocytes in Gram-stained or Papanicolaou-stained smears of endocervical mucus. MPC due to *C. trachomatis* can also produce edematous cervical ectopy (see below) and endocervical bleeding upon gentle swabbing. Unlike the endocervicitis produced by gonococcal or chlamydial infection, cervicitis caused by [HSV](#) produces ulcerative lesions on the stratified squamous epithelium of the exocervix as well as on the columnar epithelium. Yellow cervical mucus on a white swab removed from the endocervix indicates the presence of polymorphonuclear leukocytes. The mucus should be rolled thinly on a slide for Gram's staining. The presence of ≥ 20 polymorphonuclear cells per 1000 $\times$ microscopic field within strands of cervical mucus not contaminated by vaginal squamous epithelial cells or vaginal bacteria indicates endocervicitis ([Fig. 132-2](#)). Detection of intracellular gram-negative diplococci in carefully collected endocervical mucus is quite specific but $\leq 50\%$ sensitive for gonorrhea. Therefore, specific and sensitive tests for *N. gonorrhoeae* as well as *C. trachomatis* are also indicated in evaluation of MPC.

TREATMENT

Although the above criteria for [MPC](#) are neither highly specific nor highly predictive of gonococcal or chlamydial infection in many settings, current guidelines of the Centers for Disease Control and Prevention call for consideration of empirical treatment for MPC, pending test results, "for a patient who has suspected gonorrhea or chlamydial infection, if the prevalences of these infections are high in the patient population, and the patient might be difficult to locate after treatment." In this situation, therapy should include a single-dose regimen effective for gonorrhea plus treatment for chlamydial infection, as outlined in [Table 132-4](#) for treatment of urethritis. In settings where gonorrhea is much less common than chlamydial infection, initial therapy for chlamydial infection alone suffices. The etiology and potential benefit of treatment of endocervicitis not associated with gonorrhea or chlamydial infection remain undefined. Sexual partner(s) of a woman with MPC should be examined and given a regimen similar to that chosen for the woman unless results of tests for gonorrhea or chlamydial infection in either partner warrant different therapy or no therapy.

CERVICAL ECTOPY

Cervical ectopy, often mislabeled "cervical erosion," is easily confused with infectious endocervicitis. Ectopy represents the presence of the one-cell-thick columnar epithelium extending from the endocervix out onto the visible ectocervix. In ectopy, the cervical os may contain clear or slightly cloudy mucus but usually not yellow mucopus. Colposcopy shows intact epithelium. Normally found during adolescence and early adulthood, ectopy gradually recedes through the second and third decades of life, as squamous metaplasia replaces the ectopic columnar epithelium. Oral contraceptive use favors the persistence or reappearance of ectopy, while smoking apparently accelerates squamous metaplasia. Cauterization for the elimination of ectopy is not warranted. Ectopy may render the cervix more susceptible to infection with *N. gonorrhoeae*, *C.*

trachomatis, or HIV by exposing a larger area of susceptible columnar epithelium on the exocervix.

PELVIC INFLAMMATORY DISEASE See [Chap. 133](#).

ULCERATIVE GENITAL LESIONS

Genital ulceration reflects a set of important [STIs](#), most of which also sharply increase the risk of sexual acquisition and shedding of HIV. Accurate diagnosis, treatment, and prevention of these infections are high priorities. In a study of genital ulcers carried out in 1996 in 10 of the U.S. cities with the highest rates of primary syphilis, polymerase chain reaction (PCR) testing of ulcer specimens demonstrated [HSV](#) in 62% of patients, *Treponema pallidum* in 13%, and *Haemophilus ducreyi* in 12 to 20%.

In Asia and Africa, chancroid ([Fig. 132-CD1](#)) was once considered the most common type of genital ulcer, followed by primary syphilis and then genital herpes. With increased efforts to control chancroid and syphilis, together with more frequent recurrences or persistence of genital herpes attributable to the growing numbers of immunosuppressed persons with HIV infection, [PCR](#) testing of genital ulcers now clearly implicates genital herpes as the most common cause of genital ulceration in some developing countries. Lymphogranuloma venereum (LGV) ([Fig. 132-CD2](#)) and donovanosis (granuloma inguinale; [Fig. 132-CD3](#)) continue to cause genital ulceration in developing countries but rarely occur today in North America or Europe. Other causes of genital ulcer include (1) candidiasis and traumatized genital warts -- both readily recognized; (2) lesions due to genital involvement of more widespread dermatoses; and (3) cutaneous manifestations of systemic diseases, such as genital mucosal ulceration in Stevens-Johnson syndrome.

Although most genital ulcerations cannot be diagnosed confidently on clinical grounds alone, clinical findings plus epidemiologic considerations ([Table 132-6](#)) can usually guide initial management ([Table 132-7](#)) pending results of further tests. Clinicians should order a rapid serologic test for syphilis in all cases of genital ulcer and a dark-field, direct immunofluorescence, or [PCR](#) test for *T. pallidum* from all lesions except those highly characteristic of infection with [HSV](#) (i.e., those with herpetic vesicles).

Typical vesicles or pustules or a cluster of painful ulcers preceded by vesiculopustular lesions suggests genital herpes. These typical clinical presentations make detection of the virus optional; however, many patients want confirmation of the diagnosis, and differentiation of [HSV-1](#) from HSV-2 has prognostic implications, since the latter causes more frequent recurrences.

Painless, nontender, indurated ulcers with firm, nontender inguinal adenopathy suggest primary syphilis. If the results of dark-field examination and a rapid serologic test for syphilis are initially negative and the patient will comply with follow-up and sexual abstinence, the performance of two more dark-field examinations on successive days before treatment is begun will improve the sensitivity of diagnosis of syphilis, as will repeated serologic testing for syphilis 1 or 2 weeks later.

"Atypical" or clinically trivial ulcers may be more common manifestations of genital

herpes than classic vesiculopustular lesions. Specific tests for [HSV](#) in the lesions are therefore indicated ([Chap. 182](#)). Type-specific serologic tests for serum antibody to HSV-2, now commercially available, may give negative results, especially when patients present early with the initial episode of genital herpes or when HSV-1 is the cause of genital herpes (as in 15 to 30% of cases today). Furthermore, a positive test for HSV-2 antibody does not prove that the current lesions are herpetic, since nearly one-fourth of the general population of the United States becomes seropositive for HSV-2 during early adulthood. Nonetheless, a positive HSV-2 serology does enable the clinician to tell the patient that he or she has had genital herpes, should learn to recognize symptoms, should avoid sex during recurrences, and should consider use of condoms at other times.

Demonstration of *H. ducreyi* by culture (or by [PCR](#) test, when available) is most useful when ulcers are painful and purulent, especially when inguinal lymphadenopathy with fluctuance or overlying erythema is noted; if chancroid is prevalent in the community; or if the patient has recently had a sexual exposure in a chancroid-endemic area (e.g., a developing country or certain North American cities). Enlarged, fluctuant lymph nodes should be aspirated for culture or PCR tests to detect *H. ducreyi* as well as for Gram's staining and culture to rule out the presence of other pyogenic bacteria.

When genital ulcers persist beyond the natural history of initial episodes of herpes (2 to 3 weeks) or of chancroid or syphilis (up to 6 weeks) and do not resolve with syndrome-based antimicrobial therapy, then -- in addition to the usual tests for herpes, syphilis, and chancroid -- biopsy is indicated to exclude donovanosis, carcinoma, and other nonvenereal dermatoses. HIV serology should also be undertaken, since chronic, persistent genital herpes is common in AIDS.

Immediate syndrome-based treatment for acute genital ulcerations (after collection of all necessary diagnostic specimens) is often appropriate. Patients with typical initial or recurrent episodes of genital or anorectal herpes can benefit from prompt oral antiviral therapy ([Chap. 182](#)). The patient with nonvesicular ulcerative lesions who may not return for follow-up or may continue sexual activity should receive initial treatment for syphilis, together with empirical therapy for chancroid if exposed in an area where chancroid occurs or if regional lymph node suppuration is evident. In resource-poor settings lacking ready access to diagnostic tests, this approach to syndromic treatment for syphilis and chancroid has helped bring these two diseases under better control. Finally, empirical antimicrobial therapy may be indicated if ulcers persist and the diagnosis remains unclear after a week of observation despite attempts to diagnose herpes, syphilis, and chancroid.

PROCTITIS, PROCTOCOLITIS, ENTEROCOLITIS, AND ENTERITIS

Sexually acquired proctitis, with inflammation limited to the rectal mucosa, results from direct rectal inoculation of typical [STD](#) pathogens. In contrast, inflammation extending from the rectum to the colon (proctocolitis), involving both the small and the large bowel (enterocolitis), or involving the small bowel alone (enteritis) can result from ingestion of typical intestinal pathogens through oral-anal exposure during sexual contact. Anorectal pain and mucopurulent, bloody rectal discharge suggest proctitis or proctocolitis. Proctitis commonly produces tenesmus (causing frequent attempts to defecate, but not true

diarrhea) and constipation, whereas proctocolitis and enterocolitis more often cause true diarrhea. In all three conditions, anoscopy usually shows mucosal exudate and easily induced mucosal bleeding (i.e., a positive "wipe test"), sometimes with petechiae or mucosal ulcers. Exudate should be sampled for Gram's staining and other microbiologic studies. Sigmoidoscopy or colonoscopy shows inflammation limited to the rectum in proctitis or disease extending at least into the sigmoid colon in proctocolitis.

The AIDS era has brought an extraordinary shift in the clinical and etiologic spectrum of intestinal infections among homosexual men. The number of cases of the acute intestinal [STIs](#) described above has fallen as high-risk sexual behaviors have become less common in this group. At the same time, the number of AIDS-related opportunistic intestinal infections has risen rapidly, many associated with chronic or recurrent symptoms. Acquisition of *N. gonorrhoeae*, [HSV](#), or *C. trachomatis* during receptive anorectal intercourse causes most cases of infectious proctitis. Primary and secondary syphilis can also produce anal or anorectal lesions, with or without symptoms. Gonococcal or chlamydial proctitis typically involves the most distal rectal mucosa and the anal crypts and is clinically mild, without systemic manifestations. In contrast, primary proctitis due to HSV and proctocolitis due to the strains of *C. trachomatis* that cause [LGV](#) usually produce severe anorectal pain and often cause fever. Perianal ulcers and inguinal lymphadenopathy, most commonly due to HSV, can also occur in LGV or syphilis. Sacral nerve root radiculopathies, usually presenting as urinary retention, laxity of the anal sphincter, or constipation, may complicate primary herpetic proctitis. In LGV, rectal biopsy typically shows crypt abscesses, granulomas, and giant cells -- findings resembling those in Crohn's disease; such findings should always prompt rectal culture and serology for LGV, which is a curable infection. Syphilis can also produce rectal granulomas, usually in association with infiltration by plasma cells or other mononuclear cells.

Diarrhea and abdominal bloating or cramping pain without anorectal symptoms and with normal findings on anoscopy and sigmoidoscopy occur with inflammation of the small intestine (enteritis) or with proximal colitis. In homosexual men without HIV infection, enteritis is often attributable to *Giardia lamblia*. Sexually acquired proctocolitis is most often due to *Campylobacter* or *Shigella* spp.

PREVENTION AND CONTROL OF STDs

Although rates of all curable [STDs](#) fell in the United States throughout the 1990s, all other industrialized countries of comparable economic development have made greater progress. For example, Sweden has virtually eliminated the transmission of gonorrhea, syphilis, and chancroid and has achieved very low rates of HIV transmission. Elimination of syphilis as an endemic disease is now a national goal in the United States, but stronger efforts toward prevention and control of all STDs are necessary.

Prevention and control of [STDs](#) require (1) reduction of the average rate of sexual exposure through alteration of behaviors and behavioral norms among both susceptible and infected persons in all population groups; (2) reduction of the efficiency of transmission through the promotion of safer sexual practices, the use of condoms during casual or commercial sex, hepatitis B immunization, and many other approaches (e.g., early detection and treatment of other [STIs](#) to reduce the efficiency of sexual

transmission of HIV); and (3) shortening of the duration of infectivity of STDs through early detection and curative or suppressive treatment of patients and their sexual partners.

Primary care physicians usually do not screen for illicit drug use or sexual risk behaviors, even when typical patients have classic presentations for HIV infection or another [STD](#). In fact, clinicians often focus only on detection and treatment of curable STDs to reduce the duration of infectivity. They generally have relatively little training or experience in risk assessment, counseling on risk reduction, tracing and treatment of sexual contacts, or condom promotion. Financial and time constraints imposed by managed-care practice patterns may further curtail screening and prevention services. As outlined in [Fig. 132-3](#), the efforts of clinicians simply to detect and treat STDs depend in part on societal efforts to teach young people how to recognize symptoms of STDs; to motivate those with symptoms to seek care promptly; and to make such care accessible, affordable, and acceptable, especially to the young indigent patients most likely to acquire an STD.

Since many infected individuals develop no symptoms or fail to recognize and report symptoms, clinicians should routinely perform an [STI](#) risk assessment for teenagers and young adults as a selective screen. U.S. Preventive Services Task Force Guidelines recommend screening sexually active females ≥ 25 years of age for *C. trachomatis* whenever they present for health care (at least once a year); older women should be tested if they have more than one sexual partner, have begun a new sexual relationship since the previous test, or have another [STD](#) diagnosed. In the United States, widespread selective screening of young women for cervical *C. trachomatis* infection in some regions has been associated with a 50 to 60% drop in prevalence, and such screening also protects the individual woman from [PID](#). Sensitive urine-based genetic amplification tests permit expansion of screening to men and teenage boys and to women in settings where a pelvic examination is not planned or is impractical.

Although gonorrhea is now substantially less common than chlamydial infection in industrialized countries, screening tests for *N. gonorrhoeae* are still appropriate for women and teenage girls attending [STD](#) clinics and for sexually active teens and young women from areas of high gonorrhea prevalence. Routine screening of asymptomatic men for urethral gonorrhea has a very low yield in the primary care setting. However, genetic amplification tests that combine screening for *N. gonorrhoeae* and *C. trachomatis* in a single low-cost assay may facilitate the prevention and control of both infections in populations at high risk.

All patients with newly detected [STIs](#) or at high risk for STIs according to routine risk assessment as well as all pregnant women should be encouraged to undergo serologic testing for syphilis and HIV infection, with appropriate HIV counseling before and after testing. Several randomized trials have shown that *risk reduction counseling* of patients with STDs significantly lowers subsequent risk of acquiring an STD. Preimmunization serologic testing for antibody to [HBV](#) is indicated for unvaccinated persons who are known to be at high risk, such as homosexually active men and injection drug users. In most young persons, however, it is more cost-effective to vaccinate against HBV without serologic screening.

Partner notification is the process of identifying and informing partners of infected patients of possible exposure to an [STI](#) and of examining, testing, and treating partners as appropriate. In a recently summarized series of 22 reports concerning partner notification during the 1990s, index patients with gonorrhea or chlamydial infection named a mean of 0.75 to 1.6 partners, of whom one-fourth to one-third were infected; those with syphilis named 1.8 to 6.3 partners, with one-third to one-half infected; and those with HIV infection named 0.76 to 5.31 partners, with up to one-fourth infected. Persons who transmit infection or who have recently been infected and are still in the incubation period usually have no symptoms or only mild symptoms and seek medical attention only when notified of their exposure. Therefore, the clinician must encourage patients to participate in partner notification, ensure that exposed persons are notified, and guarantee confidentiality to all involved. In the United States, local health departments will usually offer assistance in partner notification, treatment, and/or counseling. It seems both feasible and most useful to notify those partners exposed within the patient's likely period of infectiousness, which is often considered the preceding 1 month for gonorrhea, 1 to 2 months for chlamydial infection, and up to 3 months for early syphilis.

Persons with a new-onset [STD](#) always have a *source* contact who gave them the infection; in addition, they may have a *secondary (spread or exposed)* contact with whom they had sex after becoming infected. The identification and treatment of these two types of contacts have different objectives. Treatment of the source contact (often a casual contact) benefits the community by preventing further transmission; treatment of the recently exposed secondary contact (typically a spouse or another steady sexual partner) prevents both the development of serious complications (such as [PID](#)) in the partner and reinfection of the index patient.

In summary, clinicians and public health agencies share responsibility for the prevention and control of [STDs](#). In the managed-care era, the role of primary care clinicians has become increasingly important in prevention as well as in diagnosis and treatment.

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133. PELVIC INFLAMMATORY DISEASE - King K. Holmes, Robert C. Brunham

DEFINITION

The term *pelvic inflammatory disease* (PID) usually refers to infection that ascends from the cervix or vagina to involve the endometrium and/or fallopian tubes. Infection can extend beyond the reproductive tract to cause pelvic peritonitis, generalized peritonitis, perihepatitis, or pelvic abscess. In rare instances, infection extends secondarily to the pelvic organs from adjacent foci of inflammation (e.g., sites of appendicitis, regional ileitis, or diverticulitis), as a result of hematogenous dissemination (e.g., of tuberculosis), or as a rare complication of certain tropical diseases (e.g., schistosomiasis). Intrauterine infection can be primary (spontaneously occurring and usually sexually transmitted) or secondary to invasive intrauterine surgical procedures (e.g., dilatation and curettage, termination of pregnancy, insertion of an intrauterine device, or hysterosalpingography) or to parturition. Endometritis or endomyometritis is particularly common following delivery by emergency cesarean section when antibiotic prophylaxis is not used.

[PID](#) is uncommon during pregnancy itself. The uterotubal junction is closed as early as the seventh week of pregnancy, and the chorioamnion becomes approximated to the endocervical os, sealing off the intrauterine cavity, at the twelfth to fifteenth week of gestation. As a consequence, ascending intrauterine infection prior to the twelfth week of gestation may be associated (as either cause or effect) with endometritis and spontaneous abortion, while ascending infection after the twelfth week may be associated with chorioamnionitis.

Spontaneously occurring [PID](#) can be of the chronic or the acute type. Chronic PID due to *Mycobacterium tuberculosis* has become uncommon in industrialized countries. However, subacute or chronic PID caused by persistent or repeated infection with *Chlamydia trachomatis* is thought to be common.

Although the clinical diagnosis of [PID](#) is imprecise, the use of endometrial biopsy together with laparoscopy provides objective evidence of a continuum progressing from cervicitis alone to endometritis, to salpingitis, and to peritonitis. In this chapter, *PID* is used to refer to the clinical syndrome that includes these conditions, while the term *salpingitis* is restricted to cases of visually or histopathologically confirmed inflammation of the fallopian tubes. The distinction between endometritis and salpingitis may be important, because long-term sequelae are much more common after salpingitis. These sequelae include infertility due to bilateral tubal occlusion, peritubal adhesions, ectopic pregnancy due to tubal damage without occlusion, chronic pelvic pain, and recurrent PID.

ETIOLOGY

The etiology of [PID](#) has varied greatly among studies for reasons related to the selection of patients, the prevalence of sexually transmitted disease (STD) pathogens at the time and place of the study, and methodology. As is summarized in [Table 133-1](#), the agents most often implicated in acute PID include those that are primary causes of cervicitis (*Neisseria gonorrhoeae* and *C. trachomatis*) and those that can be regarded as components of an altered vaginal flora.

During the 1980s, *N. gonorrhoeae* and/or *C. trachomatis* was found in 65% of women with a clinical diagnosis of PID at San Francisco General Hospital and in 85% of patients with proven salpingitis and endometritis in Seattle; in both studies, gonorrhea was nearly twice as common as chlamydial infection and dual infection was common. However, in Western Europe and parts of the United States, as gonococcal infection has come under much better control, endocervical gonococcal infection has been found in a declining proportion of women with PID, and the microbial etiology cannot be defined in a substantial proportion of cases. In general, PID is most often associated with gonorrhea where there is a high incidence of gonorrhea -- e.g., in developing countries and in indigent, inner-city populations in the United States. In several studies of women with PID, up to two-thirds of women with endocervical cultures positive for *N. gonorrhoeae* have also had endometrial, peritoneal, or tubal cultures positive for this organism. Similarly, studies of women with proven PID have shown that *C. trachomatis* can be demonstrated by culture or immunofluorescent staining in the endometrium or tubes of the majority of those who have endocervical chlamydial infection.

Anaerobic and facultative anaerobic organisms (especially *Prevotella* species, peptostreptococci, *Escherichia coli*, and group B streptococci) and genital mycoplasmas have been isolated from specimens obtained at laparoscopy from the peritoneal fluid or fallopian tubes in a varying proportion -- typically one-fourth to one-third -- of women with PID studied in the United States. These vaginal organisms can be found in association with chlamydial or gonococcal infection as well as in the absence of such infection. The importance of vaginal organisms in salpingitis has probably been overestimated in some studies in which specimens were obtained for cultures by culdocentesis or endometrial aspiration, procedures in which contamination of the aspirated specimen by vaginal flora is possible. However, specimens obtained by laparoscopy from some patients with PID have also contained anaerobic and facultative species. A compilation of seven studies of the microbial etiology of PID showed that 5 to 78% of patients had anaerobes and facultative bacteria isolated from the upper genital tract. It is extremely difficult to determine the exact microbial etiology of an individual case of PID because of the frequency of mixed infection, the difficulty in sampling the fallopian tube itself, and the complexity of the microbiologic techniques required to detect the various fastidious pathogens involved. This situation has implications for the approach to empirical antimicrobial treatment of PID.

In general, first episodes of acute PID are particularly likely to be caused by *N. gonorrhoeae* and/or *C. trachomatis*. These sexually transmitted pathogens are implicated somewhat less often in recurrent bouts of acute PID, in episodes occurring in women with intrauterine devices (IUDs), in episodes precipitated by invasive intrauterine diagnostic or therapeutic procedures (which are often associated with ascending infection caused by endogenous vaginal flora), and perhaps in HIV-associated PID.

EPIDEMIOLOGY

It has been estimated that about 850,000 cases of PID occurred in the United States each year during the mid-1970s. PID is not a reportable disease in the United States; surveillance of physicians in private practice and of hospital discharges suggests that the incidence of PID increased from the mid-1960s through the mid-1970s and may

then have decreased. Hospitalization for acute PID declined steadily from 1982 through 1997, and initial visits to physicians' offices for PID have been declining since the mid-1980s. Furthermore, the number of hospitalizations for ectopic pregnancy in the United States fell by about two-thirds from 1989 to 1997.

Acute PID is almost exclusively a disease of sexually active women. Important risk factors include a history of salpingitis or of recent vaginal douching; the use of an IUD, particularly the Dalkon shield, has also been a risk factor. In most studies, the relative risk of PID among IUD users is higher in nulliparous than in parous women and is greatest during the first few months after IUD insertion. The increased risk of PID among IUD users is evident mainly among those with multiple sex partners. In contrast, women using oral contraceptives appear to be at decreased risk of PID. Barrier methods of contraception also make PID less likely by reducing the risk of chlamydial and gonococcal infection. Tubal sterilization reduces (but does not completely eliminate) the risk of salpingitis by preventing intraluminal spread of infection into the tubes.

PATHOGENESIS

Factors cited as possibly contributing to the upward spread of gonococci and chlamydiae from the endocervix to the endometrium and endosalpinx include estrogen-dominated (thin) cervical mucus, attachment to sperm that migrate upward into the tubes, use of an IUD, vaginal douching, menstruation, and subendometrial myometrial contractions, which move particulate matter from the cervix to the fundus of the uterus between days 5 and 14 of the menstrual cycle. It is important that the onset of symptoms of *N. gonorrhoeae*-associated and *C. trachomatis*-associated PID often occurs during or soon after the menstrual period. In fallopian tube organ cultures in vitro, gonococci attach to the surface of the secretory columnar cells (but not the ciliated cells) of the endosalpinx. Gonococcal pili and perhaps other surface proteins are important in this attachment. Gonococci are then taken into the secretory cells by endocytosis. They pass through the cells -- and perhaps between cells -- and are extruded through the base of the cell into the submucosal connective tissue. Ciliary motion ceases, and then ciliated cells, although not directly invaded by gonococci, are sloughed from the mucosa -- a factor that may render the tubes more susceptible to superinfection by other organisms. It is uncertain whether this loss of ciliated cells is irreversible in vivo. Gonococcal endotoxin and peptidoglycan as well as certain cytokines (such as tumor necrosis factor) appear to be responsible for these cytotoxic effects.

C. trachomatis also infects the columnar cells of the fallopian tube but produces little damage in tubal organ cultures, perhaps because the host response is more important than directly toxic effects of bacterial products in the pathogenesis of chlamydial salpingitis. *Chlamydia*-infected epithelial cells secrete a number of proinflammatory cytokines, which are chemotactic for neutrophils and mononuclear cells. Routine endometrial biopsies from women with chlamydial mucopurulent cervicitis (MPC) show endometritis in approximately one-half of cases. Endometritis detected in this way is sometimes but not always associated with symptoms of severe abdominal pain, presentation during days 1 through 7 of the menstrual cycle, signs of uterine tenderness, and an erythrocyte sedimentation rate (ESR) of ≥ 20 mm/h. Adnexal tenderness, cervical motion tenderness, and rebound tenderness as well as leukocytosis and elevated C-reactive protein levels are all more common in women with

laparoscopic evidence of salpingitis than in those with endometritis alone. Chlamydial inclusions are demonstrable by direct immunofluorescence in columnar epithelial cells of the endometrium and endosalpinx. The endometrial biopsies usually show neutrophils infiltrating the epithelium and plasma cells infiltrating the stroma, findings also seen in gonococcal endometritis but not in the uninfected endometrium. Experimental inoculation of the fallopian tubes of lower primates with *C. trachomatis* has shown that repeated exposure to *C. trachomatis* leads to the greatest degree of tissue inflammation and damage; this finding suggests that immunopathology also underlies the pathogenesis of chlamydial disease.

The pathogenesis of [PID](#) attributable to mycoplasmas or other vaginal anaerobic or facultative organisms is less well studied. It is possible that other vaginal organisms implicated in PID often cause tubal infection in women whose tubes have already been damaged by a primary sexually transmitted pathogen (i.e., *N. gonorrhoeae* or *C. trachomatis*). The vaginal organisms implicated in PID are found in the vagina most often and in greatest concentration in bacterial vaginosis, and there is epidemiologic evidence that bacterial vaginosis itself is a predisposing factor for PID (just as poor oral hygiene is a risk factor for aspiration pneumonia).

Certain other iatrogenic factors, such as dilatation and curettage or cesarean section, can increase the risk of [PID](#) in women with endocervical gonococcal or chlamydial infection. Evidence indicates that among women undergoing cesarean section, the presence of bacterial vaginosis increases the risk of postpartum endometritis.

CLINICAL MANIFESTATIONS

Tuberculous Salpingitis Unlike nontuberculous salpingitis, genital tuberculosis often occurs in older women, many of whom are postmenopausal. Presenting symptoms include abnormal vaginal bleeding, pain (including dysmenorrhea), and infertility. Bimanual pelvic examination may be normal, though about one-quarter of these women have had adnexal masses. Endometrial biopsy shows tuberculous granulomas and provides optimal specimens for culture.

Nontuberculous Salpingitis Symptoms of nontuberculous salpingitis classically evolve from a yellow or malodorous vaginal discharge caused by [MPC](#) and/or bacterial vaginosis to midline abdominal pain and abnormal vaginal bleeding caused by endometritis and then to bilateral lower abdominal and pelvic pain caused by salpingitis, with nausea and vomiting and increased abdominal tenderness caused by peritonitis. Some patients have diffuse abdominal pain caused by generalized peritonitis or pleuritic right upper quadrant pain caused by perihepatitis. The pattern in which symptoms evolve varies from patient to patient and is also related to the etiology of the [PID](#).

The onset of [IUD](#)-associated [PID](#) is typically gradual and may be preceded by the malodorous vaginal discharge characteristic of bacterial vaginosis. The onset of gonococcal PID may be more acute than that of chlamydial PID, and PID of either etiology usually presents during the first half of the menstrual cycle.

The abdominal pain in nontuberculous salpingitis is usually described as dull or aching. In some cases, pain is lacking or is atypical, and active inflammatory changes are found

in the course of an unrelated evaluation or procedure, such as a tubal ligation or a laparoscopic evaluation for infertility. Abnormal uterine bleeding precedes or coincides with the onset of pain in ~40% of women with [PID](#), symptoms of urethritis (dysuria) occur in 20%, and symptoms of proctitis (anorectal pain, tenesmus, and rectal discharge or bleeding) are occasionally seen in women with gonococcal or chlamydial infection.

Speculum examination shows evidence of [MPC](#) (yellow endocervical discharge, easily induced endocervical bleeding) in the majority of women with gonococcal or chlamydial [PID](#). Cervical motion tenderness is produced by stretching of the adnexal attachments on the side toward which the cervix is pushed. Bimanual examination reveals uterine fundal tenderness due to endometritis and abnormal adnexal tenderness due to salpingitis that is usually, but not necessarily, bilateral. Adnexal swelling is palpable in about one-half of women with acute salpingitis, but evaluation of the adnexae in a patient with marked tenderness -- even by an experienced examiner -- is not reliable. The initial temperature is >38°C in only about one-third of patients with acute salpingitis; thus fever is not required for the diagnosis. Laboratory findings include elevation of the [ESR](#) in 75% of patients with acute salpingitis and elevation of the peripheral white blood cell count in up to 60%.

Certain clinical manifestations of acute [PID](#) have been correlated with microbial etiology. For example, the onset of salpingitis is related to menses in women with gonococcal or chlamydial infection. Women with *N. gonorrhoeae*- or *C. trachomatis*-associated salpingitis are significantly younger than women with salpingitis of other etiologies. In a Swedish study, women with *Chlamydia*-associated salpingitis had more indolent disease, with mild symptoms of significantly longer duration and less fever, than women with gonorrhea-associated salpingitis. Women with polymicrobial PID more often have tubal or pelvic abscess formation. It is suspected that, for all recognized cases of symptomatic acute chlamydial salpingitis, there is a comparable number of unrecognized cases of indolent or mildly symptomatic chlamydial salpingitis. Furthermore, it is thought that subclinical chronic or recurrent chlamydial salpingitis may be a major cause of female infertility.

Perihepatitis and Periappendicitis Symptoms of perihepatitis, including pleuritic upper abdominal pain and tenderness (usually localized to the right upper quadrant), develop in 3 to 10% of women with acute [PID](#). The onset of symptoms of perihepatitis takes place during or after the onset of symptoms of PID and may overshadow lower abdominal symptoms, thereby leading to a mistaken diagnosis of cholecystitis. In perhaps 5% of cases of acute salpingitis, early laparoscopy reveals inflammation ranging from edema and erythema of the liver capsule to exudate with fibrinous adhesions between the visceral and parietal peritoneum. When treatment is delayed and laparoscopy is performed late, dense "violin-string" adhesions are seen over the liver; chronic exertional or positional right upper quadrant pain ensues when traction is placed on the adhesions. Although perihepatitis, also known as the *Fitz-Hugh-Curtis syndrome*, was for many years specifically attributed to gonococcal PID, most cases are now attributed to chlamydial salpingitis. In patients with chlamydial salpingitis, serum titers of microimmunofluorescent antibody to *C. trachomatis* are typically much higher when perihepatitis is present than when it is absent, and it has been suggested that repeated chlamydial infections are responsible for perihepatitis.

Physical findings include right upper quadrant tenderness and usually include adnexal tenderness and cervicitis, even in patients whose symptoms are not suggestive of salpingitis. Liver function tests are nearly always normal, since inflammation is largely limited to the liver capsule and usually spares the parenchyma. Ultrasonography of the right upper quadrant is normal. The presence of [MPC](#) and pelvic tenderness in a young woman with subacute pleuritic right upper quadrant pain and normal ultrasonography of the gallbladder points to a diagnosis of perihepatitis.

Periappendicitis (appendiceal serositis without involvement of the intestinal mucosa) has been found in ~5% of patients undergoing appendectomy for suspected appendicitis and can occur as a complication of gonococcal or chlamydial salpingitis.

Influence of HIV Infection HIV infection with immunosuppression increases the risk of repeated gonococcal and chlamydial infections among repeatedly exposed female sex workers, presumably by attenuating the immune response to repeated infection. Further, among women who acquire gonococcal or chlamydial infection of the cervix, HIV infection with immunosuppression increases the likelihood of developing clinical manifestations of salpingitis. Finally, among women with salpingitis, HIV infection is associated with increased severity of salpingitis and with tuboovarian abscess requiring hospitalization and surgical drainage. However, among African women with confirmed [PID](#), those with HIV infection appear less likely to have gonorrhea or chlamydial infection than those without HIV infection, a difference suggesting that other etiologies are especially important in the immunosuppressed patient. Nonetheless, among women with HIV infection and salpingitis, the clinical response to conventional antimicrobial therapy (coupled with drainage of tuboovarian abscess, when found) has been satisfactory.

DIAGNOSIS

Early diagnosis and initiation of therapy are essential to minimize tubal scarring. A reanalysis of Westrom's cohort of Swedish women with proven salpingitis showed that those who delayed seeking care were three times more likely than those who sought care promptly to experience subsequent infertility or ectopic pregnancy. Appropriate treatment must not be withheld from patients who have an equivocal diagnosis; it is better to err on the side of overdiagnosis and overtreatment. On the other hand, it is essential to differentiate between salpingitis and other pelvic pathology, particularly surgical emergencies such as appendicitis and ectopic pregnancy.

No readily available clinical finding or laboratory test, short of laparoscopy, definitively identifies salpingitis, and routine laparoscopy to confirm suspected salpingitis is generally impractical. Most patients with acute [PID](#) have lower abdominal pain of <3 weeks' duration, pelvic tenderness on bimanual pelvic examination, and evidence of lower genital tract infection (e.g., [MPC](#)). Approximately 60% of such patients have salpingitis at laparoscopy. Among the patients with these findings, a rectal temperature >38°C, a palpable adnexal mass, and elevation of the [ESR](#) over 15 mm/h also raise the probability of salpingitis, which has been found at laparoscopy in 68% of patients with one of these additional findings, 90% of patients with two, and 96% of patients with three. However, only 17% of all patients with laparoscopy-confirmed salpingitis have had all three additional findings.

[MPC](#) is probably responsible for the presence of neutrophils in vaginal fluid in [PID](#). In a woman with pelvic pain and tenderness, demonstration of an increased number of neutrophils (30 per 1000 microscopic field in strands of cervical mucus) increases the predictive value of a clinical diagnosis of acute PID.

Several clinical features other than the presence of cervicitis also favor the diagnosis of acute [PID](#). These include onset with menses, history of recent abnormal menstrual bleeding, presence of an [IUD](#), history of salpingitis, and sexual exposure to a male with urethritis. Detection of polymorphonuclear leukocytes in pelvic peritoneal fluid aspirated by culdocentesis supports a diagnosis of suspected salpingitis. Urethritis or proctitis may occur in chlamydial or gonococcal infection but may also represent a urinary tract source or an intestinal source, respectively. Appendicitis or another disorder of the gut is favored by the early onset of anorexia, nausea, or vomiting; the onset of pain later than day 14 of the menstrual cycle; or unilateral pain limited to the right or left lower quadrant. All women in whom the diagnosis of PID is being considered should be evaluated for ectopic pregnancy. The more sensitive serum assays for human chorionic gonadotropin are usually positive when ectopic pregnancy is the diagnosis. Ultrasonography and magnetic resonance imaging (MRI) can be useful for the identification of tuboovarian or pelvic abscess. MRI or intravaginal ultrasound assessment of the tubes has been reported to show increased tubal diameter, intratubal fluid, or tubal wall thickening in cases of salpingitis.

Laparoscopy is the most specific method for diagnosis of acute salpingitis. Although laparoscopic findings may be normal if inflammation is limited to the endosalpinx or the endometrium, patients with suspected [PID](#) who have a normal laparoscopy have a better prognosis (with no sequelae at all or fewer sequelae) than patients who have abnormal laparoscopic findings. The primary and uncontested value of laparoscopy in women with lower abdominal pain is for the exclusion of other surgical problems. Some of the most common or serious problems that may be confused with salpingitis (e.g., acute appendicitis, ectopic pregnancy, corpus luteum bleeding, ovarian tumor) are unilateral. Unilateral pain or pelvic mass, though not incompatible with PID, is a strong indication for laparoscopy unless the clinical picture warrants laparotomy instead. Atypical clinical findings, such as the absence of lower genital tract infection, a missed menstrual period, a positive pregnancy test, or failure to respond to appropriate therapy, are other frequent indications for laparoscopy.

Laparoscopic criteria used for the diagnosis of salpingitis include (1) erythema of the fallopian tube, (2) edema of the fallopian tube, and (3) seropurulent exudate or fresh, easily lysed adhesions at the fimbriated end or on the serosal surface of a fallopian tube.

Endometrial biopsy is relatively sensitive and specific for the diagnosis of endometritis when the endometrial changes described above are found, and the presence of endometritis correlates well with the presence of salpingitis. Endometritis is found in at least three-fourths of women with laparoscopically confirmed salpingitis and is not found in women without [PID](#).

The etiologic diagnosis of [PID](#) can be further studied by culture or other testing of

specimens obtained by endocervical swab, endometrial aspiration, or culdocentesis or by laparoscopy or laparotomy. Endocervical swab specimens should be examined by Gram's staining for neutrophils and gram-negative diplococci and by culture or DNA amplification test for *N. gonorrhoeae*. Compared with culture, the sensitivity of Gram's staining is ~60% and the specificity is >95%. The endocervical swab specimen should also be tested for *C. trachomatis* by culture or amplification assays for chlamydial DNA or RNA. Although detection of either *N. gonorrhoeae* or *C. trachomatis* in the endocervix does not prove that either agent is also present in the upper genital tract, this finding strongly supports the diagnosis of PID. The clinical diagnosis of PID made by expert gynecologists is confirmed by laparoscopy or endometrial biopsy in only ~60% of patients but in ~90% of those who also have cultures positive for *N. gonorrhoeae* or *C. trachomatis*. There is no evidence that the isolation of anaerobes or facultative aerobes from the cervix or vagina correlates with the presence of these organisms in the upper genital tract in acute PID, but this point has not been well studied. In one study, the isolation of *Haemophilus influenzae* from the endocervix was highly correlated with this organism's recovery from the fallopian tube in cases of salpingitis. Despite the risk of contamination of endometrial specimens with components of the vaginal flora, one study showed a 2.6-fold increase in the rate of recovery of anaerobic gram-negative rods (especially *Prevotella*, black-pigmented rods, and *Fusobacterium*) by endometrial biopsy in women with endometritis compared with control women. When laparoscopy is performed, material can be obtained directly from the cul-de-sac or the fimbriated opening of the tube or by tubal aspiration if pyosalpinx is present. Such specimens should be cultured for anaerobic and facultative pathogens as well as for *N. gonorrhoeae* and *C. trachomatis*.

TREATMENT

Women with [PID](#) can be treated as either outpatients or inpatients. Over the past decade, the costs of PID treatment have declined considerably because of the increased management of patients in the ambulatory setting, with use of highly active, well-absorbed antimicrobial agents. Nonetheless, hospitalization may be necessary and should be considered when (1) the diagnosis is uncertain and surgical emergencies such as appendicitis and ectopic pregnancy cannot be excluded, (2) pelvic abscess is suspected, (3) severe illness or nausea and vomiting preclude outpatient management, (4) the patient has HIV infection, (5) the patient is assessed as unable to follow or tolerate an outpatient regimen, or (6) the patient has failed to respond to outpatient therapy. If outpatient treatment is embarked on, clinical follow-up after 48 to 72 h of antibiotic treatment should be arranged. Treatment should cover *N. gonorrhoeae*, *C. trachomatis*, gram-negative facultative bacteria (especially *E. coli* and *H. influenzae*), vaginal anaerobes, and group B streptococci. Several antimicrobial combinations do provide a broad spectrum of activity against the major pathogens in vitro, but many have not been adequately evaluated for clinical efficacy in PID ([Table 133-2](#)).

Examples of Combination Regimens with Broad Activity Against Major Pathogens in PID Recommended combination regimens for ambulatory or parenteral management of PID are presented in [Table 133-3](#).

Women managed as outpatients should receive a combined regimen with broad activity, such as ceftriaxone [250 mg intramuscularly (IM)] followed by doxycycline (100 mg by

mouth, twice a day for 14 days). Metronidazole (500 mg by mouth twice daily) can be added, if tolerated, to enhance activity against anaerobes. Alternatively, ofloxacin (400 mg twice daily) plus metronidazole (500 mg twice daily), both continued for 14 days, provide good coverage of the major pathogens.

The following two parenteral regimens have given nearly identical results in a multicenter randomized trial:

1. Doxycycline [100 mg twice a day, given intravenously (IV) or orally] plus cefotetan (2.0 g IV every 12 h) or cefoxitin (2.0 g IV every 6 h). These drugs should be continued by the IV route for at least 48 h after the patient's condition improves, then followed with doxycycline (100 mg by mouth, twice a day) to complete 14 days of therapy.
2. Clindamycin (900 mg IV every 8 h) plus gentamicin (2.0 mg/kg IV or IM followed by 1.5 mg/kg every 8 h) in patients with normal renal function. Once-daily dosing of gentamicin (with combination of the total daily dose into a single daily dose) has not been evaluated in [PID](#) but has been efficacious in other serious infections and could be substituted. Treatment with these drugs should be continued for at least 48 h after the patient's condition improves, then followed with doxycycline (100 mg orally twice a day) or with clindamycin (450 mg orally four times a day) to complete 14 days of therapy. In cases with tuboovarian abscess, many experts use oral clindamycin rather than doxycycline for continued therapy to provide better coverage for anaerobic infection.

Management of Sexual Partners Sexual partners of patients with acute [PID](#) -- particularly those who have been partners within the 1 to 2 months before the onset of symptoms of PID -- should be examined for [STDs](#) and promptly treated with a regimen effective against uncomplicated gonococcal and chlamydial infection. An important point is that ³50% of the sexual partners of women with gonococcal and/or chlamydial PID have subclinical urethral infection and may be unaware of their infection status. Treatment of PID should be considered inadequate until sexual partners have been properly evaluated and treated.

Follow-Up Hospitalized patients should show substantial clinical improvement within 3 to 5 days. Women treated as outpatients should be clinically reevaluated within 72 h. A follow-up telephone survey of women seen in an emergency room and given a prescription for 10 days of oral doxycycline for [PID](#) found that 28% never filled the prescription and 41% stopped taking medication early (after an average of 4.1 days), often because of persistent symptoms, lack of symptoms, or side effects. Women not responding favorably to ambulatory therapy should be hospitalized. After completion of treatment, tests for persistent or recurrent infection with *N. gonorrhoeae* or *C. trachomatis* should be performed if symptoms persist or recur or if the patient has not complied with therapy or has been reexposed to an untreated sex partner.

Removal of an [IUD](#) Although a beneficial impact of IUD removal on the response of acute salpingitis to antimicrobial therapy and on the risk of recurrent salpingitis has not been proven, removal of the IUD 2 or 3 days after the initiation of antimicrobial therapy seems reasonable. When an IUD is removed, contraceptive counseling is essential.

Surgery Surgery is necessary for the treatment of salpingitis only in the face of

life-threatening infection (such as rupture or threatened rupture of a tuboovarian abscess) or for drainage of an abscess. Ultrasonography and [MRI](#) are useful for diagnosing and monitoring pelvic abscesses. Conservative surgical procedures are usually sufficient. Pelvic abscesses can often be drained by posterior colpotomy, and peritoneal lavage can be used if there is generalized peritonitis.

PROGNOSIS

Among 900 women who underwent long-term follow-up for a mean period of 8 years after successful treatment of an acute episode of [PID](#) with various regimens in Sweden, late sequelae included infertility due to bilateral tubal occlusion, ectopic pregnancy due to tubal scarring without occlusion, chronic pelvic pain, and recurrent salpingitis. Chronic pain lasting >6 months was seen in 18% of patients, and infertility due to tubal occlusion in 17%; 4% of the pregnancies that did occur were ectopic, representing approximately a sixfold increase over the expected rate of ectopic pregnancies. The rate of infertility after salpingitis was found to be related to the age of the patient, the duration of symptoms when treatment was started, the severity of salpingitis (as determined by laparoscopy) at the time of diagnosis, and the number of episodes of salpingitis. The postsalpingitis risk of infertility due to tubal occlusion among sexually active women not using contraceptives was 14% at 15 to 24 years of age and 26% at 25 to 34 years of age; the risk for women of all ages combined was 11% after one episode of salpingitis, 23% after two episodes, and 54% after three or more episodes. Women with chlamydial salpingitis who developed more severe inflammatory damage to the reproductive tract had significantly increased titers of antibody to the chlamydial heat-shock protein HSP60, as did women with infertility or ectopic pregnancy following chlamydial salpingitis. The risk of infertility after treated gonococcal salpingitis appeared lower than that after chlamydial salpingitis or polymicrobial PID. A study of outcomes of PID at the University of Washington found a sevenfold increase in the risk of ectopic pregnancy and an eightfold increase in the rate of hysterectomy after PID.

In several countries, a striking relationship has also been found between infertility due to tubal occlusion and the prevalence and titer of antibody to *C. trachomatis*. Recurrent salpingitis has been seen in ~15 to 25% of women treated for salpingitis in various studies.

PREVENTION

Prevention of [PID](#) depends first on the effective control of gonococcal and chlamydial infection in the general population. Effective methods include the promotion of changes in sexual behavior and the use of barrier contraceptives together with ensuring ready access to modern methods of diagnosis of these infections and effective treatment of sex partners to control further spread. The decline in popularity of the IUD, particularly among nulliparous women, has undoubtedly helped to reduce the incidence of PID. It is also possible, but not proven, that the use of oral contraceptives and the declining proportion of women who have practiced vaginal douching since the link between douching and PID became known have contributed to lower rates of PID. A randomized controlled trial designed to determine whether selective screening for chlamydial infection reduced the risk of subsequent PID showed that women randomized to undergo screening had a 56% lower rate of PID over the following year than did women

receiving the usual care without screening. This report strongly supports risk-based screening for *Chlamydia* as a highly effective way to reduce the incidence of PID and the prevalence of post-PID sequelae.

The complications and sequelae of salpingitis are minimized by early diagnosis and prompt effective treatment. It seems logical, but is unproven, that broad-spectrum therapy effective against all of the common causes of PID offers the best outcome. Although few methodologically sound clinical trials (especially with prolonged follow-up) have been conducted, one meta-analysis showed a benefit of providing good coverage against anaerobes. One placebo-controlled study showed that concurrent anti-inflammatory therapy with prednisolone hastened the reduction of acute inflammatory changes but did not improve the end results, as measured by fertility, hysterosalpingographic findings, or chronic pain. The potential value of anti-inflammatory therapy remains to be evaluated adequately.

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SECTION 3 -CLINICAL SYNDROMES: NOSOCOMIAL INFECTIONS

134. INFECTION CONTROL IN THE HOSPITAL - *Robert A. Weinstein*

The costs of nosocomial (hospital-acquired) infections are great. It is estimated that nosocomial infections cost \$4.5 billion and contribute to 88,000 deaths annually. Although infection-control and hospital epidemiology activities have been the subjects of increasing scientific study over the past 30 years, efforts to lower infection risks have been continually challenged by the growing numbers of immunocompromised patients, antibiotic-resistant bacteria, fungal and viral superinfections, and invasive devices and procedures. Four international decennial conferences on infection control, organized by the Centers for Disease Control and Prevention (CDC), have clearly documented these formidable trends. This chapter reviews the basic surveillance and prevention activities that have been developed to deal with these problems and that form the foundation for current hospital epidemiology programs.

ORGANIZATION AND RESPONSIBILITIES OF INFECTION-CONTROL PROGRAMS

The standards of the Joint Commission on Accreditation of Healthcare Organizations require all accredited hospitals to have an active program for surveillance, prevention, and control of nosocomial infections; a multidisciplinary infection-control committee usually oversees the program. The agents of the committee are the chairperson, who is preferably an infectious disease physician, and the infection-control practitioners, who are usually trained in nursing or medical technology and in epidemiology and public health. Education of physicians in infection control and hospital epidemiology is required in infectious disease fellowship programs and is available in courses provided by professional societies, primarily the Society for Healthcare Epidemiology of America.

In the 1970s, the [CDC](#)'s extensive Study on the Efficacy of Nosocomial Infection Control found that nosocomial infection rates fell by 32% in hospitals that established programs with organized surveillance and control activities; a trained, effectual infection-control physician; and one infection-control practitioner per 250 beds. In contrast, rates in hospitals without effective programs increased by 18%. Since that study, the responsibilities and roles of hospital epidemiology programs have expanded in several directions. Diagnosis-related reimbursement has led hospital administrators to place increased emphasis on cost containment and on documentation of the cost-effectiveness of infection control. The quality-improvement movements and the Joint Commission have redirected infection-control attention, in part, beyond the mere writing of policies and procedures to improvement of the actual processes and optimization of outcomes. In a few hospitals, epidemiology programs have taken on additional pharmacoepidemiologic and antibiotic-use review responsibilities. Finally, all programs must now respond to increasing governmental regulation of hospital waste and to standards mandated by the Occupational Safety and Health Administration for protecting health care workers from occupational exposure to bloodborne pathogens and tuberculosis.

SURVEILLANCE

Traditionally, infection-control practitioners survey inpatients for nosocomial infections

(defined as those neither present nor incubating at the time of admission). Surveillance involves a review of microbiology laboratory results, "shoe-leather" epidemiology on the nursing wards, application of standardized definitions of infection, ongoing dialogue with hospital workers, and common sense. Some innovative infection-control programs have taken advantage of the increased use of computerized pharmacy, microbiology, and other databases in hospitals to create algorithm-driven surveillance activities.

Most hospitals aim surveillance at infections that (1) are associated with a high level of morbidity, e.g., intensive care unit (ICU)-related infections and nosocomial pneumonia; (2) are costly, e.g., cardiac surgical wound infections; (3) are difficult to treat, e.g., infections due to antibiotic-resistant bacteria; (4) pose recurring epidemic problems, e.g., *Clostridium difficile*-related diarrhea; and (5) are potentially preventable, e.g., vascular access-related infections. Quality-assurance activities in infection control have led to increased surveillance of the compliance of personnel with infection-control policies (e.g., monitoring of actual adherence to hand-washing recommendations).

The results of surveillance are expressed as rates; for example, 5 to 10% of patients develop nosocomial infections. Although such overall statistics are often requested of hospitals by administrators or surveyors, they have little value unless qualified by site of infection, by patient population, and by exposure to risk factors. Meaningful denominators for infection rates include the number of patients exposed to a specific risk (e.g., rates of pneumonia among patients using mechanical ventilators) and the number of intervention days (e.g., rates of pneumonia per 1000 patient-days on a ventilator).

Temporal trends in rates should be reviewed, and rates should be compared with regional and national norms. However, even comparison rates generated by the [CDC's](#) ongoing National Nosocomial Infections Surveillance System, which collects data from more than 270 hospitals that use standardized definitions of nosocomial infections, have not been validated independently and represent a nonrandom sample of hospitals. Interhospital comparisons are easily confounded by the wide range in risk factors and in severity of underlying illnesses; unless rates are adjusted for these factors, comparisons may be misleading. Unfortunately, systems for making such adjustments either are rudimentary or have not been well validated.

The ongoing analysis of an individual hospital's infection rates helps to determine whether control efforts are succeeding and where increased education and control measures should be focused. Knowledge of infection rates is also useful in discussions with the hospital administration regarding areas to which additional resources should be directed.

PREVENTION AND CONTROL MEASURES

Epidemiologic Basis and General Measures Nosocomial infections follow basic epidemiologic patterns that can help to direct prevention and control measures. Nosocomial pathogens have reservoirs, are transmitted by predictable routes, and require susceptible hosts. Reservoirs and sources exist in the inanimate environment (e.g., tap water contaminated with *Legionella*) and in the animate environment (e.g., infected or colonized health care workers, patients, and hospital visitors). The mode of

transmission most often is either cross-infection (e.g., indirect spread of pathogens from one patient to another on the inadequately washed hands of hospital personnel) or autoinoculation (e.g., aspiration of oropharyngeal flora into the lung along an endotracheal tube). Occasionally, pathogens (e.g., group A streptococci and many respiratory viruses) are spread indirectly from person to person via infectious droplets released by coughing or sneezing. Much less common -- but often devastating in terms of epidemic risk -- is true airborne spread of droplet nuclei (as in nosocomial chickenpox) or common-source spread by contaminated materials (e.g., iodophors contaminated with *Pseudomonas*). Factors that increase host susceptibility include underlying conditions and the many medical-surgical interventions and procedures that bypass or compromise normal host defenses.

Through its program, the hospital's infection-control committee must determine the general and specific measures used to control infections and must review and recommend specific antiseptics and disinfectants for hospital use. Given the prominence of cross-infection, hand washing is the single most important preventive measure in hospitals. Many studies have examined the antimicrobial activity of a wide variety of antiseptic-containing hand-washing agents. The use of such medicated agents is important before invasive procedures and possibly in [ICU](#) settings. In light of the poor general compliance with hand-washing recommendations, the importance of using any hand cleanser between patient contacts cannot be overemphasized ([Table 134-1](#)).

The fact that 25 to 50% of nosocomial infections are due to the combined effect of the patient's own flora and invasive devices highlights the importance of improvements in the use and design of such devices ([Chap. 135](#)). Intensive educational programs can be associated with at least a temporary reduction in infection rates through improved asepsis in handling and earlier removal of invasive devices, but the maintenance of such gains is often difficult. Epidemiologic studies are used increasingly to assess the value of newer devices and site-specific control measures and to debunk some traditional yet ineffective and costly measures, such as routine culturing of the environment and personnel for "pathogens."

Urinary Tract Infections Approaches to the prevention of urinary tract infections have included the use of topical meatal antimicrobials, drainage bag disinfectants, antimicrobial-coated catheters, and sealed catheter-drainage tube junctions to eliminate inadvertent breaks in the system. Because of conflicting study results, none of these measures is considered routine. Systemic antimicrobials given for other purposes decrease the risk of urinary tract infection during the first 4 days of catheterization, after which resistant bacteria or yeasts emerge as pathogens. Selective decontamination of the gut is also associated with a reduced risk. Again, however, neither approach is routine. Irrigation of catheters, with or without antimicrobials, may actually increase the risk of infection.

Pneumonia Control measures for pneumonia are aimed at the remediation of risk factors in general patient care (e.g., minimizing aspiration-prone supine positioning) and at meticulous aseptic care of respirator equipment (e.g., disinfecting or sterilizing all in-line reusable components such as nebulizers, replacing tubing circuits at intervals of >48 h -- rather than more frequently -- to lessen the number of breaks in the system,

and teaching aseptic technique for suctioning). In a large multicenter trial, sucralfate, which provides stress-ulcer prophylaxis without altering gastric pH, did not reduce the risk of ventilator-associated pneumonia, despite the theoretical advantage of lessened risk for gastric colonization by gram-negative bacilli. The benefit of selective decontamination of the oropharynx and gut with nonabsorbable antimicrobials has been controversial.

Surgical Wound Infections The most important control measures for surgical wound infections include the use of antimicrobial prophylaxis at the start of high-risk procedures, attention to technical surgical issues and operating-room asepsis (e.g., not shaving the operative site until surgery and avoiding open or prophylactic drains), and preoperative therapy for active infection. In one study, rates of postoperative infection were lower among patients who had normothermia maintained during colorectal surgery. Reporting of surveillance results to surgeons has been associated with reductions in infection rates. The increasingly extensive review of infection rates by regulatory agencies and third-party payers emphasizes the importance of stratifying rates by patient-related risk factors and of developing meaningful systems for interhospital comparisons and for wound surveillance after the patient's discharge from the hospital or clinic (when more than 50% of infections first become apparent).

Infections Related to Vascular Access and Monitoring (See also [Chap. 135](#)) Control measures for infections associated with vascular access and monitoring include the moving of peripheral or arterial catheters to a new site at specified intervals (e.g., every 72 h for peripheral intravenous catheters), which may be facilitated by use of an intravenous team; application of disposable transducers and aseptic technique for the accessing of transducers or other vascular ports; removal of "idle" catheters; and consideration of use of central venous catheters impregnated with anti-infective agents. Unresolved issues include the best frequency for the rotation of central venous catheter sites (guidewire-assisted catheter changes at the same site do not lessen infection risk); the best antiseptics for site preparation and for catheter dressing; the appropriate role for mupirocin ointment, a topical antibiotic with excellent antistaphylococcal activity, in site care; and the relative degrees of risk posed by percutaneous central catheters and by newer designs -- tunneled, totally implanted, or peripherally inserted central catheters (PICC lines). Improvements in composition of semitransparent access-site dressings and potential nursing benefits (ease of bathing and site inspection and protection of the site from secretions) favor use of such coverings.

Isolation Techniques Written policies for the isolation of infectious patients are a standard component of infection-control programs. In 1996, the [CDC](#) revised its isolation guidelines to be simpler; to recognize the importance of all body fluids, secretions, and excretions in the transmission of nosocomial pathogens; and to focus precautions on the major routes of infection transmission.

The revised guidelines contain two tiers of precautions. *Standard precautions* are designed for the care of all patients in hospitals to reduce the risk of transmission of microorganisms from both recognized and unrecognized sources of infection. These precautions include gloving, as well as hand washing, for potential contact with blood; with all other body fluids, secretions, and excretions, regardless of whether they contain visible blood; with nonintact skin; and with mucous membranes. Depending on exposure

risks, standard precautions also include use of masks, eye protection, and gowns.

In the second tier are precautions for the care of patients with suspected or diagnosed colonization or infection with transmissible pathogens. These transmission-based guidelines collapse the older category- and disease-specific isolation guidelines into three sets of precautions based on probable routes of transmission: *airborne precautions*, *droplet precautions*, and *contact precautions*. Sets of precautions may be combined for diseases that have more than one route of transmission (e.g., varicella). Potentially contagious clinical syndromes, such as acute diarrhea, are included in the revised guidelines.

Because some prevalent antibiotic-resistant pathogens, particularly vancomycin-resistant enterococci (VRE), may be present on *intact* skin of patients in hospitals, some experts recommend gloving for all contact with patients who are acutely ill and/or from high-risk units, such as [ICUs](#). In recent trials, wearing gloves did not replace the need for hand washing because hands occasionally became contaminated during wearing or removal of gloves. Some studies have suggested that use of gowns and gloves compared with routine care of patients (i.e., using neither of these barriers) decreases the risk of nosocomial infection; however, more recent evaluation suggests that gowning by personnel does not add benefit beyond that conferred by gloving and hand washing. Nevertheless, requiring increased precaution levels can improve the compliance of health care workers with isolation recommendations by 30%.

EPIDEMIC PROBLEMS

Outbreaks are always big news but probably account for fewer than 5% of nosocomial infections. The investigation and control of epidemics in hospitals require that infection-control personnel develop a case definition, confirm that an outbreak really exists (since many apparent epidemics are actually pseudo-outbreaks due to surveillance or laboratory artifacts), review aseptic practices and disinfectant use, determine the extent of the outbreak, perform an epidemiologic investigation to determine modes of transmission, work closely with microbiology personnel to culture for common sources or personnel carriers as appropriate and to type epidemiologically important isolates, and heighten surveillance to judge the effect of control measures. Control measures generally include the early reinforcement of routine aseptic practices during a search for compliance problems that may have fostered the outbreak, the ensuring of the appropriate isolation of cases (and the institution of cohort isolation and nursing if needed), and the implementation of further controls on the basis of the findings of the investigation. Examples of some potential epidemic problems follow.

Chickenpox When health care workers are exposed to chickenpox in the community or through patients with initially unrecognized infections, or when these employees work during the 24 h before developing chickenpox, infection-control practitioners institute a varicella exposure investigation and control plan. The names of exposed workers and patients are obtained; medical histories are reviewed, and (if necessary) serologic tests for immunity are conducted; physicians are notified of susceptible exposed patients; postexposure prophylaxis with varicella-zoster immune globulin (VZIG) is considered for immunocompromised or pregnant contacts (see [Table 183-1](#)); preemptive use of acyclovir is considered as an alternative strategy in some susceptible persons; and

susceptible exposed employees are furloughed during the at-risk period for disease (8 to 21 days, or 28 days if VZIG has been administered). Preexposure varicella vaccination can markedly decrease risk for susceptible employees.

Tuberculosis The resurgence of pulmonary tuberculosis in the United States since 1987 and a series of nosocomial outbreaks of infection with multidrug-resistant strains -- primarily involving patients with AIDS and their caregivers -- have led to a reevaluation of tuberculosis control. Important control measures include prompt recognition, isolation, and treatment of cases; recognition of atypical presentations (e.g., lower-lobe infiltrates without cavitation); use of negative pressure, 100% exhaust, private isolation rooms with closed doors, and six air changes per hour; use of face masks (approved by the National Institute for Occupational Safety and Health) by caregivers entering isolation rooms; possible use of high-efficiency particulate air filter units and/or ultraviolet lights for disinfecting air when other engineering controls are not feasible or reliable; and follow-up skin-testing of susceptible personnel who have been exposed to infectious patients before isolation.

Group A Streptococci The potential for a group A streptococcal outbreak should be considered when even a single nosocomial case occurs. Most outbreaks involve surgical wounds and are due to the presence of an asymptomatic carrier in the operating room. Investigation can be confounded by carriage at extrapharyngeal sites such as the rectum and vagina. Health care workers in whom carriage has been linked to nosocomial transmission of group A streptococci are removed from the patient-care setting and are not permitted to return until carriage has been eliminated by antimicrobial therapy.

Aspergillus *Aspergillus* spores are common in the environment, particularly on dusty surfaces. When hospital ceiling tiles are removed to provide access for electrical wiring or plumbing or when dusty areas are disturbed during hospital renovation, the spores become airborne. Inhalation of spores by immunosuppressed (particularly neutropenic) patients creates a risk of pulmonary and/or paranasal sinus infection and disseminated aspergillosis. Routine surveillance among neutropenic patients for infections with filamentous fungi, such as *Aspergillus* and *Fusarium*, helps hospitals to determine whether they have unduly large environmental loads of these organisms. To lower the risk, hospitals should inspect and clean air-handling equipment on a routine schedule, review all planned hospital renovations with infection-control personnel and subsequently construct appropriate barriers, remove immunosuppressed patients from renovation sites, and consider the use of high-efficiency particulate air filters for rooms housing immunosuppressed patients.

Legionella Sporadic and epidemic cases of nosocomial *Legionella* pneumonia are most often due to the contamination of potable water and predominantly affect immunosuppressed patients, particularly those receiving glucocorticoid medication. The risk varies greatly within and among geographic regions, depending on the extent of hospital hot-water contamination, on the presence or absence of high-risk patient populations, and on specific hospital practices (e.g., inappropriate use of nonsterile water in respiratory therapy equipment). Laboratory-based surveillance for nosocomial *Legionella* should be performed, and a diagnosis of legionellosis should probably be considered more often than it is. If cases are detected, environmental samples (e.g., tap

water) should be cultured. If cultures yield *Legionella* and if typing of clinical and environmental isolates reveals a correlation, eradication measures should be pursued ([Chap. 151](#)). An alternative approach is to periodically culture tap water on wards housing high-risk patients. If *Legionella* is found, a concerted effort should be made to culture samples from all patients with nosocomial pneumonia for *Legionella*.

Antibiotic-Resistant Bacteria Outbreaks of antibiotic resistance can depend on any of the following events: Darwinian selection of bacterial chromosomal mutations, spread of plasmid- and/or transposon-borne resistance among bacterial species, and (re)admission to the hospital of patients chronically infected with resistant bacteria. After the introduction of resistant strains, dissemination occurs by cross-infection on unwashed hands of caregivers or, occasionally, via personnel carriage and/or environmental contamination. Outbreak control depends on close laboratory surveillance, with early detection of problems; on the reinforcement of routine asepsis (e.g., hand washing); on the implementation of barrier precautions for all colonized and/or infected patients; on the use of patient-surveillance cultures to more fully ascertain the extent of patient colonization; and on the timely initiation of an epidemiologic investigation when rates increase. Colonized personnel who are implicated in nosocomial transmission and patients who pose a threat may be decontaminated; for example, colonization with methicillin-resistant *Staphylococcus aureus* may be controlled with oral antibiotics, including trimethoprim-sulfamethoxazole and rifampin, and with topical agents, including hexachlorophene or chlorhexidine and mupirocin. In a few [ICUs](#), selective decontamination has been used successfully as a temporary emergency control measure for outbreaks of infection due to gram-negative bacilli.

The most recent bacterial-resistance problem to plague hospitals is the emergence of [VRE](#). Initially an [ICU](#) problem, VRE have now spread onto general wards in many hospitals. VRE are particularly problematic because of a substantial "iceberg" effect (i.e., the fact that, for each individual with a clinical infection, many other patients are colonized); the occurrence of both gastrointestinal and skin colonization (reflecting fecal contamination on the skin of ill, hospitalized patients); and the propensity for these organisms to contaminate the patient's environment, which may increase the risk of cross-infection. Control of VRE requires strict attention to hand washing by personnel, concerted use of barrier precautions or cohort nursing for patients known to be colonized or infected, and emphasis on thorough cleaning of the rooms of these patients.

Spread of vancomycin resistance to *S. aureus* is a major concern. Clinical infections with methicillin-resistant *S. aureus* strains that exhibit reduced susceptibility to vancomycin have been reported in a few patients, usually in the setting of prolonged or repeated treatment with vancomycin. The detection of these strains requires augmented laboratory activities, and their identification should trigger an aggressive epidemiologic investigation and aggressive infection-control measures.

Because the excessive use of broad-spectrum antibiotics underlies many resistance problems, antibiotic-control policies ([Table 134-2](#)) must be considered a cornerstone of resistance-control efforts. Although the efficacy of antibiotic-control measures in reducing rates of antimicrobial resistance has not been proved in prospective controlled

trials, it seems worthwhile to restrict the use of particular agents to narrowly defined indications or possibly to cycle the use of antibiotic classes to limit selective pressure on the nosocomial flora.

EMPLOYEE HEALTH SERVICE ISSUES

An institution's employee health service is a critical component of its infection-control efforts. New employees should be processed through the service, where a contagious-disease history can be taken; evidence of immunity to a variety of diseases, such as hepatitis B, chickenpox, measles, and rubella, can be sought; immunizations for hepatitis B, measles, rubella, and varicella can be given as needed and a reminder about the need for yearly influenza immunization can be imparted; baseline and "booster" purified protein derivative of tuberculin skin-testing can be performed; and education about personal responsibility for infection control can be initiated. Evaluations of employees should be codified to meet the requirements of accrediting and regulatory agencies.

The employee health service must have protocols for dealing with workers who have been exposed to contagious diseases, such as those percutaneously or mucosally exposed to the blood of patients infected with HIV. Postexposure HIV prophylaxis with a combination of antiretroviral agents (e.g., zidovudine and lamivudine, with or without indinavir or nelfinavir) is recommended. Protocols are also needed for dealing with caregivers who have common contagious diseases, such as chickenpox, group A streptococcal infections, respiratory infections, and infectious diarrhea, and for those who have less common but high-visibility public health problems, such as chronic hepatitis B or C or HIV infection, for which exposure-control guidelines have been published by the [CDC](#) and by the Society for Healthcare Epidemiology of America.

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135. HOSPITAL-ACQUIRED AND INTRAVASCULAR DEVICE-RELATED INFECTIONS - Dori F. Zaleznik

Nosocomial infections are defined as infections acquired during or as a result of hospitalization. Generally, a patient who has been in the hospital for <48 h and develops an infection is considered to have been incubating the infection before hospital admission. Most infections that become manifest after 48 h are considered to be nosocomial. A patient may develop a nosocomial infection after being discharged from the hospital if the organism apparently was acquired in the hospital. Surgical wound infection developing in the weeks after hospital discharge is an example of such nosocomial infection.

INCIDENCE AND COSTS

Nosocomial infections contribute significantly to morbidity and mortality as well as to excess costs for hospitalized patients. It is estimated that 5% of patients admitted to an acute care hospital in the United States acquire a new infection, with >2 million nosocomial infections per year and an annual cost of >\$2 billion. Some authorities estimate that the odds of death are doubled for patients who develop a nosocomial infection, although clearly such factors as underlying disease and severity of illness also play an important role in outcome.

Although immunosuppressed hosts are especially vulnerable to infections acquired in a hospital, common nosocomial infections occur even in immunocompetent hosts. The National Nosocomial Infections Surveillance (NNIS) Registry has been monitoring nosocomial infection rates since 1970. Its most recent report covers the period from October 1996 through April 1998 and includes both teaching and nonteaching hospitals and both small and large facilities. The most common nosocomial infections have remained the same. Urinary tract infections (UTIs), pneumonia, and surgical-site infections (SSIs, formerly termed wound infections) are most frequent. However, primary bloodstream infections, especially those associated with intravascular devices, have increased in frequency, as have infections in medical and surgical intensive care units (ICUs) and infections caused by antimicrobial-resistant pathogens.

The potential impact of nosocomial infections is considerable when assessed in terms of incidence, morbidity, mortality, and financial burden. Analyses of these factors examine nosocomial infections as both medical and economic issues. The clinical problem facing the physician is the development of a new fever in a patient in the hospital. In the evaluation of such a patient, information about the most common categories of infection may not be sufficient. Rather, the clinician must also use clinical clues from the patient's presentation and hospitalization to diagnose a nosocomial infection.

Approach to the Patient

The evaluation of a hospitalized patient with new fever should include a careful history ([Chap. 17](#)). Particular attention should be paid to symptoms of headache, cough, abdominal pain, diarrhea, flank pain, dysuria, urinary frequency, and leg pain. Other features related to the patient's hospitalization are also important, such as the presence and type of intravenous devices, the past or current use of a urinary catheter, the

surgical procedure conducted (if any), and the new medications administered, including those for surgical prophylaxis. The physical examination should be directed at possible sources of infection and should focus particularly on the skin (with a search for rash or embolic lesions); the lungs; the abdomen (especially the right upper quadrant); the costovertebral angles; surgical wounds; the calves; and current and old intravenous access sites (for signs of phlebitis). The laboratory evaluation of all hospitalized patients with new fever should include a complete blood count with differential, a chest radiograph, and blood and urine cultures. Other diagnostic tests to consider include liver function tests, plain-film or other studies of the abdomen, routine aerobic cultures of sputum or other relevant body fluids, and (in cases of diarrhea) testing of stool for *Clostridium difficile* toxin.

CATEGORIES OF INFECTION

Pneumonia Certainly the astute clinician will question the patient thoroughly and perform a rapid comprehensive physical examination. One way to continue the approach to the hospitalized patient with a new fever is to consider potential infections that may be life-threatening, such as pneumonia. Most at risk for developing nosocomial pneumonia are patients in an [ICU](#), especially those who are intubated; patients with an altered level of consciousness, especially those with nasogastric tubes; elderly patients; patients with chronic lung disease; postoperative patients; and any of the above patients taking H₂blockers or antacids. Nosocomial pneumonia in the [NNIS](#) Registry is diagnosed 4 to 7 times per 1000 hospitalizations. Among patients on ventilators, the occurrence of pneumonia is estimated at 15 cases per 1000 ventilator days in medical and surgical ICUs. Mortality figures for nosocomial pneumonia are as high as 50%.

Oropharyngeal and gastric colonization plays a critical role in the pathogenesis of pneumonia in hospitalized patients. The oropharynx can become colonized by many species of aerobic gram-negative organisms within 48 h of the patient's hospitalization; aspiration occurs commonly during sleep and is increased by such factors as a nasogastric tube, altered consciousness, decreased gag reflex, or delayed gastric emptying. As for gastric colonization, bacterial counts in the stomach rise in the presence of medications that raise gastric pH, such as H₂blockers and antacids, as well as in malnourished, achlorhydric, and some elderly patients. The prevalence of pneumonia is reportedly two to three times higher among intubated patients receiving H₂blockers or antacids for stress-ulcer prophylaxis than among intubated patients receiving sucralfate, a medication that heals ulcers without altering gastric pH. Gastric colonization is believed to influence the development of pneumonia by retrograde colonization of the oropharynx. Ventilated patients are also at risk of developing pneumonia by exposure to bacteria leaking around the cuff of the endotracheal tube or to bacteria from nebulizers, condensate within ventilator circuits, or humidifiers.

Outside the [ICU](#), pneumonia should be suspected when a patient develops a new cough, fever, leukocytosis, sputum production, and a new infiltrate on chest x-ray. Diagnosis can be complicated in patients with congestive heart failure who have concomitant chest x-ray abnormalities or in patients with chronic sputum production. Some organisms, such as *Legionella* spp., may not be associated with peripheral leukocytosis.

In [ICU](#) patients, especially those who are intubated, the signs of pneumonia are relatively subtle, and thus the diagnosis is often relatively complex. In particular, the chest x-rays are difficult to interpret, because fluid overload, congestive heart failure, and acute respiratory distress syndrome (ARDS) are all common findings in intubated patients. Polymorphonuclear leukocytes (PMNs) are often present on Gram-stained preparations of purulent secretions from these patients. An important clue to pneumonia is a change in the output or character of these secretions. If their volume or thickness increases or their color changes, a sputum Gram's stain should be performed and pneumonia seriously considered in the differential diagnosis. Serial Gram's stains are useful, as the number of PMNs may increase substantially and the type(s) of organisms may shift with the development of pneumonia. For example, the baseline sputum sample from an intubated patient may contain about 25 PMNs per high-power field and have mixed gram-positive and gram-negative organisms of several morphologic types in moderate numbers. On the day of a new fever, the same patient may have copious amounts of more tenacious sputum with more PMNs and a predominance of enteric-appearing gram-negative rods. Even without distinct changes in the chest x-ray, this patient would be considered to have developed pneumonia. Another subtle sign of pneumonia in the intubated patient is a requirement for change in ventilator settings in the absence of fluid overload, a mechanical alteration (e.g., a shift in endotracheal tube placement), or a pneumothorax.

The major organisms of concern in nosocomial pneumonia are gram-negative aerobic bacteria. *Pseudomonas aeruginosa* was the most common isolate in the [NNIS](#) survey of [ICUs](#), with a frequency of 21%; *Staphylococcus aureus* was next most common at 20%. *Acinetobacter* has become a more common pathogen in ventilator-associated pneumonia. While surveys of organisms are useful, it is essential to know which pathogens are common in a given institution, as hospitals and especially ICUs differ in their resident flora. In some institutions, methicillin-resistant *S. aureus*, *Stenotrophomonas* (formerly *Xanthomonas*) *maltophilia*, *Flavobacterium* spp., and even *Legionella* spp. may be of particular concern. Viruses such as respiratory syncytial virus and adenovirus are receiving increased attention as etiologic agents of nosocomial pneumonia in both adults and children. In the past, viruses have been underrepresented in statistics on the agents of nosocomial pneumonia because the diagnosis of viral infection is more difficult and because many microbiology laboratories do not have the capability to isolate viruses.

Antibiotic resistance is another important issue to address in the management of a hospitalized patient. An outgrowth of the [NNIS](#) surveys is Project ICARE, which tracks antibiotic usage patterns and resistance rates in a subset of NNIS institutions. Rates of resistance are generally higher in [ICUs](#) and track with increased use of antibiotics. *P. aeruginosa*, *Enterobacter* spp., and enterococci are the pathogens of greatest concern in the development of antibiotic resistance. In addition to knowing the sensitivity patterns of the hospital flora, one must consider whether a patient has received continuous or multiple courses of antibiotic therapy. To reduce the likelihood of altering the sensitivity patterns of the patient's flora, antibiotic courses for pneumonia should be kept as short as possible, with coverage as narrow as possible for the organism(s) involved.

Bacteremia Another potentially life-threatening nosocomial infection to consider in the evaluation of the patient with a new fever is bacteremia, which is usually related to the

presence of an intravascular device ([Chap. 134](#)). While many common nosocomial infections such as pneumonia or [UTI](#) can be accompanied by bacteremia, primary bacteremia is defined by isolation of a recognized pathogen from the blood without an infection at another site. One carefully controlled study reported bloodstream infection in 2.7% of admissions to a surgical [ICU](#), with 50% mortality and a prolongation of hospitalization by 24 days in survivors.

One difficulty in assessing the significance of bacteremia is to distinguish true pathogens from contaminating skin flora. This distinction is especially important in establishing an infection of an indwelling intravascular catheter because organisms that inhabit the skin, such as coagulase-negative staphylococci, also frequently cause infection. The most common point of entry for infection related to intravascular devices is the insertion site, with spread of the infection along the outside of the device initially. Other means of entry for infecting organisms include introduction via contaminated infusates or tubing, ports, or leaking connections and hematogenous seeding of a catheter during bacteremia. While gram-negative aerobic bacilli are probably the most feared nosocomial bloodstream pathogens, the [NNIS](#) data for 1980 through 1989 showed that the isolation of these organisms had not increased in frequency over the decade. The frequency of bloodstream isolation increased the most for coagulase-negative staphylococci, with the next highest increase for *Candida* spp. Other leading causes of line-related bacteremia were *S. aureus* and enterococci. Subsequent studies confirmed these findings. Nosocomial endocarditis is an important, newly recognized entity that develops largely as a complication of invasive procedures or intravascular devices and may account for as many as 10% of cases of infective endocarditis.

Establishing an infection of an intravascular device or primary bacteremia as the cause of fever in a hospitalized patient is a diagnosis of exclusion. If a patient has a fever and signs of cutaneous involvement (erythema, induration, tenderness, or purulent drainage) at the insertion site of a catheter, full cultures should be performed, the vascular-access line removed, and the catheter tip sent for quantitative culture. Studies have correlated the growth of ≥ 15 colonies from a catheter tip with infection of the line. More commonly, the exit site does not show signs of infection, and there is considerable debate about the necessity of removing a line from a febrile patient at that point. Although line changes over a guidewire have been shown to be safe, unless another site of infection is obvious, it is generally advisable to remove the line and to change the site when a patient develops a new fever. The traditional teaching is that an infected intravenous device should be removed. In current practice, however, especially with surgically implanted intravenous catheters, a decision may be made to attempt treatment with antibiotics while leaving the catheter in place. This practice is often successful when the infecting organism is a coagulase-negative *Staphylococcus* species but is less often effective with other organisms, particularly *Candida* spp. and gram-negative bacilli. Salvage of catheters used for hemodialysis is especially important and has been successfully accomplished with infections caused by a variety of organisms.

Another controversial management issue is whether to draw blood for culture through a line. While some studies report a correlation in the 90% range between culture results for blood drawn through vascular-access lines and those for peripheral blood, the former cultures can be either false-positive or false-negative. If the line culture is positive and

no peripheral blood has been drawn, it is impossible to determine whether the patient has true bacteremia or the culture merely reflects bacteria associated with the line. Whether bacteremia is high- or low-grade and whether it is sustained or transient may influence the duration of antibiotic therapy and cannot be determined from cultures of blood specimens obtained through a line.

An area of considerable interest and controversy is the prevention of catheter-related infections through the use of intravenous devices impregnated with chlorhexidine/silver sulfadiazine or minocycline/rifampin. A meta-analysis of 11 studies found a decrease in both catheter colonization and catheter-related bacteremia with chlorhexidine/silver sulfadiazine-impregnated catheters, with associated cost savings. One multicenter prospective randomized trial directly compared the two types of catheters and found the minocycline/rifampin-impregnated catheter to be superior. Antibiotic-resistant strains were not recovered in this trial, although concern has been raised that antibiotic impregnation may increase the development of resistance. The several other studies that do not support these findings include one in which the incidence of bacteremia did not decrease with the use of chlorhexidine/silver sulfadiazine-impregnated catheters.

Surgical-Site Infection Evaluation of fever in the postoperative patient must include careful evaluation of the surgical wound. Although [SSI](#) reportedly accounts for 19% of nosocomial infections, the true incidence of postoperative wound infection is difficult to assess, particularly at a time when many patients are hospitalized for relatively short periods. In a number of studies, careful follow-up for the development of SSI after discharge -- especially observation of the wound by a trained observer, such as a nurse -- has shown the actual rates of SSI in all categories of surgery to be greater than the reported rates. SSI rates vary from 4.6 to 8.2% for nonteaching and large teaching hospitals, respectively. Rates also vary by procedure, with abdominal surgery resulting in the highest rates.

Risk factors for the development of postoperative wound infection include the presence of a drain; a long preoperative length of stay, with the rates doubling for each week of preoperative hospitalization; preoperative shaving of the field, especially if performed ³24 h beforehand; a long duration of surgery; and the presence of an untreated remote infection. Infection rates also vary with the surgeon. Perioperative antibiotic prophylaxis has been shown to decrease rates of wound infection in a number of careful studies, including those of clean surgical procedures. Antibiotic coverage after the surgical wound is closed has not been shown to provide additional benefit.

A surgical wound should be examined for localized tenderness and induration, fluctuance, drainage of purulent material, and dehiscence of sutures. Mechanical factors, as well as infection, can cause wound dehiscence. Sternal wounds following cardiac surgery are of special concern because the consequences of infection can be severe. The surface of the wound may not present an obvious cause for concern, but ongoing fevers, serous drainage, and especially the development of rocking or instability of the sternum may be sufficient cause for surgical exploration of the wound in some cases. Mediastinitis or sternal osteomyelitis is a severe complication of cardiac surgery. Wounds associated with the placement of prosthetic devices, such as mechanical joints, are also of special concern. Infection of these wounds can lead to infection of the prosthesis, and clearance of prosthetic joint infections generally requires surgical

removal of the device.

The most common pathogens causing [SSI](#) are coagulase-negative *Staphylococcus* and *S. aureus*, but antibiotic-resistant bacteria and fungi are also becoming more frequent etiologies. Early infections may be associated with organisms that produce rapid, progressive skin infection, such as group A *Streptococcus* and *Clostridium* spp. Group A *Streptococcus* has been identified in some cases of recurrent infection of saphenous-vein graft harvest sites.

Urinary Tract Infection [UTI](#), the most common type of nosocomial infection, is generally the easiest to treat and has the least severe sequelae. Four principal risk factors have been associated repeatedly with the development of UTI in hospitalized patients: female sex, prolonged urinary catheterization, lack of systemic antibiotic therapy, and breach of appropriate catheter care. The administration of systemic antibiotics to patients with urinary catheters in place for 1 to 5 days has been associated with a decrease in rates of bacteriuria. For patients with catheters in place for ≥ 6 days, however, this benefit is not observed.

The pathogenesis of catheter-associated [UTI](#) appears to differ in men and women. In women, the typical mechanism involves periurethral colonization with fecal flora and tracking of organisms up the catheter to the bladder; thus the pathogenesis resembles that of UTI in noncatheterized female patients, in whom bacteria track up the short female urethra. In contrast, periurethral colonization often cannot be demonstrated in men; most infections seem to arise from intraluminal spread of organisms to the bladder. Some organisms, such as *Proteus* and *Pseudomonas* spp., appear to facilitate the growth along the inside of the urinary catheter of a biofilm that encrusts and obstructs the flow of urine.

[UTI](#) is certainly an extremely common nosocomial infection; however, it is important to define this type of infection precisely. Especially in the evaluation of a febrile hospitalized patient, it is crucial to think carefully about all possible sources of infection and not to assume that UTI is the probable cause. In patients who have had urinary catheters in place for a number of days, fever, dysuria, frequency, leukocytosis, and especially flank pain or costovertebral angle tenderness are highly suggestive of bladder infection or pyelonephritis. In patients with fever but no other symptoms or signs referable to the urinary tract, one should look for ancillary findings suggestive of urinary tract involvement, such as white blood cells without epithelial cells in the urine sediment or leukocyte esterase or nitrite on urinalysis. A urine culture positive for a single organism should not be accepted as definitive evidence of UTI in an asymptomatic patient. While one might treat the febrile patient who has a positive urine culture with antibiotics, it is prudent to repeat the culture before the institution of therapy. Inability to recover any organism or the same organism on repeat culture, particularly if the patient does not respond to antibiotics, should raise questions about the validity of the diagnosis of UTI. In addition, isolation of two or more bacteria from a single specimen is most likely due to contamination unless there is reason to suspect a bladder diverticulum or a perinephric abscess.

Other Infectious Sources of Fever Several other types of infection may cause fever in the hospitalized patient and should be considered in the differential diagnosis of new

fever. In patients who have received antibiotics (even a single dose as surgical prophylaxis), antibiotic-associated diarrhea may develop. This condition is usually caused by the spore-forming organism *C. difficile*, which produces toxins that cause diarrhea. Some patients may appear quite toxic with this infection, with high fevers, leukocytosis, and profuse diarrhea. The organism is quite hardy and is difficult to eradicate from the hospital environment. The hands of hospital personnel have been implicated as a mode of transmission of this organism, as have electronic rectal thermometers. The colon may become colonized with *C. difficile* while the patient is in the hospital, but -- particularly if the patient is still taking antibiotics when sent home -- diarrhea may not develop until after discharge.

Unless patients are consuming foods from outside the hospital, food-borne diarrheal illness is uncommon among hospitalized patients. Thus, an extensive stool evaluation is generally not cost-effective in the management of these patients.

Other infections to consider in the hospitalized patient include decubitus ulcers, particularly in patients in chronic-care wards or confined to bed rest for prolonged periods, and sinusitis, especially in intubated patients.

NONINFECTIOUS SOURCES OF FEVER

A consideration of several common noninfectious causes of fever in hospitalized patients is part of a thorough evaluation of new fever. Drug treatment is the foremost noninfectious cause of fever. Drug fever may occur with or without an accompanying rash or eosinophilia and can be caused by a new medication or by medications the patient has been receiving for some time. Particular agents associated with drug fever include phenytoin, H₂blockers, procainamide, and antibiotics, most notably sulfonamides. Even drug-associated fevers can be quite high in some patients and may take up to 5 days to resolve after discontinuation of treatment with the offending agent. Other noninfectious causes of fever include phlebitis, often at the site of an old intravenous line and sometimes followed by suppurative thrombophlebitis with clots or septic emboli, and pulmonary emboli, especially in patients undergoing prolonged bed rest; prophylactic heparin or mechanical boots are often used to reduce the risk of pulmonary embolism in the latter patients. Other entities to consider include tissue necrosis following surgery, trauma, or burns; hematomas; pancreatitis; atelectasis; and acalculous cholecystitis.

CONCLUSION

The range of possibilities for the etiology of a new fever in a hospitalized patient is quite broad. An attention to detail, a careful history and physical examination, and a knowledge of the infections and organisms likely to cause nosocomial problems usually lead to an accurate diagnosis.

(Bibliography omitted in Palm version)

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136. INFECTIONS IN TRANSPLANT RECIPIENTS - Robert Finberg, Joyce Fingerroth

The evaluation of infections in transplant recipients involves consideration of both the donor and the recipient of the transplanted organ. Infections following transplantation are complicated by the use of drugs that are necessary to enhance the likelihood of survival of the transplanted organ but that also cause the host to be immunocompromised. Thus what might have been a latent or asymptomatic infection in an immunocompetent donor or in the recipient prior to therapy becomes a life-threatening problem when the recipient becomes immunosuppressed.

A variety of organisms have been transmitted by organ transplantation ([Table 136-1](#)). Careful attention to the sterility of the medium used to process the organ combined with meticulous microbiologic evaluation reduces rates of transmission of bacteria that may be present or grow in the organ culture medium. From 2% to >20% of donor kidneys are estimated to be contaminated with bacteria -- in most cases, with the organisms that colonize the skin or grow in the tissue culture medium used to bathe the donor kidney while it awaits implantation. The reported rate of bacterial contamination of transplanted bone marrow is as high as 17% but is most commonly ~1%. The use of enrichment columns and monoclonal-antibody depletion procedures results in a higher incidence of contamination. Approximately 2% of cryopreserved marrow and peripheral blood stem cells transfused as part of treatment for cancer are contaminated. In one series of patients receiving contaminated products, 14% had fever or bacteremia, but none died. Results of cultures performed at the time of cryopreservation and at the time of thawing were helpful in guiding therapy for the recipient.

In many transplantation centers, transmission of infections that may be latent or clinically inapparent in the donor organ has resulted in the development of specific donor-screening protocols. In addition to ordering serologic studies focusing on viruses such as herpes-group viruses [herpes simplex virus (HSV) 1, HSV-2], varicella-zoster virus (VZV), cytomegalovirus (CMV), human herpesvirus (HHV) 6, Epstein-Barr virus (EBV), HHV-8, hepatitis B and C viruses, and HIV and on parasites such as *Toxoplasma gondii*, clinicians caring for organ donors should consider assessing stool (for parasites) and skin testing for *Mycobacterium tuberculosis*. It is expected that the recipient will have been likewise assessed. This chapter considers aspects of infection unique to various transplantation settings.

INFECTIONS IN BONE MARROW AND HEMATOPOIETIC STEM CELL TRANSPLANT RECIPIENTS

Bone marrow or hematopoietic stem cell transplantation for either immunodeficiency or cancer results in a transient state of complete immune incompetence. Immediately after transplantation, both phagocytes and immune cells (T and B cells) are absent, and the host is extremely susceptible to infection. The reconstitution that follows transplantation has been likened to maturation of the immune system in neonates. The analogy does not entirely predict infections seen in bone marrow transplant (BMT) and hematopoietic stem cell transplant (HSCT) recipients, however, because the new marrow matures in an old host who has several latent infections already.

TIMING OF INFECTIONS

In the first month after bone marrow or hematopoietic stem cell transplantation, infectious complications are similar to those in granulocytopenic patients receiving chemotherapy for acute leukemia ([Chap. 85](#)). Because of the anticipated 1- to 4-week duration of neutropenia in this population, many centers give prophylactic antibiotics to patients upon initiation of chemotherapy. Prophylactic trimethoprim-sulfamethoxazole or ciprofloxacin decreases the incidence of gram-negative bacteremia among these patients.

In the second month after transplantation, a major concern (particularly in allogeneic [BMT/HSCT](#) recipients) is [CMV](#) disease ([Chap. 185](#)), which rarely has its onset earlier than 14 days after transplantation and may become evident up to 4 months after the procedure (most commonly at 1 to 3 months). In cases in which the donor marrow is depleted of T cells [to prevent graft-versus-host disease (GVHD) or eliminate a T cell tumor], the disease may be manifested earlier. Patients who receive ganciclovir (for prophylaxis, preemptive treatment, or treatment; see below) may develop CMV infection even later than 4 months after transplantation; treatment appears to delay the development of the normal immune response to CMV infection. Although CMV disease may present as isolated fever, cytopenia, or gastrointestinal disease, the foremost cause of death from CMV infection in this setting is pneumonia.

The diagnosis of pneumonia in [BMT/HSCT](#) recipients poses some special problems ([Table 85-5](#)). Because patients have undergone treatment with multiple chemotherapeutic agents and sometimes radiation, their differential diagnosis should include -- in addition to bacterial pneumonia -- [CMV](#) pneumonitis, pneumonia of other viral or fungal etiology, parasitic pneumonia, diffuse alveolar hemorrhage, and chemical- or radiation-associated pneumonitis. Since fungal disease and viruses such as respiratory syncytial virus (RSV), parainfluenza virus (types 1, 2, and 3), influenza A and B viruses, and adenovirus are also causes of pneumonia in this setting, it is important to diagnose CMV specifically (see below). *M. tuberculosis* has been an uncommon cause of pneumonia among BMT/HSCT recipients in western countries (<0.1 to 0.2%) but is common in Hong Kong (5.5%) and in countries where the prevalence of tuberculosis is high. The exposure history of the recipient is clearly critical in an assessment of posttransplantation infections.

Episodes of bacteremia due to encapsulated organisms and reactivation of [VZV](#) mark the late posttransplantation period (6 months after bone marrow reconstitution). Because of the high and prolonged risk of *Pneumocystis carinii* pneumonia (especially among patients being treated for hematologic malignancies), most patients should be maintained on prophylactic doses of trimethoprim-sulfamethoxazole starting 1 month after engraftment and continuing for at least 1 year. Such prophylaxis may also protect patients seropositive for *T. gondii*, which may cause pneumonia as well as central nervous system (CNS) lesions. The advantages of maintaining patients on daily trimethoprim-sulfamethoxazole for 1 year after transplantation include protection against *Listeria monocytogenes* and nocardial disease as well as late bacterial infections with pneumococci and *Haemophilus influenzae*, which are a consequence of the inability of the immature bone marrow to respond to polysaccharide antigens. In patients with [GVHD](#) who require prolonged or indefinite courses of steroids and other

immunosuppressive agents (e.g., cyclosporine, tacrolimus), there is a high risk of fungal infections (usually with *Candida* or *Aspergillus*), even after engraftment and resolution of neutropenia.

VIRAL INFECTIONS

[BMT/H SCT](#) recipients are susceptible to infection with a variety of viruses, including reactivation syndromes caused by most [HHVs](#) ([Table 136-2](#)) and infections caused by viruses that circulate in the community.

Herpes Simplex Virus Within the first 2 weeks after transplantation, most patients who are seropositive for [HSV](#)-1 excrete the virus in the oropharynx. The ability to isolate HSV declines with time. Administration of prophylactic acyclovir to seropositive [BMT/H SCT](#) recipients has been shown to reduce mucositis and prevent HSV pneumonia (a rare condition reported almost exclusively in BMT recipients). Both esophagitis (usually due to HSV-1) and anogenital disease (commonly induced by HSV-2) may be prevented with acyclovir prophylaxis. **For further discussion, see [Chap. 182](#).*

Varicella-Zoster Virus Reactivation of herpes zoster may occur within the first month but more commonly occurs several months after transplantation (see [Plate IID-37](#)). Reactivation rates are ~40% for allogeneic recipients and 25% for autologous recipients. Localized zoster can spread in an immunosuppressed patient. Fortunately, disseminated disease can usually be controlled with high doses of acyclovir. Because of the high incidence of dissemination of herpes zoster among patients with skin lesions, acyclovir is given prophylactically in some centers to prevent severe disease. Low doses of acyclovir (400 mg orally, three times daily) appear to be effective in preventing reactivation of [VZV](#). However, acyclovir also inhibits the development of VZV-specific immunity. Thus, its administration for only 6 months after transplantation does not prevent zoster from occurring when treatment is stopped. Some data suggest that administration of low doses of acyclovir for an entire year after transplantation is effective and may eliminate most cases of posttransplantation zoster. **For further discussion, see [Chap. 183](#).*

Cytomegalovirus The onset of [CMV](#) disease usually comes between 30 and 90 days after transplantation, when the granulocyte count is adequate but immunologic reconstitution has not occurred. CMV may cause interstitial pneumonia, bone marrow suppression, or graft failure. With the standard use of CMV-negative or filtered blood products, primary CMV infection should be a risk in allogeneic transplantation only when the donor is CMV-seropositive and the recipient is CMV-seronegative. Reactivation disease or superinfection with another strain from the donor is also common in CMV-positive recipients, and most seropositive patients who undergo bone marrow transplantation excrete CMV, with or without clinical findings. Serious CMV disease is much more common among allogeneic recipients and is often associated with [GVHD](#). In addition to pneumonia and marrow suppression (and, less often, graft failure), manifestations of CMV disease in [BMT/H SCT](#) recipients include fever with or without arthralgias, myalgias, and esophagitis. CMV ulcerations occur in both the lower and upper gastrointestinal tract, and it may be difficult to distinguish diarrhea due to GVHD from that due to CMV infection. The finding of CMV in the liver of a patient with GVHD

does not necessarily mean that CMV is responsible for hepatic enzyme abnormalities.

Management of [CMV](#) disease in [BMT/HSCT](#) recipients includes strategies directed at prophylaxis, suppression, preemptive therapy, or treatment. Prophylaxis results in a lower incidence of disease at the cost of treating many patients who otherwise would not require therapy. Because of the high fatality rate associated with CMV pneumonia in these patients and the difficulty of early diagnosis of CMV infection, prophylactic ganciclovir has been used in some centers and has been shown to abort CMV disease during the period of maximal vulnerability (from engraftment to day 120 after transplantation). The foremost problem with the administration of this drug relates to adverse effects, which include dose-related bone marrow suppression (thrombocytopenia, leukopenia, anemia, and pancytopenia). Because the frequency of CMV pneumonia is lower among autologous BMT recipients (2 to 7%) than among allogeneic BMT recipients (10 to 40%), prophylaxis in the former group will not become the rule until a less toxic antiviral agent becomes available.

Like prophylaxis, suppressive treatment, which targets patients with polymerase chain reaction evidence of [CMV](#) or CMV-positive urine cultures, entails the unnecessary treatment of many individuals (on the basis of a laboratory test that is not highly predictive of disease) with drugs that have adverse effects. Currently, because of the neutropenia associated with ganciclovir in [BMT/HSCT](#) recipients, a preemptive approach -- treatment of those patients in whose blood CMV is detected by an antigen or DNA test -- is used at most centers. This approach is almost as effective as prophylaxis or suppression and causes less toxicity. The use of the leukocyte antigen test for CMV disease (fluorescent staining of leukocytes for CMV antigens) allows earlier diagnosis but leads to treatment of more patients than would be treated on the basis of blood cultures, with a consequent increase in very late disease (>120 days after transplantation). The use of quantitative viral load assays, which are not dependent on circulating polymorphonuclear leukocytes, should permit early accurate diagnosis in the future.

Treatment of [CMV](#) pneumonia in [BMT/HSCT](#) recipients requires both intravenous immune globulin (IVIG) and ganciclovir. In patients who cannot tolerate ganciclovir, foscarnet is a useful alternative, although it may produce nephrotoxicity and electrolyte imbalance. Transfusion of CMV-specific T cells from the donor decreased viral load in a small series of patients; this result suggests that immunotherapy may play a role in the treatment of this disease in the future. **For further discussion, see [Chap. 185](#).*

Human Herpesviruses 6 and 7 [HHV-6](#), the cause of exanthem subitum in children ([Chap. 185](#)), is a ubiquitous herpesvirus that reactivates (as determined by culture of the virus from the blood) in ~50% of transplant recipients between 2 and 4 weeks after surgery. In some cases, reactivation of HHV-6 appears to be associated with neutropenia; since, like [CMV](#), this virus can be found in marrow cells, it is possible that HHV-6 reactivation is responsible for some of the neutropenia that follows bone marrow transplantation. Although encephalitis developing after transplantation has been associated with HHV-6 in cerebrospinal fluid (CSF), the causality of the association is not well defined. HHV-6 DNA is sometimes found in lung samples after transplantation. However, its role in pneumonitis is unclear. While HHV-6 has been shown to be sensitive to foscarnet (and in some instances to ganciclovir) in vitro, the efficacy of

antiviral treatment has not been well studied. Little is known about the related herpesvirus HHV-7 or its role in posttransplantation infection. **For further discussion, see Chap. 185.*

Epstein-Barr Virus Primary [EBV](#) infection can be fatal to transplant recipients; EBV reactivation can cause EBV-B cell lymphoproliferative disease (LPD), which may also be fatal to patients taking immunosuppressive drugs. The localization of EBV to B cells leads to several interesting phenomena in [BMT/HSCT](#) recipients. The marrow ablation that occurs as part of the BMT/HSCT procedure may eliminate latent EBV from the host. Infection can then be reacquired immediately after transplantation by transfer of infected donor B cells. Alternatively, transplantation from a seronegative donor may result in cure. The recipient is then at risk for a second primary infection.

[EBV-LPD](#) can develop in the recipient's B cells (if any should survive marrow ablation) but is more likely to be a consequence of outgrowth of infected donor cells. Both lytic and latent EBV replication are more likely during immunosuppression (e.g., they are associated with [GVHD](#) and the use of antibodies to T cells). Although less likely in autologous transplantation, reactivation can occur in T cell-depleted autologous recipients (e.g., patients being treated for a T cell lymphoma with marrow depletion using antibodies to T cells). EBV-LPD, which usually becomes apparent 1 to 3 months after engraftment, can cause high fevers and cervical adenopathy resembling the symptoms of infectious mononucleosis but more commonly presents as an extranodal mass. The incidence of 0.6% among allogeneic [BMT/HSCT](#) recipients contrasts with figures of ~5% for renal transplant recipients and up to 20% for cardiac transplant patients. In all cases, EBV-LPD is more likely to occur with continued immunosuppression (especially that caused by the use of antibodies to T cells and cyclosporine or tacrolimus).

[EBV](#)-specific T cells generated from the donor have been used experimentally to prevent and to treat EBV-[LPD](#) in the allogeneic recipient. Some studies indicate that EBV-LPD can be treated with antibodies to B cell surface antigens. Use of an anti-CD20 monoclonal antibody (Rituximab) to treat B cell lymphomas that express this surface protein has elicited some dramatic responses. Studies are in progress to assess efficacy in EBV-LPD, in which the involved B cells commonly bear CD20. The role of antivirals is uncertain because no available agents have been documented to have activity against latent EBV infection. Ganciclovir has been postulated to have activity on the basis of its ability to inhibit proliferation of B cells, but this activity is associated with toxicity. Both interferon and retinoic acid have been used in the treatment of EBV-LPD, as has [IVIG](#), but no large studies have assessed the efficacy of these agents. Chemotherapeutic regimens have been used as a last resort, even though patients' tolerance and long-term results have been disappointing in this setting. **For further discussion, see Chap. 184.*

Human Herpesvirus 8 The [EBV](#)-related gamma herpesvirus [HHV-8](#), which is causally associated with Kaposi's sarcoma, with primary effusion lymphoma, and sometimes with multicentric Castleman's disease, has rarely resulted in disease in [BMT/HSCT](#) recipients. The reasons may be a relatively low seroprevalence in the population and the limited duration of profound T cell suppression after bone marrow/hematopoietic stem cell transplantation. **For further discussion, see Chap. 185.*

Other (Nonherpes) Viruses Both [RSV](#) and parainfluenza viruses, particularly type 3, can cause severe or even fatal pneumonia in [BMT](#) recipients. Infections with both of these agents sometimes occur as disastrous nosocomial epidemics. Therapy with aerosolized ribavirin as well as RSV immunoglobulin or monoclonal antibody to RSV (Palivizumab) has been reported to lessen the severity of RSV disease, but there are no large studies to prove efficacy. Influenza is also seen in BMT recipients and generally mirrors the presence of infection in the community. Several drugs are available for the treatment of influenza (amantadine/rimantadine, ribavirin?) but have limited effects, primarily reducing symptoms and shortening the duration of illness. The newly approved neuraminidase inhibitors are active against both influenza A virus and influenza B virus. Their role in ameliorating disease in this patient population is unknown. Adenovirus can be isolated from BMT recipients at rates varying from 5 to 18%. Although hemorrhagic cystitis, pneumonia, and fatal disseminated infection have been reported, adenovirus infection, which (like [CMV](#) infection) usually occurs in the first or second month after transplantation, is often asymptomatic. Therapy with intravenous ribavirin is questionably effective. Cidofovir has proved effective in animal models and in case reports. Infections with parvovirus B19 (presenting as anemia or occasionally pancytopenia) and enteroviruses (sometimes fatal) can occur. Pleconaril, a newly developed capsid-binding agent, is being studied for treatment of enterovirus infection. Rotaviruses are a common cause of gastroenteritis in BMT/[HSCT](#) recipients. BK and, to a lesser extent, JC virus (polyomavirus hominis 1 and 2, respectively) are found in the urine of some transplant recipients. BK viruria may be associated with hemorrhagic cystitis. Progressive multifocal leukoencephalopathy caused by JC virus is rare among BMT/HSCT recipients compared with the rate among patients with impaired T cell function due to HIV infection. There is no known treatment for this disease; however, cidofovir and other agents are under study.

INFECTIONS IN SOLID ORGAN TRANSPLANT RECIPIENTS

Morbidity and mortality among solid organ transplant recipients have been reduced by the use of more effective antibiotics. The organisms that cause infections in recipients of solid organ transplants are different from those that infect [BMT/HSCT](#) recipients because solid organ recipients do not go through a period of neutropenia. As the transplantation procedure involves surgery, however, solid organ recipients are subject to infections at anastomotic sites and to wound infections. Compared with BMT/HSCT recipients, organ transplant patients are immunosuppressed for more prolonged periods (often permanently). Thus they are susceptible to the same organisms as patients with chronically impaired T cell immunity ([Chap. 85](#), especially [Table 85-1](#)).

During the early period (<1 month after transplantation), infections are most often caused by extracellular bacteria (staphylococci, streptococci, *Escherichia coli*, other gram-negative organisms), which often originate in surgical wound or anastomotic sites. The spectrum of infection is largely determined by the type of transplant.

In subsequent weeks, the consequences of the administration of agents that suppress cell-mediated immunity and of the acquisition or reactivation (from the transplanted organ) of viruses and parasites become apparent. [CMV](#) infection is often a problem in the first 6 months after transplantation and may present as severe systemic disease or as

an infection of the transplanted organ. [HHV-6](#) reactivation (assessed by blood culture) occurs within the first 2 to 4 weeks after transplantation and may be associated with fever and granulocytopenia.

[CMV](#) is associated not only with generalized immunosuppression but also with organ-specific, rejection-related syndromes: glomerulopathy in kidney transplant recipients, bronchiolitis obliterans in lung transplant recipients, vasculopathy in heart transplant recipients, and the vanishing bile duct syndrome in liver transplant recipients. A complex interplay between increased CMV replication and enhanced graft rejection is well established: Increasing immunosuppression leads to increased CMV replication, which is associated with graft rejection. For this reason, considerable attention has been focused on the diagnosis, treatment, and prophylaxis of CMV infection in organ transplant recipients.

Beyond 6 months after transplantation, infections characteristic of patients with defects in cell-mediated immunity -- e.g., infections with *Listeria*, *Nocardia*, various fungi, and other intracellular pathogens -- may be a problem. Elimination of these late infections will not be possible until specific tolerance to the transplanted organ can be achieved without the administration of drugs that lead to generalized immunosuppression. Meanwhile, vigilance, prophylaxis/preemptive therapy (when indicated), and rapid diagnosis and treatment of infections can be lifesaving in solid organ transplant recipients, who, unlike most [BMT](#) recipients, continue to be immunosuppressed.

Solid organ transplant recipients are susceptible to [EBV-LPD](#) from as early as 2 months to many years after transplantation. The prevalence of this complication is increased by potent and prolonged use of T cell-suppressive drugs. The condition may be reversed (in some cases) by decreasing the degree of immunosuppression. Among organ transplant patients, those with heart and lung transplants -- who receive the most intensive immunosuppressive regimens -- are most likely to develop EBV-LPD, particularly in the lungs. Although disease usually originates in recipient B cells, several cases of donor origin have been reported. There is a notable tendency for EBV-LPD to develop in the transplanted organ. High organ-specific content of B lymphoid tissues (i.e., bronchial-associated lymphoid tissue in the lung), anatomic factors (i.e., lack of access of host T cells to the transplanted organ because of disturbed lymphatics), and differences in major histocompatibility loci between the host T cells and the organ (i.e., lack of cell migration or lack of effective T cell/macrophage cooperation) may result in defective elimination of EBV-infected B cells.

INFECTIOUS COMPLICATIONS OF KIDNEY TRANSPLANTATION (See [Table 136-3](#))

Early Infections Infections developing soon after kidney transplantation are often caused by bacteria associated with skin or wound infections. Some data indicate a role for perioperative antibiotic prophylaxis, and many centers give cephalosporins or a penicillin with an aminoglycoside to decrease the risk of postoperative complications. Urinary tract infections developing soon after transplantation are usually related to anatomic alterations resulting from surgery. Such early infections may require prolonged treatment (e.g., 6 weeks of antibiotic administration for pyelonephritis). Urinary tract infections that occur >6 months after transplantation do not seem to be associated with the high rate of pyelonephritis or relapse seen with infections that occur in the first 3

months and may be treated for shorter periods.

Prophylaxis with trimethoprim-sulfamethoxazole [1 double-strength tablet (800 mg sulfamethoxazole, 160 mg trimethoprim) per day] for the first 4 months after transplantation decreases the incidence of early and middle-period infections (see below and [Table 136-4](#)).

Middle-Period Infections Because of continuing immunosuppression, kidney transplant recipients are predisposed to lung infections characteristic of those in patients with T cell deficiency (i.e., infections with intracellular bacteria, mycobacteria, nocardiae, fungi, viruses, and parasites). The high mortality associated with *Legionella pneumophila* infection ([Chap. 151](#)) led to the closing of renal transplant units in hospitals with endemic legionellosis.

About 50% of all renal transplant recipients presenting with fever 1 to 4 months after transplantation have evidence of [CMV](#) disease; CMV itself accounts for the fever in over two-thirds of cases and thus is the predominant pathogen during this period. CMV infection ([Chap. 185](#)) may also present as arthralgias or myalgias. During this period, this infection may represent primary disease (in the case of a seronegative recipient of a kidney from a seropositive donor) or may present as reactivation disease or superinfection. Patients may have atypical lymphocytosis. Unlike immunocompetent patients, however, they often do not have lymphadenopathy or splenomegaly. Therefore, clinical suspicion and laboratory confirmation are necessary for diagnosis. The clinical syndrome may be accompanied by bone marrow suppression (particularly leukopenia). CMV also causes glomerulopathy and is associated with an increased incidence of other opportunistic infections. Because of the frequency and severity of CMV disease, a considerable effort has been made to prevent and treat it in renal transplant recipients. Administration of an immune globulin preparation enriched with antibodies to CMV (CMV-Ig) decreases the incidence in the group at highest risk for severe infections (seronegative recipients of seropositive kidneys). Ganciclovir is useful for the treatment of serious CMV disease. One study showed a significant (50%) reduction in CMV disease and rejection at 6 months in patients who received prophylactic valganciclovir (an acyclovir congener) for the first 90 days after renal transplantation. If confirmed, these results will likely change practice.

Infection with the other herpes-group viruses may become evident within 6 months after transplantation or later. Early after transplantation, [HSV](#) may cause either oral or anogenital lesions that are usually responsive to acyclovir. Large ulcerating lesions in the anogenital area may lead to bladder and rectal dysfunction as well as predisposing to bacterial infection. [VZV](#) may cause fatal disseminated infection in nonimmune kidney transplant recipients, but in immune patients reactivation zoster usually does not disseminate outside the dermatome; thus disseminated VZV infection is a less fearsome complication in kidney transplantation than in bone marrow transplantation. [HHV-6](#) may reactivate and (although usually asymptomatic) may be associated with fever, rash, marrow suppression, or encephalitis.

[EBV](#) reactivation disease is more serious; it may present as an extranodal proliferation of B cells that invade the [CNS](#), nasopharynx, liver, small bowel, heart, and transplanted kidney. The disease is diagnosed by the finding of a proliferation of EBV-positive B

cells. The incidence of EBV-[LPD](#) is higher among patients given high doses of cyclosporine, tacrolimus, or other immunosuppressive agents (including anti-T cell antibodies). Fortunately, disease often regresses once immunocompetence is restored. [HHV-8](#) infection can be transmitted with the donor kidney and is associated with the development of Kaposi's sarcoma in the recipient. Kaposi's sarcoma (primary vs. reactivation of HHV-8) often appears within 1 year after transplantation, although the range of onset times is wide (1 month to ~20 years).

The papovaviruses BK and JC (polyomaviruses hominis 1 and 2) have been cultured from the urine of kidney transplant recipients (as they have from that of [BMT](#) recipients). The excretion of BK virus is associated with ureteral strictures and that of JC virus with progressive multifocal leukoencephalopathy (rare). Adenoviruses may persist with continued immunosuppression in these patients.

Kidney transplant recipients are also subject to infections with other intracellular organisms. These patients may develop pulmonary infections with *Nocardia*, *Aspergillus*, and *Mucor* as well as infections with other pathogens in which the T cell/macrophage axis plays an important role. In patients without intravenous catheters, *L. monocytogenes* is the most common cause of bacteremia³¹ month after renal transplantation. Kidney transplant recipients may develop *Salmonella* bacteremia, which can lead to endovascular infections and require prolonged therapy. Pulmonary infections with *P. carinii* are common unless the patient is maintained on trimethoprim-sulfamethoxazole prophylaxis. *Nocardia* infection ([Chap. 165](#)) may present in the skin, bones, lungs, or [CNS](#) (where it usually takes the form of single or multiple brain abscesses). *Nocardia* infection generally occurs ³¹ month after transplantation and may follow immunosuppressive treatment for an episode of rejection. Pulmonary findings are nonspecific: localized disease with or without cavities is most common, but the disease may disseminate. The diagnosis is made by culture of the organism from sputum or from the involved nodule. As with *P. carinii*, prophylaxis with trimethoprim-sulfamethoxazole appears to be efficacious in the prevention of disease. The occurrence of *Nocardia* infections >2 years after transplantation suggests that a long-term prophylactic regimen may be justified.

Toxoplasmosis can occur in seropositive patients, usually developing in the first few months after kidney transplantation. Again, trimethoprim-sulfamethoxazole is helpful in prevention. In endemic areas, histoplasmosis, coccidioidomycosis, and blastomycosis may cause pulmonary infiltrates or disseminated disease.

Late Infections Late infections (>6 months after kidney transplantation) include [CMV](#) retinitis and a variety of [CNS](#) complications. Patients (particularly those whose immunosuppression has been increased) are at risk for subacute meningitis due to *Cryptococcus neoformans*. Cryptococcal disease may present in an insidious manner (sometimes as a skin infection before the development of clear CNS findings). *Listeria* meningitis may have an acute presentation and requires prompt therapy to avoid a fatal outcome.

Patients who continue to take glucocorticoids are predisposed to infection. "Transplant elbow" is a recurrent bacterial infection in and around the elbow that is thought to result from a combination of poor tensile strength of the skin of steroid-treated patients and

steroid-induced proximal myopathy that requires patients to push themselves up with their elbows to get out of chairs. Bouts of cellulitis (usually caused by *Staphylococcus aureus*) recur until patients are provided with elbow protection.

Kidney transplant recipients are susceptible to invasive fungal infections -- such as those due to *Aspergillus* and *Rhizopus*, which may present as superficial lesions before dissemination. Mycobacterial infection (particularly that with *M. marinum*) can be diagnosed by skin examination. Infection with *Prototheca wickerhamii* (an achlorophyllic alga) has been diagnosed by skin biopsy. Warts caused by human papillomaviruses (HPVs) are a late consequence of persistent immunosuppression; local therapy is usually satisfactory.

HEART TRANSPLANTATION

Early Infections Sternal wound infection and mediastinitis are early complications of heart transplantation. An indolent course is common, with fever or a mildly elevated white blood cell count preceding the development of site tenderness or drainage. Clinical suspicion based on evidence of sternal instability and failure to heal may lead to the diagnosis. Although common residents of the skin (e.g., *S. aureus* and *S. epidermidis*) as well as gram-negative organisms (e.g., *Pseudomonas aeruginosa*) and fungi (e.g., *Candida*) are often involved, mediastinitis in these patients (in rare cases) can also be due to *Mycoplasma hominis* ([Chap. 178](#)). Since this organism requires an anaerobic environment for growth and may be difficult to see on conventional medium, the laboratory should be alerted that *M. hominis* infection is suspected. *M. hominis* mediastinitis has been cured with a combination of surgical debridement (sometimes requiring muscle-flap placement) plus clindamycin and tetracycline. Organisms associated with mediastinitis may be cultured from accompanying pericardial fluid.

Middle-Period Infections *T. gondii* ([Chap. 217](#)) resident in the heart of a seropositive donor may be transmitted to a seronegative recipient. Thus serologic screening for *T. gondii* infection is important before and in the months after cardiac transplantation. Rarely, active disease can be introduced at the time of transplantation. The overall incidence of toxoplasmosis is so high in this setting that some prophylaxis is warranted. Although alternatives are available, the most frequently used agent is trimethoprim-sulfamethoxazole, which prevents infection with *Pneumocystis*, *Nocardia*, and other bacterial pathogens. [CMV](#) has also been transmitted by heart transplantation. [CNS](#) infections can be caused by *Toxoplasma*, *Nocardia*, and *Aspergillus*. *L. monocytogenes* meningitis should be considered in heart transplant recipients with fever and headache.

[CMV](#) infection is associated with poor outcomes after heart transplantation. The virus is usually cultivable 1 to 2 months after transplantation, causes manifestations (usually fever and atypical lymphocytosis, often associated with leukopenia and thrombocytopenia) at 2 to 3 months, and produces severe disease (e.g., pneumonia) at 3 to 4 months. Seropositive recipients usually develop cultivable virus faster than patients whose primary CMV infection is a consequence of transplantation. Between 40 and 70% of patients develop symptomatic CMV disease in the form of (1) CMV pneumonia, the most likely form of CMV disease to be fatal; (2) CMV esophagitis and gastritis, sometimes accompanied by abdominal pain with or without ulcerations and

bleeding; and (3) the CMV syndrome consisting of CMV in the blood with fever, leukopenia, thrombocytopenia, and hepatic enzyme abnormalities. Ganciclovir is efficacious in the treatment of CMV infection; prophylaxis with ganciclovir or possibly with other antivirals, as described for renal transplantation, may reduce the incidence of CMV-related disease.

Late Infections [EBV](#) infection usually presents as a lymphoma-like proliferation of B cells late after heart transplantation, particularly in patients maintained on heavy immunosuppression. A subset of heart and heart-lung transplant recipients may develop early (within 2 months) fulminant EBV-[LPD](#). Treatment includes the reduction of immunosuppression if possible and the consideration of B cell antibodies (Rituximab), immunomodulatory agents, or chemotherapy, as discussed earlier under bone marrow/hematopoietic stem cell transplantation. [HHV-8](#)-associated disease, including primary effusion lymphoma, has been reported in heart transplant recipients. Prophylaxis for *P. carinii* infection is required for these patients (see below).

LUNG TRANSPLANTATION

Early Infections It is not surprising that lung transplants are predisposed to the development of pneumonia. The combination of ischemia and the resulting mucosal damage together with accompanying denervation and lack of lymph drainage probably contributes to the high rate of pneumonia (66% in one series). The prophylactic use of high doses of broad-spectrum antibiotics for the first 3 or 4 days after surgery decreases the incidence of pneumonia. Gram-negative pathogens (Enterobacteriaceae and *Pseudomonas* species) are troublesome in the first 2 weeks after surgery (the period of maximal vulnerability). Pneumonia can also be caused by *Candida* (possibly as a result of colonization of the donor lung), *Aspergillus*, and *Cryptococcus*.

Mediastinitis may occur at an even higher rate among lung transplant recipients than among heart transplant recipients and most commonly develops within 2 weeks of surgery. Pneumonitis due to [CMV](#) (which may be transmitted as a consequence of transplantation) usually presents between 2 weeks and 3 months after surgery, with primary disease occurring later than reactivation disease.

Middle-Period Infections The incidence of [CMV](#) infection, either reactivated or primary, is between 75 and 100% if either the donor or the recipient is seropositive for CMV. CMV-induced disease appears to be most severe in recipients of lung and heart-lung transplants. Whether this severity relates to the mismatch in lung antigen-presenting and host immune cells or is attributable to other (nonimmune) factors is not known. More than half of lung transplant recipients with symptomatic CMV disease have pneumonia. Difficulty in distinguishing the radiographic picture of CMV infection from organ rejection further complicates therapy. CMV can also cause bronchiolitis obliterans in lung transplants. The development of pneumonitis related to [HSV](#) has led to the prophylactic use of acyclovir. Such prophylaxis may also decrease rates of CMV disease, but ganciclovir is more active against CMV and is also active against HSV. Ganciclovir prophylaxis for CMV disease in lung transplant recipients is recommended.

Late Infections The incidence of *P. carinii* infection (which may present with a paucity of findings) is high among lung and heart-lung transplant recipients. Some form of

prophylaxis for *P. carinii* pneumonia is indicated in all organ transplant situations ([Tables 136-4](#) and [136-5](#)). Trimethoprim-sulfamethoxazole prophylaxis for 12 months after transplantation may be sufficient to prevent *P. carinii* disease in patients whose degree of immunosuppression is not increased.

As in other transplant recipients, infection with [EBV](#) may cause either a mononucleosis-like syndrome or [LPD](#). The tendency of the B cell blasts to present in the lung appears to be greater after lung transplantation than after the transplantation of other organs. Reduction of immunosuppression causes remission in some cases, but airway compression can be fatal and more rapid intervention may therefore become necessary. The approach to EBV-LPD is similar to that described in other sections.

LIVER TRANSPLANTATION

Early Infections As in other types of transplantation, early bacterial infections are a major problem after liver transplantation. Many centers administer systemic broad-spectrum antibiotics for the first 5 days after surgery, even in the absence of documented infection. However, despite prophylaxis, infectious complications are common and are correlated with the duration of the surgical procedure and the type of biliary drainage. An operation lasting >12 h is associated with an increased likelihood of infection. Patients who have a choledochojejunostomy with drainage of the biliary duct to a Roux-en-Y jejunal bowel loop have more fungal infections than those whose bile is drained via a choledochocholedochostomy with anastomosis of the donor common bile duct to the recipient common bile duct.

Peritonitis and intraabdominal abscesses are common complications of liver transplantation. Bacterial peritonitis may result from biliary leaks and primary or secondary infection after leakage of bile. Peritonitis in liver transplant recipients is often polymicrobial, commonly involving enterococci, aerobic gram-negative bacteria, staphylococci, anaerobes, or *Candida*. Only one-third of patients with intraabdominal abscesses have bacteremia. Abscesses within the first month after surgery may occur not only over the liver but also in the spleen, pericolic area, and pelvis. Treatment includes antibiotic administration and drainage as necessary.

Liver transplant patients have a high incidence of fungal infections, and the occurrence of fungal infection (often candidiasis) correlates with preoperative use of glucocorticoids, a long duration of treatment with antibacterial agents, and posttransplantation use of immunosuppressive agents.

Middle-Period Infections The development of postsurgical biliary stricture predisposes patients to cholangitis. These patients may lack the characteristic signs and symptoms of cholangitis: fever, abdominal pain, and jaundice. Alternatively, these findings may be present but may suggest graft rejection. The diagnosis of cholangitis in liver transplant recipients therefore requires documentation of bacteremia or demonstration of aggregated neutrophils in bile duct biopsy specimens. Unfortunately, invasive studies of the biliary tract (either T-tube cholangiography or endoscopic retrograde cholangiopancreatography) may themselves lead to cholangitis. For this reason, many clinicians recommend prophylaxis with antibiotics covering gram-negative organisms and anaerobes when these procedures are performed in liver transplant recipients.

Viral hepatitis is a common complication of liver transplantation ([Chap. 295](#)). Reactivation of hepatitis B and C infections, for which transplantation may be performed, is problematic. To prevent hepatitis B infection, high-dose intravenous hepatitis B immune globulin is often administered. The long-term efficacy of lamivudine (3TC) in inhibiting hepatitis B viral replication after transplantation is being studied. A combination of interferon α and ribavirin is being tested for treatment/prophylaxis of hepatitis C infection.

As in other transplantation settings, reactivation disease with herpes-group viruses is common ([Table 136-2](#)). Herpesviruses can be transmitted in donor organs. Although CMV hepatitis occurs in ~4% of liver transplant recipients, it is usually not so severe as to require retransplantation. CMV disease develops in the majority of seronegative recipients of organs from CMV-positive donors, but fatality rates are lower in liver transplant recipients than in lung or heart-lung transplant recipients. Disease due to CMV is associated with the vanishing bile duct syndrome after liver transplantation. Patients respond to treatment with ganciclovir; prophylaxis with CMV immune globulin and acyclovir or oral ganciclovir may modify disease. A role for HHV-6 in posttransplantation fever and leukopenia has been proposed. EBV-LPD after liver transplantation shows a propensity for involvement of the liver, and such disease may be of donor origin.

PANCREAS TRANSPLANTATION

Transplantation of the pancreas is complicated by early abdominal infection in ~20 to 40% of cases. To prevent contamination of the allograft with enteric bacteria and yeasts, some surgeons, instead of draining the pancreas through the bowel, drain secretions into the urinary tract or bladder. A cuff of duodenum is often used in the anastomosis between the pancreatic graft and the bladder. In addition to bicarbonate loss, this technique causes a high rate of urinary tract infection (30 to 40%) and sterile cystitis. Over the long term, bowel drainage is better tolerated. An alternative method -- the transplantation of islet cells only -- may eliminate the problems characteristically posed by wound and urinary tract sepsis in pancreas transplant recipients.

Issues related to the development of CMV infection, EBV-LPD, and infections with opportunistic pathogens in patients receiving a pancreas are similar to those in other solid organ transplant recipients.

MISCELLANEOUS INFECTIONS IN SOLID ORGAN TRANSPLANTATION

Indwelling Intravenous Catheter Infections The prolonged use of indwelling intravenous catheters for administration of medication, blood products, and nutrition is common in diverse transplantation settings and poses a risk of local and bloodstream infection. Significant insertion-site infection is most commonly caused by *S. aureus*. Bloodstream infection most frequently develops within a week of catheter placement or in patients who become neutropenic. Coagulase-negative staphylococci are the most common isolates from the blood. **For further discussion of differential diagnosis and therapeutic options, see [Chap. 85](#).*

Tuberculosis The incidence of tuberculosis occurring within 12 months after solid organ transplantation ranges broadly worldwide (0.35 to 15%), reflecting prevalences in local populations. Nonrenal transplantation, [GVHD](#) within 6 months, and intensity of immunosuppression are predictive of tuberculosis reactivation and development of disseminated disease in a host with latent disease. Tuberculosis has rarely been transmitted from the donor organ. In contrast to the low mortality in [BMT/HSCT](#) recipients, mortality in solid organ transplant patients is reported to be 29%. Isoniazid toxicity has not been a significant problem except in the liver transplantation setting.

Virus-Associated Malignancies In addition to malignancy associated with gammaherpesvirus infection ([EBV](#), [HHV-8](#)) and simple warts ([HPV](#)), transplant recipients, particularly those who require long-term immunosuppression, are more likely than the general population to develop tumors that are virus-associated or suspected of being virus-associated. The interval to tumor development is usually >1 year. Transplant recipients develop nonmelanoma skin or lip cancers that, in contrast to de novo skin cancers, have a high squamous cell-to-basal cell ratio. Whether HPV plays a major role in these lesions is being investigated. Cervical and vulvar carcinomas, quite clearly associated with HPV, develop with increased frequency in female transplant recipients. In renal transplant recipients, rates of melanoma are modestly increased and rates of cancers of the kidney and bladder are increased.

VACCINATION OF TRANSPLANT RECIPIENTS

In addition to receiving antibiotic prophylaxis, transplant recipients should be vaccinated against likely pathogens ([Table 136-6](#)). In the case of [BMT](#) recipients, optimal responses cannot be achieved until after reconstitution, despite previous immunization of both donor and recipient. Recipients of allogeneic BMTs must be reimmunized if they are to be protected against pathogens. The situation is less clear-cut in the case of autologous transplantation. T and B cells in the peripheral blood may reconstitute the response if they are transferred in adequate numbers. However, cancer patients (particularly those with Hodgkin's disease, in whom vaccination has been extensively studied) who are undergoing chemotherapy do not respond normally to immunization, and titers of antibodies to infectious agents fall more rapidly than in healthy individuals. Therefore, even immunosuppressed patients who have not had marrow transplants may need booster vaccine injections. If memory cells are specifically eliminated as part of a marrow "cleanup" procedure, it will be necessary to reimmunize the recipient with a new primary series. Optimal times for immunizations of different transplant populations are being evaluated. Immunization of household and other contacts (including health care personnel) against influenza every season is likely to benefit the patient by preventing local spread.

In the absence of compelling data as to optimal timing, it is reasonable to administer the pneumococcal and *H. influenzae* type b conjugate vaccines to both autologous and allogeneic [BMT](#) recipients 12 months after transplantation and again 12 months later (since the response to the initial vaccine dose is weak in the early posttransplantation period). These two vaccines are particularly important for patients who have undergone splenectomy. In addition, *Neisseria meningitidis* polysaccharide vaccine, diphtheria vaccine, tetanus vaccine, and inactivated polio vaccine can all be given at these same

intervals (12 and 24 months after transplantation). Some authorities recommend a new primary series for tetanus/diphtheria and inactivated polio vaccine (vaccination 12, 14, and 16 months after transplantation). Because of the risk of spread, household contacts of BMT recipients (or of patients immunosuppressed as a result of chemotherapy) should receive only inactivated polio vaccine. Live-virus measles/mumps/rubella (MMR) vaccine can be given to autologous BMT recipients 24 months after transplantation and to most allogeneic BMT recipients at the same point if they are not receiving maintenance therapy with immunosuppressive drugs and do not have ongoing [GVHD](#). The risk of spread from a household contact is lower for MMR than for polio vaccine. In patients who have active GVHD and/or are taking high maintenance doses of glucocorticoids, it may be prudent to avoid all live-virus vaccines. In the absence of detectable antibody titers, vaccination to prevent hepatitis B and hepatitis A also seems advisable.

In the case of solid organ transplant recipients, administration of all the usual vaccines and of the indicated booster doses should be completed before immunosuppression, if possible, to maximize responses. For patients taking immunosuppressive agents, the administration of pneumococcal vaccine should be repeated every 5 years. No data are available for meningococcal polysaccharide vaccine, but it is probably reasonable to administer it along with the pneumococcal vaccine or more frequently (every 3 years for persons with significant exposure risk). *H. influenzae* conjugate vaccine is safe and should be efficacious in this population; therefore, its administration before transplantation is recommended. Booster doses of this vaccine are not recommended for adults. Solid organ transplant recipients who continue to receive immunosuppressive drugs (glucocorticoids, cyclosporine) should not receive live-virus vaccines. A person in this group exposed to measles should be given immune globulin. Similarly, an immunocompromised patient who is seronegative for varicella and who comes into contact with a person who has chickenpox should be given varicella-zoster immune globulin as soon as possible (and certainly within 96 h) or, if this is not possible, started immediately on a 10- to 14-day course of acyclovir therapy. Susceptible household contacts of transplant recipients should receive live attenuated [VZV](#) vaccine, but vaccinees should avoid direct contact with the patient if a rash develops.

Immunocompromised patients who travel may benefit from some but not all vaccines. In general, they should receive any killed or inactivated vaccine preparation appropriate to the area they are visiting; this recommendation includes the vaccines for Japanese encephalitis, hepatitis A and B, poliomyelitis, meningococcal infection, and typhoid. The live typhoid vaccines are not recommended for use in most immunocompromised patients, but inactivated typhoid or the purified polysaccharide vaccine can be used. Live yellow fever vaccine should not be administered. Phenol-inactivated cholera vaccine is probably of little use in this setting. On the other hand, immunization with the purified-protein hepatitis B vaccine is indicated if patients are likely to be exposed. Patients who will reside for >6 months in areas where hepatitis B is common (Africa, Southeast Asia, the Middle East, Eastern Europe, parts of South America, and the Caribbean) should receive hepatitis B vaccine. Inactivated hepatitis A vaccine should be used in the appropriate setting ([Chap. 122](#)). If hepatitis A vaccine is not administered, travelers should consider receiving passive protection with immune globulin (the dose depending on the duration of travel in the high-risk area).

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SECTION 4 -APPROACH TO THERAPY FOR BACTERIAL DISEASES

137. TREATMENT AND PROPHYLAXIS OF BACTERIAL INFECTIONS - *Gordon L. Archer, Ronald E. Polk*

The development of drugs that prevent and cure bacterial infections is one of the twentieth century's major contributions to human longevity and quality of life. Antibacterial agents are among the most commonly prescribed drugs of any kind worldwide. Used appropriately, these drugs are lifesaving. However, their indiscriminate use drives up the cost of health care, leads to a plethora of side effects and drug interactions, and fosters the emergence of bacterial resistance, rendering previously valuable drugs useless. The rational use of antibacterial agents depends on an understanding of their mechanisms of action, pharmacokinetics, toxicities, and interactions; bacterial strategies for resistance; and bacterial susceptibility in vitro. In addition, patient-associated parameters, such as the site of infection and the immune and excretory status of the host, are critically important to appropriate therapeutic decisions. This chapter provides specific data required for making an informed choice of antibacterial agent.

MECHANISMS OF ACTION

Antibacterial agents, like all antimicrobial drugs, are directed against unique targets not present in mammalian cells. The goal is to limit toxicity to the host and maximize chemotherapeutic activity affecting invading microbes only. The mechanisms of action of the antibacterial agents to be discussed in this section are summarized in [Table 137-1](#) and are depicted in [Fig. 137-1](#).

INHIBITION OF CELL-WALL SYNTHESIS

One major difference between bacterial and mammalian cells is the presence in bacteria of a rigid wall external to the cell membrane. The wall protects bacterial cells from osmotic rupture, which would result from the cell's usual marked hyperosmolarity (by up to 20 atm) relative to the host environment. The structure conferring cell-wall rigidity and resistance to osmotic lysis in both gram-positive and -negative bacteria is peptidoglycan, a large, covalently linked sacculus that surrounds the bacterium. In gram-positive bacteria, peptidoglycan is the only layered structure external to the cell membrane and is thick (20 to 80 nm); in gram-negative bacteria, there is an outer membrane external to a very thin (1-nm) peptidoglycan layer.

Chemotherapeutic agents directed at any stage of the synthesis, export, assembly, or cross-linking of peptidoglycan lead to inhibition of bacterial cell growth and, in most cases, to cell death. Peptidoglycan is composed of (1) a backbone of two alternating sugars, *N*-acetylglucosamine and *N*-acetylmuramic acid; (2) a chain of four amino acids that extends down from the backbone (stem peptides); and (3) a peptide bridge that cross-links the peptide chains. Peptidoglycan is formed by the addition of subunits (a sugar with its five attached amino acids) that are assembled in the cytoplasm and transported through the cytoplasmic membrane to the cell surface. Subsequent cross-linking is driven by cleavage of the terminal stem-peptide amino acid. Antibacterial agents act to inhibit cell-wall synthesis in several ways, as described below.

Bacitracin, a cyclic peptide antibiotic, inhibits the conversion to its active form of the lipid carrier that moves the water-soluble cytoplasmic peptidoglycan subunits through the cell membrane to the cell exterior. Cell-wall subunits accumulate in the cytoplasm and cannot be added to the growing peptidoglycan chain.

Glycopeptides (vancomycin and teicoplanin) are high-molecular-weight antibiotics that bind to the terminal D-alanine-D-alanine component of the stem peptide while the subunits are external to the cell membrane but still linked to the lipid carrier. This binding sterically inhibits the addition of subunits to the peptidoglycan backbone.

b-Lactam antibiotics (penicillins, cephalosporins, carbapenems, and monobactams; [Table 137-2](#)), characterized by a four-membered β -lactam ring, prevent the cross-linking reaction called *transpeptidation*. The energy for attaching a peptide cross-bridge from the stem peptide of one peptidoglycan subunit to another is derived from the cleavage of a terminal D-alanine residue from the subunit stem peptide. The cross-bridge amino acid is then attached to the penultimate D-alanine by transpeptidase enzymes. The β -lactam ring of the antibiotic forms an irreversible covalent acyl bond with the transpeptidase enzyme (probably because of the antibiotic's steric similarity to the enzyme's D-alanine-D-alanine target), preventing the cross-linking reaction. Transpeptidases and similar enzymes involved in cross-linking are called *penicillin-binding proteins* (PBPs) because they all have active sites that bind β -lactam antibiotics.

Virtually all the antibiotics that inhibit bacterial cell-wall synthesis are bactericidal. That is, they eventually result in the cell's death due to osmotic lysis. However, much of the loss of cell-wall integrity following treatment with cell wall-active agents is due to the bacteria's own cell-wall remodeling enzymes (autolysins) that cleave peptidoglycan bonds in the normal course of cell growth. In the presence of antibacterial agents that inhibit cell-wall growth, autolysis proceeds without normal cell-wall repair; weakness and eventual cellular lysis occur.

INHIBITION OF PROTEIN SYNTHESIS

Most of the antibacterial agents that inhibit protein synthesis interact with the bacterial ribosome. The difference between the composition of bacterial and mammalian ribosomes gives these compounds their selectivity.

Aminoglycosides (gentamicin, kanamycin, tobramycin, streptomycin, netilmicin, neomycin, and amikacin) are a group of structurally related compounds containing three linked hexose sugars. They exert a bactericidal effect by binding irreversibly to the 30S subunit of the bacterial ribosome and blocking initiation of protein synthesis. The reason for the lethal effect of aminoglycosides (as opposed to the largely bacteriostatic effect of other protein synthesis-inhibiting antibacterial drugs, including the macrolides, the lincosamides, chloramphenicol, and tetracycline) is not completely understood. Uptake of aminoglycosides and their penetration through the cell membrane constitute an aerobic, energy-dependent process. Thus, aminoglycoside activity is markedly reduced in an anaerobic environment. *Spectinomycin*, an aminocyclitol antibiotic, also acts on the 30S ribosomal subunit but has a different mechanism of action from the

aminoglycosides and is bacteriostatic rather than bactericidal.

Macrolides (erythromycin, clarithromycin, and azithromycin) consist of a large lactone ring to which sugars are attached. They bind specifically to the 50S portion of the bacterial ribosome. After attachment of mRNA to the initiation site of the 30S ribosomal subunit (the process blocked by aminoglycosides), the 50S subunit becomes bound to the 30S component to form the 70S ribosomal complex, and protein chain elongation proceeds. Binding of macrolides to the 50S ribosomal subunit inhibits protein chain elongation.

Lincosamides (clindamycin and lincomycin), although structurally unrelated to macrolides, bind to a site on the 50S ribosome nearly identical to the binding site for macrolides. Although the mechanism and site of action of macrolides and lincosamides are similar, the number and types of bacteria against which these two groups of agents are active differ.

Chloramphenicol consists of a single aromatic ring and a short side chain. This antibiotic binds reversibly to the 50S portion of the bacterial ribosome at a site close to but not identical with the binding sites of the macrolides and lincosamides. The ribosomal binding of chloramphenicol inhibits peptide bond formation.

Tetracyclines (tetracycline, doxycycline, and minocycline) consist of four aromatic rings with various substituent groups. They interact reversibly with the bacterial 30S ribosomal subunit, blocking the binding of aminoacyl tRNA to the mRNA-ribosome complex. This mechanism is markedly different from that of the aminoglycosides, which also bind to the 30S subunit. The specificity of tetracyclines for bacteria depends both on their selectivity for bacterial ribosomes and on their requirement for active, energy-dependent transport into the bacterial cell by a system not found in mammalian cell membranes.

Mupirocin (pseudomonic acid) is produced by the bacterium *Pseudomonas fluorescens*. Its mechanism of action is unique in that it inhibits the enzyme isoleucine tRNA synthetase by competing with bacterial isoleucine for its binding site on the enzyme. Inhibition of this enzyme depletes cellular stores of isoleucine-charged tRNA and therefore leads to a cessation of protein synthesis. The antibiotic is selective for bacteria because mammalian isoleucine tRNA synthetase lacks affinity for the compound.

INHIBITION OF BACTERIAL METABOLISM

The *antimetabolites* are all synthetic compounds that interfere with bacterial synthesis of folic acid. Products of the folic acid synthesis pathway function as coenzymes for the one-carbon transfer reactions that are essential for the synthesis of thymidine, all purines, and several amino acids. Inhibition of folate synthesis leads to cessation of bacterial cell growth and, in some cases, to bacterial cell death. The principal antibacterial antimetabolites are sulfonamides (sulfisoxazole, sulfadiazine, and sulfamethoxazole) and trimethoprim.

Sulfonamides are structural analogues of *p*-aminobenzoic acid (PABA), one of the three structural components of folic acid (the other two being pteridine and glutamate). The

first step in the synthesis of folic acid is the addition of PABA to pteridine by the enzyme dihydropteroic acid synthetase. Sulfonamides compete with PABA as substrates for the enzyme. The selective effect of sulfonamides is due to the fact that bacteria synthesize folic acid, while mammalian cells cannot synthesize the cofactor and must have exogenous supplies. However, the activity of sulfonamides can be greatly reduced by the presence of excess PABA or by the exogenous addition of end products of one-carbon transfer reactions (e.g., thymidine and purines). High concentrations of the latter substances may be present in some infections as a result of tissue and white cell breakdown, compromising sulfonamide activity.

Trimethoprim is a diaminopyrimidine, a structural analogue of the pteridine moiety of folic acid. It is a competitive inhibitor of dihydrofolate reductase; this enzyme is responsible for reduction of dihydrofolic acid to tetrahydrofolic acid, the essential final component in the folic acid synthesis pathway that is necessary for all one-carbon transfer reactions. Like the sulfonamides, trimethoprim is bactericidal in the absence of thymine but is only bacteriostatic when this pyrimidine is present in high concentration. The selective antibacterial activity of trimethoprim is based on the extreme sensitivity of bacterial dihydrofolate reductase to inhibition by this drug in comparison with the mammalian enzyme. The bacterial enzyme is approximately 50,000 times more sensitive to such inhibition.

INHIBITION OF NUCLEIC ACID SYNTHESIS OR ACTIVITY

Numerous antibacterial compounds have disparate effects on nucleic acids. The *quinolones*, including nalidixic acid and its fluorinated derivatives (norfloxacin, ciprofloxacin, ofloxacin, levofloxacin, sparfloxacin, grepafloxacin, and trovafloxacin), are synthetic compounds that inhibit the activity of the A subunit of the bacterial enzyme DNA gyrase as well as topoisomerase IV. DNA gyrase and topoisomerases are responsible for negative supercoiling of DNA, an essential conformation for DNA replication in the intact cell. Inhibition of the activity of DNA gyrase and topoisomerase IV is lethal to bacterial cells. The antibiotic *novobiocin* also interferes with the activity of DNA gyrase, but it interferes with the B subunit.

Rifampin, used primarily against *Mycobacterium tuberculosis*, is also active against a variety of other bacteria. Rifampin binds tightly to the B subunit of bacterial DNA-dependent RNA polymerase, thus inhibiting transcription of DNA into RNA. Mammalian-cell RNA polymerase is not sensitive to the compound.

Nitrofurantoin, a synthetic compound, causes DNA damage. The nitrofurans, compounds containing a single five-membered ring, are reduced by a bacterial enzyme to highly reactive, short-lived intermediates that are thought to cause DNA strand breakage, either directly or indirectly.

Metronidazole, a synthetic imidazole, is active against a wide range of anaerobic bacteria and protozoa. This activity is totally dependent on the organism's anaerobic electron-transport system for energy production. In the presence of this system, the nitro group of metronidazole is reduced to a series of transiently produced, reactive intermediates that are thought to cause DNA damage. The unique redox system of anaerobes accounts for the selective antibacterial activity of metronidazole. This

compound is also a mutagen and a radiosensitizer of hypoxic mammalian cells.

ALTERATION OF CELL-MEMBRANE PERMEABILITY

The *polymyxins* (polymyxin B and colistin, or polymyxin E) are cyclic, basic polypeptides. They behave as cationic, surface-active compounds that disrupt the permeability of both the outer and the cytoplasmic membranes of gram-negative bacteria.

Gramicidin A is a polypeptide of 15 amino acids that acts as an ionophore, forming pores or channels in lipid bilayers.

MECHANISMS OF RESISTANCE

Some bacteria have *intrinsic resistance* to certain classes of antibacterial agents (e.g., obligate anaerobic bacteria to aminoglycosides and gram-negative bacteria to vancomycin). Clearly these agents can never be used alone in the treatment of infections caused by resistant bacteria. In addition, bacteria that are ordinarily susceptible to antibacterial agents can acquire resistance. *Acquired resistance* is one of the major limitations to effective antibacterial chemotherapy. Resistance can develop by mutation of resident genes or by acquisition of new genes. New genes mediating resistance are usually spread from cell to cell by way of mobile genetic elements such as plasmids, transposons, and bacteriophages. The resistant bacterial populations flourish in areas of high antimicrobial use, where they enjoy a selective advantage over susceptible populations.

The major mechanisms used by bacteria to resist the action of antimicrobial agents are inactivation of the compound, alteration or overproduction of the antibacterial target through mutation of the target protein's gene, acquisition of a new gene that encodes a drug-insensitive target, decreased permeability of the cell envelope to the agent, and active elimination of the compound from the periplasm or interior of the cell. Specific mechanisms of bacterial resistance to the major antibacterial agents are outlined below, summarized in [Table 137-1](#), and depicted in [Fig. 137-1](#).

b-LACTAMS

Bacteria develop resistance to b-lactam antibiotics by a variety of mechanisms. Most common is the destruction of the drug by b-lactamases. The b-lactamases of gram-negative bacteria are confined to the periplasm, between the inner and outer membranes, while gram-positive bacteria secrete their b-lactamases into the surrounding medium. These enzymes have a higher affinity for the antibiotic than the antibiotic has for its target. Binding results in hydrolysis of the b-lactam ring. Genes encoding b-lactamases have been found in both chromosomal and extrachromosomal locations and in both gram-positive and -negative bacteria; these genes are often on mobile genetic elements. One strategy that has been devised for circumventing resistance mediated by b-lactamases is to combine the susceptible b-lactam with an inhibitor that avidly binds the inactivating enzyme, preventing its attack on the antibiotic. Unfortunately, the inhibitors (e.g., clavulanic acid, sulbactam, and tazobactam) do not bind all classes of b-lactamase and thus cannot be depended on to prevent the

inactivation of b-lactam antibiotics by such enzymes. No b-lactam antibiotic or inhibitor has been produced that can resist all of the many b-lactamases that have been identified.

A second mechanism of bacterial resistance to b-lactam antibiotics is an alteration in PBP targets so that the PBPs have a markedly reduced affinity for the drug. While this alteration may occur by mutation of existing genes, the acquisition of new PBP genes (as in staphylococcal resistance to methicillin) or of new pieces of PBP genes (as in streptococcal, gonococcal, and meningococcal resistance to penicillin) is more important.

A final resistance mechanism is the coupling, in gram-negative bacteria, of a decrease in outer-membrane permeability with rapid efflux of the antibiotic from the periplasm to the cell exterior. Mutations of genes encoding outer-membrane proteins called *porins* decrease the entry of b-lactam antibiotics into the cell, while additional proteins form channels that actively pump b-lactams out of the cell. Resistance of Enterobacteriaceae to some cephalosporins and resistance of *Pseudomonas* spp. to cephalosporins and ureidopenicillins are the best examples of this mechanism.

VANCOMYCIN

Clinically important resistance to vancomycin was first described among enterococci in France in 1988. Vancomycin-resistant enterococci have subsequently become disseminated worldwide. The genes encoding resistance are carried on plasmids that can transfer themselves from cell to cell. Resistance is mediated by enzymes that substitute D-lactate for D-alanine on the peptidoglycan stem peptide so that there is no longer an appropriate target for vancomycin binding. This alteration does not appear to affect cell-wall integrity, however. This type of acquired vancomycin resistance is so far confined to enterococci and is seen in *Enterococcus faecium* rather than in the more common pathogen *E. faecalis*. Most clinically important staphylococci (i.e., *Staphylococcus aureus* and *S. epidermidis*) remain susceptible. However, in 1996, an isolate of *S. aureus* recovered from an infected patient in Japan was shown to be eight times less susceptible to vancomycin than were usual isolates. Since that report, an additional three *S. aureus* isolates with intermediate susceptibility to vancomycin have been recovered from infected patients in the United States, as have numerous coagulase-negative staphylococci with reduced vancomycin susceptibility. These isolates have not acquired the genes that mediate vancomycin resistance in enterococci but are mutant bacteria with markedly thickened cell walls. These mutants were apparently selected in patients who were undergoing prolonged vancomycin therapy.

AMINOGLYCOSIDES

The most common aminoglycoside resistance mechanism is inactivation of the antibiotic. Aminoglycoside-modifying enzymes, usually encoded on plasmids, transfer phosphate, adenylyl, or acetyl residues from intracellular molecules to hydroxyl or amino side groups on the antibiotic. The modified antibiotic is less active because of diminished binding to its ribosomal target. Modifying enzymes that can inactivate any of the available aminoglycosides have been found in both gram-positive and -negative bacteria.

A second aminoglycoside resistance mechanism that has been identified predominantly in clinical isolates of *Pseudomonas aeruginosa* is decreased antibiotic uptake, presumably due to alterations in the bacterial outer membrane.

MACROLIDES AND LINCOSAMIDES

Resistance in gram-positive bacteria, the usual target organisms for macrolides and lincosamides, is due to the production of an enzyme -- most commonly plasmid-encoded -- that methylates ribosomal RNA, interfering with binding of the antibiotics to their target. Methylation mediates resistance to erythromycin, clarithromycin, azithromycin, and clindamycin. Streptococci can also actively efflux these compounds.

CHLORAMPHENICOL

Most bacteria resistant to chloramphenicol produce a plasmid-encoded enzyme, chloramphenicol acetyltransferase, that inactivates the compound by acetylation.

TETRACYCLINES

The most common mechanism of tetracycline resistance in gram-negative bacteria is a plasmid-encoded active-efflux pump that is inserted into the cytoplasmic membrane and extrudes antibiotic from the cell. Resistance in gram-positive bacteria is due either to active efflux or to ribosomal alterations that diminish binding of the antibiotic to its target. Genes involved in ribosomal protection are found on mobile genetic elements.

MUPIROCIN

Although the topical compound mupirocin was relatively recently introduced into clinical use, resistance is already becoming widespread in some areas. The mechanism appears to be either mutation of the target isoleucine tRNA synthetase so that it is no longer inhibited by the antibiotic or plasmid-encoded production of a form of the target enzyme that binds mupirocin poorly.

TRIMETHOPRIM AND SULFONAMIDES

The most prevalent mechanism of resistance to trimethoprim and the sulfonamides in both gram-positive and -negative bacteria is the acquisition of plasmid-encoded genes that produce a new, drug-insensitive target -- specifically, an insensitive dihydrofolate reductase for trimethoprim and an altered dihydropteroate synthetase for sulfonamides.

QUINOLONES

Resistance to the newer fluoroquinolones emerged rapidly among *Staphylococcus* and *Pseudomonas* spp. after the introduction of these agents. The most common mechanism is the development of one or more mutations in target DNA gyrase and topoisomerase IV that prevent the antibacterial agent from interfering with the activity of the enzyme. Some gram-negative bacteria develop mutations that both decrease

outer-membrane porin permeability and cause active drug efflux from the cytoplasm. Mutations that result in active quinolone efflux are also found in gram-positive bacteria.

RIFAMPIN

Bacteria rapidly become resistant to rifampin by developing mutations in the B subunit of RNA polymerase that render the enzyme unable to bind the antibiotic. The rapid selection of resistant mutants is the major limitation to the use of this antibiotic against otherwise-susceptible staphylococci and requires that it be used in combination with another antistaphylococcal agent.

MULTIPLE ANTIBIOTIC RESISTANCE

The acquisition by one bacterium of resistance to multiple antibacterial agents is becoming increasingly common. The two major mechanisms are the acquisition of multiple unrelated resistance genes and the development of mutations in a single gene or gene complex that mediate resistance to a series of unrelated compounds. The construction of multiresistant strains by acquisition of multiple genes occurs by sequential steps of gene transfer and environmental selection in areas of high-level antimicrobial use. In contrast, mutations in a single gene can conceivably be selected in a single step. Bacteria that are multiresistant by virtue of the acquisition of new genes include hospital-associated gram-negative bacteria, enterococci, and staphylococci and community-acquired strains of salmonellae, gonococci, and pneumococci. Mutations that confer resistance to multiple unrelated antimicrobial agents occur in the genes encoding outer-membrane porins and efflux proteins of gram-negative bacteria. These mutations decrease bacterial intracellular and periplasmic accumulation of β -lactams, quinolones, tetracycline, chloramphenicol, and trimethoprim. Multiresistant bacterial isolates pose increasing problems in U.S. hospitals; strains resistant to all available antibacterial chemotherapy have already been identified.

PHARMACOKINETICS

The *pharmacokinetic profile* of an antibacterial agent refers to concentrations in serum and tissue versus time and reflects the processes of absorption, distribution, metabolism, and excretion. Important characteristics include peak and trough serum concentrations and mathematically derived parameters such as half-life, clearance, and distribution volume. Pharmacokinetic information is useful for estimating the appropriate antibacterial dose and frequency of administration, for adjusting dosages in patients with impaired excretory capacity, and for comparing one drug with another. **For further discussion of basic pharmacokinetic principles, see [Chap. 70](#).*

ABSORPTION

Data on absorption can refer to oral, intramuscular, or intravenous administration.

Oral Administration Most patients with infection are treated with oral antibacterial agents in the outpatient setting. Advantages of oral therapy over parenteral therapy include lower cost, generally fewer adverse effects (including complications of indwelling lines), and greater acceptance by patients. The percentage of an orally administered

antibacterial agent that is absorbed (i.e., the agent's *bioavailability*) ranges from as little as 10 to 20% (erythromycin and penicillin G) to nearly 100% (clindamycin, metronidazole, doxycycline, and trimethoprim-sulfamethoxazole). These differences in bioavailability are not clinically important as long as concentrations at the site of infection are sufficient to inhibit or kill the pathogen. However, therapeutic efficacy may be compromised when absorption is reduced as a result of physiologic or pathologic conditions (such as the presence of food for some drugs or the shunting of blood away from the gastrointestinal tract in patients with hypotension), drug interactions (such as that of quinolones and metal cations), or noncompliance. The oral route is usually used for patients with relatively mild infections in whom absorption is not thought to be compromised by the preceding conditions. In addition, the oral route can be used in more severely ill patients after they have responded to parenteral therapy.

Intramuscular Administration Although the intramuscular route of administration usually results in 100% bioavailability, it is not as widely used in the United States as the oral and intravenous routes, in part because of the pain often associated with intramuscular injections and the relative ease of intravenous access in the hospitalized patient. Intramuscular injection may be suitable for specific indications requiring an "immediate" and reliable effect (e.g., with long-acting forms of penicillin, including benzathine and procaine, and with single doses of ceftriaxone for uncomplicated gonococcal infection).

Intravenous Administration The intravenous route is appropriate when oral antibacterial agents are not effective against a particular pathogen, when bioavailability is uncertain, or when larger doses are required than are feasible with the oral route. After intravenous administration, bioavailability is 100%; serum concentrations are maximal at the end of the infusion. For many patients requiring long-term antimicrobial therapy, outpatient intravenous administration with the use of convenient portable pumps may be cost-effective and safe when oral therapy is not feasible. Alternatively, some oral antibacterial drugs such as fluoroquinolones are sufficiently active against Enterobacteriaceae to rival parenteral therapy; their use may allow the patient to return home from the hospital earlier or to avoid hospitalization entirely.

DISTRIBUTION

To be effective, an antibacterial agent must exceed the minimal concentration required to inhibit bacterial growth (MIC; [Chap. 121](#)). Serum concentrations usually exceed the MIC for susceptible bacteria, but since most infections are extravascular, the antibiotic must also distribute to the site of the infection. Concentrations of most antibacterials in interstitial fluid are similar to free drug concentrations in serum. However, when the infection is located in a "protected" site where penetration is poor, such as cerebrospinal fluid (CSF), the eye, the prostate, or infected cardiac vegetations, high parenteral doses or local administration for prolonged periods may be required for cure. In addition, even though an antibacterial agent may penetrate to the site of infection, its activity may be antagonized by factors in the local environment, such as an unfavorable pH or inactivation by cellular degradation products. For example, since the activity of aminoglycosides is reduced at acidic pH, the acidic environment in many infected tissues may be partly responsible for the relatively poor efficacy of aminoglycoside monotherapy. In addition, the abscess milieu reduces the activity of many antibacterial

compounds, so that surgical drainage may be required for cure.

Most bacteria that cause human infections are located extracellularly. Intracellular pathogens such as *Legionella*, *Chlamydia*, *Brucella*, and *Salmonella* may persist or cause relapse if the antibacterial agent does not enter the cell. In general, β -lactams, vancomycin, and aminoglycosides penetrate cells poorly, whereas macrolides, tetracyclines, metronidazole, chloramphenicol, rifampin, trimethoprim-sulfamethoxazole, and quinolones penetrate cells well.

METABOLISM AND ELIMINATION

Like other drugs, antibacterial agents are disposed of by hepatic elimination (metabolism or biliary elimination), by renal excretion in unchanged or metabolized form, or by a combination of the two processes. For most antibacterial drugs, metabolism leads to loss of in vitro activity, although some agents, such as cefotaxime, rifampin, and clarithromycin, have bioactive metabolites that may contribute to their overall efficacy.

The most practical application of knowing the mode of excretion of an antibacterial agent is adjustment of the dosage when elimination capability is impaired. Direct, nonidiosyncratic toxicity from antibacterial drugs most often results from failure to reduce the dosage appropriately in a patient with impaired elimination. For agents that are primarily cleared intact by glomerular filtration, drug clearance is linearly correlated with creatinine clearance. Unfortunately, for drugs whose elimination is primarily hepatic, no simple marker (such as serum creatinine) is useful for dosage adjustment in subjects with liver disease. Even in patients with severe hepatic disease, residual metabolic capability is usually sufficient to preclude accumulation and toxic effects. However, for drugs that undergo hepatic metabolism and have a narrow therapeutic index (such as chloramphenicol), alternative therapy may be warranted in patients with liver disease, since the technology for the monitoring of serum levels is not widely available.

PRINCIPLES OF ANTIBACTERIAL CHEMOTHERAPY

The choice of an antibacterial compound for a particular patient and a specific infection involves more than just a knowledge of the agent's mechanism of action and pharmacokinetic profile. The basic tenets of chemotherapy, to be elaborated below, include the following: First, material containing the infecting organism(s) should be obtained before the start of treatment so that presumptive identification can be made by microscopic examination of stained specimens and the organism can be grown for definitive identification and susceptibility testing. Second, once the organism is identified and its susceptibility to antibacterial agents is determined, the regimen with the narrowest effective spectrum should be chosen. Third, the choice of antibacterial agent is guided by the pharmacokinetic and adverse-reaction profile of active compounds, the site of infection, the immune status of the host, and evidence of efficacy from well-performed clinical trials. Finally, if all other factors are equal, the least expensive antibacterial regimen should be chosen.

SUSCEPTIBILITY OF BACTERIA TO ANTIBACTERIAL DRUGS IN VITRO

The determination of the susceptibility of the patient's infecting organism to a panel of appropriate antibacterial agents is an essential first step in devising a chemotherapeutic regimen. The details of susceptibility testing are discussed elsewhere ([Chap. 121](#)). Such testing is designed to estimate the susceptibility of a bacterial isolate to an antibacterial drug under standardized conditions that favor rapidly growing aerobic or facultative organisms and to assess bacteriostasis only. Specialized testing is required for the assessment of bactericidal antimicrobial activity; for the detection of resistance among such fastidious organisms as obligate anaerobes, *Haemophilus* spp., and pneumococci; and for the determination of resistance phenotypes with variable expression, such as resistance to methicillin or oxacillin among staphylococci.

RELATIONSHIP OF PHARMACOKINETICS AND IN VITRO SUSCEPTIBILITY TO CLINICAL RESPONSE

The relationship between the report of susceptibility in vitro and the clinical pharmacokinetics of the antibacterial agent helps predict clinical response. Bacteria are usually considered to be *susceptible* to a drug if the achievable peak serum concentration exceeds the [MIC](#) by at least fourfold. The *breakpoint* is the concentration of the antibiotic that separates susceptible from resistant bacteria ([Fig. 137-2](#)). When a majority of the isolates of a given bacterial species are inhibited at concentrations below the breakpoint, the species is considered to be within the spectrum of the antibiotic (see "Choice of Antibacterial Therapy," below).

The pharmacodynamic profile of an antibiotic is the quantitative relationship among the time course of antibiotic concentrations in serum and tissue, in vitro susceptibility, and microbial response. Three pharmacodynamic parameters quantify these relationships: the ratio of the area under the curve (AUC) for the plasma concentration vs. time curve to [MIC](#) (AUC/MIC), the ratio of the maximal serum concentration to the MIC ($C_{\max}/\text{MIC}$), and the time during a dosing interval that plasma concentrations exceed the MIC ($t > \text{MIC}$). The pharmacodynamic profile of an antibiotic class is characterized as either concentration dependent (fluoroquinolones, aminoglycosides), such that the increase in antibiotic concentration leads to a more rapid rate of bacterial death, or time dependent (β-lactams, vancomycin), such that the reduction in bacterial density is proportional to the time that concentrations exceed the MIC. For concentration-dependent antibiotics, the $C_{\max}/\text{MIC}$ or AUC/MIC ratio correlates best with the reduction in microbial density in vitro and in animal investigations. Dosing strategies attempt to maximize these ratios by the administration of a "large" dose relative to the MIC for anticipated pathogens, often at "long" intervals (relative to the serum half-life). Once-daily dosing of aminoglycoside antibiotics is the practical consequence of these relationships. In contrast, dosage strategies for time-dependent antibiotics emphasize the administration of sufficient doses at appropriate intervals to maintain serum concentrations above the MIC, typically for at least 40 to 50% of the dosing interval. The clinical implications of these relationships are in the early stages of investigation, but their elucidation should eventually result in more rational antibacterial regimens.

STATUS OF THE HOST

Various host factors must be considered in the devising of antibacterial chemotherapy.

The host's antibacterial *immune function* is of importance, particularly as it relates to opsonophagocytic function. Since the major host defense against acute, overwhelming bacterial infection is the polymorphonuclear leukocyte, patients with neutropenia must be treated aggressively and empirically with bactericidal drugs for suspected infection ([Chap. 85](#)). Likewise, patients who have deficient humoral immunity (e.g., those with chronic lymphocytic leukemia and multiple myeloma) and individuals with surgical or functional asplenia (e.g., those with sickle cell disease) should be treated empirically for infections with encapsulated organisms, especially the pneumococcus.

Pregnancy increases the risk of toxicity of certain antibacterial drugs for the mother (e.g., the hepatic toxicity of tetracycline), affects drug disposition and pharmacokinetics, and -- because of the risk of fetal toxicity -- severely limits the choice of agents for treating infections. Certain antibacterials are contraindicated in pregnancy either because their safety has not been established or because they are known to be toxic. These agents include all fluoroquinolones, clarithromycin, erythromycin estolate (but not erythromycin base), and tetracyclines. Data on the safety of many other antibacterial drugs are limited, but these drugs may be used cautiously when there is no suitable alternative and the perceived benefit outweighs the risk. These agents include the aminoglycosides, azithromycin, clindamycin, imipenem, metronidazole, trimethoprim, and vancomycin. Chloramphenicol, nitrofurantoin, and the sulfonamides are contraindicated in the third trimester but can be used cautiously in the first two trimesters.

In patients with *concomitant viral infections*, the incidence of adverse reactions to antibacterial drugs may be unusually high. For example, persons with infectious mononucleosis and those infected with HIV may react more often to ampicillin and folic acid synthesis inhibitors, respectively.

In addition, the patient's age, sex, racial heritage, and excretory status all determine the incidence and type of side effects that can be expected with certain antibacterial agents.

SITE OF INFECTION

The location of the infected site may play a major role in the choice and dose of antimicrobial drug. Patients with suspected *meningitis* should receive drugs that can cross the blood-[CSF](#) barrier; in addition, because of the relative paucity of phagocytes and opsonins at the site of infection, the agents should be bactericidal.

Chloramphenicol, one of the standard drugs used in the treatment of meningitis, is bactericidal for common organisms causing meningitis (i.e., meningococci, pneumococci, and *Haemophilus influenzae*, but *not* enteric gram-negative bacilli), is highly lipid-soluble, and enters the CSF well. However, b-lactams, the mainstay of therapy for most of these infections, do not normally reach high levels in CSF. Their efficacy is based on the increased permeability of the blood-brain and blood-CSF barriers to hydrophilic molecules during inflammation and the extreme susceptibility of most infectious organisms to even small amounts of b-lactam drug.

The vegetation, which is the major site of infection in *bacterial endocarditis*, is also a focus that is protected from normal host-defense mechanisms. Antibacterial therapy needs to be bactericidal, with the selected agent administered parenterally over a long

period and at a dose that produces serum levels at least eight times higher than the minimal bactericidal concentration (MBC) for the infecting organism. Likewise, *osteomyelitis* involves a site that is somewhat resistant to opsonophagocytic removal of infecting bacteria; furthermore, avascular bone (sequestrum) represents a foreign body that thwarts normal host-defense mechanisms. *Chronic prostatitis* is exceedingly difficult to cure because most antibiotics do not penetrate the nonfenestrated capillaries serving the prostate, especially when acute inflammation is absent. Drugs that are "ion trapped" after entering prostatic tissue, such as trimethoprim and fluoroquinolones, may be uniquely effective because of this mechanism. *Intraocular infections*, especially endophthalmitis, are difficult to treat because drug penetration into the vitreous from blood is hindered by retinal capillaries lacking fenestration. Inflammation does little to disrupt this barrier. Thus, direct injection into the vitreous is necessary in many cases. Antibiotic penetration into *abscesses* is usually poor. Even when an antibiotic does penetrate into the abscess, local conditions, such as low pH or the presence of enzymes that hydrolyze the drug, may antagonize its activity.

In contrast, *urinary tract infections*, when confined to the bladder, are relatively easy to cure, in part because of the higher concentration of most antibiotics in urine than in blood. Since blood is the usual reference fluid in defining susceptibility, even organisms found to be "resistant" to achievable serum concentrations may be susceptible to achievable urine concentrations. For drugs that are used only for the treatment of urinary tract infections, such as nitrofurantoin and methenamine salts, achievable urine concentrations are used to determine susceptibility.

COMBINATION CHEMOTHERAPY

One of the tenets of antibacterial chemotherapy is that if the infecting bacterium has been identified, the most specific chemotherapy possible should be used. The use of a single agent with a narrow spectrum of activity against the pathogen diminishes the alteration of normal flora and thus limits the overgrowth of resistant nosocomial organisms (e.g., *Candida albicans*, enterococci, *Clostridium difficile*, or methicillin-resistant staphylococci), avoids the potential toxicity of multiple-drug regimens, and reduces cost. However, certain circumstances call for the use of more than one antibacterial agent. These are summarized below.

1. *Prevention of the emergence of resistant mutants.* Spontaneous mutations occur at a detectable frequency in certain genes encoding the target proteins for some antibacterial agents. The use of these agents can eliminate the susceptible population, select out resistant mutants at the site of infection, and result in the failure of chemotherapy. Resistant mutants are usually selected when the MIC of the antibacterial agent for the infecting bacterium is close to achievable levels in serum or tissues and/or when the site of infection limits the access or activity of the agent. Among the most common examples are rifampin for staphylococci, imipenem for *Pseudomonas*, and ciprofloxacin for staphylococci and *Pseudomonas*. Small-colony variants of staphylococci resistant to aminoglycosides also emerge during monotherapy with these antibiotics. A second antibacterial agent with a mechanism of action different from that of the first is added to prevent the emergence of these resistant mutants (e.g., imipenem plus an aminoglycoside for systemic *Pseudomonas* infections). However, since resistant mutants have emerged following combination chemotherapy, this

approach is not uniformly successful.

2. *Synergistic or additive activity.* Against some bacteria, two or more agents are clearly more active than one; whether or not this is the case is usually judged on the basis of testing in vitro. Synergistic or additive activity involves a lowering of the [MIC](#) or [MBC](#) of each or all of the drugs tested in combination against a specific bacterium. In *synergy*, *each* agent is more active when combined with a second drug than it would be alone, and the drugs' combined activity is therefore greater than the sum of the individual activities of each drug. In an *additive relationship*, the combined activity of the drugs is equal to the sum of their individual activities. Among the best examples of a synergistic or additive effect, confirmed both in vitro and by animal studies, are the enhanced bactericidal activities of certain β -lactam/aminoglycoside combinations against enterococci, viridans streptococci, and *P. aeruginosa*. The synergistic or additive activity of these combinations has also been demonstrated for selected isolates of enteric gram-negative bacteria and staphylococci. The combination of trimethoprim and sulfamethoxazole has synergistic or additive activity against many enteric gram-negative bacteria. Most other antimicrobial combinations display indifferent activity (i.e., the combination is *no better* than the more active of the two agents alone), and some combinations (e.g., penicillin plus tetracycline against pneumococci) may be antagonistic (i.e., the combination is *worse* than either drug alone).

3. *Therapy directed against multiple potential pathogens.* For certain infections, either a mixture of pathogens is suspected or the patient is desperately ill with an as-yet-unidentified infection (see "Empirical Therapy," below). In these situations, the most important of the likely infecting bacteria must be covered by therapy until culture and susceptibility results become available. Examples of the former infections are intraabdominal or brain abscesses and infections of limbs in diabetic patients with microvascular disease. The latter situations include fevers in neutropenic patients, acute pneumonia from aspiration of oral flora by hospitalized patients, and septic shock or sepsis syndrome.

EMPIRICAL THERAPY

In certain situations, antibacterial therapy is begun before a specific bacterial pathogen has been identified. The choice of agent is guided by the results of studies identifying the usual pathogens at that site or in that clinical setting, by pharmacodynamic considerations, and by the resistance profile of the expected pathogens in a particular hospital or geographic area. Situations in which empirical therapy is appropriate include the following:

1. *Life-threatening infection.* Any suspected bacterial infection in a patient with a life-threatening illness should be treated presumptively. Therapy is usually begun with more than one agent and is later tailored to a specific pathogen if one is eventually identified.

2. *Treatment of infections in unhospitalized patients with no cultures performed.* In many situations, it is appropriate to treat non-life-threatening infections without obtaining cultures. These situations include outpatient infections such as community-acquired cases of pneumonia, cystitis, cellulitis or local wound infection, sinusitis, otitis, urethritis,

and prostatitis. However, if any of these infections recurs or fails to respond to initial therapy, every effort should be made to obtain cultures to guide re-treatment.

CHOICE OF ANTIBACTERIAL THERAPY

The antibacterial spectrum of specific agents and the infections for which they represent the treatment of choice are detailed below. No attempt has been made to include all the potential situations in which antibacterial agents may be used. A more detailed discussion of specific bacteria and infections that they cause can be found elsewhere in this volume.

b-LACTAMS ([Table 137-2](#))

All *penicillins* (except for the semisynthetic, penicillinase-resistant antistaphylococcal agents) are hydrolyzed by β -lactamases and are ineffective against isolates that produce these enzymes. Penicillin G has a spectrum that includes spirochetes (*Treponema pallidum*, *Borrelia*, and *Leptospira*), streptococci (groups A and B, viridans, and *Streptococcus pneumoniae*), enterococci, most *Neisseria* spp., a few staphylococci, many fastidious oral bacteria (including many *Porphyromonas* and *Prevotella* spp., streptococci, *Actinomyces*, and *Fusobacterium*), *Clostridium* spp. (except *C. difficile*), *Pasteurella multocida*, *Erysipelothrix rhusiopathiae*, and *Streptobacillus moniliformis*. However, penicillin G resistance is widespread among staphylococci; is increasing rapidly among gonococci, enterococci, and pneumococci; and is emerging among meningococci, viridans streptococci, and oral anaerobes such as *Porphyromonas* and *Prevotella*. Penicillin G is the *drug of choice* for syphilis, yaws, leptospirosis, group A and B streptococcal infections, actinomycosis, oral and periodontal infections, meningococcal meningitis and meningococemia, viridans streptococcal endocarditis, clostridial myonecrosis, tetanus, anthrax, rat-bite fever, *P. multocida* infections, and erysipeloid (*E. rhusiopathiae*).

Ampicillin extends the spectrum of penicillin G to some gram-negative rods. It is active against some isolates of *Escherichia coli*, *Proteus mirabilis*, *Salmonella*, *Shigella*, and *H. influenzae* and is one of the *drugs of choice* for susceptible organisms causing urinary tract infections, salmonellosis, *H. influenzae* meningitis and epiglottitis, and *Listeria monocytogenes* meningitis. High rates of resistance have lessened its value as empirical therapy in some situations. For example, more than 80% of isolates of *E. coli* and *P. mirabilis* are resistant in some hospitals, as are 10 to 30% of isolates of *H. influenzae*; moreover, in some outbreaks of infection due to salmonellae, all isolates are ampicillin-resistant.

The *penicillinase-resistant penicillins* are used solely for the treatment of staphylococcal infections and are the *drugs of choice* for systemic or deep staphylococcal infections caused by susceptible organisms. Unfortunately, on average, approximately 30% of *S. aureus* isolates and more than 60% of coagulase-negative staphylococcal isolates acquired in U.S. hospitals are resistant to these agents (i.e., methicillin-resistant). The spectrum of these agents also includes most of the same gram-positive bacteria that are susceptible to penicillin G.

The spectrum of the *antipseudomonal penicillins* includes the bacteria covered by

ampicillin as well as some additional nonpseudomonal enteric gram-negative bacilli. For example, piperacillin is active against many indole-positive *Proteus*, *Enterobacter*, *Klebsiella*, *Providencia*, and *Serratia* spp. However, the susceptibility of these penicillins to β -lactamase markedly limits their utility as empirical therapy when infections caused by gram-negative enteric organisms are suspected. The major use of these compounds is in the treatment of proven or suspected infections with *P. aeruginosa* and *Acinetobacter*, for which they are among the *drugs of choice*. Their relative antipseudomonal activities can be ranked as follows: piperacillin > mezlocillin/ticarcillin > carbenicillin.

The addition of *β -lactamase inhibitors* (clavulanic acid, sulbactam, or tazobactam) to ampicillin, amoxicillin, ticarcillin, or piperacillin extends the spectrum of these agents to include many organisms that are resistant by virtue of β -lactamase production. These organisms include *E. coli*, *Klebsiella* spp., all *Proteus* spp., *H. influenzae*, *Moraxella catarrhalis*, *Providencia* spp., and *Bacteroides* spp. Such combinations are also active against staphylococci that produce β -lactamase but are not methicillin-resistant. However, the efficacy of these combinations in serious staphylococcal infections has not been adequately proven. Furthermore, *Enterobacter*, *Pseudomonas*, *Acinetobacter*, and various enteric gram-negative isolates either produce β -lactamases not inhibited by these compounds or develop resistance attributable to non- β -lactamase-mediated mechanisms.

The *first-generation cephalosporins* have a spectrum that includes penicillinase-producing, methicillin-susceptible staphylococci and streptococci. While these drugs may be used when infections with gram-positive bacteria are suspected, they are *not* the drugs of choice for such infections. They have excellent activity against many isolates of *E. coli*, *Klebsiella pneumoniae*, and *P. mirabilis* and are among the *drugs of choice* in presumptive therapy for community-acquired urinary tract infections. They have no activity against *Bacteroides fragilis*, enterococci, methicillin-resistant staphylococci, *Pseudomonas*, *Acinetobacter*, *Enterobacter*, indole-positive *Proteus*, and *Serratia* and poor activity against *H. influenzae*.

The *parenteral second-generation cephalosporins* extend the gram-negative spectrum of first-generation compounds. The various second-generation agents have differing activities. Cefuroxime and cefamandole retain activity against gram-positive cocci and are also active against *H. influenzae*, *Neisseria*, some *Enterobacter* isolates, and indole-positive *Proteus* but exhibit poor activity against *B. fragilis*. Cefoxitin and cefotetan have reasonably good activity against *B. fragilis*, but cefotetan is less effective against some other *Bacteroides* spp. ([Chaps. 130](#) and [167](#)). Both of the latter drugs display poor activity against gram-positive cocci and *Enterobacter*. No second-generation cephalosporin is active against *Pseudomonas* or *Acinetobacter*.

Oral second-generation cephalosporins have fair activity against gram-positive cocci and *H. influenzae* and are widely used in outpatient therapy for otitis media, sinusitis, and lower respiratory tract infections, although cheaper agents that are equally effective are preferable. Cefixime, cefuroxime axetil, and cefpodoxime are among the *drugs of choice* for single-dose treatment of gonococcal urethritis.

Third-generation cephalosporins all have a broad spectrum of activity against enteric

gram-negative rods and are especially useful for treating hospital-acquired infections caused by multiresistant organisms. In addition, ceftazidime and cefepime have excellent antipseudomonal activity. The other third-generation cephalosporins have poor antipseudomonal activity. Since resistance to third-generation cephalosporins is increasing among all nosocomial gram-negative rods, the use of these agents should be guided by susceptibility testing. The gram-positive spectrum of the third-generation cephalosporins is variable. All are less active than first-generation cephalosporins against methicillin-susceptible staphylococci; ceftazidime has the least antistaphylococcal activity of this group. However, ceftriaxone, ceftizoxime, and cefotaxime have excellent activity against streptococci, especially *S. pneumoniae*. Ceftazidime is not recommended for treatment of streptococcal infections.

Because of its excellent gram-negative spectrum; its activity against *Haemophilus*, many *S. pneumoniae* strains, and penicillin-resistant *Neisseria*; its long serum half-life; and its high serum and CSF levels, ceftriaxone has become one of the *drugs of choice* for empirical therapy for bacterial meningitis (except that caused by *Listeria* and by highly penicillin-resistant pneumococcal strains), all gonococcal infections, salmonellosis, and typhoid fever. The third-generation cephalosporins are among the *drugs of choice* for nonpseudomonal hospital-acquired pneumonia. Cefepime is more resistant to chromosomal β -lactamase produced by *Enterobacter* spp. than are other third-generation cephalosporins. Third-generation cephalosporins have poor activity against *Bacteroides* and no activity against methicillin-resistant staphylococci, *Enterococcus*, *Acinetobacter*, or *Stenotrophomonas*.

The *carbapenems* currently available in the United States are imipenem and meropenem. Imipenem is marketed in combination with the renal dipeptidase inhibitor cilastatin, which enables imipenem to escape renal inactivation and thus to reach higher urinary levels. Imipenem and meropenem have excellent activity in vitro against virtually all bacterial pathogens except *Stenotrophomonas*, methicillin-resistant staphylococci, and *E. faecium*. Limitations to their use are their relatively low blood levels, short serum half-life, and high cost. Imipenem has dose-related central nervous system side effects that appear to be less frequent with meropenem. Resistance to imipenem and meropenem is a problem only among nosocomial isolates of *P. aeruginosa*, approximately 20% of which are resistant. Because of their broad spectrum, carbapenems can be used as empirical therapy for serious nosocomial infections thought to be caused by multiple bacterial species or multiresistant organisms. Imipenem and meropenem are often used to treat hospital-acquired infections caused by *Enterobacter* spp. because these organisms produce inducible β -lactamases that inactivate third-generation cephalosporins but not the carbapenems. The latter antibiotics are often held in reserve as therapy for nosocomial infections due to gram-negative pathogens resistant to third-generation cephalosporins.

The only *monobactam* currently available is aztreonam. This antibiotic has a spectrum limited to facultative gram-negative enteric bacilli. It has no activity against any gram-positive or anaerobic bacterium. Its gram-negative spectrum is similar to that of ceftazidime, with equally good activity against *Pseudomonas*. Its primary advantages are its theoretical ability to preserve the normal gram-positive and anaerobic flora and the lack of cross-reactive immediate hypersensitivity in patients who have had this type of reaction to other β -lactam antibiotics.

VANCOMYCIN

The spectrum of vancomycin is limited to gram-positive cocci, especially enterococci, streptococci, and staphylococci. Vancomycin serves as second-line therapy for most gram-positive bacterial infections but is the *drug of choice* for infections caused by methicillin-resistant staphylococci or *Corynebacterium jeikeium* and for serious infections in penicillin-allergic patients. Given orally (a route by which it is not absorbed), vancomycin can be used to treat antibiotic-associated pseudomembranous colitis caused by *C. difficile* in patients who have failed to respond to metronidazole, the *drug of choice*. Vancomycin has also been recommended as initial empirical therapy for presumed pneumococcal meningitis because of increasing pneumococcal resistance to penicillins and cephalosporins. Resistance to vancomycin is increasing rapidly among isolates of *E. faecium* in large hospitals, particularly in areas of high vancomycin use. In addition, *S. aureus* isolates with reduced susceptibility to vancomycin have now been detected. Because of the growing threat of vancomycin-resistant enterococci and the potential for increasing resistance among staphylococci, a national advisory committee has established guidelines for appropriate and limited use of this antibiotic.

AMINOGLYCOSIDES

The aminoglycosides are rapidly bactericidal in vitro at low concentrations, with activity limited to facultative gram-negative bacteria and staphylococci. They have no activity against anaerobic bacteria and are not effective in environments that are acidic or have a low oxygen tension. However, their spectrum includes virtually all gram-negative bacteria that are not strict anaerobes, and they are among the *drugs of choice* for any suspected gram-negative bacteremic infection, particularly in neutropenic patients. Aminoglycosides are synergistically bactericidal in combination with a penicillin for the treatment of staphylococcal, enterococcal, or viridans streptococcal endocarditis and are usually combined with a beta-lactam antibiotic for the treatment of gram-negative bacteremia. Aminoglycosides are also among the *drugs of choice* for severe infections of the upper urinary tract. The major limitations to use of aminoglycosides are their renal and otic toxicity, their diminished activity at certain sites of infection (e.g., abscesses and the central nervous system), and the resistance of target bacteria. Among the available agents, gentamicin is generally preferred because of its low cost; however, tobramycin has slightly greater activity against *P. aeruginosa*, and amikacin retains activity against many tobramycin- and gentamicin-resistant gram-negative bacteria because it is inactivated by fewer aminoglycoside-modifying enzymes. Streptomycin is still one of the *drugs of choice* in initial therapy for tularemia, plague, glanders, and brucellosis and is a second-line agent for the treatment of tuberculosis.

MACROLIDES

Erythromycin has broad-spectrum activity against gram-positive bacteria, with additional activity against *Legionella*, *Mycoplasma*, *Campylobacter*, and some *Chlamydia* isolates. It is the *drug of choice* for infections due to *Legionella*, *Campylobacter*, and *Mycoplasma* and is among the *drugs of choice* for community-acquired pneumococcal pneumonia and group A streptococcal pharyngitis in penicillin-allergic patients. However, resistance to erythromycin among group A streptococci and pneumococci is increasing

dramatically in some areas. Erythromycin also appears to be one of the *drugs of choice* for infections caused by the agent of bacillary angiomatosis (*Bartonella henselae*) in immunocompromised patients. The newer macrolides clarithromycin and azithromycin have an antibacterial spectrum similar to that of erythromycin in vitro. However, azithromycin has greater activity against *Chlamydia*. Clarithromycin, in combination with a proton pump inhibitor, has been designated a *drug of choice* for the treatment of gastric infections due to *Helicobacter pylori* (gastritis, gastric and duodenal ulcers). Both azithromycin and clarithromycin are active against nontuberculous mycobacteria, and both appear to have fewer gastrointestinal side effects than does erythromycin.

LINCOSAMIDES

The only lincosamide used in the United States is clindamycin. It shares the gram-positive coccal spectrum of erythromycin but is more active, in some cases showing bactericidal activity, against susceptible staphylococci. However, resistance among staphylococci and some streptococci, mediated by the same genes responsible for macrolide resistance, limits clindamycin's usefulness against gram-positive cocci. In general, all staphylococci resistant to erythromycin should be considered resistant to clindamycin regardless of the results of in vitro susceptibility testing. However, at least half of the streptococci resistant to erythromycin are truly susceptible to clindamycin. In these bacteria, resistance is mediated by a drug-efflux pump that removes macrolides but not lincosamides. Despite increasing resistance, clindamycin remains useful for most anaerobic infections because of its broad spectrum of activity against both gram-positive and -negative strict anaerobes. It is also a *drug of choice* for the treatment of severe, invasive group A streptococcal infections. In contrast, clindamycin, like erythromycin, has no clinically significant activity against facultative gram-negative enteric bacilli. Appropriate use is limited only by resistance or the development of pseudomembranous colitis, the major serious side effect of this drug.

CHLORAMPHENICOL

Chloramphenicol has a broad spectrum of activity against gram-positive and -negative bacteria, although plasmid-mediated resistance has diminished its effective spectrum. This antibiotic is rarely used in adult infections because of the rare idiosyncratic side effect of irreversible bone-marrow aplasia and the availability of other agents with similar activity. It remains one of the *drugs of choice* for the treatment of typhoid fever and plague and is still useful for the treatment of brucellosis and both pneumococcal and meningococcal meningitis in penicillin-allergic patients.

TETRACYCLINES

Tetracyclines have a broad spectrum of bacteriostatic activity against gram-positive and -negative bacteria and are widely used in a variety of community-acquired infections. These agents are among the *drugs of choice* for chronic bronchitis, granuloma inguinale, brucellosis (with streptomycin), tularemia, glanders, melioidosis, spirochetal infections caused by *Borrelia* (Lyme disease and relapsing fever; doxycycline), infections caused by *Vibrio vulnificus*, some *Aeromonas* infections, infections due to *Stenotrophomonas* (minocycline), plague, and ehrlichiosis (doxycycline). The tetracyclines are also used in penicillin-allergic patients for the treatment of

leptospirosis, syphilis, actinomycosis, and skin and soft-tissue infections caused by gram-positive cocci. They are among the *drugs of choice* for infections due to chlamydiae (doxycycline), rickettsiae, and ehrlichiae and for granulomatous skin infection due to *Mycobacterium marinum* (minocycline). Doxycycline is also among the drugs recommended for the treatment of community-acquired pneumonia.

SULFONAMIDES AND TRIMETHOPRIM

The folic acid synthesis inhibitors have a broad spectrum of bacteriostatic activity individually; in combination, they can be bactericidal against facultative gram-negative bacteria and staphylococci. The fixed combination of sulfamethoxazole and trimethoprim, the major folic acid synthesis inhibitors used in therapy for bacterial infections, has modest activity against some streptococci and no activity against strict anaerobes. However, resistance to the combination of sulfamethoxazole and trimethoprim is common among methicillin-resistant staphylococci and penicillin-resistant pneumococci and is increasing among *E. coli* strains that cause urinary tract infections. The individual sulfonamides are rarely used in the treatment of bacterial infections but are among the *drugs of choice* for the treatment of nocardial infections, leprosy (dapsone, a sulfone), and toxoplasmosis (sulfadiazine). Although increasing resistance has been reported among gram-negative organisms, trimethoprim-sulfamethoxazole remains one of the *drugs of choice* for the treatment of uncomplicated urinary tract infections (except for those caused by enterococci) and is widely used in the treatment of otitis media. It can be used in therapy for upper respiratory tract infections in which *S. pneumoniae*, *H. influenzae*, or *M. catarrhalis* is suspected; for gonococcal and meningococcal infections; for chancroid; and for infections thought to be caused by *Aeromonas*, *Stenotrophomonas*, *Burkholderia cepacia*, *Acinetobacter*, and *Yersinia enterocolitica*. For nosocomial infections due to *Stenotrophomonas*, trimethoprim-sulfamethoxazole is the *drug of choice*.

FLUOROQUINOLONES

The fluoroquinolones have excellent activity against most facultative gram-negative rods and variable activity against gram-positive cocci; only trovafloxacin is active against obligate anaerobes. The quinolones are the oral agents with greatest activity against *P. aeruginosa*; ciprofloxacin is the most active against this species. All the quinolones except norfloxacin are well absorbed orally; ciprofloxacin, levofloxacin, trovafloxacin, and ofloxacin are also administered as intravenous formulations. The quinolones are among the *drugs of choice* for urinary tract infections, bacterial gastroenteritis, community-acquired pneumonia, and enteric fever and may be useful in therapy for serious hospital-acquired infections caused by gram-negative organisms. While older quinolones (ciprofloxacin, ofloxacin, and norfloxacin) have limited activity against gram-positive bacteria, the newer quinolones have an expanded spectrum of activity against gram-positive cocci, including staphylococci (both methicillin-susceptible and methicillin-resistant) and streptococci (especially *S. pneumoniae*). However, because of the potential for development of severe liver toxicity, it has been recommended that trovafloxacin use be limited to hospitalized patients with serious or life-threatening infections. Quinolones can also be used as prophylaxis for persons at risk for meningococcal meningitis. However, rapid expansion in the use of quinolones should be coupled with a consideration of the potential for development of resistance among all

bacteria targeted by these drugs.

RIFAMPIN

Rifampin has been used in combinations for the treatment of serious infections due to methicillin-resistant staphylococci (e.g., coagulase-negative staphylococcal foreign-body infections). Because the spontaneous selection of rifampin-resistant mutants occurs rapidly, rifampin should never be used alone in the treatment of staphylococcal infections. Rifampin is also used for chemoprophylaxis in persons at risk of meningococcal meningitis and for the treatment of *Legionella* pneumonia.

METRONIDAZOLE

Metronidazole has a spectrum limited to anaerobic bacteria. It is one of the *drugs of choice* for the treatment of any abscess in which the involvement of obligate anaerobes is suspected (e.g., lung, brain, or intraabdominal abscesses) because of its spectrum and its ability to penetrate into the area of infection. Other antibacterial agents should be used in combination with metronidazole if facultative and aerobic pathogens are also thought to be involved. Metronidazole is the *drug of choice* for the treatment of bacterial vaginosis and antibiotic-associated pseudomembranous colitis.

URINARY TRACT ANTISEPTICS

Urinary tract antiseptics are active only in the lower urinary tract and cannot be used for the treatment of upper urinary tract or systemic infections. Their activity is limited to susceptible gram-negative enteric bacteria. The available agents in this category include nitrofurantoin and methenamine salts.

TOPICAL ANTIBACTERIAL AGENTS

Mupirocin is available only as a topical preparation for use against staphylococci and streptococci. Its major applications are for impetigo and eradication of the staphylococcal carrier state. It is the *drug of choice* for the elimination of nasal carriage of both methicillin-susceptible and methicillin-resistant staphylococci. Unfortunately, the emergence of resistance is limiting its usefulness in some hospitals.

Although their efficacy has never been well documented, topical preparations that include sulfonamides, polymyxin B, neomycin, bacitracin, gramicidin, and novobiocin in a variety of combinations are widely used as eye drops, irrigation solutions, and ointments for superficial skin infections.

ADVERSE REACTIONS

Adverse drug reactions are frequently classified by mechanism as either dose-related ("toxic") effects or unpredictable reactions. Unpredictable reactions are further categorized as either idiosyncratic or allergic. Dose-related reactions include aminoglycoside-induced nephrotoxicity, penicillin-induced seizures, and vancomycin-induced anaphylactoid reactions. Many of these reactions can be avoided by reducing dosage, limiting the duration of therapy, or reducing the frequency or rate of

administration. Adverse reactions to antibacterial agents are a common cause of morbidity, requiring alteration in therapy and additional expense, and they occasionally result in death. The elderly, often those with the more severe infections, may be especially prone to certain adverse reactions. **For further discussion of adverse drug reactions, see Chap. 71.*

b-LACTAMS

The therapeutic index for b-lactam antibiotics is broad, and dose-related adverse reactions are uncommon and largely preventable. The greatest concern is allergic reactions. All types can occur, including anaphylaxis (type 1, hypersensitivity reactions), nephritis and Coombs-positive hemolytic anemia (type 2, cytotoxic reactions), drug fever and serum sickness (type 3, immune-complex formation), contact dermatitis (type 4, cell-mediated effects), and maculopapular eruption (type 5, idiopathic reactions). Approximately 1 to 4% of treatment courses result in an allergic reaction, and approximately 0.004 to 0.015% of treatment courses result in anaphylaxis. Fewer than half the patients who claim an allergy to penicillin react to skin testing with the major and minor determinants (penicilloyl-polylysine and benzylpenicillin degradation products, respectively); those with negative skin tests only rarely react adversely to subsequent therapeutic doses. Generally, a suitable alternative to b-lactams is available for patients who have a severe allergy, and penicillin desensitization can be carefully undertaken if there is no suitable alternative. A small proportion (<2%) of persons who are allergic to penicillin react similarly when a cephalosporin is administered; thus, cephalosporins are contraindicated in patients with a history of an immediate reaction to penicillin, although they are often used in patients with a history of mild reactions. The same precaution applies to imipenem, but aztreonam is antigenically distinct and can be administered safely to the penicillin-allergic patient.

Other reactions thought to have an allergic basis include nephritis (associated with methicillin and occasionally nafcillin), hepatitis (related to oxacillin), leukopenia (following high doses of most b-lactams administered for prolonged periods), and severe skin rashes (toxic epidermal necrolysis and Stevens-Johnson syndrome). These reactions are not IgE-mediated, and skin testing is not predictive of their occurrence. For unclear reasons, most patients who have infectious mononucleosis develop a rash when given ampicillin or amoxicillin.

Miscellaneous reactions to b-lactams include gastrointestinal side effects ranging in severity from mild diarrhea (5 to 10%) to pseudomembranous colitis (<1%). Although the probability of antibiotic-associated colitis is low, a large number of cases occur because b-lactams are so commonly prescribed. Drugs excreted to a large extent through the bile, such as ampicillin and ceftriaxone, may be especially prone to cause diarrhea. The addition of clavulanic acid to amoxicillin further increases the frequency of diarrhea. Ceftriaxone, because of extremely high concentrations in bile, can cause "sludging" in the gallbladder and occasionally produces symptoms compatible with acute cholecystitis.

In high doses -- and most often in patients with renal impairment who receive an excessive dose -- penicillins (especially ticarcillin and penicillin G) can cause bleeding from impaired platelet aggregation. Ticarcillin is a disodium salt and in high doses can

cause hypokalemia and fluid overload.

Seizures are occasionally observed with β -lactams, especially penicillin G and imipenem. This reaction is most common when excessive doses relative to renal function are administered or when the patient has a history of seizures.

VANCOMYCIN

When vancomycin was first used clinically in 1956, local intolerance at the infusion site was common, as were systemic reactions, including ototoxicity and nephrotoxicity. Current formulations are of higher purity and, when proper dosage guidelines are followed, are very safe, although phlebitis can still be troublesome. The most common adverse reaction is called *red man syndrome* and is characterized by pruritus, flushing, and erythema of the head and upper torso. This anaphylactoid reaction usually follows the first dose, is dependent on dose size and infusion time, and results from vancomycin-induced release of histamine. The reaction is usually mild in adult patients who receive 1 g over 60 min and diminishes with repeated doses. If vancomycin is mistakenly given as a bolus, severe hypotension may result. In unusually sensitive patients, extending the infusion time or administering H_1 -receptor antagonists is usually effective in preventing this reaction or reducing its severity. Patients with this reaction must not be mislabeled as having an allergy to vancomycin, since vancomycin may be the only effective treatment for certain infections, such as those due to methicillin-resistant staphylococci.

Nephrotoxicity from vancomycin is mild and occurs in fewer than 5% of patients. Although some data suggest that aminoglycosides and vancomycin are synergistically nephrotoxic, this point is difficult to prove, and the simultaneous use of these agents should not be avoided if clinically indicated, as in the treatment of enterococcal endocarditis in penicillin-allergic patients.

Ototoxicity from vancomycin is rare as long as doses are appropriately reduced in patients with renal insufficiency. Other uncommon adverse reactions include leukopenia, skin rashes, and true allergy. Serum concentrations of vancomycin are of little use in predicting toxicity but may be of value in selecting dosages for patients with unstable renal function.

AMINOGLYCOSIDES

Aminoglycoside antibiotics have a narrow therapeutic index. The two most common adverse reactions are nephrotoxicity and ototoxicity. Rarely, respiratory depression is observed. Nephrotoxicity results from accumulation of the aminoglycoside in the peritubular space, with damage to the proximal tubule and a corresponding reduction in the glomerular filtration rate. The incidence of nephrotoxicity, defined as a $>0.5\%$ increase over baseline in the serum creatinine level, is approximately 5 to 10% among adult patients who receive therapy for 10 to 14 days. However, many cofactors also influence the frequency of toxicity, such as extremes of age (toxicity is uncommon among children, more common among the elderly), concomitant drug therapy, and hydration status. Nephrotoxicity is manifested clinically by a gradual rise in serum creatinine levels after a few days of therapy and is reversible if the dosage is reduced or

treatment is discontinued. Serum creatinine levels should be monitored every 3 to 5 days or more often if changes are seen. There is not an important difference among the most useful agents (gentamicin, tobramycin, and amikacin) in terms of the frequency of nephrotoxicity; streptomycin is a rare cause of nephrotoxicity. Some data suggest that once-daily administration may cause less nephrotoxicity.

Ototoxicity from aminoglycoside therapy presents as either auditory or vestibular damage. Since the aminoglycosides can destroy hair cells in the inner ear, ototoxicity may be permanent. The risk of ototoxicity increases with prolonged therapy, higher serum concentrations (especially in patients with renal impairment), hypovolemia, and concurrent treatment with other ototoxins, especially ethacrynic acid. Clinically apparent ototoxicity, manifested by diminished acuity or vestibular imbalance, is uncommon (probably occurring in <1% of cases) when the duration of therapy is kept to a minimum. With more sensitive monitoring (e.g., audiograms), asymptomatic high-tone hearing loss is more commonly noted. There are no clinically important differences among the aminoglycosides in the overall frequency of ototoxicity.

Neuromuscular depression from aminoglycosides is caused by reduced acetylcholine activity at postsynaptic membranes and can result in rare but severe respiratory depression. Risk factors include hypocalcemia, peritoneal administration, use of neuromuscular blockers, and preexisting respiratory depression. This complication can be largely avoided if the aminoglycoside is administered intravenously over 30 min or by intramuscular injection; if respiratory depression occurs, it is reversed by the administration of calcium.

Fear of toxicity should not prevent the use of aminoglycosides for a legitimate indication, since toxicity is usually mild and reversible. The value of measuring serum concentrations is controversial; these measurements are usually unnecessary when the patient is receiving once-daily therapy.

MACROLIDES

Serious adverse reactions to the macrolide antibiotics are very rare. Gastrointestinal effects, such as burning, nausea, and vomiting, are the most common adverse reactions to the macrolides; depending on dosage, these reactions may occur in up to 50% of patients, occasionally requiring early discontinuation of therapy. The mechanism is thought to be the binding of erythromycin to motilin receptors, with a consequent increase in gastrointestinal motility. Gastrointestinal side effects appear equally common for all the oral formulations and also occur with intravenous administration. Clarithromycin and azithromycin are better tolerated than erythromycin, although gastrointestinal distress is still their most common adverse effect.

Less common reactions include hepatotoxicity and ototoxicity. Hepatotoxicity is a rare, nonfatal complication that is usually associated with erythromycin estolate and presents as an allergic cholestatic jaundice. Ototoxicity is rare after oral administration but may occur in a dose-dependent pattern in up to 20% of adults who receive intravenous erythromycin (4 g/d) and have audiograms performed. Ototoxicity is usually reversible and mild. Allergic cutaneous reactions are observed in rare cases.

LINCOSAMIDES

The most common adverse effect of clindamycin is gastrointestinal distress. Diarrhea has been reported in up to 20% of patients and pseudomembranous colitis in 0.01 to 10%. The mechanism of pseudomembranous colitis is production of a toxin by *C. difficile* ([Chap. 145](#)). *C. difficile* colonizes the gastrointestinal tract and may produce a toxin when the normal flora is suppressed by clindamycin and other antibiotics, especially β -lactams. This toxin causes mucosal damage that results in cramps, pain, and diarrhea that may be bloody. Pseudomembranous colitis may follow both intravenous and oral administration and may not become manifest until after completion of therapy. Oral metronidazole or oral vancomycin is effective in treating symptomatic patients with toxin-positive stools, but some spores may survive, and relapse is frequent. Metronidazole is the *drug of choice* since oral treatment with vancomycin can select for vancomycin-resistant enterococci. Although diarrhea and pseudomembranous colitis can be caused by most antibacterial agents, the incidence in relation to the amount used appears to be highest for clindamycin. Allergic reactions (such as rashes and fever), hepatotoxicity, and neutropenia are observed only rarely.

CHLORAMPHENICOL

Chloramphenicol causes two types of bone marrow suppression: a dose-related, reversible suppression of all elements, which occurs commonly during therapy at the maximal recommended doses (4 g/d in adults), and an idiosyncratic, irreversible aplastic anemia, which occurs in approximately 1 in every 25,000 to 40,000 exposures. The irreversible form has been reported to follow all types of chloramphenicol treatment, including ocular administration, and often develops months after therapy is discontinued.

In premature neonates and infants, chloramphenicol can cause a dose-related "gray syndrome" that is characterized by cyanosis, hypotension, and death and that results from an inability of the newborn to metabolize the drug. These potentially serious toxicities and the availability of newer drugs have substantially reduced the indications for chloramphenicol.

TETRACYCLINES

Gastrointestinal effects are the most common adverse reactions to the tetracyclines. These problems may be related to a direct irritant effect, since tetracyclines can also cause esophageal ulceration when they dissolve before reaching the stomach. It is important that nighttime doses be taken with sufficient fluid. Concurrent food intake may improve tolerance, but absorption of tetracycline HCl is impaired when the drug is taken with food.

Hepatotoxicity has been reported after administration of >2 g of tetracycline intravenously and at lower doses during pregnancy. There are currently no indications for intravenous tetracycline treatment in pregnancy. All tetracyclines can cause phototoxic skin reactions; these reactions are most common with doxycycline. Other dermal reactions, including rash, are uncommon. Tetracyclines are contraindicated in children <8 years of age because of mottling of the permanent teeth; doxycycline may

be less likely than the other tetracyclines to cause this problem. Worsening of renal function in patients with preexisting renal dysfunction has been reported with use of tetracycline, although some of the increased azotemia may be due to amino acid catabolism. Doxycycline and perhaps minocycline appear to be free from these renal side effects. Alternative effective agents are nearly always available for use in patients with renal dysfunction. Minocycline can cause vertigo in up to 70% of women receiving therapeutic doses and in a lower percentage of men.

SULFONAMIDES AND TRIMETHOPRIM

The sulfonamides are generally safe, but the list of possible adverse reactions is very long. These compounds occasionally cause a number of allergic reactions, from relatively minor skin rashes (including maculopapular rashes and urticarial reactions typically appearing after a week of therapy) to severe or even life-threatening reactions such as erythema multiforme, Stevens-Johnson syndrome, and toxic epidermal necrolysis. The severe hypersensitivity reactions have occurred most commonly after treatment with the long-acting sulfonamides, such as sulfamethoxypyridazine, which are no longer used. Pyrimethamine plus sulfadoxine (Fansidar), used for malaria prophylaxis, may cause severe allergic reactions, including hepatic and hematologic toxicities, in addition to dermatologic toxicity. Photosensitivity reactions are also relatively common with sulfonamides.

Many patients infected with HIV who receive trimethoprim-sulfamethoxazole have adverse dermatologic reactions. These reactions are usually not life-threatening and appear to regress in many cases despite continuation of therapy. In high doses, trimethoprim interferes with the renal secretion of potassium. Hyperkalemia is relatively common among HIV-positive patients and is most often found after 7 days of trimethoprim-sulfamethoxazole therapy for pneumonia caused by *Pneumocystis carinii*.

Sulfonamides and trimethoprim may also cause severe hematologic complications, including agranulocytosis, hemolytic and megaloblastic anemia, and thrombocytopenia. These dose-related side effects may be greater in patients with renal insufficiency. Hemolytic anemia is most common in patients with glucose-6-phosphate dehydrogenase deficiency who take long-acting compounds; trimethoprim-sulfamethoxazole rarely causes hemolysis in such subjects. Granulocytopenia from trimethoprim-sulfamethoxazole is especially common in HIV-infected patients, occurring in 10 to 50% of this group.

Renal insufficiency, caused by crystals of the relatively insoluble acetyl metabolite, is observed primarily with the long-acting sulfonamides. Many cases of crystalluria in HIV-infected patients taking sulfadiazine for toxoplasmosis have been reported. A high level of fluid intake may prevent this complication.

It is recommended that sulfonamides not be administered to the newborn because of concerns that bilirubin may be displaced from protein-binding sites, with subsequent jaundice and kernicterus.

In addition to the preceding problems, sulfonamides may occasionally cause drug fever with serum sickness, hepatic toxicity (including necrosis), and systemic lupus

erythematosis.

FLUOROQUINOLONES

Fluoroquinolones are relatively safe; adverse reactions rarely require discontinuation of therapy. The most common reactions include gastrointestinal distress, such as nausea or diarrhea (<5%), and central nervous system effects, including insomnia and dizziness (<5%). Trovafloxacin is prone to causing dizziness, especially among women. However, of more serious concern is the report of more than 100 cases of symptomatic liver toxicity in patients receiving trovafloxacin, including 14 cases of acute liver failure strongly associated with trovafloxacin exposure. As a result, the Food and Drug Administration has recommended that this quinolone be used only in hospitalized patients with serious or life-threatening conditions in which the benefits offered by the drug outweigh its potential risks. Phototoxicity can be severe, especially with sparfloxacin. Rarely, hepatic and renal dysfunction and anaphylactoid and allergic reactions are observed. Tendon rupture has also been associated with quinolone use in rare instances. The use of these drugs is contraindicated in patients <18 years of age because of evidence in animals of cartilage damage in developing joints. In carefully selected situations in which the perceived benefits outweigh the risks (e.g., in adolescent patients with cystic fibrosis who have pulmonary exacerbations), fluoroquinolones may be useful for short-term therapy. They are contraindicated in pregnancy because of concern for the developing fetus.

RIFAMPIN

Rifampin is generally well tolerated but has several important side effects. Some patients have transient rises in hepatic aminotransferases, but these levels usually return to normal without discontinuation of the drug. Although hepatitis from rifampin itself develops only rarely, the drug is thought by some investigators to potentiate the hepatic toxicity of concomitantly administered isoniazid. Intermittent administration of rifampin (usually fewer than three times per week) has been associated with symptoms that seem to have an immunologic basis. These include flulike symptoms and (rarely) hemolysis, thrombocytopenia, shock, and renal failure. Minor gastrointestinal side effects, skin rashes, and interstitial nephritis have also been reported. Patients should be warned that rifampin and its metabolites cause secretions such as urine, tears, sweat, and saliva to turn orange and that contact lenses may be stained.

METRONIDAZOLE

Serious adverse reactions to metronidazole are uncommon. Gastrointestinal side effects such as nausea are most frequent but rarely necessitate discontinuation of therapy. Pseudomembranous colitis in association with metronidazole has been reported but is very rare. A metallic taste is relatively common, and stomatitis and glossitis are occasionally reported. Disulfiram-like reactions can occur if ethanol is ingested concurrently. Peripheral neuropathy develops in some patients, and seizures and encephalopathy have been reported after high doses and in patients with hepatic failure.

Concerns about mutagenicity and carcinogenicity from metronidazole have led to recommendations that it not be used in pregnancy (especially during the first trimester)

when alternative agents are available. Although retrospective studies have found no association between metronidazole and carcinogenesis, long-term administration of high doses should be avoided when therapeutic alternatives exist.

DRUG INTERACTIONS

Historically, clinically important interactions involving antibacterial drugs were generally of little concern, since β -lactams were the most widely used agents and rarely interacted with other drugs in a manner that affected the patient adversely. However, fluoroquinolones, macrolides, and rifampin are now more widely used, and interactions are of increasing concern. [Table 137-3](#) lists the most common and best-documented interactions of antibacterial agents with other drugs and characterizes the clinical relevance of these interactions. Coadministration of drugs paired in [Table 137-3](#) does not necessarily have clinically important adverse consequences. The result depends on the timing of administration, the dose and duration of therapy, the baseline serum concentration of the non-antibacterial drug administered, the patient's susceptibility to the pharmacologic effect of the non-antibacterial drug, and other, less-well-described cofactors. Recognition of the potential for an interaction before the administration of an antibacterial agent is crucial to the rational use of these drugs, since adverse consequences can often be prevented if the interaction is anticipated. [Table 137-3](#) is intended only to heighten awareness of the potential for an interaction. Additional sources should be consulted to identify appropriate options. **For further discussion of drug interactions, see [Chap. 70](#).*

MACROLIDES

Erythromycin and clarithromycin can inhibit the P450 enzyme CYP3A and thus the metabolism of many concurrently administered drugs, such as cisapride, theophylline, carbamazepine, terfenadine, astemizole, warfarin, and ergot alkaloids. The magnitude of the theophylline interaction is highly variable and is proportional to the dose and duration of erythromycin treatment. In contrast, cyclosporine levels predictably increase when erythromycin is administered, since CYP3A is responsible for cyclosporine metabolism. Decreased metabolism of terfenadine, astemizole, cisapride, and pimozone has been reported to cause severe cardiac dysfunction. Azithromycin has little effect on the metabolism of other drugs. In approximately 10% of patients receiving digoxin, concentrations increase when erythromycin is also given.

TETRACYCLINES

The most important interaction involving tetracyclines is the reduction in absorption when these drugs are coadministered with di- and trivalent cations, such as antacids, iron compounds, or dairy products. A similar interaction is seen with quinolones (see below). Food also adversely affects absorption of most tetracyclines. Inducers of hepatic isoenzymes, such as phenytoin and barbiturates, increase the clearance of doxycycline; although the clinical significance of this effect is unknown, use of an alternative antibiotic may be appropriate.

SULFONAMIDES

Sulfonamides may increase the hypoprothrombinemic effect of warfarin by inhibition of its metabolism and possibly by protein-binding displacement. Sulfonamides may also potentiate the effects of oral hypoglycemic agents and phenytoin through reduction in metabolism or displacement from serum protein.

FLUOROQUINOLONES

There are two clinically important drug interactions involving fluoroquinolones. First, like tetracyclines, all fluoroquinolones are chelated by divalent and trivalent cations, which prevent most of the dose from being absorbed. Second, certain fluoroquinolones (grepafloxacin, ciprofloxacin, and -- to a much lesser extent -- levofloxacin and trovafloxacin) can inhibit hepatic enzymes that metabolize theophylline, with resultant theophylline toxicity. The same mechanism accounts for increases in serum caffeine concentrations, but the clinical significance of this interaction is unknown. Scattered reports indicate that quinolones can also potentiate the nephrotoxicity of cyclosporine, exaggerate the effects of warfarin, and increase neurotoxicity when coadministered with nonsteroidal anti-inflammatory agents. However, these interactions have not been confirmed by controlled trials.

RIFAMPIN

Rifampin is an excellent inducer of many cytochrome P450 enzymes and increases the hepatic clearance of a number of drugs, including the following (with the indicated predictable outcomes): HIV-1 protease inhibitors (loss of viral suppression), oral contraceptives (pregnancy), warfarin (decreased prothrombin times), cyclosporine and prednisone (organ rejection or exacerbations of any underlying inflammatory condition), and verapamil and diltiazem (increased dosage requirements). Before rifampin is prescribed for any patient, a review of concomitant drug therapy is essential.

METRONIDAZOLE

Metronidazole can cause a disulfiram-like syndrome when alcohol is ingested; thus, patients taking metronidazole should be instructed to avoid alcohol. Inhibition of the metabolism of warfarin by metronidazole leads to significant rises in prothrombin times.

PROPHYLAXIS OF BACTERIAL INFECTIONS

Antibacterial agents are occasionally indicated for use in patients who have no evidence of infection but who have been or are expected to be exposed to bacterial pathogens under circumstances that constitute a major risk of infection. The basic tenets of antimicrobial prophylaxis are as follows: First, the risk or potential severity of infection should be greater than the risk of side effects from the antibacterial agent. Second, the antibacterial agent should be given for the shortest period necessary to prevent target infections. Third, the antibacterial agent should be given before the expected period of risk (e.g., surgical prophylaxis) or as soon as possible after contact with an infected individual (e.g., prophylaxis for meningococcal meningitis).

[Table 137-4](#) lists the major indications for antibacterial prophylaxis in adults. (The use of antibacterial agents in children to prevent rheumatic fever and otitis media under certain

circumstances is also common practice.) The table includes only those indications that are widely accepted, supported by well-designed studies, or recommended by expert panels. Prophylaxis is also used but is less widely accepted for recurrent cellulitis in conjunction with lymphedema, recurrent pneumococcal meningitis in conjunction with deficiencies in humoral immunity or CSF leaks, traveler's diarrhea, gram-negative sepsis in conjunction with neutropenia, and spontaneous bacterial peritonitis in conjunction with ascites.

The major use of antibacterial prophylaxis in the United States is for infections following surgical procedures. Antibacterial agents are administered just before the surgical procedure -- and, for long operations, during the procedure as well -- to ensure high levels in serum and tissues during surgery. The objective is to eradicate bacteria originating from the air of the operating suite, the skin of the surgical team, or the patient's own flora that may contaminate the wound. In all but colorectal surgical procedures, prophylaxis is predominantly directed against staphylococci. Prophylaxis is intended to prevent wound infection or infection of implanted devices, not all infections that may occur during the postoperative period (e.g., urinary tract infections or pneumonia). Prolonged prophylaxis merely alters the normal flora and favors infections with organisms resistant to the antibacterial agents used.

ANTIBACTERIAL COSTS AND INAPPROPRIATE USE

Use of antibacterial agents in hospitals in the United States accounts for an important percentage of all drug costs and may represent the largest expenditure for any pharmacologic class. In the outpatient setting, the costs of antibacterial drugs are second only to those of cardiovascular agents. A survey of office-based physicians found that between 1980 and 1992 there was a marked increase in the use of expensive broad-spectrum antimicrobials. It is not unusual for the purchase cost (in 2000 dollars) of a newer parenteral antibiotic to be \$1000 to \$2000 for a 10- to 14-day course of treatment. Therapy with a new oral antibiotic can easily cost \$50 to \$75. Administration costs, monitoring costs, and pharmacy charges must be added to these figures. While some newer antibacterial agents undeniably represent important advances in therapy, many newer drugs offer no advantage over older, less expensive agents.

Clinicians are understandably confused by the bewildering array of available drugs. Numerous surveys have reported that approximately 50% of antibiotic use is in some way "inappropriate." Aside from the monetary cost of unnecessary antibiotics, there are the costs of excess morbidity from adverse effects and drug interactions and the eventual costs of treating more resistant organisms. The following suggestions are intended to provide guidance through the antibiotic maze.

First, objective evidence regarding the merits of newer drugs is available through publications such as *The Medical Letter*, including the annual update of *Drugs of Choice*. Second, clinicians should become comfortable using a few drugs recommended by independent experts and should resist the temptation to use a new drug unless the merits are clear. A new antibacterial agent with a "broader spectrum and greater potency" or a "longer half-life and higher tissue levels" does not necessarily mean greater clinical efficacy. Third, the clinician must become familiar with local bacterial

susceptibility profiles. It may not be necessary to use a new drug with "improved activity against *P. aeruginosa*" if that pathogen is rarely encountered or if it retains full susceptibility to older drugs. Finally, with regard to inpatient use of antibacterial drugs, appropriate empirical treatment with one or more broad-spectrum agents may often be simplified, with use of a narrower-spectrum agent or even an oral drug, once the results of cultures and susceptibility tests become available. While there is an understandable temptation not to alter effective therapy, switching to a more specific agent, once the patient has improved clinically, does not compromise eventual outcome. If these guidelines are followed, the care of patients will not be undermined, many unnecessary complications and expenses will be avoided, and the useful life of valuable drugs will be extended.

(Bibliography omitted in Palm version)

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SECTION 5 -DISEASES CAUSED BY GRAM-POSITIVE BACTERIA

138. PNEUMOCOCCAL INFECTIONS - *Daniel M. Musher*

Streptococcus pneumoniae (the pneumococcus) was recognized as a major cause of pneumonia in the 1880s and has been a central focus of study leading to the modern understanding of humoral immunity. The name *Diplococcus pneumoniae* was assigned to the organism in 1926 on the basis of its appearance in Gram-stained sputum. In 1974, the organism was renamed *Streptococcus pneumoniae* because of its growth in chains in liquid medium. Around 1900, pneumococcal serotypes were recognized when the injection of killed organisms into a rabbit stimulated the production of serum antibody that agglutinated and caused increased capsular density of the immunizing strain as well as of some but not all other pneumococcal isolates. Ninety serotypes have now been identified, each possessing a unique polysaccharide capsule.

MICROBIOLOGY

Pneumococci are identified in the clinical laboratory as gram-positive cocci that grow in chains and are catalase-negative. They produce pneumolysin, a toxin that breaks down hemoglobin into a greenish degeneration product, thereby causing a hemolysis on blood agar. More than 98% of pneumococcal isolates are susceptible to ethylhydrocupreine (optochin), and virtually all pneumococcal colonies are dissolved by bile salts.

Peptidoglycan and teichoic acid are the principal constituents of the pneumococcal cell wall. The cell wall's integrity depends on the presence of numerous peptide side chains cross-linked by the activity of enzymes such as trans- and carboxypeptidases. β -Lactam antibiotics inactivate these enzymes by covalently binding their active site. Unique to *S. pneumoniae* and present in all strains is C (for "cell-wall") substance, a polysaccharide consisting of teichoic acid with a phosphorylcholine residue. Surface-exposed choline-binding proteins serve as a site of attachment for potential virulence factors, such as PspA, which may prevent phagocytosis. Nearly every clinical isolate of *S. pneumoniae* has a polysaccharide capsule.

There are two systems for numbering the 90 known distinct capsules of *S. pneumoniae*. In the American system, serotypes are numbered in the order in which they were identified. The strains that most frequently cause human disease were generally the earliest to be identified and thus tend to have lower numbers. The more widely accepted Danish system places serotypes into groups based on antigenic similarities; for example, Danish group 19 includes types 19F ("first recognized"), 19A, 19B, and 19C, which in the American system would be types 19, 57, 58, and 59, respectively. Serotyping was clinically relevant in the 1930s, when type-specific antisera were administered as therapy, and (although genetic typing is more specific) it has again become important for epidemiologic studies of the spread of antibiotic-resistant isolates in communities and among countries. Capsule switching has been documented and to some extent limits the epidemiologic reliability of serotyping.

EPIDEMIOLOGY

S. pneumoniae colonizes the nasopharynx and can be isolated from 5 to 10% of healthy

adults and from 20 to 40% of healthy children. Once the organisms have colonized an adult, they are likely to persist for 2 to 4 weeks but may persist for as long as 6 months. Pneumococci spread from one individual to another as a result of extensive close contact; transmission may be enhanced by poor ventilation. Day-care centers have been a site of spread, especially of penicillin-resistant strains of serotypes 6B, 14, 19F, and 23F. Outbreaks occur among adults in crowded living conditions -- e.g., in military barracks, prisons, and shelters for the homeless -- as well as among susceptible populations in settings such as nursing homes. The risk of pneumococcal pneumonia is not increased by contact in schools or workplaces (including hospitals).

The incidence of pneumococcal bacteremia is relatively high among infants up to 2 years of age and low among teenagers and young adults; rates increase beginning at around age 55. A surveillance study in South Carolina showed the incidences of pneumococcal bacteremia among infants, young adults, and persons [>]70 years of age to be 160, 5, and 70 cases per 100,000 population, respectively. Most cases of pneumococcal bacteremia in adults are due to pneumonia, and there are three to four cases of nonbacteremic pneumonia for every bacteremic case. Thus, it is estimated that there are 20 cases of pneumococcal pneumonia annually per 100,000 young adults and 280 cases annually per 100,000 persons over the age of 70. The incidence of pneumococcal bacteremia among adults exhibits a distinct midwinter peak and a striking dip in summer. In children, the incidence of bacteremia is relatively constant throughout the year except for a marked dip in midsummer. For reasons that are unclear, certain populations, including Native Americans, Native Alaskans, and African Americans, appear to be unusually susceptible to invasive pneumococcal disease. This enhanced susceptibility, not unlike that to infection with *Haemophilus influenzae*, is thought to have a genetic basis that thus far remains unelucidated.

PATHOGENETIC MECHANISMS

S. pneumoniae attaches to human nasopharyngeal cells through the specific interaction of bacterial surface adhesins, such as pneumococcal surface antigen A or choline-binding proteins, with epithelial cell receptors. Epithelial cell glycoconjugates containing the disaccharide GlcNAc₆-4Gal or asialo-GM1 glycolipid are possible binding sites. Pneumococcal phase variation, in which organisms switch between transparent and opaque, may also play a role in adherence. Upon culture, a mixed population of transparent and opaque pneumococcal colonies can be identified. Organisms from opaque colonies have relatively little peptidoglycan and relatively large capsules; those from transparent colonies have much more phosphorylcholine (which contributes to their capacity to adhere to mammalian cells) and less capsular polysaccharide. When an opaque colony is inoculated intranasally into an experimental animal, only those organisms that form transparent colonies persist. In contrast, after intraperitoneal inoculation, organisms that yield transparent colonies are rapidly cleared from the blood, while those that make opaque colonies resist clearance.

Once the nasopharynx has been colonized, infection results if the organisms are carried into anatomically contiguous areas such as the eustachian tubes or the nasal sinuses and if their clearance is hindered, for example, by mucosal edema due to allergy or viral infection. Similarly, pneumonia ensues if organisms are inhaled or aspirated into the bronchioles or alveoli and then are not cleared -- in many cases, because viral infection

or cigarette smoke or other toxic substances have increased mucus production and/or damaged ciliary action. A mechanism by which pneumococci may bind to pneumocytes after viral infection has been suggested. Pneumocytes activated by cytokines have been shown to express the receptor for platelet-activating factor. This receptor binds the phosphorylcholine residue on pneumococcal C substance, facilitating the adherence of pneumococci. Studies suggest that pneumococci may invade tissues by penetrating mucosal layers; the clinical significance of this finding remains to be determined.

Once pneumococci reach an area where they do not naturally belong, they activate complement by classic and alternative pathways and stimulate cytokine production, which leads to the attraction of polymorphonuclear neutrophils (PMNs). The polysaccharide capsule, however, renders the organisms resistant to phagocytosis. In the absence of anticapsular antibody, phagocytic cells such as alveolar macrophages have a limited capacity to ingest and kill pneumococci; a large bacterial inoculum and/or the compromise of phagocytic function allows the initiation of lung infection. Infection of the meninges, joint spaces, bones, and peritoneal cavity results from the spread of pneumococci through the bloodstream, usually but not always from a recognized focus of infection in the respiratory tract.

The capacity to cause disease reflects the ability of pneumococci to escape ingestion and killing by host phagocytic cells, on the one hand, and to stimulate an inflammatory response and damage tissues, on the other. Encapsulated pneumococci are poorly ingested and killed in vivo in the immunologically naive host or in vitro by mammalian phagocytic cells in the absence of anticapsular antibody and complement. Unencapsulated pneumococci virtually never cause invasive disease (although they can cause conjunctivitis), and mutants lacking a capsule are essentially avirulent in experimental animals. Symptoms of disease are largely attributable to the generation of an inflammatory response that may cause pain by increasing pressure (as in otitis media) or may interfere with vital bodily functions, such as oxygenation of blood (as in pneumonia) or cerebral function (as in meningitis). Cell-wall constituents of *S. pneumoniae*, including teichoic acid, C substance, and (in particular) peptidoglycan, activate complement by the alternative pathway; the reaction between cell-wall structures and antibody also activates the classic complement pathway. The result is the release of C5a, a potent attractant for [PMNs](#), into the surrounding medium. Inflammation is also facilitated by the ability of peptidoglycan to stimulate cytokine production, which activates endothelial cells to express selectin and integrin receptors for inflammatory cells. Inflammation in the central nervous system (CNS) during meningitis is a major contributor to neuronal cell injury. Pneumolysin, a thiol-activated toxin, exerts a variety of effects on ciliary cells and PMNs and also activates the classic complement pathway by direct binding of C1q. Injection of pneumolysin into the lungs of experimental animals produces the histologic features of pneumonia; in mice, immunization with this substance or challenge with genetically engineered mutants that do not produce pneumolysin is associated with a significant reduction in virulence. Autolysin may contribute to the pathogenesis of pneumococcal disease by lysing bacteria, thereby releasing their constituents and heightening the reaction with human tissues. The release of excitatory amino acids in neuronal tissue may contribute to damage caused by meningitis.

HOST DEFENSE MECHANISMS

Mechanisms of host defense may be immunologically nonspecific or specific. Nonspecific mechanisms include laminar airflow across mucous layers that filter inspired air, the glottal reflex, laryngeal closure, the cough reflex, the clearance of organisms from the lower airways by ciliated cells, and the ingestion by pulmonary macrophages and [PMNs](#) of small bacterial inocula that manage to reach alveolar spaces. Respiratory virus infection, chronic pulmonary disease, or heart failure compromises these mechanisms, predisposing to the development of pneumococcal pneumonia. Antibody to PspA and other pneumococcal constituents such as pneumolysin may be prevalent in the population and may contribute to immunity that is immunologically specific but not type specific.

Anticapsular antibody provides serotype-specific protection against pneumococcal infection. However, most healthy adults lack IgG antibody to most pneumococcal capsular polysaccharides. Antibody appears after colonization, infection, or vaccination. In the first few weeks after colonization, nonspecific mechanisms probably protect the host from infection. Thereafter, newly developed anticapsular antibody provides a high degree of specific protection. In contrast to this normal situation, adults who are at risk of aspirating pharyngeal contents and/or who have diminished mechanisms of lower airway clearance are at risk of developing pneumonia before antibody is produced. Similarly, children whose nasal mucosal membranes become acutely congested around the time of colonization are at risk of developing otitis media. Persons with a diminished capacity to form antibody remain susceptible for as long as they are colonized.

The risk of serious pneumococcal infection is greatly increased in persons with conditions that compromise IgG synthesis and/or the phagocytic function of [PMNs](#) and macrophages; this risk is also elevated in the presence of conditions associated with debilitation or malnutrition. Nearly all adults who are hospitalized for pneumococcal pneumonia have at least one of the conditions listed in [Table 138-1](#) and/or fall into one of the groups known to be at high risk on epidemiologic grounds. Prior hospitalization either predisposes to or serves as a strong marker for pneumococcal infection. The susceptibility of elderly individuals to pneumococcal pneumonia may reflect diminished clearance mechanisms as well as debilitation, malnutrition, and comorbid diseases. Although IgG responses to capsular polysaccharides, as measured by enzyme-linked immunosorbent assay (ELISA), are more or less normal in elderly persons, the functional capacity of the antibody appears to be decreased. The remarkably high incidence of pneumococcal infection -- perhaps 100-fold above baseline -- among persons with AIDS is noteworthy.

Once a pneumococcal infection has been initiated, the absence of a spleen predisposes to fulminant disease. The liver is able to remove opsonized (antibody-coated) pneumococci from the circulation; in the absence of antibody, however, only the slow passage of blood through the splenic sinuses and prolonged contact with reticuloendothelial cells in the cords of Billroth allow time for bacterial clearance. Patients without spleens may die of pneumococcal pneumonia and sepsis at such an early stage of the illness that pulmonary consolidation is not evident on x-ray before death but rather is found only at autopsy.

SPECIFIC INFECTIONS CAUSED BY *S. PNEUMONIAE*

S. pneumoniae causes infections of the middle ear, sinuses, trachea, bronchi, and lungs ([Table 138-2](#)) by direct spread from the nasopharyngeal site of colonization. Infections of the [CNS](#), heart valves, bones, joints, and peritoneal cavity usually arise by hematogenous spread; in rare cases peritoneal infection develops via ascent through the fallopian tubes. The CNS may also be infected by contiguous spread of organisms, as in patients who have a tear in the dura. Primary bacteremia -- i.e., the presence of pneumococci in the blood with no apparent source -- occurs commonly in children under 2 years of age and as a small percentage of all pneumococcal bacteremias in adults; if no therapy is given, a source may become apparent. Pleural infection results either from direct extension of pneumonia to the visceral pleura or from hematogenous spread of bacteria from a pulmonary or extrapulmonary focus; the route cannot be determined in any individual case.

Otitis Media and Sinusitis When fluid from the middle ear is cultured during acute otitis media or fluid from a paranasal sinus is cultured during acute sinusitis, *S. pneumoniae* is the most common isolate or is second only to nontypable *H. influenzae*. Whether in adults or in children, pneumococci are identified in about 40 to 50% of cases of otitis in which an etiologic agent is isolated. Prior infection by a respiratory virus or allergy is thought to contribute significantly to these pneumococcal infections by causing congestion of the openings to the eustachian tubes or the paranasal sinuses. Prospective studies of young children have shown that colonization precedes infection in most cases. For reasons that are unclear, serotypes 6B, 14, 19F, and 23F predominate both as colonizing and as infecting organisms of children; therefore, these serotypes are currently being studied most intensively for use in vaccines to be administered to young children.

Meningitis Except during outbreaks of meningococcal infection, *S. pneumoniae* is the most common etiologic agent of bacterial meningitis in adults. Because of the remarkable success of *H. influenzae* type b vaccine, *S. pneumoniae* now predominates among cases in infants and toddlers as well (but not among those in newborns). Meningitis develops either by the direct extension of infection from the sinuses or the middle ear or as a result of bacteremia with seeding of the choroid plexus. Favoring the former possibility are the association between acute otitis media and meningitis as well as the documented role of *S. pneumoniae* as the most common cause of recurrent bacterial meningitis associated with head trauma, cerebrospinal fluid (CSF) leak, and/or dural tear. Favoring the latter are the association between pneumococcal bacteremia from any source and meningitis as well as an autopsy study of temporal bone from children who died of bacterial meningitis, which yielded no evidence of extension from the middle ear.

In the meninges and subarachnoid space, pneumococcal peptidoglycan stimulates an intense inflammatory response mediated by the release of interleukin (IL)1, IL-6, C5a, tumor necrosis factor (TNF), and other proinflammatory cytokines. This inflammatory response results in raised intracranial pressure, brain edema, and decreased blood flow leading to meningismus, drowsiness, or coma. Focal neurologic signs may result from vasculitis with venous or arterial thrombosis, from cranial neuropathy due to entrapment or infarction, from local cerebritis, from subdural effusion, or from brain herniation ([Chap. 372](#)).

No distinctive clinical or laboratory feature differentiates meningitis due to *S. pneumoniae* from that due to other bacteria. Patients note the sudden onset of fever, headache, and stiffness or pain in the neck. Without treatment, there is a progression over 24 to 48 h to confusion and then obtundation. On physical examination, the patient looks acutely ill and has a rigid neck. In such cases lumbar puncture should not be delayed for computed tomography (CT) of the head unless papilledema or focal neurologic signs are evident. Typical CSF findings consist of pleocytosis (500 to 10,000 cells/uL) with a predominance of PMNs, an elevated protein level (100 to 500 mg/dL), and a decrease in glucose content. If antibiotics have not been given, large numbers of pneumococci can be seen in a Gram-stained specimen of CSF in nearly all cases, and specific therapy can be administered, although *Listeria* may be misidentified as the pneumococcus. If an effective antibiotic has already been given, the number of bacteria may be greatly decreased and microscopic examination of a Gram-stained specimen may yield negative results. In this situation, immunologic methods for the detection of pneumococcal capsule in the CSF may identify an etiologic agent in up to two-thirds of cases, although these methods have fallen out of favor. Most physicians prefer to use empirical broad-spectrum antibiotic therapy until the etiologic agent has been definitively identified and its susceptibility has been reported.

Pneumonia The distinctive symptoms and signs of pneumococcal pneumonia are (1) cough and sputum production, which reflect the proliferation of bacteria and the resulting inflammatory response in the alveoli; (2) fever; and (3) radiographic detection of an infiltrate.

Predisposing Conditions Pneumococcal pneumonia is most common at the extremes of age. Despite the undisputed role of *S. pneumoniae* as a major pathogenic bacterium for humans, the great majority of adults with pneumococcal pneumonia have underlying diseases that predispose them to infection. Otherwise-healthy military recruits involved in outbreaks of infection may be an exception to this rule; however, many of those affected have an antecedent viral-type illness that may reduce normal host resistance. In addition to prior viral respiratory illness, the most common predisposing conditions are alcoholism, malnutrition, chronic pulmonary disease of any kind, cigarette smoking, infection with HIV, diabetes mellitus, cirrhosis of the liver, anemia, prior hospitalization for any reason, renal insufficiency, and coronary artery disease (with or without recognized congestive heart failure). HIV infection is such an important predisposing factor that some authorities recommend that any young adult with pneumococcal pneumonia be tested for antibody to HIV.

Presenting Symptoms Patients often present with a preexisting respiratory condition that has distinctly deteriorated. If a viral upper respiratory illness is the predisposing factor, the patient may have felt unwell for several days, with coryza or a nonproductive cough and low-grade fever; at the time of onset of pneumonia, the temperature may rise to 38.9 to 39.4°C (102 to 103°F), and sputum production becomes prominent. In a patient who has chronic bronchitis, the sputum may increase in volume, become yellow or green and thicker than usual, and be associated with a fever that becomes progressively higher over 48 to 72 h. In a small proportion of cases, the onset of disease follows a hyperacute pattern in which the patient suddenly has a single episode of shaking chills followed by sustained fever and a cough productive of blood-tinged

sputum. This clinical picture is unfortunately called "classic," a vague term that is best avoided because many physicians believe that it means "most common," which is clearly not the case. In elderly subjects, the onset of disease may be especially insidious and may not suggest pneumonia at all. Persons in their eighties may have minimal cough, no sputum production, and no fever, instead appearing tired or confused. For the reasons noted above, the most abrupt progression of pneumococcal disease is seen in patients who have previously undergone splenectomy; these individuals may go from apparent good health to death in as little as 24 h. In pneumonia, pleuritic chest pain may result from extension of the inflammatory process to the visceral pleura; persistence of this pain, especially after the first day or two of treatment, raises concern about empyema (see "Complications," below). Nausea and vomiting or diarrhea, sometimes quite prominent, occur in up to 20% of cases. Clearly, the range of symptoms is sufficiently broad that there is no characteristic presentation to distinguish pneumococcal from other types of bacterial pneumonia (or from some types of nonbacterial pneumonia).

Physical Findings Patients with pneumococcal pneumonia usually appear ill and have a grayish, anxious appearance that differs from that of persons with viral or mycoplasmal pneumonia. Typically, the temperature is 38.9 to 39.4°C (102 to 103°F), the pulse 90 to 110 beats per minute, and the respiratory rate >20 breaths per minute. Elderly patients may have only a slight temperature elevation or be afebrile. Hypothermia is associated with increased morbidity and mortality. Herpes labialis appears in a small percentage of cases. Pain may cause diminished respiratory excursion (splinting) on the affected side. Dullness to percussion is noted in about half of cases, and vocal fremitus is increased. Breath sounds may be bronchial or tubular, and crackles are heard in most cases if enough air is being moved to generate them. Flatness to percussion at the lung base and inability to detect the expected degree of diaphragmatic motion suggest the presence of pleural fluid, which raises the possibility of empyema; the failure to assess fremitus, to distinguish dullness from flatness by percussion, or to examine for diaphragmatic excursion may leave the physician at the mercy of often ambiguous radiologic interpretations. The finding of a heart murmur, certainly if new, raises concern about endocarditis, a rare but serious complication. Hypoxia or the generalized response to pneumonia may cause the patient to be confused, but the appearance of confusion should raise concern about meningitis. Obtundation or neck stiffness should lead to an immediate consideration of this complication.

Radiographic Findings ([Fig. 138-CD1](#)) Pneumococcal pneumonia involves only one lung segment or a portion thereof in one-fourth of cases; it involves more than one segment but only one lobe or a portion thereof in another one-fourth of instances. Thus multilobar disease is seen in half of cases. Air-space consolidation is the predominant finding and is detected in 80% of cases. Air bronchogram (visualization of the air-filled bronchus against a background of consolidation in the alveoli) is evident in fewer than half of cases and is more common in bacteremic than in nonbacteremic disease. In rare instances, pneumococcal pneumonia leads to a lung abscess; a malignancy or a mixture of anaerobic and microaerophilic organisms is likely to be implicated as well. Although some pleural fluid may actually be present in half of cases, no more than 20% of patients have a sufficient volume of fluid to allow aspiration, and in only a minority of these patients is empyema documented.

General Laboratory Findings The peripheral-blood white blood cell (WBC) count exceeds 12,000/uL in the great majority of patients with pneumococcal pneumonia. However, the count is <6000/uL in 5 to 10% of persons hospitalized for pneumococcal pneumonia. Such a low count is strongly associated with lethal disease and is often but not always associated with bone marrow suppression due to alcohol ingestion. The serum bilirubin level may be modestly elevated; hypoxia, inflammatory changes in the liver, and breakdown of red blood cells in the lung are all thought to contribute to this increase. Levels of lactate dehydrogenase may be elevated. A variety of other abnormalities may be present, reflecting the contributory role of underlying diseases. **Abnormalities of pleural fluid in empyema are reviewed in Chap. 255.*

Differential Diagnosis Patients who present with community-acquired pneumonia may actually have infection of the lungs due to one of many organisms. The extensive list includes the following: *H. influenzae* or *Moraxella catarrhalis* in persons with little to predispose them other than chronic or acute inflammation of the airways; *Staphylococcus aureus* in persons who take glucocorticoids or who have major anatomic disruption of the airways; *Streptococcus pyogenes*; *Neisseria meningitidis*; anaerobic species in persons who have seizures or may have aspirated oropharyngeal or gastric secretions for some other reason; *Legionella*; *Pasteurella multocida* in dog or cat owners; gram-negative bacilli, especially in persons with severely damaged lungs who are taking glucocorticoids; viruses, especially influenza virus (in season), adenovirus, or respiratory syncytial virus; *Mycobacterium tuberculosis*; fungi, including *Pneumocystis carinii* (depending upon epidemiologic factors and the possible presence of HIV infection); *Mycoplasma*; *Chlamydia pneumoniae*, especially in older adults; and *Chlamydia psittaci* in bird owners. Many older men with lung cancer present with pneumonia, as do persons who have acute-onset inflammatory pulmonary conditions of uncertain etiology or those with pulmonary embolus and infarction. The breadth of this list vividly illustrates the difficulty of using empirical therapy for community-acquired pneumonia. Many of these diseases require evaluation, and specific therapy is available for an increasing number. Moreover, pneumococci -- perhaps the most common cause of community-acquired pneumonia -- are increasingly resistant to available antibiotics. Taken together, these factors favor precise determination of the etiology of a pneumonia syndrome whenever possible.

Diagnostic Microbiology An etiologic role for the pneumococcus in pneumonia is strongly suggested by the microscopic demonstration of large numbers of PMNs and slightly elongated gram-positive cocci in pairs and chains in the sputum (Plate VI-2). Capsules may be seen surrounding the bacterial forms. Examined areas of the slide must be free of buccal epithelial cells, which indicate the admixture of saliva with sputum; saliva may contain 10⁷ viridans streptococci per milliliter. When characteristic microscopic findings are noted, the identification of *S. pneumoniae* in sputum culture strongly indicates pneumococcal infection of the lower respiratory tract. In the absence of such microscopic findings, the identification of pneumococci by culture may be nonspecific, reflecting colonization of the upper airways. Culture is also less sensitive than microscopic examination for identifying pneumococci. Since most pneumococci do not produce distinctively mucoid colonies, their identification in the laboratory depends on the ability to select putative pneumococcal colonies for further study from among a-hemolytic streptococci of the mouth. In short, laboratory diagnosis by sputum culture relies on the quality of the specimen provided, the care with which the relevant

purulent component is separated for culture, and the assiduity with which α -hemolytic colonies are studied. These factors need to be considered when sputum cultures from patients who appear to have pneumococcal pneumonia are said to yield only "normal mouth flora" and when the medical literature describes what appear to be poor results of sputum culture. Because of the central role of microscopic examination in diagnosis, physicians may wish to view the slides with the microbiologist. Blood cultures yield *S. pneumoniae* in about 25% of cases of pneumococcal pneumonia. Modern, automated systems often yield positive blood cultures within 12 h after the sample is obtained.

Complications Empyema is the most common complication of pneumococcal pneumonia, occurring in about 2% of cases. As noted above, some fluid appears in the pleural space in a substantial proportion of cases of pneumococcal pneumonia, but this parapneumonic effusion usually reflects an inflammatory response to infection that has been contained within the lung, and its presence is self-limited. When bacteria reach the pleural space -- either hematogenously or as a result of contiguous spread, possibly across lymphatics of the visceral pleura -- empyema results. The finding of frank pus, a positive result on Gram's staining, or the presence of fluid with a pH of ≤ 7.1 indicates the need for aggressive and complete drainage, preferably by prompt insertion of a chest tube, with verification by [CT](#) that fluid has been removed. If there is no response, thoracotomy is indicated. Persistence of fever (even if low-grade) and leukocytosis after 4 or 5 days of appropriate antibiotic treatment for pneumococcal pneumonia suggests empyema. In this setting, the diagnosis is exceedingly likely if the x-ray shows the persistence of pleural fluid. At this stage, thoracotomy is often needed for cure. Aggressive drainage is likely to reduce morbidity and mortality from empyema ([Chap. 262](#)).

Other Syndromes The appearance of pneumococcal infection at other, usually sterile body sites indicates hematogenous spread, either during frank pneumonia or, in a smaller proportion of cases, from an inapparent focus of infection. A case of pneumococcal endocarditis is seen every few years at large tertiary-care hospitals. Purulent pericarditis due to this organism, occurring as a separate entity or together with endocarditis, is even rarer. Most cases of spontaneous bacterial peritonitis in children and some cases in adults are caused by *S. pneumoniae*. Peritonitis in women may be related to the use of an intrauterine contraceptive device, and pneumococcal infections of the female reproductive organs continue to be described. Septic arthritis can arise spontaneously in a natural or prosthetic joint or as a complication of rheumatoid arthritis. Osteomyelitis in adults tends to involve vertebral bones. Epidural and brain abscesses are rarely described. Cellulitis can develop and does so most often in persons who have connective tissue diseases or HIV infection. The appearance of any of these unusual pneumococcal infections in a young adult may suggest that tests for HIV infection should be undertaken.

TREATMENT

Antibiotic Susceptibility β -Lactam antibiotics, the cornerstone of therapy for serious pneumococcal infection, bind covalently to the active site and thereby block the action of the cell-membrane enzymes (endo-, trans-, and carboxypeptidases) that are responsible for cell-wall synthesis. These enzymes were identified by their reaction with radiolabeled penicillin and thus are called *penicillin-binding proteins*. In the 1960s,

virtually all clinical isolates of *S. pneumoniae* were susceptible to penicillin (i.e., were inhibited in vitro by concentrations of <0.06 ug/mL). During the past 20 years in Europe and the past 10 years or so in the United States, a steadily increasing number of pneumococcal isolates have shown some degree of resistance to penicillin. Resistance results when spontaneous mutation or acquisition of new genetic material alters penicillin-binding proteins in a manner that reduces their affinity for penicillin, thereby necessitating a higher concentration of penicillin for their saturation. The genetic information acquired also conveys resistance to other antibiotics. Mutation and selection of strains in communities in the United States -- especially in areas of high antibiotic use, such as day-care centers -- and spread of identifiable strains from other countries where antibiotics are available without prescription have contributed to the prevalence of resistance.

For most of the antibiotic era, pneumococcal susceptibility was not studied in vitro because of the organism's high degree of susceptibility to virtually all recommended antibiotics. Clearly, the situation has changed, and it seems important to study pneumococcal isolates, especially those causing invasive disease, for antibiotic susceptibility. In 1997, about 20% of pneumococcal isolates in the United States were intermediately susceptible to penicillin [minimal inhibitory concentration (MIC), 0.1 to 1.0 ug/mL] and 15% were resistant (MIC,³ 2.0 ug/mL; [Table 138-3](#)). The clinical significance of the MIC varies with the infection being treated. An intermediately resistant strain (e.g., MIC = 0.5 ug/mL) behaves as a susceptible organism when it causes pneumonia, but probably not when it causes otitis and certainly not when it causes meningitis. As a result, susceptibility may eventually be redefined on the basis of the site infected -- a concept supported by pharmacokinetic considerations and validated by outcome studies. Amoxicillin, with two- and fourfold lower MICs, appears to be more active against *S. pneumoniae* than penicillin -- thus the emerging preference for amoxicillin. Penicillin-susceptible pneumococci are susceptible to all commonly used cephalosporins. Penicillin-intermediate strains are resistant to all first- and many second-generation cephalosporins (of which cefuroxime retains the best efficacy) but are susceptible to some third-generation cephalosporins, including cefotaxime, ceftriaxone, cefepime, and cefpodoxime. One-half of highly penicillin-resistant pneumococci are also resistant to cefotaxime and ceftriaxone, a higher proportion are resistant to cefepime, and nearly all are resistant to cefpodoxime. Pneumonia caused by intermediately penicillin-resistant strains responds well to b-lactam antibiotics. Pneumonia due to fully resistant strains also responds, but probably not as reliably; data that address this issue are currently being examined. Sinusitis and otitis media caused by intermediately resistant *S. pneumoniae* do not reliably respond to therapy, and failure of therapy may be common when these conditions are caused by highly resistant pneumococcal strains.

Resistance to erythromycin extends to the new macrolides, including azithromycin and clarithromycin. This resistance will certainly affect empirical therapy for bronchitis, sinusitis, and pneumonia. In the United States, the majority of macrolide-resistant pneumococci bear the so-called M phenotype (erythromycin MIC = 1 to 8 ug/mL) and are susceptible to clindamycin. In this case, resistance is mediated by an efflux pump mechanism; it is not yet known whether M-type resistance can be overcome by clinically achievable levels of macrolides. In Europe, most macrolide resistance is due to a mutation in *ermB*, which confers high-level resistance not only to macrolides but also to

clindamycin. Rates of resistance to doxycycline among pneumococci of varying susceptibility to penicillin are similar to those observed for macrolides, whereas the overall rate of pneumococcal resistance to trimethoprim-sulfamethoxazole (25%) is sufficiently high to discourage therapy with this agent unless an isolate is known to be susceptible.

The newer fluoroquinolones remain highly effective against pneumococci, with equal efficacy against penicillin-susceptible and -resistant strains. All pneumococci are susceptible to vancomycin, although it is feared that the acquisition of vancomycin resistance by enterococci and other gram-positive bacteria may eventually lead to pneumococcal transformation to resistance. Of drugs under study, the oxazolidinones and glycopeptides appear to be most promising, with [MICs](#) for drug-resistant *S. pneumoniae* strains no higher than for penicillin-susceptible strains. Resistance to streptogramins parallels that to macrolides and limits the usefulness of these drugs for the treatment of pneumonia.

Pneumococcal susceptibility patterns vary greatly between and even within individual communities and the data are in a state of flux. It does appear, however, that the constant trend is toward more widespread resistance.

General Therapy There has been increased emphasis on outpatient therapy in patients who are at low risk (as determined by PORT score according to criteria described by the Pneumonia Outcomes Research Team; [Chap. 255](#)). This approach appears to be safe. However, if the physician is in doubt about the severity of illness, the social circumstances, or the likelihood of compliance with the prescribed antibiotic regimen, it may be best to hospitalize the patient, at least briefly.

Specific Antibiotic Therapy

Pneumonia This section will deal primarily with the treatment of pneumonia that is known to be due to *S. pneumoniae*. The broader issue of empirical therapy for community-acquired pneumonia is covered in detail elsewhere ([Chap. 255](#)). However, a few general comments on empirical therapy apply. An important problem in treating pneumonia is that, without a good sputum sample that can be Gram-stained and examined microscopically, the etiologic agent is not known at the time when treatment needs to be initiated and is not likely to become known later. Empirical therapy in such cases must be effective against *S. pneumoniae*, which remains the most likely causative agent of community-acquired pneumonia, unless epidemiologic, clinical, and radiologic findings strongly favor another etiologic entity. If a good sputum sample is obtained and only *S. pneumoniae* is visible, therapy can be focused on this organism, although additional treatment may be added for organisms that are not visualized microscopically -- e.g., influenza virus in a patient hospitalized during an influenza outbreak. Even if the pneumococcus is suspected, a certain degree of empiricism is required, because the antibiotic susceptibility of the strain involved will not be known for 1 or 2 days.

OUTPATIENT THERAPY Amoxicillin (500 mg four times daily) effectively treats all cases of pneumococcal pneumonia except those caused by the most highly penicillin-resistant isolates. Neither cefuroxime nor cefpodoxime offers any advantages

over amoxicillin since these drugs are less likely, even at high dosages, to be active against highly resistant pneumococcal strains. One of the newer fluoroquinolones in an accepted dosage for pneumonia is highly likely to be effective. Doxycycline, azithromycin, or clarithromycin will be effective in 85 to 90% of cases and clindamycin in a higher proportion. The trend toward increasing resistance to all these drugs is worrisome. Because one-fourth of all isolates are now resistant to trimethoprim-sulfamethoxazole, this agent can no longer be recommended. Since none of these therapies ensures the kind of antibiotic coverage that it would have in the past, patients should be instructed to remain in close contact with the prescribing physician, especially if there is any deterioration in their condition. It is worth noting that an outcomes study using data from the mid-1990s, when the rate of resistance was lower, showed that treatment with any of the above-mentioned antibiotics was associated with a good outcome; the majority of cases in which an etiologic agent was identified were due to *S. pneumoniae*.

INPATIENT THERAPY Pneumonia caused by penicillin-susceptible or intermediately penicillin-resistant pneumococcal isolates is readily treatable with penicillin. The dosages that follow are acceptable against intermediately resistant strains and against many or most fully resistant isolates, although they are excessive for use against susceptible isolates. Lower doses, however, cannot be recommended initially because susceptibility is not known until 24 to 72 h after treatment is begun. Patients who are sick enough to be hospitalized should be treated promptly. Most physicians favor parenteral antibiotics, although oral administration of well-absorbed drugs may be acceptable if the patient is not vomiting or hypotensive. Recommended regimens include ceftriaxone (1 to 2 g/d) or cefotaxime (1 to 2 g every 6 to 8 h). Ampicillin (1 to 2 g every 6 h) is also widely used. A quinolone or azithromycin can be given parenterally or orally. About 10 to 15% of all pneumococci are resistant to macrolides, and 1 to 2% are resistant to quinolones. Much of the resistance to macrolides among pneumococcal isolates may be overcome by the administration of azithromycin at a dosage of 500 mg on the first day and 250 mg/d thereafter. Clindamycin is effective against a higher proportion of resistant pneumococci than are the macrolides. Vancomycin is uniformly effective against pneumococci and should be used for initial therapy if there is reason to believe that a patient is infected with a strain that is resistant to the drugs listed above. As antimicrobial resistance among pneumococci evolves, updated recommendations will be issued by the Infectious Diseases Society of America, the American Thoracic Society, and the Centers for Disease Control and Prevention (CDC).

Patients with severe allergic reactions to b-lactam antibiotics should receive vancomycin (500 mg intravenously every 6 h) or a quinolone. As noted above, there have always been treatment failures unrelated to the antimicrobial susceptibility of the organism; nevertheless, the failure of a patient to respond promptly should raise the question of resistance, and vancomycin should be given until the susceptibility of the infecting strain to other drugs has been documented. Of course, evidence for loculated infections (such as empyema) and/or other causes of fever should be sought.

DURATION OF THERAPY The optimal duration of treatment for pneumococcal pneumonia is uncertain. Penicillin-susceptible strains disappear from the sputum within several hours of the first dose of penicillin, and a single dose of procaine penicillin, which results in the maintenance of an effective antimicrobial level for 24 h, was said to

cure pneumococcal pneumonia in otherwise-healthy young adults at the time when all isolates were susceptible. Most older physicians treat pneumococcal pneumonia for 5 to 10 days. In the absence of reports of therapy failure, younger physicians have tended to treat the infection for 10 to 14 days. Prolongation of therapy is a two-edged sword, especially in debilitated patients, because the risk of complications increases with each day of antibiotic treatment, particularly in the hospital setting. A few days of close observation and parenteral therapy followed by an oral antibiotic -- with the entire course of treatment continuing for no more than 5 days after the patient becomes afebrile -- may be the best approach.

Otitis Media and Acute Sinusitis Current treatment recommendations for otitis media and acute sinusitis -- conditions whose pathogenesis and microbial etiology are similar -- are based on the following points: (1) Acute otitis media is the most common infection for which antibiotics are prescribed in the United States. (2) As noted above, *S. pneumoniae* is the most likely treatable cause; taken together, *H. influenzae* and *M. catarrhalis*, many strains of which produce β -lactamases, are implicated nearly as frequently as pneumococci. (3) In the absence of diagnostic tympanocentesis, the etiologic diagnosis is nearly always presumptive. (4) Because penetration into a closed space is required, high serum levels of an effective antibiotic are required to treat otitis caused by intermediately or fully resistant pneumococci. (5) Otitis due to *S. pneumoniae* is more likely to fail to respond and to produce complications without specific therapy. (6) Antibiotics that are effective against pneumococci and yet resist β -lactamases tend to be very expensive compared with amoxicillin.

As a result of these considerations, the [CDC's](#) Otitis Media Working Group recommends that initial therapy be amoxicillin in a high dosage -- e.g., 80 mg/kg for infants and toddlers or 500 mg four times daily for adults. If this regimen fails, highly penicillin-resistant pneumococci or β -lactamase-producing bacteria may be responsible, and a course of cefpodoxime, perhaps preceded by a single parenteral dose of ceftriaxone, is recommended. Once therapy has begun, patients must be monitored closely for a response. Despite the detection (by molecular analysis) of pneumococcal DNA in middle-ear fluid, chronic serous otitis ("glue ear") is probably not due to active infection and does not require antibiotic therapy.

Meningitis A reasonable recommendation is that pneumococcal meningitis be treated initially with cefotaxime (2 g every 6 h) or ceftriaxone (1 to 2 g every 12 h) plus vancomycin (500 mg every 6 h or 1 g every 12 h). Two drugs are given initially because the cephalosporin is likely to be effective against most isolates and readily penetrates the blood-brain barrier, whereas vancomycin is uniformly effective but has a somewhat unpredictable capacity to cross the blood-brain barrier. If the isolate is shown to be penicillin-susceptible, treatment can be continued with 24 million units of penicillin every 24 h, given every 4 h in divided doses or continuously. If the isolate exhibits reduced susceptibility to penicillin but is susceptible to cefotaxime or ceftriaxone, the administration of vancomycin may be discontinued. Rifampin inhibits the bactericidal activity of β -lactam antibiotics and probably should not be added to the regimen. The total duration of therapy for pneumococcal meningitis is 10 to 14 days. Despite the central pathogenic role of inflammation in meningitis, the use of glucocorticoids or other anti-inflammatory agents is controversial, even in children, in whom most of the relevant studies have been done. Data simply do not exist on which to base an informed

decision regarding the administration of glucocorticoids or cyclooxygenase inhibitors to adults with pneumococcal meningitis ([Chap. 372](#)). Meningitis should be treated in an intensive care unit and with the participation of appropriate consultants, generally including a neurologist and a specialist in infectious diseases.

Endocarditis Pneumococcal endocarditis is associated with rapid destruction of heart valves. Vancomycin should be given pending assays for the minimal bactericidal concentrations of b-lactam antibiotics. There is no clear evidence that the addition of another antibiotic to the regimen is beneficial; aminoglycosides are somewhat synergistic and rifampin or quinolones are antagonistic with b-lactams. Endocarditis and meningitis should be treated initially in an intensive care unit, with the participation of appropriate consultants. Patients with endocarditis should probably be treated in collaboration with an infectious disease consultant, a cardiologist, and a cardiovascular surgeon.

Other Therapeutic Modalities A variety of agents that block the action of [TNF- \$\alpha\$](#) , [IL-1](#), or platelet-activating factor have conferred no benefit in and may have had a detrimental effect on pneumococcal sepsis. Similar results have been obtained with glucocorticoids.

PREVENTION

Pneumococcal vaccine contains 25 ug of capsular polysaccharide from the 23 most prevalent serotypes of *S. pneumoniae*; vaccination stimulates antibody to most serotypes in most recipients. In adults under 55 years old, protection rates are at least 85%, even 5 years or longer after vaccination. The level and duration of protection decrease with advancing age, perhaps because of a diminished avidity of the antibody for the capsular polysaccharide. As a result, persons in their eighties have 50% protection for 3 years and very little or no protection thereafter. In subgroups of the population at high risk (e.g., debilitated elderly persons and individuals with severe chronic lung disease), vaccine has not been shown conclusively to be effective. Persons who most need the vaccine because of poor IgG responses are not likely to respond to immunization with significant increases in antibody level. Nevertheless, the poor average rate of response should not deter the physician from administering vaccine to individual patients who are at increased risk of pneumococcal infection. In light of the safety, low cost, and efficacy of vaccine and the emergence of antibiotic-resistant strains, the failure to vaccinate elderly persons and individuals who have conditions predisposing to pneumococcal disease is viewed by some authorities as a missed opportunity in public health policy.

The [CDC](#)'s Immunization Practices Advisory Committee has broadened its recommendations for pneumococcal vaccination to include all persons over the age of 2 years who are at substantially increased risk of developing pneumococcal infection and/or a serious complication of such an infection. General categories included within these recommendations are as follows: (1) persons over the age of 65; (2) persons with anatomic or functional asplenia, [CSF](#) leak, diabetes mellitus, alcoholism, cirrhosis, chronic renal insufficiency, chronic pulmonary disease, or advanced cardiovascular disease; (3) persons who have an immunocompromising condition associated with increased risk of pneumococcal disease, such as multiple myeloma, lymphoma, Hodgkin's disease, HIV infection, organ transplantation, or chronic use of

glucocorticoids; (4) persons who are genetically at increased risk, such as Native Americans and Alaskans; and (5) persons who live in special environments where outbreaks are particularly likely to occur, such as nursing homes. This list should not be regarded as all-inclusive.

Recommendations regarding revaccination seem to be somewhat inconsistent. A single revaccination is advocated for persons over the age of 65. Since antibody levels decline and there is no anamnestic response, it seems more reasonable simply to recommend revaccination at 5-year intervals, especially in persons over the age of 65, who tend to have almost no local reaction, and in splenectomized patients, who are most in need. If penicillin-resistant pneumococci continue to increase in prevalence, routine immunization of children over the age of 2 years should be considered. Pneumococcal vaccine has not been useful in children < 2 years of age, who do not respond well to polysaccharide antigens. In a recent field trial, a heptavalent protein-conjugate pneumococcal polysaccharide vaccine protected infants and children against pneumococcal pneumonia, bacteremia, and meningitis; this vaccine is likely to be released for administration to young children in the next few years.

(Bibliography omitted in Palm version)

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139. STAPHYLOCOCCAL INFECTIONS - Jeffrey Parsonnet, Robert L. Deresiewicz

The staphylococci are hardy and ubiquitous colonizers of human skin and mucous membranes and were among the first human pathogens identified. They cause a variety of syndromes, including superficial and deep pyogenic infections, systemic intoxications, and urinary tract infections. Staphylococci are the leading cause of bacteremia, surgical wound infections, and infections of bioprosthetic materials in the United States; in addition, they are the second leading cause of nosocomial infections. Organisms of this genus are also a significant cause of bacterial food poisoning.

Staphylococcus aureus is the most important human pathogen in the genus. It remains a major public health concern due to its tenacity, potential destructiveness, and increasing resistance to antimicrobial agents. Although less virulent, the coagulase-negative staphylococci (CoNS), especially *S. epidermidis*, adhere avidly to prosthetic materials and are important nosocomial pathogens, especially of compromised hosts. Another CoNS species, *S. saprophyticus*, is a common cause of urinary tract infections.

TAXONOMY AND MICROBIOLOGY

Members of the genus *Staphylococcus* are nonmotile, nonsporulating gram-positive cocci, 0.5 to 1.5 μm in diameter, that occur singly and in pairs, short chains, and the irregular three-dimensional clusters from which their name is derived (Greek *staphule*, "grape-like"). Staphylococci can grow over a wide range of environmental conditions, but they grow best at temperatures between 30°C and 37°C and at a pH around neutrality. They are resistant to desiccation and to chemical disinfectants, and they tolerate NaCl concentrations up to 12%. With rare exceptions, the staphylococci are facultatively anaerobic. The more virulent staphylococci can clot plasma (coagulase-positive), while the less virulent cannot (coagulase-negative). Of the six recognized coagulase-positive staphylococcal species, the only important human pathogen is *S. aureus*, whose colonies are larger than those of *S. epidermidis*, are often pigmented (golden yellow), and are usually β -hemolytic on sheep blood agar. Twenty-eight species of CoNS are recognized. Of these, *S. epidermidis* is by far the most common nonurinary human isolate. Strains of *S. epidermidis* are typically white and nonhemolytic and may be tenaciously adherent as a result of their production of polysaccharide adhesin. *S. epidermidis* is followed in frequency by *S. hominis*, *S. haemolyticus*, and *S. warneri*. *S. lugdunensis* is increasingly recognized as a cause of serious human infection. *S. saprophyticus* is the most common staphylococcal urinary isolate.

STAPHYLOCOCCUS AUREUS

EPIDEMIOLOGY

Humans constitute the major reservoir of *S. aureus* in nature. The mucous membranes of the anterior nasopharynx are the principal site of carriage, with roughly 30% of healthy adults being so colonized at any point in time. Other common sites of colonization include the axillae, the vagina, damaged skin, and the perineum. Among postmenarcheal U.S. women, the rate of vaginal colonization by *S. aureus* ranges from

5 to 15% but rises to 30% during menses -- a change that is relevant to the pathogenesis of the toxic shock syndrome (TSS). Most adults are colonized by *S. aureus* intermittently, whereas 10 to 20% have persistent colonization and about the same percentage are never found to harbor the organism. Colonization is influenced by both microbial and host factors as well as by the nature of the competing nonstaphylococcal flora. Carriage is more common among persons with frequent staphylococcal exposure and those with habitual or chronic disruption of cutaneous epithelial integrity. Thus, colonization rates are higher among health care workers, dialysis patients, patients with type 1 diabetes, injection drug users, persons infected with HIV, and individuals with chronic dermatologic conditions. After 2 weeks in a hospital, colonization rates rise to 30 to 50%, and colonizing strains are more likely to be resistant to antibiotics.

Colonization of mucocutaneous sites is an important risk factor for staphylococcal infection. For example, surgical wound infection following cardiothoracic surgery is up to 10 times more likely among patients who harbor *S. aureus* in the nares preoperatively than among those who do not. The vast majority of postoperative wound infections of all types are caused by a strain of *S. aureus* that was present in the nares before surgery. Furthermore, presurgical clearance of carriage with topical and systemic antibiotics decreases the incidence of postoperative staphylococcal infection, but it is not yet standard practice to screen and treat patients before surgery.

PATHOGENESIS AND HOST DEFENSE

S. aureus causes two types of syndrome: *intoxications* and *infections*. The clinical manifestations of intoxications are attributable to the action of one or a few secreted products of the microorganism (toxins), and these clinical features can be reproduced by administration of the toxin(s) in the absence of the microorganism. The toxin can be produced either in vivo (as in [TSS](#) or staphylococcal scalded skin syndrome) or in a suitable vector that subsequently delivers it to the host (as in staphylococcal food poisoning). Infections, in contrast, involve bacterial proliferation, invasion or destruction of host tissues, and -- in most cases -- local and systemic inflammatory responses by the host to these events. The ability of a microorganism to infect is predicated on its ability to produce certain products that enable it to survive and prosper in the host (*virulence factors*), and *S. aureus* is particularly well-armed in this regard.

Steps in Pathogenesis The pathogenesis of staphylococcal intoxications is straightforward and involves four steps: colonization by a toxigenic strain of the bacterium, toxin production, toxin absorption, and intoxication. The pathogenesis of invasive infections is more complex and the steps are less discrete. They include colonization, invasion of the bacterium across epithelial or mucosal barriers, adherence to materials in the extracellular matrix, evasion or neutralization of host defenses, and destruction of host tissues. For both intoxications and infections, the entire process is carefully orchestrated by the bacterium in response to specific environmental conditions.

Physical Preservation of Cellular Integrity Staphylococci are robust and adaptable organisms that can survive under relatively harsh environmental conditions. A rigid cell wall confers shape and strength to the organisms. The major component of the cell wall, *peptidoglycan*, is responsible for its physical properties. Disruption of peptidoglycan

cross-linking by cell wall-active antibiotics (b-lactams or glycopeptides) renders staphylococci susceptible to lysis mediated by endogenous peptidoglycan hydrolases (*autolysins*). The osmotolerance exhibited by *S. aureus* enables the organism to grow without microbial competition in foods of low water activity and so sets the stage for contamination of food by staphylococcal enterotoxins (SEs), which cause food poisoning.

Colonization Staphylococcal colonization of the nasal mucosa is mediated by adherence of cell-surface components to host molecules (e.g., mucin carbohydrate). Factors contributing to colonization of other surfaces, such as the vaginal mucosa, are poorly understood. After colonization, the production of certain staphylococcal toxins [toxic shock syndrome toxin 1 (TSST-1), exfoliative toxins, or [SEs](#)] can ensue under the appropriate environmental conditions. Colonization may be transient or persistent; the latter condition increases the likelihood that organisms will gain access to deeper tissues, beginning the process of infection.

Invasion and Adherence to the Extracellular Matrix Staphylococci generally cannot invade through intact epithelial surfaces, which represent the primary line of antistaphylococcal defense. Invasion is facilitated by a mechanical break in the epithelium or by plugging of a gland or hair follicle. Once the epithelial barrier is breached, colonization of host tissues is facilitated by adherence of *S. aureus* to molecules present either on host cell surfaces or in the extracellular matrix. Several staphylococcal surface proteins, including protein A, function as adhesins by binding extracellular matrix molecules. These proteins have been designated "microbial-surface components recognizing adhesive matrix molecules" and may adhere to fibrinogen, fibronectin, collagen, elastin, and other serum constituents.

Destruction of Host Cells and Alteration of the Host Microenvironment A number of products of *S. aureus* alter the host environment in a way that benefits the bacterium. *Coagulase* is a secreted enzyme that binds prothrombin and thereby causes the conversion of fibrinogen to fibrin; it may aid in the establishment of an environment within host tissues that is protected from cells of the immune system or from antibiotics. *S. aureus* produces a number of *lipases*, which may enhance the organism's survival in sebaceous areas of the human body. *Hyaluronidase* hydrolyzes hyaluronic acid, a mucopolysaccharide present in extracellular ground substance; its action may facilitate the spread of the organism through the extracellular matrix to adjoining tissues. *Staphylokinase*, *thermonuclease*, and *serine protease* are other extracellular enzymatic products that may play roles in pathogenesis.

S. aureus produces a number of membrane-active toxins that probably contribute to pathogenesis by damaging host cells, although their exact role in pathogenesis remains uncertain. These toxins include α -, β -, and δ -hemolysins and the synergohymenotropic toxins (γ -hemolysin and Panton-Valentine leukocidin). *α -Hemolysin* (α -toxin) is the prototypic pore-forming toxin; it inserts into the cell membrane, creating ion-conductive channels that destroy membrane integrity. The toxin is dermonecrotic on subcutaneous injection, induces proinflammatory changes in mammalian cells, and -- in animal models -- induces many findings seen in sepsis, including hypotension and thrombocytopenia. The *synergohymenotropic toxins* are a family of bicomponent toxins. Their name derives from the fact that their two components are tropic for cell membranes and are

synergistically active against them. Like α -toxin, the synergohymenotropic toxins are pore-forming toxins. *Panton-Valentine leukocidin* is most active against polymorphonuclear cells, monocytes, and macrophages. It is dermonecrotic to rabbit skin, and strains producing it are strongly associated with human furunculosis.

Evasion of Host Defense Once staphylococci have breached mucosal or epithelial barriers, the host's immune response is directed at containing and eliminating them, principally by polymorphonuclear recruitment and phagocytic killing. The bacteria fight back by cloaking antigenic determinants on their surface, by interfering with the function of opsonins, by directly killing the phagocytes, and by developing strategies to survive within them. The pyogenic abscess, the histologic hallmark of staphylococcal infection, represents the battlefield for this encounter, in which the microorganism survives in an environment in which leukocyte function is impaired and into which antibiotics penetrate poorly. Although the abscess may contain the spread of the bacteria, patients with abscesses are symptomatic and usually require surgical drainage for relief.

Some staphylococcal components and products are direct chemoattractants for polymorphonuclear leukocytes; others provoke the release of chemoattractant cytokines that recruit phagocytic cells to the infected area. Histologic sections of early lesions typically reveal a central focus of organisms surrounded by a zone of necrotic debris, which in turn is surrounded by a zone of viable inflammatory cells. The toxic action of staphylococcal leukocidin may in part explain the zone of necrosis. After several days, fibroblasts populate the margin of the abscess and there elaborate collagen, which creates a true capsule around the abscess.

Staphylococcal cell-wall peptidoglycan activates complement, which is an important opsonin in persons lacking antibody to staphylococcal surface components. Peptidoglycan also acts as a general stimulator of inflammatory cytokine release and may thereby contribute to sepsis, but it is weaker in this regard than gram-negative lipopolysaccharide. Opsonic antibodies specific for peptidoglycan or capsule mediate phagocytosis by polymorphonuclear leukocytes and macrophages in vitro, although the role of antibody in vivo is less certain. There is considerable interstrain variation in susceptibility to opsonization, and acquired protective immunity to staphylococcal infection is generally thought *not* to develop. (In contrast, acquired antibody-mediated immunity to systemic staphylococcal intoxications does occur; e.g., >90% of healthy adults have protective antibody to the most common TSS toxin, [TSST-1](#).) Mitigating against opsonization is *protein A*, an important cell-surface component; protein A binds the Fc portion of IgG subclasses 1, 2, and 4 and thereby interferes with antibody-mediated opsonization. Polysaccharide capsule, which is produced by most clinical isolates, may also interfere with opsonization.

S. aureus has numerous defenses against killing after phagocytosis. Intracellular bacteria are usually killed rapidly by the oxidative burst within the phagosome, but staphylococcal catalase, which converts hydrogen peroxide to oxygen and water, detoxifies oxygen radicals and potentiates intracellular survival. Staphylococci are also taken up by nondedicated phagocytes, such as endothelial cells and osteoblasts, and may survive within them. One strategy for intracellular survival is the genesis of small-colony variants (SCVs). These slow-growing cells exhibit alterations in electron transport and generally produce reduced amounts of virulence determinants such as

α -toxin and coagulase. They are relatively resistant to cell wall-active antibiotics and aminoglycosides and are capable of persisting intracellularly for extended periods, thereby evading host defenses. Their slow growth also makes them less likely to be recovered in the clinical laboratory and then targeted for treatment. Their existence may in part explain the startling capacity of certain *S. aureus* infections (e.g., chronic osteomyelitis) to recrudesce after years of dormancy and the difficulty of curing infections in intravascular sites and bone. Prolonged exposure to aminoglycosides or to trimethoprim-sulfamethoxazole appears to be a risk factor for the development of SCVs.

Hosts at particular risk for staphylococcal infection include those with frequent or chronic disruptions in epithelial or mucosal integrity; those with disordered leukocyte chemotaxis, such as patients with the Chediak-Higashi or Wiskott-Aldrich syndrome; those whose phagocytes are defective in oxidative killing, as in chronic granulomatous disease; those with neutropenia or acquired functional deficiencies (e.g., deficiencies induced by exogenous glucocorticoids); and those with indwelling foreign bodies, which provide a matrix for staphylococcal adherence and biofilm formation and seriously impair phagocytic function. Patients with disorders of immunoglobulin or complement (especially C1-C4 deficiencies) are also at increased risk for *S. aureus* infection.

Superantigens The superantigens are V_{β} -restricted T cell mitogens: they bind directly and without prior processing to major histocompatibility class II molecules on the surface of antigen-presenting cells, thereby stimulating T cells on the basis of the sequence of the variable region of the β chain of the T cell receptor rather than on the basis of the epitope specified by this receptor. Accordingly, a superantigen may be able to stimulate >10% of the T cells in a given individual -- a percentage much higher than the ~1 in 10⁶ cells stimulated by conventional antigens. This massive T cell stimulation provokes an exuberant and dysregulated immune response characterized by the release of the cytokines interleukins 1 and 2, tumor necrosis factor, and interferon γ . TSS is a manifestation of this process, as some aspects of septic shock may be as well. *S. aureus* produces a number of superantigens, including the SEs, TSST-1, and possibly the exfoliative toxins (ETs). Ten SEs have been identified to date, several of which are common causative agents of staphylococcal food poisoning. The mechanism by which the SEs cause vomiting is uncertain but may involve direct neural stimulation of the autonomic nervous system rather than a local effect on the gastrointestinal mucosa. The superantigenic properties of TSST-1 and the SEs are thought to explain their ability to cause TSS, although the exact mechanism by which they craft the various clinical manifestations of TSS is uncertain. There is conflicting evidence as to whether the ETs, which cause scalded skin syndrome, are superantigens; structural data suggest that they may instead be related to the serine proteases.

Genetic Regulation of Virulence Genes The production of virulence factors by bacteria is typically tightly and coordinately regulated by genetic apparatuses that sense and respond to environmental cues. Such coordinate regulation enables an organism to rapidly tailor its repertoire of proteins to suit its changing needs, either as it passes between microenvironments or as the environment evolves around it. Many of the staphylococcal exoproteins are typical virulence factors in this regard. For example, α -, β -, and δ -hemolysins, TSST-1, staphylococcal enterotoxin B, serine protease, and thermonuclease are all produced during the late logarithmic phase of growth in batch culture, at a time when nutrients become scarce and cell density reaches saturation.

Their production is coordinately regulated and occurs reciprocally to that of the cell wall-associated staphylococcal proteins protein A and coagulase. Several genetic regulatory loci modulate these events, as do specific environmental conditions that presumably operate through those genetic loci. Foreign materials that increase the risk of [TSS](#) and conditions in food that predispose to staphylococcal food poisoning probably do so by presenting microenvironments that stimulate production of the relevant toxins. Similar events are undoubtedly operative within the environment of host tissues during infection, even in the absence of a foreign body.

Several distinct genetic loci regulate exoprotein production in *S. aureus*, the best-studied of which are *agr* (accessory gene regulator) and *sar* (staphylococcal accessory regulator). Both affect gene expression primarily at the level of transcription, and both activate the expression of secreted proteins and diminish the expression of cell wall-associated proteins during the late logarithmic phase of bacterial growth. Data suggest that *agr* may function primarily as a "quorum sensor," an apparatus that informs the bacterium of the density of staphylococci in its environment. Exoprotein regulation in *S. aureus* apparently results from a complex interplay of environmental factors and gene products.

STAPHYLOCOCCAL INTOXICATIONS

Toxic Shock Syndrome [TSS](#) is an acute, life-threatening intoxication characterized by fever, hypotension, rash, multiorgan dysfunction, and desquamation during the early convalescent period ([Fig. 139-1](#), [Fig. 139-CD6](#)). The disease was first characterized in 1978 but gained notoriety in 1980 upon the recognition of many cases among menstruating women. It is a relatively uncommon illness, with a reported annual incidence (among menstruating women) of 1 case per 100,000; it is likely, however, that the disease is substantially underreported, especially nonmenstrual cases. About half of all cases occur in settings other than menstruation and are distributed among individuals of both sexes and all ages. Menstrual and nonmenstrual cases are clinically indistinguishable. Among cases reported to the Centers for Disease Control and Prevention between 1985 and 1994, the minimum case-fatality rate was 2.5% for menstrual cases and 6.4% for nonmenstrual cases.

[TSS](#) is caused by any of several related exoproteins produced by *S. aureus*. [TSST-1](#) is the toxin most frequently implicated (causing virtually all menstrual cases), and staphylococcal enterotoxin B is the second most frequent. For illness to develop, an individual must be colonized or infected with a toxigenic strain of *S. aureus* and must lack a protective level of antibody to the toxin made by that strain. That TSS is primarily a disease of the young reflects the fact that >90% of adults have antibodies to TSS toxins.

Menstruation remains the most common setting for [TSS](#), but the disease can also complicate the use of barrier contraceptives and childbirth. Moreover, nonmenstrual TSS can ensue after superinfection of skin lesions of many types, including burns, insect bites, varicella lesions, and surgical wounds. Postoperative disease can develop from hours to weeks after any surgical procedure. Staphylococcal superinfection after influenza is a common setting for TSS, as is acute sinusitis. Overt infection with *S. aureus* is not required for the development of TSS; mere colonization with a toxigenic

strain may suffice. Accordingly, the primary site of toxin production in TSS may appear entirely benign.

[TSS](#) remains a clinically defined syndrome ([Table 139-1](#)). Patients meeting the case definition are severely ill, although milder forms of "staphylococcal toxin-mediated disease" do occur. The illness usually begins precipitously, with high fever and a complex of symptoms that may include nausea, vomiting, abdominal pain, diarrhea, muscular pain, sore throat, and headache. Dizziness is common as a manifestation of orthostatic or frank hypotension. The characteristic macular erythroderma develops over the first 2 days of illness. It is usually generalized but is sometimes locally confined; it can be evanescent or persistent. The patient's mental status is often abnormal to a degree that is out of proportion to the degree of hypotension. Conjunctival suffusion ([Fig. 18-CD3](#)), pharyngeal injection, and peripheral edema are evident in many cases; a so-called strawberry tongue develops in up to half of patients. In menstrual disease, the vaginal mucosa may be erythematous and a purulent vaginal discharge may be present, but these findings are not universal. Common laboratory abnormalities include azotemia, hypoalbuminemia, hypocalcemia, hypophosphatemia, creatine phosphokinase elevation, leukocytosis or leukopenia with a left shift, thrombocytopenia, and pyuria.

The early signs and symptoms of [TSS](#) resolve within the first few days of illness, after which complications of organ hypoperfusion, such as renal and myocardial dysfunction, fluid overload, and adult respiratory distress syndrome, dominate the picture. After about a week of illness, desquamation begins with superficial flaking of the skin of the torso, face, and extremities, which may be followed by full-thickness desquamation of the palms, soles, and digits. Common late sequelae include peripheral gangrene, reversible nail and hair loss, muscle weakness, and lingering asthenia and neuropsychiatric dysfunction.

The differential diagnosis of [TSS](#) is that of a severe febrile exanthem with hypotension. In the setting of menstruation accompanied by purulent vaginal discharge, the diagnosis may be obvious. The challenge is to recognize the less obvious cases, in which the exanthem may be fleeting, multiorgan dysfunction may be subtle, or (in nonmenstrual cases) a primary site of infection may be inapparent. Recovery of *S. aureus* supports the diagnosis, as does demonstration of toxin production by the strain and serologic susceptibility to the toxin. Other diagnoses to consider include streptococcal TSS, staphylococcal scalded skin syndrome, Kawasaki syndrome, Rocky Mountain spotted fever, leptospirosis, meningococcemia, gram-negative sepsis, exanthematous viral syndromes, and severe drug reactions. Staphylococcal TSS and streptococcal TSS ([Chap. 140](#)) can be clinically indistinguishable.

Treatment of [TSS](#) involves drainage of the site of toxin production, aggressive fluid resuscitation, and administration of antistaphylococcal antibiotics. Recent surgical wounds should be explored and irrigated, even when signs of inflammation are lacking; foreign bodies should be removed. Pressors should be used for sustained hypotension that is unresponsive to fluids. Electrolyte abnormalities, particularly hypocalcemia and hypomagnesemia, must be corrected. Penicillinase-resistant penicillins (nafcillin, oxacillin) and first-generation cephalosporins have been widely used in TSS. A growing body of clinical and laboratory evidence indicates, however, that a protein synthesis

inhibitor, such as clindamycin, might be superior to b-lactam agents. The authors recommend therapy with clindamycin (900 mg intravenously every 8 h), either alone or in combination with a b-lactam antibiotic [or vancomycin for patients perceived to be at risk for infection with methicillin-resistant *S. aureus* (MRSA)]. For a seriously ill patient in whom the diagnosis of TSS is uncertain, broad-spectrum antibiotics may be appropriate until the diagnosis is confirmed. A 14-day course of therapy -- some of which may be administered perorally -- is reasonable. Patients whose illness is severe enough to warrant vasopressors, who require mechanical ventilation, who have worsening renal function, or who have an undrainable focus of infection should be treated with intravenous immunoglobulin, which contains high levels of neutralizing antibody to TSS toxins. A single infusion of 400 mg/kg generates a protective level of antibody to [TSST-1](#) that persists for weeks. Glucocorticoids have not been shown to be of significant benefit.

Because vaginal staphylococcal carriage can be persistent or recurrent and because in more than half of all cases [TSS](#) does not elicit immunity, recurrent menstrual TSS is a concern; recurrent nonmenstrual TSS has also been reported. The risk of recurrent illness can be assessed by tests for seroconversion to [TSST-1](#). Women who do not seroconvert after acute illness (or who are not tested for antibody) should refrain indefinitely from using tampons or barrier contraceptives.

Staphylococcal Scalded Skin Syndrome This syndrome encompasses a range of cutaneous diseases of varying severity caused by [ET](#)-producing strains of *S. aureus*. The most severe form of staphylococcal scalded skin syndrome is termed *Ritter's disease* in newborns and *toxic epidermal necrolysis* (TEN) ([Fig. 18-CD5](#)) in older individuals. Milder and more common forms include *pemphigus neonatorum* and (in children and adults) *bullous impetigo* (see "Skin and Soft Tissue Infections," below). Persons >5 years old rarely develop staphylococcal TEN; those who do almost invariably have underlying disease (renal insufficiency, systemic immunosuppression). The rarity of the syndrome in adults has been ascribed to acquired immunity to the inciting toxins, to enhanced renal clearance of the toxins, and perhaps to diminished sensitivity to the action of the toxins.

Staphylococcal [TEN](#), or Ritter's disease, often begins with a nonspecific prodrome. The acute phase starts with the onset of an erythematous rash. The erythema begins in the periorbital and perioral areas and spreads to the trunk and centrifugally to the limbs. Pastia's lines may be apparent. The skin has a sandpaper texture and is often tender. Periorbital edema is common. In infants and children, fever and irritability or lethargy are common, but systemic toxicity is not. Within hours or days, wrinkling and sloughing of the epidermis begin; sloughing can be provoked by gentle stroking of the skin (Nikolsky's sign), even in areas that appear uninvolved. The denuded areas are red and glistening but not purulent, and staphylococci are not present. Exfoliation may continue in large sheets or in ragged snippets of tissue. Large, flaccid bullae may develop. As in thermal burns, significant fluid and electrolyte loss can occur at this stage, as can secondary infection. Within about 48 h, the exfoliated areas dry and secondary desquamation begins. The entire illness resolves within about 10 days. Mortality (from hypovolemia or sepsis) is ~3% among children but approaches 50% among adults. Treatment includes the administration of antistaphylococcal agents, fluid and electrolyte management, and local care to the denuded skin.

Staphylococcal Food Poisoning Between 2 and 6 h after ingestion of contaminated food, staphylococcal food poisoning begins abruptly with nausea, vomiting, crampy abdominal pain, and diarrhea. The diarrhea is usually noninflammatory and is of lower volume than that in cholera or toxigenic *Escherichia coli* infection. Fever and rash are absent, and the patient is neurologically normal. The majority of cases are self-limited and resolve between 8 and 24 h after onset. In severe cases, hypovolemia and hypotension can develop. Although most cases probably do not come to medical attention and are not diagnosed, staphylococcal intoxication is the second or third leading cause of diagnosed food poisoning in the United States.

Food poisoning is caused by the ingestion of any of the [SEs](#), which are produced by *S. aureus* in contaminated food before it is eaten. The presence of SEs in the food vector before its consumption accounts for the short incubation period of this illness. The SEs are heat stable, thus tolerating cooking conditions that kill the organisms that produced them. The disease has a high attack rate and is somewhat more common during the summer than at other times of the year. Processed meats and custard-filled baked goods are common food vectors, perhaps because staphylococci can tolerate conditions of high protein, salt, or sugar and so grow without competition in these environments. The most important epidemiologic risk factor in outbreaks of this disease is the ingestion of food that has been left at warm temperatures for prolonged periods, thereby allowing toxin production to occur before consumption. Contaminated preparation equipment and poor personal hygiene of food handlers are frequently implicated as well.

STAPHYLOCOCCAL INFECTIONS

S. aureus causes invasive disease by breaching host defense barriers, often after disruption or dysfunction of such barriers. The most common portals of entry leading to staphylococcal invasion are the skin and associated structures. A nidus for staphylococcal colonization and subsequent invasion is provided by chronic skin conditions, such as eczema and psoriasis; acute breaks in the skin, such as puncture wounds ([Fig. 139-CD1](#)), abrasions, and lacerations; and abnormalities of skin appendages, such as hair follicles and nails ([Fig. 139-CD2](#)). Colonization of the nasopharynx predisposes to respiratory tract infection after aspiration, obstruction (e.g., of a bronchus by carcinoma or of sinus ostia by trauma, edema, or polyps), or impaired ciliary function (e.g., in chronic bronchitis or acute viral infection). Intubation of the trachea provides a conduit by which upper respiratory flora, including pathogens such as *S. aureus*, can reach the lower respiratory tract.

Skin and Soft Tissue Infections *S. aureus* is the most common etiologic agent of skin and soft tissue infections ([Chap. 128](#)). Such infections are usually caused by endogenous flora -- i.e., strains of *S. aureus* that are harbored in the nares or other sites of colonization. Infection may represent a primary pathologic process, with direct invasion of skin and adjacent tissues, or a secondary process complicating preexisting lesions.

Staphylococcal infections originating in hair follicles range in severity from trivial to life-threatening. *Folliculitis* ([Fig. 128-CD3](#)) is an infection of follicular ostia; the

appearance is that of a domed yellow pustule with a narrow red margin. Infection is often self-limited, although healing may be hastened by topical antiseptics and more severe cases may benefit from topical or systemic antibiotics. A *furuncle* (often called a *boil*; [Fig. 139-CD3](#)) is a deep-seated necrotic infection of a hair follicle, most often located on the buttocks, face, or neck. Furuncles are painful and tender, and their appearance is often accompanied by fever and constitutional symptoms. Surgical drainage and systemic antibiotic treatment may hasten recovery and limit scar formation. Deep infection of a group of contiguous follicles is called a *carbuncle* ([Fig. 139-CD4](#)). This type of painful necrotic lesion occurs most commonly on the back of the neck, shoulders, hips, and thighs, typically in middle-aged or elderly men. There is intense inflammation of surrounding and underlying connective tissue, and the infection may be complicated by bacteremia. Surgical drainage and systemic antibiotic administration are indicated. *S. aureus* is also the most common cause of acute *paronychia*, infection of the lateral nail folds.

S. aureus causes *bullous impetigo* ([Fig. 128-CD1](#)), a superficial cutaneous disorder occurring predominantly in children. An epidermal split caused by [ET](#) results in the formation of 1- to 2-cm bullae containing neutrophils and organisms. *Nonbullous impetigo* is most often caused by β -hemolytic streptococci, but *S. aureus* can secondarily infect impetiginous lesions. Treatment of impetigo with a topical antibiotic, such as mupirocin, may suffice for mild and localized infection, whereas systemic therapy is indicated for widespread or severe disease or for infection accompanied by lymphadenopathy.

Cellulitis, a spreading infection of subcutaneous tissue, is occasionally caused by *S. aureus* ([Fig. 139-CD5](#)), but β -hemolytic streptococci are more common agents of this disease ([Chap. 140](#)). Secondary infection of surgical and traumatic wounds is more likely to be staphylococcal in etiology than is cellulitis arising from minor or inapparent breaks in the skin, and empiric treatment directed against both *S. aureus* and streptococci is reasonable in these settings. *Erysipelas*, the hallmark of which is a well-demarcated raised border, is a more superficial infection of the dermis and subcutaneous tissue; it is usually caused by group A streptococci and only rarely, if ever, by *S. aureus*.

Respiratory Tract Infections *S. aureus* can gain access to the lung parenchyma by two routes: aspiration of upper respiratory flora and hematogenous spread. Staphylococcal pneumonia is a relatively uncommon but severe infection, characterized clinically by chest pain, systemic toxicity, and dyspnea and pathologically by intense neutrophilic infiltration, necrosis, and abscess formation. *Pleural empyema* is a common complication and increases the already-considerable morbidity associated with this infection. Only rarely does *S. aureus* cause pneumonia without predisposing epidemiologic or host factors that favor colonization of the respiratory tract and/or that impair defense mechanisms. Residence in a chronic care facility, recent use of antibiotics, and hospitalization favor colonization -- and hence respiratory tract infection -- with *S. aureus*. Staphylococcal pneumonia most commonly follows tracheal intubation of a hospitalized patient or viral infection of the respiratory tract. Influenza virus is known both to increase respiratory colonization by *S. aureus* and to impair ciliary function (and therefore clearance of staphylococci). In a classic scenario, a patient (often elderly and/or institutionalized) develops a flulike respiratory illness and then, after several

days, deteriorates rapidly, with high fever, dyspnea, productive cough, and obtundation. The diagnosis of staphylococcal pneumonia is readily established by Gram's staining of expectorated sputum, which reveals abundant clusters of gram-positive cocci.

Hematogenous seeding of the lungs with *S. aureus* follows embolization from an intravascular nidus of infection. Common settings for septic pulmonary embolization are right-sided endocarditis (especially common among injection drug users) and septic thrombophlebitis, which is most often a complication of an indwelling venous catheter. Pneumonia is heralded by the acute onset of pleuritic chest pain and dyspnea; although diagnostic sputum may be lacking, a chest radiograph typically shows multiple nodular infiltrates, providing an important clue to both the diagnosis and the pathogenesis of disease.

Although not typically considered in the differential diagnosis of sore throat, *S. aureus* is occasionally isolated as the dominant organism from patients (especially children) with exudative *pharyngitis*. The illness may be accompanied by a scarlatiniform rash and may result in systemic toxicity (like that seen in [TSS](#)). Staphylococcal *tracheitis* may be diagnosed in children who have systemic toxicity and positive respiratory cultures but who lack pulmonary infiltrates. *S. aureus* is a prominent cause of *chronic sinusitis*, typically following the selection pressure of antimicrobial regimens that lack activity against this organism. Finally, *S. aureus* is a major etiologic agent of *sphenoid sinusitis*.

Infections of the Central Nervous System *S. aureus* gains access to structures of the central nervous system by hematogenous spread or by direct extension from contiguous structures. This organism is a prominent cause of *brain abscess*, especially as a result of embolization during mitral or aortic valve endocarditis. Such abscesses are often multiple, small, and scattered diffusely throughout the brain. Brain abscess can also develop by direct extension from frontoethmoid or sphenoid sinuses or from infected soft tissue after surgery or penetrating trauma. Patients with staphylococcal brain abscesses are more likely to have fever, meningismus, and other signs of infection than are patients with anaerobic bacterial or mixed-etiology brain abscesses. Purulent *meningitis* may accompany staphylococcal brain abscess or may develop during bacteremia in the absence of demonstrable abscesses.

S. aureus is the organism most likely to cause a variety of other space-occupying, suppurative intracranial infections. *Subdural empyema* usually develops by direct extension of osteomyelitis of the skull, after surgery or trauma, or in the setting of sinusitis. This condition may be accompanied by meningitis, epidural abscess, or intracranial phlebitis. The cardinal features of subdural empyema are fever, headache, vomiting, and signs of meningeal irritation. As the infection progresses, cerebral edema, often with infarction, may ensue and may be accompanied by alteration in mental status, seizures, and focal neurologic signs, which sometimes progress rapidly. The diagnosis should be suspected in any patient with meningeal signs and focal neurologic findings. Magnetic resonance imaging (MRI) is the diagnostic procedure of choice; lumbar puncture is contraindicated because of the danger of brainstem herniation. Early surgical drainage and treatment with an antibiotic that penetrates well into the central nervous system may be curative, although neurologic sequelae are not uncommon.

S. aureus is the most common cause of *spinal epidural abscess*, which develops most

often in association with vertebral osteomyelitis or diskitis. The diagnosis is suggested by some combination of fever, back pain, radicular pain, lower-extremity weakness, and bowel or bladder dysfunction, but the presentation is often subtle, resulting in delayed diagnosis. Patients may report only difficulty in walking or weakness, and objective findings may initially be lacking. The principal danger is the potential for necrosis of the spinal cord by compression and/or venous involvement. Early recognition of this condition is critical if long-term sequelae, such as paraplegia, are to be averted. An [MRI](#) scan of the spine establishes whether or not an epidural collection is present. Fluoroscopy- or computed tomography (CT)-guided needle aspiration may confirm the diagnosis, but an open procedure offers a higher yield. Prompt surgical decompression by laminectomy is often required for preservation of neurologic function, although a trial of antibiotic therapy alone may be considered if no focal neurologic deficits are detected at the time of diagnosis. Any deterioration in neurologic status should prompt urgent surgical intervention. The pathogenesis of *intracerebral epidural abscess* is similar to that of subdural empyema, with staphylococcal infection usually following sinusitis, craniotomy, or trauma. Clinical manifestations reflect the anatomy of the underlying osteomyelitis plus the mass effect of the abscess, cerebral edema, and (often) secondary involvement of the subdural space. Emergent surgical drainage is usually required for cure.

Finally, *S. aureus* is the most common cause of *septic intracranial thrombophlebitis*, typically following sinusitis, mastoiditis, or soft tissue infection of the face. Clinical manifestations reflect the underlying condition and the anatomic structures in contiguity with the infected vein or sinus. Focal neurologic deficits, particularly of cranial nerve function, are characteristic of cavernous sinus thrombosis. Sagittal sinus thrombosis may be manifested by leg and arm weakness and by altered mental status; infections of the lateral and petrosal sinuses also produce characteristic clinical syndromes. Intracranial phlebitis may accompany epidural abscess, subdural empyema, and meningitis and is sometimes clinically indistinguishable from other types of intracranial infection. [MRI](#) is the diagnostic procedure of choice.

Urinary Tract Infections *S. aureus* is an uncommon cause of urinary tract infection. Ascending infection almost exclusively follows instrumentation of the bladder (e.g., cystoscopy or placement of an indwelling catheter). Under other circumstances, the presence of *S. aureus* in the urine, even in low numbers, suggests staphylococcal bacteremia and hematogenous seeding of the kidneys, with or without abscess formation; staphylococcal endocarditis should be considered in this setting.

Endovascular Infections *S. aureus* is the most common cause of acute bacterial *endocarditis* of both native and prosthetic valves ([Chap. 126](#)). The organism may infect previously normal valves. Staphylococcal endocarditis presents as an acute febrile illness, rarely of more than a few weeks' duration; complications such as meningitis, brain or visceral abscess, peripheral vascular embolization, valvular incompetence with heart failure, myocardial abscess, and purulent pericarditis have often developed by the time a patient seeks medical attention. The valves most commonly involved are the mitral and/or the aortic except among injection drug users, in whom infection of the tricuspid valve is most common.

The diagnosis of endocarditis is suggested by a heart murmur and the presence of

conjunctival hemorrhages, subungual petechiae, or purpuric lesions on the distal extremities; it is readily confirmed by demonstration of high-grade bacteremia and echocardiography showing valvular vegetations. Echocardiography also helps establish which valve(s) are infected, the degree of valvular dysfunction or destruction, the quality of left ventricular function, and the presence or absence of annular or myocardial abscess. Transesophageal echocardiography (TEE) is more sensitive than transthoracic echocardiography (TTE) in detecting vegetations and abscesses, but it is also more invasive. TEE need not be performed in all cases of proven or suspected endocarditis. This approach is useful, however, in the setting of persistent bacteremia or fever (to evaluate for abscess) and in anticipation of surgery if TTE has not been sufficiently informative.

Native valve staphylococcal endocarditis carries a high mortality rate (on the order of 40%) and mandates prompt initiation of antimicrobial therapy. In addition to blood cultures and echocardiography, evaluation may include [CT](#) of the head and lumbar puncture if brain abscess or meningitis is suspected; a radionuclide study if osteomyelitis is suspected; and abdominal CT if visceral abscess is suggested by abdominal pain or persistent fever or bacteremia. Indications for valve replacement are the same as those in endocarditis caused by other organisms: persistent bacteremia (beyond 5 to 7 days of therapy), valvular dysfunction resulting in heart failure, perivalvular or myocardial abscess, or recurrent embolization. Early consultation with a cardiothoracic surgeon is advisable in all cases because of the high proportion of patients with *S. aureus* endocarditis (around half) who develop one of these complications and therefore require valve replacement, often urgently. Once there is an indication for removal of an infected valve, nothing is gained and much can be lost by delaying surgery. *S. aureus* infection of a prosthetic valve (as an early or a late complication of valve replacement) almost always requires surgery for one of the above indications.

Right-sided endocarditis, which most often develops in association with injection drug use or venous catheterization, is frequently complicated by septic pulmonary emboli but otherwise carries a lower rate of serious complications than left-sided disease. Surgery is rarely required for right-sided infection. A relatively short course of parenteral combination therapy (2 weeks) may be curative, and the prognosis is relatively good.

The propensity of *S. aureus* to adhere to and infect damaged tissues makes it the foremost cause of endovascular infections other than endocarditis. Vascular infection is a consequence of hematogenous seeding of damaged vessels, especially large arteries with atheromatous plaques, resulting in the development of a mycotic aneurysm. It may also develop by spread from a contiguous focus of infection (e.g., after vascular surgery), often resulting in an infected pseudoaneurysm, or by contamination of an intravascular device, resulting in septic phlebitis. Staphylococcal infection of an atherosclerotic artery (most commonly the abdominal aorta or iliac arteries), which may be aneurysmal to begin with, is a potentially catastrophic event. Such infections are associated with high-grade bacteremia, may result in rupture and massive hemorrhage, and require surgical resection and bypass of the infected vessel. Septic phlebitis is also associated with high-grade bacteremia and systemic toxicity but is less likely than arteritis to result in rupture. Persistent bacteremia suggests the need for surgical removal of infected thrombus or vein, but the technical difficulty of such surgery may

warrant an attempt at cure with antibiotics and anticoagulants alone.

Bacteremia A classic clinical scenario is that of a patient presenting with *S. aureus* bacteremia but without a demonstrable primary site of infection. Even in the absence of a changing murmur, peripheral embolic lesions, or a diagnostic echocardiogram, the possibility of endocarditis must be considered carefully in this situation. It is often hard to differentiate between endocarditis and bacteremia arising from another primary site; in addition, *S. aureus* may secondarily seed endovascular sites, such as heart valves or atheromatous plaques. Several criteria increase the likelihood that a patient has endocarditis as opposed to simple bacteremia: community (vs. nosocomial) acquisition of infection, absence of an apparent primary site of infection, and evidence of metastatic infection. The evaluation of a bacteremic patient should be tailored to the individual but may include an abdominal [CT](#) scan and a bone scan or gallium scan to detect an occult visceral abscess or osteomyelitis. [TEE](#) has demonstrated valvular abnormalities suggestive of endocarditis in up to one-fourth of bacteremic patients who lack clinical or [TEE](#) evidence of endocarditis. This finding has prompted some authorities to recommend TEE for all patients with staphylococcal bacteremia. The authors favor performance of this test for patients with persistent fever or bacteremia.

Complications of *S. aureus* bacteremia include abscesses of abdominal viscera, brain abscess, meningitis, septic arthritis, osteomyelitis, epidural abscess, and mycotic aneurysm. High-grade or persistent bacteremia mandates a thorough evaluation for these complications, even if a primary site of infection has been identified. The reported mortality rate for staphylococcal bacteremia ranges from 11 to 43%, with catheter-related infections carrying lower rates of complications and mortality than noncatheter infections.

Musculoskeletal Infections *S. aureus* is the most common cause of *acute osteomyelitis* ([Chap. 129](#)) in adults and one of the leading causes in children. Acute osteomyelitis develops as a result of either hematogenous seeding of bone (especially damaged bone) or direct extension from a contiguous focus of infection. The most common sites of hematogenous staphylococcal osteomyelitis in adults are the vertebral bodies; in children, the highly vascular metaphyses of long bones are most often affected. Acute osteomyelitis in adults usually presents with constitutional symptoms and pain over the affected area, often developing over several weeks or months. Leukocytosis and an elevated erythrocyte sedimentation rate or C-reactive protein level are laboratory clues to the diagnosis. Bacteremia may or may not be demonstrable. Four weeks of parenteral antibiotic therapy is usually curative.

S. aureus is also a prominent cause of *chronic osteomyelitis*, which develops at sites of previous surgery, trauma, or devascularization. In light of the hectic pace of many infections caused by *S. aureus*, chronic staphylococcal osteomyelitis can be impressively indolent; the infection may be asymptomatic for years or even decades, only to reawaken spontaneously and cause pain, sinus tract formation, and purulent drainage. A plain film of the affected area reveals bony destruction. The staphylococcal etiology of infection is best established by biopsy and culture of bone, as cultures of superficial or sinus tract drainage may yield misleading results. Cure requires surgical debridement of necrotic bone followed by a prolonged course of antibiotics. **For consensus definitions of acute and chronic osteomyelitis, see [Chap. 129](#).*

A special form of osteomyelitis is that associated with prosthetic joints or with internal or external fixation devices. Pain, fever, swelling, and decreased range of motion are cardinal features of an infected prosthesis. A plain film may suggest loosening of the prosthesis, often as radiolucency at the interface between bone and cement. *S. aureus* osteomyelitis associated with a prosthesis is infrequently cured by antibiotics alone. Persistent sepsis, persistent bacteremia, and clinical or radiologic evidence of loosening are absolute indications for removal of the prosthesis. *S. aureus* infection of fixation devices requires their removal, although this procedure may occasionally be delayed long enough to allow healing of the underlying fracture. Late relapses after apparent medical cure are not uncommon. A strategy of microbial suppression with oral antibiotics after a course of high-dose parenteral therapy is occasionally employed when removal of hardware is deemed too aggressive a measure for a particular patient.

S. aureus is a major cause of *septic arthritis* in adults ([Chap. 323](#)). Predisposing factors include injection drug use, rheumatoid arthritis, use of systemic or intraarticular steroids, penetrating trauma, and joints previously damaged by trauma or disease. Knees, hips, and sacroiliac joints are most frequently infected. In addition to parenteral antibiotics, cure requires either repeated joint aspirations -- the end points being sterilization of the joint space, a decrease in the number of leukocytes in the joint aspirate, and no reaccumulation of fluid -- or open or arthroscopic debridement and drainage. Failure to adequately drain joints infected with *S. aureus* poses a risk of permanent loss of function. *S. aureus* is also the most common cause of *septic bursitis* ([Chap. 325](#)), which most often involves bursae of the elbows, knees, and shoulders. As in arthritis, adequate drainage (via repeat aspiration, placement of a drain, or open debridement) hastens recovery and minimizes loss of function.

S. aureus infection of muscle (*pyomyositis*; [Chap. 128](#)) is relatively uncommon in temperate climates; *psoas abscess* is the most common such infection. The psoas muscle is seeded either hematogenously or by direct extension from the site of vertebral osteomyelitis; the results are pain upon extension of the hip and fever. Although formerly a cause of fever of unknown origin, psoas abscess is now relatively easy to diagnose by abdominal [CT](#) or [MRI](#). Psoas abscesses are occasionally amenable to drainage via a percutaneous catheter; if not, then surgical drainage is indicated. For reasons that are not well understood, most other cases of staphylococcal pyomyositis occur in the tropics (tropical pyomyositis); in the United States, pyomyositis is seen most often in patients with underlying conditions such as diabetes mellitus, alcoholism, immunosuppressive therapy, and hematologic malignancy.

DIAGNOSIS

The diagnosis of *S. aureus* infection is generally straightforward and is based on the isolation of the organism either from purulent material or from a normally sterile body fluid. Rarely should *S. aureus* growing from even a single blood culture be considered a contaminant. Clinical samples require no special transport media to preserve the viability of the organisms. Gram's staining of purulent material from a staphylococcal abscess invariably reveals abundant neutrophils and intra- and extracellular gram-positive cocci, which may be found singly or in pairs, tetrads, or clusters. *S. aureus* grows readily on standard laboratory media. Colonies that are catalase-positive

and coagulase- or thermonuclease-positive are identified presumptively as *S. aureus*. Commercial kits are also available for the identification of gram-positive cocci and are generally reliable for identification of *S. aureus*.

The diagnosis of staphylococcal intoxications (such as [TSS](#)) may be more difficult and may in fact rely entirely on clinical data. The contribution of the laboratory may be confirmatory -- for example, the demonstration of seroconversion to [TSST-1](#) following a compatible illness, the demonstration of toxin production in vitro by a strain isolated from a patient, or the detection of [SE](#) in a food sample.

TREATMENT

The essential elements of therapy for staphylococcal infections are drainage of purulent collections of pus, debridement of necrotic tissue, removal of foreign bodies, and administration of antimicrobial agents. The importance of adequate drainage cannot be overemphasized; all but the smallest of staphylococcal abscesses require drainage for cure. In skin and soft tissue infections, surgical drainage is occasionally all that is required for cure. It is very difficult to eradicate *S. aureus* infection in the presence of a foreign body, such as a piece of orthopedic hardware, an intravascular catheter or other device, or a pacemaker. For example, patients with catheter-associated bacteremia who are treated with antibiotics but do not have their catheters removed have been found to be six times more likely to experience a relapse or to die of their infection than are patients whose catheters are removed. Only under extraordinary circumstances should an attempt be made to cure such infections without removal of foreign material or debridement of necrotic tissue.

Antimicrobial Resistance The relentless spread of antibiotic resistance among strains of *S. aureus* is one of the great challenges facing clinicians today. Within 4 years of the introduction of penicillin G into clinical practice in 1941, β -lactamase-mediated resistance to penicillin was reported. As additional antibiotics became available in the 1950s, resistance rapidly emerged to them as well. Bacterial killing by β -lactam antibiotics depends on binding of the drugs to penicillin-binding proteins (PBPs), a group of transpeptidases that catalyze the terminal steps in peptidoglycan assembly. *S. aureus* normally produces four PBPs, all of which are inhibited by β -lactam antibiotics and several of which are essential for bacterial integrity and multiplication. Penicillin resistance in *S. aureus* is largely due to bacterial production of β -lactamase, a serine peptidase that enzymatically degrades the β -lactam ring of penicillin, thereby inactivating the drug before it can interact with the PBPs. In most communities, >90% of *S. aureus* strains produce β -lactamase and hence are resistant to penicillin.

Methicillin, the first β -lactamase-stable semisynthetic penicillin, was introduced in 1960; it took only 1 year, however, for an [MRSA](#) strain to be isolated. Classic methicillin resistance is encoded by the methicillin resistance determinant (*mec*), a 30- to 50-kb transposon-like segment of DNA that is present in MRSA strains and absent from sensitive strains. The *mecA* gene encodes a variant [PBP](#) called *PBP2 ϕ* or *PBP2a*. PBP2 ϕ has reduced affinity for β -lactam antibiotics and can substitute for the essential PBPs if they have been inactivated by β -lactams. MRSA strains are resistant to the action of all β -lactam antibiotics, including penicillins, cephalosporins, and carbapenems. Since the early 1980s, these strains have tended to be resistant to most

other antibiotics as well, including chloramphenicol, tetracyclines, and macrolides, through other resistance mechanisms. Nosocomial (as opposed to community-acquired) isolates of MRSA are especially likely to be multidrug-resistant. Classic methicillin resistance can be detected readily in the clinical microbiology laboratory by a variety of techniques. An additional mechanism of relative resistance to methicillin -- hyperproduction of β -lactamase -- has been described, but the clinical significance of this form of resistance is uncertain.

Until recently, all strains of [MRSA](#) remained susceptible to vancomycin (if nothing else), making this the drug of choice for the treatment of infections caused by suspected or proven MRSA. Unfortunately, the efficacy of vancomycin for serious *S. aureus* infections, regardless of susceptibility to other agents, is suboptimal (see "Selection of Antibiotics," below). Furthermore, resistance of *S. aureus* to vancomycin has now emerged as well (see below), making the search for new antistaphylococcal agents all the more urgent. If isolates are shown to be susceptible to clindamycin or trimethoprim-sulfamethoxazole, these agents can be effective for treatment of MRSA infection -- but again, many strains are resistant. Two newly licensed antibiotics, representing the vanguard of two new classes of drugs, may prove to be useful for treatment of infections caused by MRSA. A new antibiotic that combines two streptogramins, quinupristin and dalbavapristin, blocks protein synthesis at two ribosomal sites, resulting in a synergistic bactericidal effect on *S. aureus* and other gram-positive cocci. Linezolid, the first representative of the new oxazolidinone class of antibiotics, also demonstrates excellent activity against *S. aureus*, including multidrug-resistant strains of MRSA. Oxazolidinones are bacteriostatic, but resistance to them is unusual and there is no cross-resistance with other classes of compounds. Until data reveal the relative efficacies and toxicities of vancomycin and these new compounds, however, vancomycin remains the drug of choice for treatment of MRSA infections.

In 1996, a long-predicted monster -- *S. aureus* with decreased susceptibility to vancomycin -- finally emerged from the theoretical nightmares of microbiologists into the clinical realm. The term *vancomycin-intermediate S. aureus* (VISA) has been widely used to describe these strains, whose minimum inhibitory concentrations (MICs) of vancomycin (8 to 16 $\mu\text{g/mL}$) should theoretically confer only intermediate resistance to this agent. The clinical experience has been one of treatment failure, however. In 1997, four patients with VISA infection were reported from Japan and the United States; all died, although only one death was a direct result of the VISA infection. As of this writing, about a dozen additional cases have been reported from Asia, North America, and Europe, and the VISA genotype is already widespread in Japan. The risk factors for infection with VISA are uncertain because of the small number of reported cases, but they appear to include a history of dialysis, multiple prior courses of antibiotics (including vancomycin), admission to an intensive care unit, and prior infection with [MRSA](#).

The mechanism for decreased susceptibility to vancomycin in *S. aureus* appears to be novel and unrelated to the mechanism of resistance of vancomycin-resistant enterococci (VRE). [VISA](#) strains have unusually thick extracellular matrices that make it more difficult for vancomycin to reach its binding site at the level of the murein monomer of the cell wall. In addition, peptidoglycan from VISA appears to bind more vancomycin than does peptidoglycan from susceptible strains of *S. aureus*. These two factors create a "vancomycin sink," resulting in an increase in the [MIC](#) of vancomycin. VISA strains

may also be characterized by increased expression of [PBPs](#), slower growth, and decreased autolysis, all potentially contributing to antimicrobial resistance. Vancomycin resistance within a population of organisms is expressed in a heterogeneous manner, which may lead to difficulty in detection of the phenotype by usual susceptibility testing and may explain the failure of antimicrobial therapy in some cases. Therefore, infection with VISA should be suspected in any patient for whom seemingly appropriate therapy with vancomycin is ineffective. Removal of prosthetic material associated with infection is even more critical than usual in treating a patient infected with [MRSA](#) or VISA. Antimicrobial therapy for infections caused by VISA is discussed below (see "Selection of antibiotics").

In recent years, the percentage of staphylococcal isolates that are methicillin-resistant has risen substantially in U.S. hospitals; this trend has been driven by widespread (and often indiscriminate) antibiotic use. In some tertiary care institutions, up to 40% of *S. aureus* isolates are now resistant to methicillin, although rates of 5 to 15% are more typical. This situation is problematic for several reasons. Hospitalized patients colonized with [MRSA](#) are at increased risk (up to fourfold higher) of developing staphylococcal bacteremia than are patients colonized with methicillin-sensitive strains. The extent to which this difference reflects differences in bacterial virulence (as opposed to host factors or appropriateness of therapy) remains unclear; MRSA strains have not been shown consistently to be more virulent than sensitive strains, but the breadth of their resistance renders colonization more persistent, which in turn increases the rate of infection. In addition, higher rates of MRSA infection within an institution cause increased use of vancomycin, which contributes to the emergence of [VRE](#) and apparently puts additional pressure on MRSA to develop resistance to vancomycin as well.

A second development over the past decade has been an apparent increase in the incidence in some communities of [MRSA](#) infection among individuals without apparent risk factors for MRSA. Previously identified risk factors for MRSA include residence in a long-term-care facility; hospitalization; chronic liver, lung, or vascular disease; dialysis; malignancy; and prolonged exposure to antibiotics. An increase in community-acquired cases of MRSA infection has been reported in several locations in the United States and suggests a change in the epidemiology of infection with this organism. It is hypothesized that strains of MRSA spread from the hospital to the community, where continued exposure to antibiotics (both appropriate and unnecessary) leads to their survival advantage and persistence. These reports have been based on retrospective observations, however, and have not yet been confirmed by prospective studies. The obvious question is whether β -lactam antibiotics should continue to be used as the empirical agents of choice for community-acquired staphylococcal infections. For the time being, the incidence of MRSA infection in the community seems too low to justify more widespread use of vancomycin in this setting. There are situations, however, in which empirical use of vancomycin for community-acquired infections is justifiable and even advisable, as discussed below.

Selection of Antibiotics ([Table 139-2](#)) Although most pathogenic strains of *S. aureus* are resistant to penicillin, the development of penicillins and cephalosporins that are resistant to β -lactamase has allowed these classes of antibiotics to remain useful for treatment of most *S. aureus* infections. Nafcillin and oxacillin, both of which are

b-lactamase-resistant penicillins, are the drugs of choice for parenteral treatment of serious staphylococcal infections. Penicillin remains the drug of choice for infections caused by susceptible organisms. Drug combinations consisting of a penicillin plus a b-lactamase inhibitor are also effective but are best reserved for treatment of polymicrobial infections. Penicillin-allergic patients can usually be given a cephalosporin, although caution should be exercised if the prior adverse reaction to penicillin was anaphylaxis. Of the cephalosporins, the first-generation agents (e.g., cefazolin) are preferred for reasons related to cost and breadth of spectrum. For patients who are intolerant of all b-lactam agents, the best alternatives for parenteral administration are vancomycin and clindamycin. Dicloxacillin and cephalexin are recommended for oral treatment of minor infections or for continuation therapy; clindamycin is an alternative oral agent for most strains. Routine use of quinolones is not recommended because of the possibility of emergent resistance during therapy.

Use of vancomycin has increased dramatically over the past 20 years in response to the emergence of [MRSA](#), for which it has often been the sole therapeutic option, and the increasing number of infections caused by [CoNS](#) and other gram-positive cocci. Vancomycin's favorable pharmacokinetic properties and relatively low toxicity profile have also contributed to its widespread use. Unfortunately, increased use of vancomycin has resulted in the emergence of both [VRE](#) and [VISA](#), which are now poised to be the microbial scourges of the next decade. Equally important, however, is the fact -- often not recognized by practitioners -- that vancomycin is *less effective* than numerous other agents at our disposal. Vancomycin is generally only weakly bactericidal or even bacteriostatic for many strains of *S. aureus*. Studies of animal models have repeatedly shown vancomycin to be inferior to b-lactam agents for treatment of serious staphylococcal infections. Bacteremia is cleared more slowly in patients treated with vancomycin than in those treated with b-lactams, and clinical cure rates with vancomycin are significantly lower than with b-lactams as well. For all of these reasons, use of vancomycin should be reserved for situations in which there are no suitable alternative agents. It should not be used routinely for prophylaxis of staphylococcal infections, for empirical therapy in patients with fever and neutropenia (unless staphylococcal infection is especially likely), for decontamination of the digestive tract, for clearance of MRSA colonization, or for treatment of established gram-positive infections not due to resistant organisms. It is hoped that judicious use of vancomycin will help limit the spread of VRE and VISA and ensure that patients receive the most potent therapeutic agents.

Vancomycin remains the drug of choice for treatment of infections caused by [MRSA](#), although sensitive strains may be amenable to therapy with clindamycin or trimethoprim-sulfamethoxazole. As discussed above, two new agents, quinupristin/dalfopristin and linezolid, offer promise for treatment of MRSA, but further studies are needed before they can be recommended for routine or preferential use. Optimal therapy for infections caused by [VISA](#) is unknown. To date, all isolates of VISA have been susceptible to alternative agents. Therapeutic options include the combination of vancomycin plus a b-lactam (based largely on in vitro data), quinupristin/dalfopristin, linezolid, or one of the new quinolone antibiotics, although the potential for development of resistance to the quinolones during therapy makes their use a questionable approach for infections requiring a prolonged course of antibiotic.

In most clinical settings, no significant benefit is attained by treating *S. aureus* infections with more than one drug to which the organism is known to be susceptible. Synergy has been demonstrated in vitro for β -lactam/aminoglycoside combinations, which hasten sterilization of the blood in endocarditis. Accordingly, therapy for *S. aureus* bacteremia is often initiated with such a combination for a brief period (e.g., 5 to 7 days) -- a strategy that seems reasonable when rapid clearance of bacteremia is deemed to be critical, as in prosthetic valve endocarditis. Thereafter, the toxicity of an aminoglycoside cannot be justified. Use of rifampin in conjunction with a β -lactam antibiotic (or vancomycin) occasionally results in microbial eradication and clinical cure of otherwise refractory infections, particularly those involving foreign bodies that are judged to be unremovable or those involving avascular tissue. These successes may relate to the high level of activity of rifampin against intracellular organisms, including [SCVs](#). Chronic osteomyelitis, parameningeal infections, and septic phlebitis have all been successfully treated with rifampin plus a cell wall-active agent. Nevertheless, routine use of rifampin for serious *S. aureus* infections is not recommended because of potential added toxicity, drug interactions, and theoretical antimicrobial antagonism. Rifampin should be reserved for refractory, relapsing, or inoperable infections and should never be administered as monotherapy, which rapidly leads to resistance.

Route and Duration of Therapy Because of poor bioavailability of most oral antistaphylococcal agents, parenteral therapy should be used for infections that require high concentrations of antibiotic, such as endovascular infections, infections of poorly vascularized tissue (including abscesses), and infections of the central nervous system. Given the propensity of *S. aureus* to adhere to endovascular and devitalized or damaged tissues, high doses of antibiotics (e.g., 12 g/d of nafcillin) should be used for bacteremic infections. When high serum levels of antibiotic are required to produce adequate tissue levels (e.g., in endocarditis or osteomyelitis), the parenteral route should be used for the duration of therapy. Oral agents may suffice for the treatment of nonbacteremic infections in which high serum levels of antibiotic are not requisite (e.g., skin, soft tissue, and upper respiratory tract infections).

With the notable exceptions of bacteremia (including endocarditis) and osteomyelitis, the duration of therapy for *S. aureus* infections can be tailored to the severity of illness, the immunologic status of the host, and the response to treatment. Because antibiotics penetrate bone poorly, treatment of acute osteomyelitis in adults requires at least 4 weeks of parenteral therapy. Chronic osteomyelitis is often treated with 6 to 8 weeks of parenterally administered antibiotics followed by several months of oral therapy, especially if the adequacy of debridement is uncertain.

Acute endocarditis and other endovascular infections caused by *S. aureus* should be treated with parenteral antibiotics for 4 weeks (6 weeks in the case of prosthetic valves). Simple bacteremia, as might occur with a removable or drainable focus of infection, is curable with a shorter duration of therapy, but a 2-week course of *parenteral* therapy has traditionally been recommended *for all patients* with *S. aureus* bacteremia, even under these circumstances. The costs and effort implicit in this recommendation are apparent, but shorter courses of therapy are associated with an unacceptable rate of secondary complications. Data suggest that a 7-day course of parenteral therapy may be adequate for simple bacteremia if the clinical response to therapy is prompt, if cultures of blood obtained after 2 days of therapy are negative, and if [TEE](#) is negative for

vegetations. These data require confirmation, however. One of the more challenging aspects of treating staphylococcal bacteremia is deciding whether to administer parenteral therapy for 2 or 4 weeks. A conservative approach (one that is supported by numerous studies) dictates that 4 weeks should be standard unless specific criteria are met ([Fig. 139-2](#)).

PREVENTION AND CONTROL

Nosocomial staphylococcal outbreaks and the spread of resistant strains of *S. aureus* are serious global problems. Within an institution, the most important vector of transmission of *S. aureus* is the hands of health care workers. Patients with exposed wounds or with nasal colonization are important reservoirs of the organisms.

Transmission of *S. aureus* -- and hence the incidence of staphylococcal infection within an institution -- can be reduced most effectively by meticulous hand washing before and after contact with patients. The incidence of postoperative staphylococcal infection can be reduced by perioperative administration of an antibiotic with a favorable spectrum of activity and favorable pharmacokinetic properties, such as cefazolin, cefuroxime, or vancomycin. Elimination of nasal carriage before surgery (see below) may also prove to be effective in this regard.

More stringent infection-control measures must be taken to prevent the nosocomial spread of [MRSA](#). Such measures include assigning patients colonized or infected with MRSA to private rooms, wearing gloves for contact with contaminated wounds and mucous membranes as well as a gown if contamination of clothing is likely, and hand washing with an antiseptic soap after patient contact. Patients who are colonized but not infected with MRSA should not be treated with vancomycin merely for the sake of eliminating carriage of this organism.

Staphylococcal skin and soft tissue infections may recur once a person has been colonized with a virulent strain. In this context, therapy directed at the elimination of staphylococcal colonization may be warranted, especially for patients at particular risk for complications of infection. Use of an oral β -lactam antibiotic alone is ineffective, but combination therapy for 10 to 14 days with dicloxacillin or cephalexin (500 mg four times a day) plus rifampin (300 mg twice a day) plus mupirocin (2% ointment applied topically to both nares twice a day) is usually effective at clearing the carrier state, at least for a period of months.

COAGULASE-NEGATIVE STAPHYLOCOCCI

[CoNS](#) are a major cause of nosocomial infection and are the organisms most frequently isolated from the blood of hospitalized patients. The frequency with which they cause opportunistic infection in immunocompromised hosts attests more to the increased vulnerability of such hosts in modern medical practice than to the intrinsic virulence of the organisms. Despite the weak pathogenicity of these bacteria, the global impact of CoNS infection is considerable, including increased length and cost of hospital stay; increased use of antibiotics in general; and increased use of vancomycin in particular, which has contributed to the recent emergence of vancomycin-resistant gram-positive bacteria.

Although the variety of clinical syndromes caused by [CoNS](#) is impressive, several characteristics apply to most such infections. First, they tend to be *indolent*. There is often a long latent period between the time of contamination (e.g., of a medical device) and the onset of clinical illness; bacteremia in neutropenic patients can be an exception to this rule. Second, most CoNS infections are nosocomial in origin; important exceptions are prosthetic valve endocarditis and *S. saprophyticus* infections of the urinary tract. Third, most clinically significant infections are caused by strains of CoNS that are resistant to multiple antibiotics, including penicillins and cephalosporins. Finally, most CoNS infections are associated with a medical device of some kind, and removal of such devices is often required for cure.

EPIDEMIOLOGY AND PATHOGENESIS

[CoNS](#), particularly *S. epidermidis*, are invariable and prominent constituents of the normal human skin flora. Infection most often results from direct inoculation of a foreign body at the time it is inserted, although hematogenous seeding can also occur.

[CoNS](#) are the quintessential pathogens of medical devices. The array of virulence factors produced by CoNS is meager compared with that of the virulence factors produced by *S. aureus*, but among these few factors are substances that promote bacterial adherence to and persistence on foreign bodies. A variety of surface antigens that promote colonization of medical devices by CoNS (particularly *S. epidermidis*) have been proposed; the best-studied of these is capsular polysaccharide adhesin, which serves as the organism's capsule and promotes the initial interaction of the bacteria and a foreign body. This polysaccharide is a major component of the *S. epidermidis* biofilm, which is important in the persistence of infection; the biofilm thwarts host defenses by coating staphylococcal cells onto foreign materials and impairing phagocytic killing. CoNS appear not to make toxic exoproteins or toxins; rather, they cause disease by tenaciously persisting on foreign materials, resulting in a local and occasionally a systemic inflammatory response.

The most important risk factor for infection with [CoNS](#) is the presence of a foreign body, especially an indwelling catheter. A second major risk factor for infection is deficient phagocyte function -- especially neutropenia, which is most often an iatrogenic complication of chemotherapy for cancer but may also reflect an underlying disease process (such as leukemia). The likelihood of catheter-related infection depends upon a number of variables, including the experience and skill of the person who inserts the catheter, the length of time that a catheter is left in place, and the quality of postinsertion care of the catheter site. CoNS only rarely cause infections (other than urinary tract infections) in immunologically normal hosts and typically do so only under extenuating circumstances.

CLINICAL SYNDROMES

Because [CoNS](#) can adhere to a variety of materials, virtually all *foreign bodies* are susceptible to colonization by these organisms. CoNS are the most common pathogens complicating the use of intravenous catheters, hemodialysis shunts and grafts, cerebrospinal fluid (CSF) shunts, peritoneal dialysis catheters, pacemaker wires and electrodes, prosthetic joints, vascular grafts, and prosthetic valves. CoNS infection of

intravenous catheters may or may not be accompanied by signs of inflammation at the site of catheter insertion, and the degree of systemic toxicity (including fever) ranges from minimal to moderately severe. The diagnosis can be established by the culturing of blood drawn from the catheter and by venipuncture. Infection of CSF shunts is usually evident within several weeks of implantation. Signs of meningitis are sometimes readily apparent but more often are subtle or absent; malfunction of the shunt may be the only manifestation of shunt infection. CoNS infection of a prosthetic joint often does not become evident until long after implantation, although the inciting contamination usually occurs at the time of implantation. Infection of vascular grafts may result in the development of an aneurysm or a pseudoaneurysm, with catastrophic consequences.

[CoNS](#) are a prominent cause of *bacteremia* in immunosuppressed patients. While such infections in immunocompetent hosts are relatively benign, patients with neutropenia may have high-grade bacteremia that results in significant systemic toxicity. A serious consequence of bacteremia is the seeding of a secondary foreign body, such as a prosthetic heart valve or joint or a pacemaker.

[CoNS](#) are the organisms most commonly responsible for *prosthetic valve endocarditis*, causing the majority of infections that develop within several months of implantation as well as a substantial percentage of late infections. The syndrome is one of subacute endocarditis (thus contrasting with the syndrome produced by *S. aureus*), with an illness that is clinically indistinguishable from that caused by viridans streptococci. Infection of prosthetic valves is often complicated by valvular dysfunction secondary either to dehiscence of the sewing ring or obstruction of the valve's orifice by bulky vegetations. CoNS are a less frequent but important cause of *native valve endocarditis*, accounting for <5% of such infections and usually affecting abnormal valves.

S. saprophyticus is a common cause of *urinary tract infection* among sexually active young women, in whom it is second only to *E. coli* in frequency. Exposure to spermicide-coated condoms may increase the incidence of infection. *S. saprophyticus* produces a syndrome indistinguishable from that caused by other etiologic agents, with pyuria and symptoms of dysuria, frequency, and abdominal pain. Infection with *S. saprophyticus* is readily amenable to therapy with most agents commonly used to treat urinary tract infections. [CoNS](#) can also cause urinary tract infection in hospitalized patients who have undergone invasive procedures; such infections are especially likely to be asymptomatic and may be difficult to treat because of antimicrobial resistance.

DIAGNOSIS

Although [CoNS](#) are the most common cause of nosocomial bacteremia, they are also the most common contaminants of blood cultures; differentiation between infection and contamination often poses a challenge, with major therapeutic implications. Positive blood cultures are more likely to be "true positives" when there is a clinical illness suggestive of infection, when there is an indwelling catheter or some other risk factor for CoNS infection, and when cultures of blood drawn from multiple sites are positive for phenotypically identical organisms with the same antimicrobial susceptibility patterns. Except in the setting of neutropenia, physicians often have the luxury of awaiting the results of repeat cultures when the significance of CoNS growing from a blood culture is questionable.

TREATMENT

Removal of the foreign body (especially when it is an intravenous catheter) often constitutes adequate therapy for [CoNS](#) infection related to that device. Most infections involving a foreign body require the removal of the device -- whether a prosthetic valve, prosthetic joint, [CSF](#) shunt, vascular graft, pacemaker or defibrillator and associated hardware, or hemodialysis shunt. Cures of all such infections with antibiotics alone have been reported, however, and a patient's poor medical condition or the hazards of surgery occasionally warrant an attempt at medical cure without extirpation of the device (see below). Infections of peritoneal dialysis catheters can be cured with antibiotics alone often enough that an attempt should be made to do so. CoNS infections of central venous catheters are also amenable to medical therapy, although relapses are common. Persistent bacteremia during therapy is an absolute indication for removal of a catheter, and bacteremia after a catheter's removal suggests seeding of a secondary site.

It is difficult to make generalizations about the optimal duration of therapy for [CoNS](#) infections. In general, the duration of treatment is the same as for infection syndromes caused by other bacteria. For example, native valve endocarditis should be treated for 4 weeks, prosthetic valve endocarditis for 6. Transient bacteremia in an immunocompetent host may require no antimicrobial therapy after removal of an offending catheter. The efficacy of therapy can occasionally be enhanced by the delivery of antibiotics directly to the site of infection -- e.g., by intraventricular administration of vancomycin for central nervous system infections or by intraperitoneal administration of antibiotic for infections of peritoneal dialysis catheters.

Despite the low degree of pathogenicity of [CoNS](#), treatment of serious infections due to these organisms is often problematic because of the high percentage of strains that are resistant to commonly used antibiotics, including most oral agents. Most strains of CoNS isolated from patients in U.S. hospitals are resistant not only to penicillin but also to the penicillinase-resistant penicillins and cephalosporins. Nosocomial isolates are usually resistant to other classes of antibiotics as well. Vancomycin, to which the vast majority of CoNS remain susceptible, is of necessity the drug of choice for *empirical* therapy for serious CoNS infections. Strains proved to be susceptible to nafcillin (oxacillin) or penicillin should be treated with one of these agents or with a first-generation cephalosporin.

Synergistic combinations of antibiotics are often useful in the treatment of [CoNS](#) infections. Rifampin plays a unique role in this endeavor by virtue of its potency against most staphylococci, its excellent penetration into tissues (including those that are poorly vascularized), and the high levels it reaches within human cells and biofilm. Rifampin must be used in combination with other antibiotics because of the frequent and rapid emergence of microbial resistance to the drug when it is used alone. If an effort must be made to eradicate infection of a medical device without its removal, the concomitant use of a β -lactam antibiotic to which the organism is susceptible plus rifampin (300 mg twice daily by mouth) plus gentamicin affords the best chance for success. Vancomycin can be substituted for the β -lactam agent if so dictated by an organism's susceptibility pattern or by a patient's drug allergy.

(Bibliography omitted in Palm version)

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140. STREPTOCOCCAL AND ENTEROCOCCAL INFECTIONS - Michael R. Wessels

Many varieties of streptococci are found as part of the normal human flora colonizing the respiratory, gastrointestinal, and genitourinary tracts. Several species are important causes of human disease. Group A *Streptococcus*, or *S. pyogenes*, is responsible for streptococcal pharyngitis, one of the most common bacterial infections of school-age children, and for the postinfectious syndromes of acute rheumatic fever and poststreptococcal glomerulonephritis. Group B *Streptococcus*, or *S. agalactiae*, is the leading cause of bacterial sepsis and meningitis in newborns and a major cause of endometritis and fever in parturient women. Enterococci are important causes of urinary tract infection, nosocomial bacteremia, and endocarditis. Viridans streptococci are the most common cause of bacterial endocarditis.

Streptococci are gram-positive bacteria of spherical to ovoid shape that characteristically form chains when grown in liquid media. Most streptococci that cause human infections are facultative anaerobes, although some are strict anaerobes. Streptococci are relatively fastidious organisms, requiring enriched media for growth in the laboratory. No single scheme for classification of streptococci is entirely satisfactory. Consequently, clinicians and clinical microbiologists often identify streptococci by any of several classification systems, including hemolytic pattern, Lancefield group, species name, and common or trivial name. Many of the streptococci associated with human infection produce a zone of complete hemolysis around the bacterial colony when cultured on blood agar, a pattern known as *β* hemolysis. The *β*-hemolytic streptococci can be classified by the Lancefield system, a serologic grouping based on the reaction of specific antisera with cell-wall carbohydrate antigens of the bacteria. With rare exceptions, organisms belonging to Lancefield groups A, B, C, and G are all *β*-hemolytic streptococci, and each is associated with characteristic patterns of human infection. Other streptococci produce a zone of partial (*α*) hemolysis, often imparting a greenish appearance to the agar. These *α*-hemolytic streptococci are further identified by biochemical testing and include *S. pneumoniae*, an important cause of pneumonia, meningitis, and other infections, and several species of streptococci referred to collectively as the *viridans streptococci*, which are part of the normal oral flora and are important as agents of subacute bacterial endocarditis. Finally, some streptococci are nonhemolytic, a pattern sometimes called *γ* hemolysis. The classification of the major groups of streptococci responsible for human infections is outlined in [Table 140-1](#). Among the organisms classified serologically as group D streptococci, the enterococci are now considered to constitute a separate genus on the basis of DNA homology studies. Thus, species previously designated as *S. faecalis* and *S. faecium* have been renamed *Enterococcus faecalis* and *E. faecium*, respectively. **For further discussion of pneumococcal infections, see [Chap. 138](#).*

GROUP A STREPTOCOCCI

Lancefield's group A consists of a single species, *S. pyogenes*. As its species name implies, this organism is associated with a variety of suppurative infections. In addition, group A streptococci can trigger the postinfectious syndromes of acute rheumatic fever (which is uniquely associated with *S. pyogenes* infection; [Chap. 235](#)) and poststreptococcal glomerulonephritis ([Chap. 274](#)).

PATHOGENESIS

Group A streptococci elaborate a number of cell-surface components and extracellular products important both in the pathogenesis of infection and in the immune response of the human host. The cell wall contains a carbohydrate antigen that may be released by treatment with acid. The reaction of such acid extracts with group A-specific antiserum is the basis for the definitive identification of a streptococcal strain as *S. pyogenes*. The major surface protein of group A streptococci is M protein, which occurs in more than 100 antigenically distinct types and is the basis for the serotyping of strains with specific antisera. The M protein molecules are fibrillar structures anchored in the cell wall of the organism and extending as hairlike projections away from the cell surface. The amino acid sequence of the distal or amino-terminal portion of the M protein molecule is quite variable, accounting for the antigenic variation of the different M types, while more proximal regions of the protein are relatively conserved. A newer technique for assignment of M type to group A streptococcal isolates uses the polymerase chain reaction to amplify the variable region of the M protein gene. DNA sequence analysis of the amplified gene segment can be compared with an extensive data base [developed at the Centers for Disease Control and Prevention (CDC)] for assignment of M type. This method eliminates the need for typing sera, which are available in only a few reference laboratories. The presence of M protein on a group A streptococcal isolate correlates with its capacity to resist phagocytic killing in fresh human blood; this phenomenon appears to be due, at least in part, to the binding of plasma fibrinogen to M protein molecules on the streptococcal surface, which interferes with complement activation and deposition of opsonic complement fragments on the bacterial cell. This resistance to phagocytosis may be overcome by M protein-specific antibodies, and thus individuals with antibodies to a given M type acquired as a result of prior infection are protected against subsequent infection with organisms of the same M type but not against that with different M types.

Group A streptococci also elaborate, to varying degrees, a polysaccharide capsule composed of hyaluronic acid. The production of large amounts of hyaluronic acid capsule by certain strains lends a characteristic mucoid appearance to the bacterial colonies. The capsular polysaccharide also plays an important role in protecting the organisms from ingestion and killing by phagocytes. In contrast to M protein, the hyaluronic acid capsule is a weak immunogen, and antibodies to hyaluronate have not been shown to be important in protective immunity; the presumed explanation is the apparent structural identity between streptococcal hyaluronic acid and the hyaluronic acid of mammalian connective tissues. The capsular polysaccharide may also play a role in group A streptococcal colonization of the pharynx by binding to CD44, a hyaluronic acid-binding protein expressed on human pharyngeal epithelial cells.

Group A streptococci produce a large number of extracellular products that may be important in local and systemic toxicity and in the spread of infection through tissues. These products include streptolysins S and O, toxins that damage cell membranes and account for the hemolysis produced by the organisms; streptokinase; DNases; protease; and pyrogenic exotoxins A, B, and C. The pyrogenic exotoxins, previously known as erythrogenic toxins, cause the rash of scarlet fever. Since the mid-1980s, pyrogenic exotoxin-producing strains of group A *Streptococcus* have been linked to unusually severe invasive infections, including necrotizing fasciitis and a systemic syndrome

termed the *streptococcal toxic shock syndrome*. Several extracellular products stimulate specific antibody responses useful in the serodiagnosis of recent streptococcal infection. Tests for these antibodies are used primarily for the detection of preceding streptococcal infection in cases of suspected acute rheumatic fever or poststreptococcal glomerulonephritis.

CLINICAL MANIFESTATIONS

Pharyngitis Although seen in patients of all ages, group A streptococcal pharyngitis is one of the most common bacterial infections of childhood, accounting for 20 to 40% of all cases of exudative pharyngitis in children. It is rare among those under the age of 3. Younger children may manifest streptococcal infection with a syndrome of fever, malaise, and lymphadenopathy without exudative pharyngitis. Infection is acquired through contact with another individual carrying the organism. Respiratory droplets are the usual mechanism of spread, although other routes, including food-borne outbreaks, have been well described.

The incubation period is 1 to 4 days. Symptoms include sore throat, fever and chills, malaise, and sometimes abdominal complaints and vomiting, particularly in children. Both symptoms and signs are quite variable, ranging from mild throat discomfort with minimal physical findings to high fever and severe sore throat associated with intense erythema and swelling of the pharyngeal mucosa and the presence of purulent exudate over the posterior pharyngeal wall and tonsillar pillars. Enlarged, tender anterior cervical lymph nodes commonly accompany exudative pharyngitis.

The differential diagnosis of streptococcal pharyngitis includes the many other bacterial and viral causes of pharyngitis. Streptococcal infection is unlikely to be the cause of pharyngitis when symptoms and signs suggestive of viral infection are prominent (conjunctivitis, coryza, cough, hoarseness, or discrete ulcerative lesions of the buccal or pharyngeal mucosa). Other infections commonly producing exudative pharyngitis include infectious mononucleosis and adenovirus infection. Now rare in the United States, the pseudomembrane of diphtheria may give a similar appearance. The coryneform organism *Arcanobacterium haemolyticum* may cause pharyngitis, often in association with a scarlet fever-like rash ([Chap. 141](#)). Other causes of pharyngitis, usually without a purulent exudate, include coxsackievirus, influenza virus, mycoplasmas, and *Neisseria gonorrhoeae* and acute infection with HIV. Because of the range of clinical presentations of streptococcal pharyngitis and the large number of other agents that can produce the same clinical picture, diagnosis of streptococcal pharyngitis on clinical grounds alone is not reliable.

The throat culture remains the diagnostic "gold standard." Culture of a throat specimen that is properly collected (i.e., by vigorous rubbing of a sterile swab over both tonsillar pillars) and properly processed is the most sensitive and specific means available to make a definitive diagnosis. A rapid diagnostic kit using latex agglutination or enzyme immunoassay of swab specimens can serve as a useful adjunct to the throat culture. While precise figures on sensitivity and specificity vary among studies, the rapid diagnostic kits generally are >95% specific. Thus a positive result can be relied upon for definitive diagnosis and eliminates the need for a throat culture. However, because the rapid diagnostic tests are less sensitive than throat culture (with a relative sensitivity

ranging from 55 to 90% in comparative studies), a negative result should be confirmed with a throat culture.

TREATMENT

In the usual course of uncomplicated streptococcal pharyngitis, symptoms resolve after 3 to 5 days. The course is shortened little by treatment, which is given primarily to prevent suppurative complications and rheumatic fever. Prevention of rheumatic fever depends on eradication of the organism from the pharynx, not simply on resolution of symptoms, and requires 10 days of penicillin treatment -- either a single intramuscular dose of benzathine penicillin G or a 10-day course of oral penicillin ([Table 140-2](#)). Erythromycin may be substituted for penicillin in the treatment of individuals allergic to penicillin. Follow-up culture after treatment is no longer routinely recommended but may be warranted in selected cases, such as those involving patients or families with frequent streptococcal infections or those occurring in situations in which the risk of rheumatic fever is thought to be high (e.g., when cases of rheumatic fever have recently been reported in the community).

Complications Suppurative complications of streptococcal pharyngitis have become uncommon with the widespread use of antibiotics for most cases of symptomatic streptococcal infection. The complications result from the spread of infection from the pharyngeal mucosa to deeper tissues by direct extension or by the hematogenous or lymphatic route and may include cervical lymphadenitis, peritonsillar or retropharyngeal abscess, sinusitis, otitis media, meningitis, bacteremia, endocarditis, and pneumonia. Local complications, such as abscess formation in the peritonsillar or parapharyngeal space, should be considered in a patient with unusually severe or prolonged symptoms or localized pain associated with high fever and a toxic appearance.

Asymptomatic Carrier State Surveillance cultures have shown that up to 20% of individuals in certain populations may have asymptomatic pharyngeal colonization with group A streptococci. There are no definitive guidelines for management of these asymptomatic carriers or of asymptomatic individuals who still have a positive throat culture after a full course of treatment for symptomatic pharyngitis. A reasonable course of action is to give a single 10-day course of penicillin for symptomatic pharyngitis and, if positive cultures persist, not to re-treat unless symptoms recur. Studies of the natural history of streptococcal carriage and infection have shown that the risk both of developing rheumatic fever and of transmitting infection to others is substantially lower among asymptomatic carriers than among individuals with symptomatic pharyngitis. Therefore, overly aggressive attempts to eradicate carriage are probably not justified under most circumstances. An exception is the situation in which an asymptomatic carrier is a potential source of infection to others. Outbreaks of food-borne infection and nosocomial puerperal infection have been traced to asymptomatic carriers who may harbor the organisms in the throat, on the skin, or in the vagina or anus.

TREATMENT

In cases in which a carrier is transmitting infection to others, attempts to eradicate carriage are warranted, although data are limited on the best regimen to use to clear the organism after penicillin alone has failed. The combination of penicillin V (500 mg four

times daily for 10 days) and rifampin (600 mg twice daily for the last 4 days) has been used to eliminate pharyngeal carriage. A 10-day course of oral vancomycin (250 mg four times daily) and rifampin (600 mg twice daily) has eradicated rectal colonization. However, experience is not extensive with any regimen.

Scarlet Fever Scarlet fever consists of streptococcal infection, usually pharyngitis, accompanied by a characteristic rash. The rash arises from the effects of one of three toxins, currently designated streptococcal pyrogenic exotoxins A, B, and C and previously known as erythrogenic or scarlet fever toxins. In the past, scarlet fever was thought to reflect infection of an individual lacking toxin-specific immunity with a toxin-producing strain of group A *Streptococcus*. Susceptibility to scarlet fever was correlated with results of the Dick test. A small amount of erythrogenic toxin injected intradermally produced local erythema in susceptible individuals but elicited no reaction in those with specific immunity. Subsequent studies have suggested that development of the scarlet fever rash may reflect a hypersensitivity reaction requiring prior exposure to the toxin. For reasons that are not clear, scarlet fever has become less common in recent years, although strains of group A streptococci that produce pyrogenic exotoxins continue to be prevalent in the population.

The symptoms of scarlet fever are the same as those of pharyngitis alone ([Fig. 140-CD1](#)). The ([Fig. 140-CD2](#)) rash typically begins on the first or second day of illness over the upper trunk, spreading to involve the extremities but sparing the palms and soles. The rash is made up of minute papules, giving a characteristic "sandpaper" feel to the skin. Associated findings include circumoral pallor, "strawberry tongue" ([Fig. 140-CD3](#)) (enlarged papillae on a coated tongue, which later may become denuded), and accentuation of the rash in the skin folds (Pastia's lines). Subsidence of the rash in 6 to 9 days is followed after several days by desquamation of the palms and soles. The differential diagnosis of scarlet fever includes other causes of fever and generalized rash, such as measles and other viral exanthems, Kawasaki disease, toxic shock syndrome, and systemic allergic reactions (e.g., drug eruptions).

Skin and Soft Tissue Infections Group A streptococci -- and occasionally other streptococcal species -- cause a variety of infections involving the skin ([Fig. 140-CD4](#)), subcutaneous tissues, muscles, and fascia ([Fig. 140-CD5](#)). While several clinical syndromes, recognized according to the tissues involved, offer a useful means for classification of skin and soft tissue infections, not all cases fit exactly into a single category. The classic syndromes should be considered as general guides to predicting the level of tissue involvement in a particular patient, the probable clinical course, and the likelihood that surgical intervention or aggressive life-support will be required.

Impetigo (Pyoderma) Impetigo is a superficial infection of the skin caused primarily by group A streptococci and occasionally by other streptococci or by *Staphylococcus aureus*. Impetigo is seen most often in young children, tends to occur during the warmer months, and is more common in semitropical or tropical climates than in cooler regions. Infection is more common among children living under conditions of poor hygiene. Prospective studies have shown that colonization of unbroken skin with group A streptococci precedes the development of clinical infection. Minor trauma, such as a scratch or an insect bite, may then serve to inoculate organisms into the skin. Impetigo is best prevented, therefore, by attention to adequate hygiene. The usual sites of

involvement are the face (particularly around the nose and mouth) and the legs, although lesions may occur at other locations. Individual lesions begin as red papules, which evolve quickly into vesicular and then pustular lesions that break down and coalesce to form characteristic honeycomb-like crusts ([Plate IID-38](#)). Lesions are generally not painful, and patients do not appear ill. Fever is not a feature of impetigo and, if present, suggests either infection extending to deeper tissues or another diagnosis.

The classic presentation of impetigo usually poses little diagnostic difficulty. Cultures of impetiginous lesions often yield *S. aureus* as well as group A streptococci, but longitudinal studies have shown that, in almost all cases, streptococci can be isolated initially, with staphylococci appearing later, presumably as secondary colonizing flora. In the past, penicillin was nearly always effective against these infections; in recent years, however, penicillin treatment failures have become more common, an observation suggesting that *S. aureus* infection may have become more prominent as a cause of impetigo. *Bullous impetigo* due to *S. aureus* is distinguished from typical streptococcal infection by the presence of more extensive, bullous lesions that break down and leave thin paper-like crusts instead of the thick amber crusts of streptococcal impetigo. Other skin lesions that may be confused with impetigo include herpetic lesions -- either those of orolabial herpes simplex or those of chickenpox or zoster. Herpetic lesions can generally be distinguished by their appearance as more discrete, grouped vesicles and by a positive Tzanck test. In difficult cases, cultures of vesicular fluid should yield group A streptococci in impetigo and the responsible virus in *Herpesvirus* infections.

TREATMENT

Treatment of streptococcal impetigo is the same as that for streptococcal pharyngitis. In view of evidence that *S. aureus* has become a relatively frequent cause of impetigo, empirical regimens should cover both streptococci and *S. aureus*. For example, either dicloxacillin or cephalexin can be given at a dose of 250 mg four times daily for 10 days. Topical mupirocin ointment is also effective. Rheumatic fever (unlike pharyngitis) is not a sequela to streptococcal skin infections, although poststreptococcal glomerulonephritis may follow either skin or throat infection. The reason for this difference is not known. One hypothesis is that the immune response necessary for development of rheumatic fever occurs only after infection of the pharyngeal mucosa. In addition, the strains of group A streptococci that cause pharyngitis are generally of different M protein types than those associated with skin infections; thus the strains that cause pharyngitis may have rheumatogenic potential, while the skin-infecting strains may not.

Cellulitis Inoculation of organisms into the skin may lead to infection involving the skin and subcutaneous tissues, or *cellulitis*. The portal of entry may be a traumatic or surgical wound, an insect bite, or any other break in skin integrity. Often, no entry site is apparent.

One form of streptococcal cellulitis, *erysipelas*, is characterized by a bright red appearance of the involved skin, which forms a plateau sharply demarcated from surrounding normal skin ([Plate IID-34](#)). The lesion is warm to the touch, may be tender, and appears shiny and swollen. The skin often has a *peau d'orange* texture, which is

thought to reflect involvement of superficial lymphatics; superficial blebs or bullae may form, usually 2 or 3 days after onset. The lesion typically develops over a few hours and is associated with fever and chills. Erysipelas tends to occur in certain characteristic locations: the malar area of the face (often with extension over the bridge of the nose to the contralateral malar region) and the lower extremities. After one episode, recurrence at the same site -- sometimes years later -- is not uncommon.

Classic cases of erysipelas, with the typical features described above, are almost always due to hemolytic streptococci, usually those of group A and occasionally those of group C or G. Often, however, the appearance of streptococcal cellulitis is not sufficiently distinctive to permit a specific diagnosis on clinical grounds. The area of involvement may not be one of the typical sites for erysipelas, the lesion may be less intensely red than usual and may fade into surrounding skin, and/or the patient may appear only mildly ill. In such cases, it is prudent to broaden the spectrum of empiric antimicrobial therapy to include other pathogens, particularly *S. aureus*, that can produce cellulitis with the same appearance. Staphylococcal infection should be suspected if cellulitis develops around a wound or ulcer.

Streptococcal cellulitis tends to develop at anatomic sites in which normal lymphatic drainage has been disrupted, such as sites of prior episodes of cellulitis, the arm ipsilateral to a mastectomy and axillary lymph node dissection, a lower extremity previously involved in deep venous thrombosis or chronic lymphedema, and the leg from which a saphenous vein has been harvested for coronary artery bypass grafting. The organism may enter via a breach in the dermal barrier at a location some distance from the eventual site of clinical cellulitis. For example, some patients with recurrent episodes of leg cellulitis following saphenous vein removal stop having recurrent episodes only after treatment of tinea pedis on the affected extremity, fissures in the skin presumably having served as a portal of entry for streptococci, which then produced infection more proximally in the leg at the site of previous injury. Streptococcal cellulitis may also involve recent surgical wounds. Group A streptococci are among the few bacterial pathogens that typically produce signs of wound infection and surrounding cellulitis within the first 24 h after surgery. These wound infections are usually associated with a thin exudate and may spread rapidly, either as cellulitis in the skin and subcutaneous tissue or as a deeper tissue infection (see below). Streptococcal wound infection or localized cellulitis may also be associated with *lymphangitis*, manifested by red streaks extending proximally along superficial lymphatics from the site of infection.

TREATMENT

See [Table 140-2](#) and [Chap. 128](#).

Deep Soft Tissue Infections Necrotizing fasciitis ([Fig. 18-CD4](#)), also referred to as *hemolytic streptococcal gangrene*, is an infection involving the superficial and/or deep fascia investing the muscles of an extremity or the trunk. The source of the infection is either the skin, with organisms introduced into the tissue as a result of trauma (sometimes trivial), or the bowel flora, with organisms released during abdominal surgery or from an occult enteric source, such as a diverticular or appendiceal abscess. The site of inoculation in both forms of necrotizing fasciitis may be inapparent and is often some distance from the site of clinical involvement; e.g., the introduction of

organisms via minor trauma to the hand may be associated with clinical infection of the tissues overlying the shoulder or chest. In cases associated with the bowel flora, the infection is usually polymicrobial, involving a mixture of anaerobic bacteria (such as *Bacteroides fragilis* or anaerobic streptococci) and facultative organisms (usually gram-negative bacilli). Cases unrelated to contamination from bowel organisms are most commonly caused by group A streptococci, either alone or in combination with other organisms (most often *S. aureus*). Overall, group A streptococci are implicated in about 60% of cases of necrotizing fasciitis. The onset of symptoms is usually quite acute and is marked by severe pain at the site of involvement, malaise, fever, chills, and a toxic appearance. The physical findings, particularly early in the illness, may not be striking, with only minimal erythema of the overlying skin. Pain and tenderness are usually severe; in contrast, in more superficial cellulitis, the skin appearance is more abnormal, but pain and tenderness are only mild or moderate. As the infection progresses (often in a matter of several hours), the severity and extent of symptoms worsen, and skin changes become more evident, with the appearance of dusky or mottled erythema and edema. The marked tenderness of the involved area may evolve into anesthesia as the spreading inflammatory process produces infarction of cutaneous nerves.

Although myositis is more commonly due to *S. aureus* infection, group A streptococci occasionally produce abscesses in skeletal muscles (*streptococcal myositis*), with little or no involvement of the surrounding fascia or overlying skin. The presentation is usually subacute, but a fulminant form has been described in association with severe systemic toxicity, bacteremia, and a high mortality rate. The fulminant form may reflect the same basic disease process as that seen in necrotizing fasciitis, but with the necrotizing inflammatory process extending into the muscles themselves rather than remaining limited to the fascial layers.

TREATMENT

Once necrotizing fasciitis is suspected, early surgical exploration is both diagnostically and therapeutically indicated. Surgery reveals necrosis and inflammatory fluid tracking along the fascial planes above and between muscle groups, without involvement of the muscles themselves. The process usually extends beyond the area of clinical involvement, and extensive debridement is required. Drainage and debridement are central to the management of necrotizing fasciitis; antibiotic treatment is a useful adjunct ([Table 140-2](#)), but surgery is life-saving.

Treatment for streptococcal myositis consists of surgical drainage -- usually by an open procedure that permits evaluation of the extent of the infection and ensures adequate debridement of involved tissues -- and high-dose penicillin ([Table 140-2](#)).

Pneumonia and Empyema Group A streptococci are an occasional cause of pneumonia, generally in previously healthy individuals. The onset of symptoms may be abrupt or gradual. Pleuritic chest pain, fever, chills, and dyspnea are the characteristic symptoms. Cough is usually present but may not be prominent. Approximately one-half of patients with group A streptococcal pneumonia have an accompanying pleural effusion. In contrast to the sterile parapneumonic effusions typical of pneumococcal pneumonia, those complicating streptococcal pneumonia are almost always infected.

The empyema fluid is usually visible by chest radiography on initial presentation and may enlarge rapidly. These pleural collections should be drained early, as they tend to become loculated rapidly, resulting in a chronic fibrotic reaction that may require thoracotomy for removal.

Bacteremia, Puerperal Sepsis, and Streptococcal Toxic Shock Syndrome Group A streptococcal bacteremia is usually associated with an identifiable local infection. Bacteremia occurs rarely with otherwise uncomplicated pharyngitis, occasionally with cellulitis or pneumonia, and relatively frequently with necrotizing fasciitis. Bacteremia without an identified source raises the possibility of endocarditis, an occult abscess, or osteomyelitis. A variety of focal infections may arise secondarily from streptococcal bacteremia, including endocarditis, meningitis, septic arthritis, osteomyelitis, peritonitis, and visceral abscesses.

Group A streptococci are occasionally implicated in infectious complications of childbirth, usually endometritis and associated bacteremia. In the preantibiotic era, puerperal sepsis was commonly caused by group A streptococci, but currently it is more often caused by group B streptococci. Several nosocomial outbreaks of puerperal infection due to group A streptococci have been traced to an asymptomatic carrier, usually an individual present at the delivery of the infant. The site of carriage may be the skin, throat, anus, or vagina.

Beginning in the late 1980s, several reports described patients who had group A streptococcal infections associated with shock and multisystem organ failure. This syndrome has been called the streptococcal toxic shock syndrome because it shares certain features with staphylococcal toxic shock syndrome. In 1993, a case definition for group A streptococcal toxic shock syndrome was formulated by a group of clinicians, microbiologists, and epidemiologists in conjunction with the [CDC \(Table 140-3\)](#). The general features of the illness include fever, hypotension, renal impairment, and respiratory distress syndrome. Various types of rash have been described, but rash usually does not develop. Laboratory abnormalities include a marked shift to the left in the white blood cell differential, with many immature granulocytes; hypocalcemia; hypoalbuminemia; and thrombocytopenia, which usually becomes more pronounced on the second or third day of illness. In contrast to those with staphylococcal toxic shock, the majority of patients with the streptococcal syndrome are bacteremic. The most common associated infection is a soft tissue infection -- necrotizing fasciitis, myositis, or cellulitis -- although a variety of other associated local infections have been described, including pneumonia, peritonitis, osteomyelitis, and myometritis. Streptococcal toxic shock syndrome is associated with a mortality rate of 30%, with most deaths secondary to shock and respiratory failure. Because of its rapidly progressive and lethal course, early recognition of the syndrome is essential. Patients should be given aggressive supportive care in the form of fluid resuscitation, pressors, and mechanical ventilation in addition to antimicrobial therapy and, in cases associated with necrotizing fasciitis, surgical debridement. Exactly why certain patients develop this fulminant syndrome is not known; however, early studies of the streptococcal strains isolated from these patients demonstrated a strong association with the production of pyrogenic exotoxin A. In subsequent case series, particularly from Europe, the syndrome was also associated with strains producing exotoxin B or C.

TREATMENT

In light of the possible role of exotoxins or other streptococcal toxins in streptococcal toxic shock syndrome, treatment of the affected patients with clindamycin has been advocated by some authorities, who argue that, through its direct action on protein synthesis, clindamycin is more effective in rapidly terminating toxin production than penicillin -- a cell-wall agent. Support for this view comes from studies of an experimental model of streptococcal myositis, in which mice treated with clindamycin had a higher rate of survival than those given penicillin. Comparable data on the treatment of human infections are not available. Although clindamycin resistance in group A streptococci is uncommon (<2% among U.S. isolates), it has been documented. Thus, if clindamycin is used for initial treatment of a critically ill patient, penicillin should be given as well until the antibiotic susceptibility of the streptococcal isolate is known.

Intravenous immunoglobulin has been suggested as adjunctive therapy for streptococcal toxic shock; pooled immunoglobulin preparations are likely to contain antibodies capable of neutralizing the effects of streptococcal toxins. Anecdotal reports have suggested favorable clinical responses to intravenous immunoglobulin, but no controlled trials of this modality of therapy have yet been reported.

STREPTOCOCCI OF GROUPS C AND G

Group C and group G streptococci are β -hemolytic bacteria that occasionally cause human infections similar to those caused by group A streptococci, including pharyngitis, cellulitis and soft-tissue infections, pneumonia, bacteremia, endocarditis, and septic arthritis. Puerperal sepsis, meningitis, epidural abscess, intraabdominal abscess, urinary tract infection, and neonatal sepsis have also been reported. Group C streptococci are a common cause of infection in domesticated animals, especially horses and cattle, and some human infections have been acquired through contact with animals or through consumption of unpasteurized milk. Bacteremia and septic arthritis more frequently involve group G than group C streptococci. Group C or G streptococcal bacteremia occurs most often in patients who are elderly or chronically ill and, in the absence of an obvious local infection, is likely to reflect endocarditis. Septic arthritis, sometimes involving multiple joints, may complicate endocarditis or develop in its absence.

TREATMENT

Penicillin is the drug of choice for therapy of infections due to group C or G streptococci. Antibiotic treatment is the same as for patients with similar syndromes due to group A *Streptococcus* ([Table 140-2](#)). Patients with bacteremia or septic arthritis should receive intravenous penicillin (2 to 4 mU every 4 h). All group C and G streptococci are sensitive to penicillin; nearly all are inhibited in vitro by concentrations of ≤ 0.03 $\mu\text{g/mL}$. Occasional isolates exhibit tolerance: although inhibited by low concentrations of penicillin, they are killed only by significantly higher concentrations. The clinical significance of tolerance is unknown. Because of the poor clinical response of some patients to penicillin alone, the addition of gentamicin (1 mg/kg every 8 h for patients with normal renal function) is recommended by some authors for treatment of endocarditis or septic arthritis due to group C or G streptococci; however, combination therapy has not been shown to be

superior to treatment with penicillin alone.

Patients with joint infections often require repeated aspiration or open drainage and debridement for cure; the response to treatment may be slow, particularly in debilitated patients and those with involvement of more than one joint. Infection of prosthetic joints almost always requires removal of the prosthesis in addition to antibiotic therapy.

GROUP B STREPTOCOCCI

Identified first as a cause of mastitis in cows, streptococci belonging to Lancefield's group B have since been recognized as a major cause of sepsis and meningitis in human neonates. Group B streptococci are also a frequent cause of peripartum fever in women and an occasional cause of serious infection in nonpregnant adults. Lancefield group B consists of a single species, *S. agalactiae*, which is definitively identified with specific antiserum to the group B cell wall-associated carbohydrate antigen. A streptococcal isolate can be classified presumptively as belonging to group B on the basis of biochemical tests, including hydrolysis of sodium hippurate (in which 99% of isolates are positive), hydrolysis of bile esculin agar (in which 99 to 100% are negative), bacitracin susceptibility (in which 92% are resistant), and production of CAMP factor (in which 98 to 100% are positive). CAMP factor is a phospholipase produced by group B streptococci that results in synergistic hemolysis with b lysin produced by certain strains of *S. aureus*. Its presence can be demonstrated by cross-streaking of the test isolate and an appropriate staphylococcal strain on a blood agar plate. Group B streptococci causing human infections are encapsulated by one of nine antigenically distinct polysaccharides. The capsular polysaccharide has been shown experimentally to be important in the virulence of the organism. Antibodies to the capsular polysaccharide afford protection against group B streptococci of the same (but not of a different) capsular type.

INFECTION IN NEONATES

Two general types of group B streptococcal infection in infants are defined by the age of the patient at presentation. *Early-onset infections* occur within the first week of life, with a median age of 20 h at the onset of illness. Approximately half of these infants have signs of group B streptococcal disease at birth. The infection is acquired during or shortly before birth from organisms colonizing the maternal genital tract. Surveillance studies have shown that 5 to 40% of women are vaginal or rectal carriers of group B streptococci. Approximately 50% of infants delivered vaginally by carrier mothers become colonized, although only 1 to 2% of those colonized develop clinically evident infection. Prematurity and maternal risk factors (prolonged labor, obstetric complications, and maternal fever) are often involved. The presentation of early-onset infection is the same as that of other forms of neonatal sepsis. Typical findings include respiratory distress, lethargy, and hypotension. Essentially all infants with early-onset disease are bacteremic, one-third to one-half have pneumonia and/or respiratory distress syndrome, and one-third have meningitis.

Late-onset infections occur in infants between 1 week and 3 months of age, with a mean age at onset of 3 to 4 weeks. The infecting organism may be acquired during delivery (as in early-onset cases) or during later contact with a colonized mother,

nursery personnel, or another source. Meningitis is the most common manifestation of late-onset infection and in most cases is associated with a strain of capsular type III. Infants present with fever, lethargy or irritability, poor feeding, and seizures. The various other types of late-onset infection include bacteremia without an identified source, osteomyelitis, septic arthritis, and facial cellulitis associated with submandibular or preauricular adenitis.

TREATMENT

Penicillin is the treatment of choice for all group B streptococcal infections. Empirical broad-spectrum therapy for suspected bacterial sepsis, consisting of ampicillin and gentamicin, is generally administered until culture results become available. If cultures yield group B streptococci, many pediatricians continue to administer gentamicin, along with ampicillin or penicillin, for a few days until clinical improvement becomes evident. Infants with bacteremia or soft-tissue infection should receive penicillin at a dosage of 200,000 units/kg per day in divided doses; those with meningitis should receive 400,000 units/kg per day. Meningitis should be treated for at least 14 days because of the risk of relapse with shorter courses.

Prevention The incidence of group B streptococcal infection is unusually high among infants of women with risk factors: preterm delivery, early rupture of membranes (>24 h before delivery), prolonged labor, fever, or chorioamnionitis. Because the usual source of the organisms infecting a neonate is the mother's birth canal, efforts have been made to prevent group B streptococcal infections by the identification of high-risk carrier mothers and their treatment with various forms of antibiotic or immunoprophylaxis. Prophylactic administration of ampicillin or penicillin to such patients during delivery has been shown to reduce the risk of infection in the newborn. This approach has been hampered by the logistical difficulties of identifying colonized women before delivery, since the results of vaginal cultures early in pregnancy are poor predictors of carrier status at delivery. The [CDC](#) has suggested two alternative approaches to the prevention of neonatal group B streptococcal infection: In the first approach, women are screened for anogenital colonization at 35 to 37 weeks of pregnancy by means of a swab culture of the lower vagina and anorectum; intrapartum chemoprophylaxis is offered to carriers and is *recommended* for those carriers with any of the risk factors noted above, those anticipating multiple births, and those who have previously given birth to an infant with group B streptococcal infection. In the second approach, screening cultures need not be performed, but intrapartum chemoprophylaxis is recommended for *all* women with one or more of the risk factors noted above. The recommended regimen for chemoprophylaxis is 5 million units of penicillin G followed by 2.5 million units every 4 h until delivery. Clindamycin or erythromycin may be substituted in women allergic to penicillin.

Treatment of all pregnant women who are colonized or who have risk factors for neonatal infection will result in exposure of 15 to 25% of pregnant women and newborns to antibiotics, with the attendant risks of allergic reactions and selection for resistant organisms. Although still in the developmental stages, a group B streptococcal vaccine may ultimately offer a better solution to prevention. Because transplacental passage of maternal antibodies produces protective antibody levels in the newborn, efforts are under way to develop a vaccine against group B streptococci that can be given to

childbearing women before or during pregnancy. Results of phase 1 clinical trials of group B streptococcal capsular polysaccharide-protein conjugate vaccines suggest that a multivalent conjugate vaccine would be safe and highly immunogenic.

INFECTION IN ADULTS

The majority of group B streptococcal infections in adults are related to pregnancy and parturition. Peripartum fever, the most common manifestation, is sometimes accompanied by symptoms and signs of endometritis or chorioamnionitis (abdominal distention and uterine or adnexal tenderness). Blood cultures are often positive, as are cultures of vaginal swabs. Bacteremia is usually transitory but occasionally results in meningitis or endocarditis. Infections in adults that are not associated with the peripartum period generally involve individuals who are elderly or have some underlying chronic illness, such as diabetes mellitus or a malignancy. Among the infections that develop with some frequency in adults are cellulitis and soft tissue infection (including infected diabetic skin ulcers), urinary tract infection, pneumonia, endocarditis, and septic arthritis. Other reported infections include meningitis, osteomyelitis, and intraabdominal or pelvic abscesses.

TREATMENT

Group B streptococci are less sensitive to penicillin than group A organisms, requiring somewhat higher doses. Adults with serious localized infections (pneumonia, pyelonephritis, abscess) should receive doses in the range of 12 million units of penicillin G daily, while patients with endocarditis or meningitis should receive 18 to 24 million units per day in divided doses. Vancomycin is an acceptable alternative for patients allergic to penicillin.

ENTEROCOCCI, GROUP D STREPTOCOCCI

ENTEROCOCCI

Lancefield group D includes the enterococci, organisms now classified in a separate genus from other streptococci, and nonenterococcal group D streptococci. Enterococci are distinguished from nonenterococcal group D streptococci by their ability to grow in the presence of 6.5% sodium chloride and by the results of other biochemical tests. The enterococcal species that are significant pathogens for humans are *E. faecalis* and *E. faecium*. These organisms tend to produce infection in patients who are elderly or debilitated or in whom mucosal or epithelial barriers have been disrupted or the balance of the normal flora altered by antibiotic treatment. Urinary tract infections due to enterococci are quite common, particularly among patients who have received antibiotic treatment or undergone instrumentation of the urinary tract. Enterococci are a frequent cause of nosocomial bacteremia in patients with intravascular catheters. These organisms account for 10 to 20% of cases of bacterial endocarditis on both native and prosthetic valves. The presentation of enterococcal endocarditis is usually subacute but may be acute, with rapidly progressive valve destruction. Enterococci are frequently cultured from bile and are involved in infectious complications of biliary surgery and in liver abscesses. Moreover, enterococci are often isolated from polymicrobial infections arising from the bowel flora (e.g., intraabdominal abscesses), from abdominal surgical

wounds, and from diabetic foot ulcers. While such mixed infections are frequently cured by antimicrobials not active against enterococci, specific therapy directed against enterococci is warranted when these organisms are the predominant species or are isolated from blood cultures.

TREATMENT

Unlike other streptococci, enterococci are not reliably killed by penicillin or ampicillin alone at concentrations achieved clinically in the blood or tissues. Ampicillin reaches sufficiently high urinary concentrations to constitute adequate monotherapy for uncomplicated urinary tract infections. Because in vitro testing has shown evidence of synergistic killing of most enterococcal strains by the combination of penicillin or ampicillin with an aminoglycoside, combined therapy is recommended for enterococcal endocarditis and meningitis; the regimen is penicillin (3 to 4 million units every 4 h) or ampicillin (2 g every 4 h) plus moderate-dose gentamicin (1 mg/kg every 8 h for patients with normal renal function). Enterococcal endocarditis should be treated for a minimum of 4 weeks and for 6 weeks if symptoms have been present for ≥ 3 months or if the infection involves a prosthetic heart valve. For nonendocarditis bacteremia and other serious enterococcal infections, it is not known whether the efficacy of single-agent β -lactam therapy is improved by the addition of gentamicin, but many infectious disease specialists use combination therapy for such infections, especially in critically ill patients. Vancomycin, in combination with gentamicin, may be substituted for penicillin in allergic patients. Enterococci are resistant to all cephalosporins; therefore, this class of antibiotics should not be used for treatment of enterococcal infections.

Antimicrobial susceptibility testing should be performed routinely on enterococcal isolates from patients with serious infections, and therapy should be adjusted according to the results ([Table 140-4](#)). Most enterococci are resistant to streptomycin, and this drug should not be used for treatment of enterococcal infection unless in vitro testing of the strain indicates susceptibility. Though less widespread than streptomycin resistance, high-level resistance to gentamicin -- with a minimum inhibitory concentration (MIC) of >2000 $\mu\text{g/mL}$ -- has become common. Gentamicin-resistant enterococci should be tested for susceptibility to streptomycin; occasional gentamicin-resistant enterococci are sensitive to streptomycin. If the isolate is resistant to all aminoglycosides, treatment with penicillin or ampicillin alone may be successful. The prolonged administration (i.e., for at least 6 weeks) of high-dose ampicillin (e.g., 12 g/d) is recommended for endocarditis due to these highly resistant enterococci.

Enterococci may be resistant to penicillins via two distinct mechanisms. The first is the production of β -lactamase (mediating resistance to penicillin and ampicillin), which has been reported for *E. faecalis* isolates from several locations in the United States and other countries. Because the amount of β -lactamase produced by enterococci may be insufficient for detection by routine antibiotic susceptibility testing, isolates from serious infections should be screened specifically for β -lactamase production with use of a chromogenic cephalosporin or by another method. For the treatment of β -lactamase-producing strains, vancomycin, ampicillin/sulbactam, amoxicillin/clavulanate, or imipenem may be used in combination with gentamicin.

The second mechanism of penicillin resistance is not mediated by β -lactamase and may

be due to altered penicillin-binding proteins. This intrinsic penicillin resistance is common among *E. faecium* isolates, which routinely are more resistant to β -lactam antibiotics than are isolates of *E. faecalis*. Moderately resistant enterococci (MICs of penicillin and ampicillin, 16 to 64 $\mu\text{g/mL}$) may be susceptible to high-dose penicillin or ampicillin plus gentamicin, but strains with MICs of $\geq 200 \mu\text{g/mL}$ must be considered resistant to clinically achievable levels of β -lactam antibiotics, including imipenem. Vancomycin plus gentamicin is the recommended regimen for infections due to enterococci with high-level intrinsic resistance to β -lactams.

Vancomycin-resistant enterococci, first reported from clinical sources in the late 1980s, have become common in many hospitals. Three major vancomycin resistance phenotypes have been described: VanA, VanB, and VanC. The VanA phenotype is associated with high-level resistance to vancomycin and to teicoplanin, a related glycopeptide antibiotic not currently available in the United States. VanB and VanC strains are resistant to vancomycin but susceptible to teicoplanin, although teicoplanin resistance may develop during treatment in VanB strains. For enterococci resistant to both vancomycin and β -lactams, there are no established therapies. Regimens that have been tried with some success in individual cases or experimentally include ciprofloxacin plus rifampin plus gentamicin; ampicillin plus vancomycin (particularly if in vitro testing shows synergistic bacteriostatic activity); and chloramphenicol or tetracycline (if the strain is susceptible in vitro). Quinupristin/dalfopristin (Synercid) is a streptogramin combination with in vitro activity against *E. faecium*, including vancomycin-resistant isolates, but not against *E. faecalis* or other enterococcal species. The evidence for clinical efficacy of this agent in serious enterococcal infections is limited, and, as of this writing, quinupristin/dalfopristin is not yet licensed for use in the United States.

OTHER GROUP D STREPTOCOCCI

The main nonenterococcal group D streptococcal species that causes human infections is *S. bovis*. *S. bovis* endocarditis is often associated with neoplasms of the gastrointestinal tract -- most frequently a colon carcinoma or polyp -- but is also reported in association with other bowel lesions. When occult gastrointestinal lesions are carefully sought, abnormalities are found in $\approx 60\%$ of patients with *S. bovis* endocarditis. In contrast to the enterococci, nonenterococcal group D streptococci like *S. bovis* are reliably killed by penicillin as a single agent, and penicillin is the treatment of choice for *S. bovis* infections.

VIRIDANS AND OTHER STREPTOCOCCI

VIRIDANS STREPTOCOCCI

Consisting of multiple species of α -hemolytic streptococci, the viridans streptococci are a heterogeneous group of organisms that are important as agents of bacterial endocarditis ([Chap. 126](#)). Several species of viridans streptococci, including *S. salivarius*, *S. mitis*, *S. sanguis*, and *S. mutans*, are part of the normal flora of the mouth, where they live in close association with the teeth and gingiva. Some species contribute to the development of dental caries. The transient viridans streptococcal bacteremia induced by eating, tooth-brushing, flossing, and other sources of minor trauma, together

with adherence to biologic surfaces, is thought to account for the predilection of these organisms to cause endocarditis. Viridans streptococci are also isolated, often as part of a mixed flora, from sites of sinusitis, brain abscess, and liver abscess.

Viridans streptococcal bacteremia occurs relatively frequently in neutropenic patients, particularly after bone marrow transplantation or high-dose chemotherapy for cancer. Some of these patients develop a sepsis syndrome with high fever and shock. Risk factors for viridans streptococcal bacteremia include chemotherapy with high-dose cytosine arabinoside, prior treatment with trimethoprim-sulfamethoxazole or a fluoroquinolone, treatment with antacids or histamine antagonists, mucositis, and profound neutropenia.

The *S. milleri* group (also referred to as the *S. intermedius* or *S. anginosus* group) includes three species that cause human disease: *S. intermedius*, *S. anginosus*, and *S. constellatus*. These organisms are often considered viridans streptococci, although they differ somewhat from other viridans streptococci in both their hemolytic pattern (they may be α-, β-, or nonhemolytic) and the disease syndromes they cause. This group commonly produces suppurative infections, particularly abscesses of brain and abdominal viscera, and infections related to the oral cavity or respiratory tract, such as peritonsillar abscess, lung abscess, and empyema.

TREATMENT

Isolates from neutropenic patients with bacteremia often are resistant to penicillin; thus these patients should be treated presumptively with vancomycin until the results of susceptibility testing become available. Viridans streptococci isolated in other clinical settings usually are sensitive to penicillin.

NUTRITIONALLY VARIANT STREPTOCOCCI

Occasional isolates cultured from the blood of patients with endocarditis fail to grow when subcultured on solid media. These *nutritionally variant streptococci* require supplemental thiol compounds or active forms of vitamin B₆ (pyridoxal or pyridoxamine) for growth in the laboratory. The nutritionally variant streptococci are generally grouped with the viridans streptococci because they cause similar types of infections. However, they have been reclassified on the basis of 16S RNA sequence comparisons into a separate genus, *Abiotrophia*, with two species: *A. defectivus* and *A. adjacens*.

TREATMENT

Because treatment failure and relapse appear to be more common for cases of endocarditis due to nutritionally variant streptococci than for usual viridans streptococci, the addition of gentamicin (1 mg/kg every 8 h for patients with normal renal function) to the penicillin regimen is recommended in therapy for endocarditis due to these organisms.

OTHER STREPTOCOCCI

S. suis is an important pathogen in swine and has been reported to cause meningitis in

humans, usually in individuals with occupational exposure to pigs. Strains of *S. suis* associated with human infections have generally reacted with Lancefield group R typing serum and sometimes with group D typing serum as well. Isolates may be α- or β-hemolytic and are sensitive to penicillin. *S. iniae*, a pathogen of fish, has been associated with infections in humans who have handled live or freshly killed fish. Cellulitis of the hand is the most common form of human infection, although bacteremia and endocarditis have been reported. *Anaerobic streptococci*, or *peptostreptococci*, are part of the normal flora of the oral cavity, bowel, and vagina. Infections caused by the anaerobic streptococci are discussed in [Chap. 167](#).

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141. DIPHTHERIA, OTHER CORYNEBACTERIAL INFECTIONS, AND ANTHRAX - Randall K. Holmes

DIPHTHERIA

DEFINITION

Diphtheria is a localized infection of mucous membranes or skin caused by *Corynebacterium diphtheriae*. A characteristic pseudomembrane may be present at the site of infection ([Fig. 141-CD1](#)). Some strains of *C. diphtheriae* produce diphtheria toxin, a protein that can cause myocarditis, polyneuritis, and other systemic toxic effects. Respiratory diphtheria is usually caused by toxinogenic (*tox+*) *C. diphtheriae*, but cutaneous diphtheria is frequently caused by nontoxinogenic (*tox-*) strains.

ETIOLOGY

C. diphtheriae is an aerobic, nonmotile, nonsporulating, irregularly staining, gram-positive rod. The bacteria are club-shaped and are often arranged in clusters (*Chinese letters*) or parallel arrays (*palisades*). *C. diphtheriae* forms gray to black colonies on selective media containing tellurite. The gravis, mitis, and intermedius biotypes are distinguished by colonial morphology and laboratory tests. Both *tox+* and *tox-* strains cause infections, and *tox+* strains of all three biotypes can cause severe disease. The gene for diphtheria toxin is present in specific corynephages, and *tox-* *C. diphtheriae* can acquire the ability to produce diphtheria toxin by infection with *tox+* phages (*phage conversion*). Growth of *C. diphtheriae* under low-iron conditions that mimic the environment of host tissues induces production of diphtheria toxin and expression of systems for siderophore-dependent iron uptake and utilization of iron from heme.

IMMUNOLOGY

Treatment of diphtheria toxin with formaldehyde converts it to a nontoxic but immunogenic product (*diphtheria toxoid*). Immunization with toxoid elicits antibody (*antitoxin*) that neutralizes the toxin and prevents diphtheria. The attack rate and mortality rate for diphtheria are low in immune individuals with antitoxin titers of >0.01 unit per milliliter. Antitoxin neither prevents colonization by *C. diphtheriae* nor eradicates the *carrier state*. When most individuals in a population have protective levels of antitoxin (*herd immunity*), the carrier rate for *tox+* strains of *C. diphtheriae* falls to a low level, and the risk that susceptible individuals will be exposed to *tox+* *C. diphtheriae* decreases dramatically. Susceptible individuals may contract diphtheria if they travel to regions where the disease is present or if *tox+* strains of *C. diphtheriae* are introduced into their community.

EPIDEMIOLOGY AND IMMUNITY

Humans are the principal reservoir for *C. diphtheriae*. Transmission occurs primarily by close personal contact. The risk is greater that *C. diphtheriae* will be transmitted to susceptible individuals from patients with diphtheria than from carriers. The incubation period for respiratory diphtheria is typically 2 to 5 days and rarely up to 8 days.

Cutaneous diphtheria is usually a secondary infection whose signs develop an average of 7 days (range, 1 to >21 days) after the appearance of primary dermatologic lesions of other etiologies.

In temperate climates, diphtheria primarily involves the respiratory tract; occurs throughout the year, with a peak incidence in colder months; and is usually caused by *tox+* *C. diphtheriae*. Before immunization was introduced, diphtheria was primarily a disease of children; it affected up to 10% of individuals in this group and sometimes caused devastating epidemics. Most young infants were immune because of transplacental transfer of maternal IgG antitoxin, but children became susceptible by 6 to 12 months of age. Approximately 75% of individuals became immune by age 10 as a result of contact with *C. diphtheriae*. Mortality rates of 30 to 40% were common in untreated disease and were sometimes >50% in epidemics. Treatment with antitoxin reduced the case-fatality rate to 5 to 10%.

Routine immunization of children in the United States resulted in a progressive decrease of diphtheria from the peak of 206,939 cases (incidence rate, 191 cases per 100,000 population) in 1921 to <5 cases per year since 1980. Concomitantly, circulation of *tox+* strains of *C. diphtheriae* among the population decreased dramatically, although an endemic focus for transmission of *tox+* and *tox-* strains without clinical disease was recently identified in North Dakota. As the incidence rate of diphtheria decreased, a higher proportion of cases occurred in older persons (who were never immunized or whose immunity waned because it was not boosted by either booster doses of vaccine or contact with *C. diphtheriae*), but the case-fatality ratio remained unchanged at 5 to 10%. High rates of immunization are currently achieved by school entry (>96%), but immunization rates for younger children are substantially lower. Among adults >20 years of age, 19 to 77% are susceptible as a consequence of failure to receive periodic booster immunizations and lack of contact with *C. diphtheriae*. The most recent large diphtheria outbreak in the United States (about 1100 cases) occurred in Seattle, Washington, between 1972 and 1982. Alcoholism, low socioeconomic status, crowded living conditions, and Native-American ethnic background were significant risk factors in this outbreak.

A massive diphtheria epidemic (>157,000 cases and 5000 deaths) occurred recently in the states of the former Soviet Union and accounted for >80% of diphtheria cases reported worldwide during that interval. The epidemic began in 1990 with 1436 cases (0.49 per 100,000 population), peaked in 1995 with 50,425 cases (17.29 per 100,000 population), and waned by 1998 with 2720 cases (0.93 per 100,000 population) as the result of a mass immunization program. The progression of the epidemic was associated with emergence of a previously uncommon clonal group of *tox+* *C. diphtheriae* strains that accounted for >80% of isolates by 1994; molecular analysis of the *tox* gene from these strains demonstrated that the existing diphtheria toxoid vaccine remained appropriate for control of the epidemic by vaccination.

A majority of cases throughout this epidemic occurred in persons [≥]15 years old, and adults from 40 to 49 years old had very high incidence and death rates. In 1994, case-fatality rates varied from 2.8% in the Russian Federation to 23% in Lithuania and Turkmenistan. Factors that facilitated the spread of this epidemic included large-scale population movements, socioeconomic instability, deteriorating health infrastructure,

delayed implementation of aggressive control measures in response to the epidemic, inadequate information for physicians and the public, and frequent shortages of supplies for prevention and treatment of the disease. The most important risk factor for diphtheria in the Republic of Georgia was lack of vaccination (matched odds ratio, 19.2), but household diphtheria exposure, exposure to skin lesions, the presence of tonsils, a history of eczema, preceding fever with myalgia, sharing a bed, sharing glasses and cups, and taking a bath less often than weekly were also significant risk factors. Although small numbers of imported cases from this epidemic occurred in western European countries, none resulted in secondary transmission of diphtheria, notwithstanding a high proportion of susceptible adults in countries with imported cases. Inadequate primary immunization of children in the states of the former Soviet Union in the years preceding the epidemic, along with failure to maintain adequate immunity in adults by booster immunization, may have synergistically facilitated transmission of diphtheria and emergence of the massive epidemic in this region.

In the tropics, cutaneous diphtheria is more common than respiratory diphtheria, occurs throughout the year, and often develops as a secondary infection complicating other dermatoses. Isolates of *C. diphtheriae* from skin lesions are more often *tox*- than *tox*+. Cutaneous diphtheria is increasingly recognized in temperate climates and accounted for 86% of the 1100 cases in the Seattle epidemic of 1972 to 1982. Since 1980, cutaneous diphtheria has not been a reportable disease in the United States, and recent health statistics include only respiratory diphtheria.

During the 1990s, *tox*- strains of *C. diphtheriae* were associated with new types of infections. In the United Kingdom, these strains caused symptomatic pharyngitis, predominantly among homosexual men, that was sometimes accompanied by tonsillar exudate. In Switzerland, strains with a high potential for invasiveness were isolated from 38 intravenous drug users and shown by ribotyping to be clonally related. The latter strains caused infections of the skin (15 cases), respiratory tract (10 cases), and blood (13 cases). Among the patients with bloodstream infections, 9 had endocarditis, and 4 of these 9 patients died.

PATHOLOGY AND PATHOGENESIS

C. diphtheriae infects mucous membranes, most commonly in the respiratory tract, and also invades open skin lesions resulting from insect bites or trauma. In infections caused by *tox*+ *C. diphtheriae*, initial edema and hyperemia are often followed by epithelial necrosis and acute inflammation. Coagulation of the dense fibrinopurulent exudate produces a pseudomembrane, and the inflammatory reaction accompanied by vascular congestion extends into the underlying tissues. The pseudomembrane contains large numbers of *C. diphtheriae* organisms, but the bacteria are rarely isolated from the blood or internal organs.

Diphtheria toxin acts both locally and systemically, and the lethal dose for humans is ~0.1 ug/kg. Toxin contributes locally to pseudomembrane formation; systemically, it can cause myocarditis, neuritis, and focal necrosis in various organs, including the kidneys, liver, and adrenal glands. Changes in the myocardium include cloudy swelling of muscle fibers and interstitial edema. These changes are followed within weeks by hyaline and granular degeneration (sometimes with fatty degeneration), progressing to myolysis and

finally to the replacement of lost muscle by fibrosis. Thus, diphtheria can cause permanent cardiac damage. In diphtheritic polyneuritis, pathologic changes include patchy breakdown of myelin sheaths in peripheral and autonomic nerves, but recovery of nerve damage is the rule if the patient survives.

Diphtheria toxin is produced by *C. diphtheriae* as an extracellular polypeptide. Proteolytic cleavage forms nicked toxin consisting of fragments A and B. Fragment B binds to a plasma-membrane receptor (a precursor of a heparin-binding growth factor resembling epidermal growth factor), and the bound toxin is internalized by receptor-mediated endocytosis. Fragment A is translocated across the endosomal membrane and released into the cytoplasm, where it catalyzes the transfer of the adenosine diphosphate ribose moiety from nicotinamide adenine dinucleotide (NAD) to a modified histidine residue (diphthamide) on elongation factor 2 (EF-2), thereby inactivating EF-2 and inhibiting protein synthesis. One molecule of fragment A in the cytoplasm can kill a cell. Other metabolic alterations are secondary to inhibition of protein synthesis.

CLINICAL MANIFESTATIONS

Patients with *C. diphtheriae* in the respiratory tract are classified as diphtheria cases if pseudomembranes are present and as diphtheria carriers if pseudomembranes are absent. The disease is graded as *tonsillar* if pseudomembranes are localized to the tonsils, as *combined types* or *delayed diagnosis* if more extensive pseudomembranes are present, and as *severe* if cervical adenopathy or cervical edema is also present. Onset is often gradual, but most patients seek medical care within a few days of becoming ill. Fever of 37.8° to 38.9°C (100° to 102°F), sore throat, and weakness are the most common symptoms, while dysphagia, headache, and change of voice occur in fewer than half of patients. Neck edema and difficulty breathing are noted in £10% of patients and are associated with an increased risk of death. Systemic manifestations are due primarily to toxic effects of diphtheria toxin. Patients without toxicity exhibit discomfort and malaise associated with local infection, whereas severely toxic patients may develop listlessness, pallor, and tachycardia that can progress rapidly to vascular collapse.

Primary infection in the respiratory tract is most often tonsillopharyngeal but may also be (in decreasing order) laryngeal, nasal, and tracheobronchial. Multiple sites are frequently involved, and secondary spread of pharyngeal infection upward to the nasal mucosa or downward to the larynx and tracheobronchial tree is much more common than primary infection at those sites. Systemic toxicity is usually most severe when extensive pseudomembrane extends from the tonsils and pharynx into contiguous regions. A small percentage of patients present with malignant or "bull-neck" diphtheria, with extensive pseudomembrane formation, foul breath, massive swelling of the tonsils and uvula, thick speech, cervical lymphadenopathy, striking edematous swelling of the submandibular region and anterior neck, and severe toxicity.

In tonsillopharyngeal diphtheria, isolated spots of gray or white exudate may appear first. These spots often extend and coalesce within a day to form a confluent, sharply demarcated pseudomembrane that becomes progressively thicker, more tightly adherent to the underlying tissue, and darker gray in color. Unlike the exudate in

streptococcal pharyngitis, the diphtheritic pseudomembrane often extends beyond the margin of the tonsils onto the tonsillar pillars, palate, or uvula. Dislodging the membrane is likely to cause bleeding. Laryngeal diphtheria often presents as hoarseness and cough. Demonstration of laryngeal pseudomembrane by laryngoscopy helps distinguish diphtheria from other infectious forms of laryngitis. Patients with nasal diphtheria may present with unilateral or bilateral serosanguineous nasal discharge associated with irritation of the nares or lip. Primary or secondary diphtheritic infection occasionally involves other mucous membranes, including the conjunctiva and the membranes of the genitourinary and gastrointestinal tracts.

Cutaneous diphtheria usually presents as an infection by *C. diphtheriae* of preexisting dermatoses involving the lower extremities, upper extremities, head, or trunk. The clinical features are similar to those of other secondary cutaneous bacterial infections. In the tropics, cutaneous diphtheria may present as a primary cutaneous lesion, typically with morphologically distinct "punched-out" ulcers that are covered by necrotic slough or membrane and have well-demarcated edges.

C. diphtheriae is an occasional cause of invasive infections, including endocarditis and septic arthritis. Risk factors for such infections include preexisting cardiac abnormalities, abuse of intravenous drugs, and alcoholic cirrhosis.

COMPLICATIONS

Obstruction of the respiratory tract can be caused by extensive pseudomembrane formation and swelling early in the disease or by sloughed pseudomembrane that becomes lodged in the airways later in the disease. The risk is greater when infection involves the larynx or the tracheobronchial tree and in children because of the small size of the airways.

Myocarditis and polyneuritis are the most prominent toxic manifestations of diphtheria. The risk of each is proportional to the severity of local disease. Myocarditis occurred in 22% and neuritis in 5% of 656 hospitalized patients (54% female, 70% ≤ 15 years old) with diphtheria in the Kyrgyz Republic in 1995; 7% of patients with myocarditis and 2% of patients without myocarditis died. The median interval from hospitalization to death was 4.5 days (range, 0 to 13 days).

Bulbar dysfunction in diphtheritic neuritis typically develops during the first 2 weeks. Palatal and pharyngeal paralysis usually develops first. Swallowing is difficult, the voice is nasal, and ingested fluids may be regurgitated through the nose. Additional bulbar signs may develop over several weeks, with oculomotor and ciliary paralysis more common than facial or laryngeal paralysis. Peripheral polyneuritis typically begins from 1 to 3 months after the onset of diphtheria with proximal weakness of the extremities, which spreads distally. Paresthesia may occur, most often in a glove-and-stocking distribution. Polyneuritis usually resolves completely, with the time needed for improvement approximately equal to that elapsing from exposure to the development of symptoms.

Pneumonia occurs in more than one-half of fatal cases of diphtheria. Less common complications include renal failure, encephalitis, cerebral infarction, pulmonary

embolism, and bacteremia or endocarditis due to invasive infection by *C. diphtheriae*. Serum sickness may result from antitoxin therapy.

COURSE AND PROGNOSIS

Most cases of diphtheria develop in nonimmunized patients. The attack rate, severity of disease, and risk of complications are much lower in immunized patients. The pseudomembrane may continue to increase in size during the first day after administration of antitoxin. During the next several days to a week, it becomes softer, less adherent, and nonconfluent and eventually disappears. In the preantibiotic era, *C. diphtheriae* persisted in the throat for ~2 weeks in one-half of patients and for ³1 month in about one-fifth. Mortality increases with the severity of local disease, the extent of pseudomembrane formation, and the delay between onset of local disease and administration of antitoxin. The death rate is highest during the first week of illness; among patients with bull-neck diphtheria; among patients with myocarditis who develop ventricular tachycardia, atrial fibrillation, or complete heart block; among patients with laryngeal or tracheobronchial involvement; among infants and patients >60 years of age; and among alcoholics. Both the mortality rate and the risk of myocarditis or peripheral neuropathy are significantly lower in cutaneous diphtheria than in respiratory diphtheria.

DIAGNOSIS

A characteristic pseudomembrane on the mucosa of the tonsils, palate, oropharynx, nasopharynx, nose, or larynx suggests diphtheria but is not uniformly present. Diphtheritic pseudomembrane must be distinguished from other pharyngeal exudates, including those of group A β -hemolytic streptococcal infections, infectious mononucleosis, viral pharyngitis, fusospirochetal infection, and candidiasis. Diphtheria should be considered in patients with sore throat, cervical adenopathy or swelling, and low-grade fever, especially when these manifestations are accompanied by systemic toxicity, hoarseness, stridor, palatal paralysis, or serosanguineous nasal discharge with or without demonstrable pseudomembrane. Treatment with diphtheria antitoxin should begin as soon as the clinical diagnosis of diphtheria is made.

Definitive diagnosis of diphtheria depends on the isolation of *C. diphtheriae* from local lesions. The laboratory should be notified that diphtheria is suspected to ensure the use of selective tellurite medium appropriate for the isolation of *C. diphtheriae*. All isolates of *C. diphtheriae* should be subjected to toxicity testing. Primary isolates can be screened rapidly for the presence of the *tox* gene by the polymerase chain reaction, although occasional strains of *C. diphtheriae* that carry an inactive toxin gene give false-positive results. Biochemical tests needed to differentiate *C. diphtheriae* from corynebacteria of the normal flora (diphtheroids) require several days. Group A β -hemolytic streptococci and *Staphylococcus aureus* are also isolated frequently from patients with diphtheria.

Cutaneous diphtheria may present as a characteristic "punched-out" ulcer with a membrane, but it is more often indistinguishable from other inflammatory dermatoses. Diagnosis depends on a high degree of suspicion and on culture of cutaneous lesions on laboratory media appropriate for isolation of *C. diphtheriae*. Throat samples from all patients with cutaneous diphtheria should be cultured for *C. diphtheriae*.

TREATMENT

The decision to administer diphtheria antitoxin must be based on the clinical diagnosis of diphtheria without definitive laboratory confirmation, since each day of delay in treatment is associated with increased mortality. Because diphtheria antitoxin is produced in horses, it is necessary to inquire about possible allergy to horse serum and to perform a conjunctival or intracutaneous test with diluted antitoxin for immediate hypersensitivity. Epinephrine must be available for immediate administration to patients with severe allergic reactions. Patients with immediate hypersensitivity should be desensitized before a full therapeutic dose of antitoxin is given. The dose of diphtheria antitoxin currently recommended by the Committee on Infectious Diseases of the American Academy of Pediatrics is based on the site of the primary infection and the duration and severity of disease: 20,000 to 40,000 units for disease that has been present for ≤ 48 h and involves the pharynx or larynx; 40,000 to 60,000 units for nasopharyngeal infections; and 80,000 to 100,000 units for disease that is extensive, has been present for ≥ 3 days, or is accompanied by diffuse swelling of the neck. Antitoxin is administered intravenously by infusion in saline over 60 min to neutralize unbound toxin rapidly. The $\sim 10\%$ risk of serum sickness is acceptable because of the established therapeutic value of antitoxin in decreasing mortality from respiratory diphtheria. The risk of systemic toxicity is lower in cutaneous diphtheria than in respiratory diphtheria and must be weighed against the potential adverse effects of antitoxin treatment; authorities are not unanimous in recommending antitoxin therapy for cutaneous diphtheria.

Antibiotics have little demonstrated effect on the healing of local infection in diphtheria patients treated with antitoxin. The primary goal of antibiotic therapy for patients or carriers is therefore to eradicate *C. diphtheriae* and prevent its transmission from the patient to susceptible contacts. Erythromycin, penicillin G, rifampin, or clindamycin is recommended by most authorities. Commonly recommended regimens for the treatment of adults with respiratory diphtheria are erythromycin (500 mg four times daily, given parenterally or orally) or intramuscular procaine penicillin G (600,000 units at 12-h intervals) for 14 days. Patients with cutaneous diphtheria and carriers can be treated orally with erythromycin (500 mg four times daily) or rifampin (600 mg once daily) for 7 days. If compliance is in question, a single dose of benzathine penicillin G (1.2 to 2.4 million units intramuscularly) can be substituted. Eradication of *C. diphtheriae* should be documented by negative cultures of samples taken on two or three successive days, beginning at least 24 h after the completion of antibiotic therapy. Some authorities also recommend a repeat throat culture 2 weeks later. The small percentage of patients who continue to be infected with *C. diphtheriae* after treatment should receive an additional 10-day course of oral erythromycin or rifampin. Plasmid-mediated resistance to erythromycin of the MLS type emerged transiently in *C. diphtheriae* during the Seattle epidemic, but its frequency declined dramatically after the routine use of erythromycin was discontinued.

Patients with respiratory or cutaneous diphtheria caused by *tox+* *C. diphtheriae* or by strains of unknown toxinogenicity should be hospitalized, kept in bed initially, handled with isolation procedures appropriate for the site of infection, and given supportive care as needed. Respiratory and cardiac function must be monitored closely. Early intubation

or tracheostomy is recommended when the larynx is involved or signs of impending airway obstruction are detected. Tracheobronchial membrane can sometimes be removed mechanically via the endotracheal tube or tracheostomy. Primary or secondary pneumonia should be diagnosed and treated promptly. Sedative or hypnotic drugs that may mask respiratory symptoms are contraindicated. Close electrocardiographic monitoring, treatment of arrhythmias, and electrical pacing for heart block are essential. Congestive heart failure should be treated as described in [Chap. 232](#). Glucocorticoids do not reduce the risk of diphtheritic myocarditis or polyneuritis. Ulcerative or ecthymatous cutaneous lesions should be treated with Burow's solution applied on wet compresses after debridement of necrotic areas, and treatment for associated conditions such as pediculosis, scabies, or underlying dermatoses should be instituted. Recovery from diphtheria does not always confer active immunity, and initiation of an immunization regimen for diphtheria that is appropriate for the patient's age should be an integral part of the treatment plan.

PREVENTION

Vaccines available in the United States for immunization against diphtheria include diphtheria and tetanus toxoids and pertussis vaccine adsorbed (DTP), diphtheria and tetanus toxoids and acellular pertussis vaccine adsorbed (DTaP), diphtheria and tetanus toxoids adsorbed (DT; for pediatric use), and tetanus and diphtheria toxoids adsorbed (Td; for adult use). DTaP is preferred over DTP for primary immunization of children without contraindications, and use of DTP is no longer recommended. Td contains less diphtheria toxoid than DTP, DTaP, or DT and causes fewer adverse reactions in adults. Current guidelines for primary immunization of children and adults against diphtheria and for maintaining immunity by periodic booster doses of appropriate vaccines throughout life are summarized in [Chap. 122](#).

Close contacts of diphtheria patients should be cultured for *C. diphtheriae*, kept under surveillance for 1 week, and treated with appropriate antibiotics if cultures are positive. Previously immunized close contacts should receive an appropriate booster containing diphtheria toxoid if their last booster was given >5 years previously. If immunization status is uncertain, close contacts should receive an antibiotic regimen appropriate for carriers and a primary immunization series appropriate for their age.

OTHER CORYNEBACTERIAL INFECTIONS

DEFINITION

Medically important coryneform bacteria (formerly called *diphtheroids*) include members of the normal flora that cause opportunistic infections, human pathogens of relatively low virulence, and animal pathogens that cause occasional zoonotic infections. Reported infections caused by coryneform bacteria have increased substantially in number over the past two decades. Isolates of *C. jeikeium* and *C. urealyticum* are often resistant to multiple antibiotics.

ETIOLOGY AND LABORATORY DIAGNOSIS

Because coryneform bacteria are potential pathogens, it is important not to dismiss

them as constituents of the normal flora or as contaminants when they are found in clinical specimens. Laboratory differentiation of coryneform bacteria is important when they are isolated repeatedly, when they are recovered in pure culture or in large numbers, or when they form pigmented or hemolytic colonies.

The coryneform bacteria are a large, heterogeneous group of gram-positive, pleomorphic, irregularly staining bacilli or coccobacilli that superficially resemble *C. diphtheriae* and are difficult to identify and classify. The genus *Corynebacterium* is currently divided into three groups of species: the nonlipophilic, fermentative corynebacteria (including *C. diphtheriae*, *C. xerosis*, *C. striatum*, *C. minutissimum*, and others); the nonlipophilic, nonfermentative corynebacteria (including *C. pseudodiphtheriticum* and others); and the lipophilic corynebacteria (including *C. jeikeium*, *C. urealyticum*, and others). Coryneform bacteria also belong to many other genera (including *Actinomyces*, *Arcanobacterium*, and *Rhodococcus*) as well as to several groups that have not yet been assigned to genera and species by the U.S. Centers for Disease Control and Prevention (CDC).

ECOLOGY AND EPIDEMIOLOGY

Humans are the probable natural reservoir for *C. xerosis*, *C. pseudodiphtheriticum* (formerly *C. hofmannii*), *C. striatum*, *C. minutissimum*, *C. jeikeium* (formerly CDC group JK), *C. urealyticum* (formerly CDC group D2), and *Arcanobacterium haemolyticum* (formerly *C. haemolyticum*). Animals are the probable natural reservoir for *Actinomyces pyogenes* (formerly *C. pyogenes*; cows, sheep, pigs), *C. ulcerans* (cows, horses), and *C. pseudotuberculosis* (sheep, horses, goats, cattle). The natural reservoir for *Rhodococcus equi* (formerly *C. equi*) is soil. The ecologic niches for many other coryneform bacteria of medical importance are not well defined.

The coryneform bacteria found most frequently as components of the normal flora include *C. pseudodiphtheriticum* (pharynx, skin), *C. xerosis* (conjunctival sac, nasopharynx, skin), and *C. striatum* (anterior nares, skin). Coryneform bacteria that commonly colonize the skin of hospitalized patients include *C. jeikeium* (axilla, groin, perineum) and *C. urealyticum*. *C. jeikeium* most often colonizes patients with malignancies or severe immunodeficiency; it is also isolated from environmental sources (surfaces, air) in hospitals and from the hands of ward staff. *C. ulcerans* infections are acquired by consumption of raw milk. *C. pseudotuberculosis* infections are acquired by contact with animals or animal products or by consumption of raw milk.

PATHOGENESIS AND CLINICAL MANIFESTATIONS

C. jeikeium was recognized in 1976 as a cause of infections in immunocompromised hosts. This organism also causes infections in immunocompetent hosts, but severe infections continue to be most frequent in patients with hematologic malignancies and neutropenia. Skin colonization precedes clinical infection. Additional risk factors for nosocomial *C. jeikeium* sepsis include prolonged hospitalization, breaks in the integument, chronic intravascular catheterization, and prior treatment with broad-spectrum antibiotics. Other presentations of *C. jeikeium* infection include endocarditis, device-related infections, pulmonary infiltrates, cutaneous septic emboli, soft tissue infections, and rashes. Endocarditis due to *C. jeikeium* occurs primarily in

patients with prosthetic heart valves. *C. jeikeium* is a rare cause of central nervous system infections in patients with ventricular shunts.

C. urealyticum (formerly [CDC](#) group D2) was identified in 1985 as a significant cause of nosocomial urinary tract infections, including acute and chronic cystitis and pyelonephritis. The organism closely resembles *C. jeikeium* but differs from the latter by producing urease and failing to convert glucose to acidic metabolites. Hydrolysis of urea by urease causes alkalinization of the urine and formation of ammonium magnesium phosphate (struvite) stones. *C. urealyticum* is a cause of alkaline-encrusted cystitis in patients with preexisting bladder lesions that serve as foci for precipitation of struvite crystals. Risk factors associated with symptomatic urinary tract infections include preexisting immunosuppression, recent urologic procedures (including renal transplantation), underlying disorders of the genitourinary tract, and a history of urinary tract infections.

A. haemolyticum causes pharyngitis and chronic skin ulcers; less frequently, it causes a variety of deep tissue infections, septicemia, and endocarditis. Some 90% of *A. haemolyticum* infections occur in patients between 10 and 30 years old. *A. haemolyticum* pharyngitis in this age group is 5 to 13% as frequent as *Streptococcus pyogenes* pharyngitis. An erythematous rash is present in 30 to 67% of cases. The rash is usually scarlatiniform and most pronounced on the trunk and proximal extremities, but it sometimes resembles urticaria or erythema multiforme. Because rash is more frequent in *A. haemolyticum* infections than in *S. pyogenes* infections, *A. haemolyticum* should be considered as a possible etiology in older children and adults who present with the scarlet fever syndrome. Infection due to *A. haemolyticum* can also present as extensive pharyngeal exudate and can mimic diphtheria. *A. haemolyticum* occasionally causes peritonsillar abscess, sepsis, endocarditis, or meningitis.

C. minutissimum is frequently isolated from the lesions of erythrasma ([Fig. 141-CD2](#)), a common superficial skin infection characterized by the presence in intertriginous areas of reddish-brown, scaly, pruritic, macular patches that exhibit coral-red fluorescence under a Wood's light. The etiology of erythrasma appears to be polymicrobial; infection of the skin by *C. minutissimum* has been shown to follow the onset of maceration and scaling. Deep infections caused by *C. minutissimum* are rare and include abscesses, bacteremia, endocarditis, peritonitis, pyelonephritis, and infection of central venous catheters.

Among coryneform bacteria that cause disease in animals and occasionally in humans, *R. equi* has emerged as an important intracellular opportunistic pathogen in immunocompromised patients. Most reported cases are necrotizing pulmonary infections that resemble tuberculosis or nocardiosis in patients with severely defective cell-mediated immunity. Cases of *R. equi* infection are being diagnosed with increasing frequency in patients with AIDS.

A. pyogenes causes bovine mastitis, a disease transmitted by flies. Yearly epidemics of leg ulcers infected with *A. pyogenes* occurred among schoolchildren in Thailand between 1979 and 1984 and were postulated to have resulted from introduction of the organism into traumatic skin lesions by flies. Reported *A. pyogenes* infections in adults in Denmark have included abscesses, cystitis, intraabdominal infections, and mastoiditis.

with bacteremia.

C. ulcerans infections in humans usually present as pharyngitis and can mimic respiratory diphtheria, whereas infections caused by *C. pseudotuberculosis* typically present as suppurative granulomatous lymphadenitis. Some strains of *C. ulcerans* and *C. pseudotuberculosis* produce diphtheria toxin. Human infections by *tox*⁺ strains of *C. ulcerans* -- but not by *tox*⁺ strains of *C. pseudotuberculosis* -- have been reported, and administration of diphtheria antitoxin is therefore warranted in infections by *C. ulcerans* that are presumed on clinical grounds to be caused by toxinogenic strains.

C. pseudodiphtheriticum, a commensal of low virulence, is an uncommon cause of pneumonia in men with AIDS and of endocarditis, necrotizing tracheitis, tracheobronchitis, and urinary tract infection in patients without known immune deficiencies. Likewise, *C. xerosis* and *C. striatum* only occasionally cause human infections.

DIAGNOSIS

The clinical features of *C. jeikeium* infections are not pathognomonic. The diagnosis of these infections is based on a high index of suspicion, identification of the organism by culture in appropriate clinical specimens, and exclusion of other likely causes of infection.

C. urealyticum often goes undetected by routine urine cultures; rather, it is necessary to incubate the cultures for 24 to 48 h on blood agar or on special media. Cultivation should be prolonged in selected cases -- i.e., those involving patients (especially elderly men with preexisting genitourinary abnormalities) with alkaline urine, ammonium magnesium phosphate stones, gram-positive bacilli in the urine, or negative standard urine cultures despite clinical evidence of bacteriuria. Other microbes that can cause urinary tract infections with alkaline urine include *Proteus*, *Ureaplasma*, and some staphylococci and streptococci. Alkaline-encrusted cystitis is an anatomic diagnosis made by cystoscopy.

The differential diagnosis of *A. haemolyticum* pharyngitis with rash includes scarlet fever; rubella; staphylococcal and streptococcal toxic shock syndromes; infections caused by Epstein-Barr virus, cytomegalovirus, and enteroviruses (especially coxsackieviruses); disseminated gonococcal infection; secondary syphilis; and drug allergy. Routine diagnostic methods for throat cultures are not ideal for the detection of *A. haemolyticum*, nor is this organism detected by the rapid tests for *S. pyogenes* that are sometimes substituted for throat cultures. Pharyngitis caused by *A. haemolyticum* in adolescents and adults is likely to be underdiagnosed until improved tests for the organism are used by diagnostic laboratories.

Erythrasma is diagnosed clinically. Because of uncertainty about the etiologic role of *C. minutissimum*, culture of erythrasma lesions is not currently recommended. Pharyngitis caused by *tox*⁺ strains of *C. ulcerans* may be clinically indistinguishable from diphtheria. The presentations of infections caused by other coryneform bacteria are not usually diagnostic; cultures are required for identification of the causal organisms.

TREATMENT

Strains of *C. jeikeium* are typically resistant to most antibiotics. Vancomycin is the drug of choice for empirical treatment of infections caused by this organism, although antimicrobial susceptibility testing may reveal other antibiotic options for some isolates. For device-related *C. jeikeium* infections, removal of the infected device is usually required in addition to appropriate antibiotic therapy.

C. urealyticum is often resistant to the antibiotics used commonly for the treatment of urinary tract infections. Empirical treatment with vancomycin is appropriate pending the results of antimicrobial susceptibility testing. Several courses of antibiotic therapy may be necessary for bacteriologic cure. Patients with alkaline-encrusted cystitis require resection of the encrusted lesions in addition to antibiotic therapy.

No controlled trials of treatment for *A. haemolyticum* infections have been performed. In vitro tests usually demonstrate susceptibility to penicillins, erythromycin, azithromycin, clindamycin, doxycycline, ciprofloxacin, and vancomycin, but treatment failures have been reported with appropriate doses of penicillins. Limited data suggest that the clinical course of *A. haemolyticum* pharyngitis may be shortened by treatment with erythromycin.

Infections with *C. ulcerans* that present like diphtheria or are known to be caused by *tox+* strains should be treated like diphtheria. Oral erythromycin is usually effective for treatment of erythrasma. For infections caused by *R. equi*, vancomycin is the drug of choice. Possible alternatives include erythromycin, rifampin, aminoglycosides, and chloramphenicol; the combination of erythromycin and rifampin is attractive because of possible synergy. Penicillins should not be used, because *R. equi* rapidly develops resistance. Many weeks of antibiotic treatment, sometimes supplemented by surgical intervention, are often needed for infections caused by *R. equi*. Suppressive therapy with antibiotics should be continued indefinitely in patients with AIDS after initial treatment of infections caused by *R. equi*. Initial treatment of infections caused by other coryneform bacteria should be based on the identity of the organism and published data regarding antibiotic susceptibility. Therapy should be modified, when necessary, in light of the results of antibiotic susceptibility tests.

ANTHRAX

DEFINITION

Anthrax is an infection caused by *Bacillus anthracis* that occurs primarily in herbivores. Humans become infected when *B. anthracis* spores are introduced into the body by contact with infected animals or contaminated animal products, insect bites, ingestion, or inhalation. Aerosolized spores of *B. anthracis* have the potential for use in biological warfare or bioterrorism. Cutaneous anthrax is most common and is characterized by the development of a localized skin lesion with a central eschar surrounded by marked nonpitting edema. Inhalation anthrax (woolsorters' disease) typically involves hemorrhagic mediastinitis, rapidly progressive systemic infection, and a very high mortality rate. Gastrointestinal anthrax is rare and is associated with a high mortality rate.

ETIOLOGIC AGENT AND EPIDEMIOLOGY

B. anthracis is a large, aerobic, spore-forming, gram-positive rod that is encapsulated and nonmotile and grows in chains. Sporulation does not take place in living animals. The rectangular shape of the individual bacteria gives chains of *B. anthracis* a boxcar-like appearance. Virulent strains of *B. anthracis* are pathogenic for animals, including mice and guinea pigs. Spores of *B. anthracis* can survive for years in dry earth but are destroyed by boiling for 10 min, by treatment with oxidizing agents such as potassium permanganate or hydrogen peroxide, or by dilute formaldehyde. Most strains of *B. anthracis* are susceptible to penicillin.

Anthrax occurs worldwide and is most prevalent among domestic herbivores (including cattle, sheep, horses, and goats) and wild herbivores. Grazing animals become infected when they forage for food in areas contaminated with spores of *B. anthracis*. Anthrax in herbivores tends to be severe, with high mortality. Terminally ill animals with overwhelming bacteremic infections often bleed from the nose, mouth, and bowel, thereby contaminating soil or water with vegetative *B. anthracis* that can sporulate and persist in the environment. The carcasses of infected animals provide additional potential foci of contamination.

Humans are more resistant to anthrax than are herbivorous animals. The estimated number of human cases worldwide is 20,000 to 100,000 per year. Human cases are classified as agricultural or industrial. Agricultural cases result most often from contact with animals that have anthrax (e.g., during skinning, butchering, or dissecting), from bites of contaminated or infected flies, and (in rare instances) from consumption of contaminated meat. Industrial cases are associated with exposure to contaminated hides, goat hair, wool, or bones. Only three cases of cutaneous anthrax were reported to the [CDC](#) from 1984 through 1993, and gastrointestinal anthrax has never been documented in the United States. In an epidemic in the former Soviet Union at Sverdlovsk in 1979, cases were initially reported as cutaneous and gastrointestinal anthrax associated with contaminated meat; however, subsequent analysis of epidemiologic data and autopsy findings for most of the fatal cases established that the disease was inhalational anthrax associated with accidental airborne release of *B. anthracis* from a nearby military biological weapons facility. A massive outbreak in Zimbabwe between 1978 and the early 1980s involved more than 9700 cases of agricultural anthrax in humans. This outbreak occurred during wartime and was associated with disruption of the veterinary and medical infrastructure and cessation of veterinary anthrax vaccination programs.

PATHOGENESIS

B. anthracis can evade phagocytosis, invade the bloodstream, multiply rapidly to a high population density in vivo, and kill quickly. The poly-D-glutamic acid capsule of *B. anthracis* confers resistance to phagocytosis. Anthrax toxin consists of three different proteins called *protective antigen* (PA), *edema factor* (EF), and *lethal factor* (LF). The toxin was discovered in studies demonstrating that transfer of sterile blood from guinea pigs dying of anthrax to uninfected guinea pigs killed the recipients. PA binds to plasma membranes of target cells and is cleaved by a cellular protease into two fragments. The

larger fragment remains on the cell surface, displays a binding site for a domain that is present in both EF and LF, and serves as a specific receptor that mediates endocytic entry of EF or LF into the target cells. The catalytic activity of EF, a calmodulin-dependent adenylate cyclase, is expressed in the cytoplasm of human or animal cells that contain both calmodulin and ATP. The biologic effects of EF, which include formation of edema in anthrax lesions and inhibition of polymorphonuclear leukocyte functions, are mediated by the intracellular cyclic AMP that is produced by the enzymatic action of EF. In contrast, LF is a highly specific endopeptidase that cleaves several members of the MAP-kinase-kinase protein family and inactivates their functions in signal transduction pathways. Macrophages appear to be the principal targets of LF in animals, and intoxication of macrophages by LF is associated with production of reactive oxygen species, release of cytokines (including tumor necrosis factor α and interleukin 1 β), shock, and death.

Cutaneous anthrax is initiated when spores of *B. anthracis* are introduced into the skin through cuts or abrasions or by biting flies. The spores germinate within hours, and the vegetative cells multiply and produce anthrax toxin. The cutaneous anthrax lesion is characterized by necrosis, vascular congestion, hemorrhage, and gelatinous edema, but few leukocytes are present.

In inhalational anthrax, *B. anthracis* spores in airborne particles <5 μ m in diameter are deposited directly into the alveoli or alveolar ducts. The spores are phagocytized by alveolar macrophages, and some are carried to and germinate in mediastinal nodes. Hemorrhagic necrosis of the nodes, associated with hemorrhagic mediastinitis and overwhelming *B. anthracis* bacteremia, may develop rapidly. Secondary pneumonia sometimes occurs.

Gastrointestinal anthrax usually results from ingestion of inadequately cooked meat from animals with anthrax. Primary infection can be initiated in the intestine by organisms that survive passage through the stomach. An oropharyngeal form of the disease has also been described. Lesions in the throat or intestine are usually accompanied by hemorrhagic lymphadenitis.

B. anthracis bacteremia occurs in almost all cases of anthrax that progress to a fatal outcome.

CLINICAL MANIFESTATIONS

Approximately 95% of human cases of anthrax are the cutaneous form, and ~5% are the inhalational form. Gastrointestinal anthrax is very rare. Anthrax meningitis can occur as a complication of overwhelming *B. anthracis* bacteremia.

Cutaneous Anthrax The cutaneous lesion in anthrax is most often found on exposed areas of skin. A small red macule develops within days after inoculation of *B. anthracis* spores into skin. During the next week, the lesion typically progresses through papular and vesicular or pustular stages to the formation of an ulcer with a blackened necrotic eschar surrounded by a highly characteristic expanding zone of brawny edema. The early lesion may be pruritic, and the fully developed lesion is painless. Small satellite vesicles may surround the original lesion, and painful nonspecific regional lymphadenitis

is common. Most patients are afebrile, with mild or no constitutional symptoms; in severe cases, edema may be extensive and associated with shock. Spontaneous healing occurs in 80 to 90% of untreated cases, but edema may persist for weeks. In the 10 to 20% of untreated patients who have progressive infection, bacteremia develops and is often associated with high fever and rapid death. The differential diagnosis includes staphylococcal skin infections, tularemia, plague, and orf. Cutaneous anthrax should be considered when patients have painless ulcers associated with vesicles and edema and have had contact with animals or animal products.

Inhalational Anthrax The presenting symptoms of inhalational anthrax (woolsorters' disease) resemble those of severe viral respiratory diseases. Early diagnosis of inhalational anthrax that occurs naturally or as a consequence of biological warfare or bioterrorism is difficult. After 1 to 3 days, an acute phase supervenes, with increasing fever, dyspnea, stridor, hypoxia, and hypotension usually leading to death within 24 h. Occasionally, patients present with fulminant disease. A characteristic radiologic finding associated with hemorrhagic mediastinitis is symmetric mediastinal widening, which may provide an early clue to the diagnosis of inhalational anthrax.

Gastrointestinal Anthrax Symptoms of gastrointestinal anthrax are variable and include fever, nausea and vomiting, abdominal pain, bloody diarrhea, and sometimes rapidly developing ascites. Diarrhea is occasionally massive in volume. The major features of oropharyngeal anthrax are fever, sore throat, dysphagia, painful regional lymphadenopathy, and toxemia; respiratory distress may be evident. The primary lesion is most often located on the tonsils.

LABORATORY DIAGNOSIS

B. anthracis is present in large numbers in cutaneous lesions of anthrax and can be demonstrated by Gram's staining, direct fluorescent antibody staining, or culture unless the patient has been treated with antibiotics. A small proportion of patients with anthrax have bacteremia. Patients with anthrax meningitis have bloody spinal fluid containing large numbers of *B. anthracis* demonstrable by staining or culture. Patients with mild disease usually have normal leukocyte counts, but those with disseminated disease typically have polymorphonuclear leukocytosis. Tests for antibody to *B. anthracis* are useful in confirming the diagnosis of anthrax.

TREATMENT

Viable *B. anthracis* disappears from the lesions of cutaneous anthrax within 5 h of the initiation of treatment with parenteral penicillin G. The recommended regimen for adults is 2 million units of penicillin G at intervals of 6 h until edema subsides, with the subsequent administration of oral penicillin to complete a 7- to 10-day course. For penicillin-sensitive adults, treatment with ciprofloxacin, erythromycin, tetracycline, or chloramphenicol can be substituted. Antibiotics decrease local edema and systemic toxicity in cutaneous anthrax but do not prevent eschar formation. Cutaneous lesions should be cleaned and covered, and used dressings should be decontaminated. For inhalational or gastrointestinal anthrax, high-dose penicillin (8 to 12 million units per day in divided doses at intervals of 4 to 6 h) is recommended. A rational case can be made for passive immunization with anthrax antitoxin in addition to antibiotic therapy in

severely ill patients with anthrax, but no appropriate antitoxin is commercially available.

PREVENTION

Inhalational anthrax was essentially eliminated in England before 1940 through the development of methods to decontaminate wool and goat hair and the improvement of working conditions for handlers of animal products.

Nonliving vaccines consisting of alum-precipitated or aluminum hydroxide-adsorbed extracellular components of unencapsulated *B. anthracis* are used in the United States for military personnel, agricultural workers, veterinary personnel, and others at risk of exposure to anthrax. The major active component of these vaccines is protective antigen. Live attenuated vaccines containing spores of *B. anthracis* are used in both developed and developing countries to immunize domestic herbivores; these preparations are also used to immunize humans in Russia but not in the United States. The probable basis for attenuation of the original Pasteur spore vaccine is partial loss of a plasmid that encodes anthrax toxin. The basis for attenuation of the current Sterne spore vaccine is loss of a plasmid that encodes capsular polypeptide.

Improved anthrax vaccines for humans are needed because the current vaccines are impure and chemically complex, elicit only slow onset of protective immunity, provide incomplete protection, and cause significant adverse reactions. In addition to agricultural and industrial anthrax, the possible use of *B. anthracis* as an agent of biological warfare or bioterrorism is a stimulus for the development of an improved vaccine. Current strategies for vaccine development include purification of candidate protective antigens, expression of protective antigens in recombinant microbial vaccines, and construction of improved live attenuated strains of *B. anthracis*.

Carcasses of animals that succumb to anthrax should be buried intact or cremated. Necropsy or butchering of infected animals should be avoided because sporulation of *B. anthracis* occurs only in the presence of oxygen.

PROGNOSIS

The mortality rate is 10 to 20% for untreated cutaneous anthrax but is very low with appropriate antibiotic therapy. In contrast, the mortality rate for inhalational anthrax approaches 100%, and therapy is usually unsuccessful. The mortality rate in treated gastrointestinal anthrax is ~50%. Anthrax meningitis is usually fatal.

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142. INFECTIONS CAUSED BY *LISTERIA MONOCYTOGENES* - Anne Schuchat, Claire V. Broome

Listeria monocytogenes is a gram-positive rod that can be isolated from soil, vegetation, and many animal reservoirs. Human disease due to *L. monocytogenes* generally occurs in the setting of pregnancy or of immunosuppression caused by illness or medication. Increasing evidence suggests that a substantial portion of cases of human listeriosis are attributable to the food-borne transmission of *L. monocytogenes*. Unlike most food-borne pathogens, which cause primarily gastrointestinal illness, *L. monocytogenes* causes invasive syndromes, such as meningitis, sepsis, chorioamnionitis, and stillbirth.

ETIOLOGY

Listeriae are aerobic or facultatively anaerobic nonsporulating bacilli that grow at 1 to 45°C and typically have tumbling motility when cultured at 20 to 25°C. Characteristics that help distinguish *L. monocytogenes* from other *Listeria* spp. include the formation of a narrow zone of β hemolysis on sheep blood agar and the production of acid from glucose, maltose, L-rhamnose, and α -methyl-D-mannoside but not from D-xylose. Determination of the serotype of *L. monocytogenes* is based on somatic (O) and flagellar (H) antigens. Most cases of human disease are caused by serotypes 1/2a, 1/2b, and 4b. Molecular subtyping techniques have made it easier to discriminate among strains of *Listeria* and thus to link environmental or food isolates with clinical infections.

PATHOGENESIS

L. monocytogenes is an intracellular pathogen -- a characteristic consistent with its predilection for causing illness in persons with deficient cell-mediated immunity. The organism can be found as part of the gastrointestinal flora in healthy individuals. Lack of gastric acidity and abnormal gastrointestinal functioning may increase the risk of invasive disease following exposure to the organism in the gastrointestinal tract. The increased risk of *L. monocytogenes* infection in pregnant women may be due to both systemic and local immunologic changes associated with pregnancy. For example, local immunosuppression at the maternal-fetal interface of the placenta may facilitate intrauterine infection following transient maternal bacteremia.

The molecular pathogenesis of *L. monocytogenes* has recently been elucidated. The cell-surface protein internalin interacts with specific receptors to induce phagocytosis. Both listeriolysin O and phospholipases permit the organism to escape from the phagosome into the cytosol while avoiding intracellular killing. Through the surface protein Act A, *L. monocytogenes* uses actin-based motility to move to the cell membrane. Efficient cell-to-cell spread is accomplished by both actin filament formation and phospholipase production. Genetic determinants of these proteins have been characterized. Because the organism is adapted for both intracellular survival and direct cell-to-cell spread, it is not eliminated by antibodies.

EPIDEMIOLOGY

Long recognized as a veterinary pathogen, *L. monocytogenes* causes basilar meningitis

("circling disease") and stillbirth in sheep and cattle. The occurrence of listeriosis among humans has received increasing attention as the role of contaminated foods in the pathogenesis of epidemic listeriosis has been recognized and reports of disease associated with the expanding immunosuppressed population have accumulated.

Invasive listeriosis -- confirmed by culture of blood or cerebrospinal fluid (CSF) -- occurs in approximately 5 individuals per million population annually in the United States, for an estimated 1400 cases per year. Perinatal listeriosis complicates 9 births per 100,000. A 40% decline in incidence since the period from 1986 through 1990 may be attributable to aggressive food regulation and industrial clean-up efforts. Multistate surveillance for sporadic listeriosis suggests that 20% of infections are fatal or result in stillbirth, although higher case-fatality rates have been reported during listeriosis epidemics and were described in early series. Most cases of disease due to *L. monocytogenes* are sporadic; however, investigation of several outbreaks of listeriosis during the 1980s and 1990s demonstrated common-source food-borne transmission as a cause of human illness and showed that the incubation period for disease following consumption of contaminated food can be 2 to 6 weeks. The largest North American outbreak, which took place in Los Angeles in 1985, involved more than 100 cases and 48 deaths or stillbirths. A nationwide outbreak in France in 1992 involved 279 cases and 63 deaths. Foods implicated in outbreaks of listeriosis include contaminated coleslaw, pasteurized milk, soft cheeses, pate, ready-to-eat pork products, and hot dogs, while epidemiologic studies have implicated undercooked chicken, uncooked hot dogs, soft cheeses, and food from store delicatessen counters in sporadic disease. Listerial contamination of foods is relatively common. Among foods contaminated with the organism, those that are purchased ready to eat, are contaminated with serotype 4b, and are contaminated at a relatively high level may be the most likely to cause illness. The long incubation period associated with listeriosis contributes to the difficulty of implicating specific foods as the cause of either common-source outbreaks or sporadic cases.

Although food-borne transmission appears to be the foremost cause of epidemic and sporadic disease, several clusters of late-onset neonatal infection suggest nosocomial transmission of *L. monocytogenes*. Contaminated multiuse materials and equipment have been suggested as causes of some nosocomial clusters. Listeriosis has been reported in veterinarians and other persons in close contact with infected animals.

CLINICAL PRESENTATION

Pregnancy-associated listeriosis may occur during any stage of pregnancy, although most infections are detected during the third trimester, possibly because of failure to obtain specimens for bacterial culture earlier during gestation in instances of abortion and stillbirth. One-half to two-thirds of pregnant women with perinatal listeriosis experience a mild illness characterized by fever, myalgias, malaise, and backache, which sometimes are accompanied by diarrhea, abdominal pain, nausea, and/or vomiting during the bacteremic phase. Blood cultures should be used for diagnosis. Transplacental spread of the organism results in intrauterine infection, which can lead to chorioamnionitis, premature labor, intrauterine fetal demise, or early-onset disease of the newborn. Women with listeriosis diagnosed during pregnancy have a favorable clinical outcome after antibiotic therapy or delivery. Although often included in the differential diagnosis of recurrent spontaneous abortion, infection with *L.*

monocytogenes appears to cause fewer than 2% of stillbirths.

Neonatal listeriosis can be classified under the same categories used for group B streptococcal infection ([Chap. 140](#)), with early-onset disease evident during the first week of life and late-onset disease developing thereafter. Infants may be symptomatic at birth; most infants with early-onset disease are symptomatic by the second day of life. Aspiration of infected amniotic fluid contributes to pathogenesis. Early-onset disease may include sepsis, respiratory distress, skin lesions, and the syndrome called *granulomatosis infantisepticum*, which is characterized by disseminated abscesses involving the liver, spleen, adrenal glands, lungs, and other sites. Infants with late-onset neonatal disease are more likely than those with early-onset disease to develop meningitis. While early-onset disease is often associated with obstetric complications such as premature delivery and chorioamnionitis, late-onset disease typically affects infants born at term by uncomplicated deliveries. Infants may acquire *L. monocytogenes* during passage through the birth canal; except in several clusters of late-onset neonatal infections linked to nosocomial transmission, the pathogenesis of late-onset disease is not well understood.

Listeriosis not associated with pregnancy usually affects persons with immunosuppressive conditions, although invasive disease can also affect immunocompetent adults, particularly elderly persons. The most common underlying conditions in nonpregnant adults with listeriosis are chronic glucocorticoid therapy, solid or hematologic malignancies, diabetes mellitus, renal disease, liver disease, and AIDS. Although the prevalence of listeriosis among persons infected with HIV is much higher than that in the general population, listeriosis is a relatively uncommon opportunistic infection in AIDS.

Sepsis Clinical studies have shown that bacteremic infection without an evident focus is the most common clinical manifestation of listeriosis among immunocompromised hosts, while infection of the central nervous system (CNS) ranks second in frequency. Listerial sepsis cannot be distinguished clinically from bacteremia involving other organisms. Patients are usually febrile, often appear extremely ill, and may have prodromal symptoms including myalgia, nausea, vomiting, and diarrhea. Immunocompromised patients with listeriosis are less likely than other adults to present with CNS infection, possibly because they are more likely to have blood cultured during febrile episodes and thus to have transient listerial bacteremia recognized.

CNS Infection The most common presentation of [CNS](#) infection due to *L. monocytogenes* is meningitis, which can present as either an acute or (less often) a subacute illness. Presenting symptoms include fever, headache, and an altered level of consciousness. Examination of [CSF](#) usually reveals pleocytosis, increased protein concentrations, and normal glucose levels, although other patterns are sometimes found. Gram's stain is often unrevealing. The diagnosis is made when *L. monocytogenes* is identified on culture. Despite its name, *L. monocytogenes* is rarely associated with monocytosis of either CSF or blood. Other syndromes seen in CNS infection include meningoencephalitis; cerebritis; and brainstem, spinal cord, or intracranial abscesses. The unusual syndrome of rhombencephalitis includes asymmetric cranial-nerve palsies, altered consciousness, cerebellar signs, and motor or sensory loss. Symptoms of other nonmeningitic CNS infections include fever, ataxia,

seizures, personality changes, and coma. Nuchal rigidity is rare in nonmeningitic infections. CSF cultures may be sterile; blood cultures are usually diagnostic.

Endocarditis Like most forms of bacterial endocarditis, listerial endocarditis typically occurs in patients with prosthetic or previously damaged valves. The organism has a predilection for the left side of the heart. Endocarditis due to *L. monocytogenes* is often associated with systemic embolization.

Focal Infections Other focal infections that can follow unrecognized bacteremia include endophthalmitis, peritonitis, osteomyelitis, visceral abscess, pleuropulmonary infection, and cholecystitis. Cutaneous lesions may develop without systemic involvement and have been reported in veterinarians and poultry workers.

Recurrences Recurrent infection with *L. monocytogenes* has been reported but is rare. Many recurrences are due to the subtype responsible for the initial infection. The implication is that such recurrences result either from insufficient treatment of a focus of primary infection or from repeated exposure to a persistently contaminated source.

Gastrointestinal Illness Several common-source outbreaks of acute gastroenteritis suggest that *L. monocytogenes* can cause an acute diarrheal syndrome in persons without immunocompromising conditions. The importance of *L. monocytogenes* in sporadic diarrheal illness is unclear. Although the organism is not identified by the culture methods routinely used for stool specimens, studies using selective enrichment media for evaluation of consecutive specimens from patients hospitalized with acute diarrhea have suggested that *L. monocytogenes* is not a major cause of sporadic diarrhea.

DIAGNOSIS

Invasive listeriosis is diagnosed when the organism is cultured from a site that is usually sterile, such as blood, [CSF](#), or amniotic fluid. The organism grows readily within 36 h on routine culture media, but morphologic similarities between *Listeria* and both diphtheroids and streptococci make it necessary to use biochemical tests to identify the species. Serologic assays with whole-cell antigens have not been useful for the diagnosis of listeriosis, both because exposure to the organism (and thus the presence of antibody) may be common and because infected individuals may not produce antibody. Assays for antibody to listeriolysin O have been applied in epidemiologic investigations and, retrospectively, in the diagnosis of culture-negative [CNS](#) infection. Culture of the organism from nonsterile sites such as the vagina and rectum is not useful for clinical diagnosis, as the organism may be carried at these sites by approximately 5% of healthy individuals.

Differential diagnosis of prematurity, spontaneous abortion, or stillbirth includes infectious diseases such as group B streptococcal infection, congenital syphilis, and toxoplasmosis; pathogens such as group B streptococci and *Escherichia coli* are more common than *L. monocytogenes* as causes of meningitis and sepsis in the newborn period. Listerial infection should always be considered in the differential diagnosis of meningitis in immunosuppressed persons, particularly transplant recipients and others undergoing glucocorticoid treatment, patients with hematologic malignancy, and

HIV-infected patients. Among healthy adults, meningitis is much more likely to be caused by *Neisseria meningitidis*, *Streptococcus pneumoniae*, or viral pathogens than by *L. monocytogenes*.

TREATMENT

The treatment of choice for listeriosis is intravenous administration of either ampicillin or penicillin, often in combination with an aminoglycoside for synergy. Trimethoprim-sulfamethoxazole is bactericidal against *L. monocytogenes* and has been used successfully in the treatment of patients with penicillin allergy. *L. monocytogenes* is susceptible in vitro to penicillin G, ampicillin, erythromycin, trimethoprim-sulfamethoxazole, chloramphenicol, rifampin, tetracyclines, aminoglycosides, and imipenem. However, chloramphenicol and rifampin may antagonize the bactericidal effect of penicillins. Because *L. monocytogenes* is not sensitive to cephalosporins, these agents should not be used for single-agent empirical treatment of neonatal sepsis or of meningitis in newborns or immunocompromised hosts.

Dosages and durations of therapy have not been subjected to controlled trials. For nonpregnant adults with listeriosis, the regimen of choice is either ampicillin (12 g intravenously per day in six divided doses) or penicillin G (15 to 20 million units intravenously per day in six divided doses); for immunosuppressed patients with meningitis, some experts add gentamicin (1.3 mg/kg intravenously every 8 h) for synergy. Penicillin-allergic patients may be treated with trimethoprim-sulfamethoxazole (15/75 mg/kg intravenously per day in three equal portions every 8 h). Meningitis in an immunocompetent patient may require 2 to 3 weeks of antibiotic therapy after defervescence. Meningitis, bacteremia, endocarditis, and nonmeningitic listeriosis in immunosuppressed patients should be treated longer, probably for 4 to 6 weeks. Neonatal listeriosis can be treated with a 2-week course of ampicillin. Infants weighing <2000 g should receive 100 mg/kg per day in two equal doses during the first week of life and 150 mg/kg per day during the second week. Infants weighing ≥2000 g should receive 150 mg/kg per day in three equal doses during the first week of life and 200 mg/kg per day during the second week. The addition of an aminoglycoside should be considered for neonatal infection (gentamicin, 5 mg/kg per day in two divided doses during the first week of life; 7.5 mg/kg per day in three equal doses during the second week). For listeriosis in pregnant women, a 2-week course of ampicillin (4 to 6 g per day in four equal doses) is recommended. During the last month of pregnancy, infected women with serious penicillin allergies may be treated with erythromycin.

PROGNOSIS

Treatment of maternal bacteremia during pregnancy can prevent neonatal infection. Antibiotic therapy for the newborn can limit sequelae, although the widely disseminated disease characteristic of granulomatosis infantisepticum is frequently fatal regardless of treatment. Early-onset disease carries a higher mortality risk than late-onset infection, and immunocompromised hosts have a worse prognosis than do otherwise healthy adults with listeriosis.

PREVENTION

L. monocytogenes is frequently isolated from food; the Food and Drug Administration, the U.S. Department of Agriculture, and manufacturers are pursuing further measures to reduce *L. monocytogenes* contamination of foods that have been subjected to listericidal processing. Prevention of listeriosis requires dietary counseling of persons at increased risk of disease ([Table 142-1](#)). There is no role for the administration of prophylaxis to contacts of patients with listeriosis. Clinicians are encouraged to report cases of listeriosis to local or state health departments. Case reporting and subtyping of clinical isolates can facilitate early recognition of outbreaks and prevention of subsequent cases.

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143. TETANUS - Elias Abrutyn

DEFINITION

Tetanus is a neurologic disorder, characterized by increased muscle tone and spasms, that is caused by tetanospasmin, a powerful protein toxin elaborated by *Clostridium tetani*. Tetanus occurs in several clinical forms, including generalized, neonatal, and localized disease.

ETIOLOGIC AGENT

C. tetani is an anaerobic, motile gram-positive rod that forms an oval, colorless, terminal spore and thus assumes a shape resembling a tennis racket or drumstick. The organism is found worldwide in soil, in the inanimate environment, in animal feces, and occasionally in human feces. Spores may survive for years in some environments and are resistant to various disinfectants and to boiling for 20 min. Vegetative cells, however, are easily inactivated and are susceptible to several antibiotics (metronidazole, penicillin, and others).

Tetanospasmin is formed in vegetative cells under plasmid control. It is a single-polypeptide chain. With autolysis, the single-chain toxin is released and cleaved to form a heterodimer consisting of a heavy chain (100 kDa), which mediates binding to nerve-cell receptors and entry into these cells, and a light chain (50 kDa), which acts to block neurotransmitter release. The amino acid structures of the two most powerful toxins known, botulinum toxin and tetanus toxin, are partially homologous.

EPIDEMIOLOGY

Tetanus occurs sporadically and almost always affects nonimmunized persons, partially immunized persons, or fully immunized individuals who fail to maintain adequate immunity with booster doses of vaccine. Although tetanus is entirely preventable by immunization, the burden of disease is large worldwide. The disease is common in areas where soil is cultivated, in rural areas, in warm climates, during summer months, and among males. In countries without a comprehensive immunization program, tetanus occurs predominantly in neonates and other young children; an estimated 490,000 neonates died of tetanus worldwide in 1994 -- a reduction from 550,000 in 1993. In the United States and other nations with successful immunization programs, neonatal tetanus is rare and the disease affects other age groups and groups inadequately covered by immunization (such as nonwhites). The risk for the development of tetanus and for the most severe illness is highest among the elderly. Only 27% of persons aged 70 or older -- as opposed to 88% of 6- to 11-year-olds -- have protective antibody levels. During the years 1995 through 1997, a total of 124 cases were reported to the Centers for Disease Control and Prevention; both the overall incidence (0.15 cases per 100,000 population) and the annual average (41 cases) were the lowest ever reported in the United States. The actual burden of illness, however, was greater, because reporting is incomplete. Although the elderly customarily account for the highest proportion of cases, in 1995 through 1997 individuals ³60 years of age accounted for only 35% of cases, whereas those 20 to 59 years of age accounted for 60% and those under 20 for 5% (with one case of neonatal tetanus). The change is attributed to a decrease in incidence

in both the ≥ 60 and the <20 age groups, along with an increase among persons 20 to 59 years of age, particularly injection drug users.

In the United States, most cases of tetanus follow an acute injury, such as a puncture wound, laceration, or abrasion. Tetanus is acquired indoors or during farming, gardening, and other outdoor activities. The injury may be major but often is trivial, so that medical attention is not sought; in some instances no injury can be identified. The disease may complicate chronic conditions such as skin ulcers, abscesses, and gangrene. Tetanus is also associated with burns, frostbite, middle-ear infection, surgery, abortion, childbirth, and drug abuse, notably "skin popping." In some patients no portal of entry for the organism can be identified.

PATHOGENESIS

Contamination of wounds with spores of *C. tetani* is probably frequent. Germination and toxin production, however, take place only in wounds with low oxidation-reduction potential, such as those with devitalized tissue, foreign bodies, or active infection. *C. tetani* does not itself evoke inflammation, and the portal of entry retains a benign appearance unless infection with other organisms is present.

Toxin released in the wound binds to peripheral motor neuron terminals, enters the axon, and is transported to the nerve-cell body in the brainstem and spinal cord by retrograde intraneuronal transport. The toxin then migrates across the synapse to presynaptic terminals, where it blocks release of the inhibitory neurotransmitters glycine and γ -aminobutyric acid (GABA). The blocking of neurotransmitter release by tetanospasmin, a zinc metalloprotease, involves the cleavage of protein(s) critical to proper function of the synaptic vesicle release apparatus. With diminished inhibition, the resting firing rate of the motor neuron increases, producing rigidity. With lessened activity of reflexes that limit polysynaptic spread of impulses (a glycinergic activity), agonists and antagonists may be recruited rather than inhibited, with the consequent production of spasms. Loss of inhibition may also affect preganglionic sympathetic neurons in the lateral gray matter of the spinal cord and produce sympathetic hyperactivity and high circulating catecholamine levels. Tetanospasmin, like botulinum toxin, may block neurotransmitter release at the neuromuscular junction and produce weakness or paralysis; recovery requires sprouting of new nerve terminals.

In local tetanus, only the nerves supplying the affected muscles are involved. Generalized tetanus occurs when toxin released in the wound enters the lymphatics and bloodstream and is spread widely to distant nerve terminals; the blood-brain barrier blocks direct entry into the central nervous system. If it is assumed that intraneuronal transport times are equal for all nerves, short nerves are affected before long nerves: this fact explains the sequential involvement of nerves of the head, trunk, and extremities in generalized tetanus.

CLINICAL MANIFESTATIONS

Generalized tetanus, the most common form of the disease, is characterized by increased muscle tone and generalized spasms. The median time of onset after injury is 7 days; 15% of cases occur within 3 days and 10% after 14 days.

Typically, the patient first notices increased tone in the masseter muscles (trismus, or lockjaw). Dysphagia or stiffness or pain in the neck, shoulder, and back muscles appears concurrently or soon thereafter. The subsequent involvement of other muscles produces a rigid abdomen and stiff proximal limb muscles; the hands and feet are relatively spared. Sustained contraction of the facial muscles results in a grimace or sneer (risus sardonicus), and contraction of the back muscles produces an arched back (opisthotonos). Some patients develop paroxysmal, violent, painful, generalized muscle spasms that may cause cyanosis and threaten ventilation. These spasms occur repetitively and may be spontaneous or provoked by even the slightest stimulation. A constant threat during generalized spasms is reduced ventilation or apnea or laryngospasm. The severity of illness may be mild (muscle rigidity and few or no spasms), moderate (trismus, dysphagia, rigidity, and spasms), or severe (frequent explosive paroxysms). The patient may be febrile, although many have no fever; mentation is unimpaired. Deep tendon reflexes may be increased. Dysphagia or ileus may preclude oral feeding.

Autonomic dysfunction commonly complicates severe cases and is characterized by labile or sustained hypertension, tachycardia, dysrhythmia, hyperpyrexia, profuse sweating, peripheral vasoconstriction, and increased plasma and urinary catecholamine levels. Periods of bradycardia and hypotension may also be documented. Sudden cardiac arrest sometimes occurs, but its basis is unknown. Other complications include aspiration pneumonia, fractures, muscle rupture, deep vein thrombophlebitis, pulmonary emboli, decubitus ulcer, and rhabdomyolysis.

Neonatal tetanus usually occurs as the generalized form and is usually fatal if left untreated. It develops in children born to inadequately immunized mothers, frequently after unsterile treatment of the umbilical cord stump. Its onset generally comes during the first 2 weeks of life. Poor feeding, rigidity, and spasms are typical features of neonatal tetanus.

Local tetanus is an uncommon form in which manifestations are restricted to muscles near the wound. The prognosis is excellent.

Cephalic tetanus, a rare form of local tetanus, follows head injury or ear infection. Trismus and dysfunction of one or more cranial nerves, often the seventh nerve, are found. The incubation period is a few days and the mortality is high.

DIAGNOSIS

The diagnosis of tetanus is based entirely on clinical findings. Tetanus is unlikely if a reliable history indicates the completion of a primary vaccination series and the receipt of appropriate booster doses. Wounds should be cultured in suspected cases. However, *C. tetani* can be isolated from wounds of patients without tetanus and frequently cannot be recovered from wounds of those with tetanus. The leukocyte count may be elevated. Cerebrospinal fluid examination yields normal results. Electromyograms may show continuous discharge of motor units and shortening or absence of the silent interval normally seen after an action potential. Nonspecific changes may be evident on the electrocardiogram. Muscle enzyme levels may be raised. Serum antitoxin levels of ≥ 0.01

U/mL are considered protective and make tetanus unlikely, although cases developing despite protective antitoxin levels have been reported.

The differential diagnosis includes local conditions also producing trismus, such as alveolar abscess, strychnine poisoning, dystonic drug reactions (e.g., to phenothiazines and metoclopramide), and hypocalcemic tetany. Other conditions sometimes confused with tetanus include meningitis/encephalitis, rabies, and an acute intraabdominal process (because of the rigid abdomen). Markedly increased tone in central muscles (face, neck, chest, back, and abdomen) with superimposed generalized spasms and relative sparing of the hands and feet strongly suggests tetanus.

TREATMENT

General Measures The goals of therapy are to eliminate the source of toxin, neutralize unbound toxin, and prevent muscle spasms, monitoring the patient's condition and providing support -- especially respiratory support -- until recovery. Patients should be admitted to a quiet room in an intensive care unit, where observation and cardiopulmonary monitoring can be maintained continuously but stimulation can be minimized. Protection of the airway is vital. Wounds should be explored, carefully cleansed, and thoroughly debrided.

Antibiotic Therapy Although of unproven value, antibiotic therapy is administered to eradicate vegetative cells -- the source of toxin. The use of penicillin (10 to 12 million units intravenously, given daily for 10 days) has been recommended, but metronidazole (500 mg every 6 h or 1 g every 12 h) is preferred by some experts on the basis of this drug's excellent antimicrobial activity, a survival rate higher than that obtained with penicillin in one nonrandomized trial, and the absence of the [GABA](#) antagonistic activity seen with penicillin. Clindamycin and erythromycin are also alternatives for the treatment of penicillin-allergic patients. Additional specific antimicrobial therapy should be given for active infection with other organisms.

Antitoxin Given to neutralize circulating toxin and unbound toxin in the wound, antitoxin effectively lowers mortality; toxin already bound to neural tissue is unaffected. Human tetanus immune globulin (TIG) is the preparation of choice and should be given promptly. The dose is 3000 to 6000 units intramuscularly, usually in divided doses because the volume is large. The optimal dose is not known, however, and results from one study indicated that a 500-unit dose was as effective as higher doses. Pooled intravenous immunoglobulin may be an alternative to TIG, but the specific antitoxin concentration in this formulation is not standardized. It may be best to administer antitoxin before manipulating the wound; the value of injecting a dose proximal to the wound or infiltrating the wound is unclear. Additional doses are unnecessary because the half-life of antitoxin is long. Antibody does not penetrate the blood-brain barrier. Intrathecal administration should be considered experimental. Equine tetanus antitoxin (TAT) is not available in the United States but is used elsewhere. It is cheaper than human antitoxin, but its half-life is shorter and its administration commonly elicits hypersensitivity and serum sickness.

Control of Muscle Spasms Many agents, alone and in combination, have been used to treat the muscle spasms of tetanus, which are painful and can threaten ventilation by

causing laryngospasm or sustained contraction of ventilatory muscles. The ideal therapeutic regimen would abolish spasmodic activity without causing oversedation and hypoventilation. Diazepam, a benzodiazepine and [GABA](#) agonist, is in wide use. The dose is titrated, and large doses (≥250 mg/d) may be required. Lorazepam, with a longer duration of action, and midazolam, with a short half-life, are other options. Barbiturates and chlorpromazine are considered second-line agents. Therapeutic paralysis with a nondepolarizing neuromuscular blocking agent and mechanical ventilation may be required for the treatment of spasms unresponsive to medication or spasms that threaten ventilation. However, prolonged paralysis after the discontinuation of therapy with such agents has been described, and both the need for continued paralysis and the occurrence of complications should be assessed daily. Alternative agents include propofol, which is expensive, and dantrolene and baclofen, which are being investigated in the hope of shortening the period of therapeutic paralysis.

Respiratory Care Intubation or tracheostomy, with or without mechanical ventilation, may be required for hypoventilation due to oversedation or laryngospasm or for the avoidance of aspiration by patients with trismus, disordered swallowing, or dysphagia. The need for these procedures should be anticipated, and they should be undertaken electively and early.

Autonomic Dysfunction The optimal therapy for sympathetic overactivity has not been defined. Agents that have been considered include labetalol (ana- and b-adrenergic blocking agent that is recommended by some experts but that reportedly has caused sudden death), esmolol administered by continuous infusion (a beta blocker whose short half-life may be advantageous in the event of severe hypertension from unopposed α-adrenergic activity), clonidine (a central-acting antiadrenergic drug), and morphine sulfate. Parenteral magnesium sulfate and continuous spinal or epidural anesthesia have been used but may be more difficult to administer and monitor. The relative efficacy of these modalities has yet to be determined. Hypotension or bradycardia may require volume expansion, use of vasopressors or chronotropic agents, or pacemaker insertion.

Vaccine Patients recovering from tetanus should be actively immunized (see below) because immunity is not induced by the small amount of toxin that produces disease.

Additional Measures Additional therapeutic measures include hydration to control insensible and other fluid losses, which may be significant; the meeting of the patient's increased nutritional requirements by enteral or parenteral means; physiotherapy to prevent contractures; and administration of heparin or another anticoagulant to prevent pulmonary emboli. Bowel, bladder, and renal function must be monitored. Gastrointestinal bleeding and decubitus ulcers must be prevented, and intercurrent infection should be treated.

PREVENTION

Active Immunization All partially immunized and unimmunized adults should receive vaccine, as should those recovering from tetanus. The primary series for adults consists of three doses: the first and second doses are given 4 to 8 weeks apart, and the third dose is given 6 to 12 months after the second. A booster dose is required every 10

years and may be given at mid-decade ages -- 35, 45, and so on. Combined tetanus and diphtheria toxoid (Td) adsorbed (for adult use), rather than single-antigen tetanus toxoid, is preferred for persons >7 years of age.

Wound Management Proper wound management requires consideration of the need for (1) passive immunization with [TIG](#) and (2) active immunization with vaccine, preferably Td in persons over age 7. For clean minor wounds, Td is administered to persons who (1) have unknown tetanus immunization histories; (2) have received fewer than three doses of adsorbed tetanus toxoid; (3) have received three or more doses of adsorbed vaccine, with the last dose given >10 years previously; and (4) have received three doses of *fluid* (nonadsorbed) vaccine. The recommendations for contaminated or severe wounds are identical, except that vaccine should be given to those who have received three or more doses of adsorbed tetanus toxoid if >5 years have elapsed since the last dose. Passive immunization with TIG is not recommended for clean minor wounds but is given for all other wounds if the patient's vaccination history indicates unknown or partial immunization. The dose of TIG for passive immunization of persons with wounds of average severity is 250 units intramuscularly, which produces a protective antibody level in the serum for at least 4 to 6 weeks; the appropriate dose of [TAT](#) is 3000 to 6000 units. Vaccine and tetanus antitoxin should be administered at separate sites in separate syringes.

Neonatal Tetanus Measures aimed at preventing neonatal tetanus include maternal vaccination, even during pregnancy; efforts to increase the proportion of births that take place in the hospital; and the provision of training for nonmedical birth attendants.

PROGNOSIS

The application of methods to monitor and support oxygenation has markedly improved the prognosis in tetanus; mortality rates as low as 10% have been reported from units accustomed to handling such cases. In the United States during the period 1995 through 1997, the case-fatality rate was 11%; 11 deaths from tetanus were reported in 1990, 11 in 1991, and 9 in 1992. The outcome is poor in neonates and the elderly and in patients with a short incubation period, a short interval from the onset of symptoms to admission, or a short period from onset of symptoms to the first spasm (period of onset). Outcome is also related to the extent of prior vaccination.

The course of tetanus extends over 4 to 6 weeks, and patients may require ventilatory support for 3 weeks during this period. Increased tone and minor spasms can last for months, but recovery is usually complete.

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144. BOTULISM - Elias Abrutyn

DEFINITION

Botulism is a paralytic disease that begins with cranial nerve involvement and progresses caudally to involve the extremities. It is caused by potent protein neurotoxins elaborated by *Clostridium botulinum*. The toxins' high potency has led to consideration of their use in bioterrorism or biological warfare. Cases may be classified as (1) *food-borne botulism*, from ingestion of preformed toxin in food contaminated with *C. botulinum*; (2) *wound botulism*, from toxin produced in wounds contaminated with the organism; (3) *infant botulism*, from ingestion of spores and production of toxin in the intestine of infants; or (4) *adult infectious botulism*, a group that includes some cases in older children and adults in which disease is produced by a mechanism similar to that described for infant botulism.

ETIOLOGIC AGENT

C. botulinum, a species encompassing a heterogeneous group of anaerobic gram-positive organisms that form subterminal spores, is found in soil and marine environments throughout the world and elaborates the most potent bacterial toxin known. Organisms of types A through G have been distinguished by the antigenic specificities of their toxins; a classification system based on physiologic characteristics has also been described. Rare strains of other clostridial species -- *C. butyricum* and *C. baratii* -- have also been found to produce toxin. *C. botulinum* strains with proteolytic activity can digest food and produce a spoiled appearance; nonproteolytic types leave the appearance of food unchanged.

Of the eight distinct toxin types described (A, B, C₁, C₂, D, E, F, and G), all except C₂ are neurotoxins; C₂ is a cytotoxin of unknown clinical significance. Botulinum neurotoxin, whether ingested or produced in the intestine or a wound, enters the vascular system and is transported to peripheral cholinergic nerve terminals, including neuromuscular junctions, postganglionic parasympathetic nerve endings, and peripheral ganglia. The central nervous system is not involved. Active neurotoxin (150 kDa) is composed of a heavy chain (a 100-kDa fragment responsible for neurospecific binding and translocation into the nerve cell) and a light chain (a 50-kDa fragment responsible for intracellular catalytic activity). The steps involved in neurotoxin activity include (1) specific binding to presynaptic nerve cells at the myoneural junction, (2) internalization of the toxin inside the nerve cell in endocytic vesicles, (3) translocation of the toxin into the cytosol, and (4) proteolysis by toxin (a zinc endopeptidase) of components of the neuroexocytosis apparatus curtailing release of the neurotransmitter acetylcholine. Cure follows sprouting of new nerve terminals.

Toxin is heat-labile, but spores are highly heat-resistant; both can be inactivated under appropriate conditions (see "Prevention," below). In the gastrointestinal tract, toxin is complexed with nontoxin proteins and resists degradation.

Toxin types A, B, E, and (in rare instances) F cause human disease; type G (now called *C. argentinense*) has been associated with sudden death, but not with neuroparalytic illness, in a few patients in Switzerland; and types C and D cause animal disease.

EPIDEMIOLOGY

Human botulism occurs worldwide. In the United States, the geographic distribution of cases by toxin type parallels the distribution of organism types found in the environment. Type A predominates west of the Rocky Mountains; type B is generally distributed but is more common in the East; and type E is found in the Pacific Northwest, Alaska, and the Great Lakes area. In the United States, food-borne botulism has been associated primarily with home-canned food, particularly vegetables, fruit, and condiments, and less commonly with meat and fish. Type E outbreaks are frequently associated with fish products. Commercial products occasionally cause outbreaks, but some of these outbreaks have resulted from improper handling after purchase. Outbreaks in restaurants, schools, and private homes have been traced to uncommon sources (commercial potpies, beef stew, turkey loaf, sauteed onions, baked potatoes, and chopped garlic in oil). Food-borne botulism can occur when (1) a food to be preserved is contaminated with spores, (2) preservation does not inactivate the spores but kills other putrefactive bacteria that might inhibit the growth of *C. botulinum* and provides anaerobic conditions at a pH and temperature that allow germination and toxin production, and (3) food is not heated to a temperature that destroys toxin before being eaten.

CLINICAL MANIFESTATIONS

Food-Borne Botulism Following ingestion of food containing toxin, illness varies from a mild condition for which no medical advice is sought to very severe disease that can result in death within 24 h. The incubation period is usually 18 to 36 h but, depending on toxin dose, can extend from a few hours to several days. Symmetric descending paralysis is characteristic and can lead to respiratory failure and death. Cranial nerve involvement, which almost always marks the onset of symptoms, usually produces diplopia, dysarthria, and/or dysphagia. Weakness progresses, often rapidly, from the head to involve the neck, arms, thorax, and legs; the weakness is occasionally asymmetric. Nausea, vomiting, and abdominal pain may precede or follow the onset of paralysis. Dizziness, blurred vision, dry mouth, and very dry, occasionally sore throat are common. Patients are generally alert and oriented, but they may be drowsy, agitated, and anxious. Typically, they have no fever. Ptosis is frequent; the pupillary reflexes may be depressed, and fixed or dilated pupils are noted in half of patients. The gag reflex may be suppressed, and deep tendon reflexes may be normal or decreased. Paralytic ileus, severe constipation, and urinary retention are common.

Wound Botulism When wounds are contaminated with *C. botulinum* spores, the spores may germinate into vegetative organisms that produce toxin. This rare condition resembles food-borne illness except that the incubation period is longer, averaging about 10 days, and gastrointestinal symptoms are lacking. Wound botulism has been documented after traumatic injury involving contamination with soil; in injection drug users, for whom black-tar heroin use has been identified as a risk factor; and after cesarean delivery. The illness has occurred even after antibiotics have been given to prevent wound infection. When present, fever is probably attributable to concurrent infection with other bacteria. The wound may appear benign.

Infant Botulism In infant botulism, the most common form of the disease, toxin is produced in and absorbed from the intestine after the germination of ingested spores. The severity ranges from mild illness with failure to thrive to fulminant severe paralysis with respiratory failure and may be one cause of sudden infant death. The identification of contaminated honey as one source of spores has led to the recommendation that honey not be fed to children <12 months of age. Most cases cannot be attributed to a particular food source. The factors permitting intestinal colonization with *C. botulinum* are not fully defined, but cases usually involve infants <6 months of age; susceptibility may decrease as the normal intestinal flora develops.

Adult Infectious Botulism Rarely, botulism in adults is produced by a mechanism similar to that operative in infant botulism: intestinal colonization and toxin production. The patient may have a history of gastrointestinal disease, surgery, or recent antibiotic therapy. Toxin and organisms may be identified in the stool.

DIAGNOSIS

A diagnosis of botulism must be considered in afebrile, mentally intact patients who have symmetric descending paralysis without sensory findings. The diagnosis must be suspected on clinical grounds in the context of an appropriate history. Conditions often confused with botulism include myasthenia gravis, which may be ruled out by electromyography and antibody studies, and Guillain-Barre syndrome, which is characterized by ascending paralysis, sensory abnormalities, and elevation of the protein concentration in cerebrospinal fluid. The Fisher variant of Guillain-Barre -- a descending paralysis -- can indeed be difficult to differentiate from botulism. Other conditions that may resemble botulism include Lambert-Eaton syndrome, poliomyelitis, tick paralysis, diphtheria, and intoxications from mushrooms, medications, or chemicals. Hypermagnesemia should be considered.

The demonstration of toxin in serum by bioassay in mice is definitive, but this test may be negative, particularly in wound and infant botulism. It is performed only by specific laboratories, which can be identified through regional public health authorities. Other assays are being developed and remain experimental. The demonstration of the organism or its toxin in vomitus, gastric fluid, or stool is strongly suggestive of the diagnosis, because intestinal carriage is rare. Isolation of the organism from food without toxin is insufficient grounds for the diagnosis. Wound cultures yielding the organism are suggestive of botulism. The edrophonium chloride (Tensilon) test for myasthenia gravis may be falsely positive in botulism but is usually less dramatically positive than in the former condition. Nerve conduction velocity is normal, but compound muscle action potentials on routine nerve stimulation studies are decreased with a supramaximal stimulus, and facilitation is evident after repetitive stimulation at high frequency. Single-fiber electromyography may be helpful. The white blood cell count and erythrocyte sedimentation rate are normal.

TREATMENT

Patients should be hospitalized and monitored closely, both clinically and by spirometry, pulse oximetry, and measurement of arterial blood gases for incipient respiratory failure. Intubation and mechanical ventilation should be strongly considered when the vital

capacity is <30% of predicted, especially when paralysis is progressing rapidly and hypoxemia with absolute or relative hypercarbia is documented ([Chap. 266](#)). Serial measurements of the maximal static inspiratory pressure may be useful in predicting respiratory failure.

In food-borne illness, trivalent (types A, B, and E) equine antitoxin should be administered as soon as possible after specimens are obtained for laboratory analysis. The initiation of treatment should not await laboratory confirmation, which may take days. After testing for hypersensitivity to horse serum, a vial of antitoxin is given; repeated doses are not considered necessary. Anaphylaxis and serum sickness are risks inherent in use of the equine product, and desensitization of allergic patients may be required. If there is no ileus, cathartics and enemas may be given to purge the gut of toxin; emetics or gastric lavage can also be used if the time since ingestion is brief (only a few hours). Use of antibiotics to eliminate an intestinal source for possible continued toxin production and of guanidine hydrochloride and other drugs to reverse paralysis is of unproven value. In the United States, antitoxin as well as help in clinical management and laboratory confirmation are available at *any* time from state health departments or from the Centers for Disease Control and Prevention [at (404)639-2206; emergency number: (404)639-2888].

Treatment of infant botulism requires supportive care. Neither equine antitoxin nor antibiotics have been shown to be beneficial, and the value of human botulism immune globulin, an experimental preparation, is still being evaluated. In wound botulism, equine antitoxin is administered. The wound should be thoroughly explored and debrided, and an antibiotic such as penicillin should be given to eradicate *C. botulinum* from the site, even though the benefit of this therapy is unproven. Results of wound cultures should guide the use of other antibiotics.

Botulinum toxin is being used as therapy for strabismus, blepharospasm, and other dystonias and appears safe and effective. Generalized botulism-like weakness complicating therapy has been reported.

PROGNOSIS

Type A disease is generally more severe than type B, and mortality from botulism is higher among patients above age 60 than among younger patients. With improved respiratory and intensive care, the case-fatality rate in food-borne illness has been reduced to ~7.5% and is low in infant botulism as well. Artificial respiratory support may be required for months in severe cases. Some patients experience residual weakness and autonomic dysfunction for as long as a year after disease onset.

PREVENTION

A pentavalent vaccine (A-E) is available for use in highly exposed individuals. Spores can be inactivated by exposure to a temperature of 116° to 121°C (e.g., in steam sterilizers or pressure cookers). Toxin can be inactivated by exposure to a temperature of 100°C for 10 min. Newly identified cases should be reported immediately to public health authorities.

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145. GAS GANGRENE, ANTIBIOTIC-ASSOCIATED COLITIS, AND OTHER CLOSTRIDIAL INFECTIONS - Dennis L. Kasper, Dori F. Zaleznik

DEFINITION

Bacteria of the genus *Clostridium* are gram-positive, spore-forming, obligate anaerobes that are ubiquitous in nature. There are more than 60 recognized species of clostridia, many of which are generally considered saprophytic. Some of these species are pathogenic for humans and animals, particularly under conditions of lowered oxidation-reduction potential. Infections associated with these organisms range from localized wound contamination to overwhelming systemic disease. The four major disease categories for which clostridia are responsible are intestinal disorders, deep tissue suppurative infections, skin and soft tissue infections, and bacteremia ([Table 145-1](#)). Toxins play a major role in some of these syndromes.

ETIOLOGY

In humans, clostridia normally reside in the gastrointestinal tract and in the female genital tract, although they occasionally are isolated from the skin or the mouth. Of the known species of the genus *Clostridium*, at least 30 have been isolated from human infections. Like several other pathogenic anaerobic bacterial species, clostridia are quite aerotolerant, but they do not grow on artificial media in the presence of oxygen. Clostridia characteristically produce abundant gas in artificial media and form subterminal endospores. *C. perfringens*, one of the most important species, is encapsulated and nonmotile and rarely sporulates in artificial media; the spores can usually be destroyed by boiling. *C. tetani* and *C. botulinum* are discussed in detail in [Chaps. 143](#) and [144](#), respectively.

Clostridia are present in the normal colonic flora at concentrations of 10^9 to 10^{10} per gram. Of the 30 or more species that normally colonize humans, *C. ramosum* is the most common and is followed in frequency by *C. perfringens*. These organisms are universally present in soil at concentrations of up to 10^4 per gram. Although clostridia are gram-positive organisms, many species may appear to be gram-negative in clinical specimens or stationary-phase cultures. Therefore, the results of Gram's staining of cultures or clinical material should be interpreted with great care.

C. perfringens is the most common of the clostridial species isolated from tissue infections and bacteremias; next in frequency are *C. novyi* and *C. septicum*. In the category of enteric infections, *C. difficile* is an important cause of antibiotic-associated colitis, and *C. perfringens* is associated with food poisoning (type A) and enteritis necroticans (type C).

PATHOGENESIS

Despite the isolation of clostridial species from many serious traumatic wounds, the prevalence of severe infections due to these organisms is low. Two factors that appear to be essential to the development of severe disease are tissue necrosis and a low oxidation-reduction potential. *C. perfringens* requires about 14 amino acids and at least 6 additional growth factors for optimal growth. These nutrients are not found in

appreciable concentrations in normal body fluids but are present in necrotic tissue. When *C. perfringens* grows in necrotic tissue, a zone of tissue damage due to the toxins elaborated by the organism allows progressive growth. In contrast, when only a few bacteria leak into the bloodstream from a small defect in the intestinal wall, the organisms do not have the opportunity to multiply rapidly because blood as a medium for growth is relatively deficient in certain amino acids and growth factors. Therefore, in a patient without tissue necrosis, bacteremia is usually benign.

C. perfringens possesses at least 17 possible virulence factors, including 12 active tissue toxins and enterotoxins. This species has been divided into five types (A through E) on the basis of four major lethal toxins: a, b, e, and i. The a toxin is a phospholipase C (lecithinase) that splits lecithin into phosphorylcholine and diglyceride. This a toxin has been associated with gas gangrene and is known to be hemolytic, to destroy platelets and polymorphonuclear leukocytes (PMNs), and to cause widespread capillary damage. When injected intravenously, it causes massive intravascular hemolysis and damages liver mitochondria. The a toxin may be important in the initiation of muscle infections that may progress to gas gangrene. Experimentally, the higher the concentration of a toxin in the culture fluid, the smaller the dose of *C. perfringens* required to produce infection. The protective effect of antiserum is directly proportional to its content of antitoxin. Studies suggest that q toxin may also play an important role in pathogenesis by promoting vascular leukostasis, endothelial cell injury, and regional tissue hypoxia. The resulting perfusion defects extend the anaerobic environment and contribute to rapidly advancing tissue destruction. A characteristic pathologic finding in gas gangrene is the near absence of PMNs despite extensive tissue destruction. Experimental data indicate that both a and q toxins are essential in the leukocyte aggregation that occurs at the margins of tissue injury instead of the expected infiltration of these cells into the area of damage. Genetically altered strains induce less leukocyte aggregation when a toxin is absent and none when q toxin is missing. The other major toxins, b, e, and i, are known to increase capillary permeability.

C. difficile produces two major toxins, designated A and B. Both toxins appear to act by the same mechanism, but toxin B is 1000 times more potent. These toxins exert their effect by binding to small guanosine triphosphate-binding proteins in the Rho family within target cells. The toxins are uridine diphosphoglucose hydrolases and glucosyltransferases that glycosylate the guanosine triphosphatases, inactivating these proteins and resulting in disruption of actin. Once the actin filaments are destroyed, the cell is unable to function. Toxin B has 100-fold greater enzymatic activity than toxin A.

Diarrheal disease due to *C. difficile* is toxin-mediated. Earlier teaching about the pathogenesis of this disease centered on the overgrowth of *C. difficile* when antibiotics suppress the normal bowel flora. Actually, the mechanism is probably more complex, since many of the antibiotics that cause this disease are active against *C. difficile* as well as other members of the bowel flora and since many patients who become colonized with *C. difficile* do not develop diarrhea. Critical features in the pathogenesis of this disease include mechanisms of toxin production and the interaction of *C. difficile* with other components of the bowel flora. Some antibiotics may actually trigger toxin production by the organism. In turn, other constituents of the bowel flora may suppress or inhibit toxin production. *C. sordellii*, for example, neutralizes cytotoxin B in vitro. In addition, when antibiotics eliminate more sensitive members of the bowel flora, more

resistant organisms may produce enzymes such as β -lactamases that can inactivate antibiotics and thereby facilitate the growth of *C. difficile*.

CLINICAL MANIFESTATIONS

Intestinal Disorders

Food Poisoning *C. perfringens*, primarily type A, is the second or third most common cause of food poisoning in the United States ([Chap. 131](#)). The responsible toxin is thought to be a cytotoxin produced by more than 75% of strains isolated from cases of foodborne disease. The cytotoxin binds to a receptor on the small-bowel brush border and induces a calcium ion-dependent alteration in permeability. The associated loss of ions alters intracellular metabolism, resulting in cell death. Outbreaks generally have resulted from problems in the cooling and storage of food cooked in bulk. The food sources primarily involved are meat, meat products, and poultry. Generally, the implicated meats have been cooked, allowed to cool, and then recooked the following day, often in a stew or hash. Strains of *C. perfringens* that contaminate meat manage to survive initial cooking. During reheating, the organisms sporulate and germinate. The disease is associated with an attack rate that is often as high as 70%. Symptoms of food poisoning from type A strains develop 8 to 24 h after ingestion of foods heavily contaminated with the organism. The primary symptoms include epigastric pain, nausea, and watery diarrhea usually lasting 12 to 24 h. Fever and vomiting are uncommon. Molecular methods including ribotyping and pulsed-field gel electrophoresis have been used to detect fecal cytotoxin in outbreaks of food poisoning caused by *C. perfringens*.

C. perfringens has also been implicated in a more severe form of diarrhea than that of classic food poisoning. This more severe disease tends to occur in the elderly and has been associated with antibiotic use in hospitalized populations. In this form of disease, diarrhea is generally more profuse, of longer duration, and accompanied by abdominal pain. Blood and mucus have been detected in the feces of the affected patients. In one hospital-based study of a cluster of cases, widespread environmental contamination with *C. perfringens* spores was documented.

Enteritis necroticans Necrotizing enteritis (enteritis necroticans, or *pigbel*) is caused by toxin produced by type C strains of *C. perfringens* following ingestion of a high-protein meal in conjunction with trypsin inhibitors (e.g., in sweet potatoes) by a susceptible host who has limited intestinal proteolytic activity. This disease has been reported among children and adults in New Guinea. A similar disease, *darmbrand*, was epidemic in Germany after World War II. Clinical features of *pigbel* include acute abdominal pain, bloody diarrhea, vomiting, shock, and peritonitis; 40% of patients die. Pathologic studies reveal an acute ulcerative process of the bowel restricted to the small intestine. The mucosa is lifted off the submucosa, with the formation of large denuded areas. Pseudomembranes composed of sloughed epithelium are common, and gas may dissect into the submucosa. The source of the organisms may be the patient's own intestinal flora; cultures of ingested pork have failed to yield the organism. Antibodies to the β -toxin of *C. perfringens* have been of considerable benefit in changing the course of established disease. In a large-scale trial, children immunized with *C. perfringens* β toxoid were protected.

Neutropenic enterocolitis (typhlitis) See [Chaps. 85](#) and [167](#).

Antibiotic-associated colitis Strains of *C. difficile* that produce toxins detectable in the stool are the only identified cause of colitis induced by antibiotic use. The diagnosis of this type of colitis requires that there be no other identifiable cause of diarrhea and that the onset of symptoms occur either during antimicrobial administration or within 4 weeks after treatment with the implicated agent has been discontinued. Essentially any antibiotic can cause this syndrome; even metronidazole and vancomycin, which are used to treat the disease, have been implicated as etiologic agents in some cases. On a per-use basis, clindamycin, which was the first antibiotic described to cause this entity, is the most commonly implicated antibiotic. However, since other antibiotics are prescribed more often than clindamycin in the United States, cephalosporins are currently the antibiotics that most commonly cause *C. difficile* enterocolitis, and penicillins rank next in frequency. Diarrhea due to *C. difficile* has been reported in patients with some forms of malignancy or renal transplantation who have received tacrolimus without concomitant or previous antibiotic administration.

Antimicrobial-associated diarrhea can be divided into four categories based on the appearance of the colon: (1) normal colonic mucosa; (2) mild erythema with some edema; (3) granular, friable, or hemorrhagic mucosa; and (4) pseudomembrane formation. Most patients with antibiotic-associated diarrhea have a normal, minimally erythematous colonic mucosa with some edema. Occasionally, colitis is more severe and is characterized by a granular, friable, or hemorrhagic mucosa. Examination of stool from the affected patients may reveal large numbers of red blood cells and some leukocytes. Biopsy shows subepithelial edema with round cell infiltration of the lamina propria and focal extravasation of erythrocytes. *C. difficile* cytotoxin B has been found in 15 to 75% of stools from patients in the first three categories, which suggests that other factors are involved in the pathogenesis of antibiotic-associated diarrhea.

The most characteristic form of antibiotic-associated colitis caused by *C. difficile* is pseudomembranous colitis (PMC) ([Fig. 145-CD1](#)). More than 95% of patients with documented PMC have positive stool toxin assays. Close inspection of pseudomembranes reveals exudative, punctate, raised plaques with skip areas or edematous hyperemic mucosa. These plaques can enlarge and coalesce over large segments of intestine in the later stages of disease. The clinical spectrum of antibiotic-associated PMC is diverse. Diarrhea is the key feature; stools are usually watery, voluminous, and without gross blood or mucus. Most patients have abdominal cramps and tenderness, fever, and leukocytosis. However, the symptoms vary considerably. At one end of the spectrum are many patients with annoying diarrhea but no systemic signs or symptoms, while at the other end are those with severe systemic toxicity, fever (40° to 40.6°C, or 104° to 105°F), and peripheral white blood cell counts of up to 50,000/uL with a marked left shift. Fecal examination frequently reveals leukocytes. Without specific therapy, the course is highly variable. Some patients, particularly those with clinically mild disease, experience prompt resolution of symptoms with discontinuation of drug treatment, while others have protracted diarrhea with large stool volumes for up to 8 weeks, with resultant hypoalbuminemia and electrolyte imbalance. Severely ill patients with toxic megacolon and colonic perforation have been reported. Among patients who are severely ill mortality rates may be as high as 30%,

while in most of those with minimal symptoms disease may resolve with the discontinuation of antibiotic treatment alone. In the majority of patients, symptoms begin 4 to 10 days after antibiotic therapy is initiated. However, ~25% of patients do not develop symptoms until use of the implicated antimicrobial has been discontinued, in some instances as long as 4 weeks afterward. Some cases have been reported within hours after initiation of antibiotic therapy or after a single dose of antibiotic administered for surgical prophylaxis.

Suppurative Deep Tissue Infections Clostridia are frequently recovered from various suppurative conditions in conjunction with other anaerobic and aerobic bacteria but can also be the only organisms isolated. These suppurative conditions, which exist with severe local inflammation but usually without the characteristic systemic signs induced by clostridial toxins, include intraabdominal sepsis, empyema, pelvic abscess, subcutaneous abscess, frostbite with gas gangrene, infection of a stump in an amputee, brain abscess, prostatic abscess, perianal abscess, conjunctivitis, infection of a renal cell carcinoma, and infection of an aortic graft.

Clostridia are isolated from approximately two-thirds of patients with intraabdominal infections resulting from intestinal perforation. *C. ramosum*, *C. perfringens*, and *C. bifermentans* are the most commonly isolated species. The presence of clostridial species does not affect the clinical presentation or outcome of these infections ([Chap. 167](#)).

An association has been made between malignancy and the isolation of *C. septicum* in the absence of grossly contaminated deep traumatic wounds. A major site for such a malignancy is the gastrointestinal tract, particularly the colon. An association with leukemia or with other solid tumors has also been noted, and one case of fatal myonecrosis has been reported in a patient with ovarian cancer. Some of these patients present with *C. septicum* bacteremia; these cases have a fulminant clinical course (discussed below). Others develop localized suppurative infection in the abdomen or the abdominal wall without bacteremia. Presumably, this infection arises from a silent perforation that leads to intraabdominal abscess formation.

Clostridia have been isolated from suppurative infections of the female genital tract, particularly tuboovarian and pelvic abscesses. The major species involved has been *C. perfringens*. Most of these are mild suppurative infections without evidence of uterine gangrene. *C. perfringens* has been isolated from as many as 20% of diseased gallbladders at surgery. One clinical syndrome, emphysematous cholecystitis, is caused by clostridial species at least 50% of the time. In this syndrome, gas forms in the biliary radicles and the wall of the gallbladder. It is seen most often in diabetic patients. Although the mortality rate in this entity is higher than in more common forms of cholecystitis, there is no evidence of myonecrosis.

Clostridia are among the many organisms found in empyema fluid or isolated by transtracheal aspiration from patients with lung abscesses. There is no unique clinical clue to the presence of clostridia (as opposed to other organisms) in these infections. *C. perfringens* has been reported as a cause of empyema arising from aspiration pneumonia, pulmonary emboli, and infarction. However, the majority of cases of clostridial empyema are secondary to trauma.

Skin and Soft Tissue Infections Various categories of traumatic wound infections due to clostridia have been described: simple contamination, anaerobic cellulitis, fasciitis with or without systemic manifestations, and anaerobic myonecrosis.

Simple contamination Clostridia are cultured most often from wounds in the absence of clinical signs of sepsis. As many as 30% of battle wounds are contaminated by clostridia without signs of suppuration, and 16% of penetrating abdominal wounds yield clostridia on culture despite treatment with cephalothin and kanamycin. In cases of trauma, clostridia are isolated with equal frequency from suppurative and well-healing wounds. Thus the diagnosis of clostridial infection should be based on clinical rather than bacteriologic criteria.

Localized infection of the skin and soft tissue without systemic signs This condition, originally referred to as *anaerobic cellulitis*, is a localized infection involving the skin and soft tissue and is due to clostridia alone or with other bacteria. There are no systemic signs of toxicity, although the infection may invade locally, producing necrosis. These infections tend to be relatively indolent, spreading slowly to contiguous areas. Localized infections are relatively free of pain and edema. Perhaps because of the lack of edema, gas that is limited to the wound and the immediately surrounding tissue may be more evident than in gas gangrene. In these localized infections, gas is never found intramuscularly. Cellulitis, perirectal abscesses, and diabetic foot ulcers are typical infections from which clostridial species can be isolated. If inadequately treated, these localized infections advance by extension through subcutaneous tissue and fascial planes into muscle and may produce severe systemic disease with signs of toxemia.

A localized form of suppurative myositis has been described in heroin addicts. These patients develop local pain and tenderness in discrete areas (particularly the thigh and forearm), with the subsequent appearance of fluctuance and crepitance that require surgical drainage. The unusual aspect of these infections is that they remain localized without systemic signs of toxicity. Moreover, the affected local areas are not necessarily sites of trauma or heroin injection. Pathologic examination reveals subcutaneous abscesses, purulent myositis, and fasciitis from which clostridia are recovered in pure culture; on occasion, mixed infections involving aerobes and anaerobes are found. Wound botulism has been reported in association with the injection of black tar heroin.

Spreading cellulitis and fasciitis with systemic toxicity This condition involves diffuse spreading cellulitis and fasciitis, without myonecrosis and with only mild inflammation in muscle. Patients present with the abrupt onset of a syndrome that progresses rapidly (within hours) through the fascial planes. In cases with suppuration and gas in soft tissues as well as overwhelming toxemia, the infection is rapidly fatal. On physical examination there is subcutaneous crepitation but little localized pain. Surgery is of no proven value because there are no discretely involved tissues amenable to resection, as may be the case in myonecrosis. However, in rapidly advancing fasciitis, incision of the affected area is still the cornerstone of therapy. The initial local lesion may be quite innocuous and arises from an area involved by tumor or other infection and not by injury. The systemic toxic effects include hemolysis and injury of capillary membranes. Usually, this infection is uniformly fatal within 48 h, despite intensive therapy involving antitoxin and exchange transfusion. This syndrome is seen most commonly in patients

with carcinoma, especially of the sigmoid or the cecum. Presumably, the tumor invades the fascia, and colonic contents leak into the abdominal wall. Patients present with extreme toxicity and occasionally with total-body crepitation. The syndrome differs from necrotizing fasciitis caused by other organisms in three respects: (1) rapid mortality, (2) rapid tissue invasion, and (3) the systemic effects of the toxin, typified by massive hemolysis.

Clostridial myonecrosis (gas gangrene) ([Fig. 145-CD2](#)) Clostridial myonecrosis occurs when bacteria invade healthy muscle from adjacent traumatized muscle or soft tissue. The infection originates in a wound contaminated with clostridia. Although >30% of deep wounds are infected with clostridia, the incidence of clostridial myonecrosis is quite low. These infections occur in both military and civilian settings. An essential factor in the genesis of gas gangrene appears to be trauma, particularly involving deep muscle laceration. The entity of clostridial myonecrosis is relatively uncommon after simple, through-and-through bullet wounds without shattering of bone and is relatively common following shrapnel fragmentation wounds, particularly when deep muscle is involved. In civilian cases, gas gangrene can follow trauma, surgery, or intramuscular injection. The trauma need not be severe; however, the wound must be deep, necrotic, and without communication to the surface.

The incubation period of gas gangrene is usually short: almost always <3 days and frequently <24 h. Some 80% of cases are caused by *C. perfringens*, while *C. novyi*, *C. septicum*, and *C. histolyticum* cause most of the other cases. Typically, gas gangrene begins with the sudden onset of pain in the region of the wound, which helps to differentiate it from spreading cellulitis. Once established, the pain increases steadily in severity but remains localized to the infected area and spreads only if the infection spreads. Soon after pain develops, local swelling and edema -- accompanied by a thin, often hemorrhagic exudate -- appear. Patients frequently develop marked tachycardia, but elevation in temperature may be only minimal. Gas is usually not obvious at this early stage and may be completely absent. Frothiness of the wound exudate may be noted. The skin is tense, white, often marbled with blue, and cooler than normal. The symptoms progress rapidly; swelling, edema, and toxemia increase, and a profuse serous discharge, which may have a peculiar sweetish smell, appears. Gram's staining of the wound exudate shows many gram-positive rods with relatively few inflammatory cells.

At surgery, muscle may appear pale because of the intensity of edema, but it does not contract when probed with a scalpel. When dissected, the muscle is beefy red and nonviable and can progress to become black, friable, and gangrenous. It is important to establish a diagnosis early, preferably by frozen-section biopsy of muscle.

Despite hypotension, renal failure, and (often) body crepitation, patients with myonecrosis frequently have a heightened awareness of their surroundings until just before death, when they lapse into toxic delirium and coma. In untreated cases, as the local wounds progress, the skin becomes bronzed; bullae appear, become filled with dark red fluid, and are accompanied by dark patches of cutaneous gangrene. Gas appears in later phases ([Fig. 145-1](#)) but may not be as obvious as in anaerobic cellulitis. Jaundice is rare in wound gas gangrene (in contrast to uterine infections) and, when it does appear, is almost invariably associated with hemoglobinuria, hemoglobinemia, and

septicemia. Cases of clostridial myonecrosis without a history of trauma have been reported. These patients have bullous lesions and crepitation of the skin; they present with a rapidly worsening course that includes myonecrosis, especially of the extremities.

Bacteremia and Clostridial Sepsis The relatively common entity of transient clostridial bacteremia can arise in any hospitalized patient but is most common with a predisposing focus in the gastrointestinal tract, biliary tract, or uterus. Fever frequently resolves within 24 to 48 h without therapy. Despite the finding of clostridial bacteremia following septic abortions and the frequent isolation of clostridia from the lochia, most of the patients involved do not have evidence of sepsis. In one series of 60 patients with clostridial bacteremia, half had an infected site that could be associated with the bacteremia, while the other half had a totally unrelated illness, such as tuberculous pneumonia, meningitis, or benign gastroenteritis. By the time blood culture reports are returned, patients frequently are completely well and sometimes have been discharged. Therefore, when a blood culture is positive for clostridia, the patient must be assessed clinically rather than simply treated on the basis of the culture result.

Clostridial sepsis is an uncommon but almost invariably fatal illness following clostridial infection -- primarily that of the uterus, colon, or biliary tract. This entity must be differentiated from transient clostridial bacteremia, which is much more common. *C. perfringens* causes the majority of cases of sepsis as well as the majority of cases of transient bacteremia. *C. septicum*, *C. sordellii*, and *C. novyi* account for most of the remainder of cases. Clostridia account for 1 to 2.5% of all positive blood cultures in major hospital centers.

The majority of cases of clostridial sepsis originate from the female genital tract and follow septic abortion. Introduction of a foreign body is a common antecedent event. In the uterus, residual necrotic fetal and placental tissues and traumatized endometrium may allow the growth of clostridia. Only a small fraction of cases of septic abortion (1%) are followed by serious sepsis. In these patients, sepsis, fever, and chills begin from 1 to 3 days after the attempted abortion. The initial signs are malaise, headache, severe myalgias, abdominal pain, nausea, vomiting, and occasionally diarrhea. Frequently, a bloody or brown vaginal discharge is noted. Patients may rapidly develop oliguria, hypotension, jaundice, and hemoglobinuria. The hemolysis, which is secondary to *C. perfringens* a toxin, causes a characteristic bronzing of the skin. As in myonecrosis, the mental status of severely ill patients is characterized by increased alertness and apprehension. Local examination of the pelvis reveals foul cervical discharge, occasionally with gas. Frequently, laceration marks around the cervix or perforation of the cervical segment is evident. If the infection involves the myometrium or has spread to the adnexa, extreme tenderness, guarding, and an adnexal mass may be found.

Laboratory studies in patients with sepsis reveal an elevated white blood cell count and may show pink, hemoglobin-tinged plasma. Anemia is proportional to the degree of hemolysis, and the hematocrit may be extremely low. Platelet counts may be reduced, and there is often evidence of disseminated intravascular coagulation. Oliguria or anuria, increasingly refractory hypotension, and hemorrhage and bruising may develop.

Clostridia may enter the bloodstream from the gastrointestinal or biliary tract. This occurrence is associated with ulcerative lesions or obstruction of the small or large

intestine, necrotic or infiltrating malignancy, bowel surgery, or various abdominal catastrophes. The patient may present with an acute febrile illness, with chills and fever but no other signs of localized infection. Intravascular hemolysis occurs in as many as half of such cases. Biliary or gastrointestinal symptoms, if present, may be the only clue to the etiology. Positive blood cultures provide the definitive clue to the diagnosis.

Patients with malignant disease can also develop rapidly fatal clostridial sepsis, particularly from a gastrointestinal focus. The most common species in this setting is *C. septicum*. Characteristic signs and symptoms include fever, tachycardia, hypotension, abdominal pain or tenderness, nausea, vomiting, and (preterminally) coma. The tachycardia may be out of proportion to the fever. Only ~20 to 30% of patients develop hemolysis. A striking feature of this syndrome is the rapidity of death, which frequently occurs in <12 h.

DIAGNOSIS

The diagnosis of clostridial disease, in association with positive cultures, must be based primarily on clinical findings. Because of the presence of clostridia in many wounds, their mere isolation from any site, including the blood, does not necessarily indicate severe disease. Smears of wound exudates, uterine scrapings, or cervical discharge may show abundant large gram-positive rods as well as other organisms. Cultures should be placed in selective media and incubated anaerobically for identification of clostridia. The diagnosis of clostridial myonecrosis can be established by frozen-section biopsy of muscle.

The urine of patients with severe clostridial sepsis may contain protein and casts, and some patients may develop severe uremia. Profound alterations of circulating erythrocytes are seen in severely toxemic patients. Patients have hemolytic anemia, which develops extremely rapidly, along with hemoglobinemia, hemoglobinuria, and elevated levels of serum bilirubin. Spherocytosis, increased osmotic and mechanical red blood cell fragility, erythrophagocytosis, and methemoglobinemia have been described. Disseminated intravascular coagulation may develop in patients with severe infection. In patients with severe sepsis, Wright's or Gram's staining of a smear of peripheral blood or buffy coat may demonstrate clostridia.

X-ray examination sometimes provides an important clue to the diagnosis by revealing gas in muscles, subcutaneous tissue, or the uterus. However, the finding of gas is not pathognomonic for clostridial infection. Other anaerobic bacteria, frequently mixed with aerobic organisms, may produce gas.

The diagnosis of *C. difficile*-associated colitis is most often made by an enzyme-linked immunosorbent assay (ELISA) for toxin A. Compared with the "gold standard" tissue culture assay used primarily for the detection of toxin B, the ELISA exhibits comparable specificity and only slightly lower sensitivity (70 to 90%). The cytotoxicity assay requires a tissue culture facility, skilled laboratory technicians, and time (usually 48 h), since neutralization of the cytopathic effect with *C. sordellii* or *C. difficile* antitoxin is required before the test can be labeled positive. ELISA is more rapid and easier to perform. However, in difficult situations where the clinical diagnosis remains a possibility and ELISA results are negative, consideration should be given to requesting the cytotoxicity

assay since it may detect 5 to 10% more cases. Repeat stool testing with the same assay generally does not increase the diagnostic yield for this entity. Endoscopy, although useful in establishing the presence of [PMC](#), does not establish the etiology and should be reserved for cases with more serious disease manifestations, in which it can be used to exclude alternative diagnoses. Isolation of *C. difficile* from stool cultures is difficult. This approach should be reserved for epidemiologic studies of outbreaks since asymptomatic persons may harbor the pathogen, but production of toxin is the hallmark of disease.

TREATMENT

Traumatic wounds should be thoroughly cleansed and debrided. Traditionally, the antibiotic treatment of choice for severe clostridial infection has been penicillin G (20 million units a day in adults). Penicillin G treatment of gas gangrene has become more controversial because of increasing resistance to this drug and data obtained from animal models of infection. In a mouse model of gas gangrene, antibiotics inhibiting toxin synthesis appeared to be preferable to cell wall-active drugs; clindamycin treatment enhanced survival more than therapy with penicillin; and the combination of clindamycin and penicillin was superior to penicillin alone. For severe clostridial sepsis, clindamycin may be used at a dose of 600 mg every 6 h in combination with high-dose penicillin (3 to 4 million units every 4 h). Although no clinical trials validate this choice, it is gaining acceptance in the infectious disease community.

In cases of penicillin sensitivity or allergy, other antibiotics should be considered, but all should be tested for in vitro activity because of the occasional isolation of resistant strains. Clostridia are frequently, but not universally, susceptible in vitro to cefoxitin, carbenicillin, chloramphenicol, clindamycin, metronidazole, doxycycline, imipenem, minocycline, tetracycline, third-generation cephalosporins, and vancomycin. For severe clostridial infections, sensitivity testing should be done before an antimicrobial with unpredictable activity is used. Simple contamination of a wound with clostridia should not be treated with antibiotics. Localized skin and soft tissue infection can be managed by debridement rather than with systemic antibiotics. Drugs are required when the process extends into adjacent tissue or when fever and systemic signs of sepsis are present. Surgery is a mainstay of therapy for clostridial myonecrosis or gas gangrene. Amputation may be required for rapidly spreading infection involving a limb. Hysterectomy is required for uterine myonecrosis. Abdominal wall myonecrosis usually continues despite initial aggressive surgery and antibiotic therapy and requires repeated surgical debridement of all involved muscle.

Suppurative infections should be treated with antibiotics. Frequently, broad-spectrum antibiotics must be used because of the mixed flora involved in these infections. Aminoglycosides can be used for the aerobic gram-negative bacteria involved in mixed infections.

The use of a polyvalent gas gangrene antitoxin is still recommended by some authorities. At present, no such antitoxin is produced in the United States, and most centers have discontinued its use in the management of patients with suspected gas gangrene or clostridial postabortion sepsis because of questionable efficacy and the substantial risk of hypersensitivity to horse serum, from which the antitoxin is derived.

The use of hyperbaric oxygen in the treatment of gas gangrene is also controversial. Studies in humans are not well designed to answer questions on efficacy, but several knowledgeable authors believe that hyperbaric oxygen therapy has contributed to dramatic clinical improvement. Such therapy may, however, be associated with untoward effects due to oxygen toxicity and high atmospheric pressure. Some centers without hyperbaric chambers have reported acceptable mortality rates; thus expert surgical and medical management and control of complications are probably the most important factors in the treatment of gas gangrene. Fasciotomy should not be delayed for hyperbaric oxygen therapy.

The treatment of *C. difficile*-associated colitis requires discontinuation of therapy with the offending antimicrobial agent. In some patients, symptoms will resolve over a period of 2 weeks if the infection is left untreated. However, specific therapy shortens the duration of symptoms.

Diarrhea due to *C. difficile* should be treated with metronidazole (500 mg orally tid for 10 to 14 days). A randomized trial comparing metronidazole (250 mg qid) with oral vancomycin showed equal efficacy and relapse rates of ~9% for both regimens. Since both treatment regimens are effective, metronidazole is preferred because it is far less costly and has not been linked to the development of vancomycin-resistant enterococci. When a patient relapses, antibiotic resistance should not be inferred since it is rare; a repeat course of metronidazole is appropriate. When a patient with *C. difficile*-associated diarrhea requires continued antibiotic treatment for a serious infection such as infective endocarditis, it is often reasonable to continue therapy against *C. difficile* for the duration of the offending antibiotic treatment course and for a full 10 to 14 days following its completion. When vancomycin is used for the treatment of *C. difficile*-associated diarrhea, the starting dose should be 125 mg orally qid, although doses as high as 500 mg qid can be used if needed. When a patient requires parenteral therapy for antibiotic-associated diarrhea, intravenous metronidazole can be administered. Vancomycin is effective only if used orally; the drug is poorly absorbed after oral administration. If patients continue to have diarrhea and have signs of systemic toxicity (e.g., fever and/or leukocytosis) after 48 h of treatment with metronidazole, it is reasonable to switch to vancomycin. For especially severe disease, some experts advocate treatment with both oral vancomycin and metronidazole, although there are no trials to support this regimen.

A number of patients who respond to initial therapy present with a relapse of symptoms and a repeat positive toxin assay. Relapses following therapy are much more frequent than failures to respond to initial therapy. Most relapses occur 3 to 10 days after discontinuation of treatment. Most relapsing patients respond to a second course of antibiotics, but some go on to suffer multiple relapses. A number of options are available in this situation. Some authors report success with tapering regimens of vancomycin given daily or every other day for 1 to 2 months to avoid relapse. The resin cholestyramine binds the cytotoxin of *C. difficile* and has been used with some success to treat severe cases. Since cholestyramine also binds vancomycin, the two agents should not be used in combination. Repopulation of the normal colonic flora has also been tried in relapsing disease. Ingestion of capsules of the yeast *Saccharomyces boulardii* showed some promise in one trial; oral lactobacilli have also been used in

uncontrolled studies. The administration of intravenous immunoglobulin has been tried with success in a few children and adults with relapsing infection, although this approach is not yet considered to be recommended therapy.

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SECTION 6 -DISEASES CAUSED BY GRAM-NEGATIVE BACTERIA

146. MENINGOCOCCAL INFECTIONS - Robert S. Munford

DEFINITION

Neisseria meningitidis is the etiologic agent of two life-threatening diseases: meningococcal meningitis and fulminant meningococcemia. Meningococci also cause pneumonia, septic arthritis, pericarditis, urethritis, and conjunctivitis. Most cases are potentially preventable by vaccination.

ETIOLOGIC AGENT

N. meningitidis bacteria are gram-negative aerobic diplococci. Unlike the other neisseriae, they have a polysaccharide capsule. They are transmitted among humans, their only known habitat, via respiratory secretions. Colonization of the nasopharynx or pharynx is much more common than invasive disease.

EPIDEMIOLOGY

Meningococcal disease occurs worldwide as isolated (sporadic) cases, institution- or community-based outbreaks, and large epidemics.

Meningococci are classified into serogroups based on the antigenicity of their capsular polysaccharides. Antigenicity reflects structural differences in these polysaccharides. Five serogroups (A, B, C, Y, and W-135) are responsible for >90% of cases of meningococcal disease worldwide. Serogroup A strains, which caused most of the large epidemics of meningococcal disease during the first half of the twentieth century, are now associated with recurring epidemics in sub-Saharan Africa and other locales in the developing world. Serogroups B, C, and Y cause most cases of sporadic and epidemic meningococcal disease in industrialized countries. In the United States and Canada during the 1990s, serogroup B was the most common cause of sporadic disease, while serogroup C was a more frequent cause of outbreaks. Serogroup Y has recently been isolated from almost one-third of cases of meningococcal disease in the United States. In general, patients with serogroup Y disease are older and more likely to have a chronic underlying illness than are patients with disease caused by other serogroups. Serogroups Y and W-135 are isolated more often than the other serogroups from patients with pneumonia.

One limitation of the serogroup classification is that the genes for capsule biosynthesis can be transferred from one strain to another, with consequent changes in the capsule structure of the recipient strain and therefore in its serogroup. Other methods for tracking meningococcal strains have thus become increasingly useful. Meningococcal serotypes and subtypes are defined by antigenic differences in specific outer-membrane proteins (OMPs), whereas multilocus enzyme electrophoresis classifies bacteria into electrophoretic types (ETs). Other techniques for establishing strain identity or nonidentity are pulsed-field gel electrophoresis and amplification of bacterial genomic sequences by polymerase chain reaction. The virulent III-1 clonal complex of serogroup A was first recognized in Nepal in 1983 to 1984; it spread to Mecca, then to

sub-Saharan Africa, and subsequently to temperate Africa. Increased virulence and epidemic potential have also been ascribed to the serogroup B ET-5 complex, which was first identified in Norway in the 1970s and later caused outbreaks in Europe, Cuba, and South and North America (most recently, in the Pacific Northwest). Serogroup C ET-24 (the ET-37 complex) has caused sporadic cases and outbreaks in Canada and the United States and in some analyses has been associated with high mortality and morbidity.

Meningococcal colonization of the nasopharynx (asymptomatic carriage) can persist for months. In nonepidemic periods, ~10% of healthy individuals are colonized. Factors that predispose individuals to colonization with *N. meningitidis* include residence in the same household with a person who has meningococcal disease or is a carrier, household or institutional crowding, active or passive exposure to tobacco smoke, and a recent history of a viral upper respiratory infection. These factors have also been associated with an increased risk of meningococcal disease.

In countries with temperate climates, the attack rate for sporadic meningococcal disease is ~1 case per 100,000 persons per year. Peak disease incidence coincides with the winter peak of respiratory viral illnesses. Disease attack rates are highest among infants 3 to 9 months of age (10 to 15 cases per 100,000 infants per year). Children also have higher attack rates than adults, and there is a second peak of incidence among teenagers, in whom outbreaks have often been tied to residence in barracks, dormitories, or other crowded conditions. Although the age-specific incidence is much lower among adults (<1 case per 100,000 persons per year), approximately one-third of all cases of sporadic meningococcal disease occur in individuals ³18 years of age. During epidemics, disease incidence increases disproportionately among teenagers and young adults and during the summer and autumn.

Meningococcal disease occurs more commonly in the household contacts of primary cases. The secondary attack rate is 400 to 1000 per 100,000 household members. School-based clusters of cases have also been described; the attack rate among school contacts of cases has been estimated at 2 to 4 cases per 100,000 exposed individuals. In outbreaks on college campuses, attack rates have been highest among students living in dormitories. Most secondary cases occur within 2 weeks of the primary case, although some may develop as long as several months later. Secondary cases account for <2% of all cases reported each year in the United States.

In an outbreak, the case isolates of *N. meningitidis* are identical when they are assessed by molecular typing methods. Recent outbreaks of meningococcal disease have occurred among persons whose common exposure took place in military barracks, schools, a university campus tavern, a jail, a school bus, a disco bar, a sports club, and a hotel.

PATHOGENESIS

Meningococci that colonize the upper respiratory tract are internalized by nonciliated mucosal cells and may traverse them to enter the submucosa, from which they can make their way into the bloodstream. While meningococcal colonization occurs often in healthy humans, bloodstream infection is an infrequent event that is not essential for the

organisms' survival and spread; as is often the case, the production of human disease has no obvious evolutionary advantage for either pathogen or host. Although some strains of *N. meningitidis* are thought to cause more severe disease in humans than do other strains, the basis for this difference is not understood. Meningococci may undergo important phenotypic changes when they adapt to growth in vivo; presumed virulence traits include the antiphagocytic capsular polysaccharide, an ability to sialylate the cell wall lipooligosaccharide (LOS) so that it mimics host cell carbohydrate moieties, the secretion of IgA protease, and mechanisms for iron acquisition. The [ET-5](#) strain of serogroup B *N. meningitidis* has been associated with high case-fatality rates in some populations but not in others, however, suggesting that host factors also contribute importantly to disease pathogenesis.

A meningococcus that enters the blood from the nasopharynx and survives host defenses generally has one of two fates. If multiplication occurs slowly, the bacteria eventually seed local sites, such as the meninges, joints, or pericardium. More rapid multiplication in the blood is associated with disseminated intravascular coagulation (DIC) and shock, which usually cause symptoms before local sites become infected. There is thus a remarkable compartmentalization of bacterial growth and host inflammation in either the blood or a local site, usually the meninges.

Fulminant Meningococcemia (Purpura Fulminans) Fulminant meningococcemia is perhaps the most rapidly lethal form of septic shock experienced by humans. It differs from most other forms of septic shock by the prominence of hemorrhagic skin lesions (petechiae, purpura) and the consistent development of [DIC](#).

The dominant proinflammatory molecule in the meningococcal cell wall is the endotoxin or [LOS](#), and the outer membrane that contains it is poorly tethered to the underlying peptidoglycan. This structural peculiarity seems to account for the fact that meningococci shed LOS-containing membrane blebs as they grow. The bacteria can multiply to very high concentrations in the blood. The concentrations of endotoxin detected in the blood of patients with fulminant meningococcemia are 10- to 1000-fold greater than those found in the blood of patients with bacteremia due to other gram-negative bacteria. The bacteria and endotoxin-containing blebs stimulate monocytes, neutrophils, and endothelial cells, which then release cytokines and other mediators that can activate many distant targets, including other leukocytes and endothelial cells. In addition, meningococci can invade the vascular endothelium. When activated, the endothelium produces molecules that can be procoagulant as well as adhesive for leukocytes.

Patients with fulminant meningococcemia usually have extremely high blood levels of both proinflammatory mediators -- i.e., tumor necrosis factor (TNF) α , interleukin (IL) 1, interferon, and IL-8 -- and anti-inflammatory mediators -- i.e., IL-1 receptor antagonist (IL-1Ra), soluble IL-1 receptors, soluble TNF receptors, and IL-10. The plasma of patients with meningococcal shock can decrease the responses of normal leukocytes to stimuli such as [LOS](#); the implication is that anti-inflammatory mediators predominate in the blood.

Procoagulant, antifibrinolytic forces are predominant in the blood of patients with fulminant meningococcemia ([Fig. 146-1](#)). Monocytes express large amounts of tissue

factor. Fibrinopeptide A and thrombin-antithrombin levels are high, reflecting active clotting, while antithrombin and fibrinogen levels are low. Although the tissue factor-regulated ("extrinsic") arm of coagulation predominates, the contact system (factors XII and XI, prekallikrein, high-molecular-weight kininogen) is also activated. Striking deficiencies of antithrombin and proteins C and S can occur; studies have found a strong negative correlation between protein C activity and both the size of purpuric skin lesions and mortality. Plasminogen levels are decreased, while plasmin-antiplasmin complexes and plasminogen activator inhibitor 1 (PAI-1) levels in the blood are very high. PAI-1 levels have been correlated with mortality risk, as has a function-related polymorphism in the promoter of the PAI-1 gene.

Fibrin deposition is therefore favored both by the *procoagulant* tendency, promoted through activation of tissue factor and deficiencies of proteins C and S and antithrombin, and by an *antifibrinolytic* tendency, favored by excessive [PAI-1](#). Both platelets and leukocytes doubtless contribute to the formation of microthrombi and to the vascular injury that ensues. Thrombosis of larger vessels leads to peripheral necrosis and gangrene that may require limb or digit amputation.

None of the candidate mediators of septic shock has proven primacy ([Chaps. 38](#) and [124](#)). Numerous studies have suggested that shock and [DIC](#) are not intimately linked and that the contact arm of clotting, which is of secondary importance in the pathogenesis of DIC, plays at least a contributory role in the pathogenesis of shock. The independence of DIC and shock suggests that therapies that prevent or reverse DIC may not be helpful for patients with septic shock.

Meningitis *N. meningitidis* has a striking tropism for the meninges. Infection of the central nervous system begins in the choroid plexus or in the ependyma that lines the cerebral ventricles. Meningococci adhere to cerebral capillary endothelial cells and then enter the subarachnoid space. A vigorous local inflammatory response ensues, probably triggered by endotoxin-containing meningococcal membranes. Both bacterial growth and the inflammatory response occur within the cerebrospinal fluid (CSF), where levels of endotoxin, [IL-6](#), [TNF- \$\alpha\$](#) , IL-1b, IL-1Ra, and IL-10 exceed the concentrations found in plasma by 100- to 1000-fold. The inflammatory response is largely confined to the subarachnoid space and contiguous structures.

Patients who develop meningitis may be individuals in whom meningococci do not grow rapidly in the blood; they may have a more vigorous initial inflammatory response to invading meningococci, may have antibodies or phagocytes that slow meningococcal growth, or may lack the (unknown) factors that allow *N. meningitidis* to multiply rapidly in vivo. The prognosis of patients with meningococcal meningitis is substantially better than that of patients with fulminant meningococcemia ([Table 146-1](#)).

HOST DEFENSE MECHANISMS

Preventing meningococcal growth in blood requires bactericidal and opsonic antibodies, complement, and phagocytes ([Fig. 146-2](#)). The major bactericidal antibodies are IgM and IgG, which bind to the capsular polysaccharide. Immunity to meningococci is therefore serogroup specific. Antibodies to other surface (subcapsular) antigens may confer cross-serogroup protection. Infants are protected from meningococcal disease

during the first months of life by passively transferred maternal IgG antibodies. As maternal antibody levels wane, the attack rate increases, peaking from 3 to 9 months of age. Disease incidence declines as protective antibodies are induced by colonization with nonpathogenic bacteria that have cross-reactive antigens. In addition to *N. lactamica*, which frequently colonizes young children, some enteric bacteria have antigens that cross-react with those of meningococci. One theory relates the occurrence of meningococcal disease to the presence of high levels of IgA antibodies to meningococci, since these antibodies can block the bactericidal activity of IgM.

Complement is required for bactericidal activity and for efficient opsonophagocytosis. Individuals deficient in any of the late complement components (C5 to C9) cannot assemble the membrane attack complex needed to kill *Neisseria*. These persons typically develop less severe meningococcal disease than complement-sufficient individuals, do so at an older age, and tend to have disease due to uncommon serogroups (W-135, X, Y, Z, and 29E). Although only one-half of individuals with known late-complement-component deficiency ever experience meningococcal disease, some affected persons have several episodes. Deficiency of each of the terminal complement components is inherited in an autosomal recessive fashion. Properdin deficiency, in contrast, is X-linked; some affected males develop overwhelming meningococcal disease, an observation indicating that the alternative complement pathway is also needed for antimeningococcal host defense. The age of disease onset in properdin-deficient individuals is typically in the teens or twenties.

Activation of the classic pathway of complement by antigen-antibody complexes or of the alternative pathway by [LOS](#) or capsular polysaccharide is important for producing and maintaining C3b ([Fig. 146-2](#)). Without C3b, neither bactericidal lysis nor phagocytosis can proceed effectively. When C3b is generated, meningococcal growth is probably checked by the membrane attack complex, which produces bacterial lysis, and by robust phagocytosis. Most IgG antibodies to the meningococcal polysaccharide are of the IgG₂ isotype; a phagocytic cell defect (the FcγRIIA R131 allele) that impairs the phagocytosis of IgG₂-coated particles has been associated with more severe meningococcal disease. This allele has also been associated with a more severe clinical course in patients with late-complement-component deficiency; thus effective phagocytosis may contribute to the relatively mild meningococcal disease usually observed in these individuals.

Indirect evidence indicates that persons who mount a vigorous inflammatory response to meningococcal [LOS](#) may experience less severe disease than those whose initial response is anti-inflammatory. This observation suggests that the inflammatory response may be critical for restraining meningococcal growth. Attempts to identify disease- or severity-associated polymorphisms in cytokine genes have had inconclusive results, however.

CLINICAL MANIFESTATIONS

Upper Respiratory Tract Infections Although many patients who develop meningococcal meningitis or meningococcemia report having had throat soreness or other upper respiratory symptoms during the preceding week, it is uncertain whether these symptoms are due to infection with meningococci. Meningococcal pharyngitis is

rarely diagnosed. Adult patients with *N. meningitidis* bacteremia more often have clinically apparent disease of the respiratory tract (pneumonia, sinusitis, tracheobronchitis, conjunctivitis) than do younger patients.

Meningococcemia Most patients with meningococcal disease have both meningococcemia *and* meningitis. These conditions have a wide clinical spectrum, with many overlapping features ([Table 146-1](#)).

Approximately 10 to 30% of patients with meningococcal disease have meningococcemia without clinically apparent meningitis. Although meningococcemia is occasionally transient and asymptomatic, in most individuals it is associated with fever, chills, nausea, vomiting, and myalgias. Prostration is common. The most distinctive feature is rash ([Fig. 146-CD1](#)). Erythematous macules rapidly become petechial and, in severe cases, purpuric. Although the lesions are typically found on the trunk and lower extremities, they may also occur on the face, arms, and mucous membranes. The petechiae may coalesce into hemorrhagic bullae or may undergo necrosis and ulcerate. Patients with severe coagulopathy may develop ischemic extremities or digits, often with a sharp line of demarcation between normal and ischemic tissue.

In many patients with fulminant meningococcemia, the [CSF](#) is normal and the CSF culture is negative. Indeed, the absence of meningitis in a patient with meningococcemia is a poor prognostic sign; it suggests that the bacteria have multiplied so rapidly in the blood that meningeal seeding has not yet had time to elicit inflammation in the CSF. Most of these patients also lack evidence of an acute-phase response; i.e., the erythrocyte sedimentation rate is normal, and the C-reactive protein concentration in blood is low.

The Waterhouse-Friderichsen syndrome is a dramatic example of [DIC](#)-induced microthrombosis, hemorrhage, and tissue injury. Although overt adrenal failure is infrequently documented in patients with fulminant meningococcemia, patients may have partial adrenal insufficiency and be unable to mount the normal hypercortisolemic response to severe stress or cosyntropin stimulation. Almost all patients who die from fulminant meningococcemia have adrenal hemorrhages at autopsy.

Chronic meningococcemia is a rare syndrome of episodic fever, rash, and arthralgias that can last for weeks to months. The rash may be maculopapular; it is occasionally petechial. Splenomegaly may develop. If untreated or if treated with glucocorticoids, chronic meningococcemia may evolve into meningitis, fulminant meningococcemia, or (rarely) endocarditis.

Meningitis (See also [Chap. 372](#)) Patients with meningococcal meningitis have usually been sick for ≥ 24 h before they seek medical attention. Common presenting symptoms include nausea and vomiting, headache, neck stiffness, lethargy, and confusion. The symptoms and signs of meningococcal meningitis cannot be distinguished from those elicited by other meningeal pathogens. Many patients with meningococcal meningitis have concurrent meningococcemia, however, and petechial or purpuric skin lesions may suggest the correct diagnosis. [CSF](#) findings are consistent with those of purulent meningitis: hypoglycorrhachia, an elevated protein concentration, and a neutrophilic leukocytosis. A Gram's stain of CSF is usually positive (see "Diagnosis," below); when

this finding is unaccompanied by CSF leukocytosis, the prognosis for normal recovery is often poor.

Other Manifestations Arthritis occurs in ~10% of patients with meningococcal disease. When arthritis develops during the first few days of the patient's illness, it usually reflects direct meningococcal invasion of the joint. Arthritis that begins later in the course is thought to be due to immune complex deposition. Primary meningococcal pneumonia occurs principally in adults, often in military populations, and is most often due to serogroup Y. While meningococcal pericarditis is occasionally seen, endocarditis due to *N. meningitidis* is now exceedingly rare. Primary meningococcal conjunctivitis can be complicated by meningococcemia; systemic therapy is therefore warranted when this condition is diagnosed. Meningococcal urethritis has been reported in individuals who practice oral sex.

Complications Patients with meningococcal meningitis may develop cranial nerve palsies, cortical venous thrombophlebitis, and cerebral edema. In children subdural effusions may occur. Permanent sequelae can include mental retardation, deafness, and hemiparesis. The major long-term morbidity of fulminant meningococcemia is the loss of skin, limbs, or digits that results from ischemic necrosis and infarction.

DIAGNOSIS

Few clinical clues help the physician distinguish the patient with early meningococcal disease from patients with other acute systemic infections. The most useful clinical finding is the petechial or purpuric rash (see [Plate IID-44](#)), but it must be differentiated from the petechial lesions seen with gonococcemia (see [Plate IID-60](#)), Rocky Mountain spotted fever (see [Plate IID-45](#)), hypersensitivity vasculitis (see [Plate IIE-71](#)), endemic typhus, and some viruses. In one case series, one-half of the adults with meningococcal bacteremia had neither meningitis nor a rash.

The definitive diagnosis is established by recovering *N. meningitidis*, its antigens, or its DNA from normally sterile body fluids, such as blood, [CSF](#), or synovial fluid, or from skin lesions. Meningococci grow best on Mueller-Hinton or chocolate blood agar at 35°C in an atmosphere that contains 5 to 10% CO₂. Specimens should be plated without delay. *N. meningitidis* bacteria are oxidase-positive, gram-negative diplococci that typically utilize maltose and glucose.

A Gram's stain of [CSF](#) reveals intra- or extracellular organisms in ~85% of patients with meningococcal meningitis. The latex agglutination test for meningococcal polysaccharides is somewhat less sensitive. Reports suggest that polymerase chain reaction amplification of DNA in buffy coat or CSF samples may be more sensitive than either of these tests, and, like the latex agglutination test, this method is not affected by prior antibiotic therapy.

Throat or nasopharyngeal specimens should be cultured on Thayer-Martin medium, which suppresses the competing oral flora. Throat or nasopharyngeal cultures are recommended only for research or epidemiologic purposes, since a positive result merely confirms the carrier state and does not establish the existence of systemic disease.

TREATMENT

A third-generation cephalosporin, such as cefotaxime (2 g intravenously every 8 h) or ceftriaxone (1 g intravenously every 12 h), is preferred for initial therapy, as it may cover other bacteria (such as *Streptococcus pneumoniae* and *Haemophilus influenzae*) that can cause the same syndromes ([Chap. 372](#)). Penicillin G (4 million units intravenously every 4 h) remains an acceptable alternative in most countries; high-level penicillin resistance has been reported from Spain. In the patient who is allergic to β -lactam drugs, chloramphenicol (75 to 100 mg/kg every 6 h) is a suitable alternative; chloramphenicol-resistant meningococci have been reported from Vietnam and France, however. Although some cases of mild disease may be cured with only 2 days of treatment, most patients with meningococcal meningitis should be given antimicrobial therapy for at least 5 days. While glucocorticoid therapy for meningitis in adults is controversial, many experts administer dexamethasone, beginning if possible before antibiotic therapy is initiated ([Chap. 372](#)).

Patients with fulminant meningococcemia often experience diffuse leakage of fluid into extravascular spaces, shock, and multiple-organ dysfunction ([Chaps. 38](#) and [124](#)). Myocardial depression may be prominent. Supportive therapy has never been studied in randomized, placebo-controlled trials. Standard measures include vigorous fluid resuscitation (often requiring several liters over the first 24 h), elective ventilation, and pressors (epinephrine or dopamine). Some authorities recommend early hemodialysis or hemofiltration. Fresh frozen plasma is often given to patients who are bleeding extensively or who have severely deranged clotting parameters. Many European experts prefer to administer antithrombin to such patients. Patients with fulminant meningococcemia in whom shock persists despite vigorous fluid resuscitation should receive supplemental glucocorticoid treatment (hydrocortisone, 1 mg/kg every 6 h) pending tests of adrenal reserve. Investigational drugs for fulminant meningococcemia include bactericidal permeability-increasing (BPI) protein -- a bactericidal neutrophil protein that binds and neutralizes meningococcal [LOS](#) -- as well as several anticoagulants (activated protein C, antithrombin, and tissue factor pathway inhibitor). In a recent clinical trial, recombinant BPI protein reduced long-term complications in children with fulminant meningococcemia without definitely reducing mortality.

PROGNOSIS

When patients are first evaluated, the clinical features most strongly associated with a fatal outcome are shock, a purpuric or ecchymotic rash, a low or normal blood leukocyte count, an age ≥ 60 years, and coma. The absence of meningitis, the presence of thrombocytopenia, low blood concentrations of antithrombin or proteins S and C, high blood levels of [PAI-1](#), and a low erythrocyte sedimentation rate (or C-reactive protein level) have also been associated with increased mortality from meningococcal disease. In contrast, having received antibiotics prior to hospital admission has been associated with lower mortality in some studies.

PREVENTION

Meningococcal Polysaccharide Vaccines A single injection of quadrivalent

meningococcal polysaccharide vaccine (serogroups A, C, W-135, and Y) immunizes ~80 to 95% of immunocompetent adults. Children ³3 months of age can be vaccinated to prevent serogroup A disease, but multiple doses are required; the vaccine is otherwise ineffective in children <2 years old. The duration of vaccine-induced immunity in adults is probably <5 years. There is currently no vaccine for serogroup B; its polysaccharide is a sialic acid homopolymer that is poorly immunogenic in humans. In addition to individuals with late-complement-component or properdin deficiency, persons with sickle cell anemia, asplenia, or splenectomy should receive the quadrivalent vaccine. Vaccination is also recommended for military recruits and for individuals traveling to sub-Saharan Africa during the dry months (June to December) or to other areas with epidemic meningococcal disease. Some authorities recommend vaccination of incoming college freshmen who will live in dormitories. In general, the vaccine should be given only to persons ³2 years of age. Investigational polysaccharide-protein conjugate meningococcal vaccines appear promising; a serogroup C conjugate vaccine was licensed for use in the United Kingdom in 1999.

Screening tests for late-complement-component deficiency should be done in family members of patients who have a family history of meningococcal disease, in patients who have a recurrence, in those whose first case occurs at ³15 years of age, and in those with cases caused by serogroups other than A, B, or C.

Antimicrobial Chemoprophylaxis The attack rate for meningococcal disease among household contacts of cases is ~500-fold greater than that in the population as a whole. Close contacts of cases should receive chemoprophylaxis with rifampin (adult dosage, 600 mg orally every 12 h for four doses), ciprofloxacin (a single oral dose of 500 mg), or ofloxacin (a single oral dose of 400 mg). A single intramuscular injection of ceftriaxone (250 mg) is also effective. Close contacts include persons who live in the same household, day-care center contacts, and anyone directly exposed to the patient's oral secretions. Casual contacts are not at increased risk. Chemoprophylaxis should be administered as soon as possible after the case is identified.

Isolation Precautions The Centers for Disease Control and Prevention recommend that patients with meningococcal disease who are hospitalized be placed in respiratory isolation for the first 24 h.

Outbreak Control An organization- or community-based outbreak of meningococcal disease is defined as the occurrence of three or more cases within £3 months in persons who have a common affiliation or reside in the same area but who are not close contacts of one another; in addition, the primary disease attack rate must exceed 10 cases per 100,000 persons, and the case strains of *N. meningitidis* must be of the same molecular type. Mass vaccination should be considered when such outbreaks occur, and mass chemoprophylaxis may be used to control school- or other institution-based outbreaks. Consultation with public health authorities is recommended when such campaigns are contemplated.

(Bibliography omitted in Palm version)

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147. GONOCOCCAL INFECTIONS - Sanjay Ram, Peter A. Rice

DEFINITION

Gonorrhea is a sexually transmitted infection of epithelium and commonly manifests as cervicitis, urethritis, proctitis, and conjunctivitis. If untreated, infections at these sites can lead to local complications such as endometritis, salpingitis, tuboovarian abscess, bartholinitis, peritonitis, and perihepatitis in the female; periurethritis and epididymitis in the male; and ophthalmia neonatorum in the newborn. Disseminated gonococemia is an uncommon event whose manifestations include skin lesions, tenosynovitis, arthritis, and (in rare cases) endocarditis or meningitis.

Neisseria gonorrhoeae is a gram-negative, nonmotile, non-spore-forming organism that grows in pairs (diplococci). Each individual organism is shaped like a coffee bean, with adjacent concave sides seen on Gram's stain. Gonococci, like all other *Neisseria* spp., are oxidase positive. They are distinguished from other *Neisseria* by their ability to grow on selective media and to utilize glucose but not maltose, sucrose, or lactose.

EPIDEMIOLOGY

The incidence of gonorrhea has declined significantly in the United States, but there are still ~315,000 newly reported cases each year. Gonorrhea remains a major public health problem worldwide, is a significant cause of morbidity in developing countries, and may play a role in enhancing transmission of HIV.

Gonorrhea predominantly affects young, nonwhite, unmarried, less educated members of urban populations. The number of reported cases probably represents half of the true number of cases -- a discrepancy resulting from underreporting, self-treatment, and nonspecific treatment without a culture-proven diagnosis. The number of reported cases of gonorrhea in the United States rose from ~250,000 in the early 1960s to a high of 1.01 million in 1978. The peak recorded incidence of gonorrhea in modern times was noted in 1975, with 468 cases per 100,000 population in the United States. This peak was attributable to the interaction of several variables, including improved accuracy of diagnosis, changes in patterns of contraceptive use, and changes in sexual behavior. The incidence of the disease has since gradually declined and is currently estimated at 120 cases per 100,000, a figure that is still the highest among industrialized countries. A further decline in the overall incidence of gonorrhea in the United States over the past decade may reflect increased condom use resulting from public health efforts to curtail HIV transmission. Presently, the attack rate in the United States is highest in the 20- to 24-year age group, in which 75% of all cases occur. With adjustment for sexual experience, the risk is highest among sexually active 15- to 19-year-old women. In terms of ethnicity, rates are highest among African-Americans and lowest among persons of Asian or Pacific Island descent.

The highest incidence of gonorrhea occurs in developing countries. The exact incidence of any of the sexually transmitted diseases (STDs) is difficult to ascertain in developing countries because of limited surveillance and variable diagnostic criteria. For example, in Kenya, it was estimated in 1987 that 10% of all live births were adversely affected by STDs, and gonococcal ophthalmia neonatorum reportedly affected 4% of all live-born

infants. The median prevalence of gonorrhea in unselected populations of pregnant women has been estimated at 10% in Africa, 5% in Latin America, and 4% in Asia. Studies in Africa have clearly demonstrated that nonulcerative STDs such as gonorrhea are an independent risk factor for the transmission of HIV ([Chap. 309](#)).

Gonorrhea is transmitted from males to females more efficiently than in the opposite direction. The rate of transmission to a woman following a single unprotected sexual encounter with an infected man is on the order of 40 to 60%. Oropharyngeal gonorrhea occurs in ~20% of women who practice fellatio with infected partners. Transmission in either direction by cunnilingus is rare.

There exists in any population a small minority of individuals who have high rates of new partner acquisition. These "core-group members" or "high-frequency transmitters" are vital in sustaining [STD](#) transmission at the population level. Another instrumental factor in sustaining gonorrhea in the population is the large number of infected individuals who are asymptomatic or have minor symptoms that are ignored. These persons, unlike symptomatic individuals, do not cease sexual activity and therefore continue to transmit the disease. This situation underscores the importance of contact tracing and empirical treatment of sex partners of index cases.

PATHOGENESIS AND IMMUNOLOGY

Outer-Membrane Proteins

Pili Fresh clinical isolates of *N. gonorrhoeae* initially form piliated (fimbriated) colonies distinguishable on translucent agar. Pilus expression is rapidly switched off with unselected subculture because of rearrangements in pilus genes. This change is a basis for phase variation of gonococci. Piliated strains adhere better to cells derived from human mucosal surfaces and are more virulent in organ culture models and human inoculation experiments than nonpiliated variants. In a fallopian tube explant model, pili mediate gonococcal attachment to nonciliated columnar epithelial cells. This event initiates gonococcal phagocytosis and transport through these cells to intercellular spaces near the basement membrane or directly into the subepithelial tissue. Damage to nearby ciliated columnar epithelial cells, which is caused by the release of cytokines, results in loss of cilia and sloughing of ciliated cells and diminishes the integrity of the fallopian tube. Nonpiliated gonococci cause epithelial damage at a much slower rate. CD46 (membrane cofactor protein) is present on urogenital epithelial cells in both men and women and has been determined to be a receptor for PilC; this subunit is located at the tip of the pilus molecule and is critical in mediating adherence. Pili are also essential for genetic competence and transformation of *N. gonorrhoeae*, which permits horizontal transfer of genetic material between different gonococcal lineages in vivo.

Opacity-associated protein Another gonococcal surface protein that is important in adherence to epithelial cells is opacity-associated protein (Opa, formerly called protein II). Opa contributes to intergonococcal adhesion, which is responsible for the opaque nature of gonococcal colonies on translucent agar and the organism's adherence to a variety of eukaryotic cells, including polymorphonuclear leukocytes (PMNs). Certain Opa variants promote invasion of epithelial cells, and this effect has been linked with the ability of Opa to bind vitronectin, glycosaminoglycans, and several members of the

carcinoembryonic antigen family (CD66). Each strain of *N. gonorrhoeae* possesses as many as 11 different *opa* genes, but usually only up to 3 types are expressed at any given time. Isolates from normally sterile sites such as the fallopian tube and synovial fluid usually fail to express Opa, while isolates from mucosal sites usually form opaque colonies. Female commercial sex workers with antibodies to Opa may be less likely to develop pelvic inflammatory disease (PID) than women without such antibodies.

Porin Porin (previously designated protein I) is the most abundant gonococcal surface protein, accounting for >50% of the organism's total outer-membrane protein. Porin molecules exist as trimers that provide anion aqueous channels through the otherwise hydrophobic outer membrane. Porin shows stable interstrain antigenic variation and forms the basis for gonococcal serotyping. Two main serotypes have been identified: Por1A strains are often associated with disseminated gonococcal infection (DGI), while Por1B strains usually cause local genital infections only. DGI strains are generally resistant to the killing action of normal human serum, do not incite a significant local inflammatory response, and therefore may not cause symptoms at genital sites. These characteristics may be related to the ability of Por1A strains to bind to complement-downregulatory molecules, resulting in a diminished inflammatory response. Porin can translocate to the cytoplasmic membrane of host cells -- a process that could initiate gonococcal endocytosis and invasion. In addition, porin is an immunologic target of bactericidal and opsonophagocytic antibodies that may arise in response to immune stimulation resulting from infection or immunization with porin-containing vaccine candidates.

Other outer-membrane proteins Other notable outer-membrane proteins include H.8, a lipoprotein that is present on the surface of all gonococcal strains in high concentration and is an excellent target for antibody-based diagnostic testing, as well as transferrin-binding proteins (Tbp1 and Tbp2) and lactoferrin-binding protein, which are required for scavenging iron from transferrin and lactoferrin in vivo. Transferrin and iron have been shown to increase attachment of iron-deprived *N. gonorrhoeae* to human endometrial cells. Gonococci deficient in transferrin- and lactoferrin-binding proteins cannot establish infection in male volunteers. IgA1 protease is produced by *N. gonorrhoeae* and may protect the organism from the action of mucosal IgA.

Lipooligosaccharide Gonococcal lipooligosaccharide (LOS) consists of a lipid A and a core oligosaccharide that lacks the repeating O-carbohydrate antigenic side chain seen in other gram-negative bacteria ([Chap. 120](#)). Gonococcal LOS possesses marked endotoxic activity and contributes to the local cytotoxic effect in the fallopian tube model. LOS core sugars undergo a high degree of antigenic variation under different conditions of growth; this variation reflects genetic regulation and expression of glycotransferase genes that dictate the carbohydrate structure of LOS. These phenotypic changes may affect interactions of *N. gonorrhoeae* with elements of the humoral immune system (antibodies and complement) and may also influence direct binding of organisms to both professional and nonprofessional phagocytes (epithelial cells). For example, gonococci that are sialylated at their LOS sites bind complement factor H and downregulate the alternative pathway of complement. LOS sialylation may also mask bactericidal antibody-binding epitopes on LOS and porin and may decrease opsonophagocytosis and inhibit the oxidative burst in [PMNs](#). While sialylation of LOS confers on the bacteria the ability to attenuate the inflammatory response and evade the innate immune system,

experiments in male volunteers suggest that sialylated gonococci may be less capable of establishing infection than their unsialylated counterparts. This difference could be explained by the observation that the unsialylated terminal lactosamine residue of LOS binds to an asialoglycoprotein receptor on epithelial cells that would otherwise facilitate binding and subsequent gonococcal invasion of these cells.

Host Factors In addition to gonococcal structures that interact with epithelial cells, host factors seem to be important in mediating entry of gonococci into nonphagocytic cells. Activation of phosphatidylcholine-specific phospholipase C and acidic sphingomyelinase by *N. gonorrhoeae*, which results in the release of diacylglycerol and ceramide, is an essential requirement for the entry of *N. gonorrhoeae* into epithelial cells. Ceramide accumulation within cells leads to apoptosis, which may disrupt epithelial integrity and facilitate entry of gonococci into subepithelial tissue. Release of chemotactic factors as a result of complement activation contributes to inflammation, as does the toxic effect of [LOS](#) in provoking the release of inflammatory cytokines.

The importance of humoral immunity in host defenses against neisserial infections is best illustrated by the predisposition of persons deficient in terminal complement components (C5 through C9) to recurrent bacteremic gonococcal infections and to recurrent meningococcal meningitis or meningococcemia. Gonococcal porin induces T cell proliferative responses in persons with urogenital gonococcal disease. A significant increase in porin-specific interleukin (IL) 4-producing CD4⁺ as well as CD8⁺ lymphocytes is seen in individuals with mucosal gonococcal disease. A portion of these lymphocytes that show a porin-specific T_H2-type response could traffic to mucosal surfaces and play a role in immune protection against the disease. Few data clearly indicate that protective immunity is acquired from a previous gonococcal infection, although bactericidal and opsonophagocytic antibodies to porin and [LOS](#) may offer partial protection. On the other hand, women who are infected and acquire high levels of antibody to another outer-membrane protein, Rmp (reduction modifiable protein, formerly called protein III), may be especially likely to become reinfected with *N. gonorrhoeae* because Rmp antibodies block the effect of bactericidal antibodies to porin and LOS. Rmp shows little, if any, interstrain antigenic variation; therefore, Rmp antibodies potentially may block antibody-mediated killing of all gonococci. The mechanism of blocking has not been fully characterized, but Rmp antibodies noncompetitively inhibit binding of porin and LOS antibodies because of the proximity of these structures in the gonococcal outer membrane. Less well understood is how blocking antibody may divert complement binding to the gonococcal surface or otherwise hasten inactivation of complement. In male volunteers who have no history of gonorrhea, the net effect of these events may influence the outcome of experimental challenge with *N. gonorrhoeae*. Because Rmp bears extensive homology to enterobacterial OmpA and meningococcal class 4 proteins, it is possible that these blocking antibodies result from prior exposure to cross-reacting proteins from these species and also play a role in first-time infection with *N. gonorrhoeae*.

CLINICAL MANIFESTATIONS

Gonococcal Infection in Males Acute urethritis is the most common clinical manifestation of gonorrhea in males. The usual incubation period following exposure is 2 to 7 days, although the interval can be longer and some men remain asymptomatic.

Strains of the Por1A serotype, with nutritional requirements for arginine, hypoxanthine, and uracil (i.e., the AHU auxotype), tend to cause a greater proportion of cases of mild and asymptomatic urethritis than Por1B strains. Urethral discharge ([Fig. 147-CD1](#)) and dysuria, usually without urinary frequency or urgency, are the major symptoms. The discharge initially is scant and mucoid but becomes profuse and purulent within a day or two. The clinical manifestations of gonococcal urethritis are usually more severe and overt than those of nongonococcal urethritis, including urethritis caused by *Chlamydia trachomatis* ([Chap. 179](#)); however, exceptions are common, and it is often impossible to differentiate the causes of urethritis on clinical grounds alone. Most symptomatic males seek treatment and cease to be infectious. The remaining men, who are largely asymptomatic, accumulate in number over time and constitute about two-thirds of all infected men at any point in time. Together with men incubating the organism (who shed the organism but are asymptomatic), they serve as the source of spread of infection. Prior to the antibiotic era, symptoms of urethritis persisted for about 8 weeks. Epididymitis is now an uncommon complication, and gonococcal prostatitis occurs rarely, if at all. Other unusual local complications of gonococcal urethritis include edema of the penis due to dorsal lymphangitis or thrombophlebitis, submucous inflammatory "soft" infiltration of the urethral wall, periurethral abscess or fistulae, inflammation or abscess of Cowper's gland, and seminal vesiculitis. Balanitis may develop in uncircumcised men. After a decline in gonococcal infections among homosexual men early in the era of AIDS, a disturbing increase in gonorrhea was observed among young homosexual men in the 1990s, probably related to decreased condom use. The clinical features of anorectal and pharyngeal gonorrhea are discussed below.

Gonococcal Infections in Females

Gonococcal cervicitis Mucopurulent cervicitis is the most common [STD](#) diagnosis in American women and may be caused by *N. gonorrhoeae*, *C. trachomatis*, and other organisms. Cervicitis may coexist with candidal or trichomonal vaginitis. *N. gonorrhoeae* primarily infects the cervical os but can also infect more peripheral areas of the cervix where columnar epithelium meets stratified squamous epithelium. Except in rare instances, the vaginal mucosa, which is lined by stratified squamous epithelium, does not become infected. Bartholin's glands occasionally become infected.

Women infected with *N. gonorrhoeae* usually develop symptoms. However, the women who either remain asymptomatic or have only minor symptoms may delay in seeking medical attention. Increased vaginal discharge and dysuria (often without urgency or frequency) are the most common symptoms. Although the incubation period of gonorrhea is less well defined in women than in men, symptoms usually develop within 10 days of infection and are more acute and intense than those of chlamydial cervicitis.

The physical examination may reveal a mucopurulent discharge (mucopus) issuing from the cervical os. The examiner may check for mucopurulent discharge by swabbing a sample of mucus from the endocervix and observing its color against the white background of the swab; yellow or green mucus suggests mucopus. However, only 35% of women with gonococcal cervicitis actually have a mucopurulent discharge defined by these criteria. Since Gram's stain is not sensitive for the diagnosis of gonorrhea in women, specimens should be submitted for culture or a nonculture assay (see below). Edematous and friable cervical ectopy as well as endocervical bleeding induced by

gentle swabbing are more often seen in chlamydial infection.

N. gonorrhoeae may be recovered from the urethra and rectum of women with cervicitis, but these are rarely the sole infected sites. Urethritis in women may produce symptoms of internal dysuria, which is often attributed to "cystitis." Pyuria in the absence of bacteriuria seen on Gram's stain of unspun urine, accompanied by urine cultures that fail to yield >10⁵ colonies of bacteria usually associated with urinary tract infection, signifies the possibility of urethritis due to *C. trachomatis*. Urethral infection with *N. gonorrhoeae* may also occur in this context, but in this instance urethral cultures will usually be positive. Compression of the urethra through the anterior vaginal wall against the symphysis pubis may express urethral exudate.

Complications of gonococcal cervicitis Gonococcal infection may extend deep enough to produce dyspareunia and lower abdominal or back pain. In such cases, it is imperative to consider a diagnosis of **PID** and to administer treatment for that disease ([Chap. 133](#)). Ascending infection of the genital tract follows ~20% of cases of gonococcal cervicitis and may result in acute endometritis accompanied by abnormal menstrual bleeding, midline lower abdominal pain and tenderness, and dyspareunia. Spread to the fallopian tubes results in acute salpingitis, whose symptoms may be accompanied by signs of cervical motion tenderness and abnormal adnexal mass on pelvic examination. Patients may be febrile, and leukocytosis and an elevated erythrocyte sedimentation rate or C-reactive protein level may be detected. Co-infection with *C. trachomatis* may increase the risk of PID, which is the clinical counterpart of endometritis and salpingitis. Tubal scarring leading to infertility is the most devastating sequela of salpingitis; the increased risk of ectopic pregnancy is also significant. Prompt and appropriate antibiotic therapy for gonococcal salpingitis (prior to the development of an adnexal mass) can prevent tubal infertility in nearly all cases. Bilateral tubal damage occurs in ~20% of women with an adnexal mass. More than half of women with tubal infertility give no history of PID. These women with "silent salpingitis" may report abdominal or pelvic discomfort (such as dysmenorrhea or dyspareunia) that may be attributed to other diagnoses (such as endometriosis). Spread of infection to the pelvis may result in pelvic peritonitis characterized by nausea and vomiting. Spread of gonococci -- or, more commonly, of chlamydiae -- via the peritoneal cavity to the upper abdomen may cause perihepatitis (Fitz-Hugh-Curtis syndrome; [Chap. 133](#)).

Gonococcal vaginitis The vaginal mucosa of healthy women is lined by stratified squamous epithelium and is usually not infected by *N. gonorrhoeae*. However, gonococcal vaginitis can occur in an estrogenic women (e.g., prepubertal girls and postmenopausal women), in whom the vaginal stratified squamous epithelial layers are often thinned down to the basilar layer, which can be infected by *N. gonorrhoeae*. The intense inflammation of the vagina makes the physical (speculum and bimanual) examination extremely painful. The vaginal mucosa is red and edematous, and an abundant purulent discharge is present. Infection in the urethra and in Skene's and Bartholin's glands often accompanies gonococcal vaginitis. Inflamed cervical erosion or abscesses in nabothian cysts may also occur. Coexisting cervicitis may result in pus in the cervical os.

Anorectal Gonorrhea Because the female anatomy permits the spread of cervical exudate to the rectum, *N. gonorrhoeae* is sometimes recovered from the rectum of

women with uncomplicated gonococcal cervicitis. The rectum is the sole site of infection in only 5% of women with gonorrhea. Such women are usually asymptomatic but occasionally have acute proctitis manifested by anorectal pain or pruritus, tenesmus, purulent rectal discharge, and rectal bleeding. Among homosexual men, the frequency of gonococcal infection, including rectal infection, fell by³90% throughout the United States in the early 1980s, but a resurgence of gonorrhea among homosexual men was documented in several cities during the 1990s. Gonococcal isolates from the rectum of homosexual men tend to be more resistant than other gonococcal isolates to antimicrobials. Gonococci with multidrug resistance (*mtr*) are more resistant to bile salts and fatty acids in feces and thus are found with increased frequency in homosexual men. The *mtr* mutation involves a DNA-binding protein and results in the derepression of genes encoding an efflux mechanism of resistance. This situation may have been responsible for higher rates of treatment failure for rectal gonorrhea with older regimens consisting of penicillin or tetracyclines.

Pharyngeal Gonorrhea Pharyngeal gonorrhea is usually mild or asymptomatic, although symptomatic pharyngitis does occasionally occur with cervical lymphadenitis. The mode of acquisition is oral-genital sexual exposure, with fellatio being a more efficient means of transmission than cunnilingus. Most cases resolve spontaneously, and transmission from the pharynx to sexual contacts is rare. Pharyngeal infection almost always coexists with genital infection. Swabs from the pharynx should be plated directly onto gonococcal selective media. Because pharyngeal colonization with *N. meningitidis* needs to be differentiated from that with other *Neisseria* species, the diagnosis of pharyngeal gonorrhea is more expensive and difficult than that of anogenital gonorrhea.

Ocular Gonorrhea in Adults Ocular gonorrhea in an adult usually results from autoinoculation from an infected genital site. As in genital infection, the manifestations range from severe to occasionally mild or asymptomatic disease. The variability in clinical manifestations may result from differences in the ability of the infecting strain to elicit an inflammatory response.

Infection may result in a markedly swollen eyelid, severe hyperemia and chemosis, and a profuse purulent discharge ([Fig. 147-CD2](#)). The massively inflamed conjunctiva may be draped over the cornea and limbus. Lytic enzymes from the infiltrating [PMNs](#) occasionally cause corneal ulceration and rarely cause perforation.

Prompt recognition and treatment of this condition are of paramount importance. Gram's stain and culture of the purulent discharge establish the diagnosis. Genital cultures should also be performed.

Gonorrhea in Pregnant Women, Neonates, and Children Gonorrhea in pregnancy can have serious consequences for both the mother and the infant. Therefore, early detection and eradication of the disease in the mother are extremely important. Recognition of gonorrhea early in pregnancy also identifies a population at risk for other [STDs](#), particularly *Chlamydia* infection and syphilis. These women should be monitored closely for these infections throughout pregnancy. The incidence of gonorrhea in pregnancy ranges from rare to ~10%, depending upon the population surveyed. Salpingitis and [PID](#) can occur during the first trimester and are associated with

a high rate of fetal loss. In the second and third trimesters, the relative impermeability of the cervical mucus (under the influence of progesterone) and the obliteration of the intrauterine cavity (resulting from the attachment of the chorion to the endometrial decidua by around the twelfth week of gestation) pose physical barriers that usually prevent ascending infection. Pharyngeal infection, most often asymptomatic, may be more common during pregnancy because of altered sexual practices. Acquisition of gonococcal infection late in pregnancy can adversely affect labor and delivery as well as the well-being of the fetus. Prolonged rupture of the membranes, premature delivery, chorioamnionitis, funisitis (infection of the umbilical cord stump), and sepsis in the infant (with *N. gonorrhoeae* detected in the gastric aspirate of the newborn during delivery) are common complications of maternal gonococcal infection at term. Hazards to the fetus include spontaneous abortion, perinatal death, premature delivery, perinatal distress, and premature rupture of membranes. Other microorganisms and conditions, including *Mycoplasma hominis*, *Ureaplasma urealyticum*, *C. trachomatis*, and bacterial vaginosis, have been associated with similar complications.

The most common form of gonorrhea in neonates is *ophthalmia neonatorum*, which results from exposure to infected cervical secretions during parturition. Ocular neonatal instillation of a prophylactic agent (e.g., 1% silver nitrate eyedrops or ophthalmic preparations containing erythromycin or tetracycline) is a cost-effective measure for the prevention of ophthalmia neonatorum but is not effective for its treatment, which requires systemic antibiotics. The clinical manifestations are acute and begin 2 to 5 days after birth. A small inoculum of organisms, low virulence of the infecting strain, or partial suppression by ophthalmic prophylaxis can result in a more indolent course. Therefore, gonococcal infection must be ruled out by culture in every case of conjunctivitis in infants. An initial nonspecific conjunctivitis with a serosanguineous discharge is followed by tense edema of both eyelids, chemosis, and a profuse, thick, purulent discharge. Corneal ulcerations that result in nebulae or perforation may lead to anterior synechiae, anterior staphyloma, panophthalmitis, and blindness. Infections described at other mucosal sites in infants, including vaginitis, rhinitis, and anorectal infection, are likely to be asymptomatic. Pharyngeal colonization has been demonstrated in 35% of infants with gonococcal ophthalmia, and coughing is the most prominent symptom in these cases. Septic arthritis is the most common manifestation of systemic gonococcal infection in the newborn. The primary focus of [DGI](#) in most of these cases is uncertain. The onset usually comes at 3 to 21 days of age, and polyarticular involvement is common. Sepsis, meningitis, and pneumonia are seen in rare instances.

Any [STD](#) in children beyond the neonatal period raises the possibility of sexual abuse. In most cases of abuse, the perpetrator is a male assailant known to the child. Gonococcal vulvovaginitis is the most common manifestation of gonococcal infection in children beyond infancy. Anorectal and pharyngeal infections are common in these children and are frequently asymptomatic. The urethra, Bartholin's and Skene's glands, and the upper genital tract are rarely involved. All children with gonococcal infection should also be evaluated for *Chlamydia* infection, syphilis, and possibly HIV infection. All cases of suspected and confirmed child abuse should be reported to the appropriate social service agency in the county where the child resides.

Disseminated Gonococcal Infection [DGI](#) results from gonococcal bacteremia. In the 1970s, DGI occurred in ~0.5% to 3% of persons with untreated gonococcal mucosal

infection. The lower incidence at present is probably attributable to a decline in the prevalence of particular strains that are likely to disseminate. DGI strains resist the bactericidal action of human serum and generally do not incite inflammation at genital sites, probably because of limited generation of chemotactic factors. These strains are often of the Por1A serotype, are highly susceptible to penicillin, and have special nutritional requirements (i.e., the AHU auxotype). Menstruation is a risk factor for dissemination, and approximately two-thirds of cases of DGI are in women. In about half of affected women, symptoms of DGI begin within 7 days of onset of menses. Complement deficiencies, especially of the components involved in the assembly of the membrane attack complex (C5 through C9), predispose to neisserial bacteremia. Up to 13% of patients with DGI have complement deficiencies, and persons with more than one episode of DGI should be screened with an assay for total hemolytic complement activity.

The clinical manifestations of [DGI](#) have sometimes been classified into two stages: a bacteremic stage and a joint-localized stage with suppurative arthritis. A clear-cut progression usually is not evident. Patients in the bacteremic stage have higher temperatures, and their fever is more frequently accompanied by chills. Painful joints are common and often occur in conjunction with tenosynovitis and skin lesions.

Polyarthralgias usually include the knees, elbows, and more distal joints; the axial skeleton is generally spared. Skin lesions are seen in ~75% of patients and include papules and pustules, often with a hemorrhagic component (see [Plate IID-60; Fig. 147-CD3](#)). These lesions are usually on the extremities and number between 5 and 40. Frank arthritis, when it develops, involves one or two joints, most often (in decreasing order of frequency) the knees, wrists, ankles, and elbows. The occurrence of arthritis in the absence of signs and symptoms of the bacteremic stage has led to the suggestion that these are separate syndromes. Other joints, such as the small joints of the hands and feet and the sternoclavicular and temporomandibular joints, are occasionally involved. Most patients who develop gonococcal septic arthritis do so without prior polyarthralgias or skin lesions; in the absence of symptomatic genital infection, this disease cannot be distinguished from septic arthritis caused by other pathogens. Rarely, osteomyelitis complicates septic arthritis involving small joints.

Although it has been postulated that the initial arthritis and skin lesions are due to direct tissue invasion by *N. gonorrhoeae*, the organism has been recovered from fewer than 5% of skin lesions cultured. This low isolation rate has been attributed to either a small inoculum of infecting organisms or the fastidious growth requirements of *N. gonorrhoeae* strains that disseminate. Gonococcal antigens have been identified in "sterile" skin lesions by immunofluorescent staining techniques. There is also evidence that immune-mediated or hypersensitivity phenomena caused by gonococcal antigens account for skin lesions. Other manifestations of noninfectious dermatitis, such as nodular lesions, urticaria, and erythema multiforme, have been described. Gonococcal endocarditis, although rare today, was relatively common in the preantibiotic era, causing about one-quarter of reported cases of endocarditis. Another unusual complication of [DGI](#) is meningitis.

Gonococcal Infection in HIV-Infected Persons The association between gonorrhea and the acquisition of HIV has been demonstrated in several well-controlled studies,

mainly in Kenya and Zaire. The nonulcerative [STDs](#) enhance the transmission of HIV by three- to fivefold, possibly because of increased viral shedding in persons with urethritis or cervicitis ([Chap. 309](#)). HIV has been detected by polymerase chain reaction (PCR) more commonly in ejaculates from HIV-positive men with gonococcal urethritis than in those from HIV-positive men with nongonococcal urethritis. PCR positivity diminishes by twofold following appropriate therapy for urethritis. Not only does gonorrhea enhance the transmission of HIV; it may also increase the individual's risk for acquisition of HIV. A proposed mechanism is the significantly greater number of CD4⁺ lymphocytes and dendritic cells that can be infected by HIV in endocervical secretions of women with nonulcerative STDs than in those of women with ulcerative STDs.

DIFFERENTIAL DIAGNOSIS

The clinical features of uncomplicated gonococcal infections closely resemble those of genital infections caused by *C. trachomatis*. Although the symptoms produced by chlamydial infections tend to be milder, the two infections are often indistinguishable on clinical grounds alone. Co-infection with *N. gonorrhoeae* and *C. trachomatis* is seen in up to 40% of cases. The differential diagnosis of urethritis, epididymitis, and proctitis in men, of cervicitis in women, and of vaginitis in prepubertal girls is discussed in [Chap. 132](#); that of PID in [Chap. 133](#); and that of acute arthritis in young adults in [Chap. 323](#). The differential diagnosis of the bacteremic stage of [DGI](#) includes acute rheumatoid arthritis, sarcoidosis, erythema nodosum, drug-induced arthritis, and viral infections (e.g., hepatitis B and acute HIV infection).

LABORATORY DIAGNOSIS

A rapid diagnosis of gonococcal infection in men may be obtained by Gram's staining of urethral exudates ([Fig. 147-CD4](#)). The detection of gram-negative intracellular diplococci (GNID) is usually highly specific and sensitive in diagnosing gonococcal urethritis in symptomatic males but is only ~50% sensitive in diagnosing gonococcal cervicitis. Samples should be collected with Dacron or rayon swabs. Part of the sample should be inoculated onto a plate of modified Thayer-Martin or other gonococcal selective medium for culture. It is important to process all samples immediately because gonococci do not tolerate drying. If plates cannot be incubated immediately, they can be held safely for several hours at room temperature in candle extinction jars prior to incubation. If processing is to occur within 6 h, transport of specimens may be facilitated by the use of nonnutritive swab transport systems such as Stuart or Amies medium. For longer holding periods (e.g., when specimens for culture are to be mailed), culture media with self-contained CO₂-generating systems (such as the JEMBEC or Gono-Pak systems) may be used. Specimens should also be obtained for the diagnosis of chlamydial infection.

[PMNs](#) are often seen in the endocervix on a Gram's stain, and an abnormally increased number (³30 PMNs per field in five 1000 $\times$ oil-immersion fields) establishes the presence of an inflammatory discharge (mucopurulent cervicitis). Unfortunately, the presence or absence of [GNID](#) in cervical smears does not accurately predict which patients have gonorrhea, and the diagnosis in this setting should be made by culture. The sensitivity of a single endocervical culture is ~80 to 90%, with the precise figure depending on the quality of the medium and the adequacy of the clinical specimen. The yield can be

enhanced by culture of a second cervical specimen. If a history of rectal sex is elicited, a rectal wall swab (uncontaminated with feces) should be cultured. A presumptive diagnosis of gonorrhea cannot be made on the basis of gram-negative diplococci in smears from the pharynx, where other *Neisseria* species are components of the normal flora.

Nucleic acid probe tests are now widely used for the direct detection of *N. gonorrhoeae* in urogenital specimens. A common assay employs a nonisotopic chemiluminescent DNA probe that hybridizes specifically with gonococcal 16S ribosomal RNA. Studies assessing the utility of the nucleic acid probe system in high-risk outpatients undergoing screening for [STDs](#) have revealed that it is at least as sensitive as conventional culture techniques and may be a cost-effective alternative to culture, especially in high-risk males. A disadvantage of non-culture-based assays in general is that specimens submitted in probe-transport systems cannot be cultured subsequently. Therefore, a culture-confirmatory test is not possible, and formal antimicrobial susceptibility testing, if needed, cannot be performed. Low-cost point-of-care tests are under development for use in resource-poor settings, where specific diagnosis often gives way to syndromic management. DNA amplification techniques such as [PCR](#) may eventually prove to be equivalent to or more sensitive than culture methods.

Because of the legal implications, gonococcal infection in children must be diagnosed only with standard culture systems. Nonculture tests for gonococcal infection should not be used alone and have not been approved by the U.S. Food and Drug Administration for use with specimens obtained from the genital tract, pharynx, and rectum of infected children. Cultures should be obtained from the pharynx and anus of both girls and boys, the vagina of girls, and the urethra of boys. Cervical specimens are not recommended for prepubertal girls. For boys with a urethral discharge, a meatal specimen of the discharge is adequate for culture. Presumptive colonies of *N. gonorrhoeae* should be identified definitively by at least two independent methods (e.g., biochemical, enzyme substrate, or serologic).

Blood should be cultured in suspected cases of [DGI](#). The use of Isolator blood culture tubes may enhance the yield. The probability of positive blood cultures decreases after 48 h of illness. Synovial fluid should be inoculated into blood culture broth medium and plated onto chocolate agar rather than selective medium because this fluid is not likely to be contaminated with commensal bacteria. Gonococci are infrequently recovered from early joint effusions containing <20,000 leukocytes/uL but may be recovered from effusions containing >80,000 leukocytes/uL. The organisms are seldom recovered from blood and synovial fluid of the same patient.

TREATMENT

It is no surprise that *N. gonorrhoeae*, with its remarkable capacity to alter its antigenic structure and adapt to changes in the microenvironment, has become resistant to numerous antibiotics. The first effective agents against gonorrhea were the sulfonamides, which were introduced in the 1930s. Within a decade, antibiotic resistance emerged, resulting in treatment failures in one-third of patients. Penicillin was then employed as the drug of choice for the treatment of gonorrhea. By 1965, 42% of gonococcal isolates had developed low-level resistance to penicillin G. To prevent

treatment failures, the Centers for Disease Control and Prevention (CDC) at that time recommended doubling the dose of penicillin for the treatment of gonorrhea. Resistance due to the production of penicillinase arose later.

Gonococci become fully resistant to antibiotics either by chromosomal mutations or by acquisition of R factors (plasmids). Two types of chromosomal mutations have been described. The first type, which is drug specific, is a single-step mutation leading to high-level resistance. The second type involves mutations at several chromosomal loci that combine to determine the level as well as the pattern of resistance. Strains with mutations in chromosomal genes were first observed in the late 1950s. As recently as 1997, strains with chromosomal resistance (CMRNG) accounted for resistance to penicillin, tetracycline, or both in ~20% of strains surveyed in the United States.

b-Lactamase (penicillinase)-producing strains of *N. gonorrhoeae* (PPNG) carrying plasmids with the *Pc* determinant were seen almost simultaneously in the United States, England, western Africa, and the Philippines in the late 1970s. PPNG strains have since spread worldwide and by the early 1980s accounted for >50% of all gonococcal isolates in some parts of the developing world. The average prevalence of PPNG in the United States dropped by two-thirds after most penicillin use was discontinued and is now on the order of 4%, with higher rates reported from certain areas. *N. gonorrhoeae* strains with plasmid-borne tetracycline resistance (TRNG) can mobilize some b-lactamase plasmids, and PPNG and TRNG occur together, sometimes along with CMRNG. Penicillin, ampicillin, and tetracycline are no longer reliable agents for the treatment of gonorrhea and should not be used. Third-generation cephalosporins have remained highly effective as single-dose therapy for gonorrhea. Even though the minimal inhibitory concentrations (MICs) of ceftriaxone for certain strains may reach 0.015 to 0.125 mg/L [higher than MICs for fully susceptible strains (0.0001 to 0.008 mg/L)], these levels are greatly exceeded in blood, the urethra, and the cervix when the routinely recommended ceftriaxone and cefixime regimens are administered (see below). These regimens almost always result in an effective cure.

Quinolone-containing regimens are also recommended for treatment of gonococcal infections; the fluoroquinolones offer the advantage of antichlamydial activity when administered for 7 days. Serum concentrations following therapeutic dosages of the quinolones exceed the MIC for *N. gonorrhoeae* by ~100-fold. However, quinolone-resistant *N. gonorrhoeae* (QRNG) appeared soon after these agents were first used to treat gonorrhea, particularly in Southeast Asia. QRNG strains have been reported recently in the United States, mostly in the far western states. Alterations in DNA gyrase and topoisomerase IV have been implicated as mechanisms of fluoroquinolone resistance.

Resistance to spectinomycin, which is used as an alternative agent, has been reported, but resistance to this agent is usually not associated with resistance to other antibiotics. Therefore, spectinomycin can be reserved for use against multiresistant strains of *N. gonorrhoeae*. Nevertheless, outbreaks caused by strains resistant to spectinomycin have been documented in Korea and England when the drug was used as a primary agent to treat gonorrhea.

Although clinical isolates of *N. gonorrhoeae* vary in their antimicrobial susceptibility

patterns in different parts of the world, they remain susceptible to a wide variety of agents. Because failure of treatment can lead to continued transmission and the emergence of antibiotic resistance, the importance of adequate treatment with a regimen that the patient will adhere to cannot be overemphasized. Thus highly effective single-dose regimens have been developed for the treatment of uncomplicated gonococcal infections. The 1998 [CDC](#) treatment guidelines for gonococcal infections are summarized in [Table 147-1](#); the recommendations for uncomplicated gonorrhea apply to HIV-infected as well as HIV-uninfected patients. The third-generation cephalosporins cefixime and ceftriaxone are the mainstay of therapy for uncomplicated gonococcal infection of the urethra, cervix, rectum, or pharynx. Single doses of ciprofloxacin or ofloxacin are also effective first-line regimens. Because of resistance to fluoroquinolones in several parts of Southeast Asia, these agents can no longer be considered effective in that region. Because co-infection with *C. trachomatis* occurs frequently, initial treatment regimens must incorporate an agent (e.g., azithromycin or doxycycline) effective against chlamydial infection. Pregnant women with gonorrhea should receive concurrent treatment with a macrolide antibiotic for possible *Chlamydia* infection; doxycycline should not be used during pregnancy. A single 1-g dose of azithromycin, which is effective therapy for uncomplicated chlamydial infections, results in an unacceptably low cure rate (93%) for gonococcal infections and should not be used alone. Uncomplicated gonococcal infections in penicillin-allergic persons who cannot tolerate quinolones may be treated with a single dose of spectinomycin. Persons with uncomplicated infections who receive a recommended regimen need not return for a test of cure. Cultures for *N. gonorrhoeae* should be performed if symptoms persist after therapy with an established regimen, and any gonococci isolated should be tested for antimicrobial susceptibility.

Symptomatic gonococcal pharyngitis is more difficult to eradicate than genital infection. Few regimens result in cure rates of >90%. Persons who cannot tolerate cephalosporins or quinolones can be treated with spectinomycin, but this agent results in a cure rate of 52%. Therefore, persons given spectinomycin should have a pharyngeal culture performed 3 to 5 days after treatment as a test of cure.

Treatments for gonococcal epididymitis and PID are discussed in [Chaps. 132](#) and [133](#), respectively. Ocular gonococcal infections in older children and adults should be managed with a single dose of ceftriaxone combined with saline irrigation of the conjunctivae (both undertaken expeditiously), and patients should undergo a careful ophthalmologic evaluation that includes a slit-lamp examination.

[DGI](#) may require higher dosages and longer durations of therapy ([Table 147-1](#)). Hospitalization is indicated if the diagnosis is uncertain, if the patient has localized joint disease that requires aspiration, or if the patient cannot be relied on to comply with treatment. Open drainage is necessary only occasionally, e.g., for management of hip infections that may be difficult to drain percutaneously. Nonsteroidal anti-inflammatory agents may be indicated to alleviate pain and hasten improvement of affected joints. Gonococcal meningitis and endocarditis should be treated in the hospital with high-dose intravenous ceftriaxone (1 to 2 g every 12 h); therapy should continue for 10 to 14 days for meningitis and for at least 4 weeks for endocarditis. All persons who experience more than one episode of DGI should be evaluated for complement deficiency.

PREVENTION AND CONTROL

Condoms, if properly used, provide effective protection against the transmission and acquisition of gonorrhea as well as other infections that are transmitted to and from genital mucosal surfaces. Spermicidal preparations used with a diaphragm or cervical sponges impregnated with nonoxynol 9 offer some protection against gonorrhea and chlamydial infection. However, the frequent use of preparations that contain nonoxynol 9 is associated with mucosal disruption that paradoxically may enhance the risk of HIV infection in the event of exposure. All patients should be instructed to refer sex partners for evaluation and treatment. All sex partners of persons with gonorrhea should be evaluated and treated for *N. gonorrhoeae* and *C. trachomatis* infections if their last contact with the patient took place within 60 days before the onset of symptoms or the diagnosis of infection in the patient. If the patient's last sexual encounter was >60 days before onset of symptoms or diagnosis, the patient's most recent sex partner should be treated. Patients should be instructed to abstain from sexual intercourse until therapy is completed and until they and their sex partners no longer have symptoms. Greater emphasis must be placed on prevention by public health education, individual patient counseling, and behavior modification. Preventing the spread of gonorrhea may help reduce the transmission of HIV. No effective vaccine for gonorrhea is yet available, but efforts to test a porin vaccine candidate are under way.

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148. MORAXELLA CATARRHALIS AND OTHER MORAXELLA SPECIES - Daniel M. Musher

MORAXELLA CATARRHALIS

The gram-negative coccus now known as *Moraxella catarrhalis* has undergone three changes of name in as many decades. Originally called *Micrococcus catarrhalis*, it was renamed *Neisseria catarrhalis* in the 1960s because of its morphologic similarity to *Neisseria* spp. Then, in 1970, it was elevated to the status of a distinct genus, *Branhamella*, on the basis of DNA homology. In 1979 this organism was placed into the genus *Moraxella*, of which *Branhamella* may be a subgenus. A component of the normal bacterial flora of the upper airways, *M. catarrhalis* has been increasingly recognized as a cause of otitis media, sinusitis, and bronchopulmonary infection.

BACTERIOLOGY AND IMMUNITY

On Gram's staining, *M. catarrhalis* organisms appear as gram-negative cocci, sometimes occurring in pairs and retaining the side-by-side kidney-bean configuration of *Neisseria* ([Plate VI-1](#)). These cocci tend to retain crystal violet during the decolorizing step and may be confused with *Staphylococcus aureus*. *Moraxella* colonies grow well on blood or chocolate agar but may be overlooked because of their resemblance to *Neisseria* spp. (a major component of the normal pharyngeal flora). *Moraxella* is readily distinguishable from *Neisseria* spp. by biochemical tests.

Strains of *M. catarrhalis* show a surprising degree of homogeneity in terms of their outer-membrane proteins. Antibody to some of these proteins is generally present in serum of children over the age of 4 years; however, colonizing or disease-causing isolates may survive in serum despite this naturally present antibody and complement. Bactericidal antibody emerges following natural infection and may be directed against one or more conserved outer-membrane proteins -- a property of potential value in vaccine development. The presence of certain outer-membrane proteins is associated with virulence in mice, and antibody may be protective. These proteins are under investigation for use as vaccines.

EPIDEMIOLOGY

With repeated cultures and the use of selective media, *M. catarrhalis* can be isolated from the upper respiratory tract or saliva of 50% of healthy schoolchildren and of up to 7% of healthy adults. When conventional microbiologic techniques are used, *Moraxella* can be isolated from sputum of about 10% of persons who have chronic bronchitis and 25% of those who have bronchiectasis in the absence of acute infection. Investigators in both the northern and southern hemispheres have reported a striking seasonal variation in the isolation of this organism from clinical specimens, with a peak in late winter/early spring and a nadir in late summer/early fall. Direct contact has not been shown to contribute to community-acquired infection, but nosocomial spread of infection has been documented occasionally.

OTITIS MEDIA AND SINUSITIS

M. catarrhalis has repeatedly been shown to be the third most common bacterial isolate from middle-ear fluid of children who have otitis media, being surpassed only by *Streptococcus pneumoniae* and nontypable *Haemophilus influenzae*. Recent studies have shown that this organism is also a prominent isolate from sinus cavities in acute and chronic sinusitis.

PURULENT TRACHEOBRONCHITIS AND PNEUMONIA

M. catarrhalis causes acute exacerbations of chronic bronchitis (increased production and/or purulence of sputum), purulent tracheobronchitis (the latter also involving fever and leukocytosis), and pneumonia. The great majority of infected persons are >50 years old and have a long history of cigarette smoking and underlying chronic obstructive pulmonary disease (COPD); many have lung cancer as well. In one study, 76% of affected persons had COPD (severe in many cases), and one-third of those with COPD had lung cancer; most patients also had clinical evidence of malnutrition. In one extensive series of cases, *M. catarrhalis* pneumonia did not occur in otherwise-healthy hosts.

Symptoms of *M. catarrhalis* infection have been regarded as modest in severity. Both cough and the amount and purulence of sputum are usually increased above baseline. Chills are reported in one-quarter of patients, pleuritic pain in one-third, and malaise in 40%. Most patients have peak temperatures of <38.3°C (<101°F), and peripheral white blood cell counts are <10,000/uL in nearly one-quarter of cases. Microscopic examination of a good sputum specimen following Gram's staining regularly reveals profuse organisms, and quantitative culture yields ~ 2 × 10⁸ colony-forming units per milliliter ([Plate VI-1](#)). The radiologic appearance is variable; in one study, 43% of subjects had segmental or lobar infiltrates, and the remainder had a mixed pattern of subsegmental, segmental, interstitial, and diffuse involvement. These clinical, laboratory, and radiographic findings do not differ from those of pneumococcal or *Haemophilus* pneumonia in an older patient population. However, a far lesser degree of bloodstream invasion occurs in *M. catarrhalis* infection; in one series, none of 25 patients with *M. catarrhalis* pneumonia had bacteremia. Nevertheless, pneumonia due to *M. catarrhalis* is a marker for severe underlying disease: nearly half of patients die within 3 months of onset.

OTHER SYNDROMES

Local extension causing empyema is very uncommon, and, as might be inferred from the low rate of bacteremia, metastatic complications of *M. catarrhalis* pneumonia, such as septic arthritis, are exceedingly rare. As of 1995, 58 cases of bacteremic infection due to *M. catarrhalis* had been reported, mainly in children <10 years old or adults >60 years old; most of these patients were immunocompromised. The syndromes reported have included bacteremia with no apparent focus, pneumonia, endocarditis, and meningitis. A petechial or purpuric rash, reminiscent of that observed in meningococcal sepsis and associated with disseminated intravascular coagulation, has been described in a few cases.

TREATMENT

Treatment of *M. catarrhalis* infection with a penicillin/clavulanic acid combination seems highly appropriate. Penicillin resistance first appeared in *Branhamella* isolates in the mid-1970s and is now found in 85% of clinical isolates. Resistance is mediated by two closely related β -lactamases, BRO-1 and BRO-2, which are present in 90% and 10% of resistant isolates, respectively. These enzymes are active against penicillin, ampicillin, and amoxicillin but less so against cephalosporins, especially third-generation cephalosporins, and they bind avidly to clavulanic acid and sulbactam.

Cephalosporins, especially those of the second and third generations, are effective alternatives. Isolates in the United States are also nearly uniformly susceptible to tetracycline, erythromycin, trimethoprim-sulfamethoxazole, quinolones, and chloramphenicol, although tetracycline resistance -- perhaps due to TetB determinants -- is increasing in Europe and Asia and has been documented in the United States. A 5-day course of therapy has been shown to cure respiratory infection, although a slightly longer course may be required in sinusitis.

During the period between the identification of gram-negative cocci in a Gram-stained specimen and the final identification of the organisms by culture, the severity of the condition and the potential presence of other infecting organisms should guide antibiotic selection. For example, an exacerbation of bronchitis caused by *M. catarrhalis* might be treated with tetracycline or trimethoprim-sulfamethoxazole; however, in a patient with pneumonia, the possibility that pneumococci resistant to these agents also might be present dictates the choice of ampicillin/sulbactam or a third-generation cephalosporin, at least until culture results become available.

OTHER MORAXELLA SPECIES

Other *Moraxella* species cause a wide range of infections, including bronchitis, pneumonia, empyema, endocarditis, meningitis, conjunctivitis, urinary tract infection, septic arthritis, and wound infection. In a report on all *Moraxella* isolates submitted to the Centers for Disease Control and Prevention between 1953 and 1980, certain clinical associations were apparent ([Table 148-1](#)). *M. osloensis* and *M. nonliquefaciens*, the most commonly isolated species, were cultured from a wide range of normally sterile body sites, including blood, cerebrospinal fluid, and joints. *M. osloensis* was the *Moraxella* species most frequently isolated from blood; *M. nonliquefaciens* tended to be isolated from the ears, nose, or throat (47%) or the sputum (8%) and has since been implicated as a cause of conjunctivitis and keratitis. *M. urethralis* was isolated most often from urine and the genital tract and probably represents the *Moraxella* species implicated previously in urethritis. More than half of isolates of *M. phenylpyruvica* and *M. atlantae* were obtained from normally sterile sites. A recent study found *Moraxella* spp., including *M. catarrhalis*, in 35% of infected wounds following cat bites and in 10% of those following dog bites. The clinical features of infections due to *Moraxella* spp. other than *M. catarrhalis* and the nature of the hosts in which they occur have not been fully characterized.

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149. HAEMOPHILUS INFECTIONS - Timothy F. Murphy

HAEMOPHILUS INFLUENZAE

MICROBIOLOGY

Haemophilus influenzae was first recognized in 1892 by Pfeiffer, who erroneously concluded that the bacterium was the cause of influenza. The bacterium is a small (1- by 0.3-um) gram-negative organism of variable shape; hence, it is often described as a pleomorphic coccobacillus. In clinical specimens such as cerebrospinal fluid (CSF) and sputum, it frequently stains only faintly with phenosafranin and therefore can easily be overlooked.

H. influenzae grows both aerobically and anaerobically. Its aerobic growth requires two factors: hemin (X factor) and nicotinamide adenine dinucleotide (V factor). These requirements are used in the clinical laboratory to identify the bacterium. Six major serotypes of *H. influenzae* have been identified; designated a through f, they are based on antigenically distinct polysaccharide capsules. In addition, some strains lack a polysaccharide capsule and are referred to as *nontypable* strains. Type b and nontypable strains are the most relevant strains clinically, although encapsulated strains other than type b can cause disease. *H. influenzae* was the first free-living organism to have its entire genome sequenced.

The antigenically distinct type b capsule is a linear polymer composed of ribosyl-ribitol phosphate. Strains of *H. influenzae* type b (Hib) cause disease primarily in infants and children under the age of 6 years. Nontypable strains are primarily mucosal pathogens, although the incidence of invasive disease caused by these strains is increasing.

EPIDEMIOLOGY AND TRANSMISSION

H. influenzae is an exclusively human pathogen. The organism is spread by airborne droplets or by direct contact with secretions or fomites. Nontypable strains colonize the upper respiratory tract of up to three-fourths of healthy adults. Colonization with nontypable *H. influenzae* is a dynamic process; new strains are acquired and other strains are replaced periodically.

[Hib](#) strains colonize the nasopharynx of children at a rate of 3 to 5%; before the introduction of type b vaccine, higher rates were seen in day-care centers. The widespread use of conjugate vaccines has resulted in a striking decrease not only in the rate of nasopharyngeal colonization by Hib but also in the incidence of meningitis due to Hib. Studies in selected populations suggest the reemergence of invasive Hib infections and higher than expected rates of nasopharyngeal colonization by Hib in vaccinated children. Continued surveillance of Hib disease and colonization rates will be important in evaluating the success of current vaccination strategies.

Certain population groups have a higher incidence of invasive [Hib](#) disease than the general population. The incidence of meningitis due to Hib has been three to four times higher among black children than among white children in several studies. In some Native American groups, the incidence of invasive Hib disease is 10 times higher than

that in the general population. Although this increased incidence has not yet been accounted for, several factors may be relevant, including age at exposure to the bacterium, socioeconomic conditions, and genetic differences in the ability to mount an immune response.

PATHOGENESIS

[Hib](#) strains cause systemic disease by invasion and hematogenous spread to distant sites such as the meninges, bones, and joints. The type b polysaccharide capsule is an important virulence factor affecting the bacterium's ability to avoid opsonization and cause systemic disease.

Nontypable strains cause disease by local invasion of mucosal surfaces. Otitis media results when bacteria reach the middle ear by way of the eustachian tube. Adults with chronic bronchitis experience recurrent lower respiratory tract infection due to nontypable strains. The incidence of invasive disease caused by nontypable strains is low but increasing.

IMMUNE RESPONSE

Antibody to capsule is important in protection from infection by [Hib](#) strains. The level of (maternally acquired) serum antibody to the capsular polysaccharide, which is a polymer of polyribitol ribose phosphate (PRP), declines from birth to 6 months of age and, in the absence of vaccination, remains low until around 2 or 3 years of age. The age at the antibody nadir correlates with that of the peak incidence of type b disease. Antibody to PRP then appears partly as a result of exposure to Hib or cross-reacting antigens. Systemic Hib disease is unusual after the age of 6 years because of the presence of protective antibody. Vaccines in which PRP is conjugated to protein carrier molecules have been developed and are now used widely. These vaccines generate an antibody response to PRP in infants and are effective in preventing invasive infections in infants and children.

Since nontypable strains lack a capsule, the immune response to infection is directed at noncapsular antigens. These noncapsular antigens of *H. influenzae* have generated considerable interest as targets of the human immune response and as potential vaccine components.

CLINICAL MANIFESTATIONS

Hib The most serious manifestation of infection with [Hib](#) is meningitis. The age of peak incidence varies somewhat among populations, depending in part on the use of vaccine, but this infection primarily affects infants under 2 years of age. The clinical manifestations of meningitis caused by Hib are similar to those of meningitis caused by other bacterial pathogens. Fever and altered central nervous system function are the most common features at presentation. Nuchal rigidity may or may not be evident. Subdural effusion, the most common complication, is suspected when, despite 2 or 3 days of appropriate antibiotic therapy, the infant has seizures, hemiparesis, or continued obtundation. The overall mortality from meningitis caused by Hib is approximately 5%, and the rate of morbidity is high. Of survivors, 6% have permanent sensorineural

hearing loss, and about one-fourth have a significant handicap of some type. If more subtle handicaps are sought, up to half of survivors are found to have some neurologic sequelae, such as partial hearing loss and delay in language development.

Epiglottitis is a life-threatening infection involving cellulitis of the epiglottis and supraglottic tissues. It can lead to acute upper airway obstruction. Its unique epidemiologic features are its occurrence in an older age group (2 to 7 years old) than other [Hib](#) infections and its absence among Navajo Indians and Alaskan Eskimos. Sore throat and fever rapidly progress to dysphagia, drooling, and airway obstruction.

Cellulitis due to [Hib](#) occurs in young children. The most common location is on the head or neck, and the involved area sometimes takes on a characteristic bluish-red color. Most patients have bacteremia, and 10% have an additional focus of infection.

[Hib](#) causes *pneumonia* in infants. The infection is clinically indistinguishable from other types of bacterial pneumonia (e.g., pneumococcal pneumonia) except that Hib is more likely to involve the pleura.

Several less common invasive conditions can be important clinical manifestations of [Hib](#) infection in children. These include osteomyelitis, septic arthritis, pericarditis, orbital cellulitis, endophthalmitis, urinary tract infection, abscesses, and bacteremia without an identifiable focus. As has already been mentioned, infections due to Hib are unusual among patients older than 6 years.

Nontypable *H. influenzae* Nontypable *H. influenzae* is the second most common cause (after *Streptococcus pneumoniae*) of community-acquired bacterial pneumonia in adults. Nontypable *H. influenzae* pneumonia is especially common among patients with chronic obstructive pulmonary disease (COPD) or AIDS. The clinical features of pneumonia due to *H. influenzae* are similar to those of other types of bacterial pneumonia (including pneumococcal pneumonia). Patients present with fever, cough, and purulent sputum, usually of several days' duration. Chest radiography reveals alveolar infiltrates in a patchy or lobar distribution. Gram-stained sputum contains a predominance of small, pleomorphic, coccobacillary gram-negative bacteria.

Exacerbations of [COPD](#) caused by nontypable *H. influenzae* are characterized by increased cough, sputum production, and shortness of breath. Fever is low-grade, and no infiltrates are evident on chest x-ray.

Nontypable *H. influenzae* is one of the three most common causes of childhood otitis media (the other two being *S. pneumoniae* and *Moraxella catarrhalis*). Infants are febrile and irritable, while older children report ear pain. Symptoms of viral upper respiratory infection often precede otitis media. The diagnosis is made by pneumatic otoscopy. An etiologic diagnosis, although not routinely sought, can be established by tympanocentesis and culture of middle-ear fluid.

Nontypable *H. influenzae* also causes puerperal sepsis and is an important cause of neonatal bacteremia. These nontypable strains tend to be of biotype IV and cause invasive disease after colonizing the female genital tract.

Nontypable *H. influenzae* causes sinusitis in adults and children. In addition, the bacterium is a less common cause of various invasive infections that are reported primarily as small-series descriptions and case reports. These infections include empyema, adult epiglottitis, pericarditis, cellulitis, septic arthritis, osteomyelitis, endocarditis, cholecystitis, intraabdominal infections, urinary tract infections, mastoiditis, aortic graft infection, and bacteremia without a detectable focus.

DIAGNOSIS

The most reliable method for establishing a diagnosis of Hib infection is recovery of the organism in culture. The CSF of a patient in whom meningitis is suspected should be subjected to Gram's staining and culture. The presence of gram-negative coccobacilli in Gram-stained CSF is strong evidence for Hib meningitis. Recovery of the organism from CSF confirms the diagnosis. Cultures of other normally sterile body fluids, such as blood, joint fluid, pleural fluid, pericardial fluid, and subdural effusion, are confirmatory in other infections.

Detection of PRP is an important adjunct to culture in rapid diagnosis. Immunoelectrophoresis, latex agglutination, coagglutination, and enzyme-linked immunosorbent assay are effective in detecting PRP. These assays are particularly helpful when patients have received prior antimicrobial therapy and thus are especially likely to have negative cultures.

Before the early 1980s, nontypable strains of *H. influenzae* were frequently misidentified as Hib because of their autoagglutination when serotypes were determined in agglutination assays. Since nontypable *H. influenzae* is primarily a mucosal pathogen, it is a component of a mixed flora; this situation makes etiologic diagnosis challenging. Nontypable *H. influenzae* infection is strongly suggested by the predominance of gram-negative coccobacilli among abundant polymorphonuclear leukocytes in a Gram-stained sputum specimen from a patient in whom pneumonia or tracheobronchitis is suspected. A sputum culture is helpful when interpreted along with the results of Gram's staining. Although bacteremia is detectable in a small proportion of patients with pneumonia due to nontypable *H. influenzae*, most such patients have negative blood cultures.

A diagnosis of otitis media is based on the detection by pneumatic otoscopy of fluid in the middle ear. An etiologic diagnosis requires tympanocentesis but is not routinely sought. An invasive procedure is also required to determine the etiology of sinusitis; thus, treatment is often empirical once the diagnosis is suspected in light of clinical symptoms and sinus radiographs.

TREATMENT

Initial therapy for meningitis due to Hib should consist of a cephalosporin such as ceftriaxone or cefotaxime. For children, the dose of ceftriaxone is 75 to 100 mg/kg daily given in two doses 12 h apart. The pediatric dose of cefotaxime is 200 mg/kg daily given in four doses 6 h apart. Adult doses are 2 g every 12 h for ceftriaxone and 2 g every 4 to 6 h for cefotaxime. An alternative regimen for initial therapy is ampicillin (200 to 300 mg/kg daily in four divided doses) plus chloramphenicol (75 to 100 mg/kg daily in four

divided doses). Therapy should continue for a total of 1 to 2 weeks.

Administration of glucocorticoids to patients with [Hib](#) meningitis reduces the incidence of neurologic sequelae. The presumed mechanism is reduction of the inflammation induced by bacterial cell-wall mediators of inflammation when cells are killed by antimicrobial agents. Dexamethasone (0.6 mg/kg per day intravenously in four divided doses for 2 days) is recommended for the treatment of Hib meningitis in children over 2 months of age.

Invasive infections other than meningitis are treated with the same antimicrobial agents. For epiglottitis, the dose of ceftriaxone is 50 mg/kg daily, and the dose of cefotaxime is 150 mg/kg daily, given in three divided doses 8 h apart. Epiglottitis constitutes a medical emergency, and maintenance of an airway is critical. The duration of therapy is determined by the clinical response. A course of 1 to 2 weeks is usually appropriate.

Many infections caused by nontypable strains of *H. influenzae*, such as otitis media, sinusitis, and exacerbations of [COPD](#), can be treated with oral antimicrobial agents. Approximately 25% of nontypable strains produce β -lactamase and are resistant to ampicillin. Infections caused by ampicillin-resistant strains can be treated with a variety of agents, including trimethoprim-sulfamethoxazole, amoxicillin/clavulanic acid, various extended-spectrum cephalosporins, and newer macrolides (azithromycin and clarithromycin). Fluoroquinolones are highly active against *H. influenzae* but are not currently recommended for the treatment of children or pregnant women because of possible effects on articular cartilage.

PREVENTION

Vaccination The development of conjugate vaccines that prevent invasive infections with [Hib](#) in infants and children has been a dramatic success. Four such vaccines are licensed in the United States. In addition to eliciting protective antibody, these vaccines prevent disease by reducing pharyngeal colonization with Hib.

All children should be immunized with an [Hib](#) conjugate vaccine, receiving the first dose at approximately 2 months of age, the rest of the primary series between 2 and 6 months of age, and a booster dose at 12 to 15 months of age. Specific recommendations vary for the different conjugate vaccines. The reader is referred to the recommendations of the American Academy of Pediatrics. Currently, no vaccines are available for the prevention of disease caused by nontypable *H. influenzae*.

Chemoprophylaxis The risk of secondary disease is greater than normal among household contacts of patients with [Hib](#) disease. The attack rate is as high as 4% among susceptible infants. Therefore, all children and adults in households where there are contacts <4 years old should receive prophylaxis with oral rifampin. (This rule does not apply when all household contacts under the age of 4 years have been completely immunized with conjugate vaccine.) Children <12 years old should receive rifampin at a dose of 20 mg/kg once daily for 4 days, and adults should receive 600 mg daily for 4 days. The index case should receive rifampin before or at the time of discharge from the hospital because antimicrobial agents used for the treatment of meningitis do not reliably eradicate Hib from the nasopharynx.

When two or more cases of invasive [Hib](#) disease have occurred within 60 days at a child-care facility attended by incompletely vaccinated children, administration of rifampin to all attendees and personnel is indicated, as is recommended for household contacts. The data on secondary cases among contacts in child-care facilities following a single case are less clear. The administration of rifampin prophylaxis to contacts should be considered, but each decision should be individualized and in part based on the contacts' immunization history, the size of the center, and the extent of contact.

HAEMOPHILUS INFLUENZAE BIOGROUP AEGYPTIUS

H. influenzae biogroup *aegyptius* was formerly called *Haemophilus aegyptius* because of phenotypic characteristics distinct from those of *H. influenzae*. However, later studies involving DNA hybridization and DNA transformation demonstrated that *H. aegyptius* and *H. influenzae* are members of the same species.

H. influenzae biogroup *aegyptius* has long been associated with conjunctivitis. Moreover, this strain is now known to be the cause of Brazilian purpuric fever (BPF), which was first recognized in 1984 in the rural Brazilian town of Promissao. The sharing of many phenotypic and genotypic characteristics by the various strains of *H. influenzae* biogroup *aegyptius* that cause BPF indicates that these strains represent a clone of *H. influenzae*. The age of peak incidence of BPF is 1 to 4 years, with a range of 3 months to 8 years. The illness can occur sporadically or in outbreaks. Typically, after an episode of purulent conjunctivitis, high fever occurs in association with vomiting and abdominal pain. Within 12 to 48 h after onset, the patient develops petechiae, purpura, and peripheral necrosis and experiences vascular collapse. The characteristic laboratory features are thrombocytopenia, prolonged prothrombin time, uniformly unrevealing [CSF](#) findings, and blood cultures positive for *H. influenzae* biogroup *aegyptius*. Initial reports cited high mortality (70%), but subsequent studies have indicated that milder forms of the illness exist. Most patients have resolved or resolving purulent conjunctivitis, and culture of the conjunctiva is positive in approximately one-third of cases. BPF has been seen in several towns in Brazil and on two occasions in Australia.

HAEMOPHILUS DUCREYI

Haemophilus ducreyi is the etiologic agent of chancroid, a sexually transmitted disease characterized by genital ulceration and inguinal adenitis. *H. ducreyi* poses a significant health problem in developing countries. Although this infection is less common in the United States, its incidence has increased dramatically in the past several years. In addition to being a cause of morbidity in itself, chancroid is associated with infection with HIV because of the role of genital ulceration in the transmission of HIV.

MICROBIOLOGY

H. ducreyi is a highly fastidious coccobacillary gram-negative bacterium whose growth requires X factor (hemin). Although, in light of this requirement, the bacterium has been classified in the genus *Haemophilus*, DNA homology and chemotaxonomic studies have established substantial differences between *H. ducreyi* and other *Haemophilus* species.

Taxonomic reclassification of the organism is likely in the future but awaits further study.

The histology of the genital ulcer of chancroid is characterized by perivascular and interstitial infiltrates of macrophages and of CD4+ and CD8+ lymphocytes. The appearance is consistent with a delayed-type hypersensitivity, cell-mediated immune response. The presence of CD4+ cells and macrophages in the ulcer may explain, in part, the facilitation of transmission of HIV in patients with chancroid.

EPIDEMIOLOGY AND PREVALENCE (See also [Chap. 132](#))

Chancroid is a common cause of genital ulcers in developing countries. In the United States, chancroid is now endemic in some regions, and several large outbreaks have occurred since 1981. Recurring epidemiologic themes have been apparent in these outbreaks: (1) transmission has been predominantly heterosexual; (2) males have outnumbered females by ratios of 3:1 to 25:1; (3) prostitutes have been important in transmission of the infection; and (4) chancroid has been strongly associated with illicit drug use. The incidence of chancroid in the United States will undoubtedly increase in the coming years, and the genital ulcers associated with this infection will continue to play a role in the transmission of HIV.

CLINICAL MANIFESTATIONS

Infection is acquired as the result of a break in the epithelium during sexual contact with an infected individual. After an incubation period of 4 to 7 days, the initial lesion -- a papule with surrounding erythema -- appears ([Plate IID-54](#)). In 2 to 3 days, the papule evolves into a pustule, which spontaneously ruptures and forms a sharply circumscribed ulcer ([Fig. 132-CD1](#)) that is generally not indurated. The ulcers are painful and bleed easily; little or no inflammation of the surrounding skin is evident. Approximately half of patients develop enlarged, tender inguinal lymph nodes, which frequently become fluctuant and spontaneously rupture.

The presentation of chancroid does not usually include all of the typical clinical features and is sometimes atypical. Multiple ulcers can coalesce to form giant ulcers. Ulcers can appear and then resolve, with inguinal adenitis and suppuration following 1 to 3 weeks later; this clinical picture can be confused with that of lymphogranuloma venereum. Multiple small ulcers can resemble folliculitis. Other differential diagnostic considerations include the various infections causing genital ulceration, such as primary syphilis, condyloma latum of secondary syphilis, genital herpes, and donovanosis. In rare cases chancroid lesions become secondarily infected with bacteria; the result is extensive inflammation.

DIAGNOSIS

Clinical diagnosis of chancroid is often inaccurate, and laboratory confirmation should be attempted in suspected cases. Gram's staining of a swab of the lesion may reveal a predominance of characteristic gram-negative coccobacilli, but the presence of other bacteria often makes it difficult to interpret this result. An accurate diagnosis of chancroid relies on cultures of *H. ducreyi* from the lesion. In addition, aspiration and culture of suppurative lymph nodes should be considered. Since the organism can be

difficult to grow, the use of selective and supplemented media is necessary.

TREATMENT

Clinical isolates of *H. ducreyi* often exhibit plasmid-mediated resistance to ampicillin, chloramphenicol, tetracyclines, and sulfonamides. Nevertheless, chancroid can be treated effectively with several regimens, including (1) ceftriaxone, 250 mg intramuscularly as a single dose; (2) azithromycin, 1 g orally as a single dose; (3) erythromycin, 500 mg orally four times daily for 7 days; and (4) ciprofloxacin, 500 mg orally twice daily for 3 days. Ciprofloxacin should not be administered to pregnant or lactating women or to persons <18 years old. Any therapeutic regimen may fail; single-dose ceftriaxone has a high failure rate in HIV-positive individuals. Isolates from patients who do not respond promptly to treatment should be tested for antimicrobial susceptibility. In patients with HIV infection, healing may be slow and longer courses of treatment may be necessary. Contacts of patients with chancroid should be identified and treated whenever possible.

OTHER HAEMOPHILUS SPECIES

Haemophilus species are often recovered as components of the flora of the normal human upper respiratory tract. However, these bacteria are infrequent causes of infection because of their low pathogenic potential. *Haemophilus* species have fastidious growth requirements and are generally rather slow-growing. The species implicated in human infections include *H. parainfluenzae*, *H. aphrophilus*, and *H. paraphrophilus* ([Chap. 150](#)); *H. parahaemolyticus*; *H. haemolyticus*; and *H. segnis*. *Haemophilus* species are differentiated from one another by several characteristics, primarily their requirements for X and V factors. Species designated *para-* require V factor but not X factor for growth, whereas the others require either X and V or X only.

A variety of infections involving almost all organ systems can be caused by *Haemophilus* species. Most of these unusual manifestations have been reported as single cases and small series.

The antimicrobial susceptibility characteristics of other *Haemophilus* species are similar to those of *H. influenzae*. Some strains produce β -lactamase and are thereby resistant to ampicillin. Other strains are sensitive to ampicillin, and this agent has been used successfully to treat many infections. Alternative agents with good activity against most *Haemophilus* species include trimethoprim-sulfamethoxazole, third-generation cephalosporins, tetracycline, chloramphenicol, and aminoglycosides. Endocarditis caused by ampicillin-sensitive strains should be treated with ampicillin plus an aminoglycoside.

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150. INFECTIONS DUE TO THE HACEK GROUP AND MISCELLANEOUS GRAM-NEGATIVE BACTERIA - Dennis L. Kasper, Tamar F. Barlam

HACEK GROUP ORGANISMS

HACEK organisms are a group of fastidious, slow-growing, gram-negative bacteria whose growth requires an atmosphere of carbon dioxide. Species belonging to this group include several *Haemophilus* species, *Actinobacillus actinomycetemcomitans*, *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella kingae*. HACEK bacteria normally reside in the oral cavity and have been associated with local infections in the mouth. They are also known to cause severe systemic infections, most often bacterial endocarditis ([Chap. 126](#)).

Of the HACEK group, the *Haemophilus* species, *A. actinomycetemcomitans*, and *C. hominis* are most frequently associated with endocarditis, which can develop on either native or prosthetic valves. In large series, up to 3% of cases of infective endocarditis are attributable to HACEK organisms. The clinical course of HACEK endocarditis tends to be subacute; however, embolization is common. The overall prevalence of major emboli associated with HACEK endocarditis ranges from 28 to 60% in different series. Cultures of blood from patients with suspected HACEK endocarditis may require up to 30 days to become positive, although most are positive within the first week. Because of this slow growth, antimicrobial testing may be difficult, and strains producing β -lactamase may not be identified accurately. This factor should be considered when choosing a therapeutic regimen.

The cure rates for HACEK prosthetic valve endocarditis appear to be high. Unlike prosthetic valve endocarditis caused by other gram-negative organisms, HACEK endocarditis is often cured with antibiotic treatment alone -- i.e., without surgical intervention.

***Haemophilus* Species** *Haemophilus* species cause over half of all cases of HACEK endocarditis. *H. parainfluenzae* is most common, with *H. aphrophilus* and *H. paraphrophilus* less common. Up to 50% of patients with native valve endocarditis due to *Haemophilus* species report a history of cardiac valvular disease, 60% have been ill for <2 months before presentation, and 50% are anemic at presentation. Some 19% of these patients develop congestive heart failure. Mortality rates of up to 30% have been reported, with most deaths attributed to cerebral embolism; however, recent studies have documented mortality rates of <5%. In rare cases, *H. parainfluenzae* has been isolated from other infections, such as meningitis; brain, dental, and liver abscess; pneumonia; and septicemia.

TREATMENT

Therapy for endocarditis due to *Haemophilus* species should be based on antibiotic sensitivity testing. Empirical combination therapy with ampicillin and gentamicin, successful in prior studies, is no longer recommended because of increasing β -lactamase production by these strains. Treatment with ceftriaxone (2 g/d) is a reasonable initial approach.

Actinobacillus actinomycetemcomitans *A. actinomycetemcomitans*, another slow-growing inhabitant of the oral cavity, can be isolated from soft tissue infections and abscesses in association with *Actinomyces israelii*. About 30% of actinomycotic lesions also yield *A. actinomycetemcomitans* on culture. *A. actinomycetemcomitans* has been associated with severe destructive periodontal disease, characterized by loss of alveolar bone of the molars and incisors, in both children and adults. Patients who develop endocarditis with this organism typically have severe periodontal disease and underlying cardiac valvular damage as well as high rates of embolic phenomena. *A. actinomycetemcomitans* has been isolated from patients with brain abscess, meningitis, parotitis, osteomyelitis, urinary tract infection, pneumonia, and empyema, among other infections.

TREATMENT

Most isolates are susceptible to third-generation cephalosporins such as ceftriaxone (2 g/d), semisynthetic penicillins such as mezlocillin, trimethoprim-sulfamethoxazole, quinolones, and azithromycin. However, because of the variability among strains, susceptibility testing should be undertaken. Endocarditis should normally be treated for 4 weeks; however, prosthetic valve infections or infections in patients with complications such as embolization justify 6 weeks of therapy.

Cardiobacterium hominis *C. hominis* primarily causes endocarditis in patients with underlying valvular heart disease or with prosthetic valves. Many patients have signs and symptoms of long-standing infection before diagnosis and have evidence of arterial embolization, vasculitis, cerebrovascular accidents, immune complex glomerulonephritis, or arthritis at presentation. As in endocarditis due to other HACEK organisms, embolization, mycotic aneurysms, and congestive heart failure are frequent.

TREATMENT

Antibiotic sensitivity testing of *C. hominis* is difficult. Most cases of infection due to *C. hominis* are treated with penicillin (16 to 18 million units per day in 6 divided doses), either alone or in combination with an aminoglycoside (e.g., gentamicin, 5 to 6 mg/kg per day in 3 divided doses). The value of the aminoglycoside in this situation has not been established.

Eikenella corrodens *E. corrodens*, a fastidious, facultative gram-negative organism, is part of the endogenous flora of the mouth and nasopharynx. It is most frequently recovered from sites of infection in conjunction with other bacterial species. Clinical sources of *E. corrodens* include sites of human bite wounds (clenched-fist injuries), endocarditis, soft tissue infections of the head and neck, soft tissue infections in injection drug users, osteomyelitis, respiratory infections, chorioamnionitis, gynecologic infections associated with intrauterine devices, meningitis and brain abscesses, and visceral abscesses.

TREATMENT

E. corrodens-associated infections can be treated with ampicillin (2 g every 4 h) or with second- or third-generation cephalosporins. The organism is susceptible to the

fluoroquinolones in vitro but is resistant to metronidazole and clindamycin.

Kingella kingae *K. kingae* is a β -hemolytic, fastidious, nonmotile gram-negative rod. Because of improved microbiologic methodology, isolation of this organism is increasingly common. In young children, *K. kingae* causes septic arthritis and osteomyelitis. In several series, *K. kingae* has been the third most common cause of septic arthritis in children <24 months of age; staphylococcal and streptococcal species remain most prevalent. In children <4 years of age, there is evidence for prolonged nasopharyngeal colonization, with carriage rates of 10%. Invasive *K. kingae* infections with bacteremia are associated with stomatitis. Both *K. kingae* colonization and primary herpes -- a major cause of stomatitis -- peak in children 6 to 48 months of age. *K. kingae* bacteremia can present with a petechial rash similar to that seen with *Neisseria meningitidis* sepsis.

Infective endocarditis, unlike other infections with *K. kingae*, occurs in older children and adults. The majority of patients have preexisting valvular disease. As in endocarditis caused by the other HACEK organisms, there is a high incidence of complications, including arterial emboli, cerebrovascular accidents, tricuspid insufficiency, and congestive heart failure with cardiovascular collapse.

TREATMENT

K. kingae can be susceptible to ampicillin, second- and third-generation cephalosporins, fluoroquinolones, vancomycin, clindamycin, macrolides, and trimethoprim-sulfamethoxazole. Because of increasing β -lactamase production in *K. kingae* strains, susceptibility testing should be performed to guide therapy. Ceftriaxone (2 g/d) or ampicillin-sulbactam (3 g of ampicillin every 6 h) are both appropriate choices for initial therapy.

OTHER GRAM-NEGATIVE BACTERIA

***Acinetobacter* Species** See [Chap. 153](#).

Achromobacter xylosoxidans Previously known as *Alcaligenes xylosoxidans*, the gram-negative bacillus *Achromobacter xylosoxidans* is probably part of the endogenous intestinal flora and has been isolated from water sources. Immunocompromised hosts appear to be at increased risk for infection with this organism. Nosocomial sources to which outbreaks of infection with *A. xylosoxidans* have been attributed include contaminated intravenous fluids, pressure transducers, and disinfectants. Clinical illness has been associated with isolates from many sites, including blood (often in the setting of infected intravascular devices), urine, respiratory secretions, cerebrospinal fluid, peritoneal and pleural fluids, and endocarditic prosthetic valves. Community-acquired bacteremia with *A. xylosoxidans* usually occurs in the setting of pneumonia. Metastatic skin lesions are present in one-fifth of cases. The reported mortality rate is 67%, similar to rates for other bacteremic gram-negative pneumonias.

TREATMENT

In vitro susceptibility testing of all clinically relevant isolates is essential to the selection

of appropriate therapy.

Agrobacterium radiobacter (tumefaciens) This organism has been associated with intravascular catheter-related infections in immunocompromised hosts, especially individuals infected with HIV. Clinically important infections associated with *A. radiobacter* include prosthetic joint and prosthetic valve infections, bacteremia, peritonitis, and urinary tract infections.

TREATMENT

Antibiotic sensitivity testing is essential in the choice of therapy.

***Capnocytophaga* Species** This genus of fusiform, long, thin, gram-negative coccobacilli is facultatively anaerobic and requires an atmosphere enriched in carbon dioxide for optimal growth. *C. ochracea*, *C. gingivalis*, and *C. sputigena* are inhabitants of the healthy human oral cavity and have been isolated from the female genital tract. Their isolation has also been reported from blood, cerebrospinal fluid, and respiratory fluids (including pleural collections). These organisms have been associated with sepsis in immunocompromised hosts; particularly at risk are patients with acute myelogenous leukemia or acute lymphocytic leukemia. In the immunocompetent host, these three species probably play a role in localized juvenile periodontitis; however, they have been isolated from many other sites as well, usually as part of a polymicrobial infection. In vitro sensitivity testing of these organisms is difficult because they are slow-growing and fastidious.

C. canimorsus and *C. cynodegmi* are endogenous to the canine mouth. Patients infected with these species frequently have a history of dog bites or of exposure to dogs without scratches or bites. Asplenia, glucocorticoid therapy, and alcohol abuse are predisposing conditions and are associated with relatively fulminant infections. The interval from dog bite to presentation averages 5 days but ranges from 1 day to 1 month. *C. canimorsus* causes a wide range of infections, including severe sepsis with shock and disseminated intravascular coagulation, meningitis, endocarditis, cellulitis, and septic arthritis. In the asplenic individual who has recently sustained a dog bite, infection with this organism must be considered early because of a potentially rapid progression to death.

TREATMENT

Although penicillin has been considered first-line therapy for infections due to *C. ochracea*, *C. gingivalis*, and *C. sputigena*, an increasing number of isolates reportedly produce b-lactamase. Clindamycin (600 to 900 mg every 6 to 8 h) or drug combinations including a penicillin derivative plus a b-lactamase inhibitor -- such as ampicillin/sulbactam (1.5 to 3.0 g of ampicillin every 6 h) -- are currently recommended for empirical therapy. Penicillin (12 to 18 million units daily in 6 divided doses) is the drug of choice for infections with *C. canimorsus*. This regimen should also be given prophylactically to asplenic patients sustaining dog-bite injuries. Patients with suspected infection due to *C. canimorsus* should be treated empirically, because identification of this organism and determination of its antibiotic sensitivity can take many days. Other drugs to which *C. canimorsus* is reportedly susceptible include clindamycin, imipenem,

quinolones, and third-generation cephalosporins.

Chromobacterium violaceum This organism is rarely a human pathogen but reportedly has been responsible for life-threatening infections with severe sepsis and metastatic abscesses. A slender, slightly curved, gram-negative rod that is facultatively anaerobic, *C. violaceum* inhabits tropical water and soil and causes infection after contamination of skin wounds. Patients with defective neutrophil function (e.g., those with chronic granulomatous disease) are infected by this organism with unusual frequency. The mortality rate in the United States from infection with *C. violaceum* has been reported at >60%.

TREATMENT

C. violaceum is generally susceptible to ciprofloxacin (500 mg every 12 h orally or 400 mg every 12 h intravenously), trimethoprim-sulfamethoxazole, gentamicin, and chloramphenicol.

***Chryseobacterium* Species** *C. meningosepticum* and *C. indologenes* were previously classified as *Flavobacterium* species. *C. meningosepticum* is a ubiquitous organism and an important cause of nosocomial infections. It has been associated with outbreaks due to contaminated fluids, such as disinfectants, arterial catheter flush solutions, and aerosolized antibiotics, and with sporadic infections due to indwelling devices, vials, sink traps, feeding tubes, and other fluid-associated apparatus. Patients with nosocomial *C. meningosepticum* infection usually have underlying immunosuppression (e.g., related to malignancy). *C. meningosepticum* has been reported to cause meningitis (primarily in neonates), sepsis, endocarditis, bacteremia, soft tissue infections, and pneumonia. *C. indologenes* has caused bacteremia, sepsis, and pneumonia, typically in immunocompromised patients with indwelling devices.

TREATMENT

Antibiotic treatment should be based on susceptibility results because of the high likelihood that *C. meningosepticum* will produce β -lactamase. Early reports suggested that vancomycin might be efficacious, but more recent data refute this conclusion.

Plesiomonas shigelloides This freshwater organism is a cause of acute diarrhea ([Chap. 131](#)) and occasionally of serious extraintestinal disease. *P. shigelloides* is transmitted to humans via contaminated water or food. This motile, facultatively anaerobic gram-negative rod most often produces mild diarrhea with mucoid, bloody feces containing leukocytes. Severe extraintestinal infections have been reported, most commonly in immunocompromised hosts, and include bacteremia, cellulitis, neonatal sepsis and meningitis, and septic arthritis.

TREATMENT

There is great variability among strains in terms of antibiotic sensitivity patterns, and isolates must be tested before appropriate therapy can be selected.

***Aeromonas* Species** Five species of *Aeromonas* are known to be associated with

disease in humans, but >85% of these infections are caused by *A. hydrophila*, *A. caviae*, and *A. veronii* biovar. *sobria*. *Aeromonas* proliferates in potable and fresh water and in soil. It remains controversial whether *Aeromonas* is a cause of bacterial gastroenteritis. Although many case reports have associated *Aeromonas* with gastroenteritis, no clear outbreaks with a single isolate have been documented, no conclusive animal model exists, and asymptomatic colonization of the intestinal tract with *Aeromonas* occurs frequently. However, rare cases of hemolytic-uremic syndrome occurring after bloody diarrhea have been shown to be secondary to the presence of *Aeromonas*. In addition, identification of an enterotoxin (different from the Shiga-like toxin produced by *Escherichia coli* O157:H7) in these cases supports the hypothesis that *Aeromonas* causes gastroenteritis.

Aeromonas causes sepsis and bacteremia in infants with multiple medical problems and in immunocompromised hosts, particularly those with cancer or hepatobiliary disease. *Aeromonas* infection and sepsis can occur in trauma patients with myonecrosis or in burn patients exposed to *Aeromonas* by environmental contamination of their wounds from fresh water or soil sources. Mortality ranges from 25% for sepsis in immunocompromised adults to >90% in patients with myonecrosis. *Aeromonas* can produce skin lesions resembling the ecthyma gangrenosum lesions seen in *Pseudomonas aeruginosa* infection. These lesions are hemorrhagic vesicles surrounded by a rim of erythema with central necrosis and ulceration.

Aeromonas wound infections can occur in healthy adults who sustain minor trauma with environmental contamination, usually water-related; after severe trauma and crush injuries with sepsis and environmental exposure, usually to soil; and in nosocomial infections related to catheters, surgical incisions, or use of leeches. Other clinical manifestations include meningitis, peritonitis, pneumonia, and ocular infections.

TREATMENT

Treatment should be guided by antimicrobial susceptibility testing. *Aeromonas* species are generally susceptible to fluoroquinolones (e.g., ciprofloxacin at a dosage of 500 mg every 12 h orally or 400 mg every 12 h intravenously), trimethoprim-sulfamethoxazole at a trimethoprim dosage of 10 mg/kg per day in 3 or 4 divided doses, third-generation cephalosporins, and aminoglycosides. However, resistance is increasing.

Miscellaneous Organisms Many other gram-negative rods have been reported to cause occasional infections in hosts who are immunologically unprepared to deal with relatively avirulent organisms or who are unfortunate enough to encounter an exceptionally large inoculum. Such organisms include *Weeksellia* species; various CDC groups, such as EF-4, Ve-2 (*Flavimonas* species), IVc-2, NO-1, WO-1, and Gilardi Group WO-1; *Sphingobacterium* species; *Protomonas* species; *Ochrobactrum anthropi*; *Oligella urethralis*; and *Shewanella putrefaciens*. The reader is advised to consult subspecialty texts and references for further guidance on these organisms.

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151. **LEGIONELLA INFECTION** - Feng-Yee Chang, Victor L. Yu

DEFINITION

Legionellosis refers to the two clinical syndromes caused by bacteria of the genus *Legionella*. *Pontiac fever* is an acute, febrile, self-limited illness that has been serologically linked to *Legionella* species, whereas *Legionnaires' disease* is the designation for pneumonia caused by these species.

HISTORY

Legionnaires' disease was first recognized in 1976, when an outbreak of pneumonia took place at a hotel in Philadelphia during the American Legion Convention. Investigators from the Centers for Disease Control and Prevention (CDC) identified the causative aerobic gram-negative bacterium in lung specimens obtained from the victims at autopsy and named this organism *L. pneumophila*. Retrospective studies of stored serum samples revealed that an epidemic of Legionnaires' disease had occurred in 1957 in Austin, Minnesota. In this epidemic 78 persons were hospitalized with acute respiratory infection. Antibody determinations showed seroconversion to *L. pneumophila* in most cases.

MICROBIOLOGY

At present, the family Legionellaceae comprises 41 species with 64 serogroups. The species *L. pneumophila* causes 80 to 90% of human infections and includes at least 14 serogroups; serogroups 1, 4, and 6 are most commonly implicated in human infections. To date, 17 species other than *L. pneumophila* have been associated with human infections, among which *L. micdadei* (Pittsburgh pneumonia agent), *L. bozemanii*, *L. dumoffii*, and *L. longbeachae* are the most common.

Members of the Legionellaceae are aerobic, thin, gram-negative bacilli that do not grow on routine microbiologic media. Buffered charcoal yeast extract (BCYE) agar is the medium used to grow *Legionella*. This highly enhanced medium contains the amino acid L-cysteine, which is an absolute growth requirement for *Legionella*. Growth of the organism on BCYE medium is usually visible in 3 to 5 days at 35 to 37°C. *L. micdadei* and *L. maceachernii* produce blue colonies on BCYE medium containing bromocresol purple and bromothymol blue dyes, while the other species produce green colonies.

Antimicrobial agents, including polymyxin B, cefamandole, and vancomycin, are used in *Legionella*-selective media to suppress competing components of the microflora. Although *L. pneumophila* is relatively tolerant to these antibiotics, the drugs may inhibit the growth of other legionellae; for example, cefamandole-containing media suppress the growth of *L. micdadei*.

Traditional biochemical tests are not particularly helpful in distinguishing one *Legionella* species from another. Fatty-acid profile determination by gas-liquid chromatography and ubiquinone analysis allow identification to the genus level. The direct fluorescent antibody (DFA) test can definitively identify a number of individual species. In *L. pneumophila*, lipopolysaccharide is a prominent constituent of the outer membrane, and

the serogroup-specific antigen and antibodies detected by immunofluorescence are directed primarily at the lipopolysaccharide. Both polyclonal and monoclonal DFA reagents are commercially available. The monoclonal antibody reagent is less cross-reactive but is specific for *L. pneumophila*. Genetic analysis has been considered the definitive arbiter for the identification of individual species, with the degree of DNA sequence homology the most common criterion employed. A nucleic-acid hybridization probe reactive to *Legionella* ribosomal RNA, used with a single reagent, can identify a member of the genus within hours.

ECOLOGY AND TRANSMISSION

The natural habitats for *L. pneumophila* are aquatic bodies, including lakes and streams; *L. longbeachae* has been isolated from soil. Legionellae can survive under a wide range of environmental conditions; for example, the organisms can live for years in refrigerated water samples. Natural bodies of water contain only small numbers of legionellae. However, once the organisms enter human-constructed aquatic reservoirs (such as cooling towers or water-distribution systems), they can grow and proliferate. Factors known to enhance colonization by and amplification of legionellae include warm temperatures (25° to 42°C), stagnation, and scale and sediment. The presence of symbiotic microorganisms, including algae, amebas, ciliated protozoa, and other water-dwelling bacteria, likewise promotes growth of *L. pneumophila*.

Hot-water tanks colonized with *L. pneumophila* are significantly more likely than uncolonized tanks to be cooler (<60°C), to have a vertical configuration, to be older, and to have higher concentrations of calcium and magnesium. Vertical tanks, especially those that are electric coil-heated rather than gas-heated, have a pronounced temperature stratification and thick sediment accumulation at the bottom. Studies have shown that neither a high degree of outward cleanliness nor routine application of maintenance measures decreases the frequency or intensity of *Legionella* colonization. Thus, engineering guidelines and building codes, although often advocated as preventive measures, have relatively little impact on *Legionella* colonization.

The source of *Legionella* is water, but the mode of transmission from the environmental reservoir to the patient remains controversial. Early investigations that implicated cooling towers antedated the discovery that the organism could also exist in potable water distribution systems. It is now known that, in many outbreaks, cases of Legionnaires' disease continued to occur despite disinfection of cooling towers and the potable water supply was the actual source. Koch's postulates have been fulfilled in epidemiologic studies using molecular fingerprinting methods to link potable water sources (rather than cooling towers) to *Legionella* infection in humans. Community-acquired Legionnaires' disease has been linked to colonization of residential and industrial water supplies.

Multiple modes of transmission of *Legionella* to humans exist, including aerosolization, aspiration, and direct instillation into the lung during respiratory tract manipulations. Aspiration may be the predominant mode of transmission, but it is unclear whether *Legionella* enters the lung via oropharyngeal colonization or directly via the drinking of contaminated water. Nasogastric tubes have been linked to nosocomial Legionnaires' disease in several reports; microaspiration of contaminated water was the hypothesized

mode of transmission. Surgery with general anesthesia is a known risk factor that is consistent with aspiration. Especially compelling is the reported 30% incidence of postoperative *Legionella* pneumonia among patients undergoing head and neck surgery in a hospital with a contaminated water supply; aspiration is a recognized sequela in such cases. Studies of patients with hospital-acquired Legionnaires' disease showed that these individuals underwent endotracheal intubation significantly more often and for a significantly longer duration than patients with nosocomial pneumonia of other etiologies.

Aerosolization of legionellae by devices filled with tap water, including nebulizers and humidifiers, has caused cases of Legionnaires' disease. An ultrasonic mist machine in the produce section of a grocery store was implicated in a community outbreak. Pontiac fever has been linked to *Legionella*-containing aerosols from water-using machinery, a cooling tower, air-conditioners, and whirlpools.

EPIDEMIOLOGY

The incidence of Legionnaires' disease depends on the degree of contamination of the aquatic reservoir, the susceptibility and immune status of the persons exposed to the water from that reservoir, the intensity of exposure, and the availability of specialized laboratory tests on which the correct diagnosis can be based.

Numerous prospective studies have found *Legionella* to rank among the top four microbial causes of community-acquired pneumonia (*Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Chlamydia pneumoniae* usually ranking first, second, and third), accounting for 3 to 15% of cases. On the basis of a multihospital study of community-acquired pneumonia in Ohio, the [CDC](#) has estimated that only 3% of sporadic cases of Legionnaires' disease are correctly diagnosed. Legionellae are responsible for 10 to 50% of nosocomial pneumonias when a hospital's water system is colonized with the organisms. One situation in which the diagnosis of Legionnaires' disease should be considered is that in which the presenting patient has been hospitalized within 10 days before the onset of symptoms. In one study, a number of patients had been discharged from the hospital and readmitted with Legionnaires' disease; molecular fingerprinting showed that the isolates obtained from patients and the isolate from the hospital's water supply were similar.

The most common risk factors for Legionnaires' disease are cigarette smoking, chronic lung disease, advanced age, and immunosuppression. The disease most often develops in elderly men; this predilection is probably related to cigarette smoking. Surgery is a prominent predisposing factor in nosocomial infection, with transplant recipients at highest risk. Nosocomial cases are now being recognized among neonates and among children with immunosuppression or underlying pulmonary disease.

Pontiac fever occurs in epidemics. The high attack rate (>90%) reflects airborne transmission.

PATHOGENESIS

Legionellae enter the lungs through aspiration or direct inhalation. The organisms

possess pili that may mediate adherence to respiratory tract epithelial cells. Thus, conditions that impair mucociliary clearance, including cigarette smoking, lung disease, or alcoholism, predispose to Legionnaires' disease.

Cell-mediated immunity is the primary mechanism of host defense against *Legionella*, as it is against other intracellular pathogens, including *Mycobacterium tuberculosis*, *Listeria*, and *Toxoplasma*. Alveolar macrophages readily phagocytose legionellae. The attachment of the bacteria to phagocytes is mediated via complement receptors, which attach to the bacterial major outer-membrane protein. Binding to these receptors promotes phagocytosis but fails to trigger an oxidative burst. Although many legionellae are killed, some proliferate intracellularly until the cells rupture; the bacteria are then phagocytosed again by newly recruited phagocytes, and the cycle begins anew. Legionnaires' disease is more common and the disease manifestations are more severe in patients with depressed cell-mediated immunity, including transplant recipients, patients infected with HIV, and patients receiving glucocorticoids. The disease also occurs with unusual frequency among patients with hairy cell leukemia (which is characterized by monocyte deficiency and dysfunction) but not among patients with other types of leukemia.

The role of neutrophils in immunity appears to be minimal: neutropenic patients are not predisposed to Legionnaires' disease. Although *L. pneumophila* is susceptible to oxygen-dependent microbiologic systems in vitro, it resists killing by neutrophils.

The humoral immune system is active against *Legionella*. Type-specific IgM and IgG antibodies are measurable within weeks of infection. In vitro, antibodies promote killing of legionellae by phagocytes (neutrophils, monocytes, and alveolar macrophages). However, antibodies neither enhance lysis by complement nor inhibit intracellular multiplication within phagocytes. Immunized animals develop a specific antibody response, with subsequent resistance to *Legionella* challenge.

Some *L. pneumophila* strains are clearly more virulent than others, although the precise factors mediating virulence remain uncertain. For example, although multiple strains may colonize water-distribution systems, only a few cause disease in patients exposed to that water. At least one surface epitope of *L. pneumophila* serogroup 1 is associated with virulence. *L. pneumophila* serogroup 6 is more commonly involved in nosocomial Legionnaires' disease and is more likely to be associated with a poor outcome.

PATHOLOGY

The consistent pathologic features of Legionnaires' disease are confined to the lungs. Findings in infected lung tissue range from multifocal pneumonia with patchy lobular inflammation to extensive multilobar consolidation. Visible abscesses with central necrosis were seen in 20% of autopsied cases in one study. On histologic examination, fibrinopurulent pneumonia with intensive alveolitis and bronchiolitis is evident. Lesions of longer standing can have a nodular appearance with a central area of necrosis surrounded by macrophages and other cells. The alveoli are filled with fibrin, neutrophils, and alveolar macrophages.

Usual tissue stains, including Gram's, hematoxylin and eosin, Brown-Brenn, and

methenamine silver, do not reveal the organism. Gimenez stain can be used for imprints on fresh or fixed tissue. Dieterle's silver stain or modified Gimenez stain, although nonspecific and relatively insensitive, can be used for paraffin-fixed specimens. The DFA stain is not only specific but also the most sensitive option for visualization of the organism in tissues. Polyvalent DFA stains but not monoclonal DFA stain can be used for formalinized specimens ([Fig. 151-CD1](#)). Because the DFA stains are species and serogroup specific, false-negative results can be obtained if the incorrect reagent is used. Thus, culture is the preferred method for diagnosis based on clinical specimens.

CLINICAL AND LABORATORY FEATURES

Pontiac Fever Pontiac fever is an acute, self-limiting, flulike illness with a 24- to 48-h incubation period. Pneumonia does not develop in Pontiac fever. Malaise, fatigue, and myalgias are the most frequent symptoms, occurring in 97% of cases. Fever (usually with chills) develops in 80 to 90% of cases and headache in 80%. Other symptoms (seen in fewer than 50% of cases) include arthralgias, nausea, cough, abdominal pain, and diarrhea. Modest leukocytosis with a neutrophilic predominance is sometimes detected. Complete recovery takes place within only a few days without antibiotic therapy; a few patients may experience lassitude for many weeks thereafter. The diagnosis is established by antibody seroconversion.

Legionnaires' Disease (Pneumonia) Clinical findings that raise the possibility of Legionnaires' disease are summarized in [Table 151-1](#). Although these manifestations may provide clues to the diagnosis, prospective comparative studies have shown that they are generally nonspecific and do not serve to distinguish Legionnaires' disease from pneumonia of other etiologies. Legionnaires' disease is often included in the differential diagnosis of "atypical pneumonia," along with infection due to *Chlamydia pneumoniae*, *C. psittaci*, *Mycoplasma pneumoniae*, *Coxiella burnetii*, and some viruses. The clinical similarities among these types of pneumonia include a relatively nonproductive cough and a low incidence of grossly purulent sputum. However, the clinical manifestations of Legionnaires' disease are usually more severe than those of most "atypical" pneumonias, and the course and prognosis of *Legionella* pneumonia more resemble those of bacteremic pneumococcal pneumonia than those of pneumonia due to other "atypical" pathogens. Patients with community-acquired Legionnaires' disease are significantly more likely than patients with pneumonia of other etiologies to be admitted to an intensive care unit on presentation.

The incubation period for Legionnaires' disease is 2 to 10 days. The symptoms and signs may range from a mild cough and a slight fever to stupor with widespread pulmonary infiltrates and multisystem failure. Nonspecific symptoms -- malaise, fatigue, anorexia, and headache -- are seen early in the illness. Myalgias and arthralgias are uncommon but are unusually marked in a few patients. Upper respiratory symptoms, including coryza, are rare.

The mild cough of Legionnaires' disease is only slightly productive. Sometimes the sputum is streaked with blood. Chest pain -- either pleuritic or nonpleuritic -- can be a prominent feature and, when coupled with hemoptysis, can lead to an incorrect diagnosis of pulmonary embolism. Shortness of breath is reported by one-third to one-half of patients.

Gastrointestinal difficulties are often pronounced; abdominal pain, nausea, and vomiting affect 10 to 20% of patients. Diarrhea (watery rather than bloody) is reported in 25 to 50% of cases. The most common neurologic abnormalities are confusion or changes in mental status; however, the multitudinous neurologic symptoms reported range from headache and lethargy to encephalopathy.

Patients with Legionnaires' disease virtually always have fever. Temperatures in excess of 40.5°C (104.9°F) were recorded in 20% of the cases in one series. Relative bradycardia has been overemphasized as a useful diagnostic finding; it occurs infrequently, primarily affecting older patients with severe pneumonia. Chest examination reveals rales early in the course and evidence of consolidations as the disease progresses. Abdominal examination may reveal generalized or local tenderness.

Diarrhea and hyponatremia occur significantly more often in Legionnaires' disease than in other forms of pneumonia. Hyponatremia is most common in severe cases. The mechanism of hyponatremia does not appear to be related to inappropriate secretion of antidiuretic hormone but instead to salt and water loss. Besides hyponatremia, other laboratory abnormalities include abnormal liver function tests, hypophosphatemia, hematuria, hematologic abnormalities, and thrombocytopenia; although common, these abnormalities are not found significantly more frequently in Legionnaires' disease than in pneumonias of other etiologies.

Extrapulmonary Legionellosis Since the portal of entry for legionellae is the lung in virtually all cases, extrapulmonary manifestations usually result from bloodborne dissemination from the lung. In a prospective survey of patients with Legionnaires' disease diagnosed by isolation of the organism from sputum, legionellae were isolated from the blood by a special culture method in 38% of cases.

Legionella has been identified in the spleen, liver, or kidneys in 50% of autopsied cases of Legionnaires' disease. The organism has also been isolated from intrathoracic and inguinal lymph nodes -- a finding suggesting dissemination by lymphatic pathways. Extrapulmonary involvement, including sinusitis, peritonitis, pyelonephritis, cellulitis, and pancreatitis, has been documented predominantly in immunosuppressed patients.

The most common extrapulmonary site of legionellosis is the heart; numerous reports have described myocarditis, pericarditis, postcardiotomy syndrome, and prosthetic-valve endocarditis. Most cases have been hospital-acquired. Since many of the patients involved have not had overt pneumonia, the lung may not have been the portal of entry. Rather, in these cardiac infections, the organisms may have gained entry through a postoperative sternal wound exposed to contaminated tap water or through a mediastinal-tube insertion site.

Various other sources of or factors promoting *Legionella* infection at various extrapulmonary sites have been postulated, including the presence of foreign bodies, such as sutures and draining tubes (wound infection after cardiothoracic surgery); immersion in a Hubbard tank (superinfection of a hip wound); bloodborne dissemination from a pulmonary infection site (perirectal abscess); and ingestion of contaminated

water (peritonitis).

Chest Radiographic Abnormalities Virtually all patients with Legionnaires' disease have abnormal chest radiographs showing pulmonary infiltrates at the time of clinical presentation. In a few cases of nosocomial disease, fever and respiratory tract symptoms have preceded the appearance of the infiltrate on chest radiography. Findings on chest radiography are nonspecific and do not serve to distinguish Legionnaires' disease from pneumonias of other etiologies. Pleural effusion is evident in one-third of cases, and the diagnosis is often based on culture and antigen testing (by the method designed for use with urine) of pleural fluid obtained by thoracentesis.

In immunosuppressed patients, especially those receiving glucocorticoids, distinctive rounded nodular opacities may be seen; these lesions may expand and cavitate ([Fig. 151-1](#)). Likewise, pulmonary abscesses can occur in immunosuppressed hosts. The progression of infiltrates on chest radiography despite appropriate antibiotic therapy is common, and radiographic improvement lags behind clinical improvement by several days. Complete clearing of infiltrates requires 1 to 4 months.

DIAGNOSIS

The diagnosis of Legionnaires' disease requires special microbiologic tests ([Table 151-2](#)). The sensitivity of bronchoscopy specimens is approximately the same as that of sputum samples; if sputum is not available, bronchoscopy specimens may yield the organism. Bronchoalveolar lavage fluid gives higher yields than bronchial wash specimens. Thoracentesis should be performed if pleural effusion is found, and the fluid should be evaluated by [DFA](#) staining, culture, and the antigen test designed for use with urine.

Staining Gram's staining of material from normally sterile sites, such as pleural fluid or lung tissue, occasionally suggests the diagnosis; efforts to detect legionellae in sputum by Gram's staining typically reveal numerous leukocytes, but no organisms. When they are visualized, the organisms appear as small, pleomorphic, faint, gram-negative bacilli. *L. micdadei* organisms can be detected as weakly or partially acid-fast bacilli in clinical specimens. Modified acid-fast staining substitutes 1% sulfuric acid for the traditional 3% hydrochloric acid; the less aggressive decolorizer increases the yield of *L. micdadei*. *Legionella*-infected patients have often been treated empirically with antituberculosis medications because of false-positive acid-fast smears.

The DFA test is rapid and highly specific but is less sensitive than culture because large numbers of organisms are required for microscopic visualization. This test is more likely to be positive in advanced than in early disease.

Culture The definitive method for diagnosis of *Legionella* infection is isolation of the organism from respiratory secretions or other specimens. As has been mentioned, [BCYE](#) agar supplemented with antibiotics and dyes is the most sensitive medium, and colonies grow slowly, requiring 3 to 5 days to become grossly visible. When culture plates are overgrown with other microflora, pretreatment of the specimen with acid or heat can markedly improve the yield. *L. pneumophila* is often isolated from sputum that is not grossly or microscopically purulent; sputum containing more than 25

epithelial cells per high-power field (a finding that classically suggests contamination) may still yield *L. pneumophila*.

Antibody Detection Antibody testing of both acute- and convalescent-phase sera may be necessary. A fourfold rise in titer is diagnostic; 4 to 12 weeks are often required for the detection of an antibody response, and some patients never seroconvert. A single titer of 1:128 in a patient with pneumonia constitutes presumptive (but not definitive) evidence for Legionnaires' disease. Serology is of use primarily in epidemiologic studies. The specificity of serology for the non-*L. pneumophila* species is uncertain; there is cross-reactivity with *L. pneumophila* and some gram-negative bacilli.

Urinary Antigen The assay for *Legionella* soluble antigen in urine (Binax, Portland, ME) is rapid, relative inexpensive, easy to perform, second only to culture in terms of sensitivity, and highly specific. Its use in every clinical laboratory is recommended. The test is available only for *L. pneumophila* serogroup 1, which, as has been mentioned, causes about 80% of *Legionella* infections. Antigen in urine is detectable 3 days after the onset of clinical disease, even if specific therapy has been started; furthermore, urinary antigen persists for several weeks.

Molecular Methods Polymerase chain reaction (PCR) with DNA probes is theoretically more sensitive and specific than other methods, but results have been disappointing to date. PCR has proved useful in the identification of legionellae from environmental water specimens.

TREATMENT

Controlled evaluations of antibiotic therapy for Legionnaires' disease have never been conducted. In the 1976 American Legion outbreak, patients treated with erythromycin and tetracycline appeared to have a better outcome than those treated with other agents. These two antibiotics also exhibited intracellular activity against legionellae and were effective in animal models. The fact that *Legionella* is an intracellular pathogen provided the biologic basis for the success of erythromycin and tetracycline, given that relatively high intracellular penetration. Antibiotics capable of achieving intracellular concentrations higher than the minimal inhibitory concentration are the most likely to be efficacious in the clinical setting. The dosages for various drugs used in the treatment of *Legionella* infection are listed in [Table 151-3](#).

The newer macrolides (especially azithromycin) and quinolones are now the antibiotics of choice, displacing erythromycin. Compared with erythromycin, the newer agents azithromycin, clarithromycin, and roxithromycin have superior in vitro activity, display greater intracellular activity, and reach higher concentrations in respiratory secretions and in lung tissue. The pharmacokinetics of the newer macrolides and quinolones also allow once- or twice-daily dosing, in contrast to the four-times-daily dosing required for erythromycin. Finally, the large fluid volume required for intravenous administration, symptomatic ototoxicity, and gastrointestinal side effects have rendered erythromycin obsolete for the treatment of *Legionella* infection.

The quinolones (levofloxacin, ciprofloxacin, pefloxacin, gemifloxacin, and moxifloxacin) are more active than any of the macrolides against *Legionella* in in vitro dilution

susceptibility tests, intracellular models, and animal models. Furthermore, in open noncomparative studies of pneumonia, numerous cases of Legionnaires' disease have been successfully treated with quinolones. Quinolones are the preferred antibiotics for transplant recipients because both macrolides (except azithromycin) and rifampin interact pharmacologically with cyclosporine and tacrolimus.

Alternative agents include tetracycline and its analogues doxycycline and minocycline. Anecdotal reports have described both successes and failures with trimethoprim-sulfamethoxazole, imipenem, and clindamycin. For severely ill patients with Legionnaires' disease, the combination of rifampin plus a macrolide or a quinolone can be used for initial treatment.

Initial therapy should be given by the intravenous route. Usually, a clinical response occurs within 3 to 5 days, after which oral therapy can be substituted. The total duration of therapy in the immunocompetent host is 10 to 14 days; a longer course (3 weeks) may be appropriate for immunosuppressed patients and those with advanced disease. In the oral phase, 5 to 10 days of azithromycin is therapy sufficient.

Mortality rates for Legionnaires' disease vary, depending on the patient's underlying disease and its severity, the patient's immune status, the severity of pneumonia, and the timing of administration of appropriate antimicrobial therapy. Mortality rates are highest (80%) among immunosuppressed patients who do not receive appropriate antimicrobial therapy. With appropriate and timely antibiotic treatment, mortality from community-acquired Legionnaires' disease among immunocompetent patients ranges from 0 to 11%; without treatment, the figure may be as high as 31%. Pontiac fever requires only symptom-based treatment, not antimicrobial therapy.

PREVENTION

Routine environmental culture of the hospital water supply is recommended as an approach to the prevention of hospital-acquired Legionnaires' disease. Positive cultures from the water supply mandate the use of specialized laboratory tests (especially culture on selective media and urinary antigen assay) for patients with hospital-acquired pneumonia.

Disinfection of the water supply is now feasible. Two methods have proven reliable and cost-effective. The superheat and flush method requires heating of the water so that the distal-outlet temperature is 70 to 80°C and flushing of the distal outlets with hot water for at least 30 min. This method is ideal for emergency situations. A commercial copper and silver ionization method has proved effective in numerous hospitals. Hyperchlorination is no longer recommended because of its expense, carcinogenicity, corrosive effects on piping, and unreliable efficacy.

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152. PERTUSSIS AND OTHER *BORDETELLA* INFECTIONS - Scott A. Halperin

Pertussis is an acute infection of the respiratory tract caused by *Bordetella pertussis*. The name *pertussis* means "violent cough," which aptly describes the most consistent and prominent feature of the illness. The inspiratory sound made at the end of an episode of paroxysmal coughing gives rise to the common name for the illness, "whooping cough"; however, this feature is variable, being uncommon in infants <6 months of age and frequently absent in older children and adults. The Chinese name for pertussis is "the 100-day cough," which accurately describes the clinical course of the illness. The identification of *B. pertussis* was first reported by Bordet and Gengou in 1906, and vaccines were produced over the following two decades.

MICROBIOLOGY

Six species have been identified in the genus *Bordetella*: *B. pertussis*, *B. parapertussis*, *B. bronchiseptica*, *B. avium*, *B. holmesii*, and *B. hinzii*. *B. pertussis* infects only humans and is the most important *Bordetella* species causing human disease. *B. parapertussis* causes an illness in humans that is similar to pertussis but is typically milder; co-infections with *B. parapertussis* and *B. pertussis* have been documented. *B. bronchiseptica* is an important pathogen of domestic animals that causes kennel cough in dogs, atrophic rhinitis and pneumonia in pigs, and pneumonia in cats. Both respiratory infection and opportunistic infection are occasionally reported in humans. *B. avium* is an important cause of respiratory illness in turkeys. The remaining two species, *B. hinzii* and *B. holmesii*, have been recognized as unusual causes of bacteremia. Both of these species have been isolated from patients with sepsis, most often from those who are immunocompromised.

Bordetella species are gram-negative pleomorphic aerobic bacilli that share common genotypic characteristics. *B. pertussis* and *B. parapertussis* are the most similar of the species but differ in that *B. parapertussis* does not express the gene coding for pertussis toxin. *B. pertussis* is a slow-growing fastidious organism that requires selective medium and forms small glistening bifurcated colonies. Suspicious colonies are presumptively identified as *B. pertussis* by direct fluorescent antibody testing or by agglutination with species-specific antiserum. *B. pertussis* is further differentiated from other *Bordetella* species by biochemical and motility characteristics.

B. pertussis produces a wide array of toxins and biologically active products that are important in its pathogenesis and in immunity. Most of these virulence factors are under the control of a single genetic locus that regulates their production, resulting in antigenic modulation and phase variation. Although these processes occur both in vitro and in vivo, their importance in the pathobiology of the organism is unknown; they may play a role in intracellular persistence and person-to-person spread. The organism's most important virulence factor is *pertussis toxin*, which is composed of a B oligomer-binding subunit and an enzymatically active A protomer that ADP-ribosylates a guanine nucleotide-binding regulatory protein (G protein) in target cells, producing a variety of biologic effects. Pertussis toxin has important mitogenic activity, affects the circulation of lymphocytes, and serves as an adhesin for bacterial binding to respiratory ciliated cells. In animal models, the toxin's effects include histamine sensitization, lymphocytosis promotion, and insulin secretion. Another virulence factor is *filamentous hemagglutinin*,

a component of the cell wall and a bacterial adhesin. *Pertactin* is an outer-membrane protein and another important adhesin. *Fimbriae* are bacterial appendages that also play a role in bacterial attachment; they are the major antigens against which agglutinating antibodies are directed. These agglutinating antibodies have historically been the primary means of serotyping *B. pertussis* strains. Other virulence factors include tracheal cytotoxin, which causes respiratory epithelial damage; adenylate cyclase toxin, which impairs host immune cell function; dermonecrotic toxin, which may contribute to respiratory mucosal damage; and lipo-oligosaccharide, which has properties similar to those of other gram-negative bacterial endotoxins.

PATHOGENESIS

Infection with *B. pertussis* is initiated by attachment of the organism to the ciliated epithelial cells of the nasopharynx. Attachment is mediated by surface adhesins (e.g., pertactin and filamentous hemagglutinin), which bind to the integrin family of cell-surface proteins, probably in conjunction with pertussis toxin. The role of fimbriae in adhesion or maintenance of infection has not been fully delineated. At the site of attachment, the organism multiplies, producing a variety of other toxins that cause local mucosal damage (tracheal cytotoxin, dermonecrotic toxin). Impairment of host defense by *B. pertussis* is mediated by pertussis toxin and adenylate cyclase toxin. There is local cellular invasion, with intracellular bacterial persistence; however, systemic dissemination does not occur. Systemic manifestations (lymphocytosis) result from the effects of the toxins.

The pathogenesis of the clinical manifestations of pertussis is poorly understood. It is not known what causes the paroxysmal cough that is the hallmark of pertussis. A pivotal role for pertussis toxin has been proposed. Proponents of this position point to the efficacy of preventing clinical symptoms with a vaccine containing only pertussis toxoid. Detractors counter that pertussis toxin is not the critical factor because paroxysmal cough also occurs in patients infected with *B. parapertussis*, which does not produce pertussis toxin. It is thought that the neurologic events observed in pertussis, such as seizures and encephalopathy, are due to hypoxia from coughing paroxysms or apnea rather than to the effects of specific bacterial products. *B. pertussis* pneumonia, which occurs in up to 10% of infants with pertussis, is usually a diffuse bilateral primary infection. In older children and adults with pertussis, pneumonia is often due to secondary bacterial infection with streptococci or staphylococci.

IMMUNITY

Both humoral and cell-mediated immunity are thought to be important in pertussis. Antibodies to pertussis toxin, filamentous hemagglutinin, pertactin, and fimbriae are all protective in animal models. Pertussis agglutinins were correlated with protection in early studies of whole-cell pertussis vaccines. Serologic correlates of protection conferred by acellular pertussis vaccines have not been established, although antibody to pertactin, fimbriae, and (to a lesser degree) pertussis toxin correlated best with protection in two acellular pertussis vaccine efficacy trials. The duration of immunity after whole-cell pertussis vaccination is short-lived, with little protection remaining after 10 to 12 years. Data on the duration of protection after acellular pertussis vaccination are still being collected. Although immunity after natural infection has been said to be

lifelong, seroepidemiologic evidence suggests that it may not be and that subsequent episodes of clinical pertussis are prevented by intermittent subclinical infection.

EPIDEMIOLOGY

Pertussis is a highly communicable disease, with attack rates of 80 to 100% among unimmunized household contacts and 20% within households in well-immunized populations. The infection has a worldwide distribution, with cyclical outbreaks every 3 to 5 years (a pattern that has persisted despite widespread immunization). Pertussis occurs in all months; however, in North America, pertussis activity peaks in the summer and autumn.

Before the institution of widespread immunization programs, pertussis was one of the most common infectious causes of morbidity and death. In the United States prior to the 1940s, between 115,000 and 270,000 cases of pertussis were reported annually, with an average yearly rate of 150 cases per 100,000 population. With universal childhood immunization, the number of reported cases fell by >95%, with even more dramatic decreases in mortality. Only 1010 cases of pertussis were reported in 1976. Since that time, however, rates have slowly increased. In 1994, over 15,000 cases of pertussis were reported in the United States.

Although thought of as a disease of childhood, pertussis can affect people of all ages and is increasingly being identified as a cause of prolonged coughing illness in adolescents and adults. In unimmunized populations, pertussis incidence peaks in the preschool years, and well over half of children have the disease before reaching adulthood. In highly immunized populations such as those in North America, the peak incidence is in infants <1 year of age who have not completed the three-dose primary immunization series. Recent trends, however, show an increasing incidence of pertussis in adolescents and adults. In the United States in 1997, ~30% of patients were ≤6 months of age, 25% were adolescents, and 20% were adults. The figures for adolescents and adults are probably underestimates because of a greater degree of underrecognition and underreporting in these age groups. A number of studies of prolonged coughing illness suggest that pertussis may be the etiologic agent in 12 to 30% of adults with cough that does not improve within 2 weeks. A seroprevalence study in the United States estimated an annual incidence of pertussis of 176 cases per 100,000 healthy adults. This high incidence undoubtedly includes subclinical and mild cases that would not be readily identified as pertussis; this fact accounts for infection rates similar to those reported before the introduction of routine immunization.

Severe morbidity and mortality, however, are virtually restricted to infants. In Canada, there were 10 deaths from pertussis between 1991 and 1998; all those who died were infants ≤6 months of age. Although school-age children are the source of infection for most households, adults are the likely source for high-risk infants and may serve as the reservoir of infection between epidemic years. In developing countries, pertussis remains an important cause of infant morbidity and mortality. The World Health Organization estimated that in 1995 over 40 million people worldwide were infected by *B. pertussis* and that 355,000 children died of pertussis.

CLINICAL MANIFESTATIONS

Pertussis is a prolonged coughing illness with clinical manifestations that vary by age ([Table 152-1](#)). Classic pertussis is most often seen in preschool and school-age children, although it is not uncommon among adolescents and adults. After an incubation period averaging 7 to 10 days, an illness develops that is indistinguishable from the common cold and is characterized by coryza, lacrimation, mild cough, low-grade fever, and malaise. After 1 to 2 weeks, this *catarrhal phase* evolves into the *paroxysmal phase*: the cough becomes more frequent and spasmodic with repetitive bursts of 5 to 10 coughs, often within a single expiration. Posttussive vomiting is frequent, with a mucous plug occasionally expelled at the end of an episode. The episode may be terminated by an audible whoop, which occurs upon rapid inspiration against a closed glottis at the end of a paroxysm. During a spasm, there may be impressive neck-vein distension, bulging eyes, tongue protrusion, and cyanosis. Paroxysms may be precipitated by noise, eating, or physical contact. Between attacks, the patient's appearance is normal but increasing fatigue is evident. The frequency of paroxysmal episodes varies widely, from several per hour to 5 to 10 per day. Episodes are often worse at night and interfere with sleep. Weight loss is not uncommon as a result of interference with eating. Most complications occur during the paroxysmal stage. Fever is uncommon and suggests bacterial superinfection.

After 2 to 4 weeks, the coughing episodes become less frequent and less severe -- changes heralding the onset of the *convalescent phase*. This phase can last from 1 to 3 months and is characterized by a gradual resolution of the coughing episodes. For 6 months to a year, intercurrent viral infections may be associated with a recrudescence of paroxysmal cough.

Not all children who develop pertussis have classic disease. Although cough (typically paroxysmal) is nearly always present, whoop may occur in only half of cases. In infants, the illness may be atypical; often apnea and cyanosis are the only symptoms at presentation. Seizures, encephalopathy, and pneumonia are all more common in infants £6 months old. Pertussis-associated infant deaths due to apnea may be confused with sudden infant death syndrome. The clinical manifestations in adolescents and adults may be classic but are more often atypical. In a German study of pertussis in adults, over two-thirds had paroxysmal cough and over one-third had whoop. Adult illness in North America differs from this experience: the cough may be severe and prolonged but is less frequently paroxysmal, and a whoop is uncommon. Vomiting with cough is the best predictor of pertussis as the cause of a prolonged cough in adults. Other features predictive of the disease are a cough at night and exposure to other individuals with a prolonged coughing illness.

COMPLICATIONS

Complications are frequently associated with pertussis and are more common among infants than among older children or adults. Subconjunctival hemorrhages, abdominal and inguinal hernias, pneumothoraces, and facial and truncal petechiae can result from increased intrathoracic pressure generated by severe fits of coughing. Weight loss can follow decreased caloric intake. In a series of over 1100 children <2 years of age who were hospitalized with pertussis, 27.1% had apnea, 9.4% had pneumonia, 2.6% had seizures, and 0.4% had encephalopathy; 10 children (0.9%) died. Pneumonia is

reported in fewer than 5% of adolescents and adults and is usually caused by encapsulated organisms such as *Streptococcus pneumoniae* or *Haemophilus influenzae*; in contrast, infants develop primary *B. pertussis* pneumonia. Pneumothorax, severe weight loss, inguinal hernia, rib fracture, and cough syncope have all been reported in adolescents and adults with pertussis.

DIAGNOSIS

If the classic symptoms of pertussis are present, clinical diagnosis is not difficult. However, particularly in older children and adults, it is difficult to differentiate infections caused by *B. pertussis* and *B. parapertussis* from other respiratory tract infections on clinical grounds. Therefore, laboratory confirmation should be attempted in all cases. Lymphocytosis (absolute neutrophil count, $>10^4$ $10^9/L$) is common among young children (in whom it is unusual with other infections) but not among adolescents and adults. Culture of nasopharyngeal secretions remains the "gold standard" of diagnosis; the best specimen is collected by nasopharyngeal aspiration, in which a fine flexible plastic catheter attached to a 10-mL syringe is passed into the nasopharynx and withdrawn while gentle suction is applied. Since *B. pertussis* is highly sensitive to drying, secretions should be inoculated without delay onto appropriate media (Bordet-Gengou or Regan-Lowe) or the catheter should be flushed with a phosphate-buffered saline solution. An alternative is a nasopharyngeal culture with a calcium alginate swab; again, inoculation of culture plates should be immediate or an appropriate transport medium (such as Regan-Lowe charcoal medium) should be used. Cultures become positive by day 5 of incubation, and *B. pertussis* and *B. parapertussis* can be differentiated by agglutination with specific antisera or by direct immunofluorescence.

Nasopharyngeal cultures in untreated pertussis remain positive for a mean of 3 weeks after the onset of illness; these cultures become negative within 5 days of the institution of appropriate antimicrobial therapy. Since much of the period during which the organism can be recovered from the nasopharynx falls in the catarrhal phase, when the etiology of the infection is not suspected, there is only a small window of opportunity for culture-proven diagnosis. Cultures from infants and young children are more frequently positive than those from older children and adults; this difference may reflect earlier presentation of the former age group for medical care. The increasing availability of the polymerase chain reaction for pertussis in diagnostic laboratories is enhancing the sensitivity of the organism's detection. This method may further laboratory confirmation but does not solve problems related to the long delays in specimen procurement that often are encountered in pertussis cases. Direct fluorescent antibody tests of nasopharyngeal secretions for direct diagnosis may still be available in some laboratories but should not be used because of poor sensitivity and specificity.

As a result of the difficulties with laboratory diagnosis of pertussis in adolescents, adults, and any patient who has been symptomatic for >4 weeks, increasing attention is being given to serologic diagnosis. Enzyme immunoassays detecting IgA and IgG antibodies to pertussis toxin, filamentous hemagglutinin, pertactin, and fimbriae have been developed and assessed for their reproducibility. Two- or fourfold increases in antibody are suggestive of pertussis, although cross-reactivity of some antigens (such as filamentous hemagglutinin) among *Bordetella* species makes it difficult to depend diagnostically on seroconversion involving a single type of antibody. Late presentation

for medical care and prior immunization also complicate serologic diagnosis because the first sample obtained may in fact be a convalescent-phase specimen. Proposed criteria for serologic diagnosis based on a single serum specimen call for comparison of the patient's antibody levels with established population values; for example, a patient with serologically confirmed pertussis might be required to have a titer greater than two or three standard deviations above the mean titer for a normal population. However, at present, no antibody test is widely or commercially available, and no specific serologic criteria are universally accepted.

DIFFERENTIAL DIAGNOSIS

A child presenting with paroxysmal cough, posttussive vomiting, and whoop is likely to have an infection caused by *B. pertussis* or *B. parapertussis*; lymphocytosis increases the likelihood of a *B. pertussis* etiology. Viruses such as respiratory syncytial virus and adenovirus have been isolated from patients with clinical pertussis but probably represent co-infection. In adolescents and adults, among whom paroxysmal cough and whoop are frequently absent, the differential diagnosis of a prolonged coughing illness is more extensive. Pertussis should be suspected in anyone with a cough that does not improve within 14 days, a paroxysmal cough of any duration, or any respiratory symptoms after contact with a laboratory-confirmed case of pertussis. Other etiologies to consider include infections caused by *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, adenovirus, influenza virus, and other respiratory viruses. Use of angiotensin-converting enzyme (ACE) inhibitors, reactive airway disease, and gastroesophageal reflux disease are well-described noninfectious causes of prolonged cough in adults.

TREATMENT

Antibiotics The purpose of antibiotic therapy for pertussis is to eradicate the infecting bacteria from the nasopharynx; therapy does not substantially alter the clinical course unless given early in the catarrhal phase. Erythromycin (preferably the estolate form) is recommended at a dose of 50 mg/kg (maximum, 2 g/d) in three divided doses and reliably clears *B. pertussis* from the nasopharynx after 5 days. A dose of 1 g/d has also been shown to be effective and may be better tolerated. Although a 14-day course of therapy has been recommended to prevent relapse, one study showed that a 7-day course was equally effective. Erythromycin is also effective against other pathogens implicated in cough illness, such as *Mycoplasma* and *Chlamydia*.

Other macrolide antibiotics, such as azithromycin and clarithromycin, are active against *B. pertussis* in vitro, but data on their clinical efficacy are limited; clinical trials are under way. Trimethoprim-sulfamethoxazole (8/40 mg/kg per day in two divided doses) is recommended as an alternative for individuals who cannot use erythromycin, although good clinical data to support this recommendation are lacking. A macrolide-resistant *B. pertussis* strain has been reported from a single case in an outbreak in Arizona.

Immune Globulin Although immune globulin was used widely to treat pertussis in the preantibiotic era, evidence for its effectiveness was lacking and the commercially available product was removed from the market. There is renewed interest in the therapeutic use of immune globulin, particularly for infants who develop pertussis while

still too young to have completed their primary immunization series. A high-titer pertussis toxin immune globulin for intravenous use is undergoing clinical trials.

Supportive Care Infants have the highest rates of complication and death from pertussis; therefore, most infants and older children with severe disease should be hospitalized. Monitoring for apnea and cyanosis, administration of supplemental oxygen, management of secretions, hydration, and nutritional support are the mainstays of care. A quiet environment may decrease the stimulation that can trigger paroxysmal episodes. Assisted ventilation may be required for management of apnea or pneumonia. Use of β -adrenergic agonists and/or glucocorticoids has been advocated by some but has not been proved to be effective. Cough suppressants are not effective and play no role in the management of pertussis.

Infection Control Measures Hospitalized patients with pertussis should be placed in respiratory isolation, with the use of precautions appropriate for pathogens spread by large respiratory droplets. Isolation should continue for 5 days after initiation of erythromycin therapy or for 3 weeks (i.e., until nasopharyngeal cultures are consistently negative) in those individuals unable to tolerate antimicrobial therapy.

PREVENTION

Chemoprophylaxis Because the risk of transmission of *B. pertussis* within households is high, chemoprophylaxis is widely recommended for household contacts of pertussis cases. The effectiveness of chemoprophylaxis, although unproven, is supported by several epidemiologic studies of institutional and community outbreaks of pertussis. In the only randomized placebo-controlled study, erythromycin estolate (50 mg/kg per day in three divided doses; maximum dose, 1 g/d) was effective in reducing bacteriologically confirmed pertussis by 67%; however, there was no decrease in the incidence of clinical disease. Despite these disappointing results, many authorities continue to recommend chemoprophylaxis, particularly in households with members at high risk of severe disease (children <1 year of age). Data are not yet available on use of the newer macrolides for chemoprophylaxis.

Immunization (See also [Chap. 122](#)) The mainstay of pertussis prevention is active immunization. Pertussis vaccine has been available for over 70 years and became widely used in North America after 1940; reported cases of pertussis have since fallen by >90%. Whole-cell pertussis vaccines are prepared through the heating, chemical inactivation, and purification of whole *B. pertussis* organisms. Although effective (average efficacy estimate, 85%, with results in various studies of different products ranging from 30 to 100%), whole-cell pertussis vaccines are associated with adverse events -- both common (fever; injection site pain, erythema, and swelling; irritability) and uncommon (febrile seizures, hypotonic hyporesponsive episodes). Alleged associations of whole-cell pertussis vaccine with encephalopathy, sudden infant death syndrome, and autism, although not substantiated, have spawned an active anti-immunization lobby. The development of acellular pertussis vaccines, which are effective but less reactogenic, has greatly alleviated concerns about the inclusion of pertussis vaccine in the combined infant immunization series. In some countries (Canada, Sweden, Germany), acellular pertussis vaccines are used exclusively for childhood immunization; in the United States, acellular pertussis vaccines are now the preferred product but

whole-cell vaccine is still considered acceptable. In North America, both whole-cell and acellular pertussis vaccines are given as a three-dose primary series at 2, 4, and 6 months of age, with a reinforcing dose between 15 and 18 months of age and a booster dose at 4 to 6 years of age.

A wide variety of acellular pertussis vaccines have been developed, although not all are available in every country. All acellular pertussis vaccines currently available contain pertussis toxoid. Only one monovalent pertussis toxoid vaccine has been licensed in the United States; the remainder of the fully developed vaccines contain filamentous hemagglutinin as well as toxoid. At least four acellular pertussis vaccines also contain pertactin, and two products also contain one or more types of fimbriae. All of the licensed acellular pertussis vaccines have undergone phase 3 efficacy testing. Although differences in study design make direct comparisons difficult, an effort to standardize case definitions and the similarity of some of the studies, which used common vaccine arms to allow "bridging" of the data between studies, have permitted some general conclusions. Even though some would still disagree, most experts have concluded that two-component acellular pertussis vaccines are more effective than monocomponent vaccines and that the addition of pertactin further increases efficacy. The further addition of fimbriae appears to provide some additional protective efficacy against milder disease. In two studies, protection against pertussis by vaccines correlated best with the production of antibody to pertactin, fimbriae, and pertussis toxin.

The development of acellular pertussis vaccines has sparked interest in the potential for control of pertussis in adolescents and adults and in the possibility that pertussis control in those groups will enhance the protection of infants too young to be immunized. Whole-cell pertussis vaccine is contraindicated in individuals³⁷ years of age because of their poor toleration of possible adverse events. However, adult formulations of acellular pertussis vaccines, both alone and in combination with adult-formulation diphtheria-tetanus toxoid, have been demonstrated to be safe and immunogenic in clinical trials in adolescents and adults. Further epidemiologic studies and an efficacy study are under way to better delineate the scope of pertussis illness in adolescents and adults as well as the efficacy of a single dose of acellular pertussis vaccine. These data, along with the results of other studies characterizing the spectrum of pertussis disease in adolescents and adults, will help public health authorities and advisory committees to determine the role of adolescent and adult pertussis immunization.

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153. DISEASES CAUSED BY GRAM-NEGATIVE ENTERIC BACILLI - Thomas A. Russo

GENERAL FEATURES AND PRINCIPLES

EPIDEMIOLOGY

This chapter discusses gram-negative bacilli (GNB) belonging to the medically important genera of the family Enterobacteriaceae (*Escherichia*, *Klebsiella*, *Proteus*, *Enterobacter*, *Serratia*, *Citrobacter*, *Morganella*, *Providencia*, and *Edwardsiella*) as well as the genus *Actinobacter* from the family Neisseriaceae. These bacteria are members of normal animal and human colonic flora and/or residents of a variety of environmental habitats, including long-term-care facilities and hospitals. In healthy humans, *Escherichia coli* is the predominant species of GNB in the colonic flora. GNB (primarily *E. coli*, *Klebsiella*, and *Proteus*) only transiently colonize the oropharynx and skin. In contrast, in the long-term-care and hospital settings, a variety of GNB emerge as the dominant components of the colonizing flora of both mucosal and skin surfaces, particularly with antimicrobial use and increasing severity of disease. Acquisition of these GNB from a variety of reservoirs leads to infection.

STRUCTURE AND FUNCTION

Structurally, these organisms possess an extracytoplasmic outer membrane, a feature shared among gram-negative bacteria. The outer membrane consists of a lipid bilayer and associated proteins, lipoproteins, and polysaccharides [capsule, lipopolysaccharide (LPS)]. This structure interfaces with the environment, including the human host. A variety of components of the outer membrane are critical determinants in mediating the pathogenesis of infection and antimicrobial resistance.

INFECTIOUS SYNDROMES

Depending on both the host and the pathogen, nearly every organ and body cavity can be infected with [GNB](#). *Escherichia* and, to a lesser degree, *Klebsiella* and *Proteus* account for the majority of infections and are the most virulent pathogens of this group. However, the other genera are becoming increasingly important, particularly in long-term-care or hospitalized patients, in large part because of the organisms' innate or acquired resistance to antimicrobial agents and the increasing number of immunocompromised hosts. The mortality rate is significant in many GNB infections and correlates with the severity of illness. Especially problematic are pneumonitis and bacteremia from any source complicated by shock, which have associated mortality rates of 20 to 50%.

DIAGNOSIS

Isolation of [GNB](#) from sterile sites almost always implies infection. Their isolation from nonsterile sites, particularly from soft tissue and respiratory cultures, requires clinical correlation to differentiate colonization from infection.

TREATMENT AND PREVENTION

The antimicrobial resistance of [GNB](#) is variable and is influenced by both location and regional antibiotic use. Empirical antimicrobial choices should be based on local susceptibility patterns, but it is critical to be cognizant of emerging resistance. The acquisition of transferable plasmids that possess genes for extended-spectrum β -lactamases (ESBLs) is increasing. To date, these plasmids are most prevalent in *Klebsiella* and *E. coli*, but they have also been described (albeit less frequently) in most of the enteric GNB. The plasmids confer resistance to third-generation cephalosporins and aztreonam and frequently contain linked resistance determinants for aminoglycosides, tetracyclines, and trimethoprim-sulfamethoxazole (TMP-SMZ). In some outbreaks, strains with ESBLs also exhibit associated fluoroquinolone resistance. Derepression of inducible chromosomal β -lactamases, another important resistance mechanism, may be preexisting or may develop during therapy. This determinant confers resistance to second- and third-generation cephalosporins, to aztreonam, and often to β -lactam/ β -lactamase inhibitor combinations. Of the enteric GNB, *Enterobacter*, *Serratia*, *Citrobacter*, *Proteus vulgaris*, *Proteus penneri*, *Providencia*, and *Morganella* possess this determinant. Although relevant data are suboptimal or conflicting, combination therapy may increase antimicrobial efficacy (particularly in serious infections, such as pneumonitis) and diminish the emergence of resistance. Further, drainage of abscesses and removal of infected foreign bodies are often needed for cure. GNB are commonly part of a polymicrobial infection in which it is difficult to determine the role of each specific pathogen. Although some species are more pathogenic than others, it is usually prudent, if possible, to design an antimicrobial regimen that includes activity against all of the GNB identified, since each is capable of pathogenicity in its own right. Diligent hand washing by health care personnel and avoidance of inappropriate antimicrobial use are the two most important measures for the prevention of infection.

PATHOGENESIS

Multiple bacterial traits are required for various aspects of the pathogenesis of [GNB](#). The possession of specialized virulence genes is what defines pathogens and enables them to infect the host efficiently. As more is learned about these genes, it is becoming clear that hosts and their cognate pathogens have been coadapting throughout evolutionary history. In fact, it has been speculated that infection is just a point on the spectrum of evolutionary development between microbes and the host. At one end of this spectrum is a commensal/symbiotic interaction (e.g., mitochondria -- formerly bacteria -- within eukaryotic cells); at the other is a lethal outcome that results in a "dead-end relationship" (e.g., Ebola virus). During this host-pathogen "chess match" over time, a variety and redundancy of solutions have emerged in both pathogens and hosts that enable the partners to maintain their coexistence ([Table 153-1](#)).

Intestinal Infection Intestinal pathogenic strains of *E. coli* cause gastroenteritis by a variety of unique pathogenic mechanisms. Their virulence traits are for the most part distinct from those of *E. coli* strains that cause disease outside the bowel. This difference is not surprising in light of site-dependent differences in host environments and defense mechanisms.

Extraintestinal Infection Extraintestinal pathogenic strains of *E. coli* (ExPEC) and the

other genera discussed in this chapter cause infection outside the bowel. All are extracellular pathogens and therefore share certain pathogenic features. Innate defense systems (complement, phagocytes) and humoral immunity are the most critical host defense components. As a result, both susceptibility to and severity of infection are increased with dysfunction or deficiencies of these components (e.g., neutrophils). A given pathogen usually possesses multiple adhesins for binding to a variety of host cells (e.g., in *E. coli*: type I, Sfa/Foc, P pili). Nutrient acquisition (e.g., iron via siderophores) requires many genes that are necessary but not sufficient for pathogenesis. The ability to resist the bactericidal activity of complement and professional phagocytes in the absence of antibody (e.g., conferred by capsule or O antigen of [LPS](#)) is one of the defining traits of an extracellular pathogen. Tissue damage (e.g., hemolysis in the case of *E. coli*) may facilitate spread. However, many important virulence genes await identification, and our understanding of many aspects of the pathogenesis of [GNB](#) is in its infancy ([Chap. 120](#)). The ability to induce septic shock is another defining feature of these genera. GNB are the most common cause of this dangerous complication. The lipid A moiety of LPS and probably other bacterial factors as well (e.g., capsule) stimulate a proinflammatory host response, which, if overexuberant, results in shock ([Chap. 124](#)). Lastly, a large number of serotypes (e.g., in *E. coli*, >100 O-specific and >80 capsular antigens) exist within most genera of GNB. This antigenic variability enables immune evasion and successful recurrent infection by strains of the same species and has also impeded vaccine development ([Chap. 122](#)).

ESCHERICHIA COLI INFECTIONS

From a clinical perspective, *E. coli* can be divided into three categories: commensal strains, intestinal pathogenic (enteric or diarrheagenic) strains, and [ExPEC](#).

ETIOLOGY, EPIDEMIOLOGY, AND MANIFESTATIONS

Commensal Strains Commensal strains of *E. coli* constitute the bulk of the facultative fecal flora in most healthy humans. Such strains appear to be adapted for peaceful coexistence with the host and appear not to cause disease within the intestinal tract. Further, in humans, these microorganisms do not usually cause disease outside the intestinal tract except in the presence of precipitating factors, such as an indwelling foreign body or an impairment of host defenses. Commensal *E. coli* strains typically lack the specialized virulence traits of intestinal and [ExPEC](#) strains.

Intestinal Pathogenic Strains In contrast to commensal *E. coli*, intestinal pathogenic strains of *E. coli* are rarely encountered in the fecal flora of healthy hosts and instead appear to be essentially obligate pathogens, causing gastroenteritis or colitis whenever ingested in sufficient quantities by a naive host. At least six distinct "pathotypes" of intestinal pathogenic *E. coli* exist: (1) enterotoxigenic *E. coli* (ETEC); (2) Shiga toxin-producing *E. coli* (STEC)/enterohemorrhagic *E. coli* (EHEC); (3) enteropathogenic *E. coli* (EPEC); (4) enteroinvasive *E. coli* (EIEC); (5) enteroaggregative *E. coli* (EAEC); and (6) diffusely adherent *E. coli* (DAEC). Organisms of these pathotypes are acquired via the fecal-oral route. Transmission occurs predominantly via contaminated food and water for ETEC, STEC, EIEC, EAEC, and DAEC and by person-to-person spread for EPEC (and occasionally STEC). Humans appear to be the major reservoir (except for STEC), since the host range appears to be dictated by species-specific attachment

factors. Although there is some overlap, each pathotype possesses a unique combination of virulence traits that results in a distinctive intestinal pathogenic mechanism; however, these strains are largely incapable of causing disease outside the intestinal tract.

ETEC In tropical or developing countries, several separate episodes of ETEC infection occur in children over the first 3 years of life. The incidence of disease diminishes with age, a pattern suggesting the development of immunity. In industrialized countries, infection usually follows travel to endemic areas. ETEC is the most common cause of traveler's diarrhea ([Chap. 123](#)), being responsible for 25 to 75% of cases. Cases usually develop within the first few weeks of travel. The incidence of infection is decreased by the prudent avoidance of potentially contaminated fluids and foods. ETEC infection is uncommon in the United States, but outbreaks have taken place secondary to contamination of domestic food products. A high inoculum (10^6 to 10^{10} CFU) is needed to cause disease. After ingestion of contaminated water or food (particularly items poorly cooked, unpeeled, or unrefrigerated), the small bowel is colonized during a 1- to 7-day incubation period. Disease is mediated in part by heat-labile (LT) and/or heat-stable (STa) toxin encoded by genes present on transferable plasmids. These toxins stimulate fluid secretion via activation of adenylate cyclase (LT) and/or guanylate cyclase (STa); the result is watery diarrhea accompanied by cramps. Characteristically absent are histopathologic changes of the small bowel; mucus, blood, and inflammatory cells in stool; and fever. The disease spectrum ranges from mild illness to a life-threatening cholera-like illness. Although symptoms are usually self-limited (2 to 6 days), infection may result in significant morbidity and mortality when health care is poor and small and/or undernourished children are affected.

STEC/EHEC STEC strains constitute an emerging group of pathogens that have received substantial media attention as a result of several large outbreaks attributable to the consumption of undercooked ground beef and other foods. Serotype O157:H7 is the most prominent of the more than 30 serotypes associated with the STEC syndrome (see below). Other common serogroups include O26, O39, O103, O104, and O111. The ability to produce Shiga-like toxin (Stx2 and/or Stx1) or related toxins is the critical factor dictating whether a bacterium can cause the clinical syndrome associated with STEC. *Citrobacter* isolates that produce Stx2 and *Shigella* strains that produce related toxins can cause the same syndrome.

A combination of factors are responsible for the emergence of **STEC** disease. A number of animals, including cattle and young calves, serve as a major reservoir for these strains. Ground beef, the most common food source, is frequently contaminated during processing. Further, cattle or other animal manure used as fertilizer can contaminate produce (potatoes, lettuce, sprouts, fallen apples) and water (fecal runoff). It is estimated that $<10^3$ CFU of STEC can cause disease. Therefore, not only can low levels of food or environmental contamination (e.g., water swallowed when swimming) result in disease, but person-to-person transmission becomes an important vehicle for secondary spread (e.g., at day-care centers and in institutions). Because of the low infective dose (which is similar to that of *Shigella*), laboratory-associated infections also take place. Both outbreaks and sporadic disease occur with this group of pathogens, with a seasonal peak in the summer.

In contrast to infection with the other five pathotypes, infection with [STEC](#) occurs more frequently in developed countries, where consumption of processed foods is more common than in developing regions. FoodNet data indicate that O157 strains are the fourth most common reported cause of bacterial diarrhea in the United States (behind *Campylobacter*, *Salmonella*, and *Shigella*). Colonization of the colon and perhaps of the ileum results in symptoms after an incubation period of 3 or 4 days. Maximal disease expression requires Stx2 (produced by most O157 isolates) and/or Stx1 (more commonly produced by non-O157 isolates) as well as other virulence genes (e.g., *eaeA*). Colonic edema and an initial secretory diarrhea may develop into the syndrome's hallmark trait of grossly bloody diarrhea (detected by history or examination) in >90% of cases. Significant abdominal pain and fecal leukocytes are commonly present (70% of cases), but fever is usually absent. Occasionally, *Clostridium difficile*, *Campylobacter*, and *Salmonella* infection present in a similar fashion, as do noninfectious diseases (e.g., appendicitis, inflammatory bowel disease). STEC disease is usually self-limited, lasting 5 to 10 days. This infection can be complicated by the hemolytic-uremic syndrome (HUS), which occurs 2 to 14 days after diarrhea in 2 to 8% of cases, most often in the very young and the elderly. An estimated 50% of all cases of HUS in the United States are caused by STEC infection. This complication is probably mediated by the systemic translocation of Shiga-like toxins and subsequent cellular damage, particularly to endothelial cells in the renal and cerebral microvasculature. HUS is characterized by microangiopathic hemolytic anemia, thrombocytopenia, and renal failure. Neurologic symptoms, with or without fever, can also occur. Although mortality with dialysis support is <10%, residual renal dysfunction and neurologic sequelae may persist.

[EPEC](#) EPEC causes disease primarily in young children, including neonates. This *E. coli* group was recognized as a cause of diarrheal disease when it was found in outbreaks of infantile diarrhea (some in hospital nurseries) in industrialized countries in the 1940s and 1950s. Presently, however, infection due to EPEC is uncommon in developed countries. In contrast, EPEC is an important cause of infant diarrhea (both sporadic and epidemic) in developing countries. Breast-feeding diminishes the incidence of infection. Rapid person-to-person spread may occur. Upon colonization of the small bowel, symptoms develop after an incubation period of 1 or 2 days. Disease is not toxin-mediated. Studies have identified a variety of virulence traits responsible for adherence and a characteristic effacement of microvilli with formation of cuplike, actin-rich pedestals to which the bacteria attach. Diarrheal stool often contains mucus but not blood. Although usually self-limiting, EPEC diarrhea may persist for weeks.

[EIEC](#) EIEC is a relatively uncommon cause of diarrhea and is rarely identified in the United States, although a few food-related outbreaks have been described. In less developed countries, sporadic disease is infrequently recognized in children and travelers. EIEC shares many features with *Shigella* infection; however, unlike *Shigella*, EIEC causes disease only at a high inoculum (10^8 to 10^{10} CFU). Invasion of and replication within the colonic mucosa result in the development of symptoms after an incubation period of 1 to 3 days. Secretory diarrhea may evolve into inflammatory colitis characterized by fever, abdominal pain, tenesmus, and scant stool containing mucus, blood, and inflammatory cells. Symptoms are usually self-limited, lasting 7 to 10 days.

[EAEC](#) and [DAEC](#) These pathotypes have been described primarily in developing countries and mostly affect young children. These strains may also cause some cases

of traveler's diarrhea. A high inoculum is required for infection. In vitro, the organisms exhibit a diffuse or "stacked-brick" adherence pattern. Clinical disease has been associated with persistent diarrhea.

DIAGNOSIS

A practical approach in evaluating diarrhea is to distinguish noninflammatory from inflammatory cases ([Chap. 131](#)). [ETEC](#), [EPEC](#), [EAEC](#), and [DAEC](#) are uncommon causes of noninflammatory diarrhea in the United States. Their diagnosis requires specialized assays that are not routinely available and whose use is rarely indicated since these diseases are self-limiting. ETEC causes the majority of cases of noninflammatory traveler's diarrhea; EAEC and DAEC cause a minority of these cases. Definitive diagnosis generally is not necessary, and empirical antimicrobial treatment is a reasonable approach. If diarrhea persists with treatment, *Giardia* or *Cryptosporidium* should be sought. The diagnosis of infection with [EIEC](#), a rare cause of inflammatory diarrhea in the United States, also requires specialized assays. However, evaluation for [STEC](#), particularly when bloody diarrhea is reported or observed, is appropriate. Although screening for *E. coli* strains that do not ferment sorbitol and subsequent serotyping for O157 constitute the most common method presently used to detect STEC, testing for Shiga-like toxins or toxin genes is more sensitive, specific, and rapid. The latter approach offers another advantage: it detects both non-O157 strains and sorbitol-fermenting strains of O157, which otherwise are difficult to identify. DNA-based, enzyme-linked immunosorbent, and cytotoxicity assays are in various stages of development and will probably become the diagnostic standards in time.

Extraintestinal Pathogenic Strains From both pathogenic and clinical viewpoints, [ExPEC](#) strains are distinct from commensal and intestinal pathogenic strains of *E. coli*. ExPEC strains also make up part of the normal human fecal flora, but, in contrast to commensal strains, possess specialized genes that encode virulence factors enabling the organisms to cause extraintestinal infections ([Table 153-1](#)). ExPEC (as opposed to commensal *E. coli*) causes the majority of cases of urinary tract infection (UTI), bacteremia, and neonatal meningitis. It is likely that ExPEC also causes the majority of other extraintestinal infections due to *E. coli*. Entry into an extraintestinal site (e.g., the urinary tract or the peritoneum) -- not acquisition -- is the limiting factor for infection. All age groups, all types of hosts, and nearly every organ and site are susceptible to infection by ExPEC. Normal, previously healthy hosts infected with ExPEC can become severely ill and die. However, adverse outcomes are more prevalent in the presence of coincidental disease and abnormalities in host defenses. Typical extraintestinal infections include UTI, diverse intraabdominal infections, pneumonia (particularly in hospitalized and institutionalized patients), meningitis (mainly in neonates and patients who have undergone neurosurgery), intravascular device infection, osteomyelitis, and soft tissue infection (which usually occurs in the setting of tissue compromise). Bacteremia can accompany infection at any of these sites. Although *E. coli* is considered to be primarily a community-acquired pathogen, it is the most frequently isolated of the [GNB](#) in the ambulatory, long-term-care, and hospital settings. The scope and magnitude of infection caused by ExPEC are as great as for any other invasive bacterial pathogen. Although these isolates do not make headlines, billions of health care dollars, millions of workdays, and thousands of lives are lost to this group of pathogens each year.

Infectious Syndromes

URINARY TRACT INFECTION (UTI) The urinary tract is the site most frequently infected by [ExPEC](#). About 90% of ambulatory [UTIs](#) and 25 to 35% of long-term-care and hospital UTIs are due to *E. coli*. The majority of UTIs occur in seven epidemiologically defined groups: children <1 year of age, school-age girls, premenopausal women, men with prostatic or other causes of urinary tract obstruction, postmenopausal women, individuals with neurogenic bladders, and patients with indwelling urinary catheters. In premenopausal women, diaphragm-spermicide use, sexual activity, and a history of UTI are risk factors for infection; 20% of women with an initial infection have frequent recurrences (0.3 to >20 per year). In postmenopausal women, estrogen replacement decreases the incidence of UTI. Acceptance of the diagnosis of UTI in males (beyond the first year of life) requires clear documentation since this infection is unusual in the absence of a history of instrumentation or anal intercourse. UTI in premenopausal women alone accounts for an estimated 7 million office visits and >\$1 billion in direct medical costs annually. UTI is the second most common infection (behind lower respiratory tract infection) responsible for hospitalization.

Uncomplicated urethritis or cystitis occurs most commonly and is characterized by symptoms of dysuria, frequency, and suprapubic pain. Fever and/or back pain suggests progression to pyelonephritis. Pregnant women are at unusually high risk for this complication, which can adversely affect the outcome of pregnancy. As a result, prenatal screening for bacteriuria, with treatment when the results are positive, is the standard of care. Fever may take 5 to 7 days to resolve completely in appropriately treated patients with pyelonephritis but should fall over time. Persistently elevated or increasing fever and neutrophil counts should prompt evaluation for intrarenal or perinephric abscess and/or obstruction. Renal parenchymal damage and loss of renal function occur primarily in the setting of obstruction. Prostatic infection is generally a complication of [UTI](#) in men with a history of instrumentation and/or prostatic hypertrophy. The diagnosis and treatment of UTI are detailed in [Chap. 280](#) and are tailored according to the host, the nature and site of infection, and the local pattern of antimicrobial susceptibility.

ABDOMINAL INFECTION The abdomen is the second most frequent site of extraintestinal infection due to *E. coli*. The majority of abdominal *E. coli* infections develop outside the hospital. Any inciting event that results in disruption of the bowel mucosa (particularly the colonic mucosa) often leads to acute peritonitis (secondary peritonitis; [Chap. 130](#)). This process is usually polymicrobial, but *E. coli* is isolated in most cases. Bacteremia often complicates this acute stage of infection. Abscess formation within the peritoneum may follow the acute stage or may develop as a consequence of subclinical fecal spillage (e.g., diverticulitis, chronic appendicitis). Intraperitoneal abscesses are almost always polymicrobial, with *E. coli* as the most common [GNB](#) isolated. *E. coli* is also the GNB most often responsible for primary hepatic abscesses, hepatic abscesses in the setting of biliary disease and obstruction, septic cholangitis/cholecystitis, pancreatic abscesses, and infected pancreatic pseudocysts. This organism is the leading cause of spontaneous bacterial peritonitis, usually seen in patients who have ascites associated with cirrhosis or occasionally with malignancy. *E. coli* occasionally causes splenic abscesses and peritoneal dialysis-associated peritonitis.

([Chap. 130](#)).

PNEUMONIA *E. coli* is not usually considered a cause of pneumonia ([Chap. 255](#)). Enteric **GNB** are responsible for only 2 to 5% of cases of community-acquired pneumonia, in part because these organisms only transiently colonize the oropharynx in a minority of healthy individuals. In contrast, oral colonization with *E. coli* and other GNB increases with the severity of illness and with antibiotic use. Thus, GNB are a common cause of pneumonia acquired by residents of long-term-care institutions and are the most frequent cause of hospital-acquired pneumonia ([Chap. 135](#)), particularly in postoperative and intensive care patients. Despite significant institutional variation, *E. coli* is generally the third or fourth most commonly isolated GNB in these settings, behind *Pseudomonas* and *Klebsiella*. Regardless of the host, severe disease and high mortality rates (20 to 60%) are usually seen when GNB cause pneumonia. Tissue necrosis, probably due to cytotoxins produced by GNB, is common. Infection is usually acquired by small-volume aspiration but occasionally occurs via hematogenous spread, in which case multifocal nodular infiltrates can be seen.

MENINGITIS (See [Chap. 372](#)) **ExPEC** are a leading cause of meningitis in the first month of life. The majority of responsible strains possess the K1 capsular serotype. Outside this setting, meningitis due to *E. coli* is uncommon, occurring predominantly with cirrhosis ([Chap. 299](#)) or disruption of the meninges due to surgery or trauma.

CELLULITIS/MUSCULOSKELETAL INFECTION Infections of decubitus ulcers and the lower extremities in diabetic patients (or other hosts with neurovascular compromise) are usually polymicrobial. *E. coli* frequently contributes to infection of decubiti and occasionally to lower-extremity infections in these patients. It may occasionally cause cellulitis or burn site or surgical wound infection, particularly when the infection originates close to the perineum. Osteomyelitis secondary to contiguous spread can occur in these settings. Hematogenously acquired osteomyelitis, particularly of vertebral bodies, is more common than is appreciated, accounting for 10% of cases in some series ([Chap. 129](#)). *E. coli* occasionally causes orthopedic device-associated infection and is a rare cause of hematogenously acquired myositis. Myositis or fasciitis of the upper leg should prompt an evaluation for an abdominal source with contiguous spread.

ENDOVASCULAR INFECTION Extraintestinal isolates of *E. coli* cause a significant minority of intravascular device-associated infections ([Chap. 135](#)). Despite being one of the most common causes of bacteremia, however, *E. coli* rarely seeds native heart valves and is an uncommon cause of prosthetic valve endocarditis. Likewise, *E. coli* infections of aneurysms and vascular grafts are uncommon.

MISCELLANEOUS INFECTIONS *E. coli* can cause infection in nearly every organ and site. This organism causes a minority -- but still a significant number -- of surgical site infections (e.g., mediastinitis) and cases of complicated sinusitis. It uncommonly causes endophthalmitis.

BACTEREMIA *E. coli* bacteremia can result from extraintestinal infection of any site. The incidences of community-acquired and long-term-care/hospital-acquired bacteremia are roughly equal. Overall, it has been amply documented that *E. coli* and *Staphylococcus aureus* are the most common blood isolates (range for *E. coli*, 16 to

37%). *E. coli* is the [GNB](#) most frequently isolated from blood in the ambulatory setting and in most long-term-care and hospital settings. When *E. coli* is isolated from the blood, it is almost always clinically significant. Approximately 15% of bacteremias are complicated by septic shock. Two-thirds of bacteremias arise from the urinary tract; these infections are particularly common in the setting of pyelonephritis or obstruction (including kinked urinary catheters) or instrumentation of the urinary tract in the presence of *E. coli*. However, one should be cautious in identifying the urinary tract as the source of *E. coli* bacteremia in the absence of appropriate symptoms, despite a positive urine culture. Asymptomatic bacteriuria is common, particularly in women, even in the absence of an indwelling bladder catheter, with a prevalence of 15 to 25% after the age of 60. Therefore, occult abdominal or other sources should be considered. The abdomen is the second most common source, accounting for 25% of episodes. Although obstructive biliary tract disease (stones, tumor) and overt disruption of the bowel are responsible for many cases of *E. coli* bacteremia, some abdominal sources, such as abscesses, are remarkably silent clinically and require identification via imaging studies (e.g., computed tomography). Soft tissue, bone, and pulmonary infection are the next most frequent sources for bacteremia. As stated above, endocarditis is uncommon, occurring in only 2 of 861 bacteremias in a recent series. In the setting of chemotherapy-induced fever and neutropenia, *E. coli* is a common cause of bacteremia, usually secondary to intestinal mucositis. It is prudent in this situation, however, to exclude perirectal infection or typhlitis ([Chap. 89](#)). [ExPEC](#) strains are among the most common causes of sepsis in neonates.

Diagnosis Strains of *E. coli* that cause extraintestinal infections usually grow both aerobically and anaerobically within 24 h on standard diagnostic media and are easily identified by the clinical microbiology laboratory using standard biochemical criteria ([Chap. 121](#)). More than 90% of these strains are rapid lactose fermenters.

TREATMENT

Although *E. coli* is generally perceived as an "antibiotic-friendly" pathogen, resistance has increased over the past decade. In general, the frequency of ampicillin resistance precludes its empirical use, even in community-acquired infections. Rates of resistance to first-generation cephalosporins and [TMP-SMZ](#) in community-acquired strains are increasing in the United States (5 to 25%) and are even higher in Europe and developing countries. Not surprisingly, long-term-care and hospital isolates are more resistant than community isolates. Significant resistance (30 to 40%) to amoxicillin/clavulanic acid and piperacillin has been increasingly reported. Fortunately, resistance to second- and third-generation cephalosporins [mean rate, 3.2% according to 1998 National Nosocomial Infections Study (NNIS) data], fourth-generation cephalosporins, quinolones, monobactams (e.g., aztreonam), carbapenems (e.g., imipenem), and aminoglycosides is generally found in <10% of strains. An exception is in settings where quinolone prophylaxis is used extensively (patients with leukemia, transplant recipients); in these settings, significant quinolone resistance has emerged. Acquisition of plasmids containing [ESBLs](#) and other resistance determinants is likely to increase.

The mainstay of treatment for all diarrheal syndromes is the appropriate replacement of water and electrolytes ([Chap. 159](#)). The use of prophylactic antibiotics to prevent

traveler's diarrhea should be discouraged, especially in light of high rates of antibiotic resistance. When diarrhea is free of mucus and blood, early patient-initiated treatment with a quinolone significantly decreases the duration of illness, and the use of loperamide may halt symptoms in a few hours ([Chap. 123](#)). Treatment of [STEC](#) is controversial since antibiotics may increase the incidence of [HUS](#), perhaps via increased release of Shiga-like toxin.

INFECTIONS CAUSED BY OTHER GRAM-NEGATIVE ENTERIC BACILLI

KLEBSIELLA INFECTIONS

K. pneumoniae is the most important *Klebsiella* species medically, causing community-acquired, long-term-care, and hospital infections. *K. oxytoca* is primarily a pathogen in long-term-care and hospital settings. *K. rhinoscleromatis* and *K. ozaenae* are usually isolated from patients in tropical climates. *Klebsiella* species are broadly prevalent in the environment and colonize mucosal surfaces of mammals. In healthy humans, *K. pneumoniae* colonization rates range from 5 to 35% in the colon and from 1 to 5% in the oropharynx; the skin is usually colonized only transiently. In long-term-care facilities and hospitals, colonization occurs with *K. oxytoca* as well, and carriage rates are significant among both workers and patients. Person-to-person spread is thought to be the predominant mode of acquisition. Classically, *Klebsiella* is associated with community-acquired pneumonia, primarily in alcoholics. However, the majority of *Klebsiella* infections now occur in long-term-care facilities and hospitals. *Klebsiella* causes a spectrum of extraintestinal infections similar to that caused by *E. coli*. However, extraintestinal infections due to *Klebsiella* occur at a lower incidence in all sites except the respiratory tract. These variances in infection rates are probably due to differences in colonization and site-specific virulence traits. Antibiotic-resistant strains have been responsible for a number of nosocomial outbreaks of infection in intensive care units (ICUs) and neonatal nurseries. The most common clinical syndromes are pneumonia, [UTI](#), abdominal infection, surgical site infection, soft tissue infection, and subsequent bacteremia. *K. rhinoscleromatis* is the causative agent of rhinoscleroma, a slowly progressive (months to years) mucosal upper respiratory infection that causes necrosis and occasional obstruction of the nasal passages. *K. ozaenae* has been implicated as a cause of chronic atrophic rhinitis.

Infectious Syndromes

Pneumonia *K. pneumoniae* causes only a small proportion of cases of community-acquired pneumonia ([Chap. 255](#)). This infection occurs primarily in hosts with underlying disease, such as alcoholics, diabetics, and individuals with chronic lung disease. As in all pneumonias due to enteric [GNB](#), purulent sputum production and "airspace" disease on x-ray are typical. Presentation with earlier, less extensive infection is more common than that with the classic lobar infiltrate with a bulging fissure. Pulmonary necrosis, pleural effusion, and empyema occur with progression. Pulmonary infection in residents of long-term-care facilities and in hospitalized patients is especially frequent because of increased oropharyngeal colonization rates. Mechanical ventilation is an important risk factor.

[UTI](#) The incidence of *K. pneumoniae* UTI among healthy adults is only 1 to 2%.

However, in complicated UTIs (including those associated with indwelling bladder catheters), the incidence of *Klebsiella* infection increases to 5 to 17%.

Abdominal Infection *Klebsiella* causes a spectrum of abdominal infections similar to that caused by *E. coli* but is less frequently isolated from these infections.

Other Infections *Klebsiella* cellulitis or soft tissue infection occurs most frequently in devitalized tissue (e.g., decubitus ulcers, diabetes, burn sites) or in immunocompromised hosts. *Klebsiella* causes a significant minority of surgical site infections and nosocomial sinusitis cases as well as occasional cases of osteomyelitis contiguous to soft tissue infection, temperate myositis, and neonatal meningitis or meningitis associated with neurosurgery.

Bacteremia *Klebsiella* infection at any site can result in bacteremia. Infections of the urinary tract, respiratory tract, and abdomen each account for 15 to 30% of *Klebsiella* bacteremias. Intravascular device-related infection is another important source (5 to 15%). Surgical site infection and other miscellaneous infections account for the rest. *Klebsiella* is one of the agents that causes sepsis neonatorum and bacteremia with fever and neutropenia. Like enteric [GNB](#) in general, *Klebsiella* rarely causes endocarditis or endovascular infection.

Diagnosis Except for *K. rhinoscleromatis* and *K. ozaenae*, *klebsiellae* are readily isolated and identified by the laboratory and usually ferment lactose.

TREATMENT

K. pneumoniae and *K. oxytoca* have similar antibiotic resistance profiles. They are intrinsically resistant to ampicillin and ticarcillin. [NNIS](#) data from 1998 indicated that 10.7% of [ICU](#) patients were infected with strains resistant to third-generation cephalosporins. This increasing degree of resistance is primarily mediated by transferable plasmids containing genes that encode [ESBLs](#). In addition, these plasmids usually possess linked resistance determinants for aminoglycosides, tetracyclines, and [TMP-SMZ](#). Resistance to β -lactam/ β -lactamase inhibitor combinations and second-generation cephalosporins independent of ESBL-containing plasmids has also been increasingly described. In some outbreaks, ESBL-containing strains have displayed associated fluoroquinolone resistance. At this time, resistance to quinolones, cephamycins (e.g., cefoxitin), fourth-generation cephalosporins (e.g., cefepime), and amikacin is generally <10% but will probably increase. Carbapenems (e.g., imipenem) remain the most active antibiotic class against *Klebsiella*.

PROTEUS INFECTIONS

P. mirabilis causes 90% of *Proteus* infections. These infections occur in the community, in long-term-care facilities, and in hospitals. *P. vulgaris* and *P. penneri* are isolated primarily from infections contracted in long-term-care facilities or hospitals. *Proteus* species are part of the colonic flora of a wide variety of mammals, birds, fish, and reptiles. Their ability to generate histamine from contaminated fish has implicated these [GNB](#) in the pathogenesis of scombroid (fish) poisoning ([Chap. 131](#)). *P. mirabilis* colonizes healthy humans (prevalence, 50%), but *P. vulgaris* and *P. penneri* are isolated

primarily from individuals with underlying disease. The urinary tract is overwhelmingly the favored site of *Proteus* infection, in part because of unique pathogenic properties of the organisms. However, *Proteus* less commonly causes infection in a variety of extraintestinal sites.

Infectious Syndromes

UTI *P. mirabilis* causes only 1 to 2% of cases of **UTI** in healthy women, and *Proteus* species cause only 5% of cases of hospital-acquired UTI. However, *Proteus* is responsible for 10 to 15% of cases of complicated UTI, primarily those associated with catheterization; in the setting of long-term catheterization, their prevalence rate ranges from 20 to 45%. This high prevalence is due to the ability of *Proteus* to produce high levels of urease, which hydrolyzes urea to ammonia and results in alkalization of the urine. This situation, in turn, leads to precipitation of organic and inorganic compounds, with the formation of struvite and carbonate-apatite crystals, biofilm formation on catheters, and/or the development of calculi. *Proteus* becomes associated with the stones and usually can be eradicated only by complete stone removal. Over time, staghorn calculi may form and lead to obstruction and renal failure. Therefore, an unexplained alkaline urine should be cultured for *Proteus*, and identification of a *Proteus* species should prompt an evaluation for calculi.

Other Infections Although the majority of *Proteus* infections arise from the urinary tract, these bacteria occasionally cause pneumonia (primarily in long-term-care or hospitalized patients), nosocomial sinusitis, intraabdominal abscesses, biliary tract infection, surgical site infection, soft tissue infection (especially decubitus and diabetic ulcers), and osteomyelitis (primarily contiguous); they rarely cause temperate myositis. In addition, *Proteus* occasionally causes neonatal meningitis (with the umbilicus often implicated as the source), and cerebral abscess is a common complication.

Bacteremia The majority of *Proteus* bacteremias originate from the urinary tract; however, any of the less common sites of infection are also potential sources. Infection of intravascular devices should also be considered. Endovascular infection is rare. *Proteus* species are occasional agents of sepsis neonatorum and bacteremia with fever and neutropenia.

Diagnosis *Proteus* is readily isolated and identified by the laboratory. The majority of strains are lactose negative, and most demonstrate characteristic "swarming" motility on agar plates.

TREATMENT

P. mirabilis remains susceptible to most antimicrobial agents except tetracycline. Resistance to ampicillin and first-generation cephalosporins has been acquired by 10 to 20% of strains. Acquisition of **ESBLs** remains uncommon. *P. vulgaris* and *P. penneri* are more resistant. Resistance to ampicillin and first-generation cephalosporins is the rule for these species. Derepression of an inducible chromosomal β -lactamase (not present in *P. mirabilis*) occurs in up to 30% of strains. Imipenem, fourth-generation cephalosporins (e.g., cefepime), aminoglycosides, **TMP-SMZ**, and quinolones have excellent activity (90 to 100%).

ENTEROBACTER INFECTIONS

E. cloacae and *E. aerogenes* are responsible for most *Enterobacter* infections (65 to 75% and 15 to 25%, respectively); *E. agglomerans*, *E. sakazakii*, and *E. gergoviae* are less commonly isolated (5%, 1%, and <1%, respectively). These organisms cause primarily health care- or hospital-related infections. They are prevalent in foods, environmental sources (including health care facility equipment), and a wide variety of animals. Only a minority of healthy humans are colonized, but the percentage increases significantly in the setting of long-term care or hospitalization. Although colonization is an important prelude to infection, direct introduction via intravenous lines (e.g., contaminated intravenous fluids, pressure monitors) also occurs. Significant antibiotic resistance has developed in *Enterobacter* species and has contributed to their emergence as prominent nosocomial pathogens. Individuals who have received prior antibiotic treatment, who have comorbid disease, and who are patients in [ICUs](#) are at greatest risk for infection. *Enterobacter* causes a spectrum of extraintestinal infections similar to that described for other [GNB](#) in this chapter.

Infectious Syndromes Pneumonitis, [UTI](#) (particularly catheter-related), intravascular device-related infection, surgical wound/site infection, and abdominal infection (primarily postoperative or device-related -- e.g., biliary stents) are the most common syndromes encountered. Nosocomial sinusitis, meningitis related to neurosurgical procedures (including use of pressure monitors), osteomyelitis, and endophthalmitis after eye surgery are less frequent. *E. sakazakii* is commonly responsible for neonatal meningitis/sepsis (particularly in premature infants), and contaminated formula has been implicated as a source of this infection. Neonatal meningitis is frequently associated with brain abscesses. Bacteremia can result from infection at any of these sites. In the setting of *Enterobacter* bacteremia, contamination of intravenous fluids, blood products, catheter-flushing fluids, pressure monitors, and dialysis equipment should always be considered, particularly with epidemic infection. *Enterobacter* can also cause bacteremia in patients with fever and neutropenia. *Enterobacter* endocarditis is rare, primarily affecting abnormal native or prosthetic valves.

Diagnosis *Enterobacter* is readily isolated and identified by the laboratory. Most strains are lactose positive.

TREATMENT

Significant antimicrobial resistance exists among *Enterobacter* strains. Ampicillin and the first- and second-generation cephalosporins have little or no activity. The extensive use of third-generation cephalosporins has resulted in the selection of strains that produce high levels of β -lactamase (i.e., derepression of β -lactamase), which confers resistance to second- and third-generation cephalosporins, monobactams (e.g., aztreonam), and (frequently) β -lactam/ β -lactamase inhibitor combinations. Resistant isolates may emerge during therapy; their presence should be considered a possibility when clinical deterioration follows several days of improvement. A 34% resistance rate to third-generation cephalosporins was reported in [ICU](#) isolates in 1998 ([NNIS](#) data). Imipenem, fourth-generation cephalosporins (e.g., cefepime), aminoglycosides (amikacin > gentamicin), [TMP-SMZ](#), and quinolones have retained excellent activity (90

to 99%). However, increasing resistance to quinolones, in conjunction with the increased use of these agents, is a concern.

ACINETOBACTER INFECTIONS

A. baumannii is responsible for the majority of *Acinetobacter* infections; a minority are due to *A. calcoaceticus* and *Acinetobacter* genospecies 3 and 13TU. *Acinetobacter* is highly prevalent in the environment. It is found in most water and soil samples and has a wide habitat. *Acinetobacter* has been cultured from the moist skin of healthy humans; increased colonization of the skin and the respiratory and gastrointestinal tracts occurs in individuals in long-term-care facilities and hospitals. Reservoirs for acquisition in these settings include health care personnel, medical equipment, food, and the surrounding environment. Infections in healthy people in the community are unusual, but a few reports of pneumonia have been published. The overwhelming majority of infections are acquired in the hospital and long-term-care facilities. The spectrum of extraintestinal infections caused by *Acinetobacter* is similar to that caused by other [GNB](#). *Acinetobacter* species account for 1 to 3% of hospital-acquired infections and affect primarily immunocompromised hosts and patients with comorbid disease. [ICUs](#) are a prominent site of *Acinetobacter* infection. In some centers, the incidence of *Acinetobacter* infections, particularly those due to antibiotic-resistant strains, is increasing. Both sporadic and epidemic infection occurs, usually after the first week of hospitalization.

Infectious Syndromes The respiratory tract (particularly in ventilated patients) and intravascular devices (particularly for non-*A. baumannii* species) are the favored sites of infection. A catheterized urinary tract, postoperative sites, burn sites, biliary stents, sinuses (with tube-related ostial obstruction), and neurosurgical infections (site- or device-associated -- e.g., pressure monitors) are less common. Uncommon infections include contiguous osteomyelitis, peritonitis associated with continuous ambulatory peritoneal dialysis, and ophthalmic infection. The respiratory tract and intravascular devices are the most common sources for bacteremia.

Diagnosis On Gram's stain, *Acinetobacter* organisms usually appear as short [GNB](#) or coccobacilli. They are strictly aerobic, nonfermenting, and readily isolated and identified.

TREATMENT

Many strains of *Acinetobacter* are highly resistant to antimicrobial agents. Empirical combination therapy is prudent pending susceptibility studies. Ampicillin, aztreonam, and the first- and second-generation cephalosporins possess little or no activity against these species. The activity of mezlocillin, piperacillin, quinolones, third- and fourth-generation cephalosporins, aminoglycosides, and b-lactam/b-lactamase inhibitor combinations is variable. Imipenem is presently the most active antimicrobial (>95% sensitivity), and b-lactam/sulbactam combinations are often active. Amikacin, third- and fourth-generation cephalosporins, quinolones, and combinations consisting of a b-lactam other than sulbactam plus a b-lactamase inhibitor retain significant activity in some centers, while highly resistant strains are more common in other centers.

SERRATIA INFECTIONS

S. marcescens causes the majority of *Serratia* infections (>90%), and *S. liquefaciens* is occasionally isolated. *Serratia* are found primarily in the environment, including health care institutions and particularly in moist foci. Although strains have been isolated from a variety of animals, healthy humans are rarely colonized. In long-term-care facilities or hospitals, diverse reservoirs for the organisms include the hands of health care personnel, food, sinks, respiratory and other hospital equipment, intravenous solutions, blood products (e.g., platelets), lotions, irrigation solutions, and even disinfectants. Infection results from either direct inoculation (e.g., via intravenous fluid) or colonization (primarily of the respiratory tract) and subsequent infection. Sporadic infection is most common, but occasional epidemics and common-source outbreaks occur. The spectrum of extraintestinal infections caused by *Serratia* is similar to that for other [GNB](#). *Serratia* species account for 1 to 3% of hospital-acquired infections.

Infectious Syndromes The respiratory tract, the genitourinary tract, intravascular devices, and surgical wounds and sites are the most common sites of *Serratia* infection and sources of *Serratia* bacteremia. Soft tissue infections, including myositis, osteomyelitis, abdominal and biliary tract infection (postprocedural), contact lens-associated infection, endophthalmitis, septic arthritis (primarily with intraarticular injections), and infusion-related bacteremias occur less commonly. *Serratia* are uncommon causes of neonatal or postsurgical meningitis and bacteremia associated with fever and neutropenia. Endocarditis is rare.

Diagnosis *Serratia* are readily cultured and identified by the laboratory and are usually lactose negative. A minority of *S. marcescens* strains are red-pigmented.

TREATMENT

A high proportion of *Serratia* strains (>80%) are resistant to ampicillin and the first-generation cephalosporins. Significant resistance to ticarcillin, piperacillin, gentamicin, second- and third-generation cephalosporins, b-lactam/b-lactamase inhibitor combinations, and aztreonam has developed and may evolve during therapy. Imipenem, amikacin, cefepime, and quinolones are the most active agents, with >90% of strains susceptible.

CITROBACTER INFECTIONS

C. freundii and *C. koseri* (formerly *C. diversus*) cause the majority of human *Citrobacter* infections, which are similar epidemiologically and clinically to *Enterobacter* and *Acinetobacter* infections. *Citrobacter* organisms are commonly present in water, food, soil, and the intestinal tracts of animals. *Citrobacter* is part of the normal fecal flora in a minority of healthy humans, but colonization rates increase in long-term care facilities and hospitals -- the settings in which nearly all infections occur. *Citrobacter* species account for 1 to 2% of nosocomial infections. The affected hosts are usually immunocompromised or have comorbid disease. *Citrobacter* causes extraintestinal infections whose spectrum is similar to that described for other [GNB](#).

Infectious Syndromes The urinary tract is the site of 40 to 50% of infections due to *Citrobacter*. Less commonly infected sites include the biliary tree (particularly with

stones or obstruction), the respiratory tract, surgical sites, soft tissue (e.g., decubitus ulcers), the peritoneum, and intravascular devices. Osteomyelitis (usually contiguous), neurosurgery-related infection, and myositis occur rarely. *Citrobacter* is also an uncommon cause of neonatal meningitis; *C. koseri* accounts for 90% of cases due to this genus. A frequent and devastating complication of this infection (occurring in 50 to 80% of cases) is the development of brain abscesses. Bacteremia is most commonly due to [UTI](#), biliary or abdominal infection, or intravascular devices. *Citrobacter* is an uncommon cause of bacteremia in the setting of fever and neutropenia. Endocarditis or endovascular infection is rare.

Diagnosis *Citrobacter* species are readily isolated and identified, often as part of a polymicrobial culture; 35 to 50% of isolates are lactose positive.

TREATMENT

C. freundii is generally more resistant to antibiotics than *C. koseri*. Ampicillin and the first- and second-generation cephalosporins display poor activity against *Citrobacter*. Resistance is variable but increasing to ticarcillin, mezlocillin, piperacillin, aztreonam, quinolones, gentamicin, and third-generation cephalosporins; such resistance may evolve during therapy. β -lactamase inhibitors usually do not improve susceptibility to β -lactam agents. Imipenem, amikacin, and the fourth-generation cephalosporins are most active, with >90% of strains sensitive.

MORGANELLA AND PROVIDENCIA INFECTIONS

M. morganii (formerly *Proteus morganii*), *P. stuartii*, and (to a lesser degree) *P. rettgeri* (formerly *Proteus rettgeri*) are the members of these genera that are responsible for human infections. The epidemiologic, pathogenic, and clinical manifestations of these organisms are similar to those of *Proteus* species; however, *Morganella* and *Providencia* are almost exclusively pathogens of persons in long-term-care facilities and, to a lesser degree, hospitalized patients.

Infectious Syndromes These species are primarily urinary tract pathogens, most often associated with long-term (>30-day) catheterization. [UTI](#) in uncatheterized or short-term-catheterized individuals is uncommon. Biofilm formation or encrustation of the catheter usually develops and may lead to catheter obstruction. Likewise, infection may result in the development of struvite bladder or renal stones, which, in turn, may lead to renal obstruction and serve as foci for relapse. Other infectious syndromes occur less commonly but include surgical wound/site infections, soft tissue infection (primarily decubitus and diabetic ulcers), burn site infection, pneumonia (particularly ventilator-associated), intravascular device infection, and intraabdominal infection. Rarely, the other extraintestinal infections described for [GNB](#) also occur. Bacteremia is uncommon; although any infected site can serve as the source, the urinary tract accounts for the majority of cases, with surgical wound/site and soft tissue infections less frequently responsible.

Diagnosis *M. morganii* and *Providencia* are readily isolated and identified. Nearly all isolates are unable to ferment lactose.

TREATMENT

Morganella and *Providencia* may be highly resistant to antimicrobial agents. Ampicillin and the first-generation cephalosporins exhibit poor activity against these organisms. Variable resistance is emerging (and may evolve during therapy) against ticarcillin, mezlocillin, piperacillin, aztreonam, gentamicin, [TMP-SMZ](#), and the second- and third-generation cephalosporins and quinolones. Theb-lactamase inhibitor tazobactam (but not sulbactam or clavulanic acid) improves susceptibility to b-lactam agents somewhat. Imipenem, amikacin, and the fourth-generation cephalosporins are most active, with >90% of strains susceptible. Removal of an infected catheter or stones is critical for eradication of the organisms from the urinary tract.

EDWARDSIELLA INFECTION

E. tarda is the only member of this genus associated with human disease. This organism is found predominantly in both freshwater and marine environments and in the animals that live in these environments. Human acquisition occurs primarily during interaction with these reservoirs. *E. tarda* infection is rare in the United States; most recently reported cases are from Southeast Asia. This pathogen shares some of the clinical features of both *Salmonella* species and *Vibrio vulnificus*.

Infectious Syndromes Gastroenteritis is the predominant infectious syndrome reported (50 to 80% of infections). Self-limiting watery diarrhea is most frequent; however, cases of severe colitis responding to therapy have also been described. The most common extraintestinal infection is wound infection due to direct inoculation, which is often associated with freshwater, marine, or snake-related injuries. Other infectious syndromes appear to be due to invasion of the gastrointestinal tract and subsequent bacteremia. The majority of afflicted hosts have either liver disease or an iron-overload state (e.g., sickle cell disease). A primary bacteremic syndrome, sometimes complicated by meningitis, has been described and has a 40% case-fatality rate. Visceral (primarily hepatic) or intraperitoneal abscesses have also been reported.

Diagnosis Although *E. tarda* can readily be isolated and identified, most laboratories do not routinely identify it from stool.

TREATMENT

E. tarda is sensitive to most [GNB](#)-appropriate antimicrobial agents. Gastroenteritis is generally self-limiting, but treatment with [TMP-SMZ](#) or a quinolone may expedite its resolution. In the setting of overwhelming sepsis, quinolones, third- or fourth-generation cephalosporins, imipenem, and aminoglycosides -- alone or in combination -- are the safest choices pending susceptibility information.

INFECTIONS CAUSED BY MISCELLANEOUS GENERA

Species from genera of [GNB](#) such as *Hafnia*, *Kluyvera*, *Cedecea*, *Pantoea*, and *Ewingella* are occasionally isolated from a variety of clinical specimens, including blood, sputum, cerebrospinal fluid, joint fluid, biliary drainage, wounds, and sputum. Although their role in disease has not always been defined, these strains appear to be rare and

usually opportunistic human pathogens. The primary medical literature should be consulted for details on their potential role as infectious agents.

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154. *HELICOBACTER PYLORI* INFECTIONS - John C. Atherton, Martin J. Blaser

DEFINITION

Helicobacter pylori colonizes the human stomach and is of etiologic importance in peptic ulcer disease and gastric malignancy. Other gastric *Helicobacter* species colonize animals, some with a narrow range and others with a broad range of host species specificity. Those with broad specificity are occasionally found in humans, possibly as zoonoses. It is unclear whether *Helicobacter heilmannii* (formerly known as *Gastrospirillum hominis*), the most common of these species among isolates from humans, is associated with human disease. Numerous species of nongastric helicobacters are found in animals, and some have been isolated from human stool and gall bladder; whether these species cause disease is unknown.

ETIOLOGIC AGENT

H. pylori is a gram-negative, spiral, flagellate bacillus that naturally colonizes humans and monkeys. It is noninvasive, living in gastric mucus; a small proportion of the bacterial cells are adherent to the mucosa. Its spiral shape and flagellae render *H. pylori* motile in the mucous environment, and its efficient urease protects it against acid by catalyzing urea hydrolysis to produce buffering ammonia. In vitro, *H. pylori* is microaerophilic and slow-growing and requires complex growth media. In 1997, the complete genomic sequence of *H. pylori* was published, and this information has greatly advanced the understanding of metabolic pathways and other aspects of the organism's biology. *H. heilmannii* is a longer, more tightly coiled spiral than *H. pylori* and cannot easily be cultured in vitro at present.

EPIDEMIOLOGY

The prevalence of *H. pylori* colonization is about 30% in the United States and other developed countries as opposed to >80% in most developing countries. In the United States, prevalence varies with age; around 50% of 60-year-old persons as opposed to 25% of 30-year-old persons are colonized. Spontaneous acquisition or loss of the bacterium in adulthood is uncommon. *H. pylori* is usually acquired in childhood. (The age association is mostly due to a birth-cohort effect.) Other than age, the main risk factor for colonization is low socioeconomic status; crowding and low family income in childhood are especially strong correlates of colonization.

Humans are the only important reservoir of *H. pylori*. Members of a family may carry the same strain, and colonization is particularly common in childhood institutions. These findings imply direct person-to-person spread, but whether transmission takes place by the fecal-oral or oral-oral route is unknown. *H. pylori* DNA has been found in water sources, and indirect epidemiological evidence indicates that contaminated water may lead to human colonization in developing countries. Much research is focused on determining which of these possible routes of acquisition is most important.

CLINICAL MANIFESTATIONS

Essentially all *H. pylori*-colonized persons have gastric inflammation, but this condition

in itself is asymptomatic ([Fig. 154-1](#)). Symptoms are due to illnesses such as peptic ulceration or gastric malignancy, which develop in fewer than 10% of individuals colonized with *H. pylori*. More than 80% of peptic ulcers are related to *H. pylori* colonization, most of the remainder being due to damage caused by aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs). The main lines of evidence for an ulcer-promoting role of *H. pylori* are (1) that the presence of the organism is a risk factor for the development of ulcers, (2) that (non-NSAID-induced) ulcers rarely develop in the absence of *H. pylori*, (3) that eradication of *H. pylori* results in a dramatic drop in the rate of ulcer relapse (from about 80% to 15% in the first year), and (4) that experimental infection of gerbils and mice causes gastroduodenal injury.

Prospective case-control studies have shown that *H. pylori* colonization is a risk factor for adenocarcinomas of the stomach (other than those arising in the gastric cardia). However, persons who have had documented duodenal ulcers are less likely than other persons to develop gastric adenocarcinoma later in life; the implication is that, whereas *H. pylori* colonization increases risk for both duodenal ulcerogenesis and gastric carcinogenesis, other factors determine which disease path is taken. The presence of *H. pylori* is strongly associated with gastric lymphoma. Low-grade B-cell mucosa-associated lymphoid tissue (MALT) lymphomas, which are antigen driven, often regress following *H. pylori* eradication. Whether all such diagnosed cases represent true malignancies remains to be determined.

Most *H. pylori* colonization is asymptomatic. Whether colonization occasionally causes symptoms (nonulcer dyspepsia) in the absence of ulcers or malignancy is controversial. Some but not all trials of *H. pylori* eradication in nonulcer dyspepsia have shown a reduction of symptoms in a small proportion of patients. As there is no prospective method for identifying this small group, eradication of *H. pylori* in patients with nonulcer dyspepsia is not currently indicated.

Much interest has focused on a possible protective role for *H. pylori* in gastroesophageal reflux disease (GERD) and adenocarcinoma of the esophagus and gastric cardia. The main lines of evidence for this role are that (1) there is a temporal relationship between a falling prevalence of *H. pylori* colonization and a rising incidence of these conditions; (2) in most studies, the prevalence of *H. pylori* colonization, especially with *cagA*+strains, is lower among patients with these esophageal diseases than among control subjects; and (3) eradication of *H. pylori* often leads to the development or worsening of GERD or its symptoms. Although there are plausible mechanisms for a protective effect of *H. pylori* against these diseases, none has yet been definitively identified. Thus a causal link remains probable but unproven.

Several extra-gastrointestinal pathologies have been linked epidemiologically with *H. pylori* colonization. The most notable are ischemic heart disease and cerebrovascular disease. The associations have been found more commonly in small than in large studies, and most authorities consider them to be noncausal and due to confounding factors.

PATHOLOGY AND PATHOGENESIS

H. pylori colonization induces chronic superficial gastritis, which includes both

mononuclear and polymorphonuclear cell infiltration of the mucosa. (The term *gastritis* should be used specifically to describe histologic features; it also has been used to describe endoscopic appearances and even symptoms, neither of which have been linked to microscopic findings or to the presence of *H. pylori*.) The immune response to *H. pylori* includes both the production of antibody (local and systemic) and a cell-mediated response but is ineffective in clearing the bacterium. *H. pylori* and associated inflammation are most evident in the stomach but are also found in areas of gastric metaplasia and heterotopia (e.g., the duodenal bulb). The pattern of gastric inflammation is associated with disease risk: antral-predominant gastritis is most closely linked with duodenal ulceration and is common in the United States and other developed countries, whereas the predominant form in developing countries is pangastritis, which is epidemiologically linked with gastric ulceration and adenocarcinoma. Longitudinal analyses of gastric biopsy specimens taken years apart from the same patient show that inflammation may progress to atrophy, intestinal metaplasia, and dysplasia and then (by implication) to carcinoma. Patients with atrophic gastritis are at risk for vitamin B₁₂ deficiency and its associated hematologic and neurologic sequelae. Continuous omeprazole therapy (for example, for [GERD](#)) may speed progression to atrophy when *H. pylori* is present.

Most *H. pylori*-colonized persons do not develop clinical sequelae. That some persons develop overt disease whereas others do not is probably due to a combination of bacterial strain differences, host susceptibility to disease, and environmental factors; of these, bacterial factors are best studied.

The two major disease-associated *H. pylori* virulence factors described so far are a vacuolating cytotoxin, VacA, and a group of genes termed the *cag* pathogenicity island (*cag* Pal). VacA occurs in several forms, and its level of production varies between strains; thus, although all strains have the gene (*vacA*) encoding the protein, not all exhibit vacuolating activity in vitro. Cytotoxic strains are more commonly isolated from patients with peptic ulcer disease than from persons without ulcers. The *cag* Pal includes genes that confer enhanced virulence on *H. pylori* strains, at least partly by inducing epithelial cells to produce proinflammatory cytokines. The gene *cagA*, an imperfect marker for the *cag* Pal, is useful for epidemiologic studies because it encodes a highly immunogenic protein, CagA. Patients with peptic ulcers or gastric adenocarcinoma are more likely to have CagA antibodies than persons without these conditions. However, patients with esophageal dysplasia or adenocarcinoma or with the premalignant condition Barrett's esophagus are less likely to harbor *cagA*+strains than are *H. pylori*-positive controls. Thus, eradication of *cagA*+strains from asymptomatic persons to prevent disease is not recommended.

How does gastric *H. pylori* colonization increase risk for duodenal ulceration? One explanation is that antral *H. pylori* colonization diminishes the number of somatostatin-producing cells; somatostatin-mediated inhibition of gastrin release leads to hypergastrinemia. Individuals with antral-predominant gastritis (and thus a normally functioning acid-producing gastric corpus) develop increased acid secretion, which may increase the risk of duodenal ulceration per se or may induce gastric metaplasia in the duodenum, which becomes colonized by *H. pylori*, then inflamed, and finally ulcerated. After eradication of *H. pylori* from patients with duodenal ulcer disease, the level of acid secretion often falls.

DIAGNOSIS

Tests for *H. pylori* can be divided into two groups: invasive tests, which require upper gastrointestinal endoscopy and are based on the analysis of gastric biopsy specimens, and noninvasive tests ([Table 154-1](#)). Invasive tests are preferred for (1) the initial management of dyspeptic patients, because the decision of whether or not to eradicate *H. pylori* depends on ulcer disease status, and (2) follow-up after treatment of patients with gastric ulceration to be certain that the ulcer was not malignant. Follow-up endoscopy should be performed at least 4 weeks after cessation of all anti-*Helicobacter* drugs, since at earlier points the *H. pylori* load may be low and tests may be falsely negative. The most convenient endoscopy-based test is the biopsy urease test, in which two antral biopsy specimens are put into a gel containing urea and an indicator. The presence of *H. pylori* urease elicits a color change, which often takes place within minutes but can require up to 24 h. Histologic examination of biopsy specimens is accurate, provided that a special stain (e.g., a modified Giemsa or silver stain) permitting optimal visualization of *H. pylori* is used. Histologic study yields additional information, including the degree and pattern of inflammation, atrophy, metaplasia, and dysplasia, although these details are rarely of clinical use. Microbiologic culture is most specific but may be insensitive due to difficulty with *H. pylori* isolation. Once cultured, the identity of *H. pylori* can be confirmed by its typical appearance on Gram's stain and its positive reactions in oxidase, catalase, and urease tests. Antibiotic sensitivities also can be determined. Specimens containing *H. heilmannii* are only weakly positive in the biopsy urease test. The diagnosis is based on visualization of the characteristic long, tight spiral bacteria in histologic sections.

The simplest tests for *H. pylori* infection are serologic, involving the assessment of specific IgG levels in serum. The best of these tests are as accurate as other diagnostic methods, but many commercial tests, especially rapid office tests, perform poorly. In quantitative tests, a defined drop in antibody titer between matched serum samples taken before and 6 months after treatment (no sooner because of the slow decline in antibody titer) accurately indicates that *H. pylori* infection has been eradicated. The other major noninvasive tests are the ^{13}C and ^{14}C urea breath tests. In these simple tests, the patient drinks a labeled urea solution and then blows into a tube. The urea is labeled with either the nonradioactive isotope ^{13}C or a minute dose of the radioactive isotope ^{14}C (which exposes the patient to less radiation than a standard chest x-ray). If *H. pylori* urease is present, the urea is hydrolyzed and labeled carbon dioxide is detected in breath samples. Unlike serologic tests, urea breath tests can be used to assess the outcome of treatment 1 month after its completion and thus may replace endoscopy for this purpose in the follow-up of duodenal ulcer patients. As for endoscopic tests, all anti-*Helicobacter* drugs should be avoided in this period or the test may be falsely negative.

TREATMENT

At present, the only clear indications for treatment are *H. pylori*-related duodenal and gastric ulceration and the rare low-grade B-cell [MALT](#) lymphoma. *H. pylori* should be eradicated in patients with documented ulcer disease, whether or not the ulcers are currently active, to reduce the likelihood of relapse. At present, treatment is not

recommended for nonulcer dyspepsia or for prophylaxis against ulcers or gastric adenocarcinoma (although it may be reasonable to eradicate *H. pylori* in persons with a strong family history of gastric cancer). Reasons for avoiding treatment for these other potential indications include the expense, the induction of morbidity in otherwise healthy people, the risk of inducing widespread antibiotic resistance in *H. pylori* and in other colonizing bacteria, and the risk of inducing or worsening [GERD](#).

H. pylori is susceptible to a wide range of antibiotics in vitro, but monotherapy has been disappointing in vivo, probably because of inadequate antibiotic delivery to the full locus of colonization. Failure of monotherapy has led to the development of multidrug regimens, the most successful of which are triple and quadruple combinations that achieve *H. pylori* eradication rates of >90% in many trials and >75% in clinical practice. Current 7- to 14-day drug regimens consisting of a proton pump inhibitor and two or three antimicrobial agents often require only twice-daily dosing ([Table 154-2](#)).

The two most important goals in *H. pylori* eradication are to obtain the patient's compliance with the dosing regimen and to use drugs to which *H. pylori* has not acquired resistance. Treatment failure following minor lapses in compliance is common and often leads to acquired resistance to metronidazole or clarithromycin. To stress the importance of compliance, written instructions should be given to the patient, and minor side effects of the regimen should be explained. Resistance to metronidazole and clarithromycin is of growing concern; however, in multidrug regimens, the clinical significance of single-drug resistance is diminished. Assessment of antibiotic susceptibilities before treatment would be optimal but is not usually undertaken. In the absence of susceptibility information, a history of antibiotic use should be obtained, and, if resistance is likely, metronidazole-containing regimens should be avoided. Metronidazole resistance is common among persons who have taken the agent previously, even years earlier, for other conditions such as giardiasis or trichomoniasis. If initial *H. pylori* treatment fails, compliance should be checked and re-treatment should be based on known antibiotic susceptibilities. When this information cannot be obtained, the recommended course is quadruple therapy without clarithromycin (if a clarithromycin-containing regimen was given first) or triple therapy with omeprazole/clarithromycin/amoxicillin (if clarithromycin has not been used) ([Table 154-2](#)).

Given the high efficacy of treatment regimens, it is unclear whether the success of attempted *H. pylori* eradication should be checked. For gastric ulceration, the opportunity to retest for *H. pylori* is present at the repeat endoscopy, which is performed to evaluate healing. For duodenal ulceration, although many clinicians prefer to retest only if symptoms recur, a urea breath test or endoscopy should be performed no sooner than 1 month after treatment. This test will provide reassurance if treatment has been successful and will prompt re-treatment in cases of persistence.

Clearance of *H. heilmannii* has been described following the use of bismuth compounds alone or triple-therapy regimens. However, in the absence of trials, it is unclear whether this result represents successful treatment or natural clearance of the bacterium.

PREVENTION

Carriage of *H. pylori* has public health significance in developing countries, where gastric adenocarcinoma is a common cause of cancer death. However, *H. pylori* has co-evolved with its human host over millennia, and there may be disadvantages in preventing or eliminating colonization. For example, as has been mentioned, the absence of *H. pylori* appears to increase the risk of developing [GERD](#) and esophageal adenocarcinoma. If mass prevention were contemplated, vaccination would be preferred, and experimental immunization of animals has given promising results. However, in the United States and other developed countries, the incidences of *H. pylori* carriage, peptic ulceration, and gastric adenocarcinoma are dropping. Thus, prevention of colonization in these countries may be unnecessary or even unwise.

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155. INFECTIONS DUE TO *PSEUDOMONAS* SPECIES AND RELATED ORGANISMS

- Christopher A. Ohl, Matthew Pollack

Pseudomonas species and phylogenetically related bacteria are ubiquitous, free-living, opportunistic gram-negative pathogens. *P. aeruginosa*, the most common human pathogen in this group, is the primary subject of this chapter. Also discussed are two pathogens of increasing importance: *Burkholderia cepacia* (formerly *P. cepacia*), primarily an opportunistic pathogen, and *Stenotrophomonas maltophilia* (formerly *Xanthomonas maltophilia*), which principally infects hospitalized patients. In addition, melioidosis, a tropical systemic disease with acute and chronic manifestations caused by *B. pseudomallei* (formerly *P. pseudomallei*), will be considered.

INFECTIONS DUE TO *P. AERUGINOSA*

P. aeruginosa is a small, nonsporulating, aerobic gram-negative rod belonging to the family Pseudomonadaceae. It is motile by virtue of its single polar flagellum. More than half of all clinical isolates produce the blue-green pigment pyocyanin; this pigment is helpful in the identification of the organism and accounts for the species name *aeruginosa*, which refers to the distinctive color of copper oxide.

EPIDEMIOLOGY

P. aeruginosa is widespread in nature, inhabiting soil, water, plants, and animals (including humans). It has a predilection for moist environments. This organism occasionally colonizes the skin, external ear, upper respiratory tract, or large bowel of healthy humans. Rates of carriage are relatively low, however, except among patients who have serious underlying disease, whose host defenses have been naturally or iatrogenically compromised, who have previously received antibiotic therapy, and/or who have been exposed to the hospital environment. Under these circumstances, colonization with *P. aeruginosa* frequently precedes infection, and factors that predispose to the former also increase the likelihood of the latter.

Most *P. aeruginosa* infections are acquired in the hospital, where intensive care units account for the highest rates of infection. According to the National Nosocomial Infections Surveillance (NNIS) system, between 1992 and 1999, *P. aeruginosa* was the second most common cause of pneumonia, the fourth most common cause of urinary tract infection, and the sixth most frequent bloodstream isolate in intensive care units. Many potential reservoirs of infection have been identified in the hospital environment, including respiratory equipment, cleaning solutions, disinfectants, sinks, vegetables, flowers, endoscopes, and physiotherapy pools. Most reservoirs are associated with moisture. It is assumed that the organism is transmitted to patients via the hands of hospital personnel or via fomites. While some infecting strains of *P. aeruginosa* appear to be endemic within the hospital, others are traced to a common source associated with a specific outbreak or epidemic. Epidemiologic investigation is facilitated by serotyping (immunotyping) of strains on the basis of differences in lipopolysaccharide (LPS) structure and by the use of molecular techniques such as pulsed-field gel electrophoresis.

PATHOGENESIS

That the pathogenesis of infections due to *P. aeruginosa* is complex is evidenced by the clinical diversity of the diseases related to this organism and by the multiplicity of virulence factors it produces. *P. aeruginosa* rarely causes disease in the healthy host. Relative risk for infection is greatly increased, however, when normal cutaneous or mucosal barriers have been breached or bypassed, when immunologic defense mechanisms have been compromised, or when the protective function of the normal bacterial flora has been disrupted ([Table 155-1](#)). The ubiquity of the organism, its flexible nutritional and metabolic requirements, its environmental resiliency, and its relative resistance to antibiotics help account for the frequency and success with which it acts as an opportunistic pathogen.

Infections caused by *P. aeruginosa* usually begin with bacterial attachment and superficial colonization of cutaneous or mucosal surfaces and progress to localized bacterial invasion and damage to underlying tissues. The infection may remain anatomically localized or may spread by direct extension to contiguous structures. This process may continue with bloodstream invasion, dissemination, the systemic inflammatory-response syndrome (SIRS), multiple-organ dysfunction, and ultimately death. Not only is local infection more likely to occur in immunocompromised hosts, such as those with profound neutropenia, but it is more likely to culminate in bloodstream invasion and dissemination. [LPS](#)(endotoxin), which is a structural component of the bacterial outer membrane, is thought to play a pivotal role in the pathogenesis of the sepsis syndrome or SIRS.

The initial attachment of *P. aeruginosa* to the respiratory epithelium and other epithelial surfaces appears to be mediated by bacterial organelles called *pili* or *fimbriae* and facilitated by *alginate*, a mucoid exopolysaccharide produced by most strains of the bacterium under appropriate environmental conditions. Alginate plays an important role in colonization and infection of the respiratory tract in patients with cystic fibrosis and in the formation of biofilms within which sessile colonies of *P. aeruginosa* enjoy relative protection from host defenses and antimicrobial agents.

P. aeruginosa produces a number of extracellular virulence factors, including alkaline protease, elastase, phospholipase, cytotoxin, and exoenzymes (or exotoxins) A and S. The breakdown of host tissues by these bacterial products creates conditions conducive to enhanced bacterial proliferation, invasion, and tissue injury. Production and secretion of many of these extracellular virulence factors are under the regulatory control of a cell-to-cell signaling system that has been termed *quorum sensing*. Through lactones and other signal molecules secreted by individual *P. aeruginosa* bacteria, the entire bacterial population is able to sense its environment, communicate, and discern its own cell density. This regulatory system conceivably allows *P. aeruginosa* to produce extracellular virulence factors in a coordinated manner dependent on cell density and may give the pathogen an appreciable advantage over host defense mechanisms.

The extracellular enzyme exotoxin A -- a diphtheria-like toxin -- is produced by most clinical isolates of *P. aeruginosa*. Exotoxin A inhibits mammalian protein synthesis by transferring the adenosine diphosphate (ADP) ribose moiety of the nicotinamide adenine dinucleotide into covalent linkage with elongation factor 2, inactivating this factor's ability to catalyze the elongation step in polypeptide assembly. Another

extracellular cytotoxin, exoenzyme S, is also an adenosine diphosphate ribosyltransferase but, unlike exotoxin A, preferentially ribosylates guanosine triphosphate-binding proteins, resulting in disruption of host cell actin cytoskeletons. Exoenzyme S is one of several extracellular virulence factors of *P. aeruginosa* that is directly introduced from the bacterial cytosol into the host cell cytoplasm via a complex array of transmembrane proteins termed the *type III secretion apparatus*. This process requires direct cell contact and allows the injection of virulence factors from the bacterium into host cells without interference from humoral immune defenses.

CLINICAL MANIFESTATIONS AND DIAGNOSIS

Respiratory Tract Infections *Primary pneumonia, or nonbacteremic pneumonia*, results from aspiration of upper respiratory tract secretions; often develops in patients with chronic lung disease, congestive heart failure, or AIDS; and is most common in an intensive care setting in association with mechanical ventilator use. Fever, chills, severe dyspnea, cyanosis, productive cough, apprehension, confusion, and other signs of severe systemic toxicity characterize this acute, often life-threatening infection. Chest roentgenograms typically show bilateral bronchopneumonia with nodular infiltrates and small areas of radiolucency; pleural effusions are common; empyema is relatively uncommon; and lobar consolidation is occasionally seen. Cavitory lesions are particularly common in AIDS patients with *P. aeruginosa* pneumonia. Pathologic lesions include alveolar necrosis, focal hemorrhages, and microabscesses.

Bacteremic pneumonia due to *P. aeruginosa* begins as a respiratory infection but, in contrast to primary pneumonia, is typically associated with neutropenia, subsequent bloodstream invasion, and metastatic spread that produces characteristic lesions in the lungs and other viscera. Alveolar hemorrhage and necrosis are common. The signs and symptoms of this fulminant disease include those described for nonbacteremic pneumonia caused by this organism as well as those associated with gram-negative sepsis. Chest roentgenograms characteristically demonstrate a rapid progression from pulmonary vascular congestion to interstitial edema, then to pulmonary edema, and finally to diffuse necrotizing bronchopneumonia with cavity formation. The patient typically dies 3 or 4 days after initial presentation.

Chronic infection of the lower respiratory tract with *P. aeruginosa* is caused almost exclusively by mucoid strains, which produce alginate. Such infection is prevalent among older children and young adults with cystic fibrosis and also develops in some patients with AIDS. In patients with cystic fibrosis, mucoid strains invariably colonize and infect patients with increasing prevalence over time, contributing to the acute exacerbations and chronic progression that characterize pulmonary disease in these individuals. Airway obstruction appears to begin with bronchiolitis, which causes mucus plugging and predisposes to *P. aeruginosa* infection. The infection produces more mucus plugging, chronic suppuration, bronchiectasis, atelectasis, and ultimately fibrosis. This process progresses to pulmonary insufficiency, hypoxemia, and alterations in cardiopulmonary dynamics resulting in pulmonary hypertension and cor pulmonale.

Clinical manifestations of lower respiratory tract infections due to *P. aeruginosa* in patients with cystic fibrosis vary with the severity and duration of underlying lung disease, the frequency and intensity of acute episodes, and the presence of coinfecting

pathogens such as *B. cepacia* ([Chap. 257](#)). Soon after colonization, patients may experience recurrent upper respiratory symptoms followed by a lingering cough. Episodes of pneumonia develop later, with persistent cough between acute episodes. Eventually, patients exhibit a chronic productive cough, wheezing, diminished appetite, weight loss, growth retardation, and decreased activity. Acute exacerbations are typically accompanied by low-grade fever and heightened respiratory symptoms. Physical signs include evidence of malnutrition, an increase in anteroposterior diameter, intercostal retractions, cyanosis, inspiratory and expiratory wheezing, rhonchi, moist rales, abdominal distention, and clubbing of the fingers and toes. Laboratory abnormalities include leukocytosis with a left shift and hypoxemia with or without hypercarbia. Tests of pulmonary function demonstrate obstructive and restrictive defects. Chest roentgenograms reveal overaeration, patchy atelectasis, peribronchial fibrosis, and patchy infiltrates associated with pneumonia. In more advanced disease, there may be evidence of severe overaeration, depressed diaphragm, further increased anteroposterior diameter, extensive peribronchial infiltration, generalized bronchiectasis, and cyst formation.

Bacteremia *P. aeruginosa* remains an important cause of life-threatening bloodstream infection in immunocompromised patients. Bacteremia is frequently iatrogenic and is usually seen in hospitalized patients with various comorbid conditions ([Table 155-1](#)). Bloodstream infection can be either primary (with no identifiable source) or secondary to a discrete focus of infection.

The clinical features of *P. aeruginosa* bacteremia are similar to those of other forms of bacteremia. Common primary sites of infection include the urinary tract, gastrointestinal tract, lungs, skin and soft tissues, and intravascular foci, including indwelling central venous catheters. Fever, tachypnea, tachycardia, and prostration are common. Disorientation, confusion, or obtundation may be evident. Hypotension can progress to refractory shock. Renal failure, adult respiratory distress syndrome, and disseminated intravascular coagulation occur as complications.

Pathognomonic skin lesions termed *ecthyma gangrenosum* ([Fig. 19-CD1](#)) develop in a relatively small minority of patients with *P. aeruginosa* bacteremia. The lesions begin as small hemorrhagic vesicles surrounded by a rim of erythema and undergo central necrosis with subsequent ulceration (see [Plate IID-57C](#)). They occur singly or in small numbers on the perineum, buttocks, and extremities; in the axillae; or elsewhere. Histologically, these lesions contain numerous bacteria invading blood vessels but few inflammatory cells. Bacteria are readily visible on Gram's staining and may be cultured from aspirated material.

Endocarditis *P. aeruginosa* infects native heart valves in injection drug users as well as prosthetic heart valves. The source of *P. aeruginosa* strains infecting drug users appears to be standing water contaminating drug paraphernalia. Foreign materials mixed with heroin may cause injury to valve leaflets or mural endocardium, with resulting fibrosis and an increased risk for valve infection. Exposure of the tricuspid valve to both trauma and bacteria apparently accounts for the high incidence of tricuspid involvement in association with injection drug use.

The pulmonic, mitral, or aortic valve and the mural endocardium of either atrium may be

affected in *P. aeruginosa* endocarditis. Multiple-valve infections are common. Tricuspid or right-sided involvement is often associated with septic pulmonary emboli. Right-sided *P. aeruginosa* endocarditis usually presents subacutely, while the appearance of left-sided disease is likely to be more acute or even fulminant. Fever is virtually invariable, and murmurs are usually detectable at initial presentation or shortly thereafter. Septic pulmonary emboli associated with right-sided disease result in cough, pleuritic chest pain, sputum production, pulmonary infiltration (with or without abscess formation), and pleural effusion. Left-sided infections may present as intractable heart failure or large systemic emboli. Mycotic aneurysms, cerebritis, or brain abscess may occur; septic infarcts are occasionally found in the spleen. Skin and soft tissue manifestations, including Janeway lesions, Osler's nodes, and ecthyma gangrenosum, are relatively uncommon.

The diagnosis of *P. aeruginosa* endocarditis is based on positive blood culture in the absence of an extracardiac source; an indication of valvular dysfunction or vegetation on an echocardiogram; evidence of septic pulmonary lesions on a chest roentgenogram (in right-sided disease); and the actual demonstration of infected heart valves at the time of surgery.

Central Nervous System Infections *P. aeruginosa* infections of the central nervous system include meningitis and brain abscess. These infections follow extension from a contiguous parameningeal structure such as the ear, mastoid, or paranasal sinus; direct inoculation into the subarachnoid space or brain through head trauma, surgery, or diagnostic procedures; or bacteremic spread from infection at a distant site. Like *P. aeruginosa* infections at other anatomic sites, central nervous system infections are documented almost exclusively in patients with compromised local or systemic immune-defense mechanisms.

The clinical signs of *P. aeruginosa* meningitis, like those of other forms of acute bacterial meningitis, include fever, headache, stiff neck, confusion, and obtundation. The onset of illness may be acute or even fulminant, particularly in bacteremic patients, with a precipitous downhill course, shock, coma, and early death. In nonbacteremic patients, *P. aeruginosa* meningitis or brain abscess may present more insidiously, with a paucity of systemic symptoms. This presentation is especially common in infections resulting from recent neurosurgery, cancer of the head and neck, or direct extension from a parameningeal focus of chronic infection. Occasionally, *P. aeruginosa* meningitis runs a subacute or relapsing course that is thought to be related to the intermittent release of bacteria from a loculated site of infection.

Ear Infections *P. aeruginosa* is often found in the external auditory canal, particularly under moist conditions and in the presence of inflammation or maceration (as in "swimmer's ear"). Moreover, this organism is the predominant pathogen associated with external otitis, a usually benign inflammatory process affecting the external auditory canal. The ear is painful or merely itchy, there is a purulent discharge, and pain is elicited by pulling on the pinna. The external canal appears edematous and is filled with detritus that often prevents visualization of the tympanic membrane.

P. aeruginosa occasionally penetrates the epithelium overlying the floor of the external auditory canal at the junction between bone and cartilage and invades underlying soft

tissue. The ensuing invasive process, which involves soft tissue, cartilage, and cortical bone, is typically slow but destructive. Termed *malignant external otitis*, this condition occurs predominantly in elderly diabetic patients but is reported occasionally in infants with other underlying diseases and rarely in elderly nondiabetic patients. Virtually all cases of malignant external otitis are caused by *P. aeruginosa*. From the external ear, the infection advances to the retromandibular area or parotid space and enters the mastoid air cells and temporal bone. Advancing osteomyelitis at the base of the skull often involves the seventh, ninth, tenth, and eleventh cranial nerves. The cavernous sinus can become involved, as can the contralateral petrous apex. The middle ear is commonly spared; meningitis and brain abscess are relatively rare complications.

Otorrhea and severe otalgia are common presenting symptoms of malignant external otitis. Facial-nerve paralysis tends to occur early, while other cranial-nerve palsies appear later. There may be a loss of hearing. Constitutional symptoms such as fever and weight loss are relatively uncommon. Physical examination almost always reveals remarkable tenderness of the pinna of the ear and abnormalities of the external auditory canal, including swelling, erythema, purulent discharge, debris, and granulation tissue in the canal wall. The tympanic membrane is often hidden from view and is sometimes perforated. Inflammation may involve the pinna as well as the periauricular, retromandibular, and mastoid areas.

Peripheral leukocytosis is relatively infrequent in malignant external otitis, while the erythrocyte sedimentation rate is usually markedly elevated. Cerebrospinal fluid occasionally exhibits pleocytosis and an elevation in the protein level. Computed tomography (CT) or magnetic resonance imaging (MRI) of the mastoid or temporal bone typically reveals bony erosions and new bone formation, while the floor of the skull may have soft tissue densities associated with areas of cellulitis. In addition, technetium 99m bone scans and gallium 67 scans frequently give positive results. Cultures of samples from the external auditory canal and of surgical specimens are almost always positive for *P. aeruginosa*.

Eye Infections (See also [Chap. 28](#)) *P. aeruginosa* causes bacterial keratitis or corneal ulcer and endophthalmitis in the human eye. Keratitis due to *P. aeruginosa* may result from even minor corneal injury, which interrupts the integrity of the superficial epithelial surface and permits bacterial access to the underlying stroma. Corneal ulcer may complicate contact lens use, particularly when extended-wear soft contact lenses are involved. Contact lens solutions or the lenses themselves may be the source of the organism, which is probably inoculated into the eye at sites of minor lens-induced corneal damage. Patients who have sustained serious burns, have undergone ocular irradiation or tracheostomy, have been exposed to the intensive care environment, and/or are in a coma are also susceptible to *P. aeruginosa*-associated corneal ulcers. *P. aeruginosa* keratitis usually starts as a small central ulcer; spreads concentrically to involve a large portion of the cornea, sclera, and underlying stroma; and in some cases progresses to posterior corneal perforation.

The clinical manifestations of *P. aeruginosa* keratitis include a rapidly expanding, necrotic stromal infiltrate in the bed of an epithelial injury; surrounding epithelial edema; an anterior chamber reaction; and mucopurulent discharge adherent to the ulcer's surface. Corneal ulcer due to *P. aeruginosa* may advance rapidly to involve the entire

cornea in 2 days or may evolve subacutely over several days. Systemic symptoms are uncommon. Complications include corneal perforation, anterior chamber involvement, and endophthalmitis.

P. aeruginosa endophthalmitis is typically a rapidly progressive, sight-threatening condition that demands immediate therapeutic intervention. It may complicate penetrating injuries of the eye, intraocular surgery, hematogenous spread from other sites of *Pseudomonas* infection, or posterior perforation of corneal ulcers. Clinical manifestations may include eye pain, conjunctival hyperemia, chemosis, lid edema, decreased visual acuity, hypopyon, severe anterior uveitis, and signs of possible vitreous involvement. Panophthalmitis may result from this intraocular infection.

Bone and Joint Infections Vertebral osteomyelitis due to *P. aeruginosa* is associated with complicated urinary tract infection, genitourinary instrumentation or surgery, and injection drug use. Vertebral infections that are associated with a urinary tract source most often develop in the elderly and usually affect the lumbosacral spine. Presumably the route of infection in these patients is a shared venous plexus between the pelvis and spine. Injection drug use-related infections typically occur in younger patients and may affect the cervical or lumbosacral spine. *P. aeruginosa* vertebral osteomyelitis is usually an indolent disease. Accordingly, symptoms may develop weeks or even months before diagnosis. Back or neck pain is generally reported, while fever and systemic symptoms are relatively uncommon. Local tenderness and decreased range of motion of the affected spine are typical. Leukocytosis may be noted, the erythrocyte sedimentation rate is almost always markedly elevated, and blood cultures are sometimes positive. Roentgenograms reveal loss of bone density, narrowed intervertebral space, destruction of vertebral end plates, lytic lesions of vertebral bodies, sclerosis, and occasionally osteophyte formation. [CT](#) and [MRI](#) are the most sensitive and specific means of defining lesions. Technetium bone scans and gallium scans usually yield positive results. An etiologic diagnosis requires the culture of material obtained by needle aspiration or biopsy of the affected spine under fluoroscopic guidance; open biopsy is sometimes needed.

Sternoclavicular pyarthrosis caused by *P. aeruginosa* is another complication of injection drug use; in some cases it is associated with *P. aeruginosa* endocarditis, but more often it is not. Joint involvement is usually monoarticular, with the sternoclavicular joint more often affected than sternochondral joints. Patients present with acute or chronic pain in the anterior chest wall, often associated with fever and restricted movement of the homolateral shoulder. Physical examination reveals tenderness, erythema, and swelling over the affected joint. Leukocytosis is common, and the erythrocyte sedimentation rate is almost invariably elevated. Roentgenograms show soft tissue edema; bone demineralization; lytic lesions; and periosteal elevation of the clavicular head, rib, or sternum. Material obtained by arthrocentesis or synovial biopsy yields *P. aeruginosa* in culture.

P. aeruginosa infections of the symphysis pubis are associated with pelvic surgery and injection drug use. The symphysis pubis, like other fibrocartilaginous joints, exhibits a peculiar susceptibility to bloodborne infection with *P. aeruginosa*. Affected patients report pain in the groin, hip, thigh, and/or lower abdomen that is made worse by walking. Fever is variable, and the duration of symptoms before diagnosis ranges from days to

months. The erythrocyte sedimentation rate is markedly elevated. Roentgenography or [CT](#) shows irregularities of the pubic margins, separation of the symphysis pubis, and osteomyelitic abnormalities of the pubic rami that may be extensive. Bone scans are usually positive. Needle aspiration or biopsy is necessary to obtain material for culture. A positive culture is particularly important for the discrimination of *P. aeruginosa* infections and other pyogenic infections from osteitis pubis, which is thought to be a noninfectious condition complicating pelvic surgery, childbirth, or trauma.

P. aeruginosa osteochondritis of the foot follows puncture wounds of the foot, primarily in children. The organism infects the small joints and bones, including the proximal phalanges, metatarsals, metatarsophalangeal joints, tarsal bones, and calcaneus. On average, local pain and swelling last for several weeks, and systemic symptoms are usually lacking. There may be plantar cellulitis over the involved area or tenderness upon deep palpation. Results of roentgenograms and bone scans are generally positive. Aspiration of the affected joint frequently yields purulent material in which *P. aeruginosa* can be demonstrated by Gram's staining and by culture.

P. aeruginosa is one of the most common causative agents in a variety of other, less specific syndromes involving nonhematogenous infections of bones and joints and collectively referred to as *chronic contiguous osteomyelitis*. These infections may result, for example, from compound fractures, contamination associated with open reduction and fixation of closed fractures, sternotomy performed in conjunction with cardiac surgery, contiguous spread from infected ischemic ulcers related to peripheral vascular disease or diabetes mellitus, and cellulitis in general. The chronicity, indolence, and heterogeneity of these infections explain their varied clinical manifestations and the frequent need for complicated long-term management.

Urinary Tract Infections *P. aeruginosa* is one of the most common causes of complicated and nosocomial infections of the urinary tract. These infections may result from urinary tract catheterization, instrumentation, surgery, or obstruction; they may arise from persistent foci (e.g., the prostate or stones) and may be chronic or recurrent. The urinary tract may be a target for bloodborne infection in patients with *P. aeruginosa* bacteremia but more often is the source of bacteremia. Chronic *P. aeruginosa* infections of the urinary tract are relatively common among patients with indwelling urinary catheters, altered urinary tract anatomy secondary to diversionary procedures, and paraplegia.

The clinical features of urinary tract infections due to *P. aeruginosa* are usually indistinguishable from those of other bacterial infections. However, *P. aeruginosa* infections exhibit a propensity for persistence, chronicity, resistance to antibiotic therapy, and recurrence. More unusual forms of urinary tract involvement peculiar to *P. aeruginosa* include (1) ulcerative lesions of the renal pelvis, ureters, and bladder that cause sloughing of vesical membranes in the urine; and (2) ecthyma-like lesions of the renal cortex that are seen in association with *Pseudomonas* sepsis.

Skin and Soft Tissue Infections As indicated above, *P. aeruginosa* bacteremia may be associated with the disseminated skin lesions of ecthyma gangrenosum (see [Plate IID-57C](#)). Less common skin manifestations of *P. aeruginosa* sepsis include vesicular or pustular lesions, bullae, subcutaneous nodules, deep abscesses, and cellulitis.

Metastatic lesions of the skin or mucous membranes complicate *Pseudomonas* sepsis and occasionally produce massive necrosis or gangrene of the extremities, perineum, face, or oropharynx.

Primary *P. aeruginosa* pyoderma occurs when the skin breaks down secondary to trauma, burn injury, dermatitis, or ulcers related to peripheral vascular disease or pressure sores. Moist conditions and neutropenia may predispose to this condition. The clinical appearance of primary *P. aeruginosa* pyoderma, which frequently includes hemorrhage and necrosis, resembles that of metastatic *P. aeruginosa* skin lesions. Histologic studies document vascular invasion by bacteria in both diseases. A rare distinguishing feature of *P. aeruginosa* pyoderma is its association with a blue-green exudate and a characteristic fruity odor.

P. aeruginosa wound sepsis complicating extensive third-degree burn injuries is associated with an extremely high mortality rate. This infection results from colonization of the burn site or burn eschar, invasion of the subeschar space and underlying dermis, vascular invasion, and systemic spread. The development and progression of *P. aeruginosa* burn wound sepsis are facilitated by the injury-associated breakdown of normal skin, selection of empirical antibiotics with inadequate coverage for this pathogen, and burn-related immune defects. Local manifestations include black, dark brown, or violaceous discoloration of the burn eschar; degeneration of underlying granulation tissue, hemorrhage, and premature eschar separation; edema, hemorrhage, and necrosis of skin adjacent to the burn site; and erythematous nodular lesions in unburned skin. Systemic manifestations include fever or hypothermia and other signs of sepsis, [SIRS](#), or multiple-organ system failure. The diagnosis of *P. aeruginosa* burn sepsis is based on these local and systemic clinical manifestations and on a burn wound biopsy that reveals both $>10^5$ colony-forming units of *P. aeruginosa* per gram of tissue and histologic evidence of bacterial invasion of unburned tissue, vasculitis, or intense inflammation at the burn margin.

P. aeruginosa causes diffuse pruritic maculopapular and vesiculopustular rashes associated with exposure to contaminated hot tubs ([Fig. 128-CD4](#)), spas, whirlpools, and swimming pools. Many cases of *P. aeruginosa* dermatitis have occurred in conjunction with a common-source outbreak. At least two nosocomial common-source outbreaks -- one related to a physiotherapy pool -- have been reported. Skin rashes may be limited to areas covered by swimsuits or may be more diffuse, sparing only the head and neck. Low-grade fever or other associated symptoms are uncommon. The illness is usually self-limited, and the rash resolves without specific therapy after cessation of exposure.

***P. aeruginosa* Infections in Patients with AIDS** During the 1980s and 1990s, *P. aeruginosa* infections were increasingly associated with AIDS. The vast majority of these infections are currently seen in patients with advanced AIDS, previous opportunistic infections, and CD4+ lymphocyte counts $<100/\mu\text{L}$ (often $<50/\mu\text{L}$). The specific immunologic factors that lead to *P. aeruginosa* infections in patients with AIDS are not well understood but are speculated to be a loss of mucosal integrity, defects in cellular and humoral immunity, and qualitative leukocyte abnormalities. Of note is that the majority of *P. aeruginosa* infections in this population are community-acquired, in contrast to the nosocomial transmission documented for most *P. aeruginosa* infections.

in non-AIDS patients.

Pneumonia accounts for a substantial proportion of the *P. aeruginosa* infections in patients with AIDS. In most instances, pneumonia presents as a necrotizing infection of the pulmonary parenchyma, frequently with cavitary lesions, or as a chronic relapsing bronchopulmonary infection reminiscent of the bronchopulmonary disease seen in patients with cystic fibrosis. Also frequent are bloodstream infections, including those associated with indwelling central venous catheters, and infections of the paranasal sinuses, skin and soft tissue, and urinary tract. Bacteremia, either primary or secondary to infection at a remote site, is often recurrent, associated with high mortality, and occasionally accompanied by skin manifestations similar to those seen in non-AIDS patients.

Because *P. aeruginosa* infections occur in patients with advanced AIDS, survival after recovery from the initial infection may be limited to a few months. However, with the widespread use of highly active antiretroviral therapy and the consequent increase in CD4+ cell counts, the incidence of *P. aeruginosa* infection in patients with AIDS is likely to decline and the natural history of infection to change. For example, a few patients with recalcitrant, relapsing *P. aeruginosa* bronchopulmonary infections have reportedly experienced the resolution of infection soon after initiation of intensive antiretroviral therapy.

TREATMENT

[Table 155-2](#) lists antimicrobial agents available in the United States that are generally active against *P. aeruginosa*. [Table 155-3](#) outlines suggested antibiotic choices and an approach to therapy for selected sites of infection. The initial antibiotic selection should take into account the local patterns of antimicrobial susceptibility, while the susceptibilities of the isolate from a particular case should guide definitive antibiotic therapy.

In most severe or life-threatening infections due to *P. aeruginosa*, two antipseudomonal antibiotics to which the infecting strain is (or is likely to be) sensitive should be administered together. The benefits of this combined therapy, as determined by in vitro studies, are to increase efficacy, to achieve additive or synergistic killing, and to prevent the emergence of antibiotic resistance. Despite widespread acceptance of combination therapy for *P. aeruginosa* infections, there are few clinical data since the advent of newer β -lactam antibiotics documenting that combination therapy is more efficacious than monotherapy or that it actually forestalls the acquisition of antimicrobial resistance. Nevertheless, combination therapy continues to be recommended for most acute or fulminant infections, as outlined in [Table 155-3](#).

The appropriate duration of antibiotic therapy for disease caused by *P. aeruginosa* depends on the type, location, and severity of infection. In general, chronic infections associated with extensive tissue injury, disruption of normal anatomy, foreign or prosthetic material, or suboptimal antibiotic accessibility require therapy for weeks or even months rather than days. More acute infections may be treated aggressively but for shorter periods.

P. aeruginosa infections of the lower respiratory tract in cystic fibrosis pose a special challenge because of their long-standing nature. In general, antibiotic therapy for acute exacerbations results in short-term clinical improvement, while periodic expectant courses of antimicrobial therapy may limit disease progression. A more novel approach utilizing intermittent, cyclical administration of inhaled tobramycin has been shown to improve pulmonary function, decrease the risk of hospitalization, and reduce the density of *P. aeruginosa* in sputum of older patients with cystic fibrosis. Lung transplantation has also been employed with good results in selected cystic fibrosis patients with severe, progressive lower respiratory tract infections due to *P. aeruginosa*.

ANTIMICROBIAL RESISTANCE

Antibiotic resistance in *P. aeruginosa* is both intrinsic, as reflected by the relative paucity of antibiotics with inherent antimicrobial activity against wild-type strains, and acquired, as defined by high-level resistance to agents that would be expected to exhibit antimicrobial activity. Acquired resistance is rapidly increasing among *P. aeruginosa* isolates, particularly those associated with cystic fibrosis and with intensive care units. Escalating resistance among intensive care unit isolates is especially alarming. Data from the [NNIS](#) system show an increase in rates of resistance to imipenem and fluoroquinolones from 12% during previous years to 18.5% and 23.0%, respectively, during 1999. Factors responsible for this increase may include expanding use of immunosuppressive therapies, increased severity of illness in hospitalized patients, inadequate infection control procedures, and growing antibiotic use. Resistant organisms can be transmitted directly to patients from the hospital staff, other patients, or the environment, or they may arise de novo during therapy with any given agent. Emergence of multi-drug-resistant strains has been associated with increases in secondary bacteremia and mortality and has led in some cases to longer hospital stays and increased hospitalization costs. Therapy for patients with resistant *P. aeruginosa* infections should consist of antimicrobial agents selected on the basis of extended susceptibility testing. Increased treatment duration and surgical drainage or removal of infected tissues may be necessary.

INFECTIONS CAUSED BY OTHER *PSEUDOMONAS* SPECIES OR RELATED BACTERIA

Burkholderia cepacia *B. cepacia*, like *P. aeruginosa*, is primarily an opportunistic pathogen that is implicated in both sporadic endemic infections and occasional nosocomial outbreaks. Hospital epidemics are most frequently associated with a liquid reservoir or a moist environmental surface. Colonization by *B. cepacia* precedes infection, and distinction between the two is often difficult. *B. cepacia* has been reported to cause pneumonia, urinary tract infections, meningitis, peritonitis, surgical and burn wound infections, bacteremia, and endocarditis related to injection drug use. In addition, *B. cepacia* has been implicated as a cause of chronic lower respiratory tract infections in patients with chronic granulomatous disease, in patients with sickle cell hemoglobinopathies, and -- together with *P. aeruginosa* -- in patients with cystic fibrosis. In some patients with cystic fibrosis, the appearance of *B. cepacia* has been associated with fulminant necrotizing pneumonia, bacteremia, and a rapid downhill course.

TREATMENT

The treatment of *B. cepacia* infections is complicated by intrinsic resistance of the organism to aminoglycosides and many b-lactam agents. Although trimethoprim-sulfamethoxazole and chloramphenicol have been used successfully in the treatment of *B. cepacia* infections, resistance to these two antimicrobial agents has been reported. Carbapenems, third-generation cephalosporins, and fluoroquinolones may offer activity against sensitive strains, but relevant clinical experience is limited. Some but not all cystic fibrosis centers segregate patients infected with *B. cepacia* in an attempt to reduce horizontal transmission to uninfected patients. In addition, many centers consider lung transplantation contraindicated in these patients because of an unacceptably high mortality rate after surgery.

Stenotrophomonas maltophilia *S. maltophilia* is a ubiquitous, free-living opportunistic bacterium that has emerged as an important pathogen in hospitalized patients, particularly in cancer centers and intensive care units. Factors that lead to colonization and infection include prolonged hospitalization, malignancy, instrumentation (including urinary, peritoneal, and central venous catheterization), and prior administration of broad-spectrum antibiotics. This organism has most commonly been associated with pneumonia but also causes bacteremia, urinary tract infection, wound infection, peritonitis, cholangitis, meningitis, and (rarely) endocarditis. Acute *S. maltophilia* pneumonia -- an often devastating disease associated with bacteremia -- is being seen with increasing frequency in debilitated patients on intensive care units. Antibiotic resistance in *S. maltophilia*, based on both low outer-membrane permeability and inducible b-lactamases, is at least partly responsible for the emergence of this organism as a nosocomial pathogen under the selective pressure of antibiotic treatment.

TREATMENT

Trimethoprim-sulfamethoxazole (at a trimethoprim dose of 15 to 20 mg/kg per day for patients with normal renal function) is the drug of choice for treatment of most *S. maltophilia* infections. Alternative agents include ticarcillin/clavulanate, minocycline, and doxycycline. The third-generation cephalosporins cefoperazone and ceftazidime are occasionally active against *S. maltophilia*, but in vitro susceptibilities may not reflect clinical efficacy. The aminoglycosides and imipenem are almost always inactive. Indwelling catheters or appliances that are associated with infection should be removed.

Melioidosis Infections caused by *B. pseudomallei* constitute a broad spectrum of acute and chronic, local and systemic, clinical and subclinical disease processes collectively called *melioidosis*. *B. pseudomallei* and the infections it causes are found mainly in the tropics and are endemic in Southeast Asia and surrounding areas. *B. pseudomallei* is a free-living, small, motile, aerobic, gram-negative bacillary saprophyte normally found in soil, ponds, and rice paddies and on produce from endemic areas. It is occasionally a pathogen for animals. Humans contract the disease through soil contamination of abrasions, ingestion, or inhalation. In contrast to *B. cepacia*, *B. pseudomallei* does not establish colonization without causing infection and is rarely transmitted from person to person.

Melioidosis presents in different forms. High rates of seropositivity in endemic areas such as Vietnam, Thailand, and Malaysia suggest that many infections are clinically

inapparent. The occasional diagnosis based solely on abnormal routine chest roentgenograms represents asymptomatic pneumonitis. Acute pulmonary infections may originate in the respiratory tract or result from hematogenous spread, their severity varying from mild bronchitis to extensive necrotizing pneumonia. Onset may be sudden or gradual. Fever, productive cough, and marked tachypnea are frequent. Chest roentgenograms typically reveal upper-lobe infiltrates or thin-walled cavities that may mimic tuberculosis. Acute, localized, suppurative skin infections associated with nodular lymphangitis and regional lymphadenitis result from direct inoculation at sites of minor skin trauma. Recrudescence arising from inactive sites of infection and perhaps triggered by intercurrent illness or other events may present in an acute or chronic form.

Either acute suppurative infections or pulmonary disease may give rise to hematogenous dissemination and the acute septicemic form of melioidosis. This progression is more likely in chronically debilitated patients, such as those with diabetes mellitus or alcoholism. Septicemic patients may present with severe tachypnea, confusion, headache, pharyngitis, diarrhea, and pustular lesions of the head, trunk, and extremities. The skin may be flushed or cyanotic, signs of meningitis or arthritis may be apparent, the liver and spleen may be enlarged, and muscle tenderness may be striking. Chest roentgenograms show diffuse nodular densities that may expand, coalesce, and finally cavitate. The acute septicemic form of melioidosis usually follows a rapid downhill course, ending in early death. Mortality remains high despite optimal therapy.

The diagnosis of melioidosis should be entertained when a febrile patient who has been in an endemic area presents with an acute lower respiratory tract illness associated with tachypnea, exhibits unusual skin or subcutaneous lesions, or has a chest roentgenogram suggesting tuberculosis in the absence of sputum-associated tubercle bacilli. An etiologic diagnosis may be made by microscopic demonstration of small, irregularly staining, gram-negative rods in exudate material; by characteristic bipolar ("safety-pin") staining of organisms with methylene blue; and by a culture positive for *B. pseudomallei* and/or a fourfold or greater rise in the titer of serum antibody to the organism.

TREATMENT

The mainstay of treatment for melioidosis is antibiotic administration combined with appropriate surgical drainage of abscesses and aggressive support for patients with septicemic forms of the disease. The guidelines for antibiotic therapy are somewhat imprecise. Subclinical infection or mere seropositivity does not usually require specific therapy. Ceftazidime or imipenem appears to be the agent of choice for clinical disease, including severe infections, while trimethoprim-sulfamethoxazole, cefotaxime, and amoxicillin/clavulanate are possible alternatives. Combination therapy with ceftazidime or imipenem plus trimethoprim-sulfamethoxazole may be indicated in severe forms of melioidosis, including septicemia. Unfortunately, increasing resistance of many strains of *B. pseudomallei* to trimethoprim-sulfamethoxazole, particularly in Southeast Asia, is of concern. Patients with acute pulmonary infections who are treated with either ceftazidime or imipenem should receive antibiotics until they show definite evidence of clinical improvement (often after 10 to 30 days), at which time therapy can be switched to an oral maintenance regimen -- a combination of chloramphenicol,

trimethoprim-sulfamethoxazole, and doxycycline or the single agent amoxicillin/clavulanate -- and continued for 12 to 20 weeks. Chronic disease associated with persistently positive sputum cultures and extrapulmonary suppurative disease may require treatment for up to 1 year.

Other Species *Pseudomonas fluorescens* occasionally causes human disease; it is implicated particularly often in infections related to the administration of contaminated (stored) blood products and in pseudoinfections. Additional species that are associated only rarely with human infections include *P. putida*, *P. stutzeri*, *P. pseudoalcaligenes*, and (all formerly *Pseudomonas* species) *Burkholderia gladioli*, *B. pickettii*, *Comamonas acidovorans*, *C. testosteroni*, *Brevundimonas diminuta*, and *B. vesicularis*.

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156. SALMONELLOSIS - Cammie F. Lesser, Samuel I. Miller

The salmonellae constitute a genus of over 2300 serotypes that are highly adapted for growth in both human and animal hosts and cause a wide spectrum of disease. A subset of *Salmonella* serotypes that includes *S. typhi* and *S. paratyphi* causes enteric (typhoid) fever and is restricted to growth in human hosts. The remainder of *Salmonella* serotypes, referred to as nontyphoidal *Salmonella*, are prevalent in the gastrointestinal tracts of a broad range of animals, including mammals, reptiles, birds, and insects. Over 200 of these serotypes are pathogenic to humans; these pathogenic serotypes cause gastroenteritis and can be associated with localized infections and/or bacteremia.

ETIOLOGY

Salmonella is a large genus of gram-negative bacilli within the family Enterobacteriaceae. The nomenclature and classification of these bacteria have undergone numerous revisions, most recently in 1983 when -- on the basis of a high degree of DNA similarity between the bacterial genomes -- over 2000 bacterial strains were grouped into one species, *S. choleraesuis*. This species was further divided into seven subgroups based on host range specificity and additional DNA similarity. Almost all the strains pathogenic for humans are in subgroup 1 (*enterica* or *choleraesuis*) except for those causing rare infections (subgroups 3a and 3b). The nomenclature of this large species is quite complex. For example, the correct taxonomic name for the organism that causes enteric fever is *Salmonella choleraesuis* ssp. *choleraesuis* (or subgroup 1), serovar *typhi*. Given the cumbersome nature of this nomenclature system, a simplified system is in widespread use, in which the common species name that existed before the reclassification of the species is accepted. For example, *S. choleraesuis* ssp. *choleraesuis*, serovar *typhi*, is referred to by its common name, *S. typhi*.

The initial identification of this genus in the clinical laboratory relies on growth characteristics. Like other Enterobacteriaceae, salmonellae produce acid on glucose fermentation, reduce nitrates, and do not produce cytochrome oxidase. They are non-spore-forming and facultatively anaerobic. With few exceptions, salmonellae are motile by means of peritrichous flagella (exception: *S. gallinarum-pullorum*) and produce gas (H₂S) on sugar fermentation (exception: *S. typhi*). Since 99% of clinical isolates are lactose nonfermenters, rare clinical isolates may not be detected if a high level of suspicion is not maintained.

Salmonella can be further divided into serovars based on the detection of three major antigenic determinants: the somatic O antigen [lipopolysaccharide (LPS) cell-wall components], the surface Vi antigen (restricted to *S. typhi* and *S. paratyphi* C), and the flagellar H antigen. In general, clinical laboratories initially divide *Salmonella* into serogroups (A, B, C₁, C₂, D, and E) based on reactivity to somatic O-antigen antisera. These initial groupings provide only limited clinical information, given their high degree of cross-reactivity. Thus, additional biochemical and serologic tests are needed to determine serotype. For the epidemiologic evaluation of *Salmonella* outbreaks, specific *Salmonella* strains within serovars can be distinguished by bacteriophage typing, plasmid profile determination, and restriction length polymorphism analysis.

PATHOGENESIS

Salmonellae are transmitted to humans orally by contaminated food or water. The bacteria traverse the gastrointestinal tract, including the acidic environment of the stomach, to colonize the small intestines. In the case of enteric fever (a systemic illness), salmonellae cross the intestinal barrier, where phagocytosis by macrophages results in their dissemination throughout the reticuloendothelial system. In nontyphoidal salmonellosis, the bacteria generally cause a localized infection resulting in an influx of neutrophils to the intestines and self-limited gastroenteritis.

Numerous attempts have been made to determine the infectious dose (ID₅₀) of *Salmonella*, both in the laboratory and in the field. Controlled experiments, in which healthy volunteers were exposed to laboratory-grown strains of *S. typhi*, concluded that the ID₅₀ was 10⁶ colony-forming units (CFU); increases in the ID₅₀ corresponded to decreases in incubation time. However, analyses of salmonellosis outbreaks with a known source indicate that the ID₅₀ can be as low as 10³ CFU. Host defenses, the most important of which appears to be the acidity of the stomach, most likely account for variations in the ID₅₀. Conditions that decrease stomach acidity (an age of <1 year, antacid ingestion, or achlorhydric disease) increase susceptibility to *Salmonella* infection, as do conditions that decrease intestinal integrity (inflammatory bowel disease, history of gastrointestinal surgery, or alteration of the intestinal flora by antibiotic administration).

Once salmonellae reach the small intestine, the bacteria resist a variety of innate immune factors (including bile salts, lysozyme, complement, and cationic antimicrobial peptides) before penetrating the mucus layer. The organisms enter the intestines through phagocytic microfold or M cells overlying the Peyer's patches. Salmonellae also enter normally nonphagocytic epithelial cells by a process known as *bacteria-mediated endocytosis*, whose mechanism is not entirely clear but depends on the direct translocation of *Salmonella* proteins into the host cell cytoplasm by a specialized secretion apparatus (type III secretion).

In enteric (typhoid) fever, salmonellae (*S. typhi* or *S. paratyphi*) undergo phagocytosis by macrophages after crossing the epithelial layer of the small intestine. Once phagocytosed, the bacteria are protected from polymorphonuclear leukocytes (PMNs), the complement system, and the acquired immune response (antibodies). Salmonellae have evolved mechanisms to avoid or delay killing by macrophages. Upon phagocytosis, the bacteria form a "spacious phagosome" and alter the regulation of ~200 bacterial proteins. The best-characterized regulatory system is PhoP/PhoQ, a two-component regulon that senses changes in bacterial location and alters bacterial protein expression. The alterations mediated by PhoP/PhoQ include modifications in [LPS](#) and in the synthesis of outer-membrane proteins; these changes presumably remodel the bacteria's outer surface such that the organisms can resist microbicidal activities and possibly alter host cell signaling. PhoP/PhoQ also mediates the synthesis of divalent cationic transporters that scavenge magnesium. By a second type III secretion mechanism, salmonellae can directly translocate bacterial proteins into the macrophage, a phenomenon that is believed to promote survival within phagocytes.

After phagocytosis, salmonellae disseminate throughout the body in macrophages via

the lymphatics and colonize reticuloendothelial tissues (liver, spleen, lymph nodes, and bone marrow). During this initial incubation stage, patients are relatively asymptomatic. Signs and symptoms, including fever and abdominal pain, probably result from secretion of cytokines by macrophages when a critical number of organisms have replicated. For example, the observed hepatosplenomegaly is likely to be related to the recruitment of mononuclear cells and the development of a cell-mediated immune response to *S. typhi* colonization. The recruitment of additional mononuclear cells and lymphocytes to Peyer's patches during the several weeks after initial colonization/infection can result in marked enlargement and necrosis of the Peyer's patches, with right-lower-quadrant abdominal pain.

It is not known why *S. typhi* and *S. paratyphi* cause systemic disease while the vast majority of pathogenic *Salmonella* strains cause gastroenteritis. In contrast to enteric fever, which is characterized by an infiltration of mononuclear cells into the small-bowel mucosa, nontyphoidal *Salmonella* gastroenteritis is characterized by massive PMN infiltration into both the large- and the small-bowel mucosa. This response appears to depend on the induction of interleukin (IL) 8, a strong neutrophil chemotactic factor, which is secreted by intestinal cells. The degranulation and release of toxic substances by neutrophils may result in damage to the intestinal mucosa, causing inflammatory diarrhea.

ENTERIC (TYPHOID) FEVER

Typhoid fever is a systemic disease characterized by fever and abdominal pain caused by dissemination of *S. typhi* or *S. paratyphi*. The disease was initially called *typhoid fever* because of its clinical similarity to typhus. However, in the early 1800s, typhoid fever was clearly defined pathologically as a unique illness on the basis of its association with enlarged Peyer's patches and mesenteric lymph nodes. In 1869, given the anatomic site of infection, the term *enteric fever* was proposed as an alternative designation to distinguish typhoid fever from typhus. However, to this day, the two designations are used interchangeably.

EPIDEMIOLOGY

In contrast to other *Salmonella* serotypes, the etiologic agents of enteric fever -- *S. typhi* and *S. paratyphi* -- have no known hosts other than humans. Thus, enteric fever is transmitted only through close contact with acutely infected individuals or chronic carriers. While direct person-to-person transmission through the fecal-oral route has been documented, it is quite rare. Rather, most cases of disease result from ingestion of contaminated food or water. Health care workers occasionally acquire enteric fever after exposure to infected patients, while laboratory workers can acquire the disease after laboratory accidents.

Over the past four decades, with the advent of improvements in food handling and water/sewage treatment, enteric fever has become a rare occurrence in developed nations. Over the past 10 years, ~400 cases of typhoid fever and even fewer cases of paratyphoid fever have been reported annually in the United States. In contrast, enteric fever continues to be a global health problem, with an estimated 13 to 17 million cases worldwide resulting in ~600,000 deaths per year. Children <1 year of age appear to be

most susceptible to initial infection and to the development of severe disease.

Enteric fever is endemic in most developing regions, especially the Indian subcontinent, South and Central America, and Asia, and is related to rapid population growth, increased urbanization, inadequate human waste treatment, limited water supply, and overburdened health care systems. These conditions most likely account for the recent epidemics of typhoid fever in eastern Europe. Antibiotic resistance among salmonellae is also a rising concern and has recently been linked to antibiotic use in livestock. Many *S. typhi* strains contain plasmids encoding resistance to chloramphenicol, ampicillin, and trimethoprim -- the antibiotics that have long been used to treat enteric fever. In addition, resistance to ciprofloxacin, either chromosomally or plasmid encoded, has been observed in Asia. Morbidity and mortality are increased in outbreaks associated with antibiotic-resistant strains, presumably because of inadequate or delayed appropriate treatment.

The high worldwide prevalence of enteric fever serves as a reservoir for cases in the United States. Over 70% of U.S. cases are related to international travel within 30 days before onset. Only 3% of travelers diagnosed with enteric fever give a history of vaccination against *S. typhi* within the previous 2 years. Of U.S. cases of internationally acquired enteric fever, 80% can be linked to travel in six countries: Mexico (28%), India (25%), the Philippines (10%), Pakistan (8%), El Salvador (5%), and Haiti (4%). While the percentage of cases associated with travel to Mexico is declining, travel to the Indian subcontinent is becoming much riskier, with an incidence 18 times higher than for any other area. The recent trend toward an increased incidence of multidrug-resistant (MDR) *Salmonella* (see "Treatment," below) in developing countries is reflected by the increase in the proportion of U.S. cases caused by MDR strains from 0.6% in 1985-1989 to 12% in 1990-1994.

Almost 30% of the reported cases of enteric fever in the United States are domestically acquired. Although the majority of these cases (80%) are sporadic, large outbreaks do occur. In the most notable outbreak in the past 15 years, 47 culture-proven and 24 potential cases were linked to contaminated orange juice at a resort in New York. Evaluation of this outbreak led to the identification of a previously unknown chronic carrier. Similarly, evaluation of 25% of the 571 cases of domestically acquired enteric fever reported between 1985 and 1994 led to the identification of previously unknown chronic carriers.

CLINICAL COURSE

Enteric fever is a misnomer, in that the hallmark features of this disease -- fever and abdominal pain -- are variable. While fever is documented at presentation in >75% of cases, abdominal pain is reported in only 20 to 40%. Thus, a high index of suspicion for this potentially lethal systemic illness is necessary when a person presents with fever and a history of recent travel to a developing country.

The incubation period for *S. typhi* ranges from 3 to 21 days. This variability is most likely related to the size of the initial inoculum and the health and immune status of the host. The most prominent symptom of this systemic infection is prolonged fever (38.8° to 40.5°C, or 101.8° to 104.9°F). A prodrome of nonspecific symptoms often precedes

fever and includes chills, headache, anorexia, cough, weakness, sore throat, dizziness, and muscle pains. Gastrointestinal symptoms are quite variable. Patients can present with either diarrhea or constipation; diarrhea is more common among patients with AIDS and among children <1 year of age. As stated above, only 20 to 40% of patients present with abdominal pain, although the majority have abdominal tenderness over the course of the disease. In general, the symptoms associated with *S. typhi* are more severe than those associated with *S. paratyphi*.

Early physical findings of enteric fever include rash ("rose spots"), hepatosplenomegaly, epistaxis, and relative bradycardia. Rose spots make up a faint, salmon-colored, blanching, maculopapular rash located primarily on the trunk and chest. The rash is evident in ~30% of patients at the end of the first week and resolves after 2 to 5 days without leaving a trace. Patients can have two or three crops of lesions, and *Salmonella* can be cultured from punch biopsies of these lesions. The faintness of the rash makes it difficult to detect in dark-skinned patients. On occasion, patients who remain toxic manifest neuropsychiatric symptoms described as a "muttering delirium" or "coma vigil," with picking at bedclothes or imaginary objects.

Late complications, occurring in the third and fourth weeks of infection, are most common in untreated adults and include intestinal perforation and/or gastrointestinal hemorrhage. These complications can develop despite clinical improvement and presumably result from necrosis at the initial site of *Salmonella* infiltration in the Peyer's patches of the small intestine. Both complications are life-threatening and require immediate medical and surgical interventions, with broadened antibiotic coverage for polymicrobial peritonitis ([Chap. 130](#)) and treatment of gastrointestinal hemorrhages, including bowel resection.

Rare complications whose incidences are reduced by prompt antibiotic treatment include pancreatitis, hepatic and splenic abscesses, endocarditis, pericarditis, orchitis, hepatitis, meningitis, nephritis, myocarditis, pneumonia, arthritis, osteomyelitis, and parotitis. Despite prompt antibiotic treatment, relapse rates remain at ~10% in immunocompetent hosts.

Approximately 1 to 5% of patients with enteric fever become long-term, asymptomatic, chronic carriers who shed *S. typhi* in either urine or stool for >1 year. The incidence of chronic carriage is higher among women and among persons with biliary abnormalities (e.g., gallstones, carcinoma of the gallbladder) and gastrointestinal malignancies. The anatomic abnormalities associated with these conditions presumably allow prolonged colonization.

DIAGNOSIS

Other than a positive culture, no specific laboratory test is diagnostic for enteric fever. In 15 to 25% of cases, leukopenia and neutropenia are detectable. In the majority of cases, the white blood cell count is normal despite high fever. However, leukocytosis can develop in typhoid fever (especially in children) during the first 10 days of the illness, or later if the disease course is complicated by intestinal perforation or secondary infection. Other nonspecific laboratory results include moderately elevated values in liver function tests (aminotransferases, alkaline phosphatase, and lactate

dehydrogenase). In addition, nonspecific ST and T wave abnormalities can be seen on electrocardiograms.

The diagnostic "gold standard" is a culture positive for *S. typhi* or *S. paratyphi*. The yield of blood cultures is quite variable: it can be as high as 90% during the first week of infection and decrease to 50% by the third week. A low yield is related to low numbers of *Salmonella* (<15 organisms per milliliter) in infected patients and/or to recent antibiotic treatment. Centrifugation to isolate and culture the buffy coat, which contains abundant blood mononuclear cells associated with the bacteria, decreases time to isolation but does not affect culture sensitivity.

A diagnosis can also be based on positive cultures of stool, urine, rose spots, bone marrow, and gastric or intestinal secretions. Unlike blood cultures, bone marrow cultures remain highly (90%) sensitive despite 5 days of antibiotic therapy. Culture of intestinal secretions (best obtained by a noninvasive duodenal string test) can be positive despite a negative bone marrow culture. If blood, bone marrow, and intestinal secretions are all cultured, the yield of a positive culture is >90%. Stool cultures, while negative in 60 to 70% of cases during the first week, can become positive during the third week of infection in untreated patients. Although the majority of patients (90%) clear bacteria from the stool by the eighth week, a small percentage become chronic carriers and continue to have positive stool cultures for at least 1 year.

Several serologic tests, including the classic Widal test for "febrile agglutinins," are available; however, given high rates of false-positivity and false-negativity, these tests are not clinically useful. Polymerase chain reaction and DNA probe assays are being developed.

TREATMENT

In the preantibiotic era, the mortality rate from typhoid fever was as high as 15%. The introduction of treatment with chloramphenicol in 1948 greatly altered the disease course, decreasing mortality to <1% and the duration of fever from 14-28 days to 3-5 days. Chloramphenicol remained the standard treatment for enteric fever until the emergence of plasmid-mediated resistance to this drug in the 1970s. Given the increased mortality associated with resistance to chloramphenicol and the rare chloramphenicol-induced bone marrow toxicity, ampicillin (1 g orally every 6 h) and trimethoprim-sulfamethoxazole (TMP-SMZ; one double-strength tablet twice daily) became the mainstays of treatment.

In 1989, [MDR](#)*S. typhi* emerged. These bacteria are resistant to chloramphenicol, ampicillin, trimethoprim, streptomycin, sulfonamides, and tetracycline. Like chloramphenicol resistance, resistance to ampicillin and trimethoprim is plasmid-encoded. In 1994, 12% of *S. typhi* isolates in the United States were MDR. Thus either quinolones or third-generation cephalosporins are currently recommended for empirical antibiotic treatment. Despite efficient in vitro killing of *Salmonella*, first- and second-generation cephalosporins as well as aminoglycosides are ineffective in treating clinical infections.

Ceftriaxone (1 to 2 g intravenously or intramuscularly) for 10 to 14 days is equivalent to

oral or intravenous chloramphenicol in the treatment of susceptible strains. Preliminary studies indicate that a 5- to 7-day course of ceftriaxone is likely to be sufficient for treatment of uncomplicated cases. However, one recent report describes a ceftriaxone-resistant *Salmonella* strain isolated from a child with diarrhea and apparently acquired from antibiotic-treated cattle.

Quinolones are the only available oral antibiotics for the treatment of [MDRS. typhi](#) infections. The greatest experience has been gained for ciprofloxacin (500 mg orally twice a day for 10 days). Shorter courses of ofloxacin (10 to 15 mg/kg in divided doses twice daily for 2 to 3 days) have also been successful. However, quinolone resistance is emerging. In 1993, an outbreak of nalidixic acid-resistant *S. typhi* (NARST) infections in Vietnam was linked to chromosomal mutations in the gene encoding DNA gyrase (the target of the quinolones). NARST strains have also been isolated in India. Thus, all strains of *S. typhi* must be screened for resistance to nalidixic acid and tested for sensitivity to a clinically appropriate quinolone. Patients infected with NARST strains need to be treated with higher doses of ciprofloxacin (10 mg/kg twice a day for 10 days) or longer courses of ofloxacin (10 to 15 mg/kg in divided doses twice daily for 7 to 10 days) or with other antibiotics to which the strains are sensitive.

In cases of severe typhoid fever (fever; an abnormal state of consciousness -- i.e., delirium, obtundation, stupor, or coma -- or septic shock; and a positive culture for *S. typhi* or *S. paratyphi* A), dexamethasone treatment should be considered. In a single trial in Jakarta in the early 1980s in chloramphenicol-treated patients, treatment with dexamethasone (a single dose of 3 mg/kg followed by eight doses of 1 mg/kg, given every 6 h) decreased mortality from 56% to 10%.

The 1 to 4% of patients who develop chronic carriage of *Salmonella* can be treated for 6 weeks with an appropriate antibiotic. Treatment with oral amoxicillin, [TMP-SMZ](#), ciprofloxacin, or norfloxacin has been shown to be ~80% effective in eradicating chronic carriage of susceptible organisms. However, in cases of anatomic abnormality (e.g., biliary or kidney stones), eradication of the infection often cannot be achieved by antibiotic therapy alone but also requires surgical correction of the abnormalities.

PREVENTION AND CONTROL

Theoretically, it is possible to eliminate salmonellae that cause enteric fever since the bacteria survive only in human hosts and are spread by contaminated food and water. However, given the high prevalence of the disease in developing countries that lack good facilities for sewage disposal and water treatment, this goal is currently unrealistic. Thus, travelers to developing countries should be advised to monitor their food and water intake carefully and to consider vaccination.

Three vaccine alternatives are available: (1) a heat-killed, phenol-extracted, whole-cell vaccine (two parenteral doses); (2) Ty21a, an attenuated *S. typhi* vaccine (four oral doses); and (3) ViCPS, consisting of purified Vi polysaccharide from the bacterial capsule (one parenteral dose). In addition, an acetone-killed whole-cell vaccine is available only for use by the U.S. military. The minimal ages for vaccination with the whole-cell, Ty21a, and ViCPS vaccines are 6 years, 2 years, and 6 months, respectively. A large-scale meta-analysis of vaccine trials in populations of endemic

areas indicates that, while all three vaccines have similar efficacy for the first year, the 3-year cumulative efficacy of the whole-cell vaccine (73%) exceeds that of both Ty21a (51%) and purified Vi (55%). In addition, the heat-killed whole-cell vaccine maintains its efficacy for 5 years, while Ty21a and ViCPS most likely maintain their efficacy for 4 and 2 years, respectively. However, the whole-cell vaccine is associated with a much higher incidence of side effects than the other two vaccines: 16% of whole-cell vaccine recipients develop fever and 10% miss a day of work or school, while only 1 to 2% of persons receiving the alternative vaccines have any fever.

Although data on typhoid vaccines in travelers are limited, some evidence suggests that efficacy may be substantially lower than those for populations in endemic areas. The Centers for Disease Control and Prevention (CDC) currently recommends vaccination for persons traveling to developing countries who will have prolonged exposure to contaminated food and water or close contact with indigenous populations in rural areas. The only recommendations for domestic vaccination include people who have intimate or household contact with a chronic carrier or laboratory workers who frequently work with *S. typhi*. Given the decreased incidence of side effects and the similar short-term efficacy, the current bias is toward vaccination of travelers with either Ty21a or ViCPS.

Enteric fever is a reportable disease in the United States. This reporting system enables public health departments to track down potential source patients and thus to identify and treat chronic carriers in order to prevent further outbreaks. In addition, since 1 to 4% of patients with *S. typhi* infection become chronic carriers, it is important to monitor patients (especially those employed in child care or food handling) for chronic carriage and to treat this condition if indicated.

NONTYPHOIDAL SALMONELLOSIS

EPIDEMIOLOGY

The incidence of nontyphoidal salmonellosis has doubled in the United States over the past two decades. Currently, the [CDC](#) estimates that there are 2 million cases annually, with 500 to 2000 deaths. Although over 200 serovars of *Salmonella* are considered to be human pathogens, the majority of the reported cases in the United States is caused by *S. typhimurium* or *S. enteritidis*. The incidence of salmonellosis is highest during the rainy season in tropical climates and during the warmer months in temperate climates, coinciding with the peak in food-borne outbreaks. Morbidity and mortality associated with salmonellosis are highest among the elderly, infants, and immunocompromised individuals, including those with hemoglobinopathies and those infected with HIV or with pathogens that cause blockade of the reticuloendothelial system (e.g., patients with bartonellosis, malaria, schistosomiasis, or histoplasmosis).

Unlike *S. typhi* and *S. paratyphi*, whose only reservoir is humans, nontyphoidal salmonellosis is acquired from multiple animal reservoirs. The main mode of transmission is from food products contaminated with animal products or waste -- most commonly eggs and poultry but also undercooked meat, unpasteurized dairy products, seafood, and fresh produce.

S. enteritidis associated with chicken eggs is emerging as a major cause of food-borne disease. *S. enteritidis* causes infection of the ovaries and upper oviduct tissue of hens, resulting in contamination of the contents of eggs prior to shell deposition.

Approximately 1 in 20,000 eggs is thought to be infected with *S. enteritidis*. Between 1974 and 1994, there was a fivefold increase (from 5% to 25%) in the isolation of *S. enteritidis* from eggs in the United States; in 1998, the U.S. Department of Agriculture estimated that 80% of all salmonellosis cases were caused by infected eggs.

Eradication of *S. enteritidis* from hens has proven difficult, given that infection is spread to egg-laying hens both vertically from breeding flocks and horizontally through contact with rodents and manure. Transmission via contaminated eggs can be prevented by cooking of eggs such that the liquid yolk is solidified or through pasteurization of egg products.

Another factor in the increasing incidence of nontyphoidal salmonellosis in developed countries, including the United States, is related to the centralization of food processing and widespread distribution. For example, a 1994 outbreak of ~250,000 cases was linked to a pasteurized ice-cream premix most likely contaminated in tanker trucks that had previously carried unpasteurized eggs. Similar outbreaks have been traced to manufactured foods including pasteurized milk, infant formula, powdered-milk products, paprika-powdered potato chips, and a ready-to-eat savory snack. In addition, large outbreaks have been linked to fresh produce, including alfalfa sprouts, cantaloupe, fresh-squeezed orange juice, and sliced tomatoes, contaminated by manure or water at a single site and then broadly distributed.

A less common source of nontyphoidal *Salmonella* infections is exposure to exotic pets, especially reptiles. Fecal carriage rates in reptiles can be >90%. In the 1970s, 14% of cases of salmonellosis were attributed to small turtles; the distribution of these pets was subsequently prohibited by the U.S. Food and Drug Administration, with a resultant decline in rates of reptile-associated salmonellosis. However, since 1986, an increase in the popularity of nonbanned reptiles, including iguanas, has been followed by increases in rates of *Salmonella* infections. Other pets, including African hedgehogs, snakes, birds, rodents, baby chicks, ducklings, dogs, and cats, can also serve as potential vectors.

Antibiotic resistance is an increasing phenomenon among nontyphoidal *Salmonella* serovars. In particular, *S. typhimurium* of definitive phage type 104 (DT104) -- a serotype resistant to ampicillin, chloramphenicol, streptomycin, sulfonamides, and tetracyclines -- has become prominent in the United Kingdom. This serotype is associated with greater mortality and morbidity than other nontyphoidal *Salmonella* serotypes. Its acquisition is associated with exposure to ill farm animals and to a variety of meat products. The prevalence of *S. typhimurium* DT104 in the United States increased from 0.6% in 1979-1980 to 34% in 1996. Of concern is the isolation in the United Kingdom in 1996 of *S. typhimurium* DT104 strains resistant to ciprofloxacin (14%) or trimethoprim (24%).

CLINICAL MANIFESTATIONS

Gastroenteritis Infection with nontyphoidal *Salmonella* most often results in gastroenteritis indistinguishable from that caused by other bacterial and viral pathogens.

Nausea, vomiting, and diarrhea occur 6 to 48 h after the ingestion of contaminated food or water. Patients often experience abdominal cramping and fever (38 to 39°C, or 100.5 to 102.2°F). The diarrhea is usually characterized as loose, nonbloody stools of moderate volume. However, large-volume watery stools, bloody stools, or symptoms of dysentery do not rule out the diagnosis. Rarely, *Salmonella* causes a syndrome of pseudoappendicitis or an illness that mimics inflammatory bowel disease.

Gastroenteritis caused by nontyphoidal *Salmonella* is usually self-limited. Diarrhea resolves within 3 to 7 days and fever within 72 h. Stool cultures remain positive for 4 to 5 weeks after infection and -- in rare cases of chronic carriage (<1%) -- remain positive for >1 year. Antibiotic treatment is usually not recommended and in some studies has prolonged carriage of *Salmonella*. Neonates, the elderly, and the immunosuppressed (e.g., HIV-infected patients) with nontyphoidal *Salmonella* gastroenteritis are especially susceptible to dehydration and dissemination and may require hospitalization and antibiotic therapy.

Bacteremia and Endovascular Infections Up to 5% of patients with nontyphoidal *Salmonella* gastroenteritis have positive blood cultures, and 5 to 10% of these bacteremic persons develop localized infections. Bacteremia is particularly common and persistent among infants, the elderly, and patients with severe underlying infection or immunosuppression (e.g., transplant recipients, HIV-infected patients). *Salmonellae* have a propensity for infection of vascular sites; if >50% of three or more blood cultures are positive, an endovascular infection should be suspected. Preexisting valvular heart disease is a strong risk factor for the development of endocarditis, while atherosclerotic plaque, prosthetic grafts, and aortic aneurysms are associated with arteritis. Arteritis should be suspected in elderly patients who have a history of prolonged fever with associated back, chest, or abdominal pain preceded by gastroenteritis. Endocarditis and arteritis are rare (<1% of cases) but are associated with potentially morbid complications. Endocarditis can be complicated by cardiac valve perforation or by ring or septal abscesses, while arteritis can be associated with mycotic aneurysms, ruptured aneurysms, or vertebral osteomyelitis.

Unlike most nontyphoidal *Salmonella* serotypes, *S. choleraesuis* and *S. dublin* are frequently associated with sustained bacteremia and fever, often in the absence of a history of gastroenteritis. Similarly, these serotypes appear to be especially invasive and are often associated with metastatic infection.

Localized Infections

Intraabdominal Infections Intraabdominal infections due to nontyphoidal *Salmonella* are rare and usually manifest as hepatic or splenic abscesses or as cholecystitis. Involvement of the pancreas and adrenals and even an infected pheochromocytoma have been reported. Risk factors include anatomic abnormalities of the hepatobiliary system, including gallstones; abdominal malignancy; and sickle cell disease (especially with splenic abscesses). Eradication of the infection often requires surgical correction of anatomic abnormalities and drainage of abscesses.

Central Nervous System Infections *Salmonella* infections of the central nervous system usually manifest as meningitis, although cerebral abscesses have been found.

Meningitis is usually seen in neonates (<4 months old) and is associated with severe sequelae, including residual seizures, hydrocephalus, ventriculitis, abscess formation, subdural empyema, and permanent disability (e.g., mental retardation and paralysis).

Pulmonary Infections Nontyphoidal *Salmonella* pulmonary infections usually present as lobar pneumonia, sometimes complicated by lung abscesses, empyemas, pleural effusions, and bronchopleural fistulas. The majority of cases occur in patients with a preexisting abnormality of lung or pleura, including malignancy. Additional risk factors include sickle cell disease and glucocorticoid use. It is important to determine whether the pulmonary infection is in fact due to *Salmonella* or whether it is a secondary infection.

Urinary and Genital Tract Infections Urinary tract infections caused by nontyphoidal salmonellae present as either cystitis or pyelonephritis, usually in association with malignancy, urolithiasis, structural abnormalities, or immunosuppression (HIV infection, renal transplantation). Genital infections due to these bacteria are rare and present as ovarian and testicular abscesses, prostatitis, or epididymitis. Like other focal infections, both genital and urinary tract infections can be complicated by abscess formation.

Bone, Joint, and Soft Tissue Infections *Salmonella* osteomyelitis most commonly affects the femur, tibia, humerus, or lumbar vertebrae and is most often seen in association with sickle cell disease, hemoglobinopathies, or preexisting bone disease (e.g., fractures). Prolonged antibiotic treatment is recommended to decrease the incidence of relapse and chronic osteomyelitis. Septic arthritis occurs in the same patient population as osteomyelitis and usually presents in the knee, hip, or shoulder joints. Reactive arthritis (Reiter's syndrome) can follow *Salmonella* gastroenteritis and is seen most frequently in persons with the HLA-B27 histocompatibility antigen. *Salmonella* can cause rare soft tissue infections, usually at sites of local trauma in immunosuppressed patients.

DIAGNOSIS

Nontyphoidal *Salmonella* gastroenteritis is diagnosed when *Salmonella* is cultured from stool. All salmonellae isolated in clinical laboratories should be sent to local public health departments. In cases where there is concern about bacteremia (i.e., those including prolonged or recurrent fever), blood cultures are indicated. Once bacteremia is documented, it is important to determine whether it is high-grade (>50% of three or more blood cultures positive); if so, endovascular infection is possible and further evaluation to identify the source is indicated. In addition, depending on clinical symptoms and on whether metastatic disease is suspected, other body fluids, such as joint fluid or cerebrospinal fluid, should be cultured.

TREATMENT

Antibiotic treatment is not generally recommended for *Salmonella* gastroenteritis. The symptoms are usually self-limited and have not been demonstrated to be altered by short courses of antibiotics. In addition, in case-control and double-blind placebo-controlled trials, antibiotic treatment has been associated with increased rates of relapse and prolonged gastrointestinal carriage. Dehydration secondary to diarrhea

should be treated with fluid and electrolyte replacement.

However, preemptive antibiotic treatment should be considered in patients at increased risk for metastatic infection. These patients include neonates (probably up to 3 months of age); persons >50 years old (because of the high risk of atherosclerotic plaque or aneurysm); transplant recipients; and patients with lymphoproliferative disease, HIV infection, prosthetic joints, vascular grafts, significant joint disease, or underlying sickle cell disease. This group should receive a course of oral or intravenous antibiotics lasting for 2 or 3 days or until defervescence. Longer courses of antibiotics are not recommended because they have been associated with higher rates of chronic carriage and relapse. Rare cases of chronic nontyphoidal *Salmonella* carriage should be treated with a prolonged antibiotic course, as described above for chronic carriage of *S. typhi*.

Focal infections or life-threatening bacteremia with nontyphoidal *Salmonella* should be treated with antibiotics (at the same doses used for enteric fever). Given the increasing prevalence of antibiotic resistance, empirical therapy should include a third-generation cephalosporin and/or a quinolone. If the bacteremia is low-grade (<50% of blood cultures positive), the patient should be treated for 7 to 14 days. Patients with AIDS and *Salmonella* bacteremia should receive 1 to 2 weeks of intravenous antibiotic therapy followed by 4 weeks of oral therapy with quinolones. Patients who relapse after this regimen should receive long-term suppressive therapy with a quinolone or [TMP-SMZ](#), as indicated by bacterial sensitivities.

If the patient has an endovascular infection or endocarditis, treatment for 6 weeks with intravenous b-lactam antibiotics is indicated. Chloramphenicol treatment has been associated with high failure rates and is not recommended. Limited case reports have described the successful treatment of *Salmonella* endovascular infections with quinolones, which may prove an alternative approach in cases caused by sensitive strains. However, concern remains about the development of quinolone resistance during prolonged therapy. Surgical resection of infected aneurysms or other infected endovascular sites is often required. If surgical resection is not possible, lifelong suppressive antibiotic therapy may be indicated. For extraintestinal nonvascular infections, 2 to 4 weeks of antibiotic therapy (depending on the site) are usually recommended. In cases of chronic osteomyelitis, abscesses, and urinary or biliary tract abnormality, surgical interventions may be required in addition to prolonged antibiotic therapy to eradicate infection.

PREVENTION AND CONTROL

The incidence of nontyphoidal salmonellosis continues to rise along with rates of emergence of antibiotic-resistant strains. The increased centralization of food production plays a prominent role in the growing incidence, as one oversight can result in rapid, widespread distribution of contaminated food. Thus, it is important to monitor every step of food production, from handling of raw products to preparation of finished foods. In particular, with the increasing prevalence of *S. enteritidis* in egg-laying hens, it is recommended that pasteurized eggs be substituted for bulk-pooled eggs at all nursing homes, hospitals, and commercial food-service establishments. All cases of nontyphoidal salmonellosis should be reported to public health departments, since tracking and monitoring of these cases result in the identification of the sources of local

outbreaks and help authorities anticipate large-scale international outbreaks. Lastly, the prudent use of antimicrobial agents in both humans and animals is necessary to minimize the further emergence of antibiotic-resistant strains.

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157. SHIGELLOSIS - Gerald T. Keusch

DEFINITION

Shigellosis is an acute infectious inflammatory colitis due to one of the members of the genus *Shigella*. Although the disease is often referred to as "bacillary dysentery," many patients have only mild watery diarrhea and never develop dysenteric symptoms. Less severe illness predominates in industrialized countries such as the United States, whereas more severe, often fatal dysentery occurs in patients in developing countries.

ETIOLOGIC AGENT

Shigellae are slender, gram-negative, nonmotile bacilli and are members of the family Enterobacteriaceae and the tribe Escherichieae. They are so closely related to *Escherichia coli* that the two genera cannot be distinguished by DNA hybridization methods. In fact, *Shigella* can be thought of as a differentiated pathogenic *E. coli*. The four *Shigella* species (*S. dysenteriae*, *S. flexneri*, *S. boydii*, and *S. sonnei*) are defined on the basis of surface somatic O antigens and carbohydrate fermentation patterns. Most are lactose-negative (*S. sonnei* is a late lactose fermenter) and produce acid but not gas from glucose, resulting in a typical acid butt and alkaline slant in triple sugar iron agar without H₂S production. The genus is characterized by its ability to invade intestinal epithelial cells and to cause infection and illness in humans, even when the inoculum is small (a few hundred to a few thousand organisms).

EPIDEMIOLOGY

Worldwide, it is estimated that at least 140 million cases of shigellosis and almost 600,000 deaths due to shigellosis occur annually among children under the age of 5 years, primarily in developing countries. The organism is found everywhere in the world but is most common where poor environmental sanitation and crowding facilitate transmission from person to person. A major outbreak took place in the makeshift camps for refugees fleeing the Rwandan civil war in 1994, with thousands of cases and high mortality.

Data collected by the Centers for Disease Control and Prevention in the United States from 1967 through 1988 suggest an average annual incidence of 6 *Shigella* infections per 100,000 population, with periodic hyperendemic increases (primarily due to large outbreaks of *S. sonnei* infection) raising the rate to between 9 and 10 per 100,000. On the basis of these data, the annual number of episodes of shigellosis in the United States is estimated at 25,000 to 30,000. Incidence rates approximate 27 per 100,000 among children 1 to 4 years of age but are only 2.6 per 100,000 among persons 20 years of age or older. Cases are detected most commonly in counties with a relatively high proportion of low-income minority-group residents, including African Americans, Hispanics, and Native Americans; rates are especially high in poor urban communities, in day-care centers, and among retarded children in custodial care.

A comparison with rates among rural Guatemalan Indian children during the same period puts this disease burden into perspective. A prospective surveillance study among 321 such children revealed an annual incidence of nearly 10,000 per 100,000.

Since the description of the genus *Shigella*, major global shifts in the prevalence of its four species have been noted. Until World War I, *S. dysenteriae* type 1 was the predominant isolate, frequently causing devastating epidemics with high mortality until it was replaced by *S. flexneri*. Since World War II, however, *S. flexneri* has been steadily replaced by *S. sonnei* in the industrialized countries. The reasons for these shifts are not clear. *S. boydii*, the fourth species, has remained largely confined to the Indian subcontinent.

Shigella is highly host-adapted and is a natural pathogen only of humans and a few other primates. Transmission from person to person takes place by the fecal-oral route, generally via direct contact but sometimes through contaminated vectors such as food, water, flies, and fomites. Contaminated imported parsley from Mexico was responsible for one multistate outbreak of *S. sonnei* diarrhea. The organism can even be transmitted during participation in recreational water sports in fecally contaminated pools or lakes and can spread rapidly among confined populations in close contact -- for example, in day-care centers, in institutions for the mentally retarded, on cruise ships, or among military personnel. *Shigella* can be transmitted by anal-oral sexual practices among gay men; these cases are almost always due to *S. flexneri*. Rates of *Shigella* infection among HIV-infected individuals greatly exceed those in the non-HIV-infected population ([Chap. 309](#)).

Shigellosis is associated with a high rate of secondary household transmission. As many as 40% of children and 20% of adults who are household contacts of a case (generally a preschool child) will develop *Shigella* infection; the infection is often symptomatic in children but asymptomatic in adults, who seem to have an acquired immunity. In contrast, epidemic disease affects all ages, with clusters of severe and fatal cases in the very young and the very old. Since 1969, epidemic *S. dysenteriae* type 1 has reappeared in Latin America, in the Indian subcontinent and elsewhere in Asia, and in central and southern Africa and has been associated with relatively high mortality rates due to antimicrobial resistance and inadequate diagnosis and case management. Prolonged asymptomatic carriage is uncommon; unless there is underlying malnutrition, the organisms are generally cleared in a few weeks.

PATHOGENESIS AND PATHOLOGY

Shigellae are orally ingested and, because they survive low pH more easily than other enteric pathogens (a genetically regulated property), seem to have little difficulty in passing the gastric acid barrier. An essential step in pathogenesis is invasion of colonic epithelial cells and cell-to-cell spread of infection. This step involves initial attachment of the organism to colonic cells, entry by an endocytic mechanism in which organisms are initially encased in and then escape from plasma membrane-enclosed vesicles, and a jet propulsion-like movement to the cell membrane, from which the organism can invade the adjacent cell. This sequence of events not only provides the organism with a means to evade host defenses but also allows its effective local spread. Although invasion is initially innocuous, subsequent intracellular multiplication causes cell damage and death, ultimately resulting in characteristic mucosal ulcerations.

These events are extremely complicated and require the functions of multiple genes and

regulatory elements encoded on both the chromosome and a large 120- to 140-MDa plasmid present in all virulent shigellae as well as enteroinvasive *E. coli* (EIEC), which can cause a *Shigella*-like disease. The number of structural and regulatory genes known to be involved in pathogenesis continues to increase as the process continues to be dissected. Some of these gene products induce the phagocytosis-like uptake of the organism by causing rearrangements of the host cell's cytoskeleton. Once a single *Shigella* organism has invaded a single host cell, the entire process of bacterial escape from the phagocytic vesicle into the host cell's cytoplasm, multiplication, and cell-to-cell spread can take place without exposure of the bacterium to the extracellular milieu and to the host's defenses.

It was originally thought that shigellae invade the host across the intestinal epithelial cells; however, studies using cell culture or a rabbit-ileum in vivo model have suggested that the initial invasion may occur via the antigen-sampling M cell. The resulting limited penetration by organisms initiates an inflammatory response with neutrophil infiltration of the lamina propria, which alters the functional integrity of tight junctions between epithelial cells. These changes allow more organisms to breach the mucosal barrier at intercellular junctions and are essential for the development of illness. If neutrophil migration is directly inhibited by the treatment of animals with antibody to CD18, the escalating invasion by microorganisms does not take place.

Escape from the phagocytic vesicle is necessary for the virulence of shigellae and permits multiplication of the organisms in the cytoplasm. The multiplying organisms spread within the cytoplasm to the plasma membrane of the host cell and then from cell to cell. This spread is achieved by the polymerization of actin at the back end of the dividing bacteria (defined relative to the subsequent direction of motion). Binding and cross-linking by the host protein plastin result in a sphincter-like contraction that provides a forward propulsive force. This so-called actin motor is energized by ATP generated by a microbial-encoded ATPase called *IcsA*, which is, at the same time, phosphorylated and regulated by cyclic nucleotide-dependent protein kinases of the host. Phosphorylation may serve as a molecular host-defense mechanism to modulate virulence, limiting microbial spread.

Another important host protein involved in pathogenesis of shigellosis is the cadherin L-CAM, which is essential in the cell-to-cell spread of infection. Mutations in L-CAM alter the long finger-like protrusions induced by shigellae when they reach the plasma membrane and impair their subsequent fusion with the plasma membrane of the adjacent cell, thus inhibiting the transfer of the bacterium from one cell to another. Ultimately, the invaded host cell dies, possibly as a result of apoptosis induced by or during the process of microbial invasion.

Another property of apparent importance in virulence for *S. dysenteriae* type 1 is the ability to produce Shiga toxin, which is encoded by the iron-regulated chromosomal gene *stx*. Shiga toxin is composed of two distinct peptide subunits, each with highly conserved active regions. The first, located on the larger A subunit, is an *N*-glycosidase that hydrolyzes adenine from specific sites of ribosomal RNA of the mammalian 60S ribosomal subunit, irreversibly inhibiting protein synthesis. The second common region is a binding site on the B subunit that recognizes glycolipids of target cell membranes that terminate in a galactose-1,4-galactose disaccharide. The glycolipid Gb3,

containing a gal-gal-glu trisaccharide, is a specific receptor present on toxin-sensitive rabbit intestinal villus cells but not crypt cells, and toxin action is specific for the former.

Wild-type toxigenic *S. dysenteriae* causes more severe illness in primates than does an isogenic toxin-negative mutant. The toxin of this organism, the prototype of a family of related toxin proteins produced by enterohemorrhagic *E. coli* (EHEC), appears to play a role in the pathogenesis of microangiopathic complications, hemolytic-uremic syndrome (HUS), and thrombotic thrombocytopenic purpura: only toxin-producing shigellae and *E. coli* are associated with these systemic illnesses. Two new *Shigella* enterotoxins, ShET-1 and -2, have been described; the former is restricted almost exclusively to *S. flexneri* 2a, whereas the latter is distributed more widely (e.g., in the physiologically similar [EIEC](#)). The two enterotoxins are encoded by chromosomal and plasmid genes, respectively. Both toxins alter electrolyte transport by segments of gut in vitro and cause net fluid secretion in vivo in ligated rabbit ileal loops. Moreover, both toxins induce antibody in infected humans. However, their role (if any) in the pathogenesis of the watery diarrhea phase of shigellosis remains uncertain.

In shigellosis, the epithelial surface of the human colon shows extensive ulcerations, with an exudate consisting of desquamated colonic cells, polymorphonuclear leukocytes, and erythrocytes; the ulcerations may resemble a pseudomembrane in severely affected areas. Marked mucus depletion and increased mitotic activity are evident in the crypt regions and presumably reflect a response to the loss of surface colonic cells. The lamina propria is edematous and hemorrhagic and is infiltrated by neutrophils and plasma cells. There is also swelling of capillary and venular endothelial cells, with margination of neutrophils. At the ultrastructural level, bacteria can be seen within vesicles as well as free in the cytoplasm. Histologic examination of colon from dysenteric humans shows an alteration of mucosal endothelial cells similar to that induced by endotoxin [lipopolysaccharide (LPS)]. Shiga toxin (protein) targets endothelial cells as well, especially when toxin receptor expression is upregulated by exposure to LPS or proinflammatory cytokines. Levels of circulating LPS are high in *S. dysenteriae* type 1 infection and somewhat lower in *S. flexneri* infection, even without bacteremia. The frequency of endotoxemia in shigellosis suggests a broader role for LPS in the pathogenesis of the disease. One likely mechanism is related to the ability of LPS to induce cytokine gene transcription and the strong association of cytokine secretion and inflammation. However, bacterial invasion of the mucosa itself activates the transcription factor NF- κ B, which is involved in regulation of cytokine synthesis. Cytokine-producing cells are present in the mucosa of patients infected with *S. dysenteriae* or *S. flexneri* and in their stools as well. In fact, the number of cells producing interleukin 1, interleukin 6, interferon α , and transforming growth factor β is directly related to the severity of the inflammation. Inflammatory changes in *Shigella* infection thus appear to be components of the pathogenesis of dysentery as much as they are a consequence of the bacterial invasive process.

Epidemiologic evidence indicates that immunity develops and is serotype-specific. The precise nature of this immunity is not known. Common surface outer-membrane proteins involved in invasion elicit serum antibodies; although these are cross-reactive among *Shigella* species and serotypes, they do not seem to be protective. The serotype-specific determinants are likely to be somatic antigens, as serum antibody to [LPS](#) predicts resistance to infection, and there is evidence of IgA-mediated mucosal

responses to LPS during convalescence from shigellosis.

CLINICAL MANIFESTATIONS

Shigellosis in the United States, due primarily to *S. sonnei*, is typically an ambulatory disease, presenting as a self-limited nonbloody watery diarrhea chock full of neutrophils. The spectrum of clinical shigellosis was shown in a study in which adult volunteers ingested 10,000 organisms of *S. flexneri* type 2a. While approximately one-quarter of the volunteers never became ill, over the first 24 to 48 h ~25% developed transient fever, another 25% had fever and self-limited watery diarrhea, and the remaining 25% had fever and watery diarrhea that progressed to bloody diarrhea and dysentery. In young children in particular, the temperature can rise rapidly to 40° to 41°C and sometimes results in generalized seizures. These seizures rarely recur or result in serious sequelae. Dysentery is characterized by frequent passage (usually 10 to 30 times per day) of small-volume stools consisting of blood, mucus, and pus; this diarrhea is accompanied by abdominal cramps and tenesmus -- the painful straining with stooling that may lead to rectal prolapse, especially in young children. Severe dysentery is most likely in infection due to *S. dysenteriae* type 1, occurs less commonly with *S. flexneri*, and is least likely in *S. sonnei* infection. Patients with mild disease generally recover without specific therapy in a few days to a week. Severe shigellosis can progress to toxic dilatation and colonic perforation, which may be fatal.

Endoscopy shows the mucosa to be hemorrhagic, with mucous discharge and focal ulcerations and sometimes with overlying exudate. The majority of lesions are in the distal colon and progressively diminish in the more proximal segments of large bowel. Mild dehydration is common among patients with watery diarrhea; severe dehydration is very rare. With extensive colonic involvement, protein-losing enteropathy can occur and can have important adverse nutritional consequences, especially for already poorly nourished children.

A variety of *extraintestinal complications* of shigellosis have been described. The majority arise in patients in developing countries and are related both to the prevalence of infections due to *S. dysenteriae* type 1 and *S. flexneri* and to the poor nutritional state of the host. For example, bacteremia, thought to be relatively infrequent in the United States, develops in up to 8% of patients hospitalized for shigellosis in Dacca, Bangladesh. The causative *Shigella* species is isolated from half the patients; other Enterobacteriaceae are found in the remainder. Bacteremia is associated with higher-than-usual mortality and is more common among infants <1 year of age and among persons with protein-energy malnutrition. Persistent and clinically severe *Shigella* bacteremia has been encountered in the United States in patients with AIDS ([Chap. 309](#)).

[HUS](#) may occur with *S. dysenteriae* type 1 infection. In the United States, the more likely cause of HUS is one of the hemorrhagic colitis-causing strains of *E. coli* (such as *E. coli* O157:H7) that produce high levels of Shiga-family toxins. HUS usually develops toward the end of the first week of shigellosis, when dysentery is already resolving. Oliguria and a marked drop in hematocrit (by as much as 10% within 24 h) are the first signs and may progress to anuria with renal failure and to severe anemia with congestive heart failure, respectively. Even with advanced therapy, 5 to 10% of patients with HUS die of

the acute illness. In addition, renal damage progresses slowly over several decades in survivors, an estimated 50% of whom develop significant renal failure and most of whom require long-term dialysis or renal transplantation. Leukemoid reactions, with leukocyte counts of >50,000/uL, may occur along with HUS; thrombocytopenia (with 30,000 to 100,000 platelets/uL) is common. Profound hyponatremia and severe hypoglycemia may be documented. Central nervous system abnormalities include encephalopathic symptoms, seizures, altered consciousness, and bizarre posturing.

Less common extraintestinal manifestations include seizures in some patients and reactive arthritis in others; both of these manifestations are usually due to infection with *S. flexneri* strains. In patients expressing histocompatibility antigen HLA-B27, the full triad of Reiter's syndrome sometimes develops ([Chap. 315](#)). Pneumonia, meningitis, vaginitis (in prepubertal girls), keratoconjunctivitis, and "rose spot" rashes are rare events.

DIAGNOSIS AND LABORATORY FINDINGS

Shigellosis is the principal bacterial cause of dysentery and should be considered whenever a patient presents with bloody diarrhea. However, in the United States, because *S. sonnei* is the most common species, most patients present with fever and nonbloody watery diarrhea indistinguishable from signs caused by other bacterial or viral agents of mild to moderate diarrhea, while many patients with bloody diarrhea have [EHEC](#) as the cause. The specific diagnosis is based on culture of *Shigella* from the stool; however, diagnosis by the polymerase chain reaction is possible, and a commercial enzyme immunoassay to detect Shiga-family toxins in stool can identify most patients infected with *S. dysenteriae* type 1 or EHEC within 3 h. The yield of *Shigella* is increased if the organism is sought by stool culture when the patient has fecal leukocytes or bloody diarrhea. The organism is very labile and must be transferred quickly to plates or holding media (such as buffered glycerol saline) if it is to survive. Stool samples are preferable to swabs; when the latter are used, a rectal sample should be obtained. More than one selective medium should be used for culture -- i.e., MacConkey and one other, such as Hektoen enteric or xylose-lysine-deoxycholate. Stool cultures to diagnose nonbloody watery diarrhea have a very low yield of positives and are not cost-effective.

Serologic tests can be performed, since antibodies to somatic antigens develop early in the acute phase of disease. However, the resources for such tests are not generally available, and serologic assessments usually are used only for epidemiologic studies.

The differential diagnosis includes inflammatory colitis due to other microbial agents: [EHEC](#), [EIEC](#), *Campylobacter jejuni*, *Salmonella enteritidis*, *Yersinia enterocolitica*, *Clostridium difficile*, and the protozoan *Entamoeba histolytica*. Ulcerative colitis and Crohn's colitis are among the "noninfectious" conditions that should be considered ([Chap. 287](#)). All these infections except that due to *E. histolytica* are associated with the presence of large numbers of fecal leukocytes. Amebiasis can be diagnosed by the detection of erythrophagocytic trophozoites in the stool ([Chap. 213](#)).

Other laboratory studies are nonspecific and may disclose neutrophilic leukocytosis, anemia due to blood loss with hemorrhagic diarrhea, prerenal azotemia, or (if watery

diarrhea has been pronounced) hyperchloremic acidosis. Laboratory findings in shigellosis complicated by [HUS](#) are discussed above.

TREATMENT

The mild to moderate dehydration in shigellosis is readily corrected with oral rehydration solutions ([Chap. 159](#)). The role of antibiotic therapy is variable and depends on the organism and the severity of disease. Since *S. sonnei* infection is usually self-limited, culture results generally do not become available until the patient is better and there is little clinical need for further therapy. The use of antibiotics in severe cases with bloody diarrhea or dysentery reduces the duration of illness and can shorten the carriage state. Resistance to sulfonamides, streptomycin, chloramphenicol, and tetracyclines is almost universal, and many shigellae are now resistant to ampicillin and trimethoprim-sulfamethoxazole as well. Knowledge of the pattern of resistance in a given population, which can change with time, is useful. In the United States, multiresistant strains are most likely to be acquired during travel abroad; either ampicillin (50 to 100 mg/kg per day in children or 2 g/d in adults, in divided doses) or trimethoprim-sulfamethoxazole (8/40 mg/kg per day in children or 2 regular-strength tablets twice a day in adults, given for 5 days) is generally recommended for domestically acquired infection. Short courses of treatment (1 or 3 days) or even single doses of drugs like tetracycline and ciprofloxacin have been employed with success and may soon become the standard. Amoxicillin should *not* be substituted for ampicillin because it is not effective against shigellosis. In developing countries, where resistance to both of these drugs is commonplace, the drug of choice for the treatment of multiresistant *S. dysenteriae* type 1 infections has been nalidixic acid (55 mg/kg per day for 5 days); however, resistance to the latter agent is increasing in prevalence. The 4-fluoroquinolones (e.g., ciprofloxacin) are highly effective against all strains ([Chap. 137](#)) but are currently too costly in the developing world and are not yet approved for use in children under 17 in the United States; these drugs have caused cartilage damage in young rodents during toxicity tests, although there is no evidence for a similar effect of therapeutic doses in humans. Alternative drugs shown to be effective include oral pivamdinocillin (amdinocillin, pivoxil, pivmecillinam; still not available in the United States), azithromycin, and intravenous ceftriaxone (50 mg/kg per day for 5 days). In small-scale clinical trials, cephalexin has had no effect in limiting symptoms; single doses of ceftriaxone may be effective, but more information is needed. No antibiotic treatment is recommended for the convalescent carrier state, which usually lasts no more than several weeks. Patients with AIDS may develop chronic carriage of *Shigella* and may be subject to relapsing infection with bacteremia ([Chap. 309](#)). This cycle may be interrupted by prolonged (several weeks') treatment with a quinolone.

The role of antimotility agents such as atropine sulfate and diphenoxylate (Lomotil) and loperamide (Imodium) in the early phases of shigellosis is controversial. Loperamide, in particular, may reduce diarrhea and in one study was highly effective in combination with antimicrobials. However, these antimotility drugs are suspected of enhancing the severity of disease by delaying excretion of organisms and thus facilitating further invasion of the mucosa and complicating toxic megacolon. Therefore, they are contraindicated in infants and young children. In adults, these agents are contraindicated for use in the dysenteric phase of disease.

Treatment of complications of shigellosis often differs in developed and developing countries. For example, antibiotic-unresponsive toxic megacolon, with or without perforation, is often managed by colectomy in the United States. Surgery is less often employed in developing countries because of a lack of availability or difficulties in ileostomy management. [HUS](#) often requires dialysis. In developing countries, dialysis may be needed relatively infrequently because azotemia is slow to develop and the risk of significant hyperkalemia is often diminished by a preexisting deficiency in total-body potassium, with malnutrition and wasting of lean body mass. The management of hyponatremia, usually caused by inappropriate secretion of antidiuretic hormone (vasopressin), is governed by the severity of the condition and the symptomatic state of the patient, as outlined in [Chap. 49](#). Infusion of glucose can reverse clinical manifestations caused by hypoglycemia, and responses can be monitored by finger-stick blood glucose tests if no biochemistry laboratory is available. Optimal nutritional management is needed to correct deficiencies due to underlying malnutrition as well as the superimposed catabolic stress and protein-losing enteropathy of shigellosis. Nutritional support should begin during the acute illness and may be required for months thereafter ([Chap. 76](#)).

PREVENTION

Direct-contact transmission of shigellosis can be prevented by appropriate environmental and personal hygiene. Hand washing with soap and water, decontamination of water supplies, use of sanitary latrines or toilets, and precautions in the preparation and storage of food can all reduce the primary and secondary transmission of *Shigella* infection. In highly endemic developing countries, infants are protected during the period of exclusive breast feeding, which should be encouraged. Any measures that reduce the burden of malnutrition will also reduce the burden of shigellosis in the population. Stool precautions should be instituted for hospitalized infected patients to ensure safe disposal of infected excreta and linens, and hospital personnel must wash their hands and medical instruments (such as stethoscopes) after each contact with an infected patient. Cohorting of asymptomatic infected children, use of antibiotics to reduce infectiousness, and scrupulous attention to hygiene are usually successful in nosocomial outbreaks. Children in day care must be kept at home while clinically ill and ideally should have a negative stool culture before returning to the day-care facility. Likewise, food handlers who develop shigellosis should be culture-negative before returning to work. Antibiotic treatment is not indicated for the asymptomatic carrier state. No effective vaccine is available, although promising initial results have been reported with an *S. sonnei* [LPS](#)-protein conjugate vaccine.

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158. INFECTIONS DUE TO *CAMPYLOBACTER* AND RELATED SPECIES - Martin J. Blaser

DEFINITION

Bacteria of the genus *Campylobacter* and of the related genera *Arcobacter* and *Helicobacter* ([Chap. 154](#)) cause a variety of pyogenic infections. Although acute diarrheal illnesses are most common, these organisms may cause infections in virtually all parts of the body, especially in compromised hosts, and these infections may have late nonsuppurative sequelae. The designation *Campylobacter* comes from the Greek for "curved rod" and refers to the organism's vibrio-like morphology.

ETIOLOGY

Campylobacters are motile, non-spore-forming, curved gram-negative rods. Originally known as *Vibrio fetus*, these bacilli were reclassified as a new genus in 1973, after it was recognized that they were quite dissimilar to other vibrios. Since then, more than 15 species have been identified. These species are currently divided into three genera: *Campylobacter*, *Arcobacter*, and *Helicobacter*. Not all of the species are pathogens of humans. The human pathogens can be divided into two major groups: those that primarily cause diarrheal disease and those that cause extraintestinal infection. The principal diarrheal pathogen is *C. jejuni*, which accounts for 80 to 90% of all cases of recognized illness due to campylobacters. Other organisms that cause diarrheal disease include *C. coli*, *C. upsaliensis*, *C. lari*, and *C. fetus*. The major species causing extraintestinal illnesses is *C. fetus*; however, any of the diarrheal agents may cause systemic or localized infection as well. Neither aerobes nor strict anaerobes, these microaerophilic organisms are adapted for survival in the gastrointestinal mucous layer. This chapter will focus on *C. jejuni* and *C. fetus* as the major pathogens and prototypes for their groups; the key features of infection are listed by species (excluding *C. jejuni*, described in detail in the text below) in [Table 158-1](#).

EPIDEMIOLOGY

Campylobacters are found in the gastrointestinal tract of many animals used for food (including poultry, cattle, sheep, and swine) and of many household pets (including birds, dogs, and cats). These microorganisms usually do not cause illness in their animal hosts. In most cases, campylobacters are transmitted to humans in raw or undercooked food products or through direct contact with infected animals. In the United States and other developed countries, ingestion of contaminated poultry that has not been sufficiently cooked is the most common means of acquiring infection (50 to 70% of cases). Other modes of transmission include ingestion of raw (unpasteurized) milk or untreated water, contact with infected household pets, travel to developing countries (campylobacters being among the causes of traveler's diarrhea; [Chap. 131](#)), and (occasionally) contact with an index case who is incontinent of stool.

Campylobacter infections are not rare. Several studies indicate that, in the United States, diarrheal disease due to campylobacters is more common than that due to *Salmonella* and *Shigella* combined. Infections occur throughout the year, but their incidence peaks during summer and early autumn. Persons of all ages are affected;

however, attack rates for *C. jejuni* are highest among young children and young adults, while those for *C. fetus* are highest at the extremes of age. Systemic infections due to *C. fetus* (and to other *Campylobacter* and related species) are most common in compromised hosts. Persons at increased risk include those with AIDS, hypogammaglobulinemia, neoplasia, liver disease, diabetes mellitus, and generalized atherosclerosis as well as pregnant women. However, apparently healthy nonpregnant persons occasionally develop transient *Campylobacter* bacteremia as part of a gastrointestinal illness.

In developing countries, *C. jejuni* infections are hyperendemic, with the highest rates among children <2 years old. Infection rates fall with age, as does the illness-to-infection ratio; these observations suggest that frequent exposure to *C. jejuni* leads to the acquisition of immunity.

PATHOLOGY AND PATHOGENESIS

Many *C. jejuni* infections are subclinical, especially in partially immune hosts. Most illnesses occur within 2 to 4 days (range, 1 to 7 days) of exposure to the organism in food or water. The sites of tissue injury include the jejunum, ileum, and colon. Biopsies show an acute nonspecific inflammatory reaction, with neutrophils, monocytes, and eosinophils in the lamina propria, as well as damage to the epithelium, including loss of mucus, glandular degeneration, and crypt abscesses. Biopsy findings may be consistent with Crohn's disease or ulcerative colitis, but these "idiopathic" chronic inflammatory diseases should not be diagnosed unless infectious colitis, *specifically including* that due to infection with *Campylobacter*, has been ruled out.

The high frequency of *C. jejuni* infections and their severity and recurrence among hypogammaglobulinemic patients suggest that antibodies are important in protective immunity. The pathogenesis of infection is uncertain. Both the motility of the strain and its capacity to adhere to host tissues appear to favor disease, but classic enterotoxins and cytotoxins (although described) appear not to play any substantial role in tissue injury or disease production. The organisms have been visualized in the epithelium, albeit in low numbers. The documentation of a significant tissue response and occasionally of *C. jejuni* bacteremia further suggests that tissue invasion is clinically significant.

The pathogenesis of *C. fetus* infections is better defined. Virtually all clinical isolates of *C. fetus* possess a proteinaceous capsule-like structure (an S-layer) that renders the organism resistant to complement-mediated killing and opsonization. As a result, *C. fetus* can cause bacteremia and can seed sites beyond the intestinal tract. The ability of the organism to switch the S-layer proteins expressed, a phenomenon that results in antigenic variability, may contribute to the chronicity and high rate of recurrence of these infections in compromised hosts.

CLINICAL MANIFESTATIONS OF *C. JEJUNI* AND *C. FETUS* INFECTIONS

The clinical features of infections due to all of the *Campylobacter* and related species causing enteric disease appear to be highly similar. There is often a prodrome, with fever, headache, myalgia, and/or malaise, 12 to 48 h before the onset of diarrheal

symptoms. The most common symptoms of the intestinal phase are diarrhea, abdominal pain, and fever. The degree of diarrhea varies from several loose stools to grossly bloody stools; most patients presenting for medical attention have 10 or more bowel movements on the worst day of illness. Abdominal pain usually consists of cramping and may be the most prominent symptom. Pain usually is generalized but may become localized; *C. jejuni* infection may cause pseudoappendicitis. Fever may be the only initial manifestation of *C. jejuni* infection, a situation mimicking the early stages of typhoid fever. Febrile young children may develop convulsions. *Campylobacter* enteritis generally is self-limited; however, symptoms persist for longer than 1 week in 10 to 20% of patients seeking medical attention, and relapses occur in 5 to 10% of untreated patients.

C. fetus may cause a diarrheal illness similar to that due to *C. jejuni*, especially in normal hosts, or may cause either intermittent diarrhea or nonspecific abdominal pain without localizing signs. Sequelae are uncommon, and outcome is benign. *C. fetus* also may cause a prolonged relapsing systemic illness (with fever, chills, and myalgias) that has no obvious primary source; this manifestation is especially common in compromised hosts. Secondary seeding of an organ (e.g., meninges, brain, bone, urinary tract, or soft tissue) complicates the course, which may be fulminant. *C. fetus* infections have a tropism for vascular sites: endocarditis, mycotic aneurysm, and septic thrombophlebitis all may occur. Infection during pregnancy often leads to fetal death. *H. cinaedi* causes recurrent cellulitis with fever and bacteremia in immunocompromised hosts.

COMPLICATIONS

Except in the case of infection with *C. fetus*, bacteremia is uncommon, developing most often in immunocompromised hosts and at the extremes of age. Three patterns of extraintestinal infection have been noted: (1) transient bacteremia in a normal host with enteritis (benign course, no specific treatment needed); (2) sustained bacteremia or focal infection in a normal host (bacteremia originating from enteritis, with patients responding well to antimicrobial therapy); and (3) sustained bacteremia or focal infection in a compromised host. Enteritis may not be clinically apparent. Antimicrobial therapy, possibly prolonged, is necessary for suppression or cure of the infection.

Campylobacter infections in patients with AIDS or hypogammaglobulinemia may be severe, persistent, and extraintestinal; relapse after cessation of therapy is common. Hypogammaglobulinemic patients also may develop osteomyelitis and an erysipelas-like rash.

Local suppurative complications of infection include cholecystitis, pancreatitis, and cystitis; distant complications include meningitis, endocarditis, arthritis, peritonitis, cellulitis, and septic abortion. All are rare. Hepatitis, interstitial nephritis, and the hemolytic-uremic syndrome occasionally complicate acute infection. Reactive arthritis and other rheumatologic complaints may develop several weeks after infection, especially in persons with the HLA-B27 phenotype. Guillain-Barre syndrome follows *Campylobacter* infections uncommonly (i.e., in 1 of every 1000 to 2000 cases). For certain *C. jejuni* serotypes, such as O19, Guillain-Barre syndrome may follow 1 in every 100 to 200 cases. Because of their high incidence, it is now estimated that

Campylobacter infections may trigger 20 to 40% of all cases of Guillain-Barre syndrome.

LABORATORY FINDINGS

In patients with *Campylobacter* enteritis, peripheral leukocyte counts reflect the severity of the inflammatory process. However, stools from nearly all *Campylobacter*-infected patients presenting for medical attention in the United States contain leukocytes or erythrocytes. Fecal smears should be treated with Gram's or Wright's stain and examined in all suspected cases. When the diagnosis of *Campylobacter* enteritis is suspected on the basis of findings indicating inflammatory diarrhea (fever, fecal leukocytes), clinicians can ask the laboratory to attempt the visualization of organisms with characteristic vibrioid morphology by direct microscopic examination of stools with Gram's staining or to use phase-contrast or dark-field microscopy to identify the organisms' characteristic "darting" motility. Confirmation of the diagnosis of *Campylobacter* infection is based on identification of an isolate from cultures of stool, blood, or another site. *Campylobacter*-specific media should be used to culture stools from all patients with inflammatory or bloody diarrhea. Since all *Campylobacter* species are fastidious, they will not be isolated unless selective media or other selective techniques are used. Not all media are equally useful for isolation of the broad array of campylobacters; therefore, failure to isolate campylobacters from stool does not entirely rule out their presence. The detection of the organisms in stool almost always implies infection; there is a brief period of postconvalescent fecal carriage and no commensalism in humans. In contrast, *C. sputorum* and related organisms found in the oral cavity are commensals with rare pathogenic significance.

DIFFERENTIAL DIAGNOSIS

The symptoms of *Campylobacter* enteritis are not sufficiently unusual to distinguish this illness from that due to *Salmonella*, *Shigella*, or *Yersinia*, among other pathogens. The combination of fever and fecal leukocytes or erythrocytes is indicative of inflammatory diarrhea, and definitive diagnosis is based on culture or demonstration of the characteristic organisms on stained fecal smears. Similarly, extraintestinal *Campylobacter* illness is diagnosed by culture. Infection due to *Campylobacter* should be suspected in the setting of septic abortion and that due to *C. fetus* specifically in the setting of septic thrombophlebitis. It is important to reiterate that the presentation of *Campylobacter* enteritis may mimic that of ulcerative colitis or Crohn's disease, that *Campylobacter* enteritis is much more common than either of the latter (especially among young adults), and that biopsy may not distinguish among these entities. Thus a diagnosis of inflammatory bowel disease should not be made until *Campylobacter* infection has been ruled out, especially in persons with a history of foreign travel, significant animal contact, immunodeficiency, or practices incurring a high risk of transmission.

TREATMENT

Fluid and electrolyte replacement is central to the treatment of diarrheal illnesses ([Chap. 131](#)). Even among patients presenting for medical attention with *Campylobacter* enteritis, fewer than half will clearly benefit from specific antimicrobial therapy.

Indications for such therapy include high fever, bloody diarrhea, severe diarrhea, persistence for more than 1 week, and worsening of symptoms. A 5- to 7-day course of erythromycin (250 mg orally four times daily or -- for children -- 30 to 50 mg/kg per day, in divided doses) is the regimen of choice. Although no relevant clinical trials have been conducted, the in vitro susceptibility of *Campylobacter* species to macrolides such as clarithromycin and azithromycin suggests that these antibiotics also would be useful therapeutic agents. An alternative regimen for adults is ciprofloxacin (500 mg orally twice daily) or another fluoroquinolone for 5 to 7 days, but resistance to this class of agents is increasing. Other alternatives include tetracycline and furazolidone. Use of antimotility agents, which may prolong the duration of symptoms and has been associated with toxic megacolon and with death, is not recommended.

For systemic infections, treatment with gentamicin (1.7 mg/kg intravenously every 8 h after a loading dose of 2 mg/kg), imipenem (500 mg intravenously every 6 h), or chloramphenicol (50 mg/kg intravenously each day in three or four divided doses) should be started empirically, but susceptibility testing should then be performed. Ciprofloxacin and amoxicillin/clavulanate are alternative agents for susceptible strains. In the absence of immunocompromise or endovascular infections, therapy should be administered for 14 days. For immunocompromised patients with systemic infections due to *C. fetus* and for patients with endovascular infections, prolonged therapy (for up to 4 weeks) is usually necessary.

PROGNOSIS

Nearly all patients recover fully from *Campylobacter* enteritis, either spontaneously or after antimicrobial therapy. Volume depletion likely contributes to the few deaths that are reported. As stated above, occasional patients develop reactive arthritis or Guillain-Barre syndrome. Systemic infection with *C. fetus* is much more often fatal than that due to related species; this higher mortality reflects in part the population affected. Prognosis is dependent on the rapidity with which appropriate therapy is begun. Otherwise healthy hosts usually survive *C. fetus* infections without sequelae. Compromised hosts often have recurrent infections.

(Bibliography omitted in Palm version)

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159. CHOLERA AND OTHER VIBRIOSES - Gerald T. Keusch, Robert L. Deresiewicz, Matthew K. Waldor

Members of the genus *Vibrio* cause a number of important infectious syndromes. Classic among them is cholera, a devastating diarrheal disease caused by *V. cholerae* that has been responsible for seven global pandemics and much suffering over the past two centuries. Epidemic cholera remains a major public health concern and is dealt with at length in this chapter. Other vibrioses have also been described, including syndromes of diarrhea, soft tissue infection, or primary sepsis caused by additional named species in the genus *Vibrio*. These, too, are considered below.

All members of the genus are highly motile, facultatively anaerobic, curved gram-negative rods with one or more polar flagella. Except for *V. cholerae* and *V. mimicus*, all require salt for growth ("halophilic vibrios"). In nature, vibrios most commonly reside in tidal rivers and bays under conditions of moderate salinity. They proliferate in the summer months when water temperatures exceed 20°C. As might be expected, the illnesses they cause also increase in frequency during the warm months.

CHOLERA

DEFINITION

Cholera is an acute diarrheal disease that can, in a matter of hours, result in profound, rapidly progressive dehydration and death. Accordingly, cholera gravis (the severe form of cholera) is a much-feared disease, particularly in its epidemic presentation. Fortunately, prompt aggressive fluid repletion and supportive care can obviate the high mortality that it has historically wrought. While the term *cholera* has occasionally been applied to any severely dehydrating secretory diarrheal illness, whether infectious in etiology or not, it has generally referred to disease caused by *V. cholerae* serogroup O1. In 1992, however, a new epidemic serogroup (O139) that causes epidemic cholera emerged on the Indian subcontinent and has since killed many thousands of people.

ETIOLOGY AND EPIDEMIOLOGY

The species *V. cholerae* comprises a host of organisms classified on the basis of the carbohydrate determinants of their lipopolysaccharide (LPS) O antigens. Some 155 serogroups have now been recognized. They are divided into those that agglutinate in antisera to the O1 group antigen (*V. cholerae* O1) and those that do not (non-O1 *V. cholerae*). Although some non-O1 *V. cholerae* serogroups have occasionally caused sporadic outbreaks of diarrhea, serogroup O1 was, until the emergence of serogroup O139, the exclusive cause of epidemic cholera. *V. cholerae* O139 (also called *V. cholerae* Bengal) is discussed in greater detail below.

V. cholerae O1 exists in two biotypes, *classical* and *El Tor*, that are distinguished on the basis of a number of characteristics, including phage susceptibility and hemolysin production. Each biotype is further subdivided into two serotypes, termed *Inaba* and *Ogawa*. Serotyping is a useful tool in field epidemiologic studies. Newer molecular epidemiologic techniques, such as ribotyping and other gene-based methods, now make it possible to trace the source and origin of cholera strains from around the world.

The natural habitat of *V. cholerae* is coastal salt water and brackish estuaries, where the organism lives in close relation to plankton and where it may survive in a viable but nonculturable form. Humans become infected incidentally but, once infected, can act as vehicles for spread. Ingestion of water contaminated by human feces is the most common means of acquisition of *V. cholerae*. Consumption of contaminated food in the home, in restaurants, or from street vendors can also contribute to spread. There is no known animal reservoir. While the infectious dose is relatively high, it is markedly reduced in hypochlorhydric persons, in those using antacids, and when gastric acidity is buffered by a meal. Cholera is predominantly a pediatric disease in endemic areas, but it affects adults and children equally when newly introduced into a population. In endemic areas, the disease is more common in the summer and fall months. While this seasonality has not been explained fully, it may be due to environmental conditions that affect the multiplication of vibrios or to seasonal alterations in human behavior that affect contact with water. Asymptomatic infections are frequent and more common with the El Tor than the classical biotype. In endemic areas, children <2 years of age are less likely to develop severe cholera than are older children, perhaps because of passive immunity acquired from breast milk. For unexplained reasons, susceptibility to cholera is significantly influenced by ABO blood group status; those with type O blood are at greatest risk, while those with type AB are at least risk.

Cholera is native to the Ganges delta in the Indian subcontinent. Since 1817, seven global pandemics have occurred. The current (seventh) pandemic -- the first due to the El Tor biotype -- began in Indonesia in 1961 and spread throughout Asia as *V. cholerae* El Tor displaced the endemic classical strain in many areas. It briefly invaded Europe, but effective public health measures and the high level of sanitation combined to limit its impact. In the early 1970s, El Tor cholera exploded in Africa, causing major epidemics before becoming a persistent endemic problem. Its recent history in Africa has been punctuated by severe outbreaks, often fed by the chaos of war and genocide. Such was the case in the camps for Rwandan refugees set up in 1994 around Goma, Zaire. Tens of thousands of cases occurred and mortality was high. In 1995, the occurrence of hundreds of cases in Romania and the Black Sea states of the former Soviet Union demonstrated the potential of this organism to cause epidemics whenever public health measures break down.

Since 1973, sporadic endemic infections due to vibrios related to the seventh-pandemic strain have been recognized along the U.S. Gulf Coast of Louisiana and Texas. These infections are typically associated with the consumption of contaminated, locally harvested shellfish. Occasionally, cases in U.S. locations remote from the Gulf Coast have been linked to shipped-in Gulf Coast seafood.

Although the event was long expected, it was not until 1991 that the current cholera pandemic reached Latin America. Beginning along the Peruvian coast in January 1991, the disease was carried by fishermen to Ecuador and Colombia. It then spread in an explosive epidemic to virtually all of South and Central America and to Mexico ([Fig. 159-1](#)). About 400,000 cases were reported in the first year of the outbreak, and >1 million had been reported by the end of 1994. While the cumulative mortality rate has been <1%, the mortality rate approached 30% in the communities first affected, where a lack of familiarity with the disease led initially to the deployment of wholly ineffective

treatment. Intensive education of health care providers and of the community at large has enhanced awareness of the disease and its appropriate management and has greatly diminished mortality. As it did in Africa two decades earlier, the epidemic El Tor strain proved capable of establishing itself in inland waters rather than in its classic niche of coastal salt waters; the organism has already become endemic in many of the Latin American countries into which it was recently introduced.

Cases linked to the Latin American epidemic have occurred in the United States. For example, 11 people in New York and New Jersey were infected in two separate outbreaks in 1991 after eating boiled crabmeat illegally transported by travelers from Ecuador. Although secondary spread of this strain has not taken place in the United States, these events underscore the need for vigilance among health care professionals, even in locations remote from an epidemic.

In October 1992, a large-scale outbreak of clinical cholera occurred in the port city of Madras and surrounding towns in southern India. The etiologic agent proved to be a novel strain of *V. cholerae* belonging neither to the O1 serogroup that typically causes epidemic cholera nor to any of the 137 other serogroups known at the time. This strain spread rapidly up and down the coast of the Bay of Bengal, reaching Bangladesh in December 1992. There alone, it caused more than 100,000 cases of cholera in the first 3 months of 1993. It subsequently spread across the Indian subcontinent and to neighboring countries, affecting Pakistan, Nepal, western China, Thailand, and Malaysia by the end of 1994 ([Fig. 159-2](#)). The organism has since been designated *V. cholerae* O139 Bengal in recognition of its novel O antigen and its geographic origin.

The clinical manifestations and epidemiologic features of the disease caused by *V. cholerae* O139 Bengal are indistinguishable from those of O1 cholera. Immunity to the latter, however, is not protective against the former. Thus, although O139 Bengal cholera has been restricted almost exclusively to O1-endemic areas, it has affected patients of all ages, with most cases in adults. Moreover, populations into which *V. cholerae* O139 Bengal has been introduced have responded as virgin populations with respect to severe, lethal cholera. Because naturally acquired immunity to *V. cholerae* O1 does not cross-protect against *V. cholerae* O139 Bengal, vaccines being developed against the former are unlikely to be effective against the latter.

Like the O1 epidemics in cholera-naïve areas before it, the O139 epidemic was initially devastating. Some authorities believed that the emergence of *V. cholerae* O139 signaled the beginning of the eighth global cholera pandemic. Indeed, just as O1 El Tor replaced the classical biotype that preceded it, O139 Bengal in 1993 rapidly replaced O1 El Tor as the most common environmental isolate and the predominant cause of clinical cholera in the areas in which it had appeared ([Fig. 159-3](#)). However, by the beginning of 1994, O1 El Tor had unexpectedly resumed its dominance in Bangladesh, relegating O139 Bengal cholera to the status of a background endemic infection. In some locales O1 *V. cholerae* remains dominant; in others O139 periodically reemerges. Nevertheless, the potential for global spread of O139 Bengal was underscored by an intercontinental food-borne outbreak that occurred in early 1994 among American and British passengers on a cruise ship in Southeast Asia. Six of the 630 travelers became ill, their symptoms beginning only after they returned home.

PATHOGENESIS

In the final analysis, cholera is a toxin-mediated disease. Its characteristic watery diarrhea is due to the action of cholera toxin (CTX), a potent protein enterotoxin elaborated by the organism following its colonization of the small intestine. The bacterial properties that facilitate intestinal colonization are incompletely understood. For *V. cholerae* to colonize the small intestine and produce CTX, it must first recognize, contend with, and traverse several hostile environments. The first of these is the acidic milieu of the stomach. To elude the bactericidal effects of gastric acidity, *V. cholerae* relies, at least in part, on a relatively large inoculum size (compared to that needed for colonization by *Shigella*, for instance). The organism must next traverse the mucous layer lining the small bowel. *V. cholerae* chemotaxis and motility and a variety of proteases may allow the organism to traverse this gel covering the intestinal epithelium. Adherence to the intestinal epithelium is believed to be mediated by the toxin-coregulated pilus (TCP), so named because its synthesis is regulated in parallel with that of CTX. Studies of volunteers have established that TCP is essential for *V. cholerae* intestinal colonization. Other *V. cholerae* gene products known to be important in intestinal colonization of experimental animals include accessory colonization factors ABCD, a cell-associated hemagglutinin, iron and magnesium transport proteins, and purine and biotin biosynthesis enzymes.

[CTX](#), [TCP](#), and several other virulence factors, including accessory colonization factors and various outer-membrane proteins, are coordinately regulated by the *toxR* gene product. ToxR protein is a "master switch" that modulates the expression of virulence genes in response to signals that it senses within the environment of the host via a cascade of regulatory proteins. Coordinate regulation of virulence factor expression presumably enables the organism to tailor its repertoire of proteins to suit its needs as it passes from one microenvironment to another. Coordinate regulation of virulence gene expression by a central sensor-effect protein like ToxR has become a paradigm for similar systems that have been discovered in a wide range of pathogenic bacteria.

Once established in the human small bowel, the organism produces [CTX](#), which consists of a monomeric enzymatic moiety (the A subunit) and a pentameric binding moiety (the B subunit). The B pentamer binds to GM₁ganglioside, a glycolipid on the surface of jejunal epithelial cells that serves as the toxin receptor and makes possible the delivery of the A subunit to its cytosolic target. The activated A subunit (A₁) irreversibly transfers ADP-ribose from nicotinamide adenine dinucleotide to its specific target protein, the GTP-binding regulatory component of adenylate cyclase in intestinal epithelial cells. In this configuration, this G protein permanently upregulates the cyclase catalytic subunit; the result is the intracellular accumulation of high levels of cyclic AMP. In intestinal epithelial cells, cyclic AMP inhibits the absorptive sodium transport system in villus cells and activates the excretory chloride transport system in crypt cells, and these events lead to the accumulation of sodium chloride in the intestinal lumen. Since water moves passively to maintain osmolality, isotonic fluid accumulates in the lumen. When the volume of that fluid exceeds the capacity of the gut to resorb it, watery diarrhea results. Unless the wasted fluid and electrolytes are adequately replaced, shock (due to profound dehydration) and acidosis (due to loss of bicarbonate) follow.

Although perturbation of the adenylate cyclase pathway is the primary mechanism by

which [CTX](#) causes excess fluid secretion, it is not the only one. Increasing evidence indicates that CTX also enhances intestinal secretion via prostaglandins and/or neural histamine receptors. It is possible that the redundancy of secretory mechanisms activated by CTX accounts for the profound diarrhea and dehydration characteristic of severe cholera.

The genes encoding [CTX](#) (*ctxAB*) are part of the genome of a bacteriophage designated CTXF. The receptor for this phage on the *V. cholerae* surface is the essential *V. cholerae* intestinal colonization factor [TCP](#). Following infection of TCP+ *ctxAB*- *V. cholerae* cells, the CTXF genome stably integrates at a specific site on the *V. cholerae* chromosome. Since *ctxAB* is part of a mobile genetic element (CTXF), horizontal transfer of this bacteriophage may account for the emergence of new toxigenic *V. cholerae* serogroups. In addition, since the CTXF receptor TCP is a *V. cholerae* host colonization factor, it is possible that CTXF infection of TCP+ *ctxAB*- *V. cholerae* strains occurs primarily within the human intestine. Many of the other genes important for *V. cholerae* pathogenicity, including the genes encoding the biosynthesis of TCP, those encoding accessory colonization factors, and those regulating virulence gene expression, are clustered together on one of the two *V. cholerae* chromosomes. This cluster of virulence genes is referred to as the *V. cholerae* pathogenicity island. Similar clustering of virulence genes is found in other bacterial pathogens. It is believed that these pathogenicity islands have been acquired by horizontal gene transfer.

Molecular analysis of *V. cholerae* O139 Bengal has suggested the basis of its origin and the reasons it was able to cause an explosive epidemic of cholera. Both phenotypically and genotypically, O139 Bengal is closely related to the O1 El Tor strains of the seventh pandemic, and it seems to have arisen from them by horizontal gene transfer. It shares the virulence attributes and general pathogenic mechanisms of O1 vibrios, including possession of the same [CTX](#) prophage and the same [TCP](#). *V. cholerae* O139 Bengal is in fact virtually identical to the seventh-pandemic strains of *V. cholerae* O1 El Tor except for two important differences: production of the novel O139 [LPS](#) and of an immunologically related O-antigen polysaccharide capsule. Both of these molecules are putative virulence factors, independently enhancing colonization in a murine infection model. The ability to produce the O139 LPS is due to a replacement of a 22-kb DNA segment encoding O1 antigen biosynthesis with a 35-kb segment containing the genes encoding O139 LPS and capsule biosynthesis. Encapsulation is not a feature of O1 strains and may explain the resistance of O139 strains to human serum in vitro as well as the occasional development of O139 bacteremia.

CLINICAL MANIFESTATIONS

After a 24- to 48-h incubation period, cholera begins with the sudden onset of painless watery diarrhea that may quickly become voluminous and is often followed shortly by vomiting. In severe cases, stool volume can exceed 250 mL/kg in the first 24 h. If fluids and electrolytes are not replaced, hypovolemic shock and death ensue. Fever is usually absent. Muscle cramps due to electrolyte disturbances are common. The stool has a characteristic appearance: a nonbilious, gray, slightly cloudy fluid with flecks of mucus, no blood, and a somewhat sweet, inoffensive odor. It has been called "rice-water" stool because of its resemblance to the water in which rice has been washed. Clinical symptoms parallel volume contraction: At losses of 3 to 5% of normal body weight, thirst

develops; at 5 to 8%, postural hypotension, weakness, tachycardia, and decreased skin turgor are documented; and at >10%, oliguria, weak or absent pulses, sunken eyes (and, in infants, sunken fontanelles), wrinkled ("washerwoman") skin, somnolence, and coma are characteristic. Complications derive exclusively from the effects of volume and electrolyte depletion and include renal failure due to acute tubular necrosis. Thus, if the patient is adequately treated with fluid and salt, complications are averted and the process is self-limited, resolving in a few days.

Laboratory data usually reveal an elevated hematocrit (due to hemoconcentration) in nonanemic patients; mild neutrophilic leukocytosis; elevated levels of blood urea nitrogen and creatinine consistent with prerenal azotemia; normal sodium, potassium, and chloride levels; a markedly reduced bicarbonate level (<15 mmol/L); and an elevated anion gap (due to increases in serum lactate, protein, and phosphate). Arterial pH is usually low (about 7.2).

DIAGNOSIS

The clinical suspicion of cholera can be confirmed by the identification of *V. cholerae* in stool; however, the organism must be specifically sought. In experienced hands, it can be detected directly by dark-field microscopy on a wet mount of fresh stool, and its serotype can be discerned by immobilization with Inaba- or Ogawa-specific antiserum. Laboratory isolation of the organism requires the use of a selective medium. The best of these is thiosulfate-citrate-bile salts-sucrose (TCBS) agar, on which the organism grows as a flat yellow colony. If a delay in sample processing is expected, Carey-Blair transport medium and/or alkaline-peptone water-enrichment medium should be inoculated as well. In endemic areas there is little need for biochemical confirmation and characterization, although these tasks may be worthwhile in places where *V. cholerae* is an uncommon isolate. Standard microbiologic biochemical testing for Enterobacteriaceae will suffice for identification of *V. cholerae*. All vibrios are oxidase-positive. *V. cholerae* can be distinguished from the otherwise similar *V. mimicus* by its ability to ferment sucrose.

The yield of stool cultures for the diagnosis of *V. cholerae* infection declines late in the course of the illness or when effective antibacterial therapy is initiated. Although not generally evaluable in clinical laboratories, serum vibriocidal antibody titers can be used to confirm the diagnosis in non-cholera-endemic regions of the world. Monoclonal antibody-based diagnostic kits and methods based on the polymerase chain reaction and on DNA probes have been developed for *V. cholerae* O1 and O139 but are unlikely to become available in U.S. clinical laboratories.

TREATMENT

Cholera is simple to treat; only the rapid and adequate replacement of fluids, electrolytes, and base is required. The mortality rate for appropriately treated disease is usually <1%. However, analysis of a large outbreak of cholera among airline travelers from an endemic country to the United States revealed frequent misdiagnoses by U.S. health professionals and poor appreciation on their part of the principles of management. Compounding these problems was the general unavailability of appropriate oral fluids. Even intravenous fluid therapy typically was not optimal.

It has been proved conclusively that fluid may be given orally. This approach takes advantage of the hexose-Na⁺ cotransport mechanism to move Na⁺ across the gut mucosa together with an actively transported molecule such as glucose. Since Na⁺ losses in the stool are high, a fluid containing Na⁺ at 90 mmol/L has been recommended by the World Health Organization (WHO) ([Table 159-1](#)). This amount of Na⁺ is higher than that needed to treat diarrhea due to most other causes. The solution is safe, even for infants, if its intake is alternated with the consumption of sodium-free fluids such as breast milk or water. For the sake of simplicity, WHO advises routine use of this single solution for diarrheal disease rather than attempts to choose among multiple formulations according to etiology.

Cereal-based formulations are receiving increased attention as alternative oral rehydration solutions. Because of their lower osmolality, they may reduce stool output. A mixture with a lower sugar and salt content has also been evaluated in cholera patients, with favorable results. However, concerns have been raised over the safety of its use -- in particular, whether it could cause significant hyponatremia in patients with moderate or severe diarrhea. Because commercial oral rehydration solutions also contain concentrations of glucose and sodium lower than those of the [WHO](#) formulation, they should not yet be used routinely to treat cholera.

For initial management of severely dehydrated patients, intravenous fluid replacement is preferable, if available. Because profound acidosis (pH < 7.2) is common in this group, Ringer's lactate is the best choice among commercial products ([Table 159-2](#)). It must be used with additional potassium supplements, preferably given by mouth. The total fluid deficit in severely dehydrated patients (³10% of body weight) can be replaced safely within the first 4 h of therapy, half within the first hour. Thereafter, oral therapy can usually be initiated, with the goal of maintaining fluid intake equal to fluid output. However, patients with continued large-volume diarrhea may require prolonged intravenous treatment to keep up with gastrointestinal fluid losses. Severe hypokalemia can develop but will respond to potassium given either intravenously or orally. In the absence of adequate staff to monitor the patient's progress, the oral route of rehydration and potassium replacement is safer than the intravenous route and is physiologically regulated by thirst and urine output.

Although not necessary for cure, the use of an antibiotic to which the organism is susceptible will diminish the duration and volume of fluid loss and will hasten clearance of the organism from the stool. Single-dose tetracycline (2 g) or doxycycline (300 mg) is effective in adults but is not recommended for children under 8 years of age because of possible deposition in bone and developing teeth. Emerging drug resistance is an ever-present concern. For adults with cholera in areas where tetracycline resistance is prevalent, ciprofloxacin -- either in a single dose (30 mg/kg, not to exceed a total dose of 1 g) or in a short course (15 mg/kg bid for 3 days, not to exceed a total daily dose of 1 g) -- or erythromycin (a total of 40 mg/kg daily in three divided doses for 3 days) is a clinically effective substitute. Both drugs are highly effective in reducing total stool output, and each is significantly better than trimethoprim-sulfamethoxazole. Because of the high cost of quinolones, [WHO](#) recommends erythromycin as the first alternative to tetracycline. For children, furazolidone has been the recommended agent and trimethoprim-sulfamethoxazole the second choice. It is of note that *V. cholerae* O139 is

often resistant to both of these drugs but is susceptible to quinolones, erythromycin, tetracycline, and ampicillin (among others). Because of cost and/or toxicity issues related to the other drugs, erythromycin is a good choice for pediatric cholera, especially where O139 Bengal is present. The efficacy of single-dose erythromycin therapy for cholera has not been demonstrated.

CONTROL

In outbreaks, efforts should first be made to identify case contacts and to treat incubating carriers. Next, epidemiologic studies should be undertaken to establish the modes of transmission to define the best strategy to interrupt them. Both the establishment of rehydration centers and instruction in rehydration techniques are essential to the reduction of mortality. Immunization in these circumstances is not an effective means of control.

PREVENTION

Provision of safe water and facilities for sanitary disposal of feces, improved nutrition, and attention to food preparation and storage in the household could significantly reduce the incidence of cholera. Much effort has been devoted to the development of an effective cholera vaccine over the past two decades, with a particular focus on oral vaccine strains. Traditional killed cholera vaccine given intramuscularly provides little protection to nonimmune subjects and predictably causes adverse effects, including pain at the injection site, malaise, and fever. The vaccine's limited efficacy is at least partially due to its failure to induce a local immune response at the intestinal mucosal surface.

Two types of oral cholera vaccines are under development. The first is a killed whole-cell (WC) vaccine. Two formulations of the killed WC vaccine have been prepared: one that also contains the nontoxic B subunit of CTX (WC/BS) and one composed solely of killed bacteria. In field trials in Bangladesh, both of the killed vaccines were compared with placebo and conferred ~50% protection over a 3-year evaluation period. The protective efficacy of WC/BS was superior to that of WC during the initial 8 months of follow-up (69 versus 41%) but equivalent or inferior thereafter. Immunity was relatively sustained in persons vaccinated at an age of >5 years but was not well sustained in younger vaccinees.

The second approach is that of a live attenuated vaccine strain developed, for example, by the isolation or creation of mutants lacking active [CTX](#). Three criteria must be met in live vaccine design: The vaccine strain must induce protective immunity, it must be safe to administer, and it must be minimally reactogenic. *Safety criteria* include the vaccine strain's potential to regain virulence, either spontaneously or via horizontal gene transfer from environmental strains, as well as its potential to donate virulence genes to other strains. *Reactogenicity* refers to its potential to cause symptoms such as fever or diarrhea in vaccinees.

Strain CVD 103-HgR, an oral live cholera vaccine licensed for immunization of travelers in Europe, is derived from a classical biotype strain of *V. cholerae* by the deletion of the [CTX](#) A subunit gene and the insertion in the hemolysin gene of a mercury resistance

marker. This strain has been extensively tested in volunteers; although it is poorly excreted in the stool of human vaccinees, a single dose produces a significant increase in the titer of vibriocidal antibody in ~75% of recipients, including children between the ages of 2 and 4 years, with almost no reactogenicity. Studies in volunteers demonstrate that this vaccine is more effective against classical than against El Tor cholera. Unfortunately, in a large field trial in Indonesian children, this vaccine failed to induce protection against clinical cholera.

Other live attenuated vaccine candidate strains have been prepared from El Tor and O139 *V. cholerae*. In studies in volunteers, these vaccine strains have often exhibited significant reactogenicity whose cause (given the absence of active [CTX](#)) is unclear. Reactogenicity may result from the production of another toxic moiety (e.g., the hemagglutinin/protease or the RTX toxin) by the live attenuated strain. Alternatively, intestinal colonization itself may result in reactogenicity. These El Tor- and O139-derived live vaccine strains are therefore at least several years away from potential licensing. Because of the minimal efficacy of existing parenteral vaccines, cholera immunization is recommended for U.S. travelers only if it is mandated by the countries they plan to visit.

OTHER *VIBRIO* SPECIES

In recent years, the taxonomic, epidemiologic, pathophysiologic, and clinical features of vibrios that do not cause clinical cholera have become increasingly well understood. Ten human pathogens are currently recognized in the genus *Vibrio*. Included are species associated primarily with gastrointestinal illness (*V. parahaemolyticus*, non-O1 *V. cholerae*, *V. mimicus*, *V. fluvialis*, *V. hollisae*, and *V. furnissii*) and species associated primarily with soft tissue infections (*V. vulnificus*, *V. alginolyticus*, and *V. damsela*). In addition, *V. vulnificus* has emerged as a cause of primary sepsis in certain compromised hosts. Vibrios are abundant in coastal waters the world over and tend to concentrate in the tissues of filter-feeding mollusks. Under optimal conditions, some can double in number in as little as 9 min. Consequently, seawater and raw or undercooked shellfish are important sources of human infection ([Table 159-3](#)). Vibrios grow best at temperatures of 28°C to 44°C but not at all below 4°C or above 60°C. Most can be cultured on blood or MacConkey agar, each of which contains enough salt to support the growth of the halophilic organisms (³0.5%). As with *V. cholerae*, [TCBS](#) is the best selective medium. The species can be differentiated in the laboratory by standard biochemical tests. The most important members of the group are *V. parahaemolyticus* and *V. vulnificus*. These and selected other species are considered below in greater detail.

SPECIES ASSOCIATED PRIMARILY WITH GASTROINTESTINAL ILLNESS

V. parahaemolyticus First implicated as a cause of enteritis by Japanese workers in 1953, *V. parahaemolyticus* is now recognized as an important intestinal pathogen in many parts of the world. In one study from Japan, 24% of reported cases of food poisoning were attributed to this organism, presumably owing to the widespread consumption of raw seafood there. In the United States, *V. parahaemolyticus* has been responsible for several well-documented common-source outbreaks of diarrhea, typically linked to ingestion of undercooked or improperly handled seafood or of other

foods that have been contaminated by seawater. Most reports have come from the Atlantic Coast, the Gulf of Mexico, and Hawaii. The organism is ubiquitous in marine environments and is able to grow in saline concentrations as high as 8 to 10%. The ability to cause hemolysis on Wagatsuma agar (known as the *Kanagawa phenomenon*) is closely linked to enteropathogenicity. In one study, 96.5% of isolates from patients with diarrhea were hemolytic versus only ~1% of isolates from seawater. Hemolysis is attributed to a 42-kDa heat-stable protein, the exact pathophysiologic role of which is uncertain. The mechanism by which *V. parahaemolyticus* causes diarrhea is not clear.

V. parahaemolyticus has been associated with two distinct gastrointestinal presentations. The more common is a syndrome of watery diarrhea, accompanied in most cases by abdominal cramps, nausea, and vomiting and in about one-quarter of cases by fever and chills. The incubation period ranges from 4 h to 4 days, and the symptomatic period lasts for a median of 3 days. The vast majority of North American cases have been of this type. The less common syndrome is one of dysentery, described in India and Bangladesh and characterized by severe abdominal cramps, nausea, vomiting, and bloody or mucoid stools. Most cases of either type are self-limited and require neither antimicrobial treatment nor hospitalization. Severe infections are associated with underlying diseases, including diabetes, preexisting liver disease, iron-overload states, or immunosuppression. The occasional severe case should be treated with fluid replacement and antibiotics, as described above for cholera. Death is very rare. There are no reliable differential diagnostic features. *V. parahaemolyticus* should be considered as a possible cause in all cases of diarrhea that can be epidemiologically linked to seafood consumption or to the sea itself.

In addition to gastrointestinal disease, *V. parahaemolyticus* is a rare cause of extraintestinal infections, including wound infections, otitis, and -- very rarely -- sepsis.

Non-O1 *V. cholerae* The heterogeneous non-O1 *V. cholerae* organisms are biochemically indistinguishable from *V. cholerae* O1 on routine testing but fail to agglutinate in O1 antiserum. While technically a non-O1 vibrio, *V. cholerae* O139 Bengal is not grouped with these pathogens because of its potential to cause epidemic cholera, as detailed above. Non-O1 *V. cholerae* strains have been responsible for several well-described food-borne outbreaks of gastroenteritis as well as for sporadic cases of otitis media, wound infection, and bacteremia. About half of all U.S. isolates are obtained from stool specimens. Like other vibrios, non-O1 *V. cholerae* organisms are widely distributed in marine environments; unlike most other vibrios, however, they require only trace amounts of NaCl to survive (i.e., they are nonhalophilic). Recognized U.S. cases invariably have been associated either with the consumption of raw oysters or with recent travel, typically to Mexico. The clinical spectrum of diarrheal disease caused by non-O1 *V. cholerae* is broad and likely reflects the heterogeneous virulence attributes of the group. Occasional isolates make a protein enterotoxin very similar to [CTX](#). Others produce cytotoxins, hemolysins, or invasins.

Gastroenteritis due to non-O1 *V. cholerae* typically has an incubation period of <2 days. Stools may be copious and watery. On occasion, diarrhea may leave the patient severely dehydrated, as in cholera. Alternatively, the stools may be partly formed, less voluminous, and bloody or mucoid. Abdominal cramps, nausea, vomiting, and fever are often reported. In one series, 11% of patients were hospitalized; in another, the figure

was 50%. The duration of illness ranges from about 2 to 7 days. As in cholera, patients with significant dehydration should be treated with oral or intravenous fluids. The role of antibiotics is uncertain.

Wound infection and otitis media each account for ~10% of non-O1 *V. cholerae* isolates. Bacteremia accounts for another 20%. Patients with extraintestinal infection often have a history of occupational or recreational exposure to seawater. Bacteremia is more likely to develop in the presence of liver disease. Extraintestinal infections should be treated with antibiotics. There is a paucity of information to guide the choice of a specific agent and schedule. Most strains are sensitive in vitro to tetracycline, chloramphenicol, and other agents.

SPECIES ASSOCIATED PRIMARILY WITH SOFT TISSUE INFECTION OR BACTEREMIA

V. vulnificus Though it represents only a small minority of the *Vibrio* species found in nature (4% of Atlantic Coast isolates in one study), *V. vulnificus* is perhaps the most important cause of severe *Vibrio* infections in the United States (0.8 cases per 100,000 population in one study from Louisiana). Formerly included in the species *V. parahaemolyticus*, *V. vulnificus* was distinguished in the 1970s by its ability to ferment lactose and to cause distinct clinical syndromes. Like most vibrios, it proliferates in the warm summer months. It requires a saline environment for growth but prefers concentrations lower than those preferred by *V. parahaemolyticus* and *V. alginolyticus* (range, up to ~8%; optimal, ~1%). Infections in humans typically occur in coastal states between May and October and most often involve men over age 40. *V. vulnificus* has been linked unequivocally to two distinct syndromes: primary sepsis, typically in patients with antecedent liver disease, and primary wound infections, usually in people without underlying disease. Some authors have suggested that this organism causes gastroenteritis, but the evidence for this association is tenuous.

V. vulnificus is remarkably invasive in animal models. It is endowed with a number of virulence attributes, including an antiphagocytic capsule, serum resistance, a cytotoxin/hemolysin (the organism is Kanagawa-positive), collagenase, elastolytic protease, phospholipase, and siderophores. Its virulence, as measured by the 50% lethal dose in mice, is markedly enhanced under conditions of iron overload, a fact consonant with its propensity to infect patients with hemochromatosis.

Primary sepsis occurs most commonly in patients with cirrhosis or hemochromatosis but has also developed in patients with hematopoietic disorders or chronic renal insufficiency, in persons using immunosuppressive medications or alcohol, and (rarely) in individuals without apparent underlying disease. Most of those affected have ingested raw oysters within 2 days of onset (median incubation period, 16 h). The process begins precipitously with malaise, chills, fever (mean temperature, 39.8°C), and prostration. Hypotension develops in one-third of cases, often by the time of admission. Cutaneous manifestations, which develop in three-quarters of cases (usually by 36 h after onset), typically involve the extremities -- lower more often than upper ([Fig. 159-CD1](#)). A common sequence is the evolution of erythematous patches followed by ecchymoses, vesicles, and bullae. (Indeed, the presence of sepsis and bullous skin lesions suggests the diagnosis in an appropriate setting.) Necrosis and sloughing may occur. Laboratory

study reveals leukopenia more often than leukocytosis, thrombocytopenia, and (occasionally) elevated levels of fibrin split products. *V. vulnificus* can be cultured from blood or cutaneous lesions.

Mortality approaches 50%, with most deaths due to uncontrolled sepsis. Accordingly, prompt treatment is critical and should include empirical antibiotic administration, aggressive debridement, and general supportive care. *V. vulnificus* is sensitive to a number of antimicrobials in vitro, including tetracycline, gentamicin, and third-generation cephalosporins. No compelling clinical data from studies of humans support the preferential use of any one of these agents. Tetracycline is demonstrably superior in a murine model and on that basis is considered the drug of choice (0.5 to 1 g intravenously every 12 h), either alone or in combination with gentamicin. The duration of therapy is guided by the clinical response.

Wound infections with *V. vulnificus* can develop in patients with or without underlying disease and invariably follow contact of seawater with either a prior or a fresh wound. The incubation period is brief (4 h to 4 days; mean, 12 h). The disease begins with swelling, erythema, and -- in many cases -- intense pain around the wound. Rapidly spreading cellulitis follows, with vesicular, bullous, or necrotic lesions developing in some instances. Metastatic events do not generally occur. Fever (median temperature, 38.9°C) and leukocytosis are demonstrable in most cases. The organism can be cultured from skin lesions and occasionally from blood. Prompt antibiotic therapy and debridement are usually curative.

V. alginolyticus This species was first recognized as a human pathogen in 1973 and is now known to cause occasional wound, ear, and eye infections. It is the most salt-tolerant of the vibrios, able to grow in concentrations >10%. Most clinical isolates come from superinfected wounds, which presumably became contaminated at the beach. Infection varies in severity but is generally not serious and responds well to antibiotic therapy and drainage. A few reports have described otitis externa, otitis media, or conjunctivitis. Therapy with tetracycline is usually curative. *V. alginolyticus* is a rare cause of bacteremia in immunocompromised hosts.

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160. BRUCELLOSIS - M. Monir Madkour, Dennis L. Kasper

DEFINITION

Brucellosis is a zoonosis transmitted to humans from infected animals. Its clinical features are not disease specific. *Brucellosis* has many synonyms derived from the geographical regions in which the disease occurs (e.g., Mediterranean fever, Malta fever, Gibraltar fever, Cyprus fever); from the remittent character of its fever (e.g., undulant fever); or from its resemblance to malaria and typhoid (e.g., typhomalarial fever, intermittent typhoid).

ETIOLOGY

Human brucellosis can be caused by any of four species: *Brucella melitensis* (the most common and most virulent cause of brucellosis worldwide) is acquired primarily from goats, sheep, and camels; *B. abortus* from cattle; *B. suis* from hogs; and *B. canis* from dogs. These small aerobic gram-negative bacilli are unencapsulated, nonmotile, non-spore-forming, facultative intracellular parasites that cause lifelong infection in animals. Brucellae are killed by boiling or pasteurization of milk and milk products. They survive for up to 8 weeks in unpasteurized, white, soft cheese made from goat's milk and are not killed by freezing. The organisms remain viable for up to 40 days in dried soil contaminated with infected-animal urine, stool, vaginal discharge, and products of conception and for longer periods in damp soil.

EPIDEMIOLOGY

The global incidence of human brucellosis is not known because of the variable quality of disease reporting and notification systems in many countries. Worldwide, the only countries believed to be free of brucellosis are Norway, Sweden, Finland, Denmark, Iceland, Switzerland, the Czech and Slovak republics, Romania, the United Kingdom (including the Channel Islands), the Netherlands, Japan, Luxembourg, Cyprus, and Bulgaria; the U.S. Virgin Islands are also free of the disease. Reports indicate that, even in developed nations, the true incidence of brucellosis may be up to 26 times higher than official figures suggest. In the United States, about 200 new cases are reported every year; however, it is estimated that only 4 to 10% of cases are recognized and reported. Consumption of imported cheese, travel abroad, and occupation-related exposures are the most frequently identified sources of infection. In communities where brucellosis is endemic, the disease occurs in children and the family members of infected persons are at risk. Even in countries where animal brucellosis is controlled, the disease occasionally develops among farmers, meat-processing workers, veterinarians, and laboratory workers.

The *Brucella* organism is transmitted most commonly through the ingestion of untreated milk or milk products; raw meat (i.e., blood) and bone marrow have also been implicated. However, the organism can be contracted via inhalation during contact with animals, especially by children and by slaughterhouse, farm, and laboratory workers. Other routes of infection for at-risk workers include skin abrasion, autoinoculation, and conjunctival splashing. The organism has occasionally been transmitted from person to person through the placenta, during breast-feeding, and during sexual activity.

Aerosolized *B. melitensis* is a classic agent of biological warfare.

PATHOGENESIS AND IMMUNITY

Serum opsonizes *Brucella* organisms for ingestion by polymorphonuclear leukocytes and activated macrophages. Brucellae resist intracellular phagocytic killing by mechanisms such as the suppression of the myeloperoxide-hydrogen peroxide-halide system and the production of superoxide dismutase. The pathogen-phagocyte interaction plays a key role in determining the severity and outcome of brucellosis. The organisms surviving within and escaping from the phagocytes multiply and reach the bloodstream via the lymphatics, subsequently localizing in the liver, spleen, bones, kidneys, lymph nodes, heart valves, nervous system, and testes. In these organs, the bacteria are ingested by macrophages and survive by inhibition of phagosome-lysosome fusion. In infected tissues, inflammatory responses or noncaseating granulomas typically develop, and caseating granulomas and abscesses have been described.

Cytokines, including interleukin (IL) 1, IL-12, and tumor necrosis factor, appear to be important in host defense against *Brucella* infection. The smooth lipopolysaccharide (LPS) of *Brucella* is the major known virulence factor. Strains with rough LPS are more likely than those with smooth LPS to be lysed by nonimmune serum. In virulent strains, the foremost target for specific antibodies is the LPS. Serum IgM antibodies to LPS appear within 1 week after infection and are followed later by IgG and IgA. Titers of both IgM and IgG antibody fall after treatment, and failure of these titers to decline should prompt an evaluation for relapse or persistent infection.

CLASSIFICATION

Brucellosis is classified according to whether or not the disease is active (i.e., symptoms or progressive tissue damage and significantly raised *Brucella* agglutinin levels with or without positive cultures) and whether or not there is localized infection. The state of activity and the site of localization have a significant impact on recommended treatment. Classification of brucellosis as acute, subacute, serologic, bacteremic, or of mixed types serves no purpose in diagnosis and management.

CLINICAL MANIFESTATIONS AND COMPLICATIONS

Brucellosis is a systemic disease with protean manifestations. Its features may mimic those of other febrile illnesses. The incubation period lasts for about 1 to 3 weeks but may be as long as several months, depending on the virulence of the organisms, their route of entry, the infecting dose, and the host's preexisting health status. The onset of symptoms may be either abrupt (over 1 to 2 days) or gradual (³1 week). The most common symptoms are fever, chills, diaphoresis, headaches, myalgia, fatigue, anorexia, joint and low-back pain, weight loss, constipation, sore throat, and dry cough. Physical examination often reveals no abnormalities, and patients can look deceptively well. Some patients, in contrast, are acutely ill, with pallor, lymphadenopathy, hepatosplenomegaly, arthritis, spinal tenderness, epididymo-orchitis, rash, meningitis, cardiac murmurs, or pneumonia. The fever of brucellosis has no distinctive pattern but may exhibit diurnal variation, with normal temperatures in the morning and high

temperatures in the afternoon and evening. Localization to an organ or a system may be evident at the onset of the disease. [Table 160-1](#) lists the frequencies of key historic features, symptoms, and signs among 500 patients with brucellosis due to *B. melitensis*.

Bones and Joints Although monarticular septic arthritis occurs, 30 to 40% of patients have reactive asymmetric polyarthritis involving the knees, hips, shoulders, and sacroiliac and sternoclavicular joints. The total white cell count in synovial fluid ranges from 4000 to 40,000/mL, typically with about 60% polymorphonuclear leukocytes. The synovial fluid glucose concentration may be reduced and the protein concentration elevated; cultures of synovial fluid are positive in about 50% of cases.

Infection with *Brucella* organisms commonly causes osteomyelitis of the lumbar vertebrae, starting at the superior end plate (an area with a rich blood supply) and occasionally progressing to involve the entire vertebra, disk space, and adjacent vertebrae. Extraspinal *Brucella* osteomyelitis is rare. In *Brucella* septic arthritis and osteomyelitis, the peripheral white cell count is typically normal, while the erythrocyte sedimentation rate may be either normal or elevated.

Heart Cardiovascular complications of brucellosis include endocarditis, myocarditis, pericarditis, aortic root abscess, mycotic aneurysms, thrombophlebitis with pulmonary aneurysm, and pulmonary embolism. *Brucella* endocarditis may develop on valves previously damaged by rheumatic fever or congenital malformation but also occurs on previously normal valves. The clinical features are indistinguishable from those of endocarditis caused by other organisms ([Chap. 126](#)). Endocarditis is the leading cause of death in brucellosis, although the outcome of *Brucella* endocarditis has been more favorable in recent years because of advances in early diagnosis, antibiotic treatment, and cardiac surgery. Physicians who suspect brucellae as a cause of culture-negative endocarditis in patients with possible environmental exposure should notify the bacteriology laboratory performing the blood culture so that extended incubation, specific media, and biohazard precautions can be employed.

Respiratory Tract Brucellae can produce respiratory symptoms. A flulike illness with sore throat, tonsillitis, and dry cough is common and usually mild. Hilar and paratracheal lymphadenopathy, pneumonia, solitary or multiple pulmonary nodules, lung abscess, and empyema have been reported.

Gastrointestinal Tract and Hepatobiliary System Gastrointestinal manifestations of *Brucella* infection are generally mild and may include nausea, vomiting, constipation, acute abdominal pain, and/or diarrhea. Pathologic examination of the liver may reveal any of several changes, including noncaseating granulomas ([Fig. 160-CD1](#)), suppurative abscesses, mononuclear cell infiltration, or hyperemia of the intestinal mucosa. Acute ileitis with inflammation of Peyer's patches and colitis have been reported. Hepatic and splenic enlargement may be documented in 15 to 20% of cases, and abscesses may develop in the liver and spleen. Infected ascites, pancreatitis, and cholecystitis have been reported. Mild jaundice may be evident, with elevated levels of bilirubin and hepatic enzymes.

Genitourinary Tract The various genitourinary infections attributed to brucellae include unilateral or bilateral epididymo-orchitis, which is rarely associated with testicular

abscess. Prostatitis, seminal vesiculitis, dysmenorrhea, amenorrhea, tuboovarian abscess, salpingitis, cervicitis, acute pyelonephritis, glomerulonephritis, and massive proteinuria have also been documented. *Brucella* organisms have been cultured from the urine in up to 50% of cases of genitourinary tract infection.

Central Nervous System Neurobrucellosis is uncommon but serious and includes meningitis, meningoencephalitis, multiple cerebral or cerebellar abscesses, ruptured mycotic aneurysms, myelitis, Guillain-Barre syndrome, cranial nerve lesions, hemiplegia, sciatica, myositis, and rhabdomyolysis. Papillitis, papilledema, retrobulbar neuritis, optic atrophy, and ophthalmoplegia due to lesions in cranial nerves III, IV, and VI may occur in *Brucella* meningoencephalitis. Cerebrospinal fluid (CSF) pressure is usually elevated; the fluid may appear clear, turbid, or hemorrhagic; the protein concentration and cell count (predominantly lymphocytes) are elevated; and the glucose concentration may be either reduced or normal. In *Brucella* meningitis, which can occur at any time during the course of the disease, the organism may be cultured from the CSF.

Other Manifestations Conjunctival splashing with live attenuated *B. abortus* vaccine (S19) during animal vaccination may cause conjunctivitis, keratitis, and corneal ulcers, with progression to systemic disease in some cases. Uveitis, optic neuritis, retinopathy, retinal detachment, and endophthalmitis may result from hematogenous spread.

Skin manifestations of brucellosis are uncommon. They include maculopapular eruptions, purpura and petechiae, chronic ulcerations, multiple cutaneous and subcutaneous abscesses, discharging sinuses, superficial thrombophlebitis, erythema nodosum, and pemphigus.

Brucellosis during human pregnancy can cause abortion or intrauterine fetal death. Brucellae have been isolated from the human placenta, fetus, and newborn.

The bone marrow of *Brucella*-infected patients frequently contains noncaseating granulomas. Among the hematologic complications of brucellosis are anemia, leukopenia, and thrombocytopenia.

Endocrinologic findings reported in brucellosis include thyroiditis with abscess formation, adrenal insufficiency, and the syndrome of inappropriate secretion of antidiuretic hormone.

DIAGNOSIS

The combination of potential exposure, consistent clinical features, and significantly raised levels of *Brucella* agglutinin (with or without positive cultures of blood, body fluid, or tissues) confirms the diagnosis of active brucellosis. The organism's identity is confirmed by phage typing, DNA characterization, or metabolic profiling. Use of a CO₂ detection system (such as BACTEC; Becton Dickinson, Sparks, MD) for blood culture provides a more sensitive and rapid culture result than standard methods, with positivity usually apparent after only 2 to 5 days of incubation. Serum antibodies to *Brucella* can be detected by several methods, including standard tube agglutinins (STA), the 2-mercaptoethanol agglutination test, Coombs' test, enzyme-linked immunosorbent

assay, and polymerase chain reaction (PCR). *B. abortus* antigens, which are commonly used for serologic tests, cross-react with *B. melitensis* and *B. suis* but not with *B. canis*. The specific antigen required for assay of antibodies to *B. canis* is not commercially available. *B. canis* antibody titers can be determined in the United States at the Centers for Disease Control and Prevention in Atlanta. A false-negative result in the STA may be obtained because of the prozone phenomenon, which can be avoided by testing of sera at both low and high dilutions.

In endemic areas a *Brucella* antibody titer of 1:320 or 1:640 is significant, while in nonendemic areas an antibody titer of 1:160 is considered significant. Detection of elevated levels of antibody to *Brucella* organisms in the absence of symptoms during the screening of potential blood donors is common in endemic areas. To establish a diagnosis in these regions, clinical and serologic evaluation should be repeated after 2 to 4 weeks and a further rise in titer sought. A high titer of specific IgM suggests recent exposure, while a high titer of specific IgG suggests active disease. Lower titers of IgG may indicate past exposure or treated infection.

Cooperation and consultation with a clinical microbiology laboratory are important when brucellosis is suspected. It may be necessary to observe culture bottles for up to 6 weeks before organisms become detectable. Subcultures should be prepared on duplicate blood agar plates (with and without an atmosphere of 10% CO₂) and special media (such as a blood- or serum-enriched peptone-based medium) or with a rapid CO₂ detection system. Patients whose blood or bone marrow is cultured are positive at one site or the other in 50 to 70% of cases. The peripheral white cell count is usually normal but may be low, with relative lymphocytosis. Thrombocytopenia and disseminated intravascular coagulation may be documented. Levels of hepatic enzymes and serum bilirubin may be raised.

Radiologic investigations aimed at detecting skeletal involvement include plain radiography, bone scintigraphy, computed tomography (CT), and magnetic resonance imaging (MRI). Bone scintigraphy is more sensitive than conventional radiography in detecting areas of spinal and extraspinal involvement, particularly in the early stage of infection. CT is useful for further evaluation of spinal lesions and of the extension of infection into the spinal canal. MRI is the modality of choice for the assessment of *Brucella* spondylitis and is more sensitive than scintigraphy or CT for demonstration of the extent of disease.

Plain lateral radiography of the spine may reveal bone sclerosis, with destruction and erosion of the superior end plate anteriorly. As the disease progresses, healing with osteophyte formation and reduction of disk space may take place. In *Brucella* septic monarthrititis, plain radiography may show effusion and soft tissue swelling without bone or joint destruction. Scintigraphy may document increased uptake in sacroiliac joints or lumbar vertebrae, even when plain radiography gives normal results. [MRI](#) shows diffuse high-signal intensity of the affected vertebrae and may reveal narrowing of the spinal canal as well as loss of definition of the posterior aspect of the vertebrae.

TREATMENT

Single-agent therapy for brucellosis has now been abandoned because of the high rates

of failure and relapse and the potential development of antibiotic resistance. Relatively short courses (<8 weeks) of treatment with antibiotic combinations have similarly been associated with high rates of relapse. The combination of doxycycline and an aminoglycoside (gentamicin, streptomycin, or netilmicin) for 4 weeks followed by the combination of doxycycline and rifampin for 4 to 8 weeks is the most effective regimen. Doxycycline (which is preferred over tetracycline) is given orally in a dose of 100 mg twice daily. Gentamicin is given intramuscularly or as a slow intravenous infusion (3 to 5 mg/kg per day in divided doses every 8 h). Netilmicin (which is preferred to streptomycin) is given (intramuscularly to outpatients, intravenously to inpatients) in a dose of 2 mg/kg every 12 h; trough levels in plasma should be monitored regularly and maintained at ≤ 2 $\mu\text{g/mL}$. Streptomycin is given intramuscularly in a dose of 1 g once daily to patients under 45 years of age and in a dose of 0.5 to 0.75 g/d to older patients. Tetracycline is given orally in a dose of 250 mg every 6 h and rifampin as a single daily dose of 600 to 900 mg. An alternative regimen consists of the doxycycline/rifampin combination given for 8 to 12 weeks. The doxycycline/aminoglycoside combination is more effective than the doxycycline/rifampin combination in that rifampin reduces levels of doxycycline in plasma.

Patients with serious complications of brucellosis require urgent surgical and medical treatment. These complications include endocarditis, aortic root abscesses, mycotic aortic aneurysms, meningitis, cerebral or cerebellar abscesses, spinal or extraspinal osteomyelitis, and liver or splenic abscess. These patients should be hospitalized and given first a three-drug regimen -- i.e., oral doxycycline with intravenous aminoglycoside and rifampin -- for 4 weeks and then a two-drug regimen -- i.e., doxycycline and rifampin -- for 8 to 12 weeks. In instances of renal failure, doxycycline (at adjusted doses) can be used safely. In contrast, the use of aminoglycosides requires facilities for the monitoring of plasma levels; if such facilities are not available, then the doxycycline/rifampin combination should be administered for 8 to 12 weeks.

When used alone, fluoroquinolones (which exhibit good intracellular penetration and efficacy against *Brucella* organisms in vitro) have been associated with the development of quinolone resistance and with high rates of failure and relapse. At present, clinical data are inadequate for the formulation of recommendations regarding the combination of fluoroquinolones with doxycycline, rifampin, or streptomycin.

Third-generation cephalosporins (e.g., ceftriaxone), although active in vitro against brucellae when used alone, have also been associated with a high incidence of clinical failure and relapse. These agents may be useful in combination with other drugs for the treatment of *Brucella* meningitis.

In pregnancy, trimethoprim-sulfamethoxazole (TMP-SMZ) can be given in combination with rifampin for 8 to 12 weeks. The TMP-SMZ dosage appropriate for pregnant women is two or three tablets every 12 h (each tablet contains 80 mg of TMP and 400 mg of SMZ). Children below the age of 8 years can also be treated with rifampin and TMP-SMZ for 8 to 12 weeks, while older children should receive the same antibiotics as adults in the following doses: doxycycline, 100 mg/d orally; an aminoglycoside (gentamicin, 2 mg/kg per day in divided doses every 8 h); and rifampin, 15 mg/kg per day orally or by slow intravenous infusion. TMP-SMZ is given orally every 12 h in a dose that depends on the patient's age (birth to 6 months, 120 mg; 6 months to 6 years, 240

mg).

Within 4 to 14 days after the initiation of therapy, patients become afebrile and constitutional symptoms disappear. The enlarged liver and spleen return to their normal size within 2 to 4 weeks. An acute, intense flare-up of symptoms may follow the start of treatment, especially that with tetracyclines. This reaction is transient and does not necessitate the discontinuation of therapy. In endemic areas the coexistence of brucellosis and tuberculous spondylitis may result in a failure to respond to appropriate treatment. Treated patients whose infections are apparently cured should be followed clinically and serologically, with repeat blood cultures, every 3 to 6 months for 2 years.

PREVENTION

Efforts at prevention should be aimed at the source of infection. Immunization of animals and boiling or pasteurization of milk and milk products are important. Workers in the meat and dairy industries in the former Soviet Union, China, and France have been vaccinated; the vaccine (two injections given 2 weeks apart, each containing 1 mg of an insoluble fraction of phenol-extracted bacteria) markedly reduces the rate of infection. However, the vaccine induces fever in 6% of recipients and severe pain at the injection site in 16%. Moreover, immunity is short-lived, and vaccination should be repeated every 2 years. This vaccine is not used in the United States.

PROGNOSIS

Deaths attributable to brucellosis should be avoidable. Even before the discovery of antibiotics, the mortality rate was <2% and endocarditis was most frequently the cause of death. Morbidity due to brucellosis remains significant; its severity depends on the infecting *Brucella* species and is greatest with *B. melitensis*. Spinal damage, paraplegia, and other neurologic deficits may occur. Nerve deafness due to meningitis or secondary to treatment with streptomycin has been documented.

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161. TULAREMIA - Richard F. Jacobs

DEFINITION

Tularemia is a zoonosis caused by *Francisella tularensis*, so named in 1974 in recognition of the contributions of Edward Francis. Humans of any age, sex, or race are universally susceptible to this systemic infection. Tularemia is primarily a disease of wild animals and persists in contaminated environments, ectoparasites, and animal carriers. Human infection is incidental and usually results from interaction with biting or blood-sucking insects, wild or domestic animals, or the environment. Tularemia is common in Arkansas, Oklahoma, and Missouri, where more than 50% of the cases in the United States occur. An increasing number of cases of tularemia have been reported from the Scandinavian countries, eastern Europe, and Siberia. The illness is characterized by various clinical syndromes, the most common of which consists of an ulcerative lesion at the site of inoculation, with regional lymphadenopathy and lymphadenitis. Systemic manifestations, including pneumonia, typhoidal tularemia, and fever without localizing findings, pose a greater diagnostic challenge.

ETIOLOGY AND EPIDEMIOLOGY

F. tularensis is the etiologic agent of tularemia, which, with rare exceptions, is the only disease produced by this genus. The organism is a small, gram-negative, pleomorphic, nonmotile, non-spore-forming bacillus measuring 0.2 μm by 0.2 to 0.7 μm . Bipolar staining results in a coccoid appearance. The organism is a thinly encapsulated, nonpilated strict aerobe that invades host cells.

In nature, *F. tularensis* is a hardy organism that persists for weeks or months in mud, water, and decaying animal carcasses. Dozens of biting and blood-sucking insects, especially ticks and tabanid flies, serve as vectors. Ticks and wild rabbits are the source for most of the human cases in the endemic areas of the southeastern United States and the Rocky Mountain states. In Utah, Nevada, and California, tabanid flies are the most common vectors. Animal reservoirs include wild rabbits, squirrels, birds, sheep, beavers, muskrats, and domestic dogs and cats.

The two main biovars of *F. tularensis* -- *tularensis* (type A) and *polarctica* (type B) -- are both found in the United States. Type A produces more serious disease in humans; without treatment, the associated fatality rate is approximately 5%. Type B produces a milder, often subclinical infection that is usually contracted from water or marine mammals. Although all strains appear serologically identical, individual strains may possess varying degrees of virulence. *F. tularensis* does not produce an exotoxin, but an endotoxin similar to that of other gram-negative bacilli has been identified. The progression of illness depends upon the organism's virulence, the inoculum size, the portal of entry, and the host's immune status.

Ticks pass the organism to their offspring via a transovarian route. The organism is found in tick feces but not in large quantities in tick salivary glands. In the United States, the disease can be carried by *Dermacentor andersoni* (Rocky Mountain wood tick), *D. variabilis* (American dog tick), *D. occidentalis* (Pacific coast dog tick), and *Amblyomma americanum* (Lone Star tick). *F. tularensis* is transmitted frequently during blood meals

taken by embedded ticks following hours of attachment. It is the taking of a blood meal through a fecally contaminated field that transmits the organism. Tularemia is more common among men than among women. Person-to-person transmission is rare or nonexistent. Transmission of the organism by ticks and tabanid flies takes place mainly in the spring and summer. However, continued transmission in the winter months by trapped or hunted animals has been documented. The organism is extremely infectious. Biosafety level 2 is recommended for clinical laboratory work with material whose contamination is suspected, and biosafety level 3 is required for culture of the organism in large quantities.

PATHOGENESIS AND PATHOLOGY

The most common portal of entry for human infection is through skin or mucous membranes, either directly -- through the bite of ticks, other arthropods, or other animals -- or via inapparent abrasions. Inhalation or ingestion of *F. tularensis* can also result in infection. Although more than 10⁸ organisms are usually required to produce infection via the oral route (oropharyngeal or gastrointestinal tularemia), fewer than 50 organisms will result in infection when injected into the skin (ulceroglandular/glandular tularemia) or inhaled (pneumonia). After inoculation into the skin, the organism multiplies locally; within 2 to 5 days (range, 1 to 10 days), it produces an erythematous, tender, or pruritic papule. The papule rapidly enlarges and forms an ulcer with a black base (chancriform lesion). The bacteria spread to regional lymph nodes, producing lymphadenopathy (buboes), and, with bacteremia, may spread to distant organs.

Tularemia is characterized by mononuclear cell infiltration with pyogranulomatous pathology. The histopathologic findings can be quite similar to those in tuberculosis, although tularemia develops more rapidly. As a facultatively intracellular bacterium, *F. tularensis* can parasitize both phagocytic and nonphagocytic host cells and survive intracellularly for prolonged periods. In the acute phase of infection, the primary organs affected (skin, lymph nodes, liver, and spleen) include areas of focal necrosis, initially surrounded by polymorphonuclear leukocytes (PMNs). Subsequently, granulomas form, with epithelioid cells, lymphocytes, and multinucleated giant cells surrounded by areas of necrosis. These areas may resemble caseation necrosis but later coalesce to form abscesses.

Conjunctival inoculation can result in infection of the eye, with regional lymph node enlargement (preauricular lymphadenopathy, Parinaud's complex). Aerosolization and inhalation or hematogenous spread of organisms can result in pneumonia. In the lung, an inflammatory reaction -- including foci of alveolar necrosis and cell infiltration (initially polymorphonuclear and later mononuclear) with granulomas -- develops. Chest roentgenograms usually reveal bilateral patchy infiltrates rather than large areas of consolidation. Pleural effusions are common and may contain blood. Lymphadenopathy occurs in regions draining infected organs. Therefore, in pulmonary infection, mediastinal adenopathy may be evident, while patients with oropharyngeal tularemia develop cervical lymphadenopathy. In gastrointestinal or typhoidal tularemia, mesenteric lymphadenopathy may follow the ingestion of large numbers of organisms. The term *typhoidal tularemia* may be used to describe severe bacteremic disease, irrespective of the mode of transmission or portal of entry. Meningitis has been reported as a primary or secondary manifestation of bacteremia. Patients may also present with fever and no

localizing signs.

IMMUNOLOGY

Infection with *F. tularensis* stimulates the host to produce antibodies. However, this antibody response probably plays only a minor role in the containment of infection. In contrast, cell-mediated immunity, which develops over 2 to 4 weeks, plays a major role in containment and eradication of the infection. Macrophages, once activated, are capable of killing *F. tularensis*.

Immunospecific protection against tularemia can be afforded either by natural infection or by vaccination with live attenuated strains of *F. tularensis*. Killed vaccines, on the other hand, induce no protection against virulent *F. tularensis*. After natural infection or vaccination, serum antibodies to surface-exposed carbohydrate antigens predominate, whereas T cell determinants are located on membrane proteins beneath the bacterial capsule. T cell responses are thought to be due to priming by the organism. The anamnestic T cell response to *F. tularensis* seems to involve a multitude of microbial proteins, each with a distinct set of T cell determinants. A predominant role for CD4+ T cells is supported by the results of experiments in mice, which indicated that resistance to infection was restricted at the level of the MHC class II determinants. Humans primed to *F. tularensis* (like those primed to *Mycobacterium tuberculosis*) show a T_H1-like response. T cell proliferation is associated with the production of interleukin (IL) 2 and interferon γ but with little or no production of IL-4.

Investigations of neutrophils in cases of tularemia have suggested that [PMNs](#) are needed for defense against primary infection. PMNs may restrict the growth of *F. tularensis* before the organism becomes intracellular.

CLINICAL MANIFESTATIONS

Tularemia often starts with a sudden onset of fever, chills, headache, and generalized myalgias and arthralgias ([Table 161-1](#)). This onset takes place when the organism penetrates the skin, is ingested, or is inhaled. An incubation period of 2 to 10 days is followed by the formation of an ulcer at the site of penetration ([Fig. 161-CD1](#)), with local inflammation. The ulcer may persist for several months as organisms are transported via the lymphatics to the regional lymph nodes. These nodes enlarge ([Fig. 161-CD1](#)) and may become necrotic and suppurative. If the organism enters the bloodstream, widespread dissemination as well as signs and symptoms of endotoxemia may result.

In the United States, most patients with tularemia (75 to 85%) acquire the infection by inoculation of the skin. In adults, the most common localized form is inguinal/femoral lymphadenopathy; in children, it is cervical lymphadenopathy. About 20% of patients develop a generalized maculopapular rash, which occasionally becomes pustular. Erythema nodosum occurs infrequently. The clinical manifestations of tularemia have been divided into various syndromes, which are listed in [Table 161-2](#).

Ulceroglandular/Glandular Tularemia These two forms of tularemia account for approximately 75 to 85% of cases. The predominant form in children involves cervical or posterior auricular lymphadenopathy and is usually related to tick bites on the head and

neck. In adults, the most common form is inguinal/femoral lymphadenopathy resulting from insect and tick exposures on the lower limbs. In cases related to wild game, the usual portal of entry for *F. tularensis* is either an injury sustained while skinning or cleaning an animal carcass or a bite (usually on the hand). Epitrochlear lymphadenopathy/lymphadenitis is common in patients with bite-related injuries.

In ulceroglandular tularemia, the ulcer is erythematous, indurated, and nonhealing, with a punched-out appearance that lasts from 1 to 3 weeks. The papule may begin as an erythematous lesion that is tender or pruritic; it evolves over several days into an ulcer with sharply demarcated edges and a yellow exudate. The ulcer gradually develops a black base, and simultaneously the regional lymph nodes become tender and severely enlarged ([Fig. 161-1](#)). The affected lymph nodes may become fluctuant and drain spontaneously, but usually the condition resolves with effective treatment. Late suppuration of lymph nodes has been described in up to 25% of patients with ulceroglandular/glandular tularemia. Examination of material taken from these late fluctuant nodes after successful antimicrobial treatment has revealed sterile necrotic tissue. In 5 to 10% of patients, the skin lesion may be inapparent, with lymphadenopathy plus systemic signs and symptoms the only physical findings. This clinical syndrome is designated *glandular tularemia*. Conversely, a tick or deerfly bite on the trunk may result in an ulcer without evident lymphadenopathy.

Oculoglandular Tularemia In about 1% of patients, the portal of entry for *F. tularensis* is the conjunctiva. Usually, the organism reaches the conjunctiva through contact with contaminated fingers. The inflamed conjunctiva is painful, with numerous yellowish nodules and pinpoint ulcers. Purulent conjunctivitis with regional lymphadenopathy (preauricular, submandibular, or cervical) is evident. Because of debilitating pain, the patient may seek medical attention before regional lymphadenopathy develops. Painful preauricular lymphadenopathy is unique to tularemia and distinguishes it from cat-scratch disease, tuberculosis, sporotrichosis, and syphilis. Corneal perforation may occur.

Oropharyngeal and Gastrointestinal Tularemia Rarely, tularemia follows the ingestion of contaminated undercooked meat, the oral inoculation of *F. tularensis* from the hands in association with the skinning and cleaning of animal carcasses, or the consumption of contaminated food or water. Oral inoculation may result in acute, exudative, or membranous pharyngitis associated with cervical lymphadenopathy or in ulcerative intestinal lesions associated with mesenteric lymphadenopathy, diarrhea, abdominal pain, nausea, vomiting, and gastrointestinal bleeding. Infected tonsils become enlarged and develop a yellowish-white pseudomembrane, which can be confused with that of diphtheria. The clinical severity of gastrointestinal tularemia varies from mild, unexplained, persistent diarrhea with no other symptoms to a rapidly fulminant, fatal disease. In fatal cases, the extensive intestinal ulceration found at autopsy suggests an enormous inoculum.

Pulmonary Tularemia Tularemia pneumonia presents as variable parenchymal infiltrates that are unresponsive to treatment with β -lactam antibiotics. Tularemia must be considered in the differential diagnosis of atypical pneumonia in a patient with a history of travel to an endemic area. The disease can result from either inhalation of an infectious aerosol or spread to the lungs and pleura after bloodstream dissemination.

Inhalation-related pneumonia has been described in laboratory workers after exposure to contaminated materials and is associated with a relatively high mortality rate. Exposure to *F. tularensis* in aerosols from live domestic animals or dead wildlife (including birds) has been reported to cause pneumonia. Hematogenous dissemination to the lungs occurs in 10 to 15% of cases of ulceroglandular tularemia and in about half of cases of typhoidal tularemia. Previously, tularemia pneumonia was thought to be a disease of older patients, but as many as 10 to 15% of children with clinical manifestations of tularemia have parenchymal infiltrates detected by chest roentgenography. Patients with pneumonia usually have a nonproductive cough and may have dyspnea or pleuritic chest pain. Roentgenograms of the chest usually reveal bilateral patchy infiltrates (described as ovoid or lobar densities), lobar parenchymal infiltrates, and cavitary lesions. Pleural effusions may have a predominance of mononuclear leukocytes or [PMNs](#) and sometimes red blood cells. Empyema may develop. Patients with tularemia pneumonia can have blood cultures positive for *F. tularensis*.

Typhoidal Tularemia Once thought to represent up to 10% of all cases of tularemia, the typhoidal presentation is now considered rare in the United States. In this presentation, fever develops without apparent skin lesions or lymphadenopathy. In the absence of a history of possible contact with a vector, diagnosis can be extremely difficult. Blood cultures may be positive and patients may present with classic sepsis or septic shock in this acute systemic form of the infection. Typhoidal tularemia is usually associated with a huge inoculum or with a preexisting compromising condition. High continuous fevers, signs of endotoxemia, and severe headache are common findings. The patient may be delirious and may develop prostration and shock. If presumptive antibiotic therapy in culture-negative cases does not include an aminoglycoside, the mortality rate can approach 30%.

Other Manifestations *F. tularensis* infection has been associated with meningitis, pericarditis, hepatitis, peritonitis, endocarditis, osteomyelitis, and sepsis and septic shock with rhabdomyolysis and acute renal failure. In the rare cases of tularemia meningitis, a predominantly lymphocytic response is demonstrated in cerebrospinal fluid.

DIFFERENTIAL DIAGNOSIS

When patients in endemic areas present with fever, chronic ulcerative skin lesions, and large tender lymph nodes, a diagnosis of tularemia should be made presumptively, and confirmatory diagnostic testing and appropriate therapy should be undertaken. When the possibility of tularemia is considered in a patient with this presentation in a nonendemic area, an attempt should be made to determine whether the individual has come into contact with a potential animal vector. The level of suspicion of tularemia should be especially high in hunters, trappers, game wardens, veterinarians, laboratory workers, and individuals with a history of exposure to an insect or another animal vector. However, up to 40% of patients with tularemia have no known history of epidemiologic contact with an animal vector.

The characteristic presentation of ulceroglandular tularemia does not pose a diagnostic problem, but a less classic progression of regional lymphadenopathy or glandular

tularemia must be differentiated from other diseases. The skin lesion may resemble those seen in sporotrichosis; skin infection with *Staphylococcus aureus*, *Streptococcus pyogenes*, or *Mycobacterium marinum*; syphilis; anthrax; rat-bite fever (due to *Spirillum minus*); or rickettsiosis (scrub typhus). In the latter infections, regional lymphadenopathy is usually not as impressive as in tularemia. The lymphadenopathy of tularemia (especially glandular tularemia) must be differentiated from that of plague, lymphogranuloma venereum, and cat-scratch disease. In children, the differentiation from cat-scratch disease is made more difficult by the chronic papulovesicular lesion associated with *Bartonella henselae* infection ([Chap. 163](#)).

Oropharyngeal tularemia can resemble and must be differentiated from pharyngitis due to group A β -hemolytic streptococci, *Arcanobacterium haemolyticum*, or *Corynebacterium diphtheriae* as well as from infectious mononucleosis. Tularemia pneumonia may resemble any of the atypical pneumonias, including those due to various viruses, *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *C. psittaci*, *Legionella pneumophila*, *Coxiella burnetii*, and (occasionally) *Histoplasma capsulatum*. Typhoidal tularemia may resemble typhoid fever, other *Salmonella* bacteremias, rickettsial infections (Rocky Mountain spotted fever, ehrlichiosis), brucellosis, infectious mononucleosis, acquired toxoplasmosis, miliary tuberculosis, sarcoidosis, and hematologic or reticuloendothelial malignancies.

LABORATORY DIAGNOSIS

Direct microscopic examination of polychromatically stained tissue smears or clinical specimens reveals *F. tularensis* organisms, singly and in groups, both intra- and extracellularly. Gram's staining of clinical or biopsy material is of little value, as the small, weakly staining organisms cannot be readily distinguished from the background. An indirect fluorescent antibody test with commercially available antisera can be useful, although false-positive results due to *Legionella* spp. have been reported.

The diagnosis of tularemia is most frequently confirmed by serologic testing. In the standard tube agglutination test, a single titer of $\geq 1:160$ is interpreted as a presumptive positive result. A fourfold increase in titer between paired serum samples collected 2 to 3 weeks apart is considered diagnostic. False-negative serologic responses are obtained early in infection; up to 30% of patients infected for 3 weeks have sera that test negative. Late in infection, titers into the thousands are common, and titers of 1:20 to 1:80 may persist for years. A microagglutination test that may be as much as 100-fold more sensitive than the standard tube agglutination test has been described and is currently being used in many clinical microbiology laboratories. Enzyme-linked immunosorbent assays have proven useful for the detection of both antibodies and antigens. Analysis of urine for *F. tularensis* antigen has yielded promising results in clinical trials, but facilities for this type of analysis are not widely available. A skin test for delayed hypersensitivity to *F. tularensis* turns positive during the first week of illness and remains positive for years. The skin-test antigen, which is not commercially available, can boost titers of agglutinating antibody.

Culture and isolation of *F. tularensis* are difficult. In one study the organism was isolated in only 10% of more than 1000 human cases, 84% of which were confirmed by serology. The medium of choice is cysteine-glucose-blood agar. *F. tularensis* can be

isolated directly from infected ulcer scrapings, lymph-node biopsy specimens, gastric washings, sputum, and blood cultures. Colonies are blue-gray, round, smooth, and slightly mucoid. On media containing blood, a small zone of hemolysis usually surrounds the colony. Slide agglutination tests or direct fluorescent antibody tests with commercially available antisera can be applied directly to culture suspensions for identification.

The polymerase chain reaction (PCR) has been used to detect *F. tularensis* DNA. During a recent outbreak, a multiplex PCR was used to target 16S rRNA and to diagnose ulceroglandular tularemia with DNA extracted from wound swabs; the PCR result was positive in 29 (73%) of 40 serologically confirmed cases. However, this test has not been shown to be more sensitive than direct culture and at present remains a research tool.

TREATMENT

F. tularensis cannot be subjected to standardized antimicrobial susceptibility testing because the organism will not grow on the media used. A wide variety of antibiotics, including all β -lactam antibiotics and the newer cephalosporins, are ineffective for the treatment of this infection. Several studies indicated that third-generation cephalosporins were active against *F. tularensis* in vitro, but clinical case reports suggested a nearly universal failure rate of ceftriaxone in pediatric patients with tularemia. Although in vitro data indicate that imipenem may be active, therapy with imipenem, sulfanilamides, and macrolides is not presently recommended because of the lack of relevant clinical data. Fluoroquinolones have shown promise in terms of their relatively low toxicity and their potential for oral administration. Chloramphenicol and tetracycline have been used successfully for treatment of the acute stages of tularemia but have been associated with higher relapse rates (up to 20%) than conventionally used agents.

Streptomycin, given intramuscularly at a dose of 7.5 to 10 mg/kg every 12 h, is considered the drug of choice for adults. In severe cases, 15 mg/kg every 12 h may be used for the first 48 to 72 h. Streptomycin is also considered the drug of choice for children; the appropriate dose is 30 to 40 mg/kg daily in two divided doses administered intramuscularly. In children, after a clinical response is demonstrated at 3 to 5 days, the dose can be reduced to 10 to 15 mg/kg daily in two divided doses. Therapy is typically continued for 7 to 10 days; however, in mild to moderate cases of tularemia in which the patient becomes afebrile within the first 48 to 72 h of streptomycin treatment, a 5- to 7-day course has been successful.

Gentamicin, at a dose of 1.7 mg/kg given intravenously or intramuscularly every 8 h, is also effective. The published experience in adults consists of two reports describing, respectively, nine and eight patients who were treated effectively with gentamicin. The eight patients in one of the reports all had fever before treatment, and all eight became afebrile within 24 to 72 h. In a recent pediatric study, other symptoms, such as tender lymphadenitis and pharyngitis, also responded within 24 to 72 h of the start of gentamicin therapy.

Virtually all strains of *F. tularensis* are susceptible to streptomycin and gentamicin. In

successfully treated patients, defervescence usually occurs within 2 days, but skin lesions and lymph nodes may take 1 to 2 weeks to heal. When therapy is not initiated within the first several days of illness, defervescence may be delayed. Relapses are uncommon with streptomycin or gentamicin therapy. Late lymph-node suppuration, however, occurs in approximately 40% of children, regardless of the treatment received. These nodes have typically been found to contain sterile necrotic tissue without evidence of active infection. Patients with fluctuant nodes should receive several days of antibiotic therapy before drainage to minimize the risk to hospital personnel. Unlike streptomycin and gentamicin, tobramycin is ineffective in the treatment of tularemia and should not be used.

PROGNOSIS

If tularemia goes untreated, symptoms usually last 1 to 4 weeks but may continue for months. The mortality rate from severe untreated infection (including all cases of untreated tularemia pneumonia and typhoidal tularemia) can be as high as 30%. However, the overall mortality rate for untreated tularemia is <8%. Mortality is <1% with appropriate treatment. Poor outcomes are often associated with long delays in diagnosis and treatment. Lifelong immunity usually follows tularemia.

PREVENTION

The prevention of tularemia is based on avoidance of exposure to biting and blood-sucking insects, especially ticks and deerflies. An intradermal vaccine made from live attenuated *F. tularensis* is available from the Centers for Disease Control and Prevention. This vaccine is effective in reducing the frequency and severity of infection. Vaccination of high-risk individuals working with large quantities of cultured organisms is recommended. Others who come into contact with the organisms, such as veterinarians, hunters, or game wardens, should consider vaccination, particularly if they live in endemic areas. The avoidance of skinning wild animals, especially rabbits, and the wearing of gloves while handling animal carcasses decrease the risk of transmission. Use of insect repellents and preparations that prevent tick attachment as well as prompt removal of ticks can be helpful. Prophylaxis of tularemia has not proved effective in patients with embedded ticks or insect bites. However, in patients who are known to have been exposed to large quantities of organisms (e.g., in the laboratory) and who have incubating infection with *F. tularensis*, early treatment can prevent the development of significant clinical disease.

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162. PLAGUE AND OTHER *YERSINIA* INFECTIONS - Grant L. Campbell, David T. Dennis

PLAGUE

DEFINITION

Plague is an acute, febrile, zoonotic disease caused by infection with *Yersinia pestis*. Although human cases are infrequent and are curable with antibiotics, plague is one of the most virulent and potentially lethal bacterial diseases known. The plague bacterium occurs in widely scattered foci in Asia, Africa, and the Americas, where its usual hosts are wild and peridomestic rodents. It is transmitted to humans typically by flea bite and less commonly by direct contact with infected animal tissues or by airborne droplet. The principal clinical forms of plague are bubonic, septicemic, and pneumonic. Although most cases are now sporadic, occurring singly or in small clusters, the potential for epidemic spread remains.

ETIOLOGIC AGENT

Y. pestis is a gram-negative coccobacillus in the family Enterobacteriaceae. It is microaerophilic, nonmotile, nonsporulating, oxidase and urease negative, and biochemically unreactive. The organism is nonfastidious and infective for laboratory rodents. It grows well, if slowly, on routinely used microbiologic media (e.g., sheep blood agar, brain-heart infusion broth, and MacConkey agar). *Y. pestis* can multiply within a wide range of temperatures (-2°C to 45°C) and pH values (5.0 to 9.6), but optimal growth occurs at 28°C and at pH ~7.4. When incubated on agar plates at 37°C, colonies are pinpoint in size at 24 h and 1 to 2 mm in diameter at 48 h. The colonies are gray-white with irregular surfaces, described as having a "hammered-metal" appearance when viewed microscopically. In broth culture, *Y. pestis* grows without turbidity in clumps clinging to the sides of tubes. When stained with a polychromatic stain (e.g., Wayson or Giemsa), *Y. pestis* isolated from clinical specimens exhibits a characteristic bipolar appearance, often resembling closed safety pins. The bacterium is nonencapsulated but when grown at 30°C produces a plasmid-expressed immunogenic envelope glycoprotein, fraction 1 (F1).

HISTORIC BACKGROUND

Plague's deadly epidemic potential is notorious and well documented. The Justinian pandemic (542 to 767 A.D.) spread from central Africa to the Mediterranean littoral and thence to Asia Minor, causing an estimated 40 million deaths. The second pandemic began in central Asia, was carried to Sicily by ship from Constantinople in 1347, and swept through Europe and the British Isles in successive waves over the next four centuries. At its height, it killed as many as a quarter of the affected population and became known as the Black Death. In the third (modern) pandemic, plague appeared in Yunnan, China, in the latter half of the nineteenth century; established itself in Hong Kong in 1894; and spread by ship to Bombay in 1896 and subsequently to major port cities throughout the world, including San Francisco and several other West Coast and Gulf Coast ports in the United States. The plague bacillus was first cultured by Alexandre Yersin in Hong Kong in 1894. In 1898, Paul-Louis Simond, a French scientist

sent to investigate epidemic bubonic plague in Bombay, identified the bacillus in the tissues of dead rats and proposed transmission by rat fleas. Waldemar Haffkine, also in Bombay at that time, developed a crude vaccine.

By 1910, plague had circled the globe and established itself in rodent populations on all inhabited continents other than Australia. After 1920, however, the spread of plague was largely halted by international regulations that mandated control of rats in harbors and inspection and rat-proofing of ships. Before the third pandemic subsided, it resulted in an estimated 26 million plague cases and more than 12 million deaths, the vast majority in India. By 1950, plague outbreaks around the world had become isolated, sporadic, and manageable with modern techniques of surveillance, flea and rat control, and antimicrobial treatment of patients. Plague has nearly disappeared from cities and now occurs mostly in rural and semirural areas, where it is maintained in various rodents and their fleas. In the United States, the last outbreak of urban plague occurred in Los Angeles in 1924 and 1925, and human cases since then have resulted from animal plague exposures in rural areas of western states.

Because of its pandemic history, plague remains one of three quarantinable diseases subject to international health regulations (the other two being cholera and yellow fever). The alarm that plague is still able to evoke was highlighted by the public panic over an exaggerated international response to purported outbreaks of bubonic and pneumonic plague in India in 1994. The plague bacillus is considered to have a high potential for use in biologic terrorism; the agent is available around the world, has been "weaponized" for airborne delivery, and would be expected to cause a high primary fatality rate as well as secondary spread among an affected population.

EPIDEMIOLOGY

Y. pestis is maintained in enzootic cycles involving relatively resistant wild rodents and their fleas in mostly remote, lightly populated areas of Asia, Africa, and the Americas and in limited rural foci in extreme southeastern Europe near the Caspian Sea. Humans and other nonrodent mammals are incidental hosts. Enzootic transmission places humans at low risk, and cases are typically infrequent and sporadic. Epizootic transmission involving susceptible rodents and efficient flea vectors (both are amplifying hosts) results in local or even widespread depopulation of susceptible rodents and poses a more serious threat to humans than does enzootic transmission. In the United States, the principal epizootic hosts are various ground squirrels, prairie dogs, and chipmunks; a variety of burrowing rodents act as epizootic hosts in rural areas elsewhere in the world. *Y. pestis* occasionally spills over from wild rodents to rat species that inhabit cultivated fields and adjacent homes, villages, and towns. The organism can then be transported from towns to cities by these highly adaptable rats and their fleas. Urban plague is currently reported sporadically from a few countries such as Vietnam, Myanmar, and Madagascar.

Plague in populated areas is most likely to develop when sanitation is poor and rats are numerous -- especially the common black or roof rat (*Rattus rattus*), its close relatives, and the larger brown sewer or Norway rat (*R. norvegicus*). A high mortality rate from plague in these susceptible rat populations forces their fleas to seek alternative hosts, including humans. The cosmopolitan oriental rat flea *Xenopsylla cheopis* and (in

southern Africa and Brazil) the related species *X. brasiliensis* are efficient vectors of the plague bacillus among rats and are also efficient vectors to humans. *Y. pestis* can multiply to enormous numbers in the foregut (proventriculus) of these fleas, resulting in a bolus of organisms and clotted blood that blocks the passage of subsequent blood meals. This situation occurs only at temperatures of $\geq 28^{\circ}\text{C}$ and depends on a single protease expressed by the plasminogen activator (*pla*) gene of a 9.5-kb plasmid of *Y. pestis*. Regurgitation by a "blocked" flea while it feeds facilitates transmission of the plague bacillus to the new host.

Except for large outbreaks of pneumonic plague in Manchuria in the early part of the twentieth century, person-to-person respiratory transmission of plague during and since the third pandemic has occurred only sporadically and has been limited to clusters of close contacts of pneumonic plague patients, such as household members and caregivers. The 1994 outbreak of pneumonic plague in the city of Surat, India, although reported to be extensive, most likely involved fewer than 100 cases and 50 deaths; in 1998, a small outbreak of pneumonic plague occurred in the Ecuadorian Andes.

International health regulations require that national health authorities immediately report plague cases to the World Health Organization. From 1982 through 1996, 23,904 human plague cases and 2105 deaths (mortality, 9%) were reported by 24 countries. In the same 15-year period, the United States reported 212 plague cases (mean, 14 cases per year) and 27 deaths (mortality, 13%). Cases reported by the United States are confirmed by the plague laboratory of the Centers for Disease Control and Prevention (CDC). Animal plague occurs in 17 contiguous western states, extending from the Great Plains states and eastern Texas to the Pacific Coast; around 80% of human cases in this country now occur in New Mexico, Arizona, and Colorado and around 10% in California. Although plague in the United States is a rural disease, more than 50% of cases are thought to be caused by peridomestic exposures, especially in the southwestern states, where homes are often situated in natural surroundings that provide a favorable habitat for plague-susceptible animals (such as rock squirrels and wood rats) and their fleas. In the Sierra Nevadas of California and Nevada, epizootic plague in chipmunks and ground squirrels poses a risk to visitors in public parks. Hikers, campers, and hunters in natural areas throughout the western states are at a small but finite risk of exposure to plague, especially in the summer months.

Plague can be transmitted during the skinning and handling of carcasses of wild animals such as rabbits and hares, prairie dogs, wildcats, and coyotes. Such direct inoculation of mammal-adapted organisms is associated with primary septicemia and high mortality. Oropharyngeal plague can result from the ingestion of undercooked contaminated meat and perhaps from the manual transfer of infected fluids to the mouth during the handling of infected animal tissues.

Carnivores, including dogs and cats, can become infected with *Y. pestis* by eating infected rodents and perhaps by being bitten by fleas from infected rodents. Although clinical plague commonly develops in infected cats, it rarely does so in infected dogs, which thus do not directly expose humans to infection. However, both dogs and cats may transport infected fleas from rodent-infested areas to the home environment.

From 1950 through 1996, 387 plague cases were reported in the United States. Of the

376 evaluable cases, 322 cases (86%) presented as primary lymphadenitic (bubonic) plague, almost all of them thought to be associated with flea bites; 46 cases (12%) presented as primary septicemic plague, many of them following direct animal exposures; and 8 cases (2%) presented as primary pneumonic plague, 6 resulting from the inhalation of respiratory droplets released by infected cats and 2 from unknown sources. The last case of human-to-human plague transmission in the United States occurred in the Los Angeles outbreak of 1924/1925.

PATHOGENESIS AND PATHOLOGY

Y. pestis is highly invasive and pathogenic. The mechanisms by which the organism causes disease are incompletely understood, but both chromosome- and plasmid-encoded gene products as well as altered cell-mediated immune responses are probably involved. Three plasmids encode for a variety of known or presumed virulence factors, including the F1 envelope antigen, which confers bacterial resistance to phagocytosis by polymorphonuclear leukocytes (PMNs) in vitro; a murine exotoxin; the V antigen, which is essential for virulence, may immunocompromise the host by suppressing the synthesis of interferon γ and tumor necrosis factor α , and stimulates protective immunity in laboratory animals; pesticin, a bactericidal protein of unknown function and importance; a protease that can activate plasminogen and degrade serum complement and that is thought to play a role in the dissemination of *Y. pestis* from peripheral sites of infection; a coagulase; and a fibrinolysin. A lipopolysaccharide endotoxin, believed to be chromosomally encoded, is probably important in triggering the systemic inflammatory response syndrome and its complications.

Y. pestis organisms inoculated through the skin or mucous membranes usually invade superficial lymphatic vessels and are carried to regional lymph nodes, although direct bloodstream inoculation may take place. Mononuclear phagocytes, which can phagocytize *Y. pestis* organisms without destroying them, may play a role in dissemination of the infection to distant sites. Plague can involve almost any organ, and untreated plague generally results in widespread and massive tissue destruction. In the early stages, infected lymph nodes (buboes, [Fig. 162-1](#)) are characterized by edema and congestion without inflammatory infiltrates or apparent vascular injury. Fully developed buboes contain huge numbers of infectious plague organisms and show distorted or obliterated lymph node architecture with vascular destruction and hemorrhage, serosanguineous effusion, necrosis, and a mild neutrophilic infiltration. At this stage, the effusion often involves perinodal tissues. If several adjacent lymph nodes are involved, a boggy edematous mass can result.

Primary septicemic plague results from the direct inoculation of bacteria from infected fluids or tissues or from an infective flea bite in the apparent absence of a bubo; secondary septicemic plague occurs when lymphatic and other host defenses are breached and the plague bacillus multiplies within the bloodstream. In fatal septicemic plague, multifocal hepatic and splenic necrosis is common. Diffuse interstitial myocarditis with cardiac dilatation is sometimes found. If disseminated intravascular coagulation (DIC) ensues, vascular necrosis may lead to widespread cutaneous, mucosal, and serosal ecchymoses and petechiae. Acral gangrene sometimes develops.

Primary plague pneumonia generally begins as a lobular process and then extends by

confluence, becoming lobar and then multilobar ([Fig. 162-2](#)). Plague organisms are typically most numerous in the alveoli. Secondary plague pneumonia begins more diffusely, with organisms usually most numerous in the interstitium. In untreated cases of both primary and secondary plague pneumonia, diffuse pulmonary hemorrhage, necrosis, and neutrophilic infiltration develop.

MANIFESTATIONS

Plague is characterized by a rapid onset of fever and other systemic manifestations of gram-negative bacterial infection. If it is not quickly and correctly treated, plague can follow a toxic course, resulting in shock, multiple-organ failure, and death. In humans, the three principal forms of plague are bubonic, septicemic, and pneumonic. Bubonic plague, the most common form, is almost always caused by the bite of an infected flea but occasionally results from direct inoculation of infectious fluids. Septicemic and pneumonic plague can be either primary or secondary to metastatic spread. Unusual secondary forms include plague meningitis, endophthalmitis, and lymphadenitis at multiple sites. Primary plague pharyngitis has been documented by culture of organisms from throat swabs and can result from respiratory exposure or ingestion of contaminated meat.

Bubonic plague usually has an incubation period of 2 to 6 days, occasionally longer. Typically, the patient experiences chills; fever, with temperatures that rise within hours to 38°C; myalgias; arthralgias; headache; and a feeling of weakness. Soon -- usually within 24 h -- the patient notices tenderness and pain in one or more regional lymph nodes proximal to the site of inoculation of the plague bacillus ([Fig. 162-1](#)). Because fleas often bite the legs, femoral and inguinal nodes are most commonly involved; axillary and cervical nodes are next most commonly affected. The enlarging bubo becomes progressively painful and tender, sometimes exquisitely so. The patient usually guards against palpation and limits movement, pressure, and stretch around the bubo. The surrounding tissue often becomes edematous, sometimes markedly so, and the overlying skin may be erythematous, warm, and tense. Inspection of the skin surrounding or distal to the bubo sometimes reveals the site of a flea bite marked by a small papule, pustule, eschar, or ulcer. A list of lymphadenitic conditions that could be confused with a plague bubo would include *Staphylococcus aureus* and group A β -hemolytic streptococcal infections, cat-scratch disease, and tularemia. The bubo of plague is distinguishable from lymphadenitis of most other causes, however, by its rapid onset, its extreme tenderness, the accompanying signs of toxemia, and the absence of cellulitis or obvious ascending lymphangitis.

Treated in the uncomplicated state with an appropriate antibiotic, bubonic plague usually responds quickly, with defervescence and alleviation of other systemic manifestations over 2 to 5 days. Buboes often remain enlarged and tender for a week or more after the initiation of treatment and can become fluctuant. Without effective antimicrobial treatment, patients with typical bubonic plague manifest an increasingly toxic state of fever, tachycardia, lethargy leading to prostration, agitation and confusion, and (occasionally) convulsions and delirium. Secondary plague sepsis may result in an alarmingly rapid and refractory cascade of DIC, bleeding, shock, and organ failure. Mild forms of bubonic plague, called *pestis minor*, have been described in South America and elsewhere; in these cases, the patients are ambulatory, are only mildly febrile, and

have subacute buboes.

Septicemic plague is a progressive, overwhelming bacterial infection. Primary septicemia develops in the absence of apparent regional lymphadenitis, and the diagnosis of plague is often not suspected until preliminary blood culture results are reported to be positive by the laboratory. *Y. pestis*, however, can also be cultured from the blood of most bubonic plague patients, and bacteremia should be distinguished from septicemia, in which the patient is desperately ill and requires aggressive care. Patients with septicemic plague often present with gastrointestinal symptoms of nausea, vomiting, diarrhea, and abdominal pain, which may further confound the correct diagnosis. If not treated early with appropriate antibiotics, septicemic plague can be fulminant and fatal. In the United States in 1950 through 1996, 66 cases of septicemic plague and 18 deaths were reported, for a case-fatality rate of 27%. Petechiae, ecchymoses, bleeding from puncture wounds and orifices, and gangrene of acral parts are manifestations of [DIC](#); refractory hypotension, renal shutdown, obtundation, and other signs of shock are preterminal events. Adult respiratory distress syndrome (ARDS), which can occur at any stage of septicemic plague, is sometimes confused with other conditions, such as hantavirus pulmonary syndrome.

Of all forms of the disease, pneumonic plague develops most rapidly and is most frequently fatal. The incubation period for primary pneumonic plague is rarely longer than 1 to 4 days. The onset is most often sudden, with chills, fever, headache, myalgias, weakness, and dizziness. Pulmonary signs, including cough, sputum production, chest pain, tachypnea, and dyspnea, typically arise on the second day of illness and may be accompanied by hemoptysis, increasing respiratory distress, cardiopulmonary insufficiency, and circulatory collapse. In primary plague pneumonia, the sputum is most often watery or mucoid, frothy, and blood-tinged, but it may become frankly bloody. Pulmonary signs in primary pneumonic plague may indicate involvement of a single lobe in the early stage, with rapidly developing segmental consolidation before bronchopneumonic spread to other lobes of the same and opposite lungs. Liquefaction necrosis and cavitation may occur early in areas of consolidation and may or may not leave significant residual scarring.

Secondary plague pneumonia manifests first as diffuse interstitial pneumonitis in which sputum production is scant; since the sputum is more likely to be inspissated and tenacious in character than the sputum found in primary pneumonia, it may be less infectious. In the United States in 1950 through 1996, 39 cases of secondary pneumonic plague and 8 cases of primary pneumonic plague were reported, with no known transmission to contacts and an overall case-fatality rate of 41%. Observers in the early twentieth century remarked on the relative lack of auscultatory findings, the usual presence of toxemia, and the frequency of sudden death in patients with pneumonic plague as compared to patients with other bacterial pneumonias.

Meningitis is an unusual manifestation of plague. In the United States, there were 12 meningitis cases among the 376 evaluable plague cases reported in 1950 through 1996. All cases of meningitis were complications of bubonic plague, and all patients survived. Although meningitis may be a part of the initial presentation of plague, its onset is often delayed and is a manifestation of insufficient treatment. Recent cases in the United States have occurred during the first and second weeks of antibiotic

treatment for bubonic plague. Chronic relapsing meningeal plague over periods of weeks or even months was described in the preantibiotic era. The affected patients typically presented with fever, headache, meningismus, and pleocytosis.

Plague pharyngitis presents as fever, sore throat, cervical lymphadenitis, and headache and is often indistinguishable clinically from pharyngitis of other infectious etiologies. Caregivers working in plague-endemic areas must be alert to the possibility of plague to avoid misdiagnosis leading to delayed and/or inappropriate treatment.

LABORATORY FINDINGS AND DIAGNOSIS

Since plague is a rare disease in the United States, a high index of clinical suspicion as well as the elicitation of a thorough clinical and epidemiologic history and a careful physical examination are required for timely diagnosis and prompt institution of specific therapy. When the diagnosis of plague is delayed or missed altogether, a high case-fatality rate results; infected travelers who seek medical care after they have left endemic areas (peripatetic plague cases) are at especially high risk. Plague must be considered in the differential diagnosis of sepsis in an otherwise-healthy person who has a history of recent travel to or residence in the rural western United States. When the diagnosis of plague is being considered, close communication between clinicians and the diagnostic laboratory and between the diagnostic laboratory and a qualified reference laboratory is essential. Tests for plague are highly reliable when conducted by laboratory personnel experienced with *Y. pestis*, but such expertise is usually limited to selected reference laboratories, including state health department laboratories in some plague-endemic states and the [CDC](#) plague laboratory (Fort Collins, Colorado; tel. 970-221-6400).

When plague is suspected, specimens should be collected promptly for laboratory studies, chest roentgenograms should be obtained, and specific antimicrobial therapy should be initiated pending confirmation. Appropriate diagnostic specimens for smear and culture include citrated or heparinized whole blood from all patients with suspected plague, bubo aspirates from those with suspected buboes, sputum samples or tracheal aspirates from those with suspected pneumonic plague, and cerebrospinal fluid (CSF) from those with suspected plague meningitis. Since early buboes are often exquisitely tender and are seldom fluctuant or necrotic, these lesions usually require aspiration under local anesthesia following the injection of 1 to 2 mL of normal saline (sterile but nonbacteriostatic) into the bubo with a 20- to 22-gauge needle. A variety of appropriate culture media (including brain-heart infusion broth, sheep blood agar, and MacConkey agar) should be inoculated with a portion of each specimen. Moreover, for each specimen, at least one smear should be examined immediately with Wayson or Giemsa stain and at least one with Gram's stain; a smear should also be submitted for direct fluorescent antibody testing. An acute-phase serum specimen should be tested for antibody to *Y. pestis*; whenever possible, a convalescent-phase serum specimen collected 3 to 4 weeks later should also be tested. When a patient dies and plague is suspected, appropriate autopsy tissues for culture, direct fluorescent antibody testing, and immunohistochemical staining include buboes, all solid organs (especially liver, spleen, and lung), and bone marrow. If culture of such specimens is to be attempted, they should be sent to the laboratory either fresh or frozen on dry ice, not in preservatives or fixatives. If necessary, Cary-Blair or a similar medium can be used to

transport *Y. pestis*-infected tissues.

Laboratory confirmation of plague depends on the isolation of *Y. pestis* from cultures of body fluids or tissues. Cultures of three blood samples taken over a 45-min period before treatment will usually result in isolation of the bacterium. *Y. pestis* strains are readily distinguished from those of the closely related species *Y. pseudotuberculosis* by differences in biochemical profile, temperature-dependent susceptibility to lysis by a *Y. pestis*-specific bacteriophage, and motility. Automated bacteriologic test systems can be used to assist in the identification of isolates as *Y. pestis*, but *Y. pestis* can be misidentified (e.g., as *Y. pseudotuberculosis*) or overlooked if these systems are improperly programmed.

In the absence of *Y. pestis* isolation, plague cases can be confirmed either by the demonstration of seroconversion (a fourfold or greater titer rise) to *Y. pestis* F1 antigen in passive hemagglutination tests of acute- and convalescent-phase serum specimens or by detection of an antibody titer of >128 in a single serum sample from a patient with a plague-compatible illness who has not received plague vaccine. The specificity of a positive passive-hemagglutination test requires confirmation with the F1 antigen hemagglutination-inhibition test. A few plague patients seroconvert to F1 antigen as early as 5 days after the onset of illness. Most seroconvert between 1 and 2 weeks after onset; a few seroconvert 3 weeks or more after onset; and a few (<5%) fail to seroconvert at all. Early, specific antibiotic treatment may delay seroconversion by several weeks. After seroconversion, positive serologic titers diminish gradually over months to years. Enzyme-linked immunosorbent assays (ELISAs) for IgM and IgG antibodies to *Y. pestis* are replacing hemagglutination tests in some laboratories. Other new test methods include IgM antibody capture and competitive blocking for detection of antibody to F1.

Detection of F1 antigen in tissues or fluids by direct fluorescent antibody testing or by antigen capture is presumptive evidence of plague, as is an F1 antibody titer of >10 in a single serum sample from a patient with a plague-compatible illness who has not received plague vaccine. Visualization of characteristic bipolar bacilli in a Giemsa- or Wayson-stained smear constitutes supportive evidence of plague. Tularemia, especially the glandular, typhoidal, and pneumonic forms, can sometimes be confused clinically and epidemiologically with plague, but the results of microbiologic and serologic tests should readily distinguish these two diseases.

Patients with plague typically have white blood cell (WBC) counts of 15,000 to 25,000/uL, with a predominance of [PMNs](#) and a left shift. Leukemoid reactions with WBC counts as high as 100,000/uL can occur. Modest thrombocytopenia is usually documented, and fibrin-fibrinogen split products are often detected even in patients without frank [DIC](#). Serum levels of aminotransferases and bilirubin may be elevated. Chest roentgenograms of patients with pneumonic plague usually show patchy bronchopneumonic infiltrates as well as lobar or segmental consolidation with or without confluence ([Fig. 162-2](#)); they occasionally show cavitation. Stained sputum samples usually contain PMNs and characteristic bipolar-staining bacilli. In *Y. pestis* septicemia, visualization of the characteristic bacilli in a routine blood smear or a buffy-coat smear is an uncommon but grave prognostic sign ([Fig. 162-3](#)). In patients with plague meningitis, pleocytosis with a predominance of PMNs is the rule, and the characteristic bacilli are

usually visible in stained [CSF](#) smears.

TREATMENT

Left untreated, plague is fatal in more than 50% of cases of bubonic disease and in nearly all cases of septicemic and pneumonic disease. The overall mortality rate for plague cases in the United States since 1950 has been ~16%; deaths are almost always due to delays in seeking treatment, misdiagnosis, delays in the institution of treatment, or incorrect treatment. Rapid diagnosis and appropriate antimicrobial therapy are essential.

Guidelines for the treatment of plague are given in [Table 162-1](#). Although streptomycin is the drug of choice, gentamicin is increasingly used for the treatment of plague in the United States because of its ready availability; it is probably as effective as streptomycin, although results of controlled studies in humans have not been published. Alternative antibiotics include the tetracyclines and chloramphenicol; these agents are usually given orally with initial loading doses but may be given intravenously to critically ill patients and to patients unable to tolerate oral medication. Penicillins, cephalosporins, and macrolides are suboptimal and should not be used. Doxycycline may be as effective as other tetracyclines or even more so, but comparative evaluations have not been made. Trimethoprim-sulfamethoxazole has been used successfully to treat bubonic plague but is not considered a first-line choice. Chloramphenicol is indicated for the treatment of plague meningitis, pleuritis, endophthalmitis, and myocarditis because of its superior tissue penetration; it is used alone or in combination with streptomycin. In general, antimicrobial treatment should be continued for 10 days or for at least 3 days after the patient has become afebrile and has made a clinical recovery. Patients initially given intravenous antibiotics may be switched to oral regimens upon clinical improvement. Such improvement is usually evident 2 or 3 days after the start of treatment, even though fever may continue for several days.

Consequences of delayed treatment of plague include [DIC](#), [ARDS](#), and other complications of gram-negative sepsis. Patients with these disorders require intensive monitoring and close physiologic support, as outlined elsewhere ([Chaps. 117](#) and [265](#)). Buboes may require surgical drainage. Abscessed nodes can cause recurrent fever in patients who have apparently recovered; this relation may be occult if intrathoracic or intraabdominal nodes are involved. Although *Y. pestis* is considered to be genetically stable, a multidrug-resistant strain was isolated from a plague patient in Madagascar. This strain exhibited resistance (mediated by a transferable plasmid) to all first-line antibiotics used for treatment of plague and to the principal alternatives used for treatment and prophylaxis.

PREVENTION AND CONTROL

Persons at greatest risk for plague in the United States are those who live, work, and participate in outdoor recreational activities in areas of those western states in which plague is enzootic. Surveillance, education, and environmental management are the cornerstones of prevention and control. A network of biologists and public health specialists coordinates these activities through local and state health departments and the [CDC](#). Personal protective measures include the avoidance of areas with known

epizootic plague (which may be posted) and of sick or dead animals; the use of repellents, insecticides, and protective clothing when at risk of exposure to rodents' fleas; and the wearing of gloves when handling animal carcasses. Short-term antibiotic prophylaxis ([Table 162-2](#)) is recommended for persons known to have had direct contact with a patient with suspected or confirmed pneumonic plague and occasionally for persons who are unable to avoid an area where a plague outbreak is in progress or who may be caring for patients with plague. Patients in whom plague is suspected should be managed under isolation precautions for respiratory droplet transmission until pneumonia has been ruled out or until 48 h of specific antimicrobial therapy has been administered, after which universal precautions are adequate.

Rodent food (garbage, pet food) and habitats (brush piles, junk heaps, woodpiles) should be eliminated in domestic, peridomestic, and working environments; buildings and food stores should be rodent-proofed. The control of fleas with insecticides is a key public health measure in situations where epizootic plague activity places humans at high risk; this effort includes dusting and spraying of rodent burrows, rodent runs, and other sites where rodents and their fleas are found. In plague-endemic areas of the western United States, persons should keep their dogs and cats free of fleas and restrained. The decision to control plague by killing rodents should be left to public health authorities, and such a program should be carried out only in conjunction with effective flea control. Killing of rodents has no lasting benefit without environmental sanitation.

The previously used killed, whole-cell plague vaccine is no longer manufactured in the United States. Efforts are being made to develop improved vaccines in which the production of specific immunoprotective antibodies to *Y. pestis* is induced by recombinant antigens. In the United States, the indications for use of these newer vaccines would probably be similar to those for the previously available killed vaccine, which was mostly limited to protecting laboratory personnel who routinely worked with *Y. pestis* and some persons whose vocations brought them into regular contact with wild rodents and their fleas in areas with enzootic or epizootic plague. In addition, a vaccine might be useful in protecting selected military personnel and in responding to the possible use of *Y. pestis* as a weapon of bioterrorism.

OTHER *YERSINIA* INFECTIONS

DEFINITION

Yersiniosis is an uncommon bacterial zoonosis caused by infection with either of the two enteropathogenic *Yersinia* species: *Y. enterocolitica* or *Y. pseudotuberculosis*. Reservoir hosts of these bacteria include swine and other wild and domestic animals. These yersiniae are transmitted to humans predominantly via the oral route. Both sporadic cases and common-source outbreaks occur. The most frequent acute clinical manifestations are (1) enteritis or enterocolitis with self-limited diarrhea (especially with *Y. enterocolitica*), and (2) mesenteric adenitis and terminal ileitis (especially with *Y. pseudotuberculosis*), which can be difficult to distinguish from acute appendicitis. Septicemia and metastatic focal infections are less common. Some cases of yersiniosis are complicated by nonsuppurative, extraintestinal, inflammatory sequelae -- e.g., reactive arthritis ([Chap. 315](#)) and erythema nodosum ([Chap. 18](#)).

ETIOLOGIC AGENTS

Y. enterocolitica and *Y. pseudotuberculosis* are pleomorphic gram-negative bacilli in the family Enterobacteriaceae. They are aerobic or facultatively anaerobic, motile at 25°C, nonmotile at 37°C, oxidase negative, urease positive, able to ferment glucose, unable to ferment lactose, and usually able to reduce nitrates. They grow well, if slowly, on nonselective media (e.g., blood agar) and on most of the routine media used to select for enteric bacteria (e.g., MacConkey agar). They can multiply within a wide temperature range (-1°C to 45°C). The most clinically and epidemiologically useful methods for identifying pathogenic *Y. enterocolitica* isolates are biotyping based on biochemical profiles and serotyping according to somatic O and H antigens. Six biotypes and more than 60 serotypes of *Y. enterocolitica* are recognized. A separate serotyping system for *Y. pseudotuberculosis* (also based on somatic antigens) has distinguished six major serotypes (I through VI) and their subtypes.

EPIDEMIOLOGY

Y. enterocolitica is distributed worldwide and has been isolated from soil, fresh water, contaminated foodstuffs (e.g., meat, milk, and vegetables), and a wide variety of wild and domestic animals, including mammals, birds, amphibians, fish, and shellfish. Many serotypes isolated from environmental sources, however, evidently are not human pathogens. Most human infections have been caused by *Y. enterocolitica* serotypes O:3, O:5, O:8, and O:9, which are primarily associated with wild and domestic mammals. The incidence of these infections and their sequelae is highest in Scandinavia and some other northern European countries, but this observation may be in part an artifact of underrecognition in other countries. Because many individuals with enteric *Y. enterocolitica* infection are asymptomatic or minimally symptomatic and do not seek medical attention, reliable population-based estimates of incidence are unavailable. However, in many clinical microbiology laboratories in recent decades, *Y. enterocolitica* has been the fourth most common bacterial pathogen isolated from patients' fecal specimens, trailing *Salmonella* (the most frequently isolated), *Campylobacter*, and *Shigella* species.

All age groups are susceptible to *Y. enterocolitica* infections, but the majority of cases of enterocolitis are in children aged 1 to 4. Moreover, these infections show a modest predilection for males. Mesenteric adenitis and terminal ileitis are most common among older children and young adults. Risk factors for *Y. enterocolitica* septicemia and metastatic focal infections include chronic liver disease, malignancy, diabetes mellitus, immunosuppressive therapy, alcoholism, malnutrition, advanced age, iron overload (see below), and hemolytic anemias (including the thalassemias). The nonsuppurative sequelae of yersiniosis are most common among adults. HLA-B27 is expressed in 70 to 80% of patients who develop reactive arthritis associated with yersiniosis. HLA-B27 is not a risk factor for *Yersinia*-induced erythema nodosum; females with this condition outnumber males by 2 to 1. In Europe, *Y. enterocolitica* infections are more common in the cooler months than in warmer weather. In North America, no consistent seasonal pattern has been documented.

For several decades, serotypes O:3 and O:9 have predominated among *Y.*

enterocolitica isolates from patients in Europe. Serotype O:3 has also predominated in Canada and Japan. In the United States, serotype O:3 emerged in the 1980s to surpass serotype O:8 in frequency of isolation from patients. The incidence of *Yersinia*-induced nonsuppurative sequelae reportedly is 10 to 30% in Scandinavia and much lower in most other countries, including the United States. No convincing explanation for this observation has been confirmed, but reasonable possibilities include population genetic factors and geographic strain variation.

Common-source outbreaks of *Y. enterocolitica* enteritis have been traced to such vehicles as raw milk, contaminated pasteurized milk, and foods prepared with contaminated fresh water. In Belgium, the ingestion of ground raw pork (a regional custom) is a significant risk factor for sporadic infection with *Y. enterocolitica* serotypes O:3 and O:9. These serotypes commonly colonize the oral cavity and intestines of European swine, and *Y. enterocolitica* infection is an occupational risk of swine butchers in Europe. In the United States, sporadic cases and one outbreak of *Y. enterocolitica* O:3 infection have been associated with the preparation or ingestion of raw pork intestines (chitterlings). In some cases of yersiniosis, circumstantial evidence suggests transmission via contact with dogs and cats or their feces. Several nosocomial outbreaks of *Y. enterocolitica* infection have been described; fecal-oral transmission from person to person was suspected. Fecal-oral transmission among family members may also explain occasional secondary cases in households. In a prospective study of 50 children with *Y. enterocolitica* enteritis, fecal excretion of the organism persisted for an average of 27 days (range, 4 to 79 days) after the cessation of symptoms. A chronic carrier state, however, has not been demonstrated. *Y. enterocolitica* is a rare but often lethal cause of transfusion-associated septicemia. The explanation is that blood donors occasionally have transient, occult *Y. enterocolitica* bacteremia and that this organism can slowly multiply to high concentrations in blood refrigerated for at least 10 to 20 days.

The ecology of *Y. pseudotuberculosis* seems to parallel that of *Y. enterocolitica* closely. *Y. pseudotuberculosis* is also widespread in wild and domestic animals and is isolated from many environmental sources. Human infections with *Y. pseudotuberculosis*, however, appear to be rare. In North America and Europe, most such infections have been with serotype I, but outbreaks involving other serotypes have occurred in Japan and Scandinavia. Swine appear to be an important reservoir for pathogenic strains of *Y. pseudotuberculosis*.

PATHOGENESIS AND PATHOLOGY

Except in rare instances of transmission via contaminated blood products or direct cutaneous inoculation, the enteropathogenic yersiniae are thought to enter the host via the oral route. The 50% infectious dose in humans is uncertain but may be 10^9 . The incubation period averages 5 days (range, 1 to 11 days). Studies of animals have shown that the organisms initially invade the ileal epithelium, then are translocated via M cells into the lamina propria, and finally enter Peyer's patches, where they are able to replicate. They subsequently drain into the mesenteric lymph nodes, which undergo hyperplasia and from which the bacteria can be distributed systemically. The mesenteric lymph nodes can become intensely swollen and matted and are occasionally detected on physical examination as a tender right lower quadrant mass. Intestinal inflammation (most commonly of the distal ileum and less commonly of the ascending colon)

develops and may be accompanied by mucosal ulcerations and by the shedding of [PMNs](#) and red blood cells into the intestinal lumen. In relatively severe cases, thrombosis of mesenteric blood vessels, intestinal hemorrhage, and necrosis can occur. In patients with enteropathogenic yersinial infections who undergo exploratory laparotomy, the appendix usually is histologically normal or shows only lymphoid hyperplasia, but frank suppuration is sometimes evident.

A plasmid of ~70 kb is essential for virulence of the enteropathogenic yersiniae because it encodes at least six *Yersinia* outer-membrane proteins, some of which confer to bacterial strains such properties as cytotoxicity; resistance to phagocytosis by [PMNs](#); and the abilities to cause monocyte apoptosis (programmed cell death), to suppress the host's expression of tumor necrosis factor, to interfere with platelet aggregation and host complement activation, and to dephosphorylate host proteins. A chromosomal gene (*inv*) encodes for the surface protein invasin, which is necessary for yersinial invasion of nonphagocytic host cells (e.g., epithelial cells) in vitro and which facilitates the translocation of bacteria across the intestinal epithelium. Both *Y. enterocolitica* and *Y. pseudotuberculosis* can express at least one protein superantigen that selectively stimulates the proliferation of T cells. Many strains of *Y. enterocolitica* produce a heat-stable enterotoxin that is similar to *Escherichia coli* enterotoxin. The cell walls of *Y. enterocolitica* and *Y. pseudotuberculosis* contain a lipopolysaccharide (endotoxin). The roles of superantigens, enterotoxin, and endotoxin in the pathogenesis of yersiniosis are unclear. Some *Yersinia* strains are unable to synthesize bacterial iron chelators called *siderophores*. However, they can exploit host-chelated iron stores and the drug deferoxamine (a siderophore produced by *Streptomyces pilosus*). Therefore, iron overload (e.g., caused by hemodialysis or multiple transfusions) and deferoxamine therapy appear to be independent risk factors for *Y. enterocolitica* bacteremia (especially that involving serotypes O:3 and O:9) and to a lesser degree for *Y. pseudotuberculosis* bacteremia.

Immunogenetic factors and cell-mediated immune responses are clearly involved in the pathogenesis of reactive arthritis following infection with the enteropathogenic yersiniae. As noted above, most patients with *Yersinia*-induced reactive arthritis express HLA-B27. In addition, *Y. pseudotuberculosis* shares at least one cross-reactive epitope with HLA-B27, and *Y. enterocolitica* infection alters the expression of serologic HLA-B27 epitopes on lymphocytes and monocytes. In patients with reactive arthritis following *Y. enterocolitica* infection, yersinial antigens are commonly detectable in synovial fluid cells in the apparent absence of whole organisms. Thus, it is unknown whether the arthritis results from occult bacterial persistence through self-tolerance of HLA-B27 with a failure of cross-reactive immune responses to yersiniae, from an immune response to common antigenic determinants shared by the bacteria and host HLA-B27 (i.e., molecular mimicry), or from other mechanisms. The pathogenesis of *Yersinia*-induced erythema nodosum is obscure.

In some assays, patients with Graves' disease have an increased prevalence of serum antibodies to *Y. enterocolitica*, and the immunoglobulins of patients recovering from *Y. enterocolitica* infections react with the human thyroid-stimulating hormone receptor. However, a link between *Y. enterocolitica* infection and the subsequent development of autoimmune thyroiditis has not been convincingly demonstrated.

MANIFESTATIONS

The principal clinical manifestations of *Y. enterocolitica* infection are enteritis, enterocolitis, mesenteric adenitis, and terminal ileitis. Less common manifestations include exudative pharyngitis, septicemia, metastatic focal infections, reactive polyarthritis, and erythema nodosum. When age groups are combined, the most common presentation of *Y. enterocolitica* infection is acute diarrhea from enteritis or enterocolitis. Low-grade fever and cramping abdominal pain occur in most cases, nausea and vomiting in 15 to 40%, hematochezia in up to 30%, and a generalized maculopapular skin rash in a few cases. Diarrhea persists for an average of 2 weeks (range, 1 day to many months), during which the frequency of bowel movements diminishes. Uncommonly, enteritis or enterocolitis can be complicated by severe abdominal pain and high fever. Rare (and sometimes fatal) complications include diffuse inflammation, ulceration, hemorrhage, and necrosis of the small bowel and colon; intestinal perforation; peritonitis; ascending cholangitis; mesenteric vein thrombosis; diverticulitis; toxic megacolon; and ileocecal intussusception.

The syndrome of mesenteric adenitis and terminal ileitis without diarrhea is easily confused with appendicitis. Low-grade fever and right lower quadrant pain, tenderness, guarding, and rebound tenderness are common. During six recognized common-source outbreaks in the United States, 10% of 444 patients with symptomatic undiagnosed *Y. enterocolitica* infections underwent laparotomy for suspected appendicitis; surgical incisions became infected with *Y. enterocolitica* in a few of these cases.

Acute pharyngitis and pharyngotonsillitis, with or without cervical adenitis or intestinal illness, are less common but potentially lethal manifestations of *Y. enterocolitica* infection, particularly in adults. *Y. enterocolitica* septicemia generally presents as a severe illness with fever and leukocytosis, often with abdominal pain and jaundice and without localized signs of infection. Metastatic focal *Y. enterocolitica* infections can occur with or without clinically apparent bacteremia and can affect almost any organ system. Examples include abscess formation (e.g., in liver, spleen, kidney, lung, skeletal muscle, lymph node, or cutaneous tissue), osteomyelitis, meningitis, peritonitis, urinary tract infection, pneumonia, empyema, endocarditis, pericarditis, mycotic aneurysm, septic arthritis, suppurative conjunctivitis, panophthalmitis, Parinaud's oculoglandular syndrome, and cutaneous pustules or bullae.

In Scandinavia, the incidence of reactive arthritis following *Y. enterocolitica* infection among adults is estimated to be at least 10%. About 80% of these patients have preceding symptoms such as fever, diarrhea, or abdominal pain. Typically, these symptoms precede the arthritis by 1 week and are of short duration. The most commonly affected joints are the knees and ankles, but other joints can be involved. Typically, multiple (two to eight) joints become involved sequentially and asymmetrically over a period of a few days to 2 weeks, after which no additional joints are affected. Monoarticular arthritis occurs less commonly. In two-thirds of cases, the acute arthritis remits spontaneously within 1 to 3 months. Chronic joint disease is documented in a minority of cases. A few HLA-B27-positive patients with *Y. enterocolitica*-induced arthritis have subsequent ankylosing spondylitis, but this development is best explained by the fact that HLA-B27 is a major risk factor for each of these diseases. Mild, self-limited myocarditis accompanies about 10% of cases of *Yersinia*-induced arthritis

and can occur independently. Typical manifestations include cardiac murmurs and transient electrocardiographic abnormalities, such as prolongation of the PR interval and nonspecific ST-segment and T-wave changes. The syndrome of *Yersinia*-induced arthritis and carditis can be confused with acute rheumatic fever. In Scandinavia, erythema nodosum occurs in 15 to 20% of patients with yersiniosis, usually within a few days to 3 weeks after the onset of intestinal illness. Lesions typically are located on the lower extremities and resolve within 1 month. Less commonly reported nonsuppurative sequelae of *Y. enterocolitica* infections include reactive uveitis, iritis, conjunctivitis, urethritis, and glomerulonephritis. The complete triad of Reiter's syndrome (arthritis, conjunctivitis, and urethritis) is seen in 5 to 10% of patients with *Yersinia*-induced arthritis.

The most common clinical presentation of *Y. pseudotuberculosis* infection is fever and abdominal pain caused by mesenteric adenitis; diarrheal illness is less common than in *Y. enterocolitica* infection. Systemic manifestations, including septicemia, focal infections, reactive arthritis, and erythema nodosum, are generally similar to those associated with *Y. enterocolitica* infection. In addition, *Y. pseudotuberculosis* has been associated with a scarlet fever-like syndrome, acute interstitial nephritis, and hemolytic-uremic syndrome.

LABORATORY FINDINGS AND DIAGNOSIS

Results of routine laboratory tests in most patients with yersiniosis are nonspecific. Leukocyte counts are usually normal or slightly elevated, often with a modest left shift. Standard microbiologic methods are sufficient to isolate *Y. enterocolitica* and *Y. pseudotuberculosis* from otherwise-sterile sites, such as blood, [CSF](#), lymph node tissue, and peritoneal fluid, and from abscesses. Isolation of these organisms from feces is impeded by their slow growth and the overgrowth of normal fecal flora on culture media routinely used to select for enteric bacteria. When routine enteric media are used, the yield of yersinial isolates from feces is increased by incubation at 22 to 25°C for 48 h. The yield from feces and other grossly contaminated specimens can be further increased by the use of *Yersinia*-selective cefsulodin-Irgasan-novobiocin (CIN) agar and by cold enrichment (i.e., inoculation of feces into buffered saline and incubation at 4°C for 2 to 4 weeks, with periodic plating onto enteric media). Because bacteriologic procedures designed to isolate yersiniae from feces are not considered cost-effective, many laboratories undertake them by special request only.

The results of serologic tests can be used to support a diagnosis of yersiniosis. Agglutination tests or [ELISAs](#) are used most commonly; immunoblotting has also been used. The existence of multiple serotypes makes routine serologic tests laborious; thus these tests are generally conducted only in research laboratories or large commercial laboratories. Since these tests are experimental and are neither standardized nor well validated, and since some strains of *Yersinia* cross-react with other bacteria (e.g., *Brucella*, *Salmonella*, and *Vibrio*) and with serum from some patients with thyroiditis, results should be interpreted with caution. In typical uncomplicated cases of yersiniosis, agglutinin titers begin to rise within the first week of illness, peak in the second week, and then gradually diminish and return to normal within 3 to 6 months, although agglutinating antibody may remain detectable for several years in some cases. Because an initial serum specimen is often collected a week or more after the onset of illness,

when agglutinin titers are already high, it is usually impossible to document a fourfold or greater rise in titer between paired specimens (although a fourfold or greater fall in titer may be found). Immunohistologic techniques and polymerase chain reaction tests to detect yersinial antigens and DNA, respectively, in clinical specimens are experimental at this time.

In patients with *Yersinia*-induced reactive arthritis, synovial fluid is sterile and the leukocyte count ranges from a few hundred to 60,000/uL, with a majority of [PMNs](#). The erythrocyte sedimentation rate is often >100 mm/h. Rheumatoid factor and antinuclear antibodies are usually absent. The diagnosis of *Yersinia*-induced reactive arthritis or other nonsuppurative inflammatory sequelae can be difficult, especially when triggering infections are asymptomatic or clinically mild or occur several weeks before the diagnosis is attempted. Because the isolation of a pathogenic *Yersinia* strain from feces is the most specific diagnostic test in such cases, it should be attempted. Since culture is of limited sensitivity in this clinical setting, a high index of suspicion and positive results of serologic tests for *Y. enterocolitica* or *Y. pseudotuberculosis* are usually required for diagnosis.

TREATMENT

The effectiveness of antimicrobial agents in the treatment of yersinial enteritis, enterocolitis, mesenteric adenitis, or terminal ileitis has not been established. These conditions are usually self-limited, and their treatment is symptom-based and supportive. In uncomplicated cases, diarrhea should be treated with fluid and electrolyte replacement, with the route of delivery dependent on clinical severity. Enteric precautions are advisable for patients hospitalized with yersinial diarrhea. In general, antimicrobial treatment should be reserved for patients with septicemia, metastatic focal infections, or immunosuppression and enterocolitis. Controlled clinical comparisons of antimicrobial agents in the treatment of severe cases of yersiniosis have not yet been conducted. In such cases, drug selection should ultimately be guided by clinical response and bacterial sensitivity patterns. Clinical isolates of *Y. enterocolitica* and *Y. pseudotuberculosis* are usually susceptible in vitro to aminoglycosides, third-generation cephalosporins, chloramphenicol, quinolones, tetracyclines, and trimethoprim-sulfamethoxazole. In laboratory animals infected with enteropathogenic yersiniae, the fluoroquinolones have exerted the strongest bactericidal effects in vivo; clinical experience with these drugs against these pathogens in humans is promising but limited. Because they produce β -lactamases, isolates typically are resistant to penicillin, ampicillin, carbenicillin, and first-generation and most second-generation cephalosporins. Optimal dosages and durations of therapy have not been established. Mortality from *Y. enterocolitica* septicemia currently is ~10% despite treatment. Focal extraintestinal infections may require at least 3 weeks of therapy. No role for antimicrobial agents in the management of the nonsuppurative inflammatory manifestations of yersiniosis has been established. Patients with reactive arthritis may benefit from treatment with nonsteroidal anti-inflammatory drugs, intraarticular steroid injections, and physical therapy.

PREVENTION AND CONTROL

The importance of safe food-handling and food-preparation practices in the prevention

of yersiniosis cannot be overemphasized. Caution is particularly warranted in the case of pork and other animal products. The consumption of raw or undercooked meats, especially pork, should be avoided. Increased efforts to prevent the spread of enteric pathogens in household, pet-care, day-care, and hospital settings and in the food industry would be likely to decrease the incidence of yersiniosis. Current regulations of the U.S. Food and Drug Administration require visual inspection of packed red cell units before transfusion, with the discarding of units in which bacterial contamination is suspected on the basis of darkening (reflecting decreased oxygen saturation and hemolysis). Since the risk is minimal, more specific measures to further decrease the likelihood of transfusion of *Y. enterocolitica*-contaminated blood products (e.g., limiting the period for which red cells can be stored before transfusion) are not considered cost-effective.

Yersiniosis is not routinely reportable to public health authorities in most jurisdictions. However, clinicians who suspect a common-source outbreak (e.g., because they have documented a familial case cluster or have diagnosed the disease in several apparently unrelated patients over a short period) or some other public health threat (e.g., because they have found occult *Y. enterocolitica* bacteremia in a recent blood donor) should consult promptly with local public health officials.

(Bibliography omitted in Palm version)

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163. BARTONELLA INFECTIONS, INCLUDING CAT-SCRATCH DISEASE - Lucy Stuart Tompkins

Bartonella spp., including *B. bacilliformis*, *B. henselae*, *B. quintana*, and *B. clarridgeiae*, are tiny gram-negative bacilli that can adhere to and invade mammalian cells, including endothelial cells and erythrocytes. Previously classified as *Rochalimaea* spp. within the rickettsia group, *Bartonella* spp. have now been removed from the order Rickettsiales on the grounds that they are not obligate intracellular parasites. These agents cause a wide spectrum of clinical illnesses, including trench fever, cat-scratch disease (CSD), bacillary angiomatosis, endocarditis, Oroya fever, and verruga peruana ([Fig. 163-CD1](#)). The pathologic manifestations of *Bartonella* disease vary with the immune status of the host.

OROYA FEVER AND VERRUGA PERUANA

DEFINITION AND ETIOLOGY

Oroya fever and verruga peruana are caused by *B. bacilliformis*. Oroya fever is characterized by fever, profound anemia, and -- unless antibiotic treatment is given -- high mortality. The lesions referred to as verruga peruana may develop during the convalescent phase of Oroya fever or during chronic infection with *B. bacilliformis*. In 1885 Daniel Carrion, a Peruvian medical student, inoculated himself with blood from a patient with verruga peruana and subsequently died of Oroya fever, thus proving that both diseases are caused by a single agent.

EPIDEMIOLOGY

Infection with *B. bacilliformis* follows the bite of the sandfly vector *Phlebotomus*, an insect found in the river valleys of the Andes Mountains at altitudes of 600 to 2500 m. Oroya fever develops in nonimmune individuals who are not residents of the endemic region, whereas verruga peruana occurs in persons who apparently have been exposed in the past, including those who have recently had Oroya fever. The infection has not been acquired in the United States.

PATHOLOGY

During initial infection in the nonimmune host, *B. bacilliformis* cells adhere to erythrocytes and produce indentations in the cell membrane; the bacteria subsequently enter the erythrocytes and cause persistent deformation of the cytoskeleton. The parasitized erythrocytes are ultimately phagocytosed and destroyed. Although the life span of infected erythrocytes is markedly shortened, not all of this change can be attributed to the mechanical fragility induced by the internalization of bacteria. Decreased bone marrow erythropoiesis also contributes to anemia.

CLINICAL MANIFESTATIONS

The onset of symptoms in Oroya fever may be either insidious or abrupt, after an incubation period of approximately 3 weeks. The subacute presentation may include low-grade fever, malaise, headache, and anorexia. Sudden-onset disease commences

with high fever, chills, diaphoresis, headaches, and changes in mental status. These manifestations are followed by the sudden development of profound anemia, which is due to a marked decrease in erythrocyte numbers and is associated with macrocytic changes, poikilocytosis, Howell-Jolly bodies, nucleated erythrocytes, and immature myeloid cells. The leukocyte differential usually shifts to the left, although the total leukocyte count may be normal. The erythrocyte count may fall to extremely low levels. In eosin/thiazine-stained peripheral-blood smears, numerous microorganisms can be seen adhering to most erythrocytes.

During the acute phase, muscle and joint pain and headache may be severe; central nervous system changes include insomnia, delirium, and a decreased level of consciousness. Thrombocytopenic purpura may develop. If the patient survives, a convalescent phase ensues, characterized by the sudden disappearance of bacteria from blood smears, declining fever, and an increase in the erythrocyte count. Although much of the mortality associated with Oroya fever is due to profound anemia and toxicity, secondary bacterial infections (including salmonellosis and other enteric infections, malaria, and tuberculosis) are often an important contributing factor.

After convalescence from acute Oroya fever, verrugas may develop. These red or purple cutaneous lesions may be either tiny and sessile or large, pedunculated, and nodular. They bear a marked resemblance to the lesions of bacillary angiomatosis and to Kaposi's sarcoma.

DIAGNOSIS

During acute infection, bacteria can be cultured from the blood on agar containing rabbit blood, with incubation at 28°C. The hallmark of verruga peruana is the formation of new blood vessels (angiogenesis) at the sites of bacterial replication.

TREATMENT

Oroya fever responds to a variety of antimicrobial agents, including chloramphenicol, tetracyclines, penicillin, and streptomycin. Chloramphenicol is used most often because of its efficacy against most *Salmonella* infections (as salmonellosis may develop intercurrently). Verruga peruana may respond similarly; however, failure to respond to therapy and relapse are common and require the reinstitution of prolonged therapy.

BACILLARY ANGIOMATOSIS

DEFINITION AND ETIOLOGY

Bacillary angiomatosis was initially described as a condition occurring primarily in patients with AIDS and characterized by vascular cutaneous lesions resembling Kaposi's sarcoma. The disease can disseminate to involve virtually any organ system. Immunocompromised individuals, especially those infected with HIV, are at particularly high risk for bacillary angiomatosis, although in rare instances the patient is not obviously immunosuppressed. Both *B. henselae* and *B. quintana* (the infectious agent initially associated with trench fever) produce bacillary angiomatosis in persons with immunodeficiency.

EPIDEMIOLOGY

Acquisition of *B. henselae* has been significantly associated with exposure to young cats infested with fleas (*Ctenocephalides felis*). Because a high percentage of cats are seropositive, it has been suggested that patients with HIV infection avoid exposure to these animals. The finding that a large proportion of cats with fleas have persistent asymptomatic *B. henselae* bacteremia suggests that the domestic cat is the animal reservoir of this microorganism. The flea may serve as a transmitting vector in the cross-infection of cats, but its role in human infection is not clear. Tick-associated cases of *B. henselae* bacteremia have been reported in healthy immunocompetent individuals.

Person-to-person transmission of *B. quintana* by the human body louse (*Pediculus humanis corporis*) was documented during World War I under conditions of poor personal hygiene and sanitation. Although lice are suspected of transmission, the reservoir of *B. quintana* has not been identified.

A case-control study revealed that *B. henselae* and *B. quintana* differ significantly in terms of epidemiologic risk factors. All cases of *B. henselae* infection were associated with exposure to cats and their fleas and occurred sporadically, whereas the cases of *B. quintana* infection occurred in clusters and were associated with low socioeconomic status, homelessness, and exposure to body lice. Direct transmission of *B. henselae* from cats to their owners, presumably through cutaneous trauma, was supported by the matching DNA fingerprint patterns of isolates from the two sources.

MICROBIOLOGY

B. henselae can be demonstrated in tissue by Warthin-Starry staining. Clumps and clusters of pleomorphic bacilli appear as purple deposits in tissue stained with hematoxylin and eosin. Although the bacteria may be difficult to cultivate in the laboratory, they can eventually be isolated from cultures of blood and of material from other sites. Colonies develop after prolonged incubation (1 to 4 weeks) on blood-containing media and pit the agar; bacterial cells are gram-negative. *B. quintana* grows as a smooth, nonpitting colony on solid agar after prolonged incubation.

Classification of *B. henselae* was first accomplished when molecular techniques were used to analyze bacterial ribosomal genes extracted from tissue samples. Definitive identification of *Bartonella* spp. is based on sequence analysis of 16S ribosomal DNA.

PATHOGENESIS AND PATHOLOGY

Bacillary angiomatosis is characterized by a lobular proliferation of new blood vessels (angiogenesis) and a neutrophilic inflammatory response to myriad bacilli located within collagen-rich microscopic and macroscopic nodules. The endothelial cells lining the vascular spaces have a typical epithelioid appearance, and the lesions may resemble Kaposi's sarcoma histopathologically, although the characteristic spindle cell of the latter disease is usually absent. The bacterial and eukaryotic host factors that elicit the pathologic response are unknown.

CLINICAL MANIFESTATIONS

The skin lesions of bacillary angiomatosis (also called *epithelioid angiomatosis*) are vascular nodules, papules, or tumors ([Fig. 163-CD2](#)) that range from tiny lesions resembling cherry angiomas or pyogenic granulomas to large, pedunculated, exophytic masses ([Fig. 163-1](#)). Characteristically, the lesions are red or purple, resembling Kaposi's sarcoma; they may be surrounded by an epithelial collarette, may be located anywhere on the skin, and may involve mucous membranes. The overlying epidermis may be focally ulcerated, and the underlying bone may be invaded and destroyed.

Dissemination of *B. henselae* infection occurs primarily in patients with cellular immune defects. Clinical manifestations accompanying dissemination are often nonspecific and include persistent fever, abdominal pain, weight loss, and malaise. Although the liver, spleen, bone marrow, and lymph nodes are primarily affected, HIV-infected patients may also develop central nervous system abnormalities (including psychiatric disorders and brain lesions), which are responsive to antibiotic therapy. Skin lesions usually are not evident in disseminated infection. Involvement of the liver or spleen may produce bacillary peliosis hepatis. Patients with the latter condition may report localized pain on palpation of the abdomen. Nodular lesions of variable size can be demonstrated by computed tomography or magnetic resonance imaging, with or without contrast agents.

In a case-control study of bacillary angiomatosis (see "Epidemiology" above), only *B. henselae* was associated with hepatosplenic disease (peliosis hepatis) and displayed a predilection for the lymph nodes. *B. quintana*, in contrast, was associated with osseous and subcutaneous infection.

DIAGNOSIS

The diagnosis of bacillary angiomatosis is based primarily on the typical histopathologic findings of angiomas in association with clumps of tiny bacilli revealed by Warthin-Starry silver stain. Infection due to *B. henselae* can also be established by culture or by identification of specific DNA sequences. *B. henselae* is most easily isolated from blood through a lysis-centrifugation system. Colonies may be detected on blood-containing agar (rabbit blood is preferred) incubated with 5 to 10% CO₂ at 37°C for 2 to 4 weeks. *B. quintana* may be isolated from BACTEC (Becton Dickinson, Sparks, MD) aerobic bottles containing resin. Isolation from skin lesions and other tissues is more difficult but should be attempted when feasible. Initial reports suggested that cocultivation with endothelial cell monolayers was necessary; however, isolation by direct plating onto freshly prepared agar media has also been successful. Bacilli picked from new colonies but not subcultured may not stain, even with acridine orange; they stain weakly with safranin. Identification of *B. henselae* and *B. quintana* is based primarily on cellular fatty-acid analysis and polymerase chain reaction (PCR)-based restriction fragment length polymorphism analysis. Definitive identification of *Bartonella* spp. depends on DNA sequence analysis of 16S ribosomal RNA genes. The diagnosis of [CSD](#) (see next section) can be made by specific serologic testing that detects *B. henselae*-specific antibodies, but the sensitivity and specificity of this method in patients with cutaneous and disseminated bacillary angiomatosis have not been determined.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of cutaneous bacillary angiomatosis includes Kaposi's sarcoma, angiomas, and pyogenic granulomas. These conditions can be distinguished by histopathologic examination of biopsied material.

Cutaneous bacillary angiomatosis caused by *B. henselae* or *B. quintana* resembles verruga peruana, which is not seen outside of South America. In patients with AIDS, Kaposi's sarcoma lesions and bacillary angiomatosis may coexist.

TREATMENT

Cutaneous lesions have been treated with a wide variety of antimicrobial drugs, including macrolides, tetracyclines, and antituberculous agents; *B. henselae* is susceptible to most antibiotics in vitro. Erythromycin (2 g/d), given orally for 3 weeks, is usually effective, as are newer macrolides; however, relapse may require prolonged therapy (3 weeks to 2 months) with an antibiotic that reaches an intracellular compartment, such as a macrolide or doxycycline (200 mg/d). Patients with peliosis hepatis should be treated with intravenous antibiotics, and those with disseminated disease or bacteremia should be treated with a prolonged course (3 weeks to 2 months) of systemic antibiotic, such as a macrolide (e.g., erythromycin, 2 g/d). In a case-control study of bacillary angiomatosis, treatment with a macrolide was associated with a therapeutic response and sterile tissue samples and may have been protective, whereas treatment with trimethoprim-sulfamethoxazole, ciprofloxacin, penicillins, or cephalosporins had no protective effect. Cutaneous lesions may or may not regress spontaneously, perhaps depending on the status of the host's immunity. The safety of ciprofloxacin in pregnant or lactating women has not been established. No antimicrobial has been studied prospectively, and information on efficacy comes only from case reports.

CAT-SCRATCH DISEASE

DEFINITION AND ETIOLOGY

Typical [CSD](#) is manifested by painful regional lymphadenopathy persisting for several weeks or months after a cat scratch. Occasionally, infection may disseminate and produce more generalized lymphadenopathy and systemic manifestations, which may be confused with the manifestations of lymphoma. *B. henselae* is the causative agent of CSD. There is no evidence that *B. quintana* causes CSD, and this microbe is not carried by cats. The role of *Afipia felis* (originally proposed as the agent of CSD) is unclear inasmuch as only a few cases are associated with its isolation. *B. henselae* remains the predominant species causing typical CSD. Several reports suggest that *B. clarridgeiae* may also cause feline lymphadenopathy.

EPIDEMIOLOGY

Approximately 60% of cases of [CSD](#) in the United States occur in children. Exposure to bacteremic young cats that either are flea-infested or have been in contact with another cat carrying fleas poses a significant risk of infection. Most infections are caused by a scratch and only rare cases by a bite or by licking. Most cases occur in the warmer

months, when fleas are active. Regions of the United States where fleas are endemic have higher rates of infection. The flea may serve to transmit infection between cats; it is not known whether humans can be infected through the bite of an infected flea.

CLINICAL MANIFESTATIONS

A localized papule, progressing to a pustule that often crusts over, develops 3 to 5 days after a cat scratch ([Fig. 163-CD3](#)). Tender regional lymphadenopathy develops within 1 to 2 weeks after inoculation; by this time, the papule may have healed spontaneously. Scratches are most often sustained on the hands or face, producing epitrochlear, axillary, pectoral, and cervical lymph node involvement. The involved nodes occasionally become suppurative; bacterial superinfection with staphylococci or other cutaneous pathogens may develop. Although most patients do not have fever, systemic symptoms are frequent and include malaise, anorexia, and weight loss. Without treatment, lymphadenopathy persists for weeks or even months and may be confused with lymphatic malignancy. Other manifestations in apparently immunocompetent patients include encephalitis, seizures and coma (especially in children), meningitis, transverse myelitis, granulomatous hepatitis and splenitis, osteomyelitis, and disseminated infection. Conjunctival inoculation may cause Parinaud's oculoglandular syndrome, with conjunctivitis and preauricular lymphadenopathy.

PATHOLOGY

The histopathologic hallmark of [CSD](#) is granulomatous inflammation with stellate necrosis but no evidence of angiogenesis. Thus, infection by *B. henselae* can produce two entirely different pathologic reactions, depending on the immune status of the host: CSD or bacillary angiomatosis.

DIAGNOSIS

[CSD](#) should be suspected if the patient has a history of exposure to cats and develops lymphadenopathy and a skin lesion. The diagnosis can be confirmed by pathologic examination of the involved nodes. Tiny bacilli in clusters can sometimes be seen in biopsy samples stained with Warthin-Starry silver. The CSD skin test, in which lymph node material obtained from patients with CSD serves as an antigen, is no longer used for diagnosis because of concerns about the transmission of viral agents. A specific serologic test has been developed and may produce a positive result in 70 to 90% of patients with intact immunity. The identification of *B. henselae* 16S ribosomal RNA genes in biopsy material by PCR amplification with specific oligonucleotide primers can also be diagnostically useful; however, these methods are not yet commercially available. Cultures of lymph nodes, cerebrospinal fluid, or other tissues are rarely positive.

TREATMENT

Although [CSD](#) is generally self-limited, tender regional lymphadenopathy and systemic symptoms may be debilitating. Patients with encephalitis or other serious manifestations should be treated with antibiotics. A randomized, double-blind, placebo-controlled trial demonstrated significant clinical benefit of treatment with oral azithromycin for 5 days in

cases of typical CSD (regimen for adults weighing >100 lb: one dose of 500 mg on day 1, 250 mg on days 2 through 5). Several reports suggest that aminoglycoside treatment (e.g., intravenous gentamicin at standard doses calculated to result in therapeutic levels) is effective in patients with encephalitis and other systemic infections. The oral agents that appear to be useful are those that also are most effective for the treatment of bacillary angiomatosis; they include ciprofloxacin, doxycycline, and azithromycin. Unlike bacillary angiomatosis, CSD responds to treatment with ciprofloxacin. The necessary duration of therapy is variable.

TRENCH FEVER

DEFINITION AND ETIOLOGY

Trench fever was first described as a debilitating febrile illness associated with prolonged *B. quintana* bacteremia in soldiers fighting in Europe during World War I. Although not usually fatal, the illness accounted for substantial morbidity. In recent years, trench fever has reemerged in the United States and has been caused by either *B. henselae* -- the agent of [CSD](#) and bacillary angiomatosis -- or *B. quintana*.

EPIDEMIOLOGY

Although trench fever was thought to have disappeared from the United States, recent cases have been diagnosed in homeless persons (*B. quintana*) and in persons bitten by ticks (*B. henselae*). During World War I, trench fever was transmitted from person to person by the human body louse. Transmission by ectoparasites is suspected in the recent cases of *B. quintana* infection but has not been firmly documented. Patients with trench fever have apparently normal immune defenses.

CLINICAL MANIFESTATIONS

Trench fever is characterized by the sudden onset of headache, aseptic meningitis, persistent fever (which can be high-grade and is commonly paroxysmal), malaise, weight loss, and other nonspecific symptoms. Severe musculoskeletal pain is more common among immunocompetent than among immunocompromised patients. Bacteremia can persist for days or weeks, and relapses have followed short courses of antibiotic therapy. Localized findings are uncommon.

DIAGNOSIS

Trench fever is diagnosed by the finding of sustained bacteremia. *B. henselae* and *B. quintana* grow slowly. Colonies develop on rabbit blood agar after 1 to 4 weeks of incubation under conditions of increased CO₂. Serologic tests for this disease have not yet been standardized.

TREATMENT

A prolonged course (4 weeks) of antimicrobial therapy may be required. Agents that can cross the mammalian cell membrane are most effective, including erythromycin (2 g/d) or azithromycin (500 mg/d). Data on the efficacy of these agents come from a limited

number of case reports.

OTHER *BARTONELLA* INFECTIONS, INCLUDING CULTURE-NEGATIVE ENDOCARDITIS

The application of molecular methods to the detection of microorganisms that are difficult to cultivate in the laboratory has revealed new *Bartonella* spp. and has established *Bartonella* spp. as a cause of endocarditis cases previously classified as being of unknown etiology. *B. quintana* is the most frequently isolated *Bartonella* species in these cases. Two new species, *B. elizabethae* and *B. clarridgeiae*, as well as *B. henselae* have also been identified as agents of subacute and chronic endocarditis.

The diagnosis of *Bartonella* endocarditis is confirmed by blood cultures. Specific antibodies are produced; however, *B. quintana* infection may produce antibodies that cross-react with *Chlamydia pneumoniae*.

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164. DONOVANOSIS - Gavin Hart

Donovanosis is a chronic, progressively destructive bacterial infection of the genital region that is generally regarded as sexually transmitted. The disease has been known by many other names, the most common of which are granuloma inguinale and granuloma venereum.

ETIOLOGY

Donovanosis is caused by *Calymmatobacterium granulomatis*, an intracellular, gram-negative, pleomorphic, encapsulated (when mature) bacterium measuring 1.5 by 0.7 μm . *C. granulomatis* shares many morphologic and serologic characteristics and >99% homology at the nucleotide level with *Klebsiella* species that are pathogenic to humans. Polymerase chain reaction amplification of the *phoE* gene shows it to be closely related to that in *Klebsiella pneumoniae*, *K. rhinoscleromatis*, and *K. ozaenae*. Electron microscopy shows typical gram-negative morphology and a large capsule but no flagella. Filiform or vesicular protrusions occur on a corrugated cell wall.

EPIDEMIOLOGY

Donovanosis is endemic among Aborigines in central Australia as well as in Papua New Guinea, southeastern India, southern Africa, and the Caribbean and adjacent areas of South America. In the first half of the twentieth century, the disease was endemic in parts of the United States (with an estimated 5000 to 10,000 cases in 1947); small epidemics still occur in this country and in other developed countries. Over 70% of cases involve persons 20 to 40 years of age. The infection is predominantly sexually transmitted, but extragenital skin lesions can follow transmission from concurrent genital lesions via the fingers or through other nonsexual contact, and autoinoculation may produce new lesions from contact with adjacent skin ("kissing" lesions). Infants born to infected mothers have acquired infection at birth.

The classification of donovanosis as a sexually transmitted disease (STD) has been disputed because of cases in young children and occasionally in sexually inactive individuals, transmission by direct body contact and via inanimate intermediaries, and the low and variable prevalence of donovanosis among sexual partners (0.4 to 52%). The dominance of sexual transmission is suggested by the combined factors of lesions predominantly affecting the genitalia, the highest prevalence among persons in age and socioeconomic groups that are most often affected by STDs, and the predictable occurrence of disease in visitors to areas of endemicity following sexual exposure.

CLINICAL MANIFESTATIONS

The incubation period is usually 1 to 4 weeks but may extend to 1 year. Skin lesions have been detected in infants 6 weeks to 6 months after birth. The disease begins as one or more subcutaneous nodules that erode through the skin to produce clean, granulomatous, sharply defined, usually painless lesions ([Fig. 164-1](#)). These lesions, which bleed readily on contact, slowly enlarge. The genitalia are involved in 90% of cases, the inguinal region in 10%, and the anal region in 5 to 10%. Genital swelling, particularly of the labia, is a common feature and occasionally progresses to

pseudoelephantiasis. Phimosis and paraphimosis are common local complications, and progressive erosion of affected tissues may completely destroy the penis or other organs. Less common clinical variants include a hypertrophic form (cauliflower- or wartlike lesions), a necrotic form (destructive lesions with foul-smelling exudate, often resembling amebiasis), and a sclerotic or cicatricial form, which has a dry base with extensive scar tissue ([Fig. 132-CD3](#)).

Extragenital lesions occur in at least 6% of cases. Oral donovanosis, the most common extragenital manifestation, presents as pain or bleeding in the mouth, lesions on the lips, or extensive swelling of the gums and palate. Donovanosis may affect most bones, and sometimes many bones are affected at the same time; the tibia is involved in over 50% of such cases. Bony lesions are associated with constitutional symptoms (weight loss, fever, night sweats, and malaise) and are usually found in women. More than 50% of women have primary lesions on the cervix. Prompt pelvic examinations and early diagnosis are likely to substantially decrease the morbidity and mortality (a likely outcome in misdiagnosed spinal lesions) associated with extragenital donovanosis in women.

DIAGNOSIS

Laboratory Diagnosis The preferred method involves demonstration of typical intracellular Donovan bodies within large mononuclear cells visualized in smears prepared from lesions or biopsy specimens. With typical beefy lesions, a small piece of tissue is removed with forceps and scalpel, and a crush impression of the deep surface is made on a glass slide. The smear is air-dried, heat-fixed, and stained with Giemsa, Leishman's, or Wright's stain. For dry, flat, or necrotic lesions, a punch-biopsy specimen should be obtained from the advancing edge. This specimen can be used to prepare a smear or embedded for histologic examination (with a silver stain). Histologic examination shows epithelial proliferation, often simulating neoplasia, with a heavy inflammatory infiltrate of plasma cells, some neutrophils, and few if any lymphocytes. The large mononuclear cells are 25 to 90 μm in diameter, with a vesicular or pyknotic nucleus. Up to 20 intracytoplasmic vacuoles contain pleomorphic Donovan bodies in either young uncapsulated forms (which often resemble closed safety pins) or mature capsulated forms. *C. granulomatis* has never been grown on artificial solid media but has been cultured in chicken embryonic yolk sacs, on human monocytes, and on human epithelial (HEp-2) cells. A sensitive and specific serologic test, based on indirect immunofluorescence, has been developed.

Differential Diagnosis Condylomata lata of secondary syphilis may be confused with donovanosis; however, these lesions usually appear as white or pale moist plaques in the anogenital area, whereas the lesions of donovanosis are usually bright red. Syphilis and donovanosis frequently coexist because syphilis is usually highly prevalent in areas where donovanosis is endemic; thus positive syphilis serology does not exclude a diagnosis of donovanosis. Condylomata lata subside within 1 week of treatment with benzathine penicillin (2.4 million units), whereas donovanosis lesions remain unchanged.

The necrotic form of donovanosis may resemble squamous cell carcinoma; likewise, cervical and vulvar lesions may closely resemble carcinoma. Penile amebiasis may

resemble necrotic donovanosis but usually follows anal intercourse and is much less common than donovanosis in areas where the latter is endemic. Atypical clinical variants of chancroid, referred to as *pseudogranuloma inguinale*, have been described in patients seen at clinics in Atlanta. Disseminated donovanosis lesions of bones, particularly in the spine, can mimic tuberculosis. Lesions that produce draining sinuses near the jaw may simulate actinomycosis. The histologic findings of donovanosis must be distinguished from those of rhinoscleroma, leishmaniasis, and histoplasmosis. Genital ulcers are a risk factor for HIV acquisition in developing countries, and patients with donovanosis should be tested for HIV infection.

TREATMENT

[Table 164-1](#) shows the most effective regimens for treating donovanosis. Doxycycline is the first choice for therapy in developed countries. Erythromycin provides an effective option for pregnant patients, and azithromycin is an effective alternative that is more convenient to administer. Extensive lesions have been cured with oral azithromycin at a dosage of 500 mg/d, but the more convenient dose of 1 g weekly is also effective. Although chloramphenicol is the drug of choice in some developing countries, it is unlikely to be acceptable in developed countries because of bone marrow toxicity. Penicillin is not effective for treating donovanosis. Patients should be examined weekly, and therapy should be continued until lesions have healed (3 to 5 weeks, except in severe cases). If antibiotic therapy is stopped earlier, lesions often continue to heal, but the relapse rate is higher. If the lesions are unchanged after 2 weeks of treatment, an alternative antibiotic regimen should be used.

The treatment regimens just described are usually adequate in HIV-infected patients without immunosuppression, but an increasing failure rate has been reported in immunosuppressed patients, for whom daily administration of azithromycin is recommended if other regimens fail to elicit a response.

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SECTION 7 -MISCELLANEOUS BACTERIAL INFECTIONS

165. NOCARDIOSIS - Gregory A. Filice

The term *nocardiosis* refers to invasive disease associated with members of the genus *Nocardia*. Of the several distinctive syndromes, pneumonia and disseminated disease are most common. Others include cellulitis, lymphocutaneous syndrome, actinomycetoma, and keratitis.

MICROBIOLOGY

Nocardiae are saprophytic aerobic actinomycetes that are common worldwide in soil, where they contribute to decay of organic matter. Nocardial taxonomy is complex and incompletely understood. Seven species have been associated with human disease: *N. asteroides*, *N. brasiliensis*, *N. otitidis-caviarum* (formerly *N. caviae*), *N. farcinica*, *N. nova*, *N. transvalensis*, and *N. pseudobrasiliensis*. *N. asteroides* is the species most commonly associated with invasive disease. *N. farcinica* is less common but tends to be virulent and prone to dissemination. The new species *N. pseudobrasiliensis* accounts for most cases of invasive disease previously attributed to *N. brasiliensis*. True *N. brasiliensis* isolates are usually associated with disease limited to the skin. *N. transvalensis* is generally associated with mycetoma or, in immunosuppressed persons, with pulmonary or systemic disease.

EPIDEMIOLOGY

Approximately 1000 cases of nocardial infection are diagnosed annually in the United States, 85% of them pulmonary and/or systemic. The disease is more common among adults and males. Outbreaks, which are rare, have been associated with contamination of the hospital environment, solutions, or drug injection equipment. Person-to-person spread is not well documented. There is no known seasonality.

The risk of pulmonary or disseminated disease is greater than usual among people with deficient cell-mediated immunity, especially that associated with lymphoma, transplantation, or AIDS. In persons with AIDS, nocardiosis usually presents at a CD4+lymphocyte concentration of <250/uL. Prophylaxis with sulfamethoxazole and trimethoprim appears to reduce the risk of nocardiosis in persons with AIDS or transplanted organs. Nocardiosis has also been associated with pulmonary alveolar proteinosis, tuberculosis and other mycobacterial diseases, and chronic granulomatous disease.

N. brasiliensis, *N. asteroides*, *N. otitidis-caviarum*, and *N. transvalensis* are associated with actinomycetoma. Cases occur mainly in tropical and subtropical regions, especially those of Mexico, Central and South America, Africa, and India. The most important risk factor is frequent contact with soil or vegetable matter.

PATHOLOGY AND PATHOGENESIS

Pneumonia and disseminated disease are both thought to follow inhalation of fragmented bacterial mycelia. The characteristic histologic feature of nocardiosis is an

abscess with extensive infiltration by neutrophils and prominent necrosis. Granulation tissue usually surrounds the lesions, but extensive fibrosis or encapsulation is uncommon. Actinomycetoma is characterized by suppurative inflammation with sinus tract formation. Granules -- microcolonies composed of dense masses of bacterial filaments extending radially from a central core -- are occasionally observed in histologic preparations. They are frequently found in discharges from lesions of actinomycetoma but almost never from lesions in other forms of nocardiosis.

Nocardiae have evolved a number of properties that enable them to survive within phagocytes, including neutralization of oxidants, prevention of phagosome-lysosome fusion, and prevention of phagosome acidification. Neutrophils phagocytose the organisms and limit their growth but do not kill them efficiently. Cell-mediated immunity is important for definitive control and elimination of nocardiae.

CLINICAL MANIFESTATIONS

Respiratory Tract Disease Pneumonia is by far the most common respiratory tract nocardial disease. Nocardial pneumonia is typically subacute; symptoms have usually been present for days or weeks at presentation. The onset may be more acute in immunosuppressed patients. Cough is prominent and produces small amounts of thick, purulent sputum that is not malodorous. Fever, anorexia, weight loss, and malaise are common; dyspnea, pleuritic pain, and hemoptysis are less common. Remissions and exacerbations over several weeks are frequent.

Roentgenographic patterns are variable ([Fig. 165-CD1](#)), but some characteristics are highly suggestive. Infiltrates vary in size and are typically of at least moderate density. Single or multiple nodules are common, sometimes suggesting metastatic tumors. Infiltrates and nodules tend to cavitate. Empyema is present in one-third of cases.

Nocardiosis may spread directly from the lungs to adjacent tissues. Pericarditis, mediastinitis, and the superior vena cava syndrome have all been reported. Spread through the chest wall is rare.

Nocardial laryngitis, tracheitis, and bronchitis are much less common than pneumonia. In the major airways, disease often presents as a nodular or granulomatous mass. A few cases of sinusitis have been reported.

Nocardiae are sometimes isolated from respiratory secretions of patients without apparent nocardial disease. Most of these patients have chronic pulmonary disease with abnormal airways or parenchyma.

Extrapulmonary Dissemination In half of all cases of pulmonary nocardiosis, disease appears outside the lungs. In one-fifth of cases of disseminated disease, lung disease is not apparent. The most common site of dissemination is the brain. Other common sites include the skin and supporting structures, kidneys, bone, and muscle, but almost any organ can be involved. Peritonitis and endocarditis have been reported. The typical manifestation of extrapulmonary dissemination is a subacute abscess. A minority of abscesses outside the lungs or central nervous system (CNS) form fistulae and discharge small amounts of pus. Nocardiae have been recovered from blood in a few

cases of pneumonia or disseminated disease.

In [CNS](#) infections, brain abscesses are usually supratentorial, are often multiloculated, and may be single or multiple ([Fig. 165-1](#), 165-CD2). Brain abscesses tend to burrow into the ventricles or extend out into the subarachnoid space. The symptoms and signs are somewhat more indolent than those of other types of bacterial brain abscess. Meningitis is uncommon and is usually due to spread from a nearby brain abscess. *Nocardiae* are not easily recovered from cerebrospinal fluid (CSF).

Disease Following Transcutaneous Inoculation Disease following transcutaneous nocardial inoculation usually takes one of three forms: cellulitis, lymphocutaneous syndrome, or actinomycetoma. Cellulitis generally begins 1 to 3 weeks after a recognized breach of the skin, often with soil contamination. Subacute cellulitis with pain, swelling, erythema, and warmth develops over days to weeks. The lesions are usually firm and nonfluctuant. Disease may progress to involve underlying muscle, tendon, bones, or joints. Dissemination is rare. *N. asteroides* is common in colder climates, while *N. brasiliensis* predominates in warmer climates.

In the lymphocutaneous syndrome, there is typically a pyodermatous lesion at the site of inoculation, with central ulceration and purulent or honey-colored drainage. Subcutaneous nodules often appear along lymphatics that drain the primary lesion. The lymphangitic form closely resembles lymphocutaneous sporotrichosis ([Chap. 208](#)). Most cases of the lymphocutaneous syndrome are associated with *N. brasiliensis*.

Actinomycetoma usually begins with a nodular swelling, sometimes at a site of local trauma. Lesions typically develop on the feet or hands but may involve the posterior part of the neck, the upper back, the head, and other sites. The nodule eventually breaks down and a fistula appears. This fistula is soon accompanied by others. The fistulas tend to come and go, with new ones forming as old ones disappear. The discharge is serous or purulent, may be bloody, and often contains 0.1- to 2-mm white granules consisting of masses of mycelia. The lesions spread slowly along fascial planes to involve adjacent areas of skin, subcutaneous tissue, and bone. Over months or years, there may be extensive deformation of the affected part. Lesions involving soft tissues are only mildly painful; those affecting bones or joints are more so. Systemic symptoms are absent or minimal. Infection rarely disseminates from actinomycetoma, and lesions on the hands and feet usually cause only local disability. Lesions on the head, neck, and trunk can invade locally to involve deep organs and result in severe disability or death.

Keratitis *Nocardia* spp., usually *N. asteroides*, are uncommon causes of subacute keratitis. The infection usually follows eye trauma. Nocardial infection of lacrimal glands has been reported. Disease involving deeper eye structures is usually a manifestation of dissemination.

DIAGNOSIS

The first step in diagnosis is examination of sputum or pus for crooked, branching, beaded, gram-positive filaments 1 μ m wide and up to 50 μ m long. Most nocardiae are acid-fast in direct smears if a weak acid is used for decolorization (e.g., in the modified Kinyoun, Ziehl-Neelsen, and Fite-Faraco methods) ([Fig. 165-CD3](#)). The organisms often

take up silver stains. Nocardiae grow relatively slowly; colonies may take up to 2 weeks to appear and may not develop their characteristic appearance for up to 4 weeks. Several blood culture systems support nocardial growth. Yield is enhanced when blood cultures are incubated aerobically for up to 4 weeks and when blind subcultures are performed. Nocardial growth is so different from that of more common pathogens that the laboratory should be alerted when nocardiosis is suspected to maximize the likelihood of isolation. Since nocardiae are among the few aerobic microorganisms that use paraffin as a carbon source, paraffin baiting can be useful in isolating the organisms from mixed cultures.

In cases of pneumonia, sputum smears are often negative. Unless the diagnosis can be made in these cases by sampling lesions in other, more accessible sites, bronchoscopy or lung aspiration is usually necessary. Transtracheal aspiration should be avoided, as it frequently leads to nocardial cellulitis in tissues around the puncture wound.

In patients with nocardial pneumonia, a careful history should be obtained and a thorough physical examination performed to evaluate the possibility of dissemination. Suggestive symptoms or signs should be pursued with further diagnostic tests. Computed tomography or magnetic resonance imaging of the head, with and without contrast material, should be undertaken if signs or symptoms suggest brain involvement. Many authorities recommend brain imaging in all cases of pulmonary or disseminated disease.

When clinically indicated, [CSF](#) or urine should be concentrated and then cultured. In actinomycetoma cases, granules should be sought in the discharge. Suspect particles should be washed in saline, examined microscopically, and cultured.

Isolation of nocardiae from sputum or blood occasionally represents colonization, transient infection, or contamination. In typical cases of respiratory tract colonization, Gram-stained specimens are negative and cultures are only intermittently positive. A positive sputum culture in an immunosuppressed patient usually reflects disease. When nocardiae are isolated from an immunocompetent patient without apparent nocardial disease, the patient should be observed carefully without treatment. A patient with a host-defense defect that increases the risk of nocardiosis should usually receive antimicrobial treatment.

Nocardia spp. are difficult to differentiate from one another with standard biochemical tests, and isolates from patients with systemic or severe disease should be sent to a reference laboratory for definitive identification and antimicrobial susceptibility testing. Susceptibility results, which help differentiate species, are of less certain clinical value but sometimes guide therapy in difficult cases.

In vitro, strains of *N. farcinica* differ from most in that they are usually resistant to cephalosporins and in one-fifth of cases are resistant to imipenem. *N. pseudobrasiliensis* strains often exhibit resistance to minocycline or amoxicillin/clavulanic acid and susceptibility to ciprofloxacin or clarithromycin. *N. transvalensis* displays increased resistance to many antimicrobial agents, including amikacin, tobramycin, cefotaxime, ceftriaxone, and amoxicillin/clavulanic acid. *N. nova* isolates appear to be susceptible to ampicillin and erythromycin in vitro but also produce

b-lactamase constitutively or in the presence of ab-lactam.

Several presumptive diagnostic tests for nocardial infection have been studied, including tests for antibodies, nocardial metabolites, and nocardial DNA. None is ready for clinical use at this time.

TREATMENT

Sulfonamides are the drugs of choice for nocardiosis ([Table 165-1](#)). Initially, 6 to 8 g of sulfadiazine or sulfisoxazole per day in four divided doses should be used. After disease is controlled, 4 g/d can be used to complete therapy. In difficult cases, sulfonamide levels should be measured and dosages adjusted to keep serum levels between 100 and 150 ug/mL. The combination of sulfamethoxazole (SMZ) and trimethoprim (TMP) is probably equivalent to sulfonamides; some authorities believe that the combination may in fact be more effective, but it also poses a modestly greater risk of hematologic toxicity. At the outset, 10 to 20 mg of TMP per kg and 50 to 100 mg of SMZ per kg should be given each day in two divided doses. Later, the daily doses can be decreased to as little as 5 mg/kg and 25 mg/kg, respectively. In persons with sulfonamide allergies, desensitization usually allows continuation of therapy with these effective and inexpensive drugs.

Minocycline is the best-established alternative oral drug and should be given in doses of 100 to 200 mg twice a day. Other tetracyclines are usually ineffective. *N. nova* infections can be treated with erythromycin (500 to 750 mg four times a day) and/or ampicillin (1 g four times a day), but other *Nocardia* spp. are often resistant to both drugs. Amoxicillin (500 mg) combined with clavulanic acid (125 mg), given three times a day, has been effective in a few cases but should be avoided in cases due to *N. nova*, in which clavulanate induces b-lactamase production. Ofloxacin (400 mg twice a day) and clarithromycin (500 mg twice a day) have each been successful in a few cases.

Amikacin, the best-established parenteral drug, is given in doses of 5 to 7.5 mg/kg every 12 h. Serum levels should be monitored with prolonged therapy in patients with diminished renal function and in the elderly. Newer b-lactam antibiotics, including cefotaxime, ceftizoxime, ceftriaxone, and imipenem, are usually effective. They may be less effective in some cases caused by *N. farcinica*.

In patients receiving immunosuppressive therapy, the regimen should be continued if necessary for treatment of an underlying disease or prevention of transplant rejection. In many cases, two or more antimicrobial agents have been used to treat nocardiosis, often in combinations including a sulfonamide or minocycline. Whether such therapy is better than monotherapy is not known, and combination therapy increases the risk of toxicity.

Surgical management of nocardial disease is similar to that of other bacterial diseases. Brain abscesses should be aspirated, drained, or excised if the diagnosis is unclear, if an abscess is large and accessible, or if an abscess fails to respond to chemotherapy. Abscesses that are small or inaccessible should be treated medically; in these cases, clinical improvement should be noticeable within 1 to 2 weeks. Brain imaging should be repeated to document the resolution of lesions, although abatement on images often

lags behind clinical improvement.

Antimicrobial therapy usually suffices for nocardial actinomycetoma. In deep or extensive cases, drainage or excision of heavily involved tissue may facilitate healing, but structure and function should be preserved whenever possible.

Nocardial infections tend to relapse (particularly in patients with chronic granulomatous disease), and long courses of antimicrobial therapy are necessary. If disease is unusually extensive, if the patient is immunosuppressed, or if the response to therapy is slow, the recommendations in [Table 165-1](#) should be exceeded.

The mortality rate for pulmonary or disseminated nocardiosis outside the [CNS](#) should be <5%. CNS disease carries a higher mortality rate. Patients should be followed carefully for at least 6 months after therapy has ended. Any child with nocardiosis and no known cause of immunosuppression should undergo tests to determine the adequacy of the phagocytic respiratory burst.

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166. ACTINOMYCOSIS - Thomas A. Russo

Actinomycosis is an indolent, slowly progressive infection caused by anaerobic or microaerophilic bacteria, primarily of the genus *Actinomyces*, that colonize the mouth, colon, and vagina. Mucosal disruption may lead to infection at virtually any site in the body. The clinical presentations of actinomycosis are myriad; however, classic features include purulent foci surrounded by dense fibrosis that, over time, cross natural anatomic boundaries into contiguous structures, with the formation of fistulae and sinus tracts in some cases. In vivo growth of actinomycetes usually results in the formation of clumps called *grains* or *sulfur granules*. This infection is commonly confused with a neoplasm. Common in the preantibiotic era, actinomycosis has diminished in incidence, as has its timely recognition. Actinomycosis has been called "the most misdiagnosed disease," and it has been said that "no disease is so often missed by experienced clinicians." Thus this entity remains a diagnostic challenge. An awareness of the full spectrum of the disease will expedite its diagnosis and treatment and will minimize the unnecessary surgical interventions, morbidity, and mortality that are reported all too often.

ETIOLOGIC AGENTS

Actinomycosis is most commonly caused by *A. israelii*, *A. naeslundii*, *A. odontolyticus*, *A. viscosus*, *A. meyeri*, *A. gerencseriae*, and *Propionibacterium propionicum* are established but less common causes of the disease. Most if not all actinomycotic infections are polymicrobial. *Actinobacillus actinomycetemcomitans*, *Eikenella corrodens*, Enterobacteriaceae, and species of *Fusobacterium*, *Bacteroides*, *Capnocytophaga*, *Staphylococcus*, and *Streptococcus* are commonly isolated with actinomycetes in various combinations, depending on the site of infection. The contribution of these other species to the pathogenesis of actinomycosis is uncertain.

An increasing number of bacterial species isolated from human clinical specimens have recently been classified as *Actinomyces*. Although their role in disease has not always been defined, *A. europaeus*, *A. neuii* subspecies *neuii*, *A. neuii* subspecies *anitratus*, *A. radingae*, *A. graevenitzii*, and *A. turicensis* appear to be infrequent and often opportunistic human pathogens. The nature of the infections described to date does not clearly establish these agents as causes of the typical syndrome of actinomycosis.

EPIDEMIOLOGY

The agents of actinomycosis are members of the normal oral flora and are often cultured from the bronchi, the gastrointestinal tract, and the female genital tract. Infection occurs throughout life, with a peak incidence in the middle decades. Males have a threefold higher incidence of infection, possibly because of poorer dental hygiene and/or more frequent trauma. Likely contributing factors to the decrease in the incidence of actinomycosis since the preantibiotic era include improved dental hygiene and the initiation of antimicrobial treatment early on -- before the full development of the disease. Individuals who do not seek or have access to health care are undoubtedly at higher risk.

PATHOGENESIS AND PATHOLOGY

A vital step in the development of actinomycosis is disruption of the mucosal barrier, which allows the actinomycetes to invade beyond their endogenous habitat in the mouth, lower gastrointestinal tract, and female genitourinary tract. Local infection, subsequent extension, and (in rare instances) distant hematogenous seeding may ensue. Initial acute inflammation is followed by the characteristic chronic, indolent phase. Lesions usually appear as single or multiple indurations. Central fluctuance, with pus containing neutrophils and sulfur granules, is virtually diagnostic of this disease ([Fig. 166-1](#)). The fibrous walls of the mass are typically described as "woody." The responsible bacterial and/or host factors have not yet been identified. Once established, actinomycosis spreads contiguously in a slow progressive manner, ignoring tissue planes. Given time, sinus tracts, which can spontaneously close and reopen, will form and extend to skin, adjacent organs, or bone. These unique features of actinomycosis mimic malignancy, with which it is often confused.

Foreign bodies appear to facilitate infection. This association most frequently involves intrauterine contraceptive devices (IUCDs). In addition, an increasing number of reports have described an association of actinomycosis with HIV infection, transplantation, and chemotherapy. Ulcerative mucosal infections (e.g., by herpes simplex virus or cytomegalovirus) and abnormalities in host defenses may facilitate the development of actinomycosis in the latter settings.

CLINICAL MANIFESTATIONS

Oral-Cervicofacial Disease Actinomycosis occurs most frequently at an oral, cervical, or facial site, usually as a soft tissue swelling, abscess, or mass lesion that is often mistaken for a neoplasm. The angle of the jaw is generally involved, but a diagnosis of actinomycosis should be considered with any mass lesion or relapsing infection in the head and neck. Otitis, sinusitis, and canaliculitis can also develop. Pain, fever, and leukocytosis are variably reported. Contiguous extension to the cranium, cervical spine, or thorax is a potential sequela.

Thoracic Disease Thoracic actinomycosis usually follows an indolent progressive course, with involvement of the pulmonary parenchyma and/or the pleural space. Chest pain, fever, and weight loss are common. A cough, when present, is variably productive. The usual radiographic appearance is either a mass lesion or pneumonitis. Cavitory disease or hilar adenopathy may develop. More than 50% of cases include pleural thickening, effusion, or empyema. Rarely, pulmonary nodules or endobronchial lesions occur. Pulmonary lesions suggestive of actinomycosis may cross fissures or pleura; may involve the mediastinum, contiguous bone, or chest wall; or may be associated with a sinus tract. In the absence of these findings, thoracic actinomycosis is usually mistaken for a neoplasm or for pneumonitis due to more usual causes.

Mediastinal infection is uncommon, usually arising from thoracic extension but rarely resulting from perforation of the esophagus, from trauma, or from head and neck or abdominal disease. The structures within the mediastinum and the heart can be involved in various combinations; consequently, the possible presentations are diverse. Isolated disease of the breast has been described.

Abdominal Disease Abdominal actinomycosis poses a great diagnostic challenge. Months or years usually pass from the inciting event (e.g., appendicitis, diverticulitis, peptic ulcer disease, foreign-body perforation, bowel surgery, or ascension from IUCD-associated pelvic disease) to clinical recognition. Because of the flow of peritoneal fluid and/or the direct extension of primary disease, virtually any abdominal organ, region, or space can be involved. The disease usually presents as an abscess or a mass lesion that is often fixed to underlying tissue and mistaken for a tumor. Infiltrative disease with irregular contrast enhancement may be seen on computed tomography (CT). Sinus tracts to the abdominal wall or perianal region may develop. Recurrent disease or a wound or fistula that fails to heal (in the absence of inflammatory bowel disease) suggests actinomycosis.

Hepatic infection usually presents as single or multiple abscesses or masses. Isolated disease presumably develops via hematogenous seeding from cryptic foci. Presently available imaging and percutaneous techniques have resulted in improved diagnosis and treatment.

All levels of the urogenital tract can be infected. Renal disease usually presents as pyelonephritis and/or renal and perinephric abscess. Bladder involvement, usually due to extension of pelvic disease, may result in ureteral obstruction or fistulas to bowel, skin, or uterus.

Pelvic Disease Actinomycotic involvement of the pelvis occurs most commonly in association with an IUCD. Pelvic symptoms when an IUCD is in place or has recently been removed should prompt consideration of actinomycosis. Although the risk has not yet been quantified, it appears to be small. The disease rarely develops when the IUCD has been in place for <1 year, but the risk increases with time. Actinomycosis can also present months after the removal of the device. Symptoms are typically indolent; fever, weight loss, abdominal pain, and abnormal vaginal bleeding or discharge are the most common. The earliest stage of disease -- often endometritis -- commonly progresses to pelvic masses or a tuboovarian abscess (Fig. 166-2). Unfortunately, because the diagnosis is often delayed, a "frozen pelvis" mimicking malignancy or endometriosis can develop by the time of recognition.

An unresolved issue is whether the isolation of *Actinomyces*-like organisms (ALOs) from cultures of cervical or endometrial specimens or the detection of ALOs by immunofluorescence is correlated with IUCD-associated disease. A Papanicolaou smear may fail to detect ALOs even in the presence of active actinomycosis. Although the risk appears to be small, the consequences of infection are significant. Therefore, until more quantitative data become available, detection of ALOs or immunofluorescence-positive organisms in conjunction with symptoms that cannot be accounted for appears to warrant removal of the IUCD and -- if advanced disease is excluded -- initiation of a 14-day course of empirical treatment for possible early pelvic actinomycosis. The detection of ALOs or immunofluorescence-positive organisms in the absence of symptoms warrants education of the patient and close follow-up but not removal of the IUCD.

Central Nervous System Disease Actinomycosis of the central nervous system is rare. Single or multiple brain abscesses are most common. An abscess usually appears

on [CT](#) as a ring-enhancing lesion with a thick wall that may be irregular or nodular. Meningitis, epidural or subdural space infection, and cavernous sinus syndrome have also been described.

Musculoskeletal Infection Actinomycotic infection of the bone is usually due to adjacent soft tissue infection but may be associated with trauma (e.g., fracture of the mandible) or hematogenous spread. Because of slow disease progression, new bone formation and bone destruction are seen concomitantly. Infection of an extremity is uncommon and is usually a result of trauma. Skin, subcutaneous tissue, muscle, and bone (with periostitis or acute or chronic osteomyelitis) are involved alone or in various combinations. Cutaneous sinus tracts frequently develop.

Disseminated Disease Hematogenous dissemination of disease from any location rarely results in multiple organ involvement. The lungs and liver are most commonly affected, with the presentation of multiple nodules mimicking disseminated malignancy. The clinical presentation may be surprisingly indolent given the extent of disease.

DIAGNOSIS

The diagnosis of actinomycosis, particularly when it mimics malignancy, is rarely considered. All too often, the first mention of actinomycosis is by the pathologist after extensive surgery has been performed. Since medical therapy alone is often sufficient for cure, the challenge for the clinician is to consider the possibility of actinomycosis in time to diagnose it in the least invasive fashion and to avoid unnecessary surgery. Both fine-needle aspiration and biopsy are being used successfully to obtain clinical material for diagnosis, as are [CT](#)- and ultrasound-guided aspirations or biopsies. The diagnosis is most commonly made by microscopic identification of sulfur granules ([Fig. 166-CD1](#)); occasionally these granules, if sought, can be grossly identified from draining sinus tracts or other purulent material. Although sulfur granules are a defining characteristic of actinomycosis, granules are also found in mycetoma and botryomycosis; however, these entities can easily be differentiated from actinomycosis with appropriate histopathologic and microbiologic studies. Microbiologic identification of actinomycetes is possible in only a minority of cases and is often precluded by prior antimicrobial therapy. Therefore, for optimal yield, the avoidance of even a single dose of antibiotics is mandatory. Primary isolation usually requires 5 to 7 days but may take as long as 2 to 4 weeks. Immunofluorescence testing for *A. israelii*, *A. naeslundii*, and *P. propionicum* (available through the Centers for Disease Control and Prevention in Atlanta) has become a useful diagnostic alternative. Because these organisms are components of the normal oral and genital-tract flora, their identification in sputum, bronchial washings, and cervicovaginal secretions is of little significance in the absence of sulfur granules. *Actinomyces* can be detected in urine by means of appropriate staining and culture.

TREATMENT

Actinomycosis must be treated with high doses of antimicrobials for a prolonged period. Although therapy needs to be individualized, the intravenous administration of 18 to 24 million units of penicillin for 2 to 6 weeks, followed by oral therapy with penicillin or amoxicillin for 6 to 12 months, is a reasonable guideline for serious infections. Less extensive disease, particularly that involving the oral-cervicofacial region, may require

less intensive therapy. If therapy is extended beyond the point of resolution of measurable disease, the risk of relapse -- a clinical hallmark of this infection -- will be minimized. A similar approach is reasonable for immunocompromised patients, although refractory disease has been described in HIV-infected individuals. Antimicrobial agents whose use is supported by extensive clinical experience are listed in [Table 166-1](#). Although the role played by "companion" microbes in actinomycosis is unclear, many isolates are pathogens in their own right, and a regimen covering these organisms during the initial treatment course is reasonable. Agents whose success has been reported anecdotally are ceftriaxone, ceftizoxime, imipenem, and ciprofloxacin. Drugs that should be avoided are metronidazole, aminoglycosides, oxacillin, dicloxacillin, and cephalixin.

Combined medical-surgical therapy is still advocated by some authorities. However, an increasing body of literature now supports an initial attempt at cure with medical therapy alone, even in extensive disease. [CT](#) and magnetic resonance imaging should be used to monitor the response to therapy. Percutaneous drainage is an additional option. When a critical location is involved (e.g., the epidural space, the central nervous system) or when suitable medical therapy fails, surgical intervention may be appropriate.

(Bibliography omitted in Palm version)

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167. INFECTIONS DUE TO MIXED ANAEROBIC ORGANISMS - Dennis L. Kasper

DEFINITIONS

Anaerobic bacteria are organisms that require reduced oxygen tension for growth, failing to grow on the surface of solid media in 10% CO₂ in air. (In contrast, *microaerophilic* bacteria can grow in 10% CO₂ in air or under anaerobic or aerobic conditions, although they grow best in the presence of only a small amount of atmospheric oxygen, and *facultative* bacteria can grow in the presence or absence of air.) This chapter describes infections caused by nonsporulating anaerobic bacteria. In general, anaerobes associated with human infections are relatively aerotolerant. They can survive for as long as 72 h in the presence of oxygen, although generally they will not multiply in this environment. A far smaller number of pathogenic anaerobic bacteria (which are also part of the normal flora) die after brief contact with oxygen, even in low concentrations.

The nonsporulating anaerobic bacteria exist as components of the normal flora on the mucosal surfaces of humans and animals. The major reservoirs of these bacteria are the mouth, lower gastrointestinal tract, skin, and female genital tract. Among the constituents of the oral flora, anaerobes are the predominant commensal organisms, ranging in concentration from 10⁹/mL in saliva to 10¹²/mL in gingival scrapings. In the oral cavity, the ratio of anaerobic to aerobic bacteria ranges from 1:1 on the surface of a tooth to 1000:1 in the gingival crevice. Anaerobic bacteria are not found in appreciable numbers in the normal upper intestine until the distal ileum. In the colon, the proportion of anaerobes increases significantly, as does the overall bacterial count. For example, in the colon there are 10¹¹ to 10¹² organisms per gram of stool, with a ratio of anaerobes to aerobes of approximately 1000:1. In the female genital tract, there are approximately 10⁹ organisms per milliliter of secretions, with a ratio of anaerobes to aerobes of approximately 10:1.

Hundreds of species of anaerobic bacteria have been identified as part of the normal flora of humans. Identification of as many as 500 different anaerobic species in fecal specimens reflects the diversity of the anaerobic flora. Despite the complex array of bacteria in the normal flora, relatively few species are isolated commonly from human infection.

Anaerobic infections occur when the harmonious relationship between the host and the bacteria is disrupted. Any site in the body is susceptible to infection with these indigenous organisms when a mucosal barrier or the skin is compromised by surgery, trauma, tumor, or ischemia or necrosis, which reduce local tissue redox potentials. Because the sites that are colonized by anaerobes contain many species of bacteria, disruption of anatomic barriers allows penetration of many organisms, resulting in mixed infections involving multiple species of anaerobes combined with facultative or microaerophilic organisms. Such mixed infections are seen in the head and neck (chronic sinusitis, chronic otitis media, Ludwig's angina, and periodontal abscesses). Brain abscesses and subdural empyema are the most frequent anaerobic infections of the central nervous system. Anaerobes are responsible for pleuropulmonary diseases such as aspiration pneumonia, necrotizing pneumonia, lung abscess, and empyema. These organisms also play an important role in various intraabdominal infections, such

as peritonitis and intraabdominal and liver abscesses ([Chap. 130](#)). They are isolated frequently in female genital tract infections, such as salpingitis, pelvic peritonitis, tuboovarian abscess, vulvovaginal abscess, septic abortion, and endometritis ([Chaps. 132](#) and [133](#)). Anaerobic bacteria also are frequently found in infections of the skin, soft tissues, and bones and in bacteremia.

ETIOLOGY

The major anaerobic gram-positive cocci that produce disease are *Peptostreptococcus* spp. The major species involved in infections are *Peptostreptococcus intermedius*, *P. micros*, *P. magnus*, *P. asaccharolyticus*, *P. anaerobius*, and *P. prevotii*. Clostridia are gram-positive rods that are isolated from wounds, abscesses, sites of abdominal infection, and blood; they are discussed in [Chap. 145](#). The principal anaerobic gram-negative bacilli found in human infections are the members of the *Bacteroides* "family," which includes the *Bacteroides fragilis* group as well as *Fusobacterium*, *Prevotella*, and *Porphyromonas* spp. Another gram-negative rod, *Bilophila wadsworthia*, has been isolated from infected sites. Gram-positive anaerobic non-spore-forming bacilli are uncommon as etiologic agents of human infections. *Propionibacterium acnes*, a rare cause of foreign body infections, is one of the few non-clostridial gram-positive rods associated with infections.

The *B. fragilis* group contains the anaerobic pathogens most frequently isolated from clinical infections. Members of this group are part of the normal bowel flora; they include several distinct species, such as *B. fragilis*, *B. thetaiotaomicron*, *B. distasonis*, *B. vulgatus*, *B. uniformis*, and *B. ovatus*. (Ribosomal RNA analysis has shown that *B. distasonis* is more closely related to the genus *Porphyromonas* than it is to other *Bacteroides*.) Of this group, *B. fragilis* is the most important clinical isolate. However, *B. fragilis* is isolated from the normal fecal flora in lower numbers than other *Bacteroides* spp.

A second major group of phenotypically similar organisms is part of the indigenous oral flora. Thus these organisms are found at infected sites that can be seeded with oral microflora. Many of these species are pigment-producing bacteria previously classified as *Bacteroides melaninogenicus*. The nomenclature of this group has changed so that two distinct genera, *Prevotella* and *Porphyromonas*, are now recognized; these genera comprise several pathogenic species, including *Porphyromonas gingivalis*, *Porphyromonas asaccharolytica*, and *Prevotella oralis*. *Porphyromonas* and *Prevotella* spp. cause localized infections that can spread contiguously.

In female genital tract infections, organisms normally colonizing the vagina, such as *Prevotella bivia* and *Prevotella disiens*, are the most frequent isolates, although *B. fragilis* is not uncommon. The *Fusobacterium* species *Fusobacterium necrophorum*, *F. nucleatum*, and *F. varium*, which reside primarily in the oral cavity and the gastrointestinal tract, are also isolated from clinical infections, including necrotizing pneumonia and abscesses. *Bilophila wadsworthia* has been reported to cause serious infections, including bacteremia, necrotizing fasciitis, and abscesses; this organism is frequently resistant to several antimicrobials, including imipenem, cefoxitin, and other b-lactam agents.

Infections caused by anaerobic bacteria most frequently are due to more than one organism. These polymicrobial infections may be caused by one or several anaerobic species or by a combination of anaerobic organisms and microaerophilic or facultative bacteria acting synergistically.

Approach to the Patient

The physician must consider several points when approaching the patient with presumptive infection due to anaerobic bacteria.

1. Most of the organisms colonizing mucosal sites are harmless commensals; very few cause disease.
2. For anaerobes to cause tissue infection, they must spread beyond the normal mucosal barriers.
3. Conditions favoring the propagation of these bacteria, particularly a lowered oxidation-reduction potential, are necessary. These conditions exist at sites of trauma, tissue destruction, compromised vascular supply, and complications of preexisting infection, which produce necrosis.
4. There is a complex array of infecting flora. For example, as many as 12 different types of organisms can be isolated from a suppurative site.
5. Anaerobic organisms tend to be found in abscess cavities or in necrotic tissue. The failure of an abscess to yield organisms on routine culture is a clue that the abscess is likely to contain anaerobic bacteria. Often smears of this "sterile pus" are found to be teeming with bacteria when Gram's stain is applied. Malodorous pus suggests anaerobic infection. Although some facultative organisms, such as *Staphylococcus aureus*, are also capable of causing abscesses, abscesses in organs or deeper body tissues should call to mind anaerobic infection.
6. Gas is found in many anaerobic infections of deep tissues.
7. Some species (the best example being the *B. fragilis* group) require specific therapy. However, many synergistic infections can be cured with antibiotics directed at some but not all of the organisms involved. Antibiotic therapy, combined with debridement and drainage, disrupts the interdependent relationship among the bacteria, and some species that are resistant to the antibiotic do not survive without the coinfecting organisms.
8. Manifestations of disseminated intravascular coagulation are unusual in patients with purely anaerobic infection.

EPIDEMIOLOGY

Difficulties in the performance of appropriate cultures, contamination of cultures by aerobic bacteria or components of the normal flora, and the lack of readily available, reliable culture techniques have made it impossible to obtain accurate incidence or

prevalence data. However, anaerobic infections are encountered frequently in hospitals with active surgical, trauma, and obstetric and gynecologic services. In some centers, anaerobic bacteria, particularly *B. fragilis*, account for approximately 4% of positive blood cultures.

PATHOGENESIS

Anaerobic bacterial infections usually occur when an anatomic barrier becomes disrupted and constituents of the local flora enter a site that was previously sterile. Because of the specific growth requirements of anaerobic organisms and their presence as commensals on mucosal surfaces, conditions must arise that allow these organisms to penetrate mucosal barriers and enter tissue with a lowered oxidation-reduction potential. Therefore, tissue ischemia, trauma, surgery, perforated viscus, shock, and aspiration provide environments conducive to the proliferation of anaerobes. In the case of a perforated viscus, hundreds of species of anaerobic bacteria are spilled into the peritoneal cavity, but many of these organisms are unable to survive because the highly vascularized tissue provides a sufficiently high redox potential. The entry of oxygen into the environment results in the selection of the more aerotolerant anaerobic organisms.

The ability of an organism to adhere to host tissues is important to the establishment of infection. Some oral species adhere to crevicular epithelium in the oral cavity. *Prevotella melaninogenica* actually attaches to other microorganisms; *P. gingivalis* is a common isolate in periodontal disease. These organisms have fimbriae that facilitate attachment. Some unencapsulated *Bacteroides* strains appear to be piliated, a characteristic that may account for their ability to adhere.

The most extensively studied virulence factor of the nonsporulating anaerobes is the polysaccharide capsule of *B. fragilis*. This polysaccharide possesses distinct biologic properties, such as the ability (owing to a unique zwitterionic motif of charged sugars) to promote abscess formation. Intraabdominal abscess induction is related to the capacity of the polysaccharide to stimulate the release of cytokines, in particular interleukin 8 (IL-8) and tumor necrosis factor α (TNF- α), from resident peritoneal cells. IL-8 results in the chemotaxis of polymorphonuclear neutrophils (PMNs) into the peritoneum, where they adhere to mesothelial cells induced by TNF- α to upregulate their expression of intercellular adhesion molecule 1 (ICAM-1). PMNs adherent to ICAM-1-expressing cells probably represent the nidus for an abscess. Prophylactic or therapeutic administration of the polysaccharide or a zwitterionic mimetic to experimental animals confers protection against abscess induction following challenge with intestinal microorganisms capable of inducing abscesses. This protection is mediated by T cells controlling cytokine release, which blocks the tissue response of abscess formation. Although abscesses constitute a host response that localizes and contains infecting bacteria, abscess formation in patients with sepsis often results in severe and chronic illness that requires surgical drainage in combination with antimicrobial therapy.

Anaerobic bacteria produce a number of exoproteins that are capable of enhancing the organisms' virulence. The collagenase produced by *P. gingivalis* may enhance tissue destruction. An enterotoxin has been identified in *B. fragilis* strains associated with diarrheal disease in animals and young children. This 20-kDa zinc-dependent metalloprotease reversibly alters the morphology of the tight junctional complexes of

intestinal epithelial cells. Both *B. fragilis* and *P. melaninogenica* possess lipopolysaccharides (endotoxins) that are less biologically potent than endotoxins associated with aerobic gram-negative bacteria. This relative biologic inactivity may account for the lower frequency of disseminated intravascular coagulation and purpura in *Bacteroides* bacteremia than in facultative and aerobic gram-negative bacillary bacteremia.

CLINICAL MANIFESTATIONS

Anaerobic Infections of the Mouth, Head, and Neck (See also [Chap. 30](#)) Infections of the mouth can arise from either the supragingival or the subgingival dental plaque. Supragingival plaque formation begins with the adherence of gram-positive bacteria to the tooth surface. This form of plaque is influenced by salivary and dietary components, oral hygiene, and local host factors. Once the supragingival plaque is established, the acquisition of pathogenic bacteria and an increase in the amount of plaque are responsible for the ultimate development of gingivitis. Early bacteriologic changes in the supragingival plaque initiate an inflammatory response in the gingiva, including edema, swelling, and increased gingival fluid, and cause the development of caries and endodontic (pulp) infections. In addition, these changes contribute to the subsequent pathogenic alteration in the subgingival plaque that arises from poor or inadequate oral hygiene.

Subgingival plaque is associated with periodontal disease and disseminated infection arising from the oral cavity. Bacteria that colonize the subgingival area are primarily anaerobic. The black-pigmented gram-negative anaerobic bacilli, principally *P. gingivalis* and *P. melaninogenica*, are the most important. Infections in this area are frequently mixed and involve both anaerobic and aerobic bacteria. After establishment of local infection either in root canals or in the periodontal area, infection may extend into the mandible, causing osteomyelitis to the maxillary sinuses; or to local tissues in the submandibular or submental spaces, depending on which teeth are involved. Periodontitis also may result in spreading infection that can involve adjacent bone or soft tissues.

Gingivitis Gingivitis may become a necrotizing infection (trench mouth, Vincent's stomatitis). The onset of disease is usually sudden and is associated with tender bleeding gums, foul breath, and a bad taste. The gingival mucosa, especially the papillae between the teeth, becomes ulcerated and may be covered by a gray exudate, which is removable with gentle pressure. Patients may become systemically ill, developing fever, cervical lymphadenopathy, and leukocytosis. Occasionally, ulcerative gingivitis can spread to the buccal mucosa, the teeth, and the mandible or maxilla, resulting in widespread destruction of bone and soft tissue. This infection is termed *acute necrotizing ulcerative mucositis* (cancrum oris, noma). It destroys tissue rapidly, causing the teeth to fall out and large areas of bone -- or even the whole mandible -- to be sloughed. A strong putrid odor is frequently detected, although the lesions are not painful. The gangrenous lesions eventually heal, leaving large disfiguring defects. This infection is seen most commonly following a debilitating illness or in severely malnourished children. It has been known to complicate leukemia or to develop in individuals with a genetic deficiency of catalase.

Acute Necrotizing Infections of the Pharynx These infections usually occur in association with ulcerative gingivitis. Symptoms include an extremely sore throat, foul breath, and a bad taste accompanied by fever and a sensation of choking. Examination of the pharynx demonstrates that the tonsillar pillars are swollen, red, ulcerated, and covered with a grayish membrane that peels easily. Lymphadenopathy and leukocytosis are common. The disease may last for only a few days or, if not treated, may persist for weeks. Lesions begin unilaterally but may spread to the other side of the pharynx or the larynx. Aspiration of the infected material by the patient can result in lung abscesses. Soft tissue infection of the oral-facial area may or may not be odontogenic. *Ludwig's angina*, a periodontal infection usually arising from the tissues surrounding the third molar, may produce submandibular cellulitis that results in marked local swelling of tissues, with pain, trismus, and superior and posterior displacement of the tongue. Submandibular swelling of the neck can impair swallowing and cause respiratory obstruction. In some cases, tracheotomy may be life-saving.

Fascial Infections These infections arise from the spread of organisms originating in the upper airways to potential spaces formed by the fascial planes of the head and neck. Perimandibular space infection most commonly involves the submandibular, peritonsillar, and parapharyngeal spaces. Peritonsillar abscesses occur in association with pharyngitis. Complicated dental infections spread to the submandibular and buccal spaces. Entry of organisms by either portal can result in parapharyngeal space infections. Although there are few well-documented reports on the microbiology of these syndromes, anaerobes from the oral flora have been implicated in many cases. Fascial infections associated with *S. aureus* or *Streptococcus pyogenes* may arise from boils or impetigo, whereas anaerobes are associated with space infections either occurring spontaneously or arising from diseases of the mucous membranes or from dental manipulations.

Sinusitis and Otitis The role of anaerobic bacteria in acute sinusitis may be underestimated because of improper collection of specimens. In a study of chronic sinusitis, anaerobic bacteria were found in 52% of specimens collected during external frontoethmoidotomy or radical antrotomy. Anaerobic bacteria are much more easily implicated in chronic suppurative otitis media than in acute otitis media. Purulent exudate from chronically draining ears has been found to contain anaerobes, particularly *Bacteroides* spp., in up to 50% of cases. *B. fragilis* has been isolated from up to 28% of patients with chronic otitis media.

Complications of Anaerobic Head and Neck Infections Contiguous cranial spread of these infections may result in osteomyelitis of the skull or mandible or in intracranial infections such as brain abscess and subdural empyema. Caudal spread can produce mediastinitis or pleuropulmonary infection. Hematogenous complications may also result from anaerobic infections of the head and neck. Bacteremia, which occasionally is polymicrobial, can lead to endocarditis or other distant infections. When infections spread to produce suppurative thrombophlebitis of the internal jugular vein, a destructive syndrome (*Lemierre's*) -- with prolonged fever, bacteremia, septic emboli to both the lung and the brain, and multiple metastatic foci of suppurative infection -- may develop. This syndrome has been reported with fusobacterial septicemia following exudative pharyngitis but has been uncommon in the antimicrobial era.

Central Nervous System Infections Brain abscesses are frequently associated with anaerobic bacteria ([Chap. 372](#)). If optimal bacteriologic techniques are employed, as many as 85% of brain abscesses yield anaerobic bacteria -- most often anaerobic gram-positive cocci (especially peptostreptococci), which are followed in frequency by *Fusobacterium* and *Bacteroides* spp. Facultative or microaerophilic streptococci and coliforms often are part of a mixed infecting flora in brain abscesses.

Pleuropulmonary Infections Anaerobic pleuropulmonary infections result from the aspiration of oropharyngeal contents, often in the context of an altered state of consciousness or an absent gag reflex. Four clinical syndromes are associated with anaerobic pleuropulmonary infection produced by aspiration: simple aspiration pneumonia, necrotizing pneumonia, lung abscess, and empyema.

Aspiration Pneumonitis Aspiration pneumonitis must be distinguished from two other clinical syndromes associated with aspiration that are not of bacterial etiology. One syndrome results from aspiration of solids, usually food. Obstruction of major airways typically results in atelectasis and moderate nonspecific inflammation. Therapy consists of removal of the foreign body.

The second aspiration syndrome is more easily confused with bacterial aspiration. *Mendelson's* syndrome results from regurgitation of stomach contents and aspiration of chemical material, usually gastric juices. Pulmonary inflammation -- including the destruction of the alveolar lining, with transudation of fluid into the alveolar space -- occurs with remarkable rapidity. Typically this syndrome develops within hours, often following anesthesia when the gag reflex is depressed. The patient becomes tachypneic, hypoxic, and febrile. The leukocyte count may rise, and the chest x-ray may evolve suddenly from normal to a complete bilateral "whiteout" within 8 to 24 h. Sputum production is minimal. The pulmonary signs and symptoms can resolve quickly with symptom-based therapy or can culminate in respiratory failure, with the subsequent development of bacterial superinfection over a period of days. Antibiotic therapy is not indicated unless bacterial infection supervenes. The signs of bacterial infection include sputum production, persistent fever, leukocytosis, and clinical evidence of sepsis.

In contrast to these syndromes, bacterial aspiration pneumonia develops more slowly. It is seen in patients who are hospitalized and have a depressed gag reflex, impaired swallowing, or a tracheal or nasogastric tube; elderly patients; or those with transiently impaired consciousness in the wake of seizures, cerebrovascular accidents, or alcoholic blackouts. Patients who enter the hospital with this syndrome typically have been ill for several days and generally report low-grade fever, malaise, and sputum production. Usually the history reveals factors predisposing to aspiration, such as alcohol overdose or residence in a nursing home. Sputum characteristically is not malodorous unless the process has been under way for at least a week. A mixed bacterial flora with many PMNs is evident on Gram's staining; cultures are reliable only if contamination with the normal oral flora is avoided -- that is, if specimens are obtained by transtracheal aspiration. In general, this procedure is not indicated in the evaluation of these patients. The most commonly encountered anaerobes in these infections are pigmented and nonpigmented *Prevotella* spp., *F. nucleatum*, *Peptostreptococcus* spp., and *Bacteroides* spp. Chest x-rays show consolidation in dependent pulmonary segments: in the basilar segments of the lower lobes if the patient has aspirated while upright and in either the

posterior segment of the upper lobe (usually on the right side) or the superior segment of the lower lobe if the patient has aspirated while supine. The organisms isolated reflect the pharyngeal flora; *P. melaninogenica*, *Fusobacterium* spp., and anaerobic cocci are the most frequent isolates. The patient who aspirates in the hospital also may have a mixed infection involving enteric gram-negative rods.

Necrotizing Pneumonitis This form of anaerobic pneumonitis is characterized by numerous small abscesses that spread to involve several pulmonary segments. The process can be indolent or fulminating. This syndrome is less common than either aspiration pneumonia or lung abscess and includes features of both types of infection.

Anaerobic Lung Abscesses ([Fig. 167-CD1](#)) These abscesses result from subacute anaerobic pulmonary infection. The clinical syndrome typically involves a history of constitutional symptoms, including malaise, weight loss, fever, chills, and foul-smelling sputum, perhaps over a period of weeks ([Chap. 255](#)). Patients who develop lung abscesses characteristically have dental infection and periodontitis, but lung abscesses in edentulous patients have been reported. Abscess cavities may be single or multiple and generally occur in dependent pulmonary segments. Anaerobic abscesses must be distinguished from those associated with tuberculosis, neoplasia, and other conditions. Oral anaerobes predominate, although *B. fragilis* is isolated in up to 10% of cases. *S. aureus* may be found as well.

Empyema Empyema is a manifestation of long-standing anaerobic pulmonary infection. The clinical presentation, which includes the presence of foul-smelling sputum, resembles that of other anaerobic pulmonary infections. Patients may report pleuritic chest pain and marked chest-wall tenderness.

Empyema may be masked by overlying pneumonitis and should be considered especially in cases of persistent fever despite antibiotic therapy. Diligent physical examination and the use of ultrasound to localize a loculated empyema are important diagnostic tools. The collection of a foul-smelling exudate by thoracentesis is typical. Cultures of infected pleural fluid yield an average of 3.5 anaerobes and 0.6 facultative or aerobic bacterial species. Drainage is required. Defervescence, a return to a feeling of well-being, and resolution of the process may require several months.

Extension from a subdiaphragmatic infection also may result in anaerobic empyema. Septic pulmonary emboli may originate from intraabdominal or female genital tract infections and can produce anaerobic pneumonia.

Intraabdominal Infections Enterotoxigenic *B. fragilis* has been associated with watery diarrhea in a small number of young children and adults. In case-control studies of children with undiagnosed diarrheal disease, enterotoxigenic *B. fragilis* was isolated from significantly more children with diarrhea than children in the control group. This organism may play a role in a small proportion of childhood diarrhea cases. Neutropenic enterocolitis (typhlitis) has been associated with anaerobic infection of the cecum but -- in the setting of neutropenia ([Chap. 85](#)) -- may involve the entire bowel. Patients usually present with fever; abdominal pain, tenderness, and distension; and watery diarrhea. The bowel wall is edematous with hemorrhage and necrosis. The primary pathogen is thought by some authorities to be *C. septicum*, but other clostridia and mixed anaerobic

infections have also been implicated. More than 50% of patients developing early clinical signs can benefit from antibiotic therapy and bowel rest. Surgery is sometimes required to remove gangrenous bowel. **See Chap. 130 for a complete discussion of intraabdominal infections.*

Pelvic Infections The vagina of a healthy woman is one of the major reservoirs of anaerobic and aerobic bacteria. In the normal flora of the female genital tract, anaerobes outnumber aerobes by a ratio of approximately 10:1 and include anaerobic gram-positive cocci and *Bacteroides* spp. Anaerobes are isolated from most patients with genital tract infections not caused by a sexually transmitted pathogen. The major anaerobic pathogens are *B. fragilis*, *P. bivia*, *P. disiens*, *P. melaninogenica*, anaerobic cocci, and *Clostridium* spp. Anaerobes frequently are encountered in tuboovarian abscess, septic abortion, pelvic abscess, endometritis, and postoperative wound infection, particularly following hysterectomy. Although these infections are frequently mixed, involving both anaerobes and coliforms, pure anaerobic infections without coliform or other facultative bacterial species occur more often in pelvic than in intraabdominal sites and are characterized by drainage of foul-smelling pus or blood from the uterus, generalized uterine or local pelvic tenderness, and continued fever and chills. Suppurative thrombophlebitis of the pelvic veins may complicate the infections and lead to repeated episodes of septic pulmonary emboli.

Anaerobic bacteria have been thought to be contributing factors in the etiology of *bacterial vaginosis*. This syndrome of unknown etiology is characterized by a profuse malodorous discharge and an increase in the number of bacteria in the vagina, including *Gardnerella vaginalis*, *Prevotella* spp., *Mobiluncus* spp., peptostreptococci, and genital mycoplasmas. Anaerobic bacteria are thought to play a role in the etiology of pelvic inflammatory disease ([Chap. 133](#)), and several investigations have shown an association between bacterial vaginosis and the development of pelvic inflammatory disease.

Pelvic infections due to *Actinomyces* spp. have been associated with use of intrauterine devices ([Chap. 166](#)).

Skin and Soft Tissue Infections Injury to skin, bone, or soft tissue by trauma, ischemia, or surgery creates a suitable environment for anaerobic infections. These infections are most frequently found in sites prone to contamination with feces or with upper airway secretions -- for example, wounds associated with intestinal surgery, decubitus ulcers, or human bites. Anaerobic bacteria can be isolated in cases of crepitant cellulitis, synergistic cellulitis, or gangrene and necrotizing fasciitis ([Chaps. 128 and 145](#)). Moreover, these organisms have been isolated from cutaneous abscesses, rectal abscesses, and axillary sweat gland infections (hydradenitis suppurativa). Anaerobes are frequently cultured from foot ulcers in diabetic patients.

These soft tissue or skin infections are usually polymicrobial. A mean of 4.8 bacterial species are isolated, with a roughly 3:2 ratio of anaerobes to aerobes. The most frequently isolated organisms include *Bacteroides* spp., *Peptostreptococcus* spp., enterococci, *Clostridium* spp., and *Proteus* spp. The involvement of anaerobes in these types of infections is associated with a higher frequency of fever, foul-smelling lesions, gas in the tissues, or visible foot ulcer.

Anaerobic bacterial *synergistic gangrene* (*Meleney's gangrene*) is characterized by exquisite pain, redness, and swelling followed by induration. Erythema surrounds a central zone of necrosis. A granulating ulcer forms at the original center as necrosis and erythema extend outward. Symptoms are limited to pain; fever is not typical. These infections usually involve a combination of *Peptostreptococcus* spp. and *S. aureus*; the usual site of infection is an abdominal surgical wound or the area surrounding an ulcer on an extremity. Treatment includes surgical removal of necrotic tissue and antimicrobial administration.

Necrotizing fasciitis, a rapidly spreading destructive disease of the fascia, is usually attributed to group A streptococci but can also be caused by anaerobic bacteria, including *Peptostreptococcus* and *Bacteroides* spp. Gas may be found in the tissues. Similarly, myonecrosis can be associated with mixed anaerobic infection. *Fournier's gangrene* consists of cellulitis involving the scrotum, perineum, and anterior abdominal wall, with mixed anaerobic organisms spreading along deep external fascial planes and causing extensive loss of skin.

Bone and Joint Infections Although *actinomyces* ([Chap. 166](#)) accounts on a worldwide basis for most anaerobic infections in bone, organisms including *Peptostreptococcus* spp. or microaerophilic cocci, *Bacteroides* spp., *Fusobacterium* spp., and *Clostridium* spp. can also be found. These infections frequently arise adjacent to soft tissue infections. Hematogenous seeding of bone is uncommon. *Prevotella* and *Porphyromonas* spp. are detected in infections involving the maxilla and mandible, whereas *Clostridium* spp. have been reported as anaerobic pathogens in cases of osteomyelitis of the long bones following fracture or trauma. *Fusobacteria* have been isolated in pure culture from sites of osteomyelitis adjacent to the perinasal sinuses. *Peptostreptococcus* spp. and microaerophilic cocci have been reported as significant pathogens in infections involving the skull, mastoid, and prosthetic implants placed in bone. In patients with osteomyelitis ([Chap. 129](#)), the most reliable culture specimen is a bone biopsy sample free of normal uninfected skin and subcutaneous tissue. In patients with anaerobic osteomyelitis, a mixed flora is frequently isolated from a bone biopsy specimen.

In cases of anaerobic septic arthritis, the most common isolates are *Fusobacterium* spp. Most of the patients involved have uncontrolled peritonsillar infections progressing to septic cervical venous thrombophlebitis and resulting in hematogenous dissemination with a predilection for the joints. Unlike anaerobic osteomyelitis, anaerobic pyoarthritis in most cases is not polymicrobial and may be acquired hematogenously. Anaerobes are important pathogens in infections involving prosthetic joints; in these infections, the causative organisms (such as *Peptostreptococcus* spp. and *P. acnes*) are part of the normal skin flora.

Bacteremia Transient bacteremia is a well-known event in healthy people whose anatomic mucosal barriers have been injured (e.g., during dental extractions or dental scaling). These bacteremic episodes, which are often due to anaerobes, have no pathologic consequences. However, anaerobic bacteria are found in cultures of blood from clinically ill patients when proper culture techniques are used. *B. fragilis* is the single most common anaerobic isolate from the bloodstream.

In recent years, the rate of isolation of anaerobic bacteria from blood cultures has been decreasing. Studies from the 1970s and early 1980s found that 10 to 15% of positive blood cultures yielded anaerobes, while more recent surveys have found rates as low as 4%. The cause of this change is unknown but may be related to the administration of antibiotic prophylaxis before intestinal surgery, the earlier recognition of localized infections, and the empirical use of broad-spectrum antibiotics for presumed infection.

Once the organism has been identified, both the portal of bloodstream entry and the underlying problem that probably led to seeding of the bloodstream can often be deduced from an understanding of the organism's normal site of residence. For example, mixed anaerobic bacteremia including *B. fragilis* usually implies colonic pathology with mucosal disruption from neoplasia, diverticulitis, or some other inflammatory lesion. The initial manifestations are determined by the portal of entry and reflect the localized condition. When bloodstream invasion occurs, patients can become extremely ill, with rigors and hectic fevers ranging up to 40.6°C (105°F). The clinical picture may be quite similar to that seen in sepsis involving aerobic gram-negative bacilli. Although other complications of anaerobic bacteremia, such as septic thrombophlebitis and septic shock, have been reported, the incidence of these complications in association with anaerobic bacteremia is low. Anaerobic bacteremia is potentially fatal and requires rapid diagnosis and appropriate therapy. Mortality appears to increase with the age of the patient (with reported rates of more than 66% among patients over 60 years old), with the isolation of multiple species from the bloodstream, and with the failure to surgically remove a focus of infection.

Endocarditis and Pericarditis (See also [Chap. 126](#)) Endocarditis due to anaerobes is uncommon. However, anaerobic streptococci, which are often classified incorrectly, are responsible for this disease more frequently than is generally appreciated. Gram-negative anaerobes are unusual causes of endocarditis. Anaerobes, particularly *B. fragilis* and *Peptostreptococcus* spp., are uncommonly found in infected pericardial fluids. Anaerobic pericarditis is associated with a mortality rate of >50%.

DIAGNOSIS

Because of the time and difficulty involved in the isolation of anaerobic bacteria, diagnosis of anaerobic infections must frequently be based on presumptive evidence. Certain sites (such as avascular necrotic tissues) with lowered oxidation-reduction potential favor the diagnosis of an anaerobic infection. When infections occur in proximity to mucosal surfaces normally harboring an anaerobic flora, such as the gastrointestinal tract, female genital tract, or oropharynx, anaerobes should be considered as potential etiologic agents. A foul odor is often indicative of anaerobes, which produce certain organic acids as they proliferate in necrotic tissue. Although these odors are nearly pathognomonic for anaerobic infection, the absence of odor does not exclude an anaerobic etiology. Because anaerobes often coexist with other bacteria to cause mixed or synergistic infection, Gram's staining of exudate frequently reveals numerous pleomorphic cocci and bacilli suggestive of anaerobes. Sometimes these organisms have morphologic characteristics associated with specific species.

The presence of gas in tissues is highly suggestive, but not diagnostic, of anaerobic

infection. When cultures of obviously infected sites yield no growth, streptococci only, or a single aerobic species (such as *Escherichia coli*) and Gram's staining reveals a mixed flora, the implication is that the anaerobic microorganisms failed to grow because of inadequate transport and/or culture techniques. Failure of a patient to respond to antibiotics that are not active against anaerobes -- for example, aminoglycosides and in some circumstances penicillin, cephalosporins, or tetracyclines -- suggests anaerobic infection.

There are three critical steps in the diagnosis of anaerobic infection: (1) proper specimen collection; (2) rapid transport of the specimens to the microbiology laboratory, preferably in anaerobic transport media; and (3) proper handling of the specimens by the laboratory. Specimens must be collected by meticulous sampling of infected sites, with avoidance of contamination by the normal flora. When such contamination is likely, the specimen is unacceptable. Examples of specimens unacceptable for anaerobic culture include sputum collected by expectoration or nasal tracheal suction, bronchoscopy specimens, samples collected directly through the vaginal vault, urine collected by voiding, and feces. Specimens that can be cultured for anaerobes include blood, pleural fluid, transtracheal aspirates, pus obtained by direct aspiration from an abscess cavity, fluid obtained by culdocentesis, suprapubic bladder aspirates, cerebrospinal fluid, and lung puncture specimens.

Because even brief exposure to oxygen may kill some anaerobic organisms and result in failure to isolate them in the laboratory, air must be expelled from the syringe used to aspirate the abscess cavity, and the needle must be capped with a sterile rubber stopper. Proper precautions should be used in the handling of contaminated needles. Specimens can be injected into transport bottles containing a reduced medium or taken immediately in syringes to the laboratory for direct culture on anaerobic media. In general, swabs should not be used. If a swab must be used, it should be placed in a reduced semisolid carrying medium before transport to the laboratory. Delays in transport may lead to a failure to isolate anaerobes due to exposure to oxygen or overgrowth of facultative organisms, which may eliminate or obscure any anaerobes that are present. All clinical specimens from suspected anaerobic infections should be Gram-stained and examined for organisms with characteristic morphology ([Fig. 167-CD2](#)). It is not unusual for organisms to be observed on Gram's staining but not isolated in culture. If purulent materials are found to be sterile or organisms are seen on Gram's staining but do not grow in the culture, the involvement of anaerobes should be suspected.

TREATMENT

Successful therapy for anaerobic infections requires the administration of a combination of appropriate antibiotics, surgical resection, debridement of devitalized tissues, and drainage. Perforations must be closed promptly, closed spaces drained, tissue compartments decompressed, and an adequate blood supply established. Abscess cavities should be drained as soon as fluctuation or localization occurs. Surgery was formerly required to establish drainage; however, computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound now allow diagnostic radiologists to drain many abscess sites percutaneously.

Antibiotic Therapy and Resistance Decisions about the treatment of anaerobic infections with antibiotics are usually based on known resistance patterns in certain species, on the likelihood of encountering a given species in the case at hand, and on Gram's stain findings. Antibiotics active against *Bacteroides* spp., penicillin-resistant *Prevotella* and *Porphyromonas* spp., and *Fusobacterium* spp. can be grouped into four categories on the basis of their predicted activity against anaerobes ([Table 167-1](#)). (Nearly all the drugs listed have toxic side effects, which are described in detail in [Chap. 137](#).) In many infections, anaerobes are mixed with coliforms and other facultative organisms. The best therapeutic regimens, therefore, are usually those active against both aerobic and anaerobic bacteria. The choice of empirical antibiotics for the anaerobes in mixed infections can nearly always be made reliably, since patterns of antimicrobial susceptibility are usually predictable ([Chap. 137](#) and [Table 167-1](#)).

Antibiotic susceptibility testing of anaerobic bacteria has been difficult and controversial. Owing to the slow growth rate of many anaerobes, the lack of standardized testing methods and of clinically relevant standards for resistance, and the generally good results obtained with empirical therapy, susceptibility testing has been recommended only for the study of local or regional resistance patterns, for the prediction of the efficacy of new antibiotics, and for the management of selected patients.

Anaerobic gram-negative rods that are frequently resistant to penicillin are listed in [Table 167-2](#). Clinically important *Bacteroides* spp. are essentially all resistant to penicillin. Failures of therapy are common when documented *Bacteroides* (especially *B. fragilis*) infection is treated with penicillin or first-generation cephalosporins. The number of antimicrobial agents effective against *Bacteroides* spp. has expanded, and there are currently several useful choices ([Table 167-1](#)). In general, cure rates of >80% can be attained in patients with *Bacteroides* infection by means of appropriate antimicrobial therapy and drainage.

Resistance to metronidazole has been reported only rarely in *Bacteroides* spp. This well-tolerated drug, which reaches significant levels in serum and also can be found at high concentrations in abscess cavities, should be considered first-line therapy against *Bacteroides* infection. However, if metronidazole is used to treat mixed anaerobic and aerobic infections, it is imperative that other appropriate antibiotics be used in conjunction. Metronidazole is inactive against aerobic and facultative bacteria, *Actinomyces* spp., and *Propionibacterium* spp. The sensitivity of peptostreptococci to metronidazole is unpredictable, and penicillin remains the drug of choice.

If a patient fails to respond to one of the group 1 or group 2 drugs ([Table 167-1](#)), consideration should be given to alternative therapy and to determination of the resistance patterns among *Bacteroides* isolates. Although in vitro resistance of *Bacteroides* spp. to chloramphenicol has not been reported, this drug may not be as effective as other group 1 drugs. Ampicillin/sulbactam, ticarcillin/clavulanic acid, piperacillin/tazobactam, imipenem, and meropenem have been effective in the treatment of *B. fragilis* infection. Some newer fluoroquinolones, such as cinafloxacin, appear to be highly active against most anaerobes, including *B. fragilis*; however, ciprofloxacin and other earlier-generation quinolones should not be used as primary agents.

Treatment of Infections at Specific Sites In clinical situations, specific regimens must be tailored to the initial site of infection. The duration of therapy also depends on the infection site; the reader is referred to specific chapters on sites of infection for recommendations.

b-Lactamase production has been reported in anaerobic strains that are usually isolated from infections originating above the diaphragm. Up to 60% of clinical isolates classified as *Prevotella* or *Porphyromonas* spp., non-*B. fragilis* species of *Bacteroides*, or *Fusobacterium* spp. reportedly produce b-lactamase ([Table 167-2](#)). The clinical significance of resistance in these organisms has been suggested by studies showing clindamycin to be superior to penicillin (which for many years was considered the therapeutic gold standard) for the treatment of lung abscesses. Presumably, the success of clindamycin is attributable to a broader spectrum of activity against oral anaerobes; thus, a combination of penicillin and metronidazole or another antibiotic combination that is active against both oral anaerobes and aerobes is likely to be as effective as clindamycin. Bronchoscopy in lung abscess is indicated only to rule out airway obstruction and does not enhance drainage; in any event, it should be delayed until the antimicrobial regimen has begun to affect the disease process so that the procedure does not spread the infection. Surgery is almost never indicated because of the danger of spilling the abscess contents into the lungs.

Although most oral anaerobic infections and most cases of anaerobic pneumonia still respond to penicillin therapy, some infections due to oral organisms fail to respond to this drug, and in these cases the use of a drug that is effective against penicillin-resistant anaerobes is recommended ([Table 167-1](#)). Life-threatening infections involving the anaerobic flora of the mouth, such as space infections of the head and neck, should be treated empirically as if penicillin-resistant anaerobes are involved. Less serious infections involving the oral microflora can be treated with penicillin alone; metronidazole can be added (or clindamycin can be substituted) if the patient responds poorly to penicillin therapy. Combinations of antibiotics used to treat mixed infections of oral origin must include drugs active against the gram-positive aerobic flora of the mouth.

Chloramphenicol has been used successfully against anaerobic central nervous system infections at doses of 30 to 60 mg/kg per day, with the exact dose depending on the severity of illness. However, penicillin G and metronidazole also cross the blood-brain barrier and are bactericidal for many anaerobic organisms ([Chap. 372](#)).

Anaerobic infections arising below the diaphragm (e.g., colonic and intraabdominal infections) must be treated specifically with agents active against *Bacteroides* spp. (see [Table 167-1](#)). In intraabdominal sepsis ([Chap. 130](#)), the use of antibiotics effective against penicillin-resistant anaerobes has clearly reduced the incidence of postoperative infections and serious infectious complications. Specifically, a drug from group 1 ([Table 167-1](#)) must be included for broad-spectrum coverage. Recommended doses for commonly used group 1 drugs are given in [Table 167-3](#). Therapy for intraabdominal sepsis also must include drugs active against the gram-negative aerobic flora of the bowel. If the involvement of gram-positive bacteria such as enterococci is suspected, either ampicillin or vancomycin should be added.

Cases of anaerobic osteomyelitis in which a mixed flora is isolated from a bone biopsy specimen should be treated with a regimen that covers all the isolates. When an anaerobic organism is recognized as a major or sole pathogen infecting a joint, the duration of treatment should be similar to that used for arthritis caused by aerobic bacteria ([Chap. 323](#)). Therapy includes the management of underlying disease states, the administration of appropriate antimicrobial agents, temporary joint immobilization, percutaneous drainage of effusions, and usually the removal of infected prostheses or internal fixation devices. Surgical drainage and debridement procedures such as sequestrectomy are essential for the removal of necrotic tissue that can sustain anaerobic infections.

The outcome of anaerobic bacteremia has been shown to be significantly better in patients either initially given or switched to appropriate therapy based on known antibiotic susceptibilities.

Failure of Therapy Anaerobic infections that fail to respond to treatment or that relapse should be reassessed. Consideration should be given to additional surgical drainage or debridement. Superinfections with resistant gram-negative facultative or aerobic bacteria should be ruled out. The possibility of drug resistance must be entertained; if resistance is involved, repeated cultures may yield the pathogenic organism.

Supportive Measures Other supportive measures in the management of anaerobic infections include careful attention to fluid and electrolyte balance (since extensive local edema may lead to hypoalbuminemia); hemodynamic support for septic shock; immobilization of infected extremities; maintenance of adequate nutrition during chronic infections by parenteral hyperalimentation; relief of pain; and anticoagulation with heparin for thrombophlebitis. For patients with severe anaerobic infections of soft tissues, hyperbaric oxygen therapy is advocated by some experts, but its value has not been proven in controlled trials.

(Bibliography omitted in Palm version)

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SECTION 8 -MYCOBACTERIAL DISEASES

168. ANTIMYCOBACTERIAL AGENTS - Paul W. Wright, Richard J. Wallace, Jr.

The physician is greatly challenged to provide optimal therapy for mycobacterial illnesses because of the advent of AIDS, the increase in both drug-susceptible and multidrug-resistant tuberculosis, and the plethora of new antibiotics with antimycobacterial potential. This chapter reviews the agents used for the treatment of tuberculosis, leprosy (Hansen's disease), and diseases caused by pathogenic nontuberculous mycobacteria, including *Mycobacterium avium-intracellulare*, *M. kansasii*, the rapidly growing mycobacteria, and *M. marinum*. The use of antimycobacterial agents in patients with renal or hepatic disease and in pregnant women is summarized in [Table 168-1](#). The effects of major antimycobacterial agents on the levels, activity, and toxicity of other commonly used drugs are summarized in [Table 168-2](#).

TUBERCULOSIS

Drugs used to treat tuberculosis are classified as first-line and second-line agents. *First-line essential* antituberculous agents are the most effective and are a necessary component of any short-course therapeutic regimen. The three drugs in this category are rifampin, isoniazid, and pyrazinamide. The *first-line supplemental* agents, which are highly effective and infrequently toxic, include ethambutol and streptomycin. *Second-line* antituberculous drugs are clinically much less effective than first-line agents and much more frequently elicit severe reactions. These drugs are rarely used in therapy and then only by caregivers experienced with their use. They include para-aminosalicylic acid (PAS), ethionamide, cycloserine, kanamycin, amikacin, capreomycin, viomycin, and thiacetazone. *Newer* antituberculous drugs, which have not yet been placed in the above categories, include rifapentine, rifabutin, and the quinolones, especially ciprofloxacin, ofloxacin, and sparfloxacin.

FIRST-LINE ESSENTIAL DRUGS

Rifampin Rifampin, a semisynthetic derivative of *Streptomyces mediterranei*, is considered the most important and potent antituberculous agent. It is also active against a wide spectrum of other organisms, including some gram-positive and gram-negative bacteria, *Legionella* spp., *M. kansasii*, and *M. marinum*.

Pharmacology Rifampin is a fat-soluble complex macrocyclic antibiotic that is absorbed readily after either oral or intravenous administration. Serum levels of 10 to 20 ug/mL follow a standard oral dose of 600 mg. Rifampin distributes well throughout most body tissues, including inflamed meninges. The fact that rifampin turns body fluids (urine, saliva, sputum, tears) to a red-orange color makes it simple and inexpensive to check on a patient's compliance with therapy. Rifampin is excreted primarily through the bile and the enterohepatic circulation, while 30 to 40% of a dose is excreted via the kidneys. The drug is administered either twice weekly or daily at a dose of 600 mg for adults (10 mg/kg) and 10 to 20 mg/kg for children.

Mechanism of action Rifampin has both intracellular and extracellular bactericidal

activity. It blocks RNA synthesis by specifically binding and inhibiting DNA-dependent RNA polymerase. Susceptible strains of *M. tuberculosis* as well as *M. kansasii* and *M. marinum* are inhibited by ≤ 1 $\mu\text{g/mL}$.

Adverse effects ([Table 168-3](#)) Rifampin is generally well tolerated; the most common adverse event is gastrointestinal upset. Patients with chronic liver disease, especially those with alcoholism and the elderly, appear to be at unusually high risk for the most serious adverse reaction: hepatitis. Other adverse effects of rifampin include rash (0.8%), hemolytic anemia (<1%), thrombocytopenia, and immunosuppression of unknown clinical importance. Rifampin is a potent inducer of the hepatic microsomal enzymes and thereby decreases the half-life of a number of drugs, including digoxin, warfarin, prednisone, cyclosporine, methadone, oral contraceptives, clarithromycin, the HIV protease inhibitors, and quinidine ([Table 168-2](#)).

Resistance Resistance to rifampin results from spontaneous point mutations that alter the β subunit of the RNA polymerase (*rpoB*) gene. Studies have shown that 96% of rifampin-resistant strains have a missense mutation within a 91-bp central core region of the gene. Rifampin-resistant strains of *M. leprae* have similar mutations that alter a single serine residue (Ser-425) in the same core region of the *rpoB* gene.

Isoniazid Now considered the best antituberculous drug available after rifampin, isoniazid should be included in all tuberculosis treatment regimens unless the organism is resistant. Isoniazid is inexpensive, readily synthesized, available worldwide, highly selective for mycobacteria, and well tolerated, with only 5% of patients exhibiting adverse effects.

Mechanism of Action Isoniazid is the hydrazide of isonicotinic acid, a small water-soluble molecule that easily penetrates the cell. Its mechanism of action involves inhibition of mycolic acid cell-wall synthesis via oxygen-dependent pathways such as the catalase-peroxidase reaction. Isoniazid is bacteriostatic against resting bacilli and bactericidal against rapidly multiplying organisms, both extracellularly and intracellularly. The minimal inhibitory concentrations (MICs) of isoniazid for wild-type (untreated) strains of *M. tuberculosis* are <0.1 $\mu\text{g/mL}$, while those for *M. kansasii* are usually 0.5 to 2.0 $\mu\text{g/mL}$. The MICs of this drug for other mycobacteria are much higher.

Pharmacology Both oral and intramuscular preparations of isoniazid are readily absorbed. A 300-mg oral dose generally produces peak serum levels of 3 to 5 $\mu\text{g/mL}$. Isoniazid diffuses well throughout the body and reaches therapeutic concentrations in serum, cerebrospinal fluid (CSF), and infected tissue, including caseous granulomas. Isoniazid is metabolized in the liver via acetylation and hydrolysis; its metabolites are excreted into the urine. The rate of acetylation is genetically controlled. The recommended daily dose for the treatment of tuberculosis in the United States is 5 mg/kg for adults and 10 to 20 mg/kg for children, with a maximal daily dose of 300 mg for both groups. (Tuberculosis organizations outside the United States have recommended 5 mg/kg daily for both groups.) For intermittent therapy (usually directly observed), a maximal dose of 900 mg twice or thrice weekly is used. Even in moderate or severe renal failure, the adult dose rarely needs to be reduced below 200 mg/d.

Adverse Effects ([Table 168-3](#)) The two most important adverse effects of isoniazid

therapy are hepatotoxicity and peripheral neuropathy. Other adverse reactions are either rare or less significant and include rash (2%), fever (1.2%), anemia, acne, arthritic symptoms, a systemic lupus erythematosus-like syndrome, optic atrophy, seizures, and psychiatric symptoms. Isoniazid-associated hepatitis is idiosyncratic and increases in incidence with age. It occurs in 0.3% of treated persons under 35 years of age, 1.2% of those under 49 years of age, and 2.3% of those over 50 years of age. The risk of isoniazid-associated hepatitis is increased by daily alcohol consumption, concomitant rifampin administration, and slow isoniazid acetylation. Mortality rates from isoniazid-induced hepatitis have been reported to be 6 to 12%, but the real risk is certainly much lower: the reported rates were documented in high-risk patients who continued to take the drug despite progressive symptoms of hepatitis and without monitoring of liver enzyme levels. Liver enzymes are monitored in most settings among high-risk patients, and administration of the drug is discontinued at the onset of hepatitis. The American Thoracic Society (ATS) recommends that serum concentrations of aspartate aminotransferase (AST) or alanine aminotransferase (ALT) be determined at baseline in patients over 35 years of age who are receiving isoniazid for chemoprophylaxis, with monthly determinations thereafter. The benefit of such routine monitoring remains controversial, however. Measurement of the ALT or AST level is certainly mandatory whenever a patient notices the onset of symptoms suggestive of isoniazid-associated hepatitis (e.g., fever, anorexia, nausea, vomiting, and/or a flulike syndrome including fever and myalgias), and treatment should be discontinued until the relationship between therapy and symptoms is ascertained. Several studies have demonstrated that many patients with isoniazid intolerance can be desensitized. The ATS also recommends that discontinuation of isoniazid be strongly considered whenever an asymptomatic AST or ALT level exceeds 150 to 200 IU (three to five times the upper limit of normal) in high-risk patients whose baseline values were normal. In one study, only 11 (0.1%) of 11,141 patients had hepatotoxic reactions to isoniazid during preventive treatment.

Peripheral neuritis associated with isoniazid develops at a dose-dependent rate of 2 to 20% and probably relates to interference with pyridoxine (vitamin B₆) metabolism. This rate can be reduced to 0.2% with the prophylactic administration of 10 to 50 mg of pyridoxine daily.

Resistance Isoniazid-resistant mutants of *M. tuberculosis* occur spontaneously at a rate of 1 in 10⁵ to 10⁶ organisms. The molecular sites of isoniazid resistance have been detailed. Almost all isoniazid-resistant strains have amino acid changes in the catalase-peroxidase gene (*katG*) or a two-gene locus known as *inhA*. Missense mutations or deletion of *katG* is also associated with reduced catalase and peroxidase activity. Primary isoniazid resistance is detected in 7% of untreated patients in native U.S. populations, but the percentage is much higher in many immigrant populations.

Pyrazinamide A derivative of nicotinic acid, pyrazinamide is an important bactericidal drug used in short-course therapy for tuberculosis.

Pharmacology Pyrazinamide is well absorbed after oral administration, with a plasma concentration range of 20 to 60 µg/mL 1 to 2 h after oral ingestion of 15 to 30 mg/kg, and is well distributed throughout the body. Levels in [CSF](#) are excellent, reaching 50 to 100% of levels in serum. The serum half-life of the drug is 9 to 11 h. Pyrazinamide is

metabolized by at least two major pathways and one minor pathway in the liver; its several metabolites include pyrazinoic acid, 5-hydroxypyrazinamide, and 5-hydroxypyrazinoic acid.

Mechanism of Action Pyrazinamide is similar to isoniazid in its narrow spectrum of antibacterial activity, which essentially includes only *M. tuberculosis*. The drug is bactericidal to slowly metabolizing organisms located within the acidic environment of the phagocyte or caseous granuloma; it is active only at a pH of <6.0. Pyrazinamide is considered a prodrug and is converted by the tubercle bacillus to the active form pyrazinoic acid. The target for this compound, however, remains unknown. Susceptible strains of *M. tuberculosis* are inhibited by 20 ug/mL.

Adverse Effects ([Table 168-3](#)) At the high dosages used in the past, hepatotoxicity was a prominent complication of pyrazinamide therapy. However, at the currently recommended daily dosage of 15 to 30 mg/kg, with a maximum of 2 g (which can be given in one dose), the frequency of hepatotoxicity is no higher than that for concomitant isoniazid and rifampin therapy. Although pyrazinamide is recommended by international tuberculosis organizations for routine use in pregnancy, it is not recommended in the United States because of inadequate teratogenicity data. Hyperuricemia is a common adverse effect of pyrazinamide therapy whose incidence is probably reduced by concurrent rifampin therapy. Clinical gout is seen only rarely. Polyarthralgias are encountered fairly commonly but are not related to hyperuricemia.

Resistance Resistance to pyrazinamide is associated with loss of pyrazinamidase activity such that pyrazinamide is no longer converted to pyrazinoic acid. More than 90% of isolates with [MICs](#) of >100 ug/mL have mutations in the *pncA* gene, which encodes for pyrazinamidase. All strains of *M. bovis* are naturally resistant to pyrazinamide and have a point mutation within the *pncA* gene.

FIRST-LINE SUPPLEMENTAL DRUGS

Ethambutol A derivative of ethylenediamine, ethambutol is a water-soluble compound that is active only against mycobacteria. Susceptible species include *M. tuberculosis*, *M. marinum*, *M. kansasii*, and *M. avium-intracellulare* (MAI). Among first-line drugs, ethambutol is the least potent against *M. tuberculosis*. It is used most often with rifampin for the treatment of tuberculosis in patients who cannot tolerate isoniazid or who are thought or known to be infected with isoniazid-resistant organisms.

Mechanism of Action Ethambutol is bacteriostatic against rapidly growing mycobacteria. Its primary mechanism of action appears to be inhibition of arabinosyltransferases that mediate the polymerization of arabinose into arabinogalactan within the cell wall.

Pharmacology After oral administration, 75 to 80% of a dose of ethambutol is absorbed from the gastrointestinal tract. Serum levels peak at 2 to 4 ug/mL 2 to 4 h after a dose of 15 mg/kg. The drug's distribution throughout the body is adequate except in the [CSF](#), where it reaches only low levels. However, ethambutol can reach CSF levels up to 50% as high as peak plasma levels when administered at a daily dose of 25 mg/kg to a patient with inflamed meninges. Almost all of the dose is excreted by the kidneys within 24 h of ingestion, either unchanged or as metabolites. The usual daily adult dosage of

ethambutol is 25 mg/kg (which may be given in one dose) for the first 2 months, with a subsequent reduction to 15 mg/kg. In cases where pretreatment is necessary, the higher dose may be given for the duration. For intermittent therapy, the dosage is 50 mg/kg twice weekly or 30 mg/kg three times weekly. The dosage must be lowered for patients with renal insufficiency (a creatinine clearance rate of <25 mL/min) to prevent drug accumulation and toxicity.

Adverse Effects ([Table 168-3](#)) Ethambutol is usually well tolerated. Retrobulbar optic neuritis is the most serious adverse effect; axial or central neuritis -- the only form reported in patients taking daily doses of <30 mg/kg -- involves the papillomacular bundle of fibers and results in reduced visual acuity, central scotoma, and loss of the ability to see green. Symptoms of ocular toxicity typically develop several months after the initiation of therapy, but rapid-onset optic neuritis has been reported. The risk of optic neuritis depends on the dose and duration of therapy: this reaction develops in 5% of patients receiving a daily dose of 25 mg/kg but in fewer than 1% of patients given a daily dose of 15 mg/kg. Patients taking the lower dose should be tested at baseline and whenever there is a subjective visual change for visual acuity and red-green color discrimination. Patients taking the higher dose should be tested at baseline, monthly thereafter, and whenever there is a subjective visual change. Optic neuritis with associated visual loss is usually reversible, but recovery may take 6 months or longer.

Other adverse effects of ethambutol are infrequent. Hyperuricemia occurs but is usually asymptomatic. Optic neuritis is rare at the low dose in children; however, the use of ethambutol in very young children is problematic because visual complications are difficult to monitor.

Resistance Resistance in *M. tuberculosis* relates to missense mutations in the *embB* gene that encodes for arabinosyltransferase. Such mutations have been found in 70% of resistant strains and involve amino acid residue 306 in approximately 90% of cases. Species of nontuberculous mycobacteria that are intrinsically resistant to ethambutol have variant amino acids in this region of the gene, while susceptible species have the same amino acid sequences as *M. tuberculosis*.

Streptomycin An aminoglycoside isolated from *Streptomyces griseus*, streptomycin is available for intramuscular and intravenous administration only. In the United States, it is the least-used first-line supplemental drug for tuberculosis because of its toxicity, the difficulty in obtaining adequate [CSF](#) levels, and the inconvenience of its parenteral administration. In developing countries, however, streptomycin is frequently used because of its low cost. The drug is active against untreated strains of *M. tuberculosis*, *M. kansasii*, and *M. marinum* and against some strains of [MAI](#) at readily achievable serum levels.

Pharmacology Serum levels of streptomycin peak at 25 to 40 ug/mL after a 1.0-g dose. Streptomycin is bactericidal for rapidly dividing extracellular mycobacteria but is ineffective in the acidic environment within the macrophage. It diffuses poorly into the meninges and, in patients with meningitis, reaches [CSF](#) levels that are only 20% of serum levels.

The usual adult dose of streptomycin is 0.5 to 1.0 g (10 to 15 mg/kg) daily or five times

per week; the pediatric dose is 20 to 40 mg/kg daily, with a maximum of 1 g/d. Because streptomycin is eliminated almost exclusively by the kidneys, the dosage must be lowered and the frequency of administration reduced (to only two or three times per week) in most patients over 50 years of age and in any patient with renal impairment.

Mechanism of Action Streptomycin inhibits protein synthesis by disruption of ribosomal function.

Adverse Effects ([Table 168-3](#)) Adverse reactions to streptomycin therapy occur in 10 to 20% of patients. Ototoxicity and renal toxicity are the most common and the most serious. Renal toxicity, usually manifested as nonoliguric renal failure, is less common with streptomycin than with other frequently used aminoglycosides, such as gentamicin. Ototoxicity involves both hearing loss and vestibular dysfunction. The latter is more common and includes loss of balance, vertigo, and tinnitus. Patients receiving streptomycin must be monitored carefully for these adverse effects. Less serious reactions include perioral paresthesia, eosinophilia, rash, and drug fever.

Resistance Spontaneous resistance to streptomycin occurs in 1 in 10^5 to 10^7 organisms. In two-thirds of streptomycin-resistant strains of *M. tuberculosis*, mutations have been identified in one of two targets: a 16S rRNA gene (*rrs*) and the gene encoding ribosomal protein S12 (*rpsL*). Both targets are believed to be involved in streptomycin ribosomal binding. No mutational change has been identified in the other one-third of resistant isolates. Strains of *M. tuberculosis* that are resistant to streptomycin are not cross-resistant to capreomycin or amikacin.

SECOND-LINE DRUGS

Second-line and/or newer antituberculous agents are used either when tuberculosis is drug resistant or when first-line supplemental drugs are not available. The most important second-line drugs are discussed below in the general (descending) order of usefulness.

Capreomycin Capreomycin, a complex cyclic polypeptide antibiotic derived from *Streptomyces capreolus*, is similar to streptomycin in terms of dosing, mechanism of action, pharmacology, and toxicity. It is administered only by the intramuscular route in doses of 10 to 15 mg/kg daily or five times per week (maximal daily dose, 1 g), with peak blood levels of 20 to 40 ug/mL. After 2 to 4 months, the dosage should be reduced to 1 g two or three times a week. Cross-resistance to kanamycin and amikacin -- but not to streptomycin -- is common. After streptomycin, capreomycin is the injectable drug of choice for tuberculosis.

Amikacin and Kanamycin These well-known aminoglycosides are bactericidal to extracellular organisms. Kanamycin is rarely used because of its toxicity. Amikacin is active against *M. tuberculosis* and several of the nontuberculous species, including the rapidly growing mycobacteria, *M. scrofulaceum*, *M. leprae*, and [MAI](#). The usual adult dosage is 10 to 15 mg/kg intramuscularly or intravenously three to five times per week. Resistance to both drugs relates to a single-base-pair change at position 1408 in the 16S ribosomal RNA gene.

Para-Aminosalicylic Acid [PAS](#), a calcium or sodium salt that inhibits the growth of *M. tuberculosis* by impairing folate synthesis, is rarely indicated for the treatment of tuberculosis because of its low level of antituberculous activity and its high level of gastrointestinal toxicity (manifesting as nausea, vomiting, and diarrhea). Enteric-coated PAS granules (4 g every 8 h) may be better tolerated than other formulations and produce higher therapeutic blood levels. PAS is well absorbed after oral administration but reaches only low concentrations in the [CSF](#). The drug has a short half-life (1 h), and 80% of the dose is excreted in the urine.

Thiacetazone Also called amithiozone, thiacetazone is not available in the United States but -- because it is inexpensive and readily available -- is widely used in the developing world as a single-tablet combination with isoniazid to treat tuberculosis. The usual daily dosage is 150 mg. Thiacetazone is structurally related to isoniazid but is bacteriostatic and more toxic. The World Health Organization advises against the use of thiacetazone by HIV-infected patients because of an unacceptably high rate of severe adverse (gastrointestinal) and fatal (skin) reactions.

Viomycin A complex basic polypeptide antibiotic, viomycin has properties similar to those of capreomycin, amikacin, and kanamycin and must be administered by intramuscular injection. Ninety percent of strains of multidrug-resistant *M. tuberculosis* are inhibited by viomycin levels of 1 to 10 ug/mL. Toxic effects are more common and severe than with other polypeptide antibiotics. This drug is not available in the United States.

Ethionamide Like isoniazid and pyrazinamide, ethionamide is a derivative of isonicotinic acid. This agent is bacteriostatic against metabolizing *M. tuberculosis* and some nontuberculous mycobacteria. It is most useful in therapy for multidrug-resistant tuberculosis. However, its use is severely limited by its toxicity and frequent side effects, which include intense gastrointestinal intolerance (anorexia, vomiting, and dysgeusia), serious neurologic reactions, reversible hepatitis (5% of cases), hypersensitivity reactions, and hypothyroidism. Ethionamide is well absorbed orally and is widely distributed throughout the body at sites including the [CSF](#).

Cycloserine Cycloserine (D-4-amino-3-isoxazolidinone) is produced by *Streptomyces orchidaceus* and is active against a broad spectrum of bacteria, including *M. tuberculosis*. Cycloserine is well absorbed after oral administration and is widely distributed throughout the body fluids, including the [CSF](#). Serious side effects limit the use of this drug and include psychosis (with suicide in some cases), seizures, peripheral neuropathy, headaches, somnolence, and allergic reactions. Cycloserine should not be given to patients with epilepsy, active alcohol abuse, severe renal insufficiency, or a history of depression or psychosis.

Newer Antituberculous Drugs A number of other drugs are being evaluated for their antituberculous activity. This group includes rifabutin, rifapentine, the newer fluorinated quinolones, amoxicillin/clavulanic acid, clofazimine, clarithromycin, and rifamycins not yet approved by the U.S. Food and Drug Administration (FDA), such as KRM-1648 (benzoxazinorifamycin).

Rifabutin Rifabutin, a semisynthetic rifamycin spiropiperidyl derivative, shares many

characteristics with rifampin, including activity against *M. tuberculosis*. Rifabutin is also active against some strains of rifampin-resistant *M. tuberculosis* and is more active than rifampin against MAI and other nontuberculous mycobacteria. To date, rifabutin has been most useful in the prophylaxis of disseminated MAI infection and in the treatment of drug-resistant tuberculosis. Because it seems to exhibit more antituberculous activity than rifampin in vitro and in animals, its possible clinical advantages over rifampin are being evaluated. In a multinational trial in which either rifampin (600 mg/d) or rifabutin (150 mg/d) was administered in combination with isoniazid plus a 2-month regimen of pyrazinamide and ethambutol, the two rifamycins were equally effective and well tolerated in the treatment of newly diagnosed pulmonary tuberculosis. Rifabutin is recommended in place of rifampin for the treatment of HIV-positive individuals who are also taking a protease inhibitor.

PHARMACOLOGY The pharmacology of rifabutin is dramatically different from that of rifampin. Rifabutin is readily absorbed after a single oral dose of 300 mg and reaches peak serum levels (0.35 ug/mL) in 2 to 4 h. This lipophilic drug distributes best to tissues: tissue levels are 5 to 10 times higher than plasma levels. CSF concentrations are 30 to 70% of plasma levels in HIV-infected patients who have meningitis. The drug's slow clearance via hepatic metabolism and renal excretion results in a mean serum half-life of 45 h, which is much longer than the 3- to 5-h half-life of rifampin. Clarithromycin (but not azithromycin) and fluconazole appear to block the hepatic metabolism of rifabutin, with consequent increases in serum levels. When rifabutin is administered orally with food, its rate of absorption is slowed, but the extent of its absorption is unchanged. Adjustment of dosage is usually unnecessary in elderly patients and in patients with reduced hepatic or renal function.

MECHANISM OF ACTION In *Escherichia coli* and *Bacillus subtilis*, rifabutin inhibits DNA-dependent RNA polymerase in the same manner as rifampin. Its mode of action against mycobacteria is believed to be the same.

ADVERSE EFFECTS Most adverse effects of rifabutin are dose related and occur most frequently in patients receiving >300 mg/d. Discontinuation of therapy because of adverse drug reactions is reported in 16% of patients receiving rifabutin as opposed to 8% of those receiving a placebo. The most common symptoms are gastrointestinal; other reactions include rash, headache, asthenia, chest pain, myalgia, and insomnia. Like those taking rifampin, most patients taking rifabutin have discolored (orange to tan) urine and other body fluids. Less common adverse reactions include fever, chills, a flulike syndrome, hepatitis, *Clostridium difficile*-associated diarrhea, and a yellow skin discoloration ("pseudojaundice"). After a rifabutin dose of 450 to 600 mg in combination with clarithromycin, anterior uveitis is reported in up to 40% of patients; also common at these high doses are hyperpigmentation and the polymyalgia/arthritis syndrome. All of these conditions are reversible when treatment is discontinued. Laboratory abnormalities include neutropenia, leukopenia, thrombocytopenia, and increased levels of liver enzymes.

Rifabutin induces the hepatic cytochrome P450 enzymes but does so much less strongly than rifampin. Drugs whose metabolism is enhanced by rifabutin include anticoagulants, quinidine, oral contraceptives, sulfonamides, analgesics, dapsone, narcotics, glucocorticoids, clarithromycin, zidovudine, and cardiac glycosides.

RESISTANCE Resistance to rifabutin is attributable to the same mechanism as that to rifampin -- i.e., mutations involving the *rpoB* gene. However, of the 14 mutant *rpoB* alleles that confer resistance to rifampin, only nine confer high-level resistance to rifabutin, while the remaining five result in only small changes in rifabutin MICs, which remain ≤ 0.5 $\mu\text{g/mL}$. The MIC of rifabutin for susceptible strains of *M. tuberculosis* is low (< 0.06 $\mu\text{g/mL}$), and the drug is considered clinically active against partially resistant strains that are inhibited by plasma levels of ≤ 0.5 $\mu\text{g/mL}$. Thus rifabutin inhibits about one-quarter of rifampin-resistant strains of *M. tuberculosis*.

Rifapentine A semisynthetic cyclopentyl rifamycin antibiotic, rifapentine has received accelerated approval from the FDA for the treatment of tuberculosis. It is the first new drug approved for tuberculosis in 25 years in the United States. While similar to rifampin, rifapentine is lipophilic and longer acting -- characteristics that enhance patient compliance; the drug can be administered at a dose of 600 mg once or twice weekly. It has antibacterial activity against *M. tuberculosis* but has undergone only minimal testing against nontuberculous mycobacteria. Rifapentine has not yet been approved for the treatment of patients with HIV disease because rifapentine/rifampin monoresistance frequently develops in HIV-positive patients receiving isoniazid plus rifapentine. Like rifampin, rifapentine is active against many nonmycobacterial organisms, including *Haemophilus influenzae*, *Bordetella pertussis*, *B. parapertussis*, *Brucella* spp., *Legionella* spp., *Neisseria* spp., streptococci, and staphylococci.

In a randomized comparative study, 672 Chinese patients received isoniazid plus either rifapentine or rifampin. The isoniazid/rifapentine group had a higher relapse rate (10% versus 5%) than the isoniazid/rifampin group. Nevertheless, this disadvantage was considered acceptable in light of the lower rate of adverse effects and less frequent administration in the isoniazid/rifapentine group.

PHARMACOLOGY Food enhances the oral absorption of rifapentine, while antacids impair its absorption. After oral administration with food, this drug reaches peak serum concentrations in 5 to 6 h and achieves a steady state in 10 days. The half-life of rifapentine and its active metabolite 25-desacetyl rifapentine is approximately 13 h. The drug is highly bound to serum protein (93 to 97%), and most of the administered dose is excreted via the liver (70%). Oral clearance is more rapid in males than in females (2.51 vs 1.69 L/h), but the clinical significance of this difference is unknown.

MECHANISM OF ACTION Rifapentine exerts its bactericidal effect by inhibiting DNA-dependent RNA polymerase in susceptible bacteria. The MICs of rifapentine for rifampin-susceptible strains of *M. tuberculosis* range from 0.03 to 0.12 $\mu\text{g/mL}$.

ADVERSE EFFECTS Rifapentine demonstrates an adverse-event pattern similar to that of rifampin. Both drugs are frequently associated with hyperuricemia when administered with pyrazinamide and with elevated hepatocellular enzyme levels in 3 to 4% of patients when administered with other antituberculous agents. Liver enzyme levels should be monitored in patients receiving rifapentine who already have elevated liver enzyme concentrations or known liver disease. Like rifampin, rifapentine causes an orange-red discoloration of body fluids, including urine, saliva, and tears, and stains contact lenses.

Rifapentine induces the hepatic cytochrome P450 enzymes CYP3A4 and 2C8/9. Current induction studies suggest that its potential for drug-drug interaction may be less than that of rifampin but greater than that of rifabutin. Other drugs potentially affected by concomitant administration of rifapentine are listed in [Table 168-2](#).

Rifapentine is in category C for use in pregnancy ([Table 168-1](#)) because of its teratogenesis in rats and rabbits. There are insufficient data concerning use of this drug in pregnant and breast-feeding patients.

RESISTANCE Strains of *M. tuberculosis* resistant to rifapentine, rifampin, and rifabutin all involve spontaneous point mutations in the *rpoB* gene. All strains resistant to rifampin are also resistant to rifapentine.

Quinolones A surprisingly large number of fluorinated quinolones are being developed and studied as inhibitors of mycobacteria. Their mode of action presumably is the prevention of DNA synthesis through the inhibition of DNA gyrase. Ofloxacin, ciprofloxacin, sparfloxacin, and pefloxacin are active against many mycobacteria, including *M. tuberculosis*, *M. leprae*, *M. marinum*, *M. kansasii*, and *M. fortuitum*. These drugs are well absorbed orally, reach high serum levels, and distribute well to body tissues and fluids. While not approved for antituberculous therapy in the United States, ofloxacin -- used in combination with isoniazid and rifampin for the treatment of pulmonary tuberculosis -- has been as active and safe as ethambutol in initial trials. Adverse effects are relatively uncommon, occurring in 0.5 to 10% of cases and consisting mostly of benign reactions such as gastrointestinal intolerance, rashes, dizziness, and headache. However, more serious adverse effects are being reported and include confusion, seizures, interstitial nephritis, skin vasculitis, and acute renal failure.

Mycobacterial resistance to the fluoroquinolones develops rapidly. Its molecular basis is complex; only some strains exhibit missense mutations in the A subunit (*gyrA* gene) of DNA gyrase. Fluoroquinolone-resistant tuberculosis is a source of growing concern: 22 such cases were reported recently from New York City. Antituberculous therapy with quinolones should be reserved for patients with multidrug resistance or those who cannot tolerate first-line drugs.

LEPROSY (HANSEN'S DISEASE)

Therapy for leprosy remains difficult, especially in developing countries, because of the long course required, the high cost and low availability of most drugs, the frequency of adverse reactions to drugs, the acquisition of drug resistance, the difficulty of determining a disease end point or cure, and (given that *M. leprae* still cannot be grown in vitro) the difficulty of conducting susceptibility testing. While many drugs are active against *M. leprae*, efficacy in the treatment of leprosy has been established only for dapsone, rifampin, clofazimine, and ethionamide.

Rifampin Rifampin is considered the most active agent for the treatment of leprosy. Its worldwide use is limited only by its cost. This drug is markedly bactericidal against *M. leprae* and reduces the number of viable bacilli in the patient's tissues faster than any other available agent. Rifampin must be combined with other antileprosy drugs to

forestall resistance. For cost reasons, the drug is given at a dose of 600 mg once a month (supervised) outside the United States, but it is given daily in the United States. For details on pharmacology, adverse events, and resistance, see relevant sections under "Tuberculosis."

Dapsone Dapsone (4,4'-diaminodiphenylsulfone) inhibits bacterial folic acid synthesis. It is now considered the second drug of choice (after rifampin) in most cases of Hansen's disease because of its ready availability, low cost, and low toxicity and the susceptibility of untreated strains of *M. leprae* to very low concentrations.

Pharmacology Dapsone is well absorbed orally and distributes well throughout the body. The usual daily dosage is 100 mg for adults and 0.9 to 1.4 mg/kg for children. Plasma concentrations peak within 1 to 3 h. The median elimination half-life is 22 h. Dapsone is cleared by acetylation in the liver, with genetic variation similar to that documented for the acetylation of isoniazid. The drug is 70% bound to plasma protein. Usual daily doses produce serum concentrations of 10 to 15 ug/mL, which far exceed the [MIC](#) for *M. leprae* (0.01 to 0.001 ug/mL).

Adverse Effects Hemolysis and methemoglobinemia are common untoward reactions to dapsone. Patients should be screened for glucose-6-phosphate dehydrogenase deficiency to prevent drug-induced hemolysis. However, most patients tolerate dapsone therapy well with adequate clinical and laboratory supervision. Other side effects include gastrointestinal intolerance, headache, pruritus, peripheral neuropathies, nephritic syndrome, fever, and rash. In lepromatous and borderline lepromatous leprosy, erythema nodosum leprosum (ENL) may occur. This reaction may be difficult to distinguish from reactions of leprosy, including drug reactions and the infectious mononucleosis-like dapsone syndrome.

Clofazimine A phenazine iminoquinone dye, clofazimine is weakly bactericidal against *M. leprae*. It is useful in treating dapsone-resistant leprosy and may lessen the severity of [ENL](#). Clofazimine's mode of action is not well understood, but the drug may inhibit DNA binding. It is absorbed orally and distributed to the fatty tissues and the reticuloendothelial system. Its serum half-life is about 60 to 70 days; only a small proportion of the dose is excreted daily into the urine or bile. Bactericidal activity is very slow and is evident for about 50 days after administration. The usual adult dosage is 50 to 100 mg/d, 100 mg three times a week, or (for treatment of ENL) 300 mg/d. Untoward effects include skin discoloration and, less commonly, gastrointestinal intolerance. Clofazimine was reported to be responsible for a case of cardiotoxicity induced via ventricular arrhythmia. Even though clofazimine-resistant disease has been reported only rarely when this agent is used alone, it should be used with other effective antibiotics. Clofazimine is active in vitro against some nontuberculous mycobacterial species, including [MAI](#), *M. kansasii*, *M. simiae*, and *M. abscessus*.

Ethionamide While ethionamide (250 mg/d) has not been approved by the [FDA](#) for the treatment of leprosy, it is sometimes used in the United States in combination with rifampin (600 mg/d) to treat dapsone-resistant leprosy in patients who cannot accept the skin-depigmentation effect of clofazimine. Because resistance to ethionamide develops quickly when the drug is used alone, it must be used with other effective agents. Patients should be monitored closely for hepatotoxicity when taking ethionamide

(especially in combination with rifampin), and treatment should be discontinued if the patient's [ALT](#) levels exceed 2.5 times the normal value. Prothionamide, a congener of ethionamide that is not available in the United States, has pharmacologic properties similar to those of ethionamide and is widely used throughout the world.

Other Agents A number of other drugs exhibit significant activity against *M. leprae*, but clinical experience with these agents is lacking. Thalidomide is now approved by the [FDA](#) for treatment of [ENL](#). Although this drug may be useful in suppressing ENL, it acts as a tranquilizer, is extremely teratogenic, and should *never* be taken by anyone who is or may become pregnant. Physicians wishing to prescribe thalidomide must register with the System for Thalidomide Education and Prescription Safety (S.T.E.P.S.) at 1-888-423-5436 (Celgene Corporation).

The newer macrolide antibiotics (particularly clarithromycin), minocycline (a long-acting tetracycline), and a number of fluoroquinolones (including ofloxacin, sparfloxacin, and pefloxacin) have shown promising bactericidal activity against *M. leprae*. Ofloxacin and minocycline are being investigated with rifampin in short-course regimens for lepromatous disease. All of these newer leprosy drugs have low toxicity profiles, modes of action different from those of the established agents, and powerful bactericidal activity against *M. leprae*. However, their levels of bactericidal activity are lower than that of rifampin.

NONTUBERCULOUS MYCOBACTERIA

Although less pathogenic than *M. tuberculosis*, the nontuberculous mycobacteria can cause pulmonary, skin, bone and joint, lymph node, and soft tissue infection as well as disseminated disease in immunocompromised hosts, including patients with AIDS. [MAI](#) and *M. kansasii* are the two most common causes of nontuberculous mycobacterial pulmonary infection. Up to 40% of AIDS patients develop disseminated disease due to MAI.

Clarithromycin Clarithromycin (6-O-methylerythromycin) is a new macrolide that is similar to erythromycin in its mechanism of action. However, unlike erythromycin, it is well absorbed with or without meals and elicits little gastrointestinal intolerance at low doses. Clarithromycin distributes well into body tissues and fluids and is highly concentrated in macrophages. The drug is metabolized in the liver, and approximately 30% of a given dose is excreted in the urine. The dosage should be reduced if the creatinine clearance rate is ≤ 30 mL/min. Like erythromycin, clarithromycin binds with plasma proteins (65 to 70%) and can raise the levels of drugs such as theophylline and carbamazepine. As noted earlier, serum levels of clarithromycin are reduced by the concomitant administration of rifampin and to a lesser degree by that of rifabutin; clarithromycin treatment increases serum levels of rifabutin and some antihistamines (e.g., terfenadine), thus increasing their toxicity. Clarithromycin and (probably) azithromycin are the most active agents for the treatment of [MAI](#) infections; one of these drugs is considered an essential component of any regimens for this purpose. However, because of mutational drug resistance, clarithromycin should be given in combination with other agents, such as ethambutol and rifampin or rifabutin. The drug is also highly active against almost all other nontuberculous mycobacteria, including *M. marinum*, *M. kansasii*, *M. haemophilum*, *M. genavense*, *M. xenopi*, *M. abscessus*, *M. chelonae*, and

most isolates of *M. fortuitum*. Standard antimycobacterial doses have been 500 mg twice daily; doses of 1000 mg twice daily have been associated with increased mortality among patients with AIDS and disseminated MAI disease. The more common side effects of high doses include nausea, vomiting, a bitter taste, and (occasionally) abnormal liver-function tests. Most side effects can be minimized by reducing the dose, usually by 50%. Clarithromycin is teratogenic in laboratory animals and is in category C for use in pregnancy ([Table 168-1](#)). Mutational resistance occurs in one in 10⁸ to 10⁹ organisms and develops rapidly with monotherapy, especially that for disseminated MAI disease. Resistance results from point mutations involving adenine at positions 2058 or 2059 in the 23S ribosomal binding site.

Azithromycin Azithromycin is a macrolide that belongs to the family of azalides. It reaches much lower serum levels than clarithromycin (usually £0.5 ug/mL) but attains high tissue and macrophage concentrations and has a longer half-life, which suggests the feasibility of intermittent therapy. It is involved in few drug interactions since it does not affect the cytochrome P450 system. The usual dose is 250 to 500 mg/d. No alteration in dose is required in renal failure. The most common side effects are gastrointestinal symptoms and reversible hearing loss. Azithromycin appears to be less active than clarithromycin for both pulmonary and disseminated [MAI](#) disease. Resistance to azithromycin develops by the same mechanism as that to clarithromycin, with cross-resistance between the two macrolides.

THERAPY FOR SPECIFIC NONTUBERCULOUS MYCOBACTERIA

MAI First-line antituberculous drugs are much less active against MAI than against *M. tuberculosis*. Therapy for MAI is controversial because of the lack of controlled clinical trials. In 1990 the [ATS](#) recommended the following four-drug regimen for MAI lung disease in HIV-negative patients: 18 to 24 months of isoniazid (300 mg), rifampin (600 mg), and ethambutol (25 mg/kg, then 15 mg/kg beginning in the third month), with intermittent streptomycin. However, two subsequent events -- the demonstration of the dramatic activity of clarithromycin against both pulmonary and disseminated MAI infection and the introduction of rifabutin -- have altered the therapeutic approach to MAI infection. In the 1997 ATS recommendations for MAI lung disease, clarithromycin (500 mg twice daily) now replaces isoniazid, and rifabutin (300 mg/d) is often used in place of rifampin. Therapy for pulmonary disease is generally given until cultures have been negative for 12 months.

For disseminated disease in AIDS, one of the newer macrolides (clarithromycin or azithromycin) and ethambutol (15 mg/kg) are considered essential components of any treatment regimen, with rifabutin (300 mg) a commonly used third drug in patients not taking a protease inhibitor for their HIV infection. Other alternative drugs include ciprofloxacin, streptomycin, and amikacin. Clofazimine appears to increase mortality and should be avoided. For the prophylaxis of disseminated [MAI](#) disease, rifabutin (300 mg/d), clarithromycin (500 mg twice daily), and azithromycin (1200 mg once weekly) have all been demonstrated to be effective in controlled or comparative clinical trials.

Mycobacterium kansasii *M. kansasii* is usually susceptible to most antituberculous drugs except for pyrazinamide. Current [ATS](#) recommendations for the treatment of *M. kansasii* pulmonary disease are 18 to 24 months of daily isoniazid (300 mg), rifampin

(600 mg), and ethambutol (15 mg/kg). In patients taking protease inhibitors, rifabutin (150 mg) or clarithromycin (500 mg) twice daily should be substituted for rifampin. The potential advantages of the highly active rifabutin and the newer macrolides in immunocompetent patients have not been studied.

Rapidly Growing Mycobacteria *M. fortuitum*, *M. abscessus*, and *M. chelonae* account for more than 80% of cases of clinical disease due to rapidly growing mycobacteria. These organisms are resistant to antituberculous agents other than amikacin but are variably susceptible to several other antibiotics. Clarithromycin has dramatically changed the approach to therapy for infection with these organisms, as it inhibits all rapidly growing mycobacteria -- except for 20% of *M. fortuitum* strains and most *M. smegmatis* strains -- at concentrations of £4 ug/mL. Other drugs with good activity include amikacin (which inhibits 80 to 100% of strains), cefoxitin (80% of *M. abscessus* and *M. fortuitum* strains), doxycycline (50% of *M. fortuitum* strains), imipenem (100% of *M. fortuitum* strains, 70% of *M. chelonae* strains, and 70% of *M. abscessus* strains), the fluorinated quinolones ciprofloxacin and ofloxacin (100% of *M. fortuitum* strains), and sulfonamides (90% of *M. fortuitum* strains).

Mycobacterium marinum *M. marinum*, a photochromogen, is typically susceptible to minocycline, rifampin, ethambutol, clarithromycin, and trimethoprim-sulfamethoxazole and is resistant to isoniazid.

Mycobacterium haemophilum Infection due to *M. haemophilum* occurs most commonly as disseminated disease in immunocompromised patients with or without AIDS. This organism can cause bone and joint infection and skin infection. Isolates typically show in vitro resistance to most drugs but may be susceptible to rifampin, rifabutin, quinolones, and clarithromycin.

Mycobacterium xenopi In the United States, *M. xenopi* most often causes nosocomial infections; these infections most commonly occur in the environment of the hospital's hot-water system. In one study from Brooklyn, NY, *M. xenopi* was the second most common pathogenic nontuberculous mycobacterial species; of the 86 hospitalized patients from whom it was isolated, 41% were HIV-positive. Drug therapy for *M. xenopi* infection is difficult because in vitro drug sensitivity tests do not reliably predict clinical results. *M. xenopi* is often resistant to first-line antituberculous agents but susceptible to the newer macrolides, quinolones, streptomycin, and ethionamide.

Mycobacterium genavense *M. genavense* is a newly recognized organism that grows only in liquid media, such as Bactec 12B or 13A. This organism almost exclusively infects AIDS patients, causing disseminated disease and being isolated from blood, bone marrow, liver, lymph node, spleen, and intestinal cultures. The in vitro susceptibility profile of *M. genavense* has not been well established. Some isolates are susceptible to amikacin, clarithromycin, ofloxacin, rifampin, and rifabutin.

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169. TUBERCULOSIS - Mario C. Raviglione, Richard J. O'Brien

DEFINITION

Tuberculosis, one of the oldest diseases known to affect humans, is caused by bacteria belonging to the *Mycobacterium tuberculosis* complex. The disease usually affects the lungs, although in up to one-third of cases other organs are involved. If properly treated, tuberculosis caused by drug-susceptible strains is curable in virtually all cases. If untreated, the disease may be fatal within 5 years in more than half of cases. Transmission usually takes place through the airborne spread of droplet nuclei produced by patients with infectious pulmonary tuberculosis.

ETIOLOGIC AGENT

Mycobacteria belong to the family Mycobacteriaceae and the order Actinomycetales. Of the pathogenic species belonging to the *M. tuberculosis* complex, the most frequent and important agent of human disease is *M. tuberculosis* itself. The complex includes *M. bovis* (the bovine tubercle bacillus, once an important cause of tuberculosis transmitted by unpasteurized milk and currently the cause of a small percentage of cases in developing countries), *M. africanum* (isolated in a small proportion of cases in West and Central Africa), and *M. microti* (the "vole" bacillus, a closely related but rarely encountered organism).

M. tuberculosis is a rod-shaped, non-spore-forming, thin aerobic bacterium measuring about 0.5 μm by 3 μm . Mycobacteria, including *M. tuberculosis*, do not stain readily and are often neutral on Gram's staining. However, once stained, the bacilli cannot be decolorized by acid alcohol, a characteristic justifying their classification as acid-fast bacilli (AFB). Acid fastness is due mainly to the organisms' high content of mycolic acids, long-chain cross-linked fatty acids, and other cell-wall lipids. Microorganisms other than mycobacteria that display some acid fastness include species of *Nocardia* and *Rhodococcus*, *Legionella micdadei*, and the protozoa *Isospora* and *Cryptosporidium*. In the mycobacterial cell wall, lipids (e.g., mycolic acids) are linked to underlying arabinogalactan and peptidoglycan. This structure is responsible for the very low permeability of the cell wall and thus for the ineffectiveness of most antibiotics against the organism. Another molecule in the mycobacterial cell wall, lipoarabinomannan, is involved in the pathogen-host interaction and facilitates the survival of *M. tuberculosis* within macrophages. The several proteins characteristic of *M. tuberculosis* include those in purified protein derivative (PPD) tuberculin, a precipitate of non-species-specific molecules obtained from filtrates of heat-sterilized, concentrated broth cultures. The complete genome sequence of *M. tuberculosis* comprises about 4000 genes and has a high guanine-plus-cytosine content. A large proportion of the genes are devoted to the production of enzymes involved in lipogenesis and lipolysis and of glycine-rich proteins that are probably responsible for antigenic variations.

EPIDEMIOLOGY

Between 3.5 and 4 million new cases of tuberculosis -- all forms (pulmonary and extrapulmonary), 90% of them from developing countries -- were reported annually to the World Health Organization (WHO) in the late 1990s. However, because of a low

level of case detection and incomplete notifications in many national programs, reported cases represent only a fraction of the total. It is estimated that 8 million new cases of tuberculosis occurred worldwide in 1997, 95% of them in developing countries of Asia (5 million), Africa (1.6 million), the Middle East (0.6 million), and Latin America (0.4 million). It is also estimated that nearly 2 million deaths from tuberculosis occurred in 1997, 98% of them in developing countries.

Beginning in the mid-1980s in many industrialized countries, the number of tuberculosis case notifications, which had been falling steadily, stabilized or even began to increase. This phenomenon was first noted in the United States but was soon observed in many European countries as well. A number of factors were implicated in the resurgence of tuberculosis in the United States in 1986 through 1992 -- most notably, immigration from countries with a high prevalence of tuberculosis; infection with HIV; emergence of multidrug-resistant (MDR) tuberculosis due to strains resistant at least to isoniazid and rifampin; and social problems such as poverty, homelessness, and drug abuse. In some areas (e.g., New York City), deterioration in the public health system and dismantling of tuberculosis management services also contributed to the worsening situation. With the implementation of stronger control programs, cases began to decrease in 1993. In 1998, 18,361 cases of tuberculosis (6.8 per 100,000 population) were reported to the U.S. Centers for Disease Control and Prevention (CDC) -- a 31% decrease from the 1992 peak.

In the United States, tuberculosis is uncommon among young adults of European descent, who have only rarely been exposed to *M. tuberculosis* infection during recent decades. In contrast, because of a high risk in the past, the prevalence of *M. tuberculosis* infection is relatively high among elderly Caucasians, who remain at increased risk of developing active tuberculosis. Tuberculosis in the United States is also a disease of young adult members of the HIV-infected, immigrant, and disadvantaged/marginalized populations. Similarly, in Europe, tuberculosis has reemerged as an important public health problem, mainly as a result of cases among immigrants from high-prevalence countries.

In developing countries of Africa and Asia, tuberculosis trends over the past several decades are not entirely clear. However, in sub-Saharan African countries with reliable reporting systems, the recent spread of the HIV epidemic has been accompanied by doubling or tripling of the number of reported cases of tuberculosis during a period as short as 10 years. The growing number of young adults with *M. tuberculosis* infection has fueled the rates of active tuberculosis in many developing countries. Without greater control efforts, the annual incident cases of tuberculosis globally may increase by 40% between now and 2020.

From Exposure to Infection *M. tuberculosis* is most commonly transmitted from a patient with infectious pulmonary tuberculosis to other persons by droplet nuclei, which are aerosolized by coughing, sneezing, or speaking. The tiny droplets dry rapidly; the smallest (<5 to 10 μ m in diameter) may remain suspended in the air for several hours and may gain direct access to the terminal air passages when inhaled. There may be as many as 3000 infectious nuclei per cough. In the past, a frequent source of infection was raw milk containing *M. bovis* from tuberculous cows. Other routes of transmission of tubercle bacilli, such as through the skin or the placenta, are uncommon and of no

epidemiologic significance.

The probability of contact with a case of tuberculosis, the intimacy and duration of that contact, the degree of infectiousness of the case, and the shared environment of the contact are all important determinants of transmission. Several studies of close contacts have clearly demonstrated that tuberculosis patients whose sputum contains [AFB](#) visible by microscopy play the greatest role in the spread of infection. These patients often have cavitary pulmonary disease or tuberculosis of the respiratory tract (endobronchial or laryngeal tuberculosis) and produce sputa containing as many as 10⁵ AFB/mL. Patients with sputum smear-negative/culture-positive tuberculosis are less infectious, and those with culture-negative pulmonary disease and extrapulmonary tuberculosis are essentially noninfectious. Crowding in poorly ventilated rooms is one of the most important factors in the transmission of tubercle bacilli, since it increases the intensity of contact with a case.

In short, the risk of acquiring *M. tuberculosis* infection is determined mainly by exogenous factors. Because of delays in seeking care and in diagnosis, it is estimated that up to 20 contacts will usually be infected by each [AFB](#)-positive case before detection in high-prevalence countries.

From Infection to Disease Unlike the risk of acquiring infection with *M. tuberculosis*, the risk of developing disease after being infected depends largely on endogenous factors, such as the individual's innate susceptibility to disease and level of function of cell-mediated immunity. Clinical illness directly following infection is classified as *primary tuberculosis* and is common among children up to 4 years of age. Although this form may be severe and disseminated, it is usually not transmissible. When infection is acquired later in life, the chance is greater that the immune system will contain it, at least temporarily. The majority of infected individuals who ultimately develop tuberculosis do so within the first year or two after infection. Dormant bacilli, however, may persist for years before being reactivated to produce *secondary (or postprimary) tuberculosis*, which is often infectious. Overall, it is estimated that about 10% of infected persons will eventually develop active tuberculosis. *Reinfection* of a previously infected individual, which is probably common in areas with high rates of tuberculosis transmission, may also favor the development of disease. Molecular typing and comparison of strains of *M. tuberculosis* have suggested that up to one-third of cases of active tuberculosis in U.S. inner-city communities are due to recent transmission rather than to reactivation of latent infection.

Age is an important determinant of the risk of disease after infection. Among infected persons, the incidence of tuberculosis is highest during late adolescence and early adulthood; the reasons are unclear. The incidence among women peaks at 25 to 34 years of age. In this age group rates among women are usually higher than those among men, while at older ages the opposite is true. The risk may increase in the elderly, possibly because of waning immunity and comorbidity.

A variety of diseases favor the development of active tuberculosis. The most potent risk factor for tuberculosis among infected individuals is clearly HIV co-infection, which suppresses cellular immunity. The risk that latent *M. tuberculosis* infection will proceed to active disease is directly related to the patient's degree of immunosuppression. In a

study of HIV-infected, [PPD](#)-positive persons, this risk varied from 2.6 to 13.3 cases per 100 person-years and depended upon the CD4+ cell count. The risk of developing tuberculosis is significantly greater among HIV-infected than among HIV-uninfected hosts. Other conditions known to increase the risk of active tuberculosis among persons infected with tubercle bacilli include silicosis; lymphoma, leukemia, and other malignant neoplasms; hemophilia; chronic renal failure and hemodialysis; insulin-dependent diabetes mellitus; immunosuppressive treatment, including that administered for solid-organ transplantation; and conditions associated with malnutrition, such as gastrectomy and jejunoileal bypass surgery. Finally, the presence of old, self-healed, fibrotic tuberculous lesions constitutes a serious risk of active disease.

NATURAL HISTORY OF DISEASE

Studies conducted in various countries before the advent of chemotherapy clearly showed that untreated tuberculosis is often fatal. About one-third of patients died within 1 year after diagnosis, and one-half within 5 years. Five-year mortality among sputum smear-positive cases was 65%. Of the survivors at 5 years, about 60% had undergone spontaneous remission, while the remainder were still excreting tubercle bacilli.

The introduction of effective chemotherapy has markedly affected the natural history of tuberculosis. With proper treatment, patients have a high chance of being cured. However, improper use of antituberculosis drugs, while reducing mortality, may also result in large numbers of chronic infectious cases, often with drug-resistant bacilli.

PATHOGENESIS AND IMMUNITY

The interaction of *M. tuberculosis* with the human host begins when droplet nuclei containing microorganisms from infectious patients are inhaled. While the majority of inhaled bacilli are trapped in the upper airways and expelled by ciliated mucosal cells, a fraction (usually fewer than 10%) reach the alveoli. There, nonspecifically activated alveolar macrophages ingest the bacilli. Invasion of macrophages by mycobacteria may result in part from association of C2a with the bacterial cell wall followed by C3b opsonization of the bacteria and recognition by the macrophages. The balance between the bactericidal activity of the macrophage and the virulence of the bacillus (the latter being partially linked to the bacterium's lipid-rich cell wall and to its glycolipid capsule, both of which confer resistance to complement and free radicals of the phagocyte) determines the events following phagocytosis. The number of invading bacilli is also important.

Several genes thought to confer virulence to *M. tuberculosis* have been identified. *katG* encodes for catalase, an enzyme protective against oxidative stress; *rpoV* is the main sigma factor initiating transcription of several genes. Defects in these two genes result in loss of virulence. The *erp* gene, encoding a protein required for multiplication, also contributes to virulence. The effects of a highly virulent strain are exemplified by an outbreak of tuberculosis in two rural counties in Tennessee and Kentucky in 1994 through 1996. In this outbreak, both epidemiologic evidence of enhanced transmission with high rates of disease and accelerated growth of the strain in mice were documented.

Several observations suggest that genetic factors play a key role in innate nonimmune resistance to infection with *M. tuberculosis*. The existence of this resistance is suggested by the differing degrees of susceptibility to tuberculosis in different populations. In mice, a gene called *Nramp1* (natural resistance-associated macrophage protein 1) has a regulatory role in resistance/susceptibility to mycobacteria. The human homologue NRAMP1, cloned to chromosome 2q, may have a role in determining susceptibility to tuberculosis. In a study among West Africans, subjects heterozygous for two polymorphisms of NRAMP1 (INT4 and 3'UTR) had an apparently increased risk of tuberculosis, a finding suggesting that the susceptibility allele behaves as dominant.

In the initial stage of host-bacterium interaction, either the host's macrophages contain bacillary multiplication by producing proteolytic enzymes and cytokines or the bacilli begin to multiply. If the bacilli multiply, their growth quickly kills the macrophages, which lyse. Nonactivated monocytes attracted from the bloodstream to the site by various chemotactic factors ingest the bacilli released from the lysed macrophages. These initial stages of infection are usually asymptomatic.

About 2 to 4 weeks after infection, two additional host responses to *M. tuberculosis* develop: a tissue-damaging response and a macrophage-activating response. The *tissue-damaging response* is the result of a delayed-type hypersensitivity (DTH) reaction to various bacillary antigens; it destroys nonactivated macrophages that contain multiplying bacilli. The *macrophage-activating response* is a cell-mediated phenomenon resulting in the activation of macrophages that are capable of killing and digesting tubercle bacilli. Although both of these responses can inhibit mycobacterial growth, it is the balance between the two that determines the form of tuberculosis that will develop subsequently.

With the development of specific immunity and the accumulation of large numbers of activated macrophages at the site of the primary lesion, granulomatous lesions (tubercles) are formed. These lesions consist of lymphocytes and activated macrophages, such as epithelioid cells and giant cells. Initially, the newly developed tissue-damaging response is the only event capable of limiting mycobacterial growth within macrophages. This response, mediated by various bacterial products, not only destroys macrophages but also produces early solid necrosis in the center of the tubercle. Although *M. tuberculosis* can survive, its growth is inhibited within this necrotic environment by low oxygen tension and low pH. At this point, some lesions may heal by fibrosis and calcification, while others undergo further evolution.

Cell-mediated immunity is critical at this early stage. In the majority of infected individuals, local macrophages are activated when bacillary antigens processed by macrophages stimulate T lymphocytes to release a variety of lymphokines. These activated cells aggregate around the lesion's center and effectively neutralize tubercle bacilli without causing further tissue destruction. In the central part of the lesion, the necrotic material resembles soft cheese (*caseous necrosis*). Even when healing takes place, viable bacilli may remain dormant within macrophages or in the necrotic material for years or even throughout the patient's lifetime. These "healed" lesions in the lung parenchyma and hilar lymph nodes may later undergo calcification (*Ranke complex*).

In a minority of cases, the macrophage-activating response is weak, and mycobacterial

growth can be inhibited only by intensified [DTH](#) reactions, which lead to tissue destruction. The lesion tends to enlarge further, and the surrounding tissue is progressively damaged. At the center of the lesion, the caseous material liquefies. Bronchial walls as well as blood vessels are invaded and destroyed, and cavities are formed. The liquefied caseous material, containing large numbers of bacilli, is drained through bronchi. Within the cavity, tubercle bacilli multiply well and spread into the airways and the environment through expectorated sputum.

In the early stages of infection, bacilli are usually transported by macrophages to regional lymph nodes, from which they disseminate widely to many organs and tissues. The resulting lesions may undergo the same evolution as those in the lungs, although most tend to heal. In young children with poor natural immunity, hematogenous dissemination may result in fatal miliary tuberculosis or tuberculous meningitis.

Cell-mediated immunity confers partial protection against *M. tuberculosis*, while humoral immunity has no defined role in protection. Two types of cells are essential: macrophages, which directly phagocytize tubercle bacilli, and T cells (mainly CD4+ lymphocytes), which induce protection through the production of lymphokines.

After infection with *M. tuberculosis*, alveolar macrophages secrete a number of cytokines: interleukin (IL) 1 contributes to fever; IL-6 contributes to hyperglobulinemia; and tumor necrosis factor α (TNF- α) contributes to the killing of mycobacteria, the formation of granulomas, and a number of systemic effects, such as fever and weight loss. Macrophages are also critical in processing and presenting antigens to T lymphocytes; the result is a proliferation of CD4+ lymphocytes, which are crucial to the host's defense against *M. tuberculosis*. Qualitative and quantitative defects of CD4+ T cells explain the inability of HIV-infected individuals to contain mycobacterial proliferation. Reactive CD4+ lymphocytes produce cytokines of the T_H1 pattern and participate in MHC class II-restricted killing of cells infected with *M. tuberculosis*. T_H1 CD4+ cells produce interferon γ (IFN- γ) and IL-2 and promote cell-mediated immunity. T_H2 cells produce IL-4, IL-5, and IL-10 and promote humoral immunity. The interplay of these various cytokines and their cross-regulation determine the host's response. The role of cytokines in promoting intracellular killing of mycobacteria has not been entirely elucidated. IFN- γ may induce release of nitric oxide, and TNF- α seems also to be important. Finally, the role of other cells, such as natural killer (NK) cells, "double-negative" CD4-CD8- cells, and γ/δ T cells, in protective immunity remains unclear.

M. tuberculosis possesses various protein antigens. Some are present in the cytoplasm and cell wall; others are secreted. That the latter are more important in eliciting a T lymphocyte response is suggested by experiments documenting the appearance of protective immunity in animals after immunization with live, protein-secreting mycobacteria. Among the antigens with a potential protective role are the 30-kDa (or 85B) and the ESAT-6 antigens. Protective immunity is probably the result of reactivity to a large number of different mycobacterial antigens.

Coincident with the appearance of immunity, [DTH](#) to *M. tuberculosis* develops. This reactivity is the basis of the [PPD](#) skin test, currently the only test that reliably detects *M. tuberculosis* infection in persons without symptoms. The cellular mechanisms

responsible for PPD reactivity are related mainly to previously sensitized CD4+ lymphocytes, which are attracted to the skin-test site. There, they proliferate and produce cytokines.

In 1891, Robert Koch discovered components of *M. tuberculosis* in a concentrated liquid culture medium. Subsequently named "old tuberculin" (OT), this material was initially believed to be useful in the treatment of tuberculosis (although this idea was later disproved). It soon became clear that OT was capable of eliciting a skin reaction when injected subcutaneously into patients with tuberculosis. In 1932, Seibert and Munday purified this product by ammonium sulfate precipitation. The result was an active protein fraction known as tuberculin [PPD](#). However, the complexity and diversity of the constituents of PPD rendered its standardization difficult. PPD-S, developed by Seibert and Glenn in 1941, was chosen as the international standard. Later, the [WHO](#) and UNICEF sponsored large-scale production of a master batch of PPD, termed RT23, and made it available for general use. The greatest limitation of PPD is its lack of mycobacterial species specificity, a property that is due to the large number of proteins in this product that are highly conserved in the various species of mycobacteria.

While [DTH](#) is associated with protective immunity ([PPD](#)-positive persons being less susceptible to a new *M. tuberculosis* infection than PPD-negative persons), it by no means guarantees protection against reactivation. In fact, severe cases of active tuberculosis are often accompanied by strongly positive skin-test reactions.

CLINICAL MANIFESTATIONS

Tuberculosis is usually classified as pulmonary or extrapulmonary. Before the recognition of HIV infection, more than 80% of all cases of tuberculosis were limited to the lungs. However, up to two-thirds of HIV-infected patients with tuberculosis may have both pulmonary and extrapulmonary disease or extrapulmonary disease alone.

Pulmonary Tuberculosis Pulmonary tuberculosis can be categorized as primary or postprimary (secondary).

Primary Disease Primary pulmonary tuberculosis results from an initial infection with tubercle bacilli. In areas of high tuberculosis prevalence, this form of disease is often seen in children and is frequently localized to the middle and lower lung zones. The lesion forming after infection is usually peripheral and accompanied by hilar or paratracheal lymphadenopathy, which may not be detectable on chest radiography. In the majority of cases, the lesion heals spontaneously and may later be evident as a small calcified nodule (*Ghon lesion*).

In children and in persons with impaired immunity, such as those with malnutrition or HIV infection, primary pulmonary tuberculosis may progress rapidly to clinical illness. The initial lesion increases in size and can evolve in different ways. Pleural effusion, a frequent finding, results from the penetration of bacilli into the pleural space from an adjacent subpleural focus. In severe cases, the primary site rapidly enlarges, its central portion undergoes necrosis, and acute cavitation develops (progressive primary tuberculosis). Tuberculosis in young children is almost invariably accompanied by hilar or mediastinal lymphadenopathy due to the spread of bacilli from the lung parenchyma

through lymphatic vessels. Enlarged lymph nodes may compress bronchi, causing obstruction and subsequent segmental or lobar collapse. Partial obstruction may cause obstructive emphysema, and bronchiectasis may also develop. Hematogenous dissemination, which is common and is often asymptomatic, may result in the most severe manifestations of primary *M. tuberculosis* infection. Bacilli reach the bloodstream from the pulmonary lesion or the lymph nodes and disseminate into various organs, where they may produce granulomatous lesions. Although healing frequently takes place, immunocompromised persons (e.g., patients with HIV infection and those recovering from measles) may develop miliary tuberculosis and/or tuberculous meningitis.

Postprimary Disease Also called adult-type, reactivation, or secondary tuberculosis, postprimary disease results from endogenous reactivation of latent infection and is usually localized to the apical and posterior segments of the upper lobes, where the high oxygen concentration favors mycobacterial growth ([Fig. 169-CD1](#)). In addition, the superior segments of the lower lobes are frequently involved. The extent of lung parenchymal involvement varies greatly, from small infiltrates to extensive cavitary disease. With cavity formation, liquefied necrotic contents are ultimately discharged into the airways, resulting in satellite lesions within the lungs that may in turn undergo cavitation. Massive involvement of pulmonary segments or lobes, with coalescence of lesions, produces tuberculous pneumonia. While up to one-third of untreated patients reportedly succumb to severe pulmonary tuberculosis within a few weeks or months after onset, others undergo a process of spontaneous remission or proceed along a chronic, progressively debilitating course ("consumption"). Under these circumstances, some pulmonary lesions become fibrotic and may later calcify, but cavities persist in other parts of the lungs. Individuals with such chronic disease continue to discharge tubercle bacilli into the environment. Most patients respond to treatment, with defervescence, decreasing cough, weight gain, and a general improvement in well-being within several weeks.

Early in the course of disease, symptoms and signs are often nonspecific and insidious, consisting mainly of fever and night sweats, weight loss, anorexia, general malaise, and weakness. However, in the majority of cases, cough eventually develops -- perhaps initially nonproductive and subsequently accompanied by the production of purulent sputum. Blood streaking of the sputum is frequently documented. Massive hemoptysis may ensue as a consequence of the erosion of a fully patent vessel located in the wall of a cavity. Hemoptysis, however, may also result from rupture of a dilated vessel in a cavity (*Rasmussen's aneurysm*) or from aspergilloma formation in an old cavity. Pleuritic chest pain sometimes develops in patients with subpleural parenchymal lesions but can also result from muscle strain due to persistent coughing. Extensive disease may produce dyspnea and (occasionally) adult respiratory distress syndrome (ARDS).

Physical findings are of limited use in pulmonary tuberculosis. Many patients have no abnormalities detectable by chest examination, while others have detectable rales in the involved areas during inspiration, especially after coughing. Occasionally, rhonchi due to partial bronchial obstruction and classic amphoric breath sounds in areas with large cavities may be heard. Systemic features include fever (often low-grade and intermittent) and wasting. In some cases, pallor and finger clubbing develop. The most common hematologic findings are mild anemia and leukocytosis. Hyponatremia due to

the syndrome of inappropriate secretion of antidiuretic hormone (SIADH) has also been reported.

Extrapulmonary Tuberculosis In order of frequency, the extrapulmonary sites most commonly involved in tuberculosis are the lymph nodes, pleura, genitourinary tract, bones and joints, meninges, and peritoneum. However, virtually all organ systems may be affected. As a result of hematogenous dissemination in HIV-infected individuals, extrapulmonary tuberculosis is seen more commonly today than in the past.

Lymph-Node Tuberculosis (Tuberculous Lymphadenitis) The commonest presentation of extrapulmonary tuberculosis (being documented in more than 25% of cases), lymph-node disease is particularly frequent among HIV-infected patients. In the United States, children and women (particularly non-Caucasians) also seem to be especially susceptible. Lymph-node tuberculosis presents as painless swelling of the lymph nodes, most commonly at cervical and supraclavicular sites. Lymph nodes are usually discrete in early disease but may be inflamed and have a fistulous tract draining caseous material ([Fig. 169-CD2](#)). Systemic symptoms are usually limited to HIV-infected patients, and concomitant lung disease may or may not be present. The diagnosis is established by fine-needle aspiration or surgical biopsy. [AFB](#) are seen in up to 50% of cases, cultures are positive in 70 to 80%, and histologic examination shows granulomatous lesions. Among HIV-infected patients, granulomas are usually not seen. Differential diagnosis includes a variety of infectious conditions as well as neoplastic diseases such as lymphomas or metastatic carcinomas ([Chap. 63](#)).

Pleural Tuberculosis Involvement of the pleura is common in primary tuberculosis and results from penetration by a few tubercle bacilli into the pleural space. Depending on the extent of reactivity, the effusion may be small, remain unnoticed, and resolve spontaneously or may be sufficiently large to cause symptoms such as fever, pleuritic chest pain, and dyspnea. Physical findings are those of pleural effusion: dullness to percussion and absence of breath sounds. A chest radiograph reveals the effusion and, in no more than one-third of cases, also shows a parenchymal lesion. Thoracentesis is required to ascertain the nature of the effusion. The fluid is straw colored and at times hemorrhagic; it is an exudate with a protein concentration >50% of that in serum, a normal to low glucose concentration, a pH that is generally <7.2, and detectable white blood cells (usually 500 to 2500/mL). Neutrophils may predominate in the early stage, while mononuclear cells are the typical finding later. Mesothelial cells are generally rare or absent. [AFB](#) are very rarely seen on direct smear, but cultures may be positive for *M. tuberculosis* in up to one-third of cases. Needle biopsy of the pleura is often required for diagnosis and reveals granulomas and/or yields a positive culture in up to 70% of cases. This form of pleural tuberculosis responds well to chemotherapy and may resolve spontaneously.

Tuberculous empyema is a less common complication of pulmonary tuberculosis. It is usually the result of the rupture of a cavity, with delivery of a large number of organisms into the pleural space, or of a bronchopleural fistula from a pulmonary lesion. A chest radiograph may show pyopneumothorax with an air-fluid level. The effusion is purulent and thick and contains large numbers of lymphocytes. An acid-fast smear of pleural fluid is often found to be positive when examined by microscopy, as is culture of the pleural fluid. Surgical drainage is usually required as an adjunct to chemotherapy. Tuberculous

empyema may result in severe pleural fibrosis and restrictive lung disease.

Tuberculosis of the Upper Airways Nearly always a complication of advanced cavitary pulmonary tuberculosis ([Fig. 169-CD3](#)), tuberculosis of the upper airways may involve the larynx, pharynx, and epiglottis. Symptoms include hoarseness and dysphagia in addition to chronic productive cough. Findings depend on the site of involvement, and ulcerations may be seen on laryngoscopy. Acid-fast smear of the sputum is often positive, but biopsy may be necessary in some cases to establish the diagnosis. Cancer may have similar features but is usually painless.

Genitourinary Tuberculosis Genitourinary tuberculosis accounts for about 15% of all extrapulmonary cases, may involve any portion of the genitourinary tract, and is usually due to hematogenous seeding following primary infection. Local symptoms predominate. Urinary frequency, dysuria, hematuria, and flank pain are common presentations. However, patients may be asymptomatic and the disease discovered only after severe destructive lesions of the kidneys have developed. Urinalysis gives abnormal results in 90% of cases, revealing pyuria and hematuria. The documentation of culture-negative pyuria in acidic urine raises the suspicion of tuberculosis. An intravenous pyelogram helps in diagnosis. Calcifications and ureteral strictures are suggestive findings. Culture of three morning urine specimens yields a definitive diagnosis in nearly 90% of cases. Severe ureteral strictures may lead to hydronephrosis and renal damage.

Genital tuberculosis is diagnosed more commonly in females than in males. In females, it affects the fallopian tubes and the endometrium and may cause infertility, pelvic pain, and menstrual abnormalities. Diagnosis requires biopsy or culture of specimens obtained by dilatation and curettage. In males, tuberculosis preferentially affects the epididymis, producing a slightly tender mass that may drain externally through a fistulous tract; orchitis and prostatitis may also develop. In almost half of cases of genitourinary tuberculosis, urinary tract disease is also present. Genitourinary tuberculosis responds well to chemotherapy.

Skeletal Tuberculosis In the United States, tuberculosis of the bones and joints is responsible for about 10% of extrapulmonary cases. In bone and joint disease, pathogenesis is related to reactivation of hematogenous foci or to spread from adjacent paravertebral lymph nodes. Weight-bearing joints (spine, hips, and knees -- in that order) are affected most commonly. Spinal tuberculosis (Pott's disease or tuberculous spondylitis) often involves two or more adjacent vertebral bodies. While the upper thoracic spine is the most common site of spinal tuberculosis in children, the lower thoracic and upper lumbar vertebrae are usually affected in adults. From the anterior superior or inferior angle of the vertebral body, the lesion reaches the adjacent body, also destroying the intervertebral disk. With advanced disease, collapse of vertebral bodies results in kyphosis (*gibbus*). A paravertebral "cold" abscess may also form. In the upper spine, this abscess may track to the chest wall as a mass; in the lower spine, it may reach the inguinal ligaments or present as a psoas abscess. Computed tomography (CT) or magnetic resonance imaging (MRI) reveals the characteristic lesion and suggests its etiology, although the differential diagnosis includes other infections and tumors. Aspiration of the abscess or bone biopsy confirms the tuberculous etiology, as cultures are usually positive and histologic findings highly typical. A catastrophic

complication of Pott's disease is paraplegia, which is usually due to an abscess or a lesion compressing the spinal cord. Paraparesis due to a large abscess is a medical emergency and requires abscess drainage. Tuberculosis of the hip joints causes pain and limping; tuberculosis of the knee produces pain and swelling and sometimes follows trauma. If the disease goes unrecognized, the joints may be destroyed. Skeletal tuberculosis responds to chemotherapy, but severe cases may require surgery.

Tuberculous Meningitis and Tuberculoma Tuberculosis of the central nervous system accounts for about 5% of extrapulmonary cases. It is seen most often in young children but also develops in adults, especially those who are infected with HIV. Tuberculous meningitis results from the hematogenous spread of primary or postprimary pulmonary disease or from the rupture of a subependymal tubercle into the subarachnoid space. In more than half of cases, evidence of old pulmonary lesions or a miliary pattern is found on chest radiography. The disease may present subtly as headache and mental changes or acutely as confusion, lethargy, altered sensorium, and neck rigidity. Typically, the disease evolves over 1 or 2 weeks, a course longer than that of bacterial meningitis. Paresis of cranial nerves (ocular nerves in particular) is a frequent finding, and the involvement of cerebral arteries may produce focal ischemia. Hydrocephalus is common. Lumbar puncture is the cornerstone of diagnosis. In general, examination of the cerebrospinal fluid (CSF) reveals a high leukocyte count (usually with a predominance of lymphocytes but often with a predominance of neutrophils in the early stage), a protein content of 1 to 8 g/L (100 to 800 mg/dL), and a low glucose concentration; however, any of these three parameters can be within the normal range. [AFB](#) are seen on direct smear of CSF sediment in only 20% of cases, but repeated lumbar punctures increase the yield. Culture of CSF is diagnostic in up to 80% of cases. Imaging studies ([CT](#) and [MRI](#)) may show hydrocephalus and abnormal enhancement of basal cisterns or ependyma. If unrecognized, tuberculous meningitis is uniformly fatal. This disease responds to chemotherapy; however, neurologic sequelae are documented in 25% of treated cases, in most of which the diagnosis has been delayed. Clinical trials have demonstrated that patients treated with adjunctive glucocorticoids experience a significantly faster resolution of CSF abnormalities and elevated CSF pressure. Adjunctive glucocorticoids enhance survival and reduce the frequency of neurologic sequelae.

Tuberculoma, an uncommon manifestation of tuberculosis, presents as one or more space-occupying lesions and usually causes seizures and focal signs. [CT](#) or [MRI](#) reveals contrast-enhanced ring lesions, but biopsy is necessary to establish the diagnosis.

Gastrointestinal Tuberculosis Any portion of the gastrointestinal tract may be affected by tuberculosis. Various pathogenetic mechanisms are involved: swallowing of sputum with direct seeding, hematogenous spread, or (although rarely today) ingestion of milk from cows affected by bovine tuberculosis. The terminal ileum and the cecum are the sites most commonly involved. Abdominal pain (at times similar to that associated with appendicitis), diarrhea, obstruction, hematochezia, and a palpable mass in the abdomen are common findings at presentation. Fever, weight loss, and night sweats are also frequent. With intestinal-wall involvement, ulcerations and fistulae may simulate Crohn's disease. Anal fistulae should prompt an evaluation for rectal tuberculosis. As surgery is required in most cases, the diagnosis can be established by histologic examination and culture of specimens obtained intraoperatively.

Tuberculous peritonitis follows either the direct spread of tubercle bacilli from ruptured lymph nodes and intraabdominal organs or hematogenous seeding. Nonspecific abdominal pain, fever, and ascites should raise the suspicion of tuberculous peritonitis. The coexistence of cirrhosis ([Chap. 298](#)) in patients with tuberculous peritonitis complicates the diagnosis. In tuberculous peritonitis, paracentesis reveals an exudative fluid with a high protein content and leukocytosis that is usually lymphocytic (although neutrophils occasionally predominate). The yield of direct smear and culture is relatively low; culture of a large volume of ascitic fluid can increase the yield, but peritoneal biopsy is often needed to establish the diagnosis.

Pericardial Tuberculosis (Tuberculous Pericarditis) Due to direct progression of a primary focus within the pericardium, to reactivation of a latent focus, or to rupture of an adjacent lymph node, pericardial tuberculosis has often been a disease of the elderly in countries with low tuberculosis prevalence but develops frequently in HIV-infected patients. Case-fatality rates are as high as 40% in some series. The onset may be subacute, although an acute presentation, with fever, dull retrosternal pain, and a friction rub, is possible. An effusion eventually develops in many cases; cardiovascular symptoms and signs of cardiac tamponade may ultimately appear ([Chap. 239](#)). In the presence of effusion detected on chest radiography, tuberculosis must be suspected if the patient belongs to a high-risk population (HIV-infected, originating in a high-prevalence country), if there is evidence of previous tuberculosis or disease in other organs, or if echocardiography shows thick strands crossing the pericardial space. Diagnosis can be facilitated by pericardiocentesis under echocardiographic guidance. The pericardial fluid must be submitted for biochemical, cytologic, and microbiologic study. The effusion is exudative in nature, with a high count of leukocytes (predominantly mononuclear cells). Hemorrhagic effusion is frequent. Culture of the fluid reveals *M. tuberculosis* in about 30% of cases, while biopsy has a higher yield. Without treatment, pericardial tuberculosis is usually fatal. Even with treatment, complications may develop, including chronic constrictive pericarditis with thickening of the pericardium, fibrosis, and sometimes calcification, which may be visible on a chest radiograph. A course of glucocorticoid treatment is useful in the management of acute disease, reducing effusion, facilitating hemodynamic recovery, and thus decreasing mortality. Progression to chronic constrictive pericarditis, however, seems unaffected by such therapy.

Miliary or Disseminated Tuberculosis Miliary tuberculosis is due to hematogenous spread of tubercle bacilli. Although in children it is often the consequence of a recent primary infection, in adults it may be due to either recent infection or reactivation of old disseminated foci. Lesions are usually yellowish granulomas 1 to 2 mm in diameter that resemble millet seeds (thus the term *miliary*, coined by nineteenth-century pathologists).

Clinical manifestations are nonspecific and protean, depending on the predominant site of involvement. Fever, night sweats, anorexia, weakness, and weight loss are presenting symptoms in the majority of cases. At times, patients have a cough and other respiratory symptoms due to pulmonary involvement as well as abdominal symptoms. Physical findings include hepatomegaly, splenomegaly, and lymphadenopathy. Eye examination may reveal choroidal tubercles, which are pathognomonic of miliary tuberculosis, in up to 30% of cases. Meningismus occurs in fewer than 10% of cases.

A high index of suspicion is required for the diagnosis of miliary tuberculosis. Frequently, chest radiography ([Fig. 169-CD5](#)) reveals a miliary reticulonodular pattern (more easily seen on underpenetrated film), although no radiographic abnormality may be evident early in the course and among HIV-infected patients. Other radiologic findings include large infiltrates, interstitial infiltrates (especially in HIV-infected patients), and pleural effusion. A sputum smear is negative in 80% of cases. Various hematologic abnormalities may be seen, including anemia with leukopenia, neutrophilic leukocytosis and leukemoid reactions, and polycythemia. Disseminated intravascular coagulation has been reported. Elevation of alkaline phosphatase levels and other abnormal values in liver function tests are detected in patients with severe hepatic involvement. The [PPD](#) test may be negative in up to half of cases, but reactivity may be restored during chemotherapy. Bronchoalveolar lavage and transbronchial biopsy are more likely to permit bacteriologic confirmation, and granulomas are evident in liver or bone-marrow biopsy specimens from many patients. If it goes unrecognized, miliary tuberculosis is lethal; with proper treatment, however, it is amenable to cure.

A rare presentation seen in the elderly is *cryptic miliary tuberculosis*, which has a chronic course characterized by mild intermittent fever, anemia, and -- ultimately -- meningeal involvement preceding death. An acute septicemic form, *nonreactive miliary tuberculosis*, occurs very rarely and is due to massive hematogenous dissemination of tubercle bacilli. Pancytopenia is common in this form of disease, which is rapidly fatal. At postmortem examination, multiple necrotic but nongranulomatous ("nonreactive") lesions are detected.

Less Common Extrapulmonary Forms Tuberculosis may cause chorioretinitis, uveitis, panophthalmitis, and painful hypersensitivity-related phlyctenular conjunctivitis. Tuberculous otitis is rare and presents as hearing loss, otorrhea, and tympanic membrane perforation. In the nasopharynx, tuberculosis may simulate Wegener's granulomatosis. Cutaneous manifestations of tuberculosis include primary infection due to direct inoculation, abscesses and chronic ulcers, scrofuloderma ([Fig. 169-CD2](#)), lupus vulgaris ([Fig. 169-CD4](#)), miliary lesions, and erythema nodosum. Adrenal tuberculosis is a manifestation of advanced disease presenting as signs of adrenal insufficiency. Finally, congenital tuberculosis results from transplacental spread of tubercle bacilli to the fetus or from ingestion of contaminated amniotic fluid. This rare disease affects the liver, spleen, lymph nodes, and various other organs.

HIV-Associated Tuberculosis Tuberculosis is an important opportunistic disease among HIV-infected persons worldwide. In developing countries of Africa, Southeast Asia, and Latin America, an estimated 10 million persons were coinfectd as of 1997. In the United States, coinfection with HIV and *M. tuberculosis* is common in certain segments of the population, including drug users and some minorities. A person with skin test-documented *M. tuberculosis* infection who acquires HIV infection has a 3 to 15% annual risk of developing active tuberculosis.

The association between tuberculosis and HIV infection is supported by other epidemiologic observations. First, the rate of HIV seropositivity among patients with tuberculosis is several times higher than that among the general population: in African countries it reaches 60 to 70%. Second, marked increases in numbers of tuberculosis

cases have been reported at locations hard hit by the HIV epidemic, such as large parts of Africa (including Kenya, Tanzania, and Malawi), northern Thailand, and New York City. Globally, the proportion of tuberculosis cases associated with HIV infection reached 8% in 1997.

HIV directly attacks the critical immune mechanisms involved in protection against tuberculosis. Tuberculosis can appear at any stage of HIV infection, but its presentation varies with the stage. When cell-mediated immunity is only partially compromised, pulmonary tuberculosis presents as a typical pattern of upper lobe infiltrates and cavitation, without significant lymphadenopathy or pleural effusion. In late stages of HIV infection, a primary tuberculosis-like pattern, with diffuse interstitial or miliary infiltrates, little or no cavitation, and intrathoracic lymphadenopathy, is more common. Overall, sputum smears may be positive less frequently among tuberculosis patients with HIV infection than among those without; thus the diagnosis of tuberculosis may be unusually difficult, especially in view of the variety of HIV-related pulmonary conditions mimicking tuberculosis.

As has been mentioned, extrapulmonary tuberculosis is common among HIV-infected patients. In various series studied in the United States and many developing countries, extrapulmonary tuberculosis -- alone or in association with pulmonary disease -- has been documented in 40 to 60% of all cases in HIV co-infected individuals. The most common forms are lymphatic, disseminated, pleural, and pericardial. Mycobacteremia and meningitis are also frequent, particularly in advanced HIV disease.

The diagnosis of tuberculosis in HIV-infected patients may be difficult not only because of the increased frequency of sputum-smear negativity (up to 40% in culture-proven pulmonary cases) but also because of atypical radiographic findings, a lack of classic granuloma formation in the late stages, and negative results in [PPD](#) skin tests. Delays in treatment may prove fatal. The response to short-course chemotherapy is similar to that in HIV-seronegative patients. However, adverse effects may be more pronounced, including severe or even fatal skin reactions to amithiozone (thiacetazone).

Exacerbations in symptoms, signs, and laboratory or radiographic manifestations of tuberculosis -- termed *paradoxical reactions* -- have been associated with the administration of highly active antiretroviral treatment (HAART) regimens. The presumed pathogenesis of paradoxical reactions is an immune response to antigens released as bacilli are killed by effective chemotherapy. In patients in whom antiretroviral therapy has recently been started, paradoxical reactions may be due to improving immune function. The first priority in the management of a possible paradoxical reaction is to ensure that the clinical syndrome does not represent a failure of tuberculosis treatment or the development of another infection. Mild paradoxical reactions can be managed with symptom-based treatment. More severe manifestations may necessitate discontinuation of antiretroviral therapy. Immunomodulators, such as glucocorticoids, have been used to treat severe paradoxical reactions, although this practice has not been formally evaluated in clinical trials.

Recommendations for the prevention and treatment of tuberculosis in HIV-infected individuals have been published by the [CDC](#).

DIAGNOSIS

The key to the diagnosis of tuberculosis is a high index of suspicion. Diagnosis is not difficult with a high-risk patient -- e.g., a homeless alcoholic who presents with typical symptoms and a classic chest radiograph showing upper lobe infiltrates with cavities. On the other hand, the diagnosis can easily be missed in an elderly nursing-home resident or a teenager with a focal infiltrate.

Often, the diagnosis is first entertained when the chest radiograph of a patient being evaluated for respiratory symptoms is abnormal. If the patient has no complicating medical conditions that favor immunosuppression, the chest radiograph may show the typical picture of upper lobe infiltrates with cavitation. The longer the delay between the onset of symptoms and the diagnosis, the more likely is the finding of cavitory disease. In contrast, immunosuppressed patients, including those with HIV infection, may have "atypical" findings on chest radiography -- e.g., lower zone infiltrates without cavity formation.

AFB Microscopy A presumptive diagnosis is commonly based on the finding of AFB on microscopic examination of a diagnostic specimen such as a smear of expectorated sputum or of tissue (for example, a lymph node biopsy). Most modern laboratories processing large numbers of diagnostic specimens use auramine-rhodamine staining and fluorescence microscopy. The more traditional method -- light microscopy of specimens stained with Kinyoun or Ziehl-Neelsen basic fuchsin dyes -- is satisfactory, although more time-consuming. For patients with suspected pulmonary tuberculosis, three sputum specimens, preferably collected early in the morning, should be submitted to the laboratory for AFB smear and mycobacteriology culture. If tissue is obtained, it is critical that the portion of the specimen intended for culture not be put in formaldehyde. The use of AFB microscopy on urine or gastric lavage fluid is limited by the presence of mycobacterial commensals, which can cause false-positive results.

Mycobacterial Culture Definitive diagnosis depends on the isolation and identification of *M. tuberculosis* from a diagnostic specimen -- in most cases, a sputum specimen obtained from a patient with a productive cough. Specimens may be inoculated onto egg- or agar-based medium (e.g., Lowenstein-Jensen or Middlebrook 7H10) and incubated at 37°C under 5% CO₂. Because most species of mycobacteria, including *M. tuberculosis*, grow slowly, 4 to 8 weeks may be required before growth is detected. Although *M. tuberculosis* may be presumptively identified on the basis of growth time and colony pigmentation and morphology, a variety of biochemical tests have traditionally been used to speciate mycobacterial isolates. In today's laboratories, the use of liquid media with radiometric growth detection (e.g., BACTEC-460) and the identification of isolates by nucleic acid probes or high-pressure liquid chromatography of mycolic acids have replaced the traditional methods of isolation on solid media and identification by biochemical tests. These new methods have decreased the time required for isolation and speciation to 2 to 3 weeks. Other systems for culture on liquid media with nonradiometric detection have become available.

Nucleic Acid Amplification Several test systems based on amplification of mycobacterial nucleic acid are available. These systems permit the diagnosis of tuberculosis in as short a period as several hours. However, their applicability is limited

by low sensitivity (lower than culture) and high cost. At present, these tests are approved by the U.S. Food and Drug Administration only for species identification on [AFB](#)-positive sputa. With further improvements in performance, these tests may also be useful for the diagnosis of patients with AFB-negative pulmonary and extrapulmonary tuberculosis.

Radiographic Procedures As noted above, the initial suspicion of pulmonary tuberculosis is often based on abnormal chest radiographic findings in a patient with respiratory symptoms. Although the "classic" picture is that of upper lobe disease with infiltrates and cavities, virtually any radiographic pattern -- from a normal film or a solitary pulmonary nodule to diffuse alveolar infiltrates in a patient with [ARDS](#) -- may be seen. In the era of AIDS, no radiographic pattern can be considered pathognomonic.

PPD Skin Testing Skin testing with PPD is most widely used in screening for *M. tuberculosis* infection (see below). The test is of limited value in the diagnosis of active tuberculosis because of its low sensitivity and specificity. False-negative reactions are common in immunosuppressed patients and in those with overwhelming tuberculosis. Positive reactions are obtained when patients have been infected with *M. tuberculosis* but do not have active disease and when persons have been sensitized by nontuberculous mycobacteria ([Chap. 171](#)) or bacille Calmette-Guerin (BCG) vaccination. Although BCG vaccine is not commonly used in the United States, many immigrants will have received it. In the absence of a history of BCG vaccination, a positive skin test may provide additional support for the diagnosis of tuberculosis in culture-negative cases.

Drug Susceptibility Testing In general, the initial isolate of *M. tuberculosis* should be tested for susceptibility to the primary drugs used for treatment: isoniazid, rifampin, ethambutol, pyrazinamide, and streptomycin. In addition, drug susceptibility tests are mandatory when patients fail to respond to initial therapy or experience a relapse after the completion of treatment (see below). Susceptibility testing may be conducted directly (with the clinical specimen) or indirectly (with mycobacterial cultures) on solid or liquid medium. Results are obtained most rapidly by direct susceptibility testing on liquid medium, with an average reporting time of 3 weeks. With indirect testing on solid media, results may not be available for 8 weeks or longer. Molecular methods for the rapid identification of drug resistance are becoming available. One of the most promising uses polymerase chain reaction (PCR) for the *rpoB* gene to detect resistance to rifampin.

Additional Diagnostic Procedures Other diagnostic tests may be used when pulmonary tuberculosis is suspected. Sputum induction by ultrasonic nebulization of hypertonic saline may be useful for patients unable to produce a sputum specimen spontaneously. Frequently, patients with radiographic abnormalities that are consistent with other diagnoses (e.g., bronchogenic carcinoma) undergo fiberoptic bronchoscopy with bronchial brushings or transbronchial biopsy of the lesion. Bronchoalveolar lavage of a lung segment containing an abnormality may also be performed. In all cases, it is essential that specimens be submitted for [AFB](#) smear and mycobacterial culture. For the diagnosis of primary pulmonary tuberculosis in children, who often do not expectorate sputum, specimens from early-morning gastric lavage may yield positive cultures.

Invasive diagnostic procedures are indicated for patients with suspected extrapulmonary

tuberculosis. In addition to specimens of involved sites (e.g., [CSF](#) for tuberculous meningitis, pleural fluid and biopsy samples for pleural disease), bone marrow and liver biopsy and culture have a good diagnostic yield in disseminated (miliary) tuberculosis, particularly in HIV-infected patients, who also have a high frequency of positive blood cultures.

In some cases, cultures will be negative, but a clinical diagnosis of tuberculosis will be supported by consistent epidemiologic evidence (e.g., a history of close contact with an infectious patient), a positive [PPD](#) skin test, and a compatible clinical and radiographic response to treatment. In the United States and other industrialized countries with low rates of tuberculosis, some patients with limited abnormalities on chest radiographs and sputum positive for [AFB](#) are infected with organisms of the *M. avium* complex or *M. kansasii* ([Chap. 171](#)). Factors favoring the diagnosis of nontuberculous mycobacterial disease over tuberculosis include an absence of risk factors for tuberculosis, a negative PPD skin test, and underlying chronic obstructive pulmonary disease.

Patients with HIV-associated tuberculosis pose several diagnostic problems, as noted above in the description of clinical manifestations. Moreover, HIV-infected patients with sputum culture-positive and [AFB](#)-positive tuberculosis may present with a normal chest radiograph. Thus, in a patient with HIV infection, the finding of a normal chest radiograph does not rule out the diagnosis of pulmonary tuberculosis. An additional consideration is that, among relatively severely immunosuppressed AIDS patients in Europe and North America, *M. avium* complex disease is more common than tuberculosis, usually presenting as a disseminated disease without pulmonary parenchymal involvement.

Adjunctive Diagnostic Tests A number of methods have been evaluated as adjuncts to standard laboratory diagnosis. The most thoroughly investigated is serologic diagnosis based on detection of antibody to a variety of mycobacterial antigens. However, tests with most of the target antigens have a low predictive value when used in a population with a presumably low probability of disease. Tests aimed at detection of mycobacterial antigen by serologic methods have generally not been sufficiently sensitive to be useful. Nonspecific tests, such as the measurement of adenine deaminase in pleural fluid, have been evaluated but have not gained acceptance.

TREATMENT

Chemotherapy for tuberculosis became possible with the discovery of streptomycin in the mid-1940s. Randomized clinical trials clearly indicated that the administration of streptomycin to patients with chronic tuberculosis reduced mortality and led to cure in the majority of cases. However, monotherapy with streptomycin was frequently associated with the development of resistance to streptomycin and the attendant failure of treatment. With the discovery of para-aminosalicylic acid (PAS) and isoniazid, it became axiomatic that cure of tuberculosis required the concomitant administration of at least two agents to which the organism was susceptible. Furthermore, early clinical trials demonstrated that a long period of treatment -- i.e., 12 to 24 months -- was required to prevent the recurrence of tuberculosis.

The introduction of rifampin in the early 1970s heralded the era of effective short-course

chemotherapy, with a treatment duration of <12 months. The discovery that pyrazinamide, which was first used in the 1950s, augmented the potency of isoniazid/rifampin regimens led to the use of a 6-month course of this triple-drug regimen as standard therapy.

Drugs Five major drugs are considered the first-line agents for the treatment of tuberculosis: isoniazid, rifampin, pyrazinamide, ethambutol, and streptomycin ([Table 169-1](#)). The first four, which are usually given orally, are well absorbed, with peak serum levels at 2 to 4 h and nearly complete elimination within 24 h. These agents are recommended on the basis of their bactericidal activity (ability to rapidly reduce the number of viable organisms), their sterilizing activity (ability to kill all bacilli and thus sterilize the affected organ, measured in terms of the ability to prevent relapses), and their low rate of induction of drug resistance. Rifapentine and rifabutin, two drugs related to rifampin, are also available in the United States. **For a detailed discussion of the drugs used for the treatment of tuberculosis, see Chap. 168.*

Because of a lower degree of efficacy and a higher degree of intolerability and toxicity, a number of second-line drugs are used only for the treatment of patients with tuberculosis resistant to first-line drugs. Included in this group are the injectable drugs kanamycin, amikacin, and capreomycin and the oral agents ethionamide, cycloserine, and [PAS](#). Recently, quinolone antibiotics have become the most commonly used second-line drugs. Of available agents, ofloxacin is the most widely used, but levofloxacin and sparfloxacin are the most active, although the latter drug is associated with high rates of photosensitization. Other second-line drugs include clofazimine, amithiozone (thiacetazone, widely used with isoniazid in less wealthy countries but not marketed in North America or Europe), and amoxicillin/clavulanic acid.

Regimens Short-course regimens are divided into an initial or bactericidal phase and a continuation or sterilizing phase. During the initial phase, the majority of the tubercle bacilli are killed, symptoms resolve, and the patient becomes noninfectious. The continuation phase is required to eliminate semidormant "persisters."

The treatment regimen of choice for virtually all forms of tuberculosis in both adults and children consists of a 2-month initial phase of isoniazid, rifampin, and pyrazinamide followed by a 4-month continuation phase of isoniazid and rifampin ([Table 169-2](#)). Except for patients who seem unlikely on epidemiologic grounds to be initially infected with a drug-resistant strain, ethambutol (or streptomycin) should be included in the regimen for the first 2 months or until the results of drug susceptibility testing become available. Treatment may be given daily throughout the course or intermittently (either three times weekly throughout the course or twice weekly following an initial phase of daily therapy). A continuation phase of once-weekly rifapentine and isoniazid appears to be effective for patients who have adhered to the initial-phase treatment and have negative sputum cultures at 2 months. Intermittent treatment is especially useful for patients whose therapy is being directly observed (see below). For patients with sputum culture-negative pulmonary tuberculosis, the duration of treatment may be reduced to a total of 4 months. Pyridoxine (10 to 25 mg/d) should be added to the regimen given to persons at high risk of vitamin deficiency (e.g., alcoholics; malnourished persons; pregnant and lactating women; and patients with conditions such as chronic renal failure, diabetes, and HIV infection or AIDS, which are also associated with neuropathy).

Lack of adherence to treatment regimens is recognized worldwide as the most important impediment to cure. Moreover, the mycobacterial strains infecting patients who do not adhere to the prescribed regimen are especially likely to develop acquired drug resistance. Both patient- and provider-related factors may affect compliance. Patient-related factors include a lack of belief that the illness is significant and/or that treatment will have a beneficial effect; the existence of concomitant medical conditions (notably substance abuse); lack of social support; and poverty, with attendant joblessness and homelessness. Provider-related factors that may promote compliance include the education and encouragement of patients, the offering of convenient clinic hours, and the provision of incentives such as bus tokens.

In addition to specific measures addressing noncompliance, two other strategic approaches are used: direct observation of treatment and provision of drugs in combined formulations. Because it is difficult to predict which patients will adhere to the recommended treatment, all patients should have their therapy directly supervised, especially during the initial phase. In the United States, personnel to supervise therapy are usually available through tuberculosis control programs of local public health departments. Supervision increases the proportion of patients completing treatment and greatly lessens the chances of relapse and acquired drug resistance. Combination products (e.g., isoniazid/rifampin and isoniazid/rifampin/pyrazinamide) are available and are strongly recommended as a means of minimizing the likelihood of prescription error and of the development of drug resistance (as the result of treatment with only one agent). In some formulations of these combination products, the bioavailability of rifampin has been found to be substandard. In North America and Europe, regulatory authorities ensure that combination products are of good quality; however, this type of monitoring cannot be assumed to take place in less affluent countries. Alternative regimens for patients who exhibit drug intolerance or adverse reactions are listed in [Table 169-2](#). However, severe side effects prompting discontinuation of any of the first-line drugs and use of these alternative regimens are uncommon.

Monitoring the Response to Treatment Bacteriologic evaluation is the preferred method of monitoring the response to treatment for tuberculosis. Patients with pulmonary disease should have their sputum examined monthly until cultures become negative. With the recommended 6-month regimen, more than 80% of patients will have negative sputum cultures at the end of the second month of treatment. By the end of the third month, virtually all patients should be culture-negative. In some patients, especially those with extensive cavitory disease and large numbers of organisms, [AFB](#) smear conversion may follow culture conversion. This phenomenon is presumably due to the expectoration and microscopic visualization of dead bacilli. When a patient's sputum cultures remain positive at or beyond 3 months, treatment failure and drug resistance should be suspected (see below). A sputum specimen should be collected at the end of treatment to document cure. If mycobacterial cultures are not practical, then monitoring by [AFB](#) smear examination should be undertaken at 2, 5, and 6 months. Smears positive after 5 months are indicative of treatment failure.

Bacteriologic monitoring of patients with extrapulmonary tuberculosis is more difficult and often is not feasible. In these cases, the response to treatment must be assessed clinically.

Monitoring of the response to treatment during chemotherapy by serial chest radiographs is not recommended, as radiographic changes may lag behind bacteriologic response and are not highly sensitive. After the completion of treatment, neither sputum examination nor chest radiography is recommended for follow-up purposes. However, a chest radiograph may be obtained at the end of treatment and used for comparative purposes should the patient develop symptoms of recurrent tuberculosis months or years later. Patients should be instructed to report promptly for medical assessment should they develop any such symptoms.

During treatment, patients should be monitored for drug toxicity (see also [Table 168-3](#)). The most common adverse reaction of significance is hepatitis. Patients should be carefully educated about the signs and symptoms of drug-induced hepatitis (e.g., dark urine, loss of appetite) and should be instructed to discontinue treatment promptly and see their health care provider should these symptoms occur. Although biochemical monitoring is not routinely recommended, all adult patients should undergo baseline assessment of liver function (e.g., measurement of levels of hepatic aminotransferases and serum bilirubin). Older patients, those with histories of hepatic disease, and those using alcohol daily should be monitored especially closely (i.e., monthly), with repeated measurements of aminotransferases, during the initial phase of treatment. Up to 20% of patients have small increases in aspartate aminotransferase (up to three times the upper limit of normal) that are accompanied by no symptoms and are of no consequence. For patients with symptomatic hepatitis and those with marked (five- to sixfold) elevations in aspartate aminotransferase, treatment should be stopped and drugs reintroduced one at a time after liver function has returned to normal.

Hypersensitivity reactions usually require the discontinuation of all drugs and rechallenge to determine which agent is the culprit. Because of the variety of regimens available, it is usually not necessary -- although it is possible -- to desensitize patients. Hyperuricemia and arthralgia caused by pyrazinamide can usually be managed by the administration of acetylsalicylic acid; however, pyrazinamide treatment should be stopped if the patient develops gouty arthritis. Individuals who develop autoimmune thrombocytopenia secondary to rifampin therapy should not receive the drug thereafter. Similarly, the occurrence of optic neuritis with ethambutol and the development of eighth-nerve damage with streptomycin are indications for permanent discontinuation of these respective drugs. Other common manifestations of drug intolerance, such as pruritus and gastrointestinal upset, can generally be managed without the interruption of therapy.

Treatment Failure and Relapse As stated above, treatment failure should be suspected when a patient's sputum cultures remain positive after 3 months or when [AFB](#)smeared remain positive after 5 months. In the management of such patients, it is imperative that the current isolate be tested for susceptibility to first- and second-line agents. When the results of susceptibility testing are expected to become available within a few weeks, changes in the regimen can be postponed until that time. However, if the patient's clinical condition is deteriorating, an earlier change in regimen may be indicated. A cardinal rule in the latter situation is always to add more than one drug at a time to a failing regimen: at least two and preferably three drugs that have never been used should be added. The patient may continue to take isoniazid and rifampin along

with these new agents pending the results of susceptibility tests.

The mycobacterial strains infecting patients who experience a relapse after apparently successful treatment are less likely to have acquired drug resistance (see below) than are strains from patients in whom treatment has failed. However, if the regimen administered initially does not contain rifampin (and thus is not a short-course regimen), the probability of isoniazid resistance is high. Acquired resistance is uncommon among strains from patients who relapse after completing a short course of therapy. However, it is prudent to begin the treatment of all relapses with all five first-line drugs pending the results of susceptibility testing. In less affluent countries and other settings where facilities for culture and drug susceptibility testing are not available, a standard regimen should be used in all instances of relapse and treatment failure ([Table 169-2](#)).

Adjunctive Glucocorticoid Therapy The use of glucocorticoids for adjunctive treatment of tuberculosis is justified by their potent anti-inflammatory activity in a disease where host response plays a major role. There is a sound basis for using glucocorticoids in tuberculous meningitis and pericarditis to hasten clinical improvement (see above). Long-term benefits are limited to reduced mortality in effusive-constrictive pericarditis and decreased neurologic sequelae in meningitis. Studies have indicated that the usefulness of these agents in pleuritis may be less important than previously thought. There is not yet conclusive evidence that glucocorticoids produce benefits in patients with acute life-threatening pulmonary tuberculosis, including [ARDS](#). In general, depending upon the urgency and severity of clinical conditions, prednisone may be administered at a daily dose of 20 to 60 mg for up to 6 weeks. In meningitis, dexamethasone (up to 12 mg/d) is the preferred drug; therapy continues for 4 to 6 weeks, with gradual tapering of the dose after the first 2 weeks. Rifampin interaction with glucocorticoids, resulting in accelerated metabolism and potential adrenal crisis, must be taken into account. Caution should be exercised in the use of glucocorticoids in HIV-infected patients.

Drug-Resistant Tuberculosis Strains of *M. tuberculosis* resistant to individual drugs arise by spontaneous point mutations in the mycobacterial genome, which occur at low but predictable rates. Because there is no cross-resistance among the commonly used drugs, the probability that a strain will be resistant to two drugs is the product of the probabilities of resistance to each drug and thus is low. The development of drug-resistant tuberculosis is invariably the result of monotherapy -- i.e., the failure of the health care provider to prescribe at least two drugs to which tubercle bacilli are susceptible or of the patient to take properly prescribed therapy.

Drug-resistant tuberculosis may be either primary or acquired. *Primary* drug resistance is that in a strain infecting a patient who has not previously been treated. *Acquired* resistance develops during treatment with an inappropriate regimen. In North America and Europe, rates of primary resistance are generally low, and isoniazid resistance is most common. In the United States, while isoniazid resistance was stable at about 8% in 1993 through 1996, rates of [MDR](#) tuberculosis were declining. Resistance rates are higher among foreign-born and HIV-infected patients. Worldwide, MDR tuberculosis is a serious problem in some regions, especially in the former USSR and parts of Asia. As noted above, drug-resistant tuberculosis can be prevented by adherence to the principles of sound therapy: the inclusion of at least two bactericidal drugs to which the

organism is susceptible (in practice, four drugs are commonly given in the initial phase) and the verification that patients complete the prescribed course.

Although the 6-month regimen described in [Table 169-2](#) is highly effective for patients with initial isoniazid-resistant disease, it is prudent to extend treatment to 9 months and to include ethambutol throughout. Alternatively, rifampin, pyrazinamide, and ethambutol may be given for 6 months. For disease with high-level isoniazid resistance, isoniazid probably does not contribute to a successful outcome and can be omitted. [MDR](#) tuberculosis is more difficult to manage than is disease caused by a drug-susceptible organism, especially because resistance to other first-line drugs as well as to isoniazid and rifampin is common. For strains resistant to isoniazid and rifampin, combinations of ethambutol, pyrazinamide, and streptomycin (or, for those resistant to streptomycin as well, another injectable agent such as amikacin), given for 12 to 18 months in all and for at least 9 months after sputum culture conversion, may be effective. Many authorities would add a quinolone antibiotic to this regimen. For patients with bacilli resistant to all of the first-line agents, cure may be attained with a combination of four second-line drugs, including one injectable agent ([Table 169-2](#)). The optimal duration of treatment in this situation is not known; however, a duration of up to 24 months is recommended. For patients with localized disease and sufficient pulmonary reserve, lobectomy or pneumonectomy may be helpful. Because the management of patients with MDR tuberculosis is complicated by both social and medical factors, care of these patients should be restricted to specialists and tuberculosis control programs.

Special Clinical Situations Although comparative clinical trials of treatment for extrapulmonary tuberculosis are limited, the available evidence indicates that most forms of disease can be treated with the 6-month regimen recommended for patients with pulmonary disease. The American Academy of Pediatrics recommends that children with bone and joint tuberculosis, tuberculous meningitis, or miliary tuberculosis receive a minimum of 12 months of treatment.

Treatment for tuberculosis may be complicated by underlying medical problems that require special consideration (see also [Table 168-1](#)). As a rule, patients with chronic renal failure should not receive aminoglycosides and should receive ethambutol only if serum levels can be monitored. Isoniazid, rifampin, and pyrazinamide may be given in the usual doses in cases of mild to moderate renal failure, but the dosages of isoniazid and pyrazinamide should be reduced for all patients with severe renal failure except those undergoing hemodialysis. Patients with hepatic disease pose a special problem because of the hepatotoxicity of isoniazid, rifampin, and pyrazinamide. Patients with severe hepatic disease may be treated with ethambutol and streptomycin and, if required, with isoniazid and rifampin under close supervision. The use of pyrazinamide by patients with liver failure should be avoided. Silicotuberculosis necessitates the extension of therapy by at least 2 months. Patients with HIV infection or AIDS appear to respond well to standard 6-month therapy, although treatment may need to be prolonged if the response is suboptimal. Rifampin, a powerful inducer of hepatic microsomal enzymes, shortens the half-life of HIV protease inhibitors and therefore is contraindicated for patients receiving these drugs; instead, these individuals should be given rifabutin (150 mg/d or 300 mg twice weekly) with either indinavir or nelfinavir. Studies have shown that total systemic drug exposure, especially for rifampin, is

reduced in HIV-infected patients because of decreased bioavailability secondary to malabsorption. The clinical importance of this phenomenon remains unclear.

The regimen of choice for pregnant women (see also [Table 168-1](#)) is 9 months of treatment with isoniazid and rifampin supplemented by ethambutol for the first 2 months. When required, pyrazinamide may be given, although there are no data concerning its safety in pregnancy. Streptomycin is contraindicated because it is known to cause eighth-cranial-nerve damage in the fetus. Treatment for tuberculosis is not a contraindication to breast feeding; most of the drugs administered will be present in small quantities in breast milk, albeit at concentrations far too low to provide any therapeutic or prophylactic benefit to the child.

PREVENTION

By far the best way to prevent tuberculosis is to diagnose infectious cases rapidly and administer appropriate treatment until cure. Additional strategies include [BCG](#) vaccination and preventive chemotherapy.

BCG Vaccination BCG was derived from an attenuated strain of *M. bovis* and was first administered to humans in 1921. Many BCG vaccines are available worldwide; all are derived from the original strain, but the vaccines vary in efficacy. In fact, estimates of efficacy from randomized, placebo-controlled trials have ranged from 80% to nil. A similar range of efficacy was found in recent observational studies (case-control, historic cohort, and cross-sectional) in areas where infants are vaccinated at birth. These studies also found higher rates of efficacy in the protection of infants and young children from relatively serious forms of tuberculosis, such as tuberculous meningitis and miliary tuberculosis.

[BCG](#) vaccine is safe and rarely causes serious complications. The local tissue response begins 2 to 3 weeks after vaccination, with scar formation and healing within 3 months. Side effects -- most commonly, ulceration at the vaccination site and regional lymphadenitis -- occur in 1 to 10% of vaccinated persons. Some vaccine strains have caused osteomyelitis in approximately one case per million doses administered. Disseminated BCG infection and death have occurred in 1 to 10 cases per 10 million doses administered, although this problem is restricted almost exclusively to persons with impaired immunity, such as children with severe combined immunodeficiency syndrome (SCIDS) or adults with HIV infection. BCG vaccination induces [PPD](#) reactivity, which tends to wane with time. The presence or size of PPD skin-test reactions after vaccination does not predict the degree of protection afforded.

[BCG](#) vaccine is recommended for routine use at birth in countries with high tuberculosis prevalence. However, because of the low risk of transmission of tuberculosis in the United States and the unreliable protection afforded by BCG, the vaccine has never been recommended for general use in the United States. Currently, vaccination should be considered for [PPD](#)-negative infants and children who reside in settings where the likelihood of *M. tuberculosis* transmission and subsequent infection is high, provided no other measures can be implemented (e.g., removing the child from the source of infection). BCG vaccination may also be considered for health care workers who are employed in settings where the risk of infection by [MDR](#) strains is high despite

implementation of comprehensive tuberculosis control measures. The [CDC](#) has recommended that HIV-infected adults and children not receive BCG vaccine, although the [WHO](#) has recommended that asymptomatic HIV-infected children residing in tuberculosis-endemic areas receive BCG.

Treatment of Latent Tuberculosis Infection A major component of tuberculosis control in the United States is the treatment of selected persons with latent tuberculosis infection to prevent active disease. This intervention (formerly called preventive chemotherapy or chemoprophylaxis) is based on the results of a large number of randomized, placebo-controlled clinical trials demonstrating that a 6- to 12-month course of isoniazid reduces the risk of active tuberculosis in infected people by³90%. Analysis of available data indicates that the optimal duration of treatment is 9 to 10 months. In the absence of reinfection, the protective effect is believed to be lifelong. Clinical trials have also shown that isoniazid reduces rates of tuberculosis among [PPD](#)-positive persons with HIV infection. Studies in HIV-infected patients have demonstrated the effectiveness of a shorter course of rifampin-based treatment.

In most cases, candidates for treatment of latent tuberculosis ([Table 169-3](#)) are identified by [PPD](#) skin testing of persons in defined high-risk groups. For skin testing, 5 tuberculin units of polysorbate-stabilized PPD should be injected intradermally into the volar surface of the forearm (Mantoux method). Multipuncture tests, which may be useful for screening large populations, are not recommended for this purpose; any positive reaction to a multipuncture test must be confirmed by Mantoux testing. Reactions are read at 48 to 72 h as the transverse diameter in millimeters of induration; the diameter of erythema is not considered. In some persons, PPD reactivity wanes with time but can be recalled by a second skin test administered 1 week or more after the first (i.e., two-step testing). For persons undergoing periodic PPD skin testing, such as health care workers and individuals admitted to long-term-care institutions, initial two-step testing may preclude subsequent misclassification of persons with boosted reactions as PPD converters.

The cutoff for a positive skin test (and thus for treatment) is related both to the probability that the reaction represents true infection and to the likelihood that the individual, if truly infected, will develop tuberculosis ([Table 169-3](#)). Thus positive reactions for close contacts of infectious cases, persons with HIV infection, and previously untreated persons whose chest radiograph is consistent with healed tuberculosis are defined as an area of induration 5 mm in diameter. A 10-mm cutoff is used to define positive reactions in most other at-risk persons. For persons with a very low risk of developing tuberculosis if infected, a cutoff of 15 mm is used. Persons with a history of [BCG](#) vaccination may receive treatment, especially if BCG was given many years before.

Some [PPD](#)-negative individuals are also candidates for treatment. Infants and children who have come into contact with infectious cases should be treated and should have a repeat skin test 2 or 3 months after contact ends. Those whose test results remain negative should discontinue treatment. HIV-infected persons who have been exposed to an infectious tuberculosis patient should receive treatment regardless of the PPD test result.

Isoniazid is administered at a daily dose of 5 mg/kg (up to 300 mg/d) for 9 months. On the basis of cost-benefit analyses, a 6-month period of treatment has been recommended in the past and may be considered for HIV-negative adults with normal chest radiographs when financial considerations are important. When supervised treatment is desirable and feasible, isoniazid may be given at a dose of 15 mg/kg (up to 900 mg) twice weekly. There are two recommended alternative regimens for adults: 2 months of daily rifampin plus pyrazinamide and 4 months of daily rifampin. Although the 2-month regimen may be associated with increased drug intolerance, it may be useful in situations where long courses of isoniazid have not been feasible (e.g., jails). Either regimen should be considered for persons who are likely to have been infected with an isoniazid-resistant strain.

Contraindications to treatment with isoniazid and pyrazinamide include active liver disease. Since the major adverse reaction to these drugs is hepatitis, persons at increased risk of toxicity (e.g., those abusing alcohol daily and those with a history of liver disease) should undergo baseline and then monthly assessment of liver function during treatment. All patients should be carefully educated about hepatitis and instructed to discontinue use of the drug immediately should any symptoms develop. Moreover, patients should be seen and questioned monthly during therapy about adverse reactions and should be given no more than 1 month's supply of drug at each visit.

It may be more difficult to ensure compliance when treating persons with latent infection than when treating those with active tuberculosis. If family members of active cases are being treated, compliance and monitoring may be easier. When feasible, twice-weekly supervised therapy may increase the likelihood of completion. As in active cases, the provision of incentives may also be helpful.

BASICS OF CONTROL

The highest priority in any tuberculosis control program is the prompt detection of cases and the provision of directly observed short-course chemotherapy to all tuberculosis patients, with emphasis on the cure of sputum smear-positive cases. In addition, in low-prevalence countries with adequate resources, screening of high-risk groups (such as immigrants from high-prevalence countries and HIV-seropositive persons) is recommended. Identification of active cases of tuberculosis should be followed by treatment. [PPD](#)-positive high-risk persons should be treated for latent infection. Contact investigation is an important component of efficient tuberculosis control. In the United States, a great deal of attention has been given to the transmission of tuberculosis (particularly in association with HIV infection) in institutional settings such as hospitals, homeless shelters, and prisons. Measures to limit such transmission include respiratory isolation of persons with suspected tuberculosis until they are proven to be noninfectious (i.e., by sputum [AFB](#) smear negativity), proper ventilation in rooms of patients with infectious tuberculosis, use of ultraviolet lights in areas of increased risk of tuberculosis transmission, and periodic screening of personnel who may come into contact with known or unsuspected cases of tuberculosis. In the past, radiographic surveys, especially those conducted with portable equipment and miniature films, were advocated for case finding. Today, however, the prevalence of tuberculosis in industrialized countries is sufficiently low that "mass miniature radiography" is not

cost-effective. As mentioned above, current recommendations for the prevention and treatment of tuberculosis in HIV-infected individuals have been published by the [CDC](#).

In high-prevalence countries, tuberculosis control programs should be based on the following key elements: (1) case detection through microscopic examination of sputum from patients who present to health care facilities with cough of >3 weeks' duration; (2) administration of standard short-course chemotherapy to all sputum smear-positive patients, with direct observation of drug ingestion; (3) establishment and maintenance of a system of regular drug supply; and (4) establishment and maintenance of an effective surveillance and treatment-monitoring system allowing an analysis of treatment outcomes (e.g., cure, completion of treatment without bacteriologic proof of cure, death, treatment failure, and default) in all cases registered.

(Bibliography omitted in Palm version)

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170. LEPROSY (HANSEN'S DISEASE) - Robert H. Gelber

Leprosy, first described in ancient Indian texts from the sixth century B.C., is a nonfatal, chronic infectious disease caused by *Mycobacterium leprae*, whose clinical manifestations are largely confined to the skin, peripheral nervous system, upper respiratory tract, eyes, and testes. The unique tropism of *M. leprae* for peripheral nerves (from large nerve trunks to microscopic dermal nerves) and certain immunologically mediated reactional states are the major causes of morbidity in leprosy. The propensity of the disease, when untreated, to result in certain characteristic deformities and the recognition in most cultures that the disease is communicable from person to person have resulted historically in a profound social stigma. Today, with early diagnosis and the institution of appropriate and effective antimicrobial therapy, patients can lead productive lives in the community, and deformities and other visible manifestations can largely be prevented.

ETIOLOGY

M. leprae is an obligate intracellular bacillus (0.3 to 1 μm wide and 1 to 8 μm long) that is acid-fast, indistinguishable microscopically from other mycobacteria, ideally detected in tissue sections by a modified Fite stain, and without demonstrable strain variability. *M. leprae* produces no known toxins and is well adapted to penetrate and reside within macrophages, yet it may survive outside the body for 7 to 10 days. In untreated patients, only ~1% of *M. leprae* organisms are viable. The morphologic index, a measure of the number of acid-fast bacilli (AFB) in skin scrapings that stain uniformly bright, correlates with viability. The bacteriologic index, a logarithmic-scaled measure of the density of *M. leprae* in the dermis, may be as high as 4+ to 6+ in untreated patients, falling by one unit per year during effective therapy; the rate of fall is independent of the relative potency of effective antimicrobial therapy. A rising bacteriologic or morphologic index suggests relapse and perhaps -- if the patient is being treated -- drug resistance; the latter possibility can be confirmed or excluded in the mouse model.

The genome of *M. leprae* is only 3 million base pairs, two-thirds as large as that of *M. tuberculosis*. The bacterium's complex cell wall has a peptidoglycan backbone, which is linked to arabinogalactan and mycolic acids. Lipoarabinomannan is a key component of the cell membrane, and the outer capsule contains large amounts of an *M. leprae*-specific phenolic glycolipid (PGL-1), which is detected in serologic tests. In addition, highly conserved, usually immunogenic heat-shock proteins containing *M. leprae*-specific and mycobacterial cross-reactive epitopes are found in the cytoplasm and cell wall.

Among the mycobacteria, *M. leprae* is unique in exhibiting dopa oxidase activity and an acid-fastness that is pyridine-extractable. Although it was the first bacterium to be etiologically associated with human disease, *M. leprae* remains one of the few bacterial species that still has not been cultivated on artificial medium or tissue culture. The multiplication of *M. leprae* in mouse footpads (albeit limited, with a doubling time of ~2 weeks) has provided a means to evaluate antimicrobial agents, monitor clinical trials, and screen vaccines. *M. leprae* grows best in cooler tissues (the skin, peripheral nerves, anterior chamber of the eye, upper respiratory tract, and testes), sparing warmer areas of the skin (the axilla, groin, scalp, and midline of the back).

EPIDEMIOLOGY

Demographics Leprosy is almost exclusively a disease of the developing world, affecting areas of Asia, Africa, Latin America, and the Pacific. While Africa has the highest disease prevalence, Asia has the most cases. More than 80% of the world's cases occur in a few countries: India, China, Myanmar, Indonesia, Brazil, and Nigeria. Within endemic locales, the distribution of leprosy is quite uneven, with areas of high prevalence bordering on areas with little or no disease. In Brazil the majority of cases occur in the Amazon basin, while in Mexico leprosy is mostly confined to western states. Except as imported cases, leprosy is largely absent from the United States, Canada, and northwestern Europe. In the United States, ~4000 persons have leprosy and 100 to 200 new cases are reported annually, most of them in California, Texas, New York, and Hawaii among immigrants from Mexico, Southeast Asia, the Philippines, and the Caribbean.

The global prevalence of leprosy is difficult to assess, given that many of the locales with high prevalence lack a significant medical or public health infrastructure. Estimates range from 1.5 to 8 million affected individuals. The lower estimate includes only persons who have not completed chemotherapy, excluding those who may be physically or psychologically damaged from leprosy and who may yet relapse or develop certain immune-mediated reactions; the higher figure includes patients whose infections probably are already cured and many who have no leprosy-related deformity or disability. Although the figures on the worldwide prevalence of leprosy are debatable, it is generally agreed that the annual incidence of new cases is stable and approximates 600,000.

Leprosy is associated with poverty and rural residence. It appears not to be associated with AIDS, perhaps because of leprosy's long incubation period. Most people appear to be naturally immune to leprosy and do not develop disease manifestations following exposure. The time of peak onset is in the second and third decades of life. The most severe form, lepromatous leprosy, is twice as common among men as among women and is rarely encountered in children. The frequency of the polar forms of leprosy (see "Clinical, Histologic, and Immunologic Spectrum," below) in different countries varies widely and may in part be genetically determined; certain HLA associations are known for both lepromatous and tuberculoid leprosy (see below). In India and Africa, 90% of cases are tuberculoid; in Southeast Asia, 50% are lepromatous and 50% tuberculoid; and in Mexico, 90% are lepromatous.

Transmission *M. leprae* causes disease primarily in humans. However, in Texas and Louisiana, 15% of nine-banded armadillos are infected, and armadillo contact occasionally results in human disease. Following intravenous experimental inoculation of *M. leprae*, 60% of armadillos develop a heavy disseminated infection of the liver, spleen, lymph nodes, and skin; these experimental animals have been the source of vast quantities of *M. leprae* organisms for laboratory and clinical research.

The route of transmission of leprosy remains uncertain and may be multiple; nasal droplet infection, contact with infected soil, and even insect vectors have been considered the prime candidates. Aerosolized *M. leprae* can cause infection in

immunosuppressed mice, and a sneeze from an untreated lepromatous patient may contain $>10^{10}$ [AFB](#). Furthermore, both IgA antibody to *M. leprae* and genes of *M. leprae* -- demonstrable by polymerase chain reaction (PCR) -- have been found in the nose of individuals without signs of leprosy from endemic areas and in 19% of occupational contacts of lepromatous patients.

Several lines of evidence implicate soil transmission of leprosy. (1) In endemic countries such as India, leprosy is primarily a rural and not an urban disease. (2) *M. leprae* products have been demonstrated to be resident in soil in endemic locales. (3) Direct dermal inoculation (e.g., during tattooing) may transmit *M. leprae*, and common sites of leprosy in children are the buttocks and thighs, suggesting that microinoculation of infected soil may transmit the disease.

Evidence for insect vectors of leprosy includes the demonstration that bedbugs and mosquitoes in the vicinity of leprosaria regularly harbor *M. leprae* and that experimentally infected mosquitoes can transmit infection to mice. Skin-to-skin contact is generally not considered an important route of transmission.

In endemic countries, ~50% of leprosy patients have a history of intimate contact with an infected person (often a household member), while, for unknown reasons, leprosy patients in nonendemic locales can identify such contact only 10% of the time. Moreover, household contact with an infected lepromatous case carries an eventual risk of disease acquisition of ~10% in endemic areas as opposed to only 1% in nonendemic locales. Contact with a tuberculoid case carries a very low risk. Physicians and nurses caring for leprosy patients and the co-workers of these patients are not at risk for leprosy.

CLINICAL, HISTOLOGIC, AND IMMUNOLOGIC SPECTRUM

The incubation period prior to manifestation of clinical disease can vary between 2 and 40 years, although it is generally 5 to 7 years in duration. Leprosy presents as a spectrum of clinical manifestations that have pathologic and immunologic counterparts. The spectrum from polar tuberculoid (TT) to borderline tuberculoid (BT) to borderline lepromatous (BL) to polar lepromatous (LL) disease is associated with an evolution from localized to more generalized disease manifestations, an increasing bacterial load, and loss of *M. leprae*-specific cellular immunity. Where a patient presents on the clinical spectrum largely determines prognosis, complications, reactional states, and the intensity of antimicrobial therapy required.

Tuberculoid Leprosy At the less severe end of the spectrum is tuberculoid leprosy, which encompasses TT and BT disease. In general, these forms of leprosy result in symptoms confined to the skin and peripheral nerves. The initial lesion of tuberculoid leprosy is often a hypopigmented macule that is sharply demarcated and hypesthetic. Later, the lesions enlarge by peripheral spread, and the margins become elevated and circinate or gyrate. The central area in turn becomes atrophic and depressed. Fully developed lesions are densely anesthetic and devoid of the normal skin organs (sweat glands and hair follicles). Patients eventually have one or more asymmetrically distributed, hypopigmented, anesthetic, nonpruritic, well-defined macules, often with an erythematous or raised border ([Fig. 170-1](#); [Fig. 170-CD1](#)). Skin lesions of tuberculoid

leprosy vary in diameter from one to several centimeters and are often dry, scaly, and anhidrotic. Tuberculoid leprosy patients may also have asymmetric enlargement of one or a few peripheral nerves. Indeed, leprosy and certain rare hereditary neuropathies are the only human diseases associated with peripheral-nerve enlargement. Although any peripheral nerve may be enlarged (including small digital and supraclavicular nerves), those most commonly affected are the ulnar, posterior auricular, peroneal, and posttibial nerves, with associated hypesthesia and myopathy. At times, tuberculoid leprosy may present with only nerve-trunk involvement with no skin lesions; in such cases it is termed *neural leprosy*. TT leprosy may resolve spontaneously and is not associated with lepra reactions (see "Reactional States," below). BT leprosy does not heal spontaneously and may be associated with type 1 lepra reactions but not with erythema nodosum leprosum (ENL). TT leprosy is the most common form of the disease encountered in India and Africa but is virtually absent in Southeast Asia, where BT leprosy is frequent.

In TT leprosy the epidermis may be involved histologically, while in all other forms of leprosy the epidermis and the superficial dermis are spared, with pathology confined to the deeper dermis. On hematoxylin and eosin staining, TT and BT lesions appear as well-defined noncaseating granulomas with many lymphocytes and Langhans' giant cells. In tuberculoid leprosy, T cells breach the perineurium, and destruction of Schwann cells and axons may be evident, resulting in fibrosis of the epineurium, replacement of the endoneurium with epithelial granulomas, and occasionally caseous necrosis.[AFB](#) are generally absent or few in number. Such invasion and destruction of nerves in the dermis by T cells are pathognomonic for leprosy.

Circulating lymphocytes from patients with tuberculoid leprosy readily recognize *M. leprae* and its constituent proteins, and patients have positive lepromin skin tests (see "Diagnosis" below). In tuberculoid leprosy tissue, there is a 2:1 predominance of helper CD4⁺ over CD8⁺ T lymphocytes. Tuberculoid tissues are rich in the mRNAs of the proinflammatory T_H1 family of cytokines: interleukin (IL) 2, interferon γ (IFN- γ), and IL-12; in contrast, IL-4, IL-5, and IL-10 mRNAs are scarce.

Lepromatous Leprosy At the more severe end of the leprosy spectrum is lepromatous disease, which encompasses the LL and BL forms. The initial skin lesions of lepromatous leprosy are skin-colored or slightly erythematous papules or nodules. In time, individual lesions grow in diameter up to 2 cm; new papules and nodules then appear and may coalesce. Patients later present with symmetrically distributed skin nodules, raised plaques, or diffuse dermal infiltration ([Fig. 170-CD2](#)), which, when on the face, results in leonine facies. Late manifestations include loss of eyebrows (initially the lateral margins only; see [Plate IID-56](#)) and eyelashes, pendulous earlobes, and dry scaling skin, particularly on the feet. Almost exclusively found in western Mexico and the Caribbean is a form of lepromatous leprosy without visible skin lesions but with diffuse dermal infiltration and a demonstrably thickened dermis, termed *diffuse lepromatosis*. In lepromatous leprosy, nerve enlargement and damage tend to be symmetric, result from actual bacillary invasion, and are more insidious but ultimately more extensive than in tuberculoid leprosy. Patients with LL leprosy have symmetric acral distal peripheral neuropathy and a tendency toward symmetric nerve-trunk enlargement. They may also have signs and symptoms related to involvement of the upper respiratory tract, the anterior chamber of the eye, and the testes.

Dermatopathology in lepromatous leprosy is confined to the dermis and particularly affects the dermal appendages. Histologically, the dermis characteristically contains highly vacuolated cells (*foam cells*) otherwise found only in certain lipid-storage disorders. Indeed, on fat staining, these vacuoles are highly positive and are seen to include large amounts of *M. leprae*-associated cell wall lipids and the *M. leprae*-specific [PGL-1](#). The dermis in lepromatous leprosy contains few lymphocytes and giant cells, and granulomas are absent. In LL leprosy, bacilli are numerous in the skin (as many as 10⁹/g), where they are often found in large clumps (*globi*), and in peripheral nerves, where they initially invade Schwann cells, resulting in foamy degenerative myelination and axonal degeneration and later in Wallerian degeneration. In addition, bacilli are plentiful in circulating blood and in all organ systems except the lungs and the central nervous system. Nevertheless, patients are afebrile, and there is no evidence of major organ system dysfunction. The dermis contains more lymphocytes and fewer [AFB](#) and exhibits less vacuolization in BL than in LL leprosy.

In untreated LL patients, lymphocytes regularly fail to recognize either *M. leprae* or its protein constituents, and lepromin skin tests are negative (see "Diagnosis," below). This loss of protective cellular immunity appears to be antigen-specific, as patients are not unusually susceptible to opportunistic infections, cancer, or AIDS and maintain delayed-type hypersensitivity to *Candida*, *Trichophyton*, mumps, tetanus toxoid, and even purified protein derivative of tuberculin. At times, *M. leprae*-specific anergy is reversible with effective chemotherapy. In LL tissues, there is a 2:1 ratio of CD8⁺ to CD4⁺ T lymphocytes. LL tissues demonstrate a T_H2 cytokine profile, being rich in mRNAs for [IL-4](#), [IL-5](#), and [IL-10](#) and poor in those for [IL-2](#), [IFN- \$\gamma\$](#) , and [IL-12](#). It appears that cytokines mediate a protective tissue response in leprosy, as injection of IFN- γ or IL-2 into lepromatous lesions causes a loss of [AFB](#) and histopathologic conversion toward a tuberculoid pattern. Macrophages of lepromatous leprosy patients appear to be functionally intact; circulating monocytes exhibit normal microbicidal function and responsiveness to IFN- γ .

LL and BL patients may develop type 2 lepra reactions ([ENL](#); see "Reactional States," below), while BL patients (but not LL patients) can have type 1 lepra reactions.

Reactional States Lepra reactions comprise several common immunologically mediated inflammatory states that cause considerable morbidity. Some of these reactions precede diagnosis and the institution of effective antimicrobial therapy. Indeed, these reactions may precipitate presentation for medical attention and diagnosis; others occur after the initiation of appropriate chemotherapy. In the latter circumstances, patients often lose confidence in conventional therapy, perceiving that their leprosy is worsening. Only by warning patients of the potential for these reactions and describing their manifestations can physicians treating leprosy patients ensure continued credibility.

Type 1 Lepra Reactions (Downgrading and Reversal Reactions) These reactions occur in almost half of patients with borderline forms of leprosy but not in patients with polar disease. Manifestations include classic signs of inflammation within previously involved macules, papules, and plaques and, on occasion, the appearance of new skin lesions, neuritis, and (less commonly) fever -- generally low-grade. The nerve trunk most

commonly involved in this process is the ulnar nerve at the elbow, which may be painful and exquisitely tender. If patients with affected nerves are not treated promptly with glucocorticoids (see below), irreversible nerve damage may result in as little as 24 h. The most dramatic manifestation is footdrop, which occurs when the peroneal nerve is involved.

When type 1 lepra reactions precede the initiation of appropriate antimicrobial therapy, they are termed *downgrading reactions*, and the case becomes histologically more lepromatous; when they occur after the initiation of therapy, they are termed *reversal reactions*, and the case becomes more tuberculoid. Reversal reactions often occur in the first months or years after the initiation of therapy but may also develop several years thereafter.

Edema is the most characteristic microscopic feature of type 1 lepra lesions, whose diagnosis is primarily clinical. Reversal reactions are typified by a T_H1 cytokine profile, with an influx of CD4⁺ helper cells and increased levels of [IFN- \$\gamma\$](#) and [IL-2](#). In addition, type 1 reactions are associated with large numbers of T cells bearing α receptors -- a unique feature of leprosy.

Type 1 lepra reactions are best treated with glucocorticoids (e.g., prednisone, initially at doses of 40 to 60 mg/d). As the inflammation subsides, the glucocorticoid dose can be tapered, but steroid therapy must be continued for at least 3 months lest recurrence supervene. Because of the myriad toxicities of prolonged glucocorticoid therapy, the indications for its initiation are strictly limited to lesions whose intense inflammation poses a threat of ulceration; lesions at cosmetically important sites, such as the face; and the presence of neuritis. Mild to moderate lepra reactions that do not meet these criteria should be tolerated and glucocorticoid treatment withheld. Thalidomide is ineffective against type 1 lepra reactions; clofazimine (200 to 300 mg/d) is of questionable benefit but in any event is far less efficacious than glucocorticoids.

Type 2 Lepra Reactions (ENL) [ENL](#) occurs exclusively in patients near the lepromatous end of the leprosy spectrum, affecting nearly 50% of this group. Although ENL may precede leprosy diagnosis and initiation of therapy -- sometimes, in fact, prompting the diagnosis -- in 90% of cases it follows the institution of chemotherapy, generally within 2 years. The most common features of ENL are crops of painful erythematous papules that resolve spontaneously in a few days to a week but may recur; malaise; and fever that can be profound. However, patients may also experience symptoms of neuritis, lymphadenitis, uveitis, orchitis, and glomerulonephritis and may develop anemia, leukocytosis, and abnormal liver function tests, particularly increased aminotransferase levels. Individual patients may have either a single bout of ENL or chronic recurrent manifestations. Bouts may be either mild or severe and generalized; in rare instances, ENL results in death.

Skin biopsy of [ENL](#) papules reveals vasculitis or panniculitis, sometimes with many lymphocytes but characteristically with polymorphonuclear leukocytes as well.

Elevated levels of circulating tumor necrosis factor (TNF) have been demonstrated in [ENL](#); thus TNF may play a central role in the pathobiology of this syndrome. ENL is thought to be a consequence of immune complex deposition, given its T_H2 cytokine

profile and its high levels of [IL](#)-6 and IL-8. However, in ENL tissue, the presence of HLA Dr framework antigen of epidermal cells -- considered a marker for a delayed-type hypersensitivity response -- and evidence for higher levels of IL-2 and [IFN- \$\gamma\$](#) than are usually seen in polar lepromatous disease suggest an alternative mechanism.

Treatment must be individualized. If [ENL](#) is mild (i.e., without fever or other organ involvement, with occasional crops of only a few skin papules), it may be treated with antipyretics alone. However, in cases with many skin lesions, fever, malaise, and other tissue involvement, brief courses (1 to 2 weeks) of glucocorticoids (initially 40 to 60 mg/d) are often effective. With or without therapy, individual inflamed papules last for $\approx$ 1 week. Successful therapy is defined by the cessation of skin lesion development and the disappearance of other systemic signs and symptoms. If, despite two courses of glucocorticoid therapy, ENL appears to be recurring and persisting, treatment with thalidomide (100 to 300 mg nightly) should be initiated, with the dose depending on the initial severity of the reaction. Because even a single dose of thalidomide administered early in pregnancy may result in severe birth defects, including phocomelia, the use of this drug in the United States for the treatment of fertile females is tightly regulated and requires informed consent, prior pregnancy testing, and maintenance of birth control measures. Although the mechanism of thalidomide's dramatic action against ENL is not entirely clear, the drug's efficacy is probably attributable to its reduction of [TNF](#) levels and IgM synthesis and its slowing of polymorphonuclear leukocyte migration. After the reaction is controlled, lower doses of thalidomide (50 to 200 mg nightly) are effective in preventing relapses of ENL. Clofazimine in high doses (300 mg nightly) has some efficacy against ENL, but its use permits only a modest reduction of the glucocorticoid dose necessary for ENL control.

Lucio's Phenomenon This unusual reaction is seen exclusively in patients from the Caribbean and Mexico who have the diffuse lepromatosis form of lepromatous leprosy, most often those who are untreated. Patients with this reaction develop recurrent crops of large, sharply marginated, ulcerative lesions -- particularly on the lower extremities -- that may be generalized and, when so, are frequently fatal as a result of secondary infection and consequent septic bacteremia. Histologically, the lesions are characterized by ischemic necrosis of the epidermis and superficial dermis, heavy parasitism of endothelial cells with [AFB](#), and endothelial proliferation and thrombus formation in the larger vessels of the deeper dermis. Like [ENL](#), the Lucio reaction is probably mediated by the immune complex. Neither glucocorticoids nor thalidomide is effective against this syndrome. Optimal wound care and therapy for bacteremia are indicated. Ulcers tend to be chronic and heal poorly. In severe cases, exchange transfusion may prove useful.

Nerve Abscesses Patients with various forms of leprosy, but particularly those with the BT form, may develop abscesses of nerves (most commonly the ulnar) with an adjacent cellulitic appearance of the skin. In such conditions, the affected nerve is swollen and exquisitely tender. Although glucocorticoids may reduce signs of inflammation, rapid surgical decompression is necessary to prevent irreversible sequelae.

Complications

The Extremities Complications of the extremities in leprosy patients are primarily a consequence of neuropathy leading to insensitivity and myopathy. Insensitivity affects

fine touch, pain, and heat receptors but generally spares position and vibration appreciation. The most commonly affected nerve trunk is the ulnar nerve at the elbow, whose involvement results in clawing of the fourth and fifth fingers, loss of dorsal interosseous musculature in the affected hand, and loss of sensation in these distributions. Median nerve involvement in leprosy impairs thumb opposition and grasp, while radial nerve dysfunction, though rare in leprosy, leads to wristdrop. Tendon transfers can restore hand function but should not be performed until 6 months after the initiation of antimicrobial therapy and the conclusion of episodes of acute neuritis.

Plantar ulceration, particularly at the metatarsal heads, is probably the most frequent complication of leprosy. Plantar ulcers may become secondarily infected and lead to adjacent cellulitis and osteomyelitis. Because of the importance and critical integrity of the normal plantar fat pad, recurrent ulceration is unfortunately common once initial ulceration has occurred and the pad has been replaced by thin and less resilient fibrous scar tissue. The treatment of plantar ulceration includes debridement of devitalized and undermined tissue; discontinuation of weight-bearing, which may be accomplished by means of a total-contact cast or bed rest; and vigorous treatment of secondary infection, which most commonly is due to *Staphylococcus aureus*. Once healing takes place, walking must be limited, especially during the first week, with slow and progressive increases thereafter. Extra-depth shoes or individually fitted shoes with specially molded inserts are required to prevent recurrence.

Peroneal nerve palsies may result from leprosy itself or from one of its reactional states; the consequence is partial or complete footdrop, which causes an uneven distribution of weight on the plantar surface and hence a predilection to ulceration. Simple nonmetallic braces within the shoe may be useful, while tendon transfers can actually correct footdrop. Although uncommon, Charcot's joints, particularly of the foot and ankle, may result from leprosy.

The loss of distal digits in leprosy is a consequence of insensitivity, trauma, secondary infection, and -- in lepromatous patients -- a poorly understood and sometimes profound osteolytic process. Conscientious protection of the extremities during cooking and work and the early institution of therapy have substantially reduced the frequency and severity of distal digit loss in recent times.

The Nose In lepromatous leprosy, bacillary invasion of the nasal mucosa can result in chronic nasal congestion and epistaxis. Saline nosedrops may relieve these symptoms. Long-untreated LL leprosy may further result in destruction of the nasal cartilage, with consequent saddle-nose deformity or anosmia (more common in the preantibiotic era than at present). Nasal reconstructive procedures can ameliorate significant cosmetic defects.

The Eye Owing to cranial nerve palsies, lagophthalmus and corneal insensitivity may complicate leprosy, resulting in trauma, secondary infection, and (without treatment) corneal ulcerations and opacities. For patients with these conditions, eyedrops during the day and ointments at night provide some protection from such consequences. Furthermore, in LL leprosy, the anterior chamber of the eye is invaded by bacilli, and [ENL](#) may result in uveitis, with consequent cataracts and glaucoma. Thus leprosy is a major cause of blindness in the developing world. Slit-lamp evaluation of LL patients

often reveals "corneal beading," representing globi of *M. leprae*.

The testes *M. leprae* invades the testes, while [ENL](#) may cause orchitis. Thus males with lepromatous leprosy often manifest mild to severe testicular dysfunction, with an elevation of luteinizing and follicle-stimulating hormones, decreased testosterone, and aspermia or hypospermia in 85% of LL patients but in only 25% of BL patients. LL patients may become impotent and infertile. Impotence is sometimes responsive to testosterone replacement.

Amyloidosis Secondary amyloidosis is a complication of LL leprosy and [ENL](#) that is encountered infrequently in the antibiotic era. This complication may result in abnormalities of hepatic and particularly renal function.

DIAGNOSIS

Leprosy most commonly presents with both characteristic skin lesions and skin histopathology. Thus the disease should be suspected when a patient from an endemic area has suggestive skin lesions or peripheral neuropathy; the diagnosis should be confirmed by histopathology. In tuberculoid leprosy, lesional areas -- preferably the advancing edge -- must be biopsied because normal-appearing skin does not have pathologic features. In lepromatous leprosy, nodules, plaques, and indurated areas are optimal biopsy sites, but biopsies of normal-appearing skin are also generally diagnostic. Lepromatous leprosy is associated with diffuse hyperglobulinemia, which may result in false-positive serologic tests (e.g., VDRL, RA, ANA) and therefore can cause diagnostic confusion. On occasion, tuberculoid lesions may not (1) appear typical, (2) be hypesthetic, and (3) contain granulomas but only nonspecific lymphocytic infiltrates. In such instances, two of these three characteristics are considered sufficient for a diagnosis. It is preferable to overdiagnose leprosy rather than to allow a patient to remain untreated.

IgM antibodies to [PGL-1](#) are found in 95% of untreated lepromatous leprosy patients; the titer decreases with effective therapy. However, in tuberculoid leprosy -- the form of disease most often associated with diagnostic uncertainty owing to the absence of [AFB](#) -- patients have significant antibodies to PGL-1 only 60% of the time; moreover, in endemic locales, exposed individuals without clinical leprosy may harbor antibodies to PGL-1. Thus PGL-1 serology is of little diagnostic utility in tuberculoid leprosy. Heat-killed *M. leprae* (lepromin) has been used as a skin test reagent. It generally elicits a reaction in tuberculoid leprosy patients, may do so in individuals without leprosy, and gives negative results in lepromatous leprosy patients; consequently, it is likewise of little diagnostic value. Unfortunately, [PCR](#) of skin for *M. leprae*, although positive in LL and BL leprosy, yields negative results in 50% of tuberculoid leprosy cases, again offering little diagnostic assistance.

Included in the differential diagnosis of lesions that resemble leprosy are sarcoidosis, leishmaniasis, lupus vulgaris, lymphoma, syphilis, yaws, granuloma annulare, and various other disorders causing hypopigmentation. Sarcoidosis may result in perineural inflammation, but actual granuloma formation within dermal nerves is pathognomonic for leprosy. In lepromatous leprosy, sputum specimens may be loaded with [AFB](#) -- a finding that can be inappropriately interpreted as representing pulmonary tuberculosis.

TREATMENT

Active Agents Established agents used to treat leprosy include dapsone (50 to 100 mg/d), clofazimine (50 to 100 mg/d, 100 mg three times weekly, or 300 mg monthly), and rifampin (600 mg daily or monthly). Of these drugs, only rifampin is bactericidal. The sulfones (folate antagonists), the foremost of which is dapsone, were the first antimicrobials found to be effective for the treatment of leprosy and are still the mainstay of therapy. With sulfone treatment, skin lesions resolve and numbers of viable bacilli in the skin are reduced. Although primarily bacteriostatic, dapsone monotherapy results in only a 10% resistance-related relapse rate; after ³18 years of therapy and subsequent discontinuation, only another 10% of patients relapse, developing new, usually asymptomatic, shiny, "histoid" nodules. Dapsone is generally safe and inexpensive. Individuals with glucose-6-phosphate dehydrogenase deficiency who are treated with dapsone may develop severe hemolysis; those without this deficiency also have reduced red cell survival and a hemoglobin decrease averaging 1 g/dL. Dapsone's usefulness is limited occasionally by allergic dermatitis and rarely by the sulfone syndrome (including high fever, anemia, exfoliative dermatitis, and a mononucleosis-type blood picture). It must be remembered that rifampin induces microsomal enzymes, necessitating increased doses of medications such as glucocorticoids and oral birth control regimens. Clofazimine is often cosmetically unacceptable to light-skinned leprosy patients because it causes a red-black skin discoloration that accumulates, particularly in lesional areas, and makes the patient's diagnosis obvious to members of the community.

Other antimicrobial agents active against *M. leprae* in animal models and at the usual daily doses used in clinical trials include ethionamide/prothionamide; the aminoglycosides streptomycin, kanamycin, and amikacin (but not gentamicin or tobramycin); minocycline; clarithromycin; and several fluoroquinolones, particularly ofloxacin. Next to rifampin, minocycline, clarithromycin, and ofloxacin appear to be most bactericidal for *M. leprae*, but these drugs have not been used extensively in leprosy control programs.

Choice of Regimens Antimicrobial therapy for leprosy must be individualized, depending on the clinical/pathologic form of the disease encountered. Tuberculoid leprosy, which is associated with a low bacterial burden and a protective cellular immune response, is the easier form to treat and can be reliably cured with a finite course of chemotherapy. In contrast, lepromatous leprosy may have a higher bacillary load than any other human bacterial disease, and the absence of a salutary T cell repertoire requires prolonged or even lifelong chemotherapy. Hence, careful classification of disease prior to therapy is important. In developed countries, clinical experience with leprosy classification is limited; fortunately, however, the resources needed for skin biopsy are highly accessible and pathologic interpretation is readily available. In developing countries, clinical expertise is greater, but it may now be waning as the care of leprosy patients is integrated into general health services. In addition, access to dermatopathology services is often limited. In such instances, skin smears may prove useful, but in many locales access to the resources needed for their preparation and interpretation may also be unavailable.

A reasoned approach to the treatment of leprosy is confounded by these and several other issues:

1. Even without therapy, TT leprosy may heal spontaneously, and prolonged dapsone monotherapy (even for LL leprosy) is generally curative in 80% of cases.
2. In tuberculoid disease, there are often no bacilli found in the skin prior to therapy, and thus there is no objective measure of therapeutic success. Furthermore, despite adequate treatment, TT and particularly BT lesions often resolve little or incompletely, while relapse and late type 1 lepra reactions can be difficult to distinguish.
3. LL leprosy patients commonly harbor viable persistent *M. leprae* organisms after prolonged intensive therapy; the propensity of these organisms to initiate clinical relapse is unclear. Because relapse in LL patients after discontinuation of rifampin-containing regimens usually begins only after 7 to 10 years, follow-up over the very long term is necessary to assess ultimate clinical outcomes.
4. Even though primary dapsone resistance is exceedingly rare and multidrug therapy is generally recommended (at least for lepromatous leprosy), there is a paucity of information from experimental animals and clinical trials on the optimal combination of antimicrobials, dosing schedule, or duration of therapy.

In 1982, the World Health Organization (WHO) made recommendations for "the chemotherapy of leprosy for control programs." These recommendations came on the heels of the demonstration of the relative success of long-term dapsone monotherapy and in the context of concerns about dapsone resistance. Other complicating considerations included the limited resources available for leprosy care in the very areas where it is most prevalent and the frustration and discouragement of patients and program managers with the previous requirement for lifelong therapy for many leprosy patients. The WHO delineated for the first time a finite duration of therapy for all forms of leprosy, and -- given the prohibitive cost of daily rifampin treatment in developing countries -- encouraged the monthly administration of this agent as part of a multidrug regimen.

Over the ensuing years, these [WHO](#) recommendations have been broadly implemented, and the duration of therapy required, particularly for lepromatous leprosy, has been progressively shortened. For treatment purposes, the WHO classifies patients as paucibacillary and multibacillary. Previously, patients without demonstrable [AFB](#) in the dermis were classified as paucibacillary and those with AFB as multibacillary. Currently, owing to the perceived unreliability of skin smears in the field, patients are classified as multibacillary if they have five or more skin lesions and as paucibacillary if they have fewer than five skin lesions. The WHO recommends that paucibacillary adults be treated with 100 mg of dapsone daily and 600 mg of rifampin monthly (supervised) for 6 months ([Table 170-1](#)). Multibacillary adults should be treated with 100 mg of dapsone plus 50 mg of clofazimine daily (unsupervised) and with 600 mg of rifampin plus 300 mg of clofazimine monthly (supervised). Originally, the WHO recommended that lepromatous patients be treated for 2 years or until smears became negative (generally in ~5 years); subsequently, the acceptable course was reduced to 1 year -- a change that remains controversial in the absence of clinical trials.

Several factors, including an improved economic climate, the high relapse rates (20 to 40%, depending on the initial bacterial burden) among patients with lepromatous leprosy after [WHO](#)-recommended treatment, and the demonstrable lesional activity in fully half of tuberculoid leprosy patients after the completion of therapy, have caused many authorities to question the WHO recommendations and to favor a more intensive approach. This approach ([Table 170-1](#)) calls for tuberculoid leprosy to be treated with dapsone (100 mg/d) for 5 years and for lepromatous leprosy to be treated with rifampin (600 mg/d) for 3 years and with dapsone (100 mg/d) throughout life.

On effective antimicrobial therapy, new skin lesions and signs and symptoms of peripheral neuropathy cease appearing. Nodules and plaques of lepromatous leprosy noticeably flatten in 1 to 2 months and resolve in 1 or a few years, while tuberculoid skin lesions may disappear, improve, or remain relatively unchanged. Though the peripheral neuropathy of leprosy may improve somewhat in the first few months of therapy, rarely is it significantly ameliorated by treatment.

PREVENTION AND CONTROL

Vaccination at birth with bacille Calmette-Guerin (BCG) has proved variably effective in preventing leprosy, ranging from totally ineffective to 80% efficacious. The addition of heat-killed *M. leprae* to BCG does not increase vaccine efficacy. Because whole mycobacteria contain large amounts of lipids and carbohydrates that have proven in vitro to be immunosuppressive for lymphocytes and macrophages, *M. leprae* proteins may prove to be superior vaccines. Data from a mouse model support this possibility. Chemoprophylaxis with dapsone may reduce the number of cases of tuberculoid leprosy but not of lepromatous leprosy and hence is not recommended, even for household contacts. Because leprosy transmission appears to require close prolonged household contact, hospitalized patients need not be isolated.

(Bibliography omitted in Palm version)

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171. INFECTIONS DUE TO NONTUBERCULOUS MYCOBACTERIA - *Bernard Hirschel*

Mycobacteria are slightly curved or straight, rod-shaped or coccoid bacilli traditionally identified by the property of acid-fastness: once stained, the organisms are not easily decolorized, even with acid-alcohol, because of the composition of their cell walls. The genetic relation of mycobacteria with one another is evidenced by their ribosomal RNA sequence homology, which can be used for diagnostic purposes.

Because of the overwhelming clinical importance of tuberculosis, mycobacteriologists have distinguished the *Mycobacterium tuberculosis* complex (consisting of *M. tuberculosis*, *M. bovis*, and *M. africanum*) from all other mycobacteria. Except for *M. leprae* ([Chap. 170](#)), the other mycobacteria are referred to as atypical mycobacteria, mycobacteria other than tuberculosis (MOTT), or nontuberculous mycobacteria (NTM). The most clinically important NTM are listed and described in [Table 171-1](#). The isolation of NTM -- or the lack thereof -- from an individual patient or laboratory specimen must be interpreted with the following facts in mind:

1. Some [NTM](#) require special media and/or growth conditions. The laboratory must be alerted and cultures for acid-fast bacilli requested if the diagnosis of these infections is not to be missed.
2. [NTM](#) grow slowly. Even the so-called rapid growers take 3 to 7 days to form visible colonies on solid media, whereas slow-growing mycobacteria take weeks or do not grow at all on artificial media.
3. The slow growth of mycobacteria complicates antibiotic susceptibility testing. During prolonged incubation, antibiotics may be degraded and disappear from the culture medium. Long delays reduce the clinical usefulness of whatever results are eventually obtained.
4. Data on in vitro susceptibility correlate poorly with clinical results. For example, clarithromycin and azithromycin are highly and variably concentrated in tissues; consequently, the concentrations necessary for determining resistance are difficult to establish in vitro. Sensitivity testing, based on achievable serum levels, would have predicted that these drugs would have little efficacy in vivo; in fact, the opposite is true, both in animal models and in humans.
5. In contrast to *M. tuberculosis*, [NTM](#) are ubiquitous in the environment. Therefore, isolation of NTM from a site that is not normally sterile (such as sputum, urine, skin, or feces) does not constitute proof of disease. In Switzerland between 1983 and 1988, for example, only 23 of 513 HIV-negative patients with NTM isolates had clinically significant disease. Clusters of unusual isolates are more likely to suggest contamination -- e.g., from tap water or bronchoscopy equipment -- than to represent an epidemic of disease.

The original method for the classification of [NTM](#), developed between 1950 and 1980, depends on speed of growth, morphology, and pigmentation of colonies on solid media as well as biochemical reactions. Although reliable and inexpensive, these procedures

take a long time; a period of 12 weeks is often required for definitive identification. Of course, such delayed results are of little use in the care of patients.

The isolation of [NTM](#) from blood cultures requires the use of a special medium for lysis-centrifugation or broth culture. The lysis-centrifugation method (lysis of blood cells followed by centrifugation and plating of the pellet with the bacteria on solid medium) permits quantification of bacteremia; however, some mycobacteria (e.g., *M. genavense*) do not grow well on solid medium and will not be detected by this method. Culture in liquid broth, such as that used in the radiometric Bactec system, shortens the time needed to identify a positive culture but also precludes the study of colonial morphology and pigmentation. Molecular probes are now used for rapid identification of the most important species (*M. avium*, *M. intracellulare*, *M. goodii*, *M. kansasii*, and the *M. tuberculosis* complex) in a positive culture; a color is produced upon hybridization of the probe to specific sequences of the mycobacterial ribosome.

Twenty years ago, the field of mycobacteriology was something of a backwater. Tuberculosis was incorrectly perceived as a disappearing problem, and [NTM](#) were causing only rare and chronic diseases. AIDS, however, has brought mycobacterial infections to the forefront of clinical medicine once more. HIV and *M. tuberculosis* make a volatile mixture, and disseminated infections with NTM are extremely frequent in the advanced stages of AIDS ([Chap. 309](#)). In this setting, it is fortunate that new molecular techniques based on DNA amplification accelerate diagnosis, identify common sources of infection, and reveal new types of NTM, while new antibiotics, such as the macrolides, the rifamycins, and the fluoroquinolones, offer improved options for treatment and prevention. In addition, highly active antiretroviral therapy (HAART) has had a dramatic impact. By preventing and reversing immunodeficiency, HAART ([Chap. 309](#)) also prevents and reverses NTM infections.

NTM INFECTIONS IN AIDS AND OTHER IMMUNODEFICIENCIES

DISSEMINATED INFECTIONS

Etiology The majority of mycobacterial infections in immunocompromised hosts are caused by organisms belonging to the group referred to as the *M. avium* complex (MAC). This group has always been considered to include *M. avium* and *M. intracellulare* (designated by the abbreviation *MAI*) and in the past encompassed *M. scrofulaceum* as well (hence the abbreviation *MAIS*). With the development and marketing of diagnostic probes that distinguish *M. avium* from *M. intracellulare*, it has become clear that the vast majority of disseminated "MAC" infections in AIDS are actually caused by *M. avium*. Thus, from a microbiologic standpoint, this designation is now obsolete. However, it is still used in clinical practice and will be employed in that context herein.

M. genavense causes systemic infections similar to those caused by [MAC](#) organisms. *M. genavense* does not grow well in culture and may therefore be missed in some instances. However, in a series of nearly 200 disseminated [NTM](#) infections from Switzerland, 13% of cases were due to *M. genavense*. Other NTM, including *M. xenopi*, *M. simiae*, *M. scrofulaceum*, *M. malmoense*, and *M. celatum*, may also be involved in such cases. In addition, AIDS patients with localized NTM diseases (see "Localized

Infections," below) often have positive blood cultures (e.g., patients with skin disease due to *M. haemophilum* or with lung disease due to *M. kansasii*).

Epidemiology and Host Factors Because gastrointestinal symptoms often predominate in [NTM](#) infection and because the intestinal submucosa is intensely involved, ingestion seems logical as a primary route of infection. Many environments and animals teem with NTM (especially [MAC](#) organisms), including swamps in the southeastern United States, swine almost everywhere, piped water in New England and in Finland, and soil from potted plants in San Francisco. Birds are frequently infected with *M. genavense*. Skin-test data and humoral antibody patterns point to widespread exposure. However, a direct connection of the environment to the patient is often lacking, and it is not always clear whether strains found in the environment are pathogenic in humans. In an exhaustive study of dietary factors, patients with NTM were found to have consumed more hard cheese than controls without NTM, but no NTM could be found in samples of cheese. At present, the epidemiologic evidence is not strong enough to serve as a basis for dietary recommendations in persons at high risk of NTM infection. There is no evidence for nosocomial spread of NTM from patient to patient; however, hospital hot-water systems have been suspected as the source of isolated clusters of cases. Whereas regional variations in the environmental frequency of NTM are striking, it is difficult to correlate these variations with the frequency of NTM infection among HIV-infected patients.

Disseminated infections with [NTM](#) occur almost exclusively in severely immunosuppressed patients, usually those with AIDS. Rarely, such infections are found in patients immunosuppressed for other reasons, including transplant recipients, and patients with leukemia (in particular, hairy cell leukemia) or lymphoma. Cases in children may suggest the presence of a congenital immunodeficiency disease, such as a deficiency in the receptors for interferon γ (IFN- γ) or interleukin (IL) 12. Finally, rare cases of dissemination occur in immunocompetent patients who have extensive pulmonary disease (see "NTM Infections in Immunocompetent Patients," below).

In patients with AIDS, the risk of [NTM](#) infection correlates well with the degree of depletion of CD4⁺ lymphocytes. Disseminated NTM disease is rare among patients with >100 CD4⁺ cells per microliter; however, among patients with <10 CD4⁺ lymphocytes per microliter, the actuarial probability of having a blood culture positive for NTM reaches 40% after 1 year. [HAART](#) has greatly diminished the overall incidence of NTM disease in AIDS. Current treatment recommendations suggest starting HAART when the CD4⁺ count falls below 500/uL. At these levels, NTM disease does not occur. In extremely immunosuppressed patients who start HAART, the CD4⁺ count typically rises above 100/uL within a few months; such patients are again unlikely to develop NTM disease.

Clinical Manifestations As has already been mentioned, disseminated infection with [NTM](#) is essentially a disease of advanced immunodeficiency. In HIV-infected patients, the median CD4⁺ lymphocyte count at the time of diagnosis is ~10/uL. Certainly, other diagnoses should be considered first when a patient with symptoms suggestive of NTM infection has >100 CD4⁺ cells per microliter. Prospective monthly blood cultures have shown that NTM bacteremia often causes few or no symptoms. In clinical practice, however, cultures are not performed if the patient is asymptomatic.

Disseminated [NTM](#) infection should be suspected on the basis of prolonged fever (sometimes of varying intensity -- particularly at first -- and accompanied by night sweats) and weight loss. Signs of abdominal involvement that may be evident on computed tomography or ultrasonography include enlargement of the liver and spleen and swelling of abdominal lymph nodes, which may result in diarrhea and/or abdominal pain. Anemia and leukopenia are frequently documented; although it is tempting to relate these abnormalities to infection of bone marrow by NTM, multiple factors are usually involved.

In short, the clinical picture of infection with [NTM](#) is not distinctive. Many other conditions, including abdominal lymphoma, the HIV wasting syndrome, *Salmonella* or *Campylobacter* infection, cryptosporidiosis, or microsporidiosis, may mimic (and coexist with) disseminated NTM infection. As stated earlier, suspicion of such infection should prompt a request for blood cultures.

Diagnosis Blood cultures on special media are the cornerstone of the diagnosis of [NTM](#) infection, both in patients with organ involvement and in those without. In most symptomatic patients, the intensity of mycobacteremia is such that most or all blood cultures are positive. Therefore, the performance of multiple, repetitive cultures at short intervals is not worthwhile. Rather, in clinical practice, two or three blood cultures are sufficient. In one study, the results of prospective cultures varied, and these variations (positive followed by negative or vice versa) were unrelated to symptom status. As mentioned above, liquid cultures (e.g., the Bactec system) are likely to become positive earlier (within 7 to 14 days) and are therefore preferred to cultures on solid medium. In patients infected with *M. genavense* or *M. xenopi* and in patients being treated for [MAC](#) infection, the interval to culture positivity may be much longer. In rare cases, organ involvement in NTM infection may be found to be widespread at autopsy despite multiple negative blood cultures during life.

Because the liver and bone marrow are often involved in disseminated [NTM](#) infection, the bacteria may be visible in acid-fast-stained biopsy samples from these sites. Presumptive diagnosis by examination of a biopsied liver specimen saves time. The yield has been as high as 50% in patients with clearly abnormal values in liver function tests. However, the yield of this method has been disappointing in patients with suspected NTM infection, negative blood cultures, and normal or nearly normal results in liver function tests.

TREATMENT

Compared with *M. tuberculosis*, [NTM](#) are of low virulence. NTM tend to affect severely immunosuppressed patients, who usually have many other medical problems. Treatment is complex, relies on the use of multiple drugs with numerous adverse effects, and may interfere with antiretroviral therapy.

The drugs used for the treatment of disseminated [NTM](#) infection are different from those used against tuberculosis ([Table 171-1](#); [Chaps. 168](#) and [169](#)). In particular, isoniazid has little effect on [MAC](#) organisms. The best method for antibiotic sensitivity testing of NTM is controversial, and the question of what relation -- if any -- exists between in vitro

resistance and treatment failure remains unanswered. From the clinician's viewpoint, growth inhibition in liquid cultures is preferred to other methods of sensitivity testing because the results become available within 7 days.

The agents most active against [MAC](#) organisms are the macrolides clarithromycin and azithromycin. Both of these drugs are well absorbed from the gastrointestinal tract and well concentrated in macrophages and tissues, where their levels exceed those in plasma by more than 10-fold. Given alone, either drug can render blood cultures negative in a substantial proportion of cases. However, resistance (due to a single point mutation in the gene coding for the large ribosomal subunit) invariably develops, and [NTM](#) reappear in the bloodstream.

A majority of [MAC](#) strains are sensitive to ethambutol, ciprofloxacin, clofazimine, amikacin, rifampin, and rifabutin; that is, the concentrations of these drugs attainable in serum are inhibitory in vitro. However, none of these drugs consistently reduces the intensity of mycobacteremia when used alone. The preferred regimen for treatment of disseminated [NTM](#) infections is the combination of rifabutin (300 to 600 mg/d), clarithromycin (1 g twice daily), and ethambutol (900 mg/d). In a randomized trial, this regimen was superior to the combination of rifampin, clofazimine, ciprofloxacin, and ethambutol, with more rapid resolution of bacteremia and increased survival. The higher dose of rifabutin was more effective but frequently caused uveitis. This side effect is of special concern when rifabutin is used in combination with ritonavir, which increases the concentration of a toxic metabolite. Among the HIV protease inhibitors, indinavir and nelfinavir may be used in combination with rifabutin.

The inclusion of intravenous amikacin in multidrug regimens has not conferred additional benefit. Nonetheless, this drug may be useful in certain cases -- e.g., when resistance to clarithromycin develops or when severe gastrointestinal symptoms interfere with oral therapy. In addition, amikacin may prevent the emergence of resistance to clarithromycin when the two drugs are used concurrently.

It is not clear how long therapy needs to be administered. Older regimens did not eradicate [MAC](#), and many experts recommended lifelong treatment. Unfortunately, multidrug regimens are often poorly tolerated. In patients whose symptoms have lessened, whose blood cultures have become negative, and whose CD4+ counts have recovered to >100/uL with [HAART](#), it is reasonable to discontinue antimycobacterial treatment.

In vitro and in experimental animals, cytokines such as [IL](#)-12, granulocyte-macrophage colony-stimulating factor, and [IFN-g](#) synergistically with antibiotics against [MAC](#). In a small-scale pilot trial including seven HIV-negative patients, IFN-g was beneficial.

Encapsulation of many drugs into liposomes enhances their effect in animal models because both liposomes and [MAC](#) are ingested by macrophages. Relevant data from studies of humans are still scarce, however.

Disseminated infections caused by [NTM](#) other than MAC have been too rare for therapy to be evaluated in controlled trials. The presently recommended treatment for these infections is the same as that for disseminated MAC infections. In particular, *M.*

genavense seems to be sensitive to clarithromycin and rifabutin.

Prevention As has been discussed, disseminated infection with [MAC](#) occurs almost exclusively in persons severely immunocompromised by HIV infection. Therefore, the best approach to the prevention of MAC infections is the prevention and reversal of immunodeficiency by [HAART](#). In patients whose HIV is resistant to HAART and who are severely immunosuppressed, with CD4+ counts <100/uL, prophylaxis with rifabutin (300 mg/d), clarithromycin (500 mg once or twice daily), or azithromycin (1200 mg weekly) is likely to decrease the incidence of positive blood cultures by ~60%. Patients receiving prophylaxis have also had less fever, experienced less fatigue, and survived longer than patients not receiving prophylaxis. Although breakthrough bacteremia involving resistant organisms is a concern, this condition has not developed with rifabutin prophylaxis and is rare with clarithromycin. However, it is standard practice to rule out preexisting disseminated [NTM](#) infection (by blood culture) before starting prophylaxis.

LOCALIZED INFECTIONS

Pulmonary Disease The significance of isolation of [NTM](#) from the airways of AIDS patients merits special discussion. In general, HIV-infected patients who have NTM in sputum or bronchoalveolar lavage fluid but have little evidence of lung damage require no treatment. *M. avium* only rarely causes significant pulmonary disease in AIDS; its isolation from sputum in the absence of radiographic changes is usually without clinical significance. In contrast, the isolation of *M. kansasii* from the lung is clinically significant: this organism causes a disease -- often predominant in the upper lobes -- that resembles pulmonary tuberculosis, with fever, cough, infiltrates, and cavities. Blood cultures are often positive. Drugs active against *M. tuberculosis*, such as rifampin (600 mg/d) and isoniazid (300 mg/d), are also effective against *M. kansasii*; treatment is generally continued for 18 to 24 months, although some data suggest that 12 months may be adequate.

Skin Disease [MAC](#) organisms, which frequently cause disseminated disease with positive blood cultures in AIDS, are also rarely associated with heterogeneous skin manifestations, such as nodules, ulcers, areas of erythema, pustules, abscesses, or panniculitis. Skin biopsies and blood cultures establish the diagnosis.

M. haemophilum In contrast to [MAC](#) organisms, *M. haemophilum* has a tendency to involve the skin, bones, joints, and lungs, although most patients also have positive blood cultures. Skin lesions are nodular, may ulcerate, and are disseminated. In the absence of specific data, treatment should follow the guidelines for MAC infection.

[NTM](#) INFECTIONS IN IMMUNOCOMPETENT PATIENTS

PULMONARY DISEASE

Etiology The [NTM](#) most frequently causing pulmonary infections are *M. intracellulare*, *M. avium*, and *M. kansasii*. Many other species, such as *M. xenopi*, *M. malmoense*, and *M. interjectum*, can also be involved in these infections. Identification of the specific pathogen is important in the choice among the various therapeutic strategies. For example, *M. kansasii* responds to antituberculosis drugs, including isoniazid.

Epidemiology and Host Factors As has already been noted, [NTM](#) are ubiquitous in the environment, but their pathogenicity is low. Preexisting lung disease (e.g., chronic obstructive airway disease, cancer, previous tuberculosis, bronchiectasis, cystic fibrosis, and silicosis, with cavities and bronchiectases) is the main predisposing factor for pulmonary disease due to NTM.

Anecdotal evidence suggests that the proportion of patients without underlying lung pathology who are developing pulmonary disease due to [NTM](#) is increasing. These patients are usually elderly; many are women with pectus excavatum or scoliosis. In the latter elderly women, the lingula and the right middle lobes are particularly involved. Somewhat whimsically, the disease in these patients has been called "Lady Windermere syndrome" after the main character in Oscar Wilde's play *Lady Windermere's Fan*, who went to extremes to refrain from coughing.

Clinical Manifestations Most immunocompetent patients with pulmonary [NTM](#) infection present with chronic cough, low-grade fever, and malaise; some present with hemoptysis. These symptoms may be masked by those of the underlying disease process.

Diagnosis In contrast to the isolation of *M. tuberculosis*, of which even a single colony -- whatever its origin -- is clinically significant, the isolation of [NTM](#) from the sputum never in itself proves the existence of disease. NTM are frequently commensals and colonize both diseased and normal airways. Because treatment of NTM infection is complicated, it is important that the diagnosis be certain. The American Thoracic Society has formulated the following minimal guidelines for the diagnosis of pulmonary NTM disease: "evidence, such as an infiltrate visible on a chest roentgenogram, of disease, the cause of which has not been determined by careful clinical and laboratory studies, and...isolation of multiple colonies of the same strain of mycobacteria repeatedly, usually in the absence of other pathogens." For patients who have pulmonary infiltrates but not cavities, these criteria may not be specific enough; some patients are found to have cleared NTM after a 1-month trial of bronchial hygiene alone (inhalation of saline and bronchodilators to induce cough and sputum production). The detection by computed tomography of bronchiectases and nodular infiltrates in the same lobe may be particularly suggestive of NTM infection.

TREATMENT

Lung disease due to [NTM](#) may be managed by follow-up without treatment, by resection, or by drug therapy. No randomized trial has determined which is the best option. In retrospectively analyzed case series, patients undergoing surgery have had a better outcome than those treated only with drugs. However, selection bias has probably influenced these results since patients with extensive lung disease are poor candidates for surgery.

As in HIV-infected patients, immunocompetent patients with minimal disease do not need treatment at all. Likewise, [NTM](#) disease may present as a solitary pulmonary nodule that, once resected (to confirm or exclude a diagnosis of cancer), requires no further drug treatment.

Most other patients with pulmonary [NTM](#) disease are treated with antimicrobials; in addition, they may or may not undergo surgery. The drugs available for the treatment of infection with *M. avium* or *M. intracellulare* have already been discussed. The regimens recommended for disseminated [MAC](#) infection are preferred, although large doses of clarithromycin are often poorly tolerated by elderly patients. *M. intracellulare* may be easier to treat and eradicate than *M. avium*. In two small open studies, single-agent treatment with clarithromycin (500 mg twice daily) led to improvements detected by chest radiography and sputum culture.

Indications for surgery are difficult to establish but include a disappointing response to antibiotics, the presence of localized disease, and the absence of contraindications (especially impaired respiratory functions). Ideally, drug treatment should begin before surgery and should render the sputum negative by the time of the operation.

In contrast to [MAC](#) organisms, *M. kansasii* is predictably sensitive to antituberculosis agents. Treatment should consist of isoniazid (300 mg/d), rifampin (600 mg/d), and ethambutol (15 to 25 mg/kg per day). The optimal duration of therapy is unknown, but most patients have been treated for 18 to 24 months; 12 months may suffice. Sulfamethoxazole is recommended for the occasional patient whose infection relapses after *M. kansasii* becomes resistant to rifampin.

LYMPHADENITIS

[NTM](#) are among the causes of localized lymphadenitis. This disease occurs mostly in children between the ages of 1 and 5 years. Painless swelling of one node or a group of nodes usually affects the anterior cervical chain. Nodes may rapidly increase in size, with the formation of fistulas to the skin. *M. scrofulaceum* or [MAC](#) organisms most commonly cause NTM lymphadenitis, although many other species may be involved. Once tuberculosis has been excluded, the treatment of choice is excision without chemotherapy. When excision is dangerous because of proximity to the facial nerve, aspiration combined with chemotherapy may be effective.

SKIN DISEASE DUE TO NTM

Swimming-Pool and Fish-Tank Granuloma ([Fig. 171-CD1](#)) Between 1 week and 2 months (usually 2 to 3 weeks) after contact with contaminated tropical fish tanks, swimming pools, or saltwater fish, a small violet nodule or pustule may appear at a site of minor trauma. This lesion may evolve to form a crusted ulcer or small abscess or may remain warty. Lesions are multiple and disseminated on occasion -- particularly, but not exclusively, in immunosuppressed patients. The causative organism is *M. marinum*. The patient's clinical history, combined with the isolation of *M. marinum* after biopsy and culture, establishes the diagnosis. Lesions often heal spontaneously. In cases of persistence or dissemination, rifampin (300 to 600 mg/d) in combination with ethambutol (15 to 25 mg/kg per day), trimethoprim-sulfamethoxazole (160/800 mg twice daily), or minocycline (100 mg/d) may be tried for a period of at least 3 months. Very rarely, a similar clinical picture is produced by *M. gordonae*, a frequently isolated but usually nonpathogenic species.

Buruli Ulcer In many tropical areas throughout the world, *M. ulcerans* may cause an itching nodule on the arms or legs, which then breaks down to form a shallow ulcer of variable size ([Fig. 171-CD2](#)). The course of this condition is usually prolonged. *M. ulcerans* is difficult to culture; plates need to be incubated at low temperature. Excision constitutes the usual therapy. Treatment with rifampin, clofazimine, or trimethoprim-sulfamethoxazole has met with variable success.

NTMINFECTIONS OF SOFT TISSUE, TENDONS, BONES, AND JOINTS

Infections Linked to Injections and Surgery Occasionally, mycobacteria are isolated from nodular skin lesions of hospitalized patients, particularly those who are immunosuppressed ([Fig. 171-CD3](#)); in some instances there is associated lymphatic spread. Many cases are linked to injection; diabetic patients are at especially high risk. In ophthalmology, mycobacteria may cause keratitis and corneal ulceration after surgery or injury. Epidemics of mycobacterial infection following cardiac surgery have been linked to contaminated ice packs and contaminated porcine heart valves. These infections are usually due to *M. fortuitum*, *M. chelonae*, or *M. abscessus*, which are referred to collectively as the *M. fortuitum* complex. These are the so-called rapidly growing mycobacteria: colonies on solid medium appear 3 to 7 days after inoculation. As organisms may fail to grow at 37°C, incubation at 30 to 33°C is recommended. These mycobacteria are notoriously resistant to most antituberculosis drugs. Debridement is best combined with administration of two or three of the antibiotics mentioned in [Table 171-1](#).

Infections of Tendons, Joints, and Bones In rare cases, mycobacteria invade deep tissues after direct inoculation, via contiguous spread from superficial sites of infection, or through the bloodstream. [MAC](#) organisms and *M. ulcerans* are most often cited in these instances. *M. szulgai* seems to be involved particularly frequently in olecranon bursitis.

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SECTION 9 -SPIROCHETAL DISEASES

172. SYPHILIS - Sheila A. Lukehart

DEFINITION

Syphilis, a chronic systemic infection caused by *Treponema pallidum* subspecies *pallidum*, is usually sexually transmitted and is characterized by episodes of active disease interrupted by periods of latency. After an incubation period averaging 2 to 6 weeks, a primary lesion appears, often associated with regional lymphadenopathy. A secondary bacteremic stage, associated with generalized mucocutaneous lesions and generalized lymphadenopathy, is followed by a latent period of subclinical infection lasting many years. In about one-third of untreated cases, the tertiary stage is characterized by progressive destructive mucocutaneous, musculoskeletal, or parenchymal lesions; aortitis; or symptomatic central nervous system (CNS) disease.

ETIOLOGY

The Spirochaetales include three genera that are pathogenic for humans and for a variety of other animals: *Leptospira*, which causes human leptospirosis; *Borrelia*, which causes relapsing fever and Lyme disease; and *Treponema*, which causes the diseases known as treponematoses. The genus *Treponema* includes *T. pallidum* subspecies *pallidum*, which causes venereal syphilis; *T. pallidum* subspecies *pertenue*, which causes yaws; *T. pallidum* subspecies *endemicum*, which causes endemic syphilis or bejel; and *T. carateum*, which causes pinta ([Chap. 173](#)). Other *Treponema* species found in the human mouth, genital mucosa, and gastrointestinal tract have no proven pathogenic role in human disease. These spirochetes can be confused with *T. pallidum* on dark-field examination. An oral treponeme that is very closely related to *T. pallidum* antigenically has been found to be significantly associated with periodontitis and acute necrotizing ulcerative gingivitis; its etiologic role in these gum diseases is unknown. None of the four pathogenic treponemes has yet been cultured in quantity. Until recently, the subspecies were distinguished primarily by the clinical syndromes they produce. Recent studies have identified molecular signatures that can differentiate *T. pallidum* subspecies *pallidum* from the other pathogenic *T. pallidum* subspecies by culture-independent, polymerase chain reaction (PCR)-based methods.

T. pallidum subspecies *pallidum* (hereafter referred to simply as *T. pallidum*), a thin delicate organism with 6 to 14 spirals and tapered ends, measures 6 to 15 μm in total length and 0.2 μm in width. The cytoplasm is surrounded by a trilaminar cytoplasmic membrane, which in turn is surrounded by a delicate peptidoglycan layer providing some structural rigidity. This layer is surrounded by a lipid-rich outer membrane that contains relatively few integral membrane proteins. Six endoflagella wind around the cell body in a space between the inner cell wall and the outer membrane and may be the elements responsible for motility.

The sequencing of the genome of *T. pallidum* has yielded information about the organism's metabolic capabilities. *T. pallidum* lacks the genes required to synthesize enzyme cofactors, fatty acids, and nucleotides de novo. In addition, it lacks genes encoding the enzymes of the Krebs cycle and oxidative phosphorylation. To

compensate, the organism contains numerous genes predicted to code for transporters of amino acids, carbohydrates, and cations. In addition, the genome analyses and other studies have revealed the existence of a 12-member gene family (called *tpr*) that bears similarities to variable outer-membrane antigens of other spirochetes. Although the role of these encoded molecules has not yet been defined, the TprK antigen appears to be preferentially expressed and serves as a target of opsonic antibody.

The only known natural host for *T. pallidum* is the human. *T. pallidum* can infect many mammals, but only humans, higher apes, and a few laboratory animals regularly develop syphilitic lesions. Virulent strains of *T. pallidum* are grown and maintained in rabbits.

EPIDEMIOLOGY

Nearly all cases of syphilis are acquired by sexual contact with infectious lesions (i.e., the chancre, mucous patch, skin rash, or condyloma latum). Less common modes of transmission include nonsexual personal contact and infection in utero or following blood transfusions.

The total number of cases of syphilis reported annually in the United States fell steadily from 575,593 in 1943 to a low of 64,621 in 1987 -- an 88% decrease -- but then increased to 134,255 in 1990. The number of new cases of infectious syphilis reached a peak in 1947 and then fell to approximately 6000 in 1956; since then, a rather steady increase in infectious syphilis has been punctuated by four cycles of 7 to 10 years, each with a rapid rise and fall in incidence (with peaks in 1965, 1975, 1982, and 1990). Since 1990, the number of reported cases of infectious syphilis has again declined by >80%. In 1997, there were 8550 reported cases of primary and secondary syphilis and 46,540 cases of all stages.

The populations at highest risk for acquiring syphilis have changed. Between 1977 and 1982, approximately half of all patients with early syphilis in the United States were homosexual or bisexual men. Largely because of changing sexual practices in this population due to the AIDS epidemic, this proportion has decreased. The most recent epidemic of syphilis predominantly involved African-American heterosexual men and women and occurred largely in urban areas, where infectious syphilis has been correlated significantly with the exchange of sex for "crack" cocaine. The incidence of syphilis peaks at 15 to 34 years of age. The reported incidence is much higher among African Americans than in other ethnic groups and is higher in urban than in rural areas; 80% of infectious syphilis cases are reported from 15% of the counties in the United States.

The incidence of congenital syphilis roughly parallels that of infectious syphilis in females. The number of reported cases of congenital syphilis in infants ≤ 1 year of age was lowest (107 cases) in 1978, when infectious syphilis was most prevalent among homosexual and bisexual men. The dramatic increase in the incidence of primary and secondary syphilis among women from 1986 to 1990 resulted in a proportionate increase in the number of infants born with congenital syphilis -- to 3275 infants in 1991. The incidence of early syphilis among men and women has declined since 1991, as has the number of reported cases of congenital syphilis in infants (with 1049 cases in 1997).

It is important to note, however, that the case definition for congenital syphilis was broadened in 1989 and now includes all live or stillborn infants delivered to women with untreated or inadequately treated syphilis at delivery.

Approximately one of every two individuals named as sexual contacts of persons with infectious syphilis becomes infected. Many sexual contacts will already have developed manifestations of syphilis when they are first seen, and about 30% of apparently uninfected contacts who are examined within 30 days of exposure actually have incubating infection and will later develop infectious syphilis if not treated. Thus, the identification and "epidemiologic" treatment of all recently exposed sexual contacts constitute an important aspect of syphilis control. Also important is the identification of infected persons by serologic testing of pregnant women, persons admitted to hospitals, military inductees, and persons undergoing examination in physicians' offices. Still controversial are laws and regulations requiring routine premarital serologic testing for syphilis, where -- though national data are not available -- the yield is undoubtedly lower.

NATURAL COURSE AND PATHOGENESIS OF UNTREATED SYPHILIS

T. pallidum rapidly penetrates intact mucous membranes or microscopic abrasions in skin and within a few hours enters the lymphatics and blood to produce systemic infection and metastatic foci long before the appearance of a primary lesion. Blood from a patient with incubating or early syphilis is infectious. The generation time of *T. pallidum* during early active disease in vivo is estimated to be 30 to 33 h, and the incubation period of syphilis is inversely proportional to the number of organisms inoculated. The concentration of treponemes generally reaches at least 10^7 per gram of tissue before the appearance of a clinical lesion. On the basis of intradermal injection of graded doses of *T. pallidum* into eight volunteers, the 50% infectious dose was calculated to be 57 organisms. The median incubation period in humans (about 21 days) suggests an average inoculum of 500 to 1000 infectious organisms for naturally acquired disease. The incubation period (from inoculation until the primary lesion becomes discernible) rarely exceeds 6 weeks. Subcurative therapy during the incubation period may delay the onset of the primary lesion, but it is not certain that such treatment reduces the probability that symptomatic disease will ultimately develop.

The primary lesion appears at the site of inoculation, usually persists for 4 to 6 weeks, and then heals spontaneously. Histopathologic examination of primary lesions shows perivascular infiltration, chiefly by lymphocytes (including CD8+ and CD4+ cells), plasma cells, and macrophages, with capillary endothelial proliferation and subsequent obliteration of small blood vessels. The CD4+ infiltration displays a T_H1 -type cytokine profile consistent with the activation of macrophages. At this time *T. pallidum* is demonstrable in the chancre in spaces between epithelial cells; within invaginations or phagosomes of epithelial cells, fibroblasts, plasma cells, and the endothelial cells of small capillaries; within lymphatic channels; and in the regional lymph nodes. Phagocytosis of organisms by activated macrophages ultimately causes their destruction, which results in spontaneous resolution of the chancre.

The generalized parenchymal, constitutional, and mucocutaneous manifestations of secondary syphilis usually appear about 6 to 8 weeks after healing of the chancre,

although 15% of patients with secondary syphilis still have persisting or healing chancres. In other patients, secondary lesions may appear several months after the chancre has healed, and some patients may enter the latent stage without ever recognizing secondary lesions. The histopathologic features of secondary maculopapular skin lesions are hyperkeratosis of the epidermis; capillary proliferation with endothelial swelling in the superficial corium; and dermal papillae with transmigration of polymorphonuclear leukocytes and, in the deeper corium, perivascular infiltration by monocytes, plasma cells, and lymphocytes. Treponemes are found in many tissues, including the aqueous humor of the eye and the cerebrospinal fluid (CSF). Invasion of the [CNS](#) by *T. pallidum* occurs during the first weeks or months of infection, and CSF abnormalities are detected in as many as 40% of patients during the secondary stage. Clinical hepatitis and immune complex-induced membranous glomerulonephritis are relatively rare but recognized manifestations of secondary syphilis; liver function tests may yield abnormal results in up to a quarter of patients with early syphilis. Generalized nontender lymphadenopathy is noted in 85% of patients with secondary syphilis. The paradoxical appearance of secondary manifestations despite high titers of antibody (including immobilizing antibody) to *T. pallidum* is unexplained but may result from changes in expression of surface antigens. Secondary lesions subside within 2 to 6 weeks, and the infection enters the latent stage, which is detectable only by serologic testing. In the preantibiotic era, up to 25% of untreated patients experienced at least one generalized or localized mucocutaneous relapse, usually during the first year; therefore, identification and examination of sexual contacts are most important for patients with syphilis of <1 year's duration. Recurrent generalized rash is now rare.

In the preantibiotic era, about one-third of patients with untreated latent syphilis developed clinically apparent tertiary disease ([Fig. 128-CD5](#)); today, in industrialized countries, specific treatment and coincidental therapy for early and latent syphilis have all but eliminated tertiary disease except for sporadic cases of neurosyphilis in persons infected with HIV. In the past, the most common type of tertiary disease was the gumma, a usually benign granulomatous lesion. Today, gummas are very uncommon. Cardiovascular syphilis, now also rare, is caused by obliterative small-vessel endarteritis, usually involving the vasa vasorum of the ascending aorta and resulting in aneurysm. Asymptomatic [CNS](#) involvement is demonstrable in up to 25% of patients with late latent syphilis. The factors that contribute to the development and progression of tertiary disease are unknown.

The course of untreated syphilis was studied retrospectively in a group of nearly 2000 patients with primary or secondary disease diagnosed clinically (the Oslo Study, 1891-1951) and prospectively in 431 African-American men with seropositive latent syphilis of 3 or more years' duration (the notorious Tuskegee Study, 1932-1972). In the Oslo Study, 24% of patients developed relapsing secondary lesions within 4 years, and 28% eventually developed one or more manifestations of tertiary syphilis. Cardiovascular syphilis, including aortitis, was detected in 10% of patients, none of whom had been infected before age 15; 7% of patients developed symptomatic neurosyphilis, and 16% developed benign tertiary syphilis (gummas of the skin, mucous membranes, and skeleton). Syphilis was the primary cause of death in 15% of men and 8% of women. Cardiovascular syphilis was documented in 35% of men and 22% of women who eventually came to autopsy. In general, serious late complications were nearly twice as common among men as among women.

The Tuskegee Study showed that the death rate among untreated African-American men with syphilis (25 to 50 years old) was 17% higher than that among uninfected subjects and that 30% of all deaths were attributable to cardiovascular or [CNS](#) syphilis. By far the most important factor in increased mortality was cardiovascular syphilis. Anatomic evidence of aortitis was found in 40 to 60% of autopsied subjects with syphilis (versus 15% of control subjects), while CNS syphilis was found in only 4%. Rates of hypertension were also higher among the infected subjects. The ethical issues eventually raised by this study, begun in the preantibiotic era but continuing into the early 1970s, had a major influence on the development of current guidelines for human medical experimentation, and the history of the study may still contribute to a reluctance of some African Americans to participate as subjects in clinical research.

These two studies both showed that about one-third of patients with untreated syphilis develop clinical or pathologic evidence of tertiary syphilis, that about one-fourth die as a direct result of tertiary syphilis, and that there is additional excess mortality not directly attributable to tertiary syphilis.

MANIFESTATIONS

Primary Syphilis The typical primary chancre usually begins as a single painless papule that rapidly becomes eroded and usually becomes indurated, with a characteristic cartilaginous consistency on palpation of the edge and base of the ulcer (see [Plate IID-47, Fig. 172-CD1](#)). In heterosexual men the chancre is usually located on the penis, whereas in homosexual men it is often found in the anal canal or rectum, in the mouth, or on the external genitalia. In women, common primary sites are the cervix and labia. Consequently, primary syphilis goes unrecognized in women and homosexual men more often than in heterosexual men.

Atypical primary lesions are common. The clinical appearance depends on the number of treponemes inoculated and on the immunologic status of the patient. A large inoculum produces a dark-field-positive ulcerative lesion in nonimmune volunteers but may produce a small dark-field-negative papule, an asymptomatic but seropositive latent infection, or no response at all in individuals with a history of syphilis. A small inoculum may produce only a papular lesion, even in nonimmune individuals. Therefore, syphilis should be considered even in the evaluation of trivial or atypical dark-field-negative genital lesions. The genital lesions that most commonly must be differentiated from those of primary syphilis include traumatic superinfected lesions, lesions of herpes simplex virus infection ([Chap. 182](#)), and lesions of chancroid ([Chap. 149](#)). *Primary genital herpes* may produce inguinal adenopathy, but the nodes are tender and the lesions consist of multiple painful vesicles, which later ulcerate and are often accompanied by systemic symptoms, including fever. *Recurrent genital herpes* typically begins with a unilateral cluster of painful vesicles, usually without associated adenopathy. *Chancroid* produces painful, superficial, exudative, nonindurated ulcers, more often multiple than in syphilis (see [Plate IID-54](#)); adenopathy is common, can be either unilateral or bilateral, is tender, and may be suppurative.

Regional lymphadenopathy usually accompanies the primary syphilitic lesion, appearing within 1 week of the onset of the lesion. The nodes are firm, nonsuppurative, and

painless. Inguinal lymphadenopathy is bilateral and may occur with anal as well as with external genital chancres. Rectal chancres result in perirectal lymphadenopathy, while chancres of the cervix and vagina result in iliac or perirectal adenopathy. The chancre generally heals within 4 to 6 weeks (range, 2 to 12 weeks), but lymphadenopathy may persist for months.

Secondary Syphilis The protean manifestations of the secondary stage usually include localized or diffuse symmetric mucocutaneous lesions and generalized nontender lymphadenopathy. The healing primary chancre is still present in 15% of cases. The skin rash consists of macular, papular, papulosquamous, and occasionally pustular syphilides; often more than one form is present simultaneously. The eruption may be very subtle. Approximately 25% of patients with a discernible rash of secondary syphilis may be unaware that they have dermatologic manifestations. Initial lesions are bilaterally symmetric, pale red or pink, nonpruritic, discrete, round macules that measure 5 to 10 mm in diameter and are distributed on the trunk and proximal extremities (see [Plate IID-50](#)). After several days or weeks, red papular lesions 3 to 10 mm in diameter also appear. These lesions, which may progress to necrotic lesions (resembling pustules) in association with increasing endarteritis and perivascular mononuclear infiltration, are distributed widely, frequently involve the palms and soles (see [Plate IID-48](#)), and may occur on the face and scalp. Tiny papular *follicular syphilides* involving hair follicles may result in patchy alopecia (alopecia areata), with loss of scalp hair, eyebrows, or beard in up to 5% of cases. Progressive endarteritis obliterans and ischemia result in superficial scaling of papules (*papulosquamous syphilides*) and eventually may lead to central necrosis (*pustular syphilides*).

In warm, moist, intertriginous body areas, including the perianal area, vulva, scrotum, inner thighs, axillae, and skin under pendulous breasts, papules can enlarge and become eroded to produce broad, moist, pink or gray-white, highly infectious lesions called *condylomata lata* (see [Plate IID-49](#)); these lesions develop in 10% of patients with secondary syphilis. Superficial mucosal erosions, called *mucous patches*, occur in 10 to 15% of patients and may involve the lips, oral mucosa, tongue ([Fig. 172-1](#)), palate, pharynx, vulva and vagina, glans penis, or inner prepuce. The typical mucous patch is a painless silver-gray erosion surrounded by a red periphery. During relapses of secondary syphilis, condylomata lata are particularly common, and skin lesions tend to be asymmetrically distributed and more infiltrated, resembling skin lesions of late syphilis. These characteristics may reflect increasing cellular immunity.

Constitutional symptoms that may accompany or precede secondary syphilis include sore throat (15 to 30%), fever (5 to 8%), weight loss (2 to 20%), malaise (25%), anorexia (2 to 10%), headache (10%), and meningismus (5%). *Acute meningitis* occurs in only 1 to 2% of cases, but numbers of cells and levels of protein in [CSF](#) are increased in 30% of cases. *T. pallidum* has been recovered from CSF during primary and secondary syphilis in 30% of cases; this finding is often but not always associated with other CSF abnormalities.

Less common complications of secondary syphilis include hepatitis, nephropathy, gastrointestinal involvement (hypertrophic gastritis, patchy proctitis, ulcerative colitis, or a rectosigmoid mass), arthritis, and periostitis. Ocular findings that suggest secondary syphilis include otherwise-unexplained pupillary abnormalities, optic neuritis, and a

retinitis pigmentosa syndrome as well as the classic iritis (especially granulomatous iritis) or uveitis. The diagnosis of secondary syphilis is often considered only after the patient fails to respond to steroid therapy. Anterior uveitis has been reported in 5 to 10% of patients with secondary syphilis, and *T. pallidum* has been demonstrated in the aqueous humor from these patients. *Syphilitic hepatitis* is distinguished by an unusually high serum level of alkaline phosphatase and by a nonspecific histologic appearance that is unlike that of viral hepatitis and includes moderate inflammation with polymorphonuclear leukocytes and lymphocytes, some hepatocellular damage, and no cholestasis. *Renal involvement* produces proteinuria associated with an acute nephrotic syndrome (or rarely with hemorrhagic glomerulonephritis) and is characterized by subepithelial electron-dense deposits and glomerular immune complexes -- findings suggesting immune-complex glomerulonephritis.

Latent Syphilis Positive serologic tests for syphilis, together with a normal [CSF](#) examination and the absence of clinical manifestations of syphilis, indicate a diagnosis of latent syphilis. The diagnosis is often suspected on the basis of a history of primary or secondary lesions, a history of exposure to syphilis, or the delivery of an infant with congenital syphilis. A previous negative serologic test or a history of lesions or exposure may help establish the duration of latent infection. *Early latent* syphilis encompasses the first year after infection, while *late latent* syphilis (beginning ³¹1 year after infection in the untreated patient) is associated with relative immunity to infectious relapse and with increasing resistance to reinfection. *T. pallidum* may still seed the bloodstream intermittently during this stage. Pregnant women with latent syphilis may infect the fetus in utero. Moreover, syphilis has been transmitted through the transfusion of blood from patients with latent syphilis of many years' duration. It was previously thought that untreated late latent syphilis had three possible outcomes: (1) it could persist throughout the lifetime of the infected individual, (2) it could end in the development of late syphilis, or (3) it could end with the spontaneous cure of infection, with reversion of serologic tests to negative. It is now apparent, however, that the more sensitive treponemal antibody tests rarely, if ever, become negative without treatment. About 70% of untreated patients with latent syphilis never develop clinically evident late syphilis, but the occurrence of spontaneous cure is in doubt.

Late Syphilis The slowly progressive inflammatory disease leading to tertiary manifestations begins early during the pathogenesis of syphilis, although these manifestations may not become clinically apparent for years. Early syphilitic aortitis becomes evident soon after secondary lesions subside, and patients who develop [CSF](#) abnormalities during the early stages of syphilis appear to be at highest risk of late neurologic complications.

Asymptomatic Neurosyphilis [CNS](#) syphilis represents a continuum comprising early invasion, usually within the first weeks or months of infection, and asymptomatic involvement, which may or may not lead to neurologic manifestations. Traditionally, the diagnosis of asymptomatic neurosyphilis has been made in patients who lack neurologic symptoms and signs and who have [CSF](#) abnormalities including mononuclear pleocytosis, increased protein concentrations, or a reactive Venereal Disease Research Laboratory (VDRL) slide test. Such abnormalities are found in up to one-quarter of patients with untreated late latent syphilis, and it is these patients who are known to be at risk for neurologic complications. However, in primary and secondary syphilis, *T.*

pallidum can be isolated from CSF of 40% of patients even in the absence of other CSF abnormalities. Although the therapeutic implications of these findings in early syphilis are uncertain, it seems appropriate to conclude that even patients with early syphilis who have such findings do indeed have asymptomatic neurosyphilis and should be treated for neurosyphilis. In patients with untreated asymptomatic neurosyphilis, the overall cumulative probability of progression to clinical neurosyphilis is about 20% in the first 10 years but increases with time; the likelihood is highest among patients with the greatest degree of pleocytosis or protein elevation. Patients with untreated latent syphilis and normal CSF probably run no risk of subsequent neurosyphilis.

Symptomatic Neurosyphilis Although mixed features are common, the major clinical categories of symptomatic neurosyphilis include meningeal, meningovascular, and parenchymatous syphilis. The last category includes general paresis and tabes dorsalis. The onset of symptoms usually comes <1 year after infection for meningeal syphilis, at 5 to 10 years for meningovascular syphilis, at 20 years for general paresis, and at 25 to 30 years for tabes dorsalis. However, symptomatic neurosyphilis, particularly in the antibiotic era, often presents not as a classic picture but rather as mixed and subtle or incomplete syndromes.

Meningeal syphilis may involve either the brain or the spinal cord, and patients may present with headache, nausea, vomiting, neck stiffness, cranial nerve palsies, seizures, and changes in mental status. **Meningovascular syphilis** reflects diffuse inflammation of the pia and arachnoid together with evidence of focal or widespread arterial involvement of small, medium, or large vessels. The most common presentation is a stroke syndrome involving the middle cerebral artery of a relatively young adult; however, unlike the usual thrombotic or embolic stroke syndrome of sudden onset, meningovascular syphilis often becomes manifest after a subacute encephalitic prodrome (with headaches, vertigo, insomnia, and psychological abnormalities), which is followed by a gradually progressive vascular syndrome.

The manifestations of **general paresis** reflect widespread parenchymal damage and include abnormalities corresponding to the mnemonic *paresis*: *personality*, *affect*, *reflexes* (hyperactive), *eye* (e.g., Argyll Robertson pupils), *sensorium* (illusions, delusions, hallucinations), *intellect* (a decrease in recent memory and in the capacity for orientation, calculations, judgment, and insight), and *speech*. **Tabes dorsalis** presents as symptoms and signs of demyelination of the posterior columns, dorsal roots, and dorsal root ganglia. Symptoms include ataxic wide-based gait and footslap; paresthesia; bladder disturbances; impotence; areflexia; and loss of position, deep pain, and temperature sensations. Trophic joint degeneration (Charcot's joints) and perforating ulceration of the feet can result from loss of pain sensation. The small, irregular Argyll Robertson pupil, a feature of both tabes dorsalis and paresis, reacts to accommodation but not to light. **Optic atrophy** also occurs frequently in association with tabes.

Cardiovascular Syphilis Cardiovascular manifestations are attributable to endarteritis obliterans of the vasa vasorum, which provide the blood supply to large vessels. This condition produces medial necrosis with destruction of elastic tissue, particularly in the ascending and transverse segments of the aortic arch, resulting in uncomplicated aortitis, aortic regurgitation, saccular aneurysm, or coronary ostial stenosis. Symptoms appear from 10 to 40 years after infection. Cardiovascular complications occur more

often and at an earlier age among men than among women and may be more common among African Americans than among whites. In the preantibiotic era, symptomatic cardiovascular complications developed in about 10% of persons with late untreated syphilis, and aortic regurgitation was two to four times as common as aneurysm. However, syphilitic aortitis was demonstrated at autopsy in about one-half of African-American men with untreated syphilis.

Linear calcification of the ascending aorta on chest x-ray films suggests asymptomatic syphilitic aortitis, as arteriosclerosis seldom produces this sign. Aortic dilation and a tambour quality to the sound of aortic closure are unreliable signs of aortitis. Syphilitic aneurysms -- usually saccular, occasionally fusiform -- do not lead to dissection. Approximately 1 in 10 aortic aneurysms of syphilitic origin involves the abdominal aorta, but these aneurysms tend to occur above the renal arteries, whereas arteriosclerotic abdominal aneurysms are usually found below the renal arteries. With increasing age, the nervous system is also affected in up to 40% of patients with cardiovascular syphilis.

Late Lesions of the Eyes Iritis associated with pain, photophobia, and dimness of vision or chorioretinitis occurs not only during secondary syphilis but also as a relatively common manifestation of late syphilis. Adhesions of the iris to the anterior lens may produce a fixed pupil, not to be confused with Argyll Robertson pupil.

Late Benign Syphilis (Gumma) Gummas may be multiple or diffuse but are usually solitary lesions that range from microscopic size to several centimeters in diameter. From a histologic perspective, gummas consist of a granulomatous inflammation with a central area of necrosis surrounded by mononuclear, epithelioid, and fibroblastic cells; occasional giant cells; and perivascularitis. Although rarely demonstrated microscopically, *T. pallidum* has reportedly been recovered from these lesions. The most commonly involved sites include the skin and skeletal system, the mouth and upper respiratory tract, the larynx, the liver, and the stomach; however, any organ may be involved. Gummas of the skin produce painless and indurated nodular, papulosquamous, or ulcerative lesions that form characteristic circles or arcs, with peripheral hyperpigmentation. Gummas are usually indolent and may heal spontaneously with scarring, but they may also be explosive in onset and are often destructive. These lesions may resemble those of many other chronic granulomatous conditions, including tuberculosis and sarcoidosis, leprosy, and deep fungal infections. Skeletal gummas most frequently involve the long bones of the legs, although any bone may be affected. Trauma may predispose a specific site to involvement. Presenting symptoms usually include focal pain and tenderness. Radiographic abnormalities with advanced gummas of bone include periostitis or destructive or sclerosing osteitis. Upper respiratory gummas can lead to perforation of the nasal septum or palate. Gummatous hepatitis may produce epigastric pain and tenderness as well as low-grade fever and may be associated with splenomegaly and anemia.

The histopathology and extensive tissue necrosis associated with gummas suggest delayed hypersensitivity to *T. pallidum*. Certain individuals appear to develop an exaggerated delayed-hypersensitivity response to *T. pallidum*, which presumably is mediated by sensitized T lymphocytes and macrophages. Because the histologic changes may be suggestive but are nonspecific, the diagnosis of late benign syphilis is confirmed by serologic testing and by therapeutic trial. Treatment with penicillin results

in rapid healing of active gummatous lesions.

Congenital Syphilis Transmission of *T. pallidum* from a syphilitic woman to her fetus across the placenta may occur at any stage of pregnancy, but the lesions of congenital syphilis generally develop after the fourth month of gestation, when fetal immunologic competence begins to develop. This timing suggests that the pathogenesis of congenital syphilis depends on the immune response of the host rather than on a direct toxic effect of *T. pallidum*. The risk of infection of the fetus during untreated early maternal syphilis is estimated to be 75 to 95%, decreasing to about 35% for maternal syphilis of >2 years' duration. Adequate treatment of the mother before the 16th week of pregnancy should prevent fetal damage. Untreated maternal infection may result in a rate of fetal loss of up to 40% (with stillbirth more common than abortion because of the late onset of fetal pathology), prematurity, neonatal death, or nonfatal congenital syphilis. Among infants born alive, only fulminant congenital syphilis is clinically apparent at birth, and these babies have a very poor prognosis. The most common clinical problem is the healthy-appearing baby born to a mother with a positive serologic test. Routine serologic testing in early pregnancy is considered cost-effective in virtually all populations, even in areas with a low prenatal prevalence of syphilis. Where the prevalence of syphilis is high and when the patient is at high risk, syphilis serology should be repeated in the third trimester and at delivery.

The manifestations of congenital syphilis can be divided into three types according to their timing: (1) early manifestations, which appear within the first 2 years of life (often between 2 and 10 weeks of age), are infectious and resemble the manifestations of severe secondary syphilis in the adult; (2) late manifestations, which appear after 2 years and are noninfectious; and (3) residual stigmata. The earliest sign of congenital syphilis is usually rhinitis, or "snuffles" (23%), which is soon followed by other mucocutaneous lesions (35 to 41%). These may include bullae (syphilitic pemphigus), vesicles, superficial desquamation, petechiae, and (later) papulosquamous lesions, mucous patches, and condylomata lata. The most common early manifestations are bone changes (61%), including osteochondritis, osteitis, and periostitis. Hepatosplenomegaly (50%), lymphadenopathy (32%), anemia (34%), jaundice (30%), thrombocytopenia, and leukocytosis are common.

Neonatal congenital syphilis must be differentiated from other generalized congenital infections, including rubella, cytomegalovirus or herpes simplex virus infection, and toxoplasmosis, as well as from erythroblastosis fetalis. Neonatal death is usually due to pulmonary hemorrhage, secondary bacterial infection, or severe hepatitis.

Late congenital syphilis is that which remains untreated after 2 years of age. In perhaps 60% of cases, the infection remains subclinical; the clinical spectrum in the remainder of cases differs in certain respects from that of acquired late syphilis in the adult. For example, cardiovascular syphilis rarely develops in late congenital syphilis, whereas interstitial keratitis is much more common and occurs between the ages of 5 and 25. Other manifestations associated with interstitial keratitis are eighth-nerve deafness and recurrent arthropathy. Bilateral knee effusions are known as *Clutton's joints*. Asymptomatic neurosyphilis is present in about one-third of untreated patients, and clinical neurosyphilis occurs in one-quarter of untreated individuals over 6 years of age. Gummatous periostitis occurs between the ages of 5 and 20 and, as in nonvenereal

endemic syphilis, tends to cause destructive lesions of the palate and nasal septum.

Characteristic stigmata include *Hutchinson's teeth* -- centrally notched, widely spaced, peg-shaped upper central incisors -- and "mulberry" molars -- sixth-year molars with multiple, poorly developed cusps. The abnormal facies of patients with congenital syphilis include frontal bossing, saddle nose, and poorly developed maxillae. Saber shins, characterized by anterior tibial bowing, are rare. *Rhagades* are linear scars at the angles of the mouth and nose that are caused by secondary bacterial infection of the early facial eruption. Other stigmata include unexplained nerve deafness, old chorioretinitis, optic atrophy, and corneal opacities due to past interstitial keratitis.

LABORATORY EXAMINATIONS

Demonstration of the Organism Dark-field microscopic examination of lesion exudate is useful in evaluating moist cutaneous lesions, such as the chancre of primary syphilis or the condylomata lata of secondary syphilis. The identification of a single characteristic motile organism by a trained observer is sufficient for diagnosis. Examination of oral lesions and anal ulcers by this method is not recommended, as it is difficult to differentiate *T. pallidum* from other spirochetes that may be present.

Most syphilis is diagnosed in settings where dark-field microscopy is not available. The direct fluorescent antibody *T. pallidum* (DFA-TP) test, an alternative available at central laboratories, uses fluorescein-conjugated polyclonal antitreponemal antibody for the detection of *T. pallidum* in fixed smears prepared from suspect lesions. More sensitive [PCR](#) tests have been developed but are available only in research laboratories.

T. pallidum can be found in tissue with appropriate silver stains, although these results should be interpreted with caution because artifacts resembling *T. pallidum* are often seen. Treponemes can be demonstrated more reliably in tissue by immunofluorescent or immunohistochemical methods using specific monoclonal or polyclonal antibodies to *T. pallidum*.

Serologic Tests for Syphilis There are two types of serologic tests for syphilis: nontreponemal and treponemal. Both types of tests are reactive in persons with any treponemal infection, including yaws, pinta, and endemic syphilis.

The nontreponemal tests measure IgG and IgM directed against a cardiolipin-lecithin-cholesterol antigen complex. The most widely used nontreponemal antibody tests for syphilis are the rapid plasma reagin (RPR) test, which can be automated (ART), and the [VDRL](#) slide test. In these tests, antibody is detected by the microscopic or macroscopic flocculation of the antigen suspension. The RPR test may be more expensive than the VDRL test, but it is easier to perform and uses unheated serum; it is the test of choice for rapid serologic diagnosis in a clinic or office setting. The VDRL test, however, remains the standard for use with [CSF](#).

The [RPR](#) and [VDRL](#) tests are equally sensitive and may be used for initial screening or for quantitation of serum antibody. The titer reflects the activity of the disease. A fourfold or greater rise in titer may be seen during the evolution of early syphilis. VDRL titers usually reach 1:32 or higher in secondary syphilis. A persistent fall of two dilutions

(fourfold) or greater following treatment of early syphilis provides essential evidence of an adequate response to therapy. VDRL titers do not correspond directly to RPR titers, and sequential quantitative testing (as for response to therapy) must employ a single test.

Two standard treponemal tests are used for confirmation of reactive nontreponemal results: the fluorescent treponemal antibody-absorbed (FTA-ABS) test and the agglutination assays for antibodies to *T. pallidum*. The microhemagglutination assay for *T. pallidum* (MHA-TP) has been replaced by the Serodia TP-PA test (Fujirebio, Tokyo), which is more sensitive for primary syphilis. The *T. pallidum* hemagglutination test (TPHA) is widely used in Europe but is not available in the United States. Both the agglutination assays and the FTA-ABS tests are very specific and, when used for confirmation of positive non-treponemal tests, have a very high positive predictive value for the diagnosis of syphilis. However, even these tests give false-positive results at rates as high as 1 to 2% when used for the screening of normal populations. New enzyme-linked immunosorbent assays have also been approved as confirmatory tests.

The relative sensitivities of the [VDRL](#) test, the [FTA-ABS](#) test, and the [MHA-TP](#) in the various stages of untreated syphilis are shown in [Table 172-1](#). The nontreponemal tests are nonreactive in about one-quarter of patients presenting with primary syphilis. In early primary syphilis, the detection of antibody can be maximized either by the performance of an FTA-ABS test or simply by repetition of a VDRL test after 1 to 2 weeks if the initial VDRL result is negative. All treponemal and nontreponemal tests are reactive during secondary syphilis, and a nonreactive result virtually excludes syphilis in a patient with otherwise-compatible mucocutaneous lesions. (Fewer than 1% of patients with secondary syphilis have a VDRL test that is nonreactive or weakly reactive with undiluted serum but is positive at higher serum dilutions -- the *prozone phenomenon*.) While the nontreponemal tests will become nonreactive or will be reactive in lower titers following therapy for early syphilis, the treponemal tests often remain reactive after therapy and therefore are not helpful in determining the infection status of persons with past syphilis.

For practical purposes, most clinicians need to be familiar with the three uses of serologic tests for syphilis: (1) testing of large numbers of sera for screening or diagnostic purposes (e.g., the [RPR](#) or [VDRL](#) test), (2) quantitative measurement of the antibody titer to assess the clinical activity of syphilis or to monitor the response to therapy (e.g., the VDRL or RPR test), and (3) confirmation of the diagnosis of syphilis in a patient with a positive nontreponemal antibody test or with a suspected clinical diagnosis of syphilis (e.g., the [FTA-ABS](#) test or Serodia TP-PA).

For measurement of IgM in neonates in whom congenital syphilis is suspected, the syphilis Captia-M test (Trinity Biotech, Jamestown, NY) and the 19S IgM [FTA-ABS](#) test are available.

False-Positive Serologic Tests for Syphilis Because the antigen used in nontreponemal tests is found in other tissues, the tests may be reactive in persons without treponemal infection, although rarely do titers exceed 1:8 in such patients. In a population selected for screening because of clinical suspicion, history of exposure, or increased risk for sexually transmitted infections, fewer than 1% of reactive tests are

falsely positive. The modern [VDRL](#) and [RPR](#) tests are 97 to 99% specific, and false-positive reactions are now limited largely to those conditions listed in [Table 172-2](#). False positivity is common among persons with autoimmune disorders. The prevalence of false-positive nontreponemal tests increases with advancing age; 10% of people over 70 years of age have false-positive reactions. In the patient with a false-positive nontreponemal test, syphilis is excluded by a nonreactive treponemal test.

Evaluation for Neurosyphilis Asymptomatic involvement of the [CNS](#) is detected by examination of [CSF](#) for pleocytosis, increased protein concentration, and [VDRL](#) activity. CSF abnormalities can be demonstrated in up to 40% of cases of primary or secondary syphilis and in 25% of cases of latent syphilis. In older asymptomatic seropositive individuals, the yield of lumbar puncture is relatively low. *T. pallidum* has been recovered by CSF inoculation into rabbits from up to 30% of patients with primary or secondary syphilis but rarely from those with latent syphilis. The demonstration of *T. pallidum* in CSF is often associated with other CSF abnormalities; however, organisms can be recovered from patients with otherwise-normal CSF. Before the advent of penicillin, the risk of developing clinical neurosyphilis was roughly proportional to the intensity of CSF changes in early syphilis. CSF examination is essential in the evaluation of any seropositive patient with neurologic signs and symptoms and is recommended for all patients with untreated syphilis of unknown duration or of >1 year's duration. The possibility of asymptomatic neurosyphilis in some patients with early disease is not addressed by these recommendations. Because standard therapy with penicillin G benzathine (benzathine benzylpenicillin) for early syphilis fails to result in treponemicidal levels in the CSF, some experts advise lumbar puncture in secondary and early latent syphilis, particularly in patients with HIV infection.

In short, lumbar puncture should be performed in the evaluation of latent syphilis of >1 year's duration, in suspected neurosyphilis, and in late complications other than symptomatic neurosyphilis (since asymptomatic neurosyphilis may coexist with other late complications). [CSF](#) examination is most clearly indicated in the following situations: neurologic signs or symptoms, treatment failure, a serum reagin titer³ 1:32, HIV antibody positivity, other evidence of active syphilis (e.g., aortitis, gumma, visual or hearing changes), or plans to administer nonpenicillin therapy.

The [CSFVDRL](#) test is highly specific but relatively insensitive and may be nonreactive even in cases of progressive symptomatic neurosyphilis. The degree of sensitivity is highest in meningovascular syphilis and paresis and is lower in asymptomatic neurosyphilis and tabes dorsalis. The unabsorbed FTA test on CSF is reactive far more often than the CSF VDRL test in all stages of syphilis, but FTA reactivity may reflect passive transfer of serum antibody into the CSF. A nonreactive CSF FTA test, however, may be used to rule out neurosyphilis.

Evaluation for Syphilis in Patients Infected with HIV Because persons at highest risk for syphilis (inner-city populations, homosexually active men, and people in many developing countries) are also at increased risk for HIV infection, these two infections frequently coexist in the same patient. There is evidence that syphilis and other genital-ulcer diseases may be important risk factors for the acquisition and transmission of HIV infection.

The manifestations of syphilis may be altered in patients with concurrent HIV infection, and multiple cases of neurologic relapse following standard therapy have been reported in HIV-infected patients. *T. pallidum* has been isolated from the [CSF](#) of several patients after therapy for early syphilis with penicillin G benzathine. A recent multicenter U.S. study of early syphilis found similar therapeutic responses in persons with and without concurrent HIV infection, although the study lacked sufficient statistical power to exclude an effect of HIV and 41% of subjects were lost to follow-up. This investigation confirmed the high rate of [CNS](#) invasion in early syphilis and the persistence of *T. pallidum* after standard therapy: 11 of 43 HIV-infected patients and 21 of 88 HIV-uninfected patients had *T. pallidum* detectable in CSF before therapy; 7 of the 35 patients who underwent lumbar puncture after therapy (some HIV-infected and others uninfected) still had *T. pallidum* detectable in CSF.

The frequency of unusual clinical and laboratory manifestations of syphilis among patients co-infected with HIV is unknown. Such changes may be dependent on the stage of HIV infection and the degree of immunosuppression. There is no clear evidence that the sensitivity of serologic tests for syphilis or the serologic response to therapy in the vast majority of HIV-infected patients with early syphilis differs from the corresponding findings in patients not infected with HIV. Interpretation of serologic results should be the same for the two groups.

Persons with newly diagnosed HIV infection should be tested for syphilis. Some authorities, persuaded by reports of the persistence of *T. pallidum* in the [CSF](#) of HIV-infected persons after standard penicillin benzathine therapy for early syphilis, recommend examination of CSF for evidence of neurosyphilis for all co-infected patients, regardless of the clinical stage of syphilis, with treatment for neurosyphilis if CSF abnormalities are found or if CSF examination is not performed. Others do not recommend routine CSF examination for HIV-co-infected patients with early syphilis and believe that standard therapy is sufficient. Serologic testing after treatment is important for all patients with syphilis, particularly those also infected with HIV.

TREATMENT

Treatment of Acquired Syphilis Penicillin G is the drug of choice for all stages of syphilis. *T. pallidum* is killed by very low concentrations of penicillin G, although a long period of exposure to penicillin is required because of the unusually slow rate of multiplication of the organism. The efficacy of penicillin for syphilis remains undiminished after 50 years of use. Other antibiotics effective in syphilis include the tetracyclines, erythromycin, and the cephalosporins. Aminoglycosides and spectinomycin inhibit *T. pallidum* only in very large doses, and the sulfonamides and the quinolones are inactive.

Serum levels of penicillin G of ≥ 0.03 ug/mL for at least 7 days are considered necessary for the cure of early syphilis. Recurrence rates for a given regimen increase as infection progresses from incubating to seronegative primary to seropositive primary to secondary to late syphilis. Therefore, it is probable, but unproven, that a longer duration of therapy is required to effect cure as the infection progresses. For these reasons, some authorities use more prolonged penicillin therapy than that recommended by the U.S. Public Health Service when treating secondary, latent, or late syphilis.

The treatment regimens recommended for syphilis are summarized in [Table 172-3](#) and are discussed below.

Early Syphilis Preventive (abortive, "epidemiologic") treatment is recommended for seronegative individuals without signs of syphilis who have been exposed to infectious syphilis within the previous 3 months. Before treatment is given, every effort should be made to establish a diagnosis by examination and serologic testing. *The regimens recommended for prevention are the same as those recommended for early syphilis.*

Penicillin G benzathine is the most widely used agent for the treatment of early syphilis (including primary, secondary, and early latent syphilis), although it is more painful on injection than penicillin G procaine. A single dose of 2.4 million units cures more than 95% of cases of primary syphilis. Because the drug's efficacy in secondary syphilis may be slightly lower, some physicians administer a second dose of 2.4 million units 1 week after the initial dose at this stage of disease. Clinical relapse can follow treatment with penicillin G benzathine in patients with both HIV infection and early syphilis. Because the risk of neurorelapse may be higher in HIV-infected patients, examination of [CSF](#) from HIV-seropositive individuals with syphilis of any stage is recommended by some experts; therapy appropriate for neurosyphilis should be given if there is any evidence of [CNS](#) syphilis.

For penicillin-allergic patients with early syphilis, a 2-week course of therapy with doxycycline or tetracycline is recommended. These regimens appear to be effective, although no well-controlled studies have been performed and poor compliance may be problematic. Although ceftriaxone and azithromycin have shown activity against *T. pallidum* in animals, human trials have not been of sufficient scope to permit the recommendation of either drug for any stage of syphilis.

Late latent and late syphilis (normal CSF) If [CSF](#) abnormalities are found, the patient should be treated for neurosyphilis. The recommended treatment for late latent syphilis with normal CSF, for cardiovascular syphilis, and for late benign syphilis (gumma) is penicillin G benzathine, 2.4 million units intramuscularly once a week for 3 successive weeks (7.2 million units total). Doxycycline or tetracycline (given for 1 month) offers an untested alternative for penicillin-allergic patients with latent or late syphilis and normal CSF. The clinical response to treatment for benign tertiary syphilis is usually impressive; however, responses to therapy for cardiovascular syphilis are not dramatic because aortic aneurysm and aortic regurgitation cannot be reversed by antibiotic treatment.

Neurosyphilis The 1998 neurosyphilis treatment guidelines of the Centers for Disease Control and Prevention (CDC) are presented in [Table 172-3](#). Penicillin G benzathine, given in total doses of up to 7.2 million units to adults or 50,000 units per kilogram to infants, does not produce detectable concentrations of penicillin G in [CSF](#), and asymptomatic neurosyphilis may relapse in patients treated with 2.4 million units; the risk may be higher in HIV-infected patients. Therefore, the use of penicillin G benzathine alone for the treatment of neurosyphilis is not recommended. On the other hand, administration of intravenous penicillin G in doses of ≥ 12 million units per day for 10 days or longer is thought to ensure treponemicidal concentrations of penicillin G in CSF and occasionally cures infection in patients who fail to respond to other therapy. The

clinical response to penicillin therapy for meningeal syphilis is dramatic, but the response to treatment for parenchymal neurosyphilis is variable. In general, treatment of neurosyphilis in which damage has already been done may produce no clinical change but may arrest disease progression.

Several recent publications have reported neurologic relapse after high-dose intravenous penicillin therapy for neurosyphilis in HIV-infected patients. No alternative therapies have been explored, but careful follow-up is essential, and re-treatment is warranted in such patients.

No data support the use of antibiotics other than penicillin G for the treatment of neurosyphilis; however, some of the third-generation cephalosporins may deserve further evaluation. In patients with penicillin allergy demonstrated by skin testing, desensitization may be the best course ([Chap. 126](#)).

Management of Syphilis in Pregnancy Every pregnant woman should undergo a nontreponemal test at her first prenatal visit, and women at high risk of exposure should have a repeat test in the third trimester and at delivery. In the pregnant patient with presumed syphilis (evidenced by a reactive serology, with or without clinical manifestations) and with no history of treatment for syphilis, expeditious evaluation and initiation of treatment are essential. Therapy should be administered according to the stage of the disease, as for nonpregnant patients. Patients should be warned of the risk of a Jarisch-Herxheimer reaction, which may be associated with mild premature contractions but rarely results in premature delivery.

Penicillin is the only recommended therapy for syphilis in pregnancy. If the patient has a well-documented penicillin allergy, desensitization and penicillin treatment should be undertaken in a hospital according to the 1998 sexually transmitted diseases treatment guidelines issued by the [CDC](#). After treatment, a quantitative nontreponemal test should be repeated monthly throughout pregnancy. Treated women whose titers rise fourfold or who do not show a fourfold decrease in titer in a 3-month period should be re-treated.

Evaluation and Management of Congenital Syphilis Newborn infants of mothers with reactive [VDRL](#) or [FTA-ABS](#) tests may themselves have reactive tests, whether or not they have become infected, because of transplacental transfer of maternal IgG antibody. Rising or persistent titers indicate infection, and the infant should be treated. Neonatal IgM antibody can be detected in cord or neonatal serum with the syphilis Captia-M or 19S IgM FTA-ABS test. Alternatively, monthly quantitative nontreponemal tests may be performed on asymptomatic infants born to women treated adequately with penicillin during pregnancy.

An infant should be treated at birth if the seropositive mother has received penicillin therapy in the third trimester, inadequate penicillin treatment, or therapy with a drug other than penicillin; if her treatment status is unknown; or if the infant may be difficult to follow. It is unwise to require proof of diagnosis before treatment in such cases. The [CSF](#) should be examined to obtain baseline values before treatment. Penicillin is the only recommended drug for syphilis in infants. The penicillin dosage used for the treatment of the patient with late congenital syphilis is calculated in the same way as for the infant, until dosage based on weight reaches that used for adult neurosyphilis.

Specific recommendations for the treatment of infants are included in the [CDC's](#) 1998 guidelines.

Jarisch-Herxheimer Reaction A dramatic though usually mild reaction consisting of fever (average temperature elevation, 1.5°C), chills, myalgias, headache, tachycardia, increased respiratory rate, increased circulating neutrophil count, and vasodilation with mild hypotension may follow the initiation of treatment for syphilis. This reaction occurs in approximately 50% of patients with primary syphilis, 90% of those with secondary syphilis, and 25% of those with early latent syphilis. The onset comes within 2 h of treatment, the temperature peaks at about 7 h, and defervescence takes place within 12 to 24 h. The reaction is more delayed in neurosyphilis, with fever peaking after 12 to 14 h. In patients with secondary syphilis, erythema and edema of the mucocutaneous lesions increase; occasionally, subclinical or early mucocutaneous lesions may first become apparent during the reaction. The pathogenesis of this reaction is undefined, although recent studies have demonstrated the induction of inflammatory mediators such as tumor necrosis factor by treponemal lipoproteins. Patients should be warned to expect such symptoms, which can be managed by bed rest and aspirin. Steroid and other anti-inflammatory therapy is not required for this mild transient reaction.

Follow-Up Evaluation of Responses to Therapy The response of early syphilis to treatment should be determined by monitoring of the quantitative [VDRL](#) or [RPR](#) titer 1, 3, 6, and 12 months after treatment. More frequent serologic examination (1, 2, 3, 6, 9, and 12 months) is recommended for patients concurrently infected with HIV. Because the [FTA-ABS](#) and agglutination tests remain positive in most patients treated for seropositive syphilis, these tests are not useful in following the response to therapy. After successful treatment of seropositive first-episode primary or secondary syphilis, the VDRL titer progressively declines, becoming negative by 12 months in 40 to 75% of seropositive primary cases and in 20 to 40% of secondary cases. Patients with a history of syphilis have less rapid declines in titer and are less likely to become VDRL- or RPR-negative. If the VDRL test becomes negative or if VDRL titers drop to a fixed low value within 1 or 2 years, lumbar puncture is unnecessary since the [CSF](#) examination is almost invariably normal and there is little risk of subsequent neurosyphilis. However, if a VDRL titer $\geq 1:8$ fails to fall by at least fourfold within 12 months, if the VDRL titer rises by fourfold, or if clinical symptoms persist or recur, re-treatment is indicated. Every effort should be made to differentiate treatment failure from reinfection, and the CSF should be examined. Patients in whom treatment failure is suspected, especially those with abnormal CSF, should be treated for neurosyphilis. If the patient remains seropositive but asymptomatic after such re-treatment, no further therapy is necessary. Patients treated for late latent syphilis frequently have low initial VDRL titers and may not have a fourfold drop after therapy with penicillin; about half of these patients remain seropositive (with low titers) for years after therapy. Re-treatment is not warranted unless the titer rises or signs and symptoms of syphilis appear.

The activity of neurosyphilis correlates best with the degree of [CSF](#) pleocytosis, and this measure provides the most sensitive index of response to treatment. CSF should be examined every 6 months for 3 years after the treatment of asymptomatic or symptomatic neurosyphilis or until CSF findings return to normal. An elevated CSF cell count falls to $\leq 10/\mu\text{L}$ in 3 to 12 months in 95% of adequately treated cases and becomes normal in all cases within 2 to 4 years. Elevated levels of CSF protein fall more slowly,

and the CSF VDRL titer declines gradually over a period of several years.

Persistence of *T. pallidum* The persistence of *T. pallidum* in the aqueous humor, [CSF](#), lymph nodes, brain, inflamed temporal arteries, and other tissues after "adequate" penicillin treatment has been suggested by dark-field microscopy, immunofluorescent antibody and silver staining techniques, rabbit inoculation, and [PCR](#). Because the data on persisting treponemes are scanty, no modification of the treatment recommendations seems warranted for HIV-uninfected persons. Adherence to recommendations regarding CSF examination before the selection of therapy should minimize the possibility that *T. pallidum* will persist in the CSF.

IMMUNITY TO AND PREVENTION OF SYPHILIS

About 60% of contacts of patients with primary and secondary syphilis become infected, with lower risk in contacts exposed to early latent syphilis. The rate of development of acquired resistance to *T. pallidum* after natural or experimental infection is related to the size of the antigenic stimulus, which depends on both the size of the infecting inoculum and the duration of infection before treatment. The role of serum antibody in conferring immunity to syphilis remains controversial. Passively administered antibody prevents or delays the appearance of clinical manifestations of syphilis in the rabbit model; it does not prevent infection. Cellular immunity is considered to be of major importance in the healing of early lesions and the control of syphilitic infection. The cellular infiltration of early lesions predominantly involves T lymphocytes and macrophages. The cytokine milieu of primary and secondary lesions is of the T_H1 type, consistent with the clearance of organisms by activated macrophages. Specific antibody enhances phagocytosis and is required for macrophage-mediated killing of *T. pallidum*.

Inability to cultivate pathogenic treponemes in vitro has hindered the analysis of treponemal antigens. Attempts to induce immunity to syphilis by vaccination have shown limited promise, although repeated injection of rabbits with g-irradiated motile treponemes has conferred immunity to rechallenge. The outer membrane of *T. pallidum* contains few integral membrane proteins, and none has been definitively identified. Several newly described antigens of *T. pallidum*, including TprK, have induced partial immunity to challenge in the rabbit model, and syphilis vaccine development is being actively pursued.

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(Bibliography omitted in Palm version)

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173. ENDEMIC TREPONEMATOSES - Sheila A. Lukehart

The endemic, or nonvenereal, treponematoses are bacterial infections that are caused by close relatives of *Treponema pallidum* subspecies *pallidum*, the etiologic agent of venereal syphilis ([Chap. 172](#)). Yaws, pinta, and endemic syphilis are distinguished from venereal syphilis by mode of transmission, age of acquisition, geographic distribution, and clinical features. These infections are limited primarily to rural areas of developing nations and are seen in the United States and Europe only in recent immigrants from endemic regions. Much of our "knowledge" about the endemic treponematoses is based upon impressions and observations of health care workers who have visited endemic areas; virtually no well-designed studies of the natural history, diagnosis, or treatment of these infections have been conducted. A comparison of the treponemal infections is shown in [Table 173-1](#).

EPIDEMIOLOGY

The endemic treponematoses are chronic diseases acquired during childhood and, like syphilis, can cause severe late manifestations years after initial infection. These infections were very common in Africa, Asia, and South America when the World Health Organization (WHO) and UNICEF embarked on a highly successful mass eradication campaign. From 1952 to 1969, it is estimated that over 160 million people were examined for treponemal infections and over 50 million cases, contacts, and latent infections were treated. This categorical program is one of WHO's outstanding successes in that the prevalence of active yaws was reduced from >20% to <1% in many rural areas and endemic syphilis was eradicated in Bosnia. In the decades since the eradication programs, lack of focused surveillance and diversion of resources to other pressing needs have resulted in a resurgence of these infections in some regions, particularly in Africa. The estimated geographic distribution of the endemic treponematoses in the 1990s is shown in [Fig. 173-1](#). In the early 1980s, WHO sponsored a series of regional meetings on the endemic treponematoses, and areas of resurgent yaws morbidity were identified in West Africa (Ivory Coast, Ghana, Togo, Benin) and extending into the Central African Republic and rural Democratic Republic of Congo (formerly Zaire). The prevalence of endemic syphilis is estimated to be >10% in some regions of Mali, Niger, Burkina Faso, and Senegal. In Asia and the western Pacific, yaws is still prevalent in Indonesia, Papua New Guinea, and the Solomon Islands; cases have also been identified in Laos and Kampuchea. In the Americas, foci of yaws persist in Haiti and other Caribbean islands, Peru, Colombia, Ecuador, Brazil, Guyana, and Surinam. Pinta is limited to Central America and northern South America, where it is found rarely and only in remote villages.

MICROBIOLOGY

The etiologic agents of the endemic treponematoses are *T. pallidum* subspecies *pertenue* (yaws), *T. pallidum* subspecies *endemicum* (endemic syphilis), and *T. carateum* (pinta). These little-studied organisms are morphologically identical to *T. pallidum* subspecies *pallidum*, and no antigenic differences among the pathogenic treponemes have been identified to date. A controversy has existed about whether the treponematoses are caused by different organisms or by the same organism, with clinical manifestations and routes of transmission being defined by the climate of the

region and the culture of the population. Three of the four organisms have been placed in the same species because of their genetic similarity; the fourth (*T. carateum*) remains a separate species simply because no organisms have been available for genetic studies. However, a molecular signature has been defined that can be used to differentiate *T. pallidum* subspecies *pallidum* from *T. pallidum* subspecies *pertenue* and *T. pallidum* subspecies *endemicum*, and unpublished studies have identified a number of distinct differences in the *tpr* gene family between venereal and nonvenereal treponemes. Whether these differences are related to the different clinical courses has not yet been determined.

CLINICAL FEATURES

All of the treponemal infections are characterized by defined disease stages, with a localized primary lesion, disseminated secondary lesions, periods of latency, and possible late lesions. The stages are most clearly defined in venereal syphilis, while the primary and secondary manifestations are more frequently overlapping in yaws and endemic syphilis; the late manifestations of pinta are very mild relative to the destructive lesions of the other treponematoses. The current preference is to divide the clinical course of the endemic treponematoses into "early" and "late" stages.

The major clinical features that are thought to differ between venereal syphilis and the nonvenereal infections are the lack of congenital transmission and lack of central nervous system (CNS) involvement in the nonvenereal infections. It is not known whether these distinctions are accurate. Because of the high degree of genetic relatedness among the organisms, there is little biologic reason to think that *T. pallidum* subspecies *endemicum* and *T. pallidum* subspecies *pertenue* would be unable to cross the blood-brain barrier or to invade the placenta. These organisms obviously can disseminate from the site of primary infection to other tissues, and they can persist for decades. In this respect, they are like *T. pallidum* subspecies *pallidum*. Even if invasion of the placenta or the CNS occurs in endemic treponemal infection, there are a number of reasons that these manifestations might not have been recognized. The lack of recognized congenital infection may be due to the fact that the nonvenereal treponematoses are usually acquired during childhood. The degree of spirochetemia (the presumed source of placental and fetal infection) is greatly diminished during the latent stage, and by the time an infected girl becomes sexually mature, she would be at low risk for transplacental transmission. Neurologic involvement may not have been recognized in nonvenereal treponemal infection because of the lack of trained medical personnel in endemic regions, the lag of years to decades between acquisition of infection and possible CNS manifestations, or a low rate of symptomatic CNS disease. The lack of longitudinal studies in endemic areas makes conclusions about the natural history of these infections tenuous.

Some published evidence supports congenital transmission as well as cardiovascular, ophthalmologic, and [CNS](#) involvement in yaws. Although the case is strong, particularly for CNS involvement, most studies that have shown a relatively high incidence (average, 24.9%) of cerebrospinal fluid (CSF) abnormalities in patients with yaws were not controlled for other possible causes of CSF abnormalities, did not include treponeme-specific tests, or did not follow patients for resolution of abnormalities after antitreponemal therapy. Thus, while no firm conclusions can be drawn about the

invasion of the CNS and placenta by the non-*pallidum* treponemes, it may be erroneous to accept unquestioningly the frequently repeated belief that these organisms fail to cause such manifestations.

Yaws Also known as *pian*, *framboesia*, or *bouba*, yaws is a chronic infection that is usually acquired in childhood and is caused by *T. pallidum* subspecies *pertenue*. The disease is characterized by the development of one or several primary lesions (called the "mother yaw"), followed by the appearance of multiple disseminated skin lesions. The early lesions may persist for many months, are infectious, and usually recur several times within the early years of infection. Late manifestations are destructive and can involve skin, bone, and joints.

The infection is transmitted by direct contact with infectious lesions, and transmission may be enhanced by disruption of the skin by insect bites or abrasions. Children with open lesions and without covering clothing are most likely to transmit infection during play or group sleeping. After an average incubation period estimated at 3 to 4 weeks, the first lesion begins as a papule, usually on an extremity, and then enlarges (particularly during moist warm weather) to become papillomatous or "raspberry-like" (thus the name "framboesia") ([Fig. 173-2](#)). Regional lymphadenopathy develops, and the lesion usually heals within 6 months; dissemination is thought to occur during the early weeks and months of infection. A generalized secondary eruption, accompanied by generalized lymphadenopathy, appears either concurrent with or following the primary lesion, may take several forms (macular, papular, or papillomatous), and may become secondarily infected with other bacteria. Painful papillomatous lesions on the soles of the feet result in a painful crablike gait ("crab yaws"), and periostitis may result in nocturnal bone pain and polydactylitis. All early skin lesions are infectious, and cutaneous relapses are common during the first 5 years. Late yaws is recognized in ~10% of untreated patients and is manifested by gummas of the skin and long bone, hyperkeratoses of the palms and soles, osteitis and periostitis, and hydrarthrosis. The late gummatous lesions are characteristically very destructive and extensive. Destruction of the nose, maxilla, palate, and pharynx is termed *gangosa* and is similar to the destructive lesions seen in leprosy and leishmaniasis.

Endemic Syphilis Endemic syphilis, also called *bejel*, *siti*, *dichuchwa*, *njovera*, or *skerljevo*, is a chronic infection caused by *T. pallidum* subspecies *endemicum*. Like other endemic treponematoses, endemic syphilis is chronic and is acquired in childhood. The early lesions are primarily localized to the mucocutaneous and mucosal surfaces, and the infection may be transmitted by direct contact or by shared drinking and eating utensils. A role for insects in transmission has been suggested but is unproved. The initial lesion often goes unrecognized, and the first noticeable lesion is usually an intraoral mucous patch or a mucocutaneous lesion resembling the condylomata lata of secondary syphilis ([Fig. 173-2](#)). This eruption may last for months or even years, and treponemes can readily be demonstrated in early lesions. Periostitis and regional lymphadenopathy are common. After a variable period of latency, late manifestations may appear, including osseous and cutaneous gummas. Destructive gummas, osteitis, and gangosa are more common in endemic syphilis than in late yaws. Gummas of the nipples develop in women who have previously had endemic syphilis and who breast-feed infants with oral lesions. Thus, it appears that the late lesion may result from repeated exposure of a sensitized host.

Pinta Pinta (also called *mal del pinto*, *carate*, *azul*, or *purupuru*) is the most benign of the treponemal infections and is caused by *T. carateum*. This disease has three stages that are characterized by marked changes in skin color, but it does not appear to cause destructive lesions or to involve other tissues. Transmission occurs by direct contact, usually during late childhood. The initial papule is most often located on the extremities or face and is pruritic. After one to many months of infection, numerous disseminated secondary lesions (*pintides*) appear. These lesions are initially red but become deeply pigmented, ultimately turning a dark slate blue. The secondary lesions are infectious and highly pruritic and may persist for years. Late pigmented lesions are called *dyschromic macules* and contain treponemes. Over time, most pigmented lesions show varying degrees of depigmentation, becoming brown and eventually white and giving the skin a mottled appearance. The white achromic lesions are characteristic of the late stage.

DIAGNOSIS

Diagnosis of the endemic treponematoses is based upon clinical manifestations and, when available, serologic testing. The same tests that are used for venereal syphilis ([Chap. 172](#)) become reactive during all treponemal infections, and there is no serologic test that can discriminate among the different infections. The nonvenereal treponemal infections should be considered in the evaluation of a reactive syphilis serology in any person who has immigrated from an endemic area.

TREATMENT

The recommended therapy for patients and their contacts is benzathine penicillin at a dose of 1.2 million units intramuscularly; that for children under 10 years of age is 600,000 units. This is half the dose recommended for patients and contacts with early venereal syphilis. There have been no controlled studies to show that the lower dose is effective in stopping relapse or progression to late disease. Definitive evidence of resistance to penicillin is lacking. However, because failure to heal existing lesions and frequent relapse following treatment for yaws have been described in Papua New Guinea, some health workers have suggested doubling the recommended dose of benzathine penicillin. Solely on the basis of experience with venereal syphilis, it is thought that doxycycline, tetracycline, and erythromycin (at doses appropriate for syphilis; [Chap. 172](#)) are therapeutic alternatives for patients allergic to penicillin. A Jarisch-Herxheimer reaction ([Chap. 172](#)) may follow treatment of endemic treponematoses.

CONTROL

The endemic treponematoses can be controlled with inexpensive therapy. However, the often-remote locations of the affected populations limit availability of medical care. Although the mass treatment programs of three decades ago were widely successful, time has shown that sustained control requires vigilance in regular screening and in the investigation of outbreaks -- luxuries that are often impossible in countries with more pressing medical needs. There is concern that, as HIV spreads throughout developing countries, it may markedly affect the manifestations and transmission of the endemic

treponematoses.

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174. LEPTOSPIROSIS - Peter Speelman

Leptospirosis is an infectious disease caused by pathogenic leptospires and characterized by a broad spectrum of clinical manifestations, varying from inapparent infection to fulminant, fatal disease. In its mild form, leptospirosis may present as an influenza-like illness with headache and myalgias. Severe leptospirosis, characterized by jaundice, renal dysfunction, and hemorrhagic diathesis, is referred to as *Weil's syndrome*.

ETIOLOGIC AGENTS

Leptospires are spirochetes belonging to the order Spirochaetales and the family Leptospiraceae. Traditionally, the genus *Leptospira* comprised two species: the pathogenic *L. interrogans* and the free-living *L. biflexa*. Although seven species of pathogenic leptospires are now recognized on the basis of their DNA relatedness, it is more practical clinically and epidemiologically to use a classification based on serologic differences. The pathogenic leptospires are divided into serovars according to their antigenic composition. More than 200 serovars make up the 23 serogroups.

Leptospires are coiled, thin, highly motile organisms with hooked ends and two periplasmic flagella that permit burrowing into tissue. These organisms are 6 to 20 μm long and about 0.1 μm wide; they stain poorly but can be seen microscopically by dark-field examination and after silver impregnation staining. Leptospires require special media and conditions for growth; it may take weeks for cultures to become positive.

EPIDEMIOLOGY

Leptospirosis is a zoonosis with a worldwide distribution that affects at least 160 mammalian species. Rodents, especially rats, are the most important reservoir, although other wild mammals, dogs, fish, and birds may also harbor these microorganisms. Leptospires establish a symbiotic relationship with their host and can persist in the renal tubules for years. Some serovars are associated with particular animals -- e.g., *icterohaemorrhagiae/copenhageni* with rats, *grippotyphosa* with voles, *hardjo* with cattle, *canicola* with dogs, and *pomona* with pigs.

Transmission of leptospires may follow direct contact with urine, blood, or tissue from an infected animal or exposure to a contaminated environment; human-to-human transmission is rare. Since leptospires are excreted in the urine and can survive in water for many months, water is an important vehicle in their transmission. Epidemics of leptospirosis may result from exposure to flood waters contaminated by urine from infected animals, as has been reported from Nicaragua. Leptospirosis occurs most commonly in the tropics because the climate as well as the sometimes poor working and hygienic conditions favor the pathogen's survival.

Humans are not commonly infected with leptospires. However, in the United States, the 40 to 120 cases reported annually to the Centers for Disease Control and Prevention certainly represent a significant underestimation of the total number. Certain occupational groups are at especially high risk; included are veterinarians, agricultural workers, sewage workers, slaughterhouse employees, and workers in the fishing

industry. Such individuals may acquire leptospirosis through direct exposure to or contact with contaminated water and soil. Leptospirosis has also been recognized in deteriorating inner cities where rat populations are expanding. One report described leptospirosis in urban residents of Baltimore who were sporadically exposed to rat urine.

In western countries, recreational exposure and domestic animal contact are also prominent sources of leptospirosis. Recreational water activities, such as canoeing, windsurfing, swimming, and waterskiing, place persons at risk for leptospirosis. Sometimes the infection is acquired during travel abroad. In a recent study in the Netherlands, 14% of patients with confirmed leptospirosis had acquired the infection while traveling in tropical countries, mostly in Southeast Asia. Transmission via laboratory accidents has been reported but is rare. Occasionally, leptospirosis develops after unanticipated immersion in contaminated water (e.g., in an automobile accident). Most cases occur in men, with a peak incidence during the summer and fall in western countries and during the rainy season in the tropics.

PATHOGENESIS

The pathogenesis of leptospirosis is incompletely understood. Leptospire may enter the host through abrasions in the skin or through intact mucous membranes, especially the conjunctiva and the lining of the oro- and nasopharynx. Drinking of contaminated water may introduce leptospire through the mouth, throat, or esophagus. After entry of the organisms, leptospiremia develops, with subsequent spread to all organs. Multiplication takes place in blood and in tissues, and leptospire can be isolated from blood and cerebrospinal fluid (CSF) during the first 4 to 10 days of illness. It is not clear why the presence of leptospire in the CSF does not cause damage. All forms of leptospire can damage the wall of small blood vessels; this damage leads to vasculitis with leakage and extravasation of cells, including hemorrhages. The most important known pathogenic properties of leptospire are adhesion to cell surfaces and cellular toxicity.

Vasculitis is responsible for the most important manifestations of the disease. Although leptospire mainly infect the kidneys and liver, any organ may be affected. In the kidney, leptospire migrate to the interstitium, renal tubules, and tubular lumen, causing interstitial nephritis and tubular necrosis. Hypovolemia due to dehydration or altered capillary permeability may contribute to the development of renal failure. In the liver, centrilobular necrosis with proliferation of Kupffer cells may be found. However, severe hepatocellular necrosis is not a feature of leptospirosis. Pulmonary involvement is the result of hemorrhage and not of inflammation. Invasion of skeletal muscle by leptospire results in swelling, vacuolation of the myofibrils, and focal necrosis. In severe leptospirosis, vasculitis may ultimately impair the microcirculation and increase capillary permeability, resulting in fluid leakage and hypovolemia.

When antibodies are formed, leptospire are eliminated from all sites in the host except the eye, the proximal renal tubules, and perhaps the brain, where they may persist for weeks or months. The persistence of leptospire in the aqueous humor occasionally causes chronic or recurrent uveitis. The systemic immune response is effective in eliminating the organism but may also produce symptomatic inflammatory reactions. A rise in antibody titer coincides with the development of meningitis; this association

suggests that an immunologic mechanism is responsible.

After the start of antimicrobial treatment for leptospirosis, a Jarisch-Herxheimer reaction similar to that seen in other spirochetal diseases may develop. Although frequently described in older publications, this reaction seems to be a rare event in leptospirosis and is certainly less frequent in this infection than in other spirochetal diseases.

CLINICAL MANIFESTATIONS

It is important to try to obtain a history of exposure to contaminated materials. Serologic evidence of past inapparent infection is found in 15 to 40% of persons who have been exposed but have not become ill. In symptomatic cases of leptospirosis, clinical manifestations vary from mild to serious or even fatal. More than 90% of symptomatic persons have the relatively mild and usually anicteric form of leptospirosis, with or without meningitis. Severe leptospirosis with profound jaundice (Weil's syndrome) develops in 5 to 10% of infected individuals.

The incubation period is usually 1 to 2 weeks but ranges from 2 to 26 days. Typically, an acute leptospiremic phase is followed by an immune leptospiruric phase. The distinction between the first and second phases is not always clear, and milder cases do not always include the second phase.

Anicteric Leptospirosis Leptospirosis may present as an acute influenza-like illness, with fever, chills, severe headache, nausea, vomiting, and myalgias. Muscle pain, which especially affects the calves, back, and abdomen, is an important feature of leptospiral infection. Less common features include sore throat and rash. The patient usually has an intense headache (frontal or retroorbital) and sometimes develops photophobia. Mental confusion may be evident. Pulmonary involvement, manifested in most cases by cough and chest pain and in a few cases by hemoptysis, is not uncommon.

The most common finding on physical examination is fever with conjunctival suffusion. Less common findings include muscle tenderness, lymphadenopathy, pharyngeal injection, rash, hepatomegaly, and splenomegaly. The rash may be macular, maculopapular, erythematous, urticarial, or hemorrhagic. Mild jaundice may be present.

Most patients become asymptomatic within 1 week. After an interval of 1 to 3 days, the illness recurs in a number of cases. The start of this second (immune) phase coincides with the development of antibodies. Symptoms are more variable than during the first (leptospiremic) phase. Usually the symptoms last for only a few days, but occasionally they persist for weeks. Often the fever is less pronounced and the myalgias are less severe than in the leptospiremic phase. An important event during the immune phase is the development of aseptic meningitis. Although no more than 15% of all patients have symptoms and signs of meningitis, many patients may have [CSF](#) pleocytosis. Meningeal symptoms usually disappear within a few days but may persist for weeks. Similarly, pleocytosis generally disappears within 2 weeks but occasionally persists for months. Iritis, iridocyclitis, and chorioretinitis -- late complications that may persist for years -- can become apparent as early as the third week but often present several months after the initial illness. One epidemic of uveitis among patients with leptospirosis has been reported.

Severe Leptospirosis (Weil's Syndrome) Weil's syndrome, the most severe form of leptospirosis, is characterized by jaundice, renal dysfunction, hemorrhagic diathesis, and high mortality. This syndrome is frequently but not exclusively associated with infection due to serovar *icterohaemorrhagiae/copenhageni*. The onset of illness is no different from that of less severe leptospirosis; however, after 4 to 9 days, jaundice as well as renal and vascular dysfunction generally develop. Although some degree of defervescence may be noted after the first week of illness, a biphasic disease pattern like that seen in anicteric leptospirosis is lacking. The jaundice of Weil's syndrome, which can be profound and give an orange cast to the skin, is usually not associated with severe hepatic necrosis. Death is rarely due to liver failure. Hepatomegaly and tenderness in the right upper quadrant are usually detected. Splenomegaly is found in 20% of cases.

Renal failure may develop, often during the second week of illness. Hypovolemia and decreased renal perfusion contribute to the development of acute tubular necrosis with oliguria or anuria. Dialysis is sometimes required, although a fair number of cases can be managed without dialysis. Renal function may be completely regained.

Pulmonary involvement occurs frequently, resulting in cough, dyspnea, chest pain, and blood-stained sputum, and sometimes in hemoptysis or even respiratory failure. Hemorrhagic manifestations are seen in Weil's syndrome: epistaxis, petechiae, purpura, and ecchymoses are found commonly, while severe gastrointestinal bleeding and adrenal or subarachnoid hemorrhage are detected rarely.

Rhabdomyolysis, hemolysis, myocarditis, pericarditis, congestive heart failure, cardiogenic shock, adult respiratory distress syndrome, and multiorgan failure have all been described during severe leptospirosis.

LABORATORY AND RADIOLOGIC FINDINGS

The kidneys are invariably involved in leptospirosis ([Fig. 174-CD1](#)). Related findings range from urinary sediment changes (leukocytes, erythrocytes, and hyaline or granular casts) and mild proteinuria in anicteric leptospirosis to renal failure and azotemia in severe disease.

The erythrocyte sedimentation rate is usually elevated. In anicteric leptospirosis, peripheral leukocyte counts range from 3000 to 26,000/uL, with a left shift; in Weil's syndrome, leukocytosis is often marked. Mild thrombocytopenia occurs in up to 50% of patients and is associated with renal failure.

In contrast to patients with acute viral hepatitis, those with leptospirosis typically have elevated serum levels of bilirubin and alkaline phosphatase as well as mild increases (up to 200 U/L) in serum levels of aminotransferases. In Weil's syndrome, the prothrombin time may be prolonged but can be corrected with vitamin K. Levels of creatine phosphokinase, which are elevated in up to 50% of patients with leptospirosis during the first week of illness, may help to differentiate this infection from viral hepatitis.

When a meningeal reaction develops, polymorphonuclear leukocytes predominate

initially and the number of mononuclear cells increases later. The protein concentration in the [CSF](#) may be elevated; CSF glucose levels are normal.

In severe leptospirosis, pulmonary radiographic abnormalities are more common than would be expected on the basis of physical examination. These abnormalities most frequently develop 3 to 9 days after the onset of illness. The most common radiographic finding is a patchy alveolar pattern that corresponds to scattered alveolar hemorrhage. Radiographic abnormalities most often affect the lower lobes in the periphery of the lung fields.

DIAGNOSIS

A definite diagnosis of leptospirosis is based either on isolation of the organism from the patient or on seroconversion or a rise in antibody titer in the microscopic agglutination test (MAT). For a presumptive diagnosis of leptospirosis, an antibody titer of $\geq 1:100$ in the MAT or a positive macroscopic slide agglutination test in the presence of a compatible clinical illness is required. Antibodies generally do not reach detectable levels until the second week of illness. The antibody response can be affected by early treatment.

The macroscopic slide agglutination test with killed antigen is useful for screening but is not specific. The [MAT](#), which uses a battery of live leptospiral strains, and the enzyme-linked immunosorbent assay (ELISA), which uses a broadly reacting antigen, are the standard serologic procedures. These tests usually are available only in specialized laboratories and are used for the determination of the antibody titer and for the tentative identification of the serovar involved (thus the importance of using antigens representative of the serovars prevalent in the particular geographic area). Since cross-reactions occur frequently, however, it is often impossible to identify the infecting serovar. Serologic testing cannot be used as the basis for a decision about whether to start treatment.

In addition to the [MAT](#) and the [ELISA](#), various other tests with diagnostic value have been developed. Some tests, such as an indirect hemagglutination test, a microcapsule agglutination test, and an IgM ELISA, are commercially available. Dot-ELISA, gold immunoblot, and polymerase chain reaction techniques have been developed but are not yet used for routine diagnosis.

Leptospire can be isolated from blood and/or [CSF](#) during the first 10 days of illness and from urine for several weeks beginning at around 1 week. Cultures may become positive after 2 to 4 weeks, with a range of 1 week to 4 months. Sometimes urine cultures remain positive for months or years after the start of illness. For isolation of leptospire from body fluids or tissues, Ellinghausen-McCullough-Johnson-Harris (EMJH) medium is useful; other possibilities are Fletcher medium and Korthoff medium. Specimens can be mailed to a reference laboratory for culture, since leptospire remain viable in anticoagulated blood (heparin, EDTA, or citrate) for up to 11 days. Isolation of leptospire is important since it is the only way the infecting serovar can be correctly identified. Dark-field examination of blood or urine frequently results in misdiagnosis and should not be used.

DIFFERENTIAL DIAGNOSIS

Leptospirosis should be differentiated from other febrile illnesses associated with headache and muscle pain, such as malaria, enteric fever, viral hepatitis, dengue, *Hantavirus* infections, and rickettsial diseases. In light of the strong similarity in epidemiology and clinical presentation between leptospirosis and *Hantavirus* infections and given the reported occurrence of dual infections, it is advisable to conduct serologic testing for *Hantavirus* in cases of suspected leptospirosis. When patients have a flulike disease with disproportionately severe myalgia or aseptic meningitis, a diagnosis of leptospirosis should be considered.

TREATMENT

The effectiveness of antimicrobial therapy for the mild febrile form of leptospirosis is controversial, but such treatment is indicated for more severe forms. Treatment should be initiated as early as possible; nevertheless, contrary to previous reports, treatment started after the first 4 days of illness is effective.

For severe cases of leptospirosis, intravenous administration of penicillin G, amoxicillin, ampicillin, or erythromycin is recommended ([Table 174-1](#)). In milder cases, oral treatment with tetracycline, doxycycline, ampicillin, or amoxicillin should be considered. Although several other antibiotics, including newer cephalosporins, are highly active against leptospires in vitro, no clinical experience has yet been gained with these drugs.

In rare cases, a Jarisch-Herxheimer reaction develops within hours after the start of antimicrobial therapy (see "Pathogenesis" above). Although so far the only effective mode of management is supportive, the role of antibodies to tumor necrosis factor in the treatment of this reaction deserves further study. A beneficial effect of the use of such antibodies for the modulation of the reaction has been demonstrated in patients with louse-borne relapsing fever. Patients with severe leptospirosis and renal failure may require dialysis. Those with Weil's syndrome may need transfusions of whole blood and/or platelets. Intensive care may be necessary.

PROGNOSIS

Most patients with leptospirosis recover. Mortality is highest among patients who are elderly and those who have Weil's syndrome. Leptospirosis during pregnancy is associated with high fetal mortality. Long-term follow-up of patients with renal failure and hepatic dysfunction has documented good recovery of renal and hepatic function.

PREVENTION

Individuals who may be exposed to leptospires through their occupations or their involvement in recreational water activities should be informed about the risks. Measures for controlling leptospirosis include avoidance of exposure to urine and tissues from infected animals, vaccination of animals, and rodent control. The animal vaccine used in a given area should contain the serovars known to be present in that area. Unfortunately, some vaccinated animals still excrete leptospires in their urine. Vaccination of humans against a specific serovar prevalent in an area has been

undertaken in some European and Asian countries and has proved effective. Chemoprophylaxis with doxycycline (200 mg once a week) has appeared to be efficacious in military personnel but is indicated only in rare instances of sustained short-term exposure.

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175. RELAPSING FEVER - David T. Dennis, Grant L. Campbell

DEFINITION

The term *relapsing fever* describes two distinct borrelial disease entities: louse-borne relapsing fever (LBRF) and tick-borne relapsing fever (TBRF). Both are characterized by recurrent acute episodes of spirochetemia and fever alternating with spirochetal clearance and apyrexia.

ETIOLOGY

The worldwide distribution of relapsing spirochetal fevers was recognized in the early part of the twentieth century, and the causative agents were shown to be transmitted by lice and ticks. *Borrelia recurrentis* was identified as the cause of [LBRF](#); differing strains of borreliae causing [TBRF](#) were usually named according to the species of *Ornithodoros* tick responsible for their transmission ([Table 175-1](#)). Sequencing of both flagellin and 16S ribosomal RNA genes reveals homogeneity among LBRF strains and considerable heterogeneity between Old World and New World TBRF strains.

Relapsing-fever borreliae are gram-negative bacteria that belong to the family Spirochaetaceae. They are helical in shape and average 0.2 to 0.5 μm in width and 5 to 20 μm in length. They comprise an outer membrane, an intermediate peptidoglycan layer, and an inner cytoplasmic membrane, which encloses the protoplasmic cylinder. Periplasmic flagella (variable numbers have been described) are situated beneath the outer membrane. Relapsing-fever borreliae are slow-growing and microaerophilic; they grow best at 30 to 35°C. Both [TBRF](#) and [LBRF](#) spirochetes grow well in Barbour-Stoenner-Kelly (BSK II) medium.

Relapsing-fever borreliae are distinguished by remarkable antigenic variability and strain heterogeneity. New *Borrelia* serotypes spontaneously emerge at a high rate, resulting from a unique process of DNA rearrangement within genes located on linear plasmids. These genes code for variable major proteins (VMPs) found on the spirochete's outer-membrane surface. This antigenic variation, generated by sequential expression of previously silent *vmp* genes for serotype-specific VMPs, allows the borreliae to escape the immune response of the host and results in the relapse phenomenon characteristic of infection with these organisms. Borrelial *vmp* gene expression also varies between mammalian and arthropod hosts.

EPIDEMIOLOGY

Louse-Borne Relapsing Fever Body lice (*Pediculus humanus* var. *corporis*) become infected with *B. recurrentis* by feeding on spirochetemic humans, the only reservoirs of infection. In lice, *B. recurrentis* spirochetes are found almost exclusively in the hemolymph; humans acquire infection when infected body lice are crushed and their fluids contaminate mucous membranes or breaks in the skin (such as abrasions caused by scratching of pruritic louse bites). Spirochetes are not transmitted directly by the bite of a louse (anterior station transmission) or by inoculation of louse feces (posterior station transmission). Lice have a life span of only a few weeks, feed at frequent intervals, and survive only a few days off the human host.

[LBRF](#) has severely affected military and civilian populations disrupted by war and other disasters. During the Industrial Revolution, the disease was common among slum dwellers, prisoners, and others living in impoverished, overcrowded, and unhygienic conditions. In the first half of the twentieth century, during periods of war and famine, both LBRF and louse-borne typhus were epidemic in eastern Europe, the Balkans, and the former Soviet Union. LBRF has disappeared from its former global range as improvements have been made in standards of living, sanitation, and hygiene; it is now an important disease only in northeastern Africa, especially the highlands of Ethiopia, where an estimated 10,000 cases occur annually. In Ethiopia, the disease affects mostly homeless men crowded together in unhygienic circumstances, especially during the cool rainy season, when it is more difficult for them to change and wash their clothing. LBRF has repeatedly spilled out of Ethiopia into neighboring Somalia and Sudan, especially affecting displaced persons. LBRF does not pose a significant risk to tourists or other casual visitors but can be acquired from lice by persons (such as relief workers) in intimate contact with those affected as well as through accidental needle stick or mucocutaneous contact with infected blood.

Tick-Borne Relapsing Fever Soft ticks (*Argasidae*, *Ornithodoros* spp.) transmit [TBRF](#). The ticks become infected by feeding on spirochetemic hosts. Except for *B. duttoni* (a prominent cause of TBRF in sub-Saharan Africa), TBRF borreliae are zoonotic disease agents found naturally in rodents (rats, mice, chipmunks, and squirrels) and in lagomorphs (rabbits and hares). The spirochetes are transmitted by ticks to humans and animals via saliva and excretory fluids when the tick feeds. Infection in ticks is transmitted vertically from one stage to the next; in some species, infection is transmitted transovarially over several generations. Soft ticks are hardy and can survive for 10 years or more with only an occasional blood meal. These ticks feed painlessly, relatively quickly (for 20 to 45 min), and usually at night while hosts are sleeping. Thus patients with TBRF are often unaware of tick exposures.

[TBRF](#) borreliae are widely distributed throughout the world. Human infection with these organisms is generally underrecognized and underreported. TBRF is most highly endemic in sub-Saharan Africa but is also found in countries of the Mediterranean littoral, Middle Eastern states, southern Russia, the Indian subcontinent, and China. In the United States, this disease occurs west of the Mississippi River, especially in mountainous areas, where *B. hermsii* is the causative agent. TBRF is reported at low frequency throughout Latin America. The disease typically occurs sporadically or in small -- often familial -- clusters. Infected soft ticks may cause repeated infections among persons living or sleeping in the same dwelling. In sub-Saharan Africa, *O. moubata*, the vector of *B. duttoni*, infests native huts and rest houses, hiding in crevices of floors and walls during the day and emerging at night to feed on sleeping inhabitants. In the United States, *B. hermsii* infections most often occur during spring and summer months among persons sleeping in rustic mountain cabins. Infections of humans are sometimes precipitated by the disappearance of rodents (e.g., as a result of epizootic plague) that nest in foundations, wall spaces, and attics and that serve as the usual maintenance hosts for *O. hermsi* ticks. Outbreaks caused by *B. hermsii* have taken place among persons staying in cabins along the north rim of the Grand Canyon and in the mountains of California, Idaho, and Colorado. Rodent-infested caves in southwestern states are associated with occasional cases of relapsing fever caused by

B. turicatae.

PATHOGENESIS AND PATHOLOGY

In humans, relapsing-fever borreliae penetrate the skin or mucous membranes, multiply in the blood, and circulate in great numbers during febrile periods. The organisms also may be found in the liver, spleen, central nervous system, bone marrow, and other tissues and may be sequestered at these sites during periods of remission. The severity of disease is positively related to spirochete density in the blood. Even though the pathophysiologic manifestations of the disease resemble responses to endotoxin, and although plasma from some patients with relapsing fever coagulates *Limulus* amebocyte lysates, borreliae and other spirochetes have not been shown to express a true lipopolysaccharide (endotoxin) molecule. Infection with *B. recurrentis* does, however, activate protein mediators of inflammation, such as Hageman factor, prekallikrein, and proteins of the complement system; furthermore, a spirochetal heat-stable pyrogenic factor stimulates mononuclear phagocytes to express increased amounts of leukocyte pyrogen and thromboplastin.

The Jarisch-Herxheimer reaction in patients with [LBRF](#) is associated with a release of various cytokines into the plasma, including interleukin 6, interleukin 8, C-reactive protein, and enormous amounts of tumor necrosis factor α (TNF- α). Pretreatment of LBRF patients with antibody to TNF- α suppresses the Jarisch-Herxheimer reactions that follow penicillin treatment and reduces the plasma concentrations of certain other cytokines.

Findings at autopsy of patients with relapsing fever most often include hepatosplenomegaly and variable edema and swelling of other organs, such as the brain, lungs, and kidneys. On microscopic examination, the spleen is congested and contains multiple microabscesses composed of mononuclear cells that replace the white pulp, the myocardium displays diffuse histiocytic inflammation and interstitial edema, and the liver has areas of midzonal necrosis. Petechial hemorrhages are commonly evident over the surfaces of the meninges, pleura, heart, spleen, liver, kidneys, and mesentery. Subcapsular and parenchymal hemorrhagic infarcts of the spleen, heart, liver, and brain are sometimes grossly visible. Icterus is a common finding in severe and fatal cases of relapsing fever.

CLINICAL MANIFESTATIONS

The clinical manifestations of [LBRF](#) and [TBRF](#) are similar. The mean incubation period is 7 days (range, 2 to 18 days), and the onset of illness is sudden, with fever, headache, shaking chills, sweats, myalgias, and arthralgias. The arthralgia of relapsing fever can be severe, involving small and large joints, but there is no evidence of arthritis. Dizziness, nausea, and vomiting are common. Sleep may be difficult and is sometimes accompanied by disturbing dreams. The patient is coherent but withdrawn, thirsty, and disinterested in food and other outside stimuli. The fever is high from the first, with a usual temperature of 40°C (104°F); fever is most often irregular in pattern and is sometimes accompanied by delirium. Patients become progressively prostrate as the disease advances. The pulse is rapid and the patient is mildly tachypneic. Meningism may be found. The conjunctivae are often injected, and the patient usually exhibits

photophobia. The sclerae are sometimes icteric, most commonly in the later stages of illness. The mucous membranes are often dry, and the patient is usually dehydrated. Scattered petechiae develop on the trunk, extremities, and mucous membranes in one-third or more of patients with LBRF and in fewer patients with TBRF. A nonproductive cough is common, but chest sounds are usually normal; pleuritic pain and an accompanying pleuritic rub are sometimes noted. Cardiac findings are compatible with a high-output state; tachycardia and summation gallop are common. Tender enlargement of the spleen and liver frequently characterizes the acute phase of illness.

Epistaxis and blood-tinged sputum are common complications, and gastrointestinal and central nervous system hemorrhage can occur. Because of this coagulopathy, one [LBRF](#) outbreak in southern Sudan was thought to be viral hemorrhagic fever. Other complications of variable incidence include iridocyclitis, optic neuritis, meningitis, coma, isolated cranial-nerve palsy, pneumonitis, myocarditis, and rupture of the spleen. Infection during pregnancy can result in spontaneous abortion, stillbirth, or neonatal infection. Life-threatening complications are unusual in otherwise healthy persons given supportive care, especially if the illness is diagnosed and treated early.

Without treatment, symptoms intensify over a 2- to 7-day period (average, 5 days in [LBRF](#) and 3 days in [TBRF](#)), ending in a spontaneous crisis during which spirochetes disappear from the circulation. Treatment with one of the rapidly acting antibiotics, such as erythromycin, a tetracycline, or chloramphenicol, regularly precipitates a Jarisch-Herxheimer reaction within 1 to 4 h. The severity of this reaction is positively correlated with the density of spirochetes in the blood at the time of treatment. In the first phase of the crisis or reaction (the *chill phase*), rigors and rising fever are accompanied by an increasing metabolic rate, alveolar hyperventilation, high cardiac output, increasing peripheral vascular resistance, and decreased pulmonary arterial pressure. The body temperature commonly rises to 41°C (105.8°F). This high fever is accompanied often by agitation and confusion and sometimes by delirium. Fever can be partially controlled by the use of a cooling blanket and ice packs and by sponging of the patient with tepid water and alcohol. The chill phase terminates after 10 to 30 min, giving way to a *flush phase* characterized by a fall in body temperature, drenching sweats, and sometimes (more commonly in LBRF) a potentially dangerous fall in systemic arterial pressure and rise in pulmonary arterial pressure. Although cardiac output is maintained at high levels, the effective circulating blood volume decreases as peripheral vascular resistance falls. Vital signs must be monitored carefully during this period of the reaction, which usually lasts ≤ 8 h. Clinical and electrocardiographic evidence of myocarditis and myocardial dysfunction includes a prolonged QT_c interval, a third heart sound (S_3), elevated central venous pressure, arterial hypotension, and pulmonary edema.

The crisis is followed by a period of exhaustion, sleep, and an uneventful recovery. Not uncommonly, in the first week of convalescence, patients experience 1 or 2 days of mild fever unassociated with detectable spirochetemia. In untreated patients, spirochetemia and symptoms may recur after a period of several days or weeks (average interval to first relapse, 9 days in [LBRF](#) and 7 days in [TBRF](#)). Only one or two relapses characteristically occur in untreated patients with LBRF, whereas as many as 10 (average, three) can occur in untreated patients with TBRF. In most cases, the illness

becomes shorter and milder and the afebrile intervals longer with each relapse. Because of the great antigenic variation among *Borrelia* strains, infection confers only partial immunity, and repeated infections of the same individual have been recorded.

Diseases that should be considered in the differential diagnosis of relapsing fever or that may complicate relapsing fever include typhus fever, typhoid, nontyphoid salmonellosis, malaria, dengue and other arboviral illnesses, tuberculosis, leptospirosis, and viral hemorrhagic fevers. In the United States, the geographic distribution of Colorado tick fever overlaps that of [TBRF](#), and the two diseases have similar manifestations early in their courses.

LABORATORY FINDINGS AND DIAGNOSIS

The diagnosis of relapsing fever is confirmed most easily by the detection of spirochetes in blood, bone marrow aspirates, or cerebrospinal fluid. Motile spirochetes can be seen when fresh blood is examined by dark-field microscopy; and fixed organisms are clearly visible in Wright-, Giemsa-, or acridine orange-stained preparations of thin or dehemoglobinized thick smears of peripheral blood or buffy-coat preparations ([Fig. 175-1](#)). Organisms are found in blood taken during periods of fever preceding the crisis; smears from ³70% of patients with [LBRF](#) and from fewer patients with [TBRF](#) are positive. In reference laboratories, relapsing-fever spirochetes are cultured from blood by the inoculation of BSK II medium or by the intraperitoneal inoculation of immature laboratory mice. The detection of agglutinins against *Proteus* OX-K (Weil-Felix reaction) in convalescent-phase serum supports the diagnosis. Serum antibodies to *Borrelia* can be detected by enzyme immunoassays, but these tests are unstandardized and subject to insensitivity due to antigenic variations among strains. Serologic cross-reactions occur with other spirochetes, including *B. burgdorferi* (the agent of Lyme disease) and *Treponema pallidum*.

Other laboratory findings in relapsing fever are generally nonspecific. The leukocyte count is normal or moderately elevated, with an unremarkable cell differential. Serum bilirubin levels are generally only slightly elevated. Thrombocytopenia (mean platelet count, about 50,000/uL) is evident in patients with [LBRF](#) during the acute phase of the illness; platelet counts rebound during early convalescence. Prothrombin and partial thromboplastin times are moderately prolonged during acute illness, as are standardized bleeding times. Fibrinogen concentrations in the blood are normal, and fibrinolysis is mild or absent. Results of the Rumpel-Leede tourniquet test are negative, despite the presence of petechiae.

TREATMENT

Relapsing-fever borreliae are exquisitely sensitive to antibiotics. Treatment with erythromycin, a tetracycline, chloramphenicol, or penicillin produces rapid clearance of spirochetes and a remission of symptoms ([Table 175-2](#)). Although a single dose of erythromycin, a tetracycline, or chloramphenicol is highly effective in the treatment of [LBRF](#), less is known about the efficacy of single-dose treatment of [TBRF](#). Empirical treatment of TBRF for 7 days is therefore recommended to reduce the risk of persisting or relapsing borreliosis. For children <8 years of age and for pregnant women, erythromycin and penicillin are the preferred drugs.

The use of delayed-release intramuscular penicillin may prolong or delay the clearance of spirochetes and thereby attenuate the accompanying Jarisch-Herxheimer reaction, but this response is not predictable; furthermore, single-dose penicillin treatment sometimes results in relapse of spirochetemia and symptoms. Glucocorticoids and nonsteroidal anti-inflammatory agents do not prevent or significantly modify the cardiopulmonary disturbances of the Jarisch-Herxheimer reaction, although hydrocortisone and acetaminophen given at the same time as antibiotics reduce peak body temperature. Although pretreatment with antibody to [TNF- \$\alpha\$](#) may moderate the Jarisch-Herxheimer reaction in treated patients with [LBRF](#), its widespread use in LBRF is impractical and its use in [TBRF](#) (whose treatment is associated with a relatively mild Jarisch-Herxheimer reaction) is not warranted. Close monitoring of fluid balance, arterial and venous pressures, and myocardial function is advised in supportive management of the Jarisch-Herxheimer reaction in patients with LBRF.

The management of patients with myocardial dysfunction requires caution in the administration of intravenous fluids and, in some cases, rapid digitalization. Bleeding is not controlled by heparin, and clinical studies do not suggest that disseminated intravascular coagulopathy is important. Vitamin K and other soluble vitamins are sometimes given to counter dietary deficiencies in patients with [LBRF](#). Because postural hypotension is often pronounced during the acute phase of relapsing fever and in the early stage of recovery, patients should be assisted when arising from bed.

Untreated [LBRF](#) has a high case-fatality rate, especially among persons in otherwise poor health, such as those in famine-affected populations. The fatality rate among treated persons is usually <5%. In general, [TBRF](#) is a milder disease than LBRF: the spontaneous crisis and the Jarisch-Herxheimer reactions are less pronounced and the case-fatality rates are lower for TBRF than for LBRF.

PREVENTION AND CONTROL

[LBRF](#) can be prevented by elimination of circumstances that promote louse infestation (crowding, poverty, homelessness, poor personal hygiene), by use of practices that eliminate or reduce numbers of body lice (washing clothes, drying clothes in direct sunlight, changing clothes at frequent intervals), and by application of acaricides. Secondary complications and the spread of infection can be prevented by early case detection and treatment of infected persons and close contacts. Historically, outbreaks of LBRF have been controlled by mass delousing. In situations like those in refugee camps, individuals, their clothes, and their bedding should be deloused with appropriate acaricides, such as 0.5% permethrin dust. Impregnation of clothing with liquid permethrin, a residual acaricide, can provide long-term protection against infestation. In outbreaks of fever that involve louse-infested populations, empirical single-dose treatment with doxycycline will be effective against typhus as well as LBRF. *B. recurrentis* has a fragile life cycle and is eradicable.

[TBRF](#) can be prevented by the avoidance of rodent- and tick-infested dwellings and infested natural sites. Limiting rodent access to the foundations and attics of homes and vacation cabins and eliminating harborage for rodents in and around these dwellings reduce the potential for tick exposure. Rodents and rodent nests should be removed

from infested buildings and their surroundings. Tick harborages of infested buildings or other circumscribed sites, such as rodent burrows and nests in hollow logs surrounding dwellings and in rodent-infested caves, can be chemically treated by pest-control specialists using various acaricides, such as carbaryl, diazinon, chlorpyrifos, pyrethrins, and malathion. Persons who enter tick-infested sites can protect themselves by wearing clothing that denies ticks access to the skin, by applying repellents to exposed skin and to clothing, and by applying an acaricide containing permethrin to clothing. Reporting of suspected cases of relapsing fever to public health authorities is important so that an epidemiologic investigation and control measures can be initiated promptly.

(Bibliography omitted in Palm version)

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176. LYME BORRELIOSIS - Allen C. Steere

DEFINITION

Lyme borreliosis, a tick-transmitted spirochetal illness, usually begins with a characteristic expanding skin lesion, erythema migrans (EM; stage 1, localized infection). After several days or weeks, the spirochete may spread hematogenously to many different sites (stage 2, disseminated infection). Possible manifestations of disseminated infection include secondary annular skin lesions, meningitis, cranial or peripheral neuritis, carditis, atrioventricular nodal block, or migratory musculoskeletal pain. Months to years later (usually after periods of latent infection), intermittent or chronic arthritis, chronic encephalopathy or polyneuropathy, or acrodermatitis may develop (stage 3, persistent infection). Most patients experience early symptoms of the illness during the summer, but the infection may not become symptomatic until it progresses to stage 2 or 3. Despite regional variations, the basic stages of the illness are similar worldwide.

ETIOLOGIC AGENT

Borrelia burgdorferi, the causative agent of the disease, is a fastidious, microaerophilic bacterium. The organism contains many immunogenic proteins, including a number of differentially expressed lipoproteins, most of which are encoded by plasmid DNA. The spirochete grows best at 33°C in a complex liquid medium called Barbour, Stoenner, Kelly (BSK) medium. Culture of the organism from clinical specimens (except for biopsy samples of skin at sites of EM or acrodermatitis) has been difficult. Three groups of pathogenic *B. burgdorferi* organisms, together referred to as *B. burgdorferi sensu lato*, have been identified, and more groups surely exist. To date, North American strains have belonged to the first group, *B. burgdorferi sensu stricto*. Although all three of the identified groups have been found in Europe and Asia, most isolates there have been strains of group 2 (*B. garinii*) or group 3 (*B. afzelii*). These differences may well account for the clinical variations in the disease in different geographic regions.

EPIDEMIOLOGY

The distribution of Lyme borreliosis correlates closely with the geographic ranges of ticks of the *Ixodes ricinus* complex: *I. scapularis* (also called *I. dammini*), *I. pacificus*, *I. ricinus*, and *I. persulcatus*. *I. scapularis* is the principal vector in the northeastern United States from Massachusetts to Maryland and in the midwestern states of Wisconsin and Minnesota. Surveys in these regions have documented infection in at least 20% of *I. scapularis* ticks; most cases of Lyme disease in the United States have occurred in these areas. *I. pacificus* is the vector in the western states of California and Oregon. The disease is acquired throughout Europe (from Great Britain to Scandinavia to European Russia), where *I. ricinus* is the vector, and in Asian Russia, China, and Japan, where *I. persulcatus* is the vector. These ticks transmit other diseases that may have similar symptoms. In the United States, *I. scapularis* also transmits babesiosis and ehrlichiosis; in Europe and Asia, *I. ricinus* and *I. persulcatus* also transmit tick-borne encephalitis.

Ticks of the *I. ricinus* complex feed once during each of the three stages of their usual

2-year life cycle. Typically, larval ticks take one blood meal in the late summer, nymphs ([Fig. 176-CD1](#)) feed during the following spring and early summer, and adults feed during autumn. For *I. scapularis* in the northeast, the white-footed mouse is the preferred host of the immature larval and nymphal ticks. It is critical that both of the tick's immature stages feed on the same host, because the life cycle of the spirochete depends on horizontal transmission: in early summer from infected nymphs to mice and in late summer from infected mice to larvae, which then molt to become the infected nymphs that will begin the cycle again the following year. It is the tiny nymphal tick that is primarily responsible for transmission of the disease to humans during the early summer months. White-tailed deer, which are not involved in the life cycle of the spirochete, are the preferred host for the adult stage of *I. scapularis* and seem to be critical to the tick's survival. The adult tick occasionally transmits the spirochete to humans during the fall, but at this stage the tick is considerably larger and easier to recognize.

Lyme disease is now the most common vector-borne infection in the United States. Since surveillance was begun in 1982, more than 100,000 cases have been reported to the Centers for Disease Control and Prevention (CDC); during the 1990s, more than 10,000 new cases have been reported each summer. Cases have been noted in 48 states, but the life cycle of *B. burgdorferi* has been identified in only 19 states. Cases have occurred in association with hiking, camping, or hunting trips and with residence in wooded or rural areas. Persons of all ages and both sexes are affected.

PATHOGENESIS

To maintain its complex enzootic cycle, *B. burgdorferi* must adapt to two markedly different environments: the tick and the mammalian host. The spirochete expresses outer-surface proteins A and B (OspA and OspB) in the midgut of the tick, whereas OspC is upregulated as the organism travels to the tick's salivary gland and thence to the mammalian host.

After injection into the human skin, *B. burgdorferi* may migrate outward, producing [EM](#), and may spread hematogenously to other organs. Spread within the human host is probably facilitated through binding to the spirochete's surface by human plasminogen and urokinase-type plasminogen activator, which activates plasmin, a potent protease. The spirochete can adhere to many types of mammalian cells; it binds specifically to certain ubiquitous host integrin receptors in the extracellular matrix, to vitronectin and fibronectin, and to matrix glycosaminoglycans. *B. burgdorferi* seems to have a particular tropism for tissues of the skin, nervous system, atrioventricular node, and joints, from all of which it has been cultured, seen in histologic sections, or (more commonly) detected (via its DNA) by the polymerase chain reaction (PCR). These findings and the response of all stages of the disease to antibiotic therapy suggest that the organism persists in affected tissues throughout the illness, but the mechanisms of persistent infection are not yet clear.

The immune response in Lyme disease develops gradually. After the first several weeks of infection, mononuclear cells generally exhibit heightened responsiveness to *B. burgdorferi* antigens, and evidence of B cell hyperactivity is found, including elevated total serum IgM levels, cryoprecipitates, and circulating immune complexes. Titers of

specific IgM antibody to *B. burgdorferi* peak between the third and sixth week after disease onset. The specific IgG response develops gradually over months, with response to an increasing array of 12 or more spirochetal polypeptides and maximal expansion during the period of arthritis. The spirochete is a potent inducer of proinflammatory cytokines, including tumor necrosis factor α and interleukin 1 β . Histologic examination of all affected tissues reveals an infiltration of lymphocytes and plasma cells with some degree of vascular damage (including mild vasculitis or hypervascular occlusion), suggesting that the spirochete may have been present in or around blood vessels.

CLINICAL MANIFESTATIONS

Early Infection: Stage 1 (Localized Infection) After an incubation period of 3 to 32 days, [EM](#), which occurs at the site of the tick bite, usually begins as a red macule or papule that expands slowly to form a large annular lesion, most often with a bright red outer border and partial central clearing ([Plate IID-46](#), [Fig. 176-CD2](#)). Because of the small size of ixodid ticks, most patients do not remember the preceding tick bite. The center of the lesion sometimes becomes intensely erythematous and indurated, vesicular, or necrotic. In other instances, the expanding lesion remains an even, intense red; several red rings are found within an outside ring; or the central area turns blue before the lesion clears. Although EM can be located anywhere, the thigh, groin, and axilla are particularly common sites. The lesion is warm but not often painful. Perhaps as many as 25% of patients do not exhibit this characteristic skin manifestation.

Early Infection: Stage 2 (Disseminated Infection) Within days or weeks after the onset of [EM](#), the organism often spreads hematogenously to many sites. In these cases patients frequently develop secondary annular skin lesions similar in appearance to the initial lesion. Skin involvement is commonly accompanied by severe headache, mild stiffness of the neck, fever, chills, migratory musculoskeletal pain, arthralgias, and profound malaise and fatigue. Less common manifestations include generalized lymphadenopathy or splenomegaly, hepatitis, sore throat, nonproductive cough, conjunctivitis, iritis, or testicular swelling. Except for fatigue and lethargy, which are often constant, the early signs and symptoms of Lyme disease are typically intermittent and changing. Even in untreated patients, the early symptoms usually become less severe or disappear within several weeks.

Symptoms suggestive of meningeal irritation may develop early in Lyme disease when [EM](#) is present but usually are not associated with cerebrospinal fluid (CSF) pleocytosis or an objective neurologic deficit. After several weeks or months, about 15% of untreated patients develop frank neurologic abnormalities, including meningitis, subtle encephalitic signs, cranial neuritis (including bilateral facial palsy), motor or sensory radiculoneuropathy, mononeuritis multiplex, or myelitis -- alone or in various combinations. In the United States, the usual pattern consists of fluctuating symptoms of meningitis accompanied by facial palsy and peripheral radiculoneuropathy. Lymphocytic pleocytosis (about 100 cells per microliter) is found in CSF, often along with elevated protein levels and normal or slightly low glucose concentrations. In Europe and Asia, the first neurologic sign is characteristically radicular pain, which is followed by the development of CSF pleocytosis (called *Bannwarth's syndrome*), but meningeal or encephalitic signs are frequently absent. These early neurologic abnormalities usually

resolve completely within months, but chronic neurologic disease may occur later.

Within several weeks after the onset of illness, about 8% of patients develop cardiac involvement. The most common abnormality is a fluctuating degree of atrioventricular block (first-degree, Wenckebach, or complete heart block). Some patients have more diffuse cardiac involvement, including electrocardiographic changes indicative of acute myopericarditis, left ventricular dysfunction evident on radionuclide scans, or (in rare cases) cardiomegaly or pancarditis. Cardiac involvement usually lasts for only a few weeks but may recur. Chronic cardiomyopathy caused by *B. burgdorferi* has been reported in Europe.

During this stage, musculoskeletal pain is common. The typical pattern consists of migratory pain in joints, tendons, bursae, muscles, or bones (usually without joint swelling) lasting for hours or days and affecting one or two locations at a time.

Late Infection: Stage 3 (Persistent Infection) Months after the onset of infection, about 60% of patients in the United States who have received no antibiotic treatment develop frank arthritis. The typical pattern comprises intermittent attacks of oligoarticular arthritis in large joints (especially the knees), lasting for weeks to months in a given joint. Small joints and periarticular sites also may be affected, primarily during early attacks. The number of patients who continue to have recurrent attacks decreases each year. However, in a small percentage of cases, involvement of large joints -- usually one or both knees -- becomes chronic and may lead to erosion of cartilage and bone. These patients have a higher frequency of the class II major histocompatibility complex alleles associated with rheumatoid arthritis, particularly HLA-DRB1*0401 or *0101 alleles, than patients with brief Lyme arthritis or normal control subjects. Moreover, they may have persistent arthritis for months or even several years after the apparent eradication of spirochetes from the joints with antibiotic therapy. In these genetically susceptible individuals, autoimmunity may develop within the proinflammatory milieu of the joints because of molecular mimicry between the dominant T cell epitope of OspA and human lymphocyte function-associated antigen 1 (hLFA-1).

White cell counts in joint fluid range from 500 to 110,000/uL (average, 25,000/uL); most of these cells are polymorphonuclear leukocytes. Tests for rheumatoid factor or antinuclear antibodies usually give negative results. Examination of synovial biopsy samples reveals fibrin deposits, villous hypertrophy, vascular proliferation, microangiopathic lesions, and a heavy infiltration of lymphocytes and plasma cells.

Although less common, chronic neurologic involvement may also become apparent months or years after the onset of infection, sometimes following long periods of latent infection. The most common form of chronic central nervous system involvement is subtle encephalopathy affecting memory, mood, or sleep and often accompanied by axonal polyneuropathy manifested as either distal paresthesia or spinal radicular pain. Patients with encephalopathy frequently have evidence of memory impairment in neuropsychological tests and abnormal results in [CSF](#) analyses. In cases with polyneuropathy, electromyography generally shows extensive abnormalities of proximal and distal nerve segments. Encephalomyelitis or leukoencephalitis, a rare manifestation of Lyme borreliosis reported primarily in Europe, is a severe neurologic disorder that may include spastic paraparesis, upper motor-neuron bladder dysfunction, and lesions

in the periventricular white matter. The prolonged course of chronic neuroborreliosis following periods of latent infection is reminiscent of tertiary neurosyphilis.

Acrodermatitis chronica atrophicans ([Fig. 176-CD3](#)), the late skin manifestation of the disorder, has been associated primarily with *B. afzelii* infection in Europe and Asia. It has been observed primarily in elderly women. The skin lesions, which are usually found on the acral surface of an arm or leg, begin insidiously with reddish-violaceous discoloration; they become sclerotic or atrophic over a period of years.

DIAGNOSIS

Lyme disease is usually diagnosed by the recognition of a characteristic clinical picture with serologic confirmation. Although serologic testing may yield negative results during the first several weeks of infection, most patients have a positive antibody response to *B. burgdorferi* after that time. The limitation of serologic tests is that they do not clearly distinguish between active and inactive infection. Patients with previous Lyme disease -- particularly in cases progressing to late stages -- often remain seropositive for years, even after adequate antibiotic treatment. In addition, some patients are seropositive because of asymptomatic infection. If these individuals subsequently develop another illness, the positive serologic test for Lyme disease may cause diagnostic confusion. On the other hand, a few patients who receive inadequate antibiotic therapy during the first several weeks of infection develop subtle joint or neurologic symptoms but are seronegative. The important point is that seronegative Lyme disease is usually a mild, attenuated illness.

For serologic analysis in Lyme disease, the [CDC](#) recommends a two-step approach in which samples are first tested by enzyme-linked immunosorbent assay (ELISA) and equivocal or positive results are then tested by western blotting. During the first month of infection, both IgM and IgG responses to the spirochete should be determined, preferably in both acute- and convalescent-phase serum samples. Approximately 20 to 30% of patients have a positive response detectable in acute-phase samples, whereas about 70 to 80% have a positive response during convalescence (2 to 4 weeks later). After that time, the great majority of patients continue to have a positive IgG antibody response, and a single test (that for IgG) is usually sufficient. In persons with illness of longer than 1 month's duration, a positive IgM test result alone is likely to be false-positive; therefore, a positive IgM test should not be used to support the diagnosis in such patients. According to current criteria adopted by the CDC, an IgM western blot is considered positive if two of the following three bands are present: 23, 39, and 41 kDa. However, the combination of the 23- and 41-kDa bands may still represent a false-positive result. An IgG blot is considered positive if 5 of the following 10 bands are present: 18, 23, 28, 30, 39, 41, 45, 58, 66, and 93 kDa.

Because serologic tests do not distinguish between active and inactive infection, tests that detect the spirochete directly are being researched. *B. burgdorferi* may be cultured from skin lesions of patients with the disorder, but its culture from other sites has been a low-yield proposition. Detection of spirochetal DNA by [PCR](#) may serve as a substitute for culture in cases of Lyme arthritis. In one study, *B. burgdorferi* DNA was detected in synovial fluid samples from 75 (85%) of 88 patients and in none of 64 control samples. However, the sensitivity of PCR determinations in [CSF](#) from patients with

neuroborreliosis has been much lower. There seems to be little if any role for PCR in the detection of *B. burgdorferi* DNA in blood or urine samples.

DIFFERENTIAL DIAGNOSIS

Classic [EM](#) is a slowly expanding erythema with partial central clearing. If the lesion expands little, it may represent the red papule of an uninfected tick bite. If the lesion expands rapidly, it may represent cellulitis (e.g., streptococcal cellulitis) or an allergic reaction, perhaps to tick saliva. Patients with secondary annular lesions may be thought to have erythema multiforme, but neither the development of blistering mucosal lesions nor the involvement of the palms or soles is a feature of *B. burgdorferi* infection. In the southeastern United States, an EM-like skin lesion, sometimes with mild systemic symptoms, may be associated with *Amblyomma americanum* tick bites, but the cause of this illness has not yet been identified.

Later in the infection, the most common problem in diagnosis is to mistake Lyme disease for chronic fatigue syndrome or fibromyalgia. This difficulty is compounded by the fact that a small percentage of patients do in fact develop these chronic pain or fatigue syndromes in association with or soon after Lyme disease. Compared with Lyme disease, chronic fatigue syndrome ([Chap. 384](#)) or fibromyalgia tends to produce more generalized and disabling symptoms, including marked fatigue, severe headache, diffuse musculoskeletal pain, multiple symmetric tender points in characteristic locations, pain and stiffness in many joints, diffuse dysesthesia, difficulty with concentration, and sleep disturbances. Patients with chronic fatigue syndrome or fibromyalgia lack evidence of joint inflammation; they have normal results in neurologic tests; and they usually have a greater degree of anxiety and depression than patients with chronic neuroborreliosis.

TREATMENT

As outlined in the algorithm in [Fig. 176-1](#), the various manifestations of Lyme disease can usually be treated successfully with orally administered antibiotics; the exceptions are objective neurologic abnormalities and third-degree atrioventricular heart block, which seem to require intravenous therapy. For early Lyme disease, doxycycline is effective in men and in nonpregnant women. An advantage of this regimen is that it is also effective against the agent of human granulocytic ehrlichiosis, which is transmitted by the same tick that transmits the Lyme disease agent. Amoxicillin, cefuroxime axetil, and erythromycin or its congeners are second-, third-, and fourth-choice alternatives, respectively. In children, amoxicillin is effective (not more than 2 g/d); in cases of penicillin allergy, cefuroxime axetil or erythromycin may be used. For patients with infection localized to the skin, a 20-day course of therapy is generally sufficient; in contrast, for patients with disseminated infection, a 30-day course is recommended. Approximately 15% of patients experience a Jarisch-Herxheimer-like reaction during the first 24 h of therapy.

These oral antibiotic regimens, when given for 30 to 60 days, are effective for the treatment of Lyme arthritis. However, the response to oral therapy may be slower than that to intravenous therapy. In the small percentage of patients with arthritis in whom arthritic symptoms persist for months or even years after the apparent eradication of

spirochetes from the joints with antimicrobial therapy, treatment with anti-inflammatory agents or synovectomy may be successful.

For objective neurologic abnormalities (with the possible exception of facial palsy alone), parenteral antibiotic therapy seems to be necessary. Intravenous ceftriaxone, given for 4 weeks, is most commonly used for this purpose, but intravenous cefotaxime or intravenous penicillin G for the same duration may also be effective. In patients with high-degree atrioventricular block or a PR interval of greater than 0.3 s, intravenous therapy for at least part of the course and cardiac monitoring are recommended. Prior to the use of antibiotics for the treatment of Lyme disease, the degree of heart block was found to decrease rapidly with prednisone (40 to 60 mg/d). Although rarely used today, glucocorticoids may be of benefit in patients with complete heart block or congestive heart failure if antimicrobial therapy alone does not result in improvement within 24 h.

It is unclear how and whether asymptomatic infection should be treated, but patients with such infection are often given a course of oral antibiotics. The appropriate treatment for Lyme disease during pregnancy is also unclear. Because the risk of maternal-fetal transmission seems to be very low, standard therapy for the documented stage and manifestation of the illness may be sufficient. Relapse may follow the use of any of the antibiotic regimens for Lyme disease, and a second course of therapy may be necessary. On the other hand, in patients who develop chronic fatigue syndrome or fibromyalgia after Lyme disease, further antibiotic therapy does not seem to be of benefit.

The risk of infection with *B. burgdorferi* after a recognized tick bite is so low that antibiotic prophylaxis is not routinely indicated. However, if the tick is engorged, if follow-up is difficult, or if the patient is quite anxious, therapy with amoxicillin or doxycycline for 10 days is likely to prevent Lyme disease.

PROGNOSIS

The response to treatment is best early in the disease. Later treatment of Lyme borreliosis is still effective, but convalescence may be longer. Eventually, most patients recover with minimal or no residual deficits.

ALGORITHM FOR TESTING AND THERAPY

According to guidelines recently published by the American College of Physicians, empirical antibiotic therapy without serologic testing is recommended for patients with a high pretest probability of Lyme disease (such as those with [EM](#)); two-step testing (by [ELISA](#) and, if positive, by western blot) is recommended for patients with an intermediate pretest probability (such as those with recurrent oligoarticular arthritis); and neither testing nor treatment is recommended for patients with a low pretest probability (such as those with nonspecific symptoms of myalgias, arthralgias, or fatigue).

REINFECTION

Reinfection may occur after [EM](#) when patients are treated with antimicrobial agents. In such cases, the immune response is not adequate to provide protection from

subsequent infection. However, patients who develop an expanded immune response to the spirochete over a period of months (such as those with Lyme arthritis) have protective immunity for a period of years and do not acquire the infection again.

VACCINATION

A vaccine is now available for the prevention of Lyme disease in the United States. It consists of a recombinant outer-surface lipoprotein A (L-OspA) with adjuvant. High-titered antibody to OspA is necessary for protection. In a phase 3 efficacy trial in which subjects were given three doses of vaccine or placebo on a 0-, 1-, and 12-month schedule, vaccine efficacy in preventing definite cases of Lyme disease was 49% during the first year (after two doses) and 76% in year 2 (after three doses). Equivalent antibody titers may be obtained if the three doses are given on a 0-, 1-, and 2-month schedule. The third dose should be given in April so that the vaccine recipient will have peak antibody titers during the summer tick-transmission season. Although long-term data are not yet available, yearly booster injections may be necessary. Vaccine injection may cause a mild to moderate local or systemic reaction usually lasting for only a few days. Vaccination should be considered for individuals who live in areas that are highly endemic for the infection and who have frequent exposure to tick habitats.

(Bibliography omitted in Palm version)

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SECTION 10 -*RICKETTSIA, MYCOPLASMA, AND CHLAMYDIA*

177. RICKETTSIAL DISEASES - David Walker, Didier Raoult, J. Stephen Dumler, Thomas Marrie

The rickettsiae make up a family of gram-negative coccobacilli and short bacilli that grow strictly in eukaryotic cells. Characteristics of these organisms include their obligately intracellular location and persistence. The pathogenic rickettsiae move through mammalian reservoirs; they are transmitted by insect or tick vectors. Except for louse-borne typhus, humans are incidental hosts. Among rickettsiae, *Coxiella burnetii* (the agent of Q fever) is notorious for its ability to survive for an extended period outside of the reservoir or vector and for its extreme infectiousness: inhalation of a single microorganism can cause pneumonia. Clinical infections with rickettsiae can be classified into five general groups: (1) tick- and gamasid mite-borne, spotted fever group (SFG) rickettsial diseases; (2) flea- and louse-borne typhus group rickettsial diseases; (3) chigger-borne scrub typhus; (4) ehrlichioses; and (5) Q fever. The rickettsiae that cause spotted fevers, typhus, and scrub typhus are listed along with their vectors, geographic ranges, and associated diseases in [Table 177-1](#).

TICK- AND MITE-BORNE SPOTTED FEVERS

ROCKY MOUNTAIN SPOTTED FEVER

Rocky Mountain spotted fever (RMSF), the most severe of the rickettsial diseases, is caused by *Rickettsia rickettsii*. This organism possesses two major immunodominant surface-exposed proteins, rOmpA and rOmpB, which have species-specific conformational epitopes. rOmpA functions as an adhesin for the host cell; rOmpB, the most abundant outer-membrane protein, shares genetic sequences and limited antigens with typhus group rickettsiae. This small (0.3 µm by 1.0 µm) bacillus has a gram-negative cell wall structure; its lipopolysaccharide shares antigens mainly within the [SFG](#) and is not endotoxic in the quantities found in human infections.

Discovered in the American West in the late nineteenth century, [RMSF](#) is at present documented in 48 states, Canada, Mexico, Costa Rica, Panama, Colombia, and Brazil. It is transmitted by *Dermacentor variabilis*, the American dog tick, in the eastern two-thirds of the United States and California; by *D. andersoni*, the Rocky Mountain wood tick, in the western United States; by *Rhipicephalus sanguineus* in Mexico; and by *Amblyomma cajennense* in Central and South America. Maintained principally by transovarian transmission from one generation of ticks to the next, *R. rickettsii* can be acquired by uninfected ticks through the ingestion of a blood meal from rickettsemic small mammals.

Humans become infected during the active season of the vector tick species. In northern areas, cases occur mainly in the spring; in warmer southern states, most cases occur from May to September, although some cases are reported in the winter. Although 4% of *D. variabilis* ticks contain rickettsiae, the vast majority of these are nonpathogenic species such as *R. montana* and *R. bellii*. The likelihood of an individual tick's containing *R. rickettsii* is remote. From 1988 to 1997, the reported incidence of [RMSF](#) has been in the range of 0.16 to 0.26 cases per 100,000 population in the

United States. This rate is probably an underestimate, since the diagnosis is difficult and reporting incomplete. The 5- to 9-year-old age group has the highest incidence. The mortality rate was 20 to 25% in the preantibiotic era and now remains at about 5% because of delayed diagnosis and treatment. The case-fatality ratio is higher for males than females and increases with each decade of life above age 20.

Pathogenesis *R. rickettsii* organisms are inoculated into the dermis along with secretions of the tick's salivary glands after 36 h of feeding. Rickettsiae spread lymphohematogenously throughout the body, attach via rOmpA to the endothelial cell membrane, and induce their own engulfment. Once intracellularly located, they escape rapidly from the phagosome, replicate in the cytosol by binary fission, and spread from cell to cell, propelled by polar polymerization of the host cell's actin. The result is numerous foci of contiguous infected endothelial cells that are extensive enough to manifest clinically after a dose-dependent incubation period of approximately 1 week (range, 2 to 14 days). *R. rickettsii* is more invasive than other rickettsiae, routinely spreading to infect vascular smooth-muscle cells. Despite frequent statements to the contrary, occlusive thrombosis and ischemic necrosis are not the fundamental pathologic basis for tissue and organ injury in [RMSF](#). Instead, increased vascular permeability, with resulting edema, hypovolemia, and ischemia, is responsible. Indeed, immunohistologic studies of severely infected humans and animals have demonstrated numerous zones of infected endothelium, only a small proportion of which contain thrombi. The thrombi are usually located to one side of the lumen, which is not occluded. These hemostatic plugs appear to be an appropriate host response rather than a pathogenic process. Consumption of platelets results in thrombocytopenia in 32 to 52% of patients, but disseminated intravascular coagulation with hypofibrinogenemia is rare. Activation of platelets, generation of thrombin, and activation of the fibrinolytic system all appear to be homeostatic physiologic responses to endothelial injury.

Clinical Manifestations Early in the illness, when medical attention usually is first sought, [RMSF](#) is difficult to distinguish from many self-limiting viral illnesses. Fever, headache, malaise, myalgia, nausea, vomiting, and anorexia are the most frequent symptoms during the first 3 days. The patient becomes progressively more ill as vascular infection and injury advance. In one large series, only one-third of patients were diagnosed with presumptive RMSF early in the clinical course and treated appropriately as outpatients. In the tertiary care setting, RMSF is all too often recognized only when its late severe manifestations, developing at the end of the first week or in the second week of illness in patients without appropriate treatment, prompt admission to the intensive care unit.

The progressive nature of the infection is clearly manifested in the skin. Rash is evident in only 14% of patients on the first day of illness and in only 49% during the first 3 days. Macules (1 to 5 mm) appear first on the wrists and ankles ([Figs. 177-CD1,177-CD2](#)) and then on the remainder of the extremities and the trunk. Later, more severe vascular damage results in frank hemorrhage at the center of the maculopapule ([Fig. 177-CD3](#)), a petechia that does not disappear upon compression ([Plate IID-45](#)). This sequence of events is sometimes delayed or aborted by effective treatment. In fact, rash appears on day 6 or later in 20% of cases and does not appear at all in 9 to 16% of cases, including some with severe visceral lesions that result in death. Petechiae occur in 41 to 59% of cases, appearing on or after day 6 in 74% of cases that include a rash. Involvement of

the palms and soles, often considered diagnostically important, usually occurs relatively late in the course (after day 5 in 43% of cases) and does not occur at all in 18 to 64% of cases.

The microcirculation, both systemic and pulmonary, is the target of intracellular rickettsial infection, and the clinical manifestations reflect the ensuing vascular changes. Widespread increased vascular permeability results in edema, decreased plasma volume, hypoalbuminemia, reduced serum oncotic pressure, and prerenal azotemia. Hypotension occurs in 17% of cases. Extensive infection of the pulmonary microcirculation is associated with noncardiogenic pulmonary edema. Cardiac involvement is most frequently manifested as dysrhythmia, which is detected in 7 to 16% of cases. Pulmonary involvement, often a major factor in fatal cases, is observed in 17% of cases, of which 12% are considered to represent severe respiratory disease and 8% require mechanical ventilation.

Central nervous system (CNS) involvement is the other important determinant of the outcome of [RMSF](#). Encephalitis, presenting as confusion or lethargy, is apparent in 26 to 28% of cases. Progressively severe encephalitis manifests as stupor or delirium in 21 to 26% of cases, as ataxia in 18%, as coma in 9 to 10%, and as seizures in 8%. Cranial nerve palsy, hearing loss, severe vertigo, nystagmus, dysarthria, aphasia, unilateral corticospinal signs, ankle clonus, extensor toe signs, hyperreflexia, spasticity, fasciculations, athetosis, neurogenic bladder, hemiplegia, paraplegia, and complete paralysis have been reported. Meningoencephalitis results in cerebrospinal fluid (CSF) pleocytosis in 34 to 38% of cases; usually there are 10 to 100 cells per microliter with a mononuclear predominance, but occasionally there are more than 100 cells per microliter and a polymorphonuclear predominance. The CSF protein concentration is increased in 30 to 35% of cases, but the CSF glucose concentration is usually normal.

Renal failure, which occurs in more severely ill patients, is often reversible with rehydration. However, in the most severe cases, shock results in acute tubular necrosis-induced renal failure, which often requires hemodialysis.

Hepatic injury is manifested in 38% of cases as mildly or moderately increased serum aminotransferase concentrations and is due to focal death of individual hepatocytes, but hepatic failure does not occur. Jaundice is recognized in 8 to 9% of cases and an elevated serum bilirubin concentration in 18 to 30%. Marked hyperbilirubinemia occasionally occurs, probably as a consequence of both hemolysis and hepatocytic injury.

Bleeding is a potentially life-threatening effect of severe vascular damage. Anemia develops in 30% of cases and is severe enough to require red blood cell transfusions in 11%. Blood is detected in the stools or vomitus of 10% of patients, and death has followed massive upper gastrointestinal hemorrhage.

Other characteristic clinical laboratory findings include a normal white blood cell count with increased numbers of immature myeloid cells, increased plasma levels of proteins of the acute-phase response (C-reactive protein, fibrinogen, ferritin, and others), and hyponatremia (in 56% of cases) due to the appropriate secretion of antidiuretic hormone in response to the hypovolemic state. Skeletal muscle injury, clinically manifested as

myositis, has been documented in several individual cases by the detection of marked elevations in serum creatine kinase or of histopathologic evidence of vascular injury in skeletal muscle and multifocal rhabdomyonecrosis. Ocular involvement includes conjunctivitis in 30% of cases and retinal vein engorgement, flame hemorrhages, arterial occlusion, and papilledema with normal [CSF](#) pressure in some instances.

In untreated cases, death usually occurs 8 to 15 days after the onset of illness. A rare presentation, fulminant [RMSE](#), is fatal within 5 days after onset. This fulminant presentation has been associated with RMSF in black males who have glucose-6-phosphate dehydrogenase (G6PD) deficiency and is thought to be related to an undefined effect of hemolysis on the rickettsial infection. Although survivors of RMSF usually appear to return to their previous state of health, permanent sequelae, including neurologic deficits and amputation of gangrenous extremities, may follow severe illness.

Diagnosis The diagnosis of [RMSE](#) during the acute stage is more difficult than is generally appreciated. Clinical and epidemiologic considerations are more important than laboratory features early in the illness. The most important epidemiologic factor is a history of exposure within the 12 days preceding disease onset to a potentially tick-infested environment during a season of possible tick activity. However, only 60% of patients actually recall being bitten by a tick during the incubation period.

The differential diagnosis for early clinical manifestations of [RMSE](#) (fever, headache, and myalgia without a rash) includes influenza, enteroviral infection, infectious mononucleosis, viral hepatitis, leptospirosis, typhoid fever, gram-negative or -positive bacterial sepsis, human monocytic or granulocytic ehrlichiosis, murine typhus, sylvatic flying-squirrel typhus, and rickettsialpox. Enterocolitis may be suggested by nausea, vomiting, and abdominal pain; prominence of abdominal tenderness has resulted in exploratory laparotomy. [CNS](#) involvement may masquerade as bacterial and viral meningoencephalitis, with seizures, coma, neurologic signs, and [CSF](#) abnormalities. Cough, pulmonary signs, and chest roentgenographic opacities may lead to a diagnostic consideration of bronchitis or pneumonia.

During the first 3 days of illness, only 3% of patients exhibit the classic triad of fever, rash, and history of tick exposure. When a rash appears, a diagnosis of [RMSE](#) should certainly be considered. However, many illnesses considered in the differential diagnosis may also be associated with a rash, including rubeola, rubella, meningococemia, disseminated gonococcal infection, secondary syphilis, toxic shock syndrome, drug hypersensitivity, idiopathic thrombocytopenic purpura, thrombotic thrombocytopenic purpura, Kawasaki syndrome, and immune complex vasculitis. The converse is also true: any person in an endemic area with a provisional diagnosis of one of the above illnesses may have RMSF.

The most common serologic test for confirmation of the diagnosis is the indirect immunofluorescence assay. Between 7 and 10 days after onset, a diagnostic titer of $1:64$ is usually detectable. Latex agglutination and a solid-state enzyme immunoassay are also available commercially. Latex agglutination usually yields a diagnostic titer of $1:128$ at 7 to 9 days after onset. The sensitivity and specificity of the indirect immunofluorescence assay are 94 to 100% and 100%, respectively, and the latex agglutination test has a sensitivity of 71 to 94% and a specificity of 96 to 99%. The

performance of the solid-state immunoassay has not been reported. It is important to understand that serologic tests for [RMSF](#) are usually negative at the time of presentation for medical care and that treatment should not be delayed while a positive serologic result is awaited.

The only diagnostic test that is useful during the acute illness is immunohistologic examination (immunofluorescence or immunoenzyme staining) of a cutaneous biopsy of a rash lesion for *R. rickettsii*. Examination of a 3-mm punch biopsy of such a lesion is 70% sensitive and 100% specific. Polymerase chain reaction (PCR) amplification and detection of *R. rickettsii* DNA in peripheral blood is an insensitive approach except in the preterminal state; rickettsiae are present in large quantities in heavily infected foci of endothelial cells but in relatively low quantities in the circulation. Cultivation of rickettsiae in cell culture is technically feasible but is seldom undertaken because of biohazard and technologic concerns.

TREATMENT

The drug of choice for the treatment of both children and adults with [RMSF](#) is doxycycline, except when the patient is pregnant or allergic to the drug. Doxycycline is administered orally (or, in the presence of coma or vomiting, intravenously) at 200 mg/d in two divided doses. For children with RMSF reinfection, up to five courses of doxycycline may be administered with minimal risk of dental staining. Other regimens include oral tetracycline (25 to 50 mg/kg per day) in four divided doses. β -Lactam antibiotics, erythromycin, and aminoglycosides have no role in the treatment of RMSF, and sulfa-containing drugs are likely to exacerbate this infection. There is not enough clinical experience to comment on the use of fluoroquinolones in this setting. The most seriously ill patients are managed in intensive care units, with careful administration of fluids to achieve optimal tissue perfusion without precipitating noncardiogenic pulmonary edema. In some severely ill patients, hypoxemia requires intubation and mechanical ventilation; oliguric or anuric acute renal failure requires hemodialysis; seizures necessitate the use of antiseizure medication; anemia or severe hemorrhage necessitates transfusions of packed red blood cells; and bleeding with severe thrombocytopenia requires platelet transfusions. Heparin is not a useful component of treatment, and there is no evidence that glucocorticoids, although frequently administered, affect outcome.

Prevention Avoidance of tick bites is the only available preventive approach. Protective clothing and tick repellents, which could reduce the risk, are seldom actually used. After possible tick exposure, it is wise to inspect the body once or twice a day and remove ticks before they can inoculate rickettsiae.

MEDITERRANEAN SPOTTED FEVER (BOUTONNEUSE FEVER) AND OTHER SPOTTED FEVERS

The etiologic agent of Mediterranean spotted fever, *R. conorii*, is prevalent in southern Europe (below the 45th parallel), all of Africa, and southwestern and south-central Asia. The tick vector and reservoir is *R. sanguineus*, the dog brown tick. The name of this disease varies with the region in which it occurs; examples include Kenya tick typhus, Indian tick typhus, Israeli spotted fever, and Astrakhan spotted fever. Whatever the

designation, the disease is characterized by a high fever, rash, and -- in most geographic locales -- an inoculation eschar (*tache noire*) at the site of the tick bite. A severe form of the disease, associated with a 50% mortality rate, has been observed in patients with diabetes, alcoholism, or heart failure.

African tick-bite fever, which is caused by *R. africae* and has been recognized since the beginning of the twentieth century, was first documented in the modern era in Zimbabwe in 1992. The disease occurs in rural areas and follows bites by ticks of cattle and wild animals. *R. africae* is prevalent in *Amblyomma hebraeum* and *A. variegatum* ticks, which readily feed on humans. Cases have been confirmed not only in Zimbabwe but also in Tanzania and South Africa, and the disease is prevalent in the Caribbean islands of Guadeloupe. The incubation period is 7 days. The illness is mild and consists of headache, fever, eschar at the tick bite site, and regional lymphadenopathy. *Amblyomma* ticks often feed in groups, and several ticks may be found on one patient, with the subsequent development of multiple eschars. Rash is frequently lacking or transient and may be vesicular. African tick-bite fever is the most prevalent rickettsiosis worldwide, and, as tourism to sub-Saharan Africa increases, it is expected that more cases will be seen in non-African countries.

Rickettsia japonica causes *Japanese spotted fever* or *Oriental spotted fever*. Patients present with fever, cutaneous eruption, and an inoculation eschar. In Australia, two spotted fevers have been described. *Queensland tick typhus* is due to *R. australis* and is transmitted by *Ixodes holocyclus*. The skin rash in this disease is usually maculopapular but is sometimes vesicular, and there is an inoculation eschar. *Flinders Island spotted fever*, observed on an island close to Tasmania, is due to *R. honei*. In France, a case of *atypical Lyme disease* caused by *R. slovaca* and two cases of infection with *R. mongolotimonae* have been reported.

Diagnosis The diagnosis of these tick-borne spotted fevers is based on clinical and epidemiologic findings and is confirmed by cell-culture isolation of rickettsiae, by [PCR](#) of skin biopsies (a method not available in most laboratories), or by serology. The identification of specific species requires cross-adsorption ([Table 177-2](#)). In an endemic area, patients presenting with fever, rash, and/or a skin lesion consisting of a black necrotic area or a crust surrounded by erythema should be considered to have one of these rickettsial spotted fevers.

TREATMENT

See [Table 177-2](#).

RICKETTSIALPOX

Rickettsialpox was first described in 1946 by a general practitioner in New York City and soon afterwards was shown to be caused by a distinct species, *R. akari*. This organism was isolated from mice and their mites (*Liponyssoides sanguineus*), which maintain the organisms by transovarian transmission. *R. akari* shares lipopolysaccharide antigens with other members of the [SFG](#).

Epidemiology More than 100 cases of rickettsialpox were diagnosed annually in the

northeastern United States in the late 1940s and the 1950s, and outbreaks occurred in the Ukraine in the 1950s. However, few cases are diagnosed currently. Recently, a culture-confirmed case of rickettsialpox was documented in southern Europe. This case was initially misdiagnosed as Mediterranean spotted fever on the basis of the development of serum antibodies cross-reactive with *R. conorii*. Cases have also been reported in Arizona, Utah, and Ohio.

Clinical Manifestations A papule forms at the site of the mite bite. This lesion develops a central vesicle that becomes a 1- to 2.5-cm painless black crusted eschar surrounded by an erythematous halo ([Fig. 177-CD4](#)). Enlargement of the lymph nodes draining the region of the eschar suggests initial lymphogenous spread. After a 10-day incubation period, during which the eschar and regional lymphadenopathy frequently go unnoticed, the onset of illness is marked by malaise, chills, fever, headache, and myalgia. A macular rash appears 2 to 6 days after onset and evolves sequentially into papules, vesicles, and crusts that heal without scarring. In some cases the rash remains macular or maculopapular. Some patients suffer nausea, vomiting, abdominal pain, cough, conjunctivitis, or photophobia. Untreated rickettsialpox is not fatal, with fever lasting 6 to 10 days.

Diagnosis and Treatment See [Table 177-2](#).

FLEA- AND LOUSE-BORNE RICKETTSIAL DISEASES

ENDEMIC MURINE TYPHUS (FLEA-BORNE)

Murine typhus was postulated to be a distinct disease, with rats as the reservoir and fleas as the vector, by Maxcy in 1926. Dyer isolated the etiologic agent, *R. typhi*, from rats and fleas in 1931. By the end of World War II, murine typhus was known to be a global disease. A novel typhus group *Rickettsia* has now been shown to be maintained vertically in cat fleas and to cause human infection. This flea-transmitted species, *R. felis*, reportedly contains antigens most closely resembling typhus group rickettsiae but genetically is an [SFG](#) organism. *R. felis* has been found in 4% of cat fleas and in 33% of opossums collected in the vicinity of human murine typhus-like cases in southern Texas.

Epidemiology *R. typhi* is maintained in mammalian host/flea cycles, with rats (*Rattus rattus* and *R. norvegicus*) and the Oriental rat flea (*Xenopsylla cheopis*) as the classic zoonotic niche. Fleas acquire *R. typhi* from rickettsemic rats and carry the organism throughout the rest of their lifespan. Nonimmune rats and humans are infected when rickettsia-laden flea feces are "scratched" into pruritic bite lesions; less frequently, the flea bite itself transmits the organisms. Yet another possible route of transmission is the inhalation of aerosolized flea feces. Infected rats appear healthy, although they are rickettsemic for approximately 2 weeks.

Currently, fewer than 100 cases of endemic typhus are reported annually in the United States. These cases occur mainly in southern Texas and southern California, where the classic rat-flea cycle is absent and an opossum-cat flea (*Ctenocephalides felis*) cycle is prominent. Although *X. cheopis* fleas are inefficient at the transovarian maintenance of *R. felis*, cat fleas are highly effective at vertical transmission of this organism, whose natural occurrence has been detected in fleas in California, Texas, and Oklahoma.

Infected opossums and cat fleas as well as a case of human infection were reported from Corpus Christi, Texas, in the same environment where humans, opossums, and cat fleas are infected with *R. typhi*. Cases of endemic typhus occur year-round, mainly in warm (often coastal) areas. This infection has also been reported from Greece, Spain, and Indonesia. The peak prevalence in southern Texas is from April through June and elsewhere is during the warm months of summer and early fall. Patients seldom recall a flea bite or exposure to fleas, although exposure to animals such as cats, opossums, raccoons, skunks, and rats is reported by nearly 40% of those who are questioned.

Clinical Manifestations The incubation period of experimental murine typhus in volunteers averages 11 days, with a range of 8 to 16 days. Close observation during this period reveals prodromal symptoms of headache, myalgia, arthralgia, nausea, and malaise developing 1 to 3 days before the abrupt onset of chills and fever. Nearly all patients experience nausea and vomiting early in the illness.

The duration of untreated illness averages 12 days, with a range of 9 to 18 days. Rash is present in only 13% of patients at the time of presentation for medical care (usually about 4 days after onset of symptoms), appearing an average of 2 days later in half of the remaining patients and never appearing in the other half. The initial macular rash is often detected by careful inspection of the axilla or the inner surface of the arm. Subsequently, the rash becomes maculopapular, involving the trunk more often than the extremities; it is seldom petechial and rarely involves the face, palms, or soles. A rash is detected in only 20% of patients with dark brown or black skin.

Pulmonary involvement is frequently prominent in murine typhus; 35% of patients have a hacking, nonproductive cough, and 23% of patients who undergo chest radiography have pulmonary densities due to interstitial pneumonia, pulmonary edema, and pleural effusions. Bibasilar rales are the most common pulmonary sign. Less common clinical symptoms and signs include abdominal pain, confusion, stupor, seizures, ataxia, coma, and jaundice. Clinical laboratory studies frequently reveal anemia and leukopenia early in the course, leukocytosis late in the course, thrombocytopenia, hyponatremia, hypoalbuminemia, mildly increased serum levels of hepatic aminotransferases, and prerenal azotemia. Complications may include respiratory failure requiring intubation and mechanical ventilation, hematemesis, cerebral hemorrhage, and hemolysis (in patients with [G6PD](#) deficiency and some hemoglobinopathies). The illness is severe enough to necessitate the admission of 10% of hospitalized patients to an intensive care unit. Greater severity is generally associated with old age, underlying disease, and treatment with a sulfa drug; the case-fatality rate is 1%. In a study of children with murine typhus, 50% suffered only nocturnal fevers, feeling well enough for active daytime play.

Diagnosis and Treatment See [Table 177-2](#).

EPIDEMIC TYPHUS (LOUSE-BORNE)

Epidemic typhus due to infection with *R. prowazekii* is transmitted by the human body louse (*Pediculus humanus corporis*), which lives on clothes and is found in poor hygienic conditions (especially in jails, where the disease it causes is called *jail fever*)

and usually in cold areas. Lice acquire *R. prowazekii* when they ingest a blood meal from a rickettsemic patient. The rickettsiae multiply in the midgut epithelial cells of the louse and spill over into the louse feces. The infected louse defecates during its blood meal, and the patient autoinoculates the organisms by scratching. The fact that the louse abandons dead hosts and patients with high fever ($>40^{\circ}\text{C}$) improves its efficiency as a vector. Since the louse does not pass *R. prowazekii* to its offspring, the disease is usually spread from person to person by the louse-borne route. Lice die within 1 to 2 weeks after infection, turning red just prior to death -- hence the name *red louse disease*. This epidemic form of typhus is related to poverty, cold weather, war, and disasters and is currently prevalent in mountainous areas of Africa, South America, and Asia. A large outbreak involving 100,000 people in refugee camps in Burundi occurred in 1997, a small focus was reported in Russia for the first time in 1998, and sporadic cases have been reported from Algeria and from Peru. The global reemergence of the disease is due to proliferation of body lice. In the United States, sporadic cases of epidemic typhus are transmitted by flying-squirrel fleas. Eastern flying-squirrel (*Glaucomys volans*) lice and fleas have been found to be infected with *R. prowazekii*. The flying-squirrel fleas occasionally bite humans.

Brill-Zinsser disease is a recrudescent, mild form of epidemic typhus occurring years after the acute disease, probably as a result of immunosuppression or old age. Nathan Brill first identified recrudescent typhus in New York in 1898. In 1933 Hans Zinsser noted that more than 90% of patients with recrudescent typhus had emigrated from typhus-endemic areas of Europe. Strains of *R. prowazekii* indistinguishable from classic strains were isolated from patients with recrudescent typhus. Furthermore, *R. prowazekii* was isolated from the lymph nodes of patients undergoing elective surgery who had had typhus years earlier. Thus the typhus rickettsiae can remain dormant for years and can reactivate with waning immunity.

Clinical Manifestations After an incubation period of 1 week, the onset of illness is abrupt, with prostration, severe headache, and rapidly rising fever of 38.8 to 40.0°C (102 to 104°F). Cough is frequently prominent, occurring in 70% of patients. Myalgias are usually severe. In the outbreak in Burundi, the disease was referred to as *sutama* ("crouching"), the myalgias being so severe that patients crouched in an attempt to alleviate the pain. A rash begins on the upper trunk, usually on the fifth day, and then becomes generalized, involving all of the body except the face, palms, and soles. Initially, this rash is macular; without treatment, it becomes maculopapular, petechial, and confluent ([Fig. 177-CD5](#)). The rash is frequently absent or not detected on black skin in Africa, where 60% of patients have *spotless epidemic typhus*. Photophobia, with considerable conjunctival injection and eye pain, is frequent. The tongue may be dry, brown, and furred. Confusion and coma are common. Skin necrosis and gangrene of the digits as well as interstitial pneumonia have been noted in severe cases ([Fig. 177-CD6](#)). Untreated disease is fatal in 7 to 40% of cases, with outcome depending primarily on the condition of the host. Patients with untreated infections develop renal insufficiency and multiorgan involvement in which neurologic manifestations are frequently prominent. Overall, 12% of patients with epidemic typhus have neurologic involvement. North American *R. prowazekii* infection transmitted by flying-squirrel ectoparasites is a milder illness; whether this milder disease is due to host factors (e.g., better health status) or organism factors (e.g., attenuated virulence) is unknown.

Prevention Prevention of epidemic typhus involves control of body lice. Clothes should be changed regularly, and insecticides should be used every 6 weeks to control the louse population.

Diagnosis and Treatment See [Table 177-2](#). Epidemic typhus is sometimes misdiagnosed as typhoid fever in tropical countries.

SCRUB TYPHUS

The etiologic agent of scrub typhus is a small, obligately intracellular bacterium of the family Rickettsiaceae that differs substantially from other family members in its genetic makeup and in the composition of its cell wall (which, for example, lacks lipopolysaccharide and peptidoglycan). Consequently, this organism has been classified as a species in a separate genus, *Orientia tsutsugamushi*.

O. tsutsugamushi is maintained in nature by transovarian transmission in trombiculid mites, mainly of the genus *Leptotrombidium*. After hatching, infected larval mites (chiggers, the only stage that feeds on an animal host) inoculate organisms into the skin while feeding. Scrub typhus is found in environments that harbor the infected chiggers, particularly areas of heavy scrub vegetation -- e.g., where the forest is regrowing after being cleared and along riverbanks. Infections occur during the wet season, when the mites lay their eggs. The disease is endemic in eastern and southern Asia, northern Australia, and islands of the western Pacific Ocean. Scrub typhus is also found in tropical areas of India, Sri Lanka, Bangladesh, Myanmar, Thailand, Malaysia, Laos, Vietnam, Kampuchea, China, Taiwan, the Philippines, Indonesia, Papua New Guinea, northern Australia, and islands of the South Pacific Ocean; in temperate areas of Japan, Korea, far-eastern Russia, Tadzhikistan, the mountains of northern India, Pakistan, and Nepal; and in nontropical areas of China, such as Tibet and Shangdong Province. Those infected include indigenous rural workers, residents of suburban areas, and westerners visiting endemic areas for professional or recreational purposes. Infections are more prevalent than the number of clinical diagnoses would suggest; in some areas more than 3% of the population is infected or reinfected each month. Immunity wanes over 1 to 3 years, and there is remarkable antigenic diversity.

Clinical Manifestations The illness varies in severity from mild and self-limiting to fatal. After an incubation period of 6 to 21 days (usually 8 to 10 days), the onset of disease is characterized by fever, headache, myalgia, cough, and gastrointestinal symptoms. Some patients develop no further signs or symptoms and recover spontaneously after a few days. The classic case description includes an eschar at the site of chigger feeding, regional lymphadenopathy, and a maculopapular rash -- signs that are seldom observed in indigenous patients. Fewer than 50% of westerners develop an eschar, and fewer than 40% develop a rash (on day 4 to 6 of illness). Severe cases typically include prominent encephalitis and interstitial pneumonia as key features of vascular injury. Severe illness in persons with [G6PD](#) deficiency has been accompanied by hemolysis. The case-fatality rate for untreated classic cases is 7% but would probably be lower if all relatively mild cases (which are underdiagnosed) were included.

Diagnosis and Treatment See [Table 177-2](#). One report has described cases of scrub typhus in Thailand that do not respond to treatment with doxycycline or

chloramphenicol.

EHRlichioSES

Ehrlichiae are small, obligately intracellular bacteria with a gram-negative-type cell wall that grow in cytoplasmic vacuoles to form clusters called *morulae* ([Fig. 177-1](#)). Two distinct *Ehrlichia* species cause human infections that can be severe and frequent ([Table 177-3](#)). *E. chaffeensis*, the agent of human monocytotropic ehrlichiosis (HME), infects predominantly mononuclear phagocytic cells in tissues and blood monocytes. A member of the *E. phagocytophila* group that infects cells of myeloid lineage is the agent of human granulocytotropic ehrlichiosis (HGE). Both *E. chaffeensis* and *E. phagocytophila* are tick-borne but are transmitted by distinct vectors with little geographic overlap. *E. ewingii* is a newly recognized agent of human ehrlichiosis. Identified in four patients to date by means of a broad-range [PCR](#) assay, *E. ewingii* has previously been reported as a cause of granulocytotropic ehrlichiosis in dogs.

Ehrlichiae were discovered by veterinarians during the investigation of hemolytic anemia of cattle before 1910. Researchers thereafter discerned that "marginal points" within erythrocytes were infectious and named the agent *Anaplasma marginale*. Subsequently, several other species now known as ehrlichiae were detected as veterinary infectious agents, including *Cowdria ruminantium*, *E. canis*, *E. phagocytophila*, and *E. risticii*. In 1953, *E. sennetsu* was identified in humans with mononucleosis-like syndromes in Japan.

The current taxonomic positions are determined by nucleic acid sequences of conserved and unique genes among these species. By analysis of 16S ribosomal RNA sequences, the genus and related organisms can be divided into two major clades: the *E. canis* group (including *E. chaffeensis*) and the *E. phagocytophila* group (including *E. equi* and the agent of [HGE](#)). The *E. sennetsu* group is as distantly related to both of these clades as it is to the genus *Rickettsia* and is not tick-borne. Given the lack of transovarian transmission in ticks, the natural maintenance of the tick-borne ehrlichiae clearly depends in part upon transient or persistent infections in wild and feral mammalian reservoirs. Thus, these bacteria are propagated by horizontal transmission that relies upon a tick-mammal-tick cycle; humans are inadvertently infected when they impinge upon the natural habitats occupied by the ticks and the reservoir hosts.

HUMAN MONOCYTOTROPIC EHRlichiosis

Epidemiology Infections caused by *E. chaffeensis* have been documented in more than 500 cases reported to the Centers for Disease Control and Prevention (CDC). However, since [HME](#) is not a reportable disease in most states, this figure is a gross underestimate. Most infections have been identified in the south-central, southeastern, and mid-Atlantic states, but cases have also been recognized in California, the Pacific northwest, New England, Europe, and Africa. The vector is the Lone Star tick (*Amblyomma americanum*), which in all its life stages feeds upon white-tailed deer, a major reservoir host. Dogs have been discovered to be subclinically infected and may also be an important reservoir. Tick bites and exposures are reported by patients, frequently in rural areas and especially in the months May through July. The median age of HME patients is 44 years, and 75% of the affected individuals are male; however,

severe and fatal infections in children are also well recognized.

Clinical Manifestations *E. chaffeensis* is inoculated into the dermal blood pool created by the feeding tick and subsequently disseminates via the blood to tissues. After a median incubation period of 8 days, illness develops; only about one-third of individuals who seroconvert develop a consistent clinical illness. The classic clinical manifestations are not specific and include fever (97% of cases), headache (81%), myalgia (68%), and malaise (84%); less frequently observed are gastrointestinal involvement (nausea, vomiting, diarrhea; 25 to 68%), cough (25%), rash (36% overall, 6% at presentation) (Fig. 177-CD7), and confusion (20%). HME may be severe: 62% of patients with documented cases are hospitalized, and about 2% die. Severe complications include a toxic shock-like or septic shock-like syndrome, respiratory insufficiency and adult respiratory distress, meningoencephalitis, fulminant infection (in immunocompromised patients), severe opportunistic and nosocomial infections, and hemorrhage. Laboratory findings may be of value in the differential diagnosis; 60 to 74% of patients with HME have leukopenia (initially lymphopenia, later neutropenia), 72% have thrombocytopenia, and nearly 90% have elevations in serum levels of hepatic aminotransferases. With effective therapy, rebound lymphocytosis is common. In spite of abnormal blood counts, examinations reveal hypercellular bone marrow, and noncaseating granulomas may be present. Vasculitis is not a component of HME.

Diagnosis Because HME can be rapidly fatal, empirical antibiotic therapy should be instituted on the basis of a clinical diagnosis. This diagnosis may be suggested by fever in the setting of known tick exposure during the preceding 3 weeks, leukopenia and/or thrombocytopenia, and increased aminotransferase concentrations in serum. Morulae are rarely demonstrated in peripheral blood smears unless an intensive examination is performed; even then, an experienced microscopist is required. The active phase of HME may be diagnosed by PCR amplification of *E. chaffeensis* nucleic acids from EDTA-anticoagulated blood obtained before the start of doxycycline therapy. Retrospective serologic diagnosis requires a consistent clinical picture and detection of a fourfold increase in *E. chaffeensis* antibody titer (to³1:64) by indirect immunofluorescence in paired serum samples obtained approximately 30 days apart. It must be underscored that separate specific diagnostic tests for HME and HGE are necessary.

TREATMENT

Tetracycline is effective therapy for HME. Either tetracycline (250 to 500 mg given orally every 6 h) or doxycycline (100 mg given orally or intravenously twice daily) is associated with a lowered rate of hospitalization and a shortened duration of fever. The use of chloramphenicol is controversial, and *E. chaffeensis* is not susceptible to this drug in vitro. While a few reports document the persistence of *E. chaffeensis* in humans after the acute phase of illness, such persistence is very infrequent; most patients are cured after relatively short courses of tetracycline therapy (continuing for 3 to 5 days after defervescence).

Prevention HME is prevented by the avoidance of ticks in endemic areas. The use of protective clothing and tick repellents, careful tick searches after exposures, and prompt removal of attached ticks markedly diminish risk.

HUMAN GRANULOCYTOTROPIC EHRLICHIOSIS

Epidemiology As of 1995, approximately 150 cases of [HGE](#) had been documented in 11 states (mostly in the upper midwest and the northeast), with a distribution similar to that of Lyme disease. Most cases have been identified within the range of various *I. ricinus*-complex ticks, particularly *I. scapularis*. White-footed deer mice in the United States and red deer in Europe appear to play a role in maintaining HGE in nature. The incidence of HGE peaks in May, June, and July, but the disease may occur throughout the year in conjunction with human exposure to *Ixodes* ticks. HGE affects predominantly males (79%) and older persons (median age, 58 years).

Clinical Manifestations Because of high seroprevalence rates in endemic regions, it seems likely that only a minority of infected individuals develop clinical manifestations. The incubation period for [HGE](#) varies between 4 and 8 days, and the disease manifests as fever (94 to 100% of cases), myalgia (78 to 98%), headache (61 to 85%), and malaise (98%) -- findings suggestive of an influenza-like illness. A minority of patients develop gastrointestinal involvement, including nausea, vomiting, or diarrhea (22 to 39%); rash (2 to 11%); cough (27%); and confusion (17%). Severe complications occur most often in the elderly, but even children may be severely affected. Respiratory insufficiency, with adult respiratory distress syndrome, a toxic shock-like syndrome, and life-threatening opportunistic infections, are the most worrisome complications. Meningoencephalitis has not yet been conclusively recognized with HGE. The case-fatality rate is probably <1%, but nearly 7% of ill patients may require intensive care. As in [HME](#), laboratory findings are of great assistance; most patients develop leukopenia and/or thrombocytopenia with increased serum levels of hepatic aminotransferases. The pancytopenia observed in HGE presumably relates to sequestration or destruction of platelets and leukocytes, since the bone marrow is ordinarily normo- or hypercellular. Vasculitis is not a component of HGE. Unlike HME, HGE is not associated with granulomas. While clear evidence exists for co-infections with *Borrelia burgdorferi* and *Babesia microti*, which are transmitted by the same tick vector(s), there is little evidence of comorbidity or of a persistent or chronic phase for HGE.

Diagnosis [HGE](#) should be included in the differential diagnosis for patients who have been exposed to ticks and who develop an influenza-like illness during the season of *Ixodes* tick activity (May through December). The concurrent detection of thrombocytopenia, leukopenia, and/or elevations in serum aminotransferase activities further increases the likelihood of HGE. A substantial proportion of patients with HGE develop serologic reactions considered diagnostic of Lyme disease in the absence of clear clinical findings consistent with that diagnosis. Thus, HGE should be considered in the differential diagnosis of atypical severe presentations of Lyme disease. Although not highly sensitive, a thorough peripheral blood film examination for morulae in neutrophils may identify 20 to 75% of infections. [PCR](#) on EDTA-anticoagulated blood collected before initiation of tetracycline therapy from patients with active disease is a sensitive and specific method for early confirmation. Serodiagnosis is based mostly upon the retrospective demonstration of a fourfold increase in *E. phagocytophila* group antibody titer to a minimum of 1:80 in paired sera obtained approximately 1 month apart. IgM antibodies may be detected in many patients within the first 1.5 months after illness.

Approximately 15 to 40% of infected persons have a detectable antibody titer at presentation, but, in regions where seroprevalence is high, a single acute-phase polyvalent titer may be misleading.

TREATMENT

Doxycycline (100 mg given orally twice daily) is an effective therapeutic agent, while rifampin has been associated with clinical improvement in pregnant patients with [HGE](#). In vitro studies suggest a role for trovafloxacin, but no prospective studies of any therapy for HGE have been conducted. Most treated patients defervesce within 24 to 48 h.

Prevention Prevention of [HGE](#) requires tick avoidance. The Lyme disease vaccine offers no protection against HGE, and no other vaccine is available.

Q FEVER

Q fever results from infection with *C. burnetii*. This small gram-negative microorganism (0.2 µm by 0.7 µm) exists in two antigenic forms: phase I and phase II. When *C. burnetii* is passaged in cell cultures or embryonated eggs, its lipopolysaccharide undergoes truncation that results in an antigenic change called *phase variation*. The phase I form is extremely infectious and exists in humans and other animals. Passage in cell culture or embryonated eggs results in a shift to the phase II form, which is avirulent. The ability of *C. burnetii* to form spores allows the organism to survive in harsh environments. Indeed, it can survive for more than 40 months in skim milk at room temperature and is readily recovered from soil up to 1 month after contamination. Three different plasmids have been described in various isolates of *C. burnetii*. Q fever encompasses two broad clinical syndromes: acute and chronic infection. It is likely that the host's immune response (rather than characteristics of the infecting strain) determines whether or not chronic Q fever develops.

Epidemiology Q fever is a zoonosis. The primary sources of human infection are infected cattle, sheep, and goats. However, infected cats, rabbits, and dogs have also been shown to transmit *C. burnetii* to humans. The extensive wildlife reservoir for *C. burnetii* includes mammals, birds, and ticks. In the infected female mammal, *C. burnetii* localizes to the uterus and the mammary glands. Infection is reactivated during pregnancy, and high concentrations of *C. burnetii* are found in the placenta. At parturition, *C. burnetii* is dispersed as an aerosol, and infection follows inhalation of aerosolized organisms by a susceptible host. Infected female animals shed the organism in milk for weeks to months after parturition. In rare instances, human-to-human transmission has followed delivery of an infant to an infected woman or autopsy on an infected individual. *C. burnetii* has been transmitted via blood transfusion. Those at risk for Q fever are abattoir workers, veterinarians, and other individuals who vocationally or avocationally come into contact with infected animals. Exposure to infected newborn animals or to infected products of conception poses the highest risk. Sexual transmission has been demonstrated experimentally in mice, as has transmission during artificial insemination in cattle. Whether *C. burnetii* is sexually transmitted among humans is not yet known. While the experimental evidence on this point is contradictory, the ingestion of contaminated milk in some areas is probably a

major route of transmission to humans.

Infections due to *C. burnetii* occur in most countries. Indeed, the only areas known to be free of *C. burnetii* are New Zealand and Antarctica. The primary manifestation of acute Q fever differs from place to place: It is pneumonia in Nova Scotia (Canada) and granulomatous hepatitis in Marseille (France), while both of these manifestations are seen in the Basque country of Spain. These differences may reflect the route of infection; i.e., the ingestion of contaminated milk may result in hepatitis and the inhalation of contaminated aerosols in pneumonia.

Clinical Manifestations

Acute Q Fever The incubation period for acute Q fever ranges from 3 to 30 days. The clinical presentations include flulike syndromes, prolonged fever, pneumonia, hepatitis, pericarditis, myocarditis, meningoencephalitis, and infection during pregnancy. The symptoms of acute Q fever are nonspecific; common among them are fever, extreme fatigue, and severe headache. Other symptoms include chills, sweats, nausea, vomiting, and diarrhea, which occur in 5 to 20% of patients. Cough develops in about half of patients with Q fever pneumonia. Neurologic manifestations of acute Q fever are uncommon; however, in one outbreak in the West Midlands, United Kingdom, 23% of 102 patients had neurologic signs and symptoms as the major manifestation. A nonspecific rash may be evident in 4 to 18% of patients. The white blood cell count is usually normal. Thrombocytopenia is detected in about 25% of patients, and reactive thrombocytosis [with platelet counts of up to 1 million/uL ($1 \times 10^{12}/L$)] frequently develops during recovery. This thrombocytosis may account for cases of deep vein thrombophlebitis complicating acute Q fever in some series. Uncommon manifestations of acute Q fever include optic neuritis, extrapyramidal neurologic disease, Guillain-Barre syndrome, inappropriate secretion of antidiuretic hormone, epididymitis, orchitis, priapism, hemolytic anemia, mediastinal lymphadenopathy mimicking lymphoma, pancreatitis, erythema nodosum, and mesenteric panniculitis. Chest radiography may show an opacity that is indistinguishable from those seen in pneumonia of other etiologies ([Fig. 177-CD8](#)). Multiple rounded opacities are common; in the appropriate epidemiologic setting, they are highly suggestive of Q fever pneumonia. However, right-sided endocarditis resulting in septic pulmonary emboli can produce the same radiographic appearance.

Chronic Q Fever Chronic Q fever, which is uncommon, almost always implies endocarditis. This infection usually occurs in patients with previous valvular heart disease, immunosuppression, or chronic renal insufficiency. Fever is usually absent or low grade. Patients may have nonspecific symptoms for up to 1 year before diagnosis. Valvular vegetations have been seen in only 12% of patients with transthoracic echocardiograms, but the rate of detection may be higher with the use of transesophageal echocardiography. A high index of suspicion is necessary for a correct diagnosis. The disease should be suspected in all patients with culture-negative endocarditis. In addition, all patients with valvular heart disease and an unexplained purpuric eruption, renal insufficiency, stroke, and/or progressive heart failure should be tested for *C. burnetii* infection. Patients with chronic Q fever have hepatomegaly and/or splenomegaly. These two findings, especially in combination with positive rheumatoid factor, high erythrocyte sedimentation rate, high C-reactive protein level, and/or

increased g-globulin concentrations (up to 60 to 70 g/L), suggest this diagnosis. Other manifestations of chronic Q fever include infection of vascular prostheses, aneurysms, and bone.

Diagnosis *C. burnetii* can be isolated from buffy-coat blood samples or tissue specimens by a shell-vial technique; however, most laboratories are not permitted to attempt the isolation of *C. burnetii* since it is considered highly infectious. [PCR](#) can be used to amplify *C. burnetii* DNA from tissue or biopsy specimens. This technique can also be used on paraffin-embedded tissues. Serology, however, is the most commonly used diagnostic tool. Three techniques are available: complement fixation, indirect immunofluorescence, and enzyme-linked immunosorbent assay. Indirect immunofluorescence is sensitive and specific and is the method of choice. Rheumatoid factor should be adsorbed from the specimen before testing. An IgG titer of $\geq 1:800$ to phase I antigen is suggestive of chronic Q fever. In almost all instances of chronic Q fever, the antibody titer to phase I antigen is much higher than that to phase II antigen. The reverse is true in acute Q fever. In addition, in acute Q fever, it is usually possible to demonstrate a fourfold rise in titer between acute- and convalescent-phase serum samples.

TREATMENT

Treatment of acute Q fever with doxycycline (100 mg twice daily for 14 days) is usually successful. Quinolones are also effective. Treatment of chronic Q fever should include at least two antibiotics active against *C. burnetii*. The combination of rifampin and doxycycline has been used with success. For chronic infection, doxycycline should be given as 100 mg twice daily and rifampin as 300 mg once daily. The optimal duration of antibiotic therapy for chronic Q fever remains undetermined. We recommend a minimum of 3 years of treatment, with discontinuation only if the phase I IgA antibody titer is $\leq 1:50$ and the IgG phase I titer is $\leq 1:200$. Another therapeutic option under investigation is the combination of doxycycline (100 mg twice daily) with hydroxychloroquine (600 mg once daily). With this combination, therapy can be completed in 18 months. It is necessary to monitor hydroxychloroquine levels and to adjust the dosage to maintain a plasma concentration of 0.8 to 1.2 $\mu\text{g/mL}$. In vitro, the addition of 1 mg of hydroxychloroquine/mL renders doxycycline bactericidal for *C. burnetii*.

Prevention A vaccine has been shown to be effective in preventing Q fever in abattoir workers in Australia.

(Bibliography omitted in Palm version)

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178. MYCOPLASMA INFECTIONS - William M. McCormack

Mycoplasmas, the smallest free-living organisms known, are prokaryotes that are bounded only by a plasma membrane. Their lack of a cell wall is associated with cellular pleomorphism and resistance to cell wall-active antimicrobial agents, such as penicillins and cephalosporins. The organisms' small genome limits biosynthesis and explains the difficulties encountered with in vitro cultivation. Mycoplasmas typically colonize mucosal surfaces of the respiratory and urogenital tracts of many animal species. Sixteen species of mycoplasmas have been recovered from humans. Most are commensals. *Mycoplasma pneumoniae* causes upper and lower respiratory tract infections. *M. genitalium* and *Ureaplasma urealyticum* are established causes of urethritis and have been implicated in other genital conditions. *M. hominis* and *U. urealyticum* are part of the complex microbial flora of bacterial vaginosis.

MECHANISMS OF PATHOGENICITY

Adherence of mycoplasmas to the surface of the host cell is necessary for colonization and infection. Some pathogenic mycoplasmas are flask-shaped, with specialized tips that enhance adherence. *M. pneumoniae* adheres via a network of interactive adhesins and accessory proteins and produces hydrogen peroxide, which may cause injury to host cells. *M. hominis* metabolizes arginine, with the production of potentially cytotoxic amounts of ammonia. Ureaplasmas have been placed in a separate genus because of their unique urease activity; the metabolism of urea also produces ammonia. *M. pneumoniae* may evoke IgM autoantibodies that agglutinate human erythrocytes at 4°C. These cold agglutinins can cause anemia and other complications ([Fig. 178-CD1](#)).

MYCOPLASMA PNEUMONIAE

EPIDEMIOLOGY

M. pneumoniae causes upper and lower respiratory tract symptoms in all age groups, with the highest attack rates in 5- to 20-year-olds. The infection is acquired by inhalation of aerosols. The incubation period is 2 to 3 weeks, considerably longer than that of most other respiratory infections. Although epidemics have taken place in closed populations, such as schools and military installations, most cases occur sporadically or in families. In families, cases typically occur serially, with 2- to 3-week intervals between cases. Infections in adults are often the result of contact with children.

Infection with *M. pneumoniae* is worldwide. Cases occur throughout the year, with epidemics every few years. Some studies have noted an increase in the number of cases during the autumn months in temperate climates. Although pneumonia is the classic presentation, nonpneumonic infection is considerably more common. In very young children, most infections result only in upper respiratory symptoms, whereas children >5 and adults may have bronchitis and pneumonia.

CLINICAL PRESENTATION

After a prolonged incubation period, fever and constitutional symptoms develop along with headache and cough, both of which can be prominent and distressing. Symptoms

typically progress less rapidly than those of viral respiratory tract infections. In the minority (perhaps 5 to 10%) of infected individuals who develop tracheobronchitis or pneumonia, cough becomes more prominent. Sputum, if produced at all, is usually white and may be tinged with blood. The temperature seldom rises above 38.9 to 39.4°C (102 to 103°F). Shaking chills, myalgias, and gastrointestinal symptoms (e.g., nausea, vomiting, and diarrhea) are unusual. Chest muscle soreness may result from frequent and prolonged coughing, but true pleuritic pain is uncommon.

Pharyngeal injection is often noted. Cervical lymph node enlargement is unusual. Bullous myringitis is a unique but uncommon manifestation. As in other "atypical" pneumonias, findings on auscultation of the lung may be normal or nearly normal despite striking radiographic abnormalities. Pleural effusions develop in <20% of patients.

M. pneumoniae infection may be particularly severe in patients who have sickle cell disease and other hemoglobin S-related hemoglobinopathies. The functional asplenia seen in sickle cell disease may contribute to severe mycoplasmal disease as it does in pneumococcal infection. Severe respiratory distress and large pleural effusions may occur. Digital necrosis has been seen in patients with sickle cell disease who develop very high titers of cold agglutinins.

EXTRAPULMONARY MANIFESTATIONS

A broad array of extrapulmonary abnormalities have been associated with *M. pneumoniae* infection. Although these events are unusual, they complicate other respiratory diseases even more rarely and often provide the only clue that an otherwise unremarkable respiratory infection may be mycoplasmal.

Erythema multiforme (Stevens-Johnson syndrome; see [Plate IIE-67](#)) typically occurs in young male patients with *M. pneumoniae* infection. Other dermatologic manifestations, such as maculopapular and vesicular exanthems, erythema nodosum, and urticaria, have been reported, but none is as clearly linked to *M. pneumoniae* as is erythema multiforme.

Cardiac abnormalities reported in conjunction with *M. pneumoniae* infection include myocarditis and pericarditis, which may result in abnormalities of conduction. Of the wide variety of neurologic conditions associated with *M. pneumoniae*, most have been documented in case reports, where establishment of a cause-and-effect relationship is problematic. Central nervous system abnormalities that have been associated with *M. pneumoniae* include encephalitis, cerebellar ataxia, Guillain-Barre syndrome, transverse myelitis, and peripheral neuropathies. Arthralgias are not unusual in patients who have mycoplasmal pneumonia; mycoplasmal arthritis is rare except in patients who have hypogammaglobulinemia. Hematologic abnormalities associated with *M. pneumoniae* include hemolytic anemia and coagulopathies.

The pathogenesis of the extrapulmonary manifestations of *M. pneumoniae* infection is controversial. Occasional reports have described the identification of *M. pneumoniae* or its nucleic acids in involved tissues. The fact that most attempts at detection have been negative, however, suggests that these extrapulmonary complications have an

immunologic basis. Mycoplasmas, including *M. pneumoniae*, can nonspecifically stimulate B lymphocytes. *M. pneumoniae*-infected individuals can develop autoantibodies, including those reactive with brain, heart, and muscle.

DIAGNOSIS

Most infections with *M. pneumoniae* are not diagnosed, as they are indistinguishable from upper and lower respiratory tract infections caused by myriad other viral and bacterial pathogens. When the diagnosis is suspected, it is usually because illness is prolonged or extrapulmonary manifestations develop. The white blood cell count is generally somewhat elevated, with few immature cells. Gram's stain of sputum shows leukocytes without a predominance of any bacterial morphologic type. Since *M. pneumoniae* lacks a cell wall, it cannot be detected on Gram's stain. In patients who have pneumonia, the chest radiograph may show reticulonodular or interstitial infiltration, primarily in the lower lobes. As in other "atypical" pneumonias, radiographic abnormalities may be more prominent than would be predicted by auscultation of the chest.

M. pneumoniae can be grown on artificial media, but the process is exacting, requires special media, and takes upwards of 2 weeks. Thus, mycoplasmal cultures do not provide timely information to aid in patient management. The same, unfortunately, is true of serologic diagnosis. Specific antibodies can be detected by enzyme-linked immunoassays, indirect immunofluorescence, or complement fixation but do not develop early enough to guide decisions regarding treatment. As with most serologic tests, examination of paired acute- and convalescent-phase serum specimens is required for good sensitivity and specificity.

Cold agglutinins are nonspecific but develop within the first 7 to 10 days in more than half of patients with *M. pneumoniae* pneumonia and may be detectable when the patient presents to a health care provider. In a patient with a compatible clinical picture, a cold agglutinin titer of $\geq 1:32$ supports the diagnosis of mycoplasmal pneumonia. Cold agglutinin determinations are readily available from diagnostic laboratories. The test can also be performed at the bedside by the addition of 1 mL of the patient's blood to a tube containing anticoagulant (e.g., a tube used to collect blood for determination of prothrombin activity). Before cooling, the nonaggregated red blood cells coat the sides of the inverted tube. The blood is cooled to 4°C when the tube is placed in an ice bath for 3 to 5 min or in a standard refrigerator. In a positive test, clumps of red blood cells can be observed when the tube is inverted. Rewarming of the sample to 37°C in an incubator or by exposure to body heat should reverse the agglutination. A positive "bedside" cold agglutinin test is equivalent to a laboratory titer of $\geq 1:64$.

The lack of sensitive, specific, and timely diagnostic tests has prompted the development of a variety of antigen detection tests that do not involve serology or the cultivation of live organisms. Such tests include antigen capture, indirect enzyme immunoassays, DNA probing, and nucleic acid amplification. Since many viral and bacterial infections result in clinical presentations similar to that caused by *M. pneumoniae*, examination of specimens for single antigens is unlikely to be useful. Rather, tests that examine an individual specimen for multiple antigens are needed. Multiplex nucleic acid amplification tests that examine a single throat swab or sputum

sample for all of the most likely causative microorganisms are feasible with current technology. Prototype multiplex polymerase chain reaction (PCR) assays have already been developed. If such tests become available clinically, more precise etiologic diagnosis of upper and lower respiratory tract infections will be possible.

TREATMENT

Because most mycoplasmal infections are not specifically diagnosed, management is directed at one of two syndromes: upper respiratory tract infection or community-acquired pneumonia. Upper respiratory infections, whether caused by viruses or by *M. pneumoniae*, do not require antimicrobial treatment. Community-acquired pneumonia ([Chap. 255](#)) may be caused by bacteria such as *Streptococcus pneumoniae* and *Haemophilus influenzae* or by "atypical" agents such as *Chlamydia pneumoniae*, *Legionella pneumophila*, and *M. pneumoniae*. Recommended treatment regimens include a third-generation cephalosporin, such as intravenous ceftriaxone (1.0 g/d) or cefotaxime (1.0 g every 8 h), that is active against the conventional bacterial pathogens plus intravenous or oral erythromycin (500 mg four times a day) to cover atypical microorganisms. Newer agents that have antimicrobial activity against both conventional and atypical causes of community-acquired pneumonia may be prescribed as monotherapy. These drugs include oral clarithromycin (500 mg twice a day), intravenous or oral azithromycin (500 mg once daily), and intravenous or oral levofloxacin (500 mg once daily). Treatment of documented *M. pneumoniae* pneumonia is usually continued for 14 to 21 days.

Pneumonia due to *M. pneumoniae* is usually self-limited and is seldom life-threatening. Effective antimicrobial agents do shorten the duration of illness and, by reducing coughing, may conceivably render the patient less infectious. Although symptoms are alleviated by antimicrobial treatment, the organism usually is not eradicated. Cultures positive for *M. pneumoniae* may persist for months despite effective antimicrobial treatment. The beneficial effects, if any, of such treatment on extrapulmonary manifestations of *M. pneumoniae* infection are unknown.

GENITAL MYCOPLASMAS (See also [Chap. 132](#))

EPIDEMIOLOGY

M. hominis and *U. urealyticum* are the most prevalent genital mycoplasmas. Infants may become colonized with one or both of these organisms during passage through a colonized birth canal. Neonatal colonization tends not to persist. Only about 10% of prepubertal girls and even fewer prepubertal boys are colonized with ureaplasmas. After puberty, colonization occurs mainly as a result of sexual activity. Among adults, disadvantaged populations have higher colonization rates. Ureaplasmas can be cultured from the vaginas of ~80% of women cared for in public clinics and about half of women cared for by private obstetricians and gynecologists. Similarly, vaginal *M. hominis* is found in 50% of women attending public clinics and in ~20% of private patients. Men have somewhat lower rates of genital colonization than women. Nonetheless, both *U. urealyticum* and *M. hominis* are frequently detected in genital specimens from healthy, sexually experienced adults. Evaluation of the role of these organisms in human disease must take into account their high prevalence among healthy people.

M. fermentans colonizes both the respiratory and genital tracts in >20% of adults. There is no convincing evidence that *M. fermentans* causes human disease; although it had been implicated as a possible determinant of HIV-1 disease progression, more recent data do not support such a role. *M. genitalium* is a fastidious organism that is difficult to cultivate. [PCR](#) studies have identified the organism more successfully. Little is known about the epidemiology of *M. genitalium*.

ASSOCIATION WITH HUMAN DISEASE

Nongonococcal Urethritis *Chlamydia trachomatis* is the organism most firmly implicated in the etiology of nongonococcal urethritis (NGU). There is no doubt that both *U. urealyticum* and *M. genitalium* also cause some cases of NGU. The ubiquity of ureaplasmas among men who do not have urethritis and the difficulty of identifying *M. genitalium* do not allow precise estimation of the proportion of cases of NGU caused by each of these mycoplasmas. *U. urealyticum* and *M. genitalium* do, however, appear to cause most of the nonchlamydial cases.

Epididymitis and Prostatitis Ureaplasmas may be an occasional cause of epididymitis. *M. hominis* has not been implicated in this disease. Neither organism has been convincingly associated with prostatitis.

Pelvic Inflammatory Disease (PID) (See also [Chap. 133](#)) *M. hominis* and *U. urealyticum* are both prominent components of the complex microbial flora of bacterial vaginosis. Since bacterial vaginosis is associated with PID, it is difficult to determine whether either organism plays an independent role in this condition. Although *M. genitalium* is not associated with bacterial vaginosis, preliminary studies have linked it to PID in women who are not infected with either *Neisseria gonorrhoeae* or *C. trachomatis*.

Disorders of Reproduction Ureaplasmas have been considered as causes of involuntary infertility in both men and women, but there is no convincing evidence for such an association. These organisms have been associated with chorioamnionitis and late abortion. Given the close association of ureaplasmas with bacterial vaginosis, a condition that is strongly associated with chorioamnionitis and late abortion, it is difficult to define an independent role for ureaplasmas in this condition. In infants of very low birthweight, ureaplasmas have been shown to cause pneumonia and chronic lung disease.

Extragenital Infections Sexually acquired reactive arthritis and Reiter's disease may be triggered by ureaplasmas, although *C. trachomatis* is the usual triggering agent. Patients who have hypogammaglobulinemia may develop chronic arthritis due to ureaplasmas and some other mycoplasmal species. *M. hominis* has been identified in patients with postthoracotomy sternal wound infection and in rare instances of prosthetic heart valve and prosthetic joint infection.

DIAGNOSIS

There is seldom any reason to examine specimens from the lower genital tract (vagina, male urethra) for mycoplasmas. The ubiquity of the organisms among healthy

individuals makes a positive result uninterpretable. The organisms should be sought only in specimens from normally sterile areas, such as joint fluid with evidence of inflammation and cultures negative for conventional microorganisms.

M. hominis can replicate in many routine blood culture media without changing the appearance of the media. *M. hominis* forms nonhemolytic pinpoint colonies on blood agar; organisms cannot be visualized in gram-stained smears of these colonies. Neither *U. urealyticum* nor *M. genitalium* will grow in ordinary microbiologic media.

Microbiologic diagnosis of genital mycoplasmal infection requires specially prepared media and is beyond the capability of all but reference and research laboratories. Nucleic acid amplification tests such as [PCR](#) have been developed and may become commercially available.

TREATMENT

Ureaplasmas, *M. genitalium*, and *M. hominis* are usually susceptible to tetracyclines (e.g., doxycycline). Tetracycline-resistant ureaplasmas can be treated with erythromycin, while tetracycline-resistant strains of *M. hominis* respond to treatment with clindamycin. As noted above, a specific microbiologic diagnosis of mycoplasmal infection is seldom made. Appropriate treatment provides antimicrobial coverage for the organisms that cause the particular syndrome. Accordingly, [NGU](#) is treated with doxycycline (100 mg orally twice a day for 7 days) or azithromycin (1.0 g as a single oral dose) to provide activity against *C. trachomatis*, *U. urealyticum*, and *M. genitalium*. Recommended regimens for the treatment of [PID](#) provide antimicrobial activity against gonococci, chlamydiae, and anaerobes as well as genital mycoplasmas.

(Bibliography omitted in Palm version)

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179. CHLAMYDIAL INFECTIONS - Walter E. Stamm

The genus *Chlamydia* contains three species that infect humans: *Chlamydia psittaci*, *C. trachomatis*, and *C. pneumoniae* (formerly the TWAR agent). *C. psittaci* is widely distributed in nature, producing genital, conjunctival, intestinal, or respiratory infections in many mammalian and avian species. Genital infections with *C. psittaci* have been well characterized in several species and cause abortion and infertility. Although mammalian strains of *C. psittaci* are not known to infect humans, avian strains occasionally do so, causing pneumonia and the systemic illness known as *psittacosis*.

C. pneumoniae is a fastidious chlamydial species that appears to be a common cause of upper respiratory tract infection and pneumonia, primarily in children and young adults, and is a cause of recurrent respiratory infections in older adults. Studies have also linked *C. pneumoniae* infection to atherosclerotic cardiovascular disease and perhaps to asthma and sarcoidosis. No animal reservoir has been identified for *C. pneumoniae*; it appears to be an exclusively human pathogen spread via the respiratory route through close personal contact. To date, all strains of *C. pneumoniae* studied have been serologically homologous.

C. trachomatis is also an exclusively human pathogen and was identified as the cause of trachoma in the 1940s. Since then, *C. trachomatis* has been recognized as a major cause of sexually transmitted and perinatal infection.

Chlamydiae are obligate intracellular bacteria that are classified in their own order (Chlamydiales). They possess both DNA and RNA, have a cell wall and ribosomes similar to those of gram-negative bacteria, and are inhibited by antibiotics such as tetracycline.

A unique feature of all chlamydiae is their complex reproductive cycle. Two forms of the microorganism -- the extracellular elementary body and the intracellular reticulate body -- participate in this cycle. The elementary body is adapted for extracellular survival and is the infective form transmitted from one person to another. Elementary bodies attach to susceptible target cells (usually columnar or transitional epithelial cells) and enter the cells inside a phagosome. Within 8 h of cell entry, the elementary bodies reorganize into reticulate bodies, which are adapted to intracellular survival and multiplication. They undergo binary fission, eventually producing numerous replicates contained within the intracellular membrane-bound "inclusion body," which occupies much of the infected host cell. Chlamydial inclusions resist lysosomal fusion until late in the developmental cycle. After 24 h, the reticulate bodies condense and form elementary bodies still contained within the inclusion. The inclusion then ruptures, releasing elementary bodies from the cell to initiate infection of adjacent cells or transmission to another person.

Studies with monoclonal antibodies to and nucleotide sequencing of the major outer-membrane protein have delineated at least 20 serotypes of *C. trachomatis*. According to the classification of Wang and Grayston, strains associated with trachoma have generally been those of the A, B, Ba, and C serovars, while serovars D through K have largely been associated with sexually transmitted and perinatally acquired infections. Serovars L₁, L₂, and L₃ produce lymphogranuloma venereum (LGV) and hemorrhagic proctocolitis. The LGV strains demonstrate unique biologic behavior in that

they are more invasive than the other serovars, produce disease in lymphatic tissue, grow readily in cell culture systems and macrophages, and are fatal when inoculated intracerebrally into mice and monkeys. Non-LGV strains of *C. trachomatis* characteristically produce infections involving the superficial columnar epithelium of the eye, genitalia, and respiratory tract.

C. trachomatis has been reported as an infrequent cause of endocarditis, peritonitis, pleuritis, and possibly periappendicitis and may occasionally cause respiratory infections in older children and adults. Some immunosuppressed patients with pneumonia have had either serologic or cultural evidence of *C. trachomatis* infection, but more data are necessary to define a pathogenic role for *Chlamydia* in these patients.

SEXUALLY TRANSMITTED AND PERINATAL INFECTIONS DUE TO *C. TRACHOMATIS*

SPECTRUM OF *C. TRACHOMATIS* GENITAL INFECTIONS

Genital infections caused by *C. trachomatis* represent the most common bacterial sexually transmitted diseases (STDs) in the United States. An estimated 4 million cases occur each year. In adults, the clinical spectrum of sexually transmitted *C. trachomatis* infections parallels that of gonococcal infection. Both infections have been associated with urethritis, proctitis, and conjunctivitis in both sexes; with epididymitis in men; and with mucopurulent cervicitis (MPC) ([Fig. 179-CD1](#)), acute salpingitis, Bartholin's, and the Fitz-Hugh-Curtis syndrome (perihepatitis) in women. Moreover, both types of infection can be associated with septic arthritis. In general, however, chlamydial infections produce fewer symptoms and signs than corresponding gonococcal infections at the same anatomic site; in fact, chlamydial infections are often totally asymptomatic. Increasing evidence suggests that many chlamydial infections of the genital tract, especially in women, persist for months without producing symptoms. Simultaneous infection with *C. trachomatis* often occurs in women with cervical gonococcal infection and in heterosexual men with gonococcal urethritis.

EPIDEMIOLOGY

Infections due to *C. trachomatis* are now reportable in the United States, and national incidence data show steadily rising numbers of reported infections, undoubtedly reflecting both increased testing and increased reporting. Most testing has focused upon women to date, and thus the reported incidence is severalfold greater in women than in men; this difference likely represents a surveillance artifact.

The age of peak incidence of genital *C. trachomatis* infections, as of other sexually transmitted infections, is the late teens and early twenties. The prevalence of chlamydial urethral infection among young men is at least 3 to 5% for those seen in general medical settings or in urban high schools, >10% for asymptomatic soldiers undergoing routine physical examination, and 15 to 20% for heterosexual men seen in [STD](#) clinics. In areas where chlamydial control programs have been implemented, prevalence may be markedly reduced. In short, prevalence varies widely with the population group studied and with the geographic locale. The ratio of chlamydial to gonococcal urethritis is highest for heterosexual men and for those of high socioeconomic status and is lowest

for homosexual men and indigent populations.

The prevalence of cervical infection among women is approximately 5% for asymptomatic college students and prenatal patients in the United States, >10% for women seen in family planning clinics, and >20% for women seen in STD clinics. As in men, prevalence varies substantially by geographic locale. However, substantial prevalences (~8%) of asymptomatic chlamydial infection were recently demonstrated in young female military recruits from all parts of the United States. In this country, the prevalence of *C. trachomatis* in the cervix of pregnant women is 5 to 10 times higher than that of *Neisseria gonorrhoeae*. The prevalence of genital infection with either agent is highest among individuals who are between the ages of 18 and 24, single, and non-Caucasian (e.g., black or hispanic). Recurrent chlamydial infections occur frequently in these same risk groups, often acquired from untreated sexual partners. Oral contraceptive pill use and the presence of cervical ectopy also confer an increased risk of chlamydial infection. The proportion of infections that are asymptomatic appears to be higher for *C. trachomatis* than for *N. gonorrhoeae*, and symptomatic *C. trachomatis* infections are clinically less severe. Mild or asymptomatic chlamydial infections of the fallopian tubes nonetheless cause ongoing tubal damage and infertility. Furthermore, because the total number of *C. trachomatis* infections exceeds the total number of *N. gonorrhoeae* infections in industrialized countries, the total morbidity caused by *C. trachomatis* genital infections in these countries equals or exceeds that caused by *N. gonorrhoeae*. The prevalence of *C. trachomatis* is higher than that of *N. gonorrhoeae* in industrialized countries, in part because measures such as treatment of sex partners and routine cultures for case detection in asymptomatic individuals have been applied much more effectively to the control of gonorrhea than to the control of *C. trachomatis* infection.

PATHOGENESIS

C. trachomatis preferentially infects the columnar epithelium of the eye and the respiratory and genital tracts. The infection induces an immune response but often persists for months or years in the absence of antimicrobial therapy. Serious sequelae often occur in association with repeated or persistent infections. The precise mechanism through which repeated infection elicits an inflammatory response that leads to tubal scarring and damage in the female upper genital tract is not yet clear. One antigen, the chlamydial 60-kDa heat-shock protein, may be involved in inducing the pathologic immune response or may elicit antibodies that cross-react with human heat-shock proteins. The recent sequencing of the chlamydial genome may soon offer further insights into the pathogenic mechanisms of *C. trachomatis*.

CLINICAL MANIFESTATIONS

Nongonococcal and Postgonococcal Urethritis Nongonococcal urethritis (NGU) is a diagnosis of exclusion that is applied to men with symptoms and/or signs of urethritis who do not have gonorrhea. Postgonococcal urethritis (PGU) refers to nongonococcal urethritis developing in men 2 to 3 weeks after treatment of gonococcal urethritis with single doses of agents such as amoxicillin or cephalosporins that lack sufficient activity against chlamydiae. Since current treatment regimens for gonorrhea also include tetracycline, doxycycline, or azithromycin for possible concomitant chlamydial infection,

both the incidence of PGU and the causative role of chlamydiae in this syndrome have declined. *C. trachomatis* causes 20 to 40% of cases of NGU in heterosexual men but is less commonly isolated from homosexual men with this syndrome. The cause of most of the remaining cases is uncertain; considerable evidence suggests that *Ureaplasma urealyticum* causes many cases of NGU, while *Trichomonas vaginalis* and herpes simplex virus (HSV) cause some cases.

[NGU](#) is diagnosed by documentation of a leukocytic urethral exudate and by exclusion of gonorrhea by Gram's staining or culture. *C. trachomatis* urethritis is generally less severe than gonococcal urethritis, although in an individual patient these two forms of urethritis cannot be reliably differentiated solely on clinical grounds. Symptoms include urethral discharge (often whitish and mucoid rather than frankly purulent), dysuria, and urethral itching. Physical examination may reveal meatal erythema and tenderness and a urethral exudate that is often demonstrable only by stripping of the urethra.

At least one-third of males with *C. trachomatis* urethral infection have no demonstrable signs or symptoms of urethritis. Use of nucleic acid amplification assays on first-void urine specimens to diagnose chlamydial infections in men has facilitated more broadly based testing for asymptomatic infection in males. As a result, asymptomatic chlamydial urethritis has been demonstrated in 5 to 10% of sexually active adolescent males screened in school-based clinics or community centers. Such patients generally have first-glass pyuria (≥ 15 leukocytes per 400 $\times$ microscopic field in the sediment of first-void urine), a positive leukocyte esterase test, or an increased number of leukocytes on Gram-stained smear prepared from a urogenital swab inserted 1 to 2 cm into the anterior urethra. For the enumeration of leukocytes, the smear is first scanned at low power to identify areas of the slide containing the highest concentration of leukocytes. These areas are then examined under oil immersion (1000 $\times$). An average of four or more leukocytes in at least three of five 1000 $\times$ (oil-immersion) fields is indicative of urethritis and correlates with the recovery of *C. trachomatis*. To differentiate between true urethritis and functional symptoms among symptomatic patients or to make a presumptive diagnosis of *C. trachomatis* infection in a "high-risk" but asymptomatic man (e.g., male patients in [STD](#) clinics, sex partners of women with nongonococcal salpingitis or [MPC](#), fathers of children with inclusion conjunctivitis), the examination of an endourethral specimen for increased leukocytes is useful if specific diagnostic tests for chlamydiae are not available. Alternatively, noninvasive screening for urethritis can be accomplished by testing of a first-void urine sample for pyuria, either by microscopy or by the leukocyte esterase test. Urine can also be directly tested for chlamydiae or gonococci by DNA amplification methods, as described below.

Epididymitis *C. trachomatis* is the foremost cause of epididymitis in sexually active heterosexual men under 35 years of age, accounting for about 70% of cases. *N. gonorrhoeae* causes most of the remaining cases, and some men have simultaneous infections with both pathogens, usually accompanied by asymptomatic urethritis as defined above. In homosexual men, sexually transmitted coliform infection acquired via rectal intercourse may cause epididymitis. Coliform bacteria and *Pseudomonas aeruginosa*, usually in association with preceding urologic instrumentation or surgery, are the most common causes of epididymitis in men over 35. Men with epididymitis typically present with unilateral scrotal pain, fever, and epididymal tenderness or swelling on examination. The illness may be mild enough to treat on an outpatient basis

with oral antibiotics or severe enough to require hospitalization and parenteral therapy. Testicular torsion should be excluded promptly by radionuclide scan, Doppler flow study, or surgical exploration in a teenager or young adult who presents with acute unilateral testicular pain without urethritis. The possibility of testicular tumor or chronic infection (e.g., tuberculosis) should be excluded when a patient with unilateral intrascrotal pain and swelling does not respond to appropriate antimicrobial therapy.

Reiter's Syndrome Reiter's syndrome consists of conjunctivitis, urethritis (or cervicitis in females), arthritis, and characteristic mucocutaneous lesions ([Chap. 315](#)). *C. trachomatis* has been recovered from the urethra of up to 70% of men with untreated nondiarrheal Reiter's syndrome and associated urethritis. In the absence of overt urethritis, it is important to exclude subclinical urethritis in the men in whom this diagnosis is suspected.

The pathogenesis of Reiter's syndrome remains obscure. However, since more than 80% of affected patients have the HLA-B27 phenotype and since other mucosal infections (with *Salmonella*, *Shigella*, or *Campylobacter*, for example) produce an identical syndrome, chlamydial infection is thought to initiate an aberrant and hyperactive immune response that produces inflammation at the involved target organs in these genetically predisposed individuals. Evidence of exaggerated cell-mediated and humoral immune responses to chlamydial antigens in Reiter's syndrome supports this hypothesis. The presumptive demonstration of chlamydial elementary bodies and chlamydial DNA in the joint fluid and synovial tissue of patients with Reiter's syndrome suggests that chlamydiae may actually spread from genital to joint tissues in these patients, perhaps in macrophages.

Proctitis *C. trachomatis* strains of either the genital immunotypes D through K or the [LGV](#) immunotypes cause proctitis in homosexual men who practice receptive anorectal intercourse. In the United States, the vast majority of cases are due to immunotypes D through K and present either as asymptomatic infection or as mild proctitis not unlike gonococcal proctitis. These infections may develop in heterosexual women as well. Patients present with mild rectal pain, mucous discharge, tenesmus, and (occasionally) bleeding. Nearly all have neutrophils in their rectal Gram's stain. Anoscopy in these non-LGV cases of chlamydial proctitis reveals mild, patchy mucosal friability and mucopurulent discharge, and the disease process is limited to the distal rectum. LGV strains produce more severe ulcerative proctitis or proctocolitis that can be confused clinically with [HSV](#) proctitis (severe rectal pain, bleeding, discharge, and tenesmus) and that histologically resembles Crohn's disease in that giant cell formation and granulomas can be seen ([Chap. 287](#)). In the United States, these cases occur almost exclusively in homosexual men.

Mucopurulent Cervicitis Although many women with *C. trachomatis* infection of the cervix have no symptoms or signs, a careful speculum examination reveals evidence of [MPC](#) in 30 to 50% of cases. As is discussed more fully in [Chap. 133](#), MPC is associated with yellow mucopurulent discharge from the endocervical columnar epithelium and with ≥ 20 neutrophils per 1000 $\times$ microscopic field within strands of cervical mucus on a thinly smeared, Gram-stained preparation of endocervical exudate. Other characteristic findings include edema of the zone of cervical ectopy and a propensity of the mucosa to bleed on minor trauma -- e.g., when specimens are collected with a

swab. A Pap smear shows increased numbers of neutrophils as well as a characteristic pattern of mononuclear inflammatory cells, including plasma cells, transformed lymphocytes, and histiocytes. Cervical biopsy shows a predominantly mononuclear cell infiltrate of the subepithelial stroma, often with follicular cervicitis.

Pelvic Inflammatory Disease (PID) (See also [Chap. 133](#)) *C. trachomatis* plays an important causative role in salpingitis. Infection with *C. trachomatis* has been demonstrated in laparoscopically verified salpingitis, the organism has been recovered from the fallopian tubes in the absence of other pathogens, and serologic evidence of recent *C. trachomatis* infection has been found in women with PID. In the United States, *C. trachomatis* has been identified in the fallopian tubes or endometrium of up to 50% of women with PID, and its role as an important etiologic agent in this syndrome is well accepted.

[PID](#) occurs via ascending intraluminal spread of *C. trachomatis* from the lower genital tract. [MPC](#) is thus followed by endometritis, endosalpingitis, and finally pelvic peritonitis. Evidence of MPC is usually found in women with laparoscopically verified salpingitis. Similarly, endometritis, demonstrated by endometrial biopsy showing plasma cell infiltration of the endometrial epithelium, is documented in most women with laparoscopically verified chlamydial (or gonococcal) salpingitis. Chlamydial endometritis can also occur in the absence of clinical evidence of salpingitis: approximately 40 to 50% of women with MPC have plasma cell endometritis. Histologic evidence of endometritis has been correlated with an "endometritis syndrome" consisting of vaginal bleeding, lower abdominal pain, and uterine tenderness in the absence of adnexal tenderness. It is not known what proportion of women who have chlamydial endometritis without adnexal tenderness also have salpingitis. However, chlamydial salpingitis produces milder symptoms than does gonococcal salpingitis and may be associated with less marked adnexal tenderness. Mild adnexal or uterine tenderness in sexually active women with cervicitis suggests PID.

Infertility associated with fallopian-tube scarring has been strongly linked to antecedent *C. trachomatis* infection in serologic studies. Since many infertile women with tubal scarring and antichlamydial antibody have no history of [PID](#), it appears that subclinical tubal infection ("silent salpingitis") may produce scarring. Studies in animals and humans with salpingitis and tubal scarring suggest the continuing presence of persistent, slowly replicating chlamydial infection in tubal tissue. Ectopic pregnancy, which occurs in more than 70,000 women in the United States annually, is also thought to be related to *Chlamydia*-induced tubal scarring in many cases. While the pathogenesis of *Chlamydia*-induced tubal scarring remains poorly understood, antibodies to the chlamydial 60-kDa heat-shock protein have been correlated with tubal infertility, ectopic pregnancy, and Fitz-Hugh-Curtis syndrome (see below). Thus this antigen may initiate an immune-mediated process that ultimately damages the fallopian tube. Host genetic susceptibility, as defined by HLA type, may also play an important role.

Perihepatitis, or the Fitz-Hugh-Curtis syndrome, was originally described as a complication of gonococcal [PID](#). However, cultural and/or serologic evidence of *C. trachomatis* infection is found in three-quarters of women with this syndrome. *C. trachomatis* has also been cultured from exudate on the hepatic capsule in

laparoscopically verified cases. This syndrome should be suspected whenever a young, sexually active woman presents with an illness resembling cholecystitis (fever and right-upper-quadrant pain of subacute or acute onset). Symptoms and signs of salpingitis may be minimal. High titers of antibodies to *C. trachomatis* are generally present.

Urethral Syndrome in Women In the absence of infection with uropathogens such as coliforms or *Staphylococcus saprophyticus*, *C. trachomatis* is the pathogen most commonly isolated from college women with dysuria, frequency, and pyuria ([Chap. 280](#)). *Chlamydia* can also be isolated from the urethra of women without symptoms of urethritis, and up to 25% of female [STD](#) clinic patients with chlamydial urogenital infection have cultures positive for *C. trachomatis* from the urethra only.

***C. trachomatis* Infection in Pregnancy** *C. trachomatis* in pregnancy has been associated in some studies (but not in others) with premature delivery and with postpartum endometritis. Whether these complications are in part attributable to *C. trachomatis* is not clear.

PERINATAL INFECTIONS: INCLUSION CONJUNCTIVITIS AND PNEUMONIA

Epidemiology Studies in the United States have demonstrated that 5 to 25% of pregnant women have *C. trachomatis* infections of the cervix. In these studies, approximately one-half to two-thirds of children exposed during birth have acquired *C. trachomatis* infection. Roughly half of the infected infants (or 25% of the group exposed) have developed clinical evidence of inclusion conjunctivitis. In addition to infecting the eye, *C. trachomatis* has been isolated frequently and persistently from the nasopharynx, rectum, and vagina of such infants, occasionally for periods exceeding 1 year in the absence of treatment. Pneumonia develops in about 10% of children infected perinatally, and otitis media may in some cases result from perinatally acquired chlamydial infection.

Inclusion Conjunctivitis of the Newborn (Neonatal Chlamydial Conjunctivitis)

Neonatal chlamydial conjunctivitis has an acute onset 5 to 14 days after birth and often produces a profuse mucopurulent discharge. However, it is impossible to differentiate chlamydial conjunctivitis from other forms of neonatal conjunctivitis (such as that due to *N. gonorrhoeae*, *Haemophilus influenzae*, *Streptococcus pneumoniae*, or [HSV](#)) on clinical grounds; instead, laboratory diagnosis is required. Inclusions within epithelial cells are often detected in Giemsa-stained conjunctival smears, but these smears are considerably less sensitive than cultures, antigen detection tests, or nucleic acid hybridization tests for chlamydiae. Gram-stained smears may show gonococci or occasional small gram-negative coccobacilli in *Haemophilus* conjunctivitis, but smears should be accompanied by cultures for these agents.

Infant Pneumonia *C. trachomatis* causes a distinctive pneumonia syndrome in infants. Recent epidemiologic studies have linked chlamydial pulmonary infection in infants with increased occurrence of subacute lung disease (bronchitis, asthma, wheezing) in later childhood.

LYMPHOGRANULOMA VENEREUM

Definition [LGV](#) is a sexually transmitted infection caused by *C. trachomatis* strains of the L₁, L₂, and L₃ serovars. In the United States, most cases are caused by L₂ organisms. Acute LGV in heterosexual men is characterized by a transient primary genital lesion followed by multilocal suppurative regional lymphadenopathy. Women, homosexual men, and -- in occasional instances -- heterosexual men may develop hemorrhagic proctitis with regional lymphadenitis. Acute LGV is almost always associated with systemic symptoms such as fever and leukocytosis but is rarely associated with systemic complications such as meningoencephalitis. After a latent period of years, late complications include genital elephantiasis due to lymphatic involvement; strictures; and fistulas of the penis, urethra, and rectum.

Epidemiology [LGV](#) is usually sexually transmitted, but occasional transmission by nonsexual personal contact, fomites, or laboratory accidents has been documented. Laboratory work involving the creation of aerosols of LGV organisms (e.g., sonication, homogenization) must be conducted only with appropriate measures for biologic containment.

The peak incidence of [LGV](#) corresponds to the age of greatest sexual activity: the second and third decades of life. The worldwide incidence of LGV is falling, but the disease is still endemic and a major cause of morbidity in Asia, Africa, South America, and parts of the Caribbean. In the Bahamas, an apparent outbreak of LGV has been described in association with a concurrent increase in heterosexual infection with HIV. However, only 186 cases were reported in the United States in 1995, for a rate of 0.1 case per 100,000 population.

The frequency of infection following exposure is believed to be much lower than that for gonorrhea and syphilis. Early manifestations are recognized far more often in men than in women, who usually present with late complications. In the United States, where the reported male-to-female ratio of cases is 3.4:1, most cases have involved homosexually active men and persons returning from abroad (travelers, sailors, and military personnel). The main reservoir of infection, although it has not been directly demonstrated, is presumed to be asymptomatically infected individuals.

Clinical Manifestations In heterosexuals, a *primary genital lesion* develops from 3 days to 3 weeks after exposure. It is a small, painless vesicle or nonindurated ulcer or papule located on the penis in men and on the labia, posterior vagina, or fourchette in women. The primary lesion is noticed by fewer than one-third of men with [LGV](#) and only rarely by women. It heals in a few days without scarring and, even when noticed, is usually recognized as LGV only in retrospect. LGV strains of *C. trachomatis* have occasionally been recovered from genital ulcers and from the urethra of men and the endocervix of women who present with inguinal adenopathy; these areas may be the primary site of infection in some cases.

In women and homosexual men, *primary anal or rectal infection* develops after receptive anorectal intercourse. In women, rectal infection with [LGV](#) (or non-LGV) strains of *C. trachomatis* presumably can also arise by the contiguous spread of infected secretions along the perineum (as in rectal gonococcal infections in women) or perhaps by spread to the rectum via the pelvic lymphatics.

From the site of the primary urethral, genital, anal, or rectal infection, the organism spreads via the regional lymphatics. Penile, vulvar, or anal infection can lead to inguinal and femoral lymphadenitis. Rectal infection produces hypogastric and deep iliac lymphadenitis. Upper vaginal or cervical infection results in enlargement of the obturator and iliac nodes.

The most common presenting picture in heterosexual men is the *inguinal syndrome*, which is characterized by painful inguinal lymphadenopathy beginning 2 to 6 weeks after presumed exposure; in rare instances, the onset comes after a few months. The inguinal adenopathy is unilateral in two-thirds of cases, and palpable enlargement of the iliac and femoral nodes is often evident on the same side as the enlarged inguinal nodes ([Fig. 132-CD2](#)). The nodes are initially discrete, but progressive periadenitis results in a matted mass of nodes that becomes fluctuant and suppurative. The overlying skin becomes fixed, inflamed, and thin and finally develops multiple draining fistulas. Extensive enlargement of chains of inguinal nodes above and below the inguinal ligament ("the sign of the groove") is not specific and, although not uncommon, is documented in only a minority of cases. On histologic examination, infected nodes are initially found to have characteristic small stellate abscesses surrounded by histiocytes. These abscesses coalesce to form large, necrotic, suppurative foci. Spontaneous healing usually takes place after several months; inguinal scars or granulomatous masses of various sizes persist for life. Massive pelvic lymphadenopathy in women or homosexual men may lead to exploratory laparotomy.

As cultures and serologic tests for *C. trachomatis* are being used more often, increasing numbers of cases of [LGV](#) proctitis are being recognized in homosexual men. Such patients present with anorectal pain and mucopurulent, bloody rectal discharge. Although these patients may complain of diarrhea, they are often referring not to diarrhea but rather to frequent, painful, unsuccessful attempts at defecation (tenesmus). Sigmoidoscopy reveals ulcerative proctitis or proctocolitis, with purulent exudate and mucosal bleeding. The histopathologic findings in the rectal mucosa include granulomas with giant cells, crypt abscesses, and extensive inflammation. These clinical, sigmoidoscopic, and histopathologic findings may closely resemble those of Crohn's disease of the rectum.

Constitutional symptoms are common during the stage of regional lymphadenopathy and, in cases of proctitis, may include fever, chills, headache, meningismus, anorexia, myalgias, and arthralgias. These findings in the presence of lymphadenopathy are sometimes mistakenly interpreted as representing malignant lymphoma. Other systemic complications are infrequent but include arthritis with sterile effusion, aseptic meningitis, meningoencephalitis, conjunctivitis, hepatitis, and erythema nodosum. Chlamydiae have been recovered from the cerebrospinal fluid and in one case were isolated from the blood of a patient with severe constitutional symptoms -- a result indicating the dissemination of infection. Laboratory-acquired infections suspected of being due to the inhalation of aerosols have been associated with mediastinal lymphadenitis, pneumonitis, and pleural effusion.

Complications of untreated anorectal infection include perirectal abscess; fistula in ano; and rectovaginal, rectovesical, and ischiorectal fistulas. Secondary bacterial infection

probably contributes to these complications. Rectal stricture is a late complication of anorectal infection and usually develops 2 to 6 cm from the anal orifice -- i.e., at a site within reach on digital rectal examination. Proximal extension of the stricture for several centimeters may lead to a mistaken clinical and radiographic diagnosis of carcinoma.

A small percentage of cases of [LGV](#) in men present as chronic progressive infiltrative, ulcerative, or fistular lesions of the penis, urethra, or scrotum. Associated lymphatic obstruction may produce elephantiasis. When urethral stricture occurs, it usually involves the posterior urethra and causes incontinence or difficulty with urination.

APPROACH TO THE DIAGNOSIS AND TREATMENT OF *C. TRACHOMATIS* GENITAL INFECTIONS

Four types of laboratory procedure are available to confirm *C. trachomatis* infection: direct microscopic examination of tissue scrapings for typical intracytoplasmic inclusions or elementary bodies; isolation of the organism in cell culture; detection of chlamydial antigens or nucleic acid by immunologic or hybridization methods; and detection of antibody in serum or in local secretions.

Except in conjunctivitis, direct microscopic examination of Giemsa-stained cell scrapings for typical inclusions has an unacceptably low degree of sensitivity, and false-positive interpretations by inexperienced observers are common. Even for conjunctivitis, this approach has been replaced by direct fluorescent antibody staining of conjunctival smears to identify chlamydial elementary bodies with specific monoclonal antibodies (see below).

Cell culture techniques for isolation of *C. trachomatis* are available in most large medical centers but not in other clinical settings. In addition to limited availability, other disadvantages of cell culture include its low and variable level of sensitivity (60 to 80%), its requirement for rigorous transport conditions, and its high cost and technically demanding nature. Therefore, nonculture alternatives involving antigen detection or nucleic acid hybridization have been developed. In the direct immunofluorescent antibody (DFA) slide test, potentially infected genital or ocular secretions are smeared onto a slide, fixed, and stained with fluorescein-conjugated monoclonal antibody specific for chlamydial antigens. The observation of fluorescing elementary bodies confirms the diagnosis. Compared with culture, this test is 70 to 85% sensitive, and it is quite specific when used for confirmation of urethral, cervical, or ocular infection in high-risk patients with suspected *C. trachomatis* infection. The sensitivity and specificity of the test depend directly upon the skill of the microscopist. The apparently lower sensitivity of the test in low-risk populations, along with its relatively labor-intensive nature, limits its value as a screening tool.

Enzyme-linked immunosorbent assay (ELISA) techniques for the detection of chlamydial antigens provide another alternative to culture. The reported sensitivity and specificity of these tests for genital infections (as compared with culture) have been 60 to 80% and 97 to 99%, respectively, in high-risk populations. Sensitivities have generally been higher in cervical infection and lower in urethritis among males. Like the [DFA](#) slide test, the ELISA is less sensitive and less specific in low-prevalence populations and largely asymptomatic patients. ELISAs are better suited to screening

than is DFA because large numbers of specimens can easily be processed.

Assays with nucleic acid probes have also been developed for chlamydial diagnosis. One such test uses DNA-RNA hybridization and appears to be approximately equal to the best [ELISAs](#) in terms of sensitivity and specificity. Nucleic acid probes have also been developed for use in amplification assays such as ligase chain reaction and polymerase chain reaction (PCR). These tests are now the most sensitive chlamydial diagnostic methods available, being the first nonculture assays actually to surpass culture itself in sensitivity. In addition, the ability of these tests to detect chlamydial genes in urine with a high degree of sensitivity and specificity allows their use with urine specimens rather than with conventional urethral and cervical swabs for the first time. The use of urine specimens is particularly appealing for public-health chlamydial screening programs because of the ease of sample collection, even in community-based settings.

Serologic tests are of limited usefulness in the diagnosis of chlamydial oculogenital infections. The complement fixation test with heat-stable, genus-specific antigen has been used with some success to diagnose [LGV](#) but is insensitive in infections due to non-LGV strains of *C. trachomatis*. The microimmunofluorescence (micro-IF) test with *C. trachomatis* antigens is more sensitive but is generally available only in research laboratories. The test measures antibodies by serovar specificity and by immunoglobulin class (IgM, IgG, IgA, secretory IgA) in both serum and local secretions. Serologic diagnosis by the [micro-IF](#) test may be useful in infant pneumonia (in which high-titer IgM antibody and/or fourfold rises in titer are often demonstrated), in chlamydial salpingitis (especially Fitz-Hugh-Curtis syndrome), and in LGV. In all of these more invasive syndromes, high antibody levels are present.

[Table 179-1](#) summarizes the diagnostic tests of choice for patients with suspected *C. trachomatis* infection. With few exceptions, the most suitable method for diagnosis is demonstration of the agent by either cell culture or one of the newer nonculture techniques. Selection of the most appropriate of these tests often depends upon local availability and expertise. However, it is clear that, in most settings and for most purposes, sensitivity and specificity will be greatest with nucleic acid amplification techniques. For patients to whom medicolegal considerations may apply (victims of sexual or child abuse), cultures or nucleic acid amplification methods should always be used. Since *C. trachomatis* is an intracellular pathogen, adequate specimens for chlamydial diagnostic testing must include epithelial cells. Cultures or nonculture tests of pus are less often positive. In urethritis, a thin-shafted urogenital swab should be inserted at least 2 cm into the urethra to obtain an appropriate specimen. Although cultures of urine for chlamydiae are less sensitive than urethral cultures, studies suggest that nucleic acid amplification testing of a first-void urine specimen from men is a more sensitive and less painful diagnostic alternative to the more invasive urethral swab-based tests, culture, or antigen detection tests. The first 30 mL of voided urine should be collected for testing. When a cervical sample is collected, the external os should first be cleaned of debris and purulent material; a plastic-shafted swab should then be inserted into the cervix, rotated slowly several times, and withdrawn. For the diagnosis of urogenital (cervical or urethral) infections in women, testing of a first-void urine specimen by nucleic acid amplification methods is at least as sensitive as testing of a cervical swab. When conjunctival specimens are sought, the epithelium should be

swabbed to remove cells rather than just purulent material. All specimens for chlamydial culture should be placed immediately into transport medium and then either refrigerated (if they will reach the laboratory within 12 to 18 h) or frozen at -70°C (if longer storage is anticipated). A major advantage of the nonculture diagnostic techniques is their less rigid transport requirements; neither refrigeration nor rapid transport is needed.

From a public health viewpoint, the most effective use of chlamydial diagnostic testing has not been unequivocally established and varies with the clinical population, local resources, and laboratory expertise. Since chlamydial diagnostic testing has become more widely available and is now more sensitive and specific than in the past, its use for specific diagnosis in patients with suspected chlamydial syndromes (such as [MPC](#), [NGU](#), and [PID](#)) and their partners should be promoted. High priority should be given to the screening of asymptomatic high-risk women who would not otherwise receive treatment for presumptive chlamydial infection, especially those seen in high-risk settings (e.g., [STD](#) clinics or abortion clinics) and those with a high-risk profile (e.g., sexually active and ≤ 21 years of age, new sex partner within the preceding 2 months, or more than one current sex partner). Similar screening programs should be used to detect and treat asymptomatic urethritis in high-risk adolescent males. Where implemented, screening programs of this type have been associated with reductions in the prevalence of chlamydial infection and of its complications, such as PID.

ANTIMICROBIAL SUSCEPTIBILITY

In laboratory tests that evaluate the growth of chlamydiae in cell cultures, the tetracyclines, erythromycin, rifampin, certain fluoroquinolones (especially ofloxacin), and the macrolide azithromycin are all highly active against these organisms. Sulfonamides and clindamycin are also active against *C. trachomatis*, but to a lesser degree. Penicillin and ampicillin suppress chlamydial multiplication but do not eradicate the organism in vitro. The cephalosporins appear to be relatively ineffective against *C. trachomatis*. Streptomycin, gentamicin, neomycin, kanamycin, vancomycin, ristocetin, spectinomycin, and nystatin are not effective at concentrations inhibitory for most bacteria and fungi. There does not appear to be much strain-to-strain variation in susceptibility to antibiotics, and no clinically significant antimicrobial resistance in chlamydiae has been described. Thus antimicrobial susceptibility testing is not needed in the routine management of patients with chlamydial infection, even recurrent infection.

TREATMENT

Until the introduction of azithromycin, chlamydial infections could not be eradicated by single-dose or short-term antimicrobial regimens. In most uncomplicated infections in adults, 7 days of treatment with doxycycline or tetracycline have to be given for genital infections, but a 2-week course of therapy is recommended for complicated chlamydial infections (e.g., [PID](#), epididymitis) and at least a 3-week course for [LGV](#). Failure of treatment of genital infections with a tetracycline usually indicates poor compliance or reinfection rather than the involvement of a drug-resistant strain.

Therapy for *C. trachomatis* urethritis is more efficacious than therapy for nonchlamydial [NGU](#). *C. trachomatis* is eradicated from the urethra in nearly all cases by treatment with tetracycline hydrochloride (500 mg qid for 7 days) or doxycycline (100 mg

by mouth bid for 7 days).

Eradication of *C. trachomatis* from the cervix by tetracycline, doxycycline, and erythromycin, with doses and durations similar to those specified above for urethritis, has been demonstrated. Erythromycin base (500 mg qid for 10 to 14 days) is the regimen of choice for pregnant women with *C. trachomatis* infection. Amoxicillin (500 mg tid for 10 days) has also been used successfully in pregnant women. Tetracycline hydrochloride (500 mg qid) or doxycycline (100 mg bid) for 14 days produces clinical and microbiologic cure of epididymitis and [PID](#) associated with *C. trachomatis* infection, but in this situation a tetracycline should always be used together with a drug that is highly effective against gonorrhea.

Azithromycin is highly active against *C. trachomatis*, exhibits prolonged bioavailability, is concentrated intracellularly, and has offered the prospect of single-dose therapy for chlamydial infection for the first time. In comparative trials, a 1-g single dose of azithromycin has been as effective as 7 days of doxycycline for uncomplicated chlamydial infection. Azithromycin causes fewer adverse gastrointestinal reactions than do older macrolides such as erythromycin. The single-dose regimen of azithromycin has great appeal for the treatment of patients with uncomplicated chlamydial infection (especially those without symptoms and those with a likelihood of poor compliance) and of sexual partners of infected patients. These advantages must be weighed against the considerably greater cost of azithromycin than of doxycycline. Whenever possible, the single 1-g dose should be given as directly observed therapy. Although not approved by the U.S. Food and Drug Administration, the 1-g single-dose regimen of azithromycin appears to be safe and effective in the treatment of pregnant women.

Of the newer fluoroquinolones, ofloxacin (300 mg by mouth bid for 7 days) has been shown to be as effective as doxycycline for the treatment of chlamydial infection and appears to be safe and well tolerated. It cannot be used in pregnancy.

Treatment of Sex Partners The continued high prevalence of chlamydial infections in most parts of the United States is due primarily to the failure to diagnose -- and therefore treat -- patients with symptomatic or asymptomatic infection and their sex partners. *C. trachomatis* urethral or cervical infection has been well documented in a high proportion of the sex partners of patients with [NGU](#), epididymitis, Reiter's syndrome, salpingitis, or endocervicitis. If possible, confirmatory laboratory tests for *Chlamydia* should be undertaken in these individuals, but even those without evidence of clinical disease who have recently been exposed to proven or possible chlamydial infection (e.g., NGU) should be offered therapy.

Treatment of Neonates and Infants In neonates with conjunctivitis or infants with pneumonia, erythromycin ethylsuccinate or estolate can be given orally in a dose of 50 mg/kg per day, preferably in four divided doses, for 2 weeks. Careful attention must be given to compliance with therapy -- a frequent problem. Relapses of eye infection are common following treatment with topical erythromycin or tetracycline ophthalmic ointment and may also occur after oral erythromycin therapy. Thus follow-up cultures should be performed after treatment. Both parents should be examined for *C. trachomatis* infection and, if diagnostic testing is not readily available, should be treated with doxycycline or azithromycin.

PREVENTION

Efforts to develop a vaccine for chlamydial infection have not yet been successful. Early diagnosis and treatment shorten the duration of infectiousness of the carrier and therefore constitute primary prevention of chlamydial infection. By the early 1990s, one of the 10 regions of the United States (Region X, the Pacific Northwest) had formally undertaken a chlamydial control program involving widespread screening of women attending family planning clinics. Approximately 500,000 tests per year were conducted at 150 such clinics throughout the region in women meeting the criteria for high risk. Within 5 years, the prevalence of chlamydial infection had been reduced by >30% in this population. While other regions of the United States have now initiated similar programs, many family planning and [STD](#) clinics still do not offer chlamydial testing. The availability of highly sensitive and specific diagnostic tests that can be done with urine specimens and of single-dose therapy makes it feasible to mount an effective chlamydial control program nationwide, with screening of high-risk persons both in traditional health care settings and in novel community- and school-based settings.

TRACHOMA AND ADULT INCLUSION CONJUNCTIVITIS

DEFINITION

Trachoma is a chronic conjunctivitis associated with infection by *C. trachomatis* serovar A, B, Ba, or C. It has been responsible for an estimated 20 million cases of blindness throughout the world and remains an important cause of preventable blindness. Inclusion conjunctivitis is an acute ocular infection caused by sexually transmitted *C. trachomatis* strains (usually serovars D through K) in adults exposed to infected genital secretions and in their newborn offspring.

EPIDEMIOLOGY

Epidemiologically, two types of eye disease are caused by *C. trachomatis*. In trachoma-endemic areas where the classic eye disease is seen, transmission is from eye to eye via hands, flies, towels, and other fomites and usually involves serovar A, B, Ba, or C. In nonendemic areas, organisms of serovars D through K can be transmitted from the genital tract to the eye, usually causing only the inclusion conjunctivitis syndrome, occasionally with keratitis. Rarely, the eye disease acquired in this way progresses, with the development of pannus and scars similar to those seen in endemic trachoma. These cases may be referred to as paratrachoma to differentiate them epidemiologically from eye-to-eye-transmitted endemic trachoma.

The worldwide incidence and severity of trachoma have decreased dramatically during the past 35 years, mainly as a result of improving hygienic and economic conditions. Endemic trachoma is still the major cause of preventable blindness in northern Africa, sub-Saharan Africa, the Middle East, and parts of Asia. The endemic disease is transmitted primarily through close personal contact, particularly among young children in rural communities with limited water supplies. In endemic areas, trachoma is associated with repeated exposure and reinfection, but the infection can also become chronic and persistent. In the United States a mild form of endemic trachoma still occurs

in Mexican Americans as well as in immigrants from areas where trachoma is endemic. Acute relapse of old trachoma occasionally follows treatment with cortisone eye ointment or develops in very old persons who were exposed in their youth.

CLINICAL MANIFESTATIONS

Both endemic trachoma and adult inclusion conjunctivitis present initially as a conjunctivitis characterized by small lymphoid follicles in the conjunctiva. In regions with hyperendemic classic blinding trachoma, the disease usually starts insidiously before the age of 2 years. Reinfection is common and probably contributes to the pathogenesis of trachoma. Studies using [PCR](#) techniques indicate that chlamydial DNA is often present in the ocular secretions of patients with trachoma, even in the absence of positive cultures. Thus persistent infection may be more common than was previously thought.

The cornea becomes involved, with inflammatory leukocytic infiltrations and superficial vascularization (pannus formation). As the inflammation continues, conjunctival scarring eventually distorts the eyelids, causing them to turn inward so that the inturned lashes constantly abrade the eyeball (trichiasis and entropion); eventually the corneal epithelium is abraded and may ulcerate, with subsequent corneal scarring and blindness. Destruction of the conjunctival goblet cells, lacrimal ducts, and lacrimal gland may produce a "dry-eye" syndrome, with resultant corneal opacity due to drying (xerosis) or secondary bacterial corneal ulcers.

Communities with blinding trachoma often experience seasonal epidemics of conjunctivitis due to *H. influenzae* that contribute to the intensity of the inflammatory process. In such areas the active infectious process usually resolves spontaneously in affected persons between 10 and 15 years of age, but the conjunctival scars continue to shrink, producing trichiasis and entropion and subsequent corneal scarring in adults. In areas with milder and less prevalent disease, the process may be much slower, with active disease continuing into adulthood; blindness is rare in these cases.

Eye infection with genital *C. trachomatis* strains in sexually active young adults presents as the acute onset of unilateral follicular conjunctivitis and preauricular lymphadenopathy similar to that seen in acute adenovirus or herpesvirus conjunctivitis. If untreated, the disease may persist for 6 weeks to 2 years. It is frequently associated with corneal inflammation in the form of discrete opacities ("infiltrates"), punctate epithelial erosions, and minor degrees of superficial corneal vascularization. Very rarely, conjunctival scarring and eyelid distortion occur, particularly in patients treated for many months with topical glucocorticoids. Recurrent eye infections develop most often in patients whose sexual consorts are not treated with antimicrobials.

DIAGNOSIS

The clinical diagnosis of classic trachoma can be made if two of the following signs are present:

1. Lymphoid follicles on the upper tarsal conjunctiva
2. Typical conjunctival scarring

3. Vascular pannus

4. Limbal follicles or their sequelae, Herbert's pits

The clinical diagnosis of endemic trachoma should be confirmed by laboratory tests in children with more marked degrees of inflammation. Intracytoplasmic chlamydial inclusions are found in 10 to 60% of Giemsa-stained conjunctival smears in such populations, but isolation in cell cultures, newer antigen detection testing, or chlamydial [PCR](#) is more sensitive. Follicular conjunctivitis in adult Europeans or Americans living in trachomatous regions is rarely due to trachoma.

Sporadic cases of adult inclusion conjunctivitis must be differentiated from keratoconjunctivitis due to adenovirus or [HSV](#) and from bacterial conjunctivitis during the first 15 days after onset; later, they must be distinguished from other forms of chronic follicular conjunctivitis. Demonstration of chlamydiae by Giemsa- or immunofluorescent-stained smears, by isolation in cell cultures, or by newer nonculture tests constitutes definitive evidence of infection. Genital examination and tests for genital chlamydial infection are indicated. Serum antibody does not constitute evidence of chlamydial eye infection since many sexually active adults have acquired serum antibody from genital infection.

TREATMENT

Public health control programs for endemic trachoma have consisted of the mass application of tetracycline or erythromycin ointment to the eyes of all children in affected communities for 21 to 60 days or on an intermittent schedule. These programs also include surgical correction of intumed eyelids by a mobile surgical team that visits each locale. Single-dose azithromycin therapy is now being evaluated as an alternative method of mass antibiotic treatment for trachoma in young children and pregnant women.

Adult inclusion conjunctivitis responds well to treatment with full doses of systemic tetracycline or erythromycin for 3 weeks. Treatment of all sexual consorts of the patient simultaneously is also necessary to prevent ocular reinfection and to avoid genital disease due to chlamydial infection. Topical antibiotic treatment is not required for patients who receive systemic antibiotics.

PREVENTION

Efforts to develop a trachoma vaccine have not yet been successful. General hygienic measures associated with improved living standards are effective in the elimination of endemic trachoma. An adequate water supply for personal cleanliness may be a key factor. In some areas the reduction of numbers of flies in the household is important.

PSITTACOSIS

DEFINITION

Psittacosis is primarily an infectious disease of birds and mammals that is caused by *C. psittaci*. Transmission of infection from birds to humans results in a febrile illness characterized by pneumonitis and systemic manifestations. Inapparent infections or mild influenza-like illnesses may also occur. The term *ornithosis* is sometimes applied to infections contracted from birds other than parrots or parakeets, but *psittacosis* is the preferred generic term for all forms of the disease.

EPIDEMIOLOGY

Almost any avian species can harbor *C. psittaci*. Psittacine birds (parrots, parakeets, budgerigars) are most commonly infected, but human cases have been traced to contact with pigeons, ducks, turkeys, chickens, and many other birds. Psittacosis may be considered an occupational disease of pet-shop owners, poultry workers, pigeon fanciers, taxidermists, veterinarians, and zoo attendants. During the past 20 years, there has been an increase in incidence, with cases and outbreaks occurring primarily among employees of poultry-processing plants. It is suspected that many cases go undiagnosed and unreported. The disease appears to be especially common in England, where budgerigars are popular household pets and where restrictions on the importation of these birds have been eased.

The agent is present in nasal secretions, excreta, tissues, and feathers of infected birds. Although the disease can be fatal, infected birds frequently show only minor evidence of illness, such as ruffled feathers, lethargy, and anorexia. Asymptomatic avian carriers are common, and complete recovery may be followed by continued shedding of the organism for many months.

Psittacosis is almost always transmitted to humans by the respiratory route. On rare occasions the disease may be acquired from the bite of a pet bird. Prolonged contact is not essential for transmission of the disease; a few minutes spent in an environment previously occupied by an infected bird has resulted in human infection. In one outbreak, gardening rather than direct exposure to birds was associated with infection. A psittacosis-like agent has been transmitted among hospital personnel, with severe and sometimes fatal infections. There is evidence that these "human" strains are more virulent than avian organisms. There is no record of infection acquired by the ingestion of poultry products.

PATHOGENESIS

The psittacosis agent gains entrance to the body through the upper part of the respiratory tract, spreads via the bloodstream, and eventually localizes in the pulmonary alveoli and in the reticuloendothelial cells of the spleen and liver. Invasion of the lung probably takes place by way of the bloodstream rather than by direct extension from the upper air passages. A lymphocytic inflammatory response occurs on both the interstitial and the respiratory surfaces of the alveoli as well as in the perivascular spaces. The alveolar walls and interstitial tissues of the lung are thickened, edematous, necrotic, and occasionally hemorrhagic. Histologic examination of the affected areas reveals alveolar spaces filled with fluid, erythrocytes, and lymphocytes. The picture is not pathognomonic of psittacosis unless macrophages containing characteristic cytoplasmic inclusion bodies (Levinthal-Coles-Lillie bodies) can be identified. The respiratory

epithelium of the bronchi and bronchioles usually remains intact.

CLINICAL MANIFESTATIONS

The clinical manifestations and course of psittacosis are extremely variable. After an incubation period of 7 to 14 days or longer, the disease may start abruptly with shaking chills and fever, with temperatures ranging as high as 40.5°C (105°F); however, the onset is often gradual, with fever increasing over a 3- to 4-day period. Headache is almost always a prominent symptom; it is usually diffuse and excruciating and is often the patient's chief complaint.

Many patients present with a dry hacking cough that is usually nonproductive, but small amounts of mucoid or bloody sputum may be raised as the disease progresses. Cough may begin early in the course of the disease or as late as 5 days after the onset of fever. Chest pain, pleurisy with effusion, or a friction rub may all occur but are rare. Pericarditis and myocarditis have been reported. Most patients have a normal or slightly increased respiratory rate; marked dyspnea with cyanosis occurs only in severe psittacosis with extensive pulmonary involvement. In psittacosis, as in mycoplasmal pneumonias, the physical signs of pneumonitis tend to be less prominent than symptoms and x-ray findings would suggest. The initial examination may reveal fine sibilant rales, or clinical evidence of pneumonia may be completely lacking. Rales usually become audible and more numerous as the illness progresses. Signs of frank pulmonary consolidation are usually absent. Symptoms of upper respiratory tract infection are not prominent, although mild sore throat, pharyngitis, and cervical adenopathy are often documented; on occasion, the last may be the only manifestation of illness. Epistaxis is encountered early in the course of nearly one-fourth of cases. Photophobia is also a common complaint.

Patients often report generalized myalgia, and spasm and stiffness of the muscles of the back and neck may lead to an erroneous diagnosis of meningitis. Lethargy, mental depression, agitation, insomnia, and disorientation have been prominent features of the illness in some epidemics but not in others; delirium and stupor develop near the end of the first week in severe cases. Occasional patients are comatose when first seen, and the diagnosis of psittacosis may be elusive in these cases. Gastrointestinal manifestations such as abdominal pain, nausea, vomiting, or diarrhea are noted in some cases; constipation and abdominal distention sometimes occur as late complications. Icterus, the result of severe hepatic involvement, is a rare and ominous finding. A faint macular rash (Horder's spots) resembling the rose spots of typhoid fever has been described.

Patients without cough or other clinical evidence of respiratory involvement present with fever of unknown origin ([Chap. 125](#)). The pulse rate is slow in relation to the fever. When splenomegaly is noted in a patient with acute pneumonitis, psittacosis should be considered; the reported incidence of splenomegaly in this disease ranges from 10 to 70%. Nontender hepatic enlargement also occurs, but jaundice is rare. Thrombophlebitis is not unusual during convalescence; indeed, pulmonary infarction is sometimes a late complication and may be fatal.

In untreated cases of psittacosis, sustained or mildly remittent fever persists for 10 days

to 3 weeks or occasionally for as long as 3 months. Over this period, the respiratory manifestations gradually abate. Psittacosis contracted from parrots or parakeets is more likely to be a severe, prolonged illness than infection acquired from pigeons or barnyard fowl. Relapses occur but are rare. Occasional patients develop endocarditis, and *C. psittaci* infection should be considered in cases of culture-negative endocarditis. Secondary bacterial infections are uncommon. Immunity to reinfection is probably permanent.

LABORATORY FINDINGS

The chest x-ray in psittacosis is nonspecific and may show pneumonic lesions that are usually patchy in appearance but can be hazy, diffuse, homogeneous, lobar, atelectatic, wedge-shaped, nodular, or miliary. The white blood cell count is normal or moderately decreased in the acute phase of the disease but may rise in convalescence. The erythrocyte sedimentation rate frequently is not elevated. Transient proteinuria is common. The cerebrospinal fluid sometimes contains a few mononuclear cells but is otherwise normal. Despite hepatomegaly, the results of liver function tests are generally normal or mildly elevated.

The diagnosis can be confirmed only by isolation of the causative microorganism or by serologic studies. The agent is present in the blood during the acute phase of the disease and in the bronchial secretions for weeks or sometimes years after infection, but it is difficult to isolate. Further, the organism is hazardous to work with in the laboratory, and most clinical laboratories do not offer culture for *C. psittaci*. Thus psittacosis is most readily diagnosed by the demonstration of a rising titer of complement fixation antibody in the serum of a patient with a compatible clinical syndrome. Both an acute-phase and a convalescent-phase specimen should always be tested. *C. trachomatis*, *C. psittaci*, and *C. pneumoniae* all share a genus-specific "group" antigen, which is the basis of the complement fixation test. Thus acute infections with *C. trachomatis* or *C. pneumoniae* can also produce titer rises in this test. However, these three species have different major outer-membrane proteins that are the principal antigens in the [micro-IF](#) test. If there is doubt as to the interpretation of the complement fixation test, the micro-IF test can be used to differentiate among these antigens. The prompt initiation of treatment with tetracycline has been shown to delay an antibody rise in convalescence for several weeks or months.

DIFFERENTIAL DIAGNOSIS

A history of exposure to birds may be the only clinical basis for differentiating psittacosis from a variety of infectious and noninfectious febrile disorders. The list of pulmonary diseases that may be confused with psittacosis includes *Mycoplasma pneumoniae*, *C. pneumoniae* pneumonia, legionellosis, viral pneumonia, Q fever, coccidioidomycosis, tuberculosis, enterovirus infection, carcinoma of the lung with bronchial obstruction, and common bacterial pneumonias. In the early stages, before pneumonitis appears, psittacosis may be mistaken for influenza, typhoid fever, miliary tuberculosis, or infectious mononucleosis.

TREATMENT

The tetracyclines are consistently effective in the treatment of psittacosis. Defervescence and alleviation of symptoms usually take place within 24 to 48 h after the institution of therapy with 2 g daily in four divided doses. To avoid relapse, treatment should probably be continued for at least 7 to 14 days after defervescence. In severe cases, hospitalization and pulmonary intensive care may be indicated. Sulfonamides are not active against *C. psittaci*. Erythromycin can be used in patients allergic to or intolerant of tetracyclines.

C. PNEUMONIAE INFECTIONS

A third chlamydial species that causes disease in humans, *C. pneumoniae*, has been described in the past two decades. *C. pneumoniae* can be distinguished from the other two species on the basis of DNA hybridization and elementary body morphology. Although *C. pneumoniae* can be grown in a variety of cell cultures, it is considerably more difficult to culture than other chlamydiae, especially from clinical specimens. HL cells appear to be the most effective cell line for isolation of *C. pneumoniae*.

Knowledge of the epidemiology of *C. pneumoniae* infections has been derived primarily from serologic studies. Infections begin to occur in late childhood, achieve peak incidence in young adults, but continue throughout adult life. Seroprevalence in the many adult populations that have been tested throughout the world exceeds 40% -- a figure suggesting that *C. pneumoniae* infections are ubiquitous. Secondary episodes (reinfections) appear to occur commonly in older adults throughout life. *C. pneumoniae* also produces epidemics of pneumonia and respiratory illness, especially in close residential quarters such as military barracks. The incidence of infections outside of epidemics remains poorly defined. Transmission appears to be from person to person, probably primarily in schools and family units.

Little is known about the pathogenesis of *C. pneumoniae* infection. The infection begins in the upper respiratory tract and in many persons is a long-lived asymptomatic condition of the upper respiratory mucosal surfaces. However, in at least some individuals, the organism is transported to distant sites -- perhaps within macrophages -- since evidence exists for replication within arteries and synovial membranes of joints. A *C. pneumoniae* outer-membrane protein may induce host immune responses whose cross-reaction with human proteins results in an autoimmune reaction.

The clinical spectrum of *C. pneumoniae* infection includes acute pharyngitis, sinusitis, bronchitis, and pneumonitis, primarily in young adults. The clinical manifestations of primary infection appear to be more severe and prolonged than those of reinfection. The pneumonitis resembles that of *M. pneumoniae* pneumonia in that leukocytosis is frequently lacking and patients often have prominent antecedent upper respiratory tract symptoms, fever, nonproductive cough, a mild to moderate degree of illness, minimal findings on chest auscultation, and small segmental infiltrates on chest x-ray. In elderly patients, pneumonia due to *C. pneumoniae* can be especially severe and may necessitate hospitalization and respiratory support.

Epidemiologic studies have demonstrated an association between serologic evidence of *C. pneumoniae* infection and atherosclerotic disease of the coronary and other arteries. In addition, *C. pneumoniae* has been identified in atherosclerotic plaques by electron

microscopy, DNA hybridization, and immunocytochemistry. Recently, the organism has been recovered in culture from atheromatous plaque -- a result indicating the presence of viable replicating bacteria in vessels. Evidence from animal models supports the hypothesis that *C. pneumoniae* infection of the upper respiratory tract is followed by recovery of the organism from atheromatous lesions in the aorta and that the infection accelerates the process of atherosclerosis, especially in hypercholesterolemic animals. Antimicrobial treatment of the infected animals reverses the increased risk of atherosclerosis. In humans, two small trials in patients with unstable angina or recent myocardial infarction also suggested that antibiotics reduce subsequent untoward cardiac events. Larger trials have been initiated to determine more definitively whether antibiotics affect the risk of atherosclerosis.

Diagnosis of *C. pneumoniae* infection is currently difficult because cell culture techniques are not available for routine clinical use and nonculture tests using antigen detection methods or DNA probes have not been developed for commercial use. Acute- and convalescent-phase sera can be tested for chlamydial complement fixation antibody to make a retrospective diagnosis. However, this test does not distinguish *C. pneumoniae* infection from infection due to *C. trachomatis* or *C. psittaci*. Although controlled treatment trials have not been conducted, *C. pneumoniae* is inhibited in vitro by erythromycin and tetracycline. Recommended therapy consists of 2 g per day of either agent for 10 to 14 days. Other macrolides, such as azithromycin, and some fluoroquinolones, such as levofloxacin, also appear to be effective.

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SECTION 11 -VIRAL DISEASES

180. MEDICAL VIROLOGY - Fred Wang, Elliott Kieff

DEFINING A VIRUS

Viruses consist of a nucleic acid surrounded by one or more proteins. Some viruses also have an outer-membrane envelope. Viruses differ from other replicating organisms in that they do not have ribosomes or enzymes for high-energy phosphate generation or for protein, carbohydrate, or lipid metabolism. Viruses are obligate intracellular parasites -- that is, they require cells in order to replicate. Typically, viral nucleic acids encode proteins necessary for replicating and packaging the nucleic acids into new viral particles.

Viruses differ from viroids, prions, and virusoids. *Virusoids* are nucleic acids that depend on helper viruses to package the nucleic acids into virus-like particles. *Viroids* are simply molecules of naked, cyclical, mostly double-stranded, small RNAs and appear to be restricted to plants, in which they spread from cell to cell and are replicated by cellular RNA polymerase II. *Prions* ([Chap. 375](#)) are protein molecules that can spread from cell to cell and effect changes in the structure of their normal counterparts (cellular proteins). Prions have been implicated in neurodegenerative conditions such as Creutzfeldt-Jakob disease, kuru, and Gerstmann-Straussler disease. Prions have also been implicated in neurodegeneration associated with human infection with bovine spongiform encephalopathy ("mad cow disease").

VIRAL STRUCTURE

Viruses have from a few to 200 genes. These genes may be embodied in a single-strand or double-strand DNA genome or in a single-strand sense, a single-strand or segmented antisense, or a double-strand segmented RNA genome. Sense-strand RNA genomes can be translated directly into protein. Sense and antisense genomes are also referred to as positive-strand and negative-strand genomes, respectively. The viral nucleic acid is usually associated with one or more virus-encoded nucleoproteins in the core of the viral particle. The viral nucleic acid is almost always enclosed in a protein shell called a *capsid*. Because of the limited genetic complexity of viruses, their capsids are usually composed of multimers of identical capsomers. Capsomers are in turn composed of one or a few proteins. Capsids have icosahedral or helical symmetry. Icosahedral structures approximate spheres but have two-, three-, and fivefold axes of symmetry, while helical structures have only a twofold axis of symmetry. The entire structural unit of nucleic acid, nucleoprotein(s), and capsid is called a *nucleocapsid*. Many human viruses have a simple nucleocapsid structure. For these viruses, the outer surface of the capsid mediates contact with uninfected cells. Other viruses are more complex and have an outer envelope that is derived from membranes of the infected cell. The piece of infected cell membrane that becomes the viral envelope has usually been modified during infection by the insertion of virus-encoded glycoproteins. These glycoproteins usually mediate contact of enveloped viruses with uninfected cells. Enveloped viruses frequently have matrix or tegument proteins that fill the space between the nucleocapsid and the envelope. In general, enveloped viruses are sensitive to solvents and nonionic detergents that can disrupt the envelope, while viruses that

consist only of nucleocapsids are usually more resistant. The schematic diagram of a large and complex herpesvirus shown in [Fig. 180-1](#) illustrates the components of a complicated DNA virus. Prototypical pathogenic human viruses are listed in [Table 180-1](#). The relative sizes and structures of typical pathogenic human viruses are shown in [Fig. 180-2](#).

TAXONOMY OF PATHOGENIC HUMAN VIRUSES

As is apparent from [Table 180-1](#) and [Fig. 180-2](#), the classification of viruses into orders and families is based on nucleic acid composition, nucleocapsid size and symmetry, and envelopment status. Viruses of a single family have similar types of genomes and are morphologically similar in electron micrographs. Further subclassification into genus is dependent on similarities in epidemiology and biologic effects and on the degree of colinear nucleic acid sequence homology. In general, each human virus has a common name related to its pathologic effects or the circumstances of its discovery and a formal species name assigned by the International Committee on Taxonomy of Viruses. The latter designation consists of the name of the host followed by the family or genus of the virus and a number. This dual terminology has created a confusing situation in which viruses are referred to and referenced by either name -- e.g., varicella-zoster virus (VZV) or human herpesvirus (HHV) 3.

VIRAL INFECTION IN VITRO

STAGES OF INFECTION

At the cellular level, viral infection proceeds in stages.

Viral Interactions at the Cell Surface First, virus adsorbs to a receptor on the cell surface. Adsorption is the consequence of a molecular interaction of a viral surface protein with a molecule on the cell's plasma membrane. For example, a poliovirus capsid protein binds to a cell plasma-membrane protein of the immunoglobulin superfamily type; a rhinovirus capsid protein binds to intracellular adhesion molecule 1; an echovirus capsid protein binds to an integrin; the influenza A virus envelope hemagglutinin protein binds to sialic acid; the HIV envelope glycoprotein binds to CD4 and then engages one of several chemokine receptors that function as coreceptors for the virus; the herpes simplex virus (HSV) envelope glycoproteins bind to heparan sulfate on cell surfaces and then engage one of several immunoglobulin superfamily or tumor necrosis factor (TNF) receptors; and an Epstein-Barr virus (EBV) glycoprotein binds to the B lymphocyte complement receptor CD21. Adsorption characteristically proceeds almost as well at 4°C as at 37°C, and adsorbed virus can still be neutralized by antibody. Adsorption frequently initiates changes in virion surface proteins that result in destabilization and preparation for the next stage of entry into the cell.

After adsorption, viruses penetrate through or fuse with the cell membrane, lose their sensitivity to neutralizing antibody, and become uncoated as they enter the cytoplasm. For all viruses, penetration and uncoating result in viral nucleocapsid or nucleoprotein entry into the cytoplasm. Penetration and uncoating as well as subsequent steps in viral replication depend on the cell's energy metabolism and on biochemical changes in the cell's plasma membrane and cytoskeleton. Therefore, penetration proceeds slowly at

temperatures < 37°C. Viruses use various strategies to penetrate and enter cells. Interaction of the viral surface protein with a cellular receptor can induce changes in viral glycoproteins or capsid proteins and in cell-surface proteins, with consequent penetration. The interaction of multiple protein molecules on the viral surface with multiple molecules on the cell-surface receptor may induce receptor aggregation at the site of viral adsorption. Receptor aggregation can trigger signaling events within the cytoplasm and changes in the plasma membrane. The cell frequently misperceives that the receptor has encountered its "normal ligand," and the aggregated receptor is internalized with the attached virus through an endocytic process that involves clathrin-coated pits. Endocytosis is important in the entry of viruses as diverse as picornaviruses, influenza viruses, HIV, adenoviruses, and herpesviruses. In many cases, entry of the virus into the cytoplasm depends on acidification of the viral endosome.

One of the best-studied examples of the effect of low pH on viral penetration is influenza virus. Influenza hemagglutinin mediates adsorption, receptor aggregation, and endocytosis. In low-pH endosomes, changes in the conformation of the hemagglutinin expose amphipathic domains that interact chemically with the cell membrane and initiate fusion of the viral and cellular membranes. Data indicate that the HIV envelope glycoprotein undergoes similar conformational changes after interaction with CD4 and chemokine receptors. For influenza virus, the M2 membrane protein also plays a key role in the uncoating of the viral envelope by providing an ion channel in the envelope. Fusion of viral and cell membranes results in the mixture of viral envelope lipids and proteins with cell membrane lipids and proteins and the penetration of the influenza nucleocapsid into the cytoplasm. Little is known about the details of the fusion processes or the subsequent biochemical interactions. With more complex viruses, such as herpesviruses, different glycoproteins interact with different receptors on different cell types or on different surfaces of polarized epithelial cells. Viral glycoproteins other than the protein that mediates initial adsorption may be critical in mediating envelope fusion with cell membranes.

Viral Gene Expression and Replication After uncoating and release of viral nucleoprotein into the cytoplasm, the viral genome is transported to a site for expression and replication. In order to produce infectious progeny, viruses must (1) replicate their nucleic acid, (2) produce structural proteins, and (3) assemble the nucleic acid and proteins into progeny virions. Different viruses use different strategies and gene repertoires to accomplish these goals. DNA viruses (except for poxviruses) replicate their nucleic acid and assemble into nucleocapsid complexes in the cell nucleus. RNA viruses (except for influenza viruses) transcribe and replicate their nucleic acid and assemble entirely in the cytoplasm. Thus, the replication strategies of DNA and RNA viruses are presented separately below. Positive-strand and negative-strand RNA viruses are discussed separately. Medically important viruses of each group are used for illustrative purposes.

Positive-Strand RNA Viruses Medically important positive-strand RNA viruses include picornaviruses, flaviviruses, togaviruses, and caliciviruses. Genomic RNA from positive-strand RNA viruses is released into the cytoplasm without associated enzymes. Cell ribosomes recognize and associate with an internal ribosome entry sequence in the viral genomic RNA and translate a polyprotein that is a fusion of many or all of the viral

proteins. The viral RNA polymerase and other viral proteins are cleaved from the polyprotein by protease components of the polyprotein. Antigenomic RNA is then transcribed from the genomic RNA template. Positive-strand genomes and mRNAs are next transcribed from the antigenomic RNA by the viral RNA polymerase. Positive-strand genomic RNA is encapsidated in the cytoplasm.

Negative-Strand RNA Viruses Medically important negative-strand RNA viruses include rhabdoviruses, filoviruses, paramyxoviruses, and bunyaviruses. Negative-strand RNA virus genomes are released into the cytoplasm with an associated RNA polymerase and one or more accessory proteins. Except for influenza viruses, negative-strand RNA viruses replicate entirely in the cytoplasm. The viral RNA polymerase transcribes messenger RNAs (mRNAs) as well as full-length antigenomic RNA, which is the template for replication of genomic RNA. The mRNAs encode for RNA polymerase and accessory factors as well as for viral structural proteins. Influenza virus is an unusual negative-strand RNA virus that transcribes its mRNAs and antigenomic RNAs in the cell's nucleus. The influenza genome RNA snatches cellular mRNA cap sequences to enhance translation of viral mRNAs and uses cell splicing machinery to encode additional viral mRNAs. All negative-strand RNA viruses, including influenza viruses, assemble in the cytoplasm.

Double-Strand Segmented RNA Viruses These viruses, which are taxonomically grouped in the reovirus family, have 10 to 12 RNA segments that make up their genome. The medically important viruses in this group are rotaviruses and Colorado tick fever virus. Reovirus virions include an RNA polymerase complex. Reoviruses replicate and assemble in the cytoplasm.

DNA Viruses Medically important DNA viruses include parvoviruses, papovaviruses, human papillomaviruses (HPVs), adenoviruses, herpesviruses, and poxviruses. Other than poxviruses, most DNA viruses must get to the cell's nucleus for DNA transcription by cellular RNA polymerase II. For example, after receptor binding and fusion, herpesvirus nucleocapsids are released into the cytoplasm along with tegument proteins. The complex is then transported along microtubules to nuclear pores, and the DNA is released into the nucleus.

Transcriptional regulation and mRNA processing for nuclear DNA viruses depend on both viral and cellular proteins. For herpesviruses, the viral tegument protein can activate transcription of viral immediate-early genes, a class of genes expressed immediately after infection. Transcription of immediate-early genes requires the virus tegument protein and preexisting cellular transcription factors. One of the key preexisting cellular factors for [HSV-1](#) immediate-early gene transcription is docked in the cytoplasm in neurons; this fact may explain why HSV-1 goes into a latent state in neurons.

Transcription is often regulated into an organized cascade of viral gene expression. Herpesvirus immediate-early genes turn on the promoters for early genes. Other DNA viruses are not as dependent on transactivators encoded from the viral genome for early-gene transcription. Most early genes encode proteins that are necessary for viral DNA synthesis and for the turn-on of late-gene transcription. Late genes encode mostly viral structural proteins or viral proteins necessary for the assembly and egress of the

virus from the infected cell. Late-gene transcription is continuously dependent on DNA replication. Therefore, inhibitors of DNA replication also stop late-gene transcription.

Each DNA virus family uses unique mechanisms for replicating its DNA. Herpesvirus DNAs are linear in the virion but circularize in the infected cell. In lytic virus infection, circular herpesvirus genomes are replicated into linear concatemers through a "rolling-circle" mechanism. Herpesviruses encode a DNA polymerase and at least six other viral proteins necessary for viral DNA replication; these viruses also encode several enzymes that increase the pool of precursor deoxynucleotide triphosphates. Adenovirus genomes are linear in the virion and are replicated into complementary linear copies by a virus-encoded DNA polymerase and an initiator protein complex. The double-strand circular papovavirus genomes are replicated into progeny circular DNA molecules by cellular DNA replication enzymes. Two viral early proteins contribute to viral DNA replication and to the persistence of papovavirus DNA in latently infected cells. Other early papovavirus proteins stimulate cells to remain in cycle, thus facilitating viral DNA replication. Occasionally, [HPVs](#) integrate into the host chromosome; overexpression of viral early proteins and excessive stimulation of cellular growth result. Sometimes the consequence is the development of malignancies such as cervical cancer (see "Persistent Viral Infections and Cancer," below). Parvoviruses are the smallest DNA viruses: their genomes are half the size of the papovavirus genomes and include only two genes. Parvoviruses have negative single-strand DNA genomes. The replication of autonomous parvoviruses, such as B19, depends on cellular DNA replication and requires the virus-encoded Rep protein. Other parvoviruses, such as adeno-associated virus (AAV), are not autonomous and require helper viruses of the adenovirus or herpesvirus family for their replication. AAV has been touted as a potentially safe human gene vector because its Rep protein causes its integration at a single chromosomal site.

Poxviruses are the largest DNA viruses and are unique among these viruses in replicating and assembling in the cytoplasm. Poxviruses encode transcription factors and an RNA polymerase as well as enzymes for RNA capping and polyadenylation and for DNA synthesis. Poxvirus DNA also has a unique structure. The two strands of the double-strand linear DNA are covalently linked at the ends so that the genome is also a covalently closed single-strand circle. In addition, there are inverted repeats at the ends of the DNA. During DNA replication, the genome is cleaved within the terminal inverted repeat, and the inverted repeats self-prime complementary-strand synthesis by the virus-encoded DNA polymerase. Like herpesviruses, poxviruses encode several enzymes that increase deoxynucleotide triphosphate precursor levels and thus facilitate viral DNA synthesis.

Viruses with Both RNA and DNA Genomes Retroviruses, lentiviruses, and hepatitis B virus (HBV) are not purely RNA or DNA viruses.

Retroviruses and lentiviruses are enveloped RNA viruses with two identical sense-strand genomes and associated reverse transcriptase and integrase enzymes. Retroviruses and lentiviruses differ from all other viruses in that they reverse-transcribe themselves into partially duplicated double-strand DNA copies and then routinely integrate into the host genome as part of their replication strategy. Cellular RNA polymerase II and transcription factors regulate transcription from the integrated

provirus genome. Some retroviruses also encode for regulators of transcription and RNA processing, such as Tax and Rex in the human T-lymphotropic virus (HTLV) types I and II and Tat and Rev in HIV-1 and HIV-2. Full-length proviral transcripts are made from a promoter in the viral terminal repeat and serve as both genomic RNAs that will be packaged in the nucleocapsids and mRNAs that encode for the viral Gag protein, polymerase/integrase protein, and envelope glycoprotein. The Gag protein includes a protease that cleaves it into several components, including a viral matrix protein that coats the viral RNA. Viral RNA polymerase/integrase, matrix protein, and cellular tRNA are key components of the viral nucleocapsid. The HIV Gag protease has been an important target for inhibition of HIV replication. Remnants of simple retroviral DNA in the human genome suggest that there may be replication-competent simple human retroviruses, but little other evidence supports the existence of these viruses.

[HBV](#) replication is unique because the virus encodes and packages in the virion a reverse transcriptase and genomic RNA, which is then copied into an incomplete double-strand circular DNA genome before the virion matures in the infected cell. On entry into the cytoplasm, the virion reverse transcriptase/DNA polymerase completes DNA synthesis, and the covalently closed circular genome resides in the nucleus. Viral mRNAs are transcribed from the closed circular viral episome by cellular RNA polymerase II. A capped and polyadenylated, full-genome-length, terminally redundant transcript is packaged into virus core particles in the cytoplasm of infected cells. This RNA associates with the viral reverse transcriptase. The reverse transcriptase converts the full-length, terminally redundant, core-particle, encapsidated RNA genome into partially double-strand DNA. HBV is believed to mature by budding through the cell's plasma membrane, which has been modified by the insertion of viral surface antigen protein.

Viral Assembly and Egress For most viruses, nucleic acid and structural protein synthesis are accompanied by the assembly of protein and nucleic acid complexes. The assembly and egress of mature infectious virus mark the end of the eclipse phase of infection, during which infectious virus cannot be recovered from the infected cell. Nucleic acids from RNA viruses and poxviruses assemble into nucleocapsids in the cytoplasm. For all DNA viruses except poxviruses, viral DNA assembles into nucleocapsids in the nucleus. In general, the capsid proteins of viruses with icosahedral nucleocapsids can self-assemble into densely packed and highly ordered capsid structures. Herpesviruses require an assemblin protein as a scaffold for capsid assembly. Viral nucleic acid then spools into the assembled capsid. For herpesviruses, a full unit of the viral DNA genome is packaged into the capsid, and a capsid-associated nuclease cleaves the viral DNA at both ends. In the case of viruses with helical nucleocapsids, the protein component appears to assemble around the nucleic acid, which contributes to capsid organization.

Viruses must egress from the infected cell and not bind back to its plasma membrane. In many cases, enveloped viruses simply egress and acquire their envelope by budding through the cell's plasma membrane. Excess viral membrane glycoproteins are synthesized to saturate cell receptors and facilitate viral egress. Some viruses encode membrane proteins with enzymatic activity for receptor destruction. Influenza virus, for example, encodes a glycoprotein with neuraminidase activity, which destroys sialic acid on the infected cell's plasma membrane. Herpesvirus nucleocapsids acquire their initial

envelope by assembling in the nucleus and then budding through the nuclear membrane into the endoplasmic reticular space. The enveloped herpesvirus is then released from the cell either by maturation in cytoplasmic vesicles, which fuse with the plasma membrane and release the virus by exocytosis, or by "de-envelopment" into the cytoplasm and "re-envelopment" at the plasma membrane. In most instances, nonenveloped viruses appear to depend on the death and dissolution of the infected cell for their release.

FIDELITY OF VIRAL REPLICATION

Cells grow by doubling their genome and dividing, whereas viruses typically make large quantities of viral nucleic acid and structural proteins, and thousands of progeny may be produced from a single infected cell. Many particles partially assemble and never mature into virions. Many mature-appearing virions are imperfect and have only incomplete or nonfunctional genomes. Despite the inefficiency of assembly, a typical virus-infected cell releases 10 to 1000 infectious progeny. Some of these progeny may contain genomes that differ from those of the virus that infected the cell. Smaller, "defective" virus genomes have been noted with the replication of many RNA and DNA viruses. Virions with defective genomes can be produced in large numbers through packaging of incompletely synthesized nucleic acid. Adenovirus packaging is notoriously inefficient, and a high particle-to-infectious virus ratio may limit the amount of recombinant adenovirus that can be administered for gene therapy. Mutant viral genomes are also produced and can be of medical significance. In general, viral nucleic acid replication is more error-prone than cellular nucleic acid replication. RNA polymerases and reverse transcriptases are intrinsically more error-prone than DNA polymerases. Mutant viruses can be virulent and may preferentially cause disease through evasion of the host immune response or through resistance to antiviral drugs. Persistent hepatitis C virus (HCV) infection appears to be due to genome mutation and persistent immune escape. Changes in viral nucleic acid can also take place through infection by and recombination or reassortment between two related viruses in a single cell. While this occurrence is highly unusual, the changes could be substantial and could significantly alter virulence or epidemiology. Reassortment of an avian or mammalian influenza A hemagglutinin gene into a human influenza background is believed to play a role in the emergence of new epidemic influenza A strains.

VIRAL GENES NOT REQUIRED FOR VIRAL REPLICATION

Viruses frequently have genes encoding proteins that are not directly involved in replication or packaging of the viral nucleic acid, in virion assembly, or in regulation of the transcription of viral genes involved in those processes. Most of these proteins fall into four classes: (1) proteins that directly or indirectly alter cell growth; (2) proteins that inhibit cellular DNA, RNA, or protein synthesis so that viral mRNA can be efficiently transcribed or translated; (3) proteins that promote the cell's survival or inhibit apoptosis so that progeny virus can mature and escape from the infected cell; and (4) proteins that downregulate host inflammatory or immune responses so that virus infection can proceed in an infected person to the maximum extent consistent with virus survival and efficient transmission to a new host. More complex viruses of the poxvirus or herpesvirus family encode many proteins that serve these functions. Some of these viral proteins have motifs similar to those of cell proteins, while others are quite novel.

Virology has increasingly focused on these more sophisticated strategies evolved by viruses to permit the establishment of long-term infection in humans and other animals. These strategies often provide unique insights into the control of cell growth, cell survival, macromolecular synthesis, proteolytic processing, immune or inflammatory suppression, immune resistance, cytokine mimicry, or cytokine blockade.

HOST RANGE

The concept of host range was originally based on the cell types in which a virus replicated in tissue culture. For the most part, the host range is limited by specific cell-surface proteins required for viral adsorption or penetration. Another common basis for host-range limitation is transcription from viral promoters. Most DNA viruses depend not only on cellular RNA polymerase II and the basal components of the cellular transcription complex but also on activated components and transcriptional accessory factors, both of which differ among differentiated tissues, among cells at various phases of the cell cycle, and between resting and cycling cells.

The concept of host range for virus infection in humans includes these factors and others since (1) most viruses infect more than one cell type in vivo and (2) the virus life cycle and extent of viral replication can be affected by the differentiation and activated state of a given cell type. This point is particularly relevant for human papovavirus, herpesvirus, and lentivirus infections, in which vigorous replication during initial infection may be followed by quiescent or latent infection -- a situation that allows the virus to persist.

VIRAL CYTOPATHIC EFFECTS AND INHIBITORS OF APOPTOSIS

The replication of almost all viruses has adverse effects on the infected cell, inhibiting cellular synthesis of DNA, RNA, or proteins. This inhibitory effect probably stems from the viruses' need to prevent or limit nonspecific, innate host resistance factors, including interferon (IFN). Most commonly, viruses specifically inhibit host protein synthesis by attacking a component of the translational initiation complex -- frequently, a component that is not required for efficient translation of viral RNAs. Poliovirus protease 2A, for example, cleaves a cellular component of the complex that ordinarily facilitates translation of cell mRNAs by interacting with their 5' cap structure. Poliovirus RNA is efficiently translated without a 5' cap since it has an internal ribosome entry sequence. Influenza virus inhibits the processing of mRNA by snatching 5' cap structures from nascent cellular RNAs and using them as primers in the synthesis of viral mRNA. [HSV](#) has a virion tegument protein that inhibits cellular mRNA translation.

Apoptosis is the expected consequence of virus-induced inhibition of cellular macromolecular synthesis and viral nucleic acid replication. While the induction of apoptosis may be important for the release of some viruses (particularly nonenveloped viruses), many viruses have acquired genes or parts of genes that enable them to forestall infected-cell apoptosis. This delay may be advantageous in allowing the completion of viral replication. Adenoviruses and herpesviruses encode analogues of the cellular Bcl-2 protein, which blocks mitochondrial enhancement of proapoptotic stimuli. Poxviruses and some herpesviruses encode caspase inhibitors. Many viruses, including [HPVs](#) and adenoviruses, encode proteins that inhibit p53 or its downstream

proapoptotic effects.

VIRAL INFECTION IN VIVO

The capsid and envelope of a virus protect its genome and permit its transmission from cell to cell and to prospective hosts. Most common viral infections are spread by aerosolized particles, by ingestion of contaminated water or food, or by direct contact. In all these situations, infection begins on an epithelial or mucosal surface and spreads along it or from it to deeper tissues. Infection may then spread through the body via the bloodstream, lymphatics, or neural circuits. Parenteral inoculation also serves to transmit some viral infections among humans or from animals (including insects) to humans.

PRIMARY INFECTION

The first (primary) episode of viral infection usually lasts from several days to several weeks. During this period, the concentration of virus at sites of infection rises and then falls, usually to unmeasurable levels. The rate at which the intensity of viral infection rises and falls at a given site depends on the accessibility of that organ or tissue to both the virus and systemic immune effectors, the intrinsic ability of the virus to replicate at that site, and endogenous nonspecific and specific resistance. Typically, infections with enterovirus, mumps virus, measles virus, rubella virus, rotavirus, influenza virus, adenovirus, [HSV](#), and [VZV](#) are cleared from almost all sites within 3 to 4 weeks. Some of these viruses are especially proficient in altering or evading the innate and acquired immune responses; thus primary infection with these viruses can last for several months. Characteristically extending beyond several weeks are primary infections due to [HBV](#), [HCV](#), hepatitis D virus (HDV), [EBV](#), cytomegalovirus (CMV), HIV, [HPV](#), and molluscum contagiosum virus. For some of these viruses (e.g., HPV, HBV, HCV, HDV, and molluscum contagiosum virus), the primary phase of infection is almost indistinguishable from the persistent phase.

Disease manifestations usually arise as a consequence of viral replication at a specific site but do not necessarily correlate with levels of replication at that site. For example, the clinical manifestations of limited infection with poliovirus, enterovirus, rabies virus, measles virus, mumps virus, or [HSV](#) in neural cells are severe relative to the level of viral replication at mucosal surfaces. Similarly, significant morbidity may accompany in utero fetal infection with rubella virus or [CMV](#).

Primary infections are cleared by specific and nonspecific immune responses. Thereafter, an immunocompetent host is usually immune to the disease manifestations of reinfection by the same virus. Immunity may not prevent transient surface colonization on reexposure, or even persistent colonization.

PERSISTENT AND LATENT INFECTIONS

Relatively few viruses cause persistent or latent infections. [HBV](#), [HCV](#), rabies virus, measles virus, HIV, [HTLV](#), [HPV](#), herpesviruses, and some poxviruses are notable exceptions. The mechanisms for persistent infection vary widely. In persistent HCV infection and to a lesser extent in HIV infection, the high mutation rates in viral genome

replication significantly facilitate persistent infection, continuously yielding mutant viruses that have lost antigenic determinants to which the host has developed effective immune responses. HIV is directly immunosuppressive, depleting CD4⁺ T lymphocytes and compromising CD8⁺ cytotoxic T cell immune responsiveness. Moreover, HIV encodes a Nef protein that downmodulates major histocompatibility complex (MHC) class I expression, rendering HIV-infected cells partially resistant to immune CD8⁺ cytotoxicity. The high mutation rate and the magnitude of virus load conspire to promote persistent infection with drug-resistant HIV mutants.

In contrast, herpesviruses and papovaviruses have much lower mutation rates. Their persistence is due to their ability to establish latent infection and to reactivate from latency. In this instance, *latency* is defined as a state of infection with a full viral genome replicated by cellular DNA polymerase in conjunction with the cell genome; there is no expression of viral genes associated with lytic infection and therefore no production of infectious virus. For [HPVs](#), latently infected basal epithelial cells replicate. Some of the progeny cells provide a stable supply of latently infected basal cells, while others go on to squamous differentiation and in the process become permissive for lytic virus infection. For herpesviruses, latent infection is established in nonreplicating neural cells ([HSV](#) and [VZV](#)) and in replicating cells of early hematopoietic lineages [[EBV](#) and probably [CMV](#), [HHV-6](#), [HHV-7](#), and Kaposi's sarcoma-associated herpesvirus (KSHV, also known as [HHV-8](#))]. Reactivation from neural latency appears to be an intermittent process provoked by external stimuli, whereas reactivation from hematopoietic precursors appears to be a more continuous process. In their latent stage, HPV and herpesvirus genomes are hidden from the normal immune response. It is still not fully understood how latent and reactivated HPV and herpesvirus infections escape immediate and effective immune responses in highly immune hosts. HPV, HSV, and VZV may be somewhat protected because of their replication in middle and upper layers of the squamous epithelium. HSV and CMV are also known to encode proteins that downregulate [MHC](#) class I expression and antigenic peptide presentation on infected cells, thereby enabling these cells to escape CD8⁺ T lymphocyte cytotoxicity. Latent infection and intermittent reactivation perpetuate HPV and herpesvirus infections in human populations by allowing the viruses to persist in immune hosts and to be transmitted to the next generation of naive hosts.

Like other poxviruses, molluscum contagiosum virus cannot establish latent infection but rather causes persistent infection in hypertrophic lesions that last for months or years. This virus encodes a chemokine homologue that probably blocks inflammatory responses and an [MHC](#) class I analogue that may block cytotoxic T lymphocyte attack.

PERSISTENT VIRAL INFECTIONS AND CANCER

Persistent viral infection is estimated to be the root cause of as many as 20% of human malignancies. For the most part, cancer is an accidental and highly unusual or long-term effect of infection with oncogenic human viruses. In these malignancies, viral infection is a critical and ultimately determinative early step, and an unusual virus-infected cell undergoes the subsequent genetic changes that permit the enhanced autonomous growth and survival characteristic of a malignant cell. Most hepatocellular carcinoma is now believed to be caused by chronic inflammatory, immune, and regenerative responses to [HBV](#) or [HCV](#) infection. Epidemiologic data firmly link HBV and HCV infection

to hepatocellular carcinoma, and studies in murine experimental models indicate that chronic liver injury and repair induced by virus-encoded proteins can result in hepatocellular cancer. In rare instances, HBV DNA integrates into cellular DNA -- an event that probably contributes to the development of some tumors.

Almost all cervical carcinoma is caused by long-term persistent replication of "high-risk" genital [HPV](#) strains. An infrequent consequence of persistent HPV replication is the integration of a small fragment of the HPV genome encoding the HPV E6 and E7 proteins into chromosomal DNA. Overexpression of these proteins of HPV type 16 or 18 eliminates at least two major tumor-suppressive mechanisms in the infected cell and causes profound changes in cellular growth and survival. Nevertheless, subsequent chromosomal changes must occur over ensuing cycles of cell growth if a sufficiently malignant cell is to invade the surrounding tissues.

Similarly, long-term [EBV](#) infection and expression of the EBV oncogene LMP1 in a clone of latently infected epithelial cells appears to be a critical early step in the evolution of anaplastic nasopharyngeal carcinoma, a common malignancy in Chinese and North African populations. High-level LMP1 expression is a hallmark of many cases of Hodgkin's disease. Among younger age groups, >50% of Hodgkin's disease tumors are clonally derived from an EBV-infected cell. The [HTLV-I](#) Tax and Rex proteins appear to be critical to the initiation of cutaneous adult T cell lymphoma/leukemias that may occur long after primary HTLV-I infection.

A new [EBV](#)-related herpesvirus, [KSHV](#), was identified in a search for the postulated sexually transmitted etiologic agent of Kaposi's sarcoma in HIV-infected individuals. Molecular data confirm the presence of KSHV DNA in all Kaposi's tumors, including those associated with HIV infection, transplantation, and familial transmission.

Evidence supporting a causal role of viral infection in these malignancies includes epidemiologic data, the presence of viral DNA in all tumor cells, the ability of the viruses to transform human cells in culture, the results of in vitro assays for transforming effects of specific viral genes on cell growth, and pathologic data indicating the expression of transforming viral genes in premalignant or malignant cells in vivo.

[EBV](#) is a unique example of a human virus that relies on the normal immune response to contain the potentially unrestrained growth of infected B lymphocytes. In the initial stages of normal primary EBV infection, EBV "latently" infects B lymphocytes and expresses at least eight viral proteins that cause continuous cell proliferation. The infected cells grow indefinitely in vitro or in T cell-deficient mice. Most of the viral proteins are highly antigenic, and these virus-infected cells, which can transiently constitute 10% of the circulating B lymphocyte population, are met with an overwhelming helper and cytotoxic T cell response during primary infection. The number of virus-infected cells then falls rapidly, and the one EBV-infected cell in a million that persists does not express most of the viral proteins that cause B cell proliferation. These persisting cells are the site of normal latent infection. Breakthrough growth of the EBV-infected B lymphocytes almost never occurs in immunocompetent hosts. However, in immunosuppressed AIDS patients or organ transplant recipients, EBV-infected B lymphocytes expressing the full set of growth-transforming genes may grow, uncontrolled by the immune system, and cause self-sustained and potentially fatal

lymphoproliferative disease. Clinical investigation has resulted in novel strategies for treating these virus-induced malignancies by increasing T cell responsiveness through ex vivo expansion and readministration of EBV-specific T cells and by attacking the proliferating B cells with antibody coupled to toxins.

RESISTANCE TO VIRAL INFECTIONS

Resistance to viral infection is initially provided by factors that are not virus-specific. Physical protection is afforded by the cornified layers of the skin and by mucous secretions that continuously sweep over mucosal surfaces. Once the first cell is infected, viral infection induces [IFNs](#), which are important local resistance factors. Viral infection may also cause the release of other cytokines from infected cells. Viral protein epitopes expressed on the cell surface in the context of [MHC](#) class I and II HLA proteins attract T cells with appropriate receptors. Cytokines, inflammatory agents, and antigens released by virus-induced cell death attract inflammatory cells, dendritic cells, granulocytes, natural killer (NK) cells, and B lymphocytes to the sites of initial infection. IFNs and NK cells are particularly important in containing viral infection for the first several days. Granulocytes and macrophages are also important in the phagocytosis and degradation of viruses, especially after an initial antibody response.

Some 7 to 10 days after infection, virus-specific antibody responses, virus-specific HLA class II-restricted CD4⁺ helper T lymphocyte responses, and virus-specific HLA class I-restricted CD8⁺ cytotoxic T lymphocyte responses are detected. These responses, whose magnitude typically increases over the second and third weeks of infection, are important in rapid recovery. Between the second and third weeks of infection, the antibody type usually changes from IgM to IgG, and IgA antibody is detected at initially infected mucosal surfaces. Antibody may directly neutralize virus by binding to its surface and preventing its adsorption or penetration. Complement usually enhances virus neutralization. Antibody and complement can also lyse virus-infected cells that express viral proteins on their surface. A cell infected with an enveloped virus usually expresses viral envelope glycoprotein components on its surface and is thus rendered subject to destruction by antibodies and complement.

The antibody and CD4⁺/CD8⁺ T lymphocyte responses tend to persist for several months after primary infection. Antibody-producing lymphocytes persist in small numbers as memory cells and begin to proliferate rapidly in response to a second infection, providing an early barrier to reinfection with the same virus. Immunologic memory for T cell responses appears to be less long-lived, and redevelopment of T cell immunity may take longer than secondary antibody responses, particularly when many years have elapsed between primary infection and reexposure.

Some viruses have genes that alter innate and acquired host defenses. Adenoviruses encode small RNAs that inhibit [IFN](#) shutoff of infected-cell protein synthesis. Adenovirus E1A inhibits IFN-mediated changes in cell gene transcription. Adenovirus E3 proteins prevent [TNF](#)-induced cytolysis and block HLA class I antigen synthesis by the infected cell. [HSV](#) ICP47 and [CMV](#) US11 block class I antigen presentation. [EBV](#) encodes an interleukin (IL) 10 homologue that inhibits [NK](#) and T cell responses. Vaccinia virus B15R is an IL-1 receptor decoy. Vaccinia virus B8R is a soluble TNF receptor that blocks the effects of TNF. Vaccinia virus CrmA inhibits the ability of CD8⁺ cytotoxic cells to kill

virus-infected cells. Some poxviruses and herpesviruses encode blockers of chemokines and thereby inhibit cellular inflammatory responses. The adoption of these strategies by viruses highlights the importance of these host resistance factors in containing viral infection as well as that of redundancy in host resistance. The ultimate success of a virus requires a live host to help it disseminate.

Much has been written about the role of specific aspects of the host immune response in containment of specific virus infections. Certainly, T lymphocyte disorders are associated with severe primary and reactivated herpesvirus infections, and antibody responses are important in resistance to many RNA virus infections. However, antibody responses are also important in resistance to herpesvirus infections, as is exemplified by the utility of immunoglobulin therapy in early amelioration of these infections. T lymphocyte responses play a significant role in resistance to RNA virus infections, as is illustrated by the presence of cytotoxic T cells specific for influenza virus nucleoprotein.

Host resistance does not come without a price. Clearly, aspects of the host response contribute to the pathophysiologic manifestations and symptoms of viral infection. Inflammation at sites of viral infection can increase rates of local cell death. Immune responses to viral infection can target related epitopes on normal cells. While such effects have been demonstrated in experimental models, their role in the autoimmune manifestations of primary or recurrent human viral infections is uncertain.

INTERFERONS

All human cells can synthesize IFN- α or - β in response to viral infection. The IFN response is usually induced by the presence of double-strand viral RNA, which can be made by both RNA and DNA viruses. IFN- γ is not directly related to IFN- α or - β and is produced mainly by NK cells and by immune T lymphocytes responding to IL-12. IFN- α and - β bind to the IFN- α receptor, while IFN- γ binds to a different but related receptor. Both receptors signal through receptor-associated JAK kinases and other cytoplasmic proteins, including "STAT" proteins. These proteins are tyrosine phosphorylated by JAK kinases, translocate to the nucleus, and transactivate promoters for specific cell genes. Three types of antiviral effects are induced by IFN at the transcriptional level. The first effect is attributable to the induction of 2'-5' oligo(A) synthetases, which require double-strand RNA for their activation. Activated synthetase polymerizes oligo(A) and thereby activates RNase L, which in turn degrades single-strand RNA. The second effect takes place through the induction of PKR, a serine and threonine kinase that is also activated by double-strand RNA. PKR phosphorylates and negatively regulates the translational initiation factor eIF2- α , shutting down protein synthesis in the infected cell. A third effect is initiated through the induction of Mx proteins, a family of GTPases that is particularly important in inhibiting influenza virus and vesicular stomatitis virus replication. None of these IFN effects is directed specifically against the virus; infected-cell RNA and protein synthesis are globally inhibited. IFN probably contributes to the death of the infected cell.

DIAGNOSTIC VIROLOGY

A wide variety of methods are now used to diagnose viral infection, but serology and viral isolation in tissue culture remain the backbone of diagnostic virology. Acute- and

convalescent-phase sera with rising antibody titers to virus-specific antigens and a shift from IgM to IgG antibodies are generally accepted as diagnostic of acute viral infection. Traditionally, virus-specific antibodies have been detected by hemadsorption, hemagglutination, or indirect immunofluorescence. Immunofluorescence assays use fixed virus-infected cells as a target for serum antibodies. Hemadsorption and hemagglutination assays measure the ability of serum antibodies to the hemagglutinin proteins of RNA viruses to inhibit virus-induced adsorption or agglutination. Serologic diagnosis is based on a greater-than-fourfold rise in IgG antibody concentration when acute- and convalescent-phase serum samples are analyzed at the same time. A simultaneous fall in IgM antibody confirms recent primary viral infection. Immunofluorescence, hemadsorption, and hemagglutination assays are labor-intensive and are being replaced by enzyme-linked immunosorbent assays (ELISAs). ELISAs generally use specific viral proteins purified from virus-infected cells or produced by recombinant DNA technology. These viral antigens are attached to a solid phase, where they can be incubated with serum, washed to eliminate nonspecific antibodies, and reacted with an enzyme-linked reagent to detect human IgG or IgM antibody specifically adhering to the viral antigen on the solid phase. The amount of antibody can then be quantitated by the intensity of a color reaction mediated by the linked enzyme. ELISAs can be automated and can have enhanced sensitivity. Western blots measure antibody to multiple viral proteins simultaneously. The proteins are separated by size and transferred to an inert membrane, where they are incubated with serum antibodies. Western blots have an internal specificity control, since the level of reactivity for viral proteins can be compared with that for cellular proteins in the same sample. Western blots are a useful confirmatory test but require individual evaluation and are inherently difficult to quantitate.

Viral isolation in tissue culture is dependent on the infection of susceptible cells and amplification through viral replication in infected cells. Virus growing in tissue culture cells can frequently be identified by its effect under light microscopy. For example, [HSV](#) produces a typical cytopathic effect in rabbit kidney cells within 3 days. Other viral cytopathic effects may not be as diagnostically useful. Identification may require confirmation by staining with virus-specific monoclonal antibodies. Viruses growing in tissue culture can also be identified by hemadsorption, by interference (e.g., rubella virus-infected cells resist lysis by echovirus), or by electron microscopy (assuming that the specimen has altered cell morphology, as observed by ordinary light microscopy).

The efficiency and speed of virus identification can be enhanced by combining short-term culture with immune detection. In assays with "shell vials" of tissue culture cells growing on a coverslip, viral infection can be detected by staining of the culture with a monoclonal antibody to a specific viral protein expressed early in viral replication. Thus, virus-infected cells can be detected within hours or days of inoculation -- before the several rounds of infection that would be required to produce a visible cytopathic effect.

The sensitivity of virus isolation depends on the collection of specimens from the appropriate site and the rapid transport of these specimens in the appropriate medium to the virology laboratory. Rapid transport maintains viral viability and limits bacterial and fungal overgrowth. Lipid-enveloped viruses are generally much more sensitive to

freezing and thawing than nonenveloped viruses. The most appropriate site for culture depends on the pathogenesis of the virus in question. Nasopharyngeal, tracheal, or endobronchial aspirates are most appropriate for the identification of respiratory viruses. Sputum cultures generally are not appropriate because bacterial contamination and viscosity threaten tissue-culture cell viability. Aspirates of vesicular fluid are useful for isolation of [HSV](#) and [VZV](#). Nasopharyngeal aspirates and stool specimens may be useful when the patient has fever and a rash and an enteroviral infection is suspected. Adenoviruses can be cultured from the urine of patients with hemorrhagic cystitis. [CMV](#) can frequently be isolated from cultures of urine or buffy coat. Biopsy material can be effectively cultured when viruses infect major organs, as in HSV encephalitis or adenovirus pneumonia. Unlike serology, the isolation of a virus does not establish the time of primary infection. Many viruses persistently or intermittently colonize normal human mucosal surfaces. Saliva is not infrequently positive for herpesviruses, and 1% of normal urine samples are positive for CMV. Isolations from blood, cerebrospinal fluid (CSF), or biopsy specimens are more often diagnostic of significant virus infection.

Another method aimed at increasing the speed of viral diagnosis is direct antigen testing. Virus-infected cells obtained directly from the patient are detected by staining with virus-specific monoclonal antibodies; for example, epithelial cells obtained by nasopharyngeal aspiration can be stained with a variety of monoclonal antibodies to respiratory viruses. The Tzanck preparation used to detect multinucleated giant cells in [HSV](#)- or [VSV](#)-induced lesions was the predecessor of these direct antigen tests and can be enhanced by the use of HSV- or VZV-specific monoclonal antibodies. Similarly, monoclonal antibodies can be applied to histopathology specimens to identify virus-infected cells.

Advances in nucleic acid technology are revolutionizing diagnostic virology. The speed and sensitivity of tests that directly amplify minute amounts of viral nucleic acids present in specimens mean that detection no longer depends on viable virus and its replication. For example, amplification and detection of [HSV](#) nucleic acids leaking into the [CSF](#) of patients with HSV encephalitis can be more sensitive than culture of virus from CSF. The extreme sensitivity of these tests can be a problem, since trivial amounts of contamination can lead to false-positive results. In addition, detection of viral nucleic acids does not necessarily indicate virus-induced disease, especially in cases where viruses (e.g., herpesviruses) can cause persistent asymptomatic infection.

Measurement of the amount of viral RNA or DNA in peripheral blood is becoming an important means for determining which patients are at increased risk for virus-induced disease and for evaluating clinical responses to antiviral chemotherapy. Direct staining with [CMV](#)-specific monoclonal antibodies to quantitate virus-infected cells in the peripheral blood or CMV antigenemia can be useful in identifying which immunosuppressed patients may be at risk for CMV-induced disease. New CMV assays using nucleic acid technologies for the same purpose have been approved for clinical use. RNA viral-load measurements by nucleic acid technologies are now routinely used in AIDS patients to evaluate responses to an increasing number of antiviral agents. Viral-load measurements may also be useful for evaluating the treatment of patients with [HBV](#) and [HCV](#) infections.

The use of antiviral agents for the treatment of herpesvirus and HIV infections has been

highly effective. However, the emergence of drug-resistant HIV strains in treated patients can limit efficacy in some cases. The increased number of antiviral agents and drug classes with different viral targets has made the identification of drug-resistant viruses clinically relevant, especially for HIV infection. Drug resistance in herpesviruses is a more unusual problem.

Viral genotyping is a new and faster method for the identification of drug-resistant viruses. Rising viral loads despite antiviral chemotherapy may indicate emergence of resistant HIV strains. Resistance to reverse transcriptase or protease inhibitors has been associated with specific mutations in the reverse transcriptase or protease genes. Identification of these mutations by polymerase chain reaction amplification and nucleic acid sequencing can be clinically useful for determining which antiviral agents may still be effective. Genotyping of [HCV](#) may also help identify patients who can benefit from combination chemotherapy.

Viral phenotyping may also be useful for identifying resistant viruses associated with new or unrecognized genetic mutations. These labor-intensive assays are not routinely available, but technical advances and continued evolution of drug-resistant viruses may make these tests more clinically relevant in the near future.

IMMUNIZATION FOR THE PREVENTION OF VIRAL INFECTIONS

Viral vaccines were among the outstanding accomplishments of twentieth-century science. The scourge of smallpox has been eradicated. Poliovirus eradication may soon follow. Rabies and measles can be contained or eliminated. Excess mortality due to influenza virus epidemics can be contained, and the threat of influenza pandemics has decreased. Widespread [HBV](#) vaccination has dramatically lessened the frequency of acute and chronic hepatitis and is expected to lead to a dramatic decrease in the incidence of hepatocellular carcinoma. The ease with which some viruses are attenuated in tissue culture has led to widespread immunization against rubella, measles, mumps, and chickenpox. Recombinant DNA-based strategies will make it possible to prevent severe infections with many other viruses by using purified proteins or genetically engineered live virus vaccines. Unfortunately, there are limits to these prospects. The evolutionary divergence of HIV and [HCV](#), for example, complicates the development of highly effective immunogens for the prevention of infection with these agents. Modestly effective immunogens that incorporate multiple B and T cell epitopes may prove useful for low-level exposures.

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181. ANTIVIRAL CHEMOTHERAPY, EXCLUDING ANTIRETROVIRAL DRUGS - *Raphael Dolin*

The development of drugs for antiviral chemotherapy and chemoprophylaxis is a relatively recent but now very active area of biomedical research. Significant progress has been made in recent years on new drugs for several viral infections. Despite these advances, the field of antiviral therapy -- both the number of antiviral drugs and our understanding of their optimal use -- continues to lag behind the field of antibacterial drug treatment, in which more than 60 years of experience have now been accumulated.

The development of antiviral drugs poses several challenges. Viruses replicate intracellularly and often employ host cell enzymes, macromolecules, and organelles for synthesis of viral particles. Therefore, useful antiviral compounds must discriminate between host and viral functions with a high degree of specificity; agents without such selectivity are likely to be too toxic for clinical use.

The development of laboratory assays to assist clinicians in the appropriate use of antiviral drugs is also in its infancy. Phenotypic and genotypic assays for resistance to antiviral drugs are becoming more widely available, and correlations of laboratory results with clinical outcomes in various settings are beginning to be defined. Of particular note has been the development of highly sensitive and specific methods to measure the concentration of virus in blood (*virus load*), which permit direct assessment of the antiviral effect of a given drug regimen in the host. Virus load measurements have been useful in recognizing the risk of disease progression in patients with certain viral infections and in identifying patients in whom antiviral chemotherapy might be of greatest benefit. Like any in vitro laboratory test, these tests yield results that are highly dependent on (and likely to vary with) the laboratory techniques employed.

Information regarding the pharmacokinetics of some antiviral drugs, particularly in diverse clinical settings, is limited. Assays to measure the concentrations of these drugs, especially of their active moieties within cells, are not widely available. Thus, there are relatively few guidelines for adjusting dosages of antiviral agents to maximize antiviral activity and minimize toxicity. Clinical use of antiviral drugs must therefore be accompanied by particular vigilance with regard to unanticipated adverse effects.

Like that of other infections, the course of viral infections is profoundly affected by an interplay of the pathogen with a complex set of host defenses. The presence or absence of preexisting immunity and the ability to mount humoral and/or cell-mediated immune responses are important determinants of the outcome of viral infections. The state of the host's defenses needs to be considered when antiviral agents are utilized or evaluated.

As with any therapy, the optimal use of antiviral compounds requires a specific and timely diagnosis. For some viral infections, such as herpes zoster, the clinical manifestations are so characteristic that a diagnosis can be made on clinical grounds alone. For other viral infections, such as influenza A, epidemiologic information (e.g., the documentation of a community-wide outbreak) can be used to make a presumptive diagnosis with a high degree of accuracy. However, for most other viral infections, including herpes simplex encephalitis, cytomegaloviral infections other than retinitis, and

enteroviral infections, diagnosis on clinical grounds alone cannot be accomplished with certainty. For such infections, rapid viral diagnostic techniques are of great importance. Considerable progress has been made in recent years in the development of such tests, which are now widely available for a number of viral infections.

Despite these complexities, the efficacy of a number of antiviral compounds has been clearly established in rigorously conducted and controlled studies. As summarized in [Table 181-1](#), this chapter reviews the antiviral drugs that are currently approved or are likely to be considered for approval in the near future for use against viral infections other than those caused by HIV. Antiretroviral drugs are reviewed in [Chap. 309](#).

ANTIVIRAL DRUGS ACTIVE AGAINST RESPIRATORY INFECTIONS

AMANTADINE AND RIMANTADINE

Amantadine and the closely related compound rimantadine are primary symmetric amines. Their antiviral activity is limited to influenza A viruses, whose replication they inhibit by interfering with the uncoating of virus after infection of the cell. This interference is attributable to the agents' interaction with the influenza A M2 matrix protein, during which the ion channel function of M2 is inhibited. A substitution of a single amino acid in the M2 protein can result in a virus that is resistant to amantadine and rimantadine.

Amantadine and rimantadine have been demonstrated to be effective in the prophylaxis of influenza A in large-scale studies of young adults and in less extensive studies of children and elderly subjects. In such studies, efficacy rates of 55 to 80% in the prevention of influenza-like illness were noted, and even higher rates were reported when virus-specific attack rates were calculated. Amantadine and rimantadine have also been demonstrated to be effective in the treatment of influenza A infection in studies involving predominantly young adults and, to a lesser extent, children. Administration of these compounds within 24 to 72 h after the onset of illness has resulted in a reduction of the duration of signs and symptoms by ~50% from that in a placebo-treated group. The effect on signs and symptoms of illness is superior to that of commonly used antipyretic-analgesics. Only anecdotal reports are available concerning the efficacy of amantadine or rimantadine in the prevention or treatment of complications of influenza (e.g., pneumonia).

Amantadine and rimantadine are available only in oral formulations and are ordinarily administered to adults once or twice daily in a dose of 100 to 200 mg/d. Despite their structural similarities, the pharmacokinetics of the two compounds are different. Amantadine is not metabolized and is excreted almost entirely by the kidney, with a half-life of 12 to 17 h and peak plasma concentrations of 0.4 ug/mL. Rimantadine is extensively metabolized to hydroxylated derivatives and has a half-life of 30 h. Only 30 to 40% of an orally administered dose is recovered in the urine. The peak plasma levels of rimantadine are approximately half those of amantadine, but rimantadine is concentrated in respiratory secretions to a greater extent than amantadine. For prophylaxis, the compounds must be administered daily for the period at risk (i.e., the peak duration of the outbreak). For therapy, amantadine or rimantadine is generally administered for 5 to 7 days.

Although these compounds are generally well tolerated, 5 to 10% of amantadine recipients experience mild central nervous system side effects consisting primarily of dizziness, anxiety, insomnia, and difficulty in concentrating. These effects are rapidly reversible upon cessation of the drug's administration. At a dose of 200 mg/d, rimantadine is better tolerated than amantadine; in a large-scale study of young adults, adverse effects were no more frequent among rimantadine recipients than among placebo recipients. Seizures and worsening of congestive heart failure have also been reported in patients treated with amantadine, although a causal relationship has not been established. The dosage of amantadine should be reduced to 100 mg/d in patients with renal insufficiency [i.e., a creatinine clearance (Cr_{Cl}) rate of <50 mL/min] and in the elderly. A rimantadine dose of 100 mg/d should be used for patients with a Cr_{Cl} of <10 mL/min and in the elderly. Resistance to amantadine and rimantadine can be induced readily in vitro. The emergence and probable transmission of virus resistant to these drugs have also been noted in vivo after their use for the treatment of children or adults. In the United States, both amantadine and rimantadine are approved for the prophylaxis and treatment of influenza A in adults and for prophylaxis in children. Amantadine is also approved for the treatment of influenza A in children.

RIBAVIRIN

Ribavirin is a synthetic nucleoside analogue that inhibits a wide range of RNA and DNA viruses. The mechanism of action of ribavirin is not completely defined and may be different for different groups of viruses. Ribavirin-5'-monophosphate blocks the conversion of inosine-5'-monophosphate to xanthosine-5'-monophosphate and interferes with the synthesis of guanine nucleotides as well as that of both RNA and DNA. Ribavirin-5'-monophosphate also inhibits capping of virus-specific messenger RNA in certain viral systems. In studies demonstrating the effectiveness of ribavirin, the compound has been administered as a small-particle aerosol. It has been used to treat respiratory syncytial virus (RSV) infections in infants and -- less extensively -- to treat parainfluenza virus infections in children and influenza A and B virus infections in young adults. In infants with RSV infection who were given ribavirin by continuous aerosol for 3 to 6 days, illness and lower respiratory tract signs resolved more rapidly and arterial oxygen desaturation was less pronounced than in placebo-treated groups. Ribavirin has also had a beneficial clinical effect in infants with RSV infection who require mechanical ventilation. Aerosolized ribavirin has been administered to older children and adults with severe RSV and parainfluenza virus infections (including immunosuppressed patients), but the benefit of this treatment, if any, is unclear. In RSV infections, ribavirin is often given in combination with immunoglobulins.

Orally administered ribavirin has not been effective in the treatment of influenza A virus infections. Intravenous or oral ribavirin has reduced mortality among patients with Lassa fever; it has been particularly effective in this regard when given within the first 6 days of illness. Intravenous ribavirin has been reported to be of clinical benefit in the treatment of hemorrhagic fever with renal syndrome caused by Hantaan virus and as therapy for Argentinian hemorrhagic fever. Moreover, oral ribavirin has been recommended for the treatment and prophylaxis of Congo-Crimean hemorrhagic fever. Intravenous ribavirin is being evaluated as therapy for the hemorrhagic fever with pulmonary syndrome caused by newly described hantaviruses in the United States. Oral administration of ribavirin

reduces serum aminotransferase levels in patients with chronic hepatitis C virus (HCV) infection; since it appears not to reduce serum HCV RNA levels, the mechanism of this effect is unclear. Given in doses of 1000 to 1200 mg/d in combination with interferon (IFN) α (see below), ribavirin has been approved for the treatment of patients with chronic HCV infection.

Large doses of ribavirin administered orally (800 to 1000 mg/d) have been associated with reversible hematopoietic toxicity. This effect has not been observed with aerosolized ribavirin, apparently because little drug is absorbed systemically. Aerosolized administration of ribavirin is generally well tolerated but occasionally is associated with bronchospasm, rash, or conjunctival irritation. Aerosolized ribavirin has been licensed for treatment of [RSV](#) infection in infants and should be administered under close supervision -- particularly in the setting of mechanical ventilation, where precipitation of the drug is possible. Health care workers exposed to the drug have experienced minor toxicity, including eye and respiratory tract irritation. Because ribavirin is mutagenic, teratogenic, and embryotoxic, its use is generally contraindicated in pregnancy. Its administration as an aerosol poses a risk to pregnant health care workers.

ZANAMIVIR AND OSELTAMIVIR

Influenza viral neuraminidase is essential for release of the virus from infected cells and for its subsequent spread throughout the respiratory tract of the infected host. The enzyme cleaves terminal sialic acid residues, thus destroying the cellular receptors recognized by the viral hemagglutinin. Zanamivir, a sialic acid analogue, is a highly active and specific inhibitor of the neuraminidases of influenza A and B viruses. Oseltamivir is another neuraminidase inhibitor that is a transition-state analogue of sialic acid cleavage. Its antineuraminidase activity is similar to that of zanamivir. Oseltamivir phosphate is an ethyl ester prodrug that is converted to oseltamivir carboxylate by esterases in the liver. Both zanamivir and oseltamivir act through competitive and reversible inhibition of the active site of influenza A and B viral neuraminidases and have relatively little effect on mammalian cell enzymes. As would be expected from their different mechanisms of action, zanamivir and oseltamivir are active against strains of influenza A virus that are resistant to amantadine and rimantadine.

Zanamivir has low oral bioavailability. It is inhaled orally via a hand-held inhaler. By this route, ~15% of the dose is deposited in the lower respiratory tract, and low plasma levels of the drug are detected. Orally administered oseltamivir has an oral bioavailability of >60% and a plasma half-life of 7 to 9 h. The drug is excreted unmetabolized, primarily by the kidney.

Intranasal inhaled zanamivir is generally well tolerated. The most frequent toxicities encountered with orally administered oseltamivir are nausea, gastrointestinal discomfort, and (less commonly) vomiting. Gastrointestinal discomfort is usually transient and is less likely if the drug is administered with food. No serious clinical or laboratory toxicities have yet been reported with zanamivir or oseltamivir in clinical trials.

Inhaled zanamivir and orally administered oseltamivir have been effective in the treatment of naturally occurring influenza A or B in otherwise healthy adults. In

placebo-controlled studies, illness has been shortened by 1 to 1.5 days of therapy with either of these drugs. Once-daily inhaled zanamivir or orally administered oseltamivir provides effective prophylaxis against laboratory-documented influenza A-associated illness. The emergence of viruses resistant to zanamivir or oseltamivir appears to be infrequent in clinical studies carried out thus far.

As of this writing, zanamivir and oseltamivir have been approved by the U.S. Food and Drug Administration (FDA) for treatment of influenza in adults (and -- in the case of zanamivir -- in children³⁷ years of age) who have been symptomatic for ≤ 2 days. Indications for prophylactic use are under review.

ANTIVIRAL DRUGS ACTIVE AGAINST HERPESVIRUS INFECTIONS

ACYCLOVIR AND VALACYCLOVIR

Acyclovir is a highly potent and selective inhibitor of the replication of certain herpesviruses, including herpes simplex virus (HSV) types 1 and 2, varicella-zoster virus (VZV), and Epstein-Barr virus (EBV). It is relatively ineffective in the treatment of human cytomegalovirus (CMV) infections; however, some studies have indicated its effectiveness in the prevention of CMV-associated disease in immunosuppressed patients. Valacyclovir, the L-valyl ester of acyclovir, is converted almost entirely to acyclovir after oral administration. Valacyclovir has pharmacokinetic advantages over orally administered acyclovir: it exhibits significantly greater oral bioavailability, results in higher blood levels, and can be given less frequently than acyclovir.

The high degree of selectivity of acyclovir is related to its mechanism of action, which requires that the compound first be phosphorylated to acyclovir monophosphate. This phosphorylation occurs efficiently in herpesvirus-infected cells by means of a virus-coded thymidine kinase. In uninfected mammalian cells, little phosphorylation of acyclovir occurs, and the drug is therefore concentrated in herpesvirus-infected cells. Acyclovir monophosphate is subsequently converted by host cell kinases to a triphosphate that is a potent inhibitor of virus-induced DNA polymerase but has relatively little effect on host cell DNA polymerase. Acyclovir triphosphate can also be incorporated into viral DNA, with early chain termination.

Acyclovir is available in intravenous, oral, and topical forms, while valacyclovir is available in an oral formulation. Intravenous acyclovir is markedly effective in the treatment of mucocutaneous [HSV](#) infections in immunocompromised hosts, reducing time to healing, duration of pain, and virus shedding. When administered prophylactically during periods of intense immunosuppression (e.g., related to chemotherapy for leukemia or transplantation) and before the development of lesions, intravenous acyclovir reduces the frequency of HSV-associated disease. After prophylaxis is discontinued, HSV lesions recur. Intravenous acyclovir is also effective in the treatment of HSV encephalitis; two comparative trials have indicated that acyclovir is more effective than vidarabine for this indication (see below). Because [VZV](#) is generally less sensitive to acyclovir than is HSV, higher doses of acyclovir must be used to treat VZV infections. In immunocompromised patients with herpes zoster, intravenous acyclovir reduces the frequency of cutaneous dissemination and visceral complications and -- in one comparative trial -- was more effective than vidarabine. Acyclovir,

administered orally at doses of 800 mg five times a day, had a modest beneficial effect on localized herpes zoster lesions in both immunocompromised and immunocompetent patients. Combination of acyclovir with a tapering regimen of prednisone appeared to be more effective than acyclovir alone in terms of quality-of-life outcomes in immunocompetent herpes zoster patients over age 50. A comparative study of acyclovir (800 mg orally five times daily) and valacyclovir (1 g orally tid) in immunocompetent patients with herpes zoster indicated that the latter drug may be more effective in eliciting the resolution of zoster-associated pain. Orally administered acyclovir (600 mg five times a day) reduced complications of herpes zoster ophthalmicus in a placebo-controlled trial.

In normal children with chickenpox, acyclovir -- administered at 20 mg/kg qid, up to a maximum of 800 mg qid, within 24 h of the onset of rash -- resulted in a modest overall clinical benefit. Intravenous acyclovir has also been reported to be effective in the treatment of immunocompromised children with chickenpox.

The most widespread use of acyclovir is in the treatment of genital [HSV](#) infections. Intravenous or oral acyclovir or oral valacyclovir has shortened the duration of symptoms, reduced virus shedding, and accelerated healing when employed for the treatment of primary genital HSV infections. Oral acyclovir and valacyclovir have also had a modest effect in treatment of recurrent genital HSV infections. However, the failure of treatment of either primary or recurrent disease to reduce the frequency of subsequent recurrences has indicated that acyclovir is ineffective in eliminating latent infection. Chronic oral administration of acyclovir for periods of 1 to 6 years or longer or of valacyclovir for up to 1 year has reduced the frequency of recurrences markedly during therapy; once the drug is discontinued, lesions recur. In AIDS patients, chronic or intermittent administration of acyclovir has been associated with the development of HSV and [VZV](#) strains resistant to the action of the drug and with clinical failures. The most common mechanism of resistance is a deficiency of the virus-induced thymidine kinase. Patients with HSV or VZV infections resistant to acyclovir have frequently responded to foscarnet.

With the availability of the oral and intravenous forms, there are few indications for topical acyclovir, although treatment with this formulation has been modestly beneficial in primary genital [HSV](#) infections and in mucocutaneous HSV infections in immunocompromised hosts.

Overall, acyclovir is remarkably well tolerated and is generally free of toxicity. The most frequently encountered form of toxicity is renal dysfunction, particularly after rapid intravenous administration or with inadequate hydration. Central nervous system changes, including lethargy and tremors, are occasionally reported, primarily in immunosuppressed patients. However, whether these changes are related to acyclovir, to concurrent administration of other therapy, or to underlying infection remains unclear. Acyclovir is excreted primarily unmetabolized by the kidney, via both glomerular filtration and tubular secretion. Approximately 15% of a dose of acyclovir is metabolized to 9-[(carboxymethoxy)methyl]guanine or other minor metabolites. Reduction in dosage is indicated in patients with a Cr_{Cl} of <50 mL/min per 1.73 m². The half-life of acyclovir is ~3 h in normal adults, and the peak plasma concentration after a 1-h infusion of a dose of 5 mg/kg is 9.8 ug/mL. Approximately 22% of an orally administered acyclovir dose is

absorbed, and peak plasma concentrations of 0.3 to 0.9 ug/mL are attained after administration of a 200-mg dose. Acyclovir penetrates relatively well into the cerebrospinal fluid (CSF), with concentrations approaching half of those found in plasma.

Acyclovir causes chromosomal breakage at high doses, but its administration to pregnant women has not been associated with fetal abnormalities. Nonetheless, the potential risks and benefits of acyclovir should be carefully assessed before the drug is used in pregnancy.

Valacyclovir exhibits three to five times greater bioavailability than acyclovir. The concentration-time curve for valacyclovir, given as 1 g orally tid, is similar to that for acyclovir, given as 5 mg/kg intravenously every 8 h. The safety profiles of valacyclovir and acyclovir are similar, although thrombotic thrombocytopenic purpura/hemolytic-uremic syndrome has been reported in immunocompromised patients who have received high doses of valacyclovir. Valacyclovir is approved for the treatment of herpes zoster and for initial and recurrent episodes of genital HSV infections in immunocompetent adults as well as for suppressive treatment of genital herpes. It is being studied for use against other herpesvirus infections in various clinical settings.

CIDOFOVIR

Cidofovir is a phosphonate nucleotide analogue of cytosine. Its major use is in CMV infections, particularly retinitis, but it is active against a broad range of herpesviruses, including HSV, human herpesvirus (HHV) type 6, HHV-8, and certain other DNA viruses such as polyomaviruses, papillomaviruses, and adenoviruses. Cidofovir does not require initial phosphorylation by virus-induced kinases; the drug is phosphorylated by host cell enzymes to cidofovir diphosphate, which is a competitive inhibitor of viral DNA polymerases and, to a lesser extent, of host cell DNA polymerases. Incorporation of cidofovir diphosphate slows or terminates nascent DNA chain elongation. Cidofovir is active against HSV isolates that are resistant to acyclovir because of absent or altered thymidine kinase and against CMV isolates that are resistant to ganciclovir because of UL97 mutations. Cidofovir is usually active against foscarnet-resistant CMV, although cross-resistance to foscarnet as well as to ganciclovir has been described.

Cidofovir has poor oral availability and is administered intravenously. It is excreted primarily by the kidney and has a plasma half-life of 2.6 h. Cidofovir diphosphate's intracellular half-life of >48 h is the basis for the recommended dosing regimen of 5 mg/kg twice a week for the initial 2 weeks and then 5 mg/kg once a week. The major toxic effect of cidofovir is proximal renal tubular injury, as manifested by elevated serum creatinine levels and proteinuria. The risk of nephrotoxicity can be reduced by vigorous saline hydration and by concomitant oral administration of probenecid. Neutropenia, rashes, and gastrointestinal tolerance may also occur.

Intravenous cidofovir has been approved for the treatment of CMV retinitis in AIDS patients who are intolerant of ganciclovir or foscarnet or in whom those drugs have failed. In a controlled study, a maintenance dosage of 5 mg/kg a week administered to AIDS patients reduced the progression of CMV retinitis from that seen at 3 mg/kg.

Intravenous cidofovir has been reported anecdotally to be effective therapy for acyclovir-resistant mucocutaneous [HSV](#) infections. Likewise, topically administered cidofovir is reportedly beneficial against these infections in HIV patients; it is also being studied for the treatment of anogenital warts. Intravenous cidofovir is being evaluated as therapy for progressive multifocal leukoencephalopathy and for Kaposi's sarcoma. An ophthalmic formulation is being studied as treatment for adenoviral keratoconjunctivitis. Intravitreal cidofovir has been used to treat CMV retinitis but has been associated with significant toxicity.

FOMIVIRSEN

Fomivirsen is the first antisense oligonucleotide approved by the U.S. Food and Drug Administration (FDA) for therapy in humans. This phosphorothioate oligonucleotide, 21 nucleotides in length, inhibits [CMV](#) replication through interaction with CMV messenger RNA. Fomivirsen is complementary to messenger transcripts of the major immediate early region 2 (IE2) of CMV, which codes for proteins regulating viral gene expression. In addition to its antisense mechanism of action, fomivirsen may exert activity against CMV through inhibition of viral adsorption to cells as well as direct inhibition of viral replication. Because of its different mechanism of action, fomivirsen is active against CMV isolates that are resistant to nucleoside or nucleotide analogues, such as ganciclovir, foscarnet, or cidofovir.

Fomivirsen has been approved for intravitreal administration in the treatment of [CMV](#) retinitis in AIDS patients who have failed to respond to other treatments or cannot tolerate them. Injections of 330 mg every 2 weeks have resulted in significant reductions in the rate of progression of CMV retinitis. The major toxicity is ocular inflammation, including vitritis and iritis, which usually responds to topically administered glucocorticoids.

GANCICLOVIR

An analogue of acyclovir, ganciclovir is active against [HSV](#) and [VZV](#) and is markedly more active than acyclovir against [CMV](#). Ganciclovir triphosphate inhibits CMV DNA polymerase and can be incorporated into CMV DNA, whose elongation it eventually terminates. In HSV- and VZV-infected cells, ganciclovir is phosphorylated by virus-encoded thymidine kinases; in CMV-infected cells, it is phosphorylated by a viral kinase encoded by the UL97 gene. Ganciclovir triphosphate is present in tenfold higher concentrations in CMV-infected cells than in uninfected cells. Ganciclovir is approved for the treatment of CMV retinitis in immunosuppressed patients and for the prevention of CMV disease in transplant recipients. It is widely used for the treatment of other CMV-associated syndromes, including pneumonia, esophagogastrintestinal infections, hepatitis, and "wasting" illness.

Ganciclovir is available for intravenous or oral administration. Because its oral bioavailability is low (5 to 9%), relatively large doses (1 g tid) must be administered by this route. Oral bioavailability is enhanced if the drug is administered with food, as recommended. The serum half-life of ganciclovir is 3.5 h after intravenous administration and 4.5 h after oral administration. The drug is excreted primarily by the kidney in unmetabolized form, and its dosage should be reduced in cases of renal

failure. The most commonly employed dosage for initial therapy -- 5 mg/kg intravenously every 12 h for 14 to 21 days -- is followed by a maintenance dose of 5 mg/kg intravenously per day or 5 times per week, possibly for as long as immunosuppression persists. Oral ganciclovir is approved as an alternative to the intravenous preparation in maintenance therapy for [CMV](#) retinitis, where it appears to be somewhat less effective although more convenient than intravenous therapy. Intraocular ganciclovir, given by either intravitreal injection or intraocular implantation, has also been used to treat CMV retinitis.

Ganciclovir is effective as prophylaxis against [CMV](#)-associated disease in organ and bone marrow transplant recipients. Oral ganciclovir administered prophylactically to AIDS patients with CD4+ counts of <100/uL provided protection against the development of CMV retinitis in one large-scale study and was subsequently approved for that indication. However, the long-term benefits of this approach to prophylaxis are unestablished, and most experts do not recommend the use of oral ganciclovir for that purpose.

The administration of ganciclovir has been associated with profound bone marrow suppression, particularly neutropenia, which significantly limits the drug's use in many patients. Bone marrow toxicity is potentiated when other bone marrow suppressants, such as zidovudine, are used concomitantly.

Resistance has been noted in [CMV](#) isolates obtained after therapy with ganciclovir, especially in patients with AIDS. Such resistance may develop through a mutation in either the viral UL97 gene or the viral DNA polymerase. Ganciclovir-resistant isolates are usually sensitive to foscarnet (see below).

FAMCICLOVIR AND PENCICLOVIR

Famciclovir is the diacetyl 6-deoxyester of the guanosine analogue penciclovir. Famciclovir is well absorbed orally, with a bioavailability of 77%, and is rapidly converted by deacetylation and oxidation to penciclovir. Penciclovir's spectrum of activity and mechanism of action are similar to those of acyclovir; thus penciclovir is not active against acyclovir-resistant viruses. Penciclovir is phosphorylated initially by a virus-encoded thymidine kinase and subsequently by cellular kinases to penciclovir triphosphate, which inhibits [HSV](#)-1, HSV-2, and [VZV](#) DNA polymerases as well as hepatitis B virus (HBV). The serum half-life of penciclovir is 2 h, but the intracellular half-life of penciclovir triphosphate is 7 to 20 h -- markedly longer than that of acyclovir triphosphate. Penciclovir is eliminated primarily in the urine by both glomerular filtration and tubular secretion. The usually recommended dosage interval should be adjusted for renal insufficiency.

Clinical trials involving immunocompetent adults with herpes zoster showed that famciclovir was superior to placebo in eliciting the resolution of skin lesions and virus shedding and in shortening the duration of postherpetic neuralgia; moreover, it was at least as effective as acyclovir administered orally at a dose of 800 mg five times daily. Famciclovir was also effective in the treatment of herpes zoster in immunosuppressed patients. Clinical trials have demonstrated its effectiveness in suppression of genital [HSV](#) infections for up to 1 year and in the treatment of initial and recurrent

episodes of genital herpes. Famciclovir is effective as therapy for mucocutaneous HSV infections in HIV-infected patients. Application of a 1% penciclovir cream reduces the duration of signs and symptoms of herpes labialis in immunocompetent patients (by 0.5 to 1.0 day) and has been approved for that purpose by the [FDA](#). Famciclovir is generally well tolerated, with occasional headache, nausea, and diarrhea reported in frequencies similar to those among placebo recipients. The administration of high doses of famciclovir for 2 years was associated with an increased incidence of mammary adenocarcinomas in female rats, but the clinical significance of this effect is unknown. Intravenous penciclovir is being investigated for the treatment of mucocutaneous HSV infections in immunosuppressed patients.

FOSCARNET

Foscarnet (phosphonoformic acid) is a pyrophosphate-containing compound that potently inhibits herpesviruses, including [CMV](#). This drug inhibits DNA polymerases at the pyrophosphate binding site at concentrations that have relatively little effect on cellular polymerases. Foscarnet does not require phosphorylation to exert its antiviral activity and is therefore active against [HSV](#) and [VZV](#) isolates that are resistant to acyclovir because of deficiencies in thymidine kinase as well as against most ganciclovir-resistant strains of CMV. Foscarnet also inhibits the reverse transcriptase of HIV and is active against HIV in vivo.

Foscarnet is poorly soluble and must be administered intravenously via an infusion pump in a dilute solution over 1 to 2 h. The plasma half-life of foscarnet is 3 to 5 h and increases with decreasing renal function, since the drug is eliminated primarily by the kidneys. It has been estimated that 10 to 28% of a dose may be deposited in bone, where it can persist for months. The most common initial dosage of foscarnet -- 60 mg/kg every 8 h for 14 to 21 days -- is followed by a maintenance dose of 90 to 120 mg/kg once a day.

Foscarnet is approved for the treatment of [CMV](#) retinitis in patients with AIDS and of acyclovir-resistant mucocutaneous [HSV](#) infections. In a comparative clinical trial, the drug appeared to be about as efficacious as ganciclovir against CMV retinitis but was associated with a longer survival period, possibly because of its anti-HIV activity. Intraocular foscarnet has been used to treat CMV retinitis. Foscarnet has also been employed to treat acyclovir-resistant HSV and [VZV](#) infections as well as ganciclovir-resistant CMV infections, although resistance to foscarnet has been reported in CMV isolates obtained during therapy.

The major form of toxicity associated with foscarnet is renal impairment. Thus renal function should be monitored closely, particularly during the initial phase of therapy. Since foscarnet binds divalent metal ions, hypocalcemia, hypomagnesemia, hypokalemia, and hypo- or hyperphosphatemia can develop. Saline hydration and slow infusion appear to protect the patient against nephrotoxicity and electrolyte disturbances. Although hematologic abnormalities have been documented (most commonly anemia), foscarnet is not generally myelosuppressive and may be administered concomitantly with myelosuppressive medications such as zidovudine.

IDOXURIDINE

Idoxuridine inhibits the replication of herpesviruses and poxviruses. It was formerly used systemically to treat herpesvirus infections, but, because of associated toxicity and lack of proven efficacy, its systemic use has largely been abandoned. Topical idoxuridine is effective in the treatment of [HSV](#) keratitis, particularly in superficial infections, but has been supplanted by topically applied trifluridine and vidarabine (see below).

TRIFLURIDINE

Trifluridine is a pyrimidine nucleoside active against [HSV](#)-1, HSV-2, and [CMV](#). Trifluridine monophosphate irreversibly inhibits thymidylate synthetase, and trifluridine triphosphate inhibits viral and, to a lesser extent, cellular DNA polymerases. Because of systemic toxicity, its use is limited to topical therapy. Trifluridine is approved for treatment of HSV keratitis, for which trials have shown that it is more effective than topical idoxuridine but similarly effective to topical vidarabine. The drug has benefited some patients with HSV keratitis who have failed to respond to idoxuridine or vidarabine. Topical application of trifluridine to sites of acyclovir-resistant HSV mucocutaneous infections has also been beneficial in some cases.

VIDARABINE

Vidarabine is a purine nucleoside analogue with activity against [HSV](#)-1, HSV-2, [VZV](#), and [EBV](#). Vidarabine inhibits viral DNA synthesis through its 5¢-triphosphorylated metabolite, although its precise molecular mechanisms of action are not completely understood. Intravenously administered vidarabine has been shown to be effective in the treatment of herpes simplex encephalitis, mucocutaneous HSV infections and herpes zoster in immunocompromised patients and of neonatal HSV infections. Its use has been supplanted by intravenous acyclovir, which is more effective and easier to administer. Production of the intravenous preparation has been discontinued by the manufacturer, but vidarabine is available as an ophthalmic ointment, which is effective in the treatment of HSV keratitis.

OTHER ANTIVIRAL DRUGS

Lamivudine is a pyrimidine nucleoside analogue that is used primarily in combination therapy against HIV infection ([Chap. 309](#)). It is also active against [HBV](#) through inhibition of the viral DNA polymerase and has been approved for the treatment of chronic HBV infection. In one study, at doses of 100 mg/d for 1 year, lamivudine was well tolerated and resulted in suppression of HBV DNA levels, normalization of serum aminotransferase levels in most patients, and reduction of hepatic inflammation and fibrosis. Loss of hepatitis B e antigen (HBeAg) occurred in a minority of patients. Resistance to lamivudine developed in 15 to 36% of patients treated for 1 year and was associated with changes in the YMDD motif of HBV DNA polymerase. This is an important limitation of monotherapy with the drug. Studies of lamivudine as a component of combination therapy for hepatitis B are under way. Lamivudine also appears to be useful in the prevention or suppression of HBV infection associated with liver transplantation.

Lobucavir is a synthetic cyclobutane nucleoside analogue with activity against a broad

range of herpesviruses, HIV, and [HBV](#). It is currently under investigation in clinical trials. Its mechanism of action is through inhibition of viral DNA synthesis. Lobucavir is initially phosphorylated by virus-induced kinases, and lobucavir triphosphate is a potent inhibitor of [HSV](#), [CMV](#), and HBV DNA polymerases. Lobucavir can be administered orally or intravenously. It is excreted largely unmetabolized via the kidney and has a plasma half-life of 2 h after intravenous administration. Its oral availability is dose-dependent, ranging from 25 to 40% at doses of £200 mg and decreasing at higher doses. The preclinical toxicity profile of lobucavir appears to be similar to that of ganciclovir. However, neutropenia has been uncommonly encountered in studies in humans. Aminotransferase elevations have been documented after intravenous and (less commonly) oral administration. The most frequent adverse effects after oral administration are headache, insomnia, and gastrointestinal discomfort. In clinical trials to date, oral lobucavir has demonstrated antiviral effects in CMV- and HBV-infected patients as well as clinical benefits against HSV infections in immunocompetent patients. Short-term (1-month) oral administration of 200 mg of lobucavir bid or qid reduced serum HBV DNA levels in chronically infected patients. HBV DNA levels returned to pretreatment values when the drug was stopped, and studies of longer-term administration are under way. Lobucavir is under clinical investigation for use in HIV infection and in several herpesvirus infections.

Pleconaril is an investigational drug active in vitro against picornavirus replication, including over 90% of the most commonly isolated enterovirus types and 80% of rhinovirus serotypes. Its mechanism of action is through binding to a specific hydrophobic pocket in the viral capsid, which prevents attachment and/or uncoating of the virus. Pleconaril is poorly water soluble and is formulated as an oral suspension. After oral administration of 200- and 400-mg doses to adults, peak plasma concentrations are 1.1 and 2.4 µg/mL, respectively, and the terminal plasma half-life is 25 h. Pleconaril is generally well tolerated; the most frequently reported adverse effects are headache, nausea, diarrhea, and gastrointestinal discomfort, which have occurred at rates similar to those among placebo recipients. Orally administered pleconaril, given before and after experimental infection of healthy volunteers with coxsackievirus A21, reduced peak viral titers by >100-fold and decreased the subsequent rate of development of illness. Pleconaril treatment of adults with enteroviral meningitis decreased the overall duration of illness and headache and reduced the use of analgesics from that by placebo recipients. Clinical studies of pleconaril in other enterovirus-induced diseases are in progress.

INTERFERONS

Interferons are cytokines that exhibit a broad spectrum of antiviral activities as well as immunomodulating and antiproliferative properties. The [IFNs](#) are not available for oral administration but must be given intramuscularly, subcutaneously, or intravenously. Early studies with human leukocyte IFN demonstrated an effect in the prophylaxis of experimentally induced rhinovirus infections in humans and in the treatment of [VZV](#) infections in immunosuppressed patients. DNA recombinant technology has made available highly purified α, β, and γ IFNs that have been evaluated in a variety of viral infections. Results of such trials have confirmed the effectiveness of intranasally administered IFN in the prophylaxis of rhinovirus infections, although its use has been associated with nasal mucosal irritation. Studies have also demonstrated a beneficial

effect of intralesionally or systemically administered IFNs on genital warts. The effect of systemic administration consists primarily of a reduction in the size of lesions, and this mode of therapy may be useful in individuals who have numerous warts that cannot easily be treated by individual intralesional injection. However, lesions frequently recur after intralesional or systemic IFN therapy is discontinued.

Interferons have undergone extensive study in the treatment of chronic [HBV](#) infection. The administration of [IFN](#)-a2b (5 million units daily for 16 weeks) to patients with stable chronic HBV infection resulted in loss of markers of HBV replication, such as [HBeAg](#) and HBV DNA, in 33 to 37% of cases; 10 to 20% of patients also became negative for hepatitis B surface antigen. In >80% of patients who lose HBeAg and HBV DNA markers, serum aminotransferases return to normal levels, and both short- and long-term improvements in liver histopathology have been described. Predictors of a favorable response to therapy include low pretherapy levels of HBV DNA, high pretherapy serum levels of alanine aminotransferase (ALT), a short duration of chronic HBV infection, and active liver histopathology. Poor responses are seen in immunosuppressed patients, including those with HIV infection. Adverse effects of the above dose of IFN are common and include fever, chills, myalgia, fatigue, neurotoxicity (primarily manifested as somnolence and confusion), and leukopenia. Approximately 25% of patients receiving a daily dose of 5 million units require dose reduction, but fewer than 5% require discontinuation of therapy.

Several [IFN](#) preparations, including a2a, a2b, alfacon-1, and am1 (lymphoblastoid), have been studied as therapy for chronic [HCV](#) infections. A variety of regimens have been employed, of which the most common is IFN-a2b or -a2a at 3 million units three times per week for 12 months. A complete biochemical response, defined as a return to normal serum [ALT](#) values at the end of treatment, has been documented in ~54% of patients. In addition, liver biopsies have shown decreases in lobular and periportal inflammation. However, relapse has occurred in approximately half of all cases upon discontinuation of therapy, so that sustained responses were documented in 28% of cases. The addition of oral ribavirin to IFN-a2b -- either as initial therapy or after failure of interferon therapy alone -- resulted in significantly higher rates of sustained response (40 to 50%) than were obtained with monotherapy. Prognostic factors for a favorable response include an age of <45 years, a short duration of disease, low levels of HCV RNA, and infection with HCV genotypes other than 1. IFN alfacon, a synthetic "consensus" a interferon, appears to produce response rates similar to those elicited by IFN-a2a or -a2b and has recently been approved in the United States for the treatment of chronic hepatitis C.

Treatment of chronic hepatitis D with [IFN](#)-a apparently requires higher doses (5 million units daily or 10 million units three times per week) and is less effective than treatment of chronic hepatitis B or C. After 12 months of therapy, biochemical and virologic responses were detected in ~50% of patients, but few responses were sustained once therapy was stopped.

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SECTION 12 -DNA VIRUSES

182. HERPES SIMPLEX VIRUSES - Lawrence Corey

DEFINITION

Herpes simplex viruses (HSV-1, HSV-2; *Herpesvirus hominis*) produce a variety of infections involving mucocutaneous surfaces, the central nervous system (CNS), and -- on occasion -- visceral organs. The advent of effective chemotherapy for HSV infections has made prompt recognition of these syndromes even more clinically important than in the past.

ETIOLOGIC AGENT

The genome of [HSV](#) is a linear, double-stranded DNA molecule (molecular weight, ~100´10⁶) that encodes more than 75 gene products. The genomic structures of the two HSV subtypes are similar, and the overall sequence homology between HSV-1 and HSV-2 is ~50%. The homologous sequences are distributed over the entire genome map, and most of the polypeptides specified by one viral type are antigenically related to polypeptides of the other viral type. Many type-specific regions unique to HSV-1 and HSV-2 proteins do exist, however, and a number of them appear to be important in host immunity. These type-specific regions have been used to develop serologic assays that distinguish between the two viral subtypes. Restriction endonuclease analysis of viral DNA can be used to distinguish between the two subtypes and among strains of each subtype. The variability of nucleotide sequences from clinical strains of HSV-1 and HSV-2 is such that HSV isolates obtained from two individuals can be differentiated by restriction enzyme patterns unless the isolates are from epidemiologically related sources, such as sexual partners, mother-infant pairs, or persons involved in a common-source outbreak.

The viral genome is packaged in a regular icosahedral protein shell (capsid) composed of 162 capsomers. The outer covering of the virus is a lipid-containing membrane (envelope) derived from modified cell membrane and acquired as the DNA-containing capsid buds through the inner nuclear membrane of the host cell. Between the capsid and lipid bilayer of the envelope is the tegument. Viral replication has both nuclear and cytoplasmic phases. The initial steps of replication include attachment, fusion between the viral envelope and the cell membrane to liberate the nucleocapsid into the cytoplasm of the cell, and disassembly of the nucleocapsid to release the viral DNA. Replication of [HSV](#) is highly regulated. After fusion of the virion envelope with the host cell membrane, several viral proteins are released from the HSV virion. Some shut off host protein synthesis (by increasing cellular RNA degradation), while others "turn on" transcription of early genes of HSV replication. These early gene products, designated *a genes*, are required for synthesis of the subsequent polypeptide group, the *b polypeptides*, many of which are regulatory proteins and enzymes required for DNA replication. Most current antiviral drugs interfere with *b* proteins, such as the viral DNA polymerase enzyme. The third (*g*) class of HSV genes requires viral DNA replication for expression and constitutes most of the structural proteins specified by the virus.

After replication of the viral genome and synthesis of structural proteins, nucleocapsids

are assembled in the nucleus of the cell. Envelopment occurs as the nucleocapsids bud through the inner nuclear membrane into the perinuclear space. In some cells, viral replication in the nucleus forms two types of inclusion bodies: type A basophilic Feulgen-positive bodies that contain viral DNA and an eosinophilic inclusion body that is devoid of viral nucleic acid or protein and represents a "scar" of viral infection. Virions are then transported via the endoplasmic reticulum and the Golgi apparatus to the cell surface.

[HSV](#) infection of some neuronal cells does not result in cell death. Instead, viral genomes are maintained by the cell in a repressed state compatible with survival and normal activities of the cell, a condition called *latency*. Latency is associated with transcription of only a limited number of virus-encoded proteins. Subsequently, the viral genome may become activated, resulting in the normal pattern of regulated viral gene expression, replication, and release of HSV. The release of virus from the neuron and its subsequent entry into epithelial cells result in viral replication. This process is termed *reactivation*. Whereas infectious virus rarely can be recovered from sensory or autonomic nervous system ganglia dissected from cadavers, maintenance and growth of the neural cells in tissue culture result in production of infectious virions (*explantation*) and in subsequent permissive infection of susceptible cells (*cocultivation*). The fact that HSV replication was first detected in neurons during reactivation in vitro suggested that the neuron harbors the latent virus in vivo. Viral DNA and RNA have since been found in neural tissue at times when infectious virus cannot be isolated. Two RNA "latency-associated" transcripts that overlap the immediate early (α) gene products, called ICP-O, are found in abundance in the nuclei of latently infected neurons. These latency-associated transcripts code proteins in an antisense direction. Deletion mutants of this region that can become latent have been made. However, the efficiency of their later reactivation is reduced; thus, the antisense transcripts may play a role in maintaining rather than in establishing latency. At present, the molecular mechanisms of the latency of HSV-1 and HSV-2 are not well understood, and strategies to interrupt latency or to maintain molecular latency in neurons are not available.

PATHOGENESIS

Exposure to [HSV](#) at mucosal surfaces or abraded skin sites permits entry of the virus and initiation of its replication in cells of the epidermis and dermis. Investigators have identified several cell receptors that are ligands for HSV attachment proteins. Studies to define how these receptors influence viral replication and pathogenesis are under way. Initial HSV infection is often subclinical -- i.e., without clinically apparent lesions. Both clinical acquisition and subclinical acquisition are associated with sufficient viral replication to permit infection of either sensory or autonomic nerve endings. On entry into the neuronal cell, the virus -- or, more likely, the nucleocapsid -- is transported intraaxonally to the nerve cell bodies in ganglia. In humans, the interval from inoculation of virus in peripheral tissue to spread to the ganglia is unknown. During the initial phase of infection, viral replication occurs in ganglia and contiguous neural tissue. Virus then spreads to other mucosal skin surfaces through centrifugal migration of infectious virions via peripheral sensory nerves. This mode of spread helps explain the large surface area involved, the high frequency of new lesions distant from the initial crop of vesicles that is characteristic in patients with primary genital or oral-labial HSV infection, and the recovery of virus from neural tissue distant from neurons innervating the

inoculation site. Contiguous spread of locally inoculated virus also may take place and allow further mucosal extension of disease.

After the resolution of primary disease, infectious [HSV](#) can no longer be recovered in the ganglia. However, viral DNA can be found in 10 to 50% of ganglion cells in the anatomic region of the initial infection. Only ~1% of such cells express latency-associated transcripts of RNA detectable by current techniques. The mechanisms controlling the reactivation of HSV infection are unknown. Alterations in cellular transcripts that "maintain" latency are suspected, and several cellular protein kinases are under investigation. Experimentally, ultraviolet light, systemic and local immunosuppression, and trauma to the skin or ganglia are associated with reactivation.

Analysis of the DNA from sequentially isolated strains of [HSV](#) or from isolates from multiple infected ganglia in any one individual has revealed identical restriction endonuclease patterns in most persons. Occasionally (most frequently in immunocompromised persons), multiple strains of the same viral subtype are detected in one individual. This finding suggests that exogenous infection with different strains of the same subtype is possible although very uncommon.

IMMUNITY

Host responses to infection with [HSV](#) influence the acquisition of disease, the severity of infection, resistance to the development of latency, the maintenance of latency, and the frequency of recurrences. Both antibody-mediated and cell-mediated reactions are clinically important. Immunocompromised patients with defects in cell-mediated immunity experience more severe and more extensive HSV infections than those with deficits in humoral immunity, such as agammaglobulinemia. Experimental ablation of lymphocytes indicates that T cells play a major role in preventing lethal disseminated disease, although antibodies help reduce virus titers in neural tissue. Some of the clinical manifestations of HSV disease appear to be related to the host immune response (e.g., stromal opacities associated with recurrent herpetic keratitis). The surface viral glycoproteins have been shown to be antigens recognized by antibodies mediating neutralization and immune-mediated cytolysis (antibody-dependent cell-mediated cytotoxicity). Monoclonal antibodies specific for each of the known viral glycoproteins have, in experimental infections, conferred protection against subsequent neurologic disease or ganglionic latency. However, the use of subunit glycoprotein vaccines in humans has not been successful in reducing acquisition of infection, despite high titers of type-specific antibodies. Multiple cell populations, including natural killer cells, macrophages, a variety of T lymphocytes, and lymphokines generated by these cells, play a role in host defenses against HSV infections. In animals, passive transfer of primed lymphocytes confers protection from subsequent challenge. Maximum protection usually requires the activation of multiple T cell subpopulations, including cytotoxic T cells and T cells responsible for delayed hypersensitivity. The latter cells may confer protection by the antigen-stimulated release of lymphokines (e.g., interferons), which may have a direct antiviral effect and may activate and enhance a variety of specific and nonspecific effector cells. Increasing evidence suggests that HSV-specific CD8⁺ T cell responses are critical for clearance of virus from lesions. In addition, immunosuppressed patients with frequent and prolonged HSV lesions have fewer functional CD8⁺ T cells directed at HSV. The HSV virion contains a gene called unique

long gene no. 12 (UL-12) that can bind to the cellular transporter-activating protein TAP-1 and reduce the ability of this protein to bind HSV peptides to HLA class I, thereby reducing recognition of viral proteins by cytotoxic T cells of the host. This effect can be overcome by the addition of interferon γ , but this requires 24 to 48 h; thus, the virus has time to replicate and invade other host cells. Prior HSV-1 infection does not appear to reduce the frequency of acquisition of HSV-2 as measured by seroconversion. However, persons with prior HSV-1 infection who acquire HSV-2 appear to have a greater frequency of subclinical acquisition. These data suggest that type-specific immune responses are central to the control of HSV infection.

EPIDEMIOLOGY

Seroepidemiologic studies have documented [HSV](#) infections worldwide. Much of the humoral immune response to HSV is to type-common antigenic determinants. Serologic assays with whole-virus antigen preparations, such as complement fixation, neutralization, indirect immunofluorescence, passive hemagglutination, radioimmunoassay, and enzyme-linked immunosorbent assay, do not reliably distinguish between the two viral subtypes. Such assays are useful for differentiating uninfected (seronegative) persons from those with past HSV-1 or HSV-2 infection, but they do not reliably distinguish between the two subtypes. Serologic assays that identify antibodies to type-specific surface proteins of the two subtypes have been developed. These assays, which are based on the demonstration of antibodies to type-specific epitopes of the virus, can reliably distinguish between the human antibody responses to HSV-1 and HSV-2. The most commonly used assays are those that measure antibodies to glycoprotein G of HSV-1 (gG1) and HSV-2 (gG2). A western blot assay that can detect several HSV type-specific proteins can also be used.

Infection with [HSV](#)-1 is acquired more frequently and earlier than infection with HSV-2. More than 90% of adults have antibodies to HSV-1 by the fifth decade of life. In populations of low socioeconomic status, most persons acquire HSV-1 infection before the third decade of life.

Antibodies to [HSV](#)-2 are not detected routinely until puberty. Antibody prevalence rates correlate with past sexual activity and vary greatly among different population groups. Serosurveys indicate that nearly 22% of the United States population has antibodies to HSV-2 -- a 30% increase in the past 12 years. In most routine obstetric and family planning clinics, 25% of women have HSV-2 antibodies, although only 10% report a history of genital lesions. As many as 50% of heterosexual adults attending sexually transmitted disease clinics have antibodies to HSV-2. Antibody prevalence rates average about 5% higher among women than among men. Several studies suggest that much of this "asymptomatic" infection is largely unrecognized in that when "asymptomatic" seropositive persons are shown pictures of genital lesions, >60% subsequently identify episodes of symptomatic reactivation. Most important, these asymptomatic seropositive persons with reactivation shed virus on mucosal surfaces as frequently as those with symptomatic disease. The large reservoir of unidentified carriers of HSV-2 and the frequent asymptomatic reactivation of virus from the genital tract have fostered the continued spread of genital herpes throughout the world. HSV-2 infection is an independent risk factor for the acquisition and transmission of infection with HIV type 1. Among coinfecting persons, HIV-1 virions can be shed from herpetic

lesions of the genital region. This shedding may facilitate the spread of HIV through sexual contact.

[HSV](#) infections occur throughout the year. The incubation period ranges from 1 to 26 days (median, 6 to 8 days). Transmission can result from contact with persons with active ulcerative lesions or with persons without clinical manifestations of infection who are shedding HSV or on whose mucosal surfaces the virus is replicating. Studies using the polymerase chain reaction (PCR) have shown that HSV reactivation on mucosal surfaces is much more frequent than previously recognized. Among immunocompetent adults, HSV-2 can be isolated from the genital tract on 2 to 3% of days, and HSV DNA can be detected on 20 to 30% of days. Corresponding figures for HSV-1 in oral secretions are similar. Shedding rates are highest during the initial years of acquisition and may be as high as 30 to 50% of days during this period. Immunosuppressed patients shed HSV on mucosal sites at even higher frequency (20 to 50% of days). Daily antiviral chemotherapy can markedly reduce shedding rates. These data indicate that potential exposure to HSV from sexual or other close contact (kissing, sharing of glasses or silverware) is more common than has been thought. These shedding-rate data are consistent with the high seroprevalence of HSV infections worldwide.

CLINICAL SPECTRUM

[HSV](#) has been isolated from nearly all visceral or mucocutaneous sites. The clinical manifestations and course of HSV infection depend on the anatomic site involved, the age and immune status of the host, and the antigenic type of the virus. Primary HSV infections (i.e., first infections with either HSV-1 or HSV-2 in which the host lacks HSV antibodies in acute-phase serum) are frequently accompanied by systemic signs and symptoms, involve both mucosal and extramucosal sites, and have a longer duration of symptoms, a longer duration of virus isolation from lesions, and a higher rate of complications than recurrent episodes of disease. Both viral subtypes can cause genital and oral-facial infections, and the infections caused by the two subtypes are clinically indistinguishable. However, the frequency of reactivation of infection is influenced by anatomic site and virus type. Genital HSV-2 infection is twice as likely to reactivate and recurs 8 to 10 times more frequently than genital HSV-1 infection. Conversely, oral-labial HSV-1 infection recurs more frequently than oral-labial HSV-2 infection. Asymptomatic shedding rates follow the same pattern.

Oral-Facial Infections Gingivostomatitis ([Fig. 182-CD1](#)) and pharyngitis are the most frequent clinical manifestations of first-episode [HSV-1](#) infection, while recurrent herpes labialis ([Fig. 182-CD2](#)) is the most frequent clinical manifestation of reactivation HSV infection. HSV pharyngitis and gingivostomatitis usually result from primary infection and are most commonly seen in children and young adults. Clinical symptoms and signs, which include fever, malaise, myalgias, inability to eat, irritability, and cervical adenopathy, may last from 3 to 14 days. Lesions may involve the hard and soft palate, gingiva, tongue, lip, and facial area. HSV-1 or HSV-2 infection of the pharynx usually results in exudative or ulcerative lesions of the posterior pharynx and/or tonsillar pillars. Lesions of the tongue, buccal mucosa, or gingiva may occur later in the course in one-third of cases. Fever lasting from 2 to 7 days and cervical adenopathy are common. It can be difficult to differentiate HSV pharyngitis clinically from bacterial pharyngitis, *Mycoplasma pneumoniae* infections, and pharyngeal ulcerations of noninfectious

etiologies (e.g., Stevens-Johnson syndrome). No substantial evidence suggests that reactivation oral-labial HSV infection is associated with symptomatic recurrent pharyngitis.

Reactivation of [HSV](#) from the trigeminal ganglia may be associated with asymptomatic virus excretion in the saliva, development of intraoral mucosal ulcerations, or herpetic ulcerations on the vermilion border of the lip or external facial skin. About 50 to 70% of seropositive patients undergoing trigeminal nerve root decompression and 10 to 15% of those undergoing dental extraction develop oral-labial HSV infection a median of 3 days after these procedures.

In immunosuppressed patients, infection may extend into mucosal and deep cutaneous layers ([Fig. 182-CD3](#)). Friability, necrosis, bleeding, severe pain, and inability to eat or drink may result. The lesions of [HSV](#) mucositis are clinically similar to mucosal lesions caused by cytotoxic drug therapy, trauma, or fungal or bacterial infections. Persistent ulcerative HSV infections are among the most common infections in patients with AIDS. HSV and *Candida* infections often occur concurrently. Systemic antiviral therapy speeds the rate of healing and relieves the pain of mucosal HSV infections in immunosuppressed patients. The frequency of HSV reactivation during the early phases of transplantation or induction chemotherapy is high (50 to 90%), and prophylactic systemic antivirals such as intravenous acyclovir or penciclovir are used to reduce reactivation rates. Patients with atopic eczema also may develop severe oral-facial HSV infections (eczema herpeticum), which may rapidly come to involve extensive areas of skin and occasionally disseminate to visceral organs. Extensive eczema herpeticum has resolved promptly with the administration of intravenous acyclovir. Erythema multiforme (EM) also may be associated with HSV infections ([Plate IIE-67](#)); some evidence suggests that HSV infection is the precipitating event in ~75% of cases of cutaneous EM. HSV antigen has been demonstrated both in circulatory immune complexes and in skin lesion biopsy samples from these patients. Patients with severe HSV-associated EM are candidates for chronic suppressive oral antiviral therapy.

[HSV](#)-1 has been implicated in the etiology of Bell's palsy (flaccid paralysis of the mandibular portion of the facial nerve). Whether antiviral chemotherapy can alter the clinical course and complications of this infection is unclear.

Genital Infections First-episode primary genital herpes is characterized by fever, headache, malaise, and myalgias. Pain, itching, dysuria, vaginal and urethral discharge, and tender inguinal lymphadenopathy are the predominant local symptoms. Widely spaced bilateral lesions of the external genitalia are characteristic. Lesions may be present in varying stages, including vesicles, pustules, or painful erythematous ulcers. The cervix and urethra are involved in >80% of women with first-episode infections. First episodes of genital herpes in patients who have had prior [HSV](#)-1 infection are associated with less frequent systemic symptoms and faster healing than primary genital herpes. The clinical courses of acute first-episode genital herpes among patients with HSV-1 and HSV-2 infections are similar. However, the recurrence rates of genital disease ([Figs. 182-CD4, 182-CD5, 182-CD6](#)) differ with the viral subtype: the 12-month recurrence rates among patients with first-episode HSV-2 and HSV-1 infections are ~90% and ~55%, respectively (median number of recurrences, 4 and <1, respectively). Recurrence rates for genital HSV-2 infections vary greatly among individuals and over

time within the same individual. HSV has been isolated from the urethra and urine of men and women without external genital lesions. A clear mucoid discharge and dysuria are characteristics of symptomatic HSV urethritis. HSV has been isolated from the urethra of 5% of women with the dysuria-frequency syndrome. Occasionally, HSV genital tract disease is manifested by endometritis and salpingitis in women and by prostatitis in men. About 15% of cases of HSV-2 acquisition are associated with these nonlesional clinical syndromes, such as aseptic meningitis, cervicitis, or urethritis.

Both [HSV-1](#) and HSV-2 can cause symptomatic or asymptomatic rectal and perianal infections. HSV proctitis is usually associated with rectal intercourse. However, subclinical perianal shedding of HSV is detected both in heterosexual men and in women who report no rectal intercourse. This phenomenon is due to the establishment of latency in the sacral dermatome from prior genital tract infection, with subsequent reactivation in epithelial cells in the perianal region. Such reactivations are often subclinical. Symptoms of HSV proctitis include anorectal pain, anorectal discharge, tenesmus, and constipation. Sigmoidoscopy reveals ulcerative lesions of the distal 10 cm of the rectal mucosa. Rectal biopsies show mucosal ulceration, necrosis, polymorphonuclear and lymphocytic infiltration of the lamina propria, and (in occasional cases) multinucleated intranuclear inclusion-bearing cells. Perianal herpetic lesions are also found in immunosuppressed patients receiving cytotoxic therapy. Extensive perianal herpetic lesions and/or HSV proctitis is common among patients with HIV infection.

Herpetic Whitlow ([Fig. 182-CD7](#)) Herpetic whitlow -- [HSV](#) infection of the finger -- may occur as a complication of primary oral or genital herpes by inoculation of virus through a break in the epidermal surface or by direct introduction of virus into the hand through occupational or some other type of exposure. Clinical signs and symptoms include the abrupt onset of edema, erythema, and localized tenderness of the infected finger. Vesicular or pustular lesions of the fingertip that are indistinguishable from lesions of pyogenic bacterial infection are seen. Fever, lymphadenitis, and epitrochlear and axillary lymphadenopathy are common. The infection may recur. Prompt diagnosis (to avoid unnecessary and potentially exacerbating surgical therapy and/or transmission) is essential. Antiviral chemotherapy (to speed the healing of the process) is usually recommended (see below).

Herpes Gladiatorum [HSV](#) may infect almost any area of skin. Mucocutaneous HSV infections of the thorax, ears, face, and hands have been described among wrestlers. Transmission of these infections is facilitated by trauma to the skin sustained during wrestling. Prompt diagnosis and therapy are required to contain the spread of this infection.

Eye Infections [HSV](#) infection of the eye is the most frequent cause of corneal blindness in the United States. HSV keratitis presents with an acute onset of pain, blurring of vision, chemosis, conjunctivitis, and characteristic dendritic lesions of the cornea. Use of topical glucocorticoids may exacerbate symptoms and lead to involvement of deep structures of the eye. Debridement, topical antiviral treatment, and/or interferon therapy hastens healing. However, recurrences are common, and the deeper structures of the eye may sustain immunopathologic injury. Stromal keratitis due to HSV appears to be related to T cell-dependent destruction of deep corneal tissue. An HSV-1 epitope that is

autoreactive with T cell-targeting corneal antigens has been postulated to be a factor in this infection. Chorioretinitis, usually a manifestation of disseminated HSV infection, may occur in neonates or in patients with HIV infection. HSV and varicella-zoster virus can cause acute necrotizing retinitis as an uncommon but severe manifestation.

Central and Peripheral Nervous System Infections [HSV](#) accounts for 10 to 20% of all cases of sporadic viral encephalitis in the United States. The estimated incidence is about 2.3 cases per million persons per year. Cases are distributed throughout the year, and the age distribution appears to be biphasic, with peaks at 5 to 30 and >50 years of age. Subtype 1 virus causes >95% of cases of HSV encephalitis.

The pathogenesis of [HSV](#) encephalitis varies. In children and young adults, primary HSV infection may result in encephalitis; presumably, exogenously acquired virus enters the [CNS](#) by neurotropic spread from the periphery via the olfactory bulb. However, most adults with HSV encephalitis have clinical or serologic evidence of mucocutaneous HSV-1 infection before the onset of the CNS symptoms. In ~25% of the cases examined, the HSV-1 strains from the oropharynx and brain tissue of the same patient differ; thus some cases may result from reinfection with another strain of HSV-1 that reaches the CNS. Two theories have been proposed to explain the development of actively replicating HSV in localized areas of the CNS in persons whose ganglionic and CNS isolates are similar. Reactivation of latent HSV-1 infection in trigeminal or autonomic nerve roots may be associated with extension of virus into the CNS via nerves innervating the middle cranial fossa. HSV DNA has been demonstrated by DNA hybridization in brain tissue obtained at autopsy -- even from healthy adults. Thus, reactivation of long-standing latent CNS infection may be another mechanism for the development of HSV encephalitis.

The clinical hallmark of [HSV](#) encephalitis has been the acute onset of fever and focal neurologic (especially temporal-lobe) symptoms. Clinical differentiation of HSV encephalitis from other viral encephalitides, focal infections, or noninfectious processes is difficult. The most sensitive noninvasive method for early diagnosis of HSV encephalitis is the demonstration of HSV DNA in cerebrospinal fluid (CSF) by [PCR](#). Although titers of CSF and serum antibodies to HSV increase in most cases of HSV encephalitis, they rarely do so earlier than 10 days into the illness and therefore, while useful retrospectively, are generally not helpful in establishing an early clinical diagnosis. Demonstration of HSV antigen, HSV DNA, or HSV replication in brain tissue obtained by biopsy is highly sensitive and has a low complication rate; examination of such tissue also provides the best opportunity to identify alternative, potentially treatable causes of encephalitis. Antiviral chemotherapy reduces the rate of death from HSV encephalitis. Intravenous acyclovir is more effective than vidarabine. Even with therapy, however, neurologic sequelae are frequent, especially in persons over 35 years of age. Most authorities recommend the administration of intravenous acyclovir to patients with presumed HSV encephalitis until the diagnosis is confirmed or an alternative diagnosis is made.

[HSV](#) has been isolated from the [CSF](#) of 0.5 to 3% of patients presenting to the hospital with aseptic meningitis. HSV meningitis, which is usually seen in association with primary genital HSV infection, is an acute, self-limited disease manifested by headache, fever, and mild photophobia and lasting from 2 to 7 days. Lymphocytic pleocytosis in the

CSF is characteristic. Neurologic sequelae of HSV meningitis are rare. HSV is the most commonly identified cause of recurrent lymphocytic meningitis (Mollaret's meningitis). Demonstration of HSV antibodies in CSF or persistence of HSV DNA in CSF can establish the diagnosis. Daily administration of antiviral therapy aimed at reducing the likelihood of clinical HSV reactivation has been successful in such cases.

Autonomic nervous system dysfunction, especially of the sacral region, has been reported in association with both [HSV](#) and varicella-zoster virus infections. Numbness, tingling of the buttocks or perineal areas, urinary retention, constipation, [CSF](#) pleocytosis, and (in males) impotence may occur. Symptoms appear to resolve slowly over days to weeks. Occasionally, hypesthesia and/or weakness of the lower extremities may persist for many months. Rarely, transverse myelitis manifested by a rapidly progressive symmetric paralysis of the lower extremities or a Guillain-Barre syndrome may follow HSV infection. Similarly, peripheral nervous system involvement (Bell's palsy) or cranial polyneuritis also may be related to reactivation of HSV-1 infection. Transitory hypesthesia of the area of skin innervated by the trigeminal nerve and vestibular system dysfunction as measured by electronystagmography are the predominant signs of disease. Studies to determine whether antiviral chemotherapy may abort these signs or reduce their frequency and severity are unavailable.

Visceral Infections [HSV](#) infection of visceral organs usually results from viremia, and multiple-organ involvement is common. Occasionally, however, the clinical manifestations of HSV infection involve only the esophagus, lung, or liver. HSV esophagitis may result from direct extension of oral-pharyngeal HSV infection into the esophagus or may occur de novo by reactivation and spread of HSV to the esophageal mucosa via the vagus nerve. The predominant symptoms of HSV esophagitis are odynophagia, dysphagia, substernal pain, and weight loss. There are multiple oval ulcerations on an erythematous base with or without a patchy white pseudomembrane. The distal esophagus is most commonly involved. With extensive disease, diffuse friability may spread to the entire esophagus. Neither endoscopic nor barium examination can differentiate HSV esophagitis from *Candida* esophagitis or from esophageal ulcerations due to thermal injury, radiation, or corrosives. Endoscopically obtained secretions for cytologic examination and culture provide the most useful material for diagnosis. Systemic antiviral chemotherapy usually reduces symptoms and heals esophageal ulcerations.

[HSV](#) pneumonitis is uncommon except in severely immunosuppressed patients and may result from extension of herpetic tracheobronchitis into lung parenchyma. Focal necrotizing pneumonitis usually ensues. Hematogenous dissemination of virus from sites of oral or genital mucocutaneous disease also may occur and produce bilateral interstitial pneumonitis. Bacterial, fungal, and parasitic pathogens are commonly present in HSV pneumonitis. The mortality rate from untreated HSV pneumonia in immunosuppressed patients is high (>80%). HSV has also been isolated from the lower respiratory tract of persons with adult respiratory distress syndrome (ARDS). However, the relationship between the isolation of HSV and the pathogenesis of ARDS is unclear.

[HSV](#) is an uncommon cause of hepatitis in immunocompetent patients. HSV infection of the liver is associated with fever, abrupt elevations of bilirubin and serum aminotransferase levels, and leukopenia (<4000 white blood cells per microliter).

Disseminated intravascular coagulation also may develop.

Other reported complications of [HSV](#) infection include monarticular arthritis, adrenal necrosis, idiopathic thrombocytopenia, and glomerulonephritis. Disseminated HSV infection ([Fig. 182-CD8](#)) in immunocompetent patients is rare. In immunocompromised, burned, or malnourished patients, HSV occasionally disseminates to other visceral organs, such as the adrenal glands, pancreas, small and large intestines, and bone marrow. Rarely, primary HSV infection in pregnancy disseminates and may be associated with the death of both mother and fetus. This uncommon event is usually related to the acquisition of primary infection in the third trimester.

Neonatal HSV Infection Neonates (infants younger than 6 weeks) have the highest frequency of visceral and/or [CNS](#) infection of any [HSV](#)-infected patient population. Without therapy, the overall rate of death from neonatal herpes is 65%; fewer than 10% of neonates with CNS infection develop normally. Although skin lesions are the most commonly recognized features of disease, many infants do not develop lesions until well into the course of disease. Neonatal infection is usually acquired perinatally from contact with infected genital secretions at the time of delivery. Congenitally infected infants have been reported. In most series, 30% of neonatal HSV infections are due to HSV-1 and 70% to HSV-2. The risk of developing neonatal HSV infection is 10 times higher for an infant born to a mother who has recently acquired HSV than for other infants. Guidelines to evaluate routine serologic testing for HSV in pregnancy are being drafted and will serve as a basis on which to counsel "susceptible" (HSV-2-uninfected) women regarding the dangers of unprotected coitus and HSV-2 infection near term. Neonatal HSV-1 infections may also be acquired through postnatal contact with immediate family members who have symptomatic or asymptomatic oral-labial HSV-1 infection or through nosocomial transmission within the hospital. Antiviral chemotherapy has reduced the rate of death from neonatal herpes to 25%. However, the rate of morbidity, especially in infants with HSV-2 infection involving the CNS, is still very high.

DIAGNOSIS

Both clinical and laboratory criteria are useful for establishing the diagnosis of [HSV](#) infections. A clinical diagnosis can be made accurately when characteristic multiple vesicular lesions on an erythematous base are present. However, it is increasingly being recognized that herpetic ulcerations may clinically resemble skin ulcerations of other etiologies. Mucosal HSV infections may also present as urethritis or pharyngitis without cutaneous lesions. Thus, laboratory studies to confirm the diagnosis and to guide therapy are recommended. Staining of scrapings from the base of the lesions with Wright's, Giemsa's (Tzanck preparation; [Fig. 182-CD9](#)), or Papanicolaou's stain demonstrates characteristic giant cells or intranuclear inclusions of herpesvirus infection. These cytologic techniques are often useful as quick office procedures to confirm the diagnosis. Limitations of the cytologic method are that it does not differentiate between HSV and varicella-zoster virus infections, that it is relatively insensitive, and that the correct identification of giant cells requires experience.

[HSV](#) infection is best confirmed in the laboratory by isolation of virus in tissue culture or by demonstration of HSV antigens or DNA in scrapings from lesions. HSV causes a discernible cytopathic effect in a variety of cell culture systems, and most specimens

can be identified within 48 to 96 h after inoculation. Spin-amplified culture with subsequent staining for HSV antigen has shortened the time needed to identify HSV to <24 h. The sensitivity of viral isolation depends on the stage of lesions (with higher sensitivity in vesicular than in ulcerative lesions), on whether the patient has a first or a recurrent episode of the disease (with higher sensitivity in first than in recurrent episodes), and on whether the sample is from an immunosuppressed or an immunocompetent patient (with more antigen in immunosuppressed patients). Antigen detection procedures have approached viral isolation in terms of sensitivity in detecting HSV in genital or oral-labial lesions; however, antigen detection appears to be only ~50% as sensitive as viral isolation for the identification of HSV in cervical or salivary secretions of asymptomatic patients. [PCR](#) techniques appear to be more sensitive for HSV than viral isolation, especially for the diagnosis of [CNS](#) infections and for the detection of HSV as a cause of late-stage ulcerative lesions. Laboratory confirmation permits subtyping of the virus; information on subtype may be useful epidemiologically and may help to predict the frequency of reactivation after first-episode oral-labial or genital HSV infection.

Acute- and convalescent-phase serum can be useful in demonstrating seroconversion during primary [HSV](#)-1 or HSV-2 infection. However, only 5% of patients with recurrent mucocutaneous HSV infections have a fourfold or greater rise in titer of antibody to HSV in the interval between the collection of the first and second samples. Serologic assays, especially type-specific assays, should be used to identify asymptomatic carriers of HSV-1 or HSV-2 infection.

Several studies have shown that persons seropositive for [HSV](#)-2 to whom the clinical manifestations of HSV have been explained are able to identify symptomatic reactivations. Individuals seropositive for HSV-2 should be told about the high frequency of subclinical reactivation in mucosal surfaces not visible to the eye (e.g., cervix, urethra, perianal skin) or in microscopic ulcerations that may not be clinically symptomatic. Transmission of infection during such episodes is well established. HSV-2-seropositive persons should be educated about the high likelihood of subclinical shedding and the role condoms (male or female) may play in reducing transmission. Chronic antiviral therapy is being studied as a means of reducing the transmission of infection.

TREATMENT

Many aspects of mucocutaneous and visceral [HSV](#) infections are amenable to antiviral chemotherapy. For mucocutaneous infections, acyclovir and its congeners famciclovir and valacyclovir have been the mainstay of therapy. Several antiviral agents are available for topical use in HSV eye infections: idoxuridine, trifluorothymidine, topical vidarabine, and cidofovir. For HSV encephalitis and neonatal herpes, intravenous acyclovir is the treatment of choice.

All licensed antivirals for [HSV](#) inhibit the viral DNA polymerase. One class of drugs, typified by the drug acyclovir, is made up of substrates for the HSV enzyme thymidine kinase. Acyclovir, ganciclovir, famciclovir, and valacyclovir are all selectively phosphorylated to the monophosphate form in virus-infected cells. Cellular enzymes convert the monophosphate form of the drug to the triphosphate, which is then

incorporated into the viral DNA chain.

Acyclovir is the most frequently used agent for the treatment of [HSV](#) infections and is available in intravenous, oral, and topical formulations. Famciclovir, the oral formulation of penciclovir, is clinically effective in the treatment of a variety of HSV-1 and HSV-2 infections. Intravenous penciclovir is also available. Valacyclovir is the valyl ester of acyclovir and has greater bioavailability than acyclovir. Ganciclovir has activity against both HSV-1 and HSV-2; however, it is more toxic than acyclovir, valacyclovir, and famciclovir and is generally not recommended for the treatment of HSV infections.

All three compounds -- acyclovir, valacyclovir, and famciclovir -- have proven effective in shortening the duration of symptoms and lesions of mucocutaneous [HSV](#) infections in both immunocompromised and immunocompetent patients ([Table 182-1](#)). Intravenous and oral formulations prevent reactivation of HSV in seropositive immunocompromised patients during induction chemotherapy or in the period immediately after bone marrow or solid organ transplantation. Chronic daily suppressive therapy reduces the frequency of reactivation disease among patients with frequent genital or oral-labial herpes.

Intravenous acyclovir (30 mg/kg per day, given as a 10-mg/kg infusion over 1 h at 8-h intervals) is effective in reducing rates of death and morbidity from [HSV](#) encephalitis. Early initiation of therapy is a critical factor in outcome. The major side effect associated with intravenous acyclovir is transient renal insufficiency, usually due to crystallization of the compound in the renal parenchyma. This adverse reaction can be avoided if the medication is given slowly over 1 h and the patient is well hydrated. Because [CSF](#) levels of acyclovir average only 30 to 50% of plasma levels, the dosage of acyclovir used for treatment of [CNS](#) infection (30 mg/kg per day) is double that used for treatment of mucocutaneous or visceral disease (15 mg/kg per day).

Acyclovir-resistant strains of [HSV](#) have been identified. Most of these strains have an altered substrate specificity for phosphorylating acyclovir. Thus, cross-resistance to famciclovir and valacyclovir is usually found. Occasionally, an isolate with altered thymidine kinase (TK) specificity arises and is sensitive to famciclovir but not to acyclovir. In some patients infected with TK-deficient virus, higher doses of acyclovir are associated with clearing of lesions. In others, clinical disease progresses despite high-dose therapy. Almost all clinically significant acyclovir resistance has been seen in immunocompromised patients, and HSV-2 isolates are more often resistant than HSV-1 strains. A study by the Centers for Disease Control and Prevention indicated that ~5% of isolates from HIV-positive persons exhibit some degree of in vitro resistance to acyclovir. Isolation of HSV from persisting lesions despite adequate dosages and blood levels of acyclovir should raise the suspicion of acyclovir resistance. Therapy with the antiviral drug foscarnet is useful ([Chap. 181](#)). Because of its toxicity and cost, this drug is usually reserved for patients with extensive mucocutaneous infections. Cidofovir is a nucleotide analogue and exists as a phosphonate or monophosphate form. Most TK-deficient strains of HSV are sensitive to cidofovir. Cidofovir ointment speeds healing of acyclovir-resistant lesions. No well-controlled trials of systemic cidofovir have been reported. True TK-negative variants of HSV appear to have a reduced capacity to spread because of altered neurovirulence -- a feature important in the relatively infrequent presence of such strains in immunocompetent populations, even with increasing use of antivirals.

PREVENTION

The large reservoir of persons with asymptomatic [HSV](#)-1 and HSV-2 infections indicates that the success of efforts to control HSV disease through suppressive antiviral chemotherapy and/or educational programs will be limited. Rather, control of HSV infection will require the prevention of infection -- a goal most likely to be attained by vaccination. Several candidate vaccines are under investigation, and the prevention of HSV infection has been assigned a high public health priority.

Barrier forms of contraception, especially condoms, decrease the likelihood of transmission of [HSV](#) infection, especially during periods of asymptomatic viral excretion. When lesions are present, HSV infection may be transmitted by skin-to-skin contact despite the use of a condom. Nevertheless, the available data suggest that consistent condom use is an effective means of reducing the risk of genital HSV-2 transmission. Prevention of neonatal HSV requires the prevention of acquisition of HSV in the third trimester of pregnancy. Identification of women or couples susceptible to acquisition of HSV in pregnancy through serologic screening is receiving increasing attention, and such screening is being used with increasing frequency.

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183. VARICELLA-ZOSTER VIRUS INFECTIONS - *Richard J. Whitley*

DEFINITION

Varicella-zoster virus (VZV) causes two distinct clinical entities: varicella (chickenpox) and herpes zoster (shingles). Chickenpox, a ubiquitous and extremely contagious infection, is usually a benign illness of childhood characterized by an exanthematous vesicular rash. With reactivation of latent VZV (which is most common after the sixth decade of life), herpes zoster presents as a dermatomal vesicular rash, usually associated with severe pain.

ETIOLOGY

A clinical association between varicella and herpes zoster has been recognized for nearly 100 years. Early in the twentieth century, similarities in the histopathologic features of skin lesions resulting from varicella and herpes zoster were demonstrated. Viral isolates from patients with chickenpox and herpes zoster produced similar alterations in tissue culture -- specifically, the appearance of eosinophilic intranuclear inclusions and multinucleated giant cells. These results suggested that the viruses were biologically similar. Restriction endonuclease analyses of viral DNA from a patient with chickenpox who subsequently developed herpes zoster verified the molecular identity of the two viruses responsible for these different clinical presentations.

[VZV](#) is a member of the family Herpesviridae, sharing with other members such structural characteristics as a lipid envelope surrounding a nucleocapsid with icosahedral symmetry, a total diameter of approximately 180 to 200 nm, and centrally located double-stranded DNA that is about 125,000 bp in length.

PATHOGENESIS AND PATHOLOGY

Primary Infection Transmission is most likely to take place by the respiratory route; the subsequent localized replication of the virus at an undefined site (presumably the nasopharynx) leads to seeding of the reticuloendothelial system and ultimately to the development of viremia. Viremia in patients with chickenpox is reflected in the diffuse and scattered nature of the skin lesions and can be verified in selected cases by the recovery of [VZV](#) from the blood. Vesicles involve the corium and dermis, with degenerative changes characterized by ballooning, the presence of multinucleated giant cells, and eosinophilic intranuclear inclusions. Infection may involve localized blood vessels of the skin, resulting in necrosis and epidermal hemorrhage. With the evolution of disease, the vesicular fluid becomes cloudy because of the recruitment of polymorphonuclear leukocytes and the presence of degenerated cells and fibrin. Ultimately, the vesicles either rupture and release their fluid (which includes infectious virus) or are gradually reabsorbed.

Recurrent Infection The mechanism of reactivation of [VZV](#) that results in herpes zoster is unknown. Presumably, the virus infects the dorsal root ganglia during chickenpox, where it remains latent until reactivated. Histopathologic examination of representative dorsal root ganglia during active herpes zoster demonstrates hemorrhage, edema, and lymphocytic infiltration.

Active replication of [VZV](#) in other organs, such as the lung or the brain, can occur during either chickenpox or herpes zoster but is uncommon in the immunocompetent host. Pulmonary involvement is characterized by interstitial pneumonitis, multinucleated giant cell formation, intranuclear inclusions, and pulmonary hemorrhage. Central nervous system (CNS) infection leads to histopathologic evidence of perivascular cuffing similar to that encountered in measles and other viral encephalitides. Focal hemorrhagic necrosis of the brain, characteristic of herpes simplex virus encephalitis, is uncommon in VZV infection.

EPIDEMIOLOGY AND CLINICAL MANIFESTATIONS

Chickenpox Humans are the only known reservoir for [VZV](#). Chickenpox is highly contagious, with an attack rate of at least 90% among susceptible (seronegative) individuals. Persons of both sexes and all races are infected equally often. The virus is endemic in the population at large; however, it becomes epidemic among susceptible individuals during seasonal peaks -- namely, late winter and early spring in the temperate zone. Children between the ages of 5 and 9 are most commonly affected and account for 50% of all cases. Most other cases involve children aged 1 to 4 and those aged 10 to 14. Approximately 10% of the population of the United States over the age of 15 is susceptible to infection.

The incubation period of chickenpox ranges between 10 and 21 days but is usually between 14 and 17 days. Secondary attack rates in susceptible siblings within a household are between 70 and 90%. Patients are infectious approximately 48 h prior to the onset of the vesicular rash, during the period of vesicle formation (which generally lasts 4 to 5 days), and until all vesicles are crusted.

Clinically, chickenpox presents as a rash, low-grade fever, and malaise, although a few patients develop a prodrome 1 to 2 days before onset of the exanthem. In the immunocompetent patient, this is usually a benign illness that is associated with lassitude and with body temperatures of 37.8 to 39.4°C (100 to 103°F) of 3 to 5 days' duration. The skin lesions -- the hallmark of the infection -- include maculopapules, vesicles, and scabs in various stages of evolution ([Plate IID-36, Fig. 183-CD1](#)). These lesions, which evolve from maculopapules to vesicles over hours to days, appear on the trunk and face and rapidly spread to involve other areas of the body. Most are small and have an erythematous base with a diameter of 5 to 10 mm. Successive crops appear over a 2- to 4-day period. Lesions also can be found on the mucosa of the pharynx and/or the vagina. Their severity varies from one person to another. Some individuals have very few lesions, while others have as many as 2000. Younger children tend to have fewer vesicles than older individuals. Secondary and tertiary cases within families are associated with a relatively large number of vesicles. Immunocompromised patients -- both children and adults, particularly those with leukemia -- have lesions (often with a hemorrhagic base) that are more numerous and take longer to heal than those of immunocompetent patients. Immunocompromised individuals are also at greater risk for visceral complications, which occur in 30 to 50% of cases and are fatal 15% of the time.

The most common infectious complication of varicella is secondary bacterial superinfection of the skin, which is usually caused by *Streptococcus pyogenes* or

Staphylococcus aureus. This complication may result from excoriation of skin lesions after scratching. Gram's staining of skin lesions should help clarify the etiology of unusually erythematous and pustulated lesions.

The most common extracutaneous site of involvement in children is the [CNS](#). The syndrome of acute cerebellar ataxia and meningeal irritation generally appears around 21 days after the onset of the rash and rarely develops in the preeruptive phase. The cerebrospinal fluid (CSF) contains lymphocytes and elevated levels of protein. CNS involvement is a benign complication of [VZV](#) infection in children and generally does not require hospitalization. Aseptic meningitis, encephalitis, transverse myelitis, Guillain-Barre syndrome, and Reye's syndrome also can occur. Encephalitis is reported in 0.1 to 0.2% of children with chickenpox. Other than supportive care, no specific therapy is available for patients with CNS involvement.

Varicella pneumonia is the most serious complication following chickenpox, developing more commonly in adults (up to 20% of cases) than in children. It usually has its onset 3 to 5 days into the illness and is associated with tachypnea, cough, dyspnea, and fever. Cyanosis, pleuritic chest pain, and hemoptysis are frequent. Roentgenographic evidence of disease consists of nodular infiltrates and interstitial pneumonitis. Resolution of pneumonitis parallels improvement of the skin rash; however, patients may have persistent fever and compromised pulmonary function for weeks.

Other complications of chickenpox include myocarditis, corneal lesions, nephritis, arthritis, bleeding diatheses, acute glomerulonephritis, and hepatitis. Hepatic involvement, distinct from Reye's syndrome and usually asymptomatic, is common in chickenpox and is usually characterized by elevated levels of liver enzymes, particularly aspartate and alanine aminotransferases.

Perinatal varicella is associated with a high mortality rate when maternal disease develops within 5 days before delivery or within 48 h thereafter. Because the newborn does not receive protective transplacental antibodies and has an immature immune system, the illness may be unusually severe. The reported mortality rate has been as high as 30% in this group. Congenital varicella, with clinical manifestations of limb hypoplasia, cicatricial skin lesions, and microcephaly at birth, is extremely uncommon.

Herpes Zoster Herpes zoster, a sporadic disease, is the consequence of reactivation of latent [VZV](#) from the dorsal root ganglia. Most patients have no history of recent exposure to other individuals with VZV infection. Herpes zoster occurs at all ages, but its incidence is highest (5 to 10 cases per 1000 persons) among individuals in the sixth through the eighth decades of life. Recurrent herpes zoster is exceedingly rare except in immunocompromised hosts, especially those with AIDS.

Herpes zoster, also called shingles, is characterized by a unilateral vesicular eruption within a dermatome, often associated with severe pain. The dermatomes from T3 to L3 are most frequently involved. If the ophthalmic branch of the trigeminal nerve is involved, zoster ophthalmicus results. The factors responsible for the reactivation of [VZV](#) are not known. In children reactivation is usually benign, whereas in adults it can be debilitating. The continuum of pain from onset to resolution is known as *zoster-associated pain*. The onset of disease is heralded by pain within the dermatome

that may precede lesions by 48 to 72 h; an erythematous maculopapular rash evolves rapidly into vesicular lesions. In the normal host, these lesions may remain few in number and continue to form only for a period of 3 to 5 days. The total duration of disease is generally between 7 and 10 days; however, it may take as long as 2 to 4 weeks for the skin to return to normal. In a few patients, characteristic localization of pain to a dermatome with serologic evidence of herpes zoster has been reported in the absence of skin lesions. When branches of the trigeminal nerve are involved, lesions may appear on the face, in the mouth, in the eye, or on the tongue. In the Ramsay Hunt syndrome ([Plate IID-35](#)), pain and vesicles appear in the external auditory canal, and patients lose their sense of taste in the anterior two-thirds of the tongue while developing ipsilateral facial palsy. The geniculate ganglion of the sensory branch of the facial nerve is involved.

The most debilitating complication of herpes zoster, in both the normal and the immunocompromised host, is pain associated with acute neuritis and postherpetic neuralgia. Postherpetic neuralgia is uncommon in young individuals; however, at least 50% of patients over age 50 with zoster report some degree of pain in the involved dermatome months after the resolution of cutaneous disease. Changes in sensation in the dermatome, resulting in either hypo- or hyperesthesia, are common.

[CNS](#) involvement may follow localized herpes zoster. Many patients without signs of meningeal irritation have [CSF](#) pleocytosis and moderately elevated levels of CSF protein. Symptomatic meningoencephalitis is characterized by headache, fever, photophobia, meningitis, and vomiting. A rare manifestation of CNS involvement is granulomatous angiitis with contralateral hemiplegia, which can be diagnosed by cerebral arteriography. Other neurologic manifestations include transverse myelitis with or without motor paralysis.

Like chickenpox, herpes zoster is more severe in the immunocompromised host than in the normal individual. Lesions continue to form for over a week, and scabbing is not complete in most cases until 3 weeks into the illness. Patients with Hodgkin's disease and non-Hodgkin's lymphoma are at greatest risk for progressive herpes zoster. Cutaneous dissemination ([Plate IID-37, Fig. 183-CD2](#)) develops in about 40% of these patients. Among patients with cutaneous dissemination ([Fig. 183-CD3](#)), the risk of pneumonitis, meningoencephalitis, hepatitis, and other serious complications is increased by 5 to 10%. However, even in immunocompromised patients, disseminated zoster is rarely fatal.

Patients who have received a bone marrow transplant are at particularly high risk of [VZV](#) infection. Thirty percent of cases of posttransplantation VZV infection occur within 1 year (50% of these within 9 months); 45% of the patients involved have cutaneous or visceral dissemination. The mortality rate in this situation is 10%. Postherpetic neuralgia, scarring, and bacterial superinfection are especially frequent in VZV infections occurring within 9 months of transplantation. Among infected patients, concomitant graft-versus-host disease increases the chance of dissemination and/or death.

DIFFERENTIAL DIAGNOSIS

The diagnosis of chickenpox is not difficult. The characteristic rash and a history of recent exposure should lead to a prompt diagnosis. Other viral infections that can mimic chickenpox include disseminated herpes simplex virus infection in patients with atopic dermatitis and the disseminated vesiculopapular lesions sometimes associated with coxsackievirus infection, echovirus infection, or atypical measles. However, these rashes are more commonly morbilliform with a hemorrhagic component rather than vesicular or vesiculopustular. Rickettsialpox can be confused with chickenpox; however, it can be distinguished easily by detection of the "herald spot" at the site of the mite bite and the development of a more pronounced headache. Serologic testing is also useful in differentiating rickettsialpox from varicella.

Unilateral vesicular lesions in a dermatomal pattern should lead rapidly to the diagnosis of herpes zoster, although the occurrence of shingles without a rash has been reported. Both herpes simplex virus infections and coxsackievirus infections can cause dermatomal vesicular lesions. Supportive diagnostic virology and fluorescent staining of skin scrapings with monoclonal antibodies are helpful in ensuring the proper diagnosis. In the prodromal stage of herpes zoster, the diagnosis can be exceedingly difficult and may be made only after lesions have appeared or by retrospective serologic assessment.

LABORATORY FINDINGS

Unequivocal confirmation of the diagnosis is possible only through the isolation of [VZV](#) in susceptible tissue-culture cell lines, the demonstration of either seroconversion or a fourfold or greater rise in antibody titer between convalescent- and acute-phase serum specimens, or the detection of VZV DNA by polymerase chain reaction (PCR). A rapid impression can be obtained by a Tzanck smear, with scraping of the base of the lesions in an attempt to demonstrate multinucleated giant cells, although the sensitivity of this method is low (about 60%). PCR technology for the detection of viral DNA in vesicular fluid is available in a limited number of diagnostic laboratories. Direct immunofluorescent staining of cells from the lesion base or detection of viral antigens by other assays (such as the immunoperoxidase assay) is also useful, although these tests are not commercially available. The most frequently employed serologic tools for assessing host response are the immunofluorescent detection of antibodies to VZV membrane antigens, the fluorescent antibody to membrane antigen (FAMA) test, immune adherence hemagglutination, and enzyme-linked immunosorbent assay (ELISA). The FAMA test and the ELISA appear to be the most sensitive.

PROPHYLAXIS

While chickenpox in the otherwise healthy host is relatively benign, it can cause morbidity and death. Furthermore, the parents of a child with chickenpox often lose a significant amount of time from work. A live attenuated varicella vaccine has been licensed and is recommended for administration to all immunocompetent children and adults at risk of infection.

The immunocompromised individual is at significant risk for developing progressive varicella; modalities of prevention include passive immunization or experimental administration of the same live attenuated vaccine used in the immunocompetent child.

Immune prophylaxis can consist of the administration of specific zoster immune globulin (ZIG) derived from patients with herpes zoster, varicella-zoster immune globulin (VZIG), or the intravenous formulation of zoster immune plasma (ZIP). Both ZIG and VZIG should be given within 96 h (preferably within 72 h) of exposure to ensure efficacy. It is likely that ZIP can be given somewhat later. Indications for the administration of VZIG are summarized in [Table 183-1](#).

TREATMENT

Medical management of chickenpox in the immunologically normal host is directed toward the prevention of avoidable complications. Obviously, good hygiene includes daily bathing and soaks. Secondary bacterial infection of the skin can be avoided by meticulous skin care, particularly with close cropping of fingernails. Pruritus can be decreased with topical dressings or the administration of antipruritic drugs. Tepid water baths and wet compresses are better than drying lotions for the relief of itching. Aluminum acetate soaks for the management of herpes zoster can be both soothing and cleansing. Administration of aspirin to children with chickenpox should be avoided because of the association of aspirin derivatives with the development of Reye's syndrome. Acyclovir therapy (800 mg by mouth five times daily for 5 to 7 days) is recommended for adolescents and adults with chickenpox of ≤ 24 h duration. Likewise, acyclovir therapy may be of benefit to children <12 years of age if initiated early in the disease (<24 h) at a dose of 20 mg/kg every 6 h.

Patients with herpes zoster benefit from oral antiviral therapy, as evidenced by accelerated healing of lesions and resolution of zoster-associated pain with acyclovir, valacyclovir, or famciclovir. Acyclovir, now off patent, is administered at a dosage of 800 mg five times daily for 7 to 10 days. Famciclovir, the prodrug of penciclovir, is at least as effective as acyclovir and perhaps more so. One study showed twofold faster resolution of postherpetic neuralgia in famciclovir-treated patients with zoster than in recipients of placebo. The dose is 500 mg by mouth three times daily for 7 days. Valacyclovir, the prodrug of acyclovir, accelerates healing and resolution of zoster-associated pain more promptly than acyclovir. The dose is 1 g by mouth three times daily for 5 to 7 days. Both famciclovir and valacyclovir offer the advantage of a lower dosing frequency than acyclovir.

In the immunocompromised host, both chickenpox and herpes zoster (including disseminated disease) should be treated with intravenous acyclovir, which reduces the occurrence of visceral complications but has no effect on healing of skin lesions or pain. The dose is 10 to 12.5 mg/kg every 8 h for 7 days. Oral acyclovir therapy is not recommended for the treatment of [VZV](#) infections in immunocompromised patients. Concomitant with the administration of intravenous acyclovir, it is desirable to attempt to wean these patients from immunosuppressive treatment.

Patients with varicella pneumonia may require removal of bronchial secretions and ventilatory support. Persons with zoster ophthalmicus should be referred immediately to an ophthalmologist. Therapy for this condition consists of the administration of analgesics for severe pain and the use of atropine. Acyclovir accelerates healing.

The management of acute neuritis and/or postherpetic neuralgia can be particularly

difficult. In addition to the judicious use of analgesics, ranging from nonnarcotics to narcotic derivatives, drugs such as gabapentin, amitriptyline hydrochloride, and fluphenazine hydrochloride have been reported to be beneficial for pain relief. In one study, glucocorticoid therapy administered early in the course of localized herpes zoster significantly accelerated such quality-of-life improvements as a return to usual activity and termination of analgesia. The dose of prednisone administered orally was 60 mg/d on days 1 through 7, 30 mg/d on days 8 through 14, and 15 mg/d on days 15 through 21. This regimen is appropriate only for relatively healthy elderly persons who have moderate or severe pain at presentation. Patients with osteoporosis, diabetes mellitus, glycosuria, or hypertension may not be appropriate candidates. Glucocorticoids should not be used without concomitant antiviral therapy.

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184. EPSTEIN-BARR VIRUS INFECTIONS, INCLUDING INFECTIOUS MONONUCLEOSIS - Jeffrey I. Cohen

DEFINITION

Epstein-Barr virus (EBV) is the cause of heterophile-positive infectious mononucleosis (IM), which is characterized by fever, sore throat, lymphadenopathy, and atypical lymphocytosis. EBV is also associated with several human tumors, including nasopharyngeal carcinoma, Burkitt's lymphoma, Hodgkin's disease, and -- in patients with immunodeficiencies (including AIDS) -- B cell lymphoma. The virus, a member of the family Herpesviridae, consists of a linear, double-stranded DNA core surrounded by an icosahedral nucleocapsid and by the viral envelope, which contains glycoproteins. The two types of EBV that are widely prevalent in nature are not distinguishable by conventional serologic tests.

EPIDEMIOLOGY

[EBV](#) infections occur worldwide. These infections are most common in early childhood, with a second peak during late adolescence. By adulthood, more than 90% of individuals have been infected and have antibodies to the virus. [IM](#) is usually a disease of young adults. In lower socioeconomic groups and in areas of the world with lower standards of hygiene (e.g., developing countries), EBV tends to infect children at an early age, and symptomatic IM is uncommon. In areas with higher standards of hygiene (e.g., the United States), infection with EBV is often delayed until adulthood, and IM is more prevalent.

[EBV](#) is spread by contact with oral secretions. The virus is frequently transmitted from asymptomatic adults to infants and among young adults by transfer of saliva during kissing. Transmission by less intimate contact is rare. EBV has been transmitted by blood transfusion and by bone marrow transplantation. Studies indicate that more than 90% of asymptomatic seropositive individuals shed the virus in oropharyngeal secretions.

PATHOGENESIS

[EBV](#) is transmitted by salivary secretions. The virus infects the epithelium of the oropharynx and the salivary glands and is shed from these cells. While B cells may become infected after contact with epithelial cells, studies suggest that lymphocytes in the tonsillar crypts can be infected directly. The virus then spreads through the bloodstream. The proliferation and expansion of EBV-infected B cells along with reactive T cells during [IM](#) result in enlargement of lymphoid tissue. Polyclonal activation of B cells leads to the production of antibodies to host-cell and viral proteins. During the acute phase of IM, up to 1 in every 100 B cells in the peripheral blood is infected by EBV, while after recovery, about 1 in every million B cells is infected. During IM there is an inverted CD4+/CD8+ T cell ratio. The percentage of CD4+ T cells decreases, while there are large clonal expansions of CD8+ T cells; up to 40% of CD8+ T cells are directed against EBV antigens during acute infection. Data suggest that memory B cells, not epithelial cells, are the reservoir for EBV in the body: Shedding of EBV from the oropharynx stops but the virus persists in B cells when patients are treated with

acyclovir.

The [EBV](#) receptor (CD21), present on the surface of B cells and epithelial cells, is also the receptor for the C3d component of complement. EBV infection of epithelial cells results in viral replication and production of virions. When B cells are infected by EBV in vitro, they become transformed and can proliferate indefinitely. During latent infection of B cells, only the EBV nuclear antigens (EBNAs), latent membrane proteins (LMPs), and small EBV RNAs are expressed in vitro. EBV-transformed B cells secrete immunoglobulin; only a small fraction of cells produce virus.

Cellular immunity is more important than humoral immunity in controlling [EBV](#) infection. In the initial phase of infection, suppressor T cells, natural killer cells, and nonspecific cytotoxic T cells are important in controlling the proliferation of EBV-infected B cells. Levels of markers of T cell activation and serum interferon γ are elevated. Later in infection, HLA-restricted cytotoxic T cells that recognize [EBNAs](#) and [LMPs](#) and destroy EBV-infected cells are generated. Studies have shown that one of the late genes expressed during EBV replication, *BCRF1*, is a homologue of interleukin 10 and can inhibit the production of interferon γ by mononuclear cells in vitro. In addition, EBNA-1 inhibits antigen processing.

If T cell immunity is compromised, [EBV](#)-infected B cells may begin to proliferate. When EBV is associated with lymphoma, virus-induced proliferation is but one step in a multistep process of neoplastic transformation. In many EBV-containing tumors, LMP-1 mimics members of the tumor necrosis factor receptor family (e.g., CD40), transmitting growth-proliferating signals.

CLINICAL MANIFESTATIONS

Most [EBV](#) infections in infants and young children either are asymptomatic or present as mild pharyngitis with or without tonsillitis. In contrast, up to 75% of infections in adolescents present as [IM](#).

Signs and Symptoms The incubation period for [IM](#) in young adults is about 4 to 6 weeks. A prodrome of fatigue, malaise, and myalgia may last for 1 to 2 weeks before the onset of fever, sore throat, and lymphadenopathy. Fever is usually low-grade and is most common in the first 2 weeks of the illness; however, it may persist for over a month. Common signs and symptoms are listed along with their frequencies in [Table 184-1](#). Lymphadenopathy and pharyngitis are most prominent during the first 2 weeks of the illness, while splenomegaly is more prominent during the second and third weeks. Lymphadenopathy most often affects the posterior cervical nodes but may be generalized. Enlarged lymph nodes are frequently tender and symmetric but are not fixed in place. Pharyngitis, often the most prominent sign, can be accompanied by enlargement of the tonsils with an exudate resembling that of streptococcal pharyngitis. A morbilliform or papular rash, usually on the arms or trunk, develops in about 5% of cases. Most patients treated with ampicillin develop a macular rash ([Fig. 186-CD1](#)); this rash is not predictive of future adverse reactions to penicillins. Erythema nodosum and erythema multiforme have also been described ([Chap. 57](#)). Most patients have symptoms for 2 to 4 weeks, but malaise and difficulty concentrating can persist for months.

Symptomatic [IM](#) is uncommon in infants and young children. IM in the elderly presents relatively often as nonspecific symptoms, including prolonged fever, fatigue, myalgia, and malaise; in contrast, pharyngitis, lymphadenopathy, splenomegaly, and atypical lymphocytes are relatively rare in elderly patients.

Laboratory Findings The white blood cell count is usually elevated and peaks at 10,000 to 20,000/uL during the second or third week of illness. Lymphocytosis is usually demonstrable, with more than 10% atypical lymphocytes. The latter cells are enlarged lymphocytes that have abundant cytoplasm, vacuoles, and indentations of the cell membrane. CD8+ cells predominate among the atypical lymphocytes. Low-grade neutropenia and thrombocytopenia are common during the first month of illness. Liver function is abnormal in more than 90% of cases. Serum levels of aminotransferases and alkaline phosphatase are usually mildly elevated; the serum concentration of bilirubin is elevated in about 40% of cases.

Complications Most cases of [IM](#) are self-limited. Deaths are very rare and most often are due to central nervous system (CNS) complications, splenic rupture, upper airway obstruction, or bacterial superinfection.

When [CNS](#) complications develop, they usually do so during the first 2 weeks of [EBV](#) infection; in some patients, especially children, they are the only clinical manifestations of [IM](#). Heterophile antibodies and atypical lymphocytes may be absent. Meningitis and encephalitis are the most common neurologic abnormalities, and patients may present with headache, meningismus, or cerebellar ataxia; acute hemiplegia and psychosis have also been described. The cerebrospinal fluid (CSF) contains mainly lymphocytes, with occasional atypical lymphocytes. Most cases resolve without neurologic sequelae. Acute EBV infection has also been associated with cranial nerve palsies (especially ones involving cranial nerve VII), Guillain-Barre syndrome, acute transverse myelitis, and peripheral neuritis.

Autoimmune hemolytic anemia occurs in about 2% of cases during the first 2 weeks. In most cases the anemia is Coombs'-test positive, with cold agglutinins directed against the i red blood cell antigen. Most patients with hemolysis have mild anemia that lasts for 1 or 2 months, but some patients have severe disease with hemoglobinuria and jaundice. Nonspecific antibody responses may also include rheumatoid factor, antinuclear antibodies, anti-smooth muscle antibodies, antiplatelet antibodies, and cryoglobulins. [IM](#) has been associated with red-cell aplasia, severe granulocytopenia, thrombocytopenia, pancytopenia, and hemophagocytic syndrome. The spleen ruptures in fewer than 0.5% of cases. Splenic rupture is more common among males than among females and may be manifest as abdominal pain, referred shoulder pain, or hemodynamic compromise.

Hypertrophy of lymphoid tissue in the tonsils or adenoids can result in upper airway obstruction, as can inflammation and edema of the epiglottis, pharynx, or uvula. About 10% of patients with [IM](#) develop streptococcal pharyngitis after their initial sore throat resolves.

Other rare complications associated with acute [EBV](#) infection include hepatitis (which can

be fulminant), myocarditis or pericarditis with electrocardiographic changes, pneumonia with pleural effusion, interstitial nephritis, genital ulcerations, and vasculitis.

OTHER DISEASES ASSOCIATED WITH EBV INFECTION

[EBV](#)-associated lymphoproliferative disease has been described in patients with congenital or acquired immunodeficiency, including those with severe combined immunodeficiency or AIDS, recipients of bone marrow transplants, and recipients of organ transplants who are receiving immunosuppressive drugs (especially cyclosporine). Proliferating EBV-infected B cells infiltrate lymph nodes and multiple organs, and patients present with fever and lymphadenopathy or gastrointestinal symptoms. Pathologic studies show B cell hyperplasia or poly- or monoclonal lymphoma. The X-linked lymphoproliferative syndrome (Duncan's disease) is a recessive disorder of young boys who have a normal response to childhood infections but develop fatal lymphoproliferative disorders after infection with EBV. The gene mutated in this syndrome, SAP or SH2D1A, has been identified; its product binds to a protein that mediates interactions of B and T cells. Most patients with this syndrome die of acute [IM](#); others develop hypogammaglobulinemia, malignant B cell lymphomas, aplastic anemia, or agranulocytosis. IM has also proved fatal to some patients with no obvious preexisting immune abnormality.

Oral hairy leukoplakia ([Plate IID-42](#)) is an early manifestation of infection with HIV in adults ([Chap. 309](#)). Most patients present with raised, white corrugated lesions on the tongue (and occasionally on the buccal mucosa) that contain [EBV](#) DNA. Children infected with HIV can develop lymphoid interstitial pneumonitis; EBV DNA is often found in lung tissue from these patients.

Patients with the chronic fatigue syndrome may have titers of antibody to [EBV](#) that are elevated but are not significantly different from those in healthy EBV-seropositive adults. While some patients have malaise and fatigue that persist for weeks or months after [IM](#), persistent EBV infection is not a cause of the chronic fatigue syndrome. Chronic active EBV infection is very rare and is distinct from the chronic fatigue syndrome. The affected patients have an illness lasting more than 6 months with markedly elevated titers of antibody to EBV and evidence of organ involvement, including hepatosplenomegaly, lymphadenopathy, and pneumonitis, uveitis, or neurologic disease.

[EBV](#) is associated with several malignancies. About 15% of cases of Burkitt's lymphoma in the United States and about 90% of those in Africa are associated with EBV ([Chap. 112](#)). African patients with Burkitt's lymphoma have high levels of antibody to EBV, and their tumor tissue usually contains viral DNA. EBV-containing Burkitt's lymphoma also occurs in patients with AIDS. Anaplastic nasopharyngeal carcinoma is uniformly associated with EBV; the affected tissues contain viral DNA and antigens. Patients with nasopharyngeal carcinoma often have elevated titers of antibody to EBV ([Chap. 87](#)).

[EBV](#) has been associated with Hodgkin's disease, especially the mixed-cellularity type ([Chap. 112](#)). Patients with Hodgkin's disease often have elevated titers of antibody to EBV, and in about half of cases viral DNA and antigens are found in Reed-Sternberg cells. In some cases, EBV DNA has been detected in tonsillar carcinoma,

angioimmunoblastic lymphadenopathy, angiocentric nasal NK/T cell immunoproliferative lesions, T cell lymphoma, thymoma, gastric carcinoma, and [CNS](#) lymphoma from patients with no underlying immunodeficiency. Studies have demonstrated viral DNA in leiomyosarcomas from AIDS patients and in smooth-muscle tumors from organ transplant recipients. Virtually all CNS lymphomas in AIDS patients are associated with EBV.

DIAGNOSIS

Serologic Testing The heterophile test is used for the diagnosis of [IM](#) in children and adults ([Table 184-2](#)). Heterophile antibody is an IgM antibody that does not bind [EBV](#) proteins. In the test for this antibody, human serum is absorbed with guinea pig kidney, and the heterophile titer is defined as the greatest serum dilution that agglutinates sheep, horse, or cow erythrocytes. A titer of 40-fold or greater is diagnostic of acute EBV infection in a patient who has symptoms compatible with IM and atypical lymphocytes. Tests for heterophile antibodies are positive in 40% of patients with IM during the first week of illness and in 80 to 90% during the third week. Therefore, repeated testing may be necessary, especially if the initial test is performed early. Tests usually remain positive for 3 months after the onset of illness, but heterophile antibodies can persist for up to 1 year. These antibodies usually are not detectable in children <5 years of age, in the elderly, or in patients presenting with symptoms not typical of IM. The commercially available monospot test for heterophile antibodies is somewhat more sensitive than the classic heterophile test. False-positive results in the monospot test are more common in children and in patients with other viral infections.

[EBV](#)-specific antibody testing is used for patients with suspected acute EBV infection who lack heterophile antibodies and for patients with atypical infections. Serologic tests are particularly useful in young children, who often do not develop heterophile antibodies. Titers of IgM and IgG antibodies to viral capsid antigen (VCA) are elevated in the serum of more than 90% of patients at the onset of disease. IgM antibody to VCA is useful for the diagnosis of acute [IM](#) because it is present at elevated titers only during the first 2 months of the disease; in contrast, IgG antibody to VCA is often used to assess exposure to EBV in the past because it persists for life.

Antibodies to early antigens (EAs) are found either in a diffuse pattern in the nucleus and cytoplasm of infected cells (EA-D antibody) or restricted to the cytoplasm (EA-R antibody). These antibodies are detectable 3 to 4 weeks after the onset of symptoms in patients with [IM](#). About 70% of individuals with IM, especially those with relatively severe disease, have EA-D antibodies during the course of their illness. These antibodies usually persist for only 3 to 6 months. Levels of EA-D antibodies are also elevated in patients with nasopharyngeal carcinoma or chronic active [EBV](#) infection. EA-R antibodies are only occasionally detected in patients with IM but are often found at elevated titers in patients with African Burkitt's lymphoma or chronic active EBV infection.

IgA antibodies to [EBV](#) antigens have proved useful for the identification of patients with nasopharyngeal carcinoma and of persons at high risk for the disease. Seroconversion to [EBNA](#) positivity is also useful for the diagnosis of acute infection with EBV. Antibodies to EBNA are detectable relatively late (3 to 6 weeks after the onset of symptoms) in nearly all cases of acute EBV infection and persist for the lifetime of the patient. These

antibodies may be lacking in immunodeficient patients and in those with chronic active EBV infection.

Other Studies Detection of [EBV](#) DNA, RNA, or proteins has been valuable in demonstrating the association of the virus with various malignancies. The polymerase chain reaction has been used to detect EBV DNA in the [CSF](#) of some AIDS patients with lymphomas and to monitor the amount of EBV DNA in the blood of patients with lymphoproliferative disease. Culture of EBV from throat washings or blood is not helpful in the diagnosis of acute infection, since EBV commonly persists in the oropharynx and in B cells for the lifetime of the infected individual.

Differential Diagnosis The differential diagnosis of [IM](#) and atypical lymphocytosis includes acute infection with cytomegalovirus, *Toxoplasma*, HIV, human herpesvirus 6, and hepatitis viruses as well as drug hypersensitivity reactions. Cytomegalovirus is the most common cause of heterophile-negative mononucleosis, usually involves older patients, and is associated with a lower frequency of sore throat, splenomegaly, and lymphadenopathy than IM due to [EBV](#). Other diseases that share some of the features of IM include rubella, acute infectious lymphocytosis in children, and lymphoma or leukemia.

TREATMENT

Therapy for [IM](#) consists of supportive measures, with rest and analgesia. Excessive physical activity during the first month should be avoided to reduce the possibility of splenic rupture. If splenic rupture occurs, splenectomy is required. Glucocorticoid therapy is not indicated for uncomplicated IM and in fact may predispose to bacterial superinfection. Prednisone (40 to 60 mg/d for 2 to 3 days, with subsequent tapering of the dose over 1 to 2 weeks) has been used for the prevention of airway obstruction in patients with severe tonsillar hypertrophy, for autoimmune hemolytic anemia, and for severe thrombocytopenia. Glucocorticoids have also been used in a few selected patients with severe malaise and fever and in patients with severe [CNS](#) or cardiac disease.

Acyclovir has had no significant clinical impact on [IM](#) in controlled trials. In one study, the combination of acyclovir and prednisolone had no significant effect on the duration of symptoms of IM. Acyclovir, at a dosage of 400 to 800 mg five times daily, has been effective for the treatment of oral hairy leukoplakia (despite common relapses) and some cases of chronic active [EBV](#) disease. This agent generally has not been beneficial for patients with lymphoproliferative syndromes. When possible, therapy for EBV lymphoproliferative disease should be directed toward the reduction of immunosuppressive medication. New therapies, including the use of interferon and the infusion of donor T cells or EBV-specific cytotoxic T cells, are being studied.

The isolation of patients with [IM](#) is unnecessary. Vaccines directed against the major [EBV](#) glycoprotein have been effective in animal studies and are currently undergoing small-scale clinical trials.

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185. CYTOMEGALOVIRUS AND HUMAN HERPESVIRUS TYPES 6, 7, AND 8 - *Martin S. Hirsch*

CYTOMEGALOVIRUS

DEFINITION

Cytomegalovirus (CMV), which was initially isolated from patients with congenital cytomegalic inclusion disease, is now recognized as an important pathogen in all age groups. In addition to inducing severe birth defects, CMV causes a wide spectrum of disorders in older children and adults, ranging from an asymptomatic, subclinical infection to a mononucleosis syndrome in healthy individuals to disseminated disease in immunocompromised patients. Human CMV is one of several related species-specific viruses that cause similar diseases in various animals. All are associated with the production of characteristic enlarged cells -- hence the name *cytomegalovirus*.

[CMV](#) is a member of the β -herpesvirus group and has double-stranded DNA, a protein capsid, and a lipoprotein envelope. Like other herpesviruses, CMV demonstrates icosahedral symmetry, replicates in the cell nucleus, and can cause either a lytic and productive or a latent infection. CMV can be distinguished from other herpesviruses by certain biologic properties, such as host range and type of cytopathology induced. Viral replication is associated with the production of large intranuclear inclusions and smaller cytoplasmic inclusions. The virus appears to replicate in a variety of cell types in vivo; in tissue culture it grows preferentially in fibroblasts. Although there is little evidence that CMV is oncogenic in vivo, the virus does transform fibroblasts in rare instances, and genomic transforming fragments have been identified.

EPIDEMIOLOGY

[CMV](#) has a worldwide distribution. Approximately 1% of newborns in the United States are infected with CMV, and the percentage is higher in many less developed countries. Communal living and poor personal hygiene facilitate early spread. Perinatal and early childhood infections are common. Virus may be present in milk, saliva, feces, and urine. Transmission of CMV has been identified among young children in day-care centers and has been traced from infected toddler to pregnant mother to developing fetus. When an infected child introduces CMV into a household, 50% of susceptible family members seroconvert within 6 months.

The virus is not readily spread by casual contact but requires repeated or prolonged intimate exposure for transmission. In late adolescence and young adulthood, [CMV](#) is often transmitted sexually, and asymptomatic viral carriage in semen or cervical secretions is common. CMV antibody is present at detectable levels in nearly 100% of female prostitutes and sexually active homosexual men. Sexually active adults may harbor several strains of CMV simultaneously. Transfusion of whole blood or certain blood products containing viable leukocytes may also transmit CMV, with a frequency of 0.14 to 10% per unit transfused.

Once infected, an individual probably carries [CMV](#) for life. The infection usually remains latent. However, CMV reactivation syndromes develop frequently when T

lymphocyte-mediated immunity is compromised -- for example, after organ transplantation or in association with lymphoid neoplasms and certain acquired immunodeficiencies (in particular, infection with HIV; [Chap. 309](#)). Most primary CMV infections in organ transplant recipients ([Chap. 136](#)) result from transmission of the virus in the graft itself. In CMV-seropositive transplant recipients, infection results from reactivation of latent virus or, less commonly, from reinfection by a new strain of CMV. CMV infection may be associated with coronary artery stenosis following heart transplantation or coronary angioplasty, but this association requires further validation.

PATHOGENESIS

Congenital [CMV](#) infection can result from either primary or reactivation infection of the mother. However, clinical disease in the fetus or newborn is almost exclusively related to primary maternal infection ([Table 185-1](#)). The factors determining the severity of congenital infection are unknown; a deficient capacity to produce precipitating antibodies and to mount T cell responses to CMV is associated with relatively severe disease.

Primary infection in late childhood or adulthood is often associated with a vigorous T lymphocyte response that may contribute to the development of a mononucleosis syndrome similar to that observed following Epstein-Barr virus (EBV) infection ([Chap. 184](#)). The hallmark of such infection is the appearance of atypical lymphocytes in the peripheral blood; these cells are predominantly activated CD8+ T lymphocytes. Polyclonal activation of B cells by the virus contributes to the development of rheumatoid factors and other autoantibodies during [CMV](#) mononucleosis.

Once acquired by symptomatic or asymptomatic primary infection, [CMV](#) persists indefinitely in tissues of the host. The sites of persistent or latent infection are unclear but probably include multiple cell types and various organs. Transmission following blood transfusion or organ transplantation is due to silent infections in these tissues. Autopsy studies suggest that salivary glands and bowel may be areas of latent infection.

If the host's T cell responses become compromised by disease or by iatrogenic immunosuppression, latent virus can be reactivated to cause a variety of syndromes. Chronic antigenic stimulation in the presence of immunosuppression (for example, following tissue transplantation) appears to be an ideal setting for [CMV](#) activation and CMV-induced disease. Certain particularly potent suppressants of T cell immunity, such as antithymocyte globulin, are associated with a high rate of clinical CMV syndromes, which may follow either primary or reactivation infection. CMV may itself contribute to further T lymphocyte hyporesponsiveness, which often precedes superinfection with other opportunistic pathogens, such as *Pneumocystis carinii*. CMV and *P. carinii* are frequently found together in immunosuppressed patients with severe interstitial pneumonia. CMV may function as a cofactor to activate latent HIV infection.

PATHOLOGY

Cytomegalic cells in vivo (presumed to be infected epithelial cells) are two to four times larger than surrounding cells and often contain an 8- to 10-um intranuclear inclusion that is eccentrically placed and is surrounded by a clear halo, producing an "owl's eye"

appearance. Smaller granular cytoplasmic inclusions are demonstrated occasionally. Cytomegalic cells are found in a wide variety of organs, including salivary gland, lung, liver, kidney, intestine, pancreas, adrenal gland, and the central nervous system.

The cellular inflammatory response to infection consists of plasma cells, lymphocytes, and monocyte-macrophages. Granulomatous reactions occasionally develop, particularly in the liver. Immunopathologic reactions may contribute to [CMV](#) disease. Immune complexes have been detected in infected infants, sometimes in association with CMV-related glomerulopathies. Immune-complex glomerulopathy has been observed in some CMV-infected patients after renal transplantation.

CLINICAL MANIFESTATIONS

Congenital CMV Infection Fetal infections range from inapparent to severe and disseminated. Cytomegalic inclusion disease develops in approximately 5% of infected fetuses and is seen almost exclusively in infants born to mothers who develop primary infections during pregnancy. Petechiae, hepatosplenomegaly, and jaundice are the most common presenting features (60 to 80% of cases). Microcephaly with or without cerebral calcifications, intrauterine growth retardation, and prematurity are reported in 30 to 50% of cases. Inguinal hernias and chorioretinitis are less common. Laboratory abnormalities include elevated alanine aminotransferase levels, thrombocytopenia, conjugated hyperbilirubinemia, hemolysis, and elevated cerebrospinal fluid protein levels. The prognosis for severely infected infants is poor; the mortality rate is 20 to 30%, and few of the patients who survive escape intellectual or hearing difficulties in later years. The differential diagnosis of cytomegalic inclusion disease in infants includes syphilis, rubella, toxoplasmosis, infection with herpes simplex virus or enterovirus, and bacterial sepsis.

Most congenital [CMV](#) infections are clinically inapparent at birth. Between 5 and 25% of asymptomatically infected infants develop significant psychomotor, hearing, ocular, or dental abnormalities over the next several years.

Perinatal CMV Infection The newborn may acquire [CMV](#) at the time of delivery by passage through an infected birth canal or by postnatal contact with maternal milk or other secretions. Approximately 40 to 60% of infants who are breast-fed for longer than 1 month by seropositive mothers become infected. Iatrogenic transmission also can result from neonatal blood transfusion. Screening of blood products before they are transfused into low-birth-weight seronegative infants or into seronegative pregnant women decreases the risk of infection.

The great majority of infants infected at or after delivery remain asymptomatic. However, protracted interstitial pneumonitis has been associated with perinatally acquired [CMV](#) infection, particularly in premature infants, and occasionally has been accompanied by infection with *Chlamydia trachomatis*, *P. carinii*, or *Ureaplasma urealyticum*. Poor weight gain, adenopathy, rash, hepatitis, anemia, and atypical lymphocytosis may also be found, and CMV excretion often persists for months or years.

CMV Mononucleosis The most common clinical manifestation of [CMV](#) infection in

normal hosts beyond the neonatal period is a heterophil antibody-negative mononucleosis syndrome. This manifestation may develop spontaneously or may follow the transfusion of leukocyte-containing blood products. Although the syndrome occurs at all ages, it most often involves sexually active young adults. Incubation periods range from 20 to 60 days, and the illness generally lasts for 2 to 6 weeks. Prolonged high fevers, sometimes accompanied by chills, profound fatigue, and malaise, characterize this disorder. Myalgias, headache, and splenomegaly are frequent, but in CMV mononucleosis (as opposed to infectious mononucleosis caused by [EBV](#)), exudative pharyngitis and cervical lymphadenopathy are rare. Occasional patients develop rubelliform rashes, often after exposure to ampicillin. Less commonly observed are interstitial or segmental pneumonia, myocarditis, pleuritis, arthritis, and encephalitis. In rare cases, Guillain-Barre syndrome complicates CMV mononucleosis. The characteristic laboratory abnormality is relative lymphocytosis in peripheral blood, with more than 10% atypical lymphocytes. Total leukocyte counts may be low, normal, or markedly elevated. Although significant jaundice is uncommon, serum aminotransferase and alkaline phosphatase levels are often moderately elevated. Heterophil antibodies are absent; however, transient immunologic abnormalities are common and may include the presence of cryoglobulins, rheumatoid factors, cold agglutinins, and antinuclear antibodies. Hemolytic anemia, thrombocytopenia, and granulocytopenia complicate recovery in rare instances.

Most patients recover without sequelae, although postviral asthenia may persist for months. The excretion of [CMV](#) in urine, genital secretions, and/or saliva often continues for months or years. Rarely, CMV infection is fatal in immunocompetent hosts; even when such patients survive, they can have recurrent episodes of fever and malaise that are sometimes associated with autonomic nervous system dysfunction (e.g., attacks of sweating or flushing).

CMV Infection in the Immunocompromised Host (See also [Table 185-1](#)) CMV appears to be the most common and important viral pathogen complicating organ transplantation ([Chap. 136](#)). In recipients of kidney, heart, lung, and liver transplants, CMV induces a variety of syndromes, including fever and leukopenia, hepatitis, pneumonitis, esophagitis, gastritis, colitis, and retinitis. CMV disease may be an independent risk factor for both graft loss and death. The period of maximal risk is between 1 and 4 months after transplantation, although retinitis may be a later complication. The risk of disease appears to be greater after primary infection than after reactivation. In addition, molecular studies indicate that seropositive transplant recipients are susceptible to reinfection with donor-derived, genotypically variant CMV, and such infection often results in disease. Reactivation infection, although frequent, is less likely than primary infection to be important clinically. Clinical disease is related to various factors, such as the degree of immunosuppression; patients receiving certain immunosuppressive agents, such as antithymocyte globulin, appear to be more likely to have severe infections than those receiving other agents, such as cyclosporine. The transplanted organ is particularly vulnerable as a target for CMV infection; thus, there is a tendency for CMV hepatitis to follow liver transplantation and for CMV pneumonitis to follow lung transplantation.

[CMV](#) pneumonia occurs in 15 to 20% of bone marrow transplant recipients, with a case-fatality rate of 84 to 88%. The risk is greatest between 5 and 13 weeks after

transplantation, and the several risk factors identified include certain types of immunosuppressive therapy, acute graft-versus-host disease, older age, viremia, and seropositivity before transplantation.

[CMV](#) is recognized as an important pathogen in patients with advanced HIV infection ([Chap. 309](#)), in whom it often causes retinitis or disseminated disease, particularly when peripheral-blood CD4⁺ cell counts fall below 50 to 100/uL. As treatment for underlying HIV infection has improved, the incidence of serious CMV infections (e.g., retinitis) has decreased. However, institution of highly active antiretroviral regimens sometimes leads to acute flare-ups of CMV retinitis during the first few weeks of therapy.

Syndromes produced by [CMV](#) in the immunocompromised host often begin with prolonged fever, malaise, anorexia, fatigue, night sweats, and arthralgias or myalgias. Liver function abnormalities, leukopenia, thrombocytopenia, and atypical lymphocytosis may be observed during these episodes. The development of tachypnea, hypoxia, and unproductive cough signals respiratory involvement. Radiologic examination of the lung often demonstrates bilateral interstitial or reticulonodular infiltrates, which begin in the periphery of the lower lobes and spread centrally and superiorly; localized segmental, nodular, or alveolar patterns are less common. The differential diagnosis includes infection with *P. carinii*; infections due to other viral, bacterial, or fungal pathogens; pulmonary hemorrhage; and injury secondary to irradiation or to treatment with cytotoxic drugs.

Gastrointestinal [CMV](#) involvement may be localized or extensive and almost exclusively affects compromised hosts. Ulcers of the esophagus, stomach, small intestine, or colon may result in bleeding or perforation. CMV infection may lead to exacerbations of underlying ulcerative colitis. Hepatitis occurs frequently, particularly following liver transplantation, and CMV-associated acalculous cholecystitis and adrenalitis have been described.

[CMV](#) rarely causes meningoencephalitis in otherwise healthy individuals. Two forms of CMV encephalitis are seen in patients with AIDS. One resembles HIV encephalitis and presents as progressive dementia; the other is a ventriculoencephalitis characterized by cranial-nerve deficits, nystagmus, disorientation, lethargy, and ventriculomegaly. In immunocompromised patients, CMV can also cause subacute progressive polyradiculopathy, which is often reversible if recognized and treated promptly.

[CMV](#) retinitis is an important cause of blindness in immunocompromised patients, particularly patients with advanced AIDS ([Chap. 309](#)). Early lesions consist of small, opaque, white areas of granular retinal necrosis that spread in a centrifugal manner and are later accompanied by hemorrhages, vessel sheathing, and retinal edema (see [Plate IV-2](#)). CMV retinopathy must be distinguished from that due to other conditions, including toxoplasmosis, candidiasis, and herpes simplex virus infection.

Fatal [CMV](#) infections are often associated with persistent viremia and the involvement of multiple organ systems. Progressive pulmonary infiltrates, pancytopenia, hyperamylasemia, and hypotension are characteristic features that are frequently found in conjunction with a terminal bacterial, fungal, or protozoan superinfection. Extensive adrenal necrosis with CMV inclusions is often documented at autopsy, as is CMV

involvement of many other organs.

DIAGNOSIS

The diagnosis of [CMV](#) infection usually cannot be made reliably on clinical grounds alone. Isolation of the virus or detection of CMV antigens or DNA from appropriate clinical specimens, together with demonstration of a fourfold or greater rise in antibody titers or persistently elevated antibody titers, is the preferred diagnostic approach. Virus excretion or viremia is readily detected by culture of appropriate specimens on human fibroblast monolayers. If viral titers are high, as is frequently the case in congenital disseminated infection or in patients with AIDS, characteristic cytopathic effects may be detected within a few days. However, in some situations -- such as CMV mononucleosis -- viral titers are low, and cytopathic effects may take several weeks to appear. Many laboratories expedite diagnosis with an overnight tissue-culture method (shell vial assay) involving centrifugation and an immunocytochemical detection technique employing monoclonal antibodies to an immediate-early CMV antigen. Isolation of virus from urine or saliva does not, by itself, constitute proof of acute infection, since excretion from these sites may continue for months or years after illness. Detection of CMV viremia is a better predictor of acute infection.

Detection of [CMV](#) antigens (pp65) in peripheral-blood leukocytes or of CMV DNA in blood or tissues may hasten the diagnosis of CMV disease in certain populations, including organ transplant recipients and persons with AIDS. Such assays may yield a positive result several days earlier than culture methods. The detection of CMV DNA in cerebrospinal fluid by the polymerase chain reaction is useful in the diagnosis of CMV encephalitis or polyradiculopathy.

A variety of serologic assays are available to detect increases in titers of antibody to [CMV](#) antigens. An increased antibody level may not be detectable for up to 4 weeks after primary infection, and titers often remain high for years after infection. For this reason, single-sample antibody determinations are of no value in assessing the acuteness of infection. Detection of CMV-specific IgM is sometimes useful in the diagnosis of recent or active infection; circulating rheumatoid factors may result in occasional false-positive IgM tests.

TREATMENT

Several prophylactic measures are useful for the prevention of [CMV](#) infection in patients at high risk. The use of blood from seronegative donors or of blood that has been frozen, thawed, and deglycerolized greatly decreases the rate of transfusion-associated transmission of CMV. Similarly, matching of organ or bone marrow transplants by CMV serology, using organs only from seronegative donors for seronegative recipients, reduces rates of primary infection following transplantation. Both live attenuated and CMV subunit vaccines have been evaluated, but neither is close to approval for general use.

[CMV](#) immune globulin has been reported to reduce rates of occurrence of CMV-associated syndromes and of fungal or parasitic superinfections among seronegative renal transplant recipients. Studies in bone marrow transplant recipients

have produced conflicting results. Prophylactic acyclovir has been demonstrated to reduce rates of CMV infection and disease in certain seronegative renal transplant recipients; acyclovir is not effective in the treatment of active CMV disease, however.

Ganciclovir is a guanosine derivative that has considerably more activity against [CMV](#) than its congener acyclovir. After intracellular conversion by a viral phosphotransferase encoded by CMV gene region UL97, ganciclovir triphosphate is a selective inhibitor of CMV DNA polymerase. Several clinical studies have indicated response rates of 70 to 90% among patients with AIDS given ganciclovir for the treatment of CMV retinitis or colitis. In bone marrow transplant recipients with CMV pneumonia, ganciclovir is less effective when given alone, but it elicits a favorable clinical response 50 to 70% of the time when it is combined with CMV immune globulin. Prophylactic or suppressive ganciclovir may be useful in high-risk bone marrow or organ transplant recipients (e.g., those who are CMV-seropositive before transplantation or who are CMV culture-positive afterward). In many patients with AIDS, persistently low CD4+ cell counts, and CMV disease, clinical and virologic relapses occur promptly if treatment with ganciclovir is discontinued. Therefore, prolonged maintenance regimens are recommended for such patients. Resistance to ganciclovir is common among patients treated for more than 3 months and is usually related to mutations in the CMV UL97 gene.

Ganciclovir therapy for [CMV](#) retinitis consists of a 14- to 21-day induction course (5 mg/kg intravenously twice a day) followed by a prolonged intravenous or oral maintenance regimen. For parenteral maintenance, the dose is 5 mg/kg daily or 6 mg/kg 5 days per week. Peripheral-blood neutropenia develops in 16 to 29% of treated patients but is often ameliorated by granulocyte or granulocyte-macrophage colony-stimulating factor. Oral ganciclovir at a high dose (3 g/d) can also be used for maintenance, although the blood levels achieved are insufficient for acute induction regimens. Although progression (as assessed by funduscopy) is more rapid with oral than with intravenous ganciclovir maintenance (mean time to progression, 68 vs. 96 days; $p = .03$), the ease of administration and reduced toxicity of the oral preparation may make it an acceptable alternative for some patients who do not have sight-threatening central retinitis. The use of oral ganciclovir as prophylaxis in high-risk AIDS patients (i.e., those with CD4+ cell counts of $<100/\mu\text{L}$) has been studied in two placebo-controlled trials, with somewhat contradictory results.

Foscarnet (sodium phosphonoformate) also acts against [CMV](#) infection by inhibiting viral DNA polymerase. Because this agent does not require phosphorylation to be active, it is also effective against most ganciclovir-resistant CMV isolates. A comparative trial of foscarnet and ganciclovir in 234 patients with AIDS and CMV retinitis demonstrated equivalent activity against retinitis but longer survival (12.6 vs. 8.5 months) in the foscarnet group. Although the reasons for the latter difference are unclear, the antiretroviral activity of foscarnet and the greater use of zidovudine by foscarnet recipients are strong possibilities. Foscarnet is less well tolerated than ganciclovir and causes considerable toxicity, including renal dysfunction, hypomagnesemia, hypokalemia, hypocalcemia, genital ulcers, dysuria, nausea, and paresthesia. Moreover, foscarnet administration requires the use of an infusion pump and close clinical monitoring. With aggressive hydration and dose adjustments for renal dysfunction, the toxicity of foscarnet can be reduced. The use of foscarnet should be avoided when a saline load cannot be tolerated (e.g., in cardiomyopathy). The approved

induction regimen is 60 mg/kg every 8 h for 2 weeks, although 90 mg/kg every 12 h is equally effective and no more toxic. Maintenance infusions should deliver 90 to 120 mg/kg once daily; no oral preparation is available. Foscarnet-resistant viruses may emerge during extended therapy.

Ganciclovir may also be administered via a slow-release pellet sutured into the eye. Although this intraocular device provides good local protection, contralateral eye disease and disseminated disease are not affected, and early retinal detachment is possible. A combination of intraocular and systemic therapy may be better than the intraocular implant alone.

Cidofovir is a nucleotide analogue with a long intracellular half-life that allows intermittent intravenous administration. Induction regimens of 5 mg/kg weekly for 2 weeks are followed by maintenance regimens of 3 to 5 mg/kg every 2 weeks. Cidofovir can cause severe nephrotoxicity through dose-dependent proximal tubular cell injury; however, this adverse effect can be ameliorated somewhat by saline hydration and probenecid.

HUMAN HERPESVIRUS TYPES 6, 7, AND 8

Human herpesvirus (HHV) type 6 was first isolated in 1986 from peripheral-blood leukocytes of six persons with various lymphoproliferative disorders. The virus has a worldwide distribution, and two genetically distinct variants (HHV-6A and HHV-6B) are now recognized.

Infection with [HHV-6](#) frequently develops during infancy as maternal antibody wanes. Congenital infections have also been described. HHV-6 (mostly variant B) can cause exanthem subitum ([Fig. 18-CD2](#)) (roseola infantum), a common illness characterized by fever with subsequent rash. HHV-6 is also a major cause of febrile seizures without rash during infancy. In older age groups, HHV-6 has been associated with mononucleosis syndromes, focal encephalitis, and (in immunocompromised hosts) pneumonitis and disseminated disease. In transplant recipients, HHV-6 infection may be associated with graft dysfunction. As many as 80% of adults are seropositive for HHV-6. The virus may be transmitted by saliva and possibly by genital secretions. There is no established treatment or vaccine.

[HHV-7](#) was isolated in 1990 from T lymphocytes from the peripheral blood of a healthy 26-year-old man. Other isolates have since been obtained. It appears that the virus is frequently acquired during childhood and is frequently present in the saliva of healthy adults. No human disease has yet been definitively linked to HHV-7, although some cases of exanthem subitum ([Fig. 18-CD2](#)) and other childhood febrile illnesses have been associated with HHV-7 infection. An association has been made between HHV-7 and pityriasis rosea, but further studies must confirm this relationship.

Unique herpesvirus-like DNA sequences were reported during 1994 and 1995 in tissues derived from Kaposi's sarcoma and body cavity-based lymphoma occurring in patients with AIDS. When subjected to representational-difference analyses, more than 90% of Kaposi's sarcoma tissue samples were found to contain these sequences, whereas appropriate control tissues did not. The same herpesvirus-like DNA sequences have

been reported in Kaposi's sarcoma tissue from non-AIDS patients, in a subgroup of AIDS-related B-cell body cavity-based lymphomas, and in lymph nodes from patients with multicentric Castleman's disease (a condition also known as angiofollicular lymph node hyperplasia, giant lymph node hyperplasia, lymphoid hamartoma, and follicular lymphoreticuloma, which is especially aggressive and frequently fatal). Approximately 15% of non-Kaposi's sarcoma tissue specimens from patients with AIDS contain these herpesvirus-like sequences, which have also been found in semen from both AIDS and non-AIDS patients. The virus has been propagated in cell culture and named [HHV-8](#); it is also referred to as Kaposi's sarcoma-associated herpesvirus. Several serologic assays suggest a low rate of background positivity (0 to 29%) among HIV-negative blood donors but a high rate of positivity (>80%) among patients with Kaposi's sarcoma. The etiologic role of HHV-8 in Kaposi's sarcoma and other diseases remains to be established, although HHV-8 seroconversion during HIV infection appears to be highly predictive of the development of Kaposi's sarcoma.

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186. SMALLPOX, VACCINIA, AND OTHER POXVIRUSES - Fred Wang

Two poxviruses, smallpox virus and molluscum contagiosum virus, cause natural disease in humans, and other poxviruses are associated with zoonotic infections. Monkeypox virus and smallpox virus typically cause systemic disease with rash, whereas the other poxviruses cause localized skin lesions. Poxviruses are the only DNA viruses that replicate in the cytoplasm, where accumulated viral particles form eosinophilic inclusions, or Guarnieri bodies, visible by light microscopy. Many poxvirus genes interfere with different aspects of the host immune response and provide important insights into the pathogenesis and virulence of viral infection. Genetically engineered vaccinia and avipoxviruses offer great promise as potential vectors for vaccination against other diseases.

SMALLPOX

The last case of endemic smallpox was reported in 1977 from Somalia. In 1980 the World Health Organization officially declared that smallpox had been eliminated worldwide as a result of a global vaccination and eradication program. Important features that contributed to the unique success of this vaccine program included (1) universal interest in eliminating this costly disease with high morbidity and mortality, (2) the infection's long incubation period and low level of communicability, (3) the ease of diagnosis of skin lesions by characteristic histology or antigen detection, (4) the fact that humans were the sole reservoir of the infection, (5) the absence of a carrier state, and (6) the availability of an effective live-virus vaccine that could readily be delivered to less developed countries because of its resistance to chemicals, temperature changes, and drying. The only known remaining repositories of smallpox virus reside in two research laboratories (located in the United States and Russia), and the issue of whether these last samples should be maintained or destroyed remains controversial.

Before the eradication of smallpox, variola virus existed as two related strains: *variola major* (smallpox), with a case-mortality rate of 20 to 50%, and *variola minor* (alastrim), which caused a clinically milder form of smallpox with a mortality of <1%. The clinical presentation of smallpox is now primarily of historic note. However, the threat of biologic terrorism means that smallpox remains a remote possibility in the differential diagnosis of a vesicular exanthem. Fever and macular rash appear after an average incubation period of 12 days, with a progression to typical vesicular and pustular lesions over 1 to 2 weeks. Rash generally appears first on the face, oral mucosa, and arms, with relative sparing of the trunk. Smallpox lesions may be confused with common chickenpox (varicella-zoster) infection but tend to be more diffuse, peripheral, and uniform in their stage of development. Polymerase chain reaction promises to be more useful than traditional electron microscopy or virus isolation for confirming variola or other poxvirus infections.

VACCINIA

The origin of vaccinia virus -- the virus used for vaccination against smallpox -- is uncertain, but it was probably derived from cowpox virus, variola virus, or a hybrid of the two. It is now a laboratory virus with no natural host. Experience has proven the effectiveness of live vaccinia virus vaccine, although its efficacy and safety were not

established in controlled studies. Percutaneous administration of vaccinia virus vaccine results in protective cellular and humoral immune responses in >95% of primary vaccinees. Formation of a pustule and scab at the site of inoculation is indicative of immunity; because immunity wanes after 10 to 20 years, revaccination every 10 years is recommended for continued protection. Routine smallpox vaccination was discontinued in 1971 and has not been required for international travel since 1982. However, the development of recombinant vaccinia viruses for potential use in vaccines against other infectious agents or as immunotherapy against malignant diseases has led to the recommendation that laboratory and health care employees working directly with vaccinia virus vectors be considered for vaccination. Selected groups that may be exposed to poxviruses (e.g., some military personnel and individuals who work with animals) are also vaccinated.

The most frequent adverse complication of vaccination is inadvertent inoculation (usually autoinoculation) at other sites. More serious complications, which are more common among primary vaccinees and infants than among revaccinees and adults, include (1) generalized vaccinia in otherwise healthy individuals, which is generally self-limited; (2) eczema vaccinatum, which consists of disseminated cutaneous lesions in highly susceptible patients with eczema or other chronic skin diseases and is occasionally severe or even fatal; (3) progressive vaccinia (vaccinia necrosum), which is a severe, potentially fatal illness occurring in patients with immunodeficiency, whether congenital, acquired (e.g., via leukemia or lymphoma), iatrogenic (e.g., via chemotherapy or glucocorticoid treatment), or HIV induced; and (4) postinfectious encephalitis, which is rare (3 cases per million primary vaccinees) but can be fatal in 15 to 25% of cases and can leave 25% of patients with permanent neurologic sequelae. Since vaccinees can transmit vaccinia virus to susceptible individuals, vaccination is contraindicated if the proposed recipient or his or her household contacts have eczema, are immunocompromised, or are pregnant. Vaccinia immune globulin (0.6 mL/kg) derived from the plasma of vaccinated persons may be useful for the treatment of severe generalized vaccinia, eczema vaccinatum, progressive vaccinia, and ocular vaccinia resulting from inadvertent inoculation but is of no value for the treatment of postinfectious encephalitis.

MOLLUSCUM CONTAGIOSUM

Molluscum contagiosum is generally a benign disease characterized by pearly, flesh-colored, umbilicated skin lesions 2 to 5 mm in diameter ([Plate IID-41, Fig. 186-CD1](#)). A relative lack of inflammation and necrosis distinguishes these proliferative lesions from other poxvirus lesions. The infection can be transmitted by close contact, including sexual intercourse. Swimming pools are a common vector for transmission. Atopy and compromise of skin integrity can increase the risk of infection. Lesions can be found anywhere on the body except the palms and soles and may be associated with an eczematous rash. In most cases the disease is self-limited and has no systemic complications. Molluscum contagiosum develops especially often in association with the advanced stages of HIV infection, with a prevalence of 5 to 18% among HIV-infected patients ([Chap. 309](#)). The disease is often more generalized, severe, and persistent in AIDS patients than in other groups, frequently involving the face and upper body. Extensive molluscum contagiosum has also been reported in conjunction with other types of immunodeficiency.

The diagnosis of molluscum contagiosum can be made by histologic demonstration of cytoplasmic eosinophilic inclusions characteristic of poxvirus replication. This virus cannot be propagated in vitro, but electron microscopy and molecular studies can be used for its identification.

There is no specific systemic treatment for molluscum contagiosum, but a variety of techniques for physical ablation have been used. Molluscum contagiosum may respond to effective control of HIV infection with highly active antiretroviral therapy. Cidofovir is also being investigated for potential clinical use against molluscum contagiosum.

MONKEYPOX VIRUS AND OTHER POXVIRUSES

Monkeypox virus naturally infects nonhuman primates in the tropical rain forests of western and central Africa and can infect humans who come into direct contact with infected animals. Human disease is rare and is characterized by a systemic illness and vesicular rash similar to those of variola. A large outbreak of monkeypox occurred between February 1996 and October 1997 in central Africa, with a case-fatality ratio of 3%; a prolonged period of active cases, suggesting a potential for sustained person-to-person transmission; and a high proportion of younger patients, suggesting the possible consequences of discontinued smallpox vaccination. Clinical presentations were occasionally confused with the more common varicella-zoster virus infection.

Other poxviruses can cause localized vesicular lesions when humans come into direct contact with infected animals. These viruses include cowpox virus (rodents, cats); milkers' node virus (cows; [Fig. 186-CD2](#)); buffalopox virus (buffaloes); bovine papular stomatitis virus (cows); and orf virus ([Fig. 186-CD3](#)), which is also known as contagious pustular dermatitis virus (sheep, goats).

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187. PARVOVIRUS - Neil R. Blacklow

DEFINITION

The parvovirus group includes several species-specific viruses of animals. One parvovirus, designated B19, is known to be a human pathogen. B19 is a small (diameter, 20 to 25 nm), icosahedral, nonenveloped, single-stranded DNA virus with an outer capsid formed by two structural proteins. Individual virus particles contain DNA strands of positive or negative polarity. The virus is stable and retains infectivity after incubation at 60°C for 16 h. It has failed to grow in conventional cell culture lines and animal model systems but does replicate in vitro in erythroid progenitor cells derived from human bone marrow, umbilical cord, peripheral blood, or fetal liver sources.

During the 1980s, it was discovered that B19 causes a variety of disorders ranging from erythema infectiosum and acute arthropathy in otherwise healthy hosts to transient aplastic crisis and chronic anemia in compromised patients to fetal infection manifested by death or hydrops fetalis. Many of the severe manifestations of B19 viremia relate to the propensity of the virus to infect and lyse erythroid precursor cells in the bone marrow. The name B19 is derived from the code number of the human serum in which the virus was discovered.

PATHOGENESIS

Two studies of adult volunteers have provided a basis for understanding the pathogenesis of B19 infection, which has two phases. The first phase is characterized by viremia that develops approximately 6 days after intranasal inoculation of B19 into susceptible individuals who lack serum antibodies to the virus. The viremia lasts about 1 week; its clearance is correlated with the development of IgM antibodies to B19, which remain detectable for up to a few months. IgG antibodies develop several days later and persist indefinitely. Nonspecific systemic symptoms lasting 2 or 3 days occur early during the viremic phase; these symptoms include headache, malaise, myalgia, fever, chills, and pruritus and are accompanied by reticulocytopenia and excretion of the virus from the respiratory tract. Several days after the onset of symptoms, a clinically insignificant decline in hemoglobin concentration is noted; the decreased level is maintained for 7 to 10 days, during which time examination of bone marrow samples reveals a marked depletion of erythroid precursor cells. Transient mild lymphopenia, neutropenia, and a drop in platelet count also may be found. A second phase of illness begins around 17 or 18 days after virus inoculation (after the clearance of viremia, the cessation of viral shedding in throat secretions, and the resolution of reticulocytopenia). This illness mimics erythema infectiosum in adults, with 2 or 3 days of fine maculopapular rash accompanied by arthralgias and arthritis that last another 1 or 2 days. This phase occurs in the presence of rising serum titers of antibody to B19.

The studies just described indicate that B19 disease in the otherwise *healthy host*, manifested by self-limited erythema infectiosum and/or arthropathy, is almost certainly an immune-complex disorder. This concept is supported by the induction of erythema infectiosum through the infusion of immunoglobulins into chronically viremic patients. In contrast, B19 disease in the *compromised host* (chronic hemolytic disease or immunodeficiency syndromes) is often serious, resulting from the destruction by B19 of

erythroid precursor cells. Normal hosts can tolerate 7 to 10 days of shutoff of erythropoiesis; however, patients with hemolytic disease who require increased production of erythrocytes do not tolerate erythroid cell destruction and thus usually develop severe transient aplastic crisis. Patients who are immunodeficient may fail to clear B19 viremia, the results being persistent infection of red blood cells and chronic severe anemia. The fetus requires a higher level of red cell production than do adults and has an immature immune system; both these factors could explain B19-induced hydrops fetalis.

B19 binds specifically to a cellular receptor, erythrocyte P antigen; this specific binding explains the tropism of B19 for erythroid progenitor cells, particularly pronormoblasts and normoblasts. The few persons who lack P antigen cannot be infected with B19.

EPIDEMIOLOGY

Although B19 infections occur year-round, they appear most commonly as outbreaks of erythema infectiosum in schools during winter and spring months. Between 20 and 60% of children in outbreaks are symptomatic, and many are asymptomatically infected. Seroepidemiologic studies indicate that approximately half of adults possess serum antibodies to B19. Antibody prevalence (reflecting prior exposure and probable immunity to the virus) rises rapidly between the ages of 5 and 18 years and continues to increase with age -- a pattern probably indicating ongoing exposure during adulthood. B19 can be detected in throat swabbings, respiratory tract secretions, and serum, and its detection at these sites probably correlates with infectiousness. Thus, patients with transient aplastic crisis are highly infectious. Their infectivity has been firmly documented as the source of one well-defined nosocomial outbreak of erythema infectiosum among nurses. In contrast, individuals with erythema infectiosum are much less infectious. The usual route of viral transmission under natural conditions is unknown but may be respiratory or through direct contact. B19 can be transmitted during therapy with clotting factor concentrate, even after exposure to detergent, steam, or dry heat.

CLINICAL MANIFESTATIONS

Erythema Infectiosum (Fig. 187-CD1) Erythema infectiosum is the most common manifestation of B19 infection and occurs predominantly in children. This entity is also called *fifth disease* because it was classified in the late nineteenth century as the fifth in a series of six exanthems of childhood. Normally a mild illness, erythema infectiosum typically presents as a facial rash with a "slapped-cheek" appearance that is sometimes preceded by low-grade fever. The rash may develop quickly on the arms and legs and usually has a lacy, reticular, erythematous appearance (Plate IID-40). The trunk, palms, and soles are less commonly involved. Occasionally, the rash appears with maculopapular, morbilliform, vesicular, purpuric, or pruritic characteristics. The typical rash resolves in about a week but can recur intermittently for several weeks, particularly after stress, exercise, exposure to sunlight, bathing, or change in environmental temperature. Arthralgia and arthritis are uncommon among children but are frequent among adults, in whom the rash is often absent or nonspecific, with a lack of the characteristic facial erythema.

Arthropathy B19 infection in adults most commonly presents as acute arthralgias and arthritis, sometimes accompanied by rash. The arthritis is characteristically symmetric and peripheral, involving the wrists, hands, and knees most frequently. It normally resolves in about 3 weeks and is nondestructive. However, a small percentage of patients have arthritis persisting for months or even (in rare cases) for years. It is not known whether these individuals have persistent infection or an abnormal immune response to the virus.

Transient Aplastic Crisis B19 infection is the cause in most instances of transient aplastic crisis developing suddenly in patients with chronic hemolytic disease. Nearly all hemolytic conditions can be affected by B19 infection, including sickle cell disease, erythrocyte enzyme deficiencies, hereditary spherocytosis, thalassemias, paroxysmal nocturnal hemoglobinuria, and autoimmune hemolysis. B19-induced aplastic crisis also can occur in the setting of acute blood loss. Patients present with weakness, lethargy, pallor, and severe anemia, a syndrome often preceded by a few days of nonspecific symptoms. These patients have intense reticulocytopenia lasting 7 to 10 days, and their bone marrow contains no erythroid precursor cells despite a normal myeloid series. Transient aplastic crisis can produce life-threatening anemia and may require urgent transfusion therapy. Unlike patients with erythema infectiosum or arthropathy, those with transient aplastic crisis are viremic and can readily transmit B19 infection to other people.

Chronic Anemia in Immunodeficient Patients Immunodeficient patients may be unable to eliminate B19 infection, probably because they cannot produce adequate levels of virus-specific IgG antibodies. The result is persistent infection with destruction of erythroid precursor cells in the bone marrow and chronic transfusion-dependent anemia. This condition has been described occasionally in patients with immunodeficiency related to infection with HIV, congenital immunodeficiencies, and acute lymphoblastic leukemia during maintenance chemotherapy as well as in recipients of bone marrow, heart, liver, and renal transplants. In addition, some cases of idiopathic pure red-cell aplasia probably are caused by persistent B19 infection. B19-induced chronic anemia may be the presenting finding of an otherwise unrecognized immunodeficiency. Chronic anemia may fluctuate in intensity over time and may be cured or controlled by immunoglobulin therapy. Both the spectrum of immunodeficiencies associated with B19-induced chronic anemia and the frequency of the association remain to be determined.

Fetal and Congenital Infection Maternal B19 infections usually do not adversely affect the fetus. More often than not, in fact, the fetus remains uninfected. Therefore, couples in which the pregnant woman is infected should be counseled as to the relatively low risk of fetal infection. It is estimated that fewer than 10% of maternal B19 infections in the first 20 weeks of pregnancy lead to fetal death; when fetal death does occur, it is usually attributable to the development of nonimmune hydrops fetalis, wherein the fetus succumbs to severe anemia and congestive heart failure. In these instances, B19 can be detected in fetal tissues, with predominant infection of erythroblasts. Pregnant women with known exposure to B19 should have their serum monitored for IgM antibodies to the virus and for elevated levels of α -fetoprotein and human chorionic gonadotropin; ultrasonic examinations of the fetus for hydrops should also be conducted. Some hydropic fetuses survive B19 infection and appear normal at delivery.

Rarely, fetal infection with hydrops results in congenital anemia and hypogammaglobulinemia that is unresponsive to immunoglobulin therapy.

Possible Clinical Associations Case studies suggest a link -- as yet inconclusive -- between B19 and several rheumatic diseases, most notably rheumatoid arthritis but also vasculitis (including polyarteritis, Wegener's granulomatosis, and giant cell arteritis), lupus erythematosus, dermatomyositis, and juvenile rheumatoid arthritis. Other unproven associations include those involving multiple systems: cardiac (myocarditis), hematologic (hemophagocytic syndrome, idiopathic thrombocytopenic purpura), hepatic (fulminant hepatitis), neurologic (meningoencephalitis), and respiratory (pneumonia).

DIAGNOSIS

Diagnosis most commonly relies on measurements of B19-specific IgM and IgG antibodies, which can be detected with commercially available immunoassay kits. The virus, its DNA, or its antigens are also detected in the serum or infected tissues of some patients. Acute infection can be proven by B19-compatible symptoms and the presence of IgM antibodies or virus itself, whereas past infection is documented by IgG antibodies. Individuals with erythema infectiosum and acute arthropathy usually have IgM antibodies without detectable virus in serum. Those with transient aplastic crisis may have IgM antibodies but typically possess high titers of virus and its DNA in serum; the bone marrow of these patients shows characteristic giant pronormoblasts and hypoplasia. Immunodeficient patients with anemia often lack readily detectable antibodies but have viral particles and DNA in serum. Fetal infection may be recognized by hydrops fetalis and the presence of B19 DNA in amniotic fluid or fetal blood in association with maternal IgM antibodies to B19.

TREATMENT

Erythema infectiosum usually requires no treatment; the same is true for many cases of arthropathy. More severe cases of arthritis, particularly those involving chronic symptoms, can be treated with nonsteroidal anti-inflammatory agents. Transient aplastic crisis is usually treated with erythrocyte transfusions. In immunodeficient anemic patients, B19 infection should be treated with commercial intravenous immunoglobulin, which is known to contain IgG antibodies to B19. This therapy controls and may cure B19 infection.

PROPHYLAXIS

Prophylaxis of B19 infection with immunoglobulin should be considered for patients with chronic hemolysis or immunodeficiency and for pregnant women. The risk of infection for these persons may be reduced by hand washing before eating or after contact with respiratory or other secretions when B19 is known to be present in a community. Patients with transient aplastic crisis or chronic B19 infection (but not those with erythema infectiosum or arthropathy) pose a serious risk for nosocomial transmission of infection. They should be hospitalized in a private room with contact and respiratory isolation precautions. It is not known whether pre- or postexposure administration of immunoglobulin prevents infection. No vaccine for B19 is currently available; however, a baculovirus-infected insect cell line that expresses noninfectious immunogenic B19

capsid proteins is being evaluated to determine an optimal regimen for use as a vaccine.

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188. HUMAN PAPILLOMAVIRUSES - Richard C. Reichman

DEFINITION

Human papillomaviruses (HPVs) selectively infect the epithelium of the skin and mucous membranes. These infections may be asymptomatic, produce warts, or be associated with a variety of benign and malignant neoplasias.

ETIOLOGIC AGENT

Papillomaviruses are members of the *Papillomavirus* genus of the family Papovaviridae. They are nonenveloped, measure 50 to 55 nm in diameter, have icosahedral capsids composed of 72 capsomeres, and contain a double-stranded circular DNA genome of about 7900 base pairs. The genomic organization of all papillomaviruses is similar and consists of an early (E) region, a late (L) region, and a noncoding upstream regulatory region. Oncogenic [HPV](#) types can immortalize human keratinocytes, and this activity has been mapped to products of early genes E6 and E7. E6 protein facilitates the degradation of the p53 tumor suppressor protein, and E7 protein binds the retinoblastoma gene product and related proteins. The E1 and E2 proteins modulate viral DNA replication and regulate gene expression. The L1 gene codes for the major capsid protein, which makes up 80% of the virion mass. L2 codes for a minor capsid protein. Type-specific conformational antigenic determinants are located on the virion surface. Papillomavirus types are distinguished from one another by the degree of nucleic acid sequence homology. Distinct types share fewer than 90% of their DNA sequences in L1. More than 80 types of HPV are recognized, and individual types are associated with specific clinical manifestations ([Table 188-1](#)). HPVs are species-specific and have not been propagated in tissue culture or in common experimental animals. However, HPV types 1, 6, 11, 16, 40, and 83 have been produced in human tissues implanted in immunodeficient mice.

EPIDEMIOLOGY

There are few good studies of the incidence or prevalence of warts in well-defined human populations. Common warts (verruca vulgaris) are found in as many as 25% of some groups and are most prevalent among young children. Plantar warts (verruca plantaris) are also widely prevalent; they occur most often among adolescents and young adults. Condyloma acuminatum (anogenital warts) is one of the most common sexually transmitted diseases in the United States. [HPV](#) infection of the uterine cervix produces the squamous cell abnormalities most frequently detected on Papanicolaou smears.

Most genital [HPV](#) infections are transmitted through direct contact with infectious lesions. Close personal contact is also assumed to play a role in the transmission of most cutaneous warts; the importance of fomites in this setting is not clear. Minor trauma at the site of inoculation may facilitate transmission. Recurrent respiratory papillomatosis in young children is an uncommon disease that is acquired from maternal genital tract infection; in adults, the disease may be transmitted through orogenital sexual contact.

[HPV](#) infection has been strongly associated with the development of dysplasia and

cancer of the uterine cervix. More than 95% of cervical cancers contain DNA of oncogenic (high-risk) HPV types, such as 16, 18, and 31. HPV DNA is also present in the precursor lesions of cervical cancer, known as cervical intraepithelial neoplasias. Such lesions containing DNA of oncogenic HPV types are more likely to progress than those associated with low-risk types, such as 6 and 11. HPV DNA is transcribed in tumor tissues; many epidemiologic studies have confirmed a relation between HPV infection (with or without cofactors) and the development of cervical cancer, although most cervical HPV infections are self-limited. Infection with specific HPV types has also been associated with squamous cell carcinomas and dysplasias of the penis, anus, vagina, and vulva. In patients with epidermodysplasia verruciformis, squamous cell cancers develop frequently at sites infected with specific HPV types, including 5 and 8.

Serologic studies with virus-like particles as antigens have demonstrated type-specific antibodies in most patients with [HPV](#) genital tract infections.

CLINICAL MANIFESTATIONS

The clinical manifestations of [HPV](#) infection depend on the location of the lesions and the type of virus. Common warts ([Fig. 128-CD6](#)) usually occur on the hands as flesh-colored to brown, exophytic, hyperkeratotic papules. Plantar warts ([Fig. 188-CD1](#)) may be quite painful; they can be differentiated from calluses by paring of the surface to reveal thrombosed capillaries. Flat warts (verruca plana; [Fig. 188-CD2](#)) are most common among children and occur on the face, neck, chest, and flexor surfaces of the forearms and legs.

Anogenital warts develop on the skin and mucosal surfaces of the external genitalia and perianal areas ([Plate IID-55](#)). Among circumcised men, warts are most commonly found on the penile shaft. Lesions commonly occur at the urethral meatus and may extend proximally. Perianal warts are common among homosexual men but develop in heterosexual men as well. In women, warts appear first at the posterior introitus and adjacent labia. They then spread to other parts of the vulva and commonly involve the vagina and cervix. These lesions may be present without external warts. The differential diagnosis of anogenital warts includes condylomata lata of secondary syphilis, molluscum contagiosum, hirsutoid papillomatosis (pearly penile papules), fibroepitheliomas, and a variety of benign and malignant mucocutaneous neoplasms. Respiratory papillomatosis in young children may be life-threatening and presents as hoarseness, stridor, or respiratory distress. The disease in adults is usually milder.

Immunosuppressed patients, particularly those undergoing organ transplantation, often develop pityriasis versicolor-like lesions, from which DNA of several [HPV](#) types has been extracted. Occasionally, such lesions appear to undergo malignant transformation. Patients infected with HIV frequently have severe clinical manifestations of HPV infection and appear to be at unusually high risk for cervical and anal malignancies. HPV disease in patients with HIV infection is difficult to treat and often recurs.

Epidermodysplasia verruciformis is a rare autosomal recessive disease characterized by the inability to control [HPV](#) infection. Patients are often infected with unusual HPV types and frequently develop cutaneous squamous cell malignancies, particularly in sun-exposed areas. The lesions resemble flat warts or macules similar to those of

pityriasis versicolor.

The complications of warts include itching and occasionally bleeding. In rare cases warts become secondarily infected with bacteria or fungi. Large masses of warts may cause mechanical problems, such as obstruction of the birth canal. Dysplasias of the uterine cervix are generally asymptomatic until frank carcinoma develops. Patients with anogenital [HPV](#) disease may develop serious psychological symptoms due to anxiety or depression over this condition.

PATHOGENESIS

The incubation period of [HPV](#) disease is usually 3 to 4 months, with a range of 1 month to 2 years. All types of squamous epithelium can be infected by HPV, and the gross and histologic appearances of individual lesions vary with the site of infection and the type of virus. The replication of HPV begins with the infection of basal cells. As cellular differentiation proceeds, HPV DNA replicates and is transcribed. Ultimately, virions are assembled in the nucleus and released when keratinocytes are shed. This process is associated with proliferation of all epidermal layers except the basal layer and produces acanthosis, parakeratosis, and hyperkeratosis. Koilocytes, large round cells with pyknotic nuclei, appear in the granular layer. Histologically normal epithelium may contain HPV DNA, and residual DNA after treatment can be associated with recurrent disease.

Episomal [HPV](#) DNA is present in the nuclei of infected cells in benign lesions caused by the virus. However, in severe dysplasias and cancers, HPV DNA is generally integrated, with disruption of the E1/E2 open reading frames. This disruption leads to upregulation of E6 and E7 and subsequent interference with cellular tumor suppressor proteins.

Host defense responses to [HPV](#) infection are incompletely understood, and immune correlates of protection from infection and resolution of disease have not been established. Because patients with defects in cell-mediated immune responses, including transplant recipients and patients with HIV infection, frequently develop severe HPV disease, such responses are probably important for the control of virus replication. Histologic studies demonstrating an epidermal lymphomonocytic infiltrate in resolving warts suggest that local immunity may be of particular importance in the resolution of disease. HPV infection can also elicit a serologic response, and antibodies to the viral capsid have been found in sera from patients with anogenital warts, cutaneous warts, and respiratory papillomatosis. Antibodies to E-region proteins, most notably E7, have been detected among patients with cervical carcinoma. Vaccine studies in animals have shown that production of neutralizing antibodies can be associated with protection from papillomavirus infection.

DIAGNOSIS

Most warts that are visible to the naked eye can be diagnosed correctly by history and physical examination alone. The use of a colposcope is invaluable in assessing vaginal and cervical lesions and is helpful in the diagnosis of oral and cutaneous [HPV](#) disease as well. Papanicolaou smears prepared from cervical scrapings often show cytologic evidence of HPV infection. Persistent or atypical lesions should be biopsied and

examined by routine histologic methods. The most sensitive and specific methods of virologic diagnosis entail the use of techniques such as the polymerase chain reaction or the hybrid capture assay to detect HPV nucleic acids and to identify specific virus types. Serologic techniques to diagnose HPV infection are not helpful in individual cases and are not widely available.

TREATMENT

Decisions regarding the initiation of therapy should be made with the knowledge that currently available modes of treatment are not completely effective and some have significant side effects. In addition, treatment may be expensive, and many [HPV](#) lesions resolve spontaneously. Frequently used therapies include cryosurgery, application of caustic agents, electrodesiccation, surgical excision, and ablation with a laser. Topical antimetabolites such as 5-fluorouracil also have been used. Both failure and recurrence have been well documented with all of these methods of treatment. Cryosurgery is the initial treatment of choice for condyloma acuminatum. Topically applied podophyllum preparations as well as podofilox may also be used. Various interferon preparations have been used with modest success in the treatment of respiratory papillomatosis and condyloma acuminatum. A topically applied interferon inducer, imiquimod, is also of benefit in the treatment of condyloma acuminatum. The diagnosis and management of anogenital dysplasias and of internal anogenital warts require special skills and resources, and patients with such lesions should be referred to a qualified specialist.

No effective methods for the prevention of [HPV](#) infections are available at present other than the avoidance of contact with infectious lesions. Barrier methods of contraception may be helpful in preventing the transmission of condyloma acuminatum and other HPV-associated diseases of the genital tract. Vaccines consisting of virus-like particles can prevent papillomavirus disease in some animal models and have been shown to induce neutralizing antibodies in phase 1 studies in humans. More extensive clinical trials of these preparations are under way.

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SECTION 13 -DNA AND RNA RESPIRATORY VIRUSES

189. COMMON VIRAL RESPIRATORY INFECTIONS - *Raphael Dolin*

GENERAL CONSIDERATIONS

Acute viral respiratory illnesses are among the most common of human diseases, accounting for one-half or more of all acute illnesses. The incidence of acute respiratory disease in the United States is from 3 to 5.6 cases per person per year. The rates are highest among children under 1 year old (6.1 to 8.3 cases per year) and remain high until age 6, when a progressive decrease begins. Adults have 3 to 4 cases per person per year. Morbidity from acute respiratory illnesses accounts for 30 to 50% of time lost from work by adults and for 60 to 80% of time lost from school by children. The use of antibacterial agents to treat viral respiratory infections represents a major source of abuse of that category of drugs.

It has been estimated that two-thirds to three-fourths of cases of acute respiratory illnesses are caused by viruses. More than 200 antigenically distinct viruses from 8 different genera have been reported to cause acute respiratory illness, and it is likely that additional agents will be described in the future. The vast majority of these viral infections involve the upper respiratory tract, but lower respiratory tract disease can also develop, particularly in younger age groups and in certain epidemiologic settings.

The illnesses caused by respiratory viruses traditionally have been divided into multiple distinct syndromes, such as the "common cold," pharyngitis, croup (laryngotracheobronchitis), tracheitis, bronchiolitis, bronchitis, and pneumonia. Each of these general categories of illnesses has a certain epidemiologic and clinical profile; for example, croup occurs exclusively in very young children and has a characteristic clinical course. Some types of respiratory illnesses are more likely to be associated with certain viruses (e.g., the common cold with rhinoviruses), while others occupy characteristic epidemiologic niches (e.g., adenovirus infections in military recruits). The syndromes most commonly associated with infections with the major respiratory virus groups are summarized in [Table 189-1](#). Most respiratory viruses clearly have the potential to cause more than one type of respiratory illness, and frequently features of several types of illness are found in the same patient. Moreover, the clinical illnesses induced by these viruses are rarely sufficiently distinctive to permit an etiologic diagnosis on clinical grounds alone, although the epidemiologic setting increases the likelihood that one group of viruses rather than another is involved. In general, laboratory methods must be relied on to establish a specific viral diagnosis.

This chapter reviews viral infections caused by five of the major groups of respiratory viruses: rhinoviruses, coronaviruses, respiratory syncytial viruses, parainfluenza viruses, and adenoviruses. Influenza viruses, which are a major cause of mortality as well as morbidity, are reviewed in [Chap. 190](#). Herpesviruses, which occasionally cause pharyngitis and which also cause lower respiratory tract disease in immunosuppressed patients, are reviewed in [Chap. 182](#). Enteroviruses, which account for occasional respiratory illnesses during the summer months, are reviewed in [Chap. 193](#).

RHINOVIRUS INFECTIONS

ETIOLOGIC AGENT

Rhinoviruses are members of the Picornaviridae family, small (15 to 30 nm) nonenveloped viruses that contain a single-stranded RNA genome. In contrast to other members of the picornavirus family, such as enteroviruses, rhinoviruses are acid-labile and are almost completely inactivated at pH 3. Rhinoviruses grow preferentially at 33° to 34°C -- the temperature of the human nasal passages -- rather than at the higher temperature (37°C) of the lower respiratory tract. One hundred distinct serotypes and one subtype of rhinovirus are recognized.

EPIDEMIOLOGY

Rhinoviruses are a major cause of the common cold and have been isolated from 15 to 40% of adults with common cold-like illnesses. Overall rates of infection with rhinoviruses are higher among infants and young children and decrease with increasing age. Rhinovirus infections occur throughout the year, with seasonal peaks in early fall and spring in temperate climates. Rhinovirus infections are most often introduced into families by preschool or grade-school children younger than 6 years old. Between 25 and 70% of initial illnesses in family settings are followed by secondary cases, with the highest attack rates among the youngest siblings at home. Attack rates also increase with family size.

Rhinoviruses appear to spread through direct contact with infected secretions, usually respiratory droplets. In some studies of volunteers, transmission was most efficient by hand-to-hand contact, with subsequent self-inoculation of the conjunctival or nasal mucosa. In other studies, transmission by large- or small-particle aerosol was demonstrated. Virus also can be recovered from plastic surfaces inoculated 1 to 3 h previously; this observation suggests that environmental surfaces contribute to transmission. In studies of married couples in which neither partner had detectable serum antibody, transmission was associated with prolonged contact (122 h or more) during a 7-day period. Transmission was infrequent unless virus was recoverable from the donor's hands and nasal mucosa, at least 1000 TCID₅₀ of virus was present in nasal washes from the donor, and the donor was at least moderately symptomatic with the "cold." Despite anecdotal observations, exposure to cold temperatures, fatigue, or sleep deprivation has not been associated with increased rates of rhinovirus-induced illness in volunteers.

Infection with rhinoviruses is worldwide in distribution. By the time they reach adulthood, nearly all individuals have neutralizing antibodies to multiple serotypes, although the prevalence of antibody to any one serotype varies widely. Multiple serotypes circulate simultaneously, and generally no single serotype or group of serotypes has been more prevalent than the others.

PATHOGENESIS

Rhinoviruses infect cells through attachment to specific cellular receptors; most serotypes attach to intercellular adhesion molecule 1, while a few use low-density lipoprotein as the cellular receptor. Relatively limited information is available on the

histopathology and pathogenesis of acute rhinovirus infections in humans. Examination of biopsy specimens obtained during experimentally induced and naturally occurring illness indicates that the nasal mucosa is edematous, is often hyperemic, and -- during acute illness -- is covered by a mucoid discharge. There is a mild infiltrate with inflammatory cells, including neutrophils, lymphocytes, plasma cells, and eosinophils. Mucus-secreting glands in the submucosa appear hyperactive; the nasal turbinates are engorged, a condition that may lead to obstruction of nearby openings of sinus cavities. Several mediators, such as bradykinin, lysylbradykinin, prostaglandins, histamine, and interleukins 1, 6, and 8, have been linked to the development of signs and symptoms in rhinovirus-induced colds.

The incubation period for rhinovirus illness is short, generally 1 or 2 days. Virus shedding coincides with the onset of illness or may begin shortly before symptoms develop. The mechanisms of immunity to rhinovirus are not well worked out. In some studies, the presence of homotypic antibody has been associated with significantly reduced rates of subsequent infection and illness, but data conflict regarding the relative importance of serum and local antibody in protection from rhinovirus infection.

CLINICAL MANIFESTATIONS

The most common clinical manifestations of rhinovirus infections are those of the common cold. Illness usually begins with rhinorrhea and sneezing accompanied by nasal congestion. The throat is frequently sore, and in some cases sore throat is the initial complaint. Systemic signs and symptoms, such as malaise and headache, are mild or absent, and fever is unusual. Illness generally lasts for 4 to 9 days and resolves spontaneously without sequelae. In children, bronchitis, bronchiolitis, and bronchopneumonia have been reported; nevertheless, it appears that rhinoviruses are not major causes of lower respiratory tract disease in children. Rhinoviruses may cause exacerbations of asthma and chronic pulmonary disease in adults. The vast majority of rhinovirus infections resolve without sequelae, but complications related to obstruction of the eustachian tubes or sinus ostia, including otitis media or acute sinusitis, can develop.

DIAGNOSIS

Although rhinoviruses are the most frequently recognized cause of the common cold, similar illnesses are caused by a variety of other viruses, and the etiologic diagnosis cannot be made on clinical grounds alone. Rather, rhinovirus infection is diagnosed by isolation of the virus from nasal washes or nasal secretions in tissue culture. In practice, this procedure is rarely undertaken because of the benign, self-limited nature of the illness. Given the many serotypes of rhinovirus, diagnosis by serum antibody tests is currently impractical. Likewise, common laboratory tests, such as white cell count and sedimentation rate, are not helpful.

TREATMENT

Rhinovirus infections are generally mild and self-limited, so treatment is not usually necessary. Therapy in the form of antihistamines and nonsteroidal anti-inflammatory drugs may be beneficial in patients with particularly pronounced symptoms, and

reduction of activity is prudent in instances of significant discomfort or fatigability. Antibacterial agents should be used only if bacterial complications such as otitis media or sinusitis develop. Specific antiviral therapy is not available. Application of interferon sprays intranasally has been effective in the prophylaxis of rhinovirus infections but is also associated with local irritation of the nasal mucosa. Prevention of rhinovirus infection by antibodies directed against rhinovirus receptors or by the soluble purified receptors themselves is under study. Experimental vaccines to certain rhinovirus serotypes have been prepared, but their usefulness is questionable because of the myriad serotypes and the uncertainty about mechanisms of immunity. Thorough hand washing, environmental decontamination, and protection against autoinoculation may help to reduce rates of transmission of infection.

CORONAVIRUS INFECTIONS

ETIOLOGIC AGENT

Coronaviruses are pleomorphic, single-stranded RNA viruses that measure 80 to 160 nm in diameter. The name derives from the crownlike appearance produced by the club-shaped projections that stud the viral envelope. Coronaviruses that infect humans fall into two distinct antigenic groups (I and II), which are represented by prototype isolates 229E and OC43. Coronaviruses are fastidious and are difficult to culture in vitro. Some strains will grow only in human tracheal organ cultures rather than in tissue culture.

EPIDEMIOLOGY

Only limited seroepidemiologic studies of coronavirus infections have been conducted. Seroprevalence studies of strains 229E and OC43 have demonstrated the presence of serum antibodies at rates ranging from 12% to >80% in various populations. Overall, coronaviruses account for 10 to 20% of common colds. Coronavirus infections appear to be particularly prevalent in late fall, winter, and early spring -- times when rhinovirus infections are less common. A cyclical pattern has been suggested for outbreaks of infection with strains OC43 and 229E, with outbreaks occurring every 2 to 4 years.

CLINICAL MANIFESTATIONS

The clinical features of illness caused by coronaviruses are similar to those of illness caused by rhinoviruses. In studies of volunteers, the mean incubation period of illness induced by coronaviruses (3 days) is somewhat longer than that of illness caused by rhinoviruses, and the duration of illness is somewhat shorter (mean, 6 to 7 days). In some studies, the amount of nasal discharge was somewhat greater in colds induced by coronaviruses than in those induced by rhinoviruses. Coronaviruses have been recovered from infants with pneumonia and from military recruits with lower respiratory tract disease and have been associated with worsening of chronic bronchitis. However, the overall significance of coronaviruses in lower respiratory tract disease in humans remains unclear.

TREATMENT

The approach to the treatment of common colds caused by coronaviruses is similar to that discussed above for rhinovirus-induced illnesses. Because of uncertainty regarding the number and relative importance of coronavirus subgroups and the mechanisms of immunity, vaccines against coronaviruses have not been developed.

RESPIRATORY SYNCYTIAL VIRUS INFECTIONS

ETIOLOGIC AGENT

Respiratory syncytial virus (RSV) is a member of the Paramyxoviridae family and comprises the genus *Pneumovirus*. RSV, an enveloped virus approximately 150 to 300 nm in diameter, is so named because its replication in vitro leads to the fusion of neighboring cells into large multinucleated syncytia. The single-stranded RNA genome codes for 10 virus-specific proteins. Viral RNA is contained in a helical nucleocapsid surrounded by a lipid envelope bearing two glycoproteins: the G protein, by which the virus attaches to cells, and the F (fusion) protein, which facilitates entry of the virus into the cell by fusing host and viral membranes. RSV was once considered to be of a single antigenic type, but two distinct groups (A and B) and multiple subtypes within each group have now been described. Antigenic diversity is reflected by differences in the G protein, while the F protein is highly conserved. The epidemiologic significance of the antigenic diversity is under investigation. Both antigenic groups can circulate simultaneously in outbreaks, although the relative proportions of each vary.

EPIDEMIOLOGY

[RSV](#) is the major respiratory pathogen of young children and the foremost cause of lower respiratory disease in infants. Infection with RSV is seen throughout the world in annual epidemics that occur in late fall, winter, or spring and last up to 5 months. The virus is rarely encountered during the summer. Rates of illness are highest among infants between 1 and 6 months of age, peaking between 2 and 3 months of age. The attack rates among susceptible infants and children are extraordinarily high, approaching 100% in settings such as day-care centers where large numbers of susceptible infants are present. RSV accounts for 20 to 25% of hospital admissions of young infants and children for pneumonia and for up to 75% of cases of bronchiolitis in this age group. It has been estimated that more than half of infants who are at risk will become infected during an RSV epidemic.

In older children and adults, reinfection with [RSV](#) is frequent but disease is milder than in infancy. A common cold-like syndrome is the illness most commonly associated with RSV infection in adults. Severe lower respiratory tract disease with pneumonitis can occur in elderly (often institutionalized) adults and in patients with immunocompromising disorders or treatment, including recipients of bone-marrow and solid-organ transplants. RSV is also an important nosocomial pathogen; during an outbreak, it can infect pediatric patients and up to 25 to 50% of the staff on pediatric wards. The spread of virus among families is efficient: up to 40% of siblings may become infected when RSV is introduced into the family setting.

[RSV](#) is transmitted primarily by close contact with contaminated fingers or fomites and by self-inoculation of the conjunctiva or anterior nares. Virus also may be spread by

coarse aerosols produced by coughing or sneezing, but it is inefficiently spread by fine-particle aerosols. The incubation period is ~4 to 6 days, and virus shedding may last for 3-2 weeks in children and for shorter periods in adults.

PATHOGENESIS

Little is known about the histopathology of minor [RSV](#) infection. Severe bronchiolitis or pneumonia is characterized by necrosis of the bronchiolar epithelium and a peribronchiolar infiltrate of lymphocytes and mononuclear cells. Interstitial thickening and filling of alveolar spaces with fluid can also be found. The characteristics of the immune response to RSV are not well elucidated. Because reinfection occurs frequently and is often associated with illness, the immunity that develops after single episodes of infection clearly is not complete or long-lasting. However, the cumulative effect of multiple reinfections is to temper subsequent disease and to provide some temporary measure of protection against infection. Studies of experimentally induced disease in healthy volunteers indicate that the presence of nasal IgA neutralizing antibody correlates more closely with protection than does the presence of serum antibody. Studies in infants, however, suggest that maternally acquired antibody provides some protection from lower respiratory tract disease, although illness can be severe even in infants who have moderate levels of maternally derived serum antibody. The relatively severe disease observed in immunosuppressed patients and experimental animal models indicates that cell-mediated immunity is an important mechanism of host defense against RSV. Evidence suggests that class I MHC-restricted cytotoxic T cells may be particularly important in this regard.

CLINICAL MANIFESTATIONS

[RSV](#) infection leads to a wide spectrum of respiratory illnesses. In infants, 25 to 40% of infections result in lower respiratory tract involvement, including pneumonia, bronchiolitis, and tracheobronchitis. In this age group, illness begins most frequently with rhinorrhea, low-grade fever, and mild systemic symptoms, often accompanied by cough and wheezing. Most patients recover gradually over 1 to 2 weeks. In more severe illness, tachypnea and dyspnea develop, and eventually frank hypoxia, cyanosis, and apnea can ensue. Physical examination may reveal diffuse wheezing, rhonchi, and rales. Chest radiography shows hyperexpansion, peribronchial thickening, and variable infiltrates ranging from diffuse interstitial infiltrates to segmental or lobar consolidation. Illness may be particularly severe in children born prematurely and in those with congenital cardiac disease, bronchopulmonary dysplasia, nephrotic syndrome, or immunosuppression. One study documented a 37% mortality rate for infants with RSV pneumonia and congenital cardiac disease.

In adults, the most common symptoms of [RSV](#) infection are those of the common cold, with rhinorrhea, sore throat, and cough. Illness is occasionally associated with moderate systemic symptoms such as malaise, headache, and fever. RSV also has been reported to cause lower respiratory tract disease with fever in adults, including severe pneumonia in the elderly. RSV pneumonia can be a significant cause of morbidity and mortality in patients (particularly children) undergoing bone-marrow and solid-organ transplantation.

LABORATORY FINDINGS AND DIAGNOSIS

The diagnosis of [RSV](#) infection can be suspected on the basis of a suggestive epidemiologic setting -- that is, severe illness among infants during an outbreak of RSV in the community. Infections in older children and adults cannot be differentiated with certainty from those caused by other respiratory viruses. The specific diagnosis is established by isolation of RSV from respiratory secretions, including sputum, throat swabs, or nasopharyngeal washes. Virus is detected in tissue culture and is identified specifically through immunologic reactions detected by immunofluorescence, enzyme-linked immunosorbent assay (ELISA), or other techniques. Immunofluorescence microscopy of nasal scrapings or washings provides a rapid diagnosis. Serologic tests that depend on fourfold or greater rises in complement-fixing or neutralizing antibody titers are useful for diagnosis in older children and adults but are less sensitive in children under 4 months of age. ELISA is more sensitive than complement-fixation or neutralization tests in the detection of serum antibody. Serologic diagnosis requires comparison of acute- and convalescent-phase serum specimens and is therefore not useful during acute illness.

TREATMENT

Treatment of upper respiratory tract [RSV](#) infection is aimed primarily at the alleviation of symptoms and is similar to that for other viral infections of the upper respiratory tract. For lower respiratory tract infections, respiratory therapy, including hydration, suctioning of secretions, and administration of humidified oxygen and antibronchospastic agents, is given as needed. In severe hypoxia, intubation and ventilatory assistance may be required. Studies of infants with RSV infection who were given aerosolized ribavirin, a nucleoside analogue active in vitro against RSV, have demonstrated a beneficial effect on the resolution of lower respiratory tract illness, including alleviation of blood-gas abnormalities. Treatment with ribavirin is recommended for infants who are severely ill or who are at high risk for complications of RSV infection; included are premature infants and those with bronchopulmonary dysplasia, congenital heart disease, or immunosuppression. The effects of ribavirin in adults with RSV pneumonia have not been established. The monthly administration of human immunoglobulin with high titers of antibody to RSV (RSVIG) or of a chimeric mouse-human IgG antibody to RSV (palivizumab) has been approved as prophylaxis against RSV for children younger than 2 years of age who have bronchopulmonary dysplasia or were born prematurely.

Considerable interest exists in the development of vaccines against [RSV](#). Inactivated whole-virus vaccines have been ineffective; in one study, they actually potentiated the disease in infants. Other approaches include immunization with purified F and G surface glycoproteins of RSV or generation of stable, live attenuated virus vaccines. In settings such as pediatric wards where rates of transmission are high, barrier methods for the protection of hands and conjunctivae may be useful in reducing the spread of virus.

PARAINFLUENZA VIRUS INFECTIONS

ETIOLOGIC AGENT

Parainfluenza viruses belong to the Paramyxoviridae family, are 150 to 250 nm in diameter, are enveloped, and contain a single-stranded RNA genome. The envelope is

studded with two glycoproteins: one possesses both hemagglutinin and neuraminidase activity and the other contains fusion activity. The viral RNA genome is enclosed in a helical nucleocapsid and codes for seven or eight virus-specific proteins. All four distinct serotypes of parainfluenza viruses share certain antigens with other members of the Paramyxoviridae family, including mumps and Newcastle disease viruses.

EPIDEMIOLOGY

Parainfluenza viruses are distributed throughout the world; infection with type 4 (subtypes 4A and 4B) has been reported less widely, probably because type 4 is more difficult to grow in tissue culture. Infection is acquired in early childhood, so that by 8 years of age most children have antibodies to serotypes 1, 2, and 3. Types 1 and 2 cause epidemics during the fall, primarily in odd-numbered years. Type 3 infection has been detected during all seasons of the year, but epidemics have occurred annually in the spring.

The contribution of parainfluenza infections to respiratory disease varies with both the location and the year. In studies conducted in the United States, parainfluenza virus infections have accounted for 4.3 to 22% of respiratory illnesses in children. In adults, parainfluenza infections are generally mild and account for fewer than 5% of respiratory illnesses. The major importance of parainfluenza viruses is as a cause of respiratory illness in young children, in whom they rank second only to [RSV](#) as causes of lower respiratory tract illness. Parainfluenza virus type 1 is the most frequent cause of croup (laryngotracheobronchitis) in children, while serotype 2 causes similar, although generally less severe, disease. Type 3 is an important cause of bronchiolitis and pneumonia in infants, while illnesses associated with type 4 have generally been mild. Unlike types 1 and 2, type 3 frequently causes illness during the first month of life, when passively acquired maternal antibody is still present. Parainfluenza viruses are spread through infected respiratory secretions, primarily by person-to-person contact and/or by large droplets. The incubation period has varied from 3 to 6 days in experimental infections but may be somewhat shorter for naturally occurring disease in children.

PATHOGENESIS

Immunity to parainfluenza viruses is incompletely understood, but evidence suggests that immunity to infections with serotypes 1 and 2 is mediated by local IgA antibodies in the respiratory tract. Passively acquired serum neutralizing antibodies also confer some protection against infection with types 1, 2, and -- to a lesser degree -- 3. Studies in experimental animal models and in immunosuppressed patients suggest that cell-mediated immunity may also be important in parainfluenza virus infections.

CLINICAL MANIFESTATIONS

Parainfluenza virus infections occur most frequently among children, in whom initial infection with serotype 1, 2, or 3 is associated with an acute febrile illness 50 to 80% of the time. Children may present with coryza, sore throat, hoarseness, and cough that may or may not be croupy. In severe croup, fever persists, with worsening coryza and sore throat. A brassy or barking cough may progress to frank stridor. Most children recover over the next 1 or 2 days, although progressive airway obstruction and hypoxia

ensue occasionally. If bronchiolitis or pneumonia develops, progressive cough accompanied by wheezing, tachypnea, and intercostal retractions may occur. In this setting, sputum production increases modestly. Physical examination shows nasopharyngeal discharge and oropharyngeal injection, along with rhonchi, wheezes, or coarse breath sounds. Chest x-rays can show air trapping and occasionally interstitial infiltrates.

In older children and adults, parainfluenza infections tend to be milder, presenting most frequently as a common cold or as hoarseness, with or without cough. Lower respiratory tract involvement in older children and adults is uncommon, but tracheobronchitis in adults has been reported. Severe, prolonged, and even fatal parainfluenza infection has been reported in children and adults with severe immunosuppression, including bone-marrow and solid-organ transplant recipients.

LABORATORY FINDINGS AND DIAGNOSIS

The clinical syndromes caused by parainfluenza viruses (with the possible exception of croup in young children) are not sufficiently distinctive to be diagnosed on clinical grounds alone. A specific diagnosis is established by detection of virus in respiratory tract secretions, throat swabs, or nasopharyngeal washings. Virus is detected by growth in tissue culture (either by hemagglutination or by a cytopathic effect), by immunofluorescence of viral antigens in exfoliated cells from the respiratory tract, or by ELISA. Polymerase chain reaction assays have also been developed. Serologic diagnosis is based on a fourfold or greater rise in antibody titer, as detected by hemagglutination inhibition or by complement-fixation or neutralization tests in acute- and convalescent-phase specimens. However, as frequent heterotypic responses occur among the parainfluenza serotypes, the serotype causing illness often cannot be identified by serologic techniques alone.

Acute epiglottitis caused by *Haemophilus influenzae* type b must be differentiated from viral croup. Influenza A virus also is a common cause of croup during epidemic periods.

TREATMENT

For upper respiratory tract illness, symptoms can be treated as discussed for other viral respiratory tract illnesses. If complications such as sinusitis, otitis, or superimposed bacterial bronchitis develop, appropriate antibiotics should be administered. Mild cases of croup should be treated with bed rest and moist air generated by vaporizers. More severe cases require hospitalization and close observation for the development of respiratory distress. If acute respiratory distress develops, humidified oxygen and intermittent racemic epinephrine are usually administered. Aerosolized or systemically administered glucocorticoids are beneficial; the latter have a more profound effect. No specific antiviral therapy is available, although ribavirin is active against parainfluenza viruses in vitro and anecdotal reports describe its use clinically. Effective vaccines against parainfluenza viruses have not been developed.

ADENOVIRUS INFECTIONS

ETIOLOGIC AGENT

Adenoviruses are complex DNA viruses that measure 70 to 80 nm in diameter. Human adenoviruses belong to the genus *Mastadenovirus*, which includes at least 47 serotypes. Adenoviruses have a characteristic morphology consisting of an icosahedral shell composed of 20 equilateral triangular faces and 12 vertices. The protein coat (capsid) consists of hexon subunits with group-specific and type-specific antigenic determinants and penton subunits at each vertex primarily containing group-specific antigens. A fiber with a knob at the end projects from each penton; this fiber contains type-specific and some group-specific antigens. Human adenoviruses have been divided into six subgenera (A through F) on the basis of the homology of DNA genomes and other properties. The adenovirus genome is a linear double-stranded DNA that codes for structural and nonstructural polypeptides. The replicative cycle of adenovirus may result either in lytic infection of cells or in the establishment of a latent infection (primarily involving lymphoid cells). Some adenovirus types can induce oncogenic transformation, and tumor formation has been observed in rodents; however, despite intensive investigation, adenoviruses have not been associated with tumors in humans.

EPIDEMIOLOGY

Adenovirus infections most frequently affect infants and children. Infections occur throughout the year but are most common from fall to spring. Adenoviruses account for 3 to 5% of acute respiratory infections in children but for fewer than 2% of respiratory illnesses in civilian adults. Nearly 100% of adults have serum antibody to multiple serotypes -- a finding indicating that infection is common in childhood. Types 1, 2, 3, and 5 are the most frequent isolates from children. Certain adenovirus serotypes -- particularly 4 and 7 but also 3, 14, and 21 -- are associated with outbreaks of acute respiratory disease in military recruits in winter and spring. Adenovirus infection can be transmitted by inhalation of aerosolized virus, by inoculation of virus into conjunctival sacs, and probably by the fecal-oral route as well. Type-specific antibody generally develops after infection and is associated with protection against infection with the same serotype.

CLINICAL MANIFESTATIONS

In children, adenoviruses cause a variety of clinical syndromes. The most common is an acute upper respiratory tract infection, with prominent rhinitis. On occasion, lower respiratory tract disease, including bronchiolitis and pneumonia, also develops. Adenoviruses, particularly types 3 and 7, cause pharyngoconjunctival fever, a characteristic acute febrile illness of children that occurs in outbreaks, most often in summer camps. The syndrome is marked by bilateral conjunctivitis in which the bulbar and palpebral conjunctivae have a granular appearance. Low-grade fever is frequently present for the first 3 to 5 days, and rhinitis, sore throat, and cervical adenopathy develop. The illness generally lasts for 1 to 2 weeks and resolves spontaneously. Febrile pharyngitis without conjunctivitis also has been associated with adenovirus infection. Adenoviruses have been isolated from cases of whooping cough with or without *Bordetella pertussis*; the significance of adenovirus in that disease is unknown.

In adults, the most frequently reported illness has been acute respiratory disease caused by adenovirus types 4 and 7 in military recruits. This illness is marked by a

prominent sore throat and the gradual onset of fever, which often reaches 39°C (102.2°F) on the second or third day of illness. Cough is almost always present, and coryza and regional lymphadenopathy are frequently seen. Physical examination may show pharyngeal edema, injection, and tonsillar enlargement with little or no exudate. If pneumonia has developed, auscultation and x-ray of the chest may indicate areas of patchy infiltration.

Adenoviruses have been associated with a number of non-respiratory tract diseases, including acute diarrheal illness caused by types 40 and 41 in young children and hemorrhagic cystitis caused by types 11 and 21. Epidemic keratoconjunctivitis, caused most frequently by types 8, 19, and 37, has been associated with contaminated common sources such as ophthalmic solutions and roller towels. Adenoviruses also have been implicated in disseminated disease and pneumonia in immunosuppressed patients, including recipients of solid-organ or bone-marrow transplants and patients with AIDS. In the latter group, high-numbered and intermediate serotypes have been isolated, usually in the setting of low CD4+ counts, but their isolation frequently has not been clearly linked to disease manifestations. Adenovirus nucleic acids have been detected in myocardial cells from patients with "idiopathic" cardiomyopathies, and adenoviruses have been suggested as causative agents in some cases.

LABORATORY FINDINGS AND DIAGNOSIS

Adenovirus infection should be suspected in the epidemiologic setting of acute respiratory disease in military recruits and in certain of the clinical syndromes (such as pharyngoconjunctival fever or epidemic keratoconjunctivitis) in which outbreaks of characteristic illnesses occur. In most cases, however, illnesses caused by adenovirus infection cannot be differentiated from those caused by a number of other viral respiratory agents and *Mycoplasma pneumoniae*. A definitive diagnosis of adenovirus infection is established by culture or detection of the virus by means of ELISA or nucleic acid hybridization from sites such as the conjunctiva and oropharynx or from sputum, urine, or stool. Virus may be detected in tissue culture by cytopathic changes and specifically identified by immunofluorescence or other immunologic techniques. Adenovirus types 40 and 41, which have been associated with diarrheal disease in children, require special tissue-culture cells for isolation, and these serotypes are most commonly detected by direct ELISA of stool. Serum antibody rises can be demonstrated by complement-fixation or neutralization tests, ELISA, radioimmunoassay, or (for those adenoviruses that hemagglutinate red cells) hemagglutination inhibition tests.

TREATMENT

Only symptom-based treatment and supportive therapy are available for adenovirus infections, and no clinically useful antiviral compounds have been identified. Live vaccines have been developed against adenovirus types 4 and 7 and have been used to control illness in military recruits. These vaccines consist of live, unattenuated virus administered in enteric-coated capsules. Infection of the gastrointestinal tract with types 4 and 7 does not cause disease but stimulates local and systemic antibodies that are protective against subsequent acute respiratory disease due to those serotypes. Vaccines prepared from purified subunits of adenovirus are being investigated. Adenoviruses are also being studied as live-virus vectors for the delivery of vaccine

antigens and for gene therapy.

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190. INFLUENZA - *Raphael Dolin*

DEFINITION

Influenza is an acute respiratory illness caused by infection with influenza viruses. The illness affects the upper and/or lower respiratory tract and is often accompanied by systemic signs and symptoms such as fever, headache, myalgia, and weakness. Outbreaks of illness of variable extent and severity occur nearly every winter. Such outbreaks result in significant morbidity in the general population and in increased mortality rates among certain high-risk patients, mainly as a result of pulmonary complications.

ETIOLOGIC AGENT

Influenza viruses are members of the Orthomyxoviridae family. Influenza A and B viruses constitute one genus, and influenza C viruses make up the other. The designation of influenza viruses as type A, B, or C is based on antigenic characteristics of the nucleoprotein (NP) and matrix (M) protein antigens. Influenza A viruses are further subdivided (subtyped) on the basis of the surface hemagglutinin (H) and neuraminidase (N) antigens (see below); individual strains are designated according to the site of origin, isolate number, year of isolation, and subtype -- for example, influenza A/Sydney/5/97 (H3N2). Influenza B and C viruses are similarly designated, but H and N antigens from these viruses do not receive subtype designations, since intratypic variations in influenza B antigens are less extensive than those in influenza A viruses and may not occur with influenza C virus.

Most of the information on the molecular biology of influenza viruses has come from studies of influenza A viruses; less is known about the replicative cycle of influenza B and C viruses. Morphologically, influenza viruses A, B, and C are similar. The virions are irregularly shaped spherical particles, 80 to 120 nm in diameter, and have a lipid envelope from the surface of which the H and N glycoproteins project ([Fig. 190-1](#)). The hemagglutinin is the site by which virus binds to cell receptors, whereas the neuraminidase degrades the receptor and probably plays a role in the release of virus from infected cells after replication has taken place. Influenza viruses enter cells by receptor-mediated endocytosis, forming a virus-containing endosome. The viral hemagglutinin mediates fusion of the endosomal membrane with the virus envelope, and viral nucleocapsids are subsequently released into the cytoplasm. Antibodies to the H antigen are the major determinants of immunity to influenza virus, while those to the N antigen limit viral spread and contribute to reduction of the infection. The inner surface of the lipid envelope contains the M proteins M1 and M2, the functions of which are incompletely understood but which may be involved in virus assembly and in stabilization of the lipid envelope. The virion also contains the NP antigen, which is associated with the viral genome, as well as three polymerase (P) proteins that are essential for transcription and synthesis of viral RNA. Two nonstructural (NS) proteins of unknown function are also present in infected cells.

The genomes of influenza A and B viruses consist of eight single-stranded RNA segments, which code for the structural and nonstructural proteins. Because the genome is segmented, the opportunity for reassortment of genes during infection is

high, and reassortment occurs frequently during infection of cells with more than one influenza A virus.

EPIDEMIOLOGY

Influenza outbreaks are recorded virtually every year, although their extent and severity vary widely. Localized outbreaks take place at variable intervals, usually every 1 to 3 years. Except for the past two decades, global epidemics or pandemics have occurred approximately every 10 to 15 years since the 1918-1919 pandemic ([Table 190-1](#)).

The most extensive and severe outbreaks are caused by influenza A viruses. In part, this predominance is a result of the remarkable propensity of the H and N antigens of influenza A virus to undergo periodic antigenic variation. Major antigenic variations are referred to as *antigenic shifts*, which may be associated with pandemics and are restricted to influenza A viruses. Minor variations are called *antigenic drifts*. These antigenic changes may involve the hemagglutinin alone or both the hemagglutinin and the neuraminidase. In human infections, three major antigenic subtypes of hemagglutinins (H1, H2, and H3) and two of neuraminidases (N1 and N2) have been recognized. The hemagglutinins formerly designated as H0 and Hsw are now classified as variants of H1. An example of an antigenic shift involving both the hemagglutinin and the neuraminidase is that of 1957, when the predominant influenza A virus subtype shifted from H1N1 to H2N2; this shift resulted in a severe pandemic, with an estimated 70,000 excess deaths (i.e., deaths in excess of the number expected without an influenza epidemic) in the United States alone. In 1968, an antigenic shift involving only the hemagglutinin occurred (H2N2 to H3N2); the subsequent pandemic was less severe than that of 1957. In 1977, an H1N1 virus emerged and caused a pandemic that primarily affected younger individuals (i.e., those born after 1957). As can be seen in [Table 190-1](#), H1N1 viruses circulated from 1918 to 1956; thus, individuals born prior to 1957 would be expected to have some degree of immunity to H1N1 viruses. During most outbreaks of influenza A, a single subtype has circulated at a time. However, since 1977, H1N1 and H3N2 viruses have circulated simultaneously, resulting in outbreaks of varying severity. In some outbreaks, influenza B viruses have also circulated simultaneously with influenza A viruses.

The origin of pandemic strains is unknown. Given the marked differences between the primary structures of the hemagglutinins of different subtypes of influenza A viruses (H1, H2, and H3), it seems unlikely that antigenic shifts result from spontaneous mutations in the hemagglutinin gene. Because the segmented genome of influenza viruses may result in high rates of reassortment, it has been suggested that pandemic strains may emerge by reassortment of genes between human and animal viruses. There was concern that such reassortment might have occurred in 1997 in Hong Kong, where cases of infection caused by influenza virus A/H5N1 were detected in humans during an extensive outbreak of avian influenza A/H5N1 in poultry. However, only a few cases of A/H5N1 influenza in humans were documented, and the infection did not spread into the community. Influenza B viruses do not have an animal reservoir and do not undergo antigenic shifts, although they do undergo antigenic drift.

Pandemics provide the most dramatic evidence of the impact of influenza. However, illnesses that occur between pandemics account for greater total mortality and

morbidity, albeit over a longer period. From 1972 through the present, influenza has been associated with at least 20,000 excess deaths during more than half of the interpandemic epidemics in the United States; more than 40,000 influenza-associated deaths occurred in each of three of these epidemics. Influenza A viruses that circulate between pandemics demonstrate antigenic drifts in the H antigen. These antigenic drifts apparently result from point mutations involving the RNA segment that codes for the hemagglutinin. Epidemiologically significant strains -- that is, those with the potential to cause widespread outbreaks -- exhibit changes in amino acids in at least two of the major antigenic sites in the hemagglutinin molecule. Since two point mutations are unlikely to occur simultaneously, it is believed that antigenic drifts result from point mutations occurring sequentially during the spread of virus from person to person. Antigenic drifts have been reported nearly annually since 1977 for H1N1 viruses and since 1968 for H3N2 viruses.

Influenza A epidemics begin abruptly, peak over a 2- to 3-week period, generally last for 2 to 3 months, and often subside almost as rapidly as they began. The first indication of influenza activity in a community is an increase in the number of children with febrile respiratory illnesses who present for medical attention. This increase is followed by increases in rates of influenza-like illnesses among adults and eventually by an increase in hospital admissions for patients with pneumonia, worsening of congestive heart failure, and exacerbations of chronic pulmonary disease. Rates of absence from work and school also rise at this time. An increase in the number of deaths caused by pneumonia and influenza is generally a late observation in an outbreak. Attack rates have been highly variable from outbreak to outbreak but most commonly are in the range of 10 to 20% of the general population. During the pandemic of 1957, it was estimated that the attack rate of clinical influenza exceeded 50% in urban populations and that an additional 25% or more of individuals in these populations may have been subclinically infected with influenza A virus. Among institutionalized populations and in semiclosed settings with a large number of susceptible individuals, even higher attack rates have been reported.

Epidemics of influenza occur almost exclusively during the winter months in the temperate zones of the northern and southern hemispheres. In those locations, it is highly unusual to detect influenza A virus at other times, although serologic rises or even outbreaks have been noted rarely during warm-weather months. In contrast, influenza virus infections occur throughout the year in the tropics. Where or how influenza A virus persists between outbreaks in temperate zones is unknown. It is possible that influenza A viruses are maintained in the human population on a worldwide basis by person-to-person transmission and that large population clusters support a low level of interepidemic transmission. Alternatively, human strains may persist in animal reservoirs. Convincing evidence to support either explanation is not available. In the modern era, rapid transportation may contribute to the transmission of viruses among widespread geographic locales.

The factors that result in the inception and termination of outbreaks of influenza are incompletely understood. A major determinant of the extent and severity of an outbreak is the level of immunity in the population at risk. With the emergence of an antigenically novel influenza virus to which little or no antibody is present in a community, extensive outbreaks may occur. When the absence of antibody is worldwide, epidemic disease

may spread around the globe, resulting in a pandemic. Such pandemic waves can continue for several years, until immunity in the population reaches a high level. In the years following pandemic influenza, antigenic drifts among influenza viruses result in outbreaks of variable severity in populations with high levels of immunity to the pandemic strain that circulated earlier. This situation persists until another antigenically novel pandemic strain emerges. On the other hand, outbreaks sometimes end despite the persistence of a large pool of susceptible individuals in the population.

Occasionally, the emergence of a significantly different antigenic variant will result only in a localized outbreak. The swine influenza outbreak of 1976 in the United States, caused by an A/H1N1 virus antigenically similar to the virus that circulated in 1918-1919, may be an example, although this outbreak may have represented simply the introduction of a swine influenza virus into a crowded human population without spread beyond that setting. The cluster of human infections with influenza A/H5N1 in Hong Kong in 1997 may also be an example of this phenomenon. It has been suggested that certain viruses, such as recently circulating A/H1N1 strains, may be intrinsically less virulent and cause less severe disease than other variants, even in immunologically virgin subjects. If so, then other (undefined) factors besides the level of preexisting immunity must play a role in the epidemiology of influenza.

Influenza B virus causes outbreaks that are generally less extensive and are associated with less severe disease than those caused by influenza A virus. The hemagglutinin and neuraminidase of influenza B virus undergo less frequent and less extensive variation than those of influenza A viruses; this characteristic may account, in part, for the lesser extent of disease. Influenza B outbreaks are seen most frequently in schools and military camps, although outbreaks in institutions in which elderly individuals reside have also been noted on occasion. The most serious complication of influenza B virus infection is Reye's syndrome ([Chap. 300](#)). Influenza C virus has only infrequently been associated with human disease, although the wide prevalence of serum antibody to this virus indicates that asymptomatic infection may be common.

The morbidity and mortality caused by influenza outbreaks continue to be substantial. Most individuals who die in this setting have underlying diseases that place them at high risk for complications of influenza. Excess hospitalizations for adults with high-risk medical conditions have ranged from 20 to 1000 per 100,000 during recent outbreaks of influenza. The most prominent high-risk conditions are chronic cardiac and pulmonary diseases as well as old age. Mortality among individuals with chronic metabolic, renal, and certain immunosuppressive diseases has also been elevated, although lower than that among patients with chronic cardiopulmonary diseases. The morbidity attributable to influenza in the general population is considerable. For each of three outbreaks in the United States that were studied during the 1960s, estimated direct and indirect economic costs ranged from \$1.5 to \$3.5 billion; today such costs would obviously be much greater.

PATHOGENESIS

The initial event in influenza is infection of the respiratory epithelium with influenza virus acquired from respiratory secretions of acutely infected individuals. In all likelihood, transmission occurs via aerosols generated by coughs and sneezes, although

hand-to-hand contact, other personal contact, and even fomite transmission may take place. Experimental evidence suggests that infection by a small-particle aerosol (particle diameter, <10 μ m) is more efficient than that by larger droplets. Initially, viral infection involves the ciliated columnar epithelial cells, but it also may involve other respiratory tract cells, including alveolar cells, mucous gland cells, and macrophages. In infected cells, virus replicates within 4 to 6 h, after which infectious virus is released to infect adjacent or nearby cells. In this way, infection spreads from a few foci to a large number of respiratory cells over several hours. In experimentally induced infection, the incubation period of illness has ranged from 18 to 72 h, depending on the size of the virus inoculum. Histopathologic study reveals degenerative changes, including granulation, vacuolization, swelling, and pyknotic nuclei, in infected ciliated cells. The cells eventually become necrotic and desquamate; in some areas, previously columnar epithelium is replaced by flattened and metaplastic epithelial cells. The severity of illness is correlated with the quantity of virus shed in secretions; thus, the degree of viral replication itself may be an important factor in pathogenesis. Despite the frequent development of systemic signs and symptoms such as fever, headache, and myalgias, influenza virus has only rarely been detected in extrapulmonary sites (including the bloodstream). Evidence suggests that the pathogenesis of systemic symptoms in influenza may be related to the induction of certain cytokines, particularly tumor necrosis factor α and interleukin 6.

The host response to influenza infections involves a complex interplay of humoral antibody, local antibody, cell-mediated immunity, interferon, and other host defenses. Serum antibody responses, which can be detected by the second week after primary infection, are measured by a variety of techniques: hemagglutination inhibition (HI), complement fixation (CF), neutralization, enzyme-linked immunosorbent assay (ELISA), and antineuraminidase antibody assay. Antibodies directed against the hemagglutinin appear to be the most important mediators of immunity; in several studies, HI titers of ≥ 40 have been associated with protection from infection. Secretory antibodies produced in the respiratory tract are predominantly of the IgA class and also play a major role in protection against infection. Secretory antibody neutralization titers of ≥ 4 have also been associated with protection. A variety of cell-mediated immune responses, both antigen-specific and antigen-nonspecific, can be detected early after infection and depend on the prior immune status of the host. These responses include T-cell proliferative, T-cell cytotoxic, and natural killer cell activity. Interferons have been detected in respiratory secretions shortly after the shedding of virus has begun, and rises in interferon titers coincide with decreases in virus shedding.

The host defense factors responsible for cessation of virus shedding and resolution of illness have not been defined specifically. Virus shedding generally stops within 2 to 5 days after symptoms first appear, at a time when serum and local antibody responses often are not detectable by conventional techniques (although antibody rises may be detected earlier by use of highly sensitive techniques, particularly in individuals with previous immunity to the virus). It has been suggested that interferon, cell-mediated immune responses, and/or nonspecific inflammatory responses are important in the resolution of illness.

MANIFESTATIONS

Influenza has most frequently been described as an illness characterized by the abrupt onset of systemic symptoms, such as headache, feverishness, chills, myalgia, or malaise, and accompanying respiratory tract signs, particularly cough and sore throat. In many cases, the onset is so abrupt that patients can recall the precise time they became ill. A typical case of naturally occurring influenza is depicted in [Fig. 190-2](#). However, the spectrum of clinical presentations is wide, ranging from a mild, afebrile respiratory illness similar to the common cold (with either a gradual or an abrupt onset) to severe prostration with relatively few respiratory signs and symptoms. In most of the cases that come to a physician's attention, the patient has a fever, with temperatures of 38° to 41°C (100.4° to 105.8°F). A rapid temperature rise within the first 24 h of illness is generally followed by a gradual defervescence over a 2- to 3-day period, although, on occasion, fever may last for as long as a week. Patients report a feverish feeling and chilliness, but true rigors are rare. Headache, either generalized or frontal, is often particularly troublesome. Myalgias may involve any part of the body but are most common in the legs and lumbosacral area. Arthralgias may also develop.

Respiratory complaints often become more prominent as systemic symptoms subside. Many patients have a sore throat or persistent cough, which may last for a week or more and which is often accompanied by substernal discomfort. Ocular signs and symptoms include pain on motion of the eyes, photophobia, and burning of the eyes.

Physical findings are usually minimal in cases of uncomplicated influenza. Early in the illness, the patient appears flushed and the skin is hot and dry, although diaphoresis and mottled extremities are sometimes evident, particularly in older patients. Examination of the pharynx may yield surprisingly unremarkable results despite a severe sore throat, but injection of the mucous membranes and postnasal discharge are apparent in some cases. Mild cervical lymphadenopathy may be noted, especially in younger individuals. The results of chest examination are largely negative in uncomplicated influenza, although rhonchi, wheezes, and scattered rales have been reported with variable frequency in different outbreaks. Frank dyspnea, hyperpnea, cyanosis, diffuse rales, and signs of consolidation are indicative of pulmonary complications. Patients with apparently uncomplicated influenza have been reported to have a variety of mild ventilatory defects and increased alveolar-capillary diffusion gradients; thus, subclinical pulmonary involvement may be more frequent than is appreciated.

In uncomplicated influenza, the acute illness generally resolves over a 2- to 5-day period, and most patients have largely recovered in 1 week. In a significant minority (particularly the elderly), however, symptoms of weakness or lassitude (postinfluenza asthenia) may persist for several weeks and may prove troublesome for persons who wish to resume their full level of activity promptly. The pathogenetic basis for this asthenia is unknown, although pulmonary function abnormalities may persist for several weeks after uncomplicated influenza.

COMPLICATIONS

The most common complication of influenza is pneumonia: "primary" influenza viral pneumonia, secondary bacterial pneumonia, or mixed viral and bacterial pneumonia. Primary influenza viral pneumonia is the least common but most severe of the

pneumonic complications. It presents as acute influenza that does not resolve but instead progresses relentlessly, with persistent fever, dyspnea, and eventual cyanosis. Sputum production is generally scanty, but the sputum can contain blood. Few physical signs may be evident early in the illness. In more advanced cases, diffuse rales may be noted, and chest x-ray findings consistent with diffuse interstitial infiltrates and/or acute respiratory distress syndrome may be present. In such cases, arterial blood-gas determinations show marked hypoxia. Viral cultures of respiratory secretions and lung parenchyma, especially if samples are taken early in illness, yield high titers of virus. In fatal cases of primary viral pneumonia, histopathologic examination reveals a marked inflammatory reaction in the alveolar septa, with edema and infiltration by lymphocytes, macrophages, occasional plasma cells, and variable numbers of neutrophils. Fibrin thrombi in alveolar capillaries, along with necrosis and hemorrhage, have also been noted. Eosinophilic hyaline membranes can be found lining alveoli and alveolar ducts.

Primary influenza viral pneumonia has a predilection for individuals with cardiac disease, particularly those with mitral stenosis ([Fig. 190-CD1](#)), but has also been reported in otherwise healthy young adults as well as in older individuals with chronic pulmonary disorders. In some epidemics of influenza (notably those of 1918 and 1957), pregnancy increased the risk of primary influenza pneumonia.

Secondary bacterial pneumonia follows acute influenza. Improvement of the patient's condition over 2 to 3 days is followed by a reappearance of fever along with clinical signs and symptoms of bacterial pneumonia, including cough, production of purulent sputum, and physical and x-ray signs of consolidation. The most common bacterial pathogens in this setting are *Streptococcus pneumoniae*, *Staphylococcus aureus*, and *Haemophilus influenzae* -- organisms that can colonize the nasopharynx and that cause infection in the wake of changes in bronchopulmonary defenses. The etiology can often be determined by Gram's staining and culture of an appropriately obtained sputum specimen. Secondary bacterial pneumonia occurs most frequently in high-risk individuals with chronic pulmonary and cardiac disease and in elderly individuals. Patients with secondary bacterial pneumonia often respond to antibiotic therapy when it is instituted promptly.

Perhaps the most common pneumonic complications during outbreaks of influenza have mixed features of viral and bacterial pneumonia. Patients may experience a gradual progression of their acute illness or may show transient improvement followed by clinical exacerbation, with eventual manifestation of the clinical features of bacterial pneumonia. Sputum cultures may contain both influenza A virus and one of the bacterial pathogens described above. Patchy infiltrates or areas of consolidation may be detected by physical examination and chest x-ray. Patients with mixed viral and bacterial pneumonia generally have less widespread involvement of the lung than those with primary viral pneumonia, and their bacterial infections may respond to appropriate antibiotics. Mixed viral and bacterial pneumonia occurs primarily in patients with chronic cardiovascular and pulmonary diseases.

Other pulmonary complications associated with influenza include worsening of chronic obstructive pulmonary disease and exacerbation of chronic bronchitis and asthma. In children, influenza infection may present as croup.

In addition to the pulmonary complications of influenza, a number of extrapulmonary complications may occur. These include *Reye's syndrome*, a serious complication in children that is associated with influenza B and to a lesser extent with influenza A virus infection as well as with varicella-zoster virus infection. An epidemiologic association between Reye's syndrome and aspirin therapy for the antecedent viral infection has been noted, and the incidence of Reye's syndrome has decreased markedly with widespread warnings regarding the use of aspirin by children with acute viral respiratory infections. A detailed description of Reye's syndrome is found in [Chap. 300](#).

Myositis, rhabdomyolysis, and myoglobinuria are occasional complications of influenza infection. Although myalgias are exceedingly common in influenza, true myositis is rare. Patients with acute myositis have exquisite tenderness of the affected muscles, most commonly in the legs, and may not be able to tolerate even the slightest pressure, such as the touch of bedsheets. In the most severe cases, there is frank swelling and boggy muscles. Serum levels of creatine phosphokinase and aldolase are markedly elevated, and an occasional patient has developed renal failure from myoglobinuria. The pathogenesis of influenza-associated myositis is also unclear, although the presence of influenza virus in affected muscles has been reported.

Myocarditis and pericarditis were reported in association with influenza virus infection during the 1918-1919 pandemic; these reports were based largely on histopathologic findings, and these complications have been reported only infrequently since that time. Electrocardiographic changes during acute influenza are common among patients who have cardiac disease but have been ascribed most often to exacerbations of the underlying cardiac disease rather than to direct involvement of the myocardium with influenza virus.

Central nervous system (CNS) diseases, including encephalitis, transverse myelitis, and Guillain-Barre syndrome, have been reported during influenza. The etiologic relationship of influenza virus to such CNS illnesses remains unestablished. Toxic shock syndrome caused by *S. aureus* infection following acute influenza infection has also been reported ([Chap. 139](#)).

In addition to complications involving the specific organ systems described above, influenza outbreaks include a number of cases in which elderly and other high-risk individuals develop influenza and subsequently experience a gradual deterioration of underlying cardiovascular, pulmonary, or renal function -- changes that occasionally are irreversible and lead to death. These fatalities contribute to the overall excess mortality associated with influenza A outbreaks.

LABORATORY FINDINGS AND DIAGNOSIS

Laboratory diagnosis is accomplished during acute influenza by isolation of the virus from throat swabs, nasopharyngeal washes, or sputum. Virus is usually detected in tissue culture or less commonly is found in the amniotic cavity of chick embryos within 48 to 72 h after inoculation. The rapid viral diagnostic tests now available detect viral nucleoprotein or neuraminidase with high specificity and sensitivities of 57 to 81% compared with tissue culture. Viral nucleic acids have been detected in clinical samples by reverse transcriptase polymerase chain reaction. The type of influenza virus (A or B)

may be determined by either immunofluorescence or [HI](#) techniques, and the hemagglutinin subtype of influenza A virus (H1, H2, or H3) may be identified by HI with use of subtype-specific antisera. Serologic methods for diagnosis require comparison of antibody titers in sera obtained during the acute illness with those in sera obtained 10 to 14 days after the onset of illness and are useful primarily in retrospect. Fourfold or greater titer rises as detected by HI or [CF](#) or significant rises as measured by [ELISA](#) are diagnostic of acute infection. CF tests are generally less sensitive than other serologic techniques, but, as they detect type-specific antigens, they may be particularly useful when subtype-specific reagents are not available.

Other laboratory tests are generally not helpful in making a specific diagnosis of influenza virus infection. Leukocyte counts are variable, frequently being low early in illness and normal or slightly elevated later. Severe leukopenia has been described in overwhelming viral or bacterial infection, while leukocytosis with more than 15,000 cells/uL raises the suspicion of secondary bacterial infection.

DIFFERENTIAL DIAGNOSIS

On clinical grounds alone, an individual case of influenza may be difficult to differentiate from an acute respiratory illness caused by any of a variety of respiratory viruses or by *Mycoplasma pneumoniae*. Severe streptococcal pharyngitis or early bacterial pneumonia may mimic acute influenza, although bacterial pneumonias generally do not run a self-limited course. Purulent sputum in which a bacterial pathogen can be detected by Gram's staining is an important diagnostic feature in bacterial pneumonia. The fact that influenza occurs in characteristic outbreaks during the winter months may facilitate a clinical diagnosis. When local health authorities indicate that influenza is present in the community, an acute febrile respiratory illness can be attributed to influenza with a high degree of certainty, particularly if the typical features of abrupt onset and systemic symptoms are present.

TREATMENT

In uncomplicated cases of influenza, symptom-based therapy with acetaminophen for the relief of headache, myalgia, and fever may be considered, but the use of salicylates should be avoided in children below 18 years of age because of the possible association of salicylates with Reye's syndrome. Since cough is ordinarily self-limited, treatment with cough suppressants generally is not indicated, although codeine-containing compounds may be employed if the cough is particularly troublesome. Patients should be advised to rest and maintain hydration during acute illness and should return to full activity only gradually after the illness has resolved, especially if the illness has been severe.

Specific antiviral therapy is available for influenza: amantadine and rimantadine for influenza A and the neuraminidase inhibitors zanamivir and oseltamivir for both influenza A and influenza B. If begun within 48 h of the onset of illness, treatment with amantadine or rimantadine has reduced the duration of systemic and respiratory symptoms of influenza by ~50%. From 5 to 10% of individuals who receive amantadine experience mild [CNS](#) side effects, primarily jitteriness, anxiety, insomnia, or difficulty in concentrating. These side effects disappear promptly upon cessation of the drug.

Rimantadine appears to be equally efficacious and is associated with less frequent CNS side effects than is amantadine. In adults, the usual dose of amantadine or rimantadine is 200 mg/d for 3 to 7 days. Since both drugs are excreted via the kidney, the dose should be reduced to 100 mg/d in elderly patients and patients with renal insufficiency. Zanamivir, inhaled orally at a dose of 10 mg twice a day for 5 days, or oseltamivir, ingested orally at a dose of 75 mg twice a day for 5 days, has reduced the duration of signs and symptoms of influenza by 1 to 1.5 days if treatment is started within 2 days of the onset of illness. Zanamivir may exacerbate bronchospasm in asthmatic patients, and oseltamivir has been associated with nausea and vomiting, whose frequency can be reduced by drug administration with food. Currently, only amantadine and zanamivir are approved in the United States for treatment of children (the latter for use in children ³7 years old). Ribavirin, a nucleoside analogue with activity against a variety of viral agents, has been reported to be effective against both influenza A and influenza B virus infections when administered as an aerosol, although it is relatively ineffective when administered orally.

Studies demonstrating the therapeutic efficacy of antiviral compounds in influenza have primarily involved young adults with uncomplicated disease; it is not known whether such compounds are effective in the treatment of complications such as influenza pneumonia. Therapy for primary influenza pneumonia is directed at maintaining oxygenation and is most appropriately undertaken in an intensive care unit, with aggressive respiratory and hemodynamic support as needed. Bypass membrane oxygenators have been employed in this setting with variable results. When an acute respiratory distress syndrome develops, fluids must be administered cautiously, with close monitoring of blood gases and hemodynamic function.

Antibacterial drugs should be reserved for the therapy of bacterial complications of acute influenza, such as secondary bacterial pneumonia. The choice of antibiotics should be guided by Gram's staining and culture of appropriate specimens of respiratory secretions, such as sputum or transtracheal aspirates. If the etiology of a case of bacterial pneumonia is unclear from an examination of respiratory secretions, empirical antibiotics effective against the most common bacterial pathogens in this setting (*S. pneumoniae*, *S. aureus*, and *H. influenzae*) should be selected ([Chaps. 138, 139](#), and [149](#)).

PROPHYLAXIS

The major public health measure for prevention of influenza has been the use of inactivated influenza vaccines derived from influenza A and B viruses that circulated during the previous influenza season. If the vaccine virus and the currently circulating viruses are closely related, 50 to 80% protection against influenza would be expected. Presently available vaccines have been highly purified and are associated with few reactions. Up to 5% of individuals experience low-grade fever and mild systemic symptoms 8 to 24 h after vaccination, and up to one-third develop mild redness or tenderness at the vaccination site. Since the vaccine is produced in eggs, individuals with true hypersensitivity to egg products either should be desensitized or should not be vaccinated. Although the 1976 swine influenza vaccine appears to have been associated with an increased frequency of Guillain-Barre syndrome, influenza vaccines administered since 1976 generally have not been. Possible exceptions were noted

during the 1992-1993 and 1993-1994 influenza seasons, when there may have been an excess risk of Guillain-Barre syndrome of slightly more than one case per million among vaccine recipients. However, the overall health risk following influenza outweighs the potential risk associated with vaccination. Investigational live attenuated ("cold-adapted") influenza A and B vaccines also have been developed and have been highly effective in preventing influenza in studies in adults and children. Such vaccines are administered intranasally and stimulate local antibody production more efficiently than conventional inactivated vaccines.

The U.S. Public Health Service recommends influenza vaccination for any individual >6 months of age who is at an increased risk for complications of influenza. Included are individuals with chronic cardiovascular or pulmonary disorders (including asthma) and residents of nursing homes and other chronic-care facilities. Other populations for whom the vaccine is recommended include healthy individuals >65 years of age and individuals who have required regular medical attention for diabetes mellitus, renal disease, hemoglobinopathies, or immunosuppression. Individuals who provide care for high-risk patients or who come into frequent contact with such patients, including household members, should also receive vaccine to reduce the likelihood of transmission of infection. Vaccination is recommended for women who will be in the second or third trimester of pregnancy during the influenza season and for individuals 6 months to 18 years of age who are receiving long-term aspirin therapy and may be at risk for Reye's syndrome. Since commercially available vaccines are inactivated ("killed"), they may be administered safely to immunocompromised patients. Influenza vaccination is not associated with exacerbations of chronic nervous-system diseases such as multiple sclerosis. Vaccine should be administered early in the autumn before influenza outbreaks occur and should be repeated annually to maintain immunity against the most current influenza virus strains.

Of the vaccines currently available (inactivated whole-virus vaccine, subvirion vaccine, and purified surface-antigen vaccine), only the "split-virus" preparations (i.e., the subvirion and purified surface-antigen vaccines) should be given to children <13 years old, since the whole-virus preparations have been associated with higher rates of adverse reactions in this age group.

Studies have shown amantadine and rimantadine to be 70 to 100% effective in the prophylaxis of illness associated with influenza A virus infection. Such prophylaxis is most likely to be used for high-risk individuals who have not received influenza vaccine or in a situation where the vaccines previously administered are relatively ineffective because of antigenic changes in the circulating virus. During an outbreak, amantadine or rimantadine can be administered simultaneously with inactivated vaccine, since neither drug interferes with an immune response to the vaccine. In fact, there is evidence that the protective effects of amantadine and vaccine may be additive. Amantadine has also been employed to control nosocomial outbreaks of influenza A. For prophylaxis, administration of amantadine or rimantadine should be instituted promptly when influenza A activity is detected and must be continued daily for the duration of the outbreak. The dosage most frequently employed has been 200 mg/d for adults, but the dose should be reduced for patients with renal insufficiency and for the elderly. Viruses resistant to both amantadine and rimantadine can emerge quickly after therapy with these drugs, and the possible transmission of these resistant viruses has

been reported. The neuraminidase inhibitors zanamivir and oseltamivir have also been reported to be highly effective in the prophylaxis of influenza A and offer the advantage of efficacy against influenza B as well. They are currently under review for use as prophylaxis. As with amantadine and rimantadine, the neuraminidase inhibitors must be administered daily to maintain prophylaxis.

(Bibliography omitted in Palm version)

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SECTION 14 -RNA VIRUSES

191. THE HUMAN RETROVIRUSES - Anthony S. Fauci, Dan L. Longo

The retroviruses, which make up a large family (Retroviridae), infect mainly vertebrates. They have a unique replication cycle whereby their genetic information is encoded by RNA rather than DNA. Retroviruses contain an RNA-dependent DNA polymerase (a reverse transcriptase) that directs the synthesis of a DNA form of the viral genome after infection of a host cell. The designation *retrovirus* denotes that information in the form of RNA is transcribed into DNA in the host cell -- a sequence that overturned a central dogma of molecular biology: that information passes unidirectionally from DNA to RNA to protein. The observation that RNA was the source of genetic information in the causative agents of certain animal tumors led to a number of paradigm-shifting biologic insights regarding not only the direction of genetic-information passage but also the viral etiology of certain cancers and the concept of oncogenes as normal host genes scavenged and altered by a viral vector.

The family Retroviridae includes three subfamilies ([Table 191-1](#)): Oncovirinae, of which human T-cell lymphotropic virus (HTLV) type I is the most important in humans; Lentivirinae, of which HIV is the most important in humans; and Spumavirinae, the "foamy" viruses, named for the pathologic appearance of infected cells. A number of spumaviruses have been isolated from humans; however, they are not associated with any known disease and therefore are not discussed further in this chapter.

The wide variety of interactions of a retrovirus with its host range from completely benign events (e.g., silent carriage of endogenous retroviral sequences in the germ-line genome of many animal species) to rapidly fatal infections (e.g., exogenous infection with an oncogenic virus such as Rous sarcoma virus in chickens). The ability of retroviruses to acquire and alter the structure and function of host cell sequences has revolutionized our understanding of molecular carcinogenesis. The viruses can insert into the germ-line genome of the host cell and behave as a transposable or movable genetic element. They can activate or inactivate genes near the site of integration into the genome. They can rapidly alter their own genome by recombination and mutation under selective environmental stimuli.

Most human viral diseases occur as a consequence of either tissue destruction by the virus itself or the host's response to the virus. Although these mechanisms are operative in retroviral infections, retroviruses have additional mechanisms of inducing disease, including the malignant transformation of an infected cell and the induction of an immunodeficiency state that leads to opportunistic diseases (infections and neoplasms; [Chap. 309](#)).

STRUCTURE AND LIFE CYCLE

Despite the wide range of biologic consequences of retroviral infection, all retroviruses are similar in structure, genome organization, and mode of replication. Retroviruses are 70 to 130 nm in diameter and have a lipid-containing envelope surrounding an icosahedral capsid with a dense inner core. The core contains two identical copies of the single-stranded RNA genome. The RNA molecules are 8 to 10 kb long and are

complexed with reverse transcriptase and tRNA. Other viral proteins, such as integrase, are also components of the virion particle. The RNA has features usually found in mRNA: a cap site at the 5' end of the molecule, which is important in the initiation of mRNA translation, and a polyadenylation site at the 3' end, which influences mRNA turnover (i.e., messages with shorter polyA tails turn over faster than messages with longer polyA tails). However, the retroviral RNA is not translated; instead it is transcribed into DNA. The DNA form of the retroviral genome is called a *provirus*.

The replication cycle of retroviruses proceeds in two phases ([Fig. 191-1](#)). In the first phase, the virus enters the cytoplasm after binding to a specific cell-surface receptor (with HIV, a cell-surface coreceptor is also utilized for binding and entry); the viral RNA and reverse transcriptase synthesize a double-stranded DNA version of the RNA template; and the provirus moves into the nucleus and integrates into the host cell genome. This proviral integration is permanent. Although some animal retroviruses integrate into a single specific site of the host genome in every infected cell, the four known pathogenic human retroviruses ([HTLV-I](#), HTLV-II, HIV-1, and HIV-2) integrate randomly. This first phase of replication depends entirely on gene products in the virus. The second phase includes the synthesis and processing of viral genomes, mRNAs, and proteins using host cell machinery, often under the influence of viral gene products. Virions are assembled and released from the cell by budding from the membrane; host cell membrane proteins are frequently incorporated into the envelope of the virus. Proviral integration occurs during the S phase of the cell cycle; thus, in general, nondividing cells are resistant to retroviral infection. Only the lentiviruses are able to infect nondividing cells. Once a host is infected, it is infected for life.

Retroviral genomes include both coding and noncoding sequences ([Fig. 191-2](#)). In general, noncoding sequences are important recognition signals for DNA or RNA synthesis or processing events and are located in the 5' and 3' terminal regions of the genome. All retroviral genomes are terminally redundant, containing identical sequences called *long terminal repeats* (LTRs). The ends of the retroviral RNA genome differ slightly in sequence from the integrated retroviral DNA. In the latter, the LTR sequences are repeated in both the 5' and the 3' terminus of the virus. The LTRs contain sequences involved in initiating the expression of the viral proteins, the integration of the provirus, and the polyadenylation of viral RNAs. The primer binding site, which is critical for the initiation of reverse transcription, and the viral packaging sequences are located outside the LTR sequences. The coding regions include the *gag* (group-specific antigen, core protein), *pol* (RNA-dependent DNA polymerase), and *env* (envelope) genes. The *gag* gene encodes a precursor polyprotein that is cleaved to form three to five capsid proteins; a fraction of the Gag precursor proteins also contain a protease responsible for cleaving the Gag and Pol polyproteins. A Gag-Pol polyprotein gives rise to the protease that is responsible for cleaving the Gag-Pol polyprotein. The *pol* gene encodes three proteins: the reverse transcriptase, the integrase, and the protease. The reverse transcriptase functions to copy the viral RNA into the double-stranded DNA provirus, which can attach to the host cell DNA via the action of integrase. The protease functions to cleave the Gag-Pol polyprotein into smaller protein products. The *env* gene encodes the envelope glycoproteins: one protein that binds to specific surface receptors and determines what cell types can be infected and a smaller transmembrane protein that anchors the complex to the envelope. The cartoon in [Fig. 191-3](#) shows how the retroviral gene products make up the virus structure.

[HTLVs](#) have a region between *env* and the 3' [LTR](#) that encodes at least two proteins in overlapping reading frames; Tax, a 40-kD protein that does not bind to DNA but induces the expression of host cell transcription factors that alter host cell gene expression; and Rex, a 27-kD protein that regulates the expression of viral mRNAs. These two proteins are produced from messages that are similar but that are spliced differently from overlapping but distinct exons.

The lentiviruses in general, and HIV-1 and -2 in particular, contain a larger genome than other pathogenic retroviruses. They contain an untranslated region between *pol* and *env* that encodes portions of several proteins, varying with the reading frame into which the mRNA is spliced. Tat is a 14-kD protein that augments the expression of virus from the [LTR](#). The Rev protein of HIV-1, similar to the Rex protein of [HTLV](#), regulates RNA splicing and/or RNA transport. The Nef protein downregulates CD4, the cellular receptor for HIV; alters host T cell activation pathways; and enhances viral infectivity. The Vif protein is necessary for the proper assembly of the HIV nucleoprotein core in many types of cells; without Vif, proviral DNA is not efficiently produced in these infected cells. Vpr, Vpu (HIV-1 only), and Vpx (HIV-2 only) are viral proteins encoded by translation of the same message in different reading frames. As noted above, oncogenic retroviruses depend on cell proliferation for their replication; lentiviruses can infect nondividing cells, largely owing to effects mediated by Vpr. Vpr facilitates transport of the provirus into the nucleus and can induce other cellular changes, such as G2 growth arrest and differentiation of some target cells. Vpx is structurally related to Vpr, but its functions are not fully defined. Vpu promotes the degradation of CD4 in the endoplasmic reticulum and stimulates the release of virions from infected cells.

Retroviruses can be either exogenously acquired by infection with a virion capable of replication or transmitted in the germ line as endogenous virus. Endogenous retroviruses are often replication-defective. The human genome contains endogenous retroviral sequences, but there are no known replication-competent endogenous retroviruses in humans.

In general, viruses that contain only the *gag*, *pol*, and *env* genes either are not pathogenic or take a long time to induce disease because the pathogenesis of neoplastic transformation relies on the chance integration of the provirus at a spot in the genome that will result in the expression of a cellular gene (proto-oncogene) that becomes transforming by virtue of its unregulated expression. For example, avian leukemia virus causes B cell leukemia by inducing the expression of *myc*. Some retroviruses possess captured and altered cellular genes near their integration site, and these viral oncogenes are capable of transforming the infected host cell. Viruses that have oncogenes often have lost a portion of their genome that is required for replication. Such viruses need helper viruses to reproduce, a feature that may explain why these acute transforming retroviruses are rare in nature. All human retroviruses identified to date are exogenous and are not acutely transforming (that is, they lack a transforming oncogene).

These remarkable properties of retroviruses have led to experimental efforts to use them as vectors to insert specific genes into particular cell types, a process known as *gene therapy* or *gene transfer*. The process could be used to repair a genetic defect or

to introduce a new property that could be used therapeutically; for example, a gene (e.g., thymidine kinase) that would make a tumor cell susceptible to killing by a drug (e.g., ganciclovir) could be inserted. One source of concern about the use of retroviral vectors in humans is that replication-competent viruses might rescue endogenous retroviral replication, with unpredictable results. This concern is not merely hypothetical: The detection of proteins encoded by endogenous retroviral sequences on the surface of cancer cells implies that the genetic events leading to the cancer were able to activate the synthesis of these usually silent genes.

HUMAN T-CELL LYMPHOTROPIC VIRUS

[HTLV-I](#) was isolated in 1980 from a T-cell lymphoma cell line from a patient originally thought to have cutaneous T cell lymphoma. Later it became clear that the patient had a distinct form of lymphoma (originally reported in Japan) called *adult T cell leukemia/lymphoma* (ATL). Serologic data have determined that HTLV-I is the cause of at least two important diseases: ATL and tropical spastic paraparesis, also called *HTLV-I-associated myelopathy* (HAM). HTLV-I may also play a role in infective dermatitis and uveitis syndromes.

Two years after the isolation of [HTLV-I](#), HTLV-II was isolated from a patient with an unusual form of hairy cell leukemia that affected T cells. Although early epidemiologic studies of HTLV-II failed to reveal a consistent disease association, more recent studies suggest an association of HTLV-II with human disease (see "Associated Diseases" under "Features of HTLV-II Infection," below), particularly among injection drug users.

BIOLOGY AND MOLECULAR BIOLOGY

Because the biology of [HTLV-I](#) and that of HTLV-II are similar, the following discussion will focus on HTLV-I.

The cellular receptor for [HTLV-I](#) has not yet been identified, but it maps to chromosome 17. Generally, only T cells are productively infected, but infection of B cells and other cell types is occasionally detected. The most common outcome of HTLV-I infection is latent carriage of randomly integrated provirus in CD4⁺ T cells. HTLV-I does not contain an oncogene and does not insert into a unique site in the genome. Indeed, most infected cells express no viral gene products. The only viral gene product that is routinely expressed in tumor cells transformed by HTLV-I *in vivo* is *tax*, and even *tax* is not expressed in the tumor cells of many [ATL](#) patients. Cells transformed *in vitro*, by contrast, actively transcribe HTLV-I RNA and produce infectious virions. Most HTLV-I-transformed cell lines are the result of the infection of a normal host T cell *in vitro*. It is difficult to establish cell lines derived from authentic ATL cells.

Although *tax* does not itself bind to DNA, it does induce the expression of a wide range of host-cell gene products, including transcription factors (especially *c-rel*, *ets-1* and -2, and members of the *fos/jun* family), cytokines [e.g., interleukin (IL) 2, granulocyte-macrophage colony-stimulating factor, and tumor necrosis factor (TNF)], and membrane proteins and receptors [major histocompatibility (MHC) molecules and IL-2 receptor α]. The genes activated by *tax* are generally controlled by transcription factors of the *c-rel* and cyclic AMP response element binding (CREB) protein families. It

is unclear how this induction of host gene expression leads to neoplastic transformation. Induction of a cytokine-autocrine loop has been proposed; however, IL-2 is not the crucial cytokine. The involvement of IL-4, IL-7, and IL-15 has been proposed.

In light of the irregular expression of *tax* in [ATL](#) cells, it has been suggested that *tax* is important in the early phases of transformation but is not essential for the maintenance of the transformed state. As is clear from the epidemiology of [HTLV-I](#) infection, transformation of an infected cell is a rare event and may depend on heterogeneous second, third, or fourth genetic hits. No consistent chromosomal abnormalities have been described in [ATL](#); however, individual cases with *p53* mutations and translocations involving the T cell receptor genes on chromosome 14 have been reported. *Tax* may repress certain DNA repair enzymes, permitting the accumulation of genetic damage that would normally be repaired. However, the molecular pathogenesis of HTLV-I-induced neoplasia is not fully understood.

FEATURES OF HTLV-I INFECTION

Epidemiology [HTLV-I](#) infection is transmitted in at least three ways: from mother to child, especially in breast milk; through sexual activity, more commonly from men to women; and through the blood -- via contaminated transfusions or contaminated needles. The virus is most commonly transmitted perinatally. Compared with HIV, which can be transmitted in cell-free form, HTLV-I is less infectious, and its transmission usually requires cell-to-cell contact.

[HTLV-I](#) is endemic in southwestern Japan and Okinawa, where more than 1 million persons are infected. Antibodies to HTLV-I are present in the serum of up to 35% of Okinawans, 10% of residents of the Japanese island of Kyushu, and <1% of persons in nonendemic regions of Japan. Despite this high prevalence of infection, only about 500 cases of [ATL](#) are diagnosed in this area each year. Clusters of infection have been noted in other areas of the Orient, such as Taiwan; in the Caribbean basin, including northeastern South America; in central Africa; in Italy; in Israel; in the Arctic; and in the southeastern part of the United States.

A progressive spastic or ataxic myelopathy that develops in an individual who is [HTLV-I](#) positive (i.e., who has serum antibodies to HTLV-I) is likely to be due to direct nervous system infection with the virus; a similar disorder may result from infection with HIV or HTLV-II. In rare instances, patients with [HAM](#) are seronegative but have detectable antibody to HTLV-I in the cerebrospinal fluid (CSF).

The cumulative lifetime risk of developing [ATL](#) is 2% among [HTLV-I](#)-infected patients; a similar risk is projected for [HAM](#). The distribution of the two diseases overlaps the distribution of HTLV-I, with >95% of affected patients showing serologic evidence of HTLV-I infection. The latent period between infection and the emergence of disease is 20 to 30 years for ATL. For HAM, the median latency period is about 3.3 years (range, 4 months to 30 years). The development of ATL is rare among persons infected by blood products; however, ~20% of patients with HAM acquire HTLV-I from contaminated blood.

Associated Diseases

ATL Four clinical types of [HTLV-I](#)-induced neoplasia have been described: acute, lymphomatous, chronic, and smoldering. All of these tumors are monoclonal proliferations of CD4⁺post-thymic T cells with clonal proviral integrations and clonal T-cell receptor gene rearrangements.

About 60% of patients who develop malignancy have classic *acute* [ATL](#), which is characterized by a short clinical prodrome (~2 weeks between the first symptoms and the diagnosis) and an aggressive natural history (median survival period, 6 months). The clinical picture is dominated by rapidly progressive skin lesions, pulmonary involvement, hypercalcemia, and lymphocytosis with cells containing lobulated or "flower-shaped" nuclei (see [Plate V-40](#)). The malignant cells have monoclonal proviral integrations and express CD4, CD3, and CD25 (low-affinity [IL-2](#) receptors) on their surface. Serum levels of CD25 can be used as a tumor marker. Anemia and thrombocytopenia are rare. The skin lesions may be difficult to distinguish from those in mycosis fungoides. Lytic bone lesions, which are common, do not contain tumor cells but rather are composed of osteolytic cells, usually without osteoblastic activity. Despite the leukemic picture, bone marrow involvement is patchy in most cases.

The hypercalcemia of [ATL](#) is multifactorial; the tumor cells produce osteoclast-activating factors ([TNF- \$\alpha\$](#) , [IL-1](#), lymphotoxin) and can also produce a parathyroid hormone-like molecule. The affected patients have an underlying immunodeficiency that makes them susceptible to opportunistic infections similar to those seen in patients with AIDS ([Chap. 309](#)). The pathogenesis of the immunodeficiency is unclear. Pulmonary infiltrates in ATL patients reflect leukemic infiltration half the time and opportunistic infections with organisms such as *Pneumocystis carinii* and other fungi the other half. Gastrointestinal symptoms are nearly always related to opportunistic infection. Serum concentrations of lactate dehydrogenase (LDH) and alkaline phosphatase are often elevated. About 10% of patients have leptomeningeal involvement leading to weakness, altered mental status, paresthesia, and/or headache. Unlike other forms of central nervous system (CNS) lymphoma, ATL may be accompanied by normal [CSF](#) protein levels. The diagnosis depends on finding ATL cells in the CSF ([Chap. 112](#)).

The *lymphomatous* type of [ATL](#) occurs in ~20% of patients and is similar to the acute form in its natural history and clinical course, except that circulating abnormal cells are rare and lymphadenopathy is evident. The histology of the lymphoma is variable but does not influence the natural history. In general, the diagnosis is suspected on the basis of the patient's birthplace and the presence of skin lesions and hypercalcemia. The diagnosis is confirmed by the detection of antibodies to [HTLV-I](#) in serum.

Patients with the *chronic* form of [ATL](#) generally have normal serum levels of calcium and [LDH](#) and no involvement of the [CNS](#), bone, or gastrointestinal tract. The median duration of survival for these patients is 2 years. In some cases, chronic ATL progresses to the acute form of the disease.

Fewer than 5% of patients have the *smoldering* form of [ATL](#). In this form, the malignant cells have monoclonal proviral integration; <5% of peripheral-blood cells exhibit typical morphologic abnormalities; hypercalcemia, adenopathy, and hepatosplenomegaly do not develop; the [CNS](#), the bones, and the gastrointestinal tract are not involved; and skin

and pulmonary lesions may be present. The median survival period of this small subset of patients appears to be ³5 years.

HAM (Tropical Spastic Paraparesis) In contrast to [ATL](#), in which there is a slight predominance of male patients, [HAM](#) affects females disproportionately. [HAM](#) resembles multiple sclerosis in certain ways ([Chap. 371](#)). The onset is insidious. Symptoms include weakness or stiffness in one or both legs, back pain, and urinary incontinence. Sensory changes are usually mild, but peripheral neuropathy may develop. The disease generally takes the form of slowly progressive and unremitting thoracic myelopathy; one-third of patients are bedridden within 10 years of diagnosis, and one-half are unable to walk unassisted by this point. Patients display spastic paraparesis or paraplegia with hyperreflexia, ankle clonus, and extensor plantar responses. Cognitive function is usually spared; cranial nerve abnormalities are unusual.

Magnetic resonance imaging (MRI) reveals lesions in both the white matter and the paraventricular regions of the brain as well as in the spinal cord. Pathologic examination of the spinal cord shows symmetric degeneration of the lateral columns, including the corticospinal tracts; some cases involve the posterior columns as well. The spinal meninges and cord parenchyma contain an inflammatory infiltrate with myelin destruction.

[HTLV-I](#) is not usually found in cells of the [CNS](#) but may be detected in a small population of lymphocytes present in the [CSF](#). In general, HTLV-I replication is greater in [HAM](#) than in [ATL](#), and patients with HAM have a stronger immune response to the virus. Antibodies to HTLV-I are present in the serum and appear to be produced in the CSF of [HAM](#) patients, where titers are often higher than in the serum. The pathophysiology of HAM may involve the induction of autoimmune destruction of neural cells by T cells with specificity for viral components such as Tax or Env proteins. One theory is that susceptibility to HAM may be related to the presence of human leukocyte antigen (HLA) alleles capable of presenting viral antigens in a fashion that leads to autoimmunity. Insufficient data are available to confirm an HLA association.

Other putative HTLV-I-related diseases In areas where [HTLV-I](#) is endemic, diverse inflammatory and autoimmune diseases have been attributed to the virus, including uveitis, dermatitis, pneumonitis, rheumatoid arthritis, and polymyositis. However, a causal relationship between HTLV-I and these illnesses has not been rigorously established.

Prevention Women in endemic areas should not breast-feed their children, and blood donors should be screened for serum antibodies to [HTLV-I](#). As in the prevention of HIV infection, the practice of safe sex and the avoidance of needle sharing are important.

TREATMENT

For the small number of patients who develop [HTLV-I](#)-related disease, therapies are not curative. In patients with the acute and lymphomatous types of [ATL](#), the disease progresses rapidly. Hypercalcemia is generally controlled by glucocorticoid administration and cytotoxic therapy directed against the neoplasm. The tumor is highly responsive to combination chemotherapy that is employed against other forms of

lymphoma; however, patients are susceptible to overwhelming bacterial and opportunistic infections, and ATL relapses within 4 to 10 months after remission in most patients. The combination of interferon and zidovudine may extend survival. Because viral replication is not clearly associated with ATL progression, zidovudine is probably effective through its cytotoxic effects (as a chain-terminating thymidine analogue) rather than its antiviral effects. An experimental approach using an yttrium 90-labeled antibody to the [IL-2](#) receptor appears promising but is not widely available. Patients with the chronic or smoldering form of ATL may be managed with an expectant approach: Treat any infections, and watch and wait for signs of progression to acute disease.

Patients with [HAM](#) may obtain some benefit from the use of glucocorticoids to reduce inflammation. Antiretroviral regimens have not been effective. In one study, danazol (200 mg tid) produced significant neurologic improvement in five of six treated patients, with resolution of urinary incontinence in two cases, decreased spasticity in three, and the restoration of the ability to walk after confinement to a wheelchair in two. Physical therapy and rehabilitation are important components of management.

FEATURES OF HTLV-II INFECTION

Epidemiology [HTLV-II](#) is endemic in certain Native American tribes. It is generally considered to be a New World virus that was brought from Asia to the Americas 10,000 to 40,000 years ago during the migration of infected populations across the Bering land bridge.

The mode of transmission of [HTLV-II](#) is probably the same as that of HTLV-I (see above). HTLV-II may be less readily transmitted sexually than HTLV-I.

Studies of large cohorts of injection drug users with serologic assays that reliably distinguish [HTLV-I](#) from HTLV-II indicate that the vast majority of HTLV-positive subjects are infected with HTLV-II. The seroprevalence of HTLV in a cohort of 7841 injection drug users from drug treatment centers in Baltimore, Chicago, Los Angeles, New Jersey (Asbury Park and Trenton), New York City (Brooklyn and Harlem), Philadelphia, and San Antonio was 20.9%, with >97% of cases due to HTLV-II. The seroprevalence of HTLV-II was higher in the Southwest and the Midwest than in the Northeast. In contrast, the seroprevalence of HIV-1 was higher in the Northeast than in the Southwest or the Midwest. Approximately 3% of the cohort members were infected with both HTLV-II and HIV-1. The seroprevalence of HTLV-II increased linearly with age. Women were significantly more likely to be infected with HTLV-II than were men; the virus is thought to be more efficiently transmitted from male to female than from female to male.

Associated Diseases Although [HTLV-II](#) was isolated from a patient with a T cell variant of hairy cell leukemia, this virus has not been consistently associated with a particular disease and in fact has been thought of as "a virus searching for a disease." However, evidence is accumulating that HTLV-II may play a role in certain neurologic, hematologic, and dermatologic diseases. These data require confirmation, particularly in light of the previous confusion regarding the relative prevalences of HTLV-I and HTLV-II among injection drug users.

Prevention Avoidance of needle sharing, safe-sex practices, screening of blood (by

assays for [HTLV-I](#), which also detect HTLV-II), and avoidance of breast-feeding by infected women are important principles in the prevention of spread of HTLV-II.

HUMAN IMMUNODEFICIENCY VIRUS

HIV-1 and HIV-2 are members of the lentivirus subfamily of Retroviridae and are the only lentiviruses known to infect humans. The lentiviruses are slow-acting by comparison with viruses that cause acute infection (e.g., influenza virus) but not by comparison with other retroviruses. The features of acute primary infection with HIV resemble those of more classic acute infections. The characteristic chronicity of HIV disease is consistent with the designation *lentivirus*. **For a detailed discussion of HIV, see [Chap. 309](#).*

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192. VIRAL GASTROENTERITIS - *Harry B. Greenberg*

In less developed countries, acute infectious diarrheal disease is a leading cause of morbidity in all age groups and of mortality in infants and young children. In developed countries, acute diarrheal illness remains an important cause of morbidity among both children and adults. Two distinct groups of viruses -- the rotaviruses and the enteric caliciviruses, such as Norwalk virus -- as well as a variety of bacterial pathogens ([Chap. 131](#)) have emerged as important etiologic agents of gastroenteritis. The rotaviruses are primarily pathogens of young children. The Norwalk and related enteric caliciviruses affect adults as well as children. Several important gastrointestinal viruses are characterized in [Table 192-1](#) and depicted in [Fig. 192-1](#).

ROTA VIRUS

Classification and Characterization Rotaviruses are members of the Reoviridae family. The rotavirus virion consists of a 100-nm triple-shelled icosahedral capsid surrounding a genome composed of 11 segments of double-stranded RNA. Several genetically distinct groups of rotaviruses (groups A, B, C, etc.) have been identified, but group A strains account for the great majority of illnesses in humans. With only one exception, the rotavirus gene segments are monocistronic. The virus has two surface proteins (VP4 and VP7), both of which are involved in viral neutralization. The major internal capsid protein (VP6) is the target of cross-reactive antibody to different virus strains. This protein also appears to induce protective immunity, although the mechanism is not clear. Because rotaviruses have a segmented genome, they are capable of undergoing gene reassortment at high frequency. The role of gene reassortment in generating rotavirus antigenic diversity is not known. In immunocompetent humans and animals, rotavirus infection is characterized by replication that is localized almost exclusively in the epithelial cells of the small intestine.

Epidemiology Rotavirus infection occurs worldwide. By the age of 3 years, virtually every individual has been infected by rotaviruses at least once. Most rotavirus infections are subclinical or cause mild gastrointestinal illnesses that do not require hospitalization. The first infection is the most likely to be symptomatic; subsequent infections are often mild or asymptomatic. In areas with a temperate climate, rotavirus infection is seasonal, occurring in the colder winter months. In the United States, the annual seasonal rotavirus epidemic tends to spread from west to east, starting in California and ending in New England. In tropical areas, rotavirus infection tends to occur throughout the year, with some increase in incidence during the cooler rainy season.

Rotaviruses are the single most important cause of severe dehydrating diarrhea in infants and children <3 years old in both developed and less developed countries; they account for 25 to 50% of all cases of diarrhea requiring hospitalization or intensive rehydration therapy. During a rotavirus outbreak in a temperate climate, this percentage can be as high as 80%. In the United States, between 5 and 10% of all diarrheal episodes among children under the age of 5 years are caused by rotavirus. Infection due to rotavirus accounts for ~500,000 physician visits per year in the United States. Although severe rotavirus infections are confined primarily to infants and young children, these agents are frequently associated with mild diarrhea in adults, particularly family members of affected infants, geriatric patients, and immunocompromised hosts. They

are responsible for up to 10% of cases of traveler's diarrhea ([Chap. 131](#)). Rotaviruses also may cause occasional cases of acute and chronic diarrhea in patients with AIDS.

Rotavirus serotypes have been defined by the antigenicity of both VP4 (P serotype) and VP7 (G serotype). At least 14 distinct G serotypes of rotavirus have been described, but only four types are commonly encountered in the United States. At least 20 P serotypes have been identified to date, of which only two are common among children in this country. The relationship of the frequency of infection with these multiple serotypes to host immune status is unclear. It does appear, however, that infection with one or two serotypes induces some degree of heterotypic immunity to severe disease.

A large variety of mammalian and avian species can be infected by rotavirus, but these animal rotavirus strains do not frequently cause disease in humans. Rotaviruses are shed in very large numbers in the stool (up to 10^{10} particles per gram of feces). Although transmission presumably takes place via the fecal-oral route, rotavirus infection spreads with great efficacy in developed as well as less developed countries.

Pathophysiology Rotavirus infects and kills the mature villus tip cells of the small intestine. The mature epithelial cells are replaced by immature absorptive cells that cannot absorb carbohydrates or other nutrients efficiently. Rotavirus infection thus leads to osmotic diarrhea due to nutrient malabsorption. Changes in intracellular cyclic adenosine monophosphate or guanosine monophosphate are not involved in the etiology of rotavirus diarrhea. Rotavirus also encodes a nonstructural protein (NSP4) that appears to function as an enterotoxin during infection. It is not known whether immunity to NSP4 plays a role in protection or disease resolution.

Manifestations The manifestations of rotavirus infection range from subclinical infection through mild diarrhea to severe, occasionally fatal dehydrating illness. Most information concerning the signs and symptoms of rotavirus infection has been derived from studies of hospitalized young children. The onset of illness is usually abrupt. More than 80% of affected children develop vomiting followed by diarrhea. About one-third of hospitalized children have a temperature of $>39^{\circ}\text{C}$ ($>102.2^{\circ}\text{F}$). Mucus is commonly found in the stool, but white and red blood cells are present in the stool in fewer than 15% of cases.

Rotavirus infection frequently occurs in conjunction with respiratory tract symptoms, but there is little evidence to indicate that rotavirus replicates in the respiratory tract. Rotavirus infection has been observed in association with a wide variety of other clinical syndromes, including sudden infant death syndrome, Reye's syndrome, encephalitis, aseptic meningitis, pneumonia, exanthema subitum, Kawasaki's syndrome, necrotizing enterocolitis, intussusception, Schonlein-Henoch purpura, hemolytic-uremic syndrome, disseminated intravascular coagulation, and Crohn's disease. The etiologic relationship between these clinical syndromes and rotavirus infection is probably coincidental rather than causal. Rotavirus infection may be especially severe or even fatal in immunocompromised children.

Clinical Immunity Relative immunity to rotavirus illness is acquired early in childhood, after one or two natural infections. Subclinical infections in neonates have been shown to protect these children against severe rotavirus gastroenteritis for up to 3 years. Immunity is not complete, and adults with low levels of antibody can be symptomatically

infected. Local humoral immunity appears to be the critical determinant in protection, and cellular immune mechanisms seem to be involved as well.

Diagnosis Because rotavirus is shed in large amounts in the stool, detection is relatively easy. Various specific and highly sensitive commercial immunoassays are available to detect rotavirus antigen in fecal specimens. DNA probe diagnosis appears to be sensitive and specific, as do polymerase chain reaction (PCR)-based assays, but these detection methods have been used primarily for research purposes. No particular signs or symptoms are pathognomonic for rotavirus infection, but this infection is more frequently associated with severe dehydration than are infections caused by other enteric bacterial or viral pathogens.

TREATMENT

Although rotavirus diarrhea appears to be caused primarily by intestinal epithelial-cell lysis and death, it can be adequately managed with standard oral rehydration therapy. Only rarely is intravenous rehydration required. Since rotavirus infections have persisted in developed countries with advanced sanitation facilities and widely available clean water, it is unlikely that these infections will prove to be preventable by hygienic measures alone. Progress with a number of candidate live attenuated vaccines suggests that prevention through vaccination may be feasible. In 1998, a multivalent rotavirus vaccine was licensed for use in children under the age of 6 months in the United States. The vaccine virus was a live attenuated animal rotavirus containing genes encoding the four human G serotypes most common in the United States. The licensed vaccine was administered orally as three doses and was highly effective in preventing severe rotavirus illness. Reports of intussusception associated with the administration of rotavirus vaccine have been published; further studies are required to define the relative risk. However, because of the apparent association with intussusception, this first rotavirus vaccine has been withdrawn from use.

NORWALK AND RELATED ENTERIC CALICIVIRUSES

Classification and Characterization Various round 27- to 32-nm particles, some with clearly defined ultrastructure, have been identified in the stools of individuals with acute nonbacterial gastroenteritis. In the past, these agents have been difficult to classify because they are shed in the stool in small amounts for only a few days and have not been adapted to cell culture or to animal models. The Norwalk virus is the most extensively studied and best-characterized member of this group of enteric caliciviruses, which also includes such serologically or genetically distinct viruses as Hawaii virus, Snow Mountain virus, Southampton virus, Lordsdale virus, Mexico virus, Sapporo virus, and a number of agents described as calicivirus-like. Molecular biologic studies have shown that all these viruses have a protein structure similar to that of typical caliciviruses, with a single structural protein of ~60 kDa. The genomes of Norwalk virus and numerous related viruses have been cloned and sequenced. The genomes are plus-stranded RNA molecules of ~7.5 kb. On the basis of these molecular biologic studies, the human enteric caliciviruses can be divided into two groups: the Norwalk-like viruses (NLVs) and the Sapporo-like viruses (SLVs). The SLVs have a more typical calicivirus-like ultrastructure and cause diarrhea in young children. Unlike the NLVs, the SLVs may not cause frequent epidemics in adults.

Epidemiology Infection with [NLVs](#) is common year-round, with a clear winter peak. The seasonality of [SLVs](#) has not yet been widely studied. More than 80% of adults in both developed and less developed countries have antibodies to these viruses. Antibody is acquired at a younger age among children in less developed countries than among those in developed areas; this observation is consistent with the assumption that NLVs are spread by the fecal-oral route. In the United States, the NLVs are responsible for ~90% of all epidemics of nonbacterial gastroenteritis. These agents have been incriminated in a variety of food-borne epidemics, and transmission vehicles have included oysters, green salad, and chocolate icing. The NLVs are common causes of waterborne epidemics of gastroenteritis and have been shown to be etiologic agents in nursing home, cruise ship, and institutional (summer camp and school) outbreaks. NLVs are also responsible for a small proportion of cases of traveler's diarrhea. NLV infection is a common cause of mild to moderate childhood diarrhea.

In less developed countries, the role of [NLV](#) infection in the etiology of diarrhea in adults has not been thoroughly investigated. Preliminary studies indicate that NLVs can cause mild diarrhea in young children, but they do not appear to cause severe illness in infants in either developed or less developed countries.

Pathophysiology Information concerning the pathologic changes induced by the enteric caliciviruses is based almost entirely on a few studies of volunteers during the 1970s and 1980s. After infection with Norwalk or Hawaii virus, the architecture of the proximal small intestine is altered, with villus shortening, crypt hyperplasia, and infiltration of the lamina propria by polymorphonuclear and mononuclear cells. No changes are observed in the stomach or colon. The cells in which viral replication takes place have not been identified. The histologic alterations are accompanied by mild steatorrhea, carbohydrate malabsorption, and decreased levels of some brush border enzymes. No changes in adenylate cyclase activity have been observed.

Manifestations Norwalk illness has an incubation period of 18 to 72 h. The disease is characterized by the abrupt onset of nausea and abdominal cramps followed by vomiting and/or diarrhea. Vomiting is reported more frequently for children than for adults. Low-grade fever [$>37.5^{\circ}\text{C}$ ($>99.5^{\circ}\text{F}$)] develops in about half of affected individuals. Headache, myalgias, and abdominal pain are common. The white blood cell count is normal; rarely, there is leukocytosis with relative lymphopenia. Red and white blood cells are not found in the stool. The illness is usually mild and self-limited, lasting 24 to 48 h.

Clinical Immunity Most people in developed countries do not have long-term resistance (i.e., that lasting ≥ 2 years) to Norwalk reinfection. Short-term (several-month) immunity -- at least to homotypic challenge -- does appear to develop. In volunteers challenged with Norwalk agent, there is a paradoxical relationship between the level of antibody to [NLVs](#) and susceptibility to illness: Low levels of Norwalk antibody in the serum and intestine are associated with clinical resistance to illness. It appears, therefore, that immune mechanisms are not the primary determinants of long-term protection from NLVs. Immunity to the [SLVs](#) may be more durable.

Diagnosis, Treatment, and Prevention Enzyme-linked immunosorbent assays

and [PCR](#)-based assays have been developed for [NLVs](#) and several other 27- to 32-nm gastroenteritis agents. However, these assays are not yet widely available, and their utility for general diagnosis has not been tested. Because Norwalk illness is acute and self-limited, treatment is not usually required. In the rare case of severe vomiting or diarrhea, oral or intravenous rehydration is indicated. Because long-term immunity to the NLVs does not usually follow natural infection, the role of vaccination may be limited to specific settings (e.g., the military). Recombinant NLV particles have been produced and are capable of inducing an immune response when administered orally to volunteers.

MISCELLANEOUS ENTERIC VIRAL PATHOGENS

Enteric adenoviruses are a minor cause of diarrheal illness in infants and children, accounting for 10% of cases. These viruses differ from other adenovirus strains in a variety of ways, including neutralization serotype, restriction endonuclease digestion pattern, and ability to grow in tissue culture. The role of enteric adenovirus illness in adults or in persons in less developed countries is not known.

Several strains of antigenically distinct rotaviruses, presently called *atypical rotaviruses* or *groups B and C rotaviruses*, have been identified as the cause of occasional episodes of diarrhea in humans and animals.

Preliminary epidemiologic studies have indicated that *astroviruses* are a fairly frequent cause of mild to moderate diarrhea in young children in developed and less developed countries, accounting for about one-quarter to one-half as much illness as group A rotaviruses. Moreover, preliminary data indicate that astroviruses are a common cause of diarrhea in immunocompromised hosts, such as bone marrow transplant recipients and patients with AIDS. Astroviruses are 27 to 32 nm in diameter, have a characteristic icosahedral ultrastructure, and contain a plus-stranded RNA genome with a size of ~7.0 kb and a unique genomic organization. At least seven distinct serotypes have been identified. The current availability of sensitive and specific diagnostic assays should allow more complete assessment of the importance of these agents.

Coronaviruses are frequent causes of diarrheal disease in a variety of animals. Using electron microscopy, several investigators have identified putative coronavirus-like particles in the stools of patients with diarrhea. In most cases, however, these particles do not have the typical morphologic features of coronaviruses and may represent bacterial breakdown products or cellular fragments.

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193. ENTEROVIRUSES AND REOVIRUSES - Jeffrey I. Cohen

ENTEROVIRUSES

CLASSIFICATION AND CHARACTERIZATION

Enteroviruses are so named because of their ability to multiply in the gastrointestinal tract. Despite their name, these viruses are not a prominent cause of gastroenteritis. Enteroviruses encompass 64 human serotypes: 3 serotypes of poliovirus, 23 serotypes of coxsackievirus A, 6 serotypes of coxsackievirus B, 28 serotypes of echovirus, and enteroviruses 68 through 71.

Human enteroviruses contain a single-stranded RNA genome surrounded by an icosahedral capsid comprising four viral proteins. These viruses have no lipid envelope and are stable in acidic environments, including the stomach. They are resistant to inactivation by standard disinfectants (e.g., alcohol, detergents) and can persist for days at room temperature.

PATHOGENESIS AND IMMUNITY

Much of what is known about the pathogenesis of enteroviruses has been derived from studies of poliovirus infection. After ingestion, poliovirus is thought to infect epithelial cells in the mucosa of the gastrointestinal tract and then to spread to and replicate in the submucosal lymphoid tissue of the tonsils and Peyer's patches. The virus next spreads to the regional lymph nodes, a viremic phase ensues, and the virus replicates in organs of the reticuloendothelial system. In some cases, a second viremia occurs and the virus replicates further in various tissues, sometimes causing symptomatic disease.

It is uncertain whether poliovirus reaches the central nervous system (CNS) during viremia or whether it also spreads via peripheral nerves. Since viremia precedes the onset of neurologic disease in humans and in experimentally infected chimpanzees, it has been assumed that the virus enters the CNS via the bloodstream. The poliovirus receptor is a member of the immunoglobulin superfamily. Poliovirus infection is limited to primates, largely because of the ability of their cells to express the viral receptor. Studies demonstrating the poliovirus receptor in the end-plate region of muscle at the neuromuscular junction suggest that if the virus enters the muscle during viremia, it could travel across the neuromuscular junction up the axon to the anterior horn cells. Studies of monkeys or transgenic mice expressing the poliovirus receptor show that, after intramuscular injection, poliovirus does not reach the spinal cord if the sciatic nerve is cut. Taken together, these findings suggest that poliovirus can spread directly from muscle to the CNS by neural pathways. The receptor for echovirus types 1 and 8 is VLA-2 integrin, that for echovirus 7 is CD55, and that for coxsackievirus B is CAR (also used by adenovirus).

Poliovirus can usually be cultured from the blood 3 to 5 days after infection, before the development of neutralizing antibodies. While viral replication at secondary sites begins to slow 1 week after infection, it continues in the gastrointestinal tract. Poliovirus is shed from the oropharynx for up to 3 weeks after infection and from the gastrointestinal tract for as long as 12 weeks; immunodeficient patients can shed poliovirus for more than 1

year. During replication in the gastrointestinal tract, attenuated oral poliovirus can mutate, reverting to a more neurovirulent phenotype within a few days. The clinical significance of this increased neurovirulence is unknown.

Humoral and secretory immunity in the gastrointestinal tract is important for the control of enterovirus infections. Enteroviruses induce specific IgM, which usually persists for < 6 months, and specific IgG, which persists for life. Capsid protein VP1 is the predominant target of neutralizing antibody, which generally confers lifelong protection against subsequent disease caused by the same serotype but does not prevent infection or virus shedding. Enteroviruses also induce cellular immunity, but the importance of this mechanism in limiting infection is uncertain. Patients with impaired cellular immunity are not known to develop unusually severe disease when infected with enteroviruses. In contrast, the severe infections in patients with agammaglobulinemia emphasize the importance of humoral immunity in controlling enterovirus infections. IgA antibodies are important in reducing poliovirus replication in and shedding from the gastrointestinal tract. Breast milk contains IgA specific for enteroviruses and can protect humans from infection.

EPIDEMIOLOGY

Enteroviruses have a worldwide distribution. More than 50% of nonpoliovirus enterovirus infections and more than 90% of poliovirus infections are subclinical. When symptoms do develop, they are usually nonspecific and occur in conjunction with fever; only a minority of infections are associated with specific clinical syndromes. The incubation period for most enterovirus infections ranges from 2 to 14 days but usually is less than a week.

Enterovirus infection is more common in socioeconomically disadvantaged areas, especially in those where conditions are crowded and in tropical areas where hygiene is poor. Infection is most common among infants and young children; serious illness develops most often during the first few days of life and in older children and adults. In developing countries, where children are infected at an early age, poliovirus infection has less often been associated with paralysis; in countries with better hygiene, older children and adults are more likely to be seronegative, become infected, and develop paralysis. Passively acquired maternal antibody reduces the risk of symptomatic infection in neonates. Young children are the most frequent shedders of enteroviruses and are usually the index cases in family outbreaks. In temperate climates, enterovirus infections occur most often in the summer and fall; no seasonal pattern is apparent in the tropics.

Most enteroviruses are transmitted primarily by the fecal-oral route from fecally contaminated fingers or inanimate objects. Patients are most infectious shortly before and after the onset of symptomatic disease, when virus is present in the stool and throat. The ingestion of virus-contaminated food or water can also cause disease. Certain enteroviruses (such as enterovirus 70, which causes acute hemorrhagic conjunctivitis) can be transmitted by direct inoculation from the fingers to the eye. Airborne transmission is important for some viruses that cause respiratory tract disease, such as coxsackievirus A21. Enteroviruses can be transmitted across the placenta from mother to fetus, causing severe disease in the newborn. The transmission of

enteroviruses through blood transfusions or insect bites has not been documented. Nosocomial spread of coxsackievirus and echovirus has taken place in hospital nurseries.

DIAGNOSIS

Isolation of enterovirus in cell culture is the most common procedure for the diagnosis of infection. While cultures of stool, nasopharyngeal, or throat samples from patients with enterovirus diseases are often positive, isolation of the virus from these sites does not prove that it is directly associated with disease because these sites are frequently colonized for weeks in patients with subclinical infections. Isolation of virus from the throat is more likely to be associated with disease than isolation from the stool since virus is shed for shorter periods from the throat. Cultures of cerebrospinal fluid (CSF), serum, fluid from body cavities, or tissues are positive less frequently, but a positive result is indicative of disease caused by enterovirus. In some cases the virus can be isolated only from the blood or only from the CSF; therefore, it is important to culture multiple sites. Cultures are more likely to be positive earlier than later in the course of infection. Most human enteroviruses can be detected within a week after inoculation of cell cultures. Cultures may be negative because of the presence of neutralizing antibody, lack of susceptibility of the cells used, or inappropriate handling of the specimen. Coxsackievirus A may require inoculation into special cell-culture lines or into suckling mice.

Identification of the serotype of an enterovirus is useful primarily for epidemiologic studies and, with a few exceptions, has little clinical utility. It is important to identify serious infections with enterovirus during epidemics and to distinguish the vaccine strain of poliovirus from the other enteroviruses in the throat or in the feces. Stool and throat samples for culture as well as acute- and convalescent-phase serum specimens should be obtained from all patients with suspected poliomyelitis. In the absence of a positive [CSF](#) culture, a positive culture of stool obtained within the first 2 weeks after the onset of symptoms is most often used to confirm the diagnosis of poliomyelitis. If poliovirus is isolated, it should be sent to the Centers for Disease Control and Prevention (CDC) in Atlanta for identification as either a wild-type or a vaccine virus.

The polymerase chain reaction (PCR) has been used to amplify viral nucleic acid from [CSF](#), serum, urine, throat swabs, and tissues. A single pair of PCR primers can detect more than 92% of the serotypes that infect humans. With the proper controls, PCR of the CSF is highly sensitive (95%) and specific (>80%) and is more rapid than culture. PCR of serum is also highly sensitive and specific in the diagnosis of disseminated disease. PCR may be particularly helpful for the diagnosis and follow-up of enterovirus disease in immunodeficient patients receiving immunoglobulin therapy, whose viral cultures may be negative. Antigen detection and hybridization of enterovirus sequences in human tissues with a specific probe are additional options, but these techniques are generally less sensitive than PCR.

Serologic diagnosis of enterovirus infection is limited by the large number of serotypes and the lack of a common antigen. Demonstration of seroconversion may be useful in rare cases for confirmation of culture results, but serologic testing is usually limited to epidemiologic studies. Serum should be collected and frozen soon after the onset of

disease and again about 4 weeks later. Measurement of neutralizing titers is the most accurate method for antibody determination; measurement of complement-fixation titers is usually less sensitive. Titers of virus-specific IgM are elevated in both acute and chronic infection.

TREATMENT

Most enterovirus infections are mild and resolve spontaneously; however, intensive supportive care may be needed for cardiac, hepatic, or [CNS](#) disease. Intravenous, intrathecal, or intraventricular immunoglobulin has been used with apparent success for the treatment of chronic enterovirus meningoencephalitis and dermatomyositis in patients with hypo- or agammaglobulinemia. The disease may stabilize or resolve during therapy; however, some patients decline inexorably despite therapy. Intravenous administration of immunoglobulin with high titers of antibody to the infecting virus has been successful in the treatment of some cases of life-threatening infection in neonates, who may not have maternally acquired antibody. In one trial involving neonates with enterovirus infections, immunoglobulin containing very high titers of antibody to the infecting virus reduced rates of viremia; however, the study was too small to show a substantial clinical benefit. Oral pleconaril, a capsid-binding antiviral agent, reduced symptoms in a placebo-controlled trial of enteroviral aseptic meningitis and in a challenge study with coxsackievirus. This drug is available for compassionate use in patients with certain severe enterovirus infections. Glucocorticoids are contraindicated.

Good hand-washing practices and the use of gowns and gloves are important in limiting nosocomial transmission of enteroviruses during epidemics. Enteric precautions are indicated for 7 days after the onset of enterovirus infections.

POLIOVIRUS

Manifestations Most infections with poliovirus are asymptomatic. After an incubation period of 3 to 6 days, about 5% of patients present with a minor illness (abortive poliomyelitis) manifested by fever, malaise, sore throat, anorexia, myalgias, and headache. This condition usually resolves in 3 days. About 1% of patients present with aseptic meningitis (nonparalytic poliomyelitis). Examination of [CSF](#) reveals lymphocytic pleocytosis, a normal glucose level, and a normal or slightly elevated protein level; CSF polymorphonuclear leukocytes may be present early. In some patients, especially children, malaise and fever precede the onset of aseptic meningitis.

The least common presentation is that of paralytic disease. After one or several days, signs of aseptic meningitis are followed by severe back, neck, and muscle pain and by the rapid or gradual development of motor weakness. In some cases the disease appears to be biphasic, with aseptic meningitis followed first by apparent recovery but then (1 or 2 days later) by the return of fever and the development of paralysis; this form is more common among children than among adults. Weakness is generally asymmetric, is proximal more than distal, and may involve the legs (most commonly); the arms; or the abdominal, thoracic, or bulbar muscles. Paralysis develops during the febrile phase of the illness and usually does not progress after defervescence. Urinary retention may also occur. Examination reveals weakness, fasciculations, decreased muscle tone, and reduced or absent reflexes in affected areas. Transient hyperreflexia

sometimes precedes the loss of reflexes. Patients frequently report sensory symptoms, but objective sensory testing usually yields normal results. Bulbar paralysis may lead to dysphagia, difficulty in handling secretions, or dysphonia. Respiratory insufficiency due to aspiration, involvement of the respiratory center in the medulla, or paralysis of the phrenic or intercostal nerves may develop, and severe medullary involvement may lead to circulatory collapse. Most patients with paralysis recover some function weeks to months after infection. About two-thirds of patients have residual neurologic sequelae.

Paralytic disease is more common among older individuals, pregnant women, and persons exercising strenuously or undergoing trauma at the time of [CNS](#) symptoms. Tonsillectomy predisposes to bulbar poliomyelitis, and intramuscular injections increase the risk of paralysis in the involved limb(s).

At present, the only cases of poliomyelitis in the United States are due to live poliovirus vaccine; of the four cases reported in the United States in 1997 and 1998, three occurred in recipients of the first or second dose of oral poliovirus vaccine (OPV), and one occurred in an adult contact of a recipient of OPV. The median interval from vaccination to the onset of symptoms is usually 3 weeks. About 5% of the cases of poliomyelitis associated with vaccine occur in members of the community who have had no known direct contact with vaccinees. About 15% of all cases of vaccine-associated poliomyelitis involve immunodeficient children or adults, most of whom have hypo- or agammaglobulinemia. In these patients the median interval between vaccination and the onset of symptoms is 6 weeks, but disease can develop up to 6 months after vaccination. The risk of developing poliomyelitis after oral vaccination is estimated at 1 case per 2.5 million doses administered. The risk of developing paralytic disease after oral vaccination is about 2000 times higher among immunodeficient patients than among immunocompetent children.

The *postpolio syndrome* presents as a new onset of weakness, fatigue, fasciculations, and pain with additional atrophy of the muscle group involved during the initial paralytic disease 20 to 40 years earlier. The syndrome is more common among women and with increasing time after acute disease. The onset is insidious, and weakness occasionally extends to muscles that were not involved during the initial illness. The prognosis is generally good; progression to further weakness is usually slow, with plateau periods that range from 1 to 10 years. The postpolio syndrome is thought to be due to progressive dysfunction and loss of motor neurons that compensated for the neurons lost during the original infection and not to persistent or reactivated poliovirus infection.

Prevention and Eradication (See also [Chap. 122](#)) After a peak of 57,879 cases of poliomyelitis in the United States in 1952, the introduction of inactivated vaccine in 1955 and of oral vaccine in 1961 ultimately eradicated disease due to wild-type poliovirus in the western hemisphere. Such disease has not been documented in the United States since 1979, when cases occurred among religious groups who had declined immunization. In the western hemisphere, paralysis due to wild-type poliovirus was last documented in 1991.

In 1988, the World Health Organization adopted a resolution to eradicate poliomyelitis by the year 2000. From 1988 to 1997, the number of cases worldwide decreased by 89%, with about 6227 cases reported from 46 countries in 1998. More than 80% of the

world's cases of confirmed polio in 1998 occurred in India, Pakistan, Bangladesh, and Nigeria. Polio is a source of concern for unimmunized or partially immunized travelers to these regions. Outbreaks of polio in Europe and North America have been traced to cases imported from the Indian subcontinent. Clearly, global eradication of polio is necessary to eliminate the risk of importation of wild-type virus. Outbreaks are thought to have been facilitated by suboptimal rates of vaccination, isolated pockets of unvaccinated children, poor sanitation and crowding, improper vaccine-storage conditions, and a reduced level of response to one of the serotypes in the vaccine.

For the development of live [OPV](#) containing all three poliovirus serotypes, wild-type virus was attenuated by passage in monkey kidney cell cultures. OPV strains differ from the wild-type strains in a limited number of nucleotide changes (i.e., fewer than 60). Multiple doses are required to ensure infection and development of immunity to all three serotypes. While intramuscular injections of other vaccines (live or attenuated) can be given concurrently with OPV, unnecessary intramuscular injections should be avoided during the first month after vaccination because they increase the risk of vaccine-associated paralysis. Inactivated poliovirus vaccine is generated by formalin inactivation of the three serotypes of live poliovirus. Since 1988 an enhanced-potency inactivated poliovirus vaccine (IPV) has been available in the United States.

[OPV](#) and [IPV](#) induce antibodies that persist for at least 5 years. Both vaccines induce IgG and IgA antibodies. Compared with recipients of IPV, recipients of OPV shed less virus and less frequently develop reinfection with wild-type virus after exposure to poliovirus. Although IPV is safe and efficacious, OPV offers the advantages of ease of administration, lower cost, and induction of intestinal immunity resulting in a reduction in the risk of community transmission of wild-type virus. Because of progress toward global eradication of polio (with a reduced risk of imported cases) and the continued occurrence of cases of vaccine-associated polio, the [CDC](#) recommended in 1997 that children receive a sequential schedule of two doses of IPV followed by two doses of OPV or a four-dose schedule of IPV alone. To further reduce the risk of vaccine-associated polio, the Advisory Committee for Immunization Practices recommended (in June 1999) an all-IPV regimen for childhood poliovirus vaccination. Beginning in January 2000, children should receive IPV at 2, 4, and 6 to 18 months and 4 to 6 years of age. OPV will be used only in special circumstances: (1) for mass immunization campaigns to control outbreaks of polio; (2) for vaccination of unimmunized children who will be traveling to a polio-endemic area within 4 weeks; and (3) for children whose parents do not accept an all-IPV regimen. The latter children should receive at least two doses of IPV before receiving OPV. The risk of vaccine-associated polio should be discussed before administering OPV. Recommendations for vaccination of adults are listed in [Table 193-1](#).

COXSACKIEVIRUS, ECHOVIRUS, AND OTHER ENTEROVIRUSES

An estimated 5 to 10 million cases of symptomatic enterovirus disease occur in the United States each year. Enteroviruses are the most common cause of aseptic meningitis and nonspecific febrile illnesses of neonates. Certain clinical syndromes are more likely to be caused by certain serotypes ([Table 193-2](#)), but there is much overlap. From 1970 to 1983, 70% of enterovirus infections were caused by only 10 of the 64 human serotypes. Echoviruses 9 and 11 alone accounted for 24% of recognized

enterovirus infections; echoviruses 4, 6, and 30 and coxsackieviruses A9 and B2 through B5 accounted for 46%.

Nonspecific Febrile Illness (Summer Grippe) The most common clinical manifestation of enterovirus infection is a nonspecific febrile illness. After an incubation period of 3 to 6 days, patients present with an acute onset of fever, malaise, and headache. Occasional cases are associated with upper respiratory symptoms, and some cases include nausea and vomiting. Symptoms often last for 3 to 4 days, and most cases resolve in a week. While infections with other respiratory viruses occur more often from late fall to early spring, enterovirus febrile illness frequently occurs in the summer and early fall.

Generalized Disease of the Newborn Most serious enterovirus infections in infants develop during the first week of life, although severe disease can occur up to 3 months of age. Neonates often present with an illness resembling bacterial sepsis, with fever, irritability, and lethargy. Laboratory abnormalities include leukocytosis with a left shift, thrombocytopenia, elevated values in liver function tests, and CSF pleocytosis. The illness can be complicated by myocarditis and hypotension, fulminant hepatitis and disseminated intravascular coagulation, meningitis or meningoencephalitis, or pneumonia. It may be difficult to distinguish enterovirus infection from bacterial sepsis, although a history of a recent virus-like illness in the mother provides a clue.

Aseptic Meningitis and Encephalitis Enteroviruses are the cause of up to 90% of cases of aseptic meningitis in children and young adults in which an etiologic agent can be identified. Patients with aseptic meningitis typically present with an acute onset of fever, chills, headache, photophobia, and pain on eye movement. Nausea and vomiting are also common. Examination reveals meningismus without localizing neurologic signs; drowsiness or irritability may also be apparent. In some cases, a febrile illness may be reported that remits but returns several days later in conjunction with signs of meningitis. Other systemic manifestations may provide clues to an enteroviral cause, including diarrhea, myalgias, rash, pleurodynia, myocarditis, and herpangina. Examination of the CSF invariably reveals pleocytosis; early in the course, polymorphonuclear leukocytes may be present or even predominant, raising the possibility of bacterial or other nonviral causes of meningitis. Partially treated bacterial meningitis may be particularly difficult to exclude in some instances. A useful rule is that the CSF cell count in enteroviral meningitis shows a shift to lymphocytic predominance within 24 h of presentation, and the total count generally does not exceed 1000 cells/uL. Additional CSF findings consist of a normal glucose content and a normal or only slightly elevated (by ≤ 100 mg/mL) level of protein. Enteroviruses and mumps virus may produce a similar picture of meningitis; a low CSF glucose level suggests mumps, whereas a normal CSF glucose level and transient CSF polymorphonuclear pleocytosis suggest enterovirus infection. Symptoms ordinarily resolve within a week, although CSF abnormalities can persist for several weeks. Enteroviral meningitis is often more severe in adults than in children. Neurologic sequelae are rare, and most patients have an excellent prognosis.

Enteroviral encephalitis is much less common than enteroviral aseptic meningitis. Occasional highly inflammatory cases of enteroviral meningitis may be complicated by a mild form of encephalitis that is recognized on the basis of progressive lethargy,

disorientation, and sometimes seizures. Less commonly, severe primary encephalitis may develop. It is estimated that 10 to 20% of cases of viral encephalitis are due to enteroviruses. Immunocompetent patients generally have a good prognosis.

Patients with hypo- or agammaglobulinemia or severe combined immunodeficiency may develop chronic meningitis or encephalitis; about half of these patients have a dermatomyositis-like syndrome, with peripheral edema, rash, and myositis. They may also have chronic hepatitis. Patients may develop neurologic disease while receiving gamma globulin replacement therapy. Echoviruses (especially echovirus 11) are the most common pathogens in this situation.

Paralytic disease due to enteroviruses other than poliovirus occurs sporadically and is usually less severe than poliomyelitis. Most cases are due to enterovirus 70 or 71 or to coxsackievirus A7 or A9. Guillain-Barre syndrome is also associated with enterovirus infection. While some studies have suggested a link between enteroviruses and the chronic fatigue syndrome, most recent studies have not demonstrated such an association.

Pleurodynia (Bornholm Disease) Patients with pleurodynia present with an acute onset of fever and spasms of pleuritic chest or upper abdominal pain. Chest pain is more frequent in adults, and abdominal pain is more common in children. Paroxysms of severe, knifelike pain usually last 15 to 30 min and are associated with diaphoresis and tachypnea. Fever peaks within an hour after the onset of paroxysms and subsides when pain resolves. The involved muscles are tender to palpation, and a pleural rub may be detected. The white blood cell count and chest x-ray are usually normal. Most cases are due to coxsackievirus B and occur during epidemics. Symptoms resolve in a few days, and recurrences are rare. Treatment includes the administration of nonsteroidal anti-inflammatory agents or the application of heat to the affected muscles.

Myocarditis and Pericarditis Enteroviruses are estimated to cause up to one-third of cases of acute myocarditis. Coxsackievirus B and its RNA have been detected in pericardial fluid and myocardial tissue in some cases of acute myocarditis and pericarditis. Most cases of enteroviral myocarditis or pericarditis occur in newborns, adolescents, or young adults. More than two-thirds of patients are male. Patients often present with an upper respiratory tract infection that is followed by fever, chest pain, dyspnea, arrhythmias, and occasionally heart failure. A pericardial friction rub is documented in half of cases, and the electrocardiogram shows ST segment elevations or ST- and T-wave abnormalities. Serum levels of myocardial enzymes are often elevated. Neonates commonly have severe disease, while most older children and adults recover completely. Up to 10% of cases progress to chronic dilated cardiomyopathy. Chronic constrictive pericarditis may also be a sequela.

Exanthems Enterovirus infection is the leading cause of exanthems in children in the summer and fall. While exanthems are associated with many enteroviruses, certain types have been linked to specific syndromes. Echoviruses 9 and 16 have frequently been associated with exanthem and fever. Rashes may be discrete (rubelliform) or confluent (morbilliform), beginning on the face and spreading to the trunk and extremities. Echovirus 9 is the most common cause of rubelliform rash. Unlike the rash of rubella, the enteroviral rash occurs in the summer and is not associated with

lymphadenopathy. Roseola-like rashes develop after defervescence, with macules and papules on the face and trunk. The Boston exanthem, caused by echovirus 16, is a roseola-like rash that often affects multiple members of a family. A variety of other rashes have been associated with enteroviruses, including erythema multiforme and vesicular, urticarial, petechial, or purpuric lesions. Enanthems also occur, including lesions that resemble the Koplik's spots seen with measles.

Hand-Foot-and-Mouth Disease ([Fig. 193-CD1](#)) After an incubation period of 4 to 6 days, patients with hand-foot-and-mouth disease present with fever, anorexia, and malaise; these manifestations are followed by the development of sore throat and vesicles ([Plate IID-39](#)) on the buccal mucosa and often on the tongue and then by the appearance of tender vesicular lesions on the dorsum of the hands, sometimes with involvement of the palms. The vesicles may form bullae and quickly ulcerate. About one-third of patients also have lesions on the palate, uvula, or tonsillar pillars, and one-third have a rash on the feet (including the soles) or on the buttocks. The disease is highly infectious, with attack rates of close to 100% among young children. The lesions usually resolve in 1 week. Most cases are due to coxsackievirus A16 or enterovirus 71.

An epidemic of enterovirus 71 infection in Taiwan in 1998 resulted in thousands of cases of hand-foot-and-mouth disease or herpangina. Severe complications included CNS disease, myocarditis, and pulmonary hemorrhage. About 90% of those who died were children ≤5 years old, and these deaths were associated with pulmonary edema or pulmonary hemorrhage. CNS disease included aseptic meningitis, flaccid paralysis (similar to poliomyelitis), or rhombencephalitis with myoclonus and tremor or ataxia. The mean age of patients with CNS complications was 2.5 years, and magnetic resonance imaging in cases with encephalitis usually showed brain-stem lesions.

Herpangina Herpangina ([Fig. 193-CD2](#)) is usually caused by coxsackievirus A and presents as acute-onset fever, sore throat, dysphagia, and grayish-white papulovesicular lesions on an erythematous base that ulcerate. The lesions can persist for weeks; are present on the soft palate, anterior pillars of the tonsils, and uvula; and are concentrated in the posterior portion of the mouth. In contrast to herpes stomatitis, enteroviral herpangina is not associated with gingivitis. Acute lymphonodular pharyngitis associated with coxsackievirus A10 presents as white or yellow nodules surrounded by erythema in the posterior oropharynx. The lesions do not ulcerate.

Acute Hemorrhagic Conjunctivitis Patients with acute hemorrhagic conjunctivitis present with an acute onset of severe eye pain, blurred vision, photophobia, and watery discharge from the eye. Examination reveals edema, chemosis, and subconjunctival hemorrhage and often shows punctate keratitis and conjunctival follicles as well. Preauricular adenopathy is often found. Epidemics and nosocomial spread have been associated with enterovirus 70 and coxsackievirus A24. Systemic symptoms, including headache and fever, develop in 20% of cases, and recovery is usually complete in 10 days. The sudden onset and short duration of the illness help to distinguish acute hemorrhagic conjunctivitis from other ocular infections such as those due to adenovirus and *Chlamydia*. Paralysis has been associated with some cases of acute hemorrhagic conjunctivitis due to enterovirus 70 during epidemics.

Other Manifestations Enteroviruses are an infrequent cause of childhood pneumonia

and the common cold. Coxsackievirus B has been isolated at autopsy from the pancreas of a few children presenting with insulin-dependent diabetes mellitus; however, most attempts to isolate the virus have been unsuccessful. Other diseases that have been associated with enterovirus infection include bronchitis, bronchiolitis, croup, infectious lymphocytosis, polymyositis, acute arthritis, and acute nephritis.

REOVIRUSES

Reoviruses are double-stranded RNA viruses encompassing three serotypes. Serologic studies indicate that most humans are infected with reoviruses during childhood; however, it has been difficult to establish a definite link of reovirus infection with a particular disease. It is likely that most infections either are asymptomatic or cause very mild disease. One outbreak of reovirus infection in children resulted in minor upper respiratory tract symptoms. Reovirus is considered a rare cause of mild gastroenteritis in infants and children. Speculation regarding an association of reovirus type 3 with idiopathic neonatal hepatitis and extrahepatic biliary atresia is based on an elevated prevalence of antibody to reovirus among some of these patients, detection of viral RNA by PCR in hepatobiliary tissues in some studies, and detection of virus in the porta hepatis in one case.

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194. MEASLES (RUBEOLA) - Anne Gershon

DEFINITION

Measles (rubeola) is a highly contagious, acute, exanthematous respiratory disease with a characteristic clinical picture and pathognomonic enanthem. A successful live attenuated measles vaccine became available in 1963 in the United States and elsewhere, and measles is now an unusual disease in most developed countries where this vaccine is widely used. However, measles continues to occur sporadically in mini-epidemics in the United States, and major epidemics in developing nations make this disease a persistent cause of childhood morbidity and mortality.

ETIOLOGIC AGENT

Measles virus is a member of the genus *Morbillivirus* and the family Paramyxoviridae. It is closely related to the viruses causing canine and porcine distemper, rinderpest of cattle, and *peste des petits ruminants* of goats and sheep. There is only one antigenic type. Measles virions are pleomorphic spherical structures having a diameter of 100 to 250 nm and consisting of six proteins. The inner capsid is composed of a coiled helix of RNA and three proteins, and the outer envelope consists of a matrix protein bearing two types of short surface-glycoprotein projections or peplomers. One peplomer is a conical hemagglutinin (H) and the other a dumbbell-shaped fusion (F) protein. The genome has been sequenced, and it is thereby possible to distinguish vaccine-type measles virus from the wild type. In addition, genetic variability of wild-type measles virus occurs; eight genotypes have been identified.

EPIDEMIOLGY

Measles has a worldwide distribution; humans are the only natural hosts, although other primates can be experimentally infected. During the prevaccination era in the United States, measles epidemics occurred every 2 to 5 years in the winter and spring. In an epidemic year, roughly half a million measles cases were reported; 99% of adults had serologic evidence of previous measles infection. After the live attenuated vaccine became available, the number of cases reported to the Centers for Disease Control and Prevention (CDC) fell, with a nadir of 1497 cases in 1983. After an upsurge to more than 27,000 cases (with 89 deaths) in 1990, the disease was once more brought under control (with only 312 cases reported to the CDC in 1993), in part through the routine administration of two doses of vaccine. The foremost reason for the resurgence of measles was failure to immunize infants and young children, especially in inner-city areas. Primary vaccine failure (documented in about 5% of individuals) and secondary vaccine failure or waning immunity accounted for some cases.

In recent years the majority of cases of measles have involved preschool children. Between 1993 and 1996, fewer than 1000 cases were reported annually in the United States; in 1995 there were 309 reported cases, and in 1996 there were 508. Molecular studies indicated interruption of transmission of indigenous measles in 1993. Most cases have since resulted from international importations of the virus. Mortality is highest among children under 2 years of age and among adults. Patients with impaired cell-mediated immunity are at especially high risk for severe or even fatal measles. The

measles-associated mortality rate in the United States is about 0.3%; in developing countries, mortality frequently exceeds 1% and sometimes approaches 10%.

Measles virus is transmitted by respiratory secretions, predominantly through exposure to aerosols but also through direct contact with larger droplets. Patients are contagious from 1 or 2 days before the onset of symptoms until 4 days after the appearance of the rash. Infectivity peaks during the prodromal phase. The mean intervals from infection to onset of symptoms and to appearance of rash are 10 and 14 days, respectively.

PATHOGENESIS AND PATHOLOGY

Measles virus invades the respiratory epithelium and spreads via the bloodstream to the reticuloendothelial system, from which it infects all types of white blood cells, thereby establishing infection of the skin, respiratory tract, and other organs. Both viremia and viruria develop. Multinucleated giant cells with inclusion bodies in the nucleus and cytoplasm (Warthin-Finkeldey cells) are found in respiratory and lymphoid tissues and are pathognomonic for measles. Direct invasion of T lymphocytes and increased levels of suppressive cytokines, such as interleukin 4, may play a role in the temporary depression of cellular immunity that accompanies and transiently follows measles. The major infected cell in the blood is the monocyte. Infection of the entire respiratory tract accounts for the characteristic cough and coryza of measles and for the less frequent manifestations of croup, bronchiolitis, and pneumonia. Generalized damage to the respiratory tract, with resultant loss of cilia, predisposes to secondary bacterial infections such as pneumonia and otitis media.

Specific antibodies are not detectable before the onset of rash. Cellular immunity (consisting of cytotoxic T cells and possibly natural killer cells) plays a prominent role in host defense, and patients who are deficient in cellular immunity are at high risk for severe measles. Children with isolated agammaglobulinemia are not at increased risk. Immune reactions to the virus in the endothelial cells of dermal capillaries play a substantial role in the development of Koplik's spots (the pathognomonic enanthem) as well as in that of rash; in immunodeficient hosts, measles may be severe despite the absence of these manifestations. Measles antigens have been demonstrated in involved skin during early stages of the illness.

Pathologic changes in measles encephalitis include focal hemorrhage, congestion, and perivascular demyelination. Measles virus is rarely isolated from cerebrospinal fluid (CSF) in cases of encephalitis, which are thought to be due to the interaction of virus-infected cells with local cellular immune factors.

CLINICAL MANIFESTATIONS

Measles begins with a 2- to 4-day respiratory prodrome of malaise, cough, coryza, conjunctivitis with lacrimation, nasal discharge, and increasing fever [with temperatures as high as 40.6°C (105°F), probably reflecting secondary viremia]. At this stage of the illness, in which the rash has not yet developed, influenza may be suspected. Just before the onset of the rash, Koplik's spots appear as 1- to 2-mm blue-white spots on a bright red background. Without adequate illumination for examination, they may be overlooked. Koplik's spots are typically located on the buccal mucosa alongside the

second molars and may be extensive; they are not associated with any other infectious disease. The spots wane after the onset of rash and soon disappear. The entire buccal and inner labial mucosa may be inflamed, and the lips may be reddened.

The characteristic erythematous, nonpruritic, maculopapular rash of measles ([Fig. 194-CD1](#)) begins at the hairline and behind the ears, spreads down the trunk and limbs to include the palms and soles, and often becomes confluent. At this time, the patient is at the most severe point of the illness. By the fourth day, the rash begins to fade in the order in which it appeared. Brownish discoloration of the skin and desquamation may occur later. Fever usually resolves by the fourth or fifth day after the onset of rash; prolonged fever suggests a complication of measles. Lymphadenopathy, diarrhea, vomiting, and splenomegaly are common features. The chest x-ray may be abnormal, even in uncomplicated measles, because of the propensity of this virus to invade the respiratory tract. The entire illness usually lasts about 10 days. The disease tends to be more severe in adults than in children, with higher fever, more prominent rash, and a higher incidence of complications.

Milder forms of the illness with less intense symptoms and a milder rash, termed *modified measles*, may occur in individuals with preexisting partial immunity induced by active or passive vaccination. These patients include infants under 1 year of age who retain some proportion of passively acquired maternal antibodies. On occasion, individuals with a history of immunization may develop modified measles.

COMPLICATIONS

The complications of measles can conveniently be divided into three groups, according to the site involved: the respiratory tract, the central nervous system (CNS), and the gastrointestinal tract. Respiratory tract involvement, manifested as laryngitis, croup, or bronchitis, occurs in the majority of cases of uncomplicated measles. In young children, otitis media is the most common complication. Pneumonia is a frequent reason for hospitalization, especially of adults. The pneumonia is of viral origin in the majority of cases, but secondary bacterial infection (most commonly caused by streptococci, pneumococci, or staphylococci) also takes place with some frequency. Primary giant cell (Hecht's) pneumonia is most often documented in immunocompromised and/or malnourished patients.

Encephalographic abnormalities in the absence of symptoms of [CNS](#) disease are extremely frequent in measles. Symptomatic CNS disease, with fever, headache, drowsiness, coma, and/or seizures, occurs in about 1 case in 1000. Symptoms usually begin within days after the onset of rash but occasionally appear for the first time several weeks later. About 10% of patients do not survive acute measles encephalitis; a significant percentage of surviving patients develop permanent sequelae, such as mental retardation or epilepsy. Most cases appear to result from an immune-mediated response to myelin proteins (postinfectious encephalomyelitis) and not directly from viral infection of the CNS ([Chap. 371](#)). Rarely, transverse myelitis follows measles. Immunocompromised patients are at risk for progressive fatal encephalitis 1 to 6 months after measles; in some cases, even though prior measles has not been recognized, the virus is identified at autopsy. Subacute sclerosing panencephalitis (SSPE) -- a protracted, chronic, extremely rare form of measles encephalitis -- sometimes follows

measles and is particularly common among children who have measles before the age of 2 years ([Chap. 373](#)). SSPE has virtually disappeared in the United States as a result of widespread vaccination. Typically, progressive dementia evolves over several months. SSPE is thought to be due to a complex interaction of the host with defective measles virus. It is associated with extremely high levels of antibodies to measles virus in the blood and [CSF](#).

Gastrointestinal complications of measles include gastroenteritis, hepatitis, appendicitis, ileocolitis, and mesenteric adenitis. It is not uncommon to detect high levels of alanine and aspartate aminotransferases in the absence of gastrointestinal signs such as jaundice.

Other, rare complications include myocarditis, glomerulonephritis, and postinfectious thrombocytopenic purpura. Measles can exacerbate preexisting tuberculosis, presumably through depression of cellular immunity induced by the virus. Natural measles and immunization against measles can result in tuberculin skin-test anergy lasting for about 1 month.

ATYPICAL MEASLES

An atypical form of measles has been reported in individuals who received formalin-inactivated measles vaccine (used in the United States from 1963 through 1967 and in Canada until 1970) and subsequently were exposed to measles virus. After a several-day prodrome of fever, myalgia, and headache, the rash appears. Unlike the rash of typical measles, that of atypical measles begins peripherally and moves centrally; it can be urticarial, maculopapular, hemorrhagic, and/or vesicular. Fever is usually high and is accompanied by edema of the extremities, interstitial pulmonary infiltrates, hepatitis, and (on occasion) pleural effusion. The differential diagnosis often includes Rocky Mountain spotted fever, Henoch-Schonlein purpura, meningococcemia, drug allergy, toxic shock syndrome, and varicella. Despite the severity of atypical measles, patients invariably recover after a convalescence that may be prolonged. Measles virus is not isolated from these patients, and they do not spread the virus to others. This disease is believed to be due to hypersensitivity to measles virus induced by the inactivated vaccine. Formalin inactivation destroys the antigenicity of the F protein, antibodies to which are important in preventing spread of the virus from one cell to another. The role of cellular immunity in this process is unknown. Extremely high convalescent titers of antibody to measles virus (e.g., 1:1,000,000) are diagnostic of atypical measles. To prevent this syndrome, adults who received formalin-inactivated measles vaccine should be reimmunized with at least one dose of live attenuated measles vaccine. Since inactivated measles vaccine has not been available for more than 25 years, atypical measles has now virtually disappeared.

MEASLES IN THE IMMUNOCOMPROMISED HOST

Patients with defects in cell-mediated immunity are at risk for severe protracted and fatal measles. Included in this category are patients with congenital cellular immune defects or malignancy, recipients of immunosuppressive therapy, or persons infected with HIV. In these patients, measles may not be accompanied by a rash. Complications are primary measles (giant cell) pneumonia, progressive encephalitis beginning weeks to

months after initial infection, and (in HIV-infected patients) progression to AIDS.

MEASLES IN ADULTS

Measles is naturally a disease of childhood and, like many other viral infections, is more severe in adults than in children. About 3% of young adults with measles develop primary viral pneumonia and require hospitalization. Hepatitis and bronchospasm are more common among adults with measles than among children, and the rash is more severe and more confluent in adults. Bacterial superinfection is more common among adults, more than one-third of whom develop respiratory complications such as otitis media, sinusitis, and pneumonia. Adults may develop measles because they were never immunized or (more rarely) because their vaccine-induced immunity has waned. Very low titers of antibody to measles virus have been associated with lack of protection.

LABORATORY FINDINGS

Lymphopenia and neutropenia are common in measles and may be due to invasion of leukocytes by the virus, with subsequent cell death. Leukocytosis may herald a bacterial superinfection. Patients with measles encephalitis usually have an elevated protein concentration in [CSF](#) as well as lymphocytosis. A specific diagnosis of measles can be made quickly by immunofluorescent staining of a smear of respiratory secretions for measles antigen; monoclonal antibodies conjugated to fluorescein are commercially available for this purpose. Secretions can also be examined microscopically for multinucleated giant cells. Measles virus can be isolated from respiratory secretions or urine and rapidly identified in tissue culture with fluorescein-labeled monoclonal antibodies. Measles virus RNA has been demonstrated by diagnostic reverse-transcription polymerase chain reaction. A number of serologic tests are available for the diagnosis of measles; however, a serologic diagnosis cannot necessarily be made quickly since both acute- and convalescent-phase sera are usually tested, ideally at the same time. The older hemagglutination inhibition test has been replaced by enzyme immunoassay (EIA), which is more sensitive and simpler to perform. EIA can be used to measure specific IgM and thus to diagnose measles on the basis of an acute-phase serum sample alone. Specific IgM antibodies are detectable within 1 to 2 days after the appearance of rash, and the IgG titer rises significantly after 10 days. As already mentioned, atypical measles and [SSPE](#) are associated with extremely high titers of antibody.

DIFFERENTIAL DIAGNOSIS

Classic measles -- with Koplik's spots, cough, coryza, conjunctivitis, and a rash beginning on the head -- is easily diagnosed on clinical grounds. Modified measles is more difficult to diagnose clinically since one or more characteristic signs may be lacking. The differential diagnosis of measles includes Kawasaki's syndrome, scarlet fever, infectious mononucleosis, toxoplasmosis, drug eruption, and *Mycoplasma pneumoniae* infection. Most of these conditions can be identified by either culture or serologic assay. In the differential diagnosis of measles, attention should be paid to the current epidemiology of the disease in the community and to the patient's history of measles vaccination and foreign travel.

PREVENTION

The development of live attenuated measles vaccine by Enders and his colleagues was a milestone in American medicine. This vaccine, used in the United States for the routine immunization of children since 1963, induces seroconversion in about 95% of recipients and probably confers lifelong protection. Waning immunity to measles after immunization has been documented only on rare occasions. For the past 25 years, measles vaccine has been available as the combination vaccine measles-mumps-rubella (MMR); MMR vaccine should be administered to children between the ages of 12 and 15 months. (Vaccination at 12 months is preferred for infants whose mothers were immunized against measles in childhood. These mothers have lower antibody titers than women who have had natural measles, and their infants correspondingly have transplacental antibodies of lower titer and shorter duration.) A second dose of MMR vaccine is recommended for school-aged children at 4 to 12 years of age. This two-dose policy was developed in the late 1980s in response to measles outbreaks in the United States. Since the institution of the two-dose regimen and the increased effort to immunize all children, measles has again become an unusual disease in the United States. Regional guidelines that reflect the current local epidemiology of measles should be followed.

Older susceptible persons should also be immunized. Individuals should be considered susceptible to measles unless they have documentation of physician-diagnosed measles or of the receipt of two doses of vaccine, have laboratory evidence of measles immunity, or were born before 1957. Rarely, individuals born before 1957 develop measles, and those who are at risk of exposure to measles (e.g., health workers, teachers, and international travelers) should be tested for measles antibody and immunized if necessary. Approximately 10% of healthy vaccinees develop a fever, with temperatures up to 39.4°C (103°F), 5 to 7 days after vaccination; this fever lasts 1 to 5 days and is accompanied by a transient rash. Individuals previously immunized only with killed vaccine are considered susceptible and should receive at least one dose -- and preferably two doses -- of [MMR](#) vaccine. Transient adverse reactions in these individuals include fever, malaise, and redness and swelling at the injection site.

Because of the severity of measles in this group and the lack of reported problems following vaccination, children with asymptomatic HIV infection should receive [MMR](#) vaccine; those with severe immunosuppression (<15% CD4 lymphocytes) should not. A case of fatal measles due to vaccine-type virus was reported in a college student with AIDS. Measles vaccine is contraindicated for persons with impaired cell-mediated immunity, for pregnant women, and for persons with a history of anaphylaxis due to egg protein or neomycin. Minor illnesses, with or without fever and a history of convulsions, are not contraindications to vaccination. Vaccination should be deferred for 6 to 11 months after the receipt of immune globulin or of blood products containing antibodies and for at least 3 months after the discontinuation of immunosuppressive treatment. Vaccine failures have been ascribed to faulty storage of the preparation used, immunization of infants with preexisting (maternally derived) antibodies, and simultaneous administration of measles vaccine and immune globulin.

Children and adults who are susceptible to measles and are exposed to the disease should receive postexposure prophylaxis. Standard immune globulin, given

intramuscularly within 6 days of exposure, can exert a protective or modifying effect; the earlier it is given, the better the outcome. The dose is 0.25 mL/kg for healthy persons and 0.5 mL/kg for immunocompromised persons, with a maximum dose of 15 mL. Immune globulin is particularly strongly indicated for susceptible household contacts, especially those less than 1 year of age, and for immunocompromised persons. HIV-infected persons, particularly those with severe immunosuppression, should be given immune globulin after exposure, regardless of their measles immune status and whether or not they are receiving intravenous immunoglobulin. Vaccination within 72 h of exposure may also provide protection against clinical measles, but this strategy is contraindicated as postexposure prophylaxis for immunocompromised individuals. Vaccine and immune globulin should not be given concurrently.

TREATMENT

Therapy for measles is largely supportive and symptom-based. Patients with otitis media and pneumonia should be given standard antibiotics. Patients with encephalitis need supportive care, including observation for increased intracranial pressure. Controlled trials suggest clinical benefit from high doses of vitamin A in severe or potentially severe measles, especially in children under the age of 2 years. A dose of 50,000 IU is used for infants age 1 to 6 months, 100,000 IU for infants age 7 to 12 months, and 200,000 IU for children over 1 year. A single dose is administered on two consecutive days. Transient vomiting and headache may be associated with the administration of vitamin A. Ribavirin is effective against measles virus in vitro and may be considered for use in immunocompromised individuals.

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195. RUBELLA (GERMAN MEASLES) - Anne Gershon

DEFINITION

Rubella is an acute viral infection of children and adults that characteristically includes rash, fever, and lymphadenopathy and has a broad spectrum of other possible manifestations. However, a high percentage of rubella infections in both children and adults are subclinical. In addition, the illness can resemble a mild attack of measles (rubeola) and can cause arthritis, especially in adults. Rubella during pregnancy can lead to fetal infection, with the production of a significant constellation of malformations (*congenital rubella syndrome*) in a high proportion of infected fetuses.

ETIOLOGIC AGENT

Rubella virus, a togavirus, is the only member of the *Rubivirus* genus and is closely related to the alphaviruses. Unlike these agents, however, it does not require a vector for transmission. Moreover, there is no RNA sequence homology between rubella virus and the alphaviruses.

The rubella virion is composed of an inner icosahedral capsid of RNA and protein that is surrounded by a lipid-containing envelope with a diameter of about 60 nm. The structural proteins associated with rubella virus are E1 and E2 (transmembrane envelope glycoproteins) and C (the capsid protein that surrounds the viral RNA). Only one serotype has been identified.

EPIDEMIOLOGY

In the United States during the prevaccine era, rubella was most common in the spring and most often affected school-age children; only 80 to 90% of adults were immune; and major epidemics occurred every 6 to 9 years. The most recent epidemic in the United States occurred in 1964 to 1965, when there were more than 12 million reported cases of postnatal rubella and more than 20,000 cases of the congenital rubella syndrome. Since the introduction of live attenuated rubella vaccine in 1969, there have been no epidemics; limited outbreaks have been reported in settings where susceptible individuals come into close contact with one another (e.g., schools and workplaces). In 1996, only 213 cases of postnatally acquired rubella -- most of them in young adults -- and 2 confirmed cases of congenital rubella syndrome were reported to the Centers for Disease Control and Prevention (CDC).

Whether symptomatic or subclinical, rubella is contagious, albeit less so than measles. Its incubation period is 18 days on average, with a range of 12 to 23 days. The virus, which is spread in droplets shed in respiratory secretions, infects the respiratory tract and then the bloodstream. In postnatally acquired infections, rubella virus is shed during the prodromal phase of the illness, and shedding from the pharynx can continue for about a week after onset. Despite high titers of specific neutralizing antibodies, infants with congenital rubella may excrete rubella virus from the respiratory tract and in the urine until the age of 2 years. This excretion raises important issues related to infection control in hospital and day-care settings. Persons recently immunized with live attenuated rubella vaccine do not transmit the vaccine virus to others, although low

titers of rubella virus may be detected transiently in the pharynx.

After an attack of rubella, specific antibodies and cell-mediated immunity develop and probably play a significant role in protection against future disease. Asymptomatic reinfection at the level of the respiratory tract is common upon reexposure to the virus but is rarely if ever associated with viremia.

Rubella virus has been cultured from respiratory secretions during reinfection. Fetal infection may occur during maternal reinfection but is acknowledged to be extremely rare because of the absence of maternal viremia under these circumstances. Viremia following reinfection of individuals immunized against rubella is also rare. Thus the current level of congenital rubella in the United States is exceedingly low.

PATHOGENESIS AND PATHOLOGY

Little is known about the microscopic pathology of postnatally acquired rubella since the disease is invariably self-limited. Like that of measles, the rash of rubella is immunologically mediated; its onset coincides with the development of specific antibodies. Viremia can be demonstrated for about a week before and ends within a few days after the onset of rash.

The cause of the damage to cells and organs in congenital rubella is not well understood. Proposed mechanisms of fetal damage include mitotic arrest of cells, tissue necrosis without inflammation, and chromosomal damage. The growth of the fetus may be retarded. Other findings may include decreased numbers of megakaryocytes in the bone marrow, extramedullary hematopoiesis, and interstitial pneumonia.

CLINICAL MANIFESTATIONS

Postnatally Acquired Rubella Infection acquired after birth usually results in an extremely mild or subclinical illness. A prodromal phase is uncommon in children; adults may have more severe disease, with a brief prodrome of malaise, fever, and anorexia. The foremost symptoms of postnatally acquired rubella include posterior auricular, cervical, and suboccipital lymphadenopathy; fever; and rash. The rash often begins on the face and spreads down the body ([Fig. 195-CD1](#)). It is maculopapular but not confluent, is sometimes accompanied by mild coryza and conjunctivitis, and generally lasts for 3 to 5 days. A petechial enanthem on the soft palate, designated *Forschheimer spots*, may occur but is not specific for rubella. Fever may be absent entirely or may be present for only several days in the early phase of the illness.

Complications of postnatally acquired rubella are uncommon; bacterial superinfection is rare. One particularly troublesome complication is seen almost exclusively in women: arthritis, most frequently involving the fingers, wrists, and/or knees, develops as the rash is appearing and may take several weeks to resolve. Chronic arthritis resulting from rubella is extremely rare. Rubella virus has been isolated from joint fluid during acute rubella arthritis and from peripheral blood in chronic rubella arthritis.

Another complication of postnatally acquired rubella is hemorrhage due to both thrombocytopenia and vascular damage, which occurs in 1 of every 3000 patients.

Thrombocytopenia may last for weeks or months; it can have long-term consequences if there is bleeding into organs such as the eye or the brain.

Both children and adults may develop encephalitis after rubella; the incidence is about five times lower than that of encephalitis following measles. Adults are more likely than children to develop encephalitis; the mortality rate from this complication is 20 to 50%. Mild hepatitis is an unusual complication. Immunosuppressed patients are not at increased risk for rubella as they are for measles.

Congenital Rubella Maternal infection in early pregnancy can lead to fetal infection, with resultant congenital rubella. The classic signs of congenital rubella are cataract, heart disease, and deafness, but a myriad of other defects have been reported. These abnormalities include signs and symptoms that are transient, such as low birth weight, thrombocytopenia, hepatosplenomegaly, jaundice, and pneumonia; those that are permanent, such as deafness, pulmonic stenosis, patent ductus arteriosus, glaucoma, and cataract; and those that are developmental, such as mental retardation, diabetes mellitus, and behavioral disorders.

The most important factor in the pathogenicity of rubella virus for the fetus is gestational age at the time of infection. Maternal infection during the first trimester leads to fetal infection in about 50% of cases; maternal infection early in the second trimester leads to fetal infection in about one-third of cases. Fetal malformations not only are more common after maternal infection in the first trimester but also tend to be more severe and to involve more organ systems. While a fetus infected in the fourth week of gestation may develop many problems, one infected later (e.g., in the 20th week) may have isolated deafness as the only symptom.

DIAGNOSIS

Since postnatally acquired rubella is such a mild disease and since many cases are subclinical, diagnosis on clinical grounds can be difficult. Other diseases that may mimic rubella include toxoplasmosis, scarlet fever, modified measles, roseola, fifth disease (erythema infectiosum due to parvovirus B19), and enteroviral infection. Routine laboratory tests usually reveal leukopenia and atypical lymphocytes.

The isolation of rubella virus in cell cultures of throat samples, urine, or other secretions is difficult and expensive but is sometimes undertaken. This technique is most useful when congenital rubella is suspected. A laboratory diagnosis is more often made serologically. The most commonly used test is an enzyme-linked immunosorbent assay (ELISA) for IgG and IgM antibodies. Acute rubella is diagnosed by the documentation of a fourfold or greater rise in the titer of IgG antibodies in paired acute- and convalescent-phase serum specimens or by the detection of rubella-specific IgM antibodies in one serum specimen. However, false-negative and -positive IgM reactions are sometimes obtained. Moreover, true-positive IgM reactions can be obtained in both primary infection and reinfection. Congenital rubella is diagnosed by the isolation of rubella virus, the detection of IgM antibodies in a single serum sample, and/or the documentation of either the persistence of rubella antibodies in serum beyond 1 year of age or a rising antibody titer anytime during infancy in an unvaccinated child. Biopsied tissues and/or blood and cerebrospinal fluid have also been used for the demonstration

of rubella antigens with monoclonal antibodies and for the detection of rubella RNA by in situ hybridization and polymerase chain reaction.

PREVENTION

Live attenuated rubella vaccine was licensed in 1969, 7 years after the virus was first isolated in culture. This vaccine was developed as a strategy to prevent congenital rubella by ensuring that very few pregnant women would be susceptible and that there would be little circulating wild-type virus. Rubella vaccine induces seroconversion in more than 95% of recipients. Since its licensure, there have been no major epidemics in the United States, and the number of cases has declined by 98%. The vaccine currently licensed in the United States, RA 27/3, is propagated in human diploid cells and is more immunogenic (particularly with regard to the stimulation of secretory immunity) than previously licensed vaccines. The present vaccination strategy, developed in part when measles was not being adequately controlled, is to immunize all infants at 12 to 15 months of age with measles-mumps-rubella (MMR) vaccine and to administer a second dose at 4 to 12 years of age. Rubella vaccine may also be administered to anyone who is thought to be susceptible to the infection and is not pregnant; it is particularly important that hospital workers of either sex be immune to rubella so that nosocomial transmission is avoided. While there has been little change in the prevalence of immunity to rubella among women of childbearing age (about 80%), the incidence of congenital rubella is extremely low -- about 10 cases annually. It is likely that, although antibody may be undetectable years after immunization, protection against infection -- possibly due to cell-mediated immunity -- is the rule. At present, there is little if any evidence of significant waning of clinically important immunity to rubella with time.

On occasion, rubella vaccine may cause arthralgia or arthritis, especially in young women. Very rarely, rubella vaccination results in chronic arthritis; however, even cases of frank arthritis in vaccinees are self-limited, lasting only about 1 week.

After investigation of a series of more than 400 women who were inadvertently immunized during pregnancy and who carried their infants to term, the [CDC](#) has concluded that vaccine-type rubella virus either does not cause the congenital rubella syndrome at all or does so at an incidence too low to be detected. Nonetheless, rubella vaccine is contraindicated for use in pregnant women, and it is recommended that pregnancy be avoided for at least 3 months after rubella vaccination. It is acceptable for rubella-susceptible children whose mothers are also susceptible to be immunized, since vaccinated individuals do not shed rubella virus or transmit it to susceptible individuals. Although it is recommended that rubella vaccine not be given to immunosuppressed persons, the vaccine is given to children infected with HIV. No adverse effects of rubella vaccine have been reported in immunocompromised patients.

TREATMENT

There is no specific therapy for rubella. At one time, immune globulin was used in an effort to prevent congenital rubella when pregnant women became infected. However, since administration of immune globulin did not prevent maternal viremia, this approach was discarded. Treatment is given for symptoms such as fever, arthralgia, and arthritis.

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196. MUMPS - Anne Gershon

DEFINITION

Mumps is an acute, systemic, communicable viral infection whose most distinctive feature is swelling of one or both parotid glands. Involvement of other salivary glands, the meninges, the pancreas, and the gonads is also common.

ETIOLOGIC AGENT

Mumps virus, a paramyxovirus, is pleomorphic and has a diameter ranging from 100 to 600 nm. The virion is composed of RNA and five proteins. The RNA is surrounded by an envelope with glycoprotein projections. There are two envelope glycoproteins -- a hemagglutinin-neuraminidase (HN) and a hemolysis cell fusion antigen (F) -- as well as a matrix envelope protein (M). There are two internal components: a nucleocapsid protein (NP) and an RNA polymerase protein. There is only one antigenic type of mumps virus.

EPIDEMIOLOGY

After the introduction of mumps vaccine in 1967, the incidence of clinical mumps declined significantly in the United States. In 1968 (before widespread immunization), 185,691 cases of mumps were reported in this country. The 906 cases reported in 1995 represent a reduction in the number of cases by >99% from prevaccine levels; this is the lowest number of cases ever reported in a year. Before widespread vaccination, the incidence of mumps was highest in the winter and spring, with epidemics every 2 to 5 years. At that time, mumps was principally a disease of childhood, although today more than 50% of cases occur in young adults. Epidemics tended to occur in confined populations, such as those in schools and the military services.

The incubation period of mumps generally ranges from 14 to 18 days, with extremes of 7 and 23 days. However, because a contact may be shedding virus before the onset of clinical disease or (like one-third of patients) may have subclinical infection, the incubation period in individual cases is often uncertain. One attack of mumps usually confers lifelong immunity. Long-term immunity is also associated with immunization.

PATHOGENESIS

Mumps virus is transmitted by droplet nuclei, saliva, and fomites. Replication of the virus in the epithelium of the upper respiratory tract leads to viremia, which is followed by infection of glandular tissues and/or the central nervous system (CNS).

Little is known of the pathology of mumps since the disease is rarely fatal. The affected glands contain perivascular and interstitial mononuclear cell infiltrates with prominent edema. Necrosis of acinar and epithelial duct cells is evident in the salivary glands and in the germinal epithelium of the seminiferous tubules.

CLINICAL MANIFESTATIONS

The prodrome of mumps consists of fever, malaise, myalgia, and anorexia. Parotitis, if it develops, usually does so within the next 24 h but may be delayed for as long as a week; it is generally bilateral, although the onset on the two sides may not be synchronous and at times only one side is affected. The submaxillary and sublingual glands are involved less often than the parotid and are almost never involved alone. Swelling of the parotid is accompanied by tenderness and obliteration of the space between the ear lobe and the angle of the mandible. The patient frequently reports an earache and finds it difficult to eat, swallow, or talk. Glandular swelling increases for a few days and then gradually subsides, disappearing within a week. The orifice of Stensen's duct is commonly red and swollen. Presternal pitting edema has been described in about 5% of mumps cases, often in association with submandibular adenitis.

Other than parotitis, orchitis is the most common manifestation of mumps among postpubertal males, developing in about 20% of cases. The testis is painful and tender and is enlarged to several times its normal size; accompanying fever is common. Later, testicular atrophy develops in half of the affected men. Since orchitis is bilateral in fewer than 15% of cases, sterility after mumps is rare. Oophoritis in women -- far less common than orchitis in men -- may cause lower abdominal pain but does not lead to sterility.

Aseptic meningitis, which may develop before, during, after, or in the absence of parotitis, is a common manifestation of mumps in both children and adults. Symptoms include stiff neck, headache, and drowsiness. Pleocytosis of the cerebrospinal fluid (CSF), with up to 1000 cells/uL, may develop in up to 50% of cases of clinical mumps, but clinical signs of meningeal irritation are documented in only 5 to 25% of cases. Within the first 24 h, polymorphonuclear leukocytes may predominate in CSF, but by the second day nearly all the cells are lymphocytes. The glucose level in CSF may be abnormally low, and this finding may arouse suspicion of bacterial meningitis. Aseptic meningitis due to mumps without parotitis is indistinguishable clinically from that caused by other viruses. Mumps meningitis is almost invariably self-limited, although cranial nerve palsies have occasionally led to permanent sequelae, particularly deafness. More rarely, mumps virus may cause encephalitis, which presents as high fever with marked changes in the level of consciousness and frequently results in permanent sequelae in survivors. Other [CNS](#) problems occasionally associated with mumps include cerebellar ataxia, facial palsy, transverse myelitis, Guillain-Barre syndrome, and aqueductal stenosis leading to hydrocephalus.

Mumps pancreatitis, which may present as abdominal pain, is difficult to diagnose because an elevated serum amylase level can be associated with either parotitis or pancreatitis. Other unusual complications of mumps include myocarditis, mastitis, thyroiditis, nephritis, arthritis, and thrombocytopenic purpura. An excessive number of spontaneous abortions are associated with gestational mumps when the disease occurs during the first trimester. Mumps in pregnancy does not lead to premature birth or fetal malformations.

DIFFERENTIAL DIAGNOSIS

The diagnosis of mumps is made easily in patients with acute bilateral parotitis and a

history of recent exposure. When parotitis is unilateral or absent or when sites other than the parotid gland are involved, laboratory diagnosis is required (see below).

The myriad causes of bilateral parotid swelling other than mumps virus include infection with other viruses, such as parainfluenza virus type 3, coxsackieviruses, and influenza A virus; metabolic diseases, such as diabetes mellitus and uremia; and drugs, such as phenylbutazone and thiouracil. Unilateral parotid swelling can result from a tumor, a cyst, or a ductal obstruction due to stones or strictures. Other conditions associated with chronic parotid swelling include sarcoidosis, Sjogren's syndrome, and infection with HIV. Suppurative parotitis, usually caused by *Staphylococcus aureus*, is most often unilateral.

Other entities should be considered when manifestations consistent with mumps appear in organs other than the parotid. Testicular torsion may produce a painful scrotal mass resembling that seen in mumps orchitis. Other viruses (e.g., enteroviruses) may cause aseptic meningitis that is clinically indistinguishable from that due to mumps virus.

LABORATORY DIAGNOSIS

Mumps virus is readily isolated after inoculation of appropriate clinical specimens into a variety of host systems, such as rhesus monkey kidney cells and human embryonic lung fibroblasts. The virus can be rapidly identified by the use of cells grown in shell vials and of fluorescein-labeled monoclonal antibodies. Mumps virus may be recovered from saliva, throat, and urine during the first few days of illness and from the [CSF](#) of patients with mumps meningitis. Shedding of virus in the urine may persist for as long as 2 weeks. No particular peripheral blood cell count is characteristic of mumps.

Highly sensitive enzyme-linked immunosorbent assays are useful for diagnosis of mumps and for determination of susceptibility to the disease. Acute mumps can be diagnosed either by the examination of acute- and convalescent-phase sera for a significant increase in IgG antibody titer or by the demonstration of specific IgM in one serum specimen. Use of a skin-test antigen to assess immunity to mumps has been replaced by serologic testing.

PREVENTION

Live attenuated mumps vaccine (Jeryl Lynn strain) induces antibodies that protect against infection in more than 95% of cases. The subcutaneously administered vaccine may be given to children older than 1 year but is not recommended for younger infants because of the potential for interference by passive maternal antibodies. Mumps vaccine is usually administered as part of the measles-mumps-rubella (MMR) vaccine at the age of 12 to 15 months and again at 4 to 12 years of age. This MMR vaccine is also recommended for susceptible older children, adolescents, and adults, particularly adolescent males who have not had mumps. For these patients, either MMR or monovalent mumps vaccine may be given; two doses are preferred. Inadvertent immunization of individuals who are already immune is not associated with significant adverse reactions. Mumps vaccine is not recommended for pregnant women, for patients receiving glucocorticoids, or for other immunocompromised hosts. However, children with HIV infection who are not severely immunocompromised can safely be

immunized against mumps; MMR vaccine is usually used for this purpose ([Chap. 194](#)).

TREATMENT

Therapy for parotitis and other manifestations of mumps is symptom-based. The administration of analgesics and the application of warm or cold compresses to the parotid area may be helpful. Mumps immune globulin is of no value in the prophylaxis or treatment of established disease. Testicular pain may be minimized by the local application of cold compresses and gentle support for the scrotum. Anesthetic blocks may also be used. Neither the administration of glucocorticoids nor incision of the tunica albuginea is of proven value for the treatment of severe orchitis. Anecdotal information on a small number of patients with orchitis suggests that administration of interferon may be helpful.

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197. RABIES VIRUS AND OTHER RHABDOVIRUSES - Lawrence Corey

RABIES VIRUS

DEFINITION

Rabies is an acute viral disease of the central nervous system (CNS) that affects all mammals and that is transmitted by infected secretions, usually saliva. Most exposures to rabies are through the bite of an infected animal, but on occasion contact with a virus-containing aerosol or the ingestion or transplantation of infected tissues may initiate the disease process.

ETIOLOGY

The rabies virus is a bullet-shaped, enveloped, single-stranded RNA virus that is 75 to 80 nm in diameter and belongs to the genus *Lyssavirus* within the rhabdovirus family. The envelope glycoproteins of rabies viruses are arranged in knoblike structures that cover the surface of the virion. Genetic and phenotypic analyses of the envelope have been used to detail the molecular epidemiology and spread of unique variants within animal species. The viral glycoproteins bind to acetylcholine receptors, contribute to the neurovirulence of rabies virus, elicit neutralizing and hemagglutination-inhibiting antibodies, and stimulate cytotoxic T cell immunity. The nucleocapsid antigen induces a complement-fixing antibody as well as T helper cell reactivity. Neutralizing antibodies to the surface glycoproteins appear to be protective and are directed at conformational epitopes of the viral envelope glycoprotein. The antibodies to rabies virus used in diagnostic immunofluorescence assays are generally directed against the nucleocapsid antigens. Isolates of rabies virus from different animal species and locales differ in their antigenic and biologic properties. These variations may account for differences in virulence between isolates. Interferon is induced by rabies virus, particularly in those tissues with high virus concentrations, and may play some role in retarding progressive infection.

EPIDEMIOLOGY

Rabies is found in animals in all regions of the world except Australasia and Antarctica. Rabies exists in two epidemiologic forms: *urban rabies*, propagated chiefly by unimmunized domestic dogs and cats, and *sylvatic rabies*, propagated by skunks, foxes, raccoons, mongooses, wolves, and bats. Infection in domestic animals usually represents a "spillover" from sylvatic reservoirs of infection. Human infection occurs through contact with unimmunized domestic animals or from exposure to wild animals in locales where rabies is enzootic or epizootic. The worldwide incidence of rabies is estimated at more than 30,000 cases per year. Southeast Asia, the Philippines, Africa, the Indian subcontinent, and tropical South America are areas where the disease is especially common. In some endemic areas, 1 to 2% of autopsies yield evidence of rabies. Increased spread of terrestrial rabies (i.e., rabies in animals that walk on the ground rather than rabies in animals that fly) and increased travel to countries where urban rabies exists have made the recognition of clinical rabies and its prevention of increasing importance. While focal epidemics of terrestrial rabies have occurred in the United States and Europe, human rabies is uncommon, largely because of successful

domestic-animal vaccination programs. Since 1980, 36 human cases of rabies have been diagnosed in the United States; 58% of these cases were associated with exposure to bats, while one-third were acquired through dog bites sustained outside the United States. Most persons with proven clinical rabies in this country report no history of an animal bite. More than one-third of recent cases have been diagnosed post-mortem.

In most areas of the world, the dog is the most important vector of rabies virus for humans. However, the wolf (in eastern Europe and Arctic regions), the mongoose (in South Africa and the Caribbean), the fox (in western Europe), and the vampire bat (in Latin America) may also be prominent vectors. Although rabies in wildlife is common throughout both the developed and the undeveloped world, most cases of postexposure prophylaxis are associated with domesticated animals such as dogs and cats. In the United States, local governments are charged with initiating and maintaining programs for rabies vaccination of all dogs, cats, and ferrets. Rodents and lagomorphs are rarely infected with rabies virus. Several cases of human-to-human transmission of rabies through corneal transplantation have also been documented. Bite and nonbite exposures to infected humans could theoretically transmit rabies. Because of delayed diagnosis, postexposure prophylaxis of health care workers and close contacts of cases is common.

PATHOGENESIS

The first event in rabies is the introduction of live virus through the epidermis or onto a mucous membrane. Initial viral replication appears to occur within striated muscle cells at the site of inoculation. The peripheral nervous system is exposed at the neuromuscular and/or neurotendinous spindles of unmyelinated sensory nerve cell endings. The virus then spreads centripetally up the nerve to the [CNS](#), probably via peripheral nerve axoplasm, at a rate of ~3 mm/h. Viremia has been documented in experimental conditions but is thought not to play a role in naturally acquired disease. Once the virus reaches the CNS, it replicates almost exclusively within the gray matter and then passes centrifugally along autonomic nerves to other tissues -- the salivary glands, adrenal medulla, kidneys, lungs, liver, skeletal muscles, skin, and heart. Passage of the virus into the salivary glands and viral replication in mucinogenic acinar cells facilitate further transmission via infected saliva. The incubation period of rabies is exceedingly variable, ranging from 7 days to >1 year (mean, 1 to 2 months) and apparently depending on the amount of virus introduced, the amount of tissue involved, host defense mechanisms, and the actual distance that the virus has to travel from the site of inoculation to the CNS. Rates of infection and mortality are highest from bites on the face, intermediate from bites on the hands and arms, and lowest from bites on the legs. Cases of human rabies with an extended incubation period (2 to 7 years) have been reported, but they are rare. Host immune responses and viral strains also influence disease expression.

The neuropathology of rabies resembles that of other viral diseases of the [CNS](#): hyperemia, varying degrees of chromatolysis, nuclear pyknosis, and neuronophagia of the nerve cells; infiltration by lymphocytes and plasma cells of the Virchow-Robin space; microglial infiltration; and parenchymal areas of nerve cell destruction. In experimental animal models, adenohypophyseal infection with rabies virus, with reduction in growth

hormone and vasopressin release, is common. The most characteristic pathologic finding of rabies in the CNS is the formation of cytoplasmic inclusions called *Negri bodies* within neurons. Each eosinophilic mass measures ~10 nm and is made up of a finely fibrillar matrix and rabies virus particles. Negri bodies are distributed throughout the brain, particularly in Ammon's horn, the cerebral cortex, the brainstem, the hypothalamus, the Purkinje cells of the cerebellum, and the dorsal spinal ganglia. Negri bodies are not demonstrated in at least 20% of cases of rabies, and their absence from brain material does not rule out the diagnosis.

CLINICAL MANIFESTATIONS

The clinical manifestations of rabies can be divided into four stages: (1) a nonspecific prodrome, (2) an acute encephalitis similar to other viral encephalitides, (3) a profound dysfunction of brainstem centers that produces the classic features of rabies encephalitis, and (4) death or, in rare cases, recovery.

The prodromal period usually lasts 1 to 4 days and is marked by fever, headache, malaise, myalgias, increased fatigability, anorexia, nausea and vomiting, sore throat, and a nonproductive cough. The prodromal symptom suggestive of rabies is the complaint of paresthesia and/or fasciculations at or around the site of inoculation of virus. These sensations, which may be related to the multiplication of virus in the dorsal root ganglion of the sensory nerve supplying the area of the bite, are reported by 50 to 80% of patients.

The encephalitic phase is usually ushered in by periods of excessive motor activity, excitation, and agitation. Confusion, hallucinations, combativeness, bizarre aberrations of thought, muscle spasms, meningismus, opisthotonic posturing, seizures, and focal paralysis soon appear. Characteristically, the periods of mental aberration are interspersed with completely lucid periods, but as the disease progresses the lucid periods get shorter until the patient lapses into coma. Hyperesthesia, with excessive sensitivity to bright light, loud noise, touch, and even gentle breezes, is very common. On physical examination, the temperature may be found to be as high as 40.6°C (105°F). Abnormalities of the autonomic nervous system include dilated irregular pupils; increased lacrimation, salivation, and perspiration; and postural hypotension. Evidence of upper motor neuron paralysis, with weakness, increased deep tendon reflexes, and extensor plantar responses, is the rule. Paralysis of the vocal cords is common. Unfortunately, the presenting signs and symptoms of rabies are indistinguishable from those of other viral and neurologic diseases. Thus delays in diagnosis are frequent. The presence of hydrophobia or aerophobia (seen in about two-thirds of recent cases) increases the likelihood of antemortem diagnosis.

The manifestations of brainstem dysfunction begin shortly after the onset of the encephalitic phase. Cranial nerve involvement causes diplopia, facial palsies, optic neuritis, and the characteristic difficulty with deglutition. The combination of excessive salivation and difficulty in swallowing produces the traditional picture of "foaming at the mouth." Hydrophobia, the painful, violent, involuntary contraction of the diaphragmatic, accessory respiratory, pharyngeal, and laryngeal muscles initiated by swallowing liquids, is seen in ~50% of cases. Involvement of the amygdaloid nucleus may result in priapism and spontaneous ejaculation. The patient lapses into coma, and involvement

of the respiratory center produces an apneic death. The prominence of early brainstem dysfunction distinguishes rabies from other viral encephalitides and accounts for the rapid downhill course. The median period of survival after the onset of symptoms is 4 days, with a maximum of 20 days, unless artificial supportive measures are instituted.

If intensive respiratory support is used, a number of late complications may appear. These include inappropriate secretion of antidiuretic hormone, diabetes insipidus, cardiac arrhythmias, vascular instability, adult respiratory distress syndrome, gastrointestinal bleeding, thrombocytopenia, and paralytic ileus. Recovery is very rare and, when it occurs, gradual.

Rabies may also present as an ascending paralysis resembling the Landry/Guillain-Barre syndrome (dumb rabies, *rage tranquille*). Initially, this clinical pattern was reported most frequently among persons given postexposure rabies prophylaxis after being bitten by vampire bats. Paralytic rabies also occurs in Southeast Asia among persons with canine exposures.

The difficulty of diagnosing rabies associated with ascending paralysis is illustrated by cases of person-to-person transmission of the virus by tissue transplantation. Corneal transplants from donors who died of presumed Landry/Guillain-Barre syndrome produced clinical rabies in and caused the deaths of the recipients. Retrospective pathologic examinations of the brains of recipients demonstrated Negri bodies, and rabies virus was subsequently isolated from each donor's frozen eye.

LABORATORY FINDINGS

Early in the disease, hemoglobin values and routine blood chemistry results are normal; abnormalities develop as hypothalamic dysfunction, gastrointestinal bleeding, and other complications ensue. The peripheral white blood cell count is usually slightly elevated (12,000 to 17,000/uL) but may be normal or as high as 30,000/uL.

The specific diagnosis of rabies depends on (1) the isolation of virus from infected secretions [saliva or, rarely, cerebrospinal fluid (CSF)] or tissue (brain), (2) the serologic demonstration of acute infection, (3) the detection of viral antigen in infected tissue (e.g., corneal impression smears, skin biopsies, or brain), or (4) the detection of viral nucleic acid (RNA) by polymerase chain reaction (PCR). A reference laboratory evaluating antemortem samples can confirm rabies with high sensitivity and specificity. Isolation of virus from saliva, demonstration of viral nucleic acid in saliva, or detection of viral antigen in a nuchal skin biopsy specimen is most sensitive. Examination of corneal epithelium specimens appears less sensitive. In the unvaccinated person, demonstration of rabies antibodies in serum or CSF may be useful, although such antibodies may not appear until late in the course of disease. Samples of brain obtained at postmortem examination or brain biopsy should be subjected to (1) mouse inoculation studies for virus isolation, (2) fluorescent antibody (FA) staining for viral antigen, and (3) histologic and/or electron microscopic examination for Negri bodies or reverse transcription PCR for rabies virus RNA.

Postexposure rabies prophylaxis rarely elicits [CSF](#) neutralizing antibody to rabies virus. If present after prophylaxis, such antibody is usually found at a low titer (<1:64), whereas

CSF titers in human rabies may vary from 1:200 to 1:160,000.

DIFFERENTIAL DIAGNOSIS

There is little to distinguish rabies from other viral encephalitides. The most helpful clue to the diagnosis is a history of a bite or other salivary exposure to a potentially infected animal. As bite exposures are infrequent among U.S. cases, a history of relatively recent travel to a rabies-endemic area should be sought. Other problems to be considered in the differential diagnosis include hysterical reactions to animal bites (pseudohydrophobia), Landry/Guillain-Barre syndrome, poliomyelitis, and allergic encephalomyelitis developing in response to rabies vaccine; this last problem is usually associated with receipt of nerve tissue-derived vaccine and usually begins 1 to 4 weeks after vaccination.

TREATMENT

Postexposure Prophylaxis (See [Fig. 197-1](#)) Although rabies among humans is rare in the United States, each year ~35,000 persons receive postexposure prophylaxis. The decision to initiate postexposure prophylaxis should include the following considerations: (1) whether the individual came into physical contact with saliva or another substance likely to contain rabies virus, (2) whether rabies is known or suspected in the species and area associated with the exposure (e.g., all persons within the continental United States bitten by a bat that escapes should receive postexposure prophylaxis), and (3) the circumstances surrounding the exposure (e.g., whether the bite was provoked or unprovoked). Bites associated with the feeding of an animal are considered to have been provoked.

If rabies is known or suspected to be present in the animal species involved in a human exposure, the implicated animal should be captured if possible. Any wild animal involved in a rabies exposure; any ill, unvaccinated, or stray domestic animal involved in a rabies exposure; and any animal inflicting an unprovoked bite, exhibiting abnormal behavior, or suspected of being rabid should be humanely killed. The animal's head should be sent immediately to an appropriate laboratory for rabies [FA](#) examination. If examination of the brain by the FA technique gives negative results, it can be assumed that the saliva contains no virus, and the exposed person need not be treated. Persons exposed to wild animals that subsequently escape, that are capable of carrying rabies (bats, skunks, coyotes, foxes, raccoons, etc.), and that inhabit an area where rabies is known or suspected to be present should undergo both passive and active immunization against rabies (see below) as soon as possible after exposure.

In an area in which feline or canine rabies is not prevalent, a healthy biting dog, cat, or ferret can be confined and observed for 10 days. Persons in such an area should not begin a course of prophylaxis unless the animal develops clinical signs of rabies. If the animal becomes ill or behaves abnormally during the observation period, it should be killed for [FA](#) examination. Experimental and epidemiologic evidence suggests that animals that remain healthy for 10 days after a bite will not have transmitted rabies virus at the time of the bite. In areas of high endemicity for canine rabies, immediate examination of the animal's brain, especially in the case of a severe bite, may be warranted. Bites of rodents, rabbits, and hares almost never require antirabies

postexposure prophylaxis. Unless the exposed person can rule out a bite, scratch, or mucous membrane exposure, postexposure prophylaxis should be considered after direct contact between a human and a bat.

Postexposure prophylaxis of rabies includes rigorous cleansing and treatment of the wound and the administration of rabies vaccine together with antirabies immunoglobulin. Postexposure prophylaxis should be initiated as soon as possible after exposure. As the incubation period of rabies is quite variable, postexposure prophylaxis should be begun as long as clinical signs of rabies are not present.

1. *Wound cleansing and treatment.* Thorough cleansing and treatment of the bite wound constitute an important component of rabies prevention. The wound should be scrubbed with soap and then flushed with water. Both mechanical cleansing and chemical cleansing are important. Quaternary ammonium compounds such as 1 to 4% benzalkonium chloride, 1% cetrimonium bromide, or povidone-iodine solutions should be utilized. Tetanus toxoid and antibiotic prophylaxis should be administered as needed.

2. *Passive immunization with antirabies antiserum of either equine or human origin.* Postexposure antirabies vaccination should include the administration of both passive antibody and vaccine, except when the individual has previously received preexposure prophylaxis. Human rabies immune globulin (RIG) is preferred because equine antiserum may cause serum sickness. RIG is administered only once, at the beginning of the postexposure prophylaxis regimen. The recommended dose of RIG is 20 IU/kg. The dose of equine antiserum is 40 units/kg. The full dose should be thoroughly infiltrated into the area around the wound and into the wound itself. Any remaining portion of the dose is injected intramuscularly at a site distant from the vaccine.

3. *Active immunization with antirabies vaccine.* Three rabies vaccines are available in the United States: (1) human diploid cell vaccine (HDCV), which can be given either intramuscularly or intradermally; (2) rabies vaccine absorbed (RVA); and (3) purified chick embryo cell vaccine. The latter two vaccines are administered intramuscularly. Each vaccine is derived from a different strain of rabies virus and prepared in a slightly different formulation. The three vaccines are considered equally efficacious and safe, and any of the three can be administered in conjunction with [RIG](#). Five 1-mL doses of HDCV are given intramuscularly, preferably in the deltoid or anterolateral thigh area; the gluteal area should not be utilized. The five doses of HDCV should be administered within 28 days on the following schedule: days 0, 3, 7, 14, and 28. The World Health Organization also recommends 21- and 90-day injections. Severe reactions to these vaccines are uncommon. Immediate hypersensitivity responses, such as urticaria, have been reported in ~1 of every 650 recipients. Systemic reactions, such as fever, headache, and nausea, are generally mild and are reported in 1 to 4% of recipients. Local reactions, such as swelling, erythema, and induration at the injection site, occur in 15 to 20% of vaccinees. Guillain-Barre syndrome has been reported but appears to be quite rare.

In the developing world, several other effective rabies vaccines have been licensed and used extensively. They include vaccines made in chick embryonic cells, primary hamster cells, Vero cells, and duck embryonic cells. As some of these preparations are somewhat less immunogenic than the vaccines approved by the U.S. Food and Drug

Administration (FDA), evaluation of serum antibodies after immunization is suggested by some authorities.

The combination of [RIG](#) and [HDCV](#) elicits high titers of neutralizing antibodies in almost all recipients. Only rarely has this regimen proved unsuccessful in preventing the development of rabies. None of the patients in whom rabies was diagnosed in the United States between 1980 and 1996 had received postexposure prophylaxis. Administration of vaccine alone appears to be associated with a higher failure rate than use of the combination, especially in severe bite exposures. Because of cost, postexposure prophylaxis consisting of intradermal injections of rabies vaccine is being used increasingly in the developing world. The combination of RIG plus 0.1-mL intradermal doses of HDCV at eight sites on day 0, four sites on day 7, and one site on days 28 and 91 produces good antibody responses and has had excellent clinical results. Alternatively, the World Health Organization has approved a regimen of two 0.1-mL doses at two intradermal sites on days 0, 3, and 7 and a 0.1-mL intradermal injection at a single site on days 21 and 90. The [FDA](#) has not approved the intradermal route for postexposure prophylaxis.

Preexposure Prophylaxis Individuals at high risk of contact with rabies virus, including veterinarians, cave explorers, laboratory workers, and animal handlers, should receive preexposure prophylaxis with rabies vaccine. Three 1-mL intramuscular or three 0.1-mL intradermal injections of [HDCV](#) on days 0, 7, and 21 or 28 should be administered. Concomitant chloroquine administration interferes with the antibody response to rabies vaccine. Depending on the level of risk, serologic testing should be done at 6-month to 2-year intervals.

An immune complex reaction consisting of urticaria, arthralgia, arthritis, angioedema, and systemic symptoms has been reported in up to 6% of persons receiving intramuscular booster doses of [HDCV](#). This reaction is self-limited and appears to be associated with the presence of β -propiolactone-altered human serum albumin in the vaccine and the development of IgE antibodies to this antigen.

Persons who work in high-risk areas should undergo periodic measurement of antibodies. When neutralizing titers fall below 1:5, booster doses should be given. Booster doses may be administered as a single 1-mL intramuscular or 0.1-mL intradermal injection. Postexposure prophylaxis in individuals previously given preexposure prophylaxis consists of two intramuscular doses of [HDCV](#) on days 0 and 3. [RIG](#) is not given in these situations.

[MOKOLA VIRUS](#)

Mokola virus was first isolated from wild shrews captured in Nigeria and was shown to be related morphologically and serologically to rabies virus. The subsequent isolation of the virus from cats in South Africa suggested a wider prevalence of the agent than had previously been expected. Only two cases of clinical infection have been reported; both were in children. One patient had a nonfatal illness characterized by fever, pharyngitis, and convulsions; Mokola virus was recovered from [CSF](#). In the second patient, fever with cough and vomiting was followed within several days by drowsiness, confusion, and generalized flaccid weakness. The CSF was normal. The patient progressed to

deep coma and died within 10 days of onset. Mokola virus was isolated from the brain, and examination of histopathologic sections revealed finely granular cytoplasmic inclusions that were distinguishable from Negri bodies in many neurons.

VESICULAR STOMATITIS VIRUS

Vesicular stomatitis is a viral illness of animals that occasionally affects humans. It presents as an acute, self-limited, influenza-like disease. The disease in animals is found in the United States and South America and affects chiefly domestic cattle, horses, swine, wild deer, raccoons, skunks, and bobcats.

In animals, vesicular stomatitis is characterized by the development of vesicles on the oral mucosa, particularly the tongue; the udders; and the heels. The mode of spread is probably by direct contact; however, epidemics tend to occur in warm weather, and isolation of the virus from *Phlebotomus* sandflies in Panama and *Aedes* species in New Mexico suggests that these insects may be vectors. Two distinct serotypes, New Jersey and Indiana, have been recognized, and most outbreaks in North America have been attributed to the New Jersey strain.

In humans, vesicular stomatitis is most common among laboratory workers. In one report, three-fourths of laboratory personnel handling experimentally infected animals or manipulating the virus developed neutralizing antibodies. The disease is also transmissible, however, under natural conditions among workers having direct contact with infected animals, especially cattle. An incubation period ranging from 1 to 6 days is followed by the sudden onset of fever [with temperatures of up to 40°C (104°F)], chills, profuse sweating, myalgias, malaise, headache, and pain on ocular movement. One-third to one-half of patients have a sore throat and cervical and/or submandibular adenopathy. Small raised vesicular lesions may appear on the buccal mucosa. Conjunctivitis and coryza are evident in ~20% of cases. Occasionally, small subcorneal, intraepithelial vesicles appear on the fingers, usually in association with direct inoculation of the virus. Symptoms generally last 3 to 4 days, but occasionally the course is diphasic. Inapparent infection is common: among laboratory workers with serologic evidence of infection, only about one-half report symptoms. In some areas of Panama, 17 to 35% of the population have neutralizing antibodies to vesicular stomatitis virus.

The differential diagnosis includes hand-foot-and-mouth disease, herpangina, primary herpetic pharyngitis and other mucocutaneous syndromes, and influenza. The virus is not commonly isolated from patients. However, a rise in titer of complement-fixation and/or neutralizing antibody to vesicular stomatitis virus between acute- and convalescent-phase sera helps to confirm the diagnosis. Treatment is nonspecific.

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198. INFECTIONS CAUSED BY ARTHROPOD- AND RODENT-BORNE VIRUSES - C. J. Peters

Most viral infections that come to medical attention in office or hospital practice in the developed countries are caused by viruses that can be latent in the human host, such as the herpesviruses, or by viruses that are continuously transmitted among humans, such as measles virus, influenza virus, and HIV. However, some other viruses are transmitted in nature without regard to humans and only incidentally infect and produce disease in humans; in addition, a few agents are regularly spread among humans by arthropods. Most of these viruses either are maintained by arthropods or chronically infect rodents. Obviously, the mode of transmission is not a rational basis for taxonomic classification. Indeed, zoonotic viruses from at least seven virus families act as significant human pathogens ([Table 198-1](#)). The virus families differ fundamentally from one another in terms of morphology, replication mechanisms, and genetics. Information on a virus's membership in a family or genus is enlightening with regard to maintenance strategies, sensitivity to antivirals, and some aspects of pathogenesis but does not necessarily predict which clinical syndromes -- if any -- the virus will cause in humans.

FAMILIES OF ARTHROPOD- AND RODENT-BORNE VIRUSES

The Arenaviridae The Arenaviridae are spherical, 110- to 130-nm particles that bud from the cell's plasma membrane and utilize ambisense RNA genomes with two segments for replication. There are two main phylogenetic branches of Arenaviridae: the Old World viruses, such as Lassa fever and lymphocytic choriomeningitis (LCM) viruses, and the New World viruses, including those causing the South American hemorrhagic fevers (HFs). Arenaviruses persist in nature by chronically infecting rodents with a striking one-virus-one-rodent species relationship. These rodent infections result in long-term virus excretion and perhaps in lifelong viremia; vertical infection is common with some arenaviruses. Humans become infected through the inhalation of aerosols containing arenaviruses, which are then deposited in the terminal air passages, and probably also through close contact with rodents and their excreta, which results in the contamination of mucous membranes or breaks in the skin.

The Bunyaviridae The family Bunyaviridae includes four medically significant genera. All of these spherical viruses have three negative-sense RNA segments maturing into 90- to 120-nm particles in the Golgi complex and exiting the cell by exocytosis. Viruses of the genus *Bunyavirus* are largely mosquito-borne and have a viremic vertebrate intermediate host; many are also transovarially transmitted in their specific mosquito host. One serologic group also uses biting midges as vectors. Sandflies or mosquitoes are the vectors for the genus *Phlebovirus* (named after phlebotomus fever or sandfly fever, the best-known disease associated with the genus), while ticks serve as vectors for the genus *Nairovirus*. Viruses of both of these genera are also associated with vertical transmission in the arthropod host and with horizontal spread through viremic vertebrate hosts. The genus *Hantavirus* is unique among the Bunyaviridae in that it is not transmitted by arthropods but is maintained in nature by rodent hosts that chronically shed virus. Like the arenaviruses, the hantaviruses usually display striking virus-rodent species specificity. Hantaviruses do not cause chronic viremia in their rodent host and are transmitted only horizontally from rodent to rodent.

Other Families The Flaviviridae are positive-sense, single-stranded RNA viruses that form particles of 40 to 50 nm in the endoplasmic reticulum. The flaviviruses discussed here are from the genus *Flavivirus* and make up two phylogenetically and antigenically distinct divisions transmitted among vertebrates by mosquitoes and ticks, respectively. The mosquito-borne viruses fall into phylogenetic groups that include yellow fever virus, the four dengue viruses, and encephalitis viruses, while the tick-borne group encompasses a geographically varied spectrum of species, some of which are responsible for encephalitis or for hemorrhagic disease with encephalitis. The Reoviridae are double-stranded RNA viruses with multisegmented genomes. These 80-nm particles are the only viruses discussed in this chapter that do not have a lipid envelope and thus are insensitive to detergents. The Togaviridae have a single positive strand RNA genome and bud particles of ~60 to 70 nm from the plasma membrane. The togaviruses discussed here are all members of the genus *Alphavirus* and are transmitted among vertebrates by mosquitoes in their natural cycle. **The Filoviridae and the Rhabdoviridae are discussed in Chaps. 199 and 197, respectively.*

PROMINENT FEATURES OF ARTHROPOD- AND RODENT-BORNE VIRUSES

Although this chapter discusses the major features of selected arthropod- and rodent-borne viruses, it does not deal with more than 500 other distinct recognized zoonotic viruses, about one-fourth of which infect humans. Zoonotic viruses are undergoing genetic evolution, "new" zoonotic viruses are being discovered, and the epidemiology of zoonotic viruses is continuing to evolve through environmental changes affecting vectors, reservoirs, and humans. These zoonotic viruses are most numerous in the tropics but are also found in temperate and frigid climates. Their distribution and seasonal activity may be variable and often depend largely on ecologic conditions such as rainfall and temperature, which in turn affect the density of vectors and reservoirs and the development of infection therein.

Maintenance and Transmission Arthropod-borne viruses infect their vectors after the ingestion of a blood meal from a viremic vertebrate. The vectors then develop chronic, systemic infection as the viruses penetrate the gut and spread throughout the body. The viruses eventually reach the salivary glands during a period that is referred to as *extrinsic incubation* and that typically lasts 1 to 3 weeks in mosquitoes. At this point an arthropod is competent to continue the chain of transmission by infecting another vertebrate when a subsequent blood meal is taken. The arthropod generally is unharmed by the infection, and the natural vertebrate partner usually has only transient viremia with no overt disease. An alternative mechanism for virus maintenance in its arthropod host is transovarial transmission, which is common among members of the family Bunyaviridae.

Rodent-borne viruses such as the hantaviruses and arenaviruses are maintained in nature by chronic infection transmitted between rodents. As in arthropod-borne virus cycles, there is usually a high degree of rodent-virus specificity, and there is no overt disease in the reservoir/vector.

Epidemiology The distribution of arthropod- and rodent-borne viruses is restricted by the areas inhabited by their reservoir/vectors and provides an important clue in the differential diagnosis. [Table 198-2](#) shows the approximate geographic distribution of the

most important of these viruses. Members of each family, each genus, and even each serologically related group usually occur in each area but may not be pathogenic in all areas or may not be a commonly recognized cause of disease in all areas and so may not be included in the table. Although there is generally no overt disease in the vertebrate reservoirs, disease in nonhuman target species may be a useful diagnostic clue, and serologic testing of selected animals may be a useful way to monitor virus circulation.

Most of these diseases are acquired in a rural setting; a few have urban vectors. Seoul, sandfly fever, and Oropouche viruses are examples of urban viruses, but the most notable are yellow fever, dengue, and chikungunya viruses, which are transmitted between humans with the mosquito *Aedes aegypti* as a principal or alternate vector. A history of mosquito bite has little diagnostic significance in the individual; a history of tick bite is more diagnostically specific. Rodent exposure is often reported by persons infected with an arenavirus or a hantavirus but again has little specificity. Indeed, aerosols may infect persons who have no recollection of having even seen rodents.

Syndromes Human disease caused by arthropod- and rodent-borne viruses is often subclinical. The spectrum of possible responses to infection is wide, and our knowledge of the outcome of most of these infections is limited. The usual disease syndromes associated with these viruses have been grouped into four categories: fever and myalgia, arthritis and rash, encephalitis, and hemorrhagic fever. Although for the purposes of this discussion most viruses have been placed in a single group, the categories often overlap. For example, West Nile and Venezuelan equine encephalitis viruses are discussed as encephalitis viruses, but during epidemics they may cause many cases of milder febrile syndromes and relatively uncommon cases of encephalitis. Similarly, Rift Valley fever virus is best known as a cause of [HF](#), but the attack rates for febrile disease are far higher, and encephalitis is occasionally seen as well. [LCM](#) virus is classified as a cause of fever and myalgia because this syndrome is its most common disease manifestation and, even when central nervous system (CNS) disease occurs, it is usually mild and is preceded by fever and myalgia. Dengue virus infection is considered as a cause of fever and myalgia (dengue fever) because this is by far the most common manifestation worldwide and is the syndrome most likely to be seen in the United States; however, dengue HF is also discussed in the HF section because of its complicated pathogenesis and importance in pediatric practice in certain areas of the world.

Diagnosis Laboratory diagnosis is required in any given case, although epidemics occasionally provide clinical and epidemiologic clues on which an educated guess as to etiology can be based. For most arthropod- and rodent-borne viruses, acute-phase serum samples (collected within 3 or 4 days of onset) have yielded isolates, and paired sera have been used to demonstrate rising antibody titers by a variety of tests. Intensive efforts to develop rapid tests for [HF](#) have resulted in an antigen-detection enzyme-linked immunosorbent assay (ELISA) and an IgM-capture ELISA that can provide a diagnosis based on a single serum sample within a few hours and are particularly useful in severe cases. More sensitive reverse transcription polymerase chain reaction (RT-PCR) tests may yield diagnoses based on samples without detectable antigen and may also provide useful genetic information about the virus. Preliminary data suggest that similar tests applied to some fever-myalgia syndromes would give positive results if developed

further. Hantavirus infections differ from others discussed here in that severe acute disease is immunopathologic; patients present with serum IgM that serves as the basis for a sensitive and specific test. Every ELISA must include a control incorporating a negative antigen with each serum sample tested; the frequent failure to include such a control has resulted in numerous false-positive results in diagnostic tests.

At the time of diagnosis, patients with encephalitis are generally no longer viremic or antigenemic and usually do not have virus in cerebrospinal fluid (CSF). In this situation, the value of serologic methods and [RT-PCR](#) is being validated. IgM capture is increasingly being used for the simultaneous testing of serum and CSF. Ig [ELISA](#) or classic serology is useful in the evaluation of past exposure to the viruses, many of which circulate in areas with a minimal medical infrastructure and sometimes cause mild or subclinical infection.

The remainder of this chapter offers general descriptions of the broad syndromes caused by arthropod- and rodent-borne viruses and then addresses specific differences between diseases. It is important to remember that most of the diseases under consideration have not been studied in detail with modern medical approaches and thus available data may be incomplete or biased.

FEVER AND MYALGIA

Fever and myalgia constitute the syndrome most commonly associated with zoonotic virus infection. Many of the numerous viruses belonging to the families listed in [Table 198-1](#) probably cause this syndrome, but several viruses have been selected for inclusion in the table because of their prominent associations with the syndrome and their biomedical importance.

The syndrome typically begins with the abrupt onset of fever, chills, intense myalgia, and malaise. Patients may also report joint pains, but no true arthritis is detectable. Anorexia is characteristic and may be accompanied by nausea or even vomiting. Headache is common and may be severe, with photophobia and retroorbital pain. Physical findings are minimal and are usually confined to conjunctival injection with pain on palpation of muscles or the epigastrium. The duration of symptoms is quite variable but generally is 2 to 5 days, with a biphasic course in some instances. The spectrum of disease varies from subclinical to temporarily incapacitating.

Less constant findings include a maculopapular rash. Epistaxis may occur but does not necessarily indicate a bleeding diathesis. A minority of the cases caused by some viruses are known or suspected to include aseptic meningitis, but this diagnosis is difficult in remote areas, given the patients' photophobia and myalgia as well as the lack of opportunity to examine the [CSF](#). Although pharyngitis may be noted or radiographic evidence of pulmonary infiltrates found in some cases, these viruses are not primary respiratory pathogens. The differential diagnosis includes anicteric leptospirosis, rickettsial diseases, and the early stages of other syndromes discussed in this chapter. These diseases are often described as "flu-like," but the usual absence of cough and coryza makes influenza an unlikely confounder except at the earliest stages.

Complete recovery is generally the outcome in this syndrome, although prolonged

asthenia and nonspecific symptoms have been described in some cases, particularly after infection with [LCM](#) or dengue virus. Treatment is supportive, with aspirin avoided because of the potential for exacerbated bleeding and Reye's syndrome. Efforts at prevention are best based on vector control, which, however, may be expensive or impossible. For mosquito control, destruction of breeding sites is generally the most economically and environmentally sound approach; spraying to kill adult mosquitoes and thus to reduce their numbers transiently may have a preventive role in selected settings but has not been notably effective in the past. Measures taken by the individual to avoid the vector can be valuable. Avoiding the vector's habitat and times of peak activity, preventing the vector from entering dwellings by using screens or other barriers, judiciously applying arthropod repellents such as diethyltoluamide (DEET) to the skin, and wearing permethrin-impregnated clothing are all possible approaches, depending on the vector and its habits.

LYMPHOCYTIC CHORIOMENINGITIS

[LCM](#) is transmitted from the common house mouse (*Mus musculus*) to humans by aerosols of excreta and secretions. LCM virus, an arenavirus, is maintained in the mouse mainly by vertical transmission from infected dams. The vertically infected mouse remains viremic for life, with high concentrations of virus in all tissues. Infected colonies of pet hamsters have also served as a link to humans. LCM virus is widely used in immunology laboratories as a model of T cell function and can silently infect cell cultures and passaged tumor lines, resulting in infections among scientists and animal caretakers. Patients with LCM may have a history of residence in rodent-infested housing or other exposure to rodents. An antibody prevalence of ~5 to 10% has been reported in adults from the United States, Argentina, and endemic areas of Germany.

[LCM](#) differs from the general syndrome of fever and myalgia in that its onset is gradual. Among the conditions occasionally associated with LCM are orchitis, transient alopecia, arthritis, pharyngitis, cough, and maculopapular rash. An estimated one-fourth of patients or fewer suffer a febrile phase of 3 to 6 days and then, after a brief remission, develop renewed fever accompanied by severe headache, nausea and vomiting, and meningeal signs lasting for about a week. These patients virtually always recover fully, as do the uncommon patients with clear-cut signs of encephalitis. Recovery may be delayed by transient hydrocephalus.

During the initial febrile phase, leukopenia and thrombocytopenia are common and virus can usually be isolated from blood. During the [CNS](#) phase of the illness, virus may be found in the [CSF](#), but antibodies are present in blood. The pathogenesis of [LCM](#) is thought to resemble that following direct intracranial inoculation of the virus into adult mice; the onset of the immune response leads to T cell-mediated immunopathologic meningitis. During the meningeal phase, CSF mononuclear-cell counts range from the hundreds to the low thousands per microliter, and hypoglycorrhachia is found in one-third of cases. The IgM-capture [ELISA](#) of serum and CSF is usually positive; [RT-PCR](#) assays have been developed for application to CSF.

Infection with [LCM](#) virus should be suspected in acutely ill febrile patients with marked leukopenia and thrombocytopenia. In cases of aseptic meningitis, any of the following should suggest LCM: well-marked febrile prodrome, adult age, autumn seasonality, low

[CSF](#) glucose levels, or CSF mononuclear cell counts >1000/uL.

In pregnant women, [LCM](#) virus infection may lead to fetal invasion with consequent congenital hydrocephalus and chorioretinitis. Since the maternal infection may be mild, consisting of only a short febrile illness, antibodies to the virus should be sought in both the mother and the fetus in suspicious circumstances, particularly TORCH-negative neonatal hydrocephalus. [TORCH is a battery of tests encompassing toxoplasmosis, other conditions (congenital syphilis and viruses), rubella, cytomegalovirus, and herpes simplex virus.]

BUNYAMWERA VIRUS INFECTION

The mosquito-transmitted Bunyamwera serogroup viruses are found on every continent except Australia and Antarctica. Bunyamwera virus and its close relative Ilesha virus commonly cause febrile disease in Africa. Other related viruses are implicated in such disease in Southeast Asia (Batai virus), Europe (Calovo virus), and South America (Wyeomyia virus). In North America, Cache Valley virus has been implicated in febrile human disease and in rare instances of more serious systemic illness; the presence of serum antibodies to this virus may be associated with congenital malformations. In Central America, the closely related Fort Sherman virus causes the fever-myalgia syndrome.

GROUP C VIRUS INFECTION

The group C viruses include at least 11 agents transmitted by mosquitoes in neotropical forests. These agents are among the most common causes of arboviral infection in humans entering American jungles and cause acute febrile disease.

TAHYNA VIRUS INFECTION

This California serogroup virus (see discussion of California encephalitis, below) occurs in central and western Europe, and related viruses are emerging in Russia. The significance of Tahyna virus in human health has been well studied only in the Czech and Slovak Republics; there, the virus was found to be a prominent cause of febrile disease, in some cases causing pharyngitis, pulmonary syndromes, and aseptic meningitis. The potential for arboviruses to be unexpectedly involved in such cases in areas of high mosquito prevalence needs to be kept in mind.

ORPOUCHE FEVER

Oropouche virus is transmitted in Central and South America by a biting midge, *Culicoides paraensis*, which often breeds to high density in cacao husks and other vegetable detritus found in towns and cities. Explosive epidemics involving thousands of cases have been reported from several towns in Brazil and Peru. Rash and aseptic meningitis have been detected in a number of cases.

SANDFLY FEVER

The sandfly *Phlebotomus papatasi* transmits sandfly fever. Female sandflies may be

infected by the oral route as they take a blood meal and may transmit the virus to offspring when they lay their eggs after a second blood meal. This prominent transovarial pattern was the first to be recognized among dipterans and complicates virus control. A previous designation for sandfly fever, "3-day fever," instructively describes the brief, debilitating course associated with this essentially benign infection. There is neither a rash nor [CNS](#) involvement, and complete recovery is the rule.

Sandfly fever is found in the circum-Mediterranean area, extending to the east through the Balkans into China as well as into the Middle East and southwestern Asia. The vector is found in both rural and urban settings and is known for its small size, which enables it to penetrate standard mosquito screens and netting, and for its short flight range. Epidemics have been described in the wake of natural disasters and wars. In parts of Europe, sandfly populations and virus transmission were greatly reduced by the extensive residual spraying conducted after World War II to control malaria, and the incidence continues to be low. A common pattern of disease in endemic areas consists of high attack rates among travelers and military personnel with little or no disease in the local population, who are protected after childhood infection. In addition to the two well-characterized, non-cross-protective Sicilian and Naples virus species, more than 30 related phleboviruses are transmitted by sandflies and mosquitoes, but most are of unknown significance in terms of human health.

TOSCANA VIRUS DISEASE

Toscana virus is a *Phlebovirus* (family Bunyaviridae) transmitted primarily by the circum-Mediterranean sandfly *P. perniciosus*. The vertebrate amplifying host, if one exists, is unknown. Toscana virus infection is common during the summer among rural residents and vacationers; a number of cases have been identified in travelers returning to Germany and Scandinavia. The disease may manifest as an uncomplicated febrile illness but is often associated with aseptic meningitis, with virus isolated from the [CSF](#).

PUNTA TORO VIRUS DISEASE

Of the several phleboviruses that are associated with New World sandflies and infect humans, Punta Toro virus is the best known. The disease caused by this virus is clinically similar to but epidemiologically different from that caused by the Naples or Sicilian sandfly fever viruses. Punta Toro virus infections are sporadic and are acquired in the tropical forest, where the vectors rest on tree buttresses. Epidemics have not been reported, but antibody prevalences among inhabitants of villages in the endemic areas indicate a cumulative lifetime exposure rate of >50%.

DENGUE FEVER

All four distinct dengue viruses (dengue 1-4) have *A. aegypti* as their principal vector, and all cause a similar clinical syndrome. In rare cases, second infection with a serotype of dengue virus different from that involved in the primary infection leads to dengue [HF](#) with severe shock (see below). Sporadic cases are seen in the settings of endemic transmission and epidemic disease. Year-round transmission between latitudes 25°N and 25°S has been established, and seasonal forays of the viruses to points as far north as Philadelphia are thought to have taken place in the United States. With increasing

spread of the vector mosquito throughout the tropics and subtropics, large areas of the world have become vulnerable to the introduction of dengue viruses, particularly through air travel by infected humans, and both dengue fever and the related dengue HF are becoming increasingly common. Conditions favorable to dengue transmission exist in the southern United States, and bursts of dengue fever activity are to be expected in this region, particularly along the Mexican border, where water may be stored in containers and *A. aegypti* numbers may therefore be greatest: this mosquito, which is also an efficient vector of the yellow fever and chikungunya viruses, typically breeds near human habitation, using relatively fresh water from sources such as water jars, vases, discarded containers, coconut husks, and old tires. *A. aegypti* usually inhabits dwellings and bites during the day.

After an incubation period of 2 to 7 days, the typical patient experiences the sudden onset of fever, headache, retroorbital pain, and back pain along with the severe myalgia that gave rise to the colloquial designation "break-bone fever." There is often a macular rash on the first day as well as adenopathy, palatal vesicles, and scleral injection. The illness may last a week, with additional symptoms usually including anorexia, nausea or vomiting, marked cutaneous hypersensitivity, and -- near the time of defervescence -- a maculopapular rash beginning on the trunk and spreading to the extremities and the face. Epistaxis and scattered petechiae are often noted in uncomplicated dengue, and preexisting gastrointestinal lesions may bleed during the acute illness.

Laboratory findings include leukopenia, thrombocytopenia, and, in many cases, serum aminotransferase elevations. The diagnosis is made by IgM [ELISA](#) or paired serology during recovery or by antigen-detection ELISA or [RT-PCR](#) during the acute phase. Virus is readily isolated from blood in the acute phase if mosquito inoculation or mosquito cell culture is used.

COLORADO TICK FEVER

Several hundred cases of Colorado tick fever are reported annually in the United States. The infection is acquired between March and November through the bite of an infected *Dermacentor andersoni* tick in mountainous western regions at altitudes of 1200 to 3000 m (4000 to 10,000 ft). Small mammals serve as the amplifying host. The most common presentation consists of fever and myalgia; meningoencephalitis is not uncommon, and hemorrhagic disease, pericarditis, myocarditis, orchitis, and pulmonary presentations are also reported. Rash develops in a substantial minority of cases. The disease usually lasts 7 to 10 days and is often biphasic. The most important differential diagnostic considerations since the beginning of the twentieth century have been Rocky Mountain spotted fever and tularemia.

Infection of erythroblasts and other marrow cells by Colorado tick fever virus results in the appearance and persistence (for several weeks) of erythrocytes containing the virus. This feature, detected in smears stained by immunofluorescence, can be diagnostically helpful. The clinical laboratory detects leukopenia and thrombocytopenia.

ORBIVIRUS INFECTION

The orbiviruses encompass many human and veterinary pathogens. For example,

Orungo virus is widely transmitted by mosquitoes in tropical Africa and causes febrile disease in humans. The Kemerova complex includes the Kemerova, Lipovnik, and Tribec viruses of Russia and central Europe; these viruses are transmitted by ticks and are associated with febrile and neurologic disease.

VESICULAR STOMATITIS

See [Chap. 197](#).

ENCEPHALITIS

Arboviral encephalitis is a seasonal disease, commonly occurring in the warmer months. Its incidence varies markedly with time and place, depending on ecologic factors. The causative viruses differ substantially in terms of case-infection ratio (i.e., the ratio of clinical to subclinical infection), mortality, and residua ([Table 198-3](#)). Humans are not an important amplifier of these viruses.

All the viral encephalitides discussed in this section have a similar pathogenesis as far as is known. An infected arthropod ingests a blood meal from a human and infects the host. The initial period of viremia is thought to originate most commonly from the lymphoid system. Viremia leads to [CNS](#) invasion, presumably through infection of olfactory neuroepithelium with passage through the cribiform plate or through infection of brain capillaries and multifocal entry into the CNS. During the viremic phase, there may be little or no recognized disease except in the case of tick-borne flaviviral encephalitis, in which there may be a clearly delineated phase of fever and systemic illness. The disease process in the CNS arises partly from direct neuronal infection and subsequent damage and partly from edema, inflammation, and other indirect effects. The usual pathologic picture is one of focal necrosis of neurons, inflammatory glial nodules, and perivascular lymphoid cuffing; the severity and distribution of these abnormalities vary with the infecting virus. Involved areas display the "luxury perfusion" phenomenon, with normal or increased total blood flow and low oxygen extraction.

The typical patient presents with a prodrome of nonspecific constitutional symptoms, including fever, abdominal pain, vertigo, sore throat, and respiratory symptoms. Headache, meningeal signs, photophobia, and vomiting follow quickly. Involvement of deeper structures may be signaled by lethargy, somnolence, and intellectual deficit (as disclosed by the mental status examination or failure at serial 7 subtraction); more severely affected patients will be obviously disoriented and may be comatose. Tremors, loss of abdominal reflexes, cranial nerve palsies, hemiparesis, monoparesis, difficulty in swallowing, and frontal lobe signs are all common. Convulsions and focal signs may be evident early or may appear during the course of the disease. Some patients present with an abrupt onset of fever, convulsions, and other signs of [CNS](#) involvement. The results of human infection range from no significant symptoms through febrile headache to aseptic meningitis and finally to full-blown encephalitis; the proportions and severity of these manifestations vary with the infecting virus.

The acute encephalitis usually lasts from a few days to as long as 2 to 3 weeks, but recovery may be slow, with weeks or months required for the return of maximal recoupable function. Common complaints during recovery include difficulty

concentrating, fatigability, tremors, and personality changes. The acute illness requires management of a comatose patient who may have intracranial pressure elevations, inappropriate secretion of antidiuretic hormone, respiratory failure, and convulsions. There is no specific therapy for these viral encephalitides. The only practical preventive measures are vector management and personal protection against the arthropod transmitting the virus; for Japanese encephalitis or tick-borne encephalitis, vaccination should be considered in certain circumstances (see relevant sections below).

The diagnosis of arboviral encephalitis depends on the careful evaluation of a febrile patient with [CNS](#) disease, with rapid identification of treatable herpes simplex encephalitis, ruling out of brain abscess, exclusion of bacterial meningitis by serial [CSF](#) examination, and performance of laboratory studies to define the viral etiology. Leptospirosis, neurosyphilis, Lyme disease, cat-scratch fever, and newer viral encephalitides such as Nipah virus infection from Malaysia should be considered. The CSF examination usually shows a modest cell count -- in the tens or hundreds or perhaps a few thousand. Early in the process, a significant proportion of these cells may be polymorphonuclear leukocytes, but usually there is a mononuclear cell predominance. CSF glucose levels are usually normal. There are exceptions to this pattern of findings. In eastern equine encephalitis, for example, polymorphonuclear leukocytes may predominate during the first 72 h of disease and hypoglycorrhachia may be detected. In [LCM](#), lymphocyte counts may be in the thousands, and the glucose concentration may be diminished. Experience with imaging studies is still evolving; clearly, however, both computed tomography (CT) and magnetic resonance imaging (MRI) may be normal except for evidence of preexisting conditions or sometimes may suggest diffuse edema. Several patients with eastern equine encephalitis have had focal abnormalities, and individuals with severe Japanese encephalitis have presented with bilateral thalamic lesions that have often been hemorrhagic. Electroencephalography usually shows diffuse abnormalities and is not directly helpful.

A humoral immune response is usually detectable at or near the onset of disease. Both serum and [CSF](#) should be examined for IgM antibodies. Virus generally cannot be isolated from blood or CSF, although Japanese encephalitis virus has been recovered from CSF in severe cases. Virus can be obtained from and viral antigen is present in brain tissue, although its distribution may be focal.

CALIFORNIA, LA CROSSE, AND JAMESTOWN CANYON VIRUS ENCEPHALITIS

The isolation of California encephalitis virus established the California serogroup of viruses as a cause of encephalitis, and its use as a diagnostic antigen led to the description of many cases of "California encephalitis." In fact, however, this virus has been implicated in only a few cases of encephalitis, and the serologically related La Crosse virus is the major cause of encephalitis among viruses in the California serogroup. "California encephalitis" due to La Crosse virus infection is most commonly reported from the upper Midwest but is also found in other areas of the central and eastern United States, most often in West Virginia, Tennessee, North Carolina, and Georgia. The serogroup includes 13 other viruses, some of which may also be involved in human disease that is misattributed because of the complexity of the group's serology; these viruses include the Jamestown Canyon, snowshoe hare, Inkoo, and Trivittatus viruses, all of which have *Aedes* mosquitoes as their vector and all of which

have a strong element of transovarial transmission in their natural cycles.

The mosquito vector of La Crosse virus is *A. triseriatus*. In addition to a prominent transovarial component of transmission, a mosquito can also become infected through feeding on viremic chipmunks and other mammals as well as through venereal transmission from another mosquito. The mosquito breeds in sites such as tree holes and abandoned tires and bites during daylight hours; these findings correlate with the risk factors for cases: recreation in forested areas, residence at the forest's edge, and the presence of abandoned tires around the home. Intensive environmental modification based on these findings has reduced the incidence of disease in a highly endemic area in the Midwest. Most cases occur from July through September. The Asian tiger mosquito, *A. albopictus*, efficiently transmits the virus to mice and also transmits the agent transovarially in the laboratory; this aggressive anthropophilic mosquito has the capacity to urbanize, and its possible impact on transmission to humans is of concern.

An antibody prevalence of $\approx 20\%$ in endemic areas indicates that infection is common, but CNS disease has been recognized primarily in children < 15 years of age. The illness varies from a picture of aseptic meningitis accompanied by confusion to severe and occasionally fatal encephalitis. Although there may be prodromal symptoms, the onset of CNS disease is sudden, with fever, headache, and lethargy often joined by nausea and vomiting, convulsions (in one-half of patients), and coma (in one-third of patients). Focal seizures, hemiparesis, tremor, aphasia, chorea, Babinski's sign, and other evidence of significant neurologic dysfunction are common, but residua are not. Perhaps 10% of patients have recurrent seizures in the succeeding months. Other serious sequelae are rare, although a decrease in scholastic standing has been reported and mild personality change has occasionally been suggested. Treatment is supportive over a 1- to 2-week acute phase during which status epilepticus, cerebral edema, and inappropriate secretion of antidiuretic hormone are important concerns. Ribavirin has been used in severe cases, and a clinical trial of this drug is under way.

The blood leukocyte count is commonly elevated, sometimes reaching levels of 20,000/uL, and there is usually a left shift. CSF cell counts are typically 30 to 500/uL with a mononuclear cell predominance (although 25 to 90% of cells are polymorphonuclear in some cases). The protein level is normal or slightly increased, and the glucose level is normal. Specific virologic diagnosis based on IgM-capture assays of serum and CSF is efficient. The only human anatomic site from which virus has been isolated is the brain.

Jamestown Canyon virus has been implicated in several cases of encephalitis in adults; in these cases the disease was usually associated with a significant respiratory illness at onset. Human infection with this virus has been documented in New York, Wisconsin, Ohio, Michigan, Ontario, and other areas of North America where the vector mosquito, *A. stimulans*, feeds on its main host, the white-tailed deer.

ST. LOUIS ENCEPHALITIS

St. Louis encephalitis virus is transmitted between *Culex* mosquitoes and birds. This virus causes low-level endemic infection among rural residents of the western and central United States, where *C. tarsalis* is the vector (see "Western Equine

Encephalitis," below), but the more urbanized mosquito species *C. pipiens* and *C. quinquefasciatus* have been responsible for epidemics resulting in hundreds or even thousands of cases in cities of the central and eastern United States. Most cases occur in June through October. The urban mosquitoes breed in accumulations of stagnant water and sewage with high organic content and readily bite humans in and around houses at dusk. The elimination of open sewers and trash-filled drainage systems is expensive and may not be possible, but screening of houses and implementation of personal protective measures may be an effective approach for individuals. The rural vector is most active at dusk and outdoors; its bites can be avoided by modification of activities and use of repellents.

Disease severity increases with age: infections that result in aseptic meningitis or mild encephalitis are concentrated in children and young adults, while severe and fatal cases primarily affect the elderly. Infection rates are similar in all age groups; thus the greater susceptibility of older persons to disease is a biologic consequence of aging. The disease has an abrupt onset, sometimes following a prodrome, and begins with fever, lethargy, confusion, and headache. In addition, nuchal rigidity, hypotonia, hyperreflexia, myoclonus, and tremor are common. Severe cases can include cranial nerve palsies, hemiparesis, and convulsions. Patients often complain of dysuria and may have viral antigen in urine as well as pyuria. The overall mortality is generally ~7% but may reach 20% among patients over the age of 60. Recovery is slow. Emotional lability, difficulties in concentration and memory, asthenia, and tremor are commonly prolonged in older patients.

The [CSF](#) of patients with St. Louis encephalitis usually contains tens to hundreds of cells, with a lymphocytic predominance and a normal glucose level. Leukocytosis with a left shift is often documented.

JAPANESE ENCEPHALITIS

Japanese encephalitis virus is found throughout Asia, including far eastern Russia, Japan, China, India, Pakistan, and Southeast Asia, and causes occasional epidemics on western Pacific islands. The virus has been detected in the Torres Strait islands, and a human encephalitis case has been identified on the nearby Australian mainland. This flavivirus is particularly common in areas where irrigated rice fields attract the natural avian vertebrate hosts and provide abundant breeding sites for mosquitoes such as *C. tritaeniorhyncus*, which transmit the virus to humans. Additional amplification by pigs, which suffer abortion, and horses, which develop encephalitis, may be significant as well. Vaccination of these additional amplifying hosts may reduce the transmission of the virus. An effective, formalin-inactivated vaccine purified from mouse brain is produced in Japan and licensed for human use in the United States. It is given on days 0, 7, and 30 or -- with some sacrifice in serum neutralizing titer -- on days 0, 7, and 14. Vaccination is indicated for summer travelers to rural Asia, where the risk of clinical disease may be 0.05 to 2.1/10,000 per week. The severe and often fatal disease reported in expatriates must be balanced against the 0.1 to 1% chance of a late systemic or cutaneous allergic reaction. These reactions are rarely fatal but may be severe and have been known to begin 1 to 9 days after vaccination, with associated pruritus, urticaria, and angioedema. Live attenuated vaccines are being used in China but are not recommended in the United States at this time.

WEST NILE VIRUS INFECTION

West Nile virus is transmitted among wild birds by *Culex* mosquitoes in Africa, the Middle East, southern Europe, and Asia. It is a frequent cause of febrile disease without [CNS](#) involvement, but it occasionally causes aseptic meningitis and severe encephalitis; these serious infections are particularly common among children and the elderly. The febrile-myalgic syndrome caused by West Nile virus differs from many others by the frequent appearance of a maculopapular rash concentrated on the trunk and lymphadenopathy. Headache, ocular pain, sore throat, nausea and vomiting, and arthralgia (but not arthritis) are common accompaniments. In addition, the virus has been implicated in severe and fatal hepatic necrosis in Africa.

In 1996 West Nile virus caused more than 300 cases of [CNS](#) disease, with 10% mortality, in the Danube flood plain, including Bucharest. In 1999 the virus appeared in New York City and other areas of the northeastern United States, causing more than 60 cases of aseptic meningitis or encephalitis among humans as well as die-offs among crows, exotic zoo birds, and other avians. The encephalitis was most severe among the elderly and was often associated with notable muscle weakness and even with flaccid paralysis. The virus, thought to have been transmitted in New York City by the ubiquitous *C. pipiens* mosquito, returned to larger areas of the northeastern United States in the summer of 2000 and threatens to spread farther in the Americas via bird migration.

West Nile virus falls into the same phylogenetic group of flaviviruses as St. Louis and Japanese encephalitis viruses, as do Murray Valley and Rocio viruses. The latter two viruses are both maintained in mosquitoes and birds and produce a clinical picture resembling that of Japanese encephalitis. Murray Valley virus has caused occasional epidemics and sporadic cases in Australia. Rocio virus caused recurrent epidemics in a focal area of Brazil in 1975 to 1977 and then virtually disappeared.

CENTRAL EUROPEAN TICK-BORNE ENCEPHALITIS AND RUSSIAN SPRING-SUMMER ENCEPHALITIS

A spectrum of tick-borne flaviviruses has been identified across the Eurasian land mass. Many are known mainly as agricultural pathogens (e.g., louping ill virus in the United Kingdom). From Scandinavia to the Urals, central European tick-borne encephalitis is transmitted by *Ixodes ricinus*. Human cases occur between April and October, with a peak in June and July. A related and more virulent virus is that of Russian spring-summer encephalitis, which is associated with *I. persulcatus* and is distributed from Europe across the Urals to the Pacific Ocean. The ticks transmit the disease primarily in the spring and early summer, with a lower rate of transmission later in summer. Small mammals are the vertebrate amplifiers for both viruses. The risk varies by geographic area and can be highly localized within a given area; human cases usually follow outdoor activities or consumption of raw milk from infected goats or other infected animals.

After an incubation period of 7 to 14 days or perhaps longer, the central European viruses classically result in a febrile-myalgic phase that lasts for 2 to 4 days and is

thought to correlate with viremia. A subsequent remission for several days is followed by the recurrence of fever and the onset of meningeal signs. The [CNS](#) phase varies from mild aseptic meningitis, which is more common among younger patients, to severe encephalitis with coma, convulsions, tremors, and motor signs lasting for 7 to 10 days before improvement begins. Spinal and medullary involvement can lead to typical limb-girdle paralysis and to respiratory paralysis. Most patients recover, only a minority with significant deficits. Infections with the far eastern viruses generally run a more abrupt course. The encephalitic syndrome caused by these viruses sometimes begins without a remission and has more severe manifestations than the European syndrome. Mortality is high, and major sequelae -- most notably, lower motor neuron paralyzes of the proximal muscles of the extremities, trunk, and neck -- are common.

In the early stage of the illness, virus may be isolated from the blood. In the [CNS](#) phase, IgM antibodies are detectable in serum and/or [CSF](#). Thrombocytopenia sometimes develops during the initial febrile illness, which resembles the early hemorrhagic phase of some other tick-borne flaviviral infections, such as Kyasanur Forest disease. Other tick-borne flaviviruses are less common causes of encephalitis, including louping ill virus in the United Kingdom and Powassan virus.

There is no specific therapy for infection with these viruses. However, effective alum-adjuvanted, formalin-inactivated vaccines are produced in Austria, Germany, and Russia. Two doses of the Austrian vaccine separated by an interval of 1 to 3 months appear to be effective in the field, and antibody responses are similar when vaccine is given on days 0 and 14. Other vaccines have elicited similar neutralizing antibody titers. Since rare cases of postvaccination Guillain-Barre syndrome have been reported, vaccination should be reserved for persons likely to experience rural exposure in an endemic area during the season of transmission. Cross-neutralization for the central European and far eastern strains has been established, but there are no published field studies on cross-protection of formalin-inactivated vaccines. Because 0.2 to 4% of ticks in endemic areas may be infected, tick bites raise the issue of immunoglobulin prophylaxis. Prompt administration of high-titered specific preparations should probably be undertaken, although no controlled data are available to prove the efficacy of this measure. Immunoglobulin should not be administered late because of the risk of antibody-mediated enhancement.

POWASSAN ENCEPHALITIS

Powassan virus is a member of the tick-borne encephalitis virus complex and is transmitted by *I. cookei* among small mammals in eastern Canada and the United States, where it has been responsible for 20 recognized cases of human disease. Other ticks may transmit the virus in a wider geographic area, and there is some concern that *I. scapularis* (also called *I. dammini*), a competent vector in the laboratory, may become involved as it becomes more prominent in the United States. Patients with Powassan encephalitis -- often children -- present in May through December after outdoor exposure and an incubation period thought to be about 1 week. Powassan encephalitis is severe, and sequelae are common.

EASTERN EQUINE ENCEPHALITIS

Eastern equine encephalitis is found primarily within endemic swampy foci along the eastern coast of the United States, with a few inland foci as far removed as Michigan. Human cases present from June through October, when the bird-*Culiseta* mosquito cycle spills over into other mosquito species such as *A. sollicitans* or *A. vexans*, which are more likely to bite mammals. There is concern over the potential role of the introduced anthropophilic mosquito species *A. albopictus*, which has been found to be naturally infected and is an effective vector in the laboratory. Horses are a common target for the virus; if not vaccinated, they serve as a harbinger of human disease but probably do not play a significant role in amplification of the virus.

Eastern equine encephalitis is one of the most destructive of the arboviral conditions, with a brusque onset, rapid progression, high mortality, and frequent residua. This severity is reflected in the extensive necrotic lesions and polymorphonuclear infiltrates found at postmortem examination of the brain and the acute polymorphonuclear [CSF](#) pleocytosis often occurring during the first 1 to 3 days of disease. In addition, leukocytosis with a left shift is a common feature. A formalin-inactivated vaccine has been used to protect laboratory workers but is not generally available or applicable.

WESTERN EQUINE ENCEPHALITIS

The primary maintenance cycle for western equine encephalitis virus in the United States is between *C. tarsalis* and birds, principally sparrows and finches. Equines and humans become infected, and both species suffer encephalitis without amplifying the virus in nature. St. Louis encephalitis is transmitted in a similar cycle in the same region but causes human disease about a month earlier than the period (July through October) in which western equine encephalitis virus is active. Large epidemics of western equine encephalitis took place in the western and central United States and Canada during the 1930s to 1950s, but in recent years the disease has been uncommon. There were 41 reported cases in the United States in 1987 but only 4 reported cases from 1988 to 1995. This decline in incidence may reflect in part the integrated approach to mosquito management that has been employed in irrigation projects and the increasing use of agricultural pesticides; it almost certainly reflects the increased tendency for humans to be indoors behind closed windows at dusk, the peak period of biting by the major vector.

Western equine encephalitis virus causes a typical diffuse viral encephalitis with an increased attack rate and increased morbidity in the young, particularly children <2 years old. In addition, mortality is high among the young and the very elderly. One-third of individuals who have convulsions during the acute illness have subsequent seizure activity. Infants <1 year old -- particularly those in the first months of life -- are at serious risk of motor and intellectual damage. Twice as many males as females develop clinical encephalitis after 5 to 9 years of age; this difference may be related to greater outdoor exposure of boys to the vector but is also likely due in part to biologic differences. A formalin-inactivated vaccine has been used to protect laboratory workers but is not generally available or applicable.

VENEZUELAN EQUINE ENCEPHALITIS

There are six known types of virus in the Venezuelan equine encephalitis complex. An

important distinction is between the "epizootic" viruses (subtypes IAB and IC) and the "enzootic" viruses (subtypes ID to IF and types II to VI). The epizootic viruses have an unknown natural cycle but periodically cause extensive epidemics in equines and humans in the Americas. These epidemics rely on the high-level viremia in horses and mules that results in the infection of several species of mosquitoes, which in turn infect humans and perpetuate virus transmission. Humans also have high-level viremia but probably are not important in virus transmission. Enzootic viruses are found primarily in humid tropical forest habitats and are maintained between *Culex* mosquitoes and rodents; these viruses cause human disease but are not pathogenic for horses and do not cause epizootics.

Epizootics of Venezuelan equine encephalitis occurred repeatedly in Venezuela, Colombia, Ecuador, Peru, and other South American countries at intervals of 10 years from the 1930s until 1969, when a massive epizootic spread throughout Central America and Mexico, reaching southern Texas in 1972. Genetic sequencing of the virus from the 1969 to 1972 outbreak suggested that it originated from residual "un-inactivated" virus in veterinary vaccines. The outbreak was terminated in Texas with the use of a live attenuated vaccine (TC-83) originally developed for human use by the U.S. Army; this virus was then used for further production of inactivated veterinary vaccines. No further epizootic disease was identified until 1995 and subsequently, when additional epizootics took place in Colombia, Venezuela, and Mexico. The viruses involved in these epizootics as well as previously epizootic subtype IC viruses have been shown to be close phylogenetic relatives of known enzootic subtype ID viruses. This finding suggests that active evolution and selection of epizootic viruses are under way in northern South America.

During epizootics, extensive human infection is the rule, with clinical disease in 10 to 60% of infected individuals. Most infections result in notable acute febrile disease, while relatively few result in encephalitis. A low rate of CNS invasion is supported by the absence of encephalitis among the many infections resulting from exposure to aerosols in the laboratory or from vaccine accidents. The most recent large epizootic of Venezuelan equine encephalitis occurred in Colombia and Venezuela in 1995; of the more than 85,000 clinical cases, 4% (with a higher proportion among children than adults) included neurologic symptoms and 300 ended in death.

Enzootic strains of Venezuelan equine encephalitis virus are common causes of acute febrile disease, particularly in areas such as the Florida Everglades and the humid Atlantic coast of Central America. Encephalitis has been documented only in the Florida infections; the three cases were caused by type II enzootic virus, also called *Everglades virus*. All three patients had preexisting cerebral disease. Extrapolation from the rate of genetic change suggests that Everglades virus may have been introduced into Florida <200 years ago and that it is most closely related to the ID subtypes that appear to have given evolutionary rise to the epizootic strains active in South America.

The prevention of epizootic Venezuelan equine encephalitis depends on vaccination of horses with the attenuated TC-83 vaccine or with an inactivated vaccine prepared from that strain. Humans can be protected with similar vaccines, but the use of such products is restricted to laboratory personnel because of reactogenicity and limited availability. In addition, wild-type virus and perhaps TC-83 vaccine may have some degree of fetal

pathogenicity. Enzootic viruses are genetically and antigenically different from epizootic viruses, and protection against the former with vaccines prepared from the latter is relatively ineffective.

ARTHRITIS AND RASH

True arthritis is a common accompaniment of several viral diseases, such as rubella (caused by a non-alphavirus togavirus), parvovirus B19 infection, and hepatitis B; it is an occasional accompaniment of infection due to mumps virus, enteroviruses, herpesviruses, and adenoviruses. It is not generally appreciated that the alphaviruses are also common causes of arthritis. In fact, the alphaviruses discussed below all cause acute febrile diseases accompanied by the development of true arthritis and a maculopapular rash. Rheumatic involvement includes arthralgia alone, periarticular swelling, and (less commonly) joint effusions. Most of these diseases are less severe and have fewer articular manifestations in children than in adults. In temperate climates, these are summer diseases. No specific therapy or licensed vaccines exist.

SINDBIS VIRUS INFECTION

Sindbis virus is transmitted among birds by mosquitoes. Infections with the northern European strains of this virus (which cause, for example, Pogosta disease in Finland, Karelian fever in the independent states of the former Soviet Union, and Okelbo disease in Sweden) and with the genetically related southern African strains are particularly likely to result in the arthritis-rash syndrome. Exposure to a rural environment is commonly associated with this infection, which has an incubation period of <1 week.

The disease begins with rash and arthralgia. Constitutional symptoms are not marked, and fever is modest or lacking altogether. The rash, which lasts about a week, begins on the trunk, spreads to the extremities, and evolves from macules to papules that often vesiculate. The arthritis of this condition is multiarticular, migratory, and incapacitating, with resolution of the acute phase in a few days. Wrists, ankles, phalangeal joints, knees, elbows, and -- to a much lesser extent -- proximal and axial joints are involved. Persistence of joint pains and occasionally of arthritis is a major problem and may go on for months or even years despite a lack of deformity.

CHIKUNGUNYA VIRUS INFECTION

It is likely that chikungunya virus ("that which bends up") is of African origin and is maintained among nonhuman primates on that continent by *Aedes* mosquitoes of the subgenus *Stegomyia* in a fashion similar to yellow fever virus. Like yellow fever virus, chikungunya virus is readily transmitted among humans in urban areas by *A. aegypti*. The *A. aegypti*-chikungunya virus transmission cycle has also been introduced into Asia, where it poses a prominent health problem. The disease is endemic in rural areas of Africa, and intermittent epidemics take place in towns and cities of Africa and Asia. Chikungunya is one more reason (in addition to dengue and yellow fever) that *A. aegypti* must be controlled.

Full-blown disease is most common among adults, in whom the clinical picture may be dramatic. The brusque onset follows an incubation period of 2 to 3 days. Fever and

severe arthralgia are accompanied by chills and constitutional symptoms such as headache, photophobia, conjunctival injection, anorexia, nausea, and abdominal pain. Migratory polyarthritis mainly affects the small joints of the hands, wrists, ankles, and feet, with lesser involvement of the larger joints. Rash may appear at the outset or several days into the illness; its development often coincides with defervescence, which takes place around day 2 or day 3 of disease. The rash is most intense on the trunk and limbs and may desquamate. Petechiae are occasionally seen, and epistaxis is not uncommon, but this virus is not a regular cause of the [HF](#) syndrome, even in children. A few patients develop leukopenia. Elevated levels of aspartate aminotransferase (AST) and C-reactive protein have been described, as have mildly decreased platelet counts. Recovery may require weeks. Some older patients continue to suffer from stiffness, joint pain, and recurrent effusions for several years; this persistence may be especially common in HLA-B27 patients. An investigational live attenuated vaccine has been developed but requires further testing.

A related virus, O'nyong-nyong, caused a major epidemic of arthritis and rash involving at least 2 million people as it moved across eastern and central Africa in the 1960s. After its mysterious emergence, the virus virtually disappeared, leaving only occasional evidence of its persistence in Kenya until a transient resurgence of epidemic activity in 1997.

MAYARO FEVER

Mayaro virus is maintained in the forests of the Americas by *Haemagogus* mosquitoes and nonhuman primates. It causes a frequently endemic and sometimes epidemic infection of humans and appears to produce a syndrome resembling chikungunya.

EPIDEMIC POLYARTHRITIS (ROSS RIVER VIRUS INFECTION)

Ross River virus has caused epidemics of distinctive clinical disease in Australia since the beginning of the twentieth century and continues to be responsible for thousands of cases in rural and suburban areas annually. The virus is transmitted by *A. vigilax* and other mosquitoes, and its persistence is thought to involve transovarial transmission. No definitive vertebrate host has been identified, but several mammalian species, including wallabies, have been suggested. Endemic transmission has also been documented in New Guinea, and in 1979 the virus swept through the eastern Pacific Islands, causing hundreds of thousands of illnesses. The virus was carried from island to island by infected humans and was believed to have been transmitted among humans by *A. polynesiensis* and *A. aegypti*.

The incubation period is 7 to 11 days long, and the onset of illness is sudden, with joint pain usually ushering in the disease. The rash generally develops coincidentally or follows shortly but in some cases precedes joint pains by several days. Constitutional symptoms such as low-grade fever, asthenia, myalgia, headache, and nausea are not prominent and indeed are absent in many cases. Most patients are incapacitated for considerable periods by joint involvement, which interferes with sleeping, walking, and grasping. Wrist, ankle, metacarpophalangeal, interphalangeal, and knee joints are the most commonly involved, although toes, shoulders, and elbows may be affected with some frequency. Periarticular swelling and tenosynovitis are common, and one-third of

patients have true arthritis. Only half of all arthritis patients can resume normal activities within 4 weeks, and 10% still must limit their activity at 3 months. Occasional patients are symptomatic for 1 to 3 years but without progressive arthropathy. Aspirin and nonsteroidal anti-inflammatory drugs are effective for the treatment of symptoms.

Clinical laboratory values are normal or variable in Ross River virus infection. Tests for rheumatoid factor and antinuclear antibodies are negative, and the erythrocyte sedimentation rate is acutely elevated. Joint fluid contains 1000 to 60,000 mononuclear cells per microliter, and Ross River virus antigen is demonstrable in macrophages. IgM antibodies are valuable in the diagnosis of this infection, although they occasionally persist for years. The isolation of the virus from blood by mosquito inoculation or mosquito cell culture is possible early in the illness. Because of the great economic impact of annual epidemics in Australia, an inactivated vaccine is being developed and has been found to be protective in mice.

Perhaps because of the local interest in arboviruses in general and in Ross River virus in particular, other arthritogenic arboviruses have been identified in Australia, including Gan Gan virus, a member of the family Bunyaviridae; Kokobera virus, a flavivirus; and Barmah Forest virus, an alphavirus. The last virus is a common cause of infection and must be differentiated from Ross River virus by specific testing.

HEMORRHAGIC FEVERS

The viral **HF** syndrome is a constellation of findings based on vascular instability and decreased vascular integrity. An assault, direct or indirect, on the microvasculature leads to increased permeability and (particularly when platelet function is decreased) to actual disruption and local hemorrhage. Blood pressure is decreased, and in severe cases shock supervenes. Cutaneous flushing and conjunctival suffusion are examples of common, observable abnormalities in the control of local circulation. The hemorrhage is inconstant and is in most cases an indication of widespread vascular damage rather than a life-threatening loss of blood volume. Disseminated intravascular coagulation is occasionally found in any severely ill patient with HF but is thought to occur regularly only in the early phases of HF with renal syndrome, Crimean Congo HF, and perhaps some cases of filovirus HF. In some viral HF syndromes, specific organs may be particularly impaired, such as the kidney in HF with renal syndrome, the lung in hantavirus pulmonary syndrome, or the liver in yellow fever, but in all these diseases the generalized circulatory disturbance is critically important.

The pathogenesis of **HF** is poorly understood and varies among the viruses regularly implicated in the syndrome, which number more than a dozen. In some cases direct damage to the vascular system or even to parenchymal cells of target organs is important, whereas in others soluble mediators are thought to play the major role. The acute phase in most cases of HF is associated with ongoing virus replication and viremia. Exceptions are the hantavirus diseases and dengue HF/dengue shock syndrome (DHF/DSS), in which the immune response plays a major pathogenic role.

The **HF** syndromes all begin with fever and myalgia, usually of abrupt onset. Within a few days the patient presents for medical attention because of increasing prostration that is often accompanied by severe headache, dizziness, photophobia, hyperesthesia,

abdominal or chest pain, anorexia, nausea or vomiting, and other gastrointestinal disturbances. Initial examination often reveals only an acutely ill patient with conjunctival suffusion, tenderness to palpation of muscles or abdomen, and borderline hypotension or postural hypotension, perhaps with tachycardia. Petechiae (often best visualized in the axillae), flushing of the head and thorax, periorbital edema, and proteinuria are common. Levels of [AST](#) are usually elevated at presentation or within a day or two thereafter. Hemoconcentration from vascular leakage, which is usually evident, is most marked in hantavirus diseases and in [DHF/DSS](#). The seriously ill patient progresses to more severe symptoms and develops shock and other findings typical of the causative virus. Shock, multifocal bleeding, and [CNS](#) involvement (encephalopathy, coma, convulsions) are all poor prognostic signs.

One of the major diagnostic clues is travel to an endemic area within the incubation period for a given syndrome ([Table 198-4](#)). Except for Seoul, dengue, and yellow fever virus infections, which have urban vectors, travel to a rural setting is especially suggestive of a diagnosis of [HF](#).

Early recognition is important because of the need for virus-specific therapy and supportive measures, including prompt, atraumatic hospitalization; judicious fluid therapy that takes into account the patient's increased capillary permeability; administration of cardiogenic drugs; use of pressors to maintain blood pressure at levels that will support renal perfusion; treatment of the relatively common secondary bacterial infections; replacement of clotting factors and platelets as indicated; and the usual precautionary measures used in the treatment of patients with hemorrhagic diatheses. Disseminated intravascular coagulation should be treated only if clear laboratory evidence of its existence is found and if laboratory monitoring of therapy is feasible; there is no proven benefit of such therapy. The available evidence suggests that [HF](#) patients have a decreased cardiac output and will respond poorly to fluid loading as it is often practiced in the treatment of shock associated with bacterial sepsis. Specific therapy is available for several of the HF syndromes. In addition, several diseases considered in the differential diagnosis -- malaria, shigellosis, typhoid, leptospirosis, relapsing fever, and rickettsial disease -- are treatable and potentially lethal. Strict barrier nursing and other precautions against infection of medical staff and visitors are indicated in HF except that due to hantaviruses, yellow fever, Rift Valley fever, and dengue.

LASSA FEVER

Lassa virus is known to cause endemic and epidemic disease in Nigeria, Sierra Leone, Guinea, and Liberia, although it is probably more widely distributed in West Africa. This virus and its relatives exist elsewhere in Africa, but their health significance is unknown. Like other arenaviruses, Lassa virus is spread to humans by small-particle aerosols from chronically infected rodents and may also be acquired during the capture or eating of these animals. It can be transmitted by close person-to-person contact. The virus is often present in urine during convalescence and is suspected to be present in seminal fluid early in recovery. Nosocomial spread has occurred but is uncommon if proper sterile parenteral techniques are used. People of all ages and both sexes are affected; the incidence of disease is highest in the dry season, but transmission takes place year-round. In countries where Lassa virus is endemic, Lassa fever can be a prominent

cause of febrile disease. For example, in one hospital in Sierra Leone, laboratory-confirmed Lassa fever is consistently responsible for one-fifth of admissions to the medical wards. There are probably tens of thousands of Lassa fever cases annually in West Africa alone.

The average case has a gradual onset (among the [HF](#) agents, only the arenaviruses are typically associated with a gradual onset) that gives way to more severe constitutional symptoms and prostration. Bleeding is seen in only ~15 to 30% of cases. A maculopapular rash is often noted in light-skinned Lassa patients. Effusions are common, and male-dominant pericarditis may develop late. The fetal death rate is 92% in the last trimester, when maternal mortality is also increased from the usual 15% to 30%; these figures suggest that interruption of the pregnancy of infected women should be considered. White blood cell counts are normal or slightly elevated, and platelet counts are normal or somewhat low. Deafness coincides with clinical improvement in ~20% of cases and is permanent and bilateral in some. Reinfection may occur but has not been associated with severe disease.

High-level viremia or a high serum concentration of [AST](#) statistically predicts a fatal outcome. Thus patients with an AST level of >150 IU/mL should be treated with intravenous ribavirin. This antiviral nucleoside analogue appears to be effective in reducing mortality from rates among retrospective controls, and its only major side effect is reversible anemia that usually does not require transfusion. The drug should be given by slow intravenous infusion in a dose of 32 mg/kg; this dose should be followed by 16 mg/kg q6h for 4 days and then by 8 mg/kg q8h for 6 days.

SOUTH AMERICAN [HF](#) SYNDROMES (ARGENTINE, BOLIVIAN, VENEZUELAN, AND BRAZILIAN)

These diseases are similar to one another clinically, but their epidemiology differs with the habits of their rodent reservoirs and the interactions of these animals with humans ([Table 198-4](#)). Person-to-person or nosocomial transmission is rare but has occurred.

The basic disease resembles Lassa fever with two marked differences. First, thrombocytopenia -- often marked -- is the rule, and bleeding is quite common. Second, [CNS](#) dysfunction is much more common than in Lassa fever and is often manifest by marked confusion, tremors of the upper extremities and tongue, and cerebellar signs. Some cases follow a predominantly neurologic course, with a poor prognosis. The clinical laboratory is helpful in diagnosis since thrombocytopenia, leukopenia, and proteinuria are typical findings.

Argentine [HF](#) is readily treated with convalescent-phase plasma given within the first 8 days of illness. In the absence of passive antibody therapy, intravenous ribavirin in the dose recommended for Lassa fever is likely to be effective in all the South American HF syndromes. The transmission of the disease from men convalescing from Argentine HF to their wives suggests the need for counseling of arenavirus HF patients concerning the avoidance of intimate contacts for several weeks after recovery. A safe, effective, live attenuated vaccine exists for Argentine HF. In experimental animals, this vaccine is cross-protective against the Bolivian HF virus.

RIFT VALLEY FEVER

This mosquito-borne virus is also a pathogen of domestic animals such as sheep, cattle, and goats. It is maintained in nature by transovarial transmission in floodwater *Aedes* mosquitoes and presumably also has a vertebrate amplifier. Epizootics and epidemics occur when sheep or cattle become infected during particularly heavy rains; developing high-level viremia, these animals infect many different species of mosquitoes. Remote sensing via satellite can detect the ecologic changes associated with high rainfall that predict the likelihood of Rift Valley fever transmission; it can also detect the special depressions from which the floodwater *Aedes* mosquito vectors emerge. In addition, the virus is infectious when transmitted by contact with blood or aerosols from domestic animals or their abortuses. The slaughtered meat is not infectious; anaerobic glycolysis in postmortem tissues results in an acidic environment that rapidly inactivates Bunyaviridae such as Rift Valley fever virus and Crimean-Congo HF virus. The natural range of Rift Valley fever virus is confined to sub-Saharan Africa, where its circulation is markedly enhanced by substantial rainfall such as that which occurred during the El Nino phenomenon of 1997. The virus has also been found in Madagascar and has been introduced into Egypt, where it caused major epidemics in 1977 to 1979, 1993, and subsequently. Neither person-to-person nor nosocomial transmission has been documented.

Rift Valley fever virus is unusual in that it causes at least four different clinical syndromes. Most infections are manifested as the febrile-myalgic syndrome. A small proportion result in HF with especially prominent liver involvement. Perhaps 10% of otherwise mild infections lead to retinal vasculitis; funduscopy reveals edema, hemorrhages, and infarction, and some patients have permanently impaired vision. A small proportion of cases (<1 in 200) are followed by typical viral encephalitis. One of the complicated syndromes does not appear to predispose to another.

There is no proven therapy for any of the syndromes described above. The sensitivity of animal models of Rift Valley fever to antibody or ribavirin therapy suggests that either could be given intravenously to persons with HF. Both retinal disease and encephalitis occur after the acute febrile syndrome has ended and serum neutralizing antibody has developed -- events suggesting that only supportive care need be given. Epidemic disease is best prevented by vaccination of livestock. The established ability of this virus to propagate after an introduction into Egypt suggests that other potentially receptive areas, including the United States, should have a response ready for such an eventuality. It seems likely that this disease, like Venezuelan equine encephalitis, can be controlled only with adequate stocks of an effective live attenuated vaccine, and there are no such global stocks. A formalin-inactivated vaccine confers immunity to humans, but quantities are limited and three injections are required; this vaccine is recommended for exposed laboratory workers and for veterinarians working in sub-Saharan Africa.

CRIMEAN CONGO HF

This severe HF syndrome has a wide geographic distribution, potentially being found wherever ticks of the genus *Hyalomma* occur ([Table 198-4](#)). The propensity of these ticks to feed on domestic livestock and certain wild mammals means that veterinary

serosurveys are the most effective mechanism for the surveillance of virus circulation in a region. Human infection is acquired via a tick bite or during the crushing of infected ticks. Domestic animals do not become ill but do develop viremia; thus there is danger of infection at the time of slaughter and for a brief interval thereafter (through contact with hides or carcasses). Cases have followed sheep shearing. An epidemic in South Africa was associated with slaughter of tick-infested ostriches. Nosocomial epidemics are common and are usually related to extensive blood exposure or needle sticks.

Although generally similar to other [HF](#) syndromes, Crimean Congo HF causes extensive liver damage, resulting in jaundice in some cases. Clinical laboratory values indicate disseminated intravascular coagulation and show elevations in [AST](#), creatine phosphokinase, and bilirubin. Patients with fatal cases generally have more marked changes, even in the early days of illness, and also develop leukocytosis rather than leukopenia. Thrombocytopenia is also more marked and develops earlier in cases with a fatal outcome.

No controlled trials have been performed with intravenous ribavirin, but clinical experience and retrospective comparison of patients with ominous clinical laboratory values suggest that ribavirin is efficacious and should be given. No human or veterinary vaccines are recommended.

[HF](#) WITH RENAL SYNDROME

This disease, the first to be identified as an HF, is widely distributed over Europe and Asia; the major causative viruses and their rodent reservoirs on these two continents are Puumala virus (bank vole, *Clethrionomys glareolus*) and Hantaan virus (striped field mouse, *Apodemus agrarius*), respectively. Other potential causative viruses exist, including Dobrava virus (yellow-necked field mouse, *A. flavicollis*), which causes severe HF with renal syndrome in the Balkans. Seoul virus is associated with the Norway or sewer rat, *Rattus norvegicus*, and has a worldwide distribution through the migration of the rodent; it is associated with mild or moderate HF with renal syndrome in Asia, but in many areas of the world the human disease has been difficult to identify. Most cases occur in rural residents or vacationers; the exception is Seoul virus disease, which may be acquired in an urban or rural setting or from contaminated laboratory rat colonies. Classic Hantaan disease in Korea (Korean HF) and in rural China (epidemic HF) is most common in spring and fall and is related to rodent density and agricultural practices. Human infection is acquired primarily through aerosols of rodent urine, although virus is also present in saliva and feces. Patients with hantavirus diseases are not infectious. HF with renal syndrome is the most important form of HF today, with more than 100,000 cases of severe disease in Asia annually and milder Puumala infections numbering in the thousands as well.

Severe cases of [HF](#) with renal syndrome caused by Hantaan virus evolve in identifiable stages: the febrile stage with myalgia, lasting 3 to 4 days; the hypotensive stage, often associated with shock and lasting from a few hours to 48 h; the oliguric stage with renal failure, lasting 3 to 10 days; and the polyuric stage with diuresis and hyposthenuria.

The *febrile period* is initiated by the abrupt onset of fever, headache, severe myalgia, thirst, anorexia, and often nausea and vomiting. Photophobia, retroorbital pain, and pain

on ocular movement are common, and the vision may become blurred with ciliary body inflammation. Flushing over the face, the V area of the neck, and the back are characteristic, as are pharyngeal injection, periorbital edema, and conjunctival suffusion. Petechiae often develop in areas of pressure, the conjunctivae, and the axillae. Back pain and tenderness to percussion at the costovertebral angle reflect massive retroperitoneal edema. Laboratory evidence of mild to moderate disseminated intravascular coagulation is present. Other laboratory findings include proteinuria and an active urinary sediment.

The *hypotensive phase* is ushered in by falling blood pressure and sometimes by shock. The relative bradycardia typical of the febrile phase is replaced by tachycardia. Kinin activation is marked. The rising hematocrit reflects increasing vascular leakage. Leukocytosis with a left shift develops, and thrombocytopenia continues. Atypical lymphocytes -- which in fact are activated CD8+ and to a lesser extent CD4+ T cells -- circulate. Proteinuria is marked, and the urine's specific gravity falls to 1.010. The renal circulation is congested and compromised from local and systemic circulatory changes resulting in necrosis of tubules, particularly at the corticomedullary junction, and oliguria.

During the *oliguric phase*, hemorrhagic tendencies continue, probably in large part because of uremic bleeding defects. The oliguria persists for 3 to 10 days before renal function returns and marks the onset of the *polyuric stage*, which carries the danger of dehydration and electrolyte abnormalities.

Mild cases of [HF](#) with renal syndrome may be much less stereotypical. The presentation may include only fever, gastrointestinal abnormalities, and transient oliguria followed by hyposthenuria.

[HF](#) with renal syndrome should be suspected in patients with rural exposure in an endemic area. Prompt recognition of the disease will permit rapid hospitalization and expectant management of shock and renal failure. Useful clinical laboratory parameters include leukocytosis, which may be leukemoid and is associated with a left shift; thrombocytopenia; and proteinuria. Mainstays of therapy are the management of shock, reliance on pressors, modest crystalloid infusion, intravenous use of human serum albumin, and treatment of renal failure with prompt dialysis for the usual indications. Hydration may result in pulmonary edema, and hypertension should be avoided because of the possibility of intracranial hemorrhage. Use of intravenous ribavirin has reduced mortality and morbidity in severe cases provided treatment is begun within the first 4 days of illness. The case-fatality ratio may be as high as 15% but with proper therapy should be <5%. Sequelae have not been definitely established, but there is a correlation in the United States between chronic hypertensive renal failure and the presence of antibodies to Seoul virus.

Infections with Puumala virus, the most common cause of [HF](#) with renal syndrome in Europe, result in a much attenuated picture but the same general presentation. The syndrome may be referred to by its former name, *nephropathia epidemica*. Bleeding manifestations are found in only 10% of cases, hypotension rather than shock is usually seen, and oliguria is present in only about half of patients. The dominant features may be fever, abdominal pain, proteinuria, mild oliguria, and sometimes blurred vision or glaucoma followed by polyuria and hyposthenuria in recovery. Mortality is <1%.

The diagnosis is readily made by IgM-capture [ELISA](#), which should be positive at admission or within 24 to 48 h thereafter. The isolation of virus is difficult, but [RT-PCR](#) of a blood clot collected early in the clinical course or of tissues obtained postmortem will give positive results. Such testing is usually undertaken only if definitive identification of the infecting viral species is required or if molecular epidemiologic questions exist.

HANTAVIRUS PULMONARY SYNDROME

Hantavirus pulmonary syndrome was discovered in 1993, but retrospective identification of cases by immunohistochemistry (1978) and serology (1959) support the idea that it is a recently discovered rather than a truly new disease. The causative viruses are hantaviruses of a distinct phylogenetic lineage that is associated with the rodent subfamily Sigmodontinae. Sin Nombre virus chronically infects the deer mouse (*Peromyscus maniculatus*) and is the most important virus causing hantavirus pulmonary syndrome in the United States. The disease is also caused by a Sin Nombre virus variant from the white-footed mouse (*P. leucopus*), by Black Creek Canal virus (*Sigmodon hispidus*, the cotton rat), and by Bayou virus (*Oryzomys palustris*, the rice rat). Several other related viruses cause the disease in South America, but Andes virus is unusual in that it, alone among hantaviruses, has been implicated in human-to-human transmission. The disease is linked to rodent exposure and particularly affects rural residents living in dwellings permeable to rodent entry or working at occupations that pose a risk of rodent exposure. Each rodent species has its own particular habits; in the case of the deer mouse, these behaviors include living in and around human habitation.

The disease begins with a prodrome of about 3 to 4 days (range, 1 to 11 days) comprising fever, myalgia, malaise, and often gastrointestinal disturbances such as nausea, vomiting, and abdominal pain. Dizziness is common and vertigo occasional. Severe prodromal symptoms bring some individuals to medical attention, but patients are usually recognized as the cardiopulmonary phase begins. Typically, there is slightly lowered blood pressure, tachycardia, tachypnea, mild hypoxemia, and early radiographic signs of pulmonary edema. Physical findings in the chest are often surprisingly scant. The conjunctival and cutaneous signs of vascular involvement seen in other types of [HF](#) are absent. During the next few hours, decompensation may progress rapidly to severe hypoxemia and respiratory failure. Most patients surviving the first 48 h of hospitalization are extubated and discharged within a few days, with no apparent residua.

Management during the first few hours after presentation is critical. The goal is to prevent severe hypoxemia by oxygen therapy and, if needed, intubation and intensive respiratory management. During this period, hypotension and shock with increasing hematocrit invite aggressive fluid administration, but this intervention should be undertaken with great caution. Because of low cardiac output with myocardial depression and increased pulmonary vascular permeability, shock should be managed expectantly with pressors and modest infusion of fluid guided by the pulmonary capillary wedge pressure. Mild cases can be managed by frequent monitoring and oxygen administration without intubation. Many patients require intubation to manage hypoxemia and also develop shock. Mortality remains at ~30 to 40% with good management. The antiviral drug ribavirin inhibits the virus in vitro but did not have a

marked effect on patients treated in an open-label study.

During the prodrome, the differential diagnosis of hantavirus pulmonary syndrome is difficult, but by the time of presentation or within 24 h thereafter, a number of diagnostically helpful clinical features become apparent. Cough is not usually present at the outset but may develop later. Interstitial edema is evident on the chest x-ray. Later, bilateral alveolar edema with a central distribution develops in the setting of a normal-sized heart; occasionally, the edema is initially unilateral. Pleural effusions are often visualized. Thrombocytopenia, circulating atypical lymphocytes, and a left shift (often with leukocytosis) are almost always evident; thrombocytopenia has been a particularly important early clue. Hemoconcentration, proteinuria, and hypoalbuminemia should also be sought. Although thrombocytopenia virtually always develops and prolongation of the partial thromboplastin time is the rule, clinical evidence for coagulopathy or laboratory indications of disseminated intravascular coagulation are found in only a minority of cases, usually in severely ill patients. Severely ill patients also have acidosis and elevated serum levels of lactate. Mildly increased values in renal function tests are common, but patients with severe cases often have markedly elevated concentrations of serum creatinine; some of the viruses other than Sin Nombre virus have been associated with more kidney involvement, but few such cases have been studied. The differential diagnosis includes abdominal surgical conditions and pyelonephritis as well as rickettsial disease, sepsis, meningococcemia, plague, tularemia, influenza, and relapsing fever.

A specific diagnosis is best made by IgM testing of acute-phase serum, which has yielded positive results even in the prodrome. Tests using a Sin Nombre virus antigen detect the related hantaviruses causing the pulmonary syndrome in the Americas. Occasionally, heterologous viruses will react only in the IgG [ELISA](#), but this finding is highly suspicious given the very low seroprevalence of these viruses in normal populations. [RT-PCR](#) is usually positive when used to test blood clots obtained in the first 7 to 9 days of illness as well as tissues; this test is useful in identifying the infecting virus in areas outside the home range of the deer mouse and in atypical cases.

YELLOW FEVER

Yellow fever virus caused major epidemics in the Americas, Africa, and Europe before the discovery of mosquito transmission in 1900 led to its control through attacks on its urban vector, *A. aegypti*. Only then was it found that a jungle cycle also existed in Africa, involving other *Aedes* mosquitoes and monkeys, and that colonization of the New World with *A. aegypti*, originally an African species, had established urban yellow fever as well as an independent sylvatic yellow fever cycle in American jungles involving *Haemagogus* mosquitoes and New World monkeys. Today, urban yellow fever transmission occurs only in some African cities, but the threat exists in the great cities of South America, where reinfestation by *A. aegypti* has taken place and dengue transmission by the same mosquito is common. As late as 1905, New Orleans suffered more than 3000 cases with 452 deaths from "yellow jack." Despite the existence of a highly effective and safe vaccine, several hundred jungle yellow fever cases occur annually in South America, and thousands of jungle and urban cases occur each year in Africa.

Yellow fever is a typical [HF](#) accompanied by prominent hepatic necrosis. A period of viremia, typically lasting 3 or 4 days, is followed by a period of "intoxication." During the latter phase in severe cases, the characteristic jaundice, hemorrhages, black vomit, anuria, and terminal delirium occur, perhaps related in part to extensive hepatic involvement. Blood leukocyte counts may be normal or reduced and are often high in terminal stages. Albuminuria is usually noted and may be marked; as renal function fails in terminal or severe cases, the level of blood urea nitrogen rises proportionately. Abnormalities detected in liver function tests range from modest elevations of [AST](#) levels in mild cases to severe derangement.

Urban yellow fever can be prevented by the control of *A. aegypti*. The continuing sylvatic cycle requires vaccination of all visitors to areas of potential transmission. With few exceptions (in the very young and the elderly), reactions to vaccine are minimal; immunity is provided within 10 days and lasts for at least 10 years. An egg allergy dictates caution in vaccine administration. Although there are no documented harmful effects of the vaccine on the fetus, pregnant women should be immunized only if they are definitely at risk of yellow fever exposure. Since vaccination has been associated with several cases of encephalitis in children under 6 months of age, it should be delayed until after 12 months of age unless the risk of exposure is very high. Timely information on changes in yellow fever distribution and yellow fever vaccine requirements can be obtained from Health Information for Travelers, Centers for Disease Control and Prevention, Atlanta, GA 30333; by fax request (404-332-4565; document number 220022#); by phone (404-332-4559); or on the World-Wide Web at <http://www.cdc.gov>.

DENGUE HEMORRHAGIC FEVER/DENGUE SHOCK SYNDROME

A syndrome of [HF](#) noted in the 1950s among children in the Philippines and Southeast Asia was soon associated with dengue virus infections, particularly those occurring against a background of previous exposure to another serotype. The transient heterotypic protection after dengue virus infection is replaced within several weeks by the potential for heterotypic infection resulting in typical dengue fever (see above) or -- uncommonly -- for enhanced disease (secondary [DHF/DSS](#)). In rare instances, primary dengue infections lead to an HF syndrome, but much less is known about pathogenesis in this situation. In the past 20 years, *A. aegypti* has progressively reinvaded Latin America and other areas, and frequent travel by infected individuals has introduced multiple strains of dengue virus from many geographic areas. Thus the pattern of hyperendemic transmission of multiple dengue serotypes has now been established in the Americas and the Caribbean and has led to the emergence of DHF/DSS as a major problem there as well. Millions of dengue infections, including many thousands of cases of DHF/DSS, occur annually. The severe syndrome is unlikely to be seen in U.S. citizens since few children have the dengue antibodies that can trigger the pathogenetic cascade when a second infection is acquired.

Macrophage/monocyte infection is central to the pathogenesis of dengue fever and to the origin of [DHF/DSS](#). Previous infection with a heterologous dengue-virus serotype may result in the production of nonprotective antiviral antibodies that nevertheless bind to the virion's surface and through interaction with the Fc receptor focus secondary dengue viruses on the target cell, the result being enhanced infection. The host is also

primed for a secondary antibody response when viral antigens are released and immune complexes lead to activation of the classic complement pathway, with consequent phlogistic effects. Cross-reactivity at the T cell level results in the release of physiologically active cytokines, including interferon γ and tumor necrosis factor α . The induction of vascular permeability and shock depends on multiple factors, including the following:

1. *Presence of enhancing and nonneutralizing antibodies* -- Transplacental maternal antibody may be present in infants <9 months old, or antibody elicited by previous heterologous dengue infection may be present in older individuals. T cell reactivity is also intimately involved.
2. *Age* -- Susceptibility to [DHF/DSS](#) drops considerably after 12 years of age.
3. *Sex* -- Females are more often affected than males.
4. *Race* -- Caucasians are more often affected than blacks.
5. *Nutritional status* -- Malnutrition is protective.
6. *Sequence of infection* -- For example, serotype 1 followed by serotype 2 seems to be more dangerous than serotype 4 followed by serotype 2.
7. *Infecting serotype* -- Type 2 is apparently more dangerous than other serotypes.

In addition, there is considerable variation among strains of a given serotype, with Southeast Asian serotype 2 strains having more potential to cause [DHF/DSS](#) than others.

Dengue [HF](#) is identified by the detection of bleeding tendencies (tourniquet test, petechiae) or overt bleeding in the absence of underlying causes such as preexisting gastrointestinal lesions. Dengue shock syndrome, usually accompanied by hemorrhagic signs, is much more serious and results from increased vascular permeability leading to shock. In mild [DHF/DSS](#), restlessness, lethargy, thrombocytopenia ($<100,000/\mu\text{L}$), and hemoconcentration are detected 2 to 5 days after the onset of typical dengue fever, usually at the time of defervescence. The maculopapular rash that often develops in dengue fever may also appear in DHF/DSS. In more severe cases, frank shock is apparent, with low pulse pressure, cyanosis, hepatomegaly, pleural effusions, ascites, and in some cases severe ecchymoses and gastrointestinal bleeding. The period of shock lasts only 1 or 2 days, and most patients respond promptly to close monitoring, oxygen administration, and infusion of crystalloid or -- in severe cases -- colloid. The case-fatality rates reported vary greatly with case ascertainment and the quality of treatment; however, most DHF/DSS patients respond well to supportive therapy, and overall mortality in an experienced center in the tropics is probably as low as 1%.

A virologic diagnosis can be made by the usual means, although multiple flavivirus infections lead to a broad immune response to several members of the group, and this situation may result in a lack of virus specificity of the IgM and IgG immune responses. A secondary antibody response can be sought with tests against several flavivirus antigens to demonstrate the characteristic wide spectrum of reactivity.

The key to control of both dengue fever and [DHF/DSS](#) is the control of *A. aegypti*, which also reduces the risk of urban yellow fever and chikungunya virus circulation. Control efforts have been handicapped by the presence of nondegradable tires and long-lived plastic containers in trash repositories, insecticide resistance, urban poverty, and an inability of the public health community to mobilize the populace to respond to the need to eliminate mosquito breeding sites. Live attenuated dengue vaccines are in the late stages of development and have produced promising results in early tests. Whether vaccines can provide safe, durable immunity to an immunopathologic disease such as DHF/DSS in endemic areas is an issue that will have to be tested, but it is hoped that vaccination will reduce transmission to negligible levels.

KYASANUR FOREST DISEASE AND OMSK HEMORRHAGIC FEVER

Kyasanur Forest virus and Omsk [HF](#) virus are geographically restricted, tick-borne flaviviruses that cause a syndrome of viral HF during a wave of viremia and that may also enter the [CNS](#) to cause subsequent viral encephalitis (see discussion of tick-borne encephalitis above). There is no therapy for these infections, but an inactivated vaccine has been used in India against Kyasanur Forest disease. A new and related virus isolate has been obtained from butchers with HF in the Middle East; the implication is that there are more agents in this group.

FILOVIRUS HEMORRHAGIC FEVER

See [Chap. 199](#).

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199. FILOVIRIDAE (MARBURG AND EBOLA VIRUSES) - C. J. Peters

DEFINITION

Both Marburg virus and Ebola virus cause an acute febrile illness associated with high mortality. This illness is characterized by multisystem involvement that begins with the abrupt onset of headache, myalgias, and fever and proceeds to prostration, rash, shock, and often bleeding manifestations. Epidemics usually begin with a single case acquired from an unknown reservoir in nature and spread mainly through close contact with sick persons or their body fluids, either in the home or at the hospital.

ETIOLOGY

The family Filoviridae comprises two antigenically and genetically distinct viruses: Marburg virus and Ebola virus. Ebola virus has four readily distinguishable subtypes named for their original site of recognition (Zaire, Sudan, Cote d'Ivoire, and Reston). Except for Ebola virus subtype Reston, all the Filoviridae are African viruses that cause severe and often fatal disease in humans. The Reston virus, which has been exported from the Philippines on several occasions, has caused fatal infections in monkeys but only subclinical infections in humans. Different isolates of the four Ebola subtypes made over time and space exhibit remarkable sequence conservation, indicating marked genetic stability in their selective niche. Typical filovirus particles contain a single linear, negative-sense, single-stranded RNA arranged in a helical nucleocapsid. The virions are 790 to 970 nm in length; they may also appear in elongated, contorted forms. The lipid envelope confers sensitivity to lipid solvents and common detergents. The viruses are largely destroyed by heat (60°C, 30 min) and by acidity but may persist for weeks in blood at room temperature. The surface glycoprotein self-associates to form the virion surface spikes, which presumably mediate attachment to cells and fusion. The glycoprotein's high sugar content may contribute to its low capacity to elicit neutralizing antibodies. A smaller form of the glycoprotein, bearing many of its antigenic determinants, is produced by in vitro-infected cells and is found in the circulation in human disease; it has been speculated that this circulating soluble protein may suppress the immune response to the virion surface protein or block antiviral effector mechanisms. Both Marburg virus and Ebola virus are biosafety level 4 pathogens because of their high associated mortality rate and aerosol infectivity.

EPIDEMIOLOGY

Marburg virus was first identified in Germany in 1967, when infected African green monkeys (*Cercopithecus aethiops*) imported from Uganda transmitted the agent to vaccine-laboratory workers. Of the 25 human cases acquired from monkeys, 7 ended in death. The six secondary cases were associated with close contact or parenteral exposure. Secondary spread to the wife of one patient was documented, and virus was isolated from the husband's semen despite the presence of circulating antibodies. Subsequently, isolated cases of Marburg virus infection have been reported from eastern and southern Africa, with limited spread.

In 1999, repeated transmission of Marburg virus to workers in a gold mine in eastern Democratic Republic of Congo was documented. The secondary spread of the virus

among patients' families was more extensive than previously noted, resembling that of Ebola virus and emphasizing the importance of hygiene and proper barrier nursing in the epidemiology of these viruses in Africa.

In 1976, epidemics of severe hemorrhagic fever (550 human cases) occurred simultaneously in Zaire and Sudan, and Ebola virus was found to be the etiologic agent. Later, it was shown that different subtypes of virus -- associated with 90% and 50% mortality, respectively -- caused the two epidemics. Both epidemics were associated with interhuman spread (particularly in the hospital setting) and the use of unsterilized needles and syringes, a common practice in developing-country hospitals. The epidemics dwindled as the clinics were closed and people in the endemic area increasingly shunned affected persons and avoided traditional burial practices.

The Zaire subtype of Ebola virus recurred in a major epidemic (317 cases, 88% mortality) in Democratic Republic of Congo in 1995 and in smaller epidemics in Gabon in 1994-1996. Mortality was high, transmission to caregivers and others who had direct contact with body fluids was common, and poor hygiene in hospitals exacerbated spread. In the Congo epidemic, an index case was infected in Kikwit in January 1995. The epidemic smoldered until April, when intense nosocomial transmission forced closure of the hospitals; samples were finally sent to the laboratory for Ebola testing, which yielded positive results within a few hours. International assistance, with barrier nursing instruction and materials, was provided; nosocomial transmission ceased, hospitals reopened, and patients were segregated to prevent intrafamilial spread. The last case was reported in June 1995.

Three separate emergences of Ebola virus (subtype Zaire) were detected in Gabon from 1994 through 1996, all associated with deep forest exposure and subsequent familial and nosocomial transmission. In the 1996 episode, a physician exposed to Ebola-infected patients traveled to South Africa with a fever; a nurse who assisted in a cutdown on the physician developed Ebola hemorrhagic fever and died in spite of intensive care. The index patient was identified retrospectively on the basis of serum antibodies and virus isolation from semen. Thus, distant transport of Ebola virus is an established risk, and limited nosocomial spread is possible even under hygienic conditions.

The Reston subtype of Ebola virus was first seen in the United States in 1989, when it caused a fatal, highly transmissible disease among cynomolgus macaques imported from the Philippines and quarantined in Reston, VA, pending distribution to biomedical researchers. This and other appearances of the Reston virus have been traced to a single export facility in the Philippines, but no source in nature has been established.

Epidemiologic studies (including a specific search in the Kikwit epidemic) have failed to yield evidence for an important role of airborne particles in human disease. This lack of epidemiologic evidence is surprising and seems to conflict with the viruses' classification as biosafety level 4 pathogens based in part on their aerosol infectivity and with formal laboratory assessments showing a high degree of aerosol infectivity for monkeys. Sick humans apparently do not usually generate sufficient amounts of infectious aerosols to pose a significant hazard to those around them.

Available evidence points to a nonprimate reservoir for these viruses, but an intensive search has failed to elucidate what this reservoir might be. Speculation has centered on a possible role for bats, but that hypothesis has arisen in part merely because of the ubiquity of bats when sought in affected areas and the frustration of researchers in identifying a source of virus.

PATHOLOGY AND PATHOGENESIS

In humans and in animal models, Ebola and Marburg viruses replicate well in virtually all cell types, including endothelial cells, macrophages, and parenchymal cells of multiple organs. Viral replication is associated with cellular necrosis both in vivo and in vitro. Significant findings at the light-microscopic level include liver necrosis with Councilman bodies (intracellular inclusions that correlate with extensive collections of viral nucleocapsids), interstitial pneumonitis, cerebral glial nodules, and small infarcts. Antigen and virions are abundant in fibroblasts, interstitium, and (to a lesser extent) the appendages of the subcutaneous tissues in fatal cases; escape through small breaks in the skin or possibly through sweat glands may occur and, if so, may be correlated with the established epidemiologic risk of close contact with patients and the touching of the deceased. Inflammatory cells are not prominent, even in necrotic areas.

In addition to sustaining direct damage from viral infection, patients infected with Ebola virus (Zaire subtype) have high circulating levels of proinflammatory cytokines, which presumably contribute to the severity of the illness. In fact, the virus interacts intimately with the cellular cytokine system. It is resistant to the antiviral effects of interferon α , although this mediator is amply induced. Viral infection of endothelial cells selectively inhibits the expression of MHC class I molecules and blocks the induction of several genes by the interferons. In addition, glycoprotein expression inhibits αV integrin expression, an effect that has been shown in vitro to lead to detachment and subsequent death of endothelial cells.

Acute infection is associated with high levels of circulating virus and viral antigen. Clinical improvement takes place when viral titers decrease concomitantly with the onset of a virus-specific immune response, as detected by enzyme-linked immunosorbent assay (ELISA) or fluorescent antibody test. In fatal cases, there is usually little evidence of an antibody response and there is extensive depletion of spleen and lymph nodes. Recovery is apparently mediated by the cellular immune response: convalescent-phase plasma has little in vitro virus-neutralizing capacity and is not protective in passive transfer experiments in monkey and guinea pig models.

CLINICAL MANIFESTATIONS

After an incubation period of ~7 to 10 days (range, 3 to 16 days), the patient abruptly develops fever, severe headache, malaise, myalgia, nausea, and vomiting. Continued fever is joined by diarrhea (often severe), chest pain (accompanied by cough), prostration, and depressed mentation. In light-skinned patients (and less often in blacks), a maculopapular rash appears around day 5 to 7 and is followed by desquamation. Bleeding may begin about this time and is apparent from any mucosal site and into the skin. In some epidemics, fewer than half of patients have had overt bleeding, and this manifestation has been absent even in some fatal cases. Additional

findings include edema of the face, neck, and/or scrotum; hepatomegaly; flushing; conjunctival injection; and pharyngitis. Around 10 to 12 days after the onset of disease, the sustained fever may break, with improvement and eventual recovery of the patient. Recrudescence of fever may be associated with secondary bacterial infections or possibly with localized virus persistence. Late hepatitis, uveitis, and orchitis have been reported, with isolation of virus from semen or detection of polymerase chain reaction (PCR) products in vaginal secretions for several weeks.

LABORATORY FINDINGS

Leukopenia is common early on; neutrophilia has its onset later. Platelet counts fall below (sometimes much below) 50,000/uL. Laboratory evidence of disseminated intravascular coagulation may be found, but its clinical significance and the need for therapy are controversial. Serum levels of alanine and aspartate aminotransferases (particularly the latter) rise progressively, and jaundice develops in some cases. The serum amylase level may be elevated, and this elevation may be associated with abdominal pain suggesting pancreatitis. Proteinuria is usual; decreased kidney function is proportional to shock.

DIAGNOSIS

Most patients acutely ill with Ebola or Marburg viruses have high concentrations of virus in blood. Antigen-detection [ELISA](#) is a sensitive, robust diagnostic modality. Virus isolation and reverse transcriptase [PCR](#) are also effective and provide additional sensitivity in some cases. Patients who are recovering develop IgM and IgG antibodies that are best detected by ELISA but are also reactive in the less specific fluorescent antibody test. Skin biopsies are an extremely useful adjunct in postmortem diagnosis of Ebola and, to a lesser extent, Marburg virus infections because of the presence of large amounts of viral antigen, the relative safety of obtaining the sample, and the freedom from cold-chain requirements for formalin-fixed tissues.

TREATMENT

No virus-specific therapy is available, and, given the extensive viral involvement in fatal cases, supportive treatment may not be as useful as was once hoped. Vigorous treatment of shock should take into account the likelihood of vascular leak in the pulmonary and systemic circulation and of myocardial functional compromise. The membrane fusion mechanism of Ebola resembles that of retroviruses, and the identification of "fusogenic" sequences suggests that inhibitors of cell entry may be developed. Despite the poor neutralizing capacity of polyclonal convalescent-phase sera, phage display of immunoglobulin mRNA from convalescent bone marrow has produced monoclonal antibodies that have in vitro neutralizing capacity and mediate protection in guinea pig models.

PREVENTION

No vaccine is available, but barrier nursing precautions in African hospitals can greatly decrease the spread of the virus beyond the index case and thus prevent epidemics of filoviruses and other agents as well.

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SECTION 15 -FUNGAL AND ALGAL INFECTIONS

200. DIAGNOSIS AND TREATMENT OF FUNGAL INFECTIONS - John E. Bennett

MYCOLOGY FUNDAMENTALS

Fungi can appear microscopically as either rounded, budding forms (yeastlike organisms) or hyphae (molds; [Fig. 200-CD1](#)). Yeastlike colonies are smooth, while mold colonies are fuzzy; fungi that grow as yeasts include species of *Candida* and *Cryptococcus*, while fungi that grow as molds include species of *Aspergillus*, *Rhizopus*, and dermatophytes (ringworm fungi). The fungi that cause histoplasmosis, blastomycosis, sporotrichosis, coccidioidomycosis, and paracoccidioidomycosis are called *dimorphic* ("having two forms") because they are spherical in tissue but grow like molds when cultured at room temperature. *Candida* species other than *Candida glabrata* appear in tissue as both budding yeasts and tubular elements called *pseudohyphae*. *Pneumocystis carinii* is closer to fungi than to parasites by ribosomal sequences ([Chap. 209](#)). Because the drugs used to treat *Pneumocystis* pneumonia are also used to treat parasitic or bacterial infections, those drugs will not be discussed in this chapter.

Many fungi can form two different types of spores and are given different names, depending on the spore-bearing structures. When the spores are produced by mitosis, the fungus is said to be an *anamorph*, or to be in the imperfect state. Many fungi can have different sporulating structures in which genetic recombination occurs, often as a result of coculture with a strain of the opposite mating type. A fungus producing those distinctive spores is said to be a *teleomorph*, or to be in the perfect state. Diagnostic laboratories usually use the name of the anamorph because they do not use culture conditions that would produce the teleomorph. One exception is *Scedosporium apiospermum*, which is often observed as a teleomorph in the diagnostic laboratory and identified as *Pseudallescheria boydii*.

Most fungi that are pathogenic for humans are saprophytes in nature; they cause infection when airborne spores reach the lung or paranasal sinus or when hyphae or spores are accidentally inoculated into the skin or cornea. Acquisition of infection from another person or an animal has been reported in the case of ringworm but is very rare in other mycoses. Thus, hospitalized patients with fungal infections do not require special isolation. Most fungi infect hosts preferentially by one route and only infrequently by other routes. For example, the agents of ringworm, pityriasis versicolor, and piedra infect the epidermis and its appendages. Sporotrichosis and mycetoma usually arise from subcutaneous inoculation. Inhalation is the route of inoculation for the agents of most deep mycoses. Ingestion of fungi rarely causes infection; *Candida albicans*, a normal commensal in the mouth and intestine, reaches deeper tissues only when mucosal or cutaneous barriers are breached by disease, surgery, trauma, or catheterization. Histoplasmosis, blastomycosis, coccidioidomycosis, and paracoccidioidomycosis have been called "endemic" mycoses to emphasize their restricted geographic distribution. Some fungi, such as *Aspergillus*, are said to be opportunists in that they usually infect hosts with compromised immunity. This distinction is relative, not absolute.

Immunity after exposure to fungi may confer partial protection against reinfection. Residents of areas in which mycoses are endemic are less subject to infection than are newcomers. Predisposing factors are helpful in defining host defense. Immunoglobulin deficiencies do not appear to predispose to any mycosis, whereas neutropenia is common among patients who develop invasive aspergillosis or deep candidiasis. Cell-mediated immunity appears to be of paramount importance in most other deep mycoses.

DIAGNOSIS

Many fungi can be identified to the genus or even the species level by microscopic examination of smears or biopsy specimens. Calcofluor white staining with fluorescence microscopy is a sensitive technique for smears of sputum, bronchoalveolar lavage fluid, or pus. India ink smear remains the method of choice for detecting cryptococci in cerebrospinal fluid (CSF). *Candida* yeast cells and pseudohyphae are the only fungi that are usually gram-positive on smears. For other fungi, Gram's staining is distinctly suboptimal. For histopathology slides, Gomori methenamine silver and a neutral counterstain are preferred.

The method used has a marked effect on the rapidity and sensitivity of blood cultures for fungi except in the case of *Candida* species, which are relatively easy to grow. For most other fungi, concentration of the blood by lysis centrifugation and culture on solid medium constitute the optimal technique. Commercially available nucleic acid hybridization techniques can speed the identification of slow-growing molds, such as *Histoplasma capsulatum* and *Coccidioides immitis*. Serology has limited value, but testing of serum or [CSF](#) for cryptococcal antigen or antibody to *C. immitis* can be diagnostic. Detection of *Histoplasma* antigen in urine or serum is helpful in diagnosis and in following the results of treatment for disseminated histoplasmosis. Skin testing with fungal antigens is not useful in detecting active infection.

ANTIFUNGAL THERAPY

TOPICAL AGENTS

Imidazoles and Triazoles (See also "Systemic Antifungals," below) These synthetic compounds act by inhibiting ergosterol synthesis in the fungal cell wall and, when given topically, may cause direct damage to the fungal cytoplasmic membrane. The imidazoles available for cutaneous application include clotrimazole, econazole, ketoconazole, sulconazole, oxiconazole, and miconazole. Vaginal formulations include four imidazoles (miconazole, clotrimazole, tioconazole, and butoconazole) and one triazole (terconazole). As yet, no substantial differences in the efficacy of or local intolerance to the various topical azoles have become apparent. All are effective in the treatment of cutaneous candidiasis, tinea (pityriasis) versicolor, and mild to moderately severe ringworm of the glabrous skin. Vaginal formulations are effective for vulvovaginal candidiasis. Clotrimazole is poorly absorbed from the gastrointestinal tract, but the oral troche is useful as a topical treatment for oral and esophageal candidiasis.

Polyene Macrolide Antibiotics These broad-spectrum antifungal agents combine with sterol in the fungal cytoplasmic membrane, increasing membrane permeability.

Topically, they are not active against ringworm but are effective against candidiasis of the skin and mucous membranes. Nystatin and amphotericin B suspensions are effective in oral thrush, and vaginal troches are effective in vulvovaginal candidiasis. Both nystatin and amphotericin B are available in topical preparations for cutaneous candidiasis.

Other Topical Antifungals Ciclopirox olamine, haloprogin, terbinafine, and naftifine have the same clinical spectrum among the cutaneous mycoses as the imidazoles. Tolnaftate and undecylenic acid are effective against ringworm but not candidiasis. Keratolytic agents, such as salicylic acid, are helpful as accessory drugs for some hyperkeratotic skin lesions.

SYSTEMIC ANTIFUNGALS

Griseofulvin Griseofulvin is a useful drug in the treatment of certain kinds of ringworm; however, it is ineffective in the treatment of candidiasis. The microcrystalline and ultramicrocrystalline preparations differ in dose but not in efficacy. Absorption of both is enhanced when the drug is ingested with fat-containing foods. Griseofulvin interacts with phenobarbital and coumarin-type anticoagulants.

Terbinafine Oral terbinafine (250 mg once daily) is at least as effective as itraconazole and more effective than griseofulvin in onychomycosis and ringworm. Treatment duration ranges from 3 months for fingernails to 6 months for toenails. Gastrointestinal distress is the most common side effect. Rash, hepatitis, and pancytopenia have occurred, but serious adverse effects have been uncommon. Terbinafine decreases cyclosporine levels. Cimetidine increases and rifampin decreases terbinafine levels in blood.

Imidazoles and Triazoles

General Features The azole antifungals include imidazoles and triazoles. Fluconazole, itraconazole, and investigational azoles are all triazoles, so named because they have three nitrogens in the ring structure. This class has less impact on human hormonal synthesis and less hepatotoxicity than the only widely used systemic imidazole, ketoconazole. Itraconazole has many structural features in common with ketoconazole; however, it has a broader spectrum of activity and has largely replaced ketoconazole.

Reported interactions of itraconazole and fluconazole with other drugs are listed in [Table 200-1](#). Ketoconazole interactions (not listed) appear to be the same as those listed for itraconazole. Azole interactions with any one class of drugs, such as benzodiazepines, HMG-CoA reductase inhibitors, or drugs that decrease gastric acidity, should be considered to apply to all drugs of that class until proven otherwise. Fluconazole differs substantially from itraconazole: unlike that of itraconazole, the absorption of fluconazole is independent of food or gastric acid, and fluconazole has much less effect on the hepatic metabolism of other drugs than does itraconazole. High fluconazole blood levels engendered by azotemia or by dosages above those used in pharmacologic studies may lead to new and profound drug interactions.

All azoles have the potential for embryotoxicity and teratogenicity. In fact, it seems likely

that azoles should not be given during pregnancy without a discussion of the serious risks and possible benefits with the mother. Four infants born to mothers taking at least 400 mg of fluconazole daily for coccidioidal meningitis have had severe bone, craniofacial, or cardiac abnormalities. Similarity of these abnormalities to those in pregnant animals given fluconazole suggests that fluconazole caused the defects.

Itraconazole Itraconazole is useful in the treatment of blastomycosis, histoplasmosis, candidiasis, coccidioidomycosis, sporotrichosis, pseudallescheriasis, onychomycosis, ringworm, tinea versicolor, and some cases of aspergillosis. Its efficacy in mycoses of the central nervous system has been modest at best. Almost no bioactive drug appears in urine. Itraconazole is metabolized in the liver, with the hydroxy metabolite accounting for at least half of the antifungal activity in serum. Food increases absorption of itraconazole capsules by about threefold but substantially reduces absorption of the cyclodextrin suspension. Ability of the suspension to exert a topical as well as a systemic effect probably accounts for its improved efficacy in oropharyngeal candidiasis. The usual dosage of either oral itraconazole formulation is 100 to 200 mg once daily for oropharyngeal and esophageal candidiasis. For deep infections, itraconazole capsules are given at an initial dosage of 600 to 800 mg daily for 3 days and a subsequent dosage of 200 to 400 mg once daily continued for 6 to 12 months. Itraconazole blood levels are helpful in documenting absorption of oral itraconazole when the drug is used for the treatment of deep mycoses. An intravenous formulation is commercially available and should be considered for initial therapy in hospitalized patients in whom itraconazole absorption may be suboptimal. Immunosuppressed patients with rapidly progressing pseudallescheriasis, an infection that does not respond to amphotericin B, are candidates for intravenous itraconazole treatment.

Fluconazole This triazole can be administered in tablet form, as a suspension, or as an intravenous infusion. With a half-life of about 31 h, fluconazole can be given once a day. Approximately 80% of the drug is excreted unchanged in the urine. Patients with creatinine clearance rates of 21 to 50 mL/min and 11 to 20 mL/min should have their fluconazole doses reduced by 50 and 75%, respectively. The drug penetrates the [CSF](#) and other body fluids very well.

Nausea and abdominal distress are the most common forms of dose-limiting fluconazole toxicity. An allergic rash may develop and is particularly common among patients infected with HIV. Fatal cases of Stevens-Johnson syndrome have been described in the HIV-infected population. Alopecia commonly follows prolonged administration of 3400 mg daily but resolves when therapy is discontinued. Rare cases of anaphylaxis, hepatic necrosis, and neutropenia have been described.

Fluconazole is useful in the treatment of oropharyngeal and esophageal candidiasis in adults. A single 150-mg tablet is effective in vulvovaginal candidiasis. Catheter-acquired candidemia in the immunocompetent host responds to 400 mg of fluconazole daily in conjunction with the removal of the infected catheter. Treatment should be continued for 10 to 14 days after the patient has become afebrile. Fluconazole is also effective in initial and maintenance therapy for cryptococcal meningitis in patients with AIDS, although most of these patients should initially receive a 2-week course of intravenous amphotericin B. Patients with coccidioidal meningitis can often be given fluconazole rather than intrathecal amphotericin B as maintenance therapy.

The incidence of deep candidiasis among recipients of allogeneic bone marrow transplants can be reduced by the administration of fluconazole (400 mg daily) for 75 days after initiation of the transplantation-preparative regimen. Prophylaxis in other neutropenic patients has not appeared useful. Fluconazole (200 mg daily) reduced the incidence of cryptococcosis and mucosal candidiasis among AIDS patients whose CD4+ cell counts were <200/uL and was particularly effective among those with counts of <50/uL. However, this regimen is not recommended because it does not reduce mortality, is expensive, and can lead to drug resistance.

Fluconazole is less effective than itraconazole in blastomycosis, histoplasmosis, and sporotrichosis. The drug is not active in aspergillosis or mucormycosis.

Amphotericin B A colloidal deoxycholate complex of the polyene drug amphotericin B is available for intravenous or intrathecal administration. In-line filters with a 0.22-um pore diameter may trap some of the colloid. The catabolism of amphotericin B is extremely slow and is not influenced by renal failure, hepatic failure, or hemodialysis. The drug's penetration into [CSF](#) and vitreous humor is poor; however, the concentrations in pleural, peritoneal, and articular exudates are adequate for many mycoses. Histoplasmosis, blastomycosis, paracoccidioidomycosis, candidiasis, and cryptococcosis are the most responsive mycoses; coccidioidomycosis, extraarticular sporotrichosis, aspergillosis, and mucormycosis are less responsive; and chromoblastomycosis, mycetoma, and pseudallescheriasis respond little, if at all. The usual course is 0.5 to 0.7 mg/kg daily for 8 to 10 weeks. Infusions are generally given in 5% dextrose over 2 to 4 h.

Initial doses of amphotericin B occasionally cause marked febrile reactions that may be poorly tolerated by adult patients with limited cardiac or pulmonary function. It may be prudent to give such patients an initial 1-mg test dose followed by rapidly escalating doses, depending on tolerance. Premedication with aspirin or acetaminophen or the addition of hydrocortisone (25 mg) to the infusion decreases chills and fever. Azotemia during treatment is usual, the extent depending on the daily dose. Saline infusions have been advocated to reduce azotemia. Permanent loss of renal function is related to the total dose of amphotericin B; this condition is generally noted in adults who have received >3 g. Other side effects include anemia, hypokalemia, renal tubular acidosis, nausea, anorexia, weight loss, phlebitis, and occasionally hypomagnesemia. Intrathecal amphotericin B has been used in coccidioidal meningitis and refractory cryptococcal meningitis, although this therapy is associated with considerable toxicity.

Three lipid formulations of amphotericin B are commercially available in the United States: amphotericin B lipid complex (ABLC), amphotericin B colloidal dispersion (ABCD), and liposomal amphotericin B (L-AB). All cause less nephrotoxicity than the older amphotericin B deoxycholate complex (ABD). Acute, febrile infusion-related reactions occur with all three lipid formulations but are most severe with ABCD. The recommended duration for initial infusions of ABCD is 1 mg/kg per hour, somewhat slower than the 2-h duration of ABLC or L-AB infusions, with the intent of decreasing febrile reactions. Premedication with acetaminophen is also an option. Use of these remarkably expensive formulations should be confined to patients who cannot tolerate the nephrotoxicity of ABD. Although the lipid formulations are also approved for patients

failing to respond to ABD, there is no indication that these formulations are more effective than ABD for any mycosis.

Flucytosine Flucytosine (5-fluorocytosine) is a synthetic oral drug useful in cryptococcosis, candidiasis, and chromoblastomycosis. Within the fungal cell, flucytosine is converted to the antimetabolite 5-fluorouracil. Drug resistance appears rather rapidly when flucytosine is used alone. For this reason, the drug is generally used in combination with amphotericin B. The usual dose of flucytosine is 25 to 37.5 mg/kg every 6 h. Flucytosine is well absorbed from the gastrointestinal tract. The drug penetrates well into the [CSF](#) and is excreted unchanged in the urine. Even modest reductions in renal function may elevate flucytosine blood levels into the toxic range (³100 to 125 ug/mL). Elevated levels are associated with a significant incidence of neutropenia and thrombocytopenia and also seem to predispose to colitis, the other major toxic effect of this drug. Hepatotoxicity is idiosyncratic and uncommon. An allergic rash may develop.

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201. HISTOPLASMOSIS - John E. Bennett

ETIOLOGIC AGENT

Histoplasma capsulatum is a dimorphic fungus that grows as a mold in nature or on Sabouraud's agar at room temperature. Hyphae bear both large and small spores, which are used for identification. Nucleic acid hybridization can also be used to identify the organism in culture. *H. capsulatum* grows as a small budding yeast in host tissue and on enriched agar, such as blood cysteine glucose, at 37°C. Despite its name, the fungus is unencapsulated. Coculture of isolates with opposite mating types can produce different sporulating structures in which genetic recombination occurs. When these structures, referred to as a *teleomorph* or the *perfect state*, are seen in culture, the name *Ajellomyces capsulatus* is used.

EPIDEMIOLOGY

Infection with *H. capsulatum* has been encountered in many areas of the world but is much more frequent in certain areas. Within the United States, infection is most common in the southeastern, mid-Atlantic, and central states. Endemicity is probably contingent on the availability of proper conditions in nature for growth of the fungus. *H. capsulatum* prefers moist surface soil, particularly soil enriched by droppings of certain birds and bats. The fungus has been isolated repeatedly from such sites, and many case clusters have occurred 5 to 18 days after the exposure of groups of people to dust while (for example) cleaning dirt-floored chicken coops; raking soil beneath bird-roosting sites; exploring caves; and cleaning, remodeling, or demolishing old buildings. Skin-test reactivity in many endemic areas indicates that 80% of residents over age 16 have been exposed.

PATHOGENESIS AND PATHOLOGY

Microconidia, or small spores, of *H. capsulatum* are small enough to reach the alveoli on inhalation and are transformed there to budding forms. With time, an intense granulomatous reaction occurs. Caseation necrosis or calcification may mimic tuberculosis. In children, the primary infection usually heals completely but may leave spotty calcification in the hilar nodes or lung. Transient dissemination may leave calcified granulomas in the spleen. In adults, a rounded mass of scar tissue, with or without central calcification, may remain in the lung. This mass has been called a *histoplasma*. Previous exposure is thought to confer some protection against reinfection, but infection in persons with prior positive skin tests clearly has occurred.

In a small proportion of patients, histoplasmosis becomes a progressive, potentially fatal infection. The disease occurs either as chronic fibrocavitary pneumonia or, less commonly, as disseminated infection. Patients with either form lack a history of acute primary pulmonary histoplasmosis. Chronic pulmonary infection favors otherwise healthy males over the age of 40. A history of cigarette use or the presence of emphysema is elicited from nearly all patients with chronic progressive pulmonary histoplasmosis. An acute, rapidly fatal disseminated infection is most likely to be encountered among young children and immunosuppressed patients, including those with AIDS. A more chronic but equally lethal disseminated infection is more common

among previously healthy adults.

CLINICAL MANIFESTATIONS

The vast majority of infections are either asymptomatic or mild, and the diagnosis is elusive. Cough, fever, malaise, and chest x-ray findings of hilar adenopathy with or without one or more areas of pneumonitis are typical features. Erythema nodosum and erythema multiforme have been reported in a few outbreaks. Hilar adenopathy may cause temporary compression of the right-middle-lobe bronchus in children and young adults. Subacute pericarditis may develop, probably by extension from contiguous lymph nodes. Rarely, hilar nodes undergo a caseous, granulomatous reaction with perinodal fibrosis. Mediastinal structures become encased by progressive fibrosis, and compression of the pulmonary veins, superior vena cava, pulmonary arteries, and esophagus may take place over many years. Late in mediastinal disease, only rare nonviable *Histoplasma* cells can be found in caseous residua of lymph nodes.

Chronic pulmonary histoplasmosis is characterized by a gradual onset (over weeks or months) of increasing productive cough, weight loss, and sometimes night sweats. Chest x-ray reveals uni- or bilateral fibronodular apical infiltrates. Approximately one-third of cases stabilize or improve spontaneously early in the course. The remainder progress insidiously. Retraction and cavitation of the upper lobes occur, with spread to the apex of the lower lobes and other areas of the lung. Emphysema and bulla formation further compromise pulmonary function. Death from cor pulmonale, bacterial pneumonia, or histoplasmosis occurs after months or years.

Acute disseminated histoplasmosis may be mistaken for miliary tuberculosis ([Chap. 169](#)). Common findings include fever, emaciation, hepatosplenomegaly, lymphadenopathy, jaundice, anemia, leukopenia, and thrombocytopenia. A high index of suspicion is necessary in patients with AIDS, in whose cases there may be other explanations for the abnormalities caused by disseminated histoplasmosis ([Fig. 201-CD1](#)). All these features may be noted in chronic dissemination as well, but chronic disease tends to be more localized. Indurated ulcers of the mouth, tongue, nose, or larynx are reported in about one-fourth of cases. Other focal findings include granulomatous hepatitis, Addison's disease, gastrointestinal ulceration, endocarditis, and chronic meningitis. Chest x-ray abnormalities are evident in half of cases and characteristically consist of discrete nodules or a miliary pattern.

Infection with *H. capsulatum* var. *duboisii* is rare outside of Africa. The yeast form is larger in tissue than that of *H. capsulatum* var. *capsulatum*. Clinical manifestations resemble those of blastomycosis more than those of histoplasmosis in that skin and bone lesions are very common.

DIAGNOSIS

Culture of the etiologic organism is the preferred method for diagnosis of histoplasmosis but is often difficult. Blood cultures are best done by the lysis-centrifugation technique, with plates held at 30°C for at least 2 weeks. Approximately 15 mL of blood should be cultured from adults. Routine blood cultures in broth are generally unsuitable. Cultures of bone marrow, mucosal lesions, liver, and bronchoalveolar lavage fluid are

diagnostically useful in disseminated histoplasmosis. Sputum culture is the preferred method for the diagnosis of chronic pulmonary histoplasmosis. However, growth may require 2 to 4 weeks to become visible, and other organisms may overgrow the plate. Diagnosis based on Giemsa-stained smears of blood or bronchoalveolar lavage fluid or on methenamine silver staining of infected lung, bone marrow, lymph node, or mucosal lesions requires considerable expertise, although these techniques yield results rapidly and provide specimens that can easily be sent to a referral laboratory. Organisms may be very scanty in lesions with marked caseous necrosis. An assay for *Histoplasma* antigen in blood or urine is commercially available and is useful both for diagnosis and for monitoring of the response to therapy in patients with AIDS who have disseminated infection. Diagnosis by antigen detection requires confirmation by culture or histopathology because false-positive results have occasionally been obtained. Tests for antibody to *H. capsulatum* have been of limited value in diagnosis. Histoplasmin skin testing has proven useful in epidemiologic studies but not in clinical diagnosis. Neither skin testing nor serology has been predictive of histoplasmosis in patients infected with HIV.

TREATMENT

Acute pulmonary histoplasmosis requires no therapy. Oral itraconazole (200 mg/d) can be given to shorten the course of illness, although this effect has not been proven. Patients with mediastinal fibrosis may benefit from surgery, but their ultimate prognosis is poor. All patients with disseminated or chronic fibronodular pulmonary histoplasmosis should receive chemotherapy. Intravenous amphotericin B (0.6 mg/kg daily) is the drug of choice for the initial treatment of patients with disseminated histoplasmosis who are severely ill or immunosuppressed or whose infection involves the central nervous system; the regimen can be changed to itraconazole (200 mg twice daily) once clinical improvement is evident in these patients. Measuring itraconazole trough blood levels should be considered in those patients (e.g., patients with AIDS) who may not be absorbing the drug well ([Chap. 200](#)). Itraconazole suspension, taken fasting, is better absorbed than the capsule formulation. Fluconazole is reliably absorbed, even in patients taking drugs to block gastric acid secretion, but at doses up to 400 mg/d has been less effective in treatment of chronic pulmonary or disseminated histoplasmosis. Patients with AIDS whose disseminated histoplasmosis has responded to 10 weeks of therapy should receive itraconazole (200 mg/d) for life to prevent relapse. It remains unknown whether patients with a sustained response to highly active antiretroviral therapy can discontinue maintenance therapy with itraconazole.

Immunocompetent patients can initially be given itraconazole (200 mg twice daily) and are generally treated for 6 to 12 months. Ketoconazole (400 to 800 mg once daily) can be used instead of itraconazole for the treatment of immunocompetent patients without central nervous system disease when the lower cost is more important than the higher incidence of side effects. Alternatively, immunocompetent patients can be given a 10-week course of amphotericin B (0.5 mg/kg daily).

Long-term maintenance therapy with an azole is not recommended for patients other than those with AIDS. However, relapse of chronic pulmonary and disseminated histoplasmosis is not rare and warrants careful follow-up for 1 year after therapy.

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202. COCCIDIOIDOMYCOSIS - John E. Bennett

ETIOLOGIC AGENT

Coccidioides immitis has two forms, growing as a white fluffy mold on most culture media but as a nonbudding spherical form (a spherule) in host tissue or under special conditions. The organism reproduces in host tissue by forming small endospores within mature spherules. After rupture of the spherule, the released endospores enlarge, become spherules, and repeat the cycle. The fungus is identified by its appearance and by the formation of thick-walled, barrel-shaped spores, called *arthrospores*, in the hyphae of the mold form.

EPIDEMIOLOGY, PATHOGENESIS, AND PATHOLOGY

C. immitis is a soil saprophyte found in certain arid regions of the United States, Mexico, Central America, and South America. Within the United States, most cases of infection with *C. immitis* are acquired in California, Arizona, and western Texas. A few cases are acquired by exposure to fomites from endemic areas (e.g., in cotton bales).

Infection in humans and animals results from inhalation of wind-borne arthrospores from soil sites. This primary pulmonary infection is symptomatic in only 40% of cases, with symptoms ranging from a mild influenza-like illness to severe pneumonia. Mild self-limited infections may come to medical attention because of case clusters or hypersensitivity reactions: erythema nodosum, erythema multiforme, toxic erythema, arthralgia, arthritis, conjunctivitis, or episcleritis. Case clusters occur 10 to 14 days after a group of susceptible individuals is exposed to dust in an endemic area through such activities as archaeologic excavation, rock hunting, military maneuvers, or construction work. Windstorms can carry spores to adjacent nonendemic areas and cause case clusters. The usual course of primary pneumonia is complete healing, although an area of pneumonitis (detected on radiographs) may heal by the formation of a coinlike lesion called a *coccidioidoma*. Less commonly, a single thin-walled cavity remains as a chronic sequela in the area of consolidation. Alternatively, an area of consolidation may persist as chronic pneumonia or progress to fibronodular cavitory disease.

Pleural effusion may be the only manifestation of primary infection. Spontaneous healing of this form is common.

An uncommon but dreaded complication of coccidioidomycosis is dissemination beyond the lung and hilar lymph nodes. Dissemination is especially frequent among blacks, Filipinos, Native Americans, Mexican-Americans, pregnant women, and immunosuppressed patients, including those with AIDS.

C. immitis incites a chronic granulomatous reaction in host tissue, often with caseation necrosis. Lung and hilar node lesions may show calcification. Both IgM and IgG antibodies to *C. immitis* are induced by infection, but neither type of antibody appears to be protective. The amount of specific IgG antibody is a rough measure of the antigenic mass (i.e., of the intensity of infection), and a high titer is a poor prognostic sign. Appearance of delayed hypersensitivity to antigens of *C. immitis* is most common in clinical forms of disease with a good prognosis, such as self-limited primary pulmonary

disease. In skin tests for *Coccidioides* antigens, about half of patients with disseminated disease have negative results that portend a poor outcome.

CLINICAL MANIFESTATIONS

Symptomatic primary pulmonary infection is manifested by fever, cough, chest pain, malaise, and sometimes the hypersensitivity reactions listed above. Chest radiographs may show an infiltrate, hilar adenopathy, or pleural effusion. Mild peripheral-blood eosinophilia may be found. Spontaneous improvement begins after several days to 2 weeks of illness and usually culminates in complete recovery.

The symptoms of a chronic thin-walled cavity include cough or hemoptysis in half of cases; the other half are asymptomatic. Chronic progressive pulmonary coccidioidomycosis causes cough, sputum production, variable degrees of fever, and weight loss. The first indications of dissemination usually appear during primary infection. Reactivation with dissemination in later years occurs occasionally, especially if Hodgkin's disease, non-Hodgkin's lymphoma, renal transplantation, AIDS, or immunosuppression of some other etiology has supervened. Dissemination should be suspected when fever, malaise, hilar or paratracheal lymphadenopathy, elevated sedimentation rate, and high complement fixation titers signal abnormal persistence in patients with primary pulmonary coccidioidomycosis. With time, lesions appear in the bone, skin, subcutaneous tissue, meninges, joints, and other sites. Chronic meningitis presents as headache of indolent onset, with or without other signs of disseminated coccidioidomycosis. Cultures and smears of cerebrospinal fluid (CSF) are most often negative, but antibody is usually detectable in CSF by complement fixation. Skin lesions are indolent and maculopapular; soft tissue and bony lesions contain pus and may present as a draining sinus. Without treatment, disseminated coccidioidomycosis progresses to death over weeks to years.

Disseminated coccidioidomycosis can progress rapidly in patients with advanced HIV infection. Fever with skin or bone lesions may be the first sign. Those who present with diffuse pulmonary infiltrates have a poor prognosis. Blood cultures are positive late in the disease, if at all.

DIAGNOSIS

When coccidioidomycosis is suspected, sputum, urine, and pus should be examined for *C. immitis* by wet smear and culture. *The laboratory request should indicate clearly that coccidioidomycosis is suspected, because the mold form must be handled with extreme care to prevent infection of laboratory personnel.* On biopsy, smaller spherules must be distinguished from nonbudding forms of *Blastomyces* and *Cryptococcus*, but the appearance of the mature spherule is diagnostic.

Serologic tests are very helpful in the diagnosis of coccidioidomycosis. Latex agglutination and agar gel diffusion tests are useful in screening sera for antibody to *Coccidioides*. The complement fixation test is used for [CSF](#) determinations and for the confirmation and quantitation of serum antibody detected by screening tests. The number of cases with a positive complement fixation test depends on the severity of disease and on the laboratory performing the test. Positive tests are least common

among patients with solitary pulmonary cavities or primary pulmonary infection, while sera from patients with disseminated disease in multiple organs are nearly all positive. Seroconversion is helpful in primary pulmonary coccidioidomycosis but may not occur for up to 8 weeks after onset. A positive complement fixation test of unconcentrated CSF is diagnostic of meningitis. Rarely, a parameningeal focus causes a positive complement fixation test of CSF.

Conversion of the skin test from negative to positive (≥ 5 mm of induration at 24 or 48 h) with spherulin may take place between days 3 and 21 of symptoms in primary pulmonary coccidioidomycosis. Skin testing can be helpful in epidemiologic studies, such as investigations of case clusters or the definition of endemic areas. The utility of skin testing as a diagnostic tool is limited by the persistence of positive tests resulting from remote exposures to *Coccidioides* and by the frequency of negative skin tests among patients with either thin-walled cavities or disseminated coccidioidomycosis. A positive skin test has not predicted dissemination in HIV-infected patients. The presence of complement-fixing antibody to *C. immitis* in AIDS patients should prompt a search for active infection.

TREATMENT

Primary pulmonary coccidioidomycosis usually resolves spontaneously. Some physicians give a few weeks of treatment with intravenous amphotericin B or itraconazole to patients with unusually severe or protracted primary infection in the hope of aborting disseminated or chronic pulmonary disease.

Patients with severe or rapidly progressing disseminated coccidioidomycosis are first given intravenous amphotericin B at a dose of 0.5 to 0.7 mg/kg daily. Patients whose condition improves after 2 to 3 months of treatment with amphotericin B or who have more indolent disseminated infection are given itraconazole (200 mg twice daily) or fluconazole (400 to 600 mg/d). These oral agents are useful for long-term suppression of infection, and treatment should be continued for years. Patients with coccidioidal meningitis usually are initially given fluconazole (400 to 800 mg/d) but may require intrathecal amphotericin B. Hydrocephalus is a frequent complication of uncontrolled meningitis. Surgical debridement of bone lesions or drainage of abscesses can be helpful. The prognosis for ultimate cure of disseminated coccidioidomycosis is guarded.

Resection of chronic progressive pulmonary lesions is a helpful adjunct to chemotherapy when infection is confined to the lung and to one lobe. A single thin-walled cavity tends to close spontaneously and ordinarily is not resected. Such a cavity responds poorly to chemotherapy.

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203. BLASTOMYCOSIS - John E. Bennett

ETIOLOGIC AGENT

Blastomyces dermatitidis is a dimorphic fungus that grows at room temperature as a white or tan mold but grows within the host or at 37°C as budding, round yeastlike cells. The fungus can be identified on the basis of its appearance, its dimorphism, the small spores borne on hyphae of the mold form, or the results of nucleic acid hybridization. When isolates of the two opposite mating types are grown close together on special culture medium, such as yeast extract or soil extract agar, sporulating structures that characterize the perfect state (teleomorph), called *Ajellomyces dermatitidis*, appear.

EPIDEMIOLOGY

The infection is restricted by geography and age. Blastomycosis is uncommon in any locality, but most cases occur in the southeastern, central, and mid-Atlantic areas of the United States, with occasional cases in other localities in the United States and Canada. Cases have also been encountered in Africa, Mexico, Central America, and (rarely) South America. Most patients are between 20 and 69 years old. The male-to-female ratio is about 10:1. There is no occupational predisposition to the development of blastomycosis.

PATHOGENESIS AND PATHOLOGY

Infection with *B. dermatitidis* appears to be acquired by inhalation of the fungus from soil, decomposed vegetation, or rotting wood. Several case clusters have resulted from participation in recreational activities in wooded areas along waterways. Infection is not transmissible from person to person. The initial pulmonary infection may either heal spontaneously or become chronic. Spread to other portions of the lung, cavitation, or endobronchial lesions may be found in patients with chronic disease. Whether or not the lung lesion resolves spontaneously, infection commonly spreads hematogenously to the skin, subcutaneous tissue, bone, prostate, epididymis, or mucosa of the nose, mouth, or larynx. Less commonly, infection spreads to the brain, meninges, liver, lymph nodes, or spleen. Dissemination may not be evident for weeks or years after the appearance of the lung lesion. Progressive infection is only rarely attributable to an underlying disease, to HIV infection, or to immunosuppressive treatment. The inflammatory response includes lymphocytes, giant cells, and neutrophils. Pseudoepitheliomatous hyperplasia may be striking and may lead to a mistaken diagnosis of squamous cell carcinoma.

CLINICAL MANIFESTATIONS

A few patients have acute, self-limited pneumonia. Fever, productive cough, myalgia, and malaise usually resolve within a month. Pulmonary infiltrates clear slowly as *B. dermatitidis* disappears from the sputum.

In the vast majority of patients, blastomycosis has an indolent onset and a chronically progressive course. Fever, cough, weight loss, lassitude, skin lesions, and chest ache are common. Skin lesions favor exposed areas and enlarge over many weeks from pimples to well-circumscribed, verrucous, crusted, or ulcerated lesions ([Fig. 203-CD1](#)).

Pain and regional lymphadenopathy are minimal. Large chronic lesions may undergo central healing with scarring and contracture. Mucous membrane lesions resemble squamous cell carcinoma. Chest x-ray findings are abnormal in two-thirds of patients, with one or more pneumonic or nodular infiltrates. Calcification, hilar adenopathy, and large pleural effusions are rare. Osteolytic lesions may be found in nearly any bone and present as a cold abscess or a draining sinus. Extension to a contiguous joint may cause indolent swelling, pain, and restricted motion. Prostatic and epididymal lesions clinically resemble those of tuberculosis.

DIAGNOSIS

The diagnosis of blastomycosis is made by demonstration of the fungus in a culture of sputum, pus, or urine. An expert can diagnose blastomycosis on the basis of the appearance of the organism in wet smear or histopathologic section. The fungus may be visible in a sputum cytology smear but is easily overlooked.

TREATMENT

A few patients have developed only transitory lung lesions, but no guidelines are known to distinguish these patients from those whose disease will progress locally or disseminate. Therefore, every patient should receive treatment. Intravenous amphotericin B is the drug of choice for patients with rapidly progressive infections, severe illness, or central nervous system lesions. Skin and noncavitary lung lesions should be treated for about 8 to 10 weeks. The recommended total dose for an adult is about 2 g. Cavitary lung disease or infection extending beyond the lung and skin should be treated for about 10 to 12 weeks with ³2.5 g.

Oral itraconazole (200 mg twice daily with food) is the drug of choice for the treatment of patients who have indolent nonmeningeal blastomycosis of mild to moderate severity and who take the drug reliably. Therapy with itraconazole is continued for 6 to 12 months.

The mortality rate in appropriately treated cases is £15%.

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204. CRYPTOCOCCOSIS - John E. Bennett

ETIOLOGIC AGENT

Cryptococcosis is an infection caused by the yeastlike fungus *Cryptococcus neoformans*. This fungus reproduces by budding and forms round, yeastlike cells. Within the host and on certain culture media, a large polysaccharide capsule surrounds each yeast cell. The fungus grows well in smooth, creamy-white colonies on Sabouraud's or other simple media at 20 to 37°C. Identification of the organism is based on gross and microscopic appearance, biochemical test results, and growth at 37°C. The results of nucleic acid hybridization or the formation of brown pigment on Niger seed agar can also be used for identification.

The fungus has four capsular serotypes, designated A, B, C, and D. There are also two mating types. Coculture of opposite mating types creates a transient diploid state called *Filobasidiella neoformans* var. *neoformans* for serotypes A and D and *F. neoformans* var. *bacillispora* for serotypes B and C. Organisms not cultured under mating conditions are designated *C. neoformans* var. *neoformans* for serotypes A and D and *C. neoformans* var. *gattii* for serotypes B and C; a simple color medium distinguishes the two varieties.

EPIDEMIOLOGY

Weathered pigeon droppings commonly contain serotype A or D (*C. neoformans* var. *neoformans*). *C. neoformans* var. *gattii* has been isolated from the litter around trees of the species *Eucalyptus camaldulensis* and *E. tereticornis*. Eucalyptus isolates have so far typed as serotype B. The distribution of these eucalyptus species in Australia corresponds to the distribution of infections due to *C. neoformans* var. *gattii* in that country. The high prevalence of these trees in other subtropical climates has been postulated to explain the relative restriction of such infections to warm climates.

Cryptococcosis due to *C. neoformans* var. *neoformans* is a common complication of late infection with HIV. The incidence appears to be declining in some areas of the United States, probably as a result of highly active antiretroviral therapy and use of fluconazole for oropharyngeal candidiasis. Patients who have undergone solid-organ transplantation or glucocorticoid therapy and those with sarcoidosis are also at increased risk for infections with *C. neoformans* var. *neoformans*. Almost all such infections are caused by serotype A, although serotype D occurs in up to 20% of cases in Western Europe. Infections with var. *gattii* have been rare among AIDS patients and other immunocompromised patients, even in subtropical climates, where var. *gattii* infection occurs in previously healthy individuals.

Animals, particularly cats, can acquire cryptococcosis but have not transmitted the infection to other animals or to humans. The source from which humans acquire the infection is unknown, with the rare exception of cases acquired through a transplanted cornea, kidney, or other solid organ. Cryptococcosis is rare before puberty.

PATHOGENESIS AND PATHOLOGY

Infection is thought to be acquired by inhalation of fungus into the lungs. Pulmonary infection has a tendency toward spontaneous resolution and is frequently asymptomatic. Silent hematogenous spread to the brain leads to clusters of cryptococci in the perivascular areas of cortical gray matter, in the basal ganglia, and, to a lesser extent, in other areas of the central nervous system. The inflammatory response around these foci is usually scant. In the more chronic cases, a dense basilar arachnoiditis is typical. Lung lesions are characterized by intense granulomatous inflammation. Cryptococci are best seen in tissue by staining with methenamine silver or periodic acid-Schiff. Although a strongly positive result on mucicarmine staining of tissue is diagnostic, staining varies from intense to absent.

CLINICAL MANIFESTATIONS

Most patients have *meningoencephalitis* at the time of diagnosis. This form of the infection is invariably fatal without appropriate therapy; death occurs any time from 2 weeks to several years after the onset of symptoms. Early manifestations include headache, nausea, staggering gait, dementia, irritability, confusion, and blurred vision. Both fever and nuchal rigidity are often mild or lacking. Papilledema is evident in one-third of cases at the time of diagnosis. Cranial nerve palsies, typically asymmetric, occur in about one-fourth of cases. Other lateralized signs are rare. With progression of the infection, deepening coma and signs of brainstem compression appear. Autopsy often reveals cerebral edema in more acute cases and hydrocephalus in more chronic cases.

Pulmonary cryptococcosis causes chest pain in about 40% of patients and cough in 20%. The chest x-ray shows one or more dense infiltrates, which are often well circumscribed. Cavitation, pleural effusions, and hilar adenopathy are infrequent. Calcification is not evident, and fibrotic stranding is rarely noticeable.

Ten percent of patients with cryptococcosis have skin lesions, and the vast majority of patients with skin lesions have disseminated infection ([Fig. 204-CD1](#)). One or a few asymptomatic tiny papular lesions appear and slowly enlarge; they display a tendency toward central softening leading to ulceration. Osteolytic lesions occur in 4% of cases and usually present as a cold abscess. Rare manifestations of cryptococcosis include prostatitis, endophthalmitis, hepatitis, pericarditis, endocarditis, and renal abscess.

DIAGNOSIS

Fever and headache in a patient with AIDS or with risk factors for [HIV](#) infection suggest the possibility of cryptococcosis, toxoplasmosis, or central nervous system lymphoma. Evidence of a focal lesion on magnetic resonance imaging is unusual in cryptococcosis. Most cryptococcal cerebral mass lesions occur in patients infected with *C. neoformans* var. *gattii* who also have meningitis. In patients without AIDS, meningitis due to *C. neoformans* resembles that due to *Mycobacterium tuberculosis*, *Histoplasma capsulatum*, *Coccidioides immitis*, or metastatic cancer. Lumbar puncture is the single most useful diagnostic test. An india ink smear of centrifuged cerebrospinal fluid (CSF) sediment reveals encapsulated yeast in more than half of cases, although artifacts can cause confusion. In patients without AIDS, levels of glucose in CSF are reduced in half of all cases; protein levels are usually increased; and lymphocytic pleocytosis is usually

found. CSF abnormalities are less pronounced in patients with AIDS, although india ink smear is more often positive.

Approximately 90% of patients with cryptococcal meningoencephalitis, including all those with a positive [CSF](#) smear, have capsular antigen detectable in CSF or serum by latex agglutination. An enzyme immunoassay for cryptococcal antigen is also available. Occasional false-positive results in the above tests make culture the definitive diagnostic test and have prevented serum antigen from being a useful screening test in asymptomatic patients with AIDS. *C. neoformans* is often present in urine from patients with meningoencephalitis. Fungemia occurs in 10 to 30% of patients and is particularly common among patients with AIDS.

Pulmonary cryptococcosis mimics malignancy with regard to radiographic findings and symptoms. Sputum culture is positive in only 10% of cases, and serum antigen tests are positive in only one-third. Occasionally, *C. neoformans* appears in one or more sputum specimens as an endobronchial saprophyte. Biopsy is usually required for diagnosis.

Cutaneous cryptococcosis may be mistaken for a comedo, basal cell carcinoma, or sarcoidosis. In patients with AIDS, skin lesions may be numerous and are sometimes mistaken for molluscum contagiosum. Biopsy reveals myriad cryptococci. Osseous cryptococcosis resembles tuberculosis.

TREATMENT

Patients with AIDS and cryptococcosis are treated initially with intravenous amphotericin B (0.7 mg/kg daily) for at least 2 weeks and until their clinical condition is stable; thereafter, they receive fluconazole. The addition of flucytosine (25 mg/kg every 6 h) to amphotericin B for 2 weeks has minimal impact on morbidity and mortality. After treatment with amphotericin B, fluconazole (400 mg) is given once daily. Daily doses of 800 mg have been used with marginal changes in toxicity or efficacy. The addition of flucytosine to fluconazole increases gastrointestinal intolerance. After infection is controlled, treatment with a smaller dose of fluconazole (200 mg/d) is continued indefinitely. Itraconazole is less effective than fluconazole for maintenance therapy. It is not yet known whether patients whose CD4⁺ T lymphocyte counts have exhibited a sustained rise in response to antiretroviral therapy can safely discontinue fluconazole maintenance therapy.

In patients without AIDS, the therapeutic goal is to cure the infection, not merely to control its symptoms. A single intensive course is given until cultures from all previously positive sites (particularly [CSF](#)) become convincingly negative. Normalization of the glucose level in lumbar CSF is desirable, but complete clearing of CSF or serum antigen during therapy is not essential. Amphotericin B (0.6 to 0.7 mg/kg daily for 3-10 weeks) is the best-studied regimen. Flucytosine has been added to amphotericin B to accelerate the culture response, but grave toxicity can result unless flucytosine blood levels are kept below 100 µg/mL. Case reports have described patients without HIV infection who have responded to fluconazole or liposomal amphotericin B, but the dose and duration of treatment required to cure cryptococcal meningitis are undefined. Amphotericin B lipid complex and amphotericin B colloidal dispersion are not recommended pending further study.

Hydrocephalus may be the presenting manifestation or a later complication of cryptococcosis. Blindness, dementia, and personality change are among the other sequelae. Daily lumbar puncture or [CSF](#) shunting has been advocated -- in the hope of averting permanent blindness -- for patients with marked cerebral edema who have incipient blurred vision.

Patients with extraneural cryptococcosis most often require treatment with intravenous amphotericin B, with or without flucytosine. Observation or excision of lesions may suffice for some patients who have previously been healthy; who have a single focus in lung, skin, or bone; and who have no cryptococci in CSF, urine, or blood.

PREVENTION

Fluconazole (200 mg/d) has been shown to decrease the incidence of cryptococcosis in HIV-infected patients with CD4+ cell counts of <200/uL and particularly in those with counts of <50/uL. Weekly fluconazole has not provided this protection. Daily fluconazole has not conferred a survival advantage; in light of its cost and the currently low incidence of cryptococcosis in patients with AIDS in the United States, prophylaxis is strongly discouraged.

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205. CANDIDIASIS - John E. Bennett

ETIOLOGIC AGENTS

Candida albicans is the most common cause of mucosal candidiasis and is responsible for about half of all cases of candidemia in hospitalized patients. A small proportion of *C. albicans* isolates have been transferred to a new species, *C. dubliniensis*. *C. tropicalis*, *C. parapsilosis*, *C. guilliermondii*, *C. glabrata* (formerly *Torulopsis glabrata*), *C. krusei*, and a few other *Candida* species also cause potentially fatal bloodstream infection. Many of these non-*albicans* species can enter the bloodstream through an intravascular catheter. *Candida* species, taken together, are the fifth most common cause of nosocomial bloodstream infections in the United States.

All *Candida* species pathogenic for humans are also encountered as commensals of humans, particularly in the mouth, stool, and vagina. These species grow rapidly at 25° to 37°C on simple media as oval, budding cells. In special culture media and in tissue, hyphae or elongated branching structures called *pseudohyphae* are formed. *C. glabrata* differs from other members of the genus in that it forms no true hyphae or pseudohyphae in vitro or in infected tissue. *C. albicans* and *C. dubliniensis* can be identified presumptively by their ability to form germ tubes in serum or by the formation of thick-walled large spores called *chlamydospores*. Final identification of all *Candida* species requires biochemical tests.

PATHOGENESIS

Candidiasis is often preceded by increased colonization of the mouth, vagina, and stool with *Candida* due to broad-spectrum antibiotic therapy. Additional local and systemic factors favor infection. Oropharyngeal thrush is particularly likely to occur in neonates and in patients with diabetes mellitus, HIV infection, or dentures. Vulvovaginal candidiasis ([Chap. 132](#)) is especially common in the third trimester of pregnancy. *Candida* from the perineum can enter the urinary tract via an indwelling bladder catheter. Cutaneous candidiasis most often involves macerated skin, such as that in the diapered area of infants, under pendulous breasts, or on hands constantly in water or covered by occlusive gloves. *Candida* can pass from the colonized surface into deep tissue when the integrity of the mucosa or skin is violated, as, for example, by perforation of the gastrointestinal tract through trauma, surgery, or peptic ulceration or by mucosal damage due to cytotoxic agents used for cancer chemotherapy. Although *Candida* is not normally a resident of the skin, secretions from the mouth, rectum, or vagina as well as drainage from surgical wounds or tracheostomy sites can contaminate the hub or skin site of a catheter in an umbilical or central vein. Intravenous drug abuse or third-degree burns can also provide a skin portal for *Candida* that can lead to deep candidiasis. Once *Candida* has passed the integumentary barrier, very low birth weight (in neonates) and neutropenia or glucocorticoid therapy (in any patient) markedly compromise host defense. Hematogenous seeding is particularly evident in the retina, kidney, spleen, and liver.

CLINICAL MANIFESTATIONS

Oral thrush ([Figs. 205-CD1](#) and [205-CD2](#)) presents as discrete and confluent adherent

white plaques on the oral and pharyngeal mucosa, particularly in the mouth and on the tongue. These lesions are usually painless, but fissuring at the corners of the mouth can be painful. Unexplained oropharyngeal thrush raises the possibility of HIV infection. Oral thrush is common in acute HIV infection and becomes increasingly common as the CD4⁺ cell count falls. At CD4⁺ counts <50/uL, esophageal thrush also becomes common. HIV infection appears not to be an independent risk factor for vulvovaginal thrush.

Cutaneous candidiasis ([Figs. 205-CD3, 205-CD4](#), and [205-CD5](#)) presents as red macerated intertriginous areas, paronychia, balanitis ([Fig. 205-CD6](#)), or pruritus ani. Candidiasis of the perineal and scrotal skin may be accompanied by discrete pustular lesions on the inner aspects of the thighs. *Chronic mucocutaneous candidiasis* or *candidal granuloma* typically presents as circumscribed hyperkeratotic skin lesions, crumbling dystrophic nails, partial alopecia in areas of scalp lesions, and both oral and vaginal thrush. Systemic infection is very rare, but disfigurement of the face and hands can be severe. Other findings may include chronic epidermophytosis, dental dysplasia, and hypofunction of the parathyroid, adrenal, or thyroid gland. A variety of defects in T cell function have been described in these patients. Vulvovaginal thrush ([Chap. 132, Fig. 205-CD7](#)) causes pruritus, discharge, and sometimes pain on intercourse or urination. Speculum examination reveals an inflamed mucosa and a thin exudate, often with white curds.

Esophageal candidiasis is often asymptomatic but can cause substernal pain or a sense of obstruction on swallowing. Most lesions are in the distal third of the esophagus and appear on endoscopy as areas of redness and edema, focal white patches, or ulcers. Biopsy or brushing is required for diagnosis and for detection of concomitant infections, particularly herpes simplex in patients with hematologic malignancies and cytomegalovirus infection in AIDS patients. Esophagography (barium swallow) is diagnostically insensitive but may reveal spasm or mucosal irregularities. *Candida* esophagitis can cause bleeding and impaired alimentation. Hematogenous dissemination from the esophagus probably occurs in some neutropenic patients but is rarely reported in HIV-infected patients.

Candida can cause cystitis, pyelitis, or renal papillary necrosis in an obstructed urinary tract. When a colonized urinary tract is operated on or instrumented, candidemia may result. However, most patients with *Candida* cultured from the urine simply have bladder colonization from a Foley catheter or a sizable volume of residual urine. Contamination of a voided midstream specimen by vaginal *Candida* is also common.

Candidemia originating from an intravascular catheter may clear in the immunocompetent patient when the catheter is removed. Focal seeding of the retina can take place even if candidemia clears and the patient becomes afebrile. Unilateral or bilateral small white retinal exudates appear within 2 weeks of the onset of candidemia. Lesions may regress spontaneously or enlarge slowly. The vitreous humor becomes cloudy, and the patient notices blurring, ocular pain, or a scotoma. Retinal detachment, vitreous abscess, and extension to the anterior chamber can occur over the ensuing weeks. These retinal lesions, present in ~10% of nonneutropenic patients with candidemia, are the principal reason that systemic antifungal therapy is recommended for all patients with candidemia. Funduscopy should be performed to be certain that

retinal lesions, if present, resolve completely. Most cases with ocular involvement have occurred in nonneutropenic patients. In contrast, so-called hepatosplenic candidiasis is usually recognized in patients with acute leukemia who are recovering from profound neutropenia. This entity, better called *chronic disseminated candidiasis*, originates from intestinal seeding of the portal and venous circulation. Fever, modestly elevated serum concentrations of alkaline phosphatase, and multiple small abscesses evident on ultrasonography, magnetic resonance imaging, or computed tomography of the liver, spleen, or kidney suggest the diagnosis. During acute candidemia in neutropenic patients, small erythematous papules may appear anywhere on the skin ([Plate IID-57D](#), [Fig. 205-CD8](#)). If the patient does not expire promptly from disseminated candidiasis, the lesions will develop a necrotic center. Painful muscle lesions may also be found. Punch biopsy of a skin lesion helps distinguish this extremely grave condition from *Malassezia* folliculitis, a similar-appearing but benign condition that can involve the cape area of the chest or the extremities of a sweaty febrile patient.

Hematogenous seeding in the neutropenic patient is occasionally visible radiologically as tiny pulmonary nodules. *Candida* pneumonia, apart from hematogenous candidiasis, is very rare. Organisms seeding a native or prosthetic cardiac valve originate principally from central venous catheters; occasionally, valvular seeding is encountered in intravenous drug abusers. Emboli to large arteries, such as the iliac or femoral artery, are characteristic. Intravenous injection of impure brown heroin has caused a clinical syndrome consisting of *Candida* endophthalmitis and purulent folliculitis, sometimes accompanied by vertebral osteomyelitis. This diffuse folliculitis favors hairy areas, including the scalp and bearded facial skin.

Candida can cause indolent arthritis, most commonly of the knee, in patients who have received glucocorticoid injections into the joint, in patients who are immunosuppressed, and in low-birth-weight neonates. Prosthetic joints may become infected during implantation. Scanty growth of *Candida* from joint fluid can cause the laboratory to incorrectly dismiss the organism as a contaminant.

Hematogenous dissemination can lead to brain abscess or chronic meningitis. Diagnosis of infections of ventriculoperitoneal shunts is difficult because symptoms are indolent and cultures of lumbar fluid are usually sterile.

DIAGNOSIS

Demonstration of pseudohyphae on wet smear with confirmation by culture is the procedure of choice for diagnosing superficial candidiasis ([Fig. 205-CD9](#)). Scrapings for the smear may be obtained from skin, nails, and oral and vaginal mucosa. Culture alone is not diagnostic; however, recovery of *Candida* species from multiple superficial sites in immunosuppressed patients may portend visceral invasion.

Deeper lesions due to *Candida* may be diagnosed by histologic section of biopsy specimens or by culture of cerebrospinal fluid, blood, joint fluid, or surgical specimens. Blood cultures are useful in the diagnosis of *Candida* endocarditis and intravenous catheter-induced sepsis but are positive less often in other forms of disseminated disease. Serologic tests for antibody or antigen are not useful.

TREATMENT

Cutaneous candidiasis of macerated areas responds to measures that reduce moisture and chafing plus topical application of an antifungal agent in a nonocclusive base. Nystatin powder or a cream containing ciclopirox or an azole is useful. Clotrimazole, miconazole, econazole, ketoconazole, sulconazole, and oxiconazole are available as creams or lotions. *Candida* vulvovaginitis responds better to an azole than to nystatin suppositories. There is little difference in efficacy among miconazole, clotrimazole, tioconazole, butoconazole, and terconazole vaginal formulations. Systemic treatment of *Candida* vulvovaginitis with a single 150-mg capsule of fluconazole is more convenient than topical treatment but also poses a higher risk of adverse effects. Clotrimazole troches, used five times a day, are more effective in oral and esophageal candidiasis than nystatin suspension. Oral fluconazole (100 to 200 mg once daily) is more convenient and more effective in esophagitis than clotrimazole troches. Esophagitis not responding to fluconazole may warrant repeat endoscopy to exclude other conditions. Itraconazole suspension (100 to 200 mg/d) alleviates *Candida* esophagitis in some patients in whom fluconazole treatment fails. Amphotericin B suspension has limited use but can be tried in patients whose oropharyngeal candidiasis does not respond to azoles.

Management of recurrent oropharyngeal candidiasis in the HIV-infected patient presents special problems. Patients with CD4⁺ cell counts <100/uL who have received prolonged fluconazole therapy are at risk of developing azole resistance, requiring an increased dose to mount a response, relapsing early, and eventually failing to respond well to any dose of fluconazole. The increasing azole resistance in this population suggests that HIV-infected patients with oropharyngeal candidiasis should be treated for each individual episode and that only when episodes become intolerably frequent should weekly or daily preventive therapy be given and even then at the lowest dose required to maintain remission. In contrast, AIDS patients with *Candida* esophagitis are so prone to relapse that preventive therapy with fluconazole is recommended for all proven cases. Most HIV-infected patients with azole-resistant oropharyngeal candidiasis also have esophagitis. Nearly all patients with azole-resistant oropharyngeal or esophageal candidiasis respond to intravenous amphotericin B (0.3 to 0.5 mg/kg daily) but relapse promptly after the completion of therapy.

Bladder thrush responds to bladder irrigations with amphotericin B (50 ug/mL for 5 days). If no bladder catheter is in place, oral fluconazole can be used to control candiduria. In all forms of superficial candidiasis, relapse after successful treatment is common unless the underlying factor can be eliminated.

Intravenous amphotericin B is the drug of choice in disseminated candidiasis. The deoxycholate formulation is usually given at a dosage of 0.5 to 0.7 mg/kg daily. Open, noncomparative studies of the lipid formulations of amphotericin B have suggested that they may be useful in disseminated candidiasis, but the optimal dose and formulation remain unknown. Fluconazole in an adult dose of 100 mg/d is probably the drug of choice for chronic mucocutaneous candidiasis.

In immunocompetent patients with intravenous catheter-acquired *C. albicans* fungemia, the catheter should be removed in conjunction with the administration of either

fluconazole (400 mg/d) or amphotericin B (0.5 mg/kg daily). Patients with suppurative phlebitis of a peripheral vein should have the infected portion of the vein excised. Therapy for candidemia is continued for 2 weeks after the patient becomes afebrile. The *Candida* species involved should be considered in choosing between fluconazole and amphotericin B. *C. krusei* and *C. inconspicua* are rare causes of candidemia but are resistant to fluconazole in vitro. *C. glabrata* exhibits intermediate susceptibility to fluconazole, but too few cases have been studied to determine whether candidemia involving that species will respond as well to fluconazole as to amphotericin B. Strains of *C. lusitanae* resistant to amphotericin B but susceptible to azoles have been encountered. Intravenous amphotericin B, with or without flucytosine, is the preferred treatment for *Candida* endophthalmitis, although cures have been reported with fluconazole. Pars plana vitrectomy may facilitate diagnosis and cure when a *Candida* vitreous abscess is present. Injection of amphotericin B into the vitreous humor can also be helpful.

Injection of amphotericin B into an infected joint, pleural cavity, or peritoneum is rarely indicated. Removal of prostheses, including prosthetic joints, cardiac valves, peritoneal dialysis catheters, and central venous catheters, is usually essential. Collections of pus, such as those in the postoperative abdomen, need to be drained surgically or by percutaneous, computed tomography-guided catheterization; an exception relates to the numerous small abscesses in liver, spleen, or kidney in chronic disseminated candidiasis, which cannot be drained effectively and require prolonged antifungal therapy. In general, treatment should continue until the patient with chronic disseminated candidiasis has been afebrile and nonneutropenic for at least 2 weeks. Defects may persist on imaging studies long after cure. Relapse during another episode of neutropenia is common unless the patient is receiving amphotericin B. Repeat cytotoxic therapy or even bone marrow transplantation can be undertaken in patients with prior chronic disseminated candidiasis, but amphotericin B should be given prophylactically during neutropenia.

Fluconazole can decrease the incidence of deep candidiasis in recipients of allogeneic bone marrow transplants when 400 mg is given daily until engraftment. Although the incidence of superficial candidiasis is also decreased by fluconazole prophylaxis, superficial infection can be readily detected and treated. Aspergillosis is not prevented by prophylactic fluconazole. Studies of leukemic and other neutropenic patients have found no beneficial effect of prophylactic fluconazole.

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206. ASPERGILLOSIS - John E. Bennett

ETIOLOGIC AGENTS

Aspergillus fumigatus is the most common cause of aspergillosis, but *A. flavus*, *A. niger*, and several other species can also cause disease. *Aspergillus* is a mold with septate hyphae about 2 to 4 μm in diameter. The fungus is identified by its gross and microscopic appearance in culture.

PATHOGENESIS AND PATHOLOGY

All the common species of *Aspergillus* that cause disease in humans are ubiquitous in the environment, growing on dead leaves, stored grain, compost piles, hay, and other decaying vegetation. Inhalation of *Aspergillus* spores must be extremely common, but disease is rare. Invasion of lung tissue is confined almost entirely to immunosuppressed patients, in roughly 90% of whom two of the following three conditions will be operative: a granulocyte count in peripheral blood of $<500/\mu\text{L}$, treatment with supraphysiologic doses of adrenal glucocorticoids, and a history of treatment with cytotoxic drugs such as cyclosporine. Invasive aspergillosis is an occasional complication of AIDS. *Aspergillus* infection is characterized by hyphal invasion of blood vessels, thrombosis, necrosis, and hemorrhagic infarction. Chronic granulomatous disease of childhood also predisposes to invasive pulmonary aspergillosis, but in that situation the inflammatory response is a pyogranuloma and blood vessel invasion is rare.

Massive inhalation of *Aspergillus* spores by healthy persons can lead to acute, diffuse, self-limited pneumonitis. Epithelioid granulomas with giant cells and central pyogenic areas containing hyphae are detected in these cases. Spontaneous recovery taking several weeks is the usual course.

Aspergillus can colonize the damaged bronchial tree, pulmonary cysts, or cavities of patients with underlying lung disease. Balls of hyphae within cysts or cavities (aspergillomas), usually in the upper lobe, may reach several centimeters in diameter and may be visible on chest x-ray. Tissue invasion does not occur. The term *allergic bronchopulmonary aspergillosis* denotes the condition of patients with preexisting asthma who have eosinophilia, IgE antibody to *Aspergillus*, and fleeting pulmonary infiltrates from bronchial plugging.

CLINICAL MANIFESTATIONS

Endobronchial saprophytic pulmonary aspergillosis presents as chronic productive cough, often with hemoptysis, in a patient with prior chronic lung disease, such as tuberculosis, sarcoidosis, bronchiectasis, or histoplasmosis. *Aspergillus* may be spread from its endocavitary or endobronchial site to the pleura during the course of bacterial lung abscess or surgery. Patients reported to have chronic necrotizing *Aspergillus* pneumonia appear in most instances to have had saprophytic endobronchial colonization and a pulmonary process attributable to another disease, with or without superimposed bacterial infections. Patients with chronic pneumonia and *Aspergillus* in the sputum should be assumed to have either pneumonia of a different etiology (e.g., histoplasmosis) or *Aspergillus* pneumonia with underlying immunosuppression (e.g.,

chronic granulomatous disease or infection with HIV).

Invasive aspergillosis in the immunocompromised host presents as an acute, rapidly progressive, densely consolidated pulmonary infiltrate and is most common among patients with acute leukemia and recipients of tissue transplants. Infection progresses by direct extension across tissue planes and by hematogenous dissemination to lung, brain, and other organs. Computed tomography (CT) has been particularly valuable in suggesting the diagnosis of invasive pulmonary aspergillosis in patients with neutropenia. The earliest CT finding is one or more small pulmonary nodules. As a nodule enlarges, the dense central core of infarcted tissue becomes surrounded by edema or hemorrhage, forming a hazy rim called the *halo sign*. This rim disappears in a few days as the dense core enlarges. When bone marrow function recovers, the infarcted central core cavitates, creating the *crescent sign*. *Aspergillus* may invade immunosuppressed patients through the skin at a site of minor trauma or through the upper airway mucosa. Early lesions in the nose should be sought in patients with neutropenia who have fever and minimal epistaxis. Scarlet-red patches of the mucosa rapidly become necrotic and white, then black. Rapid extension into the adjacent paranasal sinus, orbit, or face is usual, with or without the appearance of lung lesions.

Aspergillus sinusitis in immunocompetent patients may take three forms. A ball of hyphae may form in a chronically obstructed paranasal sinus, without tissue invasion. Much less commonly, a chronic, fibrosing granulomatous inflammation associated with *Aspergillus* hyphae within tissue may begin in the sinus and spread slowly to the orbit and the brain. *Aspergillus* is also a cause of allergic fungal sinusitis, but dark-walled fungi (e.g., *Cladosporium*, *Alternaria*) are more common in this setting. Patients usually have a history of chronic allergic rhinitis, sometimes with nasal polyps, but are otherwise healthy, presenting with painless proptosis, nasal obstruction, or dull aching pain. On [CT](#) or magnetic resonance imaging, a solid soft tissue mass pushing out the lateral wall of the ethmoid sinus or the medial wall of the maxillary sinus may be detected. On sinus exploration, the mucosa is found to be thickened and inflamed but intact. Within the sinus cavity, sticky mucus with strands of neutrophils, eosinophils, Charcot-Leyden crystals, and occasional hyphae can be found.

Aspergillosis in HIV-infected patients most commonly involves the lung, presenting as fever, cough, and dyspnea. Typically, the CD4 cell count is below 50/uL. Roughly half of these patients have neutropenia or have recently been treated with glucocorticoids. Bilateral diffuse or focal pulmonary infiltrates with a tendency to cavitate constitute the most common radiologic manifestation. Well-localized, white, necrotic pseudomembranes full of hyphae or ulcers may develop in the trachea or the major bronchi. Progression of bronchitis to pneumonia is usual, but hematogenous dissemination is uncommon. Either allergic or invasive *Aspergillus* sinusitis can occur in HIV-infected patients; the allergic form can develop even at CD4 cell counts above 50/uL.

The growth of *Aspergillus* on cerumen and detritus within the external auditory canal is termed *otomycosis*. Trauma to the cornea may cause *Aspergillus* keratitis. Endophthalmitis follows the introduction of *Aspergillus* into the globe by trauma or surgery. *Aspergillus* may infect intracardiac or intravascular prostheses.

DIAGNOSIS

The repeated isolation of *Aspergillus* from sputum or the demonstration of hyphae in sputum or bronchoalveolar lavage fluid suggests endobronchial colonization or infection. Even a single isolation of *Aspergillus* from the sputum of a neutropenic patient with pneumonia, particularly a child or a nonsmoker, suggests the diagnosis of invasive aspergillosis. In patients with advanced AIDS, fever, and cough, the isolation of *Aspergillus* from respiratory secretions raises the possibility of aspergillosis and thus should prompt bronchoscopy. Fungus ball of the lung is usually detectable by chest x-ray. IgG antibody to *Aspergillus* antigens is demonstrable in the serum of many colonized patients and of virtually all patients with fungus ball.

Biopsy is usually required for the diagnosis of invasive aspergillosis of the lung, nose, paranasal sinus, bronchi, or sites of dissemination. Blood cultures are rarely positive, even in patients with infected cardiac valves (native or prosthetic). Detection of galactomannan antigen in serum suggests the diagnosis, but false-positives are frequent, particularly in children. *Aspergillus* hyphae can be identified presumptively by histology, but culture is required for confirmation and for determination of the species. Only culture can reliably distinguish aspergillosis from pseudallescheriasis; drug therapy for these two diseases differs.

TREATMENT

Patients with severe hemoptysis due to fungus ball of the lung may benefit from lobectomy. Poor pulmonary function in residual lung and dense pleural adhesions around the lesion can complicate the resection. Systemic chemotherapy is of no value in endobronchial or endocavitary aspergillosis.

Treatment with intravenous amphotericin B (1.0 to 1.5 mg/kg daily) has resulted in the arrest or cure of invasive aspergillosis when immunosuppression is not severe. Liposomal amphotericin B at daily doses of 1 to 4 mg/kg has given results that seem roughly comparable to those obtained with amphotericin B deoxycholate. Itraconazole (200 mg twice daily) is useful in some less immunosuppressed patients with indolent or slowly progressive invasive aspergillosis. Surgery is the only treatment needed for fungus ball of the sinus and for allergic fungal sinusitis. Antifungal therapy has little effect on either entity if used alone, but chronic suppressive therapy has been begun postoperatively for relapse of allergic fungal sinusitis. The prognosis for cure of invasive aspergillosis in the paranasal sinus is very poor when the patient has profound and unremitting neutropenia. The prognosis is better in less immunosuppressed patients.

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207. MUCORMYCOSIS - John E. Bennett

ETIOLOGIC AGENTS

Species of *Rhizopus*, *Rhizomucor*, and *Cunninghamella* are the most common causes of mucormycosis, but species of *Apophysomyces*, *Saksenaea*, *Mucor*, and *Absidia* also are occasionally responsible for this infection. The organism in tissue is composed of broad, rarely septate hyphae of uneven diameter (6 to 50 μ m). The organisms are inexplicably difficult to grow from infected tissue. When growth does take place, it is rapid and profuse on most media at room temperature. Identification is based on the gross and microscopic appearance of the mold.

Zygomycosis is a term that includes mucormycosis and entomophthoromycosis. The latter is a tropical infection of the subcutaneous tissue or paranasal sinuses caused by species of *Basidiobolus* and *Conidiobolus*, respectively.

EPIDEMIOLOGY AND PATHOLOGY

Rhizopus and *Rhizomucor* species are ubiquitous, appearing on decaying vegetation, dung, and foods of high sugar content. Mucormycosis is uncommon and is largely confined to patients with serious preexisting diseases. Mucormycosis originating in the paranasal sinuses and nose predominantly affects patients with poorly controlled diabetes mellitus. Patients who have undergone organ transplantation, who have a hematologic malignancy, or who are receiving long-term deferoxamine therapy are predisposed to mucormycosis of either sinus or lung. Gastrointestinal mucormycosis occurs in a variety of conditions, including uremia, severe malnutrition, and diarrheal diseases. The infection is acquired from nature, with no person-to-person spread. In all forms of mucormycosis, vascular invasion by hyphae is a prominent feature. Ischemic or hemorrhagic necrosis is the foremost histologic finding.

CLINICAL MANIFESTATIONS

Mucormycosis originating in the nose and paranasal sinuses produces a characteristic clinical picture. Low-grade fever, dull sinus pain, and sometimes nasal congestion or a thin, bloody nasal discharge are followed in a few days by double vision, increasing fever, and obtundation. Examination reveals a unilateral generalized reduction of ocular motion, chemosis, and proptosis. The nasal turbinates on the involved side may be dusky red or necrotic. A sharply delineated area of necrosis, strictly respecting the midline, may appear in the hard palate. The skin of the cheek may become inflamed. Fungal invasion of the globe or ophthalmic artery leads to blindness. Opacification of one or more sinuses is detected by computed tomography (CT) or by magnetic resonance imaging (MRI). Carotid arteriography may show invasion or obstruction of the carotid siphon. Coma is due to direct invasion of the frontal lobe. Early symptoms mimic those of bacterial sinusitis. Clouding of the sensorium may be attributed to diabetic acidosis. Cavernous sinus thrombosis may be considered when orbital invasion occurs. Without treatment, the patient may die after an interval ranging from a few days to a few weeks.

Pulmonary mucormycosis manifests as progressive severe pneumonia accompanied by

high fever and toxicity. The necrotic center of large infiltrates may cavitate. Hematogenous spread to other areas of the lung, as well as to the brain and other organs, is common. Survival beyond 2 weeks is unusual. Gastrointestinal invasion presents as one or more ulcers that tend to perforate. Hematogenous dissemination can originate from the gastrointestinal tract, lung, or paranasal sinuses. Sometimes no portal of entry can be found.

DIAGNOSIS

[CT](#) or [MRI](#) is very helpful in assessing the extent of sinusitis before surgery and in evaluating the patient afterward. CT is better for detecting bony erosion; MRI better visualizes extension into the frontal lobe or carotid artery in the siphon. Lesions of the lung and craniofacial structures are best diagnosed by biopsy and histologic section. Cultural confirmation should be attempted. Wet smear of crushed tissue can provide a rapid diagnosis. Cultures of blood and cerebrospinal fluid are negative. Smear and culture of sputum may be positive during cavitation of a lung lesion.

TREATMENT

Regulation of diabetes mellitus and a decrease in the dose of immunosuppressive drugs facilitate the treatment of mucormycosis. Extensive debridement of craniofacial lesions appears to be very important. Orbital exenteration may be required. Intravenous amphotericin B is clearly of value in craniofacial mucormycosis and should be employed in the other forms of mucormycosis as well. The maximal tolerated doses are given until progression is halted. With the deoxycholate formulation, 1 to 1.5 mg/kg daily is indicated. Therapy is continued for a total of 10 to 12 weeks. Azoles are of no value. Appropriate management results in cure of about half of craniofacial infections. The survival of patients with pulmonary, gastrointestinal, or disseminated mucormycosis is rare.

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208. MISCELLANEOUS MYCOSES AND ALGAL INFECTIONS - John E. Bennett

CHROMOBLASTOMYCOSIS ([FIG. 208-CD1](#))

This chronic subcutaneous mycosis, rarely seen in the United States, presents as a verrucoid, ulcerated, or crusted skin lesion. The disease follows the introduction of any of several fungi into subcutaneous tissue by thorns or bits of vegetation. The infection spreads over ensuing months and years to contiguous tissue, causing few symptoms. The appearance of thick-walled, dark-colored, rounded forms ("copper pennies") in histopathologic section is diagnostic. Surgical excision is the treatment of choice. Itraconazole has ameliorated some relatively small and incompletely excised lesions.

DERMATOPHYTOSIS

Definition Dermatophytosis, also known as ringworm or tinea, is a chronic fungal infection of the skin, hair, or nails.

Etiology Species of *Trichophyton*, *Microsporum*, and *Epidermophyton* are called *dermatophytes*. These organisms grow in and remain confined to the keratinous structures of the body. Other mycoses, such as candidiasis, pityriasis versicolor, and tinea nigra, sometimes include fungal invasion of keratinous structures but traditionally are not called dermatophytoses.

Pathology and Pathogenesis Dermatophyte species are referred to as *anthropophilic*, *zoophilic*, or *geophilic*, depending on whether their usual reservoir in nature appears to be humans, animals, or soil, respectively. The infectivity of organisms from all these sources is low, and outbreaks are largely confined to occasional clusters of cases of scalp infection in children. Acquisition of a dermatophytosis appears to be favored by minor trauma, maceration, and poor hygiene of the skin. Infection does not seem to confer solid immunity: Repeated infection with the same species is common, particularly with anthropophilic species. The infrequency of scalp infection among adults has been attributed to local factors rather than immunity.

Invasion of the stratum corneum by dermatophytes may cause inflammation that is either mild or (particularly with zoophilic fungi) intense. Shedding of the stratum corneum is increased by inflammation. To the extent that fungal growth cannot keep up with shedding, inflammation may help terminate infection. Conversely, infection is probably favored when shedding is reduced by treatment with glucocorticoids and cytotoxic drugs. Antifungal drugs interfere with the ability of fungal growth to keep up with shedding.

Clinical Manifestations The disease varies with the site of infection and the fungal species involved. Foot infection (athlete's foot, tinea pedis) may present as fissuring of the toe webs, scaling of the plantar surfaces, or vesicles around the toe webs and soles. Interdigital lesions may be pruritic or, when bacterial superinfection occurs, may be painful. Hand infection is less common but resembles foot infection.

Scalp dermatophytosis (tinea capitis; [Fig. 208-CD4](#)) is characterized by areas of alopecia and scaling. In so-called endothrix infection, the hair shaft breaks off at the skin

surface, leaving the hairs visible as black dots in the scalp. Some forms of scalp infection include an area of intense boggy suppuration called a *kerion*.

Dermatophytosis of the glabrous skin (tinea corporis, [Plate IID-51](#)) presents as circumscribed lesions with a wide variety of appearances, including scales, vesicles, and pustules. Inflammation may be minimal or intense. Central healing of less inflamed lesions may take place. The serpiginous border of inflammation is the source of the name *ringworm*.

Dermatophytosis of the bearded area (tinea barbae) appears as a pustular folliculitis. Onychomycosis (tinea unguium; [Fig. 208-CD5](#)) presents as a white discoloration of the nails or as thickening, chalkiness, and crumbling of the nails. Peeling and fissuring of paronychia nail folds or keratotic debris under the nail edge also may be evident.

Diagnosis Discolored hairs, scales, and keratotic debris under infected nails should be collected for KOH smear and culture. In the scraping of skin lesions, a drop of water on the skin site may keep the removed scales from flying off and thus may aid in their collection. Culture is important in distinguishing dermatophytes from *Candida* and fungal saprophytes growing in keratinaceous debris.

TREATMENT

Noninflammatory lesions of the trunk, groin, hands, and feet usually respond to twice-daily applications of clotrimazole, miconazole, ketoconazole, econazole, naftifine, terbinafine, or ciclopirox olamine cream. Hyperkeratotic lesions of the palms and soles respond slowly to these agents and may benefit from Whitfield's ointment initially to thin the keratin. Ointment should not be used between the toes, in the groin, or in the gluteal crease because maceration promotes bacterial infection.

Ringworm that is moderately severe, that is unresponsive to topical therapy, or that involves the scalp, nails, or bearded area should be treated systemically. Once-daily therapy with itraconazole (200 mg), terbinafine (250 mg), microcrystalline griseofulvin (500 mg), or ultramicrocrystalline griseofulvin (375 mg) is effective. Treatment must be continued until all infected keratin is gone. Cutting off infected hair and cleansing interdigital webs can expedite cure. Secondary bacterial infection of the foot may require soaks or antibacterial agents. The likelihood of relapse of dermatophyte foot infections may be decreased by keeping the feet clean and dry. For nail infections, itraconazole or terbinafine is preferred. In distal subungual onychomycosis, a single course of either drug results in initial improvement in half of patients, of whom half relapse. Results are better with fingernails than with toenails and for more distal rather than lateral nail involvement. To save money, itraconazole can be given as a double dose (400 mg) for 1 week each month with only marginal loss of efficacy. The duration of therapy is 2 to 3 months for fingernails and 4 to 6 months for toenails.

PROTOTHECOSIS

Prototheca species are ubiquitous achlorophyllic algae that enter the skin through contaminated wounds and cause localized infections in the olecranon bursa, skin, subcutaneous tissue, tendon sheaths, or deeper tissue. Diagnosis is based on culture or

histopathologic demonstration of sporangia with endospores in tissue. Surgical debridement and treatment with intravenous amphotericin B are useful.

FUSARIOSIS

Fusarium species can cause localized or hematogenously disseminated infection. Localized infection results from contaminated wounds. Almost all patients with hematogenously disseminated infection are severely immunosuppressed and profoundly neutropenic. Skin lesions occur in two-thirds of patients. Several painful red indurated lesions appear on the extremities or sometimes the trunk. These lesions often develop an ecchymotic center that ulcerates. A portal of infection is not usually apparent. Blood cultures have been positive in 59% of cases. Amphotericin B is probably the drug of choice for the treatment of fusariosis, but recovery depends on the diminution of neutropenia.

MALASSEZIA INFECTION (PITYRIASIS)

Malassezia furfur is part of the normal flora of the human skin but can cause tinea (pityriasis) versicolor ([Fig. 208-CD6](#)) or catheter-acquired sepsis. Tinea versicolor appears as asymptomatic, well-delineated, hyperpigmented or hypopigmented macules centered on the upper trunk and upper arms. Confluent lesions may cover large areas, making the border difficult to find. A fine "branny" scale or folliculitis is sometimes visible. When examined microscopically by KOH mount, skin sections are seen to contain characteristic round and elongated cells ([Fig. 208-CD7](#)). On inspection with Wood's light, lesions either do not fluoresce or appear yellow-green. *Erythrasma* ([Fig. 141-CD1](#)) resembles tinea versicolor but is characterized by gram-positive bacilli on smear and coral-red fluorescence. Azole creams are effective for the treatment of small areas of tinea versicolor; however, the application of selenium sulfide shampoo (Selsun) for 10 min daily, followed by showering to remove the shampoo, is more practical for large areas. Itraconazole is also effective. Catheter-acquired sepsis due to *M. furfur* develops in patients (particularly neonates) receiving intravenous lipid. The organism requires special culture conditions for growth, and the infection is cured by catheter removal.

MYCETOMA

Etiology *Actinomycetoma* refers to infection by actinomycetes of the genera *Nocardia*, *Nocardiosis*, *Streptomyces*, and *Actinomadura*. *Eumycetoma* ([Fig. 208-CD8](#)) is caused by true fungi of many different genera. The predominant agent varies with the locality.

Pathogenesis and Pathology The pathogens live in the soil and enter the skin through minor trauma. The most common site of infection is the foot. The infection runs a relentless course over many years, with destruction of contiguous bone and fascia. Grains are found in purulent foci surrounded by fibrosis and a mononuclear cell inflammatory response.

Clinical Manifestations Mycetoma is a chronic suppurative infection originating in subcutaneous tissue and characterized by the presence of grains, which are tightly clumped colonies of the causative agent. The infected site is characterized by painless

swelling, woody induration, and sinus tracts that discharge pus intermittently. Systemic symptoms do not develop, and spread to distant sites in the body does not take place.

Diagnosis Although the clinical picture is characteristic, mycetoma is sometimes confused with chronic osteomyelitis or botryomycosis. The diagnosis requires demonstration of grains in pus from the draining sinus or in biopsy sections. Many histologic sections may need to be examined to locate a grain.

TREATMENT

Actinomycetoma may respond to prolonged combination chemotherapy -- e.g., with streptomycin and either dapsone or trimethoprim-sulfamethoxazole. Eumycetoma rarely responds to chemotherapy; some cases caused by *Madurella mycetomatis* have appeared to respond to ketoconazole or itraconazole.

PARACOCCIDIOIDOMYCOSIS

Etiology Formerly called *South American blastomycosis*, this mycosis is caused by *Paracoccidioides brasiliensis*. A dimorphic fungus, *P. brasiliensis* grows as a budding yeast in tissue but may be grown as either a yeast or a mold on culture medium. The organism is identified by its gross and microscopic appearance.

Pathogenesis and Pathology Infection is thought to be acquired by inhalation of spores from environmental sources, possibly soil. Pulmonary infection produces few symptoms initially. Hematogenous spread to the mucous membranes of the mouth and nose, the lymph nodes, and other sites causes patients to seek medical attention. In fatal cases, the infection spreads to the adrenals, the gastrointestinal tract, and many other viscera.

Clinical Manifestations Common signs include indurated ulcers of the mouth, oropharynx, larynx, and nose; enlarged and draining lymph nodes; lesions of the skin and genitalia; and productive cough, weight loss, dyspnea, and sometimes fever. Paracoccidioidomycosis is acquired only in South America, Central America, and Mexico, but its extreme indolence may delay its recognition until many years after the patient has left the endemic area. Chest radiography most often shows bilateral patchy pneumonia.

Diagnosis Cultures of sputum, pus, and mucosal lesions are often diagnostic. The diagnosis can be made by smear or histologic section, although confirmation by culture is preferable. Serologic tests are useful in suggesting the diagnosis and monitoring the response to therapy.

TREATMENT

Relatively mild cases of paracoccidioidomycosis may be cured by 1 year of treatment with oral ketoconazole or itraconazole (200 to 400 mg daily). More advanced cases are treated with intravenous amphotericin B followed by itraconazole.

PHAEOPHYCOMYCOSIS

This is the name given to infections caused by fungi with dark-walled hyphae, excluding those given conventional names like chromoblastomycosis. Although an extraordinary variety of fungi and clinical syndromes are encompassed by this definition, most patients have brain abscess, subcutaneous abscess, or allergic fungal sinusitis. Most of the brain abscesses are due to *Cladophialophora bantiana*, *Ochroconis gallopavum*, *Exophiala dermatitidis*, *Bipolaris* species, and *Ramichloridium mackenziei*. Patients are previously healthy. Subcutaneous abscesses are usually single, arise at the site of minor trauma, and occur in both immunosuppressed and immunocompetent individuals. A large number of dematiaceous (dark-walled) mold species cause subcutaneous phaeohyphomycosis as well as allergic fungal sinusitis. The latter entity develops in patients with allergic rhinitis and presents as an expanding mucoid mass in one or more paranasal sinuses. The tenacious mucus contains eosinophils, Charcot-Leyden crystals, and occasional hyphae. Surgical excision of phaeohyphomycotic lesions is important. Antifungal therapy may retard recurrences but is of little value unless surgical excision has been performed.

PSEUDALLESCHERIASIS

Etiology Also called *Petriellidium boydii*, *Pseudallescheria boydii* is a mold frequently found in soil. When the fungus is isolated in the imperfect state, it is called *Scedosporium apiospermum*.

Pathogenesis and Pathology Wind-borne spores of *P. boydii*, arising from the soil, are the presumed source of infection. The fungus grows as a mold within tissue, causing necrosis and abscess formation.

Clinical Manifestations *P. boydii* resembles *Aspergillus* in its ability to colonize the endobronchial tree, to form fungus balls in the lungs or paranasal sinuses, and to invade the cornea or globe of the eye, the soft tissues, the joints, or the bones after trauma or surgery and in its propensity to invade the lungs and paranasal sinuses of immunosuppressed hosts, including patients with AIDS. Hyphae of *P. boydii* in tissue may be difficult to distinguish from those of *Aspergillus*. Infection with *P. boydii* is much less common than that with *Aspergillus*. Intravascular hyphae, a hallmark of invasive aspergillosis, are also found in pseudallescheriasis. Near-drowning in polluted water has led to severe *P. boydii* pneumonia, often with dissemination and fatal brain abscesses.

Diagnosis Demonstration of hyphae in tissue and culture confirmation are required for diagnosis.

TREATMENT

Itraconazole at the maximal tolerated doses is the regimen of choice. Surgical drainage or debridement can be helpful. The prognosis is poor.

Scedosporium prolificans, a fungus closely related to *P. boydii*, has caused infections in bones, joints, or soft tissue, usually after trauma. These infections have responded to surgical debridement. Disseminated infection with *S. prolificans* in immunosuppressed patients has been fatal. The response to treatment with all antifungal agents has been

poor.

SPOROTRICHOSIS

Etiology *Sporothrix schenckii* lives as a saprophyte on plants in many areas of the world. In nature and on culture at room temperature, the fungus grows as a mold; within host tissue or at 37°C on enriched media, it grows as a budding yeast. It is identified by its appearance in mold and yeast forms.

Pathogenesis and Pathology Infection results from the inoculation of *S. schenckii* into subcutaneous tissue through minor trauma. Nursery workers, florists, and gardeners acquire the illness from roses, sphagnum moss, and other plants. Infection may be limited to the site of inoculation (plaque sporotrichosis) or extend along proximal lymphatic channels (lymphangitic sporotrichosis). Spread beyond an extremity -- the usual site of infection -- is rare, and hematogenous dissemination from the skin remains unproven. The portal for osteoarticular, pulmonary, and other extracutaneous forms of sporotrichosis is unknown but is likely the lung.

Untreated sporotrichosis persists for months. The inflammatory response includes both the clustering of neutrophils and a marked granulomatous response with epithelioid cells and giant cells.

Clinical Manifestations In lymphangitic sporotrichosis, which is by far the most common manifestation, a nearly painless red papule forms at the site of inoculation. Over the next several weeks, similar nodules form along proximal lymphatic channels ([Fig. 208-CD9](#)). The nodules intermittently discharge small amounts of pus. Ulceration may occur. The proximal extension of these lesions, often with skip areas, is quite distinctive but may be mimicked by lesions of *Nocardia brasiliensis*, *Mycobacterium marinum*, or (in rare cases) *Leishmania brasiliensis* or *Mycobacterium kansasii*.

Plaque sporotrichosis manifests as a nontender red maculopapular granuloma confined to the site of inoculation. Osteoarticular sporotrichosis presents as mono- or polyarticular arthritis of indolent onset and progression over months or years, involving the elbows, knees, wrists, ankles, and (rarely) smaller joints of the extremities. Periarticular bone develops areas of demineralization detectable on x-ray, and draining sinuses may appear over joints and bursae. Hematogenous spread to the skin may take place during polyarticular disease, but none of the skin lesions shows lymphangitic spread. Immunosuppression, including that due to advanced infection with HIV, predisposes to hematogenous spread. Pulmonary sporotrichosis usually presents as a single chronic cavitary upper-lobe lung lesion. Chronic meningitis can develop in the absence of skin or lung lesions. *S. schenckii* is difficult to recover from cerebrospinal fluid.

Diagnosis Culture of pus, joint fluid, sputum, or a skin biopsy specimen is the preferred method of diagnosis. The appearance of *S. schenckii* in tissue is quite variable. In skin lesions, the organisms are hard to find.

TREATMENT

Cutaneous sporotrichosis can be cured with a saturated solution of potassium iodide given orally in increasing divided doses of up to 4.5 to 9 mL/d for adults, as tolerated. Gastrointestinal disturbance or acneiform rash over the cape area and face is common, but therapy should be continued for 1 month after the resolution of all lesions. Itraconazole (100 to 200 mg daily) is an effective and better-tolerated alternative. Extracutaneous sporotrichosis rarely responds to iodides, but more than half of cases have been cured by prolonged courses of intravenous amphotericin B. Itraconazole (200 mg once or twice daily) is effective in some cases of extracutaneous sporotrichosis.

TRICHOSPORONOSIS

A recent change in taxonomy of the genus *Trichosporon* has moved most of the agents causing deep infections from *T. beigelii* into the species *T. asahii*, with a few categorized as *T. mucoides*. White piedra of the scalp is caused by *T. ovoides* and that of the pubic hair by *T. inkin*. *T. cutaneum* and *T. asteroides* cause superficial infections. Most of what is currently known about *Trichosporon* infections is not species specific, so the following description refers to *T. beigelii*. *T. capitatum*, which causes disseminated infection in patients with neutropenia, was previously reclassified as *Blastoschizomyces capitatus* and will not be covered here.

T. beigelii can colonize the human gastrointestinal tract and skin and can enter the bloodstream of patients with severe neutropenia through an inapparent source. Hematogenously disseminated infection is manifested by fever and often by the development of several erythematous or purpuric tender papules anywhere on the body. Lesions can form large, tense hemorrhagic bullae. In some patients, native or prosthetic cardiac valves become infected. In tissue, hyphae and yeastlike cells are seen. Amphotericin B is probably the drug of choice for treatment, but recovery is dependent on the return of bone marrow function.

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209. PNEUMOCYSTIS CARINII INFECTION - Peter D. Walzer

DEFINITION AND DESCRIPTION

Pneumocystis carinii is an opportunistic pathogen whose natural habitat is the lung. The organism is an important cause of pneumonia in the compromised host.

Although the taxonomic status of *P. carinii* has long been controversial, molecular studies during the past decade have clearly placed the organism among the fungi. This classification is based on analysis of gene sequences for ribosomal RNA, mitochondrial proteins, and major enzymes. The cell wall of *P. carinii* contains β -1,3-glucan; drugs that inhibit β -glucan synthesis in fungi are highly active against *P. carinii* in animal models. However, in contrast to most fungi, *P. carinii* lacks ergosterol and is not susceptible to antifungal drugs that inhibit ergosterol synthesis.

Study of the basic biology of *P. carinii* has been severely hampered by the lack of a reliable in vitro cultivation system. Major developmental stages of the organism include the small (1- to 4- μ m) pleomorphic trophozoite or trophic form; the 5- to 8- μ m cyst, which has a thick cell wall and contains up to eight intracystic bodies; and the precyst, an intermediate stage. The life cycle of *P. carinii* probably involves asexual replication by the trophic form and sexual reproduction by the cyst, which ends in release of the intracystic bodies; an intracellular stage has not been identified. Ultrastructurally, *P. carinii* has a primitive organelle system, but little is known about its metabolism.

P. carinii contains two prominent antigen groups. The 95- to 140-kDa major surface glycoprotein (MSG) complex represents a family of proteins encoded by multiple genes. The MSG complex is highly immunogenic, contains shared and species-specific antigenic determinants, and exhibits protective B and T cell epitopes in animal models. The MSG complex plays a pivotal role in the host-parasite relationship in *P. carinii* infection. It facilitates adherence to host proteins via extracellular matrix proteins, surfactant proteins, and the mannose receptor; and its ability to undergo antigenic variation may represent a mechanism by which *P. carinii* evades the host immune response. The other antigen, which migrates as a band of 35 to 55 kDa, is the most common antigen recognized by the host and thus may serve as a marker of infection.

EPIDEMIOLOGY

P. carinii has a worldwide distribution among humans and has been detected in a variety of animals. The organisms found in these hosts are morphologically identical, but recent studies have revealed a high degree of genetic diversity and host specificity. Serologic surveys indicate that most healthy children have been exposed to the organism by 3 to 4 years of age. Animal model experiments have demonstrated that *P. carinii* is transmitted by the airborne route. Person-to-person transmission has been suggested by the occurrence of outbreaks of pneumocystosis among institutionalized debilitated infants and in hospitals caring for immunosuppressed patients. On the basis of animal studies, the incubation period is thought to be 4 to 8 weeks.

PATHOGENESIS AND PATHOLOGY

The host factors that predispose to the development of pneumocystosis involve defects in cellular and humoral immunity. People at risk for the disease include patients infected with HIV; persons receiving immunosuppressive therapy (particularly glucocorticoids) for cancer, organ transplantation, and other disorders; children with primary immunodeficiency diseases; and premature malnourished infants. The central role of CD4⁺ cells in host resistance to *P. carinii* has been shown by research in experimental animals and by studies that have correlated the risk of pneumocystosis in HIV-infected patients with CD4⁺ cell counts. Evidence supporting the importance of impaired humoral immunity consists of the occurrence of pneumocystosis in patients and animals with B cell defects and the beneficial effect of passively administered antibodies.

The principal host effector cells against *P. carinii* are alveolar macrophages, which ingest and kill the organism, releasing a variety of inflammatory mediators. Tumor necrosis factor and interleukin (IL) 1 are important in the early host defenses against *P. carinii*, but the role of other cytokines is less clear. Recent evidence suggests that HIV alters the mannose receptor-mediated binding and phagocytosis of *P. carinii*.

After being inhaled, *P. carinii* takes up residence in the alveoli, where it attaches tightly to type I cells but maintains an extracellular existence. In some cases, the organism remains in the host for long periods, and pneumonia develops by reactivation of latent infection; in other cases, pneumonia arises from a new bout of infection. As the immune system of the host becomes compromised, *P. carinii* organisms propagate and gradually fill the alveoli. This scenario is accompanied by a complex series of events that result in increased alveolar-capillary permeability and damage to type I cells. Surfactant abnormalities include a fall in bronchoalveolar lavage (BAL) fluid phospholipids and an increase in surfactant proteins A and D. Contributions of the host inflammatory response to lung injury are suggested by the correlation of increased IL-8 levels and neutrophil counts in BAL fluid from patients with relatively severe disease.

On lung sections stained with hematoxylin and eosin, the alveoli are filled with a typical foamy, vacuolated exudate. Severe disease may include interstitial edema, fibrosis, and hyaline membrane formation. The host inflammatory changes usually consist of hypertrophy of alveolar type II cells, a typical reparative response, and a mild mononuclear cell interstitial infiltrate. Malnourished infants display an intense plasma cell infiltrate that gave the disease its early name: interstitial plasma cell pneumonia.

CLINICAL FEATURES

Patients with *P. carinii* pneumonia develop dyspnea, fever, and nonproductive cough. Symptoms in non-HIV-infected patients often begin after the glucocorticoid dose has been tapered and typically last 1 to 2 weeks. HIV-infected patients are usually ill for several weeks or longer and have relatively subtle manifestations. However, the clinical picture in individual patients is quite variable, and a high index of suspicion and elicitation of a careful history are key factors in early detection.

Physical findings include tachypnea, tachycardia, and cyanosis, but lung auscultation reveals few abnormalities. The white blood cell count is variable and is usually governed by the patient's underlying disease. Assessment of arterial blood gases demonstrates hypoxia, an increased alveolar-arterial oxygen gradient (PA_{O2}-Pa_{O2}), and respiratory

alkalosis. There also may be changes in pulmonary function test values (diffusing capacity) and increased uptake with nuclear imaging techniques (gallium scan). Elevated serum concentrations of lactate dehydrogenase (LDH) have been reported; they probably reflect lung parenchymal damage but are not specific to *P. carinii* infection. In general, laboratory abnormalities are less severe in HIV-infected patients than in non-HIV-infected patients.

The classic findings on chest radiography consist of bilateral diffuse infiltrates beginning in the perihilar regions ([Fig. 209-1](#)), but various atypical manifestations (nodular densities, cavitary lesions) have also been reported. Patients who receive aerosolized pentamidine have an increased frequency of upper-lobe infiltrates and pneumothorax. Early in the course of pneumocystosis, the chest radiograph may be normal.

Although *P. carinii* usually remains confined to the lungs, cases of disseminated infection have occurred in both HIV-infected and non-HIV-infected patients. One risk factor for extrapulmonary spread in patients with HIV is the administration of aerosolized pentamidine. The most common sites of extrapulmonary involvement are the lymph nodes, spleen, liver, and bone marrow. Clinical manifestations range from incidental findings at autopsy to specific organ involvement. Histopathologic examination reveals the presence of *P. carinii* and the characteristic associated foamy material. Treatment for the extrapulmonary forms of pneumocystosis is the same as that for pneumonia.

DIAGNOSIS

Because the clinical picture of *P. carinii* infection can be produced by many other infectious and noninfectious agents, the diagnosis must be based on specific identification of the organism. A definitive diagnosis is made by histopathologic staining. Traditional stains have included reagents such as methenamine silver, toluidine blue, and cresyl echt violet, which selectively stain the wall of *P. carinii* cysts, and reagents such as Wright-Giemsa, which stain the nuclei of all developmental stages. Other reagents include nonspecific fluorochrome stains (calcofluor white) and Papanicolaou's stain. Immunofluorescence with monoclonal antibodies is more sensitive than histologic staining but is also more expensive. DNA amplification by the polymerase chain reaction offers the greatest sensitivity and may find a place in the routine diagnosis of *P. carinii* when commercial kits become available.

The successful diagnosis of pneumocystosis depends upon the collection of proper specimens. In general, the yield from different diagnostic procedures is higher in HIV-infected patients than in non-HIV-infected patients because of the higher organism burden in the former group. Sputum induction has gained popularity as a simple, noninvasive technique; this procedure requires trained and dedicated personnel, and its success has varied at different institutions. Fiberoptic bronchoscopy with [BAL](#), which is more sensitive than sputum induction, remains the mainstay of *P. carinii* diagnosis. This procedure also provides information about the organism burden, the host inflammatory response, and the presence of other opportunistic infections. Transbronchial biopsy and open lung biopsy, which are the most invasive procedures, are reserved for situations in which a diagnosis cannot be made by BAL.

COURSE AND PROGNOSIS

In the typical case of untreated *P. carinii* pneumonia, progressive respiratory embarrassment leads to death. Therapy is most effective when instituted early in the course of the disease, before there is extensive alveolar damage. If induced sputum is nondiagnostic and BAL cannot be performed in a timely manner, it is reasonable to begin empiric therapy with drugs active against *P. carinii*. However, this practice does not obviate the need for a specific etiologic diagnosis. With improvements in management, the case-fatality rate has been lowered to about 15% in HIV-infected patients but remains high (40%) in non-HIV-infected patients. The most widely used prognostic indicators have been the arterial oxygen pressure and the alveolar-arterial oxygen gradient. Other factors that may influence survival include neutrophil counts and IL-8 levels in BAL fluid, chest radiographic abnormalities, serum LDH and albumin levels, and the expertise of the hospital in caring for patients with HIV infection. Concurrent pulmonary infections complicate management, but the presence of cytomegalovirus usually does not affect the outcome of pneumocystosis.

TREATMENT

Trimethoprim-sulfamethoxazole (TMP-SMZ), which acts by inhibiting folic acid synthesis, is considered the drug of choice for all forms of pneumocystosis. The daily dosage, administered orally or intravenously in three or four divided doses, is 15 to 20 mg TMP/kg and 75 to 100 mg SMZ/kg. Therapy is continued for 14 days in non-HIV-infected patients and for 21 days in persons infected with HIV. Since HIV-infected patients respond more slowly than non-HIV-infected patients, it is prudent to wait at least 7 days after the initiation of treatment before concluding that therapy has failed. The addition of drugs to an existing regimen is no more effective than switching regimens and may increase the risk of toxicity. TMP-SMZ is well tolerated by non-HIV-infected patients, but more than half of HIV-infected patients experience serious adverse reactions, including fever, rash, neutropenia, thrombocytopenia, hepatitis, and hyperkalemia.

Several alternative regimens are available for the treatment of mild to moderate cases of *P. carinii* pneumonia: TMP (15 mg/kg per day orally) plus dapsone (100 mg/d orally), clindamycin (600 mg every 6 h intravenously or 300 to 450 mg every 6 h orally) plus primaquine (15 to 30 mg of base per day orally), or atovaquone alone (750 mg twice daily orally). Dapsone and primaquine should be used with caution in patients with glucose-6-phosphate dehydrogenase deficiency.

Two alternative drugs are available for the treatment of moderate to severe forms of pneumocystosis. Pentamidine, which has been used against *P. carinii* for many years, is administered as a single daily dose of 4 mg/kg by slow intravenous infusion. Pentamidine is highly toxic; its major side effects are hypotension, cardiac arrhythmias, dysglycemias, azotemia, electrolyte changes, and neutropenia. Trimetrexate is administered intravenously as a single daily dose of 45 mg/m²; in conjunction with trimetrexate therapy, folinic acid is given orally or intravenously at a dose of 20 mg/m² every 6 h to prevent bone marrow suppression.

Patients with HIV frequently experience deterioration in respiratory function shortly after receiving anti-*P. carinii* drugs. Several studies have shown that the administration of

glucocorticoids to patients with HIV and moderate to severe pneumocystosis (a P_{O_2} of ≤ 70 mmHg or a $PA_{O_2}-Pa_{O_2}$ of ≥ 35 mmHg) can prevent this problem and improve the rate of survival. The administration of steroids should be started early in the course of the illness (usually when antimicrobial drugs are begun) for maximal benefit; the recommended regimen is 40 mg of prednisone orally twice daily on days 1 to 5, 40 mg/d on days 6 to 10, and 20 mg/d on days 11 to 20. This regimen has generally proven to be safe despite concern about its effects on other opportunistic infections. The use of steroids as adjunctive therapy in HIV-infected patients with mild pneumocystosis or in non-HIV-infected patients remains to be evaluated.

PREVENTION

Primary prophylaxis is indicated for HIV-infected patients at high risk of developing pneumocystosis -- that is, those who have CD4+ cell counts of $<200/\mu\text{L}$, unexplained fever [$>37.8^\circ\text{C}$ (100°F)] for ≥ 2 weeks, or a history of oropharyngeal candidiasis. Guidelines for the administration of primary prophylaxis to other immunocompromised hosts are less clear. Secondary prophylaxis is indicated for all patients who have recovered from *P. carinii* pneumonia. Among HIV-infected patients, the risk of recurrent episodes of pneumocystosis is high and lifelong; among non-HIV-infected patients, the risk is lower and exists for as long as the immunosuppressive condition persists.

Several antimicrobial drugs are effective in preventing pneumocystosis, although some concern has been raised about possible resistance. One double-strength tablet of [TMP-SMZ](#) (160 mg TMP, 800 mg SMZ) per day is the prophylactic regimen of choice. The major limitation of TMP-SMZ treatment is the high frequency of adverse reactions among HIV-infected patients. Recommended alternative regimens include TMP-SMZ at a reduced dose (80 mg TMP, 400 mg SMZ) or frequency (3 times per week), dapsone alone at a daily oral dose of 100 mg, dapsone at a dose of 50 mg/d combined with weekly oral doses of pyrimethamine (50 mg) and folinic acid (25 mg), pentamidine at a monthly dose of 300 mg administered by Respigard nebulizer, and atovaquone at an oral dose of 1500 mg/d.

Although there are no specific recommendations for preventing the spread of *P. carinii* in health care facilities, it seems prudent to prevent direct contact between patients with pneumocystosis and other susceptible hosts.

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SECTION 16 -PROTOZOAL AND HELMINTHIC INFECTIONS: GENERAL CONSIDERATIONS

210. APPROACH TO THE PATIENT WITH PARASITIC INFECTION - *Peter F. Weller*

Because diverse parasitic organisms may infect humans, a range of factors are germane to an assessment of the possible parasitic etiology of a patient's disease. These factors include issues related to the patient's history, immune status, and presenting clinical and laboratory characteristics, especially eosinophilia. Complementing historical information, a full clinical evaluation and laboratory testing provide additional data to direct the assessment for parasitic infection. The specific tests required, ranging from standard blood biochemical assays to imaging of selected organs, are dictated by the nature of the patient's illness. Additional diagnostic testing for parasitic infections ([Chap. 211](#)) completes the evaluation.

HISTORY

Geographic History The history can provide valuable information about potential exposures to parasitic infections. A history of travel to, residence or work in, or immigration from areas of the world in which various parasites not endemic in the United States are encountered offers a clue to possible parasitic or other infectious etiologies of a patient's disease ([Chap. 123](#)). Some parasitic infections may become manifest early after a traveler's return home; paramount among these in terms of preventable mortality is malaria. If a patient has been in a region of the world where malaria is endemic (even only briefly, as in an airport layover), fever mandates a consideration of malaria, whether or not malaria chemoprophylaxis has been used. Falciparum malaria, which in a nonimmune patient may progress rapidly to serious and even life-threatening consequences ([Chap. 214](#)), is a potential medical emergency and must be considered at the initial evaluation, even if symptoms and fever patterns are suggestive of less specific flulike or gastrointestinal illness. Rarely, malaria transmission has been reported within the United States.

For patients with a history of recent travel, the onset of gastrointestinal symptoms only after return suggests protozoal diseases characterized by a 1- to 2-week delay between acquisition and appearance of symptoms -- notably, giardiasis, cyclosporiasis, and cryptosporidiosis ([Chap. 218](#)). Gastrointestinal symptoms lasting longer than a week also suggest protozoan etiologies, including giardiasis, amebiasis, and cyclosporiasis. For patients who have traveled less recently, information on the specific countries and the types of regions (urban or rural) visited and on the nature and duration of the visit, whether for general tourism or for activities related to specific occupations, is helpful in concert with presenting clinical features and hematologic and other laboratory findings. Diseases that may become manifest only some years after an individual leaves an endemic region include schistosomiasis, some forms of filariasis, strongyloidiasis, echinococcosis, and cysticercosis.

For the illnesses of patients who have never left the United States, various parasitic etiologies should be considered, depending on the presenting disease. Trichomoniasis, trichinellosis, strongyloidiasis, giardiasis, cryptosporidiosis, cyclosporiasis, echinococcosis, and pinworm are among the parasitic infections endemic in settings

within the United States. Diseases that are more frequent where there is fecal contamination of soil or other environmental sites include hookworm, ascariasis, trichuriasis, amebiasis, and strongyloidiasis; dermal exposure, as by walking barefoot on soil contaminated with parasitic larvae, predisposes residents of such an area as well as travelers to the acquisition of cutaneous larva migrans, hookworm, and strongyloidiasis.

Dietary History If more than one patient develops similar symptoms in a given situation, common-source water- or foodborne diseases (giardiasis, cryptosporidiosis, cyclosporiasis) should be considered. Waterborne infections are more likely to be acquired from surface water supplies, ranging from mountain streams to municipal reservoirs. Likewise, attention to dietary history may be helpful. Trichinellosis should be considered when the patient may have consumed contaminated pork, bear, walrus, or other meat from carnivores. Ingestion of undercooked fish predisposes to anisakiasis and to infection with other fish-dwelling nematodes, tapeworms (*Diphyllobothrium latum*), or flukes (*Nanophyetus salmincola*). Ingestion of snails or of produce contaminated with land snails can lead to infection with *Angiostrongylus cantonensis* (eosinophilic meningitis; [Chap. 219](#)). Ingestion of more exotic animal foodstuffs, including snakes, can result in the transmission of gnathostomiasis ([Chap. 219](#)). For children with a propensity for pica, ingestion of soil containing *Toxocara* eggs may lead to visceral larva migrans. Consumption of ground-grown vegetables, including those shipped in from distant fields contaminated with human feces, provides an opportunity for the ingestion of nematode eggs of *Ascaris lumbricoides* or *Trichuris trichiura*.

Other Exposure Histories An antecedent blood transfusion raises the possibility of malaria (especially that due to *Plasmodium malariae* or *P. falciparum*), babesiosis, or Chagas' disease. A history of wading or swimming in fresh water is germane to the acquisition of schistosomiasis or avian schistosome dermatitis. Fresh water may be a source of infection with free-living amebae, and these protozoa may cause meningoencephalitis or ocular infections. Arthropod vector-borne parasitic infections include malaria and lymphatic filariasis (carried by mosquitoes) and babesiosis (carried by ticks); the latter is transmitted in some regions of the United States.

Residence in an institutional setting where fecal-oral hygiene may be imperfect raises the possibility of giardiasis, cryptosporidiosis, or strongyloidiasis. Child-care centers provide opportunities for young children and their family members to acquire giardiasis, cryptosporidiosis, and pinworm infections. Trichomoniasis is transmitted sexually; giardiasis, cryptosporidiosis, amebiasis, and strongyloidiasis can be transmitted during anal intercourse or oral-anal contact.

IMMUNE STATUS

The patient's immune status is relevant in determining which parasitic infections need to be considered. In patients infected with HIV-1, especially those with depressed CD4+ lymphocyte counts, specific protozoan diseases may develop opportunistically. These infections include toxoplasmosis, isosporiasis, cyclosporiasis, cryptosporidiosis, visceral leishmaniasis, American trypanosomiasis, microsporidiosis, and infections with free-living amebae (*Acanthamoeba* and related genera). In individuals infected with human T-lymphotropic virus type 1, strongyloidiasis is a prominent consideration.

Patients who are asplenic are at risk not only for overwhelming infections due to encapsulated bacteria but also for fulminant infections caused by intraerythrocytic protozoa, including malaria and babesiosis. Patients with hypogammaglobulinemia or cystic fibrosis may develop refractory giardiasis. In patients developing symptoms of enterocolitis while receiving glucocorticoids, the possibility of an exacerbation of unsuspected strongyloidiasis or amebic colitis should be considered.

EOSINOPHILIA

Eosinophilia may offer a hematologic clue to the presence of some parasites. Only two protozoan parasites have been associated with eosinophilia: *Isospora belli* and, on occasion, *Dientamoeba fragilis*. The detection of eosinophilia generally mandates a consideration of the multicellular helminthic parasites that characteristically elicit interleukin 5-mediated eosinophilia. (Helminth-elicited eosinophilia, however, may be suppressed by glucocorticoid therapy or by intercurrent bacterial or viral infections.) The magnitude of eosinophilia tends to correlate with the extent of tissue invasion by helminths. Marked blood eosinophilia (more than 3000 eosinophils per microliter) develops during the early transpulmonary migration of intestinal nematodes, including *Ascaris* and hookworms, at a time when eggs (whose presence confirms the diagnosis) have not yet been produced in the intestinal tract.

Eosinophilia is also marked in the early stages of fluke infections, including schistosomiasis (Katayama fever), paragonimiasis, clonorchiasis, and fascioliasis; during the stage of muscle invasion in trichinellosis; during tissue migration of adult worms in loiasis and gnathostomiasis; and with heavy infections in visceral larva migrans. Eosinophilia persisting for more than a year may be indicative of hookworm infection, strongyloidiasis, visceral larva migrans (especially in children), filarial infection (including onchocerciasis, loiasis, and tropical pulmonary eosinophilia), fluke infections (including schistosomiasis, fascioliasis, clonorchiasis, and paragonimiasis), and cysticercosis. Leakage of fluids from echinococcal cysts can cause intermittent increases in eosinophilia.

Eosinophilia sometimes provides the only clue to the presence of helminthic infection and should prompt an evaluation for such infection. Serologic testing for schistosomiasis, filariasis, visceral larva migrans, and strongyloidiasis will be helpful in an assessment for some of the diseases most likely to elicit eosinophilia. Serologic evaluation for strongyloidiasis is especially important since autoinfection may permit persistence of the organisms for decades and put the patient at risk for disseminated disease if immunosuppressive glucocorticoids are later administered for any reason.

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211. LABORATORY DIAGNOSIS OF PARASITIC INFECTIONS - Charles E. Davis

The cornerstone for the diagnosis of parasitic infections is a thorough history of the patient's illness. Epidemiologic aspects of the illness are especially important because the risks of acquiring many parasites are closely related to occupation, recreation, or travel to areas of high endemicity. Without a basic knowledge of the epidemiology and life cycles of the major parasites, it is difficult to approach the diagnosis of parasitic infections systematically. Accordingly, the medical classification of important human parasites in this chapter emphasizes their geographic distribution, their transmission, and the anatomic location and stages of their life cycle in humans. The text and tables are intended to serve as a guide to the correct diagnostic procedures for the major parasitic infections and to direct the reader to other chapters that contain more comprehensive information about each infection. [Tables 211-1, 211-2,](#) and [211-3](#) summarize the geographic distributions, the anatomic locations, and the laboratory methods employed for the diagnosis of flatworm, roundworm, and protozoal infections, respectively.

In addition to selecting the correct diagnostic procedures, physicians must counsel their patients to ensure that specimens are collected properly and arrive at the laboratory promptly. For example, the diagnosis of bancroftian filariasis is unlikely to be confirmed by the laboratory unless blood is drawn near midnight, when the nocturnal microfilariae are active. Laboratory personnel and surgical pathologists should be notified in advance when a parasitic infection is suspected. Continuing interaction with the laboratory staff and the surgical pathologists increases the likelihood that parasites in body fluids or biopsy specimens will be examined carefully by the most capable individuals.

INTESTINAL PARASITES

Most helminths and protozoa exit the body in the fecal stream. The patient or the patient's attendant should be instructed to collect feces in a clean cardboard container and to record the time of collection on the container. Contamination with water, which could contain free-living protozoa, or with urine should be avoided. Fecal samples should be collected before ingestion of barium or other contrast agents for radiologic procedures and before treatment with antidiarrheal agents and antacids, because these substances change the consistency of the feces and interfere with microscopic detection of parasites. Because of the cyclic shedding of most parasites in the feces, a minimum of three samples collected on alternate days should be examined. When delays in transport to the laboratory are unavoidable or specimens must be shipped by mail, fecal samples should be kept in polyvinyl alcohol to preserve protozoal trophozoites. Refrigeration will also preserve trophozoites for a few hours and protozoal cysts and helminthic ova for several days.

Analysis of fecal samples consists of both a macroscopic and a microscopic examination. Watery or loose stools are more likely to contain protozoal trophozoites, but protozoal cysts and all stages of helminths may be found in formed feces. If adult worms or tapeworm segments are observed, they should be transported promptly to the laboratory or washed and preserved in fixative for later examination. The only tapeworm with motile segments is *Taenia saginata*, the beef tapeworm, which patients sometimes bring to the physician. Motility is an important distinguishing characteristic, because the

ova of *T. saginata* and *T. solium*, the cause of cysticercosis, are morphologically indistinguishable.

Microscopic examination of feces ([Table 211-4](#)) is not complete until direct wet mounts have been evaluated and concentration techniques as well as permanent stains have been applied. Before accepting a report of negativity for ova and parasites as final, the physician should insist that the laboratory undertake each of these procedures. Some intestinal parasites are more readily detected in material other than feces. For example, use of the string test (or one of its commercial substitutes) to sample duodenal contents is sometimes necessary to detect *Giardia lamblia*, *Cryptosporidium*, and *Strongyloides* larvae. Use of the "Scotch tape" technique to detect pinworm ova on the perianal skin sometimes also reveals ova of *T. saginata* deposited perianally when the motile segments disintegrate ([Table 211-4](#)).

Two routine solutions are used to make wet mounts for the identification of the various life stages of helminths and protozoa: physiologic saline for trophozoites, cysts, ova, and larvae and dilute iodine solution for protozoal cysts and ova. Iodine solution must never be used to examine specimens for trophozoites because it kills the parasites and thus eliminates their characteristic motility.

The two most common concentration procedures for detecting small numbers of cysts and ova are formalin-ether sedimentation and zinc sulfate flotation. The formalin-ether technique is preferable, because all parasites sediment but not all float. Slides permanently stained for trophozoites should be prepared before concentration. Additional slides stained for cysts and ova may be made from the concentrate.

In many instances, especially in the differentiation of *Entamoeba histolytica* from other amebas, identification of parasites from wet mounts or concentrates must be considered tentative. Permanently stained smears allow study of the cellular detail necessary for definitive identification. The iron-hematoxylin stain is excellent for critical work, but trichrome staining, which can be completed in 1 h, is a satisfactory alternative that also reveals parasites in specimens preserved in polyvinyl alcohol fixative.

BLOOD AND TISSUE PARASITES

Invasion of tissue by protozoa and helminths renders the choice of diagnostic techniques more difficult. For example, physicians must understand that aspiration of an amebic liver abscess rarely reveals *E. histolytica* because the trophozoites are located primarily in the abscess wall. They must remember that the urine sediment offers the best opportunity to detect *Schistosoma haematobium* in the Ethiopian youngster or the American traveler who returns from Africa with hematuria ([Table 211-5](#)). [Tables 211-1, 211-2, and 211-3](#), which offer a quick guide to the geographic distribution and anatomic locations of the major tissue parasites, should help the physician to select the appropriate body fluid or biopsy site for microscopic examination. [Tables 211-5, 211-6, and 211-7](#) provide additional information about the identification of parasites in samples from specific anatomic locations. The laboratory procedures for detection of parasites in other body fluids are similar to those used in the examination of feces. The physician should insist on wet mounts, concentration techniques, and permanent stains for all body fluids. The trichrome or iron-hematoxylin stain is satisfactory for all tissue

helminths in body fluids other than blood, but microfilarial worms and blood protozoa are more easily visualized when stained with Giemsa or Wright's stain.

The most common parasites detected in Giemsa-stained blood smears are the plasmodia, microfilariae, and African trypanosomes ([Table 211-5](#)). Most patients with Chagas' disease present in the chronic phase, when *Trypanosoma cruzi* is no longer microscopically detectable in blood smears. Wet mounts are sometimes more sensitive than stained smears for the detection of microfilariae and African trypanosomes because these active parasites cause noticeable movement of the erythrocytes in the microscopic field. Nuclepore filtration of blood facilitates the detection of microfilariae. The intracellular amastigote forms of *Leishmania* spp. and *T. cruzi* can sometimes be visualized in stained smears of peripheral blood, but aspirates of the bone marrow, liver, and spleen are the best sources for microscopic detection and culture of *Leishmania* in kala-azar and of *T. cruzi* in chronic Chagas' disease.

The diagnosis of malaria and the critical distinction among the various *Plasmodium* spp. are made by microscopic examination of stained thick and thin blood films ([Table 211-6](#); [Plates VI-3, VI-4, VI-5, VI-6, VI-7, VI-8, VI-9, VI-10, VI-11, VI-12, VI-13, VI-14, VI-15, VI-16, VI-17, VI-18, VI-19, VI-20, VI-21, VI-22, VI-23, VI-24, VI-25, VI-26, VI-27, VI-28, VI-29, VI-30, VI-31, VI-32, and VI-33](#)). Most malariologists prefer Giemsa stain because of its overall high quality, suitability for staining of both thick and thin smears, and stability in tropical climates. Wright's stain can produce high-quality thin smears and is widely used in the Americas, but it deteriorates rapidly in the tropics because its methanol base is highly hygroscopic. Specimens of capillary or venous blood should be obtained every 4 to 12 h until a diagnosis is established. The thin smear is made on clean slides exactly like a blood film for a white blood cell differential. The thick film is made by placing one drop of blood on the slide and stirring it in a circular motion to a diameter of about 2 cm. The erythrocytes in the thick film are lysed with water, but the thin film is fixed in methanol to preserve erythrocyte morphology.

Although most tissue parasites stain with the traditional hematoxylin and eosin, surgical biopsy specimens should also be stained with appropriate special stains. The surgical pathologist who is accustomed to applying silver stains for *Pneumocystis carinii* to induced sputum and transbronchial biopsies may have to be reminded to examine wet mounts and iron-hematoxylin-stained preparations of pulmonary specimens for helminthic ova and *E. histolytica*. The clinician should also be able to advise the surgeon and pathologist about optimal techniques for the identification of parasites in specimens obtained by certain specialized minor procedures ([Table 211-7](#)). For example, the excision of skin snips for the diagnosis of onchocerciasis, the collection of rectal snips for the diagnosis of schistosomiasis, and punch biopsy of skin lesions for the identification and culture of cutaneous and mucocutaneous species of *Leishmania* are simple procedures, but the diagnosis can be missed if the specimens are improperly obtained or processed.

NONSPECIFIC TESTS

Eosinophilia is a common accompaniment of infections with most of the tissue helminths; absolute numbers of eosinophils may be high in trichinosis and the migratory

phases of filariasis ([Table 211-8](#)). Intestinal helminths provoke eosinophilia only during pulmonary migration of the larval stages. Eosinophilia is not a manifestation of protozoal infections, with the possible exceptions of those due to *Isospora* and *Dientamoeba fragilis*.

Like the hypochromic, microcytic anemia of heavy hookworm infections, other nonspecific laboratory abnormalities may suggest parasitic infection in patients with appropriate geographic and/or environmental exposures. Biochemical evidence of cirrhosis or an abnormal urine sediment in an African immigrant certainly raises the possibility of schistosomiasis, and anemia and thrombocytopenia in a febrile traveler or immigrant are among the hallmarks of malaria. Computed tomography and magnetic resonance imaging also contribute to the diagnosis of infections with many tissue parasites and have become invaluable adjuncts in the diagnosis of neurocysticercosis and cerebral toxoplasmosis.

ANTIBODY AND ANTIGEN DETECTION

Useful antibody assays for many of the important tissue parasites are available; those listed in [Table 211-9](#) can be obtained from the Centers for Disease Control and Prevention (CDC) in Atlanta. The results of most serologic tests not listed in the tables and not offered by the CDC should be interpreted with caution.

The value of antibody assays is limited in the case of the filarial worms and plasmodia. The detection of antibody to plasmodia is of limited use for establishing the diagnosis of malaria in individual patients because diagnostic titers develop slowly and the tests must be sent to the CDC. Filarial antigens cross-react with those from other nematodes, and antibody assays do not distinguish between past and current infection. In contrast, a negative result in an American or European traveler virtually rules out the diagnosis of bancroftian or brugian filariasis. Promising new assays for filarial antigens and antibodies in lymphatic filariasis are not yet available in commercial kits or from the [CDC](#).

Despite these specific limitations, the restricted geographic distribution of many tropical parasites increases the diagnostic usefulness of antibody detection in travelers from industrialized countries. On the other hand, a large proportion of the world has been exposed to *Toxoplasma gondii*, and the presence of IgG antibody does not constitute proof of active disease.

Fewer antibody assays are available for the diagnosis of infection with intestinal parasites. Cross-reactivity and lack of efficient cultivation techniques, along with the ability to establish diagnoses without invasive procedures, have discouraged intensive investigation of these methods. *E. histolytica* is the major exception. Sensitive, specific serologic tests are invaluable in the diagnosis of amebiasis. Commercial kits for the detection of antigen by enzyme-linked immunosorbent assay or of whole organisms by fluorescent antibody assay are now available for several protozoan parasites ([Table 211-9](#)).

MOLECULAR TECHNIQUES

DNA hybridization with probes that are repeated many times in the genome of a specific

parasite and amplification of a specific DNA fragment by the polymerase chain reaction (PCR) are promising techniques for the diagnosis of parasitic infections. Although molecular techniques for the detection of many parasites are already being used in insect vectors, animal models, and human trials, few are available for routine use in patients at this time. The only available commercial kit is that for the identification of *Trichomonas vaginalis* by hybridization of secretions from vaginal swabs with synthetic oligonucleotide probes. The [CDC](#) will perform PCR for microsporidia, cryptosporidia, *Cyclospora*, and *E. histolytica* on stools (frozen or fixed in either potassium dichromate or ethanol) and on biopsy and bronchoalveolar lavage samples (fixed in methanol or ethanol). For *Plasmodium* and *Babesia*, PCR is performed on blood treated with EDTA or collected in IsoCode Stix (Schleicher and Schuell).

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212. THERAPY FOR PARASITIC INFECTIONS - Thomas A. Moore

Over the last few decades, the reach of some parasitic diseases such as malaria has extended because of factors such as deforestation, population shifts, global warming, and other climatic events (e.g., "El Nino"). Efforts to combat this trend are complicated by the development and spread of drug resistance among parasites and the limited introduction of new antiparasitic agents. However, significant advances toward the reduction of the burden of parasitic disease have been made. The generous donation of ivermectin and albendazole for global eradication programs has improved the health of countless individuals and offers the promise of disease eradication. The expanded use of traditionally nonparasitic agents, such as amphotericin B for visceral leishmaniasis, has offered hope against the specter of drug resistance. The introduction of newer agents, such as triclabendazole, also appears promising.

Currently recommended treatment options for most parasitic diseases of humans are listed in [Table 212-1](#). A brief summary of some of the agents can be found below; each agent is listed by its generic name. Many of the agents are approved by the Food and Drug Administration (FDA) but are considered investigational for the treatment of certain infections; these drugs are marked accordingly in [Table 212-1](#). Drugs marked in the text with an asterisk (*) are available only through the Centers for Disease Control and Prevention (CDC) Drug Service (telephone: 404-639-3670). Other drugs, marked with a dagger (+), are available only through the manufacturer; contact information for these manufacturers may be available from the CDC. Information on dosing in children and pregnant women can be obtained in the references at the end of the chapter.

Albendazole This benzimidazole derivative has recently become generally available and is active against a broad range of helminths and protozoa. All benzimidazoles act by binding to free tubulin, inhibiting the polymerization of tubulin and the microtubule-dependent uptake of glucose. In helminths, the result is the depletion of glycogen stores, but this fundamental disruption of cellular metabolism also offers treatment for a wide range of parasitic diseases. Like all benzimidazoles, albendazole is poorly absorbed from the gastrointestinal tract. However, since the active metabolite attains higher serum and cyst concentrations than mebendazole, it is more effective against echinococcal disease. Significant adverse reactions are usually limited to prolonged use and include abdominal pain and reversible hepatic dysfunction. Rarely, leukopenia and reversible alopecia occur. Albendazole is contraindicated in early pregnancy.

Artesunate and Artemether Artesunate, artemether, and the parent compound artemisinin are sesquiterpene lactones derived from the wormwood plant *Artemisia annua*. These agents have become first-line treatments for severe falciparum malaria in some areas of the world where drug resistance is a major problem. They are rapidly effective against the asexual blood forms of *Plasmodium* spp., including multidrug-resistant *P. falciparum*, but they are not active against intrahepatic forms. Their mechanism of action is not completely understood, but they are believed to act by converting to free radicals and other intermediates in the presence of intraparasitic iron; the result is alkylation of parasite proteins or membrane damage. Artemisinin derivatives presently show no cross-resistance with known antimalarials and thus are important for treating severe malaria in areas of multidrug resistance. However, long treatment

courses are required and, when these agents are used alone, recrudescence may occur. Artemisinin and its derivatives are cleared rapidly from the circulation, and their short half-lives limit their use for prophylaxis. Adverse events are usually infrequent and mild and include drug fever and contact dermatitis. Animal data suggest that neurotoxicity can develop if the compounds are administered chronically, and cerebellar dysfunction has been reported in persons treated with artesunate. These drugs are not available in the United States.

Amphotericin B Amphotericin exerts its effect by inserting itself into the cytoplasmic membrane of the organism and binding sterols, causing membrane permeability. Amphotericin B deoxycholate, a lipophilic polyene drug, is an effective agent for the treatment of leishmaniasis and amebic meningoencephalitis due to *Naegleria* spp., but its use is associated with significant nephrotoxicity and occasional allergic reactions. Three lipid-complexed formulations of amphotericin B have been released: amphotericin B colloidal dispersion (ABCD), amphotericin B lipid complex (ABLC), and liposomal amphotericin B. These agents are effective against antimony-resistant visceral leishmaniasis and offer the benefits of shorter courses and less toxicity.

Atovaquone Atovaquone is a hydroxynaphthoquinone that exerts its broad-spectrum antiprotozoal activity via inhibition of parasite mitochondrial electron transport. Although it has relatively poor bioavailability, atovaquone exhibits potent activity against toxoplasmosis when used with pyrimethamine. When combined with proguanil or doxycycline, it is effective for both treatment and prophylaxis of malaria. Side effects are rare but can include nausea and a maculopapular rash.

Azithromycin An azalide antibiotic, azithromycin has been used to treat a number of protozoal infections such as babesiosis, malaria, toxoplasmosis, and cryptosporidiosis. Azithromycin acts by inhibiting protein synthesis; in apicomplexan parasites, this inhibition occurs in the plastid. Most adverse reactions are gastrointestinal (diarrhea, nausea, abdominal pain) and uncommon and rarely require discontinuation of the drug.

Benznidazole This oral nitroimidazole derivative is used to treat acute Chagas' disease, and cure rates of 80 to 90% have been recorded. Benznidazole acts by generating oxygen radicals to which the parasite is more sensitive than mammalian cells because of a relative deficiency in antioxidant enzymes. Adverse effects are frequent and include rashes, nausea, paresthesias, and leukopenia. The safety of benznidazole in early pregnancy has not been established, but the drug should be used immediately after the first trimester to prevent congenital transmission. Benznidazole is currently unavailable in the United States.

Bithionol*Bithionol is a phenolic substance structurally related to hexachlorophene. Its antihelminthic activity is poorly understood, but the agent is believed to inhibit oxidative phosphorylation. Bithionol is no longer manufactured. A dwindling supply is available from the [CDC](#) for treatment of fascioliasis and paragonimiasis. The drug's distribution is limited to physicians treating patients with one of these diseases who are unable to use praziquantel because of previous idiosyncratic or allergic reactions or in whom a course of praziquantel has failed to eradicate infection. Significant side effects are common and include abdominal pain, diarrhea, and urticaria.

Chloroquine The best-known of the 4-aminoquinolines, chloroquine has marked, rapid schizonticidal and gametocidal activity against blood forms of *P. ovale* and *P. malariae* and against susceptible strains of *P. vivax* and *P. falciparum*. It is not active against intrahepatic forms (*P. vivax* and *P. ovale*). Chloroquine is concentrated in the acidic food vacuoles of intraerythrocytic parasites, where it reaches levels 600-fold higher than plasma levels. The drug inhibits a parasite heme polymerase that protects the parasite from the membrane-damaging byproducts of hemoglobin degradation; as a result of this inhibition, the parasite is effectively killed with its own metabolic waste. Chloroquine resistance appears to arise as a result of a decreased level of chloroquine uptake. This drug is safe for use in pregnancy. Adverse reactions include pruritus, transient headaches, and nausea.

Clindamycin This macrolide is an effective adjunct for the treatment of several protozoal infections, including toxoplasmosis, malaria, and babesiosis. The most significant potential adverse reaction is the development of *Clostridium difficile*-associated diarrhea. When administered with quinine, clindamycin is active against drug-resistant falciparum malaria; this combination is the therapy of choice for babesiosis.

Diethylcarbamazine* A piperazine derivative with a long history of successful use, this drug remains the treatment of choice for lymphatic filariasis and loiasis and has also been used for visceral larva migrans. Diethylcarbamazine exerts effects on helminths, including immobilization due to a decrease in muscle activity, disruption of microtubule formation, and alteration of helminthic surface membranes resulting in enhanced killing by the host's immune system. In addition, this agent enhances adherence properties of eosinophils. It is safe for use in pregnancy and well tolerated. Significant adverse events associated with the drug, including encephalopathy and death, are attributable to the parasite burden, which is best assessed by the degree of microfilaremia. Use in patients with onchocerciasis can precipitate a Mazzotti reaction with pruritus, fever, and arthralgias. Untoward effects produced directly by diethylcarbamazine usually involve the gastrointestinal tract and are dose-related.

Eflornithine (Difluoromethylornithine, DFMO)+ This ornithine derivative has specific activity against all stages of infection with *Trypanosoma brucei gambiense* and acts by irreversibly inhibiting ornithine decarboxylase -- an enzyme critical to the formation of polyamines, which are essential to trypanosomatids. Eflornithine readily crosses the blood-brain barrier and is excreted mainly by the kidneys. Its use is contraindicated in pregnancy. Adverse reactions, which are usually mild and reversible, include pancytopenia, diarrhea, and transient hearing loss.

Furazolidone This nitrofurantoin derivative, which is an effective alternative agent for the treatment of giardiasis, acts by damaging parasite DNA. Since it is the only agent active against *Giardia* that is available in liquid form, it is often used to treat young children. Side effects include allergic reactions, nausea, vomiting, and disulfiram-like reactions when ingested with alcohol. Because hemolytic anemia due to glutathione instability can occur, furazolidone treatment is contraindicated in mothers who are breast-feeding and in neonates.

Halofantrine An oral alternative drug for treatment of malaria due to

chloroquine-resistant *P. falciparum*, this 9-phenanthrenemethanol is one of three classes of arylaminoalcohols first identified as potential antimalarial agents by the World War II Malaria Chemotherapy Program. Its activity is believed to be similar to that of chloroquine. Halofantrine is generally well tolerated; the most commonly reported adverse effects are abdominal pain and diarrhea. The incidence of pruritus is lower than that with chloroquine. Halofantrine causes dose-related prolongation of the PR and QT intervals, and its use is contraindicated in persons who have cardiac disease or who have taken mefloquine in the preceding 3 weeks. Halofantrine treatment is also contraindicated in pregnant and lactating women. The drug is currently unavailable in the United States.

Iodoquinol This hydroxyquinoline is an effective luminal agent for the treatment of amebiasis, balantidiasis, and infection with *Dientamoeba fragilis*. Its mechanism of action is unknown. Adverse effects include headache, diarrhea, nausea, vomiting, abdominal pain, pruritus, fever, seizures, and encephalopathy. Most serious are the reactions related to prolonged high-dose therapy, which should not occur if the dosage regimens recommended in [Table 212-1](#) are followed. Because the drug contains iodine, it should be used with caution in patients with thyroid disease.

Ivermectin This derivative of avermectin is used to treat infections caused by a wide range of helminths. It is the drug of choice for the treatment of onchocerciasis, strongyloidiasis, and cutaneous larva migrans. While active against the intestinal helminths *Ascaris lumbricoides* and *Enterobius vermicularis*, it is variably effective in trichuriasis and ineffective against hookworms. Recent data suggest that ivermectin acts by opening neuromuscular membrane-associated glutamate-dependent chloride channels (unique to nematodes and arthropods) -- an event resulting in an influx of chloride ions, worm paralysis, and subsequent death by immune or other mechanisms. Ivermectin is generally safe, easy to administer, and well tolerated, but encephalopathy and occasional deaths have been reported when the drug is given to persons with high burdens of *Loa loa* microfilaremia. Ivermectin is not approved for use during pregnancy.

Mebendazole This benzimidazole derivative is widely used for treatment of intestinal helminths and exhibits activity against *Echinococcus granulosus*. Benzimidazoles block parasite microtubule assembly and glucose uptake. Because mebendazole is poorly absorbed, its incidence of side effects is low, but its usefulness in treating tissue helminths is limited. Transient abdominal pain and diarrhea sometimes occur, usually in persons with massive parasite burdens. The use of mebendazole is contraindicated in pregnancy.

Mefloquine Like quinine and chloroquine, this quinoline is active only against the asexual erythrocytic stages of malarial parasites. The mode of action of mefloquine is similar to that of chloroquine, but mefloquine is not concentrated so extensively in the food vacuole and may act on alternative targets in the parasite. Taken as a single dose, it is the preferred drug for prophylaxis of chloroquine-resistant malaria; high doses can be used for treatment. The development of drug-resistant strains of *P. falciparum* in parts of Africa and Southeast Asia is ominous, but mefloquine is still an effective drug in most of the world. It is well tolerated by most persons, but its safety in pregnancy is unknown. Adverse effects are usually dose-related and include nausea and dizziness. Psychosis and seizures occur rarely, but treatment of patients with neuropsychiatric

conditions warrants caution. Concomitant use of quinine, quinidine, or drugs causing b-adrenergic blockade may produce electrocardiographic disturbances or cardiac arrest.

Melarsoprol* This trivalent arsenical is used for the treatment of late-stage East African trypanosomiasis and is not uniformly effective. The drug enters the parasite via an adenosine transporter; resistant strains lack this transport system. Arsenicals react avidly with sulfhydryl groups on proteins and inhibit their function. This is the likely mechanism of action and the cause of the severe adverse effects commonly seen. Encephalopathy is the most serious side effect, usually occurring within 4 days of the initiation of therapy and resulting in death in 6% of recipients.

Metrifonate This organophosphorus compound has selective activity against *Schistosoma haematobium*. It is partially metabolized to 2,2-dimethyldichlorovinyl phosphate (DDVP), a highly active chemical that irreversibly inhibits the acetylcholinesterase enzyme. Schistosomal cholinesterase is more susceptible to this metabolite than is the corresponding human enzyme. Metrifonate's exact mechanism of action is uncertain, but it is believed to inhibit tegumental acetylcholine receptors that mediate glucose transport. Although the drug is well tolerated, recipients experience a transient decrease in plasma cholinesterase activity and should not be exposed to neuromuscular blocking agents or organophosphate insecticides for at least 48 h after treatment. Metrifonate's safety in pregnancy is not established. The drug is currently unavailable in the United States.

Metronidazole Of the nitroimidazoles, only metronidazole has been licensed in the United States. This drug has [FDA](#) approval only for the treatment of amebiasis and trichomoniasis, although it is currently the drug of choice for giardiasis and trichomoniasis and is an alternative agent for balantidiasis. Metronidazole is reduced by anaerobic metabolism, and the metabolite acts as an electron sink, depriving anaerobes of reducing equivalents. Covalent binding or other interactions of intermediate metabolites of metronidazole with parasite macromolecules may partly explain the efficacy of this agent. Its benefit in dracunculiasis appears to be due to a reduction in inflammation rather than to any specific antihelminthic effect. Metronidazole is generally well tolerated despite common side effects such as nausea, headache, and a metabolic aftertaste. Alcohol should be avoided due to disulfiram-like effects. Although metronidazole has not been approved or recommended for use during pregnancy, it has not been associated with birth defects.

Nifurtimox* This nitrofurantoin compound is an effective oral agent for the treatment of acute Chagas' disease. Intracellular reduction followed by auto-oxidation yielding oxygen radicals has been suggested as the mode of action of nifurtimox on *Trypanosoma cruzi* and as the basis of its toxicity in humans. Prolonged use is required, but the course may have to be interrupted due to drug toxicity, which develops in 40 to 70% of recipients. Adverse reactions are common, dose related, and reversible. They include nausea, vomiting, abdominal pain, insomnia, seizures, and polyneuritis. Nifurtimox should be avoided in early pregnancy.

Nitazoxanide+ This 5-nitrothiazole compound appears to be a safe and effective alternative agent for the treatment of cryptosporidiosis. Its mechanism of action is unknown. It is currently available only from Romark Laboratories in the United States.

Oxamniquine This tetrahydroquinoline derivative is an effective alternative agent for the treatment of schistosomiasis, although susceptibility to this drug exhibits regional variation. In treated adult schistosomes, oxamniquine produces marked tegumental alterations similar to those seen with praziquantel but less rapid (evident 4 to 8 days after treatment). Patients should be warned that their urine may have an intense orange-red color. Side effects are uncommon and usually mild, although hallucinations and seizures have been reported. Oxamniquine has not been shown to be teratogenic or embryotoxic, but its use in pregnancy has not been approved.

Paromomycin This aminoglycoside is an effective oral agent for the treatment of infections due to intestinal protozoa. Like other aminoglycosides, it is poorly absorbed after oral administration and binds to the 30S ribosomal RNA in the aminoacyl-tRNA site, resulting in inhibition of protein synthesis. Paromomycin is well tolerated and safe for use in pregnancy.

Pentamidine Isethionate This diamine is an effective alternative agent for some forms of leishmaniasis and trypanosomiasis. While its mechanism of action remains undefined, it is known to exert a wide range of effects, including interaction with trypanosomal kinetoplast DNA, interference with polyamine synthesis through a decrease in the activity of ornithine decarboxylase, and inhibition of RNA polymerase, ribosomal function, and the synthesis of nucleic acids and proteins. Adverse reactions are common and include hypotension, pancreatitis, hypoglycemia, arrhythmias, and reversible renal failure.

Praziquantel This heterocyclic prazino-isoquinoline derivative is highly active against a broad spectrum of trematodes and cestodes. It disrupts the parasite tegument, resulting in contracture with loss of adherence to host tissues and ultimately disintegration or expulsion. Drug levels are reduced by concomitantly administered glucocorticoids, but cimetidine can be used to offset this problem. Praziquantel is generally well tolerated, but seizures may result in persons with neurocysticercosis. Patients with schistosomiasis who have heavy parasite burdens may develop abdominal discomfort, nausea, headache, dizziness, and drowsiness. Although praziquantel has not been shown to be mutagenic, teratogenic, or embryotoxic, it is preferable to delay treatment until after delivery unless immediate intervention is essential. Because praziquantel is excreted in breast milk, it is recommended that women not nurse on the day(s) of drug administration or for 72 h thereafter.

Primaquine Phosphate This drug is the only agent available for eradication of the hepatic stage of malarial parasites. In order to be effective, it must be metabolized by the host. Although their parasitocidal activity remains unclear, the metabolites are believed to affect both pyrimidine synthesis and the mitochondrial electron transport chain. The major adverse effect of this drug is acute hemolysis in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency. Primaquine phosphate is otherwise well tolerated. Its use in pregnancy is contraindicated.

Proguanil (Chloroguanide) This agent, which inhibits plasmodial dihydrofolate reductase, is used with atovaquone for oral treatment of uncomplicated malaria or with chloroquine for prophylaxis in parts of Africa without widespread chloroquine-resistant *P.*

falciparum. Proguanil is quite well tolerated at the usually prescribed doses, but higher doses can produce nausea, vomiting, abdominal pain, and diarrhea. It is not available in the United States.

Pyrantel Pamoate This safe, well-tolerated, and inexpensive pyrimidine derivative depolarizes the neuromuscular junctions of most intestinal nematodes, resulting in irreversible paralysis and allowing natural expulsion of the worms with the host's feces. The drug is poorly absorbed from the gastrointestinal tract and is usually effective in a single dose. It has minimal toxicity at the oral doses used to treat intestinal helminthic infection.

Pyrimethamine When combined with short-acting sulfonamides, this diaminopyrimidine is effective in malaria, toxoplasmosis, and isosporiasis. Unlike mammalian cells, the parasites that cause these infections cannot utilize preformed pyrimidines obtained through salvage pathways but rather rely completely on de novo synthesis of pyrimidines, for which folate derivatives are essential cofactors.

Pyrimethamine-sulfadoxine is an effective adjunctive agent for the oral treatment of uncomplicated malaria due to chloroquine-resistant organisms. However, its usefulness as a first-line agent is limited by the development of resistant strains of *P. falciparum* and *P. vivax*.

When combined with sulfadiazine, pyrimethamine is the treatment of choice for toxoplasmosis, but the duration of the required course of therapy often results in the development of folate deficiency requiring folinic acid supplementation. Sulfadiazine crystals can cause hematuria, and allergic drug reactions due to sulfadiazine often require a switch to clindamycin.

Quinacrine* Quinacrine is the only drug approved by the [FDA](#) for the treatment of giardiasis. It is not commercially available but can be obtained from alternative sources through the [CDC](#) Drug Service. Quinacrine intercalates into parasite DNA and inhibits nucleic acid synthesis. Side effects are common and include nausea and vomiting, headache, and skin discoloration. Alcohol is best avoided due to a disulfiram-like effect.

Quinine and Quinidine When combined with another agent, the cinchona alkaloid quinine is effective for the oral treatment of both uncomplicated malaria due to chloroquine-resistant strains and babesiosis. Quinine acts rapidly against the asexual blood stages of all forms of human malaria. For severe malaria, only quinidine (the dextroisomer of quinine) is available in the United States. Its use requires cardiac monitoring, and dose reduction is necessary in persons with severe renal impairment. Both quinine and quinidine can produce hypoglycemia. Symptoms of cinchonism (tinnitus, headache, nausea, and visual disturbances) are dose-related and reversible.

Spiramycin+ This macrolide antibiotic is used to treat acute toxoplasmosis in pregnancy and congenital toxoplasmosis. Although not yet licensed in the United States, it is available through the [FDA](#). Complications of treatment are rare but can include life-threatening ventricular arrhythmias in neonates.

Sodium Stibogluconate* and Meglumine Antimonate These pentavalent antimony

compounds are first-line agents for the treatment of all forms of leishmaniasis. Despite their use in leishmaniasis for almost 100 years, their mechanism of action against *Leishmania* spp. remains unknown. Presumably, the compounds interfere with parasite metabolism. The drugs are taken up by the reticuloendothelial system, and their activity against *Leishmania* spp. may be enhanced by this localization. Resistance is a major problem in some areas of the world. Side effects are common and generally reversible but require temporary interruption of therapy in many cases. Chemical pancreatitis is almost universal, although it is often asymptomatic. In rare instances, pentavalent antimony compounds produce prolongation of the QT interval, and sudden death due to arrhythmia or cardiac failure has been reported. Arthralgias, myalgias, and headaches occur frequently. Since the drugs' safety in pregnancy has not been established, their use should be avoided if possible in pregnant women. These agents may be used in children >18 months of age.

Suramin* This derivative of urea is the drug of choice for the early stage of African trypanosomiasis. The drug acts by forming stable complexes with proteins, inhibiting multiple enzymes. Suramin has a variety of potentially severe side effects, including anaphylaxis, exfoliative dermatitis, paresthesias, photophobia, and renal dysfunction.

Tetracycline and Doxycycline These antibiotics are useful in the treatment of balantidiasis and *D. fragilis* infection as well as in the oral treatment of uncomplicated malaria due to chloroquine-resistant strains. The tetracyclines inhibit protein synthesis in prokaryotic ribosomes, and they probably have the same activity in parasites. Potential side effects in adults include nausea, vomiting, and photosensitivity dermatitis. Because they can impair normal development of bones and teeth, tetracyclines are contraindicated in pregnant women and children <8 years of age.

Thiabendazole This benzimidazole derivative is a potent antihelminthic agent, but its use in strongyloidiasis and other infections is limited by frequent, severe side effects. Patients most often report dizziness, headache, nausea, and vomiting. Less commonly, hepatitis and severe hypersensitivity reactions develop. The drug's mechanism of action is similar to that of other benzimidazoles. Treatment with thiabendazole is contraindicated in pregnancy.

Tinidazole This nitroimidazole is effective for the treatment of amebiasis, giardiasis, and trichomoniasis. Its mechanism of action and side effects are similar to those seen with metronidazole, but adverse events appear to be less frequent and severe with tinidazole. In addition, tinidazole is potentially curative in a single dose. This agent is currently unavailable in the United States.

Triclabendazole This benzimidazole is effective against paragonimiasis and all stages of *Fasciola hepatica*, a trematode with inherent resistance to praziquantel. The sulfoxide metabolite, which is believed to be responsible for the drug's activity, binds to fluke tubulin and disrupts microtubule-based processes. Triclabendazole is safe and well tolerated and offers single-dose cure. It is currently unavailable in the United States.

Trimethoprim-Sulfamethoxazole This synergistic antifolate compound is active against cyclosporiasis, isosporiasis, and encephalitis due to *Toxoplasma gondii*. Trimethoprim is a dihydrofolate reductase inhibitor whose effect is enhanced by

sulfamethoxazole. Adverse effects, which can be severe, are usually attributable to the sulfonamide component and involve allergic skin reactions, bone marrow suppression, and hemolysis.

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SECTION 17 -PROTOZOAL INFECTIONS

213. AMEBIASIS AND INFECTION WITH FREE-LIVING AMEBAS - Sharon L. Reed

AMEBIASIS

DEFINITION

Amebiasis is an infection with the intestinal protozoan *Entamoeba histolytica*. About 90% of infections are asymptomatic, and the remaining 10% produce a spectrum of clinical syndromes ranging from dysentery to abscesses of the liver or other organs.

LIFE CYCLE AND TRANSMISSION

E. histolytica is acquired by ingestion of viable cysts from fecally contaminated water, food, or hands. Food-borne exposure is most prevalent and is particularly likely when food handlers are shedding cysts or food is being grown with feces-contaminated soil, fertilizer, or water. Less common means of transmission include contaminated water, oral and anal sexual practices, and -- in rare instances -- direct rectal inoculation through colonic irrigation devices. Motile trophozoites are released from cysts in the small intestine and, in most patients, remain as harmless commensals in the large bowel. After encystation, infectious cysts are shed in the stool and can survive for several weeks in a moist environment. In some patients, the trophozoites invade either the bowel mucosa, causing symptomatic colitis, or the bloodstream, causing distant abscesses of the liver, lungs, or brain. The trophozoites may not encyst in patients with active dysentery, and motile hematophagous trophozoites are frequently present in fresh stools. Trophozoites are rapidly killed by exposure to air or stomach acid, however, and therefore cannot cause infection.

EPIDEMIOLOGY

About 10% of the world's population is infected with *Entamoeba*, the majority with noninvasive *Entamoeba dispar*. Amebiasis results from infection with *E. histolytica* and is the third most common cause of death from parasitic disease (after schistosomiasis and malaria). Areas of highest incidence (due to inadequate sanitation and crowding) include most developing countries in the tropics, particularly Mexico, India, and nations of Central and South America, tropical Asia, and Africa. The main groups at risk in developed countries are travelers, recent immigrants, homosexual men, and inmates of institutions.

The wide spectrum of clinical disease is caused in part by infection with the two different species of *Entamoeba*. Isolates of *E. histolytica* from patients with invasive amebiasis have unique isoenzymes, surface antigens, DNA markers, and virulence properties and now are recognized as a distinct species from the noninvasive *E. dispar*.

Most asymptomatic carriers, including homosexual men and AIDS patients, harbor *E. dispar* and have self-limited infections. These observations suggest that *E. dispar* is incapable of causing invasive disease, since *Cryptosporidium* and *Isospora belli*, which also cause only self-limited illnesses in immunocompetent people, cause devastating

diarrhea in patients with AIDS. However, host factors play a role as well. In one study, 10% of asymptomatic patients who were colonized with *E. histolytica* went on to develop amebic colitis, while the rest remained asymptomatic and cleared the infection within 1 year.

PATHOGENESIS AND PATHOLOGY

Both trophozoites ([Fig. 213-1](#)) and cysts ([Fig. 213-2](#)) are found in the intestinal lumen, but only trophozoites of *E. histolytica* invade tissue. The trophozoite is 20 to 60 μm in diameter and contains vacuoles and a nucleus with a characteristic central karyosome. In animals, depletion of intestinal mucus, diffuse inflammation, and disruption of the epithelial barrier occur before trophozoites actually come into contact with the colonic mucosa. Trophozoites attach to colonic mucus and epithelial cells by a galactose-inhibitable lectin. The earliest intestinal lesions are microulcerations of the mucosa of the cecum, sigmoid colon, or rectum that release erythrocytes, inflammatory cells, and epithelial cells. Proctoscopy reveals small ulcers with heaped up margins and normal intervening mucosa. Submucosal extension of ulcerations under viable-appearing surface mucosa causes the classic "flask-shaped" ulcer containing trophozoites at the margins of dead and viable tissues. Although neutrophilic infiltrates may accompany the early lesions in animals, human intestinal infection is marked by a paucity of inflammatory cells, probably in part because of the killing of neutrophils by trophozoites. Treated ulcers characteristically heal with little or no scarring. Occasionally, however, full-thickness necrosis and perforation occur.

Rarely, intestinal infection results in the formation of a mass lesion, or *ameboma*, in the bowel lumen. The overlying mucosa is usually thin and ulcerated, while other layers of the wall are thickened, edematous, and hemorrhagic; this condition results in exuberant formation of granulation tissue with little fibrous-tissue response.

A number of virulence factors have been linked to the ability of *E. histolytica* to invade through the interglandular epithelium. One is an extracellular cysteine proteinase that degrades collagen, elastin, secretory IgA, and the anaphylatoxins C3a and C5a. Other enzymes may disrupt glycoprotein bonds between mucosal epithelial cells in the gut. Amebas can lyse neutrophils, monocytes, lymphocytes, and cells of colonic and hepatic cell lines. The cytolytic effect of amebas appears to require direct contact with target cells and may be linked to the release of phospholipase A and pore-forming peptides.

Liver abscesses are always preceded by intestinal colonization, which may be asymptomatic. Blood vessels may be compromised early by lysis of the wall and thrombus formation. Trophozoites invade veins to reach the liver through the portal venous system. *E. histolytica* is resistant to complement-mediated lysis, a property critical to survival in the bloodstream. In contrast, *E. dispar* is rapidly lysed by complement and is thus restricted to the bowel lumen. Inoculation of amebas into the portal system of hamsters results in an acute cellular infiltrate consisting predominantly of neutrophils. Later, the neutrophils are lysed by contact with amebas, and the release of neutrophil toxins may contribute to necrosis of hepatocytes. The liver parenchyma is replaced by necrotic material that is surrounded by a thin rim of congested liver tissue. The necrotic contents of a liver abscess are classically described as "anchovy paste," although the fluid is variable in color and is composed of bacteriologically sterile

granular debris with few or no cells. Amebas, if seen, tend to be found near the capsule of the abscess.

Clinical infection does not induce immunity to recurrent colonization with *E. histolytica*, but repeated episodes of colitis or liver abscess are unusual. Antibody is not protective; titers correlate with the length of illness rather than with the severity of disease. Studies of animals suggest that cell-mediated immunity may be important for protection, although patients with AIDS appear not to be predisposed to more severe disease.

CLINICAL SYNDROMES

Intestinal Amebiasis The most common type of amebic infection is asymptomatic cyst passage. Even in highly endemic areas, most patients harbor *E. dispar*.

Symptomatic amebic colitis develops 2 to 6 weeks after the ingestion of infectious cysts. Lower abdominal pain and mild diarrhea develop gradually and are followed by malaise, weight loss, and diffuse lower abdominal or back pain. Cecal involvement may mimic acute appendicitis. Patients with full-blown dysentery may pass 10 to 12 stools per day. The stools contain little fecal material and consist mainly of blood and mucus. In contrast to those with bacterial diarrhea, fewer than 40% of patients with amebic dysentery are febrile. Virtually all patients have heme-positive stools.

More fulminant intestinal infection, with severe abdominal pain, high fever, and profuse diarrhea, is rare and occurs predominantly in children. Patients may develop toxic megacolon, in which there is severe bowel dilation with intramural air. Patients receiving glucocorticoids are at risk for severe amebiasis. Uncommonly, patients develop a chronic form of amebic colitis, which can be confused with inflammatory bowel disease. The association between severe amebiasis complications and glucocorticoid therapy emphasizes the importance of excluding amebiasis when inflammatory bowel disease is suspected. An occasional patient presents with only an asymptomatic or tender abdominal mass caused by an ameboma, which is easily confused with cancer on barium studies. A positive serologic test or biopsy can prevent unnecessary surgery in this setting. The syndrome of postamebic colitis -- persistent diarrhea following documented cure of amebic colitis -- is controversial; no evidence of recurrent amebic infection can be found, and re-treatment usually has no effect.

Amebic Liver Abscess Extraintestinal infection by *E. histolytica* most often involves the liver. Of travelers who develop an amebic liver abscess after leaving an endemic area, 95% do so within 5 months. Young patients with an amebic liver abscess are more likely than older patients to present in the acute phase with prominent symptoms of <10 days duration. Most patients are febrile and have right-upper-quadrant pain, which may be dull or pleuritic in nature and radiate to the shoulder. Point tenderness over the liver and right-sided pleural effusion are common. Jaundice is rare. Although the initial site of infection is the colon, fewer than one-third of patients with an amebic abscess have active diarrhea. Older patients from endemic areas are more likely to have a subacute course lasting 6 months, with weight loss and hepatomegaly. About one-third of patients with chronic presentations are febrile. Thus, the clinical diagnosis of an amebic liver abscess may be difficult to establish because the symptoms and signs are often nonspecific. Since 10 to 15% of patients present only with fever, amebic liver abscess

must be considered in the differential diagnosis of fever of unknown origin ([Chap. 125](#)).

Complications of Amebic Liver Abscess Pleuropulmonary involvement, which is reported in 20 to 30% of patients, is the most frequent complication of amebic liver abscess. Manifestations include sterile effusions, contiguous spread from the liver, and rupture into the pleural space. Sterile effusions and contiguous spread usually resolve with medical therapy, but frank rupture into the pleural space requires drainage. A hepatobronchial fistula may cause cough productive of large amounts of necrotic material that may contain amebas. This dramatic complication carries a good prognosis. Abscesses that rupture into the peritoneum may present as an indolent leak or an acute abdomen and require both percutaneous catheter drainage and medical therapy. Rupture into the pericardium, usually from abscesses of the left lobe of the liver, carries the gravest prognosis; it can occur during medical therapy and requires surgical drainage.

Other Extraintestinal Sites The genitourinary tract may become involved by direct extension of amebiasis from the colon or by hematogenous spread of the infection. Painful genital ulcers, characterized by a punched-out appearance and profuse discharge, may develop secondary to extension from either the intestine or the liver. Both these conditions respond well to medical therapy. Cerebral involvement has been reported in fewer than 0.1% of patients in large clinical series. Symptoms and prognosis depend on the size and location of the lesion.

DIAGNOSTIC TESTS

Laboratory Diagnosis Stool examinations, serologic tests, and noninvasive imaging of the liver are the most important procedures in the diagnosis of amebiasis. Fecal findings suggestive of amebic colitis include a positive test for heme, a paucity of neutrophils, and the presence of Charcot-Leyden crystal protein (double pyramid-shaped crystals normally found in the cytoplasm of eosinophils). The definitive diagnosis of amebic colitis is made by the demonstration of hematophagous trophozoites of *E. histolytica* ([Fig. 213-1](#)). Because trophozoites are killed rapidly by water, drying, or barium, it is important to examine at least three fresh stool specimens. Examination of a combination of wet mounts, iodine-stained concentrates, and trichrome-stained preparations of fresh stool and concentrates for cysts ([Fig. 213-2](#)) or trophozoites ([Fig. 213-1](#)) confirms the diagnosis in 75 to 95% of cases. Cultures of amebas are more sensitive but are not routinely available. If stool examinations are negative, sigmoidoscopy with biopsy of the edge of ulcers may increase the yield, but this procedure is dangerous during fulminant colitis because of the risk of perforation. Trophozoites in a biopsy specimen from a colonic mass confirm the diagnosis of ameboma, but trophozoites are rare in liver aspirates. Accurate diagnosis requires experience, since the trophozoites may be confused with neutrophils and the cysts must be differentiated morphologically from *Entamoeba hartmanni*, *Entamoeba coli*, and *Endolimax nana*, which do not cause clinical disease and do not warrant therapy. Unfortunately, the cysts of *E. histolytica* cannot be distinguished microscopically from those of *E. dispar*. Therefore, the microscopic diagnosis of *E. histolytica* can be made only by the detection of *Entamoeba* trophozoites that have ingested erythrocytes ([Fig. 213-1](#)). Diagnostic tests based on the detection of the galactose-inhibitable lectin of *E. histolytica* are now available and compare favorably with the polymerase chain reaction and with isolation in culture.

followed by isoenzyme analysis in terms of sensitivity.

Serology is an important addition to the methods used for the parasitologic diagnosis of invasive amebiasis. Kits for the performance of agar gel diffusion assays and ELISAs are commercially available, and the results of these tests are positive in more than 90% of patients with colitis, amebomas, or liver abscess. Positive results in conjunction with the appropriate clinical syndrome suggest active disease because serologic findings usually revert to negative within 6 to 12 months. Even in highly endemic areas such as South Africa, fewer than 10% of asymptomatic individuals have a positive amebic serology. The interpretation of the indirect hemagglutination test is more difficult because titers may remain positive for as long as 10 years.

Up to 10% of patients with acute amebic liver abscess may have negative serologic findings; in suspected cases with an initially negative result, testing should be repeated in a week. In contrast to carriers of *E. dispar*, most asymptomatic carriers of *E. histolytica* develop antibodies. Thus, serologic tests are helpful in assessing the risk of invasive amebiasis in asymptomatic, cyst-passing individuals in nonendemic areas. Serologic tests also should be performed in patients with ulcerative colitis before the institution of glucocorticoid therapy to prevent the development of severe colitis or toxic megacolon owing to unsuspected amebiasis.

Routine hematology and chemistry tests are usually not very helpful in the diagnosis of invasive amebiasis. About three-fourths of patients with an amebic liver abscess have leukocytosis ($>10,000$ cells/uL); this condition is particularly likely if symptoms are acute or complications have developed. Invasive amebiasis does not elicit eosinophilia. Anemia, if present, is usually multifactorial. Even with large liver abscesses, liver enzyme levels are normal or minimally elevated. The alkaline phosphatase level is most often elevated and may remain so for months. Aminotransferase elevations suggest acute disease or a complication.

Radiographic Studies Radiographic barium studies are potentially dangerous in acute amebic colitis. Amebomas are usually identified first by a barium enema, but biopsy is necessary for differentiation from carcinoma.

Radiographic techniques such as ultrasonography, computed tomography ([Fig. 213-3](#)), and magnetic resonance imaging are all useful for detection of the round or oval hypoechoic cyst. More than 80% of patients who have had symptoms for >10 days have a single abscess of the right lobe of the liver. Approximately 50% of patients who have had symptoms for <10 days have multiple abscesses. Findings associated with complications include large abscesses (>10 cm) in the superior part of the right lobe, which may rupture into the pleural space; multiple lesions, which must be differentiated from pyogenic abscesses; and lesions of the left lobe, which may rupture into the pericardium. Because abscesses resolve slowly and may increase in size in patients who are responding clinically to therapy, frequent follow-up ultrasonography may prove confusing. Complete resolution of a liver abscess within 6 months can be anticipated in two-thirds of patients, but 10% may have persistent abnormalities for a year.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of intestinal amebiasis includes bacterial diarrheas caused by *Campylobacter*; enteroinvasive *Escherichia coli*; and *Shigella*, *Salmonella*, and *Vibrio* species. Although the typical patient with amebic colitis has less prominent fever than in these other conditions as well as heme-positive stools with few neutrophils, correct diagnosis requires bacterial cultures, microscopic examination of stools, and amebic serologic testing. As has already been mentioned, amebiasis must be ruled out in any patient thought to have inflammatory bowel disease.

Because of the variety of presenting signs and symptoms, amebic liver abscess can easily be confused with pulmonary or gallbladder disease or with any febrile illness with few localizing signs, such as malaria or typhoid fever. The diagnosis should be considered in members of high-risk groups who have recently traveled outside the United States and in inmates of institutions. Once radiographic studies have identified an abscess in the liver, the most important differential diagnosis is between amebic and pyogenic abscess. Patients with pyogenic abscess typically are older and have a history of underlying bowel disease or recent surgery. Amebic serology is helpful, but aspiration of the abscess, with Gram's staining and culture of the material, may be required for differentiation of the two diseases.

TREATMENT

Intestinal Disease The drugs used to treat amebiasis can be classified according to their primary site of action. Luminal amebicides are poorly absorbed and reach high concentrations in the bowel, but their activity is limited to cysts and trophozoites close to the mucosa. Only two luminal drugs are available in the United States: iodoquinol and paromomycin ([Table 213-1](#)). Indications for the use of luminal agents include eradication of cysts in patients with colitis or a liver abscess and treatment of asymptomatic carriers. The majority of asymptomatic individuals who pass cysts are colonized with *E. dispar*, which does not warrant specific therapy. However, unless the presence of *E. dispar* can be proven by specific antigen detection tests and the lack of an antibody response, it may be prudent to treat asymptomatic individuals who pass cysts.

Tissue amebicides reach high concentrations in the blood and tissue after oral or parenteral administration. The development of nitroimidazole compounds, especially metronidazole, was a major advance in the treatment of invasive amebiasis. Patients with amebic colitis should be treated with intravenous or oral metronidazole (750 mg three times daily for 5 to 10 days). Side effects include nausea, vomiting, abdominal discomfort, and a disulfiram-like reaction. Other imidazole compounds, such as tinidazole and ornidazole, are as effective but are not available in the United States. All patients should also receive a full course of therapy with a luminal agent, since metronidazole does not eradicate cysts. Resistance to metronidazole has not been identified. Relapses are not uncommon and probably represent reinfection or failure to eradicate amebas from the bowel because of an inadequate dosage or duration of therapy.

Amebic Liver Abscess Metronidazole is the drug of choice for amebic liver abscess. The usefulness of nitroimidazoles in single-dose or abbreviated regimens is important in endemic areas where access to hospitalization is limited. With early diagnosis and therapy, mortality from uncomplicated amebic liver abscess is <1%. The second-line

therapeutic agents emetine and chloroquine should be avoided if possible because of the potential cardiovascular and gastrointestinal side effects of the former and the higher relapse rates with the latter. There is no evidence that combined therapy with two drugs is more effective than the single-drug regimen. Studies of South Africans with liver abscesses demonstrated that 72% of patients without intestinal symptoms had bowel infection with *E. histolytica*; thus, all treatment regimens should include a luminal agent to eradicate cysts and prevent further transmission. Amebic liver abscess recurs rarely.

Aspiration of Liver Abscesses More than 90% of patients respond dramatically to metronidazole therapy with decreases in both pain and fever within 72 h. Indications for aspiration of liver abscesses are (1) the need to rule out a pyogenic abscess, particularly in patients with multiple lesions; (2) the failure to respond clinically in 3 to 5 days; (3) the threat of imminent rupture; and (4) the prevention of rupture of left-lobe abscesses into the pericardium. There is no evidence that aspiration, even of large abscesses (up to 10 cm), accelerates healing. Percutaneous drainage may be successful even if the liver abscess has already ruptured. Surgery should be reserved for instances of bowel perforation and rupture into the pericardium.

PREVENTION

Amebic infection is spread by ingestion of food or water contaminated with cysts. Since an asymptomatic carrier may excrete up to 15 million cysts per day, prevention of infection requires adequate sanitation and eradication of cyst carriage. In high-risk areas, infection can be minimized by the avoidance of unpeeled fruits and vegetables and the use of bottled water. Because cysts are resistant to readily attainable levels of chlorine, disinfection by iodination (tetraglycine hydroperiodide) is recommended. There is no effective prophylaxis.

INFECTION WITH FREE-LIVING AMEBAS

EPIDEMIOLOGY

Free-living amebas of the genera *Acanthamoeba*, *Naegleria*, and *Balamuthia* are distributed throughout the world and have been isolated from a wide variety of fresh and brackish water, including that from lakes, taps, hot springs, swimming pools, and heating and air-conditioning units, and even from the nasal passages of healthy children. Encystation may protect the protozoa from desiccation and food deprivation. The persistence of *Legionella pneumophila* in water supplies may be attributable in part to chronic infection of free-living amebas, particularly *Naegleria*.

NAEGLERIA INFECTIONS

Primary amebic meningoencephalitis caused by *Naegleria fowleri* follows the aspiration of water contaminated with trophozoites or cysts or the inhalation of contaminated dust, leading to invasion of the olfactory neuroepithelium. After an incubation period of 2 to 15 days, severe headache, high fever, nausea, vomiting, and meningismus develop. Photophobia and palsies of the third, fourth, and sixth cranial nerves are common. Rapid progression to seizures and coma may follow, and most patients die within a week. Infection is most common in otherwise healthy children or young adults, who

often report recent swimming in lakes or heated swimming pools.

Diagnosis depends on the detection of motile trophozoites in wet mounts of fresh spinal fluid. Other laboratory findings resemble those for fulminant bacterial meningitis, with elevated intracranial pressure, high white blood cell counts (up to 20,000 cells/uL), and elevated protein concentrations and low glucose levels in cerebrospinal fluid. The diagnosis should be considered in any patient who has purulent meningitis without evidence of bacteria on Gram's staining, antigen detection assay, and culture. The prognosis is uniformly poor. Only four survivors, treated with high-dose amphotericin B and rifampin, have been reported. Antibodies to *Naegleria* spp. have been detected in normal adults; serologic testing is not useful in the diagnosis of acute infection.

ACANTHAMOEBA INFECTIONS

Granulomatous Amebic Encephalitis Infection with *Acanthamoeba* species follows a more indolent course and occurs typically in chronically ill or debilitated patients. Risk factors include lymphoproliferative disorders, chemotherapy, glucocorticoid therapy, lupus erythematosus, and AIDS. Infection usually reaches the central nervous system hematogenously from a primary focus in the sinuses, skin, or lungs. In the central nervous system, the onset is insidious, and the syndrome often mimics a space-occupying lesion. Altered mental status, headache, and stiff neck may be accompanied by focal findings such as cranial nerve palsies, ataxia, and hemiparesis. In the United States, cutaneous ulcers or hard nodules containing amebas were detected in 8 of 13 AIDS patients with disseminated *Acanthamoeba* infection.

Examination of the cerebrospinal fluid for trophozoites may be diagnostically helpful, but lumbar puncture may be contraindicated because of increased intracerebral pressure. Computed tomography frequently reveals cortical and subcortical lesions of decreased density consistent with embolic infarcts. In other patients, multiple enhancing lesions with edema may mimic the computed tomographic appearance of toxoplasmosis. Demonstration of the trophozoites and cysts of *Acanthamoeba* on wet mounts or in biopsy specimens establishes the diagnosis. Culture on nonnutrient agar plates seeded with *Escherichia coli* may also be helpful. Fluorescein-labeled antiserum is available from the Centers for Disease Control and Prevention (CDC) for the detection of protozoa in biopsy specimens. At least nine cases of granulomatous amebic encephalitis have been reported in patients with AIDS, in whom the disease may have an accelerated course (with survival for only 3 to 40 days) because of their difficulty in forming granulomas. Although studies in animals suggest that rifampin may be useful, the infection is almost uniformly fatal.

Keratitis The incidence of keratitis caused by *Acanthamoeba* has increased in the past 20 years, in part as a result of improved diagnosis. The first of these infections to be recognized were associated with trauma to the eye and exposure to contaminated water. At present, most infections are linked to extended-wear contact lenses. Risk factors include the use of homemade saline, the wearing of lenses while swimming, and inadequate disinfection. Since contact lenses presumably cause microscopic trauma, the early corneal findings may be nonspecific. The first symptoms usually include tearing and the painful sensation of a foreign body. Once infection is established, progression is rapid; the characteristic clinical sign is an annular, paracentral corneal

ring representing a corneal abscess. Deeper corneal invasion and loss of vision may follow.

The differential diagnosis includes bacterial, mycobacterial, and herpetic infection. The irregular polygonal cysts of *Acanthamoeba* ([Fig. 213-4](#)) may be identified in corneal scrapings or biopsy material, and trophozoites can be grown on special media. Cysts are resistant to available drugs, and the results of medical therapy have been disappointing. Some reports have suggested partial responses to propamidine isethionate eyedrops. Severe infections usually require keratoplasty.

BALAMUTHIA INFECTIONS

Balamuthia mandrillaris, a free-living ameba previously referred to as a leptomyxid ameba, is an important etiologic agent of amebic meningoencephalitis in immunocompetent hosts. The course is typically subacute, with focal neurologic signs, fever, seizures, and headaches leading to death within 1 week to several months after onset. Examination of cerebrospinal fluid reveals mononuclear pleocytosis, elevated protein levels, and normal to low glucose concentrations. Multiple hypodense lesions are usually detected with imaging studies. The diagnosis is almost always made post-mortem, and specific identification may require immunofluorescence with antibodies from the [CDC](#) to differentiate the trophozoites from *Acanthamoeba*.

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214. MALARIA AND BABESIOSIS: DISEASES CAUSED BY RED BLOOD CELL PARASITES - *Nicholas J. White, Joel G. Breman*

"Humanity has but three great enemies: Fever, famine and war; of these by far the greatest, by far the most terrible, is fever."

William Osler

MALARIA

Malaria is a protozoan disease transmitted by the bite of infected *Anopheles* mosquitoes. It is the most important of the parasitic diseases of humans, with transmission in 103 countries affecting more than 1 billion people and causing between 1 and 3 million deaths each year. Malaria has now been eradicated from North America, Europe, and Russia but, despite enormous control efforts, has resurged in many parts of the tropics. Added to this resurgence are the increasing problems of drug resistance of the parasite and insecticide resistance of the vectors. Occasional local transmission following importation of malaria has occurred recently in several southern and eastern areas of the United States and in Europe, indicating the continual danger to nonmalarious countries. Malaria remains today, as it has been for centuries, a heavy burden on tropical communities, a threat to nonendemic countries, and a danger to travelers.

ETIOLOGY AND PATHOGENESIS

Four species of the genus *Plasmodium* cause nearly all malarial infections in humans (although rare infections involve species normally affecting other primates). These are *P. falciparum*, *P. vivax*, *P. ovale*, and *P. malariae* ([Table 214-1](#)). Almost all deaths are caused by falciparum malaria. Human infection begins when a female anopheline mosquito inoculates plasmodial sporozoites from its salivary gland during a blood meal ([Fig. 214-1](#)). These microscopic motile forms of the malarial parasite are carried rapidly via the bloodstream to the liver, where they invade hepatic parenchymal cells and begin a period of asexual reproduction. By this amplification process (known as intrahepatic or preerythrocytic schizogony or merogony), a single sporozoite eventually may produce 10,000 to more than 30,000 daughter merozoites. The swollen liver cell eventually bursts, discharging motile merozoites into the bloodstream; at this point the symptomatic stage of the infection begins. In *P. vivax* and *P. ovale* infections, a proportion of the intrahepatic forms do not divide immediately but remain dormant for months to years before reproduction begins. These dormant forms, or hypnozoites, are the cause of the relapses that characterize infection with these two species.

After entry into the bloodstream, merozoites rapidly invade erythrocytes and become trophozoites. Attachment is mediated via a specific erythrocyte surface receptor. In the case of *P. vivax*, this receptor is related to the Duffy blood-group antigen Fy_a or Fy_b. Most West Africans and people with origins in that region carry the Duffy-negative FyFy phenotype and are therefore resistant to *P. vivax* malaria. During the early stage of intraerythrocytic development, the small "ring forms" of the four parasitic species appear similar under light microscopy. As the trophozoites enlarge, species-specific characteristics become evident, pigment becomes visible, and the parasite assumes an

irregular or ameboid shape. By the end of the 48-h intraerythrocytic life cycle (72 h for *P. malariae*), the parasite has consumed nearly all the hemoglobin and grown to occupy most of the red cell. Multiple nuclear divisions take place (merogony), and the red cell ruptures to release 6 to 30 daughter merozoites, each capable of invading a new red cell and repeating the cycle. The disease in human beings is caused by the direct effects of red cell invasion and destruction by the asexual parasite and the host's reaction. After a series of asexual cycles (*P. falciparum*) or immediately (*P. vivax*, *P. ovale*, *P. malariae*), some of the parasites develop into morphologically distinct long-lived sexual forms (gametocytes) that can transmit malaria.

After being ingested in the blood meal of a biting female anopheline mosquito, the male and female gametocytes form a zygote in the insect's midgut. This zygote matures into an ookinete, which penetrates and encysts in the mosquito's gut wall. The resulting oocyst expands by asexual division until it bursts to liberate myriad motile sporozoites, which then migrate in the hemolymph to the salivary gland of the mosquito to await inoculation into another human at the next feeding.

EPIDEMIOLOGY

Malaria occurs throughout most of the tropical regions of the world ([Fig. 214-2](#)). *P. falciparum* predominates in Africa, New Guinea, and Haiti; *P. vivax* is more common in Central America and the Indian subcontinent. The prevalence of these two species is approximately equal in South America, eastern Asia, and Oceania. *P. malariae* is found in most endemic areas, especially throughout sub-Saharan Africa, but is much less common than the other species mentioned. *P. ovale* is relatively unusual outside of Africa and, where it is found, comprises <1% of isolates.

The epidemiology of malaria is complex and may vary considerably even within relatively small geographic areas. Endemicity traditionally has been defined in terms of parasitemia rates or palpable-spleen rates in children 2 to 9 years of age as hypoendemic (<10%), mesoendemic (11 to 50%), hyperendemic (51 to 75%), and holoendemic (>75%). In holo- and hyperendemic areas -- e.g., certain regions of tropical Africa or coastal New Guinea, where there is intense *P. falciparum* transmission and there can be more than one human bite per infected mosquito per day -- people are infected repeatedly throughout their lives. Here, morbidity and mortality during childhood are considerable. Immunity against disease is hard won in these areas, and the young-childhood burden of disease is high; by adulthood, however, most malarial infections are asymptomatic. This situation, with frequent year-round infection, is termed *stable transmission* and generally occurs where there is holo- and hyperendemicity. In areas where transmission is low, erratic, or focal, full protective immunity is not acquired, and symptomatic disease may occur at all ages. This situation usually exists in hypoendemic areas and is termed *unstable transmission*. Even in areas with stable transmission, there is often an increased incidence coinciding with increased mosquito breeding during the rainy season. Malaria behaves like an epidemic disease in some areas, particularly those with unstable malaria, such as northern India, Sri Lanka, Southeast Asia, Ethiopia, southern Africa, and Madagascar. An epidemic can develop when there are changes in environmental, economic, or social conditions, such as heavy rains following drought or migrations (usually of refugees or workers) from a nonmalarious region to an area of high transmission; a breakdown in malaria control

and prevention services can intensify epidemic conditions. This situation usually results in considerable mortality among all age groups.

The principal determinants of the epidemiology of malaria are the number (density), the human-biting habits, and the longevity of the anopheline mosquito vectors. Not all anophelines can transmit malaria, and those that do vary considerably in their efficiency as malaria vectors. More specifically, the transmission of malaria is directly proportional to the density of the vector, the square of the number of human bites per day per mosquito, and the tenth power of the probability of the mosquito's surviving for 1 day. Mosquito longevity is particularly important, because the portion of the parasite's life cycle that takes place within the mosquito -- from gametocyte ingestion to subsequent inoculation (sporogony) -- lasts for 8 to 30 days, depending on ambient temperature; thus, to transmit malaria, the mosquito must survive for longer than 7 days. In general, at temperatures below 16 to 18°C, sporogony is not completed and transmission does not occur. Therefore, the most effective mosquito vectors of malaria are those such as *A. gambiae*, which are long-lived, occur in high densities in tropical climates, breed readily, and bite humans in preference to other animals. The entomologic inoculation rate -- the number of sporozoite-positive mosquito bites per year -- is the most common measure of malarial transmission and varies from <1 in some parts of Latin America and Southeast Asia to >300 in parts of tropical Africa.

ERYTHROCYTE CHANGES IN MALARIA

After invading an erythrocyte, the growing parasite progressively consumes and degrades intracellular proteins, principally hemoglobin. The potentially toxic heme is polymerized to biologically inert hemozoin, or malaria pigment. The parasite also alters the red cell membrane by changing its transport properties, exposing cryptic surface antigens, and inserting new parasite-derived proteins. The red cell becomes more irregular in shape, more antigenic, and less deformable.

In *P. falciparum* infections, membrane protuberances appear on the erythrocyte's surface in the second 24 h of the asexual cycle. These "knobs" extrude a high-molecular-weight, antigenically variant, strain-specific, adhesive protein (PfEMP1) that mediates attachment to receptors on venular and capillary endothelium -- an event termed *cytoadherence*. Several receptors have been identified, of which intercellular adhesion molecule 1 is probably the most important in the brain, chondroitin sulfate B in the placenta, and CD36 in most other organs. Thus the infected erythrocytes stick inside the small blood vessels. At the same stage, these *P. falciparum*-infected red cells may also adhere to uninfected red cells to form rosettes. The processes of cytoadherence and rosetting are central to the pathogenesis of falciparum malaria. They result in the sequestration of red cells containing mature forms of the parasite in vital organs (particularly the brain), where they interfere with microcirculatory flow and metabolism. Sequestered parasites continue to develop out of reach of the principal host defense mechanism: splenic processing and filtration. As a consequence, only the younger ring forms of the asexual parasites are seen in the peripheral blood in falciparum malaria, and the level of peripheral parasitemia underestimates the true number of parasites within the body. Severe malaria is also associated with reduced deformability of the uninfected erythrocytes, which compromises their passage through the partially obstructed capillaries and venules and shortens red cell survival.

In the other three "benign" malarias, sequestration does not occur, and all stages of the parasite's development are evident on peripheral blood smears. Whereas *P. vivax*, *P. ovale*, and *P. malariae* show a marked predilection for either old red cells or reticulocytes and produce a level of parasitemia seldom exceeding 2%, *P. falciparum* can invade erythrocytes of all ages and may be associated with very high levels of parasitemia.

HOST RESPONSE

Initially, the host responds to plasmodial infection by activating nonspecific defense mechanisms. Splenic immunologic and filtrative clearance functions are augmented in malaria, and the removal of both parasitized and uninfected erythrocytes is accelerated. The parasitized cells escaping splenic removal are destroyed when the schizont ruptures. The material released induces the activation of macrophages and the release of proinflammatory mononuclear cell-derived cytokines, which cause fever and exert other pathologic effects. Temperatures of 340°C damage mature parasites; in untreated infections, the effect of such temperatures is to synchronize further the parasitic cycle with eventual production of the regular fever spikes and rigors that originally served to characterize the different malarias. These regular fever patterns (tertian, every 2 days; quartan, every 3 days) are seldom seen in patients who receive prompt and effective antimalarial treatment.

The geographic distributions of sickle cell disease, thalassemia, and glucose-6-phosphate dehydrogenase (G6PD) deficiency closely resemble that of malaria before the introduction of control measures. This observation suggests that these genetic disorders confer protection against death from falciparum malaria. For example, HbA/S heterozygotes (sickle cell trait) have a sixfold reduction in the risk of dying from severe falciparum malaria. This decrease in risk appears to be related to impaired parasite growth at low oxygen tensions. In Melanesia, children with α -thalassemia appear to have more frequent malaria (both vivax and falciparum) in the early years of life, and this pattern of infection appears to protect against severe disease. In Melanesian ovalocytosis, rigid erythrocytes resist merozoite invasion and the intraerythrocytic milieu is hostile.

The specific immune response to malaria eventually controls the infection and, with exposure to sufficient strains, confers protection from high-level parasitemia and disease but not from infection. As a result of this state of infection without illness (premunition), asymptomatic parasitemia is common among adults and older children living in regions with stable and intense transmission (i.e., holo- or hyperendemic areas). Immunity is specific for both the species and the strain of infecting malarial parasite. Both humoral immunity and cellular immunity are necessary for protection, but the mechanisms of each are incompletely understood ([Fig. 214-1](#)). Immune individuals have a polyclonal increase in serum levels of IgM, IgG, and IgA, although much of this antibody is unrelated to protection. Antibodies to a variety of parasitic antigens presumably act in concert to limit in vivo replication of the parasite. In the case of falciparum malaria, the most important of these is the antigenically variant protein PfEMP1 mentioned above. Passively transferred IgG from immune adults has been shown to reduce levels of parasitemia in children, and passive transfer of maternal

antibody contributes to the relative protection of infants from severe malaria in the first months of life. This complex immunity to disease is lost when a person lives outside an endemic area for several months or longer.

Several factors retard the development of cellular immunity to malaria. These factors include the absence of major histocompatibility antigens on the surface of infected red cells, which precludes direct T cell recognition; malaria antigen-specific immune unresponsiveness; and the enormous strain diversity of malarial parasites along with the ability of the parasites to express immunodominant variant antigens on the erythrocyte surface that change during the period of infection. Strain diversity also has an impact on the heterogeneity of the humoral antibody response. Immunity to all strains is never achieved. Parasites may persist in the blood for months (or, in the case of *P. malariae*, for many years) if treatment is not given. The complexity of the immune response in malaria, the sophistication of the parasites' evasion mechanisms, and the lack of a good in vitro correlate with clinical immunity have all slowed progress toward an effective vaccine.

CLINICAL FEATURES

The first symptoms of malaria are nonspecific; the lack of a sense of well-being, headache, fatigue, abdominal discomfort, and muscle aches followed by fever are all similar to the symptoms of a minor viral illness. In some instances, a prominence of headache, chest pain, abdominal pain, arthralgia, myalgia, or diarrhea may suggest another diagnosis. Although headache may be severe in malaria, there is no neck stiffness or photophobia resembling that in meningitis. While myalgia may be prominent, it is not usually as severe as in dengue fever, and the muscles are not tender as in leptospirosis or typhus. Nausea, vomiting, and orthostatic hypotension are common. The classic malarial paroxysms, in which fever spikes, chills, and rigors occur at regular intervals, suggest infection with *P. vivax* or *P. ovale*. The fever is irregular at first (that of falciparum malaria may never become regular); the temperature of nonimmune individuals and children often rises above 40°C in conjunction with tachycardia and sometimes delirium. Although childhood febrile convulsions may occur with any of the malarias, generalized seizures are specifically associated with falciparum malaria and may herald the development of cerebral disease. Many clinical abnormalities have been described in acute malaria, but most patients with uncomplicated infections have few abnormal physical findings other than fever, malaise, mild anemia, and (in some cases) a palpable spleen. Anemia may be quite common among young children living in areas with stable transmission, particularly where there is parasite resistance to chloroquine or other drugs. Splenic enlargement is very common among otherwise-healthy individuals in malaria-endemic areas and reflects repeated infections; however, in nonimmune individuals with malaria, the spleen takes several days to become palpable. Slight enlargement of the liver is also common, particularly in young children. Mild jaundice is common in adults; it may develop in patients with otherwise uncomplicated falciparum malaria and usually resolves over 1 to 3 weeks. Malaria is not associated with a rash like those seen in meningococcal septicemia, typhus, enteric fever, viral exanthems, and drug reactions. Petechial hemorrhages in the skin or mucous membranes -- features of viral hemorrhagic fevers and leptospirosis -- develop only rarely in severe falciparum malaria.

Severe Falciparum Malaria Appropriately treated, uncomplicated falciparum malaria carries a mortality rate of ~0.1%. However, once vital organ dysfunction occurs or the proportion of erythrocytes infected increases to >3%, mortality rises steeply. The major manifestations of severe falciparum malaria are shown in [Table 214-2](#).

Cerebral Malaria Coma is a characteristic and ominous feature of falciparum malaria and, despite treatment, is associated with death rates of ~20% among adults and 15% among children. Lesser degrees of obtundation, delirium, and abnormal behavior should also be taken very seriously. The onset may be gradual or sudden following a convulsion.

Cerebral malaria manifests as diffuse symmetric encephalopathy; focal neurologic signs are unusual. Although some passive resistance to head flexion may be detected, signs of meningeal irritation are lacking. The eyes may be divergent and a pout reflex is common, but other primitive reflexes are usually absent. The corneal reflexes are preserved except in deep coma. Muscle tone may be either increased or decreased. The tendon reflexes are variable, and the plantar reflexes may be flexor or extensor; the abdominal and cremasteric reflexes are absent. Flexor or extensor posturing may be documented. Approximately 15% of patients have retinal hemorrhages; with pupillary dilatation and indirect ophthalmoscopy, this figure increases to 30 to 40%. Other abnormalities include discrete spots of retinal opacification (30 to 60%), papilledema (8% of children, rare in adults), cotton wool spots (<5%), and decolorization of a retinal vessel or segment of vessel (occasional cases). Convulsions, usually generalized and often repeated, occur in up to 50% of children with cerebral malaria. More covert seizure activity is common, particularly in children, and may manifest as repetitive tonic-clonic eye movements. Whereas adults rarely suffer neurologic sequelae, ~10% of children surviving cerebral malaria -- especially those with hypoglycemia, severe anemia, repeated seizures, and deep coma -- have some residual neurologic deficit when they regain consciousness; hemiplegia, cerebral palsy, cortical blindness, deafness, and impaired cognition and learning -- all of varying duration -- have been reported.

Hypoglycemia An important and common complication of severe malaria, hypoglycemia is associated with a poor prognosis and is particularly problematic in children and pregnant women. Hypoglycemia in malaria results from a failure of hepatic gluconeogenesis and an increase in the consumption of glucose by both host and parasite. To compound the situation, quinine and quinidine -- drugs used commonly for the treatment of severe chloroquine-resistant malaria -- are powerful stimulants of pancreatic insulin secretion. Hyperinsulinemic hypoglycemia is especially troublesome in pregnant women receiving quinine treatment. In severe disease, the clinical diagnosis of hypoglycemia is difficult: the usual physical signs (sweating, gooseflesh, tachycardia) are absent, and the neurologic impairment caused by hypoglycemia cannot be distinguished from that caused by malaria.

Lactic Acidosis Lactic acidosis commonly coexists with hypoglycemia in patients with malaria and is an important contributor to death from severe malaria. In adults, coexisting renal impairment often compounds the acidosis. Acidotic breathing, sometimes called respiratory distress, is a sign of poor prognosis. It is often followed by circulatory failure refractory to volume expansion or inotropic drugs or by respiratory arrest. The plasma concentrations of bicarbonate or lactate are the best biochemical

prognosticators in severe malaria. Lactic acidosis is caused by the combination of anaerobic glycolysis in tissues where sequestered parasites interfere with microcirculatory flow, lactate production by the parasites, and a failure of hepatic and renal lactate clearance. The prognosis of lactic acidosis is poor.

Noncardiogenic Pulmonary Edema Adults with severe falciparum malaria may develop noncardiogenic pulmonary edema even after several days of antimalarial therapy. This manifestation may also develop in otherwise-uncomplicated vivax malaria, where recovery is usual. The pathogenesis of this variant of the adult respiratory distress syndrome is unclear. The mortality rate is >80%. This condition can be aggravated by overly vigorous administration of intravenous fluid.

Renal Impairment Renal impairment is common among adults with severe falciparum malaria but rare among children. The pathogenesis of renal failure is unclear but may be related to erythrocyte sequestration interfering with renal microcirculatory flow and metabolism. Clinically and pathologically, this syndrome manifests as acute tubular necrosis; renal cortical necrosis never develops. Mortality in the initial phase of hypercatabolic acute renal failure is high; in survivors, urine flow resumes in a median of 4 days, and serum creatinine levels return to normal in a mean of 17 days ([Chap. 269](#)). Dialysis or hemofiltration considerably enhances the likelihood of a patient's survival.

Hematologic Abnormalities Anemia results from accelerated red cell destruction and removal by the spleen in conjunction with ineffective erythropoiesis. In severe malaria, both infected and uninfected red cells show reduced deformability, which correlates with prognosis and development of anemia. Splenic clearance of cells is also increased. In nonimmune individuals and in areas with unstable transmission, anemia can develop rapidly and transfusion is often required. In many areas of Africa, children may develop severe anemia as a result of repeated malarial infections. Anemia is a common consequence of antimalarial drug resistance, which results in repeated or continued infection.

Slight coagulation abnormalities are common in falciparum malaria, and mild thrombocytopenia is usual. As mentioned above, fewer than 5% of patients with severe malaria have significant bleeding with evidence of disseminated intravascular coagulation. Hematemesis, presumably from stress ulceration or acute gastric erosions, may also occur.

Liver Dysfunction Mild hemolytic jaundice is common in malaria. Severe jaundice is associated with *P. falciparum* infections, is more common among adults than among children, and results from hemolysis, hepatocyte injury, and cholestasis. When accompanied by other vital organ dysfunction (often renal impairment), liver dysfunction carries a poor prognosis. Hepatic dysfunction contributes to hypoglycemia, lactic acidosis, and impaired drug metabolism.

Other Complications Aspiration pneumonia following convulsions is an important cause of death in cerebral malaria. Chest infections and catheter-induced urinary tract infections are common among patients who are unconscious for >3 days. Septicemia may complicate severe malaria; in endemic areas *Salmonella* bacteremia has been associated specifically with *P. falciparum* infections.

Malaria in Pregnancy In hyper- and holoendemic areas, falciparum malaria in primi- and secundigravid women is associated with low birth weight (average reduction, ~170 g) and consequently increased infant and childhood mortality. In general, infected mothers in areas of stable transmission remain asymptomatic despite intense parasitization of the placenta due to sequestration of parasitized erythrocytes in the placental microcirculation. Maternal HIV infection predisposes pregnant women to a higher prevalence of malaria and parasite density and predisposes their newborns to congenital malaria infection and low birth weight.

In areas with unstable transmission of malaria, pregnant women are prone to severe infections and are particularly vulnerable to high-level parasitemia with anemia, hypoglycemia, and acute pulmonary edema. Fetal distress, premature labor, and stillbirth or low birth weight are common results. Congenital malaria occurs in fewer than 5% of newborns whose mothers are infected and is related directly to the parasite density in maternal blood and in the placenta. *P. vivax* malaria in pregnancy is also associated with a reduction in birth weight (average, 100 g), but, in contrast with the situation in falciparum malaria, this effect is greater in multigravid than in primigravid women.

Malaria in Children Most of the estimated 1 to 3 million persons who die of falciparum malaria each year are young African children. Convulsions, coma, hypoglycemia, metabolic acidosis, and severe anemia are relatively common among children with severe malaria, whereas deep jaundice, acute renal failure, and acute pulmonary edema are unusual. Severely anemic children may present with labored deep breathing, which in the past has been attributed incorrectly to "anemic congestive cardiac failure" but is in fact usually caused by metabolic acidosis, often compounded by hypovolemia. In general, children tolerate antimalarial drugs well and respond rapidly to treatment.

Transfusion Malaria Malaria can be transmitted by blood transfusion, needle-stick injury, sharing of needles by infected drug addicts, or organ transplantation. The incubation period in these settings is often short because there is no preerythrocytic stage of development. The clinical features and management of these cases are the same as for naturally acquired infections, although falciparum malaria tends to be especially severe in drug addicts. Radical chemotherapy with primaquine is unnecessary for *P. vivax* and *P. ovale* infections.

CHRONIC COMPLICATIONS OF MALARIA

Tropical Splenomegaly (Hyperreactive Malarial Splenomegaly) Chronic or repeated malarial infections produce hypergammaglobulinemia; normochromic, normocytic anemia; and, in certain situations, splenomegaly. Some residents of malaria-endemic areas in tropical Africa and Asia exhibit an abnormal immunologic response to repeated infections that is characterized by massive splenomegaly, hepatomegaly, marked elevations in serum titers of IgM and malarial antibody, hepatic sinusoidal lymphocytosis, and (in Africa) peripheral B cell lymphocytosis. This syndrome has been associated with the production of cytotoxic IgM antibodies to suppressor (CD8+) lymphocytes, antibodies to CD5+ T cells, and an increase in the ratio of CD4+ T cells to CD8+ T cells. It is believed that these events lead to uninhibited B cell production of IgM

and the formation of cryoglobulins (IgM aggregates and immune complexes). This immunologic process stimulates reticuloendothelial hyperplasia and clearance activity and eventually produces splenomegaly. Patients with hyperreactive malarial splenomegaly (HMS) present with an abdominal mass or a dragging sensation in the abdomen and occasional sharp abdominal pains suggesting perisplenitis. Anemia and some degree of pancytopenia are usually evident, but in many cases malarial parasites cannot be found in peripheral blood smears. Vulnerability to respiratory and skin infections is increased; many patients die of overwhelming sepsis. Persons with HMS who are living in endemic areas should receive antimalarial chemoprophylaxis: the results are usually good. In nonendemic areas, treatment is advised. In some cases refractory to therapy, clonal lymphoproliferation may develop and then evolve into a malignant lymphoproliferative disorder.

Quartan Malarial Nephropathy Chronic or repeated infections with *P. malariae*, and possibly other malarial species, may cause soluble immune-complex injury to the renal glomeruli, resulting in the nephrotic syndrome. Other, unidentified factors must contribute to this process since only a very small proportion of infected patients develop renal disease. The histologic appearance is that of focal or segmental glomerulonephritis with splitting of the capillary basement membrane. Subendothelial dense deposits are seen on electron microscopy, and immunofluorescence reveals deposits of complement and immunoglobulins; in samples of renal tissue from children, *P. malariae* antigens are often visible. A coarse-granular pattern of basement membrane immunofluorescent deposits (predominantly IgG3) with selective proteinuria carries a better prognosis than a fine-granular, predominantly IgG2 pattern with nonselective proteinuria. Quartan nephropathy usually responds poorly to treatment with either antimalarial agents or glucocorticoids and cytotoxic drugs.

Burkitt's Lymphoma and Epstein-Barr Virus Infection It is possible that malaria-related immunosuppression provokes infection with lymphoma viruses. Burkitt's lymphoma is strongly associated with Epstein-Barr virus. The prevalence of this childhood tumor is high in malarious areas of Africa.

DIAGNOSIS

Demonstration of the Parasite The diagnosis of malaria rests on the demonstration of asexual forms of the parasite in peripheral blood smears subjected to Romanovsky staining. Following a negative blood smear, repeat smears should be made if there is a high degree of suspicion. Giemsa at pH 7.2 is preferred; Wright's, Field's, or Leishman's stain can also be used. Both thin and thick blood smears should be examined (See [Plates VI-3, VI-4, VI-5, VI-6, VI-7, VI-8, VI-9, VI-10, VI-11, VI-12, VI-13, VI-14, VI-15, VI-16, VI-17, VI-18, VI-19, VI-20](#), and [VI-21](#) and [Plates VI-23, VI-24, VI-25, VI-26, VI-27, VI-28, VI-29, VI-30, VI-31, VI-32](#), and [VI-33](#)).

The thin blood smear should be rapidly air-dried, fixed in anhydrous methanol, and stained, and the red cells in the tail of the film should then be examined under oil immersion. The level of parasitemia is expressed as the number of parasitized erythrocytes among 1000 cells, and this figure is converted to the number of parasitized erythrocytes per microliter. Simple, sensitive, and specific antibody-based diagnostic

stick or card tests that detect *P. falciparum*-specific, histidine-rich protein (HRP) 2 or lactate dehydrogenase antigens in finger-prick blood samples have been introduced. Some of these tests carry a second antibody, which allows falciparum malaria to be distinguished from the less dangerous malarias. The relationship between parasitemia and prognosis is complex; in general, patients with $>10^5$ parasites per microliter are at increased risk of dying, but nonimmune patients may die with much lower counts and semi-immune persons may tolerate parasitemia levels many times higher with only minor symptoms. In severe malaria, a poor prognosis is indicated by a predominance of more mature *P. falciparum* parasites (i.e., $>20\%$ of parasites with visible pigment), by the presence of circulating schizonts in the peripheral blood film, or by the presence of phagocytosed malarial pigment in $>5\%$ of neutrophils. Gametocytes may remain evident for several days after treatment has begun; unless trophozoites are also visible on the blood film, their presence does not constitute evidence of drug resistance.

The thick blood film should be of uneven thickness. The smear should be dried thoroughly and stained without fixing. As many layers of erythrocytes overlie one another and are lysed during the staining procedure, the thick film has the advantage of concentrating the parasites (by 20- to 40-fold compared with a thin blood film) and thus increasing diagnostic sensitivity. Both parasites and white cells are counted, and the number of parasites per unit volume is calculated from the total leukocyte count. Alternatively, a white count of 8000/uL is assumed. A minimum of 200 white cells should be counted. Interpretation of thick films requires some experience because artifacts are common. Before a thick smear is judged to be negative, 100 to 200 fields should be examined under oil immersion. Phagocytosed malarial pigment is sometimes seen inside peripheral blood monocytes or polymorphonuclear leukocytes and may provide a clue to recent infection if malarial parasites are not detectable. After the clearance of the parasites, malarial pigment is often evident for several days in peripheral blood phagocytes, bone marrow aspirates, or smears of fluid expressed after intradermal puncture. Staining of parasites with the fluorescent dye acridine orange allows more rapid diagnosis of cases in which the level of parasitemia is low.

Laboratory Findings Normochromic, normocytic anemia is usually documented. The leukocyte count is generally low to normal, although it may be raised in very severe infections. The erythrocyte sedimentation rate, degree of plasma viscosity, and level of C-reactive protein are high. The platelet count is usually reduced to $\sim 10^5$ /uL. Severe infections may be accompanied by prolonged prothrombin and partial thromboplastin times and by severe thrombocytopenia. Levels of antithrombin III are reduced even in mild infection. In uncomplicated malaria, plasma concentrations of electrolytes, blood urea nitrogen, and creatinine are usually normal. Findings in severe malaria may include metabolic acidosis, with low plasma concentrations of glucose, sodium, bicarbonate, calcium, phosphate, and albumin together with elevations in lactate, blood urea nitrogen, creatinine, urate, muscle and liver enzymes, and conjugated and unconjugated bilirubin. Hypergammaglobulinemia is usual in immune and semi-immune subjects, and urinalysis generally gives normal results. In adults and children with cerebral malaria, the mean opening pressure at lumbar puncture is ~ 160 mm of cerebrospinal fluid (CSF); the CSF is usually normal or has a slightly elevated total protein level [<1.0 g/L (100 mg/dL)] and cell count (<20 /uL).

PREVENTION

In most of the tropics, the eradication of malaria is not yet feasible because of the widespread distribution of *Anopheles* breeding sites; the great number of infected persons; and inadequacies in resources, infrastructure, and control programs. Where possible, the disease is contained by judicious use of insecticides to kill the mosquito vector, rapid diagnosis and appropriate patient management, and administration of chemoprophylaxis to high-risk groups. Malaria researchers are intensifying their efforts to better understand parasite-human-mosquito-environmental interactions and develop more effective control and prevention interventions. Despite the enormous investment in efforts to develop a malaria vaccine, no safe, effective, long-lasting vaccine is likely to be available for general use in the near future ([Chap. 122](#)). While there is promise for one or more malaria vaccines on the more distant horizon, prevention and control measures continue to rely on antivector and drug use strategies.

Personal Protection Against Malaria Simple measures to reduce the frequency of mosquito bites in malarious areas are very important. These measures include the avoidance of exposure to mosquitoes at their peak feeding times (usually dusk and dawn, but also throughout the night) and the use of insect repellents, suitable clothing, and insecticide-impregnated bed nets. Widespread use of bed nets, particularly those treated with residual pyrethroids, reduces the incidence of malaria and has been shown to reduce mortality in western and eastern Africa.

Chemoprophylaxis ([Table 214-3](#)) Few areas of therapeutics are as controversial as antimalarial drug prophylaxis. Recommendations for prophylaxis depend on knowledge of local patterns of plasmodial drug sensitivity and the likelihood of acquiring malarial infection. Chemoprophylaxis is never entirely reliable, and malaria should always be considered in the differential diagnosis of fever in patients who have traveled to endemic areas, even if they are taking prophylactic antimalarial drugs.

Pregnant women traveling to malarious areas should be warned about the potential risks. All pregnant women at risk in endemic areas should be encouraged to attend regular antenatal clinics and should receive either prophylaxis with chloroquine or proguanil (chloroguanide) or intermittent treatment with pyrimethamine-sulfadoxine, provided there is not high-level resistance to these drugs. In addition, antimalarial prophylaxis should be considered for children between the ages of 3 months and 4 years in areas where malaria causes high childhood mortality; such prophylaxis may not be logistically or economically feasible in many countries. Children born to nonimmune mothers in endemic areas (usually expatriates moving to these areas) should receive prophylaxis from birth.

Travelers should start taking antimalarial drugs at least 1 week before departure so that any untoward reactions can be detected and therapeutic antimalarial blood concentrations will be present when needed. Antimalarial prophylaxis should continue for 4 weeks after the traveler has left the endemic area.

Mefloquine has become the antimalarial prophylactic agent of choice for much of the tropics because it is usually effective against multidrug-resistant falciparum malaria and is reasonably well tolerated. Mild nausea, dizziness, fuzzy thinking, disturbed sleep patterns, and malaise are relatively common. Approximately 1 in every 10,000 recipients

develops an acute reversible neuropsychiatric reaction manifested by confusion, psychosis, convulsions, or encephalopathy. The role of mefloquine prophylaxis in pregnancy remains uncertain; in studies in Africa, mefloquine prophylaxis was found to be effective and safe during pregnancy. However, in one study from Thailand, treatment of malaria with mefloquine was associated with an increased risk of stillbirth.

Daily administration of doxycycline is an effective alternative to mefloquine that also exhibits some causal (preerythrocytic) prophylactic activity. Doxycycline is generally well tolerated but may cause vulvovaginal thrush, diarrhea, and photosensitivity and cannot be used by children <8 years old or by pregnant women. The combination drug atovaquone-proguanil hydrochloride (3.75/1.5 mg/kg, or 250/100 mg daily, adult dose) has recently been shown to be a very effective alternate to mefloquine chemoprophylaxis in drug-resistant areas. This drug must be taken with food or a milky drink because it is highly lipophilic: it is well-tolerated by adults and children.

Chloroquine remains the drug of choice for the prevention of infection with drug-sensitive *P. falciparum* and with the other human malarial species (although chloroquine-resistant *P. vivax* has been reported from parts of eastern Asia, Oceania, and South America). Unfortunately, there are few areas of the world with chloroquine-sensitive *P. falciparum*. Chloroquine is generally well tolerated, although some patients are unable to take the drug because of malaise, headache, or (in dark-skinned patients) pruritus. A concomitant filarial infection may provoke or aggravate chloroquine-induced pruritus. Chloroquine is considered safe in pregnancy. With chronic administration for >5 years, a characteristic dose-related retinopathy may develop, but this condition is rare at the doses used for antimalarial prophylaxis. Idiosyncratic or allergic reactions are also rare. Skeletal and cardiac myopathy are potential problems with protracted prophylactic use; they occur most often at the high doses used in the treatment of rheumatoid arthritis. Neuropsychiatric reactions and skin rashes are unusual. Amodiaquine, a related aminoquinoline, is associated with a high risk of agranulocytosis (~1 person in 2000 with continuous use) and should not be used for prophylaxis.

In the past, the dihydrofolate reductase inhibitors pyrimethamine and proguanil (chloroguanide) have been administered widely, but resistant strains of both *P. falciparum* and *P. vivax* have limited their use. Whereas antimalarial quinolines such as chloroquine act on the erythrocyte stage of parasitic development, the dihydrofolate reductase inhibitors also inhibit preerythrocytic growth in the liver (causal prophylaxis) and development in the mosquito (sporonticidal activity). Proguanil is safe and well tolerated, although mouth ulceration occurs in ~8% of persons using this drug; it is considered safe for antimalarial prophylaxis in pregnancy. The prophylactic use of the combination of pyrimethamine and sulfadoxine is not recommended because of an unacceptable incidence of severe toxicity, principally exfoliative dermatitis and other skin rashes, agranulocytosis, hepatitis, and pulmonary eosinophilia. The combination of pyrimethamine with dapsone (0.2/1.5 mg/kg weekly; 25/200 mg maximum) is a second-line alternative available in some countries and can be used in areas with chloroquine-resistant *P. falciparum*. This combination is generally well tolerated; however, resistance is increasing, and dapsone may cause methemoglobinemia and allergic reactions and (at higher doses) may pose a significant risk of agranulocytosis. Primaquine (0.5 mg/kg, or 30 mg daily) has also proved safe and effective in clinical

trials in drug-resistant areas and can be considered when all other options are contraindicated. Proguanil and the pyrimethamine-dapsone combination are not available in the United States.

Because of the increasing spread and intensity of plasmodial resistance to chloroquine in Africa and other areas of the world (Fig. 214-2), the Centers for Disease Control and Prevention (CDC; <http://www.cdc.gov/travel/index.htm>), which recommends a weekly dose of mefloquine for all travelers, maintains an updated 24-h travel and malaria information audiotape that can be accessed by touch-tone telephone (888-232-3228). Regional and disease-specific documents may be requested from the CDC Fax Information Service (888-232-3299). Consultation for the evaluation of prophylaxis failures or treatment of malaria can be obtained from state and local health departments and the CDC (770-488-7788).

TREATMENT

When a patient in or from a malarious area presents with fever, thick and thin blood smears should be prepared and examined immediately to confirm the diagnosis and identify the species of infecting parasite. Repeat blood smears should be performed at least every 12 h for 2 days if the first smears are negative. Patients with severe malaria or those unable to take oral drugs should receive parenteral antimalarial therapy. If there is any doubt about the resistance status of the infecting organism, then quinine or quinidine should be given. Several drugs are available for oral treatment, and the choice of drug depends on the likely sensitivity of the infecting parasites. Despite recent evidence of chloroquine resistance in *P. vivax* (from parts of Indonesia, Oceania, and Brazil), chloroquine remains the treatment of choice for the "benign" human malarias (*P. vivax*, *P. ovale*, *P. malariae*). Characteristics of various antimalarial agents are shown in Table 214-4, and drug regimens approved by the U.S. Food and Drug Administration are detailed in Table 214-5. The availability of antimalarial drugs varies considerably between countries. Many of the drugs used to treat malaria in endemic areas are not available in temperate countries such as the United States.

Severe Malaria Because of resistance, chloroquine can no longer be relied upon in most countries for the treatment of severe malaria. The antiarrhythmic quinidine gluconate is as effective as quinine and, as it is more readily available, has replaced quinine for the treatment of malaria in the United States. The administration of quinidine must be closely monitored if dysrhythmias and hypotension are to be avoided. Total plasma levels in excess of 8 µg/mL, a QTc interval of >0.6 s, or QRS widening beyond 25% of baseline are indications for slowing infusion rates. If arrhythmia or saline-unresponsive hypotension develops, treatment with this drug should be discontinued. Quinine is safer than quinidine; cardiovascular monitoring is not required except when the recipient has cardiac disease. In some areas of Asia, the Chinese drugs derived from artemisinin (artemether and artesunate) have become first-line treatments for severe malaria. These agents are rapidly effective against multidrug-resistant falciparum malaria and are at least as effective as and safer than quinine or quinidine. They are not available in the United States.

Severe falciparum malaria constitutes a medical emergency requiring intensive nursing care and careful management. The patient should be weighed and, if comatose, placed

on his or her side. Frequent evaluation of the patient's condition is essential. Ancillary drugs such as high-dose glucocorticoids, urea, heparin, and dextran are of no value.

Parenteral antimalarial treatment should be started as soon as possible. An initial loading dose should be given so that therapeutic concentrations are reached as soon as possible. Both quinine and quinidine will cause dangerous hypotension if injected rapidly; when given intravenously, they must be administered carefully by rate-controlled infusion only. The optimal therapeutic range for quinine and quinidine in severe malaria is not known with certainty, but total plasma concentrations of 8 to 15 mg/mL for quinine and 3.5 to 8.0 mg/mL for quinidine are effective and do not cause serious toxicity. The systemic clearance and apparent volume of distribution of these alkaloids are markedly reduced and plasma protein binding is increased in severe malaria, so that the blood concentrations attained with a given dose are higher. If the patient remains seriously ill or in acute renal failure for >2 days, the maintenance doses of quinine or quinidine should be reduced by 30 to 50% to prevent toxic accumulation of the drugs. The initial doses should never be reduced. If one of the artemisinin derivatives or chloroquine is given, dose reductions are unnecessary, even in renal failure. Exchange transfusion should be considered for severely ill patients, although the precise indications for this procedure have not been agreed upon. It has been recommended that -- if safe and feasible -- exchange should be considered for parasitemia levels of 5 to 15% and is indicated for parasitemia levels of >15%. The role of prophylactic intramuscular phenobarbital in preventing convulsions in cerebral malaria also remains uncertain.

When the patient is unconscious, the blood glucose level should be measured every 4 to 6 h, and values below 2.2 mmol/L (40 mg/dL) should prompt treatment with intravenous dextrose. All patients treated with intravenous quinine or quinidine should receive a continuous infusion of 5 to 10% dextrose. The parasite count and hematocrit level should be measured every 6 to 12 h. Anemia develops rapidly; if the hematocrit falls below 20%, then whole blood (preferably fresh) or packed cells should be transfused slowly, with careful attention to circulatory status. Renal function should be checked daily. Judicious use of small doses of a diuretic to prevent fluid overload may be needed, particularly in the elderly. Children presenting with severe anemia and acidotic breathing are often hypovolemic; in this situation, resuscitation with crystalloids or blood is indicated. Accurate assessment is vital. Management of fluid balance is difficult in severe malaria, particularly in adults, because of the thin dividing line between overhydration (leading to pulmonary edema) and underhydration (contributing to renal impairment). If necessary, pulmonary artery occlusion pressures should be measured and maintained in the low-normal range. As soon as the patient can take fluids, oral therapy should be substituted for parenteral treatment.

Uncomplicated Malaria Infections due to *P. vivax*, *P. malariae*, *P. ovale*, and known sensitive strains of *P. falciparum* should be treated with oral chloroquine (total dose, 25 mg of base/kg). In Africa, chloroquine-resistant strains are usually sensitive to sulfadoxine/pyrimethamine. Where there is resistance to the latter combination, either (1) quinine plus tetracycline or doxycycline (or clindamycin) or (2) mefloquine should be used; tetracycline and doxycycline cannot be given to pregnant women or to children <8 years of age. Oral quinine is extremely bitter and regularly produces cinchonism comprising tinnitus, high-tone deafness, nausea, vomiting, and dysphoria. Compliance is poor with the required 5- to 7-day regimens of this drug. Mefloquine should be given

at a total dosage of 25 mg/kg (15 mg/kg followed 8 to 12 h later by 10 mg/kg) and, where available and approved for use, combined with artesunate or artemether (4 mg/kg per day for 3 days). Although significant resistance to mefloquine has been documented in Thailand, Burma, Vietnam, and Cambodia, this agent is usually effective against multidrug-resistant strains of *P. falciparum* outside these areas. Artemether-lumefantrine and atovaquone-proguanil are recently introduced, well-tolerated antimalarial drugs used in 3-day regimens. They are both effective against multidrug-resistant falciparum malaria.

Patients should be monitored for vomiting for 1 h after the administration of any oral antimalarial drug. Symptom-based treatment, with tepid sponging and acetaminophen administration, lowers fever and thereby reduces the patient's propensity to vomit these drugs. Minor central nervous system reactions (nausea, dizziness, sleep disturbances) are common. The incidence of serious adverse neuropsychiatric reactions to mefloquine treatment is ~1 in 1000 in Asia but may be as high as 1 in 200 among Africans and Caucasians. All the antimalarial quinolines (chloroquine, mefloquine, and quinine) exacerbate the orthostatic hypotension associated with malaria, and all are tolerated better by children than by adults. Pregnant women, young children, patients unable to tolerate oral therapy, and nonimmune subjects (e.g., travelers) with suspected malaria should be evaluated carefully and hospitalization considered. If there is any doubt as to the identity of the infecting malarial species, treatment for falciparum malaria should be given. A negative blood smear does not rule out malaria; thick blood films should be checked 1 and 2 days later to exclude the diagnosis. Nonimmune subjects receiving treatment for malaria should have daily parasite counts performed until negative thick films indicate clearance of the parasite. If the level of parasitemia does not fall below 25% of the admission value in 48 h or if parasitemia has not cleared by 7 days (and compliance is assured), drug resistance is likely and the regimen should be changed. Quinine (or quinidine) and tetracycline should be reserved for multidrug-resistant infections, but if falciparum malaria has been contracted in an area of known drug sensitivity, then treatment with chloroquine, sulfadoxine/pyrimethamine, or mefloquine is preferable because these agents are better tolerated and simpler to administer.

Primaquine (0.3 mg of base/kg; 15 mg of base, adult dose) should be given daily for 14 days to patients with *P. vivax* or *P. ovale* infections after laboratory tests for [G6PD](#) deficiency have proved negative. A dose of 22.5 to 30 mg for an adult is recommended for infections acquired in Southeast Asia and Oceania. If the patient has a mild variant of G6PD deficiency, primaquine can be given in a dose of 0.6 mg of base/kg (45 mg maximum) once weekly for 8 weeks.

PREVENTING DRUG RESISTANCE

In much of the tropics, drug-resistant *P. falciparum* is increasing in distribution, frequency, and intensity. There is a growing belief among malariologists that, to prevent resistance, falciparum malaria should no longer be treated with single drugs in endemic areas; the same rationale has been applied in the treatment of tuberculosis and HIV/AIDS. This strategy is based upon simultaneous use of two or more drugs with different modes of action: one, an artemisinin derivative (artesunate, artemether, or dihydroartemisinin), given for 3 days; and the other, a slower-acting antimalarial. In areas where *P. falciparum* is still sensitive, chloroquine is used as the second drug;

where there is low-grade chloroquine resistance (e.g., many areas of Africa), either amodiaquine or sulfadoxine/pyrimethamine can be used in combination with the artemisinin derivative. Where there is also resistance to sulfadoxine/pyrimethamine, the combinations artesunate plus mefloquine, artemether plus lumefantrine, or quinine plus tetracycline or clindamycin can be considered (although tetracycline cannot be given to pregnant women or to children <8 years of age). Atovaquone/proguanil, which is also effective against drug-resistant malaria, can also be combined with artesunate to prevent the emergence of resistance. While significant resistance to mefloquine occurs in Thailand, Burma, and Cambodia, the mefloquine/artesunate combinations are still reliably effective in these areas. The artemisinin derivatives and lumefantrine (all unlicensed in the United States) and atovaquone-proguanil are tolerated well with no significant adverse effects.

COMPLICATIONS

Acute Renal Failure If the level of blood urea nitrogen or creatinine rises despite adequate rehydration, fluid administration should be restricted to prevent volume overload. The indications for dialysis are the same as those in other forms of hypercatabolic acute renal failure ([Chap. 269](#)). Even with adequate peritoneal dialysis, secondary bacterial infections are common in the tropics, and hemodialysis or hemofiltration is preferable. Some patients pass small volumes of urine sufficient to allow control of fluid balance; these cases can be managed conservatively if other indications for dialysis do not arise. Renal function usually improves within days, but full recovery may take weeks.

Acute Pulmonary Edema Patients should be positioned at 45° and given oxygen and intravenous diuretics. Pulmonary artery occlusion pressures may be normal, indicating increased pulmonary capillary permeability. Positive pressure ventilation should be started early if the immediate measures fail ([Chap. 233](#)).

Hypoglycemia An initial slow injection of 50% dextrose (0.5 g/kg) should be followed by an infusion of 10% dextrose (0.10 g/kg per hour). The blood glucose level should be checked regularly thereafter, as recurrent hypoglycemia is common, particularly in patients receiving quinine or quinidine. In severely ill patients, hypoglycemia commonly occurs together with metabolic (lactic) acidosis and carries a poor prognosis.

Other Complications Patients who develop spontaneous bleeding should be given fresh blood and intravenous vitamin K. Convulsions should be treated with intravenous or rectal benzodiazepines and, if necessary, respiratory support. Aspiration pneumonia should be suspected in any unconscious patient with convulsions, particularly with persistent hyperventilation; intravenous antimicrobial agents and oxygen should be administered, and pulmonary toilet should be undertaken. Treatment for systemic *Salmonella* and other infections common in African children with falciparum malaria should be considered. Hypoglycemia or gram-negative septicemia should be suspected when the condition of any patient suddenly deteriorates for no obvious reason during antimalarial treatment.

BABESIOSIS

Babesiosis is a protozoan disease of animals that is transmitted by ticks; humans are infected incidentally and initially develop a nonspecific febrile illness. *Babesia* organisms enter red blood cells and resemble malarial parasites morphologically, thus posing a diagnostic problem.

ETIOLOGY AND NATURAL CYCLE

Of the more than 100 species of *Babesia*, *B. microti* and *B. divergens* are the two that cause most human infections. Ixodid (hard-bodied) ticks, in particular *Ixodes scapularis* (*I. dammini*) and *I. ricinus*, are the vectors of the parasite. Ticks ingest *Babesia* while feeding, and the parasite multiplies within the tick's gut wall. The organisms then spread to the salivary glands; their inoculation into a vertebrate host by a tick larva, nymph, or adult completes the cycle of transmission. Asexual reproduction of *Babesia* within red blood cells produces two or four parasites.

EPIDEMIOLOGY

While *Babesia* infections in wild and domestic animals are distributed globally, almost all *B. microti* infections in the United States occur along the northeastern coast, including Nantucket Island, Martha's Vineyard, and Cape Cod in Massachusetts; Block Island in Rhode Island; Long Island, Shelter Island, and Fire Island in New York; and the nearby mainland, including Connecticut. Cases also have been reported from Wisconsin, Minnesota, Virginia, Maryland, Georgia, and Mexico. *Babesia* isolates from patients in Washington and California have been characterized as WA-1-type parasites, a category that is genetically and antigenically distinct from *B. microti*. A strain isolated in Missouri differs from these isolates, suggesting that babesiosis may be an "emerging infection." The deer tick, *I. scapularis*, is the vector associated with *B. microti*. In Europe, *B. divergens* has been responsible for the majority of the 22 reported cases of babesiosis; Yugoslavia, Russia, France, the United Kingdom, and Ireland have accounted for most of these infections.

Transfusions are another source of babesiosis. In the more than 20 transfusion-associated cases reported, parasites were uncommonly detected in blood donors, but serologic testing of their blood for *Babesia* gave positive results.

Infections with *B. divergens* have occurred sporadically in previously splenectomized patients in several countries in Europe. *I. ricinus* is probably the vector in these cases, as it is for the transmission of this organism among cattle. The infected persons were predisposed to illness by their asplenic status.

I. scapularis feeds on rodents as a larva and a nymph and on deer as an adult; nymphs are abundant during the spring and summer and feed on humans readily. In some endemic areas, the seroprevalence in the human population may be >2%. This figure indicates that asymptomatic infection is more frequent than is generally thought.

CLINICAL PRESENTATION

The incubation period for *B. microti* infection is about 1 to 4 weeks. Immunosuppressed patients, splenectomized individuals, and the elderly have the most severe illness. The

clinical presentation varies widely; symptoms and signs include a gradual onset of irregular fever, chills, sweating, muscle pain, and fatigue. Mild hepatosplenomegaly and mild hemolytic anemia may develop. The level of parasitemia may exceed 10%. The illness may continue for weeks or months.

Patients infected with *B. divergens* have a more severe illness, with a rapid onset of chills, fever, nausea, vomiting, and hemolytic anemia progressing to jaundice, hemoglobinemia, and renal failure. *B. divergens* infections are often fatal.

DIAGNOSIS

Whether or not they have a history of exposure to ticks or tick bites, febrile persons living in endemic areas should have Giemsa-stained thick and thin blood films (see [Plate VI-22](#)) examined for small intraerythrocytic parasites. *B. microti* appears as a small ring form resembling *P. falciparum*. Unlike infection with *Plasmodium*, however, that with *Babesia* does not cause the production of pigment in parasites, nor are schizonts or gametocytes formed. Dividing within red blood cells, *B. microti* can form four daughter parasites attached by strands of cytoplasm; these "tetrad" forms are seen infrequently in human blood films but are a distinguishing feature. An indirect immunofluorescence antibody test is useful for the diagnosis of infection with *B. microti* but does not replace the blood smear. The serum antibody titer rises 2 to 4 weeks after the onset of illness and then wanes over 6 to 12 months; cross-reactions can occur with other species of *Babesia* and with *Plasmodium*.

About 50% of patients infected with *B. microti* have antibody to *Borrelia burgdorferi*, the agent of Lyme disease ([Chap. 176](#)); this figure varies with the geographic area. The occurrence of mixed infections is not surprising since both organisms are transmitted by *I. scapularis*. This tick species is also a potential vector of human granulocytic ehrlichiosis; the same tick may carry more than one tick-borne disease. Intraperitoneal inoculation of blood from patients with babesiosis into hamsters or gerbils results in detectable parasitemia within 2 to 4 weeks.

TREATMENT

B. microti infections in patients with intact spleens are often self-limiting without treatment, although symptoms may persist for months with or without treatment. Because silent parasitemia may have prolonged symptoms and signs, treatment is advised for all patients infected with *Babesia*. Treatment with the combination of quinine sulfate (650 mg of salt orally tid) plus clindamycin (600 mg orally tid or 1.2 g parenterally bid) for 7 to 10 days is usually effective but may not always eliminate parasites. The pediatric dose is 20 to 40 mg/kg per day for quinine sulfate and 25 mg/kg per day for clindamycin, both given in three divided doses over 7 to 10 days. Atovaquone suspension (750 mg bid) plus azithromycin (500 to 1000 mg/d) may be effective when quinine and clindamycin fail. Especially severe infections with high-level *B. microti* parasitemia in asplenic patients have been successfully treated with exchange transfusions in addition to quinine and clindamycin.

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215. LEISHMANIASIS - Barbara L. Herwaldt

OVERVIEW

DEFINITION

The term *leishmaniasis* refers collectively to various clinical syndromes caused by obligate intracellular protozoa of the genus *Leishmania* (order Kinetoplastida). Leishmaniasis is endemic in diverse ecologic settings in the tropics, the subtropics, and southern Europe that range from deserts to rain forests and from rural to periurban areas. It typically is a vector-borne zoonosis, with rodents and canids as common reservoir hosts and humans as incidental hosts. In humans, visceral, cutaneous, and mucosal leishmaniasis result from infection of macrophages throughout the mononuclear-phagocyte system, in the skin, and in the naso-oropharyngeal mucosa, respectively. Current challenges include the emergence of leishmaniasis in new geographic areas and host populations (e.g., visceral leishmaniasis in persons infected with HIV) as well as the need for field-applicable, rapid diagnostic tests and for effective, safe, and affordable oral therapies, control measures, and immunoprophylactic agents.

ETIOLOGY

The organisms that cause the various forms of leishmaniasis in humans are listed in [Table 215-1](#). Visceral leishmaniasis is typically but not exclusively caused by organisms of the *Leishmania donovani* complex; Old World cutaneous leishmaniasis by *L. tropica*, *L. major*, and *L. aethiopica*; New World (or American) cutaneous leishmaniasis by organisms of the *L. mexicana* complex and the *Viannia* subgenus; and mucosal leishmaniasis by some organisms in the *Viannia* subgenus.

LIFE CYCLE

Leishmania parasites are transmitted by the bite of female phlebotomine sandflies [genus *Phlebotomus* (Old World) or *Lutzomyia* (New World)]. As the flies attempt to feed, they regurgitate the parasite's flagellated promastigote stage into the skin of mammalian hosts. Promastigotes attach to receptors on macrophages, are phagocytized, and transform within phagolysosomes into the nonflagellated amastigote stage, which multiplies by binary fission. After rupture of infected macrophages, amastigotes are phagocytized by other macrophages. If ingested by feeding sandflies, amastigotes transform back into promastigotes, which require at least 7 days to become infective.

IMMUNOLOGY

Advances in the understanding of the immunology of leishmaniasis have made this parasitic disease the paradigm for studies of the T cell subsets and cytokines that govern resistance and susceptibility to intracellular pathogens. The paradigm is best demonstrated in murine *L. major* infection. In inbred mice, production of interferon γ (IFN- γ) by T_H1 and natural killer cells confers resistance. Interleukin (IL)12 induces naive T cells to differentiate into T_H1 cells and induces T cells and natural killer cells to produce IFN- γ . In contrast, expansion of IL-4-producing T_H2 cells mediates

susceptibility.

Not all aspects of leishmaniasis in mice, whose susceptibility to leishmanial infection is genetically determined, apply to human infection, for which the genetic determinants are being investigated. However, a consistent principle is that healing and resistance to reinfection are associated with expanding numbers of *Leishmania*-specific T_H1 cells, production of IFN- γ , and activation of macrophages to kill intracellular amastigotes. In human visceral leishmaniasis, IL-10 appears to be associated with pathology; in addition, IL-4 may contribute to progression of disease.

GENERAL DIAGNOSTIC PRINCIPLES

Definitive diagnosis of leishmaniasis requires demonstration of the parasite. To identify amastigotes by light-microscopic examination, the specimen obtained from an infected site (e.g., thin smear, histologic section) should be stained with Giemsa or another Romanovsky stain and presumptive amastigotes (2 to 4 μ m in diameter) examined under oil immersion for the presence of a nucleus and a rod-shaped kinetoplast ([Fig. 215-1](#)); the latter is a specialized mitochondrial structure that contains extranuclear DNA. Other means of parasitologic confirmation include in vitro culture (e.g., on Novy-MacNeal-Nicolle medium), animal inoculation, and use of molecular techniques that are under investigation (e.g., polymerase chain reaction).

The *Leishmania* species that infect humans are morphologically similar. They can be distinguished by isoenzyme analysis of cultured promastigotes, determination of monoclonal antibody specificity, or various molecular methods.

Indirect immunologic methods for diagnosis include serologic assays and tests for *Leishmania*-specific cell-mediated immunity (e.g., skin testing for delayed-type hypersensitivity reactions). The usefulness of such methods depends in part on the clinical syndrome (see below). Traditional serologic assays (e.g., indirect immunofluorescent antibody testing) do not reliably distinguish past from current infection, and no leishmanin skin-test preparation has been approved for use in the United States. Advances in molecular methods (e.g., production of recombinant/synthetic antigens) may lead to the development of better diagnostic techniques.

GENERAL THERAPEUTIC PRINCIPLES

For a given case of leishmaniasis, it is important to consider whether the patient's illness could result in substantial morbidity or in death and therefore requires expeditious treatment with a regimen that generally is highly effective. For decades, the pentavalent antimonial (Sbv) compounds sodium stibogluconate and meglumine antimonate have been the mainstays of antileishmanial therapy ([Table 215-2](#)). Toxicity (with such manifestations as myalgia, arthralgia, fatigue, elevated aminotransferase levels, chemical pancreatitis, and electrocardiographic abnormalities) becomes increasingly common as the course of treatment progresses but usually does not limit therapy and is reversible.

The traditional parenteral alternatives to Sbv -- amphotericin B and pentamidine

isethionate -- are generally considered more apt to induce serious or irreversible toxicity (e.g., nephrotoxicity). However, these agents are being advocated for use in some situations (see below; [Table 215-2](#)), in part because of the benefits of new formulations (e.g., lipid formulations of amphotericin B) or dosage regimens of these drugs and the decreasing effectiveness of Sb_v in some settings. Many other agents have been touted as alternatives or adjuncts to Sb_v, often on the basis of suboptimal data. Some of these agents may be useful in certain situations, with one caveat: even the results of well-conducted clinical trials are not always generalizable to the treatment of patients in other settings. Most of the nonparenteral agents evaluated to date have at best modest activity against some of the *Leishmania* species.

PREVENTION AND CONTROL

The transmission of *Leishmania* species typically is focal, with local "hot spots," in part because of the limited flight range of sandflies; these insects have a short, hopping flight style and usually remain within a few hundred meters of their breeding site. They rest in dark, moist places in habitats ranging from deserts to rain forests; peridomestic sandflies rest in debris or rubble near buildings.

Personal protective measures include the avoidance of outdoor activities when sandflies are most active (dusk to dawn); the use of mechanical barriers such as screens and bed-nets that keep out sandflies, which are about one-third the size of mosquitoes; the wearing of protective clothing; and the application of insect repellent to exposed skin. Impregnation of clothing, bed-nets, and screens with permethrin may also be useful, as may spraying of dwellings with residual-action insecticide. Vaccine strategies are being investigated. Treatment of human cases is an effective control measure only where humans are the primary reservoirs of infection. Vector control and elimination of reservoir hosts may be useful in select settings -- for example, where transmission is intra- or peridomestic.

VISCERAL LEISHMANIASIS

More than 90% of the world's cases of visceral leishmaniasis occur in Bangladesh, northeastern India (particularly Bihar State), Nepal, Sudan, and northeastern Brazil. The causative species typically are those of the *L. donovani* complex ([Table 215-1](#)). The organisms can be transmitted not only by sandflies but also congenitally and parenterally (e.g., through blood transfusions or sharing of needles). Infection begins in macrophages at the inoculation site (e.g., in dermal macrophages at the site of a sandfly bite) and disseminates throughout the mononuclear-phagocyte system in the context of both specific (i.e., to leishmanial antigens) and nonspecific (e.g., to tuberculin) anergy.

CLINICAL MANIFESTATIONS

Visceral infection can remain subclinical or become symptomatic, with an acute, subacute, or chronic course. In some settings, inapparent infections far outnumber clinically apparent ones; malnutrition is among the risk factors for the development of disease. The incubation period usually ranges from weeks to months but can be as long as years. Whereas the general term *visceral leishmaniasis* covers a broad spectrum of severity and manifestations, the term *kala-azar* (Hindi for "black fever," indicating that

the skin of some patients turns gray) generally conjures up the classic image of profoundly cachectic, febrile patients who are heavily infected with parasites and have life-threatening disease. Splenomegaly (with the spleen most often soft and nontender) typically is more impressive than hepatomegaly, and the spleen can in fact be massive; both splenomegaly and hepatomegaly result from reticuloendothelial cell hyperplasia. Peripheral lymphadenopathy is common in some settings, including Sudan.

The abnormal laboratory findings associated with advanced disease include pancytopenia -- anemia, leukopenia (neutropenia, marked eosinopenia, relative lymphocytosis and monocytosis), and thrombocytopenia -- as well as hypergammaglobulinemia (chiefly involving IgG, from polyclonal B cell activation) and hypoalbuminemia. Causes of anemia can include bone-marrow infiltration, hypersplenism, autoimmune hemolysis, and bleeding.

Some patients develop post-kala-azar dermal leishmaniasis. This syndrome is manifested by skin lesions (including macules, papules, nodules, and patches) that typically are most prominent on the face. These lesions can develop during therapy or within a few months thereafter (e.g., in East Africa) or can develop years later (e.g., in India); relapse of visceral infection can occur. Persons with persistent skin lesions can serve as reservoir hosts of infection.

Viscerotropic leishmaniasis caused by *L. tropica*, which typically is dermatropic, was recognized among U.S. soldiers who participated in Operation Desert Storm in the Persian Gulf. The affected persons had light parasite burdens and nonspecific manifestations of visceral infection (e.g., fatigue, fever, and gastrointestinal symptoms).

DIAGNOSIS

Although molecular techniques are under investigation, parasitologic diagnosis of visceral leishmaniasis has traditionally been accomplished by demonstration of the parasite on stained slides or in cultures of a tissue aspirate or a biopsy specimen (e.g., of spleen, liver, bone marrow, or lymph node). The diagnostic yield is highest for splenic aspiration (specifically, as high as 98% vs. <90% for other specimens), but this procedure can cause hemorrhage.

Patients with florid kala-azar commonly have relatively heavy parasite burdens, develop high titers of antibody to *Leishmania* (diagnostically useful but not protective), and have undetectable *Leishmania*-specific cell-mediated immunity (with leishmanin skin-test reactivity as well as lymphocyte proliferation and [IFN-γ](#) responses to leishmanial antigens typically noted only after recovery). In contrast, viscerotropic leishmaniasis can be difficult to diagnose because of a light parasite burden and a minimal antibody response. A promising noninvasive serologic method for diagnosing kala-azar uses nitrocellulose paper strips impregnated with K39, a recombinant leishmanial polypeptide; this technique is being field-tested.

The differential diagnosis of visceral leishmaniasis includes other tropical infectious diseases that cause fever or organomegaly (e.g., typhoid fever, miliary tuberculosis, brucellosis, malaria, tropical splenomegaly syndrome, and schistosomiasis) as well as diseases such as leukemia and lymphoma. Post-kala-azar dermal leishmaniasis should

be differentiated from syphilis, yaws, and leprosy.

TREATMENT

Because persons who have kala-azar generally die if not appropriately treated, highly effective therapy is essential, as is close monitoring for bleeding and intercurrent infectious conditions such as pneumonia and diarrhea. Outside of India, treatment with a pentavalent antimonial compound still is common ([Table 215-2](#)). The use of an alternative parenteral agent ([Table 215-2](#)) should be considered even for first-line therapy if unresponsiveness to Sb₃ therapy is prevalent, as it is in India, or if nonantimonial therapy would be advantageous for other reasons (e.g., toxicity profile or duration of therapy).

A major advance has been the advent of lipid formulations of amphotericin B, in which various lipids have replaced deoxycholate. These formulations, which passively target amphotericin to macrophage-rich organs, are more costly than conventional amphotericin B but are associated with less nephrotoxicity and can be given in shorter courses. Other parenteral alternatives that have merit in some settings include the aminoglycoside paromomycin (the chemical equivalent of aminosidine; not commercially available as of this writing), which has been used as monotherapy (in India) or as an adjunct to Sb₃ and pentamidine. The oral agent miltefosine is being evaluated in clinical trials and preliminarily appears to be highly effective and acceptably tolerated.

Typically, patients feel better and become afebrile during the first week of treatment. Abnormal laboratory findings and splenomegaly may take weeks or months to resolve. The best indicator of permanent cure is freedom from clinical relapse during at least 6 months of follow-up. Repeat tissue sampling is indicated if the patient's status is in question. The persistence of some parasites is not necessarily a poor prognostic indicator, whereas the apparent absence of parasites does not ensure that the patient will not have a relapse.

VISCERAL LEISHMANIASIS IN PERSONS INFECTED WITH HIV

Visceral leishmaniasis has become an important opportunistic infection among persons infected with HIV-1 in geographic areas in which both infections are endemic. To date, most dual infections have been reported from southern Europe, where *L. infantum* (of the *L. donovani* complex) is endemic. In patients infected with HIV, even relatively avirulent *Leishmania* strains can disseminate to the viscera. Clinical leishmaniasis in coinfecting patients can represent newly acquired or reactivated infection; most coinfecting patients with clinically evident leishmaniasis have fewer than 200 CD4⁺ lymphocytes per microliter. Recent data suggest that leishmaniasis is a cofactor in the pathogenesis of HIV infection; the lipophosphoglycan (a major surface molecule) of *L. donovani* induces transcription of HIV in CD4⁺ cells.

A diagnosis of visceral leishmaniasis should be considered for patients infected with HIV who have ever been in leishmaniasis-endemic areas and who have manifestations such as unexplained fever, organomegaly, anemia, or pancytopenia. Coinfecting patients can develop unusual manifestations of visceral leishmaniasis, in part because of atypical localization of the parasite (e.g., in the gastrointestinal tract).

The diagnostic sensitivity of classic serologic methods is lower in coinfecting than in immunocompetent patients (~50% vs. >90%). However, parasitologic diagnosis by noninvasive means is easier in coinfecting patients. Parasites are more commonly found in the circulating blood monocytes of these patients; the sensitivities are ~50% for a Giemsa-stained peripheral-blood smear and ~70% for culture of a buffy-coat preparation. Invasive methods of parasitologic diagnosis (e.g., microscopic examination or culture of a bone marrow aspirate) typically are highly sensitive, especially for previously untreated patients, who commonly have heavy parasite burdens.

Coinfecting patients may initially respond well to standard antileishmanial therapy, albeit with more drug toxicity than is experienced by most immunocompetent persons. However, coinfecting patients commonly have a chronic or relapsing course, seemingly irrespective of the drug regimens used for induction and suppression therapy.

CUTANEOUS LEISHMANIASIS

Cutaneous leishmaniasis has traditionally been classified as New World (American) or Old World disease. Local names for New World disease include *chiclero ulcer* ([Fig. 215-CD1](#)), *pian bois* (bush yaws), and *uta*; those for Old World disease include *oriental sore*, *bouton d'orient*, *Aleppo evil*, and *Baghdad boil*. More than 90% of the world's cases of cutaneous leishmaniasis occur in Afghanistan, Algeria, Iran, Iraq, Saudi Arabia, Syria, Brazil, and Peru. In the Americas, the leishmaniasis-endemic area extends from southern Texas to northern Argentina; the etiologic agents typically are those of the *L. mexicana* complex and the *Viannia* subgenus ([Table 215-1](#)) but also include *L. major*-like organisms and *L. chagasi*. Old World cutaneous leishmaniasis is caused by *L. tropica*, *L. major*, and *L. aethiopica* as well as by *L. infantum* and *L. donovani*.

CLINICAL MANIFESTATIONS

The incubation period for clinically evident disease typically ranges from weeks to months. The first manifestation is usually a papule at the site of the sandfly bite but can be regional lymphadenopathy (sometimes bubonic) in *L. (V.) braziliensis* infection. Most skin lesions evolve from papular to nodular to ulcerative ([Fig. 215-CD2](#)), with a central depression (which can be several centimeters in diameter) surrounded by a raised indurated border ([Fig. 215-2](#)). Some lesions persist as nodules or plaques. Multiple primary lesions, satellite lesions, regional adenopathy, sporotrichoid subcutaneous nodules, lesion pain or pruritus, and secondary bacterial infection are variably present. The infecting species, the location of the lesion, and the host's immune response are among the determinants of the clinical manifestations and chronicity of untreated lesions. For example, in the New World, lesions caused by *L. mexicana* tend to be smaller and less chronic than those caused by *L. (V.) braziliensis*; in the Old World, *L. major* tends to cause "wet" exudative lesions that are less chronic than the "dry" lesions with central crusting that are caused by *L. tropica*. The spontaneous resolution of lesions does not preclude reactivation or reinfection.

The polyparasitic and oligoparasitic ends of the spectrum of cutaneous leishmaniasis are respectively represented by the rare syndromes of diffuse cutaneous leishmaniasis (DCL) and leishmaniasis recidivans, both of which are notoriously difficult to treat. DCL,

caused by *L. aethiopica* (Old World) or by the *L. mexicana* complex (New World), develops in the context of *Leishmania*-specific anergy and is manifested by chronic, disseminated, nonulcerative skin lesions; on histopathologic examination of specimens from these lesions, abundant parasites but few lymphocytes are noted. Leishmaniasis recidivans, a hyperergic variant with scarce parasites, is usually caused by *L. tropica* and manifested by a chronic solitary lesion on the cheek that expands slowly despite central healing.

DIAGNOSIS

Dermal scrapings of debrided ulcerative lesions are useful for histologic examination, aspirates of skin lesions and lymph nodes for in vitro culture, and biopsy specimens for both examination and culture. Although examination of histologic sections of biopsy specimens can help exclude other diagnoses, amastigotes appear larger and are more easily recognizable on Giemsa-stained thin smears (e.g., smears of dermal scrapings, touch preparations of biopsy specimens). As lesions age, amastigotes become scarcer, and parasitologic confirmation becomes more difficult.

Serologic testing is an insensitive means for diagnosing cutaneous leishmaniasis; antibody titers usually are at most minimally elevated except in patients who have [DCL](#). In contrast, leishmanin skin-test reactivity usually develops during active infection in persons who have simple cutaneous or recidivans leishmaniasis but not in those who have DCL.

Cutaneous leishmaniasis is frequently confused with tropical, traumatic, and venous-stasis ulcers; foreign-body reactions; superinfected insect bites; myiasis; impetigo; fungal infections (e.g., sporotrichosis); mycobacterial infections; and other diseases (e.g., sarcoidosis, neoplasms). [DCL](#) and leishmaniasis recidivans should be differentiated from lepromatous leprosy and lupus vulgaris, respectively.

TREATMENT

Decisions about whether and how to treat cutaneous leishmaniasis should take into account whether mucosal dissemination is possible (as it is in the Americas with some organisms in the *Viannia* subgenus; [Table 215-1](#)) as well as the location (e.g., on the face), number, size, evolution, and chronicity of the cutaneous lesions. When optimal effectiveness is important, intravenous or intramuscular Sb_v therapy is recommended ([Table 215-2](#)). In studies in Colombia (predominantly with the *Viannia* subgenus), relatively short courses of treatment with pentamidine ([Table 215-2](#)) were effective (cure rate, 96%) and quite well tolerated. Thus pentamidine may be a good parenteral alternative to Sb_v. The clinical response to antileishmanial therapy begins with lessening induration; healing often continues after the end of therapy. Relapse typically is manifested by clinical reactivation at the margin of the lesion.

Although many oral agents have been touted for treatment of leishmaniasis, even those that are the most effective typically are moderately active at best and are effective only against some *Leishmania* species or strains. The oral agent miltefosine is being evaluated. In the New World, ketoconazole has some activity against *L. mexicana* and *L. (V.) panamensis* and may be more active than itraconazole (at least against the

Viannia subgenus), which is better tolerated ([Table 215-2](#)). Dapsone has looked promising in India but not in Colombia. Adjunctive immunotherapy remains highly experimental but may be useful in [DCL](#). Local or topical therapy can be considered for some cases of infection in which there is no risk of mucosal dissemination (e.g., for relatively benign lesions caused by *L. mexicana* or *L. major*). Examples of local approaches include the application of an ointment containing paromomycin and methylbenzethonium chloride (not licensed in the United States), the intralesional administration of Sb_v, heat therapy, and cryotherapy.

MUCOSAL LEISHMANIASIS

Leishmanial infection of the naso-oro-pharyngeal mucosa is a relatively rare but potentially disfiguring metastatic complication of cutaneous leishmaniasis. Mucosal disease develops despite antileishmanial cell-mediated immunity and most commonly is caused by organisms of the *Viannia* subgenus, typically *L. (V.) braziliensis* but also *L. (V.) panamensis* and *L. (V.) guyanensis*. Although mucosal disease usually becomes clinically evident within several years after the healing of the original cutaneous lesions, cutaneous and mucosal lesions can coexist or appear decades apart. Typically, the original cutaneous lesions of patients who develop mucosal disease were not treated or were suboptimally treated.

Mucosal involvement generally is manifested first by persistent unusual nasal symptoms (e.g., epistaxis), with erythema and edema of the nasal mucosa, and then by progressive, ulcerative, naso-oro-pharyngeal destruction ([Fig. 215-CD3](#)). Supportive laboratory data (e.g., a positive serologic test) are useful, but the scarcity of amastigotes makes parasitologic confirmation difficult. The differential diagnosis includes sarcoidosis, neoplasms, midline granuloma, rhinoscleroma, paracoccidioidomycosis, histoplasmosis, leprosy, syphilis, and tertiary yaws.

Treatment with a pentavalent antimonial compound is moderately effective for mild mucosal disease, whereas advanced disease may not respond to such treatment or may relapse repeatedly ([Table 215-2](#)). Amphotericin B (deoxycholate) is the best alternative drug currently available. Patients who develop signs of respiratory compromise during therapy may benefit from the concomitant administration of glucocorticoids.

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216. TRYPANOSOMIASIS - Louis V. Kirchhoff

CHAGAS' DISEASE

DEFINITION

Chagas' disease, or American trypanosomiasis, is a zoonosis caused by the protozoan parasite *Trypanosoma cruzi*. Acute Chagas' disease is usually a mild febrile illness that results from initial infection with the organism. After spontaneous resolution of the acute illness, most infected persons remain for life in the indeterminate phase of chronic Chagas' disease, which is characterized by subpatent parasitemia, easily detectable antibodies to *T. cruzi*, and an absence of symptoms. In a minority of chronically infected patients, cardiac and gastrointestinal lesions develop that can result in serious morbidity and even death.

LIFE CYCLE AND TRANSMISSION

T. cruzi is transmitted among its mammalian hosts by hematophagous triatomine insects, often called reduviid bugs. The insects become infected by sucking blood from animals or humans who have circulating parasites. Ingested organisms multiply in the gut of the triatomines, and infective forms are discharged with the feces at the time of subsequent blood meals. Transmission to a second vertebrate host occurs when breaks in the skin, mucous membranes, or conjunctivae become contaminated with bug feces that contain infective parasites. *T. cruzi* also can be transmitted by the transfusion of blood donated by infected persons, from mother to fetus, and in laboratory accidents.

PATHOLOGY

An indurated inflammatory lesion called a *chagoma* often appears at the site of the parasite's entry. Local histologic changes include the presence of parasites within leukocytes and cells of subcutaneous tissues and the development of interstitial edema, lymphocytic infiltration, and reactive hyperplasia of adjacent lymph nodes. After dissemination of the organisms through the lymphatics and the bloodstream, muscles (including the myocardium) may become heavily parasitized. The characteristic pseudocysts present in sections of infected tissues are intracellular aggregates of multiplying parasites.

The pathogenesis of chronic Chagas' disease is poorly understood. The heart is the organ most commonly affected, and changes include biventricular enlargement, thinning of the ventricular walls, apical aneurysms, and mural thrombi. Widespread lymphocytic infiltration, diffuse interstitial fibrosis, and atrophy of myocardial cells are often demonstrated, but parasites are rarely seen in myocardial tissue. Conduction-system involvement often affects the right branch and the left anterior branch of the bundle of His. In chronic Chagas' disease of the gastrointestinal tract (megadisease), the esophagus and colon may exhibit varying degrees of dilatation. On microscopic examination, focal inflammatory lesions with lymphocytic infiltration are seen, and the number of neurons in the myenteric plexus may be markedly reduced.

EPIDEMIOLOGY

T. cruzi is found only in the Americas. Wild and domestic mammals harboring *T. cruzi* and infected triatomines are found in spotty distributions from the southern United States to southern Argentina. Humans become involved in the cycle of transmission when infected vectors take up residence in the primitive wood, adobe, and stone houses common in much of Latin America. Thus, human *T. cruzi* infection is a health problem primarily among the poor in rural areas of Central and South America. Most new *T. cruzi* infections in rural settings occur in children, but the incidence is unknown because most cases go undiagnosed. Thousands of individuals also become infected every year through blood transfusions in urban areas. Several dozen patients with HIV and chronic *T. cruzi* infections who underwent acute recrudescence of the latter have been described. These patients generally presented with *T. cruzi* brain abscesses, a manifestation of the illness that does not occur in immunocompetent persons. Currently, it is estimated that 16 to 18 million people, more than a third of whom live in Brazil, are chronically infected with *T. cruzi*. Chronic Chagas' disease is a major cause of morbidity and mortality in many Latin American countries, including Mexico, since many chronically infected persons eventually develop symptomatic cardiac lesions or gastrointestinal disease.

In recent years, the rate of *T. cruzi* transmission has been decreasing in several endemic countries as a result of successful vector and blood-bank control programs. A major program in the "southern cone" nations of South America (Uruguay, Paraguay, Bolivia, Brazil, Chile, and Argentina), begun in 1991, has provided the framework for much of the progress achieved. If current trends continue, transmission will be essentially eliminated in much of the endemic range by the year 2003.

Acute Chagas' disease is rare in the United States. Five cases of autochthonous transmission and four instances of transmission by blood transfusion have been reported. Moreover, in the last 26 years, seven laboratory-acquired infections and nine imported cases of acute Chagas' disease were reported to the Centers for Disease Control and Prevention (CDC). In contrast, the prevalence of chronic *T. cruzi* infections in the United States has increased considerably in recent years. Since the mid-1970s, enormous numbers of Latin Americans have emigrated to the United States. In one study conducted in Washington, D.C., 5% of Salvadoran and Nicaraguan immigrants were found to have chronic *T. cruzi* infections. Estimates based on the latter study and on studies done in several United States blood banks put the total number of infected immigrants now living in the United States at more than 50,000. The presence of these carriers of *T. cruzi* creates a substantial risk of transmission by blood transfusion, as is evidenced by the four transfusion-associated cases just cited.

CLINICAL COURSE

The first signs of acute Chagas' disease develop at least 1 week after invasion by the parasites. When the organisms enter through a break in the skin, an indurated area of erythema and swelling (the chagoma), accompanied by local lymphadenopathy, may appear. Romana's sign -- the classic finding in acute Chagas' disease, which consists of unilateral painless edema of the palpebrae and periocular tissues -- can result when the conjunctiva is the portal of entry. These initial local signs are followed by malaise, fever, anorexia, and edema of the face and lower extremities. A morbilliform rash may also

appear. Generalized lymphadenopathy and hepatosplenomegaly may develop. Severe myocarditis develops rarely; most deaths in acute Chagas' disease are due to heart failure. Neurologic signs are not common, but meningoencephalitis has been reported. The acute symptoms resolve spontaneously in virtually all patients, who then enter the asymptomatic or indeterminate phase of chronic *T. cruzi* infection.

Symptomatic chronic Chagas' disease becomes apparent years or even decades after the initial infection. The heart is commonly involved, and symptoms are caused by rhythm disturbances, cardiomyopathy, and thromboembolism. Right bundle-branch block is the most common electrocardiographic abnormality, but other types of atrioventricular block, premature ventricular contractions, and tachy- and bradyarrhythmias occur frequently. Cardiomyopathy often results in right-sided or biventricular heart failure. Embolization of mural thrombi to the brain or other areas may take place. Patients with megaesophagus suffer from dysphagia, odynophagia, chest pain, and regurgitation. Aspiration can occur, especially during sleep, and repeated episodes of aspiration pneumonitis are common. Weight loss, cachexia, and pulmonary infection can result in death. Patients with megacolon are plagued by abdominal pain and chronic constipation, and advanced megacolon can cause obstruction, volvulus, septicemia, and death.

DIAGNOSIS

The diagnosis of acute Chagas' disease requires the detection of parasites. Microscopic examination of fresh anticoagulated blood or of the buffy coat is the simplest way to see the motile organisms. Parasites also can be seen in Giemsa-stained thin and thick blood smears. When repeated attempts to visualize the organisms are unsuccessful, mouse inoculation, culture of blood in specialized media, or xenodiagnosis can be performed. In the last technique, uninfected triatomine insects are allowed to feed on the patient's blood. When done properly, all of these methods yield positive results in a high proportion of patients with acute Chagas' disease and in at least half of those with chronic infections. Since early treatment of acute Chagas' disease is extremely important, however, the decision to initiate therapy for *T. cruzi* infection despite negative wet preparations and smears must be made on clinical and epidemiologic grounds before the results of these indirect methods become available. Serologic testing is of limited usefulness in diagnosing acute Chagas' disease.

The diagnosis of chronic Chagas' disease is made by the detection of antibodies that bind to *T. cruzi* antigens. Demonstration of the parasite is not of primary importance. Several highly sensitive serologic tests for antibodies to *T. cruzi* are used widely in Latin America, including complement-fixation and immunofluorescence tests and enzyme-linked immunosorbent assays (ELISAs). However, a persistent problem with these conventional assays is the occurrence of false-positive reactions, typically with sera from patients who have other parasitic infections or autoimmune diseases. For this reason, it is generally recommended that positivity in one assay be confirmed by two other tests and that well-characterized positive and negative comparison sera be included in each run. A highly sensitive and specific method for detecting antibodies to *T. cruzi* [approved by the Clinical Laboratory Improvement Amendment (CLIA) and available in the author's laboratory] employs immunoprecipitation of radiolabeled *T. cruzi* antigens and electrophoresis. Serodiagnostic assays that employ recombinant *T.*

cruzi proteins as target antigens are being developed, as are tests based on the amplification of *T. cruzi* DNA sequences by polymerase chain reaction. However, these tests are not yet available for general use.

TREATMENT

Therapy for Chagas' disease is unsatisfactory. Nifurtimox is the only drug active against *T. cruzi* that is available in the United States. In acute Chagas' disease, nifurtimox markedly reduces the duration of symptoms and parasitemia and decreases the mortality rate. Nevertheless, its efficacy at eradicating parasites is low. Limited studies have shown that only ~70% of acute infections are cured parasitologically by a full course of treatment. Despite its limitations, nifurtimox treatment should be initiated as early as possible in acute Chagas' disease. Moreover, when laboratory accidents occur in which it appears likely that *T. cruzi* infection could become established, nifurtimox therapy should be initiated without waiting for clinical or parasitologic indications of infection.

Common adverse effects of nifurtimox include abdominal pain, anorexia, nausea, vomiting, and weight loss. Neurologic reactions to the drug may include restlessness, disorientation, insomnia, twitching, paresthesia, polyneuritis, and seizures. These symptoms usually disappear when the dosage is reduced or treatment is discontinued. The recommended daily dosage is 8 to 10 mg/kg for adults, 12.5 to 15 mg/kg for adolescents, and 15 to 20 mg/kg for children 1 to 10 years of age. The drug should be given orally in four divided doses each day, and therapy should be continued for 90 to 120 days. Nifurtimox is available from the Drug Service of the [CDC](#) in Atlanta, Georgia (telephone number, 770-639-3670).

Benznidazole is a second agent used to treat Chagas' disease. Its efficacy is similar to that of nifurtimox, and its adverse effects include peripheral neuropathy, rash, and granulocytopenia. The recommended oral dosage is 5 mg/kg per day for 60 days. Benznidazole is used widely in Latin America.

The question of whether patients in the indeterminate or chronic symptomatic phases of Chagas' disease should be treated with nifurtimox or benznidazole has been debated for years. Studies of *T. cruzi*-infected laboratory animals and humans suggest that elimination of the parasites reduces the appearance or progression of cardiac pathology. In view of these findings, an international panel of experts has recommended that all patients infected with *T. cruzi* be treated with one drug or the other, regardless of their clinical status or the duration of infection.

The usefulness of allopurinol, fluconazole, and itraconazole for the treatment of acute Chagas' disease has been studied extensively in laboratory animals and to a lesser extent in humans. None of these drugs has exhibited a level of anti-*T. cruzi* activity that warrants its use in patients. Studies in mice have shown that recombinant interferon γ decreases the duration and severity of acute *T. cruzi* infection; however, its usefulness in persons with acute Chagas' disease has not been evaluated systematically.

Patients who develop cardiac and/or gastrointestinal disease in association with *T. cruzi* infection should be referred to appropriate subspecialists for further evaluation and

treatment. Cardiac transplantation is an option for patients with end-stage chagasic cardiopathies. Postoperative prophylaxis with nifurtimox or benznidazole should be considered because without it the immunosuppression required after surgery has been shown to result in reactivation of *T. cruzi* infection, often with serious consequences or even death.

PREVENTION

Since drug therapy is unsatisfactory and vaccines are not available, the control of *T. cruzi* transmission in endemic countries must depend on reduction of domiciliary vector populations by spraying of insecticides, improvement of housing, and education. In addition, in endemic areas, programs for the screening of donated blood for *T. cruzi* need to be expanded and improved to reduce rates of transmission by transfusion. Tourists traveling in endemic areas should avoid sleeping in dilapidated houses outside urban areas. Mosquito nets and insect repellent provide additional protection.

In the United States, the question of how best to avoid transmission of *T. cruzi* by blood transfusion is not easily resolved. Since no assay for *T. cruzi* infection has received clearance from the Food and Drug Administration (FDA) for use in blood banks, serologic screening is not yet an option. The FDA currently mandates the use of a questionnaire for identifying and deferring donors at high risk for *T. cruzi* infection. This approach may be effective and not reduce the blood supply intolerably, but it is important to bear in mind that approaches based solely on questionnaires have not been entirely successful at eliminating transfusion-associated transmission of other infectious agents.

In view of the possibly serious consequences of chronic *T. cruzi* infection, it would be prudent for all immigrants from endemic regions to be tested for evidence of infection. Identification of infected persons is also important because the implantation of pacemakers benefits some patients who develop ominous rhythm disturbances. The possibility of congenital transmission is yet another justification for screening.

Laboratory personnel should wear gloves and eye protection when working with *T. cruzi* and infected vectors.

SLEEPING SICKNESS

DEFINITION

Sleeping sickness, or human African trypanosomiasis (HAT), is caused by flagellated protozoan parasites that belong to the *T. brucei* complex and are transmitted to humans by tsetse flies. In untreated patients, the trypanosomes first cause a febrile illness that is followed months or years later by progressive neurologic impairment and death.

THE PARASITES AND THEIR TRANSMISSION

The East African (*rhodesiense*) and the West African (*gambiense*) forms of sleeping sickness are caused, respectively, by two trypanosome subspecies: *T. brucei rhodesiense* and *T. brucei gambiense*. These subspecies are morphologically

indistinguishable but cause illnesses that are epidemiologically and clinically distinct. The parasites are transmitted by blood-sucking tsetse flies of the genus *Glossina*. The insects acquire the infection when they ingest blood from infected mammalian hosts. After many cycles of multiplication in the midgut of the vector, the parasites migrate to the salivary glands. Their transmission takes place when they are inoculated during a subsequent blood meal. The injected trypanosomes multiply in the blood and other extracellular spaces and evade immune destruction in mammalian hosts for long periods by undergoing antigenic variation, in which the antigenic structure of their surface coat of glycoproteins changes periodically.

PATHOGENESIS AND PATHOLOGY

A self-limited inflammatory lesion (trypanosomal chancre) may appear a week or so after the bite of an infected tsetse fly. A systemic febrile illness then evolves as the parasites are disseminated through the lymphatics and bloodstream. Systemic [HAT](#) without central nervous system (CNS) involvement is generally referred to as *stage I disease*. In this stage, widespread lymphadenopathy and splenomegaly reflect marked lymphocytic and histiocytic proliferation and invasion of morular cells, which are plasmacytes that may be involved in the production of IgM. Endarteritis, with perivascular infiltration of both parasites and lymphocytes, may develop in lymph nodes and spleen. Myocarditis develops frequently in patients with stage I disease and is especially common in *T. b. rhodesiense* infections.

Hematologic manifestations that accompany stage [I HAT](#) include moderate leukocytosis, thrombocytopenia, and anemia. High levels of immunoglobulins, consisting primarily of polyclonal IgM, are a constant feature, and heterophile antibodies, antibodies to DNA, and rheumatoid factor are often detected. High levels of antigen-antibody complexes may play a role in the tissue damage and increased vascular permeability that facilitate dissemination of the parasites.

Stage II disease involves invasion of the [CNS](#). The presence of trypanosomes in perivascular areas is accompanied by intense infiltration of mononuclear cells. Abnormalities in cerebrospinal fluid (CSF) include increased pressure, elevated total protein concentration, and pleocytosis. In addition, trypanosomes are frequently found in CSF.

EPIDEMIOLOGY

The trypanosomes that cause sleeping sickness are found only in Africa. Approximately 50 million persons are at risk of acquiring [HAT](#), and tens of thousands of new cases occur every year. Precise data are not available because health statistics are often incomplete in the developing countries where HAT is endemic. Sleeping sickness has undergone a resurgence in recent years, with major epidemics in the Sudan, Ivory Coast, Chad, the Central African Republic, and several other endemic countries.

Humans are the only reservoir of *T. b. gambiense*, which occurs in widely distributed foci in tropical rain forests of Central and West Africa. Gambiense trypanosomiasis is primarily a problem in rural populations; tourists rarely become infected. Trypanotolerant antelope species in savanna and woodland areas of Central and East Africa are the

principal reservoir of *T. b. rhodesiense*. Cattle also can become infected but generally succumb to the parasite. Since risk results for the most part from contact with tsetse flies that feed on wild animals, humans acquire *T. b. rhodesiense* infection only incidentally, usually while working in areas where infected game and vectors are present. In addition, occasional cases occur among visitors to game parks in East Africa. During the past 22 years, 21 cases of imported [HAT](#) have been reported to the [CDC](#), most of which were caused by *T. b. rhodesiense*.

CLINICAL COURSE

A painful trypanosomal chancre appears in some patients at the site of inoculation of the parasite. Hematogenous and lymphatic dissemination (stage I disease) is marked by the onset of fever. Typically, bouts of high temperatures lasting several days are separated by afebrile periods. Lymphadenopathy is prominent in *T. b. gambiense* trypanosomiasis. The nodes are discrete, movable, rubbery, and nontender. Cervical nodes are often visible, and enlargement of the nodes of the posterior cervical triangle, or Winterbottom's sign, is a classic finding. Pruritus and maculopapular rashes are common. Inconstant findings include malaise, headache, arthralgias, weight loss, edema, hepatosplenomegaly, and tachycardia.

[CNS](#) invasion (stage II disease) is characterized by the insidious development of protean neurologic manifestations that are accompanied by progressive abnormalities in the [CSE](#). A picture of progressive indifference and daytime somnolence develops (hence the designation "sleeping sickness"), sometimes alternating with restlessness and insomnia at night. A listless gaze accompanies a loss of spontaneity, and speech may become halting and indistinct. Extrapyrarnidal signs may include choreiform movements, tremors, and fasciculations. Ataxia is frequent, and the patient may appear to have Parkinson's disease, with a shuffling gait, hypertonia, and tremors. In the final phase, progressive neurologic impairment ends in coma and death.

The most striking difference between the West African and East African trypanosomiasis is that the latter illness tends to follow a more acute course. Typically, in tourists, systemic signs of infection, such as fever, malaise, and headache, appear before the end of the trip or shortly after the return home. Persistent tachycardia unrelated to fever is common early in the course of *T. b. rhodesiense* trypanosomiasis, and death may result from arrhythmias and congestive heart failure before [CNS](#) disease develops. In general, untreated *T. b. rhodesiense* trypanosomiasis leads to death in a matter of weeks to months, often without a clear distinction between the hemolymphatic and CNS stages.

DIAGNOSIS

A definitive diagnosis of [HAT](#) requires detection of the parasite. If a chancre is present, fluid should be expressed and examined directly by light microscopy for the highly motile trypanosomes. The fluid also should be fixed and stained with Giemsa. Material obtained by needle aspiration of lymph nodes early in the course of the illness should be examined similarly. Examination of wet preparations and Giemsa-stained thin and thick films of serial blood samples is also useful. If parasites are not seen in blood, efforts should be made to concentrate the organisms; the simplest method involves the use of

quantitative buffy coat analysis tubes (QBC, Becton-Dickinson, Franklin Lakes, NJ). In these tubes, which are coated with acridine orange, the parasites are separated from blood cells by centrifugation and are easily seen under light microscopy because of the stain. The buffy coat from 10 to 15 mL of anticoagulated blood or the pellet obtained by centrifugation of the eluate from 25 to 50 mL of blood passed through a DEAE-cellulose column also can be examined. Trypanosomes may be seen in material aspirated from the bone marrow; the aspirate can be inoculated into liquid culture medium, as can blood, buffy coat, lymph node aspirates, and [CSF](#). Finally, *T. b. rhodesiense* infection can be detected by inoculation of these specimens into mice or rats, which results in patent parasitemias in a week or two. Although this method is highly sensitive for the detection of *T. b. rhodesiense*, it does not detect *T. b. gambiense* because of host specificity.

It is essential to examine [CSF](#) from all patients in whom [HAT](#) is suspected. An increase in the CSF cell count is the first abnormality to be detected; increases in opening pressure and in levels of total protein and IgM develop later. Trypanosomes may be seen in the sediment of centrifuged CSF. Any CSF abnormality in a patient in whom trypanosomes have been found at other sites must be viewed as pathognomonic for [CNS](#) involvement and thus must prompt specific treatment for CNS disease.

A number of serologic assays are available to aid in the diagnosis of [HAT](#), but their variable sensitivity and specificity mandate that decisions about treatment be based on demonstration of the parasite. These tests are of value for epidemiologic surveys.

TREATMENT

The drugs traditionally used for treatment of [HAT](#) are suramin, pentamidine, and organic arsenicals. An addition to this list is eflornithine (difluoromethylornithine), which was approved by the [FDA](#) in November 1990 for the treatment of West African trypanosomiasis. In the United States these drugs can be obtained from the [CDC](#). Therapy for HAT must be individualized on the basis of the infecting organism (*T. b. gambiense* or *T. b. rhodesiense*), the presence or absence of [CNS](#) disease, adverse reactions, and (occasionally) drug resistance. The choices of drugs for the treatment of HAT are summarized in [Table 216-1](#).

Suramin is highly effective against stage I disease. However, it can cause serious adverse effects and must be administered under the close supervision of a physician. A 100- to 200-mg intravenous test dose should be administered to detect hypersensitivity. The dosage for adults is 1 g intravenously on days 1, 3, 7, 14, and 21. The regimen for children is 20 mg/kg (maximum, 1 g) intravenously on days 1, 3, 7, 14, and 21. The drug is given by slow intravenous infusion of a freshly prepared 10% aqueous solution. Approximately 1 patient in 20,000 has an immediate, severe, and potentially fatal reaction to the drug, developing nausea, vomiting, shock, and seizures. Less severe reactions include fever, photophobia, pruritus, arthralgias, and skin eruptions. Renal damage is the most common important adverse effect of suramin. Transient proteinuria often appears during treatment. A urinalysis should be done before each dose, and treatment should be discontinued if proteinuria increases or if casts and red cells appear in the sediment. Suramin should not be given to patients with renal insufficiency.

Eflornithine is highly effective for treatment of both stages of West African trypanosomiasis. In the trials on which the [FDA](#) based its approval, this agent cured more than 90% of 600 patients with stage II disease. The recommended treatment schedule is 400 mg/kg per day intravenously in four divided doses for 2 weeks. Adverse reactions include diarrhea, anemia, thrombocytopenia, seizures, and hearing loss. The high dosage and duration of therapy required are disadvantages that make widespread use of eflornithine difficult.

Pentamidine is the alternative drug for patients with stage I [HAT](#), although some *T. b. rhodesiense* infections are unresponsive to this agent. The dose for both adults and children is 4 mg/kg per day intramuscularly or intravenously for 10 days. Frequent, immediate adverse reactions include nausea, vomiting, tachycardia, and hypotension. These reactions are usually transient and do not warrant cessation of therapy. Other adverse reactions include nephrotoxicity, abnormal liver function tests, neutropenia, rashes, hypoglycemia, and sterile abscesses.

The arsenical melarsoprol is the drug of choice for the treatment of East African trypanosomiasis with [CNS](#) involvement. Melarsoprol cures both stages of the disease and therefore is also indicated for the treatment of stage I disease in patients who fail to respond to or cannot tolerate suramin and/or pentamidine. However, because of its relatively high toxicity, melarsoprol is never the first choice for the treatment of stage I disease. The drug should be given to adults in three courses of 3 days each. The dosage is 2 to 3.6 mg/kg per day intravenously in three divided doses for 3 days followed 1 week later by 3.6 mg/kg per day, also in three divided doses and for 3 days. The latter course is repeated 10 to 21 days later. In debilitated patients, suramin is administered for 2 to 4 days before therapy with melarsoprol is initiated. An 18-mg initial dose of the latter drug, followed by progressive increases to the standard dose, has been recommended. For children, a total of 18 to 25 mg/kg should be given over 1 month. A starting dose of 0.36 mg/kg intravenously should be increased gradually to a maximum of 3.6 mg/kg at 1- to 5-day intervals, for a total of 9 or 10 doses.

Melarsoprol is highly toxic and should be administered with great care. The incidence of reactive encephalopathy has been reported to be as high as 18% in some series. Clinical manifestations of reactive encephalopathy include high fever, headache, tremor, impaired speech, seizures, and even coma and death. Treatment with melarsoprol should be discontinued at the first sign of encephalopathy but may be restarted cautiously at lower doses a few days after signs have resolved. Extravasation of the drug results in intense local reactions. Vomiting, abdominal pain, nephrotoxicity, and myocardial damage can occur.

The treatment of patients with stage II East African disease who cannot tolerate melarsoprol is problematic. The combination of the arsenical tryparsamide and suramin is one possible approach, but its efficacy is limited because suramin does not penetrate the [CNS](#) well and tryparsamide is much less effective against *T. b. rhodesiense* than it is against *T. b. gambiense*. The schedule for tryparsamide therapy is 30 mg/kg (maximum, 2 g) in a single intravenous dose every 5 days for a total of 12 doses; that for suramin treatment is 10 mg/kg intravenously every 5 days, also for a total of 12 injections. Tryparsamide can cause encephalopathy, fever, vomiting, abdominal pain, rash, tinnitus, and a variety of ocular symptoms. Alternatively, eflornithine can be

administered as outlined above to patients who cannot tolerate melarsoprol, but, as noted, its effectiveness against *T. b. rhodesiense* is variable.

PREVENTION

[HAT](#) poses complex public-health and epizootic problems in Africa. Considerable progress has been made in some areas through control programs that focus on eradication of vectors and drug treatment of infected humans; however, there is no consensus on the best approach to solving the overall problem, and major epidemics continue to occur. Individuals can reduce their risk of acquiring trypanosomiasis by avoiding areas known to harbor infected insects, by wearing protective clothing, and by using insect repellent. Chemoprophylaxis is not recommended, and no vaccine is available to prevent transmission of the parasites.

(Bibliography omitted in Palm version)

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217. TOXOPLASMA INFECTION - Lloyd H. Kasper

DEFINITION

Toxoplasmosis is the disease caused by infection with the obligate intracellular parasite *Toxoplasma gondii*. Acute infection acquired after birth may be asymptomatic but frequently results in the chronic persistence of cysts within the tissues of the host. Both acute and chronic toxoplasmosis are conditions in which the parasite is responsible for the development of clinically evident disease, including lymphadenopathy, encephalitis, myocarditis, and pneumonitis. Congenital toxoplasmosis is an infection of newborns that results from the transplacental passage of parasites from an infected mother to the fetus. These infants usually are asymptomatic at birth but later manifest a wide range of signs and symptoms, including chorioretinitis, strabismus, epilepsy, and psychomotor retardation.

ETIOLOGY

T. gondii is an intracellular coccidian that infects both birds and mammals. There are two distinct stages in the life cycle of *T. gondii*: the nonfeline and feline stages. In the nonfeline stage, tissue cysts that contain bradyzoites or sporulated oocysts are ingested by an intermediate host (e.g., a human, mouse, sheep, or pig). The cyst is rapidly digested by the acidic-pH gastric secretions. Bradyzoites or sporozoites are released, enter the small-intestinal epithelium, and transform into rapidly dividing tachyzoites. The tachyzoites can infect and replicate in all mammalian cells except red blood cells. Once attached to the host cell, the parasite penetrates the cell and forms a parasitophorous vacuole within which it divides. Parasite replication continues until the number of parasites within the cell approaches a critical mass and the cell ruptures, releasing parasites that infect adjoining cells.

As a result of this process, an infected organ soon shows evidence of cytopathology. Most tachyzoites are eliminated by means of the host's humoral and cell-mediated immune responses. Tissue cysts containing many bradyzoites develop 7 to 10 days after the systemic tachyzoite infection. These tissue cysts occur in a variety of host organs but persist principally within the central nervous system (CNS) and muscle. The development of this chronic stage completes the nonfeline portion of the life cycle. Active infection in the immunocompromised host is most likely due to the spontaneous release of encysted parasites that undergo rapid transformation into tachyzoites within the CNS.

The principal stage in the life cycle of the parasite takes place in the cat (the definitive host) and its prey. The parasite's sexual phase is defined by the formation of oocysts within the feline host. This enteroepithelial cycle begins with the ingestion of the bradyzoite tissue cysts and culminates after several intermediate stages in the production of gametes. Gamete fusion produces a zygote, which envelops itself in a rigid wall and is secreted in the feces as an unsporulated oocyst. After 2 to 3 days of exposure to air at ambient temperature, the noninfectious oocyst sporulates to produce eight sporozoite progeny. The sporulated oocyst can be ingested by an intermediate host, such as a person emptying a cat's litter box, a pig rummaging in a barnyard, or perhaps a mouse. It is in the intermediate host that the parasite completes its life cycle.

EPIDEMIOLOGY

T. gondii infects a wide range of mammals and birds. Its seroprevalence depends on the locale and the age of the population. Generally, hot arid climatic conditions are associated with a low prevalence of infection. In the United States and most European countries, the prevalence of seroconversion increases with age and exposure. For example, in the United States, 5 to 30% of individuals 10 to 19 years old and 10 to 67% of those over the age of 50 years show serologic evidence of exposure; seroprevalence increases by approximately 1% per year. In Central America, France, Turkey, and Brazil, the seroprevalence is higher.

TRANSMISSION

Oral Transmission The principal source of human *Toxoplasma* infection remains uncertain. Transmission usually takes place by the oral route and can be attributable to ingestion of either sporulated oocysts from contaminated soil or bradyzoites from undercooked meat. During acute feline infection, a cat may excrete as many as 100 million parasites per day. These very stable sporozoite-containing oocysts are highly infectious and may remain viable for many years in the soil. Humans infected during a well-documented outbreak of oocyst-transmitted infection develop stage-specific antibodies to the oocyst/sporozoite.

Children and adults also can acquire infection from tissue cysts containing bradyzoites. The ingestion of a single cyst is all that is required for human infection. Undercooking or insufficient freezing of meat is an important source of infection in the developed world. In the United States, 10 to 20% of lamb products and 25 to 35% of pork products show evidence of cysts that contain bradyzoites. The incidence in beef is much lower -- perhaps as low as 1%. Direct ingestion of bradyzoite cysts in these various meat products leads to acute infection.

Transmission via Blood or Organs In addition to oral transmission, direct transmission of the parasite by blood or organ products during transplantation takes place at a low rate. Viable parasites can be cultured from refrigerated anticoagulated blood, which may be a source of infection in individuals receiving blood transfusions. *T. gondii* infection also has been reported in kidney and heart transplant recipients who were uninfected before transplantation.

Transplacental Transmission About one-third of all women who acquire infection with *T. gondii* during pregnancy transmit the parasite to the fetus; the remainder give birth to normal, uninfected babies. Of the various factors that influence fetal outcome, gestational age at the time of infection is the most critical (see below). Few data support a role for recrudescence of maternal infection as the source of congenital disease. Thus, women who are seropositive before pregnancy usually are protected against acute infection and do not give birth to congenitally infected neonates.

The following general guidelines can be used to evaluate congenital infection. There is essentially no risk if the mother becomes infected ≥ 6 months before conception. If infection is acquired < 6 months before conception, the likelihood of transplacental

infection increases as the interval between infection and conception decreases. In pregnancy, if the mother becomes infected during the first trimester, the incidence of transplacental infection is lowest (about 15%), but the disease in the neonate is most severe. If maternal infection occurs during the third trimester, the incidence of transplacental infection is greatest (65%), but the infant is usually asymptomatic at birth. Infected infants who are normal at birth may have a higher incidence of learning disabilities and chronic neurologic sequelae than uninfected children. Only a small proportion (20%) of women infected with *T. gondii* develop clinical signs of infection. Often the diagnosis is first appreciated when routine postconception serologic tests show evidence of specific antibody.

PATHOGENESIS

Upon the host's ingestion of either tissue cysts containing bradyzoites or oocysts containing sporozoites, the parasites are released from the cysts by a digestive process. Bradyzoites are resistant to the effect of pepsin and invade the host's gastrointestinal tract. Within enterocytes, the parasites undergo morphologic transformation, giving rise to invasive tachyzoites. These tachyzoites induce a parasite-specific secretory IgA response. From the gastrointestinal tract, parasites are disseminated to a variety of organs, particularly lymphatic tissue, skeletal muscle, myocardium, retina, placenta, and the [CNS](#). At these sites, the parasite infects host cells, replicates, and invades the adjoining cells. In this fashion, the hallmarks of the infection develop: cell death and focal necrosis surrounded by an acute inflammatory response.

In the normal immune host, both the humoral and the cellular immune responses control infection; parasite virulence and tissue tropism may be strain specific. Tachyzoites are sequestered by a variety of immune mechanisms, including induction of parasitocidal antibody, activation of macrophages with radical intermediates, production of interferon (IFN- γ), and stimulation of cytotoxic T lymphocytes of the CD8⁺ phenotype. These antigen-specific lymphocytes are capable of killing both extracellular parasites and target cells infected with parasites. As tachyzoites are cleared from the acutely infected host, tissue cysts containing bradyzoites begin to appear, usually within the [CNS](#) and the retina. In the immunocompromised or fetal host, the immune factors necessary to control the spread of tachyzoite infection are lacking. This altered immune state allows the persistence of tachyzoites and gives rise to the progressive focal destruction that results in organ failure (i.e., necrotizing encephalitis, pneumonia, and myocarditis).

Persistence of infection with cysts containing bradyzoites is common in the immunocompetent host. This lifelong infection usually remains subclinical. Although bradyzoites are in a slow metabolic phase, cysts do degenerate and rupture within the [CNS](#). This degenerative process, with the development of new bradyzoite-containing cysts, is the most probable source of recrudescent infection in immunocompromised individuals and the most likely stimulus for the persistence of antibody titers in the immunocompetent host.

PATHOLOGY

Cell death and focal necrosis due to replicating tachyzoites induce an intense mononuclear inflammatory response in any tissue or cell type infected. Tachyzoites

rarely can be visualized by routine histopathologic staining of these inflammatory lesions. However, immunofluorescence staining with parasitic antigen-specific antibodies can reveal either the organism itself or evidence of antigen. In contrast to this inflammatory process caused by tachyzoites, bradyzoite-containing cysts cause inflammation only at the early stages of development, and even this inflammation may be a response to the presence of tachyzoite antigens. Once the cysts reach maturity, the inflammatory process can no longer be detected, and the cysts remain immunologically quiescent within the brain matrix until they rupture.

Lymph Nodes During acute infection, lymph node biopsy demonstrates characteristic findings, including follicular hyperplasia and irregular clusters of tissue macrophages with eosinophilic cytoplasm. Granulomas rarely are evident in these specimens. Although tachyzoites are not usually visible, they can be sought either by subinoculation of infected tissue into mice, with resultant disease, or by the polymerase chain reaction (PCR). PCR amplification of DNA fragments representing either p30 (SAG-1) or p22 (SAG-2) surface antigen or B1 antigen is an effective and sensitive assay for establishing infection of lymph node tissue by tachyzoites.

Eyes In the eye, infiltrates of monocytes, lymphocytes, and plasma cells may produce uni- or multifocal lesions. Granulomatous lesions and chorioretinitis can be observed in the posterior chamber following acute necrotizing retinitis. Other ocular complications of infection include iridocyclitis, cataracts, and glaucoma.

Central Nervous System During [CNS](#) involvement, both focal and diffuse meningoencephalitis can be documented, with evidence of necrosis and microglial nodules. Necrotizing encephalitis in patients without AIDS is characterized by small diffuse lesions with perivascular cuffing in contiguous areas. In the AIDS population, polymorphonuclear leukocytes may be present in addition to monocytes, lymphocytes, and plasma cells. Cysts containing bradyzoites frequently are found contiguous with the necrotic tissue border.

Lungs Among patients with AIDS who die of toxoplasmosis, 40 to 70% have involvement of the heart and lung. Interstitial pneumonitis can develop in the neonate and the immunocompromised patient. Thickened and edematous alveolar septa infiltrated with mononuclear and plasma cells are apparent. This inflammation may extend to the endothelial walls. Tachyzoites and bradyzoite-containing cysts have been observed within the alveolar membrane. Superimposed bronchopneumonia can be caused by other microbial agents.

Heart Cysts and aggregates of parasites in cardiac muscle tissue are evident in patients with AIDS who die of toxoplasmosis. Focal necrosis surrounded by inflammatory cells is associated with hyaline necrosis and disrupted myocardial cells. Pericarditis is associated with toxoplasmosis in some patients.

Other Sites Pathologic changes during disseminated infection are similar to those described for the lymph nodes, eyes, and [CNS](#). In patients with AIDS, the skeletal muscle, pancreas, stomach, and kidneys can be involved, with necrosis, invasion by inflammatory cells, and (rarely) the presence of tachyzoites detectable by routine staining. Large necrotic lesions may cause direct tissue destruction. In addition,

secondary effects from acute infection of these various organs, including pancreatitis, myositis, and glomerulonephritis, have been reported.

HOST IMMUNE RESPONSE

Acute *Toxoplasma* infection evokes a cascade of protective immune responses in the normal host. *Toxoplasma* enters the host at the gut mucosal level and evokes a mucosal immune response that includes the production of antigen-specific secretory IgA. Titers of serum IgA antibody directed at p30 (SAG-1) have been shown to be a useful marker of congenital and acute toxoplasmosis. Milk-whey IgA from acutely infected mothers contains a high titer of antibody to *T. gondii* and can block infection of enterocytes in vitro. In mice, IgA intestinal secretions directed at the parasite are abundant and are associated with the induction of mucosal T cells.

Within the host, *T. gondii* rapidly induces detectable levels of both IgM and IgG serum antibodies. Monoclonal gammopathy of the IgG class can occur in congenitally infected infants. IgM levels may be increased in newborns with congenital infection. The polyclonal IgG antibodies evoked by infection are parasitocidal in vitro in the presence of serum complement and are the basis for the Sabin-Feldman dye test. However, cell-mediated immunity is the major protective response evoked by the parasite during host infection. Macrophages are activated following phagocytosis of antibody-opsonized parasites. This activation can lead to death of the parasite by either an oxygen-dependent or an oxygen-independent process. If the parasite is not phagocytosed and enters the macrophage by active penetration, it continues to replicate, and this replication may represent the mechanism for transport and dissemination to distant organs. *Toxoplasma* stimulates a robust interleukin (IL) 12 response by human dendritic cells. The CD4⁺ and CD8⁺ T cell responses are antigen-specific and further stimulate the production of a variety of important lymphokines that expand the T cell and natural killer cell repertoire. *T. gondii* is a potent inducer of a T_H1 phenotype, with IL-12 and IFN- γ playing an essential role in the control of the parasites' growth in the host. Regulation of the inflammatory response is at least partially under the control of a T_H2 response that includes the production of IL-4 and IL-10 in seropositive individuals. Both asymptomatic patients and those with active infection may show a depression in the ratio of CD4⁺ to CD8⁺ lymphocytes. This shift may be correlated with a disease syndrome but is not necessarily correlated with disease outcome. Human T cell clones of both the CD4⁺ and the CD8⁺ phenotypes are cytolytic against parasite-infected macrophages. These T cell clones produce cytokines that are "microbistatic." IL-18, IL-7, and IL-15 upregulate the production of IFN- γ and may be important during acute and chronic infection. The effect of IFN- γ may be paradoxical, with stimulation of a host downregulatory response as well.

Although in patients with AIDS *T. gondii* infection is believed to be recrudescent, determination of antibody titers is not helpful in establishing reactivation. Because of the severe depletion in CD4⁺ T cells, quite frequently there is no observed increase in antibody titer during exacerbation of infection. T cells from AIDS patients with reactivation of toxoplasmosis fail to secrete both IFN- γ and IL-2. This alteration in the production of these critical immune cytokines contributes to the persistence of infection. *Toxoplasma* infection frequently develops late in the course of AIDS, when the loss of T cell-dependent protective mechanisms, particularly CD8⁺ T cells, becomes most

pronounced.

CLINICAL MANIFESTATIONS

In persons whose immune systems are intact, acute toxoplasmosis is usually asymptomatic and self-limited. This condition can go unrecognized in 80 to 90% of adults and children with acquired infection. The asymptomatic nature of this infection makes diagnosis difficult in mothers infected during pregnancy. In contrast, the wide range of clinical manifestations in congenitally infected children includes severe neurologic complications such as hydrocephalus, microcephaly, mental retardation, and chorioretinitis. If prenatal infection is severe, multiorgan failure and subsequent intrauterine fetal death can occur. In children and adults, chronic infection can persist throughout life, with little consequence to the immunocompetent host.

Toxoplasmosis in the Immunocompetent Person The most common manifestation of acute toxoplasmosis is cervical lymphadenopathy. The nodes may be single or multiple, are usually nontender, are discrete, and vary in firmness. Lymphadenopathy also may be found in suboccipital, supraclavicular, inguinal, and mediastinal areas. Generalized lymphadenopathy occurs in 20 to 30% of symptomatic patients. Between 20 and 40% of patients with lymphadenopathy also have headache, malaise, fatigue, and fever [usually with a temperature of $<40^{\circ}\text{C}$ ($<104^{\circ}\text{F}$)]. A smaller proportion of symptomatic individuals have myalgia, sore throat, abdominal pain, maculopapular rash, meningoencephalitis, and confusion. Rare complications associated with infection in the normal immune host include pneumonia, myocarditis, encephalopathy, pericarditis, and polymyositis. Symptoms associated with acute infection usually resolve within several weeks, although the lymphadenopathy may persist for some months. In a recent epidemic, toxoplasmosis was diagnosed correctly in only 3 of the 25 patients who consulted physicians. If toxoplasmosis is considered in the differential diagnosis, routine laboratory and serologic screening should be performed before node biopsy.

The results of routine laboratory studies are usually unremarkable except for minimal lymphocytosis, an elevated sedimentation rate, and a nominal increase in liver aminotransferases. Evaluation of cerebrospinal fluid (CSF) in cases with evidence of encephalopathy or meningoencephalitis shows an elevation of intracranial pressure, mononuclear pleocytosis (10 to 50 cells/mL), a slight increase in protein concentration, and (occasionally) an increase in the gamma globulin level. [PCR](#) amplification of the *Toxoplasma* DNA target sequence in the CSF may be beneficial. The CSF of chronically infected individuals is normal.

Ocular Infection Infection with *T. gondii* is estimated to cause 35% of all cases of chorioretinitis in the United States and Europe. Most ocular involvement is believed to be due to congenital infection, with a very low incidence following acquired infection. Between 1 and 3% of all patients with AIDS develop debilitating chorioretinitis due to *T. gondii*. A variety of ocular manifestations are documented, including blurred vision, scotoma, photophobia, and eye pain. Macular involvement occurs with loss of central vision, and nystagmus is secondary to poor fixation. Involvement of the extraocular muscles may lead to disorders of convergence and to strabismus. Ophthalmologic examination should be undertaken in newborns with suspected congenital infection. As the inflammation resolves, vision improves, but episodic flare-ups of chorioretinitis,

which progressively destroy retinal tissue and lead to glaucoma, are common.

The ophthalmologic examination reveals yellow-white, cotton-like patches with indistinct margins of hyperemia. As the lesions age, white plaques with distinct borders and black spots within the retinal pigment become more apparent. Lesions usually are located near the posterior pole of the retina; they may be single but are more commonly multiple. Congenital lesions may be unilateral or bilateral and show evidence of massive chorioretinal degeneration with extensive fibrosis. Surrounding these areas of involvement are a normal retina and vasculature. In patients with AIDS, retinal lesions are often large, with diffuse retinal necrosis, and include both free tachyzoites and cysts containing bradyzoites. Toxoplasmic chorioretinitis may be a prodrome to the development of encephalitis.

Infection of the Immunocompromised Person Patients with AIDS and those receiving immunosuppressive therapy for lymphoproliferative disorders are at greatest risk for developing acute toxoplasmosis. This predilection may be due either to reactivation of latent infection or to acquisition of parasites from exogenous sources such as blood or transplanted organs. In individuals with AIDS, more than 95% of cases of *Toxoplasma* encephalitis are believed to be due to recrudescent infection. In most of these cases, encephalitis develops when the CD4⁺ cell count falls below 100/uL. In the immunocompromised individual, the disease may be rapidly fatal if untreated. Thus accurate diagnosis and initiation of appropriate therapy are necessary to prevent fulminant infection.

Toxoplasmosis is a principal opportunistic infection of the [CNS](#) in persons with AIDS. Although geographic origin may be related to frequency of infection, it has no correlation with the severity of disease in the immunocompromised host. Individuals with AIDS who are seropositive for *T. gondii* are at a very high risk for developing encephalitis. In the United States, about one-third of the 15 to 40% of adult patients with AIDS who are latently infected with the parasite develop *Toxoplasma* encephalitis.

The signs and symptoms of acute toxoplasmosis in the immunocompromised patient are principally within the [CNS](#). More than 50% of patients with clinical manifestations have intracerebral involvement. Clinical findings at the time of presentation range from nonfocal to focal dysfunction. These findings include encephalopathy, meningoencephalitis, and mass lesions. Patients may present with altered mental status (75%), fever (10 to 72%), seizures (33%), headaches (56%), and focal neurologic findings (60%), including motor deficits, cranial nerve palsies, movement disorders, dysmetria, visual-field loss, and aphasia. Patients who present with evidence of diffuse cortical dysfunction develop evidence of focal neurologic disease as the infection progresses. This altered condition is due not only to the necrotizing encephalitis caused by direct invasion of the parasite but also to secondary effects, including vasculitis, edema, and hemorrhage. The onset of infection can range from an insidious process over several weeks to an acute confusional state with fulminant focal deficits, including hemiparesis, hemiplegia, visual-field defects, localized headache, and focal seizures.

Although lesions can occur anywhere within the [CNS](#), the areas most involved appear to be the brainstem, basal ganglia, pituitary gland, and corticomedullary junction. Brainstem involvement gives rise to a variety of neurologic dysfunctions, including

cranial nerve palsy, dysmetria, and ataxia. With basal ganglionic infection, patients may develop hydrocephalus, choreiform movements, and choreoathetosis. Because *Toxoplasma* usually causes encephalitis, meningeal involvement is uncommon, and thus CSF findings may be unremarkable or may include a modest increase in cell count and in protein -- but not glucose -- concentration.

Cerebral toxoplasmosis needs to be differentiated from other opportunistic infections or tumors within the CNS of those afflicted with AIDS. The differential diagnosis includes herpes simplex encephalitis, cryptococcal meningitis, progressive multifocal leukoencephalopathy, and primary CNS lymphoma. Involvement of the pituitary gland can give rise to panhypopituitarism and hyponatremia from inappropriate secretion of vasopressin (antidiuretic hormone). AIDS-dementia complex may present as cognitive impairment, attention loss, and altered memory. Brain biopsy in those patients who have been treated for *Toxoplasma* encephalitis but who continue to exhibit neurologic dysfunction often fails to identify organisms.

Autopsies of patients infected with *Toxoplasma* have demonstrated the involvement of multiple organs, including the lungs, gastrointestinal tract, pancreas, skin, eyes, heart, and liver. *Toxoplasma* pneumonia can occur and can be confused with *Pneumocystis carinii* infection. Respiratory involvement usually presents as dyspnea, fever, and a nonproductive cough and may rapidly progress to acute respiratory failure with hemoptysis, metabolic acidosis, hypotension, and (occasionally) disseminated intravascular coagulation. Histopathologic studies demonstrate necrosis and a mixed cellular infiltrate. The presence of organisms is a helpful diagnostic indicator, but organisms can also be found in healthy tissue. Infection of the heart is usually asymptomatic but can be associated with cardiac tamponade or biventricular failure. Infections of the gastrointestinal tract and the liver have been documented.

A presumptive clinical diagnosis of toxoplasmic encephalitis in patients with AIDS is based on clinical presentation, history of exposure as evidenced by positive serology, and radiologic evaluation. When these criteria are used, the predictive value is as high as 80%. More than 97% of patients with AIDS and toxoplasmosis have IgG antibody to the parasite in their sera. IgM serum antibody is usually not demonstrable. Intrathecal antibody to *T. gondii* may be present. Neuroradiologic evaluation should include double-dose contrast computed tomography (CT) of the head. By this test, single and frequently multiple contrast-enhancing lesions (<2 cm) may be identified. Magnetic resonance imaging (MRI) usually demonstrates multiple lesions and provides a more sensitive evaluation of the efficacy of therapy than does CT. Patients with primary CNS lymphoma are four times more likely than patients with *Toxoplasma* encephalitis to have solitary lesions on an MRI scan. A therapeutic trial of anti-*Toxoplasma* medications is frequently used to assess the diagnosis. Treatment of presumptive *Toxoplasma* encephalitis with pyrimethamine/clindamycin results in quantifiable clinical improvement in more than 50% of patients by day 3. By day 7, more than 90% of treated patients show evidence of improvement. In contrast, if patients fail to respond or have lymphoma, clinical signs and symptoms worsen by day 7. Patients in this category require brain biopsy with or without a change in therapy. This procedure can now be performed by a stereotactic CT-guided method that reduces the potential for complications. Brain biopsy for *T. gondii* identifies organisms in 50 to 75% of cases. Some studies indicate that PCR amplification of target genes significantly increases the

sensitivity of detection of parasites.

Congenital Toxoplasmosis Between 400 and 4000 infants born each year in the United States are affected by congenital toxoplasmosis. Infection of the placenta leads to hematogenous infection of the fetus. As has already been stated, the proportion of fetuses that become infected increases but the clinical severity of the infection declines as gestation proceeds. Persistence of the parasite can ultimately result in reactivation and further damage decades later. Factors associated with relatively severe disabilities include delayed diagnosis and initiation of therapy, neonatal hypoxia and hypoglycemia, profound visual impairment, uncorrected hydrocephalus, and increased intracranial pressure. If treated appropriately, upwards of 70% of children have normal developmental, neurologic, and ophthalmologic findings at follow-up evaluations. Treatment for 1 year with pyrimethamine and sulfonamide is tolerated with minimal toxicity (see below).

DIAGNOSIS

Tissue and Body Fluids The diagnosis of acute toxoplasmosis can be made by isolation of the parasite from blood or other body fluids after subinoculation of the sample into the peritoneal cavity of mice. Mice should be tested for organisms in the peritoneal fluid 6 to 10 days after inoculation. If no parasites are found in the mouse's peritoneal fluid, its anti-*Toxoplasma* serum titer can be evaluated 4 to 6 weeks after inoculation. Isolation of *T. gondii* from the patient's body fluids reflects acute infection, whereas isolation from biopsied tissue is an indication only of the presence of tissue cysts and should not be misinterpreted as acute toxoplasmosis. Persistent parasitemia in patients with latent, asymptomatic infection is rare. Histologic examination of lymph nodes may suggest the characteristic changes described above. Demonstration of tachyzoites in lymph nodes establishes the diagnosis of acute toxoplasmosis. Like subinoculation into mice, histologic demonstration of cysts containing bradyzoites confirms prior infection with *T. gondii* but is nondiagnostic for acute infection.

Serology The procedures just described have great diagnostic value but are limited by difficulties encountered either in the growth of parasites in vivo or in the identification of tachyzoites by histochemical methods. Serologic testing has become the routine method of diagnosis. A wide range of serologic tests that can be used to measure antibody to *T. gondii* are available commercially.

Diagnosis of acute infection with *T. gondii* can be established by detection of the simultaneous presence of IgG and IgM antibody to *Toxoplasma* in serum. The presence of circulating IgA favors the diagnosis of an acute infection. The Sabin-Feldman dye test, the indirect fluorescent antibody test, and the enzyme-linked immunosorbent assay (ELISA) all satisfactorily measure circulating IgG antibody to *Toxoplasma*. Positive IgG titers (>1:10) can be detected as early as 2 to 3 weeks after infection. These titers usually peak at 6 to 8 weeks and decline slowly to a new baseline level that persists for life. It is necessary to measure the serum IgM titer in concert with the IgG titer to better establish the time of infection. The methods currently available for this determination are the double-sandwich IgM-ELISA and the IgM-immunosorbent assay (IgM-ISAGA). Both of these assays are specific and sensitive, and their use precludes the false-positive results associated with rheumatoid factor and antinuclear antibody. The

double-sandwich IgA-ELISA is more sensitive than the IgM-ELISA for detecting congenital infection in the fetus and newborn.

The Immunocompetent Adult or Child For the patient who presents with lymphadenopathy only, a positive IgM titer is an indication of acute infection -- and an indication for therapy, if that is clinically warranted (see "Treatment" below). The serum IgM titer should be determined again in 3 weeks. An elevation in the IgG titer without an increase in the IgM titer suggests that infection is present but that it is not acute. If there is a borderline increase in either IgG or IgM, the titers should be assessed again in 3 to 4 weeks.

Ocular Toxoplasmosis Because of the congenital nature of ocular toxoplasmosis, the serum antibody titer may not correlate with the presence of active lesions in the fundus. In general, a positive IgG titer (measured in undiluted serum if necessary) in conjunction with typical lesions establishes the diagnosis. If lesions are atypical and the titer is in the low-positive range, the diagnosis is presumptive. The parasitic antigen-specific polyclonal IgG assay as well as the parasitic antigen-specific [PCR](#) may facilitate the diagnosis.

The Immunocompromised Host As discussed above, in patients with AIDS, the presence of IgG and radiologic findings consistent with toxoplasmosis are grounds for a presumptive diagnosis. Attempts to evaluate rising IgG titers or to determine whether IgM is present are not productive. Serologic evidence of infection virtually always precedes the development of *Toxoplasma* encephalitis. It is therefore important to determine the *Toxoplasma* antibody status of all patients infected with HIV. Antibody titers may range from negative to 1:1024 in patients with AIDS and *Toxoplasma* encephalitis. Fewer than 3% of patients have no demonstrable antibody to *Toxoplasma* at the time of diagnosis. Determination of the intrathecal antibody titer may be useful in identifying prior infection. [PCR](#) amplification of genetic material of the parasite found in the [CSF](#) may prove diagnostically beneficial in the future.

Patients with toxoplasmic encephalitis have focal or multifocal abnormalities demonstrable by [CT](#) or [MRI](#). These findings are not pathognomonic of *Toxoplasma* infection since 40% of [CNS](#) lymphomas are multifocal and 50% are ring-enhancing. Lesions on MRI scan are multiple and are located in both hemispheres, with the basal ganglia and corticomedullary junction most commonly involved. For both MRI and CT scans, the rate of false-negative results is approximately 10%. The finding of a single lesion on an MRI scan increases the suspicion of primary lymphoma and strengthens the argument for the performance of a brain biopsy.

Now used in some centers, SPECT (single-photon emission CT) has been touted as a definitive means of detecting or ruling out *Toxoplasma* infection when a CNS lesion is suspected. In the future, SPECT may well be widely used for this purpose.

As in other conditions, the radiologic response may lag behind the clinical response. Resolution of lesions may take from 3 weeks to 6 months. Some patients show clinical improvement despite worsening radiographic findings.

A presumptive diagnosis of *Toxoplasma* encephalitis should prompt the immediate

initiation of therapy. After 3 weeks, repeat radiologic studies should detect improvement. If glucocorticoids have been administered, radiologic studies should be repeated at the time of discontinuation to determine whether an exacerbation of disease has occurred. If the patient's clinical condition becomes worse, performance of a biopsy must be strongly considered.

Congenital Infection The issue of concern when a pregnant woman has evidence of recent *T. gondii* infection is obviously whether the fetus is infected. [PCR](#) of the amniotic fluid to detect the B1 gene of the parasite has replaced fetal blood sampling. Serologic diagnosis is based on the persistence of IgG antibody or a positive IgM titer after the first week of life (a time frame that excludes placental leak). The IgG determination should be repeated every 2 months. An increase in IgM beyond the first week of life is indicative of acute infection. However, up to 25% of infected newborns may be seronegative and have normal routine physical examinations. Thus assessment of the eye and the brain, with ophthalmologic testing, [CSF](#) evaluation, and radiologic studies, is important in establishing the diagnosis.

TREATMENT

Current therapeutic protocols are directed at folate metabolism, protein synthesis, or nucleic acid synthesis of the parasite. Pyrimethamine and trimethoprim inhibit the enzyme dihydrofolate reductase. Inhibitors of protein synthesis, including clindamycin, chlortetracycline, and azithromycin, affect growth of the parasite. Inhibitors of purine synthesis, such as arprinocid, may prove to be important. Atovaquone, which blocks pyrimidine salvage, has demonstrated activity against both *T. gondii* and *P. carinii*.

Immunologically competent adults and older children who have only lymphadenopathy do not require specific therapy unless they have persistent and severe symptoms. Patients with ocular toxoplasmosis should be treated for 1 month with pyrimethamine plus either sulfadiazine or clindamycin. Prenatal antibiotic therapy can reduce the number of infants severely affected by *Toxoplasma* infection.

Congenital Infection Congenitally infected neonates are treated with daily oral pyrimethamine (0.5 to 1 mg/kg) and sulfadiazine (100 mg/kg) for 1 year. In addition, therapy with spiramycin (100 mg/kg per day) plus prednisone (1 mg/kg per day) has been shown to be efficacious for congenital infection.

Infection in Immunocompromised Patients Patients with AIDS should be treated for acute toxoplasmosis; in the immunocompromised patient, toxoplasmosis is rapidly fatal if untreated. The mainstay of treatment for *Toxoplasma* encephalitis in immunocompromised patients is a combination regimen. Administered together for 4 to 6 weeks or until radiologic improvement is documented, pyrimethamine (a 200-mg loading dose followed by 50 to 75 mg/d) and sulfadiazine (4 to 6 g/d in four divided doses) block folic acid metabolism and reduce the parasite burden. Leucovorin (calcium folinate, 10 to 15 mg/d) is given as an adjunct to prevent the bone marrow toxicity associated with pyrimethamine. Both pyrimethamine and sulfadiazine cross the blood-brain barrier. A prominent consequence of dual therapy is the high incidence of associated toxicity (40%). Rash may develop during the first 3 weeks in up to 20% of patients but does not preclude the use of this combination. Other complications include

hematologic effects, crystalluria, hematuria, radiolucent renal stones, and nephrotoxicity. During therapy, serum levels of these drugs may be erratic, but such fluctuations have not been correlated with these complications.

Pyrimethamine and sulfadiazine are active only against the tachyzoite stage of the parasite. Thus, after immunocompromised patients complete the initial 4- to 6-week course, they must receive lifelong suppressive therapy with pyrimethamine (25 to 50 mg/d) and sulfadiazine (2 to 4 g/d). If sulfadiazine cannot be tolerated, a combination of pyrimethamine (75 mg/d) plus clindamycin (450 mg tid) can be used. It is possible that pyrimethamine (50 to 75 mg/d) is sufficient for chronic suppressive therapy.

Alternative Regimens Alternative therapies have been established because of the toxicity associated with the long-term antimicrobial therapy necessary for many individuals infected with *T. gondii*. Dapsone (diaminodiphenyl sulfone), with its longer serum half-life and decreased toxicity, is an effective alternative to sulfadiazine. Spiramycin, which has been used in Europe to treat pregnant women, reduces transplacental transmission. However, spiramycin has been ineffective as primary prophylaxis in patients with AIDS. Clindamycin is well absorbed from the gastrointestinal tract, and serum levels peak 1 to 2 h after administration. The combination of oral pyrimethamine (25 to 75 mg/d) plus intravenous clindamycin (1200 to 4800 mg/d) is effective for patients with AIDS who have *Toxoplasma* encephalitis. Toxic effects of clindamycin include nausea, vomiting, neutropenia, rash, and pseudomembranous colitis. Other macrolides that have been evaluated include roxithromycin, clarithromycin, and azithromycin. Evidence suggests that the macrolides are not beneficial by themselves, but a combination of pyrimethamine and clarithromycin appears to be effective. Atovaquone (750 mg tid or qid) is an optional agent for the treatment of individuals who are intolerant of other agents. Glucocorticoids can be used to treat intracerebral edema, but their benefit has not yet been established. It is difficult to assess the benefit of glucocorticoids when they are administered in conjunction with anti-*Toxoplasma* medication. Anticonvulsants are sometimes necessary for the treatment of seizures, but attention should be given to the potential interaction between sulfadiazine and phenytoin. A regimen of trimethoprim-sulfamethoxazole or dapsone plus pyrimethamine with leucovorin may prevent the development of *Toxoplasma* encephalitis in individuals infected with HIV who are seropositive for *T. gondii* after their CD4+ T lymphocyte count falls to 100/uL.

PREVENTION

The chances of primary infection with *Toxoplasma* can be reduced by not eating undercooked meat and by avoiding oocyst-contaminated material (i.e., a cat's litter box). Meat should be heated to 60°C or frozen to kill cysts. Hands should be washed thoroughly after work in the garden, and all fruits and vegetables should be washed. Blood intended for transfusion into *Toxoplasma*-seronegative immunocompromised individuals should be screened for antibody to *T. gondii*. Although such serologic screening is not routinely performed, seronegative women should be screened for evidence of infection several times during pregnancy if they are exposed to environmental conditions that put them at risk for infection with *T. gondii*. HIV-positive individuals should adhere closely to these preventive measures.

(Bibliography omitted in Palm version)

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218. PROTOZOAL INTESTINAL INFECTIONS AND TRICHOMONIASIS - Peter F. Weller

PROTOZOAL INFECTIONS

GIARDIASIS

Giardia lamblia is a cosmopolitan protozoal parasite that inhabits the small intestines of humans and other mammals. Giardiasis is one of the most common parasitic diseases worldwide and causes both endemic and epidemic intestinal disease and diarrhea.

Life Cycle and Epidemiology Infection follows the ingestion of the environmentally hardy cysts, which excyst in the small intestine, releasing trophozoites that multiply by binary fission, occasionally to enormous numbers. *Giardia* remains a pathogen of the proximal small bowel and does not disseminate hematogenously. Trophozoites remain free in the lumen or attach to the mucosal epithelium by means of a ventral sucking disk. As a trophozoite encounters altered conditions, it forms a morphologically distinct cyst, which is the stage of the parasite usually found in the feces. Trophozoites may be present and even predominate in loose or watery stools, but it is the resistant cyst that survives outside the body and is responsible for transmission. Cysts do not tolerate heating, desiccation, or continued exposure to feces but do remain viable for months in cold fresh water. The number of cysts excreted varies widely but can approach 10⁷ per gram of stool.

Giardia infections are common in both developed and developing countries. Ingestion of as few as 10 cysts is sufficient to cause infection in humans. Because cysts are infectious when excreted or shortly thereafter, person-to-person transmission occurs where fecal hygiene is poor. Giardiasis, as a symptomatic or an asymptomatic infection, is especially prevalent in day-care centers; person-to-person spread also takes place in other institutional settings with poor fecal hygiene and during homosexual contact. If food is contaminated with *Giardia* cysts after cooking or preparation, food-borne transmission can occur. Waterborne transmission accounts for episodic infections (e.g., in campers and other travelers) and for massive epidemics in metropolitan areas. Surface water, ranging from mountain streams to large municipal reservoirs, can become contaminated with fecally derived *Giardia* cysts; outmoded water systems are subject to cross-contamination from leaking sewer lines. The efficacy of water as a means of transmission is enhanced by the small infectious inoculum of *Giardia*, the prolonged survival of cysts in cold water, and the resistance of cysts to killing by routine chlorination methods that are adequate for controlling bacteria. Viable cysts can be eradicated from water by either boiling or filtration. In the United States, *Giardia* is a common agent identified in waterborne epidemics of gastroenteritis; it is also common in developing countries.

The importance of animal reservoirs as sources of infection for humans is unclear. *Giardia* parasites morphologically similar to those in humans are found in a large number of mammals, including beavers from reservoirs implicated in epidemics, dogs, cats, and ruminants. Although the high degree of isolate heterogeneity noted in humans is consistent with infections originating from different animal sources, animals have not been directly established as sources of human infection.

Giardiasis, like cryptosporidiosis, creates a significant economic burden because of the costs incurred in the installation of water filtration systems required to prevent waterborne epidemics, in the management of epidemics that involve large communities, and in the evaluation and treatment of endemic infections.

Pathophysiology The reasons that some, but not all, infected patients develop clinical manifestations and the mechanisms by which *Giardia* causes alterations in small-bowel function are largely unknown. Although trophozoites adhere to the epithelium, they do not cause invasive or locally destructive alterations. The lactose intolerance and significant malabsorption that develop in a minority of infected adults and children are clinical signs of the loss of brush border enzyme activities. In most infections the morphology of the bowel is unaltered, but in a few cases -- usually in chronically infected, symptomatic patients -- the histopathologic findings (including flattened villi) and the clinical manifestations resemble those of tropical sprue and gluten-sensitive enteropathy. The pathogenesis of diarrhea in giardiasis is not known.

The natural history of *Giardia* infection varies markedly. Infections may be aborted, transient, recurrent, or chronic. Parasite as well as host factors may be important in determining the course of infection and disease. Both cellular and humoral responses develop in human infections, but their precise roles in the control of infection and/or disease are unknown. Because patients with hypogammaglobulinemia commonly suffer from prolonged, severe infections that are poorly responsive to treatment, humoral immune responses appear to be important. The greater susceptibility of the young than of the old and of newly exposed persons than of chronically exposed populations also suggests that at least partial protective immunity may develop. Although no strains of the parasite that are clearly nonpathogenic have been identified, *Giardia* isolates vary biochemically and biologically. The marked biochemical differences among some isolates may help account for the different courses of infection in experimentally infected humans and animals. The surface of trophozoites is covered by a family of related cysteine-rich proteins that undergo surface antigenic variation and may contribute to prolonged and/or repeated infections.

Clinical Manifestations Disease manifestations of giardiasis range from asymptomatic carriage to fulminant diarrhea and malabsorption. Most infected persons are asymptomatic, but in epidemics the proportion of symptomatic cases may be higher. Symptoms may develop suddenly or gradually. In persons with acute giardiasis, symptoms develop after an incubation period that lasts at least 5 to 6 days and usually 1 to 3 weeks. Prominent early symptoms include diarrhea, abdominal pain, bloating, belching, flatus, nausea, and vomiting. Although diarrhea is common, upper intestinal manifestations such as nausea, vomiting, bloating, and abdominal pain may predominate. The duration of acute giardiasis is usually in excess of 1 week, although diarrhea often subsides. Individuals with chronic giardiasis may present with or without having experienced an antecedent acute symptomatic episode. Diarrhea is not necessarily prominent, but increased flatus, loose stools, sulfurous burping, and (in some instances) weight loss occur. Symptoms may be continual or episodic and can persist for years. Some persons who have relatively mild symptoms for long periods recognize the extent of their discomfort only in retrospect. Fever, the presence of blood and/or mucus in the stools, and other signs and symptoms of colitis are uncommon and

suggest a different diagnosis or a concomitant illness. Symptoms tend to be intermittent yet recurring and gradually debilitating, in contrast with the acute disabling symptoms associated with many enteric bacterial infections. Because of the less severe illness and the propensity for chronic infections, patients may seek medical advice late in the course of the illness; however, disease can be severe, resulting in malabsorption, weight loss, growth retardation, dehydration, and (in rare cases) death. A number of extraintestinal manifestations have been described, such as urticaria, anterior uveitis, and arthritis; whether these are caused by giardiasis or concomitant processes is unclear.

Giardiasis can be life-threatening in patients with hypogammaglobulinemia and is typically difficult to treat and eradicate. *Giardia* infections can complicate other preexisting intestinal diseases, such as cystic fibrosis. Although *Giardia* can cause enteric illness in patients with AIDS, neither the course of infection nor the response to treatment differs for patients with and without AIDS.

Diagnosis Giardiasis is diagnosed by the detection of parasite antigen in the feces or by the identification of cysts in the feces or of trophozoites in the feces or small intestines. Cysts are oval, measure 8 to 12 μm $\times$ 7 to 10 μm , and characteristically contain four nuclei. Trophozoites are pear-shaped, dorsally convex, flattened parasites with two nuclei and four pairs of flagella. The diagnosis is sometimes difficult to establish. Direct examination of fresh or properly preserved stools as well as concentration methods should be used. Because cyst excretion is variable and may be undetectable at times, repeated examination of stool, sampling of duodenal fluid, and biopsy of the small intestine may be required to detect the parasite. Tests for parasitic antigen in stool are at least as sensitive and specific as good microscopic examinations and are easier to perform. All of these methods occasionally yield false-negative results.

TREATMENT

Cure rates with metronidazole (250 mg tid for 5 days) are usually >80%; those with furazolidone (100 mg qid for 7 to 10 days) are somewhat lower. The latter agent is frequently used to treat children because it is available as a palatable elixir that is not bitter. Quinacrine, the first effective drug for the treatment of giardiasis, is available from a limited number of pharmacies. Albendazole (400 mg/d for 5 days) may be effective.

Patients in whom initial treatment fails can be re-treated with a longer course. Almost all patients respond to therapy and are cured, although some with chronic giardiasis experience delayed resolution of symptoms after eradication of *Giardia*. Those who remain infected after repeated treatments should be evaluated for reinfection through family members, close personal contacts, and environmental sources as well as for hypogammaglobulinemia. In cases refractory to multiple treatment courses, prolonged therapy with metronidazole (750 mg tid for 21 days) has been successful. Tinidazole, not available in the United States, is considered more effective than metronidazole or quinacrine. When children attending day-care centers infect an entire family, treatment of all infected family members, including asymptomatic carriers, may be required to prevent reinfection. Paromomycin, an oral aminoglycoside that is not well absorbed, can be given to symptomatic pregnant women, although the experience accumulated thus far is not a sufficient basis on which to judge how often this agent either eradicates

infection or ameliorates symptoms.

Prevention Although *Giardia* is extremely infectious, disease can be prevented by the exclusive consumption of noncontaminated food and water. Cooking food adequately and boiling or filtering potentially contaminated water prevent infection.

CRYPTOSPORIDIOSIS

The coccidian parasite *Cryptosporidium* is now known to cause diarrheal disease in immunocompetent human hosts and to be especially common among persons with AIDS or other forms of immunodeficiency.

Life Cycle and Epidemiology Cryptosporidiosis is acquired by the consumption of oocysts (50% infectious dose: ~132 oocysts in nonimmune individuals), which excyst to liberate sporozoites that in turn enter and infect intestinal epithelial cells. The parasite's further development involves both asexual and sexual cycles, which produce forms capable of infecting other epithelial cells and of generating oocysts that are passed in the feces. *Cryptosporidium* spp. infect a number of animals and can spread from infected animals to humans. Since oocysts are immediately infectious when passed in feces, person-to-person transmission takes place in day-care centers and among household contacts and medical providers. Waterborne transmission accounts for infections in travelers and for common-source epidemics. Oocysts are quite hardy and resist killing by routine chlorination. Both drinking water and recreational water (e.g., pools, waterslides) have been increasingly recognized as sources of infection.

Pathophysiology Although intestinal epithelial cells harbor the parasite in an intracellular vacuole, the means by which secretory diarrhea is elicited remain uncertain. No characteristic pathologic changes are found by biopsy. The distribution of infection can be spotty within the principal site of infection, the small bowel. Cryptosporidia are found in some patients in the pharynx, stomach, and large bowel and at times in the respiratory tract. Especially in patients with AIDS, involvement of the biliary tract can cause papillary stenosis, sclerosing cholangitis, or cholecystitis.

Clinical Manifestations Asymptomatic infections can occur in both immunocompetent and immunocompromised hosts. In immunocompetent persons, symptoms develop after an incubation period of about a week and consist principally of watery nonbloody diarrhea, at times in conjunction with abdominal pain, nausea, anorexia, fever, and/or weight loss. In these hosts, the illness usually subsides after 1 to 2 weeks, whereas in immunocompromised hosts, especially those with AIDS, diarrhea can be chronic, persistent, and remarkably profuse, causing clinically significant fluid and electrolyte depletion. Stool volumes may range from 1 to 25 L/d. Weight loss, wasting, and abdominal pain may be severe. Biliary tract involvement can manifest as midepigastria or right upper quadrant pain.

Diagnosis Evaluation usually starts with fecal examination for small oocysts, which are 4 to 5 μ m in diameter and are smaller than the fecal stages of most other parasites. Detection is enhanced by evaluation of stools (obtained on multiple days) by several techniques, including modified acid-fast and direct immunofluorescent stains and enzyme immunoassays. Cryptosporidia also can be identified by light and electron

microscopy at the apical surfaces of intestinal epithelium from biopsy specimens of the small bowel and, less frequently, the large bowel.

TREATMENT

To date, no chemotherapeutic agents effective against *Cryptosporidium* have been identified, although paromomycin (500 to 750 mg qid) may be partially effective for some patients infected with HIV. Improvement in immune status with antiretroviral therapy can lead to amelioration of cryptosporidiosis. Otherwise, treatment includes supportive care with replacement of fluids and electrolytes and administration of antidiarrheal agents. Biliary tract obstruction may require papillotomy or T-tube placement. Prevention requires minimizing exposure to infectious oocysts in human or animal feces. Use of submicron water filters may minimize acquisition of infection from drinking water.

ISOSPORIASIS

The coccidian parasite *Isospora belli* causes human intestinal disease. Infection is acquired by the consumption of oocysts, after which the parasite invades intestinal epithelial cells and undergoes both sexual and asexual cycles of development. Oocysts excreted in stool are not immediately infectious but must undergo further maturation. Although *I. belli* infects many animals, little is known about the epidemiology or prevalence of this parasite in humans. It appears to be most common in tropical and subtropical countries. Acute infections can begin abruptly with fever, abdominal pain, and watery nonbloody diarrhea and can last for weeks or months. In patients who have AIDS or are immunocompromised for other reasons, infections often are not self-limited but rather resemble cryptosporidiosis, with chronic, profuse watery diarrhea. Eosinophilia, which is not found in other enteric protozoan infections, may be detectable. The diagnosis is usually made by detection of the large (~25-um) oocysts in stool by modified acid-fast staining. Oocyst excretion may be low-level and intermittent; if repeated stool examinations are unrevealing, sampling of duodenal contents by aspiration or small-bowel biopsy (often with electron-microscopic examination) may be necessary.

In contrast to cryptosporidiosis, isosporiasis responds to chemotherapy. Trimethoprim-sulfamethoxazole (160/800 mg qid for 10 days and then bid for 3 weeks) has been effective; for patients intolerant of sulfonamides, pyrimethamine (50 to 75 mg/d) can be used. Relapses can occur in persons with AIDS and necessitate maintenance therapy with trimethoprim-sulfamethoxazole (160/800 mg three times a week) or combined sulfadoxine (500 mg) and pyrimethamine (25 mg) once weekly.

CYCLOSPORIASIS

Coccidian parasites of the genus *Cyclospora* have been identified as the causative organisms in diarrheal illness formerly ascribed to blue-green algal or *Cyanobacteria*-like forms. This parasite is globally distributed: illness due to *Cyclospora cayetanensis* has been reported in the United States, Asia, Africa, Latin America, and Europe. The epidemiology of this parasite has not yet been fully defined, but waterborne transmission and especially transmission in imported raspberries have been recognized.

The full spectrum of illness attributable to *Cyclospora* has not been delineated. Some patients may harbor the infection without symptoms, but many with cyclosporiasis have diarrhea, flulike symptoms, and flatulence and burping. The illness can be self-limited, can wax and wane, or (in many cases) can involve prolonged diarrhea, anorexia, and upper gastrointestinal symptoms, with sustained fatigue and weight loss in some instances. Diarrheal illness may persist for longer than a month. *Cyclospora* can cause enteric illness in patients infected with HIV, albeit at an unknown frequency.

The parasite is detectable in epithelial cells of small-bowel biopsy samples and elicits secretory diarrhea by an unknown means. The absence of fecal blood and leukocytes indicates that disease due to *Cyclospora* is not caused by destruction of the small-bowel mucosa. The diagnosis can be made by detection of spherical 8- to 10-um oocysts in the stool, although routine stool O and P examinations are not sufficient. Specific fecal examinations must be requested to detect the oocysts, which are variably acid-fast and are fluorescent when viewed with ultraviolet light microscopy. Cyclosporiasis should be considered in the differential diagnosis of prolonged diarrhea, with or without a history of travel by the patient to other countries.

Cyclosporiasis is effectively treated with trimethoprim-sulfamethoxazole (160/800 mg bid for 7 days). Patients infected with HIV, however, may experience relapses after such treatment and thus may require longer-term suppressive maintenance therapy.

MICROSPORIDIOSIS

Microsporidia are obligate intracellular spore-forming protozoa that infect many animals and cause disease in humans, especially as opportunistic pathogens in AIDS. Microsporidia are members of a distinct phylum, Microspora, which contains dozens of genera and hundreds of species. The various microsporidia are differentiated by their developmental life cycles, by ultrastructural features, and by molecular taxonomy based on ribosomal RNA. The complex life cycles of the organisms result in the production of infectious spores. Currently, six genera of microsporidia -- *Encephalitozoon*, *Pleistophora*, *Nosema*, *Vittaforma*, *Trachipleistophora*, and *Enterocytozoon* -- are recognized as causes of human disease; a seventh genus -- *Microsporidium*, which includes organisms of uncertain taxonomic status -- also causes disease in humans. Though some microsporidia are probably prevalent causes of self-limited or asymptomatic infections in immunocompetent patients, little is known of how microsporidiosis is acquired.

Microsporidiosis is most common among patients with AIDS, less common among patients with other types of immunocompromise, and rare among immunocompetent hosts. In patients with AIDS, intestinal infections with *Enterocytozoon bienersi* and *Encephalitozoon* (formerly *Septata*) *intestinalis* are increasingly recognized to contribute to chronic diarrhea and wasting; these infections are found in 10 to 40% of patients with chronic diarrhea. Both organisms have been found in the biliary tracts of patients with cholecystitis. *E. intestinalis* may also disseminate to cause fever, diarrhea, sinusitis, cholangitis, and bronchiolitis. In patients with AIDS, *E. hellem* has caused superficial keratoconjunctivitis as well as sinusitis, respiratory tract disease, and disseminated infection. Myositis due to *Pleistophora* has been documented. *Nosema*, *Vittaforma*, and *Microsporidium* have caused stromal keratitis associated with trauma in

immunocompetent patients.

Microsporidia are small gram-positive organisms with mature spores measuring 0.5 to 2 μm $\times$ 1 to 4 μm . Diagnosis of microsporidial infections in tissue often requires electron microscopy, although intracellular spores can be visualized by light microscopy with hematoxylin and eosin, Giemsa, or tissue Gram's stains. For the diagnosis of intestinal microsporidiosis, modified trichrome or chromotrope 2R-based staining and Uvitex 2B or calcofluor fluorescent staining reveal spores in smears of feces or duodenal aspirates. Definitive therapies for microsporidial infections remain to be established. For superficial keratoconjunctivitis due to *E. hellem*, topical therapy with fumagillin suspension has shown promise ([Chap. 211](#)). For enteric infections with *E. bienewisi* and *E. intestinalis* in HIV-infected patients, therapy with albendazole may be efficacious ([Chap. 211](#)).

OTHER INTESTINAL PROTOZOA

Balantidiasis *Balantidium coli* is a large ciliated protozoal parasite that can produce a spectrum of large-intestinal disease analogous to amebiasis. The parasite is widely distributed in the world. Since it infects pigs, cases in humans are more common where pigs are raised; in Muslim countries, rodents may be important carriers. Infective cysts can be transmitted from person to person and through water, but many cases are due to the ingestion of cysts derived from porcine feces in association with slaughtering, with use of pig feces for fertilizer, or with contamination of water supplies by pig feces.

Ingested cysts liberate trophozoites, which reside and replicate in the large bowel. Many patients remain asymptomatic, but some have persisting intermittent diarrhea, and a few develop more fulminant dysentery. In symptomatic individuals, the pathology in the bowel -- both gross and microscopic -- is similar to that seen in amebiasis, with varying degrees of mucosal invasion, focal necrosis, and ulceration. Balantidiasis, unlike amebiasis, does not spread hematogenously to other organs. The diagnosis is usually made by detection of the trophozoite stage in stool or sampled colonic tissue. Tetracycline (500 mg qid for 10 days) is an effective therapeutic agent.

Blastocystis hominis Infection *B. hominis*, long considered a nonpathogenic yeast, is believed by some to be a protozoan capable of causing intestinal disease, although its taxonomy and inherent pathogenicity remain uncertain. Some patients who pass *B. hominis* in their stools are asymptomatic, whereas others have diarrhea and associated intestinal symptoms. Diligent evaluation reveals other potential bacterial, viral, or protozoal causes of diarrhea in some but not all patients with symptoms. Because the pathogenicity of *B. hominis* is uncertain and because therapy for *Blastocystis* infection is neither specific nor uniformly effective, patients with prominent intestinal symptoms should be fully evaluated for other infectious causes of diarrhea. If diarrheal symptoms associated with *Blastocystis* are prominent, either metronidazole (750 mg tid for 10 days) or iodoquinol (650 mg tid for 20 days) can be used.

Dientamoeba fragilis Infection *D. fragilis* is unique among intestinal protozoa in that it has a trophozoite stage but not a cyst stage. How trophozoites survive to transmit infection is not known, but the unusually high prevalence of *D. fragilis* infection among persons with pinworm infection raises the possibility that eggs or larvae of *Enterobius* facilitate the transmission of *D. fragilis*. When symptoms develop in patients with *D.*

fragilis infection, they are generally mild and include intermittent diarrhea, abdominal pain, and anorexia. The diagnosis is made by the detection of trophozoites in stool; the lability of these forms accounts for the greater yield when fecal samples are preserved immediately after collection. Since fecal excretion rates vary, examination of several samples obtained on alternate days increases the rate of detection. Iodoquinol (650 mg tid for 20 days), paromomycin (25 to 30 mg/kg per day in three doses for 7 days), or tetracycline (500 mg qid for 10 days) is appropriate for treatment.

Sarcosporidiosis Various *Sarcocystis* spp. of coccidian parasites are widely distributed agents of infection in numerous animals. These parasites have an obligatory cycle of development involving two hosts. Sexual reproduction occurs in the intestine, with sporocysts passed in the feces; asexual multiplication leads to the development of muscle cysts. Humans can develop intestinal infections -- albeit apparently infrequently -- by ingesting muscle-stage cysts in undercooked pork or beef. While the full spectrum of the intestinal disease is not defined, a diarrheal illness can ensue, and sporocysts are found in the stool. Alternatively, ingestion of fecally derived sporocysts can lead to the development of cysts in striated or cardiac muscle. Some patients experience muscle pain and swelling, but the frequency and nature of symptoms elicited by muscle involvement are not clear, and these cysts, measuring 100 to 325 μ m, also have been found incidentally in muscle specimens. Muscle-stage infections are not followed by further spread in humans. No specific therapy exists for either intestinal or muscle-stage *Sarcocystis* infections in humans.

TRICHOMONIASIS

Various species of trichomonads can be found in the mouth (in association with periodontitis) and occasionally in the gastrointestinal tract. *Trichomonas vaginalis* -- one of the most prevalent protozoal parasites in the United States -- is a pathogen of the genitourinary tract and a major cause of symptomatic vaginitis.

Life Cycle and Epidemiology *T. vaginalis* is a pear-shaped, actively motile organism that measures about 10 by 7 μ m, replicates by binary fission, and inhabits the lower genital tract of females and the urethra and prostate of males. In the United States, it accounts for about 3 million infections per year in women. While the organism can survive for a few hours in moist environments and could be acquired by direct contact, person-to-person venereal transmission accounts for virtually all cases of trichomoniasis. Its prevalence is greatest among persons with multiple sexual partners and among those with other sexually transmitted diseases.

Clinical Manifestations Most men infected with *T. vaginalis* are asymptomatic, although some develop urethritis and a few have epididymitis or prostatitis. In contrast, infection in women, which has an incubation period of 5 to 28 days, is usually symptomatic and manifests with malodorous vaginal discharge (often yellow), vulvar erythema and itching, dysuria or urinary frequency (in 30 to 50% of patients), and dyspareunia. These manifestations, however, do not clearly distinguish trichomoniasis from other types of infectious vaginitis.

Diagnosis Detection of motile trichomonads by microscopy of wet mounts of vaginal or prostatic secretions has been the conventional means of diagnosis. Although such

microscopy provides an immediate diagnosis, its sensitivity for the detection of *T. vaginalis* is only ~50 to 60% in routine evaluations of vaginal secretions. Direct immunofluorescent antibody staining is more sensitive (70 to 90%) than wet-mount examinations. *T. vaginalis* can be recovered from the urethra of both males and females and is detectable in males after prostatic massage. Culture of the parasite is the most sensitive means of detection; however, the facilities for culture are not generally available, and detection of the organism takes 3 to 7 days.

TREATMENT

Metronidazole is the mainstay of treatment and may be given either as a single 2-g dose or as 250 mg tid for 7 days. All sexual partners must be treated concurrently to prevent reinfection, especially from asymptomatic males. Alternatives to metronidazole for treatment during pregnancy are not readily available, although use of 100-mg clotrimazole vaginal suppositories nightly for 2 weeks may cure some infections in pregnant women. Reinfection often accounts for apparent treatment failures, but strains of *T. vaginalis* exhibiting high-level resistance to metronidazole have been encountered. Treatment of these resistant infections with higher oral doses, parenteral doses, or concurrent oral and vaginal doses of metronidazole has been successful.

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SECTION 18 -HELMINTHIC INFECTIONS

219. *TRICHINELLA* AND OTHER TISSUE NEMATODES - Peter F. Weller, Leo X. Liu

Nematodes are elongated, symmetric roundworms. Parasitic nematodes of medical significance may be broadly classified as intestinal or tissue nematodes, but such a classification system is imprecise. This chapter covers trichinellosis, visceral and ocular larva migrans, cutaneous larva migrans, cerebral angiostrongyliasis, and gnathostomiasis. All are zoonotic infections caused by incidental exposure to infectious nematodes. The clinical symptoms of these infections are due largely to invasive larval stages that (except in the case of *Trichinella*) do not reach maturity in humans.

TRICHINELLOSIS

Trichinellosis develops after the ingestion of meat containing cysts of *Trichinella* -- for example, pork or other meat from a carnivore. While most infections are mild and asymptomatic, heavy infections can cause severe enteritis, periorbital edema, myositis, and (infrequently) death.

Life Cycle and Epidemiology Five species of *Trichinella* are now recognized as causes of infection in humans. Two species are distributed worldwide: *T. spiralis*, which is found in a great variety of carnivorous and omnivorous animals, and *T. pseudospiralis*, which is found in mammals and birds. *T. nativa* is present in Arctic regions and infects bears; *T. nelsoni* is found in equatorial Africa, where it is common among felid predators and scavengers such as hyenas and bush pigs; and *T. bitovi* is found in temperate areas of Europe and western Asia among carnivores but not among domestic swine.

After the consumption of trichinous meat by the host, encysted larvae are liberated by digestive acid and pepsin ([Fig. 219-1](#)). The larvae invade the small-bowel mucosa and mature rapidly into adult worms. After about 1 week, female worms release newborn larvae that migrate via the circulation to striated muscle. The larvae of all species except *T. pseudospiralis* then encyst by inducing a radical transformation in the muscle cell architecture. Although host immune responses may help to expel adult worms, they have little effect on muscle-dwelling larvae.

Human trichinellosis is most often caused by the ingestion of infected pork products and thus can occur in almost any location where the meat of domestic or wild swine is eaten. Human trichinellosis also may be acquired from the meat of other animals, including dogs (in parts of Asia and Africa), horses (in Italy and France), and bears and walrus (in northern regions). Although cattle (being herbivores) are not natural hosts of *Trichinella*, beef has been implicated in outbreaks when contaminated or adulterated with trichinous pork. Laws that prohibit the feeding of uncooked garbage to pigs have greatly reduced the transmission of trichinellosis in the United States. About 40 cases of trichinellosis are reported annually in this country, but most mild cases probably remain undiagnosed. Recent U.S. outbreaks have been attributable to undercooked ethnic pork dishes, homemade and commercial sausage, wild boar meat, and walrus meat.

Pathogenesis and Clinical Features Clinical symptoms of trichinellosis arise from the

successive phases of parasite enteric invasion, larval migration, and muscle encystment ([Fig. 219-1](#)). Most light infections (those with <10 larvae per gram of muscle) are asymptomatic, whereas heavy infections (which can involve >50 larvae per gram of muscle) can be life-threatening. Invasion of the gut by large numbers of parasites occasionally provokes diarrhea during the first week after infection. Abdominal pain, constipation, nausea, or vomiting also may be prominent. The prolonged and fulminant diarrhea noted with Arctic trichinellosis probably reflects a response to repeated infection.

Symptoms due to larval migration and muscle invasion begin to appear in the second week after infection. The migrating *Trichinella* larvae provoke a marked local and systemic hypersensitivity reaction, with fever and hypereosinophilia. Periorbital and facial edema is common, as are hemorrhages in the subconjunctivae, retina, and nail beds ("splinter" hemorrhages). A maculopapular rash, headache, cough, dyspnea, or dysphagia sometimes develops. Myocarditis with tachyarrhythmias or heart failure -- and, less commonly, encephalitis or pneumonitis -- may develop and accounts for most deaths of patients with trichinellosis.

Upon onset of larval encystment in muscle 2 to 3 weeks after infection, symptoms of myositis with myalgias, muscle edema, and weakness develop, usually overlapping with the inflammatory reactions to migrating larvae. The most commonly involved muscle groups include the extraocular muscles; the biceps; and the muscles of the jaw, neck, lower back, and diaphragm. Peaking about 3 weeks after infection, symptoms subside only gradually during a prolonged convalescence.

Laboratory Findings and Diagnosis Blood eosinophilia develops in >90% of patients with symptomatic trichinellosis and may peak at a level of >50% between 2 and 4 weeks after infection. Serum levels of IgE and muscle enzymes, including creatine phosphokinase, lactate dehydrogenase, and aspartate aminotransferase, are elevated in most symptomatic patients. Patients should be questioned thoroughly about their consumption of pork or wild-animal meat and about illness in other individuals who ate the same meat. A presumptive clinical diagnosis can be based on fevers, eosinophilia, periorbital edema, and myalgias after a suspect meal. A rise in the titer of parasite-specific antibody, which usually does not occur until after the third week of infection, confirms the diagnosis. Alternatively, a definitive diagnosis requires surgical biopsy of at least 1 g of involved muscle; the yields are highest near tendon insertions. The fresh muscle tissue should be compressed between glass slides and examined microscopically, because larvae may be overlooked by examination of routine histopathologic sections alone.

TREATMENT

Current anthelmintic drugs are ineffective against *Trichinella* larvae in muscle. Fortunately, most lightly infected patients recover uneventfully with bed rest, antipyretics, and analgesics. Glucocorticoids like prednisone (1 mg/kg daily for 5 days) are beneficial for severe myositis and myocarditis. Mebendazole and albendazole, like thiabendazole, appear to be active against enteric stages of the parasite, but their efficacy against encysted larvae has not been conclusively demonstrated.

Prevention Larvae may be killed by cooking pork until it is no longer pink or by freezing it at -15°C for 3 weeks. However, Arctic *T. nativa* larvae in walrus or bear meat are relatively resistant and may remain viable despite freezing.

VISCERAL AND OCULAR LARVA MIGRANS

Visceral larva migrans is a syndrome caused by nematodes that are normally parasitic for nonhuman host species. In humans, the nematode larvae do not typically develop into adult worms but instead migrate through host tissues and elicit eosinophilic inflammation. The most common form of visceral larva migrans is toxocariasis due to larvae of the canine ascarid *Toxocara canis* or, less commonly, the feline ascarid *T. cati*. Rare cases with eosinophilic meningoencephalitis have been caused by the raccoon ascarid *Baylisascaris procyonis*.

Life Cycle and Epidemiology The canine roundworm *T. canis* is distributed among dogs worldwide. Ingestion of infective eggs by dogs is followed by liberation of *Toxocara* larvae, which penetrate the gut wall and migrate intravascularly into the canine liver, muscle, and other tissues, where most remain in a developmentally arrested state. During pregnancy, some larvae resume migration in bitches and infect puppies prenatally (through transplacental transmission) or after birth (through suckling). Thus, in lactating bitches and puppies, larvae return to the intestinal tract and develop into adult worms, which produce eggs that are released in the feces. Humans acquire toxocariasis mainly by eating soil contaminated by puppy feces containing infective *T. canis* eggs. Visceral larva migrans is most common among children who habitually eat dirt, but most toxocaral infections are subclinical. Reported rates of *Toxocara* seropositivity range from 2% in an unselected American population to >20% among kindergarten children in the United States and England.

Pathogenesis and Clinical Features Clinical disease most commonly afflicts preschool children. After humans ingest *Toxocara* eggs, the larvae hatch and penetrate the intestinal mucosa, from which they are carried by the circulation to a wide variety of organs and tissues. The larvae invade the liver, lungs, central nervous system, and other sites, releasing toxic products and provoking intense local eosinophilic granulomatous responses. The degree of clinical illness depends on larval number and tissue distribution, reinfection, and host immune responses. Most light infections are asymptomatic and may be manifest only by blood eosinophilia. Characteristic symptoms of visceral larva migrans include fever, malaise, anorexia and weight loss, cough, wheezing, and rashes. Hepatosplenomegaly is common. These features are often accompanied by extraordinary peripheral eosinophilia, which may approach 90%. Uncommonly, seizures or behavioral disorders develop. The rare deaths in this disease are due to severe neurologic, pneumonic, or myocardial involvement.

Diagnosis In addition to prominent eosinophilia, leukocytosis and hypergammaglobulinemia are usually evident. Transient pulmonary infiltrates are apparent on chest x-rays of about half of patients with symptoms of pneumonitis. The clinical diagnosis can be confirmed by an enzyme-linked immunosorbent assay for toxocaral antibodies. Stool examination, while important in the evaluation of unexplained eosinophilia, is worthless for toxocariasis, since the larvae do not develop into egg-producing adults in humans.

The ocular form of the larva migrans syndrome occurs when *Toxocara* larvae invade the eye. An eosinophilic granulomatous mass, most commonly in the posterior pole of the retina, develops around the entrapped larva. The retinal lesion can mimic retinoblastoma in appearance, and mistaken diagnosis of the latter condition can lead to unnecessary enucleation. The spectrum of eye involvement also includes endophthalmitis, uveitis, and chorioretinitis. Unilateral visual disturbances, strabismus, and eye pain are the most common presenting symptoms. In contrast to visceral larva migrans, ocular toxocariasis usually develops in older children or young adults with no history of pica; these patients seldom have eosinophilia or visceral manifestations.

TREATMENT

The vast majority of *Toxocara* infections are self-limited and resolve without specific therapy. In patients with severe myocardial, central nervous system, or pulmonary involvement, glucocorticoids may be employed to reduce inflammatory complications. Available anthelmintic drugs, including diethylcarbamazine, mebendazole, and albendazole, have not been shown conclusively to alter the course of larva migrans. Control measures include prohibiting dog excreta in public parks and playgrounds, deworming dogs, and preventing pica in children. Treatment of ocular disease is unsatisfactory, and the role of glucocorticoids or anthelmintic drugs in management is controversial.

CUTANEOUS LARVA MIGRANS

Cutaneous larva migrans ("creeping eruption") is a serpiginous skin eruption ([Fig. 219-CD1](#)) caused by burrowing larvae of animal hookworms, usually the dog and cat hookworm *Ancylostoma braziliense*. The larvae hatch from eggs passed in dog and cat feces and mature in the soil. Humans become infected after skin contact with soil in areas frequented by dogs and cats, such as areas underneath house porches or scrub vegetation. Cutaneous larva migrans is especially prevalent among children and in regions with warm humid climates, including the southeastern United States.

After larvae penetrate the skin, erythematous lesions form along the tortuous tracts of their migration through the dermal-epidermal junction; the larvae advance several centimeters in a day. The intensely pruritic lesions may occur anywhere on the body and can be numerous if the patient has lain on the ground. Vesicles and bullae may form later. The animal hookworm larvae do not mature in humans and, without treatment, will die out after several weeks, with resolution of skin lesions. The diagnosis is made readily on clinical grounds, and a skin biopsy only rarely yields diagnostic parasite material. Symptoms can be alleviated by thiabendazole administered orally (25 mg/kg bid) or topically (10% aqueous or petroleum jelly suspension) for 2 to 5 days, by ivermectin (a single dose of 150 to 200 ug/kg), or by albendazole (200 mg bid for 2 days).

ANGIOSTRONGYLUS CANTONENSIS INFECTION

A. cantonensis, the rat lungworm, is the most common cause of human eosinophilic meningitis.

Life Cycle and Epidemiology This infection occurs principally in Southeast Asia and the Pacific Basin. *A. cantonensis* larvae produced by adult worms in the rat lung migrate to the gastrointestinal tract and are expelled with the feces. They develop into infective larvae in land snails and slugs. Humans acquire the infection by ingesting raw infected mollusks; vegetables contaminated by mollusk slime; or crabs, freshwater shrimp, and certain marine fish that have themselves eaten infected mollusks. The larvae then migrate to the brain.

Pathogenesis and Clinical Features The parasites eventually die in the central nervous system, but not before initiating pathologic consequences that, in heavy infections, can result in permanent neurologic sequelae or death. Migrating larvae cause proteolytic damage and marked local eosinophilic inflammation and hemorrhage, with subsequent necrosis and granuloma formation around dying worms. Clinical symptoms develop between 2 and 35 days after the ingestion of larvae. Patients usually present with an insidious or abrupt excruciating frontal, occipital, or bitemporal headache. Neck stiffness, nausea and vomiting, and paresthesias are also common. Fever, cranial and extraocular nerve palsies, seizures, paralysis, and lethargy are uncommon.

Laboratory Findings Examination of the cerebrospinal fluid is mandatory in suspected cases and usually reveals an elevated opening pressure, a white blood cell count of 150 to 2000/uL, and an eosinophilic pleocytosis of >20%. The protein concentration is usually elevated and the glucose level normal. The motile larvae of *A. cantonensis* are only rarely seen in the cerebrospinal fluid. Peripheral-blood eosinophilia may be mild. The diagnosis is generally based on the clinical presentation of eosinophilic meningitis together with a compatible epidemiologic history.

TREATMENT

Specific chemotherapy is not of benefit in angiostrongyliasis; larvicidal agents may actually exacerbate inflammatory brain lesions. Management consists of supportive measures, including the administration of analgesics, sedatives, and -- in severe cases -- glucocorticoids. In most patients, cerebral angiostrongyliasis has a self-limited course, and recovery is complete. The infection may be prevented by adequately cooking snails, crabs, and prawns and inspecting vegetables for mollusk infestation. Other parasitic causes of eosinophilic meningitis in endemic areas may include gnathostomiasis, paragonimiasis, schistosomiasis, and neurocysticercosis.

GNATHOSTOMIASIS

Infection of human tissues with larvae of *Gnathostoma spinigerum* can cause eosinophilic meningoencephalitis, migratory cutaneous swellings, or invasive masses of the eye and visceral organs.

Life Cycle and Epidemiology Human gnathostomiasis occurs in many countries and is notably endemic in Southeast Asia and parts of China and Japan. In nature, the mature adult worms parasitize the gastrointestinal tract of dogs and cats. First-stage larvae hatch from eggs passed into water and are ingested by *Cyclops* species (water fleas). Infective third-stage larvae develop in the flesh of many animal species (including fish,

frogs, eels, snakes, chickens, and ducks) that have eaten either infected *Cyclops* or another infected second intermediate host. Humans typically acquire the infection by eating raw or undercooked fish or poultry. The raw fish dishes of *somfak* in Thailand and *sashimi* in Japan account for most cases of human gnathostomiasis. Some cases in Thailand result from the local practice of applying frog or snake flesh as a poultice.

Pathogenesis and Clinical Features Clinical symptoms are due to the aberrant migration of a single larva into cutaneous, visceral, neural, or ocular tissues. After invasion, larval migration may cause local inflammation, with pain, cough, or hematuria accompanied by fever and eosinophilia. Painful, itchy, migratory swellings may develop in the skin, particularly in the distal extremities or periorbital area. Cutaneous swellings usually last about a week but often recur intermittently over many years. Larval invasion of the eye can provoke a sight-threatening inflammatory response. Finally, invasion of the central nervous system results in eosinophilic meningitis with myeloencephalitis, a serious complication due to ascending larval migration along a large nerve track. Patients characteristically present with agonizing radicular pain and paresthesias in the trunk or a limb, which are followed shortly by paraplegia. Cerebral involvement, with focal hemorrhages and tissue destruction, is often fatal.

Diagnosis and Treatment Cutaneous migratory swellings with marked peripheral eosinophilia, supported by an appropriate geographic and dietary history, generally constitute an adequate basis for a clinical diagnosis of gnathostomiasis. However, patients may present with ocular or cerebrospinal involvement without antecedent cutaneous swellings. In the latter case, eosinophilic pleocytosis is demonstrable (usually along with hemorrhagic or xanthochromic cerebrospinal fluid), but worms are almost never recovered from the cerebrospinal fluid. Surgical removal of the parasite from subcutaneous or ocular tissue, though rarely feasible, is both diagnostic and therapeutic. Albendazole (400 to 800 mg daily for 21 days) may be helpful. At present, cerebrospinal involvement is managed with supportive measures and generally with a course of glucocorticoids. Gnathostomiasis can be prevented by adequate cooking of fish and poultry in endemic areas.

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220. INTESTINAL NEMATODES - Peter F. Weller, Thomas B. Nutman

More than a billion people worldwide are infected with one or more species of intestinal nematodes. [Table 220-1](#) summarizes biologic and clinical features of infections due to the major intestinal parasitic nematodes. These parasites are most common in regions with poor fecal sanitation, particularly in developing countries in the tropics and subtropics but also in the United States. Although nematode infections are not usually fatal, they contribute to malnutrition and diminished work capacity. Humans may on occasion be infected with nematode parasites that ordinarily infect animals; these zoonotic infections include trichostrongyliasis, anisakiasis, capillariasis, and abdominal angiostrongyliasis.

Intestinal nematodes are roundworms; they range in length from 1 mm to many centimeters when mature ([Table 220-1](#)). Their life cycles are complex and highly varied; some species, including *Strongyloides stercoralis* and *Enterobius vermicularis*, can be transmitted directly from person to person, while others, such as *Ascaris lumbricoides*, *Necator americanus*, and *Ancylostoma duodenale*, require a soil phase for development. Because most helminth parasites do not self-replicate, the acquisition of a heavy burden of adult worms requires repeated exposure to the parasite in its infectious stage, whether larva or egg. Hence, clinical disease, as opposed to asymptomatic infection, generally develops only with prolonged residence in an endemic area. In persons with marginal nutrition, intestinal helminth infections may impair growth and development. Eosinophilia and elevated serum IgE levels are features of many helminthic infections and, when unexplained, should always prompt a search for occult helminthiasis. Significant protective immunity to intestinal nematodes appears not to develop in humans, although mechanisms of parasite immune evasion and host immune responses to these infections have not been elucidated in detail.

ASCARIASIS

A. lumbricoides is the largest intestinal nematode parasite of humans, reaching up to 40 cm in length. Most infected individuals have low worm burdens and are asymptomatic. Clinical disease arises from larval migration in the lungs or effects of the adult worms in the intestines.

Life Cycle Adult worms live in the lumen of the small intestine. Mature female *Ascaris* worms are extraordinarily fecund, each producing up to 240,000 eggs a day, which pass with the feces. Ascarid eggs, which are remarkably resistant to environmental stresses, become infective after several weeks of maturation in the soil and can remain infective for years. After infective eggs are swallowed, larvae hatched in the intestine invade the mucosa, migrate through the circulation to the lungs, break into the alveoli, ascend the bronchial tree, and return via swallowing to the small intestine, where they develop into adult worms. Between 2 and 3 months elapse between initial infection and egg production. The adult worms live for ~1 to 2 years.

Epidemiology *Ascaris* is widely distributed in tropical and subtropical regions as well as in other humid areas, including the rural southeastern United States. Transmission typically occurs through fecally contaminated soil and is due either to a lack of sanitary facilities or to the use of human manure ("night soil") as fertilizer. With their propensity

for hand-to-mouth fecal carriage, younger children in impoverished rural areas are most affected. Infection outside endemic areas, though uncommon, can occur from eggs borne on transported vegetables.

Clinical Features During the lung phase of larval migration, about 9 to 12 days after egg ingestion, patients may develop an irritating nonproductive cough and burning substernal discomfort that is aggravated by coughing or deep inspiration. Dyspnea and blood-tinged sputum are less common. Fever is usually reported, with temperatures sometimes exceeding 38.5°C (101.3°F). Eosinophilia develops during this symptomatic phase and subsides slowly over weeks. Chest x-rays may reveal evidence of eosinophilic pneumonitis (Löffler's syndrome), with round or oval infiltrates a few millimeters to several centimeters in size. These infiltrates may be transient and intermittent, clearing after several weeks. Where there is seasonal transmission of the parasite, seasonal pneumonitis with eosinophilia may develop in previously infected and sensitized hosts.

In established infections, adult worms in the small intestine usually cause no symptoms. In heavy infections, particularly in children, a large bolus of entangled worms can cause pain and small-bowel obstruction, sometimes complicated by perforation, intussusception, or volvulus. Single worms may cause disease when they migrate into aberrant sites. A large worm can enter and occlude the biliary tree, causing biliary colic, cholecystitis, cholangitis, pancreatitis, or (rarely) intrahepatic abscesses. Migration of an adult worm up the esophagus can provoke coughing and oral expulsion of the worm. In highly endemic areas, intestinal and biliary ascariasis can rival acute appendicitis and gallstones as causes of surgical acute abdomen.

Laboratory Findings Most cases of ascariasis can be diagnosed by the microscopic detection of characteristic mamillated *Ascaris* eggs (65 by 45 μ m) in fecal samples. Occasionally, patients present after passing an adult worm -- identifiable by its large size and smooth cream-colored surface -- in the stool or through the mouth or nose. During the early transpulmonary migratory phase, when eosinophilic pneumonitis occurs, larvae can be found in sputum or gastric aspirates before diagnostic eggs appear in the stool. The eosinophilia that is prominent during this early stage usually decreases to minimal levels in established infection. The large adult worms may be visualized, occasionally serendipitously, on contrast studies of the gastrointestinal tract. A plain abdominal film may reveal masses of worms in gas-filled loops of bowel in patients with intestinal obstruction. Pancreaticobiliary worms can be detected by ultrasound and endoscopic retrograde cholangiopancreatography; the latter method also has been used to extract biliary *Ascaris* worms.

TREATMENT

Ascariasis should always be treated to prevent potentially serious complications. Mebendazole or albendazole (which is considered an investigational drug by the Food and Drug Administration for this indication) is effective. These benzimidazoles are contraindicated in pregnancy and in heavy infections, in which they may provoke ectopic migration. Pyrantel pamoate and piperazine citrate are safe in pregnancy. Mild diarrhea and abdominal pain are uncommon side effects of these agents. Partial intestinal obstruction should be managed with nasogastric suction, intravenous fluid

administration, and instillation of piperazine through the nasogastric tube, but complete obstruction and its severe complications require immediate surgical intervention.

HOOKWORM

One-fourth of the world's population is infected with one of the two hookworm species (*A. duodenale* and *N. americanus*). Most infected individuals are asymptomatic. Hookworm disease develops from a combination of factors -- a heavy worm burden, a prolonged duration of infection, and an inadequate iron intake -- and results in iron-deficiency anemia and, on occasion, hypoproteinemia.

Life Cycle Adult hookworms, which are about 1 cm long, use buccal teeth (*Ancylostoma*) or cutting plates (*Necator*) to attach to the small-bowel mucosa and suck blood (0.2 mL/d per *Ancylostoma* adult) and interstitial fluid. The adult hookworms produce thousands of eggs daily. The eggs are deposited with feces in soil, where rhabditiform larvae hatch and develop over a 1-week period into infectious filariform larvae. Infective larvae penetrate the skin and reach the lungs by way of the bloodstream. There they invade alveoli and ascend the airways before being swallowed and reaching the small intestine. The prepatent period from skin invasion to appearance of eggs in the feces is about 6 to 8 weeks, but it may be longer with *A. duodenale*. Larvae of *A. duodenale*, if swallowed, can survive and develop directly in the intestinal mucosa. Adult hookworms may survive over a decade but usually live about 6 to 8 years for *A. duodenale* and 2 to 5 years for *N. americanus*.

Epidemiology *A. duodenale* is prevalent in southern Europe, North Africa, and northern Asia, and *N. americanus* is the predominant species in the western hemisphere and equatorial Africa. The two species overlap in many tropical regions, particularly Southeast Asia. In most areas, older children have the greatest incidence and intensity of hookworm infection. In rural areas where fields are fertilized with night soil, older working adults also may be heavily affected.

Clinical Features Most hookworm infections are asymptomatic. Infective larvae may provoke pruritic maculopapular dermatitis ("ground itch") at the site of skin penetration as well as serpiginous tracts of subcutaneous migration (similar to cutaneous larva migrans) in previously sensitized hosts. Larvae migrating through the lungs occasionally cause mild transient pneumonitis, but this condition develops less frequently in hookworm infection than in ascariasis. In the early intestinal phase, infected persons may develop epigastric pain (often with postprandial accentuation), inflammatory diarrhea, or other abdominal symptoms accompanied by eosinophilia. The major consequence of chronic hookworm infection is iron deficiency. Symptoms are minimal if iron intake is adequate, but marginally nourished individuals develop symptoms of progressive iron-deficiency anemia and hypoproteinemia, including weakness, shortness of breath, and skin depigmentation.

Laboratory Findings The diagnosis is established by the finding of characteristic 40- by 60- μ m oval hookworm eggs in the feces. Stool-concentration procedures may be required to detect light infections. Eggs of the two species are indistinguishable. In a stool sample that is not fresh, the eggs may have hatched to release rhabditiform larvae, which need to be differentiated from those of *S. stercoralis*. Hypochromic

microcytic anemia, occasionally with eosinophilia or hypoalbuminemia, is characteristic of hookworm disease.

TREATMENT

Hookworm infection can be eradicated with several safe and highly effective anthelmintic drugs, including mebendazole, albendazole, and pyrantel pamoate ([Chap. 212](#)). Mild iron-deficiency anemia often can be treated with oral iron alone. Severe hookworm disease with protein loss and malabsorption necessitates nutritional support and oral iron replacement along with deworming.

Ancylostoma caninum This parasite, the canine hookworm, has been identified as a cause of human eosinophilic enteritis, especially in northeastern Australia. In this zoonotic infection, adult hookworms attach to the small intestine (where they may be visualized by endoscopy) and elicit abdominal pain and intense local eosinophilia. Treatment with mebendazole (100 mg twice daily for 3 days) is effective.

STRONGYLOIDIASIS

S. stercoralis is distinguished by its ability, unusual among helminths, to replicate in the human host. This capacity permits ongoing cycles of autoinfection as infective larvae are internally produced. Strongyloidiasis can thus persist for decades without further exposure of the host to exogenous infective larvae. In immunocompromised hosts, large numbers of invasive *Strongyloides* larvae can disseminate widely and can be fatal.

Life Cycle In addition to a parasitic cycle of development, *Strongyloides* can undergo a free-living cycle of development in the soil. This adaptability facilitates the parasite's survival in the absence of mammalian hosts. Rhabditiform larvae passed in feces can transform into infectious filariform larvae either directly or after a free-living phase of development. Humans acquire strongyloidiasis when filariform larvae in fecally contaminated soil penetrate the skin or mucous membranes. The larvae then travel through the bloodstream to the lungs, where they break into the alveolar spaces, ascend the bronchial tree, are swallowed, and thereby reach the small intestine. There the larvae mature into adult worms that penetrate the mucosa of the proximal small bowel. The minute (2-mm-long) parasitic adult female worms reproduce by parthenogenesis; parasitic adult males do not exist. Eggs hatch locally in the intestinal mucosa, releasing rhabditiform larvae that migrate to the lumen and pass with the feces into soil. Alternatively, rhabditiform larvae in the bowel can develop directly into filariform larvae that penetrate the colonic wall or perianal skin and enter the circulation to repeat the migration that establishes ongoing internal reinfection. This autoinfection cycle allows strongyloidiasis to persist for decades after the host has left an endemic area.

Epidemiology *S. stercoralis* is spottily distributed in tropical areas and other hot, humid regions and is particularly common in Southeast Asia, sub-Saharan Africa, and Brazil. In the United States, the parasite is endemic in parts of the South and is found in residents of mental institutions who practice poor hygiene and in immigrants and military veterans who have lived in endemic areas abroad.

Clinical Features In uncomplicated strongyloidiasis, many patients are asymptomatic

or have mild cutaneous and/or abdominal symptoms. Recurrent urticaria, often involving the buttocks and wrists, is the most common cutaneous manifestation. Migrating larvae can elicit a pathognomonic serpiginous eruption, *larva currens* ("running larva" [Fig. 220-CD1](#)) -- a pruritic, raised, erythematous lesion that advances as rapidly as 10 cm/h along the course of larval migration. Adult parasites burrow into the duodenojejunal mucosa and can cause abdominal (usually midepigastlic) pain, which resembles peptic ulcer pain except that it is aggravated by food ingestion. Nausea, diarrhea, gastrointestinal bleeding, mild chronic colitis, and weight loss can occur. Small-bowel obstruction may develop with early, heavy infection. Pulmonary symptoms are rare in uncomplicated strongyloidiasis. Eosinophilia is common, with levels fluctuating over time.

The ongoing autoinfection cycle of strongyloidiasis is normally contained by unknown factors of the host's immune system. Abrogation of host immunity, especially with glucocorticoid therapy and much less commonly with other immunosuppressive medications, leads to hyperinfection, with the generation of large numbers of filariform larvae. Colitis, enteritis, or malabsorption may develop. In disseminated strongyloidiasis, larvae may invade not only gastrointestinal tissues and the lungs but also the central nervous system, peritoneum, liver, and kidney. Moreover, bacteremia may develop because of the entry of enteric flora through disrupted mucosal barriers. Gram-negative sepsis, pneumonia, or meningitis may complicate or dominate the clinical course. Eosinophilia is often absent in severely infected patients. Disseminated strongyloidiasis, particularly in patients with unsuspected infection who are given glucocorticoids, can be fatal. Strongyloidiasis is a frequent complication of infection with human T cell lymphotropic virus type I, but disseminated strongyloidiasis is not common among patients infected with HIV.

Diagnosis In uncomplicated strongyloidiasis, the finding of rhabditiform larvae in feces is diagnostic. The eggs are almost never detectable because they hatch in the intestine. Rhabditiform larvae are 200 to 250 μm long, with a short buccal cavity that distinguishes them from hookworm rhabditiform larvae. Single stool examinations detect only about one-third of uncomplicated infections, in which few larvae are passed. Serial examinations and the use of the agar plate detection method improve the sensitivity of stool diagnosis. In uncomplicated -- but not hyperinfection -- strongyloidiasis, stool examinations may be repeatedly negative. If stool examinations are negative, *Strongyloides* can be assayed by sampling of the duodenojejunal contents by aspiration or biopsy. An enzyme-linked immunosorbent assay for antibodies to excretory-secretory or somatic antigens of *Strongyloides* is a sensitive method of diagnosing uncomplicated infections. In disseminated strongyloidiasis, filariform larvae (550 μm long) should be sought in stool as well as in samples obtained from sites of potential larval migration, including sputum, bronchoalveolar lavage fluid, or surgical drainage fluid.

TREATMENT

Even in the asymptomatic state, strongyloidiasis must be treated because of the potential for fatal hyperinfection. Ivermectin (200 $\mu\text{g/kg}$ daily for 1 or 2 days) is more effective and better tolerated than thiabendazole (25 mg/kg bid for 2 days), whose common adverse effects include nausea, vomiting, diarrhea, dizziness, and neuropsychiatric disturbances. Because thiabendazole is not uniformly effective, stool

examinations, eosinophil counts, and monitoring of clinical symptoms should be continued after treatment. For disseminated strongyloidiasis, treatment should be extended for at least 5 to 7 days or until the parasites are eradicated.

Strongyloides fulleborni This unusual species, which has been encountered in Africa and Papua New Guinea, is thought to be transmitted from person to person and through maternal milk. *S. fulleborni* releases membranous sacs filled with eggs into the stool. Most commonly affected are infants and young children, who present with abdominal distention, respiratory distress, vomiting, or diarrhea.

TRICHURIASIS

Most infections with the whipworm *Trichuris trichiura* are asymptomatic, but heavy infections may cause gastrointestinal symptoms. Like the other soil-transmitted helminths, whipworm is distributed globally in the tropics and subtropics and is most common among poor children.

Life Cycle A broad posterior section and a thin anterior portion give *Trichuris* its characteristic whiplike shape. The adult worms reside in the colon and cecum, the anterior portions threaded into the superficial mucosa. Thousands of eggs laid daily by adult female worms pass with the feces and mature in the soil. After ingestion, infective eggs hatch in the duodenum, releasing larvae that mature before migrating to the large bowel. The entire cycle takes about 3 months, and adult worms may live for several years.

Clinical Features Tissue reactions to whipworms are mild. Most infected individuals have no symptoms or eosinophilia. Heavy infections may result in abdominal pain, anorexia, and bloody or mucoid diarrhea resembling inflammatory bowel disease. Rectal prolapse can result from massive infections in children, who often suffer from malnourishment and other diarrheal illnesses. Moderately heavy whipworm burdens also contribute to growth retardation.

Diagnosis and Treatment The characteristic 50- by 20-um lemon-shaped whipworm eggs are readily detected on stool examination. Adult worms, which are 3 to 5 cm long, occasionally can be seen on proctoscopy. Mebendazole or albendazole is safe and effective for treatment ([Chap. 212](#)).

ENTEROBIASIS (PINWORM)

E. vermicularis is more common in temperate countries than in the tropics. More than 40 million Americans, particularly schoolchildren, are estimated to be infected with pinworms.

Life Cycle and Epidemiology *Enterobius* adult worms are about 1 cm long and dwell in the bowel lumen. The gravid female worm migrates nocturnally out into the perianal region and releases up to 10,000 immature eggs. The eggs become infective within hours and are transmitted by hand-to-mouth passage. The larvae hatch and mature entirely within the intestine. This life cycle takes about 1 month, and adult worms survive for about 2 months. Self-infection results from perianal scratching and transport of

infective eggs on the hands or under the nails to the mouth. Owing to the ease of person-to-person spread, pinworm infections are common among family members and institutionalized populations.

Clinical Features Most pinworm infections are asymptomatic. Perianal pruritus is the cardinal symptom. The itching is often worse at night owing to the nocturnal migration of the female worms, and it may lead to excoriation and bacterial superinfection. Heavy infections have been claimed to cause abdominal pain and weight loss. On rare occasions, pinworms invade the female genital tract, causing vulvovaginitis and pelvic or peritoneal granulomas. Eosinophilia or elevated levels of serum IgE are rare.

Diagnosis Since pinworm eggs are not usually released in the bowel, the diagnosis cannot be made by looking for eggs in the feces. Instead, eggs deposited in the perianal region are detected by the application of clear cellulose acetate tape to the perianal region in the morning. After the tape is transferred to a microscope slide, low-power examination will reveal the characteristic pinworm eggs, which are oval, measure 55 by 25 μm , and are flattened along one side.

TREATMENT

All affected individuals should be given a dose of mebendazole or pyrantel pamoate, with treatment repeated after 10 to 14 days ([Chap. 212](#)). Treatment of household members is also advocated to eliminate asymptomatic reservoirs of potential reinfection.

TRICHOSTRONGYLIASIS

Trichostrongylus species that are normally parasites of herbivorous animals occasionally infect humans, particularly in Asia and Africa. This parasite has been termed *pseudo hookworm* because of similarities to the hookworms in life cycle and egg morphology. Humans acquire the infection by accidentally ingesting *Trichostrongylus* larvae on contaminated leafy vegetables. The larvae do not migrate in humans but mature directly into adult worms in the small bowel. These worms ingest far less blood than hookworms; most infected people are asymptomatic, but heavy infections may give rise to mild anemia and eosinophilia. *Trichostrongylus* eggs encountered on stool examination resemble those of hookworms but are larger (85 by 115 μm). Appropriate treatment consists of mebendazole or albendazole ([Chap. 212](#)).

ANISAKIASIS

Anisakiasis is a gastrointestinal infection caused by the accidental ingestion in uncooked saltwater fish of nematode larvae belonging to the family Anisakidae. The incidence of anisakiasis in the United States has increased as a result of the growing popularity of raw fish dishes. Most cases occur in Japan, the Netherlands, and Chile, where raw fish -- sushi, pickled green herring, and seiche, respectively -- are national culinary staples. Anisakid nematodes parasitize large sea mammals such as whales, dolphins, and seals. As part of a complex parasitic life cycle involving marine food chains, infectious larvae migrate to the musculature of a variety of fish. Both *Anisakis simplex* and *Pseudoterranova decipiens* have been implicated in human anisakiasis, but an identical gastric syndrome may be caused by the red larvae of eustrongylid parasites

of fish-eating birds.

When humans consume infected raw fish, live larvae may be coughed up within 48 h. Alternatively, larvae may immediately penetrate the mucosa of the stomach. Within hours, violent upper abdominal pain accompanied by nausea and occasionally vomiting ensues, mimicking an acute abdomen. The diagnosis can be established by direct visualization on upper endoscopy, outlining of the worm by contrast radiographic studies, or histopathologic examination of extracted tissue. In experienced hands, the first technique is preferable because extraction of the burrowing larvae by endoscopic technique is curative. In addition, larvae may pass to the small bowel, where they penetrate the mucosa and provoke a vigorous eosinophilic granulomatous response. Symptoms may appear 1 or 2 weeks after the infective meal, with intermittent abdominal pain, diarrhea, nausea, and fever resembling the manifestations of Crohn's disease. The diagnosis may be suggested by barium studies and confirmed by curative surgical resection of a granuloma in which the worm is embedded. Anisakid eggs are not found in the stool, since the larvae do not mature in humans. Anisakid larvae in saltwater fish are killed by cooking to 60°C, freezing at -20°C for 3 days, or commercial blast freezing, but not usually by salting, marinating, or cold smoking. No medical treatment is available; if possible, surgical or endoscopic removal should be undertaken.

CAPILLARIASIS

Intestinal capillariasis is caused by ingestion of raw fish infected with *Capillaria philippinensis*. Subsequent autoinfection can lead to a severe wasting syndrome. The disease occurs in the Philippines and Thailand and, on occasion, elsewhere in Asia. The natural cycle of *C. philippinensis* involves fish from fresh and brackish water. When humans eat infected raw fish, the larvae mature in the intestine into adult worms, which produce invasive larvae that cause intestinal inflammation and villus loss. Capillariasis has an insidious onset with nonspecific abdominal pain and watery diarrhea. If untreated, progressive autoinfection can lead to protein-losing enteropathy and severe malabsorption and ultimately to death from cachexia, cardiac failure, or superinfection. The diagnosis is established by identification of the characteristic peanut-shaped (20- by 40-um) eggs on stool examination. Severely ill patients require hospitalization and supportive therapy in addition to prolonged anthelmintic treatment with mebendazole or albendazole ([Chap. 212](#)).

ABDOMINAL ANGIOSTRONGYLIASIS

Abdominal angiostrongyliasis is found in Latin America and Africa. The zoonotic parasite *Angiostrongylus costaricensis* causes eosinophilic ileocolitis after the ingestion of contaminated vegetation. *A. costaricensis* normally parasitizes the cotton rat and other rodents, with slugs and snails serving as intermediate hosts. Humans become infected by accidentally ingesting infective larvae in mollusk slime deposited on fruits and vegetables; children are at highest risk. The larvae penetrate the gut wall and migrate to the mesenteric artery, where they develop into adult worms. Eggs deposited in the gut wall provoke an intense eosinophilic granulomatous reaction, and adult worms may cause mesenteric arteritis, thrombosis, or frank bowel infarction. Symptoms may mimic those of appendicitis, including abdominal pain and tenderness, fever, vomiting, and a palpable mass in the right iliac fossa. Leukocytosis and eosinophilia are

prominent. A barium enema may reveal ileocecal filling defects, but a definitive diagnosis is usually made surgically with partial bowel resection. Pathologic study reveals a thickened bowel wall with eosinophilic granulomas surrounding the *Angiostrongylus* eggs. In nonsurgical cases, the diagnosis rests solely on clinical grounds because larvae and eggs cannot be detected in the stool. Medical therapy for abdominal angiostrongyliasis (thiabendazole; [Chap. 212](#)) is of uncertain efficacy. Careful observation and surgical resection for severe symptoms are the mainstays of treatment.

(Bibliography omitted in Palm version)

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221. FILARIASIS AND RELATED INFECTIONS (LOIASIS, ONCHOCERCIASIS, AND DRACUNCULIASIS) - Thomas B. Nutman, Peter F. Weller

Filarial worms are nematodes that dwell in the subcutaneous tissues and the lymphatics. Eight filarial species infect humans ([Table 221-1](#)); of these, four -- *Wuchereria bancrofti*, *Brugia malayi*, *Onchocerca volvulus*, and *Loa loa* -- are responsible for most serious filarial infections. Filarial parasites, which infect an estimated 170 million persons worldwide, are transmitted by specific species of mosquitoes or other arthropods and have a complex life cycle including infective larval stages carried by insects and adult worms that reside in either lymphatic or subcutaneous tissues of humans. The offspring of adults are microfilariae, which, depending on their species, are 200 to 250 μm long and 5 to 7 μm wide, may or may not be enveloped in a loose sheath, and either circulate in the blood or migrate through the skin ([Table 221-1](#)). To complete the life cycle, microfilariae are ingested by the arthropod vector and develop over 1 to 2 weeks into new infective larvae. Adult worms live for many years, whereas microfilariae survive from 3 to 36 months.

Usually, infection is established only with repeated and prolonged exposures to infective larvae. Since the clinical manifestations of filarial diseases develop relatively slowly, these infections should be considered chronic diseases with possible long-term debilitating effects. In terms of the nature, severity, and timing of clinical manifestations, patients with filariasis who are native to endemic areas and undergo lifelong exposure may differ significantly from those who are travelers or who have recently moved to these areas. Characteristically, the disease is more acute and intense in newly exposed individuals than in natives of endemic areas.

LYMPHATIC FILARIASIS

Lymphatic filariasis is caused by *W. bancrofti*, *B. malayi*, or *B. timori*. The threadlike adult parasites reside in lymphatic channels or lymph nodes, where they may remain viable for more than two decades.

EPIDEMIOLOGY

W. bancrofti, the most widely distributed human filarial parasite, affects an estimated 115 million people and is found throughout the tropics and subtropics, including Asia and the Pacific Islands, Africa, areas of South America, and the Caribbean basin. Humans are the only definitive host for the parasite. Generally, the subperiodic form is found only in the Pacific Islands; elsewhere, *W. bancrofti* is nocturnally periodic. (Nocturnally periodic forms of microfilariae are scarce in peripheral blood by day and increase at night, whereas subperiodic forms are present in peripheral blood at all times and reach maximal levels in the afternoon.) Natural vectors for *W. bancrofti* are *Culex fatigans* mosquitoes in urban settings and anopheline or aedeian mosquitoes in rural areas.

Brugian filariasis due to *B. malayi* occurs primarily in China, India, Indonesia, Korea, Japan, Malaysia, and the Philippines. *B. malayi* also has two forms distinguished by the periodicity of microfilaremia. The more common nocturnal form is transmitted in areas of coastal rice fields, while the subperiodic form is found in forests. *B. malayi* naturally

infects cats as well as humans. *B. timori* exists only on islands of the Indonesian archipelago.

PATHOLOGY

The principal pathologic changes result from inflammatory damage to the lymphatics, which is caused by adult worms and not by microfilariae. Adult worms live in afferent lymphatics or sinuses of lymph nodes and cause lymphatic dilatation and thickening of the vessel walls. The infiltration of plasma cells, eosinophils, and macrophages in and around the infected vessels, along with endothelial and connective tissue proliferation, leads to tortuosity of the lymphatics and damaged or incompetent lymph valves. Lymphedema and chronic-stasis changes with hard or brawny edema develop in the overlying skin. These consequences of filariasis are due both to direct effects of the worms and to the immune response of the host to the parasite. These immune responses are believed to cause the granulomatous and proliferative processes that precede total lymphatic obstruction. It is thought that the vessel remains patent as long as the worm remains viable and that death of the worm leads to enhanced granulomatous reaction and fibrosis. Lymphatic obstruction results, and, despite collateralization of the lymphatics, lymphatic function is compromised.

CLINICAL FEATURES

The most common presentations of the lymphatic filariases are asymptomatic (or subclinical) microfilaremia, hydrocele, acute adenolymphangitis (ADL), and chronic lymphatic disease. In areas where *W. bancrofti* or *B. malayi* is endemic, the overwhelming majority of infected individuals have few overt clinical manifestations of filarial infection despite large numbers of circulating microfilariae in the peripheral blood. Although they may be clinically asymptomatic, virtually all persons with *W. bancrofti* or *B. malayi* microfilaremia have some degree of subclinical disease that includes microscopic hematuria and/or proteinuria, dilated (and tortuous) lymphatics (visualized by imaging), and -- in men -- scrotal lymphangiectasia (detectable by ultrasound). Despite these findings, the majority of individuals appear to remain clinically asymptomatic for years; relatively few progress to the acute and chronic stages of infection.

ADL is characterized by high fever, lymphatic inflammation (lymphangitis and lymphadenitis), and transient local edema. The lymphangitis is retrograde, extending peripherally from the lymph node draining the area where the adult parasites reside. Regional lymph nodes are often enlarged, and the entire lymphatic channel can become indurated and inflamed. Concomitant local thrombophlebitis can occur as well. In brugian filariasis, a single local abscess may form along the involved lymphatic tract and subsequently rupture to the surface. The lymphadenitis and lymphangitis involve both the upper and lower extremities in both bancroftian and brugian filariasis, but involvement of the genital lymphatics occurs almost exclusively with *W. bancrofti* infection. This genital involvement can be manifested by funiculitis, epididymitis, scrotal pain, and tenderness. In endemic areas, another type of acute disease -- dermatolymphangioadenitis (DLA) -- is recognized as a syndrome that includes high fever, chills, myalgias, and headache. Edematous inflammatory plaques clearly demarcated from normal skin are seen. Vesicles, ulcers, and hyperpigmentation may

also be noted. There is often a history of trauma, burns, radiation, insect bites, punctiform lesions, or chemical injury. Entry lesions, especially in the interdigital area, are common. DLA is often diagnosed as cellulitis.

If lymphatic damage progresses, transient lymphedema can develop into *lymphatic obstruction* and the permanent changes associated with elephantiasis. Brawny edema follows early pitting edema, and thickening of the subcutaneous tissues and hyperkeratosis occur. Fissuring of the skin develops, as do hyperplastic changes. Superinfection of these poorly vascularized tissues becomes a problem. In bancroftian filariasis, in which genital involvement is common, hydroceles may develop; in advanced stages, this condition may evolve into scrotal lymphedema and scrotal elephantiasis. Furthermore, if there is obstruction of the retroperitoneal lymphatics, the increased renal lymphatic pressure leads to rupture of the renal lymphatics and the development of chyluria, which is usually intermittent and most prominent in the morning.

The clinical manifestations of filarial infections in travelers or transmigrants who have recently entered an endemic region are distinctive. Given a sufficient number of bites by infected vectors, usually over a 3- to 6-month period, recently exposed patients can develop acute lymphatic or scrotal inflammation with or without urticaria and localized angioedema. Lymphadenitis of epitrochlear, axillary, femoral, or inguinal lymph nodes is often followed by retrogradely evolving lymphangitis. Acute attacks are short-lived and, in contrast to filarial fevers in patients native to endemic areas, are usually not accompanied by fever. With prolonged exposure to infected mosquitoes, these attacks, if untreated, become more severe and lead to permanent lymphatic inflammation and obstruction.

DIAGNOSIS

A definitive diagnosis can be made only by detection of the parasites and hence can be difficult. Adult worms localized in lymphatic vessels or nodes are largely inaccessible. Microfilariae can be found in blood, in hydrocele fluid, or (occasionally) in other body fluids. Such fluids can be examined microscopically, either directly or -- for greater sensitivity -- after concentration of the parasites by the passage of fluid through a polycarbonate cylindrical pore filter (pore size, 3 μ m) or by the centrifugation of fluid fixed in 2% formalin (Knott's concentration technique). The timing of blood collection is critical and should be based on the periodicity of the microfilariae in the endemic region involved. Many infected individuals do not have microfilaremia, and definitive diagnosis in such cases can be difficult. Assays for circulating antigens of *W. bancrofti* permit the diagnosis of microfilaremic and cryptic (amicrofilaremic) infection. Two tests are commercially available: one is an enzyme-linked immunosorbent assay (ELISA) and the other a rapid-format immunochromatographic card test. Both assays have sensitivities that range from 96 to 100% and specificities that approach 100%. There are currently no tests for circulating antigens in brugian filariasis.

Polymerase chain reaction (PCR)-based assays for DNA of *W. bancrofti* and *B. malayi* in blood have been developed. A number of studies indicate that this diagnostic method is of equivalent or greater sensitivity compared with parasitologic methods, detecting patent infection in almost all infected subjects.

In cases of suspected lymphatic filariasis, examination of the scrotum or the female breast using high-frequency ultrasound in conjunction with Doppler techniques may result in the identification of motile adult worms within dilated lymphatics. Worms may be visualized in the lymphatics of the spermatic cord in up to 80% of infected men. Live adult worms have a distinctive pattern of movement within the lymphatic vessels (termed the *filaria dance sign*).

Radionuclide lymphoscintigraphic imaging of the limbs reliably demonstrates widespread lymphatic abnormalities in both asymptomatic microfilaremic persons and those with clinical manifestations of lymphatic pathology. While of potential utility in the delineation of anatomic changes associated with infection, lymphoscintigraphy is unlikely to assume primacy in the diagnostic evaluation of individuals with suspected infection; it is principally a research tool, and the radionuclide-protein conjugates are not commercially available or approved by the U.S. Food and Drug Administration (FDA).

Eosinophilia and elevated serum concentrations of IgE and antifilarial antibody support the diagnosis of lymphatic filariasis. There is, however, extensive cross-reactivity between filarial antigens and antigens of other helminths, including the common intestinal roundworms; thus, interpretations of serologic findings can be difficult. In addition, residents of endemic areas can become sensitized to filarial antigens through exposure to infected mosquitoes without having patent filarial infections.

In acute episodes, lymphatic filariasis must be distinguished from thrombophlebitis, infection, and trauma. Retrogradely evolving lymphangitis is a characteristic feature that helps distinguish filarial lymphangitis from typically ascending bacterial lymphangitis. Chronic filarial lymphedema must be distinguished from the lymphedema of malignancy, postoperative scarring, trauma, chronic edematous states, and congenital lymphatic system abnormalities.

TREATMENT

With new definitions of clinical syndromes in lymphatic filariasis and new tools to assess clinical status (e.g., ultrasound, lymphoscintigraphy, circulating filarial antigen assays), approaches to treatment based on infection status can be considered.

Diethylcarbamazine (DEC, 6 mg/kg daily for 12 days), which has both macro- and microfilaricidal properties, remains the treatment of choice for the individual with active lymphatic filariasis (microfilaremia, antigen positivity, or adult worms on ultrasound), although albendazole (400 mg twice daily for 21 days) has demonstrated macrofilaricidal efficacy.

As has already been mentioned, a growing body of evidence indicates that, although they may be asymptomatic, virtually all persons with *W. bancrofti* or *B. malayi* microfilaremia have some degree of subclinical disease (hematuria, proteinuria, abnormalities on lymphoscintigraphy). Thus, early treatment of asymptomatic persons is recommended to prevent further lymphatic damage. For [ADL](#), supportive treatment (including the administration of antipyretics and analgesics) is recommended, as is antibiotic therapy if secondary bacterial infection is likely. Similarly, because lymphatic disease is associated with the presence of adult worms, treatment with [DEC](#) is recommended for microfilaria-negative adult-worm carriers.

In persons with chronic manifestations of lymphatic filariasis, treatment regimens that emphasize hygiene, prevention of secondary bacterial infections, and physiotherapy have gained wide acceptance for morbidity control. These regimens are similar to those recommended for lymphedema of most nonfilarial causes and known by a variety of names, including *complex decongestive physiotherapy* and *complex lymphedema therapy*. Hydroceles can be drained repeatedly or managed surgically. In patients with chronic manifestations of lymphatic filariasis, drug treatment should be reserved for cases with evidence of active infection.

The recommended course of [DEC](#) treatment (12 days; total dose, 72 mg/kg) has remained standard for many years; however, data indicate that single-dose DEC treatment with 6 mg/kg may be equally efficacious. The 12-day course provides more rapid short-term microfilarial suppression. Regimens that utilize single-dose DEC or ivermectin or combinations of single doses of albendazole and either DEC or ivermectin have all been demonstrated to have a sustained microfilaricidal effect.

Side effects of [DEC](#) treatment include fever, chills, arthralgias, headaches, nausea, and vomiting. Both the development and the severity of these reactions are directly related to the number of microfilariae circulating in the bloodstream and may represent an acute hypersensitivity reaction to the antigens being released by dead and dying parasites.

PREVENTION AND CONTROL

Avoidance of mosquito bites is usually not feasible for residents of endemic areas, but visitors should make use of insect repellent and mosquito nets. [DEC](#) can kill developing forms of filarial parasites and has been shown to be useful as a prophylactic agent in humans.

Community-based intervention is the current approach to elimination of lymphatic filariasis as a public health problem. The underlying tenet of this approach is that mass annual distribution of antimicrofilarial chemotherapy (albendazole with either [DEC](#) or ivermectin) will profoundly suppress microfilaremia. If the suppression is sustained, then transmission can be interrupted. As an added benefit, these combinations have secondary effects on gastrointestinal helminths. An alternative approach to the control of lymphatic filariasis is the use of salt fortified with DEC. Community use of DEC-fortified salt dramatically reduces microfilarial density with no apparent adverse reactions. Community education and clinical care for persons already suffering from the chronic sequelae of lymphatic filariasis are important components of filariasis control and elimination programs.

TROPICAL PULMONARY EOSINOPHILIA

Tropical pulmonary eosinophilia (TPE) is a distinct syndrome that develops in some individuals infected with lymphatic filarial species. This syndrome affects males and females at a ratio of 4:1, often during the third decade of life. The majority of cases have been reported from India, Pakistan, Sri Lanka, Brazil, and Southeast Asia.

CLINICAL FEATURES

The main features include a history of residence in filarial endemic regions, paroxysmal cough and wheezing that are usually nocturnal (and probably related to the nocturnal periodicity of microfilariae), weight loss, low-grade fever, adenopathy, and pronounced blood eosinophilia (>3000 eosinophils/uL). Chest x-rays may be normal but generally show increased bronchovascular markings; diffuse miliary lesions or mottled opacities may be present in the middle and lower lung fields. Tests of pulmonary function show restrictive abnormalities in most cases and obstructive defects in half. Total serum IgE levels (10,000 to 100,000 ng/mL) and antifilarial antibody titers are characteristically elevated.

PATHOLOGY

In [TPE](#) there is rapid clearance of microfilariae and parasite antigens from the bloodstream by the lungs, and the clinical symptoms result from allergic and inflammatory reactions elicited by the cleared parasites. In some subjects, trapping of microfilariae in other reticuloendothelial organs can cause hepatomegaly, splenomegaly, or lymphadenopathy. A prominent, eosinophil-enriched, intraalveolar infiltrate is often reported. In the absence of successful treatment, interstitial fibrosis can lead to progressive pulmonary damage.

DIFFERENTIAL DIAGNOSIS

[TPE](#) must be distinguished from asthma, Löffler's syndrome, allergic bronchopulmonary aspergillosis, allergic granulomatosis with angiitis (Churg-Strauss syndrome), the systemic vasculitides (most notably periarteritis nodosa and Wegener's granulomatosis), chronic eosinophilic pneumonia, and the idiopathic hypereosinophilic syndrome. In addition to a geographic history of filarial exposure, useful features for distinguishing TPE include wheezing that is solely nocturnal, very high levels of antifilarial antibodies, and a rapid initial response to treatment with [DEC](#).

TREATMENT

[DEC](#) is used at a dosage of 4 to 6 mg/kg of body weight per day for 14 days. Symptoms usually resolve within 3 to 7 days after the initiation of therapy. Relapse, which occurs in ~12 to 25% of cases (sometimes after an interval of years), requires re-treatment.

ONCHOCERCIASIS

Onchocerciasis ("river blindness") is caused by the filarial nematode *O. volvulus*, which infects an estimated 13 million individuals. The majority of individuals infected with *O. volvulus* live in the equatorial region of Africa extending from the Atlantic coast to the Red Sea. About 70,000 persons are infected in Guatemala and Mexico, with smaller foci in Venezuela, Colombia, Brazil, Ecuador, Yemen, and Saudi Arabia. Onchocerciasis is the second leading cause of infectious blindness worldwide.

ETIOLOGY AND EPIDEMIOLOGY

Infection in humans begins with the deposition of infective larvae on the skin by the bite

of an infected blackfly. The larvae develop into adults, which are typically found in subcutaneous nodules. About 7 months to 3 years after infection, the gravid female releases microfilariae that migrate out of the nodule and throughout the tissues, concentrating in the dermis. Infection is transmitted to other persons when a female fly ingests microfilariae from the host's skin and these microfilariae then develop into infective larvae. Adult *O. volvulus* females and males are about 40 to 60 cm and 3 to 6 cm in length, respectively. The life span of adults can be as long as 18 years, with an average of ~9 years. Because the blackfly vector breeds along free-flowing rivers and streams (particularly in rapids) and generally restricts its flight to an area within several kilometers of these breeding sites, both biting and disease transmission are most intense in these locations.

PATHOLOGY

Onchocerciasis affects primarily the skin, eyes, and lymph nodes. In contrast to that in lymphatic filariasis, the damage in onchocerciasis is elicited by microfilariae and not by adults. In the skin, there are mild but chronic inflammatory changes that can result in loss of elastic fibers, atrophy, and fibrosis. The subcutaneous nodules, or onchocercomata, consist primarily of fibrous tissues surrounding the adult worm, often with a peripheral ring of inflammatory cells. In the eye, neovascularization and corneal scarring lead to corneal opacities and blindness. Inflammation in the anterior and posterior chambers frequently results in anterior uveitis, chorioretinitis, and optic atrophy. Although punctate opacities are due to an inflammatory reaction surrounding dead or dying microfilariae, the pathogenesis of most manifestations of onchocerciasis is still unclear.

CLINICAL FEATURES

Skin Pruritus and rash are the most frequent manifestations of onchocerciasis. The pruritus can be incapacitating; the rash is typically a papular eruption that is generalized rather than localized to a particular region of the body. Long-term infection results in exaggerated and premature wrinkling of the skin, loss of elastic fibers, and epidermal atrophy that can lead to loose, redundant skin and hypo- or hyperpigmentation. Localized eczematoid dermatitis can cause hyperkeratosis, scaling, and pigmentary changes. Such lesions are often seen in the lower extremities but can be distributed more extensively.

Onchocercomata These subcutaneous nodules, which can be palpable and/or visible, contain the adult worm. In African patients, they are common over the coccyx and sacrum, the trochanter of the femur, the lateral anterior crest, and other bony prominences; in Latin American patients, they tend to develop preferentially in the upper part of the body, particularly on the head, neck, and shoulders. Nodules vary in size and characteristically are firm and not tender. It has been estimated that, for every palpable nodule, there are four deeper nonpalpable ones.

Ocular Tissue Visual impairment is the most serious complication of onchocerciasis and usually affects only those persons with moderate or heavy infections. Lesions may develop in all parts of the eye. The most common early finding is conjunctivitis with photophobia. In the cornea, punctate keratitis -- consisting of acute inflammatory

reactions surrounding dying microfilariae manifested as "snowflake" opacities -- is frequent in younger patients and resolves without apparent complications. Sclerosing keratitis occurs in 1 to 5% of infected persons and is the leading cause of onchocercal blindness in Africa. Anterior uveitis and iridocyclitis develop in ~5% of infected persons in Africa. In Latin America, complications of the anterior uveal tract (pupillary deformity) may cause secondary glaucoma. Characteristic chorioretinal lesions develop as a result of atrophy and hyperpigmentation of the retinal pigment epithelium. Constriction of the visual field and frank optic atrophy may occur.

Lymph Nodes Mild to moderate lymphadenopathy is frequent, particularly in the inguinal and femoral areas, where the enlarged nodes may hang down in response to gravity ("hanging groin"), sometimes predisposing to inguinal and femoral hernias.

Systemic Manifestations Some heavily infected individuals develop cachexia with loss of adipose tissue and muscle mass. Among adults who become blind, there is a three- to fourfold increase in the mortality rate.

DIAGNOSIS

Definitive diagnosis depends on the detection of an adult worm in an excised nodule or, more commonly, of microfilariae in a skin snip. Skin snips are obtained with a corneal-scleral punch, which collects a blood-free skin biopsy sample extending to just below the epidermis, or by lifting of the skin with the tip of a needle and excision of a small (1- to 3-mm) piece with a sterile scalpel blade. The biopsy tissue is incubated in tissue culture medium or in saline on a glass slide or flat-bottomed microtiter plate. After incubation for 2 to 4 h (or occasionally overnight in light infections), microfilariae emergent from the skin can be visualized by low-power microscopy.

Eosinophilia and elevated serum IgE levels are common but, because they occur in many parasitic infections, are not diagnostic in themselves. Assays to detect specific antibodies to *Onchocerca* and [PCR](#) to detect onchocercal DNA in skin snips are now in use in specialized laboratories and are highly sensitive and specific.

The *Mazzotti test* is a provocative technique that can be used in cases where the diagnosis of onchocerciasis is still in doubt (i.e., when skin snips and ocular examination reveal no microfilariae). A small dose of [DEC](#) (0.5 to 1.0 mg/kg) is given orally; the development or exacerbation of pruritus or rash within hours is highly suggestive of onchocerciasis.

TREATMENT

The main goals of therapy are to prevent the development of irreversible lesions and to alleviate symptoms. Surgical excision is recommended when nodules are located on the head (because of the proximity of microfilaria-producing adult worms to the eye), but chemotherapy is the mainstay of management. Ivermectin, a semisynthetic macrocyclic lactone active against microfilariae, is the first-line agent for the treatment of onchocerciasis. It is given orally in a single dose of 150 ug/kg, either yearly or semiannually. After treatment, most individuals have few or no reactions. Pruritus, cutaneous edema, and/or maculopapular rash occurs in ~1 to 10% of treated

individuals. In areas of Africa coendemic for *O. volvulus* and *L. loa*, however, ivermectin is contraindicated (as it is for pregnant or breastfeeding women) because of severe posttreatment encephalopathy seen in patients, especially children, who are heavily microfilaremic for *L. loa* (>2000 to 5000 microfilariae per milliliter). Although ivermectin treatment results in a marked drop in microfilarial density, its effect can be short-lived (<6 months in some cases). Thus, it is occasionally necessary to give ivermectin more frequently for persistent symptoms. No currently available agent kills adult *O. volvulus*.

PREVENTION

Vector control has been beneficial in highly endemic areas in which breeding sites are vulnerable to insecticide spraying, but most areas endemic for onchocerciasis are not suited to this type of control. Community-based administration of ivermectin every 6 to 12 months is now being used to interrupt transmission in endemic areas. This measure, in conjunction with vector control, has already helped reduce the prevalence of disease in endemic foci in Africa and Latin America. No drug has proven useful for prophylaxis of *O. volvulus* infection.

LOIASIS

ETIOLOGY AND EPIDEMIOLOGY

Loiasis is caused by *L. loa* (the African eye worm), which is present in the rain forests of West and Central Africa. Adult parasites (females, 50 to 70 mm long and 0.5 mm wide; males, 25 to 35 mm long and 0.25 mm wide) live in subcutaneous tissues; microfilariae circulate in the blood with a diurnal periodicity that peaks between 12:00 noon and 2:00 P.M.

CLINICAL FEATURES

Manifestations of loiasis in natives of endemic areas may differ from those in temporary residents or visitors. Among the indigenous population, loiasis is often an asymptomatic infection with microfilaremia. Infection may be recognized only after subconjunctival migration of an adult worm or may be manifested by episodic Calabar swellings, evanescent localized areas of angioedema and erythema developing on the extremities and less frequently at other sites. Nephropathy, encephalopathy, and cardiomyopathy are rare. In patients who are not residents of endemic areas, allergic symptoms predominate, episodes of Calabar swelling tend to be more frequent and debilitating, microfilaremia is rare, and eosinophilia and increased levels of antifilarial antibodies are characteristic.

PATHOLOGY

The pathogenesis of the manifestations of loiasis is poorly understood. Calabar swellings are thought to result from a hypersensitivity reaction to the adult worm.

DIAGNOSIS

Definitive diagnosis of loiasis requires the detection of microfilariae in the peripheral

blood or the isolation of the adult worm from the eye or from a subcutaneous biopsy specimen from a site of swelling developing after treatment. [PCR](#)-based assays for the detection of *L. loa* DNA in blood are now available in specialized laboratories and are highly sensitive and specific. In practice, the diagnosis must often be based on a characteristic history and clinical presentation, blood eosinophilia, and elevated levels of antifilarial antibodies, particularly in travelers to an endemic region, who are usually amicrofilaremic. Other clinical findings in the latter individuals include hypergammaglobulinemia, elevated levels of serum IgE, and elevated leukocyte and eosinophil counts.

TREATMENT

[DEC](#) (8 to 10 mg/kg per day for 21 days) is effective against both the adult and the microfilarial forms of *L. loa*, but multiple courses are frequently necessary before the disease resolves completely. In cases of heavy microfilaremia, allergic or other inflammatory reactions can take place during treatment, including central nervous system involvement with coma and encephalitis. Heavy infections can be treated initially with apheresis to remove the microfilariae and with glucocorticoids (40 to 60 mg of prednisone per day) followed by doses of DEC (0.5 mg/kg per day). If antifilarial treatment has no adverse effects, the prednisone dose can be rapidly tapered and the dose of DEC gradually increased to 8 to 10 mg/kg per day.

Albendazole and ivermectin (although not approved by the [FDA](#)) have been shown to be effective in reducing microfilarial loads. [DEC](#) (300 mg weekly) is an effective prophylactic regimen for loiasis.

STREPTOCERCIASIS

Mansonella streptocerca, found mainly in the tropical forest belt of Africa from Ghana to Zaire, is transmitted by biting midges. The major clinical manifestations involve the skin and include pruritus, papular rashes, and pigmentation changes. Many infected individuals have inguinal adenopathy, although most are asymptomatic. The diagnosis is made by detection of the characteristic microfilariae in skin snips. [DEC](#) (6 mg/kg per day in divided doses for 14 to 21 days) is effective in killing both microfilariae and adult worms. As in onchocerciasis, treatment is sometimes accompanied by urticaria, arthralgias, myalgias, headaches, and abdominal discomfort. Ivermectin at a single dose of 150 ug/kg leads to sustained suppression of microfilariae in the skin and is likely to assume primacy in the treatment of streptocerciasis.

MANSONELLA PERSTANS INFECTION

Mansonella perstans, distributed across the center of Africa and in northeastern South America, is transmitted by midges. Adult worms reside in serous cavities -- pericardial, pleural, and peritoneal -- as well as in the mesentery and the perirenal and retroperitoneal tissues. Microfilariae circulate in the blood without periodicity. The clinical and pathologic features of the infection are poorly defined. Most patients appear to be asymptomatic, but manifestations may include transient angioedema and pruritus of the arms, face, or other parts of the body (analogous to the Calabar swellings of loiasis); fever; headache; arthralgias; and right upper quadrant pain. Occasionally, pericarditis

and hepatitis occur. The diagnosis is based on the demonstration of microfilariae in blood or serosal effusions. Perstans filariasis is often associated with peripheral blood eosinophilia and antifilarial antibody elevations. Although DEC (8 to 10 mg/kg per day for 21 days) is the standard therapeutic agent, there is little evidence that it is effective. Cure is indicated by the disappearance of symptoms and eosinophilia; multiple courses of therapy are usually required. Both mebendazole (100 mg twice daily for 30 days) and albendazole (400 mg twice daily for 10 days) have been reported to be effective.

MANSONELLA OZZARDI INFECTION

The distribution of *Mansonella ozzardi* is restricted to Central and South America and certain Caribbean islands. Adult worms are rarely recovered from humans. Microfilariae circulate in the blood without periodicity. Although this organism has often been considered nonpathogenic, headache, articular pain, fever, pulmonary symptoms, adenopathy, hepatomegaly, pruritus, and eosinophilia have been ascribed to *M. ozzardi* infection. Diagnosis is made by the detection of microfilariae in peripheral blood. Ivermectin (a single dose of 6 mg) has been shown to be effective in treating this infection.

DRACUNCULIASIS (GUINEA WORM INFECTION)

ETIOLOGY AND EPIDEMIOLOGY

Dracunculiasis, caused by *Dracunculus medinensis*, is a parasitic infection whose incidence has declined dramatically because of global eradication efforts. Current estimates suggest that there are only 78,000 cases worldwide, the majority in Sudan. Humans acquire this infection when they ingest water containing infective larvae derived from *Cyclops*, a crustacean that is the intermediate host. Larvae penetrate the stomach or intestinal wall, mate, and mature. The adult male probably dies; the female *Dracunculus* develops over a year and migrates to subcutaneous tissues, usually in the lower extremity. As the thin female *Dracunculus*, ranging in length from 300 cm to 1 m, approaches the skin, a blister forms that, over days, breaks down and forms an ulcer. When the blister opens, large numbers of motile, rhabditiform larvae can be released into stagnant water; ingestion by *Cyclops* completes the life cycle.

CLINICAL FEATURES

Few or no clinical manifestations of dracunculiasis are evident until just before the blister forms, when there is an onset of fever and generalized allergic symptoms, including periorbital edema, wheezing, and urticaria. The emergence of the worm is associated with local pain and swelling. When the blister ruptures (usually as a result of immersion in water), the adult worm releases larva-rich fluid, and this release is associated with a relief of symptoms. The shallow ulcer surrounding the emerging adult worm heals over weeks to months. Such ulcers, however, can become secondarily infected, the result being cellulitis, local inflammation, abscess formation, or (uncommonly) tetanus. Occasionally, the adult worm does not emerge but becomes encapsulated and calcified.

DIAGNOSIS

The diagnosis is based on the findings developing with the emergence of the adult worm, as described above.

TREATMENT

Gradual extraction of the worm by winding of a few centimeters on a stick each day remains the common and effective practice. Worms may be excised surgically. The administration of thiabendazole (25 mg/kg twice daily for 3 days) or metronidazole (250 mg three times daily for 10 days) may relieve symptoms but has no proven activity against the worm.

PREVENTION

Prevention, which remains the only real control measure, depends on the provision of safe drinking water.

ZOONOTIC FILARIAL INFECTIONS

Dirofilariae that affect primarily dogs, cats, and raccoons and *Brugia* parasites that affect small mammals occasionally infect humans incidentally. Because humans are an abnormal host, the parasites never develop fully. Pulmonary dirofilarial infection caused by the canine heartworm *Dirofilaria immitis* generally presents in humans as a solitary pulmonary nodule. Chest pain, hemoptysis, and cough are uncommon. Infections with *D. repens* (from dogs) or *D. tenuis* (from raccoons) can cause local subcutaneous nodules in humans. Zoonotic *Brugia* infection can produce isolated lymph node enlargement. Eosinophilia levels and antifilarial antibody titers are not commonly elevated. Excisional biopsy is both diagnostic and curative; these infections usually do not respond to chemotherapy.

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222. SCHISTOSOMIASIS AND OTHER TREMATODE INFECTIONS- Adel A.F. Mahmoud

Trematodes, or flatworms, are a group of morphologically and biologically heterogeneous parasitic helminths that belong to the phylum Platyhelminthes. Human infection with trematodes occurs in many geographic areas and can cause considerable morbidity and mortality. For clinical purposes, the significant trematode infections of humans may be divided according to the tissues invaded by adult flukes: blood, biliary tree, intestines, and lungs ([Table 222-1](#)).

Trematodes share some common morphologic features, including macroscopic size (from 1 cm to several cm); dorsoventral, flattened, bilaterally symmetric bodies (adult worms); and the prominence of two suckers. Except for the schistosomes, all trematodes that parasitize humans are hermaphroditic. The life cycle of trematodes involves a definitive host (mammalian/human), in which adult worms initiate sexual reproduction, and an intermediate host (snails, fish, etc.), in which asexual multiplication of the larval forms occurs. More than one intermediate host may be necessary for some species of trematodes. Human infection is initiated either by direct penetration of intact skin or by ingestion. Upon maturation within the human host, adult flukes initiate sexual reproduction that results in egg production. Helminth ova leave the definitive host in excreta or sputum and, upon reaching suitable environmental conditions, they hatch, releasing free-living miracidia that must find a specific snail intermediate host. After asexual reproduction, cercariae are released from infected snails; these organisms either infect humans (schistosomes) or must find another intermediate host to allow encystment into metacercariae.

The host-parasite relationship in trematode infections is a product of the biologic features of these organisms: they are multicellular, undergo several developmental changes within the host, and usually result in chronic infections. In general, the distribution of worm infections in human populations is overdispersed; i.e., it follows a negative binomial mathematical relationship in which most infected individuals harbor low worm burdens while a small percentage are heavily infected. It is the heavily infected minority who are particularly prone to disease sequelae and who represent an epidemiologically significant reservoir of infection in endemic areas. It is important to appreciate that worms do not multiply within the definitive host and that they have a relatively long life span, ranging from a few months to a few years. Morbidity and mortality due to trematode infections reflect a multifactorial process that results from the tipping of a delicate balance based on the intensity of infection and the host reactions that initiate and modulate pathologic outcome. The genetics of the parasite and the human host contribute to the outcome of infection and disease. Furthermore, infections with trematodes that migrate through or reside in host tissues are associated with a moderate to high degree of peripheral blood eosinophilia; this association is of significance in protective and immunopathologic sequelae and is a useful clinical indicator of infection.

Approach to the Patient

The approach to individuals with suspected trematode infection begins with the question: Where have you been? Details of geographic history, exposure to freshwater

bodies, and indulgence in local eating habits without ensuring safety of food and drink are all essential elements in the history. The workup plan must include a detailed physical examination and tests appropriate for the suspected infection. Diagnosis is based either on detection of the relevant stage of the parasite in excreta, sputum, or (rarely) tissue samples or on sensitive and specific serologic tests. Consultation with physicians familiar with these infections or with the U.S. Centers for Disease Control and Prevention (CDC) is helpful in guiding diagnosis and selecting therapy.

BLOOD FLUKES: SCHISTOSOMIASIS

Human schistosomiasis is caused by five species of this parasitic trematode belonging to the subclass Digenea: the intestinal species *Schistosoma mansoni*, *S. japonicum*, *S. mekongi*, and *S. intercalatum* and the urinary species *S. haematobium*. Infection may cause considerable morbidity in the intestines, liver, and urinary tract, and a proportion of affected individuals die. Other schistosome species (e.g., avian species) may invade human skin but then die in subcutaneous tissue, producing only self-limiting cutaneous manifestations.

Information on the prevalence and geographic distribution of human schistosomiasis is inexact. The five species are estimated to infect 200 to 300 million people in South America, the Caribbean, Africa, the Middle East, and Southeast Asia. The total population living under conditions favoring transmission approximates double or triple that number -- a fact reflecting the public health significance of schistosomiasis.

ETIOLOGY

Human infection is initiated by penetration of intact skin with infective cercariae. These organisms are released from infected snails in freshwater bodies; they measure ~2 mm in length and possess an anterior and a ventral sucker that attaches to the skin surface and facilitates penetration. Once in the subcutaneous tissue, the organism transforms into the next stage: the schistosomula. This transformation involves morphologic, membrane, and immunologic changes, prominent among which is the transformation of the cercarial outer membrane from a trilaminar to a heptalaminar structure that is then maintained throughout the life span of the worms in humans. The transformation to a heptalaminar structure is thought to be the schistosome's main adaptive mechanism for survival in humans. Schistosomula begin their migration within 2 to 4 days via venous or lymphatic vessels, reaching the lungs and finally the liver parenchyma. Sexually mature worms descend in pairs into the venous system at specific anatomic locations: intestinal veins (*S. mansoni*, *S. japonicum*, *S. mekongi*, and *S. intercalatum*) and vesical veins (*S. haematobium*). Adult gravid females then travel against venous blood flow to small tributaries, where they deposit their ova intravascularly. Schistosome ova have specific morphologic features that can be used to differentiate species. Aided by enzymatic secretions through minipores in eggshells, ova move through the venous wall, traversing host tissues to reach the lumen of the intestinal or urinary tract, and are voided with stools or urine. Approximately 50% of ova, however, fail in their attempt to be transported to the outside environment and are either retained in host tissues locally (intestines or urinary tract) or carried by venous blood flow to the liver and other organs. Schistosome ova that reach freshwater bodies hatch, releasing free-living miracidia that seek the snail intermediate host to undergo several asexual multiplication cycles.

Finally, infective cercariae are shed from snails.

Adult schistosome worms measure ~1 to 2 cm in length. The male is slightly shorter, with a flattened body; its edges curve anteriorly to form the gynecophoral canal, in which mature adult females are usually held. The females are longer, slender, and rounded in cross-section. The precise nature of biochemical and reproductive exchanges between the two sexes is unknown, as are the regulatory mechanisms for pairing. Adult schistosomes parasitize specific sites in the host venous system. What guides adult intestinal schistosomes to branches of the superior or inferior mesenteric veins or adult *S. haematobium* worms to the vesical plexus is unknown. In addition, the evasion mechanisms by which adult worms inhibit the coagulation cascade and the effector arms of the host immune responses are not fully understood.

A systematic examination of schistosomal molecular phylogeny as well as gene structure and organization has begun. Analysis of the sequence of nuclear ribosome and internal transcribed spacer 2 and mitochondrial 16 sRNA indicates that human schistosomes may be divided into three monophyletic groups: the *S. haematobium* group, including *S. haematobium* and *S. intercalatum*; the *S. mansoni* group, including *S. mansoni* and *S. rodhaini*; and a group containing the two Asian schistosomes, *S. japonicum* and *S. mekongi*. These molecular groupings coincide with results of previous attempts to use morphologic or nucleic acid data to produce a taxonomic framework. In other studies, the schistosome genome was determined to be made up of 16 chromosomes; sexual differentiation is related to the presence of ZW chromosomes in females and ZZ chromosomes in males.

EPIDEMIOLOGY

The distribution of schistosome infection and related disease syndromes in human populations is dependent on both parasite and host factors. In endemic areas, the rate of yearly onset of new infection, or incidence, is low. Prevalence, on the other hand, starts to be appreciable by the age of 3 to 4 years and builds to a maximum that varies by endemic region (up to 100%) in the 15- to 20-year age group. Prevalence then stabilizes or decreases slightly in older age groups (>40 years). Intensity of infection (as measured by fecal or urinary egg counts, which correlate with adult worm burdens in most circumstances) follows the increase in prevalence up to the age of 15 to 20 years and then declines markedly in older age groups. This decline may reflect acquisition of resistance, or it may be due to changes in water contact patterns, since older people are exposed less. Furthermore, the unique distribution of schistosomes in human populations, which fits a negative binomial pattern (see above), may be due to heterogeneity of worm populations, with some more invasive than others; alternatively, it may be due to differences in the genetic susceptibility of host populations.

Disease due to schistosomiasis is the outcome of parasitologic, host, and additional infectious, nutritional, and environmental factors. Most of the disease syndromes relate to the presence of one or more of the parasite stages in the human host. The distribution of disease manifestations in the populations of endemic areas correlates with the intensity and duration of infection as well as with the age and genetic susceptibility of the host. Overall, disease manifestations are clinically relevant in only a small proportion of persons infected with any of the intestinal schistosomes. In contrast,

urinary schistosomiasis manifests clinically in most infected individuals.

Patients with both HIV infection and schistosomiasis have been found to excrete far fewer eggs in their stools than those infected with *S. mansoni* alone. The two groups have responded equally to treatment with praziquantel.

PATHOGENESIS AND IMMUNITY

During the invasive stage, cercaria-associated dermatitis reflects dermal and subdermal inflammatory responses -- both humoral and cell-mediated. As the parasites approach sexual maturity and the commencement of oviposition, acute schistosomiasis or Katayama fever (a serum sickness-like illness; see "Clinical Features," below) may occur. The associated antigen excess results in the formation of soluble immune complexes, which may be deposited in several tissues, initiating the sequence of pathologic events. In chronic schistosomiasis, most disease manifestations are due to eggs retained in host tissue. The granulomatous response around these ova is cell-mediated and is regulated both positively and negatively by a cascade of cytokine, cellular, and humoral responses. Granuloma formation begins with recruitment of a host of inflammatory cells in response to antigens secreted by the living organism within the ova. Cells recruited initially include phagocytes, antigen-specific T cells, and eosinophils. Fibroblasts, giant cells, and B lymphocytes predominate later. Once activated, T cells produce cytokines [such as tumor necrosis factor (TNF- α), interleukin (IL) 2, IL-4, and IL-5, which in turn activate endothelial cells] and produce specific chemokines such as monocyte chemoattractant protein 1 (MCP-1). The result is recruitment of the cellular elements that organize in the form of granulomas around parasite eggs. These lesions reach a size many times that of the eggs, thus inducing organomegaly and obstruction. Immunomodulation or downregulation of host responses to schistosome eggs plays a significant role in limiting the extent of the granulomatous lesions -- and consequently disease -- in chronically infected experimental animals or humans. The underlying mechanisms involve another cascade of regulatory cytokines (IL-10, IL-12) and idiotype antibodies. Subsequent to the granulomatous response, fibrosis sets in, resulting in more permanent disease sequelae. Because schistosomiasis is a chronic infection, the accumulation of antigen-antibody complexes results in deposits in renal glomeruli and may cause significant kidney disease.

The better-studied pathologic sequelae in schistosomiasis are those observed in liver disease. Ova that are carried by portal blood embolize to the liver. Because of their size (~150 $\times$ 60 μ m in the case of *S. mansoni*), they lodge at presinusoidal sites, where granulomas are formed. The granulomas contribute to the liver enlargement observed in infected individuals. Schistosomal hepatomegaly is also associated with certain class I and class II HLA markers; its genetic basis appears to be multigenic. Presinusoidal portal blockage causes several hemodynamic changes, including portal hypertension and associated development of portosystemic collaterals at the esophagogastric junction and other sites. Esophageal varices are most likely to break and cause repeated episodes of hematemesis. Because changes in liver hemodynamics in schistosomiasis are slow, compensatory arterialization of blood flow through the liver is established. While this compensatory mechanism may be associated with certain metabolic side effects, the retention of hepatocyte perfusion may permit the maintenance of normal liver function for several years.

After granuloma formation, the second most significant pathologic change in the liver relates to the onset of fibrosis. It is characteristically periportal (Symmers' clay-pipe stem fibrosis) but may be diffuse. Fibrosis, when diffuse, may be seen in areas of egg deposition and granuloma formation, but it is also seen in distant locations such as portal tracts. Schistosomiasis alone results in pure fibrotic lesions in the liver; cirrhosis occurs when other nutritional or infectious agents (e.g., hepatitis B or C virus) are involved. In recent years, it has been recognized that deposition of fibrotic tissue in the extracellular matrix results from the interaction of T lymphocytes with cells of the fibroblast series; several cytokines, such as IL-2, IL-4, IL-1, and transforming growth factor b(TGF-b), are known to stimulate fibrogenesis. The process may be dependent on the genetic constitution of the host. Furthermore, regulatory cytokines that can suppress fibrogenesis, such as interferon-g (IFN-g) or IL-12, may play a role in modulating the response.

While the above description focuses on granuloma formation and fibrosis of the liver, similar processes occur in urinary schistosomiasis. Granuloma formation at the lower end of the ureters obstructs urinary flow, with subsequent development of hydronephrosis and hydronephrosis. Similar lesions in the urinary bladder cause the protrusion of papillomatous structures into its cavity; these may ulcerate and/or bleed. The chronic stage of infection is associated with scarring and deposition of calcium in the bladder wall.

Immunomodulation is an essential mechanism in shaping the clinical and pathologic outcome of schistosomiasis. While most detailed immunologic analyses have been performed in experimental animals, enough evidence exists from studies in humans to delineate the suppression of T cell responses in association with active infections and a regulatory role for IL-10.

Studies on immunity to schistosomiasis, whether innate or acquired, have expanded our knowledge of the components of these responses and the target antigens. The concept of innate immunity is illustrated by the inability of avian schistosomes, which cause swimmers' itch, to reach maturity in humans. The critical question, however, is whether humans acquire immunity to schistosomes. Epidemiologic evidence suggests the onset of acquired immunity during the course of infection in young adults. Curative treatment of infection divides populations in endemic areas into those who acquire reinfection rapidly (susceptible) and those who follow a protracted course (resistant). This difference may be explained by differences in transmission, immunologic response, or genetic susceptibility. The mechanism of acquired immunity involves antibodies, complement, and several effector cells, particularly eosinophils. Furthermore, the intensity of schistosome infection has been correlated with a region in chromosome 5. In other studies, several protective schistosome antigens have been identified as vaccine candidates.

CLINICAL FEATURES

In general, disease manifestations of schistosomiasis occur in three stages, which vary not only by species but also by intensity of infection and other host factors, such as age and genetics. During the phase of cercarial invasion, a form of dermatitis may be

observed. This so-called swimmers' itch ([Fig. 222-CD1](#)) occurs most often with *S. mansoni* and *S. japonicum* infections, manifesting 2 or 3 days after invasion as an itchy maculopapular rash on the affected areas of the skin. The condition is particularly severe when humans are exposed to avian schistosomes. This form of cercarial dermatitis is seen around the freshwater lakes in the northern United States, particularly in the spring. Cercarial dermatitis is a self-limiting clinical entity. During worm maturation and at the beginning of oviposition (i.e., 4 to 8 weeks after skin invasion), acute schistosomiasis or Katayama fever -- a serum sickness-like syndrome with fever, generalized lymphadenopathy, and hepatosplenomegaly -- may develop. Individuals suffering from acute schistosomiasis show a high degree of peripheral blood eosinophilia. Parasite-specific antibodies may be detected before schistosome eggs are identified in excreta. Acute schistosomiasis has become an important clinical entity worldwide because of increased travel to endemic areas. Travelers are exposed to the parasite while swimming or wading in freshwater bodies and upon their return present with the acute manifestations of the disease. The course of acute schistosomiasis is generally benign, but deaths are occasionally reported in association with heavy exposure to schistosomes.

The main clinical manifestations of chronic schistosomiasis are species-dependent. Intestinal species (*S. mansoni*, *S. japonicum*, *S. mekongi*, and *S. intercalatum*) cause intestinal and hepatosplenic disease as well as several manifestations associated with portal hypertension. During the intestinal phase, which may begin a few months after infection and may last for years, symptomatic patients characteristically have colicky abdominal pain and bloody diarrhea. Patients may also report fatigue and an inability to perform daily routine functions and may show evidence of growth retardation. The severity of intestinal schistosomiasis is often related to the intensity of the worm burden. The disease runs a chronic course but rarely progresses to a functional level (e.g., malabsorption) or to anatomic lesions of the gut. The exception is colonic polyposis, which has been seen in some endemic areas, such as Egypt.

The hepatosplenic phase of disease manifests early (during the first year of infection, particularly in children) with enlargement of the liver due to parasite-induced granulomatous lesions. Hepatomegaly is seen in ~15 to 20% of infected individuals when whole communities in endemic areas are studied. It correlates roughly with the intensity of infection, occurs more often in children than in adults, and may be related to specific HLA haplotypes. In subsequent phases of infection, presinusoidal blockage of blood flow leads to portal hypertension and splenomegaly. Moreover, portal hypertension may lead to varices at the lower end of the esophagus and at other sites. Patients with schistosomal liver disease may have right-upper-quadrant "dragging" pain during the hepatomegaly phase, and this pain may move to the left upper quadrant as splenomegaly progresses. Bleeding from esophageal varices may, however, be the first clinical manifestation of this phase. Patients may experience repeated bleeding but seem to tolerate its impact, since an adequate total hepatic blood flow permits normal liver function for a considerable period in schistosomal hepatomegaly. In late-stage disease, typical fibrotic changes occur along with liver function deterioration and the onset of ascites, hypoalbuminemia, and defects in coagulation. Intercurrent viral infections of the liver or nutritional deficiencies may well accelerate or exacerbate the deterioration of hepatic function.

The extent and severity of intestinal and hepatic disease in schistosomiasis *mansoni* and *japonica* have been well described. While it was originally thought that *S. japonicum* might induce more severe disease manifestations because the adult worms can produce ten times more eggs than *S. mansoni*, subsequent field studies have not supported this claim. Clinical observations of individuals infected with *S. mekongi* or *S. intercalatum* have been less detailed, partly because of the far more limited geographic distribution of these organisms.

The clinical manifestations of *S. haematobium* infection occur relatively early and involve a relatively high percentage of individuals. Up to 80% of children infected with *S. haematobium* have dysuria, frequency, and hematuria, which may be terminal. Urine examination reveals blood and albumin as well as an unusually high frequency of bacterial urinary tract infection. These manifestations correlate with intense infection, the presence of urinary bladder granulomas, and subsequent ulceration. Along with the local effects of granuloma formation in the urinary bladder, obstruction of the lower end of the ureters results in hydroureter and hydronephrosis, which can be seen in 25 to 50% of infected children. As infection progresses, bladder granulomas undergo fibrosis; the result is the presence of typical sandy patches visible on cystoscopy. In many endemic areas, an association between squamous cell carcinoma of the bladder and *S. haematobium* infection has been observed. Such malignancy is detected in a younger age group than transitional cell carcinoma. In fact, *S. haematobium* has now been classified as a human carcinogen.

Significant disease may occur in other organs during chronic schistosomiasis. Most important is disease in the lungs and central nervous system; other locations, such as the skin and the genital organs, are far less frequently affected. In pulmonary schistosomiasis, embolized eggs lodge in small arterioles, producing acute necrotizing arteriolitis and granuloma formation. During *S. mansoni* and *S. japonicum* infection, schistosome eggs reach the lungs after the development of portosystemic collateral circulation; in *S. haematobium* infection, ova may reach the lungs directly via connections between the vesical and systemic circulation. After the development of arteriolitis and granuloma formation, fibrous tissue deposition is detected and leads to endarteritis obliterans, pulmonary hypertension, and cor pulmonale. This clinical entity is an uncommon presentation during chronic schistosomiasis. The most frequent symptoms are cough, fever, and dyspnea; ascites and hemoptysis are less frequently encountered. Cor pulmonale may be diagnosed radiologically on the basis of prominent right side of the heart and dilation of the pulmonary artery. Frank evidence of right-sided heart failure may be seen in late cases.

Central nervous system schistosomiasis is important but less frequent than pulmonary schistosomiasis. It characteristically occurs as cerebral disease due to *S. japonicum* infection. Migratory worms deposit eggs in the brain and induce a granulomatous response. The frequency of this manifestation among infected individuals in some endemic areas (e.g., the Philippines) is calculated at 2 to 4%. Jacksonian epilepsy due to *S. japonicum* infection is the second most common cause of epilepsy in these areas. *S. mansoni* and *S. haematobium* infections have been associated with transverse myelitis. This syndrome is thought to be due to eggs traveling to the venous plexus around the spinal cord. In schistosomiasis *mansoni*, transverse myelitis is usually seen in the chronic stage after the development of portal hypertension and portosystemic

shunts, which allow ova to travel to the spinal cord veins. This proposed sequence of events has been challenged because of a few reports of transverse myelitis occurring early in the course of *S. mansoni* infection. More information is needed to confirm these observations. During schistosomiasis haematobia, ova may travel through communication between vesical and systemic veins, resulting in spinal cord disease that may be detected at any stage of infection. Pathologic study of lesions in schistosomal transverse myelitis may reveal eggs along with necrotic or granulomatous lesions. Patients usually present with acute or rapidly progressing lower-leg weakness accompanied by sphincter dysfunction.

DIAGNOSIS

Physicians in areas not endemic for schistosomiasis face considerable diagnostic challenges. In the most common clinical presentation, a returning traveler exhibits symptoms and signs of any of the acute syndromes of schistosomiasis -- namely, cercarial dermatitis or Katayama fever. Central to correct diagnosis is a thorough inquiry into travel history and exposure to freshwater bodies, whether slow or fast running. Differential diagnosis of fever in returned travelers includes a spectrum of infections whose etiologies are viral (e.g., Dengue fever), bacterial (e.g., enteric fever, leptospirosis), rickettsial, or protozoal (e.g., malaria). In cases of Katayama fever, prompt diagnosis is essential and is based on clinical presentation, high-level peripheral blood eosinophilia, and a positive serologic assay for schistosomal antibodies. Two tests are available at the [CDC](#): the Falcon assay screening test/enzyme-linked immunosorbent assay (FAST-ELISA) and the confirmatory enzyme-linked immunoelectrotransfer blot (EITB). Both tests are highly sensitive and ~96% specific. In some instances, examination of stool or urine for ova may yield positive results.

Individuals with established infection are diagnosed by a combination of geographic history, characteristic clinical presentation, and presence of schistosome ova in excreta. The diagnosis may also be established with the serologic assays mentioned above or with those that detect circulating schistosome antigens. These assays can be applied either to blood or to other body fluids (e.g., cerebrospinal fluid). For stool examination, the Kato thick smear or any other concentration method generally identifies all but the most lightly infected individuals. Urine may be examined by microscopy of sediment or by filtration of a known volume through Nuclepore filters. Kato thick smear and Nuclepore filtration provide quantitative data on the intensity of infection, which is of value in assessing the degree of tissue damage and in monitoring the effect of chemotherapy. Finally, schistosome infection may be diagnosed by examination of tissue samples, typically rectal biopsies; other biopsy procedures (e.g., liver biopsy) are not needed, except in special circumstances.

Differential diagnosis of schistosomal hepatomegaly must include viral hepatitis of all etiologies, miliary tuberculosis, malaria, visceral leishmaniasis, ethanol abuse, and causes of hepatic and portal vein obstruction. Of patients with these conditions, only a few may present with organomegaly and relatively intact liver function. The differential diagnosis of hematuria in *S. haematobium* infection includes bacterial cystitis, tuberculosis, urinary stones, and malignancy.

TREATMENT

Treatment of schistosomiasis depends on the stage of infection and the clinical presentation. Other than topical dermatologic applications for relief of itching, no specific treatment is indicated for cercarial dermatitis caused by avian schistosomes. Therapy for acute schistosomiasis or Katayama fever needs to be adjusted appropriately for each case. While antischistosomal chemotherapy is indicated, it does not address immediate pathologic changes. In severe acute schistosomiasis, management in an acute-care setting is necessary, with supportive measures and consideration of glucocorticoid treatment. Once the acute critical phase is over, specific chemotherapy is indicated. For all individuals with infection established by either the demonstration of schistosome eggs or positive serology, treatment to eradicate the parasite should be administered. The drug of choice is praziquantel, which -- depending on the infecting species ([Table 222-2](#)) -- is administered orally as 40 or 60 mg/kg in two or three doses over a single day. Praziquantel treatment results in parasitologic cure in ~85% of cases and reduces egg counts by >90%. Few side effects have been encountered, and those that do develop usually do not interfere with completion of treatment. Other antischistosomal chemotherapeutic agents are currently considered only as alternatives when praziquantel is unavailable. The effect of antischistosomal treatment on disease manifestations varies by stage. Early hepatomegaly and bladder lesions are known to resolve following chemotherapy, but the late established manifestations, such as fibrosis, do not change. Additional management modalities are needed for individuals with other manifestations, such as hepatocellular failure or recurrent hematemesis. The use of these interventions is guided by general medical and surgical principles.

PREVENTION AND CONTROL

Since transmission of schistosomiasis is dependent on human behavior, it is theoretically possible to devise an effective preventive strategy. The geographic distribution of infections in endemic regions of the world is not clearly demarcated. It is therefore prudent for travelers to avoid contact with all freshwater bodies, irrespective of the speed of water flow or unsubstantiated claims of safety. Some topical agents, when applied to the skin, may conceivably inhibit cercarial penetration, but none of these agents is currently available. If exposure occurs, a follow-up visit with a health care provider is strongly recommended. Prevention of infection in inhabitants of endemic areas is a significant challenge. People of these regions use freshwater bodies for sanitary, domestic, recreational, and agricultural purposes. In the absence of adequate alternatives, several control measures have been used, including application of molluscicides, provision of sanitary water and means for sewage disposal, chemotherapy, and health education. Current recommendations to countries endemic for schistosomiasis emphasize the use of multiple approaches. Particularly with the advent of a single-oral-dose, safe, and effective antischistosomal agent, chemotherapy has been most successful in reducing the intensity of infection and reversing disease. The duration of this positive impact depends on transmission dynamics in a specific endemic region. The ultimate goal of research on prevention and control is the development of a vaccine. Although there are a few promising leads, this goal is probably not within reach during the next decade or so.

LIVER (BILIARY) FLUKES

Several species of biliary fluke infecting humans are particularly common in Southeast Asia and Russia. Other species are transmitted in Europe, Africa, and the Americas. On the basis of their migratory pathway in humans, these infections may be divided into the *Clonorchis* and *Fasciola* groups.

CLONORCHIASIS AND OPISTHORCHIASIS

Infection with *C. sinensis*, the Chinese or oriental fluke, is endemic among fish-eating mammals in Southeast Asia. Humans are an incidental host; the prevalence of human infection is highest in China, Vietnam, and Korea. Infection with *Opisthorchis viverrini* and *O. felineus* is zoonotic in cats and dogs. Transmission to humans occurs occasionally, particularly in Thailand (*O. viverrini*) and in Southeast Asia and eastern Europe (*O. felineus*). Data on the exact geographic distribution of these infectious agents in human populations are rudimentary.

Infection with any of these three species is established by ingestion of raw or inadequately cooked freshwater fish harboring metacercariae. These organisms excyst in the duodenum, releasing larvae that travel through the ampulla of Vater and mature into adult worms in the bile canaliculi. Mature flukes are flat and elongated, measuring 1 to 2 cm in length. The hermaphroditic worms reproduce by releasing small operculated eggs, which pass with bile into the intestines and are voided with stools. The life cycle is completed in the environment in specific freshwater snails (the first intermediate host) and encystment of metacercariae in freshwater fish.

Except for late sequelae, the exact clinical syndromes caused by clonorchiasis and opisthorchiasis are not well defined. Since most infected individuals harbor a low worm burden, many are asymptomatic. Moderate to heavy infection may be associated with vague right-upper-quadrant pain. In contrast, chronic or repeated infection is associated with manifestations such as cholangitis, cholangiohepatitis, and biliary obstruction. Cholangiocarcinoma is epidemiologically related to *C. sinensis* infection in China and to *O. viverrini* infection in northeastern Thailand. This association has resulted in the classification of these infectious agents as human carcinogens.

FASCIOLIASIS

Infections with *F. hepatica* and *F. gigantica* are worldwide zoonoses that are particularly endemic in sheep-raising countries. Human cases have been reported in South America, Europe, Africa, Australia, and the Far East. Recent estimates indicate a worldwide prevalence of 17 million cases. High endemicity has been reported in certain areas of Peru and Bolivia. In most endemic areas the predominant species is *F. hepatica*, but in Asia and Africa a varying degree of overlap with *F. gigantica* has been observed.

Humans acquire fascioliasis by ingestion of metacercariae attached to certain aquatic plants, such as watercress. Infection may also be acquired by consumption of contaminated water or ingestion of food items washed with such water. Acquisition of human infection through consumption of freshly prepared raw liver containing immature flukes has been reported. Infection is initiated when metacercariae excyst, penetrate the gut wall, and travel through the peritoneal cavity to invade the liver capsule. Adult

worms finally reach the bile ducts, where they produce large operculated eggs, which are voided in the bile and through the gastrointestinal tract to the outside environment. The flukes' life cycle is completed in specific snails (the first intermediate host) and encystment on aquatic plants.

The clinical features of fascioliasis relate to the intensity of infection, but even more to the stage of infection. Acute disease develops during the parasites' migration (1 to 2 weeks after infection) and includes fever, right-upper-quadrant pain, hepatomegaly, and eosinophilia. Computed tomography of the liver may show migratory tracks. Symptoms and signs usually subside as the parasites reach their final habitat. In individuals with chronic infection, bile duct obstruction and biliary cirrhosis are infrequently demonstrated. No relation to hepatic malignancy has been ascribed to fascioliasis.

DIAGNOSIS

The diagnosis of infection with any of the biliary flukes depends on a high degree of suspicion, the elicitation of an appropriate geographic history, and stool examination for the characteristically shaped parasite ova. Additional evidence may be obtained by documenting peripheral blood eosinophilia or imaging the liver. Serologic testing is helpful, particularly in lightly infected individuals.

TREATMENT

Drug therapy (praziquantel or triclabendazole) is summarized in [Table 222-2](#). Patients with anatomic lesions in the biliary tract or malignancy are managed according to general medical guidelines.

INTESTINAL FLUKES

Two species of intestinal flukes cause human infection in defined geographic areas worldwide. The large *Fasciolopsis buski* (adults measure 2 by 7 cm) is endemic in Southeast Asia, while the smaller *Heterophyes heterophyes* is found in the Nile Delta of Egypt and in the Far East. Infection is initiated by ingestion of metacercariae attached to aquatic plants (*F. buski*) or encysted in freshwater or brackish-water fish (*H. heterophyes*). Flukes mature in human intestines, and eggs are passed with stools. Most individuals infected with intestinal flukes are asymptomatic. In heavy *F. buski* infection, diarrhea, abdominal pain, and malabsorption may be encountered. Heavy infection with *H. heterophyes* may be associated with abdominal pain and mucous diarrhea. The diagnosis is established by detection of the characteristically shaped ova in stool samples. The drug of choice for treatment is praziquantel ([Table 222-2](#)).

LUNG FLUKES

Infection with the lung fluke *Paragonimus westermani* and related species (e.g., *P. africanus*) is endemic in many parts of the world, excluding North America and Europe. Endemicity is particularly noticeable in West Africa, Central and South America, and Asia. In nature, the reservoir hosts of *P. westermani* are wild and domestic felines. In Africa, *P. africanus* has been found in other species, such as dogs. Adult lung flukes, which are 7 to 12 mm in length, are found encapsulated in the lungs of infected persons.

In rare circumstances, flukes are found encysted in the central nervous system (cerebral paragonimiasis) or abdominal cavity. Humans acquire lung fluke infection by ingesting infective metacercariae encysted in the muscles and viscera of crayfish and freshwater crabs. In endemic areas, these crustaceans are consumed either raw or pickled. Once the organisms reach the duodenum, they excyst, penetrate the gut wall, and travel through the peritoneal cavity, diaphragm, and pleural space to reach the lungs. Mature flukes are found in the bronchioles surrounded by cystic lesions. Parasite eggs are either expectorated with sputum or swallowed and passed to the outside environment with feces. The life cycle is completed in snails and freshwater crustacea.

When maturing flukes lodge in lung tissues, they cause hemorrhage and necrosis, resulting in cyst formation. The adjacent lung parenchyma shows evidence of inflammatory infiltration, predominantly by eosinophils. Cysts usually measure 1 to 2 cm in diameter and may contain 1 or 2 worms each. With the onset of oviposition, cysts usually rupture in adjacent bronchioles -- an event allowing ova to exit from the human host. Older cysts develop thickened walls, which may undergo calcification. During the active phase of paragonimiasis, lung tissues surrounding parasite cysts may contain evidence of pneumonia, bronchitis, bronchiectasis, and fibrosis.

Pulmonary paragonimiasis is particularly symptomatic in persons with moderate to heavy infection. Productive cough with brownish sputum or frank hemoptysis associated with peripheral blood eosinophilia is usually the presenting feature. Chest examination may reveal signs of pleurisy. In chronic cases, bronchitis or bronchiectasis may predominate, but these conditions rarely proceed to lung abscess. Imaging of the lungs demonstrates characteristic features, including patchy densities, cavities, pleural effusion, and ring shadows. Cerebral paragonimiasis presents as either space-occupying lesions or epilepsy. Pulmonary paragonimiasis is diagnosed by the detection of parasite ova in sputum and/or stools. Serology is of considerable help in egg-negative cases and in cerebral paragonimiasis. The drug of choice for treatment is praziquantel ([Table 222-2](#)). Other medical or surgical management may be needed for pulmonary or cerebral lesions.

CONTROL AND PREVENTION OF TISSUE FLUKES

For residents of nonendemic areas who are visiting an endemic region, the only effective preventive measure is to avoid ingestion of local plants, fish, or crustaceans; if their ingestion is necessary, they should be washed or cooked thoroughly. Instruction on water and food preparation and consumption should be included in physicians' advice to travelers ([Chap. 123](#)). Interruption of transmission among residents of endemic areas depends on avoiding ingestion of the infective stage of the helminths and appropriate disposal of feces and sputum to prevent the hatching of eggs in the environment. These two approaches rely greatly on socioeconomic development and health education. In countries where economic progress has resulted in financial and social improvements, transmission has decreased. The third approach to control in endemic communities entails selective use of chemotherapy for individuals posing the highest risk of transmission -- i.e., those with heavy infections. The availability of praziquantel -- a broad-spectrum, safe, and effective antihelminthic agent -- provides a means for reducing the reservoirs of infection in human populations. However, the existence of most of these helminths as zoonoses in several animal species complicates control

efforts.

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223. CESTODES - A. Clinton White, Jr., Peter F. Weller

Cestodes, or tapeworms, are segmented worms. The adults reside in the gastrointestinal tract, but the larvae can be found in almost any organ. Human tapeworm infections can be divided into two major clinical groups. In one group, humans are the definitive hosts, and the adult tapeworms live in the gastrointestinal tract (*Taenia saginata*, *Diphyllobothrium*, *Hymenolepis*, and *Dipylidium caninum*). In the other, humans are intermediate hosts, and larval-stage parasites are present in the tissues. Diseases in this category include echinococcosis, sparganosis, and coenurosis. For *T. solium*, the human may be either the definitive or the intermediate host.

The ribbon-shaped tapeworm attaches to the intestinal mucosa by means of sucking cups or grooves located on the head (scolex). Behind the scolex is a short, narrow neck from which proglottids (segments) form. As each proglottid matures, it is displaced further back from the neck by the formation of new, less mature segments. The progressively elongating chain of attached proglottids, called the *strobila*, constitutes the bulk of the tapeworm. The length varies among species. In some, the tapeworm may consist of more than 1000 proglottids and may be several meters long. As each proglottid becomes gravid, eggs are released. Since eggs of the different *Taenia* species are morphologically identical, differences in the morphology of the scolex or proglottids provide the basis for diagnostic identification to the species level. Most human tapeworms require at least one intermediate host for complete larval development. After ingestion by an intermediate host, an egg releases the larval oncosphere, which penetrates the intestinal mucosa. The oncosphere migrates to tissues and develops into an encysted form known as a *cysticercus* (single scolex), a *coenurus* (multiple scolices), or a *hydatid* (cyst with daughter cysts, each containing several protoscolices). Ingestion by the definitive host of tissues containing a cyst enables a scolex to develop into a tapeworm.

TAENIASIS SAGINATA

The beef tapeworm *T. saginata* occurs in all countries where raw or undercooked beef is eaten. It is most prevalent in sub-Saharan African and Middle Eastern countries.

Etiology and Pathogenesis Humans are the only definitive host for the adult stage of *T. saginata*. This tapeworm, which can reach 8 m in length, inhabits the upper jejunum and has a scolex with four prominent suckers and 1000 to 2000 proglottids. Each gravid segment has 15 to 30 uterine branches (in contrast to 8 to 12 for *T. solium*). The eggs are indistinguishable from those of *T. solium*; each measures 30 to 40 μm and has a thick brown striated shell containing the embryo. Eggs deposited on vegetation can live for months to years until they are ingested by cattle or other herbivores. The embryo released after ingestion invades the intestinal wall and is carried to striated muscle, where it transforms into a cysticercus. When ingested in raw or undercooked beef, this form can infect humans. After the cysticercus is ingested, it takes about 2 months for an adult worm to develop.

Clinical Manifestations Patients become aware of the infection most commonly by noting passage of proglottids in their feces. The proglottids are often motile, and patients may experience perianal discomfort when proglottids are discharged. Mild

abdominal pain or discomfort, nausea, change in appetite, weakness, and weight loss can occur with *T. saginata* infection.

Diagnosis The diagnosis is made by the detection of eggs or proglottids in the stool. Eggs may also be present in the perianal area; thus, if proglottids or eggs are not found in the stool, the perianal region should be examined with use of a cellophane-tape swab (as in pinworm infection). Distinguishing *T. saginata* from *T. solium* requires examination of mature proglottids or the scolex. Serologic tests are not helpful diagnostically. Eosinophilia and elevated levels of serum IgE may be detected.

TREATMENT

A single dose of praziquantel (5 to 10 mg/kg) is highly effective.

Prevention The major method of preventing infection is the adequate cooking of beef; exposure to temperatures as low as 56°C for 5 min will destroy cysticerci. Refrigeration or salting for long periods or freezing at -10°C for 9 days also kills cysticerci in beef. General preventive measures include inspection of beef and proper disposal of human feces.

TAENIASIS SOLIUM AND CYSTICERCOSIS

The pork tapeworm *T. solium* can cause two distinct forms of infection. The form that develops depends on whether humans are infected with adult tapeworms in the intestine or with larval forms in the tissues (cysticercosis). Humans are the only definitive hosts for *T. solium*; pigs are the usual intermediate hosts, although dogs, cats, and sheep may harbor the larval forms. *T. solium* exists worldwide but is most prevalent in Latin America, Africa, South and Southeast Asia, and eastern Europe. Cysticercosis occurs in industrialized nations largely as a result of the immigration of infected persons from endemic areas.

Etiology and Pathogenesis The adult tapeworm generally resides in the upper jejunum. Its globular scolex attaches by both sucking disks and two rows of hooklets. Often only one adult worm is present, but that worm may live for years. The tapeworm, usually about 3 m in length, may have as many as 1000 proglottids, each of which produces up to 50,000 eggs. Groups of three to five proglottids are generally released and excreted into the feces, and the eggs in these proglottids are infective for both humans and animals. The eggs may survive in the environment for several months. After ingestion by the intermediate host, eggs embryonate, penetrate the intestinal wall, and are carried to many tissues, with a predilection for striated muscle of the neck, tongue, and trunk. Within 60 to 90 days, the encysted larval stage develops. These cysticerci can survive for long periods. Humans acquire infections that lead to intestinal tapeworms by ingesting undercooked pork containing cysticerci. Infections that cause human cysticercosis follow the ingestion of *T. solium* eggs, usually from fecally contaminated food. Autoinfection may occur if an individual with an egg-producing tapeworm ingests eggs derived from his or her own feces.

Clinical Manifestations Intestinal infections with *T. solium* may be asymptomatic. Epigastric discomfort, nausea, a sensation of hunger, weight loss, and diarrhea are

infrequent. Fecal passage of proglottids may be noted by patients.

In cysticercosis, the clinical manifestations are entirely different. Cysticerci can be found anywhere in the body, most commonly in the brain and the skeletal muscle. The clinical presentation of cysticercosis depends on the number and location of cysticerci as well as the extent of associated inflammatory responses or scarring. Neurologic manifestations are the most common. When inflammation surrounds cysticerci in the brain parenchyma, seizures are frequent. These seizures may be generalized, focal, or Jacksonian. Hydrocephalus results from obstruction of cerebrospinal fluid (CSF) flow by cysticerci and accompanying inflammation or by CSF outflow obstruction from arachnoiditis. Signs of increased intracranial pressure, including headache, nausea, vomiting, changes in vision, dizziness, ataxia, or confusion, are often evident. Patients with hydrocephalus may develop papilledema or display altered mental status. When cysticerci develop at the base of the brain or in the subarachnoid space, they cause chronic meningitis or arachnoiditis, communicating hydrocephalus, or strokes.

Diagnosis The diagnosis of intestinal *T. solium* infection is made by the detection of eggs or proglottids, as described for *T. saginata*. In cysticercosis, diagnosis can be difficult. A consensus conference has proposed absolute, major, minor, and epidemiologic criteria for diagnosis ([Table 223-1](#)). Diagnostic certainty is possible only with definite demonstration of the parasite (absolute criteria). This task can be accomplished by histologic observation of the parasite in excised tissue, by fundoscopic visualization of the parasite in the eye (in the anterior chamber, vitreous, or subretinal spaces), or by neuroimaging studies demonstrating cystic lesions containing a scolex. In most cases, diagnostic certainty is not possible. Instead, a clinical diagnosis is made on the basis of a combination of clinical presentation, radiographic studies, serologic tests, and exposure history.

Neuroimaging findings suggestive of neurocysticercosis constitute the primary major diagnostic criterion. These findings include cystic lesions with or without enhancement (e.g., ring enhancement), one or more calcifications (which may also have associated enhancement), or focal enhancing lesions. Cysticerci in the brain parenchyma are usually 5 to 10 mm in diameter and rounded. Cystic lesions in the subarachnoid space or fissures may enlarge up to 5 cm in diameter and may be lobulated. For cysticerci within the subarachnoid space or ventricles, the walls may be very thin and the cyst fluid is often isodense with [CSF](#). Thus, obstructive hydrocephalus or enhancement of the basilar meninges may be the only finding on computed tomography (CT) in neurocysticercosis. Cysticerci in the ventricles or subarachnoid space are usually visible to an experienced neuroradiologist on magnetic resonance imaging (MRI) or with intraventricular contrast injection. CT is more sensitive than MRI in identifying calcified lesions, whereas MRI is better for identifying cystic lesions and enhancement. Typical cigar-shaped calcifications in muscle are a second major diagnostic criterion.

The third major diagnostic criterion is detection of specific antibodies to cysticerci. While most tests employing unfractionated antigen have high rates of false-positive and -negative results, this problem can be overcome by using the more specific immunoblot assay. An immunoblot assay using lentil-lectin purified glycoproteins has >99% specificity and is highly sensitive. However, patients with single intracranial lesions or with calcifications may be seronegative. With this assay, serum samples provide greater

diagnostic sensitivity than [CSF](#). However, CSF may be useful when only unfractionated antigens are used.

Minor diagnostic criteria include the presence of subcutaneous nodules, punctate soft tissue or intracranial calcifications, clinical manifestations suggestive of neurocysticercosis (such as seizures, hydrocephalus, or altered mental status), or disappearance of lesions in conjunction with anticysticercal drug therapy. Epidemiologic criteria include current or prior residence in an endemic area, frequent travel to an endemic area, or exposure to a tapeworm carrier or household member infected with *T. solium*. Diagnosis is confirmed in patients with a combination of either two major criteria or one major criterion with two minor criteria and one epidemiologic criterion. The fulfillment of one major criterion and two other criteria or of three minor criteria with epidemiologic exposure supports a probable diagnosis. While the [CSF](#) is usually abnormal in neurocysticercosis, CSF abnormalities are not pathognomonic. Patients may have CSF pleocytosis with a predominance of lymphocytes, neutrophils, or eosinophils. The protein level in CSF may be elevated; the glucose concentration is usually normal but may be depressed.

TREATMENT

Intestinal *T. solium* infection is treated with a single dose of praziquantel (5 to 10 mg/kg). However, praziquantel can evoke an inflammatory response in the central nervous system if concomitant cryptic cysticercosis is present.

The management of neurocysticercosis focuses primarily on symptomatic treatment of seizures or hydrocephalus. Seizures can usually be controlled with anticonvulsants. If parenchymal lesions resolve without development of calcifications and patients remain free of seizures, anticonvulsant therapy can usually be discontinued after 2 years. Four placebo-controlled trials failed to identify any clinical advantage of antiparasitic drugs for parenchymal neurocysticercosis. However, trends toward faster resolution of neuroradiologic abnormalities were observed. Thus, some authorities favor use of antiparasitic drugs, including praziquantel (50 to 60 mg/kg daily in three divided doses for 15 days or 100 mg/kg in three doses given over a single day) or albendazole (15 mg/kg per day for 8 to 28 days). Both agents may exacerbate the inflammatory response around the dying parasite, exacerbating seizures or hydrocephalus. Thus, patients receiving these drugs should be carefully monitored. High-dose glucocorticoids can be used during treatment or if symptoms worsen. Since glucocorticoids induce first-pass metabolism of praziquantel and may decrease its antiparasitic effect, cimetidine should be coadministered to inhibit praziquantel metabolism.

For patients with hydrocephalus, the emergent reduction of intracranial pressure is the mainstay of therapy. In the case of obstructive hydrocephalus, this task requires either a diverting procedure, such as ventriculoperitoneal shunting, or removal of the cysticerci by craniotomy or via ventriculoscopy. Historically, shunts have usually failed. However, low failure rates have been attained with treatment with antiparasitic drugs or chronic glucocorticoids or with use of flow-sensitive shunts. In patients with subarachnoid cysts, glucocorticoids are needed to reduce arachnoiditis and accompanying vasculitis. Patients may benefit from prolonged courses of antiparasitic drugs and shunting for hydrocephalus. In patients with elevated intracranial pressure due to multiple inflamed

lesions, glucocorticoids are the mainstay of therapy, and antiparasitic drugs should be avoided until the elevated pressure resolves. For ocular and spinal medullary lesions, drug-induced inflammation may cause irreversible damage. Most patients should be managed surgically, although case reports have described cures with medical therapy.

Prevention Measures for the prevention of intestinal *T. solium* infection consist of the application to pork of precautions similar to those described above for beef with regard to *T. saginata* infection. The prevention of cysticercosis involves minimizing the opportunities for ingestion of fecally derived eggs by means of good personal hygiene, effective fecal disposal, and treatment and prevention of human intestinal infections.

ECHINOCOCCOSIS

Echinococcosis is an infection of humans caused by the larval stage of *Echinococcus granulosus*, *E. multilocularis*, or *E. vogeli*. *E. granulosus*, which produces unilocular cystic lesions, is prevalent in areas where livestock is raised in association with dogs. This tapeworm species is found in Australia, Argentina, Chile, Africa, eastern Europe, the Middle East, New Zealand, and the Mediterranean region, particularly Lebanon and Greece. *E. multilocularis*, which causes multilocular alveolar lesions that are locally invasive, is found in Alpine, sub-Arctic, or Arctic regions, including Canada, the United States, and central and northern Europe and Asia. *E. vogeli* causes polycystic hydatid disease and is found only in Central and South America. Like other cestodes, echinococcal species have both intermediate and definitive hosts. The definitive hosts are dogs that pass eggs in their feces. Cysts develop in the intermediate hosts -- sheep, cattle, humans, goats, camels, and horses for *E. granulosus* and mice and other rodents for *E. multilocularis* -- after the ingestion of eggs. When a dog ingests beef or lamb containing cysts, the life cycle is completed.

Etiology The small (5 mm long) adult *E. granulosus* worm, which lives for 5 to 20 months in the jejunum of dogs, has only three proglottids -- one immature, one mature, and one gravid. The gravid segment splits to release eggs that are morphologically similar to *Taenia* eggs and are extremely hardy. After humans ingest the eggs, embryos escape from the eggs, penetrate the intestinal mucosa, enter the portal circulation, and are carried to various organs, most commonly the liver and lungs. Larvae develop into fluid-filled unilocular hydatid cysts that consist of an external membrane and an inner germinal layer. Daughter cysts develop from the inner aspect of the germinal layer, as do germinating cystic structures called *brood capsules*. New larvae, called *protoscolices*, develop in large numbers within the brood capsule. The cysts expand slowly over a period of years.

The life cycle of *E. multilocularis* is similar except that small rodents serve as the intermediate hosts. The cyst of *E. multilocularis*, however, is quite different in that the larval form remains in the proliferative phase, the hydatid cyst is always multilocular, and vesicles progressively invade the host tissue by peripheral extension of processes from the germinal layer.

Clinical Manifestations Slowly enlarging echinococcal cysts generally remain asymptomatic until their expanding size or their space-occupying effect in an involved organ elicits symptoms. The liver and the lungs are the most common sites of these

cysts. Since a period of years elapses before cysts enlarge sufficiently to cause symptoms, they may be discovered incidentally on a routine x-ray or ultrasound study.

Patients with hepatic echinococcosis who are symptomatic most often present with abdominal pain or a palpable mass in the right upper quadrant. Compression of a bile duct or leakage of cyst fluid into the biliary tree may mimic recurrent cholelithiasis, and biliary obstruction can result in jaundice. Rupture of or episodic leakage from a hydatid cyst may produce fever, pruritus, urticaria, eosinophilia, or anaphylaxis. Pulmonary hydatid cysts may rupture into the bronchial tree or peritoneal cavity and produce cough, chest pain, or hemoptysis. Rupture of hydatid cysts may lead to multifocal dissemination of protoscolices, which can form additional cysts. Rupture can occur spontaneously or at surgery. Other presentations are due to the involvement of bone (invasion of the medullary cavity with slow bone erosion producing pathologic fractures), the central nervous system (space-occupying lesions), and the heart (conduction defects, pericarditis).

The cysts of *E. multilocularis* characteristically present as a slowly growing hepatic tumor, with progressive destruction of the liver and extension into vital structures. Patients commonly complain of upper quadrant and epigastric pain, and obstructive jaundice may be apparent. A minority of patients experience the metastasis of lesions to the lung and brain.

Diagnosis Radiographic and related imaging studies are important in detecting and evaluating echinococcal cysts. Plain films will define pulmonary cysts -- usually as rounded irregular masses of uniform density -- but may miss cysts in other organs unless there is cyst wall calcification (as occurs in the liver). [MRI](#), [CT](#), and ultrasound reveal well-defined cysts with thick or thin walls. When older cysts contain a layer of hydatid sand that is rich in accumulated scolices, these imaging methods may detect this fluid layer of different density. However, the most pathognomonic finding, if demonstrable, is that of daughter cysts within the larger cyst. This finding, like eggshell or mural calcification on CT, is indicative of *E. granulosus* infection and helps to distinguish the cyst from carcinomas, bacterial or amebic liver abscesses, or hemangiomas. CT of alveolar hydatid cysts reveals indistinct solid masses with central necrosis and plaque-like calcifications.

A specific diagnosis can be made by the examination of aspirated fluids for scoliceal hooklets, but diagnostic aspiration is not usually recommended because of the risk of fluid leakage resulting in either dissemination of infection or anaphylactic reactions. Serodiagnostic assays can be useful, although a negative test does not exclude the diagnosis of echinococcosis. Cysts in the liver elicit positive antibody responses in ~90% of cases, whereas up to 50% of individuals with cysts in the lungs are seronegative. Detection of antibody to specific echinococcal antigens by immunoblotting has the highest degree of specificity.

TREATMENT

Therapy for echinococcosis is based on considerations of the size, location, and manifestations of cysts and the overall health of the patient. Surgery has traditionally been the principal definitive method of treatment; *E. granulosus* cysts are excised, or

tissue containing *E. multilocularis* cysts is resected. Risks at surgery from leakage of fluid include anaphylaxis and dissemination of infectious scolices. The latter complication has been minimized by the instillation of scolicidal solutions such as hypertonic saline or ethanol, which may cause hypernatremia, intoxication, or sclerosing cholangitis. Albendazole, which is active against *Echinococcus*, should be administered adjunctively, beginning before resection and continuing for several weeks for *E. granulosus* and for up to 2 years for *E. multilocularis*. Percutaneous aspiration, infusion of scolicidal agents, and reaspiration (PAIR) can be used instead of surgery in many cases of cystic echinococcosis. PAIR is contraindicated for superficially located cysts (because of the risk of rupture), for cysts with multiple thick internal septal divisions (honeycombing pattern), and for cysts communicating with the biliary tree. Therapy with albendazole (15 mg/kg daily in two divided doses) should be initiated at least 4 days before the procedure and continued for at least 4 weeks afterward. Ultrasound- or CT-guided aspiration allows confirmation of the diagnosis by demonstration of protoscolices in the aspirate. Either alcohol or hypertonic saline should then be infused. Daughter cysts within the primary cyst may need to be punctured separately. In experienced hands, this approach yields rates of cure and relapse equivalent to those following surgery, with less perioperative morbidity and shorter hospitalization. Medical therapy with albendazole alone for 12 weeks to 6 months results in cure in ~30% of cases and improvement in another 50%. Many of the failures are subsequently treated successfully with PAIR or additional courses of medical therapy. Response to treatment is best assessed by serial imaging studies with attention to cyst size and consistency.

Prevention In endemic areas, echinococcosis can be prevented by administering praziquantel to infected dogs, by denying dogs access to infected animals, or by vaccinating sheep. Limitation of the number of stray dogs is helpful in reducing the prevalence of infection among humans.

HYMENOLEPIASIS NANA

Infection with *Hymenolepis nana*, the dwarf tapeworm, is the most common of all the cestode infections. *H. nana* is endemic in both temperate and tropical regions of the world. Infection is spread by fecal/oral contamination and is common among institutionalized children.

Etiology and Pathogenesis *H. nana* is the only cestode of humans that does not require an intermediate host. Both the larval and adult phases take place in the human. The adult, the smallest tapeworm parasitizing humans, is about 2 cm long and dwells in the proximal ileum. Proglottids, which are quite small and are rarely seen in the stool, release spherical eggs 30 to 44 μ m in diameter, each of which contains an oncosphere with six hooklets. The eggs are immediately infective and are unable to survive in the external environment for more than 10 days. *H. nana* can also be acquired by the ingestion of infected insects (especially larval meal-worms and larval fleas). When the egg is ingested by a new host, the oncosphere is freed and penetrates the intestinal villi, becoming a cysticercoid larva. Larvae migrate back into the intestinal lumen, attach to the mucosa, and mature over 10 to 12 days into adult worms. Eggs may also hatch before passing into the stool, causing internal autoinfection with increasing numbers of intestinal worms. Although the life span of adult *H. nana* is only about 4 to 10 weeks, the autoinfection cycle perpetuates the infection.

Clinical Manifestations *H. nana* infection, even with many intestinal worms, is usually asymptomatic. When infection is intense, anorexia, abdominal pain, and diarrhea develop.

Diagnosis Infection is diagnosed by the finding of eggs in the stool.

TREATMENT

Praziquantel (25 mg/kg once) is the treatment of choice, since it acts against both the adult worms and the cysticercoids in the intestinal villi.

Prevention Good personal hygiene and improved sanitation can eradicate the disease. Epidemics have been controlled by mass chemotherapy coupled with improved hygiene.

HYMENOLEPIASIS DIMINUTA

Hymenolepis diminuta, a cestode of rodents, occasionally infects small children, who ingest the adult worm in uncooked cereal foods contaminated by fleas and other insects in which larvae develop. Infection is usually asymptomatic and is diagnosed by the detection of eggs in the stool. Treatment with praziquantel results in cure in most cases.

DIPHYLLOBOTHRIASIS

Diphyllobothrium latum and other *Diphyllobothrium* species are found in the lakes, rivers, and deltas of the northern hemisphere, Central Africa, and Chile.

Etiology and Pathogenesis The adult worm, the longest tapeworm (up to 25 m), attaches to the ileal and occasionally to the jejunal mucosa by its suckers, which are located on its elongated scolex. The adult worm has 3000 to 4000 proglottids, which release approximately 1 million eggs daily into the feces. If an egg reaches water, it hatches and releases a free-swimming embryo that can be eaten by small freshwater crustaceans (*Cyclops* or *Diaptomus* species). After an infected crustacean containing a developed procercoid is swallowed by a fish, the larva migrates into the fish's flesh and grows into a plerocercoid, or sparganum larva. Humans acquire the infection by ingesting infected raw fish. Within 3 to 5 weeks, the tapeworm matures into an adult in the human intestine.

Clinical Manifestations Most *D. latum* infections are asymptomatic, although manifestations may include transient abdominal discomfort, diarrhea, vomiting, weakness, and weight loss. Occasionally, infection can cause acute abdominal pain and intestinal obstruction; in rare cases cholangitis or cholecystitis may be produced by migrating proglottids. Because the tapeworm absorbs large quantities of vitamin B₁₂ and interferes with ileal B₁₂ absorption, vitamin B₁₂ deficiency can develop. Up to 2% of infected patients, especially the elderly, have megaloblastic anemia resembling pernicious anemia and may exhibit neurologic sequelae of B₁₂ deficiency.

Diagnosis The diagnosis is made readily by the detection of the characteristic eggs in

the stool. The eggs possess a single shell with an operculum at one end and a knob at the other. Mild to moderate eosinophilia may be detected.

TREATMENT

Praziquantel (5 to 10 mg/kg once) is highly effective. Parenteral vitamin B₁₂ should be given if B₁₂ deficiency is manifest.

Prevention Infection can be prevented by heating fish to 54°C for 5 min or by freezing it at -18°C for 24 h. Placing fish in brine with a high salt concentration for long periods kills the eggs.

DIPYLIDIASIS

Dipylidium caninum, a common tapeworm of dogs and cats, may accidentally infect humans. Dogs, cats, and occasionally humans become infected by ingesting fleas harboring cysticercoids. Children are more likely to become infected than adults. Most infections are asymptomatic, but abdominal pain, diarrhea, anal pruritus, urticaria, eosinophilia, or passage of segments in the stool may occur. The diagnosis is made by the detection of proglottids in the stool. As in *D. latum* infection, therapy consists of praziquantel. Prevention requires anthelmintic treatment and flea control for pet dogs or cats.

SPARGANOSIS

Humans can be infected by the sparganum, or plerocercoid larva, of a diphylobothrid tapeworm of the genus *Spirometra*. Infection can be acquired by the consumption of water containing infected *Cyclops*; by the ingestion of infected snakes, birds, or mammals; or by the application of infected flesh as poultices. The worm migrates slowly in tissues, and infection commonly presents as a subcutaneous swelling. Periorbital tissues can be involved, and ocular sparganosis may destroy the eye. Surgical excision is used to treat localized sparganosis.

COENUROSIS

This rare infection of humans by the larval stage (coenurus) of the dog tapeworm *Taenia multiceps* or *T. serialis* results in a space-occupying cystic lesion. As in cysticercosis, involvement of the central nervous system and subcutaneous tissue is most common. Both definitive diagnosis and treatment require surgical excision of the lesion. Chemotherapeutic agents generally are not effective.

(Bibliography omitted in Palm version)

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PART THIRTEEN -ENDOCRINOLOGY AND METABOLISM

SECTION 1 -ENDOCRINOLOGY

327. PRINCIPLES OF ENDOCRINOLOGY - J. Larry Jameson

The management of endocrine disorders requires an understanding of such disparate areas as intermediary metabolism, reproductive physiology, bone metabolism, and growth. Accordingly, the practice of endocrinology is intimately linked to a conceptual framework for understanding hormone secretion, hormone action, and principles of feedback control systems. The endocrine system is investigated primarily by measuring hormone concentrations, thereby arming the clinician with valuable diagnostic information. Most disorders of the endocrine system are amenable to effective treatment, once the correct diagnosis is determined. Endocrine deficiency disorders are treated with physiologic hormone replacement; hormone excess conditions, usually due to benign glandular adenomas, are managed by removing tumors surgically or by reducing hormone levels medically.

SCOPE OF ENDOCRINOLOGY

The specialty of endocrinology encompasses the study of glands and the hormones they produce. The term *endocrine* was coined by Starling to contrast the actions of hormones secreted internally (endocrine) with those secreted externally (*exocrine*) or into a lumen, such as the gastrointestinal tract. The term *hormone*, derived from a Greek phrase meaning "to set in motion," aptly describes the dynamic actions of these circulating substances as they elicit cellular responses and regulate physiologic processes through feedback mechanisms.

Unlike certain other specialties in medicine, it is not possible to define endocrinology strictly along anatomic lines. The classic endocrine glands -- pituitary, thyroid, parathyroid, pancreatic islets, adrenal, and gonads -- communicate broadly with other organs through the nervous system, hormones, cytokines, and growth factors. In addition to its traditional synaptic functions, the brain produces a vast array of peptide hormones, spawning the discipline of neuroendocrinology. Through the production of hypothalamic releasing factors, the central nervous system exerts a major regulatory influence over pituitary hormone secretion ([Chap. 328](#)). The peripheral nervous system modulates adrenal medulla and pancreatic islet hormone production. The immune and endocrine systems are also intimately intertwined. The adrenal glucocorticoid, cortisol, is a powerful immunosuppressant. Cytokines and interleukins (ILs) have profound effects on the functions of the pituitary, adrenal, thyroid, and gonads. Common endocrine diseases, such as autoimmune thyroid disease and type 1 diabetes mellitus, are caused by dysregulation of immune surveillance and tolerance. Less common diseases such as polyglandular failure, Addison's disease, and lymphocytic hypophysitis also have an immunologic basis.

The interdigitation of endocrinology with physiologic processes in other specialties sometimes blurs the role of hormones. For example, hormones play an important role in maintenance of blood pressure, intravascular volume, and peripheral resistance in the cardiovascular system. The heart is the principal source of atrial natriuretic peptide,

which acts in classic endocrine fashion to induce natriuresis at a distant target organ (the kidney). Vasoactive substances such as catecholamines, angiotensin II, endothelin, and nitric oxide are involved in dynamic changes of vascular tone, in addition to their multiple roles in other tissues. Erythropoietin, a traditional circulating hormone, is made in the kidney and stimulates erythropoiesis in the bone marrow ([Chap. 104](#)). The kidney is also integrally involved in the renin-angiotensin axis ([Chap. 331](#)) and is a primary target of several hormones including parathyroid hormone (PTH), mineralocorticoids, and vasopressin. The gastrointestinal tract produces a surprising number of peptide hormones such as cholecystokinin, gastrin, secretin, and vasoactive intestinal peptide, among many others. Carcinoid and islet tumors can secrete excessive amounts of these hormones, leading to specific clinical syndromes ([Chap. 93](#)). Many of these gastrointestinal hormones are also produced in the central nervous system, where their functions remain poorly understood. As new hormones such as inhibin, ghrelin, and leptin are discovered, they become integrated into the science and practice of medicine on the basis of their functional roles rather than through their structures or mechanisms of action.

Characterization of hormone receptors frequently reveals unexpected relationships to factors in nonendocrine disciplines. The growth hormone (GH) receptor, for example, is a member of the cytokine receptor family. The G protein-coupled receptors (GPCRs), which mediate the actions of many peptide hormones, are used in numerous physiologic processes including vision, smell, and neurotransmission.

It is apparent that hormones and growth factors play an important functional role in all organ systems. Though endocrinologists are not usually involved in the administration of the hormones or growth factors used to treat diseases in other specialties (e.g., cardiology, hematology), the principles of endocrinology can be applied in these cases, thus emphasizing the impact of endocrinology across multiple disciplines.

NATURE OF HORMONES

Hormones can be divided into five major classes: (1) *amino acid derivatives* such as dopamine, catecholamines, and thyroid hormone; (2) *small neuropeptides* such as gonadotropin-releasing hormone (GnRH), thyrotropin-releasing hormone (TRH), somatostatin, and vasopressin; (3) *large proteins* such as insulin, luteinizing hormone (LH), and PTH produced by classic endocrine glands; (4) *steroid hormones* such as cortisol and estrogen that are synthesized from cholesterol-based precursors; and (5) *vitamin derivatives* such as retinoids (vitamin A) and vitamin D. A variety of *peptide growth factors*, most of which act locally, share actions with hormones. As a rule, amino acid derivatives and peptide hormone interact with cell-surface membrane receptors. Steroids, thyroid hormones, vitamin D, and retinoids are lipid-soluble and interact with intracellular nuclear receptors.

HORMONE AND RECEPTOR FAMILIES

Many hormones and receptors can be grouped into families, reflecting their structural similarities ([Table 327-1](#)). The evolution of these families generates diverse but highly selective pathways of hormone action. Recognizing these relationships allows extrapolation of information gleaned from one hormone or receptor to other family

members.

The glycoprotein hormone family, consisting of thyroid-stimulating hormone (TSH), follicle-stimulating hormone (FSH), LH, and human chorionic gonadotropin (hCG), illustrates many features of related hormones. The glycoprotein hormones are heterodimers that share the α subunit in common; the β subunits are distinct and confer specific biologic actions. The overall three-dimensional architecture of the β subunits is similar, reflecting the locations of conserved disulfide bonds that restrain protein conformation. The cloning of the β -subunit genes from multiple species suggests that this family arose from a common ancestral gene, probably by gene duplication and subsequent divergence to evolve new biologic functions.

As the hormone families enlarge and diverge, their receptors must co-evolve, if new biologic functions are to be derived. Related [GPCRs](#), for example, have evolved for each of the glycoprotein hormones. These receptors are structurally similar, and each is coupled to the G_s signaling pathway. However, there is minimal overlap of hormone binding. For example, TSH binds with high specificity to the TSH receptor but interacts weakly with the LH or the FSH receptor. Nonetheless, there can be subtle physiologic consequences of hormone cross-reactivity with other receptors. Very high levels of [hCG](#) during pregnancy stimulate the TSH receptor and increase thyroid hormone levels.

Insulin, insulin-like growth factor (IGF) I, and IGF-II share structural similarities that are most apparent when precursor forms of the proteins are compared. In contrast to the high degree of specificity seen with the glycoprotein hormones, there is moderate cross-talk among the members of the insulin/IGF family. High concentrations of an IGF-II precursor produced by certain tumors (e.g., sarcomas) can cause hypoglycemia, partly because of binding to insulin and IGF-I receptors ([Chap. 334](#)). High concentrations of insulin also bind to the IGF-I receptor, perhaps accounting for some of the clinical manifestations seen in severe insulin resistance.

Another important example of receptor cross-talk is seen with PTH and parathyroid hormone-related peptide (PTHrP) ([Chap. 341](#)). PTH is produced by the parathyroid glands, whereas PTHrP is expressed at high levels during development and by a variety of tumors. These hormones share amino acid sequence similarity, particularly in their amino-terminal regions. Both hormones bind to a single PTH receptor that is expressed in bone and kidney. Hypercalcemia and hypophosphatemia may therefore result from excessive production of either hormone, making it difficult to distinguish hyperparathyroidism from hypercalcemia of malignancy solely on the basis of serum chemistries. However, sensitive and specific assays for PTH now allow these disorders to be separated more readily.

Based on their specificities for DNA binding sites, the nuclear receptor family can be subdivided into type 1 receptors (GR, MR, AR, ER, PR) that bind steroids and type 2 receptors (TR, VDR, RAR, PPAR) that bind thyroid hormone, vitamin D, retinoic acid, or lipid derivatives. Certain functional domains in nuclear receptors, such as the zinc finger DNA-binding domains, are highly conserved. However, selective amino acid differences within this domain confer DNA sequence specificity. The hormone-binding domains are more variable, providing great diversity in the array of small molecules that can bind to

different nuclear receptors. With few exceptions, hormone binding is highly specific for a single type of nuclear receptor. One exception involves the highly related glucocorticoid and mineralocorticoid receptors. Because the mineralocorticoid receptor also binds glucocorticoids with high affinity, an enzyme (11 β -hydroxysteroid dehydrogenase) located in renal tubular cells inactivates glucocorticoids, allowing selective responses to mineralocorticoids such as aldosterone. However, when very high glucocorticoid concentrations occur, as in Cushing's syndrome, the glucocorticoid degradation pathway becomes saturated, allowing excessive cortisol levels to exert mineralocorticoid effects (sodium retention, potassium wasting). This phenomenon is particularly pronounced in ectopic adrenocorticotrophic hormone (ACTH) syndromes ([Chap. 331](#)). Another example of relaxed nuclear receptor specificity involves the estrogen receptor, which can bind an array of compounds, some of which share little structural similarity to the high-affinity ligand estradiol. This feature of the estrogen receptor makes it susceptible to activation by "environmental estrogens" such as resveratrol, octylphenol, and many other aromatic hydrocarbons. On the other hand, this lack of specificity provides an opportunity to synthesize a remarkable series of clinically useful antagonists (e.g., tamoxifen) and selective estrogen response modulators (SERMs), such as raloxifene. These compounds generate distinct conformations that alter receptor interactions with components of the transcription machinery (see below), thereby conferring their unique actions.

HORMONE SYNTHESIS AND PROCESSING

The synthesis of peptide hormones and their receptors occurs through a classic pathway of gene expression: transcription $\rightarrow$ mRNA $\rightarrow$ protein $\rightarrow$ posttranslational protein processing $\rightarrow$ intracellular sorting, membrane integration, or secretion ([Chap. 65](#)). Though endocrine genes contain regulatory DNA elements similar to those found in many other genes, their exquisite control by other hormones also necessitates the presence of specific hormone response elements. For example, the TSH genes are repressed directly by thyroid hormones acting through the thyroid hormone receptor, a member of the nuclear receptor family. Steroidogenic enzyme gene expression requires specific transcription factors such as steroidogenic factor-1 (SF-1), acting in conjunction with signals transmitted by trophic hormones (e.g., ACTH or LH). For some hormones, substantial regulation occurs at the level of translational efficiency. Insulin biosynthesis, while requiring ongoing gene transcription, is regulated primarily at the translational level in response to elevated levels of glucose or amino acids.

Many hormones are embedded within larger precursor polypeptides that are proteolytically processed to yield the biologically active hormone. Examples include: proopiomelanocortin (POMC) $\rightarrow$ ACTH; proglucagon $\rightarrow$ glucagon; proinsulin $\rightarrow$ insulin; pro-PTH $\rightarrow$ PTH, among others. In many cases, such as POMC and proglucagon, these precursors generate multiple biologically active peptides. It is provocative that hormone precursors are typically inactive, presumably adding an additional level of regulatory control. This is true not only for peptide hormones but also for certain steroids (testosterone $\rightarrow$ dihydrotestosterone) and thyroid hormone (T₄ $\rightarrow$ T₃).

Hormone precursor processing is intimately linked to intracellular sorting pathways that transport proteins to appropriate vesicles and enzymes, resulting in specific cleavage steps, followed by protein folding and translocation to secretory vesicles. Hormones

destined for secretion are translocated across the endoplasmic reticulum under the guidance of an amino-terminal signal sequence that is subsequently cleaved. Cell-surface receptors are inserted into the membrane via short segments of hydrophobic amino acids that remain embedded within the lipid bilayer. During translocation through the Golgi and endoplasmic reticulum, hormones and receptors are also subject to a variety of posttranslational modifications, such as glycosylation and phosphorylation, which can alter protein conformation, modify circulating half-life, and alter biologic activity.

Synthesis of most steroid hormones is based on modifications of the precursor, cholesterol. Multiple regulated enzymatic steps are required for the synthesis of testosterone ([Chap. 335](#)), estradiol ([Chap. 336](#)), cortisol ([Chap. 331](#)), and vitamin D ([Chap. 340](#)). This large number of synthetic steps predisposes to multiple genetic and acquired disorders of steroidogenesis (see below).

HORMONE SECRETION, TRANSPORT, AND DEGRADATION

The circulating level of a hormone is determined by its rate of secretion and its circulating half-life. After protein processing, peptide hormones ([GnRH](#), insulin, GH) are stored in secretory granules. As these granules mature, they are poised beneath the plasma membrane for imminent release into the circulation. In most instances, the stimulus for hormone secretion is a releasing factor or neural signal that induces rapid changes in intracellular calcium concentrations, leading to secretory granule fusion with the plasma membrane and release of its contents into the extracellular environment and blood stream. Steroid hormones, in contrast, diffuse into the circulation as they are synthesized. Thus, their secretory rates are closely aligned with rates of synthesis. For example, ACTH and LH induce steroidogenesis by stimulating the activity of steroidogenic acute regulatory (StAR) protein (transports cholesterol into the mitochondrion) along with other rate-limiting steps (e.g., cholesterol side-chain cleavage enzyme, CYP11A1) in the steroidogenic pathway.

Hormone transport and degradation dictate the rapidity with which a hormonal signal decays. Some hormonal signals are evanescent (e.g., somatostatin), whereas others are longer lived (e.g., TSH). Because somatostatin exerts effects in virtually every tissue, a short half-life allows its concentrations and actions to be controlled locally. Structural modifications that impair somatostatin degradation have been useful for generating long-acting therapeutic analogues, such as octreotide ([Chap. 328](#)). On the other hand, the actions of TSH are highly specific for the thyroid gland. Its prolonged half-life accounts for relatively constant serum levels, even though TSH is secreted in discrete pulses.

An understanding of circulating hormone half-life is important for achieving physiologic hormone replacement, as the frequency of dosing and the time required to reach steady state are intimately linked to rates of hormone decay. T₄, for example, has a plasma half-life of 7 days. Consequently, >1 month is required to reach a new steady state, but single daily doses are sufficient to achieve constant hormone levels. T₃, in contrast, has a half-life of 1 day. Its administration is associated with more dynamic serum levels and it must be administered two to three times per day. Similarly, synthetic glucocorticoids vary widely in their half-lives; those with longer half-lives (e.g., dexamethasone) are

associated with greater suppression of the hypothalamic-pituitary-adrenal (HPA) axis. Most protein hormones [e.g., ACTH, GH, prolactin (PRL); PTH, LH] have relatively short half-lives (<20 min), leading to sharp peaks of secretion and decay. The only accurate way to profile the pulse frequency and amplitude of these hormones is to measure levels in frequently sampled blood (every 10 min) over long durations (8 to 24 h). Because this is not practical in a clinical setting, an alternative strategy is to pool three to four samples drawn at about 30-min intervals, recognizing that pulsatile secretion makes it difficult to establish a narrow normal range. Rapid hormone decay is useful in certain clinical settings. For example, the short half-life of PTH allows the use of intraoperative PTH determinations to confirm successful removal of an adenoma. This is particularly valuable diagnostically when there is a possibility of multicentric disease or parathyroid hyperplasia, as occurs with multiple endocrine neoplasia (MEN) or renal insufficiency.

Many hormones circulate in association with serum-binding proteins. Examples include: (1) T₄ and T₃ binding to thyroxine-binding globulin (TBG), albumin, and thyroxine-binding prealbumin (TBPA); (2) cortisol binding to cortisol-binding globulin (CBG); (3) androgen and estrogen binding to sex hormone-binding globulin (SHBG) (also called testosterone-binding globulin, TeBG); (4) IGF-I and -II binding to multiple IGF-binding proteins (IGFBPs); (5) GH interactions with GH-binding protein (GHBP), a circulating fragment of the GH receptor extracellular domain; and (6) activin binding to follistatin. These interactions provide a hormonal reservoir, prevent otherwise rapid degradation of unbound hormones, restrict hormone access to certain sites (e.g., IGFBPs), and modulate the unbound, or "free," hormone concentrations. Although a variety of binding protein abnormalities have been identified, most have little clinical consequence, aside from creating diagnostic problems. For example, TBG deficiency can greatly reduce total thyroid hormone levels, but the free concentrations of T₄ and T₃ remain normal. Liver disease and certain medications can also influence binding protein levels (e.g., estrogen increases TBG) or cause displacement of hormones from binding proteins (e.g., salicylate displaces T₄ from TBG). Only free hormone is available to bind receptors and thereby elicit a biologic response. Short-term perturbations in binding proteins change the free hormone concentration, which in turn induces compensatory adaptations through feedback loops. SHBG changes in women are an exception to this self-correcting mechanism. When SHBG decreases because of insulin resistance or androgen excess, the free testosterone concentration is increased, potentially leading to hirsutism ([Chap. 53](#)). The increased free testosterone levels does not result in an adequate compensatory feedback correction because estrogen, and not testosterone, is the primary regulator of the reproductive axis.

HORMONE ACTION THROUGH RECEPTORS

Receptors for hormones are divided into two major classes -- membrane and nuclear. *Membrane receptors* primarily bind peptide hormones and catecholamines. *Nuclear receptors* bind small molecules that can diffuse across the cell membrane, such as thyroid hormone, steroids, and vitamin D. Certain general principles apply to hormone-receptor interactions, regardless of the class of receptor. Hormones bind to receptors with specificity and a high affinity that generally coincides with the dynamic range of circulating hormone concentrations. Low concentrations of free hormone (usually 10⁻¹² to 10⁻⁹ M) rapidly associate and dissociate from receptors in a bimolecular

reaction, such that the occupancy of the receptor at any given moment is a function of hormone concentration and the receptor's affinity for the hormone. Receptor numbers vary greatly in different target tissues, providing one of the major determinants of specific cellular responses to circulating hormones. For example, ACTH receptors are located almost exclusively in the adrenal cortex, and FSH receptors are found only in the gonads. In contrast, insulin and thyroid hormone receptors are widely distributed, reflecting the need for metabolic responses in all tissues.

MEMBRANE RECEPTORS

Membrane receptors for hormones can be divided into several major groups: (1) seven transmembrane [GPCRs](#), (2) tyrosine kinase receptors, (3) cytokine receptors, and (4) serine kinase receptors ([Fig. 327-1](#)). The *seven transmembrane GPCR* family binds a remarkable array of hormones including large proteins (e.g., LH, PTH), small peptides (e.g., [TRH](#), somatostatin), catecholamines (epinephrine, dopamine), and even minerals (e.g., calcium). The extracellular domains of GPCRs vary widely in size and are the major binding site for large hormones. The transmembrane-spanning regions are composed of hydrophobic α -helical domains that traverse the lipid bilayer. Like some channels, these domains are thought to circularize and form a hydrophobic pocket into which certain small ligands fit. Hormone binding induces conformational changes in these domains, transducing structural changes to the intracellular domain, which is a docking site for G proteins.

The large family of *G proteins*, so named because they bind guanine nucleotides (GTP, GDP), provides great diversity for coupling to different receptors. G proteins form a heterotrimeric complex that is composed of various α and $\beta\gamma$ subunits. The α subunit contains the guanine nucleotide-binding site and hydrolyzes GTP $\rightarrow$ GDP. The $\beta\gamma$ subunits are tightly associated and modulate the activity of the α subunit, as well as mediating their own effector signaling pathways. G protein activity is regulated by a cycle that involves GTP hydrolysis and dynamic interactions between the α and $\beta\gamma$ subunits. Hormone binding to the receptor induces GDP dissociation, allowing G_α to bind GTP and dissociate from the $\beta\gamma$ complex. Under these conditions, the G_α subunit is activated and mediates signal transduction through various enzymes such as adenylate cyclase or phospholipase C. GTP hydrolysis to GDP allows reassociation with the $\beta\gamma$ subunits and restores the inactive state. As described below, a variety of endocrinopathies result from G protein mutations or from mutations in receptors that modify their interactions with G proteins.

There are more than a dozen isoforms of the G_α subunit. G_s stimulates, whereas G_i inhibits adenylate cyclase, an enzyme that generates the second messenger, cyclic AMP, leading to activation of protein kinase A ([Table 327-1](#)). G_q subunits couple to phospholipase C, generating diacylglycerol and inositol triphosphate, leading to activation of protein kinase C and the release of intracellular calcium.

The *tyrosine kinase receptors* transduce signals for insulin and a variety of growth factors, such as [IGF-I](#), epidermal growth factor (EGF), nerve growth factor, platelet-derived growth factor, and fibroblast growth factor. The cysteine-rich extracellular ligand-binding domains contain growth factor binding sites. After ligand binding, this class of receptors undergoes autophosphorylation, inducing interactions

with intracellular adaptor proteins such as Shc and insulin receptor substrates 1 to 4. In the case of the insulin receptor, multiple kinases are activated including the Raf-Ras-MAPK and the Akt/protein kinase B pathways. The tyrosine kinase receptors play a prominent role in cell growth and differentiation as well as in intermediary metabolism.

The GH and PRL receptors belong to the *cytokine receptor* family ([Chap. 305](#)). Analogous to the tyrosine kinase receptors, ligand binding induces receptor binding to intracellular kinases -- the Janus kinases (JAKs), which phosphorylate members of the signal transduction and activators of transcription (STAT) family -- as well as other signaling pathways (Ras, PI3-K, MAPK). The activated STAT proteins translocate to the nucleus and stimulate expression of target genes ([Chap. 328](#)).

The *serine kinase receptors* mediate the actions of activins, transforming growth factor β , mullerian-inhibiting substance (MIS, also known as anti-mullerian hormone, AMH), and bone morphogenic proteins (BMPs). This family of receptors (consisting of type I and II subunits) signal through proteins termed *smads* (fusion of terms for *Caenorhabditis elegans* sma + mammalian mad). Like the [STAT](#) proteins, the smads serve a dual role of transducing the receptor signal and acting as transcription factors. The pleomorphic actions of these growth factors dictate that they act primarily in a local (paracrine or autocrine) manner. Binding proteins, such as follistatin (which binds activin and other members of this family), function to inactivate the growth factors and restrict their distribution.

NUCLEAR RECEPTORS

The family of nuclear receptors has grown to nearly 100 members, many of which are still classified as orphan receptors because their ligands, if they exist, remain to be identified ([Fig. 327-2](#)). Otherwise, most nuclear receptors are classified based on the nature of their ligands. Though all nuclear receptors ultimately act to increase or decrease gene transcription, some (e.g., glucocorticoid receptor) reside primarily in the cytoplasm, whereas others (e.g., thyroid hormone receptor) are always located in the nucleus. After ligand binding, the cytoplasmically localized receptors translocate to the nucleus.

The structures of nuclear receptors have been extensively studied, including by x-ray crystallography. The DNA binding domain, consisting of two zinc fingers, contacts specific DNA recognition sequences in target genes. Most nuclear receptors bind to DNA as dimers. Consequently, each monomer recognizes an individual DNA motif, referred to as a "half-site." The steroid receptors, including the glucocorticoid, estrogen, progesterone, and androgen receptors, bind to DNA as homodimers. Consistent with this twofold symmetry, their DNA recognition half-sites are palindromic. The thyroid, retinoid, PPAR, and vitamin D receptors bind to DNA preferentially as heterodimers in combination with retinoid X receptors (RXRs). Their DNA half-sites are arranged as direct repeats. Receptor specificity for DNA sequences is determined by (1) the sequence of the half-site, (2) the orientation of the half-sites (palindromic, direct repeat), and (3) the spacing between the half-sites. For example, vitamin D, thyroid and retinoid receptors recognize similar tandemly repeated half-sites (TAAGTCA), but these DNA repeats are spaced by three, four, and five nucleotides, respectively.

The carboxy-terminal hormone-binding domain mediates transcriptional control. For type II receptors, such as TR and RAR, co-repressor proteins bind to the receptor in the absence of ligand and silence gene transcription. Hormone binding induces conformational changes, triggering the release of co-repressors and inducing the recruitment of coactivators that stimulate transcription. Thus, these receptors are capable of mediating dramatic changes in the level of gene activity. Certain disease states are associated with defective regulation of these events. For example, mutations in the thyroid hormone receptor prevent co-repressor dissociation, resulting in a dominant form of hormone resistance ([Chap. 330](#)). In promyelocytic leukemia, fusion of RAR α to other nuclear proteins causes aberrant gene silencing and prevents normal cellular differentiation. Treatment with retinoic acid reverses this repression and allows cellular differentiation and apoptosis to occur ([Chap. 111](#)). Type 1 steroid receptors do not interact with co-repressors, but ligand binding still mediates interactions with an array of coactivators. X-ray crystallography shows that various [SERMs](#) induce distinct receptor conformations. The tissue-specific responses caused by these agents in breast, bone, and uterus appear to reflect distinct interactions with coactivators. The receptor-coactivator complex stimulates gene transcription by several pathways including (1) recruitment of enzymes (histone acetyl transferases) that modify chromatin structure, (2) interactions with additional transcription factors on the target gene, and (3) direct interactions with components of the general transcription apparatus to enhance the rate of RNA polymerase II-mediated transcription.

FUNCTIONS OF HORMONES

The functions of individual hormones are described in detail in subsequent chapters. Nonetheless, it is useful to illustrate how most biologic responses require integration of several different hormonal pathways. The physiologic functions of hormones can be divided into three general areas: (1) growth and differentiation, (2) maintenance of homeostasis, and (3) reproduction.

GROWTH

Multiple hormones and nutritional factors mediate the complex phenomenon of growth ([Chap. 328](#)). Short stature may be caused by GH deficiency, hypothyroidism, Cushing's syndrome, precocious puberty, malnutrition or chronic illness, or genetic abnormalities that affect the epiphyseal growth plates (e.g., *FGFR3* or *SHOX* mutations). Many factors (GH, [IGF-I](#), thyroid hormone) stimulate growth, whereas others (sex steroids) lead to epiphyseal closure. Understanding these hormonal interactions is important in the diagnosis and management of growth disorders. For example, delaying exposure to high levels of sex steroids may enhance the efficacy of GH treatment.

MAINTENANCE OF HOMEOSTASIS

Though virtually all hormones affect homeostasis, the most important among these are the following:

1. Thyroid hormone -- controls about 25% of basal metabolism in most tissues ([Chap. 330](#))

2. Cortisol -- exerts a permissive action for many hormones in addition to its own direct effects ([Chap. 331](#))
3. PTH -- regulates calcium and phosphorus levels ([Chap. 341](#))
4. Vasopressin -- regulates serum osmolality by controlling renal free water clearance ([Chap. 329](#))
5. Mineralocorticoids -- control vascular volume and serum electrolyte (Na⁺, K⁺) concentrations ([Chap. 331](#))
6. Insulin -- maintains euglycemia in the fed and fasted states ([Chap. 333](#))

The defense against hypoglycemia is an impressive example of integrated hormone action ([Chap. 334](#)). In response to the fasted state and falling blood glucose, insulin secretion is suppressed, resulting in decreased glucose uptake and enhanced glycogenolysis, lipolysis, proteolysis, and gluconeogenesis to mobilize fuel sources. If hypoglycemia develops (usually from insulin administration or sulfonylureas), an orchestrated counterregulatory response occurs -- glucagon and epinephrine rapidly stimulate glycogenolysis and gluconeogenesis, whereas GH and cortisol act over several hours to raise glucose levels and antagonize insulin action.

Although free water clearance is primarily controlled by vasopressin, cortisol and thyroid hormone are also important for facilitating renal tubular responses to vasopressin effects ([Chap. 329](#)). PTH and vitamin D function in an interdependent manner to control calcium metabolism ([Chap. 340](#)). PTH stimulates renal synthesis of 1,25 dihydroxyvitamin D, which increases calcium absorption in the gastrointestinal tract and enhances PTH action in bone. Increased calcium, along with vitamin D, feeds back to suppress PTH, thereby maintaining calcium balance.

Depending on the severity of a given stress and whether it is acute or chronic, multiple endocrine and cytokine pathways are activated to mount an appropriate physiologic response ([Chap. 328](#)). In severe acute stress such as trauma or shock, the sympathetic nervous system is activated and catecholamines are released, leading to increased cardiac output and a primed musculoskeletal system. Catecholamines also increase mean blood pressure and stimulate glucose production ([Chap. 72](#)). Multiple stress-induced pathways converge on the hypothalamus, stimulating several hormones including vasopressin and corticotropin-releasing hormone (CRH). These hormones, in addition to cytokines (tumor necrosis factor α , IL-2, IL-6), increase ACTH and GH production. ACTH stimulates the adrenal gland, increasing cortisol, which in turn helps to sustain blood pressure and dampen the inflammatory response. Increased vasopressin acts to conserve free water.

REPRODUCTION

The stages of reproduction include: (1) sex determination during fetal development ([Chap. 338](#)); (2) sexual maturation during puberty ([Chap. 8](#)); (3) conception, pregnancy, lactation, and child-rearing ([Chap. 336](#)), and (4) cessation of reproductive capability at

menopause. Each of these stages involves an orchestrated interplay of multiple hormones, a phenomenon well illustrated by the dynamic hormonal changes that occur during each 28-day menstrual cycle. In the early follicular phase, pulsatile secretion of LH and FSH stimulate the progressive maturation of the ovarian follicle. This results in a gradual increase of estrogen and progesterone leading to enhanced pituitary sensitivity to [GnRH](#), which, when combined with accelerated GnRH secretion, triggers the LH surge and rupture of the mature follicle. Inhibin, a protein produced by the granulosa cells, enhances follicular growth and feeds back to the pituitary to selectively suppress FSH, without affecting LH. Growth factors, such as [EGF](#) and [IGF-I](#) modulate follicular responsiveness to gonadotropins. Vascular endothelial growth factor and prostaglandins play a role in follicle vascularization and rupture.

During pregnancy, the increased production of prolactin, in combination with placentally derived steroids (e.g., estrogen and progesterone), prepares the breast for lactation ([Chap. 337](#)). Estrogens induce the production of progesterone receptors, allowing for increased responsiveness to progesterone. In addition to these and other hormones involved in lactation, the nervous system and oxytocin mediate the suckling response and milk release.

HORMONAL FEEDBACK REGULATORY SYSTEMS

Feedback control, both negative and positive, is a fundamental feature of endocrine systems. Each of the major hypothalamic-pituitary-hormone axes is governed by negative feedback, a process that maintains hormone levels within a relatively narrow range ([Chap. 328](#)). Examples of hypothalamic-pituitary negative feedback include (1) thyroid hormones on the [TRH](#)-TSH axis, (2) cortisol on the [CRH](#)-ACTH axis, (3) gonadal steroids on the [GnRH](#)-LH/FSH axis, and (4) [IGF-I](#) on the growth hormone-releasing hormone (GHRH)-GH axis ([Fig. 327-3](#)). These regulatory loops include both positive (e.g., TRH, TSH) and negative components (e.g., T₄, T₃), allowing for exquisite control of hormone levels. As an example, a small reduction of thyroid hormone triggers a rapid increase of TRH and TSH secretion, resulting in thyroid gland stimulation and increased thyroid hormone production. When the thyroid hormone reaches a normal level, it feeds back to suppress TRH and TSH, and a new steady state is attained. Feedback regulation also occurs for endocrine systems that do not involve the pituitary gland, such as calcium feedback on PTH, glucose inhibition of insulin secretion, and leptin feedback on the hypothalamus. An understanding of feedback regulation provides important insights into endocrine testing paradigms (see below).

Positive feedback control also occurs but is not well understood. The primary example is estrogen-mediated stimulation of the midcycle LH surge. Though chronic low levels of estrogen are inhibitory, gradually rising estrogen levels stimulate LH secretion. This effect, which is illustrative of an endocrine rhythm (see below), involves activation of the hypothalamic [GnRH](#) pulse generator. In addition, estrogen-primed gonadotropes are extraordinarily sensitive to GnRH, leading to a 10- to 20-fold amplification of LH release.

PARACRINE AND AUTOCRINE CONTROL

The aforementioned examples of feedback control involve classic endocrine pathways in which hormones are released by one gland and act on a distant target gland.

However, local regulatory systems, often involving growth factors, are increasingly recognized. *Paracrine regulation* refers to factors released by one cell that act on an adjacent cell in the same tissue. For example, somatostatin secretion by pancreatic islet d cells inhibits insulin secretion from nearby b cells. *Autocrine regulation* describes the action of a factor on the same cell from which it is produced. [IGF-I](#) acts on many cells that produce it, including chondrocytes, breast epithelium, and gonadal cells. Unlike endocrine actions, paracrine and autocrine control are difficult to document because local growth factor concentrations cannot be readily measured.

Anatomic relationships of glandular systems also greatly influence hormonal exposure -- the physical organization of islet cells enhances their intercellular communication; the portal vasculature of the hypothalamic-pituitary system exposes the pituitary to high concentrations of hypothalamic releasing factors; testicular seminiferous tubules gain exposure to high testosterone levels produced by the interdigitated Leydig cells; the pancreas receives nutrient information from the gastrointestinal tract; and the liver is the proximal target of insulin action because of portal drainage from the pancreas.

HORMONAL RHYTHMS

The feedback regulatory systems described above are superimposed on hormonal rhythms that are used for adaptation to the environment. Seasonal changes, the daily occurrence of the light-dark cycle, sleep, meals, and stress are examples of the many environmental events that affect hormonal rhythms. The *menstrual cycle* is repeated on average every 28 days, reflecting the time required to follicular maturation and ovulation ([Chap. 336](#)). Essentially all pituitary hormone rhythms are entrained to sleep and the *circadian cycle*, generating reproducible patterns that are repeated approximately every 24 h. The [HPA](#) axis, for example, exhibits characteristic peaks of ACTH and cortisol production in the early morning, with a nadir in the afternoon and evening. Recognition of these rhythms is important for endocrine testing and treatment. Patients with Cushing's syndrome characteristically exhibit increased midnight cortisol levels when compared to normal individuals ([Chap. 331](#)). In contrast, morning cortisol levels are similar in these groups, as cortisol is normally high at this time of day in normal individuals. The HPA axis is more susceptible to suppression by glucocorticoids administered at night as they blunt the early morning rise of ACTH. Understanding these rhythms allows glucocorticoid replacement that mimics diurnal production by administering larger doses in the morning than in the afternoon ([Chap. 331](#)).

Other endocrine rhythms occur on a more rapid time scale. Many peptide hormones are secreted in discrete bursts every few hours. LH and FSH secretion are exquisitely sensitive to [GnRH](#) pulse frequency. Intermittent pulses of GnRH are required to maintain pituitary sensitivity, whereas continuous exposure to GnRH causes pituitary gonadotrope desensitization. This feature of the hypothalamic-pituitary-gonadotrope (HPG) axis forms the basis for using long-acting GnRH agonists to treat central precocious puberty or to decrease testosterone levels in the management of prostate cancer.

It is important to be aware of the pulsatile nature of hormone secretion and the rhythmic patterns of hormone production when relating serum hormone measurements to normal values. For some hormones, integrated markers have been developed to circumvent

hormonal fluctuations. Examples include 24-h urine collections for cortisol, [IGF-I](#) as a biologic marker of GH action, and HbA1c as an index of long-term (weeks to months) blood glucose control.

Often, one must interpret endocrine data only in the context of other hormonal results. For example, parathyroid hormone levels are typically assessed in combination with serum calcium concentrations. A high serum calcium level in association with elevated PTH is suggestive of hyperparathyroidism, whereas a suppressed PTH in this situation is more likely to be caused by hypercalcemia of malignancy or other causes of hypercalcemia. Similarly, TSH should be elevated when T₄ and T₃ concentrations are low, reflecting reduced feedback inhibition. When this is not the case, it is important to consider other abnormalities in the hormonal axis, such as secondary hypothyroidism, which is caused by a defect at the level of the pituitary.

PATHOLOGIC MECHANISMS OF ENDOCRINE DISEASE

Endocrine diseases can be divided into three major types of conditions: (1) hormone excess, (2) hormone deficiency, and (3) hormone resistance ([Table 327-2](#)).

CAUSES OF HORMONE EXCESS

Syndromes of hormone excess can be caused by neoplastic growth of endocrine cells, autoimmune disorders, and excess hormone administration. Benign endocrine tumors, including parathyroid, pituitary, and adrenal adenomas, often retain the capacity to produce hormones, perhaps reflecting the fact that they are relatively well differentiated. Many endocrine tumors exhibit relatively subtle defects in their "set points" for feedback regulation. For example, in Cushing's disease, impaired feedback inhibition of ACTH secretion is associated with autonomous function. However, the tumor cells are not completely resistant to feedback, as revealed by the fact that ACTH is ultimately suppressed by higher doses of dexamethasone (e.g., high-dose dexamethasone test) ([Chap. 331](#)). Similar set point defects are also typical of parathyroid adenomas and autonomously functioning thyroid nodules.

The molecular basis of some endocrine tumors, such as the [MEN](#) syndromes (MEN-1, -2A, -2B), have provided important insights into tumorigenesis ([Chap. 339](#)). MEN-1 is characterized primarily by the triad of parathyroid, pancreatic islet, and pituitary tumors. MEN-2 predisposes to medullary thyroid carcinoma, pheochromocytoma, and hyperparathyroidism. The *MEN1* gene, located on chromosome 11q13, encodes a putative tumor-suppressor gene. Analogous to the paradigm first described for retinoblastoma, the affected individual inherits a mutant copy of the *MEN1* gene, and tumorigenesis ensues after a somatic "second hit" leads to loss of function of the normal *MEN1* gene (through deletion or point mutations).

In contrast to inactivation of a tumor-suppressor gene, as occurs in [MEN](#)-1 and most other inherited cancer syndromes, MEN-2 is caused by activating mutations in a single allele. In this case, activating mutations of the *RET* proto-oncogene, which encodes a receptor tyrosine kinase, leads to thyroid C-cell hyperplasia in childhood before the development of medullary thyroid carcinoma. Elucidation of the pathogenic mechanism has allowed early genetic screening for *RET* mutations in individuals at risk for MEN-2,

permitting identification of those who may benefit from prophylactic thyroidectomy and biochemical screening for pheochromocytoma and hyperparathyroidism.

Mutations that activate hormone receptor signaling have been identified in several [GPCRs](#) ([Table 327-3](#)). For example, activating mutations of the LH receptor causes a dominantly transmitted form of male-limited precocious puberty, reflecting premature stimulation of testosterone synthesis in Leydig cells ([Chap. 335](#)). Activating mutations in these GPCRs are located primarily in the transmembrane domains and induce receptor coupling to G_s α , even in the absence of hormone. Consequently, adenylate cyclase is activated and cyclic AMP levels increase in a manner that mimics hormone action. A similar phenomenon results from activating mutations in G_s α . When these occur early in development, they cause McCune-Albright syndrome. When they occur only in somatotropes, the activating G_s α mutations cause GH-secreting tumors and acromegaly ([Chap. 328](#)).

In autoimmune Graves' disease, antibody interactions with the TSH receptor mimic TSH action, leading to hormone overproduction ([Chap. 330](#)). Analogous to the effects of activating mutations of the TSH receptor, these stimulating autoantibodies induce conformational changes that release the receptor from a constrained state, thereby triggering receptor coupling to G proteins.

CAUSES OF HORMONE DEFICIENCY

Most examples of hormone deficiency states can be attributed to glandular destruction caused by autoimmunity, surgery, infection, inflammation, infarction, hemorrhage, or tumor infiltration ([Table 327-2](#)). Autoimmune damage to the thyroid gland (Hashimoto's thyroiditis) and pancreatic islet β cells (type 1 diabetes mellitus) are prevalent causes of endocrine disease. Mutations in a number of hormones, hormone receptors, transcription factors, enzymes, and channels can also lead to hormone deficiencies ([Table 327-3](#)).

HORMONE RESISTANCE

Most severe hormone resistance syndromes are due to inherited defects in membrane receptors, nuclear receptors, or in the pathways that transduce receptor signals ([Table 327-3](#)). These disorders are characterized by defective hormone action, despite the presence of increased hormone levels. In complete androgen resistance, for example, mutations in the androgen receptor cause genetic (XY) males to have a female phenotypic appearance, even though LH and testosterone levels are increased ([Chap. 338](#)). In addition to these relatively rare genetic disorders, more common acquired forms of functional hormone resistance include insulin resistance in type 2 diabetes mellitus, leptin resistance in obesity, and GH resistance in catabolic states. The pathogenesis of functional resistance involves receptor downregulation and postreceptor desensitization of signaling pathways; functional forms of resistance are generally reversible.

Approach to the Patient

Because endocrinology interfaces with numerous physiologic systems, there is no standard endocrine history and examination. Moreover, because most glands are

relatively inaccessible, the examination usually focuses on the manifestations of hormone excess or deficiency, as well as direct examination of palpable glands, such as the thyroid and gonads. For these reasons, it is important to evaluate patients in the context of their presenting symptoms, review of systems, family and social history, and exposure to medications that may affect the endocrine system. Astute clinical skills are required to detect subtle symptoms and signs suggestive of underlying endocrine disease. For example, a patient with Cushing's syndrome may manifest specific findings, such as central fat redistribution, striae, and proximal muscle weakness, in addition to features seen commonly in the general population, such as obesity, plethora, hypertension, and glucose intolerance. Similarly, the insidious onset of hypothyroidism -- with mental slowing, fatigue, dry skin, and other features -- can be difficult to distinguish from similar, nonspecific findings in the general population. Clinical judgment, based on knowledge of pathophysiology and experience, is required to decide when to embark on more extensive evaluation of these disorders. As described below, laboratory testing plays an essential role in endocrinology by allowing quantitative assessment of hormone levels and dynamics. Radiologic imaging tests, such as CT scan, MRI, thyroid scan, and ultrasound, are also used for the diagnosis of endocrine disorders. However, these tests are generally employed only after a hormonal abnormality has been established by biochemical testing.

Hormone Measurements and Endocrine Testing Radioimmunoassays are the most important diagnostic tool in endocrinology, as they allow sensitive, specific, and quantitative determination of steady-state and dynamic changes in hormone concentrations. Radioimmunoassays use antibodies to detect specific hormones. For many peptide hormones, these measurements are now configured as immunoradiometric assays (IRMAs), which use two different antibodies to increase binding affinity and specificity. There are many variations of these assays -- a common format involves using one antibody to capture the antigen (hormone) onto an immobilized surface and a second antibody, labeled with a fluorescent or radioactive tag, to detect the antigen. These assays are sensitive enough to detect plasma hormone concentrations in the picomolar to nanomolar range, and they can readily distinguish structurally related proteins, such as PTH from [PTHrP](#). A variety of other techniques are used to measure specific hormones, including mass spectroscopy, various forms of chromatography, and enzymatic methods; bioassays are now used rarely.

Most hormone measurements are based on plasma or serum samples. However, urinary hormone determinations remain useful for the evaluation of some conditions. Urinary collections over 24 h provide an integrated assessment of the production of a hormone or metabolite, many of which vary during the day. It is important to assure complete collections of 24-h urine samples; simultaneous measurement of creatinine provides an internal control for the adequacy of collection and can be used to normalize some hormone measurements. A 24-h urine free cortisol measurement largely reflects the amount of unbound cortisol, thus providing a reasonable index of biologically available hormone. Other commonly used urine determinations include: 17-hydroxycorticosteroids, 17-ketosteroids, vanillylmandelic acid (VMA), metanephrine, catecholamines, 5-hydroxyindoleacetic acid (5-HIAA), and calcium.

The value of quantitative hormone measurements lies in their correct interpretation in a

clinical context. The normal range for most hormones is relatively broad, often varying by a factor of two- to tenfold. The normal ranges for many hormones are gender- and age-specific. Thus, using the correct normative database is an essential part of interpreting hormone tests. The pulsatile nature of hormones and factors that can affect their secretion, such as sleep, meals, and medications, must also be considered. Cortisol values increase fivefold between midnight and dawn; reproductive hormone levels vary dramatically during the female menstrual cycle.

For many endocrine systems, much information can be gained from basal hormone testing, particularly when different components of an endocrine axis are assessed simultaneously. For example, low testosterone and elevated LH levels suggest a primarily gonadal problem, whereas a hypothalamic-pituitary disorder is likely if both LH and testosterone are low. Because TSH is a sensitive indicator of thyroid function, it is generally recommended as a first-line test for thyroid disorders. An elevated TSH level is almost always the result of primary hypothyroidism, whereas a low TSH is most often caused by thyrotoxicosis. These predictions can be confirmed by determining the free thyroxine level. Elevated calcium and PTH levels suggest hyperparathyroidism, whereas PTH is suppressed in hypercalcemia caused by malignancy or granulomatous diseases. A suppressed ACTH in the setting of hypercortisolemia, or increased urine free cortisol, is seen with hyperfunctioning adrenal adenomas.

It is not uncommon, however, for baseline hormone levels associated with pathologic endocrine conditions to overlap with the normal range. In this circumstance, dynamic testing is useful to further separate the two groups. There are a multitude of dynamic endocrine tests, but all are based on principles of feedback regulation, and most responses can be remembered based on the pathways that govern endocrine axes. *Suppression tests* are used in the setting of suspected endocrine hyperfunction. An example is the dexamethasone suppression test used to evaluate Cushing's syndrome ([Chaps. 328](#) and [331](#)). *Stimulation tests* are generally used to assess endocrine hypofunction. The ACTH stimulation test, for example, is used to assess the adrenal gland response in patients with suspected adrenal insufficiency. Other stimulation tests use hypothalamic-releasing factors such as [TRH](#), [GnRH](#), [CRH](#), and [GHRH](#) to evaluate pituitary hormone reserve ([Chap. 328](#)). Insulin-induced hypoglycemia evokes pituitary ACTH and GH responses. Stimulation tests based on reduction or inhibition of endogenous hormones are less commonly used. Examples include metyrapone inhibition of cortisol synthesis and clomiphene inhibition of estrogen feedback.

Screening and Assessment of Common Endocrine Disorders Because many endocrine disorders are prevalent in the adult population ([Table 327-4](#)), most are diagnosed and managed by general internists, family practitioners, or other primary health care providers. The high prevalence and clinical impact of certain endocrine diseases justifies vigilance for features of these disorders during routine physical examinations; laboratory screening is indicated in selected high-risk populations.

(Bibliography omitted in Palm version)

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328. DISORDERS OF THE ANTERIOR PITUITARY AND HYPOTHALAMUS - Shlomo Melmed

The anterior pituitary is often referred to as the "master gland" because, together with the hypothalamus, it orchestrates the complex regulatory functions of multiple other endocrine glands. The anterior pituitary gland produces six major hormones: (1) prolactin (PRL), (2) growth hormone (GH), (3) adrenocorticotropin hormone (ACTH), (4) luteinizing hormone (LH), (5) follicle-stimulating hormone (FSH), and (6) thyroid-stimulating hormone (TSH) ([Table 328-1](#)). Pituitary hormones are secreted in a pulsatile manner, reflecting stimulation by an array of specific hypothalamic releasing factors. Each of these pituitary hormones elicits specific responses in peripheral target tissues. The hormonal products of these peripheral glands, in turn, exert feedback control at the level of the hypothalamus and pituitary to modulate pituitary function ([Fig. 328-1](#)). Pituitary tumors cause characteristic hormone excess syndromes. Hormone deficiency may be inherited or acquired. Fortunately, efficacious treatments exist for the various pituitary hormone excess and deficiency syndromes. Nonetheless, these diagnoses are often elusive, emphasizing the importance of recognizing subtle clinical manifestations and performing the correct laboratory diagnostic tests. **For discussion of disorders of the posterior pituitary, or neurohypophysis, see [Chap. 329](#).*

ANATOMY AND DEVELOPMENT

Anatomy The pituitary gland weighs ~600 mg and is located within the sella turcica ventral to the diaphragma sella; it comprises anatomically and functionally distinct anterior and posterior lobes. The sella is contiguous to vascular and neurologic structures, including the cavernous sinuses, cranial nerves, and optic chiasm. Thus, expanding intrasellar pathologic processes may have significant central mass effects in addition to their endocrinologic impact.

Hypothalamic neural cells synthesize specific releasing and inhibiting hormones that are secreted directly into the portal vessels of the pituitary stalk. Blood supply of the pituitary gland is derived from the superior and inferior hypophyseal arteries ([Fig. 328-2](#)). The hypothalamic-pituitary portal plexus provides the major blood source for the anterior pituitary, allowing reliable transmission of hypothalamic peptide pulses without significant systemic dilution; consequently, pituitary cells are exposed to sharp spikes of releasing factors and in turn release their hormones as discrete pulses ([Fig. 328-3](#)).

The posterior pituitary is supplied by the inferior hypophyseal arteries. In contrast to the anterior pituitary, the posterior lobe is directly innervated by hypothalamic neurons (supraopticohypophyseal and tuberohypophyseal nerve tracts) via the pituitary stalk ([Chap. 329](#)). Thus, posterior pituitary production of vasopressin (antidiuretic hormone; ADH) and oxytocin is particularly sensitive to neuronal damage by lesions that affect the pituitary stalk or hypothalamus.

Pituitary Development The embryonic differentiation and maturation of anterior pituitary cells have been elucidated in considerable detail. Pituitary development from Rathke's pouch involves a complex interplay of lineage-specific transcription factors expressed in pluripotential stem cells and gradients of locally produced growth factors ([Table 328-1](#)). The transcription factor Pit-1 determines cell-specific expression

of [GH](#), [PRL](#), and [TSH](#) in somatotropes, lactotropes, and thyrotropes. Expression of high levels of estrogen receptors in cells that contain Pit-1 favors PRL expression, whereas thyrotrope embryonic factor (TEF) induces TSH expression. Pit-1 binds to GH, PRL, and TSH gene regulatory elements, as well as to recognition sites on its own promoter, providing a mechanism for perpetuating selective pituitary phenotypic stability. The transcription factor Prop-1 induces the pituitary development of Pit-1-specific lineages, as well as gonadotropes. Gonadotrope cell development is further defined by the cell-specific expression of the nuclear receptors, steroidogenic factor (SF-1) and DAX-1. Development of corticotrope cells, which express the proopiomelanocortin (POMC) gene, requires corticotropin upstream transcription element (CUTE) and the PTX-1 transcription factor. Abnormalities of pituitary development caused by mutations of Pit-1, Prop-1, SF-1, and DAX-1 result in a series of rare, selective or combined, pituitary hormone deficits.

HYPOTHALAMIC AND ANTERIOR PITUITARY INSUFFICIENCY

Hypopituitarism results from impaired production of one or more of the anterior pituitary trophic hormones. Reduced pituitary function can result from inherited disorders; more commonly, it is acquired and reflects the mass effects of tumors or the consequences of inflammation or vascular damage. These processes may also impair synthesis or secretion of hypothalamic hormones, with resultant pituitary failure ([Table 328-2](#)).

DEVELOPMENTAL AND GENETIC CAUSES OF HYPOPITUITARISM

Pituitary Dysplasia Pituitary dysplasia may result in aplastic, hypoplastic, or ectopic pituitary gland development. Because pituitary development requires midline cell migration from the nasopharyngeal Rathke's pouch, midline craniofacial disorders, such as cleft lip and palate, basal encephalocele, hypertelorism, and optic nerve hypoplasia, may be associated with pituitary dysplasia. Acquired pituitary failure in the newborn can also be caused by birth trauma, including cranial hemorrhage, asphyxia, and breech delivery.

Septo-optic Dysplasia Hypothalamic dysfunction and hypopituitarism may result from dysgenesis of the septum pellucidum or corpus callosum. Affected children have mutations in the *HESX1* gene, which is involved in early development of the ventral prosencephalon. These children exhibit cleft palate, syndactyly, ear deformities, hypertelorism, optic atrophy, micropenis, and anosmia. Pituitary dysfunction leads to diabetes insipidus, [GH](#) deficiency and short stature, and, occasionally, [TSH](#) deficiency.

Tissue-Specific Factor Mutations Several pituitary cell-specific transcription factors, such as Pit-1 and Prop-1, are critical for determining the development and function of specific anterior pituitary cell lineages. Autosomal dominant or recessive Pit-1 mutations cause combined [GH](#), [PRL](#), and [TSH](#) deficiencies. These patients present with growth failure and varying degrees of hypothyroidism. The pituitary may appear hypoplastic on magnetic resonance imaging (MRI).

Prop-1 is expressed early in pituitary development and appears to be required for Pit-1 function. Familial and sporadic *PROP1* mutations result in combined [GH](#), [PRL](#), [TSH](#), and gonadotropin deficiency, with preservation of [ACTH](#). Over 80% of these patients have

growth retardation and, by adulthood, all are deficient in TSH and gonadotropins. Because of gonadotropin deficiency, they do not enter puberty spontaneously ([Fig. 328-4](#)).

Developmental Hypothalamic Dysfunction

Kallmann Syndrome This syndrome results from defective hypothalamic gonadotropin-releasing hormone (GnRH) synthesis and is associated with anosmia or hyposmia due to olfactory bulb agenesis or hypoplasia ([Chap. 335](#)). The syndrome may also be associated with color blindness, optic atrophy, nerve deafness, cleft palate, renal abnormalities, cryptorchidism, and neurologic abnormalities such as mirror movements. Defects in the *KAL* gene, which maps to chromosome Xp22.3, prevent embryonic migration of GnRH neurons from the hypothalamic olfactory placode to the hypothalamus. Genetic abnormalities, in addition to *KAL* mutations, can also cause isolated GnRH deficiency, as autosomal recessive and dominant modes of transmission have been described. GnRH deficiency prevents progression through puberty. Males present with delayed puberty and pronounced hypogonadal features, including micropenis, probably the result of low testosterone levels during infancy ([Chap. 335](#)). Female patients present with primary amenorrhea and failure of secondary sexual development.

Kallmann syndrome and other causes of congenital [GnRH](#) deficiency are characterized by low [LH](#) and [FSH](#) levels and low concentrations of sex steroids (testosterone or estradiol). In sporadic cases of isolated gonadotropin deficiency, the diagnosis is often one of exclusion after eliminating other causes of hypothalamic-pituitary dysfunction. Repetitive GnRH administration restores normal pituitary gonadotropin responses, pointing to a hypothalamic defect.

Long-term treatment of males with human chorionic gonadotropin (hCG) or testosterone restores pubertal development and secondary sex characteristics; females can be treated with cyclic estrogen and progestin. Fertility may also be restored by the administration of subcutaneous, pulsatile [GnRH](#) using a portable infusion pump.

Laurence-Moon-Bardet-Biedl Syndrome This rare autosomal recessive disorder is characterized by mental retardation; obesity; and hexadactyly, brachydactyly, or syndactyly. Central diabetes insipidus may or may not be associated. [GnRH](#) deficiency occurs in 75% of males and half of affected females. Retinal degeneration begins in early childhood, and most patients are blind by age 30.

Frohlich Syndrome (Adipose Genital Dystrophy) A broad spectrum of hypothalamic lesions may be associated with hyperphagia, obesity, and central hypogonadism. Decreased [GnRH](#) production in these patients results in attenuated pituitary [FSH](#) and [LH](#) synthesis and release.

Prader-Willi Syndrome Chromosome 15q deletions are associated with hypogonadotropic hypogonadism, hyperphagia-obesity, chronic muscle hypotonia, mental retardation, and adult-onset diabetes mellitus ([Chap. 66](#)). Multiple somatic defects also involve the skull, eyes, ears, hands, and feet. Diminished hypothalamic oxytocin- and vasopressin-producing nuclei have been reported.

Deficient [GnRH](#) synthesis is suggested by the observation that chronic GnRH treatment restores pituitary [LH](#) and [FSH](#) release.

ACQUIRED HYPOPITUITARISM

Hypopituitarism may be caused by accidental or neurosurgical trauma; vascular events such as apoplexy; pituitary or hypothalamic neoplasms such as pituitary adenomas, craniopharyngiomas, or metastatic deposits; inflammatory disease such as lymphocytic hypophysitis; infiltrative disorders such as sarcoidosis, hemochromatosis ([Chap. 345](#)), and tuberculosis; or irradiation. It is often difficult to localize the site of hormonal dysfunction as many processes, including hemochromatosis and radiation, may affect both hypothalamic and pituitary function.

Hypothalamic Infiltration Disorders These disorders -- including those associated with sarcoidosis, histiocytosis X, amyloidosis, and hemochromatosis -- frequently involve both hypothalamic and pituitary neuronal and neurochemical tracts. Consequently, diabetes insipidus occurs in half of patients with these disorders. Growth retardation is seen if attenuated [GH](#) secretion occurs before pubertal epiphyseal closure. Hypogonadotropic hypogonadism and hyperprolactinemia are also common.

Inflammatory Lesions Pituitary damage and subsequent dysfunction can be seen with chronic infections such as tuberculosis, opportunistic fungal infections associated with AIDS, and in tertiary syphilis. Other inflammatory processes, such as granulomas or sarcoidosis, may mimic a pituitary adenoma. These lesions may cause extensive hypothalamic and pituitary damage, leading to trophic hormone failure.

Cranial Irradiation Cranial irradiation may result in long-term hypothalamic and pituitary dysfunction, especially in children and adolescents who are more susceptible to damage following whole-brain or head and neck therapeutic irradiation. The development of hormonal abnormalities correlates strongly with irradiation dosage and the time interval after completion of radiotherapy. Up to two-thirds of patients ultimately develop hormone insufficiency after a median dose of 50 Gy (5000 rad) directed at the skull base. The development of hypopituitarism occurs over 5 to 15 years and usually reflects hypothalamic damage rather than absolute destruction of pituitary cells. Though the pattern of hormone loss is variable, [GH](#) deficiency is most commonly followed by gonadotropin and [ACTH](#) deficiency. When deficiency of one or more hormones is documented, the possibility of diminished reserve of other hormones is likely. Accordingly, anterior pituitary function should be evaluated over the long term in previously irradiated patients, and replacement therapy instituted when appropriate (see below).

Lymphocytic Hypophysitis This occurs mainly in pregnant or post-partum women; it usually presents with hyperprolactinemia and [MRI](#) evidence of a prominent pituitary mass resembling an adenoma, with mildly elevated [PRL](#) levels. Pituitary failure caused by diffuse lymphocytic infiltration may be transient or permanent but requires immediate evaluation and treatment. Rarely, isolated pituitary hormone deficiencies have been described, suggesting a selective autoimmune process targeted to specific cell types. Most patients manifest symptoms of progressive mass effects with headache and visual disturbance. The erythrocyte sedimentation rate is often elevated. As the MRI image

may be indistinguishable from that of a pituitary adenoma, hypophysitis should be considered in a post-partum woman with a newly diagnosed pituitary mass before embarking on unnecessary surgical intervention. The inflammatory process often resolves after several months of glucocorticoid treatment, and pituitary function may be restored, depending on the extent of damage.

Pituitary Apoplexy Acute intrapituitary hemorrhagic vascular events can cause substantial damage to the pituitary and surrounding sellar structures. Pituitary apoplexy may occur spontaneously in a preexisting adenoma (usually nonfunctioning); postpartum (Sheehan's syndrome); or in association with diabetes, hypertension, sickle cell anemia, or acute shock. The hyperplastic enlargement of the pituitary during pregnancy increases the risk for hemorrhage and infarction. Apoplexy is an endocrine emergency that may result in severe hypoglycemia, hypotension, central nervous system (CNS) hemorrhage, and death. Acute symptoms include severe headache with signs of meningeal irritation, bilateral visual changes, ophthalmoplegia that varies, and, in severe cases, cardiovascular collapse and loss of consciousness. Pituitary computed tomography (CT) or [MRI](#) may reveal signs of intratumoral or sellar hemorrhage, with deviation of the pituitary stalk and compression of pituitary tissue.

Patients with no evident visual loss or impaired consciousness can be observed and managed conservatively with high-dose glucocorticoids. Those with significant or progressive visual loss or loss of consciousness require urgent surgical decompression. Visual recovery after surgery is inversely correlated with the length of time after the acute event. Therefore, severe ophthalmoplegia or visual deficits are indications for early surgery. Hypopituitarism is very common after apoplexy.

Empty Sella A partial or apparently totally empty sella is usually an incidental [MRI](#) finding. These patients usually exhibit normal pituitary function, implying that the surrounding rim of pituitary tissue is fully functional. Hypopituitarism, however, may develop insidiously. Pituitary masses may undergo clinically silent infarction with development of a partial or totally empty sella by cerebrospinal fluid (CSF) filling the dural herniation. Rarely, functional pituitary adenomas may arise within the rim of pituitary tissue, and these are not always visible on MRI.

PRESENTATION AND DIAGNOSIS

The clinical manifestations of hypopituitarism depend on which hormones are lost and the extent of the hormone deficiency. [GH](#) deficiency causes growth disorders in children and leads to abnormal body composition in adults (see below). Gonadotropin deficiency causes menstrual disorders and infertility in women and decreased sexual function, infertility, and loss of secondary sexual characteristics in men. [TSH](#) and [ACTH](#) deficiency usually develop later in the course of pituitary failure. TSH deficiency leads to growth retardation in children and features of hypothyroidism in children and in adults. The secondary form of adrenal insufficiency caused by ACTH deficiency leads to hypocortisolism with relative preservation of mineralocorticoid production. [PRL](#) deficiency causes failure of lactation. When lesions involve the posterior pituitary tracts, polyuria and polydipsia reflect loss of vasopressin secretion. Epidemiologic studies have documented an increased mortality rate in patients with longstanding pituitary damage, primarily from increased cardiovascular and cerebrovascular disease.

LABORATORY INVESTIGATION

Biochemical diagnosis of pituitary insufficiency is made by demonstrating low levels of trophic hormones in the setting of low target hormone levels. For example, low free thyroxine in the setting of a low or inappropriately normal [TSH](#) level suggests secondary hypothyroidism. Similarly, a low testosterone level without elevation of gonadotropins suggests hypogonadotropic hypogonadism. Provocative tests may be required to assess pituitary reserve ([Table 328-3](#)). GH responses to insulin-induced hypoglycemia, arginine, L-dopa, growth hormone-releasing hormone (GHRH), or growth hormone-releasing peptides (GHRPs) can be used to assess GH reserve. [PRL](#) and TSH responses to thyrotropin-releasing hormone (TRH) reflect lactotrope and thyrotrope function. Corticotropin-releasing hormone (CRH) administration induces [ACTH](#) release, and administration of synthetic ACTH (cortrosyn) evokes adrenal cortisol release as an indirect indicator of pituitary ACTH reserve ([Chap. 331](#)). ACTH reserve is most reliably assessed during insulin-induced hypoglycemia. However, this test should be performed cautiously in patients with suspected adrenal insufficiency because of increased risk of hypoglycemia and hypotension.

TREATMENT

Hormone replacement therapy, including glucocorticoids, thyroid hormone, sex steroids, growth hormone, and vasopressin, is usually free of complications. Treatment regimens that mimic physiologic hormone production allow for maintenance of satisfactory clinical homeostasis. Effective dosage schedules are outlined in [Table 328-4](#). Patients in need of glucocorticoid replacement require careful dose adjustments during stressful events such as acute illness, dental procedures, trauma, and acute hospitalization ([Chap. 331](#)).

HYPOTHALAMIC, PITUITARY, AND OTHER SELLAR MASSES

PITUITARY TUMORS

Pituitary adenomas are the most common cause of pituitary hormone hypersecretion and hyposecretion syndromes in adults. They account for ~10% of all intracranial neoplasms. At autopsy, up to a quarter of all pituitary glands harbor an unsuspected microadenoma (<10 mm diameter). Similarly, pituitary imaging detects small pituitary lesions in at least 10% of normal individuals.

Pathogenesis Pituitary adenomas are benign neoplasms that arise from one of the five anterior pituitary cell types. The clinical and biochemical phenotype of pituitary adenomas depend on the cell type from which they are derived and are described in detail below. Thus, tumors arising from lactotrope ([PRL](#)), somatotrope ([GH](#)), corticotrope ([ACTH](#)), thyrotrope ([TSH](#)), or gonadotrope ([LH](#), [FSH](#)) cells hypersecrete their respective hormones ([Table 328-5](#)). Plurihormonal tumors that express combinations of GH, PRL, TSH, ACTH, and the glycoprotein hormone α subunit may be diagnosed by careful immunocytochemistry or may, in fact, present with mixed clinical features of these hormonal hypersecretory syndromes. Morphologically, these tumors may arise from a single polysecreting cell type or consist of cells with mixed function within the same tumor.

Hormonally active tumors are characterized by autonomous hormone secretion with diminished responsiveness to the normal physiologic pathways of inhibition. Hormone production does not always correlate with tumor size. Small hormone-secreting adenomas may cause significant clinical perturbations, whereas larger adenomas that produce less hormone may be clinically silent and remain undiagnosed (if no central compressive effects occur). About one-third of all adenomas are clinically nonfunctioning and produce no distinct clinical hypersecretory syndrome. Most arise from gonadotrope cells and may secrete α - and β -glycoprotein hormone subunits or, very rarely, intact circulating gonadotropins. True pituitary carcinomas with documented extracranial metastases are exceedingly rare.

Almost all pituitary adenomas are monoclonal in origin, implying the acquisition of one or more somatic mutations that confer a selective growth advantage. In addition to direct studies of oncogene mutations, this idea is supported by X-chromosomal inactivation analyses of tumors in female patients heterozygous for X-linked genes. Consistent with their clonal origin, complete surgical resection of small pituitary adenomas usually cures hormone hypersecretion. Nevertheless, hypothalamic hormones, such as [GHRH](#) or [CRH](#), also enhance the mitotic activity of their respective pituitary target cells, in addition to their role in pituitary hormone regulation. Thus, patients harboring rare abdominal or chest tumors elaborating ectopic GHRH or CRH may present with somatotrope or corticotrope hyperplasia.

Several etiologic genetic events have been implicated in the development of pituitary tumors. The pathogenesis of sporadic forms of acromegaly has been particularly informative as a model of tumorigenesis. [GHRH](#), after binding to its G protein-coupled somatotrope receptor, utilizes cyclic AMP as a second messenger to stimulate [GH](#) secretion and somatotrope proliferation. A subset (~35%) of GH-secreting pituitary tumors contain mutations in Gsa (Arg 201 $\rightarrow$ Cys or His; Gln 227 $\rightarrow$ Arg). These mutations inhibit intrinsic GTPase activity, resulting in constitutive elevation of cyclic AMP, Pit-1 induction, and activation of cyclic AMP response element binding protein (CREB), thereby promoting somatotrope cell proliferation.

Characteristic loss of heterozygosity (LOH) in various chromosomes has been documented in large or invasive macroadenomas, suggesting the presence of putative tumor suppressor genes at these loci. LOH of chromosome region on 11q13, 13, and 9 is present in up to 20% of sporadic pituitary tumors including [GH](#)-, [PRL](#)-, and [ACTH](#)-producing adenomas and in some nonfunctioning tumors.

Compelling evidence also favors growth factor promotion of pituitary tumor proliferation. Basic fibroblast growth factor (bFGF) is abundant in the pituitary and has been shown to stimulate pituitary cell mitogenesis. Other factors involved in initiation and promotion of pituitary tumors include loss of negative-feedback inhibition (as seen with primary hypothyroidism or hypogonadism) and estrogen-mediated or paracrine angiogenesis. Growth characteristics and neoplastic behavior may also be influenced by several activated oncogenes, including *RAS* and pituitary tumor transforming gene (*PTTG*).

Genetic Syndromes Associated with Pituitary Tumors Several familial syndromes are associated with pituitary tumors, and the genetic mechanisms for some of these

have been unraveled ([Table 328-6](#)).

Multiple endocrine neoplasia (MEN) 1 is an autosomal dominant syndrome characterized primarily by a genetic predisposition to parathyroid, pancreatic islet, and pituitary adenomas ([Chap. 339](#)). MEN-1 is caused by inactivating germline mutations in *MENIN*, a constitutively expressed tumor-suppressor gene located on chromosome 11q13. Loss of heterozygosity, or a somatic mutation of the remaining normal *MENIN* allele, leads to tumorigenesis. About half of affected patients develop prolactinomas; acromegaly and Cushing's syndrome are less commonly encountered.

Carney syndrome is characterized by spotty skin pigmentation, myxomas, and endocrine tumors including testicular, adrenal, and pituitary adenomas. Acromegaly occurs in about 20% of patients. This autosomal dominant syndrome is associated with microsatellite alterations on chromosome 2p16.

McCune-Albright syndrome consists of polyostotic fibrous dysplasia, pigmented skin patches, and a variety of endocrine disorders, including [GH](#)-secreting pituitary tumors, adrenal adenomas, and autonomous ovarian function ([Chap. 343](#)). Hormonal hypersecretion is due to constitutive cyclic AMP production caused by inactivation of the GTPase activity of Gsa. The Gsa mutations occur postzygotically, leading to a mosaic pattern of mutant expression.

Familial acromegaly is a rare disorder in which family members may manifest either acromegaly or gigantism. The disorder is associated with [LOH](#) at a chromosome 11q13 locus distinct from that of *MENIN*.

OTHER SELLAR MASSES

Craniopharyngiomas are derived from Rathke's pouch. They arise near the pituitary stalk and commonly extend into the suprasellar cistern. These tumors are often large, cystic, and locally invasive. Many are partially calcified, providing a characteristic appearance on skull x-ray and [CT](#) images. More than half of all patients present before age 20, usually with signs of increased intracranial pressure, including headache, vomiting, papilledema, and hydrocephalus. Associated symptoms include visual field abnormalities, personality changes and cognitive deterioration, cranial nerve damage, sleep difficulties, and weight gain. Anterior pituitary dysfunction and diabetes insipidus are common. About half of affected children present with growth retardation.

Treatment usually involves transcranial or transsphenoidal surgical resection followed by postoperative radiation of residual tumor. This approach can result in long-term survival and ultimate cure, but most patients require lifelong pituitary hormone replacement. If the pituitary stalk is uninvolved and can be preserved at the time of surgery, the incidence of subsequent anterior pituitary dysfunction is significantly diminished.

Developmental failure of Rathke's pouch obliteration may lead to *Rathke's cysts*, which are small (<5 mm) cysts entrapped by squamous epithelium; these cysts are found in about 20% of individuals at autopsy. Although Rathke's cleft cysts do not usually grow and are often diagnosed incidentally, about a third present in adulthood with

compressive symptoms, diabetes insipidus, and hyperprolactinemia due to stalk compression. Rarely, internal hydrocephalus develops. The diagnosis is suggested preoperatively by visualizing the cyst wall on [MRI](#), which distinguishes these lesions from craniopharyngiomas. Cyst contents range from [CSF](#)-like fluid to mucoid material. *Arachnoid cysts* are rare and generate an MRI image isointense with cerebrospinal fluid.

Sella chordomas usually present with bony clival erosion, local invasiveness, and, on occasion, calcification. Normal pituitary tissue may be visible on [MRI](#), distinguishing chordomas from aggressive pituitary adenomas. Mucinous material may be obtained by fine-needle aspiration.

Meningiomas arising in the sellar region may be difficult to distinguish from nonfunctioning pituitary adenomas. On [MRI](#) they may be asymmetric, and on [CT](#) they may show evidence of bony erosion. Meningiomas may cause compressive symptoms.

Histiocytosis X comprises a variety of syndromes associated with foci of eosinophilic granulomas. Diabetes insipidus, exophthalmos, and punched-out lytic bone lesions (*Hand-Schuller-Christian disease*) are associated with granulomatous lesions visible on [MRI](#), as well as a characteristic axillary skin rash. Rarely, the pituitary stalk may be involved.

Pituitary metastases occur in ~3% of cancer patients. Blood-borne metastatic deposits are found almost exclusively in the posterior pituitary. Accordingly, diabetes insipidus can be a presenting feature of lung, gastrointestinal, breast, and other pituitary metastases. About half of pituitary metastases originate from breast cancer; about 25% of patients with breast cancer have such deposits. Rarely, pituitary stalk involvement results in anterior pituitary insufficiency. The [MRI](#) diagnosis of a metastatic lesion may be difficult to distinguish from an aggressive pituitary adenoma; the diagnosis may require histologic examination of excised tumor tissue. Primary or metastatic lymphoma, leukemias, and plasmacytomas also occur within the sella.

Hypothalamic hamartomas and *gangliocytomas* may arise from astrocytes, oligodendrocytes, and neurons with varying degrees of differentiation. These tumors may overexpress hypothalamic neuropeptides including [GnRH](#), [GHRH](#), or [CRH](#). In GnRH-producing tumors, children present with precocious puberty, psychomotor delay, and laughing-associated seizures. Medical treatment of GnRH-producing hamartomas with long-acting GnRH analogues effectively suppresses gonadotropin secretion and controls pubertal development. Rarely, hamartomas are also associated with craniofacial abnormalities; imperforate anus; cardiac, renal, and lung disorders; and pituitary failure (*Pallister-Hall syndrome*). Hypothalamic hamartomas are often contiguous with the pituitary, and preoperative [MRI](#) diagnosis may not be possible. Histologic evidence of hypothalamic neurons in tissue resected at transsphenoidal surgery may be the first indication of a primary hypothalamic lesion.

Hypothalamic gliomas and *optic gliomas* occur mainly in childhood and usually present with visual loss. Adults have more aggressive tumors; about a third are associated with neurofibromatosis.

Brain germ-cell tumors may arise within the sellar region. These include

dysgerminomas, which are associated with diabetes insipidus and visual loss and rarely metastasize. *Germinomas*, *embryonal carcinomas*, *teratomas*, and *choriocarcinomas* may arise in the parasellar region and produce [hCG](#). These germ-cell tumors present with precocious puberty, diabetes insipidus, visual field defects, and thirst disorders. Many patients are [GH](#)-deficient with short stature.

METABOLIC EFFECTS OF HYPOTHALAMIC LESIONS

The hypothalamus is subject to injury from mass lesions, granulomatous disorders, infections, and hemorrhage. Lesions involving the anterior and preoptic hypothalamic regions cause paradoxical vasoconstriction, tachycardia, and hyperthermia. Acute hyperthermia is usually due to a hemorrhagic insult, but poikilothermia may also occur. Central disorders of thermoregulation result from posterior hypothalamic damage. The *periodic hypothermia syndrome* comprises episodic attacks of rectal temperatures $<30^{\circ}\text{C}$, sweating, vasodilation, vomiting, and bradycardia ([Chap. 20](#)). Damage to the ventromedial nuclei by craniopharyngiomas, hypothalamic trauma, or inflammatory disorders may be associated with *hyperphagia* and *obesity*. This region appears to contain an energy-satiety center where melanocortin receptors are influenced by leptin, insulin, [POMC](#) products, and gastrointestinal peptides ([Chap. 77](#)). Median eminence involvement results in diabetes insipidus in about 50% of patients. Hypothalamic gliomas in early childhood may be associated with a diencephalic syndrome characterized by progressive severe emaciation and growth failure. Polydipsia or hypodipsia are associated with damage to central osmo-receptors located in preoptic nuclei ([Chap. 329](#)). Slow-growing hypothalamic lesions can cause increased somnolence and disturbed sleep cycles as well as obesity, hypothermia, and emotional outbursts. Lesions of the central hypothalamus may stimulate sympathetic neurons, leading to elevated serum catecholamine and cortisol levels. These patients are predisposed to cardiac arrhythmias, hypertension, and gastric erosions.

EVALUATION

Local Mass Effects Clinical manifestations of sellar lesions vary, depending on the anatomic location of the mass and direction of its extension ([Table 328-7](#)). The dorsal roof of the sella presents the least resistance to soft tissue expansion from within the confines of the sella; consequently, pituitary adenomas frequently extend in a suprasellar direction. Bony invasion may ultimately occur as well.

Headaches are common features of small intrasellar tumors, even with no demonstrable suprasellar extension. Because of the confined nature of the pituitary, small changes in intrasellar pressure stretch the dural plate; however, the severity of the headache correlates poorly with adenoma size or extension.

Suprasellar extension can lead to visual loss by several mechanisms, the most common being compression of the optic chiasm, but direct invasion of the optic nerves or obstruction of [CSF](#) flow leading to secondary visual disturbances also occur. Pituitary stalk compression by a hormonally active or inactive intrasellar mass may compress the portal vessels, disrupting pituitary access to the hypothalamic hormones and dopamine; this results in hyperprolactinemia and concurrent loss of other pituitary hormones. This "stalk section" phenomenon may also be caused by trauma, whiplash injury with

posterior clinoid stalk compression, or skull base fractures. Lateral mass invasion may impinge on the cavernous sinus and compress its neural contents, leading to cranial nerve III, IV, and VI palsies as well as effects on the ophthalmic and maxillary branches of the fifth cranial nerve ([Chap. 367](#)). Patients may present with diplopia, ptosis, ophthalmoplegia, and decreased facial sensation, depending on the extent of neural damage. Extension into the sphenoid sinus indicates that the pituitary mass has eroded through the sellar floor. Aggressive tumors may also invade the palate roof and cause nasopharyngeal obstruction, infection, and, rarely, CSF leakage. Both temporal and frontal lobes may be invaded, leading to uncinate seizures, personality disorders, and anosmia. Direct hypothalamic encroachment by an invasive pituitary mass may cause important metabolic sequelae, precocious puberty or hypogonadism, diabetes insipidus, sleep disturbances, dysthermia, and appetite disorders.

MRI Sagittal and coronal T1-weighted spin-echo MRI imaging, before and after administration of gadolinium, allow precise visualization of the pituitary gland with clear delineation of the hypothalamus, pituitary stalk, pituitary tissue and surrounding suprasellar cisterns, cavernous sinuses, sphenoid sinus, and optic chiasm. Pituitary gland height ranges from 6 mm in children to 8 mm in adults; during pregnancy and puberty, the height may reach 10 to 12 mm. The upper aspect of the adult pituitary is flat or slightly concave, but in adolescent and pregnant individuals, this surface may be convex, reflecting physiologic pituitary enlargement. The stalk should be vertical. [CT](#) scan is indicated to define the extent of bony erosion or the presence of calcification.

The soft tissue consistency of the pituitary gland is slightly heterogeneous on [MRI](#). Anterior pituitary signal intensity resembles that of brain matter on T1-imaging ([Fig. 328-5](#)). Adenoma density is usually lower than that of surrounding normal tissue on T1-weighted imaging, and the signal intensity increases with T2-weighted images. The high phospholipid content of the posterior pituitary results in a bright enhancing signal.

Sellar masses are commonly encountered as incidental findings on [MRI](#), and most of these are pituitary adenomas (incidentalomas). This finding is consistent with the observation that clinically silent pituitary microadenomas can be identified in up to 25% of pituitaries in autopsy series. In the absence of hormone hypersecretion, these small lesions can be safely monitored by MRI, which is performed annually and then less often if there is no evidence of growth. Resection should be considered for incidentally discovered macroadenomas, as about one-third become invasive or cause local pressure effects. If hormone hypersecretion is evident, specific therapies are indicated. When larger masses (>1 cm) are encountered, they should also be distinguished from nonadenomatous lesions. Meningiomas are often associated with bony hyperostosis; craniopharyngiomas may be calcified and are usually hypodense, whereas gliomas are hyperdense on T2-weighted images.

Ophthalmologic Evaluation Because optic tracts may be contiguous to an expanding pituitary mass, reproducible visual field assessment that uses perimetry techniques should be performed on all patients with sellar mass lesions that abut the optic chiasm. Loss of red perception is an early sign of optic tract pressure. Bitemporal hemianopia or superior bitemporal defects are classically observed, reflecting the location of these tracts within the inferior and posterior part of the chiasm. Early diagnosis reduces the risk of blindness, scotomas, or other visual disturbances.

Laboratory Investigation The presenting clinical features of functional pituitary adenomas (e.g., acromegaly, prolactinomas, or Cushing's disease) should guide the laboratory studies (see below). However, for a sellar mass with no obvious clinical features of hormone excess, laboratory studies are geared towards determining the nature of the tumor and assessing the possible presence of hypopituitarism. When a pituitary adenoma is suspected based on [MRI](#), initial hormonal evaluation usually includes: (1) basal [PRL](#); (2) insulin-like growth factor (IGF) I; (3) 24-h urinary free cortisol (UFC) and/or overnight oral dexamethasone (1 mg) suppression test; (4) α -subunit, [FSH](#), and [LH](#) levels; and (5) thyroid function tests. Additional hormonal evaluation may be indicated based on the results of these tests. Pending more detailed assessment of hypopituitarism, a menstrual history, testosterone level, 8 A.M. cortisol, and thyroid function tests usually identify patients with pituitary hormone deficiencies that require hormone replacement before further testing or surgery.

Histologic Evaluation Immunohistochemical staining of pituitary tumor specimens obtained at transsphenoidal surgery confirm clinical and laboratory studies and provide a histologic diagnosis when hormone studies are equivocal and in cases of clinically nonfunctioning tumors. Occasionally, ultrastructural assessment by electron microscopy is required for diagnosis.

TREATMENT

Overview Successful management of sellar masses requires accurate diagnosis as well as selection of optimal therapeutic modalities. Most pituitary tumors are benign and slow-growing. Clinical features result from local mass effects and hormonal hypo- or hypersecretion syndromes caused directly by the adenoma or as a consequence of treatment. Thus, lifelong management and follow-up are necessary for these patients.

Improved [MRI](#) technology with gadolinium enhancement for pituitary visualization, new advances in transsphenoidal surgery and in stereotactic radiotherapy (including gamma-knife radiotherapy), and novel therapeutic agents have improved pituitary tumor management. The goals of pituitary tumor treatment include normalization of excess pituitary secretion, amelioration of symptoms and signs of hormonal hypersecretion syndromes, and shrinkage or ablation of large tumor masses with relief of adjacent structure compression. Residual anterior pituitary function should be preserved and can sometimes be restored by removing tumor mass. Ideally, adenoma recurrence should be prevented.

Transsphenoidal Surgery Transsphenoidal rather than transfrontal resection is the desired surgical approach for pituitary tumors, except for the rare invasive suprasellar mass surrounding the frontal or middle fossa, the optic nerves, or invading posteriorly behind the clivus. Intraoperative microscopy facilitates visual distinction between adenomatous and normal pituitary tissue, as well as microdissection of small tumors that may not be visible by [MRI](#) ([Fig. 328-6](#)). Transsphenoidal surgery also avoids the cranial invasion and manipulation of brain tissue required by subfrontal surgical approaches. Endoscopic techniques with three-dimensional intraoperative localization have improved visualization and access to tumor tissue. The endoscopic approach is also less traumatic, as the technique is endonasal and does not require a

transsphenoidal retractor.

In addition to correction of hormonal hypersecretion, pituitary surgery is indicated for mass lesions that impinge on surrounding structures. Surgical decompression and resection are required for an expanding pituitary mass accompanied by persistent headache, progressive visual field defects, cranial nerve palsies, internal hydrocephalus, and, occasionally, intrapituitary hemorrhage and apoplexy. Repeat surgery may be required for persistent postoperative [CSF](#) leakage. Transsphenoidal surgery is sometimes used for pituitary tissue biopsy and histologic diagnosis.

Whenever possible, the pituitary mass lesion should be selectively excised; normal tissue should be manipulated or resected only when critical for effective dissection. Nonselective hemihypophysectomy or total hypophysectomy may be indicated if no mass lesion is clearly discernible, multifocal lesions are present, or the remaining nontumorous pituitary tissue is obviously necrotic. This strategy increases the likelihood of hypopituitarism and the need for lifelong hormonal replacement.

Preoperative local compression signs, including visual field defects or compromised pituitary function, may be reversed by surgery, particularly when these deficits are not long-standing. For large and invasive tumors, it is necessary to determine the optimal balance between maximal tumor resection and preservation of anterior pituitary function, especially for preserving growth and reproductive function in younger patients. Similarly, tumor invasion outside of the sella is rarely amenable to surgical cure; the surgeon must judge the risk:benefit ratio of extensive tumor resection.

Side Effects Tumor size and the degree of invasiveness largely determine the incidence of surgical complications. Operative mortality is about 1%. Transient diabetes insipidus and hypopituitarism occur in up to 20% of patients. Permanent diabetes insipidus, cranial nerve damage, nasal septal perforation, or visual disturbances may be encountered in up to 10% of patients. [CSF](#) leaks occur in 4% of patients. Less common complications include carotid artery injury, loss of vision, hypothalamic damage, and meningitis. Permanent side effects are rarely encountered after surgery for microadenomas.

Radiation Radiation is used either as a primary therapy for pituitary or parasellar masses or, more commonly, as an adjunct to surgery or medical therapy. Focused megavoltage irradiation is achieved by precise [MRI](#) localization, using a high-voltage linear accelerator and accurate isocentric rotational arcing. A major determinant of accurate irradiation is to reproduce the patient's head position during multiple visits and to maintain absolute head immobility. A total of <50 Gy (5000 rad) is given as 180-cGy (180 rad) fractions split over about 6 weeks. Stereotactic radiosurgery delivers a large single high-energy dose from a cobalt 60 source (gamma knife), linear accelerator, or cyclotron. Long-term effects of gamma-knife surgery are as yet unknown.

The role of radiation therapy in pituitary tumor management depends on multiple factors including the nature of the tumor, age of the patient, and the availability of surgical and radiation expertise. Because of its relatively slow onset of action, radiation therapy is usually reserved for postsurgical management. As an adjuvant to surgery, radiation is used to treat residual tumor and in an attempt to prevent regrowth. Irradiation offers the

only effective means for ablating significant residual tumor tissue derived from nonfunctioning tumors. [PRL](#)-, [GH](#)-, and [ACTH](#)-secreting tumor tissues are also amenable to medical therapy.

Side Effects In the short term, radiation may cause transient nausea and weakness. Alopecia and loss of taste and smell may be more long-lasting. Failure of pituitary hormone synthesis is common in patients who have undergone head and neck or pituitary-directed irradiation. More than 50% of patients develop failure of [GH](#), [ACTH](#), [TSH](#), and/or gonadotropin secretion within 10 years, usually due to hypothalamic damage. Lifelong follow-up with testing of anterior pituitary hormone reserve is therefore necessary after radiation treatment. Optic nerve damage with impaired vision due to optic neuritis is reported in about 2% of patients who undergo pituitary irradiation. Cranial nerve damage is uncommon now that radiation doses are ≤ 2 Gy (200 rad) at any one treatment session and the maximum dose is <50 Gy (5000 rad). The advent of stereotactic radiotherapy may reduce damage to adjacent structures. The cumulative risk of developing a secondary tumor after conventional radiation is 1.3% after 10 years and 1.9% after 20 years.

Medical Medical therapy for pituitary tumors is highly specific and depends on tumor type. For prolactinomas, dopamine agonists are the treatment of choice. For acromegaly and [TSH](#)-secreting tumors, somatostatin analogues and, occasionally, dopamine agonists are indicated. [ACTH](#)-secreting tumors and nonfunctioning tumors are generally not responsive to medication and require surgery and/or irradiation.

PROLACTIN

SYNTHESIS

[PRL](#) consists of 198 amino acids and has a molecular mass of 21,500 kDa; it is weakly homologous to [GH](#) and human placental lactogen (hPL), reflecting the duplication and divergence of a common GH-PRL-hPL precursor gene on chromosome 6. PRL is synthesized in lactotrotes, which comprise about 20% of anterior pituitary cells. Lactotrotes and somatotrotes are derived from a common precursor cell that may give rise to a tumor secreting both PRL and GH. Marked lactotrope cell hyperplasia develops during the last two trimesters of pregnancy and the first few months of lactation. These transient adaptive changes in the lactotrope population are induced by estrogen.

SECRETION

Fetal [PRL](#) synthesis begins at 12 weeks' gestation (about 4 weeks after [GH](#)). Normal adult serum PRL levels are about 10 to 25 ug/L in women and 10 to 20 ug/L in men. PRL secretion is pulsatile, with the highest secretory peaks occurring during rapid eye movement sleep. Peak serum PRL levels (up to 30 ug/L) occur between 4:00 and 6:00 A.M. The circulating half-life of PRL is about 50 min.

[PRL](#) is unique among the pituitary hormones in that the predominant central control mechanism is inhibitory, reflecting dopamine-mediated suppression of PRL release. This regulatory pathway is exemplified by the spontaneous PRL hypersecretion that occurs after pituitary stalk section, often a consequence of mass lesions at the skull

base.

Dopamine action is mediated by multiple receptor subtypes, each a member of the seven-transmembrane G protein-coupled receptor (GPCR) superfamily. In the pituitary, dopamine type 2 (D₂) receptors are predominant and mediate [PRL](#) inhibition. Targeted disruption (gene knockout) of the murine D₂ receptor results in hyperprolactinemia and lactotrope proliferation. Activation of D₂ receptors inhibits the cyclic AMP pathway, causing membrane hyperpolarization and closing of voltage-gated calcium channels; these events block secretory granule exocytosis by reducing intracellular free calcium. Because of the potent PRL inhibitory effects of dopamine, physiologic, pharmacologic, or pathologic alterations in dopamine action increase PRL levels. As discussed below, dopamine agonists play a central role in the management of hyperprolactinemic disorders.

[TRH](#) (pyro Glu-His-Pro-NH₂) is a hypothalamic tripeptide that releases prolactin within 15 to 30 min after intravenous injection. The physiologic relevance of TRH for [PRL](#) regulation is unclear, as it appears to primarily regulate [TSH](#) ([Chap. 330](#)). *Vasoactive intestinal peptide* (VIP) also induces PRL release, whereas glucocorticoids and thyroid hormone suppress PRL secretion.

Serum [PRL](#) levels rise after exercise, meals, sexual intercourse, minor surgical procedures, general anesthesia, acute myocardial infarction, and other forms of acute stress. PRL levels also increase significantly (~tenfold) during pregnancy and decline rapidly within 2 weeks of parturition. If breastfeeding is initiated, basal PRL levels remain elevated; suckling stimulates reflex increases in PRL levels that last for about 30 to 45 min. Breast suckling activates neural afferent pathways in the hypothalamus that induce PRL release. With time, the suckling-induced responses diminish and interfeeding PRL levels return to normal.

ACTION

The [PRL](#) receptor is a member of the type I cytokine receptor family that also includes [GH](#) and interleukin (IL) 6 receptors. Ligand binding leads to receptor dimerization followed by intracellular signaling mediated by the Janus kinase (JAK) pathway, which phosphorylates components of the signal transduction and activators of transcription (STAT) family. The STAT proteins translocate to the nucleus, where they act as transcription factors on target genes. In the breast, the lobuloalveolar epithelium proliferates in response to PRL, placental lactogens, elevated progesterone, and local paracrine growth factors; lactogenesis occurs as a result of complex multihormonal interactions ([Chap. 337](#)).

[PRL](#) acts to induce and maintain lactation, decrease reproductive function, and suppress sexual drive. These functions are geared towards ensuring that maternal lactation is sustained and not interrupted by pregnancy. PRL inhibits reproductive function at multiple levels, including suppression of hypothalamic [GnRH](#) and pituitary gonadotropin secretion, as well as impairing gonadal steroidogenesis in both female and male subjects. In the hypothalamus, PRL-mediated suppression of GnRH leads to loss of pulsatile [LH](#) secretion and abrogation of the preovulatory LH surge. In the ovary, PRL blocks folliculogenesis and inhibits granulosa cell aromatase activity, leading to

hypoestrogenism and anovulation. PRL also has a luteolytic effect, generating a shortened, or inadequate, luteal phase of the menstrual cycle. In males, attenuated LH secretion leads to low testosterone levels and decreased spermatogenesis. These hormonal changes decrease libido and reduce fertility in patients with hyperprolactinemia ([Chap. 54](#)).

[PRL](#) exerts widespread metabolic effects to ensure maintenance of sustained lactation. Gastrointestinal calcium absorption is increased, bone calcium is mobilized, bile acids are elevated, and pancreatic bcell growth is induced by PRL. Centrally, PRL acts on brain centers involved in parenting behavior, appetite stimulation, and analgesia. Concomitant with these effects, maintenance of bone mineral density is abrogated; hyperprolactinemia is associated with enhanced risk for bone loss and the long-term development of osteoporosis. PRL receptors are abundant in the osteoblasts of developing bone, and the accompanying hypoestrogenemia contributes to accelerated bone loss in hyperprolactinemic women.

HYPERPROLACTINEMIA

Etiology Hyperprolactinemia is the most common pituitary hormone hypersecretion syndrome in both males and females. [PRL](#)-secreting pituitary adenomas (prolactinomas) are the most common cause of PRL levels >100 ug/L (see below). Less pronounced PRL elevation can also be caused by microprolactinomas but is more commonly caused by drugs, pituitary stalk compression, hypothyroidism, or renal failure ([Table 328-8](#)).

Pregnancy and lactation are the important physiologic causes of hyperprolactinemia. Sleep-associated hyperprolactinemia reverts to normal within an hour of awakening. Nipple stimulation and sexual orgasm may also cause acute [PRL](#) increases. Chest wall stimulation or trauma (including chest surgery and herpes zoster) invoke the reflex suckling arc with resultant hyperprolactinemia. Chronic renal failure elevates PRL by decreasing peripheral PRL clearance. Primary hypothyroidism is associated with mild hyperprolactinemia, probably because of enhanced [TRH](#) secretion.

Lesions of the hypothalamic-pituitary region that disrupt hypothalamic dopamine synthesis, portal vessel delivery, or lactotrope responses are associated with hyperprolactinemia. Thus, hypothalamic tumors, cysts, infiltrative disorders, and radiation-induced damage cause elevated [PRL](#) levels, usually in the range of 30 to 100 ug/L. Plurihormonal adenomas (including [GH](#) and [ACTH](#) tumors) may directly hypersecrete PRL. Clinically nonfunctioning pituitary tumors commonly cause stalk pressure and hyperprolactinemia.

Drug-induced inhibition or disruption of dopaminergic receptor function results in hyperprolactinemia ([Table 328-8](#)). Thus, many antipsychotics and antidepressants cause hyperprolactinemia. Methyldopa inhibits dopamine synthesis and verapamil blocks dopamine release, also leading to hyperprolactinemia. Hormonal agents that induce [PRL](#) include estrogens, antiandrogens, and [TRH](#).

Presentation and Diagnosis Amenorrhea, galactorrhea, and infertility are the hallmarks of hyperprolactinemia in women. If hyperprolactinemia develops prior to the menarche, primary amenorrhea results. More commonly, hyperprolactinemia develops

later in life and leads to oligomenorrhea and, ultimately, to amenorrhea. Patients present with infertility, vaginal dryness, dyspareunia, and loss of libido. If hyperprolactinemia is sustained, vertebral bone mineral density can be reduced compared to age-matched controls, particularly when associated with pronounced hypoenestrogenemia. Galactorrhea is present in up to 80% of hyperprolactinemic women. Though usually bilateral and spontaneous, it may be unilateral or only expressed manually. Patients may also complain of weight gain and mild hirsutism.

In men with hyperprolactinemia, diminished libido or visual loss (from optic nerve compression) are the usual presenting symptoms. Gonadotropin suppression leads to reduced testosterone, impotence, and oligospermia. True galactorrhea is uncommon in men with hyperprolactinemia. If the disorder is longstanding, secondary effects of hypogonadism are evident, including osteopenia, reduced muscle mass, and decreased beard growth.

The diagnosis of idiopathic hyperprolactinemia is made by exclusion of known causes of hyperprolactinemia in the setting of a normal pituitary [MRI](#). Some of these patients may have small microadenomas below MRI sensitivity (~2 mm).

Laboratory Investigation Basal, fasting morning [PRL](#) levels (normally <20 ug/L) should be measured to assess hypersecretion. Because hormone secretion is pulsatile and levels vary widely in some individuals with hyperprolactinemia, it may be necessary to measure levels on several different occasions when clinical suspicion is high. Both false-positive and false-negative results may be encountered. In patients with markedly elevated PRL levels (>1000 ug/L), results may be falsely lowered because of assay artifacts; sample dilution is required to assess these high values accurately. Falsely elevated values may be caused by aggregated forms of circulating PRL, which are biologically inactive (macroprolactinemia). Hypothyroidism should be excluded by measuring [TSH](#) and T₄ levels.

TREATMENT

Treatment of hyperprolactinemia depends on the cause of elevated [PRL](#) levels. Regardless of the etiology, however, treatment should be aimed at normalizing PRL levels to alleviate suppressive effects on gonadal function, halt the galactorrhea, and preserve bone mineral density. Dopamine agonists are effective for many different causes of hyperprolactinemia (see "Treatment" for "Prolactinoma," below).

If the patient is taking a medication known to cause hyperprolactinemia, the drug should be withdrawn, if possible. For psychiatric patients who require neuroleptic agents, dose titration or the addition of a dopamine agonist can help restore normoprolactinemia and alleviate reproductive symptoms. However, dopamine agonists sometimes worsen the underlying psychiatric condition, especially at high doses. Hyperprolactinemia usually resolves after adequate thyroid hormone replacement in hypothyroid patients or after renal transplantation in patients on dialysis. Resection of hypothalamic or sellar mass lesions can reverse hyperprolactinemia caused by reduced dopamine tone. Granulomatous infiltrates rarely respond to glucocorticoid administration. In patients with irreversible hypothalamic damage, no treatment may be warranted. In up to 30% of patients with hyperprolactinemia -- with or without a visible pituitary microadenoma --

the condition resolves spontaneously.

PROLACTINOMA

Etiology and Prevalence Tumors arising from lactotrope cells account for about half of all functioning pituitary tumors, with an annual incidence of ~3/100,000 population. Mixed tumors secreting combinations of [GH](#) and [PRL](#), [ACTH](#) and PRL, and rarely [TSH](#) and PRL, are also seen. These plurihormonal tumors are usually recognized by immunohistochemistry, without apparent clinical manifestations from the production of additional hormones. Microadenomas are classified as <1 cm in diameter and do not usually invade the parasellar region. Macroadenomas are >1 cm in diameter, are locally invasive, and may impinge on adjacent structures. The female:male ratio for microprolactinomas is 20:1, whereas the gender ratio is near 1:1 for macroadenomas. Tumor size generally correlates directly with PRL concentrations; values >100 ug/L are usually associated with macroadenomas. Males tend to present with larger tumors than females, possibly because the features of hypogonadism are less readily evident. PRL levels remain stable in most patients, reflecting the slow growth of these tumors. About 5% of microadenomas progress in the long term to macroadenomas. Hyperprolactinemia resolves spontaneously in about 30% of microadenomas.

Presentation and Diagnosis Women usually present with amenorrhea, infertility, and galactorrhea. If the tumor extends outside of the sella, visual field defects or other mass effects may be seen. Men often present with impotence, loss of libido, infertility, or signs of central [CNS](#) compression including headaches and visual defects. Assuming that known physiologic and medication-induced causes of hyperprolactinemia are excluded ([Table 328-8](#)), the diagnosis of prolactinoma is likely with a [PRL](#) level >100 ug/L. PRL levels <100ug/L may be caused by microadenomas, other sellar lesions that decrease dopamine inhibition, or nonneoplastic causes of hyperprolactinemia. For this reason, an [MRI](#) should be performed in all patients with hyperprolactinemia. It is important to remember that hyperprolactinemia caused by the mass effects of nonlactotrope lesions is also corrected by treatment with dopamine agonists. Consequently, PRL suppression by dopamine agonists does not necessarily indicate that the lesion is a prolactinoma.

TREATMENT

As microadenomas rarely progress to become macroadenomas, no treatment may be needed if fertility is not desired. Estrogen replacement is indicated to prevent bone loss and other consequences of hypoestrogenemia and does not appear to increase the risk of tumor enlargement. These patients should be monitored by regular serial [PRL](#) and [MRI](#) measurements.

For symptomatic microadenomas, therapeutic goals include control of hyperprolactinemia, reduction of tumor size, restoration of menses and fertility, and improvement of galactorrhea. Dopamine agonists should be titrated to achieve maximal [PRL](#) suppression and restoration of reproductive function ([Fig. 328-7](#)). A normalized PRL level does not assure reduced tumor size. However, tumor shrinkage is not usually seen in those who do not respond with lowered PRL levels. For macroadenomas, formal visual field testing should be performed before initiating dopamine agonists. [MRI](#) and visual fields should be assessed at 6- to 12-month intervals

until the mass shrinks and annually thereafter until maximum size reduction has occurred.

Medical Oral dopamine agonists (cabergoline or bromocriptine) are the mainstay of therapy for patients with micro- or macroprolactinomas. Dopamine agonists suppress [PRL](#) secretion and synthesis as well as lactotrope cell proliferation.

Bromocriptine The ergot alkaloid bromocriptine mesylate is a dopamine receptor agonist that suppresses prolactin secretion by binding directly to lactotrope D₂dopamine receptors. Bromocriptine is used as initial therapy for both micro- and macroprolactinomas. In microadenomas the drug rapidly lowers serum prolactin levels to normal in up to 70% of patients, decreases tumor size, and restores gonadal function. In patients with macroadenomas, prolactin levels are also normalized in 70% of patients and tumor mass shrinkage (³50%) is achieved in up to 40% of patients. Mass effect symptoms, including headaches and visual disorders, usually improve dramatically within days after bromocriptine initiation; improvement of sexual function requires several weeks of treatment but may occur before complete normalization of prolactin levels. Drug withdrawal usually results in recurrent hyperprolactinemia and tumor reexpansion, with the risk of visual compromise. After initial control of [PRL](#) levels has been achieved, bromocriptine should be reduced to the lowest effective maintenance dose. In ~5% of treated patients, hyperprolactinemia may resolve and not recur when bromocriptine is discontinued after long-term treatment.

Therapy is initiated by administering a low bromocriptine dose (0.625 to 1.25 mg) at bedtime with a snack, followed by gradually increasing the dose. Most patients are successfully controlled with a daily dose of £7.5 mg (2.5 mg tid). About 20% of patients are resistant to dopaminergic treatment; they may have decreased D₂dopamine receptor numbers or a postreceptor defect. D₂receptor gene mutations in the pituitary have not been reported.

Nausea, vomiting, and postural hypotension with faintness may occur in ~25% of patients after the initial dose. These symptoms may persist in some patients. Other side effects include constipation, nasal stuffiness, dry mouth, nightmares, insomnia, and vertigo; decreasing the dose usually alleviates these problems. For the approximately 15% of patients who cannot tolerate oral bromocriptine, intravaginal administration of tablets is often efficacious.

Auditory hallucinations, delusions, and mood swings have been reported in up to 5% of patients and may be due to the dopamine agonist properties or to the lysergic acid derivative of the compound. Rare reports of leukopenia, thrombocytopenia, pleural fibrosis, cardiac arrhythmias, and hepatitis have been described.

Cabergoline An ergoline derivative, cabergoline is a long-acting dopamine agonist with high D₂receptor affinity. The drug effectively suppresses [PRL](#) for >14 days after a single oral dose and induces prolactinoma shrinkage in most patients. Cabergoline (0.5 to 1.0 mg twice weekly) achieves normoprolactinemia and resumption of normal gonadal function in ~80% of patients with microadenomas; galactorrhea improves or resolves in 90% of patients. Cabergoline normalizes PRL and shrinks ~70% of macroprolactinomas. It may also be effective in patients resistant to bromocriptine.

Adverse effects and drug intolerance are encountered less commonly than with bromocriptine.

Other dopamine agonists These include *pergolide mesylate*, an ergot derivative with dopaminergic properties; *lisuride*, an ergot derivative; and *quinagolide* (CV 205-502, Norprolac), a nonergot oral dopamine agonist with specific D₂receptor activity.

Surgery Indications for surgical debulking include dopamine resistance or intolerance and the presence of an invasive macroadenoma with compromised vision that fails to improve rapidly after drug treatment. Initial [PRL](#) normalization is achieved in about 70% of microprolactinomas after surgical resection, but only 30% of macroadenomas can be successfully resected. However, follow-up studies have shown that recurrence of hyperprolactinemia occurs in up to 20% of patients within the first year after surgery; long-term recurrence rates exceed 50% for macroadenomas. Radiotherapy for prolactinomas is reserved for patients with aggressive tumors that do not respond to maximally tolerated dopamine agonists and/or surgery.

Pregnancy The pituitary increases in size during pregnancy, reflecting the stimulatory effects of estrogen and perhaps other growth factors. About 5% of microadenomas significantly increase in size, but 15 to 30% of macroadenomas may grow during pregnancy. Bromocriptine has been used for over 25 years to restore fertility in women with hyperprolactinemia, without evidence of untoward teratogenic effects. Nonetheless, most authorities recommend strategies to minimize fetal exposure to the drug. For women taking bromocriptine who desire pregnancy, mechanical contraception should be used through three regular menstrual cycles to allow for conception timing. When pregnancy is confirmed, bromocriptine should be discontinued and [PRL](#) levels followed serially, especially if headaches or visual symptoms occur. For women harboring macroadenomas, regular visual field testing is recommended, and the drug should be reinstituted if tumor growth is apparent. Although pituitary [MRI](#) may be safe during pregnancy, this procedure should be reserved for symptomatic patients with severe headache and/or visual field defects. Alternatively, surgical decompression may be indicated if vision is threatened. Though comprehensive data support the efficacy and relative safety of bromocriptine-facilitated fertility, patients should be advised of potential unknown deleterious effects and the risk of tumor growth during pregnancy. At present, the experience with cabergoline is too limited to recommend its routine use when fertility is desired.

GROWTH HORMONE

SYNTHESIS

[GH](#) is the most abundant anterior pituitary hormone and is expressed early in fetal life (at 8 weeks' gestation). GH-secreting somatotrope cells constitute up to 50% of the total anterior pituitary cell population. Mammosomatotrope cells, which coexpress [PRL](#) with GH, can be identified using double immunostaining techniques. Somatotrope development is determined by expression of the cell-specific Pit-1 nuclear transcription factor. In addition to controlling cell differentiation, it also enhances GH gene expression. Five distinct genes on chromosome 17q22 encode GH and related proteins. The pituitary GH gene (*hGH-N*) produces two alternatively spliced products that give

rise to 22-kDa GH (191 amino acids) and a less abundant, 20-kDa GH molecule, with similar biologic activity. Placental syncytiotrophoblast cells express a GH variant (*hGH-V*) gene; the related hormone human chorionic somatotropin (HCS) is expressed by distinct members of the gene cluster. HCS shares high homology with GH yet exhibits minimal growth-promoting properties.

SECRETION

GH secretion is controlled by complex hypothalamic and peripheral factors. [GHRH](#) is a 44 amino acid hypothalamic peptide that stimulates GH synthesis and release. Synthetic agonists of the [GHRP](#) receptor stimulate GHRH and also directly stimulate GH release, but putative endogenous agonists remain incompletely characterized. *Somatostatin* (SRIF) is synthesized in the medial preoptic area of the hypothalamus and inhibits GH secretion. GHRH is secreted as discrete spikes that elicit GH pulses, whereas SRIF sets basal GH tone. SRIF is also expressed in many extrahypothalamic tissues, including the [CNS](#), gastrointestinal system, and pancreas, where it also acts to inhibit the hormone secretion. *IGF-I*, the peripheral target hormone for GH, feeds back to inhibit GH; estrogens induce GH ([Chap. 8](#)), whereas glucocorticoid excess suppresses GH release.

Two distinct surface receptors on the somatotrope regulate GH synthesis and secretion. The [GHRH](#) receptor is a [GPCR](#) that signals through the intracellular cyclic AMP pathway. Activation of this receptor stimulates somatotrope cell proliferation as well as hormone production. Inactivating mutations of the GHRH receptor cause profound dwarfism (see below). A distinct surface receptor for [GHRP](#) has also been identified. This receptor is expressed in the hypothalamus and pituitary. A natural ligand, termed *ghrelin*, binds to the GHRP receptor; it is produced in large amounts in the stomach, though its physiologic role remains unknown. Hypothalamic somatostatin binds to five distinct receptor subtypes (SSTR1 to SSTR5) that are widely expressed in different tissues, including in the pituitary. SSTR2 and SSTR5 subtypes preferentially suppress GH (and [TSH](#)) secretion.

GH secretion is pulsatile, with greater levels at night generally correlating with the onset of sleep. GH secretory rates decline markedly with age so that hormone production in middle age is about 15% of production during puberty. These changes are paralleled by an age-related decline in lean muscle mass. GH secretion is also reduced in obese individuals, though [IGF-I](#) levels are preserved, suggesting a change in the setpoint for feedback control. Elevated GH levels occur within an hour of deep sleep onset as well as after exercise, physical stress, trauma, and during sepsis. Integrated 24-h GH secretion is higher in women and is also enhanced by estrogen replacement. Using assays in common clinical use, random GH measurements are undetectable in ~50% of daytime samples obtained from healthy subjects and are undetectable in most obese and elderly subjects. Thus, single random GH measurements do not distinguish patients with adult GH deficiency from normal persons.

GH secretion is profoundly influenced by nutritional factors. Using newer ultrasensitive chemiluminescence-based GH assays with a sensitivity of 0.002 ug/L, a glucose load can be shown to suppress GH to <0.7 ug/L in female and to <0.07 ug/L in male subjects. Increased GH pulse frequency and peak amplitudes occur with chronic malnutrition or

prolonged fasting. GH is stimulated by high-protein meals and by L-arginine. GH secretion is induced by dopamine and apomorphine (a dopamine receptor agonist), as well as by α -adrenergic pathways. β -Adrenergic blockage induces basal GH and enhances [GHRH](#)- and insulin-evoked GH release.

ACTION

The pattern of [GH](#) secretion may affect tissue responses. The higher GH pulsatility observed in males, as compared to the relatively continuous GH secretion in females, may be an important biologic determinant of linear growth patterns and liver enzyme induction.

The 70-kD peripheral [GH](#) receptor protein shares structural homology with the cytokine/hematopoietic superfamily. A fragment of the receptor extracellular domain generates a soluble GH binding protein (GHBP) that interacts with GH in the circulation. The liver contains the greatest number of GH receptors. GH binding induces receptor dimerization by making distinct contact through two separate binding domains of the hormone. The dimerized receptor interacts with members of the [JAK/STAT](#) family. The activated STAT proteins translocate to the nucleus, where they modulate expression of GH-regulated target genes. GH analogues that bind to the receptor, but are incapable of mediating receptor dimerization, are potent antagonists of GH action and are being investigated for potential use in the treatment of acromegaly and diabetic microangiopathy.

[GH](#) induces protein synthesis and nitrogen retention and impairs glucose tolerance by antagonizing insulin action. GH also stimulates lipolysis, leading to increased circulating fatty acid levels, reduced omental fat mass, and enhanced lean body mass. GH promotes sodium, potassium, and water retention and elevates serum levels of inorganic phosphate. Linear bone growth occurs as a result of complex hormonal and growth factor actions, including those of [IGF-I](#). GH stimulates epiphyseal prechondrocyte differentiation. These precursor cells produce IGF-I locally and are also responsive to the growth factor.

INSULIN-LIKE GROWTH FACTORS

Though [GH](#) exerts direct effects in target tissues, many of its physiologic effects are mediated indirectly through [IGF-I](#), a potent growth and differentiation factor. The major source of circulating IGF-I is hepatic in origin. Peripheral tissue IGF-I exerts local paracrine actions that appear to be both dependent and independent of GH. Thus, GH administration induces circulating IGF-I level as well as stimulating IGF-I expression in multiple tissues.

Both [IGF-I](#) and -II are bound to one of six high-affinity circulating IGF-binding proteins (IGFBPs) that regulate IGF bioactivity. Levels of IGFBP3 are [GH](#)-dependent, and it serves as the major carrier protein for circulating IGF-I. GH deficiency and malnutrition are associated with low IGFBP3 levels. IGFBP1 and -2 regulate local tissue IGF action but do not bind appreciable amounts of circulating IGF-I.

Serum [IGF-I](#) concentrations are profoundly affected by various physiologic factors.

Levels increase during puberty, peak at 16 years, and subsequently decline by >80% during the aging process. IGF-I concentrations are higher in females than in males. Because [GH](#) is the major determinant of hepatic IGF-I synthesis, abnormalities of GH synthesis or action (e.g., pituitary failure, [GHRH](#) receptor defect, or GH receptor defect) reduce IGF-I levels. Hypocaloric states are associated with GH resistance; IGF-I levels are therefore low with cachexia, malnutrition, and sepsis. In acromegaly, IGF-I levels are invariably high and reflect a log-linear relationship with GH concentrations.

IGF-I Physiology Though [IGF-I](#) is not an approved drug, investigational studies provide insight into its physiologic effects. High doses of injected IGF-I (100 ug/kg) induce hypoglycemia, primarily because of actions through the insulin receptor. Low IGF-I doses improve insulin sensitivity in patients with severe insulin resistance and diabetes. In cachectic subjects, IGF-I infusion (12 ug/kg per hour) enhances nitrogen retention and lowers cholesterol levels. Longer-term subcutaneous IGF-I injections exert a marked anabolic effect with enhanced protein synthesis. The impact of long-term IGF-I administration on bone mineral content is as yet unclear. Although bone formation markers are induced, bone turnover may also be stimulated by IGF-I.

Side effects of [IGF-I](#) are dose-dependent. An acute overdose may result in hypoglycemia and hypotension. Fluid retention, temporomandibular jaw pain, and increased intracranial pressure are reversible. Avascular necrosis of the femoral head has been reported. Chronic excess IGF-I would presumably result in features of acromegaly.

DISORDERS OF GROWTH AND DEVELOPMENT

Skeletal Maturation and Somatic Growth Linear growth is a function of endochondral bone formation whereby cartilage is converted into bony skeleton in the long bones and vertebrae ([Chap. 340](#)). Ossification occurs within central diaphyseal and peripheral epiphyseal centers. A cartilaginous growth plate forms between the two centers, and chondrocytes proliferate within the growth plate. Linear bone growth ceases when this cartilage layer ossifies and fuses with epiphyseal and diaphyseal bone.

The growth plate is dependent on a variety of hormonal stimuli including [GH](#), [IGF-I](#), sex steroids, thyroid hormones, paracrine growth factors, and cytokines. GH directly stimulates prechondrocyte differentiation and clonal expansion, resulting in chondrocytes that express both IGF-I receptors and IGF-I protein. The growth-promoting process also requires caloric energy, amino acids, vitamins, and trace metals and consumes about 10% of normal energy production. Malnutrition impairs chondrocyte activity and reduces circulating IGF-I and [IGFBP3](#) levels.

Bone age is delayed in patients with all forms of true [GH](#) deficiency or GH receptor defects that result in attenuated GH action. Rarely, GH excess accelerates growth, particularly in the setting of delayed bone age from concomitant hypogonadism. Thyroid hormone is permissive for GH synthesis and secretion as well as for maintaining normal circulating [IGF-I](#) and binding protein levels. Bone age is delayed by thyroid hormone deficiency. Consequently, congenital or acquired hypothyroidism is associated with stunted growth, which is partially reversed by thyroid hormone replacement ([Chap. 330](#)). Elevated pubertal sex steroid levels (especially estrogen) induce the [GHRH](#)-GH-IGF-I

axis and also directly stimulate epiphyseal growth. High doses of estrogen lead to epiphyseal closure. A mutation of the estrogen receptors prevented epiphyseal closure, confirming the important role of this pathway in bone maturation. Several pathologic conditions accompanied by increased levels of sex steroids, including precocious puberty, androgen exposure (exogenous or endogenous), congenital adrenal hyperplasia, and obesity, are associated with accelerated bone maturation. Thus, children with these conditions have accelerated early growth, but end up with reduced final height. In contrast to sex steroids, glucocorticoid excess inhibits linear growth. Glucocorticoids also stimulate SRIF and inhibit peripheral GH and IGF-I receptor signaling.

Linear bone growth rates are very high in infancy and are pituitary-dependent. Mean growth velocity is ~6 cm/year in later childhood and is usually maintained within a given range on a standardized percentile chart. Peak growth rates occur during midpuberty when bone age is 12 (girls) or 13 (boys) ([Chap. 8](#)). Secondary sexual development is associated with elevated sex steroids that cause progressive epiphyseal growth plate closure.

Short stature may occur as a result of constitutive intrinsic growth defects or because of acquired extrinsic factors that impair growth ([Table 328-9](#)). Genetic disorders, including pituitary transcription factor defects, mutations in growth-related genes, and pituitary hypoplasia syndromes, may all be associated with growth delay and short stature. In general, delayed bone age in a child with short stature is suggestive of a hormonal or systemic disorder, whereas normal bone age in a short child is more likely to be caused by a genetic growth plate disorder ([Chap. 351](#)). Other bone and cartilage dysplasia syndromes are associated with specific limb-body proportion phenotypes, and some involve associated calcium disorders ([Chap. 343](#)).

Intrauterine growth retardation results in short stature and may be caused by specific congenital anomalies (e.g., IGF-I deficiency, *Russell-Silver syndrome*, chromosomal disomy) or maternal factors such as diabetes mellitus, infections, hypoxia, drug addiction, or placental dysfunction. Long-term responses of these children to GH treatment are currently being evaluated.

Turner syndrome is caused by loss of all, or part, of an X chromosome in females (XO). It is characterized by short stature, in addition to gonadal dysgenesis and other characteristic features ([Chap. 338](#)). Short stature may be improved with a combination of GH and an anabolic steroid (oxandrolone); estrogen is required to induce and sustain sexual development. *Noonan syndrome* resembles Turner syndrome phenotypically, but patients have apparently normal sex chromosomes. These patients have delayed pubertal development but not primary gonadal failure.

GH Deficiency in Children

GH Deficiency Isolated GH deficiency is characterized by short stature, micropenis, increased fat, high-pitched voice, and a propensity to hypoglycemia. The etiology of GH deficiency is not identifiable in most children with the disorder. Familial modes of inheritance are seen in one-third of these individuals and may be autosomal dominant, recessive, or X-linked, indicating that multiple genetic abnormalities can lead to GH

deficiency. About 10% of children with growth hormone deficiency have mutations in the GH-N gene. These include gene deletions and a wide range of point mutations, including some that function in a dominant negative manner (heterozygous mutations) to impair the synthesis or function of GH expressed from the normal allele. Mutations in transcription factors Pit-1 and Prop-1, which control somatotrope development, cause GH deficiency in combination with other pituitary hormone deficiencies. The diagnosis of *idiopathic GH deficiency* (IGHD) should be made only after known molecular defects have been excluded.

GHRH Receptor Mutations Recessive mutations of the GHRH receptor gene have been described in several unrelated families with severe proportionate dwarfism. The low basal GH levels in these patients cannot be stimulated by exogenous GHRH, GHRP, or insulin-induced hypoglycemia, confirming the importance of the GHRH receptor for somatotrope cell proliferation and hormonal responsiveness.

Growth Hormone Insensitivity This is caused by defects of GH receptor structure or signaling. Homozygous or heterozygous exonic and intronic mutations of the GH receptor occur mainly in the extracellular ligand-binding domain and are associated with partial or complete GH insensitivity and growth failure (*Laron syndrome*). The diagnosis of this syndrome is based on normal or high GH levels, with decreased circulating GHBP, and low IGF-I levels. Very rarely, defective IGF-I, IGF-I receptor, or IGF-I signaling defects are also encountered.

Nutritional Short Stature Caloric deprivation and malnutrition, uncontrolled diabetes, and chronic renal failure represent secondary causes of abrogated GH receptor function. These conditions also stimulate the production of proinflammatory cytokines, including tumor necrosis factor (TNF) α and ILs, which can block GH-mediated signal transduction. Children with these conditions typically exhibit features of acquired short stature with elevated GH and low IGF-I levels. Circulating GH receptor antibodies may rarely cause peripheral GH insensitivity.

Psychosocial Short Stature Emotional and social deprivation lead to growth retardation accompanied by delayed speech, discordant hyperphagia, and attenuated response to administered GH. A nurturing environment restores growth rates.

Presentation and Diagnosis Short stature is commonly encountered in clinical practice, but the criteria for biochemical diagnosis of true GH deficiency have been difficult to define. The decision to evaluate these children requires clinical judgement in association with auxologic data and family history. Short stature should be comprehensively evaluated if a patient's height is >3 SD below the mean for age or if the growth rate has decelerated. Skeletal maturation is best evaluated by measuring a radiologic bone age, which is based mainly on the degree of growth plate fusion. Final height can be predicted using standardized scales (Bayley-Pinneau or Tanner-Whitehouse) or estimated by adding 6.5 cm (boys) or subtracting 6.5 cm (girls) from the midparental height.

Laboratory Investigation Because GH secretion is pulsatile, GH deficiency is best assessed by examining the response to provocative stimuli. Random GH measurements do not distinguish normal children from those with true GH deficiency.

Adequate adrenal and thyroid hormone replacement should be assured before testing. Provocative stimuli such as exercise, insulin-induced hypoglycemia, and other pharmacologic tests normally increase GH to >7 ug/L in children. Insulin-induced hypoglycemia testing requires the blood sugar nadir to be <50% of baseline levels. This test should be performed under close supervision and is contraindicated in children with seizure disorders. IGF-I levels are not sufficiently sensitive or specific to make the diagnosis but can be useful to confirm GH deficiency; they must be controlled for age and gender. Pituitary MRI may reveal pituitary mass lesions or structural defects.

TREATMENT

Replacement therapy with recombinant GH (0.02 to 0.05 mg/kg per day subcutaneously) restores growth velocity in GH-deficient children to ~10 cm/year. If pituitary insufficiency is documented, other associated hormone deficits should be corrected -- especially adrenal steroids. GH treatment is also moderately effective for accelerating growth rates in patients with Turner syndrome and chronic renal failure.

In patients with GH insensitivity and growth retardation due to mutations of the GH receptor, treatment with IGF-I bypasses the dysfunctional GH receptor. Growth rates have been maintained for several years, and this therapy now portends improved final adult stature in this group of patients.

ADULT GH DEFICIENCY (AGHD)

This disorder is usually caused by hypothalamic or pituitary somatotrope damage. Acquired pituitary hormone deficiency follows a typical sequential pattern whereby loss of adequate GH reserve foreshadows subsequent hormone deficits. The sequential order of hormone loss is usually GH ® FSH/LH ® TSH ® ACTH. The presence of documented central hypogonadism, hypothyroidism, and/or hypoadrenalism invariably assures the presence of GH deficiency. Conversely, ~40% of patients with incipient preclinical pituitary insufficiency already manifest GH deficiency, if rigorously tested.

Presentation and Diagnosis The clinical features of AGHD include changes in body composition, lipid metabolism, and quality of life and cardiovascular dysfunction (Table 328-10). Body composition changes are common and include reduced lean body mass, increased fat mass with selective deposition of intraabdominal visceral fat, and increased waist-to-hip ratio. Hyperlipidemia, left ventricular dysfunction, hypertension, and increased plasma fibrinogen levels may also be present. Bone mineral content is reduced, with resultant increased fracture rates. Patients may exhibit social isolation, depression, and difficulty in maintaining gainful employment. Adult hypopituitarism is associated with a three-fold increased cardiovascular mortality rate in comparison to age- and sex-matched controls, and this may be due to GH deficiency.

Laboratory Investigation AGHD is rare, and in light of the nonspecific nature of associated clinical symptoms, patients appropriate for testing should be carefully selected on the basis of well-defined criteria. With few exceptions, testing should be restricted to patients with the following predisposing factors: (1) pituitary surgery, (2) pituitary or hypothalamic tumor or granulomas, (3) cranial irradiation, (4) radiologic evidence of a pituitary lesion, (5) childhood requirement for GH replacement therapy, or,

rarely, (6) unexplained low age- and sex-matched IGF-I level. The transition of the GH-deficient adolescent to adulthood requires retesting to document adult GH deficiency. Up to 20% of patients treated for childhood-onset GH deficiency are found to be GH-sufficient on repeat testing as adults.

A significant proportion (~25%) of truly GH-deficient adults have low-normal IGF-I levels. Thus, as in the evaluation of GH deficiency in children, valid age- and gender-matched IGF-I measurements provide a useful index of therapeutic responses but are not sufficiently sensitive for diagnostic purposes. AGHD is diagnosed by demonstrating a subnormal GH response (<3 ug/L) to a standard provocative test. None of the available stimulation tests provides standardized GH responses that clearly discriminate normal subjects from truly GH-deficient adults. The age-related decline of GH blurs this distinction further in elderly individuals. The most validated test to distinguish pituitary-sufficient patients from those with AGHD is insulin-induced (0.05 to 0.1 U/kg) hypoglycemia. After glucose reduction to ~40 mg/dL, most individuals experience neuroglycopenic symptoms (Chap. 334), and peak GH release occurs at 60 min and remains elevated for up to 2 h. About 90% of healthy adults exhibit GH responses >5 ug/L; AGHD is defined by a peak GH response to hypoglycemia of <3 ug/L. An attenuated GH response is observed in patients with pituitary damage, obesity, untreated hypothyroidism, depression, or chronic renal failure. Although an *insulin tolerance test* (ITT) is safe when performed under appropriate supervision, it is contraindicated in patients with diabetes, ischemic heart disease, cerebrovascular disease, or epilepsy and in elderly patients. Alternative stimulatory tests include L-dopa (500 mg orally) and intravenous arginine (30 g), GHRH (1 ug/kg), and GHRP-6 (90 ug). Combinations of these tests may evoke GH secretion in subjects not responsive to a single test.

TREATMENT

Once the diagnosis of AGHD is unequivocally established, replacement of GH may be indicated. Contraindications to therapy include the presence of an active neoplasm, intracranial hypertension, or uncontrolled diabetes and retinopathy. The starting dose of 0.15 to 0.3 mg/d should be titrated (up to a maximum of 1.25 mg/d) to maintain IGF-I levels in the mid-normal range for age- and gender-matched controls (Fig. 328-8). Women require higher doses than men, and elderly patients require less GH. Long-term GH maintenance sustains normal IGF-I levels and is associated with persistent body composition changes (e.g., enhanced lean body mass and lower body fat). High-density lipoprotein cholesterol increases, but total cholesterol and insulin levels do not change significantly. Lumbar spine bone mineral density increases, but this response is gradual (>1 year). Many patients note significant improvement in quality of life when evaluated by standardized questionnaires. The effect of GH replacement on mortality rates in GH-deficient patients is currently the subject of long-term prospective investigation.

About 30% of patients exhibit reversible dose-related fluid retention, joint pain, and carpal tunnel syndrome, and up to 40% exhibit myalgias and paresthesias. Patients receiving insulin require careful monitoring for dosing adjustments, as GH is a potent counterregulatory hormone for insulin action. Patients with type 2 diabetes mellitus initially develop further insulin resistance. However, glycemic control improves with the sustained loss of abdominal fat associated with long-term GH replacement. Headache,

increased intracranial pressure, hypertension, atrial fibrillation, and tinnitus occur rarely. Prevalence of pituitary tumor regrowth and potential progression of skin lesions are currently being assessed in long-term surveillance programs. To date, development of these potential side effects does not appear significant. For some patients, the expense of long-term GH replacement is prohibitive.

ACROMEGALY

Etiology [GH](#) hypersecretion is usually the result of somatotrope adenomas but is also rarely caused by extrapituitary lesions ([Table 328-11](#)). In addition to typical GH-secreting somatotrope adenomas, mixed mammosomatotrope tumors and acidophilic stem-cell adenomas can secrete both GH and [PRL](#). In patients with acidophilic stem-cell adenomas, features of hyperprolactinemia (hypogonadism and galactorrhea) predominate over the less clinically evident signs of acromegaly. Occasionally, mixed plurihormonal tumors are encountered that secrete [ACTH](#), the glycoprotein hormone a subunit, or [TSH](#), in addition to GH. Patients with partially empty sella may present with GH hypersecretion due to a small GH-secreting adenoma within the compressed rim of pituitary tissue; some of these may reflect the spontaneous necrosis of tumors that were previously larger. GH-secreting tumors rarely arise from ectopic pituitary tissue remnants in the nasopharynx or midline sinuses.

There are case reports of ectopic [GH](#) secretion by tumors of pancreatic, ovarian, or lung origin. Excess [GHRH](#) production may cause acromegaly because of chronic stimulation of somatotropes. These patients present with classic features of acromegaly, elevated GH levels, pituitary enlargement on [MRI](#), and pathologic characteristics of pituitary hyperplasia. The most common cause of GHRH-mediated acromegaly is a chest or abdominal carcinoid tumor. Although these tumors usually express positive GHRH immunoreactivity, clinical features of acromegaly are evident in only a minority of patients with carcinoid disease. Excessive GHRH may also be elaborated by hypothalamic tumors, usually choristomas or neuromas.

Presentation and Diagnosis Protean manifestations of [GH](#) and [IGF-I](#) hypersecretion are indolent and often are not clinically diagnosed for 10 years or more. Acral bony overgrowth results in frontal bossing, increased hand and foot size, mandibular enlargement with prognathism, and widened space between the lower incisor teeth. In children and adolescents, initiation of GH hypersecretion prior to epiphyseal long bone closure is associated with the development of pituitary gigantism ([Fig. 328-9](#)). Soft tissue swelling results in increased heel pad thickness, increased shoe or glove size, ring tightening, characteristic coarse facial features, and a large fleshy nose. Other commonly encountered clinical features include hyperhidrosis, deep and hollow-sounding voice, oily skin, arthropathy, kyphosis, carpal tunnel syndrome, proximal muscle weakness and fatigue, acanthosis nigricans, and skin tags. Generalized visceromegaly occurs, including cardiomegaly, macroglossia, and thyroid gland enlargement.

The most significant clinical impact of [GH](#) excess occurs with respect to the cardiovascular system. Coronary heart disease, cardiomyopathy with arrhythmias, left ventricular hypertrophy, decreased diastolic function, and hypertension occur in about 30% of patients. Upper airway obstruction with sleep apnea occurs in about 60% of

patients and is associated with both soft tissue laryngeal airway obstruction and central sleep dysfunction. Diabetes mellitus develops in 25% of patients with acromegaly, and most patients are intolerant of a glucose load (as GH counteracts the action of insulin). Acromegaly is associated with an increased risk of colon polyps and colonic malignancy; polyps are diagnosed in up to one-third of acromegalic patients. Overall mortality is increased about three-fold and is due primarily to cardiovascular and cerebrovascular disorders, malignancy, and respiratory disease. Unless GH levels are controlled, survival is reduced by an average of 10 years compared with an age-matched control population.

Laboratory Investigation Age- and gender-matched serum [IGF-I](#) levels are invariably elevated in acromegaly. Consequently, an IGF-I level provides a useful laboratory screening measure when clinical features raise the possibility of acromegaly. Due to the pulsatility of [GH](#) secretion, measurement of a single random GH level is not useful for the diagnosis or exclusion of acromegaly and does not correlate with disease severity. The diagnosis of acromegaly is confirmed by demonstrating the failure of GH suppression to < 1 ug/L within 1 to 2 h of an oral glucose load (75 g). About 20% of patients exhibit a paradoxical GH rise after glucose. About 60% of patients with GH-secreting tumors may exhibit paradoxical GH responses to [TRH](#) administration. [PRL](#) should be measured as it is elevated in ~25% of patients with acromegaly. Thyroid function, gonadotropins, and sex steroids may be attenuated because of tumor mass effects. Because most patients will undergo surgery with glucocorticoid coverage, tests of [ACTH](#) reserve in asymptomatic patients are more efficiently deferred until after surgery.

TREATMENT

Surgical resection of [GH](#)-secreting adenomas is the initial treatment for most patients ([Fig. 328-10](#)). Somatostatin analogues are used as adjuvant treatment for preoperative shrinkage of large invasive macroadenomas, immediate relief of debilitating symptoms, and reduction of GH hypersecretion, in elderly patients experiencing morbidity, in patients who decline surgery, or, when surgery fails, to achieve biochemical control. Irradiation or repeat surgery may be required for patients who cannot tolerate or do not respond to adjunctive medical therapy. The high rate of late hypopituitarism and the slow rate (5 to 15 years) of biochemical response are the main disadvantages of radiotherapy. Irradiation is relatively ineffective in normalizing [IGF-I](#) levels. Stereotactic ablation of GH-secreting adenomas by gamma-knife radiotherapy is promising, but long-term results are not available and the side effects have not been clearly delineated. Somatostatin analogues may be given while awaiting the full effect of radiotherapy. Systemic sequelae of acromegaly, including cardiovascular disease, diabetes, and arthritis, should also be managed aggressively. Maxillofacial surgery for mandibular repair may also be indicated.

Surgery Transsphenoidal surgical resection by an experienced surgeon is the preferred primary treatment for both microadenomas (cure rate ~70%) and macroadenomas (<50% cured). Soft tissue swelling improves immediately after tumor resection. [GH](#) levels return to normal within an hour, and [IGF-I](#) levels are normalized within 3 to 4 days. In ~10% of patients, acromegaly may recur several years after apparently successful surgery; hypopituitarism develops in up to 15% of patients.

Somatostatin Analogues Somatostatin analogues exert their therapeutic effects through SSTR2 and -5 receptors, both of which are expressed by GH-secreting tumors. Octreotide acetate is an 8-amino-acid synthetic somatostatin analogue. In contrast to native somatostatin, the analogue is relatively resistant to plasma degradation. It has a 2-h serum half-life and possesses at least 40-fold greater potency than native somatostatin to suppress GH. These properties often allow effective pharmacologic control of GH hypersecretion without prior surgery or radiotherapy. Octreotide is administered by subcutaneous injection, beginning with 50 ug tid; the dose can be gradually increased up to 1500 ug/d. Fewer than 10% of patients do not respond to the analogue. Octreotide suppresses integrated GH levels to <5 ug/L in ~70% of patients and to <2 ug/L in up to 60% of patients. It normalizes IGF-I levels in ~75% of treated patients (Fig. 328-11). Prolonged use of the analogue is not associated with desensitization, even after ³10 years of treatment. Rapid relief of headache and soft tissue swelling occurs in ~75% of patients within days to weeks of treatment initiation. Subjective clinical benefits of octreotide therapy occur more frequently than biochemical remission, and most patients report symptomatic improvement, including amelioration of headache, perspiration, obstructive apnea, and cardiac failure. Modest pituitary tumor size reduction occurs in about 40% of patients, but this effect is reversed when treatment is stopped.

Two long-acting somatostatin depot formulations, octreotide and lanreotide, are becoming the preferred medical treatment for acromegalic patients. *Sandostatin-LAR* is a sustained-release, long-acting formulation of octreotide incorporated into microspheres that sustain drug levels for several weeks after intramuscular injection. GH suppression occurs for as long as 6 weeks after a 30-mg injection; long-term monthly treatment sustains GH and IGF-I suppression and reduction of pituitary tumor size. *Lanreotide*, a slow-release depot somatostatin preparation, is a cyclic somatostatin octapeptide analogue that suppresses GH and IGF-I hypersecretion for 10 to 14 days after a 30-mg intramuscular injection. Long-term administration controls GH hypersecretion in two-thirds of treated patients and improves patient compliance because of the long interval required between drug injections.

Side Effects Somatostatin analogues are well tolerated in most patients. Adverse effects are short-lived and mostly relate to drug-induced suppression of gastrointestinal motility and secretion. Nausea, abdominal discomfort, fat malabsorption, diarrhea, and flatulence occur in one-third of patients, though these symptoms usually remit within 2 weeks. Octreotide suppresses postprandial gallbladder contractility and delays gallbladder emptying; up to 30% of patients treated long-term develop echogenic sludge or asymptomatic cholesterol gallstones. Other side effects include mild glucose intolerance due to transient insulin suppression, asymptomatic bradycardia, hypothyroxinemia, and local pain at the injection site. The cost of chronic treatment may be prohibitive.

Dopamine Agonists Bromocriptine may suppress GH secretion in some acromegalic patients, particularly those with cosecretion of PRL. High doses (³20 mg/d), administered as three to four daily doses, are usually required to lower GH, and therapeutic efficacy is modest. GH levels are suppressed to <5 ug/L in ~20% of patients, and IGF-I levels are normalized in only 10% of patients. Cabergoline also suppresses GH and decreases adenoma size when given at a relatively high dose of 0.5 mg/d. Combined treatment

with octreotide and bromocriptine induces additive biochemical control compared to either drug alone.

GH Antagonists Investigational GH analogues antagonize endogenous GH action by blocking peripheral GH binding to its receptor. Consequently, serum IGF-I levels are suppressed, potentially reducing the deleterious effects of excess endogenous GH.

Radiation External radiation therapy or high-energy stereotactic techniques are used as adjuvant therapy for acromegaly. An advantage of radiation is that patient compliance with long-term treatment is not required. Tumor mass is reduced, and GH levels are attenuated over time. However, 50% of patients require at least 8 years for GH levels to be suppressed to <5 ug/L; this suboptimal level of GH reduction is achieved in about 90% of patients after 18 years. Patients may require interim medical therapy for several years prior to attaining maximal radiation benefits. Most patients also experience hypothalamic-pituitary damage, leading to gonadotropin, ACTH, and/or TSH deficiency within 10 years of therapy.

In summary, surgery is the preferred primary treatment for GH-secreting microadenomas (Fig. 328-10). The high frequency of GH hypersecretion after macroadenoma resection usually necessitates adjuvant or primary medical therapy for these larger tumors. Patients unable to receive or respond to medical treatment can be offered radiation.

ADRENOCORTICOTROPIN HORMONE (See also Chap. 331)

SYNTHESIS

ACTH-secreting corticotrope cells constitute about 20% of the pituitary cell population. ACTH (39 amino acids) is derived from the POMC precursor protein (266 amino acids) that also generates several other peptides, including b-lipotropin, b-endorphin, met-enkephalin, a melanocyte-stimulating hormone (MSH), and corticotropin-like intermediate lobe protein (CLIP). The POMC gene is located on chromosome 2 and possesses at least three different promoter regions that account for pituitary and peripheral tissue-specific POMC expression. A proximal promoter mediates POMC expression in corticotropes. The gonads, placenta, gastrointestinal tissues, liver, kidney, adrenal medulla, lung, and lymphocytic tissue express shorter POMC transcripts derived from a downstream promoter region. Tumors arising from peripheral neuroendocrine tissues that secrete ectopic ACTH express the longer form of POMC. The POMC gene is powerfully suppressed by glucocorticoids and induced by CRH, arginine vasopressin (AVP), and gp 130 proinflammatory cytokines, including IL-6, and leukemia inhibitory factor.

CRH, a 41-amino-acid hypothalamic peptide synthesized in the paraventricular nucleus as well as in higher brain centers, is the predominant stimulator of ACTH synthesis and release. The CRH receptor is a GPCR that is expressed on the corticotrope. CRH signaling induces POMC transcription and is mediated by cyclic AMP, as well as mitogen activated protein (MAP) kinase-activator protein-1 (AP-1) cascades.

SECRETION

[ACTH](#) secretion is pulsatile and exhibits a characteristic circadian rhythm, peaking at 6 A.M. and reaching a nadir about midnight. Adrenal glucocorticoid secretion, which is driven by ACTH, follows a parallel diurnal pattern. ACTH circadian rhythmicity is determined by variations in secretory pulse amplitude rather than changes in pulse frequency. Superimposed on this endogenous rhythm, ACTH levels are increased by [AVP](#), physical stress, exercise, acute illness, and insulin-induced hypoglycemia.

Loss of cortisol feedback inhibition, as occurs in primary adrenal failure, results in extremely high [ACTH](#) levels. Glucocorticoid-mediated negative regulation of the hypothalamo-pituitary-adrenal (HPA) axis occurs as a consequence of both hypothalamic [CRH](#) suppression and direct attenuation of pituitary [POMC](#) gene expression and ACTH release. Hypothalamic [AVP](#) stimulates the protein kinase C pathway and acts synergistically with CRH to enhance ACTH production.

Acute inflammatory or septic insults activate the [HPA](#) axis through the integrated actions of proinflammatory cytokines, bacterial toxins, and neural signals. The overlapping cascade of [ACTH](#)-inducing cytokines ([TNF](#); [IL](#)-1, -2, and -6; and leukemia inhibitory factor) activates hypothalamic [CRH](#) and [AVP](#) secretion, pituitary [POMC](#) gene expression, and local paracrine pituitary cytokine networks. The resulting cortisol elevation restrains the inflammatory response and provides host protection. Concomitantly, cytokine-mediated central glucocorticoid receptor resistance impairs glucocorticoid suppression of the HPA. Thus, the neuroendocrine stress response reflects the net result of highly integrated hypothalamic, intrapituitary, and peripheral hormone and cytokine signals.

ACTION

The major function of the [HPA](#) axis is to maintain metabolic homeostasis and to mediate the neuroendocrine stress response. Peripheral and central afferent signals, which are integrated by the pituitary corticotrope cell, ultimately affect the pattern and quantity of adrenal cortisol secretion. [ACTH](#) induces cortical steroidogenesis by maintaining adrenal cell proliferation and function. The receptor for ACTH, designated *melanocortin-2 receptor*, is a [GPCR](#) that activates cyclic AMP and [MAP](#) kinase pathways; it induces steroidogenesis by stimulating a cascade of steroidogenic enzymes ([Chap. 331](#)).

ACTH DEFICIENCY

Presentation and Diagnosis Secondary adrenal insufficiency occurs as a result of pituitary ACTH deficiency. It is characterized by fatigue, weakness, anorexia, nausea, vomiting, and, occasionally, hypoglycemia (due to diminished insulin counterregulation). In contrast to primary adrenal failure, hypocortisolism associated with pituitary failure is not usually accompanied by pigmentation changes or mineralocorticoid deficiency. ACTH deficiency is commonly due to glucocorticoid withdrawal following treatment-associated suppression of the [HPA](#) axis. Isolated ACTH deficiency may occur after surgical resection of an ACTH-secreting pituitary adenoma that has suppressed the HPA axis; this phenomenon is suggestive of a surgical cure. The mass effects of other pituitary adenomas or sellar lesions may lead to ACTH deficiency, but usually in combination with other pituitary hormone deficiencies. Partial ACTH deficiency may be unmasked in the presence of an acute medical or surgical illness, when clinically

significant hypocortisolism reflects diminished ACTH reserve.

Laboratory Diagnosis Inappropriately low [ACTH](#) levels in the setting of low cortisol levels are characteristic of diminished ACTH reserve. Low basal serum cortisol levels are associated with blunted cortisol responses to ACTH provocative stimulation and impaired cortisol response to insulin-induced hypoglycemia, or testing with metyrapone or [CRH](#). **For description of provocative ACTH tests, see "Tests of Pituitary-Adrenal Responsiveness" in Chap. 331.*

TREATMENT

Glucocorticoid replacement therapy improves most features of [ACTH](#) deficiency. The total daily dose of hydrocortisone replacement should not exceed 30 mg daily, divided into two or three doses. Prednisone (5 mg each morning; 2.5 mg each evening) is longer acting and has fewer mineralocorticoid effects than hydrocortisone. Some authorities advocate lower maintenance doses in an effort to avoid cushingoid side effects. Doses should be increased several-fold during periods of acute illness or stress.

CUSHING'S DISEASE (ACTH-PRODUCING ADENOMA) (See also [Chap. 331](#))

Etiology and Prevalence Pituitary corticotrope adenomas account for 70% of patients with endogenous causes of Cushing's syndrome. However, it should be recalled that iatrogenic hypercortisolism is the most common cause of cushingoid features. Ectopic tumor [ACTH](#) production, cortisol-producing adrenal adenomas, carcinoma, and hyperplasia account for the other causes; rarely, ectopic tumor [CRH](#) production is encountered.

[ACTH](#)-producing adenomas account for about 10 to 15% of all pituitary tumors. Because the clinical features of Cushing's syndrome often lead to early diagnosis, most ACTH-producing pituitary tumors are relatively small microadenomas. However, macroadenomas are also seen, and some ACTH-secreting adenomas are clinically silent. Cushing's disease is 5 to 10 times more common in women than in men. These pituitary adenomas exhibit unrestrained ACTH secretion, with resultant hypercortisolemia. However, they retain partial suppressibility in the presence of high doses of administered glucocorticoids, providing the basis for dynamic testing to distinguish pituitary and nonpituitary causes of Cushing's syndrome.

Presentation and Diagnosis The diagnosis of Cushing's syndrome presents two great challenges: (1) to distinguish patients with pathologic cortisol excess from those with physiologic or other disturbances of cortisol production; and (2) to determine the etiology of cortisol excess, which can include iatrogenic administration of glucocorticoids, adrenal adenomas or carcinomas, pituitary adenomas, and ectopic sources of [ACTH](#) and [CRH](#).

Typical features of chronic cortisol excess include thin, brittle skin, central obesity, hypertension, plethoric moon facies, purple striae and easy bruisability, glucose intolerance or diabetes mellitus, gonadal dysfunction, osteoporosis, proximal muscle weakness, signs of hyperandrogenism (acne, hirsutism), and psychologic disturbances (depression, mania, and psychoses) ([Table 328-12](#)). Hematopoietic features of

hypercortisolism include leukocytosis, lymphopenia, and eosinopenia. Immune suppression includes delayed hypersensitivity. The protean manifestations of hypercortisolism make it challenging to decide which patients mandate formal laboratory evaluation. Certain features make pathologic causes of hypercortisolism more likely -- these include characteristic central redistribution of fat, thin skin with striae and bruising, and proximal muscle weakness. In children and in young females, early osteoporosis may be particularly prominent. The primary cause of death is cardiovascular disease, but infections and risk of suicide are also increased.

Rapid development of features of hypercortisolism associated with skin hyperpigmentation and severe myopathy suggests the possibility of ectopic sources of [ACTH](#). Hypertension, hypokalemic alkalosis, glucose intolerance, and edema are also more pronounced in these patients. Serum potassium levels <3.3 mmol/L are evident in ~70% of patients with ectopic ACTH secretion but are seen in <10% of patients with pituitary-dependent Cushing's disease.

Laboratory Investigation The diagnosis of Cushing's syndrome is based on laboratory documentation of endogenous hypercortisolism. Measurements of 24-h urine free cortisol (UFC) is a precise and cost-effective screening test. Alternatively, the failure to suppress plasma cortisol after an overnight 1-mg dexamethasone suppression test can be used to identify patients with hypercortisolism. As nadir levels of cortisol occur at night, elevated midnight samples of cortisol are suggestive of Cushing's syndrome. Basal plasma [ACTH](#) levels often distinguish patients with ACTH-independent (adrenal or exogenous glucocorticoid) from those with ACTH-dependent (pituitary, ectopic ACTH) Cushing's disease. Mean basal ACTH levels are about eight-fold higher in patients with ectopic ACTH secretion compared to those with pituitary ACTH-secreting adenomas. However, extensive overlap of ACTH levels in these two disorders precludes using ACTH to make the distinction. Instead, dynamic testing, based on differential sensitivity to glucocorticoid feedback, or ACTH stimulation in response to [CRH](#) or cortisol reduction is used to discriminate ectopic versus pituitary sources of excess ACTH ([Table 328-13](#)). Very rarely, circulating CRH levels are elevated, reflecting ectopic tumor-derived secretion of CRH and often ACTH. **For discussion of dynamic testing for Cushing's syndrome, see [Chap. 331](#).*

Most [ACTH](#)-secreting pituitary tumors are <5 mm in diameter, and about half are undetectable by sensitive [MRI](#). The high prevalence of incidental pituitary microadenomas diminishes the ability to distinguish ACTH-secreting pituitary tumors accurately by MRI.

Inferior Petrosal Venous Sampling Because pituitary [MRI](#) with gadolinium enhancement is insufficiently sensitive to distinguish small (<2 mm) pituitary [ACTH](#)-secreting adenomas from ectopic ACTH-secreting tumors that may have similar clinical and biochemical characteristics, bilateral inferior petrosal sinus ACTH sampling before and after [CRH](#) administration may be required. Simultaneous assessment of ACTH concentrations in each inferior petrosal vein and in the peripheral circulation provides a strategy for confirming and localizing pituitary ACTH production. Sampling is performed at baseline and 2, 5, and 10 min after intravenous ovine CRH (1 ug/kg) injection. An increased ratio (>2) of inferior petrosal:peripheral vein ACTH confirms pituitary Cushing's disease. After CRH injection, peak petrosal:peripheral ACTH ratios of ≥ 3

confirm the presence of a pituitary ACTH-secreting tumor. The sensitivity of this test is 99%, with very rare false-positive results. False-negative results may be encountered in patients with aberrant venous anatomic drainage. Petrosal sinus catheterizations are technically difficult, and about 0.05% of patients develop neurovascular complications. The procedure should not be performed in patients with hypertension or in the presence of a well-visualized pituitary adenoma on MRI.

TREATMENT

Selective transsphenoidal resection is the treatment of choice for Cushing's disease ([Fig. 328-12](#)). The remission rate for this procedure is ~80% for microadenomas but <50% for macroadenomas. After successful tumor resection, most patients experience a postoperative period of adrenal insufficiency that lasts for up to 12 months. This usually requires low-dose cortisol replacement, as patients experience steroid withdrawal symptoms as well as having a suppressed [HPA](#) axis. Biochemical recurrence occurs in approximately 5% of patients in whom surgery was initially successful.

When initial surgery is unsuccessful, repeat surgery is sometimes indicated, particularly when a pituitary source for [ACTH](#) is well documented. In older patients in whom growth and fertility are no longer important, hemi- or total hypophysectomy may be necessary if an adenoma is not recognized. Pituitary irradiation may be used after unsuccessful surgery, but it cures only about 15% of patients. Because radiation is slow and only partially effective in adults, steroidogenic inhibitors are used in combination with pituitary irradiation to block the adrenal effects of persistently high ACTH levels.

Mitotane (*o*, *p*-DDD) suppresses cortisol hypersecretion by inhibiting 11 β -hydroxylase and cholesterol side-chain cleavage enzymes and by destroying adrenocortical cells. Side effects of mitotane include gastrointestinal symptoms, dizziness, gynecomastia, hyperlipidemia, skin rash, and hepatic enzyme elevation. It may also lead to hypoaldosteronism. *Ketoconazole*, an imidazole derivative antimycotic agent, inhibits several P450 enzymes and effectively lowers cortisol in most patients with Cushing's disease when administered twice daily (600 to 1200 mg/d). Elevated hepatic transaminases, gynecomastia, impotence, gastrointestinal upset, and edema are common side effects. *Metyrapone* (2 to 4 g/d) inhibits 11 β -hydroxylase activity and normalizes plasma cortisol in up to 75% of patients. Side effects include nausea and vomiting, rash, and exacerbation of acne or hirsutism. Other agents include *aminoglutethimide* (250 mg tid), *trilostane* (200 to 1000 mg/d), *cyproheptadine* (24 mg/d), and IV *etomidate* (0.3 mg/kg per hour). Glucocorticoid insufficiency is a potential side effect of agents used to block steroidogenesis.

The use of steroidogenic inhibitors has decreased the need for bilateral adrenalectomy. Removal of both adrenal glands corrects hypercortisolism but may be associated with significant morbidity and necessitates permanent glucocorticoid and mineralocorticoid replacement. Adrenalectomy in the setting of residual corticotrope adenoma tissue predisposes to the development of *Nelson's syndrome*, a disorder characterized by rapid pituitary tumor enlargement and increased pigmentation secondary to high [ACTH](#) levels. Radiation therapy may be indicated to prevent the development of Nelson's syndrome after adrenalectomy.

GONADOTROPINS: [FSH](#) AND [LH](#)

SYNTHESIS AND SECRETION

Gonadotrope cells comprise about 10% of anterior pituitary cells and produce two gonadotropins -- LH and FSH. Like [TSH](#) and [hCG](#), LH and FSH are glycoprotein hormones consisting of a and b subunits. The a subunit is common to these glycoprotein hormones; specificity is conferred by the b subunits, which are expressed by separate genes.

Gonadotropin synthesis and release are dynamically regulated. This is particularly true in females, in whom the rapidly fluctuating gonadal steroid levels vary throughout the menstrual cycle. Hypothalamic [GnRH](#), a 10-amino-acid peptide synthesized in the preoptic region, regulates the synthesis and secretion of both [LH](#) and [FSH](#). GnRH is secreted in discrete pulses every 60 to 120 min, which in turn elicit LH and FSH pulses ([Fig. 328-3](#)). GnRH acts through a [GPCR](#) to stimulate phospholipase C, protein kinase C, and calcium signaling pathways. The pulsatile mode of GnRH input is essential to its action; pulses prime gonadotrope responsiveness, whereas continuous GnRH exposure induces desensitization. Based on this phenomenon, long-acting GnRH agonists are used to suppress gonadotropin levels in children with precocious puberty ([Chap. 8](#)) and in men with prostate cancer ([Chap. 95](#)) and are used in some ovulation-induction protocols to reduce endogenous gonadotropins ([Chap. 336](#)). Estrogens act at the hypothalamic and pituitary levels to control gonadotropin secretion. Chronic estrogen exposure is inhibitory, whereas rising estrogen levels, as occurs during the preovulatory surge, exert positive feedback to increase gonadotropin pulse frequency and amplitude. Progesterone slows GnRH pulse frequency but enhances gonadotropin responses to GnRH. Testosterone feedback in males also occurs at the hypothalamic and pituitary levels and partially reflects its conversion to estrogens ([Chap. 335](#)).

Though [GnRH](#) is the main regulator of [LH](#) and [FSH](#) secretion, FSH synthesis is also under separate control by the gonadal peptides inhibin and activin, which are members of the transforming growth factor b (TGF-b) family. Inhibin selectively suppresses FSH, whereas activin stimulates FSH synthesis ([Chap. 336](#)).

ACTION

The gonadotropin hormones interact with their respective [GPCRs](#) expressed in the ovary and testis, evoking germ-cell development and maturation and steroid hormone biosynthesis. In women, [FSH](#) regulates ovarian follicle development and stimulates ovarian estrogen production. [LH](#) mediates ovulation and maintenance of the corpus luteum. In men, LH induces Leydig cell testosterone synthesis and secretion and FSH stimulates seminiferous tubule development and regulates spermatogenesis.

GONADOTROPIN DEFICIENCY

Hypogonadism is the most common presenting feature of adult hypopituitarism, even when other pituitary hormones are also deficient. It is often a harbinger of hypothalamic or pituitary diseases that impair [GnRH](#) production or delivery through the pituitary stalk. As noted above, hypogonadotropic hypogonadism is a common presenting feature of

hyperprolactinemia.

A variety of inherited and acquired disorders are associated with *isolated hypogonadotropic hypogonadism* (IHH) ([Chap. 335](#)). Hypothalamic defects associated with [GnRH](#) deficiency include two X-linked disorders, Kallmann syndrome (see above) and mutations in the *DAX1* gene. GnRH receptor mutations and inactivating mutations of the [LH](#) β and [FSH](#) β subunit genes are rare causes of selective gonadotropin deficiency. Acquired forms of GnRH deficiency leading to hypogonadotropism are seen in association with anorexia nervosa ([Chap. 78](#)), stress, starvation, and extreme exercise, but may also be idiopathic. Hypogonadotropic hypogonadism in these disorders is reversed by removal of the stressful stimulus.

Presentation and Diagnosis In premenopausal women, hypogonadotropic hypogonadism presents as diminished ovarian function leading to oligomenorrhea or amenorrhea, infertility, decreased vaginal secretions, decreased libido, and breast atrophy. In hypogonadal adult males, secondary testicular failure is associated with decreased libido and potency, infertility, decreased muscle mass with weakness, reduced beard and body hair growth, soft testes, and characteristic fine facial wrinkles. Osteoporosis occurs in both untreated hypogonadal females and males.

Laboratory Investigation Central hypogonadism is associated with low or inappropriately low serum gonadotropin levels and low sex hormone concentrations (testosterone in males, estradiol in females). Three pooled serum samples drawn 20 min apart are used for accurate measurement of serum [LH](#) and [FSH](#) levels, thus allowing for the effects of hormone secretory pulses. Male patients have abnormal semen analysis.

Intravenous [GnRH](#) (100 ug) stimulates gonadotropes to secrete [LH](#) (which peaks within 30 min) and [FSH](#) (which plateaus during the ensuing 60 min). Normal responses vary according to menstrual cycle stage, age, and sex of the patient. Generally, LH levels increase about threefold, whereas FSH responses are less pronounced. In the setting of gonadotropin deficiency, a normal gonadotropin response to GnRH indicates intact gonadotrope function and suggests a hypothalamic abnormality. An absent response, however, cannot reliably distinguish pituitary from hypothalamic causes of hypogonadism. For this reason, GnRH testing usually adds little to the information gained from baseline evaluation of the hypothalamic-pituitary-gonadotrope axis, except in cases of isolated GnRH deficiency (e.g., Kallmann syndrome).

[MRI](#) examination of the sellar region and assessment of other pituitary functions are usually indicated in patients with documented central hypogonadism.

TREATMENT

In males, testosterone replacement is necessary to achieve and maintain normal growth and development of the external genitalia, secondary sex characteristics, male sexual behavior, and androgenic anabolic effects including maintenance of muscle function and bone mass. Testosterone may be administered by intramuscular injections every 1 to 4 weeks or using patches that are replaced daily ([Chap. 335](#)). Gonadotropin injections [[hCG](#) or human menopausal gonadotropin (hMG)] over 12 to 18 months are used to

restore fertility. Pulsatile [GnRH](#) therapy (25 to 150 ng/kg every 2 h), administered by a subcutaneous infusion pump, is also effective for treatment of hypothalamic hypogonadism when fertility is desired.

In premenopausal women, cyclical replacement of estrogen and progesterone maintains secondary sexual characteristics and genitourinary tract integrity and prevents premature osteoporosis and possibly coronary artery disease ([Chap. 336](#)). Gonadotropin therapy is used for ovulation induction. Follicular growth and maturation are initiated using [hMG](#) or recombinant [FSH](#); [hCG](#) is subsequently injected to induce ovulation. As in men, pulsatile [GnRH](#) therapy can be used to treat hypothalamic causes of gonadotropin deficiency.

NONFUNCTIONING AND GONADOTROPIN-PRODUCING PITUITARY ADENOMAS

Etiology and Prevalence Nonfunctioning pituitary adenomas include those that secrete little or no pituitary hormones, as well as tumors that produce too little hormone to result in recognizable clinical features. They are the most common type of pituitary adenoma and are usually macroadenomas at the time of diagnosis because clinical features are inapparent until tumor mass effects occur. Based on immunohistochemistry, most clinically nonfunctioning adenomas can be shown to originate from gonadotrope cells. These tumors typically produce small amounts of intact gonadotropins (usually [FSH](#)) as well as uncombined α and [LH](#) β and FSH β subunits. Tumor secretion may lead to elevated α and FSH β subunits and, rarely, to increased LH β subunit levels. Some adenomas express α subunits without FSH or LH. [TRH](#) administration often induces an atypical increase of tumor-derived gonadotropins or subunits.

Presentation and Diagnosis Clinically nonfunctioning tumors may present with optic chiasm pressure and other symptoms of local expansion or be incidentally discovered on an [MRI](#) performed for another indication. Menstrual disturbances or ovarian hyperstimulation rarely occur in women with large tumors that produce [FSH](#) and [LH](#). More commonly, adenoma compression of the pituitary stalk or surrounding pituitary tissue leads to attenuated LH and features of hypogonadism. Prolactin levels are usually slightly increased, also because of stalk compression. It is important to distinguish this circumstance from true prolactinomas, as most nonfunctioning tumors respond poorly to treatment with dopamine agonists.

Laboratory Investigation The goal of laboratory testing in clinically nonfunctioning tumors is to classify the type of the tumor, to identify hormonal markers of tumor activity, and to detect possible hypopituitarism. Free α subunit levels may be elevated in 10 to 15% of patients with nonfunctioning tumors. In female patients, peri- or postmenopausal basal [FSH](#) concentrations are difficult to distinguish from tumor-derived FSH elevation. Premenopausal women have cycling FSH levels, also preventing clear-cut diagnostic distinction from tumor-derived FSH. In men, gonadotropin-secreting tumors may be diagnosed because of slightly increased gonadotropins ($\text{FSH} > \text{LH}$) in the setting of a pituitary mass. Testosterone levels are usually low, despite the normal or increased LH level, perhaps reflecting reduced LH bioactivity or the loss of normal LH pulsatility. Because this pattern of hormone tests is also seen in primary gonadal failure and, to some extent, with aging ([Chap. 335](#)), the increased gonadotropins alone are insufficient for the diagnosis of a gonadotropin-secreting tumor. In the majority of patients with

gonadotrope adenomas, [TRH](#) administration stimulates LH β subunit secretion; this response is not seen in normal individuals. [GnRH](#) testing is not helpful for making the diagnosis. For nonfunctioning and gonadotropin-secreting tumors, the diagnosis usually rests on immunohistochemical analyses of resected tumor tissue, as the mass effects of these tumors usually necessitate resection.

Although acromegaly or Cushing's syndrome usually presents with unique clinical features, clinically inapparent somatotrope or corticotrope adenomas can be excluded by a normal [IGF-I](#) value and normal 24-h urinary free cortisol levels. If [PRL](#) levels are <100 ug/L in a patient harboring a pituitary mass, a nonfunctioning adenoma causing pituitary stalk compression should be considered.

TREATMENT

Asymptomatic small nonfunctioning adenomas with no threat to vision may be followed with regular [MRI](#) and visual field testing without immediate intervention. However, for larger macroadenomas, transsphenoidal surgery is the only effective way to reduce tumor size and relieve mass effects ([Fig. 328-13](#)). Although it is not usually possible to remove all adenoma tissue surgically, vision improves in 70% of patients with preoperative visual field defects. Preexisting hypopituitarism that results from tumor mass effects commonly improves or may resolve completely. Early postoperative complications include diabetes insipidus and/or inappropriate antidiuretic hormone secretion. Beginning about 6 months postoperatively, MRI scans should be performed yearly to detect tumor regrowth. Within 5 to 6 years following successful surgical resection, $\sim 15\%$ of nonfunctioning tumors recur. When substantial tumor remains after transsphenoidal surgery, adjuvant radiotherapy may be indicated to prevent tumor growth. Radiotherapy may be deferred if no postoperative residual mass is evident.

Nonfunctioning pituitary tumors respond poorly to dopamine agonist treatment, with modest tumor shrinkage occurring in $<10\%$ of patients. Although SSTR subtypes 2 and 5 have been identified on nonfunctioning pituitary adenomas, octreotide does not shrink these tumors and only modestly suppresses gonadotropin and α subunit levels. Visual improvement sometimes occurs without evident reduction of tumor size by [MRI](#), presumably reflecting relief of pressure on the optic tracts. The selective [GnRH](#) antagonist, Nal-Glu GnRH, suppresses [FSH](#) hypersecretion but has no effect on adenoma size.

THYROID-STIMULATING HORMONE

SYNTHESIS AND SECRETION

[TSH](#)-secreting thyrotrope cells comprise 5% of the anterior pituitary cell population. TSH is structurally related to [LH](#) and [FSH](#). It shares a common α subunit with these hormones but contains a specific TSH β subunit. [TRH](#) is a hypothalamic tripeptide (pyroglutamyl histidylprolinamide) that acts through a [GPCR](#) to stimulate phospholipase C, protein kinase C, and calcium pathways. TRH stimulates TSH synthesis and secretion; it also stimulates the lactotrope cell to secrete [PRL](#). TSH secretion is stimulated by TRH, whereas thyroid hormones, dopamine, SRIF, and glucocorticoids suppress TSH by overriding TRH induction.

The thyrotrope is stimulated when [TSH](#) is released from the negative feedback inhibition of thyroid hormones. Thus, thyroid damage, including surgical thyroidectomy, radiation-induced hypothyroidism, chronic thyroiditis, or prolonged goitrogen exposure, are associated with increased TSH. Long-standing untreated hypothyroidism can lead to thyrotrope hyperplasia and pituitary enlargement, which may be evident on [MRI](#).

ACTION

[TSH](#) is secreted in pulses, though the excursions are modest in comparison to other pituitary hormones because of the relatively low amplitude of the pulses and the relatively long half-life of TSH. Consequently, single determinations of TSH suffice to assess its circulating levels. TSH binds to a [GPCR](#) on thyroid follicular cells to stimulate thyroid hormone synthesis and release ([Chap. 330](#)).

TSH DEFICIENCY

Features of central hypothyroidism, due to [TSH](#) deficiency, mimic those seen with primary hypothyroidism but are generally less severe. Pituitary hypothyroidism is characterized by low basal TSH levels in the setting of low free thyroid hormone. In contrast, patients with hypothyroidism of hypothalamic origin (presumably due to a lack of endogenous [TRH](#)) may exhibit normal or even slightly elevated TSH levels. There is evidence that the TSH produced in this circumstance has reduced biologic activity because of altered glycosylation.

[TRH](#) (200 ug) injected intravenously causes a two- to threefold increase in [TSH](#) (and [PRL](#)) levels within 30 min. Although TRH testing can be used to assess TSH reserve, abnormalities of the thyroid axis can usually be detected based on basal free T₄ and TSH levels, without the need for TRH testing.

Thyroid-replacement therapy should be initiated after establishing adequate adrenal function. Dose adjustment is based on thyroid hormone levels and clinical parameters rather than the [TSH](#) level.

TSH-SECRETING ADENOMAS

[TSH](#)-producing macroadenomas are rare but are often large and locally invasive when they occur. Patients usually present with thyroid goiter and hyperthyroidism, reflecting overproduction of TSH. Diagnosis is based on demonstrating elevated serum free T₄ levels, inappropriately normal or high TSH secretion, and [MRI](#) evidence of a pituitary adenoma. An elevated free α subunit level occurs in about half of patients and supports the diagnosis of a TSH-secreting adenoma.

It is important to exclude other causes of inappropriate [TSH](#) secretion, such as resistance to thyroid hormone, an autosomal dominant disorder caused by mutations in the thyroid hormone β receptor ([Chap. 330](#)). The presence of a pituitary mass and elevated α subunit levels are suggestive of a TSH-secreting tumor. Dysalbuminemic hyperthyroxinemia syndromes, caused by various mutations in serum thyroid hormone binding proteins, are also characterized by elevated thyroid hormone levels, but with

normal rather than suppressed TSH levels. However, free thyroid hormone levels are normal in these disorders, most of which are familial.

TREATMENT

The initial therapeutic approach is to remove or debulk the tumor mass surgically, using either a transsphenoidal or subfrontal approach. Total resection is not often achieved as most of these adenomas are large and locally invasive. Normal circulating thyroid hormone levels are achieved in about two-thirds of patients after surgery. Thyroid ablation or antithyroid drugs (methimazole or propylthiouracil) can be used to reduce thyroid hormone levels. Dopamine agonists are rarely effective for suppressing [TSH](#) secretion from these tumors. However, somatostatin analogue treatment effectively normalizes TSH and a subunit hypersecretion, shrinks the tumor mass in 50% of patients, and improves visual fields in 75% of patients; euthyroidism is restored in most patients. In some patients, octreotide markedly suppresses TSH, causing biochemical hypothyroidism that requires concomitant thyroid hormone replacement. Lanreotide (30 mg intramuscularly), a long-acting somatostatin analogue (see above), effectively suppresses TSH and thyroid hormone in patients treated every 14 days.

DIABETES INSIPIDUS

**See [Chap. 329](#) for diagnosis and treatment of diabetes insipidus.*

(Bibliography omitted in Palm version)

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329. DISORDERS OF THE NEUROHYPOPHYSIS - Gary L. Robertson

The neurohypophysis, or posterior pituitary gland, is formed by axons that project from large cell bodies in the supraoptic and paraventricular nuclei of the hypothalamus to the posterior portion of the sella turcica. The neurohypophysis produces two hormones: (1) arginine vasopressin (AVP), also known as antidiuretic hormone (ADH); and (2) oxytocin. AVP acts on the renal tubules to induce water retention, leading to concentration of the urine. Oxytocin stimulates postpartum milk letdown in response to suckling. AVP deficiency causes diabetes insipidus (DI), characterized by the production of large amounts of dilute urine. Excessive or inappropriate production of AVP predisposes to hyponatremia, reflecting water retention. There are no known clinical disorders associated with oxytocin deficiency or excess.

VASOPRESSIN

ACTION

[AVP](#) is a nonapeptide composed of a six-membered disulfide ring and a tripeptide tail on which the C-terminal carboxy group is amidated ([Fig. 329-1](#)). The most important, if not the only, physiologic action of AVP is to influence the rate of water excretion by promoting concentration of urine. This antidiuretic effect is achieved by increasing the hydroosmotic permeability of cells that line the distal tubule and medullary collecting ducts of the nephron. In the absence of AVP, these cells are impermeable to water and reabsorb little, if any, of the relatively large volume of dilute filtrate that enters from the cortical nephron. Low AVP concentration results in the production of large amounts of urine (as much as 0.2 mL/kg per minute) that is maximally dilute (specific gravity and osmolality ~1.000 and 50 mmol/L, respectively), a condition known as a *water diuresis*. In the presence of AVP, the luminal surface of the cells lining the terminal collecting duct becomes selectively permeable to water, allowing it to diffuse back down the osmotic gradient created by the hypertonic renal medulla. As a result, the dilute fluid passing through the tubules is concentrated and the rate of urine flow decreases. The magnitude of this antidiuretic effect varies in direct proportion to the plasma AVP concentration. It is mediated via binding of AVP to G protein-coupled V_2 receptors on the serosal surface of the cell, activation of adenyl cyclase, and insertion into the luminal surface of water channels composed of a protein known as *aquaporin 2* ([Fig. 329-2](#)). The genes encoding the V_2 receptors and aquaporin 2 have been cloned and appear to be expressed exclusively in the distal and collecting tubules of the kidney. Nonpeptide, as well as peptide, AVP analogues with potent agonist or antagonist actions at human V_2 receptors have been developed for treating disorders of water metabolism caused by deficient or excessive production of AVP (see below).

At high concentrations, [AVP](#) also has several other actions, including contraction of smooth muscle in blood vessels in the skin and gastrointestinal tract, glycogenolysis in the liver, and potentiation of adrenocorticotrophic hormone (ACTH) release by corticotropin-releasing factor (CRF). These effects are mediated by V_{1a} or V_{1b} receptors that are coupled to phospholipase C. The genes that encode these receptors have also been cloned and sequenced and are expressed in many organs, including blood vessels, the anterior and posterior pituitary, and certain other areas of brain. Their role, if any, in human physiology/pathophysiology is still uncertain.

SYNTHESIS

[AVP](#) is synthesized via a polypeptide precursor that includes a binding protein known as *neurophysin II* and a glycosylated peptide called *copeptin*. The gene encoding the AVP precursor is located on chromosome 20 and has three exons. It is expressed in distinct subpopulations of magno- and parvocellular neurons in the supraoptic and paraventricular nuclei. Like other peptide hormones destined for secretion, newly synthesized AVP-neurophysin II precursor is translocated from the cytosol to the endoplasmic reticulum, where the signal peptide is removed and the prohormone folds and oligomerizes before moving through the Golgi apparatus to the neurosecretory vesicles; there it is transported down the axons and further cleaved to AVP, neurophysin II, and copeptin. Stimulation of the neurons results in an influx of calcium, fusion of the neurosecretory vesicle with the cell membrane, and extrusion of its contents into the systemic circulation.

SECRETION

The secretion of [AVP](#) is regulated primarily by the "effective" osmotic pressure of body fluids. This control is mediated by specialized cells, known as *osmoreceptors*, which appear to be located in the anteromedial hypothalamus near the supraoptic nucleus. These osmoreceptors are extremely sensitive to small changes in the plasma concentration of sodium and certain other effective solutes such as mannitol but show little or no response to other solutes such as urea or glucose. They appear to have inhibitory as well as stimulatory components that function in concert to regulate AVP secretion around a specific set point. Thus, when plasma osmolality/sodium are depressed to a certain minimum or threshold level of ~280 mosmol/kg or 135 meq/L, respectively, plasma AVP is suppressed to low or undetectable levels and a water diuresis ensues. Conversely, when plasma osmolality/sodium rise above this "threshold," plasma AVP rises steeply in direct proportion, reaching a concentration sufficient to produce a maximum antidiuresis when plasma osmolality/sodium reach ~295 mosmol/kg and 143 meq/L. However, the exact "set" and "sensitivity" of this osmoregulatory system vary appreciably from person to person, apparently as a result of genetic influences, and change during pregnancy, the menstrual cycle, and normal aging; they can also be altered or disrupted by various pathologic conditions.

[AVP](#) secretion can also be influenced by acute changes in blood volume or pressure. This baroregulation is mediated largely by neuronal afferents that originate in transmural pressure receptors of the cardiac atria, aorta, and carotid arteries; project via the vagus and glossopharyngeal nerves to the nucleus tractus solitarius of the brain stem; and then ascend to the paraventricular and supraoptic nuclei of the hypothalamus. These pathways regulate AVP release by maintaining a tonic inhibitory tone that decreases when blood volume or pressure falls by >10 to 20%. This baroregulatory system is probably of minor importance in the physiology of AVP secretion because the hemodynamic changes required to affect it are larger than those usually occurring in the course of normal activities. Moreover, moderate hemodynamic stimuli do not disrupt or override the osmoregulatory system but do lower its threshold or set point by an amount proportional to the magnitude of the hypovolemia or hypotension. However, the baroregulatory system undoubtedly plays an important role in AVP secretion in patients

with large, acute disturbances of hemodynamic function.

[AVP](#) secretion can also be stimulated by a variety of other nonosmotic variables including nausea, acute hypoglycemia, glucocorticoid deficiency, smoking, and, possibly, hyperangiotensinemia. The emetic stimuli are extremely potent since they typically elicit immediate, 50- to 100-fold increases in plasma AVP, even when the nausea is transient and unassociated with vomiting or other symptoms. They appear to act via the emetic center in the medulla and can be completely blocked by treatment with antiemetics such as fluphenazine. There is no evidence that pain or other noxious stresses have any effect on AVP unless they elicit a vasovagal reaction with its associated nausea and hypotension.

METABOLISM

From the venous circulation, [AVP](#) distributes rapidly into a space roughly equal in size to the extracellular fluid volume. It is cleared irreversibly from this space with a $t_{1/2}$ of 10 to 30 min. Most AVP clearance is due to degradation in the liver and kidneys. Urinary clearance of the hormone is normally much less than creatinine clearance but can vary as much as tenfold, depending on individual differences and changes in total solute clearance. Therefore, measurement of urinary AVP excretion rates are not always a reliable indicator of changes in secretion or plasma levels of the hormone. During pregnancy, the metabolic clearance of AVP is increased three- to fourfold due to placental production of an N-terminal peptidase.

THIRST

Though [AVP](#) regulates the effective osmotic pressure of body fluids by varying the rate of urinary free-water excretion, it cannot reduce insensible or urinary water output below a certain minimum obligatory level. Thus, an additional mechanism -- thirst -- is essential to prevent hypertonic dehydration. Like AVP, thirst is regulated primarily by an osmostat that is located in the anteromedial hypothalamus and is able to detect very small changes in the plasma concentration of sodium and certain other effective solutes. It functions like the AVP osmostat except that it appears to be "set" slightly higher. This arrangement ensures that thirst, polydipsia, and dilution of body fluids do not occur until dehydration and the resultant rise in plasma osmolality start to exceed the defensive capacity of the antidiuretic mechanism.

OXYTOCIN

Oxytocin is also a nonapeptide and differs from [AVP](#) only at positions 3 and 8 ([Fig. 329-1](#)). However, it has relatively little antidiuretic effect and seems to act mainly on mammary ducts to facilitate milk letdown during nursing ([Chap. 337](#)). It may also help to initiate or facilitate labor by stimulating contraction of uterine smooth muscle, but it is not yet clear if this action is physiologic or necessary for normal delivery. Both the mammary and uterine effects are mediated by a G protein-coupled receptor that is linked to phospholipase C. Antagonists for this receptor have been developed and tested in humans for possible use in treating premature labor.

Oxytocin is also synthesized via a macromolecular precursor that is encoded by a gene

located on chromosome 20, very near the [AVP](#) gene. However, it differs in orientation and size and encodes only a signal peptide, the hormone, and its associated neurophysin. Oxytocin is also expressed in different magnocellular neurons than AVP and, in humans, is not subject to any of the same regulatory influences. Indeed, with the possible exception of nipple stimulation in the postpartum period, no other stimuli are known to consistently induce release of the hormone in humans. Plasma oxytocin is not increased during pregnancy or at the initiation of labor, although the latter condition may be facilitated by upregulation of oxytocin receptors. The distribution and clearance of oxytocin are similar to those of AVP. Oxytocin is also degraded by the liver and kidneys and an N-terminal peptidase produced by the placenta.

DEFICIENCIES OF VASOPRESSIN SECRETION AND ACTION

DIABETES INSIPIDUS

Clinical Characteristics Decreased secretion or action of [AVP](#) usually manifests as [DI](#), a syndrome characterized by the production of abnormally large volumes of dilute urine. The 24-h urine volume is >50 mL/kg body weight and the osmolality is <300 mmol/kg. The polyuria produces symptoms of urinary frequency, enuresis, and/or nocturia, which may disturb sleep and cause mild daytime fatigue or somnolence. It is also associated with thirst and a commensurate increase in fluid intake (polydipsia). Clinical signs of dehydration are uncommon unless fluid intake is impaired.

Etiology Deficient secretion of [AVP](#) can be primary or secondary. The primary form usually results from agenesis or irreversible destruction of the neurohypophysis and is variously referred to as *neurohypophyseal DI*, *neurogenic DI*, *pituitary DI*, *cranial DI*, or *central DI*. It can be caused by a variety of congenital, acquired, or genetic disorders but almost half the time it is idiopathic ([Table 329-1](#)). The genetic form of neurohypophyseal [DI](#) is usually transmitted in an autosomal dominant mode and is caused by diverse mutations in the coding region of the AVP-neurophysin II gene. The mutant precursor cannot be processed properly or efficiently and eventually destroys the neuron, thereby accounting for the dominant mode of transmission and the delayed onset of the disorder. An X-linked recessive form also occurs. A primary deficiency of plasma AVP can also result from increased metabolism by an N-terminal aminopeptidase produced by the placenta. It is referred to as *gestational DI* since the signs and symptoms manifest during pregnancy and usually remit several weeks after delivery. However, a subclinical deficiency in AVP secretion can often be demonstrated in the nonpregnant state in these individuals, indicating that damage to the neurohypophysis may also contribute to the AVP deficiency. Finally, a primary deficiency of AVP can also result from malformation or destruction of the neurohypophysis by a variety of diseases or toxins ([Table 329-1](#)).

Secondary deficiencies of [AVP](#) result from inhibition of secretion by excessive intake of fluids. They are referred to as *primary polydipsia* and can be divided into three subcategories. One of them, called *dipsogenic DI*, seems to be caused by an inappropriate increase in thirst caused by a reduction in the "set" of the osmoregulatory mechanism. It sometimes occurs in association with multifocal diseases of the brain such as neurosarcoid, tuberculous meningitis, or multiple sclerosis but is often idiopathic. The second subtype, called *psychogenic polydipsia*, is not associated with

thirst, and the polydipsia seems to be a feature of psychosis. The third subtype, which may be referred to as *iatrogenic polydipsia*, results from recommendations of health professionals or the popular media to increase fluid intake for its presumed preventive or therapeutic benefits for other disorders.

Primary deficiencies in the antidiuretic action of [AVP](#) result in *nephrogenic DI* ([Table 329-1](#)). It can be genetic, acquired, or caused by exposure to various drugs. The genetic form is usually transmitted in an X-linked mode and is caused by mutations in the coding region of the V_2 receptor gene. An autosomal recessive form is caused by mutations in the gene encoding the aquaporin protein that forms the water channels in the distal nephron.

Secondary deficiencies in the antidiuretic response to [AVP](#) result from polyuria per se. They appear to be caused by washout of the medullary concentration gradient and/or suppression of aquaporin function. They usually resolve 24 to 48 h after the polyuria is corrected but often complicate interpretation of certain acute tests commonly used for differential diagnosis.

Pathophysiology When the net secretion or antidiuretic effect of [AVP](#) is decreased by >80 to 85%, the amount of hormone produced under basal conditions is insufficient to concentrate the urine and the rate of output increases exponentially to symptomatic levels. If the AVP defect is primary (e.g., the patient has pituitary, gestational, or nephrogenic [DI](#)), the polyuria results in a small (1 to 2%) decrease in body water and a commensurate increase in plasma osmolality and sodium concentration that stimulate thirst and a compensatory increase in water intake. As a result, *overt physical or laboratory signs of dehydration do not develop unless the patient also has a defect in thirst (see below) or fails to drink for some other reason.*

The severity of the defect in antidiuretic function varies markedly among patients with pituitary, gestational, or nephrogenic [DI](#). In some, the deficiencies in [AVP](#) secretion or action are so severe that basal urine output approximates the maximum (10 to 15 mL/min); even an intense stimulus such as nausea or severe dehydration does not increase plasma AVP enough to concentrate the urine. In others, however, the deficiency in AVP secretion or action is less pronounced, and a modest stimulus such as a few hours of fluid deprivation, smoking, or a vasovagal reaction increases plasma AVP sufficiently to produce a profound antidiuresis. The maximum urine osmolality achieved in these patients is usually less than normal, largely because their maximal concentrating capacity is temporarily impaired by chronic polyuria per se. However, in a few patients with partial pituitary or nephrogenic DI, it can reach levels as high as 800 mosmol/kg ([Fig. 329-3](#)).

In primary polydipsia, the pathogenesis of the polydipsia and polyuria is the reverse of that in neurohypophyseal, nephrogenic, and gestational [DI](#). Thus, the excessive intake of fluids slightly increases body water, thereby reducing plasma osmolality, [AVP](#) secretion, and urinary concentration. The latter results in a compensatory increase in urinary free-water excretion that varies in direct proportion to intake. Therefore, clinically appreciable overhydration is uncommon unless the compensatory water diuresis is impaired by a drug or disease that stimulates or mimics endogenous AVP.

In the dipsogenic form of primary polydipsia, fluid intake is excessive because the osmotic threshold for thirst appears to be reset to the left, often well below that for [AVP](#) release. As a result, thirst is abnormally increased and cannot be completely relieved because plasma AVP is suppressed and an offsetting water diuresis develops before plasma osmolality is reduced sufficiently to eliminate the dipsogenic stimulus. Typically, therefore, patients with dipsogenic [DI](#) present with complaints of chronic thirst, polydipsia, and polyuria indistinguishable from those in patients with pituitary, gestational, or nephrogenic [DI](#). When deprived of fluids or subjected to some other acute osmotic or nonosmotic stimulus, they invariably increase plasma AVP normally, but the resultant increase in urine concentration is usually subnormal because their renal capacity to concentrate the urine is also blunted by chronic polyuria. Thus, their antidiuretic response to these stimuli may be indistinguishable from that in patients with partial pituitary, partial gestational, or partial nephrogenic [DI](#) ([Fig. 329-3](#)).

Differential Diagnosis When symptoms of urinary frequency, enuresis, nocturia, and/or persistent thirst are present, causes other than polyuria should be excluded. A 24-h urine output > 50 mL/kg per day (>3500 mL in a 70-kg man) is suspicious for [DI](#). If the osmolality of the 24-h urine is >300 mosmol/kg, the patient has a solute diuresis and should be evaluated for uncontrolled diabetes mellitus or other less common causes of excessive solute excretion. However, if the 24-h urine osmolality is <300 mosmol/kg, the patient has a water diuresis and should be evaluated further to determine which type of [DI](#) is present.

In differentiating between the various types of [DI](#), the history, physical examination, and routine laboratory tests may be helpful but are rarely sufficient because few, if any, of the findings are pathognomonic. Except in the rare patient who is clearly dehydrated under basal conditions of *ad libitum* fluid intake, this evaluation should begin with a *fluid deprivation test*. To minimize patient discomfort, avoid excessive dehydration, and maximize the information obtained, the test should be started in the morning and water balance should be monitored closely with hourly measurements of body weight, plasma osmolality and/or sodium concentration, and urine volume and osmolality.

If fluid deprivation does not result in urine concentration (osmolality >300 mosmol/kg, specific gravity >1.010) before body weight decreases by 5% or plasma osmolality/sodium exceed the upper limit of normal, primary polydipsia and a partial defect in [AVP](#) secretion or action are largely excluded ([Fig. 329-3](#)). In these patients, severe pituitary or nephrogenic [DI](#) can usually be distinguished by administering desmopressin (DDAVP, 0.03 ug/kg subcutaneously or intravenously) and repeating the measurement of urine osmolality 1 to 2 h later. An increase of >50% indicates severe pituitary [DI](#), whereas a smaller or absent response is strongly suggestive of nephrogenic [DI](#).

However, these indirect criteria are not useful for diagnosis in patients who concentrate their urine during fluid deprivation, because the changes in urine osmolality are remarkably similar in primary polydipsia and partial pituitary and partial nephrogenic [DI](#) ([Fig. 329-3](#)). In this situation, the safest and most reliable way to differentiate these conditions is to measure plasma or urine [AVP](#) collected before and during the fluid deprivation test and analyze the result in relation to the concurrent plasma or urine osmolality ([Fig. 329-4](#)). This approach invariably differentiates partial

nephrogenic DI from partial pituitary DI and primary polydipsia. It also differentiates pituitary DI from primary polydipsia if the hormone is measured when plasma osmolality or sodium is clearly above the normal range. However, the requisite level of hypertonic dehydration is difficult to produce by fluid deprivation alone when urine concentration occurs. Therefore, it is usually necessary to add an infusion of hypertonic (3%) saline and repeat the AVP measurements when plasma osmolality rises to >300 mmol/kg ($\text{Na}^+ > 145$ mmol/L). This endpoint is usually reached within 30 to 120 min if the hypertonic saline is infused at a rate of 0.1 mL/kg per minute and the fluid deprivation is maintained.

The differential diagnosis of [DI](#) may also be facilitated by magnetic resonance imaging (MRI) of the pituitary and hypothalamus. In most healthy adults and children, the posterior pituitary emits a hyperintense signal in T1 weighted mid-sagittal images. This "bright spot" is almost invariably absent or abnormally small in patients with pituitary DI but is present in 80 to 90% of those with primary polydipsia. Thus, the presence of a normal bright spot virtually excludes pituitary DI, whereas its absence supports but does not prove this diagnosis. Therefore, the MRI findings must be interpreted with caution and only in conjunction with other diagnostic studies based on assays of [AVP](#) or the differential responses to treatment.

TREATMENT

The signs and symptoms of uncomplicated pituitary [DI](#) can be eliminated completely by treatment with DDAVP ([Fig. 329-5](#)). It is a synthetic analogue of [AVP](#) ([Fig. 329-1](#)) that acts selectively at V_2 receptors to increase urine concentration and decrease urine flow in a dose-dependent manner. However, it is more resistant to degradation than AVP and has a three- to fourfold longer duration of action. This property makes it particularly useful in the treatment of gestational DI or pituitary DI during pregnancy. DDAVP can be given by intravenous or subcutaneous injection, nasal inhalation, or oral tablet. The doses required to control pituitary DI completely vary widely, depending on the patient and the route of administration. However, they usually range from 1 to 2 μg qd or bid by injection, 10 to 20 μg bid or tid by nasal spray, and 100 to 400 μg bid or tid orally. The onset of action is rapid, ranging from as little as 15 min after injection to 60 min after oral administration. When given in doses sufficient to completely normalize urinary osmolality and flow, DDAVP produces a slight (1 to 3%) increase in total-body water and a commensurate decrease in plasma osmolality and sodium concentration that rapidly eliminates thirst and polydipsia. Consequently, water balance is maintained and hyponatremia does not develop unless the patient has an associated abnormality in the osmoregulation of thirst or ingests/receives excessive amounts of fluid for some other reason. Fortunately, abnormal thirst occurs in $<10\%$ of patients with pituitary DI, and the other causes of excessive intake can usually be eliminated by educating the patient about the risks of drinking for reasons other than thirst. Therefore, most patients with pituitary DI can take desmopressin in doses sufficient to maintain a normal urine output continuously and do not need to endure the inconvenience and discomfort of allowing intermittent escape to prevent water intoxication.

Pituitary [DI](#) can also be treated with chlorpropamide (Diabinese). The mechanism of its antidiuretic action is uncertain but may involve potentiation of the effect of small amounts of [AVP](#) or direct activation of the V_2 receptor. In patients with severe as well as

partial pituitary DI, doses of chlorpropamide similar to those used in the treatment of diabetes mellitus (125 to 500 mg once daily) increase urine concentration and decrease urine flow, thirst, and polydipsia in a manner similar to DDAVP. The antidiuresis is usually less rapid and smaller than that produced by DDAVP but is almost always sufficient to reduce urine output by 30 to 70%. Moreover, its antidiuretic effect can be enhanced appreciably by cotreatment with a thiazide diuretic. The ability of water loading to reduce the antidiuretic effect of chlorpropamide makes it particularly useful in the treatment of patients who have pituitary DI and abnormal thirst since it is less likely than DDAVP to produce water intoxication. However, unlike DDAVP, chlorpropamide can have other side effects including hypoglycemia, which can be precipitated by severe reductions in caloric intake or heavy exercise, and it exhibits a disulfuram (Antabuse)-like reaction to ethanol. Chlorpropamide is contraindicated in the treatment of gestational DI because its teratogenicity is unknown.

Primary polydipsia cannot be treated with DDAVP in the usual way because a sustained inhibition of the compensatory water diuresis almost invariably results in the development of water intoxication within 24 to 48 h. This complication can also be caused by administration of a thiazide diuretic, smoking, or other nonosmotic stimuli to endogenous [AVP](#) secretion. Iatrogenic polydipsia can often be corrected by patient counseling; however, there is no effective treatment for either psychogenic or dipsogenic [DI](#). In the latter, nocturia or nocturnal enuresis can often be controlled safely by administering a single small dose of DDAVP at bedtime. If the dose is adjusted carefully to provide no more than 8 to 10 h of antidiuresis, it will not result in water intoxication, because patients with dipsogenic, as well as other forms of DI, tend to drink less fluid at night than during the day. Family or other caregivers of patients with psychogenic or dipsogenic DI should also be warned about the hazards of water intoxication caused by a variety of diseases or drugs that can stimulate or mimic the antidiuretic effects of endogenous AVP (see below).

The symptoms and signs of nephrogenic [DI](#) are not affected by treatment with DDAVP or chlorpropamide but may be reduced by treatment with a thiazide diuretic and/or amiloride in conjunction with a low-sodium diet. Inhibitors of prostaglandin synthesis (e.g., indomethacin) are also effective in many patients.

ADIPSIC HYPERNATREMIA

Clinical Characteristics Adipsic hypernatremia is characterized by chronic or recurrent hypertonic dehydration and a deficient [AVP](#) response to osmotic stimulation. Despite their dehydration, the patients have little or no thirst and may even resist efforts to increase their oral intake of fluids. The hypernatremia varies in severity and is associated with commensurate signs of hypovolemia such as tachycardia, postural hypotension, azotemia, hyperuricemia, and hypokalemia. Muscle weakness, pain, rhabdomyolysis, hyperglycemia, hyperlipidemia, and acute renal failure may also occur. Most patients remain conscious unless they have severe hyperglycemia and/or hypertonicity or go on to develop hyponatremia as a result of excessive rehydration.

Etiology Adipsic hypernatremia is caused by agenesis or destruction of the hypothalamic osmoreceptors that normally regulate thirst and [AVP](#) secretion. The osmoreceptor deficiency can usually be traced to an identifiable congenital or acquired

disease in the hypothalamus but is sometimes idiopathic ([Table 329-2](#)). The neurohypophysis and its other regulatory afferents are usually spared; an [MRI](#) typically shows a normal posterior pituitary bright spot, and the AVP response to nonosmotic stimuli is also normal. Occasionally, the neurohypophysis is also affected, resulting in a combined defect in water balance that is particularly severe and difficult to manage.

Pathophysiology Lack of thirst and failure to drink enough water to replenish renal and extrarenal losses decrease total-body water and increase plasma osmolality/sodium. Plasma renin activity and aldosterone secretion also increase, and plasma potassium falls due to increased urinary excretion. The severity, frequency, and speed with which hypertonic dehydration develops vary markedly from patient to patient, or from time to time in the same patient, owing largely to differences in the rate of insensible and/or renal loss.

The osmoregulation of [AVP](#) secretion is also impaired in nearly all patients with adipsic hypernatremia ([Fig. 329-6](#)). This deficiency is obvious when the hormone is measured in the presence of hypertonic dehydration but is rarely severe enough to produce [DI](#). During rehydration, however, some patients exhibit a further decrease in their plasma AVP and develop DI before their hypernatremia is fully corrected. In other patients, the osmoregulatory deficiency appears to be complete because basal plasma AVP remains relatively fixed, irrespective of whether plasma osmolality and sodium are above, within, or below the normal range. Thus, if overhydrated, these patients do not mount a compensatory water diuresis and quickly develop a hyponatremic syndrome that is clinically and biochemically indistinguishable from acute syndrome of inappropriate antidiuretic hormone, which is commonly referred to as SIADH (see below). In all but a few patients, the abnormality of AVP secretion is limited to the osmoregulatory system since the hormone responds normally to all nonosmotic stimuli, such as nausea.

Differential Diagnosis Adipsic hypernatremia should be distinguished from the hypernatremia that results from various other causes. These distinctions can usually be made from the history, physical examination, and routine laboratory tests. If a conscious patient denies thirst and/or does not drink vigorously in the presence of significant hypernatremia, the diagnosis of hypodipsia or adipsia can be made with confidence. This diagnosis is supported by laboratory evidence of hypovolemia (azotemia, hypokalemia, hyperuricemia, hyperreninemia) and a relative deficiency of plasma [AVP](#). Close monitoring of these variables and urine osmolality during rehydration is useful for differentiating the patients who develop [DI](#) or SIADH in response to forced hydration. If the patient is obtunded or otherwise unable to answer questions or drink at the time of presentation, the possibility of adipsic hypernatremia can be evaluated after treatment by assessing the thirst and plasma AVP response to a controlled fluid deprivation-hypertonic saline infusion test similar to that described for evaluation of DI.

TREATMENT

Adipsic hypernatremia should be treated by administering water by mouth, if the patient is alert, or 0.45% saline by vein, if the patient is obtunded or uncooperative. The number of liters of free water that will be required to correct the deficit (*DFW*) can be estimated from body weight in kg (*BW*) and the serum sodium concentration in mmol/L (*S_{Na}*) by the formula $DFW = 0.5BW [(S_{Na} - 140)/140]$. If serum glucose (*S_{Glu}*) is elevated, the

measured S_{Na} should be corrected (S_{Na}^*) by the formula $S_{Na}^* = S_{Na} + [(S_{Glu} - 90)/36]$. This amount plus an allowance for continuing insensible and urinary losses should be given over a 24- to 48-h period. If [DI](#) is present or develops during rehydration, DDAVP should also be given in standard doses to minimize urinary losses. If hyperglycemia and/or hypokalemia are present, insulin and/or potassium supplements should be given. These variables plus urine output and plasma urea/creatinine should be monitored closely during treatment for signs of emerging DI, SIADH, or acute renal failure.

Once the acute fluid and electrolyte imbalances are corrected, an [MRI](#) of the brain and tests of anterior pituitary function should be performed. A long-term management plan to prevent or minimize recurrence of the fluid and electrolyte imbalance should also be developed, including a practical method that the patient can use to regulate fluid intake in accordance with day-to-day variations in water balance. The most effective way to accomplish these objectives is to prescribe DDAVP or chlorpropamide to completely control [DI](#), if it is present, and teach the patient how to use day-to-day changes in body weight as a guide for adjusting fluid intake. Prescribing a constant fluid intake is less satisfactory because it does not take into account the large, uncontrolled variations in insensible loss that inevitably occur.

EXCESS VASOPRESSIN SECRETION AND ACTION

HYPONATREMIA (See also [Chap. 49](#))

Clinical Characteristics Excessive secretion or action of [AVP](#) results in the production of decreased volumes of more highly concentrated urine. If not accompanied by a commensurate reduction in fluid intake, the reduced suppressibility of AVP results in water retention and a decrease in plasma osmolality/sodium. If the hyponatremia develops gradually or has been present for more than a few days, it may be asymptomatic. However, if it develops acutely, it is almost always accompanied by symptoms and signs of water intoxication that may include mild headache, confusion, anorexia, nausea, vomiting, coma, and convulsions. Severe hyponatremia may be lethal. Depending on the cause of the increased antidiuresis, osmotically inappropriate thirst and/or fluid intake and other disturbances of fluid and electrolyte balance may also be present.

Etiology Osmotically inappropriate antidiuresis can be caused by a primary defect in [AVP](#) secretion or action or can be secondary to a recognized nonosmotic stimulus such as hypovolemia, hypotension, or glucocorticoid deficiency ([Table 329-3](#)). The primary forms are generally referred to as SIADH or euvolemic (type III) hyponatremia. They have many different causes, including ectopic production of AVP by lung cancer or other neoplasms, eutopic release by various diseases or drugs, and exogenous administration of AVP, DDAVP, or large doses of oxytocin. The ectopic forms result from abnormal and presumably unregulated expression of the AVP-NP II gene by primary or metastatic malignancies. They do not usually remit unless the ectopic source is eliminated. The eutopic forms manifest most often in patients with acute infections or strokes, but the mechanisms by which these diseases disrupt osmoregulation are not known. A form of acute or chronic hyponatremia very similar to SIADH can also result from stimulation of AVP secretion by protracted nausea or isolated glucocorticoid deficiency. In these patients the excess AVP secretion can be corrected quickly and

completely by specific treatments (antiemetics or glucocorticoids) that are not useful in other forms of SIADH.

The secondary forms of osmotically inappropriate antidiuresis are usually divided into two groups: type I (hypervolemic) and type II (hypovolemic) hyponatremia. Type I occurs in sodium-retaining, edema-forming states such as congestive heart failure, cirrhosis, or nephrosis and is thought to be due to a reduction in "effective" blood volume. Type II occurs in sodium-depleted states such as severe gastroenteritis, diuretic abuse, or mineralocorticoid deficiency and is probably due to a reduction in blood volume and/or pressure.

Pathophysiology In SIADH, interference with the osmotic suppression of [AVP](#) results in significant expansion and dilution of body fluids only if water intake exceeds the rate of insensible and urinary output. These abnormalities in water intake often result from an associated defect in the osmoregulation of thirst but can also be due to psychogenic or iatrogenic factors, including the administration of intravenous fluids.

In SIADH, the defect in the osmoregulation of antidiuretic function can take any of four distinct forms ([Fig. 329-6](#)). In one of them, [AVP](#) secretion remains fully responsive to changes in plasma osmolality/sodium, but the threshold or set point of the osmoregulatory system is abnormally low. Patients with this kind of downward resetting of the osmostat differ from those with the other types of osmoregulatory defect in that they are able to maximally suppress plasma AVP and dilute their urine if their fluid intake is high enough to reduce their plasma osmolality/sodium to the new set point. Another, smaller subgroup (about 10% of the total) do not have a demonstrable defect in the osmoregulation of AVP ([Fig. 329-6](#)). Thus, their inappropriate antidiuresis may be due to other abnormalities such as enhanced renal sensitivity to the antidiuretic effect of normally low levels of AVP or activation of aquaporin 2 water channels by a mechanism that is independent of AVP and V_2 receptors.

The extracellular volume expansion that results from excessive retention of water in SIADH also produces an increase in atrial natriuretic hormone, suppression of plasma renin activity, and a compensatory increase in urinary sodium excretion that serves to reduce the hypervolemia but aggravates the hyponatremia. Thus, hyponatremia is due to a decrease in total-body sodium as well as an increase in total-body water. The acute retention of water and fall in plasma sodium also cause a rise in intracellular volume. The resultant brain swelling increases intracranial pressure and probably causes the acute symptoms of water intoxication. After several days, this intracellular volume expansion may be reduced by inactivation or elimination of intracellular solutes, resulting in the remission of symptoms that often occur with hyponatremia of this duration.

In type I (edematous) or type II (hypovolemic) hyponatremia, the osmotic inhibition of [AVP](#) and urine concentration is counteracted by a hemodynamic stimulus that results from a substantial reduction in effective or absolute blood volume. In both cases, the inadequate suppression of AVP appears to be due to downward resetting of the osmostat. The resultant antidiuresis is usually enhanced by decreased distal delivery of filtrate that results from increased reabsorption of sodium in proximal nephrons secondary to the hypovolemia. If it is not associated with a commensurate reduction in

water intake, the marked reduction in urine output that ensues also leads to expansion and dilution of body fluids with symptoms of hyponatremia. This attenuates, but does not completely eliminate, the antidiuresis because the amount of water retained is usually insufficient to fully correct the effective or absolute hypovolemia. Unlike SIADH, therefore, plasma renin activity is elevated, causing secondary hyperaldosteronism and hypokalemia. The disturbance in salt and water balance that underlies the hyponatremia also differs from SIADH in that total-body sodium as well as water is increased in type I, whereas both are decreased in type II.

Differential Diagnosis When unexplained symptoms or signs consistent with water intoxication are present, serum sodium should be measured. If it is low and the reduction cannot be accounted for by an increase in plasma glucose or other solutes such as mannitol (e.g., plasma osmolality is also low), the type of hypotonic hyponatremia present can be determined by estimating extracellular fluid volume from the history, physical examination, and routine chemistries. If these findings are ambiguous or contradictory, measuring the rate of urinary sodium excretion or plasma renin activity may be helpful. These measurements can be misleading, however, if SIADH is stable or resolving or if the patient has type II hyponatremia due to a primary defect in renal conservation of sodium, surreptitious diuretic abuse, or hyporeninemic hypoaldosteronism. The latter may be suspected if serum potassium is elevated instead of low as is usually seen in types I and II hyponatremia. Measurements of plasma [AVP](#) are currently of no diagnostic value since they exhibit the same wide variation in abnormalities in all three types of hyponatremia. In patients who fulfill the clinical criteria for SIADH, plasma cortisol should also be measured to rule out unsuspected secondary adrenal insufficiency. If this is normal and there is no other obvious cause for SIADH, a careful search for occult lung cancer should also be undertaken.

TREATMENT

In acute SIADH, the keystone to treatment of hyponatremia is to restrict total fluid intake to less than the sum of insensible losses and urinary output. Total intake should include the water derived from food (300 to 500 mL/d). Because insensible losses in adults usually approximate 500 mL/d total discretionary intake (all water in liquid form) should be at least 500 mL less than urinary output. If achieved, this deficit usually reduces body water and increases serum sodium by about 1 to 2% per day. If more rapid correction of the hyponatremia is desired to eliminate severe symptoms or signs, the fluid restriction can be supplemented by intravenous infusion of hypertonic (3%) saline. This treatment has the advantage of correcting the sodium deficiency that is partly responsible for the hyponatremia as well as producing a solute diuresis that serves to remove some of the excess water. However, if the hyponatremia has been present for more than 24 to 48 h and is corrected too rapidly, the saline infusion also has the potential to produce central pontine myelinolysis, an acute, potentially fatal neurologic syndrome characterized by quadriparesis, ataxia, and abnormal extraocular movements ([Chap. 376](#)). The following guidelines appear to minimize, if not eliminate, the risk of this complication: the 3% saline should be infused at a rate ≤ 0.05 mL/kg body weight per minute; the effect should be monitored continuously by STAT measurements of serum sodium at least once every hour; and the infusion should be stopped as soon as serum sodium increases by 12 mmol/L or to 130 mmol/L, whichever comes first. Urinary output should also be monitored continuously since spontaneous remission of the SIADH can result in an

acute water diuresis that greatly accelerates the rate of rise in serum sodium produced by fluid restriction and 3% saline infusion.

In chronic SIADH, the hyponatremia can be minimized or eliminated by treatment with demeclocycline, 150 to 300 mg orally three or four times a day, or fludrocortisone, 0.05 to 0.2 mg orally twice a day. The effect of the demeclocycline manifests in 7 to 14 days and is due to production of a reversible form of nephrogenic^{[DI](#)}. Potential side effects include phototoxicity and azotemia. The effect of fludrocortisone also manifests in 1 to 2 weeks and is partly due to increased retention of sodium and possibly inhibition of thirst. It also increases urinary potassium excretion, which may require replacement through dietary adjustments or supplements. Fludrocortisone may induce hypertension, occasionally necessitating discontinuation of the treatment.

One or more nonpeptide^{[AVP](#)}antagonists that block the antidiuretic effect of AVP may soon be approved for use in the United States. Preliminary studies with these antagonists in acute or chronic SIADH indicate that they produce a dose-dependent increase in urinary free-water excretion, which, if combined with a modest restriction of fluid intake, gradually reduces body water and corrects the hyponatremia without any recognized adverse effect. Thus, they may become the treatment of choice for those forms of SIADH in which there is inappropriate secretion of AVP that cannot be corrected by other, more specific therapy such as antiemetics or glucocorticoids.

When an SIADH-like syndrome is due to protracted nausea and vomiting or isolated glucocorticoid deficiency, all abnormalities can be corrected quickly and completely by giving an antiemetic or hydrocortisone. As with other treatments, care must be taken to ensure that serum sodium does not rise too quickly or too far.

In type I hyponatremia, the only treatment currently available is severe fluid restriction, administration of urea or mannitol to produce a solute diuresis, and/or administration of cardiotonics or serum albumin to correct the effective hypovolemia. None of these treatments is particularly effective, and some (e.g., administration of mannitol or albumin) carry significant risks. Infusion of hypertonic saline is contraindicated because it worsens the sodium retention and edema and may precipitate cardiovascular decompensation. However, preliminary studies indicate that the^{[AVP](#)}antagonists may be almost as effective and safe in type I hyponatremia as they are in SIADH. Thus, they may become the treatment of choice for this form of hyponatremia also.

In type II hyponatremia, the defect in^{[AVP](#)}secretion and water balance can usually be corrected easily and quickly by stopping the loss of sodium and water and/or replacing the deficits by mouth or intravenous infusion of normal or hypertonic saline. As with the treatment of other forms of hyponatremia, care must be taken to ensure that plasma sodium does not increase too rapidly. Fluid restriction or administration of AVP antagonists is contraindicated as they would only aggravate the underlying volume depletion and could result in cardiovascular decompensation.

(Bibliography omitted in Palm version)

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330. DISORDERS OF THE THYROID GLAND - J. Larry Jameson, Anthony P. Weetman

The thyroid gland produces two related hormones, thyroxine (T₄) and triiodothyronine (T₃) ([Fig. 330-1](#)). These hormones play a critical role in cell differentiation during development and help to maintain thermogenic and metabolic homeostasis in the adult. Thyroid hormones act through nuclear hormone receptors to modulate gene expression. Disorders of the thyroid gland result primarily from autoimmune processes that either stimulate the overproduction of thyroid hormones (*thyrotoxicosis*) or cause glandular destruction and underproduction of thyroid hormones (*hypothyroidism*). In addition, neoplastic processes in the thyroid gland can lead to benign nodules and various forms of thyroid cancer.

ANATOMY AND DEVELOPMENT

The thyroid gland is located in the neck, anterior to the trachea, between the cricoid cartilage and the suprasternal notch. The thyroid (Greek *thyreos*, shield, plus *eidos*, form) consists of two lobes that are connected by an isthmus. It is normally 12 to 20 g in size, highly vascular, and soft in consistency. Four parathyroid glands, which produce parathyroid hormone ([Chap. 341](#)), are located in the posterior region of each pole of the thyroid. The recurrent laryngeal nerves traverse the lateral borders of the thyroid gland and must be identified during thyroid surgery to avoid vocal cord paralysis.

The thyroid gland develops from the floor of the primitive pharynx during the third week of gestation. The gland migrates from the foramen cecum, at the base of the tongue, along the thyroglossal duct to reach its final location in the neck. This feature accounts for the rare ectopic location of thyroid tissue at the base of the tongue (lingual thyroid), as well as for the presence of thyroglossal duct cysts along this developmental tract. Thyroid hormone synthesis normally begins at about 11 weeks' gestation.

The parathyroid glands migrate from the third (inferior glands) and fourth (superior glands) pharyngeal pouches before becoming embedded in the thyroid gland. Neural crest derivatives from the ultimobranchial body give rise to thyroid medullary C cells that produce calcitonin, a calcium-lowering hormone. The C cells are interspersed throughout the thyroid gland, although their density is greatest in the juncture of the upper one-third and lower two-thirds of the gland.

Thyroid gland development is controlled by a series of developmental transcription factors. Thyroid transcription factor (TTF) 1 (also known as NKX2A), TTF-2 (also known as FKHL15), and paired homeobox-8 (PAX-8) are expressed selectively, but not exclusively, in the thyroid gland. In combination, they orchestrate thyroid cell development and the induction of thyroid-specific genes such as thyroglobulin (Tg), thyroid peroxidase (TPO), the sodium iodide symporter (NIS), and the thyroid-stimulating hormone receptor (TSH-R). Mutations in these developmental transcription factors or their downstream target genes are rare causes of thyroid agenesis or dysmorphogenesis and can cause congenital hypothyroidism ([Table 330-1](#)). Congenital hypothyroidism is common enough (approximately 1 in 3000 to 4000 newborns) that neonatal screening is now performed in most industrialized countries (see below). Though the underlying causes of most cases of congenital hypothyroidism

are unknown, early treatment with thyroid hormone replacement precludes potentially severe developmental abnormalities.

The mature thyroid gland contains numerous follicles composed of thyroid follicular cells that surround secreted colloid, a proteinaceous fluid that contains large amounts of thyroglobulin, the protein precursor of thyroid hormones ([Fig. 330-2](#)). The thyroid follicular cells are polarized -- the basolateral surface is apposed to the bloodstream and an apical surface faces the follicular lumen. Increased demand for thyroid hormone, usually signaled by thyroid-stimulating hormone (TSH) binding to its receptor on the basolateral surface of the follicular cells, leads to [Tg](#) reabsorption from the follicular lumen and proteolysis within the cell to yield thyroid hormones for secretion into the bloodstream.

REGULATION OF THE THYROID AXIS

[TSH](#), secreted by the thyrotrope cells of the anterior pituitary, plays a pivotal role in control of the thyroid axis and serves as the most useful physiologic marker of thyroid hormone action. TSH is a 31-kDa hormone composed of α and β subunits; the α subunit is common to the other glycoprotein hormones [luteinizing hormone, follicle-stimulating hormone, human chorionic gonadotropin (hCG)], whereas the TSH β subunit is unique to TSH. The extent and nature of carbohydrate modification are modulated by thyrotropin-releasing hormone (TRH) stimulation and influence the biologic activity of the hormone. TSH has been produced recombinantly and is approved for use in the detection of residual thyroid cancer (see "Follow-up Whole-Body Scanning and Thyroglobulin Determinations," below).

The thyroid axis is a classic example of an endocrine feedback loop. Hypothalamic [TRH](#) stimulates pituitary production of [TSH](#), which, in turn, stimulates thyroid hormone synthesis and secretion. Thyroid hormones feed back negatively to inhibit TRH and TSH production ([Fig. 330-2](#)). The "set-point" in this axis is established by TSH, the level of which is a sensitive and specific marker of thyroid function. TRH is the major positive regulator of TSH synthesis and secretion. TRH acts through a seven-transmembrane G protein-coupled receptor (GPCR) that activates phospholipase C to generate phosphatidylinositol turnover and the release of intracellular calcium. Peak TSH secretion occurs ~15 min after administration of exogenous TRH. Dopamine, glucocorticoids, and somatostatin suppress TSH but are not of major physiologic importance except when these agents are administered in pharmacologic doses. Reduced levels of thyroid hormone increase basal TSH production and enhance TRH-mediated stimulation of TSH. High thyroid hormone levels rapidly and directly suppress TSH and inhibit TRH-mediated stimulation of TSH, indicating that thyroid hormones are the dominant regulator of TSH production. Like other pituitary hormones, TSH is released in a pulsatile manner and exhibits a diurnal rhythm; its highest levels occur at night. However, these TSH excursions are modest in comparison to those of other pituitary hormones, in part because TSH has a relatively long plasma half-life (50 min). Consequently, single measurements of TSH are adequate for assessing its circulating level. TSH is measured using immunoradiometric assays that are highly sensitive and specific. These assays are capable of distinguishing between normal and suppressed TSH values, thus allowing TSH to be used for the diagnosis of hyperthyroidism (low TSH) as well as hypothyroidism (high TSH).

THYROID HORMONE SYNTHESIS, METABOLISM, AND ACTION

THYROID HORMONE SYNTHESIS

Thyroid hormones are derived from **Tg**, a large iodinated glycoprotein. After secretion into the thyroid follicle, Tg is iodinated on selected tyrosine residues that are subsequently coupled via an ether linkage. Reuptake of Tg into the thyroid follicular cell allows proteolysis and the release of T₄ and T₃.

Iodine Metabolism and Transport Iodide uptake is a critical first step in thyroid hormone synthesis. Ingested iodine is bound to serum proteins, particularly albumin. Unbound iodine is excreted in the urine. The thyroid gland extracts iodine from the circulation in a highly efficient manner. For example, 10 to 25% of radioactive tracer (e.g., ¹²³I) is taken up by the normal thyroid gland over 24 h; this value can rise to 70 to 90% in Graves' disease.

Iodide uptake is mediated by the Na⁺/I⁻-symporter (NIS), which is expressed at the basolateral membrane of thyroid follicular cells. NIS is most highly expressed in the thyroid gland but is also expressed at low levels in the salivary glands, lactating breast, and placenta. The iodide transport mechanism is highly regulated, allowing adaptation to variations in dietary supply. Low iodine levels increase the amount of NIS and stimulate uptake, whereas high iodine levels suppress NIS expression and uptake. The selective expression of the NIS in the thyroid allows isotopic scanning, treatment of hyperthyroidism, and ablation of thyroid cancer with radioisotopes of iodine, without significant effects on other organs. Mutation of the *NIS* gene is a rare cause of congenital hypothyroidism, underscoring its importance in thyroid hormone synthesis.

Iodine deficiency is prevalent in many mountainous regions and in central Africa, central South America, and northern Asia. In areas of relative iodine deficiency, there is an increased prevalence of goiter and, when deficiency is severe, hypothyroidism and cretinism. *Cretinism* is characterized by mental and growth retardation and occurs when children who live in iodine-deficient regions are not treated with iodine or thyroid hormone to restore normal thyroid hormone levels during early childhood. These children are often born to mothers with iodine deficiency, suggesting that maternal thyroid hormone deficiency worsens the condition. Concomitant selenium deficiency may also contribute to the neurologic manifestations of cretinism. Iodine supplementation of salt, bread, and other food substances has markedly reduced the prevalence of cretinism. Unfortunately, however, iodine deficiency remains the most common cause of preventable mental deficiency, often because of resistance to the use of food additives or the cost of supplementation. In addition to overt cretinism, mild iodine deficiency can lead to subtle reduction of IQ. Iodine intake is assessed by determination of excretion in a 24-h urine collection. Oversupply of iodine, through supplements or foods enriched in iodine (e.g., shellfish, kelp), is associated with an increased incidence of autoimmune thyroid disease. The recommended average daily intake of iodine is 150 µg/d for adults, 90 to 120 µg/d for children, and 200 µg/d for pregnant women.

Organification, Coupling, Storage, Release After iodide enters the thyroid, it is

trapped and transported to the apical membrane of thyroid follicular cells where it is oxidized in an organification reaction that involves [TPO](#) and hydrogen peroxide. The reactive iodine atom is added to selected tyrosyl residues within [Tg](#), a large (660 kDa) dimeric protein consisting of 2769 amino acids. The iodotyrosines in Tg are then coupled via an ether linkage in a reaction that is also catalyzed by TPO. Either T₄ or T₃ can be produced by this reaction, depending on the number of iodine atoms present in the iodotyrosines. After coupling, Tg is taken back into the thyroid cell where it is processed in lysosomes to release T₄ and T₃. Uncoupled mono- and diiodotyrosines (MIT, DIT) are deiodinated by the enzyme dehalogenase, thereby recycling any iodide that is not converted into thyroid hormones.

Disorders of thyroid hormone synthesis are rare causes of congenital hypothyroidism. The vast majority of these disorders are due to recessive mutations in [TPO](#) or [Tg](#), but defects have also been identified in the [TSH-R](#), NIS, the pendrin anion transporter, hydrogen peroxide generation, and in dehalogenase. Because of the biosynthetic defect, the gland is incapable of synthesizing adequate amounts of hormone, leading to increased [TSH](#) and a large goiter.

TSH Action TSH regulates thyroid gland function through the [TSH-R](#), a seven-transmembrane [GPCR](#). The TSH-R is coupled to the α subunit of stimulatory G protein (G_{sa}) and activates adenylyl cyclase, leading to increased production of cyclic AMP. TSH also stimulates phosphatidylinositol turnover by activating phospholipase C. The functional role of the TSH-R has been underscored by naturally occurring mutations. Recessive loss-of-function mutations are a rare cause of thyroid hypoplasia and congenital hypothyroidism. Dominant gain-of-function mutations cause sporadic or familial nonautoimmune hyperthyroidism that is characterized by goiter, thyroid cell hyperplasia, and autonomous function. Most of these activating mutations involve in amino acid substitutions in the transmembrane domain of the receptor. They are thought to mimic conformational changes in the receptor similar to those induced by TSH binding or the interactions of thyroid-stimulating immunoglobulins (TSI) in Graves' disease. Activating TSH-R mutations also occur as somatic events and lead to clonal selection and expansion of the affected thyroid follicular cell (see below).

Factors that Influence Hormone Synthesis and Release [TSH](#) is the dominant hormonal regulator of thyroid gland growth and function. However, a variety of growth factors, most produced locally in the thyroid gland, also influence thyroid hormone synthesis. These include insulin-like growth factor I (IGF-I), epidermal growth factor, transforming growth factorb (TGF-b), endothelins, and various cytokines. The quantitative roles of these factors are not well understood, but they are important in selected disease states. In acromegaly, for example, increased levels of growth hormone and IGF-I are associated with goiter and predisposition to multinodular goiter. Certain cytokines and interleukins (ILs) produced in association with autoimmune thyroid disease induce thyroid growth, whereas others lead to apoptosis. As noted above, iodine is an important regulator of thyroid function. For example, iodine deficiency increases thyroid blood flow and stimulates uptake by the NIS. Excess iodide transiently inhibits thyroid iodide organification, a phenomenon known as the *Wolff-Chaikoff effect*. In individuals with a normal thyroid, the gland escapes from this inhibitory effect and iodide organification resumes; the suppressive action of high iodide may persist, however, in patients with underlying autoimmune thyroid disease.

THYROID HORMONE TRANSPORT AND METABOLISM

Serum Binding Proteins T₄ is secreted from the thyroid gland in at least 20-fold excess over T₃ ([Table 330-2](#)). Both hormones circulate bound to plasma proteins, including thyroxine-binding globulin (TBG), transthyretin (TTR, formerly known as thyroxine-binding prealbumin, or TBPA), and albumin. The functions of serum-binding proteins are to increase the pool of circulating hormone, delay hormone clearance, and perhaps to modulate hormone delivery to selected tissue sites. The concentration of TBG is relatively low (1 to 2 mg/dL), but because of its high affinity for thyroid hormones (T₄>T₃), it carries about 80% of the bound hormones. Albumin has relatively low affinity for thyroid hormones but has a high plasma concentration (~3.5 g/dL), and it binds up to 10% of T₄ and 30% of T₃. TTR also carries about 10% of T₄ but little T₃.

When the effects of the various binding proteins are combined, approximately 99.98% of T₄ and 99.7% of T₃ are protein-bound. Because T₃ is less tightly bound than T₄, the amount of free T₃ is greater than free T₄, even though there is less total T₃ in the circulation. The unbound, or free, concentrations of the hormones are ~2 × 10⁻¹¹M for T₄ and ~6 × 10⁻¹²M for T₃, which roughly correspond to the thyroid hormone receptor binding constants for these hormones (see below). Only the free hormone is biologically available to tissues. Therefore, homeostatic mechanisms that regulate the thyroid axis are directed towards maintenance of normal concentrations of free hormones.

Dysalbuminemic Hyperthyroxinemia A number of inherited and acquired abnormalities affect thyroid hormone binding proteins. X-linked [TBG](#) deficiency is associated with very low levels of total T₄ and T₃. However, because free hormone levels are normal, patients are euthyroid and [TSH](#) levels are normal. The importance of recognizing this disorder is to avoid efforts to normalize total T₄ levels, as this leads to thyrotoxicosis and is futile because of rapid hormone clearance in the absence of TBG. TBG levels are elevated by estrogen because of increased sialylation and delayed TBG clearance. Consequently, in women who are pregnant or taking estrogen-containing contraceptives, elevated TBG increases total T₄ and T₃ levels; however, free T₄ and T₃ levels are normal. Mutations in TBG, [TTR](#), and albumin that increase binding affinity for T₄ and/or T₃ cause disorders known as *euthyroid hyperthyroxinemia* or *familial dysalbuminemic hyperthyroxinemia* (FDH) ([Table 330-3](#)). These disorders are usually dominantly transmitted and result in increased total T₄ and/or T₃, but free hormone levels are normal. The familial nature of the disorders, and the fact that TSH levels are normal rather than suppressed, should suggest the diagnosis. Free hormone levels (ideally measured by dialysis) are normal in FDH. The diagnosis can be confirmed, if necessary, by using tests that measure the affinities of radiolabeled hormone binding to specific transport proteins or by performing DNA sequence analyses of the abnormal transport protein genes.

Certain medications, such as salicylates and salicylate, can displace thyroid hormones from circulating binding proteins. Though these drugs transiently perturb the thyroid axis by increasing free thyroid hormone levels, [TSH](#) is suppressed until a new steady state is reached, thereby restoring euthyroidism. Circulating factors associated with acute illness may also displace thyroid hormone from binding proteins (see "Sick Euthyroid Syndrome," below).

Deiodinases In many respects, T_4 may be thought of as a precursor for the more potent T_3 . T_4 is converted to T_3 by the deiodinase enzymes ([Fig. 330-1](#)). Type I deiodinase, which is located primarily in thyroid, liver, and kidney, has a relatively low affinity for T_4 . Type II deiodinase has a higher affinity for T_4 and is found primarily in the pituitary gland, brain, brown fat, and thyroid gland. The presence of type II deiodinase allows it to regulate T_3 concentrations locally, a property that may be important in the context of levothyroxine (T_4) replacement. Type II deiodinase is also regulated by thyroid hormone -- hypothyroidism induces the enzyme, resulting in enhanced $T_4 \rightarrow T_3$ conversion in tissues such as brain and pituitary. $T_4 \rightarrow T_3$ conversion may be impaired by fasting, systemic illness or acute trauma, oral contrast agents, and a variety of medications (e.g., propylthiouracil, propranolol, amiodarone, glucocorticoids). Type III deiodinase inactivates T_4 and T_3 and is the most important source of reverse T_3 (rT_3).

THYROID HORMONE ACTION

Nuclear Thyroid Hormone Receptors Thyroid hormones act by binding to nuclear receptors, termed *thyroid hormone receptors* (TRs) α and β . Both TR α and TR β are expressed in most tissues, but their relative levels of expression vary among organs; TR α is particularly abundant in brain, kidney, gonads, muscle, and heart, whereas TR β expression is relatively high in the pituitary and liver. Both receptors are variably spliced to form unique isoforms. The TR β 2 isoform, which has a unique amino terminus, is selectively expressed in the hypothalamus and pituitary, where it appears to play a role in feedback control of the thyroid axis. The TR α 2 isoform contains a unique carboxy terminus that prevents thyroid hormone binding; it may function to block the action of other TR isoforms.

The TRs contain a central DNA-binding domain and a C-terminal ligand-binding domain. They bind to specific DNA sequences, termed *thyroid response elements* (TREs), in the promoter regions of target genes ([Fig. 330-3](#)). The activated receptor can either stimulate gene transcription (e.g., myosin heavy chain α) or inhibit transcription (e.g., TSH β -subunit gene), depending on the nature of the regulatory elements in the target gene. The receptors bind as homodimers or as heterodimers with retinoic acid X receptors (RXRs) ([Chap. 327](#)).

Thyroid hormones bind with similar affinities to TR α and TR β . However, T_3 is bound to its receptors with about 10 to 15 times greater affinity than T_4 , which explains its increased hormonal potency. Though T_4 is produced in excess of T_3 , receptors are occupied mainly by T_3 , reflecting $T_4 \rightarrow T_3$ conversion by peripheral tissues, greater T_3 bioavailability in the plasma, and receptors' greater affinity for T_3 . After binding to TRs, thyroid hormone induces conformational changes in the receptors that modify its interactions with accessory transcription factors. In the absence of thyroid hormone binding, the aporeceptors bind to corepressor proteins that inhibit gene transcription. Hormone binding dissociates the corepressors and allows the recruitment of coactivators that enhance transcription. The discovery of TR interactions with corepressors explains the fact that TR silences gene expression in the absence of hormone binding. Consequently, hormone deficiency has a profound effect on gene expression because it causes active gene repression as well as loss of hormone-induced stimulation. This concept has been corroborated by the finding that targeted deletion of the TR genes in

mice has a less pronounced phenotypic effect than hormone deficiency.

Thyroid Hormone Resistance Resistance to thyroid hormone (RTH) is an autosomal dominant disorder characterized by elevated free thyroid hormone levels and inappropriately normal or elevated [TSH](#). Individuals with RTH do not, in general, exhibit signs and symptoms that are typical of hypothyroidism, apparently because hormone resistance is compensated by increased levels of thyroid hormone. The clinical features of RTH can include goiter, attention deficit disorder, mild reduction in IQ, delayed skeletal maturation, tachycardia, and impaired metabolic responses to thyroid hormone.

The disorder is caused by mutations in the [TR](#) receptor gene. These mutations, located in restricted regions of the ligand-binding domain, cause loss of receptor function. However, because the mutant receptors retain the capacity to dimerize with [RXRs](#), bind to DNA, and recruit corepressor proteins, they function as antagonists of the remaining, normal TR β and TR α receptors. This property, referred to as "dominant negative" activity, explains the autosomal dominant mode of transmission. The diagnosis is suspected when free thyroid hormone levels are increased without suppression of [TSH](#). Similar hormonal abnormalities are common in other affected family members, though the TR β mutation arises de novo in about 20% of patients. DNA sequence analysis of the TR β gene provides a definitive diagnosis. [RTH](#) must be distinguished from other causes of euthyroid hyperthyroxinemia (e.g., familial dysalbuminemic hyperthyroxinemia) and inappropriate secretion of TSH by TSH-secreting pituitary adenomas ([Chap. 328](#)). In most patients, no treatment is indicated; the importance of making the diagnosis is to avoid inappropriate treatment of mistaken hyperthyroidism and to provide genetic counseling.

PHYSICAL EXAMINATION

In addition to the examination of the thyroid itself, the physical examination should include a search for signs of abnormal thyroid function and the extrathyroidal features of ophthalmopathy and dermopathy (see below). Examination of the neck begins by inspecting the seated patient from the front and side, and noting any surgical scars, obvious masses, or distended veins. The thyroid can be palpated with both hands from behind or the examiner can face the patient, using the thumbs to palpate each lobe. Most often it is best to use a combination of these methods, especially in cases of doubt or when there are small nodules. The patient's neck should be slightly flexed to relax the neck muscles. After locating the cricoid cartilage, the isthmus can be identified and followed laterally to locate either lobe (the right lobe is normally slightly larger than the left). By asking the patient to swallow sips of water, thyroid consistency can be better appreciated as the gland moves beneath the examiner's fingers.

Features to be noted include thyroid size, consistency, nodularity, and any tenderness or fixation. An estimate of thyroid size (normally 12 to 20 g) should be made, and a drawing is often the best way to record findings. However, ultrasound is the method of choice when it is important to determine thyroid size accurately. The size, location, and consistency of any nodules should also be depicted. A bruit over the gland indicates increased vascularity, as occurs in hyperthyroidism. If the lower borders of the thyroid lobes are not clearly felt, a goiter may be retrosternal. Large retrosternal goiters can cause venous distention over the neck and difficulty breathing, especially when the

arms are raised (Pemberton's sign). With any central mass above the thyroid, the patient should be asked to stick out his or her tongue, as thyroglossal cysts then move upward. The thyroid examination is not complete without assessment for lymphadenopathy in the supraclavicular and cervical regions of the neck.

LABORATORY EVALUATION

MEASUREMENT OF THYROID HORMONES

The enhanced sensitivity and specificity of *TSH* assays have greatly improved laboratory assessment of thyroid function. Because *TSH* levels change dynamically in response to alterations of free T_4 and T_3 , a logical approach to thyroid testing is to determine first whether *TSH* is suppressed, normal, or elevated. With rare exceptions (see below), a normal *TSH* level excludes a primary abnormality of thyroid function. This strategy depends on the use of immunoradiometric assays (IRMAs) for *TSH* that are sensitive enough to discriminate between the lower limit of the reference range and the suppressed values that occur with thyrotoxicosis. Extremely sensitive (fourth generation) assays can detect *TSH* levels ≤ 0.004 mU/L, but for practical purposes assays sensitive to ≤ 0.1 mU/L are sufficient. The widespread availability of the *TSH* IRMA has rendered the *TRH* stimulation test virtually obsolete, as the failure of *TSH* to rise after an intravenous bolus of 200 to 400 μ g *TRH* has the same implications as a suppressed basal *TSH* measured by IRMA.

The finding of an abnormal *TSH* level must be followed by measurements of circulating thyroid hormone levels to confirm the diagnosis of hyperthyroidism (suppressed *TSH*) or hypothyroidism (elevated *TSH*). Radioimmunoassays are widely available for serum *total T₄* and *total T₃*. T_4 and T_3 are highly protein-bound, and numerous factors (illness, medications, genetic factors) can influence protein binding. It is useful, therefore, to measure the free or unbound hormone levels, which correspond to the biologically available hormone pool. Two direct methods are used to measure *free thyroid hormones*: (1) free thyroid hormone competition with radiolabeled T_4 (or an analogue) for binding to a solid-phase antibody, and (2) physical separation of the free hormone fraction by ultracentrifugation or equilibrium dialysis. Though early free hormone immunoassays suffered from artifacts, newer assays agree well with the results of the more technically demanding and expensive physical separation methods. An indirect method to estimate free thyroid hormone levels is to calculate the free T_3 or free T_4 index from the total T_4 or T_3 concentration and the *thyroid hormone binding ratio* (THBR). The latter is derived from the *T₃-resin uptake test*, which determines the distribution of radiolabeled T_3 between an absorbent resin and the unoccupied thyroid hormone binding proteins in the sample. The binding of the labeled T_3 to the resin is increased when there is reduced unoccupied protein binding sites (e.g., *TBG* deficiency) or increased total thyroid hormone in the sample; it is decreased under the opposite circumstances. The product of THBR and total T_3 or T_4 provides the *free T₃ or T₄ index*. In effect, the index corrects for anomalous total hormone values caused by abnormalities in hormone-protein binding.

Total thyroid hormone levels are elevated when *TBG* is increased due to estrogens (pregnancy, oral contraceptives, hormone replacement therapy, tamoxifen), and decreased when *TBG* binding is decreased (androgens, the nephrotic syndrome).

Genetic disorders and acute illness can also cause abnormalities in thyroid hormone binding proteins, and various drugs (phenytoin, carbamazepine, salicylates, and nonsteroidal anti-inflammatory drugs) can interfere with thyroid hormone binding. Because free thyroid hormone levels are normal and the patient is euthyroid in all of these circumstances, assays that measure free hormone are preferable to those for total thyroid hormones.

For most purposes, the free T_4 level is sufficient to confirm thyrotoxicosis, but 2 to 5% of patients have only an elevated T_3 level (T_3 toxicosis). Thus, free T_3 levels should be measured in patients with a suppressed [TSH](#) but normal free T_4 levels. Free T_3 levels are normal in about 25% of patients with hypothyroidism and provide little useful information in this setting.

There are several clinical conditions in which the use of [TSH](#) as a screening test may be misleading, particularly without simultaneous free T_4 determinations. Any severe nonthyroidal illness can cause abnormal TSH levels (see below). Although hypothyroidism is the most common cause of an elevated TSH level, rare causes include a TSH-secreting pituitary tumor ([Chap. 328](#)), thyroid hormone resistance, and assay artifact. Conversely, a suppressed TSH level, particularly <0.1 mU/L, usually indicates thyrotoxicosis but may also be seen during the first trimester of pregnancy (due to hCG secretion), after treatment of hyperthyroidism (because TSH remains suppressed for several weeks), and in response to certain medications (e.g., high doses of glucocorticoids or dopamine). Importantly, secondary hypothyroidism, caused by hypothalamic-pituitary disease, is associated with a variable (low to high-normal) TSH level, which is inappropriate for the low free T_4 level. Thus, *TSH should not be used to assess thyroid function in patients with suspected or known pituitary disease.*

Tests for the end-organ effects of thyroid hormone excess or depletion, such as estimation of basal metabolic rate, tendon reflex speed, or serum cholesterol, are not useful as clinical determinants of thyroid function.

TESTS TO DETERMINE THE ETIOLOGY OF THYROID DYSFUNCTION

Autoimmune thyroid disease is detected most easily by measuring circulating antibodies against [TPO](#) and [Tg](#). As antibodies to Tg alone are rare, it is reasonable to measure only TPO antibodies. About 5 to 15% of euthyroid women and up to 2% of euthyroid men have thyroid antibodies; such individuals are at increased risk of developing thyroid dysfunction. Almost all patients with autoimmune hypothyroidism, and up to 80% of those with Graves' disease, have TPO antibodies, usually at high levels.

[TSI](#) are antibodies that stimulate the [TSH-R](#) in Graves' disease. They can be measured in bioassays or indirectly in assays that detect antibody binding to the receptor. The main use of these assays is to predict neonatal thyrotoxicosis caused by high maternal levels of TSI in the last trimester of pregnancy.

Serum Tg levels are increased in all types of thyrotoxicosis except thyrotoxicosis factitia. The main role for [Tg](#) measurement, however, is in the follow-up of thyroid cancer patients. After total thyroidectomy and radioablation, Tg levels should be undetectable; measurable levels (>1 to 2 ng/mL) suggest incomplete ablation or recurrent cancer.

RADIOIODINE UPTAKE AND THYROID SCANNING

The thyroid gland selectively transports radioisotopes of iodine (^{123}I , ^{125}I , ^{131}I) and $^{99\text{m}}\text{Tc}$ pertechnetate, allowing thyroid imaging and quantitation of radioactive tracer fractional uptake.

Graves' disease is characterized by an enlarged gland and increased tracer uptake that is distributed homogeneously. Toxic adenomas appear as focal areas of increased uptake, with suppressed tracer uptake in the remainder of the gland. In toxic multinodular goiter, the gland is enlarged -- often with distorted architecture -- and there are multiple areas of relatively increased or decreased tracer uptake. Subacute thyroiditis is associated with very low uptake because of follicular cell damage and [TSH](#) suppression. *Thyrotoxicosis factitia*, caused by self-administration of thyroid hormone, is also associated with low uptake.

Although the use of fine-needle aspiration (FNA) biopsy has diminished the use of thyroid scans in the evaluation of solitary thyroid nodules, the functional features of thyroid nodules have some prognostic significance. So-called cold nodules, which have diminished tracer uptake, are usually benign. However, these nodules are more likely to be malignant (~5 to 10%) than so-called hot nodules, which are almost never malignant.

Thyroid scanning is also used in the follow-up of thyroid cancer. After thyroidectomy and ablation using ^{131}I , there is diminished radioiodine uptake in the thyroid bed, allowing the detection of metastatic thyroid cancer deposits that retain the ability to transport iodine. Whole-body scans using 111 to 185 MBq (3 to 5 mCi) ^{131}I are typically performed after thyroid hormone withdrawal to raise the [TSH](#) level or after the administration of recombinant human TSH.

THYROID ULTRASOUND

Ultrasonography is used increasingly to assist in the diagnosis of nodular thyroid disease, a reflection of the limitations of the physical examination and improvements in ultrasound technology. Using 10-MHz instruments, spatial resolution and image quality are excellent, allowing the detection of nodules and cysts >3 mm. In addition to detecting thyroid nodules, ultrasound is useful for monitoring nodule size, for guiding [FNA](#) biopsies, and for the aspiration of cystic lesions. Ultrasound is also used in the evaluation of recurrent thyroid cancer, including possible spread to cervical lymph nodes.

AUTOIMMUNE BASIS OF THYROID DISEASE

PREVALENCE

Thyroid autoimmunity can cause several forms of thyroiditis and may lead to hypothyroidism as well as Graves' disease. Focal thyroiditis is present in 20 to 40% of autopsy cases and is associated with serologic evidence of autoimmunity, particularly the presence of [TPO](#) antibodies. These antibodies are 4 to 10 times more common in otherwise healthy women than men. About 5% of women experience self-limited *postpartum (silent) thyroiditis* in the months after pregnancy, often with transient clinical

symptoms. This condition is associated with the presence of TPO antibodies ante-partum. Up to 20% of women with an episode of postpartum thyroiditis develop permanent hypothyroidism 5 to 10 years after delivery. Autoimmune-mediated hypothyroidism affects about 5 to 10% of middle-aged and elderly women, depending on diagnostic criteria and geographic location. Graves' disease is about one-tenth as common as hypothyroidism and tends to occur in younger individuals. Although seemingly diverse, these disorders have many pathophysiologic features in common, and patients may progress from one state to the other as the autoimmune process changes.

SUSCEPTIBILITY FACTORS

As with most autoimmune disorders, susceptibility is determined by a combination of genetic and environmental factors. The concordance rate for Graves' disease in monozygotic twins is 20 to 30%. The risk of autoimmune thyroid disease is increased among siblings, who may exhibit features of either Graves' disease or autoimmune hypothyroidism. The autoimmune polyglandular syndrome type 2 ([Chap. 339](#)) involves the occurrence of autoimmune thyroid dysfunction with other autoimmune diseases (type 1 diabetes mellitus, Addison's disease, pernicious anemia, vitiligo). Shared genetic factors are likely in this group of autoimmune disorders.

HLA-DR3 is the best documented genetic risk factor for Graves' disease and autoimmune hypothyroidism in Caucasians, though different HLA associations exist for other racial groups, such as the Japanese and Chinese. A weak association with polymorphisms in the T cell regulatory gene CTLA-4 has been found in several racial groups. Other loci, including a region on chromosome 18q21, may be linked to Graves' disease as well as to several other autoimmune disorders such as type 1 diabetes mellitus, rheumatoid arthritis, and systemic lupus erythematosus (SLE). The female preponderance of thyroid autoimmunity is most likely due to the influence of sex steroids. Some studies suggest an association between antecedent major life events and Graves' disease, but a causal role for stress in the autoimmune process remains to be clearly established. Smoking is a minor risk factor for Graves' disease but a major risk factor for the development of ophthalmopathy. There is no convincing evidence for a role of infection in susceptibility, except for the congenital rubella syndrome, which is associated with a high frequency of autoimmune hypothyroidism. Viral thyroiditis does not induce subsequent autoimmune thyroid disease.

HUMORAL FACTORS

The thyrotoxicosis of Graves' disease is caused by [TSH-R](#)-stimulating immunoglobulins that bind to the receptor and mimic the action of [TSH](#). These [TSI](#) can cross the placenta and cause *transient neonatal thyrotoxicosis*, a phenomenon that complicates 1 to 2% of pregnancies in women with active or previous Graves' disease. However, the autoimmune response against the TSH-R can also result in antibodies that block TSH function, causing hypothyroidism. Stimulating and blocking antibodies bind to separate epitopes on the receptor. TSH-R blocking antibodies are found in about 20% of Asian patients with autoimmune hypothyroidism and are associated with thyroid atrophy; blocking antibodies are less common in Caucasians. Patients may have a mixture of TSH-R antibodies, and thyroid function can oscillate between hyperthyroidism and

hypothyroidism as stimulating or blocking antibodies become dominant. Predicting the course of disease in such individuals is difficult, and close monitoring of thyroid function is required. Assays that measure the binding of antibodies to the receptor by competition with radiolabeled TSH [TSH-binding inhibiting immunoglobulins (TBII)] provide no information about functional effects and are used primarily to demonstrate the presence of TSH-R antibodies in atypical patients. Bioassays measure antibody-mediated stimulation of cyclic AMP production in cultured thyroid cells or cells transfected with the TSH-R. The use of these assays does not generally alter clinical management.

Antibodies to [Tg](#) and [TPO](#), readily measured by immunofluorescence, hemagglutination, enzyme-linked immunosorbent assay, or radioimmunoassay, are clinically useful markers of thyroid autoimmunity, as discussed above. Any pathogenic effect is likely to be restricted to a secondary role in amplifying an ongoing autoimmune response. For instance, T cell- or cytokine-mediated injury to thyroid follicles could expose the enzyme on the apical border of follicles to TPO antibodies, which may then bind to the autoantigen and fix complement. There is evidence for intrathyroidal complement activation in both Graves' disease and autoimmune hypothyroidism. Tg antibodies do not fix complement, but could be involved in antibody-dependent, natural killer cell-mediated cytotoxicity. The NIS is also a target of autoantibody production in up to one-third of patients with autoimmune thyroid disease, but the functional consequences, if any, have not been established.

CELL-MEDIATED FACTORS

Activated circulating T cells are increased in autoimmune thyroid disease, and the gland is infiltrated with CD4+ and CD8+ T cells. The latter are believed to mediate perforin-dependent cytotoxicity, leading ultimately to thyroid cell destruction. In addition, thyroid cells undergo apoptosis through cytokine-mediated upregulation of Fas and possibly Fas ligand. Cytokines produced by the infiltrating immune cells also induce expression of thyroid cell-surface molecules that lead to: (1) engagement by immune cells (e.g., adhesion molecules, HLA class I and II molecules); (2) induction of cytokine secretion by the thyroid cells themselves; (3) production of nitric oxide; and (4) reduction of thyroid hormone production through inhibition of [TSH-R](#), [TPO](#), and [Tg](#) synthesis. Administration of high concentrations of cytokines for therapeutic purposes [especially interferon (IFN) α] is associated with increased autoimmune thyroid disease, presumably via mechanisms similar to those that occur in sporadic autoimmune disease.

Cytokines appear to play a major role in thyroid-associated ophthalmopathy. There is infiltration of the extraocular muscles by activated T cells; the release of cytokines results in fibroblast activation and increased synthesis of glycosaminoglycans that trap water, thereby leading to characteristic muscle swelling. Late in the disease, there is fibrosis and only then do the muscle cells show evidence of injury. Orbital fibroblasts may be uniquely sensitive to cytokines, perhaps explaining the anatomic localization of the immune response. Though the pathogenesis of thyroid-associated ophthalmopathy remains unclear, there is mounting evidence that expression of the [TSH-R](#) may provide an important orbital autoantigen. In support of this idea, injection of TSH-R into certain strains of mice induces autoimmune hyperthyroidism, as well as features of ophthalmopathy. A variety of autoantibodies against orbital muscle and fibroblast

antigens have been detected in patients with ophthalmopathy, but these antibodies most likely arise as a secondary phenomenon, dependent on T cell-mediated autoimmune responses.

HYPOTHYROIDISM

Iodine deficiency remains the most common cause of hypothyroidism worldwide. In areas of iodine sufficiency, autoimmune disease (Hashimoto's thyroiditis) and iatrogenic causes (treatment of hyperthyroidism) are most common ([Table 330-4](#)).

CONGENITAL HYPOTHYROIDISM

Prevalence Hypothyroidism occurs in about 1 in 3000 to 4000 newborns. It may be transient, especially if the mother has [TSH-R](#) blocking antibodies or has received antithyroid drugs, but permanent hypothyroidism occurs in the majority. Neonatal hypothyroidism is due to thyroid gland dysgenesis in 85%, inborn errors of thyroid hormone synthesis in 10 to 15%, and is TSH-R antibody-mediated in 5% of affected newborns. The developmental abnormalities are twice as common in girls. Mutations that cause congenital hypothyroidism are being increasingly recognized, but the vast majority remain idiopathic ([Table 330-1](#)).

Clinical Manifestations The majority of infants appear normal at birth, and <10% are diagnosed based on clinical features, which include prolonged jaundice, feeding problems, hypotonia, enlarged tongue, delayed bone maturation, and umbilical hernia. Importantly, permanent neurologic damage results if treatment is delayed. Typical features of adult hypothyroidism may also be present ([Table 330-5](#)).

Diagnosis and Treatment Because of the severe neurologic consequences of untreated congenital hypothyroidism, neonatal screening programs have been established in developed countries ([Chap. 68](#)). These are generally based on measurement of [TSH](#) or T_4 levels in heel-prick blood specimens. When the diagnosis is confirmed, T_4 is instituted at a dose of 10 to 15 $\mu\text{g/kg}$ per day and the dosage is adjusted by close monitoring of TSH levels. T_4 requirements are relatively great during the first year of life, and a high circulating T_4 level is usually needed to normalize TSH. Early treatment with T_4 results in normal IQ levels, but subtle neurodevelopmental abnormalities may be detected in those with the most severe hypothyroidism at diagnosis or when treatment is suboptimal.

AUTOIMMUNE HYPOTHYROIDISM

Classification Autoimmune hypothyroidism may be associated with a goiter (Hashimoto's, or *goitrous thyroiditis*) or, at the later stages of the disease, minimal residual thyroid tissue (*atrophic thyroiditis*). Because the autoimmune process gradually reduces thyroid function, there is a phase of compensation during which normal thyroid hormone levels are maintained by a rise in [TSH](#). Though some patients may have minor symptoms, this state is called *subclinical hypothyroidism*. Later, free T_4 levels fall and TSH levels rise further; symptoms become more readily apparent at this stage (usually $\text{TSH} > 10 \text{ mU/L}$), which is referred to as *clinical hypothyroidism (overt hypothyroidism)*.

Prevalence The mean annual incidence rate of autoimmune hypothyroidism is up to 4 per 1000 women and 1 per 1000 men. It is more common in certain populations, such as the Japanese, probably as a consequence of genetic factors and chronic exposure to a high-iodine diet. The mean age at diagnosis is about 60 years, and the prevalence of overt hypothyroidism increases with age. Subclinical hypothyroidism is found in 6 to 8% of women (10% over the age of 60) and 3% of men. The annual risk of developing clinical hypothyroidism is about 4% when subclinical hypothyroidism is associated with positive [TPO](#) antibodies.

Pathogenesis In Hashimoto's thyroiditis, there is a marked lymphocytic infiltration of the thyroid with germinal center formation, atrophy of the thyroid follicles accompanied by oxyphil metaplasia, absence of colloid, and mild to moderate fibrosis. In atrophic thyroiditis, the fibrosis is much more extensive, lymphocyte infiltration is less pronounced, and thyroid follicles are almost completely absent. Atrophic thyroiditis likely represents the end stage of Hashimoto's thyroiditis rather than a distinct disorder. Autoimmune features are similar in both types of hypothyroidism, though [TSH-R](#) blocking antibodies may be more frequent in Asian patients with atrophic thyroiditis. The mechanisms that result in thyroid follicular destruction are predominantly T cell mediated, but antibodies may also contribute to thyroid dysfunction by complement fixation or inhibition of thyroid cell function (see "Autoimmune Basis of Thyroid Disease," above).

Clinical Manifestations The main clinical features of hypothyroidism are summarized in [Table 330-5](#). The onset is usually insidious, and the patient may become aware of symptoms only when euthyroidism is restored. Patients with Hashimoto's thyroiditis may present because of goiter rather than symptoms of hypothyroidism. The goiter may not be large but is usually irregular and firm in consistency. It is often possible to palpate a pyramidal lobe, normally a vestigial remnant of thyroglossal duct. Rarely, uncomplicated Hashimoto's thyroiditis is associated with pain.

Patients with atrophic thyroiditis, or the late stage of Hashimoto's thyroiditis, present with symptoms and signs of hypothyroidism. The skin is dry, and there is decreased sweating, thinning of the epidermis, and hyperkeratosis of the stratum corneum. Increased dermal glycosaminoglycan content traps water, giving rise to skin thickening without pitting (*myxedema*). Typical features include a puffy face with edematous eyelids and nonpitting pretibial edema ([Figs. 330-4, 330-CD1](#) and [330-CD2](#)). There is pallor, often with a yellow tinge due to carotene accumulation. Nail growth is retarded, and hair is dry, brittle, difficult to manage, and falls out easily. In addition to diffuse alopecia, there is thinning of the outer third of the eyebrows.

Other common features include constipation and weight gain (despite a poor appetite). In contrast to popular perception, the weight gain is usually modest and due mainly to fluid retention in the myxedematous tissues. Libido is decreased in both sexes, and there may be oligomenorrhea or amenorrhea in long-standing disease, but menorrhagia is also common. Fertility is reduced and the incidence of miscarriage is increased. Prolactin levels are often modestly increased ([Chap. 328](#)) and may contribute to alterations in libido and fertility as well as causing galactorrhea.

Myocardial contractility and pulse rate are reduced, leading to a reduced stroke volume

and bradycardia. Increased peripheral resistance may be accompanied by hypertension, particularly diastolic. Blood flow is diverted from the skin, producing the cool extremities. Pericardial effusions occur in up to 30% of patients but rarely compromise cardiac function. Though alterations in myosin heavy chain isoform expression have been documented, cardiomyopathy is unusual. Fluid may also accumulate in other serous cavities and in the middle ear, giving rise to conductive deafness. Pulmonary function is generally normal, but dyspnea may be due to pleural effusion, impaired respiratory muscle function, diminished ventilatory drive, or sleep apnea.

Carpal tunnel and other entrapment syndromes are common, as is impairment of muscle function with stiffness, cramps, and pain. On examination, there may be slow relaxation of tendon reflexes ([Video 330-1](#)) and pseudomyotonia. Memory and concentration are impaired. Rare neurologic problems include reversible cerebellar ataxia, dementia, psychosis, and myxedema coma. *Hashimoto's encephalopathy* is a rare and distinctive syndrome associated with myoclonus and slow-wave activity on electroencephalography, which can progress to confusion, coma, and death. It is steroid-responsive and may occur in the presence of autoimmune thyroiditis, without hypothyroidism. The hoarse voice and occasionally clumsy speech of hypothyroidism are due to fluid accumulation in the vocal cords and tongue.

The features described above are due to a shortage of thyroid hormone. However, autoimmune hypothyroidism may be associated with signs or symptoms of other autoimmune diseases, particularly vitiligo, pernicious anemia, Addison's disease, alopecia areata, and type 1 diabetes mellitus. Less common associations include celiac disease, dermatitis herpetiformis, chronic active hepatitis, rheumatoid arthritis, [SLE](#), and Sjogren's syndrome. Thyroid-associated ophthalmopathy, which usually occurs in Graves' disease (see below), occurs in about 5% of patients with autoimmune hypothyroidism.

Autoimmune hypothyroidism is uncommon in children and usually presents with slow growth and delayed facial maturation. The appearance of permanent teeth is also delayed. Myopathy, with muscle swelling, is more common than in adults. In most cases, puberty is delayed, but precocious puberty sometimes occurs. There may be intellectual impairment if the onset is before 3 years and the hormone deficiency is severe.

Laboratory Evaluation A summary of the investigations used to determine the existence and cause of hypothyroidism is provided in [Fig. 330-5](#). A normal [TSH](#) level excludes primary (but not secondary) hypothyroidism. If the TSH is elevated, a free T_4 level is needed to confirm the presence of clinical hypothyroidism, but free T_4 is inferior to TSH when used as a screening test, as it will not detect subclinical or mild hypothyroidism. Circulating free T_3 levels are normal in about 25% of patients, reflecting adaptive responses to hypothyroidism. T_3 measurements are therefore not indicated.

Once clinical or subclinical hypothyroidism is confirmed, the etiology is usually easily established by demonstrating the presence of [TPO](#) antibodies, which are present in 90 to 95% of patients with autoimmune hypothyroidism. [TBI](#) can be found in 10 to 20% of patients, but these determinations are not needed routinely. If there is any doubt about

the cause of a goiter associated with hypothyroidism, [FNA](#) biopsy can be used to confirm the presence of autoimmune thyroiditis. Other abnormal laboratory findings in hypothyroidism may include increased creatine phosphokinase, elevated cholesterol and triglycerides, and anemia (usually normocytic or macrocytic). Except when accompanied by iron deficiency, the anemia and other abnormalities gradually resolve with thyroxine replacement.

Differential Diagnosis An asymmetric goiter in Hashimoto's thyroiditis may be confused with a multinodular goiter or thyroid carcinoma, even when thyroid antibodies are present. Ultrasound can be used to show the presence of a solitary lesion or a multinodular goiter, rather than the heterogeneous thyroid enlargement typical of Hashimoto's thyroiditis. [FNA](#) biopsy is useful in the investigation of focal nodules. Other causes of hypothyroidism are discussed below but rarely cause diagnostic confusion ([Table 330-4](#)).

OTHER CAUSES OF HYPOTHYROIDISM

Iatrogenic hypothyroidism is a common cause of hypothyroidism and can often be detected by screening before symptoms develop. In the first 3 to 4 months after radioiodine treatment, transient hypothyroidism may occur due to reversible radiation damage rather than to cellular destruction. Low-dose thyroxine treatment can be withdrawn if recovery occurs. Because [TSH](#) levels are suppressed by hyperthyroidism, free T_4 levels are a better measure of thyroid function than TSH in the months following radioiodine treatment. Mild hypothyroidism after subtotal thyroidectomy may also resolve after several months, as the gland remnant is stimulated by increased TSH levels.

Iodine deficiency is responsible for endemic goiter and cretinism but is an uncommon cause of adult hypothyroidism unless the iodine intake is very low or there are complicating factors, such as the consumption of thiocyanates in cassava or selenium deficiency. Though hypothyroidism due to iodine deficiency can be treated with thyroxine, public health measures to improve iodine intake should be advocated to eliminate this problem. Iodized salt or bread or the use of a single bolus of oral or intramuscular iodized oil have all been used successfully.

Paradoxically, chronic iodine excess can also induce goiter and hypothyroidism. The intracellular events that account for this effect are unclear, but individuals with autoimmune thyroiditis are especially susceptible. Iodine excess is responsible for the hypothyroidism that occurs in up to 13% of patients treated with amiodarone (see below). Other drugs, particularly lithium, may also cause hypothyroidism.

Secondary hypothyroidism is usually diagnosed in the context of other anterior pituitary hormone deficiencies; isolated [TSH](#) deficiency is very rare ([Chap. 328](#)). TSH levels may be low, normal, or even slightly increased in secondary hypothyroidism; the latter is due to secretion of immunoactive but bioinactive forms of TSH. The diagnosis is confirmed by detecting a low free T_4 level. The goal of treatment is to maintain free T_4 levels in the upper half of the reference range, as TSH levels cannot be used to monitor therapy.

TREATMENT

Clinical Hypothyroidism If there is no residual thyroid function, the daily replacement dose of levothyroxine is usually 1.5 ug/kg body weight (typically 100 to 150 ug). In many patients, however, lower doses suffice until residual thyroid tissue is destroyed. In patients who develop hypothyroidism after the treatment of Graves' disease, there is often underlying autonomous function, necessitating lower replacement doses (typically 75 to 125 ug/d).

Adult patients under 60 without evidence of heart disease may be started on 50 to 100ug levothyroxine (T₄) daily. The dose is adjusted on the basis of [TSH](#) levels, with the goal of treatment being a normal TSH, ideally in the lower half of the reference range. TSH responses are gradual and should be measured about 2 months after instituting treatment or after any subsequent change in levothyroxine dosage. The clinical effects of levothyroxine replacement are often slow to appear. Patients may not experience full relief from symptoms until 3 to 6 months after normal TSH levels are restored. Adjustment of levothyroxine dosage is made in 12.5- or 25-ug increments if the TSH is high; decrements of the same magnitude should be made if the TSH is suppressed. Patients with a suppressed TSH of any cause, including T₄ overtreatment, have an increased risk of atrial fibrillation and reduced bone density.

Although dessicated animal thyroid preparations (thyroid extract USP) are available, they are not recommended as potency and composition vary between batches. Interest in using levothyroxine combined with liothyronine (triiodothyronine, T₃) has been revived, based on studies suggesting that patients feel better when taking the T₄/T₃ combination compared to T₄ alone. However, a long-term benefit from this combination is not established. There is no place for liothyronine alone as long-term replacement, because the short half-life necessitates three or four daily doses and is associated with fluctuating T₃ levels.

Once full replacement is achieved and [TSH](#) levels are stable, follow-up measurement of TSH is recommended at annual intervals and may be extended to every 2 to 3 years, if a normal TSH is maintained over several years. It is important to ensure ongoing compliance, however, as patients do not feel any difference after missing a few doses of levothyroxine, sometimes leading to self-discontinuation.

In patients of normal body weight who are taking 3200 ug of levothyroxine per day, an elevated [TSH](#) level is often a sign of poor compliance. This is also the likely explanation for fluctuating TSH levels, despite a constant levothyroxine dosage. Such patients often have normal or high free T₄ levels, despite an elevated TSH, because they remember to take medication for a few days before testing; this is sufficient to normalize T₄ but not TSH levels. It is important to consider variable compliance, as this pattern of thyroid function tests is otherwise suggestive of disorders associated with inappropriate TSH secretion ([Table 330-3](#)). Because T₄ has a long half-life (7 days), patients who miss doses can be advised to take up to three doses of the skipped tablets at once. Other causes of increased levothyroxine requirements must be excluded, particularly malabsorption (e.g., celiac disease, small-bowel surgery) and drugs that interfere with T₄ absorption or clearance such as cholestyramine, ferrous sulfate, calcium supplements, lovastatin, aluminum hydroxide, rifampicin, amiodarone, carbamazepine, and phenytoin.

Subclinical Hypothyroidism By definition, subclinical hypothyroidism refers to biochemical evidence of thyroid hormone deficiency in patients who have few or no apparent clinical features of hypothyroidism. There are no generally accepted guidelines for the treatment of subclinical hypothyroidism. As long as excessive treatment is avoided, there is little risk in correcting a slightly increased [TSH](#), and some patients likely derive modest clinical benefit from treatment. Moreover, there is some risk that patients will progress to overt hypothyroidism, particularly when [TPO](#) antibodies are present. Treatment is administered by starting with a low dose of levothyroxine (25 to 50 ug/d) with the goal of normalizing TSH.

Special Treatment Considerations Rarely, levothyroxine replacement is associated with pseudotumor cerebri in *children*. Presentation appears to be idiosyncratic and occurs months after treatment is begun. Women with a history or high risk of hypothyroidism should ensure that they are euthyroid prior to conception and during early pregnancy as maternal hypothyroidism may adversely affect fetal neural development. [TSH](#) and free T₄ levels should be measured once pregnancy is confirmed and at the beginning of the second and third trimesters. The dose of levothyroxine may need to be increased by ³50% during pregnancy and returned to previous levels after delivery. In the *elderly*, especially patients with known coronary artery disease, the starting dose of levothyroxine is 12.5 to 25 ug/d with similar increments every 2 to 3 months until TSH is normalized. In some patients it may be impossible to achieve full replacement, despite optimal antianginal treatment. *Emergency surgery* is generally safe in patients with untreated hypothyroidism, although routine surgery in a hypothyroid patient should be deferred until euthyroidism is achieved.

Myxedema coma still has a high mortality rate, despite intensive treatment. Clinical manifestations include reduced level of consciousness, sometimes associated with seizures, as well as the other features of hypothyroidism ([Table 330-5](#)). Hypothermia can reach 23°C (74°F). There may be a history of treated hypothyroidism with poor compliance, or the patient may be previously undiagnosed. Myxedema coma almost always occurs in the elderly and is usually precipitated by factors that impair respiration, such as drugs (especially sedatives, anesthetics, antidepressants), pneumonia, congestive heart failure, myocardial infarction, gastrointestinal bleeding, or cerebrovascular accidents. Sepsis should also be suspected. Exposure to cold may also be a risk factor. Hypoventilation, leading to hypoxia and hypercapnia, plays a major role in pathogenesis; hypoglycemia and dilutional hyponatremia also contribute to the development of myxedema coma.

Levothyroxine can initially be administered as a single intravenous bolus of 500 ug, which serves as a loading dose. Although further levothyroxine is not strictly necessary for several days, it is usually continued at a dose of 50 to 100 ug/d. If a suitable intravenous preparation is not available, the same initial dose of levothyroxine can be given by nasogastric tube (though absorption may be impaired in myxedema). An alternative is to give liothyronine (T₃) intravenously or via nasogastric tube, in doses ranging from 10 to 25 ug every 8 to 12 h. This treatment has been advocated because T₄→T₃ conversion is impaired in myxedema coma. However, excess liothyronine has the potential to provoke arrhythmias. Another commonly used option is to combine levothyroxine (200 ug) and liothyronine (25 ug) as a single, initial intravenous bolus

followed by daily treatment with levothyroxine (50 to 100 ug/d) and liothyronine (10 ug every 8 h).

Supportive therapy should be provided to correct any associated metabolic disturbances. External warming is indicated only if the temperature is $<30^{\circ}\text{C}$, as it can result in cardiovascular collapse ([Chap. 17](#)). Space blankets should be used to prevent further heat loss. Parenteral hydrocortisone (50 mg every 6 h) should be administered, as there is impaired adrenal reserve in profound hypothyroidism. Any precipitating factors should be treated, including the early use of broad-spectrum antibiotics, pending the exclusion of infection. Ventilatory support with regular blood gas analysis is usually needed during the first 48 h. Hypertonic saline or intravenous glucose may be needed if there is hyponatremia or hypoglycemia; hypotonic intravenous fluids should be avoided because they may exacerbate water retention secondary to reduced renal perfusion and inappropriate vasopressin secretion. The metabolism of most medications is impaired, and sedatives should be avoided if possible or used in reduced doses. Blood levels should be monitored, when available, to guide medication dosage.

THYROTOXICOSIS

Thyrotoxicosis is defined as the state of thyroid hormone excess and is not synonymous with *hyperthyroidism*, which is the result of excessive thyroid function. However, the major etiologies of thyrotoxicosis are hyperthyroidism caused by Graves' disease, toxic multinodular goiter, and toxic adenomas. Other causes are listed in [Table 330-6](#).

GRAVES' DISEASE

Epidemiology Graves' disease accounts for 60 to 80% of thyrotoxicosis, though the prevalence varies among populations, depending mainly on iodine intake (high iodine intake is associated with an increased prevalence of Graves' disease). Graves' disease occurs in up to 2% of women but is one-tenth as frequent in men. The disorder rarely begins before adolescence and typically occurs between 20 and 50 years of age, though it also occurs in the elderly.

Pathogenesis The hyperthyroidism of Graves' disease is caused by [TSI](#) that are directed to the [TSH-R](#) (see "Autoimmune Basis of Thyroid Disease," below). Other thyroid autoimmune responses coexist in these patients, and therefore there is no direct correlation between the levels of TSI and thyroid hormones. The extrathyroidal manifestations of Graves' disease -- i.e., ophthalmopathy and dermopathy -- are due to immunologically mediated activation of fibroblasts in the extraocular muscles and skin, with accumulation of glycosaminoglycans, leading to the trapping of water and edema. Later, fibrosis becomes prominent. The fibroblast activation is caused by cytokines ([IFN- \$\gamma\$](#) , tumor necrosis factor, [IL-1](#)) derived from locally infiltrating T cells and macrophages.

Clinical Manifestations Signs and symptoms include features that are common to any cause of thyrotoxicosis ([Table 330-7](#)) as well as those specific for Graves' disease. The clinical presentation depends on the severity of thyrotoxicosis, the duration of the disease, individual susceptibility to excess thyroid hormone, and the age of the patient. In the elderly, features of thyrotoxicosis may be subtle or masked, and patients may

present mainly with fatigue and weight loss, leading to *apathetic hyperthyroidism*.

Thyrotoxicosis may cause unexplained weight loss, despite an enhanced appetite, and is due to the increased metabolic rate. Weight gain occurs in 5 to 10% of patients, however, as a result of increased food intake. Other prominent features include hyperactivity, nervousness, and irritability, ultimately leading to a sense of easy fatigability in some patients. Insomnia and impaired concentration are common; apathetic thyrotoxicosis may be mistaken for depression in the elderly. Fine tremor is a very frequent finding, best elicited by asking patients to stretch out the fingers and feeling the fingertips with the palm. Common neurologic manifestations include hyperreflexia, muscle wasting, and proximal myopathy without fasciculation. Chorea is a rare feature. Thyrotoxicosis is sometimes associated with a form of hypokalemic periodic paralysis; this disorder is particularly common in Asian males with thyrotoxicosis.

The most common cardiovascular manifestation is sinus tachycardia, often associated with palpitations and sometimes due to supraventricular tachycardia. The high cardiac output produces a bounding pulse, widened pulse pressure, and an aortic systolic murmur, and can lead to worsening of angina or heart failure in the elderly or those with preexisting heart disease. Atrial fibrillation is more common in patients >50. Treatment of the thyrotoxic state alone reverts atrial fibrillation to normal sinus rhythm in fewer than half of patients, suggesting the existence of an underlying cardiac problem in the remainder.

The skin is usually warm and moist, and the patient complains of sweating and heat intolerance, particularly during warm weather. Palmar erythema; onycholysis; and, less commonly, pruritus, urticaria, and diffuse hyperpigmentation may be evident. Hair texture may become fine, and a diffuse alopecia occurs in up to 40% of patients, persisting for months after restoration of euthyroidism. Gastrointestinal transit time is decreased, leading to increased stool frequency, often with diarrhea and occasionally mild steatorrhea. Women frequently experience oligomenorrhea or amenorrhea; in men there may be impaired sexual function and, rarely, gynecomastia. The direct effect of thyroid hormones on bone resorption leads to osteopenia in long-standing thyrotoxicosis; mild hypercalcemia occurs in up to 20% of patients, but hypercalcuria is more common. There is a small increase in fracture rate in patients with a previous history of thyrotoxicosis.

In Graves' disease the thyroid is usually diffusely enlarged to two to three times its normal size. The consistency is firm, but less so than in multinodular goiter. There may be a thrill or bruit due to the increased vascularity of the gland and the hyperdynamic circulation.

Lid retraction, causing a staring appearance, can occur in any form of thyrotoxicosis and is the result of sympathetic overactivity. However, Graves' disease is associated with specific eye signs that comprise *Graves' ophthalmopathy* ([Fig. 330-6A](#) and [330-CD2](#)). This condition is also called *thyroid-associated ophthalmopathy*, as it occurs in the absence of Graves' disease in 10% of patients. Most of these individuals have autoimmune hypothyroidism or thyroid antibodies. The onset of Graves' ophthalmopathy occurs within the year before or after the diagnosis of thyrotoxicosis in 75% of patients

but can sometimes precede or follow thyrotoxicosis by several years, accounting for some cases of euthyroid ophthalmopathy.

Many patients with Graves' disease have little clinical evidence of ophthalmopathy. However, the enlarged extraocular muscles typical of the disease, and other subtle features, can be detected in almost all patients when investigated by ultrasound or computed tomography (CT) imaging of the orbits. Unilateral signs are found in up to 10% of patients. The earliest manifestations of ophthalmopathy are usually a sensation of grittiness, eye discomfort, and excess tearing. About a third of patients have proptosis, best detected by visualization of the sclera between the lower border of the iris and the lower eyelid, with the eyes in the primary position. Proptosis can be measured using an exophthalmometer. In severe cases, proptosis may cause corneal exposure and damage, especially if the lids fail to close during sleep. Periorbital edema, scleral injection, and chemosis are also frequent. In 5 to 10% of patients, the muscle swelling is so severe that diplopia results, typically but not exclusively when the patient looks up and laterally. The most serious manifestation is compression of the optic nerve at the apex of the orbit, leading to papilledema, peripheral field defects, and, if left untreated, permanent loss of vision.

Many scoring systems have been used to gauge the extent and activity of the orbital changes in Graves' disease. The NO SPECS scheme is an acronym derived from the following classes of eye change:

0= No signs or symptoms

1 =Only signs (lid retraction or lag), no symptoms

2 =Soft tissue involvement (periorbital edema)

3 =Proptosis (>22 mm)

4 =Extraocular muscle involvement (diplopia)

5 =Corneal involvement

6= Sight loss

Although useful as a mnemonic, the NO SPECS scheme is inadequate to describe the eye disease fully, and patients do not necessarily progress from one class to another. When Graves' eye disease is active and severe, referral to an ophthalmologist is indicated and objective measurements are needed, such as lid fissure width; corneal staining with fluorescein; and evaluation of extraocular muscle function (e.g., Hess chart), intraocular pressure and visual fields, acuity, and color vision.

Thyroid dermopathy occurs in <5% of patients with Graves' disease ([Fig. 330-6B](#)), almost always in the presence of moderate or severe ophthalmopathy. Although most frequent over the anterior and lateral aspects of the lower leg (hence the term *pretibial myxedema*), skin changes can occur at other sites, particularly after trauma. The typical lesion is a noninflamed, indurated plaque with a deep pink or purple color and an

"orange-skin" appearance. Nodular involvement can occur, and the condition can rarely extend over the whole lower leg and foot, mimicking elephantiasis. *Thyroid acropachy* refers to a form of clubbing found in <1% of patients with Graves' disease (Fig. 330-6C). It is so strongly associated with thyroid dermopathy that an alternative cause of clubbing should be sought in a Graves' patient without coincident skin and orbital involvement.

Laboratory Evaluation Investigations used to determine the existence and cause of thyrotoxicosis are summarized in Fig. 330-7. In Graves' disease, the TSH level is suppressed and free and total thyroid hormone levels are increased. In 2 to 5% of patients (and more in areas of borderline iodine intake), only T₃ is increased (T₃toxicosis). The converse state of T₄toxicosis, with elevated total and free T₄ and normal T₃ levels, is occasionally seen when hyperthyroidism is induced by excess iodine, providing surplus substrate for thyroid hormone synthesis. Measurement of TPO antibodies is useful in differential diagnosis, but assays for TSH-R antibodies are not usually needed. Associated abnormalities that may cause diagnostic confusion in thyrotoxicosis include elevation of bilirubin, liver enzymes, and ferritin. Microcytic anemia and thrombocytopenia occur less often.

Differential Diagnosis Diagnosis of Graves' disease is straightforward in a patient with biochemically confirmed thyrotoxicosis, diffuse goiter on palpation, ophthalmopathy, positive TPO antibodies, and often a personal or family history of autoimmune disorders. For patients with thyrotoxicosis who lack these features, the most reliable diagnostic method is a radionuclide (^{99m}Tc, ¹²³I, or ¹³¹I) scan of the thyroid, which will distinguish the diffuse, high uptake of Graves' disease from nodular thyroid disease, destructive thyroiditis, ectopic thyroid tissue, and factitious thyrotoxicosis. In secondary hyperthyroidism due to a TSH-secreting pituitary tumor, there is also a diffuse goiter. The presence of a nonsuppressed TSH level, and the finding of a pituitary tumor on CT or magnetic resonance imaging (MRI) scan readily identify such patients.

Clinical features of thyrotoxicosis can mimic certain aspects of other disorders including panic attacks, mania, pheochromocytoma, and the weight loss associated with malignancy. The diagnosis of thyrotoxicosis can be easily excluded if the TSH level is normal. A normal TSH also excludes Graves' disease as a cause of diffuse goiter.

Clinical Course Clinical features generally worsen without treatment; mortality was 10 to 30% before the introduction of satisfactory therapy. Some patients with mild Graves' disease experience spontaneous relapses and remissions. Rarely, there may be fluctuation between hypo- and hyperthyroidism due to changes in the functional activity of TSH-R antibodies. About 15% of patients who enter remission after treatment with antithyroid drugs develop hypothyroidism 10 to 15 years later as a result of the destructive autoimmune process. The clinical course of ophthalmopathy does not follow that of the thyroid disease. Ophthalmopathy typically worsens over the initial 3 to 6 months, followed by a plateau phase over the next 12 to 18 months, with spontaneous improvement, particularly in the soft tissue changes. However, the course is more fulminant in up to 5% of patients, requiring intervention in the acute phase if there is optic nerve compression or corneal ulceration. Diplopia may appear late in the disease due to fibrosis of the extraocular muscles. Some studies suggest that radioiodine treatment for hyperthyroidism worsens the eye disease in a small proportion of patients (especially smokers). Antithyroid drugs or surgery have no adverse effects on the

clinical course of ophthalmopathy. Thyroid dermopathy, when it occurs, usually appears 1 to 2 years after the development of Graves' hyperthyroidism; it may improve spontaneously.

TREATMENT

The *hyperthyroidism* of Graves' disease is treated by reducing thyroid hormone synthesis, using antithyroid drugs, or by reducing the amount of thyroid tissue with radioiodine (^{131}I) treatment or subtotal thyroidectomy. Antithyroid drugs are the predominant therapy in many centers in Europe and Japan, whereas radioiodine is more often the first line of treatment in North America. These differences reflect the fact that no single approach is optimal and that patients may require multiple treatments to achieve remission.

The main *antithyroid drugs* are the thionamides, such as propylthiouracil, carbimazole, and the active metabolite of the latter, methimazole. All inhibit the function of [TPO](#), reducing oxidation and organification of iodide. These drugs also reduce thyroid antibody levels by mechanisms that remain unclear, and they appear to enhance rates of remission. Propylthiouracil inhibits deiodination of T_4 to T_3 . However, this effect is of minor benefit, except in the most severe thyrotoxicosis, and is offset by the much shorter half-life of this drug (90 min) compared to methimazole (6 h).

There are many variations of antithyroid drug regimens. The initial dose of carbimazole or methimazole is usually 10 to 20 mg every 8 or 12 h, but once-daily dosing is possible after euthyroidism is restored. Propylthiouracil is given at a dose of 100 to 200 mg every 6 to 8 h, and divided doses are usually given throughout the course. Lower doses of each drug may suffice in areas of low iodine intake. The starting dose of antithyroid drugs can be gradually reduced (titration regimen) as thyrotoxicosis improves. Alternatively, high doses may be given combined with levothyroxine supplementation (block-replace regimen) to avoid drug-induced hypothyroidism. Initial reports suggesting superior remission rates with the block-replace regimen have not been reproduced in several other trials. The titration regimen is often preferred to minimize the dose of antithyroid drug and provide an index of treatment response.

Thyroid function tests and clinical manifestations are reviewed 3 to 4 weeks after starting treatment, and the dose is titrated based on free T_4 levels. Most patients do not achieve euthyroidism until 6 to 8 weeks after treatment is initiated. [TSH](#) levels often remain suppressed for several months and therefore do not provide a sensitive index of treatment response. The usual daily maintenance doses of antithyroid drugs in the titration regimen are 2.5 to 10 mg of carbimazole or methimazole and 50 to 100 mg of propylthiouracil. In the block-replace regimen, the initial dose of antithyroid drug is held constant and the dose of levothyroxine is adjusted to maintain normal free T_4 levels.

Maximum remission rates (up to 30 to 50% in some populations) are achieved by 18 to 24 months. For unclear reasons, remission rates appear to vary in different geographic regions. Patients with severe hyperthyroidism and large goiters are most likely to relapse when treatment stops, but outcome is difficult to predict. All patients should be followed closely for relapse during the first year after treatment and at least annually thereafter.

The common side effects of antithyroid drugs are rash, urticaria, fever, and arthralgia (1 to 5% of patients). These may resolve spontaneously or after substituting an alternative antithyroid drug. Rare but major side effects include hepatitis, an [SLE](#)-like syndrome, and, most importantly, agranulocytosis (<1%). It is essential that antithyroid drugs are stopped and not restarted if a patient develops major side effects. Patients should be given written instructions regarding the symptoms of possible agranulocytosis (e.g., sore throat, fever, mouth ulcers) and the need to stop treatment pending a complete blood count to confirm that agranulocytosis is not present. Management of agranulocytosis is described in [Chap. 109](#). Most physicians do not prospectively monitor blood counts, as the onset of agranulocytosis is idiosyncratic and abrupt.

Propranolol (20 to 40 mg every 6 h) or longer acting beta blockers, such as atenolol, may be useful to control adrenergic symptoms, especially in the early stages before antithyroid drugs take effect. Anticoagulation with warfarin should be considered in all patients with atrial fibrillation. If digoxin is used, increased doses are often needed in the thyrotoxic state.

Radioiodine causes progressive destruction of thyroid cells and can be used as initial treatment or for relapses after a trial of antithyroid drugs. There is a small risk of thyrotoxic crisis (see below) after radioiodine, which can be avoided by pretreatment with antithyroid drugs for at least a month before treatment. Antecedent treatment with antithyroid drugs should be considered in all elderly patients, or in those with cardiac problems, to deplete thyroid hormone stores before administration of radioiodine. Antithyroid drugs must be stopped 3 to 5 days before radioiodine administration to achieve optimum iodine uptake.

Efforts to calculate an optimal dose of radioiodine that achieves euthyroidism, without a high incidence of relapse or progression to hypothyroidism, have not been successful. Some patients inevitably relapse after a single dose because the biologic effects of radiation vary between individuals, and hypothyroidism cannot be uniformly avoided even using accurate dosimetry. A practical strategy is to give a fixed dose based on clinical features, such as the severity of thyrotoxicosis, the size of the goiter (increases the dose needed), and the level of radioiodine uptake (decreases the dose needed).¹³¹ dosage generally ranges between 185 MBq (5 mCi) to 555 MBq (15 mCi). Incomplete treatment or early relapse is more common in males and in patients <40 years of age. Many authorities favor an approach aimed at thyroid ablation (as opposed to euthyroidism), given that levothyroxine replacement is straightforward and most patients ultimately progress to hypothyroidism over 5 to 10 years anyway, frequently with some delay in the diagnosis of hypothyroidism.

Certain radiation safety precautions are necessary in the first few days after radioiodine treatment, but the exact guidelines vary depending on local protocols. In general, patients need to avoid close, prolonged contact with children and pregnant women for several days because of possible transmission of residual isotope and excessive exposure to radiation emanating from the gland. Rarely there may be mild pain due to radiation thyroiditis 1 to 2 weeks after treatment. Hyperthyroidism can persist for 2 to 3 months before radioiodine takes full effect. For this reason, β -adrenergic blockers or antithyroid drugs can be used to control symptoms during this interval. Persistent

hyperthyroidism can be treated with a second dose of radioiodine, usually 6 months after the first dose. The risk of hypothyroidism after radioiodine depends on the dosage but is at least 10 to 20% in the first year and 5% per year thereafter. Patients should be informed of this possibility before treatment and require close follow-up during the first year and annual thyroid function testing thereafter.

Pregnancy and breast feeding are absolute contraindications to radioiodine treatment, but patients can conceive safely 6 to 12 months after treatment. The presence of severe ophthalmopathy requires caution, and some authorities advocate the use of prednisone, 40 mg/d, at the time of radioiodine treatment, tapered over 2 to 3 months to prevent exacerbation of ophthalmopathy. The overall risk of cancer after radioiodine treatment in adults is not increased, but many physicians avoid radioiodine in children and adolescents because of the theoretical risks of malignancy.

Subtotal thyroidectomy is an option for patients who relapse after antithyroid drugs and prefer this treatment to radioiodine. Some experts recommend surgery in young individuals, particularly when the goiter is very large. Careful control of thyrotoxicosis with antithyroid drugs, followed by potassium iodide (3 drops SSKI orally tid) is needed prior to surgery to avoid thyrotoxic crisis and to reduce the vascularity of the gland. The major complications of surgery -- i.e., bleeding, laryngeal edema, hypoparathyroidism, and damage to the recurrent laryngeal nerves -- are unusual when the procedure is performed by highly experienced surgeons. Recurrence rates in the best series are <2%, but the rate of hypothyroidism is only slightly less than that following radioiodine treatment.

The titration regimen of antithyroid drugs should be used to manage Graves' disease in *pregnancy*, as blocking doses of these drugs produce fetal hypothyroidism. Propylthiouracil is usually used because of relatively low transplacental transfer and its ability to block $T_4 \rightarrow T_3$ conversion. Also, carbimazole and methimazole have been associated with rare cases of fetal *aplasia cutis*. The lowest effective dose of propylthiouracil should be given, and it is often possible to stop treatment in the last trimester since [TSH-R](#) antibodies tend to decline in pregnancy. Nonetheless, the transplacental transfer of these antibodies rarely causes *fetal thyrotoxicosis* or *neonatal thyrotoxicosis*. Poor intrauterine growth, a fetal heart rate of >160 beats/min, and high levels of maternal TSH-R antibodies should suggest this complication. Antithyroid drugs given to the mother can be used to treat the fetus and may be needed for 1 to 3 months after delivery, until the maternal antibodies disappear from the baby's circulation. The post-partum period is a time of major risk for relapse of Graves' disease. Breast feeding is safe with low doses of antithyroid drugs. Graves' disease in *children* is best managed with antithyroid drugs, often given as a prolonged course of the titration regimen. Surgery may be indicated for severe disease. Radioiodine can also be used in children, though most experts defer this treatment until adolescence or later.

Thyrotoxic crisis, or *thyroid storm*, is rare and presents as a life-threatening exacerbation of hyperthyroidism, accompanied by fever, delirium, seizures, coma, vomiting, diarrhea, and jaundice. The mortality rate due to cardiac failure, arrhythmia, or hyperthermia is ~30%, even with treatment. Thyrotoxic crisis is usually precipitated by acute illness (e.g., stroke, infection, trauma, diabetic ketoacidosis), surgery (especially on the thyroid), or radioiodine treatment of a patient with partially treated or untreated

hyperthyroidism. Management requires intensive monitoring and supportive care, identification and treatment of the precipitating cause, and measures that reduce thyroid hormone synthesis. Large doses of propylthiouracil (600-mg loading dose and 200 to 300 mg every 6 h) should be given orally or by nasogastric tube or per rectum; the drug's inhibitory action on $T_4 \rightarrow T_3$ conversion makes it the agent of choice. One hour after the first dose of propylthiouracil, stable iodide is given to block thyroid hormone synthesis via the Wolff-Chaikoff effect (the delay allows the antithyroid drug to prevent the excess iodine from being incorporated into new hormone). A saturated solution of potassium iodide (5 drops SSKI every 6 h), or ipodate or iopanoic acid (0.5 mg every 12 h), may be given orally. (Sodium iodide, 0.25 g intravenously every 6 h is an alternative but is not generally available.) Propranolol should also be given to reduce tachycardia and other adrenergic manifestations (40 to 60 mg orally every 4 h; or 2 mg intravenously every 4 h). Although other β -adrenergic blockers can be used, high doses of propranolol have been documented to decrease $T_4 \rightarrow T_3$ conversion, and the doses can be easily adjusted. Caution is needed to avoid acute negative inotropic effects, but controlling the heart rate is important, as some patients develop a form of high-output heart failure. Additional therapeutic measures include glucocorticoids (e.g., dexamethasone, 2 mg every 6 h), antibiotics if infection is present, cooling, and intravenous fluids.

Ophthalmopathy requires no active treatment when it is mild or moderate, as there is usually spontaneous improvement. General measures include meticulous control of thyroid hormone levels, advice about cessation of smoking, and an explanation of the natural history of ophthalmopathy. Discomfort can be relieved with artificial tears (e.g., 1% methylcellulose) and the use of dark glasses with side frames. Periorbital edema responds to a more upright sleeping position. Corneal exposure during sleep can be avoided by taping the eyelids shut. Minor degrees of diplopia improve with prisms fitted to spectacles. Severe ophthalmopathy, with optic nerve involvement or chemosis resulting in corneal damage, is an emergency requiring joint management with an ophthalmologist. Short-term benefit can be gained in about two-thirds of patients by the use of high-dose glucocorticoids (e.g., prednisone, 40 to 80 mg daily), sometimes combined with cyclosporine. Glucocorticoid doses are tapered by 5 mg every 1 to 2 weeks, but the taper often results in reemergence of congestive symptoms. Pulse therapy with intravenous methylprednisolone (1 g of methylprednisolone in 250 mL of saline infused over 2 h daily for 1 week) followed by an oral regimen is also used. Once the eye disease has stabilized, surgery may be indicated for relief of diplopia and correction of the appearance of the eyes. Orbital decompression can be achieved by removing bone from any wall of the orbit, thereby allowing displacement of fat and swollen extraocular muscles. The transantral route is used most often, as it requires no external incision. Proptosis recedes an average of 5 mm, but there may be residual or even worsened diplopia. Alternatively, retroorbital tissue can be decompressed without removal of bony tissue. External beam radiotherapy of the orbits has been used for many years, but the objective evidence that this therapy is beneficial remains equivocal.

Thyroid dermopathy does not usually require treatment but can cause cosmetic problems or interfere with the fit of shoes. Surgical removal is not indicated. Treatment consists of topical, high-potency glucocorticoid ointment under an occlusive dressing. Octreotide may be beneficial.

OTHER CAUSES OF THYROTOXICOSIS

Destructive thyroiditis (subacute or silent thyroiditis) typically presents with a short thyrotoxic phase due to the release of preformed thyroid hormones and catabolism of [Tg](#) (see "Subacute Thyroiditis," below). True hyperthyroidism is absent, as demonstrated by a low radionuclide uptake. Circulating Tg and [fT₄](#) levels are usually increased. Other causes of thyrotoxicosis with low or absent thyroid radionuclide uptake include *thyrotoxicosis factitia*; iodine excess and, rarely, ectopic thyroid tissue, particularly teratomas of the ovary (*struma ovarii*); and functional metastatic follicular carcinoma. Whole-body radionuclide studies can demonstrate ectopic thyroid tissue, and *thyrotoxicosis factitia* can be distinguished from destructive thyroiditis by the clinical features and low levels of Tg. Amiodarone treatment is associated with thyrotoxicosis in up to 10% of patients, particularly in areas of low iodine intake.

TSH-secreting pituitary adenoma is a rare cause of thyrotoxicosis. It can be identified by the presence of an inappropriately normal or increased [TSH](#) level in a patient with hyperthyroidism, diffuse goiter, and elevated free T₄ and T₃ levels ([Chap. 328](#)). Elevated levels of the α subunit of TSH, released by the TSH-secreting adenoma, support this diagnosis, which can be confirmed by demonstrating the pituitary tumor on [CT](#) or [MRI](#) scan. A combination of transsphenoidal surgery, sella irradiation, and octreotide may be required to normalize TSH, as many of these tumors are large and locally invasive at the time of diagnosis. Radioiodine or antithyroid drugs can be used to control thyrotoxicosis.

Thyrotoxicosis caused by *toxic multinodular goiter* and *hyperfunctioning solitary nodules* is discussed below.

THYROIDITIS

A clinically useful classification of thyroiditis is based on the onset and duration of disease ([Table 330-8](#)).

ACUTE THYROIDITIS

Acute thyroiditis is rare and is due to suppurative infection of the thyroid. In children and young adults, the most common cause is the presence of a piriform sinus, a remnant of the fourth branchial pouch that connects the oropharynx with the thyroid. Such sinuses are predominantly left sided. A long-standing goiter and degeneration in a thyroid malignancy are risk factors in the elderly. The patient presents with thyroid pain, often referred to the throat or ears, and a small, tender goiter that may be asymmetric. Fever, dysphagia, and erythema over the thyroid are common, as are systemic symptoms of a febrile illness and lymphadenopathy.

The differential diagnosis of *thyroid pain* includes subacute or, rarely, chronic thyroiditis, hemorrhage into a cyst, malignancy including lymphoma, and, rarely, amiodarone-induced thyroiditis or amyloidosis. However, the abrupt presentation and clinical features of acute thyroiditis rarely cause confusion. The erythrocyte sedimentation rate (ESR) and white cell count are usually increased, but thyroid function is normal. [FNA](#) biopsy shows infiltration by polymorphonuclear leukocytes; culture of the sample can identify the organism. Caution is needed in immunocompromised patients

as fungal or *Pneumocystis* thyroiditis can occur in this setting. Antibiotic treatment is guided initially by Gram stain and subsequently by cultures of the FNA biopsy. Surgery may be needed to drain an abscess, which can be localized by [CT](#) scan or ultrasound. Tracheal obstruction, septicemia, retropharyngeal abscess, mediastinitis, and jugular venous thrombosis may complicate acute thyroiditis but are uncommon with prompt use of antibiotics.

SUBACUTE THYROIDITIS

This is also termed *de Quervain's thyroiditis*, *granulomatous thyroiditis*, or *viral thyroiditis*. Many viruses have been implicated, including mumps, coxsackie, influenza, adenoviruses, and echoviruses, but attempts to identify the virus in an individual patient are often unsuccessful and do not influence management. The diagnosis of subacute thyroiditis is often overlooked because the symptoms can mimic pharyngitis. The peak incidence occurs at 30 to 50 years, and women are affected three times more frequently than men.

Pathophysiology The thyroid shows a characteristic patchy inflammatory infiltrate with disruption of the thyroid follicles and multinucleated giant cells within some follicles. The follicular changes progress to granulomas accompanied by fibrosis. Finally, the thyroid returns to normal, usually several months after onset. During the initial phase of follicular destruction, there is release of [Tg](#) and thyroid hormones, leading to increased circulating free T_4 and T_3 and suppression of [TSH](#) ([Fig. 330-8](#)). During this destructive phase, radioactive iodine uptake is low or undetectable. After several weeks, the thyroid is depleted of stored thyroid hormone and a phase of hypothyroidism typically occurs, with low free T_4 (and sometimes T_3) and moderately increased TSH levels. Radioactive iodine uptake returns to normal or is even increased as a result of the rise in TSH. Finally, thyroid hormone and TSH levels return to normal as the disease subsides.

Clinical Manifestations The patient usually presents with a painful and enlarged thyroid, sometimes accompanied by fever. There may be features of thyrotoxicosis or hypothyroidism, depending on the phase of the illness. Malaise and symptoms of an upper respiratory tract infection may precede the thyroid-related features by several weeks. In other patients, the onset is acute, severe, and without obvious antecedent. Though the patient typically complains of a sore throat, examination reveals a small goiter that is exquisitely tender, and asymmetry is common. Pain is often referred to the jaw or ear. Complete resolution is the usual outcome, but permanent hypothyroidism can occur, particularly in those with coincidental thyroid autoimmunity. A prolonged course over many months, with one or more relapses, occurs in a small percentage of patients.

Laboratory Evaluation As depicted in [Fig. 330-8](#), thyroid function tests characteristically evolve through three distinct phases over about 6 months: (1) thyrotoxic phase, (2) hypothyroid phase, and (3) recovery phase. In the thyrotoxic phase, T_4 and T_3 levels are increased, reflecting their discharge from the damaged thyroid cells, and [TSH](#) is suppressed. The T_4/T_3 ratio is greater than in Graves' disease or thyroid autonomy, in which T_3 is often disproportionately increased. The diagnosis is confirmed by a high [ESR](#) and low radioiodine uptake. Serum [IL-6](#) levels increase during the thyrotoxic phase. The white blood cell count may be increased, and thyroid

antibodies are negative. If the diagnosis is in doubt, [FNA](#) biopsy may be useful, particularly to distinguish unilateral involvement from bleeding into a cyst or neoplasm.

TREATMENT

Relatively large doses of aspirin (e.g., 600 mg every 4 to 6 h) or nonsteroidal anti-inflammatory drugs are sufficient to control symptoms in most cases. If this treatment is inadequate, or if the patient has marked local or systemic symptoms, glucocorticoids should be given. The usual starting dose is 40 to 60 mg prednisone, depending on severity. The dose is gradually tapered over 6 to 8 weeks, in response to improvement in symptoms and the [ESR](#). If a relapse occurs during glucocorticoid withdrawal, treatment should be started again and withdrawn more gradually. In these patients, it is useful to wait until the radioactive iodine uptake normalizes before stopping treatment. Thyroid function should be monitored every 2 to 4 weeks using [TSH](#) and free T₄ levels. Symptoms of thyrotoxicosis improve spontaneously but may be ameliorated by β -adrenergic blockers; antithyroid drugs play no role in treatment of the thyrotoxic phase. Levothyroxine replacement may be needed if the hypothyroid phase is prolonged, but doses should be low enough (50 to 100 μ g daily) to allow TSH-mediated recovery.

SILENT THYROIDITIS

Painless thyroiditis, or "*silent*" *thyroiditis*, occurs in patients with underlying autoimmune thyroid disease. It has a clinical course similar to that of subacute thyroiditis, except that there is little or no thyroid tenderness. The condition occurs most frequently 3 to 6 months after pregnancy and is then termed *post-partum thyroiditis*. Typically, patients have a brief phase of thyrotoxicosis, lasting 2 to 4 weeks, followed by hypothyroidism for 4 to 12 weeks, and then resolution; often, however, only one phase is apparent. As in subacute thyroiditis, the radioactive iodine uptake is initially suppressed. In addition to the painless goiter, silent thyroiditis can be distinguished from subacute thyroiditis by the normal [ESR](#) and the presence of [TPO](#) antibodies. Glucocorticoid treatment is not indicated for silent thyroiditis. Severe thyrotoxic symptoms can be managed with a brief course of propranolol, 20 to 40 mg three or four times daily. Thyroxine replacement may be needed for the hypothyroid phase but should be withdrawn after 6 to 9 months, as recovery is the rule. Annual follow-up thereafter is recommended, as a proportion of these individuals develop permanent hypothyroidism.

DRUG-INDUCED THYROIDITIS

Patients receiving [IFN- \$\alpha\$](#) , [IL-2](#), or amiodarone may develop painless thyroiditis. [IFN- \$\alpha\$](#) , which is used to treat chronic hepatitis B or C, causes thyroid dysfunction in up to 5% of treated patients. It has been associated with painless thyroiditis, hypothyroidism, and Graves' disease. [IL-2](#), which has been used to treat various malignancies, has also been associated with thyroiditis and hypothyroidism, though fewer patients have been studied. For discussion of amiodarone, see "Amiodarone Effects on Thyroid Function," below.

CHRONIC THYROIDITIS

The most common cause of chronic thyroiditis is *Hashimoto's thyroiditis*, an autoimmune disorder that often presents as a firm or hard goiter of variable size (see above). *Riedel's thyroiditis* is a rare disorder that typically occurs in middle-aged women. It presents with an insidious, painless goiter with local symptoms due to compression of the esophagus, trachea, neck veins, or recurrent laryngeal nerves. Dense fibrosis disrupts normal gland architecture and can extend outside the thyroid capsule. Despite these extensive histologic changes, thyroid dysfunction is uncommon. The goiter is hard, nontender, often asymmetric and fixed, leading to suspicion of a malignancy. Diagnosis requires open biopsy as [FNA](#) biopsy is usually unhelpful. Treatment is surgical and directed to relief of compressive symptoms. There is an association between Riedel's thyroiditis and idiopathic fibrosis at other sites (retroperitoneum, mediastinum, biliary tree, lung, and orbit).

SICK EUTHYROID SYNDROME

Any acute, severe illness can cause abnormalities of circulating [TSH](#) or thyroid hormone levels in the absence of underlying thyroid disease, making these measurements potentially misleading. The major cause of these hormonal changes is the release of cytokines. Unless a thyroid disorder is strongly suspected, the routine testing of thyroid function should be avoided in acutely ill patients.

The most common hormone pattern in sick euthyroid syndrome (SES) is a decrease in total and free T_3 levels (low T_3 syndrome) with normal levels of T_4 and [TSH](#). The magnitude of the fall in T_3 correlates with the severity of the illness. T_4 conversion to T_3 via peripheral deiodination is impaired, leading to increased reverse T_3 (rT_3). Despite this effect, decreased clearance rather than increased production is the major basis for increased rT_3 . Also, T_4 is alternately metabolized to the hormonally inactive T_3 sulfate. It is generally assumed that this low T_3 state is adaptive, as it can be induced in normal individuals by fasting. Teleologically, the fall in T_3 may provide a mechanism for limiting catabolism in starved or ill patients.

Very sick patients may have a fall in total T_4 and T_3 levels (low T_4 syndrome). This state has a poor prognosis. A key factor in the fall in T_4 levels is altered binding to [TBG](#). Free T_4 assays usually demonstrate a normal free T_4 level in such patients, depending on the assay method used. Fluctuation in [TSH](#) levels also creates challenges in the interpretation of thyroid function in sick patients. TSH levels may range from <0.1 to >20 mU/L; these alterations reverse after recovery, confirming the absence of underlying thyroid disease. A rise in cortisol or administration of glucocorticoids may provide a partial explanation for decreased TSH levels. However, the exact mechanisms underlying the subnormal TSH seen in 10% of sick patients and the increased TSH seen in 5% remain unclear.

Any severe illness can induce changes in thyroid hormone levels, but certain disorders exhibit a distinctive pattern of abnormalities. Acute liver disease is associated with an initial rise in total (but not free) T_3 and T_4 levels, due to [TBG](#) release; these levels become subnormal with progression to liver failure. A transient increase in total and free T_4 levels, usually with a normal T_3 level, is seen in 5 to 30% of acutely ill psychiatric patients. [TSH](#) values may be transiently low, normal, or high in these patients. In the early stage of HIV infection, T_3 and T_4 levels rise, even if there is weight loss. T_3 levels fall

with progression to AIDS, but TSH levels usually remain normal. Renal disease is often accompanied by low T_3 concentrations, but with normal rather than increased rT_3 levels, due to an unknown factor that increases uptake of rT_3 into the liver.

The diagnosis of the [SES](#) is challenging. Historic information may be limited, and patients often have multiple metabolic derangements. Useful features to consider include previous history of thyroid disease and thyroid function tests, evaluation of the severity and time course of the patient's acute illness, documentation of medications that may affect thyroid function or thyroid hormone levels, and measurements of rT_3 together with free thyroid hormones and [TSH](#). The diagnosis of SES is frequently presumptive, given the clinical context and pattern of laboratory values; only resolution of the test results with clinical recovery can clearly establish this disorder. Treatment of SES with thyroid hormone (T_4 and/or T_3) is controversial, but most authorities recommend monitoring the patient's thyroid function tests during recovery, without administering thyroid hormone, unless there is historic or clinical evidence suggestive of hypothyroidism. Sufficiently large randomized controlled trials using thyroid hormone are unlikely to resolve this therapeutic controversy in the near future, because clinical presentations and outcomes are highly variable.

AMIODARONE EFFECTS ON THYROID FUNCTION

Amiodarone is a commonly used type III antiarrhythmic agent ([Chap. 230](#)). It is structurally related to thyroid hormone and contains 39% iodine by weight. Thus, typical doses of amiodarone (200 mg/d) are associated with very high iodine intake, leading to >40-fold increases in plasma and urinary iodine levels. Moreover, because amiodarone is stored in adipose tissue, high iodine levels persist for >6 months after discontinuation of the drug. Amiodarone inhibits deiodinase activity, and its metabolites function as weak antagonists of thyroid hormone action. Amiodarone has the following multiple effects on thyroid function: (1) acute, transient changes in thyroid function; (2) hypothyroidism in patients susceptible to the inhibitory effects of a high iodine load; and (3) thyrotoxicosis that may be caused by at least three mechanisms -- a Jod-Basedow effect from the iodine load in the setting of multinodular goiter, a thyroiditis-like condition, and possibly induction of autoimmune Graves' disease.

The initiation of amiodarone treatment is associated with a transient decrease of T_4 levels, reflecting the inhibitory effect of iodine on T_4 release. Soon thereafter, most individuals escape from iodide-dependent suppression of the thyroid (Wolff-Chaikoff effect), and the inhibitory effects on deiodinase activity and thyroid hormone receptor action become predominant. These events lead to the following pattern of thyroid function tests: increased T_4 , decreased T_3 , increased rT_3 , and a transient increase of [TSH](#) (up to 20 mU/L). TSH levels normalize or are slightly suppressed after about 1 to 3 months.

The incidence of hypothyroidism from amiodarone varies geographically, apparently correlating with iodine intake. Hypothyroidism occurs in up to 13% of amiodarone-treated patients in iodine-replete countries, such as the United States, but is less common (<6% incidence) in areas of lower iodine intake, such as Italy or Spain. The pathogenesis appears to involve an inability of the thyroid to escape from the high iodine load. Consequently, amiodarone-associated hypothyroidism is more common in

women and individuals with positive [TPO](#) antibodies. It is usually unnecessary to discontinue amiodarone for this side effect, as levothyroxine can be used to normalize thyroid function. [TSH](#) levels should be monitored, because T_4 levels are often increased for the reasons described above.

The management of amiodarone-induced thyrotoxicosis (AIT) is complicated by the fact that there are several causes of thyrotoxicosis and because the increased thyroid hormone levels exacerbate underlying arrhythmias and coronary artery disease. Amiodarone treatment causes thyrotoxicosis in 10% of patients living in areas of low iodine intake and in 2% of patients in regions of high iodine intake. There are two major forms of AIT. Type 1 AIT is associated with an underlying thyroid abnormality (preclinical Graves' disease or nodular goiter). Thyroid hormone synthesis becomes excessive as a result of increased iodine exposure (Jod-Basedow phenomenon). Type 2 AIT occurs in individuals with no intrinsic thyroid abnormalities and is the result of drug-induced lysosomal activation leading to destructive thyroiditis with histiocyte accumulation in the thyroid. Mild forms of type 2 AIT can resolve spontaneously or can occasionally lead to hypothyroidism. Color-flow doppler thyroid scanning shows increased vascularity in type 1 but decreased vascularity in type 2 AIT; [IL-6](#) levels are markedly raised in type 2 but only slightly increased in type 1 AIT. Thyroid scans are difficult to interpret in this setting, because the high endogenous iodine levels diminish tracer uptake. However, the presence of normal or increased uptake favors type 1 AIT.

In amiodarone-induced thyrotoxicosis the drug should be stopped, if possible, though this is often impractical because of the underlying cardiac disorder. Discontinuation of amiodarone will not have an acute effect because of its storage and prolonged half-life. High doses of antithyroid drugs can be used in type 1 [AIT](#) but are often ineffective. Potassium perchlorate, 200 mg every 6 h, has been used to reduce thyroidal iodide content. Perchlorate treatment has been associated with agranulocytosis, though the risk appears relatively low with short-term use. Glucocorticoids, administered as for subacute thyroiditis, are beneficial in type 2 AIT. Lithium blocks thyroid hormone release and can provide modest benefit. Near-total thyroidectomy rapidly decreases thyroid hormone levels and may be the most effective long-term solution, if the patient can undergo the procedure safely.

THYROID FUNCTION IN PREGNANCY

Three factors alter thyroid function in pregnancy: (1) the transient increase in [hCG](#) during the first trimester, which stimulates the [TSH-R](#); (2) the estrogen-induced rise in [TBG](#) during the first trimester, which is sustained during pregnancy; and (3) increased urinary iodide excretion, which can cause impaired thyroid hormone production in areas of marginal iodine sufficiency. Women with a precarious iodine intake (<50 $\mu\text{g}/\text{d}$) are most at risk of developing a goiter during pregnancy, and iodine supplementation should be considered to prevent maternal and fetal hypothyroidism and the development of neonatal goiter.

The rise in circulating [hCG](#) levels during the first trimester is accompanied by a reciprocal fall in [TSH](#) that persists into the middle of pregnancy. This appears to reflect weak binding of hCG, which is present at very high levels, to the [TSH-R](#). Rare individuals have been described with variant TSH-R sequences that enhance hCG binding and TSH-R

activation. Occasionally these hCG-induced changes in thyroid function result in transient gestational hyperthyroidism and/or *hyperemesis gravidarum*, a condition characterized by severe nausea and vomiting and risk of volume depletion. Antithyroid drugs are rarely needed, and parenteral fluid replacement usually suffices until the condition resolves.

Maternal hypothyroidism occurs in 2 to 3% of women of child-bearing age and is associated with increased risk of developmental delay in the offspring. Thyroid hormone requirements are increased by 25 to 50 ug/d during pregnancy.

GOITER AND NODULAR THYROID DISEASE

Goiter refers to an enlarged thyroid gland. Biosynthetic defects, iodine deficiency, autoimmune disease, and nodular diseases can each lead to goiter, though by different mechanisms. Biosynthetic defects and iodine deficiency are associated with reduced efficiency of thyroid hormone synthesis, leading to increased [TSH](#), which stimulates thyroid growth as a compensatory mechanism to overcome the block in hormone synthesis. Graves' disease and Hashimoto's thyroiditis are also associated with goiter. In Graves' disease, the goiter results mainly from the [TSH-R](#)-mediated effects of [TSI](#). The goitrous form of Hashimoto's thyroiditis occurs because of acquired defects in hormone synthesis, leading to elevated levels of TSH and its consequent growth effects. Lymphocytic infiltration and immune system-induced growth factors also contribute to thyroid enlargement in Hashimoto's thyroiditis. Nodular disease is characterized by the disordered growth of thyroid follicles, often combined with the gradual development of fibrosis. The management of goiter differs in patients depending on the etiology, and the detection of thyroid enlargement on physical examination should prompt further evaluation to identify its cause.

Nodular thyroid disease is common, occurring in about 3 to 7% of adults when assessed by physical examination. Using more sensitive techniques, such as ultrasound, it is present in >25% of adults. Thyroid nodules may be solitary or multiple, and they may be functional or nonfunctional.

DIFFUSE NONTOXIC (SIMPLE) GOITER

Etiology and Pathogenesis When diffuse enlargement of the thyroid occurs in the absence of nodules and hyperthyroidism, it is referred to as a *diffuse nontoxic goiter*. This is sometimes called *simple goiter*, because of the absence of nodules, or *colloid goiter*, because of the presence of uniform follicles that are filled with colloid. Worldwide, diffuse goiter is most commonly caused by iodine deficiency and is termed *endemic goiter* when it affects >5% of the population. In nonendemic regions, *sporadic goiter* occurs, and the cause is usually unknown. Thyroid enlargement in teenagers is sometimes referred to as *juvenile goiter*. In general, goiter is more common in women than men, probably because of the greater prevalence of underlying autoimmune disease and the increased iodine demands associated with pregnancy.

In *iodine-deficient areas*, thyroid enlargement reflects a compensatory effort to trap iodide and produce sufficient hormone under conditions in which hormone synthesis is relatively inefficient. Somewhat surprisingly, [TSH](#) levels are usually normal or only slightly

increased, suggesting increased sensitivity to TSH or activation of other pathways that lead to thyroid growth. Iodide appears to have direct actions on thyroid vasculature and may indirectly affect growth through vasoactive substances such as endothelins and nitric oxide. Endemic goiter is also caused by exposure to environmental *goitrogens* such as cassava root, which contains a thiocyanate, vegetables of the Cruciferae family (e.g., brussels sprouts, cabbage, and cauliflower), and milk from regions where goitrogens are present in grass. Though relatively rare, inherited defects in thyroid hormone synthesis also lead to a diffuse nontoxic goiter. These involve abnormalities at each step in hormone synthesis including iodide transport (NIS), [Tg](#) synthesis, organification and coupling ([TPO](#)), and the regeneration of iodide (dehalogenase).

Clinical Manifestations and Diagnosis If thyroid function is preserved, most goiters are asymptomatic. Spontaneous hemorrhage into a cyst or nodule may cause the sudden onset of localized pain and swelling. Examination of a diffuse goiter reveals a symmetrically enlarged, nontender, generally soft gland without palpable nodules. Goiter is defined, somewhat arbitrarily, as a lateral lobe with a volume greater than the thumb of the individual being examined. If the thyroid is markedly enlarged, it can cause tracheal or esophageal compression. These features are unusual, however, in the absence of nodular disease and fibrosis. *Substernal goiter* may obstruct the thoracic inlet. *Pemberton's sign* refers to symptoms of faintness with evidence of facial congestion and external jugular venous obstruction when the arms are raised above the head, a maneuver that draws the thyroid into the thoracic inlet. Respiratory flow measurements and [CT](#) or [MRI](#) should be used to evaluate substernal goiter in patients with obstructive signs or symptoms.

Thyroid function tests should be performed in all patients with goiter to exclude thyrotoxicosis or hypothyroidism. It is not unusual, particularly in iodine deficiency, to find a low total T₄, with normal T₃ and [TSH](#), reflecting enhanced T₄→T₃ conversion. A low TSH, particularly in older patients, suggests the possibility of thyroid autonomy or undiagnosed Graves' disease, causing subclinical thyrotoxicosis. [TPO](#) antibodies may be useful to identify patients at increased risk of autoimmune thyroid disease. Low urinary iodine levels (<100 ug/L) support a diagnosis of iodine deficiency. Thyroid scanning is not generally necessary but will reveal increased uptake in iodine deficiency and most cases of dyshormonogenesis. Ultrasound is not generally indicated in the evaluation of diffuse goiter, unless a nodule is palpable on physical examination.

TREATMENT

Iodine or thyroid hormone replacement induces variable regression of goiter in iodine deficiency, depending on how long it has been present and the degree of fibrosis that has developed. For other causes of nontoxic diffuse goiter, levothyroxine can be used in an attempt to reduce goiter size. Because of the possibility of underlying thyroid autonomy, caution should be exercised when instituting suppressive thyroxine therapy, particularly if the baseline [TSH](#) is in the low-normal range. In younger patients, the dose can be started at 100 ug/d and adjusted to suppress the TSH into the low-normal but detectable range. Treatment of elderly patients should be initiated at 50 ug/d. The efficacy of suppressive treatment is greater in younger patients and in those with soft goiters. Significant regression is usually seen within 3 to 6 months of treatment; after this time it is unlikely to occur. In older patients, and in those with some degree of

nodular disease or fibrosis, fewer than one-third demonstrate significant shrinkage of the goiter. Surgery is rarely indicated for diffuse goiter. Exceptions include documented evidence of tracheal compression or obstruction of the thoracic inlet, which are more likely to be associated with substernal multinodular goiters (see below). Subtotal or near-total thyroidectomy for these or cosmetic reasons should be performed by an experienced surgeon to minimize complication rates, which occur in up to 10% of cases. Surgery should be followed by mild suppressive treatment with levothyroxine to prevent regrowth of the goiter. Radioiodine reduces goiter size by about 50% in the majority of patients. It is rarely associated with transient acute swelling of the thyroid, which is usually inconsequential unless there is severe tracheal narrowing. If not treated with levothyroxine, patients should be followed after radioiodine treatment for the possible development of hypothyroidism.

NONTOXIC MULTINODULAR GOITER

Etiology and Pathogenesis Depending on the geographic region and the sensitivity of the methods used to detect the disorder, multinodular goiter (MNG) is common, occurring in between 1 and 12% of the population. MNG is more common in women than men and increases in prevalence with age. It is more common in iodine-deficient regions but also occurs in regions of iodine sufficiency, reflecting multiple genetic, autoimmune, and environmental influences on the pathogenesis.

Individual patients exhibit wide variation in nodule size. Histology reveals a spectrum of morphologies ranging from hypercellular regions to cystic areas filled with colloid. Fibrosis is often extensive, and areas of hemorrhage or lymphocytic infiltration may be seen. Using molecular techniques, most nodules within a [MNG](#) are polyclonal in origin, suggesting a hyperplastic response to locally produced growth factors and cytokines. [TSH](#), which is usually not elevated, may play a permissive or contributory role. Monoclonal lesions also occur within a MNG, reflecting mutations in genes that confer a selective growth advantage to the progenitor cell.

Clinical Manifestations Most patients with nontoxic [MNG](#) are asymptomatic and, by definition, euthyroid. MNG typically develops over many years and is detected on routine physical examination or because an individual notices an enlargement in the neck. If the goiter is large enough, it can ultimately lead to compressive symptoms including difficulty swallowing, respiratory distress (tracheal compression), or plethora (venous congestion), but these symptoms are uncommon. Symptomatic MNGs are usually extraordinarily large and/or develop fibrotic areas that cause compression. Sudden pain in a MNG is often caused by hemorrhage into a nodule but should raise the possibility of invasive malignancy. Hoarseness, reflecting laryngeal nerve involvement, also suggests malignancy.

Diagnosis On examination, thyroid architecture is distorted and multiple nodules of varying size can be appreciated. Substernal goiter is suggested by Pemberton's sign. Because many nodules are deeply embedded in thyroid tissue or reside in posterior or substernal locations, it is not possible to palpate all nodules. [ATSH](#) level should be measured to exclude subclinical hyper- or hypothyroidism, but thyroid function is usually normal. Tracheal deviation is common, but compression must usually exceed 70% of the tracheal diameter before there is significant airway compromise. Pulmonary function

testing can be used to assess the functional effects of compression and to detect tracheomalacia, which characteristically causes inspiratory stridor. [CT](#) or [MRI](#) can be used to evaluate the anatomy of the goiter and the extent of substernal extension, which is often much greater than is apparent on physical examination. A barium swallow may reveal the extent of esophageal obstruction. [MNG](#) does not appear to predispose to thyroid carcinoma or to more aggressive carcinoma. For this reason, and because it is not possible to biopsy all nodular lesions, thyroid biopsies should only be performed if malignancy is suspected because of a dominant or enlarging nodule.

TREATMENT

Most nontoxic [MNGs](#) can be managed conservatively. T_4 suppression is rarely effective for reducing goiter size and introduces the risk of thyrotoxicosis, if there is underlying autonomy or if it develops during treatment. If levothyroxine is used, it should be started at low doses (50 ug) and advanced gradually while monitoring the [TSH](#) level to avoid excessive suppression. Contrast agents and other iodine-containing substances should be avoided because of the risk of inducing the *Jod-Basedow effect*, characterized by enhanced thyroid hormone production by autonomous nodules. Radioiodine is being used with increasing frequency because it often decreases goiter size and may selectively ablate regions of autonomy. Dosage of ^{131}I depends on the size of the goiter and radioiodine uptake but is usually about 3.7 MBq (0.1 mCi) per gram of tissue, corrected for uptake [typical dose, 370 to 1070 Mbq (10 to 29 mCi)]. Repeat treatment may be needed. It is possible to achieve a 40 to 50% reduction in goiter size in most patients. Earlier concerns about radiation-induced thyroid swelling and tracheal compression have diminished as recent studies have shown this complication to be rare. When acute compression occurs, glucocorticoid treatment or surgery may be needed. Radiation-induced hypothyroidism is less common than occurs after treatment for Graves' disease. However, posttreatment autoimmune thyrotoxicosis may occur in up to 5% of patients treated for nontoxic MNG. Surgery remains highly effective but is not without risk, particularly in older patients with underlying cardiopulmonary disease.

TOXIC MULTINODULAR GOITER

The pathogenesis of toxic [MNG](#) appears to be similar to that of nontoxic MNG, the major difference being the presence of functional autonomy in toxic MNG. The molecular basis for autonomy in toxic MNG remains unknown. As in nontoxic goiters, many nodules are polyclonal, while others are monoclonal and vary in their clonal origins. Genetic abnormalities known to confer functional autonomy, such as activating [TSH-R](#) or G_s mutations (see below), are not usually found in the autonomous regions of toxic MNG goiter.

In addition to features of goiter, the clinical presentation of toxic [MNG](#) includes subclinical hyperthyroidism or mild thyrotoxicosis. The patient is usually elderly and may present with atrial fibrillation or palpitations, tachycardia, nervousness, tremor, or weight loss. Recent exposure to iodine, from contrast dyes or other sources, may precipitate or exacerbate thyrotoxicosis. The [TSH](#) level is low. The T_4 level may be normal or minimally increased; T_3 is often elevated to a greater degree than T_4 . Thyroid scan shows heterogeneous uptake with multiple regions of increased and decreased uptake; 24-h uptake of radioiodine may not be increased.

TREATMENT

The management of toxic [MNG](#) is challenging. Antithyroid drugs, often in combination with beta blockers, can normalize thyroid function and address clinical features of thyrotoxicosis. This treatment, however, often stimulates the growth of the goiter, and, unlike in Graves' disease, spontaneous remission does not occur. Radioiodine can be used to treat areas of autonomy, as well as to decrease the mass of the goiter. Usually, however, some degree of autonomy remains, presumably because multiple autonomous regions emerge as soon as others are treated. Nonetheless, a trial of radioiodine should be considered before subjecting patients, many of whom are elderly, to surgery. Surgery provides definitive treatment of underlying thyrotoxicosis as well as goiter. Patients should be rendered euthyroid using antithyroid drugs before operation.

HYPERFUNCTIONING SOLITARY NODULE

A solitary, autonomously functioning thyroid nodule is referred to as *toxic adenoma*. The pathogenesis of this disorder has been unraveled by demonstrating the functional effects of mutations that stimulate the [TSH-R](#) signaling pathway. Most patients with solitary hyperfunctioning nodules have acquired somatic, activating mutations in the TSH-R ([Fig. 330-9](#)). These mutations, located primarily in the receptor transmembrane domain, induce constitutive receptor coupling to G_{sa} , increasing cyclic AMP levels and leading to enhanced thyroid follicular cell proliferation and function. Less commonly, somatic mutations are identified in G_{sa} . These mutations, which are similar to those seen in McCune-Albright syndrome ([Chap. 336](#)) or in a subset of somatotrope adenomas ([Chap. 328](#)), impair GTP hydrolysis, also causing constitutive activation of the cyclic AMP signaling pathway. In most series, activating mutations in either the TSH-R or the G_{sa} subunit genes are identified in >90% of patients with solitary hyperfunctioning nodules.

Thyrotoxicosis is usually mild. The disorder is suggested by the presence of the thyroid nodule, which is generally large enough to be palpable, and by the absence of clinical features suggestive of Graves' disease or other causes of thyrotoxicosis. A thyroid scan provides a definitive diagnostic test, demonstrating focal uptake in the hyperfunctioning nodule and diminished uptake in the remainder of the gland, as activity of the normal thyroid is suppressed.

TREATMENT

Radioiodine ablation is usually the treatment of choice. Because normal thyroid function is suppressed, ^{131}I is concentrated in the hyperfunctioning nodule with minimal uptake and damage to normal thyroid tissue. Relatively large radioiodine doses [e.g., 370 to 1110 MBq (10 to 29.9 mCi) ^{131}I] have been shown to correct thyrotoxicosis in about 75% of patients within 3 months. Hypothyroidism occurs in <10% of patients over the next 5 years. Surgical resection is also effective and is usually limited to enucleation of the adenoma or lobectomy, thereby preserving thyroid function and minimizing risk of hypoparathyroidism or damage to the recurrent laryngeal nerves. Medical therapy using antithyroid drugs and beta blockers can normalize thyroid function but is not an optimal long-term treatment. Ethanol injection under ultrasound guidance has been used

successfully in some centers to ablate hyperfunctioning nodules. Repeated injections (often more than 5 sessions) are required but reduce nodule size. Normal thyroid function can be achieved in most patients using this technique.

BENIGN NEOPLASMS

The various types of benign thyroid nodules are listed in [Table 330-9](#). These lesions are common (5 to 10% adults) and often multiple, particularly when assessed by sensitive techniques such as ultrasound. The risk of malignancy is very low for *macrofollicular adenomas* and *normofollicular adenomas*. *Microfollicular*, *trabecular*, and *Hurthle cell variants* raise greater concern, partly because the histology is more difficult to interpret. About one-third of palpable nodules are *thyroid cysts*. These may be recognized by their ultrasound appearance or based on aspiration of large amounts of pink or straw-colored fluid (colloid). Many are mixed cystic/solid lesions, in which case it is desirable to aspirate cellular components under ultrasound or harvest cells after cytopspin of cyst fluid. Cysts frequently recur, even after repeated aspiration, and may require surgical excision if they are large or if the cytology is suspicious. Sclerosis has been used with variable success but is often painful and may be complicated by infiltration of the sclerosing agent.

The treatment approach for benign nodules is similar to that for [MNG](#). [TSH](#) suppression with levothyroxine decreases the size of about 30% of nodules and may prevent further growth. The TSH level should be suppressed into the low-normal range, assuming there are no contraindications; alternatively, nodule size can be monitored without suppression. If a nodule has not decreased in size after 6 to 12 months of suppressive therapy, treatment should be discontinued as little benefit is likely to accrue from long-term treatment.

THYROID CANCER

Thyroid carcinoma is the most common malignancy of the endocrine system. Malignant tumors derived from the follicular epithelium are classified according to histologic features. Differentiated tumors, such as papillary thyroid cancer (PTC) or follicular thyroid cancer (FTC), are often curable, and the prognosis is good for patients identified with early-stage disease. In contrast, anaplastic thyroid cancer (ATC) is aggressive, responds poorly to treatment, and is associated with a bleak prognosis.

The incidence of thyroid cancer (~9/100,000 per year) increases with age, plateauing after about age 50 ([Fig. 330-10](#)). Age is also an important prognostic factor -- thyroid cancer at young age (<20) or in older persons (>65) is associated with a worse prognosis. Thyroid cancer is twice as common in women as men, but male sex is associated with a worse prognosis. Additional important risk factors include a history of childhood head or neck irradiation, large nodule size (≥ 4 cm), evidence for local tumor fixation or invasion into lymph nodes, and the presence of metastases ([Table 330-10](#)).

Several unique features of thyroid cancer facilitate its management: (1) thyroid nodules are readily palpable, allowing early detection and biopsy by [FNA](#); (2) iodine radioisotopes can be used to diagnose (^{123}I) and treat (^{131}I) differentiated thyroid cancer, reflecting the unique uptake of this anion by the thyroid gland; and (3) serum markers allow the

detection of residual or recurrent disease, including the use of [Tg](#) levels for [PTC](#) and [FTC](#) and calcitonin for medullary thyroid cancer (MTC).

CLASSIFICATION

Thyroid neoplasms can arise in each of the cell types that populate the gland, including thyroid follicular cells, calcitonin-producing C cells, lymphocytes, and stromal and vascular elements, as well as metastases from other sites ([Table 330-9](#)). The American Joint Committee on Cancer (AJCC) has designated a staging system using the TNM classification ([Table 330-11](#)). Several other classification and staging systems are also widely used, some of which place greater emphasis on histologic features or risk factors such as age or gender.

PATHOGENESIS AND GENETIC BASIS

Radiation Early studies of the pathogenesis of thyroid cancer focused on the role of external radiation, which predisposes to chromosomal breaks, presumably leading to genetic rearrangements and loss of tumor-suppressor genes. External radiation of the mediastinum, face, head, and neck region was administered in the past to treat an array of conditions including acne and enlargement of the thymus, tonsils, and adenoids. Radiation exposure increases the risk of benign and malignant thyroid nodules, is associated with multicentric cancers, and shifts the incidence of thyroid cancer to an earlier age group. Radiation from nuclear fallout also predisposes to thyroid cancer. Children seem more predisposed to the effects of radiation than adults. Of note, radiation derived from ¹³¹I therapy appears to contribute little, if any, increased risk of thyroid cancer.

TSH and Growth Factors Thyroid growth is regulated primarily by [TSH](#) but also by a variety of growth factors and cytokines. Many differentiated thyroid cancers express TSH receptors and, therefore, remain responsive to TSH. This observation provides the rationale for T₄ suppression of TSH in patients with thyroid cancer. Residual expression of TSH receptors also allows TSH-stimulated uptake of ¹³¹I therapy (see below).

Oncogenes and Tumor-Suppressor Genes Thyroid cancers are monoclonal in origin, consistent with the idea that they originate as a consequence of mutations that confer a growth advantage to a single cell. In addition to increased rates of proliferation, some thyroid cancers exhibit impaired apoptosis and features that enhance invasion, angiogenesis, and metastasis ([Chap. 83](#)). By analogy with the model of multistep carcinogenesis proposed for colon cancer ([Chap. 81](#)), thyroid neoplasms have been analyzed for a variety of genetic alterations, but without clear evidence of an ordered acquisition of somatic mutations as they progress from the benign to the malignant state. On the other hand, certain mutations are relatively specific for thyroid neoplasia, some of which correlate with histologic classification ([Table 330-12](#)). For example, activating mutations of the [TSH-R](#) and the G_{sα} subunit are associated with autonomously functioning nodules. Though these mutations induce thyroid cell growth, this type of nodule is almost always benign. A variety of rearrangements involving the *RET* gene on chromosome 10 bring this receptor tyrosine kinase under the control of other promoters, leading to receptor overexpression. *RET* rearrangements occur in 20 to 40% of [PTCs](#) in different series and were observed with increased frequency in tumors developing after

the Chernobyl radiation disaster. Rearrangements in PTC have also been observed for another tyrosine kinase gene, *TRK1*, which is located on chromosome 1. To date, the identification of PTC with *RET* or *TRK1* rearrangements has not proven useful for predicting prognosis or treatment responses. *RAS* mutations are found in about 20 to 30% of thyroid neoplasms, including adenomas as well as PTC and [FTC](#), suggesting that these mutations do not strongly affect tumor phenotype. Loss of heterozygosity (LOH), consistent with deletions of tumor-suppressor genes, is particularly common in [FTC](#), often involving chromosomes 3p or 11q. Mutations of the tumor suppressor, p53, appear to play an important role in the development of [ATC](#). Because p53 plays a role in cell cycle surveillance, DNA repair, and apoptosis, its loss may contribute to the rapid acquisition of genetic instability as well as poor treatment responses ([Chap. 82](#)). The role of other tumor-suppressor genes in thyroid cancer is under investigation ([Table 330-12](#)).

[MTC](#), when associated with multiple endocrine neoplasia (MEN) type 2, harbors an inherited mutation of the *RET* gene. Unlike the rearrangements of *RET* seen in [PTC](#), the mutations in MEN-2 are point mutations that induce constitutive activity of the tyrosine kinase ([Chap. 339](#)). MTC is preceded by hyperplasia of the C cells, raising the likelihood that as-yet-unidentified "second hits" lead to cellular transformation. A subset of sporadic MTC contain somatic mutations that activate *RET*.

WELL-DIFFERENTIATED THYROID CANCER

Papillary [PTC](#) is the most common type of thyroid cancer, accounting for 70 to 90% of well-differentiated thyroid malignancies. Microscopic PTC is present in as many as 25% of thyroid glands at autopsy, but most of these lesions are very small (several millimeters) and are not clinically significant. Characteristic cytologic features of PTC help make the diagnosis by [FNA](#) or after surgical resection; these include psammoma bodies, cleaved nuclei with an "orphan-Annie" appearance caused by large nucleoli, and the formation of papillary structures.

[PTC](#) tends to be multifocal and to invade locally within the thyroid gland as well as through the thyroid capsule and into adjacent structures in the neck. It has a propensity to spread via the lymphatic system but can metastasize as well, particularly to bone and lung. Because of the relatively slow growth of the tumor, a significant burden of pulmonary metastases may accumulate, sometimes with remarkably few symptoms. The prognostic implication of lymph node spread is debated. Lymph node involvement by thyroid cancer can be remarkably well tolerated but probably increases the risk of recurrence and mortality, particularly in older patients. The staging of PTC by the TNM system is outlined in [Table 330-11](#). Most papillary cancers are identified in the early stages (>80% stages I or II) and have an excellent prognosis, with survival curves similar to expected survival ([Fig. 330-11A](#)). Mortality is markedly increased in stage IV disease (distant metastases), but this group comprises only about 1% of patients. The treatment of PTC is described below.

Follicular The incidence of [FTC](#) varies widely in different parts of the world; it is more common in iodine-deficient regions. [FTC](#) is difficult to diagnose by [FNA](#) because the distinction between benign and malignant follicular neoplasms rests largely on evidence of invasion into vessels, nerves, or adjacent structures. [FTC](#) tends to spread by

hematogenous routes leading bone, lung, and central nervous system metastases. Mortality rates associated with FTC are less favorable than for [PTC](#), in part because a larger proportion of patients present with stage IV disease ([Fig. 330-11B](#)). Poor prognostic features include distant metastases, age >50 years, primary tumor size >4 cm, Hurthle cell histology, and the presence of marked vascular invasion.

TREATMENT

Surgery All well-differentiated thyroid cancers should be surgically excised. In addition to removing the primary lesion, surgery allows accurate histologic diagnosis and staging, and multicentric disease is commonly found in the contralateral lobe. Lymph node spread can also be assessed at the time of surgery, and involved nodes can be removed. Recommendations about the extent of surgery vary for stage I disease, as survival rates are similar for lobectomy and near-total thyroidectomy. Lobectomy is associated with a lower incidence of hypoparathyroidism and injury to the recurrent laryngeal nerves. However, it is not possible to monitor [Tg](#) levels or to perform whole-body ¹³¹I scans in the presence of the residual lobe. Moreover, if final staging or subsequent follow-up indicates the need for radioiodine scanning or treatment, repeat surgery is necessary to remove the remaining thyroid tissue. The authors favor near-total thyroidectomy in almost all patients; complication rates are acceptably low if the surgeon is highly experienced in the procedure. This approach, in combination with postsurgical radioablation of the remnant thyroid tissue, facilitates the use of radioiodine scanning and Tg determinations to assess disease recurrence.

TSH Suppression Therapy As most tumors are still [TSH](#)-responsive, levothyroxine suppression of TSH is a mainstay of thyroid cancer treatment. Though TSH suppression clearly provides therapeutic benefit, there are no prospective studies that identify the optimal level of TSH suppression. A reasonable goal is to suppress TSH to as low as possible without subjecting the patient to unnecessary side effects from excess thyroid hormone, such as atrial fibrillation, osteopenia, anxiety, and other manifestations of thyrotoxicosis. For patients at low risk of recurrence, TSH should be suppressed into the low but detectable range (0.1 to 0.5 IU/L). For patients at high risk of recurrence, or with known metastatic disease, complete TSH suppression is indicated, if there are no strong contraindications to mild thyrotoxicosis. In this instance, free T₄ or free T₃ levels must also be monitored to avoid excessive treatment.

Radioiodine Treatment Well-differentiated thyroid cancer still incorporates radioiodine, though less efficiently than normal thyroid follicular cells. Radioiodine uptake is determined primarily by expression of the NIS and is stimulated by [TSH](#), requiring expression of the [TSH-R](#). The retention time for radioactivity is influenced by the extent to which the tumor retains differentiated functions such as iodide trapping and organification. After near-total thyroidectomy, substantial thyroid tissue remains, particularly in the thyroid bed and surrounding the parathyroid glands. Consequently, ¹³¹I ablation is necessary to eliminate remaining normal thyroid tissue and may treat residual tumor cells.

Indications The use of therapeutic doses of radioiodine remains an area of controversy in thyroid cancer management. Postoperative thyroid ablation and radioiodine treatment of known residual [PTC](#) or [FTC](#) reduce recurrence rates. ¹³¹I ablation of remaining normal

thyroid tissue also facilitates the detection of recurrent disease, using either whole-body iodine scanning or measurements of [Tg](#). For tumors that take up iodine, ^{131}I treatment can reduce or eliminate residual disease with relatively little associated toxicity. However, it is not clear that prophylactic radioiodine treatment reduces mortality for patients at relatively low risk. Most patients with stage 1 PTC with primary tumors <1.5 cm in size can usually be managed safely with thyroxine suppression, without radiation treatment, as the risk of recurrence and mortality is very low. For patients with larger papillary tumors, spread to the adjacent lymph nodes, FTC, or evidence of metastases, thyroid ablation and radioiodine treatment are generally indicated.

^{131}I thyroid ablation and treatment As noted above, the decision to use ^{131}I for thyroid ablation should be coordinated with the surgical approach, as radioablation is much more effective when there is minimal remaining normal thyroid tissue. A typical strategy is to treat the patient for several weeks postoperatively with liothyronine (25 ug bid or tid), followed by thyroid hormone withdrawal. Ideally, the [TSH](#) level should increase to >50 IU/L over about 3 to 4 weeks. The level to which TSH rises is dictated largely by the amount of normal thyroid tissue remaining postoperatively. A scanning dose of ^{131}I [usually 148 to 185 MBq (4 to 5 mCi)] will reveal the amount of residual tissue and provides guidance about the dose needed to accomplish ablation. A maximum outpatient dose of 1110 MBq (29.9 mCi) ^{131}I can be administered in the United States, though ablation is often more complete using greater doses [1850 to 2775 MBq (50 to 75 mCi)]. In patients with known residual cancer, the larger doses ensure thyroid ablation and may destroy remaining tumor cells. A whole-body scan following the high-dose radioiodine treatment is useful to identify possible metastatic disease.

Follow-up whole-body thyroid scanning and thyroglobulin determinations An initial whole-body scan should be performed about 6 months after surgery and thyroid ablation. The strategy for follow-up management of thyroid cancer has been altered by the availability of recombinant human [TSH](#) (rhTSH) to stimulate ^{131}I uptake and by the improved sensitivity of [Tg](#) assays to detect residual or recurrent disease. A scheme for using either rhTSH or thyroid hormone withdrawal for thyroid scanning is summarized in [Fig. 330-12](#). After thyroid ablation, rhTSH can be used to stimulate ^{131}I uptake without subjecting patients to thyroid hormone withdrawal and its associated symptoms of hypothyroidism and the risk of prolonged TSH-stimulated tumor growth. This approach is recommended for patients predicted to be at low risk of disease recurrence, since rhTSH is not currently approved for use in conjunction with therapeutic doses of ^{131}I . Alternatively, in patients who are likely to require ^{131}I treatment, the traditional approach of thyroid hormone withdrawal can be used to increase TSH. This involves switching patients from levothyroxine (T_4) to the more rapidly cleared hormone, liothyronine (T_3), thereby allowing TSH to increase more quickly. If residual disease is detected on the initial whole-body scan [148 to 185 MBq (4 to 5 mCi)], a larger treatment dose, usually between 2775 and 5550 MBq (75 and 150 mCi), can be administered depending on the degree of residual uptake and assessment of cancer risk. Because TSH stimulates Tg levels, Tg measurements should be obtained after administration of rhTSH or when TSH levels have risen after thyroid hormone withdrawal. If the initial whole-body scan is negative and Tg levels are low, a repeat scan should be performed 1 year later. If still negative, the patient can be managed with suppressive therapy and measurements of Tg every 6 to 12 months. If a second follow-up scan is negative, no further scanning may be necessary if the patient is at low risk and there is no clinical or laboratory

evidence of recurrence. Many authorities advocate radioiodine treatment for scan-negative, Tg-positive (Tg >5 to 10 ng/mL) patients, as many derive therapeutic benefit from a large dose of ¹³¹I.

In addition to radioiodine, external beam radiotherapy is also used to treat specific metastatic lesions, particularly when they cause bone pain or threaten neurologic injury (e.g., vertebral metastases).

ANAPLASTIC AND OTHER FORMS OF THYROID CANCER

Anaplastic Thyroid Cancer As noted above, [ATC](#) is a poorly differentiated and aggressive cancer. The prognosis is poor, and most patients die within 6 months of diagnosis. Because of the undifferentiated state of these tumors, radioiodine uptake is usually negligible but can be used therapeutically if there is residual uptake. Chemotherapy has been attempted with multiple agents, including anthracyclines and paclitaxel, but is usually futile. External radiation therapy can be attempted and continued if tumors are responsive.

Thyroid Lymphoma Lymphoma in the thyroid gland often arises in the background of Hashimoto's thyroiditis. A rapidly expanding thyroid mass should suggest the possibility of this diagnosis. Diffuse large cell lymphoma is the most common type in the thyroid. Biopsies reveal sheets of lymphoid cells that can be difficult to distinguish from small cell lung cancer or [ATC](#). These tumors are often highly sensitive to external radiation. Surgical resection should be avoided as initial therapy because it may spread disease that is otherwise localized to the thyroid. If staging indicates disease outside of the thyroid, treatment should follow guidelines used for other forms of lymphoma ([Chap. 112](#)).

MEDULLARY THYROID CARCINOMA

[MTC](#) can be sporadic or familial and accounts for about 5 to 10% of thyroid cancers. There are three familial forms of MTC: [MEN-2A](#), [MEN-2B](#), and familial MTC without other features of MEN ([Chap. 339](#)). In general, MTC is more aggressive in [MEN-2B](#) than in [MEN-2A](#), and familial MTC is more aggressive than sporadic MTC. Elevated serum calcitonin provides a marker of residual or recurrent disease. It is reasonable to test all patients with MTC for *RET* mutations, as genetic counseling and testing of family members can be offered to those individuals who test positive for mutations.

The management of [MTC](#) is primarily surgical. Unlike tumors derived from thyroid follicular cells, these tumors do not take up radioiodine. External radiation treatment and chemotherapy may provide palliation in patients with advanced disease ([Chap. 339](#)).

Approach to the Patient

Palpable thyroid nodules are found in about 5% of adults, though the prevalence varies considerably worldwide. Given this high prevalence rate, it is common for the practitioner to identify and evaluate thyroid nodules. The main goal of this evaluation is to identify, in a cost-effective manner, the small subgroup of individuals with malignant lesions.

As described above, nodules are more common in iodine-deficient areas, in women, and with aging. Most palpable nodules are >1 cm in diameter, but the ability to feel a nodule is influenced by its location within the gland (superficial versus deeply embedded), the anatomy of the patient's neck, and the experience of the examiner. More sensitive methods of detection, such as thyroid ultrasound and pathologic studies, reveal thyroid nodules in >20% of glands. These findings have led to much debate about how to detect nodules and which nodules to investigate further. Most authorities still rely on physical examination to detect thyroid nodules, reserving ultrasound for monitoring nodule size or as an aid in thyroid biopsy.

It is important to distinguish whether a patient presents with a solitary thyroid nodule or a prominent nodule in the context of a [MNG](#), as the incidence of malignancy is greater in solitary nodules. An approach to the evaluation of a solitary nodule is outlined in [Fig. 330-13](#). Most patients with thyroid nodules have normal thyroid function tests. Nonetheless, thyroid function should be assessed by measuring a [TSH](#) level, which may be suppressed by one or more autonomously functioning nodules. If the TSH is suppressed, a radionuclide scan is indicated to determine if the identified nodule is "hot," as lesions with increased uptake are almost never malignant and [FNA](#) is unnecessary. Otherwise, FNA biopsy should be the first step in the evaluation of a thyroid nodule. FNA has good sensitivity and specificity when performed by physicians familiar with the procedure and when the results are interpreted by experienced cytopathologists. The technique is particularly accurate for detecting [PTC](#). The distinction of benign and malignant follicular lesions is often not possible using cytology alone.

In several large studies, [FNA](#) biopsies yield the following findings: 70% benign, 10% malignant or suspicious for malignancy, and 20% nondiagnostic or yielding insufficient material for diagnosis. Characteristic features of malignancy mandate surgery. A diagnosis of follicular neoplasm also warrants surgery, as benign and malignant lesions cannot be distinguished based on cytopathology or frozen section. The management of patients with benign lesions is more variable. Many authorities advocate [TSH](#) suppression, whereas others monitor nodule size without suppression. With either approach, thyroid nodule size should be monitored, either by palpation or ultrasound. Repeat FNA is indicated if a nodule enlarges, and most authorities recommend a second biopsy within 2 to 5 years to confirm the benign status of the nodule.

Nondiagnostic biopsies occur for many reasons, including a fibrotic reaction with relatively few cells available for aspiration, a cystic lesion in which cellular components reside along the cyst margin, or a nodule that may be too small for accurate aspiration. For these reasons, ultrasound-guided [FNA](#) is useful when the FNA is repeated. Ultrasound is also increasingly used for initial biopsies in an effort to enhance nodule localization and the accuracy of sampling.

The evaluation of a thyroid nodule is stressful for most patients. They are concerned about the possibility of thyroid cancer, whether verbalized or not. It is constructive, therefore, to review the diagnostic approach and to reassure patients when malignancy is not found. When a suspicious lesion or thyroid cancer is identified, an explanation of

the generally favorable prognosis and available treatment options should be provided.

(Bibliography omitted in Palm version)

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331. DISORDERS OF THE ADRENAL CORTEX - Gordon H. Williams, Robert G. Dluhy

BIOCHEMISTRY AND PHYSIOLOGY

The adrenal cortex produces three major classes of steroids: (1) glucocorticoids, (2) mineralocorticoids, and (3) adrenal androgens. Consequently, normal adrenal function is important for modulating intermediary metabolism and immune responses through glucocorticoids; blood pressure, vascular volume, and electrolytes through mineralocorticoids; and secondary sexual characteristics (in females) through androgens. The adrenal axis plays an important role in the stress response by rapidly increasing cortisol levels. Adrenal disorders include hyperfunction (Cushing's syndrome) and hypofunction (adrenal insufficiency) as well as a variety of genetic abnormalities of steroidogenesis.

STEROID NOMENCLATURE

Steroids contain as their basic structure a cyclopentenoperhydrophenanthrene nucleus consisting of three 6-carbon hexane rings and a single 5-carbon pentane ring ([Fig. 331-1](#)). The carbon atoms are numbered in a sequence beginning with ring A. Adrenal steroids contain either 19 or 21 carbon atoms. The C₁₉steroids have methyl groups at C-18 and C-19. C₁₉steroids with a ketone group at C-17 are termed *17-ketosteroids*; C₁₉steroids have predominantly androgenic activity. The C₂₁steroids have a 2-carbon side chain (C-20 and C-21) attached at position 17 and methyl groups at C-18 and C-19; C₂₁steroids with a hydroxyl group at position 17 are termed *17-hydroxycorticosteroids*. The C₂₁steroids have either glucocorticoid or mineralcorticoid properties.

BIOSYNTHESIS OF ADRENAL STEROIDS

Cholesterol, derived from the diet and from endogenous synthesis, is the substrate for steroidogenesis. Uptake of cholesterol by the adrenal cortex is mediated by the low-density lipoprotein (LDL) receptor. With long-term stimulation of the adrenal cortex by adrenocorticotrophic hormone (ACTH), the number of [LDL](#) receptors increases. The three major adrenal biosynthetic pathways lead to the production of glucocorticoids (cortisol), mineralocorticoids (aldosterone), and adrenal androgens (dehydroepiandrosterone). Separate zones of the adrenal cortex synthesize specific hormones ([Fig. 331-2](#)). This zonation is accompanied by the selective expression of the genes encoding the enzymes unique to the formation of each type of steroid: aldosterone synthase is normally expressed only in the outer (glomerulosa) cell layer, whereas 17-hydroxylase is expressed only in the (inner) fasciculata-reticularis cell layers, which are the sites of cortisol and androgen biosynthesis, respectively.

STEROID TRANSPORT

Cortisol occurs in the plasma in three forms: free cortisol, protein-bound cortisol, and cortisol metabolites. *Free cortisol* is physiologically active hormone that is not protein-bound and, can act therefore, directly on tissue sites. Normally, <5% of circulating cortisol is free. Only the unbound cortisol and its metabolites are filterable at

the glomerulus. Increased quantities of free steroid are excreted in the urine in states characterized by hypersecretion of cortisol, because the unbound fraction of plasma cortisol rises. Plasma has two cortisol-binding systems. One is a high-affinity, low-capacity α_2 -globulin termed *transcortin* or *cortisol-binding globulin* (CBG), and the other is a low-affinity, high-capacity protein, *albumin*. The binding affinity of CBG for cortisol is reduced in areas of inflammation, thus increasing the local concentration of free cortisol. When the concentration of cortisol exceeds 700 nmol/(25 ug/dL), part of the excess binds to albumin, and a greater proportion than usual circulates unbound. The CBG level is increased in high-estrogen states (e.g., pregnancy, oral contraceptive administration). The rise in CBG is accompanied by a parallel rise in *protein-bound cortisol*, with the result that the plasma cortisol concentration is elevated. However, the free cortisol level probably remains normal, and manifestations of glucocorticoid excess are absent. Most synthetic glucocorticoid analogues bind less efficiently to CBG (~70% binding). This may explain the propensity of some synthetic analogues to produce cushingoid effects at low doses. *Cortisol metabolites* are biologically inactive and bind only weakly to circulating plasma proteins.

Aldosterone is bound to proteins to a smaller extent than cortisol, and an ultrafiltrate of plasma contains as much as 50% of the circulating concentration of aldosterone.

STEROID METABOLISM AND EXCRETION

Glucocorticoids The daily secretion of cortisol ranges between 40 and 80 μ mol (15 and 30 mg, 8-10 mg/m²), with a pronounced circadian cycle. The plasma concentration of cortisol is determined by the rate of secretion, the rate of inactivation, and the rate of excretion of free cortisol. The liver is the major organ responsible for steroid inactivation. A major enzyme regulating cortisol metabolism is 11 β -hydroxysteroid dehydrogenase (11 β -HSD). There are two isoforms: 11 β -HSD I is primarily expressed in the liver and acts as a reductase, converting the inactive cortisone to the active glucocorticoid, cortisol; the 11 β -HSD II isoform is expressed in a number of tissues and converts cortisol to the inactive metabolite cortisone. The oxidative reaction of 11 β -HSD I is increased in hyperthyroidism.

Mineralocorticoids In normal individuals with a normal salt intake, the average daily secretion of aldosterone ranges between 0.1 and 0.7 μ mol (50 and 250 ug). During a single passage through the liver, >75% of circulating aldosterone is normally inactivated by ring A reduction and conjugation with glucuronic acid because it is only weakly bound to proteins. However, under certain conditions, such as congestive failure, this rate of inactivation is reduced.

Adrenal Androgens The major androgen secreted by the adrenal is dehydroepiandrosterone (DHEA) and its sulfuric acid ester (DHEAS). From 15 to 30 mg of these compounds is secreted daily. Smaller amounts of androstenedione, 11 β -hydroxyandrostenedione, and testosterone are secreted. [DHEA](#) is the major precursor of the urinary 17-ketosteroids. Two-thirds of the urine 17-ketosteroids in the male are derived from adrenal metabolites, and the remaining one-third comes from testicular androgens. In the female, almost all urine 17-ketosteroids are derived from the adrenal.

Steroids diffuse passively through the cell membrane and bind to intracellular receptors ([Chap. 327](#)). Glucocorticoids and mineralocorticoids bind with nearly equal affinity to the mineralocorticoid receptor (MR). However, only glucocorticoids bind to the glucocorticoid receptor (GR). After the steroid binds to the receptor, the steroid-receptor complex is transported to the nucleus, where it binds to specific sites on steroid-regulated genes, altering levels of transcription. Some actions of glucocorticoids (e.g., anti-inflammatory effects) are mediated by GR-mediated inhibition of other transcription factors, such as activating protein-1 (AP-1) or nuclear factor kappa-B (NFKB), which normally stimulate the activity of various cytokine genes. Because cortisol binds to the MR with the same affinity as aldosterone, mineralocorticoid specificity is achieved by local metabolism of cortisol to the inactive compound cortisone by 11b-[HSDII](#). The glucocorticoid effects of other steroids, such as high-dose progesterone, correlate with their relative binding affinities for the GR. Inherited defects in the GR cause glucocorticoid resistance states. Individuals with GR defects have high levels of cortisol but do not have manifestations of hypercortisolism.

In addition to the classic genomic effects, which are mediated by steroids binding to cytosolic receptors, evidence is accumulating that mineralocorticoids also have acute, nongenomic effects, presumably by activating a cell-surface receptor yet to be identified. This effect uses a G-protein signaling pathway; among the actions is modification of the sodium-hydrogen exchanger. This effect has been demonstrated in both epithelial and nonepithelial cells, e.g., myocytes and leukocytes.

[ACTH](#)PHYSIOLOGY

ACTH and a number of other peptides (lipotropins, endorphins, and melanocyte-stimulating hormones) are processed from a larger precursor molecule of 31,000 mol wt -- pro-opiomelanocortin (POMC) ([Chap. 328](#)). POMC is made in a variety of tissues, including brain, anterior and posterior pituitary, and lymphocytes. The constellation of POMC-derived peptides secreted depends on the tissue. ACTH, a 39-amino acid peptide, is synthesized and stored in basophilic cells of the anterior pituitary. The *N*-terminal 18-amino acid fragment of ACTH has full biologic potency, and shorter *N*-terminal fragments have partial biologic activity. Release of ACTH and related peptides from the anterior pituitary gland is stimulated by corticotropin-releasing hormone (CRH), a 41-amino acid peptide produced in the median eminence of the hypothalamus ([Fig. 331-3](#)). Urocortin, a neuropeptide related to CRH, also binds to CRH receptors. Urocortin mimics many of the central effects of CRH (e.g., appetite suppression, anxiety), but its role in ACTH regulation is unclear. Some related peptides such as b-lipotropin (b-LPT) are released in equimolar concentrations with ACTH, suggesting that they are cleaved enzymatically from the parent POMC before or during the secretory process. However, b-endorphin levels may or may not correlate with circulating levels of ACTH, depending on the nature of the stimulus. The functions and regulation of secretion of the related peptides derived from POMC are poorly understood.

The major factors controlling [ACTH](#) release include [CRH](#), the free cortisol concentration in plasma, stress, and the sleep-wake cycle ([Fig. 331-3](#)). The plasma level of ACTH varies during the day as a result of its pulsatile secretion, and it follows a circadian pattern with a peak just prior to waking and a nadir before sleeping. If a new sleep-wake cycle is

adopted, the pattern changes over several days to conform to it. ACTH and cortisol levels also increase in response to eating. Stress (e.g., pyrogens, surgery, hypoglycemia, exercise, and severe emotional trauma) causes the release of CRH and arginine vasopressin (AVP) and activation of the sympathetic nervous system. These changes in turn enhance ACTH release, acting individually or in concert. For example, AVP release acts synergistically with CRH to amplify ACTH secretion; CRH also stimulates the locus coeruleus/sympathetic system. Stress-related secretion of ACTH abolishes the circadian periodicity of ACTH levels but is, in turn, suppressed by prior high-dose glucocorticoid administration. The normal pulsatile, circadian pattern of ACTH release is regulated by CRH; this mechanism is the so-called open feedback loop. CRH secretion, in turn, is influenced by hypothalamic neurotransmitters. For example, serotonergic and cholinergic systems stimulate the secretion of CRH and ACTH; there is contradictory evidence regarding the inhibitory effects of α -adrenergic agonists and γ -aminobutyric acid (GABA) on CRH release. In addition, there may be direct pituitary effects of these neurotransmitters. There is also evidence for peptidergic regulation of ACTH release. For example, β -endorphin and enkephalin inhibit the secretion of ACTH, whereas vasopressin and angiotensin II augment it. The immune system also influences the hypothalamic-pituitary-adrenal axis ([Fig. 331-4](#)). For example, inflammatory cytokines [tumor necrosis factor (TNF)- α , interleukin (IL)-1 α , IL-1 β , and IL-6] produced by monocytes increase ACTH release by stimulating secretion of CRH and/or AVP. Finally, ACTH release is regulated by the level of free cortisol in plasma. Cortisol decreases the responsiveness of pituitary corticotrophic cells to CRH; the response of the [POMC](#) mRNA to CRH is also inhibited by glucocorticoids. In addition, glucocorticoids inhibit the locus coeruleus/sympathetic system and CRH release. The latter servomechanism establishes the primacy of cortisol in the control of ACTH secretion. The inhibition of ACTH occurs in two phases: (1) an early fast feedback, mediated via the MR, which lasts <10 min and depends on both the rate of increase of glucocorticoid levels and the specific glucocorticoid administered; and (2) a time-dependent, delayed feedback, likely mediated by the GR, which is probably due to inhibition of synthesis of the precursor protein. The suppression of ACTH secretion that results in adrenal atrophy following *prolonged* glucocorticoid therapy is caused primarily by suppression of hypothalamic CRH release, as exogenous CRH administration in this circumstance produces a rise in plasma ACTH. Cortisol also exerts feedback effects on higher brain centers (hippocampus, reticular system, and septum) and perhaps on the adrenal cortex ([Fig. 331-4](#)).

The biologic half-life of [ACTH](#) in the circulation is <10 min. The action of ACTH is also rapid; within minutes of its release, the concentration of steroids in the adrenal venous blood increases. ACTH stimulates steroidogenesis via activation of the membrane-bound adenylyl cyclase. Adenosine-3',5'-monophosphate (cyclic AMP), in turn, activates protein kinase enzymes, thereby resulting in the phosphorylation of proteins that activate steroid biosynthesis.

RENIN-ANGIOTENSIN PHYSIOLOGY (See also [Chap. 246](#))

Renin is a proteolytic enzyme that is produced and stored in the granules of the juxtaglomerular cells surrounding the afferent arterioles of glomeruli in the kidney. Renin exists in both active and inactive forms. Whether the inactive form is a precursor ("prorenin") or is a product formed after release is uncertain. Renin acts on the basic

substrate angiotensinogen (a circulating α_2 -globulin made in the liver) to form the decapeptide angiotensin I ([Fig. 331-5](#)). Angiotensin I is then enzymatically transformed by angiotensin-converting enzyme (ACE), which is present in many tissues (particularly the pulmonary vascular endothelium), to the octapeptide angiotensin II by the removal of the two C-terminal amino acids. Angiotensin II is a potent pressor agent and exerts its action by a direct effect on arteriolar smooth muscle. In addition, angiotensin II stimulates production of aldosterone by the zona glomerulosa of the adrenal cortex; the heptapeptide angiotensin III may also stimulate aldosterone production. The two major classes of angiotensin receptors are termed *AT1* and *AT2*; *AT1* may exist as two subtypes *a* and *b*. Most of the effects of angiotensins II and III are mediated by the *AT1* receptor. Angiotensinases rapidly destroy angiotensin II (half-life, approximately 1 min), while the half-life of renin is more prolonged (10 to 20 min). In addition to circulating renin-angiotensin, many tissues have a local renin-angiotensin system and the ability to produce angiotensin II. These tissues include the uterus, placenta, vascular tissue, heart, brain, and, particularly, the adrenal cortex and kidney. Although the role of locally generated angiotensin II is not established, it may be involved in the growth and modulation of function of the adrenal cortex and vascular smooth muscle.

The amount of renin released reflects the combined effects of four interdependent factors. The *juxtaglomerular cells*, which are specialized myoepithelial cells that cuff the afferent arterioles, act as miniature pressure transducers, sensing renal perfusion pressure and corresponding changes in afferent arteriolar perfusion pressures. For example, under conditions of a reduction in circulating blood volume, there is a corresponding reduction in renal perfusion pressure and, therefore, in afferent arteriolar pressure ([Fig. 331-5](#)). This change is perceived by the juxtaglomerular cells as a decreased stretch exerted on the afferent arteriolar walls. The juxtaglomerular cells then release more renin into the renal circulation. This results in the formation of angiotensin I, which is converted in the kidney and peripherally to angiotensin II by [ACE](#). Angiotensin II influences sodium homeostasis via two major mechanisms: it changes renal blood flow so as to maintain a constant glomerular filtration rate, thereby changing the filtration fraction of sodium, and it stimulates the adrenal cortex to release aldosterone. Increasing plasma levels of aldosterone enhance renal sodium retention and thus result in expansion of the extracellular fluid volume, which, in turn, dampens the stimulus for renin release. In this context, the renin-angiotensin-aldosterone system regulates volume by modifying renal hemodynamics and tubular sodium transport.

A second control mechanism for renin release is centered in the *macula densa* cells, a group of distal convoluted tubular epithelial cells directly apposed to the juxtaglomerular cells. They may function as chemoreceptors, monitoring the sodium (or chloride) load presented to the distal tubule, and such information may be conveyed to the juxtaglomerular cells, where appropriate modifications in renin release take place. Under conditions of increased delivery of filtered sodium to the macula densa, increasing release of renin decreases the glomerular filtration rate, thereby reducing the filtered load of sodium.

The *sympathetic nervous system* regulates the release of renin in response to assumption of the upright posture. The mechanism is either a direct effect on the juxtaglomerular cell to increase adenyl cyclase activity or an indirect effect on either the juxtaglomerular or the macula densa cells via vasoconstriction of the afferent arteriole.

Finally, circulating factors influence renin release. Increased dietary intake of *potassium* decreases, and decreased potassium intake increases, renin release. The significance of these effects is unclear. *Angiotensin II* exerts negative feedback control on renin release that is independent of alterations in renal blood flow, blood pressure, or aldosterone secretion. *Atrial natriuretic peptides* also inhibit renin release. Thus, the control of renin release involves both *intrarenal* (pressor receptor and macula densa) and *extrarenal* (sympathetic nervous system, potassium, angiotensin, etc.) mechanisms. Steady-state renin levels reflect all these factors, with the intrarenal mechanism predominating.

GLUCOCORTICOID PHYSIOLOGY

The division of adrenal steroids into glucocorticoids and mineralocorticoids is arbitrary in that most glucocorticoids have some mineralocorticoid-like properties. The descriptive term *glucocorticoid* is used for adrenal steroids whose predominant action is on intermediary metabolism. Their overall actions are directed at enhancing the production of the high-energy fuel, glucose, and reducing all other metabolic activity not directly involved in that process. Sustained activation, however, results in a pathophysiologic state, e.g., Cushing's syndrome. The principal glucocorticoid is cortisol (hydrocortisone). The effect of glucocorticoids on intermediary metabolism is mediated by the [GR](#). Physiologic effects of glucocorticoids include the regulation of protein, carbohydrate, lipid, and nucleic acid metabolism. Glucocorticoids raise the blood glucose level by acting as an insulin antagonist and by suppressing the secretion of insulin, thereby inhibiting peripheral glucose uptake, which promotes hepatic glucose synthesis (gluconeogenesis) and increases hepatic glycogen content. The actions on protein metabolism are mainly catabolic in effect, resulting in an increase in protein breakdown and nitrogen excretion. In large part, these actions reflect a mobilization of glycogenic amino acid precursors from peripheral supporting structures, such as bone, skin, muscle, and connective tissue, due to protein breakdown and inhibition of protein synthesis and amino acid uptake. Hyperaminoacidemia also facilitates gluconeogenesis by stimulating glucagon secretion. Glucocorticoids act directly on the liver to stimulate the synthesis of certain enzymes, such as tyrosine aminotransferase and tryptophan pyrrolase. Glucocorticoids regulate fatty acid mobilization by enhancing the activation of cellular lipase by lipid-mobilizing hormones (e.g., catecholamines and pituitary peptides).

The actions of cortisol on protein and adipose tissue vary in different parts of the body. For example, pharmacologic doses of cortisol can deplete the protein matrix of the vertebral column (trabecular bone), whereas long bones (which are primarily compact bone) are affected only minimally; similarly, peripheral adipose tissue mass decreases, whereas abdominal and interscapular fat expand.

Glucocorticoids have anti-inflammatory properties, which are probably related to effects on the microvasculature and to suppression of inflammatory cytokines. In this sense, glucocorticoids modulate the immune response via the so-called immune-adrenal axis ([Fig. 331-4](#)). This "loop" is one mechanism by which a stress, such as sepsis, increases adrenal hormone secretion, and the elevated cortisol level in turn suppresses the immune response. For example, cortisol maintains vascular responsiveness to

circulating vasoconstrictors and opposes the increase in capillary permeability during acute inflammation. Glucocorticoids cause a leukocytosis due to release from the bone marrow of mature cells as well as to inhibition of their egress through the capillary wall. Glucocorticoids produce a depletion of circulating eosinophils and of lymphoid tissue, specifically T cells, by causing a redistribution from the circulation into other compartments. Thus, cortisol impairs cell-mediated immunity. Glucocorticoids also inhibit the production and action of the mediators of inflammation, such as the lymphokines and prostaglandins. These actions occur via the [GR](#) and are blocked by inhibitors of RNA and protein synthesis. Glucocorticoids inhibit the production and action of interferon by T lymphocytes and the production of [IL-1](#) and IL-6 by macrophages. The antipyretic action of glucocorticoids may be explained by the effect on IL-1, which appears to be an endogenous pyrogen ([Chap. 17](#)). Glucocorticoids also inhibit the production of T cell growth factor (IL-2) by T lymphocytes. Glucocorticoids reverse macrophage activation and antagonize the action of migration-inhibiting factor (MIF), leading to reduced adherence of macrophages to vascular endothelium. Glucocorticoids inhibit prostaglandin and leukotriene production by inhibiting the activity of phospholipase A₂, thus blocking release of arachidonic acid from phospholipids. Finally, glucocorticoids inhibit the production and inflammatory effects of bradykinin, platelet-activating factor, and serotonin. It is probably only at pharmacologic dosages that antibody production is reduced and lysosomal membranes are stabilized, the latter effect suppressing the release of acid hydrolases.

Cortisol levels respond within minutes to stress, whether physical (trauma, surgery, exercise), psychological (anxiety, depression), or physiologic (hypoglycemia, fever). The reasons why elevated glucocorticoid levels protect the organism under stress are not understood, but in conditions of glucocorticoid deficiency, such stresses may cause hypotension, shock, and death. Consequently, in individuals with adrenal insufficiency, glucocorticoid administration should be increased during stress.

Cortisol has major effects on body water. It helps regulate the extracellular fluid volume by retarding the migration of water into cells and by promoting renal water excretion, the latter effect mediated by suppression of vasopressin secretion, by an increase in the rate of glomerular filtration, and by a direct action on the renal tubule. The consequence is to prevent water intoxication by increasing solute-free water clearance. Glucocorticoids also have weak mineralocorticoid-like properties, and high doses promote renal tubular sodium reabsorption and increased urine potassium excretion. Glucocorticoids also can influence behavior; emotional disorders may occur with either an excess or a deficit of cortisol. Last, cortisol suppresses the secretion of pituitary [POMC](#) and its derivative peptides ([ACTH](#), b-endorphin, and [b-LPT](#)) and the secretion of hypothalamic [CRH](#) and vasopressin.

MINERALOCORTICOID PHYSIOLOGY

Mineralocorticoids are major regulators of extracellular fluid volume and the major determinant of potassium metabolism. These effects are mediated by the binding of aldosterone to the [MR](#) in target tissues, primarily the kidney. Volume is regulated through a direct effect on the collecting duct, where aldosterone causes an increase in sodium retention and an increase in potassium excretion. The reabsorption of sodium ions causes a fall in the transmembrane potential, thus enhancing the flow of positive

ions, such as potassium, out of the cell into the lumen. The reabsorbed sodium ions are transported out of the tubular epithelium into the renal interstitial fluid and from there into the renal capillary circulation. Water passively follows the transported sodium.

Because the concentration of hydrogen ion is greater in the lumen than in the cell, hydrogen ion is also actively secreted. Mineralocorticoids also act on the epithelium of the salivary ducts, sweat glands, and gastrointestinal tract to cause reabsorption of sodium in exchange for potassium.

When normal individuals are given aldosterone, an initial period of sodium retention is followed by natriuresis, and sodium balance is reestablished after 3 to 5 days. As a result, edema does not develop. This process is referred to as the *escape phenomenon*, signifying an "escape" by the renal tubules from the sodium-retaining action of aldosterone. While renal hemodynamic factors may play a role in the escape, the level of atrial natriuretic peptide also increases. However, it is important to realize that there is no escape from the potassium-losing effects of mineralocorticoids.

There are additional nonclassic effects of mineralocorticoids, primarily on nonepithelial cells. These effects are likely genomic and therefore mediated through activation of the cytosolic MR, but they do not include a modification of sodium-potassium homeostasis. They are probably mediated by mineralocorticoids modifying the expression of several collagen genes and/or genes controlling tissue growth factors, e.g., transforming growth factor β (TGF- β) and plasminogen activator inhibitor (PAI-1). The resultant effects lead to microangiopathy, necrosis (acutely), and fibrosis in a variety of tissues, e.g., heart, kidney, and vasculature. Increased levels of aldosterone are not necessary to produce this damage; rather, an imbalance between the level of aldosterone and the volume and/or sodium balance state appears to be the critical factor.

Three primary mechanisms control aldosterone release -- the renin-angiotensin system, potassium, and [ACTH \(Table 331-1\)](#). The renin-angiotensin system controls extracellular fluid volume via regulation of aldosterone secretion ([Fig. 331-5](#)). In effect, the renin-angiotensin system maintains the circulating blood volume constant by causing aldosterone-induced sodium retention during volume deficiency and by decreasing aldosterone-dependent sodium retention when volume is ample.

Potassium ion directly stimulates aldosterone secretion, independent of the circulating renin-angiotensin system, which it suppresses ([Fig. 331-5](#)). In addition to potassium's direct effect, it also modifies aldosterone secretion indirectly by activating the local renin-angiotensin system in the zona glomerulosa. This effect can be blocked by the administration of [ACE](#) inhibitors that reduce the local production of angiotensin II and thereby reduce the acute aldosterone response to potassium. An increase in serum potassium of as little as 0.1 mmol/L increases plasma aldosterone levels under certain circumstances. Oral potassium loading therefore increases aldosterone secretion, excretion, and plasma levels.

Physiologic amounts of [ACTH](#) stimulate aldosterone secretion acutely, but this action is not sustained unless ACTH is administered in a pulsatile fashion. Most studies relegate ACTH to a minor role in the control of aldosterone. For example, subjects receiving high-dose glucocorticoid therapy, and with presumed complete suppression of ACTH,

have normal aldosterone secretion in response to sodium restriction.

Prior dietary intake of both potassium and sodium can alter the magnitude of the aldosterone response to acute stimulation. This effect results from a change in the expression and activity of aldosterone synthase. Increasing potassium intake or decreasing sodium intake sensitizes the response of the glomerulosa cells to acute stimulation by [ACTH](#), angiotensin II, and/or potassium. Thus, regulation of aldosterone secretion occurs both early and late in its synthetic pathway.

Neurotransmitters (dopamine and serotonin) and some peptides, such as atrial natriuretic peptide, α -melanocyte-stimulating hormone (α -MSH), and β -endorphin, also participate in the regulation of aldosterone secretion ([Table 331-1](#)). Thus, the control of aldosterone secretion involves both stimulatory and inhibitory factors.

ANDROGEN PHYSIOLOGY

Androgens regulate male secondary sexual characteristics and can cause virilizing symptoms in women ([Chap. 53](#)). Adrenal androgens have a minimal effect in males whose sexual characteristics are predominately determined by gonadal steroids (testosterone). In females, however, several androgen-like effects, e.g., sexual hair, are largely mediated by adrenal androgens. The principal adrenal androgens are [DHEA](#), androstenedione, and 11-hydroxyandrostenedione. DHEA and androstenedione are weak androgens and exert their effects via conversion to the potent androgen testosterone in extraglandular tissues. DHEA also has poorly understood effects on the immune and cardiovascular systems. Adrenal androgen formation is regulated by [ACTH](#), not by gonadotropins. It follows that adrenal androgens are suppressed by exogenous glucocorticoid administration.

LABORATORY EVALUATION OF ADRENOCORTICAL FUNCTION

A basic assumption is that measurements of the plasma or urinary level of a given steroid reflects the rate of adrenal *secretion* of that steroid. However, urine *excretion* values may not truly reflect the secretion rate because of improper collection or altered metabolism. Plasma levels reflect the level of secretion only at the time of measurement. The plasma level (*PL*) depends on two factors: the secretion rate (*SR*) of the hormone and the rate at which it is metabolized, i.e., its metabolic clearance rate (*MCR*). These three factors can be related as follows:

BLOOD LEVELS

Peptides The plasma levels of [ACTH](#) and angiotensin II can be measured by immunoassay techniques. Basal ACTH secretion shows a circadian rhythm, with lower levels in the early evening than in the morning. However, ACTH is secreted in a pulsatile manner, leading to rapid fluctuations superimposed on this circadian rhythm. Angiotensin II levels also vary diurnally and are influenced by dietary sodium and potassium intakes and posture. Both upright posture and sodium restriction elevate angiotensin II levels.

Most clinical determinations of the renin-angiotensin system, however, involve measurements of peripheral *plasma renin activity* (PRA) in which the renin activity is gauged by the generation of angiotensin I during a standardized incubation period. This method depends on the presence of sufficient angiotensinogen in the plasma as substrate. The generated angiotensin I is measured by radioimmunoassay. The PRA depends on the dietary sodium intake and on whether the patient is ambulatory. In normal humans, the PRA shows a diurnal rhythm characterized by peak values in the morning and decreases in activity in the afternoon. An alternative approach is to measure plasma active renin, which is easier and not dependent on endogenous substrate concentration. PRA and active renin correlate very well on low-sodium diets ($r = 0.85$ to 0.9) but less well on high-sodium diets.

Steroids Cortisol and aldosterone are both secreted episodically, and levels generally vary during the day, with peak values in the morning and low levels in the evening. In addition, the plasma level of aldosterone, but not of cortisol, is increased by dietary potassium loading, by sodium restriction, or by assumption of the upright posture. Measurement of the sulfate conjugate of [DHEA](#) may be a useful index of adrenal androgen secretion, as little DHEA sulfate is formed in the gonads and because the half-life of DHEA sulfate is 7 to 9 h. However, DHEA sulfate levels reflect both DHEA production and sulfatase activity.

URINE LEVELS

For the assessment of glucocorticoid secretion, the urine *17-hydroxycorticosteroid* assay has been replaced by measurement of urinary free cortisol. Elevated levels of urinary free cortisol correlate with states of hypercortisolism, reflecting changes in the levels of unbound, physiologically active circulating cortisol. Normally, the rate of excretion is higher in the daytime (7 A.M. to 7 P.M.) than at night (7 P.M. to 7 A.M.).

Urinary *17-ketosteroids* originate in either the adrenal gland or the gonad. In normal women, 90% of urinary 17-ketosteroids is derived from the adrenal, and in men 60 to 70% is of adrenal origin. Urine 17-ketosteroid values are highest in young adults and decline with age.

A carefully timed urine collection is a prerequisite for all excretory determinations. Urinary creatinine should be measured simultaneously to determine the accuracy and adequacy of the collection procedure.

STIMULATION TESTS

Stimulation tests are useful in the diagnosis of hormone deficiency states.

Tests of Glucocorticoid Reserve Within minutes after administration of [ACTH](#), cortisol levels increase. This responsiveness can be used as an index of the functional reserve of the adrenal gland for production of cortisol. Under maximal ACTH stimulation, cortisol secretion increases tenfold, to 800 $\mu\text{mol/d}$ (300 mg/d), but maximal stimulation can be achieved only with prolonged ACTH infusions.

A screening test (the so-called rapid [ACTH](#) stimulation test) involves the administration of 25 units (0.25 mg) of cosyntropin intravenously or intramuscularly and measurement of plasma cortisol levels before and 30 and 60 min after administration; the test can be performed at any time of the day. The most clear-cut criterion for a normal response is a stimulated cortisol level of >500 nmol/L (>18 ug/dL), and the minimal stimulated normal increment of cortisol is >200 nmol/L (>7 ug/dL) above baseline. Severely ill patients with elevated basal cortisol levels may show no further increases following acute ACTH administration.

Tests of Mineralocorticoid Reserve and Stimulation of the Renin-Angiotensin System Stimulation tests use protocols designed to create a programmed volume depletion, such as sodium restriction, diuretic administration, or upright posture. A simple, potent test consists of severe sodium restriction and upright posture. After 3 to 5 days of a 10-mmol/d sodium intake, rates of aldosterone secretion or excretion should increase two- to threefold over the control values. Supine morning plasma aldosterone levels are usually increased three- to sixfold, and they increase a further two- to fourfold in response to 2 to 3 h of upright posture.

When the dietary sodium intake is normal, stimulation testing requires the administration of a potent diuretic, such as 40 to 80 mg furosemide, followed by 2 to 3 h of upright posture. The normal response is a two- to fourfold rise in plasma aldosterone levels.

SUPPRESSION TESTS

Suppression tests to document hypersecretion of adrenal hormones involve measurement of the target hormone response after standardized suppression of its tropic hormone.

Tests of Pituitary-Adrenal Suppressibility The [ACTH](#) release mechanism is sensitive to the circulating glucocorticoid level. When blood levels of glucocorticoid are increased in normal individuals, less ACTH is released from the anterior pituitary and less steroid is produced by the adrenal gland. The integrity of this feedback mechanism can be tested clinically by giving a glucocorticoid and judging the suppression of ACTH secretion by analysis of urine steroid levels and/or plasma cortisol and ACTH levels. A potent glucocorticoid such as dexamethasone is used, so that the agent can be given in an amount small enough not to contribute significantly to the pool of steroids to be analyzed.

The best *screening* procedure is the overnight dexamethasone suppression test. This involves the measurement of plasma cortisol levels at 8 A.M. following the oral administration of 1 mg dexamethasone the previous midnight. The 8 A.M. value for plasma cortisol in normal individuals should be <140 nmol/L (5 ug/dL).

The definitive test of adrenal suppressibility consists in administering 0.5 mg dexamethasone every 6 h for two successive days while collecting urine over a 24-h period for determination of creatinine and free cortisol and/or measuring plasma cortisol levels. In a patient with a normal hypothalamic-pituitary [ACTH](#) release mechanism, a fall in the urine free cortisol to <80 nmol/d (30 ug/d) or of plasma cortisol to <140 nmol/L (5 ug/dL) is seen on the second day of administration.

A normal response to either suppression test implies that the glucocorticoid regulation of [ACTH](#) and its control of the adrenal glands is physiologically normal. However, an isolated abnormal result, particularly to the overnight suppression test, does not in itself imply pituitary and/or adrenal disease.

Tests of Mineralocorticoid Suppressibility These tests rely on an expansion of extracellular fluid volume, which should decrease circulating plasma renin activity and decrease the secretion and/or excretion of aldosterone. Various tests differ in the rate at which extracellular fluid volume is expanded. One convenient suppression test involves the intravenous infusion of 500 mL/h of normal saline solution for 4 h, which normally suppresses plasma aldosterone levels to <220 pmol/L (<8 ng/dL) on a sodium-restricted diet or to <140 pmol/L (<5 ng/dL) on a normal sodium intake. Alternatively, a high-sodium diet can be administered for 3 days with 0.2 mg fludrocortisone twice daily. Aldosterone excretion is measured on the third day and should be <28 nmol/d (10 ug/d). These tests should not be performed in potassium-depleted individuals since they carry a risk of precipitating hypokalemia.

TESTS OF PITUITARY-ADRENAL RESPONSIVENESS

Stimuli such as insulin-induced hypoglycemia, [AVP](#), and pyrogens cause the release of [ACTH](#) from the pituitary by an action on higher neural centers or on the pituitary itself. Insulin-induced hypoglycemia is particularly useful, because it stimulates the release of both growth hormone and ACTH. In this test, regular insulin (0.05 to 0.1 U/kg body weight) is given intravenously as a bolus to reduce the fasting glucose level to at least 50% below basal. The normal cortisol response is a rise to more than 500 nmol/L (18 ug/dL).

One of the best ways to test the integrity of the pituitary-adrenal axis is the metyrapone test. Metyrapone inhibits 11 β -hydroxylase in the adrenal. As a result, the conversion of 11-deoxycortisol (compound S) to cortisol is impaired, causing 11-deoxycortisol to accumulate in the blood and the blood level of cortisol to decrease ([Fig. 331-2](#)). The hypothalamic-pituitary axis responds to the declining cortisol blood levels by releasing more [ACTH](#). Note that assessment of the response depends on both an intact hypothalamic-pituitary axis and an intact adrenal gland.

Although modifications of the original metyrapone test have been described, we believe the best involves administering 750 mg of the drug by mouth every 4 h over a 24-h period and comparing the control and postmetyrapone plasma levels of 11-deoxycortisol, cortisol, and [ACTH](#). In normal individuals, plasma 11-deoxycortisol levels should exceed 210 nmol/L (7 ug/dL) and ACTH levels should exceed 17 pmol/L (75 pg/mL) following metyrapone administration. The metyrapone test does not accurately reflect ACTH reserve if subjects are ingesting exogenous glucocorticoids or drugs that accelerate the metabolism of metyrapone (e.g., phenytoin).

A direct and selective test of the pituitary corticotrophs can be achieved with [CRH](#). The bolus injection of ovine CRH (corticotropin-releasing hormone; 1 ug/kg body weight) stimulates secretion of [ACTH](#) and [b-LPH](#) in normal human subjects within 15 to 60 min. In normal individuals, the mean increment in ACTH is 9 pmol/L (40 pg/mL). However, the

magnitude of the ACTH response is less than that produced by the insulin tolerance test, which implies that additional factors (such as vasopressin) augment stress-induced increases in ACTH secretion.

Although the rapid [ACTH](#) stimulation test is useful for the diagnosis of primary adrenal insufficiency, normal cortisol responsiveness may be seen in some patients with a partial ACTH deficit and absence of adrenal atrophy. These patients have an inadequate pituitary ACTH reserve and fail to increase ACTH secretion in response to a stress such as surgery or hypoglycemia. Because the use of a bolus of exogenous ACTH does not invariably exclude a diagnosis of secondary adrenocortical insufficiency, direct tests of pituitary ACTH reserve (metyrapone test, insulin tolerance testing) may be required in the appropriate clinical setting. Alternatively, ACTH at a physiologic dose (1 ug), the so-called low-dose ACTH test, may be used. Abnormal response is similar to the rapid ACTH test. However, levels need to be measured at 30 min, and the ACTH needs to be injected directly intravenously because it can be absorbed to plastic tubing. On the other hand, the rapid ACTH test can distinguish between primary and secondary adrenal insufficiency, because aldosterone secretion is preserved in secondary adrenal failure by the renin-angiotensin system and potassium. Cosyntropin (25 units) is given intravenously or intramuscularly, and plasma cortisol and aldosterone levels are measured before and 30 and 60 min after administration. The cortisol response is abnormal in both groups, but patients with secondary insufficiency show an increase in aldosterone levels by at least 140 pmol/L (5 ng/dL). No aldosterone response is seen in patients in whom the adrenal cortex is destroyed.

HYPERFUNCTION OF THE ADRENAL CORTEX

Excess cortisol is associated with Cushing's syndrome; excess aldosterone causes aldosteronism; and excess adrenal androgens cause adrenal virilism. These syndromes do not always occur in the "pure" form but may have overlapping features.

CUSHING'S SYNDROME

Etiology Cushing described a syndrome characterized by truncal obesity, hypertension, fatigability and weakness, amenorrhea, hirsutism, purplish abdominal striae, edema, glucosuria, osteoporosis, and a basophilic tumor of the pituitary. As awareness of this syndrome has increased, the diagnosis of Cushing's syndrome has been broadened into the classification shown in [Table 331-2](#). Regardless of etiology, all cases of endogenous Cushing's syndrome are due to increased production of cortisol by the adrenal. In most cases the cause is *bilateral adrenal hyperplasia* due to hypersecretion of pituitary [ACTH](#) or ectopic production of ACTH by a nonpituitary source. The incidence of pituitary-dependent adrenal hyperplasia is three times greater in women than in men, and the most frequent age of onset is the third or fourth decade. Most evidence indicates that the primary defect is the de novo development of a pituitary adenoma, as tumors are found in >90% of patients with pituitary-dependent adrenal hyperplasia. Alternatively, the defect may occasionally reside in the hypothalamus or in higher neural centers, leading to release of [CRH](#) inappropriate to the level of circulating cortisol. The consequence would be that a higher level of cortisol is required to reduce ACTH secretion to normal. This primary defect leads to hyperstimulation of the pituitary, resulting in hyperplasia or tumor formation. In surgical series, most individuals with

hypersecretion of pituitary ACTH are found to have a microadenoma (<10 mm in diameter; 50% are ≤5 mm in diameter), but a pituitary macroadenoma (>10 mm) or diffuse hyperplasia of the corticotrophic cells may be found. In some studies, the recurrence rate is >20%. Unfortunately, it may be difficult to distinguish between recurrence and inadequate primary therapy. Traditionally, only an individual who has an ACTH-producing pituitary tumor is defined as having *Cushing's disease*.

Nonpituitary tumors may secrete polypeptides that are biologically, chemically, and immunologically indistinguishable from either [ACTH](#) or [CRH](#) and that cause bilateral adrenal hyperplasia ([Chap. 100](#)). The ectopic production of CRH results in clinical, biochemical, and radiologic features indistinguishable from those caused by hypersecretion of pituitary ACTH. The typical signs and symptoms of Cushing's syndrome may be absent or minimal with ectopic ACTH production, and hypokalemic alkalosis is a prominent manifestation. Most of these cases are associated with the primitive small cell (oat cell) type of bronchogenic carcinoma or with tumors of the thymus, pancreas, or ovary; medullary carcinoma of the thyroid; or bronchial adenomas. The onset of Cushing's syndrome may be sudden, particularly in patients with carcinoma of the lung, and this feature accounts in part for the failure of these patients to exhibit the classic manifestations. On the other hand, patients with carcinoid tumors or pheochromocytomas have longer clinical courses and usually exhibit the typical cushingoid features. The ectopic secretion of ACTH is also accompanied by the accumulation of ACTH fragments in plasma and by elevated plasma levels of ACTH precursor molecules. Because such tumors may produce large amounts of ACTH, baseline steroid values are usually markedly elevated, and increased skin pigmentation may be present. Indeed, hyperpigmentation in patients with Cushing's syndrome almost always points to an extraadrenal tumor, either in an extracranial location or within the cranium.

Approximately 20 to 25% of patients with Cushing's syndrome have an adrenal neoplasm. These tumors are usually unilateral, and about half are malignant. Occasionally, patients have biochemical features both of pituitary [ACTH](#) excess and of an adrenal adenoma. These individuals usually have *nodular hyperplasia* of both adrenal glands often the result of prolonged ACTH stimulation in the absence of a pituitary adenoma. Two additional entities cause nodular hyperplasia: a familial disorder in children or young adults (so-called pigmented micronodular dysplasia; see below) and an abnormal cortisol response to gastric inhibitory polypeptide or luteinizing hormone, probably secondary to expression of receptors for these hormones in the adrenal cortex.

The most common cause of Cushing's syndrome is *iatrogenic* administration of steroids for a variety of reasons. Although the clinical features bear some resemblance to those seen with adrenal tumors, these patients are usually distinguishable on the basis of history and laboratory studies.

Clinical Signs, Symptoms, and Laboratory Findings Many of the signs and symptoms of Cushing's syndrome follow logically from the known action of glucocorticoids ([Table 331-3](#)). Mobilization of peripheral supportive tissue causes muscle weakness and fatigability, osteoporosis, broad violaceous cutaneous striae, and easy bruisability. The latter signs are secondary to weakening and rupture of collagen

fibers in the dermis. Osteoporosis may cause collapse of vertebral bodies and pathologic fractures of other bones. Decreased bone mineralization is particularly pronounced in children. Increased hepatic gluconeogenesis and insulin resistance can cause impaired glucose tolerance. Overt diabetes mellitus occurs in <20% of patients, who probably are individuals with a predisposition to this disorder. Hypercortisolism promotes the deposition of adipose tissue in characteristic sites, notably the upper face (producing the typical "moon" facies), the interscapular area (producing the "buffalo hump"), and the mesenteric bed (producing "truncal" obesity) ([Fig. 331-6](#)). Rarely, episternal fatty tumors and mediastinal widening secondary to fat accumulation occur. The reason for this peculiar distribution of adipose tissue is not known, but it is associated with insulin resistance and/or elevated insulin levels. The face appears plethoric, even in the absence of any increase in red blood cell concentration ([Fig. 331-CD1](#)). Hypertension is common, and emotional changes may be profound, ranging from irritability and emotional lability to severe depression, confusion, or even frank psychosis. In women, increased levels of adrenal androgens can cause acne, hirsutism, and oligomenorrhea or amenorrhea. Some signs and symptoms in patients with hypercortisolism -- i.e., obesity, hypertension, osteoporosis, and diabetes -- are nonspecific and therefore are less helpful in diagnosing the condition. On the other hand, easy bruising, typical striae, myopathy, and virilizing signs (although less frequent) are, if present, more suggestive of Cushing's syndrome ([Table 331-3](#)).

Except in iatrogenic Cushing's syndrome, plasma and urine cortisol levels are variably elevated. Occasionally, hypokalemia, hypochloremia, and metabolic alkalosis are present, particularly with ectopic production of [ACTH](#).

Diagnosis The diagnosis of Cushing's syndrome depends on the demonstration of increased cortisol production and failure to suppress cortisol secretion normally when dexamethasone is administered ([Chap. 328](#)). Once the diagnosis is established, further testing is designed to determine the etiology ([Fig. 331-7](#) and [Table 331-4](#)).

For initial screening, the overnight dexamethasone suppression test is recommended (see above). In difficult cases (e.g., in obese patients), measurement of a 24-h urine free cortisol also can be used as a screening test. A level >140 nmol/d (50 ug/d) is suggestive of Cushing's syndrome. The definitive diagnosis is then established by failure of urinary cortisol to fall to less than <25 nmol/d (10 ug/d) or of plasma cortisol to fall to <140 nmol/L (5 ug/dL) after a standard low-dose dexamethasone suppression test (0.5 mg every 6 h for 48 h). Owing to circadian variability, plasma cortisol and, to a certain extent, [ACTH](#) determinations are not meaningful when performed in isolation, but the absence of the normal fall of plasma cortisol at midnight is consistent with Cushing's syndrome.

The task of determining the etiology of Cushing's syndrome is complicated by the fact that all the available tests lack specificity and by the fact that the tumors producing this syndrome are prone to spontaneous and often dramatic changes in hormone secretion (periodic hormonogenesis). No test has a specificity >95%, and it may be necessary to use a combination of tests to arrive at the correct diagnosis. A useful step to distinguish patients with an [ACTH](#)-secreting pituitary microadenoma or hypothalamic-pituitary dysfunction from those with other forms of Cushing's syndrome is to determine the response of cortisol output to administration of high-dose dexamethasone (2 mg every 6

h for 2 days). An alternative 8-mg, overnight high-dose dexamethasone test has been developed; however, this test has a lower sensitivity and specificity than the standard test. When the diagnosis of Cushing's syndrome is clear-cut on the basis of baseline urinary and plasma assays, the high-dose dexamethasone suppression test may be used without performing the preliminary low-dose suppression test. The high-dose suppression test provides close to 100% specificity if the criterion used is suppression of urinary free cortisol by >90%. Occasionally, in individuals with bilateral nodular hyperplasia and/or ectopic [CRH](#) production, steroid output is also suppressed. Failure of low- and high-dose dexamethasone administration to suppress cortisol production ([Table 331-4](#)) is usual in patients with adrenal hyperplasia secondary to an ACTH-secreting pituitary macroadenoma or an ACTH-producing tumor of nonendocrine origin and in those with adrenal neoplasms.

Plasma [ACTH](#) levels can be useful in distinguishing the various causes of Cushing's syndrome, particularly in separating ACTH-dependent from ACTH-independent causes. In general, measurement of plasma ACTH is useful in the diagnosis of ACTH-independent etiologies of the syndrome, since most adrenal tumors cause low or undetectable ACTH levels [<2 pmol/L (10 pg/mL)]. Furthermore, ACTH-secreting pituitary macroadenomas and ACTH-producing nonendocrine tumors usually result in elevated ACTH levels. In the ectopic ACTH syndrome, ACTH levels may be elevated to >110 pmol/L (500 pg/mL), and in most patients the level is >40 pmol/L (200 pg/mL). In Cushing's syndrome as the result of a microadenoma or pituitary-hypothalamic dysfunction, ACTH levels range from 6 to 30 pmol/L (30 to 150 pg/mL) [normal, <14 pmol/L (<60 pg/mL)], with half of values falling in the normal range. However, the main problem with the use of ACTH levels in the differential diagnosis of Cushing's syndrome is that ACTH levels may be similar in individuals with hypothalamic-pituitary dysfunction, pituitary microadenomas, ectopic [CRH](#) production, and ectopic ACTH production (especially carcinoid tumors) ([Table 331-4](#)).

Because of these difficulties, several additional tests have been advocated, such as the metyrapone and [CRH](#) infusion tests. The rationale underlying these tests is that steroid hypersecretion by an adrenal tumor or the ectopic production of [ACTH](#) will suppress the hypothalamic-pituitary axis so that inhibition of pituitary ACTH release can be demonstrated by either test. Thus, most patients with pituitary-hypothalamic dysfunction and/or a microadenoma have an increase in steroid or ACTH secretion in response to metyrapone or CRH administration, whereas most patients with ectopic ACTH-producing tumors do not. Most pituitary macroadenomas also respond to CRH, but their response to metyrapone is variable. However, false-positive and -negative CRH tests can occur in patients with ectopic ACTH and pituitary tumors.

The main diagnostic dilemma in Cushing's syndrome is to distinguish those instances due to microadenomas of the pituitary and/or pituitary-hypothalamic dysfunction from those due to tumors (e.g., carcinoids or pheochromocytoma) that produce [CRH](#) and/or [ACTH](#) ectopically. The clinical manifestations are similar unless the ectopic tumor produces other symptoms, such as diarrhea and flushing from a carcinoid tumor or episodic hypertension from a pheochromocytoma. Sometimes, one can distinguish between ectopic and pituitary ACTH production by using metyrapone or CRH tests, as noted above. In these situations, computed tomography (CT) of the pituitary gland is usually normal. Magnetic resonance imaging (MRI) with the enhancing

agent gadolinium may be better than CT for this purpose but demonstrates pituitary microadenomas in only half of patients with Cushing's disease. In subjects with negative imaging studies, selective petrosal sinus venous sampling for ACTH is employed in some centers. Demonstration of an ACTH gradient between the petrosal sinus and peripheral blood localizes the source of ACTH overproduction to the pituitary gland but does not distinguish pituitary-dependent adrenal hyperplasia from pituitary hyperplasia secondary to a tumor producing CRH. CRH levels should be measured in the peripheral blood prior to petrosal sinus sampling. In centers where petrosal sinus sampling is performed frequently, it has proved useful for distinguishing pituitary and nonpituitary sources of ACTH excess. However, the catheterization procedure is technically difficult, and complications have occurred.

The diagnosis of a *cortisol-producing adrenal adenoma* is suggested by disproportionate elevations in baseline urine free-cortisol levels with only modest changes in urinary 17-ketosteroids or plasma [DHEA](#)sulfate. Adrenal androgen secretion is usually reduced in these patients owing to the cortisol-induced suppression of [ACTH](#) and subsequent involution of the androgen-producing zona reticularis.

The diagnosis of *adrenal carcinoma* is suggested by a palpable abdominal mass and by *markedly* elevated baseline values of *both* urine 17-ketosteroids and plasma [DHEA](#)sulfate. Plasma and urine cortisol levels are variably elevated. Adrenal carcinoma is usually resistant to both [ACTH](#) stimulation and dexamethasone suppression. Elevated adrenal androgen secretion often leads to virilization in the female. Estrogen-producing adrenocortical carcinoma usually presents with gynecomastia in men and dysfunctional uterine bleeding in women. These adrenal tumors secrete increased amounts of androstenedione, which is converted peripherally to the estrogens estrone and estradiol ([Chap. 337](#)). Adrenal carcinomas that produce Cushing's syndrome are often associated with elevated levels of the intermediates of steroid biosynthesis (especially 11-deoxycortisol), suggesting inefficient conversion of the intermediates to the final product. This feature also accounts for the characteristic increase in 17-ketosteroids. Approximately 20% of adrenal carcinomas are not associated with endocrine syndromes and are presumed to be nonfunctioning or to produce biologically inactive steroid precursors. In addition, the excessive production of steroids is not always clinically evident (e.g., androgens in adult men).

Differential Diagnosis

Pseudo-Cushing's Syndrome Problems in diagnosis include patients with obesity, chronic alcoholism, depression, and acute illness of any type. Extreme *obesity* is uncommon in Cushing's syndrome; furthermore, with exogenous obesity, the adiposity is generalized, not truncal. On adrenocortical testing, abnormalities in patients with exogenous obesity are usually modest. Basal urine steroid excretion levels in obese patients are also either normal or slightly elevated. Some patients have elevated conversion of secreted cortisol into excreted metabolites. Urinary and blood cortisol levels are usually normal, and the diurnal pattern in blood and urine levels is normal. Patients with *chronic alcoholism* and those with *depression* share similar abnormalities in steroid output: modestly elevated urine cortisol, blunted circadian rhythm of cortisol levels, and resistance to suppression using the overnight dexamethasone test. In contrast to alcoholic subjects, depressed patients do not have signs and symptoms of

Cushing's syndrome. Following discontinuation of alcohol and/or improvement in the emotional status, results of steroid testing usually return to normal. One or more of three tests have been used to differentiate mild Cushing's syndrome and pseudo-Cushing's syndrome. The serum cortisol level following the standard 2-day low-dose dexamethasone test has very high sensitivity and specificity. While the [CRH](#) test alone is less useful, in combination with the low-dose dexamethasone test, there is nearly complete discrimination between these two conditions. Finally, a midnight cortisol level obtained in awake patients may have similar predictive value as the low-dose dexamethasone test if a cut-off of 210 nmol/L (7.5 ug/dL) is used. Patients with *acute illness* often have abnormal results on laboratory tests and fail to exhibit pituitary-adrenal suppression in response to dexamethasone, since major stress (such as pain or fever) interrupts the normal regulation of [ACTH](#) secretion. *Iatrogenic Cushing's syndrome*, induced by the administration of glucocorticoids or other steroids such as megestrol that bind to the glucocorticoid receptor, is indistinguishable by physical findings from endogenous adrenocortical hyperfunction. The distinction can be made, however, by measuring blood or urine cortisol levels in a basal state; in the iatrogenic syndrome these levels are low secondary to suppression of the pituitary-adrenal axis. The severity of iatrogenic Cushing's syndrome is related to the total steroid dose, the biologic half-life of the steroid, and the duration of therapy. Also, individuals taking afternoon and evening doses of glucocorticoids develop Cushing's syndrome more readily and with a smaller total daily dose than do patients taking morning doses only. The enzymatic disposition and binding of administered steroids differ among patients.

Radiologic Evaluation for Cushing's Syndrome The preferred radiologic study for visualizing the adrenals is a [CT](#) scan of the abdomen ([Fig. 331-8](#)). CT is of value both for localizing adrenal tumors and for diagnosing bilateral hyperplasia. All patients believed to have hypersecretion of pituitary [ACTH](#) should have a pituitary [MRI](#) scan with the contrast agent gadolinium. Even with this technique, small microadenomas may be undetectable; alternatively, false-positive masses due to cysts or nonsecretory lesions of the normal pituitary may be imaged. In patients with ectopic ACTH production, chest CT is a useful first step.

Evaluation of Asymptomatic Adrenal Masses With abdominal [CT](#) scanning, many incidental adrenal masses (so-called incidentalomas) are discovered. This is not surprising, since 10 to 20% of subjects at autopsy have adrenocortical adenomas. The first step in evaluating such patients is to determine whether the tumor is functioning by means of appropriate screening tests, e.g., measurement of 24-h urine catecholamines and metabolites and serum potassium and assessment of adrenal cortical function by dexamethasone-suppression testing. However, 90% of incidentalomas are nonfunctioning. If an extraadrenal malignancy is present, there is a 30 to 50% chance that the adrenal tumor is a metastasis. If the primary tumor is being treated and there are no other metastases, it is prudent to obtain a fine-needle aspirate of the adrenal mass to establish the diagnosis. In the absence of a known malignancy the next step is unclear. The probability of adrenal carcinoma is <0.01 percent, the vast majority of adrenal masses being benign adenomas. Features suggestive of malignancy include large size (a size >4 to 6 cm suggests carcinoma); irregular margins; and inhomogeneity, soft tissue calcifications visible on CT ([Fig. 331-8](#)), and findings characteristic of malignancy on a chemical-shift [MRI](#) image. If surgery is not performed, a repeat CT scan should be obtained in 3 to 6 months. Fine needle aspiration is not useful

to distinguish between benign and malignant primary adrenal tumors.

TREATMENT

Adrenal Neoplasm When an adenoma or carcinoma is diagnosed, adrenal exploration is performed with excision of the tumor. Adenomas may be resected using laparoscopic techniques. Because of the possibility of atrophy of the contralateral adrenal, the patient is treated pre- and postoperatively as if for total adrenalectomy, even when a unilateral lesion is suspected, the routine being similar to that for an Addisonian patient undergoing elective surgery (see [Table 331-8](#)).

Despite operative intervention, most patients with adrenal carcinoma die within 3 years of diagnosis. Metastases occur most often to liver and lung. The principal drug for the treatment of adrenocortical carcinoma is mitotane (*o,p*-DDD), an isomer of the insecticide DDT. This drug suppresses cortisol production and decreases plasma and urine steroid levels. Although its cytotoxic action is relatively selective for the glucocorticoid-secreting zone of the adrenal cortex, the zona glomerulosa may also be inhibited. Because mitotane also alters the extraadrenal metabolism of cortisol, plasma and urinary cortisol levels must be assessed to titrate the effect. The drug is usually given in divided doses three to four times a day, with the dose increased gradually to tolerability (usually <6 g daily). At higher doses, almost all patients experience side effects, which may be gastrointestinal (anorexia, diarrhea, vomiting) or neuromuscular (lethargy, somnolence, dizziness). All patients treated with mitotane should receive long-term glucocorticoid maintenance therapy, and, in some, mineralocorticoid replacement is appropriate. In approximately one-third of patients, both tumor and metastases regress, but long-term survival is not altered. In many patients, mitotane only inhibits steroidogenesis and does not cause regression of tumor metastases. Osseous metastases are usually refractory to the drug and should be treated with radiation therapy. Mitotane can also be given as adjunctive therapy after surgical resection of an adrenal carcinoma, although there is no evidence that this improves survival. Because of the absence of a long-term benefit with mitotane, alternative chemotherapeutic approaches based on platinum therapy have been used. However, there are no data presently available indicating a prolongation of life.

Bilateral Hyperplasia Patients with hyperplasia have a relative or absolute increase in [ACTH](#) levels. Since therapy would logically be directed at reducing ACTH levels, the ideal primary treatment for ACTH- or [CRH](#)-producing tumors, whether pituitary or ectopic, is surgical removal. Occasionally (particularly with ectopic ACTH production) surgical excision is not possible because the disease is far advanced. In this situation, "medical" or surgical adrenalectomy may correct the hypercortisolism.

Controversy exists as to the proper treatment for bilateral adrenal hyperplasia when the source of the [ACTH](#) overproduction is not apparent. In some centers, these patients (especially those who suppress after the administration of a high-dose dexamethasone test) undergo surgical exploration of the pituitary via a transsphenoidal approach in the expectation that a microadenoma will be found. However, in most circumstances selective petrosal sinus venous sampling is recommended, and the patient is referred to an appropriate center if the procedure is not available locally. If a microadenoma is not found at the time of exploration, total hypophysectomy may be needed. Complications

of transsphenoidal surgery include cerebrospinal fluid rhinorrhea, diabetes insipidus, panhypopituitarism, and optic or cranial nerve injuries.

In other centers, total adrenalectomy is the treatment of choice. The cure rate with this procedure is close to 100%. The adverse effects include the certain need for lifelong mineralocorticoid and glucocorticoid replacement and a 10 to 20% probability of a pituitary tumor developing over the next 10 years (Nelson's syndrome; [Chap. 328](#)). Many of these tumors require surgical therapy. It is uncertain whether they arise de novo in these patients or were present prior to adrenalectomy but were too small to be detected. Periodic radiologic evaluation of the pituitary gland by [MRI](#) as well as serial [ACTH](#) measurements should be performed in all individuals after bilateral adrenalectomy for Cushing's disease. Such pituitary tumors may become locally invasive and impinge on the optic chiasm or extend into the cavernous or sphenoid sinuses.

Except in children, pituitary irradiation is rarely used as primary treatment, being reserved rather for postoperative tumor recurrences. In some centers, high levels of gamma radiation can be focused on the desired site with less scattering to surrounding tissues by using stereotactic techniques. Side effects of radiation include ocular motor palsy and hypopituitarism. There is a long lag time between treatment and remission, and the remission rate is usually <50%.

Finally, in occasional patients in whom a surgical approach is not feasible, "medical" adrenalectomy may be indicated ([Table 331-5](#)). Inhibition of steroidogenesis may also be indicated in severely cushingoid subjects prior to surgical intervention. Chemical adrenalectomy may be accomplished by the administration of the inhibitor of steroidogenesis ketoconazole (600 to 1200 mg/d). In addition, mitotane (2 or 3 g/d) and/or the blockers of steroid synthesis aminoglutethimide (1 g/d) and metyrapone (2 or 3 g/d) may be effective either alone or in combination. Mitotane is slow to take effect (weeks). Mifepristone, a competitive inhibitor of the binding of glucocorticoid to its receptor, may be a treatment option. Adrenal insufficiency is a risk with all these agents, and replacement steroids may be required.

ALDOSTERONISM

Aldosteronism is a syndrome associated with hypersecretion of the mineralocorticoid aldosterone. In *primary* aldosteronism the cause for the excessive aldosterone production resides within the adrenal gland; in *secondary* aldosteronism the stimulus is extraadrenal.

Primary Aldosteronism In the original case of excessive and inappropriate aldosterone production, the disease was the result of an *aldosterone-producing adrenal adenoma* (Conn's syndrome). Most cases involve a unilateral adenoma, which is usually small and may occur on either side. Rarely, primary aldosteronism is due to an adrenal carcinoma. Aldosteronism is twice as common in women as in men, usually occurs between the ages of 30 and 50, and is present in approximately 1% of unselected hypertensive patients. However, the prevalence may be as high as 10%, depending on the criteria and study population. Most of this difference is not secondary to the prevalence of patients with an aldosteronoma but rather because of the inclusion of

those with bilateral hyperplasia. In many patients with clinical and biochemical features of primary aldosteronism, a solitary adenoma is not found at surgery. Instead, these patients have *bilateral cortical nodular hyperplasia*. In the literature, this disease is also termed *idiopathic hyperaldosteronism*, and/or *nodular hyperplasia*. The cause is unknown.

Signs and Symptoms Hypersecretion of aldosterone increases the renal distal tubular exchange of intratubular sodium for secreted potassium and hydrogen ions, with progressive depletion of body potassium and development of hypokalemia. Most patients have diastolic hypertension, which may be very severe, and headaches. The hypertension is probably due to the increased sodium reabsorption and extracellular volume expansion. *Potassium depletion* is responsible for the muscle weakness and fatigue and is due to the effect of potassium depletion on the muscle cell membrane. The polyuria results from impairment of urinary concentrating ability and is often associated with polydipsia.

Electrocardiographic and roentgenographic signs of left ventricular enlargement are, in part, secondary to the hypertension. However, the left ventricular hypertrophy is disproportionate to the level of blood pressure when compared to individuals with essential hypertension, and regression of the hypertrophy occurs even if blood pressure is not reduced after removal of an aldosteronoma. Electrocardiographic signs of potassium depletion include prominent U waves, cardiac arrhythmias, and premature contractions. In the absence of associated congestive heart failure, renal disease, or preexisting abnormalities (such as thrombophlebitis), edema is characteristically absent. However, structural damage to the cerebral circulation, retinal vasculature, and kidney occurs more frequently than would be predicted based on the level and duration of the hypertension. Proteinuria may occur in as many as 50% of patients with primary aldosteronism, and renal failure occurs in up to 15%. Thus, it is probable that excess aldosterone production induces cardiovascular damage independent of its effect on blood pressure.

Laboratory Findings Laboratory findings depend on both the duration and the severity of potassium depletion. An overnight concentration test often reveals impaired ability to concentrate the urine, probably secondary to the hypokalemia. Urine pH is neutral to alkaline because of excessive secretion of ammonium and bicarbonate ions to compensate for the metabolic alkalosis.

Hypokalemia may be severe (<3 mmol/L) and reflects body potassium depletion, usually >300 mmol. In mild forms of primary aldosteronism, potassium levels may be normal. *Hypernatremia* is due to sodium retention, a concomitant water loss from polyuria, and a resetting of the osmostat. Metabolic alkalosis and elevation of serum bicarbonate are a result of hydrogen ion loss into the urine and migration into potassium-depleted cells. The alkalosis is perpetuated by potassium deficiency, which increases the capacity of the proximal convoluted tubule to reabsorb filtered bicarbonate. If hypokalemia is severe, serum magnesium levels are also reduced.

Diagnosis The diagnosis is suggested by persistent hypokalemia in a nonedematous patient with a normal sodium intake who is not receiving potassium-wasting diuretics (furosemide, ethacrynic acid, thiazides). If hypokalemia occurs in a hypertensive patient

taking a potassium-wasting diuretic, the diuretic should be discontinued and the patient should be given potassium supplements. After 1 to 2 weeks, the potassium level should be remeasured, and if hypokalemia persists, the patient should be evaluated for a mineralocorticoid excess syndrome ([Fig. 331-9](#)).

The criteria for the diagnosis of primary aldosteronism are (1) diastolic hypertension without edema, (2) hyposecretion of renin (as judged by low plasma renin activity levels) that fails to increase appropriately during volume depletion (upright posture, sodium depletion), and (3) hypersecretion of aldosterone that does not suppress appropriately in response to volume expansion.

Patients with primary aldosteronism characteristically *do not have edema*, since they exhibit an "escape" phenomenon from the sodium-retaining aspects of mineralocorticoids. Rarely, pretibial edema is present in patients with associated nephropathy and azotemia.

The estimation of plasma renin activity is of limited value in separating patients with primary aldosteronism from those with hypertension of other causes. Although failure of plasma renin activity to rise normally during volume-depletion maneuvers is a criterion for a diagnosis of primary aldosteronism, suppressed renin activity also occurs in about 25% of patients with essential hypertension.

Although a renin measurement alone lacks specificity, the ratio of serum aldosterone to plasma renin activity is a very useful screening test. A high ratio (>30), when aldosterone is expressed as ng/dL and plasma renin activity as ng/mL per hour, strongly suggests autonomy of aldosterone secretion. Aldosterone levels need to be >500 pmol/L (>15 ng/dL) and the salt intake not be restricted in making this assessment. Ultimately, it is necessary to demonstrate a lack of aldosterone suppression to diagnose primary aldosteronism ([Fig. 331-9](#)). The autonomy exhibited in these patients refers only to the resistance to suppression of secretion during volume expansion; aldosterone can and does respond in a normal or above-normal fashion to the stimulus of potassium loading or [ACTH](#) infusion.

Once hyposecretion of renin and failure of aldosterone secretion suppression are demonstrated, aldosterone-producing adenomas should be localized by abdominal [CT](#) scan, using a high-resolution scanner as many aldosteronomas are <1 cm in size. If the CT scan is negative, percutaneous transfemoral bilateral adrenal vein catheterization with adrenal vein sampling may demonstrate a two- to threefold increase in plasma aldosterone concentration on the involved side. In cases of hyperaldosteronism secondary to cortical nodular hyperplasia, no lateralization is found. It is important for samples to be obtained simultaneously if possible and for cortisol levels to be measured to ensure that false localization does not reflect dilution or an [ACTH](#)- or stress-induced rise in aldosterone levels. In a patient with an adenoma, the aldosterone/cortisol ratio lateralizes to the side of the lesion.

Differential Diagnosis Patients with hypertension and hypokalemia may have either primary or secondary hyperaldosteronism ([Fig. 331-10](#)). A useful maneuver to distinguish between these conditions is the measurement of plasma renin activity. Secondary hyperaldosteronism in patients with accelerated hypertension is due to

elevated plasma renin levels; in contrast, patients with primary aldosteronism have suppressed plasma renin levels. Indeed, in patients with a serum potassium concentration of <2.5 mmol/L, a high ratio of plasma aldosterone to plasma renin activity in a random sample is usually sufficient to establish the diagnosis of primary aldosteronism without additional testing.

Primary aldosteronism must also be distinguished from other *hypermineralocorticoid states*. Nonaldosterone mineralocorticoid states will have suppressed plasma renin activity but low aldosterone levels. The most common problem is to distinguish between hyperaldosteronism due to an adenoma and that due to idiopathic bilateral nodular hyperplasia. This distinction is of importance because hypertension associated with idiopathic hyperplasia is usually not benefited by bilateral adrenalectomy, whereas hypertension associated with aldosterone-producing tumors is usually improved or cured by removal of the adenoma. Although patients with idiopathic bilateral nodular hyperplasia tend to have less severe hypokalemia, lower aldosterone secretion, and higher plasma renin activity than do patients with primary aldosteronism, differentiation is impossible solely on clinical and/or biochemical grounds. An anomalous postural decrease in plasma aldosterone and elevated plasma 18-hydroxycorticosterone levels are present in most patients with a unilateral lesion. However, these tests are also of limited diagnostic value in the individual patient, because some adenoma patients have an increase in plasma aldosterone with upright posture, so-called renin-responsive aldosteronoma. A definitive diagnosis is best made by radiographic studies, including bilateral adrenal vein catheterization, as noted above.

In a few instances, hypertensive patients with hypokalemic alkalosis have adenomas that secrete deoxycorticosterone (DOC). Such patients have reduced plasma renin activity levels, but aldosterone levels are either normal or reduced, suggesting the diagnosis of mineralocorticoid excess due to a hormone other than aldosterone. Several inherited disorders have clinical features similar to those of primary aldosteronism (see below).

TREATMENT

Primary aldosteronism due to an adenoma is usually treated by surgical excision of the adenoma. Where possible a laparoscopic approach is favored. However, dietary sodium restriction and the administration of an aldosterone antagonist, e.g., spironolactone, are effective in many cases. Hypertension and hypokalemia are usually controlled by doses of 25 to 100 mg spironolactone every 8 h. In some patients medical management has been successful for years, but chronic therapy in men is usually limited by side effects of spironolactone such as gynecomastia, decreased libido, and impotence.

When idiopathic bilateral hyperplasia is suspected, surgery is indicated only when significant, symptomatic hypokalemia cannot be controlled with medical therapy, e.g., by spironolactone, triamterene, or amiloride. Hypertension associated with idiopathic hyperplasia is usually not benefited by bilateral adrenalectomy.

Secondary Aldosteronism *Secondary aldosteronism* refers to an appropriately increased production of aldosterone in response to activation of the renin-angiotensin system ([Fig. 331-10](#)). The production rate of aldosterone is often higher in patients with

secondary aldosteronism than in those with primary aldosteronism. Secondary aldosteronism usually occurs in association with the accelerated phase of hypertension or on the basis of an underlying edema disorder. Secondary aldosteronism in pregnancy is a normal physiologic response to estrogen-induced increases in circulating levels of renin substrate and plasma renin activity and to the antialdosterone actions of progestogens.

Secondary aldosteronism in hypertensive states is due either to a primary overproduction of renin (primary reninism) or to an overproduction of renin secondary to a decrease in renal blood flow and/or perfusion pressure ([Fig. 331-10](#)). Secondary hypersecretion of renin can be due to a narrowing of one or both of the major renal arteries by atherosclerosis or by fibromuscular hyperplasia. Overproduction of renin from both kidneys also occurs in severe arteriolar nephrosclerosis (malignant hypertension) or with profound renal vasoconstriction (the accelerated phase of hypertension). The secondary aldosteronism is characterized by hypokalemic alkalosis, moderate to severe increases in plasma renin activity, and moderate to marked increases in aldosterone levels ([Chap. 246](#)).

Secondary aldosteronism with hypertension can also be caused by rare renin-producing tumors (primary reninism). These patients have the biochemical characteristics of renal vascular hypertension, but the primary defect is renin secretion by a juxtaglomerular cell tumor. The diagnosis can be made by demonstration of normal renal vasculature and/or demonstration of a space-occupying lesion in the kidney by radiographic techniques and documentation of a unilateral increase in renal vein renin activity. Rarely, these tumors arise in tissues such as the ovary.

Secondary aldosteronism is present in many forms of *edema*. The rate of aldosterone secretion is usually increased in patients with edema caused by either cirrhosis or the nephrotic syndrome. In congestive heart failure, elevated aldosterone secretion varies depending on the severity of cardiac failure. The stimulus for aldosterone release in these conditions appears to be *arterial hypovolemia* and/or hypotension. Thiazides and furosemide often exaggerate secondary aldosteronism via volume depletion; hypokalemia and, on occasion, alkalosis can then become prominent features. On occasion secondary hyperaldosteronism occurs without edema or hypertension (Bartter's and Gitelman's syndromes, see below).

SYNDROMES OF ADRENAL ANDROGEN EXCESS

Adrenal androgen excess results from excess production of [DHEA](#) and androstenedione, which are converted to testosterone in extraglandular tissues; elevated testosterone levels account for most of the virilization. Adrenal androgen excess may be associated with the secretion of greater or smaller amounts of other adrenal hormones and may, therefore, present as "pure" syndromes of virilization or as "mixed" syndromes associated with excessive glucocorticoids and Cushing's syndrome. **For further discussion of hirsutism and virilization, see [Chap. 53](#).*

HYPOFUNCTION OF THE ADRENAL CORTEX

Cases of adrenal insufficiency can be divided into two general categories: (1) those

associated with primary inability of the adrenal to elaborate sufficient quantities of hormone, and (2) those associated with a secondary failure due to inadequate [ACTH](#) formation or release ([Table 331-6](#)).

PRIMARY ADRENOCORTICAL DEFICIENCY (ADDISON'S DISEASE)

The original description of Addison's disease -- "general languor and debility, feebleness of the heart's action, irritability of the stomach, and a peculiar change of the color of the skin" -- summarizes the dominant clinical features. Advanced cases are usually easy to diagnose, but recognition of the early phases can be a real challenge.

Incidence Primary insufficiency is relatively rare, may occur at any age, and affects both sexes equally. Because of the common therapeutic use of steroids, secondary adrenal insufficiency is relatively common.

Etiology and Pathogenesis Addison's disease results from progressive destruction of the adrenals, which must involve >90% of the glands before adrenal insufficiency appears. The adrenal is a frequent site for chronic granulomatous diseases, predominantly tuberculosis but also histoplasmosis, coccidioidomycosis, and cryptococcosis. In early series, tuberculosis was responsible for 70 to 90% of cases, but the most frequent cause now is *idiopathic* atrophy, and an autoimmune mechanism is probably responsible. Rarely, other lesions are encountered, such as adrenoleukodystrophy, bilateral hemorrhage, tumor metastases, HIV, cytomegalovirus (CMV), amyloidosis, adrenomyeloneuropathy, familial adrenal insufficiency, or sarcoidosis.

Although half of patients with idiopathic atrophy have circulating adrenal antibodies, autoimmune destruction is probably secondary to cytotoxic T lymphocytes. Specific adrenal antigens to which autoantibodies may be directed include 21-hydroxylase (CYP21A2) and side chain cleavage enzyme but the significance of these antibodies in the pathogenesis of adrenal insufficiency is unknown. Some antibodies cause adrenal insufficiency by blocking the binding of [ACTH](#) to its receptors. Some patients also have antibodies to thyroid, parathyroid, and/or gonadal tissue ([Chap. 339](#)). There is also an increased incidence of chronic lymphocytic thyroiditis, premature ovarian failure, type 1 diabetes mellitus, and hypo- or hyperthyroidism. The presence of two or more of these autoimmune endocrine disorders in the same person defines the polyglandular autoimmune syndrome type II. Additional features include pernicious anemia, vitiligo, alopecia, nontropical sprue, and myasthenia gravis. Within families, multiple generations are affected by one or more of the above diseases. Type II polyglandular syndrome is the result of a mutant gene on chromosome 6 and is associated with the HLA alleles B8 and DR3.

The combination of parathyroid and adrenal insufficiency and chronic mucocutaneous moniliasis constitutes type I polyglandular autoimmune syndrome. Other autoimmune diseases in this disorder include pernicious anemia, chronic active hepatitis, alopecia, primary hypothyroidism, and premature gonadal failure. There is no HLA association; this syndrome is inherited as an autosomal recessive trait. It is caused by mutations in autoimmune polyendocrinopathy candidiasis ectodermal dystrophy (APECED) located on chromosome 21q22.3. The gene encodes a transcription factor thought to be

involved in lymphocyte function. The type I syndrome usually presents during childhood, whereas the type II syndrome is usually manifested in adulthood.

Clinical suspicion of adrenal insufficiency should be high in patients with AIDS ([Chap. 309](#)). [CMV](#) regularly involves the adrenal glands (so-called CMV necrotizing adrenalitis), and involvement with *Mycobacterium avium-intracellulare*, *Cryptococcus*, and Kaposi's sarcoma has been reported. Adrenal insufficiency in AIDS patients may not be manifest, but tests of adrenal reserve frequently give abnormal results. When interpreting tests of adrenocortical function, it is important to remember that medications such as rifampin, phenytoin, ketoconazole, megace, and opiates may cause or potentiate adrenal insufficiency. Adrenal hemorrhage and infarction occur in patients on anticoagulants and in those with circulating anticoagulants and hypercoagulable states, such as the antiphospholipid syndrome.

There are several rare genetic causes of adrenal insufficiency that present primarily in infancy and childhood (see below).

Clinical Signs and Symptoms Adrenocortical insufficiency caused by gradual adrenal destruction is characterized by an insidious onset of fatigability, weakness, anorexia, nausea and vomiting, weight loss, cutaneous and mucosal pigmentation, hypotension, and occasionally hypoglycemia ([Table 331-7](#)). Depending on the duration and degree of adrenal hypofunction, the manifestations vary from mild chronic fatigue to fulminating shock associated with acute destruction of the glands, as described by Waterhouse and Friderichsen.

Asthenia is the cardinal symptom. Early it may be sporadic, usually most evident at times of stress; as adrenal function becomes more impaired, the patient is continuously fatigued, and bed rest is necessary.

Hyperpigmentation may be striking or absent. It commonly appears as a diffuse brown, tan, or bronze darkening of parts such as the elbows or creases of the hand and of areas that normally are pigmented such as the areolae about the nipples. Bluish-black patches may appear on the mucous membranes. Some patients develop dark freckles, and irregular areas of vitiligo may paradoxically be present. As an early sign, tanning following sun exposure may be persistent.

Arterial hypotension with postural accentuation is frequent, and blood pressure may be in the range of 80/50 or less.

Abnormalities of gastrointestinal function are often the presenting complaint. Symptoms vary from mild anorexia with weight loss to fulminating nausea, vomiting, diarrhea, and ill-defined abdominal pain, which may be so severe as to be confused with an acute abdomen. Patients may have personality changes, usually consisting of excessive irritability and restlessness. Enhancement of the sensory modalities of taste, olfaction, and hearing is reversible with therapy. Axillary and pubic hair may be decreased in women due to loss of adrenal androgens.

Laboratory Findings In the early phase of gradual adrenal destruction, there may be no demonstrable abnormalities in the routine laboratory parameters, but adrenal reserve

is decreased -- that is, while basal steroid output may be normal, a subnormal increase occurs after stress. Adrenal stimulation with [ACTH](#) uncovers abnormalities in this stage of the disease, eliciting a subnormal increase of cortisol levels or no increase at all. In more advanced stages of adrenal destruction, serum sodium, chloride, and bicarbonate levels are reduced, and the serum potassium level is elevated. The hyponatremia is due both to loss of sodium into the urine (due to aldosterone deficiency) and to movement into the intracellular compartment. This extravascular sodium loss depletes extracellular fluid volume and accentuates hypotension. Elevated plasma vasopressin and angiotensin II levels may contribute to the hyponatremia by impairing free water clearance. Hyperkalemia is due to a combination of aldosterone deficiency, impaired glomerular filtration, and acidosis. Basal levels of cortisol and aldosterone are subnormal and fail to increase following ACTH administration. Mild to moderate hypercalcemia occurs in 10 to 20% of patients for unclear reasons. The electrocardiogram may show nonspecific changes, and the electroencephalogram exhibits a generalized reduction and slowing. There may be a normocytic anemia, a relative lymphocytosis, and a moderate eosinophilia.

Diagnosis The diagnosis of adrenal insufficiency should be made only with [ACTH](#) stimulation testing to assess adrenal reserve capacity for steroid production (see above for ACTH test protocols). In brief, the best screening test is the cortisol response 60 min after 250 ug of cosyntropin given intramuscularly or intravenously. Cortisol levels should exceed 495 nmol/L (18 ug/dL). If the response is abnormal, then primary and secondary adrenal insufficiency can be distinguished by measuring aldosterone levels from the same blood samples. In secondary, but not primary, adrenal insufficiency the aldosterone increment will be normal [≥ 150 pmol/l (5 ng/dL)]. Furthermore, in primary adrenal insufficiency, plasma ACTH and associated peptides ([b-LPT](#)) are elevated because of loss of the usual cortisol-hypothalamic-pituitary feedback relationship, whereas in secondary adrenal insufficiency, plasma ACTH values are low or "inappropriately" normal ([Fig. 331-11](#)).

Differential Diagnosis Since weakness and fatigue are common, diagnosis of early adrenocortical insufficiency may be difficult. However, the combination of mild gastrointestinal distress, weight loss, anorexia, and a suggestion of increased pigmentation makes it mandatory to perform [ACTH](#) stimulation testing to rule out adrenal insufficiency, particularly before steroid treatment is begun. Weight loss is useful in evaluating the significance of weakness and malaise. Racial pigmentation may be a problem, but a *recent* and progressive *increase* in pigmentation is usually reported by the patient with gradual adrenal destruction. Hyperpigmentation is usually absent when adrenal destruction is rapid, as in bilateral adrenal hemorrhage. The fact that hyperpigmentation occurs with other diseases may also present a problem, but the appearance and distribution of pigment in adrenal insufficiency are usually characteristic. When doubt exists, measurement of ACTH levels and testing of adrenal reserve with the infusion of ACTH provide clear-cut differentiation.

TREATMENT

All patients with adrenal insufficiency should receive specific hormone replacement. Like diabetics, these patients require careful education about the disease. Replacement therapy should correct both glucocorticoid and mineralocorticoid deficiencies.

Hydrocortisone (cortisol) is the mainstay of treatment. The dose for most adults (depending on size) is 20 to 30 mg/d. Patients are advised to take glucocorticoids with meals or, if that is impractical, with milk or an antacid, because the drugs may increase gastric acidity and exert direct toxic effects on the gastric mucosa. To simulate the normal diurnal adrenal rhythm, two-thirds of the dose is taken in the morning, and the remaining one-third is taken in the late afternoon. Some patients exhibit insomnia, irritability, and mental excitement after initiation of therapy; in these, the dosage should be reduced. Other situations that may necessitate smaller doses are hypertension and diabetes mellitus. Obese individuals and those on anticonvulsive medications may require increased dosages. Measurements of plasma [ACTH](#) or cortisol or of urine cortisol levels do not appear to be useful in determining optimal glucocorticoid dosages.

Since the replacement dosage of hydrocortisone does not replace the mineralocorticoid component of the adrenal hormones, mineralocorticoid supplementation is usually needed. This is accomplished by the administration of 0.05 to 0.1 mg fludrocortisone per day by mouth. Patients should also be instructed to maintain an ample intake of sodium (3 to 4 g/d).

The adequacy of mineralocorticoid therapy can be assessed by measurement of blood pressure and serum electrolytes. Blood pressure should be normal and without postural changes; serum sodium, potassium, creatinine, and urea nitrogen levels should also be normal. Measurement of plasma renin levels may also be useful in titrating the dose.

In female patients with adrenal insufficiency, androgen levels are also low. Thus, some physicians believe that daily replacement with 25 to 50 mg of [DHEA](#) orally may improve quality of life and skeletal density.

Complications of glucocorticoid therapy, with the exception of gastritis, are *rare* at the dosages recommended for treatment of adrenal insufficiency. Complications of mineralocorticoid therapy include hypokalemia, hypertension, cardiac enlargement, and even congestive heart failure due to sodium retention. Periodic measurements of body weight, serum potassium level, and blood pressure are useful. All patients with adrenal insufficiency should carry medical identification, should be instructed in the parenteral self-administration of steroids, and should be registered with a medical alerting system.

Special Therapeutic Problems During periods of intercurrent illness, especially in the setting of fever, the dose of hydrocortisone should be doubled. With severe illness it should be increased to 75 to 150 mg/d. When oral administration is not possible, parenteral routes should be employed. Likewise, before surgery or dental extractions, supplemental glucocorticoids should be administered. Patients should also be advised to increase the dose of fludrocortisone and to add salt to their otherwise normal diet during periods of strenuous exercise with sweating, during extremely hot weather, and with gastrointestinal upsets such as diarrhea. A simple strategy is to supplement the diet one to three times daily with salty broth (1 cup of beef or chicken bouillon contains 35 mmol of sodium). For a representative program of steroid therapy for the patient with adrenal insufficiency who is undergoing major surgery, see [Table 331-8](#). This schedule is designed so that on the day of surgery it will mimic the output of cortisol in normal individuals undergoing prolonged major stress (10 mg/h, 250 to 300 mg/d). Thereafter, if the patient is improving and is afebrile, the dose of hydrocortisone is tapered by 20 to

30% daily. Mineralocorticoid administration is unnecessary at hydrocortisone doses >100 mg/d because of the mineralocorticoid effects of hydrocortisone at such dosages.

SECONDARY ADRENOCORTICAL INSUFFICIENCY

[ACTH](#) deficiency causes *secondary* adrenocortical insufficiency; it may be a selective deficiency, as is seen following prolonged administration of excess glucocorticoids, or it may occur in association with deficiencies of multiple pituitary hormones (panhypopituitarism) ([Chap. 328](#)). Patients with secondary adrenocortical hypofunction have many symptoms and signs in common with those having primary disease but are *characteristically not hyperpigmented*, since ACTH and related peptide levels are low. In fact, plasma ACTH levels distinguish between primary and secondary adrenal insufficiency, since they are elevated in the former and decreased to absent in the latter. Patients with total pituitary insufficiency have manifestations of multiple hormone deficiencies. An additional feature distinguishing primary adrenocortical insufficiency is the *near-normal level of aldosterone secretion* seen in pituitary and/or isolated ACTH deficiencies ([Fig. 331-11](#)). Patients with pituitary insufficiency may have hyponatremia, which can be dilutional or secondary to a subnormal increase in aldosterone secretion in response to severe sodium restriction. However, severe *dehydration*, *hyponatremia*, and *hyperkalemia* are characteristic of severe mineralocorticoid insufficiency and favor a diagnosis of primary adrenocortical insufficiency.

Patients receiving long-term steroid therapy, despite physical findings of Cushing's syndrome, develop adrenal insufficiency because of prolonged pituitary-hypothalamic suppression and adrenal atrophy secondary to the loss of endogenous [ACTH](#). These patients have two deficits, a loss of adrenal responsiveness to ACTH and a failure of pituitary ACTH release. They are characterized by low blood cortisol and ACTH levels, a low baseline rate of steroid excretion, and abnormal ACTH and metyrapone responses. Most patients with steroid-induced adrenal insufficiency eventually recover normal hypothalamic-pituitary-adrenal responsiveness, but recovery time varies from days to months. The rapid ACTH test provides a convenient assessment of recovery of hypothalamic-pituitary-adrenal function. Because the plasma cortisol concentrations after injection of cosyntropin and during insulin-induced hypoglycemia are usually similar, the rapid ACTH test assesses the integrated hypothalamic-pituitary-adrenal function (see "Tests of Pituitary-Adrenal Responsiveness," above). Some investigators suggest using the low-dose (1 ug) ACTH test for suspected secondary ACTH deficiency. Additional tests to assess pituitary ACTH reserve include the standard metyrapone and insulin-induced hypoglycemia tests.

Glucocorticoid therapy in patients with secondary adrenocortical insufficiency does not differ from that for the primary disorder. Mineralocorticoid therapy is usually not necessary, as aldosterone secretion is preserved.

ACUTE ADRENOCORTICAL INSUFFICIENCY

Acute adrenocortical insufficiency may result from several processes. On the one hand, *adrenal crisis* may be a rapid and overwhelming intensification of chronic adrenal insufficiency, usually precipitated by sepsis or surgical stress. Alternatively, acute hemorrhagic destruction of both adrenal glands can occur in previously well subjects. In

children, this event is usually associated with septicemia with *Pseudomonas* or meningococcemia (Waterhouse-Friderichsen syndrome). In adults, anticoagulant therapy or a coagulation disorder may result in bilateral adrenal hemorrhage. Occasionally, bilateral adrenal hemorrhage in the newborn results from birth trauma. Hemorrhage has been observed during pregnancy, following idiopathic adrenal vein thrombosis, and as a complication of venography (e.g., infarction of an adenoma). The third and most frequent cause of acute insufficiency is the rapid withdrawal of steroids from patients with adrenal atrophy owing to chronic steroid administration. Acute adrenocortical insufficiency may also occur in patients with congenital adrenal hyperplasia or those with decreased adrenocortical reserve when they are given drugs capable of inhibiting steroid synthesis (mitotane, ketoconazole) or of increasing steroid metabolism (phenytoin, rifampin).

Adrenal Crisis The long-term survival of patients with adrenocortical insufficiency depends largely on the prevention and treatment of adrenal crisis. Consequently, the occurrence of infection, trauma (including surgery), gastrointestinal upsets, or other stresses necessitates an immediate increase in hormone. In untreated patients, preexisting symptoms are intensified. Nausea, vomiting, and abdominal pain may become intractable. Fever may be severe or absent. Lethargy deepens into somnolence, and hypovolemic vascular collapse ensues. In contrast, patients previously maintained on chronic glucocorticoid therapy may not exhibit dehydration or hypotension until they are in a preterminal state, since mineralocorticoid secretion is usually preserved. In all patients in crisis, a precipitating cause should be sought.

TREATMENT

Treatment is directed primarily toward repletion of circulating glucocorticoids and replacement of the sodium and water deficits. Hence an intravenous infusion of 5% glucose in normal saline solution should be started with a bolus intravenous infusion of 100 mg hydrocortisone followed by a continuous infusion of hydrocortisone at a rate of 10 mg/h. An alternative approach is to administer a 100-mg bolus of hydrocortisone intravenously every 6 h. However, only continuous infusion maintains the plasma cortisol constantly at stress levels [>830 nmol/L (30 ug/dL)]. Effective treatment of hypotension requires glucocorticoid replacement and repletion of sodium and water deficits. If the crisis was preceded by prolonged nausea, vomiting, and dehydration, several liters of saline solution may be required in the first few hours. Vasoconstrictive agents (such as dopamine) may be indicated in extreme conditions as adjuncts to volume replacement. With large doses of steroid, e.g., 100 to 200 mg hydrocortisone, the patient receives a maximal mineralocorticoid effect, and supplementary mineralocorticoid is superfluous. Following improvement, the steroid dosage is tapered over the next few days to maintenance levels, and mineralocorticoid therapy is reinstituted if needed ([Table 331-8](#)).

HYPOALDOSTERONISM

Isolated aldosterone deficiency accompanied by normal cortisol production occurs in association with hyporeninism, as an inherited biosynthetic defect, postoperatively following removal of aldosterone-secreting adenomas, during protracted heparin or heparinoid administration, in preterminal disease of the nervous system, and in severe

postural hypotension.

The feature common to all forms hypoaldosteronism is the inability to increase aldosterone secretion appropriately in response to salt restriction. Most patients have unexplained hyperkalemia, which often is exacerbated by restriction of dietary sodium intake. In severe cases, urine sodium wastage occurs at a normal salt intake, whereas in milder forms, excessive loss of urine sodium occurs only with salt restriction.

Most cases of isolated hypoaldosteronism occur in patients with a deficiency in renin production (so-called hyporeninemic hypoaldosteronism), most commonly in adults with diabetes mellitus and mild renal failure and in whom hyperkalemia and metabolic acidosis are out of proportion to the degree of renal impairment. Plasma renin levels fail to rise normally following sodium restriction and postural changes. The pathogenesis is uncertain. Possibilities include renal disease (the most likely), autonomic neuropathy, extracellular fluid volume expansion, and defective conversion of renin precursors to active renin. Aldosterone levels also fail to rise normally after salt restriction and volume contraction; this effect is probably related to the hyporeninism, since biosynthetic defects in aldosterone secretion usually cannot be demonstrated. In these patients, aldosterone secretion increases promptly after [ACTH](#) stimulation, but it is uncertain whether the magnitude of the response is normal. On the other hand, the level of aldosterone appears to be subnormal in relationship to the hyperkalemia.

Hypoaldosteronism can also be associated with high renin levels and low or elevated levels of aldosterone (see below). Severely ill patients may also have hyperreninemic hypoaldosteronism; such patients have a high mortality rate (80%). Hyperkalemia is not present. Possible explanations for the hypoaldosteronism include adrenal necrosis (uncommon) or a shift in steroidogenesis from mineralocorticoids to glucocorticoids, possibly related to prolonged [ACTH](#) stimulation.

Before the diagnosis of isolated hypoaldosteronism is considered for a patient with hyperkalemia, "pseudohyperkalemia" (e.g., hemolysis, thrombocytosis) should be excluded by measuring the *plasma* potassium level. The next step is to demonstrate a normal cortisol response to [ACTH](#) stimulation. Then, the response of renin and aldosterone levels to stimulation (upright posture, sodium restriction) should be measured. Low renin and aldosterone levels establish the diagnosis of hyporeninemic hypoaldosteronism. A combination of high renin levels and low aldosterone levels is consistent with an aldosterone biosynthetic defect or a selective unresponsiveness to angiotensin II. Finally, there is a condition that clinically and biochemically mimics hypoaldosteronism with elevated renin levels. However, the aldosterone levels are not low but high -- so-called pseudohypoaldosteronism. This inherited condition is caused by a mutation in the epithelial sodium channel (see below).

TREATMENT

The treatment is to replace the mineralocorticoid deficiency. For practical purposes, the oral administration of 0.05 to 0.15 mg fludrocortisone daily should restore electrolyte balance if salt intake is adequate (e.g., 150 to 200 mmol/d). However, patients with hyporeninemic hypoaldosteronism may require higher doses of mineralocorticoid to correct hyperkalemia. This need poses a potential risk in patients with hypertension,

mild renal insufficiency, or congestive heart failure. An alternative approach is to reduce salt intake and to administer furosemide, which can ameliorate acidosis and hyperkalemia. Occasionally, a combination of these two approaches is efficacious.

GENETIC CONSIDERATIONS

Glucocorticoid Diseases

Congenital Adrenal Hyperplasia Congenital adrenal hyperplasia (CAH) is the consequence of recessive mutations that cause one of several distinct enzymatic defects (see below). Because cortisol is the principal adrenal steroid regulating [ACTH](#) elaboration and because ACTH stimulates adrenal growth and function, a block in cortisol synthesis may result in the enhanced secretion of adrenal androgens and/or mineralocorticoids depending on the site of the enzyme block. In severe congenital virilizing hyperplasia, the adrenal output of cortisol may be so compromised as to cause adrenal deficiency despite adrenal hyperplasia.

[CAH](#) is the most common adrenal disorder of infancy and childhood ([Chap. 338](#)). Partial enzyme deficiencies can be expressed after adolescence, predominantly in women with hirsutism and oligomenorrhea but minimal virilization. Late-onset adrenal hyperplasia may account for 5 to 25% of cases of hirsutism and oligomenorrhea in women, depending on the population.

ETIOLOGY Enzymatic defects have been described in 21-hydroxylase (CYP21A2), 17 α -hydroxylase/17,20-Lyase (CYP17), 11 β -hydroxylase (CYP11B1), and in (3 β -[HSD](#)2) ([Fig. 331-2](#)). Although the cDNAs for these enzymes have been cloned, the diagnosis of specific enzyme deficiencies with genetic techniques is not practical for routine use. CYP21A2 deficiency is closely linked to the HLA-B locus of chromosome 6 so that HLA typing and/or DNA polymorphism can be used to detect the heterozygous carriers and to diagnose affected individuals in some families ([Chap. 306](#)). The clinical expression in the different disorders is variable, ranging from virilization of the female (CYP21A2) to feminization of the male (3 β -HSD2) ([Chap. 338](#)).

Adrenal virilization in the female at birth is associated with ambiguous external genitalia (*female pseudohermaphroditism*). Virilization probably begins after the fifth month of intrauterine development. At birth there may be enlarged genitalia in the male infant and enlargement of the clitoris, partial or complete fusion of the labia, and sometimes a urogenital sinus in the female. If the labial fusion is nearly complete, the female infant has external genitalia resembling a penis with hypospadias. In the *postnatal* period, CAH is associated with virilization in the female and isosexual precocity in the male. The excessive androgen levels result in accelerated growth, so that bone age exceeds chronologic age. Because epiphyseal closure is hastened by excessive androgens, growth stops, but truncal development continues, the characteristic appearance being a short child with a well-developed trunk.

The most common form of [CAH](#) (95% of cases) is a result of impairment of CYP21A2. In addition to cortisol deficiency, aldosterone secretion is decreased in approximately one-third of the patients. Thus, with CYP21A2 deficiency, adrenal virilization occurs with or without a salt-losing tendency due to aldosterone deficiency ([Fig. 331-2](#)).

CYP11B1 deficiency causes a "hypertensive" variant of [CAH](#). Hypertension and hypokalemia occur because of the impaired conversion of 11-deoxycorticosterone to corticosterone, resulting in the accumulation of 11-deoxycorticosterone, a potent mineralocorticoid. The degree of hypertension is variable. Increased shunting again occurs into the androgen pathway.

CYP17 deficiency is characterized by hypogonadism, hypokalemia, and hypertension. This rare disorder causes decreased production of cortisol and shunting of precursors into the mineralocorticoid pathway with hypokalemic alkalosis, hypertension, and suppressed plasma renin activity. Usually, 11-deoxycorticosterone production is elevated. Because CYP17 hydroxylation is required for biosynthesis of both adrenal androgens and gonadal testosterone and estrogen, this defect is associated with sexual immaturity, high urinary gonadotropin levels, and low urinary 17-ketosteroid excretion. Female patients have primary amenorrhea and lack of development of secondary sexual characteristics. Because of deficient androgen production, male patients have either ambiguous external genitalia or a female phenotype (*male pseudohermaphroditism*). Exogenous glucocorticoids can correct the hypertensive syndrome, and treatment with appropriate gonadal steroids results in sexual maturation.

With 3b-[HSD2](#) deficiency, conversion of pregnenolone to progesterone is impaired, so that the synthesis of both cortisol and aldosterone is blocked, with shunting into the adrenal androgen pathway via 17a-hydroxypregnenolone and [DHEA](#). Because DHEA is a weak androgen, and because this enzyme deficiency is also present in the gonad, the genitalia of the male fetus may be incompletely virilized or feminized. Conversely, in the female, overproduction of [DHEA](#) may produce partial virilization.

DIAGNOSIS The diagnosis of [CAH](#) should be considered in infants having episodes of acute adrenal insufficiency or salt-wasting or with hypertension. The diagnosis is further suggested by the finding of hypertrophy of the clitoris, fused labia, or a urogenital sinus in the female or of isosexual precocity in the male. In infants and children with a CYP21A2 defect, increased urine 17-ketosteroid excretion and increased plasma [DHEA](#)sulfate levels are typically associated with an increase in the blood levels of 17-hydroxyprogesterone and the excretion of its urinary metabolite pregnanetriol. Demonstration of elevated levels of 17-hydroxyprogesterone in amniotic fluid at 14 to 16 weeks of gestation allows prenatal detection of affected female infants.

The diagnosis of a *salt-losing form* of [CAH](#) due to defects in CYP21A2 is suggested by episodes of acute adrenal insufficiency with hyponatremia, hyperkalemia, dehydration, and vomiting. These infants and children often crave salt and have laboratory findings indicating deficits in both cortisol and aldosterone secretion.

With the *hypertensive form* of [CAH](#) due to CYP11B1 deficiency, 11-deoxycorticosterone and 11-deoxycortisol accumulate. The diagnosis is confirmed by demonstrating increased levels of 11-deoxycortisol in the blood or increased amounts of tetrahydro-11-deoxycortisol in the urine. Elevation of 17-hydroxyprogesterone levels does not imply a coexisting CYP21A2 deficiency.

Very high levels of urine [DHEA](#) with low levels of pregnanetriol and of cortisol metabolites

in urine are characteristic of children with 3b-[HSD2](#) deficiency. Marked salt-wasting may also occur.

Adults with *late-onset adrenal hyperplasia* (partial deficiency of CYP21A2, CYP11B1, or 3b-[HSD2](#)) are characterized by normal or moderately elevated levels of urinary 17-ketosteroids and plasma [DHEA](#)sulfate. A high basal level of a precursor of cortisol biosynthesis (such as 17-hydroxyprogesterone, 17-hydroxypregnenolone, or 11-deoxycortisol), or elevation of such a precursor after [ACTH](#) stimulation, confirms the diagnosis of a partial deficiency. Measurement of steroid precursors 60 min after bolus administration of ACTH is usually sufficient. Adrenal androgen output is easily suppressed by the standard low-dose (2 mg) dexamethasone test.

TREATMENT

Patients with [CAH](#) have a fundamental defect of cortisol deficiency with resultant excessive [ACTH](#) secretion, producing hyperplasia of the adrenal glands and causing additional shunting into the precursor steroid pathways. Therapy in these patients consists of daily administration of glucocorticoids to suppress pituitary ACTH secretion. Because of its cost and intermediate half-life, prednisone is the drug of choice except in infants, in whom hydrocortisone is usually used. In adults with late-onset adrenal hyperplasia, the smallest single bedtime dose of a long- or intermediate-acting glucocorticoid that suppresses pituitary ACTH secretion should be administered. The amount of steroid required by children with CAH is approximately 1 to 1.5 times the normal cortisol production rate of 27 to 35 μmol (10 to 13 mg) of cortisol per square meter of body surface per day and is given in divided doses two or three times per day. The dosage schedule is governed by repetitive analysis of the urinary 17-ketosteroids, plasma [DHEA](#)sulfate, and/or precursors of cortisol biosynthesis. Skeletal growth and maturation must also be monitored closely, as overtreatment with glucocorticoid replacement therapy retards linear growth.

Receptor Mutations Much less common than [CAH](#) are three syndromes secondary to mutation(s) in a key receptor involved in adrenal function. *Isolated glucocorticoid deficiency* is a rare autosomal recessive disease secondary to a mutation in the [ACTH](#) receptor. Usually mineralocorticoid function is normal. Adrenal insufficiency is manifest within the first 2 years of life usually as hyperpigmentation, convulsions, and/or frequent episodes of hypoglycemia. In some patients the adrenal insufficiency is associated with achalasia and alacrima -- Allgrove's, or triple A, syndrome. However, in some triple A syndrome patients, no mutation in the ACTH receptor has been identified, suggesting that a distinct genetic abnormality causes this syndrome. *Adrenal hypoplasia congenita* is a rare X-linked disorder caused by a mutation in the *DAX1* gene located on the X chromosome. This gene encodes an orphan nuclear receptor that plays an important role in the development of the adrenal cortex and also the hypothalamic-pituitary-gonadal axis. Thus, patients present with signs and symptoms secondary to deficiencies of all three major adrenal steroids -- cortisol, aldosterone, and adrenal androgens -- as well as gonadotropin deficiency. Finally a rare cause of hypercortisolism without cushingoid stigmata is *primary cortisol resistance* due to mutations in the glucocorticoid receptor. The resistance is incomplete because patients do not exhibit signs of adrenal insufficiency. Thus, these three rare inherited disorders have in common an elevated ACTH. However, the clinical manifestations range from no

evidence of adrenal insufficiency to only cortisol deficiency (similar to secondary adrenal insufficiency) to a clinical picture indistinguishable from classic Addison's disease.

Miscellaneous Conditions Adrenoleukodystrophy causes severe demyelination and early death in children, and adrenomyeloneuropathy is associated with a mixed motor and sensory neuropathy with spastic paraplegia in adults; both disorders are associated with elevated circulating levels of very long chain fatty acids and cause adrenal insufficiency. Autosomal recessive mutations in the steroidogenic acute regulatory (STAR) protein gene cause congenital lipoid adrenal hyperplasia ([Chap. 338](#)), which is characterized by adrenal insufficiency and defective gonadal steroidogenesis. Because STAR mediates cholesterol transport into the mitochondrion, mutations in the protein cause massive lipid accumulation in steroidogenic cells, ultimately leading to cell toxicity. Thus, these three rare inherited disorders have in common an elevated ACTH. However, the clinical manifestations range from no evidence of adrenal insufficiency to only cortisol deficiency (similar to secondary adrenal insufficiency) to a clinical picture indistinguishable from classic Addison's disease.

MINERALOCORTICOID DISEASES

Some forms of [CAH](#) have a mineralocorticoid component (see above). Others are caused by a mutation in other enzymes or ion channels important in mediating or mimicking aldosterone's action.

Hypermineralocorticoidism

Low Plasma Renin Activity Rarely, hypermineralocorticoidism is due to a defect in cortisol biosynthesis, specifically 11- or 17-hydroxylation. [ACTH](#) levels are increased, with a resultant increase in the production of the mineralocorticoid 11-deoxycorticosterone. Hypertension and hypokalemia can be corrected by glucocorticoid administration. The definitive diagnosis is made by demonstrating an elevation of precursors of cortisol biosynthesis in the blood or urine or by direct demonstration of the genetic defect.

Glucocorticoid administration can also ameliorate hypertension or produce normotension even though a hydroxylase deficiency cannot be identified ([Fig. 331-9](#)). These patients have normal to slightly elevated aldosterone levels that do not suppress in response to saline but do suppress in response to 2 days of dexamethasone (2 mg/d). The condition is inherited as an autosomal dominant trait and is termed *glucocorticoid-remediable aldosteronism* (GRA). This entity is secondary to a chimeric gene duplication whereby the 11-hydroxylase gene promoter (which is under the control of [ACTH](#)) is fused to the aldosterone synthase coding sequence. Thus, aldosterone synthase activity is ectopically expressed in the zona fasciculata and is regulated by ACTH, in a fashion similar to the regulation of cortisol secretion. Screening for this defect is best performed by assessing the presence or absence of the chimeric gene. Because the abnormal gene may be present in the absence of hypokalemia, its frequency as a cause of hypertension is unknown. Individuals with suppressed plasma renin levels and juvenile-onset hypertension or a history of early-onset hypertension in first-degree relatives should be screened for this disorder. Early hemorrhagic stroke also occurs in GRA-affected individuals.

Glucocorticoid-remediable hyperaldosteronism documented by genetic analysis may be treated with glucocorticoid administration or antimineralocorticoids, e.g., spironolactone, triamterene, or amiloride. Glucocorticoids should be used only in small doses to avoid inducing iatrogenic Cushing's syndrome. A combination approach is often necessary.

High Plasma Renin Activity Bartter's syndrome is characterized by severe hyperaldosteronism (hypokalemic alkalosis) with moderate to marked increases in renin activity and hypercalciuria, but normal blood pressure and no edema; this disorder usually begins in childhood. Renal biopsy shows juxtaglomerular hyperplasia. The pathogenesis involves a defect in the renal conservation of sodium or chloride. The renal loss of sodium is thought to stimulate renin secretion and aldosterone production. Hyperaldosteronism produces potassium depletion, and hypokalemia further elevates prostaglandin production and plasma renin activity. In some cases, the hypokalemia may be potentiated by a defect in renal conservation of potassium. Increased production of prostaglandins is probably not a primary abnormality, since administration of inhibitors of prostaglandin synthesis reverses the features only temporarily ([Chap. 276](#)). Bartter's syndrome is caused by a mutation in the renal Na-K-2Cl co-transporter gene.

Gitelman's syndrome is an autosomal recessive trait characterized by renal salt wasting and as a result, as in Bartter's syndrome, activation of the renin-angiotensin-aldosterone system. As a consequence affected individuals have low blood pressure, low serum potassium, low serum magnesium, and high serum bicarbonate. In contrast to Bartter's syndrome, urinary calcium excretion is reduced. Gitelman's syndrome results from loss-of-function mutations of the renal thiazide-sensitive Na-Cl co-transporter.

Increased Mineralocorticoid Action Liddle's syndrome is a rare autosomal dominant disorder that mimicks hyperaldosteronism. The defect is in the genes encoding the β or γ subunits of the epithelial sodium channel. Both renin and aldosterone levels are low, owing to the constitutively activated sodium channel and the resulting excess sodium reabsorption in the renal tubule.

A rare autosomal recessive cause of hypokalemia and hypertension is 11 β -[HSDII](#) deficiency, in which cortisol cannot be converted to cortisone and hence binds to the [MR](#) and acts as a mineralocorticoid. This condition, also termed *apparent mineralocorticoid excess syndrome*, is caused by a defect in the gene encoding the renal isoform of this enzyme, 11 β -HSD II. Patients can be identified either by documenting an increased ratio of cortisol to cortisone in the urine or by genetic analysis. Patients with the 11 β -HSD deficiency syndrome can be treated with small doses of dexamethasone. Although dexamethasone is a potent glucocorticoid that suppresses [ACTH](#) and endogenous cortisol production, it binds less well to the mineralocorticoid receptor than does cortisol.

The ingestion of candies or chewing tobacco containing certain forms of licorice produces a syndrome that mimics primary aldosteronism. The component of such agents that causes sodium retention is glycyrrhizinic acid, which inhibits the 11 β -[HSDII](#) and hence allows cortisol to act as a mineralocorticoid and cause sodium retention, expansion of the extracellular fluid volume, hypertension, depressed plasma renin levels, and suppressed aldosterone levels. The diagnosis is established or excluded by

a careful history.

Decreased Mineralocorticoid Production or Action In patients with these conditions, disorders of aldosterone biosynthesis or action are associated with high renin levels, salt wasting, and hyperkalemia. The aldosterone levels may be low or elevated. In patients with a deficiency in aldosterone biosynthesis, the transformation of corticosterone into aldosterone is impaired, owing to a mutation in the aldosterone synthase (CYP11B2) gene. These patients have low to absent aldosterone secretion, elevated plasma renin levels, and elevated levels of the intermediates of aldosterone biosynthesis (corticosterone and 18-hydroxycorticosterone).

Pseudohypoaldosteronism type I (PHA-I) is an autosomal recessive disorder that is seen in the neonatal period and is characterized by salt wasting, hypotension, hyperkalemia, and high renin and aldosterone levels. In contrast to the gain-of-function mutations in the epithelial sodium channel (ENaC) in Liddle's syndrome, mutations in PHA-I result in loss of ENaC function.

NONSPECIFIC CLINICAL USE OF ADRENAL STEROIDS

The widespread use of glucocorticoids emphasizes the need for a thorough understanding of the metabolic effects of these agents. Before adrenal hormone therapy is instituted, the expected gains should be weighed against undesirable effects.

HOW SERIOUS IS THE DISORDER?

In a patient who has unexplained shock or in whom other measures have failed, the physician need not hesitate to employ high-dose steroid therapy. In contrast, one should exercise restraint in administering steroids to a patient with early rheumatoid arthritis for whom physiotherapy, anti-inflammatory agents, disease-modifying agents, and general medical care have not been tried ([Chap. 312](#)).

HOW LONG WILL GLUCOCORTICOID THERAPY BE REQUIRED?

The use of intravenous steroids for 24 to 48 h for a life-threatening situation such as status asthmaticus or pseudotumor cerebri has few or no contraindications, in contrast to the initiation of chronic steroid therapy for asthma, arthritis, or psoriasis. In the latter instances, the almost certain development of some degree of Cushing's syndrome must be weighed against the potential benefit. These side effects should be minimized by a careful choice of steroid preparations, alternate-day or interrupted therapy; the use of topical steroids, e.g., inhaled, intranasal, or dermal, whenever possible; and the judicious use of supplementary adjuvants.

WHICH PREPARATION IS BEST?

Several considerations should be taken into account in deciding which steroid preparation to use.

1. *The biologic half-life.* The rationale behind alternate-day therapy is to decrease the metabolic effects of the steroids for a significant part of each 48 h period while still

producing a pharmacologic effect durable enough to be effective. Too long a half-life would defeat the first purpose, and too short a half-life would defeat the second. In general, the more potent the steroid, the longer its biologic half-life.

2. *The importance of the mineralocorticoid effects of the steroid.* Most synthetic steroids have less mineralocorticoid effect than hydrocortisone ([Table 331-9](#)).

3. *The biologically active form of the steroid.* Cortisone and prednisone have to be converted to biologically active metabolites before anti-inflammatory effects can occur. Because of this, in a condition for which steroids are known to be effective and when an adequate dose has been given without response, one should consider substituting hydrocortisone or prednisolone for cortisone or prednisone.

4. *The cost of the medication.* This is a serious consideration if chronic administration is planned. Prednisone is the least expensive of available steroid preparations.

5. *The type of formulation.* Topical steroids have the distinct advantage over oral steroids in reducing the likelihood of systemic side effects. In addition, some inhaled steroids have been designed to minimize side effects by increasing their hepatic inactivation if they are swallowed ([Chap. 252](#)). However, all topical steroids can be absorbed into the systemic circulation.

EVALUATION OF PATIENTS PRIOR TO INITIATING STEROID THERAPY (See [Table 331-10](#))

Chronic Infection Three issues demand attention: (1) Any active infection, particularly tuberculosis, should be identified. (2) If tuberculosis is present, steroid therapy should be employed only in conjunction with antituberculous chemotherapy. The chest film and tuberculin test provide baseline information for future comparison. Since high-dose steroids may impair the tuberculin reaction, serial chest roentgenograms may be necessary. (3) Infection due to opportunistic pathogens should be constantly considered in patients on steroid therapy, especially when combined with other immunosuppressive agents.

Diabetes Mellitus Prolonged glucocorticoid therapy may unmask or aggravate diabetes mellitus. The presence of diabetes mellitus or the demonstration of impaired glucose tolerance may affect the decision to institute glucocorticoid therapy.

Osteoporosis Patients receiving long-term steroid therapy are at risk for osteoporosis. Indeed, osteoporosis with vertebral fractures or compression is a dreaded complication for patients at high risk (postmenopausal women, elderly men, patients with restricted physical activity, and especially organ transplant patients who are maintained on high doses to prevent rejection). Alternate-day or interrupted steroid therapy does not prevent the risk of this complication ([Table 331-11](#)). Adjunctive therapies with antiresorptive agents, such as parenteral or oral bisphosphonates, have been shown to be effective in treating the osteoporosis ([Chap. 342](#)). Bone mineral density should be assessed periodically with dual-energy X-ray absorptiometry (DEXA) scans.

Peptic Ulcer, Gastric Hypersecretion, or Esophagitis In conventional doses

(equivalent to 15 mg prednisone per day), glucocorticoids probably do not cause peptic ulceration; whether higher doses cause ulcer disease is not established and probably depends on duration, dose of treatment, and predisposing factors such as hypoalbuminemia or cirrhosis and also whether subjects are concomitantly ingesting nonsteroidal anti-inflammatory agents (NSAIDs). However, even at conventional doses, patients with a history of ulcer may experience aggravation of symptoms while receiving glucocorticoids. Consequently, all individuals with a positive history or with known risk factors should be managed with a vigorous antiulcer program (antacids, H₂receptor antagonists, or ATPase inhibitors) along with glucocorticoids. *The development of anemia in a patient receiving glucocorticoids should suggest gastrointestinal bleeding as a cause.*

Hypertension or Cardiovascular Disease The capacity of many adrenal steroid preparations to promote sodium retention makes it necessary to exercise caution when using them in patients with preexisting hypertension or cardiovascular or renal disease. The use of preparations with minimal sodium-retaining activity, restriction of dietary sodium intake, and the use of diuretic agents and supplementary potassium salts minimize the mineralocorticoid effects of steroids. However, hypertension may be exacerbated by steroid-induced increases in renin substrate and angiotensin II levels and by reduction in vasodilator prostaglandin production. Steroids also accelerate atherogenesis by induction of hypertension, glucose intolerance, and unfavorable lipid profiles. Glucocorticoid-associated lipid abnormalities include hypertriglyceridemia, hypercholesterolemia, and increased [LDL](#) cholesterol levels.

Psychological Difficulties Steroid therapy may cause psychological disturbances. In general, these disturbances correlate better with the patient's personality than with the dose of hormone, although larger doses cause more serious reactions. There is no reliable method of predicting the psychological reaction to steroid therapy; moreover, previous tolerance of steroids does not necessarily ensure the safety of subsequent courses. Likewise, untoward psychological reactions on one occasion do not invariably mean that the patient will respond unfavorably to a second course. However, patients with depressive symptoms during a first course of steroids may benefit from prophylactic treatment prior to a second course.

Sleeplessness is common and can be minimized by using the shorter-acting steroids and by prescribing the total daily amount as a single early-morning dose.

ALTERNATE-DAY STEROID THERAPY

The most effective way to minimize the cushingoid effects of glucocorticoids is to administer the total 48-h dose as a *single dose of intermediate-acting steroid* in the morning, *every other day*. If symptoms of the underlying disorder can be controlled by this technique, it offers distinct advantages. Three considerations deserve mention: (1) The alternate-day schedule may be approached through transition schedules that allow the patient to adjust gradually; (2) supplementary nonsteroid medications may be needed on the "off" day to minimize symptoms of the underlying disorder; and (3) many symptoms that occur during the off day (e.g., fatigue, joint pain, muscle stiffness or tenderness, and fever) may represent relative adrenal insufficiency rather than exacerbation of the underlying disease.

The alternate-day approach capitalizes on the fact that cortisol secretion and plasma levels normally are highest in the early morning and lowest in the evening. The normal pattern is mimicked by administering an intermediate-acting steroid in the morning (7 to 8 A.M.) ([Table 331-9](#)).

Initially, the steroid program often requires daily or more frequent doses of steroid to achieve the desired anti-inflammatory or immunity-suppressing action. *Only after this desired effect is achieved is an attempt made to switch to an alternate-day program.* A number of schedules can be used for transferring from a daily to an alternate-day program. The key points to be considered are flexibility in arranging a program and the use of supportive measures on the off day. One may attempt a gradual transition to the alternate-day schedule rather than an abrupt changeover. One approach is to keep the steroid dose constant on one day and gradually reduce it on the alternate day. Alternatively, the steroid dose can be increased on one day and reduced on the alternate day. In any case, it is important to anticipate that some increase in pain or discomfort may occur in the 36 to 48 h following the last dose.

WITHDRAWAL OF GLUCOCORTICOIDS FOLLOWING LONG-TERM USE

It is possible to reduce gradually and eventually to discontinue a daily steroid dose, but under most circumstances withdrawal of steroids should be initiated by first implementing an alternate-day schedule. Patients who have been on an alternate-day program for a month or more experience less difficulty during termination regimens. The dosage is gradually reduced and finally discontinued after a replacement dosage has been reached (e.g., 5 to 7.5 mg prednisone). Complications rarely ensue unless undue stress is experienced, and patients should understand that for 1 year or longer after withdrawal from long-term high-dose steroid therapy, supplementary hormone should be given in the event of a serious infection, operation, or injury. A useful strategy in patients with symptoms of adrenal insufficiency on a tapering regimen is to measure plasma cortisol levels prior to the steroid dose. A level <140 nmol/L (5 ug/dL) indicates continuing suppression of the pituitary-adrenal axis and implies that a more cautious tapering of steroids is indicated.

In patients on high-dose daily steroid therapy, it is advised to reduce dosage to approximately 20 mg prednisone daily as a single morning dose before beginning the transition to alternate-day therapy. If a patient cannot tolerate an alternate-day program, consideration should be given to the possibility that the patient has developed primary adrenal insufficiency.

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332. PHEOCHROMOCYTOMA - Lewis Landsberg, James B. Young

Pheochromocytomas produce, store, and secrete catecholamines. They are usually derived from the adrenal medulla but may develop from chromaffin cells in or about sympathetic ganglia (extraadrenal pheochromocytomas or paragangliomas). Related tumors that secrete catecholamines and produce similar clinical syndromes include chemodectomas derived from the carotid body and ganglioneuromas derived from the postganglionic sympathetic neurons.

The clinical features are due predominantly to the release of catecholamines and, to a lesser extent, to the secretion of other substances. Hypertension is the most common sign, and hypertensive paroxysms or crises, often spectacular and alarming, occur in over half the cases.

Pheochromocytoma occurs in approximately 0.1% of the hypertensive population but is, nevertheless, an important correctable cause of high blood pressure. Indeed, it is usually curable if properly diagnosed and treated, but it may be fatal if undiagnosed or mistreated. Postmortem series indicate that most pheochromocytomas are unsuspected clinically, even when the tumor is related to the fatal outcome.

PATHOLOGY

Location and Morphology In adults, approximately 80% of pheochromocytomas are unilateral and solitary, 10% are bilateral, and 10% are extraadrenal. In children, a fourth of tumors are bilateral, and an additional fourth are extraadrenal. Solitary lesions inexplicably favor the right side. Although pheochromocytomas may grow to large size (over 3 kg), most weigh less than 100 g and are less than 10 cm in diameter. The tumors are highly vascular.

The tumors are made up of large, polyhedral, pleomorphic chromaffin cells. Less than 10% of these tumors are malignant. As with other endocrine tumors, malignancy cannot be determined from the histologic appearance; tumors that contain large numbers of aneuploid or tetraploid cells, as determined by flow cytometry, are more likely to recur. Local invasion of surrounding tissues or distant metastases indicate malignancy.

Extraadrenal Pheochromocytomas Extraadrenal pheochromocytomas usually weigh 20 to 40 g and are <5 cm in diameter. Most are located within the abdomen in association with the celiac, superior mesenteric, and inferior mesenteric ganglia. Approximately 10% are in the thorax, 1% are within the urinary bladder, and <3% are in the neck, usually in association with the sympathetic ganglia or the extracranial branches of the ninth or tenth cranial nerves.

Catecholamine Synthesis, Storage, and Release Pheochromocytomas synthesize and store catecholamines by processes resembling those of the normal adrenal medulla ([Chap. 72](#)). Little is known about the mechanisms of catecholamine release from pheochromocytomas, but changes in blood flow and necrosis within the tumor may be the cause in some instances. These tumors are not innervated, and catecholamine release does not result from neural stimulation. Pheochromocytomas also store and secrete a variety of peptides, including endogenous opioids, adrenomedullin,

endothelin, erythropoietin, parathyroid hormone-related protein, neuropeptide Y, and chromagranin A ([Chap. 72](#)). These peptides contribute to the clinical manifestations in selected cases, as noted below.

Epinephrine, Norepinephrine, and Dopamine Most pheochromocytomas contain and secrete both norepinephrine and epinephrine, and the percentage of norepinephrine is usually greater than in the normal adrenal. Most extraadrenal pheochromocytomas secrete norepinephrine exclusively. Rarely, pheochromocytomas produce epinephrine alone, particularly in association with multiple endocrine neoplasia (MEN). Although epinephrine-producing tumors may cause a preponderance of metabolic and beta-receptor effects, in general the major catecholamine secreted cannot be predicted from the clinical presentation. Increased production of dopamine and homovanillic acid (HVA) is uncommon with benign lesions but may occur with malignant pheochromocytoma.

FAMILIAL PHEOCHROMOCYTOMA

In approximately 5% of cases, pheochromocytoma is inherited as an autosomal dominant trait either alone or in combination with other abnormalities such as [MEN](#) type 2a (Sipple's syndrome) or type 2b (mucosal neuroma syndrome) ([Chap. 339](#)), von Hippel-Lindau's retinal cerebellar hemangioblastomosis, or von Recklinghausen's neurofibromatosis. Bilateral adrenal pheochromocytomas are common in the familial syndromes; within MEN kindreds, over half of pheochromocytomas are bilateral. A familial syndrome should be suspected in any patient with bilateral pheochromocytomas.

GENETIC CONSIDERATIONS

Several molecular genetic abnormalities have been identified in the familial pheochromocytoma syndromes. The [MEN](#) 2A and B syndromes are associated with abnormalities in the RET protooncogene located in pericentromeric region of chromosome 10 ([Chap. 339](#)). These mutations result in the constitutive activation of the receptor tyrosine kinase exposing adrenal medullary chromaffin cells and parafollicular C cells to hyperplasia and rendering them susceptible to malignant transformation. The RET mutations are located in the extracellular domain in MEN 2A and in the intracellular portion of the receptor in families with the MEN 2B syndrome. Interestingly, mutations at specific sites in the RET protooncogene are highly predictive of pheochromocytoma. The different phenotypic manifestations of the syndrome in different families, therefore, reflect differences in the specific mutations.

In the von Hippel-Lindau (VHL) syndrome, mutation of one copy of the VHL tumor suppressor gene is associated with the development of tumors characteristic of the syndrome including pheochromocytomas. Loss of function of the VHL tumor suppressor gene promotes tumor formation by mechanisms that are incompletely understood but may involve mRNA transcript elongation. In the VHL syndrome, the frequency of pheochromocytoma varies considerably in different kindreds. As in the [MEN](#) 2 syndromes, certain VHL mutations are highly associated with the development of pheochromocytoma. Of further interest is the recent finding that the VHL mutation has been identified in some kindreds with familial pheochromocytoma as the sole

manifestation without other clinical evidence of the VHL syndrome. Missense mutations, as opposed to deletions, insertions, or non-sense mutations, appear to be more commonly associated with pheochromocytoma. There is also a high incidence of germ-line VHL mutations in patients with thoracic extraadrenal pheochromocytomas.

Interestingly, neither the RET protooncogene nor the VHL mutation occurs commonly as a somatic mutation in sporadic pheochromocytomas. Screening apparently sporadic cases, however, may uncover a germ-line mutation and lead to the identification of an involved family that was unsuspected on clinical grounds.

CLINICAL FEATURES

Pheochromocytoma occurs at all ages but is most common in young to midadult life. Some series show a slight female preponderance. Most patients come to medical attention as a result of hypertensive crisis, paroxysmal symptoms suggestive of seizure disorder or anxiety attacks, or hypertension that responds poorly to conventional treatment. Less commonly, unexplained hypotension or shock in association with surgery or trauma will suggest the diagnosis. Most patients have hypertension in association with headaches, excessive sweating, and/or palpitations.

Hypertension Hypertension is the most common manifestation. In approximately 60% of cases the hypertension is sustained, although significant blood pressure lability is usually present, and half of patients with sustained hypertension have distinct crises or paroxysms. The other 40% have blood pressure elevations only during an attack. The hypertension is often severe, occasionally malignant, and may be resistant to treatment with standard antihypertensive drugs.

Paroxysms or Crises The paroxysm or crisis occurs in over half of patients. In an individual patient, the symptoms are often similar with each attack. The paroxysms may be frequent or sporadic, occurring at intervals as long as weeks or months. With time, the paroxysms usually increase in frequency, duration, and severity.

The attack usually has a sudden onset. It may last from a few minutes to several hours or longer. Headache, profuse sweating, palpitations, and apprehension, often with a sense of impending doom, are common. Pain in the chest or abdomen may be associated with nausea and vomiting. Either pallor or flushing may occur during the attack. The blood pressure is elevated, often to alarming levels, and the elevation is usually accompanied by tachycardia.

The paroxysm may be precipitated by any activity that displaces the abdominal contents. In some cases a particular stimulus may induce an attack in a characteristic fashion, but in others no clearly defined precipitating event can be found. Although anxiety may accompany the attacks, mental or psychological stress does not usually provoke a crisis.

Other Distinctive Clinical Features Symptoms and signs of an increased metabolic rate, such as profuse sweating and mild to moderate weight loss, are common. Orthostatic hypotension is a consequence of diminished plasma volume and blunted sympathetic reflexes. Both these factors predispose the patient with unsuspected

pheochromocytoma to hypotension or shock during surgery or trauma. Secretion of the hypotensive peptide adrenomedullin may contribute to the hypotension in some patients.

Cardiac Manifestations Sinus tachycardia, sinus bradycardia, supraventricular arrhythmias, and ventricular premature contractions all have been noted. Angina and acute myocardial infarction may occur even in the absence of coronary artery disease. A catecholamine-induced increase in myocardial oxygen consumption and, perhaps, coronary spasm may play a role in these ischemic events. Electrocardiographic changes, including nonspecific ST-T wave changes, prominent U waves, left ventricular strain patterns, and right and left bundle branch blocks may be present in the absence of demonstrable ischemia or infarction. Cardiomyopathy, either congestive with myocarditis and myocardial fibrosis or hypertrophic with concentric or asymmetric hypertrophy, may be associated with heart failure and cardiac arrhythmias. Multiorgan system failure with noncardiogenic pulmonary edema may be the presenting manifestation. Elevated levels of amylase originating from damaged pulmonary endothelium and abdominal pain may suggest acute pancreatitis, although serum lipase levels are normal.

Carbohydrate Intolerance Over half of patients have impaired carbohydrate tolerance due to suppression of insulin and stimulation of hepatic glucose output. The impaired glucose tolerance rarely requires treatment with insulin and disappears after removal of the tumor.

Hematocrit The elevated hematocrit is secondary to diminished plasma volume. Rarely, production of erythropoietin by the tumor may cause a true erythrocytosis.

Other Manifestations Hypercalcemia has been attributed to the ectopic secretion of parathyroid hormone-related protein. Fever and an elevated erythrocyte sedimentation rate have been reported in association with the production of interleukin 6. Elevated temperature more commonly reflects catecholamine-mediated increases in metabolic rate and diminished heat dissipation secondary to vasoconstriction. Polyuria is an occasional finding, and rhabdomyolysis with myoglobinuric renal failure may result from extreme vasoconstriction with muscle ischemia.

Pheochromocytoma of the Urinary Bladder Pheochromocytoma in the wall of the urinary bladder may result in typical paroxysms in relation to micturition. The location in the bladder wall is responsible for the occurrence of symptoms while the tumors are quite small, and, consequently, catecholamine excretion may be normal or minimally elevated. Hematuria is present in over half of patients, and the tumor can often be visualized at cystoscopy.

Adverse Drug Interactions Severe and occasionally fatal paroxysms have been induced by opiates, histamine, adrenocorticotropin, saralasin, and glucagon. These agents appear to release catecholamines directly from the tumor. Indirect-acting sympathomimetic amines, including methyldopa (when administered intravenously), may cause an increase in blood pressure by releasing catecholamines from the augmented stores within nerve endings. Drugs that block neuronal uptake of catecholamines, such as tricyclic antidepressants or guanethidine, may enhance the

physiologic effects of circulating catecholamines. Indeed, all medications should be considered carefully and administered cautiously in patients with known or suspected pheochromocytoma.

Associated Diseases Pheochromocytoma is associated with medullary carcinoma of the thyroid in the [MEN](#) syndrome types 2a and 2b and with hyperparathyroidism in MEN 2a ([Chap. 339](#)). Hypercalcemia, resolving after tumor resection, also has been described in the absence of parathyroid disease, as described above. Individuals at risk for MEN 2a and 2b should be screened periodically for pheochromocytoma by assay of a 24-h urine sample for catecholamines, including measurement of epinephrine. Pheochromocytoma should be excluded or removed before thyroid or parathyroid surgery.

The association of pheochromocytoma and neurofibromatosis is not common. Nevertheless, since incomplete forms of neurofibromatosis may be associated with pheochromocytoma, minor manifestations such as cafe au lait spots, vertebral abnormalities, or kyphoscoliosis should increase the suspicion of pheochromocytoma in a patient with hypertension. The incidence of pheochromocytoma in some kindreds with von Hippel-Lindau disease may be as high as 10 to 25%. Many of these are unsuspected clinically and diagnosed on a computed tomography (CT) scan or at postmortem.

The incidence of cholelithiasis is 15 to 20%. Cushing's syndrome is a rare association, usually a consequence of ectopic secretion of adrenocorticotrophic hormone by the pheochromocytoma or, less commonly, by a coexistent medullary carcinoma of the thyroid.

Diagnosis The diagnosis is established by the demonstration of increased excretion of catecholamines or catecholamine metabolites. The diagnosis can usually be made by the analysis of a single 24-h urine sample, provided the patient is hypertensive or symptomatic at the time of collection.

Biochemical Tests The assays employed include those for vanillylmandelic acid (VMA), the metanephrines, and unconjugated or "free" catecholamines ([Chap. 72](#)). The VMA assay is both less sensitive and less specific than assays of metanephrines or catecholamines. Accuracy of diagnosis is improved when two of three determinations are employed. The following considerations apply to all the urinary tests: (1) Despite claims for the adequacy of determinations made on random urine samples, analysis of a full 24-h urine sample is preferable. Creatinine should also be determined to assess the adequacy of collection. (2) Where possible, the collection should be made when the patient is at rest, on no medication, and without recent exposure to radiographic contrast media. When it is not practical to discontinue all medications, drugs known specifically to interfere with these assays (as noted below) should be avoided. (3) The urine should be acidified and refrigerated during and after collection. (4) With high-quality assays, dietary restrictions are minimal and should be specified by the laboratory performing the analyses. (5) Although most patients with pheochromocytoma excrete increased amounts of catecholamines and catecholamine metabolites at all times, the yield is increased in patients with paroxysmal hypertension if a 24-h urine collection is initiated during a crisis.

Free Catecholamines The upper limit of normal for total urinary catecholamines is between 590 and 885 nmol (100 and 150 ug) per 24 h. In most patients with pheochromocytoma, values in excess of 1480 nmol (250 ug) per day are obtained. Measurement of epinephrine is often of value, since increased epinephrine excretion [over 275 nmol (50 ug) per 24 h] is usually due to an adrenal lesion and may be the only abnormality in cases associated with [MEN](#). False-positive increases in catecholamine excretion result from exogenous catecholamines and related drugs such as methyldopa, levodopa, labetalol, and sympathomimetic amines, which may elevate catecholamine excretion for up to 2 weeks. Endogenous catecholamines from stimulation of the sympathoadrenal system also may increase urinary catecholamine excretion. Relevant clinical situations that cause such increases include hypoglycemia, strenuous exertion, central nervous system disease with increased intracranial pressure, severe hypoxia, and clonidine withdrawal.

Metanephrines and VMA In most laboratories, the upper limit of normal is 7 umol (1.3 mg) of total metanephrines and 35 umol (7.0 mg) of [VMA](#) excretion per 24 h. In most patients with pheochromocytoma, the increase in these urinary metabolites is considerable, often to more than three times the normal range. Metanephrine excretion is increased by exogenous and endogenous catecholamines and by treatment with monoamine oxidase inhibitors; propranolol may cause a spurious increase in metanephrine excretion, since a propranolol metabolite interferes in the commonly used spectrophotometric assay. VMA is less affected by endogenous and exogenous catecholamines but is spuriously increased by a variety of drugs, including carbidopa. VMA excretion is decreased by monoamine oxidase inhibitors.

Plasma Catecholamines Measurement of plasma catecholamines has a limited application. The care required in obtaining basal levels ([Chap. 72](#)) and the satisfactory results with urinary determinations make measurement of plasma catecholamines unnecessary in most cases. Plasma catecholamine levels are affected by the same drugs and physiologic perturbations that increase urinary catecholamine excretion. In addition, α - and β -adrenergic receptor blocking agents may elevate plasma catecholamines by impairing clearance.

When the clinical features suggest pheochromocytoma and the urinary assay results are borderline, measurement of plasma catecholamines may be worthwhile. Markedly elevated basal levels of total catecholamines support the diagnosis, although approximately one-third of patients with pheochromocytoma have normal or slightly elevated basal values. The usefulness of plasma catecholamine determinations may be increased by agents that suppress sympathetic nervous system activity. Clonidine and ganglionic blocking agents ([Chap. 72](#)) reduce plasma catecholamine levels in normal subjects and in patients with essential hypertension. These drugs have little effect on catecholamine levels in patients with pheochromocytoma. In patients with elevated or borderline basal catecholamine values, failure to suppress plasma or urinary levels with clonidine supports the diagnosis of pheochromocytoma.

Pharmacologic Tests Reliable methods for the measurement of catecholamines and catecholamine metabolites in urine have rendered obsolete both the provocative and adrenolytic tests, which are nonspecific and entail considerable risk. A modified version

of the adrenolytic test may be of some use, however, as a therapeutic trial in a patient in hypertensive crisis with features suggestive of pheochromocytoma. A positive response to phentolamine (5-mg bolus following a test dose of 0.5 mg) is a reduction in blood pressure of at least 35/25 mmHg that peaks after 2 min and persists for 10 to 15 min. The pharmacologic response is never diagnostic, and biochemical confirmation is essential. Provocative tests in normotensive patients are potentially dangerous and rarely indicated. However, a glucagon provocative test may be of use in patients with paroxysmal hypertension and nondiagnostic basal catecholamine levels. Glucagon has a negligible effect on blood pressure or plasma catecholamine levels in normal or hypertensive subjects. In patients with pheochromocytoma, on the other hand, glucagon may increase both blood pressure and circulating catecholamine levels. The elevation in plasma catecholamine concentration, moreover, may occur without a blood pressure response. It must be emphasized, however, that life-threatening pressor crises have occurred after administration of glucagon to patients with pheochromocytoma, so the test should never be performed casually. Careful continuous monitoring of the blood pressure is required, intravenous access must be adequate, and phentolamine must be at hand to terminate the test if a significant pressor reaction ensues.

Differential Diagnosis Since the manifestations of pheochromocytoma can be protean, the diagnosis must be considered and excluded in many patients with suggestive clinical features. In patients with essential hypertension and "hyperadrenergic" features such as tachycardia, sweating, and increased cardiac output, and in patients with anxiety attacks associated with blood pressure elevations, analysis of a 24-h urine collection is usually decisive in excluding the diagnosis. Repeated determinations on urine collected during attacks may be necessary, however, before the diagnosis can be excluded with certainty. The clonidine suppression and glucagon stimulation tests may be helpful in excluding the diagnosis in difficult cases. Pressor crises associated with clonidine withdrawal and the use of cocaine or monoamine oxidase inhibitors ([Chap. 72](#)) may mimic the paroxysms of pheochromocytoma. Factitious crises may be produced by self-administration of sympathomimetic amines in psychiatrically disturbed patients.

Intracranial lesions, particularly posterior fossa tumors or subarachnoid hemorrhage, may cause hypertension and increased excretion of catecholamines or catecholamine metabolites. While this is most common in patients with an obvious neurologic catastrophe, the possibility of subarachnoid or intracranial hemorrhage secondary to pheochromocytoma should be considered. Diencephalic or autonomic epilepsy may be associated with paroxysmal spells, hypertension, and increased plasma catecholamine levels. This rare entity may be difficult to distinguish from pheochromocytoma, but an aura, an abnormal electroencephalogram, and a beneficial response to anticonvulsant medications will often suggest the proper diagnosis.

TREATMENT

Preoperative Management The induction of stable α -adrenergic blockade is the basis of preoperative management and provides the foundation for successful surgical treatment. Once the diagnosis is established, the patient should be placed on phenoxybenzamine to induce a long-lived, noncompetitive α -receptor blockade. The usual initial dose is 10 mg every 12 h with increments of 10 to 20 mg added every few days until the blood pressure is controlled and the paroxysms disappear. Because of the

long duration of action, the therapeutic effects are cumulative, and the optimal dose must be achieved gradually with careful monitoring of supine and upright blood pressures. Most patients require between 40 and 80 mg phenoxybenzamine per day, although 200 mg or more may be necessary. Phenoxybenzamine should be administered for at least 10 to 14 days prior to surgery. Over this time, the combination of α -receptor blockade and a liberal salt intake will restore the contracted plasma volume to normal. Before adequate α -adrenergic blockade with phenoxybenzamine is achieved, paroxysms may be treated with oral prazosin or noncompetitive intravenous phentolamine. Selective α_1 antagonists have been employed for preoperative preparation, but their role in preparative management should be limited to the treatment of individual paroxysms. They may be useful as antihypertensive agents in patients with suspected pheochromocytoma while workup is in progress, since they are usually better tolerated than phenoxybenzamine and will prevent serious pressor crises if pheochromocytoma is present. Nitroprusside, calcium channel blocking agents, and possibly angiotensin-converting enzyme inhibitors reduce blood pressure in patients with pheochromocytoma. Nitroprusside may also be useful in the treatment of pressor crises.

β -Adrenergic receptor blocking agents should be given only after α blockade has been induced, since administration of such agents by themselves may cause a paradoxical increase in blood pressure by antagonizing β -mediated vasodilation in skeletal muscle. β blockade is usually initiated when tachycardia develops during the induction of α -adrenergic blockade. Low doses often suffice, and a reasonable starting dose is 10 mg propranolol three to four times per day, increased as needed to control the pulse rate. β blockade is effective for catecholamine-induced arrhythmias, particularly those potentiated by anesthetic agents.

Preoperative Localization of the Tumor Surgical removal of pheochromocytoma is facilitated if the location of the tumor or tumors can be established preoperatively. Once pheochromocytoma is diagnosed, localization should be undertaken while the patient is being prepared for surgery. [CT](#) or magnetic resonance imaging (MRI) of the adrenals is usually successful in identifying intraadrenal lesions. Extraadrenal tumors within the chest can frequently be identified by conventional chest films or CT. MRI is useful in identifying extraadrenal tumors in the abdomen. If these studies are negative, abdominal aortography (once α -adrenergic blockade is complete) may identify extraadrenal pheochromocytomas in the abdomen, since these lesions are often supplied by a large aberrant artery. If aortography, CT, and MRI fail to localize the lesion, venous sampling at different levels of the inferior and superior vena cava may reveal catecholamine gradients in the region drained by the tumor; this area may then be restudied by selective angiography or scanning by CT or MRI. An additional localization technique involves a radionuclide scintiscan after administration of the radiopharmaceutical [^{131}I]metaiodobenzylguanidine (MIBG). This agent is concentrated by the amine uptake process and produces an external scintigraphic image at the site of the tumor. This type of scanning may be useful in characterizing lesions discovered by CT when biochemical confirmation is indeterminate, as well as in localizing extraadrenal pheochromocytomas. Percutaneous fine-needle aspiration of chromaffin tumors is contraindicated; indeed, pheochromocytoma should be considered before any adrenal lesions are aspirated.

Surgery Surgical treatment of pheochromocytoma is best performed in centers with experience in the preoperative, anesthetic, and intraoperative management of pheochromocytoma. In experienced hands, surgical mortality is <2 or 3%.

Monitoring during the surgical procedure should include continuous recording of arterial pressure and central venous pressure as well as electrocardiography; in the presence of cardiac disease or if congestive failure has been present, pulmonary capillary wedge pressure should be monitored. Adequate fluid replacement is crucial. Intraoperative hypotension responds better to volume replacement than to vasoconstrictors. Hypertension and cardiac arrhythmias are most likely during induction of anesthesia, intubation, and manipulation of the tumor. Intravenous phentolamine is usually sufficient to control the blood pressure, but nitroprusside may be required. Propranolol may be given in the treatment of tachycardia or ventricular ectopy.

Pheochromocytoma in Pregnancy Spontaneous labor and vaginal delivery in unprepared patients are usually disastrous for mother and fetus. In early pregnancy, the patient should be prepared with phenoxybenzamine, and the tumor should be removed as soon as the diagnosis is confirmed. The pregnancy need not be terminated, but the operative procedure itself may result in spontaneous abortion. In the third trimester, treatment with adrenergic blocking agents should be undertaken; when the fetus is of sufficient size, cesarean section may be followed by extirpation of the tumor. Although the safety of adrenergic blocking drugs in pregnancy is not established, these agents have been administered in several cases without obvious adverse effect. Antepartum diagnosis and treatment lowers the maternal death rate to that approaching nonpregnant pheochromocytoma patients; fetal death rate, however, remains elevated.

Unresectable and Malignant Tumors In cases of metastatic or locally invasive tumor in patients with intercurrent illness that precludes surgery, long-term medical management is required. When the manifestations cannot be adequately controlled by adrenergic blocking agents, the concomitant administration of metyrosine may be required. This agent inhibits tyrosine hydroxylase, diminishes catecholamine production by the tumor, and often simplifies chronic management. Malignant pheochromocytoma frequently recurs in the retroperitoneum, and it metastasizes most commonly to bone and lung. Although these malignant tumors are resistant to radiotherapy, combination chemotherapy has had limited success in controlling them. Use of ¹³¹I-MIBG has had limited success in the treatment of malignant pheochromocytoma, due to poor uptake of the radioligand.

PROGNOSIS AND FOLLOW-UP

The 5-year survival rate after surgery is usually over 95%, the recurrence rate is <10%. After successful surgery, catecholamine excretion returns to normal in about 2 weeks and should be measured to ensure complete tumor removal. Catecholamine excretion should be assessed at the reappearance of suggestive symptoms or yearly if the patient remains asymptomatic. For malignant pheochromocytoma, the 5-year survival rate is <50%.

Complete removal cures the hypertension in approximately three-fourths of patients. In the remainder, hypertension recurs but is usually well controlled by standard

antihypertensive agents. In this group, either underlying essential hypertension or irreversible vascular damage induced by catecholamines may cause the persistence of the hypertension.

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333. DIABETES MELLITUS - Alvin C. Powers

Diabetes mellitus (DM) comprises a group of common metabolic disorders that share the phenotype of hyperglycemia. Several distinct types of DM exist and are caused by a complex interaction of genetics, environmental factors, and life-style choices. Depending on the etiology of the DM, factors contributing to hyperglycemia may include reduced insulin secretion, decreased glucose usage, and increased glucose production. The metabolic dysregulation associated with DM causes secondary pathophysiologic changes in multiple organ systems that impose a tremendous burden on the individual with diabetes and on the health care system. In the United States, DM is the leading cause of end-stage renal disease, nontraumatic lower extremity amputations, and adult blindness. With an increasing incidence worldwide, DM will likely continue to be a leading cause of morbidity and mortality for the foreseeable future.

CLASSIFICATION

Recent advances in the understanding of the etiology and pathogenesis of diabetes have led to a revised classification ([Table 333-1](#)). Although all forms of [DM](#) are characterized by hyperglycemia, the pathogenic mechanisms by which hyperglycemia arises differ widely. Some forms of DM are characterized by an absolute insulin deficiency or a genetic defect leading to defective insulin secretion, whereas other forms share insulin resistance as their underlying etiology. Recent changes in classification reflect an effort to classify DM on the basis of the pathogenic process that leads to hyperglycemia, as opposed to criteria such as age of onset or type of therapy ([Fig. 333-1](#)).

The two broad categories of [DM](#) are designated type 1 and type 2. Type 1A DM results from autoimmune beta cell destruction, which usually leads to insulin deficiency. Type 1B DM is also characterized by insulin deficiency as well as a tendency to develop ketosis. However, individuals with type 1B DM lack immunologic markers indicative of an autoimmune destructive process of the beta cells. The mechanisms leading to beta cell destruction in these patients are unknown. Relatively few patients with type 1 DM fall into the type 1B idiopathic category; many of these individuals are either African-American or Asian in heritage.

Type 2 [DM](#) is a heterogeneous group of disorders usually characterized by variable degrees of insulin resistance, impaired insulin secretion, and increased glucose production. Distinct genetic and metabolic defects in insulin action and/or secretion give rise to the common phenotype of hyperglycemia in type 2 DM (see below). The identification of distinct pathogenic processes in type 2 DM has important potential therapeutic implications, as pharmacologic agents that target specific metabolic derangements become available.

Two features of the current classification of [DM](#) diverge from previous classifications. First, the terms *insulin-dependent diabetes mellitus* (IDDM) and *noninsulin-dependent diabetes mellitus* (NIDDM) are obsolete. These previous designations reflected the observation that most individuals with type 1 DM (previously IDDM) have an absolute requirement for insulin treatment, whereas many individuals with type 2 DM (previously NIDDM) do not require insulin therapy to prevent ketoacidosis. However, because many

individuals with type 2 DM eventually require insulin treatment for control of glycemia, the use of the latter term generated considerable confusion.

A second difference is that age is no longer used as a criterion in the new classification system. Although type 1DM most commonly develops before the age of 30, an autoimmune beta cell destructive process can develop at any age. In fact, it is estimated that between 5 and 10% of individuals who develop DM after age 30 have type 1A DM. Likewise, although type 2 DM more typically develops with increasing age, it also occurs in children, particularly in obese adolescents.

OTHER TYPES OF DM

Other etiologies forDMinclude specific genetic defects in insulin secretion or action, metabolic abnormalities that impair insulin secretion, and a host of conditions that impair glucose tolerance ([Table 333-1](#)). *Maturity onset diabetes of the young* (MODY) is a subtype of DM characterized by autosomal dominant inheritance, early onset of hyperglycemia, and impairment in insulin secretion (discussed below). Mutations in the insulin receptor cause a group of rare disorders characterized by severe insulin resistance.

DMcan result from pancreatic exocrine disease when the majority of pancreatic islets (>80%) are destroyed. Several endocrinopathies can lead to DM as a result of excessive secretion of hormones that antagonize the action of insulin. Notable within this group are acromegaly and Cushing's disease, both of which may present with DM. Viral infections have been implicated in pancreatic islet destruction, but are an extremely rare cause of DM. Congenital rubella greatly increases the risk for DM; however, most of these individuals also have immunologic markers indicative of autoimmune beta cell destruction.

GESTATIONAL DIABETES MELLITUS (GDM)

Glucose intolerance may develop and first become recognized during pregnancy. Insulin resistance related to the metabolic changes of late pregnancy increases insulin requirements and may lead to hyperglycemia or impaired glucose tolerance. GDM is seen in approximately 4% of pregnancies in the United States; most women revert to normal glucose tolerance post-partum but have a substantial risk (30 to 60%) of developingDM later in life.

EPIDEMIOLOGY

The worldwide prevalence ofDM has risen dramatically over the past two decades. It is projected that the number of individuals with DM will continue to increase in the near future. Between 1976 and 1994, for example, the prevalence of DM among adults in the United States increased from 8.9% to 12.3%. These findings, based on national epidemiologic data, include individuals with a diagnosis of DM and those with undiagnosed DM (based on identical diagnostic criteria). Likewise, prevalence rates of impaired fasting glucose (IFG) increased from 6.5% to 9.7% over the same period. Although the prevalence of both type 1 and type 2 DM is increasing worldwide, the prevalence of type 2 DM is expected to rise more rapidly in the future because of

increasing obesity and reduced activity levels.

There is considerable geographic variation in the incidence of both type 1 and type 2 [DM](#). For example, Scandinavia has the highest rate of type 1 DM (in Finland, incidence is 35/100,000 per year). The Pacific Rim has a much lower rate (in Japan and China, incidence is 1 to 3/100,000 per year) of type 1 DM; Northern Europe and the United States share an intermediate rate (8 to 17/100,000 per year). Much of the increased risk of type 1 DM is believed to reflect the frequency of high-risk HLA alleles among ethnic groups in different geographic locations.

The prevalence of type 2 [DM](#) and its harbinger, impaired glucose tolerance (IGT), is highest in certain Pacific islands, intermediate in countries such as India and the United States, and relatively low in Russia and China. This variability is likely due to both genetic and environmental factors. There is also considerable variation in DM prevalence among different ethnic populations within a given country.

In 1998, approximately 16 million individuals in the United States met the diagnostic criteria for [DM](#). This represents ~6% of the population. About 800,000 individuals in the United States develop DM each year. The vast majority of these (>90%) have type 2 DM. The number of people with DM increases with the age of the population, ranging from an incidence of ~1.5% in individuals from 20 to 39 years to ~20% of individuals >75 years. The incidence of DM is similar in men and women throughout most age ranges but is slightly greater in men >60 years. The prevalence of DM is approximately twofold greater in African Americans, Hispanic Americans, and Native Americans than in non-Hispanic whites, and the onset of type 2 DM occurs, on average, at an earlier age in the former groups than in the non-Hispanic white population. The incidence of type 2 DM in these ethnic groups is rapidly increasing. The reasons for these differences are not yet clear.

DIAGNOSIS

Revised criteria for diagnosing [DM](#) have been issued by consensus panels of experts from the National Diabetes Data Group and the World Health Organization ([Table 333-2](#)). The revised criteria reflect new epidemiologic and metabolic evidence and are based on the following premises: (1) the spectrum of fasting plasma glucose (FPG) and the response to an oral glucose load varies in normal individuals, and (2) DM defined as the level of glycemia at which diabetes-specific complications are noted and not on the level of glucose tolerance from a population-based viewpoint. For example, the prevalence of retinopathy in Native Americans (Pima Indian population) begins to increase at a FPG > 6.4 mmol/L (116 mg/dL) ([Fig. 333-2](#)).

Glucose tolerance is classified into three categories based on the [FPG](#): (1) FPG < 6.1 mmol/L (110 mg/dL) is considered normal; (2) FPG $\geq$ 6.1 mmol/L (110 mg/dL) but < 7.0 mmol/L (126 mg/dL) is defined as [IFG](#); and (3) FPG $\geq$ 7.0 mmol/L (126 mg/dL) warrants the diagnosis of [DM](#). IFG is a new diagnostic category defined by the Expert Committee on the Diagnosis and Classification of Diabetes Mellitus. It is analogous to [IGT](#), which is defined as plasma glucose levels between 7.8 and 11.1 mmol/L (140 and 200 mg/dL) 2 h after a 75-g oral glucose load ([Table 333-2](#)). Individuals with IFG or IGT are at substantial risk for developing type 2 DM and cardiovascular disease in the future,

though they may not meet the criteria for DM.

The revised criteria for the diagnosis of [DM](#) emphasize the [FPG](#) as the most reliable and convenient test for diagnosing DM in asymptomatic individuals. A random plasma glucose concentration ≥ 11.1 mmol/L (200 mg/dL) accompanied by classic symptoms of DM (polyuria, polydipsia, weight loss) is sufficient for the diagnosis of DM ([Table 333-2](#)). Oral glucose tolerance testing, although still a valid mechanism for diagnosing DM, is not recommended as part of routine screening.

Some investigators have advocated the hemoglobin A1c (HbA1c) as a diagnostic test for [DM](#). Though there is a strong correlation between elevations in the plasma glucose and the HbA1c (discussed below), the relationship between the [FPG](#) and the HbA1c in individuals with normal glucose tolerance or mild glucose intolerance is less clear, and the test is not universally standardized or available.

The diagnosis of [DM](#) has profound implications for an individual from both a medical and financial standpoint. Thus, the health care provider must be certain that these criteria are completely satisfied before assigning the diagnosis of DM to an individual. The revised criteria also allow for the diagnosis of DM to be withdrawn in situations where the [FPG](#) no longer exceeds these criteria. Abnormalities on screening tests for diabetes should be repeated before making a definitive diagnosis of DM, unless acute metabolic derangements or a markedly elevated plasma glucose are present ([Table 333-2](#)).

SCREENING

Widespread use of the [FPG](#) as a screening test for type 2 [DM](#) is strongly encouraged because: (1) a large number of individuals who meet the current criteria for DM are unaware that they have the disorder, (2) epidemiologic studies suggest that type 2 DM may be present for up to a decade before diagnosis, and (3) as many as 50% of individuals with type 2 DM have one or more diabetes-specific complications at the time of their diagnosis. The Expert Committee suggests screening all individuals >45 years every 3 years and screening asymptomatic individuals with additional risk factors ([Table 333-3](#)) at an earlier age. In contrast to type 2 DM, it is rare for an individual to have a long asymptomatic period of hyperglycemia prior to the diagnosis of type 1 DM. A number of immunologic markers for type 1 DM are becoming available (discussed below), but their use is currently discouraged pending the identification of clinically beneficial interventions for individuals at high risk for developing type 1 DM.

INSULIN BIOSYNTHESIS, SECRETION, AND ACTION

BIOSYNTHESIS

Insulin is produced in the beta cells of the pancreatic islets. It is initially synthesized as a single-chain 86-amino-acid precursor polypeptide, preproinsulin. Subsequent proteolytic processing removes the aminoterminal signal peptide, giving rise to proinsulin. Proinsulin is structurally related to insulin-like growth factors I and II, which bind weakly to the insulin receptor ([Chap. 327](#)). Cleavage of an internal 31-residue fragment from proinsulin generates the C peptide and the A (21 amino acids) and B (30 amino acids) chains of insulin, which are connected by disulfide bonds. The mature insulin molecule

and C peptide are stored together and cosecreted from secretory granules in the beta cells. Because the C peptide is less susceptible than insulin to hepatic degradation, it is a useful marker of insulin secretion and allows discrimination of endogenous and exogenous sources of insulin in the evaluation of hypoglycemia ([Chap. 334](#)). Human insulin is now produced by recombinant DNA technology; structural alterations at one or more residues are useful for modifying its physical and pharmacologic characteristics (see below).

SECRETION

Glucose is the key regulator of insulin secretion by the pancreatic beta cell, although amino acids, ketones, various nutrients, gastrointestinal peptides, and neurotransmitters also influence insulin secretion. Glucose levels >3.9 mmol/L (70 mg/dL) stimulate insulin synthesis, primarily by enhancing protein translation and processing, as well as inducing insulin secretion. Glucose stimulates insulin secretion through a series of regulatory steps that begin with transport into the beta cell by the GLUT2 glucose transporter ([Fig. 333-3](#)). Glucose phosphorylation by glucokinase is the rate-limiting step that controls glucose-regulated insulin secretion.

Further metabolism of glucose-6-phosphate via glycolysis generates ATP, which inhibits the activity of an ATP-sensitive K^+ channel. This channel is a complex of two separate proteins, one of which is the receptor for certain oral hypoglycemics (e.g., sulfonylureas, meglitinides); the other subunit is an inwardly rectifying K^+ channel protein. Inhibition of this K^+ channel induces beta cell membrane depolarization, opening of voltage-dependent calcium channels (leading to an influx of calcium), and stimulation of insulin secretion. Careful studies of insulin secretory profiles reveal pulsatile pattern of hormone release, with small secretory bursts occurring about every 10 min, superimposed upon greater amplitude oscillations of about 80 to 150 min. Meals or other major stimuli of insulin secretion induce large (four- to fivefold increase versus baseline) bursts of insulin secretion that usually last for 2 to 3 h before returning to baseline. Derangements in these normal secretory patterns are one of the earliest signs of beta cell dysfunction in [DM](#) (see below).

ACTION

Once insulin is secreted into the portal vein, ~50% is removed and degraded by the liver. Unextracted insulin enters the systemic circulation and binds to its receptor in target sites. The insulin receptor belongs to the tyrosine kinase class of membrane-bound receptors ([Chap. 327](#)). Insulin binding to the receptor stimulates intrinsic tyrosine kinase activity, leading to receptor autophosphorylation and the recruitment of intracellular signaling molecules, such as insulin receptor substrates (IRS) 1 and 2 ([Fig. 333-4](#)). These and other adaptor proteins initiate a complex cascade of phosphorylation and dephosphorylation reactions, ultimately resulting in the widespread metabolic and mitogenic effects of insulin. As an example, activation of the phosphatidylinositol-3 ϕ -kinase (PI-3 kinase) pathway stimulates translocation of glucose transporters (e.g., GLUT4) to the cell surface, an event that is crucial for glucose uptake by skeletal muscle and fat. Activation of other insulin receptor signaling pathways induces glycogen synthesis, protein synthesis, lipogenesis, and regulation of various genes in insulin-responsive cells.

Glucose homeostasis reflects a precise balance between hepatic glucose production and peripheral glucose uptake and utilization. Insulin is the most important regulator of this metabolic equilibrium, but the effects of other pathways including neural input, metabolic signals, and hormones (e.g., glucagon) result in integrated control of glucose supply and utilization ([Chap. 334](#); [Fig. 334-1](#)). In the fasting state, low insulin levels promote hepatic gluconeogenesis and glycogenolysis to prevent hypoglycemia. Low insulin levels decrease glycogen synthesis, reduce glucose uptake in insulin-sensitive tissues, and promote mobilization of stored precursors. Reduced insulin levels are also permissive in allowing glucagon to stimulate glycogenolysis and gluconeogenesis by the liver and renal medulla. These processes are of critical importance to ensure an adequate glucose supply for the brain. Postprandially, a large glucose load elicits a rise in insulin and fall in glucagon, leading to a reversal of these processes. The major portion of postprandial glucose is utilized by skeletal muscle. Other tissues, most notably the brain, utilize glucose in an insulin-independent fashion.

PATHOGENESIS

TYPE 1 DM

Type 1A [DM](#) develops as a result of the synergistic effects of genetic, environmental, and immunologic factors that ultimately destroy the pancreatic beta cells. The temporal development of type 1A DM is shown schematically as a function of beta cell mass in [Fig. 333-5](#). Individuals with a genetic susceptibility have normal beta cell mass at birth but begin to lose beta cells secondary to autoimmune destruction that occurs over months to years. This autoimmune process is thought to be triggered by an infectious or environmental stimulus and to be sustained by a beta cell-specific molecule. In the majority of individuals, immunologic markers appear after the triggering event but before diabetes becomes clinically overt. Beta cell mass then begins to decline, and insulin secretion becomes progressively impaired, although normal glucose tolerance is maintained. The rate of decline in beta cell mass varies widely among individuals, with some patients progressing rapidly to clinical diabetes and others evolving more slowly. Features of diabetes do not become evident until a majority of beta cells are destroyed (~80%). At this point, residual functional beta cells still exist but are insufficient in number to maintain glucose tolerance. The events that trigger the transition from glucose intolerance to frank diabetes are often associated with increased insulin requirements, as might occur during infections or puberty. Following the initial clinical presentation of type 1A DM, a "honeymoon" phase may ensue during which time glycemic control is achieved with modest doses of insulin or, rarely, insulin is not needed. However, this fleeting phase of endogenous insulin production from residual beta cells disappears as the autoimmune process destroys the remaining beta cells, and the individual becomes completely insulin deficient.

GENETIC CONSIDERATIONS

The genetic contributions to type 1A [DM](#) involve multiple genes. The development of the disease appears to require inheritance of a sufficient complement of genes to confer susceptibility to the disorder. The concordance of type 1A DM in identical twins ranges between 30 and 70%, indicating that additional modifying factors must be involved in

determining whether diabetes develops. The major susceptibility gene for type 1A DM is located in the HLA region on chromosome 6. Polymorphisms in the HLA complex appear to account for 40 to 50% of the genetic risk of developing type 1A DM. This region contains genes that encode the class II MHC molecules, which present antigen to helper T cells and thus are involved in initiating the immune response ([Chaps. 305,306,307](#)). The ability of class II MHC molecules to present antigen is dependent on the amino acid composition of their antigen-binding sites. Amino acid substitutions may influence the specificity of the immune response by altering the binding affinity of different antigens for the class II molecules.

Most individuals with type 1A [DM](#) have the HLA DR3 and/or DR4 haplotype. Refinements in genotyping of HLA loci have shown that the haplotypes DQA1*0301, DQB1*0302 and DQA1*501, DQB1*0201 have the strongest association with type 1A DM. These haplotypes are present in 40% of children with type 1A DM as compared to 2% of the normal U.S. population.

In addition to MHC class II associations, at least 17 different genetic loci may contribute susceptibility to type 1A [DM](#). For example, polymorphisms in the promoter region of the insulin gene appear to account for ~10% of the predisposition to type 1A DM. Genes that confer protection against the development of the disease also exist. For example, the haplotype DQA1*0102, DQB1*0602 is present in 20% of the U.S. population but is extremely rare in individuals with type 1A DM (<1%).

Although type 1A [DM](#) is clearly associated with certain predisposing genotypes, most individuals with these haplotypes do not develop diabetes. In addition, most individuals with type 1A DM do not have a first-degree relative with this disorder. Nevertheless, the risk of developing type 1A DM for relatives of individuals with the disease is considerably higher compared to the risk for the general population.

Autoimmune Factors Although other islet cell types [alpha cells (glucagon-producing), delta cells (somatostatin-producing) or PP cells (pancreatic polypeptide-producing)] are functionally and embryologically similar to beta cells and express most of the same proteins as beta cells, they are inexplicably spared from the autoimmune process. Pathologically, the pancreatic islets are infiltrated with lymphocytes (in a process termed *insulitis*). After all beta cells are destroyed, the inflammatory process abates, the islets become atrophic, and immunologic markers disappear. Studies of the insulitis and autoimmune process in humans and animal models of type 1A [DM](#) (NOD mouse and BB rat) have identified the following abnormalities in both the humoral and cellular arms of the immune system: (1) islet cell autoantibodies; (2) activated lymphocytes in the islets, peripancreatic lymph nodes, and systemic circulation; (3) T lymphocytes that proliferate when stimulated with islet proteins; and (4) release of cytokines within the insulitis. Beta cells seem to be particularly susceptible to the toxic effect of some cytokines (tumor necrosis factor α , interferon γ , and interleukin 1). The precise mechanisms of beta cell death are not known but may involve formation of nitric oxide metabolites, apoptosis, and direct CD8⁺ T cell cytotoxicity. Islet autoantibodies are not thought to be involved in the destructive process, as these antibodies do not generally react with the cell surface of islet cells and are not capable of transferring diabetes mellitus to animals.

Pancreatic islet molecules targeted by the autoimmune process include insulin, glutamic

acid decarboxylase (GAD; the biosynthetic enzyme for the neurotransmitter GABA), ICA-512/IA-2 (homology with tyrosine phosphatases), and phogrin (insulin secretory granule protein). Other less clearly defined autoantigens include an islet ganglioside and carboxypeptidase H. With the exception of insulin, none of the autoantigens are beta cell specific, which raises the question of how the beta cells are selectively destroyed. Current theories favor initiation of an autoimmune process directed at one beta cell molecule, which then spreads to other islet molecules as the immune process destroys beta cells and creates a series of secondary autoantigens. The beta cells of individuals who develop type 1A [DM](#) do not differ from beta cells of normal individuals, since transplanted islets are destroyed by a recurrence of the autoimmune process of type 1A DM.

Immunologic Markers Islet cell autoantibodies (ICAs) are a composite of several different antibodies directed at pancreatic islet molecules such as [GAD](#), insulin, IA-2/ICA512, and an islet ganglioside and serve as a marker of the autoimmune process of type 1A [DM](#). Testing for ICAs can be useful in classifying the type of DM as type 1A and in identifying nondiabetic individuals at risk for developing type 1A DM. ICAs are present in the majority of individuals (>75%) diagnosed with new-onset type 1A DM, in a significant minority of individuals with newly diagnosed type 2 DM, and occasionally in individuals with [GDM](#) (<5%). ICAs are present in 3 to 4% of first-degree relatives of individuals with type 1A DM. In conjunction with impaired insulin secretion on intravenous glucose tolerance testing, they predict a >50% risk of developing type 1A DM within 5 years. Without this impairment in insulin secretion, the presence of ICAs predicts a 5-year risk of <25%. Based on these data, the risk of a first-degree relative developing type 1A DM is relatively low, and even ICA-positive individuals are not destined to develop diabetes. At present, the ICAs are used predominantly as a research tool and not in clinical practice, in part because of the technically demanding nature of the assay but also because no treatments have been proven to prevent the occurrence or progression of type 1A DM.

Environmental Factors Numerous environmental events have been proposed to trigger the autoimmune process in genetically susceptible individuals; however, none have been conclusively linked to diabetes. Identification of an environmental trigger has been difficult because the event may precede the onset of [DM](#) by several years ([Fig. 333-5](#)). Putative environmental triggers include viruses (coxsackie and rubella most prominently), early exposure to bovine milk proteins, and nitrosourea compounds. Epidemiologic studies have noted an association between bovine milk intake and type 1A DM; studies are ongoing to investigate a possible relationship between exposure to bovine milk and the autoimmune process of type 1A DM.

Prevention of Type 1A DM A number of interventions have successfully delayed or prevented diabetes in animal models. Some interventions have targeted the immune system directly (immunosuppression, selective T cell subset deletion, induction of immunologic tolerance to islet proteins), whereas others have prevented islet cell death by blocking cytotoxic cytokines or increasing islet resistance to the destructive process. Though results in animal models are promising, most of these interventions have not been successful in preventing type 1A [DM](#) in humans. Clinical trials of several interventions are underway in the United States and Europe. The Diabetes Prevention Trial -- type 1 is being conducted to determine whether administering insulin to

individuals at high risk for developing type 1A DM can induce immune tolerance and alter the autoimmune process of type 1A DM.

TYPE 2 DM

Type 2DM is a heterogeneous disorder with a complex etiology that develops in response to genetic and environmental influences. Central to the development of type 2 DM are insulin resistance and abnormal insulin secretion. Although controversy remains regarding the primary defect, most studies support the view that insulin resistance precedes insulin secretory defects.

GENETIC CONSIDERATIONS

Type 2DM has a strong genetic component. Although the major genes that predispose to this disorder have yet to be identified, it is clear that the disease is polygenic and multifactorial. Various genetic loci contribute to susceptibility, and environmental factors (such as nutrition and physical activity) further modulate phenotypic expression of the disease. The concordance of type 2 DM in identical twins is between 70 and 90%. Individuals with a parent with type 2 DM have an increased risk of diabetes; if both parents have type 2 DM, the risk in offspring may reach 40%. Insulin resistance, as demonstrated by reduced glucose utilization in skeletal muscle, is present in many nondiabetic, first-degree relatives of individuals with type 2 DM. However, definition of the genetic abnormalities of type 2 DM remains a challenge because the genetic defect in insulin secretion or action may not manifest itself unless an environmental event or another genetic defect, such as obesity, is superimposed.

The identification of individuals with mutations in various molecules involved in insulin action (e.g., the insulin receptor and enzymes involved in glucose homeostasis) has been useful for characterizing key steps in insulin action. However, mutations in these molecules account for a very small fraction of type 2DM. Likewise, genetic defects in proteins involved in insulin secretion have not been found in most individuals with type 2 DM. Genome-wide scanning for mutations or polymorphisms associated with type 2 DM is being used in an effort to identify genes associated with type 2 DM.

Pathophysiology Type 2DM is characterized by three pathophysiologic abnormalities: impaired insulin secretion, peripheral insulin resistance, and excessive hepatic glucose production. Obesity, particularly visceral or central, is very common in type 2 DM. Insulin resistance associated with obesity augments the genetically determined insulin resistance of type 2 DM. Adipocytes secrete a number of biologic products (leptin, tumor necrosis factor α , free fatty acids) that modulate processes such as insulin secretion, insulin action, and body weight and may contribute to the insulin resistance. In the early stages of the disorder, glucose tolerance remains normal, despite insulin resistance, because the pancreatic beta cells compensate by increasing insulin output. As insulin resistance and compensatory hyperinsulinemia progress, the pancreatic islets become unable to sustain the hyperinsulinemic state. IGT, marked by elevations in postprandial glucose, then develops. A further decline in insulin secretion and an increase in hepatic glucose production lead to overt diabetes with fasting hyperglycemia. Ultimately, beta cell failure may ensue.

Metabolic Abnormalities

Insulin Resistance This is caused by the decreased ability of insulin to act effectively on peripheral target tissues (especially muscle and liver) and is a prominent feature of type 2DM. This resistance is relative, since supernormal levels of circulating insulin will normalize the plasma glucose. Insulin dose-response curves exhibit a rightward shift, indicating reduced sensitivity, and a reduced maximal response, indicating an overall decrease in maximum glucose utilization (30 to 60% lower than normal individuals). Resistance to the action of insulin impairs glucose utilization by insulin-sensitive tissues and increases hepatic glucose output -- both effects contributing to the hyperglycemia of diabetes. Increased hepatic glucose output predominantly accounts for increased FPG levels, whereas decreased peripheral glucose usage results in postprandial hyperglycemia. In skeletal muscle, there is a greater impairment in nonoxidative glucose usage (glycogen formation) than in oxidative glucose metabolism through glycolysis. Glucose usage in insulin-independent tissues is not decreased in type 2 DM.

The precise molecular mechanism of insulin resistance in type 2DM has yet to be elucidated. Insulin receptor levels and tyrosine kinase activity in skeletal muscle are reduced, but these alterations are most likely secondary to hyperinsulinemia and are not a primary defect. Therefore, postreceptor defects are believed to play the predominant role in insulin resistance (Fig. 333-4). Polymorphisms in IRS-1 may be associated with glucose intolerance, raising the possibility that polymorphisms in various postreceptor molecules may combine to create an insulin-resistant state.

A current focus for the pathogenesis of insulin resistance focuses on a PI-3 kinase signaling defect, which causes reduced translocation of GLUT4 to the plasma membrane, among other abnormalities. Of note, not all insulin signal transduction pathways are resistant to the effects of insulin (e.g., those controlling cell growth and differentiation). Consequently, hyperinsulinemia may actually increase the insulin action through these pathways.

Another emerging theory proposes that elevated levels of free fatty acids, a common feature of obesity, may contribute to the pathogenesis of type 2DM in several different ways. Free fatty acids can impair glucose utilization in skeletal muscle, promote glucose production by the liver, and impair beta cell function.

Impaired Insulin Secretion Insulin secretion and sensitivity are interrelated (Fig. 333-6). In type 2DM, insulin secretion initially increases in response to insulin resistance in order to maintain normal glucose tolerance. Initially, the insulin secretory defect is mild and selectively involves glucose-stimulated insulin secretion. The response to other nonglucose secretagogues, such as arginine, is preserved. Eventually, the insulin secretory defect progresses to a state of grossly inadequate insulin secretion. Some endogenous insulin production continues, but the amount secreted is less than the amount secreted by normal individuals at the same plasma glucose concentration.

The reason(s) for the decline in insulin secretory capacity in type 2DM is unclear. Despite the assumption that a second genetic defect -- superimposed upon insulin resistance -- leads to beta cell failure, intense genetic investigation has so far excluded

mutations in islet candidate genes. Islet amyloid polypeptide or amylin is cosecreted by the beta cell and likely forms the amyloid fibrillar deposit found in the islets of individuals with longstanding type 2 DM. Whether such islet amyloid deposits are a primary or secondary event is not known. The metabolic environment may also impact islet function negatively. For example, chronic hyperglycemia paradoxically impairs islet function ("glucose toxicity") and leads to a worsening of hyperglycemia. Improvement in glycemic control is often associated with improved islet function. In addition, elevation of free fatty acid levels ("lipotoxicity") also worsens islet function.

Increased Hepatic Glucose Production The liver maintains plasma glucose during periods of fasting through glycogenolysis and gluconeogenesis using substrates derived from skeletal muscle and fat (alanine, lactate, glycerol, and fatty acids). Insulin promotes the storage of glucose as hepatic glycogen and suppresses gluconeogenesis. In type 2DM, insulin resistance in the liver arises from the failure of hyperinsulinemia to suppress gluconeogenesis, which results in fasting hyperglycemia and decreased glucose storage by the liver in the postprandial state. Increased hepatic glucose production occurs early in the course of diabetes, though likely after the onset of insulin secretory abnormalities and insulin resistance in skeletal muscle.

Insulin Resistance Syndromes It is likely that the insulin resistance condition comprises a spectrum of disorders, with hyperglycemia representing one of the most readily diagnosed features. *Syndrome X* is a term used to describe a constellation of metabolic derangements that includes insulin resistance, hypertension, dyslipidemia, central or visceral obesity, endothelial dysfunction, and accelerated cardiovascular disease. Epidemiologic evidence supports hyperinsulinemia as a marker for coronary artery disease risk, though an etiologic role has not been demonstrated.

A number of forms of severe insulin resistance may be associated with a phenotype similar to that in type 2DM or IGT (Table 333-1). *Acanthosis nigricans* and signs of hyperandrogenism (hirsutism, acne, and oligomenorrhea) are common physical features. In addition to rare genetic syndromes seen in early childhood, two distinct syndromes of severe insulin resistance have been described in adults: (1) type A, which affects young women and is characterized by severe hyperinsulinemia, obesity, and features of hyperandrogenism; and (2) type B, which affects middle-aged women and is characterized by severe hyperinsulinemia, features of hyperandrogenism, and autoimmune disorders. Individuals with the type A insulin resistance syndrome have an undefined defect in the insulin signaling pathway; individuals with the type B insulin resistance syndrome have autoantibodies directed at the insulin receptor. These receptor autoantibodies may block insulin binding or may stimulate the insulin receptor, leading to intermittent hypoglycemia.

Polycystic ovary syndrome (PCOS) is a common disorder that affects premenopausal women and is characterized by chronic anovulation and hyperandrogenism. Insulin resistance is seen in a significant subset of women with PCOS, and the disorder substantially increases the risk for type 2DM, independent of the effects of obesity. Both metformin and thiazolidinediones may attenuate hyperinsulinemia, ameliorate hyperandrogenism, and induce ovulation, but are not approved for this indication.

Prevention Because type 2DM is preceded by a period of IGT, a number of life-style

modifications and pharmacologic agents have been suggested to prevent or delay its onset. Individuals with a strong family history or those at high risk for developing DM should be strongly encouraged to maintain a normal body mass index and to engage in regular physical activity. Beyond this general advice, however, there are no specific interventions proven to prevent type 2 DM. Clinical trials of various interventions in individuals with IGT or early DM are underway in the United States and worldwide.

MODY: GENETICALLY DEFINED, MONOGENIC FORMS OF DIABETES MELLITUS

Several monogenic forms of [DM](#) have recently been identified. [MODY](#) comprises a phenotypically and genetically heterogeneous subtype of DM. Onset of the disease typically occurs between the ages of 10 and 25. Five different variants of MODY, due to mutations in genes encoding islet cell transcription factors or glucokinase ([Fig. 333-3](#)), have been identified so far, and all are transmitted as autosomal dominant disorders ([Table 333-1](#)). MODY 2, the most common variant, is caused by mutations in the glucokinase gene. Glucokinase catalyzes the formation of glucose-6-phosphate from glucose, a reaction that is important for glucose sensing by the beta cells and for glucose utilization by the liver. As a result of glucokinase mutations, higher glucose levels are required to elicit insulin secretory responses, thus altering the set point for insulin secretion. MODY 1, MODY 3, and MODY 5 are caused by mutations in the hepatocyte nuclear transcription factors HNF-4a, HNF-1a, and HNF-1b, respectively. As their names imply, these transcription factors are expressed in the liver but also in other tissues, including the pancreatic islets. The mechanisms by which such mutations lead to DM is not well understood, but it is likely that these factors affect islet development or the transcription of genes that are important in stimulating insulin secretion. MODY 4 is a rare variant caused by mutations in the insulin promoter factor (IPF-1), which is a transcription factor that regulates both pancreatic development and insulin gene transcription. Homozygous inactivating mutations lead to pancreatic agenesis, whereas heterozygous mutations result in early-onset DM. Studies of populations with type 2 DM suggest that mutations in the glucokinase gene and various islet cell transcription factors do not account for ordinary type 2 DM. Nevertheless, elucidation of the molecular genetics underlying these rare forms of DM has been important in identifying critical steps in the control of pancreatic beta cell function.

COMPLICATIONS OF DM

ACUTE COMPLICATIONS

Diabetic ketoacidosis (DKA) and nonketotic hyperosmolar state (NKHS) are acute complications of diabetes. DKA is seen primarily in individuals with type 1 [DM](#), and NKHS is seen in individuals with type 2 DM. Both disorders are associated with absolute or relative insulin deficiency, volume depletion, and altered mental status. DKA and NKHS exist along a continuum of hyperglycemia, with or without ketosis. The metabolic similarities and differences in DKA and NKHS are highlighted in [Table 333-4](#). Both disorders are associated with potentially serious complications if not promptly diagnosed and treated.

DIABETIC KETOACIDOSIS

Clinical Features The symptoms and physical signs of [DKA](#) are listed in [Table 333-5](#). DKA may be the initial symptom complex that leads to a diagnosis of type 1DM, but more frequently it occurs in individuals with established diabetes. Nausea and vomiting are often prominent, and their presence in an individual with diabetes warrants laboratory evaluation for DKA. Abdominal pain may be severe and sometimes suggests acute pancreatitis or ruptured viscus. Hyperglycemia leads to glucosuria, volume depletion, tachycardia, and possibly hypotension. Kussmaul respirations and an acetone odor on the patient's breath (both secondary to metabolic acidosis) are classic signs of the disorder. Lethargy and central nervous system depression may evolve into coma with severe DKA. Cerebral edema, an extremely serious complication of DKA, is seen most frequently in children. Signs of infection, which may precipitate DKA, should be sought on physical examination, even in the absence of fever.

Pathophysiology [DKA](#) results from insulin deficiency combined with counterregulatory hormone excess (glucagon, catecholamines, cortisol, and growth hormone). Both insulin deficiency and glucagon excess, in particular, are necessary for DKA to develop. The hyperglycemia of DKA results from increased hepatic glucose production (gluconeogenesis and glycogenolysis) and impaired peripheral glucose utilization. The decreased ratio of insulin to glucagon promotes gluconeogenesis, glycogenolysis, and ketone body formation in the liver, as well as increasing substrate delivery from fat and muscle (free fatty acids, amino acids) to the liver.

The combination of insulin deficiency and hyperglycemia reduces the hepatic level of fructose-2,6-phosphate, which alters the activity of phosphofructokinase and fructose-1,6-bisphosphatase. Glucagon excess decreases the activity of pyruvate kinase, whereas insulin deficiency increases the activity of phosphoenolpyruvate carboxykinase. These hepatic changes shift the handling of pyruvate toward glucose synthesis and away from glycolysis. Glycogenolysis is promoted by the increased levels of glucagon and catecholamines in the face of low insulin levels. Insulin deficiency also reduces levels of the GLUT4 glucose transporter, which impairs glucose uptake into skeletal muscle and fat and reduces intracellular glucose metabolism ([Fig. 333-4](#)).

Ketosis results from a marked increase in free fatty acid release from adipocytes, with a resulting shift toward ketone body synthesis in the liver. Reduced insulin levels, in combination with elevations in catecholamines and growth hormone, lead to an increase in lipolysis and release of free fatty acids. Normally, these free fatty acids are converted to triglycerides or very low density lipoproteins (VLDL) in the liver, but in [DKA](#), hyperglucagonemia alters hepatic metabolism to favor ketone body formation, through activation of the enzyme carnitine palmitoyltransferase I. This enzyme is crucial for regulating fatty acid transport into the mitochondria, where beta oxidation and conversion to ketone bodies occurs. At physiologic pH, ketone bodies exist as ketoacids, which are neutralized by bicarbonate. As bicarbonate stores are depleted, metabolic acidosis ensues. Increased lactic acid production also contributes to the acidosis. The increased free fatty acids result in increased triglyceride production and increased hepatic production of VLDL. VLDL clearance is also reduced because the activity of insulin-sensitive lipoprotein lipase is decreased. Hypertriglyceridemia may be severe enough to cause pancreatitis.

[DKA](#) can be precipitated by inadequate levels of plasma insulin for a variety of reasons

([Table 333-5](#)). Most commonly, DKA is precipitated when relatively insufficient insulin is available when insulin requirements increase, as might occur during a concurrent illness. Failure to augment insulin therapy appropriately by the patient or health care team compounds the problem. Occasionally, complete omission of insulin by the patient or health care team (in a hospitalized patient with type 1DM) precipitates DKA. Patients using insulin infusion devices with short-acting insulin have a greater potential for DKA, since even a brief interruption in insulin delivery (e.g., mechanical malfunction) quickly leads to insulin deficiency.

Laboratory Abnormalities and Diagnosis The timely diagnosis of [DKA](#) is crucial and allows for prompt initiation of therapy. DKA is characterized by hyperglycemia, ketosis, and metabolic acidosis (increased anion gap) along with a number of secondary metabolic derangements ([Table 333-4](#)). Serum bicarbonate is frequently <10 mmol/L, and arterial pH ranges between 6.8 and 7.3, depending on the severity of the acidosis. Despite a total-body potassium deficit, the serum potassium at presentation is typically at the high end of the normal range or mildly elevated, secondary to the acidosis. Total-body stores of sodium, chloride, phosphorous, and magnesium are also reduced in DKA, but are not accurately reflected by their levels in the serum. Elevated blood urea nitrogen (BUN) and serum creatinine levels reflect intravascular volume depletion. Interference from acetoacetate may falsely elevate the serum creatinine measurement. Leukocytosis, hypertriglyceridemia, and hyperlipoproteinemia are commonly found as well. Hyperamylasemia may suggest a diagnosis of pancreatitis, especially when accompanied by abdominal pain. However, in DKA the amylase is usually of salivary origin and thus is not diagnostic of pancreatitis.

The measured serum sodium is reduced as a consequence of the hyperglycemia [1.6 meq (1.6 mmol/L) reduction in serum sodium for each 100 mg/dL (5.6 mmol/L) rise in the serum glucose]. A normal serum sodium in the setting of [DKA](#) indicates a more profound water deficit. In "conventional" units, the calculated serum osmolality [$2 \times$ (serum sodium + serum potassium) + plasma glucose (mg/dL)/ 18 + [BUN](#)/ 2.8] is mildly to moderately elevated, though to a lesser degree than that found in [NKHSH](#) hyperosmolar state (see below).

In [DKA](#), the ketone body, b-hydroxybutyrate, is synthesized at a threefold greater rate than acetoacetate; however, the latter ketone body is preferentially detected by a commonly used ketosis detection reagent (nitroprusside). Serum ketones are present at significant levels (usually positive at serum dilution of 1:8 or greater). The nitroprusside tablet, or stick, is often used to detect urine ketones; certain medications such as captopril or penicillamine may cause false-positive reactions. Serum or plasma assays for b-hydroxybutyrate more accurately reflect the true ketone body level.

The metabolic derangements of [DKA](#) exist along a spectrum, beginning with mild acidosis with moderate hyperglycemia evolving into more severe findings. The degree of acidosis and hyperglycemia do not necessarily correlate closely, as a variety of factors determine the level of hyperglycemia (oral intake, urinary glucose loss). Ketonemia is a consistent finding in DKA and distinguishes it from simple hyperglycemia.

TREATMENT

The management of [DKA](#) is outlined in [Table 333-6](#). After initiating intravenous fluid replacement and insulin therapy, the agent or event that precipitated the episode of DKA should be sought and aggressively treated. If the patient is vomiting or has altered mental status, a nasogastric tube should be inserted to prevent aspiration of gastric contents. Central to successful treatment of DKA is careful patient monitoring and frequent reassessment to ensure that the patient and the metabolic derangements are improving. A comprehensive flow sheet should record chronologic changes in vital signs, fluid intake and output, and laboratory values as a function of insulin administered.

After the initial bolus of normal saline, replacement of the sodium and free water deficit is carried out over the next 24 h (fluid deficit is often 3 to 5 L). When hemodynamic stability and adequate urine output are achieved, intravenous fluids should be switched to 0.45% saline at a rate of 200 to 300 mL/h, depending on the calculated volume deficit. The change to 0.45% saline helps reduce the trend toward hyperchloremia later in the course of [DKA](#). Alternatively, initial use of lactated Ringer's intravenous solution may reduce the hyperchloremia that commonly occurs with normal saline.

A bolus of intravenous or intramuscular insulin (10 to 20 units) should be administered immediately ([Table 333-6](#)), and subsequent treatment should provide continuous and adequate levels of circulating insulin. Intravenous administration is preferred, because it assures rapid distribution and allows adjustment of the infusion rate as the patient responds to therapy. Intravenous insulin should be continued until the acidosis resolves and the patient is metabolically stable. As the acidosis and insulin resistance associated with [DKA](#) resolve, the insulin infusion rate can be decreased (to 1 to 4 units/h). Intermediate or long-acting insulin, in combination with subcutaneous regular insulin, should be administered as soon as the patient resumes eating, as this facilitates transition to an outpatient insulin regimen and reduces length of hospital stay. It is crucial to continue the insulin infusion until adequate insulin levels are achieved by the subcutaneous route. Even relatively brief periods of inadequate insulin administration in this transition phase may allow for DKA relapse.

Hyperglycemia usually improves at a rate of 4.2 to 5.6 mmol/L (75 to 100 mg/dL per hour) as a result of insulin-mediated glucose disposal, reduced hepatic glucose release, and rehydration. The latter reduces catecholamines, increases urinary glucose loss, and expands the intravascular volume. The decline in the plasma glucose within the first 1 to 2 h may be more rapid and is mostly related to volume expansion. When the plasma glucose reaches 13.9 mmol/L (250 mg/dL), glucose should be added to the 0.45% saline infusion to maintain the plasma glucose in the 11.1 to 13.9 mmol/L (200 to 250 mg/dL) range, and the insulin infusion should be continued. Ketoacidosis begins to resolve as insulin reduces lipolysis, increases peripheral ketone body use, suppresses hepatic ketone body formation, and promotes bicarbonate regeneration. However, the acidosis and ketosis resolve at a slower rate than does the hyperglycemia. As ketoacidosis improves, β -hydroxybutyrate is converted to acetoacetate. Ketone body levels may appear to increase if measured by laboratory assays that use the nitroprusside reaction, which only detects acetoacetate and acetone levels. The improvement in acidosis and anion gap, a result of bicarbonate regeneration and decline in ketone bodies, is reflected by a rise in the serum bicarbonate level and the

arterial pH. Depending on the rise of serum chloride, the anion gap (but not bicarbonate) will normalize. A hyperchloremic acidosis [serum bicarbonate of 15 to 18 mmol/L (15 to 18 meq/L)] often follows successful treatment and is minimized by the use of hypotonic intravenous solutions. This gradually resolves as the kidney regenerates bicarbonate and excretes chloride.

Potassium stores are depleted in [DKA](#) [estimated deficit 3 to 5 mmol/kg (3 to 5 meq/kg)], but the serum potassium may be normal or even elevated at the time of presentation. During treatment with insulin and fluids, various factors contribute to the development of hypokalemia. These include insulin-mediated potassium transport into cells, resolution of the acidosis (which also promotes potassium entry into cells), and urinary loss of potassium salts of organic acids. Thus, potassium repletion should commence as soon as adequate urine output and a normal serum potassium are documented. If the initial serum potassium level is elevated, then potassium repletion should be delayed until the potassium falls into the normal range. Inclusion of 20 to 40 meq of potassium in each liter of intravenous fluid is reasonable, but additional potassium supplements may also be required. To reduce the amount of chloride administered, potassium phosphate or acetate can be substituted for the chloride salt. The goal is to maintain the serum potassium >3.5 mmol/L (3.5 meq/L).

Despite a bicarbonate deficit, bicarbonate replacement is not usually necessary or advisable. In fact, theoretical arguments suggest that bicarbonate administration and rapid reversal of acidosis may impair cardiac function, impair tissue oxygenation, and promote hypokalemia. The results of most clinical trials do not support the routine use of bicarbonate replacement. In the presence of severe acidosis (arterial pH < 7.0 or hypotension unresponsive to fluid resuscitation), some physicians administer bicarbonate [50 to 150 mmol/L (meq/L) of sodium bicarbonate in 250 mL of 0.45% saline over 1 to 2 h until the serum bicarbonate rises to approximately 10 mmol/L (meq/L)]. Hypophosphatemia may result from increased glucose usage, but randomized clinical trials have not demonstrated that phosphate replacement is beneficial in [DKA](#). If the serum phosphate is < 0.32 mmol/L (1.0 mg/dl), then phosphate supplement should be considered and the serum calcium monitored. Hypomagnesemia may develop during DKA therapy and may also require supplementation.

With appropriate therapy, the mortality of [DKA](#) is low (<5%) and is related more to the underlying or precipitating event, such as infection or myocardial infarction. The major nonmetabolic complication of DKA therapy is cerebral edema, which most often develops in children as DKA is resolving. The etiology and optimal therapy for cerebral edema are not well established, but overreplacement of free water should be avoided. Venous thrombosis and adult respiratory distress syndrome occasionally complicate DKA.

Following successful treatment of [DKA](#), the physician and patient should review the sequence of events that led to DKA to prevent future recurrences. Foremost is patient education about the symptoms of DKA, its precipitating factors, and the management of diabetes during a concurrent illness. During illness or when oral intake is compromised, patients should: (1) frequently measure the capillary blood glucose; (2) measure urinary ketones when the serum glucose >16.5 mmol/L (300 mg/dL); (3) drink fluids to maintain hydration; (4) continue or increase insulin; and (5) seek medical attention if dehydration,

persistent vomiting, or uncontrolled hyperglycemia develop. In this way, early DKA can be detected and treated appropriately on an outpatient basis.

NONKETOTIC HYPEROSMOLAR STATE

Clinical Features [NKHS](#) is most commonly seen in elderly individuals with type 2 [DM](#). Its most prominent features include polyuria; orthostatic hypotension; and a variety of neurologic symptoms that include altered mental status, lethargy, obtundation, seizure, and possibly coma. The prototypical patient is a mildly diabetic, elderly individual with a several week history of polyuria, weight loss, and diminished oral intake that culminates in mental confusion, lethargy, or coma. The physical examination reflects profound dehydration and hyperosmolality and reveals hypotension, tachycardia, and altered mental status. Notably absent are symptoms of nausea, vomiting, and abdominal pain and the Kussmaul respirations characteristic of [DKA](#). NKHS is often precipitated by a serious, concurrent illness such as myocardial infarction or stroke. Sepsis, pneumonia, and other serious infections are frequent precipitants and should be sought thoroughly. In addition, a debilitating condition (prior stroke or dementia) or social situation that compromises water intake may contribute to the development of the disorder. Finally, the development of NKHS can be associated with the use of certain medications (thiazide diuretics, glucocorticoids, phenytoin).

Pathophysiology Insulin deficiency and inadequate fluid intake are the underlying causes of [NKHS](#). Insulin deficiency increases hepatic glucose production (through glycogenolysis and gluconeogenesis) and impairs glucose utilization in skeletal muscle (see above discussion under [DKA](#)). Hyperglycemia induces an osmotic diuresis that leads to profound intravascular volume depletion, which is exacerbated by inadequate fluid replacement. The absence of ketosis in NKHS is not completely understood. Presumably, the insulin deficiency is only relative and less severe than in DKA. Lower levels of counterregulatory hormones and free fatty acids have been found in NKHS than in DKA in some studies. It is also possible that the liver is less capable of ketone body synthesis or that the insulin/glucagon ratio does not favor ketogenesis.

Laboratory Abnormalities and Diagnosis The laboratory features in [NKHS](#) are summarized in [Table 333-4](#). Most notable are the marked hyperglycemia [plasma glucose may be >55.5 mmol/L (1000 mg/dL)], hyperosmolality (>350 mosmol/L), and prerenal azotemia. The measured serum sodium may be normal or slightly low despite the marked hyperglycemia. The corrected serum sodium is usually increased [add 1.6 meq to measured sodium for each 5.6 mmol/L (100 mg/dL) rise in the serum glucose]. In contrast to [DKA](#), acidosis and ketonemia are absent or mild. A small anion gap metabolic acidosis may be present secondary to increased lactic acid. Moderate ketonuria, if present, is secondary to starvation.

TREATMENT

Volume depletion and hyperglycemia are prominent features of both [NKHS](#) and [DKA](#). Consequently, therapy of these disorders involves several shared elements ([Table 333-6](#)). In both disorders, careful monitoring of the patient's fluid status, laboratory values, and insulin infusion rate is crucial. Underlying or precipitating problems should be aggressively sought and treated. In NKHS, the volume depletion, free water deficit,

and hyperosmolality are greater than in DKA. The patient with NKHS is usually older, more likely to have mental status changes, and thus more likely to have a life-threatening precipitating event with accompanying comorbidities. Even with proper treatment, NKHS has a substantially higher mortality than DKA (up to 50% in some clinical series).

Fluid replacement should initially stabilize the hemodynamic status of the patient (1 to 3 L of 0.9% normal saline over the first 2 to 3 h). Because the fluid deficit in [NKHS](#) is accumulated over a period of days to weeks, the rapidity of reversal of the hyperosmolar state must balance the need for free water repletion and the observation that too rapid a reversal may worsen neurologic function. If the serum sodium is $>150\text{mmol/L}$ (150meq/L), 0.45% saline should be used. After hemodynamic stability is achieved, the intravenous fluid administration is directed at reversing the free water deficit using hypotonic fluids (0.45% saline initially then 5% dextrose in water, D₅W). The calculated free water deficit (which averages 9 to 10 L) should be reversed over the next 1 to 2 days (infusion rates of 200 to 300 mL/h of hypotonic solution). Potassium repletion is usually necessary and should be dictated by repeated measurements of the serum potassium. In patients taking diuretics, the potassium deficit can be quite large and may be accompanied by magnesium deficiency. Hypophosphatemia may occur during therapy and can be improved by using KPO₄ and beginning nutrition.

As in [DKA](#), rehydration and volume expansion lower the plasma glucose initially, but insulin is eventually required. In [NKHS](#), patients tend to be more sensitive to insulin than in DKA and dose requirements are not usually as large. A reasonable regimen for NKHS begins with an intravenous insulin bolus of 5 to 10 units followed by intravenous insulin at a constant infusion rate (3 to 7 units/h). As in DKA, glucose should be added to intravenous fluid when the plasma glucose falls to 13.9mmol/L (250mg/dL), and the insulin infusion rate should be decreased to 1 to 2 units/h. The insulin infusion should be continued until the patient has resumed eating and can be transferred to a subcutaneous insulin regimen. The patient should be discharged from the hospital on insulin, though some patients can later undergo a trial of oral glucose-lowering agents.

CHRONIC COMPLICATIONS

The chronic complications of [DM](#) affect many organ systems and are responsible for the majority of morbidity and mortality associated with the disease. Chronic complications can be divided into vascular and nonvascular complications ([Table 333-7](#)). The vascular complications of DM are further subdivided into microvascular (retinopathy, neuropathy, nephropathy) and macrovascular complications (coronary artery disease, peripheral vascular disease, cerebrovascular disease). Nonvascular complications include problems such as gastroparesis, sexual dysfunction, and skin changes. This division is rather arbitrary since it is likely that multiple pathogenic processes are involved in all forms of complications.

The risk of chronic complications increases as a function of the duration of hyperglycemia; they usually become apparent in the second decade of hyperglycemia. Since type 2 [DM](#) may have a long asymptomatic period of hyperglycemia, many individuals with type 2 DM have complications at the time of diagnosis.

The microvascular complications of both type 1 and type 2 [DM](#) result from chronic hyperglycemia. Randomized, prospective clinical trials involving large numbers of individuals with type 1 or type 2 DM have conclusively demonstrated that a reduction in chronic hyperglycemia prevents or reduces retinopathy, neuropathy, and nephropathy. Other incompletely defined factors also modulate the development of complications. For example, despite longstanding DM, some individuals never develop nephropathy or retinopathy. Many of these patients have glycemic control that is indistinguishable from those who develop microvascular complications. Because of these observations, it is suspected that a genetic susceptibility for developing particular complications exists. However, the genetic loci responsible for these susceptibilities have not yet been identified.

Evidence implicating a causative role for chronic hyperglycemia in the development of macrovascular complications is less conclusive, but some results suggest a role for chronic hyperglycemia in the development of macrovascular disease. For example, coronary heart disease events and mortality are two to four times greater in patients with type 2 [DM](#). These events correlate with fasting and postprandial plasma glucose levels as well as with the HbA1c. Other factors (dyslipidemia and hypertension) also play important roles in macrovascular complications.

MECHANISMS OF COMPLICATIONS

Although chronic hyperglycemia is an important etiologic factor leading to complications of [DM](#), the mechanism(s) by which it leads to such diverse cellular and organ dysfunction is unknown. Three major theories, which are not mutually exclusive, have been proposed to explain how hyperglycemia might lead to the chronic complications of DM ([Fig. 333-7](#)).

One hypothesis is that increased intracellular glucose leads to the formation of advanced glycosylation end products (AGEs) via the nonenzymatic glycosylation of cellular proteins. Nonenzymatic glycosylation results from the interaction of glucose with amino groups on proteins. AGEs have been shown to cross-link proteins (e.g., collagen, extracellular matrix proteins), accelerate atherosclerosis, promote glomerular dysfunction, reduce nitric oxide synthesis, induce endothelial dysfunction, and alter extracellular matrix composition and structure. The serum level of AGEs correlates with the level of glycemia, and these products accumulate as glomerular filtration rate declines.

A second hypothesis proposed to explain how chronic hyperglycemia leads to complications of [DM](#) is based on the observation that hyperglycemia increases glucose metabolism via the sorbitol pathway. Intracellular glucose is predominantly metabolized by phosphorylation and subsequent glycolysis, but when intracellular glucose is increased, some glucose is converted to sorbitol by the enzyme aldose reductase. Increased sorbitol concentrations affect several aspects of cellular physiology (decreased myoinositol, altered redox potential) and may lead to cellular dysfunction. However, testing of this theory in humans, using aldose reductase inhibitors, has not demonstrated beneficial effects on clinical endpoints of retinopathy, neuropathy, or nephropathy.

A third hypothesis proposes that hyperglycemia increases the formation of diacylglycerol leading to activation of certain isoforms of protein kinase C (PKC), which, in turn, affect a variety of cellular events that lead to [DM](#)-related complications. For example, PKC activation by glucose alters the transcription of genes for fibronectin, type IV collagen, contractile proteins, and extracellular matrix proteins in endothelial cells and neurons in vitro. Growth factors appear to play an important role in DM-related complications. Vascular endothelial growth factor (VEGF) is increased locally in diabetic proliferative retinopathy and decreases after laser photocoagulation. Transforming growth factor b (TGF-b) is increased in diabetic nephropathy and appears to stimulate basement membrane production of collagen and fibronectin by mesangial cells. Other growth factors, such as platelet-derived growth factor, epidermal growth factor, insulin-like growth factor I, growth hormone, basic fibroblast growth factor, and even insulin, have been suggested to play a role in DM-related complications.

Although hyperglycemia serves as the initial trigger for complications of diabetes, it is still unknown whether the same pathophysiologic processes are operative in all complications or whether certain processes predominate in certain organs. Finally, oxidative stress and free radical generation, as a consequence of the hyperglycemia, may also promote the development of complications.

GLYCEMIC CONTROL AND COMPLICATIONS

The Diabetes Control and Complications Trial (DCCT) provided definitive proof that reduction in chronic hyperglycemia can prevent many of the early complications of type 1 [DM](#). This large multicenter clinical trial randomized over 1400 individuals with type 1 DM to either intensive or conventional diabetes management, and then evaluated the development of retinopathy, nephropathy, and neuropathy. Individuals in the intensive diabetes management group received multiple administrations of insulin each day along with intense educational, psychological, and medical support. Individuals in the conventional diabetes management group received twice daily insulin injections and quarterly nutritional, educational, and clinical evaluation. The goal in the former group was normoglycemia; the goal in the latter group was prevention of symptoms of diabetes. Individuals in the intensive diabetes management group achieved a substantially lower HbA1c (7.2%) than individuals in the conventional diabetes management group (HbA1c of 9.0%).

Results from the [DCCT](#) demonstrated that improvement of glycemic control reduced nonproliferative and proliferative retinopathy (47% reduction), microalbuminuria (39% reduction), clinical nephropathy (54% reduction), and neuropathy (60% reduction). Improved glycemic control also slowed the progression of early diabetic complications. There was a nonsignificant trend in reduction of macrovascular events. The results of the DCCT predicted that individuals in the intensive diabetes management group would gain 7.7 additional years of sight, 5.8 additional years free from end-stage renal disease (ESRD), and 5.6 years free from lower extremity amputations. If all complications of [DM](#) were combined, individuals in the intensive diabetes management group would experience 15.3 more years of life without significant microvascular or neurologic complications of DM as compared to individuals who received standard therapy. This translates into an additional 5.1 years of life expectancy for individuals in the intensive diabetes management group. The benefit of the improved glycemic control during the

DCCT persisted even after the study concluded and glycemic control worsened.

The benefits of an improvement in glycemic control occurred over the entire range of HbA1c values ([Fig. 333-8](#)), suggesting that at any HbA1c level, an improvement in glycemic control is beneficial. Therefore, there is no threshold beneath which the HbA1c can be reduced and the complications of [DM](#) prevented. The clinical implication of this finding is that the goal of therapy is to achieve an HbA1c level as close to normal as possible, without subjecting the patient to excessive risk of hypoglycemia.

Considerable debate has emerged as to whether the [DCCT](#) findings are applicable to individuals with type 2 [DM](#), in whom insulin resistance, hyperinsulinemia, and obesity predominate. Concerns have been raised that therapies associated with weight gain and additional insulin therapy may worsen underlying insulin resistance and hyperinsulinemia. Despite these concerns, most available data support extrapolation of the results of the DCCT to individuals with type 2 DM.

The United Kingdom Prospective Diabetes Study (UKPDS) studied the course of >5000 individuals with type 2 [DM](#) for >10 years. This complex and important study utilized multiple treatment regimens and monitored the effect of intensive glycemic control and risk factor treatment on the development of diabetic complications. Newly diagnosed individuals with type 2 DM were randomized to (1) intensive management using various combinations of insulin, a sulfonylurea, or metformin; or (2) conventional therapy using dietary modification and pharmacotherapy with the goal of symptom prevention. In addition, individuals were randomly assigned to different antihypertensive regimens. Individuals in the intensive treatment arm achieved an HbA1c of 7.0%, compared to a 7.9% HbA1c in the standard treatment group. The UKPDS demonstrated that each percentage point reduction in HbA1c was associated with a 35% reduction in microvascular complications, a 25% reduction in DM-related deaths, and a 7% reduction in all-cause mortality. As in the [DCCT](#), there was a continuous relationship between glycemic control and development of complications. Although there was no statistically significant effect of glycemic control on cardiovascular complications, there was a 16% reduction in fatal and nonfatal myocardial infarctions.

One of the major findings of the [UKPDS](#) was the observation that strict blood pressure control significantly reduced both macro- and microvascular complications. In fact, the beneficial effects of blood pressure control were greater than the beneficial effects of glycemic control. Lowering blood pressure to moderate goals (144/82 mmHg) reduced the risk of [DM](#)-related death, stroke, microvascular end points, retinopathy, and heart failure (risk reductions between 32 and 56%). Improved glycemic control did not conclusively reduce (nor worsen) cardiovascular mortality but was associated with improvement with lipoprotein risk profiles, such as reduced triglycerides and increased high-density lipoprotein (HDL).

Similar reductions in the risks of retinopathy and nephropathy were also seen in a small trial of lean Japanese individuals with type 2 [DM](#) randomized to either intensive glycemic control or standard therapy with insulin (Kumamoto study). These results demonstrate the effectiveness of improved glycemic control in individuals of different ethnicity with a presumably different etiology of DM (i.e., phenotypically different from those in the [DCCT](#) and [UKPDS](#)).

The findings of the [DCCT](#), [UKPDS](#), and Kumamoto study support the idea that chronic hyperglycemia plays a causative role in the pathogenesis of diabetic microvascular complications. These landmark studies prove the value of metabolic control and emphasize the importance of (1) intensive glycemic control in all forms of [DM](#), and (2) early diagnosis and strict blood pressure control in type 2 DM.

OPHTHALMOLOGIC COMPLICATIONS OF DIABETES MELLITUS

[DM](#) is the leading cause of blindness between the ages of 20 and 74 in the United States. The gravity of this problem is highlighted by the finding that individuals with DM are 25 times more likely to become legally blind than individuals without DM. Blindness is primarily the result of progressive diabetic retinopathy and clinically significant macular edema. Diabetic retinopathy is classified into two stages: nonproliferative and proliferative. *Nonproliferative diabetic retinopathy* usually appears late in the first decade or early in the second decade of the disease and is marked by retinal vascular microaneurysms, blot hemorrhages, and cotton wool spots (see [Plate IV-15](#)). Mild nonproliferative retinopathy progresses to more extensive disease, characterized by changes in venous vessel caliber, intraretinal microvascular abnormalities, and more numerous microaneurysms and hemorrhages. The pathophysiologic mechanisms invoked in nonproliferative retinopathy include loss of retinal pericytes, increased retinal vascular permeability, alterations in retinal blood flow, and abnormal retinal microvasculature, all of which lead to retinal ischemia.

The appearance of neovascularization in response to retinal hypoxia is the hallmark of *proliferative diabetic retinopathy*. These newly formed vessels may appear at the optic nerve and/or macula and rupture easily, leading to vitreous hemorrhage, fibrosis, and ultimately retinal detachment. Not all individuals with nonproliferative retinopathy develop proliferative retinopathy, but the more severe the nonproliferative disease, the greater the chance of evolution to proliferative retinopathy within 5 years. This creates a clear opportunity for early detection and treatment of diabetic retinopathy (discussed below). In contrast, *clinically significant macular edema* may appear when only nonproliferative retinopathy is present. Fluorescein angiography is often useful to detect macular edema, which is associated with a 25% chance of moderate visual loss over the next 3 years.

Duration of [DM](#) and degree of glycemic control are the best predictors of the development of retinopathy. Nonproliferative retinopathy is found in almost all individuals who have had DM for >20 years (25% incidence with 5 years, and 80% incidence with 15 years of type 1 DM). Although there is genetic susceptibility for retinopathy, it confers less influence on the development of retinopathy than either the duration of DM or the degree of glycemic control.

TREATMENT

The most effective therapy for diabetic retinopathy is prevention. Intensive glycemic control will greatly delay the development or slow the progression of retinopathy in individuals with either type 1 or type 2 [DM](#). Paradoxically, during the first 6 to 12 months of improved glycemic control, established diabetic retinopathy may transiently worsen.

Fortunately, this progression is temporary, and in the long term, improved glycemic control is associated with less diabetic retinopathy. Individuals with known retinopathy should be considered candidates for prophylactic photocoagulation when initiating intensive therapy. Once advanced retinopathy is present, improved glycemic control imparts less benefit, though adequate ophthalmologic care can prevent most blindness.

Equally as important as glycemic control are regular, comprehensive eye examinations for all individuals with [DM](#). Most diabetic eye disease can be successfully treated if detected early. Routine, nondilated eye examinations by the primary care provider or diabetes specialist are *inadequate* to detect diabetic eye disease properly. The treatment of diabetic eye disease requires an ophthalmologist experienced in these disorders. Laser photocoagulation is very successful in preserving vision. Proliferative retinopathy is usually treated with panretinal laser photocoagulation, whereas macular edema is treated with focal laser photocoagulation. Although exercise has not been conclusively shown to worsen proliferative diabetic retinopathy, most ophthalmologists advise individuals with advanced diabetic eye disease to limit physical activities associated with repeated Valsalva maneuvers. Aspirin therapy (650 mg/d) does not appear to influence the natural history of diabetic retinopathy, but studies of other antiplatelet agents are under way.

RENAL COMPLICATIONS OF DIABETES MELLITUS

Diabetic nephropathy is the leading cause of [ESRD](#) in the United States and a leading cause of [DM](#)-related morbidity and mortality. Proteinuria in individuals with DM is associated with markedly reduced survival and increased risk of cardiovascular disease. Individuals with diabetic nephropathy almost always have diabetic retinopathy also.

Like other microvascular complications, the pathogenesis of diabetic nephropathy is related to chronic hyperglycemia (Fig. 334-7). The mechanisms by which chronic hyperglycemia leads to [ESRD](#), though incompletely defined, involve the following: interaction of soluble factors (growth factors, angiotensin II, endothelin, [AGEs](#)), hemodynamic alterations in the renal microcirculation (glomerular hyperfiltration, increased glomerular capillary pressure), and structural changes in the glomerulus (increased extracellular matrix, basement membrane thickening, mesangial expansion, fibrosis). Some of these effects may be mediated through angiotensin receptors. Smoking accelerates the decline in renal function.

The natural history of diabetic nephropathy is shown schematically in [Fig. 333-9](#) and is characterized by a fairly predictable pattern of events. Although this sequence of events was defined for individuals with type 1 [DM](#), a similar pattern is also likely in type 2 DM. Glomerular hyperfusion and renal hypertrophy occur in the first years after the onset of DM and are reflected by an increased glomerular filtration rate (GFR). During the first 5 years of DM, thickening of the glomerular basement membrane, glomerular hypertrophy, and mesangial volume expansion occur as the GFR returns to normal. After 5 to 10 years of type 1 DM, ~40% of individuals begin to excrete small amounts of albumin in the urine (microalbuminuria). *Microalbuminuria* is defined as 30 to 300 mg/d in a 24-h collection or 30 to 300 ug/mg creatinine in a spot collection. The appearance of microalbuminuria (incipient nephropathy) in type 1 DM is a very important predictor of progression to overt proteinuria (>300 mg/d). Blood pressure may rise slightly at this

point but usually remains in the normal range. Once overt proteinuria is present, there is a steady decline in GFR, and ~50% of individuals reach [ESRD](#) in 7 to 10 years. The early pathologic changes and albumin excretion abnormalities are reversible with normalization of plasma glucose. However, once nephropathy becomes overt, the pathologic changes are likely irreversible.

The nephropathy that develops in type 2 [DM](#) differs from that of type 1 DM in the following respects: (1) microalbuminuria or overt nephropathy may be present when type 2 DM is diagnosed, reflecting its long asymptomatic period; (2) hypertension more commonly accompanies microalbuminuria or overt nephropathy in type 2 DM; and (3) microalbuminuria may be less predictive of progression to overt nephropathy in type 2 DM. Finally, it should be noted that albuminuria in type 2 DM may be secondary to factors unrelated to DM, such as hypertension, congestive heart failure, prostate disease, or infection.

Other renal problems may also occur in individuals with [DM](#). Type IV renal tubular acidosis (hyporeninemic hypoaldosteronism) occurs in many individuals with DM. These individuals develop a propensity to hyperkalemia, which may be exacerbated by medications [especially angiotensin-converting enzyme (ACE) inhibitors]. Patients with DM are predisposed to radiocontrast-induced nephrotoxicity. Individuals with DM undergoing radiographic procedures with contrast dye should be well hydrated before and after dye exposure, and the serum creatinine should be monitored for several days following the procedure.

TREATMENT

The optimal therapy for diabetic nephropathy is prevention. As part of comprehensive diabetes care, microalbuminuria should be detected at an early stage when effective therapies can be instituted. The recommended strategy for detecting microalbuminuria is outlined in [Fig. 333-10](#). Interventions effective in slowing progression from microalbuminuria to overt nephropathy include: (1) near normalization of glycemia, (2) strict blood pressure control, and (3) administration of [ACE](#) inhibitors.

Improved glycemic control reduces the rate at which microalbuminuria appears and progresses in both type 1 and type 2 [DM](#). However, once overt nephropathy exists, it is unclear whether improved glycemic control will slow progression of renal disease. During the phase of declining renal function, insulin requirements may fall as the kidney is a site of insulin degradation. Furthermore, glucose-lowering medications (sulfonylureas and metformin) may accumulate and are contraindicated in renal insufficiency.

Many individuals with type 1 or type 2 [DM](#) develop hypertension. Numerous studies in both type 1 and type 2 DM demonstrate the effectiveness of strict blood pressure control in reducing albumin excretion and slowing the decline in renal function. Blood pressure should be maintained at <130/85 mmHg in diabetic individuals without proteinuria. A slightly lower blood pressure (120/80) should be targeted for individuals with microalbuminuria or overt nephropathy. Treatment of hypertension is discussed below.

[ACE](#) inhibitors reduce the progression of overt nephropathy in individuals with type 1 or

type 2DM and should be prescribed in individuals with type 1 or type 2 DM and microalbuminuria. After 2 to 3 months of therapy, measurement of proteinuria should be repeated and the drug dose increased until either the albuminuria disappears or the maximum dose is reached. If an ACE inhibitor has an unacceptable side-effect profile (hyperkalemia, cough, and renal insufficiency), angiotensin II receptor blockers and calcium channel blockers (phenylalkylamine class) are alternatives. However, their efficacy in slowing the fall in glomerular filtration rate is not proven. Blood pressure control with any agent is extremely important, but a drug-specific benefit in diabetic nephropathy, independent of blood pressure control, has been shown only for ACE inhibitors.

A consensus panel of the American Diabetes Association (ADA) suggests modest restriction of protein intake in diabetic individuals with microalbuminuria (0.8 g/kg per day, which is the adult Recommended Daily Allowance, and about 10% of the daily caloric intake). Protein intake should be restricted further in individuals with overt diabetic nephropathy (0.6 g/kg per day), though conclusive proof of the efficacy of protein restriction is lacking.

Nephrology consultation should be considered after the diagnosis of early nephropathy. Once overt nephropathy ensues, the likelihood of ESRD is very high. As compared to nondiabetic individuals, hemodialysis in patients with DM is associated with more frequent complications, such as hypotension (autonomic neuropathy, loss of reflex tachycardia), more difficult vascular access, and accelerated progression of retinopathy. Survival after the onset of ESRD is shorter in the diabetic population compared to nondiabetics with similar clinical features. Atherosclerosis is the leading cause of death in diabetic individuals on dialysis, and hyperlipidemia should be aggressively treated. Renal transplantation from a living-related donor is the preferred therapy but requires chronic immunosuppression. Combined pancreas-kidney transplant offers the promise of normoglycemia but requires substantial expertise.

NEUROPATHY AND DIABETES MELLITUS

Diabetic neuropathy occurs in approximately 50% of individuals with long-standing type 1 and type 2DM. It may manifest as polyneuropathy, mononeuropathy, and/or autonomic neuropathy. As with other complications of DM, the development of neuropathy correlates with the duration of diabetes and glycemic control. Because the clinical features of diabetic neuropathy are similar to those of other neuropathies, the diagnosis of *diabetic* neuropathy should be made only after other possible etiologies are excluded (Chap. 378).

Polyneuropathy/Mononeuropathy The most common form of diabetic neuropathy is distal symmetric *polyneuropathy*. It most frequently presents with distal sensory loss. Hyperesthesia, paresthesia, and pain also occur. Any combination of these symptoms may develop as neuropathy progresses. Physical examination reveals sensory loss, loss of ankle reflexes, and abnormal position sense. Paresthesia is characteristically perceived as a sensation of numbness, tingling, sharpness, or burning that begins in the feet and spreads proximally. Neuropathic pain develops in some of these individuals, occasionally preceded by improvement in their glycemic control. Pain typically involves the lower extremities, is usually present at rest, and worsens at night. Both an acute

(lasting <12 months) and a chronic form of painful diabetic neuropathy have been described. As diabetic neuropathy progresses, the pain subsides and eventually disappears, and a sensory deficit in the lower extremities persists.

Diabetic polyradiculopathy is a syndrome characterized by severe disabling pain in the distribution of one or more nerve roots. It may be accompanied by motor weakness. Intercostal or truncal radiculopathy causes pain over the thorax or abdomen. Involvement of the lumbar plexus or femoral nerve may cause pain in the thigh or hip and may be associated with muscle weakness in the hip flexors or extensors (diabetic amyotrophy). Fortunately, diabetic polyradiculopathies are usually self-limited and resolve over 6 to 12 months.

Mononeuropathy (dysfunction of isolated cranial or peripheral nerves) is less common than polyneuropathy in [DM](#) and presents with pain and motor weakness in the distribution of a single nerve. A vascular etiology is favored, but the pathogenesis is unknown. Involvement of the third cranial nerve is most common and is heralded by diplopia. Physical examination reveals ptosis and ophthalmoplegia with normal papillary constriction to light. Sometimes cranial nerves IV, VI, or VII (Bell's palsy) are affected. Peripheral mononeuropathies or simultaneous involvement of more than one nerve (mononeuropathy multiplex) may also occur.

Autonomic Neuropathy Individuals with long-standing type 1 or 2 [DM](#) may develop signs of autonomic dysfunction involving the cholinergic, noradrenergic, and peptidergic (peptides such as pancreatic polypeptide, substance P, etc.) systems. DM-related autonomic neuropathy can involve multiple systems, including: the cardiovascular, gastrointestinal, genitourinary, sudomotor, and metabolic systems. Autonomic neuropathies affecting the cardiovascular system cause a resting tachycardia and orthostatic hypotension. Reports of sudden death have also been attributed to autonomic neuropathy. Gastroparesis and bladder-emptying abnormalities are also likely related to the autonomic neuropathy seen in DM (discussed below). Hyperhidrosis of the upper extremities and anhidrosis of the lower extremities result from sympathetic nervous system dysfunction. Anhidrosis of the feet can promote dry skin with cracking, which increases the risk of skin ulceration. Autonomic neuropathy may reduce counterregulatory hormone release, leading to an inability to sense hypoglycemia appropriately (*hypoglycemia unawareness*; [Chap. 334](#)), thereby subjecting the patient to the risk of severe hypoglycemia and complicating efforts to improve glycemic control.

TREATMENT

Treatment of diabetic neuropathy is less than satisfactory. Improved glycemic control should be pursued and will improve nerve conduction velocity, but the symptoms of diabetic neuropathy may not necessarily improve. Efforts to improve glycemic control may be confounded by autonomic neuropathy and hypoglycemia unawareness. Avoidance of neurotoxins (alcohol), supplementation with vitamins for possible deficiencies (B₁₂, B₆, folate; [Chap. 75](#)), and symptomatic treatment are the mainstays of therapy. Aldose reductase inhibitors do not currently offer significant symptomatic relief. Loss of sensation in the foot places the patient at risk for ulceration and its sequelae; consequently, prevention of such problems is of paramount importance. Since the pain of acute diabetic neuropathy may resolve over the first year, analgesics may be

discontinued as progressive neuronal damage from [DM](#) occurs. Chronic, painful diabetic neuropathy is difficult to treat but may respond to tricyclic antidepressants (amitriptyline, desipramine, nortriptyline), gabapentin, nonsteroidal anti-inflammatory agents (avoid in renal dysfunction), and other agents (mexilitine, phenytoin, carbamazepine, capsaicin cream). Referral to a pain management center may be necessary.

Therapy of orthostatic hypotension secondary to autonomic neuropathy is difficult. A variety of agents have limited success (fludrocortisone, midodrine, clonidine, octreotide, and yohimbine) but have significant side effects. Nonpharmacologic maneuvers (adequate salt intake, avoidance of dehydration and diuretics, and lower extremity support hose) may offer some benefit.

GASTROINTESTINAL/GENITOURINARY DYSFUNCTION

Long-standing type 1 and 2 [DM](#) may affect the motility and function of gastrointestinal (GI) and genitourinary systems. The most prominent GI symptoms are delayed gastric emptying (gastroparesis) and altered small- and large-bowel motility (constipation or diarrhea). *Gastroparesis* may present with symptoms of anorexia, nausea, vomiting, early satiety, and abdominal bloating. Nuclear medicine scintigraphy after ingestion of a radiolabeled meal is the best study to document delayed gastric emptying, but noninvasive "breath tests" following ingestion of a radiolabeled meal are under development. Though parasympathetic dysfunction secondary to chronic hyperglycemia is important in the development of gastroparesis, hyperglycemia itself also impairs gastric emptying. Nocturnal diarrhea, alternating with constipation, is a common feature of DM-related GI autonomic neuropathy. In type 1 DM, these symptoms should also prompt evaluation for celiac sprue because of its increased frequency. Esophageal dysfunction in long-standing DM is common but usually asymptomatic.

Diabetic autonomic neuropathy may lead to genitourinary dysfunction including cystopathy, erectile dysfunction, and female sexual dysfunction (reduced sexual desire, dyspareunia, reduced vaginal lubrication). Symptoms of diabetic cystopathy begin with an inability to sense a full bladder and a failure to void completely ([Chap. 48](#)). As bladder contractility worsens, bladder capacity and the postvoid residual increase, leading to symptoms of urinary hesitancy, decreased voiding frequency, incontinence, and recurrent urinary tract infections. Diagnostic evaluation includes cystometry and urodynamic studies.

Erectile dysfunction and retrograde ejaculation are very common in [DM](#) and may be one of the earliest signs of diabetic neuropathy. Erectile dysfunction, which increases in frequency with the age of the patient and the duration of diabetes, may occur in the absence of other signs of diabetic autonomic neuropathy.

TREATMENT

Current treatments for these complications of [DM](#) are inadequate. Improved glycemic control should be a primary goal, as some aspects (neuropathy, gastric function) may improve as near-normoglycemia is achieved. Smaller, more frequent meals that are easier to digest (liquid) and low in fat and fiber may minimize symptoms of gastroparesis. Cisapride (10 to 20 mg before each meal) is probably the most effective

medication but has been removed from use in the U.S. market except under special circumstances. Other agents with some efficacy include dopamine agonists (metoclopramide, 5 to 10 mg, and domperidone, 10 to 20 mg, before each meal) and bethanechol (10 to 20 mg before each meal). Erythromycin interacts with the motilin receptor and may promote gastric emptying. Diabetic diarrhea in the absence of bacterial overgrowth is treated symptomatically with loperamide but may respond to clonidine at higher doses (0.6 mg tid) or octreotide (50 to 75 ug tid subcutaneously). Treatment of bacterial overgrowth with antibiotics is sometimes useful ([Chap. 286](#)).

Diabetic cystopathy should be treated with timed voiding or self-catherization. Medications (bethanechol) are inconsistently effective. The drug of choice for erectile dysfunction is sildenafil, but the efficacy in individuals with [DM](#) is slightly lower than in the nondiabetic population ([Chap. 51](#)). Sexual dysfunction in women may be improved with use of vaginal lubricants, treatment of vaginal infections, and systemic or local estrogen replacement.

CARDIOVASCULAR MORBIDITY AND MORTALITY

Cardiovascular disease is increased in individuals with type 1 or type 2 [DM](#). The Framingham Heart Study revealed a marked increase in several cardiovascular diseases in DM including peripheral vascular disease, congestive heart failure, coronary artery disease, myocardial infarction, and sudden death (risk increase from one- to fivefold). The American Heart Association recently designated DM as a major risk factor for cardiovascular disease (same category as smoking, hypertension, and hyperlipidemia). Because of the extremely high frequency of underlying cardiovascular disease in individuals with diabetes (especially in type 2 DM), evidence of atherosclerotic vascular disease should be sought in the individual with diabetes who has symptoms suggestive of cardiac ischemia, peripheral or carotid arterial disease, a resting electrocardiogram indicative of prior infarction, plans to initiate an exercise program, proteinuria, or two other cardiac risk factors (ADA recommendations). The absence of chest pain ("silent ischemia") is common in individuals with diabetes, and a thorough cardiac evaluation is indicated in individuals undergoing major surgical procedures.

The increase in morbidity and mortality appears to relate to the synergism of hyperglycemia with other cardiovascular risk factors. For example, after controlling for all known cardiovascular risk factors, type 2 [DM](#) increases the cardiovascular death rate by twofold in men and fourfold in women. Risk factors for macrovascular disease in diabetic individuals include dyslipidemia, hypertension, obesity, reduced physical activity, and cigarette smoking. Additional risk factors specific to the diabetic population include microalbuminuria, gross proteinuria, an elevation in serum creatinine, and altered platelet function. Insulin resistance, as reflected by elevated serum insulin levels, is associated with an increased risk of cardiovascular complications in individuals with and without DM. Individuals with insulin resistance and type 2 DM have elevated levels of plasminogen activator inhibitors (especially PAI-1) and fibrinogen, which enhances the coagulation process and impairs fibrinolysis, thus favoring the development of thrombosis.

Despite proof that improved glycemic control reduces microvascular complications

in [DM](#), it is possible that macrovascular complications may be unaffected or even worsened by such therapy. Concerns about the anabolic and atherogenic potential of insulin remain, since in nondiabetic individuals, higher serum insulin levels (indicative of insulin resistance) are associated with a greater risk of cardiovascular morbidity and mortality. In the [DCCT](#), the number of cardiovascular events did not differ between the standard and intensively treated groups. However, the duration of DM in these individuals was relatively short, and the total number of events was very low. An improvement in the lipid profile of individuals in the intensive group [lower total and low-density lipoprotein (LDL) cholesterol, lower triglycerides] suggested that intensive therapy may reduce the risk of cardiovascular morbidity and mortality associated with DM. In the [UKPDS](#), improved glycemic control did not conclusively reduce cardiovascular mortality. Importantly, treatment with insulin and the sulfonylureas did not appear to increase the risk of cardiovascular disease in individuals with type 2 DM, refuting prior claims about the atherogenic potential of these agents.

In addition to coronary artery disease, cerebrovascular disease is increased in individuals with [DM](#) (threefold increase in stroke). Individuals with DM have an increased incidence of congestive heart failure (diabetic cardiomyopathy). The etiology of this abnormality is probably multifactorial and includes factors such as myocardial ischemia from atherosclerosis, hypertension, and myocardial cell dysfunction secondary to chronic hyperglycemia.

TREATMENT

In general, the treatment of coronary disease is no different in the diabetic individual ([Chap. 244](#)), though overall prognosis after myocardial infarction is worse in the diabetic population. Revascularization procedures for coronary artery disease, including percutaneous transluminal coronary angioplasty (PTCA) and coronary artery bypass grafting (CABG), are less efficacious in the diabetic individual. Initial success rates of PTCA in diabetic individuals are similar to those in the nondiabetic population, but diabetic patients have higher rates of restenosis and lower long-term patency and survival rates. Perioperative mortality from CABG is not altered in [DM](#), but both short- and long-term survival are reduced. Recent trials indicate that diabetic individuals with multivessel coronary artery disease or who recently suffered a Q-wave myocardial infarction have better long-term survival with CABG than PTCA.

Results of studies investigating the effect of intensive diabetes management on survival rates and cardiovascular events after myocardial infarction have been conflicting. In the face of conflicting data, the [ADA](#) has emphasized the importance of glycemic control and aggressive cardiovascular risk modification in all individuals with [DM](#). Despite past trepidation about using beta blockers in individuals who have diabetes, these agents clearly benefit diabetic patients after myocardial infarction, analogous to the benefit in nondiabetic individuals. [ACE](#) inhibitors may also be particularly beneficial in reducing mortality after myocardial infarction in patients with DM.

Antiplatelet therapy reduces cardiovascular events in individuals with [DM](#) who have coronary artery disease. Current recommendations by the [ADA](#) suggest the use of aspirin as secondary prevention of additional coronary events. Although data demonstrating efficacy in primary prevention of coronary events are lacking, antiplatelet

therapy should be considered, especially in diabetic individuals with other coronary risk factors such as hypertension, smoking, or hyperlipidemia. The aspirin dose (81 to 325 mg) is the same as that in nondiabetic individuals. Aspirin therapy does not have detrimental effects on renal function or hypertension, nor does it influence the course of diabetic retinopathy or maculopathy.

Cardiovascular Risk Factors

Dyslipidemia Individuals with [DM](#) may have several forms of dyslipidemia ([Chap. 344](#)). Because of the additive cardiovascular risk of hyperglycemia and hyperlipidemia, lipid abnormalities should be aggressively detected and treated as part of comprehensive diabetes care ([Fig. 333-11](#)). The most common pattern of dyslipidemia is hypertriglyceridemia and reduced [HDL](#) cholesterol levels. DM itself does not increase levels of [LDL](#), but the small dense LDL particles found in type 2 DM are more atherogenic because they are more easily glycosylated and susceptible to oxidation.

According to guidelines of the [ADA](#) and the American Heart Association, the lipid profile in diabetic individuals without cardiovascular disease (primary prevention) should be: [LDL](#) < 3.4 mmol/L (130 mg/dL); [HDL](#) > 0.9 mmol/L (35 mg/dL) in men and >1.2 mmol/L (45 mg/dL) in women; and triglycerides < 2.3 mmol/L (200 mg/dL). In diabetic individuals with cardiovascular disease, the LDL goal is < 2.6 mmol/L (100 mg/dL). Because of the risk of cardiovascular disease in diabetic individuals, many authorities recommend that optimal lipid levels for all individuals with [DM](#) (with or without cardiovascular disease) should be: LDL < 2.6 mmol/L (100 mg/dL), HDL > 1.15 mmol/L (45 mg/dL) in men and > 1.41 mmol/L (55 mg/dL) in women; and triglycerides < 2.3 mmol/L (200 mg/dL). The ADA recommends dietary modification in diabetic individuals without cardiovascular disease and a LDL cholesterol of 2.6 to 3.3 mmol/L (100 to 129 mg/dL). If multiple cardiovascular risk factors are present, the goal should be a LDL < 2.6 mmol/L (100 mg/dL) even without known cardiovascular disease.

Almost all studies of diabetic dyslipidemia have been performed in individuals with type 2 [DM](#) because of the greater frequency of dyslipidemia in this form of diabetes. Interventional studies have shown that the beneficial effects of [LDL](#) reduction are similar in the diabetic and nondiabetic populations. Large prospective trials of primary and secondary intervention for coronary heart disease have included a small number of individuals with type 2 DM, and subset analyses have consistently found that reductions in LDL reduce cardiovascular events and morbidity in individuals with DM ([Fig. 333-CD1](#)). Most clinical trials used HMG CoA reductase inhibitors, although a fibric acid derivative was also beneficial in one trial. No prospective studies have addressed similar questions in individuals with type 1 DM.

Based on the guidelines provided by the [ADA](#) and the American Heart Association, the order of priorities in the treatment of hyperlipidemia is: (1) lower the [LDL](#) cholesterol, (2) raise the [HDL](#) cholesterol, and (3) decrease the triglycerides. A treatment strategy depends on the pattern of lipoprotein abnormalities ([Fig. 333-11](#)). Initial therapy for all forms of dyslipidemia should include dietary changes, as well as the same life-style modifications recommended in the nondiabetic population (smoking cessation, control of blood pressure, weight loss, increased physical activity). The dietary recommendations for individuals with [DM](#) are similar to those advocated by the National Cholesterol

Education Program ([Chap. 344](#)) and include an increase in monounsaturated fat and carbohydrates and a reduction in saturated fats and cholesterol. Though viewed as important, the response to dietary alterations is often modest [<0.6 -mmol/L (<25 -mg/dL) reduction in the LDL]. Improvement in glycemic control will lower triglycerides and have a modest beneficial effect on raising HDL. Most medications that improve glycemic control are useful in lowering triglycerides and may raise the HDL slightly. Though fibric acid derivatives have some efficacy and are well tolerated, nicotinic acid may worsen glycemic control and increase insulin resistance; thus, niacin is relatively contraindicated in diabetic patients on oral glucose-lowering agents. As noted above, HMG CoA reductase inhibitors have proven benefit in patients with DM, even with modest elevations in LDL. Combination therapy with an HMG CoA reductase inhibitor and fibric acid derivative may be useful but increases the possibility of myositis. Bile acid-binding resins should not be used if hypertriglyceridemia is present.

Hypertension Hypertension can accelerate other complications of [DM](#), particularly cardiovascular disease and nephropathy. Hypertension therapy should first emphasize life-style modifications such as weight loss, exercise, stress management, and sodium restriction ([Chap. 35](#)). Antihypertensive agents should be selected based on the advantages and disadvantages of the therapeutic agent in the context of an individual patient's risk factor profile. [ACE](#) inhibitors are glucose- and lipid-neutral and thus positively impact the cardiovascular risk profile. For example, captopril actually improves insulin resistance, reduces [LDL](#) slightly, and increases [HDL](#) slightly. In one study of nondiabetic individuals, the ACE inhibitor ramipril reduced the risk of developing type 2 DM. Other effective agents include α -adrenergic blockers (prazosin, terazosin, doxazosin), calcium channel blockers, beta blockers (both β_1 selective and nonselective), thiazide diuretics (hydrochlorothiazide and its derivatives), central adrenergic antagonists (clonidine, methyldopa), and vasodilators (minoxidil, hydralazine). DM-related considerations include the following:

1. α -Adrenergic blockers slightly improve insulin resistance and positively impact the lipid profile, whereas beta blockers and thiazide diuretics can increase insulin resistance, negatively impact the lipid profile, and slightly increase the risk of developing type 2 diabetes.
2. Beta blockers, often questioned because of the potential masking of hypoglycemic symptoms, are effective agents and hypoglycemic events are rare when cardioselective (β_1) agents are used.
3. Central adrenergic antagonists and vasodilators are lipid- and glucose-neutral.
4. Sympathetic inhibitors and α -adrenergic blockers may be associated with orthostatic hypotension in the diabetic individual with autonomic neuropathy.
5. Calcium channel blockers are glucose- and lipid-neutral, and some evidence suggests that they reduce cardiovascular morbidity and mortality in type 2 DM, particularly in elderly patients with systolic hypertension.

If microalbuminuria or overt albuminuria is present, the optimal antihypertensive agent is an [ACE](#) inhibitor. If albumin excretion is normal, then an ACE inhibitor or other

antihypertensive agent may be used. Low-dose diuretics and beta blockers are sometimes preferred as initial agents because of their clear efficacy in the nondiabetic population. Since hypertension is often difficult to control with a single agent (especially in type 2DM), multiple antihypertensive agents are usually required when blood pressure goals (<130/85 mmHg) are not achieved. In this setting, long-acting calcium channel antagonists should be considered as additional, or second-line, agents, as these drugs appear to provide protection against cardiovascular events. ACE inhibitors are contraindicated in pregnant diabetic patients and those anticipating pregnancy. Because of the high prevalence of atherosclerotic disease in individuals with DM, the possibility of renovascular hypertension should be considered when the blood pressure is not readily controlled.

LOWER EXTREMITY COMPLICATIONS

DM is the leading cause of nontraumatic lower extremity amputation in the United States. Foot ulcers and infections are also a major source of morbidity in individuals with DM. The reasons for the increased incidence of these disorders in DM are complex and involve the interaction of several pathogenic factors: neuropathy, abnormal foot biomechanics, peripheral vascular disease, and poor wound healing. The peripheral sensory neuropathy interferes with normal protective mechanisms and allows the patient to sustain major or repeated minor trauma to the foot, often without knowledge of the injury. Disordered proprioception causes abnormal weight bearing while walking and subsequent formation of callus or ulceration. Motor and sensory neuropathy leads to abnormal foot muscle mechanics and to structural changes in the foot (hammer toe, claw toe deformity, prominent metatarsal heads). Autonomic neuropathy results in anhidrosis and altered superficial blood flow in the foot, which promote drying of the skin and fissure formation. Peripheral vascular disease and poor wound healing impede resolution of minor breaks in the skin, allowing them to enlarge and to become infected.

Approximately 15% of individuals with DM develop a foot ulcer, and a significant subset of those individuals will at some time undergo amputation (14 to 24% risk with that ulcer or subsequent ulceration). Risk factors for foot ulcers or amputation include: male sex, diabetes >10 years' duration, peripheral neuropathy, abnormal structure of foot (bony abnormalities, callus, thickened nails), peripheral vascular disease, smoking, and history of previous ulcer or amputation. Glycemic control is also a risk factor -- each 2% increase in the HbA1c increases the risk of a lower extremity ulcer by 1.6 times and the risk of lower extremity amputation by 1.5 times.

TREATMENT

The optimal therapy for foot ulcers and amputations is prevention through identification of high-risk patients, education of the patient, and institution of measures to prevent ulceration. High-risk patients should be identified during the routine foot examination performed on all patients with DM (see "Ongoing Aspects of Comprehensive Diabetes Care," below). Patient education should emphasize: (1) careful selection of footwear, (2) daily inspection of the feet to detect early signs of poor-fitting footwear or minor trauma, (3) daily foot hygiene to keep the skin clean and moist, (4) avoidance of self-treatment of foot abnormalities and high-risk behavior (e.g., walking barefoot), and (5) prompt consultation with a health care provider if an abnormality arises. Patients at high risk for

ulceration or amputation may benefit from evaluation by a foot care specialist. Interventions directed at risk factor modification include orthotic shoes and devices, callus management, nail care, and prophylactic measures to reduce increased skin pressure from abnormal bony architecture. Attention to other risk factors for vascular disease (smoking, dyslipidemia, hypertension) and improved glycemic control are also important.

Despite preventive measures, foot ulceration and infection are common and represent a potentially serious problem. Due to the multifactorial pathogenesis of lower extremity ulcers, management of these lesions must be multidisciplinary and often demands expertise in orthopedics, vascular surgery, endocrinology, podiatry, and infectious diseases. The plantar surface of the foot is the most common site of ulceration. Cellulitis without ulceration is also frequent and should be treated with antibiotics that provide broad-spectrum coverage, including anaerobes (see below).

An infected ulcer is a clinical diagnosis, since superficial culture of any ulceration will likely find multiple possible bacterial pathogens. The infection surrounding the foot ulcer is often the result of multiple organisms (gram-positive and -negative cocci and anaerobes), and gas gangrene may develop in the absence of clostridial infection. Cultures taken from the debrided ulcer base or from purulent drainage are most helpful. Wound depth should be determined by inspection and probing with a blunt-tipped sterile instrument. Plain radiographs of the foot should be performed to assess the possibility of osteomyelitis in chronic ulcers that have not responded to therapy. Nuclear medicine bone scans may be helpful, but overlying subcutaneous infection is often difficult to distinguish from osteomyelitis. Indium-labeled white cell studies are more useful in determining if the infection involves bony structures or only soft tissue, but they are technically demanding. Magnetic resonance imaging of the foot may be the most specific modality, although distinguishing bony destruction due to osteomyelitis from destruction secondary to Charcot arthropathy is difficult. If surgical debridement is necessary, bone biopsy and culture usually provide the answer.

Osteomyelitis is best treated by a combination of prolonged antibiotics and debridement of infected bone. The possible contribution of vascular insufficiency should be considered in all patients. Noninvasive blood-flow studies are often unreliable in [DM](#), and angiography may be required, recognizing the risk of contrast-induced nephrotoxicity. Peripheral vascular bypass procedures are often effective in promoting wound resolution and in decreasing the need for amputation of the ischemic limb.

A growing number of possible treatments for diabetic foot ulcers exist, but they have yet to demonstrate clear efficacy in prospective, controlled trials. A recent consensus statement from the [ADA](#) identified six interventions with demonstrated efficacy in diabetic foot wounds: (1) off-loading, (2) debridement, (3) wound dressings, (4) appropriate use of antibiotics, (5) revascularization, and (6) limited amputation. Off-loading is the complete avoidance of weight bearing on the ulcer, which removes the mechanical trauma that retards wound healing. Bed rest and a variety of orthotic devices limit weight bearing on wounds or pressure points. Surgical debridement of neuropathic wounds is important and effective, but clear efficacy of other modalities for wound cleaning (enzymes, soaking, whirlpools) is lacking. Dressings promote wound healing by creating a moist environment and protecting the wound. Antiseptic agents and topical antibiotics

should be avoided. Referral for physical therapy, orthotic evaluation, and rehabilitation may be useful once the infection is controlled.

Mild or non-limb-threatening infections can be treated with oral antibiotics (cephalosporin, clindamycin, amoxicillin/clavulanate, and fluoroquinolones), surgical debridement of necrotic tissue, local wound care (avoidance of weight bearing over the ulcer), and close surveillance for progression of infection. More severe ulcers may require intravenous antibiotics as well as bed rest and local wound care. Urgent surgical debridement may be required. Intravenous antibiotics should provide broad-spectrum coverage directed toward *Staphylococcus aureus*, streptococci, gram-negative aerobes, and anaerobic bacteria. Initial antimicrobial regimens include cefotetan, ampicillin/sulbactam, or the combination of clindamycin and a fluoroquinolone. Severe infections, or infections that do not improve after 48 h of antibiotic therapy, require expansion of antimicrobial therapy to treat methicillin-resistant *S. aureus* (e.g., vancomycin) and *Pseudomonas aeruginosa*. If the infection surrounding the ulcer is not improving with intravenous antibiotics, reassessment of antibiotic coverage and reconsideration of the need for surgical debridement or revascularization are indicated. With clinical improvement, oral antibiotics and local wound care can be continued on an outpatient basis with close follow-up. As infection improves, a comprehensive assessment of modifiable risk factors for foot ulceration should be performed and should involve health professionals with expertise in podiatry, orthotics, vascular surgery, and orthopedics.

New information about wound biology has led to a number of new technologies (e.g., living skin equivalents and growth factors such as basic fibroblast growth factor) that may prove useful. Recombinant platelet-derived growth factor has some benefit and complements the basic therapies of off-loading, debridement, and antibiotics. Hyperbaric oxygen has been used, but rigorous proof of efficacy is lacking.

INFECTIONS

Individuals with [DM](#) exhibit a greater frequency and severity of infection. The reasons for this increase include incompletely defined abnormalities in cell-mediated immunity and phagocyte function associated with hyperglycemia, as well as diminished vascularization secondary to long-standing diabetes. Hyperglycemia likely aids the colonization and growth of a variety of organisms (*Candida* and other fungal species). Many common infections are more frequent and severe in the diabetic population, whereas several rare infections are seen almost exclusively in the diabetic population. Examples of this latter category includes rhinocerebral mucormycosis and malignant otitis externa, which is usually secondary to *P. aeruginosa* infection in the soft tissue surrounding the external auditory canal. Malignant otitis externa begins with pain and discharge and may progress rapidly to osteomyelitis and meningitis. These infections should be sought, in particular, in patients presenting with [NKHS](#).

Pneumonia, urinary tract infections, and skin and soft tissue infections are all more common in the diabetic population. In general, the organisms that cause pulmonary infections are similar to those found in the nondiabetic population; however, gram-negative organisms, *S. aureus*, and *Mycobacterium tuberculosis* are more frequent pathogens. Urinary tract infections (either lower tract or pyelonephritis) are the

result of common bacterial agents such as *Escherichia coli*, though several yeast species (*Candida* and *Torulopsis glabrata*) are commonly observed. Complications of urinary tract infections include emphysematous pyelonephritis and emphysematous cystitis. Bacteriuria occurs frequently in individuals with diabetic cystopathy. Susceptibility to furunculosis, superficial candidal infections, and vulvovaginitis is increased. Poor glycemic control is a common denominator in individuals with these infections. Diabetic individuals have an increased rate of colonization of *S. aureus* in the skin folds and nares. Diabetic patients also have a greater risk of postoperative wound infections.

DERMATOLOGIC MANIFESTATIONS

The most common skin manifestations of [DM](#) are protracted wound healing and skin ulcerations. Diabetic dermopathy, sometimes termed *pigmented pretibial papules*, or "diabetic skin spots," begins as an erythematous area and evolves into an area of circular hyperpigmentation ([Fig. 333-CD2](#)). These lesions result from minor mechanical trauma in the pretibial region and are more common in elderly men with DM. Bullous diseases (shallow ulcerations or erosions in the pretibial region) are also seen. *Necrobiosis lipoidica diabetorum* is a rare disorder of DM that predominantly affects young women with type 1 DM, neuropathy, and retinopathy. It usually begins in the pretibial region as an erythematous plaque or papules that gradually enlarge, darken, and develop irregular margins, with atrophic centers and central ulceration. They may be painful. *Acanthosis nigricans* (hyperpigmented velvety plaques seen on the neck or extensor surfaces) is sometimes a feature of severe insulin resistance and accompanying diabetes ([Fig. 333-CD3](#)). Generalized or localized *granuloma annulare* (erythematous plaques on the extremities or trunk) and *scleredema* (areas of skin thickening on the back or neck at the site of previous superficial infections) are more common in the diabetic population. *Lipoatrophy* and *lipohypertrophy* can occur at insulin injection sites but are unusual with the use of human insulin. Xerosis and pruritus are common and are relieved by skin moisturizers.

Approach to the Patient

[DM](#) and its complications produce a wide range of symptoms and signs; those secondary to acute hyperglycemia may occur at any stage of the disease, whereas those related to chronic complications begin to appear during the second decade of hyperglycemia. Individuals with previously undetected type 2 DM may present with chronic complications of DM at the time of diagnosis. The history and physical examination should assess for symptoms or signs of acute hyperglycemia and should screen for the chronic complications and conditions associated with DM.

History A complete medical history should be obtained with special emphasis on [DM](#)-relevant aspects such as weight, family history of DM and its complications, risk factors for cardiovascular disease, prior medical conditions, exercise, smoking, and ethanol use. Symptoms of hyperglycemia include polyuria, polydipsia, weight loss, fatigue, weakness, blurry vision, frequent superficial infections (vaginitis, fungal skin infections), and slow healing of skin lesions after minor trauma. Metabolic derangements relate mostly to hyperglycemia (osmotic diuresis, reduced glucose entry into muscle) and to the catabolic state of the patient (urinary loss of glucose and

calories, muscle breakdown due to protein degradation and decreased protein synthesis). Blurred vision results from changes in the water content of the lens and resolves as the hyperglycemia is controlled.

In a patient with established [DM](#), the initial assessment should also include special emphasis on prior diabetes care, including the type of therapy, prior HbA1c levels, self-monitoring blood glucose results, frequency of hypoglycemia, presence of DM-specific complications, and assessment of the patient's knowledge about diabetes. The chronic complications may afflict several organ systems, and an individual patient may exhibit some, all, or none of the symptoms related to the complications of DM (see above). In addition, the presence of DM-related comorbidities should be sought (cardiovascular disease, hypertension, dyslipidemia).

PHYSICAL EXAMINATION

In addition to a complete physical examination, special attention should be given to [DM](#)-relevant aspects such as weight or body mass index, retinal examination, orthostatic blood pressure, foot examination, peripheral pulses, and insulin injection sites. Careful examination of the lower extremities should seek evidence of peripheral neuropathy, calluses, superficial fungal infections, nail disease, and foot deformities (such as hammer or claw toes and Charcot foot) in order to identify sites of potential skin ulceration. Vibratory sensation (128-MHz tuning fork at the base of the great toe) and the ability to sense touch with a monofilament (5.07, 10-g monofilament) are useful to detect moderately advanced diabetic neuropathy. Since dental disease is more frequent in DM, the teeth and gums should also be examined.

Classification of DM in an Individual Patient The etiology of diabetes in an individual with new-onset disease can usually be assigned on the basis of clinical criteria. Individuals with type 1 [DM](#) tend to have the following characteristics: (1) onset of disease prior to age 30; (2) lean body habitus; (3) requirement of insulin as the initial therapy; (4) propensity to develop ketoacidosis; and (5) an increased risk of other autoimmune disorders such as autoimmune thyroid disease, adrenal insufficiency, pernicious anemia, and vitiligo. In contrast, individuals with type 2 DM often exhibit the following features: (1) develop diabetes after the age of 30; (2) are usually obese (80% are obese, but elderly individuals may be lean); (3) may not require insulin therapy initially; and (4) may have associated conditions such as insulin resistance, hypertension, cardiovascular disease, dyslipidemia, or polycystic ovary syndrome. In type 2 DM, insulin resistance is often associated with abdominal obesity (as opposed to hip and thigh obesity) and hypertriglyceridemia. Although most individuals diagnosed with type 2 DM are older, the age of diagnosis appears to be declining in some ethnic groups, and there is a marked increase among overweight teenagers. On the other hand, some individuals (<10%) with the phenotypic appearance of type 2 DM do not have absolute insulin deficiency but have autoimmune markers suggestive of type 1 DM. Thus, despite the revised classification of DM, it remains difficult to categorize some patients unequivocally. Individuals who deviate from the clinical profile of type 1 and type 2 DM, or who have other associated defects such as deafness, pancreatic exocrine disease, and other endocrine disorders, should be classified accordingly ([Table 333-1](#)).

Laboratory Assessment The laboratory assessment should first determine whether

the patient meets the diagnostic criteria for [DM](#) ([Table 333-2](#)) and should then assess the degree of glycemic control (HbA1c, discussed below). In addition to the standard laboratory evaluation, the patient should be screened for DM-associated conditions (e.g., microalbuminuria, dyslipidemia, thyroid dysfunction). Individuals at high risk for cardiovascular disease should be screened for asymptomatic coronary artery disease by appropriate cardiac stress testing, when indicated.

The classification of the type of [DM](#) does not usually require laboratory assessments. Serum insulin or C-peptide measurements do not clearly distinguish type 1 from type 2 DM at the time of diabetes onset; a low C-peptide level merely confirms a patient's need for insulin. Conversely, many individuals with new-onset type 1 DM retain some C-peptide production. Measurement of islet cell antibodies at the time of diabetes onset may be useful if the type of DM is not clear based on the characteristics discussed above, but this knowledge does not usually alter therapy, which is based primarily on empirical metabolic features.

LONG-TERM TREATMENT

OVERALL PRINCIPLES

The goals of therapy for type 1 or type 2 [DM](#) are to: (1) eliminate symptoms related to hyperglycemia, (2) reduce or eliminate the long-term microvascular and macrovascular complications of DM, and (3) allow the patient to achieve as normal a life-style as possible. To reach these goals, the physician should identify a target level of glycemic control for each patient, provide the patient with the educational and pharmacologic resources necessary to reach this level, and monitor/treat DM-related complications. Symptoms of diabetes usually resolve when the plasma glucose is <11.1 mmol/L (200 mg/dL), and thus most DM treatment focuses on achieving the second and third goals.

The care of an individual with either type 1 or type 2 [DM](#) requires a multidisciplinary team. Central to the success of this team are the patient's participation, input, and enthusiasm, all of which are essential for optimal diabetes management. Members of the health care team include the primary care provider and/or the endocrinologist or diabetologist, a certified diabetes educator, and a nutritionist. In addition, when the complications of DM arise, subspecialists (including neurologists, nephrologists, vascular surgeons, cardiologists, ophthalmologists, and podiatrists) with experience in DM-related complications are essential.

A number of names are sometimes applied to different approaches to diabetes care, such as intensive insulin therapy, intensive glycemic control, and "tight control." The current chapter, however, will use the term *comprehensive diabetes care* to emphasize the fact that optimal diabetes therapy involves more than plasma glucose management. Though glycemic control is central to optimal diabetes therapy, comprehensive diabetes care of both type 1 and type 2 [DM](#) should also detect and manage DM-specific complications and modify risk factors for DM-associated diseases.

In addition to assessing the physical aspects of the patient with [DM](#), the physician and members of the diabetes management team should consider social, family, financial, cultural, and employment-related issues that may have an impact on diabetes care.

With this information, the physician can work with the patient and his or her family to establish therapeutic goals and design a comprehensive and feasible plan for optimal diabetes care.

EDUCATION OF THE PATIENT ABOUT DM, NUTRITION, AND EXERCISE

Patient participation is an essential component of comprehensive diabetes care. The patient with type 1 or type 2 [DM](#) should receive education about nutrition, exercise, care of diabetes during illness, and medications to lower the plasma glucose. Along with improved compliance, patient education allows individuals with DM to assume greater responsibility for their care. Patient education should be viewed as a continuing process with regular visits for reinforcement; it should *not* be a process that is completed after one or two visits to a nurse educator or nutritionist.

Diabetes Education The diabetes educator is a health care professional (nurse, dietitian, or pharmacist) with specialized patient education skills who is certified in diabetes education (indicating demonstrated skills in diabetes knowledge and education and certification by the American Association of Diabetes Educators). The educator is a vital member of the comprehensive diabetes care program and educates the patient about a number of issues important for optimal diabetes care, including self-monitoring of blood glucose; urine ketone monitoring (type 1 [DM](#)); insulin administration; guidelines for diabetes management during illnesses; management of hypoglycemia; foot and skin care; diabetes management before, during, and after exercise; and risk factor-modifying activities.

Nutrition *Medical nutrition therapy* (MNT) is a term used by the [ADA](#) to describe the optimal coordination of caloric intake with other aspects of diabetes therapy (insulin, exercise, weight loss). Historically, nutrition has imposed restrictive, complicated regimens on the patient. Current practices have greatly changed, though many patients and health care providers still view the diabetic diet as monolithic and static. For example, modern MNT now includes foods with sucrose and seeks to modify other risk factors such as hyperlipidemia and hypertension rather than focusing exclusively on weight loss in individuals with type 2 [DM](#). Like other aspects of DM therapy, MNT must be adjusted to meet the goals of the individual patient. Furthermore, MNT education is an important component of comprehensive diabetes care and should be reinforced by regular patient education. In general, the components of optimal MNT are similar for individuals with type 1 or type 2 DM ([Table 333-8](#)).

The goal of [MNT](#) in the individual with type 1 [DM](#) is to coordinate and match the caloric intake, both temporally and quantitatively, with the appropriate amount of insulin. MNT in type 1 DM and self-monitoring of blood glucose must be integrated to define the optimal insulin regimen. MNT must be flexible enough to allow for exercise, and the insulin regimen must allow for deviations in caloric intake. An important component of MNT in type 1 DM is to minimize the weight gain often associated with intensive diabetes management.

The goals of [MNT](#) in type 2 [DM](#) are slightly different and address the greatly increased prevalence of cardiovascular risk factors (hypertension, dyslipidemia, obesity) and disease in this population. The majority of these individuals are obese, and weight loss

is still strongly encouraged and should remain an important goal. Medical treatment of obesity is a rapidly evolving area and is discussed in [Chap. 77](#). Hypocaloric diets and modest weight loss often result in rapid and dramatic glucose lowering in individuals with new-onset type 2 DM. Nevertheless, numerous studies document that long-term weight loss is uncommon. Therefore, current MNT for type 2 DM should emphasize modest caloric reduction, increased physical activity, and reduction of hyperlipidemia and hypertension. Increased consumption of soluble, dietary fiber may improve glycemic control in individuals with type 2 DM.

Exercise Exercise is an integral component of comprehensive diabetes care that can have multiple positive benefits (cardiovascular benefits, reduced blood pressure, maintenance of muscle mass, reduction in body fat, weight loss, etc.). For individuals with type 1 or type 2 [DM](#), exercise is also useful for lowering plasma glucose (during and following exercise) and increasing insulin sensitivity.

Despite its benefits, exercise presents several challenges for individuals with [DM](#) because they lack the normal glucoregulatory mechanisms. Skeletal muscle is a major site for metabolic fuel consumption in the resting state, and the increased muscle activity during vigorous, aerobic exercise greatly increases fuel requirements. Individuals with type 1 DM are prone to either hyperglycemia or hypoglycemia during exercise, depending on the preexercise plasma glucose, the circulating insulin level, and the level of exercise-induced catecholamines. If the insulin level is too low, the rise in catecholamines may increase the plasma glucose excessively, promote ketone body formation, and possibly lead to ketoacidosis. Conversely, if the circulating insulin level is excessive, this relative hyperinsulinemia may reduce hepatic glucose production (decreased glycogenolysis, decreased gluconeogenesis) and increase glucose entry into muscle, leading to hypoglycemia.

To avoid exercise-related hyper- or hypoglycemia, individuals with type 1 [DM](#) should: (1) monitor blood glucose before, during, and after exercise; (2) delay exercise if blood glucose is >14 mmol/L (250 mg/dL), <5.5 mmol/L (100 mg/dL), or if ketones are present; (3) eat a meal 1 to 3 h before exercise and take supplemental carbohydrate feedings at least every 30 min during vigorous or prolonged exercise; (4) decrease insulin doses (based on previous experience) before exercise and inject insulin into a nonexercising area; and (5) learn individual glucose responses to different types of exercise and increase food intake for up to 24 h after exercise, depending on intensity and duration of exercise. In individuals with type 2 DM, exercise-related hypoglycemia is less common but can occur in individuals taking either insulin or sulfonylureas.

Because asymptomatic cardiovascular disease appears at a younger age in both type 1 and type 2 [DM](#), formal exercise tolerance testing may be warranted in diabetic individuals with any of the following: age ≥ 35 years, long-standing type 1 DM (>20 to 25 years' duration), microvascular complications of DM (retinopathy, microalbuminuria, or nephropathy), peripheral vascular disease, other risk factors of coronary artery disease, or autonomic neuropathy. Untreated proliferative retinopathy is a relative contraindication to vigorous exercise, since this may lead to vitreous hemorrhage or retinal detachment.

MONITORING THE LEVEL OF GLYCEMIC CONTROL

Optimal monitoring of glycemic control involves plasma glucose measurements by the patient and an assessment of long-term control by the physician (measurement of HbA1c and review of the patient's self-measurements of plasma glucose). These measurements are complementary: the patient's measurements provide a picture of short-term glycemic control, whereas the HbA1c reflects average glycemic control over the previous 2 to 3 months. Integration of both measurements provides an accurate assessment of the glycemic control achieved.

Self-Monitoring of Blood Glucose Self-monitoring of blood glucose (SMBG) is the standard of care in diabetes management and allows the patient to monitor his or her blood glucose at any time. In SMBG, a small drop of blood and an easily detectable enzymatic reaction allow measurement of the capillary plasma glucose. By combining glucose measurements with diet history, medication changes, and exercise history, the physician and patient can improve the treatment program.

The frequency of [SMBG](#) measurements must be individualized and adapted to address the goals of diabetes care as defined by the patient and the health care provider. Individuals with type 1 [DM](#) should routinely measure their plasma glucose four to eight times per day to estimate and select mealtime boluses of short-acting insulin and to modify long-acting insulin doses. Most individuals with type 2 DM require less frequent monitoring, though the optimal frequency of SMBG has not been clearly defined. Individuals with type 2 DM who are on oral medications should utilize SMBG as a means of assessing the efficacy of their medication and diet. Since plasma glucose levels fluctuate less in these individuals, one to two SMBG measurements per day (or fewer) may be sufficient. Individuals with type 2 DM who are on insulin should utilize SMBG more frequently than those on oral agents.

Two devices for continuous blood glucose monitoring have been recently approved by the U.S. Food and Drug Administration (FDA). The Glucowatch uses iontophoresis to assess glucose in interstitial fluid, whereas the Minimed device uses an indwelling subcutaneous catheter to monitor interstitial fluid glucose. Both devices utilize immobilized glucose oxidase to generate electrons in response to changing glucose levels. Though clinical experience with these devices is limited, they perform well in clinical trials and appear to provide useful short-term information about the patterns of glucose changes as well as an enhanced ability to detect hypoglycemic episodes.

Although urine glucose testing does not provide an accurate assessment of glycemic control, urine ketones are a sensitive indicator of early diabetic ketoacidosis and should be measured in individuals with type 1 DM when the plasma glucose is consistently >16.7 mmol/L (300 mg/dL); during a concurrent illness; or with symptoms such as nausea, vomiting, or abdominal pain.

Assessment of Long-Term Glycemic Control Measurement of glycated hemoglobin is the standard method for assessing long-term glycemic control. When plasma glucose is consistently elevated, there is an increase in nonenzymatic glycation of hemoglobin; this alteration reflects the glycemic history over the previous 2 to 3 months, since erythrocytes have an average life span of 120 days. There are numerous laboratory methods for measuring the various forms of glycated hemoglobin, and these have

significant interassay variations. Because of its superior specificity and reliability, the HbA1c assay performed by the high-performance liquid chromatography (HPLC) method has become the standard reference method for most glyated hemoglobin measurements. Since glyated hemoglobin measurements are usually compared to prior measurements, it is essential for the assay results to be comparable. Depending on the assay methodology for HbA1c, hemoglobinopathies, hemolytic anemia, and uremia may interfere with the HbA1c result.

Glyated hemoglobin or HbA1c should be measured in all individuals with [DM](#) during their initial evaluation and as part of their comprehensive diabetes care. As the primary predictor of long-term complications of DM, the HbA1c should mirror, to a certain extent, the short-term measurements of [SMBG](#). These two measurements are complementary in that recent intercurrent illnesses may impact the SMBG measurements but not the HbA1c. Likewise, postprandial and nocturnal hyperglycemia may not be detected by the SMBG of fasting and preprandial capillary plasma glucose but will be reflected in the HbA1c. When measured by [HPLC](#), the HbA1c approximates the following mean plasma glucose values: an HbA1c of 6% is 6.6 mmol/L (120 mg/dL), 7% is 8.3 mmol/L (150 mg/dL), 8% is 10.0 mmol/L (180 mg/dL), etc. [A 1% rise in the HbA1c translates into a 1.7-mmol/L (30 mg/dL) increase in the mean glucose.] The degree of glycation of other proteins, such as albumin, has been used as an alternative indicator of glycemic control when the HbA1c is inaccurate (hemolytic anemia, hemoglobinopathies). The fructosamine assay (using albumin) is an example of an alternative measurement of glycemic control and reflects the glycemic status over the 2 to 4 prior weeks. Current consensus statements do not favor the use of alternative assays of glycemic control, as there are no studies to indicate whether such assays accurately predict the complications of DM.

TREATMENT

Establishment of a Target Level of Glycemic Control Because the complications of [DM](#) are related to glycemic control, normoglycemia or near normoglycemia is the desired, but often elusive, goal for most patients. However, normalization of the plasma glucose for long periods of time is extremely difficult, as demonstrated by the [DCCT](#). Regardless of the level of hyperglycemia, improvement in glycemic control will lower the risk of diabetes complications ([Fig. 333-8](#)).

The target for glycemic control (as reflected by the HbA1c) must be individualized, and the health care provider should establish the goals of therapy in consultation with the patient after considering a number of medical, social, and life-style issues. Some important factors to consider include the patient's age, ability to understand and implement a complex treatment regimen, presence and severity of complications of diabetes, ability to recognize hypoglycemic symptoms, presence of other medical conditions or treatments that might alter the response to therapy, life-style and occupation (e.g., possible consequences of experiencing hypoglycemia on the job), and level of support available from family and friends.

The [ADA](#) has established suggested glycemic goals based on the premise that glycemic control predicts development of [DM](#)-related complications. In general, the target HbA1c should be <7.0% ([Table 333-9](#)). Other consensus groups (such as the Veterans

Administration) have suggested HbA1c goals that take into account the patient's life expectancy at the time of diagnosis and the presence of microvascular complications. Such recommendations strive to balance the financial and personal costs of glycemic therapy with anticipated benefits (reduced health care costs, reduced morbidity). One limitation to this approach is that the onset of hyperglycemia in type 2 DM is difficult to ascertain and likely predates the diagnosis. Furthermore, though the life expectancy can be predicted for a patient population, the physician must treat an individual patient; consequently, the target HbA1c must be individualized to accommodate these other considerations.

Type 1 Diabetes Mellitus

General Aspects Comprehensive diabetes care should be instituted in all individuals with type 1DM and should involve attention to nutrition, exercise, and risk factor management in addition to insulin administration. TheADA recommendations for fasting and bedtime glycemic goals and HbA1c targets are summarized in Table 333-9. The goal is to design and implement insulin regimens that mimic physiologic insulin secretion. Because individuals with type 1 DM lack endogenous insulin production, administration of basal, exogenous insulin is essential for regulating glycogen breakdown, gluconeogenesis, lipolysis, and ketogenesis. Likewise, postprandial insulin replacement should be appropriate for the carbohydrate intake and promote normal glucose utilization and storage.

Intensive Management Intensive diabetes management is defined by theADA as "...a mode of treatment for the person withDM that has the goal of achieving euglycemia or near-normal glycemia using all available resources to accomplish this goal." These resources include thorough and continuing patient education, comprehensive recording of plasma glucose measurements and nutrition intake by the patient, and a variable insulin regimen that matches glucose intake and insulin dose. Insulin regimens usually include multiple-component insulin regimens, multiple daily injections (MDI), or insulin infusion devices (all discussed below).

The benefits of intensive diabetes management and improved glycemic control include a reduction in the microvascular complications ofDM and a possible delay or reduction in the macrovascular complications of DM. From a psychological standpoint, the patient experiences greater control over his or her diabetes and often notes an improved sense of well-being, greater flexibility in the timing and content of meals, and the capability to alter insulin dosing with exercise. In addition, intensive diabetes management in pregnancy reduces fetal malformation and morbidity. Intensive diabetes management is also strongly encouraged in newly diagnosed patients with type 1 DM because it may prolong the period of C-peptide production, which may result in better glycemic control and a reduced risk of serious hypoglycemia.

Although intensive management confers impressive benefits, it is also accompanied by significant personal and financial costs and is therefore not appropriate for all individuals. It requires a combination of dedication, persistence, and motivation on the part of the patient, as well as medical, educational, nursing, nutritional, and psychological expertise on the part of the diabetes management team. Circumstances in which intensive diabetes management should be strongly considered are listed in

[Table 333-10.](#)

Insulin Preparations Current insulin preparations are generated by recombinant DNA technology and consist of the amino acid sequence of human insulin. Animal insulin (beef or pork) is no longer used. Human insulin has been formulated with distinctive pharmacokinetics to mimic physiologic insulin secretion ([Table 333-11](#)). In the United States, all insulin is formulated as U-100 (100 units/mL), whereas in some other countries it is available in other units (e.g., U-40 = 40 units/mL). One short-acting insulin formulation, lispro, is an insulin analogue in which the 28th and 29th amino acids (lysine and proline) on the insulin B chain have been reversed by recombinant DNA technology. This insulin analogue has full biologic activity but less tendency toward subcutaneous aggregation, resulting in more rapid absorption and onset of action and a shorter duration of action. These characteristics are particularly advantageous for allowing entrainment of insulin injection and action to rising plasma glucose levels following meals, although improvement in HbA1c values have not been found consistently. The shorter duration of action also appears to be associated with a decreased number of hypoglycemic episodes, primarily because the decay of lispro action corresponds better to the decline in plasma glucose after a meal. Insulin glargine is a long-acting biosynthetic human insulin that differs from normal insulin in that asparagine is replaced by glycine at amino acid 21, and two arginine residues are added to the C-terminus of the B chain. Compared to NPH insulin, the onset of insulin glargine action is later, the duration of action is longer (~24 h), and there is no pronounced peak. A lower incidence of hypoglycemia, especially at night, was reported in one trial with insulin glargine when compared to NPH insulin. Since glargine has only recently approved, clinical experience is limited. Additional insulin analogues are currently under development.

Basal insulin requirements are provided by intermediate (NPH or lente) or long-acting (ultralente or glargine) insulin formulations. These are usually combined with short-acting insulin in an attempt to mimic physiologic insulin release with meals. Although mixing of intermediate and short-acting insulin formulations is common practice, this mixing may alter the insulin absorption profile (especially those of short-acting insulins). For example, the absorption of regular insulin is delayed when mixed for even short periods of time (<5 min) with lente or ultralente insulin, but not when mixed with NPH insulin. Lispro absorption is delayed by mixing with NPH but not ultralente. Insulin glargine should not be mixed with other insulins. The miscibility of human regular and NPH insulin allows for the production of combination insulins that contain 75% NPH and 25% regular (75/25), 70% NPH and 30% regular (70/30), or equal mixtures of NPH and regular. These combinations of insulin are more convenient for the patient but prevent adjustment of only one component of the insulin formulation. The alteration in insulin absorption when the patient mixes different insulin formulation should not discourage the patient from mixing insulin. However, the following guidelines should be followed: (1) mix the different insulin formulations in the syringe immediately before injection (inject within 2 min after mixing); (2) if possible, do not store insulin as a mixture; and (3) follow the same routine in terms of insulin mixing and administration to standardize the physiologic response to injected insulin.

Insulin Regimens Representations of the various insulin regimens that may be utilized in type 1DM are illustrated in [Fig. 333-12](#). Although the insulin profiles are depicted as

"smooth," symmetric curves, there is considerable patient-to-patient variation in the peak and duration. In all regimens, long-acting insulins (NPH, lente, ultralente, or glargine insulin) supply basal insulin, whereas prandial insulin is provided by either regular or lispro insulin. Lispro should be injected just before a meal; regular insulin is given 30 to 45 min prior to a meal.

A shortcoming of current insulin regimens is that injected insulin immediately enters the systemic circulation, whereas endogenous insulin is secreted into the portal vein. Thus, exogenous insulin administration exposes the liver to subphysiologic insulin levels. No insulin regimen reproduces the precise insulin secretory pattern of the pancreatic islet. However, the most physiologic regimens entail more frequent insulin injections, greater reliance on short-acting insulin, and more frequent capillary plasma glucose measurements. In general, individuals with type 1DM require 0.5 to 1.0 U/kg per day of insulin divided into multiple doses. Initial insulin-dosing regimens should be conservative; approximately 40 to 50% of the insulin should be given as basal insulin. A single daily injection of insulin is not appropriate therapy in type 1 DM.

One commonly used regimen consists of twice-daily injections of an intermediate insulin (NPH or lente) mixed with a short-acting insulin before the morning and evening meal (Fig. 333-12A). Such regimens usually prescribe two-thirds of the total daily insulin dose in the morning (with about two-thirds given as intermediate-acting insulin and one-third as short-acting) and one-third before the evening meal (with approximately one-half given as intermediate-acting insulin and one-half as short-acting). The drawback to such a regimen is that it enforces a rigid schedule on the patient, in terms of daily activity and the content and timing of meals. Although it is simple and effective at avoiding severe hyperglycemia, it does not generate near-normal glycemic control in most individuals with type 1DM. Moreover, if the patient's meal pattern or content varies or if physical activity is increased, hyperglycemia or hypoglycemia may result. Moving the intermediate insulin from before the evening meal to bedtime may avoid nocturnal hypoglycemia and provide more insulin as glucose levels rise in the early morning (so-called dawn phenomenon). The insulin dose in such regimens should be adjusted based onSMBGresults with the following general assumptions: (1) the fasting glucose is primarily determined by the prior evening intermediate-acting insulin; (2) the pre-lunch glucose is a function of the morning short-acting insulin; (3) the pre-supper glucose is a function of the morning intermediate-acting insulin; and (4) the bedtime glucose is a function of the pre-supper, short-acting insulin.

Multiple-component insulin regimens refer to the combination of basal insulin; preprandial short-acting insulin; and changes in short-acting insulin doses to accommodate the results of frequentSMBG, anticipated food intake, and physical activity. Sometimes also referred to as *multiple daily injections*, such regimens offer the patient maximal flexibility in terms of life-style and the best chance for achieving near normoglycemia. One such regimen, shown inFig. 333-12B, consists of a basal insulin with ultralente twice a day and preprandial lispro. The lispro dose is based on individualized algorithms that integrate the preprandial glucose and the anticipated carbohydrate intake. An alternative multiple-component insulin regimen consists of bedtime intermediate insulin, a small dose of intermediate insulin at breakfast (20 to 30% of bedtime dose), and preprandial short-acting insulin. There are numerous variations of these regimens that can be optimized for individual patients. Frequent

SMBG (four to 8 times per day) is absolutely essential for these types of insulin regimens.

Continuous subcutaneous insulin infusion (CSII) is another multiple-component insulin regimen ([Fig. 333-12C](#)). Sophisticated insulin infusion devices are now available that can accurately deliver small doses of insulin (microliters per hour). For example, multiple basal infusion rates can be programmed to: (1) accommodate nocturnal versus daytime basal insulin requirement, (2) alter infusion rate during periods of exercise, or (3) select different waveforms of insulin infusion. A preprandial insulin ("bolus") is delivered by the insulin infusion device based on instructions from the patient, which follow individualized algorithms that account for preprandial plasma glucose and anticipated carbohydrate intake. These devices require a health professional with considerable experience with insulin infusion devices and very frequent patient interactions with the diabetes management team. Insulin infusion devices present unique challenges, such as infection at the infusion site, unexplained hyperglycemia because the infusion set becomes obstructed, or diabetic ketoacidosis if the pump becomes disconnected. Since most physicians use lispro insulin in CSII, the extremely short half-life of this insulin quickly leads to insulin deficiency if the delivery system is interrupted. Essential to the safe use of infusion devices is thorough patient education about pump function and frequent [SMBG](#).

Type 2 Diabetes Mellitus

General Aspects The goals of therapy for type 2 [DM](#) are similar to those in type 1: improved glycemic control with near normalization of the HbA1c. While glycemic control tends to dominate the management of type 1 DM, the care of individuals with type 2 DM must also include attention to the treatment of conditions associated with type 2 DM (obesity, hypertension, dyslipidemia, cardiovascular disease) and detection/management of DM-related complications ([Fig. 333-13](#)). DM-specific complications may be present in up to 20 to 50% of individuals with newly diagnosed type 2 DM. Reduction in cardiovascular risk is of paramount importance as this is the leading cause of mortality in these individuals.

Diabetes management should begin with [MNT](#) (discussed above). An exercise regimen to increase insulin sensitivity and promote weight loss should also be instituted. After MNT and increased physical activity have been instituted, glycemic control should be reassessed; if the patient's glycemic target is not achieved after 3 to 4 weeks of MNT, pharmacologic therapy is indicated. Pharmacologic approaches to the management of type 2 [DM](#) include both oral glucose-lowering agents and insulin; most physicians and patients prefer oral glucose-lowering agents as the initial choice. Any therapy that improves glycemic control reduces "glucose toxicity" to the islet cells and improves endogenous insulin secretion.

Glucose-Lowering Agents Recent advances in the therapy of type 2 [DM](#) have generated considerable enthusiasm for oral glucose-lowering agents that target different pathophysiologic processes in type 2 DM. Based on their mechanisms of action, oral glucose-lowering agents are subdivided into agents that increase insulin secretion, reduce glucose production, or increase insulin sensitivity ([Table 333-12](#)). Oral glucose-lowering agents (with the exception of α -glucosidase inhibitors) are ineffective

in type 1 DM and should not be used for glucose management of severely ill individuals with type 2 DM. Insulin is sometimes the initial glucose-lowering agent.

INSULIN SECRETAGOGUES Insulin secretagogues stimulate insulin secretion by interacting with the ATP-sensitive potassium channel on the beta cell ([Fig. 333-1](#)). These drugs are most effective in individuals with type 2DM of relatively recent onset (<5 years), who have endogenous insulin production and tend to be obese. At maximum doses, first-generation sulfonylureas are similar in potency to second-generation agents but have a longer half-life, a greater incidence of hypoglycemia, and more frequent drug interactions ([Table 333-13](#)). Thus, second-generation sulfonylureas are generally preferred. An advantage to a more rapid onset of action is better coverage of the postprandial glucose rise, but the shorter half-life of such agents requires more than once-a-day dosing. Sulfonylureas reduce both fasting and postprandial glucose and should be initiated at low doses and increased at 1- to 2-week intervals based on [SMBG](#). In general, sulfonylureas increase insulin acutely and thus should be taken shortly before a meal; with chronic therapy, though, the insulin release is more sustained. Repaglinide is not a sulfonylurea but also interacts with the ATP-sensitive potassium channel. Because of its short half-life, it is usually given with or immediately before each meal to reduce meal-related glucose excursions.

Insulin secretagogues are well tolerated in general. All of these agents, however, have the potential to cause profound and persistent hypoglycemia, especially in elderly individuals. Hypoglycemia is usually related to delayed meals, increased physical activity, alcohol intake, or renal insufficiency. Individuals who ingest an overdose of these agents develop prolonged and serious hypoglycemia and should be monitored closely in the hospital ([Chap. 334](#)). Most sulfonylureas are metabolized in the liver to compounds that are cleared by the kidney. Thus, their use in individuals with significant hepatic or renal dysfunction is not advisable. Weight gain, a common side effect of sulfonylurea therapy, results from the increased insulin levels and improvement in glycemic control. Some sulfonylureas have significant drug interactions with other medications such as alcohol, warfarin, aspirin, ketoconazole, α -glucosidase inhibitors, and fluconazole. Despite prior concerns that use of sulfonylureas might increase cardiovascular risk, recent trials have refuted this claim.

BIGUANIDES Metformin is representative of this class of agents. It reduces hepatic glucose production through an undefined mechanism and may improve peripheral glucose utilization slightly ([Table 333-12](#)). Metformin reduces fasting plasma glucose and insulin levels, improves the lipid profile, and promotes modest weight loss. The initial starting dose of 500 mg once or twice a day can be increased to 850 mg tid or 1000 mg bid. Because of its relatively slow onset of action and gastrointestinal symptoms with higher doses, the dose should be escalated every 2 to 3 weeks based on [SMBG](#) measurements. The major toxicity of metformin, lactic acidosis, can be prevented by careful patient selection. Metformin should not be used in patients with renal insufficiency [serum creatinine >133 $\mu\text{mol/L}$ (1.5 mg/dL) in men or >124 $\mu\text{mol/L}$ (1.4 mg/dL) in women, with adjustments for age], any form of acidosis, congestive heart failure, liver disease, or severe hypoxia. Metformin should be discontinued in patients who are seriously ill, in patients who can take nothing orally, and in those receiving radiographic contrast material. Insulin should be used until metformin can be restarted. Though well tolerated in general, some individuals develop gastrointestinal side effects

(diarrhea, anorexia, nausea, and metallic taste) that can be minimized by gradual dose escalation. Because the drug is metabolized in the liver, it should not be used in patients with liver disease or heavy ethanol intake.

α-GLUCOSIDASE INHIBITORS α-Glucosidase inhibitors (acarbose and miglitol) reduce postprandial hyperglycemia by delaying glucose absorption; they do not affect glucose utilization or insulin secretion ([Table 333-12](#)). Postprandial hyperglycemia, secondary to impaired hepatic and peripheral glucose disposal, contributes significantly to the hyperglycemic state in type 2DM. These drugs, taken just before each meal, reduce glucose absorption by inhibiting the enzyme that cleaves oligosaccharides into simple sugars in the intestinal lumen. Therapy should be initiated at a low dose (25 mg of acarbose or miglitol) with the evening meal and may be increased to a maximal dose over weeks to months (50 to 100 mg for acarbose or 50 mg for miglitol with each meal). The major side effects (diarrhea, flatulence, abdominal distention) are related to increased delivery of oligosaccharides to the large bowel and can be reduced somewhat by gradual upward dose titration. α-Glucosidase inhibitors may increase levels of sulfonylureas and increase the incidence of hypoglycemia. Simultaneous treatment with bile acid resins and antacids should be avoided. These agents should not be used in individuals with inflammatory bowel disease, gastroparesis, or a serum creatinine >177 μmol/L (2.0 mg/dL). This class of agents is not as potent as other oral agents in lowering the HbA1c but is unique in that it reduces the postprandial glucose rise even in individuals with type 1 DM.

THIAZOLIDINEDIONES Thiazolidinediones represent a new class of agents that reduce insulin resistance. These drugs bind to a nuclear receptor (peroxisome proliferator-activated receptor, PPAR-γ) that regulates gene transcription. The PPAR-γ receptor is found at highest levels in adipocytes but is expressed at lower levels in many other insulin-sensitive tissues. Agonists of this receptor promote adipocyte differentiation and may reduce insulin resistance in skeletal muscle indirectly. Thiazolidinediones reduce the fasting plasma glucose by improving peripheral glucose utilization and insulin sensitivity ([Table 333-12](#)). Circulating insulin levels decrease with use of the thiazolidinediones, indicating a reduction in insulin resistance. Although direct comparisons are not available, the two currently available thiazolidinediones appear to have similar efficacy; the therapeutic range for pioglitazone is 15 to 45 mg/d in a single daily dose and for rosiglitazone is 2 to 8 mg/d -- once a day at lower doses and bid at higher doses. The ability of thiazolidinediones to influence other features of the insulin resistance syndrome is under investigation.

The prototype of this class of drugs, troglitazone, was withdrawn from the U.S. market after reports of hepatotoxicity and an association with an idiosyncratic liver reaction that sometimes led to hepatic failure. The two other thiazolidinediones, rosiglitazone and pioglitazone, thus far do not appear to induce the liver abnormalities seen with troglitazone. However, long-term experience with the newer agents is limited. Consequently, the [FDA](#) recommends measurement of liver function tests prior to initiating therapy with a thiazolidinedione and at regular intervals (every two months for the first year and then periodically). The thiazolidinediones raise [LDL](#) and [HDL](#) slightly and lower triglycerides by 10 to 15%, but the clinical significance of these changes is not known. Thiazolidinediones are associated with minor weight gain (1 to 2 kg), a small reduction in the hematocrit, and a mild increase in plasma volume. Cardiac function is not

affected, but the incidence of peripheral edema is increased. They are contraindicated in patients with liver disease or congestive heart failure (class III or IV).

Thiazolidinediones have been shown to induce ovulation in premenopausal women with polycystic ovary syndrome (see "Insulin Resistance Syndromes," above). Women should be warned about the risk of pregnancy, since the safety of thiazolidinediones in pregnancy is not established.

INSULIN THERAPY IN TYPE 2 DM Modest doses of insulin are quite efficacious in controlling hyperglycemia in newly diagnosed type 2DM. Insulin should be considered as the initial therapy in type 2 DM, particularly in lean individuals or those with severe weight loss, in individuals with underlying renal or hepatic disease that precludes oral glucose-lowering agents, or in individuals who are hospitalized or acutely ill. Insulin therapy is ultimately required by a substantial number of individuals with type 2 DM because of the progressive nature of the disorder and the relative insulin deficiency that develops in patients with long-standing diabetes.

Because endogenous insulin secretion continues and is capable of providing some coverage of mealtime caloric intake, insulin is usually initiated in a single dose of intermediate-acting insulin (0.3 to 0.4 U/kg per day), given either before breakfast or just before bedtime (or ultralente at bedtime). Since fasting hyperglycemia and increased hepatic glucose production are prominent features of type 2DM, bedtime insulin is more effective in clinical trials than a single dose of morning insulin. Some physicians prefer a relatively low, fixed starting dose of intermediate-acting insulin (~15 to 20 units in the morning and 5 to 10 units at bedtime) to avoid hypoglycemia. The insulin dose may then be adjusted in 10% increments as dictated by SMBG results. Both morning and bedtime intermediate insulin may be used in combination with oral glucose-lowering agents (biguanides, α -glucosidase inhibitors, or thiazolidinediones).

CHOICE OF INITIAL GLUCOSE-LOWERING AGENT Though insulin is an effective primary therapy for type 2DM, most patients and physicians currently prefer oral glucose-lowering drugs as the initial pharmacologic approach. The level of hyperglycemia should influence the initial choice of therapy. Assuming maximal benefit of MNT and increased physical activity has been realized, patients with mild to moderate hyperglycemia [fasting plasma glucose <11.1 to 13.9 mmol/L (200 to 250 mg/dL)] often respond well to a single oral glucose-lowering agent. Patients with more severe hyperglycemia [fasting plasma glucose >13.9 mmol/L (250 mg/dL)] may respond partially but are unlikely to achieve normoglycemia with oral monotherapy. Nevertheless, many physicians prefer a stepwise approach that starts with a single agent and adds a second agent to achieve the glycemic target (see "Combination Therapy," below). Some physicians begin insulin in individuals with severe hyperglycemia [fasting plasma glucose >13.9 to 16.7 mmol/L (250 to 300 mg/dL)]. This approach is based on the rationale that more rapid glycemic control will reduce "glucose toxicity" to the islet cells, improve endogenous insulin secretion, and possibly allow oral glucose-lowering agents to be more effective. If this occurs, the insulin may be discontinued.

Insulin secretagogues, biguanides, α -glucosidase inhibitors, thiazolidinediones, and insulin are approved for monotherapy of type 2DM. Although each class of oral glucose-lowering agents has unique advantages and disadvantages, certain generalizations apply: (1) insulin secretagogues, biguanides, and thiazolidinediones

improve glycemic control to a similar degree (1 to 2% reduction in HbA1c) and are more effective than α -glucosidase inhibitors; (2) assuming a similar degree of glycemic improvement, no clinical advantage to one class of drugs has been demonstrated, and any therapy that improves glycemic control is beneficial; (3) insulin secretagogues and α -glucosidase inhibitors begin to lower the plasma glucose immediately, whereas the glucose-lowering effects of the biguanides and thiazolidinediones are delayed by several weeks to months; (4) not all agents are effective in all individuals with type 2 DM (primary failure); (5) biguanides, α -glucosidase inhibitors, and thiazolidinediones do not directly cause hypoglycemia; and (6) most individuals will eventually require treatment with more than one class of oral glucose-lowering agents, reflecting the progressive nature of type 2 DM.

Considerable clinical experience exists with sulfonylureas and metformin because they have been available for several decades. It is assumed that the α -glucosidase inhibitors and thiazolidinediones, which are newer classes of oral glucose-lowering drugs, will reduce DM-related complications by improving glycemic control, although long-term data are not yet available. The thiazolidinediones are theoretically attractive because they target a fundamental abnormality in type 2 DM, namely insulin resistance. However, these agents are currently more costly than others and require liver function monitoring.

A reasonable treatment algorithm for initial therapy proposes either a sulfonylurea or metformin as initial therapy because of their efficacy, known side-effect profile, and relatively low cost (Fig. 333-14). Metformin has the advantage that it promotes mild weight loss, lowers insulin levels, improves the lipid profile slightly, and may have a lower secondary failure rate. However, there is no difference in response rate or degree of glycemic control when metformin and sulfonylureas are compared in randomized, prospective clinical trials. Based on SMBG results and the HbA1c, the dose of either the sulfonylurea or metformin should be increased until the glycemic target is achieved. α -Glucosidase inhibitors and thiazolidinediones are alternative, initial agents (Fig. 333-14).

When used as monotherapy, approximately one-third of individuals will reach their target glycemic goal with either a sulfonylurea or metformin. Approximately 25% of individuals will not respond to sulfonylureas or metformin; under these circumstances, the drug usually should be discontinued. Some individuals respond to one agent but not the other. The remaining individuals treated with either sulfonylureas or metformin alone will exhibit some improvement in glycemic control but will not achieve their glycemic target and should be considered for combination therapy.

COMBINATION THERAPY WITH GLUCOSE-LOWERING AGENTS A number of combinations of therapeutic agents are successful in type 2 DM, and the dosing of agents in combination is the same as when the agents are used alone. Because mechanisms of action of the first and second agents are different, the effect on glycemic control is usually additive. Commonly used regimens include: (1) insulin secretagogue with metformin or thiazolidinedione, (2) sulfonylurea with α -glucosidase inhibitor, and (3) insulin with metformin or thiazolidinedione. The combination of metformin and a thiazolidinedione is also effective and complementary. If adequate control is not achieved with two oral agents, bedtime insulin or a third oral agent may be added stepwise. However, long-term experience with any triple combination is lacking, and

experience with two-drug combinations is relatively limited.

Insulin becomes required as type 2DM enters the phase of relative insulin deficiency (as seen in long-standing DM) and is signaled by inadequate glycemic control on one or two oral glucose-lowering agents. Insulin can be used in combination with any of the oral agents in patients who fail to reach the glycemic target. For example, a single dose of intermediate-acting insulin at bedtime is effective in combination with metformin. As endogenous insulin production falls further, multiple injections of intermediate-acting and short-acting insulin regimens are necessary to control postprandial glucose excursions. These combination regimens are identical to the intermediate- and short-acting combination regimens discussed above for type 1 DM. Since the hyperglycemia of type 2 DM tends to be more "stable," these regimens can be increased in 10% increments every 2 to 3 days using SMBG results. The daily insulin dose required can become quite large (1 to 2 units/kg per day) as endogenous insulin production falls and insulin resistance persists. Individuals who require >1 unit/kg per day of intermediate-acting insulin should be considered for combination therapy with metformin or a thiazolidinedione. The addition of a thiazolidinedione can reduce insulin requirements in some individuals with type 2 DM, while maintaining or even improving glycemic control.

Intensive diabetes management ([Table 333-10](#)) is a treatment option in type 2 patients who cannot achieve optimal glycemic control and are capable of implementing such regimens. A recent study from the Veterans Administration found that intensive diabetes management is not associated with a greater degree of side effects (hypoglycemia, weight gain) than standard insulin therapy. The effect of higher insulin levels associated with intensive diabetes management on the prognosis of diseases commonly associated with type 2DM (cardiovascular disease, hypertension) is still debated. In selected patients with type 2 DM, insulin pumps improve glycemic control and are well tolerated.

Emerging Therapies Whole pancreas transplantation (conventionally performed concomitantly with a renal transplant) may normalize glucose tolerance and is an important therapeutic option in type 1 diabetes, though it requires substantial expertise and is associated with the side effects of immunosuppression. Pancreatic islet transplantation has been plagued by limitations in pancreatic islet isolation and graft survival, but recent advances in specific immunomodulation have greatly improved the results. Islet transplantation is an area of active clinical investigation.

Advances in molecular biology and new insights into normal mechanisms of glucose homeostasis have led to a number of emerging therapies for diabetes and its complications. For example, glucagon-like peptide 1, a potent insulin secretagogue, may be efficacious in type 2DM. Inhaled insulin and additional insulin analogues are in advanced stages of clinical trials. Aminoguanidine, an inhibitor of the formation of advanced glycosylation end products, and inhibitors of protein kinase C may reduce the complications of DM. Closed-loop pumps that infuse the appropriate amount of insulin in response to changing glucose levels are potentially feasible now that continuous glucose-monitoring technology has been developed.

COMPLICATIONS OF THERAPY FOR DIABETES MELLITUS

As with any therapy, the benefits of efforts directed towards glycemic control must be weighed against the risks of treatment. Side effects of intensive treatment include an increased frequency of serious hypoglycemia, weight gain, increased economic costs, and greater demands on the patient. In the [DCCT](#), quality of life was very similar in the intensive therapy and standard therapy groups. The most serious complication of therapy for [DM](#) is hypoglycemia ([Chap. 334](#)). Weight gain occurs with most (insulin, insulin secretagogues, thiazolidinediones) but not all (metformin and α -glucosidase inhibitors) therapies that improve glycemic control due to the anabolic effects of insulin and the reduction in glucosuria. In the DCCT, individuals with the greatest weight gain exhibited increases in [LDL](#) cholesterol and triglycerides as well as increases in blood pressure (both systolic and diastolic) similar to those seen in individuals with type 2 DM and insulin resistance. These effects could increase the risk of cardiovascular disease in intensively managed patients. As discussed previously, improved glycemic control is sometimes accompanied by a transient worsening of diabetic retinopathy or neuropathy.

ONGOING ASPECTS OF COMPREHENSIVE DIABETES CARE

The morbidity and mortality of [DM](#)-related complications can be greatly reduced by timely and consistent surveillance procedures ([Table 333-14](#)). These screening procedures are indicated for all individuals with DM, but numerous studies have documented that most individuals with diabetes do not receive comprehensive diabetes care. Screening for dyslipidemia and hypertension should be performed annually. In addition to routine health maintenance, individuals with diabetes should also receive the pneumococcal and tetanus vaccines (at recommended intervals) and the influenza vaccine (annually).

An annual comprehensive eye examination should be performed by a qualified optometrist or ophthalmologist. If abnormalities are detected, further evaluation and treatment require an ophthalmologist skilled in diabetes-related eye disease. Because many individuals with type 2 [DM](#) have had asymptomatic diabetes for several years before diagnosis, a consensus panel from the [ADA](#) recommends the following ophthalmologic examination schedule: (1) individuals with onset of DM at <29 years should have an initial eye examination within 3 to 5 years of diagnosis, (2) individuals with onset of DM at >30 years should have an initial eye examination at the time of diabetes diagnosis, and (3) women with DM who are contemplating pregnancy should have an eye examination prior to conception and during the first trimester.

An annual foot examination should: (1) assess blood flow, sensation, and nail care; (2) look for the presence of foot deformities such as hammer or claw toes and Charcot foot; and (3) identify sites of potential ulceration. Calluses and nail deformities should be treated by a podiatrist; the patient should be discouraged from self-care of even minor foot problems.

An annual microalbuminuria measurement is advised in individuals with type 1 or type 2 [DM](#) and no protein on a routine urinalysis ([Fig. 333-10](#)). If the urinalysis detects proteinuria, the amount of protein should be quantified by standard urine protein measurements. If the urinalysis was negative for protein in the past, microalbuminuria should be the annual screening examination. Routine urine protein measurements do

not detect low levels of albumin excretion. Screening should commence 5 years after the onset of type 1 DM and at the time of onset of type 2 DM.

SPECIAL CONSIDERATIONS IN DIABETES MELLITUS

PSYCHOSOCIAL ASPECTS

As with any chronic, debilitating disease, the individual with [DM](#) faces a series of challenges that affect all aspects of daily life. The individual with DM must accept that he or she may develop complications related to DM. Even with considerable effort, normoglycemia can be an elusive goal, and solutions to worsening glycemic control may not be easily identifiable. The patient should view him- or herself as an essential member of the diabetes care team and not as someone who is cared for by the diabetes team. Emotional stress may provoke a change in behavior so that individuals no longer adhere to a dietary, exercise, or therapeutic regimen. This can lead to the appearance of either hyper- or hypoglycemia. Depression and eating disorders (in women) are more common in individuals with type 1 or type 2 DM ([Chap. 78](#)).

MANAGEMENT IN THE HOSPITALIZED PATIENT

Virtually all medical and surgical subspecialties may be involved in the care of hospitalized patients with diabetes. General anesthesia, surgery, and concurrent illness raise the levels of counterregulatory hormones (cortisol, growth hormone, catecholamines, and glucagon), and infection may lead to transient insulin resistance. These factors increase insulin requirements by increasing glucose production and impairing glucose utilization and thus may worsen glycemic control. On the other hand, the concurrent illness or surgical procedure may prevent the patient with [DM](#) from eating normally and may promote hypoglycemia. Glycemic control should be assessed (with HbA1c) and, if feasible, should be optimized prior to surgery. Electrolytes, renal function, and intravascular volume status should be assessed as well. The extremely high prevalence of asymptomatic cardiovascular disease in individuals with DM (especially in type 2 DM) may require preoperative cardiovascular evaluation.

The goals of diabetes management during hospitalization are avoidance of hypoglycemia, optimization of glycemic control, and transition back to the outpatient diabetes treatment regimen. Attention to each stage in this process requires integrating information regarding the plasma glucose, diabetes treatment regimen, and clinical status of the patient. For example, some surgical procedures utilizing local anesthesia or epidural anesthesia may have minimal effects on glycemic control. If the patient is eating soon after the procedure and there is no disruption of the patient's regular meal plans, then glycemic control is usually maintained.

The physician caring for an individual with diabetes in the perioperative period, during times of infection or serious physical illness, or simply when fasting for a diagnostic procedure must monitor the plasma glucose vigilantly, adjust the diabetes treatment regimen, and provide glucose infusion as needed. Several different treatment regimens (intravenous or subcutaneous insulin regimens) can be employed successfully. Individuals with type 1 [DM](#) require continued insulin administration to maintain the levels of circulating insulin necessary to prevent [DKA](#). Prolongation of a surgical procedure or

delay in the recovery room is not uncommon and may result in periods of insulin deficiency. Even relatively brief periods without insulin may lead to mild DKA. Individuals with type 1 DM who are undergoing general anesthesia and surgery, or who are seriously ill, should receive continuous insulin, either through an intravenous insulin infusion or by subcutaneous administration of a reduced dose of long-acting insulin. Short-acting insulin alone is insufficient.

Individuals with type 2DM can be managed with either insulin infusion or a reduced dose of subcutaneous insulin. Oral glucose-lowering agents are discontinued at the time a combined insulin/glucose infusion is started. Oral agents such as sulfonylureas, metformin, acarbose, and thiazolidinediones are not useful in regulating the plasma glucose in clinical situations where the insulin requirements and glucose intake are changing rapidly. Moreover, these oral agents may be dangerous if the patient is fasting (e.g., hypoglycemia with sulfonylureas). Metformin should be withheld when radiographic contrast media will be given or if severe congestive heart failure, acidosis, or declining renal function is present.

Insulin infusions can effectively control plasma glucose in the perioperative period and when the patient is unable to take anything by mouth. The absorption of subcutaneous insulin may be variable in such situations because of changes in blood flow. The physician must consider carefully the clinical setting in which an insulin infusion will be utilized, including whether adequate ancillary personnel are available to monitor the plasma glucose frequently and whether they can adjust the insulin infusion rate, either based on an algorithm or in consultation with the physician. The initial rate for an insulin infusion may range from 0.5 to 5 units/h, depending on the degree of insulin resistance and the clinical situation. Based on hourly capillary glucose measurements, the insulin infusion rate is adjusted to maintain the plasma glucose within the desired range [5.6 to 11.1 mmol/L (100 to 200 mg/dL)]. Glucose infusion, initiated at the time the patient begins fasting, should be adjusted to deliver the equivalent of 50 to 150 mL of D₅W/h until the patient is reliably taking nutrition orally. The insulin infusion can be temporarily discontinued if hypoglycemia occurs and may be resumed at a lower infusion rate once the plasma glucose exceeds 5.6 mmol/L (100 mg/dL).

Insulin infusion is the preferred method for managing patients with type 1DM in the perioperative period or when serious concurrent illness is present. Individuals with type 2 DM can be managed with an insulin infusion, but subcutaneous insulin in reduced doses can be used effectively as well. If the diagnostic or surgical procedure is brief and performed under local or regional anesthesia, a reduced dose of subcutaneous, long-acting insulin may suffice. This approach facilitates the transition back to the long-acting insulin after the procedure. The dose of long-acting insulin should be reduced by 30 to 40%, and short-acting insulin is either held or, likewise, reduced by 30 to 40%. Glucose should be infused to prevent hypoglycemia.

Total Parenteral Nutrition (See [Chap. 76](#)) Total parenteral nutrition (TPN) greatly increases insulin requirements. In addition, individuals not previously known to have DM may become hyperglycemic during TPN and require insulin treatment. Intravenous insulin infusion is the preferred treatment for hyperglycemia, and rapid titration to the required insulin dose is done most efficiently using a separate insulin infusion. After the total insulin dose has been determined, insulin may be added directly to the TPN

solution. Often, individuals receiving either TPN or enteral nutrition receive their caloric loads continuously and not at "meal times"; consequently, subcutaneous insulin regimens must be adjusted.

GLUCOCORTICOIDS

Glucocorticoids increase insulin resistance, decrease glucose utilization, increase hepatic glucose production, and impair insulin secretion. These changes lead to a worsening of glycemic control in individuals with [DM](#) and may precipitate diabetes in other individuals ("steroid-induced diabetes"). The effects of glucocorticoids on glucose homeostasis are dose-related, usually reversible, and most pronounced in the postprandial period. If the fasting plasma glucose is near the normal range, oral diabetes agents (sulfonylureas and acarbose) may be sufficient to reduce hyperglycemia. If the fasting plasma glucose >11.1 mmol/L (200 mg/dL), oral agents are usually not efficacious and insulin therapy is required. Short-acting insulin may be required to supplement long-acting insulin in order to control postprandial glucose excursions.

REPRODUCTIVE ISSUES

Reproductive capacity in either men or women with [DM](#) appears to be normal. Menstrual cycles may be associated with alterations in glycemic control in women with DM. Pregnancy is associated with marked insulin resistance; the increased insulin requirements often precipitate DM and lead to the diagnosis of [GDM](#). Glucose, which at high levels is a teratogen to the developing fetus, readily crosses the placenta, but insulin does not. Thus, hyperglycemia or hypoglycemia from the maternal circulation may stimulate insulin secretion in the fetus. The anabolic and growth effects of insulin may result in macrosomia. GDM complicates approximately 4% of pregnancies in the United States. The incidence of GDM is greatly increased in certain ethnic groups, including African Americans and Hispanic Americans, consistent with a similar increased risk of type 2 DM. Current recommendations advise screening for glucose intolerance between weeks 24 and 28 of pregnancy in women with high risk for GDM (≥25 years; obesity; family history of DM; member of an ethnic group such as Hispanic American, Native American, Asian American, African American, or Pacific Islander). Therapy for GDM is similar to that for individuals with pregnancy-associated diabetes and involves [MNT](#) and insulin, if hyperglycemia persists. Oral glucose-lowering agents have not been approved for use during pregnancy. With current practices, the morbidity and mortality of the mother with GDM and the fetus are no different from those in the nondiabetic population. Individuals who develop GDM are at marked increased risk for developing type 2 DM in the future and should be screened periodically for DM. After delivery, glucose homeostasis should be reassessed in the mother. Most individuals with GDM revert to normal glucose tolerance, but some will continue to have overt diabetes or impairment of glucose tolerance. In addition, children of women with GDM appear to be at risk for obesity and glucose intolerance and have an increased risk of diabetes beginning in the later stages of adolescence.

Pregnancy in individuals with known [DM](#) requires meticulous planning and adherence to strict treatment regimens. Intensive diabetes management and normalization of the HbA1c are the standard of care for individuals with existing DM who are planning

pregnancy. The crucial period of glycemic control is extremely early following fertilization. The risk of fetal malformations is increased 4 to 10 times in individuals with uncontrolled DM at the time of conception. The goals are normal plasma glucose during the preconception period and throughout the periods of organ development in the fetus.

LIPODYSTROPHIC DM (See also [Chap. 354](#))

Lipodystrophy, or the loss of subcutaneous fat tissue, may be generalized in certain genetic conditions such as leprechaunism. Generalized lipodystrophy is associated with severe insulin resistance and is often accompanied by acanthosis nigricans and dyslipidemia. Localized lipodystrophy associated with insulin injections has been reduced considerably by the use of human insulin.

Protease Inhibitors and Lipodystrophy Protease inhibitors used in the treatment of HIV disease ([Chap. 309](#)) have been associated with a centripetal accumulation of fat (visceral and abdominal area), accumulation of fat in the dorsocervical region, loss of extremity fat, decreased insulin sensitivity (elevations of the fasting insulin level and reduced glucose tolerance on intravenous glucose tolerance testing), and dyslipidemia. Although many aspects of the physical appearance of these individuals resemble Cushing's syndrome, derangements in cortisol secretion have not been found consistently and do not appear to account for this appearance. Although some individuals have [IGT](#), diabetes is not a common feature. The possibility remains that this is related to HIV infection by some undefined mechanism, since some features of the syndrome were observed before the introduction of protease inhibitors. Therapy for HIV-related lipodystrophy is not well established.

(Bibliography omitted in Palm version)

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334. HYPOGLYCEMIA - Philip E. Cryer

Hypoglycemia occurs most commonly as a result of treating patients with diabetes mellitus. However, a number of other disorders, including insulinoma, large mesenchymal tumors, end-stage organ failure, alcoholism, endocrine deficiencies, postprandial reactive hypoglycemic conditions, and inherited metabolic disorders, are also associated with hypoglycemia ([Table 334-1](#)). Hypoglycemia is sometimes defined as a plasma glucose level <2.5 to 2.8 mmol/L (<45 to 50 mg/dL). However, as discussed below, the glucose thresholds for hypoglycemia-induced symptoms and physiologic responses vary widely, depending on the clinical setting. Therefore, *Whipple's triad* provides an important framework for making the diagnosis of hypoglycemia: (1) symptoms consistent with hypoglycemia, (2) a low plasma glucose concentration, and (3) relief of symptoms after the plasma glucose level is raised. Hypoglycemia can cause significant morbidity and can be lethal, if severe or prolonged; it should be considered in any patient who presents with confusion, altered level of consciousness, or seizures.

SYSTEMIC GLUCOSE BALANCE AND COUNTERREGULATION

Glucose is an obligate metabolic fuel for the brain under physiologic conditions. By contrast, other organs can use fatty acids, in addition to glucose, to generate energy. The brain cannot synthesize glucose and stores only a few minutes' supply as glycogen. It therefore requires a continuous supply of glucose, which is delivered by facilitated diffusion from arterial blood. As the plasma glucose concentration falls below the physiologic range, blood-to-brain glucose transport becomes insufficient for adequate brain energy metabolism and functioning. It is therefore not surprising that redundant physiologic mechanisms prevent or rapidly correct hypoglycemia.

Plasma glucose levels are maintained within a narrow range, usually between 3.3 and 8.3 mmol/L (60 and 150 mg/dL), despite wide variation in food intake and activity level. This delicate balance requires dynamic regulation of glucose influx into the circulation as glucose utilization in various tissues can change rapidly. The diet is normally a major source of glucose. However, between meals or during fasting, serum glucose levels are maintained primarily by the breakdown of glycogen in the liver and by gluconeogenesis ([Fig. 334-1](#)). In most people, hepatic glycogen stores are sufficient to maintain plasma glucose levels for 8 to 12 h, but this time period can be shorter if glucose demand is increased by exercise or if glycogen stores are depleted by illness or starvation.

As glycogen stores are depleted, glucose is generated by gluconeogenesis, which occurs primarily in the liver but also in the kidney. Gluconeogenesis requires a coordinated supply of precursors from liver, muscle, and adipose tissue. Muscle provides lactate, pyruvate, alanine, and other amino acids; triglycerides in adipose tissue are broken down into glycerol, which is a precursor for gluconeogenesis, and free fatty acids, which generate acetyl CoA for gluconeogenesis and provide an alternative fuel source to tissues other than brain.

The balance of glucose production and its uptake and utilization in peripheral tissues is exquisitely regulated by a network of hormones, neural pathways, and metabolic signals ([Chap. 333](#)). Among the factors that control glucose production and utilization, insulin

plays a dominant and pivotal role. In the fasting state, insulin is suppressed, allowing increased gluconeogenesis in the liver and the kidney and enhancing glucose generation by the breakdown of liver glycogen. Low insulin levels also reduce glucose uptake and utilization in peripheral tissues and allow lipolysis and proteolysis to occur, leading to the release of precursors for gluconeogenesis and providing alternative energy sources. In the fed state, insulin release from the pancreatic cells reverses these processes. Glycogenolysis and gluconeogenesis are inhibited, thereby reducing hepatic and renal glucose output; peripheral glucose uptake and utilization are enhanced; lipolysis and proteolysis are restrained; and energy storage is promoted by the conversion of substrates into glycogen, triglycerides, and proteins. Other hormones including glucagon, epinephrine, growth hormone, and cortisol play less important roles in the control of glucose flux during normal physiologic circumstances. However, as described below, these hormones are critically important in the response to hypoglycemia.

As glucose levels approach, and ultimately enter, the hypoglycemic range, a characteristic sequence of *counterregulatory hormone responses* occurs. Glucagon is the first and most important of these responses. It promotes glycogenolysis and gluconeogenesis. Epinephrine can also play an important role in the acute response to hypoglycemia, particularly when glucagon is insufficient. It, too, stimulates glycogenolysis and gluconeogenesis as well as limiting glucose utilization by insulin-sensitive tissues. When hypoglycemia is prolonged, growth hormone and cortisol also reduce glucose utilization and support its production.

The glucose thresholds at which various counterregulatory hormone responses occur are quite similar in healthy subjects ([Table 334-2](#)). Nonetheless, these thresholds are dynamic and can be influenced by recent metabolic events. A person with poorly controlled diabetes can have symptoms of hypoglycemia at higher-than-normal glucose levels. Recurrent hypoglycemia, as may occur in individuals with diabetes or in the setting of an insulinoma, shifts thresholds for symptoms and counterregulatory responses to lower glucose levels.

CLINICAL MANIFESTATIONS

Symptoms of hypoglycemia can be divided into two categories, neuroglycopenic and neurogenic (or autonomic) responses. Neuroglycopenic symptoms are the direct result of central nervous system neuronal glucose deprivation. They include behavioral changes, confusion, fatigue, seizure, loss of consciousness, and, if hypoglycemia is severe and prolonged, death. Hypoglycemia-induced autonomic responses include adrenergic symptoms such as palpitations, tremor, and anxiety as well as cholinergic symptoms such as sweating, hunger, and paresthesia. Adrenergic symptoms are mediated by norepinephrine released from sympathetic postganglionic neurons and the release of epinephrine from the adrenal medullae. Increased sweating is mediated by cholinergic sympathetic nerve fibers. Patients with diabetes mellitus learn to recognize the characteristic symptoms of hypoglycemia, but these are less familiar to individuals with other causes of hypoglycemia. Symptoms may be less pronounced with repeated hypoglycemic episodes (see below).

Common signs of hypoglycemia include pallor and diaphoresis. Heart rate and the

systolic blood pressure are typically raised, but these findings may not be prominent. The neuroglycopenic manifestations are valuable, albeit nonspecific, signs. Transient focal neurologic deficits occur occasionally.

CAUSES

Hypoglycemia is traditionally classified as *postprandial* or *fasting*. However, in the clinical setting, hypoglycemia most commonly results from the treatment of diabetes. This topic is therefore addressed before considering the other causes of hypoglycemia.

HYPOGLYCEMIA IN DIABETES

Frequency and Impact Were it not for hypoglycemia, diabetes would be rather easy to treat by administering enough insulin (or any effective drug) to lower plasma glucose concentrations to, or below, the normal range. Because of imperfections in all current insulin-replacement regimens, individuals with type 1 diabetes are at ongoing risk for periods of relative hyperinsulinemia with resultant hypoglycemia. Those attempting to achieve near-normal glycemic control may experience several episodes of asymptomatic or symptomatic hypoglycemia each week. Plasma glucose levels may be <2.8 mmol/L (<50 mg/dL) as much as 10% of the time. At least 25% of such patients suffer an episode of severe, temporarily disabling hypoglycemia, often with seizure or coma, in a given year. Although seemingly complete recovery from the latter is the rule, the possibility of persistent cognitive deficits has been raised, but permanent neurologic defects are rare. About 2 to 4% of deaths associated with type 1 diabetes are estimated to result from hypoglycemia. Fear of hypoglycemia also can lead to disabling psychosocial morbidity.

Hypoglycemia is a less frequent problem in type 2 diabetes but occurs nevertheless in those treated with insulin or sulfonylureas. Transient, mild hypoglycemia may be seen with the shorter-acting sulfonylureas and repaglinide, which also acts by enhancing insulin secretion. Patients taking the long-acting sulfonylureas, chlorpropamide and glyburide, occasionally experience episodes of severe hypoglycemia that may last up to 24 to 36 h.

Conventional Risk Factors Insulin excess is the primary determinant of risk from iatrogenic hypoglycemia. Relative or absolute insulin excess occurs when: (1) insulin (or oral agent) doses are excessive, ill timed, or of the wrong type; (2) the influx of exogenous glucose is reduced, as during an overnight fast or following missed meals or snacks; (3) insulin-independent glucose utilization is increased, as during exercise; (4) insulin sensitivity is increased, as occurs with effective intensive therapy, in the middle of the night, late after exercise, or with increased fitness or weight loss; (5) endogenous glucose production is reduced, as following alcohol ingestion; and (6) insulin clearance is reduced, as in renal failure. However, analyses of the Diabetes Control and Complications Trial (DCCT) indicate that these conventional risk factors explain only a minority of episodes of severe iatrogenic hypoglycemia and that other causes are involved in the majority of episodes.

Hypoglycemia-Associated Autonomic Failure It is now clear that inadequate physiologic counterregulatory and behavioral responses greatly compound the problem

of hypoglycemia caused by insulin excess. Hypoglycemia-associated autonomic failure has two main components: (1) reduced counterregulatory hormone responses, which result in impaired glucose generation; and (2) hypoglycemia unawareness, which precludes appropriate behavioral responses, such as eating.

Defective Glucose Counterregulation The counterregulatory hormone response is fundamentally altered in all people with established (e.g., absent C peptide) type 1 diabetes. As the patient becomes totally insulin-deficient over the first few months or years of the disease, circulating insulin levels are no longer tightly coordinated with glucose levels and are a passive function of administered insulin. Thus, insulin levels do not always decline as glucose levels fall; the first defense against hypoglycemia is lost. Over the same time frame, the glucagon response to falling glucose levels diminishes, and the second defense against hypoglycemia is lost. The cause of defective glucagon production by the pancreatic islet α cells is unknown, but it is tightly linked to the loss of insulin production by the β cells. It is a functional abnormality rather than an absolute deficiency of glucagon, as responses to stimuli other than hypoglycemia are intact. The third defense against hypoglycemia is compromised when the epinephrine response to hypoglycemia is reduced. In contrast to the absent glucagon response, epinephrine deficiency is a threshold abnormality; an epinephrine response can still be elicited, but a lower plasma glucose concentration is required. This threshold shift is largely the result of recent antecedent hypoglycemia, although an additional anatomic component may be present as well in patients affected by classic diabetic autonomic neuropathy. The development of a reduced epinephrine response is a critical pathophysiologic event. Prospective studies have shown that patients with combined deficiencies of glucagon and epinephrine suffer severe hypoglycemia at rates 25-fold or greater than individuals with absent glucagon but intact epinephrine responses.

Hypoglycemia Unawareness Hypoglycemia unawareness refers to loss of the warning symptoms of hypoglycemia that normally alert individuals to the presence of hypoglycemia and prompt them to eat to abort the episode. Under these circumstances, the first manifestation of hypoglycemia is neuroglycopenia, and it is often too late for patients to treat themselves. Like defective counterregulation, the presence of hypoglycemia unawareness has been shown in prospective studies to be associated with a high frequency of severe hypoglycemia.

The interplay of factors involved in hypoglycemia-associated autonomic failure in type 1 diabetes, and consequent hypoglycemia unawareness, is summarized in [Fig. 334-2](#). Periods of relative or absolute therapeutic insulin excess, in the setting of absent glucagon responses, lead to episodes of iatrogenic hypoglycemia. These episodes, in turn, cause reduced autonomic (including adrenomedullary) responses to falling glucose concentrations. These impaired autonomic responses result in reduced symptoms of impending hypoglycemia (e.g., hypoglycemia unawareness) and, because epinephrine responses are reduced in the setting of absent glucagon responses, impaired physiologic defense against developing hypoglycemia. Thus, a vicious cycle of recurrent hypoglycemia is created and perpetuated. The syndrome of hypoglycemia unawareness and the reduced epinephrine component of defective glucose counterregulation are reversible after as little as 2 weeks of scrupulous avoidance of hypoglycemia. This involves a shift of glycemic thresholds back to higher plasma glucose concentrations.

Hypoglycemia Risk Factor Reduction It is possible to minimize the risk of hypoglycemia by applying the principles of modern therapy -- patient education and empowerment, frequent self-monitoring of blood glucose, flexible insulin (and other drug) regimens, rational glycemic goals, and ongoing professional guidance and support. With respect to the latter, the issue of hypoglycemia needs to be addressed in every patient contact. If hypoglycemia is a recognized problem, first consider each of the conventional risk factors summarized earlier and recommend the appropriate adjustments of medications, diet, and life-style. Nonselective beta blockers may attenuate the recognition of hypoglycemia and they impair glycogenolysis; a relatively selective β_1 -antagonist (e.g., metoprolol or atenolol) is preferable when a beta blocker is indicated. One should consider the issue of compromised glucose counterregulation. Although it is possible to test for this abnormality using a low-dose insulin infusion test, this is not practical. A diagnosis of hypoglycemia unawareness can usually be made from the history. It should be remembered that hypoglycemia unawareness implies that previous episodes of hypoglycemia have occurred, whether these are documented or not. If low glucose levels are not apparent from the patient's self-monitoring log, one should suspect hypoglycemia during the night. The presence of clinical hypoglycemia unawareness makes defective glucose counterregulation quite likely. A 2 to 3 week period of conscientious avoidance of hypoglycemia is advisable.

REACTIVE HYPOGLYCEMIA

The postprandial (reactive) hypoglycemia occurs only after meals, and hypoglycemia is self-limited. Postprandial hypoglycemia occurs in children with certain rare enzymatic defects in carbohydrate metabolism such as hereditary fructose intolerance and galactosemia ([Chap. 350](#)). Reactive hypoglycemia also occurs in some individuals who have undergone gastric surgery that results in the rapid passage of food from the stomach to the small intestine. This type of *alimentary hypoglycemia* causes a rapid postprandial rise in plasma glucose levels and the release of gut incretins, which induce an exuberant insulin response and subsequent hypoglycemia. Administration of α -glucosidase inhibitor, which delays carbohydrate digestion and thus glucose absorption from the intestine, can be considered for treatment of reactive hypoglycemia, although its efficacy remains to be established in controlled trials.

If postprandial symptoms occur as an idiopathic disorder, caution should be exercised before labeling a person with the diagnosis of hypoglycemia. Indeed, a self-diagnosis of hypoglycemia has often been reinforced by the finding of a "low" venous glucose concentration late after glucose ingestion. An oral glucose tolerance test should not be used in this setting. Plasma glucose falls as low as 2.4 mmol/L (43 mg/dL) after a 100-g glucose load in 5% of normal asymptomatic individuals, making it difficult to identify hypoglycemia based on the results of this test. The diagnosis of postprandial hypoglycemia requires documentation of Whipple's triad after a typical mixed meal. The cause of repetitive postprandial symptoms in certain individuals is unknown, but they may be particularly sensitive to the normal autonomic responses that follow ingestion of a meal.

FASTING HYPOGLYCEMIA

There are many causes of fasting hypoglycemia ([Table 334-1](#)). In addition to insulin and

sulfonylureas used in the treatment of diabetes, ethanol use is a relatively common cause of hypoglycemia. Sepsis and renal failure are often complicated by hypoglycemia, but it is less common in other critical illnesses. Endocrine deficiencies, non- β -cell tumors, and endogenous hyperinsulinemia (including that caused by an insulinoma) are rare causes of hypoglycemia. Enzymatic metabolic errors that cause hypoglycemia are also rare but are being recognized more frequently in infants and children ([Chaps. 350 and 352](#)).

Drugs In contrast to the sulfonylureas and benzoic acid derivatives (e.g., repaglinide), other oral hypoglycemic agents -- biguanides (e.g., metformin), α -glucosidase inhibitors (e.g., acarbose, miglitol), and thiazolidinediones (e.g., troglitazone, rosiglitazone, pioglitazone) -- do not act by stimulating insulin secretion. Therefore, with these agents, insulin levels usually decrease appropriately as plasma glucose levels fall. Nonetheless, these drugs can contribute to hypoglycemia in other ways. Treatment with α -glucosidase inhibitor alters the management of hypoglycemia; pure glucose should be used rather than ingestion of complex carbohydrates. Thiazolidinediones, as well as metformin, can predispose to hypoglycemia in patients receiving combined treatment with insulin or an insulin secretagogue.

Ethanol blocks gluconeogenesis but not glycogenolysis. Thus, alcohol-induced hypoglycemia typically occurs after a several-day ethanol binge during which the person eats little food, thereby causing glycogen depletion. Hypoglycemia in this setting can be profound, with mortality rates as high as 10%. Blood ethanol levels correlate poorly with plasma glucose concentrations at the time of diagnosis, as hypoglycemia occurs late in the sequence and often precludes further alcohol consumption.

Pentamidine, which is used to treat *Pneumocystis* pneumonia and other parasitic infections, is toxic to the pancreatic β cell. It causes insulin release initially, with hypoglycemia in about 10% of treated patients, and predisposes to the development of diabetes mellitus later. Quinine also stimulates insulin secretion. However, the relative contribution of hyperinsulinemia to the pathogenesis of hypoglycemia in quinine-treated patients who are critically ill with malaria is debated. Salicylates and sulfonamides can cause hypoglycemia but do so rarely. There are reports of hypoglycemia attributed to nonselective β -adrenergic antagonists (e.g., propranolol) and a variety of other drugs.

Critical Illness Rapid and extensive hepatic destruction (e.g., severe toxic hepatitis) causes fasting hypoglycemia because the liver is the major site of endogenous glucose production. The mechanism of hypoglycemia reported in patients with cardiac failure is unknown but likely involves hepatic congestion. Although the kidneys are a source of glucose production, it is perhaps too simplistic to attribute hypoglycemia in people with renal failure to this mechanism alone. The clearance of insulin is reduced substantially in renal failure, and reduced mobilization of gluconeogenic precursors has been reported.

Sepsis is sometimes complicated by hypoglycemia, which is multifactorial in origin. There is impaired endogenous glucose production, perhaps the result of hepatic hypoperfusion, and increased glucose utilization, which is induced by cytokines in macrophage-rich tissues such as the liver, spleen, and ileum and in muscle. Nutrition is also often inadequate in the setting of sepsis. Hypoglycemia can be seen with

prolonged starvation, perhaps as a result of the loss of whole-body fat stores and the subsequent depletion of gluconeogenic precursors (e.g., amino acids), which necessitate increased glucose utilization.

Endocrine Deficiencies Neither cortisol nor growth hormone is critical to the prevention of acute hypoglycemia, at least in adults. Nonetheless, hypoglycemia can occur with prolonged fasting in patients with untreated primary adrenocortical failure (Addison's disease) or hypopituitarism. Anorexia and weight loss are typical features of chronic cortisol deficiency and likely result in glycogen depletion with increased reliance on gluconeogenesis. Cortisol deficiency is associated with low levels of gluconeogenic precursors, suggesting that substrate-limited gluconeogenesis, in the setting of glycogen depletion, is the cause of the impaired ability to tolerate fasting in cortisol-deficient individuals. Growth hormone deficiency can cause hypoglycemia in young children. In addition to extended fasting, high rates of glucose utilization (e.g., during exercise, pregnancy) or low rates of glucose production (e.g., following alcohol ingestion) can precipitate hypoglycemia in adults with hypopituitarism. Cortisol and growth hormone secretion should be evaluated in patients with fasting hypoglycemia when the history suggests pituitary or adrenal disease and when other causes of hypoglycemia are not apparent.

As discussed earlier, the combined loss of counterregulatory glucagon and epinephrine responses plays a central role in the pathogenesis of hypoglycemia in diabetes mellitus. However, hypoglycemia is not a feature of the epinephrine-deficient state that results from bilateral adrenalectomy when glucocorticoid replacement is adequate, nor does it occur during pharmacologic adrenergic blockage when other glucoregulatory systems are intact. There are case reports of fasting hypoglycemia attributed to isolated glucagon or epinephrine deficiency, although hyperinsulinemia was not excluded convincingly in the neonatal cases and other counterregulatory defects may have contributed in the adults. Thus, the regular assessment of glucagon and epinephrine secretion is not warranted.

Non- β -Cell Tumors Fasting hypoglycemia, often termed *non-islet cell tumor hypoglycemia*, occurs in some patients with large mesenchymal or other tumors (e.g., hepatoma, adrenocortical tumors, carcinoids). The glucose kinetic patterns resemble those of hyperinsulinism, but insulin secretion is suppressed appropriately during hypoglycemia. In most instances, hypoglycemia is due to overproduction of an incompletely processed form of insulin-like growth factor (IGF)II. Although total IGF-II levels are not consistently elevated, circulating free IGF-II levels are high. Hypoglycemia may result from IGF-II actions through the insulin or IGF-I receptors. Because of negative-feedback suppression of growth hormone secretion, IGF-I levels tend to be low, causing an increased IGF-II to IGF-I ratio.

Endogenous Hyperinsulinism Hypoglycemia due to excessive endogenous insulin secretion can be caused by: (1) a primary pancreatic islet β -cell disorder, typically a β -cell tumor (insulinoma), sometimes multiple insulinomas, or, especially in infants or young children, a functional β -cell disorder without an anatomic correlate; (2) a β -cell secretagogue, often a sulfonylurea, and, theoretically, a β -cell-stimulating autoantibody; (3) an autoantibody to insulin; or (4) perhaps ectopic insulin secretion. None of these disorders is common. Endogenous hyperinsulinism is more likely in an overtly well

individual without other apparent causes of hypoglycemia such as a relevant drug history, critical illness, endocrine deficiencies, or a non- β -cell tumor. Accidental, surreptitious, or even malicious administration of a sulfonylurea or insulin should also be considered in such individuals.

The fundamental pathophysiologic feature of endogenous hyperinsulinism is the failure of insulin secretion to fall to very low rates during hypoglycemia. This is assessed by measuring insulin, proinsulin, and C peptide, which is derived from the processing of proinsulin. The critical diagnostic findings are a plasma insulin concentration ≥ 36 pmol/L (≥ 6 uU/mL) and a plasma C-peptide concentration ≥ 0.2 nmol/L (≥ 0.6 ng/mL) when the plasma glucose concentration is ≤ 2.5 mmol/L (≤ 45 mg/dL) in the fasting state with symptoms of hypoglycemia. Insulin and C-peptide levels do not need to be absolutely high (e.g., relative to euglycemic normal values) but only inappropriately high in the setting of fasting hypoglycemia. Plasma proinsulin concentrations are also inappropriately high, particularly in patients with an insulinoma. Sulfonylureas, because they stimulate insulin secretion, result in a pattern of glucose, insulin, and C-peptide levels that is indistinguishable from that produced by a primary β cell disorder. The measurement of sulfonylureas in plasma or urine distinguishes these conditions. Antibodies to insulin produce *autoimmune hypoglycemia* following the transition from the postprandial to the postabsorptive state, as insulin slowly dissociates from the antibodies. Total and free plasma insulin concentrations are inappropriately high. The distinguishing finding is the presence of circulating antibodies to insulin, but the need to measure these routinely is debated, since autoimmune hypoglycemia appears to be rare. Autoantibodies to the insulin receptor are another rare cause of hypoglycemia and usually occur in the context of other autoimmune diseases. A few cases of apparent ectopic secretion of insulin (from a non- β -cell tumor) have been reported.

Insulinoma and Other Primary β Cell Disorders Insulinomas are rare, but because approximately 90% are benign, they are a treatable cause of potentially fatal hypoglycemia. The yearly incidence is estimated to be 1 in 250,000. About 60% of cases occur in women. The median age at presentation is 50 years in sporadic cases, but it usually presents in the third decade when associated with multiple endocrine neoplasia type 1 ([Chap. 339](#)). Insulinomas arise within the substance of the pancreas in $>99\%$ of cases and are usually small (1 to 2 cm). About 5 to 10% of insulinomas are malignant, as evidenced by the presence of metastases.

Insulinomas almost always come to clinical attention because of hypoglycemia rather than mass effects. As noted earlier, unusually low plasma glucose concentrations may be required to produce symptoms and signs of hypoglycemia because recurrent hypoglycemia shifts the glycemic thresholds. Although symptomatic hypoglycemia can be seen after an overnight fast, it often follows exercise. Rarely, symptomatic hypoglycemia occurs following meals, but most such patients have evidence of fasting hypoglycemia as well.

Octreotide scans localize about half of insulinomas. Arteriography has been used extensively in the past, but false-negative and false-positive results occur, and it is generally preferable to use less invasive computed tomography (CT) or magnetic resonance imaging (MRI) scans, which detect 45 to 75% of tumors. Preoperative ultrasound is of value in some patients. Intraoperative ultrasonography has high

sensitivity and may localize tumors not identified by palpation. Surgical resection of a solitary insulinoma is generally curative. Diazoxide, which inhibits insulin secretion, and the somatostatin analogue, octreotide, can be used to treat hypoglycemia in patients with unresectable insulinomas.

Factitious Hypoglycemia Factitious hypoglycemia, caused by malicious or self-administration of insulin or ingestion of a sulfonylurea, shares many clinical and laboratory features with insulinoma. It is most common among health care workers, patients with diabetes or their relatives, and people with a history of other factitious illnesses. When this diagnosis is suspected, it is useful to seek previous medical records, which may reveal admissions for similar episodes as well as relevant laboratory data. In individuals taking exogenous insulin, factitious hypoglycemia can be distinguished from insulinoma by the presence of high insulin levels without a concomitant increase in the C-peptide level, which is suppressed by the exogenous insulin. As noted above, sulfonylureas stimulate endogenous insulin and can therefore be detected only by measuring drug levels in plasma or urine. Factitious or surreptitious hypoglycemia should be considered in every patient requiring a fasting test for hypoglycemia. In addition to laboratory tests, observing the behavior of the patient may help make this diagnosis.

Approach to the Patient

In addition to recognition and documentation of hypoglycemia, and often urgent treatment, diagnosis of the hypoglycemic mechanism is critical for choosing a treatment that prevents, or at least minimizes, recurrent hypoglycemia. A diagnostic algorithm is shown in [Fig. 334-3](#).

Recognition and Documentation Urgent treatment is often necessary in patients with suspected hypoglycemia. Nevertheless, blood should be drawn, whenever possible, before the administration of glucose to allow documentation of the plasma glucose level. Convincing documentation of hypoglycemia requires the fulfillment of Whipple's triad. Thus, *the ideal time to test the plasma glucose is during an episode associated with hypoglycemic symptoms*. A normal plasma glucose concentration measured when the patient is free of symptoms does not exclude hypoglycemia at the time of earlier symptoms. When the cause of hypoglycemia is obscure, additional assays should include glucose, insulin, C peptide, sulfonylurea levels, cortisol, and ethanol.

Hypoglycemia is sometimes detected serendipitously. A distinctly low plasma glucose measurement in a person without a history of corresponding symptoms raises the possibility of a laboratory error caused by ongoing metabolism of glucose by the formed elements of the blood after the sample is drawn. This type of artifactually low glucose level is particularly likely when leukocyte, erythrocyte, or platelet counts are abnormally high, but it can also occur if separation of the plasma or serum from the formed elements is delayed.

Diagnosis of the Hypoglycemic Mechanism In an adult patient with documented hypoglycemia, a plausible hypoglycemic mechanism and further diagnostic evaluation can be guided by the history, physical examination, and available laboratory data ([Fig. 334-3](#)). Relevant historic elements include: drug history, particularly hypoglycemic

agents or alcohol use; relevant critical illness (hepatic, renal, or cardiac failure, sepsis, or inanition); previous gastric surgery associated with postprandial hypoglycemia; features suggestive of cortisol or growth hormone deficiency; inherited enzyme deficiencies associated with hypoglycemia; or features of a non- β -cell tumor. Absent these, one must consider medication error, endogenous hyperinsulinism, or surreptitious sulfonylurea or insulin administration. In the absence of documented spontaneous hypoglycemia, overnight fasting, or food deprivation during observation in the outpatient setting, will sometimes elicit hypoglycemia and allow diagnostic evaluation. If these maneuvers do not reveal hypoglycemia, and there is a high degree of clinical suspicion, an extended fast lasting up to 72 h is often required to make these diagnoses. This procedure should be performed in the hospital with careful supervision and should be terminated if the plasma glucose drops to <2.5 mmol/L (<45 mg/dL) and the patient has symptoms. It is essential to draw blood samples for appropriate tests before administering glucose or allowing the patient to eat.

Urgent Treatment Oral treatment with glucose tablets or glucose-containing fluids, candy, or food is appropriate if the patient is able and willing to take these. A reasonable initial dose is 20 g of glucose. If neuroglycopenia precludes oral feedings, parenteral therapy is necessary. Intravenous glucose (25 g) should be given using a 50% solution followed by a constant infusion of 5 or 10% dextrose. If intravenous therapy is not practical, subcutaneous or intramuscular glucagon can be used, particularly in people with type 1 diabetes mellitus. Because it acts primarily by stimulating glycogenolysis, glucagon is ineffective in glycogen-depleted individuals (e.g., those with alcohol-induced hypoglycemia). These treatments raise plasma glucose concentrations only transiently, and patients should be encouraged to eat as soon as practical in order to replete glycogen stores.

Prevention of Recurrent Hypoglycemia Prevention of recurrent hypoglycemia requires an understanding of the hypoglycemic mechanism. Offending drugs can be discontinued or their doses reduced. It should be remembered that hypoglycemia caused by sulfonylureas may recur after a period of many hours or days. Underlying critical illnesses can often be treated. Cortisol and growth hormone can be replaced, if deficient. Surgical, radiotherapeutic, or chemotherapeutic reduction of a non- β -cell tumor can alleviate hypoglycemia, even if the tumor cannot be cured; glucocorticoid or growth hormone administration may also reduce hypoglycemic episodes in such patients. Surgical resection of an insulinoma is often curative; medical therapy with diazoxide or octreotide can be used if resection is not possible and in patients with a nontumor primary β cell disorder. The treatment of autoimmune hypoglycemia (e.g., with a glucocorticoid) is more problematic, but this disorder is often self-limited. Failing these treatments, frequent feedings and avoidance of fasting may be required. Uncooked cornstarch at bedtime or an overnight infusion of intragastric glucose may be necessary in some patients.

(Bibliography omitted in Palm version)

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335. DISORDERS OF THE TESTES - James E. Griffin, Jean D. Wilson

The testes produce sperm and the steroid hormones that regulate male sexual function. Both processes are under complex feedback control by the hypothalamic-pituitary system so that the testes have biosynthetic and regulatory features similar to those of the ovary and the adrenal. Testicular hormones are also responsible for the formation of the basic male phenotype during embryogenesis ([Chap. 338](#)). Disorders that affect testicular function are common. Infertility occurs in about 5% of men; Klinefelter syndrome (XXY) occurs in 1 in 500 men and often escapes diagnosis; and various disorders cause hypogonadism, a condition that can be treated by hormone replacement. The testes are also a site of malignancies, most of which are highly responsive to radiation and/or chemotherapy ([Chap. 96](#)).

PHYSIOLOGY AND REGULATION OF TESTICULAR FUNCTION

The testis consists of two components -- clusters of interstitial or Leydig cells, where androgenic steroids are synthesized, and a system of spermatogenic tubules for the production and transport of sperm. The components are regulated by the pituitary gonadotropins -- luteinizing hormone (LH), which stimulates Leydig cell function, and follicle-stimulating hormone (FSH), which controls Sertoli cell function and spermatogenesis ([Fig. 335-1](#)).

GONADOTROPIN REGULATION AND TESTICULAR FUNCTION

Gonadotropin-releasing hormone (GnRH), also called luteinizing hormone-releasing hormone (LHRH), regulates the production of the gonadotropins, [LH](#) and [FSH](#) ([Chap. 328](#)). Because hypothalamic GnRH is secreted in discrete pulses, the plasma concentrations of LH, FSH, and testosterone are not constant, but fluctuate in a pulsatile pattern that mirrors the secretion of GnRH ([Fig. 335-2](#)). Pulsatile secretion is most apparent for LH because it has a relatively short plasma half-life by comparison to FSH; pulsatile secretion of testosterone in response to LH is also apparent, although the pulses are dampened because of the need to stimulate steroid synthesis and secretion by the Leydig cell. In contrast to women, in whom the frequency and amplitude of LH pulses vary during the menstrual cycle, the frequency of LH pulses in adult men is relatively constant at about one pulse every 1 to 2 h.

Testosterone secretion is regulated primarily by pituitary [LH](#). [FSH](#) may augment testosterone secretion by stimulating a Sertoli-cell derived factor that enhances testosterone production. Testosterone feeds back to regulate the hypothalamic-pituitary production of LH. It decreases hypothalamic GnRH pulse frequency and diminishes pituitary sensitivity to GnRH, leading to lower LH levels. Although the pituitary can convert testosterone to dihydrotestosterone and to estrogens, testosterone itself is the primary regulator of gonadotropin secretion by the pituitary. Under ordinary circumstances, LH secretion is exquisitely sensitive to the feedback effects of testosterone, with almost complete suppression after the administration of amounts of exogenous androgen that approximate the normal daily secretory rate of testosterone (~20 μmol or 6 mg). However, prolonged elevation of plasma LH (as in testicular deficiency) renders the pituitary less sensitive to negative feedback control by androgen.

FSH is regulated by **GnRH** and also by the gonadal peptides -- inhibins and activins. Inhibins A and B are heterodimeric proteins (composed of α and β subunits) that selectively suppress FSH without affecting **LH**; activins are homodimeric proteins (composed of α and β subunits) that selectively stimulate FSH production. Inhibin B, which is the major form of inhibin in the male, is produced by the Sertoli cell and provides feedback control of FSH production. Activins, which are produced in the pituitary as well as the gonad, stimulate FSH production through an autocrine-paracrine mechanism.

The interlocking system in which two pituitary hormones (**LH** and **FSH**) regulate testicular function provides a precise dual-control mechanism in which hormonal signals from Leydig cells and the spermatogenic tubules feed back on the hypothalamic-pituitary system to regulate their own function ([Fig. 335-1](#)).

THE LEYDIG CELL

Testosterone Synthesis The biochemical pathway by which the 27-carbon sterol cholesterol is converted to androgens and estrogens is depicted in [Fig. 335-3](#). Cholesterol, which can be either synthesized de novo in the Leydig cell or derived from plasma lipoproteins, is converted to testosterone as the result of five enzymatic reactions: (1) cholesterol side chain cleavage (CYP11A1); (2) 3 β -hydroxysteroid dehydrogenase/isomerase 2 (3 β -HSD2); (3) 17 α -hydroxylase (CYP17); (4) 17,20-lyase (CYP17); and (5) 17 β -hydroxysteroid dehydrogenase 3 (17 β -HSD3). Both the 17 α -hydroxylase and the 17,20-lyase reactions are catalyzed by a single cytochrome CYP17; post-translational modification (phosphorylation) of the enzyme and the presence of enzyme cofactors confers 17,20-lyase activity, thereby allowing androgen synthesis in the testis and zona reticularis of the adrenal gland. The first four reactions take place in the adrenal as well as the testis. The rate-limiting process in testosterone synthesis is the delivery of cholesterol by the steroid acute regulatory (StAR) protein to the inner mitochondrial membrane where it can undergo side chain cleavage by CYP11A1 to form pregnenolone. **LH** from the pituitary stimulates the activity of StAR protein and the enzymes in the steroid pathway. Additional steroids including estradiol are synthesized in small amounts in the Leydig cell.

TESTOSTERONE SECRETION AND TRANSPORT

Only about 70 nmol (20 μ g) of testosterone is stored in the normal testes, so the total hormone content turns over about 200 times each day to provide the average of 17 to 20 μ mol (5 to 6 mg) that is secreted into plasma in normal young men ([Fig. 335-4](#)). Testosterone is transported in plasma bound to protein, largely to albumin and to a specific transport protein, sex hormone-binding globulin (SHBG), also called testosterone-binding globulin (TeBG). The bound and unbound fractions in plasma are in dynamic equilibrium, only ~1 to 3% being unbound. Because of rapid dissociation from albumin, the fraction of circulating testosterone available for entry into tissues (bioavailable testosterone) approximates the sum of the free and albumin-bound fractions or about half the total plasma level.

Peripheral Metabolism of Androgens Testosterone serves as a circulating precursor (or prohormone) for the formation of two other hormones that mediate many of the

physiologic processes involved in androgen action ([Fig. 335-3](#)). Testosterone can be 5 α -reduced to dihydrotestosterone, which is responsible for many of the differentiative, growth-promoting, and functional aspects of male sexual differentiation and virilization. Circulating testosterone (and androstenedione) also can be converted to estrogens by aromatase (CYP19) in extraglandular tissues ([Fig. 335-4](#)). All estrone production [averaging about 240 nmol (66 ug) per day] can be accounted for by formation from circulating precursors. The mean estradiol production is approximately 170 nmol (45 ug) per day; ~35% is derived from circulating testosterone, 50% is derived from the estrone, and 15% is secreted directly by the testes. When gonadotropin levels are elevated, estradiol secretion by the testes increases. Thus, the physiologic effects of testosterone are the result of the combined actions of testosterone itself plus those of the active androgen and estrogen metabolites of the parent molecule.

The 5 α -reduced and estrogenic metabolites can exert local (paracrine) actions in the tissues in which they are formed or enter the circulation and act as hormones at other sites. Circulating dihydrotestosterone is formed principally in the androgen target tissues; estrogen formation takes place in many tissues, the most significant being adipose tissue. The overall rate of extraglandular estrogen formation increases with age and with increased mass of adipose tissue.

Testosterone and its active metabolites are catabolized in the liver and excreted predominantly in the urine, approximately half in the form of urinary 17-ketosteroids (primarily androsterone and etiocholanolone) and half as polar metabolites (diols, triols, and conjugates).

Androgen Action The major functions of androgen are formation of the male phenotype during sexual differentiation, regulation of [LH](#) secretion, and induction of sexual maturation at puberty. The cellular process by which androgens perform these functions is schematized in [Fig. 335-5](#). Testosterone enters the cell by passive diffusion and can be converted to dihydrotestosterone by either steroid 5 α -reductase 1 or 2; 5 α -reductase 2 is responsible for dihydrotestosterone formation in most androgen target tissues. Testosterone or dihydrotestosterone is then bound to the androgen-receptor protein in the nucleus. The hormone-receptor complex binds to specific DNA sequences to regulate the transcription of messenger RNA and, ultimately, the synthesis of cellular proteins. The androgen receptor, which is encoded by a gene on the long arm of the X chromosome, contains 917 amino acids and has a molecular mass of about 110 kDa. A polymorphic region in the amino terminus of the receptor, which contains a variable number of glutamine repeats, appears to modify the activity of the receptor. The androgen receptor is similar in structure to other steroid hormone receptors and has distinct hormone-binding, DNA-binding, and transcriptional regulatory domains ([Chap. 327](#)). Estradiol acts by a similar mechanism but has its own distinct estrogen receptors a and b ([Chap. 336](#)).

Although testosterone and dihydrotestosterone bind to the same receptor, their physiologic roles differ. The testosterone-receptor complex regulates gonadotropin secretion, spermatogenesis, and the virilization of the wolffian ducts during sexual differentiation ([Chap. 338](#)), whereas the dihydrotestosterone-receptor complex is responsible for external virilization during embryogenesis and for most androgen actions during sexual maturation and adult sexual life. The mechanism by which two hormones

can interact with the same receptor but have different physiologic effects is not well understood. However, dihydrotestosterone binds to the receptor much more tightly than does testosterone, and hence its formation serves to amplify the hormonal signal.

THE SEMINIFEROUS TUBULES AND SPERMATOGENESIS

Spermatogenesis is dependent on both pituitary [FSH](#) and androgen production by the adjacent Leydig cells ([Fig. 335-1](#)). The function of FSH in gametogenesis has been clarified by rare, naturally occurring mutations in the *FSHb* gene and in the FSH receptor. Females with these mutations are hypogonadal and infertile because ovarian follicles do not mature, whereas males with mutations in the FSH pathway exhibit variable degrees of impaired spermatogenesis. Thus, while FSH is not absolutely required for spermatogenesis, it increases the number and maturation of sperm. The major site of FSH action is the Sertoli cell, which regulates germ cell proliferation and maturation in the seminiferous tubules. Androgen, which reaches very high concentrations locally in the testis, appears to be essential for spermatogenesis, acting through receptors located in the seminiferous tubules. Several cytokines and growth factors are also involved in the regulation of spermatogenesis by paracrine and autocrine mechanisms. The normal adult testes produce >100 million sperm per day.

The Sertoli cell cannot synthesize steroid hormones de novo but can convert testosterone that diffuses from adjacent Leydig cells to estradiol and to dihydrotestosterone. The Sertoli cell also produces inhibin B. Damage to the seminiferous tubules (e.g., by radiation) reduces inhibin B production, causing a selective increase in [FSH](#).

ASSESSMENT OF TESTICULAR FUNCTION

LEYDIG CELL FUNCTION

History and Physical Examination The assessment of Leydig cell function and androgen status should include inquiry about the presence of developmental abnormalities of the urogenital tract; the timing and extent of sexual maturation at puberty; the rate of beard growth; and the current libido, sexual function, strength, and energy. Inadequate Leydig cell function or androgen action during embryogenesis may cause hypospadias, cryptorchidism, or microphallus. If Leydig cell failure occurs before puberty, sexual maturation will not occur, and the individual will develop the features termed *eunuchoidism*, including an infantile amount and distribution of body hair, poor development of skeletal muscles, and delayed closure of the epiphyses, so that the arm span is more than 5 cm greater than the height and the lower body segment (heel to pubic bone) is more than 5 cm longer than the upper body segment (pubic bone to crown). Detection of postpubertal Leydig cell failure requires a high index of suspicion and appropriate laboratory assessment because some functions that require androgens for initiation continue unabated when Leydig cell failure occurs, and functions that eventually regress may do so very slowly. For example, the frequency of shaving may not decrease for months or years because of slow decline in the rate of beard growth once established. Furthermore, decreased sexual function in adult men may be caused by nonendocrine as well as endocrine factors ([Chap. 51](#)).

Plasma Testosterone and Dihydrotestosterone Levels Plasma testosterone is measured by immunoassay. Testosterone is secreted into plasma in a pulsatile fashion every 60 to 90 min ([Fig. 335-2](#)). A single random testosterone sample provides a result within $\pm 20\%$ of the true mean value only two-thirds of the time; a pool of three samples spaced 15 to 20 min apart provides a more accurate assessment. The plasma testosterone level in normal adult men ranges from 10 to 35 nmol/L (3 to 10 ng/mL). However, in some normal men with long interpulse intervals of LH, testosterone levels can transiently fall below this normal range, emphasizing the importance of using pooled or repeated samples before making a diagnosis of testosterone deficiency. In adult men, plasma testosterone levels also vary somewhat throughout the day and at different times of the year. In young adult men the plasma testosterone level is ~30% higher in the morning than in the evening. Estimation of [SHBG](#) concentration by radioimmunoassay is sometimes useful in the interpretation of total plasma testosterone levels. Bioavailable testosterone in plasma can be estimated by measuring the non-SHBG-bound fraction of testosterone.

The plasma testosterone level is slightly higher in prepubertal boys than in girls, ranging in both from 0.2 to 0.7 nmol/L (0.05 to 0.2 ng/mL). The rise in plasma testosterone level at the start of male puberty begins as a result of sleep-related nocturnal gonadotropin surges, so that levels of plasma testosterone and [LH](#) are initially higher at night than during the day. Random daytime levels of plasma testosterone increase gradually as puberty progresses and reach adult levels at about age 17.

Dihydrotestosterone is also measured by immunoassay. In young men the plasma dihydrotestosterone level is about one-tenth the value for testosterone, averaging ~2 nmol/L (0.6 ng/mL). In older men with benign prostatic hyperplasia, plasma dihydrotestosterone levels average ~3 nmol/L (0.9 ng/mL).

Urinary 17-Ketosteroids The measurement of urinary 17-ketosteroids is not a valid way to assess testicular function because testosterone contributes only ~40% of urinary 17-ketosteroids in men, the bulk being derived from adrenal androgens.

Plasma [LH](#) Plasma LH is also measured by immunoassay. Dual-site immunometric assays have largely replaced radioimmunoassays. Because LH is secreted in a pulsatile fashion, assay of a pool of three samples drawn 15 to 20 min apart, as described above, provides a value approaching the true mean. In early puberty, plasma LH secretion increases only during sleep, but in the adult the pulsatile secretion is of similar magnitude during sleep and waking periods. The normal plasma LH values should be established for a given laboratory with an appropriate reference standard. A low plasma testosterone level can be interpreted correctly only if plasma LH is measured simultaneously, and, likewise, the "appropriateness" of a given plasma LH value must be interpreted in relation to the plasma testosterone level. For example, a low plasma testosterone level coupled with a low LH level implies hypothalamic or pituitary disease, whereas the finding of a low plasma testosterone level and a high LH level suggests primary testicular insufficiency.

Response to Gonadotropin Stimulation Leydig cell function is difficult to assess before puberty, when both [LH](#) and testosterone levels are low, but it is possible to measure response of plasma testosterone to gonadotropin stimulation as an index of

Leydig cell capacity. Normal prepubertal boys respond to 3 to 5 days of injection of 1000 to 2000 IU of human chorionic gonadotropin (hCG) with an increase in the plasma testosterone level to ~7 nmol/L (2 ng/mL); the response increases with the initiation of puberty and peaks in early puberty.

Response to GnRH Before puberty, there is minimal response of plasma LH and FSH to the administration of GnRH because the pituitary has not been "primed" by previous exposure to GnRH or gonadal steroids. After pubertal development, the LH response to acute administration of GnRH increases, while the FSH response is less robust. The amount of LH released after acute administration of GnRH probably reflects the amount of stored hormone in the pituitary. When 100 ug GnRH is given subcutaneously or intravenously to normal men, LH levels usually increase four- to fivefold, with the peak level at 30 min. However, the range of response is broad, and some normal men exhibit less than a doubling of LH levels. In primary testicular failure, measurement of basal LH is usually sufficient, and assessment of GnRH response is of little aid in diagnosis. Since men with either pituitary or hypothalamic disease can have a normal or an abnormal LH response to acute administration of GnRH, a normal response does not clearly distinguish these causes of gonadal deficiency. A subnormal response is, however, of value in establishing that an abnormality exists, even though the site of the defect is not clearly determined. If pulsatile GnRH or daily infusions of GnRH for a week lead to the development of a normal acute LH response, a hypothalamic etiology is likely.

SEMINIFEROUS TUBULE FUNCTION

Examination of the Testes Evaluation of the testes is an essential portion of the physical examination. The prepubertal testis measures about 2 cm in length and 2 mL in volume and grows during puberty to reach the adult proportions by age 16. When damage to the seminiferous tubules occurs before puberty, the testes are small and firm, whereas the testes are usually soft after postpubertal damage (the capsule, once enlarged, does not contract to its previous size). Testes in adult men average 4.6 cm in length (range, 3.5 to 5.5 cm), corresponding to a volume of 12 to 25 mL, and the seminiferous tubules account for ~60% of testicular mass. Advanced age does not influence testicular size, so the significance of small testes in the adult is the same at all ages. Asian men have smaller testes than western Europeans, independent of differences in body size. Because of its possible causal role in infertility, the presence of varicocele should be sought by palpation with the patient standing.

Semen Analysis Seminal fluid is analyzed on samples obtained by masturbation into a glass container after 24 to 36 h of abstinence. Analysis should be performed within an hour of collection. The normal ejaculate volume is 2 to 6 mL. Immediately after ejaculation, the seminal fluid coagulates, followed in 15 to 30 min by liquefaction. Motility should be assessed in undiluted seminal fluid; >60% of the sperm should be motile and of normal morphology. The normal range for sperm density is generally considered to be >20 million per milliliter, with a total count of >60 million per ejaculate, but the definition of a minimally adequate ejaculate is not clear. Some men with low sperm counts are nevertheless fertile. This uncertainty as to the lower level of sperm density, percent motility, and percent normal forms in fertile semen stems from two issues. First, the seminal fluid is routinely evaluated by tests that do not assess the

functional capacity of sperm. Second, many factors produce temporary aberrations in sperm count; in men with semen of equivocal quality, it is necessary to examine three or more ejaculates to determine whether the abnormal findings are permanent.

Plasma FSH Plasma FSH, as measured by immunoassay, usually correlates inversely with spermatogenesis. When damage to the germinal epithelium is severe, plasma levels of inhibin B fall and plasma levels of FSH increase.

Testicular Biopsy Testicular biopsy is useful in some patients with oligospermia and azoospermia both as an aid in diagnosis and as an indication of the feasibility of treatment. For example, normal findings on testicular biopsy and a normal FSH level in an azoospermic man suggest obstruction of the vas deferens, which may be correctable surgically. In some men with severe oligospermia, testicular biopsy allows retrieval of sperm for intracytoplasmic sperm injections (ICSI) into oocytes ([Chap. 54](#)).

ESTROGENIC FUNCTION

Examination of the Breasts Breast enlargement (gynecomastia) is the most consistent feature of feminizing states in men ([Chap. 337](#)). Gynecomastia is due to an increase in both glandular and adipose tissue. The presence of gynecomastia should be sought by examining the sitting patient, using the fingers to grasp glandular tissue. Early or minimal breast enlargement may be missed if the breast is palpated with the flat of the hand while the patient is supine. In obese men it is important to try to define the rim of the glandular tissue where it meets the adipose tissue of the chest wall.

Plasma Estrogen As discussed above, most of the estradiol and all of the estrone produced in normal men is formed by extraglandular aromatization of circulating androgens. The plasma level of estradiol usually is <180 pmol/L (50 pg/mL) in normal men; the plasma estrone level is somewhat higher but usually is <300 pmol/L (80 pg/mL). Elevations in estrogen production and estrogen plasma levels can be due to increases in plasma precursors (liver or adrenal disease), to increased extraglandular aromatization (obesity), or to increased production by the testes (testicular tumors, androgen resistance, gonadotropin stimulation).

PHASES OF NORMAL TESTICULAR FUNCTION

The phases of male sexual life can be defined in terms of the plasma testosterone value ([Fig. 335-6](#)). In the male embryo the production of testosterone by the testes commences at about 7 weeks of gestation and is stimulated in part by placental hCG. Shortly thereafter, plasma testosterone attains a high level that is maintained until late in gestation. The level then falls so that at the time of birth the plasma testosterone level is only slightly higher in males than in females. Shortly after birth, a transient increase of pituitary gonadotropins raises the plasma testosterone level in the male infant for ~3 months, before hormone levels again decrease to low levels by age 6 months to 1 year. The significance of the rise in plasma testosterone level during the first year of life is not certain. However, in other primates neonatal activation of the hypothalamic-pituitary-testicular axis is important for subsequent normal pubertal development. The testosterone concentration then remains low (but slightly higher in boys than in girls) until the onset of puberty, when it begins to rise in boys, reaching

adult levels by about age 17. The level of bioavailable testosterone remains constant until the 40s when it begins to decline at a rate of ~1.2% per year; the level of [SHBG](#) increases by ~1.2% per year so that there is little decline in total testosterone until the later decades of life. During the third, or adult, phase of male sexual life, sperm production becomes sufficient to allow reproduction to take place. The physiologic events during these various phases differ, as do the pathologic consequences of derangements in testicular function. Male sexual differentiation during embryogenesis is considered in [Chap. 338](#).

ABNORMALITIES OF TESTICULAR FUNCTION

PUBERTY

The control of puberty is poorly understood and may reside in the hypothalamic-pituitary system, the testes, or the adrenals ([Chap. 8](#)). Before the onset of puberty, gonadotropin secretion by the pituitary is low, but prepubertal castration causes a rise in plasma gonadotropin levels. This finding suggests that before puberty the negative feedback control of gonadotropin secretion is exquisitely sensitive to the small amount of circulating testosterone. The onset of puberty is heralded by sleep-associated surges in gonadotropin secretion. Later in puberty the rises in [LH](#) and [FSH](#) levels persist throughout the day. Thus, with maturation, the hypothalamic-pituitary system becomes less sensitive to negative feedback control, and the consequences are higher mean plasma levels of testosterone and gonadotropins, maturation of the testes, and the onset of spermatogenesis. The rise in gonadotropin secretion is the consequence of an increase both in [GnRH](#) secretion and in the sensitivity of the pituitary to GnRH.

The somatic changes at the time of puberty are secondary to the rise in plasma testosterone, which induces the growth of the accessory organs of male reproduction (the penis, the prostate, the seminal vesicles, and the epididymides). Accelerated linear growth is accompanied by growth of muscle and connective tissue, which account for the major portion of nitrogen retention at puberty. The principal androgen-sensitive muscles are those of the pectoral region and the shoulder. The characteristic hair growth of male puberty involves development of mustache and beard; regression of the scalp line; appearance of body, extremity, and perianal hair; and extension of the pubic hair upward into a diamond-shaped pattern. Growth of axillary and pubic hair is initiated under the control of adrenal androgens and is promoted by testicular androgens. The larynx enlarges, and the vocal cords thicken, resulting in a lowering of the pitch of the voice. Hemoglobin levels increase by ~1 g/dL. These various androgen-mediated growth and maturation processes reach some limiting value, so that once puberty is completed, the administration of pharmacologic doses of androgen has no further effect. The entire process is heralded by testicular enlargement beginning at age 11 to 12 and is usually completed within 5 years, although some aspects of virilization, such as growth of the chest hair, may continue over a decade or more.

The events of normal male puberty are variable in onset, duration, and sequence. The central issue in dealing with disorders of puberty is separating true absence or precocity from the extremes of normal variation. The use of staging criteria that correlate developmental and anatomic landmarks with chronologic age is useful in making this distinction ([Chap. 8](#)).

Sexual Precocity Premature development of sexual characteristics that are phenotypically appropriate -- i.e., virilization in boys -- is termed *isosexual precocity*. *Heterosexual precocity* refers to feminizing syndromes in boys.

Isosexual precocity Sexual development before age 9 in boys is generally considered abnormal. *True precocious puberty* or *complete isosexual precocity* occurs when both virilization and spermatogenesis are premature. *Precocious pseudopuberty* or *incomplete isosexual precocity* refers to virilization unaccompanied by spermatogenesis. This distinction is blurred in practice, because pure virilizing syndromes may cause activation of gonadotropin secretion secondarily and thus be followed by development of spermatogenesis. Furthermore, local androgen production in the testis, as in Leydig cell tumors, can cause local areas of spermatogenesis and limited sperm production around the tumor. We therefore prefer a two-part classification: virilizing syndromes (in which hypothalamic-pituitary activity is appropriate for age) and premature activation of the hypothalamic-pituitary system.

Virilizing syndromes can result from Leydig cell tumors, [hCG](#)-secreting tumors, adrenal tumors, congenital adrenal hyperplasia (most commonly 21-hydroxylase deficiency), androgen administration, or Leydig cell hyperplasia. In these disorders plasma testosterone levels are inappropriately elevated for the age. Leydig cell tumors are rare in children but should be suspected when the testes are asymmetric in size ([Chap. 96](#)). Virilizing adrenal tumors mainly secrete androstenedione and dehydroepiandrosterone, some of which is converted to testosterone; consequently, they cause increased 17-ketosteroid excretion. Glucocorticoid administration does not reduce 17-ketosteroid excretion to normal in patients with testicular or adrenal tumors, in contrast to the prompt decrease that occurs after such treatment in patients with congenital adrenal hyperplasia. Congenital adrenal hyperplasia leads to elevated 17-hydroxyprogesterone levels and, as a consequence, elevated androgen levels ([Chaps. 331](#) and [338](#)). When this disorder is treated with glucocorticoids, true precocious puberty can then result if the increased androgen levels have caused sufficient hypothalamic maturation.

Gonadotropin-independent sexual precocity in boys may occur as a result of autonomous Leydig cell hyperplasia in the absence of a Leydig cell tumor. The disorder can occur sporadically or can be inherited as a male-limited autosomal disorder either from affected fathers or from mothers who are unaffected carriers. It is due to point mutations in the [LH](#) receptor that cause constitutive activation of the receptor in the absence of LH. Virilization usually begins by age 2. Testosterone levels are elevated, often to the adult male range; however, immunoreactive and bioactive LH levels and the LH response to [GnRH](#) are prepubertal. In the past many of these boys were mistakenly thought to have true precocious puberty because spermatogenesis may be present.

Premature activation of the hypothalamic-pituitary system Central precocious puberty may be "idiopathic" or due to central nervous system (CNS) tumors, infections, or injuries. Early hypothalamic-pituitary activation typically is associated with features of normal puberty, i.e., sleep-related gonadotropin secretion, elevated plasma bioactive [LH](#), and enhanced gonadotropin response to [GnRH](#). Since the diagnosis of idiopathic true precocious puberty is one of exclusion, patients may later prove to have been misclassified and to have a CNS abnormality. With improved means of diagnosis, such

as magnetic resonance imaging, delays in diagnosis will probably be less frequent.

Management of sexual precocity due to steroid- or gonadotropin-producing tumors, congenital adrenal hyperplasia, or [CNS](#) abnormality is directed toward the primary disease. In boys with Leydig cell hyperplasia, attempts have been made to lower plasma testosterone with medroxyprogesterone acetate or ketoconazole, or to blunt hormone action with spironolactone, but treatment remains suboptimal. Idiopathic true precocious puberty and true precocious puberty due to inoperable [CNS](#) lesions are treated with long-acting [GnRH](#) analogue therapy, which inhibits gonadotropin production and testosterone synthesis, reversing pubertal maturation and decreasing the rate of skeletal development.

Heterosexual precocity Feminization in prepubertal boys can result from absolute or relative increases in estrogen due to a variety of causes ([Chap. 337](#)).

Delayed or Incomplete Puberty Separating failure of puberty from variants of normal development is one of the most difficult problems in endocrinology. Some boys fail to show the normal spurt of growth and sexual development at the usual time but eventually commence puberty by age 16 or older. Adolescence may then either progress rapidly, or slow pubertal development and growth may continue until age 20 to 22. Many men with delayed onset of puberty attain heights within the normal adult range. The history may reveal that a parent or sibling had a similar pattern of development. Panhypopituitarism and hypothyroidism can cause pubertal failure ([Chaps. 328](#) and [330](#)). Absent puberty also can result from primary testicular disease; this diagnosis is suspected on the basis of low plasma testosterone levels and elevated [FSH](#) and [LH](#) levels. Hereditary androgen resistance (in which plasma testosterone and LH levels are both high) usually causes male pseudohermaphroditism but in mild form may be manifested by absent or incomplete puberty ([Chap. 338](#)).

Most boys with absent puberty have low plasma levels of both testosterone and gonadotropins; in these boys it is necessary to distinguish delayed puberty from isolated gonadotropin deficiency or idiopathic *hypogonadotropic hypogonadism* (*Kallman syndrome*). The manifestations of isolated gonadotropin deficiency vary from boys with eunuchoidal features and testes of prepubertal size to those with partial [LH](#) and/or [FSH](#) deficiency and some degree of testicular enlargement and pubertal development. Anosmia or hyposmia is caused by abnormal development of the olfactory tracts (which share progenitor cells with [GnRH](#) neurons) and is characteristically seen in Kallmann syndrome. X-linked *adrenal hypoplasia congenita (AHC)* is characterized by primary adrenal insufficiency, which usually presents in infancy, and hypogonadotropic hypogonadism, caused by deficient GnRH production and abnormal gonadotrope function. Congenital hypogonadotropic hypogonadism is frequently associated with cryptorchidism and a prepubertal manifestation can be micropenis, in which the size of the penis is below the fifth percentile for the age.

The pathogenesis of hypogonadotropic hypogonadism can involve several distinct abnormalities of [GnRH](#) formation or action. Some cases are inherited as an X-linked recessive trait associated with defects in a neural cell adhesion molecule (KAL) involved in the migration of GnRH neurons into the olfactory bulb. Other causes involve autosomal dominant disorders with variable expressivity; rare autosomal recessive

cases are due to mutations that impair the GnRH receptor. Serum [FSH](#) and [LH](#) levels are usually below the normal male range, and plasma testosterone levels are low for age. The secretion of other pituitary hormones is normal. The administration of pulsatile GnRH corrects the endocrine abnormalities and initiates spermatogenesis in patients with GnRH deficiency; all patients respond to gonadotropin replacement. If untreated, these patients usually remain in the prepubertal state indefinitely.

It is particularly difficult to distinguish hypogonatropic hypogonadism from delayed puberty in boys of early or midpubertal age; the presence of microphallus, anosmia, or a family history may suggest the diagnosis. In the absence of such evidence, observation through the teenage years may be required before it becomes clear whether a patient has delayed puberty or a permanent form of hypogonadotropic hypogonadism. Since delayed puberty is associated with a decreased bone mass, therapy should not be delayed too long. In some cases the response of plasma [LH](#) to [GnRH](#) stimulation may be helpful in suggesting that puberty is imminent.

ADULTHOOD

At the completion of puberty, plasma testosterone levels reach the adult level of 10 to 35 nmol/L (3 to 10 ng/mL) throughout the day, plasma gonadotropins are in the normal adult range, and sperm production is sufficient to allow reproduction. The adult pattern of hypothalamic-pituitary-gonadal regulation is sustained in the normal man for more than 40 years. However, the system is subject to a variety of influences, at the level of both the testes and the hypothalamic-pituitary system. Spermatogenesis is exquisitely sensitive to alterations in scrotal temperature, and brief increases in either systemic or local temperature (as in a hot bath) can be followed by temporary decreases in sperm production. The system also is influenced by diet, drugs, alcohol, environmental agents, and psychological stress, any of which may cause temporary decreases in sperm count.

Persistent abnormalities of testicular function in adult men can be due to hypothalamic-pituitary disorders ([Chap. 328](#)), testicular defects, or abnormalities of sperm transport. Certain of these conditions tend to affect Leydig cell function or spermatogenesis selectively, but most impair both androgenization and fertility ([Table 335-1](#)). The interlocking of Leydig cell function and fertility is due to the dependence of spermatogenesis on androgen. Even a partial decrease in testosterone production can cause infertility. Certain conditions (hyperprolactinemia, radiation therapy, cyclophosphamide therapy, autoimmunity, paraplegia, androgen resistance) can cause either isolated infertility or a combined defect in testicular function.

Hypothalamic-Pituitary Disorders Disorders of the hypothalamus and pituitary can impair the secretion of gonadotropins either as one manifestation of a generalized disease of the anterior pituitary ([Chap. 328](#)) or as an isolated defect. In the latter case the cause is usually hypogonadotropic hypogonadism, in which secretion of both [LH](#) and [FSH](#) are impaired. This disorder usually is congenital but may be acquired. Alternatively, gonadotropin secretion can be altered by factors other than hypothalamic-pituitary pathology. For example, elevation of plasma cortisol, as in the *Cushing syndrome*, can depress LH secretion independent of a space-occupying lesion of the pituitary. Critical illness also suppresses plasma gonadotropin levels. Some patients with uncontrolled *congenital adrenal hyperplasia* have elevated levels of

adrenal androgens, suppressed gonadotropin secretion, and consequent infertility. Likewise, the use of *androgens* for purposes other than replacement therapy can inhibit gonadotropin secretion and impair sperm production (see below). *Hyperprolactinemia* (as the consequence either of pituitary adenomas or of drugs such as phenothiazines) can cause combined Leydig cell and seminiferous tubule dysfunction, presumably due to inhibition of LH and FSH secretion by prolactin. Occasionally, impaired fertility in hyperprolactinemia is associated with normal gonadotropin and androgen levels and is presumed to result from direct inhibition of spermatogenesis by prolactin.

Hemochromatosis usually impairs testicular function as the result of effects on the pituitary; less often it affects the testis directly ([Chap. 345](#)). In some conditions, testosterone levels may be decreased in association with normal LH levels, and the mechanism is less clear. Men with massive obesity have decreased levels of [SHBG](#) and of total and bioavailable testosterone, which return toward normal with weight loss. Obesity may also contribute to the decreased testosterone levels in the subset of such men with Pickwickian syndrome ([Chap. 263](#)). Some men with temporal lobe seizures also have hypogonadotropic hypogonadism.

Testicular Defects Abnormalities of testicular function in the adult man can be grouped into several categories: developmental and structural defects of the testes, acquired testicular defects, and disorders secondary to systemic disease.

Developmental abnormalities The *Klinefelter syndrome* (XXY, both the classic and the mosaic forms) and the *XX male syndrome* are usually not recognized until after the time of expected puberty ([Chap. 338](#)). Some developmental defects cause infertility in the presence of normal androgen production. These include varicocele, germinal cell aplasia, deletions or mutations of the azoospermia factor (*AZF*) genes on the Y chromosome, and cryptorchidism. *Varicocele* may be of etiologic importance in as much as one-third of all cases of male infertility. It is caused by retrograde flow of blood into the internal spermatic vein that eventuates in progressive, often palpable dilation of the peritesticular pampiniform plexus of veins. Varicocele occurs in ~10 to 15% of men in the general population and in 20 to 40% of men with infertility. It is thought to result from incompetence of the valve between the internal spermatic vein and the renal vein and is more common on the left side (85%). Unilateral varicocele increases the blood flow and the temperature of both testes as a result of the extensive anastomoses of the venous systems. The increased scrotal (and testicular) temperature is believed to be the cause of the poor-quality semen and infertility (the testes no longer are 2°C cooler than the abdominal cavity). The findings on semen analysis are usually nonspecific, with all parameters showing some abnormality. Surgical repair of varicocele results in fertility in about half of men, with the best results (70% pregnancy rate) in those whose preoperative sperm counts are >10 million per milliliter.

Some patients with *germinal cell aplasia* (the Sertoli cell-only syndrome) have a positive family history and may constitute a specific group in whom the germinal epithelium is missing with resulting azoospermia; plasma testosterone and [LH](#) values are normal, and plasma [FSH](#) levels are elevated. Other patients with identical histologic and clinical findings have androgen resistance or a history of viral orchitis or cryptorchidism; microdeletions of one or more genes (e.g., *Deleted in Azoospermia*, *DAZ*) on the Y chromosome have been documented in 10 to 20% of men with azoospermia or oligospermia (many of whom have germinal cell aplasia), depending on the criteria used

for selection.

Unilateral *cryptorchidism*, even when corrected before puberty, is associated with abnormal semen in many individuals, indicating that the testes can be bilaterally abnormal even in unilateral cryptorchidism.

The *immotile cilia syndrome* is an autosomal recessive defect characterized by immotility or poor motility of the cilia of the airways and of the sperm. Kartagener's syndrome is a subgroup of the immotile cilia syndrome associated with situs inversus, chronic sinusitis, and bronchiectasis ([Chap. 256](#)). The structural abnormality leading to impaired motility of cilia can usually be defined by the electron-microscopic appearance showing defects in the dynein arms, spokes, or microtubule doublets. Cilia from epithelia and sperm tails exhibit the same defects, but the pulmonary manifestations may be minor. Other less well understood structural defects can cause immotility of sperm without involvement of cilia in the lung.

Acquired testicular defects Acquired testicular failure in the adult man can be due to *viral orchitis*. The responsible viruses include mumps virus, echovirus, lymphocytic choriomeningitis virus, and group B arboviruses. The orchitis is due to actual infection of the tissue by virus rather than to indirect effects of infection. Orchitis occurs in as many as one-fourth of adult men with mumps; in about two-thirds the orchitis is unilateral, and in the remainder it is bilateral. Orchitis usually develops a few days after the onset of parotitis but may precede it. The testis may return to normal size and function or undergo atrophy. Atrophy is believed to be due both to direct effects of the virus on the seminiferous tubules and to ischemia secondary to pressure and edema within the taut tunica albuginea. Semen analysis returns to normal in three-fourths of men with unilateral involvement and in only one-third of men with bilateral orchitis. Atrophy is usually perceptible within 1 to 6 months after the acute illness, and the degree of atrophy is not necessarily proportional to the severity of the acute orchitis. Unilateral atrophy occurs in about one-third of patients, and bilateral atrophy occurs in about one-tenth.

Trauma, including torsion, can also cause secondary atrophy of the testes. The exposed position of the testes in the scrotum renders them susceptible to both thermal and physical trauma -- particularly in men with hazardous occupations.

The testes are sensitive to *radiation damage*; decreased secretion of testosterone appears to be a consequence of diminished testicular blood flow. Doses >200 mGy (20 rad) cause increases in plasma [FSH](#) and [LH](#) levels and damage to the spermatogonia. After about 800 mGy (80 rad), oligospermia or azoospermia develops, and higher doses may obliterate the germinal epithelium, except for occasional stem and Sertoli cells. Fractionated radiation may have a more profound effect than single-dose radiation. Recovery of sperm density occurs in a dose-related fashion, and complete recovery of sperm density may require as long as 5 years. Permanent infertility can occur after radiation therapy for malignant lymphoma despite shielding of the testes. Permanent androgen deficiency in adult men is uncommon after therapeutic radiation; however, most boys given direct testicular radiation therapy for acute lymphoblastic leukemia have permanently low plasma testosterone levels. Sperm banking should be considered in patients before they undergo radiation treatment or chemotherapy.

In general, *drugs* interfere with testicular function in one of four ways -- inhibition of testosterone synthesis, blockade of androgen action, enhancement of estrogen levels, or direct inhibition of spermatogenesis. Spironolactone and ketoconazole block the synthesis of androgen by interfering with the late steps in androgen biosynthesis. Spironolactone and cimetidine compete with androgen for binding to the androgen receptor and thus block androgen action in target cells. Testosterone levels may be low, and estradiol levels may be elevated in persons using marijuana, heroin, or methadone, although the exact reasons are unclear. Alcohol, when consumed in excess for prolonged periods, causes decreased plasma testosterone levels, independent of liver disease or malnutrition. Elevated plasma estradiol and decreased plasma testosterone levels may occur in men taking digitalis.

Antineoplastic and chemotherapeutic agents commonly interfere with spermatogenesis. Cyclophosphamide causes azoospermia or extreme oligospermia within a few weeks after the initiation of therapy. Cessation of therapy is followed by a return of spermatogenesis within 3 years in about half of patients. Combination chemotherapy for acute leukemia, Hodgkin's disease, and other malignancies also may impair Leydig cell function. In pubertal boys this impairment is manifested by decreased serum testosterone and elevated [LH](#) levels; in adult men testosterone levels do not decline, and the impaired Leydig cell function may be detected only as an enhanced LH response to [GnRH](#). The alkylating agents in the chemotherapeutic regimens seem to be responsible for Leydig cell toxicity.

Because of the toxic effects of many physical and chemical agents on spermatogenesis, the occupational and recreational history should be carefully evaluated in all men with infertility. Known environmental hazards include microwaves, ultrasound, and chemicals such as the nematocide dibromochloropropane, cadmium, and lead. In some populations, sperm density is said to have declined by as much as 40% in the past 50 years, and it has been postulated that environmental estrogens or antiandrogens may be responsible.

Testicular failure also occurs as a part of *polyglandular autoimmune insufficiency* ([Chap. 339](#)). Sperm antibodies can cause isolated male infertility. In some instances these antibodies are secondary phenomena resulting from duct obstruction or vasectomy. *Granulomatous diseases* can destroy the testes, and testicular atrophy occurs in 10 to 20% of men with lepromatous leprosy owing to direct invasion of the tissue by the mycobacteria. The tubules are involved initially, followed by endarteritis and destruction of Leydig cells.

Testicular abnormalities associated with systemic disease In *cirrhosis of the liver*, a combined testicular and pituitary abnormality leads to decreased testosterone production independent of the direct toxic effects of ethanol. Although the plasma [LH](#) level is elevated, the level may be below the expected range given the degree of androgen deficiency. This situation most likely results from the inhibition of LH secretion by estrogen in patients with chronic liver disease. Increased estrogen production results from impaired hepatic extraction of adrenal androstenedione and subsequent increased extraglandular conversion to estrone and estradiol. In effect, estrogen precursors are shunted to sites of extraglandular aromatization. Testicular atrophy and gynecomastia

are present in about half of men with cirrhosis, and many such men are impotent. Successful liver transplantation reverses the effects of cirrhosis on the pituitary-testicular axis.

In chronic *renal failure*, androgen synthesis and sperm production decrease despite elevated plasma gonadotropins. The elevated [LH](#) level is due to increased production and reduced clearance but does not restore normal testosterone production. In addition, about one-fourth of men with renal failure have hyperprolactinemia; the role of hyperprolactinemia in decreasing testosterone production is unclear. Low testosterone coupled with normal or increased plasma estrogen levels cause gynecomastia in about half of men on chronic hemodialysis, and about half of men on dialysis have decreased libido and/or impotence. Improvement in testosterone production with hemodialysis is incomplete, but successful transplantation may return testicular function to normal.

Men with *sickle cell anemia* usually have impaired secondary sexual development, and testicular atrophy is present in one-third of them. The defect may be at either the testicular or the hypothalamic-pituitary level. Abnormalities in Leydig cell function, frequently accompanied by decreased sperm density, have been noted in a variety of chronic systemic diseases, including protein-energy *malnutrition*, advanced *Hodgkin's disease* and *cancer* before chemotherapy, and *amyloidosis*. Most of these disorders cause a lowered plasma testosterone level coupled with a normal to increased plasma [LH](#) level, suggesting combined hypothalamic-pituitary and testicular defects. Similar hormone changes occur after *surgery*, *myocardial infarction*, and severe *burns* and thus may be a nonspecific effect of illness.

In HIV-infected men, elevation of gonadotropins (a compensated state of hypogonadism) may precede the development of overt hypogonadism, but 35 to 50% of men with AIDS eventually develop low testosterone levels. Elevation of [SHBG](#) levels may partially mask the fall in testosterone. Some of the hormonal changes in this disorder are likely nonspecific and related to severe illness. Whether testosterone deficiency contributes to the muscle wasting and weight loss characteristic of this disorder is unclear, but androgen replacement therapy may increase muscle and lean body mass.

Sperm density can decrease temporarily after *acute febrile illness* in the absence of a change in testosterone production. Infertility in men with *celiac disease* is associated with a hormonal pattern typical of androgen resistance, namely, elevated testosterone and [LH](#) levels. *Neurologic* diseases associated with altered testicular function include myotonic dystrophy, spinobulbar muscular atrophy, and paraplegia. In myotonic dystrophy, small testes may be associated with impairment of both spermatogenesis and Leydig cell function. Spinobulbar muscular atrophy is caused by an expansion of the glutamine repeat sequences in the amino-terminal region of the androgen receptor; this expansion impairs function of the androgen receptor, but it is unclear how the alteration is related to the neurologic manifestations. Men with spinobulbar muscular atrophy often have as a late manifestation underandrogenization and infertility and the hormonal features of androgen resistance ([Chap. 338](#)). *Spinal cord lesions* that cause paraplegia lead to a temporary decrease in testosterone levels and may cause persistent defects in spermatogenesis; some patients retain the capacity for penile erection and ejaculation.

Androgen resistance Defects of the androgen receptor cause resistance to the action of androgen, usually associated with defective male phenotypic development, infertility, and underandrogenization ([Chap. 338](#)). Mutations of the androgen receptor that cause mild androgen resistance can cause infertility due to oligo- or azoospermia in otherwise phenotypically normal men.

Impairment of Sperm Transport Disorders of sperm transport may cause infertility in as many as 6% of infertile men with normal virilization. Obstruction of the ejaculatory system may be unilateral or bilateral, congenital or acquired. In men with unilateral obstruction, infertility may result from antisperm antibodies. Congenital defects of the vas deferens can occur as an isolated abnormality associated with absence of the seminal vesicles (and consequently absence of fructose in the ejaculate), in men whose mothers received *diethylstilbestrol* during pregnancy, and in men with *cystic fibrosis*. Furthermore, congenital bilateral absence of the vas deferens can be due to mutations in the cystic fibrosis conductance regulator (*CFTR*) gene; some of these mutations are distinct from those associated with the more typical pulmonary and gastrointestinal manifestations of cystic fibrosis. Acquired obstructive azoospermia can occur at the level of the epididymis in association with chronic infections of the paranasal sinuses and lungs and with tuberculosis, leprosy, and gonorrhea.

Empirical Therapy of Male Infertility Disorders for which there are logical or effective treatments (genital tract obstruction, sperm autoimmunity, gonadotropin deficiency) account for only 10% of infertile men, and pregnancies are infrequent when the male partner has genital tract obstruction or sperm autoimmunity. Severe oligospermia/azoospermia from other causes accounts for about one-fourth of cases of male infertility and has largely been considered untreatable. The other two-thirds of infertile men have a partial reduction in semen parameters and subfertility of a variable degree, and in this group spontaneous fertility may occur in untreated men (as high as 25% in one year). In the past various empirical therapies (e.g., testosterone rebound, gonadotropins, antiestrogens) have been tried without success. The only successful empirical therapy for men with mild to moderate defects in semen quality is in vitro fertilization. However, standard in vitro fertilization does not provide a good outcome in the presence of severe semen abnormalities, such as a sperm density of <5 million per milliliter, poor motility, and many abnormal forms. For such men the technique of intracytoplasmic sperm injection (ICSI) has been a major advance; indeed fertilization and pregnancy rates with this technique are similar to those for standard in vitro techniques in couples with fallopian tube pathology, e.g., a 50 to 70% fertilization rate and a 30% pregnancy rate per cycle. This technique is sometimes successful with spermatozoa recovered from testicular biopsies in men with azoospermia. *[The management of male infertility is discussed in Chap. 54.](#)

Fertility Control in Men (See also [Chap. 54](#)) A variety of approaches to fertility control in men have been tried, including use of the condom as an effective barrier that also prevents sexually transmitted diseases. Vasectomy, which involves transection or ligation of the vas deferens, has a high success rate and can be performed on an outpatient basis. The time required for azoospermia to occur after the operation depends on the number of sperm in the terminal vas deferens and ejaculatory ducts at the time of surgery, but it is usually less than 40 days. Azoospermia should be documented in each case to prove effectiveness. No deleterious effects on either

testosterone production or the hypothalamic-pituitary axis have been documented. Despite reports of immune-complex-associated accelerated atherosclerosis in vasectomized nonhuman primates, there does not appear to be any association between vasectomy and atherosclerosis in men. Vasectomy should be recommended only for men requesting permanent sterilization. Only about 30 to 40% of men subjected to vasovasostomy for reanastomosis of the vas subsequently achieve fertility. Suppression of gonadotropins with long-acting [GnRH](#) analogues or GnRH antagonists causes marked reduction in sperm counts but requires concomitant androgen replacement. The efficacy and acceptance of this approach to male contraception remain to be established.

GONADAL FUNCTION DURING AGING

Beginning at about age 40, mean plasma bioavailable testosterone concentrations decline gradually; about 40% of elderly men have low bioavailable testosterone levels. Although statistically lower than the levels in young men, the concentrations of total testosterone usually remain within the normal range, even in elderly men. The cause of the reduced testosterone level is likely a decreased number of Leydig cells. In older men seminiferous tubule function and sperm production also usually decline. Plasma [LH](#) and [FSH](#) levels are often slightly elevated, consistent with a decline in gonadal function. An increase in the conversion of androgen to estrogen in peripheral tissues results in a decrease in the effective ratio of androgen to estrogen. These latter hormonal changes may play a role in the development of prostatic hyperplasia and in the development of gynecomastia in aging men ([Chaps. 95](#) and [337](#)). Male sexual function gradually declines after early adulthood, but there is no convincing evidence that hormonal changes have any direct bearing on changes in sexual function with age in healthy men.

Prostatic Hyperplasia See [Chap. 95](#).

Cancer of the Prostate See [Chap. 95](#).

DISORDERS OF ALL AGES

Testicular Tumors (See also [Chap. 96](#)) Low levels of [hCG](#) are present in normal testes may be elevated in persons with testicular tumors. Indeed, an elevated plasma level of the b subunit of hCG (hCG-b) is a sensitive and specific marker of tumor activity in some men with germ cell tumors. Plasma levels of hCG-b are elevated in all men with choriocarcinoma, in one-third of those with embryonal carcinomas and teratocarcinomas, and rarely in those with seminomas. Changes in hCG-b levels correlate with response to therapy.

Testicular tumors can cause elevated estradiol and testosterone levels by at least two mechanisms: (1) Trophoblastic, Leydig, and Sertoli cell tumors produce both hormones autonomously; pituitary gonadotropin secretion and hormone production by the uninvolved portions of the testes are depressed, and azoospermia is common; (2) hCG secretion by the tumors can increase estradiol and testosterone production in the unaffected areas of the testes; azoospermia is uncommon with such tumors. When estrogens and androgens are formed (directly or indirectly) by the tumors, feminization, virilization, or no obvious change may result, depending on the hormones produced and

the age of the patient. α -Fetoprotein can provide another cellular marker of testicular tumor activity.

Gynecomastia See [Chap. 337](#).

TREATMENT

Androgens

Pharmacologic Preparations When testosterone is taken by mouth, it is absorbed into the portal blood and degraded promptly by the liver, so that only insignificant amounts reach the systemic circulation; when administered parenterally, testosterone is rapidly absorbed from the injection vehicle and rapidly degraded. As a consequence, effective androgen therapy requires the administration of either a slowly absorbed form of testosterone (dermal patches or micronized oral testosterone) or modified analogues. Chemical modifications either retard absorption or catabolism, or enhance the androgenic potency, so that full effects can be achieved at a lower blood level of drug. Three types of modification have had widespread clinical application ([Fig. 335-7](#)): (1) esterification of the 17 β -hydroxyl group, (2) alkylation at the 17 α position, and (3) alteration of the ring structure, particularly by substitutions at the 2, 9, and 11 positions. Most pharmacologic agents actually have combinations of ring structure alterations and either 17 α -alkylation or esterification of the 17 β -hydroxyl group. Esterification decreases the polarity of the molecule so that the steroid is more soluble in the fat vehicles used for injection, leading to slower release into the circulation. Most esters must be injected parenterally. The larger the acid esterified, the slower the release and the more prolonged the action. Esters such as testosterone cypionate and testosterone enanthate can be injected every 1 to 3 weeks, the usual regimen being 200 mg of either ester intramuscularly every 2 weeks. Because the esters are hydrolyzed before the hormones act, therapy can be monitored by assaying the plasma testosterone level at various times after administration.

The oral effectiveness of 17 α -alkylated androgens (such as methyltestosterone and methandrostenolone) is due to slower hepatic catabolism, which allows the alkylated derivatives to reach the systemic circulation. For this reason, 17 α -methyl or -ethyl substitution is a feature of most orally active androgens. Unfortunately, all 17 α -alkylated steroids can cause abnormal liver function, and for this reason they have a limited role in therapy.

Other alterations of the ring structure have been adopted empirically; some slow the rate of inactivation, others enhance the potency of a given molecule, and some alter the conversion to other active metabolites. For example, the potency of fluoxymesterone may be due to the fact that, unlike most androgens, it is a poor precursor for conversion to estrogens in peripheral tissues.

Three transdermal preparations are available in which a testosterone-loaded patch is applied to the skin each day. One is a scrotal patch (Testoderm) that contains no permeation enhancers, which may irritate the skin, but it has a low rate of acceptance. The other two systems (Androderm, Testosterone TTS) are applied to the trunk, arms, or thighs. Each patch provides physiologic testosterone levels that mimic the normal

diurnal variation with higher levels in the morning hours. High rates of dermatologic problems have been reported with the Androderm transdermal system.

Side Effects of Androgens All androgens carry the risk of inducing virilization in women. Early manifestations include acne, coarsening of the voice, hirsutism, and menstrual irregularities. If treatment is discontinued as soon as these effects develop, the manifestations may slowly subside. Long-term side effects such as male-pattern baldness, marked hirsutism, voice changes, and hypertrophy of the clitoris are largely irreversible. At physiologic replacement doses, testosterone esters have few toxic effects in mature men. At supraphysiologic doses, however, gonadotropin secretion is inhibited, the testes shrink, and the sperm count falls (indeed, androgen abuse can be associated with low sperm counts that may persist for 9 months or longer after cessation of the steroid). In some older men, testosterone therapy may cause polycythemia (hematocrit > 52%); in men predisposed to obstructive sleep apnea, androgen therapy may initiate or worsen symptoms. In older men, the presence of benign prostatic hyperplasia is not a contraindication for androgen therapy, but such men should be screened for prostate cancer before initiating androgen replacement ([Chap. 95](#)).

The so-called toxic side effects vary among the different agents and with the clinical setting in which they are used. Retention of a limited amount of sodium is an inevitable consequence of androgen therapy and may lead to edema in patients with underlying heart disease or renal failure, or when androgens are administered in enormous amounts. Although androgens do not cause malignancy, they may promote the growth of and intensify pain from carcinomas of the prostate and breast in men.

The feminizing side effects of androgen therapy in men are poorly understood. Testosterone (but not 5 α -reduced androgens) can be converted (aromatized) in extraglandular tissues to estradiol. The most common manifestation of feminization is the development of gynecomastia. Such breast enlargement is common in children given androgens, possibly because of a greater capacity to convert androgens to estrogens in childhood. The administration of testosterone esters to men results in an increase in plasma estrogen levels, but in men with normal liver function gynecomastia usually develops only after use of high doses.

All 17 α -alkylated androgens can produce liver function abnormalities such as elevated plasma levels of alkaline phosphatase and conjugated bilirubin. The incidence of clinical liver disease probably depends on the previous integrity of the liver, but jaundice may occur in the absence of preexisting liver disease. 17 α -Alkylated drugs also increase the levels of a variety of plasma proteins that are synthesized in the liver. The most serious complications of 17 α -alkylated androgens are peliosis hepatis (blood-filled cysts in the liver) and hepatoma. These disorders were initially described in patients with aplastic anemia, many of whom had Fanconi anemia, itself a predisposing factor for the development of malignancy. However, both lesions can occur after administration of substituted androgens for other indications, including use by athletes. These tumors may either follow a benign course after discontinuation of the drugs or be rapidly fatal.

One indication for 17 α -alkylated androgens is in the treatment of hereditary angioedema in which the desired therapeutic benefit (increased level of the inhibitor of the first

component of complement) may actually be an effect of the 17-alkylated side chain rather than of the parent androgen. As a consequence, weak androgens such as danazol are effective in this disorder ([Fig. 335-7](#)). Danazol is also used in the management of endometriosis ([Chap. 52](#)).

Replacement Therapy The aim of androgen therapy in hypogonadal men is to restore or bring to normal male secondary sexual characteristics (beard, body hair, external genitalia) and male sexual behavior and to mimic the hormonal effects on somatic development (hemoglobin, muscle mass, nitrogen balance, and epiphyseal closure). Since an assay for plasma testosterone is available for monitoring therapy, the treatment of androgen deficiency is almost universally successful. The parenteral administration of a long-acting testosterone ester such as 100 to 200 mg testosterone enanthate at 1- to 2-week intervals results in a sustained increase in plasma testosterone to the normal male range. Alternatively, testosterone may be administered transdermally. Testosterone patches, which are available in different doses, are replaced daily. If hypogonadism is primary and of long duration (as in the Klinefelter syndrome), suppression of plasma [LH](#) to the normal range may not occur for many weeks, if at all. There is considerable variability in the relation between plasma testosterone and male sexual behavior, but in cases of postpubertal testicular failure (even of many years duration), normal sexual activity usually is resumed after adequate replacement. Androgen administration does not restore spermatogenesis in hypogonadal states, but the volume of the ejaculate (derived largely from the prostate and seminal vesicles) and other male secondary sex characteristics return to normal. The effects of endogenous androgen on hemoglobin, nitrogen retention, and skeletal development are also reproduced.

In men of all ages in whom hypogonadism developed before expected puberty (such as men with isolated gonadotropin deficiency), it is appropriate to bring plasma testosterone into the adult range slowly. When therapy is commenced at the time of expected puberty in such men, the normal events of puberty proceed in the usual fashion. If therapy is delayed until after the time of usual puberty, the degree to which normal virilization will occur is variable, but many patients undergo a relatively complete anatomic and functional maturation. Intermittent low-dose androgen therapy is indicated in prepubertal hypogonadal boys with micropallus to bring the external genitalia into the normal range. If such patients are monitored closely and given androgens for only short periods, therapy usually has no adverse effects on somatic growth.

In boys of pubertal age with either isolated gonadotropin deficiency or primary testicular disease, the usual practice is to institute androgen therapy between the ages of 12 and 14 years, depending on the subjective need for sexual development. The initial administration of small doses of testosterone esters followed by a gradual increase to 100 to 150 mg/m² of body surface area every 1 to 3 weeks should result in a normal pubertal growth spurt. The time from the start of treatment to the appearance of secondary sex characteristics is variable. Penile development, deepening of the voice, and the appearance of other secondary sexual characteristics usually commence during the first year of treatment. In normal boys, puberty extends over several years, and treatment designed to replicate normal development does not shorten the process greatly.

Testosterone exerts its full action only in the presence of a balanced hormonal environment and, particularly, in the presence of adequate levels of growth hormone. Consequently, prepubertal boys with coexisting growth hormone and androgen deficiency respond poorly to androgens unless growth hormone is given simultaneously.

Pharmacologic Uses Androgens have been used for a variety of disorders unassociated with hypogonadism in the hope that potential benefits from the nonvirilizing actions of the agents (such as increases in nitrogen retention, muscle mass, and hemoglobin) would outweigh any deleterious actions of the drugs. The most common nonreplacement uses of androgen have been attempts to improve nitrogen balance in catabolic states (e.g., AIDS), self-administration by athletes to increase muscle mass and/or athletic performance, attempts to enhance erythropoiesis in refractory anemias (including the anemia of renal failure), treatment of hereditary angioedema and endometriosis, and management of growth retardation of various etiologies. Most of the expected benefits in these disorders have not been realized for two reasons. First, modest pharmacologic doses of androgens have little physiologic effect in men when superimposed on normal testicular androgen, and in women the virilizing side effects of androgens are formidable. Second, no androgen has been devised that exhibits only the nonvirilizing effects of the hormone. This conclusion is not surprising in view of the fact that the physiologic actions of androgens are mediated by a single, high-affinity receptor ([Fig. 335-5](#)).

The most pervasive form of androgen abuse is by male athletes in the expectation that muscle development and athletic performance will be improved. In controlled studies using modest pharmacologic doses (two to four times the usual replacement doses), these agents do not improve performance consistently. However, at the doses frequently taken by athletes (which sometimes exceed 10 times the replacement dose), androgens do enhance nitrogen balance and muscle mass; since the drugs have multiple side effects at high doses, these benefits do not outweigh the risks associated with androgen abuse in man, and the use of androgens by female athletes is associated with disfiguring virilization. Thus this practice cannot be condemned too harshly. The only established indications for androgen therapy outside of male hypogonadism are in selected patients with anemia due to bone marrow failure or hereditary angioedema and as an adjunct to growth hormone therapy.

Gonadotropins Gonadotropin therapy is used to establish or restore fertility in patients with gonadotropin deficiency of any cause. Several gonadotropin preparations are available. Human menopausal gonadotropin (hMG) (purified from the urine of postmenopausal women) contains 75 IU [FSH](#) and 75 IU [LH](#) per vial. [hCG](#) (purified from the urine of pregnant women) has little FSH activity and resembles LH in its ability to stimulate testosterone production by Leydig cells. Because of the expense of hMG, treatment is usually begun with hCG alone, and hMG is added later to promote the FSH-dependent stages of spermatid development. A high ratio of LH to FSH activity and treatment for 6 to 18 months may be necessary to bring about maturation of the prepubertal testes. Recombinant human FSH is also available and has been used mainly for ovulation induction in women. Trials are underway to examine its effects on spermatogenesis in men with hypogonadotropic hypogonadism. Once spermatogenesis is restored with combined FSH and LH therapy, hCG alone is often sufficient to maintain spermatogenesis.

The dose of [hCG](#) required to maintain a normal testosterone level varies from 1000 to 5000 IU weekly. A number of regimens have been used to induce maturation of spermatogenesis. Most involve starting with 2000 IU hCG three or more times a week until most of the clinical parameters, including plasma testosterone levels, are normal. [hMG](#) (usually one ampule) is then added three times a week to complete the development of spermatogenesis. The length of therapy required to restore spermatogenesis may be as long as 12 months.

[GnRH](#) and GnRH Analogues GnRH (gonadorelin) is available for endocrine testing and is used by some physicians for chronic therapy of the infertility of hypogonadotropic hypogonadism. In the latter instance, it is necessary to administer GnRH in frequent boluses (25 to 200 ng/kg of body weight every 2 h) with the use of portable infusion pumps, analogous to those used for insulin administration. In general, pulsatile GnRH does not appear to be more efficacious than gonadotropin in returning sperm counts to normal. GnRH analogues (leuprolide, nafarelin, histrelin) are available for the suppression of gonadotropin secretion, leading to hypogonadism. In prostatic cancer, testicular androgen production can be blocked by monthly injection of 7.5 mg leuprolide in depot form.

(Bibliography omitted in Palm version)

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336. DISORDERS OF THE OVARY AND FEMALE REPRODUCTIVE TRACT - Bruce R. Carr, Karen D. Bradshaw

The ovary is the source of ova for reproduction and of the hormones that regulate female sexual life. The anatomic structure, response to hormonal stimuli, and secretory capacity of the ovary vary at different periods of life. This chapter will review normal ovarian physiology as a background for understanding ovarian abnormalities and will consider other disorders of the female reproductive tract.

DEVELOPMENT, STRUCTURE, AND FUNCTION OF THE OVARY

EMBRYOLOGY

During the third week of gestation, the primordial germ cells differentiate from the endoderm lining the yolk sac at the caudal end of the embryo. The germ cells migrate to the genital ridge adjacent to the mesonephric kidney by the fifth week of gestation and undergo mitotic division. The gonads exist in an undifferentiated state until the seventh week of fetal life, at which time the primitive ovary can be distinguished from the testis ([Chap. 338](#)). Estrogen formation in the ovary commences between weeks 8 and 10, and by 10 to 11 weeks of gestation, oogonia in the ovarian cortex begin developing into primary oocytes. The ovary contains a finite number of germ cells, the number peaking at about 7 million oogonia by the fifth to sixth month of gestation. Subsequently, the germ cells decrease in number through a process of atresia so that only 1 million remain at birth, 400,000 are present at menarche, and only a few remain at menopause. Two normal X chromosomes are required for development of the ovary; in individuals with a 45,X karyotype, ovarian development occurs, but the rate of atresia is accelerated so that only a fibrous streak remains at birth ([Chap. 338](#)).

After the oogonia cease to proliferate, meiosis commences, continues until the diplotene stage of the first meiotic division is completed, and then is arrested until the onset of ovulation at puberty. From the fifth month of fetal life, the primordial follicle consists of the primary oocyte arrested in meiosis, a single surrounding layer of granulosa cells, and a basement membrane that separates the primordial follicle from surrounding stromal (interstitial) tissues.

PUBERTAL MATURATION

The final maturation of ovarian follicles commences during puberty. The two major hormones that regulate follicular development are the pituitary gonadotropins -- follicle-stimulating hormone (FSH) and luteinizing hormone (LH) ([Fig. 336-1](#)). During the second trimester of fetal development, the plasma gonadotropins rise to levels similar to those at menopause. This peak in gonadotropin levels may be responsible for the simultaneous peak in oocyte replication. After the second trimester, the hypothalamic-pituitary axis (the so-called gonadostat) becomes functional and is sensitive to negative feedback by steroid hormones, particularly estrogen and progesterone produced in the placenta. The levels of circulating gonadotropins consequently decrease, and gonadotropins are almost undetectable at the time of birth. In the neonate, concomitant with the decrease in estrogen and progesterone levels caused by separation from the placenta, there is a rebound increase in gonadotropin

secretion for the first few months of life. With continued maturation of the hypothalamic-pituitary system, the gonadostat becomes exquisitely sensitive to negative feedback by low levels of circulating steroid hormones, and plasma gonadotropins again decrease.

As the time of puberty nears, a decrease in the sensitivity of the gonadostat allows for increased secretion of [FSH](#) and [LH](#), possibly secondary to increased episodic or pulsatile secretion of gonadotropin-releasing hormone (GnRH) by the hypothalamus ([Chap. 328](#)). A sleep-induced, pulsatile pattern of LH secretion then ensues, the first step in the development of a cyclic pattern of gonadotropin secretion ([Fig. 336-1](#)). The increase in estrogen secretion exerts a positive feedback, which leads to an exaggeration of the pulsatile release of LH and eventually to menarche and ovulation, after which plasma gonadotropin concentrations reach adult values, which are similar during day and night. After the menopause, plasma gonadotropin levels rise, then plateau 5 to 10 years after menopause and remain fairly constant until the eighth to ninth decade of life, when the levels may fall. Although ovarian function is regulated primarily by LH and FSH, the ovary is a source of peptide and protein hormones and growth factors such as inhibin and activin that may play a role in ovarian function and regulation. The production of inhibin by the mature ovary accounts, in part, for the relative reduction in FSH that is seen during the reproductive years ([Fig. 336-1](#)).

With puberty the sensitivity of the hypothalamic-pituitary centers to circulating steroid hormones is decreased, GnRH release by the hypothalamus increases, gonadotropin secretion by the pituitary is enhanced, ovarian estrogen secretion increases, and the anatomic changes of puberty ensue. At age 10 to 11, the first secondary sexual characteristics begin to appear in girls, namely, development of the breast buds (thelarche), followed by the development of pubic hair (pubarche), and later by the development of axillary hair (adrenarche). The growth of pubic and axillary hair is believed to be initiated by adrenal androgens, the levels of which begin to rise at approximately 6 to 8 years of age. A growth spurt ensues, and peak growth rate is attained by age 12.

The culmination of puberty is the onset of predictable, cyclic menses. The average time between the beginning of breast development and the onset of menses (menarche) is 2 years. During the first few years after menarche, menstrual cycles are often irregular and unpredictable due to anovulation. The age of menarche is variable and is influenced by socioeconomic and genetic factors and by general health. In the United States, the mean age of menarche is believed to have decreased at a rate of 3 to 4 months per decade over the past 100 years and is now around 12 years, a change believed to be due to improved nutrition. A body weight of around 48 kg or some critical combination of weight, body water, and body fat is associated with development of hypothalamic insensitivity to feedback by steroids that leads to increased secretion of gonadotropins and finally to menarche. Obese girls have earlier menarche than girls with normal weights. In contrast, active participation in sports or ballet, malnutrition, and chronic debilitating disease can delay menarche.

MATURE OVARY

Morphology The anatomic components and function of the adult ovary are illustrated

schematically in [Fig. 336-2](#). Under the influence of gonadotropins, a group of primary follicles are recruited, and by day 6 to 8 of the menstrual cycle, one follicle becomes mature or "dominant," a process characterized by accelerated growth of granulosa cells and enlargement of the fluid-filled antrum. The recruited follicles not destined to ovulate undergo degeneration, similar to the atresia that occurs in other follicles during embryogenesis. Just prior to ovulation, meiosis resumes in the ovum of the dominant follicle, and the first meiotic division results in formation of the first polar body. The antrum rapidly enlarges (up to 10 to 25 mm in size), follicular fluid increases in amount, and the follicular surface thins and forms a conical stigma. Ovulation from the dominant follicle occurs some 16 to 23 h after the [LH](#) peak and some 24 to 38 h after the onset of the LH surge as the result of rupture of the follicular wall at the area of the stigma. The ovum is then expelled together with a mass of surrounding granulosa cells called *cumulus cells*. The rupture is believed to result from the action of hydrolytic enzymes on the surface of the follicle, possibly under the control of prostaglandins. The second meiotic division occurs after the egg is fertilized by a sperm, and the second polar body is then extruded. The formation of the *corpus luteum* begins in the retained remnant of the ovulated follicle; the remaining granulosa and theca cells increase in size and accumulate lipids and a yellow pigment, lutein, to become "luteinized." After a period of 14 ± 2 days (the functional life of the corpus luteum), the corpus luteum begins to atrophy, to be replaced in time by a fibrous scar, the *corpus albicans*. The factors that limit the life span of the human corpus luteum are not known, but if pregnancy occurs, the corpus luteum persists under the influence of placental or chorionic gonadotropins, and progesterone is produced by the corpus luteum for the support of pregnancy.

Hormone Formation

Steroid Hormones Like other steroid hormones, ovarian steroids are derived from cholesterol ([Fig. 336-3](#)). The ovary can synthesize cholesterol de novo and can also utilize cholesterol obtained from circulating lipoproteins as substrate for steroid hormone formation ([Fig. 336-4](#)). Virtually all ovarian cells are believed to possess the complete complement of enzymes required for the synthesis of estradiol from cholesterol ([Fig. 336-3](#)); however, different cell types in the ovary contain different amounts of these enzymes so that the main steroids produced differ in different compartments. For example, the corpus luteum forms mainly progesterone and 17-hydroxyprogesterone, whereas theca and stromal cells convert cholesterol to androstenedione and testosterone. Granulosa cells are particularly rich in the aromatase enzyme responsible for estrogen synthesis and utilize as substrates for this process androgens synthesized in the granulosa cells and the adjacent theca cells.

The principal sites of action of [LH](#) and [FSH](#) are also illustrated in [Figs. 336-3](#) and [336-4](#). LH acts primarily to regulate the early steps in steroid hormone biosynthesis, namely, the transport of cholesterol into the mitochondria by steroidogenic acute regulatory (StAR) protein and its conversion to pregnenolone. FSH acts mainly to regulate the final process by which androgens are aromatized to estrogens. As a consequence, LH enhances substrate flow and the formation of androgens and/or progesterone in the absence of FSH, whereas FSH action is impeded in the absence of LH because of diminished substrate for aromatization.

ESTROGENS Naturally occurring estrogens are 18-carbon steroids characterized by an

aromatic A ring, a phenolic hydroxyl group at C-3, and either a hydroxyl group (estradiol) or a ketone (estrone) at C-17 ([Fig. 336-3](#)). (For the numbering of the steroid ring, see [Fig. 335-1](#).) The principal estrogen secreted by the ovary and the most potent estrogen is estradiol. Estrone is also produced by the ovary, but most estrone is formed by extraglandular conversion of androstenedione in peripheral tissues. Estriol (16-hydroxyestradiol), the main estrogen in urine, arises from the 16-hydroxylation of estrone and estradiol. Catechol estrogens are formed by hydroxylation of estrogens at the C-2 or C-4 position and may act as the intracellular mediators of some estrogen action. Estrogens promote development of the secondary sexual characteristics in women and cause uterine growth, thickening of the vaginal mucosa, thinning of the cervical mucus, and development of the ductule system of the breasts. Estrogens also alter lipid profiles and exert vascular effects that help prevent cardiovascular disease. The mechanism of estrogen action in target tissues is similar to that for other steroid hormones and involves binding to a nuclear steroid receptor -- either estrogen receptor (ER)a or ERb -- and enhancement of the transcription of messenger RNA, which in turn causes increased protein synthesis in the cell cytoplasm ([Chap. 327](#)). These receptors have specific tissue site expression and bind various estrogens with different affinities, thereby conferring selective actions.

PROGESTERONE Progesterone, a 21-carbon steroid ([Fig. 336-3](#)), is the principal hormone secreted by the corpus luteum and is responsible for progestational effects, i.e., induction of secretory activity in the endometrium of the estrogen-primed uterus in preparation for implantation of the fertilized egg. Progesterone also induces a decidual reaction in endometrium. Other effects include inhibition of uterine contractions, an increase in the viscosity of cervical mucus, glandular development of the breasts, and an increase in basal body temperature (thermogenic effect).

ANDROGENS The ovary synthesizes a variety of 19-carbon steroids, including dehydroepiandrosterone, androstenedione, testosterone, and dihydrotestosterone, principally in stromal and thecal cells. The major ovarian 19-carbon steroid is androstenedione ([Fig. 336-3](#)), part of which is secreted into the plasma and part of which is converted to estrogen in granulosa cells or to testosterone in the interstitium. Androstenedione can also be converted to testosterone and estrogens in peripheral tissues. Only testosterone and dihydrotestosterone are true androgens that interact with the androgen receptor and induce virilizing signs in women ([Chaps. 53](#) and [335](#)).

Other Hormones *Inhibin* is secreted in two forms (A and B) by the follicle and inhibits the release of [FSH](#) by the hypothalamic-pituitary unit. *Activin* is also secreted by the follicle and may enhance FSH secretion as well as having local effects on ovarian steroidogenesis. *Follistatin* is an activin-binding protein that attenuates the actions of activin and other members of the transforming growth factor (TGF) family.

Some ovarian hormones play an uncertain role in human physiology. *Relaxin*, a polypeptide hormone produced by the human corpus luteum and by the decidua, causes softening of the cervix and loosening of the symphysis pubis in preparation for parturition in animals. *Oxytocin*, *vasopressin*, and other hypothalamic and pituitary hormones are also present in granulosa and/or luteal cells, but their function in these cells is unknown. *Follicle regulatory protein* (FRP), found in human follicular fluid, inhibits granulosa secretion and growth. *Gonadocrinins*, peptides purified from rat

follicular fluid, stimulate the release of both [FSH](#) and [LH](#) from the pituitary in vitro and in vivo. Granulosa cells secrete *oocyte maturation inhibitor* (OMI), a factor that prevents premature ovulation. In addition, in the gonads of both sexes a *meiosis-inducing substance* (MIS) triggers the onset of meiosis, an event that occurs earlier in ovarian than in testicular development. Local growth factors [including insulin-like growth factors (IGFs) 1 and 2 and [TGFa](#) and -b] may also influence steroid secretion by the ovary.

The Normal Menstrual Cycle The menstrual cycle is divided into a follicular or proliferative phase and a luteal or secretory phase ([Fig. 336-5](#)). The secretion of [FSH](#) and [LH](#) is fundamentally under negative feedback control by ovarian steroids (particularly estradiol) and by inhibin (which selectively suppresses FSH), but the response of gonadotropins to different levels of estradiol varies. FSH secretion is inhibited progressively as estrogen levels increase -- typical negative feedback. In contrast, LH secretion is suppressed maximally by sustained low levels of estrogen and is enhanced by a rising level of estradiol -- positive feedback. Feedback of estrogen involves both the hypothalamus and pituitary. Negative feedback suppresses [GnRH](#) and inhibits gonadotropin production. Positive feedback is associated with an increased frequency of GnRH secretion and enhanced pituitary sensitivity to GnRH.

The length of the menstrual cycle is defined as the time from the onset of one menstrual bleeding episode to onset of the next. In women of reproductive age, the cycle averages 28 ± 3 days and the mean duration of flow is 4 ± 2 days. Longer menstrual cycles (usually characterized by anovulation) occur at menarche and near the onset of menopause. At the end of a cycle plasma levels of estrogen and progesterone fall, and circulating levels of [FSH](#) increase. Under the influence of FSH, follicular recruitment results in development of the follicle that will be dominant during the next cycle.

After the onset of menses, follicular development continues, but [FSH](#) levels decrease. Approximately 8 to 10 days prior to the midcycle [LH](#) surge, plasma estradiol levels begin to rise as the result of estradiol formation by the granulosa cells of the dominant follicle. During the second half of the follicular phase, LH levels also begin to rise (owing to positive feedback). Just before ovulation, estradiol secretion reaches a peak and then falls. Immediately thereafter, a further rise in the plasma level of LH mediates the final maturation of the follicle, followed by follicular rupture and ovulation 16 to 23 h after the LH peak. The rise in LH is accompanied by a smaller increase in the level of plasma FSH, the physiologic significance of which is unclear. The plasma progesterone level also begins to rise just prior to midcycle and facilitates the positive feedback action of estradiol on LH secretion.

At the onset of the luteal phase, plasma gonadotropins decrease and plasma progesterone increases. A secondary rise in estrogens causes further gonadotropin suppression. Near the end of the luteal phase, progesterone and estrogen levels fall, and [FSH](#) levels begin to rise to initiate the development of the next follicle (usually in the contralateral ovary) and the next menstrual cycle. Inhibin A levels are low in the follicular phase but reach a peak in the luteal phase. Inhibin B levels, in contrast, are increased in the follicular phase and low in the luteal phase.

The endometrium lining the uterine cavity undergoes marked alterations in response to the changing plasma levels of ovarian hormones ([Fig. 336-5](#)). Concurrent with the

decrease in plasma estrogen and progesterone and the decline of corpus luteum function in the late luteal phase, intense vasospasm occurs in the spiral arterioles supplying blood to the endometrium, causing ischemic necrosis, endometrial desquamation, and bleeding. This vasospasm is caused by locally synthesized prostaglandins. The onset of bleeding marks the first day of the menstrual cycle. By the fourth to fifth day of the cycle, the endometrium is thin. During the proliferative phase, glandular growth of the endometrium is mediated by estrogen. After ovulation, increased progesterone levels lead to further thickening of the endometrium, but the rapid growth slows. The endometrium then enters the secretory phase, characterized by tortuosity of the glands, curling of the spiral arterioles, and glandular secretion. As corpus luteum function begins to wane in the absence of conception, the sequence of events leading to menstruation is again set into action.

Biphasic changes in basal body temperature are characteristic of the ovulatory cycle and are mediated by alterations in progesterone levels ([Fig. 336-5](#)). An increase in basal body temperature by 0.3 to 0.5°C begins after ovulation, persists during the luteal phase, and returns to the normal baseline (36.2 to 36.4°C) after the onset of the subsequent menses.

Cellular Interactions in the Ovary during the Normal Cycle [LH](#) stimulates thecal cells surrounding the follicle to form androgens, and androstenedione diffuses across the basement membrane of the follicle into granulosa cells, where it is aromatized to estrogen ([Figs. 336-3](#) and [336-4](#)).

The increase of [FSH](#) late in the preceding menstrual cycle stimulates growth and recruitment of the primary follicles by enhancing granulosa cell proliferation, resulting ultimately in the formation of the dominant follicle. This function of FSH is underscored by the fact that only primary follicles are seen in patients with mutations in FSH or the FSH receptor. In the granulosa cells, FSH also stimulates estrogen synthesis. Enhanced secretion of estradiol causes an increase in the number of estradiol receptors and further proliferation of granulosa cells. In the late follicular phase, FSH, in concert with estradiol, causes induction of [LH](#) receptors on the granulosa cells. LH acts via these receptors to increase progesterone secretion at midcycle. The amount of progesterone formed by the follicle is believed to be limited by the availability of cholesterol to serve as substrate for steroidogenesis and by the fact that most of the progesterone is converted to androstenedione by thecal cells. Prior to ovulation, the granulosa cells of the follicle are bathed in follicular fluid but have limited access to circulating blood and consequently to plasma low-density lipoprotein (LDL). As depicted in [Fig. 336-4](#), the granulosa cells become vascularized after ovulation, and plasma cholesterol is made available to serve as the major substrate for progesterone synthesis by the corpus luteum. Thus, increased progesterone synthesis by the corpus luteum is the consequence of increased substrate availability. The peak in progesterone secretion by the corpus luteum occurs 8 days after ovulation at the time of maximal vascularization of the granulosa cells.

MENOPAUSE

The *menopause* is defined as the final episode of menstrual bleeding in women. However, the term is used commonly to refer to the time interval that encompasses the

transitional period between the reproductive years up to and after the last episode of menstrual bleeding. During this period, there is a progressive loss of ovarian function and a variety of endocrine, somatic, and psychological changes.

The median age of women at the time of cessation of menstrual bleeding is 50 to 51 years. Since the life expectancy of women is close to 80 years, approximately one-third of life occurs after cessation of reproductive function. Preceding the menopause, the pattern of menstrual cycles is variable, but the interval between menses usually becomes shorter, as follicular recruitment is hastened by increases in [FSH](#). Day 3 FSH and E_2 levels are often elevated. Ovulatory cycles continue for some period of time, then anovulation becomes common.

The menopause is the consequence of the exhaustion of ovarian follicles. The decrease in the number of ova begins in intrauterine life; by the time of the menopause, few ova remain, and these appear to be nonfunctional. Only a small number of ova are lost as the result of ovulation during reproductive life; the majority are lost by atresia. The cessation of follicular development results in decreased production of estradiol, inhibin, and other hormones, which causes a loss of negative feedback on the hypothalamic-pituitary centers. In turn, the levels of plasma gonadotropins increase, with [FSH](#) levels rising earlier and higher than [LH](#) levels ([Figs. 336-1](#) and [336-6](#)). The higher concentration of FSH than LH in postmenopausal women may result from the decrease in inhibin secretion by the ovary, from the fact that FSH is cleared from plasma less rapidly than LH, and possibly from the loss of positive feedback on LH production by estradiol.

The ovaries of postmenopausal women are small and wrinkled, and the residual cells are predominantly stromal. Estrogen and androgen levels in plasma are reduced but not absent ([Fig. 336-6](#)). Before the menopause, plasma androstenedione is derived almost equally from the adrenals and the ovaries; after menopause the ovarian contribution ceases so that the plasma levels of androstenedione fall by 50% ([Fig. 336-6](#)). However, the menopausal ovary continues to secrete testosterone, presumably formed in stromal cells.

Circulating estrogens in the ovulating woman are derived from two sources. Some 60% of mean estrogen formation during the menstrual cycle is in the form of estradiol, formed primarily by ovaries, and the remainder is estrone, formed mainly in extraglandular tissues from androstenedione. After menopause, extraglandular estrogen formation is the major pathway for estrogen synthesis. Because adipose tissue is a major site of extraglandular estrogen production, peripheral estrogen formation may actually be enhanced in obese postmenopausal women, so that total estrogen production rates may be as great or greater than in premenopausal women. The predominant estrogen formed is estrone rather than estradiol.

The most common menopausal symptoms are vasomotor instability (hot flashes), atrophy of the urogenital epithelium and skin, decreased size of the breasts, and osteoporosis. Approximately 40% of menopausal women develop symptoms serious enough to seek medical assistance.

The pathogenesis of the hot flash is uncertain. There is a close temporal relationship

between the onset of the hot flash and pulses of [LH](#) secretion, which reflect hypothalamic secretion of [GnRH](#). Alterations in catecholamine, prostaglandin, endorphin, or neurotensin metabolism may play a role in conjunction with low estrogen production. Symptoms associated with the hot flash, including nervousness, anxiety, irritability, and depression, may or may not be caused entirely by estrogen deficiency.

The decrease in size of the tissues of the female reproductive tract and breasts in the menopause is due to estrogen deficiency. The vaginal mucosa and the endometrium usually become thin and atrophic (although endometrial hyperplasia occurs in one-fifth of postmenopausal women).

Osteoporosis is one of the dread afflictions of aging, and there is a close relationship between estrogen deprivation and its development. Approximately one-fourth of aging women and one-tenth of elderly men sustain a vertebral or hip fracture between the ages of 60 and 90, and the incidence is highest in elderly white women. Such fractures are a major cause of loss of independence, death and morbidity, and the fracture-related mortality increases from <10% in the 60- to 64-year age group to >30% in patients over 80 ([Chap. 342](#)). Many factors affect the development of osteoporosis, including ethnic origin, diet, activity, smoking, and general health, and estrogen deprivation is of particular importance. White and Asian postmenopausal women are more predisposed to osteoporosis and its consequences because bone mass in this group is lower prior to menopause, so loss in bone density has more severe consequences. Further evidence that osteoporosis is a disease of estrogen deprivation is suggested by the early development of osteoporosis in women with premature menopause due to either natural causes or surgical castration. After the menopause women experience an increase in the incidence of cardiovascular disease as the result of a decrease in the level of high-density lipoprotein (HDL) cholesterol as well as the effects of hypoestrogenism on vascular endothelium and reactivity.

LABORATORY AND CLINICAL ASSESSMENT OF HORMONAL STATUS

The hormonal status of women can usually be assessed by history and physical examination. In general, the presence of secondary sexual characteristics such as normal female breast development indicates adequate estrogen secretion in the past, and the presence of regular, predictable, cyclic menses implies that ovulation and the production of gonadotropins, estrogen, progesterone, and androgens are adequate and that the outflow tract is intact. Such a history may be more valuable than laboratory tests in evaluating ovarian hormone status. However, laboratory tests provide valuable ancillary information in the evaluation of women with endocrine dysfunction or infertility ([Chap. 54](#)).

PITUITARY GONADOTROPINS

Plasma gonadotropins are assessed by radioimmunoassay (RIA), fluoroimmunoassay (FIA), or immunoradiometric assay (IRMA). Because both [FSH](#) and [LH](#) are secreted in a pulsatile manner, the results obtained from a single serum sample may be difficult to interpret. Consequently, multiple samples taken at 20-min intervals over 2 h may be pooled to obtain a mean value. Serum gonadotropin measurements are of the most use in evaluating women with suspected ovarian failure and in supporting the diagnosis of

polycystic ovarian syndrome (PCOS) and hypogonadotropic hypogonadism. The normal ranges for serum LH and FSH in ovulating women are 0.8 to 57 and 1.4 to 21 IU/L, respectively. FSH levels that are persistently >40 IU/L are diagnostic of ovarian failure, and an LH value <0.8 IU/L suggests hypogonadotropic hypogonadism. In practice, however, gonadotropin values may be equivocal and must be interpreted in light of the remainder of the findings.

OVARIAN HORMONES

The mean plasma levels and production rates of the principal ovarian hormones are presented in [Table 336-1](#).

Estrogen The presence of normal secondary sexual characteristics implies that estrogen production was adequate in the past. The current estrogen status can be estimated by pelvic examination. The presence of a moist, rugated vagina with copious, clear, thin cervical mucus that can be stretched and that exhibits arborization or ferning when spread on a slide is strong evidence of adequate estrogen production. Cytologic demonstration of mature vaginal epithelial cells and abundant cornified squamous epithelial cells with pyknotic nuclei confirms the presence of adequate estrogen levels.

The progesterone-withdrawal test provides a functional assessment of the endometrium, outflow tract, and estrogen status. If menses appear within a week to 10 days after the end of a trial of medroxyprogesterone acetate (10 mg by mouth once or twice a day for 5 days) or after a single intramuscular injection of progesterone (100 mg), then prior estrogen priming was adequate to allow withdrawal bleeding.

Owing to its variable level in plasma during the normal cycle and the difficulty of estimating the day of the cycle in women with abnormal cycles, the measurement of estrogen levels in plasma or urine is of little use in the routine assessment of estrogen status. Measurement of plasma estradiol is useful during attempts to induce ovulation with gonadotropins to prevent the development of the ovarian hyperstimulation syndrome and is used along with ultrasound assessment to monitor follicular growth in women who are to undergo in vitro fertilization.

Progesterone Cyclic, predictable menses also imply that adequate progesterone is secreted during the luteal phase of the menstrual cycle. Assessment of progesterone is useful to detect ovulation and to evaluate the adequacy of the luteal phase in infertile women. Several functional assays of progesterone can be used. The least expensive and most useful is the daily measurement of basal body temperature throughout a cycle. Owing to the thermogenic properties of progesterone, a normal biphasic monthly curve showing a temperature elevation lasting for approximately 2 weeks after ovulation is a valid indication of progesterone secretion during the luteal phase ([Fig. 336-5](#)). The presence of viscous cervical mucus that does not stretch or fern and of predominantly intermediate cells on vaginal cytology or demonstration of a secretory epithelium in an endometrial biopsy during the luteal phase on days 20 to 22 of the cycle provides additional assessment of progesterone secretion. In addition, serum progesterone can be measured to assess the function of the corpus luteum.

Androgen Under normal conditions, the ovary secretes androstenedione, testosterone,

and dehydroepiandrosterone. In conditions of androgen excess, hirsutism and/or virilization are common. The evaluation of androgen excess is discussed in [Chap. 53](#).

DIAGNOSIS OF PREGNANCY

Pregnancy is usually recognized on the basis of the history and physical examination. That is, a woman with previously cyclic, predictable menses develops amenorrhea accompanied by breast tenderness, malaise, lassitude, and nausea, and on physical examination the uterus is soft and enlarged.

Assays of placental products facilitate the diagnosis of pregnancy. Human chorionic gonadotropin (hCG) is secreted by the trophoblastic cells of the placenta into the maternal plasma and excreted in the urine. Assays of the hCG content of serum or urine use antibodies against hCG and make it possible to detect pregnancies 8 to 10 days after ovulation, before the first missed menstrual period and long before pregnancy can be diagnosed by clinical assessments. Assay of the β subunit of hCG in serum or urine makes it possible to differentiate between excess LH and hCG, an important distinction in evaluating women with trophoblastic disease such as hydatidiform mole or choriocarcinoma. Sensitive and specific hCG-based pregnancy tests are now available for testing by patients at home.

DISORDERS OF OVARIAN FUNCTION

PREPUBERTAL YEARS

Puberty is said to be *precocious* if breast budding begins before age 8 or if menarche occurs before age 9. Those disorders in which the developing sexual characteristics are appropriate for the genetic and gonadal sex -- i.e., feminization in girls or virilization in boys -- are termed *isosexual precocity*, whereas *heterosexual precocity* occurs when sexual characteristics are not in accord with the genetic sex, namely, virilization in girls or feminization in boys. Pubertal disorders of boys are described in [Chap. 335](#).

Isosexual Precocious Puberty Isosexual precocious puberty in girls can be divided into three major categories ([Table 336-2](#)).

True Precocious Puberty True precocious puberty is characterized by an early but otherwise normal sequence of pubertal development, including increased secretion of gonadotropins and ovulatory menstrual cycles. Constitutional or idiopathic precocious puberty accounts for 90% of cases. In these individuals, no cause for the premature maturation of the central nervous system-hypothalamic-pituitary axis can be identified, and the diagnosis is confirmed by finding an adult pattern of LH and FSH release on a GnRH stimulation test. As many as half these individuals have abnormal findings on electroencephalograms. Premature appearance of secondary sexual characteristics and of ovulatory cycles with the accompanying risk of fertility may cause significant emotional disturbance. Therefore, prompt initiation of therapy is imperative. GnRH analogues suppress gonadotropins and inhibit estrogen synthesis, thereby blocking precocious puberty; they may also prevent premature closure of the epiphyses and the resulting short stature.

About 10% of cases are due to organic brain diseases, including brain tumors (hypothalamic gliomas, astrocytomas, ependymomas, germinomas, and hamartomas), encephalitis, meningitis, hydrocephalus, head injury, tuberous sclerosis, and neurofibromatosis. It is essential to distinguish this group of patients from those with the idiopathic disorder, and patients whose disorder is designated as idiopathic occasionally prove to have such tumors. Fortunately, most patients with organic lesions serious enough to cause precocious puberty have obvious neurologic signs and symptoms. Evaluation of all patients with precocious puberty should include, at a minimum, skull films and computed tomography (CT) or magnetic resonance imaging (MRI) of the brain. The success of treatment depends on the nature of the lesion, but surgical and radiation treatment of well-localized tumors is occasionally successful.

A rare cause of isosexual precocity is congenital adrenal hyperplasia due to 21-hydroxylase deficiency in girls in whom treatment is delayed until 4 to 8 years of age. After initiation of glucocorticoid replacement, such individuals may undergo isosexual precocious puberty ([Chap. 331](#)).

Precocious Pseudopuberty Precocious pseudopuberty occurs when girls undergo feminization as a consequence of enhanced estrogen formation but do not ovulate or develop cyclic menses. Ovarian cysts or tumors that secrete estrogen (granulosa-theca cell tumors) are the most frequent cause of precocious pseudopuberty. Granulosa-theca cell tumors associated with intestinal polyps and pigmentation of the mucous membranes occur in the Peutz-Jeghers syndrome. Other ovarian tumors that secrete estrogens (or androgens that can be converted to estrogens at extraglandular sites) include dysgerminomas, teratomas, cystadenomas, and ovarian carcinomas ([Chap. 97](#)). Ovarian tumors can usually be detected by rectoabdominal examination or by sonography, [CT](#), [MRI](#), and/or laparoscopy. Ovarian teratomas and choriocarcinomas and other carcinomas that secrete [hCG](#) do not cause precocious puberty in girls unless they also secrete estrogen ([hCG](#) or [LH](#) in the absence of [FSH](#) does not induce ovarian estrogen production). Rarely, feminizing tumors of the adrenal cause isosexual precocious puberty by direct formation of estrogens or by secretion of weak androgens, which are converted to estrogens in extraglandular tissues.

Other causes of precocious pseudopuberty include the following:

1. The McCune-Albright syndrome (polyostotic fibrous dysplasia) is due to an activating mutation in the G-protein, Gsa, that occurs during embryogenesis, leading to a mosaic pattern of expression in various tissues. It is characterized by cafe au lait spots, cystic fibrous dysplasia of bones, and sexual precocity. In the ovary, the Gsa mutation mimics the action of [FSH](#), leading to autonomous follicle development and estrogen formation. Occasionally, this disorder leads to true precocious puberty ([Chap. 343](#)).
2. Primary hypothyroidism is occasionally associated with enhanced secretion of [FSH](#), inducing ovarian estrogen secretion. High levels of thyroid-stimulating hormone (TSH) caused by hypothyroidism may also stimulate the FSH receptor.
3. The Russell-Silver syndrome, or congenital asymmetry, is associated with short stature and precocious feminization.

4. Estrogen-containing medications, including use of estrogen-containing creams for diaper rash or the ingestion of meat from estrogen-treated animals or poultry or of any estrogen by mouth, can cause this disorder.

Incomplete Isosexual Precocity This term is used to describe the premature development of a single pubertal event and encompasses several entities. Breast budding prior to age 7 (*premature thelarche*) without other evidence of estrogen secretion and without premature bone maturation is believed to be due to a transient increase in estrogen secretion or to a temporary increase in sensitivity to the small amounts of circulating estrogens formed prior to puberty. Usually, the disorder is self-limited and resolves spontaneously. Occasionally, axillary hair and/or pubic hair (*premature adrenarche* and *premature pubarche*) appear without any other secondary sexual development. The phenomenon is associated with adrenal androgen secretion in the range of normal puberty and can be distinguished from syndromes of virilization by the absence of clitoromegaly. It requires no treatment, and patients enter puberty at about the average time.

Heterosexual Precocity Virilization in a prepubertal female is usually due to congenital adrenal hyperplasia or to androgen secretion by an ovarian or adrenal tumor. The manifestations of virilization are described in [Chaps. 53](#) and [331](#). Virilization in girls with congenital adrenal hyperplasia usually takes place in a background of variable sexual ambiguity (see [Chap. 338](#)).

Evaluation of Sexual Precocity The evaluation of sexual precocity involves a careful history and physical examination, including rectoabdominal examination, abdominal sonography, determination of bone age, and [GnRH](#) stimulation test, and measurement of thyroid hormones, [TSH](#), and gonadotropins (and androgen or estrogen levels when appropriate). [MRI](#) and/or [CT](#) scans should be obtained if a neurologic disorder is suspected and no evidence of ovarian or adrenal tumor is found.

REPRODUCTIVE YEARS

Disorders of the Menstrual Cycle

Abnormal Uterine Bleeding Between menarche and the menopause, almost every woman experiences one or more episodes of abnormal uterine bleeding, here defined as any bleeding pattern that differs in frequency, duration, or amount from the pattern observed during a normal menstrual cycle. A variety of descriptive terms (such as *menorrhagia*, *metrorrhagia*, and *menometrorrhagia*) have been used to characterize patterns of abnormal uterine bleeding. A more logical approach is to divide abnormal uterine bleeding into those patterns associated with ovulatory cycles and those associated with anovulatory cycles.

Ovulatory Cycles Normal menstrual bleeding with ovulatory cycles is spontaneous, regular, cyclic, and predictable and is frequently associated with discomfort (*dysmenorrhea*). Deviations from this pattern associated with cycles that are still regular and predictable are most often due to organic disease of the outflow tract. For example, regular but prolonged and excessive bleeding episodes unassociated with bleeding dyscrasias (hypermenorrhea or menorrhagia) can result from abnormalities of the uterus

such as submucous leiomyomas, adenomyosis, or endometrial polyps. Regular, cyclic, predictable menstruation characterized by spotting or light bleeding is termed *hypomenorrhea* and is due to obstruction of the outflow tract as from intrauterine synechiae or scarring of the cervix. Intermenstrual bleeding between episodes of regular, ovulatory menstruation is also often due to cervical or endometrial lesions. An exception to the association between organic disease and abnormal uterine bleeding is the occurrence of regular menstruation more frequently than 21 days apart (*polymenorrhea*). Such cycles may be a normal variant.

Anovulatory Cycles Uterine bleeding that is unpredictable with respect to amount, onset, and duration and usually painless is described as *dysfunctional uterine bleeding*. This disorder is not due to abnormalities of the uterus but rather to chronic anovulation and occurs when there is interruption of the normal sequence of follicular and luteal phases under the influence of a dominant follicle and its resulting corpus luteum. As discussed above, uterine bleeding in ovulatory cycles is due to progesterone withdrawal and requires that the endometrium first be primed with estrogen. (When castrates or postmenopausal women are given progesterone, withdrawal bleeding usually does not occur.)

Dysfunctional uterine bleeding can occur in women who have a transient disruption of the synchronous hypothalamic-pituitary-ovarian patterns necessary for ovulatory cycles, most often at the extremes of the reproductive life -- in the early menarche and in the perimenopausal period -- but also after temporary stress or intercurrent illness.

Primary dysfunctional uterine bleeding can result from three disorders.

1. *Estrogen withdrawal bleeding* occurs when estrogen is given to a castrated or postmenopausal woman and then withdrawn. As in other types of dysfunctional uterine bleeding, this form of menstrual bleeding is usually painless.

2. *Estrogen breakthrough bleeding* occurs when there is continuous estrogen stimulation of the endometrium not interrupted by cyclic progesterone secretion and withdrawal. This is the most common type of dysfunctional uterine bleeding and is usually due to anovulation associated with chronic acyclic estrogen production, as in women with [PCOS](#). Such women may have histories of irregular, unpredictable menses, oligomenorrhea, or amenorrhea (see below). Alternatively, estrogen breakthrough bleeding can occur in hypogonadal women given estrogens chronically rather than intermittently and in women with estrogen-secreting tumors of the ovary. Estrogen breakthrough bleeding may be profuse and is unpredictable with respect to duration, amount of flow, and time of occurrence. The endometrium is typically thin because its repair between episodes of bleeding is incomplete.

3. *Progesterone breakthrough bleeding* occurs in the presence of abnormally high ratios of progesterone to estrogen, e.g., in women using continuous low-dose oral contraceptives.

The approach to a patient with dysfunctional uterine bleeding begins with a careful history of menstrual patterns and prior hormonal therapy. Since not all urogenital tract bleeding is from the uterus, rectal, bladder, and vaginal or cervical sources must be

excluded by physical examination. If the bleeding is from the uterus, a pregnancy-related disorder such as abortion or ectopic pregnancy must be ruled out.

TREATMENT

Once the diagnosis of dysfunctional uterine bleeding is established, a rational approach to management is as follows: During a first episode of dysfunctional bleeding the patient can simply be observed, provided the bleeding is not copious and no evidence of bleeding dyscrasia is present. If bleeding is moderately severe, control can be achieved with relatively high dose estrogen oral contraceptives for 3 weeks. Alternatively, a regimen of three or four low-dose oral contraceptive pills per day for 1 week followed by tapering to the usual dosage for up to 3 weeks is also effective. If uterine bleeding is more severe, hospitalization, bed rest, and intramuscular injections of estradiol valerate (10 mg) and hydroxyprogesterone caproate (500 mg) or intravenous or intramuscular conjugated estrogens (25 mg) usually control the bleeding. After initial treatment, iron replacement should be instituted, and recurrence can be prevented by cyclic oral contraceptives for 2 to 3 months (or more if pregnancy is not desired). Alternatively, menses can be induced every 2 to 3 months with medroxyprogesterone acetate, 10 mg by mouth once or twice a day for 10 days. If hormone therapy fails to control uterine bleeding, an endometrial biopsy, hysteroscopy, or dilatation and curettage may be required for diagnosis and therapy. Indeed, uterine sampling should be performed prior to hormone therapy in women at risk for endometrial cancer (i.e., in women who are approaching the age of menopause or are massively obese); endometrial cancer is rare in ovulatory women of reproductive age.

Amenorrhea An acceptable definition of amenorrhea is failure of menarche by age 15, irrespective of the presence or absence of secondary sexual characteristics, or the absence of menstruation for 6 months in a woman with previous periodic menses. However, women who do not fulfill these criteria should be evaluated if (1) the patient and/or her family are greatly concerned, (2) no breast development has occurred by age 13, or (3) any sexual ambiguity or virilization is present ([Chap. 338](#)). Amenorrhea is commonly categorized as either primary (the woman has never menstruated) or secondary (when menstruation has been present for a variable period of time in the past and has ceased). However, some disorders can cause either primary or secondary amenorrhea. For example, most women with gonadal dysgenesis have primary amenorrhea, but some have a few follicles and ovulate for short periods so that pregnancy occurs rarely. Furthermore, patients with chronic anovulation ([PCOS](#)) usually have secondary amenorrhea but on occasion have primary amenorrhea. For these reasons, categorization of amenorrhea into primary and secondary types is less helpful than a classification based on the underlying physiologic derangements: (1) anatomic defects, (2) ovarian failure, and (3) chronic anovulation with or without estrogen present.

ANATOMIC DEFECTS Anatomic or structural defects of the genital tract can preclude menstrual bleeding. Starting from the caudal end of the female genital tract, labial fusion is often associated with disorders of sexual development, particularly female pseudohermaphroditism (congenital adrenal hyperplasia or exposure to maternal androgens in utero; [Chap. 338](#)). Congenital defects of the vagina, imperforate hymen, and transverse vaginal septae can also cause amenorrhea. These women frequently have accumulation of menstrual blood behind the obstruction and may have cyclic,

predictable episodes of abdominal pain.

More severe mullerian anomalies include mullerian agenesis (the Mayer-Rokitansky-Kuster-Hauser syndrome; [Chap. 338](#)), second in frequency only to gonadal dysgenesis as a cause of primary amenorrhea. It can be caused by mutations in the genes encoding anti-mullerian hormone (AMH) or its receptor (AMHR). Women with this syndrome have a 46,XX karyotype, female secondary sex characteristics, and normal ovarian function, including cyclic ovulation, but have absence or hypoplasia of the vagina. The uterus usually consists of only rudimentary bicornuate cords, but if the uterus contains endometrium, cyclic abdominal pain and accumulation of blood may occur, as in other forms of outlet obstruction. One-third of women with this syndrome have abnormalities of the urogenital tract, and one-tenth have skeletal anomalies, usually involving the spine. The major diagnostic problem is distinguishing mullerian agenesis from complete testicular feminization, in which 46,XY genetic males with testes differentiate as phenotypic women but with a blind vaginal pouch and no uterus. Women with testicular feminization have feminized breasts but a paucity of pubic and axillary hair. The disorder is X-linked and is caused by mutations in the androgen receptor that result in profound resistance to the action of testosterone ([Chap. 338](#)). Testicular feminization can be diagnosed by demonstrating a male level of serum testosterone and a 46,XY karyotype, whereas demonstration of a 46,XX karyotype, the biphasic basal body temperature curve characteristic of ovulation, and elevated levels of progesterone during the luteal phase establish the diagnosis of mullerian agenesis.

A rare cause of absence of the uterus in 46,XY phenotypic women who are sexually infantile is the so-called testicular regression syndrome or testicular agenesis ([Chap. 338](#)).

Other abnormalities of the uterus that cause amenorrhea include obstruction due to scarring or stenosis of the cervix, often resulting from surgery, electrocautery, laser therapy, or cryosurgery. Such destruction of the endometrium (Asherman's syndrome) usually follows vigorous curettage for postpartum hemorrhage or after therapeutic abortion complicated by infection. This diagnosis is confirmed by hysterosalpingography or by direct visual examination of the endometrial scarring or synechiae using a hysteroscope.

Treatment of disorders of the outflow tract is surgical.

OVARIAN FAILURE Primary ovarian failure is associated with elevated plasma gonadotropin levels and can result from several causes. The most frequent cause is *gonadal dysgenesis*, in which the germ cells are absent and the ovary is replaced by a fibrous streak ([Chaps. 65](#) and [338](#)). Women with gonadal dysgenesis can be divided into two broad groups on the basis of chromosomal karyotype. The most common type is due to deletion of genetic material in the X chromosomes and accounts for about two-thirds of cases of gonadal dysgenesis. A 45,X karyotype is found in about half of women with this disorder, and most have somatic defects, including short stature, webbed neck, shield chest, and cardiovascular defects, collectively termed the *Turner phenotype*. The remainder of women with X chromosome abnormalities have chromosomal mosaicism with or without associated structural abnormalities of the X. The most common form of mosaicism is 45,X/46,XX. Gonadal tumors are rare in 45,X

patients, but gonadal malignancies may occur in women with chromosomal mosaicism involving the Y chromosome. Therefore, chromosomal analysis should be performed in all cases of amenorrhea associated with ovarian failure, and the streak gonad should be removed if a Y chromosome is present. One means of identifying the presence of a Y chromosome is to amplify the sex-determining regions of the Y chromosome (SRY) by means of the polymerase chain reaction ([Chap. 338](#)). Approximately 90% of women with gonadal dysgenesis due to partial or complete deletion of the X never have menstrual bleeding, and the remaining 10% have sufficient follicles to experience menses and, rarely, fertility; the menstrual and reproductive lives of such individuals are invariably brief.

One-tenth of individuals identified as having bilateral streak gonads have a normal 46,XX or 46,XY karyotype and are said to have *pure gonadal dysgenesis*. These individuals have either normal or above-average stature, owing to failure of estrogen-mediated epiphyseal closure in the presence of a normal chromosomal constitution. Pure gonadal dysgenesis does not constitute a phenotypic or chromosomally homogeneous disorder ([Chap. 338](#)). Occasional women with a 46,XY karyotype develop signs of virilization, including clitoromegaly, and have an increased incidence of tumors in the gonadal streaks; as a consequence, gonadal streaks should be removed prophylactically, as discussed above, when a Y chromosome is present. Approximately two-thirds of women with 46,XX gonadal dysgenesis experience no menses, while the remainder have one or more menstrual episodes and are occasionally fertile.

Other causes of ovarian failure and amenorrhea include deficiency of the *CYP17* gene that encodes 17 α -hydroxylase and 17,20-lyase activities, premature ovarian failure, the resistant-ovary syndrome, and ovarian failure secondary to chemotherapy or radiation therapy for malignancy. *17 α -Hydroxylase deficiency* is characterized by primary amenorrhea, sexual infantilism, and hypertension, the latter due to increased production of desoxycorticosterone (DOC); whereas women with *17,20-lyase deficiency* have primary amenorrhea and sexual infantilism with normal blood pressure ([Chaps. 331 and 338](#)). The diagnosis of *premature ovarian failure* or *premature menopause* is applied to women who cease menstruating before age 40. The ovaries in such women are similar to the ovaries of postmenopausal women, containing few or no follicles as the result of accelerated follicular atresia. Premature ovarian failure due to ovarian antibodies may be one component of polyglandular failure, together with adrenal insufficiency, hypothyroidism, and other autoimmune disorders ([Chap. 339](#)). A rare form of ovarian failure is the *resistant-ovary syndrome*, in which the ovaries contain many follicles that are arrested in development prior to the antral stage, possibly because of resistance to the action of [FSH](#) in the ovary. A subset of these individuals have mutations in FSH or its receptor. To differentiate this disorder from the 46,XX variety of pure gonadal dysgenesis, which is also associated with sexual immaturity, it is necessary to perform ovarian biopsy. However, it is not clinically useful to make this distinction, since the conventional treatment of infertility in both conditions is usually unsuccessful. Women with ovarian failure who desire pregnancy have been treated with hormone replacement and transfer of donor embryos to the uterine cavity or fallopian tubes.

Chronic Anovulation At least 80% or more of gynecologic endocrine disorders result

from chronic anovulation. Women with chronic anovulation fail to ovulate spontaneously but may ovulate with appropriate therapy. The ovaries of such women do not secrete estrogen in a normal cyclic pattern; it is clinically useful to differentiate those women who produce enough estrogen to have withdrawal bleeding after progestogen therapy from those who do not; the latter often have hypothalamic-pituitary dysfunction.

CHRONIC ANOVULATION WITH ESTROGEN PRESENT Women with chronic anovulation who experience withdrawal bleeding after progestogen administration are said to be in a state of "estrus" due to the acyclic production of estrogen, largely estrone, by extraglandular aromatization of circulating androstenedione. This disorder is commonly termed *polycystic ovarian syndrome* ([PCOS](#)) and is characterized by infertility, hirsutism, obesity, and amenorrhea or oligomenorrhea. When spontaneous uterine bleeding occurs in women with PCOS, it is unpredictable as to time of onset, duration, and amount; on occasion the bleeding can be severe. The dysfunctional uterine bleeding is usually due to estrogen breakthrough (see above).

The disorder, as originally described by Stein and Leventhal, was characterized by enlarged, polycystic ovaries, but it is now known to be associated with a variety of pathologic findings in the ovaries, only some of which result in enlargement and none of which are pathognomonic. The most common finding is a white, smooth, sclerotic ovary with a thickened capsule, multiple follicular cysts in various stages of atresia, a hyperplastic theca and stroma, and rare or absent corpora albicans. Other ovaries have hyperthecosis in which the ovarian stroma is hyperplastic and may contain lipid-laden luteal cells. Thus, the diagnosis of [PCOS](#) is a clinical one, based on the coexistence of chronic anovulation and varying degrees of androgen excess. The fundamental defect that causes PCOS is unknown, and it is likely to have several distinct causes.

In most women with [PCOS](#), menarche occurs at the expected time, but uterine bleeding is unpredictable in onset, duration, and amount. Amenorrhea ensues after a variable period, although primary amenorrhea occurs in some women. Signs of androgen excess (hirsutism) usually become evident around the time of menarche. One scenario suggests that this disorder originates as an exaggerated adrenarche in obese girls ([Fig. 336-7](#)). The combination of elevated levels of adrenal androgens and obesity leads to increased formation of extraglandular estrogen. This estrogen exerts a positive feedback on [LH](#) secretion and negative feedback on [FSH](#) secretion, resulting in a ratio of LH to FSH levels in plasma that is characteristically greater than 2. The increased LH levels can then lead to hyperplasia of the ovarian stroma and theca cells and increased androgen production, which in turn provides more substrate for peripheral aromatization and perpetuates the chronic anovulation. In the advanced stage of the disorder, the ovary is the major site of androgen production, but the adrenal may continue to secrete excess androgen as well. The greater the obesity, the more strongly this sequence would be perpetuated because more androgen is converted to estrogen by adipose tissue stromal cells, which in turn exaggerates inappropriate LH release by positive feedback. Ovarian follicles from women with PCOS have low aromatase activity, but normal aromatase can be induced by treatment with FSH. An association exists between PCOS/hyperthecosis, virilization, acanthosis nigricans, and insulin resistance; in the ovary, insulin may interact via the insulin-like growth factor receptors to enhance androgen synthesis in insulin-resistant states.

TREATMENT

Treatment of [PCOS](#) is directed toward interrupting the self-perpetuating cycle and can be accomplished in several ways, such as by decreasing ovarian androgen secretion (by wedge resection or the use of oral contraceptive agents), decreasing peripheral estrogen formation (by weight reduction), or enhancing [FSH](#) secretion [by administration of clomiphene, human menopausal gonadotropin (hMG), [GnRH](#) (gonadorelin) by portable infusion pump, or purified FSH (urofollitropin)]. The choice of therapy depends on the clinical findings and the needs of the patient. An attempt at weight reduction is appropriate in all who are obese. If the woman is not hirsute and does not desire pregnancy, periodic withdrawal menses can be induced with medroxyprogesterone acetate 10 days per month; such treatment prevents the development of endometrial hyperplasia. If the woman is hirsute and does not desire pregnancy, the ovarian (and possibly the adrenal) component of androgen production can be suppressed with combined estrogen-progestogen oral contraceptive agents. Combined oral contraceptives are also indicated if prolonged or excessive menstrual bleeding is present. Once androgen excess is controlled, treatment of previously existing hair growth by shaving, depilatories, or electrolysis may be indicated ([Chap. 53](#)). If pregnancy is desired, ovulation must be induced. The insulin-sensitizing drugs metformin and troglitazone improve fertility in women with PCOS. Clomiphene promotes ovulation in three-fourths of cases, or ovulation can be induced with hMG, urofollitropin, or gonadorelin ([Chap. 54](#)). Pretreatment with GnRH analogues prior to use of hMG, urofollitropin, or gonadorelin may improve the rates of ovulation and pregnancy. Women with PCOS are at increased risk of ovarian hyperstimulation after treatment with gonadotropins. They also experience increased rates of spontaneous abortion. An alternative therapy is ovarian drilling by laser or cautery performed at laparoscopy in women in whom hormonal therapy is not effective; however, the procedure is associated with a high incidence of ovarian adhesions.

Chronic anovulation with estrogen present also may occur with tumors of the ovary. These include granulosa-theca cell tumors, Brenner tumors, cystic teratomas, mucous cystadenomas, and Krukenberg tumors ([Chap. 97](#)). Such tumors can either secrete excess estrogen themselves or produce androgens that are aromatized in extraglandular sites. Chronic anovulation and the clinical features of [PCOS](#) result. Occasionally, areas of the ovary not involved with tumors show the characteristic histologic changes of PCOS. Other causes of chronic anovulation with estrogen present include adrenal production of excess androgen (usually adult-onset adrenal hyperplasia due to partial 21-hydroxylase deficiency) and various thyroid disorders.

CHRONIC ANOVULATION WITH ESTROGEN ABSENT Women with chronic anovulation who have low or absent estrogen production and do not experience withdrawal bleeding after progestogen treatment usually have hypogonadotropic hypogonadism due to disease of either the pituitary or the central nervous system.

Isolated hypogonadotropic hypogonadism associated with defects of smell (olfactory bulb defects) is known as the *Kallmann syndrome* ([Chaps. 328](#) and [335](#)). Affected women are sexually infantile and have a defect in the synthesis and/or release of [GnRH](#). Hypothalamic lesions that impair GnRH production and cause hypogonadotropic hypogonadism include craniopharyngioma, germinoma (pinealoma), glioma,

Hand-Schuller-Christian disease, teratomas, endodermal-sinus tumors, tuberculosis, sarcoidosis, and metastatic tumors that cause suppression or destruction of the hypothalamus. Central nervous system trauma and irradiation can also cause hypothalamic amenorrhea and deficiencies in secretion of growth hormone, adrenocorticotrophic hormone (ACTH), vasopressin, and thyroid hormone.

More commonly, gonadotropin deficiency leading to chronic anovulation is believed to arise from functional disorders of the hypothalamus or higher centers. A history of a stressful event in a young woman is frequent. For example, chronic anovulation can begin suddenly in a woman who leaves home for the first time or experiences the death of a loved one. Gonadotropin and estrogen levels are in the low to low-normal range as compared with normal women in the early follicular phase of the cycle. In addition, rigorous exercise, such as jogging or ballet, and diets that result in excessive weight loss may lead to chronic anovulation, particularly in girls with a history of prior menstrual irregularity. The amenorrhea in these women does not appear to be due to weight loss alone but to a combination of a decrease in body fat and chronic stress. An extreme form of weight loss with chronic anovulation occurs in anorexia nervosa. Anorexia nervosa is characterized by the development in a young woman of amenorrhea with associated severe weight loss, distorted attitudes toward eating and weight gain, and distorted body image. In anorexia nervosa amenorrhea can precede, follow, or coincide with weight loss ([Chap. 78](#)). During successful therapy, gonadotropin changes recapitulate those observed during normal puberty ([Fig. 336-1](#)).

In addition, chronic debilitating diseases such as end-stage kidney disease, malignancy, and malabsorption are believed to lead to development of hypogonadotropic hypogonadism via a hypothalamic mechanism.

Treatment of chronic anovulation due to hypothalamic disorders includes ameliorating the stressful situation, decreasing exercise, and correcting weight loss if appropriate. These women appear to be susceptible to the development of osteoporosis; estrogen replacement therapy is recommended to induce and maintain normal secondary sexual characteristics and prevent bone loss in those who do not desire pregnancy, and gonadotropin or gonadorelin therapy is indicated when pregnancy is desired (see "Treatment," below). When appropriate, therapy is directed at the primary disease of the hypothalamus.

Disorders of the pituitary can lead to the estrogen-deficient form of chronic anovulation by at least two mechanisms -- direct interference with gonadotropin secretion by lesions that either obliterate or interfere with the gonadotropic cells (chromophobe adenomas, Sheehan's syndrome) or inhibition of gonadotropin secretion in association with excess prolactin (prolactinoma). *Pituitary tumors* make up approximately 10% of all intracranial tumors and may secrete no hormone, one hormone, or more than one hormone ([Chap. 328](#)). Prolactin levels are elevated in 50 to 70% of patients with pituitary tumors, either because of prolactin secretion by the tumor itself (in the case of prolactinomas) or because the tumor mass interferes with the normal hypothalamic inhibition of prolactin secretion.

Prolactinomas can be divided into microadenomas (<10 mm in diameter) and macroadenomas (>10 mm). Prolactin excess associated with low levels of [LH](#)

and [FSH](#) constitutes a specific subtype of hypogonadotropic hypogonadism. One-tenth or more of amenorrheic women have increased levels of serum prolactin, and more than half of women with both galactorrhea and amenorrhea have elevated prolactin levels. The amenorrhea is most often associated with decreased or absent estrogen production, but prolactin-secreting tumors on occasion are associated with normal ovulatory menses or chronic anovulation with estrogen present. The increased frequency of diagnosis of prolactin-secreting adenomas is probably due to several factors, including increased awareness, improved radiographic detection methods, and availability of radioimmunoassays for prolactin. Since in older autopsy series a 9 to 23% prevalence of pituitary adenomas was observed in asymptomatic women, the clinical and prognostic significance of small microadenomas in asymptomatic individuals is unclear. However, when tumors of any size are associated with symptoms of amenorrhea or galactorrhea, particularly when visual field defects or severe headaches are present, bromocriptine therapy or neurosurgical evaluation is indicated. In the latter half of pregnancy, prolactin-secreting pituitary tumors may expand, leading to headaches, compression of the optic chiasm, bitemporal hemianopsia, and blindness. Therefore, before inducing ovulation for the purposes of achieving pregnancy, it is mandatory to exclude the presence of a pituitary tumor. **The evaluation, differential diagnosis, and management of hyperprolactinemia are described in [Chap. 328](#).*

Large pituitary tumors such as null cell adenomas -- whether or not hyperprolactinemia is present -- are likely to be associated with deficiency of hormones in addition to gonadotropins ([Chap. 328](#)).

Craniopharyngiomas, which are thought to arise from remnants of Rathke's pouch, account for 3% of intracranial neoplasms, occur most frequently in the second decade of life, and may extend into the suprasellar region. Many of these tumors calcify and can be diagnosed by conventional skull films. Patients often present with sexual infantilism, delayed puberty, and amenorrhea due to gonadotropin deficiency; secretion of [TSH](#), [ACTH](#), growth hormone, and vasopressin may also be impaired.

Panhypopituitarism can occur spontaneously, be caused by mutations in transcription factors (Pit1; Prop1) involved in pituitary gland development, result from surgical or radiation treatment of pituitary adenomas, or develop after postpartum hemorrhage (Sheehan's syndrome). Patients with the latter disorder characteristically have failure to lactate or ovulate, loss of genital and axillary hair, hypothyroidism, and adrenal insufficiency ([Chap. 328](#)).

Evaluation of Amenorrhea A general scheme for the evaluation of women with amenorrhea is given in [Fig. 336-8](#). On physical examination, attention should be given to three features: (1) the degree of maturation of the breasts, pubic and axillary hair, and external genitalia; (2) the current estrogen status; and (3) the presence or absence of a uterus. Pregnancy should be excluded in all women with amenorrhea; it is prudent to perform a suitable pregnancy screening test even when the history and physical examination are not suggestive. Once that is done, the cause of amenorrhea can frequently be diagnosed clinically. For example, Asherman's syndrome is suggested by a history of curettage in a woman who previously menstruated; in women with primary amenorrhea and sexual infantilism, the essential differential diagnosis is between gonadal dysgenesis and hypopituitarism, and the diagnosis of gonadal dysgenesis

(Turner's syndrome) or of anatomic defects of the outflow tract (mullerian agenesis, testicular feminization, and cervical stenosis) is frequently suggested on the basis of physical findings. When a specific cause is suspected, it is appropriate to proceed directly to confirm the diagnosis (obtaining a chromosomal karyotype or measurement of plasma gonadotropins). It is also useful to measure serum prolactin and [FSH](#) levels during the initial evaluation.

Estrogen status is evaluated by determining if the vaginal mucosa is moist and rugated and if the cervical mucus can be stretched and shown to fern upon drying. If these criteria are indeterminate, a progestational challenge is indicated, most often the administration of 10 mg medroxyprogesterone acetate by mouth once or twice daily for 5 days or 100 mg progesterone in oil intramuscularly. (It should be emphasized that progestogen should never be administered until pregnancy is excluded.) If estrogen levels are adequate (and the outflow tract is intact), menstrual bleeding should occur within 1 week of ending the progestogen treatment. If withdrawal bleeding occurs, the diagnosis is chronic anovulation with estrogen present, usually caused by [PCOS](#).

If no withdrawal bleeding or only minimal vaginal spotting occurs, the nature of the subsequent workup depends on the results of the initial prolactin assay. If plasma prolactin is elevated, or if galactorrhea is present, radiography of the pituitary should be undertaken. When the plasma prolactin level is normal in an anovulatory woman with estrogen absent and with elevated [FSH](#) levels, the diagnosis is ovarian failure. If the gonadotropins are in the low or normal range, the diagnosis is either hypothalamic-pituitary disorder or an anatomic defect of the outflow tract. As indicated previously, the diagnosis of outflow tract disorder is usually suspected or established on the basis of the history and physical findings. When the physical findings are not clear-cut, it is useful to administer cyclic estrogen plus progestogen (1.25 mg oral conjugated estrogens per day for 3 weeks, with 10 mg medroxyprogesterone acetate added for the last 7 to 10 days of estrogen treatment), followed by 10 days of observation. If no bleeding occurs, the diagnosis of Asherman's syndrome or another anatomic defect of the outflow tract is confirmed by hysterosalpingography or hysteroscopy. If withdrawal bleeding occurs following the estrogen-progestogen combination, the diagnosis of chronic anovulation with estrogen absent (functional hypothalamic amenorrhea) is suggested. Radiologic evaluations of the pituitary-hypothalamic areas may be indicated in the latter cases -- irrespective of the prolactin level -- because of the danger of overlooking a pituitary-hypothalamic tumor and because the diagnosis of functional hypothalamic amenorrhea is one of exclusion ([Chap. 328](#)).

Infertility Infertility, the failure to become pregnant after 1 year of unprotected intercourse, affects approximately 10 to 15% of couples and is a common reason for seeking gynecologic assistance ([Chap. 54](#)). Male factors account for 40% of infertility problems ([Chap. 335](#)). In women, failure of ovulation accounts for 30% of cases; pelvic factors, such as tubal disease or endometriosis, account for half; and a cervical factor is implicated in about one-tenth. In 10 to 20% of infertile women no etiology is found.

Medical Aspects of Pregnancy (See also [Chap. 7](#)) The possibility of pregnancy should be considered in all women of reproductive age who are evaluated for medical illness or considered for surgery. Procedures such as x-ray exposure, drugs, and

anesthetics may be harmful to the developing fetus, and a variety of medical problems may worsen during pregnancy, including hypertension; diseases of the heart, lungs, kidney, and liver; and metabolic and endocrine disorders. Abnormal vaginal bleeding or amenorrhea during the reproductive years should prompt consideration of a complication of pregnancy, such as incomplete abortion, ectopic pregnancy, or trophoblastic disease (hydatidiform mole or choriocarcinoma). Women who present with these complications of pregnancy often have histories of abdominal pain and vaginal bleeding and may have evidence of intraabdominal hemorrhage.

Choriocarcinoma is a particular problem because of its protean manifestations. Half these malignancies follow pregnancies complicated by hydatidiform mole, and the remainder occur after spontaneous abortion, ectopic pregnancy, or normal deliveries. Patients may present with intraabdominal bleeding due to rupture of the uterus, liver, or ovary, with pulmonary manifestations (cough, hemoptysis, pleuritic pain, dyspnea, and respiratory failure) or gastrointestinal symptoms, usually chronic blood loss or melena. In addition, patients can present with cerebral metastases or renal involvement. The diagnosis can be established by demonstrating an elevated level of the b subunit of [hCG](#) in plasma. Treatment and cure are possible with chemotherapeutic agents (dactinomycin and/or methotrexate). **The manifestations of choriocarcinoma in men are discussed in [Chap. 96](#).*

Ovarian Tumors See [Chap. 97](#)

TREATMENT

Progestogens The major use of progestogens is in conjunction with estrogen to ensure the full maturation of the endometrium, both in combination birth control pills and in the therapy of hypogonadal states. In certain circumstances, however, progestogen therapy is appropriate by itself -- to induce a progestational effect on the estrogen-primed endometrium (in diagnostic tests for the evaluation of amenorrhea), to inhibit pituitary gonadotropins for contraception (the progestogen-only birth control pill or progestogen-containing implants), for prophylaxis to prevent hyperplasia in [PCOS](#), for palliation in cases of endometrial and breast carcinoma, and for treatment of endometriosis. Even when a direct progestational effect is desired, the available oral drugs substitute a synthetic derivative for the naturally occurring hormone. Oral progestogens include medroxyprogesterone acetate, megestrol acetate, norethindrone, norgestrel, and micronized progesterone. Parenteral agents include progesterone in oil, medroxyprogesterone acetate suspension, and 17-hydroxyprogesterone caproate. Vaginal progesterone suppositories are used for treatment of luteal-phase defects, and progestogen implants are available for long-term contraception.

The most common undesirable side effect is breakthrough bleeding, which occurs when progestogens are used continuously. Other complications include nausea, vomiting, and hirsutism. Abnormal liver function is a side effect of those derivatives with alkyl substitution in the 17a position. Synthetic progestogens are contraindicated if pregnancy is known or suspected, because of the risk of birth defects.

Estrogens Estrogens are used for the treatment of gonadal failure, the control of fertility, and the management of dysfunctional uterine bleeding. However, none of the

available oral or parenteral hormones replaces the pattern of circulating estradiol levels characteristic of the normally cycling, premenopausal woman ([Fig. 336-5](#)). Estrogens that can be given by mouth are either nonsteroidal agents (such as diethylstilbestrol) that mimic the action of estradiol, estrogen conjugates that must be hydrolyzed before they become active (usually estrone sulfate from pregnant mare's urine), or estrogen analogues that cannot be metabolized to estradiol (mestranol, quinestrol;[Fig. 336-9](#)). Even when micronized estradiol is given orally, it is rapidly converted in the body to estrone. Because oral therapy neither replaces nor mimics the daily secretory pattern of the deficient hormone, such therapy must be viewed as a pharmacologic substitution rather than a physiologic replacement. Likewise, the use of parenteral estrogens rarely mimics the physiologic situation. Parenteral preparations of conjugated estrogens, like the oral derivatives, are poor precursors of estradiol, and estradiol esters (estradiol benzoate and valerate) rarely result in plasma estradiol levels that mimic the normal monthly secretory cycle of the hormone. Transdermal estradiol results in constant levels of blood estradiol and is effective in the treatment of menopausal symptoms. Estrogen vaginal rings and creams can be used for local treatment of vaginal atrophy, but systemic absorption is variable. The side effects of estrogen substitution differ at various times of life.

Hypoestrogenism In women with decreased estrogen production, whether due to disease of the ovaries (gonadal dysgenesis) or to hypogonadotropic hypogonadism, treatment with cyclic estrogens should be instituted at the time of expected puberty to induce the development and maintenance of female secondary sexual characteristics and to prevent osteoporosis. The most commonly used medications are conjugated estrogens (0.625 to 1.25 mg/d by mouth) together with medroxyprogesterone acetate (2.5 mg/d or 5 to 10 mg during the last several days of monthly estrogen treatment to prevent development of endometrial hyperplasia). Alternatively, oral contraceptives may be given ([Chap. 54](#)). Abnormal bleeding in women receiving estrogen replacement mandates histologic evaluation of the endometrium. Such substitution therapy or the use of oral contraceptives may also be used for the purpose of suppressing pituitary gonadotropins, as in women with [PCOS](#), in whom the major therapeutic aim is suppression of ovarian androgen production.

Temporary administration of estrogens in larger quantities (up to two times the usual adult maintenance dose) may be necessary to induce the full development of secondary sexual characteristics in girls and for the control of menopausal symptoms. Even larger doses of parenteral estrogens (10 mg estradiol valerate or 25 mg conjugated estrogen) in conjunction with progestogen may be required in some instances of dysfunctional uterine bleeding. In addition to the potential long-term side effects of all estrogens (see below), high doses may cause nausea, vomiting, and edema.

Contraceptives See [Chap. 54](#)

Estrogen Treatment of the Menopause The use of estrogens in postmenopausal women is based on evidence that they may relieve some of the complications of the postmenopausal state, including osteoporosis, and some manifestations of aging itself. In some parts of the United States, as many as half of women in the menopausal age group used one or more forms of estrogen replacement for a median period of 5 years.

The menopause is not, however, a state of simple estrogen deprivation, as some estrogens continue to be produced. It is instead a state of altered estrogen metabolism; the predominant estrogen becomes estrone, which is formed by extraglandular conversion of prehormone, rather than estradiol secreted by the ovary. As is true for all estrogen therapy, the estrogen treatment of the menopause is actually a pharmacologic substitution of one or another estrogen analogue for estradiol rather than a physiologic replacement of the missing steroid. The estrogens available for replacement therapy include conjugated estrogens, estrogen substitutes (diethylstilbestrol), synthetic estrogen (ethinyl estradiol or derivatives), micronized estradiol, estrogen-containing vaginal creams, and estrogen-containing dermal patches. Selective [ER](#) modulators (e.g., raloxifene) have estrogenic activity in the bone and the cardiovascular system but are antiestrogenic in breast and uterus. Raloxifene binds to the ER, with a conformational change in the domain of the receptor involved in transcription. Regimens associated with a low risk of complications include the following: (1) cyclic estrogen therapy in the lowest effective dose for 25 days per month or continuous estrogens given each day of the month, (2) estrogens plus the addition of progestogen during the last 10 to 14 days of estrogen therapy, (3) low-dose continuous progestogen plus estrogen given daily, and (4) daily selective ER modulators (SERM).

The most clear-cut early benefit of estrogen therapy in the menopause is the relief of vasomotor instability (hot flashes) and of atrophy of the urogenital epithelium and skin. Estrogen therapy ameliorates these symptoms in most cases. When estrogen therapy is intended to treat hot flashes alone, it should be continued for only a few years, since hot flashes tend to diminish after 3 to 4 years in untreated women. Selective [ER](#) modulators have no effect on hot flashes.

Several lines of evidence indicate that routine estrogen therapy is beneficial in preventing the complications of menopausal osteoporosis, especially in high-risk women (i.e., thin white women; [Chap. 342](#)). First, in women undergoing premature menopause, the incidence and complication rates of osteoporosis are increased, and long-term estrogen replacement appears to be beneficial. Second, estrogen therapy has short-term positive effects on calcium balance and long-term beneficial effects on bone density. Third, in women given estrogen therapy, the incidence of fractures is decreased.

Of the potential side effects, the possibility of an increased risk of endometrial carcinoma is perhaps most worrisome. The relative risk of developing endometrial adenocarcinoma in estrogen users is between six and eight times the risk in nonusers. This risk increases with increasing duration and dosage of estrogen but is largely negated in women given combination estrogen-progestogen therapy. Despite the large body of evidence linking endometrial carcinoma and estrogen use, the increased incidence primarily involves low-grade malignancies that may be difficult to distinguish histologically from hyperplasia. These forms of malignancy have little effect on life expectancy.

Apprehension concerning worsening of hypertension and thromboembolic disease appears to be due to reports of the effects of estrogen-progestogen oral contraceptives during the reproductive years and not to estrogen use in postmenopausal women. There is no conclusive evidence that low-dose estrogen therapy after menopause

increases the incidence or the severity of breast cancer or hypertension, but the risk for venous thromboembolism is slightly increased. Low-dose estrogen treatment after menopause does not appear to influence the development of atherosclerosis, myocardial infarction, or stroke. Strong evidence suggests that, in fact, estrogens may decrease the incidence of death from myocardial infarction. There is a slightly increased risk for the development of gallbladder disease with postmenopausal estrogen use.

A reasonable approach to the postmenopausal use of estrogens is as follows: (1) For long-term use, estrogens should be given in the minimal effective doses (0.625 mg conjugated estrogen orally or 1.0 mg micronized estradiol or transdermal estradiol 0.05 to 1.0 mg in a formulation that is changed every 3.5 days or every week). For women with an intact uterus, it is the practice in some clinics to give estrogens alone for 15 days and estrogen plus a daily progestogen dose for the remainder of the month. The most common regimens involve continuous estrogen plus low-dose continuous progestogen. (2) Such replacement therapy is indicated routinely in women undergoing premature menopause (surgically induced or spontaneous). (3) Estrogen therapy is also indicated routinely in women of all ages who have severe hot flashes or symptomatic atrophy of the urogenital epithelium. Hot flashes rarely persist for longer than 7 years, so, if therapy is given for this purpose, its duration can be limited. (4) In women who have had a hysterectomy, the potential benefits of treatment appear to outweigh the dangers, and in such women cyclic or continuous estrogen without progestogen is recommended. Whether estrogens should be given routinely to all women with intact uteri is an unsettled question, but the authors prescribe it in the absence of contraindications in hopes of ameliorating osteoporosis and reducing the risk of cardiovascular disease. (5) Raloxifene is given as a 60-mg tablet daily when the goal is to provide protection against bone loss without incurring additional risk of estrogen-dependent breast cancer. Each woman receiving estrogens must be monitored indefinitely at yearly intervals.

Induction of Ovulation See [Chap. 54](#)

OTHER DISORDERS OF THE FEMALE REPRODUCTIVE TRACT

VULVA

Most disorders of the vulva are due to venereal disease, most commonly syphilis (painless chancre), condylomata acuminata (venereal warts), and herpes vulvitis (painful ulcers; [Chap. 132](#)). All other lesions of the vulva, particularly in older women, must be biopsied. Early biopsy of cancer of the vulva is mandatory, because when it becomes symptomatic (pruritus and bleeding), it has often progressed to an advanced stage.

VAGINA

Infections of the vagina usually present as vaginal discharge and pruritus. The most frequent organisms are *Trichomonas*, *Candida albicans*, and *bacterial vaginalis* ([Chap. 132](#)). The diagnosis is made by microscopic examination of the discharge, and appropriate therapy can be instituted using vaginal or oral antibiotics.

Abnormalities of the vagina and cervix in female offspring of women given

diethylstilbestrol during pregnancy include adenosis of the vagina and structural abnormalities of the vagina, cervix, and uterus; the risk of developing a rare vaginal cancer (adenocarcinoma, clear cell type) is increased (2 per 10,000 exposed women). Periodic examination of women at risk should begin at age 12 to 14, and reexamination should be done after any episode of abnormal bleeding.

CERVIX

Preinvasive lesions of the cervix (also known as *cervical intraepithelial neoplasia*) and invasive carcinoma of the cervix can be detected reliably by obtaining a Papanicolaou (Pap) smear.

Evaluation of the Pap Smear The incidence of invasive cervical cancer has declined as a result of Pap smear screening. In the United States, approximately 2 to 3 million abnormal Pap smears are found each year. Most represent low-grade lesions but require appropriate follow-up. The follow-up of abnormal Pap smears requires an understanding of the Bethesda system for evaluating such smears (see below) and of the limitations of cytologic screening systems. Further evaluation may require repeat cytologic examination, colposcopy, or both.

Current Screening Recommendations Risk factors for cervical neoplasia include a history of multiple sexual partners, coitus beginning at an early age, a history of infection with human papilloma virus (HPV), infection with HIV or another immunosuppressed state, and a history of cancer of the lower genital tract. Cervical cancer screening is recommended annually beginning at 18 years of age or when the woman becomes sexually active, if earlier than age 18. "Less frequent" screening is performed when three consecutive, negative, satisfactory annual Pap smears have been obtained or if the woman is in a low-risk category. There is no upper age limit for screening, because the prevalence of invasive cancer shows a linear increase with age, most of these cancers being diagnosed after age 50. Even after hysterectomy, annual screening should be performed if there is a history of abnormal Pap smears or other lower genital tract neoplasia.

The Bethesda System of Cytologic Examination Pap smears are evaluated in regard to the adequacy of the specimen (satisfactory for evaluation, satisfactory but limited, or unsatisfactory for evaluation because of a stated reason), the general diagnosis (normal or abnormal), and a descriptive diagnosis if the smear is abnormal. The descriptive diagnoses include benign cellular changes, reactive cellular changes, and epithelial cell abnormalities, the latter including (1) atypical squamous cells of undetermined significance (ASCUS); (2) low-grade squamous intraepithelial lesion (LSIL), which is further categorized to include HPV infection, cervical intraepithelial neoplasia (CIN 1), and high-grade squamous intraepithelial lesion (HSIL, which is itself subdivided into CIN 2 and CIN 3); and (3) squamous cell carcinoma.

Guidelines for the Management of Women with Abnormal Pap Smears For ASCUS smears that are unqualified or suggest a reactive process, a repeat smear should be obtained every 4 to 6 months for 2 years until three consecutive negative smears have been obtained. For ASCUS smears that are unqualified but have severe inflammation, any specific cause should be treated, and the smear should be repeated in 2 to 3

months; because invasive carcinoma can be obscured by severe inflammation, clinical evaluation is mandatory. For postmenopausal women not using hormone replacement, a course of topical estrogen should be given before the test is repeated. For LSIL smears, the Pap test is repeated every 4 to 6 months for 2 years until three consecutive negative smears have been obtained; treatment of [HPV](#) is of no established benefit, and there is a high rate of regression of LSIL, so that in compliant, low-risk individuals, the outcome is usually favorable. If LSIL is persistent, colposcopy with directed biopsy is performed, and endocervical curettage is undertaken if a specific diagnosis is made by biopsy. Cervical cone biopsy or loop electrosurgical excision procedures are performed for higher-grade lesions such as HSIL. If cervical cancer is diagnosed by biopsy, clinical staging is performed, and the patient is treated with radiation therapy or surgery.

UTERUS

Only 40% of cases of endometrial adenocarcinoma are detected by Pap smear. In women at high risk for endometrial carcinoma (because of obesity, a history of chronic anovulatory cycles, diabetes mellitus, hypertension, or estrogen treatment), yearly endometrial sampling should be performed. Measurement of endometrial thickness by sonography can indicate which patients are at risk for endometrial pathology. Endometrial thickness <5 mm is rarely associated with either hyperplasia or cancer. Low-dose oral estrogen therapy rarely causes breakthrough or withdrawal bleeding in postmenopausal women. Therefore, irrespective of whether the patient is using estrogen therapy, the occurrence of postmenopausal bleeding makes it mandatory to obtain a tissue diagnosis by either endometrial sampling or curettage to exclude endometrial cancer.

One of the most common disorders of the uterus and the most frequent tumor of women (one of four women affected) is the uterine leiomyoma, or fibroid tumor. Three-fourths of women with leiomyoma are asymptomatic, and the diagnosis is made on routine pelvic examination. When the tumor is associated with excessive menstrual blood loss, is large or fast-growing, or causes significant pelvic pain ([Chap. 52](#)), the preferred treatment is hysterectomy if there is no desire for further childbearing. In young women, myomectomy is sometimes indicated when infertility or repeated fetal wastage is a manifestation or where future childbearing is desired.

FALLOPIAN TUBES AND OVARIES

Infectious pelvic inflammatory disease is a common disorder of the fallopian tubes and usually becomes symptomatic after a menstrual period; the symptoms include fever, chills, abdominal pain, and vaginal discharge, and pelvic tenderness on physical examination is common. The initiating organism most often is *Chlamydia trachomatis* or *Neisseria gonorrhoeae*, but tuboovarian abscess and sterility are probably caused by mixed aerobic and anaerobic superinfections and require wide-spectrum antibiotic treatment ([Chap. 133](#)).

Endometriosis is a benign disorder characterized by the presence and proliferation of endometrial tissue (stroma and glands) outside the endometrial cavity. The clinical manifestations are variable. Endometriosis occurs most commonly between the ages of 30 and 40 and is found incidentally at the time of surgery in approximately one-fifth of all

gynecologic operations. The fertility rate is reduced in affected women. The disorder usually involves the posterior cul-de-sac or the ovaries and can give rise to ovarian enlargement (endometriomas), although it may involve distant sites (lung, umbilicus). The major symptom is pelvic pain, characteristically dysmenorrhea ([Chap. 52](#)). However, the frequency and severity of symptoms correlate poorly with the extent of disease. Other manifestations include dyspareunia, pain with defecation, and infertility. The characteristic physical findings are multiple tender nodules palpable along the uterosacral ligament at the time of rectal-vaginal examination, a posteriorly fixed uterus, or enlarged, cystic ovaries. The diagnosis can only be confirmed by direct visualization, usually at diagnostic laparoscopy. Treatment depends on the degree of involvement and the desires of the patient and includes observation for mild disease with no associated infertility or pain, hormonal suppressive therapy, conservative surgery by laparoscopy or laparotomy if fertility is desired, or removal of the uterus, tubes, and ovaries in severe disease. Endometriosis is rare after the menopause.

Any adnexal mass that persists for more than 6 weeks or is larger than 6 cm must be evaluated. Although ovarian cysts and neoplasms are the most common pelvic adnexal masses (see above), tumors of the fallopian tubes, uterus, gastrointestinal tract, or urinary tract should also be considered. Sonography or radiographic evaluation is often helpful in identifying the nature of the adnexal mass prior to surgical exploration.

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337. ENDOCRINE DISORDERS OF THE BREAST - Jean D. Wilson

The breasts are the site of fatal and preventable cancer in women and provide clues to underlying systemic illness in both men and women. Consequently, examination of the breasts is an important part of the physical examination. It is the duty of every physician to distinguish the abnormal from the normal at the earliest possible stage and to seek referral if there is any doubt. **For discussion of cancer of the breast, see [Chap. 89](#).*

ENDOCRINE CONTROL OF THE BREAST

There is no histologic or functional difference in the breasts of prepubertal boys and girls, but a profound sexual dimorphism in breast development ensues at the time of puberty. The endocrine control of female breast development is illustrated in [Fig. 337-1](#). Growth of the female breast at puberty is mediated primarily by estradiol, which induces the enlargement, division, and elongation of the tubular duct system and maturation of the nipples. Administration of estrogen to men is equally effective in this regard. To produce true alveolar development at the ends of the ducts, however, the synergistic action of progesterone is required. Within the gland a variety of mediators influence epithelial cell division and differentiation, including stimulatory factors such as the insulin-like growth factors, transforming growth factor α , and epidermal growth factor and inhibitory factors such as transforming growth factor β . Once the anatomic development of the ducts and alveoli is complete, the continued action of estrogen and progesterone is not required for lactation itself.

The endocrine control of milk formation is complex, requiring, in addition to appropriate priming by estrogen and progesterone, lactogenic hormones and the permissive action of glucocorticoid, insulin, thyroxine, and, in some species, growth hormone. There are two lactogenic hormones: (1) human placental lactogen (hPL, or chorionic somatomammotropin) and (2) prolactin. hPL is secreted in large amounts by the placenta during the latter part of gestation and prepares the breast for milk production. It disappears from the maternal (and fetal) circulation shortly after termination of pregnancy. The secretion of pituitary prolactin ([Chap. 328](#)) rises during pregnancy and plays the critical role in the initiation and maintenance of lactation in the puerperium. During late pregnancy and lactation, 60 to 80% of the anterior pituitary may consist of prolactin-secreting lactotrope cells, reflecting the stimulatory effects of estrogen on these cells.

Unlike most pituitary hormones, prolactin secretion is controlled predominantly by tonic inhibition. Under basal conditions inhibitory hypothalamic hormones, the most important being dopamine, are delivered from the central nervous system to the pituitary via the hypothalamic portal system and inhibit the release of prolactin into the blood ([Chap. 328](#)). Most factors that influence prolactin secretion do so by affecting the synthesis or release of dopamine. Basal prolactin levels in the mother fall after delivery, but prolactin secretion is enhanced by stimulation of the breasts, such as the act of nursing (the so-called sucking reflex), a phenomenon that is probably mediated by the reflex release of oxytocin, which acts as a prolactin-releasing factor. Prolactin binds to specific receptors on the cell surface of the breast acinar cells and activates the JAK-STAT signal transduction cascade to stimulate the synthesis of casein, whey acidic protein, and other milk constituents. In the postgestational state, the normal lactating woman

forms about a liter of milk per day containing 38 g fat, 70 g lactose, and 12 g protein. Lactation can be suppressed by the administration of estrogens or diethylstilbestrol, which inhibit milk production by direct effects on the breast, or dopamine agonists such as bromocriptine, which inhibit prolactin secretion by the pituitary. Alternatively, if a woman does not nurse or use breast pumps postpartum, lactation usually ceases in 1 to 2 weeks.

GALACTORRHEA

Galactorrhea refers to the nonpuerperal discharge of milk-containing fluid from the breast. The definition of exactly what constitutes galactorrhea is not always clearly defined in the literature. According to the studies of Friedman and Goldfein, breast secretions are absent in normal, regularly menstruating nulligravid women. However, breast secretions can be demonstrated in a fourth of normal women who have been pregnant in the past; thus breast secretions in small amounts may be of no clinical significance in these instances. Spontaneous leakage of milk from the breasts is usually of more significance than milk that must be expressed. When the secretion is milky or white, it is safe to assume that it contains fat, casein, and lactose and is in fact milk; the concentration of milk constituents may increase after repeated sampling. When the secretion is brown or greenish, it rarely contains normal milk constituents and consequently may not result from an underlying endocrinopathy. Bloody discharges may be due to neoplasms of the breast. With these issues in mind, galactorrhea can be defined as inappropriate production of milk that is persistent or worrisome to the patient, recognizing that in some instances no underlying pathology may be demonstrated.

Since the action of a lactogenic hormone is necessary for the initiation of milk production, it is logical to consider galactorrhea as a consequence of deranged prolactin physiology. However, as indicated above, a complex hormonal milieu is necessary for lactation. Milk production does not take place in many instances in which prolactin is elevated, both in men and in women who have not been exposed to the necessary hormonal environment. As a consequence, hyperprolactinemia is more common than galactorrhea. Furthermore, while enhanced prolactin secretion is necessary for the initiation of lactation, continued production can be maintained in the presence of minimally or intermittently elevated prolactin levels so that basal plasma prolactin levels are not always elevated in patients with galactorrhea. In some such women prolactin levels may be elevated during sleep or with stimulation of the nipple; in others, hyperprolactinemia may have been present transiently. Perhaps the strongest evidence for a critical role for prolactin in galactorrhea is the fact that administration of dopaminergic agents that suppress plasma prolactin levels corrects galactorrhea even when the basal plasma prolactin levels are normal.

Differential Diagnosis Galactorrhea can be due to failure of the normal hypothalamic inhibition of prolactin release, to increased prolactin-releasing factor(s), or to autonomous prolactin secretion by tumors ([Table 337-1](#)). Pituitary stalk section, whether traumatic or secondary to the mass effects of sellar tumors, results in increases in prolactin secretion due to interruption in the delivery of dopamine to the pituitary. Likewise, many drugs that influence the central nervous system (CNS) (including virtually all psychotropic agents, methyldopa, reserpine, and antiemetics) enhance prolactin release, presumably by inhibiting synthesis, release, or action of dopamine.

Estrogens increase prolactin secretion, but estrogen withdrawal (as in the discontinuation of oral contraceptives) may also trigger the onset of galactorrhea. CNS diseases outside the pituitary can cause galactorrhea presumably by interfering with the production or delivery of dopamine to the pituitary (CNS sarcoidosis, craniopharyngioma, pinealoma, encephalitis, meningitis, hydrocephalus, hypothalamic tumors).

In primary hypothyroidism, galactorrhea results from the enhanced production of thyrotropin-releasing hormone (TRH), which also stimulates prolactin release; thyroid hormone replacement corrects the galactorrhea. A similar mechanism, involving enhanced secretion of oxytocin, may cause the galactorrhea that follows breast surgery or breast trauma.

Enhanced prolactin release can also occur from pituitary or nonpituitary tumors. Three types of pituitary tumors ([Chap. 328](#)) can cause galactorrhea: (1) pure prolactin-secreting micro- or macroadenomas, (2) mixed tumors that secrete both growth hormone and prolactin and cause acromegaly with galactorrhea, and (3) large null cell adenomas. The latter may interfere with the delivery of dopamine to the pituitary, either by mass effects on the hypothalamus or by compressing the pituitary stalk. Excess growth hormone secretion, in the absence of hyperprolactinemia, on occasion causes galactorrhea. Rarely, prolactin is secreted by bronchogenic carcinomas, and hydatidiform moles and choriocarcinomas may secrete placental lactogen.

In series involving several hundred patients with galactorrhea, a pituitary tumor was identified in about one-fourth, other known causes were identified in another fourth or fifth, and the remaining half fell into the idiopathic category. Many of the latter group ultimately developed prolactin-secreting pituitary tumors, some probably had subtle disorders of hypothalamic function, and in others a drug-related cause may have been missed. The fact remains that no satisfactory diagnosis is reached in many patients. When menses are normal, the likelihood of establishing a cause for galactorrhea is poor.

Galactorrhea is unusual in men, even in the presence of profound elevations of plasma prolactin; when it does occur, it is usually upon the background of a feminizing state (see below).

Diagnostic Evaluation If hyperprolactinemia is present, the evaluation is fundamentally that of a pituitary tumor once drug causes and hypothyroidism are excluded ([Chap. 328](#)). Even when a specific cause cannot be identified and the diagnosis of idiopathic galactorrhea is made by exclusion, it is necessary to remember that pituitary tumors may subsequently become manifest. The higher the prolactin values and the more persistent the galactorrhea, the greater is the likelihood of such a development.

TREATMENT

Breast binders can be effective in patients with mild galactorrhea of unknown etiology, presumably by preventing stimulation of the nipple and the consequent perpetuation of lactation. The aim of treatment in other instances is to correct the elevated prolactin

level, and treatment of a pituitary tumor, cessation of causative drugs, or correction of hypothyroidism is often followed by the disappearance of galactorrhea. Dopamine agonists that suppress plasma prolactin have been used to treat idiopathic hyperprolactinemia, prolactin-secreting tumors of the pituitary ([Chap. 328](#)), and even normoprolactinemic galactorrhea. These drugs suppress lactation and may cause resumption of menstrual cycles (and even fertility) in women with amenorrhea and galactorrhea.

GYNECOMASTIA

Enlargement of the male breast, or *gynecomastia*, can be a normal physiologic phenomenon at certain times of life or the result of several pathologic states ([Table 337-2](#)). A central issue in the evaluation of breast tissue in adult men is the separation of the normal from the abnormal, as gynecomastia can be an important indicator of underlying disease. It is sometimes difficult to distinguish true breast tissue enlargement from adipose tissue (*lipomastia*). True glandular tissue is often palpable, especially around the areolae, as it is firmer and contains cordlike features that are distinct from the texture of adipose tissue. In difficult cases, true gynecomastia can be identified by mammography or ultrasonography. In this discussion, we shall assume that any palpable breast tissue in men (except for the three physiologic states see below) can be due to an underlying endocrinopathy and deserves, at a minimum, a limited evaluation.

Early gynecomastia is characterized by proliferation in the breast of both the fibroblastic stroma and the duct system, which elongates, buds, and duplicates. As gynecomastia persists, progressive fibrosis and hyalinization are associated with regression of epithelial proliferation and, eventually, a decrease in the number of ducts. When the cause of the gynecomastia is corrected early in the course, resolution occurs by reduction in size and epithelial content with gradual disappearance of the ducts, leaving hyaline bands that eventually disappear.

Growth of the breast in men, as in women, is mediated by estrogen and results from an decrease in the ratio of active androgen to estrogen. As described in [Chap. 335](#), estradiol formation in normal men occurs principally by the conversion of circulating androgen to estrogen in extraglandular tissues; the normal ratio of the two hormones in plasma is about 300:1. Growth of the breast ensues in men when the normal ratio decreases as the result of diminished testosterone production or action, enhanced estrogen formation, or both processes occurring simultaneously.

Physiologic Gynecomastia In the *newborn* transient enlargement of the breast is due to the action of maternal and/or placental estrogens. The enlargement usually disappears in a few weeks but may persist longer. *Adolescent* gynecomastia is common at some time during puberty. The median age of onset is 14; it is often asymmetric or transiently unilateral, frequently tender, and it regresses so that by age 20 only a small number of men have palpable vestiges of glandular tissue in one or both breasts. Although the origin of the excess estrogen has not been identified, the onset of gynecomastia correlates with the increase in adrenal androgens at adrenarche. In addition, the luteinizing hormone (LH) stimulation of androgen synthesis by the Leydig cell in early puberty may be associated with transient elevations of plasma estradiol so that the ratio of potent androgen to estrogen is low prior to the completion of puberty.

Gynecomastia of aging also occurs in 40% or more of otherwise healthy elderly men. A likely explanation is the increase with age in the conversion of androgens to estrogens in extraglandular tissues. Abnormal liver function or drug therapy may be contributing causes in such men.

Pathologic Gynecomastia Pathologic gynecomastia can result from one of three basic mechanisms: deficiency in testosterone production or action (with or without a secondary increase in estrogen production), increase in estrogen production, or drugs ([Table 337-2](#)). Most of the disorders that cause primary and secondary testicular failure are discussed in [Chap. 335](#). The fact that deficient testosterone production can cause gynecomastia is illustrated by the syndrome of congenital anorchia in which normal (or slightly low) estradiol levels, in the presence of profoundly decreased testosterone production, results in florid gynecomastia. Decreased testosterone production is also responsible for gynecomastia in some men with Klinefelter syndrome or testicular failure of other causes. In the disorders of androgen resistance, such as testicular feminization, deficient androgen action and increased testicular estrogen production are both present.

A primary increase in estrogen production can result from a variety of causes. Increased secretion of testicular estrogen may result from elevations in plasma gonadotropins, as in cases of aberrant production of chorionic gonadotropin by testicular tumors or by bronchogenic carcinomas, from the ovarian elements in the gonads of men with true hermaphroditism, or as the result of formation by testicular tumors (particularly Leydig and Sertoli cell tumors). Increased conversion of androgen to estrogens in extraglandular tissues can be due either to increased availability of substrate (androstenedione) for extraglandular estrogen formation (congenital adrenal hyperplasia, hyperthyroidism, and most feminizing adrenal tumors), or to diminished catabolism of androstenedione (liver disease) so that estrogen precursors are shunted to aromatase in peripheral sites. Extraglandular aromatase can be increased in tumors of the liver or adrenal gland and rarely as a result of an inherited disorder manifested by gynecomastia in affected males and macromastia in females.

Drugs can cause gynecomastia by several mechanisms. Many drugs either act directly as estrogens or cause an increase in plasma estrogen activity (e.g., in men receiving diethylstilbestrol for prostatic carcinoma and in transsexuals in preparation for sex-change operations). Boys and young men are particularly sensitive to estrogen and can develop gynecomastia after the use of dermal ointments containing estrogen or after the ingestion of milk or meat from estrogen-treated animals. The gynecomastia of digitalis ingestion is usually attributed to an estrogen-like side effect of the drug, but it occurs most commonly in men with abnormal liver function. A second mechanism of drug-induced gynecomastia is illustrated by clomiphene and human chorionic gonadotropin (hCG), which cause enhanced testicular secretion of estrogen. Other drugs cause gynecomastia by interfering with testosterone synthesis (ketoconazole and alkylating agents) and/or testosterone action, for instance, by blocking the binding of androgen to its receptor protein in target tissues (spironolactone and cimetidine). Finally, drugs that cause gynecomastia by mechanisms that have not been defined include busulfan, ethionamide, isoniazid, methyldopa, tricyclic antidepressants, penicillamine, omeprazole, calcium channel blocking agents, angiotensin-converting enzyme inhibitors, metoclopramide, antiretroviral agents, diazepam, marijuana, and heroin. In some instances, the feminization is due to effects of drugs on liver function.

Treatment with growth hormone can cause gynecomastia even in prepubertal boys, suggesting that growth hormone itself or one of the insulin-like growth factors has a direct effect on the breast.

Diagnostic Evaluation The evaluation of patients with gynecomastia should include: (1) a careful drug history; (2) measurement and examination of the testes (if both are small, a chromosomal karyotype should be obtained; if the testes are asymmetric, a workup for testicular tumor should be instituted); (3) evaluation of liver function; and (4) endocrine workup to include measurement of serum androstenedione or 24-h urinary 17-ketosteroids (usually elevated in feminizing adrenal states), measurement of plasma estradiol and [hCG](#) (helpful if elevated but usually normal), and measurement of plasma [LH](#) and testosterone. If LH is high and testosterone is low, the diagnosis is usually testicular failure; if LH and testosterone are both low, the diagnosis is most likely increased primary estrogen production (e.g., a Sertoli cell tumor of the testis), provided hypogonadotropic hypogonadism has been excluded; and if both LH and testosterone are elevated, the diagnosis is either an androgen-resistance state or a gonadotropin-secreting tumor.

A satisfactory diagnosis can be made in only half or fewer of patients referred for gynecomastia. This implies either that the diagnostic techniques are not sufficiently refined to recognize mild disturbances, that many causes of gynecomastia are as yet undefined, that the causes may be transient and difficult to diagnose, or that gynecomastia may in some instances be normal rather than due to a pathologic state. Because of the problem of separating the normal from the abnormal, gynecomastia should probably be worked up routinely only if the drug history is negative, the breast is tender (indicating rapid growth), or the breast mass is >4 cm in diameter. However, the decision to perform an endocrine evaluation depends on the clinical context. For example, gynecomastia associated with signs of underandrogenization should always be evaluated. A firm or hard breast mass should raise suspicion of male breast cancer ([Chap. 89](#)).

TREATMENT

When the primary cause can be identified and corrected, the breast enlargement usually subsides promptly and eventually disappears. For example, androgen replacement therapy may produce dramatic improvement in men with testicular insufficiency. However, if the gynecomastia is of long duration (and fibrosis has replaced the original ductal hyperplasia), correction of the primary defect may not be followed by resolution. In such instances and when the primary cause cannot be corrected, surgery is the only effective therapy. Indications for surgery include several psychological and/or cosmetic problems, continued growth or tenderness, or suspected malignancy. Although the relative risk of carcinoma of the breast is increased in men with gynecomastia, it is rare nevertheless. Prophylactic radiation of the breasts prior to the institution of diethylstilbestrol or estrogen therapy is effective in preventing gynecomastia and has a low complication rate. In patients who have painful gynecomastia and who are not candidates for other therapy, treatment with antiestrogens such as tamoxifen may be indicated.

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338. DISORDERS OF SEXUAL DIFFERENTIATION - Jean D. Wilson, James E. Griffin

Sexual differentiation is a sequential and ordered process. *Chromosomal sex*, established at the moment of fertilization, determines *gonadal sex*, and gonadal sex, in turn, causes the development of *phenotypic sex*, in which the male or female urogenital tract is formed ([Fig. 338-1](#)). A disturbance of any step in this process during embryogenesis may impair sexual differentiation ([Table 338-1](#)). Known causes of such disorders include environmental insults as in the ingestion of a virilizing drug during pregnancy, nonfamilial aberrations of the sex chromosomes as in 45,X gonadal dysgenesis, birth defects of multifactorial etiology as in most cases of hypospadias, and disorders due to single gene mutations as in the testicular feminization syndrome.

Limitations of knowledge make it necessary to make empirical assignments as to the type of derangement in some disorders, but specific diagnoses can usually be made as the result of genetic, endocrine, phenotypic, and chromosomal assessment. As a consequence, gender assignment in the newborn is usually appropriate, even in extreme instances of ambiguous genitalia.

NORMAL SEXUAL DIFFERENTIATION

The first event in sexual differentiation is the establishment of chromosomal sex, the heterogametic sex (XY) being male and the homogametic sex (XX) female. The embryos of both sexes then develop in an identical fashion until approximately 40 days of gestation.

The second phase of sexual differentiation is the conversion of the indifferent gonad into a testis or an ovary. No matter how many X chromosomes are present (as in 47,XXY, 48,XXXY, etc.), a testis will develop as long as a normal Y chromosome is present. Differentiation of the indifferent gonad into a testis is initiated by the actions of a single gene on the short arm of the Y chromosome (*SRY*); expression of an *SRY* transgene into female mice causes them to develop as males. The gene encodes a DNA-binding protein, but the mechanism by which *SRY* promotes testicular development remains poorly defined. At least four additional genes are also necessary for normal testicular development: (1) the Wilms' tumor-related gene (*WT1*), (2) steroidogenic factor 1 (*SF1*), (3) *SRY*-related HMG-box 9 (*SOX9*), and (4) dosage-sensitive sex reversal-adrenal hypoplasia congenita critical region on the X chromosome gene 1 (*DAX1*). These genes each encode putative transcription factors that regulate the expression of genes necessary for gonadal survival; mutations in *SRY* or any of the four downstream genes impair testicular development. It remains unclear if there are analogous "ovarian-determining genes," or if ovarian development is a default pathway in the absence of testicular determination genes. Mutations in some of the genes noted above (e.g., *SF1*) also prevent normal ovarian development.

The final process in sexual differentiation, the translation of gonadal sex into phenotypic sex (formation of the male or female urogenital tracts), is the consequence of the type of gonad formed and the endocrine secretions of the fetal gonads. The internal urogenital tract is derived from the wolffian and mullerian ducts that exist side by side in early embryos of both sexes ([Fig. 338-1A](#)). In the male the wolffian ducts give rise to the

epididymides, vasa deferentia, and seminal vesicles, and the mullerian ducts disappear. In the female the mullerian ducts are converted into the fallopian tubes, uterus, and upper vagina, and the wolffian ducts regress. The external genitalia and urethra in the two sexes develop from common anlage -- the urogenital sinus and the genital tubercle, folds, and swellings ([Fig. 338-1B](#)). The urogenital sinus gives rise to the prostate and prostatic urethra in the male and to the urethra and lower portion of the vagina in the female. The genital tubercle gives rise to the glans penis in the male and the clitoris in the female. The urogenital swellings become the scrotum or the labia majora, and the urethral folds form the labia minora or fuse to form the shaft of the penis and the male urethra.

In the absence of the testes, as in the normal female or in the male embryo castrated prior to the onset of gonadal differentiation, phenotypic sex develops along female lines. Thus, masculinization of the fetus is induced by hormones from the fetal testes, whereas female development does not require the presence of the ovary. Phenotypic sex normally conforms to chromosomal sex. That is, chromosomal sex determines gonadal sex, and gonadal sex, in turn, controls phenotypic sex.

Formation of the male phenotype is vested in the action of three hormones. Two -- antimullerian hormone (AMH) and testosterone -- are secreted by the fetal testis. AMH [also termed *mullerian-inhibiting substance* (MIS)] is a protein that suppresses the mullerian ducts and prevents development of the uterus and fallopian tubes in the male. Testosterone acts directly to virilize the wolffian duct and is the precursor for the third embryonic male hormone, dihydrotestosterone ([Chap. 335](#)), which promotes development of the male urethra and prostate and formation of the penis and scrotum. Testosterone and dihydrotestosterone induce formation of the male urogenital tract during fetal life by acting through the same nuclear androgen receptor by which they act in postembryonic life ([Chap. 335](#)).

The secretion of testosterone by the fetal testes approaches a maximum by the tenth week of gestation, and formation of the sexual phenotypes is largely completed by the end of the first trimester. During the latter phases of gestation, the ovarian follicles develop and the vagina matures in the female, and testicular descent and phallic growth take place in the male.

DISORDERS OF CHROMOSOMAL SEX

Disorders of chromosomal sex ([Table 338-2](#)) occur when the number or structure of the X or Y chromosomes is abnormal ([Chap. 66](#)).

KLINEFELTER SYNDROME

Clinical Features Klinefelter syndrome is characterized by small, firm testes, azoospermia, gynecomastia, and elevated levels of plasma gonadotropins in men with two or more X chromosomes. The common karyotype is either a 47,XXY chromosomal pattern (the classic form) or 46,XY/47,XXY mosaicism. It is the most frequent major abnormality of sexual differentiation, the incidence being around 1 in 500 men.

Prepubertally, the testes are small but otherwise appear normal. After puberty, the

disorder is manifest as infertility, gynecomastia, or occasionally underandrogenization ([Table 338-3](#)). Hyalinization of the seminiferous tubules and azoospermia are consistent features of the 47,XXY variety. The small, firm testes are usually <2 cm and always <3.5 cm in length (corresponding to 2- and 12-mL volume, respectively). Longer legs cause increased mean height. Gynecomastia is common and ordinarily develops during adolescence, is generally bilateral and painless, and may become disfiguring ([Chap. 337](#)). Obesity and varicose veins occur in one-third to one-half, and leg ulcers are associated with deficiency of plasminogen activator inhibitor-1. The diagnosis is made most frequently in boys with developmental delay and/or learning disabilities and social maladjustment; abnormalities of thyroid function, diabetes mellitus, and pulmonary disease are also common. The risk of breast cancer is 20 times that of normal men (but only about a fifth that in women). Most have male psychosexual orientation and function sexually as normal men.

About 10% of the patients have the mosaic form, as estimated by chromosomal karyotypes on peripheral blood leukocytes. The frequency of this variant may be underestimated, since chromosomal mosaicism can be present in the testes when the peripheral leukocyte karyotype is normal. The mosaic form is usually not as severe as the 47,XXY variety, and the testes may be normal ([Table 338-3](#)). The endocrine abnormalities are less severe, and gynecomastia and azoospermia are less common, and occasional mosaic individuals are fertile. In some the diagnosis may not be suspected because the manifestations are so mild.

Approximately 30 additional variants of Klinefelter syndrome have been described, including those with uniform cell lines (such as 48,XXYY, 48,XXXY, and 49,XXXXY) and a variety of mosaicisms of the X chromosome with or without associated structural abnormalities of the X. In general, the greater the chromosomal abnormality the more severe the manifestations.

Pathophysiology The classic form is due to meiotic nondisjunction of the chromosomes during gametogenesis ([Fig. 338-2](#)). About 40% of the responsible meiotic nondisjunctions occur during spermatogenesis, and 60% occur during oogenesis. Advanced maternal age is a predisposing factor. The mosaic form is thought to result from chromosomal mitotic nondisjunction after fertilization of the zygote and can take place in either a 46,XY zygote ([Fig. 338-2](#)) or a 47,XXY zygote. The latter situation, double nondisjunction (meiotic and mitotic), may be the usual cause and thus explain why the mosaic form is less frequent than the classic disorder.

Levels of plasma follicle stimulating hormone (FSH) and luteinizing hormone (LH) are usually high; FSH shows the best discrimination because of the consistent damage to the seminiferous tubules. The plasma testosterone level averages half normal but may overlap the normal range. Mean plasma estradiol levels are elevated; early, estradiol secretion by the testes may increase in response to the elevated plasma LH level, but the testicular secretion of estradiol and testosterone eventually declines. Elevated plasma estradiol late in the course is probably due both to decreased metabolic clearance and increased conversion of testosterone to estradiol in extragonadal tissues. The net result both early and late is a variable degree of feminization and virilization. Feminization, including gynecomastia, depends on the ratio of circulating estrogen to androgen (relative or absolute), and individuals with low plasma testosterone and high

plasma estradiol levels are more likely to develop gynecomastia ([Chap. 337](#)). Men with untreated Klinefelter syndrome may have enlarged pituitary glands, presumably due to hyperplasia of the gonadotrophs due to inadequate testosterone feedback.

TREATMENT

In men with some sperm production, or in whom spermatids can be recovered from testicular biopsy, fertility is possible with in vitro fertilization ([Chap. 335](#)). Gynecomastia should be treated surgically. Some underandrogenized men benefit from supplemental androgen, particularly in those with decreased bone density, but such treatment may worsen the gynecomastia, presumably by providing increased substrate for estrogen formation in peripheral tissues. Testosterone should be injected in the form of testosterone cypionate or testosterone enanthate or administered via the transdermal route ([Chap. 335](#)). Following the administration of testosterone, the plasma [LH](#) level returns to normal only after several months, if at all.

XX MALE SYNDROME

A 46,XX karyotype is found in approximately 1 in 20,000 phenotypic males. The findings resemble those in Klinefelter syndrome: the testes are small and firm (generally <2 cm), gynecomastia is frequent, the penis is normal to small in size, azoospermia and hyalinization of the seminiferous tubules are usual, no female urogenital structures are present, psychosexual identification is male, mean plasma testosterone level is low, plasma estradiol level is elevated, and plasma gonadotropin levels are high. Affected individuals differ from typical Klinefelter patients only in that average height is less than in normal men, the incidence of cognitive problems is not increased, and the incidence of hypospadias is increased.

The majority of XX males have Y-related DNA (e.g., detected by polymerase chain reaction of the *SRY* gene); thus an X-Y or Y-autosome translocation appears to be the common cause. Some 46,XX males are negative for all Y-specific DNA, suggesting that their disorder is due to mutation in a downstream, autosomal or X-linked gene involved in development of the testes. The management is similar to that of Klinefelter syndrome.

GONADAL DYSGENESIS (TURNER SYNDROME)

Clinical Features Gonadal dysgenesis is characterized by primary amenorrhea, sexual infantilism, short stature, multiple congenital anomalies, and bilateral streak gonads in phenotypic women with any of several defects of the X chromosome. This condition should be distinguished from (1) mixed gonadal dysgenesis in which a unilateral testis and a contralateral streak gonad may be present; (2) pure gonadal dysgenesis in which bilateral streak gonads are associated with a normal 46,XX or 46,XY karyotype, normal stature, and primary amenorrhea; and (3) the Noonan syndrome, an autosomal dominant disorder in both sexes characterized by webbed neck, short stature, congenital heart disease, cubitus valgus, and other congenital defects despite normal karyotypes and normal gonads.

The incidence is estimated at 1 in 3000 newborn females; the prenatal incidence may be as high as 2% of all human conceptuses, only a small fraction of whom survive to

term. The diagnosis is made either at birth because of the associated anomalies or at puberty when amenorrhea and failure of sexual development are noted in conjunction with the associated anomalies. Gonadal dysgenesis is the most common cause of primary amenorrhea, accounting for a third of such patients. The external genitalia are unambiguously female but remain immature, and there is no breast development unless exogenous estrogen is given. The fallopian tubes and uterus are immature, and bilateral streak gonads are present in the broad ligaments. Primordial germ cells are present transiently during embryogenesis but disappear because of an accelerated rate of atresia ([Chap. 336](#)). After the age of expected puberty, these streaks lack identifiable follicles and ova and consist of fibrous tissue that is indistinguishable from normal ovarian stroma.

The somatic abnormalities primarily involve the skeleton and connective tissue. Lymphedema of the hands and feet, webbing of the neck, low hairline, redundant skin folds on the back of the neck, a shieldlike chest with widely spaced nipples, and growth retardation are features that suggest the diagnosis in infancy. Micrognathia, epicanthal folds, prominent low-set or deformed ears, a fishlike mouth, and ptosis may be present. Short fourth metacarpals are present in half, and 10 to 20% have coarctation of the aorta. In adults, the average height rarely exceeds 150 cm. Associated conditions include renal malformations, pigmented nevi, hypoplastic nails, tendency to keloid formation, perceptive hearing loss, unexplained hypertension, glucose intolerance, and autoimmune thyroid disease.

Pathophysiology About half have a 45,X karyotype, approximately one-fourth have mosaicism with no structural abnormality (46,XX/45,X), and the remainder have a structurally abnormal X chromosome with or without mosaicism ([Chap. 66](#)). The mechanism of chromosome loss is unknown and may occur during gametogenesis in either parent or as a mitotic error during one of the early cleavage divisions of the fertilized zygote ([Fig. 338-2](#)). Short stature and other somatic features result from haploinsufficiency of one or more genes encoded on the short arm of the X chromosome. Streak gonads result when genetic material is missing from either the long or short arm of the X. In individuals with mosaicism or structural abnormalities of the X, the phenotype on average is less severe than in the 45,X variety. In some patients with hypertrophy of the clitoris, an unidentified fragment of a chromosome is present and is assumed to be an abnormal Y; gonadoblastoma may develop in the streak gonads in this subset of patients. The Y-linked genes that predispose to gonadoblastoma are distinct from *SRY* because XY women with *SRY* deletions or mutations are also at risk for such tumors. Rarely, familial transmission of gonadal dysgenesis can be the result of a balanced X-autosome translocation ([Chap. 66](#)). Analysis of chromosomal karyotype is necessary to establish the diagnosis and to identify the group with Y chromosomal elements and hence a chance of developing malignancy in the streak gonads.

After the time of expected puberty, pubic and axillary hair remain sparse, the breasts are infantile, and no menses occur. Serum [FSH](#) is elevated in infancy, falls during midchildhood to the normal range, and increases to castrate levels at the age of 9 or 10. At this time, the serum [LH](#) level is also elevated, and plasma estradiol levels are low [<40 pmol/L (<10 pg/mL)]. Approximately 2% of 45,X and 12% of mosaic women have sufficient residual follicles to allow some menstruation, and occasionally minimally

affected women become pregnant; the reproductive life in such individuals is brief.

TREATMENT

At the anticipated time of puberty, replacement therapy with estrogen should be instituted to induce maturation of the breasts, labia, vagina, uterus, and fallopian tubes ([Chap. 336](#)). Linear growth and bone maturation rates usually double during the first year of treatment with estradiol, but the eventual height rarely approaches the predicted height. Combination therapy with oxandrolone and/or growth hormone accelerates growth and increases final height. Streak gonads should be removed in all women who are virilized or have Y-chromosome sequences.

MIXED GONADAL DYSGENESIS

Clinical Features Mosaicism for a Y-bearing cell line, usually the 45,X/46,XY karyotype, is responsible for most instances of mixed gonadal dysgenesis. Affected individuals usually have a testis on one side and a streak gonad on the other, but bilateral dysgenetic testes or bilateral streak gonads may be present. The incidence is unknown, but in most hospitals the disorder is the second most common cause of ambiguous genitalia in the neonate after congenital adrenal hyperplasia.

The phenotype varies depending on the proportion of XY cells and their distribution. About two-thirds of such children are reared as females. Many have ambiguous genitalia, including phallic enlargement, a urogenital sinus, and some labioscrotal fusion. In most the testis is intraabdominal; individuals with a testis in the inguinal or scrotal position are usually reared as males. A uterus, vagina, and at least one fallopian tube are almost invariably present.

The prepubertal testis appears relatively normal. The postpubertal testis contains abundant Leydig cells, but the seminiferous tubules lack germinal elements and contain only Sertoli cells. The streak gonad, a thin, pale, elongated structure located either in the broad ligament or along the pelvic wall, is composed of ovarian stroma. At puberty the testis secretes androgen, causing virilization and phallic enlargement. Feminization is rare; when it occurs, estrogen secretion from a gonadal tumor should be suspected.

Approximately a third of these individuals exhibit somatic features of 45,X gonadal dysgenesis. Approximately two-thirds have the 45,X/46,XY karyotype, and the remainder have a 46,XY karyotype or a variant mosaicism. The origin of 45,X/46,XY mosaicism is best explained by the loss of a Y chromosome during an early mitotic division of an XY zygote similar to the postulated loss of the X chromosome in the 46,XY/47,XXY mosaicism shown in [Fig. 338-2](#).

Pathophysiology It is assumed (but has been difficult to prove) that the 46,XY cell line stimulates testicular differentiation, whereas the 45,X stem leads to the development of the contralateral streak gonad. Both masculinization and mullerian duct regression in utero are incomplete. Since Leydig cell function may be that of a normal male at puberty, inadequate virilization in utero may be due to delayed development of a testis that is ultimately capable of normal Leydig cell function.

TREATMENT

For the older child or adult in whom gender is established prior to diagnosis, the central consideration is the possibility of tumor development in the gonads, which can occur prior to puberty. The overall incidence of seminomas and gonadoblastomas may be as high as 25%. Such tumors occur most frequently in subjects with a female phenotype who lack the somatic features of 45,X gonadal dysgenesis and are more common in testes than in the streak gonad. When the diagnosis is established in phenotypic females, prophylactic gonadectomy should be performed because gonadal tumors may occur in childhood and because the testes secrete androgen at puberty and thus cause virilization. Such women, like those with gonadal dysgenesis, are then given estrogen to induce and maintain feminization.

When the diagnosis is established in phenotypic males during late childhood or in adults, the management is more complicated. Men with mixed gonadal dysgenesis are infertile (no germinal elements are present in the testes) and have a high risk of developing gonadal tumors. In deciding which testes can be safely conserved the following observations apply: (1) tumors develop in scrotal streak gonads but not in scrotal testes, (2) tumors that develop in intraabdominal testes are always associated with ipsilateral müllerian duct structures, and (3) tumors in streak gonads are always associated with tumors in the contralateral abdominal testis. Based on these observations, it is recommended that (1) all streak gonads should be removed, (2) scrotal testes should be preserved, and (3) intraabdominal testes should be excised unless they can be relocated in the scrotum and are not associated with ipsilateral müllerian duct structures. Reconstructive surgery of the phallus should be performed when appropriate.

When the diagnosis is established in early infancy and the genitalia are ambiguous, gender assignment is usually female, and resection of the phallus and gonadectomy can be performed in infancy, sometimes in one procedure. If the decision is for male gender assignment, the same criteria apply as to which testes should be removed as in older males.

TRUE HERMAPHRODITISM

Clinical Features True hermaphroditism is a condition in which both an ovary and a testis or one or more gonads with features of both (ovotestis) are present. To justify the diagnosis, both types of gonadal epithelium must be documented histologically, the presence of ovarian stroma without oocytes not being sufficient. The incidence is unknown, but more than 400 cases have been reported. Three categories are recognized: (1) one-fifth are bilateral -- testicular and ovarian tissue (ovotestes) on each side; (2) two-fifths are unilateral -- an ovotestis on one side and an ovary or a testis on the other; and (3) the remainder are lateral -- a testis on one side and an ovary on the other.

The external genitalia exhibit all gradations of the male-to-female spectrum. Two-thirds of affected individuals are sufficiently masculinized to be reared as males, but fewer than one-tenth have normal male external genitalia; most have hypospadias and incomplete labioscrotal fusion. Two-thirds of phenotypic females have an enlarged

clitoris, and most have a urogenital sinus. Differentiation of the internal ducts usually corresponds to the adjacent gonad. Although an epididymis usually develops adjacent to a testis, the vas deferens is usually incomplete. Of the patients with an ovotestis, three-fourths have an epididymis, two-thirds have a fallopian tube, one-tenth have a vas deferens, and one-tenth have both a vas deferens and a fallopian tube. The uterus may be hypoplastic or unicornuate. The ovary is usually in the normal position, but the testis or ovotestis may be found at any level along the route of testicular descent, frequently associated with an inguinal hernia. Testicular tissue is present in the scrotum or the labioscrotal fold in one-third, in the inguinal canal in one-third, and in the abdomen in one-third.

Variable feminization and virilization ensue at puberty; three-fourths develop gynecomastia, and about half menstruate. In phenotypic men, menstruation may cause cyclic hematuria. Ovulation occurs in approximately one-fourth and is more common than spermatogenesis. In men, ovulation may cause testicular pain. Fertility has been reported in women and more rarely in men. Congenital malformations of other systems are unusual.

Pathophysiology About two-thirds of individuals have a 46,XX karyotype, a tenth have a 46,XY karyotype, and the remainder are chimeras or mosaics in whom a Y cell line is present. The mechanism responsible for the abnormal gonadal development is unknown. Only 10% of 46,XX true hermaphrodites are *SRY* positive, presumably the consequence of mosaicism or translocation of a portion of the Y chromosome; the remainder are believed to result from gain-of-function mutations in downstream genes involved in *SRY* action. On occasion, multiple sibs with a 46,XX karyotype are affected, possibly the result of an autosomal or X-linked mutation.

Because corpora lutea are frequently present in the ovaries, it is presumed that the female neuroendocrine axis functions normally in such individuals. Feminization (gynecomastia and menstruation) is the result of secretion of estradiol by ovarian tissue. In masculinized patients, secretion of androgen predominates, and some produce sperm.

TREATMENT

When the diagnosis is made in early infancy, gender assignment depends on the anatomic features. In older children and adults, gonads and internal duct structures that are contradictory to the predominant phenotype (and the gender of rearing) should be removed, and the external genitalia should be modified when appropriate. Gonadal tumors are rare but have been reported in true hermaphrodites who carry Y chromosome sequences. Consequently, the possibility of future tumor development must be taken into account when the decision is made to preserve gonadal tissue.

DISORDERS OF GONADAL SEX

Disorders of gonadal sex result when chromosomal sex is normal but differentiation of the gonads is abnormal. Thus, gonadal sex does not correspond to chromosomal sex.

PURE GONADAL DYSGENESIS

Clinical Features Pure gonadal dysgenesis is a disorder in which phenotypic females with gonads and genitalia characteristic of gonadal dysgenesis (bilateral streaks, infantile uterus and fallopian tubes, and sexual infantilism) have normal height, few if any somatic anomalies, and either a normal 46,XX or 46,XY karyotype. This disorder is about one-tenth as common as gonadal dysgenesis. It is genetically distinct but cannot be distinguished clinically from those instances of gonadal dysgenesis with minimal somatic abnormalities. The height is normal or greater than normal, some individuals being >170 cm. Estrogen levels vary from profound deficiency typical of 45,X gonadal dysgenesis to some breast development and menses that terminate in an early menopause. About 40% have some feminization. Axillary and pubic hair is scanty, and the internal genitalia consist of müllerian derivatives only. In both the 46,XX and the 46,XY forms the disorder prevents differentiation of ovary or testis, respectively; the development of the female phenotype and the elevation of gonadotropin secretion are due to failure of gonadal development.

Tumors may develop in the streak gonads, particularly dysgerminoma or gonadoblastoma in the 46,XY disorder. Such tumors may be heralded by the development of virilizing signs or a pelvic mass.

Pathophysiology Although chromosomal mosaicisms have been described under this nosology, the designation here is restricted to women with uniform 46,XX or 46,XY karyotypes. (Those with mosaicism are variants of gonadal dysgenesis or mixed gonadal dysgenesis, as described above.) The rationale for this restricted definition is based on the fact that both the XX and XY varieties can result from single gene mutations that are presumed to involve gene(s) essential for gonadal development. Several sibships have been reported in which more than one individual is affected with the 46,XX disorder, frequently the result of consanguineous matings, suggesting an autosomal recessive inheritance.

The 46,XY variety may occur in families; in some the disorder appears to be inherited as an X-linked trait, and in others the pattern suggests a male-limited autosomal recessive inheritance. About 15% of 46,XY women have either a deletion or a mutation in the *SRY* coding sequence. Other instances could be due to mutations in *SRY* outside the coding sequence, in other genes that influence *SRY* expression, or in the downstream genes that are controlled by *SRY*. Indeed, mutations in several genes that are downstream of *SRY* are now known to be a cause of the dysgenetic testes syndrome (also termed *dysgenetic male pseudohermaphroditism*; see below).

TREATMENT

The management of the estrogen deficiency is identical to that in gonadal dysgenesis; namely, appropriate estrogen replacement therapy is initiated at the time of expected puberty and maintained in adult life ([Chap. 336](#)). Because of the high frequency of gonadal tumors in the 46,XY variety, the streak gonads should be removed once the diagnosis is made; development of virilizing signs is indication for immediate surgery. The natural history of the gonadal tumors is uncertain, but the prognosis after surgical removal is usually good.

DYSGENETIC TESTES

Individuals with these disorders are genetic males with disorders of testicular development that vary from streak gonads similar to those in gonadal dysgenesis to less severe defects. The disorders are frequently associated with systemic abnormalities, because many of the genes involved are also involved in the development of other tissues. The first of these genes to be identified was the Wilms' tumor gene *WT1*; mutations of this gene cause two disorders -- the Denys-Drash and Frasier syndromes. *Denys-Drash syndrome* is an autosomal dominant disorder characterized by development of Wilms' tumors in males with a spectrum of gonadal defects ranging from streak gonads to less severely affected testes; urogenital defects include diffuse mesangial sclerosis of the kidneys. The underlying mutations in the zinc finger region of WT-1 are believed to inhibit the function of the wild-type protein and hence act as dominant negative mutations. Patients with *Frasier syndrome* have gonadal dysgenesis, impaired virilization, and focal sclerosis of the kidney but do not develop Wilms' tumors. Mutations in intron 9 of the *WT1* gene that cause Frasier syndrome interfere with the synthesis of specific splice variants of *WT1*.

A second downstream gene that is essential for differentiation of the testes is *SF1* (also known as *FTZF1*), a member of the nuclear hormone receptor superfamily. SF-1 regulates the expression of many genes involved in adrenal and gonadal development and steroidogenesis, as well as the *AMH* gene. Heterozygous mutation of this autosomal gene has been associated with 46,XY gonadal dysgenesis with adrenal insufficiency.

Another downstream gene is *SOX9*, a close relative of *SRY* that maps to chromosome 17q. This gene is expressed in high levels in the testes, where it is believed to be a key regulator of male differentiation. Heterozygous mutations of this gene cause 46,XY gonadal dysgenesis and skeletal abnormalities (*campomelic dysplasia*).

46,XY gonadal dysgenesis is also associated with duplication of the short arm of the X chromosome (Xp21), a phenomenon termed *dosage-sensitive sex reversal*. Loss-of-function mutations of the *DAX1* gene in this region of the X chromosome are associated with adrenal hypoplasia congenita and hypogonadotrophic hypogonadism. The DAX-1 protein inhibits the expression of SF-1-regulated genes, providing a potential mechanism by which an excess of DAX-1 could cause gonadal dysgenesis in genetic males with Xp21 duplications.

THE ABSENT TESTES SYNDROME (ANORCHIA, TESTICULAR REGRESSION, GONADAL AGENESIS, AGONADISM)

Clinical Features A spectrum of phenotypes occurs in 46,XY males with absent or rudimentary testes but in whom unequivocal evidence exists that endocrine function of the testis (e.g., consistent müllerian duct regression and variable testosterone synthesis) was present at some time during embryonic life. In pure gonadal dysgenesis, in contrast, no evidence can be inferred for gonadal function during embryonic development. The manifestations vary from complete failure of virilization to incomplete virilization of the external genitalia to otherwise normal men with bilateral anorchia.

The purest form is represented by 46,XY females with absent testes, sexual infantilism, and absence of both mullerian duct derivatives and accessory organs of male reproduction. Such individuals differ from those with 46,XY pure gonadal dysgenesis in that no gonadal remnant can be identified, including no streak gonad and no mullerian derivatives. Testicular failure must have occurred, therefore, between the onset of [AMH](#) synthesis and the onset of testosterone secretion (e.g., after development of Sertoli cells but before the onset of Leydig cell function).

In others, testicular failure occurred later in gestation, and these individuals may constitute problems in gender assignment. In some, failure of mullerian regression is more pronounced than failure of testosterone secretion, but none exhibit normal mullerian development. In those with more extensive virilization, the external genitalia are phenotypically male, but rudimentary oviducts and vasa deferentia may coexist internally.

At the final extreme is the syndrome of bilateral anorchia in which phenotypic men have absence of mullerian structures and gonads but male wolffian duct derivatives and external genitalia. Microphallus implies that failure of testosterone secretion occurred late in embryogenesis after anatomic development of the male urethra was complete. Gynecomastia may or may not be present.

Pathophysiology The pathogenesis is not understood. Testicular regression could be the result of mutant genes, teratogen, or trauma, and the disorder may well be heterogeneous in origin. Several instances of agonadism have occurred in the same family, some unilateral and others bilateral. Some individuals in whom no testes can be identified at laparotomy have blood testosterone values above the castrate range, presumably derived from remnant testes.

TREATMENT

The management of the two extremes is clear-cut. Sexually infantile, phenotypic females should be given adequate estrogen to ensure appropriate feminization, and any coexisting vaginal agenesis should be treated by surgical or medical means. Likewise, phenotypic males with anorchia should be given androgen replacement to allow normal male secondary sexual development. Individuals with incomplete virilization or ambiguous external genitalia demonstrate a more complex problem and require careful assessment to determine appropriate gender assignment, hormonal therapy at the time of expected puberty, and surgical correction of the external genitalia when appropriate.

DISORDERS OF PHENOTYPIC SEX

Disorders of phenotypic sex occur in 46,XX or 46,XY individuals with appropriate gonadal sex but in whom development of the urogenital tract is inappropriate for the chromosomal/gonadal sex.

FEMALE PSEUDOHERMAPHRODITISM

Female pseudohermaphroditism occurs in 46,XX women with bilateral ovaries but with variable virilization of the urogenital tract because of androgen excess during fetal life.

Congenital Adrenal Hyperplasia

Clinical Features The pathways by which glucocorticoids are synthesized in the adrenal gland and androgens are formed in the testis and adrenal are summarized in [Fig. 338-3](#). Three reactions are common to both pathways (cholesterol side chain cleavage, 3 β -hydroxysteroid dehydrogenase/isomerase, and 17 α -hydroxylase); impairment of any of these reactions results in deficiency of glucocorticoid and androgen synthesis and consequently causes both congenital adrenal hyperplasia (due to enhanced ACTH levels) and defective virilization of the male embryo (male pseudohermaphroditism). Two reactions are involved exclusively in androgen synthesis (17,20-lyase and 17 β -hydroxysteroid dehydrogenase); deficiency in either results in pure male pseudohermaphroditism with normal glucocorticoid synthesis. Deficiency of either of the terminal two enzymes of glucocorticoid synthesis (21-hydroxylase and 11 β -hydroxylase) impairs formation of hydrocortisone; the compensatory increase in ACTH secretion causes adrenal hyperplasia and a secondary increase in androgen formation that virilizes the female and induces precocious masculinization in the male.

The major features of congenital adrenal hyperplasia are listed in [Table 338-4](#). The *adrenal insufficiency* can be equally severe and life-threatening in both sexes and is described in [Chap. 331](#). Some defects in steroidogenesis cause female pseudohermaphroditism, and some cause male pseudohermaphroditism. (3 β -hydroxysteroid dehydrogenase/isomerase deficiency can cause either male or female pseudohermaphroditism, but since incomplete virilization of the male is more common, the disorder will be discussed under male pseudohermaphroditism.)

Congenital adrenal hyperplasia due to classic 21-hydroxylase deficiency is the most common cause of ambiguous genitalia in the newborn, with an incidence of between 1 in 5000 and 1 in 15,000 live births in Europe and the United States; it may or may not be associated with mineralocorticoid deficiency (salt loss) ([Table 338-4](#)). Virilization is usually apparent at birth in the female and within the first 2 to 3 years of life in the male. Manifestations in females include hypertrophy of the clitoris with ventral chordee, partial fusion of the labioscrotal folds, and variable virilization of the urethra. The uterus, fallopian tubes, and ovaries are normal, and the wolffian ducts do not virilize, probably because adrenal function begins relatively late in embryogenesis. The external appearance of an affected female newborn is similar to that of a male with bilateral cryptorchidism and hypospadias. The labioscrotal folds are bulbous and rugated and resemble a scrotum. Rarely, the virilization is so severe that development of a complete penile urethra and prostate results in errors in sex assignment at birth. Radiography following the injection of radiopaque dye into the external genital orifice is helpful in demonstrating the presence of vagina, uterus, and (sometimes) fallopian tubes. Occasionally, virilization of the female is slight or absent at birth and becomes evident in later infancy, adolescence, or adulthood (the so-called nonclassic or late-onset form of the disorder). In both sexes, rapid somatic maturation results in premature epiphyseal closure and a short adult height. The untreated female with the classic disorder grows rapidly during the first year of life and has progressive virilization. At the time of expected puberty there is a failure of normal female sexual development and absence of menstruation.

Since male phenotypic differentiation is normal, the condition is usually not recognized at birth in boys in the absence of adrenal insufficiency. However, early maturation of the external genitalia, development of secondary sex characteristics, coarsening of the voice, frequent erections, and excessive muscular development are noticeable during the first few years of life. Virilization in the male can follow two patterns. Excessive adrenal androgens can inhibit gonadotropin production so that the testes remain infantile in size despite the acceleration of masculinization. Such untreated adult men are capable of erection and ejaculation but have no spermatogenesis. Alternatively, adrenal androgen secretion can induce premature maturation of the hypothalamic-pituitary axis and initiate a true precocious puberty including early maturation of spermatogenesis ([Chap. 335](#)). The untreated male is also subject to the development of ACTH-dependent "tumors" of the adrenal rest cells of the testes.

In classic 21-hydroxylase deficiency, which accounts for about 95% of congenital adrenal hyperplasia, decreased production of hydrocortisone leads to increased release of ACTH, enlargement of the adrenal glands, and partial or complete compensation of the defect in the secretion of hydrocortisone. In about half, the enzyme defect appears to be partial, and cortisol secretion is normal. This form is termed *simple virilizing*. When deficiency of the enzyme is more complete, the so-called salt-losing form of 21-hydroxylase deficiency, production of cortisol and aldosterone is inadequate, leading to severe salt wastage with anorexia, vomiting, volume depletion, and vascular collapse within the first few weeks of life. In untreated patients, there is overproduction of the cortisol precursors prior to the 21-hydroxylase step, causing an increase in plasma progesterone and 17-hydroxyprogesterone. These steroids are weak aldosterone antagonists at the receptor level; in the compensated state aldosterone production increases to attempt to maintain normal sodium balance. Increased substrate availability is also responsible for the enhanced androgen synthesis and hence for the virilization.

Female pseudohermaphroditism also occurs in 11 β -hydroxylase deficiency. In this disorder, a block in hydroxylation at the 11-carbon results in the accumulation of 11-deoxycortisol and deoxycorticosterone (DOC), a potent salt-retaining hormone that causes hypertension rather than salt loss. The clinical features that stem from glucocorticoid deficiency and androgen excess are similar to those in 21-hydroxylase deficiency.

Pathophysiology Both disorders are due to autosomal recessive mutations. The carrier frequency for CYP21A2 deficiency is about 1 in 50. Because the gene is located on the sixth chromosome close to the HLA-B locus, heterozygous carriers and homozygotes within a given family can be identified on the basis of the HLA haplotype. At the molecular level the mutations that give rise to 21-hydroxylase deficiency are highly polymorphic; indeed, partial gene deletions (10 to 30%), conversion of the gene from a functional state to a form that is not transcribed normally (10%), and point mutations (60 to 75%) have been characterized in the disorder. The classic disorder is due to mutations that severely impair enzyme activity, and less severe mutations cause the nonclassic, or late-onset, variety. 11 β -hydroxylase activity is encoded by two genes on chromosome 8; mutations of the *CYP11B1* gene are responsible for this disorder. The *CYP11B2* gene encodes aldosterone synthase. A late-onset variant of 11 β -hydroxylase deficiency exists but has not been characterized at the molecular level.

Urinary excretion of 17-ketosteroids and of the metabolites that accumulate proximal to the enzymatic blocks is increased. Plasma ACTH is elevated. In CYP21A2 deficiency, 17-hydroxyprogesterone accumulates in blood and is excreted predominantly as pregnanetriol. In CYP11B1 deficiency, 11-deoxycortisol accumulates in blood and is excreted predominantly as tetrahydrocortexolone. **For additional discussion of the endocrine pathology, see Chap. 331.*

TREATMENT

Gender assignment should correspond to the chromosomal and gonadal sex, and appropriate surgical correction of the external genitalia should be undertaken as early as possible. This is of importance because appropriately treated men and women are capable of fertility. However, if the correct diagnosis is made late (after 3 years of age), gender assignment should be changed only after careful consideration of the psychosexual background.

Treatment with appropriate glucocorticoids prevents the consequences of hydrocortisone deficiency, arrests the rapid virilization, and prevents premature somatic and epiphyseal maturation. The suppression of the abnormal steroid secretion corrects the hypertension in CYP11B1 deficiency and in both disorders allows normal onset of menses and development of female secondary sex characteristics. In males, glucocorticoid therapy suppresses adrenal androgens and results in normal gonadotropin secretion, testicular development, and spermatogenesis. The usual maintenance dose of hydrocortisone is 10 to 20 mg/m² per day, given in three divided doses. However, the dose must be adjusted on an individual basis to optimize ACTH suppression while avoiding glucocorticoid side effects, which include growth retardation as well as Cushingoid features. Measurements of plasma 17-hydroxyprogesterone, androstenedione, ACTH, and renin levels have all been used to assess adequacy of replacement therapy. In severe CYP21A2 deficiency associated with salt loss or elevated plasma renin activity, treatment with mineralocorticoids is also indicated, and plasma renin levels should be monitored to assess the adequacy of mineralocorticoid replacement. Trials are underway to assess the potential use of antiandrogens (e.g., spironolactone, cyproterone acetate, flutamide) or aromatase inhibitors (e.g., letrozole, testolactone) as adjunctive therapy that may allow reductions in glucocorticoid dose. Treatment of affected fetuses in utero (beginning at 4 to 6 weeks) has been accomplished by administering dexamethasone (which crosses the placenta) to the mother. Though this treatment can reduce the extent of virilization in some girls, it is associated with maternal side effects, and the long-term consequences have not been established.

Other Causes of Female Pseudohermaphroditism Placental aromatase deficiency due to mutations in the gene encoding aromatase (*CYP19*) causes virilization of female embryos because of defective conversion of androgens to estrogens in the placenta and the secondary increase in testosterone levels in the fetus; in postnatal life *CYP19* deficiency in women causes hirsutism and development of polycystic ovaries. Female pseudohermaphroditism can also occur in babies born to mothers with virilizing tumors of the ovary (e.g., arrhenoblastomas or luteomas of pregnancy) and, rarely, to mothers with virilizing adrenal tumors. In the past, the administration to pregnant women of progestogens with androgenic side effects (such as 17 α -ethinyl-19-nor-testosterone) to

prevent abortion resulted in masculinization of female fetuses.

Developmental Disorders of Mullerian Ducts (Congenital Absence of the Vagina, Mullerian Agenesis)

Clinical Features Congenital hypoplasia or absence of the vagina in combination with abnormal or absent uterus (the Mayer-Rokitansky-Kuster-Hauser syndrome) is second to gonadal dysgenesis as a cause of primary amenorrhea. Most patients are ascertained after the time of expected puberty because of failure to menstruate. The height is normal, and the breasts, axillary and pubic hair, and habitus are feminine in character. The uterus can vary from almost normal, lacking only a conduit to the introitus, to the characteristic rudimentary bicornuate cords with or without a lumen. In some patients cyclic abdominal pain indicates that sufficient functional endometrium is present to result in retrograde menstruation and/or hematometra.

About one-third have abnormal kidneys, most commonly agenesis, ectopy, fused kidneys of the horseshoe type, or solitary ectopic kidneys in the pelvis. Skeletal abnormalities are present in one-tenth; most involve the spine, and limb and rib defects account for the rest. Specific abnormalities include wedge vertebrae, fused rudimentary or asymmetric vertebral bodies, supernumerary vertebrae, and the Klippel-Feil syndrome (congenital fusion of the cervical spine, short neck, low posterior hairline, and painless limitation of cervical movement).

Pathophysiology The karyotype is 46,XX. Familial occurrence has been described, and the pattern of inheritance in most is consistent with a sex-limited autosomal dominant mutation. Sporadic cases may represent new mutations of the type responsible for the familial disorder or be multifactorial in etiology. In the familial cases, expressivity is variable; some have skeletal or renal abnormalities only, and some have other abnormalities of mullerian derivatives such as a double uterus. Bilateral renal aplasia in stillborn infants is commonly associated with absence of the uterus and vagina. Thus, the family history should be probed for isolated skeletal and renal abnormalities and for stillbirths that might result from congenital absence of both kidneys. Ovarian function is normal, and successful pregnancies have occurred after corrective vaginal surgery in patients with a normal uterus.

TREATMENT

Vaginal agenesis can be treated by surgical or nonsurgical means. Surgical repair generally utilizes a split-thickness skin graft around a solid rubber mold to create an artificial vagina. Medical therapy involves the repeated application of pressure against the vaginal dimple with a simple dilator to force development of adequate vaginal depth. In view of complication rates of 5 to 10% in surgical series, surgery should be reserved for patients in whom a well-formed uterus is present and the possibility of fertility exists. Frequent coitus or instrumental dilatation is essential for maintaining patency of neovaginas formed by either technique.

MALE PSEUDOHERMAPHRODITISM

Defective virilization of the 46,XY embryo (male pseudohermaphroditism) can result

from defects in androgen synthesis, defects in androgen action, defects in mullerian duct regression, and uncertain causes.

Abnormalities in Androgen Synthesis

Clinical Features Enzymatic defects that result in defective testosterone synthesis ([Fig. 338-3](#)) account for only about a fifth of cases of male pseudohermaphroditism ([Tables 338-4](#) and [338-5](#)). Each of the defects blocks a step in the conversion of cholesterol to testosterone. Three enzymatic reactions are common to the synthesis of other adrenal hormones as well: cholesterol side chain cleavage, 3 β -hydroxysteroid dehydrogenase/isomerase, and 17 α -hydroxylase (CYP17). Consequently, their deficiency results in congenital adrenal hyperplasia ([Table 338-4](#)) as well as male pseudohermaphroditism. Two others (17,20-lyase and 17 β -hydroxysteroid dehydrogenase 3) are unique to the pathway of androgen synthesis, and their deficiency results only in male pseudohermaphroditism. Since androgens are obligatory precursors of estrogens, synthesis of estrogen is also low in all but the terminal defect (17 β -hydroxysteroid dehydrogenase 3 deficiency).

Adrenal dysfunction is described in [Chap. 331](#), and the present discussion concerns the abnormal sexual development. In genetic males with defective testosterone synthesis, absence of the uterus and fallopian tubes indicates that mullerian duct inhibition was normal during embryogenesis. Masculinization of the urogenital tract and external genitalia and virilization at puberty vary from almost normal to absent, and, therefore, the manifestations vary from men with mild hypospadias to phenotypic women who prior to puberty resemble women with complete testicular feminization. This heterogeneity is the consequence of varying severity of the enzymatic defects, varying effects of the steroids that accumulate proximal to the various metabolic blocks, and the presence of alternative enzymatic pathways in some disorders. In patients with partial defects in whom the plasma testosterone level is normal, the diagnosis can only be made by measuring the steroids that accumulate proximal to the metabolic block.

Congenital lipoid adrenal hyperplasia is an autosomal recessive disorder in which virtually no urinary steroids (either 17-ketosteroids or 17-hydroxycorticoids) can be detected. Since the defect blocks the conversion of cholesterol to pregnenolone, a step catalyzed by the cholesterol side chain cleavage enzyme (CYP11A1), it was originally assumed that the defect must involve this enzyme. However, the disorder is instead caused by mutations in the *steroidogenic acute regulatory (StAR)* gene, which encodes the protein that transports cholesterol from the cytosol to the inner mitochondrial membrane in the adrenal and gonads. Manifestations of the disorder include salt wasting and profound adrenal insufficiency, and most affected individuals die during infancy. At autopsy, the adrenals and testes are enlarged and infiltrated with lipid. Affected males are incompletely masculinized, whereas affected female infants have normal genital development because lipid accumulation in the ovary does not occur until there is follicular development and active steroidogenesis.

3 β -Hydroxysteroid dehydrogenase/isomerase 2 deficiency causes varying failure of masculinization and the development of a vagina in male infants. Female infants may be modestly virilized at birth due to the weak androgenic potency of dehydroepiandrosterone, the major steroid secreted. If the enzyme is absent in both the

adrenal and testis, no urinary steroids contain a D₄-3-keto configuration, whereas in patients in whom the defect is partial or affects only the testis, the urine may contain normal or elevated levels of D₄-3-ketosteroids. Most patients have marked salt wasting and profound adrenal insufficiency, and long-term survival in untreated cases occurs only in states of partial deficiency. Minimally affected males may experience an otherwise normal male puberty except for profound gynecomastia. In these boys, a low-normal blood testosterone level is accompanied by elevated D₅precursor steroids. 3β-Hydroxysteroid dehydrogenase activity is catalyzed by more than one isoenzyme. The type 2 isoenzyme is expressed in adrenals and gonads; the disorder described above is due to any of several mutations in this enzyme. The coding sequence of this gene is said to be normal in several individuals with the late-onset variant of the disease, the pathophysiology of which is unclear.

17α-Hydroxylase-17,20-lyase (CYP17) deficiency impairs the introduction of the 17-hydroxyl and the scission of the C-17,20 carbon bond that convert pregnenolone and progesterone to dehydroepiandrosterone and androstenedione, respectively. These reactions are mediated by a single enzyme CYP17, which is encoded on chromosome 10; it remains unclear why both reactions occur in the ovary and testis, whereas in the adrenal 17-hydroxyprogesterone is largely converted to glucocorticoids and mineralocorticoids rather than the 19-carbon steroids. Of note, some patients appear to have selective impairment of either 17α-hydroxylase or 17,20-lyase activity. These mutations have identified enzyme domains proposed to undergo posttranslational modification and selective interactions with cofactors that switch enzymatic activity. Whatever the mechanism, the consequences of 17α-hydroxylase and 17,20-lyase deficiencies are different.

17α-Hydroxylase deficiency is characterized by hypogonadism, absence of secondary sex characteristics, hypokalemic alkalosis, hypertension, and virtually undetectable hydrocortisone secretion in phenotypic women. Formation of both corticosterone and DOC by the adrenal is elevated, and urinary 17-ketosteroids are low. Aldosterone secretion is low due to high plasma DOC and depressed angiotensin levels and returns to normal after suppressive doses of glucocorticoids. In 46,XX individuals, amenorrhea, absent sexual hair, and hypertension are common, but the phenotype is that of a normal prepubertal woman. In males, the deficiency results in defective virilization that varies from complete male pseudohermaphroditism to ambiguous genitalia with perineoscrotal hypospadias and, in some, gynecomastia. Adrenal insufficiency does not develop, since the secretion of both corticosterone (a weak glucocorticoid) and DOC (a mineralocorticoid) is elevated. Hypertension and hypokalemia are prominent (even in the neonatal period) and remit after suppression of the DOC secretion by glucocorticoid replacement. A variety of point mutations, deletions, and insertions in the *CYP17* gene have been characterized in affected individuals.

17,20-Lyase deficiency in males is associated with normal function of the adrenal cortex and a variable pattern of male pseudohermaphroditism. In the majority there is genital ambiguity at birth, with some virilization at the time of expected puberty. Rare 46,XY patients have had a female phenotype and no virilization at the time of expected puberty. The disorder has been recognized in one 46,XX woman with sexual infantilism. Mutations of *CYP17* that cause this disorder involve an area of the gene that is known to encode a binding site for the redox-partner of the enzyme.

17b-Hydroxysteroid dehydrogenase 3 (17b-HSD-3) deficiency involves the final step in testosterone biosynthesis, reduction of the 17-keto group of androstenedione. This is the most common enzymatic defect in testosterone synthesis. Affected 46,XY males usually have a female phenotype with a blind-ending vagina and absence of müllerian derivatives, but inguinal or abdominal testes and virilized wolffian duct structures are present. At the time of expected puberty, both virilization (with phallic enlargement and development of facial and body hair) and a variable degree of feminization take place. In some untreated patients, reversal of gender behavior from female to male occurs at puberty. Plasma testosterone level may be in the low-normal range, making it essential to document elevation in plasma androstenedione to make the diagnosis. Isoenzymes encoded by several different genes possess 17b-hydroxysteroid dehydrogenase activity, but the isoenzyme 3 is expressed in the testes. A variety of mutations have been characterized in the 17b-HSD-3 gene in affected individuals.

Pathophysiology These various disorders are inherited as autosomal recessive traits. The pattern of steroid secretion and excretion depends on the site of the various metabolic blocks ([Fig. 338-3](#)). In general, gonadotropin secretion is high, and many individuals with incomplete defects are able to compensate so that the steady-state levels of testosterone may be normal or almost normal.

In rare cases of male pseudohermaphroditism, testosterone formation is deficient for reasons other than a single enzyme defect in androgen synthesis. These include disorders in which Leydig cell agenesis is due to autosomal recessive loss-of-function mutations of the [LH](#) receptor or to the secretion of a biologically inactive LH molecule. In addition, as described above, several disorders, including familial 46,XY pure gonadal dysgenesis, sporadic dysgenetic testes, and the absent testis syndrome, are characterized by deficient testosterone production due to abnormal gonadal development.

TREATMENT

Therapy with glucocorticoids and in some instances mineralocorticoids is indicated in those disorders causing adrenal hyperplasia. The management of the genital abnormalities depends on the individual case. Fertility has not been reported, and its consideration does not enter into sex assignment. In genetic females there is no problem (except in diagnosis), in that affected individuals are raised appropriately as females and estrogen therapy is begun at the time of expected puberty to promote development of female secondary sex characteristics. Whether newborn males with ambiguous genitalia should be raised as males or females depends on the anatomic defect; in general, the more severely affected should be raised as females, and corrective surgery of the genitalia and removal of the testes should be undertaken as early as possible. In such women estrogen therapy is begun at the appropriate age to allow development of normal female secondary sex characteristics. In individuals raised as males, corrective surgery is indicated for any coexisting hypospadias, and plasma androgens should be monitored at the time of expected puberty to determine whether testosterone therapy is appropriate.

Abnormalities in Androgen Action Several disorders of male phenotypic development

result from abnormalities of androgen action. The spectrum of phenotypes is described in [Tables 338-4](#) and [338-5](#). In these disorders, testosterone formation and mullerian regression are normal, but male development is impaired because of resistance to androgen action in target tissues.

Steroid 5 α -Reductase 2 Deficiency This autosomal recessive disorder is characterized by (1) severe perineoscrotal hypospadias; (2) a blind vaginal pouch of variable size opening either into the urogenital sinus or into the urethra; (3) testes with normal epididymides, vasa deferentia, and seminal vesicles, and termination of the ejaculatory ducts in the blind-ending vagina; (4) a female habitus with normal axillary and pubic hair but without female breast development; (5) the absence of uterus and fallopian tubes; (6) normal male plasma testosterone; and (7) masculinization to a variable degree at the time of puberty.

The realization that virilization during embryogenesis is defective only in the urogenital sinus and the external genitalia provided insight into the fundamental abnormality. Testosterone, the androgen secreted by the fetal testis, is responsible for conversion of the wolffian duct into the epididymis, vas deferens, and seminal vesicle, whereas dihydrotestosterone mediates virilization of the urogenital sinus and the external genitalia. Consequently, impairment of dihydrotestosterone formation in a male embryo would be expected to cause the phenotype in this disorder -- normal male wolffian duct derivatives with defective masculinization of the external genitalia and urogenital sinus. Since testosterone itself regulates LH secretion ([Chap. 335](#)), plasma LH level is normal or minimally elevated. As a result, testosterone and estrogen production rates are those of normal men, and gynecomastia does not develop.

The fact that 5 α -reductase 2 enzyme is deficient in this disorder was established by assay of biopsied tissues and cultured fibroblasts from affected individuals. Deletions or point mutations in the gene encoding steroid 5 α -reductase 2 have been identified in most families studied. Approximately 40% are compound heterozygotes.

Receptor Disorders The androgen receptor is a member of the steroid/thyroid family of receptors with steroid-binding, DNA-binding, and functional domains and is encoded by a gene on the long arm of the X chromosome. Mutations of this gene impair receptor function and hence impair male phenotypic differentiation and/or virilization.

CLINICAL FEATURES *Complete testicular feminization* (also called *complete androgen insensitivity*) is a common form of male pseudohermaphroditism; estimates of frequency vary from 1 in 20,000 to 1 in 64,000 male births. It is the third most common cause of primary amenorrhea after gonadal dysgenesis and congenital absence of the vagina. The features are characteristic. Namely, a woman is ascertained either because of inguinal hernia (prepubertal) or primary amenorrhea (postpubertal). The development of the breasts, the habitus, and the distribution of body fat are female in character so that most have a truly feminine appearance. Axillary and pubic hair is absent or scanty, but some vulval hair is usually present. Scalp hair is that of a normal woman, and facial hair is absent. The external genitalia are unambiguously female, and the clitoris is normal. The vagina is short and blind-ending and may be absent or rudimentary. All internal genitalia are absent except for testes that contain normal Leydig cells and seminiferous tubules without spermatogenesis. The testes may be located in the abdomen, along the

course of the inguinal canal, or in the labia majora. Occasionally, remnants of mullerian or wolffian duct origin are present in the paratesticular fascia or in fibrous bands extending from the testis. Patients tend to be rather tall, and bone age is normal. Psychosexual development is unmistakably female with regard to behavior, outlook, and maternal instincts.

The major complication of undescended testes in this disorder, as in all forms of cryptorchidism, is the development of tumors ([Chap. 96](#)). Since affected individuals undergo normal pubertal growth and feminize at the time of expected puberty and since testicular tumors rarely develop until after puberty, it is usual to delay gonadectomy until after the time of expected puberty. Prepubertal gonadectomy is indicated if the testes are present in the inguinal region or labia majora and result in discomfort or hernia formation. (If hernia repair is indicated prepubertally, most physicians prefer to remove the testes at the same time to limit the number of operative procedures.) If the testes are removed prepubertally, estrogen therapy is required at the appropriate age to ensure normal growth and breast development. Postpubertal gonadectomy causes menopausal symptoms and other evidence of estrogen withdrawal, and suitable estrogen replacement is indicated ([Chap. 336](#)).

Incomplete testicular feminization is about one-tenth as frequent as the complete form. In this disorder there is minor virilization of the external genitalia (partial fusion of the labioscrotal folds and/or some degree of clitoromegaly), normal pubic hair, and mixed virilization and feminization at the time of expected puberty. The vagina is short and blind-ending, but in contrast to the complete form, the wolffian duct derivatives are often partially developed. Since women with the incomplete disorder virilize at the time of expected puberty, gonadectomy should be performed before the expected time of puberty in all prepubertal patients with clitoromegaly or posterior labial fusion.

Reifenstein syndrome (also called *partial androgen insensitivity*) is the term applied to forms of incomplete male pseudohermaphroditism initially described by a number of eponyms (Reifenstein syndrome, Gilbert-Dreyfus syndrome, Lubs syndrome). These syndromes are mutations that partially impair the function of the androgen receptor. The most common phenotype is a man with perineoscrotal hypospadias and gynecomastia, but the spectrum of defective virilization in affected families ranges from men with azoospermia to phenotypic women with pseudovaginas. Axillary and pubic hair is normal, but chest and facial hair is scanty. Cryptorchidism is common, the testes are small, and azoospermia is present. Some have defects in wolffian duct derivatives such as absence or hypoplasia of the vas deferens. Since the psychological development in most is unequivocally male, the hypospadias and cryptorchidism should be corrected surgically. The treatment of the gynecomastia is surgical removal.

A disorder of the androgen receptor that is not actually a form of male pseudohermaphroditism is manifested as *infertility and/or undervirilization in phenotypic men*. Some such individuals are minimally affected members of families with Reifenstein syndrome in whom azoospermia is the only manifestation of the receptor defect. The *undervirilized fertile male* is an even more subtle manifestation of androgen receptor defect. In these families, affected men have gynecomastia and undervirilization, and some are fertile. More commonly, however, individuals with negative family histories present with male infertility with or without undervirilization.

PATHOPHYSIOLOGY The karyotype is 46,XY, and the mutation is X-linked. The frequency of a positive family history varies from about two-thirds of patients with testicular feminization and Reifenstein syndrome to only occasional patients with the infertile male syndrome. The disorder in subjects with a negative family history is believed to be the result of new mutations.

Hormone dynamics are similar in all disorders of the androgen receptor. Plasma testosterone levels and rates of testosterone production by the testes are normal or high. Elevated testosterone production is caused by the high mean plasma level of [LH](#), which, in turn, is due to defective feedback regulation caused by resistance to the action of androgen at the hypothalamic-pituitary level. Elevated LH concentration is responsible also for the increased estrogen production by the testes ([Chap. 337](#)). (In normal men, most estrogen is derived from peripheral formation from circulating androgens, but when the plasma LH level is elevated, the testes secrete increased amounts of estrogen into the circulation.) Thus resistance to the feedback regulation of LH secretion by circulating androgen results in elevated plasma LH levels, and this, in turn, results in the enhanced secretion of both testosterone and estradiol by the testes. Gonadotropin levels rise even higher (and menopausal symptoms may develop) when the testes are removed, indicating that gonadotropin secretion is under partial feedback control. Presumably, in the steady state and in the absence of an androgen effect, estrogen alone regulates LH secretion, a control purchased at the expense of an elevated plasma estrogen concentration for a male. The hormonal changes in the infertile male syndrome are similar to those in the other receptor disorders but less marked. Some men with this syndrome do not have an elevation of plasma LH or plasma testosterone level.

Feminization in these disorders is the result of two interlocking phenomena. First, androgens and estrogens have antagonistic effects, and in the absence of androgen action, the cellular effect of estrogen is unopposed. Second, the testicular production of estradiol is greater than that of the normal male (although less than that of the normal female). Variable degrees of androgen resistance and enhanced estradiol production result in different degrees of defective virilization and enhanced feminization in the four clinical syndromes.

Each of these syndromes is the result of defects in the androgen receptor. In most families, the fundamental defect is due to point mutations in the coding sequence leading to premature termination codons or to amino acid substitutions in the hormone-binding domain. Such mutations impair receptor function to variable degrees. Some families with clinical syndromes typical of an androgen receptor disorder have normal androgen binding in fibroblasts; in most, point mutations in the DNA-binding domain of the androgen receptor are responsible for the androgen resistance.

TREATMENT

Individuals with 5 α -reductase deficiency who are raised as females but elect at the time of expected puberty to change social sex to male or who are raised from the first as males should be monitored carefully and given supplemental androgens, preferably those such as nandrolone decanoate that do not require 5 α -reduction for activation,

when virilization is incomplete. Fertility has been reported in such an individual. Individuals with 5 α -reductase deficiency who continue to function as females should be gonadectomized, given feminizing doses of estrogens indefinitely, and receive surgical correction of the introitus when appropriate. The management of subjects with androgen receptor defects depends on the phenotypic manifestations. Women with testicular feminization should be castrated (preferably after the completion of the pubertal growth spurt and the feminization of the breasts) to prevent tumor development in the testes and receive estrogen replacement to maintain feminization, prevent hot flashes, and protect the bones; shallow vaginal depth can usually be treated medically with the Frank technique. Men with the Reifenstein phenotype should have surgical correction of the hypospadias and may require surgery for gynecomastia; supplemental androgen therapy in these men rarely improves the incomplete virilization.

Persistent Mullerian Duct Syndrome Men with this uncommon disorder have testes and male phenotypic development and in addition have fallopian tubes and a uterus. In some, one or both testes are descended and the uterus and ipsilateral fallopian tube are in the inguinal canal or scrotum; both testes and fallopian tubes may be present in the hernia sac or can be drawn into it. In others, the testes are located high in the abdomen and hernias are not present. In both types, the vasa deferentia are embedded in the wall of the uterus, a feature that complicates surgical procedures designed to preserve potential fertility. Most individuals have uninformative family histories, but in some the condition is inherited as an autosomal recessive trait. Because the external genitalia are well developed and the subjects masculinize normally at puberty, it is assumed that during the critical stage of embryonic differentiation the fetal testes produce a normal amount of androgen. However, mullerian regression does not occur. Two types of mutations have been described in this disorder. In one the gene that encodes [AMH](#) is defective, and blood levels of AMH are usually low or undetectable; in the other the AMH receptor is defective, and blood levels of AMH are elevated. To minimize the chance of tumor development and to maintain virilization, orchiopexy should be performed. Malignancy in the uterus or vagina has not been described, and because the vasa deferentia are closely associated with the broad ligaments, the uterus and vagina should be left in place to avoid disruption of the vasa deferentia during removal and consequently to preserve possible fertility.

Developmental Defects of the Male Genitalia

Hypospadias Hypospadias is a congenital anomaly in which the urethra terminates in an abnormal position along the ventral midline of the penis at some site between the normal urethral meatus and the perineum. This malformation occurs in 0.5 to 0.8% of male births in the United States and is often associated with ventral contraction and bowing of the penis (chordee). It is common to categorize hypospadias as glandular (involving the glans penis), penile, or perineoscrotal. Since androgens control penile development, hypospadias is generally assumed to result from some unidentified defect in androgen formation or action during embryogenesis. Indeed, hypospadias occurs in most disorders of male sexual differentiation, and a rare cause of hypospadias is maternal ingestion of progestational agents early in pregnancy. However, the known causes (single-gene defects, chromosomal abnormalities, and maternal drug ingestion) account for only about one-fourth of cases, and the etiology of most is unknown. The management is surgical.

Cryptorchidism The control of testicular descent is poorly understood, both in regard to the nature of the forces that cause the movement and to the hormonal factors that regulate the process. In anatomic terms, testicular descent can be divided into three phases: (1) transabdominal movement of the testis from its site of origin above the kidney to the inguinal ring, (2) formation of the opening in the inguinal canal (processus vaginalis) through which the testis exits the abdominal cavity, and (3) actual movement of the testis through the inguinal canal to its permanent site in the scrotum. This process occurs over a 6- to 7-month period during gestation, beginning at about the sixth week, and is not completed in some until after birth. Impairment at any stage in this process can impair descent of one or both testes. About 3% of full-term and 30% of premature male infants have at least one cryptorchid testis at birth, but descent is usually completed within the first few weeks of life, so that the incidence of failure of descent by 6 to 9 months of age is only 0.6 to 0.7%. It is this latter category of maldescent that requires intervention.

Permanent cryptorchidism can be classified as intraabdominal (10%), canalicular (in the inguinal canal) (20%), high scrotal (40%), or obstructed (30%), in which maldescent is due to a physical barrier between the inguinal pouch and the inlet of the scrotum. These disorders must be distinguished from the temporarily retracted normal testis.

The cryptorchid testis functions poorly after puberty, but the extent to which maldescent is the result of an abnormality of the testis or the cause of abnormal function is unknown. Two general theories have been advanced as to the etiology -- inadequate intraabdominal pressure and deficient endocrine function of the testis either because of deficient testosterone synthesis or inadequate formation of [AMH](#). Indeed, defects that result in inadequate development of intraabdominal pressure or inadequate development of the testes can cause cryptorchidism. As in hypospadias, however, the known causes of cryptorchidism constitute only a small fraction of the cases, and the etiology in most remains to be identified. Two complications of cryptorchidism are important; spermatogenesis cannot occur at the temperature of the abdominal cavity, and it is necessary to correct the process as early as possible to allow possible fertility. The fact that infertility is common in men who have been treated for unilateral as well as bilateral cryptorchidism suggests that maldescent is usually the consequence rather than the cause of the testicular malfunction. There is also a greater frequency of malignancy in undescended testes, which should be surgically corrected for this reason ([Chap. 96](#)).

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339. DISORDERS AFFECTING MULTIPLE ENDOCRINE SYSTEMS- Steven I. Sherman, Robert F. Gagel

NEOPLASTIC DISORDERS AFFECTING MULTIPLE ENDOCRINE ORGANS

Several distinct genetic disorders predispose to endocrine gland neoplasia and cause hormone excess syndromes ([Table 339-1](#)). DNA-based genetic testing is now available for these disorders, but effective management requires an understanding of endocrine neoplasia and the range of clinical features that may be manifest in an individual patient.

MULTIPLE ENDOCRINE NEOPLASIA (MEN) TYPE 1

Clinical Manifestations MEN 1, or Wermer's syndrome, is characterized by neoplasia of parathyroid, pituitary, and pancreatic islet cells ([Table 339-1](#)). The syndrome is inherited as an autosomal dominant trait, so that each child of an affected parent has a 50% chance of inheriting the predisposing gene.

Several features of the pathogenesis of [MEN](#) 1 have important implications for its management. Though each tumor is derived from a single cell (clonal in origin), any endocrine cell within the affected organs can become transformed. Hyperplasia is the initiating lesion, followed later by adenomatous or carcinomatous changes.

Consequently, the disease process within a single organ is multicentric. Neoplasia in one organ may affect the progression of disease in another organ. For example, the ectopic production of hypothalamic releasing hormones by a pancreatic tumor may stimulate the growth of a pituitary tumor. Because MEN 1 generally evolves over a 30- to 40-year period, the manifestations depend in part on when the disorder is identified.

Hyperparathyroidism is the most common manifestation of [MEN](#) 1. Hypercalcemia may be present during the teenage years, and most individuals are affected by age 40 ([Fig. 339-1](#)). Screening for hyperparathyroidism involves measurement of either an albumin-adjusted or ionized serum calcium level. The diagnosis is established by demonstrating elevated levels of serum calcium and intact parathyroid hormone. Manifestations of hyperparathyroidism in MEN 1 do not differ substantially from those in sporadic hyperparathyroidism and include calcium-containing kidney stones, bone abnormalities, and gastrointestinal and musculoskeletal complaints ([Chap. 341](#)).

Other familial disorders associated with hypercalcemia include familial parathyroid hyperplasia, familial adenomatous hyperparathyroidism, and familial hypocalciuric hypercalcemia (FHH). Calcium excretion is usually elevated in the patient with [MEN](#) 1 or other forms of primary hyperparathyroidism and low in FHH. Another distinguishing feature is that the serum calcium level is rarely elevated at birth in patients with MEN 1 but is frequently elevated in newborns with FHH. Differentiation of hyperparathyroidism of MEN 1 from other forms of familial primary hyperparathyroidism is usually based on family history, histologic features of resected parathyroid tissue, and, sometimes, long-term observation to determine whether other manifestations of MEN 1 develop. FHH is due to inactivating mutations of the calcium sensor, a transmembrane G protein-coupled receptor found in parathyroid tissue and kidney ([Chap. 341](#)).

Parathyroid hyperplasia is the common cause of hyperparathyroidism in [MEN](#) 1,

although single and multiple adenomas have been described. Hyperplasia of one or more parathyroid glands is common in younger patients; adenomas are usually found in older patients or those with long-standing disease.

Neoplasia of the pancreatic islets is the second most common manifestation of [MEN 1](#) and tends to occur in parallel with hyperparathyroidism ([Fig. 339-1](#)). Increased pancreatic islet cell hormones include pancreatic polypeptide (75 to 85%), gastrin [60%; Zollinger-Ellison syndrome (ZES)], insulin (25 to 35%), vasoactive intestinal peptide (VIP) (3 to 5%; Verner-Morrison or watery diarrhea syndrome), glucagon (5 to 10%), and somatostatin (1 to 5%). The tumors rarely produce adrenocorticotropin (ACTH), corticotropin-releasing hormone (CRH), growth hormone-releasing hormone (GHRH), calcitonin gene products, neurotensin, gastric inhibitory peptide, and others. Many of the tumors produce more than one peptide. The pancreatic neoplasms differ from the other components of MEN 1 in that approximately one-third of the tumors display malignant features, including hepatic metastases ([Chap. 93](#)).

Pancreatic islet cell tumors are diagnosed by identification of a characteristic clinical syndrome, hormonal assays with or without provocative stimuli, or radiographic techniques. One approach involves annual screening of people at risk with measurement of basal and meal-stimulated levels of pancreatic polypeptide to identify the tumors as early as possible; the rationale of this screening strategy is the concept that surgical removal of islet cell tumors at an early stage will be curative. Other approaches to screening include measurement of serum gastrin and pancreatic polypeptide levels every 2 to 3 years, with the rationale that pancreatic neoplasms will be detected at a later stage but can be managed medically, if possible, or by surgery. High-resolution, early-phase computed tomography (CT) scanning provides the best noninvasive technique for identification of these tumors, but intraoperative ultrasonography is the most sensitive method for detection of small tumors.

[ZES](#) is caused by excessive gastrin production and occurs in more than half of [MEN 1](#) patients with pancreatic islet cell tumors ([Fig. 339-1](#)) ([Chap. 93](#)). Clinical features include increased gastric acid production, recurrent peptic ulcers, diarrhea, and esophagitis. The ulcer diathesis is refractory to conservative therapy such as antacids. The diagnosis is made by finding increased gastric acid secretion, elevated basal gastrin levels in serum [generally >115 pmol/L (200 pg/mL)], and an exaggerated response of serum gastrin to either secretin or calcium. Other causes of elevated serum gastrin levels, such as achlorhydria, treatment with H₂receptor antagonists or omeprazole, retained gastric antrum, small-bowel resection, gastric outlet obstruction, and hypercalcemia, should be excluded. Gastrin-producing carcinoid-like tumors are frequently present in the duodenal wall.

Insulinoma causes hypoglycemia in about one-third of [MEN 1](#) patients with pancreatic islet cell tumors ([Fig. 339-1](#)). The tumors may be benign or malignant (25%). The diagnosis can be established by documenting hypoglycemia during a short fast with simultaneous inappropriate elevation of serum insulin and C-peptide levels. More commonly, it is necessary to subject the patient to a supervised 72-h fast to provoke hypoglycemia ([Chap. 334](#)). Large insulinomas may be identified by [CT](#) scanning; small tumors not detected by radiographic techniques may be localized by selective arteriographic injection of calcium into each of the arteries that supply the pancreas and

sampling the hepatic vein for insulin to determine the anatomic region containing the tumor. Intraoperative ultrasonography can also be used to localize these tumors, but preoperative calcium injection data are helpful in guiding the subtotal pancreatectomy if multiple or no abnormalities are detected by intraoperative ultrasonography.

Glucagonoma in occasional [MEN 1](#) patients causes a syndrome of hyperglycemia, skin rash (necrolytic migratory erythema), anorexia, glossitis, anemia, depression, diarrhea, and venous thrombosis. In about half of these patients the plasma glucagon level is high, leading to its designation as the *glucagonoma syndrome*, although elevation of plasma glucagon level in MEN 1 patients is not necessarily associated with these symptoms. The glucagonoma syndrome may represent a complex interaction between glucagon overproduction and the nutritional status of the patient.

The *Verner-Morrison* or *watery diarrhea syndrome* consists of watery diarrhea, hypokalemia, hypochlorhydria, and metabolic acidosis. The diarrhea can be voluminous and is almost always found in association with an islet cell tumor, prompting use of the term *pancreatic cholera*. However, the syndrome is not restricted to pancreatic islet tumors and has been observed with carcinoids or other tumors. This syndrome is believed to be due to overproduction of [VIP](#), although plasma VIP levels may not be elevated. Hypercalcemia may be induced by the effects of VIP on bone as well as by hyperparathyroidism.

Pituitary tumors occur in more than half of patients with [MEN 1](#) and tend to be multicentric, making them difficult to resect ([Chap. 328](#)). Prolactinomas are most common ([Fig. 339-1](#)) and are diagnosed by finding serum prolactin levels >200 ug/L, with or without a pituitary mass evident by magnetic resonance imaging (MRI). Values <200 ug/L may be due to a prolactin-secreting neoplasm or to compression of the pituitary stalk by a different type of pituitary tumor. Acromegaly due to excessive growth hormone (GH) production is the second most common syndrome caused by pituitary tumors in MEN 1 ([Chap. 328](#)) but can rarely be due to production of [GHRH](#) by an islet cell tumor. Cushing's disease can be caused by [ACTH](#)-producing pituitary tumors or by ectopic production of ACTH or [CRH](#) by other tumors in the MEN 1 syndrome. Diagnosis of pituitary Cushing's disease is generally best accomplished by a high-dose dexamethasone suppression test or by petrosal venous sinus sampling for ACTH after intravenous injection of CRH ([Chap. 328](#)). Differentiation of a primary pituitary tumor from an ectopic CRH-producing tumor may be difficult because the pituitary is the source of ACTH in both disorders; documentation of CRH production by a pancreatic islet or carcinoid tumor may be the only method of proving ectopic CRH production. Adrenal cortical tumors are found in almost one-half of gene carriers but are rarely functional; malignancy in the cortical adenomas is uncommon.

Unusual manifestations of MEN 1 The rare carcinoid tumors in MEN 1 are of the foregut type and are derived from thymus, lung, stomach, or duodenum; they may metastasize or be locally invasive. These tumors usually produce serotonin, calcitonin, or [CRH](#); the typical carcinoid syndrome with flushing, diarrhea, and bronchospasm is rare ([Chap. 93](#)). Subcutaneous or visceral lipomas and cutaneous leiomyomas may also be present but rarely undergo malignant transformation. Skin angiofibromas or collagenomas are seen in most patients with MEN 1 when carefully sought.

GENETIC CONSIDERATIONS

[MEN1](#) is transmitted as an autosomal dominant trait, reflecting the fact that the *MEN1* gene, located on chromosome 11q13, encodes a tumor suppressor protein termed *menin* ([Fig. 339-2](#)). Affected individuals typically harbor a germline mutation in *MEN1* and acquire a "second hit" in the normal gene as a result of another mutation or, more commonly, loss of the portion of chromosome 11 that contains the *MEN1* locus ([Chap. 81](#)). Though the function of menin is not well understood, it is a nuclear protein that interacts with a transcriptional factor, Jun D, suggesting a role in cell growth control. Several missense mutations in menin prevent its interaction with Jun D.

MEN1 gene mutations are found in >90% of families with the syndrome ([Fig. 339-2](#)). Genetic testing can be performed in individuals at risk for the development of [MEN 1](#), particularly when the specific mutation is known. The value of genetic testing for this disorder, in contrast to MEN 2 (see below), is debated because predisposed individuals must still be screened repeatedly using endocrine tests. A negative genetic analysis will, however, exclude disease with near 100% certainty in kindreds with a known mutation; for this reason, genetic testing is likely to gain favor as it becomes more widely available. A significant percentage of sporadic parathyroid, islet cell, and carcinoid tumors also have loss or mutation of *MEN1*. It is presumed that these mutations are somatic and occur in a single cell, leading to subsequent transformation.

TREATMENT

Almost everyone who inherits a mutant *MEN1* gene develops at least one clinical manifestation of the syndrome. Most develop hyperparathyroidism, 80% develop pancreatic islet cell tumors, and more than half develop pituitary tumors. For most of these tumors, initial surgery is not curative and patients frequently require multiple surgical procedures and surgery on two or more endocrine glands during a lifetime. For this reason, it is essential to establish clear goals for management of these patients rather than to recommend surgery casually each time a tumor is discovered. Ranges for acceptable management are discussed below.

Hyperparathyroidism Individuals with serum calcium levels >3.0 mmol/L (12 mg/dL), evidence of calcium nephrolithiasis or renal dysfunction, neuropathic or muscular symptoms, or bone involvement (including osteopenia) should undergo parathyroid exploration. In [MEN 1](#) an additional criterion for parathyroid surgery is hypercalcemia associated with elevated gastrin levels, because elevated serum calcium may stimulate gastrin production and [ZES](#), a condition that may be improved by return of calcium levels to normal. There is less agreement regarding the necessity for parathyroid exploration in individuals who do not meet these criteria, and observation may be appropriate in the MEN 1 patient with asymptomatic hyperparathyroidism.

When parathyroid surgery is indicated in [MEN 1](#), all parathyroid tissue should be identified and removed at the time of primary operation, and parathyroid tissue should be implanted in the nondominant forearm. Thymectomy should also be performed because of the potential for later development of malignant carcinoid tumors. If reoperation is necessary, transplanted tissue can be resected under local anesthesia with titration of tissue removal to return the serum calcium level to normal. A less

desirable approach is to remove 3 to 3½ parathyroid glands from the neck, carefully marking the location of residual tissue so that the remaining tissue can be located easily during subsequent surgery.

Pancreatic Islet Tumors (See [Chap. 93](#) for discussion of pancreatic islet tumors not associated with [MEN 1](#).) Two features of pancreatic islet cell tumors in MEN 1 complicate the management. First, the pancreatic islet cell tumors are multicentric, malignant about a third of the time, and cause death in 10 to 20% of patients. Second, removal of all pancreatic islets to prevent malignancy causes diabetes mellitus, a disease with severe long-term complications. These features make it difficult to formulate clear-cut guidelines, but some general concepts appear to be valid. First, islet cell tumors producing insulin, glucagon, [VIP](#), [GHRH](#), or [CRH](#) should be resected because medical therapy is generally ineffective. Second, gastrin-producing islet cell tumors that cause [ZES](#) are frequently multicentric. Recent experience suggests that a high percentage of ZES in MEN 1 is caused by duodenal wall tumors and that resection of these tumors improves the cure rate. Treatment with H₂receptor antagonists (cimetidine or ranitidine) and the H⁺,K⁺-ATPase inhibitors (omeprazole or lansoprazole) provides an alternative to surgery for control of ulcer disease in patients with multicentric tumors or with hepatic metastases. Third, in families in which there is a high incidence of malignant islet cell tumors that cause death, total pancreatectomy at an early age may be justified to prevent malignancy.

Management of metastatic islet cell carcinoma is unsatisfactory. Hormonal abnormalities can sometimes be controlled. For example, [ZES](#) can be treated with H₂receptor antagonists or H⁺,K⁺-ATPase inhibitors; the somatostatin analogue, octreotide, is useful in the management of carcinoid and the watery diarrhea syndrome. Bilateral adrenalectomy may be required for ectopic [ACTH](#) syndrome if medical therapy is ineffective ([Chap. 331](#)). Islet cell carcinomas frequently metastasize to the liver but may grow slowly. Hepatic artery embolization or chemotherapy (5-fluorouracil, streptozocin, chlorozotocin, doxorubicin, or dacarbazine) may reduce tumor mass, control symptoms of hormone excess, and prolong life; however, these treatments are never curative.

Pituitary Tumors Treatment of prolactinomas with dopamine agonists (bromocriptine, cabergoline, or quinagolide) usually returns the serum prolactin level to normal and prevents further tumor growth ([Chap. 328](#)). Surgical resection of a prolactinoma is rarely curative but may relieve mass effects. Transsphenoidal resection is appropriate for neoplasms that secrete [ACTH](#), [GH](#), or the α-subunit of the pituitary glycoprotein hormones. Octreotide reduces tumor mass in one-third of GH-secreting tumors and reduces GH and insulin-like growth factor I levels in >75% of patients. Radiation therapy may be useful for large or recurrent tumors.

Improvements in the management of [MEN 1](#), particularly islet cell and pituitary tumors, have improved outcome in these patients substantially. As a result, other neoplastic manifestations, such as carcinoid syndrome, are now seen with increased frequency.

MULTIPLE ENDOCRINE NEOPLASIA TYPE 2

Clinical Manifestations Medullary thyroid carcinoma (MTC) and pheochromocytoma

are associated in two major syndromes: [MEN](#) type 2A and MEN type 2B ([Table 339-1](#)). MEN 2A is the combination of MTC, hyperparathyroidism, and pheochromocytoma. Three subvariants of MEN 2A are familial medullary thyroid carcinoma (FMTc), MEN 2A with cutaneous lichen amyloidosis, and MEN 2A with Hirschsprung disease. MEN type 2B is the combination of MTC, pheochromocytoma, mucosal neuromas, intestinal ganglioneuromatosis, and marfanoid features.

Multiple Endocrine Neoplasia Type 2A [MTC](#) is the most common manifestation. This tumor usually develops in childhood, beginning as hyperplasia of the calcitonin-producing cells (C cells) of the thyroid. MTC is typically located at the junction of the upper one-third and lower two-thirds of each lobe of the thyroid, reflecting the high density of C cells in this location; tumors >1 cm in size are frequently associated with local or distant metastases. Measurement of the serum calcitonin level after calcium or pentagastrin injection makes it possible to diagnose this disorder when the likelihood of metastasis is low (see below).

Pheochromocytoma occurs in approximately 50% of patients with [MEN](#) 2A and causes palpitations, nervousness, headaches, and sometimes sweating ([Chap. 332](#)). About half the tumors are bilateral, and >50% of patients who have had unilateral adrenalectomy develop a pheochromocytoma in the contralateral gland within a decade. A second feature of these tumors is a disproportionate increase in the secretion of epinephrine relative to norepinephrine. Capsular invasion is common, but malignant behavior is uncommon.

Hyperparathyroidism occurs in 15 to 20% of patients, with the peak incidence in the third or fourth decade. The manifestations of hyperparathyroidism do not differ from those in other forms of primary hyperparathyroidism ([Chap. 341](#)), with nephrolithiasis being common. Diagnosis is established by finding hypercalcemia, hypophosphatemia, hypercalciuria, and an inappropriately high serum level of intact parathyroid hormone. Multiglandular parathyroid hyperplasia is the most common histologic finding, although with long-standing disease adenomatous changes may be superimposed on hyperplasia.

Multiple Endocrine Neoplasia Type 2B The association of [MTC](#), pheochromocytoma, mucosal neuromas, and a marfanoid habitus is designated [MEN](#) 2B. MTC in MEN 2B develops earlier and is more aggressive than in MEN 2A. Metastatic disease has been described prior to 1 year of age, and death commonly occurs in the second or third decade of life. However, the prognosis is not invariably bad even in patients with metastatic disease, as evidenced by a number of multigenerational families with this disease.

Pheochromocytoma occurs in more than half of [MEN](#) 2B patients and does not differ from that in MEN 2A. Hypercalcemia is rare in MEN 2B, and there are no well-documented examples of hyperparathyroidism.

The mucosal neuromas and marfanoid body habitus are the most distinctive features and are recognizable in childhood. Neuromas are present on the tip of the tongue, under the eyelids, and throughout the gastrointestinal tract and are true neuromas, distinct from neurofibromas. Children may present with gastrointestinal symptoms,

including increased gas, intermittent obstruction, and diarrhea caused by neuromas.

GENETIC CONSIDERATIONS

Mutations of the *RET* proto-oncogene have been identified in 93 to 95% of patients with **MEN 2** ([Fig. 339-3](#)). *RET* encodes a tyrosine kinase receptor that is normally activated by glial cell line-derived neurotrophic factor. *RET* mutations induce constitutive activity of the receptor, explaining the autosomal dominant transmission of the disorder.

Naturally occurring mutations localize to two regions of the RET tyrosine kinase receptor. The first is a cysteine-rich extracellular domain; point mutations in the coding sequence for one of five cysteines (codons 609, 611, 618, 620, or 634) cause amino acid substitutions that induce receptor dimerization and activation in the absence of its ligand. Codon 634 mutations occur in 80% of **MEN 2A** kindreds and are most commonly associated with classic MEN 2A features ([Figs. 339-3](#) and [339-2](#)); an arginine substitution at this codon accounts for half of all MEN 2A mutations. All reported families with MEN 2A and cutaneous lichen amyloidosis are consistently associated with a codon 634 mutation. Mutations of codons 609, 611, 618, or 620 occur in 10 to 15% of MEN 2A kindreds and are more commonly associated with **FMTC** ([Fig. 339-3](#)). Mutations in codons 609, 618, and 620 have also been identified in MEN 2A and in the Hirschsprung variant ([Fig. 339-3](#)).

The second region of the RET tyrosine kinase that is mutated in **MEN 2** is in the substrate recognition pocket at codon 918 ([Fig. 339-3](#)). This activating mutation is present in approximately 95% of patients with MEN 2B and accounts for 10 to 15% of all *RET* proto-oncogene mutations in MEN 2. Mutations of codon 883 and 22 have also been identified in a few patients with MEN 2B.

From 3 to 5% of kindreds with **FMTC** have no identifiable mutation of either of these regions. In a few such kindreds mutations of codons 768, 790, 791, 804, and 891 have been identified ([Fig. 339-3](#)).

Somatic mutations (found only in the tumor and not transmitted in the germline) of the *RET* proto-oncogene have been identified in sporadic **MTC**; 25 to 35% of sporadic tumors have codon 918 mutations, and somatic mutations in codons 630, 768, and 804 have also been identified ([Fig. 339-3](#)). Germline mutations of the *RET* proto-oncogene are present in about 6% of patients with apparent sporadic MTC, indicating that other family members may be at risk for the disease.

TREATMENT

Screening for Multiple Endocrine Neoplasia Type 2 Death from **MTC** can be prevented by early thyroidectomy. The identification of *RET* proto-oncogene mutations and the application of DNA-based molecular diagnostic techniques to identify these mutations has simplified the screening process. During the initial evaluation of a kindred, a *RET* proto-oncogene analysis should be performed on an individual with proven **MEN 2A**. Establishment of the specific mutation in a kindred facilitates the subsequent analysis of other family members. Each family member at risk should be tested twice for the presence of the specific mutation; the second analysis should be

performed on a new DNA sample and, ideally, in a second laboratory to exclude sample mix-up or technical error (see endocr06.mda.uth.tmc.edu for a list of laboratory testing sites). Individuals in a kindred with a known mutation who have two normal analyses can be excluded from further screening.

There is general consensus that children with codon 883, 918, and 922 mutations, or those associated with [MEN 2B](#), should have a total thyroidectomy and central lymph node dissection (level VI) performed during the first months of life or soon after identification of the syndrome. If local metastasis is discovered, a more extensive lymph node dissection (levels II to V) is generally indicated. In children with codon 611, 618, 620, 630, 634, and 891 mutations, thyroidectomy should be performed before the age of 6 years because of reports of local metastatic disease in children this age. Finally, there are kindreds with codon 609, 768, 790, 791, and 804 mutations where the phenotype of [MTC](#) appears to be less aggressive. In these kindreds, and in those with rare mutations, two management approaches have been suggested in association with genetic counseling: (1) perform a total thyroidectomy with or without central node dissection at some arbitrary age (perhaps 6 to 12 years of age), or (2) continue annual or biannual provocative testing for calcitonin release with performance of total thyroidectomy with or without central neck dissection when the test becomes abnormal. The pentagastrin test involves measurement of serum calcitonin basally and 2, 5, 10, and 15 min after a bolus injection of 5 ug pentagastrin per kilogram body weight. Patients should be warned before injection of epigastric tightness, nausea, warmth, and tingling of extremities and reassured that the symptoms will last approximately 2 min. The recent unavailability of pentagastrin in the United States has led to use of a short calcium infusion, performed by obtaining a baseline serum calcitonin and then infusing 150 mg calcium salt intravenously over 10 min with measurement of serum calcitonin at 5, 10, 15, 30 min after initiation of the infusion.

The *RET* proto-oncogene analysis should be performed in patients with suspected [MEN 2B](#) to detect codon 883, 918, and 922 mutations, especially in newborn children where the diagnosis is suspected but the clinical phenotype is not fully developed. Other family members at risk for MEN 2B should also be tested because the mucosal neuromas can be subtle and not always identified. In the rare families with proven germline transmission of MTC but no identifiable *RET* proto-oncogene mutation, annual pentagastrin or calcium-pentagastrin testing should be performed on members at risk.

Annual screening for pheochromocytoma in subjects with germline *RET* mutations should be performed by measuring basal plasma or 24-h urine catecholamines and metanephrines. The goal is to identify a pheochromocytoma before it causes significant symptoms or is likely to cause sudden death, an event most commonly associated with large tumors. Although there are kindreds with [FMTC](#) and specific *RET* mutations in which no pheochromocytomas have been identified ([Fig. 339-3](#)), it is not clear that a large enough experience has been gained to exclude pheochromocytoma screening in these individuals. Radiographic studies, such as [MRI](#) or [CT](#) scans, are generally reserved for individuals with abnormal screening tests or with symptoms suggestive of pheochromocytoma ([Chap. 332](#)). Women should be tested during pregnancy because undetected pheochromocytoma can cause maternal death during childbirth.

Measurement of serum calcium and parathyroid hormone levels every 2 to 3 years

provides an adequate screen for hyperparathyroidism, except in those families in which hyperparathyroidism is a prominent component, where measurements should be made annually.

Treatment of Medullary Thyroid Carcinoma [MTC](#) is a multicentric disorder. Total thyroidectomy with a central lymph node dissection should be performed in children who carry the mutant genes. Incomplete thyroidectomy leaves the possibility of later transformation of residual long-term C cells. The goal of early therapy is cure, and a strategy that does not accomplish this goal is short-sighted. Long-term follow-up studies indicate an excellent outcome with approximately 90% of children free of disease 15 to 20 years after surgery. In contrast, 15 to 25% of patients in whom the diagnosis is made on the basis of a palpable thyroid nodule die from the disease within 15 to 20 years.

In adults with [MTC](#) >1 cm in size, metastases to regional lymph nodes are common. Total thyroidectomy with central lymph node dissection and selective dissection of other regional chains provide the best chance for cure. In patients with extensive local metastatic disease in the neck, external radiation may prevent local recurrence or reduce tumor mass but is not curative. Chemotherapy with combinations of adriamycin, vincristine, cyclophosphamide, and dacarbazine may provide palliation.

Treatment of Pheochromocytoma The long-term goal for management of pheochromocytoma is to prevent death and cardiovascular complications. Improvements in radiographic imaging of the adrenals make direct examination of the apparently normal contralateral gland during surgery less important, and the rapid evolution of laparoscopic surgery has simplified management of early pheochromocytoma. The major question is whether to remove both adrenal glands or to remove only the affected adrenal at the time of primary surgery. Issues to be considered in making this decision include the possibility of malignancy (<15 reported cases), the likelihood of developing pheochromocytoma in the apparently unaffected gland over an 8- to 10-year period, and the risks of adrenal insufficiency caused by removal of both glands (at least two deaths related to adrenal insufficiency in [MEN 2](#) patients). Most clinicians recommend removing only the affected gland. If both adrenals are removed, glucocorticoid and mineralocorticoid replacement is mandatory. An alternative approach is to remove the pheochromocytoma and adrenal medulla, leaving the adrenal cortex behind. This approach is usually successful and eliminates the necessity for steroid hormone replacement, although the pheochromocytoma recurs in some.

Treatment of Hyperparathyroidism Hyperparathyroidism has been managed by one of two approaches. Removal of $3\frac{1}{2}$ glands with maintenance of the remaining half gland in the neck is the usual procedure. In families in whom hyperparathyroidism is a prominent manifestation (almost always associated with a codon 634 *RET* mutation) and recurrence is common, total parathyroidectomy with transplantation of parathyroid tissue into the nondominant forearm is preferred. This approach is discussed above in the context of hyperparathyroidism associated with [MEN 1](#).

OTHER GENETIC TUMOR SYNDROMES

A number of mixed syndromes exist in which the neoplastic associations differ from those in [MEN 1](#) or 2 ([Table 339-1](#)).

The cause of von Hippel-Lindau (VHL) syndrome, the association of central nervous system tumors, renal cell carcinoma, pheochromocytoma, and islet cell neoplasms, is mutations in the *VHL* tumor-suppressor gene. Germline-inactivating mutations of the *VHL* gene cause tumor formation when there is additional loss or somatic mutation of the normal *VHL* allele in brain, kidney, pancreatic islet, or adrenal medullary cells. A specific subset of mutations is more common in families with pheochromocytomas.

The molecular defect in type 1 neurofibromatosis inactivates neurofibromin, a cell membrane-associated protein that normally activates a GTPase. Inactivation of this protein impairs GTPase and causes continuous activation of p21 Ras and its downstream tyrosine kinase pathway. Endocrine tumors also form in less common neoplastic genetic syndromes. These include Cowden's disease, Carney complex, familial acromegaly, and familial carcinoid syndrome.

IMMUNOLOGIC SYNDROMES AFFECTING MULTIPLE ENDOCRINE ORGANS

When immune dysfunction affects two or more endocrine glands and other nonendocrine immune disorders are present, the *polyglandular autoimmune* (PGA) *syndromes* should be considered. The PGA syndromes are classified as two main types: the type I syndrome starts in childhood and is characterized by mucocutaneous candidiasis, hypoparathyroidism, and adrenal insufficiency; the type II, or *Schmidt syndrome*, is more likely to present in adults and most commonly comprises adrenal insufficiency, thyroiditis, and type 1 diabetes mellitus. However, the type II syndrome is heterogeneous and may consist of autoimmune thyroid disease along with a variety of other autoimmune endocrine disorders ([Table 339-2](#)).

POLYGLANDULAR AUTOIMMUNE SYNDROME TYPE I

[PGA](#) type I is usually recognized in the first decade of life and requires two of three components for diagnosis: mucocutaneous candidiasis, hypoparathyroidism, and adrenal insufficiency. Mineralocorticoids and glucocorticoids may be lost simultaneously or sequentially. This disorder is also called *autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy* (APECED). Other endocrine defects can include gonadal failure, hypothyroidism, anterior hypophysitis, and, less commonly, destruction of the β cells of the pancreatic islets and development of insulin-dependent (type 1) diabetes mellitus. Additional features include hypoplasia of the dental enamel, ungual dystrophy, tympanic membrane sclerosis, vitiligo, keratopathy, and gastric parietal cell dysfunction resulting in pernicious anemia. Some patients develop autoimmune hepatitis, malabsorption (variably attributed to intestinal lymphangiectasia, IgA deficiency, bacterial overgrowth, or hypoparathyroidism), asplenism, achalasia, and cholelithiasis ([Table 339-2](#)). At the outset, only one organ may be involved, but the number increases with time so that patients eventually manifest two to five components of the syndrome.

Most patients initially present with oral candidiasis in childhood; it is poorly responsive to treatment and relapses frequently. Chronic hypoparathyroidism usually occurs before adrenal insufficiency develops. More than 60% of postpubertal women develop premature hypogonadism. The endocrine components, including adrenal insufficiency

and hypoparathyroidism, may not develop until the fourth decade, making continued surveillance necessary.

Type [IPGA](#) syndrome allows no HLA associations and is inherited as an autosomal recessive trait. The responsible gene, designated as either *APECED* or *AIRE*, encodes a transcription factor that is expressed in thymus and lymph nodes; a variety of different mutations have been reported.

POLYGLANDULAR AUTOIMMUNE SYNDROME TYPE II

[PGA](#) type II is characterized by two or more of the endocrinopathies listed in [Table 339-2](#). Most often these include primary adrenal insufficiency, Graves' disease or autoimmune hypothyroidism, type 1 diabetes mellitus, and primary hypogonadism. Because adrenal insufficiency is relatively rare, it is frequently used to define the presence of the syndrome. Among patients with adrenal insufficiency, type 1 diabetes mellitus coexists in 52% and autoimmune thyroid disease occurs in 69%. However, many patients with antimicrosomal and antithyroglobulin antibodies never develop abnormalities of thyroid function. Thus, increased antibody titers alone are poor predictors of future disease. Other associated conditions include hypophysitis, celiac disease, atrophic gastritis, and pernicious anemia. Vitiligo, caused by antibodies against the melanocyte (see [Plate IIA-11](#)), and alopecia are less common than in the type I syndrome. Mucocutaneous candidiasis does not occur. A few patients develop a late-onset, usually transient hypoparathyroidism caused by antibodies that compete with parathyroid hormone for binding to the parathyroid hormone receptor. Up to 25% of patients with myasthenia gravis, and an even higher percentage who have myasthenia and a thymoma, have PGA type II ([Chap. 380](#)).

The type II syndrome is familial in nature but does not exhibit a characteristic Mendelian pattern of transmission. Like many of the individual autoimmune endocrinopathies, certain HL-DR3 and HLA-DR4 alleles increase disease susceptibility; several different genes probably contribute to the expression of this syndrome.

A variety of autoantibodies are seen in [PGA](#) type II, including antibodies directed against: (1) thyroid antigens such as thyroid peroxidase, thyroglobulin, or the thyroid stimulating hormone (TSH) receptor; (2) adrenal side chain cleavage enzyme, steroid 21-hydroxylase, or [ACTH](#) receptor; and (3) pancreatic islet glutamic acid decarboxylase or the insulin receptor, among others. The roles of cytokines such as interferon and cell-mediated immunity are unclear.

DIAGNOSIS

The clinical manifestations of adrenal insufficiency often develop slowly, may be difficult to detect, and can be fatal if not diagnosed and treated appropriately. Thus, prospective screening should be performed routinely in all patients and family members at risk for [PGA](#) types I and II. The most effective screening test for adrenal disease is a cosyntropin stimulation test ([Chap. 331](#)). A fasting blood glucose level can be obtained to screen for hyperglycemia. Additional screening tests should include measurements of [TSH](#), luteinizing hormone, follicle-stimulating hormone, and, in men, testosterone levels. In families with suspected type I PGA syndrome, calcium and phosphorus levels

should be measured. These screening studies should be performed every 1 to 2 years up to about age 50 in families with PGA type II syndrome and until about age 40 in patients with type I syndrome. Screening measurements of autoantibodies against potentially affected endocrine organs are of uncertain prognostic value. The differential diagnosis of PGA syndrome should include the DiGeorge syndrome (hypoparathyroidism due to glandular agenesis and mucocutaneous candidiasis), Kearns-Sayre syndrome (hypoparathyroidism, primary hypogonadism, type 1 diabetes mellitus, and panhypopituitarism), Wolfram's syndrome (congenital diabetes insipidus and diabetes mellitus), and congenital rubella (type 1 diabetes mellitus and hypothyroidism).

TREATMENT

With the exception of Graves' disease, the management of each of the endocrine components of the disease involves hormone replacement and is covered in detail in the chapters on adrenal, thyroid, gonadal, and parathyroid disease ([Chaps. 330,331,335,336](#), and [341](#)). One aspect of therapy deserves special emphasis. Namely, primary hypothyroidism can mask adrenal insufficiency by prolonging the half-life of cortisol; consequently, administration of thyroid hormone to a patient with unsuspected adrenal insufficiency can precipitate adrenal crisis. Thus, all patients with hypothyroidism in the context of [PGA](#) syndrome should be screened for adrenal disease and, if it is present, be treated with glucocorticoids prior to or concurrently with thyroid hormone therapy.

OTHER AUTOIMMUNE ENDOCRINE SYNDROMES

Insulin Receptor Antibodies Rare insulin-resistance syndromes occur in patients who develop antibodies that block the interaction of insulin with its receptor. Conversely, other classes of anti-insulin receptor antibodies can activate the receptor and can cause hypoglycemia; this disorder should be considered in the differential diagnosis of fasting hypoglycemia ([Chap. 334](#)).

Patients with insulin receptor antibodies and acanthosis nigricans are often middle-aged women who acquire insulin resistance in association with other autoimmune disorders such as systemic lupus erythematosus or Sjogren's syndrome. Vitiligo, alopecia, Raynaud's phenomenon, and arthritis may also be seen. Other autoimmune endocrine disorders, including thyrotoxicosis, hypothyroidism, and hypogonadism, occur rarely. Acanthosis nigricans, a velvety, hyperpigmented, thickened skin lesion, is prominent on the dorsum of the neck and other skin fold areas in the axillae or groin and often heralds the diagnosis in these patients. However, acanthosis nigricans also occurs in patients with obesity or polycystic ovarian syndrome, in which insulin resistance appears to be due to a postreceptor defect; thus acanthosis nigricans itself is not diagnostic of the immunologic form of insulin resistance.

Some patients with acanthosis nigricans have mild glucose intolerance, with a compensatory increase in insulin secretion that is only detected when insulin levels are measured. Others have severe diabetes mellitus requiring massive doses of insulin (several thousand units per day) to lower the blood glucose levels. The nature of the antibodies determines the manifestations; though insulin resistance is more common,

fasting hypoglycemia can result from insulinomimetic antibodies.

Ataxia telangiectasia is an autosomal recessive disorder caused by mutations in *ATM*, a gene involved in cellular responses to ionizing radiation and oxidative damage ([Chap. 364](#)). This disorder is characterized by ataxia, telangiectasia, immune abnormalities, and an increased incidence of malignancies. Insulin-resistant diabetes mellitus occurs and is associated with anti-insulin antibodies.

Autoimmune Insulin Syndrome with Hypoglycemia This disorder typically occurs in patients with other autoimmune disorders and is caused by polyclonal insulin-binding autoantibodies that bind to endogenously synthesized insulin. If the insulin dissociates from the antibodies several hours or more after a meal, hypoglycemia can result. Most cases of the syndrome have been described from Japan, and there may be a genetic component. In plasma cell dyscrasias such as multiple myeloma, the plasma cells may produce monoclonal antibodies against insulin and cause hypoglycemia by a similar mechanism.

Antithyroxine Antibodies and Hypothyroidism Circulating autoantibodies against thyroid hormones in patients with both immune thyroid disease and plasma cell dyscrasias such as Waldenstrom's macroglobulinemia can bind thyroid hormones, decrease their biologic activity, and result in primary hypothyroidism. In other patients the antibodies simply interfere with thyroid hormone immunoassays and cause false elevations or decreases in measured hormone levels.

Crow-Fukase Syndrome The features of this syndrome are highlighted by an acronym that emphasizes its important features: *polyneuropathy, organomegaly, endocrinopathy, M-proteins, and skin changes* (POEMS). The most important feature is a severe, progressive sensorimotor polyneuropathy associated with a plasma cell dyscrasia. Localized collections of plasma cells (plasmacytomas) can cause sclerotic bone lesions and produce monoclonal IgG or IgA proteins. Endocrine manifestations include amenorrhea in women and impotence and gynecomastia in men, hypogonadism, hyperprolactinemia, type 2 diabetes mellitus, primary hypothyroidism, and adrenal insufficiency. Skin changes include hyperpigmentation, thickening of the dermis, hirsutism, and hyperhidrosis. Hepatomegaly and lymphadenopathy occur in about two-thirds of patients, and splenomegaly is seen in about one-third. Other manifestations include increased cerebrospinal fluid pressure with papilledema, peripheral edema, ascites, pleural effusions, glomerulonephritis, and fever. Five-year survival is about 60%.

The systemic nature of the disorder may cause confusion with other connective tissue diseases. The endocrine manifestations suggest an autoimmune basis of the disorder, but circulating antibodies against endocrine cells have not been demonstrated. Increased serum and tissue levels of interleukin 6, interleukin 1b, vascular endothelial growth factor, and tumor necrosis factor α are present, but the pathophysiologic basis for the [POEMS](#) syndrome is uncertain. Therapy directed against the plasma cell dyscrasia such as local radiation of bony lesions, chemotherapy, plasmapheresis, and treatment with all-*trans* retinoic acid may result in endocrine improvement.

MISCELLANEOUS DISORDERS WITH ENDOCRINE MANIFESTATIONS

A variety of other clinical and genetic disorders are associated with multiple endocrine manifestations are summarized in [Table 339-3](#).

(Bibliography omitted in Palm version)

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SECTION 2 -DISORDERS OF BONE AND MINERAL METABOLISM

340. INTRODUCTION TO BONE AND MINERAL METABOLISM - Michael F. Holick, Stephen M. Krane

BONE STRUCTURE AND METABOLISM (See also [Chap. 343](#))

Bone is a dynamic tissue that is remodeled constantly throughout life. The arrangement of compact and cancellous bone provides a strength and density suitable for mobility. In addition, bone provides a reservoir for calcium, magnesium, phosphorus, sodium, and other ions necessary for homeostatic functions. The skeleton is highly vascular and receives about 10% of the cardiac output.

The extracellular components of bone consist of a solid mineral phase in close association with an organic matrix, of which 90 to 95% is type I collagen ([Chap. 351](#)). The noncollagenous portion of the organic matrix contains proteins derived from serum (albumin and α_2 -HS glycoproteins), proteins containing α -carboxyglutamic acid (GLA) [bone GLA protein (BGP), *osteocalcin*, and a matrix GLA protein], the glycoprotein *osteonectin*, the phosphoprotein *osteopontin*, sialoproteins, *thrombospondin*, and other less well characterized proteins. Some of these proteins may function in initiating mineralization and in binding of the mineral phase to the matrix. The mineral phase is made up of calcium and phosphate and is best characterized as a poorly crystalline hydroxyapatite. The mineral phase of bone is deposited initially in intimate relation to the collagen fibrils and is found in specific locations in the "holes" between the collagen fibrils. This architectural arrangement of mineral and matrix results in a two-phase material well suited to withstand mechanical stresses.

Osteoblasts synthesize and secrete the organic matrix. Mineralization of the matrix, both in trabecular bone and in osteons of compact cortical bone (haversian systems), begins soon after the matrix is secreted (primary mineralization) but is not completed until after several weeks (secondary mineralization). Osteoblasts are derived from cells of mesenchymal origin ([Fig. 340-1A](#)). Although relatively little is known about the controls of osteoblast development, two genes have been shown to be important: core-binding factor A1 (*CBFA1*) and Indian hedgehog (*Ihh*). *CBFA1* is a transcription factor and a homologue of the *Drosophila* factor, runt, and is expressed specifically in osteoblast progenitors and regulates the expression of several osteoblast-specific genes including osteopontin, bone sialoprotein, type I collagen, osteocalcin, and receptor-activator of NF κ B (RANK) ligand. *CBFA1* expression is regulated, in part, by bone morphogenetic proteins (BMPs). *Cbfa1*-deficient mice are devoid of osteoblasts. Mice with a functional deletion of *Cbfa1* (*Cbfa1*^{-/-}) have a cartilaginous skeleton but no osteoblasts and no bone, whereas mice with a deletion of only one allele (*Cbfa1*^{+/-}) do have an osseous skeleton but have a delay in intramembranous bone formation of some cranial bones and the clavicles. The latter abnormalities are similar to those in the human disorder cleidocranial dysplasia, which maps to the locus that corresponds to *Cbfa1*.

The growth factor [Ihh](#) also plays a critical role in osteoblast development, as evidenced by the fact that *Ihh*-deficient mice lack osteoblasts in bone formed by endochondral ossification. Numerous other growth-regulatory factors affect osteoblast function,

including transforming growth factor (TGF) β types I and II, acidic fibroblast growth factor (aFGF) and basic fibroblast growth factor (bFGF), platelet-derived growth factor (PDGF), and insulin-like growth factors (IGFs) I and II. Active osteoblasts are characterized by their location and morphology; the presence of a specific skeletal form of alkaline phosphatase; the presence of receptors for parathyroid hormone (PTH) and 1,25-dihydroxyvitamin D [$1,25(\text{OH})_2\text{D}$]; and the ability to synthesize specific matrix proteins, such as type I collagen, osteocalcin, and osteopontin. As an osteoblast secretes matrix, which is then mineralized, the cell becomes an *osteocyte*, still connected with its blood supply through a series of canaliculi. Osteocytes are thought to be the mechanosensors in bone that communicate signals to surface osteoblasts and their progenitors through the canalicular network.

Resorption of bone is carried out mainly by *osteoclasts*, multinucleated cells that are formed by fusion of cells derived from hematopoietic stem cells related to the mononuclear phagocyte series. Multiple factors regulating osteoclast development have been identified ([Fig. 340-1B](#)). Macrophage colony stimulating factor (M-CSF) plays a critical role at several steps in the pathway that ultimately leads to fusion of osteoclast progenitor cells to form multinucleated, active osteoclasts. Discovery of the [RANK](#) signaling pathway provides new insight into the pathway that links osteoblast and osteoclast development ([Fig. 343-2](#)). RANK ligand is expressed on the surface of osteoblast progenitors and stromal fibroblasts. In a process involving cell-cell interactions, it binds to the RANK receptor on osteoclast progenitors, stimulating a signal transduction cascade that leads to osteoclast differentiation and activation. Alternatively, a soluble decoy receptor, referred to as *osteoprotegerin* (OPG), can bind RANK ligand and inhibit osteoclast differentiation. Several growth factors and cytokines, including interleukins (IL) 1, 6, and 11, tumor necrosis factor (TNF), interferon, and M-CSF, modulate the osteoclast differentiation and function. Osteoclasts are also regulated indirectly by osteoblasts and adjacent stromal fibroblasts in the marrow. For example, [PTH](#) receptors are not found on mature osteoclasts, and PTH increases osteoclastic bone resorption by first acting on osteoblasts or stromal fibroblasts. $1,25(\text{OH})_2\text{D}$ receptors are found in precursor cells that can differentiate into monocytes or osteoclasts, and $1,25(\text{OH})_2\text{D}$ also promotes differentiation along the osteoclast pathway.

In the embryo and in the growing child, bone develops by remodeling and replacing previously calcified cartilage (endochondral bone formation) or is formed without a cartilage matrix (intramembranous bone formation). The [PTH](#)/PTHrP receptor plays a central role in the control of chondrocyte differentiation at growth plates ([Chap. 341](#)). [Ihh](#) production by growth plate chondrocytes stimulates the production of PTH-related peptide (PTHrP), which slows the differentiation of chondrocytes. This pathway creates a local feedback system as *Ihh* is suppressed by the actions of PTHrP. Consistent with this mechanism, mice with null mutations of the PTH/PTHrP receptor or PTHrP exhibit growth plate chondrodysplasia that reflects accelerated differentiation of proliferating chondrocytes. In humans, homozygous inactivating mutations of the PTH/PTHrP receptor cause Blomstrand's chondrodysplasia.

New bone, whether formed in infants or in adults during repair, has a relatively high ratio of cells to matrix and is characterized by coarse fiber bundles of collagen that are interlaced and randomly dispersed (woven bone). In adults, the more mature bone is

organized with fiber bundles regularly arranged in parallel or concentric sheets (lamellar bone). In long bones, deposition of lamellar bone in a concentric arrangement around blood vessels forms the haversian systems. Growth in length of bones is dependent on proliferation of cartilage cells and on the endochondral sequence at the growth plate. Growth in width and thickness is accomplished by formation of bone at the periosteal surface and by resorption at the endosteal surface, with the rate of formation exceeding that of resorption. In adults, after the epiphyses close, growth in length and endochondral bone formation cease, except for some activity in the cartilage cells beneath the articular surface. Even in adults, however, remodeling of bone (remodeling of haversian systems as well as trabecular bone) continues throughout life. In adults, ~4% of the surface of trabecular bone (such as iliac crest) is involved in active resorption, whereas 10 to 15% of trabecular surfaces is covered with osteoid. Radioisotope studies indicate that as much as 18% of the total skeletal calcium is deposited and removed each year. Thus, bone is an active metabolizing tissue that requires an intact blood supply.

The response of bone to fractures, infection, and interruption of blood supply and to expanding lesions is relatively limited. Dead bone must be resorbed, and new bone must be formed, a process carried out in association with growth of new blood vessels into the involved area. In injuries that disrupt the organization of the tissue, such as a fracture in which apposition of fragments is poor or when motion exists at the fracture site, the progenitor stromal cells differentiate into cells with functional capacities different from those of osteoblasts, and varying amounts of fibrous tissue and cartilage are formed. When there is good apposition with fixation and little motion at the fracture site, repair occurs predominantly by formation of new bone without other scar tissue.

Remodeling of bone occurs along lines of force modulated by the mechanical stresses to which it is subjected. The signals from these mechanical stresses are sensed by osteocytes, which then transmit other signals to osteoclasts (or their precursors) or osteoblasts (or their precursors). A bowing deformity increases new bone formation at the concave surface and resorption at the convex surface, seemingly designed to produce the strongest mechanical structure. Expanding lesions in bone, such as tumors, induce resorption at the surface in contact with the tumor. Even in a disorder as architecturally disruptive as Paget's disease, remodeling is dictated by mechanical forces. Thus, the plasticity of bone is due to the interaction of cells with each other and with the environment.

The cycle of bone resorption and formation is a highly orchestrated process carried out by the basic multicellular unit (BMU), composed of a group of osteoclasts and osteoblasts ([Fig. 340-2](#)). Osteoclast-mediated resorption of bone takes place in scalloped spaces (*Howship's lacunae*) where the osteoclasts are attached through a specific $\alpha_v\beta_3$ integrin to components of the bone matrix such as osteopontin. This clear zone contains contractile proteins. The resorbing end of the cell forms a specialized ruffled border, which is in contact with the bone. Proteins, including a specialized proton-pump ATPase, are found in the ruffled border membrane and contribute to the acid environment, which solubilizes the mineral phase. In addition to the proton pump, carbonic anhydrase (type II isoenzyme) is required to maintain the acid pH. Other features of active osteoclasts include expression of the proto-oncogene *c-src*, tartrate-resistant acid phosphatase, cell-surface receptors for calcitonin, sodium pumps

of the kidney type, a bicarbonate/chloride exchanger of the band 3 family, and an ability to resorb mineralized bone. The bone matrix is resorbed in the acid environment adjacent to the ruffled border by proteinases that act at low pH, such as cathepsin K, following solubilization of the mineral phase.

Bone formation involves deposition of an organic matrix by osteoblasts followed by mineralization. The mineral phase is composed of calcium and phosphorus, and the concentration of these ions in the plasma and extracellular fluid (ECF) influences the rate at which mineral is formed. In vitro, mineralization can proceed and crystals of hydroxyapatite can grow at concentrations of calcium and phosphorus similar to those in an ultrafiltrate of plasma. The calcium phosphate solid phase at the inception of mineralization is brushite ($\text{CaHPO}_4 \times 2\text{H}_2\text{O}$). As mineralization progresses, the solid phase is a poorly crystalline hydroxyapatite with a relatively low (~1.2) calcium/phosphate molar ratio. With age and maturation, the perfection of the crystals and the calcium/phosphate ratio increase. Fluoride ions, when incorporated into the mineral phase, decrease the proportion of amorphous calcium phosphate and enhance the crystal structure.

There is a limit for the concentration of calcium and phosphorus ions in the [ECF](#) below which mineralization does not occur. A "solubility product" for bone mineral is difficult to calculate because (1) the mineral phase itself is of variable composition, and (2) the various components in ECF that regulate this solubility product are not known. Nevertheless, when the concentrations of calcium and phosphorus in ECF are excessive, a mineral phase may form in areas (e.g., soft tissues) that are not normally mineralized.

Collagens from a variety of sources can catalyze the nucleation of a mineral phase of calcium and phosphorus from solutions of these ions. The organization of collagen probably influences the amount and type of mineral phase formed in bone. The primary structures of type I collagen in skin and bone tissues are similar. There are differences, however, in posttranslational modifications of type I collagen, such as hydroxylations; glycosylations; and the type, number, and distribution of intermolecular cross-links ([Chap. 351](#)). In addition, the holes in the packing structure of the collagen are larger in mineralized collagen of bone and dentin than in unmineralized collagens such as tendon. Single amino-acid substitutions in the helical portion of either the $\alpha 1$ or $\alpha 2$ chain of type I collagen due to mutations in the *COL1A1* or *COL1A2* genes in osteogenesis imperfecta disrupt the organization of bone, and this indicates the importance of the fibrillar matrix in the structure of bone. At the time of their discovery, it was thought that some of the non-collagenous bone proteins -- e.g., osteocalcin (bone-GLA protein), osteonectin, and osteopontin -- played a role in mineralization, although such a role has not been established. Osteocalcin, a product exclusively of osteoblasts, is measurable by immunoassays in normal human serum, and its levels correlate with other measurements of bone formation. Matrix-GLA protein (MGP), a component of bone as well as non-osseous tissues, acts to inhibit mineralization in the non-osseous tissues. Thus, functional deletion of the MGP gene in mice results in massive soft tissue calcification, particularly in arterial walls.

Alkaline phosphatase is a marker for osteoblasts, and cellular levels of this enzyme correlate with rates of bone formation. Although mineralization defects occur in

individuals with mutations that decrease alkaline phosphatase activity (hypophosphatasia), the function of alkaline phosphatase in mineralization is not completely understood. Other circulating markers of bone formation include osteocalcin and type I procollagen C-terminal peptides. Urinary markers for bone resorption are hydroxyproline, hydroxylysine and its glycosides, and the bone-specific hydroxypyridinium collagen cross-links ([Chap. 342](#)). Inorganic pyrophosphate is a potent inhibitor of mineralization at levels below those necessary to bind calcium ions.

CALCIUM METABOLISM

A total of 1 to 2 kg of calcium is present in the average adult, >98% of it in the skeleton. The calcium of the mineral phase at the surface of the crystals is in equilibrium with that in the [ECF](#), but only a minor fraction of the total pool (~0.5%) is exchangeable. In normal adults plasma levels range from 2.2 to 2.6 mmol/L (8.8 to 10.4 mg/dL). The calcium in plasma is present as three forms: free ions, ions bound to plasma proteins, and, to a small extent, diffusible complexes. The concentration of free calcium ions, averaging 1.2 mmol/L (4.8 mg/dL), influences many cellular functions and is subjected to tight hormonal control, especially through [PTH](#) ([Chap. 341](#)). The concentration of serum proteins is an important determinant of calcium ion concentration; most calcium ion is bound to albumin. Ionized calcium can be measured directly with the use of calcium-specific electrodes. If ionized calcium cannot be measured, certain approximations can be used to estimate the protein-bound and ionized fractions. One formula that approximates the amount of calcium bound to protein is

A simplified correction is sometimes used to assess whether the total serum calcium concentration is abnormal when serum proteins are low. The correction is to add 1 mg/dL to the serum calcium level for every 1 g/dL by which the serum albumin level is below 4.0 g/dL. If the serum calcium level, for example, is 7.8 mg/dL (a subnormal value) and the serum albumin level is only 3.0 g/dL, then the stated serum calcium level is corrected by adding 1 mg/dL; the corrected value of 8.8 mg/dL is within the normal range.

The concentration of calcium ions in the [ECF](#) is kept constant by processes that constantly add and remove calcium. Calcium enters the plasma via absorption from the intestinal tract and resorption of ions from the bone mineral. Calcium leaves the ECF via secretion into the gastrointestinal tract (~100 to 200 mg/d), urinary excretion (~50 to 300 mg/d), deposition in bone mineral, and losses in sweat (up to 100 mg/d). Bone resorption and formation are tightly coupled, with approximately 12 mmol (500 mg) calcium entering and leaving the skeleton daily ([Fig. 340-3](#)). Calcium ions inside the cell mediate a variety of cellular functions. The level of free calcium in the cell is very low, approximately 0.1 $\mu\text{mol/L}$; thus, the gradient between plasma and intracellular free calcium is about 10,000 to 1. This gradient is tightly regulated by various channels and ion pumps.

The average dietary calcium intake for most adults in the United States is approximately 15 to 20 mmol/d (0.6 to 0.8 g/d). However, with heightened awareness of the role of adequate calcium intake for the prevention of osteoporosis, many adults on

supplements have an intake of 20 to 37 mmol/d (0.8 to 1.5 g/d). Less than half of dietary calcium is absorbed in adults. Calcium absorption increases during periods of rapid growth in children, in pregnancy, and in lactation and decreases with advancing age. Most of the calcium is absorbed in the proximal small intestine, and the efficiency of absorption decreases in the more distal intestinal segments. Both active transport and diffusion-limited absorption are involved; the former is more important in the upper intestine and the latter in the lower intestine. Both processes are influenced by vitamin D (see below). All forms of calcium in the diet are not equally absorbed; calcium as the chloride is probably absorbed more efficiently than that in other preparations. Secretion of calcium into the intestinal lumen is constant and independent of absorption. If calcium availability in the diet is low [<12 mmol/d (500 mg/d)], a positive calcium balance requires an efficiency of absorption >30 to 40%.

The urinary calcium excretion of normal adults having an average calcium intake ranges between 2.5 and 10 mmol/d (100 and 400 mg/d). When the dietary calcium level is <5 mmol/d (200 mg/d), urinary calcium excretion is usually <5 mmol/d (200 mg/d). However, in most normal individuals, wide variations in dietary intake have little effect on urinary calcium. Hence, when the diet is low in calcium, the relative inefficiency of renal calcium conservation leads to a negative calcium balance unless calcium absorption is maximal ([Fig. 340-3](#)).

The amount of calcium in the urine is small compared with that filtered by the glomerulus [~ 150 to 250 mmol/d (6 to 10 g/d)] because the rates of reabsorption of the filtered calcium are high. Reabsorption takes place predominantly in the proximal tubule ($\sim 60\%$) and in Henle's loop ($\sim 25\%$) and to a small extent in the distal tubule. The calcium-sensing receptor also plays a role in renal calcium excretion, though the mechanisms that regulate its function have not been fully defined ([Chap. 341](#)). The excretion of other electrolytes affects the urinary excretion of calcium. For example, urinary calcium is usually proportional to urinary sodium; sulfate also increases calcium excretion.

A deficiency of [PTH](#) or vitamin D, intestinal disease, or severe dietary calcium deprivation may provide challenges to calcium homeostasis that cannot be compensated adequately by renal calcium conservation, resulting in a negative calcium balance. Increased bone resorption may protect against [ECF](#) calcium depletion even in states of chronic negative calcium balance but only at the expense of progressive bone loss.

PATHOPHYSIOLOGY

A decrease in the concentration of free calcium ions in plasma results in increased neuromuscular irritability and tetany. This syndrome is characterized by peripheral and perioral paresthesia, carpal spasm, pedal spasm, anxiety, seizures, bronchospasm, laryngospasm, Chvostek's sign, Trousseau's sign, and Erb's sign, and lengthening of the QT interval of the electrocardiogram. In infants tetany may be manifested only by irritability and lethargy. The level of calcium ions that determines which features of tetany will be manifested varies among individuals. Tetany is also influenced by other components of the [ECF](#); e.g., hypomagnesemia and alkalosis lower whereas hypokalemia and acidosis raise the threshold for tetany.

Increases in total serum calcium concentration are usually accompanied by increases in free calcium levels and may be associated with anorexia, nausea, vomiting, constipation, hypotonia, depression, and occasionally lethargy and coma. Persistent hypercalcemia, especially when accompanied by normal or elevated levels of serum phosphate, may cause ectopic deposition of a solid phase of calcium and phosphate in walls of blood vessels, connective tissue about the joints, gastric mucosa, cornea, and renal parenchyma. Hypercalcemia per se alters renal function in addition to the pathologic effects of calcium phosphate deposition.

PHOSPHORUS METABOLISM

Phosphorus is a major component of bone and of all other tissues and in some form is involved in almost all metabolic processes, including energy storage, membrane transport, membrane composition, and signal transduction. About 600 g of phosphorus is present in the normal adult, of which 85% is present in the crystalline structure of the skeleton.

In plasma from fasting subjects, most of the phosphorus is present as inorganic orthophosphate in concentrations of approximately 0.75 to 1.45 mmol/L (2.5 to 4.5 mg/dL). In contrast to calcium, of which ~50% is bound, only ~12% of the phosphorus in plasma is bound to proteins. Free HPO_4^{2-} and NaHPO_4 normally account for ~75% of the total phosphorus, and free H_2PO_4 accounts for ~10%. Since so many species are present, depending on pH and other factors, concentrations are usually expressed in terms of elemental phosphorus, in units of mmol/L or mg/dL. The serum phosphorus, however, can vary based on age; young children have almost twice the serum phosphorus as adults due to the need for rapid skeletal mineralization. Postmenopausal women also have higher circulating phosphorus levels. After ingesting a meal containing carbohydrate, there is a decrease in serum phosphorus levels [by 0.3 to 0.5 mmol/dL (1.0 to 1.5 mg/dL)] in response to the increase in insulin secretion, which enhances cellular phosphorus uptake and utilization. An increase in serum pH will decrease serum phosphorus, whereas a decrease in pH increases phosphorus concentration. There is a circadian variation in phosphorus concentration even during a 24-h fast: the nadir occurs between 9:00 A.M. and noon followed by an increase to a plateau in the afternoon and another small peak after midnight.

Phosphorus is plentiful in the diet. Common sources include dairy products, meats, eggs, and carbonated beverages that contain phosphoric acid. Approximately 60 to 70% of phosphorus is passively absorbed in the small intestine ([Fig. 340-4](#)). $1,25(\text{OH})_2\text{D}$ enhances phosphorus absorption along the entire small intestine, with the highest efficiency in the jejunum and ileum. Chronic low phosphorus intake (<2 mg/kg of body weight per day) decreases serum phosphorus levels. Low serum phosphorus stimulates the renal production of $1,25(\text{OH})_2\text{D}$, which, in turn, increases the efficiency of intestinal absorption up to 80 to 90%. $1,25(\text{OH})_2\text{D}$ also decreases [PTH](#) secretion and, thereby, reduces renal tubular loss of phosphorus.

The major control of phosphorus balance is exerted by the kidney. Approximately 90% of phosphorus in the circulation is filtered through the glomerulus and is largely absorbed by the proximal tubule such that only 10 to 15% of the filtered load is normally

excreted. Urinary phosphorus excretion is reflective of dietary intake. Phosphorus absorption in the proximal tubule is coupled with sodium absorption. The primary regulation of phosphorus metabolism occurs in the distal convoluted tubule, and this mechanism is independent of sodium reabsorption. Volume expansion and decreased sodium reabsorption increase phosphorus clearance.

HYPOPHOSPHATEMIA

Causes Although there are many potential causes for hypophosphatemia ([Table 340-1](#)), the most common etiologies include: (1) decreased intestinal phosphorus absorption, either due to vitamin D deficiency or the presence of a phosphorus-binding antacid; (2) urinary losses that are PTH- or alcohol-mediated; and (3) a shift of phosphorus from extracellular to intracellular compartments due to exogenous administration of insulin or consumption of nutrients that stimulate insulin release (e.g., carbohydrates). Increased renal clearance of phosphorus occurs in primary hyperparathyroidism, vitamin D deficiency, vitamin D-resistant and D-dependent rickets, hyperglycemic states, and oncogenic osteomalacia. In vitamin D deficiency, serum phosphorus is low because of decreased intestinal absorption as well as secondary hyperparathyroidism, which increases phosphorus losses in the urine. In X-linked hypophosphatemic rickets, there is a genetic defect in the *PHEX* gene, which encodes a neutral endopeptidase presumed to degrade the phosphaturia hormone known as *phosphatonin*. The disorder is associated with a severe renal leak of phosphorus into the urine. In addition, there is a defect in hypophosphatemia-mediated stimulation of 25(OH)D-1 α -hydroxylase, resulting in decreased intestinal phosphorus absorption. Acidosis and hyperglycemic states associated with polyuria also cause excessive phosphorus loss in the urine. Ketoacidosis enhances intracellular and organic phosphorus degradation, thereby releasing large amounts of inorganic phosphorus into the circulation that is cleared into the urine. In ketosis, the serum phosphorus is often normal because of the continuous shift of phosphorus from intracellular to extracellular pools. However, when the ketosis is corrected, hypophosphatemia is apparent because of the return of phosphorus into the intracellular compartment ([Chap. 333](#)). A severe, acquired form of hypophosphatemia, *oncogenic osteomalacia*, is associated with vascular, mesenchymal tumors such as hemangiopericytomas but occasionally also with small cell lung cancer, prostate cancer, and other malignant tumors. It is likely that these tumors secrete a substance similar or identical to phosphatonin. The phosphorus levels in these patients are usually extremely low [0.4 to 0.5 mmol/L (1.2 to 1.5 mg/dL)], and the 1,25(OH)₂D levels are low or undetectable. The disorder is associated with severe fatigue, muscle weakness, and unrelenting bone discomfort.

Alcohol abuse is the most common cause of severe hypophosphatemia, which is caused by poor dietary intake of phosphorus, ethanol-enhanced urinary excretion of inorganic phosphorus, the use of calcium- or aluminum-containing antacids, and vomiting. Hypophosphatemia may transiently worsen with refeeding. Alcoholics may also have associated calcium and vitamin D deficiency and secondary hyperparathyroidism, which enhances phosphorus-wasting in the urine. Alcoholic ketoacidosis induces marked phosphaturia. Intense hyperventilation for prolonged periods may depress serum phosphorus levels due to associated alkalosis. Rapid correction of chronic respiratory acidosis has also been associated with hypophosphatemia and can lead to diaphragm weakness and an exacerbation of

respiratory failure. Advanced leukemia with blast crisis (leukocyte counts usually $>100,000$) may cause severe hypophosphatemia; the likely cause is a rapid uptake of phosphorus into the rapidly dividing cells.

Laboratory and Clinical Findings Serum phosphorus levels should be determined in a fasting state. Mild hypophosphatemia is not usually associated with clinical symptoms. In severe hypophosphatemia [<0.3 mmol/L (<1.0 mg/dL)], multiple organ systems are affected. Patients become irritable, apprehensive, and hyperventilate, resulting in complaints of muscle weakness, numbness, and paresthesia. In the most severe form, they are confused or obtunded and suffer from seizures and coma, which can ultimately lead to death. This metabolic encephalopathy is often associated with slowing of the electroencephalogram.

Phosphorus is essential for muscle function because of the need for large amounts of ATP and creatine phosphate. Patients with severe hypophosphatemia often complain of fatigue, muscle weakness, myalgia, and myopathy. Hypophosphatemia can cause rhabdomyolysis, which is particularly common in chronic alcoholics or during alcohol withdrawal. Rhabdomyolysis can be precipitated during treatment for diabetic ketoacidosis or by hyperalimentation or refeeding in a malnourished patient. Cardiomyopathy can also occur, resulting in reduced cardiac output, impaired pressor responsiveness to catecholamines, hypotension, and ventricular arrhythmias. Restoration of phosphorus deficits can result in prompt reversal. Severe muscle weakness can lead to respiratory insufficiency.

Erythrocytes and leukocytes are highly dependent on phosphorus for their function. Chronic hypophosphatemia decreases 2,3-bisphosphoglycerate and ATP, enhancing oxygen dissociation from hemoglobin and leading to tissue hypoxia. Hypophosphatemia causes impaired phagocytosis and opsonization and, therefore, increases susceptibility to bacterial and fungal infections.

Chronic hypophosphatemia causes a mineralization defect of the skeleton. In children, this causes rickets. In adults, chronic hypophosphatemia (often due to vitamin D deficiency) causes osteomalacia (see below). Patients with severe renal phosphorus-wasting and severe chronic hypophosphatemia may have marked fatigue, muscle weakness, and severe bone pain, especially of their long bones and ribcage.

TREATMENT

Mild hypophosphatemia usually resolves spontaneously when the underlying cause is corrected. Oral phosphorus replacement is sufficient if serum phosphorus is >0.3 mmol/L (1 mg/dL) and the patient is asymptomatic. Milk is an excellent source of phosphorus as it contains 1 g of inorganic phosphorus per liter. Carbonated beverages that contain phosphoric acid provide another source of phosphorus, especially for patients with lactase deficiency. Pharmaceutical preparations of phosphorus, such as Neutraphos or KPhos, contain sodium and potassium salts of phosphate. Depending on the degree of hypophosphatemia, up to 3 g/d can be given in four to six divided doses per 24 h. These doses usually do not cause diarrhea; >5 g/d will induce diarrhea.

For severe hypophosphatemia, with serum phosphorus levels <0.2 to 0.3 mmol/L (<0.5

to 1.0 mg/dL),³³ g/d of phosphorus may be required over several days to replete body stores. In patients with severe symptomatic hypophosphatemia who are unable to eat, intravenous phosphorus can be given, up to 1 g in 1 L of fluid over 8 to 12 h. Some caution is necessary when giving phosphorus intravenously because of the potential for precipitating soft tissue calcification. A serum calcium $\times$ serum phosphorus product >70 markedly increases the risk of soft tissue calcification and nephrocalcinosis. Patients with chronic hypophosphatemia caused by inherited or acquired renal phosphorus leak require vigilance when receiving high doses of oral phosphorus. Transiently elevated serum phosphorus levels can decrease ionized calcium levels, resulting in chronic stimulation of the parathyroid gland and leading to autonomous, persistent hyperplasia of the parathyroid glands. Thus, it is best to give frequent divided doses of phosphorus (four to six times a day), equaling a total of 2 to 3 g/d.

Phosphorus should not be given intramuscularly or subcutaneously because it can cause soft tissue necrosis and severe discomfort. Intravenous sodium or potassium phosphate, 15 mmol (0.465 g) of elemental phosphorus given in 100 mL of 0.9% saline over 60 min, elevates serum phosphorus levels by an average of 0.6 to 1.2 mmol/L (1.75 to 3.8 mg/dL).

HYPERPHOSPHATEMIA

In adults, hyperphosphatemia is defined as a serum phosphorus level >1.6 mmol/L (5 mg/dL). In children, this level is much higher. The most common causes of hyperphosphatemia are acute and chronic renal failure ([Table 340-2](#)). In renal failure, the loss of tubular function impairs phosphorus excretion. This results in a cascade of events that can also affect calcium and phosphorus metabolism. The increase in serum phosphorus levels reduces serum calcium levels and the production of $1,25(\text{OH})_2\text{D}$, leading to decreased intestinal calcium absorption and secondary hyperparathyroidism. Patients with pseudohypoparathyroidism and tumoral calcinosis also have decreased renal phosphorus clearance that results in hyperphosphatemia. Hypothyroidism reduces renal phosphorus excretion and may increase circulating concentrations of phosphorus. Vitamin D intoxication, due to excessive ingestion of either vitamin D or one of its analogues, can cause hyperphosphatemia along with hypercalcemia. Severe hypothermia, crush injuries, nontrauma rhabdomyolysis, tumoral calcinosis, and cytotoxic therapy of hematologic malignancies such as acute lymphoblastic leukemia can be associated with hyperphosphatemia. The serum phosphorus level can be artifactually elevated due to hemolysis of the blood sample. Thrombocytosis and multiple myeloma can cause spuriously elevated serum phosphorus levels due to thrombocytolysis.

Laboratory and Clinical Findings A rapid elevation of serum phosphorus can cause hypocalcemia and symptoms of neuromuscular irritability and tetany. Chronic hyperphosphatemia in association with normocalcemia can result in nephrocalcinosis and soft tissue calcification.

TREATMENT

In addition to treating the underlying disorder, dietary phosphorus intake should be limited by restricting carbonated beverages containing phosphoric acid and decreasing

milk and dairy product consumption. The dietary intake of phosphorus should be between 600 and 1000 mg a day with modest protein restriction. For control of chronic hyperphosphatemia, usually in patients with chronic renal failure, oral aluminum hydroxide or aluminum carbonate gels are indicated. Prolonged use of aluminum-containing compounds is not recommended because of aluminum toxicity causing adynamic bone disease, proximal myopathy, encephalopathy, and anemia. When hyperphosphatemia is due to vitamin D intoxication, calcium salts are contraindicated because the high efficiency of calcium absorption can lead to severe hypercalcemia, soft tissue calcification, and nephrocalcinosis.

MAGNESIUM METABOLISM

Magnesium is the most abundant intracellular divalent cation. It is an essential cofactor for a multitude of enzymatic reactions that are important for the generation of energy from ATP. Approximately 30% of magnesium in the serum is protein-bound, 55% is ionized, and the remaining 15% is complexed. Like calcium, magnesium is bound to albumin, and it is the ionized fraction that is important for physiologic processes including neuromuscular function and maintenance of cardiovascular tone.

The serum concentration of magnesium is tightly regulated within a narrow range of approximately 0.7 to 1.1 mmol/L (1.4 to 2.2 meq/L)(1.7 to 2.6 mg/dl) as a result of the efficient absorption of dietary magnesium by the small intestine and conservation of magnesium in the kidney. About 30% of dietary magnesium is absorbed in the small intestine, but this fraction increases markedly when intake is substantially reduced. Approximately 96% of filtered magnesium is reabsorbed along the nephron, and only 4% is excreted into the urine. Because there is no regulation of magnesium absorption in the distal tubule and because magnesium reabsorption is very efficient, an increase in distal delivery increases magnesium loss in the urine.

HYPOMAGNESEMIA

Although magnesium deficiency is a common clinical problem, serum magnesium levels are often overlooked or not measured in patients at risk for the disorder. Approximately 10% of patients admitted to city hospitals are hypomagnesemic, and up to 65% of patients in intensive care units may be magnesium-deficient. Hypomagnesemia is caused primarily by renal or gastrointestinal losses or decreased efficiency of intestinal magnesium absorption ([Table 340-3](#)). Reduced renal reabsorption due to loop diuretics and alcohol use is a common cause of hypomagnesemia. Because magnesium excretion is tightly coupled to sodium and calcium excretion, intravenous fluid therapy and volume-expanded states, such as primary hyperaldosteronism, may result in hypomagnesemia. Hypercalcemia and hypercalciuria decrease tubular reabsorption of magnesium. Osmotic diuresis in diabetes mellitus is one of the more common causes of hypomagnesemia.

Vomiting and nasogastric suctioning can cause severe magnesium depletion because intestinal tract fluids contain ~0.5 mmol/L (1.2 mg/dL)(1 meq/L). Fluid loss from diarrhea can contain as much as 7.4 mmol/L (18 mg/dL)(15 meq/L). Consequently, ulcerative colitis, Crohn's disease, and intestinal or biliary fistulas can result in magnesium depletion. Hypomagnesemia is prevalent in alcoholics. Ethanol causes a transient loss

of magnesium in the urine. In most alcoholics, however, the magnesium deficit is modest. A more profound fall in serum magnesium levels may occur during alcohol withdrawal, where the decrease is associated with falls in levels of serum phosphate and potassium, probably due to shifts of these ions into intracellular compartments. The use of loop diuretics, as well as aminoglycosides, cisplatin, cyclosporine, and amphotericin B can increase renal loss of magnesium.

The clinical manifestations of hypomagnesemia are similar to those of severe hypocalcemia. The signs and symptoms of hypomagnesemia include muscle weakness, prolonged PR and QT intervals, and cardiac arrhythmias. Positive Chvostek's and Trousseau's signs indicative of hypocalcemia are often positive in hypomagnesemic patients as well; carpopedal spasm can also occur with hypomagnesemia. Magnesium is important for effective [PTH](#) secretion as well as the renal and skeletal responsiveness to PTH; thus, hypomagnesemia is often associated with hypocalcemia due to impaired PTH secretion and function ([Chap. 341](#)).

Low serum magnesium levels <0.7 mmol/L (1.8 mg/dL)(1.5 meq/L) are indicative of magnesium deficiency. For mild deficiency, oral magnesium replacement is effective. The major side effect is diarrhea. Symptoms often occur when the serum magnesium is <0.5 mmol/L (1.2 mg/dL)(1.0 meq/L). This level is indicative of significantly depleted total-body magnesium stores. Because most magnesium resides in the intracellular space, the total-body magnesium deficit is often ~ 200 mmol (4800 mg) by the time serum levels fall to <0.5 mmol/L (1.2 mg/dL)(1.0 meq/L). Parenteral magnesium administration is usually needed under these circumstances. Two grams of magnesium sulfate [8.0 mmol (192 mg)(16.2 meq) of magnesium] can be given intravenously, with a cumulative dose up to 24 mmol (576 mg)(48 meq) over 24 h. Alternatively, a 50% solution of 2 g of magnesium sulfate can be given every 8 h intramuscularly although these injections can be painful. Patients with severe hypomagnesemia and associated seizures or acute arrhythmias can be given 4 to 8 mmol (96 to 192 mg)(8 to 16 meq) of magnesium as an intravenous injection over 5 to 10 min, followed by 24 mmol/d (576 mg/d)(48 meq/d).

A normal serum magnesium concentration attained after acute magnesium repletion is not necessarily indicative of repletion of the total-body magnesium stores. Restoration of urinary magnesium excretion is a better indicator of magnesium repletion. Once urinary magnesium excretion increases, the body stores are usually replenished. Patients who have chronic magnesium loss from intestinal or renal sources may require continued oral magnesium supplementation on a daily basis of up to 12.5 mmol/d (300 mg/d) in divided doses. Patients with renal failure need to be monitored carefully to prevent hypermagnesemia.

HYPERMAGNESEMIA

Hypermagnesemia is rare but can be seen in renal failure when patients are taking magnesium-containing antacids, laxatives, enemas, or infusions ([Table 340-4](#)). It can also be seen in acute rhabdomyolysis.

The most readily detected clinical sign of hypermagnesemia is the disappearance of deep tendon reflexes. Neuromuscular symptoms include depressed respiration and

apnea due to paralysis of the voluntary muscles, prolonged PR intervals, and increased QRS duration and QT interval; complete heart block and cardiac arrest can occur. Hypocalcemia may occur because hypermagnesemia depresses [PTH](#) secretion and induces an end-organ resistance to PTH similar to the effect seen in hypomagnesemia.

Treatment includes stopping the antacid or other preparations that contain large amounts of magnesium. The excess magnesium is quickly excreted by the kidney. Renal failure patients may require dialysis against a low magnesium bath. For severe hypermagnesemia with associated life-threatening complications, intravenous calcium in doses of 100 to 200 mg (elemental) over 5 to 10 min will antagonize the toxic effects of magnesium.

VITAMIN D

Vitamin D is a hormone rather than a classic vitamin, since with adequate exposure to sunlight, no dietary supplements are needed. Vitamin D exerts its physiologic effects on bone, intestine, kidney, and the parathyroid glands to modulate calcium and phosphorus metabolism. The active principle of vitamin D is synthesized under metabolic control via successive hydroxylations in the liver and kidney and is transported through the blood to its main target tissues (the small intestine and bone), where it regulates calcium homeostasis.

PHOTOBIOGENESIS

Vitamin D₃ is a derivative of 7-dehydrocholesterol (provitamin D₃), the immediate precursor of cholesterol. When skin is exposed to sunlight or certain artificial light sources, the ultraviolet radiation enters the epidermis and causes transformation of 7-dehydrocholesterol to vitamin D₃. Wavelengths between 290 and 315 nm are absorbed by the conjugated double bonds at C₅ and C₇ of 7-dehydrocholesterol to produce previtamin D₃ ([Fig. 340-5](#)). Vitamin D₃ is made in the skin from the previtamin for many hours after a single sun exposure ([Fig. 340-5](#)). Once vitamin D₃ is synthesized, it is translocated from the epidermis into the circulation by the vitamin D-binding protein. Melanin in the skin competes with 7-dehydrocholesterol for ultraviolet photons and thus can limit the synthesis of previtamin D₃. The photochemical isomerization of previtamin D₃ and vitamin D₃ to biologically inert products appears to be more important in preventing excessive production of previtamin D₃ and vitamin D₃ during prolonged exposure to the sun.

Aging decreases the capacity of the skin to produce vitamin D₃; this capacity is reduced more than fourfold after age 70. Topical sunscreens can reduce or prevent cutaneous production of vitamin D₃ by absorbing the solar radiation responsible for previtamin D₃ synthesis in the skin. Other factors that affect the cutaneous synthesis of vitamin D₃ include altitude, geographic location, time of day, and area exposed. Latitude has profound effects on the cutaneous synthesis of vitamin D₃. As the zenith angle of the sun increases with approaching winter, more of the high-energy ultraviolet photons responsible for formation of the previtamin are absorbed by the ozone layer. In an area such as Boston (42°N), the absorption of these photons is so complete that essentially no vitamin D₃ is made in the skin between the months of November through February.

When the entire body is exposed to sufficient sunlight to cause mild erythema, the increase in the blood vitamin D is approximately equivalent to consuming oral doses of 10,000 to 25,000 international units (1 IU = 0.025 ug) of vitamin D. Only when skin irradiation is insufficient to produce the required quantities of vitamin D₃ is dietary supplementation needed to prevent skeletal mineralization defects. The fortification of milk and some cereals with either crystalline vitamin D₂ ([Fig. 340-5](#)) or vitamin D₃ should prevent rickets and osteomalacia. A survey of the vitamin D content in milk from the United States and western Canada revealed, however, that 71% did not contain 80 to 120% of the amount of vitamin D on the label and that ~15% of skim milk did not contain detectable vitamin D.

In 1997, the Food and Nutrition Board for the Institute of the Medicine recommended 200 IU/d as the adequate intake of vitamin D for neonates, children, and adults up to 50 years. For adults 51 to 70 and >71 years, the committee recommended 400 and 600 IU/d, respectively. In the absence of adequate sunlight exposure, all children and adults require at least 400 to 600 IU/d.

METABOLISM

In the liver, vitamin D is metabolized to 25-hydroxyvitamin D [25(OH)D] by hepatic mitochondrial and/or microsomal enzyme(s) ([Fig. 340-5](#)). 25(OH)D is one of the major circulating metabolites, and its half-life is about 21 days. The concentrations of 25(OH)D and some of its metabolites in the serum are measured using competitive binding assays. The normal serum 25(OH)D concentration varies among different laboratories from 20 to 200 nmol/L (8 to 80 ng/mL). Individuals exposed to excessive sunlight may have concentrations of 25(OH)D up to 250 nmol/L (100 ng/mL) without adverse effects on calcium metabolism. The serum 25(OH)D levels usually reflect both 25-hydroxyvitamin D₂ [25(OH)D₂] and 25-hydroxyvitamin D₃ [25(OH)D₃]. The ratio of these two 25-hydroxylated derivatives depends on the relative amounts of vitamins D₂ or D₃ present in the diet and the amount of previtamin D₃ produced by exposure to sunlight.

The hepatic 25-hydroxylation of vitamin D is regulated by a product feedback mechanism. This regulation, however, is not tight; an increase in dietary intake or endogenous production of vitamin D₃ increases 25(OH)D levels in the serum. The levels can rise to >1200 nmol/L (500 ng/mL) when the intake of vitamin D is excessive. Serum 25(OH)D levels are reduced in severe chronic liver disease ([Table 340-5](#)). 25(OH)D is probably not biologically active at physiologic levels in vivo but is active in vitro at high concentrations.

After formation in the liver, 25(OH)D is bound by the vitamin D-binding protein and transported to the kidney for an additional stereospecific hydroxylation on either C₁ or C₂₄ ([Fig. 340-5](#)). The kidney plays a pivotal role in the metabolism of 25(OH)D to the biologically active metabolite. The renal mitochondrial 25(OH)D-1-hydroxylase activity is enhanced by hypocalcemia to increase the rate of conversion of 25(OH)D to 1,25(OH)₂D. Hypocalcemia may not control this hydroxylation directly, however. Any decrease in the serum concentration of calcium below normal is a stimulus for increased secretion of PTH, which increases the synthesis of 1,25(OH)₂D in the renal proximal convoluted tubule. The renal production of 1,25(OH)₂D enhances the effects of PTH in

lowering circulating concentrations (and presumably renal intracellular concentrations) of phosphate ([Fig. 340-6](#)). 1,25(OH)₂D also influences the renal metabolism of 25(OH)D by diminishing 25(OH)D-1α-hydroxylase activity and enhancing the metabolism of 25(OH)D to 24R,25-dihydroxyvitamin D [24,25(OH)₂D].

24,25(OH)₂D is normally present in serum at a concentration of 1 to 10 nmol/L (0.5 to 5.0 ng/mL). 24,25(OH)₂D is also a substrate for renal 25(OH)D-1α-hydroxylase and is converted to 1α,24R,25-trihydroxyvitamin D [1,24,25(OH)₃D], which, in turn, is metabolized to the biologically inactive substance calcitroic acid ([Fig. 340-5](#)). Cultured cells that possess nuclear receptors for 1,25(OH)₂D, such as chondrocytes, skin keratinocytes and fibroblasts, and intestinal and melanoma cells, also metabolize 25(OH)D to 24,25(OH)₂D. Studies of the vitamin D-24-hydroxylase null mice indicate that the major role of 24-hydroxylation is in the regulation of levels of 1,25(OH)₂D.

PHYSIOLOGY

1,25(OH)₂D, produced by the kidney and the placenta, is the only known important metabolite of vitamin D; the potential roles of other metabolites have not been clarified. 1,25(OH)₂D bound to a vitamin D-binding protein is delivered to various target organs, where the free form is taken up by cells and transported to a specific nuclear receptor protein. The vitamin D receptor (VDR) belongs to the nuclear receptor superfamily of steroid-retinoid-thyroid hormone-vitamin D transcription regulatory factors ([Chap. 327](#)). The VDR interacts with the retinoic acid X receptor (RXR) to form a heterodimeric (RXR-VDR) complex that binds to specific DNA sequences, termed the *vitamin D response elements* (VDREs). After 1,25(OH)₂D binds to the receptor, it induces conformational changes that result in the recruitment of a multitude of transcriptional coactivators that stimulate the transcription of target genes. In the intestine, the activated VDR stimulates calcium-binding protein synthesis; in bone, it stimulates production of osteocalcin, osteopontin, and alkaline phosphatase. 1,25(OH)₂D also may have nonnuclear effects on its target tissues; 1,25(OH)₂D increases the transport of calcium from the extracellular to intracellular space, and it can mobilize calcium from intracellular calcium pools and enhance phosphatidylinositol metabolism. In the intestine, the net effect of 1,25(OH)₂D is to stimulate calcium and phosphate transport from the lumen of the small intestine into the circulation ([Fig. 340-6](#)). The effect of 1,25(OH)₂D on the enhancement of bone resorption is synergistic with that of PTH. Mature osteoclasts do not possess receptors for either PTH or 1,25(OH)₂D. Both PTH and 1,25(OH)₂D interact with their specific receptors on osteoblasts or stromal fibroblasts to induce the production of [RANK](#) ligand on the osteoblast's cell surface. As described above, the RANK ligand interacts with the RANK receptor on immature osteoclasts, stimulating immature osteoclastic precursors to differentiate into mature osteoclasts. The role of 1,25(OH)₂D in the renal handling of calcium and phosphorus remains uncertain. Whatever the role of extraintestinal VDRs may be, the compelling evidence is that the phenotype of VDR null mice is corrected in the setting of normal mineral ion homeostasis. Thus the skeletal consequences of VDR ablation are the result of impaired intestinal calcium absorption and/or the accompanying secondary hyperparathyroidism and hypophosphatemia.

Receptors for 1,25(OH)₂D are also present in cells not classically considered target organs for this hormone, including skin, breast, pituitary, parathyroids, pancreatic beta

cells, gonads, brain, skeletal muscle, circulating monocytes, and activated B and T lymphocytes. Although its physiologic role in these cells remains to be determined, 1,25(OH)₂D inhibits proliferation of keratinocytes and fibroblasts, stimulates terminal differentiation of keratinocytes, induces monocytes to produce interleukin (IL)1 and to differentiate into macrophages and osteoclast-like cells, inhibits the production of [PTH](#), and inhibits the production of IL-2 and immunoglobulin by activated T and B lymphocytes, respectively.

In addition, a variety of tumor cell lines, including lines derived from breast carcinomas, melanomas, and promyeloblasts, possess receptors for 1,25(OH)₂D. Tumor cell lines that have 1,25(OH)₂D receptors respond to the hormone by decreasing the rate of proliferation and enhancing differentiation. For example, when malignant receptor-positive human promyelocytic cells (HL-60) are exposed to 1,25(OH)₂D, the cells mature into functioning macrophages within 1 week. Although calcitriol [1,25(OH)₂D] is not useful for the treatment of leukemia, the antiproliferative effects of calcitriol and its analogue calcipotriene provide the rationale for their use in the treatment of psoriasis.

1,25(OH)₂D regulates [PTH](#) synthesis by negative feedback ([Fig. 340-6](#)). This effect is the rationale for giving 1,25(OH)₂D₃ and its less calcemic-inducing analogue 19-nor-1,25-dihydroxyvitamin D₃ ([Fig. 340-7](#)), to lower circulating levels of PTH in patients with chronic renal failure ([Chap. 341](#)).

The principal physiologic mechanism regulating the production of 1,25(OH)₂D appears to involve changes in serum extracellular calcium concentrations that result in reciprocal changes in secretion of [PTH](#), the latter controlling, possibly through actions on serum or tissue phosphorus levels, the rate of 1,25(OH)₂D production. Other factors that enhance 1,25(OH)₂D production include estrogen, prolactin, and growth hormone. Humans adapt to increased calcium requirements during growth, pregnancy, and lactation by increasing the efficiency of intestinal calcium absorption, possibly by enhancing 25(OH)D-1α-hydroxylase activity. During the first two trimesters of pregnancy, the levels of 1,25(OH)₂D increase in proportion to the concentration of the vitamin D-binding protein; levels of free 1,25(OH)₂D do not change. During the last trimester, the need for calcium for mineralization of the fetal skeleton is met by an increase in the concentrations of free 1,25(OH)₂D and enhanced maternal intestinal calcium absorption.

Most measurements of circulating 1,25(OH)₂D in various physiologic or pathologic states utilize a receptor/competitive binding assay. Serum levels of vitamin D and 25(OH)D vary with the season and with vitamin D intake, whereas levels of 1,25(OH)₂D appear to be unaltered by seasonal variation, by increases in dietary vitamin D, or by exposure to sunlight ([Table 340-6](#)); as long as vitamin D supplies and circulating concentrations of 25(OH)D are sufficient, metabolic influences control the renal 25(OH)D-1α-hydroxylase to ensure a closely regulated circulating concentration of 1,25(OH)₂D. The serum concentration of 1,25(OH)₂D ranges from 40 to 160 pmol/L (16 to 65 pg/mL), and its serum half-life is from 3 to 6 h.

PHARMACOLOGY

Casual exposure to sunlight provides most people with adequate vitamin D. In elderly

individuals, exposure of hands, face, and arms to a suberythral dose of sunlight two to three times a week is usually adequate. A variety of over-the-counter vitamin preparations contain 400 IU of either vitamin D₂(ergocalciferol) or vitamin D₃(cholecalciferol). More potent preparations of vitamin D (calciferol) are available in capsule and tablet form (50,000 IU), as oil (500,000 IU/mL), and in oral solution (8000 IU/mL). A single oral dose of 50,000 IU of vitamin D₂ increases the circulating concentration of vitamin D from <25 nmol/L (10 ng/mL) to 130 to 260 nmol/L (50 to 100 ng/mL) within 12 to 24 h; the plasma half-life is about 2 days. Serum concentrations of 25(OH)D and 1,25(OH)₂D are not changed by these doses of vitamin D. For treatment of vitamin D deficiency, 50,000 IU of vitamin D once a week for 8 weeks raises the circulating concentration of 25(OH)D into the normal range; in the presence of secondary hyperparathyroidism, the circulating concentrations of 1,25(OH)₂D can increase to supranormal levels [up to 600 pmol/L (250 pg/mL)]. 25(OH)D₃(calcifediol) available in capsules containing either 20 or 50 ug may be useful in treating vitamin D deficiency [low 25(OH)D concentrations] in patients with severe liver dysfunction. Pharmacologic doses are used to treat disorders of 25(OH)D metabolism; in pharmacologic doses, 25(OH)D₃ is believed to act via interaction with the [VDR](#). Calcitriol is available in capsules containing 0.25 or 0.5 ug and as a solution for intravenous use (1.0 and 2.0 ug/mL). Calcitriol is efficacious in a variety of disorders ([Chap. 341](#)), but even low doses can cause hypercalcemia, leading to attempts to develop analogues with less calcemic activity. Two such calcitriol analogues have been approved in the United States for the treatment of renal osteodystrophy; 19-nor-1,25-dihydroxyvitamin D₂, and 24-epi-1,25-dihydroxyvitamin D₂ ([Fig. 340-7](#)). 1 α -Hydroxyvitamin D₃[1(OH)D₃] is a potent 1,25(OH)₂D₃ agonist that is used in Europe and Japan. The structure of this analogue is identical to that of the natural renal hormone with the exception that it lacks a C₂₅OH. In humans, this analogue is rapidly metabolized by the liver to 1,25(OH)₂D₃. Topical preparations of calcitriol (3 ug/g) in Europe and calcipotriene (50 ug/g) in Europe and the United States are used for the treatment of psoriasis. When applied over a large surface area, both can potentially cause hypercalcemia and hypercalciuria. Oral calcitriol is also effective for psoriasis and psoriatic arthritis.

When vitamin D is chemically manipulated to rotate the A ring through 180°, the C₃b-OH assumes a geometric position that mimics the C₁a-OH ([Fig. 340-7](#)). These compounds, called *pseudo-1 α -hydroxyvitamin D analogues*, include the clinically useful dihydrotachysterol (DHT). This analogue is less effective in stimulating intestinal calcium transport on a weight basis than either vitamin D or 1,25(OH)₂D. Because it does not require 1 α -hydroxylation to be active on intestinal calcium transport, it is 3 to 10 times more potent than vitamin D in disease states that impair renal 25(OH)D-1 α -hydroxylase, such as hypoparathyroidism and chronic renal failure. Dihydrotachysterol is efficiently metabolized in the liver to 25-hydroxy-DHT, which is the biologically active form.

RICKETS AND OSTEOMALACIA

Rickets and osteomalacia are disorders in which mineralization of the organic matrix of the skeleton is defective. These disorders are caused by a number of different conditions associated with vitamin D deficiency or resistance ([Table 340-7](#)). In *rickets*, the growing skeleton is involved; defective mineralization occurs both in the bone and cartilaginous matrix of the growth plate. The term *osteomalacia* is usually used for this mineralization disorder in the adults in whom the epiphyseal growth plates are closed.

For normal skeletal mineralization, sufficient calcium and phosphate must be present at the mineralization sites. In addition, intact metabolic and transport functions of osteoblasts and chondrocytes and adequate production of cross-linked collagen matrix are required. In cartilage, the initial mineral phase is enclosed in membrane-bound extracellular vesicles. If the osteoblast continues to produce matrix components that cannot be mineralized adequately, rickets or osteomalacia results. A characteristic feature of these disorders is therefore an increase in osteoid volume and thickness (the latter being normally <12 to 14 μm) and a decrease in calcification of the mineralization front. This can be detected in unmineralized sections by the fluorescence of previously ingested tetracyclines or by special stains. The inadequate mineralization of the matrix of cartilage in growing children leads to a widening of the epiphyseal plates of the long bones due to a disorganization of the otherwise highly ordered columns of hypertrophied cartilage cells. In addition, the poorly mineralized long bones are incapable of withstanding usual mechanical stresses and tend to undergo bowing deformities. Growth of the epiphyseal plates is diminished, stunting the growth of the long bones. Osteomalacia also compromises the architectural structure and strength of the skeleton in adults, causing an increase in fractures.

PATHOPHYSIOLOGY

A large number of disorders are associated with rickets or osteomalacia, primarily through alterations of vitamin D nutrition or metabolism or because of phosphate wasting ([Table 340-7](#)). *Hypovitaminosis D* results from inadequate endogenous production of vitamin D₃ in the skin, from insufficient dietary supplementation, and/or from an inability of the small intestine to absorb adequate amounts of the vitamin from the diet. Resistance to the effects of vitamin D can result from (1) use of drugs that antagonize vitamin D action, (2) alterations in the metabolism of vitamin D, or (3) deficient or defective receptors for 1,25(OH)₂D. The consequences of hypovitaminosis D include (1) disturbances of mineral ion metabolism and secretion of [PTH](#), and (2) mineralization defects in the skeleton (e.g., rickets in children, osteomalacia in adults). With an adequate glomerular filtration rate (GFR), the main changes are hypophosphatemia, normal or near-normal serum calcium levels, increased levels of PTH, and low levels of 25(OH)D ([Table 340-5](#)).

With regard to calcium metabolism, lack of vitamin D action leads to insufficient intestinal calcium absorption and hypocalcemia. The latter stimulates the secretion of [PTH](#) (secondary hyperparathyroidism), which enhances calcium release from bone, decreases calcium clearance by the kidney, and tends to blunt the hypocalcemia; as a consequence, most patients have a normal or low-normal serum calcium level. (Late in the course of untreated hypovitaminosis D, severe hypocalcemia develops.) Hypophosphatemia is more marked than hypocalcemia, especially in early stages of the deficiency. The efficiency of intestinal phosphate absorption is also decreased. The increased secretion of PTH, although partially effective in minimizing hypocalcemia, leads to urinary phosphate wasting because of decreased renal tubular reabsorption. This latter effect is the most significant factor in causing hypophosphatemia. Aging decreases the responsiveness of the renal 25(OH)D-1-hydroxylase to PTH, decreasing circulating levels of 1,25(OH)₂D and contributing to decreased calcium absorption in the elderly.

Although the conversion of vitamin D to 25(OH)D is impaired in severe chronic liver disease, there is not a strong correlation between low serum 25(OH)D levels and osteopenia. Patients with nephrotic syndrome with >4 g/d of proteinuria may have low 25(OH)D levels owing to loss in the urine of the vitamin D-binding protein with its associated tightly bound 25(OH)D. Circulating levels of 25(OH)D may also be decreased when the metabolism of 25(OH)D to 1,25(OH)₂D is increased, as in sarcoidosis and hyperparathyroidism. Chronic anticonvulsant therapy can also cause the development of osteomalacia or rickets; mineralization defects are worse in patients receiving multiple drug therapy and when vitamin D intake or exposure to sunlight is inadequate. Anticonvulsant drugs have multiple effects on calcium metabolism. Phenobarbital induces hepatic microsomal enzymes, alters the kinetics of the vitamin D-25-hydroxylase, and stimulates bile secretion, resulting in decreased serum concentrations of vitamin D and 25(OH)D. Phenytoin and phenobarbital inhibit intestinal calcium transport and bone mineral mobilization, independent of effects on vitamin D metabolism.

Glucocorticoids in high doses cause osteoporosis but do not induce osteomalacia and rickets ([Chap. 342](#)). Glucocorticoids directly inhibit vitamin D-mediated intestinal calcium absorption and bone mineral mobilization. Patients receiving glucocorticoids chronically may have depressed circulating levels of 1,25(OH)₂D; the mechanism(s) is unknown. Glucocorticoids also exert direct effects to induce apoptosis of osteoblasts and osteocytes.

A genetic defect in the hepatic 25-hydroxylation of vitamin D has not been described, but in one inherited disorder of calcium and bone metabolism, renal production of 1,25(OH)₂D is defective because of recessive mutations 25(OH)D-1 α -hydroxylase activity. In this syndrome of pseudovitamin D-deficient rickets (also known as vitamin D-dependent rickets type I), renal production of 1,25(OH)₂D is impaired, circulating levels of 1,25(OH)₂D are low, but the therapeutic response to physiologic doses of calcitriol (0.25 to 1.0 ug/d) is normal. In patients with a similar phenotype, pseudovitamin D-resistant rickets (vitamin D-dependent rickets type II), mutations in the vitamin D receptor impair its function by altering the binding of the hormone to the receptor or by altering the binding of the receptor heterodimer complex to DNA. Individuals with this disorder have high circulating levels of 1,25(OH)₂D, but the hormone is ineffective because of the receptor defect. Alopecia is another feature of this disorder, suggesting a role for the [VDR](#) in hair follicle development.

Inherited forms of phosphate wasting disorders include X-linked hypophosphatemic rickets and autosomal dominant hypophosphatemic rickets. The gene responsible for the autosomal form is unknown but has been mapped to chromosome 12p13. The X-linked disorder is caused by mutations in the *PHEX* (phosphate-regulating gene with homology to endopeptidases on the X-chromosome) gene. The *PHEX* gene is postulated to encode an osteoblast protein that inactivates phosphatonin, a phosphaturic factor. Consequently, inactivating mutations result in phosphate wasting. The gene is expressed in heterozygotes, suggesting that the disorder is caused by haploinsufficiency. In patients with X-linked hypophosphatemic rickets, serum concentrations of 1,25(OH)₂D are normal or low. Since hypophosphatemia is a potent stimulator of the renal 25(OH)D-1 α -hydroxylase, levels of 1,25(OH)₂D should be high,

which suggests the existence of a functional defect in the 25(OH)D-1 α -hydroxylase in this disorder. Therefore, the combination of calcitriol and phosphate supplements is more effective than therapy with phosphate supplements alone.

Patients with hypocalcemia due to hypoparathyroidism or pseudohypoparathyroidism have lower-than-normal mean serum concentrations of 1,25(OH) $_2$ D, although individual values may be in the normal range. In these patients, small replacement doses of calcitriol (0.25 to 1.0 μ g/d) are effective even when the serum 25(OH)D concentrations are elevated. These observations indicate that absent or ineffective action of [PTH](#) decreases the activity of renal 25(OH)D-1 α -hydroxylase. It is not known whether serum 1,25(OH) $_2$ D levels would be restored if the hyperphosphatemia were controlled.

Patients with tumor-induced (oncogenic) osteomalacia have low levels of serum phosphorus and 1,25(OH) $_2$ D. Some of these tumors, especially malignant carcinomas, produce PTHrP ([Chap. 341](#)), leading to hypercalcemia as well as hypophosphatemia. Other tumors, particularly more benign neoplasms of vascular or mesenchymal origin, may be responsible for severe hypophosphatemia in the presence of normocalcemia. The mechanism for inhibition of 1,25(OH) $_2$ D synthesis remains unknown; after removal of the tumor, however, the serum phosphorus and 1,25(OH) $_2$ D levels return to normal.

It has been suggested that alteration of vitamin D receptor levels in target tissues (such as intestine) could affect calcium and bone metabolism, and bone mineral density appears to be associated with specific polymorphisms of the [VDR](#) gene. These polymorphisms affect the noncoding intervening DNA sequences (introns) or the coding sequence in a way that does not alter the amino acid sequence of the VDR. Some reports suggest that polymorphisms involving the endonucleases Bsm-I and Taq-I (bb, TT) are associated with higher bone mineral density; other studies have not confirmed these findings. The genetic contribution of VDR polymorphic variants to bone mineral density, as well as a number of other diseases with which they have been associated, require additional, larger-scale studies.

CLINICAL FEATURES

The clinical manifestations of rickets are the result of skeletal deformities, susceptibility to fractures, weakness and hypotonia, and disturbances in growth. As the disorder progresses, particularly that associated with vitamin D deficiency, children are unable to walk without support due to the skeletal deformities in the lower limbs and severe muscle weakness ([Fig. 340-8](#)). Abnormal parietal flattening and frontal bossing develops in the skull. The calvariae are softened (*craniotabes*) and sutures may be widened. Prominence of the costochondral junctions is called the *rachitic rosary* and the indentation of the lower ribs at the site of attachment of the diaphragm is known as *Harrison's groove*. If untreated, deformities of the pelvis and extremities progress, with bowing being particularly common in the tibia, femur, radius, and ulna. For women, the flattening of the pelvis increases the risk of maternal and infant morbidity and mortality during childbirth. Fractures are frequent, dental eruption is often delayed, and enamel defects are common.

The presentation of osteomalacia in adults is usually more insidious. Bone pain and muscle weakness are common complaints and may be overlooked as indicators of

vitamin D deficiency. It is estimated in the United States and Europe that >40% of the adult population over the age of 50 are vitamin D deficient. Although the standard lower limit of the normal range for 25(OH)D is 10 ng/mL, several studies suggest that the cutoff for vitamin D deficiency should be increased to 20 ng/mL. This suggestion is based on [PTH](#) responses to various serum levels of 25(OH)D. For example, elevated levels of PTH are frequently seen in individuals with 25(OH)D levels between 10 and 20 ng/mL, and administration of vitamin D (50,000 units of vitamin D once a week for 8 weeks) lowers PTH levels. Defects in skeletal mineralization may accompany these disturbances in vitamin D and mineral metabolism.

Pain in the hips may result in an antalgic gait. Muscle weakness is often associated with osteomalacia but is difficult to distinguish from hesitancy to move because of skeletal pain. Proximal weakness may mimic that of primary muscle disorders and contribute to the waddling gait. Many factors, including secondary hyperparathyroidism, hypophosphatemia, and vitamin D deficiency, contribute to the myopathy. Fractures of the involved bones may occur with minimal trauma. When the ribs are involved, severe deformities may develop in the thorax, and the collapse of the vertebral bodies may produce loss of height.

RADIOLOGIC FEATURES

In rickets, the most prominent radiologic alteration is evident at the growth plate (physis) which is increased in thickness, cupped, and hazy at the metaphyseal border owing to decreased calcification of the hypertrophic zone and inadequate mineralization of the primary spongiosa. The trabecular pattern of the metaphysis is abnormal, the cortices of the diaphysis may be thinned, and the shafts may be bowed.

In osteomalacia, a decrease in bone density is usually associated with loss of trabeculae and thinning of the cortices. Radiologic and bone densitometric changes are indistinguishable from those in osteoporosis ([Chap. 342](#)). Trabecular patterns may be blurred, producing a homogeneous "ground glass" appearance. Radiolucent bands, ranging from a few millimeters to several centimeters in length and usually oriented perpendicular to the surface of the bones, suggest the presence of osteomalacia. They are particularly common at the inner aspects of the femur (especially near the femoral neck), in the pelvis, in the outer edge of the scapula, in the upper fibula, and in the metatarsals. These radiolucent bands, called *pseudofractures* or *Looser's zones*, occur most often where major arteries cross the bones and are thought to be due to the pulsation of these vessels in the undermineralized area ([Fig. 340-9](#)). Increased rather than decreased density of bones may be observed in patients who have renal tubular disorders rather than vitamin D deficiency and may produce a striking thickening of the cortices and a trabecular pattern of the spongy bone. Despite the increase in bone mass per unit volume, the trabeculae are covered with thickened osteoid seams typical of osteomalacia. Similar findings may occur in patients with chronic renal failure. The reason for the hyperostosis is unknown; the bone is architecturally abnormal and is subject to fracture with minimal trauma.

LABORATORY FINDINGS

Changes in serum concentration of calcium, inorganic phosphorus, 25(OH)D, and

1,25(OH)₂D vary with the different disorders. In vitamin D deficiency, whether due to dietary lack, inadequate sunlight exposure, or intestinal malabsorption, serum calcium levels are normal or low, whereas phosphorus and 25(OH)D levels are consistently low, the latter usually <15 nmol/L (<10 ng/mL) depending on the assay used. In contrast, levels of 1,25(OH)₂D may be normal or elevated owing to secondary hyperparathyroidism and the fact that circulating levels of 1,25(OH)₂D are 1000-fold less than those of 25(OH)D. Only when vitamin D deficiency is chronic and severe is hypocalcemia observed. It may be sufficiently severe to produce tetany. Mild acidosis and generalized aminoaciduria result from secondary hyperparathyroidism. Patients with renal tubular disorders have normal serum calcium levels and hypophosphatemia. Other laboratory findings such as glucosuria, aminoaciduria, acidosis, and hypouricemia reflect variable degrees of disturbance of proximal tubular function or are features of underlying disease (e.g., low plasma ceruloplasmin in Wilson's disease or abnormalities of immunoglobulins in multiple myeloma). In chronic renal failure, hyperphosphatemia and hypocalcemia are usually accompanied by normal 25(OH)D and low 1,25(OH)₂D levels. In nephrotic syndrome, serum 25(OH)D levels can be low owing primarily to urinary losses of vitamin D binding protein-bound 25(OH)D. Serum phosphorus levels are also normal or elevated in hypophosphatasia. Markers of bone resorption increase when secondary hyperparathyroidism and excessive bone resorption are associated with the defect in mineralization. Alkaline phosphatase levels in serum are usually elevated in rickets and osteomalacia.

TREATMENT

In rickets and osteomalacia due to dietary absence of vitamin D or inadequate exposure to sunlight, vitamin D₂(ergocalciferol) or vitamin D₃(cholecalciferol) is given orally in doses of 800 to 4000 IU (0.02 to 0.1 mg) daily for 6 to 12 weeks, followed by daily supplements of 200 to 600 IU, which are adequate to prevent the development of the disorder in otherwise normal persons. In elderly persons with vitamin D deficiency, the administration of 50,000 IU vitamin D by mouth once each week for 8 weeks raises the serum levels of 25(OH)D into the mid-normal range. In infants and children, such treatment causes improvement in muscle tone and strength, an increase in serum calcium and phosphorus levels, and a decrease in alkaline phosphatase levels after several weeks. Radiologic evidence of healing appears within weeks, and healing may be complete by a few months. Calcium supplements and larger initial doses of vitamin D may be necessary in infants and children with tetany. In adults with nutritional osteomalacia, healing of pseudofractures may be evident within 3 to 4 weeks after therapy with as little as 2000 IU (0.5 mg) vitamin D daily. Healing is usually complete by 6 months.

Patients with osteomalacia due to intestinal malabsorption do not respond to small doses of vitamin D. In the presence of active steatorrhea, daily oral doses of vitamin D of 50,000 to 100,000 IU (1.25 to 2.5 mg) and large doses of calcium (e.g., 15 g calcium lactate or 4 g calcium carbonate orally per day) may be required. In some instances, oral vitamin D is ineffective, and the parenteral route is required (e.g., 10,000 IU/d intramuscularly). Another approach is the use of artificial ultraviolet B radiation or exposure to sunlight in addition to supplemental calcium. Small doses of calcitriol (0.5 to 1.0 µg daily) are also usually effective in this form of osteomalacia. Inorganic phosphate therapy is not indicated either in deficiency or in intestinal malabsorption of the vitamin,

since hypocalcemia will develop and intestinal calcium absorption will remain inadequate. In all patients in whom large doses of vitamin D are used, serum calcium and 25(OH)D levels should be monitored periodically. Semiquantitative urinary calcium measurements are inadequate.

In patients treated with multiple anticonvulsant agents, it is usually necessary to continue the drugs while adding 1000 IU/d of vitamin D and to monitor levels of serum calcium and serum 25(OH)D until a therapeutic response (evidence of radiologic healing, improvement in symptoms) is obtained.

Treatment of rickets and osteomalacia in the presence of renal tubular disorders is more difficult. Oral supplements of inorganic phosphate in divided doses of phosphorus (as elemental P), 1.0 to 3.6 g/d (50 mg/kg body weight per day for children), and calcitriol, 0.5 to 2.0 ug/d (30 ng/kg body weight per day for children), constitute the best regimen to restore skeletal growth and heal the bone disease. Patients with nephrotic syndrome and low serum 25(OH)D levels benefit from modest vitamin D supplementation (800 to 1000 IU/d). Small doses of calcitriol are equally effective in treating hypocalcemia and osteodystrophy resulting from chronic renal failure. The recommended initial dose of calcitriol is 0.25 ug/d. If after 2 to 4 weeks on this dose the biochemical parameters are unaltered, the dose is increased by 0.25 ug/d every 2 to 4 weeks until a satisfactory clinical biochemical response (including elevation of serum calcium levels and decrease in [PTH](#) levels) is obtained. The usual dose is 0.5 to 1.0 ug/d. Calcitriol may also be administered intravenously (1.0 to 2.5 ug three times weekly) to patients on dialysis, particularly to treat refractory osteitis fibrosa. Serum calcium levels should be monitored frequently during the first 1 to 2 months of therapy and less frequently once a stable dose has been established.

In patients who have had rickets in childhood, the abnormal mechanical stress of severe deformities may contribute to the development of degenerative joint disease, particularly in the hips and knees. Osteotomies at the proper time after healing may prevent this complication and the requirement for more extensive arthroplasties later in life.

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341. DISEASES OF THE PARATHYROID GLAND AND OTHER HYPER- AND HYPOCALCEMIC DISORDERS - John T. Potts, Jr

The four parathyroid glands are located posterior to the thyroid gland. They produce parathyroid hormone (PTH), which is the primary regulator of calcium physiology. PTH acts directly on bone, where it induces calcium resorption, and on the kidney, where it stimulates calcium reabsorption and synthesis of 1,25-dihydroxyvitamin D [$1,25(\text{OH})_2\text{D}$], a hormone that stimulates gastrointestinal calcium absorption. Serum PTH levels are tightly regulated by a negative feedback loop. Calcium, acting through the calcium-sensing receptor, and vitamin D, acting through its nuclear receptor, inhibit PTH synthesis and release. Understanding the hormone pathways that regulate calcium levels and bone metabolism is essential for effective diagnosis and management of a wide array of hyper- and hypocalcemic disorders.

Hyperparathyroidism, characterized by excess production of [PTH](#), is a common cause of hypercalcemia and is usually the result of autonomously functioning adenomas or hyperplasia. Surgery for this disorder is highly effective and has been shown recently to reverse some of the deleterious effects of long-standing PTH excess on bone density. Hypercalcemia of malignancy is also common and is usually due to the overproduction of parathyroid hormone-related peptide (PTHrP) by cancer cells. The similarities in the biochemical characteristics of hyperparathyroidism and hypercalcemia of malignancy, first noted by Albright in 1941, are now known to reflect the actions of PTH and PTHrP through the same G protein-coupled (GPC) PTH/PTHrP receptor.

Clarification over the past decade of genetic influences on parathyroid gland and bone cell function helps in constructing a logical approach to hyper- and hypocalcemic disorders. Advances that have occurred include elucidation of the genetic basis of multiple endocrine neoplasia (MEN) types 1 and 2, familial hypocalciuric hypercalcemia (FHH), the different forms of pseudohypoparathyroidism (PHP), Jansen's syndrome, disorders of vitamin D synthesis and action, and the molecular events associated with parathyroid gland neoplasia. The advent of new drugs, including bisphosphonates and selective estrogen receptor modulators (SERMs), offers new avenues for the treatment and prevention of metabolic bone disease. PTH analogues are promising therapeutic agents for the treatment of postmenopausal or senile osteoporosis, and calcimimetic agents, which act through the calcium-sensing receptor, may provide new approaches for PTH suppression.

PARATHYROID HORMONE

PHYSIOLOGY

The primary function of [PTH](#) is to maintain the extracellular fluid (ECF) calcium concentration within a narrow normal range. The hormone acts directly on bone and kidney and indirectly on intestine through its effects on synthesis of $1,25(\text{OH})_2\text{D}$ to increase serum calcium concentrations; in turn, PTH production is closely regulated by the concentration of serum ionized calcium. This feedback system is the critical homeostatic mechanism for maintenance of ECF calcium. Any tendency toward hypocalcemia, as might be induced by calcium-deficient diets, is counteracted by an increased secretion of PTH. This in turn (1) acts to increase the rate of dissolution of

bone mineral, thereby increasing the flow of calcium from bone into blood; (2) reduces the renal clearance of calcium, returning more of the calcium filtered at the glomerulus into ECF; and (3) increases the efficiency of calcium absorption in the intestine. Immediate control of blood calcium is probably due to effects of the hormone on bone and, to a lesser extent, on renal calcium clearance. Maintenance of steady-state calcium balance, on the other hand, probably results from the effects of $1,25(\text{OH})_2\text{D}$ on calcium absorption ([Chap. 340](#)). The renal actions of the hormone are exerted at multiple sites and include inhibition of phosphate transport (proximal tubule), increased reabsorption of calcium (distal tubule), and stimulation of the renal $25(\text{OH})\text{D}-1\alpha$ -hydroxylase. As much as 12 mmol (500 mg) calcium is transferred between the ECF and bone each day (a large amount in relation to the total ECF calcium pool), and PTH has a major effect on this transfer. The homeostatic role of the hormone can preserve calcium concentration in blood acutely at the cost of bone destruction.

[PTH](#) has multiple actions on bone, some direct and some indirect. It increases the rate of calcium release from bone into blood acutely; PTH-mediated changes in bone calcium release can be seen within minutes. The chronic effects of PTH are to increase the number of bone cells, both osteoblasts and osteoclasts, and to increase the remodeling of bone; these effects are apparent within hours after the hormone is given and persist for hours after PTH is withdrawn. Continuous exposure to elevated levels of PTH for days (as in hyperparathyroidism or long-term infusions in animals) leads to increased osteoclast-mediated bone resorption. However, the administration of PTH intermittently over days in animals or osteoporotic patients leads to a net stimulation of bone formation rather than bone breakdown. Striking increases, especially in trabecular bone in the spine and hip, have been reported with the use of PTH in combination with estrogen. PTH as monotherapy caused a highly significant reduction in fracture incidence in a worldwide placebo-controlled trial.

Osteoblasts (or stromal cell precursors), which have [PTH](#) receptors, are crucial to this bone-forming effect of PTH; osteoclasts, which appear to lack PTH receptors, mediate bone breakdown. PTH-mediated stimulation of osteoclasts is believed to be indirect, acting in part through cytokines released from osteoblasts to activate osteoclasts; in experimental studies of bone resorption in vitro, osteoblasts must be present for PTH to activate osteoclasts to resorb bone. The nature of the cytokines that stimulate osteoclasts is a subject of major interest. Insulin-like growth factor 1, interleukin 6, granulocyte-macrophage colony stimulating factor, and possibly other agents are candidates, but the definitive messenger(s) has not been determined. Direct cell-to-cell contact between osteoblasts (stromal cells) and osteoclast precursors is also key to osteoclast function. Cell-associated ligands and receptors, as well as soluble decoy receptors, are involved in these interactions ([Chap. 340](#)).

STRUCTURE

[PTH](#) is an 84-amino-acid single-chain peptide. The amino acid sequence of PTH has been characterized in multiple mammalian species, revealing marked conservation in the amino-terminal portion, which is critical for many biologic actions of the molecule. Synthetic fragments of the amino-terminal sequence as small as 1-14 residues are sufficient to activate the major receptor (see below). Biologic roles for the

carboxyl-terminal region of PTH are under investigation; a separate receptor may exist for this region of the molecule. Fragments shortened or modified at the amino terminus still bind to the PTH receptor but lose the capacity to stimulate biologic responses. For example, the peptide composed of sequences 7-34 is a competitive inhibitor of active hormone binding to receptors in vitro but is a weak inhibitor in vivo.

BIOSYNTHESIS, SECRETION, AND METABOLISM

Synthesis Parathyroid cells have multiple methods of adapting to increased needs for PTH production. Most rapid (within minutes) is secretion of preformed hormone in response to hypocalcemia. Second, within hours, changes in gene activity and increased PTH mRNA are induced by sustained hypocalcemia. Finally, protracted challenge leads within days to cellular replication to increase gland mass.

PTH is initially synthesized as a larger molecule (preproparathyroid hormone, consisting of 115 amino acids), which is then reduced in size by a second cleavage (proparathyroid hormone, 90 amino acids) before secretion as the 84-amino-acid peptide. The hydrophobic regions of the preproparathyroid hormone serve a role in guiding transport of the polypeptide from sites of synthesis on polyribosomes through the endoplasmic reticulum to secretory granules. In one kindred with hypoparathyroidism, a mutation in the preprotein region of the gene disrupts this critical hydrophobic sequence and interferes with hormone secretion.

Studies with cloned and expressed PTH genes in vitro have demonstrated DNA regions involved in transcriptional control, including sites for interaction and regulation by the vitamin D receptor, as well as sites through which ambient calcium regulates transcription. Suppression of PTH gene activity at the transcriptional level by calcium is nearly maximal at physiologic concentrations; hypercalcemia results in no significant change. Hypocalcemia, however, increases transcriptional activity within hours. $1,25(\text{OH})_2\text{D}_3$ strongly suppresses PTH gene transcription, though not when chronic hypocalcemia is induced experimentally in animals. In patients with renal failure, however, intravenous administration of supraphysiologic levels of $1,25(\text{OH})_2\text{D}_3$ or analogues of the active metabolite can dramatically suppress PTH overproduction, which is sometimes difficult to control due to severe secondary hyperparathyroidism. Control over hormone stores is exerted by variation in the rates of proteolytic destruction of preformed hormone under the control of ECF calcium; high calcium increases and low calcium inhibits the proteolytic destruction of hormone stores. Regulation of hormone precursor processing and proteolytic destruction of preformed hormone (posttranslational regulation of hormone production) is an important mechanism for mediating rapid (minutes) changes in hormone availability.

Regulation of PTH Secretion PTH secretion increases steeply to a maximum value of five times the basal rate of secretion as calcium concentration falls from normal to the range of 1.9 to 2.0 mmol/L (7.5 to 8.0 mg/dL) (measured as total calcium). The ionized fraction of blood calcium is the important determinant of hormone secretion. Magnesium may influence hormone secretion in the same direction as calcium. It is unlikely, however, that physiologic variations in magnesium concentration affect PTH secretion. Severe intracellular magnesium deficiency impairs PTH secretion (see below).

The level of [ECF](#) calcium controls [PTH](#) secretion by interaction with a calcium sensor, a [GPCR](#) for which Ca^{2+} ions act as the ligand (see below). This receptor is a member of a distinctive subfamily of the GPCR superfamily that is characterized by a large extracellular domain suitable for "clamping" the small-molecule ligand. Stimulation of the receptor by high calcium levels leads to suppression of PTH secretion. The intracellular signals generated by the active receptor appear to be inositol triphosphate (IP_3) and diacylglycerol (DAG) formed by activation of phospholipase. The receptor is present in parathyroid glands and the calcitonin-secreting cells (C cells) of the thyroid, brain, and kidney. Genetic evidence has revealed a key biologic role for the calcium-sensing receptor in parathyroid gland responsiveness to calcium and, unexpectedly, in renal calcium clearance. Point mutations associated with loss of function cause a syndrome resembling hyperparathyroidism ([FHH](#)) but with hypocalciuria. On the other hand, gain-of-function mutations cause a form of hypocalcemia resembling hypoparathyroidism (see below).

Metabolism The secreted form of [PTH](#) is indistinguishable by immunologic criteria and by molecular size from the 84-amino-acid peptide (PTH 1-84) extracted from glands. However, much of the immunoreactive material found in the circulation is smaller than the extracted or secreted hormone. The principal circulating fragments of immunoreactive hormone lack a portion of the critical amino-terminal sequence required for biologic activity and, hence, are biologically inactive fragments (so-called middle- and carboxyl-terminal fragments). Much of the proteolysis of hormone occurs in the liver and kidney. However, fragments corresponding to the middle- and carboxyl-terminal portions have also been detected in effluent blood from the parathyroids and in the peripheral circulation; there is no convincing evidence, however, for circulating amino-terminal fragments. Peripheral metabolism of PTH does not appear to be regulated by physiologic states (high versus low calcium, etc.); hence peripheral metabolism of hormone, although responsible for rapid clearance of secreted hormone, appears to be a high-capacity, metabolically invariant catabolic process.

The rate of clearance of the secreted 84-amino-acid peptide from blood is more rapid than the rate of clearance of the biologically inactive fragment(s) corresponding to the middle- and carboxyl-terminal regions of [PTH](#). Consequently, the interpretation of PTH immunoassays is influenced by the nature of the peptide fragments detected by the antibodies. Before the introduction of double-antibody assays designed to detect intact, biologically active hormone, most immunoassays also measured biologically inert long-lived fragments. Changes in the rate of production or clearance of fragments therefore alter the concentration of immunoreactive hormone.

Although the problems inherent in [PTH](#) measurements have been largely circumvented by use of double-antibody assays that detect only the intact molecule, new evidence has revealed the existence of a hitherto unappreciated larger PTH fragment that may affect the interpretation of most currently available double-antibody assays as well. A large amino-terminally truncated form of PTH, possibly PTH(7-84), is present in normal and uremic individuals in addition to PTH(1-84). The concentration of the putative 7-84 fragment relative to that of intact PTH(1-84) is higher with induced hypercalcemia (e.g., in uremic patients) than in eucalcemic or hypocalcemic conditions. The large fragment almost certainly cannot have (on the basis of structure-activity studies with PTH discussed above) much, if any, of the biologic potency of PTH. The suggestion that the

PTH(7-84)-like fragment might act as an inhibitor of PTH action remains to be clarified. The identification of this fragment has clinical significance, particularly in renal failure, as efforts to prevent secondary hyperparathyroidism by a variety of measures (vitamin D analogues, higher calcium intake, and phosphate-lowering strategies) may have led to oversuppression of biologically active intact PTH when the presence of the amino-terminally truncated PTH was not appreciated. The role, if any, of excessive PTH suppression due to inaccurate measurement of PTH in adynamic bone disease in renal failure (see below) is unknown. Newer assays with extreme amino-terminal epitopes are being studied intensively.

PARATHYROID HORMONE-RELATED PROTEIN

The paracrine factor termed *PTHrP* is responsible for most instances of hypercalcemia of malignancy, a syndrome that resembles hyperparathyroidism. Many different cell types produce [PTHrP](#), including brain, pancreas, heart, lung, mammary tissue, placenta, endothelial cells, and smooth muscle. In fetal animals, PTHrP directs transplacental calcium transfer, and high concentrations of PTHrP are produced in mammary tissue and secreted into milk. Human and bovine milk, for example, contain very high concentrations of the hormone; the biologic significance of the latter is unknown. PTHrP may also play a role in uterine contraction and other biologic functions, still being clarified in other tissue sites.

[PTH](#) and [PTHrP](#), although distinctive products of different genes, exhibit considerable functional and structural homology ([Fig. 341-1](#)) and may have evolved from a shared ancestral gene. The structure of the gene for human PTHrP, however, is more complex than that of PTH, containing multiple exons and multiple sites for alternate splicing patterns during formation of the mRNA. Protein products of 141, 139, and 173 amino acids are produced, and other molecular forms may result from tissue-specific degradation at accessible internal cleavage sites. The biologic roles of these various molecular species and the nature of the circulating forms of PTHrP are unclear. It is uncertain whether PTHrP circulates at any significant level in normal human adults; as a paracrine factor, PTHrP may be produced, act, and be destroyed locally within tissues. In adults PTHrP appears to have little influence on calcium homeostasis, except in disease states, when large tumors, especially of the squamous cell type, lead to massive overproduction of the hormone ([Fig. 341-1](#)).

PTH AND PTHRP HORMONE ACTION

Because [PTHrP](#) shares a significant homology with [PTH](#) in the critical amino terminus, it binds to and activates the PTH/PTHrP receptor, indistinguishably from effects seen with PTH. The 500-amino-acid PTH/PTHrP receptor (also known as the PTH1 receptor) belongs to a subfamily of [GPCR](#) that includes those for glucagon, secretin, and vasoactive intestinal peptide. The extracellular regions are involved in hormone binding, and the intracellular domains, after hormone activation, bind G protein subunits to transduce hormone signaling into cellular responses through stimulation of second messengers ([Fig. 341-2](#)). A second PTH receptor (PTH2 receptor) is expressed in brain, pancreas, and several other tissues. Its amino acid sequence and the pattern of its binding and stimulatory response to PTH and PTHrP differ from those of the PTH1 receptor. The PTH/PTHrP receptor responds equivalently to PTH and PTHrP, whereas

the PTH2 receptor responds only to PTH. The endogenous ligand and the physiologic significance of this receptor are not completely defined.

The PTH1 and PTH2 receptors can be traced backward in evolutionary time to fish. The zebrafish PTH1 and PTH2 receptors exhibit the same selective responses to [PTH](#) and [PTHrP](#) as do the human PTH1 and PTH2 receptors. The evolutionary conservation of structure and function suggests unique biologic roles for these receptors. Recently, a 39-amino-acid hypothalamic peptide, tubular infundibular peptide (TIP-39), has been characterized and is a likely natural ligand of the PTH2 receptor.

G proteins of the G_s class link the [PTH/PTHrP](#) receptor to adenylate cyclase, an enzyme that generates cyclic AMP, leading to activation of protein kinase A. Coupling to G proteins of the G_q class links hormone action to phospholipase C, an enzyme that generates inositol phosphates (e.g., [IP₃](#)) and [DAG](#), leading to activation of protein kinase C and intracellular calcium release ([Fig. 341-2](#)). Studies using the cloned PTH/PTHrP receptor confirm that it can be coupled to more than one G protein and second-messenger kinase pathway, apparently explaining the multiplicity of pathways stimulated by PTH. Incompletely characterized second-messenger responses may be independent of phospholipase C or adenylate cyclase stimulation (the latter, however, is the strongest and best characterized second messenger signaling pathway for PTH).

The details of the biochemical steps by which an increased intracellular concentration of cyclic AMP, [IP₃](#), [DAG](#), and intracellular Ca^{2+} lead to ultimate changes in [ECF](#) calcium and phosphate ion translocation or bone cell function are unknown. Stimulation of protein kinases (A and C) and calcium transport channels is associated with a variety of hormone-specific tissue responses. These responses include inhibition of phosphate and bicarbonate transport, stimulation of calcium transport, and activation of renal 1 α -hydroxylase in the kidney. The responses in bone include effects on collagen synthesis; increased alkaline phosphatase, ornithine decarboxylase, citrate decarboxylase, and glucose-6-phosphate dehydrogenase activities; DNA, protein, and phospholipid synthesis; and calcium and phosphate transport. Ultimately, these biochemical events lead to an integrated hormonal response in bone turnover and calcium homeostasis.

[PTH](#) also activates Na^+/Ca^{2+} exchanges in renal distal tubular sites and stimulates translocation of preformed calcium transport channels, moving them from the interior to the apical surface to mediate increased tubular uptake of calcium. PTH-dependent stimulation of phosphate excretion (blocking reabsorption -- the opposite effect from actions on calcium in the kidney) involves the sodium-dependent phosphate cotransporter, NPT-2, lowering its apical membrane content (and therefore function). Similar shifts may be involved in other renal tubular transport effects of PTH.

PTHrP exerts important developmental influences on fetal bone development and in adult physiology. A homozygous knockout of the PTHrP gene (or the gene for the [PTH](#) receptor) in mice causes a lethal deformity in which animals are born with severe skeletal deformities resembling chondrodysplasia ([Fig. 341-3](#)).

CALCITONIN (See also [Chap. 339](#))

Calcitonin is a hypocalcemic peptide hormone that in several mammalian species acts as the physiologic antagonist to [PTH](#). Calcitonin seems to be of limited physiologic significance in humans, at least in calcium homeostasis, as contrasted with a clearly definable role in calcium metabolism in many other mammalian species. Calcitonin is of medical significance, however, because of its role as a tumor marker in sporadic and hereditary cases of medullary carcinoma and its medical use as an adjunctive treatment in severe hypercalcemia and in Paget's disease of bone.

The hypocalcemic activity of calcitonin is accounted for primarily by inhibition of osteoclast-mediated bone resorption and secondarily by stimulation of renal calcium clearance. These effects are mediated by receptors on osteoclasts and renal tubular cells. Calcitonin exerts additional effects through receptors present in brain, gastrointestinal tract, and the immune system. The hormone, for example, exerts analgesic effects directly on cells in the hypothalamus and related structures, possibly by interacting with receptors for related peptide hormones, such as calcitonin gene-related peptide (CGRP) or amylin. The latter ligands have specific high-affinity receptors and also can bind to and trigger calcitonin receptors. The calcitonin receptors are homologous in structure to the [PTH/PTHrP](#) receptor.

The thyroid is the major source of the hormone, and the cells involved in calcitonin synthesis arise from neural crest tissue. During embryogenesis, these cells migrate into the ultimobranial body, derived from the last branchial pouch. In submammalian vertebrates, the ultimobranial body constitutes a discrete organ, anatomically separate from the thyroid gland; in mammals, the ultimobranial gland fuses with and is incorporated into the thyroid gland.

The naturally occurring calcitonins consist of a peptide chain of 32 amino acids. There is considerable sequence variability among species. Calcitonin from salmon is 10 to 100 times more potent than mammalian forms in lowering serum calcium in animals. Calcitonin is synthesized as a precursor molecule that is four times larger than calcitonin itself. Analysis of the sequence of the coding portions of the gene for rat calcitonin indicates that at least two peptides flank calcitonin. It is likely (by analogy with the common precursor for adrenocorticotrophic hormone and endorphin) that these peptides, of still uncharacterized biologic function, are released along with calcitonin.

There are two calcitonin genes, a and b, located on chromosome 11 in the general region of the b-globulin and [PTH](#) genes; the transcriptional control of these genes is complex. Two different mRNA molecules are transcribed from the a gene; one is translated into the precursor for calcitonin, and the other message is translated into an alternative product, [CGRP](#). CGRP is synthesized wherever the calcitonin mRNA is expressed, e.g., in medullary carcinoma of the thyroid. The b, or CGRP-2, gene is transcribed into the mRNA for CGRP in the central nervous system (CNS); this gene does not produce calcitonin, however. CGRP has cardiovascular actions and may serve as a neurotransmitter or play a developmental role in the CNS.

The secretion of calcitonin is under the direct control of blood calcium. The circulating level of calcitonin in humans is lower than that in many other species. In humans, changes in calcium and phosphate metabolism are not seen despite extreme variations in calcitonin production; no definite effects are attributable to calcitonin deficiency (totally

thyroidectomized patients receiving only replacement thyroxine) or excess (patients with medullary carcinoma of the thyroid, a calcitonin-secreting tumor) ([Chap. 339](#)). Although there are no obvious abnormalities in calcium metabolism in patients with elevated calcitonin levels, bone remodeling is chronically suppressed. Calcitonin has been a useful pharmacologic agent to suppress bone resorption in Paget's disease ([Chap. 343](#)), has had limited use in the treatment of osteoporosis ([Chap. 342](#)), and is useful in early phases of treatment of severe hypercalcemia (see below).

HYPERCALCEMIA

Hypercalcemia can be a manifestation of a serious illness such as malignancy or can be detected coincidentally by laboratory testing in a patient with no obvious illness. The number of patients recognized with asymptomatic hypercalcemia, usually hyperparathyroidism, increased in the late twentieth century but is now declining somewhat, perhaps due to decreased use of routine blood calcium measurements or for other unknown reasons.

Whenever hypercalcemia is confirmed, a definitive diagnosis must be established. Although hyperparathyroidism, a frequent cause of asymptomatic hypercalcemia, is a chronic disorder in which manifestations, if any, may be expressed only after months or years, hypercalcemia can also be the earliest manifestation of malignancy, the second most common cause of hypercalcemia in the adult. The causes of hypercalcemia are numerous ([Table 341-1](#)), but hyperparathyroidism and cancer account for 90% of cases.

Before undertaking a workup, it is essential to be sure that true hypercalcemia, not a false-positive laboratory test, is present. A false-positive diagnosis of hypercalcemia is usually the result of inadvertent hemoconcentration during blood collection or elevation in serum proteins such as albumin. Hypercalcemia is a chronic problem, and it is cost-effective to obtain several serum calcium measurements; these tests need not be in the fasting state.

Clinical features are helpful in differential diagnosis. Hypercalcemia in an adult who is asymptomatic is usually due to primary hyperparathyroidism. In malignancy-associated hypercalcemia the disease is usually not occult; rather, symptoms of malignancy bring the patient to the physician, and hypercalcemia is discovered during the workup. In such patients the interval between detection of hypercalcemia and death is often <6 months. Accordingly, if an asymptomatic individual has had hypercalcemia or some manifestation of hypercalcemia, such as kidney stones, for >1 or 2 years, it is unlikely that malignancy is the cause. Nevertheless, differentiating primary hyperparathyroidism from *occult* malignancy can occasionally be difficult, and careful evaluation is required, particularly when the duration of the hypercalcemia is unknown. Hypercalcemia not due to hyperparathyroidism or malignancy can result from excessive vitamin D action, high bone turnover from any of several causes, or from renal failure ([Table 341-1](#)). Dietary history and a history of ingestion of vitamins or drugs are often helpful in diagnosing some of the less frequent causes. PTH immunoassays based on double-antibody methods serve as the principal laboratory test in differential diagnosis.

Hypercalcemia from any cause can result in fatigue, depression, mental confusion,

anorexia, nausea, vomiting, constipation, reversible renal tubular defects, increased urination, a short QT interval in the electrocardiogram, and, in some patients, cardiac arrhythmias. There is a variable relation from one patient to the next between the severity of hypercalcemia and the symptoms. Generally, symptoms are more common at calcium levels >2.9 to 3 mmol/L (11.5 to 12.0 mg/dL), but some patients, even at this level, are asymptomatic. When the calcium level is >3.2 mmol/L (13 mg/dL), calcification in kidneys, skin, vessels, lungs, heart, and stomach occurs and renal insufficiency may develop, particularly if blood phosphate levels are normal or elevated due to impaired renal function. Severe hypercalcemia, usually defined as >3.7 to 4.5 mmol/L (15 to 18 mg/dL) can be a medical emergency; coma and cardiac arrest can occur.

Except in malignancy-associated hypercalcemia, acute management of the hypercalcemia is usually successful prior to definitive therapy. The type of treatment is based on the severity of the hypercalcemia and the nature of associated symptoms.

PRIMARY HYPERPARATHYROIDISM

Natural History and Incidence Primary hyperparathyroidism is a generalized disorder of calcium, phosphate, and bone metabolism due to an increased secretion of [PTH](#). The elevation of circulating hormone usually leads to hypercalcemia and hypophosphatemia. There is great variation in the manifestations. Patients may present with multiple signs and symptoms, including recurrent nephrolithiasis, peptic ulcers, mental changes, and, less frequently, extensive bone resorption. However, with greater awareness of the disease and wider use of multiphasic screening tests, including blood calcium assays, the diagnosis is frequently made in patients who have no symptoms and minimal, if any, signs of the disease other than hypercalcemia and elevated levels of PTH. The manifestations may be subtle, and the disease may have a benign course for many years or a lifetime. Rarely, hyperparathyroidism develops or worsens abruptly and causes severe complications, such as marked dehydration and coma, so-called hypercalcemic parathyroid crisis.

The annual incidence of the disease is estimated to be as high as 0.2% in patients >60 , with an estimated prevalence, including undiscovered asymptomatic patients, of $\approx 1\%$. The disease has a peak incidence between the third and fifth decades but occurs in young children and in the elderly.

Etiology

Solitary Adenomas The cause of hyperparathyroidism is one or more hyperfunctioning glands. The traditional view has been that a single abnormal gland is the cause in approximately 80% of patients; the abnormality in the gland is usually a benign neoplasm or adenoma and rarely a parathyroid carcinoma. Some surgeons and pathologists report that the enlargement of multiple glands is common; double adenomas are reported. In approximately 15% of patients, all glands are hyperfunctioning; *chief cell parathyroid hyperplasia* is usually hereditary and frequently associated with other endocrine abnormalities.

Multiple Endocrine Neoplasia Hereditary hyperparathyroidism can occur without other endocrine abnormalities but is usually part of a *multiple endocrine neoplasia* syndrome

([Chap. 339](#)). [MEN](#) 1 (Wermer's syndrome) consists of hyperparathyroidism and tumors of the pituitary and pancreas, often associated with gastric hypersecretion and peptic ulcer disease (Zollinger-Ellison syndrome). MEN 2A is characterized by pheochromocytoma and medullary carcinoma of the thyroid, as well as hyperparathyroidism; MEN 2B has additional associated features such as multiple neuromas but usually lacks hyperparathyroidism. Each of these MEN syndromes is transmitted in an autosomal dominant manner.

Pathology Adenomas are most often located in the inferior parathyroid glands, but in 6 to 10% of patients, parathyroid adenomas may be located in the thymus, the thyroid, the pericardium, or behind the esophagus. Adenomas are usually 0.5 to 5 g in size but may be as large as 10 to 20 g (normal glands weigh 25 mg on average). Chief cells are predominant in both hyperplasia and adenoma. The adenoma is sometimes encapsulated by a rim of normal tissue. With chief cell hyperplasia, the enlargement may be so asymmetric that some involved glands appear grossly normal. If generalized hyperplasia is present, however, histologic examination reveals a uniform pattern of chief cells and disappearance of fat even in the absence of an increase in gland weight. Thus, microscopic examination of biopsy specimens of several glands is essential to interpret findings at surgery. When an adenoma is present, the other glands are usually normal and contain a normal distribution of all cell types (rather than only chief cells) and normal amounts of fat.

Parathyroid carcinoma is usually not aggressive in character. Long-term survival without recurrence is common if at initial surgery the entire gland is removed without rupture of the capsule. Recurrent parathyroid carcinoma is usually slow-growing with local spread in the neck, and surgical correction of recurrent disease may be feasible. Occasionally, however, parathyroid carcinoma is more aggressive, with distant metastases (lung, liver, and bone) found at the time of initial operation. It may be difficult to appreciate initially that a primary tumor is carcinoma; increased numbers of mitotic figures and increased fibrosis of the gland stroma may precede invasion. The diagnosis of carcinoma is often made in retrospect. Hyperparathyroidism from a parathyroid carcinoma may be indistinguishable from other forms of primary hyperparathyroidism; a potential clue to the diagnosis, however, is provided by the degree of calcium elevation. Calcium values of 3.5 to 3.7 mmol/L (14 to 15 mg/dL) are frequent with carcinoma and may alert the surgeon to remove the abnormal gland with care to avoid capsular rupture.

GENETIC CONSIDERATIONS

Defects Associated with Hyperparathyroidism As in many other types of neoplasia, two fundamental types of genetic defects have been identified in parathyroid gland tumors: (1) overactivity of protooncogenes, and (2) loss of function of tumor suppressor genes. The former, by definition, can lead to uncontrolled cellular growth and function by activation (gain-of-function mutation) of a single allele of the responsible gene, whereas the latter requires loss of function of both allelic copies.

Mutations in the *MENIN* gene locus on chromosome 11q13 are responsible for causing [MEN](#) 1; the normal allele of this gene fits the definition of a tumor suppressor gene. A mutation of one allele is inherited; loss of the other allele via somatic cell mutation leads to monoclonal expansion and tumor development in tissues such as the

parathyroids. In approximately 20% of sporadic parathyroid adenomas, the *MENIN* locus on chromosome 11 appears to be deleted, implying that the same defect responsible for MEN 1 can also cause the sporadic disease ([Fig. 341-4A](#)). Consistent with the Knudson hypothesis for two-step neoplasia in certain inherited cancer syndromes ([Chap. 81](#)), the earlier onset of hyperparathyroidism in the hereditary syndromes reflects the statistical probability of only one mutational event triggering the monoclonal outgrowth. In sporadic adenomas, typically occurring later in life, two different somatic events must occur before the *MENIN* gene is silenced.

The *MENIN* gene codes for a novel protein consisting of 610 amino acids. The protein has a nuclear localization signal and appears to interact with the transcription factor Jun D. Most of the mutations are clearly of the inactivating type (nonsense, deletions); there is not, however, a good correlation between clinical features in different kindreds and the specific mutation detected (e.g., penetrance or age of onset of pituitary or pancreatic tumors). This is in contrast to the correlation between genotype and phenotype in [MEN 2](#) (see below).

Other presumptive antioncogenes involved in hyperparathyroidism include a gene mapped to chromosome 1p seen in 40% of sporadic parathyroid adenomas and a gene mapped to chromosome Xp11 in patients with secondary hyperparathyroidism and renal failure, who progressed to "tertiary" hyperparathyroidism, now known to reflect monoclonal outgrowths within previously hyperplastic glands.

The *Rb* gene, a tumor suppressor gene located on chromosome 13q14, was initially associated with retinoblastomas but has since been implicated in many other forms of neoplasia including parathyroid carcinoma. Allelic deletion (with a presumed point mutation in the second allele) has been identified in all parathyroid carcinomas examined; there is also an abnormal staining pattern of the protein product of the gene. Allelic deletion is also seen in 10% of parathyroid adenomas, although the abnormal staining pattern of the Rb protein is not seen. Other gene loci on chromosome 13 may be involved in addition to the *Rb* locus.

There are two rare syndromes associated with hyperparathyroidism that involve one or more genes located on chromosome 1q. The hereditary hyperparathyroidism jaw tumor (HPT-JT) syndrome shows an autosomal dominant inheritance pattern; the jaw tumors are benign, but the parathyroid pathology may involve carcinoma as well as adenoma. Parathyroid carcinoma may also appear in the other syndrome, familial isolated primary hyperparathyroidism (FIPH). Both syndromes have been mapped through linkage studies to the chromosome 1q21-q31 region. Certain findings have led to speculation that this chromosome region might contain a protooncogene rather than an antioncogene.

In some parathyroid adenomas, activation of a protooncogene has been identified ([Fig. 341-4B](#)). A reciprocal translocation involving chromosome 11 has been identified that juxtaposes the *PTH* gene promoter upstream of a gene product termed *PRAD-1*, a cyclin D protein that plays a key role in normal cell division. This translocation is found in as many as 15% of parathyroid adenomas, usually in larger tumors. Targeted overexpression of cyclin D₁ in the parathyroid glands of transgenic mice causes the development of hyperparathyroidism, consistent with the role of this cell cycle control

protein in parathyroid neoplasia.

A mutated protooncogene, *RET*, is involved in each of the clinical variants of [MEN2](#) ([Chap. 339](#)). *RET* encodes a tyrosine kinase-type receptor; specific mutations lead to constitutive activity of the receptor, thereby explaining the autosomal dominant mode of transmission and the relatively early onset of neoplasia.

Signs and Symptoms Half or more of patients with hyperparathyroidism are asymptomatic. In series in which patients are followed without operation, as many as 80% are classified as without symptoms. Manifestations of hyperparathyroidism involve primarily the kidneys and the skeletal system. Kidney involvement, due either to deposition of calcium in the renal parenchyma or to recurrent nephrolithiasis, was present in 60 to 70% of patients prior to 1970. With earlier detection, renal complications occur in <20% of patients in many large series. Renal stones are usually composed of either calcium oxalate or calcium phosphate. In occasional patients, repeated episodes of nephrolithiasis or the formation of large calculi may lead to urinary tract obstruction, infection, and loss of renal function. Nephrocalcinosis may also cause decreased renal function and phosphate retention.

The distinctive bone manifestation of hyperparathyroidism is *osteitis fibrosa cystica*, which in series reported 50 years ago occurred in 10 to 25% of patients. In recent years, osteitis fibrosa cystica is very rare in primary hyperparathyroidism, probably due to the increased incidence of mild disease. Histologically, the pathognomonic features are an increase in the giant multinucleated osteoclasts in scalloped areas on the surface of the bone (Howship's lacunae) and a replacement of the normal cellular and marrow elements by fibrous tissue. X-ray changes include resorption of the phalangeal tufts and replacement of the usually sharp cortical outline of the bone in the digits by an irregular outline (subperiosteal resorption).

With the use of multiple markers of bone turnover, such as formation indices (bone-specific alkaline phosphatase, osteocalcin, and type I procollagen peptides) and bone resorption indices (including hydroxypyridinium collagen cross-links and telopeptides of type I collagen), increased skeletal turnover is detected in essentially all patients with established hyperparathyroidism.

Computed tomography (CT) scan and dual-energy x-ray absorptiometry (DEXA) of the spine provide reproducible quantitative estimates (within a few percent) of spinal bone density ([Chap. 342](#)). Similarly, cortical bone density in the extremities can be quantified by single-photon densitometry, usually of the distal radius at a site chosen to be primarily cortical. Studies reveal that cortical bone density is reduced while cancellous bone density, especially in the spine, is relatively preserved. Serial studies in patients who choose to be followed without surgery have indicated that in the majority there is little further change over a number of years, consistent with laboratory data indicating relatively unchanged blood calcium and [PTH](#) levels. After an initial loss of bone mass in patients with mild asymptomatic hyperparathyroidism, a new equilibrium may be reached, with bone density and biochemical manifestations of the disease remaining relatively unchanged. This clinical course has led to the recommendations (discussed below) that asymptomatic patients may be safely followed with medical supervision. Certain recent findings have raised questions about the impact of asymptomatic

hyperparathyroidism on the skeleton, however. In one careful, long-term study, parathyroidectomy led to improvements in bone density in the spine and hip in 10 to 15% of patients; the improved bone density has been maintained for a number of years of follow-up. Another group compared fracture incidence in a large cohort of hyperparathyroid patients followed for years without surgery versus those seen in an age- and sex-matched control population. The incidence of fractures of the spine, wrist, and ribs was significantly increased in the hyperparathyroid group (although there were no data available on bone density).

In symptomatic patients, dysfunctions of the central nervous system, peripheral nerve and muscle, gastrointestinal tract, and joints also occur. An awareness of the signs and symptoms of hyperparathyroidism may give the initial clue to the diagnosis. It has been reported that severe neuropsychiatric manifestations may be reversed by parathyroidectomy; it remains unclear, in the absence of controlled studies, whether this improvement has a defined cause-and-effect relationship. Generally, the fact that hyperparathyroidism is common in elderly patients, in whom there are often other problems, suggests the possibility that such coexisting problems as hypertension, renal deterioration, and depression may not be parathyroid-related and suggests caution in recommending parathyroid surgery as a cure for these conditions.

Neuromuscular manifestations may include proximal muscle weakness, easy fatigability, and atrophy of muscles and may be so striking as to suggest a primary neuromuscular disorder. The distinguishing feature is the complete regression of neuromuscular disease after surgical correction of the hyperparathyroidism.

Gastrointestinal manifestations are sometimes subtle and include vague abdominal complaints and disorders of the stomach and pancreas. Again, cause and effect are unclear. In [MEN 1](#) patients with hyperparathyroidism, duodenal ulcer may be the result of associated pancreatic tumors that secrete excessive quantities of gastrin (Zollinger-Ellison syndrome). Pancreatitis has been reported in association with hyperparathyroidism, but the incidence and the mechanism are not established.

Chondrocalcinosis and pseudogout are said to be sufficiently frequent in hyperparathyroidism that screening of such patients is warranted. Occasionally, pseudogout is the initial manifestation.

Diagnosis The diagnosis is typically made by detecting an elevated immunoreactive [PTH](#) level in a patient with asymptomatic hypercalcemia (see "Differential Diagnosis: Special Tests," below). Serum phosphate is usually low but may be normal, especially if renal failure has developed. Hypophosphatemia is a less specific diagnostic finding than hypercalcemia for two reasons: (1) phosphate levels are influenced by dietary intake, diurnal variations, and other factors; to be useful, samples must be obtained in the morning under fasting conditions; and (2) patients with severe hypercalcemia of any cause may have a low serum phosphate.

Many tests based on renal responses to excess [PTH](#) (renal calcium and phosphate clearance; blood phosphate, chloride, magnesium; urinary or nephrogenous cyclic AMP) were used in earlier decades. These tests have low specificity for hyperparathyroidism and are therefore not cost-effective; they have been replaced by PTH immunoassays.

TREATMENT

Medical Treatment Management of hyperparathyroidism involves two separate issues. The critical question is whether the disease should be treated surgically. If severe hypercalcemia [3.7 to 4.5 mmol/L (15 to 18 mg/dL)] is present, surgery is mandatory as soon as the diagnosis can be confirmed by a [PTH](#) immunoassay. However, in most patients with hyperparathyroidism, hypercalcemia is mild and does not require urgent surgical or medical treatment.

Several hundred patients have been closely followed without surgery in attempts to define the natural history of the disease and the benefits of surgery versus the risks of medical observation. Large-scale randomized, prospective clinical trials have not been undertaken, however. Rather, the long-term effects of hyperparathyroidism have been assessed in patients who do not have kidney stones, osteitis fibrosa cystica, or other clear-cut symptoms. Progressive loss of bone mass is a worrisome problem in women who face the problem of age-dependent and estrogen-deficient bone loss in the absence of hyperparathyroidism. The principal concern is that such patients, even though asymptomatic, will suffer sufficient bone loss due to [PTH](#) excess to make them more vulnerable to developing symptomatic osteoporosis.

The National Institutes of Health held a Consensus Conference on Management of Asymptomatic Hyperparathyroidism in 1991. *Asymptomatic hyperparathyroidism* was defined as documented (presumptive) hyperparathyroidism without signs or symptoms attributable to the disease. The consensus was that patients <50 should undergo surgery, given the long surveillance that would be required. Patients >50 are appropriate for medical monitoring if certain criteria are met and the patients wish to avoid surgery. Guidelines for recommending surgery in patients with asymptomatic hyperparathyroidism include the following:

1. Elevation of serum calcium, >0.25 to 0.40 mmol/L (1 to 1.6 mg/dL) above the upper limit of normal for the test laboratory.
2. History of life-threatening hypercalcemia, such as an episode induced by dehydration and recurring illness.
3. Reduction of age-matched creatinine clearance by >30% without a known cause. Presence of kidney stones detected by abdominal radiograph even if they are asymptomatic.
4. Elevation of 24-h urinary calcium excretion >400 mg.
5. Reduction of bone mass more than 2 standard deviations below normal using one of several noninvasive methods.

Other considerations that favor surgery include concern that consistent follow-up would be unlikely or that coexistent illness would complicate management. More recent data indicated that a subgroup of patients had selective vertebral osteopenia out of proportion to bone loss at other sites and responded to surgery with striking restoration

of bone mass (average >20%), suggesting that such patients might also be recommended for surgery. Asymptomatic patients should be monitored regularly. Surgical correction of hyperparathyroidism can always be undertaken when indicated, since the success rate is high (>90%), mortality is low, and morbidity is minimal. The goals of monitoring are early detection of worsening hypercalcemia, deteriorating bone or renal status, or other complications of hyperparathyroidism.

The consensus panel did not make a recommendation as to estrogen use in patients for whom surgery was not elected because there was insufficient cumulative experience with such therapy to balance theoretical risks (breast and endometrial cancer) versus benefits. New medical therapies may change the approach to the disease in the future. Raloxifene (Evista), the first of the [SERMS](#), has been shown to have many of the bone protective effects of estrogen in osteoporotic subjects yet at the same time lowers the incidence of breast cancer; use of this agent has not yet been reported in a series of hyperparathyroid patients, however. Early experience with calcimimetics, drugs that selectively stimulate the calcium sensor and suppress [PTH](#) secretion, indicates that these agents decrease PTH levels for several hours after a single dose.

European investigators have reported serious cardiovascular complications in patients with hyperparathyroidism that reverse, at least in part, after surgery. They also found an increased incidence of malignancy and an absolute increase in age-adjusted mortality due to hyperparathyroidism. These reports clearly describe experiences in a group of patients with more advanced disease, at least based on laboratory tests such as blood [PTH](#) levels, than the patients followed in the United States. In fact, a recent long-term epidemiologic study of a large cohort of patients in the United States, with a milder form of the disease (fitting the criteria for medical surveillance listed above), had an age-adjusted mortality no different than that of euparathyroid patients and no increased incidence of malignancy.

Surgical Treatment Parathyroid exploration is challenging and is best undertaken by an experienced surgeon with the help of an experienced pathologist. Certain features help in predicting the pathology (e.g., multiple abnormal glands in familial cases). However, some critical decisions regarding management can be made only during the operation. The examination by frozen section of tissue removed at surgery helps direct the subsequent course of the operation.

As discussed above, there are many unresolved issues to consider in surgery for this disease. At the extreme of conservatism, the surgical approach is based on the view that typically only one gland (the adenoma) is abnormal. If an enlarged gland is found, a normal gland should be sought. In this view, if a biopsy of a normal-sized second gland confirms its histologic (and presumed functional) normality, no further exploration, biopsy, or excision is needed. At the other extreme is the minority viewpoint that all four glands be sought and that most of the total parathyroid tissue mass should be removed. The concern with the former approach is that the recurrence rate of hyperparathyroidism may be high if a second abnormal gland is missed; the latter approach could involve unnecessary surgery and an unacceptable rate of hypoparathyroidism. The majority viewpoint, judged by surgical reviews, is in favor of conservative surgery, i.e., removal of what is usually only one enlarged gland but only after four-gland exploration to eliminate the possibility that more than one gland is abnormal. When normal glands are found in

association with one enlarged gland, excision of the single adenoma usually leads to cure or symptom-free disease, although long-term follow-up studies are limited.

Surgical management has been enhanced recently by the use of preoperative ^{99m}Tc sestamibi scans to predict the location of an abnormal gland and intraoperative sampling of PTH before and at 5- to 15-min intervals after removal of a suspected adenoma to confirm a rapid fall (>50%) in PTH levels. In several centers, a combination of preoperative sestamibi imaging, cervical block anesthesia, minimal surgical incision, and intraoperative PTH measurements has allowed successful outpatient surgical management with a clear-cut cost benefit compared to general anesthesia and more extensive neck surgery. The use of these minimally invasive approaches requires clinical judgment to select patients unlikely to have multiple gland disease (e.g., MEN or secondary hyperparathyroidism).

Multiple gland hyperplasia, as predicted in familial cases, poses more difficult questions of surgical management. Once a diagnosis of hyperplasia is established, all the glands must be identified. Two schemes have been proposed for surgical management. One is that three glands be totally removed and the fourth gland be partially excised; care is taken to leave a good blood supply for the remaining gland. Other surgeons advocate total parathyroidectomy with immediate transplantation of a portion of a removed, minced parathyroid gland into the muscles of the forearm, with the view that surgical excision is easier from the ectopic site in the arm if there is recurrent hyperfunction. When parathyroid carcinoma is encountered, the tissue should be widely excised; care must be taken to avoid rupture of the capsule to prevent local seeding of tumor cells.

In a minority of cases, if no abnormal parathyroid glands are found in the neck, the issue of further exploration must be decided. There are documented cases of five or six parathyroid glands and of unusual locations for adenomas, such as in the mediastinum. A variety of techniques have been developed to aid in the preoperative localization of the abnormal parathyroid tissue. Usually these techniques are reserved for patients with initial unsuccessful neck explorations, since the combined success of the localization techniques is not better than that of an experienced parathyroid surgeon in finding the abnormal tissue at the first operation. Noninvasive or minimally invasive techniques include ultrasound, CT scan of the neck and mediastinum, and differential scanning after technetium-sestamibi administration.

When a second parathyroid exploration is indicated, the minimally invasive techniques such as ultrasound, CT scan, and isotope scanning should probably be combined with venous sampling and/or selective digital arteriography in one of the centers specializing in these techniques. Intraoperative monitoring of PTH levels by rapid PTH immunoassays may be useful in guiding the surgery, especially in patients who are reexplored after an initial unsuccessful operation. At one center, long-term cures have been achieved with selective embolization or injection of large amounts of contrast material into the end-arterial circulation feeding the parathyroid tumor.

A decline in serum calcium occurs within 24 h after successful surgery; usually blood calcium falls to low-normal values for 3 to 5 days until the remaining parathyroid tissue resumes hormone secretion. Severe postoperative hypocalcemia is likely only if osteitis fibrosa cystica is present or if injury to all the normal parathyroid glands occurs during

surgery.

In general, patients who do not have symptomatic bone disease or a large deficit in bone mineral and who have good renal and gastrointestinal function have few problems with postoperative hypocalcemia. The extent of postoperative hypocalcemia varies with the surgical approach. If all glands are biopsied, hypocalcemia may be transiently symptomatic and more prolonged. Hypocalcemia is more likely to be symptomatic after second parathyroid explorations, particularly when normal parathyroid tissue was removed at the initial operation and when the manipulation and/or biopsy of the remaining normal glands is more extensive in the search for the missing adenoma.

Patients with hyperparathyroidism have efficient intestinal calcium absorption due to the increased levels of $1,25(\text{OH})_2\text{D}$ stimulated by [PTH](#) excess. Once hypocalcemia signifies successful surgery, patients can be put on a high-calcium intake or be given oral calcium supplements. Despite mild hypocalcemia, most patients do not require parenteral therapy. If the serum calcium falls to <2 mmol/L (8 mg/dL), *and if the phosphate level rises simultaneously*, the possibility that surgery has caused hypoparathyroidism must be considered. Coexistent hypomagnesemia should be checked for, as it interferes with PTH secretion and causes functional hypoparathyroidism (see below). Parenteral calcium replacement at a low level should be instituted when hypocalcemia is symptomatic. Such indications include a general sense of anxiety and positive Chvostek and Trousseau signs coupled with serum calcium consistently <2 mmol/L (8 mg/dL). For parenteral therapy, calcium (gluconate or chloride) solutions are prepared at a concentration of 1 mg/mL in 5% dextrose in water. The rate and duration of intravenous therapy are determined by the severity of the symptoms and the response of the serum calcium. An infusion of 0.5 to 2 (mg/kg)/h or 30 to 100 mL/h of a 1-mg/mL solution usually suffices to relieve symptoms. Usually, parenteral therapy is required for only a few days. If symptoms worsen or if parenteral calcium is needed for >2 to 3 days, therapy with a vitamin D analogue and/or oral calcium (2 to 4 g/d) should be started (see below). It is cost-effective to use calcitriol (doses of 0.5 to 1.0 $\mu\text{g/d}$) because of the rapidity of onset and rapidity of cessation of action, in comparison to other forms of vitamin D (see below). A rise in blood calcium after several months of vitamin D replacement may indicate restoration of parathyroid function to normal. It is also appropriate to monitor serum PTH serially to estimate gland function in such patients.

Magnesium deficiency may also complicate the postoperative course. Magnesium deficiency impairs the secretion of [PTH](#), and so hypomagnesemia should be corrected whenever detected. Magnesium chloride is effective by mouth, but this compound is not widely available. Repletion is usually parenteral. Because the depressant effect of magnesium on central and peripheral nerve functions does not occur at levels <2 mmol/L (normal range 0.8 to 1.2 mmol/L), parenteral replacement can be given rapidly. A cumulative dose as great as 0.5 to 1 mmol/kg of body weight can be administered if severe hypomagnesemia is present; often, however, total doses of 20 to 40 mmol are sufficient. The magnesium is given either as an intravenous infusion over 8 to 12 h or in divided doses intramuscularly (magnesium sulfate, USP).

OTHER PARATHYROID-RELATED CAUSES OF HYPERCALCEMIA

Lithium Therapy Lithium, used in the management of bipolar depression and other psychiatric disorders, causes hypercalcemia in approximately 10% of treated patients. The hypercalcemia is dependent on continued lithium treatment, remitting and recurring when lithium is stopped and restarted. The parathyroid adenomas reported in some hypercalcemic patients with lithium therapy may reflect the presence of an independently occurring parathyroid tumor; a permanent effect of lithium on parathyroid gland growth need not be implicated as most patients have complete reversal of hypercalcemia when lithium is stopped. However, long-standing stimulation of parathyroid cell replication by lithium may predispose to development of adenomas (as is documented in secondary hyperparathyroidism and renal failure).

The presence of hypercalcemia does not correlate with plasma lithium level, but the frequency with which hypercalcemia occurs is sufficiently high to support a causal relationship between lithium and the hypercalcemia, particularly the dependence of the hypercalcemia on the continuation of the lithium. At the levels achieved in blood in treated patients, lithium can be shown in vitro to shift the [PTH](#) secretion curve in response to calcium to the right; i.e., higher calcium levels are required to lower PTH secretion, probably acting at the calcium sensor (see below). It is logical to assume that this effect can cause elevated PTH levels and consequent hypercalcemia in otherwise normal individuals. If persistent hypercalcemia is detected during lithium therapy, it may be necessary to try alternative medication for the underlying psychiatric illness. Parathyroid surgery should not be recommended unless hypercalcemia and elevated PTH levels persist after lithium is discontinued.

GENETIC DISORDERS CAUSING HYPERPARATHYROID-LIKE SYNDROMES

Familial Hypocalciuric Hypercalcemia [FHH](#) (also called *familial benign hypercalcemia*) is inherited as an autosomal dominant trait. Affected individuals are discovered because of asymptomatic hypercalcemia. This disorder and Jansen's disease (discussed below) are variants of hyperparathyroidism. FHH involves excessive secretion of [PTH](#), whereas Jansen's disease is caused by excessive biologic activity of the PTH receptor in target tissues. Neither disorder, however, involves a primary growth disorder of the parathyroids.

The pathophysiology of [FHH](#) is now understood. The primary defect is abnormal sensing of the blood calcium by the parathyroid gland and renal tubule, causing inappropriate secretion of [PTH](#) and excessive renal reabsorption of calcium ([Fig. 341-5](#)). The calcium sensor is a member of the third family of [GPCR](#) (type C or III) and is located on chromosome 3. The receptor responds to the [ECF](#) calcium concentration, suppressing PTH secretion through second messenger signaling, thereby providing negative-feedback regulation of PTH secretion. More than 20 different mutations in the calcium-sensing receptor have been identified in patients with FHH ([Fig. 341-6](#)). These mutations lower the capacity of the sensor to bind calcium, and the mutant receptors function as though blood calcium levels are low; excessive secretion of PTH occurs from an otherwise normal gland. Approximately two-thirds of patients with FHH have mutations within the protein-coding region of the gene. The remaining one-third of kindreds may have mutations in the gene promoter or in other regions of the genome identified through mapping studies (e.g., chromosome 19).

Even before elucidation of the pathophysiology of [FHH](#), abundant clinical evidence served to separate the disorder from primary hyperparathyroidism. Patients with primary hyperparathyroidism have <99% renal calcium reabsorption, whereas most patients with FHH have >99% reabsorption. The hypercalcemia in FHH is often detectable in affected members of the kindreds in the first decade of life, whereas hypercalcemia rarely occurs in patients with primary hyperparathyroidism or the [MEN](#) syndromes who are <10. [PTH](#) may be elevated in FHH, but the values are usually normal or lower for the same degree of calcium elevation than in patients with primary hyperparathyroidism. Parathyroid surgery in a few patients with FHH led to permanent hypoparathyroidism, but hypocalciuria persisted nevertheless, establishing that hypocalciuria, therefore, is not PTH-dependent (now known to be due to the abnormal calcium sensor in the kidney).

Few clinical signs or symptoms are present in patients with [FHH](#), and other endocrine abnormalities are not present. Most patients are detected as a result of family screening after hypercalcemia is detected in a proband. In those patients inadvertently operated upon, the parathyroids appeared normal or moderately hyperplastic. Parathyroid surgery is not appropriate, nor, in view of the lack of symptoms, does medical treatment seem needed to lower the calcium. Calcimimetic agents that bind to the calcium sensor and elevate the set point are under investigation.

One striking exception to the rule against parathyroid surgery in this syndrome is the occurrence, usually in consanguineous marriages (due to the rarity of gene), of a homozygous or compound heterozygote state, resulting in complete loss of the calcium sensor function. In this condition, neonatal severe hypercalcemia, total parathyroidectomy is mandatory.

Jansen's Disease Mutations in the [PTH/PTHrP](#) receptor have been identified as responsible for this rare autosomal dominant syndrome ([Fig. 341-7](#)). Because the mutations lead to constitutive receptor function, one abnormal copy of the mutant receptor is sufficient to cause the disease, thereby accounting for its dominant mode of transmission. The disorder leads to short-limbed dwarfism due to abnormal regulation of the bone growth plate. In adult life, there are numerous abnormalities in bone, including multiple cystic resorptive areas resembling those seen in severe hyperparathyroidism. Hypercalcemia and hypophosphatemia with undetectable or low PTH levels are typically seen. The pathogenesis of the disease has been confirmed by transgenic experiments in which targeted expression of the mutant receptor to the growth plate emulated several features of the disorder.

MALIGNANCY-RELATED HYPERCALCEMIA

Clinical Syndromes and Mechanisms of Hypercalcemia Hypercalcemia due to malignancy is common (occurring with 10 to 15% of certain types of tumor, such as lung carcinoma), often severe and difficult to manage, confusing as to etiology, and sometimes difficult to distinguish from primary hyperparathyroidism. Although malignancy is often clinically obvious, hypercalcemia can occasionally be due to an occult tumor. Previously, hypercalcemia associated with malignancy was thought to be due to local invasion and destruction of bone by tumor cells; many cases are now known to result from the elaboration by the malignant cells of humoral mediators of hypercalcemia. [PTHrP](#) is the responsible humoral agent in most cases.

The histologic character of the tumor is more important than the extent of skeletal metastases in predicting hypercalcemia. Small cell carcinoma (oat cell) and adenocarcinoma of the lung, although the most common lung tumors associated with skeletal metastases, rarely cause hypercalcemia. By contrast, as many as 10% of patients with squamous cell carcinoma of the lung develop hypercalcemia. Histologic studies of bone in patients with squamous cell or epidermoid carcinoma of the lung, in sites invaded by tumor as well as areas remote from tumor invasion, reveal bone remodeling, including osteoclastic and osteoblastic activity. In contrast, minimal skeletal metabolic activation occurs even if there are extensive skeletal metastases of small cell (oat cell) carcinoma.

Two main mechanisms of hypercalcemia are operative in cancer hypercalcemia. Many solid tumors associated with hypercalcemia, particularly squamous cell and renal tumors, produce and secrete humoral factors that cause increased bone resorption and mediate the hypercalcemia through systemic actions on the skeleton. Alternatively, direct bone marrow invasion occurs with hematologic malignancies such as leukemia, lymphoma, and multiple myeloma. Lymphokines and cytokines produced by cells involved in the marrow response to the tumors promote resorption of bone through local destruction. Several hormones, hormone analogues, cytokines, and growth factors have been implicated as the result of clinical assays, in vitro tests, or chemical isolation. In some lymphomas, typically B cell lymphomas, there is an increased blood level of $1,25(\text{OH})_2\text{D}$, which is probably produced by lymphocytes. The etiologic factor produced by activated normal lymphocytes and by myeloma and lymphoma cells, termed *osteoclast activation factor*, now appears to represent the biologic action of several different cytokines, probably interleukin 1 and lymphotoxin or tumor necrosis factor.

The more common mechanism, humoral hypercalcemia of malignancy, occurs with cancers of the lung and kidney, in particular, in which bone metastases are absent, minimal, or not detectable clinically. The clinical picture resembles primary hyperparathyroidism (hypophosphatemia accompanies hypercalcemia), and elimination or regression of the primary tumor leads to disappearance of the hypercalcemia. The disorder is due to secretion by the tumors of the [PTH](#)-like factor, [PTHrP](#), that activates the PTH/PTHrP receptor (see above).

As in hyperparathyroidism, patients with the humoral hypercalcemia of malignancy have elevated urinary nephrogenous cyclic AMP excretion, hypophosphatemia, and increased urinary phosphate clearance. However, in humoral hypercalcemia of malignancy, immunoreactive [PTH](#) is undetectable or suppressed, making the differential diagnosis easier. Other features of the disorder differ from those of true hyperparathyroidism. Patients may have high, rather than low, renal calcium clearance (relative to serum calcium when compared to true hyperparathyroidism, unlike the expected elevation) and low to normal levels of $1,25(\text{OH})_2\text{D}$. The reason that the humoral syndrome differs from hyperparathyroidism in these parameters is unclear since the biologic actions of PTH and [PTHrP](#) are presumably exerted through the same receptor. Other cytokines elaborated by the malignancy may be responsible for these variations from hyperparathyroidism. In some patients with the humoral hypercalcemia of malignancy, osteoclastic resorption is unaccompanied by an osteoblastic or bone-forming response, implying inhibition of the normal coupling of bone formation and

resorption. Thus the interaction of more than one substance may determine whether hypercalcemia develops in a particular patient.

Several different assays (single- or double-antibody, different epitopes) have been developed to detect [PTHrP](#). Most data indicate that circulating PTHrP levels are undetectable (or low) in normal individuals, elevated in most cancer patients with the humoral syndrome, and high in human milk. Despite the discovery of PTHrP, identifying the etiologic mechanisms in cancer hypercalcemia is often complex. For example, in breast carcinoma (metastatic to bone) and in a distinctive type of T cell lymphoma/leukemia initiated by human T cell lymphotropic virus I, hypercalcemia is caused by direct local lysis of bone as well as by a humoral mechanism involving excess production of PTHrP.

Diagnostic Issues Levels of [PTH](#) measured by the double-antibody technique are undetectable or extremely low in tumor hypercalcemia, as would be expected with the mediation of the hypercalcemia by a factor other than PTH (the hypercalcemia suppresses the normal parathyroid glands). In a patient with minimal symptoms referred for hypercalcemia, low or undetectable PTH and elevated [PTHrP](#) levels would focus attention on occult malignancy.

Ordinarily, the diagnosis of cancer hypercalcemia is not difficult because tumor symptoms are prominent when hypercalcemia is detected. Indeed, hypercalcemia may be noted incidentally during the workup of a patient with known or suspected malignancy. Clinical suspicion that malignancy is the cause of the hypercalcemia is heightened when there are other paraneoplastic signs or symptoms, such as weight loss, fatigue, muscle weakness, or unexplained skin rash, or when symptoms specific for a particular tumor are present. Squamous cell tumors are most frequently associated with hypercalcemia, particularly tumors of the lung, kidney, head and neck, and urogenital tract. Radiologic examinations can focus on these areas when clinical evidence is unclear. Bone scans with technetium-labeled bisphosphonate are useful for detection of osteolytic metastases; the sensitivity is high, but specificity is low; results must be confirmed by conventional x-rays to be certain that areas of increased uptake are due to osteolytic metastases per se. Bone marrow biopsies are helpful in patients with anemia or abnormal peripheral blood smears.

TREATMENT

Treatment of the hypercalcemia of malignancy is first directed to control of the tumor; reduction of tumor mass usually corrects hypercalcemia. If a patient has severe hypercalcemia yet has a good chance for effective tumor therapy, treatment of the hypercalcemia should be vigorous while awaiting the results of definitive therapy. If hypercalcemia occurs in the late stages of a tumor that is resistant to therapy, the treatment of the hypercalcemia should be judicious as high calcium levels can have a mild sedating effect. Standard therapies for hypercalcemia (discussed below) are applicable to patients with malignancy.

VITAMIN D-RELATED HYPERCALCEMIA

Hypercalcemia caused by vitamin D can be due to excessive ingestion or abnormal

metabolism of the vitamin. Abnormal metabolism of the vitamin is usually acquired in association with a widespread granulomatous disorder. Vitamin D metabolism is carefully regulated, particularly the activity of renal 1 α -hydroxylase, the enzyme responsible for the production of 1,25(OH) $_2$ D ([Chap. 340](#)). The regulation of 1 α -hydroxylase and the normal feedback suppression by 1,25(OH) $_2$ D seem to work less well in infants than in adults and to operate poorly, if at all, in sites other than the renal tubule; these phenomena explain the occurrence of hypercalcemia secondary to excessive 1,25(OH) $_2$ D $_3$ production in infants with Williams' syndrome (see below) and in adults with sarcoidosis or lymphoma.

Vitamin D Intoxication Chronic ingestion of 50 to 100 times the normal physiologic requirement of vitamin D (amounts >50,000 to 100,000 U/d) is usually required to produce significant hypercalcemia in normal individuals. An upper limit of dietary intake of 2000 U/d (50 μ g/d) in adults is now recommended because of concerns about potential toxic effects of cumulative supraphysiologic doses. Vitamin D excess increases intestinal calcium absorption and, if severe, also increases bone resorption.

Hypercalcemia in vitamin D intoxication is due to an excessive biologic action of the vitamin, perhaps the consequence of increased levels of 25(OH)D rather than increased levels of the usual active metabolite 1,25(OH) $_2$ D (the latter is frequently not elevated in vitamin D intoxication). 25(OH)D has definite, if low, biologic activity in intestine and bone. The production of 25(OH)D is less tightly regulated than is the production of 1,25(OH) $_2$ D. Hence concentrations of 25(OH)D are elevated several-fold in patients with excess vitamin D intake.

The diagnosis is substantiated by documenting elevated levels of 25(OH)D >100 ng/mL. Hypercalcemia is usually controlled by restriction of dietary calcium intake and appropriate attention to hydration. These measures, plus discontinuation of vitamin D, usually lead to resolution of hypercalcemia. However, vitamin D stores in fat may be substantial, and vitamin D intoxication may persist for weeks after vitamin D ingestion is terminated. Such patients are responsive to glucocorticoids, which in doses of 100 mg/d of hydrocortisone or its equivalent, usually return serum calcium levels to normal over several days; severe intoxication may require intensive therapy.

Sarcoidosis and Other Granulomatous Diseases In patients with sarcoidosis and other granulomatous diseases, such as tuberculosis and fungal infections, excess 1,25(OH) $_2$ D is synthesized in macrophages or other cells in the granulomas. Indeed, increased 1,25(OH) $_2$ D levels have been reported in anephric patients with sarcoidosis and hypercalcemia. Macrophages obtained from granulomatous tissue convert 25(OH)D to 1,25(OH) $_2$ D at an increased rate. There is a positive correlation in patients with sarcoidosis between 25(OH)D levels (reflecting vitamin D intake) and the circulating concentrations of 1,25(OH) $_2$ D, whereas normally there is no increase in 1,25(OH) $_2$ D with increasing 25(OH)D levels due to multiple feedback controls on renal 1 α -hydroxylase ([Chap. 340](#)). The usual regulation of active metabolite production by calcium or PTH does not operate in these patients; hypercalcemia does not lead to a reduction in the blood levels of 1,25(OH) $_2$ D in patients with sarcoidosis. Clearance of 1,25(OH) $_2$ D from blood may be decreased in sarcoidosis as well. PTH levels are usually low and 1,25(OH) $_2$ D levels are elevated, but primary hyperparathyroidism and sarcoidosis may coexist in some patients.

Management of the hypercalcemia can often be accomplished by avoiding excessive sunlight exposure and limiting vitamin D and calcium intake. Presumably, however, the abnormal sensitivity to vitamin D and abnormal regulation of $1,25(\text{OH})_2\text{D}$ synthesis will persist as long as the disease is active. Alternatively, glucocorticoids in the equivalent of ≈ 100 mg/d of hydrocortisone control hypercalcemia. Glucocorticoids appear to act by blocking excessive production of $1,25(\text{OH})_2\text{D}$ as well as the response to it in target organs.

Idiopathic Hypercalcemia of Infancy This rare disorder, usually referred to as *Williams' syndrome*, is an autosomal dominant disorder characterized by multiple congenital development defects, including supraaortic stenosis, mental retardation, and an elfin facies, in association with hypercalcemia due to abnormal sensitivity to vitamin D. The syndrome was first recognized in England after the fortification of milk with vitamin D. Levels of $1,25(\text{OH})_2\text{D}$ are elevated, ranging from 46 to 120 nmol/L (150 to 500 pg/mL). The mechanism of the abnormal sensitivity to vitamin D and of the increased circulating levels of $1,25(\text{OH})_2\text{D}$ is still unclear. Studies suggest that mutations involving the elastin locus and perhaps other genes on chromosome 7 may play a role in the pathogenesis.

HYPERCALCEMIA ASSOCIATED WITH HIGH BONE TURNOVER

Hyperthyroidism As many as 20% of hyperthyroid patients have high-normal or mildly elevated serum calcium concentrations; hypercalciuria is even more common. The hypercalcemia is due to increased bone turnover, with bone resorption exceeding bone formation. Severe calcium elevations are not typical, and the presence of such suggests a concomitant disease such as hyperparathyroidism. Usually, the diagnosis is obvious, but signs of hyperthyroidism may occasionally be occult, particularly in the elderly ([Chap. 330](#)). Hypercalcemia is managed by treatment of the hyperthyroidism.

Immobilization Immobilization is a rare cause of hypercalcemia in adults in the absence of an associated disease but may cause hypercalcemia in children and adolescents, particularly after spinal cord injury and paraplegia or quadriplegia. With resumption of ambulation, the hypercalcemia in children usually returns to normal.

The mechanism appears to involve a disproportion between bone formation and bone resorption. Hypercalciuria and increased mobilization of skeletal calcium can develop in normal volunteers subjected to extensive bed rest, although hypercalcemia is unusual. Immobilization of an adult with a disease associated with high bone turnover, such as Paget's disease, may cause hypercalcemia.

Thiazides Administration of benzothiadiazines (thiazides) can cause hypercalcemia in patients with high rates of bone turnover, such as patients with hypoparathyroidism treated with high doses of vitamin D. Traditionally, thiazides are associated with aggravation of hypercalcemia in primary hyperparathyroidism, but this effect can be seen in other high-bone-turnover states as well. The mechanism of thiazide action is complex. Chronic thiazide administration leads to reduction in urinary calcium; the hypocalciuric effect appears to reflect the enhancement of proximal tubular resorption of sodium and calcium in response to sodium depletion. Some of this renal effect is due to

augmentation of [PTH](#) action and is more pronounced in individuals with intact PTH secretion. However, thiazides cause hypocalciuria in hypoparathyroid patients on high-dose vitamin D and oral calcium replacement if sodium intake is restricted. This finding is the rationale for the use of thiazides as an adjunct to therapy in hypoparathyroid patients, as discussed below. Thiazide administration to normal individuals causes a transient increase in blood calcium (usually within the high-normal range) that reverts to preexisting levels after a week or more of continued administration. If hormonal function and calcium and bone metabolism are normal, homeostatic controls are reset to counteract the calcium-elevating effect of the thiazides. In the presence of hyperparathyroidism or increased bone turnover from another cause, homeostatic mechanisms are ineffective. The abnormal effects of the thiazide on calcium metabolism disappear within days of cessation of the drug.

Vitamin A Intoxication Vitamin A intoxication is a rare cause of hypercalcemia and is most commonly a side effect of dietary faddism ([Chap. 75](#)). Calcium levels can be elevated into the 3 to 3.5 mmol/L (12 to 14 mg/dL) range after the ingestion of 50,000 to 100,000 units of vitamin A daily (10 to 20 times the minimum daily requirement). Typical features of severe hypercalcemia include fatigue, anorexia, and, in some, severe muscle and bone pain. Excess Vitamin A intake is presumed to increase bone resorption.

The diagnosis can be established by history and by measurement of vitamin A levels in serum, which may be severalfold above normal. Occasionally, skeletal x-rays reveal periosteal calcifications, particularly in the hands. Withdrawal of the vitamin is usually associated with prompt disappearance of the hypercalcemia and reversal of the skeletal changes. As in vitamin D intoxication, administration of 100 mg/d hydrocortisone or its equivalent leads to a rapid return of the serum calcium to normal.

HYPERCALCEMIA ASSOCIATED WITH RENAL FAILURE

Severe Secondary Hyperparathyroidism Secondary hyperparathyroidism occurs when partial resistance to the metabolic actions of [PTH](#) leads to excessive production of the hormone. Parathyroid gland hyperplasia occurs because resistance to the normal level of PTH leads to hypocalcemia, which, in turn, is a stimulus to parathyroid gland enlargement. This concept is supported by studies of the treatment of patients treated with bisphosphonates, which block the skeletal resorptive response. Because a portion of PTH secretion by each parathyroid cell is not suppressible by any degree of elevation of blood calcium, larger glands (more cells) have a higher hormone output at the hypercalcemic end of the dose-response curve.

Secondary hyperparathyroidism occurs not only in patients with renal failure but also in those with osteomalacia due to multiple causes ([Chap. 340](#)), including deficiency of vitamin D action, and [PHP](#) (deficient response to [PTH](#) at the level of the receptor). Hypocalcemia seems to be the common denominator in initiating secondary hyperparathyroidism. Only in patients with renal failure, however, is hypercalcemia sometimes encountered despite appropriate medical management regimens (see below). Primary and secondary hyperparathyroidism can be distinguished conceptually by the autonomous growth of the parathyroid glands in primary hyperparathyroidism (presumably irreversible) and the adaptive response of the parathyroids in secondary

hyperparathyroidism (typically reversible). In fact, reversal over weeks from an abnormal pattern of secretion, presumably accompanied by involution of parathyroid gland mass to normal, occurs in patients who have been treated effectively to reverse the resistance to PTH (such as with calcium and vitamin D in osteomalacia).

Patients with secondary hyperparathyroidism may develop bone pain, ectopic calcification, and pruritus. The bone disease seen in patients with secondary hyperparathyroidism and renal failure is termed *renal osteodystrophy*. Osteomalacia (predominantly due to vitamin D and calcium deficiency) and/or osteitis fibrosa cystica (excessive [PTH](#) action on bone) may occur.

Two other skeletal disorders are associated with long-term dialysis in patients with renal failure. Aluminum deposition (see below) is associated with an osteomalacia-like picture. The other entity is a low-bone-turnover state termed "aplastic" or "adynamic" bone disease; [PTH](#) levels are lower than in typical secondary hyperparathyroidism. It is believed that the condition is caused, at least in part, by excessive PTH suppression, which may be even greater than previously appreciated in light of evidence that some of the immunoreactive PTH detected by most commercially available PTH assays is not the full-length biologically active molecule (as discussed above).

TREATMENT

Medical therapy to reverse secondary hyperparathyroidism includes reduction of excessive blood phosphate by restriction of dietary phosphate, the use of nonabsorbable antacids, and careful, selective addition of calcitriol (0.25 to 2.0 ug/d); calcium carbonate is preferred over aluminum-containing antacids to prevent aluminum toxicity. Intravenous calcitriol, administered as several pulses each week, helps control secondary hyperparathyroidism. Aggressive but carefully administered medical therapy can often, but not always, reverse hyperparathyroidism and its symptoms and manifestations.

Occasional patients develop severe manifestations of secondary hyperparathyroidism, including hypercalcemia, pruritus, extraskeletal calcifications, and painful bones, despite aggressive medical efforts to suppress the hyperparathyroidism. [PTH](#) hypersecretion no longer responsive to medical therapy, a state of severe hyperparathyroidism in patients with renal failure that requires surgery, has been referred to as *tertiary hyperparathyroidism*. Parathyroid surgery is necessary to control this condition. Based on genetic evidence from examination of tumor samples in these patients, the emergence of autonomous parathyroid function is due to a monoclonal outgrowth of one or more previously hyperplastic parathyroid glands.

Aluminum Intoxication Aluminum intoxication (and often hypercalcemia as a complication of medical treatment) may occur in patients on chronic dialysis; manifestations include acute dementia and unresponsive and severe osteomalacia. Bone pain, multiple nonhealing fractures, particularly of the ribs and pelvis, and a proximal myopathy may occur. Hypercalcemia develops when these patients are treated with vitamin D or calcitriol because of impaired skeletal responsiveness. Aluminum is present at the site of osteoid mineralization, osteoblastic activity is minimal, and calcium incorporation into the skeleton is impaired. Prevention is accomplished by avoidance of

aluminum excess in the dialysis regimen; treatment of established disease involves mobilizing aluminum through the use of the chelating agent deferoxamine ([Chap. 348](#)).

Milk-Alkali Syndrome The milk-alkali syndrome is due to excessive ingestion of calcium and absorbable antacids such as milk or calcium carbonate. It is much less frequent since nonabsorbable antacids and other treatments became available for peptic ulcer disease. However, the increased use of calcium carbonate in the management of osteoporosis has led to reappearance of the syndrome. Several clinical presentations -- acute, subacute, and chronic -- have been described, all of which feature hypercalcemia, alkalosis, and renal failure. The chronic form of the disease, termed *Burnett's syndrome*, is associated with irreversible renal damage. The acute syndromes reverse if the excess calcium and absorbable alkali are stopped.

Individual susceptibility is important in the pathogenesis, as many patients are treated with calcium carbonate alkali regimens without developing the syndrome. One variable is the fractional calcium absorption as a function of calcium intake. Some individuals absorb a high fraction of calcium, even with intakes as high as 2 g or more of elemental calcium per day, instead of reducing calcium absorption with high intake, as occurs in most normal individuals. Resultant mild hypercalcemia after meals in such patients is postulated to contribute to the generation of alkalosis. Development of hypercalcemia causes increased sodium excretion and some depletion of total-body water. These phenomena and perhaps some suppression of endogenous [PTH](#) secretion due to mild hypercalcemia lead to increased bicarbonate resorption and to alkalosis in the face of continued calcium carbonate ingestion. Alkalosis per se selectively enhances calcium resorption in the distal nephron, thus aggravating the hypercalcemia. The cycle of mild hypercalcemia® bicarbonate retention® alkalosis® renal calcium retention® severe hypercalcemia perpetuates and aggravates hypercalcemia and alkalosis as long as calcium and absorbable alkali are ingested.

DIFFERENTIAL DIAGNOSIS: SPECIAL TESTS

Differential diagnosis of hypercalcemia is best achieved by using clinical criteria, but the immunoassay for [PTH](#) is especially useful in distinguishing among major causes ([Fig. 341-8](#)). The clinical features that deserve emphasis are the presence or absence of symptoms or signs of disease and evidence of chronicity. If one discounts fatigue or depression, >90% of patients with primary hyperparathyroidism have *asymptomatic hypercalcemia*; symptoms of malignancy are usually present in cancer-associated hypercalcemia. Disorders other than hyperparathyroidism and malignancy cause <10% of cases of hypercalcemia, and some of the nonparathyroid causes are associated with clear-cut manifestations such as renal failure.

Hyperparathyroidism is the likely diagnosis in patients with *chronic hypercalcemia*. If hypercalcemia has been manifest for >1 year, malignancy can usually be excluded as the cause. A striking feature of malignancy-associated hypercalcemia is the rapidity of the course, whereby signs and symptoms of the underlying malignancy are evident within months of the detection of hypercalcemia. A careful history of dietary supplements and drug use may suggest intoxication with vitamins D or vitamin A or the use of thiazides.

Although clinical considerations are helpful in arriving at the correct diagnosis of the cause of hypercalcemia, appropriate laboratory testing is essential for definitive diagnosis. The immunoassay for [PTH](#) should separate hyperparathyroidism from all other causes of hypercalcemia. Patients with hyperparathyroidism have elevated PTH levels despite hypercalcemia, whereas patients with malignancy and the other causes of hypercalcemia (except for disorders mediated by PTH such as lithium-induced hypercalcemia) have levels of hormone below normal or undetectable. Assays based on the double-antibody method for PTH exhibit very high sensitivity (especially if serum calcium is simultaneously evaluated) and specificity for the diagnosis of primary hyperparathyroidism ([Fig. 341-9](#)).

In summary, [PTH](#) values are elevated in >90% of parathyroid-related causes of hypercalcemia, undetectable or low in malignancy-related hypercalcemia, and undetectable or normal in vitamin D-related and high-bone-turnover causes of hypercalcemia. In view of the specificity of the PTH immunoassay and the high frequency of hyperparathyroidism in hypercalcemic patients, it is cost-effective to measure the PTH level in all hypercalcemic patients unless malignancy or a specific nonparathyroid disease is obvious. False-positive PTH assay results are rare. There are very rare reports of ectopic production of excess PTH by nonparathyroid tumors. Immunoassays for [PTHrP](#) are helpful in diagnosing certain types of malignancy-associated hypercalcemia. Although [FHH](#) is parathyroid-related, the disease should be managed distinctively from hyperparathyroidism. Clinical features and the low urinary calcium excretion can help make the distinction. Because the incidence of malignancy and hyperparathyroidism both increase with age, they can coexist as two independent causes of hypercalcemia.

1,25(OH)₂D levels are elevated in many (but not all) patients with primary hyperparathyroidism. In other disorders associated with hypercalcemia, concentrations of 1,25(OH)₂D are low or, at the most, normal. However, this test is of low specificity and is not cost-effective, as not all patients with hyperparathyroidism have elevated 1,25(OH)₂D levels, and not all nonparathyroid hypercalcemic patients have suppressed 1,25(OH)₂D. Measurement of 1,25(OH)₂D is, however, critically valuable in establishing the cause of hypercalcemia in sarcoidosis and certain B cell lymphomas.

A useful general approach is outlined in [Fig. 341-8](#). If the patient is *asymptomatic* and there is evidence of *chronicity* to the hypercalcemia, hyperparathyroidism is almost certainly the cause. If [PTH](#) levels (usually measured at least twice) are elevated, the clinical impression is confirmed and little additional evaluation is necessary. If there is only a short history or no data as to the duration of the hypercalcemia, *occult malignancy* must be considered; if the PTH levels are not elevated, then a thorough workup must be undertaken for malignancy, including chest x-ray, [CT](#) of chest and abdomen, and bone scan. Immunoassays for [PTHrP](#) may be especially useful in such situations. Attention should also be paid to clues for underlying hematologic disorders such as anemia, increased plasma globulin, and abnormal serum immunoelectrophoresis; bone scans can be negative in some patients with metastases, such as in multiple myeloma. Finally, if a patient with chronic hypercalcemia is asymptomatic and malignancy therefore seems unlikely on clinical grounds, but PTH values are not elevated, it is useful to search for other chronic causes of hypercalcemia, such as occult sarcoidosis.

TREATMENT

Hypercalcemic States The approach to medical treatment of hypercalcemia varies with its severity. Mild hypercalcemia, <3.0 mmol/L (12 mg/dL), can be managed by hydration. More severe hypercalcemia [levels of 3.2 to 3.7 mmol/L (13 to 15 mg/dL)] must be managed aggressively; above that level, hypercalcemia can be life-threatening and requires emergency measures. By using a combination of approaches, the serum calcium concentration can be decreased by 0.7 to 2.2 mmol/L (3 to 9 mg/dL) within 24 to 48 h in most patients, enough to relieve acute symptoms, prevent death from hypercalcemic crisis, and permit diagnostic evaluation. Therapy can then be directed at the underlying disorder -- the second priority.

Hypercalcemia develops because of excessive skeletal calcium release, increased intestinal calcium absorption, or inadequate renal calcium excretion. Understanding the particular pathogenesis helps guide therapy. For example, hypercalcemia in patients with malignancy is primarily due to excessive skeletal calcium release and is, therefore, minimally improved by restriction of dietary calcium. On the other hand, patients with vitamin D hypersensitivity or vitamin D intoxication have excessive intestinal calcium absorption, and restriction of dietary calcium is beneficial. Decreased renal function or [ECF](#) depletion decreases urinary calcium excretion. In such situations, rehydration may rapidly reduce or reverse the hypercalcemia, even though increased bone resorption persists. As outlined below, the more severe the hypercalcemia, the greater the number of combined therapies that should be used. Rapid acting (hours) approaches -- rehydration, forced diuresis, and calcitonin -- can be used with the most effective antiresorptive agents, such as bisphosphonates (since severe hypercalcemia usually involves excessive bone resorption).

Hydration, Increased Salt Intake, Mild and Forced Diuresis The first principle of treatment is to restore normal hydration. Many hypercalcemic patients are dehydrated because of vomiting, inanition, and/or hypercalcemia-induced defects in urinary concentrating ability. The resultant drop in glomerular filtration rate is accompanied by an additional decrease in renal tubular sodium and calcium clearance. Restoring a normal [ECF](#) volume corrects these abnormalities and increases urine calcium excretion by 2.5 to 7.5 mmol/d (100 to 300 mg/d). Increasing urinary sodium excretion to 400 to 500 mmol/d increases urinary calcium excretion even further than simple rehydration. After rehydration has been achieved, saline can be administered or furosemide or ethacrynic acid can be given twice daily to depress the tubular reabsorptive mechanism for calcium (care must be taken to prevent dehydration). The combined use of these therapies can increase urinary calcium excretion to ≈ 12.5 mmol/d (500 mg/d) in most hypercalcemic patients. Since this is a substantial percentage of the exchangeable calcium pool, the serum calcium concentration usually falls 0.25 to 0.75 mmol/L (1 to 3 mg/dL) within 24 h. Precautions should be taken to prevent potassium and magnesium depletion; calcium-containing renal calculi are a potential complication.

Under life-threatening circumstances, the preceding approach can be pursued more aggressively, giving as much as 6 L isotonic saline (900 mmol sodium) daily plus furosemide or equivalent in doses up to 100 mg every 1 to 2 h or ethacrynic acid in doses to 40 mg every 1 to 2 h. Urinary calcium excretion may exceed 25 mmol/d (1000

mg/d), and the serum calcium may decrease by 31 mmol/L (4 mg/dL) within 24 h. Depletion of potassium and magnesium is inevitable unless replacements are given; pulmonary edema can be precipitated. The potential complications can be reduced by careful monitoring of central venous pressure and plasma or urine electrolytes; catheterization of the bladder may be necessary. This treatment approach should be supplemented with agents to block bone resorption. Though these agents do not become effective for several days, forced diuresis is difficult to sustain even in patients with good cardiopulmonary and renal function.

Bisphosphonates The bisphosphonates are analogues of pyrophosphate, with high affinity for bone, especially in areas of increased bone turnover, where they are powerful inhibitors of bone resorption. These bone-seeking compounds are stable in vivo because phosphatase enzymes cannot hydrolyze the central carbon-phosphorus-carbon bond. The bisphosphonates are concentrated in areas of high bone turnover and are taken up by and inhibit osteoclast action; the mechanism of action is complex. Bisphosphonates alter osteoclast proton pump function or impair the release of acid hydrolases into the extracellular lysosomes contiguous with mineralized bone. They may also inhibit the differentiation of monocyte-macrophage precursors into osteoclasts and possibly have effects on osteoblasts as well. The bisphosphonate molecules that contain amino groups in the side chain structure (see below) interfere with prenylation of proteins and can lead to cellular apoptosis. The highly active non-amino group-containing bisphosphonates are also metabolized to cytotoxic products.

The initial bisphosphonate widely used in clinical practice, etidronate, was effective but had several disadvantages, including the capacity to inhibit bone formation as well as blocking resorption. Subsequently, a number of second-generation compounds have become the mainstays of antiresorptive therapy for treatment of hypercalcemia. The most widely used, pamidronate, is a potent inhibitor of osteoclast-mediated skeletal resorption yet does not cause mineralization defects at ordinary doses. Several additional bisphosphonates (alendronate, tiludronate, and risedronate) are potent and also have a highly favorable ratio of blocking resorption versus inhibiting bone formation. Though the bisphosphonates have similar structures, the routes of administration, efficacy, toxicity, and side effects vary. The potency of the compounds for inhibition of bone resorption varies a thousandfold in the order of etidronate, tiludronate, pamidronate, alendronate, and risedronate. Oral alendronate is approved for the therapy of osteoporosis in the United States, and in Europe oral preparations of these bisphosphonates are used in the chronic treatment of hypercalcemia. Only the intravenous use of pamidronate is approved for this purpose in the United States; between 30 and 90 mg pamidronate, given as a single intravenous dose over a few hours, returns serum calcium to normal within 24 to 48 h with an effect that lasts for weeks in 80 to 100% of patients.

Pamidronate causes low-grade fever in as many as 20% of patients, likely related to release of cytokines from osteoclasts, monocytes, and macrophages ([Table 341-2](#)). This effect is usually seen only with the initial doses. Etidronate causes hyperphosphatemia through a direct renal mechanism, whereas hypophosphatemia is seen after therapy with other bisphosphonates. Overall, second-generation bisphosphonates are now the agents of choice in severe hypercalcemia, particularly that

associated with malignancy. Zoledronate, a third-generation bisphosphonate, is claimed to be 100 to 800 times more potent than pamidronate and to normalize calcium more quickly and for longer periods.

Calcitonin Calcitonin acts within a few hours of its administration, through receptors on osteoclasts, to block bone resorption and, in addition, to increase urinary calcium excretion by inhibition of renal tubular calcium reabsorption. However, calcitonin leads to variable and usually minimal lowering of calcium. Tachyphylaxis, a known phenomenon with this drug, may explain the variable results. However, in life-threatening hypercalcemia, calcitonin can be used effectively within the first 24 h in combination with rehydration and saline diuresis while waiting for more sustained effects from a simultaneously administered bisphosphonate such as pamidronate. Usual doses of calcitonin are 2 to 8 U/kg of body weight intravenously, subcutaneously, or intramuscularly every 6 to 12 h.

Other Therapies *Plicamycin* (mithramycin), which inhibits bone resorption, has been a useful therapeutic agent but is now little used because of the effectiveness of bisphosphonates. Plicamycin must be given intravenously, either as a bolus injection or by slow infusion. The usual dose is 25 ug/kg body weight. Major side effects include thrombocytopenia, hepatocellular necrosis with increased lactic acid dehydrogenase (LDH) and aspartate aminotransferase (AST) levels, and decreased levels of clotting factors with resultant epistaxis, bruising, hemorrhage, and bleeding gums.

Gallium nitrate exerts a hypocalcemic action by inhibiting bone resorption and altering the structure of bone crystals. Major disadvantages include the 5-day duration of infusion and the potential for nephrotoxicity and relatively shorter duration of action than bisphosphonates. Accordingly it is not often used because of superior alternatives.

Glucocorticoids increase urinary calcium excretion and decrease intestinal calcium absorption when given in pharmacologic doses, but they also cause negative skeletal calcium balance. In normal individuals and in patients with primary hyperparathyroidism, glucocorticoids neither increase nor decrease the serum calcium concentration. In patients with hypercalcemia due to certain osteolytic malignancies, however, glucocorticoids may be effective as a result of antitumor effects. The malignancies in which hypercalcemia responds to glucocorticoids include multiple myeloma, leukemia, Hodgkin's disease, other lymphomas, and carcinoma of the breast, at least early in the course of the disease. Glucocorticoids are also effective in treating hypercalcemia due to vitamin D intoxication and sarcoidosis. In all the preceding situations, the hypocalcemic effect develops over several days, and the usual glucocorticoid dosage is 40 to 100 mg prednisone (or its equivalent) daily in four divided doses. The side effects of chronic glucocorticoid therapy may be acceptable in some circumstances.

Dialysis is often the treatment of choice for hypercalcemia complicated by renal failure, which is difficult to manage. Peritoneal dialysis with calcium-free dialysis fluid can remove 5 to 12.5 mmol (200 to 500 mg) of calcium in 24 to 48 h and lower the serum calcium concentration by 0.7 to 3 mmol/L (3 to 12 mg/dL). Large quantities of phosphate are lost during dialysis, and serum inorganic phosphate concentrations usually fall, thus aggravating hypercalcemia. Therefore, the serum inorganic phosphate concentration should be measured after dialysis, and phosphate supplements should be added to the

diet or to dialysis fluids if necessary.

Phosphate therapy, oral or intravenous, has a limited role in certain circumstances. Patients with primary hyperparathyroidism are frequently hypophosphatemic, and hypercalcemia of other causes also may be complicated by hypophosphatemia. Hypophosphatemia decreases the rate of calcium uptake into bone, increases intestinal calcium absorption, and directly and indirectly stimulates bone breakdown. These effects aggravate hypercalcemia, and correcting hypophosphatemia lowers the serum calcium concentration. The usual treatment is 1 to 1.5 g phosphorus per day for several days, given in four divided doses to minimize the chances of developing hyperphosphatemia. Such therapy has been administered for prolonged periods in selected patients. It is generally believed, but not established, that toxicity does not occur if therapy is limited to restoring serum inorganic phosphate concentrations to normal.

Raising the serum inorganic phosphate concentration above normal decreases serum calcium levels, sometimes strikingly. Intravenous phosphate is one of the most dramatically effective treatments available for severe hypercalcemia but is toxic and even dangerous (fatal hypocalcemia). For these reasons, it is used rarely and only in severely hypercalcemic patients with cardiac or renal failure. A phosphate phosphorus dose of 31500 mg intravenously over 6 to 8 h leads to a prompt decrease in serum calcium of as much as 1.2 to 2.5 mmol/L (5 to 10 mg/dL) in patients with initially normal serum inorganic phosphate concentrations. This therapy should be employed only in extreme emergencies. Inorganic phosphate is commercially available for oral use in liquid, powder, and capsule form and as a liquid for intravenous use. It is important to calculate doses in terms of phosphate phosphorus.

Summary The various therapies for hypercalcemia are listed in [Table 341-2](#). The choice depends on the underlying disease, the severity of the hypercalcemia, the serum inorganic phosphate level, and the renal, hepatic, and bone marrow function. Mild hypercalcemia [≤ 3 mmol/L (12 mg/dL)] can usually be managed by hydration. Severe hypercalcemia [3.7 mmol/L (15 mg/dL)] requires rapid correction. Calcitonin should be given for its rapid, albeit short-lived, blockade of bone resorption, and intravenous pamidronate should be administered, although its onset of action is delayed for 1 to 2 days. In addition, for the first 24 to 48 h, aggressive sodium-calcium diuresis with intravenous saline and large doses of furosemide and ethacrynic acid following initial hydration should be initiated, but only if appropriate monitoring is available and cardiac and renal function are adequate. Otherwise, dialysis may be necessary. Intermediate degrees of hypercalcemia between 3.0 and 3.7 mmol/L (12 and 15 mg/dL) should be approached with vigorous hydration and then the most appropriate selection for the patient of the combinations used with severe hypercalcemia.

HYPOCALCEMIA

PATHOPHYSIOLOGY OF HYPOCALCEMIA: CLASSIFICATION BASED ON MECHANISM

Chronic hypocalcemia is less common than hypercalcemia; causes include chronic renal failure, hereditary and acquired hypoparathyroidism, vitamin D deficiency, [PHP](#),

and hypomagnesemia.

Critically ill patients may have transient hypocalcemia with severe sepsis, burns, acute renal failure, and extensive transfusions with citrated blood. Acute hypocalcemia with certain medications is usually transient and may produce no symptoms. Although as many as half of patients in an intensive care setting are reported to have calcium concentrations <2.1 mmol/L (8.5 mg/dL), $<10\%$ have a reduction in ionized calcium. Patients with severe sepsis may have a decrease in ionized calcium (true hypocalcemia), but in other severely ill individuals, hypoalbuminemia is the primary cause of the reduced total calcium concentration. Alkalosis increases calcium binding to proteins, and in this setting direct measurements of ionized calcium should be made.

Medications such as protamine, heparin, and glucagon may cause transient hypocalcemia. These forms of hypocalcemia are usually not associated with tetany and resolve with improvement in the overall medical condition. The hypocalcemia after repeated transfusions of citrated blood usually resolves quickly.

Patients with *acute pancreatitis* have hypocalcemia that persists during the acute inflammation and varies in degree with the severity of the pancreatitis. The cause of hypocalcemia remains unclear. [PTH](#) values are reported to be low, normal, or elevated, and both resistance to PTH and impaired PTH secretion have been postulated. Occasionally, a chronic low total calcium and low ionized calcium concentration are detected in an elderly patient without obvious cause and with a paucity of symptoms; the pathogenesis is unclear.

Chronic hypocalcemia, however, is usually symptomatic and requires treatment. Neuromuscular and neurologic manifestations of chronic hypocalcemia include muscle spasms, carpopedal spasm, facial grimacing, and, in extreme cases, laryngeal spasm and convulsions. Respiratory arrest may occur. Increased intracranial pressure occurs in some patients with long-standing hypocalcemia, often in association with papilledema. Mental changes include irritability, depression, and psychosis. The QT interval on the electrocardiogram is prolonged, in contrast to its shortening with hypercalcemia. Arrhythmias occur, and digitalis effectiveness may be reduced. Intestinal cramps and chronic malabsorption may occur. Chvostek's or Trousseau's sign can be used to confirm latent tetany.

The classification of hypocalcemia shown in [Table 341-3](#) is based on the premise that [PTH](#) is responsible for minute-to-minute regulation of plasma calcium concentration and, therefore, that the occurrence of hypocalcemia must mean a failure of the homeostatic action of PTH. Failure of the PTH response can occur due to hereditary or acquired parathyroid gland failure, if PTH is ineffective in target organs, or if the action of the hormone is overwhelmed by the loss of calcium from the [ECF](#) at a rate faster than it can be replaced.

PTH ABSENT

Whether hereditary or acquired, hypoparathyroidism has a number of common components. Acute and chronic symptoms of untreated hypocalcemia are shared by both types of hypoparathyroidism, although the onset of hereditary hypoparathyroidism

is more gradual and is often associated with other developmental defects. Basal ganglia calcification and extrapyramidal syndromes are more common and earlier in onset in hereditary hypoparathyroidism. In earlier decades, acquired hypoparathyroidism secondary to surgery in the neck was more common than hereditary hypoparathyroidism, but the frequency of surgically induced parathyroid failure has diminished as a result of improved surgical techniques that spare the parathyroid glands and increased use of nonsurgical therapy for hyperthyroidism. [PHP](#), an example of ineffective PTH action rather than a failure of parathyroid gland production, may share several features with hypoparathyroidism, including extraosseous calcification and extrapyramidal manifestations such as choreoathetotic movements and dystonia.

Papilledema and raised intracranial pressure may occur in both hereditary or acquired hypoparathyroidism, as do chronic changes in fingernails and hair and lenticular cataracts, the latter usually reversible with treatment of hypocalcemia. Certain skin manifestations, including alopecia and candidiasis, are characteristic of hereditary hypoparathyroidism associated with autoimmune polyglandular failure ([Chap. 339](#)).

Hypocalcemia associated with hypomagnesemia is associated with both deficient [PTH](#) release and impaired responsiveness to the hormone. Patients with hypocalcemia secondary to hypomagnesemia have absent or low levels of circulating PTH, indicative of diminished hormone release despite maximum physiologic stimulus by hypocalcemia. Plasma PTH levels return to normal with correction of the hypomagnesemia. Thus hypoparathyroidism with low levels of PTH in blood can be due to hereditary gland failure, acquired gland failure, or acute but reversible gland dysfunction (hypomagnesemia).

Genetic Abnormalities and Hereditary Hypoparathyroidism Hereditary hypoparathyroidism can occur as an isolated entity without other endocrine or dermatologic manifestations (idiopathic hypoparathyroidism); more typically, it occurs in association with other abnormalities such as defective development of the thymus or failure of function of other endocrine organs such as the adrenal, thyroid, or ovary ([Chap. 339](#)). Idiopathic and hereditary hypoparathyroidism are often manifest within the first decade but may appear later.

A rare form of hypoparathyroidism associated with defective development of both the thymus and the parathyroid glands is termed the *DiGeorge syndrome* (DGS), or the *velocardiofacial syndrome* (VCFS). Congenital cardiovascular, facial, and other developmental defects are present, and most patients die in early childhood with severe infections, hypocalcemia and seizures, or cardiovascular complications. Some survive into adulthood, and milder, incomplete forms occur. Most cases are sporadic, but an autosomal dominant form involving microdeletions of chromosome 22q11.2 has been described. Smaller deletions in this region are seen in incomplete forms of the DGS syndrome, appearing in childhood or adolescence, that are manifest primarily by parathyroid gland failure.

Hypoparathyroidism can occur in association with a complex hereditary autoimmune syndrome involving failure of the adrenals, the ovaries, the immune system, and the parathyroids in association with recurrent mucocutaneous candidiasis, alopecia, vitiligo, and pernicious anemia ([Chap. 339](#)). The responsible gene on chromosome 21q22.3 has

been identified. The protein product, which resembles a transcription factor, has been termed the *AIRE* (autoimmune regulator). A stop codon mutation occurs in many Finnish families with the disorder, commonly referred to as *polyglandular autoimmune type 1 deficiency*.

Gain-of-function mutations in the calcium-sensing receptor cause *autosomal dominant hypocalcemia* (ADH). These mutations induce constitutive receptor functions that lead to features that are the inverse of [FHH](#). The activated receptor suppresses [PTH](#), leading to hypocalcemia; receptor activation in the kidney results in excessive renal calcium excretion. Recognition of the syndrome is important because efforts to treat the hypocalcemia of these patients with vitamin D analogues and increased oral calcium exacerbate the already excessive urinary calcium secretion (several grams or more per 24 h), leading to irreversible renal damage from stones and ectopic calcification.

Hypoparathyroidism is seen in two disorders associated with mitochondrial dysfunction and myopathy, one termed the *Kearns-Sayre syndrome* (KSS), with ophthalmoplegia and pigmentary retinopathy, and the other termed the *MELAS syndrome*, mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes. Mutations or deletions in mitochondrial genes have been identified ([Chap. 67](#)).

The two other rare forms of hypoparathyroidism with other multisystem developmental abnormalities follow either an autosomal dominant pattern, with deafness and/or renal dysplasia, or an autosomal recessive pattern, with growth retardation and dysmorphic features.

Hereditary hypoparathyroidism occurs also as an isolated entity without any other defects. The pattern of inheritance varies and includes autosomal dominant, autosomal recessive, and X-linked inheritance patterns. In one family in which the disorder is transmitted as an autosomal dominant trait, a structural abnormality in the [PTH](#) gene has been identified. A defect in the signal sequence needed for processing of the hormone impairs PTH secretion. In another kindred with autosomal recessive inheritance, the mutant allele in the first intron of the PTH gene causes a splicing defect in mRNA production. An X-linked recessive form of hypoparathyroidism has been described in males from two kindreds that are probably related. The locus of the defect has been located to chromosome Xq26-q27.

Acquired Hypoparathyroidism *Acquired chronic hypoparathyroidism* is usually the result of inadvertent surgical removal of all the parathyroid glands; in some instances, not all the tissue is removed, but the remainder undergoes vascular supply compromise secondary to fibrotic changes in the neck after surgery. In the past, the most frequent cause of acquired hypoparathyroidism was surgery for hyperthyroidism. Hypoparathyroidism now usually occurs after surgery for hyperparathyroidism when the surgeon, facing the dilemma of removing too little tissue and thus not curing the hyperparathyroidism, removes too much. Parathyroid function may not be totally absent in all patients with postoperative hypoparathyroidism.

Even rarer causes of acquired chronic hypoparathyroidism include radiation-induced damage subsequent to radioiodine therapy of hyperthyroidism and glandular damage in patients with hemochromatosis or hemosiderosis after repeated blood transfusions.

Infection may involve one or more of the parathyroids but usually does not cause hypoparathyroidism because all four glands are rarely involved.

Transient hypoparathyroidism is frequent following surgery for hyperparathyroidism. After a variable period of hypoparathyroidism, normal parathyroid function may return due to hyperplasia or recovery of remaining tissue. Occasionally, recovery occurs months after surgery.

TREATMENT

Treatment of acquired and hereditary hypoparathyroidism involves replacement with vitamin D or $1,25(\text{OH})_2\text{D}_3$ (calcitriol) combined with a high oral calcium intake. In most patients, blood calcium and phosphate levels are satisfactorily regulated, but some patients show resistance and a brittleness with a tendency to alternate between hypocalcemia and an overshoot hypercalcemia. For many patients, vitamin D in doses of 1 to 3 mg/d (40,000 to 120,000 U/d) combined with ^{31}g elemental calcium is satisfactory. The wide dosage range reflects the variation encountered from patient to patient; precise regulation of each patient is required. Compared to typical daily requirements in euparathyroid patients of 200 U/d, the high dose of vitamin D reflects the reduced conversion of vitamin D to $1,25(\text{OH})_2\text{D}_3$. Many physicians now use 0.5 to 1.0 μg of calcitriol in management of such patients, especially if they are difficult to control. When vitamin D (because of storage in fat) is withdrawn, weeks are required for the disappearance of the biologic effects, compared with a few days for calcitriol, which has a rapid turnover.

Oral calcium and vitamin D restore the overall calcium-phosphate balance but do not reverse the lowered urinary calcium reabsorption typical of hypoparathyroidism. Therefore, care must be taken to avoid excessive urinary calcium excretion after vitamin D and calcium replacement therapy; otherwise, kidney stones can develop. Thiazide diuretics lower urine calcium by as much as 100 mg/d in hypoparathyroid patients on vitamin D, provided they are maintained on a low-sodium diet. Use of thiazides seems to be of benefit in mitigating hypercalciuria and easing the daily management of these patients.

Hypomagnesemia Severe hypomagnesemia is associated with hypocalcemia ([Chap. 340](#)). Restoration of the total-body magnesium deficit leads to rapid reversal of hypocalcemia. There are at least two causes of the hypocalcemia -- impaired [PTH](#) secretion and reduced responsiveness to PTH.

Hypomagnesemia is generally classified as primary or secondary; primary hypomagnesemia is due to hereditary defects in intestinal absorption or renal reabsorption of magnesium. Secondary hypomagnesemia, a more common condition, occurs on a nutritional basis or as a result of acquired intestinal or renal disorders. The most common causes of the secondary disorder are chronic alcoholism with poor nutritional intake, intestinal malabsorption syndromes, and parenteral nutrition when magnesium replacement is omitted.

The effects of magnesium on [PTH](#) secretion are similar to those of calcium; hypermagnesemia suppresses and hypomagnesemia stimulates PTH secretion. The

effects of magnesium on PTH secretion are normally of little significance, however, because the calcium effects dominate. Greater change in magnesium than in calcium is needed to influence hormone secretion. Nonetheless, hypomagnesemia might be expected to increase hormone secretion. It is therefore surprising to find that severe hypomagnesemia is associated with blunted secretion of PTH. The explanation for the paradox is that severe, chronic hypomagnesemia leads to intracellular magnesium deficiency, which interferes with secretion and peripheral responses to PTH. The mechanism of the cellular abnormalities caused by hypomagnesemia is unknown, although effects on adenylate cyclase (for which magnesium is a cofactor) have been proposed.

Serum magnesium must usually fall below 0.4 mmol/L (1.0 mg/dL) to cause hypocalcemia. [PTH](#) levels are undetectable or inappropriately low despite the stimulus of severe hypocalcemia, and acute repletion of magnesium leads to a rapid increase in PTH level. Serum phosphate levels are often not elevated, in contrast to the situation with acquired or idiopathic hypoparathyroidism, probably because phosphate deficiency is a frequent accompaniment of hypomagnesemia.

Diminished peripheral responsiveness to [PTH](#) also occurs in some patients, as documented by subnormal response in urinary phosphorus and urinary cyclic AMP excretion after administration of exogenous PTH to patients who are hypocalcemic and hypomagnesemic. Both blunted PTH secretion and lack of renal response to administered PTH can occur in the same patient. When acute magnesium repletion is undertaken, the restoration of PTH levels to normal or supranormal may precede restoration of normal serum calcium by several days.

TREATMENT

Repletion of magnesium cures the condition, and attention must be given to restoring the intracellular deficiency, which may be considerable. Repletion should be parenteral. After intravenous magnesium administration, serum magnesium may return transiently to the normal range, but unless replacement therapy is adequate serum magnesium will again fall. If renal function is normal, urinary magnesium excretion is a useful indicator of magnesium repletion, as magnesium is retained by the kidney until the deficiency is corrected. Intracellular deficits can be ≥ 50 mmol. Parenteral administration of 10 to 14 mmol magnesium usually reverses the signs of magnesium deficiency, but greater amounts may occasionally be required if the deficit is large. If the cause of the hypomagnesemia is renal magnesium wasting, magnesium may have to be given chronically to prevent recurrence ([Chap. 340](#)).

[PTH](#) INEFFECTIVE

PTH is ineffective when the hormone receptor-guanyl nucleotide-binding protein complex is defective ([PHP](#), discussed below), when PTH action to promote calcium absorption from the diet is impaired because of vitamin D deficiency or because vitamin D is ineffective (receptor or synthesis defects), or in chronic renal failure in which the calcium-elevating action of PTH is impaired.

Typically, hypophosphatemia is more severe than hypocalcemia in vitamin D deficiency

states because of the increased secretion of [PTH](#), which, although only partly effective in elevating blood calcium, is capable of promoting phosphaturia.

[PHP](#), on the other hand, has a pathophysiology different from the other disorders of ineffective [PTH](#) action. PHP resembles hypoparathyroidism (in which PTH synthesis is deficient) and is manifested by hypocalcemia and hyperphosphatemia. The cause of the disorder is defective hormone activation of guanyl nucleotide-binding proteins, resulting in failure of PTH to increase intracellular cyclic AMP (see below).

Chronic Renal Failure Improved medical management of chronic renal failure and/or a more indolent course of the renal disease now allow many patients to survive long enough to develop features of renal osteodystrophy. Phosphate retention and impaired production of $1,25(\text{OH})_2\text{D}$ are the principal factors that cause calcium deficiency, secondary hyperparathyroidism, and bone disease. The uremic state also causes impairment of intestinal absorption by mechanisms other than defects in vitamin D metabolism. Nonetheless, treatment with supraphysiologic amounts of vitamin D or calcitriol corrects the impaired calcium absorption.

Hyperphosphatemia in renal failure lowers blood calcium levels by several mechanisms, including extraosseous deposition of calcium and phosphate, impairment of the bone-resorbing action of [PTH](#), and reduction in $1,25(\text{OH})_2\text{D}$ production by remaining renal tissue. Low levels of $1,25(\text{OH})_2\text{D}$ due to hyperphosphatemia and destruction of renal tissue and are critical in the development of hypocalcemia.

TREATMENT

Therapy of chronic renal failure ([Chap. 270](#)) involves appropriate management of patients prior to dialysis and adjustment of regimens once dialysis is initiated. Attention should be paid to restriction of phosphate in the diet; use of calcium-containing salts as phosphate-binding antacids is preferable, rather than aluminum, to avoid the problem of aluminum intoxication; provision of an adequate calcium intake by mouth, usually 1 to 2 g/d; and supplementation with 0.25 to 1.0 ug/d calcitriol. Each patient must be monitored closely. The aim of therapy is to restore normal calcium balance to prevent osteomalacia and secondary hyperparathyroidism. Reduction of hyperphosphatemia and restoration of normal intestinal calcium absorption by calcitriol can improve blood calcium levels and reduce the manifestations of secondary hyperparathyroidism. Since adynamic bone disease can occur in association with low [PTH](#) levels, it is important to avoid excessive suppression of the parathyroid glands while recognizing the beneficial effects of controlling the secondary hyperparathyroidism. These patients should probably be closely monitored with PTH assays that detect only the full-length PTH 1-84 to avoid interference by biologically inactive amino-terminally truncated PTH.

Vitamin D Deficiency due to Inadequate Diet and/or Sunlight Vitamin D deficiency due to inadequate intake of dairy products enriched with vitamin D, lack of vitamin supplementation, and reduced sunlight exposure in the elderly, particularly during winter in northern latitudes, is more common in the United States than previously recognized. Biopsies of bone in elderly patients with hip fracture (documenting osteomalacia) and abnormal levels of vitamin D metabolites, [PTH](#), calcium, and phosphate indicate that vitamin D deficiency may occur in as many as 25% of elderly patients, particularly in

areas where there is little ambient sunlight. Concentrations of 25(OH)D are low or low-normal in these patients. Quantitative histomorphometry of bone biopsy specimens reveals widened osteoid seams consistent with osteomalacia. PTH hypersecretion compensates for the tendency for the blood calcium to fall but also induces renal phosphate wasting and results in osteomalacia.

Treatment involves adequate replacement with vitamin D and calcium until the deficiencies are corrected. Severe hypocalcemia rarely occurs in moderately severe vitamin D deficiency of the elderly, but vitamin D deficiency must be considered in the differential diagnosis of mild hypocalcemia.

Defective Vitamin D Metabolism

Anticonvulsant therapy Anticonvulsant therapy with any of several agents induces acquired vitamin D deficiency by increasing the conversion of vitamin D to inactive compounds. The more marginal the vitamin D intake in the diet, the more likely that anticonvulsant therapy will lead to abnormal mineral and bone metabolism ([Chap. 340](#)).

Although 1,25(OH)₂D levels are lower in patients treated with chronic anticonvulsants than in the normal population, there is a great deal of variation. The greater prevalence of the disorder in some European populations and in the mentally retarded may reflect the lower vitamin D intake of those groups. Restoration of bone mineral mass and reversal of hypocalcemia can be accomplished with vitamin D replacement plus oral calcium. Administration of 50,000 units of vitamin D monthly may be preventive if anticonvulsant therapy needs to be given chronically.

Vitamin D-dependent rickets type I Rickets can be due to *resistance to the action of* vitamin D as well as to vitamin D deficiency. Vitamin D-dependent rickets type I, previously termed *pseudo-vitamin D-resistant rickets*, differs from true vitamin D-resistant rickets (vitamin D-dependent rickets type II, see below) in that it is less severe and the biochemical and radiographic abnormalities can be reversed with appropriate doses of the vitamin or the active metabolite, 1,25(OH)₂D₃.

Clinical features include hypocalcemia, often with tetany or convulsions, hypophosphatemia, secondary hyperparathyroidism, and osteomalacia, often associated with skeletal deformities and increased alkaline phosphatase. Physiologic amounts of calcitriol cure the disease ([Chap. 340](#)). This finding fits with the pathophysiology as the disorder, which is autosomal recessive, is now known to be caused by a series of mutations in the gene for the 25(OH)D-1α-hydroxylase. Over 20 different mutations have been identified. All patients have both alleles inactivated, but often the genetic pattern is that of a compound heterozygote. Response to high doses of vitamin D or calcifediol, as noted in prior years, is probably due to direct effects of 25(OH)D at high levels. Treatment begins with 1 to 2 ug/d calcitriol, but maintenance is satisfactory with physiologic doses of calcitriol (0.5 to 1.0 ug/d). Careful adjustment of calcitriol dose is required, particularly during growth periods.

Vitamin D Ineffective

Intestinal Malabsorption Mild hypocalcemia, secondary hyperparathyroidism, severe

hypophosphatemia, and a variety of nutritional deficiencies occur with gastrointestinal diseases. Hepatocellular dysfunction can lead to reduction in 25(OH)D levels, as in portal or biliary cirrhosis of the liver, and malabsorption of vitamin D and its metabolites, including 1,25(OH)₂D, may occur in a variety of bowel diseases, hereditary or acquired. Hypocalcemia itself can lead to steatorrhea, due to deficient production of pancreatic enzymes and bile salts. Depending on the disorder, vitamin D or its metabolites can be given parenterally, guaranteeing adequate blood levels of active metabolites.

Vitamin D-dependent rickets type II Vitamin D-dependent rickets type II results from end-organ resistance to the active metabolite 1,25(OH)₂D₃. The clinical features resemble those of the type I disorder and include hypocalcemia, hypophosphatemia, secondary hyperparathyroidism, and rickets. A clear distinction is partial or total alopecia in type II. Plasma levels of 1,25(OH)₂D are at least three times normal, in keeping with the refractoriness of the end organs. Some patients respond to very high doses of vitamin D or vitamin D metabolites (e.g., 17 to 20 ug/d calcitriol). Earlier suggestions that there were both receptor and postreceptor defects are incorrect. All of the genetically characterized phenotypes have mutations in the gene for the vitamin D receptor. Nineteen mutations have been identified that affect different regions of the receptor primarily in the DNA binding domain (with normal ligand binding: these were the so-called postreceptor defects detected by earlier indirect methods) or in the ligand-binding domain (classified previously as receptor negative).

Pseudohypoparathyroidism [PHP](#) is a hereditary disorder characterized by symptoms and signs of hypoparathyroidism, typically in association with distinctive skeletal and developmental defects. The hypoparathyroidism is due to a deficient end-organ response to [PTH](#). Hyperplasia of the parathyroids, a response to hormone resistance, causes elevation of PTH levels. Studies, both clinical and basic, have clarified some aspects of this syndrome, including the variable clinical spectrum, the pathophysiology, the genetic defects, and the inheritance.

A working classification of the various forms of [PHP](#) is given in [Table 341-4](#). The classification scheme is based on the signs of ineffective [PTH](#) action (low calcium and high phosphate), urinary cyclic AMP response to exogenous PTH, the presence or absence of *Albright's hereditary osteodystrophy* (AHO), and assays of the concentration of the G_s subunit of the adenylate cyclase enzyme. Using these criteria, there are four types: PHP type I, subdivided into a and b categories; PHP-II; and pseudopseudohypoparathyroidism (PPHP).

PHP-Ia and PHP-Ib Individuals with [PHP](#)-I, the most common of the disorders, show a deficient urinary cyclic AMP response to administration of exogenous [PTH](#). Patients with PHP-I are divided into type a, who have reduced amounts of G_s in vitro assays with erythrocytes, and type b, with normal amounts of G_s in erythrocytes. There is a third type (PHP-Ic, reported in a few patients) that differs from PHP-Ia only in having normal erythrocyte levels of G_s despite having [AHO](#), hypocalcemia, and decreased urinary cyclic AMP responses to PTH (presumably with a post-G_s defect in adenylyl cyclase stimulation).

Most patients show characteristic features of [AHO](#), consisting of short stature, round face, skeletal anomalies (brachydactyly), and heterotopic calcification. Patients have low

calcium and high phosphate levels, as with true hypoparathyroidism. [PTH](#) levels, however, are elevated, reflecting resistance to hormone action.

Amorphous deposits of calcium and phosphate are found in the basal ganglia in about half of patients. The defects in metacarpal and metatarsal bones are sometimes accompanied by short phalanges as well, possibly reflecting premature closing of the epiphyses. The typical findings are short fourth and fifth metacarpals and metatarsals. The defects are usually bilateral. Exostoses and radius curvus are frequent. Impairments in olfaction and taste and unusual dermatoglyphic abnormalities have been reported.

PPHP The initial view that the defect responsible for [PHP](#)-Ia was simply the deficiency of G_s subunits was temporarily confounded by the subsequent discovery that the same 50% reduction in G_s subunits was seen in patients with [PPHP](#), who have typical features of the hereditary osteodystrophy syndrome despite normal serum calcium levels and normal response of urinary cyclic AMP to exogenous [PTH](#).

Multiple defects have now been identified in the *GNAS-1* gene in [PHP](#)-Ia and [PPHP](#) patients. This gene, which is located on chromosome 20q13, encodes the stimulatory G protein subunit G_{sa} , among other products (see below). Mutations include abnormalities in splice junctions associated with deficient mRNA production and point mutations that result in a protein with defective function as well as the 50% reduction in G_{sa} levels in erythrocytes.

Detailed analyses of disease transmission in affected kindreds have clarified many features of [PHP](#)-Ia, [PPHP](#), and [PHP](#)-Ib ([Fig. 341-10](#)). The former two entities, traced through multiple kindreds, have an inheritance pattern consistent with gene imprinting -- only females, not males, can transmit the full disease with hypocalcemia -- and [PHP](#) and [PPHP](#) do not coexist in the same generation. The phenomenon of gene imprinting involves selective inactivation of either the maternal or the paternal allele ([Chap. 65](#)). In the case of the G_{sa} gene, it is paternally imprinted (silenced) so that the disease [PHP](#)-Ia is never inherited from the father carrying the defective allele but only from the mother. On the other hand, the defective allele is not imprinted or silenced in all tissues. It seems possible, therefore, that the [AHO](#) phenotype recognized in [PPHP](#) as well as [PHP](#)-Ia reflects haplotype insufficiency. In the renal cortex, however, it is postulated that only the maternal allele is normally active, such that lack of activity from a defective paternal allele is not of consequence. This explains the occurrence in [PHP](#)-Ia of hypocalcemia, hyperphosphatemia, and other stigmata such as variable resistance to other hormones (if similar tissue-specific imprinting occurs in other organs). Strong evidence favoring this overall hypothesis comes from gene knockout studies in the mouse (ablating exon 2 of the gene). Mice inheriting the mutant allele from the female had undetectable G_{sa} protein in renal cortex and were hypocalcemic and resistant to renal actions of [PTH](#). Offspring inheriting the mutant allele from the male showed no evidence of [PTH](#) resistance or hypercalcemia.

The complex mechanisms that control the *GNAS-1* gene also contribute to challenges involved in unraveling the pathogenesis of these disorders. Alternative splicing patterns produce three different transcripts that encode distinct proteins. In addition to G_{sa} , this gene encodes a second protein product with a unique NH₂-terminus (the XL exon); XL_{as}

includes exons 2-13. It is unknown whether this protein can function as a stimulatory G protein, but the mRNA encoding it is expressed in numerous endocrine tissues and is transcribed from only the paternal allele. A third transcript is transcribed from only the maternal allele and encodes the protein product, NESP55, which contains no homology with XL α s or G α s.

PHP-Ib, lacking the **AHO** phenotype, shares with PHP-Ia the resistance to **PTH** action and a blunted urinary cyclic AMP response to administered PTH, a standard test for hormone resistance ([Table 341-4](#)). PHP-Ib patients, however, show normal levels of G α s in erythrocytes. Bone responsiveness may be excessive rather than blunted in PHP-Ib compared to PHP-Ia patients, based on case reports that have emphasized an osteitis fibrosa-like pattern in some PHP patients who lack the AHO phenotype. The inheritance patterns in PHP-Ib kindreds are clearly consistent with paternal imprinting and lack male transmission of symptomatic disease; gene cloning studies have narrowed the responsible region to chromosome 20, close to -- if not within -- the *GNAS-1* gene locus. Elucidation of the responsible genetic and pathogenetic mechanisms in this disorder may further illuminate the function of the complex *GNAS-1* gene and the role of its products in hormonal signaling.

PHP-II refers to patients with hypocalcemia and hyperphosphatemia who have a normal urinary cyclic AMP response to **PTH**. These patients are assumed to have a defect in the response to PTH at a locus distal to cyclic AMP production, although at least some patients may instead have occult vitamin D deficiency.

The diagnosis of these hormone-resistant states can usually be made without difficulty when there is a positive family history for developmental defects and/or the presence of developmental anomalies, including brachydactyly, in association with the signs and symptoms of hypoparathyroidism. In all categories -- **PHP-Ia**, **-Ib**, and **-II** -- serum **PTH** levels are elevated, particularly when patients are hypocalcemic. However, patients with PHP-Ib or PHP-II do not have phenotypic abnormalities, only hypocalcemia with high PTH levels, confirming hormone resistance. In PHP-Ib, the response of urinary cyclic AMP to the administration of exogenous PTH is blunted. Levels of G α s subunits in erythrocyte membranes are, however, normal in those with PHP-Ib. The diagnosis of PHP-II is more complex, in that cyclic AMP responses in urine are, by definition, normal. Since vitamin D deficiency itself can dissociate phosphaturic and urinary cyclic AMP responses to exogenous PTH, vitamin D deficiency must be excluded before the diagnosis of PHP-II can be entertained.

TREATMENT

Treatment of **PHP** is similar to that of hypoparathyroidism, except that the doses of vitamin D and calcium are usually lower than those required in true hypoparathyroidism, presumably because the defect in PHP is only partial because of imprinting in specific tissues (renal cortex vs. renal medulla). Variability in response makes it necessary to establish the optimal regimen for each patient, based on maintaining the appropriate blood calcium level and urinary calcium excretion.

PTH Overwhelmed Occasionally, loss of calcium from the ECF is so severe that PTH cannot compensate. Such situations include acute pancreatitis and severe, acute

hyperphosphatemia, often in association with renal failure, conditions in which there is rapid efflux of calcium from extracellular fluid. Severe hypocalcemia can occur quickly; PTH rises in response to hypocalcemia but does not return blood calcium to normal.

Severe, Acute Hyperphosphatemia Severe hyperphosphatemia is associated with extensive tissue damage or cell destruction ([Chap. 340](#)). The combination of increased release of phosphate from muscle and impaired ability to excrete phosphorus because of renal failure causes moderate to severe hyperphosphatemia, the latter causing calcium loss from the blood and mild to moderate hypocalcemia. Hypocalcemia is usually reversed with tissue repair and restoration of renal function as phosphorus and creatinine values return to normal. There may even be a mild hypercalcemic period in the oliguric phase of renal function recovery. This sequence, severe hypocalcemia followed by mild hypercalcemia, reflects widespread deposition of calcium in muscle and subsequent redistribution of some of the calcium to the [ECF](#) after return of phosphate levels to normal.

Other causes of hyperphosphatemia include hypothermia, massive hepatic failure, and hematologic malignancies, either because of high cell turnover of malignancy or because of cell destruction by chemotherapy.

TREATMENT

Treatment is directed toward lowering of blood phosphate by the administration of phosphate-binding antacids or dialysis, often needed for the management of renal failure. Although calcium replacement may be necessary if hypocalcemia is severe and symptomatic, calcium administration during the hyperphosphatemic period tends to increase extraosseous calcium deposition and aggravate tissue damage. The levels of $1,25(\text{OH})_2\text{D}$ may be low during the hyperphosphatemic phase and return to normal during the oliguric phase of recovery.

Osteitis Fibrosis after Parathyroidectomy Severe hypocalcemia after parathyroid surgery is less common now that osteitis fibrosa cystica is an infrequent manifestation of hyperparathyroidism. When osteitis fibrosa cystica is severe, however, bone mineral deficits can be large. After parathyroidectomy, hypocalcemia can persist for days if calcium replacement is inadequate. Treatment may require parenteral administration of calcium; addition of calcitriol and oral calcium supplementation is sometimes needed for weeks to a month or two until bone defects are filled (which, of course, is of therapeutic benefit in the skeleton), making it possible to discontinue parenteral calcium and/or reduce the amount.

DIFFERENTIAL DIAGNOSIS OF HYPOCALCEMIA

Care must be taken to ensure that true hypocalcemia is present; in addition, acute transient hypocalcemia can be a manifestation of a variety of severe, acute illnesses, as discussed above. *Chronic hypocalcemia*, however, can usually be ascribed to a few disorders associated with absent or ineffective [PTH](#). Important clinical criteria include the duration of the illness, signs or symptoms of associated disorders, and the presence of features that suggest a hereditary abnormality. A nutritional history can be helpful in recognizing a low intake of vitamin D and calcium in the elderly, and a history of

excessive alcohol intake may suggest magnesium deficiency.

Hypoparathyroidism and [PHP](#) are typically lifelong illnesses, usually (but not always) appearing by adolescence; hence a recent onset of hypocalcemia in an adult is more likely due to nutritional deficiencies, renal failure, or intestinal disorders that result in deficient or ineffective vitamin D. Neck surgery, even long past, however, can be associated with a delayed onset of postoperative hypoparathyroidism. A history of seizure disorder raises the issue of anticonvulsive medication. Developmental defects, particularly in childhood and adolescence, may point to the diagnosis of PHP. Rickets and a variety of neuromuscular syndromes and deformities may indicate ineffective vitamin D action, either due to defects in vitamin D metabolism or to vitamin D deficiency.

A pattern of *low calcium with high phosphorus* in the absence of renal failure or massive tissue destruction almost invariably means hypoparathyroidism or [PHP](#). A *low calcium and low phosphorus* points to absent or ineffective vitamin D, thereby impairing the action of [PTH](#) on calcium metabolism (but not phosphate clearance). The relative ineffectiveness of PTH in vitamin D deficiency, anticonvulsant therapy, gastrointestinal disorders, and hereditary defects in vitamin D metabolism leads to secondary hyperparathyroidism as a compensation. The relatively unopposed action of the excess PTH on renal tubule phosphate transport, which is less dependent on vitamin D than calcium transport, accounts for renal phosphate wasting and hypophosphatemia.

Exceptions to these patterns may occur. Most forms of hypomagnesemia are due to long-standing nutritional deficiency as seen in chronic alcoholics. Despite the fact that the hypocalcemia is principally due to an acute absence of [PTH](#), phosphate levels are usually low, rather than elevated as in hypoparathyroidism. Chronic renal failure is often associated with hypocalcemia and hyperphosphatemia, despite secondary hyperparathyroidism.

Diagnosis is usually established by application of the [PTH](#) immunoassay, tests for vitamin D metabolites, and measurements of the urinary cyclic AMP response to exogenous PTH. In hereditary and acquired hypoparathyroidism and in severe hypomagnesemia, PTH is either undetectable or in the normal range. This finding in a hypocalcemic patient is supportive of hypoparathyroidism, as distinct from ineffective PTH action, in which even mild hypocalcemia is associated with elevated PTH levels. Hence a failure to detect elevated PTH levels establishes the diagnosis of hypoparathyroidism; elevated levels suggest the presence of secondary hyperparathyroidism, as found in many of the situations in which the hormone is ineffective due to associated abnormalities in vitamin D action. Assays for 25(OH)D and 1,25(OH)₂D can be helpful. Low or low normal 25(OH)D indicates vitamin D deficiency due to lack of sunlight, inadequate vitamin D intake, or intestinal malabsorption. A low level of 1,25(OH)₂D in the presence of elevated concentrations of PTH suggests ineffective PTH action in disorders such as chronic renal failure, severe vitamin D deficiency, vitamin D-dependent rickets type I, and [PHP](#). Recognition that mild hypocalcemia, rickets, and hypophosphatemia are due to anticonvulsant therapy is made by history.

TREATMENT

Hypocalcemic States The management of hypoparathyroidism, [PHP](#), chronic renal failure, and hereditary defects in vitamin D metabolism involves the use of vitamin D or vitamin D metabolites and calcium supplementation. Vitamin D itself is the least expensive form of vitamin D replacement and is frequently used in the management of uncomplicated hypoparathyroidism and some disorders associated with ineffective vitamin D action. When vitamin D is used prophylactically, as in the elderly or in those with chronic anticonvulsant therapy, there is a wider margin of safety than with the more potent metabolites. However, most of the conditions in which vitamin D is administered chronically for hypocalcemia require amounts 50 to 100 times the daily replacement dose because the formation of $1,25(\text{OH})_2\text{D}$ is deficient. In such situations, vitamin D is no safer than the active metabolite because intoxication can occur with high-dose therapy (because of storage in fat). Calcitriol is more rapid in onset of action and also has a short biologic half-life.

Vitamin D (5 ug/d) or calcifediol and lower doses of calcitriol (0.25 to 1.0 ug/d) are required to prevent rickets in normal individuals. In contrast, 1 to 3 mg (1000 to 3000 ug of vitamin D₂ or D₃) is typically required in hypoparathyroidism; doses of calcifediol are also high (several hundred micrograms per day). The dose of calcitriol is unchanged in hypoparathyroidism, since the defect is in hydroxylation by the $25(\text{OH})\text{D}-1\alpha$ -hydroxylase.

Patients with hypoparathyroidism should be given 2 to 3 g elemental calcium by mouth each day. The two agents, vitamin D or calcitriol and oral calcium, can be varied independently. If hypocalcemia alternates with episodes of hypercalcemia in more brittle patients with hypoparathyroidism, administration of calcitriol and use of thiazides, as discussed above, may make management easier.

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342. OSTEOPOROSIS - Robert Lindsay, Felicia Cosman

Osteoporosis, characterized by decreased bone strength, is prevalent among postmenopausal women but also occurs in men and women with underlying conditions or major risk factors associated with bone demineralization. Its chief clinical manifestations are vertebral and hip fractures. Osteoporosis affects >10 million individuals in the United States, but only 10 to 20% are diagnosed and treated.

DEFINITION

Osteoporosis is defined as a reduction of bone mass (or density) or the presence of a fragility fracture. This reduction in bone tissue is accompanied by deterioration in the architecture of the skeleton, leading to a markedly increased risk of fracture. Osteoporosis is defined operationally as a bone density that falls 2.5 standard deviations (SD) below the mean -- also referred to as a *T-score* of -2.5. Those who fall at the lower end of the young normal range (a *T-score* of >1 SD below the mean) have low bone density and are considered to be at increased risk of osteoporosis.

EPIDEMIOLOGY

In the United States, as many as 8 million women and 2 million men have osteoporosis (*T-score* <-2.5), and an additional 18 million individuals have bone mass levels that put them at increased risk of developing osteoporosis (e.g., bone mass *T-score* <-1.0). Osteoporosis occurs more frequently with increasing age as bone tissue is progressively lost. In women, the loss of ovarian function at menopause (typically after age 50) precipitates rapid bone loss such that most women meet the criteria for osteoporosis by age 70.

The epidemiology of fractures follows similar trends as the loss of bone density. Fractures of the distal radius increase in frequency before age 50 and plateau by age 60, with only a modest age-related increase thereafter. In contrast, incidence rates for hip fractures double every 5 years after age 70 ([Fig. 342-1](#)). This distinct epidemiology may be related to the way people fall as they age, with fewer falls on an outstretched hand. At least 1.5 million fractures occur each year in the United States as a consequence of osteoporosis. As the population continues to age, the total number of fractures will continue to escalate.

About 300,000 hip fractures occur each year in the United States, most of which require hospital admission and surgical intervention. The probability that a 50-year-old white individual will have a hip fracture during his or her lifetime is 14% for women and 5% for men; the risk for African Americans is much lower (about half these rates). Hip fractures are associated with a high incidence of deep vein thrombosis and pulmonary embolism (20 to 50%) and a mortality rate between 5 and 20% during the few months after surgery.

There are about 500,000 vertebral crush fractures per year in the United States. Only a fraction of these are recognized clinically, since many are relatively asymptomatic and are identified incidentally during radiography for other purposes ([Fig. 342-2](#)). Vertebral fractures rarely require hospitalization but are associated with long-term morbidity and a

slight increase in mortality. Multiple fractures lead to height loss (often of several inches), kyphosis, and secondary pain and discomfort related to altered biomechanics of the back. Thoracic fractures can be associated with restrictive lung disease, whereas lumbar fractures are associated with abdominal symptoms including distention, early satiety, and constipation.

Approximately 200,000 wrist fractures occur in the United States each year. Fractures of other bones also occur with osteoporosis, which is not surprising given that bone loss is a systemic phenomenon. Fractures of the pelvis and proximal humerus are clearly associated with osteoporosis. Although some fractures are clearly the result of major trauma, the threshold for fracture is reduced for an osteoporotic bone ([Fig. 342-3](#)). A list of common risk factors for osteoporotic fractures is summarized in [Table 342-1](#). Prior fractures, a family history of osteoporotic fractures, and low body weight are each independent predictors of fracture. Chronic diseases that increase the risk of falling or frailty, including dementias, Parkinson's disease, and multiple sclerosis, also increase fracture risk.

In the United States and Europe, osteoporosis-related fractures are more common among women than men, presumably due to a lower peak bone mass as well as postmenopausal bone loss in women. However, this gender difference in bone density and age-related increase in hip fractures is not as apparent in some other cultures, possibly due to genetics, physical activity level, or diet.

PATHOPHYSIOLOGY

Bone Remodeling Osteoporosis results from bone loss due to normal age-related changes in bone remodeling as well as extrinsic and intrinsic factors that exaggerate this process. These changes may be superimposed on a low peak bone mass. Consequently, the bone remodeling process is fundamental for understanding the pathophysiology of osteoporosis ([Chap. 340](#)). The skeleton increases in size by linear growth and by apposition of new bone tissue on the outer surfaces of the cortex ([Fig. 342-4](#)). This latter process is the phenomenon of modeling, which also allows the long bones to adapt in shape to the stresses placed upon them. Increased sex hormone production at puberty is required for maximum skeletal maturation, which reaches maximum mass and density in early adulthood. Nutrition and lifestyle also play an important role in growth, though genetic factors are the major determinants of peak skeletal mass and density. Numerous genes control skeletal growth, peak bone mass, and body size, but it is likely that separate genes control skeletal structure and density. Heritability estimates of 50 to 80% for bone density and size have been derived based on twin studies. Though peak bone mass is often lower among individuals with a family history of osteoporosis, association studies of candidate genes [vitamin D receptor; Type I collagen, the estrogen receptor (ER), interleukin (IL) 6; and insulin-like growth factor (IGF) I] have not been consistently replicated. Linkage studies suggest that a genetic locus on chromosome 11 is associated with high bone mass.

Once peak skeletal mass has been attained, the process of remodeling becomes the principal metabolic activity of the skeleton. This process has three primary functions: (1) to repair microdamage within the skeleton, (2) to maintain skeletal strength, and (3) to supply calcium from the skeleton to maintain serum calcium. Acute demands for calcium

involve osteoclast-mediated resorption as well as calcium transport by osteocytes. The activation of remodeling may be induced by microdamage to bone due to excessive or accumulated stress.

Bone remodeling is also regulated by several circulating hormones, including estrogens, androgens, vitamin D, and parathyroid hormone (PTH), as well as locally produced growth factors such as IGF-I and -II, transforming growth factor (TGF) β , parathyroid hormone-related peptide (PTHrP), ILs, prostaglandins, tumor necrosis factor (TNF), and osteoprotegerin ligand (Fig. 342-5). Additional influences include nutrition (particularly calcium intake) and physical activity level. The end result of this remodeling process is that the resorbed bone is replaced by an equal amount of new bone tissue. Thus, the mass of the skeleton remains constant after peak bone mass is achieved in adulthood. After age 30 to 45, however, the resorption and formation processes become imbalanced, and resorption exceeds formation. This imbalance may begin at different ages and varies at different skeletal sites; it becomes exaggerated in women after menopause. Excessive bone loss can be due to an increase in osteoclastic activity and/or a decrease in osteoblastic activity. In addition, an increase in remodeling activation frequency can magnify the small imbalance seen at each remodeling unit.

In trabecular bone, if the osteoclasts are sufficiently aggressive to penetrate trabeculae, they leave no template for new bone formation to occur and, consequently, may cause rapid bone loss. In cortical bone, increased activation of remodeling creates more porous bone. The effect of this increased porosity on cortical bone strength may be modest if the overall diameter of the bone is not changed. However, decreased apposition of new bone on the periosteal surface coupled with increased endocortical resorption of bone decreases the biomechanical strength of long bones. Even a slight exaggeration in normal bone loss patterns increases the risk of osteoporotic fracture.

Calcium Nutrition Peak bone mass may be impaired by inadequate calcium intake during growth, thereby leading to increased risk of osteoporosis in later life. During the adult phase of life, calcium deprivation induces secondary hyperparathyroidism and an increase in the rate of remodeling. PTH stimulates the hydroxylation of vitamin D in the kidney, leading to increased levels of 1,25-dihydroxyvitamin D [$1,25(\text{OH})_2\text{D}$] and enhanced gastrointestinal calcium absorption. PTH also reduces renal calcium loss. Though these are appropriate short-term homeostatic responses for improving calcium economy, the long-term effects are detrimental to the skeleton because of the ongoing imbalance at remodeling sites.

Total daily calcium intakes of <400 mg are likely to be detrimental to the skeleton, but there is more doubt about intakes in the 600- to 800-mg range, which is the average intake among adults in the United States. The recommended daily required intake of 1000 to 1200 mg for adults accommodates population heterogeneity in controlling calcium balance (Chap. 73).

Vitamin D (See also Chap. 340) Severe vitamin D deficiency causes rickets in children or osteomalacia in adults. There is accumulating evidence that vitamin D deficiency may be more prevalent than previously thought, particularly among individuals at increased risk, such as the elderly; those living in northern latitudes; and in individuals with poor nutrition, malabsorption, or chronic liver or renal disease. Modest vitamin D deficiency

leads to compensatory secondary hyperparathyroidism and is an important risk factor for osteoporosis and fractures. Some studies have shown that >50% of inpatients on a general medical service exhibit biochemical features of vitamin D deficiency, including increased levels of PTH and alkaline phosphatase and lower levels of ionized calcium. In women living in northern latitudes, it has been shown that vitamin D levels decline during the winter months. This is associated with a striking seasonal bone loss, reflecting increased bone turnover. Treatment with vitamin D and calcium supplementation prevents this seasonal effect on bone metabolism. Reduced fracture rates have also been documented among individuals in northern latitudes who have greater vitamin D intake and have higher 25-hydroxyvitamin D [25(OH)D] levels (see below).

Estrogen Status Estrogen deficiency probably causes bone loss by two distinct but interrelated mechanisms: (1) activation of new bone remodeling sites, and (2) exaggeration of the imbalance between bone formation and resorption. The change in activation frequency causes a transient bone loss until a new steady state between resorption and formation is achieved. The remodeling imbalance, however, results in a permanent decrement in mass that can only be corrected by a remodeling event during which bone formation exceeds resorption. In addition, the very presence of more remodeling sites in the skeleton increases the probability that trabeculae will be penetrated, thereby eliminating the template upon which new bone can be formed and accelerating the loss of bony tissue.

The most frequent estrogen-deficient state is the cessation of ovarian function at the time of menopause, which occurs on average at the age of 51. Thus, with current life expectancy, an average woman will spend about 30 years without ovarian supply of estrogen. The mechanism by which estrogen deficiency causes bone loss is summarized in [Fig. 342-5](#). Marrow cells (macrophages, monocytes, osteoclast precursors, mast cells) as well as bone cells (osteoblasts, osteocytes, osteoclasts) express [ERs](#) α and β . The net effect of estrogen deficiency is increased osteoclast recruitment and perhaps activity. Estrogen may also play an important role in determining the life span of bone cells by controlling the rate of apoptosis. Thus, in situations of estrogen deprivation, the life span of osteoblasts may be decreased whereas the longevity of osteoclasts is increased.

Since remodeling is initiated at the surface of bone, it follows that trabecular bone -- which has a considerably larger surface area (80% of the total) than cortical bone -- will be preferentially affected by estrogen deficiency. Fractures occur earliest at sites where trabecular bone contributes most to bone strength; consequently, vertebral fractures are the most common early consequence of estrogen deficiency.

Physical Activity Inactivity, such as prolonged bed rest or paralysis, results in significant bone loss. Concordantly, athletes have higher bone mass than the general population. These changes in skeletal mass are most marked when the stimulus begins during growth and before the age of puberty. Adults are less capable than children of increasing bone mass following restoration of physical activity. Epidemiologic data support the beneficial effects on the skeleton of chronic high levels of physical activity. Fracture risk is lower in rural communities and in countries where physical activity is maintained into old age. However, when exercise is initiated during adult life, the effects

of moderate exercise are modest, with a bone mass increase of 1 to 2%. It is argued that more active individuals are less likely to fall and are more capable of protecting themselves upon falling, thereby reducing fracture risk.

Chronic Disease Various genetic and acquired diseases are associated with an increase in the risk of osteoporosis ([Table 342-2](#)). Mechanisms that contribute to bone loss are unique for each disease and typically result from multiple factors including nutrition, reduced physical activity levels, and factors that affect bone-remodeling rates.

Medications A large number of medications used in clinical practice have potentially detrimental effects on the skeleton ([Table 342-3](#)). *Glucocorticoids* are a common cause of medication-induced osteoporosis. It is often not possible to determine the extent to which osteoporosis is related to the glucocorticoid or to other factors, as treatment is superimposed on the effects of the primary disease, which may itself be associated with bone loss (e.g., rheumatoid arthritis). Excessive doses of thyroid hormone can accelerate bone remodeling and result in bone loss.

Other medications have less detrimental effects upon the skeleton than pharmacologic doses of glucocorticoids. *Anticonvulsants* are thought to increase the risk of osteoporosis, although many affected individuals have concomitant vitamin D insufficiency, as anticonvulsants that induce the cytochrome P450 system alter vitamin D metabolism. Patients undergoing transplantation are at high risk for rapid bone loss and fracture not only from glucocorticoids but also from treatment with other *immunosuppressants*, such as cyclosporine and tacrolimus (FK506). In addition, these patients often have underlying metabolic abnormalities, such as hepatic or renal failure, that predispose to osteopenia.

Cigarette Consumption The use of cigarettes over a long period has detrimental effects on bone mass. These effects may be mediated directly, by toxic effects on osteoblasts, or indirectly by modifying estrogen metabolism. On average, cigarette smokers reach menopause 1 to 2 years earlier than the general population. Cigarette smoking also has secondary effects on bone growth, such as illness, frailty, decreased exercise, and the need for additional medications (e.g., glucocorticoids for lung disease).

MEASUREMENT OF BONE MASS

Several noninvasive techniques are now available for estimating skeletal mass or density. These include dual-energy x-ray absorptiometry (DXA), single-energy x-ray absorptiometry (SXA), quantitative computed tomography (CT), and ultrasound.

[DXA](#) is a highly accurate x-ray technique that has become the standard for measuring bone density in most centers. Though it can be used for measurements of any skeletal site, clinical determinations are usually made of the lumbar spine and hip. Portable DXA machines have been developed that measure the heel (calcaneus), forearm (radius and ulna), or finger (phalanges), and DXA can also be used to measure body composition. In the DXA technique, two x-ray energies are used to estimate the area of mineralized tissue, and the mineral content is divided by the area, which partially corrects for body size. However, this correction is only partial since DXA is a two-dimensional scanning

technique and cannot estimate the depths or posteroanterior length of the bone. Thus, small people tend to have lower-than-average bone mineral density (BMD). Bone spurs, which are frequent in osteoarthritis, tend to falsely increase bone density of the spine. Because DXA instrumentation is provided by several different manufacturers, the output varies in absolute terms. Consequently, it has become standard practice to relate the results to "normal" values using T-scores, which compare individual results to those in a young population that is matched for race and gender. Alternatively, Z-scores compare individual results to those of an age-matched population that is also matched for race and gender. Thus, a 60-year-old woman with a Z-score of -1 (1 SD below mean for age) could have a T-score of -2.5 (2.5 SD below mean for a young control group) ([Fig. 342-6](#)).

[CT](#) is used primarily to measure the spine, and peripheral CT is used to measure bone in the forearm or tibia. Research into the use of CT for measurement of the hip is ongoing. The results obtained from CT are different from all others currently available since this technique specifically analyzes trabecular bone and can provide a true density (mass of bone per unit volume) measurement. However, CT remains expensive, involves greater radiation exposure, and is less reproducible.

Ultrasound is used to measure bone mass by calculating the attenuation of the signal as it passes through bone or the speed with which it traverses the bone. It is unclear whether ultrasound assesses bone quality, but this may be an advantage of the technique. Because of its relatively low cost and mobility, ultrasound is amenable for use as a screening procedure.

All of these techniques for measuring [BMD](#) have been approved by the U.S. Food and Drug Administration (FDA) based upon their capacity to predict fracture risk. The hip is the preferred site of measurement in most individuals, since it directly assesses bone mass at an important fracture site. When hip measurements are performed by [DXA](#), the spine can be measured at the same time. In younger individuals, such as perimenopausal women, spine measurements may be the most sensitive indicator of bone loss.

When to Measure Bone Mass Clinical guidelines developed by the National Osteoporosis Foundation recommend bone mass measurements in postmenopausal women, assuming they have risk factors for osteoporosis in addition to age, gender, and estrogen deficiency. The guidelines further recommend that bone mass measurement be considered in *all* women by age 60 to 65. Criteria approved for Medicare reimbursement of [BMD](#) are summarized in [Table 342-4](#).

When to Treat Based Upon Bone Mass Results The guidelines developed by the National Osteoporosis Foundation suggest that patients be considered for treatment when [BMD](#) > 2.5 SD below the mean value for young adults (T-score $\leq$ -2.5). Treatment should also be considered in women with risk factors in addition to menopause, if measurement of BMD of the hip gives a T score < -2.0. Because the fracture risk increases continuously as T-scores decline, there is no critical threshold and treatment decisions must be individualized. Cost-benefit analyses in this area are changing rapidly because of the availability of new drugs [e.g., bisphosphonates, selective estrogen receptor modulators (SERMs)] and the results of trials [e.g., the Heart and

Estrogen-Progestin Replacement Study (HERS)] examining the long-term cardiovascular effects of hormone replacement therapy (HRT).

Approach to the Patient

The perimenopausal transition is a good opportunity to initiate discussion about risk factors for osteoporosis and to consider indications for a [BMD](#) test. A careful history and physical examination should be performed to identify risk factors for osteoporosis. As noted above, a low Z-score increases the suspicion of a secondary disease. Height loss >2.5 to 3.8 cm (1 to 1.5 in.) is an indication for radiography to rule out asymptomatic vertebral fractures, as is the presence of significant kyphosis or back pain, particularly if it began after menopause. For patients who present with fractures, it is important to ensure that the fractures are truly due to trauma or osteoporosis and not to secondary underlying malignancy. Usually this is clear on routine radiography, but on occasion, [CT](#), magnetic resonance imaging, or radionuclide scans may be helpful. Severe unremitting back pain also raises the suspicion of other causes such as malignancy (especially myeloma).

Routine Laboratory Evaluation There is no established algorithm for the evaluation of women presenting with osteoporosis. A general evaluation that includes complete blood count, serum calcium, and perhaps urine calcium is helpful for identifying selected secondary causes of low bone mass, particularly for women with fractures or very low Z-scores. An elevated serum calcium level suggests hyperparathyroidism or malignancy, whereas a reduced serum calcium level may reflect malnutrition and osteomalacia. In the presence of hypercalcemia, a serum [PTH](#) level differentiates between hyperparathyroidism (PTH-) and malignancy (PTH⁺), and a high [PTHrP](#) level can help document the presence of humoral hypercalcemia of malignancy ([Chap. 341](#)). A low urine calcium (<50 mg/24 h) suggests osteomalacia, malnutrition, or malabsorption; a high urine calcium (>300 mg/24 h) is indicative of hypercalciuria and must be investigated further. Hypercalciuria occurs primarily in three situations: (1) a renal calcium leak, which is more frequent in males with osteoporosis; (2) absorptive hypercalciuria, which can be idiopathic or associated with increased 1,25(OH)₂D in granulomatous disease; or (3) hematologic malignancies or conditions associated with excessive bone turnover such as Paget's disease, hyperparathyroidism, and hyperthyroidism.

When there is clinical suspicion of hyperthyroidism or Cushing's syndrome, thyroid stimulating hormone (TSH) or urinary free cortisol levels should be measured. When bowel disease, malabsorption, or malnutrition is suspected, serum albumin, cholesterol, and a complete blood count should be checked. Asymptomatic malabsorption might be suspected if there is anemia (macrocytic -- vitamin B₁₂ or folate deficiency; or microcytic -- iron deficiency), or low serum cholesterol or urinary calcium levels. If these or other features suggest malabsorption, further evaluation is required. Asymptomatic celiac sprue with selective malabsorption is not uncommon; the diagnosis requires antigliadin and antiendomysial antibody tests and often a small-bowel biopsy. A trial of a gluten-free diet may be confirmatory ([Chap. 286](#)).

Myeloma can masquerade as generalized osteoporosis, although it more commonly presents with bone pain and characteristic "punched-out" lesions on radiography.

Serum and urine electrophoresis and evaluation for light chains in urine are required to exclude this diagnosis. A bone marrow biopsy may be required to rule out myeloma (in patients with equivocal electrophoretic results) and can also be used to exclude mastocytosis, leukemia, and other marrow infiltrative disorders, such as Gaucher's disease.

Bone Biopsy Although the use of bone biopsy is rarely required today, it remains an important tool in clinical research. Tetracycline labeling of the skeleton allows determination of the rate of remodeling as well as evaluation for other metabolic bone diseases. The current use of [BMD](#) tests, in combination with hormonal evaluation and biochemical markers of bone remodeling, has largely replaced bone biopsy.

Biochemical Markers Several biochemical tests are now available that provide an index of the overall rate of bone remodeling ([Table 342-5](#)). Biochemical markers are usually characterized as those related primarily to *bone formation* or *bone resorption*. These tests measure the overall state of bone remodeling at a single point in time. Clinical use of these tests has been hampered by biologic variability (in part related to circadian rhythm) as well as to analytical variability.

For the most part, remodeling markers do not predict rates of bone loss well enough to use this information clinically. However, markers of bone resorption may help in the prediction of fracture risk, particularly in older individuals. In women ³⁶⁵ years, when bone density results are greater than the usual treatment thresholds noted above, a high level of bone resorption should prompt consideration of treatment. The primary use of biochemical markers is for monitoring the response to treatment. With the introduction of antiresorptive therapeutic agents, bone remodeling declines rapidly, with the fall in resorption occurring earlier than the fall in formation. Inhibition of bone resorption is maximal within 3 to 6 months. Thus, measurement of bone resorption prior to initiating therapy and 4 to 6 months after starting therapy provides an earlier estimate of patient response than does bone densitometry. A decline in resorptive markers can be ascertained after treatment with bisphosphonates and [HRT](#); this effect is less marked after treatment with either raloxifene or intranasal calcitonin. A biochemical marker response to therapy is particularly useful for asymptomatic patients and helps to ensure long-term compliance. When agents that stimulate bone formation become available, bone remodeling markers may be useful to help select therapy. However, since all current therapeutic approaches reduce bone turnover, this strategy currently has little value.

TREATMENT

Management of Osteoporotic Fractures Treatment of the patient with osteoporosis frequently involves management of acute fractures as well as treatment of the underlying disease. Hip fractures almost always require surgical repair if the patient is to become ambulatory again. Depending on the location and severity of the fracture, condition of the neighboring joint, and general status of the patient, procedures may include open reduction and internal fixation with pins and plates, hemiarthroplasties, and total arthroplasties. These surgical procedures are followed by intense rehabilitation in an attempt to return patients to their prefracture functional level. Long bone fractures often require either external or internal fixation. Other fractures (e.g., vertebral, rib, and

pelvic fractures) are usually managed with only supportive care, requiring no specific orthopedic treatment.

Only ~25 to 30% of vertebral compression fractures present with sudden-onset back pain. For acutely symptomatic fractures, treatment with analgesics is required, including nonsteroidal anti-inflammatory agents and/or acetaminophen, sometimes with the addition of a narcotic agent (codeine or oxycodone). A few small, randomized clinical trials have demonstrated that calcitonin may reduce pain related to acute vertebral compression fracture. A recently developed, but still experimental, technique involves percutaneous injection of artificial cement (polymethylmethacrylate) into the vertebral body (vertebroplasty or kyphoplasty); this has been reported to offer significant immediate pain relief in the majority of patients. Short periods of bed rest may be helpful for pain management, but, in general, early mobilization is recommended as it helps prevent further bone loss associated with immobilization. Occasionally, use of a soft elastic-style brace may facilitate earlier mobilization. Muscle spasms often occur with acute compression fractures and can be treated with muscle relaxants and heat treatments.

Severe pain usually resolves within 6 to 10 weeks. Chronic pain is probably not bony in origin; instead, it is related to abnormal strain on muscles, ligaments, and tendons and to secondary facet-joint arthritis associated with alterations in thoracic and/or abdominal shape. Chronic pain is difficult to treat effectively and may require analgesics, sometimes including narcotic analgesics. Frequent intermittent rest in a supine or semireclining position is often required to allow the soft tissues, which are under tension, to relax. Back-strengthening exercises (paraspinal) may be beneficial. Heat treatments help relax muscles and reduce the muscular component of discomfort. Various physical modalities, such as ultrasound and transcutaneous nerve stimulation, may be beneficial in some patients. Pain also occurs in the neck region, not as a result of compression fractures (which almost never occur in the cervical spine as a result of osteoporosis) but because of chronic strain associated from trying to elevate the head in a person with a severe thoracic kyphosis.

Multiple vertebral fractures are often associated with psychological symptoms, not always commonly appreciated. The changes in body configuration and back pain can lead to marked loss of self-image and a secondary depression. Altered balance, precipitated by the kyphosis and the anterior movement of the body's center of gravity, leads to a fear of falling, a consequent tendency to remain indoors, and the onset of social isolation. These symptoms can sometimes be alleviated by family support and/or psychotherapy. Medication may be necessary when depressive features are present.

Management of the Underlying Disease

Risk Factor Reduction Patients should be thoroughly educated to reduce the likelihood of any risk factors associated with bone loss and falling. Medications should be reviewed to ensure that any glucocorticoid medication is truly indicated and is being given in doses as low as possible. For those on thyroid hormone replacement, [TSH](#) testing should be performed to ensure that an adequate, but not excessive, dose is being used, as excess can be associated with increased bone loss. In patients who smoke, efforts should be made to facilitate smoking cessation. Reducing

risk factors for falling also includes alcohol abuse treatment and a review of the medical regimen for any drugs that might be associated with orthostatic hypotension and/or sedation, including hypnotics and anxiolytics. If nocturia occurs, the frequency should be reduced, if possible (e.g., by decreasing or modifying diuretic use), as arising in the middle of sleep is a common precipitant of a fall. Patients should be instructed about environmental safety with regard to eliminating exposed wires, curtain strings, slippery rugs, and mobile tables. Avoiding stocking feet on wood floors, checking carpet condition (particularly on stairs), and providing good light in paths to bathrooms and outside the home are good preventive measures. Treatment for impaired vision is recommended, particularly a problem with depth perception, which is specifically associated with increased falling risk. Elderly patients with neurologic impairment (e.g., stroke, Parkinson's disease, Alzheimer's disease) are particularly at risk of falling and require specialized supervision and care.

Nutritional Recommendations

CALCIUM A large body of data indicates that optimal calcium intake reduces bone loss and suppresses bone turnover. Recommended intakes from a recent report from the Institute of Medicine are shown in [Table 342-6](#). The National Health and Nutritional Evaluation Studies (NHANES) have consistently documented that average calcium intakes fall considerably short of these recommendations. The preferred source of calcium is from dairy products and other foods, but many patients require additional calcium supplementation. Food sources of calcium are dairy products (milk, yogurt, and cheese) and fortified foods such as certain cereals, waffles, snacks, juices, and crackers. Some of these fortified foods contain as much calcium per serving as milk.

If a calcium supplement is required, it should be taken in doses of 600 mg at a time, as the calcium absorption fraction decreases at higher doses. Calcium supplements should be calculated based on the elemental calcium content of the supplement, not the weight of the calcium salt ([Table 342-7](#)). Calcium supplements containing carbonate are best taken with food since they require acid for solubility. Calcium citrate supplements can be taken at any time.

Several controlled clinical trials of calcium plus vitamin D have confirmed reductions in clinical fractures, including fractures of the hip (~20 to 30% risk reduction). All recent studies of pharmacologic agents have been conducted in the context of calcium replacement ($\pm$ vitamin D). Thus, it is standard practice to ensure an adequate calcium and vitamin D intake in patients with osteoporosis, whether they are receiving additional pharmacologic therapy or not.

Although side effects from supplemental calcium are minimal, individuals with a history of kidney stones should have a 24-h urine calcium determination before starting increased calcium to avoid hypercalciuria. Furthermore, a thiazide-containing diuretic might be indicated in some patients to increase renal tubular calcium reabsorption and to reduce urine calcium levels.

VITAMIN D Vitamin D is synthesized in skin under the influence of heat and ultraviolet light ([Chap. 340](#)). However, large segments of the population do not obtain sufficient vitamin D to maintain what is now considered an adequate supply [serum 25(OH)D

consistently >15 to 20 ng/mL]. Since vitamin D supplementation at doses that would achieve these serum levels is safe and inexpensive, it is now routine to recommend supplemental vitamin D. The Institute of Medicine recommends daily intakes of 200 IU for adults <50 years of age, 400 IU for those from 50 to 70 years, and 600 IU for those >70 years. Multivitamin tablets usually contain 400 IU, and many calcium supplements also contain vitamin D.

OTHER NUTRIENTS Other nutrients such as salt and caffeine may have modest effects on calcium excretion or absorption. Adequate vitamin K status is required for optimal carboxylation of osteocalcin. States in which vitamin K nutrition or metabolism is impaired, such as with long-term coumadin therapy, have been associated with reduced bone mass.

Magnesium is abundant in foods, and magnesium deficiency is quite rare in the absence of a serious chronic disease. Magnesium supplementation may be warranted in patients with inflammatory bowel disease, celiac sprue, chemotherapy, severe diarrhea, malnutrition, or alcoholism. Phytoestrogens may impact skeletal health, although the degree of this effect is unclear. Dietary phytoestrogens, which are derived primarily from soy products and legumes (e.g., garbanzo beans, chickpeas, and lentils), are insufficiently potent to justify their use in place of a pharmacologic agent in the treatment of osteoporosis.

Patients with hip fracture are often frail and relatively malnourished. Some data suggest an improved outcome in such patients when they are provided calorie and protein supplementation.

Exercise Exercise in young individuals increases the likelihood that they will attain the maximal genetically determined peak bone mass. Meta-analyses of studies performed in postmenopausal women indicate that weight-bearing exercise prevents bone loss but does not appear to result in substantial bone gain. When the exercise is discontinued, any effects on bone mass wane. It is important to note, however, that exercise also has beneficial effects on neuromuscular function. Exercise can improve coordination, balance, and strength and thereby reduce the risk of falling, as well as the severity of injury upon a fall. Therefore, the beneficial effects of exercise on muscle mass and reduced risk of falling justify its recommendation for all age groups. A walking program is a practical way to start. Other activities such as dancing, racquet sports, cross-country skiing, and use of gym equipment are also recommended, depending on the patient's personal preference. Even women who cannot walk benefit from swimming or water exercises, not so much for the effects on bone, which are quite minimal, but because of effects on muscle. Exercise habits should be consistent, optimally at least three times a week.

Pharmacologic Therapies Until fairly recently, estrogen treatment, either by itself or in concert with a progestin, was the primary therapeutic agent for prevention or treatment of osteoporosis. Over the past 5 years, a number of new drugs have appeared, and more are expected in the near future. Some are agents that specifically treat osteoporosis (bisphosphonates, calcitonin); others, such as tissue-selective estrogens (or [SERMs](#)), have broader effects. The availability of these drugs allows therapy to be tailored to the needs of an individual patient. The evidence supporting the effectiveness

of each remedy is variable, in part because these treatments are new.

Estrogens A large body of clinical trial data indicates that various types of estrogens (conjugated equine estrogens, estradiol, estrone, esterified estrogens, ethinyl estradiol, and mestranol) reduce bone turnover, prevent bone loss, and induce small increases in bone mass of the spine, hip, and total body. The effects of estrogen are seen in women with natural or surgical menopause and in late postmenopausal women with or without established osteoporosis. Estrogens are efficacious when administered orally, buccally, vaginally, percutaneously, subcutaneously, and transdermally. For both oral and transdermal routes of administration, combined estrogen/progestin preparations are now available in many countries, obviating the problem of taking two tablets or using a patch and oral progestin. One large study, referred to as PEPI (Postmenopausal Estrogen/ Progestin Intervention Trial), indicated that C-21 progestins alone do not augment the effect of estrogen on bone mass ([Fig. 342-7](#)).

DOSE OF ESTROGEN For oral estrogens, the recommended dose is 0.3 mg/d for esterified estrogens, 0.625 mg/d for conjugated equine estrogens, and 5 ug/d for ethinyl estradiol. For transdermal estrogen, the commonly used dose supplies 50 ug estradiol per day, but a lower dose may be appropriate for some individuals. Dose-response data are not available for other routes of administration.

FRACTURE DATA In contrast to the body of clinical trial data evaluating the effects of estrogen on bone mass, its effects on fracture occurrence have been less well studied. Epidemiologic databases indicate that women who take estrogen replacement have a 50% reduction, on average, of osteoporotic fractures, including hip fractures. The beneficial effect of estrogen is greatest among those who start replacement early and continue the treatment; the benefit wanes after discontinuation such that there is no residual protective effect against fracture by 10 years after discontinuation. There are no clinical trial data confirming that estrogen administration reduces the risk of hip fracture. In fact, the [HERS](#) trial of women with established coronary artery disease showed no reduction in the risk of hip or clinical fractures in the estrogen-progestin arm relative to the placebo group. These data may not be definitive, however, since the women were not chosen for osteoporosis risk and were at unknown risk of osteoporotic fracture. Furthermore, radiographic vertebral fractures were not assessed in this study. One clinical study which looked at all nonvertebral fractures suggested a reduction in HRT-treated women.

A few clinical trials have evaluated spine fracture occurrence as an outcome with estrogen therapy. One that used high doses of estrogen (2.5 mg conjugated equine estrogen per day) indicated marked vertebral fracture reduction in estrogen-treated women. Several other small studies, using lower estrogen doses, have consistently shown that estrogen treatment reduces the incidence of vertebral compression fracture. The ongoing Women's Health Initiative will provide additional data about the effects of estrogen on the risk of other osteoporosis-related fractures.

Long-term estrogen use may be associated with an increase in the risk of venous thromboembolism and gallbladder, uterine, and breast cancer; in observational studies, estrogens have been associated with a significant reduction in myocardial infarction, although this was not so in HERS. The WHI will provide further information.

MODE OF ACTION Two subtypes of [ERs](#), α and β , have been identified in bone and other tissues. Cells of monocyte lineage express both ER α and β , as do osteoblasts. Estrogen-mediated effects vary depending on the receptor type. Using ER knockout mouse models, elimination of ER α produces a modest reduction in bone mass, whereas ER β null animals had very little abnormality, except greater cortical bone mass. A male patient with a homozygous mutation of ER α had markedly decreased bone density as well as abnormalities in epiphyseal closure, confirming the important role of ER α in bone biology. The mechanism of estrogen action in bone is an area of active investigation ([Fig. 342-5](#)). Though data are conflicting, estrogens appear to inhibit osteoclasts directly. However, the majority of estrogen (and androgen) effects on bone resorption are mediated indirectly through paracrine factors produced by osteoblasts. These actions include: (1) increasing [IGF-1](#) and [TGF- \$\beta\$](#) , and (2) suppressing [IL-1](#) (α and β), [IL-6](#), [TNF- \$\alpha\$](#) , and osteocalcin synthesis. The consequence of these effects is primarily to decrease bone resorption.

Progestins In women with a uterus, daily progestin or cyclical progestins at least 12 days per month are prescribed in combination with estrogens to reduce the risk of uterine cancer. Medroxyprogesterone acetate and norethindrone acetate blunt the high-density lipoprotein response to estrogen, but micronized progesterone does not. Neither medroxyprogesterone acetate nor micronized progesterone appears to have an independent effect on bone; at lower doses, norethindrone acetate might have an additive benefit. On breast tissue, progestins may increase the risk of breast cancer, though this is by no means definite.

Tissue-Selective Estrogens, or SERMs Two [SERMs](#) are currently being used in postmenopausal women: raloxifene, which is approved for prevention and treatment of osteoporosis, and tamoxifen, which is approved for the prevention and treatment of breast cancer.

Tamoxifen reduces bone turnover and bone loss in postmenopausal women compared to placebo groups. These findings support the concept that tamoxifen acts as an estrogenic agent in bone. There are limited data on the effect of tamoxifen on fracture risk, but the Breast Cancer Prevention study indicated a possible reduction in clinical vertebral, hip, and Colles' fractures. The major benefit of tamoxifen is on breast cancer occurrence. The breast cancer prevention trial indicated that tamoxifen administration over 4 to 5 years reduced the incidence of new invasive and noninvasive breast cancer by approximately 45% in women at increased risk of breast cancer. The incidence of [ER](#)-positive breast cancers was reduced by 65%.

Raloxifene (60 mg/d) has effects on bone turnover and bone mass that are very similar to those of tamoxifen, indicating that this agent is also estrogenic on the skeleton. The effect of raloxifene on bone density (+1.4 to 2.8% versus placebo in the spine, hip, and total body) is somewhat less than that seen with standard doses of estrogens. Raloxifene reduces the occurrence of vertebral fracture by 30 to 50%, depending on the subpopulation.

Raloxifene, like tamoxifen and estrogen, has effects throughout other organ systems. The most positive effect appears to be a reduction in invasive breast cancer (mainly

decreased [ER](#)-positive) occurrence of about 70% in women who take raloxifene compared to placebo. In contrast to tamoxifen, raloxifene is not associated with an increase in the risk of uterine cancer or benign uterine disease. Raloxifene increases the occurrence of hot flashes. Although raloxifene reduces serum total and low-density lipoprotein cholesterol, lipoprotein(a), and fibrinogen, no studies including cardiovascular disease or cerebrovascular disease endpoints are available.

MODE OF ACTION OF SERMS All [SERMs](#) bind to the [ER](#), but each agent produces a unique receptor conformation. As a result, specific coactivator or corepressor proteins are bound to the receptor ([Chap. 327](#)), resulting in differential effects on gene transcription that vary according to other transcription factors present in the cell. Another aspect of selectivity is the affinity of each SERM for the different ER α and ER β subtypes, which are expressed differentially in various tissues. These tissue-selective effects of SERMs offer the possibility of tailoring estrogen therapy to best meet the needs and risk factor profile of an individual patient.

Bisphosphonates Both alendronate and risendronate are approved for the prevention and treatment of postmenopausal osteoporosis and treatment of steroid-induced osteoporosis. Risedronate is also approved for the prevention of steroid-induced osteoporosis.

Alendronate has been shown to have dramatic effects in patients with osteoporosis, decreasing bone turnover and increasing bone mass in the spine by up to 8% versus placebo and by 6% versus placebo in the hip ([Fig. 342-8](#)). Multiple trials have evaluated the effect of alendronate on fracture occurrence. The Fracture Intervention Trial provided evidence in over 2000 women with prevalent vertebral fractures that daily alendronate treatment (5 mg/d for 2 years and 10 mg/d for 9 months afterwards) reduces vertebral fracture risk by about 50%, multiple vertebral fractures by up to 90%, and hip fractures by up to 50% ([Fig. 342-9](#)). Several subsequent trials have confirmed these findings. For example, in a study of >1900 women with low bone mass treated with alendronate (10 mg/d) versus placebo, the incidence of all nonvertebral fractures was reduced by ~47% after just 1 year.

Alendronate (5 to 10 mg/d) should be given with a full glass of water before breakfast, as bisphosphonates are poorly absorbed. Because of the potential for esophageal irritation, alendronate is contraindicated in patients who have stricture or inadequate emptying of the esophagus. It is recommended that patients remain upright for at least 30 min after taking the medication to avoid esophageal irritation. Cases of esophagitis, esophageal ulcer, and esophageal stricture have been described, but the incidence appears to be low. In clinical trials, overall gastrointestinal symptomatology was no different with alendronate compared to placebo.

Risedronate produces a dramatic reduction in bone turnover and an increase in bone mass. Controlled clinical trials have demonstrated >40% reduction in vertebral fracture risk over 3 years, accompanied by a 33% reduction in clinical nonspine fractures. Reports from several studies show a 40% reduction in hip fracture in patients with osteoporosis, with a somewhat greater effect in patients with prevalent vertebral fractures. Patients should take risedronate (5.0 mg orally) with a full glass of plain water [0.18 to 0.25 L (6 to 8 oz)], to facilitate delivery to the stomach, and should not lie down

for 30 min after taking the drug. The incidence of gastrointestinal side effects in these trials with risedronate was similar to that of placebo.

Etidronate was the first bisphosphonate to be approved, initially for use in Paget's disease and hypercalcemia. This agent has also been used in osteoporosis trials of smaller magnitude than those performed for alendronate and risedronate. Etidronate probably has some efficacy against vertebral fracture when given as an intermittent cyclical regimen (2 weeks on, 2 1/2 months off).

MODE OF ACTION Bisphosphonates are structurally related to pyrophosphates, compounds that are incorporated into bone matrix. Through mechanisms that remain to be fully elucidated, bisphosphonates specifically impair osteoclast function and reduce osteoclast number, in part by the induction of apoptosis. Recent evidence suggests that the nitrogen-containing bisphosphonates also inhibit protein prenylation, one of the end products in the mevalonic acid pathway. This effect disrupts intracellular protein trafficking and may ultimately lead to apoptosis. Bisphosphonates have very long retention in the skeleton and may exert long-term effects.

Calcitonin Calcitonin is a polypeptide hormone produced by the thyroid gland ([Chap. 341](#)). Its physiologic role is unclear as no skeletal disease has been described in association with calcitonin deficiency or calcitonin excess. Calcitonins are approved by the [FDA](#) for Paget's disease, hypercalcemia, and osteoporosis in women >5 years past menopause.

Injectable calcitonin produces small increments in bone mass of the lumbar spine. However, difficulty of administration and frequent reactions, including nausea and facial flushing, make general use limited. In 1995, a nasal spray containing calcitonin (200 IU/d) was approved for treatment of osteoporosis in postmenopausal women. Several studies indicate that nasal calcitonin produces small increments in bone mass and a small reduction in new vertebral fractures in calcitonin-treated patients versus those on calcium alone.

Calcitonin is not indicated for prevention of osteoporosis and is not sufficiently potent to prevent bone loss in early postmenopausal women. As mentioned above, calcitonin might have an analgesic effect on bone pain, both in the subcutaneous and possibly the nasal form.

MODE OF ACTION Calcitonin suppresses osteoclast activity by direct action on the osteoclast calcitonin receptor. Osteoclasts exposed to calcitonin cannot maintain their active ruffled border, which normally maintains close contact with underlying bone. Calcitonin also affects osteoclast mobility and the movement of enzyme-containing cytoplasmic granules.

Experimental Agents

PARATHYROID HORMONE Endogenous [PTH](#) is an 84-amino-acid peptide that is largely responsible for calcium homeostasis ([Chap. 341](#)). Although chronic elevation of PTH, as occurs in hyperparathyroidism, is associated with bone loss (particularly cortical bone), PTH can also exert anabolic effects on bone. Consistent with this, some

observational studies have indicated that mild elevations in PTH are associated with maintenance of trabecular bone mass. On the basis of these findings, preclinical and early clinical studies have been performed using an exogenous PTH analogue (1-34 PTH). The first randomized controlled trial in postmenopausal women showed that PTH, when superimposed on ongoing estrogen therapy, produced substantial increments in bone mass (13% over a 3-year period compared to estrogen alone). This increment in bone mass was also associated with a reduction in risk of vertebral compression deformity ([Fig. 342-10](#)). More recent studies have confirmed the ability of combined treatment with estrogen and PTH to induce striking increases in bone mass.

[PTH](#) use may be limited by its mode of administration, which currently requires subcutaneous injection. Alternative modes of delivery are being investigated, including transdermal and inhalation routes. The optimal frequency of administration also remains to be established, and it is possible that PTH might also be effective when used in high doses, 1 month out of every 3.

MODE OF ACTION Exogenously administered [PTH](#) appears to have direct actions on osteoblast activity, with biochemical and histomorphometric evidence of de novo bone formation early in response to PTH, prior to activation of bone resorption. Subsequently, PTH activates bone remodeling but still appears to favor bone formation over bone resorption. PTH stimulates [IGF-I](#) and collagen production and appears to increase osteoblast number by inhibiting apoptosis and stimulating replication.

FLUORIDE Fluoride has been available for many years and is a potent stimulator of osteoprogenitor cells when studied in vitro. It has been used in multiple osteoporosis studies with conflicting results, in part related to use of varying doses and preparations. Fluoride produces marked effects on bone mass, especially in the spine, where gains of around 10% per year have been observed. However, despite increments in bone mass, there is no consistent effect of fluoride on vertebral or nonvertebral fracture, which might actually increase when high doses of fluoride are used. Furthermore, animal data suggest that there is reduced biomechanical strength when fluoride is incorporated into bone as fluoroapatite, with excess osteoid accumulation and evidence of woven rather than lamellar bone formation, especially at high doses. For these reasons, fluoride remains an experimental agent, despite its long history and multiple studies.

OTHER POTENTIAL ANABOLIC AGENTS Several small studies of growth hormone (GH), alone or in combination with other agents, have not shown consistent or substantial positive effects on skeletal mass. Many of these studies are relatively short-term, and the effects of GH and the [IGFs](#) are still under investigation. Anabolic steroids, mostly derivatives of testosterone, act primarily as antiresorptive agents to reduce bone turnover but may also stimulate osteoblastic activity. Effects on bone mass remain unclear but appear weak, in general, and use is limited by masculinizing side effects. Several recent observational studies suggest that the statin drugs, currently used to treat hypercholesterolemia, may be associated with increased bone mass and reduced fractures, but there are not clinical trial data.

Nonpharmacologic Approaches Protective pads worn around the outer thigh, which cover the trochanteric region of the hip can prevent hip fractures in elderly residents in nursing homes. The use of hip protectors is limited largely by compliance and comfort,

but new devices are being developed that may circumvent these problems and provide adjunctive treatments.

Treatment Monitoring There are currently no well-accepted guidelines for monitoring treatment of osteoporosis. Because most osteoporosis treatments produce small or moderate bone mass increments on average, it is reasonable to consider [BMD](#) as a monitoring tool. As with any biologic or assay determination, there is precision error with repeated measurements. Changes must exceed ~4% in the spine and 6% in the hip to be considered significant in any individual. The hip is the preferred site due to larger surface area and greater reproducibility. Medication-induced increments may require several years to produce changes of this magnitude (if they do at all). Consequently, it can be argued that BMD should not be repeated at intervals <2 years. Only significant BMD reductions should prompt a change in medical regimen, as it is expected that many individuals will not show responses greater than the detection limits of the current measurement techniques.

Biochemical markers of bone turnover may prove useful for treatment monitoring, but there is currently little hard evidence to support this concept; it remains unclear which endpoint is most useful. If bone turnover markers are used, a determination should be made before starting therapy and repeated³⁴ months after therapy is initiated. In general, a change in bone turnover markers must be 30 to 40% lower than the baseline to be significant because of the biologic and technical variability in these tests. A positive change in biochemical markers and/or bone density can be useful to help patients adhere to treatment regimens.

GLUCOCORTICOID-INDUCED OSTEOPOROSIS

Osteoporotic fractures are a well-characterized consequence of the hypercortisolism associated with Cushing's syndrome. However, the therapeutic use of glucocorticoids is by far the most common form of glucocorticoid-induced osteoporosis. Glucocorticoids are widely used in the treatment of a variety of disorders, including chronic lung disorders, rheumatoid arthritis and other connective tissue diseases, inflammatory bowel disease, and posttransplantation. Osteoporosis and related fractures are serious side effects of chronic glucocorticoid therapy. Because the effects of glucocorticoids on the skeleton are often superimposed upon the consequences of aging and menopause, it is not surprising that women and the elderly are most frequently affected. The skeletal response to steroids is remarkably heterogeneous, however, and even young, growing individuals treated with glucocorticoids can present with fractures.

The risk of fractures depends on the dose and duration of glucocorticoid therapy. Thus, cumulative dose is an important determinant of fracture risk. Bone loss is more rapid during the early months of treatment, and trabecular bone is more severely affected than cortical bone. Bone loss has been documented with the use of oral prednisone at doses that are generally considered to be less than replacement levels, and the lower threshold is not known. High-dose inhaled glucocorticoids can produce systemic effects on the skeleton, as can intraarticular injections. Alternate-day delivery does not appear to ameliorate the skeletal effects of glucocorticoids. The prevalence of vertebral fractures in asthmatic patients treated for 1 year with glucocorticoids is 11%, and increased risk of fractures has been demonstrated in most other disease states treated

with glucocorticoids.

Pathophysiology Glucocorticoids increase bone loss by multiple mechanisms including: (1) inhibition of osteoblast function and potential increase in osteoblast apoptosis, resulting in impaired synthesis of new bone; (2) stimulation of bone resorption, probably as a secondary effect; (3) impairment of the absorption of calcium across the intestine, probably by a vitamin D-independent effect; (4) increase of urinary calcium loss and induction of some degree of secondary hyperparathyroidism; (5) reduction of adrenal androgens and suppression of ovarian and testicular secretion of estrogens and androgens; and (6) potential induction of glucocorticoid myopathy, which may exacerbate effects on skeletal and calcium homeostasis, as well as increase the risk of falls.

Evaluation of the Patient Because of the prevalence of glucocorticoid-induced osteopenia, it is important to evaluate the status of the skeleton in all patients being initiated on or already receiving long-term glucocorticoid therapy. Modifiable risk factors should be identified, including those for falls. Examination should include height and muscle strength testing. Laboratory evaluation should include an assessment of 24-h urinary calcium. A task force of the American College of Rheumatology recommends that all patients who are being initiated on glucocorticoids and patients already on long-term (>6 months) glucocorticoids have measurement of bone mass at both the spine and hip using [DXA](#). If only one skeletal site can be measured, it is best to assess the spine in individuals <60 years and the hip for those >60 years.

Prevention Bone loss caused by glucocorticoids can be prevented, and the risk of fractures significantly reduced. Strategies must include using the lowest dose of glucocorticoid for disease management. Topical and inhaled routes of administration are preferred, where appropriate. Risk factor reduction is important, including smoking cessation, limitation of alcohol consumption, and participation in weight-bearing exercise, where appropriate. All patients should receive an adequate calcium and vitamin D intake from the diet or from supplements.

TREATMENT

Only bisphosphonates have been demonstrated in large clinical trials to reduce the risk of fractures in patients being treated with glucocorticoids. Risedronate has been shown to prevent bone loss and to reduce vertebral fracture risk by about 70%. Similar beneficial effects are observed in studies of alendronate and etidronate. Controlled trials of [HRT](#) have shown bone-sparing effects, and calcitonin also has some protective effect. Thiazides reduce urine calcium loss, but their role in prevention of fractures is unclear.

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343. PAGET'S DISEASE AND OTHER DYSPLASIAS OF BONE - Stephen M. Krane, Alan L. Schiller

PAGET'S DISEASE OF BONE

Paget's disease of bone (osteitis deformans) is characterized by excessive resorption of bone by osteoclasts, followed by the replacement of normal marrow by vascular, fibrous connective tissue. At some stage and to a variable degree, the resorbed bone is replaced by coarse-fibered, dense trabecular bone organized in haphazard fashion. The irregular and often rapid deposition of this new bone, to a great extent still lamellar, results in an increase in the number of prominent, irregular cement lines that give the bone its characteristic "mosaic" pattern. Most lesions show both excessive resorption and chaotic new bone formation. The disorder is usually focal but may be widespread.

INCIDENCE

The prevalence is difficult to determine because the disease is often asymptomatic. It is frequently detected when roentgenograms are obtained for other reasons or because of a high level of alkaline phosphatase on routine blood screening. On the basis of autopsy examination, the incidence is estimated to be about 3% in individuals over age 40; the likelihood of occurrence increases with age. The incidence varies in different parts of the world. Radiologic surveys indicate that the frequency in adults is <1% in the United States, Great Britain, and Australia. In India, Japan, the Middle East, and Scandinavia, the disease is rare.

ETIOLOGY

The cause is unknown. Some of the manifestations can be suppressed with glucocorticoids, salicylates, and cytotoxic drugs, but there is no convincing evidence that the fundamental lesion is inflammatory. Intranuclear inclusions have been found by electron microscopy in osteoclasts in pagetic bone. Some of the inclusions resemble nucleocapsids of viruses belonging to the measles group. Indirect immunofluorescence studies using antibodies to measles virus suggest that the inclusions are indeed measles virus nucleocapsids. The presence of mutations in specific regions of the viral genome is consistent with persistent infection. In some individuals with Paget's disease, osteoclasts and bone marrow mononuclear cells contain nucleocapsids of respiratory syncytial virus alone or in addition to nucleocapsids of measles virus; in some areas of Britain, canine distemper virus sequences have been identified in pagetic bone cells. Thus, different paramyxoviruses may have roles in the initiation or propagation of Paget's disease. Further evidence supporting the potential role of measles virus in the pathogenesis of the excessive bone resorption in Paget's disease has been obtained from studies of osteoclast precursors in vitro. Normal bone marrow-derived CD34+ cells transduced with a measles virus nucleocapsid gene differentiate into abnormal multinucleated osteoclasts that can resorb bone.

There is renewed interest in genetic factors that might be important in the predisposition to and pathogenesis of Paget's disease. Several large kindreds have been identified in which Paget's disease affects two or more generations with a pattern of inheritance consistent with autosomal dominant transmission. A rare Paget's disease-like disorder,

familial expansile osteolysis, has been mapped to chromosome 18q21-22; Paget's disease was mapped to the same locus in several families. However, in other families with Paget's disease, there is no linkage to 18q21-22, indicating genetic heterogeneity. In some sporadic osteosarcomas, there is constitutional loss of heterozygosity mapped to the same region of chromosome 18, and the rare sarcomas associated with Paget's disease (see below) also exhibit loss of heterozygosity in this region. It is of great interest that the gene within the 18q21-22 locus (*TNFRSF11A* gene) responsible for familial expansile osteolysis encodes the receptor activator of NfκB (RANK), a member of the tumor necrosis factor (TNF) superfamily crucial for osteoclast differentiation (see below). The mutation results in constitutive activation of RANK. It is unknown whether similar mutations occur in some individuals with Paget's disease.

PATHOPHYSIOLOGY

The characteristic feature is increased resorption of bone accompanied by an increase in bone formation. In the early phase, bone resorption predominates (e.g., in the variant *osteoporosis circumscripta*) and the bones are very vascular. This has been termed the *osteoporotic, osteolytic, or destructive phase* of disease. Body calcium balance may be negative. Commonly, the excessive resorption is followed closely by formation of new pagetic bone. In this so-called mixed phase of the disease, the rate of bone formation is so geared to that of bone resorption that the magnitude of the increase in bone turnover is not reflected in the overall calcium balance. As the activity decreases, the resorptive rate may decline progressively relative to formation, eventually leading to development of hard, dense, less vascular bone (the so-called *osteoplastic or sclerotic phase*) and a positive calcium balance. The rates of bone turnover may be increased enormously in the early phases of the disease, occasionally more than 20 times normal.

Increased generation and overactivity of osteoclasts are considered the major abnormality. The osteoclasts are larger than normal and contain multiple pleomorphic nuclei. Increased numbers of osteoclast-like multinucleated cells are generated from hematopoietic precursors in long-term marrow cultures from individuals with Paget's disease. Production of interleukin (IL) 6 by the pagetic bone (marrow) cells is increased, and the cells are more sensitive than normal to the pro-resorptive effects of $1,25(\text{OH})_2\text{D}_3$.

The calcification rate is characteristically increased in pagetic bone. Bone turnover correlates with the increased plasma level of bone alkaline phosphatase, which is higher in Paget's disease than in any other condition except for hereditary hyperphosphatasia. Although increased bone resorption enhances release of calcium and phosphate ions from bone, the plasma concentrations of these ions are usually normal, presumably because of mineral deposition in new bone and because of feedback control of parathyroid hormone secretion. The concentration of phosphate in the plasma is normal or slightly elevated. When the imbalance between bone formation and resorption favors resorption, as after prolonged immobilization or fractures, urinary calcium excretion may be increased and on occasion hypercalcemia may occur. If, on the other hand, bone formation exceeds resorption (relatively uncommon), circulating levels of parathyroid hormone may be increased. Significant increases in trabecular bone resorption and osteoid surfaces in normal bone from patients with Paget's disease may be due to compensatory, secondary hyperparathyroidism. Resorption involves both the organic

and mineral phases of bone. While the inorganic ions of the mineral phase are reutilized for bone formation, amino acids such as hydroxyproline and hydroxylysine and the hydroxypyridinium cross-link compounds are released during resorption of the collagen matrix of bone and are not reutilized for collagen biosynthesis. The increased urinary excretion of small peptides containing hydroxyproline reflects increased bone resorption. The pyridinium cross-link compounds pyridinoline (Pyr) and deoxypyridinoline (D-Pyr) are released from bone collagen during osteoclastic bone resorption and can be measured in urine by several commercial assays. The C- and N-telopeptide measurements are also useful for monitoring therapy, but measurements of serum alkaline phosphatase activity alone are usually sufficient.

RADIOLOGIC CHANGES

The pelvic bones are most commonly involved, followed by the femur, skull, tibia, lumbosacral spine, dorsal spine, clavicles, and ribs; small bones are not as frequently diseased. The lytic phase of the disease may be overlooked except when it occurs in the skull as osteoporosis circumscripta, with areas of sharply demarcated radiolucency in the frontal, parietal, and occipital bones. In the long bones, the lytic areas are usually first seen at one end and progress toward the other end with a V-shaped advancing edge. The lesion may cause expansion of the cortex and exhibit features suggesting malignancy. Usually lysis is followed by a zone of increased density, representing the new bone formation of the mixed phase of the disease. In general, the bone enlarges with an irregularly widened cortex in a coarse, striated pattern and with increased density, occasionally focal in distribution. Perpendicular lines of radiolucency (cortical infractions) are frequent and occur on the convex side of bowed long bones, particularly the femur and tibia. The remodeling of the pagetic bone usually follows the lines of stress produced by muscle pull or gravity, accounting for the characteristic lateral bowing of the femur or anterior bowing of the tibia and the tendency for most of the dense bone to be deposited on the concave side of the bowed bone. In the mixed stage, there is enlargement and thickening of the skull, especially of the outer table, with irregular, spotty areas of increased density ([Fig. 343-1](#)). The changes in the pelvis reflect the varying degrees of bone resorption and new bone formation and are frequently accompanied by a characteristic thickening of the pelvic brim. In the sclerotic phase of the disease, the bone may show uniform increase in density, often in the absence of striations. This feature is common in the facial bones but occasionally occurs in the vertebrae, where a homogeneous, dense pattern gives an "ivory" appearance similar to that typical of Hodgkin's disease, although the involved vertebrae in Hodgkin's disease are not enlarged. Computed tomography (CT) and magnetic resonance imaging (MRI) are useful in defining atypical lesions, particularly when neoplastic involvement is suspected. Technetium 99m diphosphonate bone scans are useful in documenting the extent of disease when therapy is contemplated or to confirm the diagnosis when radiologic findings are inconclusive.

CLINICAL MANIFESTATIONS

The clinical presentation is a function of the extent of the disease, the particular bones involved, and the presence of complications. Many patients are asymptomatic. In these individuals the disorder is discovered during radiologic examination of the pelvis or spine for another problem or because of the finding of an elevated level of plasma alkaline

phosphatase. Other individuals may gradually become aware of a swelling or deformity of a long bone or develop a disturbance in gait due to unequal length of the lower extremities. Enlargement of the skull is often not noticed by the patients, except by awareness of increasing hat size. Facial pain and headache are initial complaints in some; backache and leg pain are common. The pain is usually dull but may be shooting or knifelike. Back pain is most common in the lumbar region and may radiate into the buttocks or lower extremities. This pain can be due to the pagetic process itself, to distortion of articular facets, or to secondary osteoarthritis. Pain in the lower extremities may be associated with the transverse cortical infarctions along the convex lateral surface of the femur or the anterior surface of the tibia. New lytic lesions detected on bone scan may be the most painful. Pain may also be due to involvement of the hip joint resembling degenerative joint disease, which is characterized by narrowing of the joint space, bony lipping at the margin of the acetabulum, and deepening of the acetabulum. Angioid streaks may be present in the retina. Hearing loss can be due to direct involvement of the ossicles of the inner ear, involvement of bone in the region of the cochlea, or impingement on the eighth cranial nerve in the auditory foramen. More serious neurologic complications can result from overgrowth of bone at the base of the skull (platybasia) and compression of the brainstem. Compression of the spinal cord can cause paraplegia, particularly with involvement of the mid-dorsal spine. Pathologic fractures of vertebrae may also produce spinal cord lesions.

COMPLICATIONS

Blood flow may be markedly increased in extremities involved with Paget's disease. There is proliferation of blood vessels in pagetic bone, but anatomic and functional studies have not confirmed the presence of arteriovenous fistulas. Although blood flow is increased in bone, cutaneous vasodilatation in the pagetic extremities accounts for the increased warmth noted clinically. When the disease is widespread, involving one-third or more of the skeleton, the increased blood flow raises *cardiac output* and rarely leads to high-output heart failure. However, heart disease in individuals with Paget's disease is usually due to the same conditions that occur in other patients of similar age. *Pathologic fracture* may occur at any stage but is more common in the destructive phase of the disease. In the weight-bearing bones fractures are often incomplete, multiple, and on the convex side of the bone. They may occur spontaneously or after slight trauma. Though many lesions heal with no major disability, more serious fractures also occur. Complete fractures are often transverse as if the bone was snapped like a piece of chalk.

There is no characteristic level of urinary calcium excretion, but it tends to be higher when the resorptive phase predominates, possibly accounting for the somewhat higher incidence of *urinary stones* in these patients. *Hyperuricemia* and gout are common in men with Paget's disease, and calcific peri-arthritis may occur.

Sarcoma is the dread complication. The incidence is probably $\leq 1\%$, although higher incidence has been noted in series that include many patients with polyostotic involvement. The sarcomas most frequently arise in the femur, humerus, skull, facial bones, and pelvis and rarely in the vertebrae. Pagetic osteosarcomas are lytic in appearance on radiographs, in contrast to the sclerotic appearance of radiation-induced osteosarcomas. The tumors are multicentric in about 20% of patients. Fibrosarcomas

and chondrosarcomas have also been found. Pain and swelling are the common complaints that lead to recognition of the sarcomas. The extent and character of the neoplastic involvement are established by [CT](#) and/or [MRI](#). In occasional patients, an "explosive rise" of the phosphatase level may accompany the growth of the sarcoma, whereas in patients with limited Paget's disease, phosphatase levels may be only slightly elevated and give no clue to the development of the malignant lesion. The prognosis is poor following the development of sarcomas, and ablative surgery is rarely successful. In contrast to the successful treatment of some osteosarcomas in children, chemotherapy has little effect on survival of patients with pagetic osteosarcomas. Reparative granulomas resembling giant cell tumors may cause local destruction, but they do not metastasize.

TREATMENT

Most patients require no treatment, since the disease is localized and does not cause symptoms. Indications for therapy include persistent pain in involved bones, neural compression, rapidly progressive deformity resulting in disabling disturbance of posture and/or gait, high-output congestive heart failure, hypercalcemia, severe hypercalciuria with or without formation of renal stones, repeated fractures or nonunion, and preparation for major orthopedic surgery. Nonsteroidal anti-inflammatory drugs, such as one of the COX2 inhibitors or acetaminophen, may be useful to relieve pain, especially that involving the hip joints. Patients with severe hip or knee pain, unrelieved by analgesics and not responsive to therapy with agents that inhibit bone resorption, are candidates for total joint replacement. Results of joint replacement are often excellent, although patients with Paget's disease have an increased risk of ectopic bone formation around the operative site. Osteotomies are also useful in patients with bowing deformities of the tibia. In patients who have undergone surgical procedures, early ambulation and adequate fluid intake are important to prevent the development of hypercalciuria and hypercalcemia.

Potent bisphosphonates can inhibit bone resorption and are usually well tolerated. They appear to act by adsorbing to the surface of the calcium/phosphate mineral phase of bone and inhibiting osteoclast function. Bisphosphates are chemically stable analogues of inorganic pyrophosphate and are available in two classes: nitrogen-containing and non-nitrogen-containing. The non-nitrogen-containing bisphosphates (e.g., clodronate and etidronate) are metabolically incorporated into nonhydrolyzable analogues of ATP that may inhibit ATP-mediated reactions. The nitrogen-containing bisphosphates (e.g., pamidronate, alendronate, and risedronate) do not form ATP compounds, but they do inhibit enzymes in the mevalonate pathway, particularly enzymes involved in the synthesis of farnesyl pyrophosphate and geranylgeranyl pyrophosphate. The latter compounds are involved in protein prenylation reactions.

The first bisphosphate available for treatment of Paget's disease in the United States, editronate, was moderately effective in alleviating symptoms but did not decrease biochemical indices to the normal range. Editronate also inhibits mineralization of bone and produces osteomalacia. The newer bisphosphonates such as alendronate, pamidronate, risedronate, and tiludronate are more potent than etidronate and do not produce mineralization defects. Consequently, editronate is no longer indicated for treatment of Paget's disease. In the United States, pamidronate is approved for

intravenous use, and alendronate and risedronate are approved for oral administration.

The bisphosphonates as a class are poorly absorbed from the gastrointestinal tract. Alendronate should be given orally with water after an overnight fast 30 to 60 min before breakfast; the dose is 40 mg/d for 6 months. Risedronate is administered at 30 mg/d for 2 to 3 months. Gastric irritability and rarely esophageal ulcerations may occur. Several regimens are advocated for the intravenous administration of pamidronate. For example, pamidronate is used intravenously as an infusion of 30 mg/d over 3 to 4 h in 5% glucose in water or normal saline on three or four successive days. Responses are usually rapid, with decreases in urinary excretion of hydroxyproline and pyridinium cross-link compounds within days to weeks, followed by a fall in levels of serum alkaline phosphatase. Flulike symptoms accompanied by fever may occur but usually subside rapidly.

Patients given bisphosphonates should also be given daily calcium supplements of 1 to 1.5 g and approximately 400 IU of vitamin D. Clinical and biochemical improvement often lasts for more than a year after bisphosphonate therapy. Clinical evaluation and assessment of alkaline phosphatase levels at 3-month intervals are useful for assessing the need for retreatment. Radiographs at 6-month intervals may be indicated for evaluation of lytic lesions, which usually heal with these agents.

Calcitonin therapy has largely been replaced by bisphosphonates for primary treatment of severe disease, but calcitonin may still be useful in patients who cannot tolerate alendronate or risedronate because of gastrointestinal side effects or who prefer to avoid intravenous therapy with pamidronate. The administration of porcine, salmon, and human *calcitonins* for prolonged periods decreases plasma alkaline phosphatase and urinary hydroxyproline excretion. Treatment with calcitonin variably decreases bone pain due to suppression of the pagetic lesion as well as to an independent, centrally mediated analgesic effect. The calcitonins are probably most useful in patients with pain in areas of pagetic involvement not due to associated joint disease. The dose of salmon calcitonin is 50 to 100 MRC units daily given subcutaneously. In most cases, it is possible to reduce the dose to three times weekly. Some patients develop a sensation of warmth and/or nausea 30 min to several hours after injection. Nasal spray formulations of calcitonin can be administered at doses of 200 IU/d. Cytotoxic drugs such as plicamycin and dactinomycin no longer have a place in therapy.

Although the bisphosphonates and calcitonins act primarily to decrease bone resorption, the rate of new bone formation subsequently falls. As a result, the state of high bone turnover is shifted to a state of lower turnover, where rates of formation and resorption are still apparently geared to each other. In this lower turnover state, collagen fibers of the bone matrix are deposited in a more orderly fashion similar to normal bone.

HYPEROSTOSIS

A number of disease states have in common an increase in the mass of bone per unit volume (*hyperostosis*) ([Table 343-1](#)). This increase in bone mass is often associated with disturbance in the architecture of the tissue. The additional bone may be located at the periosteum, within the compact bone of the cortex, or in the trabeculae of the cancellous regions. In some diseases, the increase in bone mass may be spotty, as in

osteopoikilosis, whereas in others, most of the skeleton may be involved, as in the malignant form of osteopetrosis in children. The increase in mass is usually not due to an excessive amount of mineral relative to matrix, except in disorders where islands of calcified cartilage may persist such as osteopetrosis. In some diseases, such as the osteosclerosis of untreated renal insufficiency, bone mass and radiodensity may be increased, even though the new bone formed is poorly mineralized and contains widened osteoid seams.

Although hyperostosis is usually due to decreased numbers of osteoclasts or altered osteoclast function, dysfunction of osteoblasts can also occur. For example, an engineered null mutation of the osteocalcin gene in mice results in a higher bone mass due to increased bone formation without change in bone resorption. Infection of newborn mice also produces an osteopetrosis-like phenotype in which osteoblast progenitors appear to induce increased bone formation. In human osteopetrosis of the relatively benign and sporadic type, viral nucleocapsid particles have been found in osteoclasts, and it is possible that viral infection accounts for the excessive bone mass.

OSTEOPETROSIS

Also known as Albers-Schonberg or marble bone disease, osteopetrosis is clinically, biochemically, and genetically heterogeneous. Although osteopetrosis has many causes, a defect in bone resorption is always the underlying mechanism.

Several inherited forms of osteopetrosis occur in rodents, some of which can be cured by bone marrow transplantation from a normal littermate and are probably due to stem cell defects. The osteopetrosis in *op/op* mice and in *tl/tl* toothless rats is not cured by bone marrow transplantation, however. These animals have few osteoclasts, and those that are present appear to be defective. The *op/op* mice have a defect in the coding region of the gene for macrophage colony stimulating factor (M-CSF). The skeletal defects in these animals and in *tl/tl* rats can be reversed by treatment with M-CSF. Another form of osteopetrosis has been produced in mice by targeted disruption of the *c-src* gene, which is normally expressed at high levels in osteoclasts. These *src*^{-/-} mice still have osteoclasts on bone surfaces, but they fail to form a ruffled border at the bone-resorbing surface. Disruption of the *c-fos* gene results in osteopetrosis in which osteoclasts are absent.

Important advances have also been made in understanding the interactions between osteoblasts/stromal cells and hemopoietic osteoclast precursor cells that lead to osteoclastogenesis ([Fig. 343-2](#)). A novel member of the [TNF](#) receptor superfamily, referred to as *osteoprotegerin* (OPG; also known as *osteoclastogenesis inhibitory factor*, OCIF) functions as a soluble decoy receptor that binds, and presumably neutralizes, [RANK](#) ligand, a transmembrane ligand expressed on osteoblasts/stromal cells. RANK ligand binds to RANK, a transmembrane receptor on hemopoietic osteoclast precursor cells, to activate osteoclast differentiation and function ([Chap. 340](#)). The RANK receptor binds to intracellular signaling molecules called *TNF receptor-associated factors* (TRAFs) that activate NFκB, a transcription factor known to be required for normal osteoclast function. Genetic models in mice are beginning to unravel this complex signaling pathway. Expression of a soluble version of RANK ligand stimulates osteoclast differentiation. Overexpression of OPG in transgenic mice leads to

osteopetrosis, apparently by blocking RANK ligand. Mice deficient in RANK lack osteoclasts and develop severe osteopetrosis (as well as T cell immunologic defects). TRAF6-deficient mice also develop osteopetrotic features because of defective osteoclast function. It is clear, based on these and other lines of evidence, that the RANK ligand/OPG/RANK/TRAF/NF κ B pathway plays a pivotal role in the control of osteoclast development and function.

In humans the infantile autosomal recessive form of osteopetrosis, until recently, has been of unknown cause. It is a severe bone disease that is usually fatal within the first decade of life. Osteoclasts are usually present in normal or increased numbers. In addition, since bone resorption is markedly suppressed, it has been assumed that the defect is not in genes responsible for osteoclast differentiation but in those responsible for osteoclast function. This form of osteopetrosis is manifested in utero and progresses after birth with anemia, hepatosplenomegaly, hydrocephalus, cranial nerve involvement, and death, often due to infections. Transplantation of bone marrow from allogeneic donors to provide normal osteoclast precursor cells has been successful in several patients, in whom osteopetrotic bone was repopulated with donor osteoclasts that produced radiologic and/or bone-biopsy evidence of bone resorption. Nearly 100 bone marrow transplantations have been reported over the past 15 years. If the transplants are successful, markers of donor cells can be found in bone resorbing areas and skeletal improvement persists for years. Although restoration of visual acuity usually does not occur with successful transplantation, this is the only means of approaching cure even in mild cases. The genetic defect in about half of the subjects studied has now been identified. The gene in the human disease was mapped to chromosome 11q13, a region that contains several potential candidate genes. In mice, introduction of a null mutation in one of the genes in this region, *Tcirg1*, that encodes the osteoclast-specific (*OC116*) subunit of the vacuolar proton pump ([V]-type H⁺-ATPase) responsible for acidification of the extracellular compartment adjacent to the brush border, results in osteopetrosis with abundant osteoclasts. Furthermore a deletion of the 5' portion of the gene is the cause for the defect in *oc/oc* mice with spontaneous osteopetrosis. In approximately half of the human subjects with the autosomal recessive form of osteopetrosis so far examined, missense, frameshift, or potential splicing mutations have just been identified in the homologous gene, *TC1RG1*. Thus, mutations in *TG1RG1* are a frequent, although not the sole, cause of this form of osteopetrosis.

Less fulminant forms of osteopetrosis occur in older children and adults. In some the disorder appears to be sporadic, and in others the osteopetrosis is inherited as an autosomal dominant trait and progresses with age; anemia is not as severe, neurologic abnormalities are not as frequent, and recurrent pathologic fractures are the main feature. Although the disorder is most common in infants and children, the diagnosis may be made in adults when roentgenograms are obtained because of fractures or unrelated diseases.

An "intermediate" form of autosomal recessive osteopetrosis has been described in kindreds in which the skeletal abnormality is associated with renal tubular acidosis and cerebral calcification. This form is compatible with long survival and is associated with profound impairment of the activity of one of the isoenzymes of carbonic anhydrase (carbonic anhydrase II). Carbonic anhydrase II is a major component of the system that generates the acid environment adjacent to the ruffled border of the osteoclast.

Deficiency of the enzyme impairs bone resorption. The defect in remodeling results in disorganization of bone structure, with thickened cortices and lack of funnelization of metaphyses. Despite increased density, the bone may be abnormal mechanically and can fracture readily. Osteomalacia or rickets is sometimes a component of osteopetrosis in children.

Roentgenograms reveal uniformly dense, sclerotic bone, often with no distinction between the cortical and cancellous regions ([Fig. 343-3](#)). In the severe infantile form, there is persistence of the primary spongiosa with central calcified cartilage cores surrounded by woven bone. Osteoclasts may be increased in number but do not function normally due to the acidification defect that results from the mutated vacuolar proton pump. In other forms of osteopetrosis, there may be different morphologic abnormalities such as loss of ruffled borders. The variability may reflect heterogeneity in this syndrome, as in the osteopetrosis in rodents. The long bones are usually involved, with increased density along the entire shaft. The metaphyses have a characteristic clubbed or splayed appearance. Alternating horizontal bands of increased and decreased density in the long bones and vertebrae suggest that the defect is intermittent during periods of growth. The skull, pelvis, ribs, and other bones may be involved. The phalanges and the distal humerus are usually spared.

Encroachment of bone on the marrow cavity, particularly in the severe infantile disorder, is associated with anemia of the myelophthitic type with extramedullary hematopoiesis in liver, spleen, and lymph nodes and enlargement of these organs. Neurologic abnormalities caused by encroachment on cranial nerves include optic atrophy, nystagmus, papilledema, exophthalmos, and impairment of extraocular muscles. Facial paralysis and deafness are frequent; trigeminal lesions and anosmia are less common. In infants, macrocephaly, hydrocephalus, and convulsions may occur, and infections such as osteomyelitis are frequent. Renal tubular acidosis is a feature of the osteopetrosis associated with carbonic anhydrase II deficiency.

In the less severe forms, about half of patients have no symptoms, and the disorder is discovered incidentally on roentgenograms. Others present with fractures, bone pain, osteomyelitis, and cranial nerve palsies.

Fractures may occur with trivial trauma. Healing of such fractures is usually slow but satisfactory. When the disease is manifested first in adult life, fractures may be the only clinical problem. Levels of calcium and alkaline phosphatase in the plasma are usually normal in adults, but hypophosphatemia and moderate hypocalcemia may occur in children. Serum acid phosphatase levels are usually increased.

As mentioned, in children with severe osteopetrosis, bone marrow transplantation from allogeneic donors or HLA-identical siblings has resulted in histologic and radiologic increases in bone resorption and variable improvement in anemia, vision, hearing, and growth and development. Unfortunately, it is not always possible to find appropriate donors, or patients may not be good candidates for bone marrow transplantation. In some patients with the lethal forms of the disorder, calcitriol therapy is associated with the appearance of osteoclasts with normal ruffled borders and other evidence of increased bone resorption.

PYKNODYSOSTOSIS

Pyknodysostosis is an autosomal recessive form of osteosclerosis that superficially resembles osteopetrosis. It is a form of short-limbed dwarfism associated with bone fragility and a tendency to fracture with minimal trauma. Nevertheless, life span is usually normal. In addition to a generalized increase in bone density, features include short stature; separated cranial sutures; hypoplasia of the mandible; kyphoscoliosis and deformities of the trunk; persistence of deciduous teeth; progressive acroosteolysis of the terminal phalanges; high, arched palate; proptosis; blue sclerae; and a pointed, beaked nose. Patients usually present because of frequent fractures. The disorder is caused by mutations in a gene on chromosome 1q21 that encodes cathepsin K, a cysteine protease that is expressed in normal osteoclasts. Null mutations in the cathepsin K gene in mice result in a phenotype with many features of pyknodysostosis. Osteoclasts are present but do not function normally since there is no proteinase secreted into the area adjacent to the ruffled border where bone collagen resorption normally takes place.

OSTEOMYELOSCLEROSIS

In osteomyelosclerosis, the marrow cells are replaced by diffuse fibroplasia, occasionally accompanied by osseous metaplasia and increased skeletal density on roentgenograms. In early stages woven bone may be found in intratrabecular locations, whereas in more advanced stages, woven bone is observed in the medulla. The disorder is probably a phase in the course of the myeloproliferative disorders and is characteristically accompanied by extramedullary hematopoiesis.

Hyperostosis corticalis generalisata (van Buchem's disease) is characterized by osteosclerosis of the skull (base and calvaria), lower jaw, clavicles, and ribs and thickening of the diaphyseal cortices of the long and short bones. Alkaline phosphatase levels in the serum are elevated, and the disorder may be due to increased formation of bone of normal structure. The major manifestations are due to neural compression and consist of optic atrophy, facial paralysis, and perception deafness. In *hyperostosis generalisata with pachydermia* (Uehlinger), the sclerosis is due to increased formation of subperiosteal spongy bone and involves the epiphyses, metaphyses, and diaphyses. Pain, swelling of joints, and thickening of the skin of the lower arms are common.

HEREDITARY HYPERPHOSPHATASIA

This disorder is characterized by structural deformities of the skeleton, with increased thickness of the calvaria, increased density at the base of the skull, and widening and loss of normal architecture of the shafts and the epiphyses of the long and short bones. The failure to deposit normal bone and the haphazard orientation of lamellae suggest active remodeling that resembles that of Paget's disease. Osteoclasts with multiple nuclei characteristic of Paget's disease and the typical "mosaic" pattern of faceted units of lamellar bone are not found, however. Levels of plasma alkaline phosphatase and urinary excretion of hydroxyproline peptides and other collagen-degradation products are increased. The disorder is apparently inherited as an autosomal recessive trait. Treatment with bisphosphonates or calcitonin therapy may be of value.

PROGRESSIVE DIAPHYSEAL DYSPLASIA (CAMURATI-ENGELMANN DISEASE)

This is an autosomal dominant disorder in which a symmetric thickening and increased diameter of the diaphyses of long bones occurs, particularly in the femur, tibia, fibula, radius, and ulna. Pain over affected areas, fatigue, abnormal gait, and muscle wasting are the major manifestations. Serum alkaline phosphatase levels may be elevated, and, on occasion, hypocalcemia and hyperphosphatemia may be found. Other abnormalities include anemia, leukopenia, and an elevated erythrocyte sedimentation rate. Linkage studies have localized a candidate gene (*DPD1*) to chromosome 19q13.2. Clinical and biochemical improvement may result from the use of glucocorticoids.

MELORHEOSTOSIS

This rare, sporadic condition usually begins in childhood and is characterized by a slowly progressive linear hyperostosis in one or more bones of one limb, usually in a lower extremity. All segments of the bone may be involved, with sclerotic areas that have a "flowing" distribution. The involved limb is often extremely painful. Soft tissue masses, not connected to bone, are often mineralized and are composed of osseous or cartilaginous tissue. Other types of soft tissue masses are associated with joint contractures or consist of fibrofatty, lymphatic, or vascular tissue.

OSTEPOIKILOSIS

This is a benign autosomal dominant trait usually discovered by chance. In some families, the occurrence of melorheostosis suggests that these disorders may involve the same genetic locus. Osteopoikilosis is characterized by dense spots of trabecular bone <1 cm in diameter, usually of uniform density, located in the epiphyses and adjacent parts of the metaphyses. All bones may be involved except the skull, ribs, and vertebrae.

HYPEROSTOSIS FRONTALIS INTERNA

This is an abnormality of the inner table of the frontal bones of the skull consisting of smooth, rounded enostoses covered by dura and projecting into the cranial cavity. These enostoses are usually <1 cm at their greatest diameter and do not extend posteriorly beyond the coronal suture. The abnormality is found almost exclusively in women who are frequently obese, hirsute, and may have a variety of neuropsychiatric complaints (Morgagni-Stewart-Morel syndrome). The disorder also occurs in women with no obvious illness or particular associated disease. The finding in the skull may be a manifestation of a generalized metabolic disorder.

FIBROUS DYSPLASIA (MCCUNE-ALBRIGHT SYNDROME)

The bony lesions of fibrous dysplasia are characterized by proliferation of fibroblast-like cells that in some areas have features of osteoblasts, with production of an extracellular matrix that may be calcified and have the appearance of woven bone. In other areas the cells have features of chondrocytes and produce a cartilage-like extracellular matrix. The lesions of fibrous dysplasia are usually focal and have a radiolucent appearance; they may be monostotic or polyostotic. The disorder occurs with equal frequency in both

sexes. Some individuals have distinctive areas of skin pigmentation and precocious puberty (McCune-Albright syndrome) ([Chap. 336](#)). These diverse manifestations are the consequence of postzygotic mutations in the gene encoding the regulatory G_{sa} proteins.

INCIDENCE

The monostotic form is the most common type of fibrous dysplasia. The lesions can be asymptomatic, can be associated with local pain, or predispose to pathologic fracture. Most of the lesions are in the ribs or in the craniofacial bones, especially the maxillae. Other bones that may be affected include metaphyseal or diaphyseal portions of the proximal femurs or tibias. Monostotic fibrous dysplasia is most often diagnosed in patients between 20 and 30 years of age. There are usually no associated skin lesions. Approximately one-quarter of the individuals with the polyostotic form have more than half the skeleton involved by disease. One side of the body may be affected, and the lesions may be distributed segmentally in a limb, particularly in the lower extremities. Craniofacial lesions are present in approximately half of patients with the polyostotic form. Whereas the monostotic form is usually detected in young adults, fractures and skeletal deformities occur in childhood in the polyostotic form; early-onset disease is generally more severe. Lesions, especially monostotic lesions, can become quiescent at puberty and worsen during pregnancy. McCune-Albright syndrome (polyostotic fibrous dysplasia, multiple café au lait spots, and sexual precocity) is more common (10:1) in females. Short stature is due to premature closure of the epiphyses.

PATHOPHYSIOLOGY

Histologically, the lesions contain benign-appearing fibroblastic tissue arranged in a loose whorled pattern ([Fig. 343-4](#)). Malignant transformation of either monostotic or polyostotic fibrous dysplasia occurs with a frequency of <1%. The malignant change is usually detected in the third or fourth decade in individuals who have had lesions first identified in childhood. In about one-third of the cases the neoplasms arise in previously irradiated lesions. Ossifying fibroma of long bones is a peculiar fibroosseous cortical lesion that may be a variant of fibrous dysplasia. It is most common in the tibial shaft in teenagers. Although benign, the lesion has a tendency to recur if not adequately excised.

Fibrous dysplasia and McCune-Albright syndrome represent a phenotypic spectrum of disorders caused by activating mutations in the *GNAS1* gene, which encodes the G_{sa} protein. Because these postzygotic mutations occur at different stages in early development, the extent and type of tissues affected by the mutations are variable, explaining the mosaic pattern of skin and bone changes. The mutations occur in regions (e.g., Arg 201) of G_{sa} that selectively inhibit its GTPase activity. Because the GTP-bound form of the regulatory protein confers its active state ([Chap. 341](#)), the mutations confer constitutive stimulation of the cyclic AMP-protein kinase A signal transduction pathway. The mutations in *GNAS1* are also found in patients with fibrous dysplasia without manifestations of the McCune-Albright syndrome. Thus, in the bony lesions these mutations result in abnormalities in osteoblastic differentiation and the production of abnormal bone. In addition, there is an associated increase in osteoclastic bone resorption that provides a rationale for therapy with the bisphosphonate, pamidronate. Other tissues in which growth control and function are strongly regulated by

G_{sα}protein-coupled receptors are particularly susceptible to the mutations. In addition to bone (parathyroid hormone receptor) and skin (melanocyte-stimulating hormone receptor), various endocrine glands, including the ovary (follicle-stimulating hormone receptor), thyroid (thyroid-stimulating hormone receptor), adrenal (ACTH-receptor), and pituitary (growth hormone-releasing hormone receptor), are commonly affected by the G_{sα}mutations. It is of interest that the genetic abnormality in Albright's hereditary osteodystrophy (pseudohypoparathyroidism) is the opposite of that found in the McCune-Albright syndrome. In the former, alterations in G_{sα}function or expression result in *deficient* activity and decreased responsiveness to hormones that function through cyclic AMP-mediated signal transduction pathways ([Chap. 341](#)).

RADIOLOGIC CHANGES

The roentgenographic appearance of the lesions is that of a radiolucent area with a well-delineated, smooth or scalloped border, typically associated with focal thinning of the cortex of the bone ([Fig. 343-5](#)). Fibrous dysplasia can cause bones to become larger than normal, a feature characteristic of Paget's disease as well. The "ground-glass" appearance is due to the thin spicules of calcified woven bone. Deformities can include coxa vara, shepherd's-crook deformity of the femur, bowing of the tibia, Harrison's grooves, and protrusio acetabuli. Involvement of facial bones, usually with lesions of increased radiodensity, may create a leonine appearance (*leontiasis ossea*). Fibrous dysplasia of the temporal bones can cause progressive loss of hearing and obliteration of the external ear canal. Advanced skeletal age in girls is correlated with sexual precocity but can occur in boys without sexual precocity. Occasionally, a focus of fibrous dysplasia may undergo cystic degeneration, with an enormous distortion of the shape of the bone, and mimic the so-called aneurysmal bone cyst.

CLINICAL MANIFESTATIONS

The clinical course is variable. Skeletal lesions are usually detected because of localized pain, deformities, or fractures. Other symptoms ascribable to bone involvement are headache, seizures, cranial nerve abnormalities, hearing loss, narrowing of the external ear canal, or even spontaneous scalp hemorrhages if there is craniofacial bone disease. On rare occasions the onset of sexual precocity is the first clinical manifestation of the McCune-Albright syndrome. Serum calcium and phosphorus values are usually normal. In approximately one-third of patients with the polyostotic form, bone turnover is increased, as reflected by high levels of serum alkaline phosphatase and increased urinary excretion of collagen breakdown products. In some patients high cardiac output resembles that in extensive Paget's disease. Widespread disease does not usually develop when the disease is mild at the outset.

The cutaneous pigmentation in most patients with McCune-Albright syndrome consists of isolated dark-brown to light-brown macules that tend to be located on one side of the midline ([Fig. 343-6](#)). The border is usually, although not always, irregular or jagged ("coast of Maine"), in contrast to the smooth borders of the pigmented macules of neurofibromatosis ("coast of California"). As a rule, there are fewer than six of the lesions, which range in size from 1 cm to very large lesions, covering areas such as the back, buttocks, or sacral regions. When the lesions are in the scalp, the overlying hair may be more deeply pigmented. Localized alopecia is associated with osteomas of the

skin, and such lesions tend to overlie skeletal lesions. The pigmentation also tends to be on the same side as the skeletal lesions and actually to overlie them. Occasionally, neurofibromatosis and fibrous dysplasia coexist.

Sexual precocity occurs more commonly in girls than in boys. Premature vaginal bleeding, breast development, and growth of axillary and pubic hair are the usual features. Sexual precocity is due to autonomous end-organ activity, not to pituitary or hypothalamic dysfunction. Thus, girls have high estrogen levels and low or undetectable gonadotropins. The characteristic pigmented macules are usual but not invariable. Hyperthyroidism occurs with increased frequency, and rare associations include Cushing's syndrome, acromegaly, pulmonary lesions, and soft tissue myxomas. Hypophosphatemic osteomalacia may also accompany fibrous dysplasia and resembles the disorder associated with other skeletal and nonskeletal tumors.

Although the lytic lesions of fibrous dysplasia resemble the brown tumors of hyperparathyroidism, the age of the patient, normal calcium levels, increased density of bone in the skull, and areas of cutaneous pigmentation identify the former condition. Fibrous dysplasia and hyperparathyroidism may coexist, however. Neurofibromas may involve bone and produce cutaneous pigmentation as well as nodules in the skin. The pigmented macules of neurofibromatosis are more numerous and more widely distributed than in fibrous dysplasia, usually have smooth borders, and tend to involve areas such as the axillary folds. Other lesions that have roentgenographic features similar to those of isolated fibrous dysplasia are unicameral bone cysts, aneurysmal bone cysts, and nonossifying fibromas. Leontiasis ossea is most often due to fibrous dysplasia, although other disorders may also produce this appearance, such as craniometaphyseal dysplasia, hyperphosphatasia, and, in adults, Paget's disease.

TREATMENT

Fibrous dysplasia is not curable. The skeletal lesions, however, can be improved by orthopedic procedures such as casting, osteotomy with internal fixation, curettage, and bone grafting, depending on the lesion and the age of the patient. Indications for such procedures include progressive deformity, nonunion of fractures, and pain unresponsive to conservative treatment. Calcitonin may be effective in treatment of widespread disease associated with bone pain and high serum alkaline phosphatase levels. Pamidronate (0.5 to 1 mg/kg per day intravenously for 2 to 3 days) at 6-month or yearly intervals has been shown to reduce bone pain with refilling of osteolytic lesions in about half of patients and to decrease elevated levels of serum alkaline phosphatase and urinary hydroxyproline excretion. Precocious puberty does not respond to long-acting gonadotropin-releasing hormone (GnRH) analogues, consistent with the autonomous function of the gonads. Aromatase inhibitors, such as testolactone (22 mg/kg per day), have been used to block estrogen production, but with limited efficacy. Promising initial results have been seen with estrogen antagonists, such as tamoxifen. In addition to preventing pubertal progression, blockade of estrogen action is helpful to prevent early epiphyseal closure and short stature.

OTHER DYSPLASIAS OF BONE AND CARTILAGE

A variety of diseases of bone and cartilage have been called *dystrophies* or *dysplasias*.

The *osteochondrodysplasias* are heritable disorders of connective tissue characterized by primary abnormalities of cartilage that lead to disturbances in cartilage and bone growth and development. They comprise several hundred distinct entities, which can be distinguished on the basis of clinical, genetic, and radiologic features. The molecular defects in a number of these disorders have been identified utilizing positional cloning and screening of candidate genes. Several of the disorders are due to mutations in collagen genes. **For discussion of chondrodysplasias, see [Chap. 351](#).*

SPONDYLOEPIPHYSEAL DYSPLASIA

The *spondyloepiphyseal dysplasias* are disorders in which abnormalities of growth occur in various bones, including the vertebrae, pelvis, carpal and tarsal bones, and the epiphyses of tubular bones. On the basis of roentgenographic findings, this group can be divided into (1) those with generalized platyspondyly, (2) those with multiple epiphyseal dysplasias, and (3) those with epiphysometaphyseal dysplasias. *Morquio's syndrome*, in which there is a defect in degradation of glycosaminoglycans (therefore, a "mucopolysaccharidosis"), is inherited as an autosomal recessive trait and is associated with corneal opacities, dental defects, variable disturbances in intellect, and increased urinary excretion of keratosulfate; it belongs in the first group ([Chap. 349](#)). Other forms of spondyloepiphyseal dysplasia, some of which are accounted for by defects in type II collagen, may not be recognized until late in childhood or young adult life. Flat vertebral bodies are associated with other abnormalities in shape and alignment. The disordered development of the capital femoral epiphyses leads to irregularities in shape and flattening of the femoral heads and early onset of osteoarthritis of the hips.

ACHONDROPLASIA

This disorder is among the more common types of dwarfism (1 in 15,000 to 1 in 40,000 live births). It is inherited as an autosomal dominant trait, although most cases are sporadic and due to new fibroblast growth factor receptor 3 (FGFR3) mutations (see below). The appearance of short limbs, particularly the proximal portions, with a normal trunk is characteristically accompanied by a large head, a saddle nose, and an exaggerated lumbar lordosis. The length of the spine is almost always normal. Features of the disorder are usually recognizable at birth. Those who survive infancy usually have normal mental and sexual development, and life span may be normal. Spinal deformity nevertheless may lead to a cord compression and nerve root encroachment, especially in those with kyphoscoliosis. Homozygous achondroplasia is a more serious disorder and a cause of neonatal death.

The most common mutation responsible for achondroplasia substitutes an arginine for glycine in the transmembrane domain of [FGFR3](#) and causes a gain-of-function, implying that fibroblast growth factor normally acts via the FGFR3 to inhibit chondrocyte proliferation in the growth plate. The abnormal proliferation at the growth plate, leaving other areas relatively unaffected in the tubular bones, causes production of short bones that are proportionately thick. Formation and maturation of the secondary ossification centers and articular cartilage are not disturbed. Appositional growth at the metaphysis continues, with resulting flare in this region of the bone; intramembranous bone formation at the periosteum is normal. Consistent with the inhibitory role of the FGFR3, a null mutation of the *Fgfr3* gene in mice causes increased growth in the physis.

Mutations in other domains of the *FGFR3* gene have been described in thanatophoric dysplasia, the most severe and lethal dysplasia. In several types of the so-called craniosynostosis syndromes (Pfeiffer, Crouzon, Jackson-Weiss, and Apert syndromes), mutations have been identified in the *FGFR1* or *FGFR2* genes.

Pseudoachondroplasia clinically resembles achondroplasia with respect to the limb deformities but there are no skull abnormalities. Affected individuals have mutations in the gene encoding a non-collagenous component of cartilage called cartilage oligomeric matrix protein (COMP). Mutations in the *COMP* gene have also been described in one of the less severe forms of multiple epiphyseal dysplasia (EDM1).

ENCHONDROMATOSIS (DYSCHONDROPLASIA, OLLIER'S DISEASE)

This is also a disorder of the growth plate in which the hypertrophic cartilage is not resorbed and ossified normally. It results in masses of cartilage with disorderly arrangement of the chondrocytes showing variable proliferative and hypertrophic changes. These masses are located in the metaphyses in close association with the growth plate in children but may be diaphyseal in teenagers and young adults. The disorder is usually recognized in childhood by the appearance of deformities or retardation in growth. The most common sites of involvement are the ends of long bones, usually in the region where rate of growth is most marked. The pelvis is often involved, but ribs, sternum, and skull are seldom affected. There is a tendency toward unilateral involvement. Chondrosarcoma develops occasionally in the enchondromata. The association of enchondromatosis and cavernous hemangiomas in the soft tissues including the skin is known as *Maffucci's syndrome*. Both Ollier's disease and Maffucci's syndrome have been associated with other primary malignancies as diverse as granulosa cell tumor of the ovary and cerebral gliomas.

MULTIPLE EXOSTOSES (DIAPHYSEAL ACLASIS OR OSTEOCHONDROMATOSIS)

This is a disorder of the metaphysis, transmitted in an autosomal dominant manner, in which areas of the growth plate become displaced, presumably by growing through a defect in the perichondrium, or so-called ring of Ranvier. The spongiosa forms within the mass as vessels invade the cartilage. Therefore, the diagnostic radiographic finding is the direct continuity of the mass to the marrow cavity of the parent bone with absence of underlying cortex. Usually the growth of these exostoses ceases when growth of the adjacent plate ceases. The lesions may be solitary or multiple and are usually located in the metaphyseal areas of long bones, with the apex of the exostosis directed toward the diaphysis. Often the lesions produce no symptoms, but occasionally, interference with the function of a joint or tendon or compression of nerves may result. Dwarfism may occur. The metacarpals may be shortened, resembling those seen in Albright's hereditary osteodystrophy. Multiple exostoses are sometimes seen in patients with pseudohypoparathyroidism.

An exostosis may suddenly begin to enlarge long after growth should have ceased, and rarely, chondrosarcomas may develop from the cartilage cap of an exostosis. Although pregnancy may stimulate growth of an exostosis that clinically may mimic malignancy, the lesion merely undergoes exuberant endochondral ossification and cartilage hyperplasia without malignant changes.

Multiple inactivating mutations or deletions have been identified in the *EXT1* or *EXT2* genes in patients with hereditary multiple exostoses. The *EXT* genes probably function normally as a tumor suppressor, and mutations in *EXT* could contribute both to the development of the exostoses and to the malignant transformation to chondrosarcoma that sometimes occurs. Mutations in *EXT* genes apparently cause abnormal processing the cytoskeletal proteins in chondrocytes.

(Bibliography omitted in Palm version)

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SECTION 3 -DISORDERS OF INTERMEDIARY METABOLISM

344. DISORDERS OF LIPOPROTEIN METABOLISM - *Henry N. Ginsberg,, Ira J. Goldberg*

Lipoproteins are macromolecular complexes that carry hydrophobic plasma lipids, particularly cholesterol and triglyceride, in the plasma. More than half of the coronary heart disease (CHD) in the United States is attributable to abnormalities in the levels and metabolism of plasma lipids and lipoproteins. Some premature CHD is due to mutations in major genes involved in lipoprotein metabolism. However, elevated lipoprotein levels in most patients with CHD reflect the adverse impact of a sedentary lifestyle, excess body weight, and diets high in total and saturated fat superimposed on a genetic background that confers susceptibility to increased circulating lipids. A large body of evidence indicates that lifestyle changes and drug treatment strategies that correct hyperlipidemias reduce CHD risk ([Chap 242](#)). More than 70 clinical trials examining the effects of cholesterol reduction have been reported, including several large-scale studies using the potent cholesterol-lowering HMG-CoA reductase inhibitors (also known as statins). These studies unequivocally demonstrate that lowering low-density lipoprotein (LDL) cholesterol reduces fatal and nonfatal heart attacks ([Table 344-1](#)).

This chapter focuses on the major lipid disorders, including both the dyslipoproteinemias caused by single-gene defects and the disorders that are likely to be multifactorial in origin. A practical approach is provided to assist in the identification, evaluation, and treatment of patients with increased risk of [CHD](#).

LIPID AND LIPOPROTEIN TRANSPORT

LIPOPROTEIN STRUCTURE

Lipoproteins are spherical particles made up of hundreds of lipid and protein molecules. They are smaller than red blood cells and visible only by electron microscopy. However, when the larger, triglyceride-rich lipoproteins are present in high concentration, plasma can appear turbid or milky to the naked eye. The major lipids of the lipoproteins are cholesterol, triglycerides, and phospholipids. Triglycerides and the esterified form of cholesterol (cholesteryl esters) are nonpolar lipids that are insoluble in aqueous environments (hydrophobic) and comprise the core of the lipoproteins. Phospholipids and a small quantity of free (unesterified) cholesterol, which are soluble in both lipid and aqueous environments (amphipathic), cover the surface of the particles, where they act as the interface between the plasma and core components. A family of proteins, the apolipoproteins, also occupies the surface of the lipoproteins; the apolipoproteins play crucial roles in the regulation of lipid transport and lipoprotein metabolism.

Lipoproteins have been classified on the basis of their densities into five major classes: (1) chylomicrons, (2) very low density lipoproteins (VLDL), (3) intermediate-density lipoproteins (IDL), (4) [LDL](#), and (5) high-density lipoproteins (HDL). The physical-chemical characteristics of the major lipoprotein classes are presented in [Table 344-2](#).

APOLIPOPROTEINS

The apolipoproteins (apos) provide structural stability to the lipoproteins and determine the metabolic fate of the particles upon which they reside. They were named in an arbitrary alphabetical order and, for the purposes of this discussion, will be described in relation to their association with lipoprotein classes ([Table 344-3](#)).

There are two forms of apo B -- apo B100 and apo B48. Apo B100 is the major apolipoprotein of [VLDL](#), [IDL](#), and [LDL](#), comprising approximately 30, 60, and 95% of the protein in these lipoproteins, respectively. Apo B100 has a molecular mass of about 545 kDa and is synthesized in the liver. It is essential for the assembly and secretion of VLDL from the liver and is the ligand for the removal of LDL by the LDL receptor. The LDL receptor is a cell-surface protein that binds and internalizes lipoproteins that contain apo B100 or apo E. The LDL receptor binding domain of apo B100 is the sequence between amino acids 3200 and 3600, a region that is absent in apo B48.

Apo B48 is essential for the assembly and secretion of chylomicrons. Apo B48 is encoded by the same gene and messenger ribonucleic acid (mRNA) as Apo B100. However, the mRNA is edited in an unusual way: A cytidine deaminase in the intestine changes a cytidine to a uridine in base 6666 of the apo B100 mRNA to produce a stop codon so that apo B48 contains only the N-terminal 48% of the full-length apo B100. In contrast, the apo B100 mRNA in human liver is not edited. The role of apo B48 in the metabolism of chylomicrons in plasma is unclear. Individuals with mutations that interfere with the normal synthesis of apo B have absent or very low levels of chylomicrons, [VLDL](#), [IDL](#) and [LDL](#).

The apolipoproteins of the C series are synthesized in the liver and are present in all plasma lipoproteins (trace amounts in [LDL](#)). Individual apo Cs have different metabolic roles, but all inhibit the removal of plasma chylomicrons and [VLDL](#) remnants by the liver. Overexpression of apo CI in transgenic mice inhibits the uptake of chylomicron and VLDL remnants by the liver. Apo CI under- or overexpression has not been described in humans. Apo CII is an essential activator of the enzyme lipoprotein lipase (LPL), which hydrolyzes triglycerides in chylomicrons and VLDL; individuals lacking apo CII have severe hypertriglyceridemia. Apo CIII inhibits LPL, and apo CIII overexpression in transgenic mice causes severe hypertriglyceridemia. Humans who lack apo CIII have accelerated rates of VLDL triglyceride lipolysis.

Apo E is synthesized mainly in hepatocytes but is also made in other cells, including macrophages, neurons, and glial cells. It is found in chylomicrons, [IDL](#), [VLDL](#), and [HDL](#) and mediates the uptake of these lipoproteins in the liver by both the [LDL](#) receptor and the LDL receptor-related protein (LRP). Apo E also binds to heparin-like proteoglycan molecules on the surface of all cells. There are three major apo E alleles: E2, E3, and E4; these isoforms differ in sequence at two positions and have frequencies of about 0.12, 0.75, and 0.13, respectively, in the general population. Apo E2 binds to the LDL receptor with lower affinity than apo E3 or E4. Individuals who are homozygous for apo E2 may develop severe hyperlipidemia (type III dysbetalipoproteinemia); complete absence of apo E increases plasma levels of chylomicron and VLDL remnants and causes early atherosclerosis.

Apo AI, apo AII, and apo AIV are found primarily on [HDL](#). Apo AI and apo AII are synthesized in the small intestine and the liver; apo AIV is made only in the intestine. Apo AI comprises about 70 to 80% of the protein of HDL and plays a critical structural role in HDL particles. Individuals with a profound deficiency of apo AI also lack HDL. Apo AI activates the enzyme lecithin:cholesterol acyltransferase (LCAT), which esterifies free cholesterol in plasma. Plasma levels of HDL cholesterol and apo AI are inversely related to risk for [CHD](#), and some patients with apo AI deficiency develop early, severe atherosclerosis. Transgenic mice that overexpress human apo AI are resistant to atherosclerosis. Apo AII is the second most abundant apoprotein in HDL, but its function has not been determined; transgenic mice that overexpress apo AII have high plasma levels of both HDL cholesterol and triglycerides but may be susceptible to atherosclerosis. Apo AII knockout mice have low levels of HDL, indicating that apo AII is also necessary for the integrity of HDL particles. Apo AIV, a minor component of HDL and chylomicrons may play a role in the activation of LCAT.

Apoprotein(a), a large glycoprotein that shares a high degree of sequence homology with plasminogen, is made by hepatocytes and is secreted into plasma where it forms a covalent linkage with the apo B100 of [LDL](#) to form lipoprotein(a). The physiologic role of lipoprotein(a) is not known, but elevated levels are associated with an increased risk for atherosclerosis.

ENZYMES INVOLVED IN LIPOPROTEIN METABOLISM

[LPL](#) is synthesized in fat and muscle, secreted into the interstitial space, transported across endothelial cells, and bound to proteoglycans on the luminal surfaces in the adjacent capillary beds. LPL mediates the hydrolysis of the triglycerides of chylomicrons and [VLDL](#) to generate free fatty acids and glycerol. The free fatty acids diffuse into adjacent tissues to be burned for energy or stored as fat. Most circulating LPL is associated with [LDL](#). Insulin stimulates the synthesis and secretion of LPL; reduced LPL activity in diabetes mellitus can lead to impaired triglyceride clearance. Homozygotes for mutations that impair LPL have severe hypertriglyceridemia that usually manifests in childhood (type I hyperlipidemia). Heterozygotes for LPL defects have mild to moderate fasting hypertriglyceridemia but may have marked hypertriglyceridemia after consuming a high-fat meal. LPL is also expressed in macrophages, including cholesterol ester-laden macrophages (foam cells) in atherosclerotic lesions. In this setting, secreted LPL may associate with LDL, causing retention of the lipoprotein in the subendothelial space.

Hepatic triglyceride lipase (HTGL), a member of a family of enzymes that includes [LPL](#) and pancreatic lipase, is synthesized in the liver and interacts with lipoproteins in hepatic sinusoids. HTGL removes triglycerides from [VLDL](#) remnants ([IDL](#)), thus promoting the conversion of VLDL to [LDL](#). It may also play a role in the clearance of chylomicron remnants and in the conversion of [HDL](#)₂ to HDL₃ in the liver by hydrolyzing the triglyceride and phospholipid in HDL (see below). Severe hypertriglyceridemia in individuals with genetic deficiency of HTGL is due to the accumulation of chylomicron and VLDL remnants in plasma. In contrast to most patients with hypertriglyceridemia, however, individuals with HTGL deficiency have normal levels of HDL.

[LCAT](#) is synthesized in the liver and secreted into plasma where it is bound

predominantly to [HDL](#). LCAT mediates the transfer of linoleate from lecithin to free cholesterol on the surface of HDL to form cholesteryl esters that are then transferred to [VLDL](#) and eventually [LDL](#). Apo A1 is a cofactor for esterification of free cholesterol by LCAT. Deficiency of LCAT can be caused by mutations in the enzyme or in Apo A1. LCAT deficiency causes low levels of cholesteryl esters and HDL, and it can lead to corneal clouding and renal insufficiency.

Cholesteryl ester transfer protein (CETP) is synthesized primarily in the liver and circulates in plasma in association with [HDL](#). CETP mediates the exchange of cholesteryl esters from HDL with triglyceride from chylomicrons or [VLDL](#). This exchange can explain much of the inverse relationship between plasma levels of triglycerides and HDL cholesterol. [LDL](#) cholesteryl ester can also be exchanged with triglyceride from chylomicrons and VLDL, leading to the generation of small, dense LDL. Individuals who are homozygotes for mutations in the CETP gene have marked elevations of HDL cholesterol and apo A1. Heterozygotes for these mutations have slight elevations of HDL, indicating that CETP plays an important role in the removal of cholesteryl esters from HDL.

Phospholipid transfer protein (PLTP) is synthesized in the liver and lung. The production of mature [HDL](#) particles depends on PLTP, which provides phospholipid to the enlarging particles.

TRANSPORT OF EXOGENOUS (DIETARY) LIPIDS

Exogenous lipid transport in chylomicrons and chylomicron remnants is depicted in [Fig. 344-1A](#). In western societies, where individuals ordinarily consume 50 to 100 g of fat and 0.5 g of cholesterol during three or four meals, transport of dietary fats is essentially continual. Normolipidemic individuals dispose of most dietary fat in the bloodstream within 8 h of the last meal, but some individuals with dyslipidemia, particularly those with elevated fasting levels of [VLDL](#) triglyceride, have measurable levels of intestinally derived lipoproteins in the circulation as long as 24 h after the last meal.

In the intestinal mucosa dietary triglyceride and cholesterol are incorporated into the core of nascent chylomicrons. The surface coat of the chylomicron is composed of phospholipid, free cholesterol, apo B48, apo A1, apo AII, and apo AIV. The chylomicron, essentially a fat droplet containing 80 to 95% triglycerides, is secreted into lacteals and transported to the circulation via the thoracic duct. In the plasma, apo C proteins are transferred to the chylomicron from [HDL](#). Apo CII mediates hydrolysis of triglycerides by activating [LPL](#) on capillary endothelial cells in fat and muscle. After the triglyceride core has been hydrolyzed, apo CII and apo CIII recirculate back to HDL. The addition of apo E allows the chylomicron remnant to bind first to heparan sulfate proteoglycans within the space of Disse and then to hepatic LDL receptors and/or LDL receptor-related protein. As a consequence, dietary triglyceride is delivered to adipocytes and muscle cells as fatty acids, and dietary cholesterol is taken up by the liver where it can be used for bile acid formation, incorporated into membranes, resecreted as lipoprotein cholesterol back into the circulation, or excreted as cholesterol into bile. Dietary cholesterol also regulates endogenous hepatic cholesterol synthesis.

Abnormal transport and metabolism of chylomicrons may predispose to atherosclerosis,

and postprandial hyperlipidemia may be a risk factor for [CHD](#). Chylomicrons and their remnants can be taken up by cells of the vessel wall, including monocyte-derived macrophages that migrate into the vessel wall from plasma. Cholesteryl ester accumulation by these macrophages transforms them into foam cells, the earliest cellular lesion of the atherosclerotic plaque ([Chap. 241](#)). If the postprandial levels of chylomicrons or their remnants are elevated, or if their removal from plasma is prolonged, cholesterol delivery to the artery wall may be increased.

TRANSPORT OF ENDOGENOUS LIPIDS

The endogenous lipid transport system, which conveys lipids from the liver to peripheral tissues and from peripheral tissues back to the liver, can be separated into two subsystems: the apo B100 lipoprotein system ([VLDL](#), [IDL](#), and [LDL](#)) and the apo A1 lipoprotein system ([HDL](#)).

The Apo B100 Lipoprotein System (See [Fig. 344-1B](#)) In the liver, triglycerides are made from fatty acids that are either taken up from plasma or synthesized de novo within the liver. Cholesterol can also be synthesized by the liver or delivered to the liver via lipoproteins, particularly chylomicron remnants. These core lipids are packaged together with apo B100 and phospholipids into [VLDL](#) and secreted into plasma where apos C1, CII, CIII, and E are added to the nascent VLDL particles. Triglycerides make up the bulk of the VLDL (55 to 80% by weight), and the size of the VLDL is determined by the amount of triglyceride available. Hence, very large triglyceride-rich VLDL are secreted in situations where excess triglycerides are synthesized, such as in states of caloric excess, in diabetes mellitus, and after alcohol consumption. Small VLDL are secreted when fewer triglycerides are available. Although VLDL are the principal hepatic lipoprotein secreted by most individuals, VLDL and cholesteryl ester-enriched [IDL](#) and/or [LDL](#)-like particles may be secreted by the liver in individuals with combined hyperlipidemia (see below).

In the plasma, triglycerides are hydrolyzed by [LPL](#) and [VLDL](#) particles are converted to VLDL remnants ([IDL](#)). In contrast to chylomicron remnants, VLDL remnants can either enter the liver or give rise to [LDL](#). Larger VLDL particles carry more triglycerides and are likely to be removed directly from plasma without being converted to LDL; apo E in the VLDL remnants binds to the LDL receptor to mediate removal from the plasma. Smaller, more dense VLDL particles are more efficiently converted to LDL; apo E and [HTGL](#) play important roles in this process. Individuals with deficiency of either apo E2 or HTGL accumulate IDL in plasma. Apo B100 is the only protein that remains on the surface of the LDL particle.

The half-life of [LDL](#) in plasma is determined principally by the availability (or "activity") of LDL receptors. Most plasma LDL is taken up by the liver, and the remainder is delivered to peripheral tissues, including the adrenals and gonads, which utilize cholesterol as a precursor for steroid hormone synthesis. The adrenals have the highest concentration of LDL receptors per cell in the body. Overall, about 70 to 80% of LDL catabolism occurs via LDL receptors, and the remainder is removed by fluid endocytosis and possibly by other receptors.

The [LDL](#) receptor, a glycoprotein with a molecular mass of approximately 160 kDa, is

present on the surfaces of nearly all cells in the body. Goldstein and Brown characterized the molecular genetics and cell biology of the LDL receptor and defined its role in cholesterol metabolism. They showed that cholesterol delivered to the cytoplasm by LDL regulates both the rate of cholesterol synthesis in the liver and the number of LDL receptors on the surface of hepatocytes. LDL receptor synthesis is mediated by sterol response element regulatory proteins (SREBPs). These transcription factors are activated in the absence of cholesterol, proteolytically cleaved, and transferred from the endoplasmic reticulum into the nucleus where they stimulate LDL receptor gene expression. Though the LDL receptor is a major factor in determining plasma LDL cholesterol levels, the rates of entry of [VLDL](#) into plasma and the efficiency with which VLDL is converted to LDL also influence steady-state LDL concentrations in plasma.

Increased levels of plasma [LDL](#) cholesterol and apo B100 are risk factors for atherosclerosis. Normal LDL does not cause foam cell formation when incubated with cultured macrophages or smooth-muscle cells. But, when LDL undergoes lipid peroxidation, it becomes a ligand for alternative, scavenger receptors that are present on endothelial cells and macrophages. Uptake of modified (oxidized) lipoproteins by these receptors in macrophages results in formation of cholesterol-laden foam cells. In addition to inducing foam cell formation, oxidized LDL acts in the vessel wall to stimulate the secretion of cytokines and growth factors by endothelial cells, smooth-muscle cells, and monocyte-derived macrophages ([Chap. 242](#)). The consequence is recruitment of more monocytes to the lesion and proliferation of smooth-muscle cells, which synthesize and secrete increased amounts of extracellular matrix, such as collagen. The critical role of LDL in atherosclerosis has been confirmed in genetically altered mice. Although mice are normally resistant to atherosclerosis, increased plasma levels of remnant lipoproteins or LDL lead to atherosclerosis in this species.

The role of [VLDL](#) in atherogenesis is less certain. The major reason for this uncertainty derives from the inverse relationship between elevated levels of triglyceride-rich lipoproteins and reduced levels of the antiatherogenic [HDL](#) cholesterol. It is possible, for example, that hypertriglyceridemia may not be directly atherogenic but rather the surrogate of other lipoprotein abnormalities. If postprandial hyperlipidemia is a risk factor for [CHD](#), individuals who have normal fasting plasma triglyceride levels but develop postprandial hypertriglyceridemia after consumption of a fat load would be misclassified as "normal" in studies in which only fasting blood samples are analyzed. It is clear that cholesteryl ester-enriched VLDL, isolated from cholesterol-fed animals, can be taken up by receptors on macrophages and smooth-muscle cells and cause foam cell formation. These cholesteryl ester-laden VLDLs are enriched in apo E and are probably representative of VLDL remnants. Thus, the risk of atherosclerosis from hypertriglyceridemia and elevated VLDL levels may be determined by the level of cholesteryl ester-enriched VLDL remnants. The atherogenic potential of [IDL](#) is probably similar to that of VLDL remnants.

Apo AI-Containing Lipoproteins (See [Fig. 344-1C](#)) In contrast to atherogenic apo B lipoproteins, the apo AI-containing [HDL](#) appear to be antiatherogenic. In fact, in some studies, HDL cholesterol levels are as strong an indicator of protection from [CHD](#) as [LDL](#) cholesterol levels are an indicator of risk. Although a great deal is known about the HDL transport system, the mechanism by which these lipoproteins protect against

atherosclerosis is poorly defined.

[HDL](#) particles are formed in plasma from the coalescence of individual phospholipid-apolipoprotein complexes. Apo AI appears to be the crucial, structural apoprotein for HDL, and apo AI/phospholipid complexes probably fuse with other phospholipid vesicles that contain apo AII and apo AIV to form the various types of HDL. The C apoproteins can be added to HDL after their secretion as phospholipid complexes or by their transfer from triglyceride-rich lipoproteins. This may involve the action of [PLTP](#). These small, cholesterol-poor HDL particles are heterogeneous in size and content and are referred to as HDL₃. Free cholesterol is transferred from cell membranes to HDL₃; a cholesterol transporter called ABC1 mediates this important first step in reverse cholesterol transport. Free cholesterol in HDL₃ is converted to cholesteryl ester by [LCAT](#), and the cholesteryl ester moves into the core of the HDL. Formation of cholesteryl ester increases the capacity of the HDL₃ to accept more free cholesterol and enlarge to form the more buoyant class of HDL particles termed *HDL₂*. HDL₂ can be metabolized by two pathways: (1) cholesteryl esters can be transferred from HDL₂ to apo B lipoproteins or cells, or (2) the entire HDL₂ particle can be removed from plasma. The transfer of cholesteryl ester from HDL to triglyceride-rich apo B lipoproteins (chylomicrons and [VLDL](#) in the fed and fasted states, respectively) is mediated by [CETP](#). Triglyceride is transferred to HDL in this process and is a substrate for lipolysis by [LPL](#) and/or [HTGL](#). As a result, HDL₂ is converted back into HDL₃. When the apo B lipoproteins are removed by the liver, reverse cholesterol transfer is complete. HDL cholesteryl ester may also be transferred selectively to cells via interaction of HDL with the scavenger receptor B-1, a receptor expressed by hepatocytes and steroid-producing cells. HDL-mediated reverse cholesterol transport (from peripheral tissues to the liver) is thought to be the primary mechanism by which HDL protects against atherosclerosis.

Rarely, low plasma [HDL](#) is due to a genetic deficiency of one of the structural components of HDL (such as apo AI). However, low HDL cholesterol levels are usually the secondary consequence of increased plasma levels of [VLDL](#) and [IDL](#) (or chylomicrons and their remnants). Mutations in the *ABC1* gene (see above) are associated with Tangier's disease, a rare form of low HDL. Low levels of HDL cholesterol and apo AI may increase atherosclerosis risk by any of several mechanisms. HDL could remove cholesterol from foam cells in atherosclerotic lesions or protect [LDL](#) from oxidative modification. Alternatively, the atherosclerotic risk of low HDL may be due to the commonly associated elevations of apo B-containing lipoproteins, which accept HDL cholesteryl esters and deliver cholesteryl esters to the vessel wall.

THE HYPERLIPOPROTEINEMIAS (See [Table 344-4](#))

HYPERCHOLESTEROLEMIA

Elevated levels of fasting plasma total cholesterol in the presence of normal levels of triglycerides are almost always associated with increased concentrations of plasma [LDL](#) cholesterol (type IIa), as LDL carries about 65 to 75% of total plasma cholesterol. A rare individual with markedly elevated [HDL](#) cholesterol may also have increased plasma total cholesterol levels. Elevations of LDL cholesterol can result from single-gene defects, polygenic disorders, or from the secondary effects of other disease

states.

Familial Hypercholesterolemia (FH) FH is a codominant genetic disorder that occurs in the heterozygous form in approximately 1 in 500 individuals. FH is due to mutations in the gene for the [LDL](#) receptor and is genetically heterogeneous, >200 different mutations in the gene having been described. Plasma levels of LDL cholesterol are elevated at birth and remain so throughout life. In untreated adults, total cholesterol levels range from 7 to 13 mmol/L (275 to 500 mg/dL). Plasma triglyceride levels are typically normal, and [HDL](#) cholesterol levels are normal or reduced. As would be expected of a disorder with decreased numbers of LDL receptors, the fractional clearance of LDL apo B is reduced. LDL production is increased because the liver secretes more [VLDL](#) and [IDL](#) and more IDL particles are converted to LDL rather than taken up by the hepatic LDL receptors. FH heterozygotes usually develop severe atherosclerosis in early or middle age. *Tendon xanthomas*, which are due to both intracellular and extracellular deposits of cholesterol, most commonly involve the Achilles tendons and the extensor tendons of the knuckles; they are found in about 75% of adults with FH ([Fig. 344-CD1](#)). *Tuberous xanthomas*, which are softer, painless nodules on the elbows and buttocks, and *xanthelasmas*, which are barely elevated deposits of cholesterol on the eyelids, are common in heterozygous FH ([Figs. 344-CD2](#) and [344-CD3](#)). [CHD](#) develops in men by the fourth decade of life or earlier.

The homozygous form of [FH](#) occurs in 1 out of 1 million individuals and is associated with a marked increase of plasma cholesterol levels (>13 mmol/L; >500 mg/dL), large xanthelasmas, and prominent tendon and planar xanthomas. These individuals have severe, premature [CHD](#) that can be manifested in childhood.

Familial Defective Apo B100 This autosomal dominant disorder is a phenocopy of [FH](#) and is due to a missense mutation at amino acid 3500 that reduces the affinity of [LDL](#) for the LDL receptor and, thus, impairs LDL catabolism. The prevalence and manifestations of both the heterozygous and homozygous forms are similar to those produced by mutations of the LDL receptor.

Polygenic Hypercholesterolemia Most moderate hypercholesterolemia [plasma cholesterol levels between 6.5 and 9 mmol/L (240 and 350 mg/dL)] is polygenic in origin. Multiple genes interact with environmental factors to contribute to the hypercholesterolemia, and both overproduction and reduced catabolism of [LDL](#) are thought to play roles in the pathophysiology. The severity is probably affected by the consumption of saturated fat and cholesterol, age, and the level of physical activity. Plasma triglyceride and [HDL](#) cholesterol levels are usually normal. These individuals are at increased risk of atherosclerosis. Tendon xanthomas are not present. Genes involved in cholesterol and bile acid metabolism may be involved in the pathogenesis.

HYPERTRIGLYCERIDEMIA

The diagnosis of hypertriglyceridemia is made by determining plasma lipids after an overnight fast. Because of the less certain association of triglycerides with [CHD](#) (compared to [LDL](#) cholesterol), plasma concentrations greater than the 90th or 95th percentile for age and sex have been used to define hypertriglyceridemia. Some studies show, however, that plasma triglyceride levels >130 to 150 mg/dL are

associated with low [HDL](#) cholesterol levels and small, dense LDL particles. Furthermore, a meta-analysis of several prospective population studies confirms that triglyceride concentrations are independent predictors of CHD risk. Isolated elevations of plasma triglycerides can be due to increased levels of [VLDL](#) (type IV) or combinations of VLDL and chylomicrons (type V). Rarely, only chylomicron levels are elevated (type I). Plasma is usually clear when triglyceride levels are <4.5 mmol/L (<400 mg/dL) and cloudy when levels are higher and VLDL (and/or chylomicron) particles become large enough to scatter light. When chylomicrons are present, a creamy layer floats to the top of plasma after refrigeration for several hours. Tendon xanthomas and xanthelasmas do not occur with isolated hypertriglyceridemia, but eruptive xanthomas (small orange-red papules) ([Fig. 344-CD4](#)) can appear on the trunk and extremities when triglyceride levels are >11.5 mmol/L (>1000 mg/dL) (i.e., when chylomicronemia is present). At these high levels of triglycerides, the retinal vessels can appear to be orange-yellow in color (lipemia retinalis). Pancreatitis is the major risk associated with plasma triglyceride concentrations >11 mmol/L (>1000 mg/dL).

Elevations in plasma triglycerides are usually associated with increased synthesis and secretion of [VLDL](#) triglycerides by the liver. Hepatic triglyceride synthesis is regulated by substrate flow (the availability of free fatty acids), energy balance (the level of glycogen stores in the liver), and hormonal status (the balance between insulin and glucagon). Obesity, excessive consumption of simple sugars and saturated fats, inactivity, alcohol consumption, and insulin resistance are commonly associated with hypertriglyceridemia. In most of these situations, increased free fatty acid flux from adipose tissue to the liver stimulates the assembly and secretion of VLDL. When VLDL triglyceride levels are markedly elevated [>11.5 mmol/L (>1000 mg/dL)], [LPL](#) may be saturated so that an acquired LPL deficiency develops during the postprandial period even if there is no underlying genetic disorder. The addition of chylomicrons to the circulation may cause dramatic increases in plasma triglycerides.

Familial Hypertriglyceridemia Familial hypertriglyceridemia appears to be transmitted as an autosomal dominant disorder, though the underlying mutation(s) have not been identified. The pathophysiology is complex: both reduced catabolism of triglyceride-rich lipoproteins and overproduction of [VLDL](#) have been reported. Elevated levels of fasting plasma triglycerides in the range of 2.3 to 8.5 mmol/L (200 to 750 mg/dL) are usually associated with increased levels of VLDL triglycerides only. When VLDL triglyceride levels are markedly elevated (regardless of etiology), chylomicron triglycerides can also be present, even after a 14-h fast. A 20-year follow-up of individuals with familial hypertriglyceridemia demonstrated a moderate increase in [CHD](#) risk.

Familial Lipoprotein Lipase Deficiency This autosomal recessive disorder is due to the severe impairment or absence of [LPL](#), leading to massive accumulation of chylomicrons in plasma. Manifestations begin in infancy and include pancreatitis, eruptive xanthomas, hepatomegaly, splenomegaly, foam cell infiltration of the bone marrow, and, when the level of triglycerides is >11 mmol/L (1000 mg/dL), lipemia retinalis. Atherosclerosis is not accelerated. The diagnosis is suspected by finding a creamy layer (chylomicrons) at the top of plasma that has incubated overnight at 4°C ; it is confirmed by demonstrating that LPL levels in plasma do not increase after the administration of heparin (which normally releases LPL from endothelial surfaces). Manifestations recede dramatically when patients are placed on fat-free diets.

[LPL](#) levels are within the normal range in most patients with moderate hypertriglyceridemia [2.8 to 5.6 mmol/L (250 to 500 mg/dL)]. Heterozygous mutations in the LPL gene are present in 5 to 10% of hypertriglyceridemic individuals; LPL activity may be reduced by 20 to 50% in these individuals. Heterozygotes for LPL deficiency may also present with severe hypertriglyceridemia if they have poorly controlled diabetes, are pregnant, consume excessive quantities of alcohol, take exogenous estrogen, or are obese.

Familial Apoprotein CII Deficiency This rare autosomal recessive disorder causes a functional deficiency of [LPL](#) and clinical manifestations similar to those of familial LPL deficiency. Deficiency of apoprotein CII impairs hydrolysis of chylomicrons and [VLDL](#) so that either, or both, lipoproteins accumulate in blood. The diagnosis is suspected in children or adults with recurrent attacks of pancreatitis and confirmed by demonstrating the absence of apo CII on gel electrophoresis and that plasma transfusion (which contains abundant apo CII) causes a dramatic fall in plasma triglycerides. Heterozygotes have half-normal levels of apo CII, may have mild elevations of triglycerides, and are asymptomatic. Dietary fat restriction should be life-long.

Hepatic Lipase Deficiency Total deficiency of [HTGL](#) is a rare autosomal recessive disorder that impairs the final catabolism and/or remodeling of small [VLDL](#) and [IDL](#). Subjects with HTGL deficiency often have elevated levels of VLDL remnants; [HDL](#)₂ levels may be elevated because HTGL participates in the conversion of HDL₂ to HDL₃. HTGL activity is frequently increased in hypertriglyceridemic individuals, but the meaning of this association is unclear.

HYPERCHOLESTEROLEMIA WITH HYPERTRIGLYCERIDEMIA

Concomitant hypercholesterolemia and hypertriglyceridemia occurs in two disorders -- familial combined hyperlipidemia (FCHL) and dysbetalipoproteinemia.

Familial Combined Hyperlipidemia [FCHL](#) is transmitted as an autosomal dominant disorder. Probands (the initial case discovered within a family) typically have combined hyperlipidemia, isolated hypertriglyceridemia, or isolated elevated levels of [LDL](#) cholesterol. The diagnosis requires documentation at some time of combined hyperlipidemia in the proband or, if the proband has isolated hypercholesterolemia or hypertriglyceridemia, the various lipid phenotypes in first-degree relatives at risk. The lipoprotein phenotype in affected individuals may change over time. The underlying defect in this disorder is not known, though mutations or polymorphisms in the [LPL](#) gene and in the gene cluster for apo AI, apo CIII, and apo AIV may contribute to the disorder in some families. Insulin resistance is present in many individuals with FCHL; the link may result from increased free fatty acid flux driving assembly and secretion of apo B100 lipoproteins.

[FCHL](#) is associated with increased secretion of [VLDL](#) particles, as determined by the flux of VLDL apo B. The lipoprotein patterns associated with the disorder are most likely determined by genetic polymorphisms in genes that regulate the metabolism of VLDL. For example, if the affected individual also has a defect in [LPL](#), hypertriglyceridemia will be present. Since the hydrolysis of VLDL triglycerides also regulates the generation

of [LDL](#) in plasma, individuals with FCHL who have inefficient catabolism of VLDL may also have reduced levels of LDL cholesterol and high VLDL cholesterol. Finally, individuals with FCHL who synthesize normal quantities of triglycerides and secrete VLDL that carries normal amounts of triglyceride generate increased numbers of LDL particles and present with isolated elevations of plasma LDL cholesterol. These variations in VLDL catabolism, together with additional genetic heterogeneity and environmental variability, form the basis for the variable phenotype in this disorder. FCHL may occur in as many as 0.5 to 1.0% of Americans and is the most common familial lipid disorder in survivors of myocardial infarction. The increased risk for atherosclerosis is due to the presence of increased numbers of small, atherogenic VLDLs and the conversion of VLDL to the more atherogenic [IDL](#) and LDL. Persons with FCHL usually have clear plasma and do not have xanthomas or xanthelasma.

Dysbetalipoproteinemia This rare disorder affects 1 in 10,000 persons and is due to homozygosity for apo E2, the binding-defective form of apo E. Because apo E plays a crucial role in the catabolism of chylomicron and [VLDL](#) remnants, affected individuals have elevations in both VLDL triglyceride and VLDL cholesterol, and chylomicron remnants are present in fasting plasma. The ratio of total cholesterol to triglyceride approximates 1.0, and the ratio of VLDL cholesterol to triglyceride is greater than 0.25. [LDL](#) and [HDL](#) cholesterol levels are usually low. Although 1% of the population is homozygous for apo E2, most have normal plasma triglyceride and cholesterol levels. Thus, a second defect in lipid metabolism must be present in the 0.01% of individuals with dysbetalipoproteinemia. These individuals may have tuberous xanthomas and deposits of cholesterol in the palmar creases (striae palmaris); the latter, appearing as yellow-orange lines, are specific for dysbetalipoproteinemia. The risk for atherosclerosis and its complications is increased, with onset in the fourth and fifth decades. The incidence of peripheral vascular disease is higher than in [FH](#).

REDUCED HDL CHOLESTEROL

Low levels of [HDL](#) cholesterol can be defined as <0.9 mmol/L (<35 mg/dL) in men and <1 to 1.2 mmol/L (<40 to 45 mg/dL) in women. Low concentrations of HDL cholesterol are usually associated with coexistent hypertriglyceridemia, though "primary hypoalphalipoproteinemia" has been identified in both individuals and families. The relationship between hypertriglyceridemia and low HDL levels probably derives from: (1) [CETP](#)-mediated transfer of cholesteryl ester from the core of HDL to [VLDL](#); (2) shift of surface components, particularly phospholipids apo CII, and apo CIII, from HDL to VLDL; and (3) increased fractional catabolism of the cholesteryl ester-poor apoAI that results from the first two processes. The complexity of the relationship between HDL and triglyceride levels is highlighted by the fact that HDL levels do not return to normal when fasting plasma triglycerides are reduced in most persons with hypertriglyceridemia and low HDL cholesterol levels. Low HDL is clinically silent, and the plasma is usually clear (it can be cloudy or creamy if there is concomitant hypertriglyceridemia).

Primary hypoalphalipoproteinemia refers to the state where [HDL](#) cholesterol concentrations are markedly reduced but plasma triglyceride concentrations are normal. Many individuals with this phenotype have had hypertriglyceridemia in the past or have an older (or more obese) first-degree relative who has both low HDL and increased triglyceride levels. Hence, both family studies and long-term follow-up may be required

to identify individuals with primary reductions in HDL cholesterol. Rare mutations have been described in the apo A1 gene that lead to reductions in apo A1 synthesis or increases in catabolism. One mutation that is common in Italy, apo A1-Milano, is associated with a high fractional clearance rate of apo A1 but is not associated with increased risk for atherosclerosis.

Some rare genetic disorders of lipid metabolism are summarized in [Table 344-5](#).

SECONDARY CAUSES OF HYPERLIPOPROTEINEMIA (See [Table 344-6](#))

Diabetes Mellitus Diabetes can affect lipid and lipoprotein metabolism through several mechanisms ([Chap. 333](#)). In type 1 diabetes mellitus (DM) (formerly called insulin-dependent diabetes mellitus), plasma lipids are usually normal when control of diabetes with insulin is adequate. In diabetic ketoacidosis, hypertriglyceridemia can be severe due to increases in both [VLDL](#) and chylomicrons. These abnormalities are associated with overproduction of VLDL and [LPL](#) deficiency secondary to insulinopenia. They usually improve with tight control of the diabetes. In type 2 DM (formerly called non-insulin-dependent diabetes mellitus), insulin resistance and obesity combine to cause mild to moderate hypertriglyceridemia and low [HDL](#) cholesterol levels. In general, this pattern of dyslipidemia is due to overproduction of VLDL. LDL cholesterol is usually normal in type 2 DM, though the LDLs are small, dense, and perhaps more atherogenic. Treatment of type 2 DM and weight reduction improve, but usually do not completely correct, the dyslipidemia (particularly the low HDL cholesterol levels). Therapy of hyperlipidemia should not be delayed in patients with type 2 DM, as they are at increased risk for [CHD](#). It is recommended that patients with diabetes should be treated as if they already have CHD, i.e., the treatment goal is to reduce their LDL to <2.6 mmol/L (<100 mg/dL) ([Fig. 344-2](#)).

Hypothyroidism Hypothyroidism accounts for about 2% of all cases of hyperlipidemia and is second only to [DM](#) as a cause of secondary hyperlipidemia. Levels of [LDL](#) cholesterol can be elevated, even in patients with subclinical disease in whom thyroid-stimulating hormone (TSH) levels are elevated but other thyroid function tests are normal. Hypertriglyceridemia can occur if obesity is present. Hypothyroidism is also associated with increased levels of [HDL](#) cholesterol, probably because of reduced [HTGL](#) activity. Correction of hypothyroidism reverses the lipid abnormalities.

Renal Disease Renal disease causes a wide range of lipid abnormalities. The nephrotic syndrome can be accompanied by elevations in [LDL](#), [VLDL](#), or both. The severity of the hyperlipidemia correlates with the degree of hypoproteinemia. Renal failure is associated with hypertriglyceridemia and low [HDL](#) cholesterol concentrations.

Ethanol The metabolism of ethanol enhances the level of NADH in the liver which, in turn, stimulates the synthesis of fatty acids and their incorporation into triglycerides. Moderate ethanol consumption raises plasma [VLDL](#) levels, with the degree of elevation dependent on the baseline level. Severe hypertriglyceridemia and pancreatitis usually develop on the background of a genetic hyperlipidemia and heavy alcohol intake. Because ethanol also stimulates the synthesis of apo A1 and inhibits [CETP](#), ethanol-associated hypertriglyceridemia is usually accompanied by normal or elevated levels of [HDL](#) cholesterol.

Liver Disease Primary biliary cirrhosis and extrahepatic biliary obstruction can cause hypercholesterolemia and elevated levels of plasma phospholipids associated with increased levels of an abnormal lipoprotein (lipoprotein X; [Chap. 299](#)) and [LDL](#). Severe liver injury often leads to a decrease in levels of both cholesterol and triglyceride. Acute hepatitis can cause elevated levels of [VLDL](#) and impairment of [LCAT](#) formation.

AIDS Use of protease inhibitor therapies has been associated with a generalized metabolic syndrome that includes hypertriglyceridemia, alterations in fat distribution, and occasionally type 2 [DM](#) ([Chap. 309](#)).

DIAGNOSIS

Although the initial indication of an abnormality in lipoprotein metabolism is via blood measurements of triglyceride and cholesterol, the disorders are due to abnormalities of specific lipoproteins. Thus, lipoprotein analysis should assess [VLDL](#), [LDL](#), and [HDL](#) levels. Direct measurements of plasma LDL require laborious centrifugation techniques. However, LDL cholesterol concentrations can be estimated indirectly in individuals with triglyceride levels <4.5 mmol/L (<400 mg/dL) by subtracting the HDL and VLDL cholesterol from the total plasma cholesterol. HDL cholesterol is determined after chemical precipitation of VLDL and LDL. VLDL cholesterol is estimated to be the plasma triglyceride level divided by five. Therefore

where all values are measured in milligrams per deciliter.

In persons with triglyceride levels >4.5 mmol/L (>400 mg/dL), the ratio of triglyceride to cholesterol in [VLDL](#) is >5 , and this equation cannot be used to calculate the plasma [LDL](#) cholesterol level. The other disorder that is not detected with this method is dysbetalipoproteinemia because the ratio of triglyceride to cholesterol in the VLDL is $<<5$. In these two situations, direct measurement of LDL cholesterol must be performed in ultracentrifuged plasma. Commercial methods for the measurement of "direct LDL" are available. Although these methods appear to be precise and accurate, the measured values for LDL cholesterol are 0.06 to 0.17 mmol/L (5 to 15 mg/dL) less than estimated LDL because the estimated value is actually the combination of [IDL](#) and LDL. If a "direct LDL" measurement is used, the National Cholesterol Education Program (NCEP) guidelines (based on estimated LDL) must be adjusted before therapeutic decisions are made.

Because plasma triglyceride levels rise and both [HDL](#) and [LDL](#) cholesterol levels fall modestly after a fat-containing meal (due to the action of [CETP](#)), it is preferable to measure plasma lipids after a 12-h fast. Measuring cholesterol levels alone will not detect individuals with isolated low HDL; screening for [CHD](#) should therefore include measurement of HDL. Because serum lipid levels vary from day to day, at least two to three measurements should be made days or weeks apart before initiating therapy. Some experts advocate the use of total cholesterol/HDL ratios as a better assessment of individual risk. This is a reasonable approach provided both the patient and physician are aware that the treatment goal is to reduce LDL. In addition, rare patients with very

high or very low levels of both LDL and HDL have ratios that are not interpretable on the basis of population studies. Although some laboratories offer measurements of individual apoproteins (e.g., apo B100 and apo AI), or size estimates of LDL, this information is not generally helpful in decision-making. Measurement of lipoprotein (a) levels can provide an indication of risk that cannot be gleaned from lipid measurements. Lipoprotein electrophoresis is not useful except for the diagnosis of dysbetalipoproteinemia, a diagnosis that otherwise requires ultracentrifugation methods. Apo E genotyping is also helpful in the diagnosis of dysbetalipoproteinemia (although rarely the disorder can be due to other defects in the apo E gene).

Both [LDL](#) and [HDL](#) cholesterol levels are temporarily decreased for several weeks after myocardial infarction or acute inflammatory states but can be accurately measured if blood is obtained within 8 h of the event.

Approach to the Patient

Elevated LDL Cholesterol Treatment of elevated LDL cholesterol can have either of two aims -- *primary prevention* of the complications of atherosclerosis or *secondary treatment* after complications have occurred. The rationale for primary prevention is based on the large body of data linking elevated levels of LDL cholesterol with increased [CHD](#) risk and an impressive body of clinical and experimental data demonstrating that reducing LDL cholesterol slows progression and may actually induce regression of atherosclerotic lesions ([Chap. 242](#)). Both primary and secondary intervention trials indicate that total mortality can be reduced when the LDL cholesterol is lowered ([Table 344-1](#)). A meta-analysis of four randomized trials (4S, CARE, AFCAPS/TexCAPS, LIPID) comparing HMG-CoA reductase inhibitors to control included 30,817 participants and found that HMG-CoA reductase inhibitor treatment was associated with: (1) a 20% decrease in total cholesterol, a 28% decrease in LDL cholesterol, a 13% decrease in triglycerides, and a 5% increase in [HDL](#) cholesterol; (2) a 31% decrease in major coronary events and a 21% decrease in all-cause mortality; (3) similar risk reduction in women and men; and (4) no effect on noncardiovascular mortality. Unexpectedly, the risk of stroke was also reduced 19 to 32% by HMG-CoA reductase inhibitor treatment, even though previous observational studies show a relatively weak association between cholesterol level and stroke risk.

Dietary Alterations A fundamental starting point for both primary prevention and secondary treatment involves counseling to modify diet, exercise, smoking, and other life-style factors that increase the risk of CHD. The typical American diet derives about 35% of its calories from fat (14 to 15% from saturated fat) and contains 400 to 500 mg/d of cholesterol. Individuals with hyperlipidemia should be encouraged to eat a diet lower in cholesterol and saturated fat. The [NCEP](#) Step 1 diet, which is recommended for all Americans above age 2, provides 30% of calories from fat, <10% of calories from saturated fat, and <300 mg/d of cholesterol ([Table 344-7](#)). Carbohydrate is the typical nutrient used to replace fat in patients with isolated hypercholesterolemia. In general, whole-milk dairy products, egg yolks, meats, palm oil, and coconut oil should be replaced with fresh fruits and vegetables, complex carbohydrates (especially whole-grain products), and low-fat dairy products. Shellfish are low in fat content and, except for shrimp, also have low cholesterol levels; shrimp, in moderation, is acceptable. Portion size needs to be stressed; the protein and fat-rich portion of meat in

a given meal should be <115 g (4 oz), the size of a deck of cards. Substitutions with any food low in saturated fat such as bran, nuts, and olive oil will have positive effects on [LDL](#). Hydrogenation of vegetable oils increases the saturation of the fatty acids. In particular, trans-fatty acids, mainly found in commercially hydrogenated vegetable oils, raise LDL and can lower [HDL](#) cholesterol levels. Use of stanol-containing margarines has, by contrast, lowered LDL cholesterol about 5 to 10% by blocking cholesterol absorption in the small intestine. When further diet therapy is indicated, the NCEP Step 2 diet provides 30% of calories as fat but <7% of calories from saturated fat, and <200 mg/d of cholesterol. After changing from the average American diet to the Step 1 diet, the LDL cholesterol usually drops 8 to 10%; an additional reduction of 5 to 7% can be achieved by advancing to the Step 2 diet. There is, however, great individual variability in diet responsiveness, and several values should be obtained before judging the efficacy of any diet treatment.

Primary Prevention The [NCEP](#) Adult Treatment Panel recommends measuring plasma cholesterol in all adults older than age 20 at least every 5 years. Ideally, this testing involves a lipoprotein profile to allow better risk stratification. Primary prevention goals include [LDL](#) cholesterol <3.36 mmol/L (<130 mg/dL), triglycerides <1.7 mmol/L (<150 mg/dL), and [HDL](#) cholesterol >1.03 mmol/L (>40 mg/dL) for men and >1.29 mmol/L (>50 mg/dL) for women. In individuals without [DM](#) or known CHD, treatment recommendations for primary prevention are outlined in [Fig. 344-2](#). Assessment of risk factors in addition to LDL cholesterol is an essential part of this decision-making process. Risk factors include: (1) family history of premature [CHD](#) (<55 years in a male parent or sibling or <65 in female relatives), (2) hypertension (even if it is controlled with medications), (3) cigarette smoking (>10 cigarettes per day), (4) DM, and (5) low HDL [<0.9 mmol/L (<35 mg/dL)]. In addition, because CHD is more prevalent in older individuals, age (men >45 years, women >55 years, or younger women with premature menopause without estrogen replacement) is also an important risk factor. HDL cholesterol >1.6 mmol/L (>60 mg/dL) is a negative risk factor, i.e., one other risk factor can be negated by a high HDL cholesterol level.

In individuals with fewer than two risk factors, life-style modifications alone and follow-up testing may be used if [LDL](#) is <4.14 mmol/L (<160 mg/dL). For those with LDL >4.91 mmol/L (>190 mg/dL), drug treatment is indicated. If two or more risk factors are present, drug treatment in addition to life-style modifications should be instituted if LDL cholesterol is >3.36 mmol/L (>130 mg/dL). HMG-CoA reductase inhibitors are first-line medications for most patients; niacin and resins are second-line treatments (see below).

Secondary Prevention The [NCEP](#) guidelines are stringent for the secondary treatment of patients with [CHD](#). Patients with CHD should be screened for lipid abnormalities during and after their initial diagnoses. A goal of lowering plasma [LDL](#) concentrations to <2.6 mmol/L (<100 mg/dL) is advocated for such individuals as well as for patients with [DM](#) ([Fig. 344-2](#)). As described below, this requires modifications of diet in addition to the use of one or more medications.

If a patient with [CHD](#) has only a modestly elevated [LDL](#) cholesterol level [e.g., <3.36 mmol/L (<130 mg/dL)], a 4- to 6-week period of Step 1 diet therapy can precede the addition of drugs. In such a patient, moving to the Step 2 diet, which provides the same total fat but <7% of calories from saturated fat, can be useful. If, however, the LDL

cholesterol is >3.36 mmol/L (>130 mg/dL), drug therapy should be instituted along with diet therapy ([Fig. 344-2](#)).

High Triglycerides and Low HDL The evidence that treatment to reduce plasma triglyceride levels or increase levels of HDL cholesterol leads to long-term health benefits is less compelling than that for treatment of high LDL levels. Two recent clinical trials have shown, however, that lowering triglycerides (using fibric acids) or lowering LDL (HMG-CoA reductase inhibitors) decreases CHD events in these patients. There have been no intervention trials in which only increases in HDL cholesterol concentrations have been achieved. Beneficial effects of niacin have been attributed, in part, to its HDL-raising effect and its action to reduce triglycerides and LDL. Even with drugs that primarily affect LDL cholesterol levels, such as bile acid-binding resins and HMG-CoA reductase inhibitors, some of the benefits achieved may be related to increases in HDL cholesterol levels.

In patients with isolated elevations of triglyceride levels or with hypertriglyceridemia and high LDL cholesterol, life-style modifications should be introduced as described above, and weight reduction should be strongly encouraged if obesity is present. Fat intake should be decreased, but the concomitant increase in carbohydrate intake may raise triglyceride and lower HDL cholesterol levels. If this occurs, replacing some of the saturated fat with monounsaturated fat, which will not raise LDL cholesterol, may be valuable. Severe hypertriglyceridemia and hyperchylomicronemia require very low fat diets, avoidance of free sugars, and decreased alcohol intake. Patients with genetic LPL deficiency are instructed to prepare their food using medium-chain triglycerides, which are not incorporated into chylomicrons. Fish oils decrease triglyceride synthesis, and high doses may be used for severe hypertriglyceridemia.

The management of hypertriglyceridemia focuses on the associated LDL and HDL concentrations as guidelines for therapy. Thus, the overall risk profile can be used to set goals for LDL cholesterol, using a low HDL level (commonly associated with hypertriglyceridemia) as a concomitant major risk factor for atherosclerosis. However, when triglyceride levels are >5.6 mmol/L (>500 mg/dL), the risk of developing pancreatitis increases, and a direct focus on lowering triglycerides is recommended. Thus, triglyceride levels >5.6 mmol/L (>500 mg/dL) are generally treated with drugs, whereas lower levels [2.2 to 5.6 mmol/L (200 to 500 mg/dL)] are not treated unless other CHD risk factors are present ([Fig. 344-2](#)).

TREATMENT

Three classes of lipid-lowering agents are recommended as first-line therapy against hypercholesterolemia: (1) the HMG-CoA reductase inhibitors; (2) niacin; and (3) the bile acid-binding resins ([Table 344-8](#)). Fibrin acid derivatives are second-line agents for hypercholesterolemia and are most effective for lowering triglycerides.

HMG-CoA Reductase Inhibitors This class of drugs, which include lovastatin, simvastatin, pravastatin, fluvastatin, atorvastatin, and cerivastatin, inhibits the rate-limiting step in hepatic cholesterol biosynthesis (the conversion of HMG-CoA to mevalonate), causing an increase in LDL receptor levels in hepatocytes and enhanced receptor-mediated clearance of LDL cholesterol from the circulation. At usual doses, the

HMG-CoA reductase inhibitors decrease total cholesterol by 20 to 30% and LDL cholesterol by 25 to 40%. Larger reductions may be achieved with higher doses. Treatment with reductase inhibitors often reduces triglycerides by 10 to 20%, possibly due to reduced secretion of [VLDL](#) by the liver. Higher doses of more potent reductase inhibitors, which can lower LDL cholesterol by 45 to 60%, can lower triglycerides by 30 to 45%. [HDL](#) cholesterol levels rise about 5 to 10%. In comparison with other lipid-lowering agents, HMG-CoA reductase inhibitors are relatively free of side effects. Mild, transient elevations in liver enzymes occur with all of the agents at the highest doses, but elevations in serum aminotransferases to more than three times the upper limits of normal occur in <2% of patients. Therapy should be discontinued when elevations of this magnitude occur. A rare but potentially serious adverse effect of HMG-CoA reductase inhibitors is myopathy, manifest by muscle pain with elevation of serum creatine phosphokinase (CPK). This occurs in <1% of patients treated with reductase inhibitors alone but is more common (about 2 to 3%) when used in combination with gemfibrozil, niacin, or cyclosporine.

Niacin The mechanism of action of niacin is not fully understood, but it appears to inhibit the secretion of lipoproteins containing apo B100 from the liver. Niacin decreases both total and [LDL](#) cholesterol approximately 15 to 25%, reduces [VLDL](#) levels by 25 to 35%, and raises [HDL](#) cholesterol levels by as much as 15 to 25%. Thus, niacin exerts favorable changes on the three major lipoproteins (VLDL, LDL, and HDL). Efficacy of monotherapy was confirmed in a long-term secondary prevention trial in which niacin significantly reduced the incidence of myocardial infarction. An even longer-term follow-up of that study (15 years total) showed an 11% decrease in all-cause mortality among patients randomized to niacin. Because of its ability to reduce VLDL synthesis, niacin is also a first-line drug for treatment of hypertriglyceridemia.

Niacin is safe, having been in use for almost 30 years, but unpleasant side effects, including cutaneous flushing with or without pruritus, may limit patient acceptability. The cutaneous symptoms tend to subside after several weeks and may be minimized by initiating therapy at low doses or by administering aspirin 30 min before the niacin dose. Less common adverse effects include elevations of liver enzymes, gastrointestinal distress, impaired glucose tolerance, and elevated serum uric acid levels with or without gouty arthritis. Liver enzymes may be elevated in 3 to 5% of patients on full doses of niacin (>2 g/d). Because of its propensity to worsen the control of blood sugar, niacin should be used with caution in patients with [DM](#). Niaspan, an intermediate-release form of niacin, appears to exhibit lipid-altering activity similar to regular niacin.

Bile Acid-Binding Resins Cholestyramine and colestipol have been in use as lipid-lowering agents for almost three decades. These drugs interfere with reabsorption of bile acids in the intestine, resulting in a compensatory increase in bile acid synthesis and upregulation of [LDL](#) receptors in hepatocytes. The bile acid sequestrants are useful in the treatment of patients with elevated levels of LDL cholesterol and normal triglycerides. Sequestrants produce dose-dependent decreases on the order of 15 to 25% in total cholesterol and of 20 to 35% in LDL cholesterol. The agents cause modest increases in [HDL](#) cholesterol. A limitation of the sequestrants is their tendency to raise triglyceride levels through compensatory increases in hepatic synthesis of [VLDL](#); they should not be given to hypertriglyceridemic individuals. Bile acid-binding resins are efficacious and safe and are recommended for young adult men and premenopausal

women with moderate cholesterol elevations. Patient compliance is low, in part because of the need to dissolve these powdered agents in fluid; the availability of colestipol as a tablet may alleviate this problem. Gastrointestinal side effects include constipation, bloating, and gas.

Combination Therapy Combinations of bile acid-binding resins and reductase inhibitors are effective for the treatment of severe, isolated elevations of [LDL](#) cholesterol. Combinations of reductase inhibitors and niacin, or resins and niacin, are useful for the treatment of high LDL and low [HDL](#) cholesterol levels, though the former combination carries an increased risk of myositis (2 to 3%). If triglyceride and LDL levels are both elevated (HDL is usually reduced as well), resins and niacin are an excellent combination, with resins and gemfibrozil (see below) as an alternative. The combination of a reductase inhibitor and gemfibrozil can be useful when LDL cholesterol is very high in the face of concomitant hypertriglyceridemia, but the risk of myositis (about 2 to 3%) must be considered. Combinations of reductase inhibitors with either niacin or gemfibrozil might best be reserved for patients with [CHD](#) and combined hyperlipidemia.

LDL Apheresis In patients with homozygous [FH](#) and in ordinary FH patients who respond poorly to diet and drug therapy or who cannot tolerate drugs, apheresis at 7- to 14-day intervals can cause profound lowering of LDL cholesterol levels. Diet and drug regimens are continued during treatment. This approach should be considered for patients with few therapeutic options.

Fibric Acids Gemfibrozil and fenofibrate stimulate the activity of a liver transcription factor termed *PPARα* that increases [LPL](#) activity and production of apo AI. Moreover, these drugs reduce [VLDL](#) triglyceride entry into plasma and reduce synthesis of apo CIII, which might improve LPL-induced lipolysis or reduce VLDL secretion. Stimulation of peroxisomal fatty acid oxidation by fibrates may also contribute to the triglyceride-lowering actions. Gemfibrozil and fenofibrate treatment is associated with 25 to 40% reductions in plasma triglyceride levels. Postprandial triglyceride levels, which are linked to fasting concentrations, are also reduced. HDL cholesterol levels increase 5 to 15% with fibrate treatment. Fibric acids and a low-fat diet are particularly useful in the treatment of dysbetalipoproteinemia and are first-line therapy for this disorder except in postmenopausal women, who should initially be given estrogen replacement (if not contraindicated).

Significant increases in [LDL](#) cholesterol can accompany otherwise potentially beneficial falls in triglycerides and increases in [HDL](#) cholesterol during fibrate therapy. Such rises may require a change to another drug or addition of a second agent.

In the short term, these drugs are well tolerated; mild gastrointestinal distress in the form of epigastric pain is the major side effect. Elevations of liver enzymes occur in 2 to 3% of patients but do not usually require cessation of treatment. Rarely, hepatitis can occur. Fibrates appear to make the bile more lithogenic, and long-term use is probably associated with a twofold increase in gallstone formation. Myopathy with myositis is a rare occurrence with the fibrates, either alone or in combination with HMG CoA reductase inhibitors.

Fish Oils Large doses of omega-3 fatty acids reduce triglyceride levels by diminishing

their production. In the United States, omega-3 fatty acid capsules contain 40 to 60% omega-3 fatty acids; the rest of the fatty acids are omega-6. Therefore, to consume 2 to 4 g of omega-3 fatty acids, an individual must take 5 to 10 capsules per day.

HYPOCHOLESTEROLEMIA

A low total cholesterol concentration [<2.6 mmol/L (<100 mg/dL)] in an adult can be due to rare hereditary traits or secondary to a number of diseases. As described earlier, mutations in the gene for apo B100 that disrupt synthesis or produce truncated forms of apo B100 are associated with hypobetalipoproteinemia. These mutations are inherited as codominant traits; heterozygotes have plasma cholesterol levels in the range of 1.3 to 2.6 mmol/L (50 to 100 mg/dL), with reduced [LDL](#) cholesterol levels but normal plasma [HDL](#) cholesterol levels. Heterozygotes are asymptomatic, whereas hypolipoproteinemia homozygotes (or compound heterozygotes) have even lower total and LDL cholesterol concentrations and may have malabsorption of fats and fat-soluble vitamins similar to that in abetalipoproteinemia.

Abetalipoproteinemia ([Table 344-5](#)) is a rare, autosomal recessive disorder in which there are mutations in the microsomal triglyceride transfer protein (MTP) gene. Individuals who are homozygous for this disorder have total cholesterol levels <1.3 mmol/L (<50 mg/dL) and essentially no [VLDL](#), [IDL](#), [LDL](#), or chylomicrons. Because dietary fat and vitamins A and E are transported from the intestine in chylomicrons, these patients may have malabsorption of fat and fat-soluble vitamins. Vitamin E deficiency in infancy and early childhood can result in neurologic problems ([Chap. 75](#)). If vitamin replacement is adequate, individuals with abetalipoproteinemia can live normal, healthy lives.

Moderately low levels of total cholesterol may also be associated with extreme reductions in [HDL](#) cholesterol. As noted above, these are almost always secondary to mutations in the gene for apo AI and a lack of apo AI in plasma.

A number of systemic diseases can cause low cholesterol concentrations. Malnutrition, often associated with alcoholism or gastrointestinal disease, can cause low levels of total and [LDL](#) cholesterol. Hyperthyroidism, particularly when severe, can reduce cholesterol levels. Patients with uncontrolled AIDS may have total cholesterol levels <2.1 mmol/L (<80 mg/dL), usually associated with severe wasting, diarrhea, and a poor prognosis. Several neoplasms, particularly those involving the hematopoietic system, are associated with hypocholesterolemia. Patients with acute and chronic myelogenous leukemia and myeloid metaplasia with splenomegaly can have severe reductions in both LDL and [HDL](#) levels. Other diseases with concomitant splenomegaly, including lipid storage diseases such as Gaucher's disease and Niemann-Pick disease, can cause very low LDL and HDL cholesterol concentrations due to increased lipoprotein catabolism.

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345. HEMOCHROMATOSIS - Lawrie W. Powell, Kurt J. Isselbacher

DEFINITION

Hemochromatosis is a common disorder of iron storage in which an appropriate increase in intestinal iron absorption results in deposition of excessive amounts of iron in parenchymal cells with eventual tissue damage and impaired function of organs, especially the liver, pancreas, heart, joints, and pituitary. The disease was termed *hemochromatosis* and the iron-storage pigment was called *hemosiderin* because it was believed that the pigment was derived from the blood. The terms *hemosiderosis* and *siderosis* are often used to describe the presence of stainable iron in tissues, but tissue iron must be quantified to assess body iron status (see below and [Chap. 105](#)).

Hemochromatosis implies potentially severe progressive iron overload leading to fibrosis and organ failure. Cirrhosis of the liver, diabetes mellitus, arthritis, cardiomyopathy, and hypogonadotropic hypogonadism are common manifestations.

Although there is debate about definitions, it seems logical to use the following terminology:

1. *Hereditary or genetic hemochromatosis*: This disorder is most often caused by inheritance of a mutant *HFE* gene, which is tightly linked to the HLA-A locus on chromosome 6p (see below). The genetic disease can be recognized during its early stages when iron overload and organ damage are minimal. At this stage the disease is best referred to as *early or precirrhotic hemochromatosis* ([Fig. 345-1](#)).

2. *Secondary iron overload*: Tissue injury usually occurs secondary to an iron-loading anemia such as thalassemia or sideroblastic anemia, in which increased erythropoiesis is ineffective. In the acquired iron-loading disorders, massive iron deposits in parenchymal tissues can lead to the same clinical and pathologic features as in hemochromatosis.

PREVALENCE

Hemochromatosis is one of the most common genetic diseases, although its prevalence varies in different ethnic groups. It is most common in populations of northern European extraction in whom approximately 1 in 10 persons are heterozygous carriers and 0.3 to 0.5% are homozygotes. However, expression of the disease is modified by several factors, especially dietary iron intake, blood loss associated with menstruation and pregnancy, and blood donation. The clinical expression of the disease is 5 to 10 times more frequent in men than in women. Nearly 70% of affected patients develop the first symptoms between ages 40 and 60. The disease is rarely evident before age 20, although with family screening (see below) and periodic health examinations, asymptomatic subjects with iron overload can be identified, including young menstruating women. A recent study in a European non-blood bank population revealed that 30% of homozygous individuals did not have evidence of iron overload. Thus, the penetrance of the mutation is variable.

GENETIC BASIS AND MODE OF INHERITANCE

The gene involved in the most common form of hemochromatosis was cloned in 1996 and is termed *HFE*. A homozygous G→A mutation resulting in a cysteine to tyrosine substitution at position 282 (C282Y) is the most common mutation. It was identified in 85 to 100% of patients with hereditary hemochromatosis in populations of northern European descent but was found in only 60% of cases from Mediterranean populations (e.g., southern Italy). A second, relatively common *HFE* mutation has also been identified. This results in an amino acid substitution of histidine to aspartic acid at position 63 (H63D). Some compound heterozygotes (e.g., one copy each of C282Y and H63D) have increased body iron stores. Thus, *HFE*-associated hemochromatosis is inherited as an autosomal recessive trait; heterozygotes have no, or minimal, increase in iron stores. However, in some cases this slight increase in hepatic iron acts as a cofactor that aggravates other diseases such as porphyria cutanea tarda (PCT) and nonalcoholic steatohepatitis.

Mutations in other genes, currently unidentified, are responsible for non-*HFE* associated hemochromatosis, including juvenile hemochromatosis, which affects subjects in the second and third decade of life ([Table 345-1](#)).

PATHOGENESIS

Normally, the body iron content of 3 to 4 g is maintained such that intestinal mucosal absorption of iron is equal to iron loss. This amount is approximately 1 mg/d in men and 1.5 mg/d in menstruating women. In hemochromatosis, mucosal absorption is inappropriate to body needs and amounts to 4 mg/d or more. The progressive accumulation of iron causes an early elevation in plasma iron, an increased saturation of transferrin, and progressive elevation of plasma ferritin level ([Fig. 345-1](#)).

The *HFE* gene encodes a 343 amino acid protein that is structurally related to MHC class I proteins. The basic defect in hemochromatosis is a lack of cell surface expression of HFE (due to the C282Y mutation). The normal (wild type) HFE protein forms a complex with β_2 -microglobulin and transferrin, and the C282Y mutation completely abrogates this interaction. As a result, the mutant HFE protein remains trapped intracellularly, reducing transferrin receptor-mediated iron uptake by the intestinal crypt-cell. This is postulated to upregulate the divalent metal transporter (DMT-1) on the brush border of the villous cells, leading to inappropriately increased intestinal iron absorption. In advanced disease, the body may contain 20 g or more of iron that is deposited mainly in parenchymal cells of the liver, pancreas, and heart. Iron may be increased 50- to 100-fold in the liver and pancreas and 5- to 25-fold in the heart. Iron deposition in the pituitary causes hypogonadotropic hypogonadism in both men and women. Tissue injury may result from disruption of iron-laden lysosomes, from lipid peroxidation of subcellular organelles by excess iron, or from stimulation of collagen synthesis by activated stellate cells.

Secondary iron overload with deposition in parenchymal cells occurs in chronic disorders of erythropoiesis, particularly in those due to defects in hemoglobin synthesis or ineffective erythropoiesis such as sideroblastic anemia and thalassemia ([Chap. 106](#)). In these disorders, the absorption of iron is increased. Moreover, these patients require blood transfusions and are also frequently treated inappropriately with iron. [PCT](#), a disorder characterized by a defect in porphyrin biosynthesis ([Chap. 346](#)), is also

sometimes associated with excessive parenchymal iron deposits. The magnitude of the iron load in PCT is usually insufficient to produce tissue damage. However, recent reports have found that many patients with PCT also have mutations in the *HFE* gene, and some have associated hepatitis C infection. Although the relationship among these disorders remains to be clarified, iron overload accentuates the inherited enzyme deficiency in PCT and should be avoided along with other agents (alcohol, estrogens, haloaromatic compounds) that may exacerbate PCT. Another cause of hepatic parenchymal iron overload is hereditary aceruloplasminemia. In this disorder, impairment of iron mobilization due to deficiency of ceruloplasmin (a ferroxidase) causes iron overload in hepatocytes.

Alcoholic subjects with end-stage chronic liver disease may have increased tissue iron stores of the degree seen in hemochromatosis. The increased iron may be caused by cell death and uptake of the released iron, as well as by hemolysis associated with spur-cell anemia ([Chap. 108](#)). Hemochromatosis in a heavy drinker can be distinguished from alcoholic liver disease by the presence of the C282Y mutation.

Excessive iron ingestion over many years rarely results in hemochromatosis. An important exception has been reported in South Africa among groups who brew fermented beverages in vessels made of iron. Hemochromatosis has on occasion been described in apparently normal subjects who have taken medicinal iron over many years, but such individuals probably have a genetic disorder.

The common denominator in all patients with hemochromatosis is *excessive amounts of iron in parenchymal tissues*. Parenteral administration of iron in the form of blood transfusions or iron preparations results predominantly in reticuloendothelial cell iron overload. This appears to lead to less tissue damage than iron loading of parenchymal cells.

PATHOLOGY

At autopsy, the enlarged nodular liver and pancreas are rusty in color. Histologically, iron is increased in amount in many organs, particularly in the liver, heart, and pancreas, and to a lesser extent in the endocrine glands. The epidermis of the skin is thin, and melanin is increased in the cells of the basal layer. Deposits of iron are present around the synovial lining cells of the joints.

In the liver of patients with hemochromatosis, parenchymal iron is in the form of ferritin and hemosiderin. In the early stages these deposits are seen in the periportal parenchymal cells, especially within lysosomes in the pericanalicular cytoplasm of the hepatocytes. This stage progresses to perilobular fibrosis and eventually to deposition of iron in bile duct epithelium, Kupffer cells, and fibrous septa. In the advanced stage, a macronodular or mixed macro- and micronodular cirrhosis develops.

CLINICAL MANIFESTATIONS

Initial symptoms include weakness, lassitude, weight loss, change in skin color, abdominal pain, loss of libido, and symptoms of diabetes mellitus. Hepatomegaly, increased pigmentation, spider angiomas, splenomegaly, arthropathy, ascites, cardiac

arrhythmias, congestive heart failure, loss of body hair, testicular atrophy, and jaundice are prominent in advanced disease.

The *liver* is usually the first organ to be affected, and hepatomegaly is present in more than 95% of symptomatic patients. Hepatic enlargement may exist in the absence of symptoms or of abnormal liver function tests. Indeed, over half of patients with symptomatic hemochromatosis have little laboratory evidence of functional impairment of the liver, in spite of hepatomegaly and fibrosis. Loss of body hair, palmar erythema, testicular atrophy, and gynecomastia are common. Manifestations of portal hypertension and esophageal varices occur less commonly than in cirrhosis from other causes. Hepatocellular carcinoma develops in about 30% of patients with cirrhosis, and it is the most common cause of death in treated patients; hence the importance of early diagnosis and therapy. Its incidence increases with age, is more common in men, and occurs almost exclusively in cirrhotic patients. Splenomegaly occurs in approximately half of symptomatic cases.

Excessive skin pigmentation is present in over 90% of symptomatic patients at the time of diagnosis. The characteristic metallic or slate gray hue is sometimes referred to as bronzing and results from increased melanin and iron in the dermis. Pigmentation usually is diffuse and generalized, but it may be more pronounced on the face, neck, extensor aspects of the lower forearms, dorsa of the hands, lower legs, genital regions, and in scars.

Diabetes mellitus occurs in about 65% of patients and is more likely to develop in those with a family history of diabetes, suggesting that direct damage to the pancreatic islets by iron deposition occurs in combination with a genetic predisposition. The management is similar to that of other forms of diabetes, although pronounced insulin resistance is more common in association with hemochromatosis. Late complications are the same as seen in other causes of diabetes mellitus.

Arthropathy develops in 25 to 50% of patients. It usually occurs after age 50, but may occur as a first manifestation, or long after therapy. The joints of the hands, especially the second and third metacarpophalangeal joints, are usually the first joints involved, a feature that helps to distinguish the chondrocalcinosis associated with hemochromatosis from the idiopathic form ([Chap. 322](#)). A progressive polyarthritis involving wrists, hips, ankles, and knees also may ensue. Acute brief attacks of synovitis may be associated with deposition of calcium pyrophosphate (chondrocalcinosis or pseudogout), mainly in the knees. Radiologic manifestations include cystic changes of the subchondral bones, loss of articular cartilage with narrowing of the joint space, diffuse demineralization, hypertrophic bone proliferation, and calcification of the synovium. The arthropathy tends to progress despite removal of iron by phlebotomy. Although the relation of these abnormalities to iron metabolism is not known, the fact that similar changes occur in other forms of iron overload suggests that iron is directly involved.

Cardiac involvement is the presenting manifestation in about 15% of patients. The most common manifestation is congestive heart failure, which occurs in about 10% of young adults with the disease, especially those with juvenile hemochromatosis. Symptoms of congestive failure may develop suddenly, with rapid progression to death if untreated. The heart is diffusely enlarged and may be misdiagnosed as idiopathic cardiomyopathy

if other overt manifestations are absent. Cardiac arrhythmias include premature supraventricular beats, paroxysmal tachyarrhythmias, atrial flutter, atrial fibrillation, and varying degrees of atrioventricular block.

Hypogonadism occurs in both sexes and may antedate other clinical features. Manifestations include loss of libido, impotence, amenorrhea, testicular atrophy, gynecomastia, and sparse body hair. These changes are primarily the result of decreased production of gonadotropins due to impairment of hypothalamic-pituitary function by iron deposition; however, primary testicular dysfunction may be seen in some cases. Adrenal insufficiency, hypothyroidism, and hypoparathyroidism may also occur.

DIAGNOSIS

The association of (1) hepatomegaly, (2) skin pigmentation, (3) diabetes mellitus, (4) heart disease, (5) arthritis, and (6) hypogonadism should suggest the diagnosis. However, a parenchymal iron overload of comparatively short duration or modest degree may exist with none or only some of these manifestations [e.g., in young subjects ([Fig. 345-1](#))]. Therefore, a high index of suspicion is needed to make the diagnosis early. This is particularly important because treatment before there is permanent organ damage can reverse the iron toxicity and restore life expectancy to normal (see below).

The history should be particularly detailed in regard to disease in other family members, alcohol ingestion, iron intake, and ingestion of large doses of ascorbic acid, which promotes iron absorption ([Chap. 75](#)). Appropriate tests should be performed to exclude iron deposition due to hematologic disease. The presence of liver, pancreatic, cardiac, and joint disease should be confirmed by physical examination, roentgenography, and standard function tests of these organs. The degree of increase in total-body iron stores should be assessed with particular attention to an increase in parenchymal iron concentration, with or without tissue damage.

The methods available for assessing parenchymal iron stores include (1) measurement of serum iron and the percent saturation of transferrin (or the unsaturated iron-binding capacity); (2) measurement of serum ferritin concentration; (3) liver biopsy with measurement of the iron concentration and calculation of the hepatic iron index ([Table 345-2](#)), (4) estimation of chelatable iron stores following the administration of deferoxamine; and (5) computed tomography (CT) and/or magnetic resonance imaging (MRI) of the liver. Each has its advantages and limitations. The serum iron level and percent saturation of transferrin are elevated early in the course, but their specificity is reduced by significant false-positive and false-negative rates. For example, serum iron concentration may be increased in patients with alcoholic liver disease without iron overload; in this situation, however, the hepatic iron index is usually not increased as in hemochromatosis ([Table 345-1](#)). In otherwise healthy persons, a fasting serum transferrin saturation greater than 50% is abnormal and suggests homozygosity for hemochromatosis.

The serum ferritin concentration is usually a good index of body iron stores, whether decreased or increased. In fact, an increase of 1 ug/L in serum ferritin level reflects an

increase of about 65 mg in body stores. In most untreated patients with hemochromatosis, the serum ferritin level is greatly increased ([Fig. 345-1](#) and [Table 345-1](#)). However, in patients with inflammation and hepatocellular necrosis, serum ferritin levels may be elevated out of proportion to body iron stores due to increased release from tissues. A repeat determination of serum ferritin should therefore be carried out after acute hepatocellular damage has subsided, e.g., in alcoholic liver disease. Ordinarily, the combined measurements of the percent transferrin saturation and serum ferritin level provide a simple and reliable screening test for hemochromatosis, including the precirrhotic phase of the disease. If either of these tests is abnormal, genetic testing for hemochromatosis should be performed ([Fig. 345-2](#)).

The role of liver biopsy in the diagnosis and management of hemochromatosis is being reassessed as a result of the widespread availability of genetic testing for the C282Y mutation. The absence of severe fibrosis can be accurately predicted in most patients using clinical and biochemical variables. Thus, there is virtually no risk of severe fibrosis in a C282Y homozygous subject with: (1) serum ferritin level less than 1000 ug/L; (2) normal serum alanine amino transaminase values; (3) no hepatomegaly; and (4) no excess alcohol intake. However, it should be emphasized that liver biopsy is the only reliable method for establishing or excluding the presence of hepatic cirrhosis, which is the critical factor determining prognosis and the risk of developing hepatocellular carcinoma. Biopsy also permits histochemical estimation of tissue iron and measurement of hepatic iron concentration. Increased density of the liver due to iron deposition can be demonstrated by [CT](#) or [MRI](#). A retrospective assessment of body iron storage is also provided by performing weekly phlebotomy and calculating the amount of iron removed before iron stores are exhausted (1 mL blood = approximately 0.5 mg iron).

SCREENING FOR HEMOCHROMATOSIS

When the diagnosis of hemochromatosis is established, it is important to counsel and screen other family members ([Chap. 68](#)). Asymptomatic as well as symptomatic family members with the disease usually have an increased saturation of transferrin and an increased serum ferritin concentration. These changes occur even before the iron stores are greatly increased ([Fig. 345-1](#)). All first-degree relatives of patients with hemochromatosis should be tested for the C282Y and H63D mutations and advised appropriately. In affected individuals, it is important to confirm or exclude the presence of cirrhosis, and begin therapy as early as possible. When children of a proband are affected, a homozygote-heterozygote mating is most likely.

The role of population screening for hemochromatosis is controversial. Hemochromatosis fulfills the criteria established by the World Health Organization for population screening, and DNA testing could, in principle, be performed along with other neonatal tests. However, because iron overload does not develop until the second, third, or fourth decades, and the degree of penetrance is still uncertain, screening by phenotypic expression is more practical at present ([Fig. 345-2](#)).

TREATMENT

The therapy of hemochromatosis involves removal of the excess body iron and

supportive treatment of damaged organs. Iron removal is best begun by weekly or twice-weekly phlebotomy of 500 mL. Although there is an initial modest decline in the volume of packed red blood cells to about 35 mL/dL, the level stabilizes after several weeks. The plasma transferrin saturation remains increased until the available iron stores are depleted. In contrast, the plasma ferritin concentration falls progressively, reflecting the gradual decrease in body iron stores. Since one 500-mL unit of blood contains 200 to 250 mg iron and about 25 g iron should be removed, weekly phlebotomy may be required for 1 or 2 years. When the transferrin saturation and ferritin level become normal, phlebotomies are performed at appropriate intervals to maintain levels within the normal range. The measurements promptly become abnormal with iron reaccumulation. Usually one phlebotomy every 3 months will suffice.

Chelating agents such as deferoxamine, when given parenterally, remove 10 to 20 mg iron per day, which is much less than that mobilized by once-weekly phlebotomy. Phlebotomy is also less expensive, more convenient, and safer for most patients. However, chelating agents are indicated when anemia or hypoproteinemia is severe enough to preclude phlebotomy. Subcutaneous infusion of deferoxamine using a portable pump is the most effective means of administration.

The management of hepatic failure, cardiac failure, and diabetes mellitus is similar to conventional therapy for these conditions. Loss of libido and change in secondary sex characteristics are partially relieved by parenteral testosterone or gonadotropin therapy ([Chap. 335](#)).

PROGNOSIS

The principal causes of death in untreated patients are cardiac failure (30%), hepatocellular failure or portal hypertension (25%), and hepatocellular carcinoma (30%).

Life expectancy is improved by removal of the excessive stores of iron and maintenance of these stores at near-normal levels. The 5-year survival rate with therapy increases from 33 to 89%. With repeated phlebotomy, the liver and spleen decrease in size, liver function improves, pigmentation of skin decreases, and cardiac failure may be reversed. Diabetes improves in about 40%, but removal of excess iron has little effect on hypogonadism or arthropathy. Hepatic fibrosis may decrease, but cirrhosis is irreversible. End-stage liver disease can be treated with orthotopic liver transplantation, but the results are suboptimal unless excess iron stores are first corrected. Hepatocellular carcinoma usually occurs as a late sequela in patients who are cirrhotic at presentation. The apparent increase in its incidence in treated patients is probably related to their increased life span. Hepatocellular carcinoma does not appear to develop if the disease is treated in the precirrhotic stage. Indeed, the life expectancy of homozygotes treated before the development of cirrhosis is normal.

The importance of family screening and early therapy cannot be emphasized too strongly. Asymptomatic subjects detected by family studies should have phlebotomy therapy if iron stores are moderately to severely increased. Assessment of iron stores at appropriate intervals is also important. With this management approach, most manifestations of the disease can be prevented.

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346. THE PORPHYRIAS - Robert J. Desnick

The porphyrias are inherited or acquired disorders of specific enzymes in the heme biosynthetic pathway ([Fig. 346-1](#)). These disorders are classified as either *hepatic* or *erythropoietic* depending on the primary site of overproduction and accumulation of the porphyrin precursor or porphyrin ([Tables 346-1](#) and [346-2](#)), although some have overlapping features. The major manifestations of the hepatic porphyrias are neurologic, including neuropathic abdominal pain, neuropathy, and mental disturbances, whereas the erythropoietic porphyrias characteristically cause cutaneous photosensitivity. The reason for neurologic involvement in the hepatic porphyrias is poorly understood. Cutaneous sensitivity to sunlight is due to the fact that excitation of excess porphyrins in the skin by long-wave ultraviolet light leads to cell damage, scarring, and deformation. Steroid hormones, drugs, and nutrition influence the production of porphyrin precursors and porphyrins, thereby precipitating or increasing the severity of some porphyrias. Thus, the porphyrias are multifactorial genetic disorders, in which environmental, physiologic, and genetic factors interact to cause disease ([Chap. 68](#)).

Because many symptoms of the porphyrias are nonspecific, diagnosis is often delayed. Laboratory testing is required to confirm or exclude the various types of porphyria ([Table 346-2](#)). Urinary δ -aminolevulinic acid (ALA) and porphobilinogen (PBG) are easily quantitated by chemical methods; the urinary porphyrin isomers can be separated and quantitated by high-performance liquid chromatography. The diagnostic profile of accumulated precursors and/or porphyrins in each disorder can also be defined by extraction and thin-layer chromatography of fecal porphyrins. However, a definite diagnosis requires demonstration of the specific enzyme deficiency or gene defect. The isolation and characterization of the cDNAs encoding the heme biosynthetic enzymes have permitted identification of the genetic basis of each porphyria. Molecular genetic analyses now make it possible to provide precise heterozygote identification and prenatal diagnoses in families with known mutations or with informative polymorphisms.

HEME BIOSYNTHESIS

The first and last three enzymes in the heme biosynthetic pathway are located in the mitochondrion, whereas the other four are in the cytosol ([Fig. 346-1](#)). The first enzyme, δ -aminolevulinic acid synthase ([ALA synthase](#)), catalyzes the condensation of glycine, activated by pyridoxal phosphate and succinyl coenzyme A, to form ALA. In the liver, this rate-limiting enzyme can be induced by a variety of drugs, steroids, and other chemicals. Distinct erythroid-specific and nonerythroid (e.g., housekeeping) forms of ALA synthase are encoded by separate genes; defects in the erythroid form cause X-linked sideroblastic anemia (XLSA).

The second enzyme, δ -aminolevulinic acid dehydratase ([ALA dehydratase](#)), catalyzes the condensation of two molecules of ALA to form [PBG](#). Four molecules of PBG condense to form the tetrapyrrole uroporphyrinogen (URO) III by a two-step process catalyzed by hydroxymethylbilane (HMB) synthase (also known as PBG deaminase or URO I synthase) and URO III synthase. HMB synthase catalyzes the head-to-tail condensation of four PBG molecules by a series of deaminations to form the linear tetrapyrrole HMB. URO synthase catalyzes the rearrangement and rapid cyclization of HMB to form the asymmetric, physiologic, octacarboxylate porphyrinogen URO III isomer.

The fifth enzyme in the pathway, [URO](#) decarboxylase, catalyzes the sequential removal of the four carboxyl groups from the acetic acid side chains of URO III to form coproporphyrinogen (COPRO) III, a tetracarboxylate porphyrinogen. This compound then enters the mitochondrion, where COPRO oxidase, the sixth enzyme, catalyzes the decarboxylation of two of the four propionic acid groups to form the two vinyl groups of protoporphyrinogen (PROTO) IX, a dicarboxylate porphyrinogen. Next, PROTO oxidase oxidizes PROTO IX to protoporphyrin IX by the removal of six hydrogen atoms. The product of the reaction is a porphyrin (oxidized form), in contrast to the preceding tetrapyrrole intermediates, which are porphyrinogens (reduced forms). Finally, ferrous iron is inserted into protoporphyrin IX to form heme, a reaction catalyzed by the eighth enzyme in the pathway, ferrochelatase (also known as heme synthetase or protoheme ferredoxin).

Each of the heme biosynthetic enzymes is encoded by a separate gene. Full-length human cDNAs for each of the enzymes, including those for both forms of [ALA](#) synthase, have been isolated and sequenced, and the chromosomal locations of the genes have been identified ([Table 346-3](#)).

REGULATION OF HEME BIOSYNTHESIS

About 85% of the heme produced in the body is synthesized in erythroid cells to provide heme for hemoglobin; most of the remainder is produced in the liver, where the biosynthetic pathway is under negative feedback control ([Chap. 104](#)). "Free" heme in the liver regulates the synthesis and mitochondrial translocation of the housekeeping form of [ALA](#) synthase. Heme represses the synthesis of the ALA synthase mRNA and interferes with the transport of the enzyme from the cytosol into mitochondria. ALA synthase is increased by many of the same chemicals that induce the cytochrome P450 enzymes in the endoplasmic reticulum of the liver. Because most of the heme in the liver is used for the synthesis of cytochrome P450 enzymes, hepatic ALA synthase and the cytochrome P450s are regulated in a coordinated fashion.

Different regulatory mechanisms control production of heme for hemoglobin. The erythroid-specific [ALA](#) synthase encoded on the X chromosome is expressed at higher levels than the hepatic enzyme, and an erythroid-specific control mechanism regulates iron transport into erythroid cells. During erythroid differentiation, the activities of the heme biosynthetic enzymes are increased.

THE HEPATIC PORPHYRIAS

The acute hepatic porphyrias are characterized by the rapid onset of neurologic manifestations. During the acute attack, individuals have markedly elevated plasma and urinary concentrations of the porphyrin precursors [ALA](#) and [PBG](#), which originate from the liver.

ALA DEHYDRATASE-DEFICIENT PORPHYRIA

This is a rare autosomal recessive disorder that has been described in only a few patients. Onset and severity of the disease are variable, presumably depending on the

amount of residual ALA dehydratase activity. Treatment and prevention of the neurologic complications are the same as for other acute porphyrias (see below).

Clinical Features The clinical presentation is variable. The first reported cases were in two unrelated German men who had clinical onset during adolescence of abdominal pain and neuropathy, resembling acute intermittent porphyria (AIP; see below). A Swedish infant presented with failure to thrive and required transfusions and parenteral nutrition. Presumably, the earlier age of onset and more severe manifestations reflect a more complete enzyme deficiency. A Belgian man developed an acute motor polyneuropathy and polycythemia at age 63. Recently a Japanese woman was described who had her first acute attack and the syndrome of inappropriate secretion of antidiuretic hormone at age 69.

Diagnosis Patients have increased urinary levels of [ALA](#) and coproporphyrin. ALA dehydratase activity in erythrocytes is <5% of normal. Because either succinylacetone (which accumulates in hereditary tyrosinemia and is structurally similar to ALA) or lead can inhibit ALA dehydratase, increase urinary excretion of ALA, and cause manifestations that resemble those of the acute porphyrias, lead intoxication and hereditary tyrosinemia (fumarylacetoacetase deficiency) should be considered in the differential diagnosis of ALA dehydratase-deficient porphyria. Immunologic studies in the reported cases demonstrated the presence of nonfunctional enzyme proteins that cross-reacted with anti-ALA dehydratase antibodies. DNA analysis revealed different missense mutations that resulted in the amino acid substitutions G133R and V275M in the infantile-onset patient, and R240W and A274T in a juvenile-onset patient.

Heterozygotes are clinically asymptomatic and do not excrete increased levels of [ALA](#), but they can be detected by demonstration of intermediate levels of erythrocyte ALA dehydratase activity or by demonstrating a specific mutation in the *ALA dehydratase* gene. Prenatal diagnosis of this disorder has not been achieved but should be possible by determination of the ALA dehydratase activity in cultured chorionic villi or amniocytes.

TREATMENT

Treatment is similar to that of [AIP](#) (see below). The severely affected infant was supported by hyperalimentation and periodic blood transfusions. Continued failure to thrive led to liver transplantation, which did not improve the hematologic manifestations.

ACUTE INTERMITTENT PORPHYRIA

This hepatic porphyria is an autosomal dominant condition resulting from the half-normal level of [HMB](#) synthase (also termed [PBG](#) deaminase) activity. The disease is widespread but is especially common in Scandinavia and perhaps Great Britain. The enzyme deficiency can be demonstrated in most heterozygous individuals, but clinical expression is highly variable. Activation of the disease is related to environmental or hormonal factors, such as drugs, diet, and steroid hormones, which can precipitate the manifestations. Attacks can be prevented by avoiding known precipitating factors.

Clinical Features Most heterozygotes remain clinically asymptomatic (latent) unless

exposed to factors that increase the production of porphyrins. Endogenous and exogenous gonadal steroids, porphyrinogenic drugs, alcohol ingestion, and low-calorie diets, usually instituted for weight loss, are common precipitating factors. [Table 346-4](#) lists the major drugs that are harmful in [AIP](#) [and also in hereditary coproporphyria (HCP) and variegate porphyria (VP)] and some drugs and anesthetic agents known to be safe. More extensive lists of drugs considered harmful or safe are available (see the bibliography), but information is incomplete for many of them. Attacks also can be provoked by infections and by surgery.

Because the neurovisceral symptoms rarely occur before puberty and are often nonspecific, a high index of suspicion is required to make the diagnosis. The disease can be disabling but is rarely fatal. Abdominal pain, the most common symptom, is usually steady and poorly localized but may be cramping. Ileus, abdominal distention, and decreased bowel sounds are common. However, increased bowel sounds and diarrhea may occur. Abdominal tenderness, fever, and leukocytosis are usually absent or mild because the symptoms are neurologic rather than inflammatory. Nausea, vomiting, constipation, tachycardia, hypertension, mental symptoms, pain in the limbs, head, neck, or chest, muscle weakness, sensory loss, dysuria, and urinary retention are characteristic. Tachycardia, hypertension, restlessness, tremors, and excess sweating are due to sympathetic overactivity.

The peripheral neuropathy is due to axonal degeneration (rather than demyelination) and primarily affects motor neurons. Significant neuropathy does not occur with all acute attacks; abdominal symptoms are usually more prominent. Motor neuropathy affects the proximal muscles initially, more often in the shoulders and arms. The course and degree of involvement are variable. Deep tendon reflexes may be normal or hyperactive but are usually decreased or absent with advanced neuropathy. Motor weakness can be asymmetric and focal and may involve cranial nerves. Sensory changes such as paresthesia and loss of sensation are less prominent. Progressive muscle weakness can lead to respiratory and bulbar paralysis and death when diagnosis and treatment are delayed. Sudden death may result from sympathetic overactivity and cardiac arrhythmia.

Mental symptoms such as anxiety, insomnia, depression, disorientation, hallucinations, and paranoia can occur in acute attacks. Seizures can be due to neurologic effects or to hyponatremia. Treatment of seizures is difficult because virtually all antiseizure drugs (except bromides) may exacerbate [AIP](#) (clonazepam may be safer than phenytoin or barbiturates). Hyponatremia results from hypothalamic involvement and inappropriate vasopressin secretion or from electrolyte depletion due to vomiting, diarrhea, poor intake, or excess renal sodium loss. Persistent hypertension and impaired renal function may occur. When an attack resolves, abdominal pain may disappear within hours, and paresis begins to improve within days and may continue to improve over several years.

Diagnosis [ALA](#) and [PBG](#) levels are increased in plasma and urine during acute attacks. Urinary PBG excretion is usually 220 to 880 $\mu\text{mol/d}$ (50 to 200 mg/d) [normal, 0 to 18 $\mu\text{mol/d}$ (0 to 4 mg/d)], and urinary ALA excretion is 150 to 760 $\mu\text{mol/d}$ (20 to 100 mg/d) [normal, 8 to 53 $\mu\text{mol/d}$ (1 to 7 mg/d)]. The excretion of these compounds generally decreases with clinical improvement, particularly after hematin infusions (see below). A normal urinary PBG level effectively excludes [AIP](#) as a cause for current symptoms.

Fecal porphyrins are usually normal or minimally increased in AIP, in contrast to [HCP](#) and [VP](#). Most asymptomatic ("latent") heterozygotes with [HMB](#) synthase deficiency have normal urinary excretion of ALA and PBG. Therefore, measurement of HMB synthase in erythrocytes is useful to confirm the diagnosis and to screen asymptomatic family members.

The enzyme deficiency is detectable in erythrocytes from most [AIP](#) heterozygotes (*classic AIP*). Note that the activity is higher in young erythrocytes and may increase into the normal range in AIP when erythropoiesis is increased due to a concurrent condition. However, patients with the rare erythroid form of AIP (*erythroid, or variant, AIP*) have normal enzyme levels in erythrocytes and deficient activity in nonerythroid tissues (see below). The erythroid and housekeeping forms of [HMB](#) synthase are encoded by a single gene, which has two promoters: one promoter generates the ubiquitously expressed housekeeping mRNA; the other promoter transcribes the erythroid-specific mRNA. Several deletions and over 150 different point mutations have been found in the coding region of the gene in unrelated AIP families ([Fig. 346-2](#)). These mutations alter the kinetic properties and/or stability of the mutant enzymes or create premature termination codons. Mutations that cause erythroid AIP variants with half-normal enzyme in nonerythroid tissues, but normal activity in erythrocytes, include point mutations in the initiation methionine codon (which prevent translation) or in the 5' donor splice site of intron 1 (which cause abnormal splicing of the HMB synthase transcript).

Heterozygotes can be identified using various polymorphic sites in the [HMB](#) synthase gene. Efforts are now under way to identify the specific mutations in the *HMB synthase* gene in all [AIP](#) families; this information will make it possible to identify all heterozygotes in affected families and to advise them to avoid the factors that cause acute attacks. The prenatal diagnosis of a fetus at risk can be made with cultured amniotic cells or chorionic villi.

TREATMENT

During acute attacks, narcotic analgesics may be required for abdominal pain, and phenothiazines are useful for nausea, vomiting, anxiety, and restlessness. Chloral hydrate can be given for insomnia, and benzodiazepines are probably safe in low doses, if a minor tranquilizer is required. Although intravenous glucose (at least 300 g/d) can be effective in acute attacks of porphyria, a more complete parenteral nutritional regimen may be beneficial if oral feeding is not possible for a prolonged period. However, intravenous heme is more effective than glucose in reducing porphyrin precursor excretion and probably leads to more rapid recovery. The response to heme therapy is reduced if therapy is delayed. Therefore, 3 to 4 mg of heme, in the form of hematin (Abbott Laboratories), heme albumin, or heme arginate (Leiras Oy, Turku, Finland), may be infused daily for 4 days beginning as soon as possible after onset of an attack. Heme arginate and heme albumin are chemically stable and are less likely than hematin to produce phlebitis or an anticoagulant effect. The rate of recovery from an acute attack depends on the degree of neuronal damage and may be rapid (1 to 2 days) with prompt therapy. Recovery from severe motor neuropathy may require months or years. Identification and avoidance of inciting factors can hasten recovery from an attack and prevent future attacks. Multiple inciting factors may contribute to a

symptomatic episode. Frequent clear-cut cyclical attacks occur in some women and can be prevented with a long-acting gonadotropin-releasing hormone analogue (this indication is not approved by the U.S. Food and Drug Administration) ([Chap. 336](#)).

PORPHYRIA CUTANEA TARDA

Porphyria cutanea tarda (PCT), the most common of the porphyrias, can be sporadic (type I) or familial (types II and III) and can also develop after exposure to halogenated aromatic hydrocarbons. Hepatic [URO](#) decarboxylase is deficient in all types of PCT. In type I PCT, URO decarboxylase activity is normal in erythrocytes. In type II PCT, an autosomal dominant disorder, the enzyme is deficient in erythrocytes and other tissues. In type III PCT, deficiency of the enzyme is limited to the liver. Deficient hepatic URO decarboxylase and a porphyrin pattern resembling PCT can be produced by exposure of normal individuals to a number of halogenated aromatic hydrocarbons. Hepatoerythropoietic porphyria (HEP) is an autosomal recessive form of porphyria that results from marked systemic deficiency of URO decarboxylase activity.

Clinical Features Cutaneous photosensitivity is the major clinical feature. Neurologic manifestations are not observed. Fluid-filled vesicles and bullae develop on sun-exposed areas such as the face, the dorsa of the hands and feet, the forearms, and the legs. The skin in these areas is friable, and minor trauma may lead to the formation of bullae. The appearance of small white plaques, termed *milia*, may precede or follow vesicle formation. Bullae and denuded skin heal slowly and are subject to infection. Other features include hypertrichosis and hyperpigmentation, especially of the face, and thickening, scarring, and calcification resembling the cutaneous changes of systemic sclerosis.

A number of factors contribute to the development of hepatic [URO](#) decarboxylase deficiency, including excess alcohol, iron, and estrogens. The importance of excess hepatic iron as a precipitating factor is underscored by the finding that the incidence of the common hemochromatosis-causing mutations, *HFE* C282Y and H63D, are increased in patients with types I and II [PCT](#) ([Chap. 345](#)). Various chemicals can also induce PCT; an epidemic of PCT occurred in eastern Turkey in the 1950s as a consequence of wheat contaminated with the fungicide hexachlorobenzene. PCT also occurs after exposure to other chemicals, including di- and trichlorophenols and 2,3,7,8-tetrachlorodibenzo-(*p*)-dioxin (TCDD, dioxin). Patients with PCT characteristically have liver damage and are at risk for hepatocellular carcinoma. These carcinomas do not produce porphyrins.

[HEP](#) resembles congenital erythropoietic porphyria (CEP) and usually presents with blistering skin lesions, hypertrichosis, scarring, and red urine in infancy or childhood.

Diagnosis Porphyrins are increased in the liver, plasma, urine, and stool. The urinary [ALA](#) level may be slightly increased, but the [PBG](#) level is normal. Urinary porphyrins consist mostly of uroporphyrin and 7-carboxylate porphyrin, with lesser amounts of coproporphyrin and 5- and 6-carboxylate porphyrins. Plasma porphyrins are also increased in a pattern that resembles that in urine. Isocoproporphyrins are increased in feces and sometimes in plasma and urine. The finding of increased isocoproporphyrins is diagnostic for a deficiency of hepatic [URO](#) decarboxylase.

Type II [PCT](#) and [HEP](#) can be distinguished by finding decreased [URO](#) decarboxylase in erythrocytes. URO decarboxylase activity in liver, erythrocytes, and cultured skin fibroblasts in type II PCT is approximately 50% of normal in affected individuals and in family members with latent disease. In HEP, the URO decarboxylase activity is markedly deficient, with typical levels of 3 to 10% of normal. Several point mutations have been identified in the coding region of the *URO decarboxylase* gene from unrelated type II PCT and HEP patients ([Fig. 346-3](#)). Excess hepatic iron contributes to development of sporadic and familial forms of PCT. As noted above, coinheritance of *HFE* mutations that cause hemochromatosis increases susceptibility to PCT-precipitating factor. In the familial forms (types II and III), iron inhibits the residual normal enzyme, so that enzymatic activity in liver is <50% of normal. In type I PCT the decreased hepatic URO decarboxylase activity is not accompanied by a decrease in the amount of enzyme protein, suggesting that the enzyme is present in an inactive form; hepatic URO decarboxylase activity gradually increases after a remission is induced by phlebotomy.

TREATMENT

Alcohol, estrogens, iron supplements, and, if possible, any drugs that may exacerbate the disease should be discontinued, but this step does not always lead to improvement. A complete response can almost always be achieved by repeated phlebotomy to reduce hepatic iron. A unit (450 mL) of blood can be removed every 1 to 2 weeks. Because iron overload is not marked in most cases, remission may occur after only five or six phlebotomies. Hemoglobin levels or hematocrits and serum ferritin should be followed closely to prevent development of iron deficiency and anemia. After remission, continued phlebotomy may not be needed even if ferritin levels return to normal. Relapses are treated by additional phlebotomy.

[PCT](#) can also be treated with chloroquine or hydroxychloroquine, both of which complex with the excess porphyrins and promote their excretion. Small doses (e.g., 125 mg chloroquine phosphate twice weekly) should be given, because standard doses can induce transient, sometimes marked increases in photosensitivity and hepatocellular damage. Hepatic imaging can diagnose or exclude complicating hepatocellular carcinoma. Treatment of PCT in patients with end-stage renal disease is facilitated by administration of erythropoietin.

HEREDITARY COPROPORPHYRIA

[HCP](#) is an autosomal dominant form of hepatic porphyria that results from half-normal levels of [COPRO](#) oxidase activity. Photosensitivity may occur. A few cases of homozygous HCP have been reported.

Clinical Features [HCP](#) is influenced by the same factors that cause attacks in [AIP](#). The disease is latent before puberty, and symptoms are more common in women. Neurovisceral symptoms and other manifestations are virtually identical to those of AIP. Photosensitivity may resemble that in [PCT](#) and [VP](#). Cutaneous lesions may begin in childhood in rare homozygous cases.

Diagnosis Coproporphyrin is markedly increased in the urine and feces in symptomatic disease and sometimes when there are no symptoms. Urinary [ALA](#) and [PBG](#) levels are increased during acute attacks but may return to normal when symptoms resolve. Although the diagnosis can be confirmed by measuring [COPRO](#) oxidase activity, these assays are not widely available and require cells other than erythrocytes.

TREATMENT

Neurologic symptoms are treated as in [AIP](#) (see above). Phlebotomy and chloroquine are ineffective when cutaneous lesions are present.

VARIEGATE PORPHYRIA

[VP](#), a hepatic porphyria that results from the deficient activity of [PROTO](#) oxidase, is transmitted in an autosomal dominant manner and can present with neurologic symptoms, photosensitivity, or both.

Clinical Features Neurovisceral signs and symptoms develop after puberty and are similar to those of [AIP](#) or [HCP](#) (see above). Attacks are provoked by the same drugs, steroids, and nutritional factors that are detrimental in AIP. Skin manifestations are more common than in HCP but usually occur apart from the neurovisceral symptoms. Because the skin lesions in [VP](#), HCP, and [PCT](#) are not distinguishable by clinical examination or biopsy, these conditions must be diagnosed by assay of porphyrins and porphyrin precursors in blood, urine, and feces.

[VP](#) is particularly common in South Africa, where 3 of every 1000 whites have the disorder. Most are descendants of a couple who emigrated from Holland to South Africa in 1688. Homozygous VP is associated with photosensitivity, neurologic symptoms, and developmental disturbances, including growth retardation, in infancy or childhood; all cases had increased erythrocyte levels of zinc protoporphyrin, a characteristic finding in all homozygous porphyrias so far described.

Dual porphyria, the simultaneous occurrence of [VP](#) and familial [PCT](#), has been documented in several kindreds. *Chester porphyria* was described in a large British family in which individuals had acute porphyric attacks and deficiency of both [PROTO](#) oxidase and [HMB](#) synthase. Photosensitivity was not observed. It is unclear whether Chester porphyria is a variant of VP or [AIP](#).

Diagnosis When [VP](#) is symptomatic, levels of fecal protoporphyrin and coproporphyrin and of urinary coproporphyrin are increased. Urinary [ALA](#) and [PBG](#) levels are increased during acute attacks. Plasma levels of porphyrins are increased, particularly when there are cutaneous lesions. VP can be distinguished rapidly from all other porphyrias by examining the fluorescence emission spectrum of porphyrins in plasma at neutral pH. This test is particularly useful for differentiating VP from [PCT](#).

Assays of [PROTO](#) oxidase activity in cultured fibroblasts or lymphocytes are not widely available. Some latent cases of [VP](#) can be diagnosed by measurement of fecal porphyrins in relatives of VP patients.

TREATMENT

Acute attacks are treated with hematin as in [AIP](#). Other than avoiding sun exposure, there are few effective measures for treating the skin lesions. b-Carotene, phlebotomy, and chloroquine are not helpful.

THE ERYTHROPOIETIC PORPHYRIAS

In the erythropoietic porphyrias, porphyrins from bone marrow erythrocytes and plasma are deposited in the skin and lead to cutaneous photosensitivity.

X-LINKED SIDEROBLASTIC ANEMIA

[XLSA](#) results from the deficient activity of the erythroid form of [ALA](#) synthase and is associated with ineffective erythropoiesis, weakness, and pallor.

Clinical Features Typically, males with [XLSA](#) develop refractory hemolytic anemia, pallor, and weakness during infancy. They have secondary hypersplenism, become iron overloaded, and can develop hemosiderosis. The severity depends on the level of residual erythroid [ALA](#) synthase activity and on the responsiveness of the specific mutation to pyridoxal 5¢-phosphate supplementation (see below). Peripheral blood smears reveal a hypochromic, microcytic anemia with striking anisocytosis, poikilocytosis, and polychromasia; the leukocytes and platelets appear normal. Hemoglobin content is reduced, and the mean corpuscular volume and mean corpuscular hemoglobin concentration are decreased. Patients with milder, late-onset disease have been reported recently.

Diagnosis Bone marrow examination reveals hypercellularity with a left shift and megaloblastic erythropoiesis with an abnormal maturation. A variety of Prussian blue-staining sideroblasts are observed. Levels of urinary porphyrin precursors and of both urinary and fecal porphyrins are normal. The level of erythroid [ALA](#) synthase is decreased in bone marrow, but this enzyme is difficult to measure in the presence of the normal ALA synthase housekeeping enzyme. Definitive diagnosis requires the demonstration of mutations in the *erythroid ALA synthase* gene.

TREATMENT

The severe anemia may respond to pyridoxine supplementation. This cofactor is essential for [ALA](#) synthase activity, and mutations in the pyridoxine binding site of the enzyme have been found in several responsive patients. Cofactor supplementation may make it possible to eliminate or reduce the frequency of transfusion. Unresponsive patients may be transfusion-dependent and require chelation therapy.

CONGENITAL ERYTHROPOIETIC PORPHYRIA

[CEP](#) is an autosomal recessive disorder, also known as *Gunther's disease*, that is due to the markedly deficient activity of [URO](#) synthase; it is associated with hemolytic anemia and cutaneous lesions. CEP is characterized by accumulation of uroporphyrin I and coproporphyrin I isomers.

Clinical Features Severe cutaneous photosensitivity begins in early infancy. The skin over sun-exposed areas is friable, and bullae and vesicles are prone to rupture and infection. Skin thickening, focal hypo- and hyperpigmentation, and hypertrichosis of the face and extremities are characteristic. Secondary infection of the cutaneous lesions can lead to disfigurement of the face and hands. Porphyrins are deposited in teeth and in bones. As a result, the teeth are reddish brown and fluoresce on exposure to long-wave ultraviolet light. Hemolysis is probably due to the marked increase in erythrocyte porphyrins and leads to splenomegaly. Adults with a milder form of the disease have been described.

Diagnosis Uroporphyrin and coproporphyrin (mostly type I isomers) accumulate in the bone marrow, erythrocytes, plasma, urine, and feces. The diagnosis should be confirmed by demonstration of markedly deficient [URO](#) synthase activity. The disease can be detected in utero by measuring porphyrins in amniotic fluid and URO synthase activity in cultured amniotic cells or chorionic villi. Molecular analyses of the mutant alleles from over 20 unrelated patients have revealed the presence of gene rearrangements, an mRNA processing defect, and several point mutations that cause amino acid substitutions.

TREATMENT

The transfusion of sufficient blood to suppress erythropoiesis is effective but results in iron overload. Splenectomy may reduce hemolysis and decrease transfusion requirements. Protection from sunlight and from minor skin trauma is important. b-Carotene may be of some value. Complicating bacterial infections should be treated promptly. Recently, bone marrow transplantation has proven effective in several transfusion-dependent children, providing the rationale for stem-cell gene therapy.

ERYTHROPOIETIC PROTOPORPHYRIA

Erythropoietic protoporphyria (EPP) is an autosomal dominant disorder due to the partial deficiency of ferrochelatase activity. Protoporphyrin accumulates in erythroid cells and plasma and is excreted in bile and feces. EPP is the most common erythropoietic porphyria and, after [PCT](#), the second most common porphyria.

Clinical Features Skin photosensitivity usually begins in childhood. The skin manifestations differ from those of other porphyrias. Vesicular lesions are uncommon. Redness, swelling, burning, and itching can develop within minutes of sun exposure and resemble angioedema. Symptoms may seem out of proportion to the visible skin lesions. Sparse vesicles and bullae occur in 10% of cases. Chronic skin changes may include lichenification, leathery pseudovesicles, labial grooving, and nail changes. Severe scarring is rare, as are pigment changes, friability, and hirsutism.

The primary source of excess protoporphyrin is the bone marrow reticulocyte. Erythrocyte protoporphyrin is free (not complexed with zinc) and is mostly bound to hemoglobin. In plasma, protoporphyrin is bound to albumin. Hemolysis and anemia are usually absent or mild.

Liver function is usually normal, but in some patients accumulation of protoporphyrin causes chronic liver disease that can progress to liver failure and death. The hepatic complications are often preceded by increasing levels of erythrocyte and plasma protoporphyrin and probably result, in part, from protoporphyrin accumulation in the liver. Protoporphyrin is insoluble, forms crystalline structures in liver cells, and can decrease hepatic bile flow. Gallstones composed at least in part of protoporphyrin occur in some patients.

Some obligate heterozygotes are asymptomatic and have little or no increase in erythrocyte protoporphyrin. Thus there is phenotypic variation in this disease.

Diagnosis Protoporphyrin levels are increased in bone marrow, circulating erythrocytes, plasma, bile, and feces. Urinary levels of porphyrin and porphyrin precursors are normal. Ferrochelatase activity in cultured lymphocytes or fibroblasts is decreased.

TREATMENT

Oral β -carotene (120 to 180 mg/d) improves tolerance to sunlight in many patients. The dosage may need to be adjusted to maintain serum carotene levels in the recommended range of 10 to 15 $\mu\text{mol/L}$ (600 to 800 $\mu\text{g/dL}$). Mild skin discoloration due to carotenemia is the only significant side effect. The beneficial effects of β -carotene may involve quenching of singlet oxygen or free radicals. Unfortunately, this drug is less effective in other forms of porphyria associated with photosensitivity.

Treatment of hepatic complications is difficult. However, cholestyramine and other porphyrin absorbents such as activated charcoal may interrupt the enterohepatic circulation of protoporphyrin and promote its fecal excretion, leading to some improvement. Splenectomy may be helpful when the disease is accompanied by hemolysis and significant splenomegaly. Caloric restriction and drugs or hormones that may induce the heme pathway or impair hepatic excretory function should be avoided. Iron deficiency should be prevented or treated. Transfusions or intravenous heme therapy may suppress erythroid and hepatic protoporphyrin production and are sometimes beneficial. Liver transplantation has been carried out in some patients with severe liver complications.

(Bibliography omitted in Palm version)

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347. DISORDERS OF PURINE AND PYRIMIDINE METABOLISM - Robert L. Wortmann

Purines and pyrimidines are the bases that, when linked to sugars (ribose or deoxyribose) and phosphate groups, create the nucleic acids that comprise the building blocks of RNA and DNA. The main purine bases are adenine and guanine; the pyrimidine bases include cytosine, thymine, and uracil. In addition, purines participate in diverse cellular functions, including intracellular energy metabolism (e.g., ATP), cell signaling pathways (e.g., GTP), and intercellular communication (e.g., adenosine). The nucleotides therefore serve fundamental roles in the replication of genetic material, gene transcription, protein synthesis, and cellular metabolism. Disorders that involve abnormalities of nucleotide metabolism range from relatively common diseases such as hyperuricemia and gout, in which there is increased production or impaired excretion of a metabolic end product of purine metabolism (uric acid), to rare enzyme deficiencies that affect purine and pyrimidine synthesis or degradation. Understanding these biochemical pathways has led, in some instances, to the development of specific forms of treatment, such as the use of allopurinol to reduce uric acid production.

URIC ACID METABOLISM

Uric acid is the final breakdown product of purine degradation in humans. It is a weak acid with pK_a s of 5.75 and 10.3. Urates, the ionized forms of uric acid, predominate in plasma extracellular fluid and synovial fluid, with approximately 98% existing as monosodium urate at pH 7.4. Monosodium urate is easily dialyzed from plasma. Binding of urate to plasma proteins has little physiologic significance.

Plasma is saturated with monosodium urate at a concentration of 415 $\mu\text{mol/L}$ (6.8 mg/dL) at 37°C. At higher concentrations, plasma is therefore supersaturated, creating the potential for urate crystal precipitation. However, precipitation sometimes does not occur even at plasma urate concentrations as high as 4800 $\mu\text{mol/L}$ (80 mg/dL), perhaps because of the presence of solubilizing substances in plasma.

Uric acid is more soluble in urine than in water, possibly because of the presence of urea, proteins, and mucopolysaccharides. The pH of urine greatly influences its solubility. At pH 5.0, urine is saturated with uric acid at concentrations ranging from 360 to 900 $\mu\text{mol/L}$ (6 to 15 mg/dL). At pH 7.0, saturation is reached at concentrations between 9480 and 12,000 $\mu\text{mol/L}$ (158 and 200 mg/dL). Ionized forms of uric acid in urine include mono- and disodium, potassium, ammonium, and calcium urates.

Although purine nucleotides are synthesized and degraded in all tissues, urate is produced only in tissues that contain xanthine oxidase, primarily the liver and small intestine. The amount of urate in the body is the net result of the amount produced and the amount excreted ([Fig. 347-1](#)). Urate production varies with the purine content of the diet and the rates of purine biosynthesis, degradation, and salvage. Normally, two-thirds to three-fourths of urate is excreted by the kidneys, and most of the remainder is eliminated through the intestines. A four-component model describes the renal handling of uric acid in humans and includes: (1) glomerular filtration, (2) tubular reabsorption, (3) secretion, and (4) postsecretory reabsorption ([Fig. 347-2](#)). Approximately 8 to 12% of urate filtered by the glomeruli is excreted in the urine as uric acid. After filtration, 98 to

100% of the urate is reabsorbed; about half the reabsorbed urate is secreted back into the proximal tubule, and about 40% of that is again reabsorbed.

Serum urate levels vary with age and sex. Most children have serum urate concentrations of 180 to 240 $\mu\text{mol/L}$ (3.0 to 4.0 mg/dL). Levels begin to rise during puberty in males but remain low in females until menopause. Although the cause of this gender variation is not completely understood, it is due in part to a higher excretion of urate in females. Mean serum urate values of adult men and premenopausal women are 415 and 360 $\mu\text{mol/L}$ (6.8 and 6.0 mg/dL), respectively. After menopause, values for women increase to approximate those of men. In adulthood, concentrations rise steadily over time and vary with height, body weight, blood pressure, renal function, and alcohol intake.

HYPERURICEMIA

Hyperuricemia can result from increased production or decreased excretion of uric acid or from a combination of the two processes. When sustained hyperuricemia exists, plasma and extracellular fluids are supersaturated with respect to urate, and total body urate is increased. Sustained hyperuricemia predisposes some individuals to develop clinical manifestations including gouty arthritis ([Chap. 322](#)) and renal dysfunction (see below).

Hyperuricemia may be defined as a plasma (or serum) urate concentration $>420 \mu\text{mol/L}$ (7.0 mg/dL). This definition is based on physicochemical, epidemiologic, and disease-related criteria. Physicochemically, hyperuricemia is the concentration of urate in the blood that exceeds the solubility limits of monosodium urate in plasma, 415 $\mu\text{mol/L}$ (6.8 mg/dL). In epidemiologic studies, hyperuricemia is defined as the mean plus 2 standard deviations of values determined from a randomly selected healthy population. When measured in unselected individuals, 95% have serum urate concentrations $<420 \mu\text{mol/L}$ (7.0 mg/dL). Finally, hyperuricemia can be defined in relation to the risk of disease. The risk of developing gouty arthritis or urolithiasis increases with urate levels $>420 \mu\text{mol/L}$ (7.0 mg/dL) and escalates in proportion to the degree of elevation. Hyperuricemia is present in between 2.0 and 13.2% of ambulatory adults and somewhat more frequently in hospitalized individuals.

CAUSES OF HYPERURICEMIA

Hyperuricemia may be classified as primary or secondary depending on whether the cause is innate or is the result of an acquired disorder ([Table 347-1](#)). However, it is more useful to classify hyperuricemia in relation to the underlying pathophysiology, i.e., whether it results from increased production, decreased excretion, or a combination of the two ([Fig. 347-1](#), [Table 347-2](#)).

Increased Urate Production Diet provides an exogenous source of purines and, accordingly, contributes to the serum urate in proportion to its purine content. Strict restriction of purine intake reduces the mean serum urate level by about 60 $\mu\text{mol/L}$ (1.0 mg/dL) and urinary uric acid excretion by approximately 1.2 mmol/d (200 mg/d). Because about 50% of ingested RNA purine and 25% of ingested DNA purine appear in the urine as uric acid, foods high in nucleic acid content have a significant effect on the

serum urate level. Such foods include liver, "sweetbreads" (i.e., thymus and pancreas), kidney, and anchovy.

Endogenous sources of purine production also influence the serum urate level ([Fig. 347-3](#)). De novo purine biosynthesis, the formation of a purine ring from nonring structures, is an 11-step process that results in formation of inosine monophosphate (IMP). The first step combines phosphoribosylpyrophosphate (PRPP) and glutamine and is catalyzed by amidophosphoribosyltransferase (amidoPRT). The rates of purine biosynthesis and urate production are determined, for the most part, by this enzyme. AmidoPRT is regulated by the substrate PRPP, which drives the reaction forward, and by the end products of biosynthesis (IMP and other ribonucleotides), which provide feedback inhibition. A secondary regulatory pathway is the salvage of purine bases by hypoxanthine phosphoribosyltransferase (HPRT). HPRT catalyzes the combination of the purine bases hypoxanthine and guanine with PRPP to form the respective ribonucleotides IMP and guanosine monophosphate (GMP). Increased salvage activity thus retards de novo synthesis by reducing PRPP levels and increasing concentrations of inhibitory ribonucleotides.

Serum urate levels are closely coupled to the rates of de novo purine biosynthesis, which is driven in part by the level of [PRPP](#), as evidenced by two inborn errors of purine metabolism. Both increased PRPP synthetase activity and [HPRT](#) deficiency are associated with overproduction of purines, hyperuricemia, and hyperuricaciduria (see below for clinical descriptions). An X-linked disorder that causes an increase in activity of the enzyme PRPP synthetase leads to increased PRPP production and accelerated de novo biosynthesis. PRPP is a substrate and allosteric activator of [amidoPRT](#), the first enzyme in the de novo pathway. HPRT deficiency is also X-linked and enhances urate biosynthesis in two ways. PRPP is accumulated as a result of decreased utilization in the salvage pathway and, in turn, provides increased substrate for amidoPRT and de novo biosynthesis. In addition, decreased formation of the nucleoside monophosphates, [IMP](#) and [GMP](#), via the salvage pathway impairs feedback inhibition on amidoPRT, further enhancing de novo biosynthesis.

Accelerated purine nucleotide degradation can also cause hyperuricemia, i.e., with conditions of rapid cell turnover, proliferation, or cell death, as in leukemic blast crises, cytotoxic therapy for malignancy, hemolysis, or rhabdomyolysis. Nucleic acids released from cells are hydrolyzed by the sequential activities of nucleases and phosphodiesterases, forming nucleoside monophosphates, which in turn are degraded to nucleosides, bases, and urate. Hyperuricemia can result from excessive degradation of skeletal muscle ATP after strenuous physical exercise or status epilepticus and in glycogen storage diseases types III, V, and VII ([Chap. 350](#)). The hyperuricemia of myocardial infarction, smoke inhalation, and acute respiratory failure may also be related to accelerated breakdown of ATP.

Decreased Uric Acid Excretion Over 90% of individuals with sustained hyperuricemia have a defect in the renal handling of uric acid. In hyperuricemia with gout the renal defect is evidenced by a lower than normal ratio of urate clearance to glomerular filtration rate (or urate to insulin clearance rate) over a wide range of filtered loads. As a result, gouty individuals excrete approximately 40% less uric acid than nongouty individuals for any given plasma urate concentration. Uric acid excretion increases in

gouty and nongouty individuals when plasma urate levels are raised by purine ingestion or infusion, but in those with gout, plasma urate concentrations must be 60 to 120 $\mu\text{mol/L}$ (1 to 2 mg/dL) higher than normal to achieve equivalent uric acid excretion rates.

Altered uric acid excretion could theoretically result from decreased glomerular filtration, decreased tubular secretion, or enhanced tubular reabsorption. Decreased urate filtration does not appear to cause primary hyperuricemia but does contribute to the hyperuricemia of renal insufficiency. Although hyperuricemia is invariably present in chronic renal disease, the correlation between serum creatinine, urea nitrogen, and urate concentration is poor. Uric acid excretion per unit of glomerular filtration rate increases progressively with chronic renal insufficiency, but tubular secretory capacity tends to be preserved, tubular reabsorptive capacity is reduced, and extrarenal clearance of uric acid increases as renal damage becomes more severe.

Decreased tubular secretion of urate causes the secondary hyperuricemia of acidosis. Diabetic ketoacidosis, starvation, ethanol intoxication, lactic acidosis, and salicylate intoxication are accompanied by accumulations of organic acids (β -hydroxybutyrate, acetoacetate, lactate, or salicylates) that compete with urate for tubular secretion. Hyperuricemia may be due to enhanced reabsorption of uric acid distal to the site of secretion. This mechanism is known to be responsible for the hyperuricemia of extracellular volume depletion that occurs with diabetes insipidus or diuretic therapy.

Combined Mechanisms Both increased urate production and decreased uric acid excretion may contribute to hyperuricemia. Individuals with a deficiency of glucose-6-phosphatase, the enzyme that hydrolyzes glucose-6-phosphate to glucose, are hyperuricemic from infancy and develop gout early in life ([Chap. 350](#)). Increased urate production results from accelerated ATP degradation during fasting or hypoglycemia. In addition, the lower levels of nucleoside monophosphates decrease feedback inhibition of [amidoPRT](#), thereby accelerating de novo biosynthesis. Glucose-6-phosphatase-deficient individuals may also develop hyperlacticacidemia, which blocks uric acid excretion by decreasing tubular secretion.

Patients with hereditary fructose intolerance caused by fructose-1-phosphate aldolase deficiency also develop hyperuricemia by both mechanisms. In homozygotes, vomiting and hypoglycemia after fructose ingestion can lead to hepatic failure and proximal renal tubular dysfunction. Ingestion of fructose, the substrate for the enzyme, causes accumulation of fructose-1-phosphate. This action results in ATP depletion, accelerated purine nucleotide catabolism, and hyperuricemia. Both lactic acidosis and renal tubular acidosis contribute to urate retention. Heterozygous carriers develop hyperuricemia, and perhaps one-third develop gout. The heterozygous state has a prevalence of 0.5 to 1.5%, suggesting that fructose-1-phosphate aldolase deficiency may be a relatively common cause of familial gout.

Alcohol also promotes hyperuricemia by both mechanisms. Excessive alcohol consumption accelerates hepatic breakdown of ATP and increases urate production. Alcohol consumption can also induce hyperlacticacidemia, which blocks uric acid secretion. The higher purine content in some alcoholic beverages such as beer may also be a factor.

EVALUATION OF HYPERURICEMIA

Hyperuricemia does not necessarily represent a disease, nor is it a specific indication for therapy. Rather, the finding of hyperuricemia is an indication to determine its cause. The decision to treat depends on the cause and the potential consequences of the hyperuricemia in each individual.

Quantification of uric acid excretion can be used to determine whether hyperuricemia is caused by overproduction or decreased excretion. On a purine-free diet, men with normal renal function excrete <3.6 mmol/d (600 mg/d). Thus, the hyperuricemia of individuals who excrete uric acid above this level while on a purine-free diet is due to purine overproduction, whereas it is due to decreased excretion in those who excrete lower amounts on the purine-free diet. If the assessment is performed while the patient is on a regular diet, the level of 4.2 mmol/d (800 mg/d) can be used as the discriminating value. With renal insufficiency, less urate is filtered in the glomeruli and less uric acid appears in the urine. Consequently, a lower 24-h urinary uric acid value in the presence of renal insufficiency does not necessarily rule out urate overproduction, but an elevated value provides strong evidence of urate overproduction. Spuriously high values can occur if a uricosuric agent is being taken at the time of urine collection. Glucocorticoids, ascorbic acid, salicylates in doses >2 g/d, and other agents that promote urate excretion interfere with the interpretation of results.

Assessment of the ratio of uric acid to creatinine (or the ratio of uric acid clearance to creatinine clearance) in spot or random urine samples is not a reliable method to screen for urate overproduction. However, this is a useful tool for evaluating individuals with acute renal failure suspected of having acute uric acid nephropathy (see below).

Pyrazinamide, which has a suppressive action on tubular secretion, can be used to investigate presecretory reabsorption of uric acid. Probenecid, an agent that inhibits postsecretory reabsorption, can be used to evaluate tubular secretion and postsecretory reabsorption.

COMPLICATIONS OF HYPERURICEMIA

The most recognized complication of hyperuricemia is *gouty arthritis*. In the general population the prevalence of hyperuricemia ranges between 2.0 and 13.2%, and the prevalence of gout is between 1.3 and 3.7%. The higher the serum urate level, the more likely an individual is to develop gout. In one study, the incidence of gout was 4.9% for individuals with serum urate concentrations >540 $\mu\text{mol/L}$ (9.0 mg/dL) compared with 0.5% for those with values between 415 and 535 $\mu\text{mol/L}$ (7.0 and 8.9 mg/dL). The complications of gout correlate with both the duration and severity of hyperuricemia. **For further discussion of gout, see Chap. 322.*

Hyperuricemia also causes several renal problems: (1) nephrolithiasis; (2) urate nephropathy, a rare cause of renal insufficiency attributed to monosodium urate crystal deposition in the renal interstitium; and (3) uric acid nephropathy, a reversible cause of acute renal failure resulting from deposition of large amounts of uric acid crystals in the renal collecting ducts, pelvis, and ureters.

Nephrolithiasis Uric acid nephrolithiasis occurs most commonly, but not exclusively, in individuals with gout. In gout, the prevalence of nephrolithiasis correlates with the serum and urinary uric acid levels, reaching approximately 50% with serum urate levels of 770 $\mu\text{mol/L}$ (13 mg/dL) or urinary uric acid excretion $>6.5 \text{ mmol/d}$ (1100 mg/d).

Uric acid stones can develop in individuals with no evidence of arthritis, only 20% of whom are hyperuricemic. Uric acid can also play a role in other types of kidney stones. Some nongouty individuals with calcium oxalate or calcium phosphate stones have hyperuricemia or hyperuricaciduria. Uric acid may act as a nidus on which calcium oxalate can precipitate or lower the formation product for calcium oxalate crystallization.

Urate Nephropathy Urate nephropathy, sometimes referred to as *urate nephrosis*, is a late manifestation of severe gout and is characterized histologically by deposits of monosodium urate crystals surrounded by a giant cell inflammatory reaction in the medullary interstitium and pyramids. The disorder is now rare and cannot be diagnosed in the absence of gouty arthritis. The lesions may be clinically silent or cause proteinuria, hypertension, and renal insufficiency.

Uric Acid Nephropathy This reversible cause of acute renal failure is due to precipitation of uric acid in renal tubules and collecting ducts that causes obstruction to urine flow. Uric acid nephropathy develops following sudden urate overproduction and marked hyperuricaciduria. Factors that favor uric acid crystal formation include dehydration and acidosis. This form of acute renal failure occurs most often during an aggressive "blastic" phase of leukemia or lymphoma prior to or coincident with cytolytic therapy but has also been observed in individuals with other neoplasms, following epileptic seizures, and after vigorous exercise with heat stress. Autopsy studies have demonstrated intraluminal precipitates of uric acid, dilated proximal tubules, and normal glomeruli. The initial pathogenic events are believed to include obstruction of collecting ducts with uric acid and obstruction of distal renal vasculature.

If recognized, uric acid nephropathy is potentially reversible. Appropriate therapy has reduced the mortality from about 50% to practically nil. Serum levels cannot be relied on for diagnosis because this condition has developed in the presence of urate concentrations varying from 720 to 4800 $\mu\text{mol/L}$ (12 to 80 mg/dL). The distinctive feature is the urinary uric acid concentration. In most forms of acute renal failure with decreased urine output, urinary uric acid content is either normal or reduced, and the ratio of uric acid to creatinine is <1 . In acute uric acid nephropathy the ratio of uric acid to creatinine in a random urine sample or 24-h specimen is >1 , and a value that high is essentially diagnostic.

TREATMENT

Asymptomatic Hyperuricemia Hyperuricemia is present in approximately 5% of the population and in up to 25% of hospitalized individuals. The vast majority are asymptomatic with regard to their hyperuricemia and are at no clinical risk because of it. Elevated serum urate concentrations have been associated with insulin resistance, obesity, hypertension, dyslipidemia (sometimes referred to as syndrome X), and atherosclerotic disease. However, urate does not appear to have a causal role in the development of coronary heart disease or death from cardiovascular disease. In the

past, the association of hyperuricemia with cardiovascular disease and renal failure led to the use of urate-lowering agents for people with asymptomatic hyperuricemia. This practice is no longer recommended with the exception of individuals receiving cytolytic therapy for neoplastic disease, in which treatment is given in an effort to prevent uric acid nephropathy.

Hyperuricemic individuals are at risk to develop gouty arthritis, especially those with higher serum urate levels. However, treatment of asymptomatic hyperuricemia to prevent the first attack of gouty arthritis is not indicated because most hyperuricemic people never develop gout. Furthermore, neither structural kidney damage nor tophi are identifiable before the first attack. Reduced renal function cannot be attributed to asymptomatic hyperuricemia, and treatment of asymptomatic hyperuricemia does not alter the progression of renal dysfunction in patients with renal disease. Although nephrolithiasis is common in gouty patients, and a number of individuals with nephrolithiasis are hyperuricemic, increased risk of stone formation in people with asymptomatic hyperuricemia is not established.

Thus, because treatment with antihyperuricemic agents entails inconvenience, cost, and potential toxicity, routine treatment of asymptomatic hyperuricemia cannot be justified other than for prevention of acute uric acid nephropathy. In addition, routine screening for asymptomatic hyperuricemia is not recommended. If hyperuricemia is diagnosed, however, the cause should be determined. Causal factors should be corrected if the condition is secondary, and associated problems such as hypertension, hypercholesterolemia, diabetes mellitus, and obesity should be treated.

Symptomatic Hyperuricemia (See [Chap. 322](#) for treatment of gout)

Nephrolithiasis Antihyperuricemic therapy is recommended for the individual who has both gouty arthritis and either uric acid- or calcium-containing stones, both of which may occur in association with hyperuricaciduria. Regardless of the nature of the calculi, fluid ingestion should be sufficient to produce a daily urine volume >2 L. Alkalinization of the urine with sodium bicarbonate or acetazolamide may be justified to increase the solubility of uric acid. Specific treatment of uric acid calculi requires reducing the urine uric acid concentration with allopurinol. Allopurinol administration decreases the serum urate concentration and the urinary excretion of uric acid in the first 24 h, with a maximum reduction occurring within 2 weeks. The average effective dose of allopurinol is 300 mg/d. Allopurinol can be given once a day because of the long half-life (18 h) of its active metabolite oxypurinol. The drug is effective in patients with renal insufficiency, but the dose should be reduced. Allopurinol is also useful in reducing the recurrence of calcium oxalate stones in gouty patients and in nongouty individuals with hyperuricemia or hyperuricaciduria. Potassium citrate (30 to 80 mmol/d orally in divided doses) is an alternative therapy for patients with uric acid stones alone or mixed calcium/uric acid stones. Allopurinol is also indicated for the treatment of 2,8-dihydroxyadenine kidney stones.

Uric Acid Nephropathy Uric acid nephropathy is often preventable, and immediate, appropriate therapy has greatly reduced the mortality rate. Vigorous intravenous hydration and diuresis with furosemide dilute the uric acid in the tubules and promote urine flow to ³100 mL/h. The administration of acetazolamide, 240 to 500 mg every 6 to

8 h, and sodium bicarbonate, 89 mmol/L, intravenously enhances urine alkalinity and thereby solubilizes more uric acid. It is important to ensure that the urine pH remains >7.0 and to watch for circulatory overload. In addition, antihyperuricemic therapy in the form of allopurinol in a single dose of 8 mg/kg is administered to reduce the amount of urate that reaches the kidney. If renal insufficiency persists, subsequent daily doses should be reduced to 100 to 200 mg because oxypurinol, the active metabolite of allopurinol, accumulates in renal failure. Despite these measures, hemodialysis may be required.

HYPOURICEMIA

Hypouricemia, defined as a serum urate concentration <120 $\mu\text{mol/L}$ (2.0 mg/dL) can result from decreased production of urate, increased excretion of uric acid, or a combination of both mechanisms. It occurs in <0.2% of the general population and <0.8% of hospitalized individuals. Hypouricemia causes no symptoms or pathology and therefore requires no therapy. It is, however, a sign of potential pathology, and its cause should be determined.

Most hypouricemia results from increased renal uric acid excretion. The finding of normal amounts of uric acid in a 24-h urine collection in an individual with hypouricemia is evidence for a renal cause. Medications with uricosuric properties ([Table 347-3](#)) include aspirin (at doses >2.0 g/d), x-ray contrast materials, and glycerylguaiacolate. Total parenteral hyperalimentation can also cause hypouricemia, possibly a result of the high glycine content of the infusion formula. Other causes of increased urate clearance include conditions such as neoplastic disease, hepatic cirrhosis, diabetes mellitus, and inappropriate secretion of vasopressin; defects in renal tubular transport such as primary Fanconi syndrome and Fanconi syndromes caused by Wilson's disease, cystinosis, multiple myeloma, and heavy metal toxicity; and isolated congenital defects in the bidirectional transport of uric acid.

Hypouricemia from decreased production of urate is accompanied by very low urinary uric acid levels. Accumulation of other purine nucleosides and bases may occur depending on the specific defect. Individuals treated with allopurinol and some patients with neoplastic disease or severe hepatic dysfunction are hypouricemic and excrete increased quantities of hypoxanthine and xanthine in the urine. Xanthine oxidase deficiency can be inherited or acquired. Inherited forms include isolated xanthine oxidase deficiency and combined xanthine oxidase and sulfite oxidase deficiencies. Both cause hypouricemia and xanthinuria. Affected individuals excrete essentially no uric acid and may develop xanthine nephrolithiasis. Individuals with purine nucleoside phosphorylase deficiency, an inborn error of metabolism causing T cell-deficient immune dysfunction, are hypouricemic and excrete increased quantities of guanosine, deoxyguanosine, inosine, and deoxyinosine in the urine.

INBORN ERRORS OF PURINE METABOLISM (See also [Table 347-4](#), [Fig. 347-3](#))

HPRT DEFICIENCY

A complete deficiency of HPRT, the Lesch-Nyhan syndrome, is characterized by hyperuricemia, self-mutilative behavior, choreoathetosis, spasticity, and mental

retardation. A partial deficiency of HPRT, the Kelley-Seegmiller syndrome, is associated with hyperuricemia but no central nervous system manifestations. In both disorders, the hyperuricemia results from urate overproduction and can cause uric acid crystalluria, nephrolithiasis, obstructive uropathy, and gouty arthritis. Early diagnosis and appropriate therapy with allopurinol can prevent or eliminate all the problems attributable to hyperuricemia but have no effect on the behavioral or neurologic abnormalities.

[HPRT](#) catalyzes the reaction that combines [PRPP](#) and the purine bases hypoxanthine and guanine to form the respective nucleoside monophosphate [IMP](#) or [GMP](#) and pyrophosphate. The enzyme is encoded by a single gene located on the X chromosome in region q26-q27. Consequently, affected males are hemizygous for the trait and inherit the mutant allele from their asymptomatic mother, who is a carrier, or are the result of spontaneous gene mutations. The deficiency state is generally the result of point mutations, small deletions or insertions, or endoduplication of exons rather than major gene alterations.

INCREASED PRPP SYNTHETASE ACTIVITY

Cells from individuals with increased [PRPP](#) synthetase activity contain elevated levels of PRPP. The high substrate content drives de novo purine synthesis, causing overproduction of uric acid. Like the [HPRT](#) deficiency states, PRPP synthetase overactivity is X-linked and results in gouty arthritis and uric acid nephrolithiasis. Nerve deafness occurs in some families.

ADENINE PHOSPHORIBOSYLTRANSFERASE (APRT) DEFICIENCY

Individuals with a deficiency of APRT develop kidney stones composed of 2,8-dihydroxyadenine. APRT catalyzes the conversion of adenine to adenosine monophosphate (AMP). In the absence of APRT, adenine is converted by xanthine oxidase to 2,8-dihydroxyadenine, which is insoluble in urine. Reports of 2,8-dihydroxyadenine stones are rare, most likely because of its chemical similarity to uric acid. Analysis by x-ray powder diffraction is necessary for correct identification. Because this technique is rarely employed, many 2,8-dihydroxyadenine stones are incorrectly called uric acid. The consequence of this misidentification is not deleterious, as allopurinol therapy is the correct treatment for each type of stone.

[APRT](#) deficiency is inherited as an autosomal recessive trait. Caucasians with the disorder have a complete deficiency (type I), whereas Japanese subjects have some measurable enzyme activity (type II). Expression of the defect is similar in the two populations, as is the frequency of the heterozygous state (0.4 to 1.1 per 100).

HEREDITARY XANTHINURIA

A deficiency of xanthine oxidase causes all purine in the urine to occur in the form of hypoxanthine and xanthine. About two-thirds of deficient individuals are asymptomatic. The remainder develop kidney stones composed of xanthine. A very small number of symptomatic individuals also have myopathy or recurrent polyarteritis. Xanthinuria appears to be inherited in an autosomal recessive pattern.

In a second form of inherited xanthinuria, the deficiency of xanthine oxidase is associated with a deficiency of sulfite oxidase. Neurologic symptoms attributable to the sulfite oxidase deficiency predominate over those of xanthinuria in individuals with the combined deficiency.

MYOADENYLATE DEAMINASE DEFICIENCY

Adenylate deaminase ([AMP](#)deaminase) catalyzes the conversion of AMP to [IMP](#) with the release of ammonia and is an integral component of the purine nucleotide cycle, which plays an important role in skeletal muscle energy metabolism. Both primary (inherited) and secondary (acquired) forms of myoadenylate deaminase deficiency have been described. Myoadenylate deaminase is the only activity affected in the inherited form, whereas other muscle enzymes (creatine kinase and myokinase) are also decreased in the acquired deficiencies. In contrast, mRNA abundance is low in muscle from patients with acquired deficiencies, suggesting a different molecular basis for this form.

The primary form is inherited as an autosomal recessive trait. Clinically, this form does not appear to cause disease, and most individuals with this defect may be asymptomatic. Another explanation for the myopathy should be sought in symptomatic patients with this deficiency. The acquired deficiency occurs in association with a wide variety of neuromuscular disease, including muscular dystrophies, neuropathies, inflammatory myopathies, and collagen vascular diseases.

ADENYLOSUCCINATE LYASE DEFICIENCY

Adenylosuccinate lyase participates in the synthesis of purine nucleotides in two ways. It catalyzes the conversion of succinylaminoimidazole carboxamide ribotide (SAICAR) to aminoimidazole carboxamide ribotide (AICAR) in the de novo pathway and in the conversion of [AMP](#)succinate to AMP in the purine nucleotide cycle. Deficiency of this enzyme is due to an autosomal recessive trait and causes profound psychomotor retardation, seizures, and other movement disorders. All individuals with this deficiency are mentally retarded, and most are autistic.

ADENOSINE DEAMINASE DEFICIENCY AND PURINE NUCLEOSIDE PHOSPHORYLASE DEFICIENCY See [Chap. 308](#).

PYRIMIDINE DISORDERS

The pyrimidine, cytidine, is found in both DNA and RNA; it is a complementary base pair for guanine. Thymidine is found only in DNA where it is paired with adenine. Uridine is found only in RNA and can pair with either adenine or guanine in RNA secondary structures. Pyrimidines can be synthesized by a de novo pathway ([Fig. 347-4](#)) or reused in a salvage pathway. More than 25 different enzymes are involved in pyrimidine synthesis, salvage, and degradation pathways. Nonetheless, disorders of pyrimidine metabolism are rare. They are more difficult to recognize than purine disorders because of heterogeneous phenotypes and the absence of readily detected biochemical markers. Three disorders of pyrimidine metabolism are discussed below.

OROTIC ACIDURIA

Hereditary orotic aciduria is an autosomal recessive disorder caused by mutations in a bifunctional enzyme, uridine-5¢-monophosphate (UMP) synthase, which converts orotic acid to UMP in the de novo synthesis pathway ([Fig. 347-4](#)). The same protein encodes two distinct enzymatic activities. The disorder is characterized by hypochromic megaloblastic anemia that is unresponsive to vitamin B₁₂ and folic acid, growth retardation, and neurologic abnormalities. Increased excretion of orotic acid causes crystalluria and obstructive uropathy. Replacement of uridine (100 to 200 mg/kg per day) corrects the anemia, reduces orotic acid excretion, and improves the other sequelae of the disorder.

PYRIMIDINE 5¢-NUCLEOTIDASE DEFICIENCY

Pyrimidine 5¢-nucleotidase catalyzes the removal of the phosphate group from pyrimidine ribonucleoside monophosphates (cytidine-5¢-monophosphate or [UMP](#)) ([Fig. 347-4](#)). Deficiency of this enzyme is transmitted as a recessive trait and causes hemolytic anemia with prominent basophilic stippling of erythrocytes. The accumulation of pyrimidines or cytidine diphosphate choline (CDPC) is thought to induce hemolysis. The enzyme deficiency alters nucleoside composition and thereby generates a characteristic ultraviolet spectrum in deproteinized erythrocytes. There is no specific treatment.

DIHYDROPYRIMIDINE DEHYDROGENASE DEFICIENCY

Dihydropyrimidine dehydrogenase (DPD) is the rate-limiting enzyme in the pathway of uracil and thymine degradation ([Fig. 347-4](#)). Deficiency of this enzyme causes excessive urinary excretion of uracil and thymine. DPD deficiency is transmitted in a recessive manner and causes nonspecific cerebral dysfunction with convulsive disorders, motor retardation, and mental retardation. A splice donor site mutation, which causes deletion of exon 14, accounts for 52% of mutant alleles. No specific treatment is available. DPD is also involved in the degradation of 5-fluorouracil (5-FU), a chemotherapeutic agent that inhibits thymidylate synthase. Consequently, deficiency of this enzyme is associated with 5-FU neurotoxicity.

MEDICATION EFFECTS ON PYRIMIDINE METABOLISM

In addition to the role of [DPD](#) in [5-FU](#) degradation (see above), other medications can influence pyrimidine metabolism. Leflunomide, which is used to treat rheumatoid arthritis, inhibits de novo pyrimidine synthesis by inhibiting dihydroorotate dehydrogenase, resulting in an antiproliferative effect on T cells. Allopurinol, an inhibitor of xanthine oxidase and purine synthesis, also inhibits orotidine-5¢-phosphate decarboxylase, a step in [UMP](#) synthesis. Consequently, allopurinol use is associated with increased excretion of orotidine and orotic acid; there are no known clinical effects of this inhibition.

(Bibliography omitted in Palm version)

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348. WILSON'S DISEASE - I. Herbert Scheinberg

Wilson's disease is an inherited disorder of copper metabolism in individuals with two mutant *ATP7B* genes. Impairment of the normal excretion of hepatic copper results in toxic accumulation of the metal in liver, brain, and other organs. The disease occurs in every ethnic and geographic population, with a worldwide prevalence of ~1 in 30,000, and a heterozygous carrier frequency of ~1 in 90.

NATURAL HISTORY

The average concentrations of ceruloplasmin and hepatic copper are indistinguishable in normal neonates and in patients with Wilson's disease. In normal infants, however, the ceruloplasmin concentration increases and the hepatic copper concentration falls to adult levels during the first 3 months of life. In infants with Wilson's disease, the neonatal deficiency of ceruloplasmin and excess of hepatic copper persist indefinitely. Clinical manifestations are rare before age 6, occur most frequently in mid-adolescence, and eventually develop in all untreated patients.

In about half of patients any of four types of hepatic disturbances may herald the clinical onset. *Acute hepatitis* is usually self-limited, is often mistaken for viral hepatitis or infectious mononucleosis, and may be forgotten later in life. *Parenchymal liver disease* may persist after acute hepatitis or may develop insidiously without prior acute disease into a histologic and clinical picture indistinguishable from chronic active hepatitis and cirrhosis. In other patients *cirrhosis* may develop insidiously after a lapse of decades with no prior sign or symptom of liver disease. *Fulminant hepatitis*, generally fatal, is characterized by progressive jaundice, ascites, encephalopathy, hypoalbuminemia, hypoprothrombinemia, moderately elevated plasma levels of liver enzymes, and Coombs-negative hemolytic anemia.

In most other patients neurologic or psychiatric disturbances are the first clinical signs and are always accompanied by Kayser-Fleischer rings ([Plate III-16](#)). These golden deposits of copper in Descemet's membrane of the cornea do not interfere with vision but indicate that copper has been released from the liver and has probably caused brain damage. If a patient with frank neurologic or psychiatric disease does not have Kayser-Fleischer rings when examined by a trained observer using a slit lamp, the diagnosis of Wilson's disease can be excluded. Rarely, Kayser-Fleischer rings may be accompanied by sunflower cataracts.

The neurologic manifestations include resting and intention tremors, spasticity, rigidity, chorea, drooling, dysphagia, and dysarthria. Babinski responses may be present, and abdominal reflexes are often absent. Inexplicably -- in view of the ubiquity of copper excess in the brain -- sensory changes never occur, except for headache.

Psychiatric disturbances are present in most patients with neurologic symptoms. Schizophrenia, manic-depressive psychoses, and classic neuroses may occur, but the commonest disturbances are bizarre behavioral patterns that defy classification. Improvement in the psychiatric state can occur with pharmacologic reduction of the copper excess, but psychotherapy and additional pharmacotherapy may be required.

In about 5% of patients the clinical onset reflects neither a hepatic nor a central nervous system disturbance. The first manifestation may be primary or secondary amenorrhea or repeated and unexplained spontaneous abortions, perhaps due to excess free copper in intrauterine secretions. Kayser-Fleischer rings may occasionally first be discovered during routine ophthalmologic examination.

PATHOGENESIS

The metabolic defect in Wilson's disease is an inability to maintain a near-zero balance of copper. Dietary copper is generally in excess of the small amount that is essential to life. Normally, any excess absorbed copper is excreted by the liver; in patients with Wilson's disease, copper accumulates in the liver, reaching a mean level of about 1000 ug/g dry weight -- 40 times normal.

Fatty infiltration of the hepatic parenchyma and nuclear glycogen deposits are the earliest findings by light microscopy ([Fig. 348-1](#)). With electron microscopy, characteristic mitochondrial abnormalities appear to be specific for Wilson's disease. Later, necrosis, inflammation, fibrosis, bile duct proliferation, and cirrhosis ensue. Abnormalities in liver chemistries, particularly elevations in aminotransferases, may be seen at any stage. The capacity of hepatocytes to store copper is eventually exceeded, and copper is released into blood and taken up into extrahepatic tissues with disastrous effects in the brain ([Table 348-1](#)).

With magnetic resonance imaging, the effects of copper toxicity in the brain are seen most frequently in the lenticular nuclei and less commonly in the pons, medulla, thalamus, cerebellum, and cerebral cortex. Opalski and Alzheimer type II cells are present early in the course, although neither is specific for Wilson's disease, and neuronal necrosis and cavitation develop later.

An increased copper concentration in the kidney produces little, if any, structural change and usually does not alter renal function. Microscopic hematuria and/or minimal proteinuria occur occasionally; and nephrocalcinosis, renal calculi, and renal tubular acidosis are rare. Pathologic effects in other organs and tissues are minor.

GENETIC CONSIDERATIONS

The autosomal recessive Wilson disease gene, *ATP7B*, and the X-linked Menkes disease gene, *ATP7A*, are membrane-bound, P-type, copper-transporting ATPases containing 6 copper-binding sites ([Chap. 353](#)). The amino acid sequences of these genes are 54% identical. In liver the Wilson protein incorporates copper ions into apoceruloplasmin to form ceruloplasmin whose catabolism is accompanied by biliary excretion of its copper ions. In fetal liver the Menkes protein incorporates copper ions into apoceruloplasmin to form fetal ceruloplasmin whose catabolism is accompanied by hepatic retention of its copper ions. In Wilson's disease, ceruloplasmin that is synthesized in the presence of the *ATP7B* mutant is catabolized like fetal ceruloplasmin, leading to hepatic retention of its copper ions.

DIAGNOSIS

The diagnosis is easy provided it is suspected. Wilson's disease should be considered in any patient younger than 40 years with an unexplained disorder of the central nervous system, signs or symptoms of hepatitis, chronic active hepatitis, unexplained persistent elevations of serum aminotransferase, hemolytic anemia in the presence of hepatitis, or unexplained cirrhosis and in any patient who has a relative with Wilson's disease.

The diagnosis is confirmed by the demonstration of either (1) a serum ceruloplasmin level <20 mg/dL and Kayser-Fleischer rings or (2) a serum ceruloplasmin level <20 mg/dL and a concentration of copper in a liver biopsy sample >250 ug/g dry weight. Most symptomatic patients excrete >100 ug copper per day in urine and have histologic abnormalities on liver biopsy.

TREATMENT

Treatment consists of removing and detoxifying the deposits of copper as rapidly as possible and must be instituted once the diagnosis is secure whether the patient is ill or asymptomatic. Penicillamine is administered orally in an initial dose of 1 g daily in a single or divided doses at least 30 min before and 2 h after eating. Because penicillamine has an antipyridoxine effect, 25 mg/d of pyridoxine is also given. In ~10% of patients sensitivity to penicillamine develops early, making it necessary to monitor the body temperature and skin daily. White blood cell and platelet counts should be assessed and urinalysis performed several times during the first month of treatment. Penicillamine should be discontinued and replaced by trientine, if rash, fever, leukopenia, thrombocytopenia, lymphadenopathy, or proteinuria develops, or if neurologic worsening accompanies the institution of penicillamine and persists for a week or more.

After therapy with penicillamine has been successfully instituted, the patient should be seen at 1- to 3-month intervals to assess the effectiveness of therapy and monitor for late drug toxicity. The history and the physical examination should focus on hepatic, neurologic, and psychiatric signs and symptoms. Slit-lamp examination of the corneas should be performed by an ophthalmologist if neurologic or psychiatric disturbances appear or worsen. White blood cell and platelet counts, transaminase levels, albumin, bilirubin, and free serum copper (total serum copper minus ceruloplasmin-bound copper) should be measured, the aim being a concentration of free copper less than ~ 0.2 umol/dL (10 ug/dL). A persistent concentration >0.4 umol/dL (20 ug/dL) indicates that the dose of penicillamine is too low or that the patient is noncompliant. For patients who are asymptomatic or who have improved maximally after several years on 1 g/d of penicillamine, the usual effective maintenance dose is 0.75 g/d taken 45 min before breakfast.

At any time, even after years of uneventful penicillamine administration, granulocytopenia, thrombocytopenia, the nephrotic syndrome, Goodpasture's syndrome, systemic lupus erythematosus, severe arthralgias, myasthenia, mammary gigantism, or elastosis perforans serpiginosa may supervene. Except for transient thrombocytopenia or granulocytopenia, these reactions mandate the replacement of penicillamine with trientine.

The dose of trientine is 1g/d on an empty stomach. Most patients find it convenient to take four 250-mg capsules, delaying breakfast for about an hour. Pyridoxine need not be given. Although the only reported toxic reaction to trientine is sideroblastic anemia, the same clinical procedures and laboratory determinations should be performed during its administration as are used during penicillamine therapy. Except for systemic lupus erythematosus and, occasionally, elastosis perforans serpiginosa, the other late penicillamine-induced toxic reactions disappear or improve with trientine therapy. Moreover, trientine is as effective therapeutically as penicillamine.

Zinc acetate or gluconate are effective as maintenance therapy, at doses of 150 mg/d of elemental zinc, for patients who are asymptomatic or have improved maximally on penicillamine or trientine. Zinc must *not*, however, be given together with penicillamine or trientine, both of which can chelate zinc and form complexes that are therapeutically ineffective.

Treatment must be continued for life. Inadequate treatment or interruption of therapy can be fatal or cause irreversible relapse. Indeed, of 11 patients who voluntarily discontinued penicillamine after years of successful treatment, 8 died after an average of 2.6 years of noncompliance. In contrast, of 13 patients in whom trientine was substituted because of an adverse reaction to penicillamine, 1 died accidentally, 5 were lost to follow-up, and 7 are alive and well 11 to 23 years later.

Prophylactic treatment of more than 100 asymptomatic patients with a documented diagnosis of Wilson's disease has shown that continual therapy with penicillamine or trientine can maintain the asymptomatic state indefinitely. Several such patients have been treated with penicillamine for >30 years.

Patients with severe neurologic disease who do not improve with penicillamine or trientine therapy that has reduced serum free copper to <0.3 $\mu\text{mol/dL}$ (15 $\mu\text{g/dL}$) may benefit significantly from treatment with dimercaprol. Intramuscular injections of 3 mL (containing 300 mg of dimercaprol) are given on five successive weekdays for 4 weeks. Treatment is interrupted for 1 week, and a second 4-week course is given. If there is neurologic improvement, additional courses are given as long as improvement continues. If no improvement is seen after two courses, it is unlikely that additional courses will be effective.

The simultaneous occurrence of fulminant hepatitis and Coombs-negative hemolytic anemia may be the initial clinical manifestation of Wilson's disease or may occur in a noncompliant patient. The syndrome is almost always fatal, usually within a week or two, unless liver transplantation is performed. Transplantation is also indicated if progressive hepatic insufficiency occurs despite adequate treatment with penicillamine or trientine.

More than 150 women with Wilson's disease treated with penicillamine and more than 20 women treated with trientine have had successful and uneventful pregnancies.

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349. LYSOSOMAL STORAGE DISEASES - Gregory A. Grabowski

GENERAL FEATURES

Lysosomes are heterogeneous subcellular organelles containing specific hydrolyases that allow targeted processing or degradation of proteins, nucleic acids, carbohydrates, and lipids. There are >30 different lysosomal storage diseases, and they vary greatly in the types of metabolic abnormalities that occur as well as in their clinical manifestations. Nonetheless, these disorders are considered together because they share a related pathophysiology that involves the accumulation of specific macromolecules within cells that normally process large amounts of these substrates. The disorders are classified based on the nature of the stored material and include mucopolysaccharidoses (MPS), gangliosidoses, glycosphingolipidoses, glycoproteinosis, mucopolipidoses, leukodystrophies, and lipid storage disorders ([Table 349-1](#)). The most prevalent lysosomal storage diseases in adults are Fabry disease, Gaucher disease, and Niemann Pick disease (NPD).

Lysosomal storage diseases should be considered in the differential diagnosis of patients with neurologic or muscular degeneration, unexplained hepatomegaly or splenomegaly, or skeletal dysplasias and deformations. Physical findings are disease-specific, and definitive diagnosis is made by enzyme assays.

PHYSIOLOGY OF LYSOSOMES

Lysosomal biogenesis is a continuous process that involves ongoing synthesis of lysosomal hydrolases, membrane constitutive proteins, and new membranes. Lysosomes originate from the fusion of *trans*-golgi network (TGN) vesicles with late endosomes. Progressive vesicular acidification accompanies the maturation of TGN vesicles, which contain various hydrolases, into lysosomes. Early endosomes have an internal pH of ~6.0 to 6.2; late endosomes and lysosomes have a pH of ~5.5 to 6.0 and ~5, respectively. This gradient facilitates the pH-dependent dissociation of receptors and ligands [e.g., the mannose-6-phosphate (M6P) receptor and M6P-containing oligosaccharides] as well as activating lysosomal hydrolase function. This dynamic system, which has supplanted the static view of the lysosome, is consistent with the presence of heterogeneous populations of similar organelles whose contents differ significantly in time and location within the cell.

The accurate sorting, targeting, and activation of lysosomal enzymes is essential for maintaining normal cellular function. Abnormalities at several steps along the biosynthetic pathway can impair enzyme activation and lead to a lysosomal storage disorder. After cleavage of the hydrophobic signal peptide in the endoplasmic reticulum (ER) membrane, *N*-glycosylation occurs cotranslationally in the lumen of the ER. Complex oligosaccharide modifications occur during transit through the Golgi. The [M6P](#) modification of high-mannose oligosaccharide chains of many soluble lysosomal hydrolases occurs early in this process. Defects of this modification result in inappropriate extracellular secretion of most soluble lysosomal hydrolases, leading to severe phenotypes (I-cell disease). Lysosomal integral or associated membrane proteins (LIMPS or LAMPS) are sorted to the membrane or interior of the lysosome by several different signals. Phosphorylation, sulfation, additional proteolytic processing,

and macromolecular assembly of heteromers occur concurrently and are critical to enzyme function. Defects in the latter can result in multiple enzyme/protein deficiencies.

PATHOGENESIS OF LYSOSOMAL STORAGE DISEASES

The final common pathway for lysosomal storage diseases is the accumulation of specific macromolecules within tissues and cells that normally have a high flux of these substrates. The majority of lysosomal enzyme deficiencies result from point mutations or genetic rearrangements at a locus that encodes a single lysosomal hydrolase. However, some mutations cause deficiencies of several different lysosomal hydrolases by altering the enzymes/proteins involved in targeting, active site modifications, or macromolecular association or trafficking. All are inherited as autosomal recessive disorders, except Hunter ([MPS II](#)) and Fabry diseases, which are X-linked. Lysosomal distortion, which is caused by substrate accumulation, probably has significant pathologic consequences. However, abnormal amounts of metabolites may also have pharmacologic effects important to disease pathophysiology.

For many lysosomal diseases, the accumulated substrates are endogenously synthesized within particular tissue sites of pathology. Other diseases have greater exogenous substrate supplies; that is, they are delivered by low-density lipoprotein receptor-mediated uptake in Fabry and cholesteryl ester storage diseases or by phagocytosis in Gaucher disease type 1.

The concept of the *threshold hypothesis* has important implications for disease classification, pathophysiology, and treatment. It implies a threshold of enzyme activity below which disease develops. Consequently, small changes in enzyme activity near the threshold can lead to or prevent disease. A critical element of this model is that enzymatic activity can be challenged by changes in substrate flux based on genetic background, cell turnover, recycling, or metabolic demands. Thus, a set level of residual enzyme may be adequate for substrate in some tissues or cells, but not in others. For some lysosomal storage diseases [e.g., metachromatic leukodystrophy (MLD), Tay-Sachs disease, Gaucher disease], genotype-phenotype correlations may help to predict the severity of clinical consequences associated with certain levels of enzyme activity. Defining enzyme activity thresholds may also be useful for predicting dose-response relationships for treatment and for the evaluation of exogenous enzyme replacement therapy.

SPECIFIC DISORDERS

MUCOPOLYSACCHARIDOSES (MPS)

The various forms of [MPS](#) result from deficiencies of lysosomal enzymes needed for glycosaminoglycan (GAG) catabolism. GAGs are long-chain, complex carbohydrates that are linked to proteins in connective tissue. They are components of proteoglycans and include: chondroitin-4-sulfate, chondroitin-6-sulfate, heparan sulfate, dermatan sulfate, keratan sulfate, and hyaluronic acid. The particular accumulated GAG is determined by the specific enzyme deficiency. GAG accumulation in various tissues produces a spectrum of clinical features that can include skeletal abnormalities, corneal clouding, organomegaly, joint stiffness, hernias, short stature, and, in some disorders,

mental retardation. Vacuolated lymphocytes in the peripheral smear and excessive urinary GAGs are typical findings. Clinical and laboratory features overlap among the MPS diseases and are not diagnostic. The MPS diseases occur with individual frequencies of about 1/50,000 to 1/100,000 in most populations.

The diagnosis is established by specific enzyme assays or, when known, DNA mutation analysis. Prenatal diagnosis is conducted most frequently with cultured amniotic cells. Mutation studies for carriers can be performed, if the mutation is known.

Treatment of the [MPS](#) diseases requires comprehensive, multisystem evaluations. Symptomatic therapies currently include corneal transplantation, correction of nerve entrapment, and heart valve replacement. Physical therapy for joint contractures is needed in all but MPS IV. In MPS III variants, psychotropic drugs are used to control behavior. For patients with MPS IH and IV, cervical myelopathy can be prevented by prospective cervical spinal fusion. The efficacy of bone marrow transplantation and enzyme replacement is under investigation.

MPS IH (Hurler Disease) This is a severe autosomal recessive disorder that results from numerous different mutations of α -L-iduronidase. Progressive mental retardation, hepatosplenomegaly, skeletal malformations, and cardiopulmonary compromise typically lead to death during the first decade. Affected individuals appear normal at birth but exhibit accelerated growth and mild coarsening of facial features in the first year. Subsequently, there is slowing of growth, leading to short stature. In the first 2 years, clinical diagnosis is suggested by hepatosplenomegaly, corneal clouding, coarse features, large tongue, joint stiffness, and characteristic dysostosis multiplex on skeletal x-rays. Instability of the cervical vertebral bodies can lead to paralysis, particularly with subluxation on hyperextension. Developmental delay is apparent by 12 to 28 months, with subsequent slow mental regression. Additional features of this multisystem disease include hearing loss, chronic respiratory infections, valvular heart disease, and brain ventricular enlargement. The latter occurs from involvement of the arachnoid granulations.

MPS (Scheie Disease) and MPS I H/S (Hurler-Scheie Disease) These [MPS](#) variants are less severe than MPS IH (Hurler disease). They result from allelic mutations in the α -L-iduronidase gene, presumably with a less severe effect on enzyme function. Patients with MPS IS can survive into late adulthood with normal intelligence, though with severe progressive skeletal disease that resembles osteoarthritis. Bone marrow transplantation in MPS IH, if instituted before substantial central nervous system CNS involvement, has shown therapeutic promise. Preliminary intravenous enzyme administration has led to improvement in hepatosplenomegaly and connective tissue involvement.

MPS II (Hunter Syndrome) This is an X-linked recessive disorder that results from deletions and point mutations in the gene encoding iduronate sulfatase. Clinically, [MPS](#) IH and II are similar, though corneal clouding is absent in MPS II. Clinical manifestations range from severe [CNS](#) and visceral involvement with death in late childhood to milder forms with normal CNS function and survival into adulthood. Bone marrow transplantation has not been successful for treating the severe variants, and experience in the less severe variants is too limited to permit conclusions. Enzyme therapy trials are

imminent.

MPS IIIA, IIIB, IIIC, and IIID (the Sanfilippo Syndromes) These autosomal recessive disorders are caused by various enzymatic deficiencies as summarized in [Table 349-1](#). Skeletal defects and hepatosplenomegaly are less pronounced in this group of [MPS](#) variants, though progressive behavioral problems, mental retardation, and seizures are present. Affected patients can survive into the third or fourth decade with progressive [CNS](#) disease.

MPS IV (Morquio Syndrome) These [MPS](#) variants are autosomal recessive disorders characterized by severe skeletal diseases that resemble the spondyloepiphyseal dysplasias. There is extreme shortening of the trunk due to multiple vertebral collapses. The long bones are relatively spared. Joint laxity can lead to osteoarthritis-like destruction of the joints. Upper cervical spinal cord compression due to atlantoaxial instability predisposes to subluxation and paralysis. Many patients have mitral valve insufficiency that can be functionally significant. The A and B variants are distinguished clinically by more severe skeletal disease, *N*-acetylgalactosamine-6-sulfate sulfatase deficiency in A, than *inb*-galactosidase defects in B. Enzyme therapy trials will begin in the near future.

MPS VI (Maroteaux-Lamy Disease) Mutations in the arylsulfatase B gene cause this autosomal recessive disorder. Although clinically variable, the general phenotype resembles Hurler disease. Intelligence is normal, and the life span can extend beyond three decades. Cardiac valvular disease and progressive pulmonary hypertension are frequent causes of death. Bone marrow transplantation may be useful in diminishing these manifestations.

GM₂GANGLIOSIDOSES

The Tay-Sachs and Sandhoff disease variants are caused by defects *inb*-hexosaminidase (Hex) A and/or B. Hex A is a heteromeric protein with a and *b* chains, whereas Hex B contains only *b* chains. The a and *b* chains are encoded by different genes. Infantile, juvenile, and adult-onset variants are distinguished by age at onset and rate of progression.

In addition to other clinical manifestations described below, specific neurologic features, such as the retinal cherry red spot, suggest the diagnosis of Tay-Sachs and Sandhoff diseases. Diagnosis is confirmed by [Hex](#) A and/or B levels in blood plasma or nucleated cells. Screening for Tay-Sachs disease carriers in the Ashkenazi Jewish population is recommended.

Tay-Sachs Disease About 1 in 30 Ashkenazi Jews is a carrier for Tay-Sachs disease, which is caused by total [Hex](#) A deficiency. The infantile form is a fatal neurodegenerative disease that is characterized by macrocephaly, loss of motor skills, increased startle reaction, and macular pallor with cherry red spot on retinal examination. The juvenile-onset form presents with ataxia and dementia, with death by age 10 to 15 years. The adult-onset disorder is characterized by clumsiness in childhood; progressive motor weakness in adolescence; and additional spinocerebellar, lower motor neuron symptoms, and dysarthria in adulthood. Intelligence declines slowly, and psychosis is

also common.

Sandhoff Disease Sandhoff disease is nearly identical to Tay-Sachs disease, though hepatosplenomegaly and bony dysplasias are present in the former. The later onset variants are characterized by progressive visceral and [CNS](#) disease. Treatment is supportive.

NEUTRAL GLYCOSPHINGOLIPID LIPID STORAGE DISORDERS

Fabry Disease This is an X-linked disorder that results from a variety of mutations in the α -galactosidase gene. This enzyme cleaves the terminal α -galactosyl moiety from globotriaosylceramide (trihexosylceramide, THC), a key step in glycosphingolipid metabolism. Clinically, the disease manifests with angiokeratomas (telangiectatic skin lesions); hypohidrosis; corneal and lenticular opacities; acroparesthesia; and small-vessel disease of the kidney, heart, and brain. The estimated prevalence of hemizygous males with Fabry disease is 1/40,000.

The angiokeratomas and acroparesthesia may appear in childhood and lead to early diagnosis, if suspected. Angiokeratomas are punctate, dark red to blue-black, flat or slightly raised, usually symmetric, and do not blanch with pressure. They range from barely visible to several millimeters in diameter and have a tendency to increase in size and number with age. Characteristically, they are most dense between the umbilicus and knees -- "the bathing suit area" -- but may occur anywhere, including the mucosal surfaces. Corneal and lenticular lesions, detectable on slit-lamp examination, are present in affected men and ~70% of heterozygous women. Tortuosity of the conjunctival and retinal vessels is common. The acroparesthesia can be debilitating in childhood and adolescence, with a tendency to decrease after the third decade. Episodic agonizing, burning pain of the hands, feet, and proximal extremities can last from minutes to days and can be precipitated by exercise, fatigue, or fever. Abdominal pain can resemble that from appendicitis or renal colic.

Casts and microscopic hematuria can occur early, whereas proteinuria, isosthenuria, and progressive renal dysfunction occur in the second to fourth decades. Progressive renal failure occurs and requires transplantation. Hypertension, left ventricular hypertrophy, anginal chest pain with or without myocardial ischemia or infarction, and congestive heart failure can occur in the third to fourth decades. Leg lymphedema without hypoproteinemia and episodic diarrhea also occur. Death is due to renal failure or cardiovascular or cerebrovascular disease in untreated patients. Variants with residual α -galactosidase activity may have late-onset manifestations limited to the cardiovascular system that resemble hypertrophic cardiomyopathies. Heterozygous females may exhibit some of these clinical manifestations but usually not the severe organ involvement.

Acroparesthesia, hypohidrosis or anhidrosis, angiokeratomas, and the typical corneal and lenticular lesions provide a presumptive diagnosis in males. Angiokeratomas are not diagnostic, however, and also occur in Fordyce scrotal angiokeratoma and several other lysosomal storage diseases.

Phenytoin and carbamazepine diminish the chronic and episodic acroparesthesia.

Chronic hemodialysis or kidney transplantation can be lifesaving in patients with renal failure. Initial enzyme therapy results are promising.

Gaucher Disease This is an autosomal recessive disorder that results from defective activity of acidb-glucosidase; >175 mutations have been described. This enzyme cleaves glucosylceramide, the parent compound of many glycosphingolipids and related glucolipids. Disease variants are classified based on the absence or presence and severity of neuronopathic involvement.

Type 2 Gaucher disease is a rare, severe [CNS](#) disease that leads to death by 2 years of age; it will not be addressed here.

Type 3 Gaucher disease has highly variable manifestations in the [CNS](#) and viscera. It can present in early childhood with rapidly progressive, massive visceral disease and slowly progressive to static CNS involvement; in adolescence with dementia; or in early adulthood with rapidly progressive, uncontrollable myoclonic seizures and mild visceral disease. Variants that span this spectrum also occur. Visceral disease in type 3 is nearly identical to that in type 1 but is generally more severe (see below). Early CNS findings may be limited to defects in lateral gaze tracking, which may remain static for decades. Mental retardation can be slowly progressive or static. This variant is most frequent among individuals of Swedish descent.

Type 1 Gaucher disease is a highly variable nonneuronopathic disease that can present in childhood to adulthood with slowly to rapidly progressive visceral disease. There is marked variability in age at onset and degree and progression of visceral involvement. In general, earlier diagnoses are associated with worse prognosis. The average age at diagnosis is ~20 years in Caucasian populations and somewhat younger in other groups. This pattern of presentation is distinctly bimodal, however, with peaks at <10 to 15 years and at ~25 years. Younger patients tend to have a greater degree of hepatosplenomegaly and accompanying blood cytopenias. In contrast, the older group has a greater tendency for chronic bone disease. Hepatosplenomegaly occurs in virtually all symptomatic patients and can be minor or massive. Accompanying anemia and thrombocytopenia are variable and are not linearly related to liver or spleen volume. Severe liver dysfunction is unusual, though minor liver function abnormalities are common. Splenic infarctions can resemble an acute abdomen. Pulmonary hypertension and alveolar Gaucher cell accumulation are uncommon, but life-threatening, and can occur at any age.

Though it is more common in adult patients, clinically evident skeletal disease in children can be devastating, resulting in massive destruction of the axial and peripheral skeleton. All patients with Gaucher disease have nonuniform infiltration of bone marrow by lipid-laden macrophages, termed *Gaucher cells*. This can lead to marrow packing with subsequent infarction, ischemia, necrosis, and cortical bone destruction. Bone marrow involvement spreads from proximal to distal in the limbs and can involve the axial skeleton extensively, causing vertebral collapse. In addition to bone marrow involvement, bone remodeling is defective, with loss of total bone calcium leading to osteopenia, osteonecrosis, avascular infarction, and vertebral compression fractures and spinal cord involvement. Aseptic necrosis of the femoral head is common, as is fracture of the femoral neck. The mechanism by which diseased bone marrow

macrophages interact with osteoclasts and/or osteoblasts to lead to this complex bone disease is not well understood.

Affected patients experience chronic, ill-defined bone pain that can be debilitating and poorly correlated with radiographic findings. These are treated symptomatically. Some patients have one or more "bone crises" in their lifetimes that are associated with localized, excruciating pain, and, on occasion, local erythema, fever, and leukocytosis. Some patients have frequent crises, whereas other patients experience only one. Any bone can be involved, though the femurs and vertebral bodies are affected most often. These crises represent acute infarctions of bone, as evidenced in nuclear scans by localized absent uptake of pyrophosphate agents. X-rays are usually negative initially but may show lytic lesions 4 to 6 months after the acute phase. Osteomyelitis should be excluded by appropriate cultures. Bone cultures should be obtained only under sterile operating room conditions to minimize the chance of seeding an infection.

The diagnosis of Gaucher disease is established by demonstrating decreased acidb-glucosidase activity (0 to 20% of normal) in nucleated cells. The enzyme is not present in bodily fluids. The sensitivity of enzyme testing is poor for detecting heterozygous carriers; molecular testing is preferred when the mutations are known. The disease frequency varies from about 1 in 1000 in Ashkenazi Jews to <1 in 100,000 in some other populations. About 1 in 12 to 15 Ashkenazi Jews carries a Gaucher disease allele. Four common mutations account for ~90 to 95% of the mutations in affected patients: N370S (1226G), 84GG (a G insertion at cDNA position 84), L444P (1448C), and IVS-2 (an intron 2 splice junction mutation).

Genotype/phenotype studies indicate a significant correlation, though not absolute, between disease type and severity and the acid b-glucosidase genotype. For example, the most common mutation in the Ashkenazi Jewish population (N370S) shares a 100% association (to date) with nonneuronopathic, type 1 Gaucher disease, possibly because the N370S enzyme retains significant activity. The N370S/N370S and N370S/other mutant allele genotypes are associated with later onset/less severe and earlier onset/severe disease, respectively. The other alleles are L444P (very low activity), 84GG (null), or IVS-2 (null), and rare/private or uncharacterized alleles. As many as 50 to 60% of N370S/N370S patients are discovered as asymptomatic family members. The N370S/other mutant allele genotypes have disease onset about 2 decades earlier than those with N370S/N370S. The L444P/L444P patients almost always have life-threatening to very severe/early-onset disease, and many, though not all, develop [CNS](#) involvement in the first 2 decades of life. Some patients with this genotype have lethal neuronopathic disease at <1 year (type 2). Thus, this genotype is prognostic of very severe disease, with or without obvious CNS involvement.

Symptomatic management of the blood cytopenias and joint replacement surgeries continue to have important roles in the treatment of affected patients. However, enzyme therapy is currently the treatment of choice in significantly affected patients. Cerezyme, a recombinantly produced mannose-terminated (macrophage-targeted) acidb-glucosidase, has proved highly efficacious and safe in diminishing the hepatosplenomegaly and improving bone marrow involvement and hematologic findings.

Niemann-Pick Disease [NPD](#) is an autosomal recessive trait that occurs as several variants. Types A and B result from defects in acid sphingomyelinase; various mutations have been detected at this locus. Other variants, including NPD C, result from defective transport of cholesterol across the lysosomal membrane. Only NPD variants A and B will be considered here.

[NPD](#) A and B are distinguished primarily by an early age of onset and progressive [CNS](#) disease in A. NPD A typically has onset in the first 6 months, with rapidly progressive CNS deterioration, spasticity, failure to thrive, and massive hepatosplenomegaly. In contrast, NPD B has a later, more variable onset and progression of hepatosplenomegaly, with eventual development of cirrhosis and hepatic replacement by foam cells. Affected patients develop progressive pulmonary disease with dyspnea, hypoxemia, and a reticular infiltrative pattern on chest x-ray. Foam cells are present in alveoli, lymphatic vessels, and pulmonary arteries. Progressive hepatic or lung disease with associated bronchopneumonia, pulmonary hypertension, cor pulmonale, and decreased diffusion capacities lead to demise in adolescence to early adulthood.

The diagnosis is established by markedly decreased (1 to 10% of normal) sphingomyelinase activity in nucleated cells. Enzyme assays to detect [NPD](#) A or B carriers are unreliable. In families with known mutations, the molecular defect in heterozygotes can be identified by DNA analysis.

There is no specific treatment for [NPD](#). The efficacy of hepatic or bone marrow transplantation has not been proven. Clinical trials using enzyme therapy are anticipated to begin soon.

THE LEUKODYSTROPHIES

Globoid cell leukodystrophy and metachromatic leukodystrophy variants are autosomal recessive disorders that primarily involve [CNS](#) white matter and myelinated peripheral nervous system tracts.

Globoid Cell Leukodystrophy (GCL) The GCL variants are due to mutations in the galactosylceramidase gene. The term *globoid cell* is derived from the presence of characteristic multinucleated cells filled with galactosylceramide in the brains of affected patients. Infantile GCL (Krabbe disease) is a rapidly progressive, fatal disorder; patients succumb in the first 2 years. Juvenile and adult variants present with more slowly progressive dementia. For all variants, manifestations are confined to the [CNS](#) and peripheral nervous system. Diagnosis is confirmed by demonstrating defective galactosylceramidase activity in nucleated cells. Treatment is supportive, but bone marrow transplantation has shown some promise in the later onset variants.

Metachromatic Leukodystrophy [MLD](#) is due to defects in arylsulfatase A and accumulation of its substrate, galactosylceramide sulfate or sulfatide. The late-infantile form presents in the second year with progressive regression of developmental milestones and intellectual development. The disease is fatal in the first decade. The juvenile and adult forms have variable manifestations and present with gait disturbances, ataxia, mental regression, peripheral neuropathy, and/or seizures. In

adults, behavioral disturbances, psychosis, and dementia tend to predominate. These later onset diseases may respond to bone marrow transplantation.

The diagnosis is established by demonstrating a deficiency of arylsulfatase A in nucleated cells. Homozygosity for a null allele (splicing defect in intron 2) produces severe infantile disease, whereas P426L homozygotes develop adult-onset disease. Compound heterozygotes, such as null and P426L alleles, have juvenile-onset variants. The diagnosis of [MLD](#) is complicated by the presence of a very frequent (10 to 15%) "pseudodeficiency allele" for arylsulfatase A. Although in vitro activity with synthetic substrates is deficient, cleavage of sulfatide is low-normal in vivo. Compound heterozygotes for the pseudodeficiency allele and true MLD alleles therefore appear to have deficient arylsulfatase A activity, diagnostic of MLD, but these individuals are not affected by the disease. MLD-causing mutations also occur on the background of the pseudodeficient allele, emphasizing the importance of careful genetic testing.

GLYCOGEN STORAGE DISEASE TYPE II (See also [Chap. 350](#))

Glycogen storage disease type II is an autosomal recessive disorder due to defects in acid α -glucosidase that lead to lysosomal glycogen accumulation. Numerous mutations have been found at this locus in affected patients. Skeletal and cardiac muscles are primarily involved. The infantile form (Pompe disease) is a fatal disorder characterized by hypertrophic cardiomegaly, macroglossia, and hypotonia due to glycogen accumulation in muscle. The juvenile form has progressive proximal muscle weakness, including impairment of respiratory function. Adult patients have phenotypes resembling slowly progressive muscular dystrophies. Prenatal diagnosis can be performed. Treatment is supportive, and enzyme trials are underway.

MUCOLIPIDOSES

Mucopolidoses II (I-cell) and III (pseudo-Hurler polydystrophy) are rare autosomal recessive diseases. Both are caused by defective targeting of lysosomal hydrolases that require the [M6P](#) signal for sorting to the lysosome. As a result, >20 enzymes are secreted out of the cell and their substrates accumulated in specific cell types.

N-acetylglucosamine-1-phosphotransferase activity, which is necessary for developing the M6P signal, is defective in these diseases. I-cell disease has a phenotype similar to [MPS](#) IH, whereas mucopolidosis-III is more similar to MPS IH/S; mental retardation is a feature of both. Diagnosis is suspected based on the characteristic phenotype and is established by demonstrating greatly elevated serum levels of lysosomal enzymes, as well as their deficiency in cells. Specific enzyme assays can also be performed. Carrier detection is possible but is not straightforward. Prenatal diagnosis can be performed using amniotic fluid and cells in at-risk families. Treatment is supportive.

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350. GLYCOGEN STORAGE DISEASES AND OTHER INHERITED DISORDERS OF CARBOHYDRATE METABOLISM - Yuan-Tsong Chen

Carbohydrate synthesis and degradation play a vital role in cellular function by providing the energy required for most metabolic processes. The carbohydrates to be discussed include three monosaccharides: glucose, galactose, and fructose, and a polysaccharide, glycogen; the relevant biochemical pathways involved in the metabolism of these carbohydrates are shown in [Fig. 350-1](#). Glucose is the principle substrate of energy metabolism in humans. Metabolism of glucose generates ATP via glycolysis and mitochondrial oxidative phosphorylation. A continuous source of glucose from dietary intake, gluconeogenesis, and degradation of glycogen maintain normal blood glucose levels. Sources of glucose in our diet are obtained by ingesting polysaccharides, primarily starch, and disaccharides including lactose, maltose, and sucrose. Galactose and fructose are two other monosaccharides that provide fuel for cellular metabolism; however, their role as fuel sources is much less significant than that of glucose. Galactose is derived from lactose (galactose+ glucose), which is found in milk and milk products. If necessary, galactose can be incorporated into glycogen and thus becomes a source of glucose. Galactose is also an important component for certain glycolipids, glycoproteins, and glycosaminoglycans. The two dietary sources of fructose are sucrose (fructose + glucose), a commonly used sweetener, and fructose itself, which is found in fruits, vegetables, and honey.

This chapter is devoted to the inherited disorders of carbohydrate metabolism caused by defects in enzymes or transport proteins involved in glycogen metabolism, gluconeogenesis, and glycolysis ([Table 350-1](#)). Defects in glycogen metabolism typically cause an accumulation of glycogen in the tissues; hence, the name *glycogen storage diseases*. The defects in gluconeogenesis or glycolytic pathways including galactose and fructose metabolism do not usually result in glycogen accumulation.

Clinical manifestations of the various disorders of carbohydrate metabolism differ markedly. The symptoms range from harmless to lethal. Unlike disorders of lipid metabolism, mucopolysaccharidoses, or other storage diseases, dietary therapy has been effective in many of the carbohydrate disorders. Almost all the genes responsible for the inherited defects of carbohydrate metabolism have been cloned, and mutations have been identified. Advances in our understanding of the molecular basis of these diseases are being used to improve diagnosis and management, and some of these disorders are candidates for early trials of gene therapy.

Glycogen, the storage form of glucose in animal cells, is composed of glucose residues joined in straight chains by α 1-4 linkages and branched at intervals of 4 to 10 residues with α 1-6 linkages. The tree-like molecule can have a molecular weight of many millions and may aggregate to form structures recognizable by electron microscopy. In muscle, glycogen forms *b* particles, which are spherical and contain up to 60,000 glucose residues. Each *b* particle contains a covalently linked protein called *glycogenin*. Liver contains *b* particles and rosettes of glycogen called *a* particles, which appear to be aggregated *b* particles.

The primary function of glycogen varies in different tissues. In skeletal muscle, stored glycogen is a source of fuel that is used for short-term, high-energy consumption during

muscle activity; in the brain, the small amount of stored glycogen is used during brief periods of hypoglycemia or hypoxia as an emergency supply of energy. In contrast, the liver takes up glucose from the bloodstream after a meal and stores it as glycogen. When blood glucose levels start to fall, the liver converts glycogen back into glucose and releases it into the blood for use by tissues such as brain and erythrocytes that cannot store significant amounts of glycogen.

Glycogen storage diseases are inherited disorders that affect glycogen metabolism. Disorders in virtually every enzyme involved in the synthesis or degradation of glycogen and its regulation cause some type of glycogen storage disease ([Fig. 350-1](#)) in which glycogen is abnormal in quantity, quality, or both. Excluded from this chapter are those conditions in which tissue glycogen accumulation is secondary, such as overtreatment of diabetes mellitus with insulin or administration of pharmacologic amounts of glucocorticoids.

Historically, the glycogen storage diseases were categorized numerically in the order in which the enzymatic defects were identified. They can also be classified by the organs involved and clinical manifestations, the system followed in this chapter ([Table 350-1](#)).

Because liver and muscle have abundant glycogen, they are the most commonly and seriously affected tissues. The hepatic glycogen storage diseases can be divided into two groups, with some overlap. The first is characterized by hepatomegaly and hypoglycemia. Because carbohydrate metabolism in the liver controls plasma glucose levels, the disorders of hepatic glycogen degradation and glucose release cause fasting hypoglycemia. Diseases in this group include glucose-6-phosphatase deficiency (type I), debranching enzyme deficiency (type III), liver phosphorylase deficiency (type VI), phosphorylase kinase deficiency (type IX), glycogen synthase deficiency (type 0), and glucose transporter-2 defects (type XI). The second group, characterized by cirrhosis of the liver and hepatomegaly, is associated with accumulation of abnormal forms of glycogen, which may be the cause of the hepatocellular injury. This group is represented by branching enzyme deficiency (type IV).

The role of glycogen in muscle is to provide substrates for the generation of sufficient ATP for muscle contraction. The muscle glycogen storage diseases can also be divided into two groups. The first is a muscle-energy disorder characterized by muscle pain, exercise intolerance, myoglobinuria, and susceptibility to fatigue. This group includes type V (McArdle disease), a muscle phosphorylase deficiency, and deficiencies of phosphofructokinase (type VII), phosphoglycerate kinase, phosphoglycerate mutase, lactate dehydrogenase, fructose 1,6-bisphosphate aldolase A, and pyruvate kinase. Some of these latter enzyme deficiencies are associated with a compensated hemolysis, suggesting a more generalized defect in glucose metabolism. The second group of muscle disorders is characterized by progressive skeletal muscle weakness and atrophy and/or cardiomyopathy; it includes a lysosomal enzyme deficiency (acid- α -glucosidase, type II) and deficiency of cardiac-specific phosphorylase kinase. Some glycogen storage diseases such as debranching enzyme deficiency (type IIIa) and branching enzyme deficiency (type IV) involve both muscle and liver.

The overall frequency of all forms of glycogen storage disease is approximately 1 in 20,000 live births; most are inherited as autosomal recessive traits, but

phosphoglycerate kinase deficiency and one form of phosphorylase kinase deficiency are X-linked disorders. The most common childhood disorders are glucose-6-phosphatase deficiency (type I), lysosomal acid-glucosidase deficiency (type II), debrancher deficiency (type III), and liver phosphorylase kinase deficiency (type IX). The most common adult disorder is myophosphorylase deficiency (type V, or McArdle disease). In the past, the prognosis for many glycogen storage diseases was guarded. However, early diagnosis and better management have improved the survival rates, and many affected children are now adults.

GLYCOGEN STORAGE DISEASES: LIVER GLYCOGENOSES

DISORDERS WITH HEPATOMEGALY AND HYPOGLYCEMIA

Type I Glycogen Storage Disease (Glucose-6-Phosphatase or Translocase Deficiency, von Gierke Disease) Type I glycogen storage disease is due to a defect in glucose-6-phosphatase in liver, kidney, and intestinal mucosa. It can be divided into two subtypes: type Ia, in which the glucose-6-phosphatase enzyme is defective, and type Ib, which is due to a defect in the translocase that transports glucose-6-phosphate across the microsomal membrane. The defects in both subtypes lead to inadequate conversion in the liver of glucose-6-phosphate to glucose and thus make affected individuals susceptible to fasting hypoglycemia.

GENETIC CONSIDERATIONS

Type I glycogen storage disease is an autosomal recessive disorder. Both types Ia and Ib disease have been reported in many ethnic groups, but type Ia is rarely seen in blacks. The structural gene for glucose-6-phosphatase is located on chromosome 17q21; three common mutations (R83C, 130X, Q347X) are responsible for 70% of the known disease alleles. The structural gene for glucose 6-phosphate translocase is located on chromosome 11q23; two mutations, G339C and 1211delCT, appear to be prevalent in Caucasian patients, while W118R appears to be most common in Japanese patients. Carrier detection and prenatal diagnosis are possible with the use of molecular techniques.

Clinical and laboratory findings Persons with type I disease may develop hypoglycemia and lactic acidosis during the neonatal period, but, more commonly, they present at 3 to 4 months of age with hepatomegaly and/or hypoglycemia. These children often have doll-like faces with fat cheeks, relatively thin extremities, short stature, and a protuberant abdomen that is due to massive hepatomegaly; the kidneys are enlarged, but the spleen and heart are of normal size.

The hallmarks of the disease are hypoglycemia, lactic acidosis, hyperuricemia, and hyperlipidemia. Hypoglycemia and lactic acidosis can develop after a short fast. Hyperuricemia is present in young children, but gout rarely develops before puberty. Despite hepatomegaly, liver enzymes are usually normal or near normal. Intermittent diarrhea may occur (the mechanism is not known). Easy bruising and epistaxis are associated with a prolonged bleeding time as a result of impaired platelet aggregation/adhesion.

Hypertriglyceridemia may cause the plasma to appear "milky," and cholesterol and phospholipids are also elevated. The lipid abnormality resembles type IV hyperlipidemia and is characterized by increased levels of very low-density lipoprotein (VLDL); low-density lipoprotein (LDL); increased levels of apolipoproteins B, C, and E; and normal or reduced levels of apolipoproteins A and D. The hepatocytes are distended by glycogen and fat with large and prominent lipid vacuoles. There is little associated fibrosis.

All these findings apply to both types Ia and Ib disease, but type Ib has the additional feature of recurrent bacterial infections due to neutropenia and impaired neutrophil function. Oral and intestinal mucosa ulcerations are common, and inflammatory bowel disease may occur.

Long-term complications Although type I glycogen storage disease mainly affects the liver, multiple organ systems are also involved. Gout usually becomes symptomatic around puberty as a result of the long-term hyperuricemia. Puberty is often delayed, but fertility appears to be normal. Hypertriglyceridemia causes an increased risk of pancreatitis, but premature atherosclerosis has not been documented. Impaired platelet aggregation may reduce the risk of atherosclerosis.

By the second or third decade of life, most patients with type I glycogen storage disease develop hepatic adenomas that can hemorrhage and, in rare cases, may become malignant. Other complications include pulmonary hypertension and osteoporosis.

Renal disease is a late complication, and almost all patients older than 20 years have proteinuria. Many have hypertension, kidney stones, nephrocalcinosis, and altered creatinine clearance. Glomerular hyperfiltration, increased renal plasma flow, and microalbuminuria can occur before the onset of gross proteinuria. In young patients, hyperfiltration and hyperperfusion may be the only signs of renal abnormalities. With advanced renal disease, focal segmental glomerulosclerosis and interstitial fibrosis are evident on biopsy. In some patients, renal function deteriorates and progresses to failure, requiring dialysis or transplantation. Other abnormalities in renal function include amyloidosis, Fanconi-like syndrome, and distal renal tubular acidification defect. The increases in renal perfusion and maternal blood volume that normally occur in pregnancy can exacerbate renal problems. In addition, hypoglycemia may also become more difficult to control.

Diagnosis The diagnosis of type I disease can be suspected on the basis of clinical presentation and abnormal plasma lactate and lipid values. In addition, administration of glucagon or epinephrine causes little or no rise in blood glucose but increases lactate levels significantly. Before the glucose 6-phosphatase and glucose 6-phosphate translocase genes were cloned, a definitive diagnosis required a liver biopsy to demonstrate a deficiency. Gene-based mutation analysis now provides a noninvasive way of diagnosis for most patients with types Ia and Ib disease.

TREATMENT

Treatment is designed to maintain normal blood glucose levels and is achieved by continuous nasogastric infusion of glucose or oral administration of uncooked cornstarch. Nasogastric drip feeding in early infancy may consist of an elemental enteral

formula or may contain only glucose to maintain normoglycemia during the night; frequent feedings with a high-carbohydrate content are given during the day.

Uncooked cornstarch acts as a slow-release form of glucose and can be given at a dose of 1.6 g/kg every 4 h for infants younger than 2 years. As the child grows older, the cornstarch regimen can be changed to every 6 h, and it can be given by mouth as a liquid (1:2, weight:volume) at a dose of 1.75 to 2.5 g/kg of body weight. Because fructose and galactose cannot be converted to free glucose, their dietary intake should be restricted, and dietary supplements of multivitamins and calcium are required. Allopurinol is given to lower the levels of uric acid. In patients with type Ib disease, granulocyte and granulocyte-macrophage colony stimulating factors have been used successfully to correct the neutropenia, decrease the severity of bacterial infection, and improve the chronic inflammatory bowel disease.

Before surgery, the bleeding status of the patient should be evaluated, and good metabolic control should be established. Prolonged bleeding time can be corrected by the administration of a constant intravenous glucose infusion for 24 to 48 h before surgery. Vasopressin can be given during surgery to reduce bleeding complications, and normal glucose levels should be maintained throughout surgery.

Prognosis In the past, many patients with type I glycogen storage disease died, and the prognosis was guarded for those who survived. The long-term complications discussed above occur mostly in adults whose disease was not adequately treated during childhood. Early diagnosis and initiation of effective treatment have improved the outcome, but it is not known if all long-term complications can be avoided through good metabolic control.

Type III Glycogen Storage Disease (Debrancher Deficiency, Limit Dextrinosis)

Type III glycogen storage disease is caused by a deficiency of glycogen debranching enzyme. Debranching enzyme and phosphorylase are responsible for complete degradation of glycogen; when debranching enzyme is defective, glycogen breakdown is incomplete, and an abnormal glycogen accumulates that has short outer chains and resembles limit dextrin.

GENETIC CONSIDERATIONS

The type III glycogenoses are inherited as autosomal recessive traits. The disease has been reported in many different ethnic groups, and the frequency is relatively high in non-Ashkenazi Jews of North African descent. The gene for debranching enzyme is located on chromosome 1p21. At least 20 different mutations that cause type III disease have been identified. Two mutations (17delAG and Q6X), both located in exon 3 at amino acid codon 6, are exclusively found in the subtype IIIb. Carrier detection and prenatal diagnosis are possible with DNA-based linkage or mutation analysis.

Clinical and laboratory findings Deficiency of glycogen debranching enzyme causes hepatomegaly, hypoglycemia, short stature, variable skeletal myopathy, and cardiomyopathy. The disorder usually involves both liver and muscle and is termed *type IIIa glycogen storage disease*. However, in about 15% of patients, the disease appears to involve only the liver and is classified as *type IIIb*.

During infancy and childhood, the disease may be almost indistinguishable from type I disease because hepatomegaly, hypoglycemia, hyperlipidemia, and growth retardation are common features of both. Splenomegaly may be present, but the kidneys are not enlarged in type III disease. Remarkably, hepatomegaly and hepatic symptoms in most patients with type III disease improve with age and usually disappear after puberty. However, progressive liver cirrhosis with failure may occur and seems especially common in Japanese patients.

In patients with muscle involvement (type IIIa), muscle weakness is usually minimal during childhood but can become severe during the third or fourth decade of life, as evidenced by slowly progressive weakness and muscle wasting. Electromyographic (EMG) changes are consistent with a widespread myopathy, and nerve conduction may be abnormal. Ventricular hypertrophy is frequent, but overt cardiac dysfunction is rare. Hepatic symptoms may be so mild that the diagnosis is not made until adulthood, when neuromuscular disease becomes manifest. Polycystic ovaries appear to be a common finding in female patients; fertility, however, does not seem to be affected.

Hypoglycemia, hyperlipidemia, and elevated liver transaminases occur in children. In contrast to type I disease, fasting ketosis is prominent, and blood lactate and uric acid concentrations are usually normal. The administration of glucagon 2 h after a carbohydrate meal causes a normal rise of blood glucose, but after an overnight fast glucagon may provoke no change in blood glucose. Serum creatine kinase levels can sometimes be used to identify patients with muscle involvement, but normal levels do not rule out muscle enzyme deficiency.

The histology of the liver is characterized by a universal distention of hepatocytes by glycogen and by the presence of fibrous septa. The fibrosis and the paucity of fat distinguish type III from type I glycogenosis. The fibrosis can range from minimal periportal fibrosis to micronodular cirrhosis.

Diagnosis In type IIIa glycogen storage disease, deficient debranching enzyme activity can be demonstrated in liver, skeletal muscle, and heart. In contrast, patients with type IIIb have debranching enzyme deficiency in the liver but not in muscle. In the past, definitive assignment of subtype required enzyme assays in both liver and muscle. DNA-based analyses now provide a noninvasive way of subtyping these disorders in most patients.

TREATMENT

Dietary management of type III disease is less demanding than that of type I. If hypoglycemia is present, frequent high-carbohydrate meals with cornstarch supplements or nocturnal gastric drip feedings are usually effective. A high-protein diet during the day plus overnight protein enteral infusion may be tried in patients with myopathy, but it is not established whether such a regimen is effective. Patients do not need to restrict dietary intake of fructose and galactose, as do those with type I disease.

Prognosis Liver symptoms improve with age and usually disappear after puberty. Cirrhosis of the liver may occur later in life. In type IIIa disease, muscle weakness and

atrophy worsen during adulthood.

Type VI Glycogen Storage Disease [Liver Phosphorylase Deficiency (Hers Disease)] The number of patients with enzymatically documented liver phosphorylase deficiency is small. It appears that patients with liver phosphorylase deficiency have a benign course. These patients present with hepatomegaly and growth retardation early in childhood. Hypoglycemia, hyperlipidemia, and hyperketosis are usually mild if present. Plasma lactic acid and uric acid levels are normal. The heart and skeletal muscles are not involved. The hepatomegaly and growth retardation improve with age and usually disappear at puberty. Treatment is symptomatic. A high-carbohydrate diet and frequent feeding are effective in preventing hypoglycemia, but most patients require no specific treatment. The liver phosphorylase gene is located on chromosome 14q21. A splicing site mutation in intron 13 has been identified in a large Mennonite kindred, and four other mutations have been found in patients with different ethnic backgrounds.

Type IX Glycogen Storage Disease (Liver Phosphorylase Kinase Deficiency) Defects of phosphorylase kinase cause a heterogeneous group of glycogenoses. The heterogeneity is due to the complexity of the phosphorylase kinase enzyme complex. It consists of four subunits (α, β, γ, and δ), each encoded by different genes (X chromosome as well as autosomes) that are differentially expressed in various tissues. Phosphorylase kinase deficiency can be divided into several subtypes on the basis of the gene/subunit involved, the tissues that are primarily affected, and the mode of inheritance.

Subtypes of Phosphorylase Kinase Deficiency

X-LINKED LIVER PHOSPHORYLASE KINASE DEFICIENCY X-linked liver phosphorylase kinase deficiency is one of the most common liver glycogenoses. Phosphorylase kinase activity may also be deficient in erythrocytes and leukocytes but is normal in muscle. Typically, a child between the ages of 1 and 5 years presents with growth retardation and hepatomegaly. Levels of cholesterol, triglycerides, and liver enzymes are mildly elevated. Ketosis may occur after fasting. Lactic and uric acid levels are normal. Hypoglycemia is mild, if present. The rise in the blood glucose level after the administration of glucagon is normal. Hepatomegaly and abnormal blood chemistries gradually return to normal with age. Most adults achieve a normal final height and are practically asymptomatic, despite a persistent phosphorylase kinase deficiency.

Liver histology shows glycogen-distended hepatocytes. The accumulated glycogen (α particles, rosette form) has a frayed or burst appearance and is less compact than in type I or type III disease. Fibrous septa and low-grade inflammatory changes may be present.

The structural gene for the liver isoform of the phosphorylase kinase α-subunit is located on chromosome Xp22, and mutations of this gene have been found in the disorder. Subtle mutations tend to retain the phosphorylase kinase activity in blood cells, whereas nonsense mutations cause enzyme deficiency in both liver and blood cells.

AUTOSOMAL LIVER AND MUSCLE PHOSPHORYLASE KINASE DEFICIENCY An autosomal recessive form of liver and muscle phosphorylase kinase deficiency has

been reported in several patients. As in the X-linked form of the disorder, hepatomegaly and growth retardation are the predominant symptoms in early childhood. Some patients also exhibit muscle hypotonia and have reduced activity of phosphorylase kinase in muscle. This form of the phosphorylase kinase deficiency is caused by mutations in the β -subunit of the gene located on chromosome 16q12-13.

AUTOSOMAL LIVER PHOSPHORYLASE KINASE DEFICIENCY In contrast to the benign course of X-linked phosphorylase kinase deficiency, patients with the autosomal recessive form of liver phosphorylase kinase deficiency have more severe phenotypes and often develop liver cirrhosis. This form of phosphorylase kinase deficiency is due to mutations in the testis/liver isoform of the α -subunit gene located on chromosome 16p.

MUSCLE-SPECIFIC PHOSPHORYLASE KINASE DEFICIENCY Muscle-specific phosphorylase kinase deficiency causes cramps and myoglobinuria on exercise or progressive muscle weakness and atrophy. The activity of the enzyme is decreased in muscle but normal (when determined) in liver and blood cells. There is no hepatomegaly or cardiomegaly. The disorder may be due to mutation in the muscle isoform of the α -subunit located on the X chromosome.

CARDIAC-SPECIFIC PHOSPHORYLASE KINASE DEFICIENCY Several sporadic cases of cardiac-specific phosphorylase kinase deficiency have been reported. All patients died during infancy from cardiac failure due to massive glycogen deposition in the myocardium. The molecular basis has not been defined.

Diagnosis Definitive diagnosis of phosphorylase kinase deficiency requires demonstration of the enzymatic defect in affected tissues. Although phosphorylase kinase can be measured in leukocytes and erythrocytes, the enzyme has many tissue-specific isozymes, and the diagnosis can be missed without studies of the liver, muscle, or heart.

TREATMENT

The treatment for liver phosphorylase or phosphorylase kinase deficiency is based on symptoms. A high-carbohydrate diet and frequent feedings are effective in preventing hypoglycemia, but most patients require no specific treatment. Prognosis is usually good; adult patients have normal stature and minimal hepatomegaly. There is no treatment for the fatal form of isolated cardiac phosphorylase kinase deficiency.

Type 0 Glycogen Storage Disease (Glycogen Synthase Deficiency) Strictly speaking, type 0 is not a glycogen storage disease, as the deficiency of the enzyme leads to decreased glycogen stores. The patients present in early infancy with morning drowsiness and fatigue and sometimes convulsions associated with hypoglycemia and hyperketonemia. There is no hepatomegaly or hyperlipidemia. Prolonged hyperglycemia and elevated lactate levels with normal insulin levels after administration of glucose suggest a possible diagnosis of glycogen synthase deficiency. Definitive diagnosis requires a liver biopsy to measure the enzyme activity. Treatment is symptomatic and involves frequent feedings rich in protein and nighttime supplements of uncooked cornstarch to alleviate hypoglycemia. Prognosis is good as patients survive to adulthood with a resolution of hypoglycemia except during pregnancy. Mutations in the liver

glycogen synthase gene (located on chromosome 12p12.2) that cause glycogen synthase deficiency have been identified.

Type XI Glycogen Storage Disease (Hepatic Glycogenosis with Renal Fanconi Syndrome, Fanconi-Bickel Syndrome) This rare autosomal recessive disease is caused by defects in the facilitative glucose transporter 2 (GLUT-2) which transports glucose in and out of hepatocytes, pancreatic cells, and the baso-lateral membranes of intestinal and renal epithelial cells. The disease is characterized by proximal renal tubular dysfunction, impaired glucose and galactose utilization, and accumulation of glycogen in liver and kidney.

GENETIC CONSIDERATIONS

The low prevalence of Fanconi-Beckel syndrome (fewer than 100 cases reported worldwide) is underscored by the fact that consanguinity is found in 70% of the patients with a detectable [GLUT-2](#) mutation. The gene for GLUT-2 is located on chromosome 3q26 and most mutations detected so far cause premature termination of translation.

Clinical and laboratory findings The affected child presents in the first year of life with failure to thrive, rickets, and a protuberant abdomen due to liver and kidney enlargement. Laboratory findings include glucosuria, phosphaturia, generalized aminoaciduria, bicarbonate wasting, hypophosphatemia, increased serum alkaline phosphatase levels, and radiologic findings of rickets. Mild fasting hypoglycemia and hyperlipidemia may be present. Liver transaminases, plasma lactate, and uric acid levels are usually normal. Oral galactose or glucose tolerance tests show intolerance to these sugars, which may be caused by the functional loss of [GLUT-2](#), which prevents liver uptake of these sugars. Tissue biopsies show marked accumulation of glycogen in hepatocytes and proximal renal tubular cells, presumably due to the altered transport of glucose out of these organs.

TREATMENT

There is no specific therapy. Growth retardation persists through adulthood. Symptomatic replacement of water, electrolytes, and vitamin D, restriction of galactose intake, and a diabetes mellitus-like diet, presented in frequent and small meals with a cornstarch supplement, may improve growth.

DISORDERS ASSOCIATED WITH LIVER CIRRHOSIS

Type IV Glycogen Storage Disease (Branching Enzyme Deficiency, Amylopectinosis, or Andersen Disease) Deficiency of branching enzyme activity results in accumulation of an abnormal glycogen with poor solubility. The disease is referred to as *type IV glycogen storage disease or amylopectinosis*, because the abnormal glycogen has fewer branch points, more 1-4 linked glucose units, and longer outer chains, resulting in a structure resembling amylopectin.

GENETIC CONSIDERATIONS

Type IV glycogen storage disease is a rare autosomal recessive disease. Prenatal

diagnosis is available with use of cultured amniocytes or chorionic villi to measure the level of enzymatic activity. The glycogen branching enzyme gene is located on chromosome 3p12. Both hepatic and neuromuscular forms of the disease are caused by mutations in the same branching enzyme gene; its characterization in individual patients may be useful in predicting the clinical course.

Clinical and laboratory findings This disorder is clinically variable. The most common form is characterized by progressive cirrhosis of the liver and is manifest in the first 18 months of life as hepatosplenomegaly and failure to thrive. The cirrhosis progresses to cause portal hypertension, ascites, esophageal varices, and liver failure that leads to death by age 5. Less frequently, patients survive without progression of liver disease.

Tissue deposition of amylopectin-like materials can be demonstrated in liver, heart, muscle, skin, intestine, brain, spinal cord, and peripheral nerve. The histologic findings in the liver are characterized by both micronodular cirrhosis and faintly stained basophilic inclusions in the hepatocytes. The inclusions consist of coarsely clumped, stored material that is periodic acid Schiff-positive and partially resistant to diastase digestion. Electron microscopy shows, in addition to the conventional c and b glycogen particles, an accumulation of fibrillar aggregations typical of amylopectin. Definitive diagnosis requires demonstration that branching enzyme activity is deficient in liver, muscle, cultured skin fibroblasts, or leukocytes.

A neuromuscular form of type IV glycogen storage disease has also been reported. Patients with this disease may present (1) at birth with severe hypotonia, muscle atrophy, and neuronal involvement and die during the neonatal period; (2) in late childhood with myopathy or cardiomyopathy; or (3) as adults with diffuse central and peripheral nervous system dysfunction accompanied by accumulation of polyglucosan bodies in the nervous system (so-called adult polyglucosan body disease). Definitive diagnosis of the adult disease requires assay of the branching enzyme in leukocytes or nerve biopsy, as the deficiency is limited to those tissues.

TREATMENT

There is no specific treatment for type IV glycogen storage disease. For progressive hepatic failure, liver transplantation has been performed. However, caution should be taken in selecting patients for liver transplantation because a nonprogressive hepatic form of the disease exists, and extra hepatic manifestations of the disease may occur after transplantation.

GLYCOGEN STORAGE DISEASES: MUSCLE GLYCOGENOSES

DISORDERS WITH MUSCLE-ENERGY IMPAIRMENT

Type V Glycogen Storage Disease (Muscle Phosphorylase Deficiency, McArdle Disease) Deficiency of muscle phosphorylase is the prototype muscle-energy disorder. Deficiency of this enzyme in muscle limits ATP generation by glycogenolysis and results in glycogen accumulation.

GENETIC CONSIDERATIONS

Type V glycogen storage disease is an autosomal recessive disorder that does not appear to have ethnic predilection. The gene for muscle phosphorylase is located on chromosome 11q13. The most common mutation in patients in the United States is a nonsense mutation that changes an arginine to a stop codon (R49X), and the most common mutation in the Japanese is deletion of a single codon (F708). These features allow DNA-based diagnosis and carrier detection in these two populations.

Clinical and laboratory findings Symptoms usually develop first in adulthood and are characterized by exercise intolerance with muscle cramps. Two types of activity tend to cause symptoms: (1) brief exercise of great intensity, such as sprinting or carrying heavy loads; and (2) less intense but sustained activity, such as climbing stairs or walking uphill. Moderate exercise, such as walking on level ground, can be performed by most patients for long periods. Many patients experience a characteristic "second wind" phenomenon; if they rest briefly at the first appearance of muscle pain, they can resume exercise with more ease. About half of patients report burgundy-colored urine after exercise, the consequence of myoglobinuria secondary to rhabdomyolysis. Intense myoglobinuria after vigorous exercise may cause renal failure. Although most patients are diagnosed in the second or third decade, many report weakness and lack of endurance since childhood. In rare cases, EMG findings may suggest an inflammatory myopathy, and the diagnosis can be confused with polymyositis.

The level of serum creatine kinase is usually elevated at rest and increases more after exercise. Exercise also increases the levels of blood ammonia, inosine, hypoxanthine, and uric acid. The latter abnormalities are attributed to accelerated recycling of muscle purine nucleotides in the face of insufficient ATP production.

Diagnosis Lack of an increase in blood lactate and exaggerated blood ammonia elevations after an ischemic exercise test are indicative of muscle glycogenosis and suggest a defect in the conversion of glycogen or glucose to lactate. The abnormal exercise response, however, is not limited to type V disease and can occur with other defects in glycogenolysis or glycolysis, such as deficiencies of muscle phosphofructokinase or debranching enzyme (when the test is done after fasting). Definitive diagnosis is made by enzymatic assay in muscle tissue or by mutation analysis of the myophosphorylase gene.

TREATMENT

In general, avoidance of strenuous exercise can prevent major episodes of rhabdomyolysis. Exercise tolerance can be augmented by aerobic training or by ingestion of glucose or fructose. A high-protein diet may increase exercise endurance in some patients. Longevity does not appear to be affected.

Type VII Glycogen Storage Disease (Muscle Phosphofructokinase Deficiency, Tarui Disease) Type VII disease is caused by a deficiency of muscle phosphofructokinase, which catalyzes the conversion of fructose-6-phosphate to fructose-1,6-diphosphate and is a key regulatory enzyme of glycolysis.

Phosphofructokinase is composed of three isozyme subunits (M, muscle; L, liver; and P,

platelet), which are encoded by different genes and are differentially expressed in tissues. Skeletal muscle contains only M isozyme, and red blood cells contain a hybrid of L and M forms. Type VII disease is due to defective M isoenzyme, which causes complete enzyme deficiency in muscle and partial deficiency in red blood cells.

GENETIC CONSIDERATIONS

Type VII glycogen storage disease is inherited as an autosomal recessive trait. The disease appears to be rare, and most reported patients are either Ashkenazi Jews or Japanese. The gene for the M isoenzyme is located on chromosome 12q13.3. In Ashkenazi Jews, 95% of mutant alleles are either a splicing defect or a nucleotide deletion.

Clinical and laboratory findings The features are similar to those in type V disease, namely, early onset of fatigue and pain with exercise. Vigorous exercise causes severe muscle cramps and myoglobinuria. However, several features of type VII disease are distinctive. (1) Exercise intolerance is usually evident in childhood, is more severe than in type V disease, and may be associated with nausea and vomiting. (2) A compensated hemolysis occurs as evidenced by an increased level of serum bilirubin and reticulocyte count. (3) Hyperuricemia is common and becomes more marked after exercise. (4) An abnormal glycogen resembling amylopectin is present in muscle fibers; it is periodic acid Schiff-positive and resistant to diastase digestion. (5) Exercise intolerance is particularly acute after meals rich in carbohydrate because glucose cannot be utilized in muscle and because the ingested glucose inhibits lipolysis and thus deprives muscle of fatty acid and ketone substrates. In contrast, patients with type V disease can metabolize glucose derived from either liver glycogenolysis or exogenous glucose. Indeed, glucose infusion improves exercise tolerance in patients with type V disease.

Two rare type VII variants have been reported. One begins in infancy with hypotonia and limb weakness, and a rapidly progressive myopathy leads to death by age 4. The other occurs in adults and is characterized by a slowly progressive, fixed muscle weakness rather than by cramps and myoglobinuria.

Diagnosis The M isoenzyme defect must be demonstrated in muscle, red blood cells, or cultured skin fibroblasts by biochemical or histochemical techniques.

TREATMENT

There is no specific treatment. Avoidance of strenuous exercise prevents acute attacks of muscle cramps and myoglobinuria.

Other Muscle Glycogenoses with Muscle-Energy Impairment Five additional enzyme defects produce muscle glycogenoses, namely, deficiencies in phosphoglycerate kinase, phosphoglycerate mutase, lactate dehydrogenase, fructose 1,6-bisphosphate aldolase A, and pyruvate kinase. All five enzymes affect terminal glycolysis, and deficiency causes muscle-energy impairment similar to that in type V and type VII disease. The failure of blood lactate to increase in response to exercise can be used to separate muscle glycogenoses from disorders of lipid metabolism, such as carnitine palmitoyl transferase II deficiency and very long chain acyl-coenzyme A

dehydrogenase deficiency, which also cause muscle cramps and myoglobinuria. Muscle glycogen levels may be normal in the disorders affecting terminal glycolysis, and definitive diagnosis is made by assaying the enzymatic activity in muscle.

DISORDERS WITH PROGRESSIVE SKELETAL MUSCLE MYOPATHY AND/OR CARDIOMYOPATHY

Type II Glycogen Storage Disease (Acid α -1,4 Glucosidase Deficiency, Pompe Disease) (See also [Chap. 349](#)) Type II disease is caused by a deficiency of lysosomal acid α -1,4 glucosidase (acid maltase), an enzyme responsible for the degradation of glycogen in lysosomal vacuoles. It is characterized by the accumulation of glycogen in lysosomes as opposed to its accumulation in cytoplasm in the other glycogenoses.

GENETIC CONSIDERATIONS

Pompe disease is an autosomal recessive disorder and does not appear to have an ethnic predilection. The gene for acid α -glucosidase is on chromosome 17q25. A splice site mutation (IVS1-13TG) is common in patients with adult-onset disease. Prenatal diagnosis with amniocytes or chorionic villi is available.

Clinical and laboratory findings The disorder encompasses a range of phenotypes, each including myopathy but differing in age of onset, organ involvement, and clinical severity. The most severe is the infantile-onset disease with cardiomegaly, hypotonia, and death before 1 year of age. Infants appear normal at birth but soon develop generalized muscle weakness with feeding difficulties, macroglossia, hepatomegaly, and congestive heart failure due to a hypertrophic cardiomyopathy. Electrocardiographic findings include a high-voltage QRS complex and a shortened PR interval. Death usually occurs from cardiorespiratory failure.

The juvenile or late-childhood form is characterized by skeletal muscle manifestations, usually without cardiac involvement, and a slowly progressive course. The juvenile form typically presents as delayed motor milestones (if age of onset is early enough) and difficulty in walking; these manifestations are followed by swallowing difficulties, proximal muscle weakness, and respiratory muscle involvement. This form can cause death before the end of the second decade.

An adult form of type II disease presents as a slowly progressive myopathy without cardiac involvement and has its onset between the second and seventh decades. The clinical picture is dominated by slowly progressive proximal muscle weakness with truncal involvement. The pelvic girdle, paraspinal muscles, and the diaphragm are most seriously affected. The initial symptoms may be respiratory insufficiency manifested by somnolence, morning headache, orthopnea, and exertional dyspnea.

Laboratory findings include elevated levels of serum creatine kinase, aspartate transaminase, and lactate dehydrogenase, particularly in infants. Muscle biopsy shows the presence of vacuoles that stain positively for glycogen, and muscle acid phosphatase is increased, presumably from a compensatory increase of lysosomal enzymes. Electron microscopy reveals the glycogen accumulation. EMG reveals myopathic features with irritability of muscle fibers and pseudomyotonic discharges.

Serum creatine kinase is not always elevated in adults and, depending on the muscle biopsied or tested, muscle histology or EMG may not be abnormal. It is prudent to examine affected muscle.

Diagnosis Diagnosis can be established by demonstration of the absence or reduced levels of acid α -glucosidase activity in muscle or cultured skin fibroblasts. Deficiency is usually more severe in the infantile form than in the juvenile and adult disorders.

TREATMENT

No effective treatment for the infantile form is currently available. Enzyme replacement is a promising therapy for this fatal lysosomal storage disease, and clinical trials are ongoing to test its safety and efficacy. A high-protein diet may be useful for the juvenile and adult forms. Nocturnal ventilatory support can improve the quality of life.

DISORDERS OF GALACTOSE METABOLISM

GALACTOSE 1-PHOSPHATE URIDYL TRANSFERASE DEFICIENCY GALACTOSEMIA

"Classic" galactosemia is a serious disease with an early onset of symptoms; the incidence is 1 in 60,000. The newborn infant normally receives up to 20% of caloric intake as lactose, which consists of glucose and galactose. Without the transferase, the infant is unable to metabolize galactose 1-phosphate ([Fig. 350-1](#)), the accumulation of which results in injury to parenchymal cells of the kidney, liver, and brain.

GENETIC CONSIDERATIONS

Galactosemia caused by transferase deficiency is inherited as an autosomal recessive trait; there are several enzymatic variants. The Duarte variant exhibits diminished red cell enzyme activity that is usually of no clinical significance. Some African-American patients have milder symptoms despite the absence of measurable transferase activity in erythrocytes; these patients retain 10% of the enzyme activity in liver and intestinal mucosa. Most Caucasian patients have no detectable enzyme activity in any of these tissues. The gene for galactose-1 phosphate uridyl transferase is located on chromosome 9p13. In African Americans, 48% of the mutant alleles are represented by the S135L substitution, perhaps accounting for the milder phenotype. In the Caucasian population, 70% of mutant alleles are represented by the Q188R substitution. Carrier testing and prenatal diagnosis can be carried out by direct enzyme analysis of amniocytes or chorionic villi. DNA-based testing can also be performed.

Clinical and laboratory findings The clinical manifestations of uridyl transferase deficiency are myriad, necessitating a high index of suspicion if the diagnosis is to be made. Clinical features may include: jaundice, hepatomegaly, vomiting, hypoglycemia, convulsions, lethargy, irritability, feeding difficulties, poor weight gain, aminoaciduria, cataracts, vitreous hemorrhage, hepatic cirrhosis, ascites, splenomegaly, or mental retardation. Patients with galactosemia are at increased risk for *Escherichia coli* neonatal sepsis; the onset of sepsis often precedes the diagnosis of galactosemia. When the diagnosis is not made at birth, damage to the liver (cirrhosis) and brain

(mental retardation) becomes increasingly severe and irreversible. For this reason, routine neonatal screening tests for galactosemia have been instituted in many parts of the world.

Diagnosis The preliminary diagnosis of galactosemia is made by demonstrating a reducing substance (by Clinitest) in urine specimens collected while the patient is receiving human or cow's milk or another formula containing lactose. The reducing substance found in urine, which is negative in a glucose oxidase test, can be identified as galactose with chromatography or an enzymatic test specific for galactose. Definitive diagnosis requires the demonstration of deficient activity of galactose-1-phosphate uridyl transferase in erythrocytes or other tissues, which also exhibit increased concentrations of galactose-1-phosphate.

TREATMENT

Because of widespread newborn screening for galactosemia, patients are now being identified and treated early. Elimination of galactose from the diet reverses growth failure, renal, and hepatic dysfunction. Cataracts regress and most patients have no impairment of eyesight. Early diagnosis and treatment have improved the prognosis of galactosemia; on long-term follow-up, however, patients still have ovarian failure manifest as primary or secondary amenorrhea, as well as developmental delay and learning disabilities, which increase in severity with age. In addition, most patients have speech disorders and a smaller number demonstrate poor growth and impaired motor function and balance (with or without overt ataxia). The relative control of galactose-1-phosphate levels does not always correlate with long-term outcome, suggesting that other factors, such as uridine diphosphate (UDP)-galactose deficiency (a donor for galacto-lipids and proteins), may be responsible for some of the metabolic consequences of the disease.

GALACTOKINASE DEFICIENCY

In contrast to the multiple systems that are affected in uridyl transferase deficiency, cataracts are usually the sole manifestation of galactokinase deficiency. The affected infant is otherwise asymptomatic. This disorder is characterized by elevated blood galactose levels with normal uridyl transferase activity and an absence of galactokinase activity in erythrocytes. Treatment is dietary restriction of galactose intake. Mutations leading to galactokinase deficiency have been identified in the gene coding for galactokinase (located on chromosome 17q24).

URIDINE DIPHOSPHATE GALACTOSE 4-EPIMERASE (UDP GAL 4-EPIMERASE) DEFICIENCY

The abnormally accumulated metabolites in this disorder are very much like those seen in uridyl transferase deficiency; however, there is also an increase in cellular UDP galactose. There are two distinct forms of epimerase deficiency. A benign form was discovered incidentally as a result of the neonatal screening program. Affected persons in this case are healthy; the enzyme deficiency is limited to leukocytes and erythrocytes, without deranged metabolism in other tissues, and no treatment is required. The second form of epimerase deficiency is severe with clinical manifestations resembling uridyl

transferase deficiency. Additional features include hypotonia and nerve deafness. The enzyme deficiency is generalized, and clinical symptoms respond to restriction of dietary galactose. Although this form of galactosemia is very rare, it must be considered in a symptomatic patient who has normal transferase activity. The gene for epimerase is located on chromosome 1p35-36; mutations responsible for both forms of the epimerase deficiency have been identified.

DISORDERS OF FRUCTOSE METABOLISM

DEFICIENCY OF FRUCTOKINASE (BENIGN FRUCTOSURIA)

This condition is not associated with any clinical manifestations. It is an incidental finding, usually made through the detection of fructose as a reducing substance in the urine. No treatment is necessary.

DEFICIENCY OF FRUCTOSE 1,6-BISPHOSPHATE ALDOLASE (ALDOLASE B) (HEREDITARY FRUCTOSE INTOLERANCE)

This is a severe disease of infants that appears with the ingestion of fructose-containing food. It is caused by deficiency of fructose 1,6-bisphosphate aldolase B activity in the liver, kidney, and intestine. The enzyme catalyzes the hydrolysis of fructose 1-phosphate and fructose 1,6-bisphosphate into the 3-carbon sugars, dihydroxyacetone phosphate, glyceraldehyde-3-phosphate, and glyceraldehyde ([Fig. 350-1](#)). Deficiency of this enzyme activity causes rapid accumulation of fructose 1-phosphate and initiates severe toxic symptoms when exposed to fructose.

GENETIC CONSIDERATIONS

The true incidence of hereditary fructose intolerance is not known but may be as high as 1 in 23,000. The gene for aldolase B is on chromosome 9q22.3. Several mutations causing hereditary fructose intolerance have been identified. A single missense mutation, which results in substitution of proline for alanine at position 149, is the most common mutation identified in northern Europeans. This mutation plus two other point mutations account for approximately 80 to 85% of hereditary fructose intolerance in Europe and the United States. The diagnosis of hereditary fructose intolerance can thus be made in most cases by direct DNA analysis. Prenatal diagnosis should be possible from either amniocentesis or chorionic villi with the use of DNA for mutational or linkage analysis.

Clinical and laboratory findings Patients with fructose intolerance are healthy and asymptomatic until fructose or sucrose (table sugar) is ingested (usually from fruit, fruit juice, or sweetened cereal). Clinical manifestations may resemble those of galactosemia and include jaundice, hepatomegaly, vomiting, lethargy, irritability, and convulsions. Laboratory findings include prolonged clotting time, hypoalbuminemia, elevation of bilirubin and transaminases, and proximal renal tubular dysfunction. If the disease is not diagnosed and intake of the noxious sugar persists, hypoglycemic episodes recur, and liver and kidney failure progress, eventually leading to death.

Diagnosis Suspicion of the enzyme deficiency is suggested by the presence of a reducing substance in the urine during an attack. The diagnosis is supported by an

intravenous fructose tolerance test, which will cause a rapid decline of serum phosphate, followed by blood glucose, and a subsequent rise of uric acid and magnesium. An oral tolerance test should not be performed as patients may become acutely ill. Definitive diagnosis is made by assay of fructaldolase B activity in the liver.

TREATMENT

Treatment consists of the complete elimination of all sources of sucrose, fructose and sorbitol from the diet. With this treatment, liver and kidney dysfunction improve, and catch-up growth is common. Intellectual development is usually unimpaired. As the patient matures, symptoms become milder, even after fructose ingestion, and the long-term prognosis is good. Owing to dietary avoidance of sucrose, affected patients have few dental caries.

FRUCTOSE 1,6-DIPHOSPHATASE DEFICIENCY

Fructose 1,6-diphosphatase deficiency is a defect in gluconeogenesis. The disease is characterized by life-threatening episodes of acidosis, hypoglycemia, hyperventilation, convulsions, and coma. These episodes are triggered by febrile infections and gastroenteritis when oral food intake decreases. Laboratory findings include low blood glucose, high lactate and uric acid levels, and a blood gas picture of metabolic acidosis. In contrast to hereditary fructose intolerance, there is usually no aversion to sweets, and renal tubular and liver functions are normal. Treatment of acute attacks consists of correction of hypoglycemia and acidosis by intravenous infusion; the response is usually rapid. Later, avoidance of fasting and elimination of fructose and sucrose from the diet prevent further episodes. For long-term prevention of hypoglycemia, a slowly released carbohydrate such as cornstarch is useful. Patients who survive childhood seem to develop normally.

Diagnosis The diagnosis is established by demonstrating enzyme deficiency in either the liver or an intestinal biopsy specimen. The enzyme defect may also be demonstrated in leukocytes in some cases. The gene coding for fructose 1,6-diphosphatase is located on chromosome 9q22. In patients with known mutations, carrier detection and prenatal diagnosis are possible by DNA-based testing.

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351. INHERITED DISORDERS OF CONNECTIVE TISSUE - Darwin J. Prockop, Helena Kuivaniemi, Gerard Tromp, Leena Ala-Kokko

Heritable disorders that involve the major connective tissues of the body such as bone, skin, cartilage, blood vessels, and basement membranes are among the most common genetic diseases in human beings. Here we will focus primarily on those disorders that can have severe manifestations, are relatively common, and are sufficiently understood at the molecular level to provide useful paradigms: osteogenesis imperfecta (OI), the Ehlers-Danlos syndrome (EDS), chondrodysplasias (CDs), the Marfan syndrome (MFS), epidermolysis bullosa (EB), and the Alport syndrome (AS).

THE CHALLENGE OF CLASSIFYING THE DISEASES

The original classification of connective tissue diseases was based on the pattern of inheritance, the cluster of signs and symptoms, the histologic changes in tissues, and limited information about the molecular defects involved. This classification included about a dozen types and subtypes for [OI](#), about the same number for the [EDS](#), and over 150 for the [CDs](#). Several limitations in these original classifications are now apparent. One is that the same mutation does not always produce the same disease phenotype in terms of severity of the condition or its clinical course. Such phenotypic variation occurs in many genetic diseases, including the connective tissue disorders, in which some members of a family are severely affected, whereas others with the same mutation have a mild disorder.

Most patients with classic features of a severe connective tissue disease have a mutation in a gene or genes coding for a single protein. For example, the majority of patients with [OI](#) have a mutation in one of the two genes coding for type I procollagen. Similarly, most patients with [MFS](#) have mutations in a gene for fibrillin. For other disease categories, the situation is more complex. In [EDS](#), for example, the type IV variant is usually caused by mutations in the gene for type III procollagen, the type VI variant by defects in the gene for lysyl hydroxylase, and the type VII variant by defects that impair the processing of type I procollagen to type I collagen.

Classifications of these disorders also tend to overemphasize the etiologic differences between severe genetic diseases that are apparent in infants and the more common diseases that appear much later in life. Single-gene defects can cause subsets of late-onset diseases such as osteoporosis, aneurysms, and osteoarthritis. For example, a small subset of patients with postmenopausal osteoporosis have mutations in the genes for procollagen I similar to the mutations in the same genes that produce lethal variants of [OI](#). Likewise, a subset of patients with familial aortic aneurysms have mutations in the gene for procollagen III similar to the mutations in the same gene that cause lethal variants of type IV [EDS](#), and occasional patients with osteoarthritis have mutations in the gene for procollagen II similar to the mutations in the same gene that cause lethal [CDs](#). There is disagreement as to the best diagnosis for such patients in that some investigators feel that after a mutation similar to those seen in the early-onset diseases is identified, the patients should be reclassified as having mild forms of OI, EDS, or CD, even though they do not have definitive evidence of the early-onset diseases or seek medical attention until adulthood. Therefore, the category of diseases referred to as *inherited disorders of connective tissue* may have to be expanded as we

obtain additional information about more common diseases.

DEFINITION AND COMPOSITION OF CONNECTIVE TISSUES

Connective tissues are composed of specific macromolecules, many of which are also constituents of the lung, the kidney, the walls of blood vessels, the vitreous gel of the eye, and the synovial fluid. Indeed, most organs and tissues contain small amounts of the same macromolecules assembled into membranes and septa. Therefore, virtually all structures contain some connective tissue.

The distinguishing feature of connective tissues is that the component macromolecules are assembled into an insoluble extracellular matrix ([Table 351-1](#)). The macromolecules include at least 19 different types of collagens, the related fibrous proteins known as *elastin* and *fibrillin*, a series of proteoglycans, and components whose structure and function are only partially defined.

Differences in the connective tissues of bone, skin, and cartilage are in part explained by differences in the content of specific components ([Table 351-1](#)). For example, tendons and ligaments consist primarily of type I collagen fibrils and small amounts of other components that help organize the type I fibrils into fibers and fiber bundles. Cartilage consists primarily of fibrils of type II collagen in the form of arcades that are distended by highly charged proteoglycans. The extracellular matrix of the aorta contains collagens that provide tensile strength and elastin that provides elasticity. Differences among the connective tissues also depend on the three-dimensional organization of the molecular components. The type I collagen fibrils in tendon are packed into thick, parallel bundles of fibers, whereas type I collagen fibrils in skin are randomly oriented. In cortical bone, helical arrays of type I collagen fibrils are deposited around haversian canals.

BIOSYNTHESIS OF CONNECTIVE TISSUE

Connective tissues form primarily by a process of self-assembly, in which a molecule of the correct size, shape, and surface properties binds to other molecules with the same or similar structure in a spontaneous but ordered manner. The molecular mechanisms and driving forces are similar to those involved in crystal formation.

Collagen Synthesis The self-assembly of connective tissue is illustrated by the assembly of collagen into fibrils. The collagen molecule that forms fibrils is a long, thin rod consisting of three polypeptide chains wrapped into a rigid, ropelike triple helix ([Fig. 351-1](#)). The molecule has a triple-helical conformation, because each of the three chains has a simple, repetitive amino acid sequence of about 1000 amino acids in which glycine (Gly) appears as every third amino acid. Therefore, the sequence of each chain can be designated as $(\text{-Gly-X-Y})_{333}$, where X and Y represent amino acids other than glycine. To fold into a triple helix, every third amino acid in a chain must be glycine, the smallest amino acid, since this residue must fit in a sterically restricted space where the three chains of the triple helix come together. Many of the X- and Y-position amino acids are proline and hydroxyproline, which provide rigidity to the triple helix. The remaining amino acids form clusters of hydrophobic and charged regions on the surface of the molecule that direct how one molecule spontaneously binds to other

collagen molecules and thereby self-assembles into the large collagen fibrils in tissues (see [Fig. 351-1](#)).

More than 19 different collagens have been identified. Most are minor constituents that probably have highly specialized functions. The fibrillar collagens are found in tissues as long, highly ordered fibrils with a characteristic banding pattern by electron microscopy. Type I collagen, the most abundant, is found as cross-striated fibrils in a large number of tissues ([Table 351-1](#)). It is composed of two identical chains called $\alpha 1(I)$ and a third called $\alpha 2(I)$. Type II collagen, another fibrillar collagen of cartilage, is composed of three identical chains called $\alpha 1(II)$. Type III collagen is found in small amounts in many tissues that contain type I collagen and in large amounts in large blood vessels; it is composed of three identical chains called $\alpha 1(III)$. The nonfibrillar collagens are similar to the fibrillar collagens in that they contain -Gly-X-Y- sequences of amino acids that form triple-helical domains, but they also contain large globular domains. Self-assembly of most of the nonfibrillar collagens usually involves binding between the globular domains to form networks. For example, the type IV collagen in basement membranes self-assembles into a complex three-dimensional network that provides a diffusion barrier in the renal glomerulus and pulmonary alveolus. The network also provides support for epithelial and endothelial cells in these tissues and in skin, the gastrointestinal tract, and blood vessels. Some nonfibrillar collagens bind to the surface of fibrils formed by the more abundant collagens and alter the lateral growth of the fibrillar collagens or prevent the fibrils from coalescing into fiber bundles.

Because the molecules or monomers fibrillar collagens spontaneously self-assemble into fibrils, they are first synthesized as larger and more soluble precursors called *procollagens* and composed of pro α chains. As the pro α chains of procollagen are synthesized on ribosomes, the free ends move into the cisternae of the rough endoplasmic reticulum ([Fig. 351-1](#)). Hydrophobic signal peptides at the N termini are cleaved, and additional posttranslational reactions begin. Proline residues in the Y position of the repeating -Gly-X-Y- sequences are converted to hydroxyproline by prolyl hydroxylase in a reaction requiring ascorbic acid. Lysine residues in the Y position are similarly hydroxylated to hydroxylysine by lysyl hydroxylase. Many of the hydroxylysine residues are glycosylated with galactose or with galactose and glucose. A large mannose-rich oligosaccharide is assembled on the C-terminal propeptide of each chain. The association of the pro α chains is directed by the structure and the surface properties of the globular C-propeptides. After the C-propeptides assemble correctly, the structure is locked in place by the formation of interchain disulfide bonds. Posttranslational modifications of the pro α chains continue until each chain acquires about 100 hydroxyproline residues. Then a few of the hydroxyproline-rich -Gly-X-Y- sequences at the C terminus of the protein fold into a triple helix. The short region of triple helix becomes a nucleus for self-assembly of the triple helix of the whole protein, much like a nucleus for crystallization, in that the triple-helical conformation in one -Gly-X-Y- sequence induces the next -Gly-X-Y- sequence to fold into the same conformation. As a result, the conformation is propagated in a zipper-like fashion from the C terminus to the N terminus of the molecule, and the entire α -chain domain becomes a continuous triple helix. The protein then passes from the rough endoplasmic reticulum to other compartments and is secreted. The requirement for ascorbic acid in the hydroxylation of prolyl residues explains why wounds fail to heal in scurvy ([Chap. 75](#)). If sufficient proline residues are not converted to hydroxyproline, collagen cannot

fold into a triple helix that is stable at body temperature. The abnormal protein accumulates in the cisternae of the rough endoplasmic reticulum and is slowly degraded.

After secretion, procollagen is processed to collagen by cleavage of the N-propeptides by procollagen N-proteinase and of the C-propeptides by procollagen C-proteinase. The processing of type I procollagen by the two proteinases converts the precursor to type I collagen and thereby decreases the solubility of the protein about 1000-fold. The 1000-fold decrease in solubility provides the entropic energy that drives the spontaneous self-assembly of the collagen into fibrils. Collagen monomers first assemble into a nucleus that grows by addition of monomers as determined by the structure of the nucleus. The nucleus and the final fibril can be assembled spontaneously from a single kind of collagen such as type I or type II. Fibrils can also be assembled as copolymers in which two or more collagens are incorporated into the same fibril simultaneously. Alternatively, a second collagen or a proteoglycan can bind to the surface of a growing fibril or to a fully formed fibril and thereby influence the final structure and the functional properties of the fibrils. The final structure of the fibrils in tissues is also influenced by the pressure and tensions on the fibrils, particularly after their tips are inserted into muscle and bone. The tension on tendons, for example, probably makes the initial thin fibrils coalesce into large fiber bundles.

Self-assembled collagen fibers have considerable tensile strength, which is increased by cross-linking reactions that form covalent bonds between a chains in one molecule and a chains in adjacent molecules. The first step in cross-linking is oxidation by lysyl oxidase of amino groups on a few lysine or hydroxylysine residues to form aldehydes that interact to form stable covalent bonds.

During growth and development, the collagen fibrils in all tissues undergo repeated synthesis, degradation, and resynthesis. The degradation of collagen fibers in tissues is initiated by specific collagenases found in leukocytes, fibroblasts, synovial cells, or related cell types. The collagenases cleave the collagen molecule at a point about three-quarters of the distance from its N terminus. The cleavage apparently triggers unfolding of the molecules on the surface of a fibril and further degradation by other proteinases.

Collagen fibers in most tissues of normal adults undergo very little metabolic turnover. One exception to this is the collagen fibrils that are degraded and resynthesized as part of the continual remodeling of bone. Although the collagen in many adult tissues is metabolically stable, the rate of turnover changes under some circumstances. In starvation, a large fraction of the collagen in skin and other connective tissues is degraded, thus providing amino acids for gluconeogenesis. Large losses of collagen also occur in most connective tissues during immobilization or prolonged periods of low-gravitational stress. In rheumatoid arthritis, pannus invasion causes a rapid degradation of collagen in the articular cartilage. Glucocorticoids decrease the collagen content of most connective tissues, including bone, by decreasing the rate of collagen synthesis. Decreases in collagen weaken tissues. In many pathologic states, however, collagen is deposited in excess. With injury to tissue, inflammation is usually followed by increased deposition primarily of type I collagen fibrils in the form of fibrotic tissue and scars. The deposition of collagen fibrils during the repair process is largely irreversible

and is a major feature of the pathologic changes in hepatic cirrhosis, pulmonary fibrosis, atherosclerosis, and nephrosclerosis and in the scarring in skin and ligaments after surgery or trauma.

Elastin Synthesis Elastin assembly appears to be closely related to that of collagen, since a few of the prolines in the protein are hydroxylated to hydroxyproline by prolyl hydroxylase. The elastin monomer, however, is a single polypeptide that does not fold into a defined three-dimensional structure and is not synthesized as a larger precursor molecule. Instead, it is slowly secreted from cells into extracellular compartments, where it forms amorphous deposits around previously deposited microfibrils. The elastin deposits then become covalently cross-linked through oxidation of lysine residues to aldehydes by the same lysyl oxidase that initiates the cross-linking of collagen. The microfibrils in elastin deposits are largely composed of fibrillin, a large protein that forms beadlike strands.

Proteoglycan Synthesis The synthesis of proteoglycans begins in the cisternae of the rough endoplasmic reticulum with assembly of a core protein that then undergoes modification by sugar and sulfate transferases that generate large side chains of glycosaminoglycans. At least 30 proteoglycans have been identified by differences in the structures of their core proteins. The major proteoglycan of cartilage, called *aggrecan*, has a core protein of about 2000 amino acids to which are bound multiple side chains of chondroitin sulfate and keratan sulfate, polysaccharides consisting of highly charged and repetitive disaccharide sequences. After secretion from cells, the aggrecan monomer binds to a smaller protein called a *link protein*. The complex of core protein and link protein then spontaneously binds to a long chain of hyaluronic acid to form a huge copolymer called a *proteoglycan aggregate*. The highly charged proteoglycan aggregate binds water and small ions and thereby provides a large swelling pressure and resiliency to cartilage. Smaller proteoglycans such as decorin, biglycan, and fibromodulin have smaller core proteins with fewer and different polysaccharide side chains. They do not form large aggregates with hyaluronate but bind to fibrils of collagen or fibronectin and may thereby help regulate the assembly or spatial orientation of fibrils. One group of small proteoglycans known as *syndecans* binds to the plasma membranes of cells and may have a role in cell migration along fibrils or in signal transduction.

The assembly of bone follows much the same principles as the assembly of other connective tissues ([Chap. 340](#)). The first step is deposition of osteoid tissue that consists largely of type I collagen fibrils. Mineralization of osteoid occurs by steps that are still incompletely defined; proteins such as osteopontin and osteocalcin probably bind to the collagen fibrils and chelate calcium to initiate mineralization. Small proteoglycans such as decorin or fibromodulin may also play a role.

MUTATIONS THAT PRODUCE DISEASES OF CONNECTIVE TISSUES

Because of the large number of tissue-specific macromolecules present in connective tissues, a large number of gene-protein systems are candidates for mutations that might cause disease of the tissues.

The most complete data on mutations causing heritable disorders of connective tissue

are available on [OI](#). Most patients with severe OI (types II and III) have mutations in either the gene for the $\alpha 1(I)$ chain or the gene for the $\alpha 2(I)$ chain of type I procollagen (the COL1A1 and COL1A2 genes). In patients with mild disease, many of the mutations decrease expression of protein from one allele of the genes. Most of the mutations in patients with severe OI cause synthesis of a structurally abnormal but partially functional α chain. Mutations that cause synthesis of structurally abnormal α chains include partial gene deletions, partial gene duplications, and RNA splicing mutations. The most common mutations, however, cause the substitution of single amino acids with bulky side chains for the glycine residues that appear as every third amino acid in the triple helix of a α chain ([Fig. 351-2](#)). The structurally abnormal α chains exert their effects primarily through one of three mechanisms ([Fig. 351-1](#)). First, the presence of an abnormal α chain in a procollagen molecule containing two normal α chains can prevent folding of the protein into a triple-helical conformation and lead to degradation of the whole molecule in a process called *procollagen suicide*. Similar dominant negative mutations are seen with other multisubunit proteins. The net result of procollagen suicide is accumulation of the unfolded protein in the rough endoplasmic reticulum of cells and a reduction in the amount of collagen available for fibril assembly. Second, the presence of one abnormal α chain in a procollagen molecule can interfere with cleavage of the N-propeptide from the protein. The persistence of the N-propeptide on a fraction of the molecules interferes with the self-assembly of normal collagen so that thin and irregular collagen fibrils are formed. Third, the substitution of a bulkier amino acid for glycine can produce a change in the conformation of the molecule and result in the assembly of collagen fibrils that are abnormally branched or abnormally thick and short. Also, copolymerization of the mutated collagen with normal collagen can slow fibril assembly and decrease the total amount of collagen incorporated into fibrils.

Over 300 mutations in the two genes for type I procollagen have been found in patients with [OI](#) ([Fig. 351-2](#)). Initially, there was concern that many of the mutations might be neutral variations in the structure of the genes and not the cause of the disease phenotypes. However, the causal relationship between most of the mutations and the disease has been established by several kinds of evidence: (1) DNA linkage studies in families with mild variants of OI showed that specific mutated alleles were coinherited with the disease phenotypes. (2) Probands with lethal variants of OI were shown to have new mutations not found in the normal parents or in only a few cells from a mosaic parent (see below). (3) Studies with cultured skin fibroblasts from patients demonstrated that the mutations either produced specific disruptions in the biosynthesis of type I procollagen or caused synthesis of a type I procollagen that formed abnormal collagen fibrils ([Fig. 351-1](#)). (4) The mutations in probands with OI were not found in normal alleles for the genes. (5) Expression of several of the mutated genes for type I procollagen in transgenic mice generated disease phenotypes similar to those seen in patients who inherited the mutated genes ([Fig. 351-3](#)).

The data on mutations in type I procollagen that cause [OI](#) have been used as a paradigm for defining other mutations in other collagen and procollagen genes that cause other disorders of connective tissue. For example, similar mutations in the gene for type III procollagen occur in patients with the type IV variant of [EDS](#), which causes early death because of rupture of the aorta or other hollow organs ([Fig. 351-4](#)). Also, similar mutations in the gene for type II procollagen (COL2A1) are found in a number of

patients with [CDs](#) ([Fig. 351-5](#)). In addition, transgenic mice expressing mutated genes for type II procollagen develop phenotypes resembling the CDs. Similar mutations in the gene for type VII collagen (COL7A1) are found in patients with the dystrophic form of [EB](#), and mutations in the genes for type IV collagen are found in many patients with [AS](#). As discussed below, the paradigm for defining the consequences of mutations in procollagen genes also helps explain findings on mutations in a fibrillin gene that cause [MFS](#) and mutations in keratin genes that cause the simplex variant of EB.

Several generalizations can be made about mutations in collagen genes. One is that unrelated patients rarely have the same mutation in the same gene. Another is that mutations that cause the most severe disease are usually new mutations in one allele that occur either during the generation of the germline in one of the parents or during meiosis in the fertilized egg ([Chap. 65](#)). Still another generalization is that most mild variants are caused by mutations that are specific or "private" to a given family. Indeed, the number of recurrent mutations in structural genes are so infrequent that there are, in effect, no common mutations responsible for the disorders in unrelated patients and no "hot spots" that contain most of the mutations.

Another general trend is that similar mutations in the same gene can produce different disease syndromes in terms of both severity and the major tissues involved. One reason for heterogeneity in pathologic manifestations is that different regions of a large molecule may be more important for its function in some connective tissues than in others. For example, some regions of the type I collagen molecule may be essential for the binding of mineralizing proteins in bone so that mutations in these regions cause fragile bones but do not impair function in skin and other nonmineralizing tissues. It is more difficult, however, to explain how the same mutation can produce a severe phenotype in some and a mild phenotype in other members of the same family. Such phenotypic variation appears to be characteristic of [OI](#), where some subjects are short and have multiple fractures from minor trauma, whereas others in the same family can be of normal stature and free of fractures. In the past, such phenotypic variation was explained by undefined variations in the genetic background of different family members. Studies in transgenic mice, however, demonstrated similar phenotypic variation with expression of a mutated collagen gene in an inbred strain of mice in whom the genetic background is uniform. Therefore, the phenotypic variation is probably caused by undefined stochastic or chance events during embryonic and fetal development. Although dramatic phenotypic variations are relatively rare in OI and related disorders, it is important to consider in counseling families about the consequences of inherited mutations.

OSTEOGENESIS IMPERFECTA

[OI](#) is an inherited disorder that causes a generalized decrease in bone mass (osteopenia) and makes the bones brittle. The disorder is frequently associated with blue sclerae, dental abnormalities (dentinogenesis imperfecta), progressive hearing loss, and a positive family history. The most severe forms cause death in utero, at birth, or shortly thereafter. The course of mild and moderate forms is more variable. Some patients appear normal at birth and become progressively worse. Some have multiple fractures in infancy and childhood, improve after puberty, and fracture more frequently later in life. Women are particularly prone to fracture during pregnancy and after

menopause. A few women from families with mild variants of OI do not develop fractures until after menopause, and their disease may be difficult to distinguish from postmenopausal osteoporosis.

Classification The most common classification for OI was developed by Sillence ([Table 351-2](#)). Type I, the mildest form, is inherited as an autosomal dominant trait. Most patients have distinctly blue sclerae. Type I is subdivided into types IA and IB depending on whether or not dentinogenesis imperfecta is present. Type II is lethal in utero or shortly after birth. Radiographic criteria can be used to subdivide type II into five groups, with subgroup 1 showing the most severe changes and subgroup 5 the least. Types III and IV OI are intermediate in severity between types I and II. They differ from type I because of lesser severity and because the sclerae are only slightly bluish in infancy and white in adulthood. Type III differs from type IV in that it tends to become more severe with age. Also, type III can be inherited either as an autosomal recessive or autosomal dominant trait, whereas type IV is always dominant. The clinical courses are variable, and the mode of inheritance in types III and IV may be difficult to ascertain because many patients have sporadic mutations and because many couples with one child severely affected by OI do not have additional children. For these and related reasons, the distinction between type IV OI and other severe variants of OI may not be helpful. Therefore, it may be sufficient to classify patients simply as mild (type I), lethal (type II), and moderately severe (type III).

Incidence Type I OI has a frequency of about 1 in 30,000. Type II OI has a reported incidence at birth of about 1 in 60,000, but the incidence of the three severe forms recognizable at birth (types II, III, and IV) may be as high as 1 in 20,000.

Skeletal Changes In type I OI, the fragility of bones may be severe enough to limit physical activity or so mild that individuals are unaware of any disability. Radiographs of the skull of patients with mild disease may show a mottled appearance because of small islands of irregular ossification. In type II OI, bones and other connective tissues are so fragile that massive injuries can occur in utero or during delivery ([Fig. 351-3](#)). Ossification of many bones is frequently incomplete. Continuously beaded or broken ribs and crumpled long bones (accordion femora) may be present. For unclear reasons, the long bones may be either thick or thin. In types III and IV, multiple fractures from minor physical stress can produce severe deformities. Kyphoscoliosis can impair respiration, cause cor pulmonale, and predispose to pulmonary infections. The appearance on radiographs of "popcorn-like" deposits of mineral on the ends of long bones is an ominous sign. Progressive neurologic symptoms may result from basilar compression and communicating hydrocephalus.

In all forms of OI, bone mineral density in unfractured bone is decreased. However, the degree of osteopenia may be difficult to evaluate because recurrent fractures limit exercise and thereby worsen the decrease in bone mass. Surprisingly, fractures appear to heal normally.

Ocular Changes The sclerae can be normal, slightly bluish, or bright blue. The color is probably caused by a thinness of the collagen layers of the sclerae that allows the choroid layers to be seen. Blue sclerae, however, are an inherited trait in some families who do not have increased bone fragility.

Dentinogenesis Imperfecta The teeth may be normal, moderately discolored, or grossly abnormal. The enamel generally appears normal, but the teeth may have a characteristic amber, yellowish brown, or translucent bluish gray color because of improper deposition or deficiency of dentin. The deciduous teeth are usually smaller than normal, whereas permanent teeth are frequently bell-shaped and restricted at the base. In some patients, the teeth readily fracture and need to be extracted. The defect in dentin is directly attributable to the fact that normal dentin is rich in type I collagen. Similar tooth defects, however, can be inherited without any evidence of [OI](#).

Hearing Loss Hearing loss usually begins during the second decade of life and occurs in over 50% of subjects over age 30. The loss can be conductive, sensorineural, or mixed and varies in severity. The middle ear usually exhibits maldevelopment, deficient ossification, persistence of cartilage in areas that are normally ossified, and abnormal calcium deposits.

Associated Features Other connective tissue involvement can include thin skin that scars extensively, joint laxity with permanent dislocations indistinguishable from those of [EDS](#), and, occasionally, cardiovascular manifestations such as aortic regurgitation, floppy mitral valves, mitral incompetence, and fragility of large blood vessels. For unknown reasons, some patients develop a hypermetabolic state with elevated serum thyroxine levels, hyperthermia, and excessive sweating.

Molecular Defects Most patients with [OI](#) have mutations in one of the two genes that encode type I procollagen. Over 90% of patients with type I OI and blue sclerae have mutations in the *proa1(I)* gene that decrease the steady-state levels of the mRNA for *proa1(I)* chains and decrease the rates of synthesis of *proa1(I)* chains relative to those for *proa2(I)* chains. In more severe forms (types II, III, and IV), the effects of mutations that cause synthesis of abnormal *proa* chains are amplified by the three mechanisms discussed above ([Fig. 351-1](#)). Mutations that change the structure of the protein near the N-proteinase cleavage site cause accumulation of a partially processed procollagen and produce lax joints similar to those in type VII [EDS](#) that is caused by mutations in the gene for the N-proteinase. Mutations that change the structure in the middle or near the C terminus of the molecule tend to cause severe or lethal variants of OI. It is difficult, however, to correlate the site or nature of the mutation and the clinical phenotype ([Fig. 351-2](#)). Most patients are heterozygotes with mutations in a single allele, but rare patients are homozygotes with two mutated alleles for *proa1(I)* or *proa2(I)* chains.

Mosaicism in Germ-Line Cells and in Somatic Cells Most lethal [OI](#) is the result of new autosomal dominant mutations. The frequency of a second child with lethal OI in the same family, however, is about 7% because of germ-line mosaicism in one of the parents. The presence of germ-line mosaicism has been demonstrated in several fathers of patients with type II OI by demonstrating the mutated gene in a fraction of their sperm. Apparently normal parents of children with severe OI may also have somatic cell mosaicism in which the mutated allele is present in a fraction of somatic cells such as fibroblasts, leukocytes, and hair root cells. Because of the possibility of germ-line mosaicism, asymptomatic parents of a child with severe OI should be counseled that recurrence can occur.

Diagnosis The diagnosis is usually made on the basis of clinical criteria. The presence of fractures together with blue sclerae, dentinogenesis imperfecta, or family history of the disease is usually sufficient to make the diagnosis. Other causes of pathologic fractures must be excluded, including the battered child syndrome, nutritional deficiencies, malignancies, and other inherited disorders such as [CDs](#) and hypophosphatasia ([Table 351-3](#)). X-rays usually reveal a decrease in bone density that can be verified by photon or x-ray absorptiometry. There is no consensus, however, as to whether the diagnosis can be made by microscopy of bone. A molecular defect in type I procollagen can be demonstrated in half or more of patients by incubating skin fibroblasts with radioactive amino acids and then analyzing the pro α chains by polyacrylamide gel electrophoresis. The analysis detects decreases in the rate of synthesis of pro α 1(I) chains relative to pro α 2(I) chains, abnormally long pro α chains, abnormally short pro α chains, and pro α chains with abnormal posttranslational modification because of an amino acid substitution that impairs folding of the triple helix. The mutations themselves can be defined in most patients by sequencing of genomic DNA. Because each proband and family usually has a "private" mutation, extensive analysis of about 10,000 bases in each of the two genes is required to identify the exact mutation. After a mutation in a type I procollagen gene is identified, a test based on the polymerase chain reaction can be used to screen family members at risk and for prenatal diagnosis.

TREATMENT

Many patients with [OI](#) have successful careers despite severe deformities. Those with mild disorder may need little treatment when fractures decrease after puberty, but women require special attention during pregnancy and after menopause, when fractures again increase. More severely affected children require a comprehensive program of physical therapy, surgical management of fractures and skeletal deformities, and vocational education.

Many of the fractures are only slightly displaced and have little soft tissue swelling. Therefore, they can be treated with minimal support or traction for a week or two followed by a light cast. If fractures are relatively painless, physical therapy can be initiated early. A judicious amount of exercise prevents loss of bone mass secondary to physical inactivity. Some physicians advocate insertion of steel rods into long bones to correct limb deformities; the risk/benefits and cost/benefits of such procedures are difficult to evaluate. Aggressive conventional intervention is usually warranted for pneumonia and cor pulmonale. For severe hearing loss, stapedectomy or replacement of the stapes with a prosthesis may be successful. Moderately to severely affected patients should be evaluated periodically to anticipate possible neurologic problems. About half of children have a substantial increase in growth when given growth hormone. Treatment with bisphosphonates to decrease bone loss has been introduced on an experimental basis. Initial results are promising, but the long-term effects of decreasing bone resorption are unknown. Also, a clinical trial has been initiated to use stromal cells from bone marrow that can differentiate into osteoblasts after systemic infusion. In the first phase of the trial, three children with severe [OI](#) (type III) showed clinical improvement after marrow ablation and transplantation of whole bone marrow from an HLA-compatible sibling.

A program for careful orthotic management developed by Bleck and a program for compressive management developed by Marini are useful. Counseling and emotional support are important for patients and parents, and lay organizations in some countries provide help in these areas. Prenatal ultrasonography will detect severely affected fetuses at about 16 weeks of pregnancy. Diagnosis by demonstrating synthesis of abnormal procollagen chains or by DNA sequencing can be carried out in chorionic villus biopsies at 8 to 12 weeks of pregnancy.

EHLERS-DANLOS SYNDROME

[EDS](#) is characterized by hyperelasticity of the skin and hypermobile joints.

Classification Five types of [EDS](#) were initially defined based primarily on the extent to which the skin, joints, and other tissues are involved, but the classification has now been extended ([Table 351-4](#)). Type I is the classic, severe form of the disease, with both severe joint hypermobility and skin that is velvety in texture, hyperextensible, and easily scarred. Type II is similar to type I but milder. In type III joint hypermobility is more prominent than skin changes. In type IV the skin changes are more prominent than joint changes. However, type IV patients are predisposed to sudden death from rupture of large blood vessels or other hollow organs. Type V is similar to type II but is inherited as an X-linked trait. Type VI is characterized by scoliosis, ocular fragility, and a cone-shaped deformity of the cornea (keratoconus). Type VII is characterized by marked joint hypermobility that is difficult to distinguish from type III except by the specific molecular defects in the processing of type I procollagen to collagen. Type VIII is distinguished by periodontal changes. Types IX, X, and XI were defined on the basis of preliminary biochemical and clinical data, but these classifications have not proven useful. Because of overlapping signs and symptoms, many patients and families with some of the features of EDS cannot be assigned to any of the defined types.

Incidence The incidence of [EDS](#) is difficult to establish, largely because patients with mild skin or joint symptoms rarely seek medical attention. It is also difficult to define the normal range of variation for joint mobility or skin elasticity. The incidence may be about 1 in 5000 births, although a higher value has been reported for blacks. Types I, II, and III account for most diagnoses.

Skin The changes vary from thin and velvety skin to skin that is either dramatically hyperextensible ("rubber man" syndrome) or easily torn or scarred. Type I patients develop characteristic "cigarette-paper" scars. In type IV extensive scars and hyperpigmentation develop over bony prominences, and the skin may be so thin that subcutaneous blood vessels are visible. In type VIII the skin is more fragile than hyperextensible, and it heals with atrophic, pigmented scars. Easy bruisability occurs in several types of [EDS](#).

Ligament and Joint Changes Laxity and hypermobility of joints vary from mild to unreducible dislocations of hips and other large joints. In mild forms patients learn to reduce dislocations themselves and to avoid them by limiting physical activity. In more severe forms, surgical repair may be required. Some patients have progressive difficulty with age, but severe joint laxity is compatible with a normal life span.

Associated Changes Mitral valve prolapse and hernias occur, particularly in type I. Pes planus and mild to moderate scoliosis are common. Extreme joint laxity and repeated dislocations may lead to degenerative arthritis. In type VI the eye may rupture with minimal trauma, and kyphoscoliosis can cause respiratory impairment. Sclerae may be blue in type VI.

Molecular Defects Mutations in two of the three genes for type V collagen have been found in patients with types I and II [EDS](#). The mutations include glycine substitutions in the triple-helical domain, RNA splicing mutations, exon skipping mutations, a small deletion of 7 bp, and a substitution of serine for cysteine in the C-propeptide. Mutations in both $\alpha 1(V)$ and $\alpha 2(V)$ chain are found in patients with type I EDS, but to date only mutations in $\alpha 1(V)$ chain have been found in patients with type II EDS. Electron microscopy of skin from some patients with types I, II, or III EDS are consistent with mutations in a low-abundance collagen such as types III or V that either copolymerize with or bind to the surface of type I fibrils. However, irregular fibrils are not seen in all patients, and similar irregular fibrils can be seen in normal skin.

Most patients with type IV [EDS](#) have a defect either in the synthesis or structure of type III procollagen, a finding consistent with the fact that these patients are prone to spontaneous rupture of the aorta and intestines, tissues rich in type III collagen. The thinness and scarring of skin are more difficult to explain, since type III constitutes a small fraction of the collagen in skin ([Table 351-1](#)). The >50 mutations identified in the type III procollagen gene include partial gene deletions, RNA splicing mutations, and single-base mutations that cause substitution of glycine by amino acids with bulkier side chains ([Fig. 351-4](#)). In effect, most of the mutations lead to synthesis of abnormal but partially functional $\alpha 1(III)$ chains that produce procollagen suicide or alter fibril formation by the same mechanisms that amplify the effects of mutations in the genes for type I procollagen. Similar mutations in type III procollagen can cause aortic aneurysms in some individuals without other evidence of EDS type IV, [MFS](#), or other inherited disorders of connective tissue.

Type VI [EDS](#) is caused by mutations in the gene that encodes lysyl hydrolase. In one series, all the patients were homozygous or compound heterozygotes for the mutated genes, and all the mutations caused profound deficiency of lysyl hydroxylase, a decrease in the hydroxylysine content of collagen, and a decrease in the cross-links in collagen fibers. The decrease in cross-links is explained by the observation that some cross-links are less stable if formed from lysine instead of hydroxylysine.

Type VII [EDS](#) is due to a defect in the conversion of procollagen to collagen caused either by mutations that make type I procollagen resistant to cleavage by procollagen N-proteinase or by mutations that decrease the activity of the enzyme. The type VIIA mutations alter the cleavage site in the $\alpha 1(I)$ chain, and the type VIIB mutations alter the cleavage site in the $\alpha 2(I)$ chain. Both types are dominantly inherited. Type VIIC is caused by mutations that decrease the activity of procollagen N-proteinase and is inherited as an autosomal recessive trait. In all three forms of type VII EDS, the persistence of the N-propeptide causes the formation of collagen fibrils that are thin and irregular. Since most patients do not have clinical osteopenia, the thin and irregular fibrils apparently suffice for the mineralization of bone but do not provide the necessary tensile strength for ligaments and joint capsules. However, some patients fracture easily

and are difficult to distinguish from variants of OI.

The cause of type VIII [EDS](#) is unknown. Type IX is a disorder of copper transport. The syndrome, also referred to as *Menkes' syndrome*, is due to an X-linked defect and is associated with cutis laxa, hypopigmentation, unusual hair ("kinky"), vascular aneurysms, neurologic degeneration, and mental retardation. Mutations in a gene coding for a copper-transporting ATPase cause the disease ([Chaps. 348](#) and [353](#)). Type X EDS may be caused by defects in fibronectin, but no specific mutations have been defined.

Diagnosis The diagnosis is based on clinical criteria. Biochemical assays and gene analyses for known molecular defects in [EDS](#) are difficult and time-consuming, but specific diagnostic tests should be available in the future for families in which the genes at fault have been defined.

TREATMENT

There is no specific therapy. Surgical repair and tightening of joint ligaments require careful evaluation of individual patients, as the ligaments frequently do not hold sutures. Patients with easy bruisability should be evaluated for other bleeding disorders. Patients with type IV [EDS](#) and members of their families should probably be evaluated at regular intervals by sonography and related techniques for early detection of aneurysms. Surgical repair of aneurysms may be difficult because of increased friability of tissues, and there is limited experience with elective surgery in such patients. Also, women with type IV EDS should be counseled about the increased risk of uterine rupture, bleeding, and other complications of pregnancy.

CHONDRODYSPLASIAS (See also [Chap. 343](#))

The [CDs](#) are inherited skeletal disorders that cause dwarfism and abnormal body proportions. The category also includes some individuals with normal stature and body proportions who have features such as ocular changes or cleft palate that are common in more severe CDs. Many patients develop degenerative joint changes, and mild CD in adults may be difficult to differentiate from primary generalized osteoarthritis. Some authors refer to the disorders as "skeletal dysplasias," but CD is a more widely used term.

Classification Over 150 distinct types and subtypes have been defined ([Table 351-5](#)) based on criteria such as "bringing death" (thanatophoric), causing "twisted" bones (diastrophic), affecting metaphyses (metaphyseal), affecting epiphyses (epiphyseal), and producing histologic changes such as an apparent increase in the fibrous material in the epiphyses (fibrochondrogenesis). Also, a number of eponyms are based on the first or most comprehensive case reports. Severe forms of the diseases produce gross distortions of most cartilaginous structures and of the eye. Mild forms are more difficult to classify. Among the features are cataracts, degeneration of the vitreous and retinal detachment, high forehead, hypoplastic facies, cleft palate, short extremities, and gross distortions of the epiphyses, metaphyses, and joint surfaces.

Incidence Data on the frequency of most [CDs](#) are not available, but the incidence of the

Stickler syndrome may be as high as 1 in 10,000. Therefore, the diseases are probably among the more common heritable disorders of connective tissue.

Molecular Defects The first mutations shown to cause [CDs](#) were in the COL2A1 gene for type II collagen, the most abundant protein in cartilage. A number of mutations in this gene have now been reported in variants of CD ranging from mild to lethal ([Fig. 351-5](#)). A large fraction of patients with lethal CDs, a smaller number of patients with moderately severe CDs, and about 2% of families with early-onset generalized osteoarthritis have mutations in the same gene. However, similar phenotypes can also be caused by mutations in other genes, including genes for three other collagens, additional components of the cartilage matrix, growth factors, growth factor receptors, and transcription factors (see [Table 351-6](#) for selected examples). The number of mutated genes reported does not necessarily reflect the incidence of such mutations in the diseases themselves but rather the complexity of the genes and the technical difficulties in searching the complete gene for mutations. Also, it reflects the availability of large families for DNA linkage analysis and the vigor with which investigators have pursued their interest in a given gene. It is likely that mutations in additional genes will be found.

Mutations in the COL2A1 gene were first found in patients with severe [CDs](#) characterized by gross deformities of bones and cartilage such as spondyloepiphyseal dysplasia congenita, hypochondrogenesis/achondrogenesis II, and the Kniest syndrome. However, mutations in the COL2A1 gene have been found in a few families in which few if any symptoms are present in childhood but in which joint stiffness, joint pain, and degenerative changes of osteoarthritis develop in midlife. The mutations in the COL2A1 gene are similar to the mutations in the genes for types I and III procollagens ([Fig. 351-5](#)), and the correlations between genotype and the severity of the phenotype are equally difficult. In addition, mutations that change a codon for a Y-position amino acid in the -Gly-X-Y- repeat sequence from an arginine to cystine were found in families with early-onset osteoarthritis and minimal evidence of CDs. Stickler syndrome and related syndromes are caused by mutations in three different genes: the COL2A1 gene for type II collagen and the COL11A1 and COL11A2 genes for type XI collagen. A series of mutations that introduce premature terminal signals in the COL2A1 gene lead to classic Stickler syndrome. However, some patients with classic Stickler syndrome have glycine substitutions in COL11A1. RNA splicing mutations in the COL11A1 gene are found in patients with the Marshall syndrome, which is similar to classic Stickler syndrome but with milder eye changes and more severe hearing loss. Patients classified as having nonocular Stickler syndrome have RNA splicing mutations in the COL11A2 gene.

Many individuals with the Schmid metaphyseal [CD](#), characterized by short stature, coxa vara, flaring metaphyses, and waddling gait, have mutations in the gene for the type X collagen, a short, network-forming collagen found primarily in the hypertrophic zone of endochondral cartilage.

Mutations in the receptor for fibroblast growth factor 3 (FGFR-3) are present in most patients with achondroplasia, the most common cause of short-limbed dwarfism accompanied by macrocephaly and dysplasias of the metaphyses of long bones ([Table 351-6](#)). The same single-base mutation in the gene that converts glycine to arginine at position 380 is present in >90% of patients. Most patients represent sporadic new mutations, and this nucleotide change must be one of the most common recurring

mutations in the human genome. The mutation causes unregulated signal transduction through the receptor and inappropriate development of cartilage. Mutations that alter other domains of FGFR-3 have been found in patients with the more severe disorders hypochondroplasia and thanatophoric dysplasia and in a few families with a variant of craniosynostosis. However, most patients with craniosynostosis appear to have mutations in the related gene FGFR-2 gene.

Mutations in the gene for the cartilage oligomeric matrix protein (COMP) have been found in patients with multiple epiphyseal dysplasia or pseudoachondroplasia, and in related syndromes characterized by short limbs and degenerative arthritis. However, some families with multiple epiphyseal dysplasia had a mutation in the gene for $\alpha 2(\text{IX})$ or $\alpha 3(\text{IX})$ chain of type IX collagen (COL9A2 and COL9A3). All the known mutations in these two type IX collagen genes in patients with multiple epiphyseal dysplasia cause splicing out of the codons of exon 3. A mutation in the COL9A2 gene was also found in patients with the common condition of sciatica and herniations of vertebral discs. About 4% of Finnish probands with the phenotype had a single base substitution that converted a codon for glutamate to tryptophan in $\alpha 2(\text{IX})$ chain of type IX collagen.

Diagnosis The diagnosis of severe forms of [CD](#) is made on the basis of the physical appearance, x-ray findings, histologic changes, and clinical course ([Table 351-5](#)).

TREATMENT

No definitive therapy is available. Symptomatic treatment is directed to secondary features such as degenerative arthritis. Many patients require joint replacement surgery and corrective surgery for cleft palate. The eyes should be monitored carefully for the development of cataracts and for the need for laser therapy to prevent retinal detachment. Patients should probably be advised to avoid obesity and contact sports. Counseling for the psychological problems of short stature is critical, and support groups have formed in many countries. Ultrasonography is sometimes successful for prenatal diagnosis but less frequently than with [OI](#). Specific DNA tests are available for the [CDs](#) caused by mutations in the genes for types II, X, and XI collagens.

MARFAN SYNDROME

Severe [MFS](#) is characterized by a triad of features: (1) long, thin extremities frequently associated with other skeletal changes, such as loose joints and arachnodactyly; (2) reduced vision as the result of dislocations of the lenses (ectopia lentis); and (3) aortic aneurysms that typically begin at the base of the aorta.

Classification The clinical diagnosis is frequently problematic because some affected members of families with [MFS](#) present with only one or two features of the typical clinical triad. Also, many patients present with one or two of the features of MFS without a family history, apparently because they represent sporadic mutations. Therefore, it is frequently difficult to determine on the basis of clinical data alone whether a patient with ectopia lentis or the characteristic body habitus of MFS is at risk for developing a life-threatening aortic aneurysm. The new DNA diagnostic tests for mutations in the fibrillin-1 and fibrillin-2 genes can probably resolve most, but not all, of these problems.

Most patients who are prone to develop an aortic aneurysm as a component of MFS can be identified by detection of mutations in the fibrillin-1 gene. Some of these patients develop aortic aneurysms because of a mutation in the fibrillin-1 gene without the skeletal or ocular changes characteristic of MFS. Patients with the rarer form of MFS that is characterized by contractural arachnodactyly instead of loose joints can usually be identified by detection of a mutation in the fibrillin-2 gene that is similar in structure to the gene for fibrillin-1. Preliminary data suggest that patients with mutations in the fibrillin-2 gene are not prone to develop aneurysms. However, affected members of some rare families with a mutation in the fibrillin-1 gene also do not develop aortic aneurysms, even though they may show the skeletal or ocular changes. Therefore, the DNA tests are most helpful if: (1) a mutation is detected in either of the two genes, and (2) informative data are available on the clinical symptoms that the same mutation produces in the patient's family or in other families with similar clinical features.

Incidence and Inheritance [MFS](#) has an incidence of about 1 in 10,000 in most racial and ethnic groups. The disorder is inherited as an autosomal dominant trait; at least one-fourth of patients do not have an affected parent, and their cases are probably due to new mutations.

Skeletal Changes Patients are usually tall compared with other members of the same family and have long limbs. The ratio of the upper segment (top of the head to top of the pubic ramus) to the lower segment (top of the pubic ramus to the floor) is usually 2 SDs below mean for age, race, and sex. The fingers and hands are long and slender and have a spider-like appearance (arachnodactyly). Many patients have severe chest deformities, including depression (pectus excavatum), protrusion (pectus carinatum), or asymmetry. Scoliosis is frequent and usually accompanied by kyphosis. High-arched palate and high pedal arches or pes planus are common. A few patients have severe joint hypermobility similar to [EDS](#).

Cardiovascular Changes Cardiovascular abnormalities are the major source of morbidity and mortality ([Chap. 247](#)). Mitral valve prolapse develops early in life and in about one-quarter progresses to mitral valve regurgitation of increasing severity because of redundancy of the leaflets, stretching of the chordae tendineae, and dilatation of the valvulae annulus. Dilatation of the root of the aorta and the sinuses of Valsalva are characteristic and ominous features of the disease that can develop at any age and in rare instances may be detected by echocardiography in utero. The rate of dilatation is unpredictable, but it can lead to aortic regurgitation, dissection of the aorta, and rupture. Dilatation is probably accelerated by physical and emotional stress, as well as by pregnancy.

Ocular Changes Dislocations of the lens may be readily apparent, but diagnosis usually requires pupillary dilatation and slit-lamp examination. The displacement is usually not progressive but may contribute to the formation of cataracts. The ocular globe is frequently elongated, most patients are myopic, and some develop retinal detachment. A few patients have lattice degeneration and retinal tears; most have adequate vision.

Associated Changes Striae may occur over the shoulders and buttocks. Otherwise the skin is normal. A number of patients develop spontaneous pneumothorax. Inguinal and

incisional hernias are common. Marked dilatation of the dural sac is seen frequently in computed tomography scans, but the condition is usually asymptomatic. Patients are typically thin with little subcutaneous fat, but adults may develop centripetal obesity.

Molecular Defects Most patients with the classic features of [MFS](#) are heterozygotes for mutations in a gene on chromosome 15 that encodes fibrillin-1, a glycoprotein of 350 kDa that is a major component of elastin-associated microfibrils. These microfibrils are abundant in large blood vessels and the suspensory ligaments of the lens. Mutations in the fibrillin-1 gene include missense, nonsense, in-frame deletions, and RNA splicing mutations. Many of the mutations are single amino acid substitutions in the epidermal growth factor-like domains of the molecule that may be involved with calcium binding. Mutations in the fibrillin-2 gene that cause the MFS variant characterized by contractures appear to follow a similar pattern. As with most genetic diseases, the nature and location of mutations in the genes are only an approximate guide to the severity of the phenotype unless the same mutation has been seen in other members of the same family or in similar unrelated patients. However, there is a clustering of mutations in the middle portion of the molecule of fibrillin-1 encoded by exons 23 to 32 that causes the most severe phenotype, referred to as *neonatal lethal MFS*. The function of fibrillin has not been defined, but the data suggest that fibrillin self-assembles into a fibrillar structure and that the conformation and surface properties of the entire molecule are critical for normal assembly. Therefore, the functional consequences of mutations that change the amino acid sequence of fibrillin may be similar to the effects of mutations that change the conformation of a fibrillar collagen.

Diagnosis The diagnosis is easily established if the patient and other members of the family have dislocated lenses, aortic dilatation, and long and thin extremities together with kyphoscoliosis or other chest deformities. The diagnosis is frequently made if ectopia lentis and an aneurysm of the ascending aorta occur in the absence of a Marfan habitus or a positive family history. All patients in whom the diagnosis is suspected should have a slit-lamp examination and an echocardiogram. Also, homocystinuria ([Table 351-3](#)) should be ruled out by a negative cyanide-nitroprusside test for disulfides in the urine ([Chap. 352](#)). A few patients with types I, II, and III [EDS](#) have ectopia lentis but lack the Marfan habitus and instead have characteristic skin changes not present in [MFS](#). Patients with familial aortic aneurysms tend to develop aneurysms at the base of the abdominal aorta. The location of the aneurysms, however, is somewhat variable, and the high incidence of aortic aneurysms in the general population (1 in 100) makes the differential diagnosis difficult unless other features of MFS are clearly present. A few families with familial aortic aneurysms have mutations in the gene for type III procollagen ([Fig. 351-4](#)).

TREATMENT

There is no established treatment, but several investigators have recommended use of propranolol or other β -adrenergic blocking agents to lower blood pressure and thereby delay or prevent aortic dilatation. Surgical replacement of the aorta, aortic valve, and mitral valve has been successful in some patients, and all patients should be followed carefully with echocardiography and other techniques for evaluation of cardiovascular changes ([Chap. 247](#)). Patients should probably be advised of the risks of severe physical and emotional stress and of pregnancy.

The scoliosis tends to be progressive and should be treated by mechanical bracing and physical therapy if $>20^\circ$ or by surgery if it progresses to $>45^\circ$. Estrogen has been tried in girls with scoliosis, but the results are inconclusive. Dislocated lenses rarely require surgical removal, but patients should be followed closely for retinal detachment.

Diagnostic tests based on detection of fibrillin defects in cultured skin fibroblasts or DNA analysis of the gene are now available from several laboratories.

DISEASES RELATED TO ELASTIN

As may be expected from the role of elastin in maintaining the elasticity of skin, mutations in the elastin gene cause *cutis laxa*, a rare and heterogeneous group of disorders characterized by skin that is both lax and inelastic. Three different frame-shift mutations were found in three unrelated families with dominant forms of the disease. Surprisingly, other mutations in the elastin gene produce phenotypes that primarily involve the aorta, whose elasticity also depends on the presence of elastin. A large deletion that includes the elastin gene and probably several adjacent genes causes the *Williams syndrome*, characterized by supravalvular aortic stenosis, growth retardation, characteristic facies, and an unusual mental phenotype of low intelligence quotient together with a high degree of sociability.

EPIDERMOLYSIS BULLOSA

[EB](#) consists of a group of similar disorders in which the skin and related epithelial tissues break and blister as the result of minor trauma. As with most heritable disorders of connective tissues, the clinical manifestations range from lethal to mild.

Classification Four types of EB are defined on the basis of the level at which blistering occurs: [EB](#) simplex for blistering in the epidermis, EB hemidesmosomal for fissures between keratinocytes and between keratinocytes and the basal lamina, EB junctional for blistering in the dermal-epidermal junction, and EB dystrophica for blistering in the dermis ([Table 351-7](#)).

Incidence The incidence of [EB](#) in the United States is estimated to be 1 in 50,000.

Molecular Defects The molecular basis of several specific variants of [EB](#) has been defined. A series of patients with EB simplex were found to have mutations in either keratin 14 or keratin 5, two of the major keratins in basal epithelial cells. Patients with the related syndrome, epidermolytic ichthyosis, have mutations in keratin 1 and keratin 10. The new disease phenotype of hemidesmosomal EB has three clinical variants that are caused by mutations in one of four genes ([Table 351-7](#)): (1) A generalized atrophic and benign form of EB is caused by mutations in the COL17A1 gene for type XVII collagen; (2) a variant with EB associated with pyloric atresia and other intestinal abnormalities is caused by mutations in either the gene for the $\alpha 6$ integrin (ITG A6) or the gene for the $\beta 4$ integrin (ITG B4); and (3) another variant characterized by relatively mild blistering at birth but associated with late-onset muscular dystrophy is caused by mutations in the gene for plectin (PLEC-1). Junctional EB is caused by mutations in any one of three genes for laminin (LAMA-3, LAMB-3, LAMC-2). The most severe dystrophic

form of EB is caused by mutations in the gene for type VII collagen (COL7A1).

Diagnosis The diagnosis is based on skin that readily breaks and forms blisters. [EB](#) simplex and EB hemidesmosomal are generally milder than EB junctional or EB dystrophica. EB dystrophica variants usually cause large and prominent scars. Precise classification within subtypes usually requires electron microscopy. The treatment is symptomatic.

ALPORT SYNDROME (See also [Chap. 275](#))

[AS](#) is an inherited disorder characterized by hematuria. Four forms of the disease are now recognized: (1) classic AS, which is inherited as an X-linked disorder with hematuria, sensorineural deafness, and conical deformation of the anterior surface of the lens (lenticonus); (2) a subtype of the X-linked form associated with diffuse leiomyomatosis; (3) an autosomal recessive form; and (4) an autosomal dominant form. The two autosomal forms can cause renal disease without deafness or lenticonus.

Incidence The incidence of [AS](#) is about 1 in 10,000 in the general population and as high as 1 in 5000 in some ethnic groups. About 80% of AS patients have the X-linked variant.

Molecular Defects Electron microscopy of kidneys from patients with classic [AS](#) demonstrates that the glomerular basement membrane is up to five times thicker than normal and that the lamina densa is distorted and split. The X-linked and autosomal recessive forms are caused primarily by mutations in genes for the $\alpha 3(\text{IV})$, $\alpha 4(\text{IV})$, $\alpha 5(\text{IV})$, or $\alpha 6(\text{IV})$ chains of type IV collagen, a major component of basement membranes. The type IV collagen in most membranes consists primarily of $\alpha 1(\text{IV})$ and $\alpha 2(\text{IV})$ chains folded into a large, rodlike molecule with globular ends and a long triple-helical domain that is interrupted by short sequences that do not form triple helices. The molecules self-assemble through both the globular ends and the long triple-helical domain to form a complex three-dimensional network. The four additional α chains of type IV collagen are minor components of basement membranes, similar in structure, and are probably incorporated into the same or similar molecules. The six genes for the proteins are arranged in tandem pairs on different chromosomes in a head-to-head orientation and with overlapping promoters, i.e., the $\alpha 1(\text{IV})$ and $\alpha 2(\text{IV})$ genes are head-to-head on chromosome 13q34, the $\alpha 3(\text{IV})$ and $\alpha 4(\text{IV})$ genes are on chromosome 2q35-37, and the $\alpha 5(\text{IV})$ and $\alpha 6(\text{IV})$ genes are on chromosome Xq22. An X-linked variant is caused by mutations in the COL4A5 gene, and the X-linked variant associated with leiomyomatosis is caused by deletions that involve both the COL4A5 gene and the nearby COL4A6 gene. The autosomal recessive variants are caused by mutations in either the COL4A3 or COL4A4 genes. The mutations responsible for the autosomal dominant variants are still unknown, but they have been mapped to the same locus as the COL4A3 and COL4A4 genes.

Diagnosis The diagnosis of classic [AS](#) is based on X-linked inheritance of hematuria, sensorineural deafness, and lenticonus. Because of the X-linked transmission, women are usually less severely affected than men and are generally underdiagnosed. The hematuria progresses to nephritis and may cause renal failure in late adolescence in affected males and at older ages in some women. The sensorineural deafness is

primarily in the high-tone range. It can frequently be detected only by an audiogram and is usually not progressive. The lenticonus can occur without nephritis but is generally considered to be pathognomonic of classic AS.

TREATMENT

There is no known treatment, but renal transplantation is usually successful.

(Bibliography omitted in Palm version)

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352. INHERITED DISORDERS OF AMINO ACID METABOLISM AND STORAGE -

Nicola Longo

All polypeptides and proteins are polymers of amino acids. Eight amino acids, referred to as *essential*, cannot be synthesized by humans and must be obtained from dietary sources. The others are formed endogenously. Although most of the body's amino acids are "tied up" in proteins, small intracellular pools of *free* amino acids are in equilibrium with extracellular reservoirs in plasma, cerebrospinal fluid, and the lumina of the gut and kidney. Physiologically, amino acids are more than mere "building blocks" of proteins. Some (glycine, glutamate, γ -aminobutyric acid) are neurotransmitters. Others (phenylalanine, tyrosine, tryptophan, glycine) are precursors of hormones, coenzymes, pigments, purines, or pyrimidines. Each has a unique degradative pathway by which its nitrogen and carbon components are used for the synthesis of other amino acids, carbohydrates, and lipids.

More than 70 disorders of amino acid metabolism are now known. The catabolic and storage defects (approximately 60) discussed in this chapter far outnumber the transport abnormalities (approximately 10) considered in [Chap. 353](#). Each of these disorders is rare -- the incidences range from 1 in 10,000 for cystinuria or phenylketonuria to 1 in 200,000 for homocystinuria or alkaptonuria. Collectively, however, they occur in perhaps 1 in 500 to 1 in 1000 live births. Almost all are transmitted as autosomal recessive traits.

The features of inherited disorders of amino acid catabolism are summarized in [Table 352-1](#). In general, these disorders are named for the compound that accumulates to highest concentration in blood (*-emias*) or urine (*-urias*). For many conditions (often called *aminoacidopathies*), the parent amino acid is found in excess; for others, generally referred to as *organic acidemias*, products in the catabolic pathway accumulate. Which compound(s) accumulates depends, of course, on the site of the enzymatic block, the reversibility of the reactions proximal to the lesion, and the availability of alternative pathways of metabolic "runoff." For some amino acids, such as the sulfur-containing or branched-chain molecules, defects have been described at nearly each step in the catabolic pathway. For others, only small numbers of defective reactions have been described. Biochemical and genetic heterogeneity is common. Four distinct forms of hyperphenylalaninemia, seven forms of homocystinuria, and seven types of methylmalonic acidemia are recognized. Such heterogeneity reflects the presence of an even larger array of molecular defects.

The manifestations of these conditions differ widely ([Table 352-1](#)). Some, such as sarcosinemia or hyperprolinemia, produce no clinical consequences. At the other extreme, complete deficiency of ornithine transcarbamylase or of branched-chain keto acid dehydrogenase is lethal in the untreated neonate. Central nervous system (CNS) dysfunction, in the form of developmental retardation, seizures, alterations in sensorium, or behavioral disturbances, is present in more than half the disorders. Protein-induced vomiting, neurologic dysfunction, and hyperammonemia occur in many disorders of urea cycle intermediates. Metabolic ketoacidosis, often accompanied by hyperammonemia, is a frequent presenting finding in the disorders of branched-chain amino acid metabolism. Occasional disorders produce focal tissue or organ involvement such as liver disease, renal failure, cutaneous abnormalities, or ocular lesions.

The clinical manifestations in many of these conditions can be prevented or mitigated if diagnosis is achieved early and appropriate treatment (i.e., dietary protein or amino acid restriction or vitamin supplementation) is instituted promptly. For this reason, several aminoacidopathies and organic acidemias are routinely screened in newborns using an array of chemical and microbiologic techniques. Once a presumptive diagnosis is made, confirmation can be provided by direct enzyme assay on extracts of leukocytes, erythrocytes, or cultured fibroblasts. DNA-based testing is possible for several disorders including phenylketonuria, ornithine transcarbamylase deficiency, citrullinemia, gyrate atrophy of the retina, propionic acidemia, and methylmalonic acidemia. As additional mutations are defined, DNA-based analysis may allow better predictions of outcome and improved therapeutic plans.

Several of these disorders (including branched-chain ketoaciduria, isovaleric acidemia, propionic acidemia, methylmalonic acidemia, homocystinuria, cystinosis, phenylketonuria, ornithine transcarbamylase deficiency, citrullinemia, argininosuccinic aciduria) can be diagnosed prenatally by chemical analysis of amniotic fluid or by chemical, enzymatic, or DNA-based studies of fresh or cultured amniotic fluid cells. In addition to predicting genotype and alleviating parental anxiety, prenatal diagnosis has led to improved treatment of affected newborns.

The focus of this chapter is on selected disorders that illustrate the principles, properties, and problems presented by the disorders of amino acid metabolism.

THE HYPERPHENYLALANINEMIAS

DEFINITION

The hyperphenylalaninemias ([Table 352-1](#)) result from impaired conversion of phenylalanine to tyrosine. The most common and clinically important is *phenylketonuria*, which is characterized by an increased concentration of phenylalanine in blood, increased concentrations of phenylalanine and its by-products (notably phenylpyruvate, phenylacetate, phenyllactate, and phenylacetylglutamine) in urine, and severe mental retardation if untreated in infancy.

GENETIC CONSIDERATIONS

Each of the hyperphenylalaninemias results from reduced activity of phenylalanine hydroxylase. In humans, this complete enzyme system is expressed only in liver. Phenylalanine and molecular oxygen are substrates, and a reduced pteridine, tetrahydrobiopterin, is a cofactor ([Fig. 352-1](#)). Tyrosine and dihydrobiopterin are the products of this catalytic system, the latter being reconverted to tetrahydrobiopterin by two enzymes, pterin-4a-carbinolamine dehydratase and dihydropteridine reductase.

Abnormalities in phenylalanine metabolism are autosomal recessive traits that occur in about 1 in 10,000 births. Phenylketonuria type I is widely distributed among whites and Orientals. It is rare in blacks. Phenylalanine hydroxylase activity in obligate heterozygotes is low but higher than in affected homozygotes. Adult heterozygous carriers are clinically well but can be identified by an increased ratio of phenylalanine/tyrosine in plasma in the semifasting state. They may have transient

cognitive impairment after phenylalanine loads. Hyperphenylalaninemias are caused by mutations in the gene encoding phenylalanine hydroxylase (*PAH*) or in genes encoding enzymes involved in tetrahydrobiopterin synthesis or recycling. In the vast majority of patients, mutations occur in the *PAH* gene (causing phenylketonuria type I), and >300 mutations have been identified. Mutations causing a complete impairment of enzyme activity, such as the R408W, are associated with a more severe outcome requiring stringent dietary restriction of phenylalanine. Mutations causing a less complete deficiency of the enzyme, such as the I65T, are associated with milder forms of phenylketonuria.

In fewer than 2% of patients with phenylketonuria, mutations occur in other genes, including dihydrobiopterin reductase (*DHPR*) (30%), 6-pyruvoyl-tetrahydropterin synthase (*6-PTS*) (60%), GTP cyclohydrolase I (*GTP-CH*) (5%), and pterin-4a-carbinolamine dehydratase (*PCD*) (5%). In these cases, the impairment in phenylalanine hydroxylation results from tetrahydrobiopterin deficiency due to blocks in the pathway by which tetrahydrobiopterin is synthesized from GTP (phenylketonuria type III and malignant hyperphenylalaninemia) or deficiency of dihydropterine reductase (phenylketonuria type II), the enzyme that regenerates tetrahydrobiopterin from dihydrobiopterin ([Fig. 352-1](#)). Tyrosine hydroxylase and tryptophan hydroxylase also require tetrahydrobiopterin. Their products (L-dopa and 5-hydroxytryptophan) are essential for the synthesis of neurotransmitters. Heterozygotes for these conditions do not have hyperphenylalaninemia, but carriers of mutations in GTP cyclohydrolase have a peculiar form of dystonia, transmitted as a dominant trait with variable expressivity and higher penetrance in females; it is exquisitely responsive to levodopa. Neurotransmitter levels are not altered in transient hyperphenylalaninemia (sometimes called *transient phenylketonuria*), which has been described in some patients with pterin-4a-carbinolamine dehydratase deficiency.

Etiology and Pathogenesis Phenylalanine accumulation in blood and urine and reduced tyrosine formation are direct consequences of the impaired hydroxylation. In untreated phenylketonuria and in its tetrahydrobiopterin-deficient variants, plasma concentrations of phenylalanine become sufficiently high [1 mmol/L (16 mg/dL)] to activate alternative pathways of metabolism and lead to formation of phenylpyruvate, phenylacetate, phenyllactate, and other derivatives that are rapidly cleared by the kidney and excreted in urine. The severe brain damage is due to several consequences of phenylalanine accumulation: competitive inhibition of transport of other amino acids required for protein synthesis, impaired polyribosome formation or stabilization, reduced synthesis and increased degradation of myelin, and inadequate formation of norepinephrine and serotonin. Phenylalanine is a competitive inhibitor of tyrosinase, a key enzyme in the pathway of melanin synthesis. This block, plus reduced availability of the melanin precursor tyrosine, accounts for the hypopigmentation of hair and skin.

Clinical Manifestations No abnormalities are apparent at birth, but untreated children with classic phenylketonuria fail to attain early developmental milestones, develop microcephaly, and demonstrate progressive impairment of cerebral function. Hyperactivity, seizures, and severe mental retardation are major clinical problems later in life. Electroencephalographic abnormalities; "mousy" odor of skin, hair, and urine (due to phenylacetate accumulation); and a tendency to hypopigmentation and eczema complete the devastating clinical picture. In contrast, affected children who are detected

at birth and treated promptly show none of these abnormalities. Children with tetrahydrobiopterin deficiency, however, suffer a worse clinical course. Seizures appear early, followed by progressive cerebral and basal ganglia dysfunction (rigidity, chorea, spasms, hypotonia). Most succumb to secondary infection within a few years despite early diagnosis and vigorous treatment.

A number of women with phenylketonuria who have been treated since infancy have reached adulthood and become pregnant. If maternal phenylalanine levels are not strictly controlled before and during pregnancy, their offspring are at risk, even if they are heterozygous, for *maternal phenylketonuria*. Affected children have microcephaly and an increased risk of congenital defects. After birth, these children have severe neurodevelopmental delay and growth retardation.

Diagnosis Plasma phenylalanine concentrations are usually normal at birth in the hyperphenylalaninemias but rise rapidly after institution of protein feedings. To prevent mental retardation, diagnosis and initiation of dietary treatment of classic phenylketonuria must occur before the child is 3 weeks of age. For this reason, most newborns in North America and Europe are screened by determinations of blood phenylalanine concentration using the Guthrie bacterial inhibition assay. Abnormal values are confirmed using quantitative analysis of plasma amino acids. Prenatal diagnosis of type I phenylketonuria is now feasible using DNA-based tests that detect specific mutations or polymorphic markers that are linked to the *PAH* gene.

In newborns with type I phenylketonuria, plasma levels depend on the amount of phenylalanine in the diet and the degree of impairment of phenylalanine hydroxylase. Dietary phenylalanine restriction is usually instituted if blood phenylalanine levels are >250 $\mu\text{mol/L}$ (4 mg/dL). Careful monitoring of these infants reveals the degree of phenylalanine hydroxylase impairment and dictates the degree of dietary phenylalanine restriction.

Deficiency of tetrahydrobiopterin, which occurs in 1 to 2% of newborns with increased blood phenylalanine, is excluded by screening the urinary pteridine profile and by assay of dihydropteridine reductase activity on dried blood specimens. Dihydropteridine reductase deficiency and the blocks in tetrahydrobiopterin synthesis can be detected in utero using assays on cultured amniocytes. Tetrahydrobiopterin deficiency manifests with hyperphenylalaninemia and progressive neurologic impairment despite prompt dietary restriction of phenylalanine.

TREATMENT

Phenylketonuria is the first inherited metabolic disease in which a strategy of reducing the accumulation of the offending metabolite prevented the dire clinical consequences. This was accomplished by a special diet low in phenylalanine and supplemented with tyrosine. Tyrosine becomes an essential amino acid in phenylalanine hydroxylase deficiency. Sufficient phenylalanine is provided for new protein synthesis and normal growth. This amount varies with age and requires frequent adjustments, especially early in life. Ordinarily, plasma phenylalanine concentrations are maintained between 120 and 360 $\mu\text{mol/L}$ (2 and 6 mg/dL). Such diet therapy must be instituted during the first 3 weeks of life. Even then, modest [CNS](#) dysfunction may occur with more deleterious

mutations or after excess protein intake. Because uncontrolled hyperphenylalaninemia results in brain damage, dietary restriction should be continued and monitored indefinitely, recognizing that transient phenylketonuria may not require lifelong therapy. An enteric-coated formulation of phenylalanine ammonia lyase, which degrades phenylalanine in the gut, is under investigation for its potential to reduce the strain imposed by the special diet.

Children with tetrahydrobiopterin deficiency deteriorate despite dietary phenylalanine restriction. Such patients may be helped, however, by a regimen in which dietary phenylalanine restriction is combined with tetrahydrobiopterin supplements, levodopa, 5-hydroxytryptophan, and carbidopa. Finally, the deleterious consequences of maternal phenylketonuria can be minimized by continuing lifelong phenylalanine-restricted diets in females with phenylketonuria and assuring strict phenylalanine restriction prior to conception and throughout gestation.

THE HOMOCYSTINURIAS (HYPERHOMOCYSTEINEMIAS)

The homocystinurias are seven biochemically and clinically distinct disorders ([Table 352-1](#)), each characterized by increased concentration of the sulfur-containing amino acid homocystine in blood and urine. The most common form results from reduced activity of cystathionine- β -synthase (CBS), an enzyme in the transsulfuration pathway that converts methionine to cysteine ([Fig. 352-2](#)). The other forms are the result of impaired conversion of homocysteine to methionine, a reaction catalyzed by homocysteine:methyltetrahydrofolate methyltransferase (also known as *methionine synthase*) and two essential cofactors, methyltetrahydrofolate and methylcobalamin (methyl-vitamin B₁₂). Depending on the underlying disorder, some patients show chemical and, in some instances, clinical improvement following administration of specific vitamin supplements (pyridoxine, folate, or cobalamin) ([Chap. 75](#)). In classic homocystinuria, the levels of free homocystine in plasma increase and result in homocystinuria. *Hyperhomocysteinemia* refers to increased total plasma concentration of homocysteine in the sulfhydryl and disulfide form, free and protein-bound. Hyperhomocysteinemia, in the absence of significant homocystinuria, is found in individuals who are heterozygous or homozygous for certain genetic defects that impair folate or vitamin B₁₂ metabolism or cause cystathionine synthase deficiency. Changes of homocysteine levels are also observed with increasing age; in postmenopausal women; in patients with renal failure, hypothyroidism, leukemias, or psoriasis; and during therapy with drugs such as methotrexate, nitrous oxide, isoniazid, and some antiepileptic agents.

Homocysteine acts as an atherogenic and thrombophilic agent. An increase in total plasma homocysteine represents an independent risk factor for coronary, cerebrovascular, and peripheral arterial disease as well as for deep-vein thrombosis ([Chap. 241](#)). Homocysteine is synergistic with hypertension and smoking, and it is additive with other risk factors that predispose to peripheral arterial disease. In addition, hyperhomocysteinemia and folate and vitamin B₁₂ deficiency have been associated with an increased risk of neural tube defects in pregnant women.

CYSTATHIONINE- β -SYNTHASE DEFICIENCY

Definition Deficiency of this enzyme leads to increased concentrations of methionine and homocystine in body fluids and to decreased concentrations of cysteine and cystine. Clinical hallmarks include dislocation of optic lenses (usually downward and medially), mental retardation, marfanoid habitus, osteoporosis, and thrombotic vascular disease.

GENETIC CONSIDERATIONS

The sulfur atom of the essential amino acid methionine is transferred to cysteine by the transsulfuration pathway ([Fig. 352-2](#)). In one of these steps, homocysteine condenses with serine to form cystathionine. This reaction is catalyzed by the pyridoxal phosphate-dependent enzyme [CBS](#). Heterogenous mutations in the *CBS* gene are present in different families. The G307S mutation is associated with lack of response to pyridoxine, whereas the I278T mutation correlates with pyridoxine-responsiveness and a milder clinical phenotype. Homocystinuria is common in Ireland (1 in 60,000 births) but rare elsewhere (<1 in 200,000 births).

Etiology and Pathogenesis Homocysteine and methionine accumulate in cells and body fluids; cysteine synthesis is impaired, resulting in reduced concentrations of this amino acid and its disulfide form, cystine. In approximately half of patients, synthase activity in liver, brain, leukocytes, and cultured fibroblasts is undetectable. In the remaining patients, tissues retain 1 to 5% of normal activity, and this residual activity can often be stimulated by pyridoxine supplementation.

Homocysteine interferes with the normal cross-linking of collagen, an effect that likely plays an important role in the ocular, skeletal, and vascular complications. Altered collagen in the suspensory ligament of the optic lens and in bone matrix may account for the dislocated lenses and osteoporosis. Similarly, interference with normal ground substance metabolism in vascular walls may predispose to the arterial and venous thrombotic diathesis. Increased platelet adhesiveness may result from homocysteine accumulation, thereby contributing to the thrombotic occlusive disease so often observed. Recurrent cerebrovascular accidents secondary to thrombotic disease may account for the mental retardation, but direct chemical effects on cerebral cell metabolism have not been excluded.

Clinical Manifestations More than 80% of homozygotes for complete [CBS](#) deficiency develop dislocated optic lenses. This abnormality usually appears by 3 to 4 years of age and often results in glaucoma and impaired visual acuity. Mental retardation occurs in about half of such patients, often accompanied by ill-defined behavioral disturbances. Osteoporosis is a common radiologic finding (seen in two-thirds of patients by age 15) but rarely causes clinical disease. Life-threatening vascular complications, probably initiated by damage to vascular endothelium, are the major cause of morbidity and mortality. Occlusion of coronary, renal, and cerebral arteries with attendant tissue infarction can occur during the first decade of life. Nearly a fourth of patients die of vascular disease before age 30. These vascular complications seem to be exacerbated by angiographic procedures. Importantly, pyridoxine-responsive patients have milder clinical manifestations in all regards and may escape newborn screening and present with ectopia lentis or premature vascular occlusion. Heterozygous carriers for *CBS* deficiency (about 1 in 70 in the population) may have hyperhomocysteinemia, with an

increased risk for premature coronary, peripheral, and cerebral occlusive vascular disease.

Diagnosis The cyanide-nitroprusside test is a simple way of demonstrating increased excretion of sulfhydryl-containing compounds in urine. This is confirmed by measurement of free plasma methionine and homocystine. Plasma methionine tends to be increased in synthase-deficient patients and normal or low in those with other causes of homocystinuria and impaired methionine formation (see below). Diagnostic confirmation depends on measurements of [CBS](#) activity in tissue extracts or cells cultured from patients. Heterozygotes can be identified by measurement of peak serum homocystine after an oral methionine load (100 mg/kg) and by measurement of tissue synthase activity.

TREATMENT

As with classic phenylketonuria, effective treatment depends on early diagnosis. A number of infants diagnosed in the newborn period have been treated successfully with methionine-restricted, cystine-supplemented diets. In approximately half of patients, oral pyridoxine (25 to 500 mg/d) produces a fall in plasma and urinary methionine and homocystine and an increase in cystine concentration in body fluids. This effect probably reflects a modest increase in [CBS](#) activity in cells of patients in whom the defect is characterized either by reduced affinity for cofactor or by accelerated degradation of mutant enzyme. Vitamin supplementation at these doses is apparently harmless and should be tried in all patients. Folate deficiency should be prevented by adequate supplementation. Betaine has also been effective in reducing homocystine levels in pyridoxine-unresponsive patients.

5,10-METHYLENETETRAHYDROFOLATE REDUCTASE (MTFR) DEFICIENCY

Definition Hyperhomocysteinemia with normal or decreased methionine levels is caused by deficiency of MTFR, the enzyme involved in the synthesis of 5-methyltetrahydrofolate, a cofactor in the enzymatic formation of methionine from homocysteine ([Fig. 352-2](#)). [CNS](#) dysfunction and premature vascular occlusion may occur.

Genetic Basis and Pathogenesis 5-Methyltetrahydrofolate:homocysteine methyltransferase (methionine synthase) catalyzes the conversion of homocysteine to methionine. A primary defect in [MTFR](#) activity results, secondarily, in deficient methyltransferase activity and impaired conversion of homocysteine to methionine. This series of reactions is critical to normal DNA and RNA synthesis. Methionine deficiency and impaired nucleic acid synthesis may contribute to [CNS](#) dysfunction, while homocystine accumulation may predispose to thrombosis.

Hyperhomocysteinemia is inherited as an autosomal recessive trait and is caused by mutations in the [MTFR](#) (*MTHFR*) gene, which is located on chromosome 1p36.3. A thermolabile variant form of this enzyme, which has reduced activity, may be a common cause of hyperhomocysteinemia associated with increased risk of vascular disease in young adults.

Clinical Manifestations More than 30 patients with homocystinuria due to [MTFR](#) deficiency have been reported. The most severely affected have developmental retardation and cerebral atrophy early in life. Others have behavioral disturbances (catatonia) during the second decade or mild retardation. The severity of the clinical manifestations reflects the severity of the reductase deficiency.

Diagnosis Increased concentrations of free homocystine in body fluids with normal or decreased concentrations of methionine suggest severe [MTFR](#) deficiency. Total plasma homocysteine levels slightly above the normal range suggest milder dysfunction of this enzyme. Serum folate concentration is low in some patients. Confirmation requires direct MTFR assays in cultured fibroblasts.

TREATMENT

Therapeutic experience is limited. Folate, vitamin B₁₂, methionine, or betaine supplementation decrease homocystine urinary excretion and improve the clinical manifestations in some patients.

DEFICIENCY OF COBALAMIN (VITAMIN B₁₂) COENZYME SYNTHESIS

Definition Five other forms of homocystinuria also reflect impaired conversion of homocysteine to methionine. The primary defects in these entities, however, are in the synthesis of methylcobalamin, a cobalamin (vitamin B₁₂) coenzyme required by methionine synthase (MS) ([Fig. 352-2](#)). In some, methylmalonic acid accumulates in body fluids because of impaired synthesis of a second coenzyme, adenosylcobalamin, required for isomerization of methylmalonyl coenzyme A (CoA) to succinyl CoA. These disorders are designated cblC, -D, -E, -F, and -G.

Etiology and Pathogenesis As with [MTFR](#) deficiency, each disorder impairs remethylation of homocysteine. Since methylcobalamin is required for methyl-group transfer from methyltetrahydrofolate to homocysteine, impaired cobalamin metabolism leads to deficient methyltransferase activity. The defects responsible for impaired synthesis of methylcobalamin involve one of several steps in lysosomal or cytosolic activation of the vitamin precursor ([Fig. 352-2](#)). Somatic cell genetic studies indicate that each of the five abnormalities (cblC to -G) is distinct and imply that all are inherited as autosomal recessive traits.

Clinical Manifestations More than 45 patients -- mostly children -- with these defects in cobalamin metabolism have been described. Although clinical manifestations vary, abnormalities include developmental delay, dementia, spasticity, megaloblastic anemia, and pancytopenia. It is not possible to define a specific clinical syndrome for each of the defects in cobalamin metabolism.

Diagnosis Homocystinuria, homocysteinemia, and hypomethioninemia are the chemical hallmarks. Methylmalonic acidemia, too, has been noted in those defects resulting from defective synthesis of both cobalamin coenzymes. These findings may also be present in juvenile- or adult-onset pernicious anemia in which intestinal cobalamin absorption is impaired. Measurement of serum cobalamin concentrations, low in pernicious anemia and normal in patients with defective conversion of cobalamin vitamin to coenzymes,

helps in the differential diagnosis. Definitive diagnosis depends on demonstrating impaired coenzyme synthesis in cultured cells.

TREATMENT

Treatment of affected children with hydroxycobalamin injections (1 to 2 mg/QD) and betaine supplements decreases homocystine and methylmalonate excretion; the hematologic and neurologic deficits have also been diminished to variable degrees in some patients. Intervention early in life seems to offer the best long-term prognosis.

ALKAPTONURIA

Definition Alkaptonuria is a rare disorder of tyrosine catabolism in which deficiency of homogentisate 1,2-dioxygenase (also known as *homogentisic acid oxidase*) leads to excretion of large amounts of homogentisic acid in urine and accumulation of oxidized homogentisic acid pigment in connective tissues (*ochronosis*). After many years, ochronosis produces a distinctive form of degenerative arthritis.

Genetic Basis and Pathogenesis Alkaptonuria was the first human disease shown to be inherited as an autosomal recessive trait. Affected homozygotes occur with a frequency of about 1 in 200,000. Heterozygous carriers are clinically well and excrete no homogentisic acid in urine, even after loading doses of tyrosine. The gene for homogentisate 1,2-dioxygenase (*HGD*) maps to chromosome 3q21-q23 and encodes a 445 amino acid protein expressed not only in liver and kidney but also in small intestine, colon, and prostate. Expression in this latter organ is consistent with accumulation of black calculi of homogentisic acid in the prostate of patients with alkaptonuria, sometimes requiring prostatectomy.

Patients have minimally increased concentrations of homogentisic acid in blood because it is rapidly cleared by the kidney. However, homogentisic acid accumulates in cells and body fluids. Its oxidized polymers bind to collagen, leading to the progressive deposition of a gray to bluish-black pigment. The mechanism(s) by which this deposition causes degenerative changes in cartilage, intervertebral disk, and other connective tissues is unknown but may involve direct chemical irritation, impaired collagen cross-linking, disturbed articular chondrocyte metabolism, or some combination of factors.

Clinical Manifestations Alkaptonuria may go unrecognized until middle life when degenerative joint disease develops. Prior to this time, the tendency of the patient's urine to darken on standing may go unnoticed, as may slight pigmentation of the sclerae and ears. The latter manifestations are generally the earliest external evidence of the disorder and develop after age 20 to 30. Foci of gray-brown scleral pigment and generalized darkening of the concha, anthelix, and finally, helix of the ear are typical. Ear cartilages may be irregular and thickened. *Ochronotic arthritis* is heralded by pain, stiffness, and some limitation of motion of the hips, knees, and shoulders. Acute arthritis may resemble rheumatoid arthritis, but small joints are usually spared. Limitation of motion and ankylosis of the lumbosacral spine are common late manifestations. Pigmentation of heart valves, larynx, tympanic membranes, and skin occurs, and occasional patients develop pigmented renal or prostatic calculi. Degenerative

cardiovascular disease may be increased in older patients.

Diagnosis A patient whose urine darkens to blackness on standing must be suspected of having alkaptonuria, but this may not be observed with the use of modern plumbing conditions. The diagnosis is usually made from the triad of degenerative arthritis, ochronotic pigmentation, and urine that turns black upon alkalinization. Homogentisic acid in urine may be identified presumptively by other tests: after addition of ferric chloride, a purple-black color is observed; treatment with Benedict's reagent yields a brown color; addition of a saturated silver nitrate solution produces an intermediate black color. These screening tests can be confirmed by chromatographic, enzymatic, or spectrophotometric determinations of homogentisic acid. X-rays of the lumbar spine show degeneration and dense calcification of the intervertebral disks and narrowing of the intervertebral spaces (bamboo-like appearance).

TREATMENT

There is no specific treatment for ochronotic arthritis. Joint manifestations might be mitigated if homogentisic acid accumulation and deposition could be curbed by dietary restriction of phenylalanine and tyrosine, but the long-term nature of the disease discourages such therapeutic attempts. Ascorbic acid impedes oxidation and polymerization of homogentisic acid in vitro, but the efficacy of this form of treatment has not been established. Symptomatic treatment is similar to that for osteoarthritis ([Chap. 321](#)).

CYSTINOSIS

Definition Cystinosis is a rare disorder characterized by the intralysosomal accumulation of free cystine in body tissues. This results in the appearance of cystine crystals in the cornea, conjunctiva, bone marrow, lymph nodes, leukocytes, and internal organs. Three variants have been identified: an infantile (nephropathic) form leading to the Fanconi syndrome and renal insufficiency in the first decade, a juvenile (intermediate) form in which renal disease is manifest during the second decade, and an adult (benign) form characterized by deposition of cystine in the cornea but not in the kidney.

GENETIC CONSIDERATIONS

All types are inherited as autosomal recessive traits. The gene for the infantile form of nephropathic cystinosis encodes an integral membrane protein, which is a putative lysosomal cystine transporter. The gene is located on chromosome 17p13 and is designated *CTNS*. It is highly expressed in the pancreas, kidney, skeletal muscle, placenta, and heart. A common 65-kb deletion accounts for the majority of patients of European ancestry with infantile cystinosis. The juvenile- and adult-onset forms of nephropathic cystinosis are allelic with the infantile form. Obligate heterozygotes have intracellular cystine concentrations intermediate between those of normal persons and affected patients, but they are free of clinical abnormalities.

The basic defect involves impaired efflux of cystine from lysosomes rather than an abnormality in cystine catabolism. Lysosomal cystine efflux is an active, ATP-dependent

process. The cystine content of tissues may be more than 100 times normal in the infantile form and more than 30 times normal in the adult form. Intracellular cystine in lysosomes does not exchange with other intra- or extracellular pools of this amino acid. Neither plasma nor urinary concentrations of cystine are particularly elevated. Cystine accumulation in the kidney causes renal insufficiency in the infantile and juvenile forms. Patchy depigmentation and degeneration of the peripheral retina occur in the infantile and juvenile forms. Cystine crystals may also be deposited in the cornea, ocular conjunctiva, or uvea.

Clinical Manifestations In the infantile form, abnormalities are usually apparent by 6 to 10 months of age. Growth retardation, vomiting, fever, vitamin D-resistant rickets, polyuria, dehydration, and metabolic acidosis are prominent. Generalized proximal tubular dysfunction (the Fanconi syndrome) leads to hyperphosphaturia and hypophosphatemia, renal glycosuria, generalized amino aciduria, low plasma carnitine, hypouricemia, and often hypokalemia. Death due to uremia or intercurrent infection usually occurs before age 10. Ocular manifestations are prominent. Photophobia is usually demonstrable within the first years of life due to cystine deposits in the cornea, and retinal degeneration may appear even earlier. Hypothyroidism, insulin-dependent diabetes mellitus, and delayed puberty are often observed in older patients.

In contrast, patients with the adult form have only ocular abnormalities. Photophobia, headache, and burning or itching of the eyes are major complaints. Glomerular and tubular function and the integrity of the retina are preserved. The findings in the juvenile variant fall between these extremes. Ocular and renal manifestations do not become significant until the second decade. The renal lesion, albeit milder than that in the infantile form, eventually leads to renal insufficiency.

Diagnosis Cystinosis must be considered in any child with vitamin D-resistant rickets, the Fanconi syndrome, or glomerular insufficiency. Hexagonal or rectangular cystine crystals can be detected in the cornea (by slit-lamp examination), in leukocytes from peripheral blood or bone marrow, or in biopsies of rectal mucosa. Diagnosis is confirmed by measurement of cystine in leukocytes. The infantile form has been diagnosed prenatally by the demonstration of increased cystine content in cultured amniotic fluid cells.

TREATMENT

The adult form is benign and requires no treatment. Symptomatic treatment of renal disease in the infantile or juvenile form includes maintenance of adequate fluid intake to prevent dehydration; correction of the metabolic acidosis; administration of supplementary calcium, phosphate, and vitamin D to heal the rickets; and carnitine supplements (100 mg/kg per day) to correct the increased urinary losses. Specific therapy with the free thiol cysteamine slows the progression of renal dysfunction and improves growth. This compound acts in lysosomes by forming a mixed disulfide with cysteine, allowing it to be transported out of the organelle by an unrelated transporter not affected by the disease. Treatment is more effective if initiated before the patient is 2 years of age. Eye drops containing cysteamine can remove corneal crystals but requires frequent applications (10 to 14 times a day).

Children with nephropathic cystinosis and end-stage renal disease benefit from kidney transplantation. Patients who tolerate the procedure and do not develop immunologic problems have return of kidney function toward normal. The transplanted kidneys have not developed the functional abnormalities typical of cystinosis (i.e., the Fanconi syndrome or glomerular insufficiency). Patients may, however, continue to accumulate cystine in the cornea and other organs (thyroid, brain, and muscle).

PRIMARY HYPEROXALURIA

Definition Primary hyperoxaluria is the designation for two rare autosomal recessive disorders characterized by chronic excessive urinary excretion of oxalic acid and by calcium oxalate nephrolithiasis and nephrocalcinosis. Typically, patients with both forms develop renal insufficiency early in life and die of uremia. Calcium oxalate deposits are widespread in renal and extrarenal tissues, causing a condition referred to as *oxalosis*.

GENETIC CONSIDERATIONS

The metabolic basis for the primary hyperoxalurias involves pathways of glyoxylate metabolism. In type I hyperoxaluria, urinary excretion of oxalate and of the oxidized and reduced forms of glyoxylate is increased. The excessive synthesis of these substances results from a block in glyoxylate metabolism. The primary defect in most patients is deficiency of the hepatic peroxisomal enzyme alanine:glyoxylate amino transferase. The gene (*AGXT*) for this enzyme maps to 2q36-37, and several distinct mutations have been defined in patients with type I hyperoxaluria. Some of these mutations misdirect the enzyme to mitochondria and render it nonfunctional.

In type II hyperoxaluria, L-glyceric acid is excreted in excess along with oxalate. In this condition, activity of D-glyceric acid dehydrogenase, which catalyzes the reduction of hydroxypyruvate to D-glyceric acid in the catabolic pathway of serine metabolism, is absent in leukocytes (and presumably other tissues). The accumulated hydroxypyruvate is instead reduced by lactic dehydrogenase to the L-isomer of glycerate, which is excreted in the urine. The same enzyme possesses glyoxylate reductase activity, and its deficiency promotes the oxidation of glyoxylate to oxalate, thus causing the formation of increased oxalate.

Stone formation, nephrocalcinosis, and oxalosis are due to insolubility of calcium oxalate. Extrarenal deposits of oxalate are prominent in the heart, walls of arteries and veins, male urogenital tract, and bone, particularly in type I hyperoxaluria.

Clinical Manifestations Nephrolithiasis and oxalosis may be manifest during the first year of life. Most patients experience renal colic or hematuria between ages 2 and 10 and succumb to uremia before age 20. With the onset of uremia, patients may develop severe peripheral arterial spasm and necrosis with resulting vascular insufficiency. Oxalate excretion falls as renal failure worsens. In patients with delayed onset of symptoms, survival to age 50 or 60 has been reported, despite recurrent nephrolithiasis. Type II hyperoxaluria is a milder disease with less involvement of extrarenal organs and delayed impairment of kidney function.

Diagnosis Oxalate excretion in normal children or adults is <0.5 mmol (60 mg) per 1.73

m² surface area per day. Patients with type I or type II hyperoxaluria excrete two to four times this amount. Distinction between the two types depends on the identification of the other organic acids that identify them: glycolic acid in type I and L-glyceric acid in type II. Since patients with pyridoxine deficiency or chronic ileal disease may excrete excessive amounts of oxalate, these conditions must be excluded.

TREATMENT

There is no satisfactory treatment. Increasing the volume of urine can transiently reduce urinary oxalate concentration. Large doses of pyridoxine (100 mg/d) may reduce urinary oxalate in some patients, but long-term effects are not dramatic. A diet high in phosphate content seems to reduce the frequency of attacks of renal colic, but oxalate excretion is unaffected. Combined liver-kidney transplantation can correct the enzyme deficiency and replace the damaged organs. Liver transplantation seems promising in patients diagnosed before the onset of kidney failure.

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353. INHERITED DEFECTS OF MEMBRANE TRANSPORT - Nicola Longo

Specific membrane transporters mediate the passage of a wide variety of substances across plasma cell membranes. Classes of substrates represented include amino acids, sugars, cations, anions, vitamins, and water. The disorders considered in this chapter have three features in common: each is characterized by a specific defect in the transport of one or more compounds; each is inherited as a dominant or recessive trait, implying that variation in a single genetic locus is involved; and each is presumed to reflect a primary alteration in a specific membrane protein. Many of these defects have been well characterized physiologically and genetically. Inherited defects impairing the transport of amino acids, hexoses, and chloride are discussed here as examples of the abnormalities encountered; others are considered elsewhere in this text.

The number of inherited disorders of membrane transport continues to increase with the identification of new transporters and the clarification of the molecular basis of diseases with previously unknown pathophysiology. The first transport disorders identified affected the gut or the kidney, but transport processes are now proving essential for the normal function of every organ. Mutations in transporter molecules have been demonstrated in disorders of the heart, muscle, brain, and endocrine and sensory organs (see examples in [Table 353-1](#)). In some cases, the same phenotype can be caused by mutations in different genes (*nonallelic heterogeneity*), often because they encode interacting proteins. In other cases, distinct mutations in the same gene (*allelic heterogeneity*) can cause different diseases depending on the degree of functional inactivation caused by the mutation, the presence of dominant-negative effects, or paradoxical activation of function.

DISORDERS OF AMINO ACID TRANSPORT

As listed in [Table 353-1](#), 10 disorders of amino acid transport have been described. Five (cystinuria, dibasic aminoaciduria, Hartnup disease, iminoglycinuria, and dicarboxylic aminoaciduria) show transport abnormalities for structurally related amino acids, thereby implying the existence of group-specific membrane receptors or carriers. With the exception of iminoglycinuria and dicarboxylic aminoaciduria, the defects have important clinical consequences. The remaining five disorders affect the transport of only one amino acid, implying the existence of substrate-specific transport systems. Each of these conditions affects transport in the kidney, gut, or both; none has been shown to alter transport in other tissues.

CYSTINURIA

Definition Cystinuria, the most common inborn error of amino acid transport, is characterized by impaired renal tubular reabsorption and excessive urinary excretion of the dibasic amino acids lysine, arginine, ornithine, and cystine. A similar transport defect exists in the intestinal mucosa. Because cystine is the least soluble of the naturally occurring amino acids, its overexcretion predisposes to the formation of renal, ureteral, and bladder stones. Such stones are responsible for the signs and symptoms of the disorder.

GENETIC CONSIDERATIONS

Cystinuria is among the most common inborn errors, with a frequency of 1 in 10,000 to 1 in 15,000 in many ethnic groups. The disorder is transmitted as an autosomal recessive trait and results from impaired function of membrane carrier proteins in the apical brush border of proximal renal tubule and small intestinal cells.

There are three genetic variants of cystinuria. The urinary excretion patterns and renal clearance abnormalities in each type are similar in homozygotes, but the three variants are distinguished by studies of intestinal transport in homozygotes and urinary excretion patterns in heterozygotes. Type I homozygotes lack mediated intestinal transport of cystine, lysine, arginine, and ornithine; heterozygotes have normal urinary amino acid excretion patterns. Type II homozygotes lack mediated lysine transport in the gut but retain some capacity for cystine transport; heterozygotes have moderately increased urinary excretion of each of the four amino acids. Type III homozygotes retain some capacity for mediated intestinal transport of the four involved substrates; heterozygotes have modestly increased urinary lysine and cystine. The gene for type I cystinuria (*SLC3A1*) encodes solute carrier family 3 and maps to chromosome 2p16.3. The other two types of cystinuria (types II and III) are caused by mutations in *SLC7A9*, which maps to chromosome 19q13 and encodes the light subunit needed for the correct processing of *SLC3A1*.

Clinical Manifestations Massive excretion of cystine and the other dibasic amino acids occurs in homozygotes with classic cystinuria. Cystine stones account for 1 to 2% of all urinary tract calculi but are the most common cause of stones in children. The maximum solubility of cystine in the physiologic urinary pH range of 4.5 to 7.0 is about 1200 $\mu\text{mol/L}$ (300 mg/L). Since affected homozygotes regularly excrete 2400 to 7200 μmol (600 to 1800 mg) daily, crystalluria and stone formation are a constant threat. Stone formation usually becomes manifest in the second or third decade but may occur in the first year of life or as bladder calculi at birth. Symptoms and signs are those typical of urolithiasis: hematuria, flank pain, renal colic, obstructive uropathy, and infection ([Chap. 279](#)). Recurrent urolithiasis may lead to progressive renal insufficiency.

Diagnosis The presence of cystine in a urinary tract stone is pathognomonic of cystinuria. However, because half the stones in cystinuric individuals are of mixed composition, and because the cystine core in as many as 10% may not be detected, a urinary nitroprusside test should be performed on all patients with urolithiasis to exclude this diagnosis. The nitroprusside test is also positive (appearance of a cherry red color) in some heterozygotes for cystinuria and in patients with hypercystinuria, homocystinuria, and mercaptolactate-cysteine disulfiduria. When cystine content exceeds 1000 $\mu\text{mol/L}$ (250 mg/L), cystine crystals may be seen in the sediment of acidified, concentrated, chilled urine. These hexagonal crystals are pathognomonic of cystine overexcretion in patients not taking sulfonamides.

Diagnostic confirmation of cystinuria depends on demonstration of the characteristic amino acid excretion pattern in the urine. Selective overexcretion of cystine, lysine, arginine, and ornithine can be demonstrated by qualitative and quantitative chromatography. Quantitation is important for differentiating heterozygotes from homozygotes and for following free cystine excretion during therapy.

TREATMENT

Management is aimed at preventing cystine crystal formation by reducing the concentration of cystine in urine. This aim is accomplished by increasing urinary volume and by maintaining an alkaline urine pH. Fluid ingestion in excess of 4 L/d is essential, and 5 to 7 L/d is optimal. Urinary cystine concentration should be <1000 to 1200 $\mu\text{mol/L}$ (250 to 300 mg/L). The daily fluid ingestion necessary to maintain this dilution of excreted cystine should be spaced over 24 h, with one-third of the total volume ingested between bedtime and 2 to 3 A.M. Stones can be prevented and even dissolved by such hydration. It must be made clear to individuals with cystinuria that water is a necessary drug for them. Solubility of cystine in urine rises sharply above pH 7.5, and urinary alkalization can be therapeutic in some situations. Vigorous administration of sodium bicarbonate, acetazolamide, and polycitrates is required to maintain a persistently alkaline pH, but this measure introduces the danger of inducing formation of calcium oxalate, calcium phosphate, and magnesium ammonium phosphate stones and of producing nephrocalcinosis.

Another treatment involves administration of penicillamine, which undergoes sulfhydryl-disulfide exchange with cystine to form the mixed disulfide of penicillamine and cysteine. Since this disulfide is more than 50 times as soluble as cystine, penicillamine (1 to 3 g/d) reduces free cystine excretion markedly, thereby preventing new stone formation and promoting dissolution of existing calculi. Unfortunately, side effects include acute serum sickness, agranulocytosis, pancytopenia, immune glomerulitis, and the Goodpasture syndrome. Thus its use should be reserved for patients who fail to respond to hydration alone or who are in a high-risk category (one remaining kidney, renal insufficiency). Tiopronin (α -mercaptopropionylglycine, 800 to 1200 mg/d in four divided doses) has a mechanism of action similar to that of penicillamine, has lower toxicity, and is a suitable alternative. Captopril, a sulfhydryl-containing antihypertensive agent, has limited efficacy to reduce cystine excretion. When medical management fails, urologic surgery is required, but it should be a last resort as cystine stones reform more easily in scarred epithelium. Small (<1.5 cm) cystine stones can be treated with extracorporeal shock wave lithotripsy. Ureteroscopic removal is effective for ureteral stones, while larger or branched cystine stones require percutaneous nephrostolithotomy, sometimes associated with other procedures. All these procedures may produce smaller fragments, which can cause severe renal colic. Occasional patients progress to renal failure and require kidney transplantation.

DIBASIC AMINOACIDURIA

This disorder is characterized by a defect in renal tubular reabsorption of the three dibasic amino acids lysine, arginine, and ornithine but *not* cystine. There are two variants, transmitted as autosomal recessive traits. In the common form of dibasic aminoaciduria (type II), also known as *lysineric protein intolerance*, homozygotes show defective intestinal transport of dibasic amino acids as well as exaggerated renal losses. It is most common in Finland (1 in 60,000) and is rare elsewhere. The transport defect affects basolateral rather than luminal membrane transport and is associated with impairment of the urea cycle. The defective gene (*SLC7A7*) in this condition maps to chromosome 14q11.2 and encodes a unique membrane transporter, y^+LAT , that

associates with the cell-surface glycoprotein 4F2 heavy chain to form the complete sodium-independent transporter γ^+L . The requirement for multiple gene products in the formation of this dimeric transporter probably explains part of the intrafamilial variability observed in lysinuric protein intolerance.

Manifestations are related to the losses of ornithine, arginine, and lysine. Affected patients present in childhood with hepatosplenomegaly, protein intolerance, and episodic ammonia intoxication. Older patients may present with severe osteoporosis, impairment of kidney function, or interstitial changes in the lungs. Plasma concentrations of lysine, arginine, and ornithine are reduced, whereas urinary excretion of lysine and orotic acid are increased. Hyperammonemia may develop after the ingestion of protein loads or with infections, probably due to insufficient amounts of arginine and ornithine to maintain proper function of the urea cycle. The clinical features have been attributed to the hyperammonemia and to insufficient amounts of lysine to support protein synthesis during growth.

Type I dibasic aminoaciduria has been described in a large French-Canadian kindred. Type I patients have profound mental retardation without hyperammonemia and protein intolerance. The condition also differs from type II by the presence of a modest excess of dibasic amino acids in the urine of asymptomatic heterozygotes. Type I disease may involve the same transport system as that impaired in the more common type II dibasic aminoaciduria.

TREATMENT

Dietary protein should be restricted in conjunction with supplementation of citrulline (2 to 8 g/d), a neutral amino acid that fuels the urea cycle when metabolized to arginine and ornithine. Carnitine supplements may improve growth by sparing lysine and by enhancing fatty acid oxidation. Pulmonary disease responds to glucocorticoids in some patients.

HARTNUP DISEASE

Pellagra-like skin lesions, variable neurologic manifestations, and neutral or aromatic aminoaciduria characterize this disease. Alanine, serine, threonine, valine, leucine, isoleucine, phenylalanine, tyrosine, tryptophan, glutamine, asparagine, and histidine are excreted in urine in quantities 5 to 10 times normal, and intestinal transport of these same amino acids is defective. The clinical manifestations result from nutritional deficiency of the essential amino acid tryptophan, caused by its intestinal and renal malabsorption. Manifestations are episodic, related in part to metabolic demands for tryptophan. Only a small fraction of patients with the chemical findings of this disorder develop a pellagra-like syndrome, implying that manifestations depend on other factors in addition to the transport defect.

Hartnup disease is inherited as an autosomal recessive trait, and the gene has been mapped to chromosome 11q13. Homozygotes occur with a frequency of about 1 in 24,000 births. Heterozygotes exhibit no clinical or chemical abnormalities. In patients with Hartnup disease, the renal and intestinal transport defect for tryptophan leads to niacin deficiency. Tryptophan metabolism leads to the synthesis of niacin and

nicotinamide-adenine dinucleotide and supplies about half the daily niacin needs. The transport defect likely reflects abnormalities of a group-specific system for neutral amino acids. Some residual reabsorptive capacity persists for each involved amino acid. This suggests that they are transported by other carrier systems as well, a conclusion supported by the identification of patients with substrate-specific transport defects for tryptophan, methionine, and histidine.

The diagnosis of Hartnup disease should be suspected in any patient with clinical features of pellagra without a history of dietary niacin deficiency ([Chap. 75](#)). The neurologic and psychiatric manifestations range from attacks of cerebellar ataxia to mild emotional lability to frank delirium and are usually accompanied by exacerbations of the erythematous, eczematoid skin rash. Fever, sunlight, stress, and sulfonamide therapy provoke clinical relapses. Diagnosis is made by detection of the neutral aminoaciduria, which does not occur in dietary niacin deficiency. Treatment is directed at niacin repletion and includes a high-protein diet and daily nicotinamide supplementation (50 to 250 mg). Tryptophan ethyl esters can also bypass the absorption defect.

IMINOGLYCINURIA

This benign autosomal recessive trait is characterized by excessive urinary excretion of glycine and the imino acids proline and hydroxyproline. Homozygotes occur with a frequency of about 1 in 16,000. The enhanced excretion of glycine, proline, and hydroxyproline reflects a defect in the tubular transport system shared by these three compounds. An intestinal transport defect may also be present. This suggests that more than one mutation may lead to iminoglycinuria, a thesis corroborated by the demonstration that obligate heterozygotes from some, but not all, families have glycinuria. No consistent clinical abnormalities have been reported in homozygotes that are usually detected by urinary amino acid screening programs.

DICARBOXYLIC AMINOACIDURIA

Selective urinary loss and exaggerated endogenous renal clearance of glutamic and aspartic acids have been described in two unrelated children. Intestinal absorption of these dicarboxylic amino acids was impaired in one. This patient suffered from recurrent hypoglycemia; the other was asymptomatic.

SUBSTRATE-SPECIFIC DEFECTS IN AMINO ACID TRANSPORT

Rare pedigrees exist in which individuals have defective renal tubular reabsorption and/or impaired intestinal absorption of a single free amino acid ([Table 353-1](#)). These disorders, each apparently inherited as an autosomal recessive trait, suggest that transport of amino acids is catalyzed by substrate-specific as well as group-specific transport mechanisms. Examples include hypercystinuria, lysinuria, histidinuria, and selective malabsorption of methionine or tryptophan.

DISORDERS OF HEXOSE TRANSPORT

D-Glucose is the major carbohydrate used by the cell for energy production and many other anabolic purposes. A number of transporter proteins work together to maintain

glucose homeostasis in the intact organism by coordinating absorption and utilization of D-glucose by all cells in the body. Two main classes of glucose transporters have been identified in humans: active Na⁺-glucose cotransporters (SGLT) and the facilitative glucose transporters (GLUT). Na⁺-glucose cotransporters actively concentrate glucose inside intestinal and renal cells using the electrochemical potential of Na⁺ as their energy source. Defects in this class of transporters cause renal glycosuria (SGLT2) and intestinal glucose-galactose malabsorption (SGLT1). Facilitative glucose transporters allow glucose to enter cells using its own concentration gradient. This process is essential for the delivery of glucose to the cell for energy production. Defects in facilitative transporters cause the glucose-transporter protein syndrome (GLUT1) and the Fanconi-Bickel syndrome (GLUT2) and may be involved in one subtype of glycogen storage disorder (GLUT7).

DISORDERS OF CONCENTRATIVE GLUCOSE TRANSPORTERS

Renal glycosuria is characterized by the urinary excretion of glucose at normal concentrations of blood glucose. The renal tubular malabsorption is specific for glucose. Unlike generalized tubular dysfunction, other compounds such as phosphate and amino acids are transported normally. The genetic basis for this condition is not known at present. The condition is benign, but occasionally glycosuria may be severe enough to cause polyuria and polydipsia. Even more rarely, dehydration or ketosis may develop under conditions of stress such as pregnancy or starvation.

In normal persons, glucose is present in the glomerular filtrate at a concentration equal to that in plasma water and is reabsorbed throughout the proximal renal tubule by a sodium-dependent, phlorizin-inhibitable transport process. Reabsorptive capacity exceeds normal plasma glucose concentration. The plasma concentration at which filtered glucose begins to escape proximal tubular reabsorption is usually around 10 mmol/L (200 mg/dL). Maximal renal absorptive capacity is exceeded at a filtered load of around 2 mmol (325 mg)/min per 1.73 m² body surface area, and this value is defined as the tubular maximum for glucose (TmG).

Two patterns of glycosuria are recognized: type A, characterized by a reduced tubular maximum reabsorptive capacity, and type B, showing a reduced threshold for glycosuria, an increased "splay" in the titration curve, and a normal **TmG**. Renal glycosuria occurs in homozygotes with either of these recessively inherited mutations and in compound heterozygotes with these presumably allelic mutations. Modest reduction in renal threshold or TmG is present in heterozygotes in some families; modest glycosuria occurs in such family members when plasma glucose is elevated. In the few patients studied, renal glycosuria was not associated with impaired intestinal transport.

Glucose-galactose malabsorption is characterized by profuse, watery diarrhea in infants fed milk or foods containing lactose, sucrose, glucose, or galactose. The primary defect involves the sodium-hexose cotransporter in the intestinal and renal brush border. A specific defect in intestinal absorption of glucose and galactose can be demonstrated by oral tolerance tests that produce little or no increase in plasma glucose or galactose. Active D-glucose and D-galactose transport is absent in affected children, and intermediate transport capacity is present in their parents. Fructose-containing or

carbohydrate-free formulas are well tolerated. Treatment with a glucose- and galactose-free diet leads to resolution of symptoms in childhood. Although the basic transport defect is present throughout life, most patients show an improved tolerance for glucose and galactose with age.

A number of these patients have renal glycosuria at normal plasma glucose concentrations. Renal titration studies generally demonstrate a reduced threshold for glucose reabsorption (type B renal glycosuria) and a normal [TmG](#). Urinary glucose loss is not as severe as in isolated renal glycosuria. This finding suggests the presence of multiple glucose transport proteins in the kidney. One, whose gene remains to be identified, is responsible for the bulk of glucose reabsorption in the proximal convoluted tubule and is believed to be abnormal in renal glycosuria. Another transporter, SGLT1, is shared by glucose and galactose and is responsible for the reabsorption of the least traces of glucose in the late proximal straight tubule. Its function is abnormal in glucose-galactose malabsorption, and heterogeneous mutations have been found in the *SGLT1* gene on chromosome 22q. In glucose-galactose malabsorption, as in renal glycosuria, transport of sugars in other tissues is normal, reflecting the multiplicity and tissue specificity of hexose transporters.

DISORDERS OF FACILITATIVE GLUCOSE TRANSPORTERS

At least five different facilitative glucose transporters (GLUT1, -2, -3, -4, and -7) mediate the influx and efflux of glucose in mammalian cells. Disease-causing mutations have been identified in two of these transporters (GLUT1 and GLUT2). The tissue specificity and redundancy of facilitative glucose transporters in different tissues helps to explain the clinical manifestations that result from their defective function. De novo mutations in the gene encoding the ubiquitous GLUT1 transporter cause the glucose-transporter protein syndrome. Patients present with seizures, developmental delay, and acquired microcephaly. These patients have normal blood glucose concentration but markedly decreased concentration of glucose in their cerebrospinal fluid. GLUT1 is the predominant glucose transporter in the blood-brain barrier. Haploinsufficiency reduces the transfer of glucose through the blood-brain barrier, restricting the energy supply to the brain. This defect is expressed in other cells, such as erythrocytes and fibroblasts, that have less stringent energy requirements or express additional glucose transporters, preventing cellular damage and clinical sequelae. Therapy in the glucose-transporter protein syndrome consists of using a ketogenic diet to deliver alternative fuels to the brain.

The GLUT2 transporter is expressed mainly in liver, pancreaticb cells, and in the basolateral membrane of gut and renal tubular cells. Truncating mutations in the *GLUT2* transporter gene have been identified in the autosomal recessive disorder Fanconi-Bickel syndrome. Patients present early in life with failure to thrive and polydipsia, with prominent glycosuria and aminoaciduria, rickets, fasting hypoglycemia with ketonuria, and prolonged postprandial hyperglycemia. Glycogen is accumulated in the liver and kidney, reflecting the inability to release glucose through the GLUT2 transporter. This results in fasting hypoglycemia and ketonuria and in generalized renal tubular dysfunction. The prolonged postprandial hyperglycemia is due to decreased sugar uptake by the liver and by the pancreaticb cell, the latter resulting in defective insulin synthesis and release. Affected patients do not develop diabetes because

human pancreatic b cells have alternative glucose transporters (GLUT1 and GLUT3) that can partially compensate for the absence of GLUT2 transporters. Therapy consists of symptomatic replacement of the renal losses of water and electrolytes, vitamin D replacement, and a diet plan consisting of frequent meals rich in complex carbohydrates to prevent hypoglycemia, analogous to the treatment of patients with glycogen storage diseases ([Chap. 350](#)).

DEFECTIVE ANION TRANSPORT: CHLORIDORRHEA

This rare, autosomal recessive disease results from impairment of active transport of chloride in the ileum and colon. Absence of chloride-bicarbonate ion exchange causes profound symptoms even before birth (polyhydramnios and absence of meconium). Massive watery diarrhea is apparent from the first days of life. This fluid loss, with its attendant impairment of electrolyte homeostasis, is life-threatening. A hypokalemic, hypochloremic, hyponatremic metabolic alkalosis develops with dehydration and secondary hyperaldosteronism. Fecal fluid contains an excess of chloride ion over the sum of the accompanying cations sodium and potassium. Fecal chloride concentration always exceeds 90 mmol/L when volume and serum electrolyte disturbances are corrected, and this chloridorrhea is diagnostic. Renal chloride transport is normal. Decreased urine chloride results from the kidney's attempts to conserve salt and water. The defective gene in this condition, called *DRA* (for downregulated in adenoma), maps to chromosome 7q and encodes an anion transporter, which is expressed only in the gastrointestinal tract. A deletion of the Val 317 codon in the *DRA* gene is responsible for the Finnish form of congenital chloride diarrhea.

Treatment requires adequate, lifelong repletion of electrolyte and fluid losses. Exact replacement of water, sodium chloride, and potassium chloride can prevent the growth and psychomotor retardation and the development of progressive renal damage. The renal lesion, with hyalinized glomeruli, juxtaglomerular hyperplasia, calcifications, and arteriolar changes, is probably a result of chronic volume depletion. Treatment of hyperreninemia and hypokalemia with prostaglandin inhibitors may reduce renal damage but does not alter intestinal symptoms or the need for chronic sodium chloride repletion. Omeprazole, while not decreasing the need for adequate oral replacement of electrolytes, may decrease stool output and improve the social life of patients.

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354. THE LIPODYSTROPHIES AND OTHER PRIMARY DISORDERS OF ADIPOSE TISSUE - Abhimanyu Garg

The distribution and quantity of adipose tissue are controlled by multiple factors, including genetic background, diet, hormones, and exercise. The lipodystrophies are a heterogeneous group of adipose tissue disorders characterized by a selective loss of body fat ([Tables 354-1](#) and [354-2](#)). Patients with lipodystrophies have a propensity to develop insulin resistance, hypertriglyceridemia, diabetes mellitus, and fatty liver.

FAMILIAL OR GENETIC LIPODYSTROPHIES

CONGENITAL GENERALIZED LIPODYSTROPHY (BERARDINELLI-SEIP SYNDROME)

Clinical Features The primary diagnostic features of congenital generalized lipodystrophy (CGL) include a near-total lack of body fat and a marked muscular appearance from birth ([Fig. 354-1A](#)). On careful physical examination, however, fat can be detected in the palms and soles; with magnetic resonance imaging (MRI), normal amounts of fat can also be visualized in the orbits, scalp, perineum, and juxtaarticular and epidural regions (where the cushioning or protective functions of adipose tissue are critical). MRI studies also reveal a near-complete absence of metabolically active adipose tissue from most subcutaneous areas, intraabdominal and intrathoracic regions, and bone marrow.

Children exhibit accelerated linear growth and advanced bone age, but plasma levels of growth hormone or insulin-like growth factor I (IGF-I) are normal. Their basal metabolic rate is relatively high. It is unclear, however, whether hypermetabolism results from a primary increase in sympathetic nervous system activity or whether it is a compensatory response to protect against excessive heat loss due to extreme lack of body fat. Other features include acanthosis nigricans, prominent umbilicus or hernia, and an acromegalic appearance with coarse facial features and large hands and feet. Occasionally, excess body hair and hyperhidrosis have been noted. Fatty liver has been noted during infancy and can lead to cirrhosis and its complications. Liver, spleen, and kidney enlargement can cause abdominal protuberance. A few patients develop hypertrophic cardiomyopathy, but it rarely leads to heart failure.

Postpubertal women may have clitoromegaly, mild hirsutism, polycystic ovaries, and oligomenorrhea. Successful pregnancy in affected women is rare, whereas affected males have normal reproductive potential. Penile enlargement may be noted in childhood. After puberty, the skeleton appears sclerotic and focal lytic lesions develop in the appendicular bones. Some patients develop goiter.

Metabolic Abnormalities Patients with [CGL](#) have markedly elevated fasting serum insulin and C-peptide concentrations, as well as extreme insulin resistance. Diabetes mellitus appears during the pubertal years, and pancreatic pathology reveals severe amyloidosis of the pancreatic islets with loss of β cells. Plasma leptin concentrations are low, as expected in view of the reduced adipose tissue. Fasting plasma free fatty acid concentrations are normal. Hypertriglyceridemia may be observed during childhood, and patients can develop chylomicronemia, eruptive xanthomas, and acute pancreatitis. Low

concentrations of high-density lipoprotein (HDL) cholesterol are also common.

Course Patients with [CGL](#) are at risk of early mortality from cirrhosis and its complications, acute pancreatitis, or diabetic nephropathy. They also may develop diabetic retinopathy. Despite long-standing hypertriglyceridemia and hyperglycemia, atherosclerotic vascular complications are rare.

GENETIC CONSIDERATIONS

Approximately 120 patients of various ethnic backgrounds have been reported with this autosomal recessive form of lipodystrophy. The absence of fat could result from agenesis, failure of differentiation of preadipocytes, or an inability of mature adipocytes to synthesize and/or store triglycerides. Numerous candidate genes have been excluded, including the insulin receptor, β_3 -adrenergic receptor, fatty acid binding protein 2, [IGF-I](#) receptor, insulin receptor substrate-1, hormone-sensitive lipase, leptin, and peroxisome proliferator-activated receptor γ . Genome-wide linkage analysis of 17 families revealed genetic heterogeneity, but a candidate *CGL1* gene was localized to chromosome 9q34; the defective *CGL* gene(s) have yet to be identified.

FAMILIAL PARTIAL LIPODYSTROPHY

Dunnigan Variety

Clinical Features Patients with familial partial lipodystrophy, Dunnigan variety (FPLD) appear normal during childhood. During puberty, however, these patients begin to lose subcutaneous fat from the limbs and trunk, while exhibiting "increased muscularity" ([Fig. 354-1B](#)). Many patients accumulate excess fat in the face and neck, often resulting in a double chin, supraclavicular humps, and round face. Occasionally, fat accumulates in the axillae. Labia majora appear prominent in women. [MRI](#) demonstrates excess fat inside the abdomen and in the intermuscular fasciae. Bone marrow and fat in certain mechanical locations, such as the orbits and joints, are normal. Acanthosis nigricans, hirsutism, menstrual abnormalities, and polycystic ovaries are infrequent. Hepatomegaly due to fatty liver is common, but progression to cirrhosis has not been reported.

Metabolic Abnormalities Patients with [FPLD](#) have mild to moderate insulin resistance, and diabetes mellitus develops, usually after the second decade. Patients have low serum [HDL](#) cholesterol and can develop severe hypertriglyceridemia. Fasting plasma free fatty acid concentrations may be elevated.

Course Major causes of morbidity and mortality in patients with [FPLD](#) include coronary heart disease, other atherosclerotic vascular complications, and acute pancreatitis.

GENETIC CONSIDERATIONS

This rare autosomal dominant disorder has been reported in 35 Caucasian families and one Indian family comprising approximately 200 affected individuals. The *FPLD* locus has been mapped to chromosome 1q21-22. Recently, several missense mutations in the gene encoding the nuclear envelope protein lamin A/C (*LMNA*) have been found to be responsible for [FPLD](#). Alternative splicing of *LMNA* produces Lamin A and C,

members of the intermediate filament multigene family. All mutations causing typical FPLD cluster in exon 8 of *LMNA*, affecting the globular C-terminal tail of the Lamin A/C protein except one in exon 11 that only affects Lamin A and causes an atypical mild FPLD. The loss of subcutaneous adipose tissue in the limbs and trunk in FPLD may be due to adipocyte apoptosis and degeneration. Fat accumulation in the face and neck may be a secondary phenomenon, since it is not always present.

Kobberling Variety Characteristic features include loss of fat from the limbs with preservation of facial fat; truncal subcutaneous fat may be excessive. Most patients have hypertriglyceridemia and diabetes mellitus. Still unknown are the pattern of inheritance, age of onset, and whether this disorder is a distinct entity or a variant of the Dunnigan variety. Only a few women from two small pedigrees and four sporadic cases have been reported; it is not yet clear whether men can also be affected.

Mandibuloacral Dysplasia Variety This autosomal recessive disorder is characterized by short stature, high-pitched voice, mandibular and clavicular hypoplasia, dental abnormalities, acroosteolysis, stiff joints, and ectodermal defects. A few patients have also exhibited loss of limb fat. Insulin resistance and diabetes mellitus are rare.

OTHER TYPES

An autosomal dominant type of generalized lipodystrophy with acromegaloid features has been reported in a pedigree from Brazil. The onset of lipodystrophy occurred after 18 years of age. In another form of lipodystrophy, marked loss of subcutaneous fat from the limbs, face, palms, and soles, but excess subcutaneous fat in the neck and trunk has been noted.

ACQUIRED LIPODYSTROPHIES

ACQUIRED GENERALIZED LIPODYSTROPHY (LAWRENCE SYNDROME)

This form of lipodystrophy has been reported in approximately 50 patients and is characterized by a generalized disappearance of fat, mostly during childhood or adolescence. It is three times more common in females than males.

Clinical Features Fat loss affects the face, neck, trunk, and extremities and usually occurs over several months or years; superficial veins and muscles become prominent ([Fig. 354-1C](#)). Fat loss can include the palms and soles. In some, the onset of the disorder was reported after infections such as varicella, measles, pertussis, diphtheria, pneumonia, osteomyelitis, parotitis, infectious mononucleosis, or hepatitis. In others, lipodystrophy starts with painful, purple-brown subcutaneous nodules that leave depressed areas with loss of subcutaneous fat. Adipose tissue may show infiltration with lymphocytes, mononuclear macrophages, fat-cell necrosis, and fat-filled macrophages; this infiltration is consistent with a type of acute panniculitis. Almost one-third of these patients develop acanthosis nigricans, and some have mild hirsutism. Hepatomegaly due to fatty infiltration is a consistent finding and can lead to cirrhosis. Splenomegaly has also been reported.

Metabolic Abnormalities Ketosis-resistant diabetes mellitus usually occurs after the

onset of lipodystrophy, and metabolic abnormalities are similar to those in [CGL](#). Severely hyperglycemic patients may have elevated plasma free fatty acids.

Pathogenesis It is not yet known whether preceding infections play a causal role in this disorder. Some patients reportedly develop autoimmune diseases, including childhood dermatomyositis, juvenile rheumatoid arthritis, Hashimoto's thyroiditis, vitiligo, hemolytic anemia, or chronic active hepatitis. Autoantibodies against adipocyte membranes have been reported; it seems likely that antibody- and/or cell-mediated adipocyte lysis causes fat loss in these patients.

ACQUIRED PARTIAL LIPODYSTROPHY (BARRAQUER-SIMONS SYNDROME)

This form of lipodystrophy affects females three times more often than males and has been reported in about 200 patients. The onset usually occurs during childhood or adolescence. Fat loss typically affects the face, neck, upper limbs, thorax, and upper abdomen, and there is increased fat deposition in the hips and lower extremities ([Fig. 354-1D](#)).

Clinical Features Fat loss occurs gradually over 1 to 2 years and initially affects the face; other areas are affected later. In most cases, the lower abdomen, hips, and lower extremities are spared. In general, patients do not develop insulin resistance and other metabolic abnormalities, acanthosis nigricans, hirsutism, or menstrual problems. Approximately one-third of patients develop mesangiocapillary glomerulonephritis, usually 10 years after disease onset. Systemic lupus erythematosus and other autoimmune diseases have also been reported, including childhood dermatomyositis, thyroiditis, pernicious anemia, celiac disease, dermatitis herpetiformis, rheumatoid arthritis, Sjogren's syndrome, temporal arteritis, and leukocytoclastic vasculitis. In addition, many patients have serum antinuclear and anti-double-stranded DNA antibodies.

Pathogenesis C3 nephritic factor (C3NeF), a polyclonal IgG immunoglobulin, can be detected in the serum of up to 90% of these patients. Serum C3 is universally low, but C1q, C4, C5, C6, factor B, and properdin concentrations are usually normal (which suggests activation of the alternative complement pathway). Loss of fat may be due to C3NeF-induced lysis of adipocytes that express factor D. C3NeF also binds and inactivates factor H, which can induce glomerulonephritis by mechanisms similar to those seen in genetic factor H deficiency.

HIV-1 PROTEASE INHIBITOR-INDUCED LIPODYSTROPHY

Highly active antiretroviral therapy (HAART) therapy for HIV, a combination which includes HIV-1 protease inhibitors, is associated with the development of lipodystrophy in the majority of patients after 18 months to 2 years of treatment ([Chap. 309](#)). It is characterized by marked reduction in subcutaneous fat from the face, trunk, and limbs, resulting in an appearance of "increased muscularity." Excess fat may also accumulate around the neck (double chin and buffalo hump) and inside the abdomen. Patients are prone to develop insulin resistance, diabetes mellitus, and hypertriglyceridemia. It is unclear whether this disorder is caused by a side effect of one or more of the drugs (most likely a protease inhibitor) or by a metabolic response to dramatic reduction of

viral load. Hormonal causes, such as hypercortisolism, have been excluded.

LOCALIZED LIPODYSTROPHIES

These disorders are characterized by a loss of subcutaneous adipose tissue from small areas or parts of a limb. Fat loss may occur secondary to injections of insulin, glucocorticoids, antibiotics, iron dextrans, or diphtheria/pertussis/tetanus vaccine. Repeated pressure against any body part, such as the thigh or chin, can cause lipodystrophy. In some patients, acute panniculitis causes localized lipodystrophy without progressing further. *Centrifugal lipodystrophy* begins in the abdomen, groin, and axillae of children under the age of 3, and eventually spreads to involve the entire abdomen. The surrounding areas show slightly erythematous and scaly changes with an accumulation of lymphocytes and histiocytes on histology. Complete or partial improvement occurs spontaneously after 8 to 10 years.

TREATMENT

Patients with lipodystrophies have cosmetic problems that warrant judicious treatment. Facial reconstruction can be accomplished with free flaps, transposition of facial muscle, and silicone or other implants. In acquired partial lipodystrophy, adipose tissue transplantation from the thigh to face lasts for only 2 to 5 years. In [FPLD](#), excess fat in the face and neck may require liposuction or lipectomy. Etretinate and fish oil have improved acanthosis nigricans in some patients with generalized lipodystrophy.

Dietary fat should be restricted for patients with severe hypertriglyceridemia. Reduced energy intake and increased physical activity can mitigate insulin resistance. In children, however, enough energy should be provided to allow for normal growth and development. Medium chain triglycerides have been reported to benefit some patients with acquired generalized lipodystrophy.

In patients with [CGL](#), diabetes control may require extremely high doses of insulin. Oral hypoglycemic agents may also be used ([Chap. 333](#)). Glycemic control can mitigate dyslipidemia and prevent diabetic complications. Severe hypertriglyceridemia should be treated with fibrates and/or omega-3 polyunsaturated fatty acids. Niacin worsens glycemic control and should not be used. Estrogens should be avoided because they may accentuate hypertriglyceridemia and can cause acute pancreatitis.

LIPOMATOSIS

MULTIPLE SYMMETRIC LIPOMATOSIS (MADELUNG DISEASE)

This type of lipomatosis affects men 4 to 15 times more frequently than women. It is characterized by a symmetric, progressive growth of nonencapsulated subcutaneous adipose tissue, primarily in the neck (bull neck with buffalo hump and double chin) and supraclavicular and shoulder regions. Fat may also accumulate in the trunk and proximal limbs, though the distal arms and legs are spared. Rarely, laryngeal, tracheal, or vena caval compression may occur from deep lipomatous infiltration in the neck and mediastinum. Many patients also have peripheral neuropathy; hypertriglyceridemia and hyperuricemia are uncommon. Serum [HDL](#) cholesterol levels are usually elevated, and

diabetes mellitus has not been reported.

Most of these patients have a preceding history of heavy ethanol intake. The underlying mechanisms and predisposing factors for the disorder, however, remain unknown. The lipomatous tissue contains small adipocytes (but not brown fat) with increased lipoprotein lipase activity and reduced catecholamine-stimulated lipolysis. In several families and in some sporadic cases, mitochondrial DNA mutations A-to-G and G-to-A transitions at nucleotides 8344 and 8363, respectively, or deletions in the tRNA_{Lys} gene have been reported. Most of these patients had multiple, discrete, and encapsulated lipomas in the neck and trunk, which is distinct from the features of typical patients with multiple symmetric lipomatosis. These patients also have associated peripheral neuropathy, myopathy, cerebellar ataxia, myoclonus, or hearing loss. Mitochondrial DNA mutations have not been found in many patients with typical multiple symmetric lipomatosis.

Surgical resection may be required to relieve compression or for cosmetic reasons. Cessation of alcohol intake does not result in regression but may slow growth rate.

OTHER FORMS OF LIPOMATOSIS

Mediastinal lipomatosis is characterized by local overgrowth of adipose tissue in the mediastinum. It occurs in patients with Cushing's syndrome and can occasionally cause tracheal compression.

Pelvic lipomatosis is characterized by overgrowth of pelvic fat, causing bladder dysfunction (frequency, dysuria, and nocturia), constipation, and lower abdominal pain. Bilateral ureteral obstruction may also occur. The male:female ratio is 18:1. The etiology is not known, but the condition may result from a localized manifestation of obesity. Surgery may be needed to relieve urinary tract obstruction.

Epidural lipomatosis occurs in obese patients or in those receiving exogenous steroid therapy. Fat deposition most often occurs in the thoracic or lumbar spine, causing back pain, radicular pain, or spinal cord compression. Laminectomy may be indicated for cord compression. Weight loss or discontinuation of steroid therapy may also be helpful.

ADIPOSIS DOLOROSA (DERCUM DISEASE)

This is a rare disease of unknown etiology that mainly affects obese postmenopausal women (female:male ratio, 30:1). It is characterized by the presence of multiple circumscribed or diffuse painful subcutaneous fat deposits on the trunk and limbs, particularly near the knees. Patients also report weakness, fatigue, and emotional lability. Relief of pain is difficult; intravenous lidocaine, glucocorticoids, surgical excision, and liposuction are sometimes helpful.

ACUTE PANNICULITIS

A variety of systemic diseases including collagen vascular diseases such as systemic lupus erythematosus and scleroderma are associated with *acute panniculitis*, or *nodular fat necrosis* ([Chap. 311](#)). Panniculitis may also occur as a manifestation of

lymphoproliferative disorders ([Chap. 112](#)).

Disseminated fat necrosis is usually associated with acute pancreatitis or pancreatic carcinoma. It may be caused by the release of pancreatic enzymes into the circulation ([Chap. 304](#)).

HORMONAL EFFECTS ON ADIPOSE DISTRIBUTION

A variety of hormones influence the distribution of adipose tissue. Growth hormone, for example, reduces truncal fat but can increase fat in the palms and soles. Insulin enhances lipogenesis and fat storage. Thyroid hormones increase metabolic rate, including energy expenditure by fat tissue. Estrogens induce fat accumulation in the hips, legs, breasts and other subcutaneous regions. Glucocorticoids redistribute adipose tissue from peripheral to central locations. In Cushing's syndrome, characteristic features include buffalo hump, increased supraclavicular and truncal fat.

(Bibliography omitted in Palm version)

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